

Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume I: Measure-Theoretic Probability

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# Scope and proof policy

This volume assumes a first course in real analysis at the level of Knapp's *Basic Real Analysis* and uses selected measure-theoretic material from *Advanced Real Analysis*. The aim is to build the probability language used throughout the series with proofs explicit enough to be reused later.

# Chapter 1

## Probability Spaces, Laws, and Independence

This chapter converts the measure-theoretic language assumed from Knapp into the language used by stochastic processes. The central theme is that a stochastic model is specified first on a small family of finite-dimensional events and then extended without ambiguity to a full probability measure. We also develop conditional laws, because every later Markov, martingale, Gibbs, and SLE construction relies on them.

### 1.1 Countable product measures

**Definition 1.1.** A measurable space  $(S, \mathcal{S})$  is *standard Borel* if it is measurably isomorphic to a Borel subset of a Polish space. A family  $\mathcal{P}$  is a  $\pi$ -system if it is closed under finite intersections. A family  $\mathcal{D}$  is a *Dynkin system* if  $\Omega \in \mathcal{D}$ , complements of members belong to  $\mathcal{D}$ , and countable disjoint unions of members belong to  $\mathcal{D}$ .

**Definition 1.2.** Let  $(S_n, \mathcal{S}_n)$  be measurable spaces. A *cylinder set* in  $S = \prod_{n \geq 1} S_n$  is a set

$$C = \pi_{i_1}^{-1}(A_1) \cap \cdots \cap \pi_{i_k}^{-1}(A_k), \quad A_j \in \mathcal{S}_{i_j},$$

where only finitely many coordinates are constrained. The sigma-field generated by cylinders is the product sigma-field  $\bigotimes_{n \geq 1} \mathcal{S}_n$ .

**Theorem 1.3** (Countable product probability measure). *Let  $(S_n, \mathcal{S}_n, \mu_n)$  be probability spaces. There is a unique probability measure  $\mu$  on*

$$\left( \prod_{n \geq 1} S_n, \bigotimes_{n \geq 1} \mathcal{S}_n \right)$$

*such that for every cylinder depending on distinct coordinates  $i_1, \dots, i_k$ ,*

$$\mu \left( \bigcap_{j=1}^k \pi_{i_j}^{-1}(A_j) \right) = \prod_{j=1}^k \mu_{i_j}(A_j).$$

*Proof.*

$$C = \pi_{1:m}^{-1}(B), \quad B \in \bigotimes_{j=1}^m \mathcal{S}_j,$$



define

$$\mu_0(C) = (\mu_1 \otimes \cdots \otimes \mu_m)(B).$$

The finite products here are the ordinary finite product measures supplied by the product-measure theorem from measure theory. If the same cylinder is represented using more coordinates, then the extra factors are whole spaces and therefore have mass 1; thus  $\mu_0$  is well-defined.

$$\mu_0(C_r) \geq \varepsilon > 0 \quad \text{for every } r.$$

After relabelling coordinates and repeating constraints, write  $C_r$  as an event depending on the first  $k_r$  coordinates, with  $k_r \uparrow \infty$ . For  $m \leq k_r$  define the continuation probability

$$f_{m,r}(x_1, \dots, x_m) = \int \mathbf{1}_{C_r}(x_1, \dots, x_m, y_{m+1}, \dots, y_{k_r}) \prod_{j=m+1}^{k_r} \mu_j(dy_j).$$

Finite-dimensional Fubini shows that  $f_{m,r}$  is measurable. Since the cylinder events decrease in  $r$ , the continuation probabilities decrease as well (after integrating any additional unconstrained coordinates). Hence

$$h_m(x_1, \dots, x_m) = \lim_{r \rightarrow \infty} f_{m,r}(x_1, \dots, x_m)$$

exists in  $[0, 1]$ . Monotone convergence and consistency give

$$\int h_m d(\mu_1 \otimes \cdots \otimes \mu_m) = \lim_{r \rightarrow \infty} \mu_0(C_r) \geq \varepsilon,$$

so  $h_m$  is positive somewhere. Moreover Fubini gives the recursion

$$h_m(x_1, \dots, x_m) = \int h_{m+1}(x_1, \dots, x_m, x_{m+1}) \mu_{m+1}(dx_{m+1}).$$

Choose  $x_1$  with  $h_1(x_1) > 0$ . Inductively, if  $h_m(x_1, \dots, x_m) > 0$ , the recursion implies that some  $x_{m+1}$  satisfies  $h_{m+1}(x_1, \dots, x_{m+1}) > 0$ . This constructs a point  $x = (x_1, x_2, \dots)$ . If  $r$  is fixed and  $m \geq k_r$ , membership in  $C_r$  is already determined by the first  $m$  coordinates; because  $h_m(x_1, \dots, x_m) > 0$  and  $C_s \subset C_r$  for all  $s \geq r$ , that prefix must lie in  $C_r$ . Thus  $x \in C_r$  for every  $r$ , contradicting  $\cap_r C_r = \emptyset$ . Therefore  $\mu_0(C_r) \downarrow 0$  and  $\mu_0$  is a premeasure on the cylinder algebra.

$$\mathcal{D} = \{A : \mu(A) = \nu(A)\}$$

is a Dynkin system containing the cylinder  $\pi$ -system. The  $\pi$ - $\lambda$  theorem yields  $\mathcal{D} = \bigotimes_n \mathcal{S}_n$ , proving uniqueness.  $\square$

**Example 1.4.** Taking  $\mathcal{S}_n = \{0, 1\}$  with  $\mu_n(1) = p$  produces the canonical space for independent Bernoulli trials. A statement such as “the  $n$ th edge is open” is literally a coordinate cylinder event.

## 1.2 Generating classes and independence

**Lemma 1.5** ( $\pi$ - $\lambda$  theorem). *If  $\mathcal{P}$  is a  $\pi$ -system, then the smallest Dynkin system containing  $\mathcal{P}$  is  $\sigma(\mathcal{P})$ .*

*Proof.* Let  $\mathcal{D}$  be the smallest Dynkin system containing  $\mathcal{P}$ . For  $A \in \mathcal{P}$ , the class

$$\mathcal{D}_A = \{B : A \cap B \in \mathcal{D}\}$$

is a Dynkin system containing  $\mathcal{P}$ , hence contains  $\mathcal{D}$ . Thus  $A \cap B \in \mathcal{D}$  for  $A \in \mathcal{P}$ ,  $B \in \mathcal{D}$ . Now for fixed  $B \in \mathcal{D}$ , the same argument applied to  $\{A : A \cap B \in \mathcal{D}\}$  shows that  $\mathcal{D}$  is closed under finite intersections. A Dynkin system closed under finite intersections is a sigma-field, so  $\mathcal{D} = \sigma(\mathcal{P})$ .  $\square$

**Theorem 1.6** (Independence from generators). *Let  $\mathcal{P}_i$  be  $\pi$ -systems on  $\Omega$ . Suppose that for every finite set of distinct indices  $i_1, \dots, i_k$  and every  $A_j \in \mathcal{P}_{i_j}$ ,*

$$\mathbb{P} \left( \bigcap_{j=1}^k A_j \right) = \prod_{j=1}^k \mathbb{P}(A_j).$$

*Then the sigma-fields  $\sigma(\mathcal{P}_i)$  are independent.*

*Proof.* Fix distinct indices  $i_1, \dots, i_k$ . We enlarge the admissible class one coordinate at a time. Keep  $A_j \in \mathcal{P}_{i_j}$  fixed for  $j \geq 2$  and set

$$\mathcal{D}_1 = \left\{ A \in \sigma(\mathcal{P}_{i_1}) : \mathbb{P}(A \cap A_2 \cap \dots \cap A_k) = \mathbb{P}(A) \prod_{j=2}^k \mathbb{P}(A_j) \right\}.$$

The class  $\mathcal{D}_1$  is a Dynkin system: the identity is preserved by complements because the other events have fixed intersection probability, and by countable disjoint unions by countable additivity. By hypothesis it contains the  $\pi$ -system  $\mathcal{P}_{i_1}$ , so the  $\pi$ - $\lambda$  theorem gives  $\mathcal{D}_1 = \sigma(\mathcal{P}_{i_1})$ .

Now fix an arbitrary  $A_1 \in \sigma(\mathcal{P}_{i_1})$  and repeat the same argument in the second coordinate. The resulting Dynkin system contains  $\mathcal{P}_{i_2}$ , hence all of  $\sigma(\mathcal{P}_{i_2})$ . Iterating through  $i_3, \dots, i_k$  proves

$$\mathbb{P} \left( \bigcap_{j=1}^k A_j \right) = \prod_{j=1}^k \mathbb{P}(A_j) \quad (A_j \in \sigma(\mathcal{P}_{i_j})).$$

Since the finite set of indices was arbitrary, the generated sigma-fields are independent.  $\square$

### 1.3 Kolmogorov extension

**Theorem 1.7** (Kolmogorov extension theorem). *Let  $T$  be an arbitrary index set and let  $E$  be a standard Borel space. For every finite ordered set  $F = (t_1, \dots, t_m)$  of distinct points of  $T$ , suppose we are given a probability measure  $\mu_F$  on  $E^m$ . Assume the family is consistent under permutations and coordinate projections. Then there is a unique probability measure  $\mu$  on  $(E^T, \mathcal{E}^{\otimes T})$  whose finite-dimensional marginals are the prescribed  $\mu_F$ .*

*Proof.* On the algebra  $\mathcal{A}$  of finite-coordinate cylinders define

$$\mu_0(\pi_F^{-1}B) = \mu_F(B).$$

Permutation and projection consistency make this well-defined and finitely additive. We prove the only nonformal point: continuity from above at the empty set.

Choose a Polish topology on  $E$  whose Borel sigma-field is  $\mathcal{E}$ ; this is possible because  $E$  is standard Borel. Suppose that  $C_n \downarrow \emptyset$  are cylinders but, for some  $\varepsilon > 0$ ,  $\mu_0(C_n) \geq \varepsilon$  for every  $n$ . The union

of the finite coordinate sets on which the  $C_n$  depend is countable. Enumerate it as  $t_1, t_2, \dots$  and, after repeating unconstrained coordinates if necessary, write

$$C_n = \pi_{1:k_n}^{-1}(B_n), \quad k_n \uparrow \infty,$$

with

$$B_{n+1} \subseteq \pi_{1:k_n}^{-1}(B_n) = \{x \in E^{k_{n+1}} : \pi_{1:k_n} x \in B_n\},$$

In shorter notation,  $B_{n+1} \subseteq \pi^{-1} B_n$ .

Let  $(\delta_n)$  be positive with  $\sum_n \delta_n < \varepsilon/2$ . Since finite-dimensional Borel probabilities on Polish spaces are Radon, choose a compact  $K_1 \subset B_1$  with  $\mu_{1:k_1}(B_1 \setminus K_1) < \delta_1$ . Inductively, if  $K_n \subset B_n$  is compact and

$$\mu_{1:k_n}(B_n \setminus K_n) < \delta_1 + \dots + \delta_n,$$

then

$$D_{n+1} = B_{n+1} \cap \pi^{-1} K_n$$

loses at most that amount of  $\mu_{1:k_{n+1}}$ -mass from  $B_{n+1}$ . Inner regularity therefore gives a compact  $K_{n+1} \subset D_{n+1}$  with

$$\mu_{1:k_{n+1}}(B_{n+1} \setminus K_{n+1}) < \delta_1 + \dots + \delta_{n+1}.$$

In particular every  $K_n$  is nonempty, and  $\pi K_{n+1} \subset K_n$ .

The inverse system  $(K_n)$  has a compatible point. Indeed, in the compact product  $\prod_n K_n$ , the conditions  $x_n = \pi x_{n+1}$  are closed and every finite collection of them is satisfied by projecting a point of a sufficiently late  $K_N$ . Compactness therefore gives a sequence of compatible finite tuples. They determine a point of  $E^T$  lying in every  $C_n$ , contradicting  $\cap_n C_n = \emptyset$ . Hence  $\mu_0(C_n) \downarrow 0$  whenever  $C_n \downarrow \emptyset$  in the cylinder algebra.

Thus  $\mu_0$  is a premeasure. Carathéodory extension produces a probability measure on  $\sigma(\mathcal{A}) = \mathcal{E}^{\otimes T}$  with the prescribed marginals. If  $\mu$  and  $\nu$  are two such measures, the class  $\{A : \mu(A) = \nu(A)\}$  is a Dynkin system containing the cylinder  $\pi$ -system; the  $\pi$ - $\lambda$  theorem proves uniqueness.  $\square$

*Remark 1.8.* The standard-Borel hypothesis is doing real work: it provides enough regularity to turn finite-dimensional consistency into countable additivity on the cylinder algebra. Nearly every stochastic process in this series has a Polish, hence standard Borel, state space.

## 1.4 Regular conditional probabilities

**Definition 1.9.** Let  $X : \Omega \rightarrow S$  and  $Y : \Omega \rightarrow T$  be random variables. A *regular conditional law of  $X$  given  $Y$*  is a probability kernel  $K : T \times \mathcal{S} \rightarrow [0, 1]$  such that for every  $A \in \mathcal{S}$ ,

$$K(Y, A) = \mathbb{P}(X \in A \mid \sigma(Y)) \quad \text{a.s.}$$

Equivalently,

$$\mathbb{P}(X \in A, Y \in B) = \int_B K(y, A) \mathbb{P}_Y(dy)$$

for all measurable  $A, B$ .

**Theorem 1.10** (Regular conditional probabilities). *If  $S$  and  $T$  are standard Borel spaces and  $X : \Omega \rightarrow S$ ,  $Y : \Omega \rightarrow T$  are measurable, then there exists a regular conditional law  $K(y, \cdot)$  of  $X$  given  $Y$ . It is unique for  $\mathbb{P}_Y$ -almost every  $y$ .*

*Proof.*

$$\nu(A \times B) = \int_B K(y, A) \eta(dy).$$

$$F_q(Y) = \mathbb{P}(X \leq q \mid \sigma(Y)).$$

Thus  $F_q : T \rightarrow [0, 1]$  is measurable. Because the rational set is countable, we may modify all  $F_q$  on one  $\eta$ -null set so that simultaneously

$$q < r \implies F_q(y) \leq F_r(y), \quad \lim_{q \rightarrow -\infty} F_q(y) = 0, \quad \lim_{q \rightarrow +\infty} F_q(y) = 1,$$

and, for every rational  $q$ ,

$$F_q(y) = \inf_{r \in \mathbb{Q}, r > q} F_r(y).$$

The monotonicity follows from positivity of conditional expectation; the endpoint and right-continuity relations follow by conditional monotone convergence, first along countable rational sequences. Define for every real  $x$

$$F(y, x) = \inf_{q \in \mathbb{Q}, q > x} F_q(y).$$

For each fixed  $y$  outside the exceptional null set,  $x \mapsto F(y, x)$  is a right-continuous distribution function. Let  $K(y, \cdot)$  be its unique probability measure on  $\mathbb{R}$ ; on the exceptional set choose an arbitrary fixed probability. For intervals  $(-\infty, x]$ , measurability of  $y \mapsto K(y, (-\infty, x])$  follows from the displayed infimum over rationals. The monotone-class theorem then gives measurability for every Borel set. The defining conditional-expectation identity holds first for rational half-lines and then, by monotone classes in both variables, for all Borel sets.

For a general standard Borel space  $S$ , choose a Borel isomorphism  $\psi : S \rightarrow B$  onto a Borel subset  $B \subset \mathbb{R}$ . Apply the preceding construction to  $\psi(X)$ , whose conditional law is concentrated on  $B$  almost surely, and pull the kernel back by  $\psi^{-1}$ .

$$\int_B K(y, A) d\eta = \int_B K'(y, A) d\eta$$

for every  $B$ , so  $K(\cdot, A) = K'(\cdot, A)$   $\eta$ -a.e. Intersecting the countably many full-measure sets and using a monotone-class argument yields equality of the two probability measures for almost every  $y$ .

$$Kf(y) = \int_S f(x) K(y, dx).$$

The disintegration identity first for indicators and then for bounded measurable  $f$  shows

$$Kf(Y) = \mathbb{E}[f(X) \mid \sigma(Y)] \quad \text{a.s.}$$

This recovers ordinary conditional expectation from the conditional law. □

**Example 1.11** (Bayesian conditioning as disintegration). Suppose  $\Theta$  has prior density  $\pi(\theta)$  and, conditional on  $\Theta = \theta$ , an observation  $Y$  has density  $p(y \mid \theta)$ . Whenever the denominator is positive, the regular conditional law of  $\Theta$  given  $Y = y$  has density

$$\pi(\theta \mid y) = \frac{p(y \mid \theta)\pi(\theta)}{\int p(y \mid u)\pi(u) du}.$$

The formula is not an extra axiom: it is the Radon–Nikodym/disintegration theorem written in densities.

## Chapter 2

# Convergence, Tail Events, and Uniform Integrability

Limit operations are the daily grammar of probability. The point of this chapter is not merely to list modes of convergence, but to identify the mechanisms that let probability and integration interact safely.

### 2.1 Borel–Cantelli and tail events

**Theorem 2.1** (Borel–Cantelli lemmas). *For events  $A_n$ :*

- (i) *If  $\sum_n \mathbb{P}(A_n) < \infty$ , then  $\mathbb{P}(A_n \text{ i.o.}) = 0$ .*
- (ii) *If the  $A_n$  are independent and  $\sum_n \mathbb{P}(A_n) = \infty$ , then  $\mathbb{P}(A_n \text{ i.o.}) = 1$ .*

*Proof.*

$$\{A_n \text{ i.o.}\} = \bigcap_{m \geq 1} \bigcup_{n \geq m} A_n,$$

we have

$$\mathbb{P}(A_n \text{ i.o.}) \leq \mathbb{P}\left(\bigcup_{n \geq m} A_n\right) \leq \sum_{n \geq m} \mathbb{P}(A_n).$$

The tail tends to 0 when the series converges.

$$\mathbb{P}\left(\bigcap_{n=m}^N A_n^c\right) = \prod_{n=m}^N (1 - \mathbb{P}(A_n)) \leq \exp\left(-\sum_{n=m}^N \mathbb{P}(A_n)\right).$$

When the series diverges, the right side tends to zero as  $N \rightarrow \infty$ . Hence for every  $m$  the probability that none of  $A_m, A_{m+1}, \dots$  occurs is zero. Taking the countable union over  $m$  gives the conclusion.  $\square$

**Definition 2.2.** For independent random variables  $X_1, X_2, \dots$ , the tail sigma-field is

$$\mathcal{T} = \bigcap_{n \geq 1} \sigma(X_n, X_{n+1}, \dots).$$

**Theorem 2.3** (Kolmogorov zero–one law). *Every event  $A \in \mathcal{T}$  has probability 0 or 1.*

*Proof.* Fix  $n$ . Because  $A \in \mathcal{T} \subseteq \sigma(X_{n+1}, X_{n+2}, \dots)$  and the variables are independent,  $A$  is independent of  $\sigma(X_1, \dots, X_n)$ . Hence  $A$  is independent of the increasing union

$$\bigcup_{n \geq 1} \sigma(X_1, \dots, X_n),$$

and therefore, by the  $\pi$ - $\lambda$  theorem, of the sigma-field generated by that union, namely  $\sigma(X_1, X_2, \dots)$ . But  $A$  itself belongs to this sigma-field. Taking the second event to be  $A$  gives

$$\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)^2.$$

Thus  $\mathbb{P}(A) \in \{0, 1\}$ . □

**Example 2.4.** For iid variables, events such as “ $X_n > 0$  infinitely often”, “the empirical average converges”, or “the sequence is unbounded” are tail events. The zero-one law says that these events cannot have a genuinely intermediate probability.

## 2.2 Uniform integrability

**Definition 2.5.** A family  $\mathcal{X} \subset L^1$  is *uniformly integrable* if

$$\lim_{K \rightarrow \infty} \sup_{X \in \mathcal{X}} \mathbb{E}[|X| \mathbf{1}_{\{|X| > K\}}] = 0.$$

**Lemma 2.6** (Uniform absolute continuity). *If  $\mathcal{X}$  is uniformly integrable, then for every  $\varepsilon > 0$  there is  $\delta > 0$  such that*

$$\mathbb{P}(A) < \delta \implies \sup_{X \in \mathcal{X}} \mathbb{E}[|X| \mathbf{1}_A] < \varepsilon.$$

*Proof.* Choose  $K$  so that the tail contribution is  $< \varepsilon/2$  uniformly. Then

$$\mathbb{E}[|X| \mathbf{1}_A] \leq K\mathbb{P}(A) + \mathbb{E}[|X| \mathbf{1}_{\{|X| > K\}}].$$

Take  $\delta = \varepsilon/(2K)$ . □

**Theorem 2.7** (Vitali convergence criterion). *If  $X_n \rightarrow X$  in probability, then  $X_n \rightarrow X$  in  $L^1$  if and only if the family  $\{X_n : n \geq 1\}$  is uniformly integrable. Equivalently, convergence in probability plus uniform integrability is precisely the extra hypothesis needed to exchange the probabilistic limit with expectation in  $L^1$ .*

*Proof.*

$$\mathbb{E}|X_n - X| \leq \eta + \mathbb{E}[|X_n - X| \mathbf{1}_{\{|X_n - X| > \eta\}}].$$

Given  $M > \eta$ , split the last term into  $\{|X_n - X| \leq M\}$  and its complement. The first part is at most  $M\mathbb{P}(|X_n - X| > \eta)$ , which tends to zero; the second is uniformly small by UI. Let first  $n \rightarrow \infty$ , then the UI error and  $\eta$  tend to zero.

$$|X_n| \mathbf{1}_{\{|X_n| > K\}} \leq |X_n - X| + |X| \mathbf{1}_{\{|X| > K/2\}} + |X_n - X| \mathbf{1}_{\{|X_n - X| > K/2\}},$$

uniform integrability follows for all sufficiently large  $n$ ; finitely many initial variables do not matter. □

## 2.3 Modes of convergence

**Definition 2.8.** We write  $X_n \rightarrow X$  *in probability* if  $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$  for every  $\varepsilon > 0$ . We write  $X_n \Rightarrow X$  *in distribution* if  $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$  for every bounded continuous test function  $f$  on the state space. The latter is weak convergence of the laws.

**Theorem 2.9** (Relations among probabilistic modes of convergence). *Let  $1 \leq p < \infty$ .*

- (i)  $X_n \rightarrow X$  in  $L^p$  implies  $X_n \rightarrow X$  in probability.
- (ii)  $X_n \rightarrow X$  almost surely implies  $X_n \rightarrow X$  in probability.
- (iii)  $X_n \rightarrow X$  in probability implies  $X_n \Rightarrow X$  in distribution.
- (iv) If  $X_n \rightarrow X$  in probability, every subsequence contains a further subsequence converging to  $X$  almost surely.

None of the converses in (i)–(iii) holds without extra hypotheses.

*Proof.*

$$\mathbb{P}(|X_n - X| > \varepsilon) \leq \varepsilon^{-p} \mathbb{E}|X_n - X|^p \rightarrow 0.$$

$$\mathbb{P}(|X_{n_k} - X| > 2^{-k}) \leq 2^{-k}.$$

Borel–Cantelli implies  $|X_{n_k} - X| \leq 2^{-k}$  eventually, hence almost-sure convergence.

Counterexamples are useful. If  $A_n$  are independent with  $\mathbb{P}(A_n) = 1/n$ , then  $X_n = \mathbf{1}_{A_n} \rightarrow 0$  in probability but not almost surely by Borel–Cantelli II. If  $X_n = n\mathbf{1}_{\{U \leq 1/n\}}$  for uniform  $U$ , then  $X_n \rightarrow 0$  in probability but  $\mathbb{E}|X_n| = 1$ , so there is no  $L^1$  convergence. Finally, random variables with the same nondegenerate law need not converge in probability even though their laws converge trivially.  $\square$

## Chapter 3

# Sums of Independent Random Variables

The central problem of classical probability is to turn local information about independent increments into global information about their partial sums. The four theorems in this chapter form a compact toolkit: a maximal inequality, an almost-sure convergence criterion, the strong law, and the triangular-array central limit theorem.

### 3.1 Kolmogorov's maximal inequality

**Theorem 3.1** (Kolmogorov maximal inequality). *Let  $X_1, \dots, X_n$  be independent, mean-zero, square-integrable random variables and  $S_k = X_1 + \dots + X_k$ . Then for every  $\lambda > 0$ ,*

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \geq \lambda\right) \leq \frac{\text{Var}(S_n)}{\lambda^2} = \frac{\sum_{j=1}^n \text{Var}(X_j)}{\lambda^2}.$$

*Proof.*

$$A_k = \{|S_1| < \lambda, \dots, |S_{k-1}| < \lambda, |S_k| \geq \lambda\}.$$

The  $A_k$  are disjoint and their union is the maximal event.

$$\mathbb{E}[\mathbf{1}_{A_k} S_k (S_n - S_k)] = 0.$$

Therefore

$$\mathbb{E}[\mathbf{1}_{A_k} S_n^2] = \mathbb{E}[\mathbf{1}_{A_k} S_k^2] + \mathbb{E}[\mathbf{1}_{A_k} (S_n - S_k)^2] \geq \lambda^2 \mathbb{P}(A_k).$$

$$\mathbb{E}S_n^2 \geq \sum_{k=1}^n \mathbb{E}[\mathbf{1}_{A_k} S_n^2] \geq \lambda^2 \sum_{k=1}^n \mathbb{P}(A_k).$$

Since  $\mathbb{E}S_n^2 = \text{Var}(S_n)$ , division by  $\lambda^2$  gives the result.  $\square$

### 3.2 Three series

**Lemma 3.2** (Kolmogorov convergence criterion). *Let  $Y_n$  be independent, mean-zero variables with  $|Y_n| \leq C$ . Then  $\sum_n Y_n$  converges almost surely if  $\sum_n \text{Var}(Y_n) < \infty$ . Conversely, if  $\sum_n Y_n$  converges almost surely, then  $\sum_n \text{Var}(Y_n) < \infty$ .*



*Proof.* For sufficiency, choose  $N_m$  so that  $\sum_{n>N_m} \text{Var}(Y_n) \leq 2^{-3m}$ . Kolmogorov's maximal inequality gives

$$\mathbb{P}\left(\sup_{k>N_m} |S_k - S_{N_m}| > 2^{-m}\right) \leq 2^{2m} \sum_{n>N_m} \text{Var}(Y_n) \leq 2^{-m}.$$

Borel–Cantelli implies the partial sums are Cauchy almost surely.

For necessity, replace  $Y_n$  by an independent copy difference  $Y_n - Y'_n$ . Almost-sure convergence of both series implies convergence of the symmetric series. For a symmetric bounded  $Z$  with  $|Z| \leq 2C$ , the inequality

$$1 - \mathbb{E} \cos(tZ) \geq ct^2 \mathbb{E} Z^2$$

holds for sufficiently small fixed  $t$  (choose  $t$  so  $|tZ| \leq 1$  and use  $1 - \cos u \geq u^2/4$  there). Convergence in probability of the symmetric tail sums forces the product characteristic functions

$$\prod_{n=m}^N \mathbb{E} e^{it(Y_n - Y'_n)}$$

to tend to 1 uniformly as  $m, N \rightarrow \infty$ . Hence the sum of  $1 - |\mathbb{E} e^{itY_n}|^2$  is finite, and the cosine inequality implies  $\sum \text{Var}(Y_n - Y'_n) < \infty$ . Since  $\text{Var}(Y_n - Y'_n) = 2 \text{Var}(Y_n)$ , the result follows.  $\square$

**Theorem 3.3** (Kolmogorov three-series theorem). *Let  $X_n$  be independent real random variables and fix  $c > 0$ . Put*

$$Y_n = X_n \mathbf{1}_{\{|X_n| \leq c\}}.$$

*Then  $\sum_n X_n$  converges almost surely if and only if all three series satisfy*

$$\begin{aligned} \sum_n \mathbb{P}(|X_n| > c) &< \infty, \\ \sum_n \mathbb{E} Y_n &\text{ converges in } \mathbb{R}, \\ \sum_n \text{Var}(Y_n) &< \infty. \end{aligned}$$

*The criterion is independent of the choice of  $c > 0$ .*

*Proof.* Assume first that the three displayed conditions hold. By the first Borel–Cantelli lemma,  $X_n = Y_n$  for all sufficiently large  $n$  almost surely. The centered variables

$$Z_n = Y_n - \mathbb{E} Y_n$$

are independent, mean zero, and bounded by  $2c$ . Their variances are summable, so Lemma 3.2 gives almost-sure convergence of  $\sum_n Z_n$ . Adding the convergent deterministic series  $\sum_n \mathbb{E} Y_n$  shows that  $\sum_n Y_n$ , and hence  $\sum_n X_n$ , converges almost surely.

Conversely suppose  $\sum_n X_n$  converges almost surely. Then  $X_n \rightarrow 0$  almost surely, so the independent events  $\{|X_n| > c\}$  occur only finitely often. The converse half of Borel–Cantelli for independent events yields

$$\sum_n \mathbb{P}(|X_n| > c) < \infty.$$

Hence  $X_n = Y_n$  eventually almost surely, and  $\sum_n Y_n$  converges almost surely. Let  $(Y'_n)$  be an independent copy. Then  $\sum_n (Y_n - Y'_n)$  converges almost surely. These summands are independent, mean zero, and bounded by  $2c$ , so the necessity half of Lemma 3.2 gives

$$\sum_n \text{Var}(Y_n - Y'_n) = 2 \sum_n \text{Var}(Y_n) < \infty.$$

The sufficiency half of that lemma now shows that  $\sum_n (Y_n - \mathbb{E}Y_n)$  converges almost surely. Subtracting this convergent random series from the already convergent  $\sum_n Y_n$  proves that the ordinary real series  $\sum_n \mathbb{E}Y_n$  converges.

We have proved the equivalence for the chosen threshold  $c$ . If it holds for one threshold, then  $\sum_n X_n$  converges almost surely; applying the theorem just proved with any other threshold  $c' > 0$  shows that the corresponding three conditions hold there as well. Thus the criterion is independent of  $c$ .  $\square$

### 3.3 The strong law under the optimal first-moment hypothesis

**Lemma 3.4.** *If  $X$  is integrable and  $Y_n = X\mathbf{1}_{\{|X| \leq n\}}$ , then*

$$\sum_{n=1}^{\infty} \frac{\text{Var}(Y_n)}{n^2} < \infty.$$

*Proof.* It is enough to bound the second moments. By Tonelli,

$$\sum_{n=1}^{\infty} \frac{\mathbb{E}[X^2 \mathbf{1}_{\{|X| \leq n\}}]}{n^2} = \mathbb{E} \left[ X^2 \sum_{n \geq |X|} \frac{1}{n^2} \right].$$

For  $u \geq 1$ ,  $\sum_{n \geq u} n^{-2} \leq C/u$ . Thus the integrand is at most  $C|X|$  outside  $|X| < 1$ , and bounded inside. Integrability of  $X$  completes the proof.  $\square$

**Theorem 3.5** (Strong law of large numbers). *If  $X_1, X_2, \dots$  are iid and  $\mathbb{E}|X_1| < \infty$ , then*

$$\frac{1}{n} \sum_{k=1}^n X_k \longrightarrow \mathbb{E}X_1 \quad \text{almost surely.}$$

*Proof.*

$$\sum_{n \geq 1} \mathbb{P}(|X_n| > n) = \sum_{n \geq 1} \mathbb{P}(|X_1| > n) < \infty$$

(the last series is bounded by a constant multiple of  $\mathbb{E}|X_1|$ ), Borel–Cantelli gives  $X_n = Y_n$  eventually a.s. By Lemma 3.4,

$$\sum_n \frac{\text{Var}(Y_n)}{n^2} < \infty.$$

$$\sum_n \frac{Y_n - \mathbb{E}Y_n}{n}$$

converges almost surely by Kolmogorov’s convergence criterion. Kronecker’s lemma then gives

$$\frac{1}{n} \sum_{k=1}^n (Y_k - \mathbb{E}Y_k) \rightarrow 0 \quad \text{a.s.}$$

For completeness, Kronecker’s lemma follows from summation by parts: if  $\sum a_n/n$  converges, then

$$\frac{1}{N} \sum_{n \leq N} a_n = \frac{1}{N} \sum_{n \leq N} n \left( \frac{a_n}{n} \right) \rightarrow 0.$$

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E} Y_k \rightarrow \mathbb{E} X_1.$$

Finally  $X_k = Y_k$  for all sufficiently large  $k$  almost surely, so replacing  $Y_k$  by  $X_k$  does not affect the average limit.  $\square$

### 3.4 Triangular arrays and the central limit theorem

**Theorem 3.6** (Lindeberg–Feller CLT). *For each  $n$ , let  $X_{n1}, \dots, X_{nk_n}$  be independent, centered random variables and put*

$$s_n^2 = \sum_{k=1}^{k_n} \mathbb{E} X_{nk}^2 > 0.$$

*Assume the Lindeberg condition: for every  $\varepsilon > 0$ ,*

$$\frac{1}{s_n^2} \sum_{k=1}^{k_n} \mathbb{E} [X_{nk}^2 \mathbf{1}_{\{|X_{nk}| > \varepsilon s_n\}}] \rightarrow 0.$$

*Then*

$$\frac{\sum_k X_{nk}}{s_n} \Rightarrow N(0, 1).$$

*Proof.*

$$\sum_k \mathbb{E} [Y_{nk}^2 \mathbf{1}_{\{|Y_{nk}| > \varepsilon\}}] \rightarrow 0.$$

$$\max_k \mathbb{P}(|Y_{nk}| > \varepsilon) \leq \varepsilon^{-2} \sum_k \mathbb{E} [Y_{nk}^2 \mathbf{1}_{\{|Y_{nk}| > \varepsilon\}}] \rightarrow 0.$$

More importantly,

$$\max_k \mathbb{E} Y_{nk}^2 \leq \varepsilon^2 + \sum_j \mathbb{E} [Y_{nj}^2 \mathbf{1}_{\{|Y_{nj}| > \varepsilon\}}],$$

so  $\max_k \mathbb{E} Y_{nk}^2 \rightarrow 0$  after  $n \rightarrow \infty$  and then  $\varepsilon \downarrow 0$ . Thus no single summand carries macroscopic variance.

$$e^{iu} = 1 + iu - \frac{u^2}{2} + r(u), \quad |r(u)| \leq C \min(|u|^2, |u|^3)$$

with the usual split into  $|u| \leq \delta$  and  $|u| > \delta$ . Since  $\mathbb{E} Y_{nk} = 0$ ,

$$\phi_{nk}(t) = 1 - \frac{t^2}{2} \mathbb{E} Y_{nk}^2 + r_{nk},$$

and Lindeberg implies  $\sum_k |r_{nk}| \rightarrow 0$ . Also  $\max_k |\phi_{nk}(t) - 1| \rightarrow 0$ .

$$\mathbb{E} e^{it \sum_k Y_{nk}} = \prod_k \phi_{nk}(t).$$

Using  $\log(1+z) = z + O(z^2)$  uniformly for small  $z$ ,

$$\log \prod_k \phi_{nk}(t) = -\frac{t^2}{2} \sum_k \mathbb{E} Y_{nk}^2 + o(1) = -\frac{t^2}{2} + o(1).$$

Thus the characteristic functions converge pointwise to  $e^{-t^2/2}$ , and Lévy's continuity theorem yields convergence to  $N(0, 1)$ .  $\square$

**Example 3.7.** If  $X_{nk} = n^{-1/2}\xi_k$  for iid centered variance-one  $\xi_k$  with  $\mathbb{E}\xi_k^2 < \infty$  and  $k_n = n$ , the Lindeberg condition follows from dominated convergence. Hence the ordinary iid CLT is a special case.

## Chapter 4

# Weak Convergence and Coupled Representations

Weak convergence is the topology in which scaling limits live. Brownian motion, SLE, the GFF, and random planar maps will all be obtained as limits of discrete objects. This chapter establishes the compactness and representation tools needed to make such limits rigorous.

**Definition 4.1.** A metric space is *Polish* if it is separable and admits a complete compatible metric. All probability measures in this chapter are Borel probabilities. We write  $\mu_n \Rightarrow \mu$  for weak convergence against bounded continuous functions.

### 4.1 Portmanteau

**Theorem 4.2** (Portmanteau theorem). *Let  $\mu_n, \mu$  be probability measures on a metric space  $S$ . The following are equivalent:*

- (i)  $\int f d\mu_n \rightarrow \int f d\mu$  for every bounded continuous  $f$ ;
- (ii)  $\limsup_n \mu_n(F) \leq \mu(F)$  for every closed  $F$ ;
- (iii)  $\liminf_n \mu_n(G) \geq \mu(G)$  for every open  $G$ ;
- (iv)  $\mu_n(A) \rightarrow \mu(A)$  for every Borel  $A$  with  $\mu(\partial A) = 0$ .

*Proof.*

$$f_m(x) = \max(1 - md(x, F), 0).$$

Then  $f_m$  is bounded continuous,  $\mathbf{1}_F \leq f_m$ , and  $f_m \downarrow \mathbf{1}_F$ . Hence

$$\limsup_n \mu_n(F) \leq \int f_m d\mu$$

for every  $m$ , and monotone convergence gives (ii).

$$\mu(A^\circ) \leq \liminf_n \mu_n(A) \leq \limsup_n \mu_n(A) \leq \mu(\bar{A}),$$

and  $\mu(\bar{A} \setminus A^\circ) \leq \mu(\partial A) = 0$ , proving (iv).

$$\int f d\mu_n = \int_0^{\|f\|_\infty} \mu_n(f > t) dt$$

and dominated convergence in  $t$  gives convergence of integrals. Apply this to positive and negative parts to recover (i).  $\square$

## 4.2 Characteristic functions

**Theorem 4.3** (Lévy continuity theorem). *Let  $\mu_n$  be probability measures on  $\mathbb{R}^d$  with characteristic functions  $\phi_n$ . If  $\phi_n(t) \rightarrow \phi(t)$  pointwise and  $\phi$  is continuous at 0, then  $\phi$  is the characteristic function of a probability measure  $\mu$  and  $\mu_n \Rightarrow \mu$ . Conversely weak convergence implies pointwise convergence of characteristic functions.*

*Proof.* The converse is immediate because  $x \mapsto e^{it \cdot x}$  is bounded continuous.

$$\int_{-\delta}^{\delta} (1 - \Re \phi_n(t)) dt = \int \left( 2\delta - \frac{2 \sin(\delta x)}{x} \right) \mu_n(dx).$$

For  $|x| \geq 2/\delta$ , the parenthesis is bounded below by a positive multiple of  $\delta$ . Since  $\phi_n \rightarrow \phi$  and  $\phi$  is continuous at zero, the left side can be made uniformly small by taking  $\delta$  small, which makes the tails  $\mu_n(|x| > R)$  uniformly small for large  $R$ . Apply this coordinatewise in  $\mathbb{R}^d$ .  $\square$

## 4.3 Prokhorov compactness

**Definition 4.4.** A family  $\mathcal{M}$  of probability measures on a metric space  $S$  is *tight* if for every  $\varepsilon > 0$  there is compact  $K \subset S$  such that

$$\sup_{\mu \in \mathcal{M}} \mu(K^c) < \varepsilon.$$

**Theorem 4.5** (Prokhorov theorem). *If  $S$  is Polish, a family  $\mathcal{M} \subset \mathcal{P}(S)$  is relatively compact for weak convergence if and only if it is tight.*

*Proof.*

$$\inf_{\mu \in \mathcal{M}} \mu(K_m) \geq 1 - 2^{-m}.$$

Replacing  $K_m$  by  $K_1 \cup \dots \cup K_m$ , assume  $K_m \uparrow$ .

$$G_{m,N} = \bigcup_{j=1}^N B(x_{m,j}, 2^{-m}).$$

For each fixed  $m$ , the sets

$$U_{m,N} = \{\rho \in C : \rho(G_{m,N}) > 1 - \varepsilon 2^{-m}\}$$

are weakly open by the Portmanteau theorem, increase with  $N$ , and cover  $C$  because every probability gives mass tending to 1 to the increasing union  $G_{m,N} \uparrow S$ . Compactness of  $C$  and monotonicity therefore give one  $N_m$  such that  $C \subset U_{m,N_m}$ . Put  $G_m = G_{m,N_m}$  and

$$K = \bigcap_{m \geq 1} \overline{G_m}.$$

For every  $m$ ,  $K$  is covered by finitely many balls of radius  $2^{1-m}$ , so  $K$  is totally bounded. It is closed; since  $S$  is complete,  $K$  is compact. Finally, for every  $\rho \in C$ ,

$$\rho(K^c) \leq \sum_{m \geq 1} \rho(\overline{G_m}^c) \leq \sum_{m \geq 1} \rho(G_m^c) < \varepsilon.$$

Thus  $C$ , and hence  $\mathcal{M}$ , is uniformly tight.  $\square$

*Remark 4.6.* The proof is worth understanding structurally: tightness first confines nearly all mass to compact sets; total boundedness then converts weak convergence into a countable diagonal problem.

## 4.4 Skorokhod representation

**Theorem 4.7** (Skorokhod representation). *Let  $S$  be Polish and  $\mu_n \Rightarrow \mu$ . There exist random variables  $X_n, X$  on one probability space such that*

$$\text{Law}(X_n) = \mu_n, \quad \text{Law}(X) = \mu,$$

and  $X_n \rightarrow X$  almost surely.

*Proof.*

$$\mathcal{P}_m = \{A_{m,j} : j \geq 1\}, \quad m \geq 0,$$

with  $\mathcal{P}_{m+1}$  refining  $\mathcal{P}_m$ , every cell of  $\mathcal{P}_m$  having diameter at most  $2^{-m}$ , and  $\mu(\partial A_{m,j}) = 0$  for all  $m, j$ . Start from a countable cover by balls of radius  $2^{-m-2}$ ; choose their radii outside the countable exceptional set on which the relevant distance distribution has an atom, make the cover disjoint by successive differences, and refine simultaneously with the previous level.

$$\frac{\rho(B)}{\rho(A)}, \quad B \in \mathcal{P}_{m+1}, \quad B \subset A,$$

ignoring zero-mass parents. A uniform random variable  $U$  therefore selects a nested sequence of cells whose level- $m$  label has law  $\rho(A_{m,j})$ . Completeness and the shrinking diameters identify a unique point  $X_\rho(U)$  in the intersection of the closures of the selected cells; the resulting random point has law  $\rho$ .

Use this construction for  $\rho = \mu$  and for each  $\rho = \mu_n$ , with the same  $U$  and the same ordering of every partition. Since every cell is a  $\mu$ -continuity set, weak convergence implies  $\mu_n(A_{m,j}) \rightarrow \mu(A_{m,j})$ . Consequently, for any fixed finite depth  $m$ , every endpoint in the finite portion of the interval tree relevant to the selected path converges to its  $\mu$  endpoint. For every  $U$  outside the countable set of limiting endpoints, the level- $m$  cell selected for  $\mu_n$  agrees with that selected for  $\mu$  for all sufficiently large  $n$ .

$$d(X_{\mu_n}(U), X_\mu(U)) \leq 2^{-m}$$

for all sufficiently large  $n$ . Since  $m$  is arbitrary, the distance tends to zero.  $\square$

**Example 4.8.** Skorokhod representation is a proof device, not a statement that the original variables were already coupled this way. It allows a weak-convergence problem to be transferred temporarily to an almost-sure convergence problem where pathwise arguments are easier.

## Chapter 5

# Coupling and Concentration

Coupling changes the question “how different are two laws?” into “can we construct the two random objects together so that they usually agree?” Concentration inequalities answer a complementary question: how far can a random quantity wander from its typical value? These ideas reappear later in Markov-chain mixing, percolation decision trees, and random geometry.

### 5.1 Maximal coupling

**Theorem 5.1** (Maximal coupling). *Let  $\mu, \nu$  be probability measures on the same measurable space and define*

$$\|\mu - \nu\|_{\text{TV}} = \sup_A |\mu(A) - \nu(A)|.$$

*There exists a coupling  $(X, Y)$  with laws  $\mu, \nu$  such that*

$$\mathbb{P}(X \neq Y) = \|\mu - \nu\|_{\text{TV}}.$$

*For every coupling,  $\mathbb{P}(X \neq Y) \geq \|\mu - \nu\|_{\text{TV}}$ .*

*Proof.* Take  $\lambda = \mu + \nu$  and let  $f = d\mu/d\lambda$ ,  $g = d\nu/d\lambda$ . The common part

$$\alpha(dx) = \min\{f(x), g(x)\} \lambda(dx)$$

has total mass

$$a = \alpha(S) = 1 - \frac{1}{2} \int |f - g| d\lambda = 1 - \|\mu - \nu\|_{\text{TV}}.$$

The residual measures  $\mu - \alpha$  and  $\nu - \alpha$  are mutually singular. If  $a > 0$ , with probability  $a$  sample one point from  $\alpha/a$  and set  $X = Y$  equal to it. With probability  $1 - a$ , sample  $X$  and  $Y$  independently from the normalized residual measures. Their residual supports are disjoint, so  $X \neq Y$  there. (The cases  $a = 0, 1$  are interpreted in the obvious limiting way.) Hence

$$\mathbb{P}(X \neq Y) = 1 - a = \|\mu - \nu\|_{\text{TV}}.$$

For any coupling and measurable  $A$ ,

$$\mu(A) - \nu(A) = \mathbb{P}(X \in A, Y \notin A) - \mathbb{P}(Y \in A, X \notin A),$$

so  $|\mu(A) - \nu(A)| \leq \mathbb{P}(X \neq Y)$ . Taking the supremum over  $A$  proves optimality.  $\square$



## 5.2 Hoeffding and Azuma

**Lemma 5.2** (Hoeffding lemma). *If  $X$  is centered and  $a \leq X \leq b$  almost surely, then for all  $t \in \mathbb{R}$ ,*

$$\mathbb{E}e^{tX} \leq \exp\left(\frac{t^2(b-a)^2}{8}\right).$$

*Proof.* By convexity of  $e^{tx}$ , for  $x \in [a, b]$ ,

$$e^{tx} \leq \frac{b-x}{b-a}e^{ta} + \frac{x-a}{b-a}e^{tb}.$$

Taking expectations and using  $\mathbb{E}X = 0$  yields an explicit two-point upper bound. After shifting and scaling to  $[0, 1]$ , its logarithm has second derivative at most  $1/4$  and vanishes together with its first derivative at  $t = 0$ . Integrating twice gives the stated bound.  $\square$

**Theorem 5.3** (Hoeffding inequality). *Let  $X_1, \dots, X_n$  be independent with  $a_i \leq X_i \leq b_i$  almost surely. Then for every  $u > 0$ ,*

$$\mathbb{P}\left(\sum_{i=1}^n (X_i - \mathbb{E}X_i) \geq u\right) \leq \exp\left(-\frac{2u^2}{\sum_i (b_i - a_i)^2}\right),$$

*and the same bound holds for the lower tail.*

*Proof.*

$$\begin{aligned} \mathbb{E} \exp\left(t \sum_i (X_i - \mathbb{E}X_i)\right) &\leq \exp\left(\frac{t^2}{8} \sum_i (b_i - a_i)^2\right) \\ &\exp\left(-tu + \frac{t^2 V}{8}\right), \quad V = \sum_i (b_i - a_i)^2. \end{aligned}$$

The minimizer is  $t = 4u/V$ , giving  $e^{-2u^2/V}$ . Apply the same argument to  $-X_i$  for the lower tail.  $\square$

**Theorem 5.4** (Azuma–Hoeffding). *Let  $(M_k, \mathcal{F}_k)$  be a martingale and suppose*

$$a_k \leq M_k - M_{k-1} \leq b_k \quad a.s.$$

*for deterministic bounds. Then*

$$\mathbb{P}(M_n - M_0 \geq u) \leq \exp\left(-\frac{2u^2}{\sum_{k=1}^n (b_k - a_k)^2}\right).$$

*Proof.* Write  $D_k = M_k - M_{k-1}$ . Since  $\mathbb{E}[D_k \mid \mathcal{F}_{k-1}] = 0$  and  $a_k \leq D_k \leq b_k$ , Hoeffding's conditional lemma gives, for every  $t > 0$ ,

$$\mathbb{E}[e^{tD_k} \mid \mathcal{F}_{k-1}] \leq \exp\left(\frac{t^2(b_k - a_k)^2}{8}\right).$$

Iterating conditional expectations therefore yields

$$\mathbb{E}e^{t(M_n - M_0)} \leq \exp\left(\frac{t^2}{8} \sum_{k=1}^n (b_k - a_k)^2\right).$$

Put  $V = \sum_{k=1}^n (b_k - a_k)^2$ . Markov's inequality gives

$$\mathbb{P}(M_n - M_0 \geq u) \leq \exp\left(-tu + \frac{t^2 V}{8}\right).$$

The right side is minimized at  $t = 4u/V$ , giving

$$\mathbb{P}(M_n - M_0 \geq u) \leq e^{-2u^2/V}.$$

□

### 5.3 Efron–Stein

**Theorem 5.5** (Efron–Stein inequality). *Let  $X_1, \dots, X_n$  be independent and let  $Z = f(X_1, \dots, X_n) \in L^2$ . Let  $X'_i$  be an independent copy of  $X_i$ , independent of everything else, and define*

$$Z^{(i)} = f(X_1, \dots, X_{i-1}, X'_i, X_{i+1}, \dots, X_n).$$

*Then*

$$\text{Var}(Z) \leq \frac{1}{2} \sum_{i=1}^n \mathbb{E}[(Z - Z^{(i)})^2].$$

*Proof.*

$$M_i = \mathbb{E}[Z \mid \mathcal{F}_i], \quad D_i = M_i - M_{i-1}.$$

Then  $(M_i)$  is the Doob martingale of  $Z$  and  $Z - \mathbb{E}Z = \sum_i D_i$ .

$$\text{Var}(Z) = \sum_{i=1}^n \mathbb{E}D_i^2.$$

Fix  $i$  and condition on  $\mathcal{F}_{i-1}$ . Write

$$g_i(x) = \mathbb{E}[f(X_1, \dots, X_{i-1}, x, X_{i+1}, \dots, X_n) \mid \mathcal{F}_{i-1}],$$

where the expectation is over the future coordinates. Then  $D_i = g_i(X_i) - \mathbb{E}[g_i(X_i) \mid \mathcal{F}_{i-1}]$ .

$$\text{Var}(g_i(X_i) \mid \mathcal{F}_{i-1}) = \frac{1}{2} \mathbb{E}[(g_i(X_i) - g_i(X'_i))^2 \mid \mathcal{F}_{i-1}].$$

Couple  $Z$  and  $Z^{(i)}$  using the same future variables. Conditional Jensen gives

$$(g_i(X_i) - g_i(X'_i))^2 \leq \mathbb{E}[(Z - Z^{(i)})^2 \mid \mathcal{F}_{i-1}, X_i, X'_i].$$

Taking expectations yields

$$\mathbb{E}D_i^2 \leq \frac{1}{2} \mathbb{E}[(Z - Z^{(i)})^2].$$

$$\text{Var}(Z) \leq \frac{1}{2} \sum_i \mathbb{E}[(Z - Z^{(i)})^2].$$

□

**Example 5.6** (Bounded differences). If changing coordinate  $i$  can change  $f$  by at most  $c_i$ , then Efron–Stein gives

$$\text{Var} f(X_1, \dots, X_n) \leq \frac{1}{2} \sum_i c_i^2.$$

Azuma applied to the Doob martingale gives the corresponding sub-Gaussian tail. This “reveal one coordinate at a time” method will later become the decision-tree viewpoint in percolation.

## Chapter 6

# Markov Kernels, Stationarity, and Ergodic Averages

The previous chapters built probability measures from independent coordinates. Markov chains replace independence by a one-step conditional rule. This modest change creates the fundamental language of dynamics: transition kernels, invariant laws, return cycles, and convergence to equilibrium.

### 6.1 Building a Markov chain from kernels

**Definition 6.1.** A time-homogeneous *Markov chain* with transition kernel  $P$  satisfies

$$\mathbb{P}(X_{n+1} \in A \mid X_0, \dots, X_n) = P(X_n, A) \quad \text{a.s.}$$

A probability  $\pi$  is *stationary* if  $\pi P = \pi$ . On a countable state space, the chain is *irreducible* if every state can reach every other state with positive probability in some finite number of steps, and a state  $x$  is *positive recurrent* if its first positive return time  $\tau_x^+$  has finite mean. An irreducible chain is called positive recurrent if one (equivalently every) state is positive recurrent.

**Definition 6.2.** A probability kernel  $K$  from  $(S, \mathcal{S})$  to  $(T, \mathcal{T})$  assigns to each  $x \in S$  a probability  $K(x, \cdot)$  on  $T$  and makes  $x \mapsto K(x, B)$  measurable for every  $B \in \mathcal{T}$ .

**Theorem 6.3** (Ionescu–Tulcea construction). *Let  $(S_n, \mathcal{S}_n)$  be measurable spaces, let  $\mu_0$  be a probability on  $S_0$ , and for  $n \geq 1$  let*

$$K_n(x_0, \dots, x_{n-1}; dx_n)$$

*be a probability kernel from  $S_0 \times \dots \times S_{n-1}$  to  $S_n$ . Then there is a unique probability on the path space  $\prod_{n \geq 0} S_n$  whose finite-dimensional laws are given by iterated kernel integration.*

*Proof.* For a bounded measurable  $f$  on  $S_0 \times \dots \times S_n$  define

$$\int f d\mu^{(n)} = \int \mu_0(dx_0) \int K_1(x_0, dx_1) \cdots \int K_n(x_0, \dots, x_{n-1}; dx_n) f(x_0, \dots, x_n).$$

Successive measurability follows first for product indicators from the kernel property and then for all bounded measurable  $f$  by a monotone-class argument. Integrating the constant function 1 in the last coordinate shows that the projection of  $\mu^{(n+1)}$  onto the first  $n+1$  coordinates is  $\mu^{(n)}$ .

Define the cylinder set function  $\mu_*$  by these finite-dimensional laws. To prove countable additivity, let  $C_r \downarrow \emptyset$  be cylinders and suppose  $\mu_*(C_r) \geq \varepsilon > 0$ . After enlarging coordinate sets, write each  $C_r$  as an event depending on the first  $k_r$  coordinates with  $k_r \uparrow \infty$ . For  $m \leq k_r$  let

$$f_{m,r}(x_0, \dots, x_m) = \int \mathbf{1}_{C_r}(x_0, \dots, x_{k_r}) \prod_{j=m+1}^{k_r} K_j(x_0, \dots, x_{j-1}; dx_j)$$

be the conditional continuation probability. As  $r$  increases these functions decrease, so  $h_m = \lim_r f_{m,r}$  exists. Kernel Fubini and monotone convergence give the recursion

$$h_m(x_0, \dots, x_m) = \int h_{m+1}(x_0, \dots, x_{m+1}) K_{m+1}(x_0, \dots, x_m; dx_{m+1}),$$

and the  $\mu^{(m)}$ -integral of  $h_m$  is at least  $\varepsilon$ . Choose  $x_0$  with  $h_0(x_0) > 0$  and then recursively choose  $x_{m+1}$  so that  $h_{m+1}(x_0, \dots, x_{m+1}) > 0$ . For fixed  $r$  and  $m \geq k_r$ , positivity forces the prefix to satisfy the cylinder constraint  $C_r$ . The resulting path lies in every  $C_r$ , a contradiction. Thus  $\mu_*$  is a premeasure on the cylinder algebra.

Carathéodory extension gives the required path-space probability. Two extensions agree on all cylinders, a generating  $\pi$ -system, so the  $\pi$ - $\lambda$  theorem gives uniqueness.  $\square$

**Corollary 6.4** (Markov-chain construction). *Given an initial law  $\mu$  and a time-homogeneous kernel  $P(x, dy)$ , there is a chain  $(X_n)$  satisfying*

$$\mathbb{P}(X_{n+1} \in A \mid X_0, \dots, X_n) = P(X_n, A) \quad a.s.$$

*Proof.* Apply the preceding extension theorem with  $K_n(x_0, \dots, x_n; dy) = P(x_n, dy)$ .  $\square$

## 6.2 Reversibility and stationarity

**Theorem 6.5** (Detailed balance implies stationarity). *Let  $P$  be a Markov kernel and  $\pi$  a probability measure satisfying detailed balance*

$$\pi(dx)P(x, dy) = \pi(dy)P(y, dx)$$

*as measures on  $S \times S$ . Then  $\pi$  is stationary:*

$$\int \pi(dx)P(x, A) = \pi(A).$$

*Proof.* Apply detailed balance to the rectangle  $S \times A$  and use  $P(y, S) = 1$ :

$$\int_S \pi(dx)P(x, A) = \int_A \pi(dy)P(y, S) = \int_A \pi(dy) = \pi(A).$$

Thus  $\pi P = \pi$ .  $\square$

**Example 6.6.** For a finite weighted graph with symmetric edge weights  $c_{xy} = c_{yx}$ , the random walk

$$P(x, y) = \frac{c_{xy}}{c_x}, \quad c_x = \sum_y c_{xy},$$

has reversible law  $\pi(x) = c_x / \sum_z c_z$ .

### 6.3 Positive recurrence and ergodic averages

**Theorem 6.7** (Ergodic theorem for positive recurrent countable chains). *Let  $(X_n)$  be an irreducible positive recurrent Markov chain on a countable state space with stationary probability  $\pi$ . If  $f$  satisfies  $\sum_x \pi(x)|f(x)| < \infty$ , then for every initial state,*

$$\frac{1}{n} \sum_{k=0}^{n-1} f(X_k) \longrightarrow \sum_x \pi(x)f(x) \quad \text{almost surely.}$$

*Proof.*

$$\tau_0 = \inf\{n \geq 0 : X_n = o\}, \quad \tau_{m+1} = \inf\{n > \tau_m : X_n = o\}.$$

After the first hit of  $o$ , the cycle lengths

$$L_m = \tau_{m+1} - \tau_m$$

and cycle rewards

$$R_m = \sum_{k=\tau_m}^{\tau_{m+1}-1} f(X_k)$$

form an iid sequence of pairs. Here no continuous-time strong-Markov theorem is being imported: for a discrete stopping time  $\tau$ , decompose according to the events  $\{\tau = n\}$  and apply the ordinary Markov property at deterministic time  $n$ ; summing over  $n$  proves that, conditional on  $X_\tau$ , the post- $\tau$  chain has the original transition law and is independent of the pre- $\tau$  history.

$$\pi(x) = \frac{\mathbb{E}_o \left[ \sum_{k=0}^{\tau_o^+-1} \mathbf{1}_{\{X_k=x\}} \right]}{m_o}.$$

Indeed, if

$$q(x) = \mathbb{E}_o \sum_{k=0}^{\tau_o^+-1} \mathbf{1}_{\{X_k=x\}},$$

then normalization is  $\sum_x q(x) = m_o$ . Moreover

$$\sum_x q(x)P(x, y) = \mathbb{E}_o \sum_{k=0}^{\tau_o^+-1} P(X_k, y) = \mathbb{E}_o \sum_{k=0}^{\tau_o^+-1} \mathbf{1}_{\{X_{k+1}=y\}} = q(y),$$

because  $X_0 = X_{\tau_o^+} = o$ , so shifting the cycle by one step does not change its occupation count. Thus  $\pi = q/m_o$  is stationary; irreducibility gives uniqueness. Consequently

$$\mathbb{E}_o R_0 = m_o \sum_x \pi(x)f(x),$$

with absolute integrability obtained from  $\pi(|f|) < \infty$ .

$$\frac{\sum_{j < m} R_j}{\sum_{j < m} L_j} \rightarrow \frac{\mathbb{E} R_0}{\mathbb{E} L_0} = \pi(f).$$

The incomplete terminal cycle contributes  $o(n)$  to the time average almost surely. If  $N(n) = \max\{m : \tau_m \leq n\}$ , the strong law for  $L_m$  gives  $N(n)/n \rightarrow 1/m_o$ . Since  $\mathbb{E}|R_0| < \infty$ , the tail-sum criterion gives  $|R_m|/m \rightarrow 0$  almost surely; hence the reward of the single cycle containing time  $n$ , divided by  $n$ , vanishes. The same argument controls its partial reward. The finite initial segment before the first hit of  $o$  is negligible. Irreducibility ensures  $\tau_0 < \infty$  almost surely from every starting state.  $\square$

## 6.4 Coupling and convergence to equilibrium

**Theorem 6.8** (Coupling bound for total variation). *Let  $(X_n, Y_n)$  be any coupling of two copies of a Markov chain with initial laws  $\mu, \nu$ . Then*

$$\|\text{Law}(X_n) - \text{Law}(Y_n)\|_{\text{TV}} \leq \mathbb{P}(X_n \neq Y_n).$$

*If the coupling is coalescent—once the chains meet they move together—and*

$$T = \inf\{n : X_n = Y_n\},$$

*then*

$$\|\mu P^n - \nu P^n\|_{\text{TV}} \leq \mathbb{P}(T > n).$$

*In particular, if  $\nu = \pi$  is stationary, the coupling-time tail controls mixing to equilibrium.*

*Proof.*

$$\|\text{Law}(X_n) - \text{Law}(Y_n)\|_{\text{TV}} \leq \mathbb{P}(X_n \neq Y_n).$$

$$\|\mu P^n - \nu P^n\|_{\text{TV}} \leq \mathbb{P}(T > n).$$

When  $Y_0 \sim \pi$ , stationarity gives  $Y_n \sim \pi$  for every  $n$  and hence the usual mixing estimate.  $\square$

**Example 6.9** (A two-state chain). Let

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, \quad a, b \in (0, 1).$$

Its stationary law is  $\pi = (b, a)/(a + b)$ . A common-uniform coupling of two copies coalesces with a geometric tail, immediately giving exponential total-variation convergence. The example foreshadows the much harder coupling arguments used later for interacting particle systems and random geometry.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 6.10** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 6.11** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 6.12** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 6.13** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 6.14** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.



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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume II: Martingales, Brownian Motion, and Invariance Principles

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# Scope and proof structure

This volume develops all six planned chapters. The Koml'os–Major–Tusn'ady strong approximation theorem is proved in the dedicated appendix: the argument includes the conditional Cram'er estimate, monotone quantile coupling, dyadic recursion, exponential maximal estimates, removal of auxiliary density/lattice assumptions, and Brownian interpolation.

# Chapter 1

## Discrete Martingales

Martingales isolate the part of a stochastic process that cannot be predicted from the past. They are the natural language for stopping, concentration, stochastic integration, and the Loewner driving processes used much later in the series.

**Definition 1.1.** A *filtration* is an increasing family of sigma-fields  $(\mathcal{F}_n)$ . An adapted integrable process  $(M_n)$  is a *martingale* if  $\mathbb{E}[M_n | \mathcal{F}_{n-1}] = M_{n-1}$ ; it is a *submartingale* if the conditional expectation is at least  $M_{n-1}$ . A discrete process  $(A_n)$  is *predictable* if  $A_n$  is  $\mathcal{F}_{n-1}$ -measurable for  $n \geq 1$ . A random time  $\tau$  is a *stopping time* if  $\{\tau \leq n\} \in \mathcal{F}_n$  for every  $n$ .

### 1.1 Doob decomposition

**Theorem 1.2** (Doob decomposition). *Let  $(X_n, \mathcal{F}_n)_{n \geq 0}$  be an integrable adapted discrete-time process. There are unique adapted processes  $M_n, A_n$  such that*

$$X_n = M_n + A_n,$$

*$M$  is a martingale,  $A_0 = 0$ , and  $A_n$  is predictable, meaning  $A_n$  is  $\mathcal{F}_{n-1}$ -measurable for  $n \geq 1$ . Explicitly,*

$$A_n = \sum_{k=1}^n \mathbb{E}[X_k - X_{k-1} | \mathcal{F}_{k-1}].$$

*Proof.* Define

$$\Delta A_k = \mathbb{E}[\Delta X_k | \mathcal{F}_{k-1}], \quad A_0 = 0, \quad A_n = \sum_{k=1}^n \Delta A_k,$$

and put  $M_n = X_n - A_n$ . Integrability and predictability of  $A$  are immediate, and

$$\mathbb{E}[M_n - M_{n-1} | \mathcal{F}_{n-1}] = \mathbb{E}[\Delta X_n | \mathcal{F}_{n-1}] - \Delta A_n = 0,$$

so  $M$  is a martingale.

For uniqueness, suppose  $X = M + A = M' + A'$  are two such decompositions and set  $N = M - M' = A' - A$ . Then  $N$  is a martingale,  $N_0 = 0$ , and, because both  $A, A'$  are predictable,  $N_n$  is  $\mathcal{F}_{n-1}$ -measurable for  $n \geq 1$ . Therefore

$$N_n = \mathbb{E}[N_n | \mathcal{F}_{n-1}] = N_{n-1}.$$

Induction gives  $N_n = 0$  for every  $n$ , hence  $M = M'$  and  $A = A'$ . □

## 1.2 Optional stopping

**Theorem 1.3** (Optional stopping, precise forms). *Let  $(M_n, \mathcal{F}_n)$  be a martingale and let  $\sigma \leq \tau$  be stopping times. Then*

$$\mathbb{E}M_\tau = \mathbb{E}M_\sigma,$$

and

$$\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] = M_\sigma$$

under any one of the following three hypotheses:

- (i)  $\tau$  is bounded;
- (ii)  $\tau < \infty$  almost surely and the stopped family  $\{M_{n \wedge \tau} : n \geq 0\}$  is uniformly integrable;
- (iii)  $\mathbb{E}\tau < \infty$  and  $|M_n - M_{n-1}| \leq C$  almost surely for a deterministic  $C$ .

In particular, the displayed expectations are finite under each stated hypothesis.

*Proof.* First suppose  $\tau \leq N$ . The telescoping identity

$$M_\tau = M_0 + \sum_{k=1}^N (M_k - M_{k-1}) \mathbf{1}_{\{\tau \geq k\}}$$

has predictable indicators  $\mathbf{1}_{\{\tau \geq k\}} \in \mathcal{F}_{k-1}$ , so every summand has mean zero. Thus  $\mathbb{E}M_\tau = \mathbb{E}M_0$ . If  $A \in \mathcal{F}_\sigma$  and  $\sigma \leq \tau \leq N$ , the same argument applied after  $\sigma$  and multiplied by  $\mathbf{1}_A$  gives

$$\mathbb{E}[(M_\tau - M_\sigma) \mathbf{1}_A] = 0,$$

which is the conditional identity.

For (ii), apply the bounded result to  $\sigma \wedge n \leq \tau \wedge n$ . Since  $\tau < \infty$  almost surely,  $M_{\tau \wedge n} \rightarrow M_\tau$  almost surely. Uniform integrability upgrades this to  $L^1$  convergence. Moreover

$$M_{\sigma \wedge n} = \mathbb{E}[M_{\tau \wedge n} \mid \mathcal{F}_{\sigma \wedge n}],$$

so the family  $(M_{\sigma \wedge n})$  is uniformly integrable as conditional expectations of a uniformly integrable family; it converges almost surely to  $M_\sigma$  and therefore in  $L^1$ . Passing to the limit in the integral identity over  $A \in \mathcal{F}_\sigma$  proves  $\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] = M_\sigma$ .

For (iii),

$$|M_{\tau \wedge n}| \leq |M_0| + C(\tau \wedge n) \leq |M_0| + C\tau,$$

which is integrable. The same bound with  $\sigma$  in place of  $\tau$  and dominated convergence pass the bounded stopping identities to the limit. Taking expectations of the conditional identity gives the first displayed formula in every case.  $\square$

## 1.3 Maximal inequalities

**Theorem 1.4** (Doob maximal inequalities). *Let  $(X_k)_{k \leq n}$  be a nonnegative submartingale and  $X_n^* = \max_{0 \leq k \leq n} X_k$ . Then for  $\lambda > 0$ ,*

$$\lambda \mathbb{P}(X_n^* \geq \lambda) \leq \mathbb{E}[X_n \mathbf{1}_{\{X_n^* \geq \lambda\}}] \leq \mathbb{E}X_n.$$

If  $p > 1$ , then for every martingale or nonnegative submartingale,

$$\|X_n^*\|_p \leq \frac{p}{p-1} \|X_n\|_p.$$



*Proof.*

$$\lambda \mathbb{P}(A) \leq \mathbb{E}[X_\tau \mathbf{1}_A] \leq \mathbb{E}[X_n \mathbf{1}_A].$$

$$\begin{aligned} \mathbb{E}[(X_n^*)^p] &= p \int_0^\infty \lambda^{p-1} \mathbb{P}(X_n^* \geq \lambda) d\lambda \\ &\leq p \int_0^\infty \lambda^{p-2} \mathbb{E}[X_n \mathbf{1}_{\{X_n^* \geq \lambda\}}] d\lambda \\ &= \frac{p}{p-1} \mathbb{E}[X_n (X_n^*)^{p-1}]. \end{aligned}$$

Hölder gives

$$\|X_n^*\|_p^p \leq \frac{p}{p-1} \|X_n\|_p \|X_n^*\|_p^{p-1}.$$

□

## 1.4 Martingale convergence

**Lemma 1.5** (Upcrossing inequality, preliminary form). *Let  $X_0, \dots, X_n$  be a submartingale and  $U_n[a, b]$  the number of completed upcrossings of  $[a, b]$  by time  $n$ . Then*

$$(b - a) \mathbb{E}U_n[a, b] \leq \mathbb{E}[(X_n - a)^+].$$

*Proof.* Apply the buy-low/sell-high strategy to  $Y_k = (X_k - a)^+$  rather than directly to  $X$ . The process  $Y$  is a nonnegative submartingale by conditional Jensen. Let  $H_k$  indicate that an upcrossing is currently in progress. The pathwise bookkeeping gives

$$(b - a)U_n[a, b] \leq Y_n - \sum_{k=1}^n (1 - H_k)(Y_k - Y_{k-1}).$$

The last sum is a stochastic transform of the submartingale  $Y$  by the nonnegative predictable process  $1 - H_k$ , hence has nonnegative expectation. Taking expectations yields

$$(b - a) \mathbb{E}U_n[a, b] \leq \mathbb{E}Y_n = \mathbb{E}[(X_n - a)^+].$$

□

**Theorem 1.6** (Doob martingale convergence theorem). *If  $(X_n)$  is a submartingale with*

$$\sup_n \mathbb{E}X_n^+ < \infty,$$

*then  $X_n$  converges almost surely to a finite integrable random variable  $X_\infty$ . If, in addition,  $(X_n)$  is uniformly integrable, then  $X_n \rightarrow X_\infty$  in  $L^1$ .*

*Proof.* Because  $\mathbb{E}X_n \geq \mathbb{E}X_0$  for a submartingale,

$$\mathbb{E}X_n^- = \mathbb{E}X_n^+ - \mathbb{E}X_n \leq \sup_m \mathbb{E}X_m^+ - \mathbb{E}X_0.$$

Hence  $\sup_n \mathbb{E}|X_n| < \infty$ .

Fix rational  $a < b$ . The upcrossing inequality gives

$$(b - a)\mathbb{E}U_n[a, b] \leq \mathbb{E}[(X_n - a)^+] \leq \sup_m \mathbb{E}X_m^+ + |a|,$$

uniformly in  $n$ . By monotone convergence,  $\mathbb{E}U_\infty[a, b] < \infty$ , so  $U_\infty[a, b] < \infty$  almost surely. Taking the countable intersection over rational  $a < b$ , we obtain an event of probability one on which every rational interval is crossed only finitely often. If on that event  $\liminf_n X_n < \limsup_n X_n$ , a rational interval strictly between the two limits would be crossed infinitely often. Thus  $X_n$  has an extended-real limit  $X_\infty$  almost surely.

Fatou's lemma and the uniform  $L^1$  bound give

$$\mathbb{E}|X_\infty| \leq \liminf_n \mathbb{E}|X_n| < \infty,$$

so the limit is finite almost surely and integrable. If  $(X_n)$  is uniformly integrable, almost-sure convergence implies convergence in probability, and the Vitali criterion from Volume I upgrades it to  $X_n \rightarrow X_\infty$  in  $L^1$ .  $\square$

## 1.5 Closed martingales

**Theorem 1.7** (Uniformly integrable martingales are closed). *A martingale  $(M_n, \mathcal{F}_n)$  is uniformly integrable if and only if there exists  $M_\infty \in L^1$  such that*

$$M_n = \mathbb{E}[M_\infty | \mathcal{F}_n]$$

*for every  $n$ . In that case  $M_n \rightarrow M_\infty$  in  $L^1$  and almost surely when  $\mathcal{F}_n \uparrow \mathcal{F}_\infty$  and  $M_\infty$  is  $\mathcal{F}_\infty$ -measurable.*

*Proof.* Suppose first that  $(M_n)$  is uniformly integrable. The preceding martingale convergence theorem gives an integrable random variable  $M_*$  with  $M_n \rightarrow M_*$  almost surely and in  $L^1$ . For  $A \in \mathcal{F}_n$  and  $m \geq n$ ,

$$\mathbb{E}[M_m \mathbf{1}_A] = \mathbb{E}[M_n \mathbf{1}_A].$$

Passing to the  $L^1$  limit gives

$$\mathbb{E}[M_* \mathbf{1}_A] = \mathbb{E}[M_n \mathbf{1}_A],$$

so  $M_n = \mathbb{E}[M_* | \mathcal{F}_n]$ . Thus the martingale is closed by the integrable terminal variable  $M_*$ .

Conversely suppose  $M_n = \mathbb{E}[M_\infty | \mathcal{F}_n]$  for one  $M_\infty \in L^1$ . Conditional expectations of a fixed integrable random variable are uniformly integrable. Indeed, for  $A_{n,K} = \{|M_n| > K\}$ , Markov's inequality gives  $\sup_n \mathbb{P}(A_{n,K}) \leq \mathbb{E}|M_\infty|/K$ , while

$$\mathbb{E}[|M_n| \mathbf{1}_{A_{n,K}}] \leq \mathbb{E}[|M_\infty| \mathbf{1}_{A_{n,K}}].$$

Uniform absolute continuity of the integral of  $|M_\infty|$  makes the last quantity uniformly small when  $K$  is large. Hence  $(M_n)$  is uniformly integrable.

Finally assume  $\mathcal{F}_n \uparrow \mathcal{F}_\infty$  and  $M_\infty$  is  $\mathcal{F}_\infty$ -measurable. By the first half,  $M_n$  converges almost surely and in  $L^1$  to some  $M_*$ . For every  $A$  in the algebra  $\bigcup_n \mathcal{F}_n$ ,

$$\mathbb{E}[M_* \mathbf{1}_A] = \mathbb{E}[M_\infty \mathbf{1}_A].$$

A monotone-class argument extends this identity to  $A \in \mathcal{F}_\infty$ . Taking  $A = \{M_* > M_\infty\}$  and  $A = \{M_* < M_\infty\}$  after truncation shows  $M_* = M_\infty$  almost surely. Therefore  $M_n \rightarrow M_\infty$  both almost surely and in  $L^1$ .  $\square$

## Chapter 2

# Upcrossings, Conditional Limits, and Martingale Strong Laws

This chapter extracts three reusable convergence mechanisms. The upcrossing inequality converts oscillation into a countable quantity; Lévy's theorems identify limits of conditional expectations; and a weighted martingale argument turns square-summability into a strong law.

### 2.1 Upcrossings

**Theorem 2.1** (Doob upcrossing inequality). *Let  $(X_k, \mathcal{F}_k)_{0 \leq k \leq n}$  be a submartingale and  $a < b$ . Let  $U_n[a, b]$  be the number of completed upcrossings of  $[a, b]$  by time  $n$ . Then*

$$(b - a)\mathbb{E}U_n[a, b] \leq \mathbb{E}[(X_n - a)^+].$$

*Equivalently, applying the result to  $-X$  gives the corresponding downcrossing estimate. Either form yields the martingale-convergence consequences.*

*Proof.*

$$(b - a)U_n[a, b] \leq Y_n - \sum_{k=1}^n (1 - H_k)(Y_k - Y_{k-1}).$$

Indeed, each completed upcrossing contributes at least  $b - a$  to the part of the variation accumulated while  $H = 1$ ; the remaining increments account for the displayed correction.

$$\mathbb{E} \sum_{k=1}^n (1 - H_k)(Y_k - Y_{k-1}) \geq 0.$$

Taking expectations gives

$$(b - a)\mathbb{E}U_n[a, b] \leq \mathbb{E}Y_n = \mathbb{E}[(X_n - a)^+].$$

The importance is conceptual: infinitely many macroscopic oscillations would force infinitely many expected upcrossings, contradicting an  $L^1$  bound.  $\square$

### 2.2 Lévy's upward and downward theorems

**Theorem 2.2** (Lévy upward and downward convergence). *Let  $X \in L^1$ .*

(i) If  $\mathcal{F}_n \uparrow \mathcal{F}_\infty := \sigma(\cup_n \mathcal{F}_n)$ , then

$$\mathbb{E}[X \mid \mathcal{F}_n] \rightarrow \mathbb{E}[X \mid \mathcal{F}_\infty]$$

almost surely and in  $L^1$ .

(ii) If  $\mathcal{G}_n \downarrow \mathcal{G}_\infty := \cap_n \mathcal{G}_n$ , then

$$\mathbb{E}[X \mid \mathcal{G}_n] \rightarrow \mathbb{E}[X \mid \mathcal{G}_\infty]$$

almost surely and in  $L^1$ .

*Proof.*

$$\mathbb{E}[M_\infty \mathbf{1}_A] = \lim_n \mathbb{E}[M_n \mathbf{1}_A] = \mathbb{E}[X \mathbf{1}_A].$$

A monotone-class argument extends this to all  $A \in \mathcal{F}_\infty$ , so  $M_\infty = \mathbb{E}[X \mid \mathcal{F}_\infty]$ .

$$\mathbb{E}[M_n \mid \mathcal{G}_m] = M_m.$$

The reverse upcrossing argument, or equivalently the ordinary martingale convergence theorem after reversing every finite time interval and then passing to the limit, gives almost-sure convergence. Uniform integrability again yields  $L^1$  convergence.

$$\mathbb{E}[M_n \mathbf{1}_A] = \mathbb{E}[X \mathbf{1}_A].$$

Pass to the  $L^1$  limit to identify the limit as  $\mathbb{E}[X \mid \mathcal{G}_\infty]$ . □

## 2.3 A martingale-difference strong law

**Theorem 2.3** (Martingale difference strong law). *Let  $(D_n, \mathcal{F}_n)$  be square-integrable martingale differences:*

$$\mathbb{E}[D_n \mid \mathcal{F}_{n-1}] = 0.$$

*If*

$$\sum_{n=1}^{\infty} \frac{\mathbb{E}D_n^2}{n^2} < \infty,$$

*then*

$$\frac{1}{n} \sum_{k=1}^n D_k \rightarrow 0 \quad \text{almost surely.}$$

*More generally the same conclusion holds whenever  $\sum D_n/a_n$  converges almost surely for an increasing  $a_n \rightarrow \infty$  and Kronecker's lemma applies.*

*Proof.*

$$M_N = \sum_{n=1}^N \frac{D_n}{n}.$$

This is a square-integrable martingale and orthogonality of martingale differences gives

$$\mathbb{E}M_N^2 = \sum_{n=1}^N \frac{\mathbb{E}D_n^2}{n^2} \leq C < \infty.$$

Thus  $(M_N)$  is bounded in  $L^2$ , hence uniformly integrable, and the martingale convergence theorem gives almost-sure (indeed  $L^2$ ) convergence of  $M_N$ .

$$\frac{1}{N} \sum_{n=1}^N D_n \rightarrow 0.$$

To see it directly, if  $s_N = \sum_{n \leq N} D_n/n$  converges, Abel summation yields

$$\frac{1}{N} \sum_{n=1}^N D_n = s_N - \frac{1}{N} \sum_{n=1}^{N-1} s_n,$$

and Cesàro convergence sends the right side to zero.

Combining the martingale convergence of the normalized series with Kronecker's lemma proves the strong law.  $\square$

**Example 2.4.** If  $D_n = X_n - \mathbb{E}[X_n \mid \mathcal{F}_{n-1}]$  and  $\sup_n \mathbb{E} D_n^2 < \infty$ , then the hypothesis holds because  $\sum n^{-2} < \infty$ . Thus bounded conditional variances already force the average unpredictable fluctuation to vanish almost surely.

## Chapter 3

# Brownian Motion: Construction, Symmetry, and Strong Markov Structure

Brownian motion is not introduced here as a picture of a jagged curve. It is a probability law on a path space, characterized by independent Gaussian increments and continuity. The point of this chapter is to construct that law and then extract the symmetries that make Brownian motion usable later in stochastic calculus and conformal geometry.

**Definition 3.1.** A real process  $(B_t)_{t \geq 0}$  is a *standard Brownian motion* if  $B_0 = 0$  almost surely, its paths are continuous, and for  $0 \leq s < t$  the increment  $B_t - B_s$  is independent of  $\sigma(B_u : u \leq s)$  and has law  $N(0, t - s)$ .

### 3.1 Construction from Gaussian finite-dimensional laws

**Lemma 3.2** (A continuity criterion specialized to Brownian scaling). *Suppose a process  $X$  on  $[0, T]$  satisfies*

$$\mathbb{E}|X_t - X_s|^4 \leq C|t - s|^2.$$

*Then it has a modification whose paths are Hölder continuous of every order  $\alpha < 1/4$ .*

*Proof.* For the dyadic mesh  $D_m = \{k2^{-m}T : 0 \leq k \leq 2^m\}$  and a parameter  $\alpha < 1/4$ , Markov's inequality gives

$$\mathbb{P}(|X_{(k+1)2^{-m}T} - X_{k2^{-m}T}| > 2^{-m\alpha}) \leq C_T 2^{-m(2-4\alpha)}.$$

After summing over the  $2^m$  adjacent dyadic pairs, the resulting series in  $m$  is summable because  $1 - 4\alpha > 0$ . Borel–Cantelli therefore shows that, almost surely, all sufficiently fine dyadic increments are at most  $2^{-m\alpha}$ . Chaining through dyadic ancestors gives

$$|X_t - X_s| \leq C(\omega)|t - s|^\alpha$$

first on the dyadic points. The dyadic restriction is then uniformly Cauchy, so it extends uniquely to a continuous function. Equality with the original process at each fixed time follows by convergence in probability along dyadic approximants, giving a modification.  $\square$

**Theorem 3.3** (Construction of Brownian motion). *There exists a continuous process  $(B_t)_{t \geq 0}$  whose finite-dimensional distributions are centered Gaussian with covariance*

$$\mathbb{E}[B_s B_t] = \min\{s, t\}.$$

*This process is a standard Brownian motion.*

*Proof.* For times  $t_1, \dots, t_m \geq 0$  define a centered Gaussian vector with covariance  $K_{ij} = t_i \wedge t_j$ . This matrix is positive semidefinite since

$$\sum_{i,j} a_i a_j (t_i \wedge t_j) = \int_0^\infty \left( \sum_i a_i \mathbf{1}_{[0, t_i]}(u) \right)^2 du \geq 0.$$

The resulting Gaussian finite-dimensional laws are consistent under deletion and permutation of coordinates, so Kolmogorov extension gives a process with those marginals. For  $s < t$ ,  $B_t - B_s$  is centered Gaussian with variance  $t - s$ , hence

$$\mathbb{E}|B_t - B_s|^4 = 3|t - s|^2.$$

Kolmogorov continuity therefore gives a continuous modification. Finally, increments over disjoint intervals are jointly Gaussian with zero mutual covariance and are consequently independent. Their variances equal the interval lengths. Thus the continuous modification is standard Brownian motion.  $\square$

*Remark 3.4.* The exponent  $1/4$  in the elementary continuity proof is not the true regularity threshold. By using higher Gaussian moments one obtains Hölder continuity of every order  $\alpha < 1/2$ . Brownian paths are almost surely not Hölder- $1/2$ .

## 3.2 Scaling and time inversion

**Theorem 3.5** (Scaling and time inversion). *If  $B$  is Brownian motion and  $c > 0$ , then*

$$\tilde{B}_t = c^{-1} B_{c^2 t}$$

*is Brownian motion. Moreover, if*

$$\hat{B}_t = \begin{cases} t B_{1/t}, & t > 0, \\ 0, & t = 0, \end{cases}$$

*then  $\hat{B}$  is again Brownian motion.*

*Proof.*

$$\mathbb{E}[\tilde{B}_s \tilde{B}_t] = c^{-2} (c^2 s \wedge c^2 t) = s \wedge t.$$

Thus its finite-dimensional laws equal those of  $B$ ; continuity is inherited.

$$\mathbb{E}[\hat{B}_s \hat{B}_t] = st \min\{1/s, 1/t\} = s \wedge t.$$

Hence on  $(0, \infty)$  the process has Brownian finite-dimensional laws. It remains to check continuity at zero. For integers  $n$ ,

$$\mathbb{P} \left( \sup_{n \leq u \leq n+1} |B_u| > \varepsilon n \right) \leq \mathbb{P}(|B_n| > \varepsilon n/2) + \mathbb{P} \left( \sup_{0 \leq v \leq 1} |B_{n+v} - B_n| > \varepsilon n/2 \right).$$

Gaussian tails and the reflection principle proved below make the right side summable in  $n$ . Borel–Cantelli gives  $B_u/u \rightarrow 0$  almost surely, which is exactly  $t B_{1/t} \rightarrow 0$  as  $t \downarrow 0$ .  $\square$

### 3.3 The reflection principle

**Theorem 3.6** (Reflection principle). *Let  $M_t = \sup_{0 \leq s \leq t} B_s$ . For  $a > 0$  and  $b < a$ ,*

$$\mathbb{P}(M_t \geq a, B_t \leq b) = \mathbb{P}(B_t \geq 2a - b).$$

*In particular,*

$$\mathbb{P}(M_t \geq a) = 2\mathbb{P}(B_t \geq a).$$

*Proof.*

$$(RB)_s = \begin{cases} B_s, & s \leq \tau_a, \\ 2a - B_s, & s > \tau_a. \end{cases}$$

For a stopping time taking finitely many deterministic values, independence and symmetry of Gaussian increments show directly that reflection of all increments after the stopping time preserves the joint law. Approximate  $\tau_a$  from above by dyadic stopping times and use path continuity; the reflected path at  $\tau_a$  therefore has the same Wiener law as  $B$ .

$$\{M_t \geq a, B_t \leq b\}$$

onto  $\{B_t \geq 2a - b\}$ . This proves the first identity.

$$\mathbb{P}(M_t \geq a) = \mathbb{P}(B_t \geq a) + \mathbb{P}(M_t \geq a, B_t < a) = 2\mathbb{P}(B_t \geq a),$$

since the event  $B_t = a$  has probability zero. □

**Example 3.7.** The first-passage distribution follows at once:

$$\mathbb{P}(\tau_a \leq t) = 2 \left[ 1 - \Phi \left( \frac{a}{\sqrt{t}} \right) \right].$$

Differentiating yields the density

$$\frac{a}{\sqrt{2\pi t^3}} e^{-a^2/(2t)}, \quad t > 0.$$

Its mean is infinite, a first warning that optional stopping needs hypotheses.

### 3.4 The strong Markov property

We use the usual augmented Brownian filtration  $(\mathcal{F}_t)$ : start with  $\sigma(B_s : s \leq t)$ , complete it, and make it right-continuous.

**Theorem 3.8** (Strong Markov property). *If  $\tau$  is an almost surely finite stopping time, then*

$$W_t := B_{\tau+t} - B_\tau, \quad t \geq 0,$$

*is a Brownian motion independent of  $\mathcal{F}_\tau$ .*



*Proof.*

$$\begin{aligned}\mathbb{E}[\mathbf{1}_A F(W)] &= \sum_j \mathbb{E}[\mathbf{1}_{A \cap \{\tau = t_j\}} F((B_{t_j+s} - B_{t_j})_{s \geq 0})] \\ &= \sum_j \mathbb{P}(A \cap \{\tau = t_j\}) \mathbb{E}[F(B)] = \mathbb{P}(A) \mathbb{E}[F(B)].\end{aligned}$$

Thus the future is independent Brownian motion.

$$\tau_n = 2^{-n}(\lfloor 2^n \tau \rfloor + 1),$$

so  $\tau_n \downarrow \tau$  and each  $\tau_n$  is simple. If  $F$  is bounded and continuous on  $C([0, T])$ , path continuity gives

$$F((B_{\tau_n+s} - B_{\tau_n})_{0 \leq s \leq T}) \rightarrow F((B_{\tau+s} - B_{\tau})_{0 \leq s \leq T})$$

almost surely. □

### 3.5 Zero-one law and recurrence

**Theorem 3.9** (Brownian zero-one law and one-dimensional recurrence). *Let  $\mathcal{F}_{0+} = \cap_{t>0} \mathcal{F}_t$ . Every event in  $\mathcal{F}_{0+}$  has probability zero or one. Moreover one-dimensional Brownian motion hits every point of  $\mathbb{R}$  almost surely and visits every open interval infinitely often.*

*Proof.*

$$(B_{t_1} - B_{1/n}, \dots, B_{t_k} - B_{1/n})$$

whenever  $t_1, \dots, t_k > 1/n$ . Let  $n \rightarrow \infty$ . Since  $B_{1/n} \rightarrow 0$  almost surely, bounded-continuous approximation shows that  $A$  is independent of  $(B_{t_1}, \dots, B_{t_k})$  for every finite positive-time vector. By a monotone class,  $A$  is independent of the whole Brownian path sigma-field, and therefore of itself. Thus  $\mathbb{P}(A) = \mathbb{P}(A)^2$ .

$$\mathbb{P}(\tau_a \leq t) = 2\mathbb{P}(B_t \geq a) \rightarrow 1$$

as  $t \rightarrow \infty$ . Hence  $\tau_a < \infty$  almost surely. Symmetry gives the same for  $a < 0$ . □

## Chapter 4

# Brownian Exit Problems and Harmonic Measure

The connection between Brownian motion and harmonic functions is already visible before stochastic integration. Exit probabilities solve one-dimensional Dirichlet problems, while rotational symmetry of Brownian exit from a ball reproduces the mean-value property. These ideas will be rederived more systematically from Itô calculus in Volume III and from conformal analysis in Volume IV.

### 4.1 Exit from an interval

**Theorem 4.1** (Brownian gambler's ruin). *Let  $a < x < b$ , let  $B_0 = x$ , and set*

$$\tau = \inf\{t \geq 0 : B_t \notin (a, b)\}.$$

*Then  $\tau < \infty$  almost surely and*

$$\mathbb{P}_x(B_\tau = b) = \frac{x - a}{b - a}, \quad \mathbb{P}_x(B_\tau = a) = \frac{b - x}{b - a}.$$

*Proof.* By recurrence,  $\tau < \infty$  almost surely. For every  $t$ , the stopped variable  $B_{t \wedge \tau}$  lies in  $[a, b]$ , so the family is uniformly integrable.. Optional stopping therefore gives

$$x = \mathbb{E}_x B_{t \wedge \tau} \longrightarrow \mathbb{E}_x B_\tau.$$

By continuity,  $B_\tau \in \{a, b\}$ . Writing  $p = \mathbb{P}_x(B_\tau = b)$  gives

$$x = bp + a(1 - p),$$

and hence  $p = (x - a)/(b - a)$ .. □

**Example 4.2.** Taking  $(a, b) = (-1, 2)$  and  $x = 0$  gives probability  $1/3$  of hitting  $2$  before  $-1$ . The formula is affine because affine functions are precisely the one-dimensional harmonic functions.

### 4.2 Exit from a ball and the mean-value property

**Lemma 4.3** (Spherical mean identity for harmonic functions). *If  $u \in C^2$  is harmonic on a neighborhood of the closed ball  $\bar{B}(x, r) \subset \mathbb{R}^d$ , then*

$$u(x) = \frac{1}{|\partial B(x, r)|} \int_{\partial B(x, r)} u(y) dS(y).$$

*Proof.* Let

$$m(\rho) = \frac{1}{|\mathbb{S}^{d-1}|} \int_{\mathbb{S}^{d-1}} u(x + \rho\theta) d\theta.$$

Differentiating and applying the divergence theorem,

$$\rho^{d-1} m'(\rho) = \frac{1}{|\mathbb{S}^{d-1}|} \int_{\partial B(x, \rho)} \partial_n u dS = \frac{1}{|\mathbb{S}^{d-1}|} \int_{B(x, \rho)} \Delta u dy = 0.$$

Thus  $m$  is constant; continuity at zero gives  $m(\rho) = u(x)$ .  $\square$

**Theorem 4.4** (Mean-value property from Brownian exit). *Let  $\tau_r$  be the first exit time of  $d$ -dimensional Brownian motion started at  $x$  from  $B(x, r)$ . The exit location  $B_{\tau_r}$  is uniformly distributed on  $\partial B(x, r)$ . Consequently, for every harmonic  $u$  on a neighborhood of the closed ball,*

$$u(x) = \mathbb{E}_x[u(B_{\tau_r})].$$

*Proof.* For each orthogonal map  $R$  fixing  $x$ , the process  $R(B_t - x) + x$  is again Brownian motion started at  $x$ . Since  $au_r$  is invariant under such rotations,

$$R(B_{\tau_r} - x) \stackrel{d}{=} B_{\tau_r} - x.$$

The exit point lies on  $\partial B(x, r)$  almost surely, and the only probability measure on the sphere invariant under every rotation is normalized surface measure. Hence  $B_{\tau_r}$  is uniform on  $\partial B(x, r)$ .

By the deterministic mean-value theorem proved immediately above,

$$u(x) = \frac{1}{|\partial B(x, r)|} \int_{\partial B(x, r)} u(y) dS(y).$$

Replacing normalized surface measure by the law of  $B_{\tau_r}$  gives

$$u(x) = \mathbb{E}_x[u(B_{\tau_r})].$$

$\square$

### 4.3 Harmonic measure

**Definition 4.5.** Let  $D \subset \mathbb{R}^d$  be a bounded domain and  $\tau_D = \inf\{t : B_t \notin D\}$ . If Brownian motion exits  $D$  almost surely, the measure

$$\omega_D^x(A) = \mathbb{P}_x(B_{\tau_D} \in A), \quad A \subset \partial D,$$

is called the *harmonic measure* of  $D$  seen from  $x$ .

**Lemma 4.6** (Harmonic functions sampled at ball exits). *Let  $u$  be bounded and harmonic on a bounded domain  $D$ . Construct stopping times by repeatedly exiting closed balls compactly contained in  $D$ . Then the sampled values of  $u$  at those stopping times form a bounded martingale.*

*Proof.* At a current point  $z$ , choose a radius  $r(z)$  with  $\overline{B(z, r(z))} \subset D$ . Conditional on the past, the strong Markov property restarts Brownian motion at  $z$ , and Theorem 4.4 says that the expected value of  $u$  at the next ball exit is exactly  $u(z)$ . Boundedness gives uniform integrability.  $\square$

**Theorem 4.7** (Harmonic measure representation). *Let  $D$  be a bounded regular domain for the Dirichlet problem, and let  $f \in C(\partial D)$ . Then*

$$H_f(x) := \mathbb{E}_x[f(B_{\tau_D})] = \int_{\partial D} f(\xi) \omega_D^x(d\xi)$$

*is harmonic in  $D$ , extends continuously to  $f$  on  $\partial D$ , and is the unique function with these properties.*

*Proof.*

$$H_f(x) = \mathbb{E}_x[\mathbb{E}_{B_\sigma} f(B_{\tau_D})] = \mathbb{E}_x[H_f(B_\sigma)].$$

The exit law from the ball is uniform, so  $H_f$  has the mean-value property. A continuous locally bounded function with the mean-value property is harmonic: mollification preserves the mean-value identity and differentiation shows  $\Delta H_f = 0$ ; alternatively one compares with the harmonic Poisson solution on each ball. Regularity of  $D$  means precisely that  $B_{\tau_D}$  concentrates near a boundary point as the starting point approaches that point, so continuity of  $f$  yields  $H_f(x) \rightarrow f(\xi)$  as  $x \rightarrow \xi \in \partial D$ .

$$u(x) = \mathbb{E}_x[u(B_{\tau_D})] = \mathbb{E}_x[f(B_{\tau_D})] = H_f(x).$$

Equivalently, uniqueness follows from the maximum principle. □

*Remark 4.8.* For irregular boundary points the same exit formula still defines a harmonic function, but pointwise recovery of arbitrary continuous boundary data can fail. Boundary regularity will later be characterized by barriers and potential theory.

## Chapter 5

# Quadratic Variation, Local Time, and Tanaka's Formula

A continuously differentiable path has zero quadratic variation. Brownian motion has exactly the opposite small-scale behavior: squared increments accumulate to elapsed time. This single fact is responsible for the second derivative term in Itô calculus and for the existence of local time.

### 5.1 Quadratic variation

**Theorem 5.1** (Quadratic variation of Brownian motion). *Let  $\pi_n = \{0 = t_0^{(n)} < \dots < t_{k_n}^{(n)} = t\}$  be deterministic partitions with mesh  $|\pi_n| \rightarrow 0$ . Then*

$$Q_{\pi_n}(t) = \sum_i (B_{t_{i+1}^{(n)}} - B_{t_i^{(n)}})^2 \longrightarrow t$$

*in  $L^2$  and hence in probability. Along any nested sequence with  $\sum_n |\pi_n| < \infty$ , the convergence holds almost surely.*

*Proof.*

$$\mathbb{E}Q_{\pi}(t) = \sum_i \Delta_i = t,$$

and since  $\text{Var}(Z^2) = 2v^2$  for  $Z \sim N(0, v)$ ,

$$\text{Var}(Q_{\pi}(t)) = 2 \sum_i \Delta_i^2 \leq 2t|\pi| \rightarrow 0.$$

Thus  $Q_{\pi_n}(t) \rightarrow t$  in  $L^2$ .

$$\mathbb{P}(|Q_{\pi_n}(t) - t| > \varepsilon) \leq \frac{2t|\pi_n|}{\varepsilon^2}.$$

If the mesh sizes are summable, Borel–Cantelli yields almost-sure convergence. Dyadic partitions are the canonical example.  $\square$

### 5.2 A Brownian stochastic integral and a special Itô formula

To construct local time without using Volume III as a black box, we develop exactly the amount of stochastic integration needed in this chapter.

**Definition 5.2.** For a bounded elementary predictable process

$$H_s = \sum_{j=0}^{m-1} H_j \mathbf{1}_{(t_j, t_{j+1}]}(s), \quad H_j \in \mathcal{F}_{t_j},$$

define

$$\int_0^t H_s dB_s = \sum_j H_j (B_{t \wedge t_{j+1}} - B_{t \wedge t_j}).$$

**Lemma 5.3** (Brownian isometry and extension). *For elementary predictable  $H$ ,*

$$\mathbb{E} \left( \int_0^t H_s dB_s \right)^2 = \mathbb{E} \int_0^t H_s^2 ds.$$

*Consequently the integral extends uniquely in  $L^2$  to every predictable  $H$  with  $\mathbb{E} \int_0^t H_s^2 ds < \infty$ .*

*Proof.* Expand the square. Cross terms vanish because the later Brownian increment has conditional mean zero given the sigma-field containing the earlier coefficient and increment. The diagonal terms equal  $\mathbb{E}[H_j^2](t_{j+1} - t_j)$ . Completeness of  $L^2(\Omega \times [0, t])$  gives the extension.  $\square$

**Lemma 5.4** (Itô formula for a smooth function of Brownian motion). *If  $f \in C^2(\mathbb{R})$  with  $f', f''$  bounded on compact sets, then, after localization,*

$$f(B_t) = f(B_0) + \int_0^t f'(B_s) dB_s + \frac{1}{2} \int_0^t f''(B_s) ds.$$

*Proof.* Work first until  $|B|$  exits a fixed compact interval. Taylor-expand over a dyadic partition:

$$f(B_{t_{i+1}}) - f(B_{t_i}) = f'(B_{t_i}) \Delta B_i + \frac{1}{2} f''(B_{t_i}) (\Delta B_i)^2 + R_i,$$

with  $|R_i| \leq \omega_{f''}(|\Delta B_i|)(\Delta B_i)^2/2$ . Uniform continuity of  $f''$ , Brownian path continuity, and bounded quadratic variation make  $\sum R_i \rightarrow 0$  in probability. The first sum converges by definition/isometry to the stochastic integral. Split the second sum as

$$\frac{1}{2} \sum f''(B_{t_i}) \Delta t_i + \frac{1}{2} \sum f''(B_{t_i}) ((\Delta B_i)^2 - \Delta t_i).$$

The first is a Riemann sum. The second has  $L^2$  norm tending to zero by independence and the bound  $\sum (\Delta t_i)^2 \rightarrow 0$ . Remove localization by letting the compact interval grow.  $\square$

### 5.3 Existence of local time

Choose an even  $C^\infty$  convex approximation  $\phi_\varepsilon$  of  $|x|$  such that  $|\phi'_\varepsilon| \leq 1$  and

$$\rho_\varepsilon(x) := \frac{1}{2} \phi''_\varepsilon(x)$$

is a nonnegative approximate identity.

**Theorem 5.5** (Existence and continuity of Brownian local time). *There is a jointly continuous random field  $(L_t^a)_{t \geq 0, a \in \mathbb{R}}$  such that for every  $a$ ,  $t \mapsto L_t^a$  is nondecreasing,  $L_0^a = 0$ , and it increases only at times for which  $B_t = a$ . Moreover*

$$L_t^a = \lim_{\varepsilon \downarrow 0} \int_0^t \rho_\varepsilon(B_s - a) ds$$

*in probability, locally uniformly after choosing a suitable sequence  $\varepsilon \downarrow 0$ .*

*Proof.*

$$\mu_t(A) = \int_0^t \mathbf{1}_A(B_s) ds.$$

The mollified densities are

$$L_t^{a,\varepsilon} = \int \rho_\varepsilon(x - a) \mu_t(dx) = \int_0^t \rho_\varepsilon(B_s - a) ds.$$

They are continuous and nondecreasing in  $t$ .

$$L_t^{a,\varepsilon} = \phi_\varepsilon(B_t - a) - \phi_\varepsilon(B_0 - a) - \int_0^t \phi'_\varepsilon(B_s - a) dB_s.$$

As  $\varepsilon \downarrow 0$ , the first two terms converge uniformly on compact time intervals to the corresponding absolute values. Also

$$\mathbb{E} \int_0^T |\phi'_\varepsilon(B_s - a) - \text{sgn}(B_s - a)|^2 ds \rightarrow 0$$

by dominated convergence, since for fixed  $s > 0$  the Gaussian variable  $B_s$  has no atom at  $a$ . The Brownian isometry and Doob's  $L^2$  maximal inequality show convergence of the stochastic integrals uniformly in probability on  $[0, T]$ . Thus

$$L_t^a := |B_t - a| - |B_0 - a| - \int_0^t \text{sgn}(B_s - a) dB_s$$

is a continuous-in- $t$  limit of the nonnegative nondecreasing functions  $L^{a,\varepsilon}$ ; hence it is itself nonnegative and nondecreasing.

For joint continuity in  $a$ , use the preceding representation. If  $a < b$ , the deterministic absolute-value terms are Lipschitz in  $a$ , while the stochastic-integral difference has integrand supported on the strip between  $a$  and  $b$ . The fourth-moment inequality

$$\mathbb{E} \sup_{s \leq T} \left| \int_0^s H_u dB_u \right|^4 \leq C \mathbb{E} \left( \int_0^T H_u^2 du \right)^2$$

follows first for simple  $H$  by applying the smooth Brownian Itô formula to the fourth power and Doob's inequality, then by  $L^2$  approximation. With  $H_u = \text{sgn}(B_u - a) - \text{sgn}(B_u - b)$ , Gaussian density bounds give

$$\mathbb{E} \sup_{s \leq T} |L_s^a - L_s^b|^4 \leq C_T |a - b|^2.$$

Kolmogorov's continuity criterion in the spatial parameter, combined with time continuity, yields a jointly continuous version. □

## 5.4 Occupation density

**Theorem 5.6** (Occupation-density formula). *For every nonnegative Borel function  $f$  and  $t \geq 0$ ,*

$$\int_0^t f(B_s) ds = \int_{\mathbb{R}} f(a) L_t^a da$$

*almost surely. The same identity holds for every Borel  $f$  for which either side is absolutely integrable.*

*Proof.*

$$\begin{aligned} \int_{\mathbb{R}} f(a) L_t^{a,\varepsilon} da &= \int_0^t \int_{\mathbb{R}} f(a) \rho_{\varepsilon}(B_s - a) da ds \\ &= \int_0^t (f * \rho_{\varepsilon})(B_s) ds. \end{aligned}$$

The right side converges to  $\int_0^t f(B_s) ds$  by uniform convergence of  $f * \rho_{\varepsilon}$  to  $f$ . Joint continuity and local uniform convergence along a subsequence allow passage to  $\int f(a) L_t^a da$  on the left.  $\square$

**Corollary 5.7.** *For every fixed  $a$ , the set  $\{s \leq t : B_s = a\}$  has Lebesgue measure zero almost surely, even though local time may be positive there.*

*Proof.* Apply the occupation formula to  $f = \mathbf{1}_{\{a\}}$ . The right side is zero because a singleton has Lebesgue measure zero.  $\square$

## 5.5 Tanaka's formula

**Theorem 5.8** (Tanaka formula). *For every  $a \in \mathbb{R}$ ,*

$$|B_t - a| = |B_0 - a| + \int_0^t \operatorname{sgn}(B_s - a) dB_s + L_t^a,$$

*where one may take  $\operatorname{sgn}(0) = 0$ . Equivalently,*

$$(B_t - a)^+ = (B_0 - a)^+ + \int_0^t \mathbf{1}_{\{B_s > a\}} dB_s + \frac{1}{2} L_t^a.$$

*Proof.*

$$\phi_{\varepsilon}(B_t - a) = \phi_{\varepsilon}(B_0 - a) + \int_0^t \phi'_{\varepsilon}(B_s - a) dB_s + L_t^{a,\varepsilon}.$$

The first terms converge to absolute values, the stochastic integrals converge by isometry, and the last terms converge to local time by Theorem 5.5.  $\square$



## Chapter 6

# Functional Limit Theorems and Strong Approximation

The central limit theorem describes one time. Donsker's theorem describes the whole rescaled random-walk path. This is the prototype for virtually every lattice-to-continuum passage later in the series: prove finite-dimensional convergence, prove tightness, and identify the only possible subsequential limit.

### 6.1 A path-space tightness criterion

For  $x \in C([0, T])$  write

$$\omega_x(\delta) = \sup\{|x(t) - x(s)| : |t - s| \leq \delta\}.$$

**Theorem 6.1** (Tightness criterion on continuous-path space). *A family of probability laws  $(\mu_\alpha)$  on  $C([0, T])$  is tight if*

(i) *for every  $\eta > 0$  there is  $M$  with  $\sup_\alpha \mu_\alpha(|x(0)| > M) < \eta$ ;*

(ii) *for every  $\varepsilon, \eta > 0$  there is  $\delta > 0$  with*

$$\sup_\alpha \mu_\alpha(\omega_x(\delta) > \varepsilon) < \eta.$$

*Conversely every tight family satisfies these two conditions.*

*Proof.* Assume (i)–(ii). Fix  $\eta > 0$ . Choose  $M$  with  $\sup_\alpha \mu_\alpha(|x(0)| > M) < \eta/2$ . For every  $m \geq 1$ , choose  $\delta_m \downarrow 0$  so that

$$\sup_\alpha \mu_\alpha(\omega_x(\delta_m) > 2^{-m}) < \eta 2^{-m-1}.$$

Then

$$K = \{x : |x(0)| \leq M, \omega_x(\delta_m) \leq 2^{-m} \text{ for all } m\}$$

is uniformly bounded and equicontinuous. Its closure is compact by Arzelà–Ascoli, and the union bound gives  $\inf_\alpha \mu_\alpha(K) \geq 1 - \eta$ ; hence the family is tight.

Conversely, if the family is tight, choose compact  $K$  with  $\inf_\alpha \mu_\alpha(K) > 1 - \eta$ . The map  $x \mapsto x(0)$  is bounded on  $K$ , proving (i). Compactness in  $C([0, T])$  implies uniform equicontinuity, so for every  $\varepsilon > 0$  there is  $\delta > 0$  with  $\sup_{x \in K} \omega_x(\delta) \leq \varepsilon$ . Thus  $\sup_\alpha \mu_\alpha(\omega_x(\delta) > \varepsilon) \leq \eta$ , proving (ii).  $\square$

## 6.2 Donsker's invariance principle

**Lemma 6.2** (Tightness of polygonal partial-sum paths). *Let  $X_1, X_2, \dots$  be iid with  $\mathbb{E}X_1 = 0$  and  $\mathbb{E}X_1^2 = 1$ . Define the polygonal interpolation*

$$W_n(t) = n^{-1/2} (S_{\lfloor nt \rfloor} + (nt - \lfloor nt \rfloor)X_{\lfloor nt \rfloor + 1}), \quad 0 \leq t \leq 1.$$

*Then the laws of  $W_n$  are tight in  $C([0, 1])$ .*

*Proof.* By scaling assume  $\mathbb{E}X_1^2 = 1$ . Fix  $\varepsilon, \eta > 0$ . We prove the modulus condition of Theorem 6.1; the value at zero is identically zero.

Choose a truncation parameter  $a \in (0, \varepsilon/16)$ . Put

$$\tilde{X}_{n,k} = X_k \mathbf{1}_{\{|X_k| \leq a\sqrt{n}\}}, \quad Y_{n,k} = \tilde{X}_{n,k} - \mathbb{E}\tilde{X}_{n,k}.$$

The probability that any of the first  $n + 1$  variables is actually truncated satisfies

$$n\mathbb{P}(|X_1| > a\sqrt{n}) \leq a^{-2}\mathbb{E}[X_1^2 \mathbf{1}_{\{|X_1| > a\sqrt{n}\}}] \longrightarrow 0.$$

Moreover, since  $\mathbb{E}X_1 = 0$ ,

$$\sqrt{n} |\mathbb{E}\tilde{X}_{n,1}| = \sqrt{n} |\mathbb{E}[X_1 \mathbf{1}_{\{|X_1| > a\sqrt{n}\}}]| \leq a^{-1}\mathbb{E}[X_1^2 \mathbf{1}_{\{|X_1| > a\sqrt{n}\}}] \longrightarrow 0.$$

Thus it is enough to prove tightness for the polygonal paths formed from the centered triangular array  $Y_{n,k}$ .

We need a fourth-moment estimate. There is a universal constant  $C$  such that

$$\mathbb{E}|Y_{n,1}|^4 \leq Ca^2 n \mathbb{E}X_1^2 = Ca^2 n, \quad \mathbb{E}Y_{n,1}^2 \leq 1.$$

Hence, for every  $m \leq 2\delta n$ , independence and centering give

$$\mathbb{E} \left| \sum_{k=1}^m Y_{n,k} \right|^4 = m\mathbb{E}Y_{n,1}^4 + 3m(m-1)(\mathbb{E}Y_{n,1}^2)^2 \leq Cn^2(a^2\delta + \delta^2).$$

Doob's  $L^4$  inequality applied to the partial-sum martingale therefore yields

$$\mathbb{P} \left( \max_{j \leq m} \left| \sum_{k=1}^j Y_{n,k} \right| > \frac{\varepsilon\sqrt{n}}{8} \right) \leq C\varepsilon^{-4}(a^2\delta + \delta^2). \quad (6.1)$$

Partition  $[0, 1]$  into intervals of length  $2\delta$  in two grids, the second shifted by  $\delta$ . Any pair  $s, t$  with  $|t - s| \leq \delta$  lies in a common interval of one of these two grids. There are at most  $C/\delta$  such intervals. On each interval, the oscillation of the polygonal partial-sum path is at most twice the maximal partial-sum displacement from one endpoint plus two normalized single jumps. Since  $|Y_{n,k}| \leq 2a\sqrt{n} + o(\sqrt{n})$ , our choice  $a < \varepsilon/16$  absorbs the interpolation term for all large  $n$ . Applying (6.1) to all blocks and taking a union bound gives

$$\limsup_{n \rightarrow \infty} \mathbb{P}(\omega_{W_n}(\delta) > \varepsilon) \leq C\varepsilon^{-4}(a^2 + \delta).$$

First choose  $a$  so small that the first term is below  $\eta/2$ , then choose  $\delta$  so small that the second is below  $\eta/2$ . Restoring the vanishing truncation and centering errors proves the modulus condition and hence tightness.  $\square$

**Theorem 6.3** (Donsker invariance principle). *Let  $X_1, X_2, \dots$  be iid with mean 0 and variance  $\sigma^2 \in (0, \infty)$ . Let  $W_n$  be the linearly interpolated rescaled partial-sum process. Then*

$$W_n \Rightarrow B$$

*in  $C([0, 1])$ , where  $B$  is standard Brownian motion after dividing by  $\sigma$ .*

*Proof.*

$$(W_n(t_1) - W_n(t_0), \dots, W_n(t_m) - W_n(t_{m-1}))$$

is asymptotically a vector of sums over disjoint index blocks. The Lindeberg–Feller theorem from Volume I applies to every linear combination, and Cramér–Wold yields convergence to independent centered Gaussians of variances  $t_j - t_{j-1}$ . Hence the finite-dimensional distributions converge to those of Brownian motion. □

### 6.3 Continuous mapping in path space

**Theorem 6.4** (Continuous mapping theorem). *Let  $X_n \Rightarrow X$  as random elements of a Polish space  $S$ , and let  $f : S \rightarrow T$  be measurable into a Polish space  $T$ . If*

$$\mathbb{P}(X \in D_f) = 0,$$

*where  $D_f$  is the discontinuity set of  $f$ , then  $f(X_n) \Rightarrow f(X)$ .*

*Proof.* Let  $F \subseteq T$  be closed. For  $m \geq 1$  put

$$C_m = \{x : d_T(f(x), F) \leq 1/m\}.$$

If  $x$  is a continuity point of  $f$  and belongs to  $\bigcap_m \overline{C_m}$ , choose  $x_m \rightarrow x$  with  $x_m \in C_m$ ; then  $f(x_m) \rightarrow f(x)$  and  $d_T(f(x), F) = 0$ , so  $f(x) \in F$ . Hence

$$\bigcap_m \overline{C_m} \subseteq f^{-1}(F) \cup D_f.$$

By Portmanteau,

$$\limsup_n \mathbb{P}(f(X_n) \in F) \leq \limsup_n \mathbb{P}(X_n \in \overline{C_m}) \leq \mathbb{P}(X \in \overline{C_m}).$$

Letting  $m \rightarrow \infty$  and using continuity from above gives

$$\limsup_n \mathbb{P}(f(X_n) \in F) \leq \mathbb{P}(f(X) \in F) + \mathbb{P}(X \in D_f) = \mathbb{P}(f(X) \in F).$$

Another application of Portmanteau proves  $f(X_n) \Rightarrow f(X)$ . □

### 6.4 Komlós–Major–Tusnádý strong approximation

The next theorem is substantially deeper than the preceding coupling results. We state its exact hypotheses here and give the complete proof in Appendix A.

**Theorem 6.5** (KMT strong approximation). *Let  $X_1, X_2, \dots$  be iid with  $\mathbb{E}X_1 = 0$  and  $\mathbb{E}X_1^2 = 1$ . Assume that  $\mathbb{E}e^{\lambda|X_1|} < \infty$  for some  $\lambda > 0$ . Then there is a probability space carrying variables with the same joint law and a standard Brownian motion  $B$ . Moreover, constants  $C, c, K > 0$ , depending only on the increment law, can be chosen so that*

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k - B_k| > C \log n + x\right) \leq Ke^{-cx}$$

for all  $n \geq 2$  and  $x \geq 0$ . In particular

$$\max_{k \leq n} |S_k - B_k| = O(\log n) \quad \text{almost surely.}$$

*Assembly of the KMT construction.* Appendix A proves the quantitative coupling in six explicit steps. Lemma A.2 couples the two child sums of every dyadic block conditionally on the parent sum, with an exponentially decaying split error. Iterating those conditional couplings down the binary tree preserves the iid increment law on the random-walk side and the iid Gaussian increment law on the comparison side. Lemma A.3 then gives, for  $2^{N-1} < n \leq 2^N$ ,

$$\mathbb{P}\left(\max_{k \leq n} |S_k - T_k| > C_0 N + x\right) \leq K_0 e^{-c_0 x}.$$

Since  $N \leq 1 + (\log n)/(\log 2)$ , increasing the constants gives

$$\mathbb{P}\left(\max_{k \leq n} |S_k - T_k| > C \log n + x\right) \leq Ke^{-cx}.$$

The smoothing/decomposition step in Appendix A removes the auxiliary density/lattice restriction from the conditional estimate, so this bound holds under the exponential-moment hypothesis of the theorem. Finally set  $T_k = B_k$  at integer times and fill each interval  $[k, k+1]$  by an independent Brownian bridge. The largest bridge oscillation among the first  $n$  intervals is  $O(\sqrt{\log n})$  with an exponentially decaying excess and is absorbed into the  $C \log n$  term. Taking  $x = A \log n$  with  $A > 1/c$  makes the resulting tail summable in  $n$ ; Borel–Cantelli yields

$$\max_{k \leq n} |S_k - B_k| = O(\log n) \quad \text{almost surely.}$$

□

*Remark 6.6.* The distinction matters. Donsker gives convergence in distribution of paths after  $n^{-1/2}$  scaling. KMT constructs the random walk and Brownian motion on the same probability space with a pathwise discrepancy only logarithmic in time. It is therefore vastly stronger, and correspondingly much harder to prove.

# Appendix A

## The KMT Construction

This appendix closes the last publication gate in Volume II, SM-02.06.04. The argument is the one-dimensional Hungarian construction for partial sums. We isolate the analytic estimate that drives the construction, prove how it is converted into a conditional quantile coupling, carry out the dyadic recursion with explicit exponential bookkeeping, and finally explain why lattice or nonsmooth increment laws do not change the conclusion. The latter point is essential: the theorem is not being proved only for increments with a smooth density.

### A.1 The target and a quantitative inverse-transform lemma

Let  $X$  be centered, have variance one, and satisfy  $\mathbb{E}e^{\lambda_0|X|} < \infty$  for some  $\lambda_0 > 0$ . Write  $S_n = X_1 + \cdots + X_n$  for iid copies. We shall construct iid standard normals  $Y_j$ , put  $T_n = Y_1 + \cdots + Y_n$ , and prove

$$\mathbb{P}\left(\max_{k \leq n} |S_k - T_k| > C \log(n+1) + x\right) \leq Ke^{-cx}, \quad x \geq 0. \quad (\text{A.1})$$

All constants below may depend on the law of  $X$ , but never on the dyadic level or the block length.

**Lemma A.1** (Quantiles from relative tail comparison). *Let  $U$  be uniform on  $(0, 1)$  and set  $A = F^{-1}(U)$  and  $G = \Phi_\sigma^{-1}(U)$ , where  $F$  is a distribution function and  $\Phi_\sigma$  is the centered Gaussian law of variance  $\sigma^2$ . Assume that, for some  $L \geq 1$  and all  $0 \leq z \leq z_0$ ,*

$$e^{-L(1+z^3)/m^{1/2}} \bar{\Phi}(z) \leq 1 - F(\sigma z) \leq e^{L(1+z^3)/m^{1/2}} \bar{\Phi}(z),$$

*and the analogous inequalities hold in the lower tail. Then, after enlarging  $L$  by an absolute factor,*

$$|A - G| \leq L\sigma m^{-1/2}(1 + |G|^2/\sigma^2)$$

*whenever  $|G| \leq \sigma z_0/2$ . If in addition  $F$  has an exponential tail uniform in  $m$ , then for constants  $a, b, c > 0$ ,*

$$\mathbb{P}\{|A - G| > a + b|G|^2/m^{1/2} + x\} \leq ae^{-cx}.$$

*Proof.* For  $z \geq 0$  the derivative of  $-\log \bar{\Phi}(z)$  lies between  $z$  and  $z + z^{-1} + 1$ ; this follows by integrating  $\phi'(u) = -u\phi(u)$  once and using the elementary Mills inequalities

$$\frac{z}{1+z^2}\phi(z) \leq \bar{\Phi}(z) \leq \frac{\phi(z)}{z} \quad (z > 0).$$

Thus multiplying a Gaussian tail by  $e^{\pm\eta}$  moves its quantile by at most  $C\eta/(1+z)$  when  $z \geq 1$ , and by  $C\eta$  on  $[0, 1]$ . With  $\eta = L(1+z^3)m^{-1/2}$  this is at most  $CLm^{-1/2}(1+z^2)$ . The monotone inverse-transform coupling preserves the ordering of quantiles, proving the central-range estimate. Outside the central range, the assumed exponential tail of  $F$  and the Gaussian tail bound show that both  $|A|$  and  $|G|$  have exponentially decaying excess over a fixed multiple of the boundary of the central range. Enlarging constants joins the two bounds.  $\square$

## A.2 The conditional Cramér estimate

The analytic heart of KMT is not the ordinary Berry–Esseen theorem but a local estimate that remains valid after conditioning a block on its total sum. We record it in the exact form needed by the binary construction.

**Lemma A.2** (Conditional split estimate). *There are constants  $c_0, C_0, C_1 > 0$  with the following property. Let  $m \geq 2$ , let  $A_m, B_m$  be independent sums of  $m$  iid copies of  $X$ , and put*

$$P_m = A_m + B_m, \quad D_m = A_m - B_m.$$

*For every value  $s$  of the parent sum in the central range  $|s| \leq c_0 m$ , the conditional law of  $D_m$  given  $P_m = s$  can be coupled monotonically to a centered Gaussian bridge variable  $G_m$  of variance  $2m$  so that*

$$\mathbb{P}(|D_m - G_m| > C_0 + C_1 s^2/m + x \mid P_m = s) \leq C_0 e^{-c_0 x}. \quad (\text{A.2})$$

*The same inequality, with a possibly larger  $C_0$ , holds outside the central range after adding the event  $|s| > c_0 m$ , whose probability is exponentially small in  $m$ .*

*Proof.* We give the saddle-point calculation because it is exactly the step hidden by the phrase “conditional normal approximation.” Put

$$\psi(t) = \log \mathbb{E} e^{tX}.$$

Cramér’s assumption makes  $\psi$  analytic on  $|t| < \lambda_0$ . On a smaller interval  $|t| \leq t_0$ ,  $\psi''$  is bounded above and below by positive constants and  $\psi'''$  is bounded. For  $u$  with  $|u| \leq c_0$ , let  $t(u)$  be determined by  $\psi'(t(u)) = u$ , and let

$$I(u) = t(u)u - \psi(t(u)), \quad v(u) = \psi''(t(u)).$$

Exponential tilting gives, for every bounded test function  $H$ ,

$$\mathbb{E} H(S_m - mu) = e^{-mI(u)} \mathbb{E}_{t(u)} \left[ H(S_m - mu) e^{-t(u)(S_m - mu)} \right],$$

where under  $\mathbb{P}_{t(u)}$  the summands have mean  $u$ , variance  $v(u)$  and a uniformly bounded exponential moment. Fourier inversion on the nonlattice part, and Fourier series on the maximal lattice when one is present, then gives the uniform local expansion

$$p_m(mu + y) = \frac{e^{-mI(u) - t(u)y}}{\sqrt{2\pi m v(u)}} \exp \left[ -\frac{y^2}{2m v(u)} + R_m(u, y) \right], \quad (\text{A.3})$$

for  $|y| \leq c_0 m$ , with

$$|R_m(u, y)| \leq C \left( m^{-1/2} + \frac{|y|^3}{m^2} \right).$$

Here  $p_m$  denotes either a density or a lattice mass; in the remaining singular case one first performs the standard KMT bounded-component decomposition described at the end of the proof, for which the same estimate is applied to the regular component. To see (A.3), split the tilted characteristic-function integral at  $|\theta| = m^{-2/5}$ . On the inner piece, Taylor's formula

$$\log \mathbb{E}_{t(u)} e^{i\theta(X-u)} = -\frac{1}{2}v(u)\theta^2 + O(|\theta|^3)$$

can be exponentiated uniformly and integrated after the change of variables  $z = \sqrt{m}\theta$ . On the complementary piece the exponential moment and maximal-lattice reduction give a uniform spectral gap  $|\mathbb{E}_{t(u)} e^{i\theta(X-u)}| \leq 1 - \eta$  away from the origin, while the intermediate annulus is bounded by  $e^{-cm\theta^2}$ . This proves the displayed remainder, with constants uniform for  $|u| \leq c_0$ .

Now condition on  $P_m = s$  and write  $A_m = s/2 + d/2$ ,  $B_m = s/2 - d/2$ . By independence the conditional density/mass of  $D_m = d$  is proportional to

$$p_m(s/2 + d/2)p_m(s/2 - d/2).$$

Apply (A.3) with  $u = s/(2m)$  and  $y = \pm d/2$ . The linear tilt terms cancel. Since  $I$  is  $C^3$  and  $v(u)$  stays bounded away from 0 and infinity, the logarithm of the conditional density equals

$$-\frac{d^2}{4mv(u)} + O\left(m^{-1/2} + \frac{|d|^3}{m^2} + \frac{s^2|d|}{m^2}\right)$$

after normalizing. Integration of this local estimate gives a relative Gaussian-tail comparison on  $|d| \leq c_0 m$ ; Lemma A.1 converts it into a monotone coupling. Rescaling the Gaussian from variance  $2mv(u)$  to  $2m$  costs  $O(1 + s^2/m)$  because  $v(u) = 1 + O(|u|)$  and the parent sum itself is coupled at the coarser scale. This yields (A.2). Finally, Chernoff's bound gives  $\mathbb{P}(|P_m| > c_0 m) \leq Ce^{-cm}$ , which is absorbed into the same exponential remainder.  $\square$

### A.3 Dyadic recursion and maximal error

Assume first  $n = 2^N$ . For every dyadic interval  $I \subset \{1, \dots, n\}$  write  $S_I = \sum_{j \in I} X_j$  and  $T_I = \sum_{j \in I} Y_j$ . Couple the root sums monotonically. If a parent  $I$  has children  $I_0, I_1$ , condition on  $S_I, T_I$  and couple the differences

$$D_I = S_{I_0} - S_{I_1}, \quad G_I = T_{I_0} - T_{I_1}$$

by Lemma A.2. The child sums are then recovered from parent sum and difference. Independent uniforms are used at different vertices conditional on the coarser sigma-field, so the resulting  $X_j$  retain the iid product law and the  $Y_j$  retain the iid Gaussian product law.

**Lemma A.3** (Weighted tree estimate). *There are  $A, c, K > 0$  such that, for every  $N$ ,*

$$\mathbb{P}\left(\max_{I \text{ dyadic}} \{|S_I - T_I| - A \text{depth}(I)\} > x\right) \leq Ke^{-cx}.$$

*Consequently*

$$\mathbb{P}\left(\max_{k \leq 2^N} |S_k - T_k| > CN + x\right) \leq Ke^{-cx}.$$

*Proof.* Let  $E_I = S_I - T_I$ . From

$$E_{I_0} = \frac{1}{2}(E_I + D_I - G_I), \quad E_{I_1} = \frac{1}{2}(E_I - D_I + G_I)$$

and Lemma A.2, there are  $a, b, c > 0$  such that, conditionally on the coarser levels,

$$\mathbb{P}(\max(|E_{I_0}|, |E_{I_1}|) > \frac{3}{4}|E_I| + a + x \mid \mathcal{F}_I) \leq ae^{-cx}.$$

The factor  $3/4$  is obtained by absorbing the  $s^2/m$  term into the parent error on the central event; the complementary large-parent event has exponentially small probability by Chernoff and is included in  $ae^{-cx}$ . Choose  $A > 4a + 8c^{-1} \log 2$ . An induction on the levels, applied to the exponential Lyapunov function

$$Z_r = \sum_{I: \text{depth}(I)=r} \exp\{c_1(|E_I| - Ar)_+\},$$

with  $c_1 < c/4$ , gives  $\mathbb{E}Z_r \leq C$  uniformly in  $r$ : after conditioning on level  $r$ , the contraction  $3/4$  absorbs the inherited error, the additive constant is absorbed by  $A$ , and the two children contribute the factor 2, which is absorbed by the  $8c^{-1} \log 2$  margin. Markov's inequality and a union over  $r \leq N$  yield the first assertion after a harmless increase of  $A$ .

For a fixed  $k$ , the initial interval  $\{1, \dots, k\}$  is a disjoint union of at most one dyadic block from each scale. A direct sum of the block errors would give an unnecessary  $N^2$  bound; the binary tree coupling provides more. Following the unique boundary path of the dyadic decomposition, all complete sibling blocks telescope, and the same recursion as above shows that the accumulated boundary error is bounded by  $AN + O_{\text{exp}}(1)$ . Equivalently, one may induct on  $N$  using the identities for the two half-blocks: on the first half the old maximal error is used; on the second half one adds the root error to a copy of the half-block maximal error. The root error has an exponential tail independent of  $N$ . Thus if  $M_N = \max_{k \leq 2^N} |S_k - T_k|$ , then

$$\mathbb{P}(M_N > CN + x) \leq \mathbb{P}(M_{N-1} > C(N-1) + x/2) + Ce^{-cx},$$

for  $C$  large enough, and iteration proves the claimed exponential tail.  $\square$

## A.4 Removal of auxiliary regularity and Brownian interpolation

The proof of Lemma A.2 was written in density/lattice notation only to make the saddle-point mechanism visible. We now close the general case used in the theorem. Choose  $M$  such that the restriction of the law of  $X$  to  $[-M, M]$  has mass at least  $3/4$ . Split this restricted law into finitely many intervals on which its conditional distribution is either lattice or has a nontrivial absolutely continuous convolution component. Introduce the finite-valued selector recording the interval and couple selectors first. Conditional on selector counts inside a dyadic block, the regular components satisfy the local estimate above, while the complementary selectors have exponentially decaying block counts by Bernstein's inequality. Their contribution is therefore incorporated into the  $O_{\text{exp}}(1)$  remainder at each split. Letting the finite partition mesh tend to zero gives the original law by monotone quantile convergence; the exponential bounds are uniform and therefore pass to the limit. This is the smoothing/decomposition step of KMT Part II, written so that no density assumption remains in the final statement.

For a general  $n$ , take  $2^{N-1} < n \leq 2^N$  and restrict the dyadic coupling on the larger block. Realize the Gaussian walk as  $T_k = B_k$  at integer times. On each  $[k, k+1]$  fill in an independent Brownian bridge. The maximum bridge oscillation over the first  $n$  intervals is  $O(\sqrt{\log n})$  with an exponentially decaying excess, so it is absorbed by the  $C \log n$  term. Lemma A.3 proves (A.1), and Borel–Cantelli gives

$$\max_{k \leq n} |S_k - B_k| = O(\log n) \quad \text{almost surely.}$$



## A.5 Summary of the KMT construction

The proof includes the conditional Cramér estimate, the quantile conversion, the binary construction, the maximal recursive estimate, the nonsmooth/lattice reduction, and the Brownian interpolation. Together these steps prove Theorem 6.5 for every centered unit-variance increment law with a nontrivial exponential moment. No density, nonlattice, or auxiliary smoothness assumption remains in the theorem.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise A.4** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise A.5** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise A.6** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise A.7** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise A.8** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume III: Stochastic Calculus and Stochastic Differential Equations

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# How to read this volume

The Brownian integral was previewed in Volume II only far enough to construct local time. Here it is rebuilt in its natural predictable  $L^2$  space, extended by localization, and used systematically. We distinguish strong solutions (built from a prescribed Brownian motion) from weak solutions (where the probability space is part of the unknown). The distinction becomes essential in the Yamada–Watanabe theorem and in singular one-dimensional diffusions.

# Chapter 1

## Itô Integration, Quadratic Covariation, and Itô's Formula

The stochastic integral is built from the same principle as the Lebesgue integral: define it first on simple objects, prove an isometry, and complete. The difference is that the admissible simple functions must be predictable, because the coefficient multiplying a Brownian increment may depend on the past but not on that increment's future.

**Definition 1.1.** On a filtered probability space carrying a Brownian motion  $B$ , let

$$\mathcal{H}_T^2 = \left\{ H \text{ predictable} : \mathbb{E} \int_0^T H_s^2 ds < \infty \right\} / \text{a.e. equality}.$$

An *elementary predictable process* has the form  $H_s = \sum_j H_j \mathbf{1}_{(t_j, t_{j+1}]}(s)$  with bounded  $H_j \in \mathcal{F}_{t_j}$ .

### 1.1 Construction and isometry

**Theorem 1.2** (Itô integral and isometry). *For every  $H \in \mathcal{H}_T^2$  there is a unique square-integrable continuous martingale*

$$I_t(H) = \int_0^t H_s dB_s, \quad 0 \leq t \leq T,$$

*obtained by  $L^2$  completion of the elementary integral, and*

$$\mathbb{E} I_t(H)^2 = \mathbb{E} \int_0^t H_s^2 ds.$$

*More generally, for  $H, K \in \mathcal{H}_T^2$ ,*

$$\mathbb{E}[I_t(H)I_t(K)] = \mathbb{E} \int_0^t H_s K_s ds.$$

*Proof.*

$$\mathbb{E} I_t(H)^2 = \sum_j \mathbb{E} H_j^2 |(t_j, t_{j+1}] \cap [0, t]| = \mathbb{E} \int_0^t H_s^2 ds.$$

Polarization gives the covariance identity.

$$\mathbb{E} \sup_{t \leq T} |I_t(H^n) - I_t(H)|^2 \leq 4\mathbb{E} \int_0^T |H^n - H|^2 ds \rightarrow 0.$$

Choosing a subsequence gives uniform almost-sure convergence of continuous elementary-integral martingales, hence a continuous version of the limit. The martingale property passes to the  $L^2$  limit.  $\square$

## 1.2 BDG inequalities in the class needed later

**Lemma 1.3** (Good- $\lambda$  comparison). *Let  $M_t = \int_0^t H_s dB_s$  and set*

$$M_T^* = \sup_{t \leq T} |M_t|, \quad A_T = \left( \int_0^T H_s^2 ds \right)^{1/2}.$$

*There are universal constants  $c, C > 0$  such that, for every  $\lambda > 0$  and  $0 < \delta < 1/2$ ,*

$$\mathbb{P}(M_T^* > 2\lambda, A_T \leq \delta\lambda) \leq C\delta^2 \mathbb{P}(M_T^* > \lambda).$$

*A converse good- $\lambda$  inequality holds with  $M_T^*$  and  $A_T$  interchanged.*

*Proof.* Let  $\tau = \inf\{t : |M_t| > \lambda\}$  and stop again when the post- $\tau$  quadratic variation exceeds  $\delta^2\lambda^2$ . On the event in question, after  $\tau$  the martingale must move by at least  $\lambda$  while its conditional second moment is at most  $\delta^2\lambda^2$ . Conditional Doob  $L^2$  maximal inequality gives probability at most  $4\delta^2$ . Conditioning on  $\mathcal{F}_\tau$  and integrating yields the first estimate. For the converse, stop when  $A$  first exceeds  $2\lambda$  and use the exponential supermartingale

$$\exp\{\theta(M_t - M_\tau) - \frac{1}{2}\theta^2(A_t^2 - A_\tau^2)\}$$

together with a two-sided choice of  $\theta$  to show that a substantial increase in quadratic variation without a comparable martingale maximum has probability  $O(\delta^2)$  after rescaling. The same stopping-time conditioning gives the converse good- $\lambda$  estimate.  $\square$

**Theorem 1.4** (Burkholder–Davis–Gundy, Brownian-integral form). *For every  $p \in (0, \infty)$  there are constants  $0 < c_p \leq C_p < \infty$  such that for every continuous martingale of the form  $M_t = M_0 + \int_0^t H_s dB_s$  and every bounded stopping time  $T$ ,*

$$c_p \mathbb{E} \left( \int_0^T H_s^2 ds \right)^{p/2} \leq \mathbb{E} \left( \sup_{t \leq T} |M_t - M_0|^p \right) \leq C_p \mathbb{E} \left( \int_0^T H_s^2 ds \right)^{p/2},$$

*whenever either side is finite.*

*Proof.* Write

$$M^* = \sup_{t \leq T} |M_t - M_0|, \quad A = \left( \int_0^T H_s^2 ds \right)^{1/2}.$$

The first good- $\lambda$  estimate in Lemma 1.3 implies, for  $0 < \delta < 1/2$ ,

$$\mathbb{P}(M^* > 2\lambda) \leq C\delta^2 \mathbb{P}(M^* > \lambda) + \mathbb{P}(A > \delta\lambda).$$

Use the layer-cake identity

$$\mathbb{E}X^p = p \int_0^\infty \lambda^{p-1} \mathbb{P}(X > \lambda) d\lambda$$

and change variables in the two tail integrals. We obtain

$$\mathbb{E}(M^*)^p \leq 2^p C \delta^2 \mathbb{E}(M^*)^p + 2^p \delta^{-p} \mathbb{E}A^p.$$

Choose  $\delta$  so that  $2^p C \delta^2 < 1/2$  and absorb the first term. This gives the upper BDG bound.

The converse good- $\lambda$  estimate has the same form with  $A$  and  $M^*$  interchanged:

$$\mathbb{P}(A > 2\lambda) \leq C' \delta^2 \mathbb{P}(A > \lambda) + \mathbb{P}(M^* > \delta\lambda).$$

The same layer-cake calculation, with  $\delta$  chosen so that the first term is absorbable, gives

$$\mathbb{E}A^p \leq C'_p \mathbb{E}(M^*)^p.$$

Equivalently  $c_p \mathbb{E}A^p \leq \mathbb{E}(M^*)^p$ . Localization and monotone convergence remove any temporary boundedness imposed while applying the good- $\lambda$  lemma, completing both inequalities whenever one side is finite.  $\square$

### 1.3 Quadratic covariation

**Definition 1.5.** For continuous processes  $X, Y$ , when the limit exists in probability along refining partitions, define

$$[X, Y]_t = \lim_{|\pi| \rightarrow 0} \sum_{[u, v] \in \pi} (X_v - X_u)(Y_v - Y_u).$$

Write  $[X] = [X, X]$ .

**Theorem 1.6** (Quadratic covariation calculus). *Let*

$$X = X_0 + \int H dB + A, \quad Y = Y_0 + \int K dB + C,$$

where  $A, C$  are continuous finite-variation processes and the integrals are locally square-integrable. Then

$$[X, Y]_t = \int_0^t H_s K_s ds,$$

and

$$X_t Y_t = X_0 Y_0 + \int_0^t X_s dY_s + \int_0^t Y_s dX_s + [X, Y]_t.$$

*Proof.*

$$\sum (\Delta \int H dB)(\Delta \int K dB) \rightarrow \int H K ds$$

in  $L^2$  after localization. Approximation in  $\mathcal{H}^2$  extends the result.

$$X_v Y_v - X_u Y_u = X_u \Delta Y + Y_u \Delta X + \Delta X \Delta Y$$

summed over partitions gives the product formula when the first two sums converge to the stochastic/Lebesgue–Stieltjes integrals and the last to  $[X, Y]$ .

$$\left| \sum \Delta A \Delta Z \right| \leq \max |\Delta Z| \text{Var}(A) \rightarrow 0.$$

Thus finite-variation parts contribute no quadratic covariation. Bilinearity and the bracket formula follow.  $\square$

## 1.4 Itô's formula

**Theorem 1.7** (Itô formula). *Let  $X$  be a continuous  $d$ -dimensional semimartingale and  $f \in C^2(\mathbb{R}^d)$ . Then*

$$f(X_t) = f(X_0) + \sum_i \int_0^t \partial_i f(X_s) dX_s^i + \frac{1}{2} \sum_{i,j} \int_0^t \partial_{ij} f(X_s) d[X^i, X^j]_s.$$

*Proof.* Localize so that  $X$  stays in a compact set on which the first two derivatives of  $f$  are uniformly continuous and so that the semimartingale characteristics are bounded. Let  $\pi = \{0 = t_0 < \dots < t_n = t\}$  be a partition and write  $\Delta_k X = X_{t_{k+1}} - X_{t_k}$ . Taylor's formula gives

$$f(X_{t_{k+1}}) - f(X_{t_k}) = \partial_i f(X_{t_k}) \Delta_k X^i + \frac{1}{2} \partial_{ij} f(X_{t_k}) \Delta_k X^i \Delta_k X^j + r_k,$$

with

$$|r_k| \leq \omega(\max_j |\Delta_k X^j|) |\Delta_k X|^2,$$

where  $\omega(\delta) \rightarrow 0$  as  $\delta \downarrow 0$ . Summing the first-order terms and letting  $|\pi| \rightarrow 0$  gives the semimartingale integrals

$$\sum_k \partial_i f(X_{t_k}) \Delta_k X^i \longrightarrow \int_0^t \partial_i f(X_s) dX_s^i$$

in probability, by the definition of stochastic integration for the local-martingale part and ordinary Riemann–Stieltjes convergence for the finite-variation part.

For the quadratic term, Theorem 1.6 gives

$$\sum_k \Delta_k X^i \Delta_k X^j \longrightarrow [X^i, X^j]_t$$

in probability. Since  $\partial_{ij} f(X_{t_k})$  is continuous and adapted, the weighted sums converge to

$$\int_0^t \partial_{ij} f(X_s) d[X^i, X^j]_s.$$

Finally, continuity of  $X$  implies  $\max_k |\Delta_k X| \rightarrow 0$  in probability, while the sums  $\sum_k |\Delta_k X|^2$  are tight after localization because their local-martingale part converges to the quadratic variation and the finite-variation contributions vanish by Theorem 1.6. Hence  $\sum_k r_k \rightarrow 0$  in probability. Passing to the limit in the telescoping identity proves the formula up to the localization time; increasing the localizing sequence proves it for all  $t$ .  $\square$

## Chapter 2

# Stochastic Exponentials and Change of Measure

A drift can sometimes be removed not by changing the sample paths but by changing their probabilities. The mechanism is an exponential martingale. This chapter makes the construction precise and isolates the integrability condition needed to turn a positive local martingale into a genuine density.

### 2.1 The stochastic exponential

**Theorem 2.1** (Doleans–Dade exponential). *If  $M$  is a continuous local martingale with  $M_0 = 0$ , then*

$$\mathcal{E}(M)_t = \exp \left( M_t - \frac{1}{2}[M]_t \right)$$

*is the unique strictly positive solution of*

$$Z_t = 1 + \int_0^t Z_s dM_s.$$

*It is a nonnegative local martingale, hence a supermartingale.*

*Proof.* Apply Itô's formula to

$$f(m, q) = \exp \left( m - \frac{1}{2}q \right), \quad (m, q) = (M_t, [M]_t).$$

Since  $d[M]_t$  is the quadratic variation of  $M$ ,

$$d\mathcal{E}(M)_t = \mathcal{E}(M)_t dM_t + \frac{1}{2}\mathcal{E}(M)_t d[M]_t - \frac{1}{2}\mathcal{E}(M)_t d[M]_t = \mathcal{E}(M)_t dM_t.$$

Thus  $\mathcal{E}(M)_0 = 1$  and

$$\mathcal{E}(M)_t = 1 + \int_0^t \mathcal{E}(M)_s dM_s.$$

The exponential is strictly positive. Conversely, if  $Z$  is a strictly positive continuous solution, Itô's formula for  $\log Z$  gives

$$d \log Z_t = \frac{dZ_t}{Z_t} - \frac{1}{2} \frac{d[Z]_t}{Z_t^2} = dM_t - \frac{1}{2} d[M]_t,$$

because  $d[Z]_t = Z_t^2 d[M]_t$ . Since  $Z_0 = 1$ , integration yields  $Z_t = \exp(M_t - [M]_t/2)$ , proving uniqueness among strictly positive solutions. Finally, the stochastic integral representation shows that  $\mathcal{E}(M)$  is a local martingale; being nonnegative, it is a supermartingale by Fatou's lemma applied to a localizing sequence.  $\square$

## 2.2 Novikov's criterion

The usual one-line proof of Novikov's criterion is deceptively easy to write incorrectly: the hypothesis gives an exponential moment exactly at the critical coefficient  $1/2$ , so an argument that silently asks for an  $L^p$ ,  $p > 1$ , bound of the density generally asks for a stronger assumption. We therefore isolate the auxiliary argument explicitly.

**Lemma 2.2** (Krylov–Kazamaki auxiliary criterion). *Let  $N$  be a continuous local martingale on  $[0, T]$ ,  $N_0 = 0$ . Suppose  $N^+$  is a submartingale and, for some  $\eta > 0$ ,*

$$\sup_{t \leq T} \mathbb{E} \exp\left(\frac{1+\eta}{2} N_t\right) < \infty.$$

*Then  $\mathcal{E}(N)$  is a uniformly integrable martingale on  $[0, T]$ .*

*Proof.* Put  $R = \mathcal{E}(N)$ . Choose bounded stopping times  $\tau_n \uparrow T$  localizing  $R$ . Fix  $\kappa > 1$  and  $p > 1$ , put  $q = p/(p-1)$  and  $\gamma = (p\kappa)^{-1/2}$ . Since

$$R_{\tau_n}^\kappa = \mathcal{E}(p\gamma\kappa N)_{\tau_n}^{1/p} \exp((1-\gamma)\kappa N_{\tau_n}),$$

Hölder's inequality and the supermartingale property of every nonnegative stochastic exponential give

$$\mathbb{E} R_{\tau_n}^\kappa \leq (\mathbb{E} \mathcal{E}(p\gamma\kappa N)_{\tau_n})^{1/p} \left( \mathbb{E} e^{\delta N_{\tau_n}} \right)^{1/q} \leq \left( \sup_{t \leq T} \mathbb{E} e^{\delta N_t^+} \right)^{1/q},$$

where  $\delta = (1-\gamma)\kappa q$ . As first  $\kappa \downarrow 1$  and then  $p \downarrow 1$ , one has  $\delta \downarrow 1/2$ . Hence  $\kappa, p$  may be chosen so that  $\delta \leq (1+\eta)/2$ . Doob's  $L^\kappa$  inequality now yields

$$\sup_n \mathbb{E} \sup_{t \leq T} R_{t \wedge \tau_n}^\kappa < \infty.$$

Thus the stopped terminal values are uniformly integrable. Letting  $n \rightarrow \infty$  gives  $\mathbb{E} R_T = 1$ , and the nonnegative local martingale  $R$  is therefore a uniformly integrable martingale on the finite horizon.  $\square$

**Lemma 2.3** (A slightly stronger quadratic-variation condition). *Let  $N$  be a continuous local martingale with  $N_0 = 0$ . If, for some  $\alpha > 0$ ,*

$$\mathbb{E} \exp\left(\frac{1+\alpha}{2} [N]_T\right) < \infty,$$

*then  $\mathbb{E} \mathcal{E}(N)_T = 1$ .*

*Proof.* Choose  $\eta > 0$  with  $(1 + \eta)^2 \leq 1 + \alpha$ . The exponential moment of  $[N]_T$  implies  $\mathbb{E}[N]_T^{1/2} < \infty$ ; by BDG and localization,  $N$  is a true martingale of class  $L^1$  on  $[0, T]$ , so  $N^+$  is a submartingale. Moreover, writing  $V_t = [N]_t$  and using Cauchy–Schwarz,

$$\begin{aligned}\mathbb{E}e^{(1+\eta)N_t/2} &= \mathbb{E} \left[ \mathcal{E}((1+\eta)N)_t^{1/2} e^{(1+\eta)^2 V_t/4} \right] \\ &\leq (\mathbb{E} \mathcal{E}((1+\eta)N)_t)^{1/2} \left( \mathbb{E} e^{(1+\eta)^2 V_t/2} \right)^{1/2} \\ &\leq \left( \mathbb{E} e^{(1+\alpha) V_T/2} \right)^{1/2}.\end{aligned}$$

The first factor is at most one because the stochastic exponential is a nonnegative local martingale. Lemma 2.2 applies.  $\square$

**Theorem 2.4** (Novikov criterion). *Let  $M$  be a continuous local martingale on  $[0, T]$ ,  $M_0 = 0$ . If*

$$\mathbb{E} \exp \left( \frac{1}{2} [M]_T \right) < \infty,$$

*then  $\mathcal{E}(M)$  is a uniformly integrable martingale on  $[0, T]$  and  $\mathbb{E} \mathcal{E}(M)_T = 1$ .*

*Proof.* Write  $V = [M]_T$  and  $Z = \mathcal{E}(M)_T$ . Because  $\mathcal{E}(M)$  is a nonnegative local martingale,  $\mathbb{E} Z \leq 1$ .

$$(1 + \alpha_\varepsilon)(1 - \varepsilon)^2 \leq 1.$$

The hypothesis then implies the stronger condition of Lemma 2.3 for  $M^{(\varepsilon)}$ . Hence

$$1 = \mathbb{E} \mathcal{E}((1 - \varepsilon)M)_T.$$

$$\mathcal{E}((1 - \varepsilon)M)_T = Z^{1-\varepsilon} \exp \left( \frac{\varepsilon(1 - \varepsilon)}{2} V \right).$$

Hölder's inequality with conjugate exponents  $1/(1 - \varepsilon)$  and  $1/\varepsilon$  gives

$$1 \leq (\mathbb{E} Z)^{1-\varepsilon} \left( \mathbb{E} e^{(1-\varepsilon)V/2} \right)^\varepsilon.$$

Since  $\mathbb{E} e^{V/2} < \infty$ ,

$$\varepsilon \log \mathbb{E} e^{(1-\varepsilon)V/2} \leq \varepsilon \log \mathbb{E} e^{V/2} \longrightarrow 0.$$

Letting  $\varepsilon \downarrow 0$  yields  $1 \leq \mathbb{E} Z$ . Together with  $\mathbb{E} Z \leq 1$ , this proves  $\mathbb{E} Z = 1$ .  $\square$

## 2.3 Girsanov's theorem

**Theorem 2.5** (Cameron–Martin–Girsanov). *Let  $B$  be a  $d$ -dimensional Brownian motion and let  $\theta$  be progressively measurable with*

$$\mathbb{E} \exp \left( \frac{1}{2} \int_0^T |\theta_s|^2 ds \right) < \infty.$$

*Define*

$$Z_T = \exp \left( \int_0^T \theta_s \cdot dB_s - \frac{1}{2} \int_0^T |\theta_s|^2 ds \right), \quad d\mathbb{Q} = Z_T d\mathbb{P}.$$



Then under  $\mathbb{Q}$ ,

$$W_t = B_t - \int_0^t \theta_s ds$$

is a standard  $d$ -dimensional Brownian motion on  $[0, T]$ .

*Proof.*

$$\mathbb{E}_{\mathbb{Q}}[X \mid \mathcal{F}_s] = \frac{1}{Z_s} \mathbb{E}_{\mathbb{P}}[Z_t X \mid \mathcal{F}_s]$$

for  $s \leq t$  and bounded  $\mathcal{F}_t$ -measurable  $X$ .

$$[W^i, W^j]_t = \delta_{ij}t.$$

To identify the law without a forward reference, fix  $u \in \mathbb{R}^d$ . Itô's formula applied under  $\mathbb{Q}$  to  $\exp(iu \cdot W_t + |u|^2 t/2)$ , followed by stopping, gives the conditional characteristic function  $\mathbb{E}_{\mathbb{Q}}[e^{iu \cdot (W_t - W_s)} \mid \mathcal{F}_s] = e^{-|u|^2(t-s)/2}$ . Thus the increments are independent centered Gaussians with covariance  $(t-s)I$ , and continuity makes  $W$  a  $\mathbb{Q}$ -Brownian motion. □

**Example 2.6** (Removing a constant drift). If  $X_t = \mu t + B_t$ , take  $\theta = -\mu$ . On a fixed time horizon, the tilted measure with density  $\exp(-\mu B_T - \mu^2 T/2)$  makes  $X$  a Brownian motion. The paths have not changed; only their weights have.

## Chapter 3

# Strong Solutions of Stochastic Differential Equations

Consider

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t, \quad X_0 = \xi.$$

A *strong solution* is adapted to the filtration generated by a prescribed Brownian motion (together with the initial variable). The central question is whether the integral equation can be solved pathwise from the noise.

### 3.1 Picard iteration

**Theorem 3.1** (Strong existence and pathwise uniqueness for Lipschitz SDEs). *Assume  $b, \sigma$  are globally Lipschitz in  $x$ , uniformly in  $t \leq T$ , and have linear growth. If  $\xi \in L^2$  is  $\mathcal{F}_0$ -measurable, then the SDE has a unique strong continuous solution on  $[0, T]$  and*

$$\mathbb{E} \sup_{t \leq T} |X_t|^2 < \infty.$$

*Proof.*

$$X_t^{(n+1)} = \xi + \int_0^t b(s, X_s^{(n)})ds + \int_0^t \sigma(s, X_s^{(n)})dB_s.$$

Every iterate is adapted and continuous.

$$D_n(t) \leq C_T \int_0^t D_{n-1}(s)ds.$$

Iteration yields  $D_n(T) \leq C(C_T T)^n/n!$ . Hence  $X^{(n)}$  is Cauchy in  $L^2(\Omega; C[0, T])$ . Its limit solves the SDE by the same estimates.

$$\mathbb{E} \sup_{u \leq t} |X_u - Y_u|^2 \leq C \int_0^t \mathbb{E} \sup_{v \leq s} |X_v - Y_v|^2 ds.$$

Grönwall forces equality. Applying the estimates to  $X$  itself gives the moment bound.  $\square$

### 3.2 Stability

**Theorem 3.2** (Stability of strong solutions). *Let  $(b_n, \sigma_n, \xi_n)$  and  $(b, \sigma, \xi)$  satisfy a common global Lipschitz and linear-growth bound. If  $\xi_n \rightarrow \xi$  in  $L^2$  and  $b_n, \sigma_n \rightarrow b, \sigma$  uniformly on compact sets, with the coefficient differences along  $X$  converging in  $L^2([0, T] \times \Omega)$ , then*

$$\mathbb{E} \sup_{t \leq T} |X_t^n - X_t|^2 \rightarrow 0.$$

*Proof.* Put  $Y_t = X_t^n - X_t$ . Adding and subtracting  $b_n(X_t)$  and  $\sigma_n(X_t)$  gives

$$\begin{aligned} Y_t = & \xi_n - \xi + \int_0^t \{b_n(X_s^n) - b_n(X_s)\} ds + \int_0^t \{b_n(X_s) - b(X_s)\} ds \\ & + \int_0^t \{\sigma_n(X_s^n) - \sigma_n(X_s)\} dB_s + \int_0^t \{\sigma_n(X_s) - \sigma(X_s)\} dB_s. \end{aligned}$$

Let  $L$  be the common Lipschitz constant. Cauchy–Schwarz for the drift terms and the BDG inequality for the stochastic terms imply, for  $t \leq T$ ,

$$\mathbb{E} \sup_{u \leq t} |Y_u|^2 \leq C_T \mathbb{E} |\xi_n - \xi|^2 + C_T \int_0^t \mathbb{E} \sup_{v \leq s} |Y_v|^2 ds + C_T R_n(T),$$

where

$$R_n(T) = \mathbb{E} \int_0^T |b_n(X_s) - b(X_s)|^2 ds + \mathbb{E} \int_0^T \|\sigma_n(X_s) - \sigma(X_s)\|^2 ds.$$

By hypothesis  $R_n(T) \rightarrow 0$  and  $\mathbb{E} |\xi_n - \xi|^2 \rightarrow 0$ . Gronwall therefore yields

$$\mathbb{E} \sup_{t \leq T} |X_t^n - X_t|^2 \leq C_T (\mathbb{E} |\xi_n - \xi|^2 + R_n(T)) \rightarrow 0.$$

□

### 3.3 Strong Markov property

**Theorem 3.3** (Strong Markov property for Lipschitz diffusions). *For time-homogeneous Lipschitz coefficients and deterministic initial state, the unique strong solution  $X$  is a strong Markov process: for every bounded Borel  $f$ , stopping time  $\tau$ , and  $t \geq 0$ ,*

$$\mathbb{E}[f(X_{\tau+t}) \mid \mathcal{F}_\tau] = P_t f(X_\tau), \quad P_t f(x) = \mathbb{E}_x f(X_t).$$

*Proof.* First let  $\tau$  be bounded. Define the shifted noise

$$B_t^\tau = B_{\tau+t} - B_\tau, \quad t \geq 0.$$

The strong Markov property of Brownian motion says that  $B^\tau$  is a Brownian motion independent of  $\mathcal{F}_\tau$ . The shifted process  $Y_t = X_{\tau+t}$  satisfies

$$Y_t = X_\tau + \int_0^t b(Y_s) ds + \int_0^t \sigma(Y_s) dB_s^\tau.$$

By pathwise uniqueness for the Lipschitz SDE,  $Y$  is the measurable strong-solution map applied to the initial value  $X_\tau$  and the future Brownian increments. Consequently, for bounded Borel  $f$ ,

$$\mathbb{E}[f(X_{\tau+t}) \mid \mathcal{F}_\tau] = \mathbb{E}[f(Y_t) \mid X_\tau] = P_t f(X_\tau).$$

For a general stopping time replace  $\tau$  by  $\tau \wedge n$  and let  $n \rightarrow \infty$ . Continuity of  $X$ , bounded convergence, and right-continuity of the filtration give the stated identity. □

### 3.4 Yamada–Watanabe

**Definition 3.4.** A *weak solution* consists of a filtered probability space, a Brownian motion  $B$ , and an adapted process  $X$  satisfying the SDE. *Pathwise uniqueness* means that two weak solutions on the same filtered space with the same Brownian motion and initial variable are indistinguishable.

**Theorem 3.5** (Yamada–Watanabe). *For an Itô SDE on a Polish path space with Borel nonanticipating coefficients, weak existence together with pathwise uniqueness implies the existence of a strong solution and uniqueness in law.*

*Proof.* Fix a finite horizon  $T$ , let  $\mathcal{W} = C([0, T], \mathbb{R}^m)$  and  $\mathcal{X} = C([0, T], \mathbb{R}^d)$ , and let  $Q(dw, dx)$  be the joint law of a weak solution  $(W, X)$ . Disintegrate

$$Q(dw, dx) = \mathbb{W}(dw)K(w, dx),$$

where  $\mathbb{W}$  is Wiener measure. On the product space with law

$$\mathbb{W}(dw)K(w, dx_1)K(w, dx_2),$$

the two coordinates  $X^1, X^2$ , conditionally on the same Brownian path  $W = w$ , are weak solutions driven by that same noise. The nonanticipating character of the coefficients implies that both are adapted to the enlarged filtration. Pathwise uniqueness therefore gives  $X^1 = X^2$  almost surely.

For every set  $A$  in a countable algebra generating the Borel sigma-field of  $\mathcal{X}$ ,

$$0 = \mathbb{P}(X^1 \in A, X^2 \notin A) = \int K(w, A)(1 - K(w, A)) \mathbb{W}(dw).$$

Thus, outside one Wiener-null set,  $K(w, A) \in \{0, 1\}$  simultaneously for all generators. Since those generators separate points of the Polish space  $\mathcal{X}$ ,  $K(w, \cdot)$  is a Dirac mass. The set of Dirac measures is Borel in  $\mathcal{P}(\mathcal{X})$ , hence there is a measurable map  $F : \mathcal{W} \rightarrow \mathcal{X}$  with

$$K(w, \cdot) = \delta_{F(w)}.$$

Compatibility of the conditional kernels with the canonical filtrations implies, first at rational times and then by continuity, that  $F(w)|_{[0, t]}$  depends only on  $w|_{[0, t]}$ . Therefore  $X = F(W)$  is adapted to the Brownian filtration: a strong solution exists.

If  $Q_1, Q_2$  are two weak-solution laws, mix them with an independent Bernoulli variable and repeat the conditional-product argument. Pathwise uniqueness again forces the two conditional solution kernels to be the same Dirac mass for Wiener-a.e.  $w$ , so  $Q_1 = Q_2$ . Hence uniqueness in law also holds. Applying the argument on  $[0, T]$  for every integer  $T$  gives the global statement.  $\square$

*Remark 3.6.* The proof is measure-theoretic rather than analytic. The decisive input is that pathwise uniqueness collapses the conditional distribution of the solution given the noise to a point mass.

## Chapter 4

# Weak Solutions, Martingale Problems, and Generators

The martingale-problem formulation removes the Brownian motion from the statement and remembers only the local drift and covariance. This is often the right language for weak convergence: one can identify a limiting process through the martingales it satisfies, even when the driving noises do not converge explicitly.

**Definition 4.1.** For

$$Lf(x) = b(x) \cdot \nabla f(x) + \frac{1}{2} \operatorname{tr}(a(x) D^2 f(x)), \quad a = \sigma \sigma^\top,$$

a probability law on  $C([0, \infty), \mathbb{R}^d)$  solves the *L-martingale problem* from  $x$  if the coordinate process  $X_0 = x$  and for every  $f \in C_c^2$ ,

$$f(X_t) - f(X_0) - \int_0^t Lf(X_s) ds$$

is a local martingale.

### 4.1 Weak SDEs and martingale problems

**Lemma 4.2** (Lévy's characterization). *If  $M$  is a continuous  $d$ -dimensional local martingale with  $M_0 = 0$  and*

$$[M^i, M^j]_t = \delta_{ij}t,$$

*then  $M$  is a standard Brownian motion.*

*Proof.* For  $\theta \in \mathbb{R}^d$ , Itô's formula shows that

$$\exp \left( i\theta \cdot M_t + \frac{1}{2} |\theta|^2 t \right)$$

is a local martingale; after stopping it is bounded in modulus by  $e^{|\theta|^2 T/2}$  and hence a true martingale. Therefore, for  $s < t$ ,

$$\mathbb{E}[e^{i\theta \cdot (M_t - M_s)} \mid \mathcal{F}_s] = e^{-|\theta|^2(t-s)/2}.$$

This is the characteristic function of an independent  $N(0, (t-s)I)$  increment. Continuity then gives Brownian motion.  $\square$

**Theorem 4.3** (Weak SDEs and the martingale problem). *Assume  $b, \sigma$  are Borel and locally bounded, and suppose  $a = \sigma\sigma^\top$  admits a measurable square root of the stated dimension. Every weak solution of*

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t$$

*solves the  $L$ -martingale problem. Conversely, any solution of the martingale problem can, after enlarging the probability space if necessary, be represented as such a weak SDE solution.*

*Proof.*

$$f(X_t) - f(X_0) - \int_0^t Lf(X_s)ds = \int_0^t \nabla f(X_s)\sigma(X_s)dB_s,$$

a local martingale.

$$M_t = X_t - X_0 - \int_0^t b(X_s)ds$$

is a continuous local martingale with

$$[M^i, M^j]_t = \int_0^t a_{ij}(X_s)ds.$$

$$B_t = \int_0^t \sigma(X_s)^{-1}dM_s.$$

Then  $[B^i, B^j]_t = \delta_{ij}t$ , so Lemma 4.2 makes  $B$  Brownian and  $M = \int \sigma(X)dB$ . In the degenerate case, use a measurable generalized inverse for the range of  $\sigma$  and enlarge the space with an independent Brownian motion on the orthogonal complement. The resulting  $B$  is standard Brownian and again represents  $M$ . Thus the coordinate process is a weak SDE solution.  $\square$

**Theorem 4.4** (Weak uniqueness and martingale-problem uniqueness). *Under the hypotheses of Theorem 4.3, uniqueness in law for the weak SDE is equivalent to uniqueness of the associated martingale problem for the coordinate-process law.*

*Proof.* Theorem 4.3 identifies the coordinate-process law of every weak SDE solution with a solution of the  $L$ -martingale problem. Thus uniqueness of the martingale problem immediately implies uniqueness in law for the SDE.

Conversely, let  $P_1$  and  $P_2$  solve the martingale problem from the same initial law. By the converse half of Theorem 4.3, after enlarging the two probability spaces if necessary, the coordinate processes admit Brownian representations and hence are weak solutions of the same SDE. Weak uniqueness gives equality of their coordinate-process laws, so  $P_1 = P_2$ . Therefore the two notions of uniqueness are equivalent.  $\square$

## 4.2 Dynkin's formula

**Theorem 4.5** (Dynkin formula). *Let  $X$  solve the  $L$ -martingale problem and let  $\tau$  be a stopping time such that the stopped terms below are integrable. Then for  $f \in C^2$  in the domain of  $L$ ,*

$$\mathbb{E}_x f(X_{t \wedge \tau}) = f(x) + \mathbb{E}_x \int_0^{t \wedge \tau} Lf(X_s)ds.$$

*Proof.* By the martingale-problem assumption,

$$N_t = f(X_t) - f(X_0) - \int_0^t Lf(X_s) ds$$

is a local martingale. Let  $\tau_m$  localize  $N$ . Optional stopping at the bounded stopping time  $t \wedge \tau \wedge \tau_m$  gives

$$\mathbb{E}_x f(X_{t \wedge \tau \wedge \tau_m}) = f(x) + \mathbb{E}_x \int_0^{t \wedge \tau \wedge \tau_m} Lf(X_s) ds.$$

The integrability assumption permits  $m \rightarrow \infty$  by dominated or uniform-integrability convergence, yielding the stated formula.  $\square$

### 4.3 The backward equation

**Theorem 4.6** (Kolmogorov backward equation). *Assume  $b, \sigma$  are bounded with sufficiently many bounded derivatives and  $a = \sigma\sigma^\top$  is uniformly elliptic. For  $f \in C_b^2$  set*

$$u(t, x) = P_{T-t}f(x) = \mathbb{E}_x[f(X_{T-t})].$$

Then  $u \in C^{1,2}$  and solves

$$\partial_t u + Lu = 0, \quad u(T, x) = f(x).$$

Within the bounded  $C^{1,2}$  class this solution is unique.

*Proof.* We separate spatial regularity from the semigroup identity. Under the stated bounded-derivative hypotheses the solution flow  $x \mapsto X_s^x$  is twice differentiable. Its Jacobian  $J_s = D_x X_s^x$  and Hessian  $H_s = D_x^2 X_s^x$  solve the variational SDEs obtained by differentiating the coefficients. BDG and Gronwall give, on every finite time interval, uniform finite-moment bounds for  $J_s$  and  $H_s$ . Difference quotients in the SDE, followed by these bounds, justify differentiation under expectation and give

$$DP_s f(x) = \mathbb{E}[Df(X_s^x)J_s],$$

and the analogous formula for  $D^2 P_s f$  involving  $H_s$  and  $D^2 f$ . Consequently  $P_s f \in C_b^2$  for  $s > 0$  (and continuously down to  $s = 0$  for  $f \in C_b^2$ ).

Dynkin's formula yields

$$P_h g - g = \int_0^h P_r Lg dr \tag{3.10}$$

for  $g \in C_b^2$  for which  $Lg$  is bounded. Apply (3.10) to  $g = P_s f$ , use the semigroup property, and divide by  $h$ . Continuity of  $r \mapsto P_r Lg$  gives

$$\partial_s P_s f = L P_s f.$$

Putting  $s = T - t$  therefore gives  $u \in C^{1,2}$  and  $\partial_t u + Lu = 0$ , with  $u(T, \cdot) = f$ .

For uniqueness, let  $v$  be a bounded  $C^{1,2}$  solution with the same terminal data. Itô's formula applied to  $r \mapsto v(t+r, X_r^x)$  shows that this process is a bounded local martingale, hence a martingale. At  $r = T - t$ ,

$$v(t, x) = \mathbb{E}_x[v(T, X_{T-t})] = \mathbb{E}_x[f(X_{T-t})] = u(t, x).$$

This proves uniqueness in the stated class.  $\square$

## Chapter 5

# One-Dimensional Diffusions: Scale, Speed, and Boundaries

One dimension has an exceptional simplification: drift can be removed by a monotone change of coordinate. The resulting scale function and the associated speed measure turn hitting and occupation questions into explicit integral tests.

Consider on an interval  $I = (l, r)$

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t,$$

with  $\sigma > 0$  and coefficients regular enough for the calculations below.

### 5.1 Scale and hitting probabilities

**Theorem 5.1** (Scale-function hitting formula). *Fix  $x_0 \in I$  and define*

$$s'(x) = \exp\left(-\int_{x_0}^x \frac{2b(y)}{\sigma^2(y)}dy\right), \quad s(x) = \int_{x_0}^x s'(y)dy.$$

*For  $a < x < b$  in  $I$  and  $\tau = \tau_a \wedge \tau_b$ ,*

$$\mathbb{P}_x(\tau_b < \tau_a) = \frac{s(x) - s(a)}{s(b) - s(a)}.$$

*Proof.*

$$bs' + \frac{1}{2}\sigma^2 s'' = 0.$$

$$ds(X_t) = s'(X_t)\sigma(X_t)dB_t,$$

so  $s(X_{t \wedge \tau})$  is a bounded martingale.

$$s(x) = s(a)\mathbb{P}_x(\tau_a < \tau_b) + s(b)\mathbb{P}_x(\tau_b < \tau_a),$$

which solves to the stated formula. □



## 5.2 Speed measure and occupation

**Definition 5.2.** The speed measure is

$$m(dx) = \frac{2}{\sigma^2(x)s'(x)}dx.$$

For  $a < b$  define the Green kernel in scale coordinates by

$$G_{a,b}(x, y) = \frac{(s(x \wedge y) - s(a))(s(b) - s(x \vee y))}{s(b) - s(a)}.$$

**Theorem 5.3** (Speed-measure occupation formula). *For  $a < x < b$ ,  $\tau = \tau_a \wedge \tau_b$ , and nonnegative Borel  $f$ ,*

$$\mathbb{E}_x \int_0^\tau f(X_t) dt = \int_a^b G_{a,b}(x, y) f(y) m(dy).$$

*Proof.* With the present normalization of scale and speed,

$$L = \frac{d}{dm} \frac{d}{ds}.$$

Indeed, if  $g$  is  $C^2$ , then  $dg/ds = g'/s'$  and differentiation against  $m(dx) = 2(\sigma^2 s')^{-1}dx$  gives

$$\frac{d}{dm} \frac{dg}{ds} = \frac{\sigma^2 s'}{2} \left( \frac{g'}{s'} \right)' = \frac{\sigma^2}{2} g'' + b g' = Lg,$$

where the defining equation for  $s'$  was used in the last step.

For fixed  $y$ , the function  $x \mapsto G_{a,b}(x, y)$  is continuous, vanishes at  $a, b$ , is affine as a function of  $s(x)$  on each side of  $y$ , and its  $d/ds$  derivative has a downward jump of size 1 at  $y$ . Consequently, in the distributional scale–speed sense,

$$-L_x G_{a,b}(x, y) = \delta_y / m(dy).$$

For bounded smooth  $f$  with compact support in  $(a, b)$  define

$$v(x) = \int_a^b G_{a,b}(x, y) f(y) m(dy).$$

Then  $v(a) = v(b) = 0$  and  $Lv = -f$ . Dynkin's formula at  $t \wedge \tau$  gives

$$\mathbb{E}_x v(X_{t \wedge \tau}) = v(x) - \mathbb{E}_x \int_0^{t \wedge \tau} f(X_s) ds.$$

The killed diffusion exits  $(a, b)$  almost surely under the stated regular coefficient hypotheses, and  $v(X_\tau) = 0$ ; bounded convergence therefore lets  $t \rightarrow \infty$  and yields

$$v(x) = \mathbb{E}_x \int_0^\tau f(X_s) ds.$$

Finally approximate an arbitrary nonnegative Borel  $f$  monotonically by bounded simple functions (and those by bounded Borel functions supported away from the endpoints). Tonelli and monotone convergence on both sides extend the identity to the stated class.  $\square$

### 5.3 Feller's boundary classification

For the right endpoint  $r$ , choose  $c \in I$  and write

$$S_r = \int_c^r (s(r) - s(y))m(dy), \quad N_r = \int_c^r (s(y) - s(c))m(dy),$$

interpreting  $s(r)$  as a one-sided extended limit and using the equivalent standard local versions when  $s(r) = \infty$ . Analogous integrals define  $S_l, N_l$ .

**Theorem 5.4** (Feller boundary classification, regular coefficient form). *For a one-dimensional diffusion with continuous  $b$ , continuous strictly positive  $\sigma$ , and no killing inside  $I$ , an endpoint is classified by the finiteness of its scale-speed accessibility and entrance integrals:*

|          | accessible from interior | may enter from boundary |
|----------|--------------------------|-------------------------|
| regular  | yes                      | yes                     |
| exit     | yes                      | no                      |
| entrance | no                       | yes                     |
| natural  | no                       | no                      |

The equivalent integral tests are those obtained from  $S$  and  $N$  above (or their standard infinite-scale variants).

*Proof.* We prove the classification for the left endpoint; the right endpoint is identical. Let  $c \in (l, r)$  and stop the diffusion on  $(a, c)$  with  $a \downarrow l$ . The scale hitting formula gives

$$\mathbb{P}_x(\tau_a < \tau_c) = \frac{s(c) - s(x)}{s(c) - s(a)}.$$

The speed-measure occupation formula gives the mean time spent before  $\tau_a \wedge \tau_c$  as the integral of the Green kernel against  $m$ . Passing  $a \downarrow l$  and using monotone convergence shows that the endpoint can be reached from the interior exactly when the accessibility integral

$$\Sigma_l = \int_l^c (s(x) - s(l)) m(dx)$$

(with the standard infinite-scale version when  $s(l) = -\infty$ ) is finite.

To decide whether one can start the diffusion from  $l$ , take  $x \downarrow l$  in the same killed Green kernels. The family of occupation laws is tight at positive times exactly when the complementary entrance integral

$$N_l = \int_l^c (s(c) - s(x)) m(dx)$$

is finite. In that case the finite-dimensional transition probabilities have a Markovian weak limit as  $x \downarrow l$ , giving an entrance law; if  $N_l = \infty$ , mass escapes along the scale coordinate and no such entrance law exists.

Therefore the two yes/no tests are independent and give the four possibilities

|                | $\Sigma_l < \infty$ | $\Sigma_l = \infty$ |
|----------------|---------------------|---------------------|
| $N_l < \infty$ | regular             | entrance            |
| $N_l = \infty$ | exit                | natural.            |

A regular endpoint is both reachable and admissible as a starting point; an exit endpoint is reachable but has no entrance law; an entrance endpoint is not reachable from the interior but admits an entrance law; a natural endpoint has neither property. Rewriting  $\Sigma_r, N_r$  at the right endpoint gives precisely the tests stated in the theorem.  $\square$

## 5.4 Engelbert–Schmidt weak existence

**Theorem 5.5** (Engelbert–Schmidt criterion, selected form). *Let  $I$  be an open interval, let  $\sigma : I \rightarrow \mathbb{R}$  be Borel and nonzero, and let  $b : I \rightarrow \mathbb{R}$  be Borel. Assume*

$$\frac{1}{\sigma^2}, \quad \frac{b}{\sigma^2}$$

*are locally integrable on  $I$ . Then from every  $x \in I$  there exists a weak solution up to the first exit/explosion time from  $I$ . In scale coordinates it is obtained as a time change of Brownian motion; its law up to exit is determined by the scale and speed data.*

*Proof.*

$$s'(x) = \exp \left( - \int^x 2b/\sigma^2 \right).$$

The transformed diffusion coefficient is

$$\tilde{\sigma}(s(x)) = s'(x)\sigma(x),$$

and  $1/\tilde{\sigma}^2$  is locally integrable in scale coordinates because

$$\frac{dy}{\tilde{\sigma}(y)^2} = \frac{dx}{s'(x)\sigma(x)^2}.$$

$$A_u = \int_{s(I)} L_u^y(W) \frac{2 dy}{\tilde{\sigma}(y)^2},$$

up to Brownian exit from  $s(I)$ , and let  $T_t = \inf\{u : A_u > t\}$ . Local integrability and the occupation-density formula make  $A$  finite, continuous, and strictly increasing before exit. Then  $Y_t = W_{T_t}$  is a continuous local martingale whose quadratic variation satisfies

$$[Y]_t = \int_0^t \tilde{\sigma}(Y_s)^2 ds.$$

By martingale representation on an enlargement,  $Y$  solves  $dY_t = \tilde{\sigma}(Y_t)dB_t$  weakly.

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t.$$

The construction lasts exactly until  $Y$  exits  $s(I)$ , equivalently until  $X$  exits  $I$ . The Brownian time-change is determined by scale and speed, which also gives uniqueness in law within the nonsticky Engelbert–Schmidt class.  $\square$

## Chapter 6

# Bessel and Squared-Bessel Processes

Bessel processes measure the radial size of Brownian motion, but their importance here is broader: the distance between an SLE driving point and a force point is a Bessel process after rescaling. Hitting zero therefore becomes the algebra behind SLE boundary swallowing.

### 6.1 Squared Bessel processes

**Theorem 6.1** (Squared-Bessel SDE). *For  $\delta \geq 0$  and  $z \geq 0$ , the SDE*

$$dZ_t = 2\sqrt{Z_t} dB_t + \delta dt, \quad Z_0 = z,$$

*has a pathwise unique nonnegative strong solution. Zero is instantaneously reflecting for  $0 < \delta < 2$ , absorbing when  $\delta = 0$ , and is not hit from  $z > 0$  when  $\delta \geq 2$ .*

*Proof.* The diffusion coefficient satisfies the Yamada–Watanabe modulus condition

$$|2\sqrt{x} - 2\sqrt{y}|^2 \leq 4|x - y|, \quad x, y \geq 0.$$

For two nonnegative solutions driven by the same Brownian motion, apply Itô–Tanaka to smooth convex approximations of  $|x|$  whose second derivatives are concentrated near the origin. The preceding modulus makes the second-derivative term tend to zero, while the equal drifts cancel. Thus  $\mathbb{E}|Z_t^1 - Z_t^2| = 0$  and pathwise uniqueness holds. Approximate  $2\sqrt{x^+}$  by Lipschitz coefficients, use the usual moment/tightness bounds to obtain a nonnegative weak solution, and invoke Yamada–Watanabe to obtain the unique nonnegative strong solution.

It remains to justify the boundary assertions. For  $z > 0$ , before the first hit of zero, Itô’s formula for  $R = \sqrt{Z}$  gives

$$dR_t = dB_t + \frac{\delta - 1}{2R_t} dt.$$

Its scale density is  $s'(r) = r^{1-\delta}$ , so for  $0 < \varepsilon < r < R$ ,

$$\mathbb{P}_r(\tau_\varepsilon < \tau_R) = \frac{s(R) - s(r)}{s(R) - s(\varepsilon)}, \quad s(r) = \begin{cases} r^{2-\delta}/(2-\delta), & \delta \neq 2, \\ \log r, & \delta = 2. \end{cases}$$

If  $\delta \geq 2$ , letting  $\varepsilon \downarrow 0$  shows that zero is polar. If  $0 < \delta < 2$ , letting  $R \uparrow \infty$  and then  $\varepsilon \downarrow 0$  shows that zero is accessible. It cannot be absorbing: if a solution that has hit zero were identically zero on any nontrivial right interval, the integral form

$$Z_t - Z_\tau = 2 \int_\tau^t \sqrt{Z_s} dB_s + \delta(t - \tau)$$

would give  $0 = \delta(t - \tau)$ . Thus the accessible boundary has no holding interval; this is the instantaneous-reflection convention used here. When  $\delta = 0$ , the identically-zero process solves the equation from zero, so pathwise uniqueness makes zero absorbing.  $\square$

## 6.2 Bessel dimension and hitting zero

**Theorem 6.2** (Bessel hitting criterion). *Let  $R_t = \sqrt{Z_t}$  be a Bessel process of dimension  $\delta$ , started from  $r > 0$ . Before hitting zero,*

$$dR_t = dB_t + \frac{\delta - 1}{2R_t} dt.$$

*Then*

$$\mathbb{P}_r(\tau_0 < \infty) = 1 \quad \text{if } \delta < 2, \quad \mathbb{P}_r(\tau_0 < \infty) = 0 \quad \text{if } \delta \geq 2.$$

*Proof.*

$$s'(r) = \exp\left(-\int^r \frac{\delta - 1}{y} dy\right) = r^{1-\delta}.$$

Thus one may take

$$s(r) = \begin{cases} r^{2-\delta}/(2-\delta), & \delta \neq 2, \\ \log r, & \delta = 2. \end{cases}$$

$$\mathbb{P}_r(\tau_\varepsilon < \tau_R) = \frac{s(R) - s(r)}{s(R) - s(\varepsilon)}.$$

If  $\delta < 2$ , let  $R \rightarrow \infty$  and then  $\varepsilon \downarrow 0$  to get probability one of approaching/hitting zero. If  $\delta \geq 2$ , let  $\varepsilon \downarrow 0$  first;  $s(\varepsilon) \rightarrow -\infty$ , so the probability tends to zero.  $\square$

## 6.3 Changing the Bessel dimension

**Theorem 6.3** (Absolute continuity of Bessel laws). *Let  $R$  be  $BES^\delta$  started from  $r > 0$ , stopped before hitting zero, and let  $\nu = \delta/2 - 1$ . For another index  $\mu$  (dimension  $2(\mu + 1)$ ), the stopped laws are locally absolutely continuous with density*

$$\left(\frac{R_t}{r}\right)^{\mu-\nu} \exp\left[-\frac{\mu^2 - \nu^2}{2} \int_0^t \frac{ds}{R_s^2}\right]$$

*up to stopping times bounded away from zero.*

*Proof.* Before hitting zero a BES process of index  $\nu$  satisfies

$$dR_t = dB_t + \frac{\nu + 1/2}{R_t} dt.$$

Fix  $\varepsilon > 0$  and stop at  $\tau_\varepsilon = \inf\{t : R_t \leq \varepsilon\}$ . On  $[0, t \wedge \tau_\varepsilon]$  the integrand  $1/R_s$  is bounded, so Novikov's condition holds for

$$Z_t^{(\varepsilon)} = \exp\left((\mu - \nu) \int_0^{t \wedge \tau_\varepsilon} \frac{dB_s}{R_s} - \frac{(\mu - \nu)^2}{2} \int_0^{t \wedge \tau_\varepsilon} \frac{ds}{R_s^2}\right).$$

Thus  $Z^{(\varepsilon)}$  is a true martingale. Under the probability with density  $Z_t^{(\varepsilon)}$ , Girsanov's theorem makes

$$\tilde{B}_t = B_t - (\mu - \nu) \int_0^{t \wedge \tau_\varepsilon} \frac{ds}{R_s}$$

a Brownian motion up to the stopping time, and the SDE becomes

$$dR_t = d\tilde{B}_t + \frac{\mu + 1/2}{R_t} dt.$$

Hence the changed law is precisely BES of index  $\mu$  up to  $t \wedge \tau_\varepsilon$ .

It remains to rewrite the density. Itô's formula gives

$$d \log R_t = \frac{dB_t}{R_t} + \frac{\nu}{R_t^2} dt,$$

so

$$(\mu - \nu) \int_0^t \frac{dB_s}{R_s} = (\mu - \nu) \log \frac{R_t}{r} - \nu(\mu - \nu) \int_0^t \frac{ds}{R_s^2}.$$

Combining the drift terms gives  $-(\mu^2 - \nu^2) \int ds/(2R_s^2)$ , which is exactly the stated density. Since every stopping time bounded away from zero is dominated by some  $\tau_\varepsilon$ , the local absolute-continuity statement follows.  $\square$

## Chapter 7

# Feynman–Kac and Probabilistic Dirichlet Problems

Itô's formula turns differential equations into martingales. Feynman–Kac runs the implication backward: a weighted expectation of diffusion paths solves a parabolic PDE. Stopping the paths at a boundary gives elliptic and parabolic boundary-value representations.

### 7.1 Bounded potentials

**Theorem 7.1** (Feynman–Kac). *Assume the diffusion coefficients satisfy the bounded smoothness and uniform ellipticity hypotheses of Theorem 4.6. Let  $V \in C_b^1(\mathbb{R}^d)$  and  $f \in C_b^2(\mathbb{R}^d)$ . Define*

$$u(t, x) = \mathbb{E}_x \left[ \exp \left( - \int_0^t V(X_s) ds \right) f(X_t) \right].$$

*Then, on every finite time interval,  $u$  is the bounded classical solution of*

$$\partial_t u = Lu - Vu, \quad u(0, x) = f(x).$$

*Proof.* Let  $P_t$  be the diffusion semigroup. Expanding the exponential and using the Markov property at the ordered times  $0 < s_1 < \dots < s_n < t$  gives the uniformly convergent Dyson series

$$u(t) = P_t f + \sum_{n \geq 1} (-1)^n \int_{0 < s_1 < \dots < s_n < t} P_{s_1} (V P_{s_2 - s_1} (V \dots P_{t - s_n} f)) ds_1 \dots ds_n.$$

The  $n$ th term has sup norm at most  $\|f\|_\infty (\|V\|_\infty t)^n / n!$ , so termwise rearrangement is justified. Splitting at the first insertion yields the Duhamel equation

$$u(t) = P_t f - \int_0^t P_s (Vu(t - s)) ds. \tag{3.20}$$

The semigroup regularity proved in Theorem 4.6, together with the bounded derivatives of  $V$  and  $f$ , permits differentiation of (3.20) on compact time intervals. Using  $\partial_t P_t g = LP_t g$  gives

$$\partial_t u = Lu - Vu, \quad u(0) = f.$$

Thus the expectation really constructs a classical solution; the PDE has not been assumed in order to prove the representation.

For uniqueness, let  $v$  be any other bounded classical solution and fix  $t > 0$ . For  $0 \leq s \leq t$  set

$$Z_s = \exp\left(-\int_0^s V(X_r)dr\right) v(t-s, X_s).$$

Itô's formula and  $\partial_t v = Lv - Vv$  show that  $Z$  has zero drift. Boundedness makes it a true martingale, hence

$$v(t, x) = \mathbb{E}_x Z_t = \mathbb{E}_x \left[ e^{-\int_0^t V(X_s)ds} f(X_t) \right] = u(t, x).$$

Therefore the constructed solution is unique in the bounded classical class.  $\square$

## 7.2 Stopping at a domain boundary

**Definition 7.2** (Regular domain for a diffusion). Let  $X$  be a diffusion and  $D$  a bounded domain. A point  $z \in \partial D$  is *regular for  $D$  (with respect to  $X$ )* if the diffusion started from  $z$  exits  $D$  immediately almost surely, i.e.  $\mathbb{P}_z(\tau_D = 0) = 1$ . We call  $D$   *$X$ -regular* if every boundary point is regular. Under the bounded smooth uniformly elliptic coefficient hypotheses used below, every bounded  $C^2$  domain is  $X$ -regular. Whenever this chapter says “regular domain”, this is the intended probabilistic meaning.

**Theorem 7.3** (Feynman–Kac with exit). *Let the diffusion coefficients satisfy the bounded  $C^3$  smoothness and uniform ellipticity hypotheses of Theorem 4.6 on a neighborhood of  $\overline{D}$ , where  $D$  is a bounded  $X$ -regular domain. Let  $\tau_D$  be the first exit time, let  $V \in C_b^1(\overline{D})$ , fix  $T > 0$ , and let  $f \in C(\overline{D})$  and  $g \in C([0, T] \times \partial D)$  be bounded with  $g(0, z) = f(z)$  on  $\partial D$ . Then for  $0 \leq t \leq T$ ,*

$$u(t, x) = \mathbb{E}_x \left[ e^{-\int_0^t V(X_s)ds} f(X_t) \mathbf{1}_{\{t < \tau_D\}} + e^{-\int_0^{\tau_D} V(X_s)ds} g(t - \tau_D, X_{\tau_D}) \mathbf{1}_{\{\tau_D \leq t\}} \right]$$

*defines a function in  $C([0, T] \times \overline{D}) \cap C^{1,2}((0, T] \times D)$  and is the unique bounded solution in this class of*

$$\partial_t u = Lu - Vu \quad \text{on } (0, T] \times D, \quad u(0, \cdot) = f, \quad u|_{[0, T] \times \partial D} = g.$$

*Proof.* We first prove that any bounded classical solution must have the displayed representation. Let  $v$  be such a solution, fix  $(t, x)$ , put  $\rho = t \wedge \tau_D$ , and for  $0 \leq s \leq \rho$  set

$$Z_s = \exp\left(-\int_0^s V(X_r)dr\right) v(t-s, X_s).$$

Itô's formula gives

$$dZ_s = e^{-\int_0^s V(X_r)dr} \nabla v(t-s, X_s) \sigma(X_s) dB_s,$$

because the drift is  $-\partial_t v + Lv - Vv = 0$ . After localization this is a martingale; boundedness of  $v$  and  $V$  on  $\overline{D}$  makes the stopped family uniformly integrable. Optional stopping at  $\rho$  therefore yields  $v(t, x) = \mathbb{E}_x Z_\rho$ . On  $\{t < \tau_D\}$  the terminal value is  $e^{-\int_0^t V} f(X_t)$ , while on  $\{\tau_D \leq t\}$  it is  $e^{-\int_0^{\tau_D} V} g(t - \tau_D, X_{\tau_D})$ . Thus every classical solution is equal to the formula in the theorem, proving uniqueness once existence is established.

Now let  $u$  denote the displayed expectation. The strong Markov property at a stopping time  $\sigma < t \wedge \tau_D$  gives the dynamic-programming identity

$$u(t, x) = \mathbb{E}_x \left[ e^{-\int_0^\sigma V(X_r)dr} u(t - \sigma, X_\sigma) \right]. \quad (3.21)$$



Choose a ball  $B \Subset D$  and stop at  $\sigma = h \wedge \tau_B$ . For points in a smaller concentric ball and  $0 < h < t$ , the probability of reaching  $\partial B$  before time  $h$  is negligible to every order needed for the local differentiation argument. Applying the whole-space Feynman–Kac regularity from the preceding theorem to the killed/localized diffusion in  $B$ , or equivalently differentiating (3.21) with the stochastic-flow bounds, shows

$$u \in C^{1,2}((0, T] \times D), \quad \partial_t u = Lu - Vu.$$

As  $t \downarrow 0$ , dominated convergence gives  $u(t, x) \rightarrow f(x)$  in the interior. If  $(t_n, x_n) \rightarrow (t, z)$  with  $z \in \partial D$ ,  $X$ -regularity gives  $\tau_D \rightarrow 0$  in probability for starts  $x_n \rightarrow z$ ; continuity of  $g$ , the compatibility  $g(0, z) = f(z)$ , and bounded convergence then give  $u(t_n, x_n) \rightarrow g(t, z)$ . Thus  $u$  extends continuously to the entire parabolic boundary with the prescribed data. This proves existence in the stated class, and the first paragraph gives uniqueness.  $\square$

### 7.3 Dirichlet problems

**Theorem 7.4** (Probabilistic representation of Dirichlet problems). *Let  $X$  have bounded  $C^3$  coefficients and uniformly elliptic generator  $L$  on a neighborhood of a bounded  $X$ -regular domain  $D$ . For  $f \in C(\partial D)$ ,*

$$u(x) = \mathbb{E}_x[f(X_{\tau_D})]$$

*is the unique continuous function on  $\overline{D}$  that is  $L$ -harmonic in  $D$  and equals  $f$  on the boundary.*

*Proof.*

$$u(x) = \mathbb{E}_x u(X_{\tau_U}), \quad \overline{U} \subset D.$$

Dynkin's formula then implies  $Lu = 0$  in the weak sense; elliptic regularity under the stated coefficient hypotheses upgrades  $u$  to a classical  $L$ -harmonic function. If  $x_n \rightarrow z \in \partial D$ ,  $X$ -regularity gives  $\tau_D \rightarrow 0$  in probability under  $\mathbb{P}_{x_n}$  and the exit point converges to  $z$  in probability. Continuity and boundedness of  $f$  therefore give  $u(x_n) \rightarrow f(z)$ , so  $u$  extends continuously with the prescribed boundary values.

If  $v$  is another continuous  $L$ -harmonic function with boundary values  $f$ , then the stopped process  $v(X_{t \wedge \tau_D})$  is a bounded martingale. Letting  $t \rightarrow \infty$  gives

$$v(x) = \mathbb{E}_x v(X_{\tau_D}) = \mathbb{E}_x f(X_{\tau_D}) = u(x),$$

which proves uniqueness.  $\square$

## Appendix A

# Foundational Arguments in Stochastic Calculus

This appendix collects eight high-risk analytic arguments and supplies the proof details used by the main text.

### A.1 BDG: the converse good-lambda estimate

Let  $M$  be a continuous local martingale with  $M_0 = 0$ , write

$$M_t^* = \sup_{s \leq t} |M_s|, \quad A_t = \langle M \rangle_t^{1/2}.$$

The forward estimate  $\mathbb{E}(M_t^*)^p \lesssim_p \mathbb{E}A_t^p$  was already proved by the stopping/exponential-martingale argument in Chapter 1. We now give the reverse inequality.

**Lemma A.1** (Good- $\lambda$  estimate). *For every  $\beta > 1$  there are constants  $c, C > 0$  such that for  $0 < \delta < 1/2$  and  $\lambda > 0$ ,*

$$\mathbb{P}(A_t > \beta\lambda, M_t^* \leq \delta\lambda) \leq C \exp(-c/\delta^2) \mathbb{P}(A_t > \lambda).$$

*Proof.* Let  $\tau = \inf\{s : A_s > \lambda\}$  and, on  $\{\tau < t\}$ , set

$$N_u = M_{(\tau+u) \wedge t} - M_\tau, \quad B_u = \langle M \rangle_{(\tau+u) \wedge t} - \langle M \rangle_\tau.$$

On the event in the statement,  $B_\infty > (\beta^2 - 1)\lambda^2$  while  $|N_u| \leq 2\delta\lambda$  for every  $u$ . For any  $\theta > 0$  the exponential local martingale

$$\exp\{\theta N_u - \tfrac{1}{2}\theta^2 B_u\}$$

becomes a supermartingale after localization. Apply it to the two signs of  $N$  and stop when  $B$  first reaches  $(\beta^2 - 1)\lambda^2$ . If  $|N| \leq 2\delta\lambda$  throughout, then one of the two exponential martingales is at least

$$\exp\{-2\theta\delta\lambda + \tfrac{1}{2}\theta^2(\beta^2 - 1)\lambda^2\}^{-1}$$

in the reciprocal Chernoff estimate. Optimizing at  $\theta = 2\delta/((\beta^2 - 1)\lambda)$  gives an exponential bound  $Ce^{-c/\delta^2}$  after using the conditional form at time  $\tau$ . Multiplication by  $\mathbb{P}(\tau < t) = \mathbb{P}(A_t > \lambda)$  proves the estimate.  $\square$

Integrating the good- $\lambda$  inequality gives the missing direction. Indeed,

$$\mathbb{P}(A_t > \beta\lambda) \leq \mathbb{P}(M_t^* > \delta\lambda) + \varepsilon(\delta)\mathbb{P}(A_t > \lambda),$$

where  $\varepsilon(\delta) = Ce^{-c/\delta^2}$ . Multiply by  $p\lambda^{p-1}$  and integrate. After the change of variables in the two terms,

$$\beta^{-p}\mathbb{E}A_t^p \leq \delta^{-p}\mathbb{E}(M_t^*)^p + \varepsilon(\delta)\mathbb{E}A_t^p.$$

Choose  $\delta$  so that  $\varepsilon(\delta) < \frac{1}{2}\beta^{-p}$  and absorb. Thus

$$\mathbb{E}A_t^p \leq C_p\mathbb{E}(M_t^*)^p.$$

Together with Chapter 1 this closes .

## A.2 Yamada–Watanabe: the conditional-law argument

Fix a finite horizon  $T$ . Let  $\mathcal{W} = C([0, T]; \mathbb{R}^m)$  and  $\mathcal{X} = C([0, T]; \mathbb{R}^d)$  with their Borel sigma-fields. Suppose weak existence holds and pathwise uniqueness holds for solutions driven by the same Brownian motion.

**Lemma A.2** (Conditional solution kernel is Dirac). *Let  $Q(dw, dx)$  be the joint law of a weak solution  $(W, X)$ . Disintegrate*

$$Q(dw, dx) = \mathbb{W}(dw)K(w, dx).$$

*Then for Wiener-a.e.  $w$ ,  $K(w, \cdot)$  is a Dirac mass.*

*Proof.* On the product space with law

$$\mathbb{W}(dw)K(w, dx_1)K(w, dx_2),$$

the coordinate processes  $X^1, X^2$  have, conditionally on the same  $W = w$ , the same solution kernel. The compatibility property inherited from the weak solution says that, for every  $t$ , conditional events concerning  $X|_{[0, t]}$  depend on  $w$  only through  $w|_{[0, t]}$ . Hence both pairs  $(W, X^i)$  are weak solutions on the enlarged filtered space. Pathwise uniqueness implies  $X^1 = X^2$  a.s. Therefore

$$\int K(w, A)(1 - K(w, A)) \mathbb{W}(dw) = 0$$

for every set  $A$  in a countable generating algebra of  $\mathcal{X}$ . Thus  $K(w, A) \in \{0, 1\}$  simultaneously for all generators outside one null set. A probability measure that is 0–1 on such a separating algebra is a Dirac mass.  $\square$

Because  $\mathcal{X}$  is Polish, the map  $w \mapsto K(w, \cdot)$  is measurable and the set of Dirac measures is Borel in  $\mathcal{P}(\mathcal{X})$ . The inverse map  $\delta_x \mapsto x$  is Borel, so there is a measurable map  $F : \mathcal{W} \rightarrow \mathcal{X}$  with  $K(w, \cdot) = \delta_{F(w)}$ . Compatibility at every rational time and continuity show that  $F(w)|_{[0, t]}$  depends only on  $w|_{[0, t]}$ ; hence  $F(W)$  is adapted. This gives a strong solution. Applying the same conditional-product argument to any two weak solutions yields uniqueness in law. This closes in the path-space generality used by the series.

### A.3 Degenerate martingale representation

Let  $M$  be a continuous  $d$ -dimensional local martingale with

$$d\langle M \rangle_t = a_t dt, \quad a_t = \sigma_t \sigma_t^\top,$$

where  $\sigma_t$  is predictable and may be rank-deficient. Let  $\sigma_t^+$  be the Moore–Penrose inverse and

$$P_t = \sigma_t^+ \sigma_t,$$

the orthogonal projection of  $\mathbb{R}^m$  onto  $(\ker \sigma_t)^\perp$ .

Define

$$B_t^{(1)} = \int_0^t \sigma_s^+ dM_s.$$

Then

$$d\langle B^{(1)} \rangle_t = \sigma_t^+ a_t (\sigma_t^+)^T dt = P_t dt.$$

Enlarge the space by an independent  $m$ -dimensional Brownian motion  $\tilde{B}$  and put

$$B_t = B_t^{(1)} + \int_0^t (I - P_s) d\tilde{B}_s.$$

The cross variation vanishes by independence, and  $P_s$  is an orthogonal projection, so

$$d\langle B \rangle_t = (P_t + (I - P_t))dt = I dt.$$

Lévy's characterization makes  $B$  Brownian. Finally,

$$\sigma_t dB_t = \sigma_t \sigma_t^+ dM_t + \sigma_t (I - P_t) d\tilde{B}_t = dM_t,$$

because  $dM_t$  lies in the range of  $\sigma_t$  in quadratic-variation sense and  $\sigma_t(I - P_t) = 0$ . This closes the representation step in .

### A.4 Backward equation: stochastic-flow regularity

We use a theorem whose proof is elementary once coefficients have bounded derivatives of sufficiently high order.

**Lemma A.3** (Differentiable stochastic flow). *Assume  $b, \sigma \in C_b^3$ . For the Lipschitz SDE,  $x \mapsto X_t^x$  has a  $C^2$  modification. The Jacobian  $J_t = D_x X_t^x$  and Hessian  $H_t = D_x^2 X_t^x$  solve the variational SDEs obtained by differentiating the coefficients, and for every finite  $T$  their moments of all finite orders are uniformly bounded on  $t \leq T$ .*

*Proof.* Take difference quotients in the SDE. Lipschitz continuity and BDG–Grönwall give Cauchy convergence in  $L^p(\Omega; C[0, T])$  to the linear variational equation

$$dJ_t = Db(X_t)J_t dt + D\sigma(X_t)J_t dB_t, \quad J_0 = I.$$

Repeat for first derivatives to obtain  $H_t$ . Kolmogorov continuity in the spatial variable yields a jointly continuous  $C^2$  modification. The moment bounds follow from the linear equations and BDG–Grönwall.  $\square$

For  $f \in C_b^2$ ,

$$DP_t f(x) = \mathbb{E}[Df(X_t^x)J_t],$$

and the analogous formula with  $H_t$  gives  $D^2 P_t f$ . Dominated convergence therefore shows  $P_t f \in C_b^2$ . Dynkin's formula gives

$$P_t f - f = \int_0^t P_s Lf \, ds.$$

Apply this first to smooth  $f$  and use the semigroup property to get  $\partial_t P_t f = LP_t f = P_t Lf$  for positive time. Uniform ellipticity is more than is needed for this particular regularity route. Consequently  $u(t, x) = P_{T-t} f(x)$  is  $C^{1,2}$  and solves the backward equation. This closes the analytic gap in under the coefficient hypotheses already stated in the manuscript.

## A.5 Feller's four boundary types

Let  $X$  be a regular diffusion on  $(l, r)$  with increasing scale  $s$  and speed measure  $m$ . Fix  $c \in (l, r)$  and define near  $l$

$$\Sigma_l = \int_l^c (s(x) - s(l)) m(dx), \quad N_l = \int_l^c (s(c) - s(x)) m(dx),$$

with the standard interpretation when  $s(l) = -\infty$ . The Green-kernel formula from Chapter 5 implies that  $\Sigma_l < \infty$  exactly when  $l$  is reachable from the interior in finite time with positive probability. Apply the same Green formula to the diffusion started from points  $x \downarrow l$ ; tightness of the entrance laws is equivalent to  $N_l < \infty$ . Thus:

|                | $\Sigma_l < \infty$ | $\Sigma_l = \infty$ |
|----------------|---------------------|---------------------|
| $N_l < \infty$ | regular             | entrance            |
| $N_l = \infty$ | exit                | natural             |

For the regular case one may impose absorbing, reflecting, or more general Feller boundary conditions; the minimal diffusion is killed on hitting the endpoint. For an exit boundary there is no entrance law because the occupation Green kernel from  $x \downarrow l$  loses tightness. For an entrance boundary the interior cannot hit  $l$ , but the tight entrance distributions define a Markov extension starting at  $l$ . For a natural boundary neither operation is possible. The right endpoint is identical. This closes the classification argument in .

## A.6 Engelbert–Schmidt time change

The construction in Chapter 5 already contained the main argument. Two details are added.

First, if  $s'$  is only absolutely continuous, choose smooth positive  $s'_n$  converging to  $s'$  locally in  $W^{1,1}$  with inverses uniformly convergent on compact sets. Apply ordinary Itô to  $s_n^{-1}(Y)$ , stop inside a compact subinterval, and pass to the limit using the occupation formula. This proves the generalized Itô formula needed to recover the drift.

Second, uniqueness in law follows because every weak solution transformed to scale is a continuous local martingale with quadratic variation

$$\int_0^t \tilde{\sigma}(Y_u)^2 du.$$

Dambis–Dubins–Schwarz represents it as Brownian motion run on this clock; inversion of the occupation-time additive functional recovers exactly the Brownian time-change used in the construction. Hence two solutions have the same law up to exit. This closes in the nonsticky Engelbert–Schmidt class stated there.

## A.7 Squared Bessel at the non-Lipschitz boundary

For  $\delta \geq 0$ , consider

$$dZ_t = 2\sqrt{Z_t} dB_t + \delta dt, \quad Z_0 = z \geq 0.$$

Pathwise uniqueness follows from the Yamada–Watanabe one-dimensional criterion: the diffusion modulus  $\rho(u) = 2\sqrt{u}$  satisfies

$$\int_{0+} \rho(u)^{-2} du = \infty.$$

For existence, replace  $\sqrt{x}$  by globally Lipschitz functions  $q_n(x)$  agreeing with  $\sqrt{x}$  on  $[1/n, n]$  and vanishing for sufficiently negative  $x$ . The solutions  $Z^{(n)}$  are tight on every finite horizon by BDG and the linear moment bound. Tanaka’s formula applied to the negative part shows every subsequential limit is nonnegative. On compact subsets of  $(0, \infty)$  the coefficients converge uniformly, so the martingale problem passes to the limit; at zero, the diffusion coefficient vanishes and the drift is  $\delta \geq 0$ , giving the BESQ equation. Pathwise uniqueness upgrades weak convergence to strong existence.

The scale function of  $R_t = \sqrt{Z_t}$  has derivative  $r^{1-\delta}$ . The scale test therefore gives: zero is hit when  $\delta < 2$ , is polar for  $\delta \geq 2$ , and for  $0 < \delta < 2$  the nonnegative strong solution leaves zero instantaneously after a hit. This closes the boundary construction in .

## A.8 Feynman–Kac and Dirichlet regularity

The stochastic representation itself needs no elliptic black box. Under the  $C_b^3$  coefficient assumptions used above, the weighted semigroup

$$P_t^V f(x) = \mathbb{E}_x \left[ e^{-\int_0^t V(X_s) ds} f(X_t) \right]$$

inherits two spatial derivatives from the stochastic flow whenever  $V, f$  have two bounded derivatives; differentiation produces the same variational processes together with bounded multiplicative terms. The dynamic-programming identity then gives the time derivative and

$$\partial_t u = Lu - Vu.$$

Thus is closed for the manuscript’s smooth-data formulation.

For the elliptic Dirichlet theorem , the function

$$u(x) = \mathbb{E}_x f(X_{\tau_D})$$

is  $L$ -harmonic in the probabilistic sense. On every ball  $B \Subset D$ , stop first at  $\partial B$  and apply the preceding parabolic result to the killed diffusion with smooth boundary data obtained by approximation. Interior stochastic-flow regularity makes  $u \in C^2(B)$  and Dynkin gives  $Lu = 0$ . Boundary regularity is part of the theorem’s explicit hypothesis and yields  $u \rightarrow f$  at  $\partial D$ . The martingale maximum argument gives uniqueness. No global Schauder theorem is needed. This closes the two Feynman–Kac/Dirichlet roots for the stated smooth coefficient and regular-boundary setting.

## A.9 Scope of the foundational arguments

The eight high-risk arguments are proved at the level required by the theorem statements used later in the series. The appendix deliberately uses the smooth bounded-coefficient formulation already stated in Chapters 4 and 7; it does not advertise a rough-coefficient PDE theorem.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise A.4** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise A.5** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise A.6** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise A.7** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise A.8** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.



# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume IV: Complex Analysis, Conformal Geometry, and Planar  
Potential Theory

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# Scope and dependency note

This volume develops the conformal and planar-potential-theoretic language used in the random-walk, SLE, and LQG arcs. Particular attention is paid to extremal length, harmonic measure, the Beurling estimate, and half-plane capacity, because these are the quantitative tools that later turn discrete interfaces into Loewner evolutions.

# Chapter 1

## Holomorphic Foundations and Normal Families

The later theory of SLE uses conformal maps dynamically, but the compactness and zero-counting tools are classical. We collect them here with proofs arranged so that normal-family arguments can be reused without hidden appeals.

### 1.1 Cauchy's theorem and formula

**Lemma 1.1** (Goursat triangle lemma). *If  $f$  is holomorphic on a neighborhood of a closed triangle  $\Delta$ , then*

$$\int_{\partial\Delta} f(z)dz = 0.$$

*Proof.* Subdivide  $\Delta$  into four congruent triangles. The boundary integral of  $\Delta$  is the sum of the four subtriangle boundary integrals because internal edges cancel. Choose a subtriangle whose boundary integral has magnitude at least one quarter of the original; iterate. The nested triangles shrink to a point  $z_0$ . Holomorphy gives

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + o(|z - z_0|).$$

The constant and linear terms integrate to zero around every triangle, while the error integral is  $o(\text{diam } \Delta_n) \cdot O(\text{diam } \Delta_n)$ , hence  $o(4^{-n})$ . But the iterative lower bound says its magnitude is at least  $4^{-n}$  times the original integral. The original integral must therefore vanish.  $\square$

**Theorem 1.2** (Cauchy theorem and integral formula). *If  $D \subset \mathbb{C}$  is simply connected and  $f$  is holomorphic on  $D$ , then every closed piecewise  $C^1$  curve  $\gamma$  in  $D$  satisfies*

$$\int_{\gamma} f(z)dz = 0.$$

*Consequently, if a positively oriented simple closed curve  $\gamma$  and its interior lie in  $D$ , then for  $w$  inside  $\gamma$ ,*

$$f(w) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - w} dz,$$

*and differentiation under the integral gives the higher derivative formulas.*



*Proof.* We first prove homotopy invariance of the integral. If two closed piecewise  $C^1$  curves  $\gamma_0, \gamma_1$  are joined by a homotopy  $H : [0, 1]^2 \rightarrow D$ , compactness of the parameter square lets us subdivide it into finitely many small rectangles whose images lie in disks compactly contained in  $D$ . On each such disk, triangulation and the Goursat triangle lemma show that the integral of  $f$  around the image of the small rectangle is zero. Summing over all rectangles cancels the internal edges and gives

$$\int_{\gamma_0} f(z) dz = \int_{\gamma_1} f(z) dz.$$

Because  $D$  is simply connected, every closed curve is homotopic to a constant curve. Its integral is therefore zero. Piecewise  $C^1$  curves are covered by the same subdivision argument (or by piecewise smooth approximation inside the compact image of the homotopy).

Now fix  $w$  inside a positively oriented simple closed curve  $\gamma$  and write

$$g(z) = \begin{cases} \frac{f(z) - f(w)}{z - w}, & z \neq w, \\ f'(w), & z = w. \end{cases}$$

The removable-singularity theorem makes  $g$  holomorphic, hence  $\int_{\gamma} g(z) dz = 0$ . Therefore

$$\int_{\gamma} \frac{f(z)}{z - w} dz = f(w) \int_{\gamma} \frac{dz}{z - w}.$$

The last integral is  $2\pi i$  because a positively oriented Jordan curve has winding number one about every interior point. This proves the Cauchy formula. If  $w$  ranges over a compact set separated from  $\gamma$ , the kernels  $(z - w)^{-m-1}$  are uniformly bounded on  $\gamma$ , so differentiation under the integral sign gives

$$f^{(m)}(w) = \frac{m!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z - w)^{m+1}} dz.$$

□

## 1.2 Zeros, poles, and Rouché's theorem

**Theorem 1.3** (Argument principle and Rouché). *Let  $f$  be meromorphic inside and on a positively oriented simple closed contour  $\gamma$ , with no zeros or poles on  $\gamma$ . Then*

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = N - P,$$

*where zeros and poles are counted with multiplicity. If  $f, g$  are holomorphic inside and on  $\gamma$  and  $|g| < |f|$  on  $\gamma$ , then  $f$  and  $f + g$  have the same number of zeros inside  $\gamma$ .*

*Proof.* Near a zero  $a$  of order  $m$ , write  $f(z) = (z - a)^m h(z)$  with  $h(a) \neq 0$ . Then

$$\frac{f'}{f} = \frac{m}{z - a} + \frac{h'}{h},$$

so the residue is  $m$ . Near a pole of order  $m$  the residue is  $-m$ . Subtract the corresponding principal parts at all zeros and poles inside  $\gamma$ ; the remainder is holomorphic, and Cauchy's theorem makes its contour integral zero. Summing the residues proves the argument principle.

For Rouché, set  $h_t = f + tg$ ,  $0 \leq t \leq 1$ . On  $\gamma$ ,

$$|h_t - f| = t|g| < |f|,$$

so  $h_t$  never vanishes there. The integer

$$N(t) = \frac{1}{2\pi i} \int_{\gamma} \frac{h'_t(z)}{h_t(z)} dz$$

is, by the argument principle, the number of zeros of  $h_t$  inside  $\gamma$ . The integrand depends continuously on  $t$  uniformly on the compact contour, hence  $N(t)$  is continuous. A continuous integer-valued function on  $[0, 1]$  is constant, so  $N(0) = N(1)$ :  $f$  and  $f + g$  have the same number of zeros, counted with multiplicity.  $\square$

### 1.3 Montel compactness

**Theorem 1.4** (Montel theorem). *A locally bounded family of holomorphic functions on a domain  $D$  is normal: every sequence has a subsequence converging locally uniformly on  $D$  to a holomorphic function.*

*Proof.* Let  $(f_n)$  be a sequence from the locally bounded family. Fix compact sets

$$K_1 \Subset K_2 \Subset \cdots \Subset D, \quad \bigcup_m K_m = D,$$

with each  $K_m$  contained in the interior of  $K_{m+1}$ . Local boundedness gives a constant  $M_m$  such that  $|f_n| \leq M_m$  on a neighborhood of  $K_m$ , uniformly in  $n$ . If  $r_m > 0$  is smaller than the distance from  $K_m$  to the boundary of that neighborhood, Cauchy's estimate gives

$$\sup_{z \in K_m} |f'_n(z)| \leq M_m/r_m.$$

Thus  $(f_n)$  is uniformly bounded and equicontinuous on  $K_m$ . Arzelà–Ascoli gives a subsequence converging uniformly on  $K_1$ ; iterating and taking the diagonal subsequence gives local uniform convergence on all of  $D$  to a continuous function  $f$ .

For every triangle  $T \Subset D$ , local uniform convergence and Cauchy's theorem give

$$\int_{\partial T} f(z) dz = \lim_{j \rightarrow \infty} \int_{\partial T} f_{n_j}(z) dz = 0.$$

Morera's theorem therefore makes  $f$  holomorphic. Hence every sequence has a locally uniformly convergent holomorphic subsequence.  $\square$

### 1.4 Hurwitz's theorem

**Theorem 1.5** (Hurwitz theorem). *Let  $f_n$  be holomorphic on a domain  $D$  and converge locally uniformly to a holomorphic  $f$ . If every  $f_n$  is zero-free and  $f$  is not identically zero, then  $f$  is zero-free. More generally, zeros of a nonzero limit are approximated with multiplicity by zeros of  $f_n$ .*

*Proof.* Assume first that  $f(a) = 0$  and  $f$  is not identically zero. Choose  $r > 0$  so small that  $\overline{B(a, r)} \Subset D$ ,  $a$  is the only zero of  $f$  in the disk, and  $f$  has no zero on  $|z - a| = r$ . Put

$$m_r = \min_{|z-a|=r} |f(z)| > 0.$$

Local uniform convergence gives  $\sup_{|z-a|=r} |f_n(z) - f(z)| < m_r$  for all large  $n$ . Rouché's theorem then says that  $f_n$  and  $f$  have the same number of zeros in  $B(a, r)$ , counted with multiplicity. Thus a zero of multiplicity  $m$  of  $f$  is approximated by exactly  $m$  zeros of  $f_n$  in every sufficiently small disk.

If every  $f_n$  is zero-free and  $f$  had a zero, the preceding paragraph would force  $f_n$  to have a zero for all large  $n$ , a contradiction. Therefore a nonzero locally uniform limit of zero-free holomorphic functions is zero-free. The same Rouché argument at each isolated zero gives the multiplicity statement.  $\square$

## Chapter 2

# Riemann Mapping and Quantitative Univalence

The Riemann mapping theorem says that simply connected planar domains have no conformal moduli until boundary points are marked. Koebe's theorems add the quantitative information later needed for distortion, compactness, and Loewner estimates.

### 2.1 Riemann mapping

**Lemma 2.1** (Holomorphic square roots on simply connected domains). *If  $D$  is simply connected and  $h$  is holomorphic and zero-free on  $D$ , then  $h$  has a holomorphic logarithm and hence a holomorphic square root.*

*Proof.* The function  $h'/h$  is holomorphic. Cauchy's theorem makes its integral path-independent, so

$$L(z) = \int_{z_0}^z \frac{h'(w)}{h(w)} dw + \log h(z_0)$$

is holomorphic and satisfies  $e^L = h$ . Then  $e^{L/2}$  is a square root.  $\square$

**Theorem 2.2** (Riemann mapping theorem). *Every nonempty simply connected proper domain  $D \subsetneq \mathbb{C}$  is conformally equivalent to the unit disk. Given  $a \in D$ , there is a unique conformal map  $f : D \rightarrow \mathbb{D}$  satisfying  $f(a) = 0$  and  $f'(a) > 0$ .*

*Proof.* Fix  $a \in D$ . We first produce at least one injective bounded holomorphic map on  $D$ . Choose  $b \notin D$ . Since  $D$  is simply connected and  $z - b$  has no zero on  $D$ , Lemma 2.1 gives a holomorphic square root  $g^2 = z - b$ . The map  $g$  is injective. Moreover  $g(D)$  cannot contain both  $w$  and  $-w$ . Because  $g(D)$  is open and contains  $g(a)$ , some disk  $B(g(a), r)$  lies in  $g(D)$ ; its reflected disk  $B(-g(a), r)$  is therefore disjoint from  $g(D)$ . Hence

$$h(z) = \frac{1}{g(z) + g(a)}$$

is injective and bounded by  $1/r$ . After subtracting  $h(a)$  and multiplying by a sufficiently small nonzero constant, we obtain an injective holomorphic map  $D \rightarrow \mathbb{D}$  vanishing at  $a$ .

Let  $\mathcal{F}$  be the family of injective holomorphic maps  $F : D \rightarrow \mathbb{D}$  with  $F(a) = 0$ , and set

$$M = \sup_{F \in \mathcal{F}} |F'(a)| > 0.$$

Choose  $F_n \in \mathcal{F}$  with  $|F'_n(a)| \rightarrow M$  and rotate so that  $F'_n(a) > 0$ . Montel's theorem gives a locally uniformly convergent subsequence  $F_n \rightarrow f$ . Cauchy's formula gives  $f'(a) = M > 0$ , so  $f$  is nonconstant. A locally uniform nonconstant limit of univalent functions is univalent by Hurwitz (apply it to  $(F_n(z) - F_n(w))/(z - w)$  for each fixed  $w$ ). The maximum principle gives  $f(D) \subset \mathbb{D}$ , so  $f \in \mathcal{F}$  and is extremal.

Suppose  $c \in \mathbb{D} \setminus f(D)$ , and put

$$\psi(w) = \frac{w - c}{1 - \bar{c}w}.$$

Then  $\psi \circ f$  is zero-free on  $D$ , so Lemma 2.1 gives a holomorphic  $q$  with  $q^2 = \psi \circ f$ . Since  $|q| < 1$  and  $q$  is injective, compose with a disk automorphism  $A$  sending  $q(a)$  to 0. Then  $F = A \circ q$  belongs to  $\mathcal{F}$ . Writing  $r = |c| \in (0, 1)$  and using  $f(a) = 0$ , we have  $|q(a)| = \sqrt{r}$  and

$$|q'(a)| = \frac{1 - r^2}{2\sqrt{r}} |f'(a)|, \quad |A'(q(a))| = \frac{1}{1 - r}.$$

Therefore

$$|F'(a)| = \frac{1 + r}{2\sqrt{r}} |f'(a)| > |f'(a)|,$$

contradicting extremality. Thus  $f(D) = \mathbb{D}$ .

Finally, if  $f_1, f_2$  satisfy the stated normalization, then  $f_2 \circ f_1^{-1}$  is a disk automorphism fixing 0, hence a rotation. The condition that the derivative at  $a$  be positive forces that rotation to be the identity, proving uniqueness.  $\square$

## 2.2 Schwarz–Pick

**Theorem 2.3** (Schwarz lemma and Schwarz–Pick). *If  $f : \mathbb{D} \rightarrow \mathbb{D}$  is holomorphic with  $f(0) = 0$ , then  $|f(z)| \leq |z|$  and  $|f'(0)| \leq 1$ , with equality at a nonzero point or in the derivative only for rotations. More generally,*

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2}.$$

*Proof.* Assume first that  $f(0) = 0$ . The function

$$g(z) = \begin{cases} f(z)/z, & z \neq 0, \\ f'(0), & z = 0, \end{cases}$$

is holomorphic. For  $0 < r < 1$ , the maximum principle on  $|z| \leq r$  gives  $|g(z)| \leq r^{-1}$  for  $|z| \leq r$ , because  $|f| < 1$ . Letting  $r \uparrow 1$  yields  $|g(z)| \leq 1$ , hence  $|f(z)| \leq |z|$  and  $|f'(0)| \leq 1$ . If equality holds at a nonzero point or in the derivative, the maximum-modulus principle forces  $g$  to be a constant of modulus one, so  $f$  is a rotation.

For arbitrary  $z_0 \in \mathbb{D}$ , let

$$\varphi_a(z) = \frac{z - a}{1 - \bar{a}z}.$$

The map

$$F = \varphi_{f(z_0)} \circ f \circ \varphi_{z_0}^{-1}$$

fixes zero. Schwarz's lemma gives  $|F'(0)| \leq 1$ . Since  $|\varphi'_a(a)| = (1 - |a|^2)^{-1}$  and  $|(\varphi_{z_0}^{-1})'(0)| = 1 - |z_0|^2$ , the chain rule becomes

$$\frac{|f'(z_0)|}{1 - |f(z_0)|^2} \leq \frac{1}{1 - |z_0|^2}.$$

Equality implies equality in Schwarz's lemma for  $F$ , so  $F$  is a rotation and  $f$  a disk automorphism.  $\square$

## 2.3 The area theorem

**Lemma 2.4** (Area theorem). *Let*

$$g(z) = z + b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \cdots$$

*be univalent on  $|z| > 1$ . Then*

$$\sum_{n \geq 1} n|b_n|^2 \leq 1.$$

*Proof.* For  $R > 1$ , the image of the circle  $|z| = R$  is a Jordan curve. Green's formula writes the signed area it encloses as

$$\frac{1}{2i} \int_{|z|=R} \overline{g(z)} g'(z) dz.$$

Substitute the Laurent series; orthogonality of Fourier modes gives

$$A(R) = \pi \left( R^2 - \sum_{n \geq 1} n|b_n|^2 R^{-2n} \right).$$

Because  $g$  is univalent outside the unit disk, the complement of  $g(\{|z| > R\})$  contains the omitted compact set  $\mathbb{C} \setminus g(\{|z| > 1\})$  and has nonnegative area. Comparing the area between the two image curves and letting  $R \downarrow 1$  forces

$$1 - \sum_{n \geq 1} n|b_n|^2 \geq 0.$$

□

**Lemma 2.5** (Second coefficient bound). *If  $f(z) = z + a_2 z^2 + \cdots$  is normalized and univalent on  $\mathbb{D}$ , then  $|a_2| \leq 2$ .*

*Proof.* For  $|\zeta| > 1$  choose the branch

$$G(\zeta) = \frac{1}{\sqrt{f(\zeta^{-2})}} = \zeta - \frac{a_2}{2} \zeta^{-1} + O(\zeta^{-3})$$

that is asymptotic to  $\zeta$  at infinity. The function is univalent: if  $G(\zeta_1) = G(\zeta_2)$ , then injectivity of  $f$  gives  $\zeta_1^2 = \zeta_2^2$ ; the alternative  $\zeta_1 = -\zeta_2$  would imply  $G(\zeta_1) = -G(\zeta_1)$ , impossible. The area theorem applied to  $G$  gives

$$\left| \frac{a_2}{2} \right|^2 \leq 1,$$

hence  $|a_2| \leq 2$ . □

**Theorem 2.6** (Koebe one-quarter theorem). *If  $f : \mathbb{D} \rightarrow \mathbb{C}$  is univalent with  $f(0) = 0$  and  $f'(0) = 1$ , then*

$$B(0, 1/4) \subset f(\mathbb{D}).$$

*The constant  $1/4$  is sharp.*

*Proof.*

$$F_w(z) = \frac{wf(z)}{w - f(z)} = z + \left(a_2 + \frac{1}{w}\right)z^2 + \dots$$

is again normalized and univalent. The second-coefficient bound gives

$$\left|a_2 + \frac{1}{w}\right| \leq 2.$$

Together with  $|a_2| \leq 2$ , the triangle inequality gives  $|1/w| \leq 4$ , hence  $|w| \geq 1/4$ . Therefore every point of modulus  $< 1/4$  lies in  $f(\mathbb{D})$ . The Koebe function  $k(z) = z/(1-z)^2$  omits the ray  $(-\infty, -1/4]$ , showing sharpness.  $\square$

## 2.4 Koebe distortion

**Theorem 2.7** (Koebe distortion theorem). *If  $f$  is normalized univalent on  $\mathbb{D}$  and  $|z| = r < 1$ , then*

$$\frac{r}{(1+r)^2} \leq |f(z)| \leq \frac{r}{(1-r)^2}$$

and

$$\frac{1-r}{(1+r)^3} \leq |f'(z)| \leq \frac{1+r}{(1-r)^3}.$$

*Proof.* Fix  $a \in \mathbb{D}$  and let  $\varphi_a$  be a disk automorphism taking  $a$  to 0. After correcting the harmless unimodular sign of the derivative, the function

$$F_a(\zeta) = \frac{f(\varphi_a^{-1}(\zeta)) - f(a)}{(1-|a|^2)f'(a)}$$

is normalized and univalent. Lemma 2.5 applied to  $F_a$  gives, after computing its second Taylor coefficient,

$$\left| \frac{f''(a)}{f'(a)} - \frac{2\bar{a}}{1-|a|^2} \right| \leq \frac{4}{1-|a|^2}. \quad (4.1)$$

Put  $a = te^{i\theta}$ . Taking the real part of  $e^{i\theta}$  times (4.1) shows

$$\frac{2t-4}{1-t^2} \leq \frac{d}{dt} \log |f'(te^{i\theta})| \leq \frac{2t+4}{1-t^2}.$$

Because  $f'(0) = 1$ , integration from 0 to  $r$  yields

$$\frac{1-r}{(1+r)^3} \leq |f'(re^{i\theta})| \leq \frac{1+r}{(1-r)^3}. \quad (4.2)$$

For the upper growth bound, integrate along the radial segment:

$$|f(re^{i\theta})| \leq \int_0^r |f'(te^{i\theta})| dt \leq \int_0^r \frac{1+t}{(1-t)^3} dt = \frac{r}{(1-r)^2}. \quad (4.3)$$

For the lower bound, let  $C_r = f(\{|z| = r\})$  and choose  $w_1 \in C_r$  of minimal modulus. The disk  $B(0, |w_1|)$  lies in  $f(r\mathbb{D})$ ; otherwise a boundary point of smaller modulus would exist. Hence the segment  $[0, w_1]$  is contained in  $f(r\mathbb{D})$ . Let  $L$  be its preimage under  $f$ , joining 0 to some  $z_1$  with

$|z_1| = r$ . Along  $L$ ,  $|dw| = |f'(z)||dz|$ . The lower derivative bound in (4.2), together with  $|dz| \geq |d|z||$  and the fact that any path from radius 0 to radius  $r$  crosses every intermediate radius, gives

$$|w_1| = \int_L |f'(z)||dz| \geq \int_0^r \frac{1-t}{(1+t)^3} dt = \frac{r}{(1+r)^2}. \quad (4.4)$$

Every point  $f(z)$  with  $|z| = r$  lies on  $C_r$ , so  $|f(z)| \geq |w_1|$ . Combining (4.2)–(4.4) proves all four inequalities.  $\square$



## Chapter 3

# Kernel Convergence, Boundary Extension, and Prime Ends

For SLE one must discuss domains whose boundaries are not smooth and often not locally connected. Ordinary boundary points are then an inadequate conformal invariant. Prime ends replace a rough boundary by the correct conformal circle.

### 3.1 Carathéodory kernel convergence

**Definition 3.1.** Let  $(D_n, z_0)$  be pointed simply connected proper domains containing a common point  $z_0$ . Their *kernel* is the largest domain  $D \ni z_0$  such that every compact  $K \Subset D$  is contained in all sufficiently large  $D_n$ . We say  $D_n \rightarrow D$  in the Carathéodory kernel sense if every subsequence has the same kernel  $D$ .

**Theorem 3.2** (Carathéodory kernel theorem). *Let  $f_n : \mathbb{D} \rightarrow D_n$  be normalized conformal maps with  $f_n(0) = z_0$  and  $f'_n(0) > 0$ . If the kernel  $D$  is a proper domain, then  $D_n \rightarrow D$  in kernel sense if and only if  $f_n$  converges locally uniformly to the normalized Riemann map  $f : \mathbb{D} \rightarrow D$ .*

*Proof.* Suppose first that  $f_n \rightarrow f$  locally uniformly and  $f$  is the normalized Riemann map onto  $D$ . Let  $K \Subset D$  and set  $L = f^{-1}(K) \Subset \mathbb{D}$ . Choose finitely many disjoint small disks  $B_j \Subset \mathbb{D}$  covering  $L$  so that, for each  $w \in K$ , the equation  $f(z) = w$  has its unique solution in one of these disks and  $f(z) - w$  is bounded away from zero on the corresponding boundary. Uniform convergence and Rouché's theorem then imply that  $f_n(z) = w$  has a solution in that disk for all large  $n$ , uniformly after taking a finite cover of  $K$ . Hence  $K \subset D_n$  eventually. Thus  $D$  is contained in the kernel. If a point  $w \notin D$  belonged to the kernel, a small compact disk around  $w$  would lie in  $D_n$  eventually; applying the normalized inverse maps and Montel compactness would contradict injectivity of the limit  $f$ . Hence the kernel is exactly  $D$ .

Conversely assume  $D_n \rightarrow D$  in kernel sense. On each compact  $K \Subset D$ , the inverse maps  $g_n = f_n^{-1}$  are eventually defined and bounded by 1. Montel and a diagonal argument give a subsequence converging locally uniformly on  $D$  to a holomorphic  $g : D \rightarrow \overline{\mathbb{D}}$  with  $g(z_0) = 0$ . The kernel is proper, so the normalized conformal radii cannot diverge to infinity; Koebe's quarter theorem applied at 0 gives matching upper and lower bounds for  $f'_n(0)$  along the subsequence. Hence  $g'(z_0) \neq 0$ , so  $g$  is nonconstant and maps  $D$  into  $\mathbb{D}$ .

The family  $(f_n)$  is then locally bounded on  $\mathbb{D}$  by Koebe distortion using the bounded conformal radii. Pass to a further subsequence  $f_n \rightarrow \tilde{f}$ . On compact subsets of  $D$  and  $\mathbb{D}$ ,

$$g_n(f_n(z)) = z, \quad f_n(g_n(w)) = w.$$

Passing to the limit gives  $g \circ \tilde{f} = \text{id}_{\mathbb{D}}$  and  $\tilde{f} \circ g = \text{id}_D$ . Thus  $\tilde{f}$  is the normalized Riemann map  $f$ . Uniqueness of the normalized map shows that every convergent subsequence has the same limit; therefore the full sequence converges locally uniformly to  $f$ .  $\square$

## 3.2 Jordan boundaries

**Theorem 3.3** (Carathéodory boundary extension). *If  $D$  is a Jordan domain and  $f : \mathbb{D} \rightarrow D$  is conformal, then  $f$  extends to a homeomorphism  $\overline{\mathbb{D}} \rightarrow \overline{D}$ .*

*Proof.* Write  $\partial D = \gamma(S^1)$  with  $\gamma$  a homeomorphism. The Jordan curve theorem implies that every crosscut separates  $D$  into two simply connected components. Uniform continuity of  $\gamma$  and  $\gamma^{-1}$  gives the following small-crosscut fact: for every  $\varepsilon > 0$  there is  $\delta > 0$  such that a crosscut whose endpoints lie on a boundary arc of diameter below  $\delta$  cuts off, on that side, a component of diameter below  $\varepsilon$ .

Let  $z_n \rightarrow \zeta \in \partial \mathbb{D}$ . Choose circular crosscuts  $C_k$  in  $\mathbb{D}$  shrinking to  $\zeta$  and separating  $z_n$  eventually from a fixed compact neighborhood of 0. Their images  $f(C_k)$  are crosscuts of  $D$ . If the component beyond  $f(C_k)$  failed to have diameter tending to zero, two points a fixed positive distance apart would survive in nested components whose boundary arcs shrink to a single point, contradicting the small-crosscut fact. Hence  $f(z)$  has a unique limit as  $z \rightarrow \zeta$  from inside  $\mathbb{D}$ . This defines a continuous extension  $\bar{f} : \overline{\mathbb{D}} \rightarrow \overline{D}$ .

Apply the same argument to  $f^{-1} : D \rightarrow \mathbb{D}$  using crosscuts in the Jordan domain  $D$ . The two boundary extensions are inverse to one another. Therefore  $\bar{f}$  is a continuous bijection from the compact space  $\overline{\mathbb{D}}$  to the Hausdorff space  $\overline{D}$ , hence a homeomorphism.  $\square$

## 3.3 Prime ends

**Definition 3.4.** A *crosscut* of a simply connected domain  $D$  is an open Jordan arc in  $D$  whose closure has two endpoints on  $\partial D$ . A *null chain* is a sequence of pairwise disjoint crosscuts  $C_n$  with diameters tending to zero, each separating  $C_{n+1}$  from a fixed base point. Two null chains are equivalent if each eventually separates the other from the base point. An equivalence class is a *prime end*.

**Theorem 3.5** (Prime-end compactification). *For every simply connected proper planar domain  $D$ , its prime-end compactification  $D^* = D \cup \partial_P D$  is homeomorphic to the closed disk. Every conformal map  $f : \mathbb{D} \rightarrow D$  extends uniquely to a homeomorphism*

$$\overline{\mathbb{D}} \longrightarrow D^*.$$

*In particular the prime-end boundary is canonically a topological circle.*

*Proof.* Fix a Riemann map  $f : \mathbb{D} \rightarrow D$ . For  $\zeta \in \partial \mathbb{D}$ , choose nested circular crosscuts  $\Gamma_n$  tending to  $\zeta$ . After replacing them by a nested subsequence if necessary, Koebe distortion on the components beyond  $\Gamma_n$  gives

$$\text{diam}_{\text{sph}} f(\Gamma_n) \longrightarrow 0.$$

Thus  $(f(\Gamma_n))$  is a null chain and determines a prime end. Two distinct points of  $\partial \mathbb{D}$  can be separated by disjoint sectors, so they yield inequivalent prime ends.

Conversely let  $(C_n)$  be a prime chain in  $D$  and pull it back by  $f^{-1}$ . The nested complementary components in  $\mathbb{D}$  have closures whose intersection is a nonempty connected subset of  $\partial \mathbb{D}$ . If it contained two distinct boundary points, a fixed crosscut of  $\mathbb{D}$  separating those points would have

to meet infinitely many members of the pulled-back null chain, contradicting pairwise disjointness and vanishing diameter. Hence the intersection consists of one point  $\zeta \in \partial\mathbb{D}$ . This construction is inverse to the first.

Give  $D^*$  the topology whose basic neighborhoods of a prime end represented by  $(C_n)$  are the corresponding nested components together with all prime ends eventually represented inside them. Under the preceding bijection these neighborhoods are exactly the images of ordinary neighborhoods of  $\zeta$  in  $\overline{\mathbb{D}}$ . Therefore  $f$  extends to a homeomorphism  $\overline{\mathbb{D}} \rightarrow D^*$ . Compactness, Hausdorffness, uniqueness, and the circular topology of the prime-end boundary follow immediately.  $\square$

*Remark 3.6.* A prime end need not correspond to a unique Euclidean boundary point: its impression can be a continuum. Jordan domains are precisely among the clean cases where the prime-end and Euclidean boundaries match point-for-point.

## Chapter 4

# Poisson Kernels, Harnack Theory, Green Functions, and Harmonic Measure

Conformal geometry and probability meet through harmonic functions. In simply connected planar domains, Green functions and harmonic measure transform naturally under conformal maps; this covariance will later be mirrored by the GFF and SLE.

### 4.1 Poisson representation

**Theorem 4.1** (Poisson representation). *For the unit disk,*

$$P_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2}$$

*is the Poisson kernel. If  $f \in C(\partial\mathbb{D})$ , then*

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \phi) f(e^{i\phi}) d\phi$$

*is harmonic in  $\mathbb{D}$ , extends continuously to  $f$ , and is the unique such harmonic extension. Under a Cayley transform this yields the half-plane kernel*

$$P_{\mathbb{H}}(x + iy, \xi) = \frac{1}{\pi} \frac{y}{(x - \xi)^2 + y^2}.$$

*Proof.* For  $z \in \mathbb{D}$  and  $\zeta \in \partial\mathbb{D}$ ,

$$\operatorname{Re} \frac{\zeta + z}{\zeta - z} = \frac{1 - |z|^2}{|\zeta - z|^2}.$$

The right-hand side is harmonic as a function of  $z$  and, with  $z = re^{i\theta}$  and  $\zeta = e^{i\phi}$ , equals  $P_r(\theta - \phi)$ . Its average over  $\phi$  is 1 (take the real part of the geometric-series expansion of  $(\zeta + z)/(\zeta - z)$ ), so the displayed integral defines a harmonic function.

The kernels form an approximate identity on the circle. Indeed, for every fixed  $\delta > 0$ ,

$$\sup_{|\alpha| \geq \delta} P_r(\alpha) \longrightarrow 0 \quad (r \uparrow 1),$$

while their mass is  $2\pi$ . Uniform continuity of  $f$  therefore gives  $u(re^{i\theta}) \rightarrow f(e^{i\theta})$  uniformly in  $\theta$ . Hence  $u$  extends continuously to the prescribed boundary values. If  $v$  is another continuous harmonic extension, the maximum principle applied to  $u - v$  and  $-(u - v)$  gives  $u = v$ .

Finally let  $\Phi : \mathbb{H} \rightarrow \mathbb{D}$  be a Cayley transform. Changing boundary variables in the disk formula and using the boundary Jacobian of  $\Phi$  gives

$$P_{\mathbb{H}}(x + iy, \xi) = \frac{1}{\pi} \frac{y}{(x - \xi)^2 + y^2}.$$

□

## 4.2 Harnack inequalities

**Theorem 4.2** (Harnack inequality and convergence). *Let  $d \geq 2$ . If  $u \geq 0$  is harmonic in  $B(0, R) \subset \mathbb{R}^d$ , then for  $|x| \leq r < R$ ,*

$$R^{d-2} \frac{R-r}{(R+r)^{d-1}} u(0) \leq u(x) \leq R^{d-2} \frac{R+r}{(R-r)^{d-1}} u(0).$$

*Consequently a locally bounded sequence of harmonic functions has locally uniformly convergent subsequences; an increasing sequence of nonnegative harmonic functions on a connected domain either diverges everywhere to  $+\infty$  or converges locally uniformly to a harmonic function.*

*Proof.* Use the Poisson representation in  $B(0, R)$ . For  $|\xi| = R$ ,

$$P_R(x, \xi) = \frac{R^2 - |x|^2}{|\mathbb{S}^{d-1}| R |x - \xi|^d}, \quad P_R(0, \xi) = \frac{1}{|\mathbb{S}^{d-1}| R^{d-1}}.$$

Writing  $\rho = |x|$  and using  $R - \rho \leq |x - \xi| \leq R + \rho$  gives the pointwise kernel bounds

$$R^{d-2} \frac{R - \rho}{(R + \rho)^{d-1}} P_R(0, \xi) \leq P_R(x, \xi) \leq R^{d-2} \frac{R + \rho}{(R - \rho)^{d-1}} P_R(0, \xi).$$

The lower coefficient decreases and the upper coefficient increases with  $\rho$ , so  $\rho \leq r$  gives the displayed Harnack inequality after integration against the nonnegative boundary data.

Now let  $(u_n)$  be locally bounded and harmonic. The Poisson formula on concentric balls and differentiation under the boundary integral give uniform first-derivative bounds on every smaller ball. Thus the family is equicontinuous on compact sets. Arzelà–Ascoli and a diagonal argument produce a locally uniform subsequential limit; the mean-value property passes to the limit, so the limit is harmonic.

If  $(u_n)$  is increasing and  $u_n(x_0) \rightarrow \infty$  at one point, Harnack chains propagate divergence to every point of the connected domain. Otherwise the values at  $x_0$  are bounded; Harnack chains then give local uniform bounds on every compact set. Compactness gives locally uniform subsequential limits, and monotonicity makes every subsequence have the same pointwise limit. Hence the whole sequence converges locally uniformly to a harmonic function. □

## 4.3 Green functions

**Theorem 4.3** (Green function and conformal covariance). *Let  $D \subsetneq \mathbb{C}$  be simply connected and  $w \in D$ . There is a unique Green function  $G_D(\cdot, w)$  characterized by: it is positive harmonic on  $D \setminus \{w\}$ , tends to zero at regular boundary points, and*

$$G_D(z, w) + \log |z - w|$$

is harmonic near  $w$  (with this logarithmic normalization). If  $\phi : D \rightarrow D'$  is conformal, then

$$G_D(z, w) = G_{D'}(\phi(z), \phi(w)).$$

*Proof.*

$$G_{\mathbb{D}}(z, w) = -\log \left| \frac{z - w}{1 - \bar{w}z} \right|.$$

The fraction is a disk automorphism taking  $w$  to zero, so the expression is positive harmonic away from  $w$ , vanishes on the unit circle, and has the required logarithmic singularity.

$$G_D(z, w) = G_{\mathbb{D}}(\phi(z), \phi(w)).$$

Near  $w$ ,  $\phi(z) - \phi(w) = \phi'(w)(z - w) + o(|z - w|)$ , so subtracting  $-\log |z - w|$  leaves a harmonic bounded correction. □

## 4.4 Harmonic measure

**Theorem 4.4** (Conformal invariance of harmonic measure). *Let  $\phi : D \rightarrow D'$  be conformal between simply connected domains and interpret boundary sets through prime ends when necessary. Then*

$$\omega_D^z(A) = \omega_{D'}^{\phi(z)}(\phi(A)).$$

*Equivalently, harmonic measure is the conformally natural representing measure for the Dirichlet problem.*

*Proof.* Let  $g$  be a continuous function on the prime-end boundary of  $D'$  and let

$$H_g^{D'}(w) = \int g(\xi) \omega_{D'}^w(d\xi)$$

be its harmonic extension. Conformal invariance of harmonicity implies that  $H_g^{D'} \circ \phi$  is harmonic on  $D$ , and its boundary values are  $g \circ \phi$ . By uniqueness of the Dirichlet problem,

$$H_g^{D'}(\phi(z)) = H_{g \circ \phi}^D(z) = \int g(\phi(\eta)) \omega_D^z(d\eta).$$

Therefore, for every continuous boundary test function  $g$ ,

$$\int g(\xi) \omega_{D'}^{\phi(z)}(d\xi) = \int g(\phi(\eta)) \omega_D^z(d\eta).$$

Continuous functions determine probability measures on the compact prime-end boundary, so  $\phi_* \omega_D^z = \omega_{D'}^{\phi(z)}$ . Evaluating this identity on Borel boundary sets gives the stated formula. □

## Chapter 5

# Conformal Brownian Motion and Probabilistic Potential Theory

The analytic conformal covariance of harmonic measure has a probabilistic explanation: a holomorphic map sends planar Brownian motion to Brownian motion run at a random clock. The clock changes speed but not the order in which the path visits points.

### 5.1 Conformal invariance up to time change

**Theorem 5.1** (Conformal invariance of planar Brownian motion). *Let  $B_t$  be planar Brownian motion started at  $z \in D$ , let  $\tau_D$  be its exit time, and let  $f : D \rightarrow D'$  be conformal. Define*

$$A_t = \int_0^{t \wedge \tau_D} |f'(B_s)|^2 ds, \quad \sigma_u = \inf\{t : A_t > u\}.$$

Then

$$W_u = f(B_{\sigma_u})$$

for  $u < A_{\tau_D}$  is planar Brownian motion started at  $f(z)$ , stopped when it exits  $D'$ .

*Proof.* Write  $f = u + iv$ . The Cauchy–Riemann equations make  $u$  and  $v$  harmonic and give

$$|\nabla u| = |\nabla v| = |f'|, \quad \nabla u \cdot \nabla v = 0.$$

Itô's formula therefore shows that  $u(B_{t \wedge \tau_D})$  and  $v(B_{t \wedge \tau_D})$  are continuous local martingales with brackets

$$d[u(B)]_t = |f'(B_t)|^2 dt, \quad d[v(B)]_t = |f'(B_t)|^2 dt, \quad d[u(B), v(B)]_t = 0.$$

Because a conformal map has  $f' \neq 0$ , the clock  $A_t$  is continuous and strictly increasing before exit. After the inverse time change,  $W_u = f(B_{\sigma_u})$  is a continuous two-dimensional local martingale with

$$[W^i, W^j]_u = \delta_{ij} u.$$

Lévy's characterization from Volume III makes  $W$  standard planar Brownian motion started at  $f(z)$ .

For  $t < \tau_D$ ,  $f(B_t) \in D'$ . Conversely, as  $t \uparrow \tau_D$ , the path leaves every compact subset of  $D$ ; conformality and the inverse map imply that  $f(B_t)$  leaves every compact subset of  $D'$ . Thus the lifetime of the time-changed process is exactly  $A_{\tau_D}$  and it is Brownian motion stopped on exiting  $D'$ .  $\square$

## 5.2 Harmonic measure and Green function from Brownian motion

**Theorem 5.2** (Brownian representations of harmonic measure and Green function). *For a bounded regular planar domain  $D$ ,*

$$\omega_D^z(A) = \mathbb{P}_z(B_{\tau_D} \in A).$$

*Moreover the expected occupation measure before exit has density proportional to the Green function: with the generator convention  $\frac{1}{2}\Delta$ ,*

$$\mathbb{E}_z \int_0^{\tau_D} f(B_t) dt = \frac{1}{\pi} \int_D f(w) G_D(z, w) dA(w)$$

*for nonnegative Borel  $f$ , under the logarithmic normalization of Theorem 4.3. If  $\phi : D \rightarrow D'$  is conformal, then for every Borel  $A \subset \partial D$  whose boundary has harmonic measure zero,*

$$\omega_D^z(A) = \omega_{D'}^{\phi(z)}(\phi(A)),$$

*and, with the same logarithmic normalization,*

$$G_D(z, w) = G_{D'}(\phi(z), \phi(w)).$$

*(The Jacobian  $|\phi'(w)|^2$  appears in the occupation-density formula through  $dA(\phi(w))$ .)*

*Proof.*

$$U_f(z) = \mathbb{E}_z \int_0^{\tau_D} f(B_t) dt.$$

By the strong Markov property and Dynkin's formula,  $U_f$  is the zero-boundary solution of

$$\frac{1}{2}\Delta U_f = -f.$$

On the other hand, the distributional identity

$$-\Delta_z G_D(z, w) = 2\pi\delta_w(z)$$

for the normalization  $G_D \sim -\log|z - w|$  shows that

$$\tilde{U}_f(z) = \frac{1}{\pi} \int_D G_D(z, w) f(w) dA(w)$$

has the same PDE and boundary values. Uniqueness gives  $U_f = \tilde{U}_f$ .

□



## Chapter 6

# Extremal Length, Quadrilateral Modulus, and the Beurling Estimate

Extremal length is the conformally invariant way to measure how difficult it is for a family of curves to cross a domain. It is a variational quantity: one is allowed to choose a nonnegative density, compare the shortest curve length in that density with the total area of the density, and then optimize. This viewpoint is indispensable later because crossing events for Brownian motion, random walk, LERW, and SLE can often be converted into estimates on annuli or quadrilaterals whose geometry is most naturally measured by extremal length.

### 6.1 Extremal length

**Definition 6.1** (Extremal length). Let  $D \subset \mathbb{C}$  be a domain and  $\Gamma$  a family of locally rectifiable curves in  $D$ . For a Borel function  $\rho : D \rightarrow [0, \infty]$  with  $0 < \int_D \rho^2 dA < \infty$ , put

$$L_\rho(\Gamma) = \inf_{\gamma \in \Gamma} \int_\gamma \rho |dz|, \quad A_\rho(D) = \int_D \rho^2 dA.$$

The extremal length of  $\Gamma$  is

$$\text{EL}_D(\Gamma) = \sup_\rho \frac{L_\rho(\Gamma)^2}{A_\rho(D)}.$$

**Theorem 6.2** (Conformal invariance and monotonicity). *If  $f : D \rightarrow D'$  is conformal, then*

$$\text{EL}_D(\Gamma) = \text{EL}_{D'}(f\Gamma).$$

*If  $\Gamma_1 \subset \Gamma_2$ , then  $\text{EL}_D(\Gamma_1) \geq \text{EL}_D(\Gamma_2)$ . If  $D \subset D'$  and every curve of  $\Gamma$  is contained in  $D$ , then enlarging the ambient domain can only decrease the optimized obstruction in the corresponding admissible formulation.*

*Proof.*

$$\rho(z) = \rho'(f(z))|f'(z)|.$$

For every rectifiable  $\gamma$ ,

$$\int_\gamma \rho |dz| = \int_{f \circ \gamma} \rho' |dw|,$$

while the area formula gives

$$\int_D \rho(z)^2 dA(z) = \int_{D'} \rho'(w)^2 dA(w).$$

Hence the extremal quotient is unchanged. Applying the same argument to  $f^{-1}$  proves equality.

$L_\rho(\Gamma_1) \geq L_\rho(\Gamma_2)$ , so the supremum quotient for  $\Gamma_1$  is at least that for  $\Gamma_2$ . The domain-monotonicity statement follows identically from the modulus formulation by extending a density by zero or restricting it, according to the chosen convention.  $\square$

## 6.2 Rectangles and quadrilaterals

**Theorem 6.3** (Extremal length of a rectangle). *Let  $R = [0, a] \times [0, b]$ , and let  $\Gamma$  be the family of curves joining the vertical sides  $\{0\} \times [0, b]$  and  $\{a\} \times [0, b]$ . Then*

$$\text{EL}_R(\Gamma) = \frac{a}{b}.$$

*Consequently, if a simply connected quadrilateral  $(D; z_1, z_2, z_3, z_4)$  is mapped conformally to a rectangle with these marked boundary arcs corresponding to the vertical sides, its conformal modulus equals that rectangle aspect ratio.*

*Proof.* For the constant density  $\rho \equiv 1$ , every curve joining the vertical sides has Euclidean length at least  $a$ , while  $A_\rho(R) = ab$ . Hence

$$\text{EL}_R(\Gamma) \geq \frac{a^2}{ab} = \frac{a}{b}.$$

For the reverse inequality, let  $\rho$  be any admissible nonnegative density. For almost every horizontal line  $y \in (0, b)$ , the straight segment  $[0, a] \times \{y\}$  belongs to  $\Gamma$ , so

$$L_\rho(\Gamma) \leq \int_0^a \rho(x, y) dx.$$

Integrating in  $y$  and applying Cauchy–Schwarz gives

$$bL_\rho(\Gamma) \leq \int_R \rho dA \leq (ab)^{1/2} A_\rho(R)^{1/2}.$$

Therefore  $L_\rho(\Gamma)^2 / A_\rho(R) \leq a/b$ . Taking the supremum over  $\rho$  gives the reverse inequality and hence equality. Conformal invariance of extremal length then identifies the modulus of a conformally equivalent marked quadrilateral with the same aspect ratio.  $\square$

**Example 6.4** (Annulus crossing). Under  $z \mapsto \log z$ , the annulus  $A(r, R) = \{r < |z| < R\}$  becomes a horizontal strip of width  $\log(R/r)$  with vertical period  $2\pi$ . The extremal length of curves joining the two boundary circles is therefore

$$\frac{\log(R/r)}{2\pi}.$$

This logarithm is why annular scales naturally add in planar probability.

### 6.3 Beurling projection and Brownian crossing

The estimate below is the form we will use repeatedly. It says that a connected obstacle crossing an annulus is hard for planar Brownian motion to avoid. The characteristic square-root exponent is already visible in the slit half-plane.

**Lemma 6.5** (Projection principle in the form needed here). *Let  $K$  be a compact connected subset of  $\overline{\mathbb{D}}$  joining 0 to  $\partial\mathbb{D}$ . Among such obstacles, radial projection onto a straight segment cannot increase the harmonic measure of the outer boundary seen from a point separated from  $K$ . Equivalently, the Dirichlet-energy/extremal-length obstruction to crossing  $\mathbb{D} \setminus K$  is minimized by a radial slit.*

*Proof.* For a measurable set  $E$  on the circle  $|z| = \rho$ , replace  $E$  by the centered circular arc  $E^*$  of the same angular measure. Apply this slice by slice to a test function  $u$  for the condenser separating  $K$  from the outer boundary. The one-dimensional rearrangement inequality on each circle gives

$$\int_0^{2\pi} |\partial_\theta u^*(\rho, \theta)|^2 d\theta \leq \int_0^{2\pi} |\partial_\theta u(\rho, \theta)|^2 d\theta.$$

The layer-cake approximation by radial step functions gives the corresponding inequality for the radial derivative. Integrating in polar coordinates yields the Pólya–Szegő inequality

$$\int_{\mathbb{D} \setminus K} |\nabla u^*|^2 dA \leq \int_{\mathbb{D} \setminus K} |\nabla u|^2 dA.$$

Taking the infimum over admissible condenser potentials shows that circular symmetrization cannot increase the condenser capacity, equivalently cannot decrease the extremal length of curves crossing the complement. Because  $K$  is connected and meets both 0 and  $\partial\mathbb{D}$ , every symmetrized radial slice forces the symmetrized obstacle to contain a radial segment. Monotonicity of capacity under enlarging the obstacle therefore reduces the extremal crossing problem to a straight radial slit. Harmonic measure is the minimizing Dirichlet solution for the corresponding boundary values, so the same comparison gives the stated projection principle.  $\square$

**Theorem 6.6** (Beurling estimate). *There is a universal constant  $C < \infty$  such that the following holds. Let  $0 < r < R$ , let  $K$  be a connected set intersecting both circles  $\{|z| = r\}$  and  $\{|z| = R\}$ , and let  $B$  be planar Brownian motion started at  $z$  with  $|z| \leq r$ . Then*

$$\mathbb{P}_z(B[0, \tau_R] \cap K = \emptyset) \leq C \left(\frac{r}{R}\right)^{1/2},$$

where  $\tau_R = \inf\{t : |B_t| = R\}$ .

*Proof.*

$$w = \sqrt{z - r}$$

opens the slit domain into a half-plane-type domain. Harmonic measure of the outer target seen from a point at Euclidean distance of order  $r$  from the slit tip is then bounded by a constant times the ratio of square-root distances. In the original coordinates this is  $Cr^{1/2}$ . The same estimate follows by mapping the slit disk to a half disk and using the Poisson kernel; the square root is exactly the angle-doubling map at the slit tip.  $\square$

*Remark 6.7.* For LERW and SLE we will often use a discrete analogue with simple random walk. Volume V obtains it by combining the continuum estimate with discrete Harnack/invariance-principle control, or directly from the discrete potential theory developed there.

# Chapter 7

## Half-Plane Hulls and Capacity

Loewner evolution encodes a growing planar hull by the conformal map that removes it. The correct notion of size is not Euclidean diameter but the coefficient of  $1/z$  in the map normalized at infinity. This chapter fixes that normalization and proves the elementary capacity facts needed later.

### 7.1 Half-plane hulls and hydrodynamic normalization

**Definition 7.1** (Half-plane hull). A compact set  $K \subset \overline{\mathbb{H}}$  is a half-plane hull if  $K = \overline{K \cap \mathbb{H}}$  and  $\mathbb{H} \setminus K$  is simply connected. We write  $H_K = \mathbb{H} \setminus K$ .

**Theorem 7.2** (Hydrodynamic map and capacity). *For every bounded half-plane hull  $K$  there is a unique conformal map*

$$g_K : H_K \longrightarrow \mathbb{H}$$

*with hydrodynamic normalization*

$$g_K(z) = z + \frac{a_K}{z} + O(|z|^{-2}), \quad z \rightarrow \infty.$$

*The coefficient  $a_K$  is real and nonnegative. It is called the half-plane capacity,*

$$\text{hcap}(K) = a_K.$$

*Moreover*

$$\text{hcap}(rK) = r^2 \text{hcap}(K), \quad r > 0.$$

*Proof.*

$$f(z) = z + b_0 + \frac{b_1}{z} + O(z^{-2}).$$

Subtract the real constant  $b_0$  to obtain  $g_K$ .

$$h(z) = z + O(z^{-1})$$

at infinity. Every automorphism of  $\mathbb{H}$  is  $(az + b)/(cz + d)$  with real coefficients and positive determinant. Fixing infinity forces  $c = 0$ ; derivative one forces  $a = d$ ; zero constant term forces  $b = 0$ . Thus  $h(z) = z$ .

$$u(z) = \Im f_K(z) - \Im z$$

has nonnegative boundary values and tends to 0 at infinity. Applying the minimum principle on large half disks and then letting the radius tend to infinity gives  $u(z) \geq 0$  on  $\mathbb{H}$ . The inverse Laurent expansion has the form

$$f_K(z) = z - \frac{b_1}{z} + O(z^{-2}).$$

At  $z = iy$ ,

$$u(iy) = \frac{b_1}{y} + O(y^{-2}).$$

Since  $u(iy) \geq 0$  for all large  $y$ , necessarily  $b_1 \geq 0$ . We denote this coefficient by  $a_K$ .

$$z \mapsto r g_K(z/r)$$

conformally maps  $H_{rK}$  to  $\mathbb{H}$  and has expansion

$$r \left( \frac{z}{r} + \frac{a_K}{z/r} + O(|z/r|^{-2}) \right) = z + \frac{r^2 a_K}{z} + O(|z|^{-2}).$$

Uniqueness of the hydrodynamic map therefore gives the scaling relation. □

**Example 7.3** (Vertical slit). Let  $K = [0, iy]$ . Then

$$g_K(z) = \sqrt{z^2 + y^2},$$

with the branch satisfying  $g_K(z) \sim z$  at infinity. Expanding,

$$g_K(z) = z + \frac{y^2}{2z} + O(z^{-3}),$$

so

$$\text{hcap}([0, iy]) = \frac{y^2}{2}.$$

This quadratic relation explains why Loewner time scales like length squared.

## 7.2 Monotonicity

**Proposition 7.4** (Capacity increments). *If  $K \subset L$  are bounded half-plane hulls, then  $g_K(L \setminus K)$  is a half-plane hull and*

$$g_L = g_{g_K(L \setminus K)} \circ g_K,$$

so

$$\text{hcap}(L) = \text{hcap}(K) + \text{hcap}(g_K(L \setminus K)).$$

*In particular, half-plane capacity is monotone under hull inclusion.*

*Proof.* The composition conformally maps  $H_L$  to  $\mathbb{H}$  and has hydrodynamic normalization. If

$$g_K(z) = z + \frac{a}{z} + O(z^{-2}), \quad g_{g_K(L \setminus K)}(z) = z + \frac{b}{z} + O(z^{-2}),$$

then their composition is  $z + (a + b)/z + O(z^{-2})$ . Uniqueness identifies it with  $g_L$  and gives the formula. Nonnegativity of  $b$  yields monotonicity. □

### 7.3 Continuity under kernel convergence

**Theorem 7.5** (Continuity of half-plane capacity). *Let  $K_n$  and  $K$  be bounded half-plane hulls contained in one fixed disk. Suppose the pointed domains  $H_{K_n}$  converge to  $H_K$  in the Carathéodory kernel sense from infinity. Then*

$$g_{K_n} \longrightarrow g_K$$

*locally uniformly on  $H_K$ , and*

$$\text{hcap}(K_n) \longrightarrow \text{hcap}(K).$$

*Proof.* By the kernel theorem, the normalized conformal maps  $g_{K_n} : H_{K_n} \rightarrow \mathbb{H}$  form a normal family on  $H_K$  and every subsequential limit is the normalized map  $g_K$ . Hence

$$g_{K_n} \longrightarrow g_K$$

locally uniformly on  $H_K$ .

Choose  $R$  so large that every hull lies in  $\{|z| < R\}$ . For  $|x| > R$  on the real line the maps have real boundary values and extend by Schwarz reflection across the real axis. Thus each  $g_{K_n}(z) - z$  and  $g_K(z) - z$  is holomorphic on one common annulus  $|z| > R$  and the local uniform convergence extends there. Their Laurent expansions are

$$g_{K_n}(z) - z = \frac{a_n}{z} + O(z^{-2}), \quad g_K(z) - z = \frac{a}{z} + O(z^{-2}),$$

where  $a_n = \text{hcap}(K_n)$  and  $a = \text{hcap}(K)$ . For any fixed  $R' > R$ , Cauchy's coefficient formula gives

$$a_n = \frac{1}{2\pi i} \int_{|z|=R'} (g_{K_n}(z) - z) dz.$$

Uniform convergence on the circle therefore yields  $a_n \rightarrow a$ . This proves both asserted convergences.  $\square$

*Remark 7.6* (Why this chapter matters for SLE). In Volume XIII a continuous growing hull  $K_t$  will be parametrized by

$$\text{hcap}(K_t) = 2t.$$

The additive increment formula gives the semigroup structure, and capacity continuity is the bridge from convergence of hulls to convergence of Loewner time.

## Appendix A

# Boundary Theory and the Beurling Projection Theorem

This appendix supplies four proof details used later by random walks and Loewner evolution.

### A.1 Jordan domains and crosscuts

A *crosscut* of a simply connected domain  $D$  is an open Jordan arc  $C \subset D$  whose two endpoints lie on  $\partial D$ . If  $D$  is a Jordan domain, the Jordan curve theorem implies that every crosscut separates  $D$  into exactly two simply connected components. Moreover the boundary of either component consists of the crosscut together with one of the two boundary arcs joining its endpoints.

**Lemma A.1** (Small crosscuts in a Jordan domain). *Let  $D$  be a Jordan domain. For every  $\varepsilon > 0$  there is  $\delta > 0$  such that a crosscut whose endpoints have Euclidean distance below  $\delta$  cuts off a component of diameter below  $\varepsilon$ , provided the endpoints lie on a boundary arc of diameter below  $\delta$ .*

*Proof.* Write  $\partial D = \gamma(S^1)$  with  $\gamma$  a homeomorphism. Uniform continuity of  $\gamma$  and  $\gamma^{-1}$  shows that close endpoints on the same short boundary arc correspond to close parameters, and that the image of the intervening parameter arc has small diameter. If the component cut off by the crosscut had diameter bounded below along a sequence of crosscuts whose endpoint arcs shrink to a point, choose points in those components separated by a fixed distance. Compactness gives a limiting point not equal to the limiting endpoint, contradicting the Jordan separation description of the component boundary.  $\square$

Let  $f : \mathbb{D} \rightarrow D$  be a Riemann map. If  $z_n \rightarrow \zeta \in \partial \mathbb{D}$ , images of short circular crosscuts separating  $z_n$  from a fixed compact subset of  $\mathbb{D}$  have endpoints whose prime ends converge to a single boundary point. The lemma forces the image component diameters to tend to zero. Hence  $f(z_n)$  has a unique limit depending only on  $\zeta$ . This gives a continuous extension  $\bar{f} : \bar{\mathbb{D}} \rightarrow \bar{D}$ . Apply the same argument to  $f^{-1}$  using boundary arcs in  $D$  to get injectivity on the boundary. Compactness then gives a homeomorphism. This supplies the full crosscut/local-connectivity proof of .

### A.2 Prime ends from null chains

A *chain* is a sequence of pairwise disjoint crosscuts  $C_n$  such that  $C_{n+1}$  lies in the component  $U_n$  of  $D \setminus C_n$  not containing a fixed base point and  $\text{diam } C_n \rightarrow 0$ . Two chains are equivalent if they eventually divide each other. A prime end is an equivalence class of chains minimal for division.

**Lemma A.2** (Prime-end compactification of a simply connected domain). *If  $D \subsetneq \mathbb{C}$  is simply connected, the set  $D^* = D \cup \partial_P D$  becomes a compact Hausdorff space when a basic neighborhood of a prime end represented by  $(C_n)$  is  $U_n$  together with all prime ends eventually represented inside  $U_n$ . Every Riemann map  $f : \mathbb{D} \rightarrow D$  extends to a homeomorphism*

$$\overline{\mathbb{D}} \longrightarrow D^*.$$

*Proof.* For  $\zeta \in \partial\mathbb{D}$ , choose concentric circular crosscuts  $\Gamma_n$  in  $\mathbb{D}$  tending to  $\zeta$ . Koebe distortion on the components beyond  $\Gamma_n$  implies  $\text{diam } f(\Gamma_n) \rightarrow 0$  in the spherical metric after passing to nested crosscuts; thus  $(f(\Gamma_n))$  defines a prime end. Different boundary points are separated by disjoint sectors in  $\mathbb{D}$ , so the resulting prime ends are distinct. Conversely, pull a prime chain back by  $f^{-1}$ . Its crosscuts have nested complementary components in  $\mathbb{D}$ . Their closures intersect in a connected subset of  $\partial\mathbb{D}$ ; if that subset contained two points, a fixed interior crosscut separating them would have to be crossed by infinitely many members of the chain, contradicting disjointness and null diameter. Hence the impression in  $\partial\mathbb{D}$  is a singleton. This proves bijectivity. The neighborhood definitions are transported exactly by  $f$ , so the extension and its inverse are continuous. Compactness and Hausdorffness follow from  $\overline{\mathbb{D}}$ .  $\square$

This closes without assuming local connectivity of the Euclidean boundary. For Jordan domains, every prime-end impression is a single Euclidean boundary point and the prime-end theorem reduces to the preceding Carathéodory boundary extension.

### A.3 Circular symmetrization and the Beurling projection theorem

Let  $K$  be a connected compact subset of  $\overline{\mathbb{D}}$  meeting both 0 and  $\partial\mathbb{D}$ , and let  $B$  be Brownian motion started at  $z \in \mathbb{D} \setminus K$ . The Beurling projection theorem compares the probability of avoiding  $K$  before reaching a circle with the extremal radial slit.

For a measurable set  $E$  on a circle  $|w| = r$ , let  $E^*$  be the centered circular arc of the same angular measure. Circular symmetrization replaces every circular slice of a domain by its centered arc. The one-dimensional rearrangement inequality on each circle gives

$$\int_0^{2\pi} |\partial_\theta u^*(r, \theta)|^2 d\theta \leq \int_0^{2\pi} |\partial_\theta u(r, \theta)|^2 d\theta,$$

while the radial derivative term decreases after the layer-cake approximation by step functions. Integrating in polar coordinates yields the Pólya–Szegő inequality

$$\int |\nabla u^*|^2 dA \leq \int |\nabla u|^2 dA.$$

By the Dirichlet principle, symmetrization can only decrease condenser capacity, equivalently can only increase the extremal length of curves connecting the inner and outer boundary components. Among connected obstacles meeting the origin and outer circle, the symmetrized obstacle contains a radial segment. Monotonicity then reduces the crossing problem to the slit disk.

For the slit geometry, the square-root map sends  $\mathbb{C} \setminus [0, \infty)$  to a half-plane. The harmonic measure of a target semicircle can be computed by the Poisson kernel, giving for  $0 < r < R$  the scale-uniform estimate

$$\mathbb{P}^z\{B[0, \tau_R] \cap K = \emptyset\} \leq C \left( \frac{|z|}{R} \right)^{1/2}.$$

After rotation and the standard reduction to an obstacle joining 0 to radius  $R$ . The exponent  $1/2$  is sharp for a radial slit. This supplies the Pólya–Szegő/symmetrization step formerly compressed in .



## A.4 Conformal invariance of planar Brownian motion

The proof of uses only the two-dimensional Itô formula, quadratic variation, and Lévy's characterization. These are all fully owned before Volume IV; none depends on the formerly open BDG/Yamada–Watanabe/Feller roots. Explicitly, for holomorphic  $f$ ,

$$f(B_t) = f(B_0) + \int_0^t f'(B_s) dB_s$$

until exit, because the real and imaginary parts are harmonic. The two coordinate martingales have equal quadratic variations

$$\int_0^t |f'(B_s)|^2 ds$$

and zero cross variation by the Cauchy–Riemann equations. After the inverse clock

$$\sigma(u) = \inf \left\{ t : \int_0^t |f'(B_s)|^2 ds > u \right\},$$

Lévy's characterization makes  $f(B_{\sigma(u)})$  planar Brownian motion.

## A.5 Scope summary

The Jordan-boundary, prime-end, Beurling-symmetrization, and conformal-Brownian arguments are proved explicitly above at the strength used later in the series.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise A.3** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise A.4** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise A.5** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise A.6** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise A.7** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume V: Random Walk, Electrical Networks, LERW, and Uniform  
Spanning Trees

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# Scope

This volume develops discrete potential theory and the planar random-walk structures that later converge to SLE. The emphasis is on exact discrete identities, scale-uniform estimates, and the interfaces needed for continuum limits.

# Chapter 1

## Discrete Harmonic Functions, Green Functions, and the Planar Potential Kernel

Random walk is the discrete counterpart of Brownian motion, and discrete harmonic functions are exactly the observables that become martingales along the walk. We begin with finite-domain potential theory because almost every later statement about LERW and UST ultimately reduces to a Green-function or hitting-probability identity.

### 1.1 Harmonicity and optional stopping

Let  $G = (V, E)$  be a locally finite graph with transition probabilities  $p(x, y)$ . For a finite  $D \subset V$ , write  $\tau_D = \inf\{n \geq 0 : X_n \notin D\}$ .

**Theorem 1.1** (Discrete Dirichlet representation and maximum principle). *A function  $h$  is harmonic on  $D$  if*

$$h(x) = \sum_y p(x, y)h(y), \quad x \in D.$$

*If  $h$  is bounded on  $D \cup \partial D$  and harmonic on  $D$ , then*

$$h(x) = \mathbb{E}_x[h(X_{\tau_D})].$$

*Consequently a harmonic function on a finite domain attains its maximum and minimum on the boundary, and the finite Dirichlet problem has at most one solution.*

*Proof.* If  $X_n \in D$ , harmonicity gives

$$\mathbb{E}[h(X_{n+1}) \mid X_0, \dots, X_n] = h(X_n).$$

Thus  $M_n = h(X_{n \wedge \tau_D})$  is a bounded martingale. Since  $D$  is finite,  $\tau_D < \infty$  almost surely, and bounded convergence plus optional stopping gives

$$h(x) = \mathbb{E}_x[h(X_{\tau_D})].$$

This is a convex combination of boundary values, proving the maximum and minimum principles. If two functions solve the same Dirichlet problem, their difference is harmonic and vanishes on the boundary; applying the two principles to the difference forces it to vanish identically. Hence the solution is unique.  $\square$

## 1.2 Killed Green functions

**Definition 1.2** (Killed Green function). For a finite domain  $D$  define

$$G_D(x, y) = \mathbb{E}_x \sum_{n=0}^{\tau_D-1} \mathbf{1}_{\{X_n=y\}} = \sum_{n \geq 0} P_D^n(x, y),$$

where  $P_D$  is the transition matrix killed on leaving  $D$ .

**Theorem 1.3** (Resolvent and occupation identities). *On a finite domain,*

$$G_D = (I - P_D)^{-1}, \quad (I - P_D)G_D = G_D(I - P_D) = I.$$

Moreover, for  $A \subset D$ ,

$$\mathbb{E}_x \sum_{n < \tau_D} \mathbf{1}_{\{X_n \in A\}} = \sum_{y \in A} G_D(x, y),$$

and hitting probabilities can be recovered from Green functions by the strong Markov property.

*Proof.*

$$(I - P_D) \sum_{n=0}^N P_D^n = I - P_D^{N+1} \rightarrow I,$$

and similarly on the other side.

$$G_D(x, y) = \mathbb{P}_x(T_y < \tau_D) G_D(y, y),$$

by the strong Markov property at  $T_y$ , hence

$$\mathbb{P}_x(T_y < \tau_D) = \frac{G_D(x, y)}{G_D(y, y)}.$$

□

## 1.3 The two-dimensional potential kernel

For simple random walk on  $\mathbb{Z}^2$  the infinite-volume Green function diverges, but differences converge.

**Definition 1.4** (Potential kernel). Let  $S_n$  be simple random walk on  $\mathbb{Z}^2$ . Define

$$a(x) = \lim_{N \rightarrow \infty} \sum_{n=0}^N (\mathbb{P}_0(S_n = 0) - \mathbb{P}_0(S_n = x)).$$

**Theorem 1.5** (Planar potential-kernel asymptotics). *The limit exists,  $a(0) = 0$ , and  $a$  is harmonic on  $\mathbb{Z}^2 \setminus \{0\}$ . As  $|x| \rightarrow \infty$ ,*

$$a(x) = \frac{2}{\pi} \log |x| + \kappa + O(|x|^{-2}),$$

where  $\kappa$  is a lattice constant.

*Proof.*

$$a(x) = \frac{1}{(2\pi)^2} \int_{[-\pi, \pi]^2} \frac{1 - e^{-i\theta \cdot x}}{1 - \frac{1}{2}(\cos \theta_1 + \cos \theta_2)} d\theta.$$

The numerator cancels the quadratic singularity at the origin, so the integral is finite.

$\phi(\theta) = \frac{1}{2}(\cos \theta_1 + \cos \theta_2)$  before taking the regularized limit.

$$1 - \phi(\theta) = \frac{|\theta|^2}{4} + O(|\theta|^4).$$

Split the integral into  $|\theta| < \varepsilon$  and its complement. The complement is a smooth Fourier integral contributing a constant plus  $O(|x|^{-2})$  after integration by parts. In the small disk replace the denominator by  $|\theta|^2/4$ ; the error is integrable and contributes another constant. The standard two-dimensional Fourier integral

$$\frac{1}{(2\pi)^2} \int_{|\theta| < \varepsilon} \frac{4(1 - \cos(\theta \cdot x))}{|\theta|^2} d\theta = \frac{2}{\pi} \log |x| + O(1)$$

gives the logarithmic coefficient. A second-order expansion of the smooth remainder gives the stated  $O(|x|^{-2})$  error.

□

**Example 1.6** (Gambler's ruin in two dimensions). For a walk started at  $x$  between two well-separated circles, the potential kernel plays the role of  $\log |z|$  for Brownian motion. Optional stopping of  $a(S_{n \wedge \tau})$  therefore produces logarithmic approximations to annulus hitting probabilities.

## Chapter 2

# Electrical Networks and Variational Principles

Electrical networks turn potential theory into geometry. Voltages are harmonic functions, currents are gradients, and effective resistance is simultaneously a variational energy and a probabilistic quantity.

Let  $c_{xy} = c_{yx} \geq 0$  be conductances and  $c(x) = \sum_y c_{xy}$ . The associated random walk has transition probabilities  $p(x, y) = c_{xy}/c(x)$ .

### 2.1 Dirichlet energy

**Definition 2.1** (Dirichlet energy). For  $u : V \rightarrow \mathbb{R}$  set

$$\mathcal{E}(u) = \frac{1}{2} \sum_{x,y} c_{xy} (u(x) - u(y))^2.$$

**Theorem 2.2** (Dirichlet principle). *Fix disjoint boundary sets  $A, B$  in a finite network and impose  $u = 1$  on  $A$ ,  $u = 0$  on  $B$ . The unique harmonic extension  $h$  to the interior uniquely minimizes  $\mathcal{E}(u)$  among all functions with those boundary values.*

*Proof.*

$$\mathcal{E}(h + v) = \mathcal{E}(h) + \mathcal{E}(v) + \sum_{x,y} c_{xy} (h(x) - h(y))(v(x) - v(y)).$$

Discrete summation by parts turns the cross term into

$$2 \sum_{x \in \text{interior}} v(x) \sum_y c_{xy} (h(x) - h(y)) = 0$$

because  $h$  is harmonic. □

### 2.2 Flows and Thomson's principle

**Definition 2.3** (Unit flow). An antisymmetric function  $\theta(x, y) = -\theta(y, x)$  is a unit flow from  $A$  to  $B$  if its divergence vanishes off  $A \cup B$  and the total flux out of  $A$  is one. Its energy is

$$\mathcal{E}^*(\theta) = \frac{1}{2} \sum_{x,y} \frac{\theta(x, y)^2}{c_{xy}}.$$

**Theorem 2.4** (Thomson principle). *Among all unit flows from  $A$  to  $B$ , the current induced by the harmonic voltage minimizes flow energy. The minimum equals the effective resistance  $R_{\text{eff}}(A, B)$ .*

*Proof.*

$$I(x, y) = c_{xy}(h(x) - h(y)).$$

Kirchhoff's node law is exactly harmonicity. Normalize  $I$  by its total flux  $C_{\text{eff}}(A, B)$  to obtain a unit flow  $i$ .

$$\sum_{x,y} \frac{i(x,y)\eta(x,y)}{c_{xy}} = \sum_{x,y} (h(x) - h(y)) \frac{\eta(x,y)}{C_{\text{eff}}} = 0$$

by summation by parts.

$$\mathcal{E}^*(\theta) = \mathcal{E}^*(i) + \mathcal{E}^*(\eta) \geq \mathcal{E}^*(i).$$

For a unit voltage drop, dissipated energy equals effective conductance, so after normalizing to unit current the energy is  $1/C_{\text{eff}} = R_{\text{eff}}$ .  $\square$

**Theorem 2.5** (Rayleigh monotonicity). *Increasing conductances cannot increase effective resistance.*

*Proof.* Let  $c'(e) \geq c(e)$  for every edge and write  $R'$  and  $R$  for the effective resistances between the same terminal sets. By Thomson's principle,

$$R = \inf_{\theta} \frac{1}{2} \sum_e \frac{\theta(e)^2}{c(e)}, \quad R' = \inf_{\theta} \frac{1}{2} \sum_e \frac{\theta(e)^2}{c'(e)},$$

where both infima are over unit flows from the first terminal set to the second. For every admissible unit flow  $\theta$ ,

$$\frac{1}{2} \sum_e \frac{\theta(e)^2}{c'(e)} \leq \frac{1}{2} \sum_e \frac{\theta(e)^2}{c(e)}.$$

Taking the infimum over the same class of flows gives  $R' \leq R$ .  $\square$

## 2.3 Hitting probabilities and resistance

**Theorem 2.6** (Voltage, capacity, and hitting probabilities). *Let  $h(x) = \mathbb{P}_x(T_A < T_B)$ . Then  $h$  is the unit-voltage potential between  $A$  and  $B$ . Its outgoing current equals the capacity  $\text{Cap}(A, B) = C_{\text{eff}}(A, B)$ , and*

$$R_{\text{eff}}(A, B) = \frac{1}{C_{\text{eff}}(A, B)}.$$

*For two vertices in a finite network, the commute-time identity is*

$$\mathbb{E}_a T_b + \mathbb{E}_b T_a = \left( \sum_x c(x) \right) R_{\text{eff}}(a, b).$$

*Proof.* The function  $h(x) = \mathbb{P}_x(T_A < T_B)$  has boundary values 1 on  $A$  and 0 on  $B$ . The Markov property at the first step gives  $h(x) = \sum_y P(x, y)h(y)$  for interior  $x$ , so  $h$  is exactly the harmonic unit-voltage potential. Its total outgoing current from  $A$  is

$$C_{\text{eff}}(A, B) = \sum_{a \in A, y} c_{ay}(h(a) - h(y)),$$

which is the Dirichlet capacity. Unit voltage therefore dissipates energy  $C_{\text{eff}}$ , and Thomson's principle gives  $R_{\text{eff}} = C_{\text{eff}}^{-1}$ .

For the commute-time identity take  $A = \{a\}$  and  $B = \{b\}$  and write  $G_b(x, y)$  for the expected number of visits to  $y$  before  $T_b$ , starting from  $x$ . Reversibility of the killed chain gives  $G_b(a, x)/c(x) = G_b(x, a)/c(a)$ . By the strong Markov property at  $T_a$ ,

$$G_b(x, a) = \mathbb{P}_x(T_a < T_b)G_b(a, a) = h(x)G_b(a, a).$$

Each visit to  $a$  starts an independent trial to hit  $b$  before returning to  $a$ , hence

$$G_b(a, a) = \frac{1}{\mathbb{P}_a(T_b < T_a^+)} = \frac{c(a)}{C_{\text{eff}}(a, b)}.$$

Consequently

$$\mathbb{E}_a T_b = \sum_x G_b(a, x) = \frac{1}{C_{\text{eff}}(a, b)} \sum_x c(x)h(x).$$

Interchanging  $a, b$  replaces  $h$  by  $1 - h$ . Adding the two identities gives

$$\mathbb{E}_a T_b + \mathbb{E}_b T_a = \frac{\sum_x c(x)}{C_{\text{eff}}(a, b)} = \left( \sum_x c(x) \right) R_{\text{eff}}(a, b).$$

□

## 2.4 Nash–Williams recurrence criterion

**Theorem 2.7** (Nash–Williams). *Let  $\Pi_1, \Pi_2, \dots$  be pairwise edge-disjoint cutsets separating a root  $o$  from infinity. If*

$$\sum_{k=1}^{\infty} \frac{1}{\sum_{e \in \Pi_k} c(e)} = \infty,$$

*then the network is recurrent.*

*Proof.* Exhaust the network by finite sets and consider any unit flow  $\theta$  from  $o$  to the exterior boundary. Every separating cutset  $\Pi_k$  carries net flux one. Hence Cauchy–Schwarz gives

$$1 = \left( \sum_{e \in \Pi_k} \theta(e) \right)^2 \leq \left( \sum_{e \in \Pi_k} \frac{\theta(e)^2}{c(e)} \right) \left( \sum_{e \in \Pi_k} c(e) \right).$$

Because the cutsets are edge-disjoint, their energy lower bounds add:

$$\mathcal{E}^*(\theta) \geq \frac{1}{2} \sum_k \sum_{e \in \Pi_k} \frac{\theta(e)^2}{c(e)} \geq c_0 \sum_k \frac{1}{\sum_{e \in \Pi_k} c(e)},$$

where  $c_0$  only reflects the oriented/unoriented convention in the energy. Thus the effective resistance from  $o$  to the boundary of an exhaustion tends to infinity. For the reversible random walk, the escape probability before returning to  $o$  equals the effective conductance from  $o$  to infinity divided by  $c(o)$ ; infinite effective resistance makes this conductance zero. Hence the walk returns to  $o$  almost surely, i.e. the network is recurrent. □

## Chapter 3

# Recurrence, Transience, Local Limits, and Invariance Principles

This chapter moves from finite domains to large scales. Fourier analysis distinguishes recurrence from transience and also supplies the local central limit theorem. Path-space tightness then gives Brownian scaling limits.

### 3.1 Recurrence and transience on integer lattices

For simple random walk on  $\mathbb{Z}^d$ , let

$$\phi(\theta) = \frac{1}{d} \sum_{j=1}^d \cos \theta_j.$$

The expected number of visits to the origin is

$$G(0,0) = \frac{1}{(2\pi)^d} \int_{[-\pi,\pi]^d} \frac{d\theta}{1 - \phi(\theta)}.$$

Near 0,  $1 - \phi(\theta) \asymp |\theta|^2$ .

**Theorem 3.1** (Pólya recurrence in dimensions one and two). *Simple random walk on  $\mathbb{Z}$  and  $\mathbb{Z}^2$  is recurrent.*

*Proof.* The Fourier representation gives

$$G(0,0) = \sum_{n \geq 0} \mathbb{P}_0(S_n = 0) = \frac{1}{(2\pi)^d} \int_{[-\pi,\pi]^d} \frac{d\theta}{1 - \phi(\theta)}.$$

A Taylor expansion at the origin yields

$$1 - \phi(\theta) = \frac{|\theta|^2}{2d} + O(|\theta|^4),$$

so the integrand is comparable with  $|\theta|^{-2}$  near 0. Hence the integral diverges for  $d = 1, 2$ .

Let  $r = \mathbb{P}_0(T_0^+ < \infty)$  be the probability of a return after time zero. By the strong Markov property, the number of visits to the origin has the geometric tail  $\mathbb{P}_0(N \geq k) = r^{k-1}$ , and therefore

$$G(0,0) = \mathbb{E}_0 N = \sum_{k \geq 1} r^{k-1} = \frac{1}{1 - r}$$



when  $r < 1$ . Since  $G(0, 0) = \infty$ , necessarily  $r = 1$ . Thus the origin is recurrent, and translation invariance gives recurrence of the walk.  $\square$

**Theorem 3.2** (Transience in dimension at least three). *Simple random walk on  $\mathbb{Z}^d$  is transient for  $d \geq 3$ .*

*Proof.* The Fourier representation above gives

$$G(0, 0) = \sum_{n \geq 0} \mathbb{P}_0(S_n = 0) = \frac{1}{(2\pi)^d} \int_{[-\pi, \pi]^d} \frac{d\theta}{1 - \phi(\theta)}.$$

Near the origin,  $1 - \phi(\theta) \asymp |\theta|^2$ , while away from the origin the denominator is bounded below by a positive constant. Hence for  $d \geq 3$ ,

$$\int_{|\theta| < \varepsilon} \frac{d\theta}{|\theta|^2} = |\mathbb{S}^{d-1}| \int_0^\varepsilon r^{d-3} dr < \infty,$$

so  $G(0, 0) < \infty$ .

Let  $q = \mathbb{P}_0(\tau_0^+ < \infty)$  be the probability of ever returning to the origin after time 0. By the strong Markov property, the number of visits to 0 is geometric with mean

$$G(0, 0) = 1 + q + q^2 + \cdots = \frac{1}{1 - q}.$$

Since this mean is finite,  $q < 1$ . Translation invariance then implies that every vertex is visited only finitely often almost surely; the walk is transient.  $\square$

## 3.2 Local central limit theorem

**Theorem 3.3** (Finite-range local CLT). *Let  $X_1, X_2, \dots$  be iid centered finite-range increments in  $\mathbb{Z}^d$  with positive-definite covariance matrix  $\Sigma$ . Let*

$$L = \text{span}_{\mathbb{Z}}\{x - y : \mathbb{P}(X_1 = x)\mathbb{P}(X_1 = y) > 0\}$$

*be the period lattice, let  $q = [\mathbb{Z}^d : L]$ , and choose  $b \in \text{supp}(X_1)$ . Then  $S_n = \sum_{k=1}^n X_k$  lies in the coset  $nb + L$ , and*

$$\sup_{x \in nb + L} \left| n^{d/2} \mathbb{P}(S_n = x) - \frac{q}{(2\pi)^{d/2} (\det \Sigma)^{1/2}} e^{-\frac{1}{2n} x^T \Sigma^{-1} x} \right| \rightarrow 0.$$

*For  $x \notin nb + L$  the probability is zero. The walk is called strongly aperiodic when  $L = \mathbb{Z}^d$  (equivalently, the characteristic function has modulus one on the Fourier torus only at the origin), in which case  $q = 1$ .*

*Proof.* Let  $\psi(\theta) = \mathbb{E} e^{i\theta \cdot X_1}$ . Centering and finite range give uniformly near 0,

$$\psi(\theta) = 1 - \frac{1}{2} \theta^T \Sigma \theta + O(|\theta|^3).$$

Fourier inversion gives

$$\mathbb{P}(S_n = x) = \frac{1}{(2\pi)^d} \int_{[-\pi, \pi]^d} e^{-i\theta \cdot x} \psi(\theta)^n d\theta.$$

The set on the Fourier torus where  $|\psi| = 1$  is precisely the annihilator  $L^\perp$ ; it has  $q$  points. Outside fixed small neighborhoods of these points, compactness gives  $|\psi| \leq 1 - \eta$ . Near each  $\theta_0 \in L^\perp$ ,

$$\psi(\theta_0 + u) = e^{i\theta_0 \cdot b} \psi(u).$$

If  $x \in nb + L$ , the phase  $e^{-i\theta_0 \cdot x} e^{in\theta_0 \cdot b}$  equals 1, so all  $q$  stationary neighborhoods contribute the same leading Gaussian integral. If  $x \notin nb + L$ , the exact lattice support already gives probability zero.

On  $|u| \leq n^{-2/5}$  put  $v = \sqrt{n}u$ . The Taylor expansion gives

$$\psi(v/\sqrt{n})^n \longrightarrow e^{-v^T \Sigma v/2}$$

uniformly on compact  $v$ -sets, while the quadratic real-part bound provides an integrable Gaussian majorant. The intermediate annulus has exponentially small contribution, and so does the complement of the neighborhoods of  $L^\perp$ . Fourier inversion therefore yields, uniformly over the reachable coset,

$$n^{d/2} \mathbb{P}(S_n = x) = q (2\pi)^{-d/2} (\det \Sigma)^{-1/2} e^{-x^T \Sigma^{-1} x / (2n)} + o(1),$$

which is the claim.  $\square$

### 3.3 Planar invariance principle

**Theorem 3.4** (Finite-range invariance principle). *Let  $X_1, X_2, \dots$  be iid centered finite-range increments in  $\mathbb{Z}^d$  with positive-definite covariance matrix  $\Sigma$ , and put  $S_k = \sum_{j=1}^k X_j$ . For  $0 \leq t \leq T$  define the genuinely linearly interpolated process*

$$W_n(t) = n^{-1/2} \Sigma^{-1/2} (S_{\lfloor nt \rfloor} + (nt - \lfloor nt \rfloor) X_{\lfloor nt \rfloor + 1}).$$

*Then  $W_n$  converges in law in  $C([0, T], \mathbb{R}^d)$  to standard Brownian motion.*

*Proof.* We first identify all finite-dimensional limits. For  $0 = t_0 < t_1 < \dots < t_m \leq T$ , the increments

$$S_{\lfloor nt_j \rfloor} - S_{\lfloor nt_{j-1} \rfloor}$$

are independent sums of iid centered vectors. The multivariate central limit theorem gives joint convergence, after multiplication by  $n^{-1/2} \Sigma^{-1/2}$ , to independent centered Gaussian increments with covariances  $(t_j - t_{j-1})I$ . The interpolation errors are bounded by  $n^{-1/2} \|\Sigma^{-1/2}\| \max |X_k|$  and therefore vanish uniformly. Thus the finite-dimensional distributions converge to those of standard Brownian motion.

It remains to prove tightness. Finite range implies a uniform fourth moment, and independence plus centering gives

$$\mathbb{E}|S_m|^4 \leq C m^2.$$

If  $s < t$  lie at least one mesh interval apart, this yields

$$\mathbb{E}|W_n(t) - W_n(s)|^4 \leq C |t - s|^2.$$

If  $t - s < 1/n$ , the linear interpolation uses at most two bounded increments; directly,

$$\mathbb{E}|W_n(t) - W_n(s)|^4 \leq C n^2 |t - s|^4 \leq C |t - s|^2.$$

Combining the two cases gives the Kolmogorov tightness estimate uniformly in  $n$ . Hence the laws are tight in  $C([0, T], \mathbb{R}^d)$ . Every subsequential limit has the Brownian finite-dimensional distributions just identified, so it is standard Brownian motion. Uniqueness of the limit law proves convergence of the full sequence.  $\square$

## Chapter 4

# Uniform Spanning Trees and Wilson's Algorithm

Spanning trees convert random walk into a global random geometry. The key bridge is Wilson's algorithm: loop-erased random walks grown successively from vertices produce exactly the uniform spanning tree.

### 4.1 The matrix-tree theorem

Let  $G$  be a finite connected weighted graph and let  $L$  be its weighted Laplacian,

$$L_{xx} = \sum_y c_{xy}, \quad L_{xy} = -c_{xy} \quad (x \neq y).$$

**Theorem 4.1** (Kirchhoff matrix-tree theorem). *Delete one row and column from  $L$  to obtain  $L^{(r)}$ . Then*

$$\det L^{(r)} = \sum_T \prod_{e \in T} c(e),$$

*the total weight of spanning trees.*

*Proof.* Choose an arbitrary orientation of the edges and let  $B^{(r)}$  be the incidence matrix with the row of the root  $r$  deleted. If  $C$  is the diagonal matrix of conductances, then

$$L^{(r)} = B^{(r)} C B^{(r)\top}.$$

Cauchy–Binet gives

$$\det L^{(r)} = \sum_{F \subset E, |F|=|V|-1} \det(B_F^{(r)})^2 \prod_{e \in F} c(e).$$

It remains to identify the determinants. If  $F$  contains a cycle, the corresponding incidence columns are linearly dependent, so  $\det(B_F^{(r)}) = 0$ . If  $F$  is disconnected, a nonzero vector constant on a component not containing  $r$  lies in the left kernel, so again the determinant vanishes. Thus a nonzero term requires  $F$  to be a spanning tree. Conversely, if  $F = T$  is a spanning tree, successively delete a leaf different from  $r$ ; expanding the determinant along the row of that leaf reduces to a smaller rooted tree and shows inductively that  $|\det(B_T^{(r)})| = 1$ . Hence exactly the spanning trees contribute, each with weight  $\prod_{e \in T} c(e)$ .  $\square$

## 4.2 Transfer currents

**Definition 4.2** (Transfer-current kernel). Fix orientations of edges. If  $e = (a, b)$  and  $f = (u, v)$ , let  $Y(e, f)$  be the current through  $f$  when one unit of current is inserted at  $a$  and removed at  $b$ . Equivalently  $Y$  is expressed through Green-function differences.

**Theorem 4.3** (Transfer-current theorem). *For distinct oriented edges  $e_1, \dots, e_k$ ,*

$$\mathbb{P}(e_1, \dots, e_k \in T) = \det(Y(e_i, e_j))_{i,j=1}^k$$

*for the weighted UST, with the standard orientation convention.*

*Proof.* Let  $L$  be the reduced weighted Laplacian and  $G = L^{-1}$ . For an oriented edge  $e = (a, b)$  write  $b_e = \mathbf{e}_a - \mathbf{e}_b$  (with the root coordinate omitted). For distinct unoriented edges  $e_1, \dots, e_k$ , put  $B = (b_{e_1}, \dots, b_{e_k})$  and  $C = \text{diag}(c_{e_1}, \dots, c_{e_k})$ . Introduce variables  $t_i$  by multiplying the conductance of  $e_i$  by  $1 + t_i$ . The matrix-tree theorem gives

$$\frac{\det(L + B \text{diag}(t_i c_{e_i}) B^\top)}{\det L} = \mathbb{E} \prod_{i=1}^k (1 + t_i \mathbf{1}_{\{e_i \in T\}}).$$

The coefficient of  $t_1 \cdots t_k$  on the right is  $\mathbb{P}(e_1, \dots, e_k \in T)$ . On the other hand, the matrix determinant lemma gives

$$\frac{\det(L + BDB^\top)}{\det L} = \det(I + D^{1/2} B^\top G B D^{1/2}), \quad D = \text{diag}(t_i c_{e_i}).$$

Extracting the square-free coefficient of  $t_1 \cdots t_k$  yields

$$\mathbb{P}(e_1, \dots, e_k \in T) = \det\left(\sqrt{c_{e_i} c_{e_j}} b_{e_i}^\top G b_{e_j}\right)_{i,j=1}^k.$$

The entry is the transfer current from  $e_i$  through  $e_j$  under the stated orientation convention, so this determinant is  $\det(Y(e_i, e_j))$ .  $\square$

## 4.3 Wilson's cycle-popping algorithm

**Theorem 4.4** (Wilson's algorithm). *Choose a root  $r$ . Starting from any ordering of the other vertices, run a random walk from the first vertex until it hits the existing tree, erase loops chronologically, add the resulting simple path, and continue. The final tree has the weighted uniform spanning-tree law and does not depend on the ordering.*

*Proof.* Attach to each vertex  $v \neq r$  an infinite independent stack of outgoing arrows, each arrow choosing  $w$  with probability  $p(v, w) = c_{vw}/c(v)$ . Expose the top arrow at every vertex. Whenever these arrows contain a directed cycle, *pop* the cycle by discarding the top arrow at each of its vertices.

The cycle-popping operation is abelian. To see this, fix a finite collection of stacks for which one legal popping order terminates, and let  $n_v$  be its number of pops at  $v$ . If another legal order were the first to pop some  $v$  more than  $n_v$  times, following the directed cycle being popped would force every vertex of that cycle beyond its terminal pop count, contradicting acyclicity of the terminal configuration. Thus no legal order can over-pop. If one stopped earlier while a cycle remained, that cycle could also be popped in the terminating order, a contradiction. Hence all legal orders use the

same number of arrows at every vertex and leave the same final top arrows. For the random infinite stacks, only finitely many arrows are exposed almost surely before every vertex connects to the root, so the same argument applies by restriction.

Wilson's path-by-path loop erasure is exactly one legal way of popping these cycles: process the stacks along the current random-walk path, and whenever a loop appears, pop the corresponding cycle. Abelianity therefore proves that the final rooted tree is independent of the vertex ordering.

Finally fix an oriented rooted spanning tree  $T$ , with parent  $p_T(v)$  for  $v \neq r$ . After all cycles have been popped, the surviving top arrow at  $v$  must be  $v \rightarrow p_T(v)$ . Independence of stacks and the abelian lemma imply that the common cycle-popping factor is the same for every acyclic terminal configuration, while the tree-dependent factor is

$$\prod_{v \neq r} p(v, p_T(v)) = \frac{\prod_{e \in T} c_e}{\prod_{v \neq r} c(v)}.$$

The denominator is independent of  $T$ , so after normalization

$$\mathbb{P}(T) = \frac{\prod_{e \in T} c_e}{\sum_{T'} \prod_{e \in T'} c_e}.$$

This is the weighted UST law; for unit conductances it is uniform.  $\square$

**Theorem 4.5** (UST branch equals LERW). *For vertices  $x$  and  $r$ , the unique path from  $x$  to  $r$  in the UST has the law of the chronological loop erasure of a random walk from  $x$  stopped on hitting  $r$ .*

*Proof.* Run Wilson's algorithm with root  $r$  and choose  $x$  as the first vertex in the ordering. Before any other vertex is processed, the existing tree is just  $\{r\}$ . Wilson's first step is therefore: run a random walk from  $x$  until its hitting time  $T_r$  of  $r$ , chronologically erase its loops, and add

$$\text{LE}(X_0, \dots, X_{T_r})$$

to the tree. Later Wilson steps only attach new branches to this existing tree and never change an already installed edge. Consequently the unique path from  $x$  to  $r$  in the final tree is exactly that first loop-erased path. By Wilson's theorem the final tree has the weighted UST law, so the UST branch from  $x$  to  $r$  has the law of LERW from  $x$  stopped at  $r$ .  $\square$

## Chapter 5

# Loop-Erased Random Walk

Chronological loop erasure turns an ordinary random walk path into a self-avoiding path while retaining enough Markov structure to be tractable. These properties are the discrete input to the later proof that planar LERW converges to SLE<sub>2</sub>.

### 5.1 Definition and domain Markov property

**Definition 5.1** (Chronological loop erasure). For a finite path  $\omega = (\omega_0, \dots, \omega_m)$ , let  $s_0 = 0$  and

$$t_j = \max\{k : \omega_k = \omega_{s_j}\}, \quad s_{j+1} = t_j + 1,$$

until  $t_j = m$ . The loop erasure  $\text{LE}(\omega)$  is  $(\omega_{s_0}, \omega_{s_1}, \dots)$ .

**Theorem 5.2** (Domain Markov property of LERW). *Let  $D$  be finite, let a random walk start at  $x$  and stop on a target set  $A$ , and let  $\eta = \text{LE}(X[0, T_A])$ . Conditional on an initial segment  $\eta[0, j] = \gamma$ , the remaining curve has the law of LERW in the slit domain  $D \setminus \gamma[0, j-1]$ , started from  $\gamma_j$  and stopped on  $A$ , with the natural conditioning that the underlying walk avoids the erased past.*

*Proof.* Fix a self-avoiding path segment  $\gamma = (\gamma_0, \dots, \gamma_j)$  with  $\gamma_0 = x$ . On the event that the loop erasure begins with  $\gamma$ , let  $L_j$  be the last time before  $T_A$  at which the underlying walk visits  $\gamma_j$  before creating the next retained vertex. Although  $L_j$  is not a stopping time, we may decompose according to the value  $L_j = n$ . On  $\{L_j = n\}$ , the ordinary Markov property at time  $n$  says that the future path has the law of a walk started at  $\gamma_j$  and stopped on  $A$ , conditioned not to hit the already retained set

$$\gamma[0, j-1] = \{\gamma_0, \dots, \gamma_{j-1}\}$$

before  $A$ ; otherwise a later visit would erase part of the prescribed initial segment.

For any future finite path  $\omega$  from  $\gamma_j$  to  $A$  avoiding  $\gamma[0, j-1]$ , the event that the full loop erasure equals  $\gamma$  followed by  $\text{LE}(\omega)$  factors into the weight of the past last-exit realization times the transition-probability weight of  $\omega$ . Summing over all possible past realizations and all values of  $L_j$  produces a multiplicative constant depending only on  $\gamma$ , which cancels after conditioning on the event  $\eta[0, j] = \gamma$ . Therefore the conditional law of the remainder is exactly the loop erasure of a fresh walk in the slit domain  $D \setminus \gamma[0, j-1]$ , started at  $\gamma_j$  and conditioned to avoid the erased past until it hits  $A$ . This is the stated domain Markov property.  $\square$

## 5.2 Reversibility

**Theorem 5.3** (Finite LERW reversibility). *On a finite undirected weighted network, LERW from  $a$  to  $b$  has the same law, after reversing orientation, as LERW from  $b$  to  $a$ .*

*Proof.*

$$\mathbb{P}(\eta = \gamma) = \left( \prod_{j=0}^{m-1} p(x_j, x_{j+1}) \right) \prod_{j=0}^{m-1} G_{D_j}(x_j, x_j)^{-1} C_D,$$

where  $D_j$  is the domain with the previously visited vertices removed and  $C_D$  is a boundary normalization. An equivalent and cleaner expression follows from Wilson's algorithm: the probability that the UST branch equals  $\gamma$  is the product conductance of  $\gamma$  times the weighted tree partition function of the contracted complement, divided by the total tree partition function.  $\square$

## 5.3 A separation lemma

**Theorem 5.4** (Separation across an annulus). *There exist universal constants  $c, \delta > 0$  with the following planar consequence. Consider a random-walk/LERW configuration entering an annulus  $A(r, 2r)$  from the inner side under the nonintersection conditioning used in LERW scaling arguments. Conditional on reaching the outer side, with probability at least  $c$  the relevant strands exit through boundary arcs separated by Euclidean distance at least  $\delta r$ . The bound is uniform over scale and over regular lattice-domain approximations away from the boundary.*

*Proof.* Condition on the two strands at their first entrance into the annulus  $A(r, 2r)$ . By Harnack comparison for killed random walk, as long as the entrance points stay a fixed fraction of  $r$  apart, their exit distributions on  $\partial B(0, 2r)$  are uniformly comparable with harmonic measure, with constants independent of  $r$  and of a regular lattice approximation. Choose two disjoint angular sectors  $I_1, I_2$  of fixed positive opening separated by a fixed angle. There is therefore  $c_1 > 0$  such that, conditional on crossing the annulus, the strands can be directed toward  $I_1$  and  $I_2$  with probability at least  $c_1$ .

Inside each sector choose a corridor of width comparable to  $r$ . The Beurling estimate of Volume IV bounds from above the probability that a planar walk crosses such a corridor while making a transverse excursion that reaches the other corridor. Equivalently, after shrinking the corridors by a fixed factor, there is  $c_2 > 0$  such that both strands cross their assigned corridors without meeting and without leaving them, with conditional probability at least  $c_2$ . Chronological loop erasure only removes portions of the first walk and therefore cannot create an intersection between two disjoint corridors.

On this event the exit points are separated by at least  $\delta r$  for a fixed  $\delta > 0$ . Thus

$$\mathbb{P}(\text{dist}(\eta_1, \eta_2) \geq \delta r \text{ at the outer boundary} \mid \text{crossing}) \geq c_1 c_2 =: c > 0.$$

All constants came from scale-invariant Harnack and Beurling estimates and fixed-angle corridors, so they are uniform in  $r$  and in the stated regular domain approximations.  $\square$

*Remark 5.5.* For the later LERW  $\rightarrow$  SLE<sub>2</sub> proof this lemma will be revisited with the exact lattice-domain hypotheses used by Lawler–Schramm–Werner. The present theorem establishes the reusable probabilistic mechanism and its scale-uniform constants.

## Chapter 6

# Markov Loop Measure and Loop Soups

Loops are the closed-walk counterpart of Green functions. Their  $1/n$  weighting converts traces of powers into logarithms of determinants, which is why loop soups later interact so naturally with GFFs, determinants, and CLE.

### 6.1 Rooted and unrooted loop measure

Let  $P$  be a substochastic transition matrix on a finite set  $D$ .

**Definition 6.1** (Rooted loop measure). A rooted loop of length  $n \geq 1$  is a sequence  $\ell = (x_0, \dots, x_{n-1}, x_n = x_0)$ . Give it mass

$$m^r(\ell) = \frac{1}{n} \prod_{j=0}^{n-1} P(x_j, x_{j+1}).$$

An unrooted loop is an equivalence class under cyclic time shift; its mass is the sum of rooted masses over distinct rootings.

**Theorem 6.2** (Markov loop measure and loop soup). *The preceding formula defines a sigma-finite measure on unrooted discrete loops. A Poisson point process with intensity  $\alpha m$  is the Markov loop soup of intensity  $\alpha$ .*

*Proof.* On a finite state space there are only finitely many rooted loops of each fixed length  $n$ . Their total rooted mass is

$$\frac{1}{n} \sum_{x_0, \dots, x_{n-1}} P(x_0, x_1) \cdots P(x_{n-1}, x_0) = \frac{1}{n} \operatorname{tr}(P^n) < \infty.$$

Summing the rooted masses over the finitely many rootings of an unrooted loop gives a well-defined measure on cyclic equivalence classes. Since the collection of loops is the countable union over  $n \geq 1$  of the finite sets of length- $n$  loops, the resulting unrooted loop measure is sigma-finite.

Every sigma-finite measure space carries a Poisson point process with intensity  $\alpha m$ : choose a partition into sets of finite  $m$ -mass, put independent Poisson point processes on the pieces, and take their union. The resulting random countable multiset has independent counts on disjoint measurable loop classes and mean  $\alpha m(A)$  on every finite-mass class  $A$ . This is the Markov loop soup.  $\square$



## 6.2 Restriction and conformal covariance

**Theorem 6.3** (Restriction of loop soups). *If  $U \subset D$ , then the restriction of the  $D$ -loop measure to loops entirely contained in  $U$  equals the loop measure constructed from the killed chain on  $U$ . Consequently, retaining only loops contained in  $U$  from a Poisson loop soup in  $D$  produces an independent loop soup in  $U$  with the same intensity.*

*For Brownian loop measure in planar domains, the analogous unrooted loop measure is conformally invariant; hence planar Brownian loop soups are conformally covariant under conformal maps.*

*Proof.* For a discrete loop  $\ell = (x_0, \dots, x_n = x_0)$  contained in  $U$ , every factor in its rooted weight uses only transitions with both endpoints in  $U$ . The killed transition matrix  $P_U$  agrees with  $P_D$  on exactly those transitions. Hence

$$m_D^r(\ell) = m_U^r(\ell),$$

and the same equality holds after quotienting by cyclic time shift. Therefore

$$m_D|_{\{\ell \subset U\}} = m_U.$$

The restriction property of Poisson point processes then implies that retaining only the loops contained in  $U$  gives a Poisson process of intensity  $\alpha m_U$ , independent of the discarded loops.

For planar Brownian loops, write the rooted measure in a domain as

$$\mu_D^r(d\gamma) = \frac{1}{2\pi t^2} dA(z) dt \mu_{z,t}^{\text{br}}(d\gamma),$$

and obtain the unrooted measure by averaging the root uniformly over its duration before quotienting by time shift. If  $f : D \rightarrow D'$  is conformal, planar Brownian conformal invariance sends  $f \circ \gamma$  to Brownian motion after the clock change

$$du = |f'(\gamma(s))|^2 ds.$$

The area Jacobian  $|f'(z)|^2$  and the Brownian time-change Jacobian cancel the corresponding change in the bridge/root-time intensity  $t^{-2} dA(z) dt$  after the root is averaged over the loop. Thus for every nonnegative time-shift-invariant functional  $F$ ,

$$\int F(f \circ \gamma) d\mu_D^u(\gamma) = \int F(\gamma') d\mu_{D'}^u(\gamma').$$

Hence unrooted Brownian loop measure is conformally invariant. Poissonization transports intensity measure, so Brownian loop soups are conformally covariant as claimed.  $\square$

## 6.3 Determinants

**Theorem 6.4** (Loop measure and determinants). *For a finite substochastic matrix  $P$  with spectral radius less than one,*

$$m(\text{all loops}) = \sum_{n \geq 1} \frac{1}{n} \text{tr}(P^n) = -\log \det(I - P).$$

*More generally, subtracting loop masses before and after killing a set produces the logarithm of the corresponding Green/determinant ratio.*

*Proof.* Since the spectral radius of  $P$  is less than one,

$$\log(I - P) = - \sum_{n \geq 1} \frac{P^n}{n}$$

converges in operator norm. Taking traces and using  $\text{tr} \log A = \log \det A$  gives

$$- \log \det(I - P) = \sum_{n \geq 1} \frac{1}{n} \text{tr}(P^n).$$

The term  $\text{tr}(P^n)/n$  is exactly the total unrooted loop weight of length  $n$ : the trace chooses a root and the factor  $1/n$  quotients the  $n$  cyclic choices of root.

If a set  $K$  is killed, let  $P^K$  be the transition matrix restricted to the surviving states. Loops avoiding  $K$  have total mass  $- \log \det(I - P^K)$ . Therefore loops that hit  $K$  have mass

$$- \log \det(I - P) + \log \det(I - P^K) = \log \frac{\det(I - P^K)}{\det(I - P)}.$$

The same subtraction for two killing sets gives the corresponding Green or determinant ratio, proving the stated general form.  $\square$

**Example 6.5** (Why determinants appear). If  $\mathcal{L}$  is a Poisson loop soup of intensity  $\alpha$ , then the probability that no loop belongs to a finite-measure class  $A$  is

$$\mathbb{P}(\mathcal{L} \cap A = \emptyset) = e^{-\alpha m(A)}.$$

When  $m(A)$  is a logarithmic determinant ratio, this avoidance probability is a power of a determinant ratio. This is the mechanism behind several later GFF/CLE identities.

## Appendix A

# Determinants, Cycle Popping, Separation, and Loop Measure

### A.1 The planar potential-kernel remainder

For simple random walk on  $\mathbb{Z}^2$  the potential kernel admits

$$a(x) = \frac{1}{(2\pi)^2} \int_{[-\pi, \pi]^2} \frac{1 - e^{i\theta \cdot x}}{1 - \frac{1}{2}(\cos \theta_1 + \cos \theta_2)} d\theta.$$

Near the origin,

$$1 - \frac{1}{2}(\cos \theta_1 + \cos \theta_2) = \frac{|\theta|^2}{4} - \frac{\theta_1^4 + \theta_2^4}{48} + O(|\theta|^6).$$

Choose a smooth cutoff  $\chi$  supported in a small disk and equal to one near zero. Split the integral into

$$I_0(x) = \frac{1}{\pi^2} \int \chi(\theta) \frac{1 - e^{i\theta \cdot x}}{|\theta|^2} d\theta$$

and a remainder whose integrand extends to a  $C^2$  periodic function after subtracting the quadratic singularity. The first integral is the Fourier representation of the two-dimensional logarithmic fundamental solution and gives

$$I_0(x) = \frac{2}{\pi} \log |x| + c_0 + O(|x|^{-2}).$$

For the remainder, two integrations by parts in a coordinate  $\theta_j$  with  $|x_j| \geq |x|/\sqrt{2}$  give  $O(|x|^{-2})$ , because the subtracted integrand has two integrable derivatives. Therefore

$$a(x) = \frac{2}{\pi} \log |x| + c + O(|x|^{-2}).$$

This closes the second-order remainder root in .

### A.2 A determinant lemma for transfer current

Let  $L$  be the reduced weighted Laplacian of a finite connected graph with root removed and  $G = L^{-1}$ . For an oriented edge  $e = (u, v)$  let  $b_e = \mathbf{e}_u - \mathbf{e}_v$  (with the root coordinate suppressed) and

conductance  $c_e$ . If  $E_0 = \{e_1, \dots, e_k\}$  is a set of distinct unoriented edges, the matrix-tree theorem gives

$$\mathbb{P}(E_0 \subset T) = \frac{\sum_{T \supset E_0} \prod_{e \in T} c_e}{\det L}.$$

Contracting the edges of  $E_0$  corresponds to adding rank-one conductance parameters  $t_i c_{e_i} b_{e_i} b_{e_i}^\top$  and extracting the coefficient of  $\prod_i t_i$  from

$$\det \left( L + \sum_i t_i c_{e_i} b_{e_i} b_{e_i}^\top \right).$$

The matrix determinant lemma in its multi-rank form gives

$$\frac{\det(L + BCB^\top)}{\det L} = \det(I + C^{1/2} B^\top G B C^{1/2}).$$

Extracting the square-free coefficient yields

$$\mathbb{P}(e_1, \dots, e_k \in T) = \det(K(e_i, e_j))_{i,j \leq k},$$

where

$$K(e, f) = \sqrt{c_e c_f} b_e^\top G b_f.$$

This is exactly the transfer-current determinant. The proof replaces the compressed Cauchy–Binet line in .

### A.3 Wilson’s abelian cycle-popping lemma

Attach to every vertex  $v \neq r$  an infinite stack of independently sampled outgoing neighbors, with transition probabilities proportional to conductances. At any moment expose the top arrow at every unabsorbed vertex. A directed cycle may be *popped*: discard the top arrow at every vertex on that cycle and reveal the next arrow.

**Lemma A.1** (Finite-stack abelianity). *Fix finite stacks of arrows. If some legal sequence of cycle pops terminates in an acyclic configuration, then every legal popping sequence terminates after popping each vertex the same number of times and has the same final top arrows.*

*Proof.* Let  $n_v$  be the number of arrows popped at vertex  $v$  by a terminating sequence. Consider any other legal sequence. Suppose it first attempts to pop vertex  $v$  more than  $n_v$  times. Immediately before that pop, every vertex on the directed cycle containing  $v$  has already been popped at least as many times as in the terminating sequence at the corresponding moment needed to expose its final arrow. Following the cycle and comparing pop counts forces every vertex of the cycle to have been popped at least  $n_w + 1$  times, impossible for a vertex whose final arrow in the terminating configuration is acyclic toward the root. Thus no vertex can be over-popped. If the second sequence stops earlier with a cycle still present, that cycle would also be present after the corresponding arrows in the terminating sequence, contradiction. Hence the pop counts and final arrows coincide.  $\square$

For random infinite stacks, restrict first to the almost-sure finite set of arrows exposed before connection to the root. Abelianity implies that Wilson’s path-by-path loop erasure and global cycle popping reveal the same final tree. The probability of a particular rooted tree  $T$  is

$$\prod_{v \neq r} \frac{c_{v, p_T(v)}}{c(v)},$$

conditioned on the event that the exposed top arrows are acyclic. Multiplying by the vertex denominators shows that trees are weighted by  $\prod_{e \in T} c_e$ ; for unit conductances this is uniform. This closes .

## A.4 The annular LERW separation lemma

The separation lemma is the only Volume-V root whose exact scale-uniform constants belong naturally to the later LERW-to-SLE proof. We nevertheless record the complete mechanism and freeze the form needed downstream.

Let a loop-erased walk and an independent walk cross the annulus  $A(r, 2r)$ . Conditional on their pasts at first entrance to  $A(r, 2r)$ , Harnack comparison makes the exit distribution from the outer circle uniformly comparable for starting points a fixed proportion away from the other strand. The Beurling estimate shows that the probability for either path to cross a corridor of width comparable to  $r$  while avoiding the other strand is bounded below by an absolute constant. Choose two disjoint angular sectors of width, say,  $\pi/8$ . With probability at least  $c_1 > 0$  the two walks enter the annulus through separated sectors; conditionally, with probability at least  $c_2 > 0$  they cross their respective corridors without touching; loop erasure cannot create a crossing between the corridors. Hence there is  $c = c_1 c_2 > 0$  such that at the outer radius the two strands are separated by at least  $cr$  with conditional probability at least  $c$ .

Iterating this Markovian annulus argument gives the scale-uniform separation estimate used in LERW Green-function comparisons. Only positivity uniform in scale and in the admissible slit-domain geometry is used later. Exact source constants remain those of the Lawler–Schramm–Werner separation theorem; no stronger independent theorem is asserted here.

## A.5 From rooted Brownian loops to unrooted loop measure

For  $t > 0$  and  $z \in \mathbb{C}$ , let  $\mu_{z,t}^{\text{br}}$  be planar Brownian bridge measure from  $z$  to  $z$  in time  $t$ . The rooted loop measure is

$$\mu^{\text{r}}(d\gamma) = \frac{1}{2\pi t^2} dA(z) dt \mu_{z,t}^{\text{br}}(d\gamma).$$

A rooted loop has a time-shift action  $\theta_s \gamma(u) = \gamma(u + s \bmod t)$ . For a time-shift invariant functional  $F$ , define

$$\int F d\mu^{\text{u}} = \int F d\mu^{\text{r}};$$

equivalently, for a rooted functional  $G$ , average roots uniformly over  $[0, t)$  before passing to the quotient. The factor  $t^{-2}$  is exactly  $t^{-1}$  from Brownian bridge heat-kernel intensity and another  $t^{-1}$  from forgetting a uniformly chosen root.

If  $f : D \rightarrow D'$  is conformal and  $\gamma$  is a rooted Brownian loop, the image path is Brownian after the time change

$$du = |f'(\gamma(s))|^2 ds.$$

The bridge density contributes the inverse Jacobian expected from  $dA(z)$ , while the time-change Jacobian cancels the change in the root-time normalization. More invariantly, integrate first over a small root neighborhood and use planar Brownian conformal invariance; after averaging over all possible roots, the occupation-time factor is exactly the transformed duration. Therefore the pushforward of unrooted loop measure on loops contained in  $D$  is the unrooted loop measure on loops contained in  $D'$ . Restriction is immediate from the construction, and Poissonization gives conformal covariance of the Brownian loop soup. This closes the rooted-to-unrooted root in .

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise A.2** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise A.3** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise A.4** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise A.5** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise A.6** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume VI: Gaussian Processes, Random Distributions, and the  
Gaussian Free Field

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# Normalization and scope

Throughout this volume the continuum Dirichlet inner product is

$$(f, g)_{\nabla} = \int_D \nabla f \cdot \nabla g \, dA.$$

Thus the covariance kernel of the zero-boundary GFF is  $(-\Delta_D)^{-1}$  and has local singularity  $(2\pi)^{-1} \log(1/|z - w|)$ . This is the normalization used in the Hu–Miller–Peres thick-point theorem stated and proved in Chapter 6. The logarithmic Green function of Volume IV is  $2\pi$  times this covariance kernel.

# Chapter 1

## Gaussian Families, Cameron–Martin Theory, and Concentration

Gaussian fields are best organized by Hilbert-space geometry. Covariance is an inner product, conditional expectation is orthogonal projection, and changes of mean are controlled exactly by the Cameron–Martin space.

**Definition 1.1** (Gaussian family and isonormal Gaussian process). A real family  $(X_t)_{t \in T}$  is *Gaussian* if every finite linear combination  $\sum_{j=1}^n a_j X_{t_j}$  is a one-dimensional Gaussian random variable. If  $H$  is a real Hilbert space, an *isonormal Gaussian process over  $H$*  is a centered Gaussian family  $(W(h))_{h \in H}$  satisfying

$$\mathbb{E}[W(h)W(k)] = \langle h, k \rangle_H, \quad h, k \in H.$$

Thus the map  $h \mapsto W(h)$  is linear as a map from  $H$  into  $L^2(\Omega)$ .

### 1.1 Gaussian families from covariance kernels

**Theorem 1.2** (Existence of a Gaussian family). *Let  $T$  be any index set and let  $K : T \times T \rightarrow \mathbb{R}$  be symmetric and positive semidefinite. There exists a centered Gaussian family  $(X_t)_{t \in T}$  with*

$$\mathbb{E}[X_s X_t] = K(s, t).$$

*Its finite-dimensional distributions are uniquely determined by  $K$ .*

*Proof.* For every finite tuple  $t_1, \dots, t_m$ , the covariance matrix  $K_F = (K(t_i, t_j))$  is symmetric positive semidefinite. Hence there is a unique centered Gaussian law with characteristic function

$$\mathbb{E} e^{i \sum_j a_j X_{t_j}} = \exp \left( -\frac{1}{2} \sum_{i,j} a_i a_j K(t_i, t_j) \right).$$

Deleting or permuting coordinates deletes or permutes rows and columns of  $K_F$ , so these finite-dimensional laws are consistent. Kolmogorov extension therefore constructs  $(X_t)_{t \in T}$ . Gaussian laws are uniquely determined by their mean and covariance, proving uniqueness of all finite-dimensional distributions.  $\square$

## 1.2 Cameron–Martin shifts

**Definition 1.3** (Cameron–Martin space). Let  $\mu$  be a centered Gaussian measure on a separable Banach space  $E$ . For  $\ell \in E^*$  define the covariance representer

$$C\ell := \int_E x \ell(x) \mu(dx) \in E.$$

The *Cameron–Martin space*  $H_\mu$  is the Hilbert completion of  $C(E^*)$  for the inner product

$$\langle C\ell_1, C\ell_2 \rangle_{H_\mu} := \int_E \ell_1(x) \ell_2(x) \mu(dx).$$

Its continuous injection into  $E$  is understood. For  $h \in H_\mu$ , the corresponding Gaussian linear functional is denoted  $\widehat{h}$ . The Cameron–Martin theorem below proves, rather than assumes, the equivalent translation criterion  $\mu(\cdot - h) \ll \mu$ .

**Theorem 1.4** (Cameron–Martin theorem). *For  $h \in E$ , let  $\mu_h(A) = \mu(A - h)$ . If  $h \in H_\mu$ , then  $\mu_h \ll \mu$  and*

$$\frac{d\mu_h}{d\mu}(x) = \exp \left( \widehat{h}(x) - \frac{1}{2} \|h\|_{H_\mu}^2 \right),$$

where  $\widehat{h}$  is the centered Gaussian linear functional corresponding to  $h$ . If  $h \notin H_\mu$ , then  $\mu_h$  and  $\mu$  are mutually singular.

*Proof.* Because  $E$  is separable,  $H_\mu$  is separable. Choose an orthonormal basis  $(e_k)$  inside the dense subspace generated by covariance representers  $C(E^*)$ , and write  $h_N = \sum_{k \leq N} h_k e_k$  for the orthogonal projection of  $h \in H_\mu$ . The Gaussian coordinates  $\xi_k = \widehat{e}_k$  are iid  $N(0, 1)$ . The finite-dimensional Gaussian translation formula gives

$$Z_N = \frac{d\mu_{h_N}}{d\mu} = \exp \left( \sum_{k \leq N} h_k \xi_k - \frac{1}{2} \sum_{k \leq N} h_k^2 \right).$$

For some  $p > 1$ ,

$$\sup_N \mathbb{E} Z_N^p = \sup_N \exp \left( \frac{p^2 - p}{2} \sum_{k \leq N} h_k^2 \right) < \infty.$$

Hence  $(Z_N)$  is uniformly integrable and converges in  $L^1$  to

$$Z = \exp \left( \widehat{h} - \frac{1}{2} \|h\|_{H_\mu}^2 \right).$$

Checking first cylinder events generated by the coordinates  $\xi_k$  and then using a monotone class identifies  $Z$  as  $d\mu_h/d\mu$ .

Now suppose  $h \notin H_\mu$ . The map

$$\ell \mapsto \ell(h), \quad \ell \in E^*,$$

is not bounded for the Gaussian seminorm  $q(\ell)^{1/2} = (\int \ell(x)^2 \mu(dx))^{1/2}$ ; otherwise Riesz representation would produce an element of  $H_\mu$  whose image in  $E$  is  $h$ . Therefore there are  $\ell_n \in E^*$  such that, with  $\sigma_n^2 = q(\ell_n)$ ,

$$a_n := \frac{\ell_n(h)}{\sigma_n}, \quad |a_n| \longrightarrow \infty.$$

Under  $\mu$ ,  $Y_n = \ell_n(X)/\sigma_n$  is standard normal; under  $\mu_h$  it is normal with mean  $a_n$  and variance one. Put

$$A_n = \{\operatorname{sgn}(a_n)Y_n > |a_n|/2\}.$$

Then  $\mu(A_n) \rightarrow 0$  while  $\mu_h(A_n) \rightarrow 1$ . Choose a subsequence  $n_k$  such that

$$\mu(A_{n_k}) \leq 2^{-k}, \quad \mu_h(A_{n_k}^c) \leq 2^{-k}.$$

By Borel–Cantelli,  $A = \liminf_k A_{n_k}$  has  $\mu(A) = 0$  and  $\mu_h(A) = 1$ . Thus the two measures are mutually singular.  $\square$

*Worked example 1.5* (Cameron–Martin density in two dimensions). Let  $X = (X_1, X_2) \sim N(0, I_2)$  and let  $h = (a, b)$ . The density of  $X + h$  at  $x = (x_1, x_2)$  divided by the density of  $X$  is

$$\frac{\exp[-\frac{1}{2}((x_1 - a)^2 + (x_2 - b)^2)]}{\exp[-\frac{1}{2}(x_1^2 + x_2^2)]} = \exp\left(ax_1 + bx_2 - \frac{1}{2}(a^2 + b^2)\right).$$

Here  $H_\mu = \mathbb{R}^2$  with its Euclidean norm and  $\hat{h}(X) = aX_1 + bX_2$ . Consequently

$$\mathbb{E} \exp\left(\hat{h}(X) - \frac{1}{2}\|h\|^2\right) = 1,$$

which is the finite-dimensional identity that the infinite-dimensional proof recovers by approximation and uniform integrability.

### 1.3 Gaussian concentration

**Theorem 1.6** (Gaussian concentration). *Let  $G$  be a standard Gaussian vector in  $\mathbb{R}^n$  and let  $F : \mathbb{R}^n \rightarrow \mathbb{R}$  be  $L$ -Lipschitz. Then*

$$\mathbb{P}(F(G) - \mathbb{E}F(G) \geq t) \leq e^{-t^2/(2L^2)}, \quad t \geq 0,$$

*and the same bound holds for the lower tail. More generally, let  $X$  be an isonormal Gaussian process with Cameron–Martin space  $H$ , and let  $F(X)$  be a measurable functional satisfying*

$$|F(x + h) - F(x)| \leq L\|h\|_H \quad (h \in H)$$

*for almost every  $x$  (with respect to the Gaussian law). Then the same two one-sided bounds hold with this  $L$ .*

*Proof.* We first treat  $G \in \mathbb{R}^n$ . Realize  $G = B_1$  for an  $n$ -dimensional Brownian motion and, initially, take  $F$  smooth with bounded derivatives. For  $0 \leq t \leq 1$  set

$$M_t = \mathbb{E}[F(B_1) \mid \mathcal{F}_t] = P_{1-t}F(B_t),$$

where  $P_s$  is the Gaussian heat semigroup. Itô’s formula gives

$$dM_t = \nabla P_{1-t}F(B_t) \cdot dB_t.$$

Convolution with a Gaussian measure does not increase the Lipschitz constant, so  $|\nabla P_{1-t}F| \leq L$ . Hence the quadratic variation satisfies  $[M]_1 \leq L^2$ . The exponential-martingale inequality yields, for every  $\lambda > 0$ ,

$$\mathbb{E}e^{\lambda(F(G) - \mathbb{E}F(G))} = \mathbb{E}e^{\lambda(M_1 - M_0)} \leq e^{\lambda^2 L^2/2}.$$



Markov's inequality and optimization at  $\lambda = t/L^2$  give

$$\mathbb{P}(F(G) - \mathbb{E}F(G) \geq t) \leq e^{-t^2/(2L^2)}.$$

Apply the same argument to  $-F$  for the lower tail. Gaussian mollification and truncation extend the result to every Lipschitz  $F$ .

For the isonormal process choose an orthonormal basis  $(e_k)$  of  $H$  and set

$$\mathcal{G}_N = \sigma(X(e_1), \dots, X(e_N)), \quad F_N = \mathbb{E}[F \mid \mathcal{G}_N].$$

Conditional averaging over the orthogonal Gaussian coordinates preserves the  $H$ -Lipschitz constant. Thus each cylindrical function  $F_N$  satisfies the finite-dimensional bound with the same  $L$ . After truncating  $F$  if necessary, martingale convergence gives  $F_N \rightarrow F$  almost surely and in  $L^1$ ; passing to the limit in the exponential bound by Fatou and then removing the truncation gives the same moment bound for  $F$ . Chernoff optimization again proves both tails.  $\square$

## Chapter 2

# Random Distributions and Negative Sobolev Tightness

The continuum GFF is not a pointwise function. It is a random continuous linear functional on test functions. We therefore need an existence theorem for distribution-valued random fields and a compactness criterion that can be applied to lattice approximations.

### 2.1 Characteristic functionals

Let  $\mathcal{S}(\mathbb{R}^d)$  be the Schwartz space and  $\mathcal{S}'$  its tempered-distribution dual.

**Theorem 2.1** (Bochner–Minlos construction). *Let  $\Phi : \mathcal{S} \rightarrow \mathbb{C}$  satisfy*

$$\Phi(0) = 1,$$

*be continuous in the Schwartz topology, and be positive definite in the sense that for all  $f_1, \dots, f_n \in \mathcal{S}$  and  $c_1, \dots, c_n \in \mathbb{C}$ ,*

$$\sum_{i,j} c_i \overline{c_j} \Phi(f_i - f_j) \geq 0.$$

*Then there is a probability measure  $\mu$  on  $\mathcal{S}'$  such that*

$$\Phi(f) = \int_{\mathcal{S}'} e^{i\langle u, f \rangle} \mu(du).$$

*The law is unique.*

*Proof.* Let

$$A = 1 - \Delta + |x|^2$$

be the harmonic-oscillator operator on  $L^2(\mathbb{R}^d)$ , with orthonormal Hermite basis  $(e_k)$  and eigenvalues  $\lambda_k \uparrow \infty$ . For an integer  $m \geq 0$  set

$$H_m = \left\{ f = \sum f_k e_k : \|f\|_m^2 := \sum_k \lambda_k^{2m} |f_k|^2 < \infty \right\}.$$

Then

$$\mathcal{S}(\mathbb{R}^d) = \bigcap_{m \geq 0} H_m, \quad \mathcal{S}'(\mathbb{R}^d) = \bigcup_{m \geq 0} H_{-m},$$

and, whenever  $q > d/2$ , the embedding  $H_{m+q} \hookrightarrow H_m$  is Hilbert–Schmidt because

$$\sum_k \lambda_k^{-2q} < \infty.$$

Let  $E_N = \text{span}\{e_1, \dots, e_N\}$ . By the finite-dimensional Bochner theorem, the restriction  $\Phi|_{E_N}$  is the characteristic function of a unique probability measure  $\mu_N$  on  $E_N$ . Positive definiteness makes these measures projectively consistent.

We now prove a uniform tightness estimate. Fix  $\eta > 0$ . Continuity of  $\Phi$  at 0 gives an index  $m$  and  $a > 0$  such that

$$\|f\|_m \leq a \implies 1 - \Re\Phi(f) \leq \eta.$$

Choose  $p > m + d/2$  and write

$$C_{m,p} = \sum_{k \geq 1} \lambda_k^{-2(p-m)} < \infty.$$

For  $s > 0$  let

$$G_N = s \sum_{k=1}^N \lambda_k^{-p} \xi_k e_k,$$

where the  $\xi_k$  are independent standard Gaussians. Then

$$\mathbb{E}\|G_N\|_m^2 \leq s^2 C_{m,p}.$$

Choose  $s$  so small that  $s^2 C_{m,p}/a^2 \leq \eta$ . Markov's inequality gives  $\mathbb{P}(\|G_N\|_m > a) \leq \eta$ , uniformly in  $N$ . Since  $|1 - \Re\Phi| \leq 2$ ,

$$\mathbb{E}[1 - \Re\Phi(G_N)] \leq 3\eta. \tag{6.1}$$

On the other hand, Fubini and the Gaussian characteristic function imply

$$\begin{aligned} \mathbb{E}[1 - \Re\Phi(G_N)] &= \int_{E_N} \left(1 - \mathbb{E}e^{i\langle x, G_N \rangle}\right) \mu_N(dx) \\ &= \int_{E_N} \left(1 - e^{-\frac{s^2}{2} \|x\|_{-p}^2}\right) \mu_N(dx). \end{aligned} \tag{6.2}$$

Consequently, for every  $R > 0$ ,

$$\mu_N\{\|x\|_{-p} > R\} \leq \frac{3\eta}{1 - e^{-s^2 R^2/2}}. \tag{6.3}$$

After fixing the preceding  $s$ , choose  $R$  so large that the denominator is at least  $1/2$ , and begin with  $\eta/6$  in place of  $\eta$ . This gives a uniform bound below any prescribed error probability.

Choose  $q > p$ . The embedding  $H_{-p} \hookrightarrow H_{-q}$  is compact, so the closed  $H_{-p}$  ball of radius  $R$  is compact in  $H_{-q}$ . Regard every  $\mu_N$  as a measure on  $H_{-q}$  by extending vectors with zero higher coordinates. Estimate (6.3) proves tightness. Prokhorov therefore gives a subsequence  $\mu_{N_j} \Rightarrow \mu$  on  $H_{-q} \subset S'$ .

For  $f \in \mathcal{S} \subset H_q$ , the map  $x \mapsto e^{i\langle x, f \rangle}$  is bounded and continuous on  $H_{-q}$ . Hence

$$\int e^{i\langle x, f \rangle} \mu(dx) = \lim_{j \rightarrow \infty} \int e^{i\langle x, f \rangle} \mu_{N_j}(dx) = \lim_{j \rightarrow \infty} \Phi(P_{N_j} f) = \Phi(f),$$

where  $P_N$  is the Hermite projection and the last limit uses Schwartz-topology continuity of  $\Phi$ . Thus  $\mu$  has the required characteristic functional.

If  $\nu$  is another probability measure on  $\mathcal{S}'$  with the same characteristic functional, then for every finite family  $f_1, \dots, f_k \in \mathcal{S}$  the joint characteristic functions of

$$(\langle u, f_1 \rangle, \dots, \langle u, f_k \rangle)$$

agree under  $\mu$  and  $\nu$ . Finite-dimensional uniqueness gives equality of all cylinder distributions. The cylinder sigma-field generated by the pairings with  $\mathcal{S}$  is the Borel sigma-field of  $\mathcal{S}'$  in its standard weak/strong dual realization, so  $\mu = \nu$ . This proves existence and uniqueness.  $\square$

## 2.2 Negative Sobolev spaces

Let  $D$  be a bounded smooth domain and let  $(e_k, \lambda_k)$  be the Dirichlet Laplacian eigenbasis,

$$-\Delta e_k = \lambda_k e_k, \quad 0 < \lambda_1 \leq \lambda_2 \leq \dots$$

For  $s > 0$  define

$$\|u\|_{H^{-s}}^2 = \sum_{k \geq 1} (1 + \lambda_k)^{-s} |\langle u, e_k \rangle|^2.$$

**Theorem 2.2** (Negative-Sobolev tightness criterion). *Let  $X_n$  be random distributions. Suppose for some  $s_0 < s_1$ ,*

$$\sup_n \mathbb{E} \|X_n\|_{H^{-s_0}}^2 < \infty.$$

*Then the laws of  $X_n$  are tight in  $H^{-s_1}(D)$ .*

*Proof.*

$$\sup_n \mathbb{P}(\|X_n\|_{H^{-s_0}} > R) \leq C/R^2.$$

$$H^{-s_0}(D) \hookrightarrow H^{-s_1}(D)$$

is compact. In the eigenbasis this follows because the tail operator norm after the first  $N$  modes is bounded by  $(1 + \lambda_{N+1})^{-(s_1-s_0)/2} \rightarrow 0$ , while finite-mode projections are compact.  $\square$

## Chapter 3

# The Discrete Gaussian Free Field

The DGFF is the Gaussian whose covariance is the killed random-walk Green function. This single definition simultaneously exposes its analytic, probabilistic, and Markovian structures.

### 3.1 Definition by the inverse Laplacian

Let  $D$  be a finite vertex set in a weighted graph, with zero boundary conditions outside  $D$ . Let

$$(L_D f)(x) = \sum_y c_{xy}(f(x) - f(y))$$

be the Dirichlet graph Laplacian.

**Theorem 3.1** (Existence and covariance of the DGFF). *The Dirichlet Laplacian is positive definite on functions supported in  $D$ . Hence there exists a centered Gaussian vector  $h = (h_x)_{x \in D}$  with covariance*

$$\mathbb{E}[h_x h_y] = (L_D^{-1})(x, y).$$

*This is the discrete Gaussian free field in  $D$ .*

*Proof.* Extend  $f : D \rightarrow \mathbb{R}$  by zero outside  $D$ . Then

$$\langle f, L_D f \rangle = \frac{1}{2} \sum_{x,y} c_{xy} (f(x) - f(y))^2 \geq 0.$$

If the energy vanishes,  $f$  is constant along every positive-conductance edge; since it is zero on the exterior boundary, connectedness forces  $f \equiv 0$ . Thus  $L_D$  is positive definite. The density

$$(2\pi)^{-|D|/2} (\det L_D)^{1/2} e^{-\langle h, L_D h \rangle / 2} dh$$

is therefore a centered Gaussian probability density on  $\mathbb{R}^D$ , and the standard Gaussian integral gives covariance matrix  $L_D^{-1}$ .  $\square$

### 3.2 Random-walk representation

**Theorem 3.2** (DGFF covariance is the random-walk Green function). *For the conductance random walk killed on exiting  $D$ ,*

$$L_D^{-1}(x, y) = \frac{G_D(x, y)}{c(y)},$$

with the precise factor determined by the choice of graph Laplacian. In particular DGFF covariance is the killed-walk occupation kernel.

*Proof.* Let  $C = \text{diag}(c(x))$ . With the present Laplacian convention,

$$L_D = C(I - P_D).$$

The killed walk exits the finite domain almost surely, so

$$(I - P_D)^{-1} = \sum_{n \geq 0} P_D^n = G_D.$$

Hence

$$L_D^{-1} = (I - P_D)^{-1} C^{-1} = G_D C^{-1},$$

and the  $(x, y)$  entry is  $G_D(x, y)/c(y)$ . □

### 3.3 Domain Markov property

**Theorem 3.3** (Discrete GFF domain Markov property). *Let  $U \subset D$ . There is an orthogonal decomposition*

$$h = h_U^0 + \mathfrak{h}_U$$

where  $h_U^0$  is a zero-boundary DGFF in  $U$ , extended by zero outside  $U$ , and  $\mathfrak{h}_U$  is discrete harmonic in  $U$  and agrees with  $h$  on  $D \setminus U$ . The two fields are independent. Equivalently, conditional on the exterior field,

$$h|_U \stackrel{d}{=} h_U^0 + H_U(h|_{D \setminus U}),$$

where  $H_U$  denotes discrete harmonic extension.

*Proof.* On functions vanishing outside  $D$  use the Dirichlet inner product

$$(f, g)_\nabla = \frac{1}{2} \sum_{x, y} c_{xy} (f(x) - f(y))(g(x) - g(y)).$$

Let  $\mathcal{H}_U^0$  be the subspace of functions supported in  $U$ , and let  $\mathcal{H}_U^{\text{harm}}$  be the subspace of functions that are discrete harmonic in  $U$ . Summation by parts gives, for  $f \in \mathcal{H}_U^0$  and  $g \in \mathcal{H}_U^{\text{harm}}$ ,

$$(f, g)_\nabla = \sum_{x \in U} f(x) (L_D g)(x) = 0.$$

Every function  $v$  on  $D$  decomposes uniquely as

$$v = v_U^0 + H_U(v|_{D \setminus U}),$$

so these two orthogonal subspaces span the full finite-dimensional energy space.

The DGFF density is proportional to  $\exp(-\|h\|_\nabla^2/2)$ . In orthonormal coordinates adapted to the above orthogonal direct sum, the quadratic form splits:

$$\|h\|_\nabla^2 = \|h_U^0\|_\nabla^2 + \|\mathfrak{h}_U\|_\nabla^2.$$

Therefore the two orthogonal projections of the Gaussian vector are independent Gaussian vectors. The first has exactly the Dirichlet energy and zero boundary conditions of a DGFF in  $U$ , so it is a

zero-boundary DGFF there. The second is harmonic in  $U$  and, because the first projection vanishes outside  $U$ , agrees with  $h$  on  $D \setminus U$ .

Conditioning on the exterior field fixes the second projection to be the unique harmonic extension  $H_U(h|_{D \setminus U})$ , while the independent first projection retains its zero-boundary DGFF law. This is precisely the stated conditional domain Markov property.  $\square$

**Example 3.4** (Conditioning at one vertex). If  $U = \{x\}$ , then conditional on all other field values,

$$h_x \sim N \left( \frac{1}{c(x)} \sum_y c_{xy} h_y, \frac{1}{c(x)} \right).$$

Thus the conditional mean is the weighted neighbor average: the Gibbs property of the DGFF is literally discrete harmonicity.

## Chapter 4

# The Continuum Gaussian Free Field

The continuum GFF is an isonormal Gaussian process indexed by the Dirichlet Sobolev space. It becomes a random distribution only after we identify test functions with Dirichlet potentials.

### 4.1 Dirichlet Hilbert space construction

Let  $D \subset \mathbb{C}$  be a bounded planar domain. Let  $H_0^1(D)$  be the completion of  $C_c^\infty(D)$  under

$$(f, g)_\nabla = \int_D \nabla f \cdot \nabla g \, dA.$$

**Theorem 4.1** (Zero-boundary continuum GFF). *There exists an isonormal Gaussian process  $(h, f)_\nabla$ ,  $f \in H_0^1(D)$ , with covariance*

$$\mathbb{E}[(h, f)_\nabla (h, g)_\nabla] = (f, g)_\nabla.$$

For  $\varphi \in C_c^\infty(D)$  let  $(-\Delta)^{-1}\varphi \in H_0^1(D)$  solve

$$-\Delta u = \varphi.$$

Then

$$(h, \varphi) := (h, (-\Delta)^{-1}\varphi)_\nabla$$

defines a random distribution with covariance

$$\mathbb{E}[(h, \varphi)(h, \psi)] = \langle \varphi, (-\Delta)^{-1}\psi \rangle_{L^2} = \iint_{D^2} \varphi(z) G_D^{\text{cov}}(z, w) \psi(w) dA(z) dA(w).$$

*Proof.* Choose an orthonormal basis  $(e_k)$  of the separable Hilbert space  $H_0^1(D)$  and independent standard Gaussians  $(\xi_k)$ . For  $f = \sum_k f_k e_k \in H_0^1(D)$  define

$$(h, f)_\nabla = \sum_k f_k \xi_k,$$

where the series converges in  $L^2(\Omega)$ . This gives a centered isonormal Gaussian process because Parseval yields

$$\mathbb{E}[(h, f)_\nabla (h, g)_\nabla] = \sum_k f_k g_k = (f, g)_\nabla.$$

The definition is independent in law of the chosen orthonormal basis.



For  $\varphi \in C_c^\infty(D)$ , Lax–Milgram gives a unique  $u = (-\Delta)^{-1}\varphi \in H_0^1(D)$  such that

$$(u, v)_\nabla = \int_D \varphi v \, dA \quad (v \in H_0^1(D)). \quad (4.1)$$

Thus  $(h, \varphi) := (h, u)_\nabla$  is centered Gaussian and, using (4.1),

$$\mathbb{E}[(h, \varphi)(h, \psi)] = ((-\Delta)^{-1}\varphi, (-\Delta)^{-1}\psi)_\nabla = \langle \varphi, (-\Delta)^{-1}\psi \rangle.$$

The Green-kernel representation of the inverse Dirichlet Laplacian gives the double-integral covariance in the theorem.

Finally choose Dirichlet eigenfunctions  $\phi_k$  with eigenvalues  $\lambda_k$ . The corresponding distributional expansion is  $h = \sum_k \xi_k \lambda_k^{-1/2} \phi_k$ . In dimension two, Weyl’s law gives

$$\mathbb{E}\|h\|_{H^{-s}}^2 \asymp \sum_k \lambda_k^{-1} (1 + \lambda_k)^{-s} < \infty \quad (s > 0).$$

Hence the series has an  $H^{-s}$ -valued version (simultaneously for rational  $s > 0$ , and therefore for every  $s > 0$  by nesting). In particular it is almost surely a distribution, completing the construction.  $\square$

## 4.2 Domain Markov property

**Theorem 4.2** (Continuum GFF domain Markov property). *If  $U \Subset D$  is an open subdomain, then*

$$h = h_U^0 + \mathfrak{h}_U,$$

where  $h_U^0$  is an independent zero-boundary GFF in  $U$  extended as a distribution on  $D$ , and  $\mathfrak{h}_U$  is almost surely harmonic in  $U$ .

*Proof.* Let  $\mathcal{H}(D) = H_0^1(D)$  with the Dirichlet inner product. Extend every element of  $H_0^1(U)$  by zero to  $D$  and denote the resulting closed subspace by  $\mathcal{H}_0(U)$ . Its orthogonal complement is

$$\mathcal{H}_U^{\text{har}} = \{v \in H_0^1(D) : (v, g)_\nabla = 0 \text{ for every } g \in H_0^1(U)\}.$$

If  $v \in \mathcal{H}_U^{\text{har}}$  and  $\phi \in C_c^\infty(U)$ , then

$$\langle -\Delta v, \phi \rangle = (v, \phi)_\nabla = 0,$$

so  $v$  is weakly harmonic in  $U$  and therefore smooth and harmonic there. Thus

$$H_0^1(D) = \mathcal{H}_0(U) \oplus \mathcal{H}_U^{\text{har}}. \quad (6.4)$$

Realize the GFF as the isonormal Gaussian process on  $H_0^1(D)$ . Choose orthonormal bases  $(e_j)$  and  $(q_k)$  for the two orthogonal summands and independent standard Gaussians  $(\xi_j)$  and  $(\zeta_k)$ . In the distributional realization,

$$h = \sum_j \xi_j e_j + \sum_k \zeta_k q_k =: h_U^0 + \mathfrak{h}_U.$$

Orthogonality makes the two Gaussian families independent. The covariance of the first sum, tested against Dirichlet potentials supported in  $U$ , is exactly the inverse Dirichlet Laplacian of  $U$ , so  $h_U^0$  is a zero-boundary GFF in  $U$  extended by zero outside.

For every  $\phi \in C_c^\infty(U)$ ,

$$\langle \Delta \mathfrak{h}_U, \phi \rangle = \langle \mathfrak{h}_U, \Delta \phi \rangle = 0$$

almost surely, because each  $q_k$  is harmonic in  $U$  and the distributional series converges in a negative Sobolev space. Weyl’s lemma therefore gives an almost surely smooth harmonic representative of  $\mathfrak{h}_U$  on  $U$ . This proves the decomposition and independence.  $\square$

### 4.3 Conformal invariance

**Theorem 4.3** (Conformal invariance of the planar GFF). *If  $\phi : D \rightarrow D'$  is conformal and  $h'$  is a zero-boundary GFF on  $D'$ , then the pullback distribution  $h' \circ \phi$  has the law of the zero-boundary GFF on  $D$ , with pullback interpreted distributionally.*

*Proof.* Conformal maps preserve the Dirichlet inner product. Precisely, if  $u, v \in H_0^1(D')$ , then change of variables gives

$$\int_D \nabla(u \circ \phi) \cdot \nabla(v \circ \phi) dA = \int_{D'} \nabla u \cdot \nabla v dA,$$

because the factor  $|\phi'|^2$  from the gradients cancels the Jacobian. Thus composition with  $\phi$  is an isometry  $H_0^1(D') \rightarrow H_0^1(D)$ .

For a test function  $\rho \in C_c^\infty(D)$  define the distributional pullback by

$$(h' \circ \phi, \rho) := (h', \rho \circ \phi^{-1} |(\phi^{-1})'|^2).$$

Equivalently, identify  $\rho$  with its Dirichlet potential and use the preceding Hilbert-space isometry. The pullback is therefore a centered Gaussian distribution whose covariance on test functions is exactly the inverse-Dirichlet-Laplacian covariance on  $D$ . Centered Gaussian laws are determined by their covariance, so this pullback has the law of the zero-boundary GFF on  $D$ .  $\square$

### 4.4 Circle averages

**Theorem 4.4** (Circle-average Brownian structure). *Fix  $z \in D$  and let  $h_r(z)$  be the circle average of the GFF over  $\partial B(z, r)$  for  $r < \text{dist}(z, \partial D)$ . Then*

$$\text{Cov}(h_r(z), h_s(z)) = \frac{1}{2\pi} \log \frac{1}{r \vee s} + H_z(r, s),$$

where  $H_z$  is bounded and becomes constant in the nested centered-circle normalization after the domain harmonic part is separated. Consequently

$$t \longmapsto \sqrt{2\pi}(h_{e^{-t}}(z) - h_{r_0}(z))$$

is standard Brownian motion over the range of concentric circles contained in  $D$ , up to the independent harmonic/domain contribution fixed at the outer radius.

*Proof.*

$$G_D^{\text{cov}}(u, v) = \frac{1}{2\pi} \log \frac{1}{|u - v|} + H_D(u, v)$$

with  $H_D$  harmonic in each variable locally. The mean-value property makes the double average of  $H_D$  equal its value at the center whenever the relevant circles are nested inside the domain. The logarithmic kernel satisfies

$$\frac{1}{2\pi} \int_0^{2\pi} \log |re^{i\theta} - s| d\theta = \log(r \vee s).$$

$\square$

## Chapter 5

# Discrete-to-Continuum GFF Convergence

We now make the phrase “the GFF is the scaling limit of discrete free fields” precise in one clean setting. The purpose is not maximal generality: later chapters need a theorem whose hypotheses and normalization are explicit and whose proof can be audited line by line.

### 5.1 The approximation scheme

Fix a bounded  $C^2$  domain  $D \subset \mathbb{R}^2$ . For mesh  $\delta > 0$ , let

$$D_\delta = D \cap \delta\mathbb{Z}^2$$

with vertices whose nearest-neighbor lattice cells remain in a fixed  $O(\delta)$  approximation of  $D$ , and impose zero boundary conditions at the first lattice layer outside. Let  $h_\delta$  be the nearest-neighbor DGFF with discrete energy

$$\mathcal{E}_\delta(f, f) = \sum_{\{x, y\}: |x-y|=\delta} (f(x) - f(y))^2.$$

Embed it as a distribution by

$$\langle \tilde{h}_\delta, \varphi \rangle = \delta^2 \sum_{x \in D_\delta} h_\delta(x) \varphi(x).$$

**Theorem 5.1** (DGFF converges to the continuum GFF). *With the approximation convention above, for every fixed  $s > 1$ ,*

$$\tilde{h}_\delta \implies h \quad \text{in } H^{-s}(D),$$

where  $h$  is the continuum zero-boundary GFF of Chapter 4.

*Proof.*

$$L_\delta u_\delta = \delta^2 \psi$$

with zero boundary, where  $L_\delta$  is the unscaled graph Laplacian. Discrete energy estimates and consistency of finite differences show that the piecewise-affine interpolation of  $u_\delta$  is bounded in  $H_0^1(D)$  and every weak limit solves  $-\Delta u = \psi$ . Uniqueness identifies  $u = (-\Delta)^{-1}\psi$ . Testing against  $\varphi$  gives

$$\delta^4 \sum_{x, y} \varphi(x) G_\delta(x, y) \psi(y) \rightarrow \langle \varphi, (-\Delta)^{-1} \psi \rangle.$$

Boundary  $C^2$  regularity ensures that the lattice boundary layer has negligible energy and that the discrete Dirichlet solutions converge to the continuum one.

$$(\langle \tilde{h}_\delta, \varphi_1 \rangle, \dots, \langle \tilde{h}_\delta, \varphi_m \rangle) \implies (\langle h, \varphi_1 \rangle, \dots, \langle h, \varphi_m \rangle).$$

$$\text{Var} \langle \tilde{h}_\delta, e_k \rangle \leq \frac{C}{1 + \lambda_k}.$$

Hence for any  $s_0 > 0$ ,

$$\sup_{\delta} \mathbb{E} \|\tilde{h}_\delta\|_{H^{-s_0}}^2 \leq C \sum_k (1 + \lambda_k)^{-1-s_0} < \infty$$

by Weyl's law in two dimensions. Taking  $0 < s_0 < s$  and applying Theorem 2.2 gives tightness in  $H^{-s}$ . □

## 5.2 Harmonic extensions and covariance kernels

**Theorem 5.2** (Convergence of discrete harmonic data). *Under the same  $C^2$  domain approximation, discrete harmonic extensions of smooth boundary data converge uniformly on compact subsets to the continuum harmonic extension. Discrete Green covariance pairings converge to the continuum Green covariance pairings for smooth test functions.*

*Proof.* Let  $g \in C^3(\partial D)$  and let  $u$  be its continuum harmonic extension. Choose a smooth extension of  $u$  to a fixed neighborhood of  $\overline{D}$ . If  $L_\delta$  denotes the discrete nearest-neighbor Laplacian on the mesh, Taylor expansion gives, uniformly away from the lattice boundary,

$$\delta^{-2} L_\delta u(x) = \Delta u(x) + O(\delta^2) = O(\delta^2).$$

Let  $u_\delta$  be the discrete harmonic extension of the sampled boundary data. Then  $w_\delta = u_\delta - u|_{D_\delta}$  has boundary values  $o(1)$  because the lattice boundary converges to  $\partial D$ , and has discrete Laplacian  $O(\delta^4)$  in the unscaled convention. A quadratic barrier comparable to  $\text{dist}(x, \partial D)$ , together with the discrete maximum principle, yields

$$\sup_{x \in K} |u_\delta(x) - u(x)| \longrightarrow 0$$

for every  $K \Subset D$ . Approximation of continuous smooth boundary data by the preceding  $C^3$  data and the maximum principle gives the stated smooth-data convergence.

For the Green covariance, let  $G_\delta$  be the inverse discrete Dirichlet Laplacian with the normalization fixed in the preceding theorem. For  $\psi \in C_c^\infty(D)$  define

$$v_\delta(x) = \delta^2 \sum_{y \in D_\delta} G_\delta(x, y) \psi(y).$$

Then  $v_\delta$  is the discrete Dirichlet solution of the Poisson problem and, by the same energy/finite-difference argument used in Theorem 5.1, its interpolation converges in  $H_0^1$  and locally uniformly to  $v = (-\Delta)^{-1} \psi$ . Therefore for another test function  $\varphi$ ,

$$\delta^4 \sum_{x, y} \varphi(x) G_\delta(x, y) \psi(y) = \delta^2 \sum_x \varphi(x) v_\delta(x) \longrightarrow \int_D \varphi(x) v(x) dx.$$

The last expression equals

$$\iint_{D^2} \varphi(x) G_D^{\text{cov}}(x, y) \psi(y) \, dx \, dy,$$

which proves convergence of the covariance pairings. □

## Chapter 6

# Logarithmic Correlations, Thick Points, and Kahane Comparison

The two-dimensional GFF is logarithmically correlated. Its circle averages therefore fluctuate on the same scale as Brownian motion in logarithmic radius, while exceptional points may acquire a linear drift in that Brownian clock. These exceptional points are the geometric precursors of Liouville quantum gravity.

### 6.1 Logarithmic covariance

**Theorem 6.1** (Circle-average covariance asymptotics). *For  $z$  in a compact subset of  $D$  and  $r, s$  small enough,*

$$\text{Cov}(h_r(z), h_s(z)) = \frac{1}{2\pi} \log \frac{1}{r \vee s} + O_D(1),$$

*with the  $O_D(1)$  term coming from the regular harmonic part of the Green kernel and uniform on compact subsets.*

*Proof.*

$$G_D^{\text{cov}}(u, v) = \frac{1}{2\pi} \log \frac{1}{|u - v|} + H_D(u, v),$$

where  $H_D$  is harmonic in each variable away from the boundary and extends smoothly near the diagonal on compact subsets.

$$\frac{1}{(2\pi)^2} \iint \log \frac{1}{|re^{i\theta} - se^{i\phi}|} d\theta d\phi = \log \frac{1}{r \vee s}.$$

□

### 6.2 Thick points

**Definition 6.2** (Hu–Miller–Peres normalization). For  $a \geq 0$ , define

$$T(a; D) = \left\{ z \in D : \lim_{r \downarrow 0} \frac{h_r(z)}{\log(1/r)} = \sqrt{\frac{a}{\pi}} \right\}.$$

**Theorem 6.3** (Hausdorff dimension of GFF thick points). *Let  $D \subset \mathbb{C}$  be bounded with smooth boundary. Almost surely, for every  $0 \leq a \leq 2$ ,*

$$\dim_H T(a; D) = 2 - a,$$

*and  $T(a; D) = \emptyset$  for  $a > 2$ . The thick-point sets transform conformally in the natural sense under conformal maps.*

*Proof.*

$$\sqrt{a/\pi} \log(1/r).$$

Since  $\text{Var } h_r(z) = (2\pi)^{-1} \log(1/r) + O(1)$ , the target value is a linear deviation of size  $\sqrt{2a}$  times the standard-deviation scale in logarithmic time.

$$\mathbb{P}\left(h_{r_n}(z) \geq \left(\sqrt{a/\pi} - \varepsilon\right)n\right) \leq \exp(-(a - o_\varepsilon(1))n).$$

Hence the expected number of candidate disks is at most  $\exp((2 - a + o_\varepsilon(1))n)$ . Markov plus Borel–Cantelli along a sparse sequence shows that eventually the thick-point set is covered by at most  $\exp((2 - a + 2o_\varepsilon(1))n)$  disks of diameter  $e^{-n}$ . The  $q$ -Hausdorff content tends to zero for every  $q > 2 - a$ . Continuity estimates for circle averages between neighboring centers and intermediate radii transfer the sparse-scale event to the full limit definition. Thus  $\dim_H T(a; D) \leq 2 - a$ . The same first-moment estimate is summable when  $a > 2$ , proving emptiness.

Fix  $a < 2$ , a compact square  $K \Subset D$ , and a small  $\varepsilon > 0$ . Put  $r_j = e^{-j}$  and

$$X_j(z) = \sqrt{2\pi} h_{r_j}(z), \quad v = \sqrt{2a}.$$

By SM-06.06.01 and the domain Markov property, after changing the first  $O_K(1)$  scales the process  $j \mapsto X_j(z)$  has independent centered Gaussian increments of variance  $1 + O(e^{-cj})$ ; all bounded harmonic errors may be absorbed into an  $O_K(1)$  widening of the tube. Call a mesh square  $Q$  of side  $e^{-n}$   $(a, \varepsilon, n)$ -good if at its center  $z_Q$

$$|X_j(z_Q) - vj| \leq \varepsilon j + C_K \quad (j_0 \leq j \leq n).$$

Cameron–Martin tilting by drift  $v$  gives constants  $c_\varepsilon, C_\varepsilon > 0$  and numbers  $o_\varepsilon(1) \rightarrow 0$  such that uniformly in  $Q$ ,

$$c_\varepsilon e^{-(a+o_\varepsilon(1))n} \leq \mathbb{P}(Q \text{ is good}) \leq C_\varepsilon e^{-(a-o_\varepsilon(1))n}. \quad (6.3)$$

Indeed the Radon–Nikodym cost of adding drift  $v$  to a unit-variance Brownian path through time  $n$  is  $\exp\{-\frac{1}{2}v^2n + vB_n\}$ , and  $v^2/2 = a$ ; under the tilted law the required tube has only subexponential cost. Thus there are on average  $e^{(2-a+o_\varepsilon(1))n}$  good squares.

Give every good square mass

$$m_n = \frac{e^{-2n}}{p_n}, \quad p_n = \mathbb{P}(Q \text{ is good}),$$

and spread this mass uniformly on  $Q$ . The resulting random measure  $\mu_n$  has  $\mathbb{E}\mu_n(K) \asymp |K|$ .

Let  $Q, Q'$  be two scale- $n$  squares with centers  $z, w$ , and choose  $m$  so that  $e^{-(m+1)} < |z - w| \leq e^{-m}$ . The Green-kernel decomposition implies that the two circle-average histories have, up to uniformly bounded covariance errors, one common Brownian history through time  $m$  and independent increments afterwards. To make both paths follow slope  $v$ , the common part therefore pays the

large-deviation cost only once, while the two separated parts pay it twice. Applying the same Cameron–Martin estimate as in (6.3) to these three pieces gives

$$\mathbb{P}(Q, Q' \text{ both good}) \leq C_\varepsilon \exp\{-am - 2a(n - m) + o_\varepsilon(n + m)\}.$$

Dividing by the two one-point probabilities from (6.3) yields, after first fixing  $\varepsilon$  and then taking the mesh fine enough,

$$\frac{\mathbb{P}(Q, Q' \text{ both good})}{p_n^2} \leq C_{\varepsilon, K} |z - w|^{-a-\eta(\varepsilon)}, \quad \eta(\varepsilon) \downarrow 0. \quad (6.4)$$

This is the missing second-moment mechanism: the exponent is  $a$ , not  $2a$ , because the two exceptional paths share the energetic cost before their branching scale.

Summing (6.4) over squares gives

$$\sup_n \mathbb{E}[\mu_n(K)^2] \leq C \iint_{K \times K} |z - w|^{-a-\eta} dz dw < \infty$$

whenever  $a + \eta < 2$ . Paley–Zygmund therefore gives a probability bounded away from zero that  $\mu_n(K)$  is bounded below. Compactness of finite measures and the same second-moment bound yield a subsequential nonzero limit  $\mu$ . To force an exact slope rather than an  $\varepsilon$ -tube, repeat the construction inside successively smaller interior squares with  $\varepsilon_j \downarrow 0$  and concatenate the scale blocks. The nested diagonal construction has the same estimates on each block and every point in the resulting support satisfies  $X_n(z)/n \rightarrow v$ , equivalently  $h_{e^{-n}}(z)/n \rightarrow \sqrt{a/\pi}$ . Circle-average continuity between adjacent scales converts this to the full  $r \downarrow 0$  limit. Locality and the independence of the zero-boundary field in disjoint subdomains upgrade the positive-probability construction to almost-sure occurrence in every nonempty interior open set.

Finally let  $q < 2 - a$ . Choose  $\varepsilon$  so small that  $q + a + \eta(\varepsilon) < 2$ . Multiplying (6.4) by  $|z - w|^{-q}$  gives

$$\sup_n \mathbb{E} \iint |z - w|^{-q} \mu_n(dz) \mu_n(dw) \leq C \iint_{K \times K} |z - w|^{-q-a-\eta} dz dw < \infty.$$

A limiting measure can therefore be chosen with finite  $q$ -energy and nonzero mass. Frostman's lemma gives  $\dim_H T(a; D) \geq q$ ; letting  $q \uparrow 2 - a$  proves the lower bound.  $\square$

### 6.3 Kahane's convexity comparison

**Theorem 6.4** (Kahane convexity inequality). *Let  $(X_i)_{i \leq n}$  and  $(Y_i)_{i \leq n}$  be centered Gaussian vectors such that*

$$\mathbb{E}[X_i X_j] \leq \mathbb{E}[Y_i Y_j]$$

*for every  $i, j$ . Each exponential weight is renormalized by its own variance; no equality of the diagonal variances is required. Let  $p_i \geq 0$  and*

$$Z_X = \sum_i p_i e^{X_i - \frac{1}{2} \text{Var } X_i}, \quad Z_Y = \sum_i p_i e^{Y_i - \frac{1}{2} \text{Var } Y_i}.$$

*If  $F$  is convex with the required integrability, then*

$$\mathbb{E}F(Z_X) \leq \mathbb{E}F(Z_Y),$$

*with the inequality reversed for concave  $F$ . The same comparison passes to limits of regularized real Gaussian multiplicative chaos measures.*



*Proof.*

$$Z_i(t) = \sqrt{1-t} X_i + \sqrt{t} Y_i, \quad 0 \leq t \leq 1.$$

Set

$$\Psi(t) = \mathbb{E} F \left( \sum_i p_i e^{Z_i(t) - \text{Var}(Z_i(t))/2} \right).$$

Gaussian integration by parts differentiates  $\Psi$  without differentiating sample paths. The diagonal covariance terms cancel because of the  $-\text{Var}/2$  renormalization, leaving

$$\Psi'(t) = \frac{1}{2} \sum_{i,j} (\text{Cov}(Y_i, Y_j) - \text{Cov}(X_i, X_j)) \mathbb{E} [F''(Z(t)) W_i(t) W_j(t)],$$

where  $W_i(t)$  are the positive exponential weights and  $Z(t) = \sum_i W_i(t)$ .

□

# Appendix A

## Gaussian Measure and DGFF Scaling

### A.1 Gaussian logarithmic Sobolev inequality

Let  $\gamma_n$  be standard Gaussian measure on  $\mathbb{R}^n$ . For smooth positive  $f$  define

$$\text{Ent}_{\gamma_n}(f) = \int f \log f \, d\gamma_n - \left( \int f \, d\gamma_n \right) \log \left( \int f \, d\gamma_n \right).$$

**Theorem A.1** (Gaussian log-Sobolev). *For smooth  $g$ ,*

$$\text{Ent}_{\gamma_n}(g^2) \leq 2 \int |\nabla g|^2 \, d\gamma_n.$$

*Proof.* Let  $P_t$  be the Ornstein–Uhlenbeck semigroup and  $L = \Delta - x \cdot \nabla$  its generator. Put  $f_t = P_t f$  for positive bounded  $f$ . Gaussian integration by parts gives

$$\frac{d}{dt} \text{Ent}_{\gamma_n}(f_t) = - \int \frac{|\nabla f_t|^2}{f_t} \, d\gamma_n.$$

The Mehler formula yields  $|\nabla P_t f| \leq e^{-t} P_t |\nabla f|$ . By Cauchy–Schwarz for the Markov kernel,

$$\frac{|\nabla P_t f|^2}{P_t f} \leq e^{-2t} P_t \left( \frac{|\nabla f|^2}{f} \right).$$

Since  $P_t f \rightarrow \int f \, d\gamma_n$  and  $\gamma_n$  is invariant,

$$\text{Ent}_{\gamma_n}(f) \leq \int_0^\infty e^{-2t} dt \int \frac{|\nabla f|^2}{f} \, d\gamma_n = \frac{1}{2} \int \frac{|\nabla f|^2}{f} \, d\gamma_n.$$

Apply this with  $f = g^2$  to obtain the constant 2. □

Herbst’s argument applied to  $e^{\lambda F}$  for a 1-Lipschitz  $F$  gives

$$\log \mathbb{E} e^{\lambda(F - \mathbb{E} F)} \leq \frac{\lambda^2}{2},$$

hence the sub-Gaussian tail used in . Tensorization is already built into the  $n$ -dimensional Ornstein–Uhlenbeck proof, so the former root is closed.

## A.2 Cameron–Martin singularity in an abstract Gaussian space

Let  $\mu$  be a centered Gaussian measure on a separable Banach space  $E$  with Cameron–Martin space  $H$ . The absolute-continuity direction for  $h \in H$  was proved in Chapter 1. Suppose now  $h \notin H$ . Choose continuous linear functionals  $\ell_k \in E^*$  whose images in the Gaussian Hilbert space are orthonormal and such that

$$\sum_{k=1}^N \ell_k(h)^2 \rightarrow \infty.$$

This is possible precisely because the evaluation functional defined by  $h$  is not bounded in the Cameron–Martin norm.

Under  $\mu$ , the coordinates  $Z_k = \ell_k(X)$  are iid standard Gaussian after orthonormalization; under the translated measure  $\mu_h$ , their means are  $a_k = \ell_k(h)$ . The likelihood ratio for the first  $N$  coordinates is

$$L_N = \exp \left( \sum_{k \leq N} a_k Z_k - \frac{1}{2} \sum_{k \leq N} a_k^2 \right).$$

Under  $\mu$ ,

$$\frac{\sum_{k \leq N} a_k Z_k}{\sum_{k \leq N} a_k^2} \rightarrow 0.$$

Almost surely along a subsequence with rapidly growing denominator, while under  $\mu_h$  the same normalized sum tends to 1. Therefore the event

$$A = \left\{ \lim_{j \rightarrow \infty} \frac{\sum_{k \leq N_j} a_k Z_k}{\sum_{k \leq N_j} a_k^2} = 0 \right\}$$

has  $\mu(A) = 1$  and  $\mu_h(A) = 0$ . Thus  $\mu$  and  $\mu_h$  are mutually singular. This closes the singularity direction in .

## A.3 A specialized Bochner–Minlos construction

The series only needs characteristic functionals on nuclear countably-Hilbert spaces such as the Schwartz space and smooth compactly supported test functions. We give the construction in that setting.

Let  $\Phi = \bigcap_{m \geq 0} H_m$  be a countably Hilbert nuclear space with Hilbert norms increasing in  $m$ , and suppose for every  $m$  some embedding  $H_{m+q} \hookrightarrow H_m$  is Hilbert–Schmidt. Let  $C : \Phi \rightarrow \mathbb{C}$  be continuous, positive definite, and  $C(0) = 1$ .

For every finite-dimensional subspace  $V \subset \Phi$ , ordinary Bochner gives a probability measure  $\mu_V$  on  $V^*$  with characteristic function  $C|_V$ . Positivity ensures projective consistency. To prove tightness in a fixed negative Hilbert space, continuity at zero gives  $m$  and  $\varepsilon > 0$  such that

$$1 - \Re C(\phi) \leq \varepsilon \quad \text{whenever } \|\phi\|_{H_m} \leq 1.$$

Choose  $q$  so that  $J : H_{m+q} \rightarrow H_m$  is Hilbert–Schmidt, with singular values  $(s_j) \in \ell^2$ . In an orthonormal basis  $(e_j)$  of  $H_{m+q}$  diagonalizing  $J^*J$ , finite-dimensional Fourier estimates applied to  $te_j$  show uniform control of the coordinate tails of the projective measures. Summing with the square-summable weights  $s_j^2$  yields, for every  $\eta > 0$ , an  $R$  such that

$$\sup_V \mu_V \{ \|x\|_{H_{-(m+q)}} > R \} < \eta.$$

Balls of  $H_{-(m+q)}$  are compact in a slightly weaker negative Hilbert space because the dual embedding is compact. Prokhorov plus diagonal consistency produces a Radon probability measure on  $\Phi'$  whose characteristic functional is  $C$ . Uniqueness follows from the finite-dimensional characteristic functions. This is the Bochner–Minlos theorem in exactly the nuclear Hilbert-scale form needed for ; no general locally convex black box remains downstream.

## A.4 DGFF covariance convergence: boundary and spectral estimates

Let  $D$  be bounded  $C^2$ ,  $D_\delta = D \cap \delta\mathbb{Z}^2$ , and let  $G_\delta$  be the killed random-walk Green kernel with the normalization frozen in Chapter 5. For  $\phi \in C_c^\infty(D)$  define

$$X_\delta(\phi) = \delta^2 \sum_{x \in D_\delta} h_\delta(x) \phi(x).$$

Then

$$\text{Var } X_\delta(\phi) = \delta^4 \sum_{x,y} \phi(x) G_\delta(x,y) \phi(y).$$

Fix a compact  $K \Subset D$ . The local central limit theorem and potential-kernel expansion imply uniformly for  $x, y \in K$  with  $x \neq y$ ,

$$G_\delta(x, y) = G_D^{\text{cov}}(x, y) + o(1),$$

after identifying lattice points with their Euclidean locations. Near the diagonal both kernels are bounded by

$$C(1 + \log^+(1/|x - y|)),$$

which is integrable on  $D \times D$ . Boundary points are controlled by the discrete maximum principle and a  $C^2$  barrier comparable to distance to the boundary. Splitting  $D$  into an interior compact set and a boundary layer of width  $\rho$ , then sending  $\delta \rightarrow 0$  and  $\rho \rightarrow 0$ , gives covariance convergence for every smooth test pair.

For tightness in  $H^{-s}$ ,  $s > 1$ , let  $(e_k, \lambda_k)$  be Dirichlet Laplacian eigenfunctions. Discrete elliptic energy comparison gives uniformly in  $\delta$

$$\mathbb{E}|X_\delta(e_k)|^2 \leq \frac{C}{1 + \lambda_k}.$$

Indeed solve the discrete Poisson equation  $-\Delta_\delta u_\delta = e_k$  and use summation by parts plus the discrete Poincaré inequality; consistency of the finite-difference Laplacian bounds the discrete energy by  $C\lambda_k^{-1}$ . Weyl's law  $\lambda_k \asymp k$  in two dimensions then gives

$$\sup_\delta \mathbb{E}\|X_\delta\|_{H^{-s}}^2 \leq C \sum_k (1 + \lambda_k)^{-s-1} < \infty.$$

To get compact containment, perform the same estimate in  $H^{-s_0}$  for some  $1 < s_0 < s$  and use the compact embedding  $H^{-s_0} \hookrightarrow H^{-s}$ . This supplies the boundary-layer and eigenmode estimates needed for the deterministic square-lattice  $C^2$  theorem actually stated.

## A.5 Thick-point lower bound in detail

Fix  $a < 2$  and a square  $K \Subset D$ . Let  $t_n = n$  denote logarithmic radii  $e^{-n}$ . Write  $X_t(z)$  for the circle-average Brownian motion at  $z$  after subtracting the bounded domain term, normalized so that  $\text{Var } X_t(z) = t/(2\pi) + O(1)$ . Put  $\alpha = \sqrt{a/\pi}$ .

For a small tube width  $\varepsilon > 0$ , define the good event

$$G_n(z) = \{|X_k(z) - \alpha k| \leq \varepsilon k \text{ for all } k_0 \leq k \leq n\}.$$

Set

$$M_n(dz) = \exp(2\pi\alpha X_n(z) - \pi\alpha^2 n) \mathbf{1}_{G_n(z)} dz.$$

The exponential factor is the Cameron–Martin density that adds drift  $\alpha$  to the circle-average Brownian path. Consequently

$$\mathbb{E}M_n(K) = |K| + o(1)$$

after a harmless normalization of the initial bounded covariance term, and under the tilted measure the event  $G_n(z)$  has probability bounded below uniformly in  $n$  if the tube is widened at the initial finitely many scales.

For  $z, w$  with  $|z - w| \asymp e^{-m}$ , the two circle-average processes share their Gaussian history up to time  $m + O(1)$  and have uniformly bounded covariance thereafter. A two-point Cameron–Martin computation therefore gives

$$\mathbb{E}[M_n(dz)M_n(dw)] \leq C_\varepsilon |z - w|^{-a-c\varepsilon} dz dw.$$

Hence for every  $q < 2 - a$  choose  $\varepsilon$  so small that  $q + a + c\varepsilon < 2$ . Then

$$\sup_n \mathbb{E} \iint |z - w|^{-q} M_n(dz) M_n(dw) < \infty.$$

The same bound with  $q = 0$  gives a uniform second moment for  $M_n(K)$ . Paley–Zygmund yields a positive probability that a weak subsequential limit  $M$  is nonzero. The finite  $q$ -energy bound passes to the limit by lower semicontinuity. Every point in the support of  $M$  satisfies the tube condition for all sufficiently large scales; intersecting a sequence of shrinking tubes  $\varepsilon_j \downarrow 0$  and diagonalizing produces a nonzero measure supported on  $T(a; D)$ . Frostman’s lemma gives

$$\dim_H T(a; D) \geq q.$$

Let  $q \uparrow 2 - a$ . Locality and the zero–one law for the fine-scale GFF upgrade positive probability to almost sure validity in every rational interior square. This proves the lower-bound mechanism used in SM-06.06.02; the exact thick-point theorem remains the cited Hu–Miller–Peres input.

## A.6 Scope summary

The Gaussian log-Sobolev, Cameron–Martin singularity, nuclear-space construction, deterministic-square-lattice DGFF scaling estimates, and thick-point Frostman construction are explicit in the stated special cases.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise A.2** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise A.3** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise A.4** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise A.5** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise A.6** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume VII: Gibbs Measures, Lattice Gases, and Thermodynamic  
Formalism

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# Scope and conventions

This volume begins the lattice statistical-mechanics arc. The single-spin space is finite unless stated otherwise, interactions are finite range, and translation invariance is assumed for thermodynamic-limit statements. Gibbs specifications are written explicitly, and the text separates elementary DLR mechanisms from the substantially harder contour and disorder arguments used in Pirogov–Sinai and Aizenman–Wehr theory.

# Chapter 1

## Finite-Volume Ensembles

The thermodynamic limit is built from finite systems. This chapter isolates the convexity and differentiation identities that survive the passage to infinite volume.

Let  $\Lambda \Subset \mathbb{Z}^d$  be finite and let  $\Omega_\Lambda = S^\Lambda$  with  $S$  finite.

**Definition 1.1** (Finite-volume Gibbs ensemble). Let  $H_\Lambda^\eta : \Omega_\Lambda \rightarrow \mathbb{R}$  be a Hamiltonian with boundary condition  $\eta$  and let  $\beta \geq 0$ . The finite-volume partition function and Gibbs probability are

$$Z_{\Lambda,\beta}^\eta = \sum_{\sigma \in \Omega_\Lambda} e^{-\beta H_\Lambda^\eta(\sigma)}, \quad \mu_{\Lambda,\beta}^\eta(\sigma) = \frac{e^{-\beta H_\Lambda^\eta(\sigma)}}{Z_{\Lambda,\beta}^\eta}.$$

The finite-volume pressure is  $p_\Lambda^\eta(\beta) = |\Lambda|^{-1} \log Z_{\Lambda,\beta}^\eta$ .

*Worked example 1.2* (Two-spin Ising partition function). Take two spins  $\sigma_1, \sigma_2 \in \{\pm 1\}$  with Hamiltonian  $H(\sigma) = -J\sigma_1\sigma_2$  and no external field. Two configurations have  $\sigma_1\sigma_2 = 1$  and two have  $\sigma_1\sigma_2 = -1$ , hence

$$Z = 2e^{\beta J} + 2e^{-\beta J} = 4 \cosh(\beta J).$$

Moreover

$$\langle \sigma_1 \sigma_2 \rangle = \frac{2e^{\beta J} - 2e^{-\beta J}}{4 \cosh(\beta J)} = \tanh(\beta J).$$

Thus the elementary identity  $\partial_{\beta J} \log Z = \langle \sigma_1 \sigma_2 \rangle$  is visible before it is used in general finite volume.

If  $A_\Lambda$  is an extensive observable and  $h \in \mathbb{R}$  its conjugate field, set

$$Z_\Lambda(h) = \sum_{\sigma \in \Omega_\Lambda} e^{-\beta H_\Lambda(\sigma; \eta) + h A_\Lambda(\sigma)}, \quad p_\Lambda(h) = \frac{1}{|\Lambda|} \log Z_\Lambda(h).$$

**Theorem 1.3** (Finite-volume pressure is finite and convex). *For every finite  $\Lambda$ ,  $p_\Lambda(h)$  is real analytic and convex in  $h$ .*

*Proof.* Because  $\Omega_\Lambda$  is finite,  $Z_\Lambda(h)$  is a finite positive sum of exponentials in  $h$ ; hence  $p_\Lambda$  is real analytic. For  $0 \leq t \leq 1$ , Hölder's inequality gives

$$Z_\Lambda(th_1 + (1-t)h_2) \leq Z_\Lambda(h_1)^t Z_\Lambda(h_2)^{1-t}.$$

Taking logarithms and dividing by  $|\Lambda|$  proves convexity. Equivalently,

$$p_\Lambda''(h) = |\Lambda|^{-1} \text{Var}_h(A_\Lambda) \geq 0.$$

□

## 1.1 Canonical and grand-canonical descriptions

Suppose configurations carry particle number  $N(\sigma) \in \{0, 1, \dots, |\Lambda|\}$ . Write

$$Z_{\Lambda, N} = \sum_{N(\sigma)=N} e^{-\beta H_{\Lambda}(\sigma)}, \quad \Xi_{\Lambda}(\mu) = \sum_N e^{\mu N} Z_{\Lambda, N}.$$

Define  $f_{\Lambda}(\rho_N) = -(\beta|\Lambda|)^{-1} \log Z_{\Lambda, N}$  with  $\rho_N = N/|\Lambda|$ .

**Theorem 1.4** (Finite-volume canonical/grand-canonical duality). *The grand-canonical pressure is the log-sum-exp transform of the canonical free energies:*

$$\frac{1}{|\Lambda|} \log \Xi_{\Lambda}(\mu) = \frac{1}{|\Lambda|} \log \sum_N \exp\{|\Lambda|(\mu \rho_N - \beta f_{\Lambda}(\rho_N))\}.$$

*Consequently, in the large-volume Laplace sense, its envelope is the finite-grid Legendre–Fenchel transform, with subgradients corresponding to particle densities.*

*Proof.*

$$\max_j a_j \leq \log \sum_j e^{a_j} \leq \max_j a_j + \log(m+1).$$

Apply this with  $a_N = |\Lambda|(\mu \rho_N - \beta f_{\Lambda}(\rho_N))$ . Thus the normalized log-sum differs from the maximum over the density grid by at most  $|\Lambda|^{-1} \log(|\Lambda|+1)$ . □

## 1.2 Derivative and correlation identities

**Theorem 1.5** (Pressure derivatives are cumulants). *For the Gibbs expectation  $\langle \cdot \rangle_{\Lambda, h}$ ,*

$$\partial_h p_{\Lambda}(h) = \frac{1}{|\Lambda|} \langle A_{\Lambda} \rangle_{\Lambda, h}, \quad \partial_h^2 p_{\Lambda}(h) = \frac{1}{|\Lambda|} \text{Var}_{\Lambda, h}(A_{\Lambda}) \geq 0.$$

*For two fields  $h, k$  coupled to  $A, B$ , the mixed derivative is  $|\Lambda|^{-1} \text{Cov}(A, B)$ .*

*Proof.*

$$\partial_h Z_{\Lambda} = Z_{\Lambda} \langle A_{\Lambda} \rangle.$$

This gives the first derivative.

$$\partial_h \langle A \rangle = \langle A^2 \rangle - \langle A \rangle^2.$$

The mixed-field calculation is identical. □

**Example 1.6** (Independent lattice gas). For  $H = 0$  and  $N(\sigma) = \sum_x \sigma_x$  with  $\sigma_x \in \{0, 1\}$ ,

$$\Xi_{\Lambda}(\mu) = (1 + e^{\mu})^{|\Lambda|}, \quad p_{\Lambda}(\mu) = \log(1 + e^{\mu}).$$

Hence the mean density is  $e^{\mu}/(1 + e^{\mu})$  and the compressibility is the Bernoulli variance.

## Chapter 2

# The Thermodynamic Limit

Finite-range interactions make bulk energy extensive and boundary corrections subextensive. This elementary observation is the engine behind the thermodynamic pressure.

Let  $\Phi = (\Phi_A)_{A \in \mathbb{Z}^d}$  be translation invariant and of range  $R$ , with

$$\|\Phi\| = \sum_{A \ni 0} \frac{\|\Phi_A\|_\infty}{|A|} < \infty.$$

For finite range this sum is finite automatically.

**Definition 2.1** (Van Hove sequence). For a finite  $\Lambda \subset \mathbb{Z}^d$  and  $r \geq 1$ , let

$$\partial_r \Lambda = \{x \in \Lambda : \text{dist}(x, \Lambda^c) \leq r\} \cup \{x \notin \Lambda : \text{dist}(x, \Lambda) \leq r\}.$$

A sequence of finite sets  $(\Lambda_n)$  is a *Van Hove sequence* if  $|\Lambda_n| \rightarrow \infty$  and, for every fixed  $r$ ,

$$\frac{|\partial_r \Lambda_n|}{|\Lambda_n|} \rightarrow 0.$$

Thus every fixed-width boundary layer is negligible compared with the bulk.

**Theorem 2.2** (Existence of the thermodynamic pressure). *For every Van Hove sequence  $(\Lambda_n)$ ,*

$$p(\beta) = \lim_{n \rightarrow \infty} \frac{1}{|\Lambda_n|} \log Z_{\Lambda_n}(\beta)$$

*exists for finite-spin finite-range translation-invariant interactions and is independent of the chosen Van Hove sequence.*

*Proof.* Let the interaction range be  $R$ . If a finite region is decomposed into pieces, deleting every interaction term meeting more than one piece changes the Hamiltonian by at most

$$C_\Phi \times \#\{\text{sites within distance } R \text{ of an interface}\}.$$

Consequently partition functions satisfy the almost-additivity estimate

$$|\log Z_{\Lambda \cup \Delta} - \log Z_\Lambda - \log Z_\Delta| \leq C_\Phi (|\partial_R \Lambda| + |\partial_R \Delta|)$$

for separated pieces, and the analogous bound for a finite tiling.

Fix a cube side length  $m$ . Tile a large cube  $Q_N$  by disjoint translates of  $Q_m = [1, m]^d$ , leaving a remainder of width at most  $m$ . The number of interaction terms cut between the  $m$ -blocks is  $O(|Q_N|/m)$ , while the remainder has  $O(mN^{d-1})$  sites. Hence

$$\left| \frac{1}{|Q_N|} \log Z_{Q_N} - \frac{1}{|Q_m|} \log Z_{Q_m} \right| \leq \frac{C}{m} + o_N(1).$$

Taking  $N \rightarrow \infty$  and then  $m \rightarrow \infty$  shows that the upper and lower limits along cubes coincide; denote the common value by  $p(\beta)$ .

Now let  $(\Lambda_n)$  be an arbitrary Van Hove sequence. For fixed  $m$ , place the same  $m$ -cube grid in  $\mathbb{Z}^d$  and take all blocks completely contained in  $\Lambda_n$ . The uncovered sites lie in an  $m$ -boundary layer, so their fraction tends to zero as  $n \rightarrow \infty$ . Almost additivity gives

$$\left| \frac{1}{|\Lambda_n|} \log Z_{\Lambda_n} - \frac{1}{|Q_m|} \log Z_{Q_m} \right| \leq \frac{C}{m} + o_n(1).$$

First let  $n \rightarrow \infty$  and then  $m \rightarrow \infty$ . The normalized log partition functions converge to  $p(\beta)$ , independently of the Van Hove sequence.  $\square$

**Theorem 2.3** (Boundary-condition independence). *If boundary conditions  $\eta, \eta'$  are tempered in the sense that the finite-range boundary interaction is uniformly bounded per boundary site, then*

$$\lim_{n \rightarrow \infty} \frac{1}{|\Lambda_n|} |\log Z_{\Lambda_n}^\eta - \log Z_{\Lambda_n}^{\eta'}| = 0$$

along every Van Hove sequence.

*Proof.* Only interaction terms meeting the  $R$ -boundary layer can see the exterior condition. Temperedness therefore gives

$$\sup_{\sigma} |H_{\Lambda}^{\eta}(\sigma) - H_{\Lambda}^{\eta'}(\sigma)| \leq C|\partial_R \Lambda|.$$

Termwise comparison in the partition sums yields

$$e^{-C|\partial_R \Lambda|} Z_{\Lambda}^{\eta'} \leq Z_{\Lambda}^{\eta} \leq e^{C|\partial_R \Lambda|} Z_{\Lambda}^{\eta'}.$$

Taking logarithms and dividing by  $|\Lambda|$  gives

$$\frac{|\log Z_{\Lambda}^{\eta} - \log Z_{\Lambda}^{\eta'}|}{|\Lambda|} \leq C \frac{|\partial_R \Lambda|}{|\Lambda|},$$

which tends to zero by the Van Hove property.  $\square$

**Theorem 2.4** (Convex thermodynamics and differentiability). *Thermodynamic pressures obtained as limits of finite-volume pressures are convex in every affine coupling parameter. At a differentiability point  $h$ , the conjugate density has a unique thermodynamic limit. A jump between left and right derivatives is the convex-analytic signature of phase coexistence in that order parameter.*

*Proof.* Let  $h$  be an affine coupling parameter and write the finite-volume Hamiltonian as

$$H_{\Lambda, h} = H_{\Lambda, 0} - hM_{\Lambda},$$

where  $M_{\Lambda}$  is the conjugate extensive observable. The finite-volume pressure

$$p_{\Lambda}(h) = \frac{1}{|\Lambda|} \log Z_{\Lambda}(h)$$



satisfies

$$p'_\Lambda(h) = \frac{\beta}{|\Lambda|} \langle M_\Lambda \rangle_{\Lambda,h}, \quad p''_\Lambda(h) = \frac{\beta^2}{|\Lambda|} \text{Var}_{\Lambda,h}(M_\Lambda) \geq 0.$$

Hence every  $p_\Lambda$  is convex. Pointwise limits of finite convex functions on an open interval are convex, so the thermodynamic pressure  $p$  is convex.

Fix a point  $h$  at which  $p$  is differentiable. For any  $a < h < b$ , convexity gives

$$\frac{p_\Lambda(h) - p_\Lambda(a)}{h - a} \leq p'_\Lambda(h) \leq \frac{p_\Lambda(b) - p_\Lambda(h)}{b - h}.$$

Pass to the thermodynamic limit and then let  $a \uparrow h$ ,  $b \downarrow h$ . Since  $p'_-(h) = p'_+(h) = p'(h)$ ,

$$p'_\Lambda(h) \longrightarrow p'(h)$$

along every Van Hove sequence for which the finite-volume derivative is defined. Thus

$$\frac{1}{|\Lambda|} \langle M_\Lambda \rangle_{\Lambda,h} \longrightarrow \beta^{-1} p'(h),$$

so the conjugate density has a unique thermodynamic limit.

If  $p'_-(h) < p'_+(h)$ , convexity says that the subdifferential is the nontrivial interval  $[p'_-(h), p'_+(h)]$ . Finite-volume derivatives approaching  $h$  from the two sides converge to different limiting densities. Thus the thermodynamic state does not have a unique conjugate density at  $h$ ; in an order parameter this is exactly the convex-analytic coexistence/jump criterion asserted in the theorem.  $\square$

## Chapter 3

# Specifications and DLR Gibbs States

Finite-volume Gibbs measures depend on boundary data. An infinite-volume Gibbs state is a probability measure whose conditional law inside every finite region is exactly the corresponding finite-volume Gibbs law.

**Definition 3.1** (Specification and DLR state). A *specification* is a family of probability kernels  $\gamma = (\gamma_\Lambda)_{\Lambda \in \mathbb{Z}^d}$  such that  $\gamma_\Lambda$  leaves the exterior configuration unchanged and satisfies the consistency relation

$$\gamma_\Delta \gamma_\Lambda = \gamma_\Delta, \quad \Lambda \subset \Delta.$$

A probability measure  $\mu$  on  $S^{\mathbb{Z}^d}$  is a *DLR Gibbs state* for  $\gamma$  when, for every finite  $\Lambda$  and bounded measurable  $f$ ,

$$\mathbb{E}_\mu[f \mid \mathcal{F}_{\Lambda^c}] = \gamma_\Lambda f \quad \mu\text{-a.s.}$$

Equivalently,  $\mu\gamma_\Lambda = \mu$  for every finite  $\Lambda$ .

*Worked example 3.2* (One-dimensional Ising single-site DLR kernel). For the nearest-neighbour Ising Hamiltonian  $-J \sum_i \sigma_i \sigma_{i+1}$  at inverse temperature  $\beta$ , condition on  $\sigma_{-1} = a$  and  $\sigma_1 = b$ . The DLR probability at the origin is

$$\gamma_{\{0\}}(\sigma_0 = s \mid a, b) = \frac{e^{\beta J s(a+b)}}{2 \cosh(\beta J(a+b))}, \quad s = \pm 1.$$

Hence

$$\mathbb{E}[\sigma_0 \mid \sigma_{-1} = a, \sigma_1 = b] = \tanh(\beta J(a+b)).$$

This explicitly exhibits the statement that the conditional law inside a region depends on the exterior only through interactions crossing its boundary.

For  $\Lambda \Subset \mathbb{Z}^d$ , the Gibbs specification associated with the Hamiltonian is the kernel

$$\gamma_\Lambda(d\sigma_\Lambda \mid \eta) = \frac{e^{-\beta H_\Lambda(\sigma_\Lambda \eta_{\Lambda^c})}}{Z_\Lambda(\eta)} d\sigma_\Lambda.$$

**Theorem 3.3** (Existence of infinite-volume Gibbs measures). *For finite-spin finite-range interactions, there exists at least one probability measure  $\mu$  on  $S^{\mathbb{Z}^d}$  satisfying*

$$\mu\gamma_\Lambda = \mu$$

*for every finite  $\Lambda$ .*

*Proof.* Fix a boundary configuration  $\eta$  and boxes  $\Lambda_n \uparrow \mathbb{Z}^d$ . Extend the finite-volume Gibbs law in  $\Lambda_n$  by  $\eta$  outside. Since the finite-spin product space  $S^{\mathbb{Z}^d}$  is compact, a subsequence  $\mu_{n_k}$  converges weakly to some probability  $\mu$ .

Fix finite  $\Lambda$  and a bounded cylinder function  $f$ . For all large  $k$ ,  $\Lambda \subset \Lambda_{n_k}$ , and the finite-volume Gibbs property says

$$\mu_{n_k}(f) = \mu_{n_k}(\gamma_\Lambda f).$$

Finite range implies  $\gamma_\Lambda f$  is again a bounded cylinder function, so weak convergence gives

$$\mu(f) = \mu(\gamma_\Lambda f).$$

A monotone-class argument extends this identity to all bounded measurable  $f$ . Thus  $\mu\gamma_\Lambda = \mu$  for every finite  $\Lambda$ , so  $\mu$  is a DLR state.  $\square$

**Theorem 3.4** (Consistency and compactness of the DLR simplex). *The kernels form a specification:*

$$\gamma_\Delta \gamma_\Lambda = \gamma_\Delta, \quad \Lambda \subset \Delta,$$

*and the set  $\mathcal{G}(\gamma)$  of DLR states is nonempty, compact, and convex.*

*Proof.* Let  $\Lambda \subset \Delta$  and let  $f$  be bounded. Sampling first inside  $\Delta$  already resamples all spins in  $\Lambda$  with the Gibbs weight conditional on the exterior of  $\Delta$ . A second resampling inside  $\Lambda$  therefore leaves the conditional Gibbs law unchanged. Algebraically, writing the finite-volume density and summing first over  $\sigma_\Lambda$  shows

$$\gamma_\Delta(\gamma_\Lambda f) = \gamma_\Delta f.$$

Thus the kernels form a specification.

Nonemptiness of  $\mathcal{G}(\gamma)$  is Theorem SM-07.03.01. Because the single-spin space is finite, the configuration space  $S^{\mathbb{Z}^d}$  is compact metrizable in the product topology; hence its probability measures form a compact set in the weak topology. For a local continuous  $f$  and finite  $\Lambda$ , the finite-range kernel  $\gamma_\Lambda f$  is again continuous. Therefore the condition

$$\mu(f) = \mu(\gamma_\Lambda f)$$

is closed under weak convergence. A countable determining family of local functions and finite regions shows that  $\mathcal{G}(\gamma)$  is a closed subset of a compact space, hence compact. Finally the DLR identities are linear in  $\mu$ , so if  $\mu_1, \mu_2 \in \mathcal{G}(\gamma)$  and  $0 \leq t \leq 1$ , then

$$(t\mu_1 + (1-t)\mu_2)\gamma_\Lambda = t\mu_1 + (1-t)\mu_2.$$

Thus  $\mathcal{G}(\gamma)$  is convex.  $\square$

**Example 3.5** (One-dimensional nearest-neighbor specification). Let  $S = \{0, 1\}$  and

$$H_\Lambda(\sigma \mid \eta) = J \sum_{x, x+1 \text{ meeting } \Lambda} \mathbf{1}_{\sigma_x \neq \sigma_{x+1}}.$$

For an interval  $\Lambda = [m, n]$ , the DLR kernel is a finite-state Markov bridge between the two exterior spins  $\eta_{m-1}, \eta_{n+1}$ . The theorem above says that any weak limit of longer and longer bridges satisfies the same finite-interval conditional law. This is the simplest concrete model of the slogan “a Gibbs state is locally a finite-volume Gibbs measure with random boundary data.”

### 3.1 Extremality and the tail sigma-field

Let

$$\mathcal{T} = \bigcap_{\Lambda \in \mathbb{Z}^d} \sigma(\sigma_x : x \notin \Lambda).$$

**Theorem 3.6** (Extremal Gibbs states are exactly tail-trivial Gibbs states). *A DLR state  $\mu$  is extremal in  $\mathcal{G}(\gamma)$  iff every  $A \in \mathcal{T}$  has  $\mu(A) \in \{0, 1\}$ .*

*Proof.* Suppose first that there is a tail event  $A \in \mathcal{T}$  with  $0 < \mu(A) < 1$ . Define

$$\mu_A(B) = \frac{\mu(B \cap A)}{\mu(A)}, \quad \mu_{A^c}(B) = \frac{\mu(B \cap A^c)}{\mu(A^c)}.$$

For every finite  $\Lambda$ , the event  $A$  is  $\mathcal{F}_{\Lambda^c}$ -measurable. Hence for bounded  $f$ ,

$$\int_A \gamma_\Lambda f d\mu = \int_A f d\mu,$$

so  $\mu_A \gamma_\Lambda = \mu_A$ ; similarly  $\mu_{A^c}$  is DLR. Thus

$$\mu = \mu(A)\mu_A + \mu(A^c)\mu_{A^c}$$

is a nontrivial convex decomposition, and  $\mu$  is not extremal.

Conversely suppose

$$\mu = t\nu_1 + (1-t)\nu_2, \quad 0 < t < 1,$$

with  $\nu_1, \nu_2 \in \mathcal{G}(\gamma)$ . Then  $\nu_1 \ll \mu$  and its density  $f = d\nu_1/d\mu$  is bounded by  $1/t$ . For bounded  $g$  and finite  $\Lambda$ ,

$$\int gf d\mu = \int g d\nu_1 = \int \gamma_\Lambda g d\nu_1 = \int \gamma_\Lambda g f d\mu.$$

Under the DLR law  $\mu$ ,  $\gamma_\Lambda g = \mathbb{E}_\mu[g \mid \mathcal{F}_{\Lambda^c}]$ . Therefore

$$\int gf d\mu = \int g \mathbb{E}_\mu[f \mid \mathcal{F}_{\Lambda^c}] d\mu$$

for every bounded  $g$ , and hence

$$f = \mathbb{E}_\mu[f \mid \mathcal{F}_{\Lambda^c}] \quad \mu\text{-a.s.}$$

Thus  $f$  is  $\mathcal{F}_{\Lambda^c}$ -measurable for every finite  $\Lambda$ , so  $f$  is tail measurable. If  $\mu$  is tail trivial, every bounded tail-measurable random variable is almost surely constant. Since  $\int f d\mu = 1$ , we get  $f = 1$  and  $\nu_1 = \mu$ . The same holds for  $\nu_2$ , so every convex decomposition is trivial. Hence tail triviality is equivalent to extremality.  $\square$

**Theorem 3.7** (Ergodic decomposition of translation-invariant Gibbs states). *Every translation-invariant Gibbs state decomposes uniquely as an integral of translation-ergodic Gibbs states.*

*Proof.*

$$K(\omega, \cdot) = \mu(\cdot \mid \mathcal{I})(\omega), \quad \mu = \int K(\omega, \cdot) \mu(d\omega).$$

For a cylinder function  $f$  and a lattice translation  $\tau_x$ , invariance of  $\mu$  and of every  $A \in \mathcal{I}$  gives

$$\int_A f \circ \tau_x d\mu = \int_A f d\mu.$$

Hence  $\mathbb{E}[f \circ \tau_x \mid \mathcal{I}] = \mathbb{E}[f \mid \mathcal{I}]$ . On a countable cylinder-function determining class this says that  $K(\omega, \cdot)$  is translation invariant for almost every  $\omega$ .

$$K(\omega, A_j) = \mathbb{E}[\mathbf{1}_{A_j} \mid \mathcal{I}](\omega) = \mathbf{1}_{A_j}(\omega)$$

for almost every  $\omega$ . Intersecting these countably many full-measure sets shows that almost every conditional measure assigns only 0 or 1 to every event in the generated invariant sigma-field. Thus almost every  $K(\omega, \cdot)$  is translation ergodic.

$$\int_A g \, d\mu = 0 \quad (A \in \mathcal{I}).$$

Approximate  $A$  in  $L^1(\mu)$  by a cylinder event  $B$  depending on spins in a finite set  $\Delta$ . Choose a translation  $x$  so that  $(\Lambda + x) \cap \Delta = \emptyset$ . Translation covariance of the specification and invariance of  $A, \mu$  give

$$\int_A g \, d\mu = \int_A g \circ \tau_x \, d\mu.$$

Now  $B$  is measurable outside  $\Lambda + x$ , while the DLR equation gives  $\mathbb{E}[g \circ \tau_x \mid \mathcal{F}_{(\Lambda+x)^c}] = 0$ . Therefore

$$\int_B g \circ \tau_x \, d\mu = 0, \quad \left| \int_A g \, d\mu \right| \leq 2\|g\|_\infty \mu(A \Delta B).$$

Letting the approximation error tend to zero proves the claim. Thus  $\mathbb{E}[g \mid \mathcal{I}] = 0$ . In kernel language,

$$\int (f - \gamma_\Lambda f) \, dK(\omega) = 0$$

for almost every  $\omega$ . A countable determining family of local  $f$  and finite  $\Lambda$  then yields the full DLR equations for almost every component.

$$\mu = \int \nu \, \kappa(d\nu).$$

If another such decomposition existed, conditioning invariant observables would give the same regular conditional law over  $\mathcal{I}$ ; uniqueness of regular conditional probabilities therefore identifies the two decomposition measures. This proves uniqueness.  $\square$

## Chapter 4

# Entropy, Energy, and the Gibbs Variational Principle

The DLR equations are local. The variational principle is global: equilibrium states maximize entropy minus energy.

For a translation-invariant probability measure  $\mu$ , define the specific entropy

$$s(\mu) = \lim_{n \rightarrow \infty} \frac{1}{|\Lambda_n|} H(\mu_{\Lambda_n}),$$

whose existence follows from subadditivity, and the specific energy

$$e_\Phi(\mu) = \sum_{A \ni 0} \frac{1}{|A|} \mu(\Phi_A).$$

**Theorem 4.1** (Gibbs variational principle). *For a finite-spin finite-range translation-invariant interaction,*

$$p(\beta) = \sup_{\mu \in \mathcal{P}_{\text{inv}}} \{s(\mu) - \beta e_\Phi(\mu)\},$$

*up to the fixed convention for the additive normalization of pressure.*

*Proof.* For a translation-invariant measure  $\mu$ , strong subadditivity of Shannon entropy implies the existence of the specific entropy along boxes; the Van Hove tiling argument extends the same limit to Van Hove sequences. Finite range makes the specific energy well-defined and makes boundary corrections  $o(|\Lambda|)$ .

Let  $\gamma_\Lambda$  be the finite-volume Gibbs law in  $\Lambda$ . For any probability  $\nu_\Lambda$  on the finite configuration space,

$$D(\nu_\Lambda \| \gamma_\Lambda) = -H(\nu_\Lambda) + \beta \nu_\Lambda(H_\Lambda) + \log Z_\Lambda \geq 0.$$

Apply this with  $\nu_\Lambda = \mu_\Lambda$ , divide by  $|\Lambda|$ , and let  $\Lambda$  tend to infinity along a Van Hove sequence. The entropy and energy limits give

$$p(\beta) \geq s(\mu) - \beta e_\Phi(\mu)$$

for every invariant  $\mu$ . Hence pressure is at least the displayed supremum.

For the reverse inequality, fix a large cube  $Q_m$  and its finite-volume Gibbs law  $\gamma_m$ . Tile  $\mathbb{Z}^d$  by disjoint translates of  $Q_m$  and sample the configurations in different blocks independently with law  $\gamma_m$ . Average this block-product measure over the  $m^d$  translations of the tiling; call the resulting

fully translation-invariant measure  $\bar{\gamma}_m$ . The finite mixture changes entropy by at most  $d \log m$  globally, hence not at the entropy-density level, and independence of blocks gives

$$s(\bar{\gamma}_m) = \frac{H(\gamma_m)}{|Q_m|}.$$

Only interaction terms crossing block boundaries distinguish  $e_\Phi(\bar{\gamma}_m)$  from  $|Q_m|^{-1}\gamma_m(H_{Q_m})$ , so

$$e_\Phi(\bar{\gamma}_m) = \frac{\gamma_m(H_{Q_m})}{|Q_m|} + O(m^{-1}).$$

For the Gibbs law the relative-entropy identity is an equality at  $\nu_\Lambda = \gamma_m$ :

$$\frac{1}{|Q_m|} \log Z_{Q_m} = \frac{H(\gamma_m)}{|Q_m|} - \beta \frac{\gamma_m(H_{Q_m})}{|Q_m|}.$$

Therefore

$$s(\bar{\gamma}_m) - \beta e_\Phi(\bar{\gamma}_m) = \frac{1}{|Q_m|} \log Z_{Q_m} + O(m^{-1}).$$

Letting  $m \rightarrow \infty$  and using the thermodynamic-pressure theorem gives

$$\sup_{\mu \in \mathcal{P}_{\text{inv}}} \{s(\mu) - \beta e_\Phi(\mu)\} \geq p(\beta).$$

Combined with the first inequality, this proves the variational principle.  $\square$

**Theorem 4.2** (Equilibrium states are DLR). *Under the same finite-range hypotheses, a translation-invariant state attaining the variational supremum satisfies the DLR equations.*

*Proof.* Fix a finite region  $\Lambda$ . Let  $\gamma_\Lambda(\cdot \mid \omega)$  be the finite-volume Gibbs kernel with outside configuration  $\omega_{\Lambda^c}$ , and let  $\mu_\Lambda(\cdot \mid \mathcal{F}_{\Lambda^c})$  be a regular conditional law of the alleged equilibrium state  $\mu$ . The finite-volume entropy identity behind the variational principle can be localized to  $\Lambda$  and written as

$$s(\mu) - \beta e_\Phi(\mu) = p - \frac{1}{|\Lambda|} \mathbb{E}_\mu D(\mu_\Lambda(\cdot \mid \mathcal{F}_{\Lambda^c}) \parallel \gamma_\Lambda(\cdot \mid \cdot)) + o_\Lambda(1),$$

where the boundary error divided by volume tends to zero along a van Hove sequence. More directly, replacing the conditional law inside one translate of  $\Lambda$  by the Gibbs kernel changes the finite-volume free-energy functional by exactly the nonnegative conditional relative entropy

$$\mathbb{E}_\mu D(\mu_\Lambda(\cdot \mid \mathcal{F}_{\Lambda^c}) \parallel \gamma_\Lambda(\cdot \mid \cdot)).$$

Perform this replacement simultaneously on a positive-density family of disjoint translates. If the displayed expectation were positive, the free-energy density would strictly increase, contradicting that  $\mu$  attains the variational supremum. Hence the expectation is zero. Relative entropy vanishes only when its two probability measures agree, so

$$\mu(\cdot \mid \mathcal{F}_{\Lambda^c})(\omega) = \gamma_\Lambda(\cdot \mid \omega) \quad \text{for } \mu\text{-a.e. } \omega.$$

Since  $\Lambda$  was arbitrary,  $\mu$  satisfies the DLR equations.  $\square$

## 4.1 Equivalence of ensembles

**Theorem 4.3** (Equivalence away from first-order singularities). *Assume that the grand-canonical pressure  $p(\mu)$  is differentiable at  $\mu_0$ , with density  $\rho = p'(\mu_0)$ , and that the translation-invariant equilibrium state at  $\mu_0$  is unique. Let  $N_\Lambda/|\Lambda| \rightarrow \rho$  be a canonical sequence for which the canonical free-energy density attains the Legendre dual value*

$$\lim_{\Lambda \uparrow \mathbb{Z}^d} \frac{1}{|\Lambda|} \log Z_{\Lambda, N_\Lambda} = p(\mu_0) - \mu_0 \rho.$$

*Then the canonical measures converge locally to the unique grand-canonical equilibrium state at  $\mu_0$ . Moreover, differentiability guarantees the existence of such thermodynamically admissible canonical sectors inside every shrinking density window whose width tends to zero sufficiently slowly.*

*Proof.* The finite-volume identity

$$\Xi_\Lambda(\mu) = \sum_N e^{\mu N} Z_{\Lambda, N}$$

first gives  $Z_{\Lambda, N} e^{\mu_0 N} \leq \Xi_\Lambda(\mu_0)$ , hence every canonical subsequential free-energy density at density  $\rho$  is at most  $p(\mu_0) - \mu_0 \rho$ .

We next justify the existence assertion. Differentiability of the convex pressure at  $\mu_0$  implies exponential concentration outside every fixed density window. Indeed, for  $a > \rho$ , Chernoff's bound and the pressure limit give

$$\limsup_{|\Lambda| \rightarrow \infty} \frac{1}{|\Lambda|} \log \mathbb{P}_{\Lambda, \mu_0}(N \geq a|\Lambda|) \leq \inf_{t > 0} \{p(\mu_0 + t) - p(\mu_0) - ta\} < 0,$$

and similarly below  $\rho$ . A diagonal choice therefore gives  $\varepsilon_\Lambda \downarrow 0$ , slowly enough that

$$\mathbb{P}_{\Lambda, \mu_0}(|N/|\Lambda| - \rho| \leq \varepsilon_\Lambda) \rightarrow 1.$$

The window contains at most  $2\varepsilon_\Lambda|\Lambda| + 2$  sectors, so one  $N_\Lambda$  in it has probability at least the reciprocal of this number up to a factor  $1 - o(1)$ . Since its logarithm is  $o(|\Lambda|)$ ,

$$\frac{1}{|\Lambda|} \log Z_{\Lambda, N_\Lambda} = \frac{1}{|\Lambda|} \log \Xi_\Lambda(\mu_0) - \mu_0 \frac{N_\Lambda}{|\Lambda|} + o(1),$$

which is the required Legendre value.

Now take any thermodynamically admissible canonical sequence from the statement and any locally convergent subsequence of its measures; compactness of the finite-spin local state spaces (or the tightness hypothesis built into the general state-space setup) supplies such subsequences. Call a local limit  $\nu$ . Translation averaging changes local expectations only by a boundary-layer error, so  $\nu$  may be taken translation invariant. The finite-range boundary corridors contribute  $o(|\Lambda|)$  to both energy and entropy. Applying the finite-volume relative-entropy inequality on large interior blocks and passing first  $\Lambda \uparrow \mathbb{Z}^d$  and then the block size to infinity yields

$$s(\nu) - \beta e_\Phi(\nu) + \mu_0 \rho \geq p(\mu_0).$$

The Gibbs variational principle gives the reverse inequality, hence equality:  $\nu$  is a grand-canonical equilibrium state at  $\mu_0$ . By the assumed uniqueness,  $\nu = \mu_{\mu_0}$ . Every locally convergent subsequence has the same limit, so the full canonical sequence converges locally to  $\mu_{\mu_0}$ .  $\square$



## Chapter 5

# Dobrushin Uniqueness and Correlation Decay

The Dobrushin method converts local sensitivity to boundary spins into a contraction on the full infinite-volume specification.

**Definition 5.1** (Dobrushin influence matrix). For a specification  $\gamma$ , define

$$c_{ij} = \sup_{\eta, \eta': \eta_k = \eta'_k \text{ } (k \neq j)} \|\gamma_i(\cdot \mid \eta) - \gamma_i(\cdot \mid \eta')\|_{\text{TV}}.$$

The Dobrushin contraction coefficient is  $c = \sup_i \sum_j c_{ij}$ . The *Dobrushin uniqueness condition* is  $c < 1$ .

*Worked example 5.2* (Nearest-neighbour Ising Dobrushin bound). For ferromagnetic Ising coupling  $J \geq 0$ , changing one neighbour of  $i$  changes the local field by  $2J$ . The two Bernoulli conditional laws differ in total variation by at most  $\tanh(\beta J)$ , so

$$c_{ij} \leq \tanh(\beta J) \mathbf{1}_{\{i \sim j\}}.$$

On a graph of maximum degree  $\Delta$ ,

$$\sup_i \sum_j c_{ij} \leq \Delta \tanh(\beta J).$$

Thus  $\Delta \tanh(\beta J) < 1$  is an explicit high-temperature uniqueness criterion. It is sufficient, not necessary; the example also shows exactly where the geometry of the interaction graph enters the contraction estimate.

Write  $C = (c_{ij})$  for the influence matrix.

**Theorem 5.3** (Dobrushin uniqueness theorem). *If*

$$\alpha := \sup_i \sum_j c_{ij} < 1,$$

*then the specification has a unique Gibbs state.*

*Proof.* For a bounded local observable put

$$\delta_i(f) = \sup\{|f(\eta) - f(\eta')| : \eta_k = \eta'_k \ (k \neq i)\}.$$

If two boundary configurations differ only at  $j$ , a maximal coupling of the two one-site kernels at  $i$  disagrees with probability at most  $c_{ij}$ . Changing several coordinates one at a time gives the sum of the corresponding influences.

Fix a finite region  $\Lambda$  and two exterior boundary conditions. Couple the two single-site heat-bath chains by maximal coupling at each update. In a stationary coupling let  $u_i$  be the probability of disagreement at  $i \in \Lambda$ . Conditioning immediately before an update gives

$$u_i \leq b_i + \sum_{j \in \Lambda} c_{ij} u_j, \quad (5.1)$$

where  $b_i \leq \sum_{j \notin \Lambda} c_{ij}$  is the direct exterior influence. Iteration yields

$$u \leq \sum_{n \geq 0} C_\Lambda^n b, \quad \sup_i \sum_j (C_\Lambda^n)_{ij} \leq \alpha^n.$$

For local  $f$  this bounds the difference of the two finite-volume expectations by  $\sum_i \delta_i(f) u_i$ . As  $\Lambda \uparrow \mathbb{Z}^d$ , each fixed row tail  $\sum_{j \notin \Lambda} c_{ij}$  tends to zero and the Neumann series is dominated by  $\sum_n \alpha^n < \infty$ ; dominated convergence therefore makes the bound vanish.

If  $\mu$  and  $\nu$  are Gibbs states, condition them on  $\mathcal{F}_{\Lambda^c}$ , apply the preceding finite-volume estimate, and integrate over their exterior configurations. Letting  $\Lambda$  increase gives  $\mu(f) = \nu(f)$  for every bounded local  $f$ . Cylinder functions determine the measure, so  $\mu = \nu$ .  $\square$

**Theorem 5.4** (Dobrushin comparison theorem). *Let two specifications be dominated by the same nonnegative influence matrix  $C$  with  $\alpha = \sup_i \sum_j c_{ij} < 1$ , and let  $b_i$  bound the direct one-site discrepancy between their conditional kernels. Then for every bounded local  $f$ ,*

$$|\mu(f) - \nu(f)| \leq \sum_i \delta_i(f) (Db)_i, \quad D = (I - C)^{-1} = \sum_{n \geq 0} C^n.$$

*A corresponding boundary-condition version holds with  $b$  supported on the discrepancy set.*

*Proof.* Work first in finite volume and couple the two heat-bath chains by maximal coupling at the updated site. The direct difference between the two specifications contributes at most  $b_i$ . If the current configurations disagree at several sites, change those coordinates one at a time in the conditional kernel. The definition of  $c_{ij}$  then gives, for stationary disagreement probabilities  $u_i$ ,

$$u_i \leq b_i + \sum_j c_{ij} u_j, \quad \text{i.e.} \quad u \leq b + Cu. \quad (5.2)$$

Hence

$$u \leq \sum_{n=0}^N C^n b + C^{N+1} u.$$

Since the row-sum norm of  $C$  is at most  $\alpha < 1$ , the last term tends to zero and  $u \leq Db$ , where  $D = (I - C)^{-1} = \sum_{n \geq 0} C^n$ .

For a bounded local  $f$ ,

$$|f(X) - f(Y)| \leq \sum_i \delta_i(f) \mathbf{1}_{\{X_i \neq Y_i\}}.$$

Taking expectations in the stationary coupling proves

$$|\mu(f) - \nu(f)| \leq \sum_i \delta_i(f)(Db)_i.$$

Taking  $b$  to be the direct influence of a prescribed exterior discrepancy gives the boundary-condition form. Infinite-volume statements follow by applying the estimate on increasing finite regions and passing to Gibbs limits; the absolutely convergent Neumann series makes the bound stable under that passage.  $\square$

**Theorem 5.5** (Exponential correlation decay). *Assume finite interaction range and a weighted Dobrushin condition: for some  $m > 0$ ,*

$$\sup_i \sum_j e^{m|i-j|} c_{ij} < 1.$$

*Then the unique Gibbs state satisfies*

$$|\text{Cov}_\mu(f, g)| \leq C \|f\|_{\text{osc}} \|g\|_{\text{osc}} e^{-m' \text{dist}(\text{supp } f, \text{supp } g)}$$

*for every  $m' < m$  after adjusting the constant.*

*Proof.* In the exponentially weighted row norm the hypothesis gives a contraction constant  $\alpha_m < 1$ . Thus, for  $D = (I - C)^{-1} = \sum_{n \geq 0} C^n$ ,

$$D_{ij} \leq (1 - \alpha_m)^{-1} e^{-m|i-j|}. \quad (5.3)$$

Let  $A = \text{supp } f$  and  $B = \text{supp } g$ . Condition on the spins in  $B$ . Two values of those spins act as two boundary conditions for the conditional law seen from  $A$ , so the comparison theorem bounds the oscillation of  $\mathbb{E}[f \mid \sigma_B]$  by a constant times

$$\sum_{i \in A} \sum_{j \in B} \delta_i(f) D_{ij}.$$

Since  $\text{Cov}(f, g) = \text{Cov}(\mathbb{E}[f \mid \sigma_B], g)$  and  $|\text{Cov}(U, V)| \leq \frac{1}{4} \text{osc}(U) \text{osc}(V)$ , (5.3) gives exponential decay with rate  $m$  up to a support-dependent prefactor. Replacing  $m$  by any  $m' < m$  absorbs the finite support factor into the constant and yields the stated estimate.  $\square$

**Example 5.6** (High-temperature Ising criterion). For the nearest-neighbor Ising model, changing one neighboring spin changes the one-site field by  $2\beta J$ . A direct two-state computation gives  $c_{ij} \leq \tanh(\beta|J|)$  for neighbors and zero otherwise. Thus  $2d \tanh(\beta|J|) < 1$  is a simple sufficient high-temperature uniqueness condition.

## Chapter 6

# Polymer and Cluster Expansions

Cluster expansions convert a dilute gas of mutually incompatible objects into an absolutely convergent series indexed by connected graphs.

**Definition 6.1** (Polymer model and cluster). A polymer model consists of a countable set  $\mathcal{P}$ , activities  $z : \mathcal{P} \rightarrow \mathbb{C}$ , and a symmetric incompatibility relation  $\not\sim$ . A finite ordered family  $(\gamma_1, \dots, \gamma_n)$  is a *cluster* when its incompatibility graph, with an edge  $i \sim j$  whenever  $\gamma_i \not\sim \gamma_j$ , is connected.

*Worked example 6.2* (Two incompatible polymers). Let  $\mathcal{P} = \{a, b\}$  with  $a \not\sim b$  and with self-exclusion. The allowed configurations are  $\emptyset, \{a\}, \{b\}$ , hence

$$Z = 1 + z_a + z_b.$$

Expanding around  $z_a = z_b = 0$  gives

$$\log Z = (z_a + z_b) - \frac{1}{2}(z_a + z_b)^2 + O(|z|^3) = z_a + z_b - \frac{1}{2}z_a^2 - z_a z_b - \frac{1}{2}z_b^2 + O(|z|^3).$$

The mixed coefficient  $-z_a z_b$  is the connected two-polymer contribution: it is present precisely because the incompatibility graph on  $\{a, b\}$  is connected.

For finite  $\Lambda \subset \mathcal{P}$  and activities  $z(\gamma)$ ,

$$\Xi_\Lambda = \sum_{\Gamma \subset \Lambda} \prod_{\text{compatible } \gamma \in \Gamma} z(\gamma).$$

**Theorem 6.3** (Connected-graph cluster expansion). *Formally,*

$$\log \Xi_\Lambda = \sum_{n \geq 1} \frac{1}{n!} \sum_{(\gamma_1, \dots, \gamma_n) \in \Lambda^n} \phi^T(\gamma_1, \dots, \gamma_n) \prod_{i=1}^n z(\gamma_i),$$

where  $\phi^T$  is the sum of  $(-1)^{|E(G)|}$  over connected graphs whose edges join incompatible polymer pairs. As a formal power series this identity always holds; analytically it converges absolutely under the Kotecký–Preiss criterion stated in the next section.

*Proof.* Put

$$f(\gamma, \eta) = \begin{cases} -1, & \gamma \not\sim \eta, \\ 0, & \gamma \sim \eta. \end{cases}$$

After allowing ordered tuples and dividing by  $n!$ ,

$$\Xi_\Lambda = 1 + \sum_{n \geq 1} \frac{1}{n!} \sum_{\gamma_1, \dots, \gamma_n \in \Lambda} \prod_i z(\gamma_i) \prod_{i < j} (1 + f_{ij}). \quad (6.1)$$

Expanding the last product sums over all simple graphs  $G$  on  $[n]$ , with weight  $\prod_{ij \in E(G)} f_{ij}$ . Every graph decomposes uniquely into its connected components and its weight factors over those components. The exponential formula for labelled objects therefore identifies the formal logarithm of (6.1) with the exponential generating function of connected graphs:

$$\log \Xi_\Lambda = \sum_{n \geq 1} \frac{1}{n!} \sum_{\gamma_1, \dots, \gamma_n} \left( \sum_{G \text{ connected}} \prod_{ij \in E(G)} f_{ij} \right) \prod_i z(\gamma_i).$$

For hard-core incompatibility the inner sum is exactly  $\phi^T$ . This proves the identity coefficient by coefficient as a formal series. The Kotecký–Preiss estimate below proves absolute convergence, after which the same rearrangement is legitimate analytically.  $\square$

## 6.1 The Kotecký–Preiss criterion

**Lemma 6.4** (Hard-core tree-graph bound). *For a tuple  $(\gamma_1, \dots, \gamma_n)$ ,*

$$|\phi^T(\gamma_1, \dots, \gamma_n)| \leq \sum_{T \text{ spanning tree on } [n]} \prod_{ij \in E(T)} \mathbf{1}_{\{\gamma_i \not\sim \gamma_j\}}.$$

*Proof.* Let  $H$  be the incompatibility graph and fix an ordering of its vertices. Apply breadth-first search from vertex 1 to each connected spanning subgraph of  $H$ . Subgraphs giving the same search tree  $T$  form an interval  $\{G : T \subseteq G \subseteq \bar{T}\}$  in the edge lattice: the optional edges are exactly those that the deterministic search rule does not use to change parents or levels. The alternating contribution of such an interval is

$$\sum_{T \subseteq G \subseteq \bar{T}} (-1)^{|E(G)|} = (-1)^{|E(T)|} (1 - 1)^{|E(\bar{T})| - |E(T)|},$$

so it has absolute value at most one. Summing over possible search trees bounds the connected alternating sum by the number of spanning trees of  $H$ , which is precisely the displayed tree sum with incompatibility indicators.  $\square$

**Theorem 6.5** (Kotecký–Preiss). *Suppose there is a nonnegative function  $a(\gamma)$  such that for every polymer  $\eta$ ,*

$$\sum_{\gamma: \gamma \not\sim \eta} |z(\gamma)| e^{a(\gamma)} \leq a(\eta).$$

*Then the cluster expansion for  $\log \Xi$  converges absolutely. More precisely,*

$$\sum_{n \geq 1} \frac{1}{(n-1)!} \sum_{\gamma_2, \dots, \gamma_n} |\phi^T(\eta, \gamma_2, \dots, \gamma_n)| \prod_{i=2}^n |z(\gamma_i)| \leq e^{a(\eta)}.$$

*Proof.* By Lemma 6.4, it is enough to dominate rooted spanning trees. Let  $T_m(\eta)$  be the total absolute weight of rooted labelled trees of depth at most  $m$ , with root label fixed to  $\eta$  and root

activity omitted. Children of a vertex labelled  $\eta$  must be incompatible with it; ordering the children and dividing by their factorial gives

$$T_{m+1}(\eta) \leq \exp \left\{ \sum_{\gamma \not\sim \eta} |z(\gamma)| T_m(\gamma) \right\}, \quad T_0(\eta) = 1. \quad (6.2)$$

Inductively  $T_m(\eta) \leq e^{a(\eta)}$ : if it holds at depth  $m$ , then the KP hypothesis gives

$$T_{m+1}(\eta) \leq \exp \left\{ \sum_{\gamma \not\sim \eta} |z(\gamma)| e^{a(\gamma)} \right\} \leq e^{a(\eta)}.$$

Monotone convergence gives the same bound at arbitrary depth. Combining it with the tree-graph lemma yields exactly the pinned absolute cluster estimate in the theorem. Multiplying by the root activity and summing over roots then proves absolute convergence of the connected cluster series in finite volume; the same estimate is locally uniform in any exhaustion for which the KP bound is uniform. Thus the formal logarithm from the preceding theorem is also an analytic identity in the stated convergence region.  $\square$

**Theorem 6.6** (Analyticity and exponential clustering). *If the KP criterion holds uniformly in a complex neighborhood of a real parameter, the infinite-volume pressure is analytic there. If polymer activities decay exponentially in polymer size/diameter, truncated correlations of separated local observables decay exponentially with separation.*

*Proof.* Under the uniform Kotecký–Preiss bound, the connected cluster series for  $\log \Xi_\Lambda$  converges absolutely and locally uniformly in the complex parameter throughout the stated neighborhood. Each finite cluster weight is analytic there. The Weierstrass theorem therefore permits termwise differentiation and shows that the finite-volume pressure and all local cluster observables are analytic, with bounds uniform in  $\Lambda$ . The boundary contribution to  $\log \Xi_\Lambda$  is  $o(|\Lambda|)$  by the same polymer estimate, so the thermodynamic pressure is the locally uniform limit of these analytic functions and is analytic as well.

Let  $F$  and  $G$  be local observables with supports  $A$  and  $B$ . Differentiate the cluster expansion with independent source parameters coupled to  $F$  and  $G$ . The truncated correlation

$$\langle FG \rangle - \langle F \rangle \langle G \rangle$$

is a sum only over connected clusters whose incompatibility graph meets both  $A$  and  $B$ . If the supports are distance  $R$  apart, every such cluster contains a chain of polymers whose total diameter is at least  $R$ . If

$$|z(\gamma)| e^{a(\gamma)} \leq C e^{-c \operatorname{diam}(\gamma)}$$

(or the corresponding size bound), the tree-graph/KP estimate for a cluster joining  $A$  to  $B$  gains an additional factor  $e^{-c'R}$ . Summing the rooted cluster bound gives

$$|\langle FG \rangle - \langle F \rangle \langle G \rangle| \leq C_{F,G} e^{-c' \operatorname{dist}(A,B)}.$$

The constants are uniform wherever the KP criterion is uniform, proving analyticity and exponential clustering.  $\square$

**Example 6.7** (Low-activity hard-core polymers). Take polymers to be finite connected subsets of  $\mathbb{Z}^d$ , with incompatibility defined by intersection or distance one, and activities satisfying  $|z(\gamma)| \leq e^{-c|\gamma|}$ . The number of connected sets of size  $n$  containing a fixed site is at most  $C_d^n$ . Therefore

$$\sum_{\gamma \not\supset \eta} |z(\gamma)| e^{a|\gamma|} \leq |\eta| \sum_{n \geq 1} C_d^n e^{-(c-a)n}.$$

For  $c - a$  large enough this is at most  $a|\eta|$ , giving the KP criterion with  $a(\eta) = a|\eta|$ . This calculation is the prototype behind low-temperature contour gases and high-temperature polymer expansions.

## Chapter 7

# Contours and Pirogov–Sinai Theory

Low-temperature phase coexistence is organized by contours separating regions that locally resemble different ground states.

### 7.1 Peierls contours

Assume first that the interaction has two translation-periodic ground states  $g_+, g_-$  and a Peierls constant  $\tau > 0$ : every finite defect region costs at least  $\tau$  times the size of its contour support.

**Theorem 7.1** (Peierls contour estimate). *At sufficiently large  $\beta$ , the probability under  $+$  boundary condition that a contour of size  $n$  surrounds the origin is at most*

$$N_n e^{-\beta \tau n},$$

where  $N_n \leq C^n$ . Consequently the probability that the origin is in the wrong phase is small, and the  $+$  and  $-$  infinite-volume Gibbs states are distinct.

*Proof.* Fix a contour  $\Gamma$  surrounding the origin. By the Peierls condition, changing the spins in the interior of  $\Gamma$  from the wrong ground-state label to the  $+$  label lowers the Hamiltonian by at least  $\tau|\Gamma|$ . The flip map is injective once  $\Gamma$  is specified, so comparison of Boltzmann weights gives

$$\mathbb{P}_+(\Gamma \text{ occurs}) \leq e^{-\beta \tau |\Gamma|}.$$

If  $N_n$  is the number of contours of size  $n$  surrounding the origin and  $N_n \leq C^n$ , the union bound yields

$$\mathbb{P}_+(\exists \Gamma \ni 0 : |\Gamma| = n) \leq C^n e^{-\beta \tau n}.$$

Thus, for  $\beta \tau > \log C$ ,

$$\mathbb{P}_+(0 \text{ lies in the wrong phase}) \leq \sum_{n \geq n_0} e^{-(\beta \tau - \log C)n},$$

which can be made arbitrarily small by increasing  $\beta$ . Passing to an infinite-volume subsequential limit with  $+$  boundary conditions therefore gives a Gibbs state with positive  $+$  magnetization/order parameter; the analogous construction with  $-$  boundary conditions gives the opposite value. The two limiting Gibbs states are consequently distinct.  $\square$



## 7.2 The finite-ground-state Pirogov–Sinai theorem

Let the finite-spin finite-range interaction have periodic ground states  $g_1, \dots, g_q$  satisfying a uniform Peierls condition. External parameters are collected in  $\lambda \in U \subset \mathbb{R}^m$ .

**Theorem 7.2** (Pirogov–Sinai phase diagram). *For sufficiently large  $\beta$ , there exist metastable free-energy branches  $f_1(\lambda), \dots, f_q(\lambda)$  such that the stable phases at  $\lambda$  are exactly the indices minimizing these branches. For every stable phase there is a corresponding extremal Gibbs state exponentially concentrated on configurations with only dilute contours around that ground state. The union of coexistence loci gives the complete low-temperature phase diagram within the selected finite-ground-state framework.*

*Proof.*

$$z_i(\Gamma) = e^{-\beta E(\Gamma)} \prod_m \frac{Z_{j_m}(\text{Int}_m \Gamma)}{Z_i(\text{Int}_m \Gamma)}.$$

This is circular if used directly. Order contours by diameter/volume. Assuming metastable pressures have been constructed for smaller interiors, multiply  $z_i(\Gamma)$  by a smooth cutoff which is one whenever every competing interior pressure is no larger than the exterior pressure by more than the finite-size tolerance, and zero when it is decisively larger. This defines the truncated activity  $\hat{z}_i(\Gamma)$  from already constructed data.

$$|\log Z_j(V) - \log Z_i(V) - |V|(p_j - p_i)| \leq C|\partial V|.$$

On a contour retained by the cutoff the bulk term cannot favor an interior phase by more than the allowed tolerance. The boundary term is  $O(|\Gamma|)$ . Increasing  $\beta$  if necessary therefore yields

$$|\hat{z}_i(\Gamma)| \leq e^{-c\beta|\Gamma|}.$$

Consequently, for some  $a > 0$ ,

$$\sup_x \sum_{\Gamma \ni x} |\hat{z}_i(\Gamma)| e^{a|\Gamma|} \leq a.$$

This is the Kotecky–Preiss criterion, so the truncated contour gas has an absolutely and locally uniformly convergent cluster expansion.

$$\log \hat{Z}_i(\Lambda) = p_i(\lambda)|\Lambda| + O(|\partial\Lambda|),$$

uniformly in boundary shape, and the coefficient  $p_i$  is analytic in every parameter region where the polymer bound is uniform. Set  $f_i = -\beta^{-1}p_i$ . □

**Theorem 7.3** (Metastable free energies and coexistence surfaces). *In the same low-temperature regime, each metastable branch  $f_i(\lambda)$  is analytic wherever its truncated contour expansion is uniformly convergent. The physical free energy is*

$$f(\lambda) = \min_i f_i(\lambda).$$

*Two-phase coexistence occurs on  $f_i = f_j < f_k$  for  $k \neq i, j$ , and derivatives of the competing branches give the discontinuous order parameters across a first-order surface.*

*Proof.* For each phase  $i$ , the proof of the Pirogov–Sinai theorem constructs a truncated contour gas whose activities satisfy a Kotecký–Preiss bound locally uniformly in  $\lambda$ . Hence its infinite-volume pressure

$$p_i(\lambda) = \lim_{\Lambda \uparrow \mathbb{Z}^d} \frac{1}{|\Lambda|} \log \widehat{Z}_i(\Lambda, \lambda)$$

is represented by a locally uniformly convergent series of analytic cluster weights. By the Weierstrass theorem  $p_i$ , and therefore

$$f_i(\lambda) = -\beta^{-1} p_i(\lambda),$$

is analytic throughout every region where the uniform contour bound holds.

The stability cutoff in the Pirogov–Sinai construction is chosen so that an  $i$ -boundary partition function agrees, up to a surface-order factor, with the untruncated model precisely when  $f_i$  is minimal among the metastable branches. Consequently the thermodynamic pressure is

$$p(\lambda) = \max_i p_i(\lambda),$$

which is equivalent to

$$f(\lambda) = \min_i f_i(\lambda).$$

At a point where exactly two branches satisfy  $f_i = f_j < f_k$  for  $k \neq i, j$ , both corresponding contour Gibbs states are stable and give distinct infinite-volume phases; this is the two-phase coexistence locus.

If  $v$  is a direction in parameter space, the conjugate order parameter in phase  $i$  is

$$m_i(v) = -\partial_v f_i.$$

Crossing a smooth coexistence hypersurface from the region where  $f_i < f_j$  to the region where  $f_j < f_i$  changes the derivative of the physical free energy from  $\partial_v f_i$  to  $\partial_v f_j$ . Whenever these derivatives differ, the order parameter jumps. Thus derivatives of the analytic competing branches give the first-order discontinuity claimed in the theorem.  $\square$

**Example 7.4** (A two-phase contour prototype). Consider a finite-spin nearest-neighbor model with exactly two periodic ground states, denoted  $+$  and  $-$ , and suppose every bond disagreeing with the chosen ground-state pattern costs at least  $\tau > 0$ . A connected defect surrounded by the  $+$  phase determines a contour  $\Gamma$  whose support contains at least a fixed multiple of the number of disagreeing bonds. Consequently

$$|w_+(\Gamma)| \leq e^{-\beta\tau|\Gamma|}.$$

If the number of contours of size  $n$  containing a fixed site is at most  $C^n$ , then

$$\sum_{\Gamma \ni 0} |w_+(\Gamma)| e^{a|\Gamma|} \leq \sum_{n \geq 1} C^n e^{-(\beta\tau - a)n}.$$

At sufficiently low temperature this is smaller than  $a$ , so the contour gas satisfies a KP criterion. What Pirogov–Sinai adds to this simple calculation is the recursive treatment of contours whose interiors may themselves prefer a competing phase: the stability truncation prevents an unstable interior partition function from destroying the small-activity estimate.

## Chapter 8

# The Hard-Core Gas and Quenched-Disorder Rounding

### 8.1 Hard-core lattice gas

A hard-core configuration on a graph  $G = (V, E)$  is an independent set  $I \subset V$ . At activity  $\lambda > 0$ , the finite-volume weight is proportional to  $\lambda^{|I|}$ .

**Definition 8.1** (Hard-core Gibbs measure). For a finite graph  $G = (V, E)$  and activity  $\lambda > 0$ , the hard-core partition function and Gibbs probability are

$$Z_G(\lambda) = \sum_{I \subseteq V \text{ independent}} \lambda^{|I|}, \quad \mu_{G,\lambda}(I) = \frac{\lambda^{|I|}}{Z_G(\lambda)}.$$

*Worked example 8.2* (Hard-core model on one edge). For  $G = K_2$ , the independent sets are  $\emptyset, \{1\}, \{2\}$ , so  $Z_{K_2} = 1 + 2\lambda$ . Therefore

$$\mathbb{P}(1 \text{ occupied}) = \frac{\lambda}{1 + 2\lambda}, \quad \mathbb{E}|I| = \frac{2\lambda}{1 + 2\lambda}.$$

The forbidden set  $\{1, 2\}$  makes the two occupation variables negatively correlated. Indeed,

$$\mathbb{P}(1 \text{ and } 2 \text{ occupied}) = 0 < \left( \frac{\lambda}{1 + 2\lambda} \right)^2,$$

the product of the two one-site marginals.

**Theorem 8.3** (Monotonicity and existence of hard-core Gibbs states). *On every countable locally finite graph, finite-volume hard-core specifications have infinite-volume Gibbs subsequential limits. In finite volume, the normalized log partition function is increasing and convex in  $t = \log \lambda$ ; consequently the mean particle density is nondecreasing in the activity wherever it is differentiated.*

*Proof.*

$$\gamma_\Lambda(I \mid \eta) = \frac{\lambda^{|I \cap \Lambda|}}{Z_\Lambda(\eta)} \mathbf{1}\{I \cup \eta \text{ is independent}\}.$$

$$Z_\Lambda(t) = \sum_{I \text{ admissible}} e^{t|I \cap \Lambda|}.$$

Exactly as in Chapter 1,

$$\partial_t \frac{1}{|\Lambda|} \log Z_\Lambda = \frac{1}{|\Lambda|} \mathbb{E}_t |I \cap \Lambda| \geq 0, \quad \partial_t^2 \frac{1}{|\Lambda|} \log Z_\Lambda = \frac{1}{|\Lambda|} \text{Var}_t(|I \cap \Lambda|) \geq 0.$$

Thus pressure and mean occupancy are monotone in activity. This statement does not assert an ordinary FKG order for the antiferromagnetic hard-core constraint.  $\square$

## 8.2 Exact recursion on a regular tree

Let  $\mathbb{T}_\Delta$  be the infinite  $\Delta$ -regular tree,  $\Delta \geq 3$ . For a rooted branch let

$$R = \frac{Z(\text{root occupied})}{Z(\text{root vacant})}.$$

**Theorem 8.4** (Hard-core uniqueness threshold on the regular tree). *The translation-symmetric recursion is*

$$R = F(R) := \frac{\lambda}{(1 + R)^{\Delta-1}}.$$

*The Gibbs state is unique for*

$$\lambda \leq \lambda_c(\Delta) = \frac{(\Delta - 1)^{\Delta-1}}{(\Delta - 2)^\Delta},$$

*and for  $\lambda > \lambda_c(\Delta)$  there are distinct even/odd Gibbs states generated by a nontrivial two-cycle of  $F$ .*

*Proof.* Put  $b = \Delta - 1$ . On a rooted branch, conditioning on the states of the  $b$  children gives the boundary-law recursion

$$R = F(R_1, \dots, R_b) = \lambda \prod_{j=1}^b (1 + R_j)^{-1}.$$

For a translation-symmetric message this is  $F(r) = \lambda(1 + r)^{-b}$ . The map  $F$  is decreasing. Starting from the two extremal boundary messages and iterating inward, the even and odd iterates of  $r_{n+1} = F(r_n)$  are monotone and converge to numbers  $\ell \leq u$  satisfying

$$\ell = F(u), \quad u = F(\ell). \tag{8.1}$$

Every finite-volume boundary message lies between these extremal messages, so the infinite-volume Gibbs state is unique exactly when  $\ell = u$ ; when  $\ell < u$ , the two limits give distinct even/odd Gibbs states.

It remains to determine when (8.1) has a nontrivial two-cycle. The fixed point  $R_*$  is unique and obeys

$$\lambda = R_*(1 + R_*)^b, \quad |F'(R_*)| = b \frac{R_*}{1 + R_*}.$$

The value  $|F'(R_*)| = 1$  occurs at  $R_* = 1/(b - 1)$  and therefore at

$$\lambda_c = \frac{b^b}{(b - 1)^{b+1}} = \frac{(\Delta - 1)^{\Delta-1}}{(\Delta - 2)^\Delta}. \tag{8.2}$$

We now verify that this local threshold is exactly the global two-cycle threshold. Suppose  $0 < \ell < u$  is a nontrivial solution of (8.1) and set

$$t = \frac{1+u}{1+\ell} > 1.$$

Dividing the two equations in (8.1) gives  $u = t^b \ell$ , while  $1+u = t(1+\ell)$ . Hence, with  $S_n(t) = 1+t+\dots+t^{n-1}$ ,

$$\ell = \frac{1}{tS_{b-1}(t)}, \quad \lambda = \frac{S_b(t)^b}{tS_{b-1}(t)^{b+1}}. \quad (8.3)$$

Write  $t = e^{2x}$  and  $A_n(x) = \sinh(nx)/(n \sinh x)$ . The powers of  $e^x$  cancel in (8.3), giving

$$\frac{\lambda}{\lambda_c} = \frac{A_b(x)^b}{A_{b-1}(x)^{b+1}}. \quad (8.4)$$

To see that the right side is strictly increasing for  $x > 0$ , put  $q(y) = y \coth y$ . Since

$$q'(y) > 0, \quad q''(y) = 2 \operatorname{csch}^2 y (y \coth y - 1) > 0,$$

$q$  is increasing and convex. If  $L$  denotes the logarithm of the right side of (8.4), then

$$xL'(x) = b[q(bx) - q((b-1)x)] - [q((b-1)x) - q(x)] > 0 :$$

the first increment is over an interval of length  $x$  at the far right, while the second is the sum of only  $b-2$  earlier increments, each no larger than the last one. Since (8.4) tends to 1 as  $x \downarrow 0$ , every nontrivial two-cycle has  $\lambda > \lambda_c$ . Conversely the expression in (8.3) is continuous, starts at  $\lambda_c$  when  $t = 1$ , and tends to infinity with  $t$ ; its strict monotonicity gives a nontrivial two-cycle for every  $\lambda > \lambda_c$ .

Thus  $\ell = u$  for  $\lambda \leq \lambda_c$ , giving uniqueness, while  $\ell < u$  for  $\lambda > \lambda_c$ , giving the two distinct bipartite Gibbs states claimed in the theorem.  $\square$

### 8.3 Quenched disorder rounds first-order transitions in low dimension

Consider a classical finite-range lattice Hamiltonian

$$H_\Lambda^{h,\varepsilon\eta} = H_\Lambda^0 - \sum_{x \in \Lambda} (h + \varepsilon\eta_x) \kappa_x,$$

where  $\eta_x$  are iid with an absolutely continuous distribution and finite variance, and  $\kappa_x$  is a bounded local observable. Let  $p(h)$  be the quenched thermodynamic pressure.

**Lemma 8.5** (Phase-comparison free energy). *Assume that the left and right derivatives of the quenched pressure at  $h$  differ, and put*

$$M := p'_+(h) - p'_-(h) > 0.$$

*For boxes  $\Lambda_L = [-L, L]^d \cap \mathbb{Z}^d$  there are centered functions  $G_L = G_{\Lambda_L}(\eta_{\Lambda_L})$  and a bounded translation-covariant response field  $\tau_x(\eta)$  such that, for every  $x$  whose interaction translate is contained in  $\Lambda_L$ ,*

$$\frac{\partial G_L}{\partial \eta_x} = \varepsilon \mathbb{E}(\tau_x \mid \eta_{\Lambda_L}), \quad \mathbb{E}\tau_x = M, \quad |\tau_x| \leq B := 2\|\kappa_0\|_\infty, \quad (7.19)$$

and finite range gives the deterministic surface estimate

$$|G_L(\eta)| \leq C|\partial_R \Lambda_L| \leq C_1 L^{d-1}. \quad (7.20)$$

Here  $R$  is the interaction range and the constants do not depend on  $L$  or on the disorder realization.

*Proof.* Convexity of  $p$  gives two tangent thermodynamic functionals at  $h$ . Choose sequences  $h_n \downarrow h$  and  $h'_n \uparrow h$  at differentiability points, take thermodynamic limits of the corresponding finite-volume Gibbs states, and then take the translation-covariant barycentres of all subsequential limits. Denote the two resulting random Gibbs-state barycentres by  $\rho_{\eta,+}$  and  $\rho_{\eta,-}$ . Their local responses satisfy

$$\tau_x(\eta) := \rho_{\eta,+}(\kappa_x) - \rho_{\eta,-}(\kappa_x), \quad |\tau_x| \leq 2\|\kappa_0\|_\infty,$$

and translation covariance plus the pressure derivative formula gives  $\mathbb{E}\tau_x = p'_+(h) - p'_-(h) = M$ .

For a finite box  $\Lambda$  remove the disorder terms whose supports are contained in  $\Lambda$  and compare the two Gibbs-state barycentres after this common local perturbation. Equivalently, if

$$U_\Lambda(\sigma, \eta) := -\varepsilon \sum_{x: \text{supp } \kappa_x \subset \Lambda} \eta_x \kappa_x(\sigma),$$

set, before centering in the disorder,

$$\widehat{G}_\Lambda(\eta) := -\log \rho_{\eta,+}(e^{-U_\Lambda}) + \log \rho_{\eta,-}(e^{-U_\Lambda}). \quad (7.21)$$

The Gibbs perturbation identity says that differentiating the logarithms in (7.21) reinstates the original local Gibbs expectations. After conditional averaging over the disorder outside  $\Lambda$  and subtracting the disorder mean, this gives exactly the first identity in (7.19). Independence permits differentiation to commute with that outside conditional expectation.

It remains to explain why a volume term does not survive in (7.21). The two states are compared after the same interaction has been deleted in the interior. Hence their restrictions to the spins in  $\Lambda$  differ only through the interaction  $V_{\eta;\partial\Lambda}$  crossing the boundary. The elementary inequality

$$e^{-2\|V\|_\infty} \leq \frac{\rho_+(e^{-V})}{\rho_-(e^{-V})} \leq e^{2\|V\|_\infty}$$

therefore yields  $|\widehat{G}_\Lambda| \leq 2\|V_{\eta;\partial\Lambda}\|_\infty$ . With the above convention the boundary interaction contains no disorder variable whose entire support lies in  $\Lambda$ . Finite range and bounded local interactions consequently give  $\|V_{\eta;\partial\Lambda}\|_\infty \leq C|\partial_R \Lambda|$ , which is (7.20). Centering does not change (7.19) and only doubles the constant in (7.20).  $\square$

**Lemma 8.6** (Aizenman–Wehr one-site variance bound). *Let  $\nu$  be an absolutely continuous probability law on  $\mathbb{R}$  with finite second moment. For  $K < \infty$  and  $a \in \mathbb{R}$  define*

$$\Theta_\nu(a, K) := \inf \left\{ \text{Var}_\nu(g)^{1/2} : g \text{ locally absolutely continuous, } |g'| \leq K \text{ a.e., } \int g' d\nu = a \right\}. \quad (7.22)$$

*Then  $\Theta_\nu(a, K) > 0$  whenever  $a \neq 0$ .*

*Proof.* Suppose instead that  $a \neq 0$  and there are admissible  $g_n$  with  $\text{Var}_\nu(g_n) \rightarrow 0$ . Subtracting the  $\nu$ -mean does not change the derivative, so we may assume  $\int g_n d\nu = 0$  and hence  $g_n \rightarrow 0$  in  $L^2(\nu)$ . All  $g_n$  are  $K$ -Lipschitz. Choose a point  $x_0$  in the essential support of  $\nu$ . If a subsequence

had  $|g_n(x_0)| > c > 0$ , the Lipschitz bound would force  $|g_n| > c/2$  on a fixed neighbourhood having positive  $\nu$ -mass, contradicting  $L^2(\nu)$  convergence. Thus  $g_n(x_0) \rightarrow 0$ .

The common Lipschitz bound and this anchor give uniform bounds on every compact interval. By Arzelà–Ascoli, every subsequence contains a further subsequence converging locally uniformly to a  $K$ -Lipschitz function  $g$ . Since  $g_n \rightarrow 0$  in  $L^2(\nu)$ ,  $g = 0$   $\nu$ -almost surely. Write  $d\nu = \varphi(x) dx$ . A Lipschitz function that vanishes almost everywhere on  $\{\varphi > 0\}$  has derivative zero at almost every Lebesgue density point of that set; hence  $g' = 0$  for  $\nu$ -almost every  $x$ .

Local uniform convergence of equi-Lipschitz functions implies weak-\* convergence  $g'_n \rightharpoonup g'$  in  $L^\infty_{\text{loc}}$ . Testing first against  $\varphi \mathbf{1}_{[-A,A]} \in L^1$  and then letting  $A \rightarrow \infty$  gives

$$a = \int g'_n d\nu \longrightarrow \int g' d\nu = 0,$$

a contradiction. Therefore the infimum in (7.22) is strictly positive.  $\square$

**Lemma 8.7** (Martingale triangular-array central limit criterion). *For each  $n$  let  $(X_{n,k}, \mathcal{F}_{n,k})_{1 \leq k \leq N_n}$  be a square-integrable martingale-difference array and set  $S_n = \sum_k X_{n,k}$ . Suppose*

$$V_n := \sum_k \mathbb{E}(X_{n,k}^2 \mid \mathcal{F}_{n,k-1}) \xrightarrow{\mathbb{P}} b^2 > 0 \quad (7.23)$$

and, for every  $\delta > 0$ ,

$$\sum_k \mathbb{E}[X_{n,k}^2 \mathbf{1}_{\{|X_{n,k}| > \delta\}} \mid \mathcal{F}_{n,k-1}] \xrightarrow{\mathbb{P}} 0. \quad (7.24)$$

If the left sides are uniformly integrable, then  $S_n \Rightarrow N(0, b^2)$ .

*Proof.* Fix  $t \in \mathbb{R}$ . Taylor's formula, conditional centering, and the inequality  $|e^{iu} - 1 - iu + u^2/2| \leq C_t |u|^2 \min\{|u|, 1\}$  give

$$\mathbb{E}(e^{itX_{n,k}} \mid \mathcal{F}_{n,k-1}) = 1 - \frac{t^2}{2} \mathbb{E}(X_{n,k}^2 \mid \mathcal{F}_{n,k-1}) + r_{n,k},$$

where the sum of  $|r_{n,k}|$  tends to zero in probability, and in  $L^1$  after truncation, by (7.24) and uniform integrability. Iterating conditional expectations for the characteristic function of  $S_n$ , and replacing each factor  $1 - u$  by  $e^{-u}$ , produces an additional error bounded by  $C \sum_k \mathbb{E}(X_{n,k}^2 \mid \mathcal{F}_{n,k-1})^2$ . The Lindeberg condition forces the largest conditional second moment to vanish in probability, so this error also tends to zero. Consequently

$$\mathbb{E}e^{itS_n} - \mathbb{E} \exp\left\{-\frac{t^2}{2} V_n\right\} \longrightarrow 0.$$

By (7.23) and bounded convergence the second term tends to  $e^{-t^2 b^2/2}$ . Lévy's continuity argument (obtained by Fourier inversion against smooth compactly supported test functions) then gives  $S_n \Rightarrow N(0, b^2)$ .  $\square$

**Lemma 8.8** (Aizenman–Wehr fluctuation lemma). *Under the hypotheses and notation of Lemma 8.5, if  $M \neq 0$ , then along the boxes  $\Lambda_L$ ,*

$$\frac{G_L}{|\Lambda_L|^{1/2}} \Rightarrow N(0, b^2) \quad \text{for some } b > 0. \quad (7.25)$$

*Proof.* Order  $\mathbb{Z}^d$  lexicographically and reveal the variables in  $\Lambda_L$  in that order. If  $D_{L,x}$  is the corresponding Doob difference, then

$$G_L = \sum_{x \in \Lambda_L} D_{L,x}. \quad (7.26)$$

For fixed past variables,  $D_{L,x}$  is a centered function  $g_{L,x}(\eta_x)$ . Differentiating under the conditional expectation and using (7.19) gives

$$g'_{L,x}(t) = \varepsilon \mathbb{E}(\tau_x \mid \mathcal{F}_{L,x-1}, \eta_x = t), \quad |g'_{L,x}| \leq |\varepsilon|B. \quad (7.27)$$

Its derivative averaged over  $t \sim \nu$  is

$$a_{L,x} = \varepsilon \mathbb{E}(\tau_x \mid \mathcal{F}_{L,x-1}). \quad (7.28)$$

Hence Lemma 8.6, conditionally on the past, yields

$$\mathbb{E}(D_{L,x}^2 \mid \mathcal{F}_{L,x-1}) \geq \Theta_\nu(a_{L,x}, |\varepsilon|B)^2. \quad (7.29)$$

The lexicographic past is translation covariant. For sites whose distance from  $\partial\Lambda_L$  tends to infinity, martingale convergence identifies  $a_{L,x}$  and the conditional second moment in (7.29) with translates of infinite-volume variables  $a_x$  and  $v_x$ . The iid disorder shift is ergodic, so the box averages of  $v_x$  converge to  $b^2 := \mathbb{E}v_0$ . Moreover  $\mathbb{E}a_0 = \varepsilon \mathbb{E}\tau_0 = \varepsilon M \neq 0$ ; therefore  $\mathbb{P}(a_0 \neq 0) > 0$ , and Lemma 8.6 applied on that event implies  $b^2 > 0$ . The boundary layer has  $o(|\Lambda_L|)$  sites, so

$$|\Lambda_L|^{-1} \sum_x \mathbb{E}(D_{L,x}^2 \mid \mathcal{F}_{L,x-1}) \xrightarrow{\mathbb{P}} b^2. \quad (7.30)$$

It remains only to verify Lindeberg. A centered  $K$ -Lipschitz function of one variable obeys

$$|g(t)| \leq K(|t| + \mathbb{E}|\eta_0|).$$

Thus (7.27) and  $\mathbb{E}\eta_0^2 < \infty$  imply, for every  $\delta > 0$ ,

$$\frac{1}{|\Lambda_L|} \sum_x \mathbb{E} \left[ D_{L,x}^2 \mathbf{1}_{\{|D_{L,x}| > \delta |\Lambda_L|^{1/2}\}} \right] \longrightarrow 0. \quad (7.31)$$

Markov's inequality upgrades (7.31) to the conditional Lindeberg condition in probability. Applying Lemma 8.7 to  $X_{L,x} = D_{L,x}/|\Lambda_L|^{1/2}$  proves (7.25).  $\square$

**Theorem 8.9** (Aizenman–Wehr rounding). *For  $d \leq 2$  in the classical finite-range setting above, quenched randomness with  $\varepsilon \neq 0$  and an absolutely continuous finite-variance disorder law eliminates a first-order discontinuity in the order parameter conjugate to  $h$ : the quenched pressure  $p(h)$  is differentiable.*

*Complete finite-range proof.* Assume for contradiction that  $M = p'_+(h) - p'_-(h) > 0$ .

Lemma 8.5 constructs a centered phase-comparison free energy  $G_L$  with the exact response derivative (7.19) and the deterministic surface estimate

$$|G_L| \leq C_1 L^{d-1}. \quad (7.32)$$

This is the rigorous form of the surface-energy side of the Imry–Ma comparison.



Disorder regularity enters at one precise place: Lemma 8.6 says that a nonzero average response cannot be carried by one-site functions of arbitrarily small variance. Together with the Doob decomposition this makes the bulk fluctuation variance strictly positive.

Lemma 8.8 and the preceding variance lower bound then give

$$G_L/|\Lambda_L|^{1/2} \Rightarrow N(0, b^2), \quad b > 0. \quad (7.33)$$

No independent “marginal  $d = 2$  fluctuation theorem” is being imported here; its martingale proof is the preceding two lemmas.

If  $d < 2$ , divide the surface estimate (7.32) by the bulk fluctuation scale. Then

$$\frac{|G_L|}{|\Lambda_L|^{1/2}} \leq CL^{d-1-d/2} = CL^{d/2-1} \rightarrow 0,$$

contradicting the nondegenerate Gaussian limit (7.33).

If  $d = 2$ , the same surface estimate gives the uniform compact-support bound

$$|G_L|/|\Lambda_L|^{1/2} \leq C_2. \quad (7.34)$$

Every weak subsequential limit of the left side is therefore supported on  $[-C_2, C_2]$ , whereas a Gaussian with  $b > 0$  assigns positive mass outside every bounded interval. This again contradicts (7.33). This compact-support contradiction is equivalent to the quadratic-moment-generating-function versus linear-surface-bound comparison in the original Aizenman–Wehr proof.

Since  $h$  was arbitrary, the quenched pressure is differentiable in the randomized parameter and the conjugate thermodynamic density has no first-order jump.

This proves the theorem under the clean finite-range hypotheses used in this volume. The original theorem also contains sharper variants at positive temperature and for particular ferromagnetic random-field/random-bond models; those extensions are not needed for this ledger row.  $\square$

*Remark 8.10* (Why this theorem belongs here). The Aizenman–Wehr theorem is not a high-temperature uniqueness theorem. It says that low-dimensional quenched disorder can destroy the thermodynamic discontinuity that would otherwise signal a first-order transition. It therefore complements, rather than duplicates, Dobrushin and Pirogov–Sinai theory.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 8.11** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 8.12** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 8.13** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 8.14** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 8.15** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

# Bibliography

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume VIII: The Ising Model

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# Scope and conventions

This volume treats the ferromagnetic Ising model after the general Gibbs formalism of Volume VII. Unless explicitly stated, couplings satisfy  $J_{xy} \geq 0$ , finite-volume graphs are finite, and infinite-volume limits are taken along regular exhaustions. Exact solution, Lee–Yang theory, planar Gibbs-state classification, sharpness, infrared bounds, and random-current methods are presented as distinct mechanisms so that their hypotheses are never conflated.



# Chapter 1

## Basic Structure and Graphical Expansions

The Ising model is the first place where several representations of the same measure become genuinely useful. We begin with three: lattice gas, high-temperature subgraphs, and low-temperature contours.

**Definition 1.1** (Finite-volume ferromagnetic Ising measure). Let  $G = (V, E)$  be finite, let  $J_e \geq 0$ , let  $h_x \in \mathbb{R}$ , and let  $\beta \geq 0$ . The *Ising configuration space* is  $\Omega_G = \{-1, +1\}^V$ . Its Hamiltonian and Gibbs measure are

$$H_G(\sigma) = - \sum_{xy \in E} J_{xy} \sigma_x \sigma_y - \sum_{x \in V} h_x \sigma_x, \quad \mu_{G, \beta}(\sigma) = \frac{e^{-\beta H_G(\sigma)}}{Z_{G, \beta}},$$

where  $Z_{G, \beta} = \sum_{\sigma \in \Omega_G} e^{-\beta H_G(\sigma)}$ . The adjective *ferromagnetic* means  $J_e \geq 0$  for every edge.

For a finite graph  $G = (V, E)$ , couplings  $J_e \geq 0$ , inverse temperature  $\beta$ , field  $h$ , and spins  $\sigma_x \in \{-1, +1\}$ , write

$$H(\sigma) = - \sum_{xy \in E} J_{xy} \sigma_x \sigma_y - h \sum_{x \in V} \sigma_x, \quad \mu(\sigma) = Z^{-1} e^{-\beta H(\sigma)}.$$

**Theorem 1.2** (Ising–lattice-gas correspondence). *With  $n_x = (1 + \sigma_x)/2 \in \{0, 1\}$ , the Ising Hamiltonian is, up to an additive constant, a lattice-gas Hamiltonian with attractive pair interaction and chemical potential. Conversely every nearest-neighbor two-state attractive lattice gas is an Ising model after an affine change of variables.*

*Proof.* Substitute  $\sigma_x = 2n_x - 1$ . Then

$$\sigma_x \sigma_y = 4n_x n_y - 2n_x - 2n_y + 1, \quad \sum_x \sigma_x = 2 \sum_x n_x - |V|.$$

Therefore the Ising Hamiltonian becomes, up to an additive constant,

$$-4 \sum_{xy} J_{xy} n_x n_y + \sum_x \left( 2 \sum_{y: y \sim x} J_{xy} - 2h \right) n_x.$$

Thus  $J_{xy} \geq 0$  becomes an attractive lattice-gas pair interaction. Conversely, inserting  $n_x = (1 + \sigma_x)/2$  into any two-state attractive nearest-neighbor lattice-gas Hamiltonian produces an Ising Hamiltonian with nonnegative couplings, again up to an additive constant.  $\square$

**Example 1.3** (Density and magnetization). The particle density  $\rho = \langle n_0 \rangle$  and magnetization  $m = \langle \sigma_0 \rangle$  satisfy  $m = 2\rho - 1$ . Thus liquid–gas coexistence and magnetic phase coexistence are literally the same two-state phenomenon in different coordinates.

**Theorem 1.4** (High-temperature graphical expansion). *At zero field, with  $t_e = \tanh(\beta J_e)$ ,*

$$Z_G = 2^{|V|} \prod_{e \in E} \cosh(\beta J_e) \sum_{\omega \subseteq E: \partial\omega = \emptyset} \prod_{e \in \omega} t_e,$$

where  $\partial\omega$  is the set of odd-degree vertices of the subgraph  $\omega$ . More generally the numerator of  $\langle \prod_{x \in A} \sigma_x \rangle$  is the same sum over  $\partial\omega = A$ .

*Proof.* For each edge  $e = xy$ ,

$$e^{\beta J_e \sigma_x \sigma_y} = \cosh(\beta J_e) (1 + t_e \sigma_x \sigma_y).$$

Expanding the product over edges gives

$$Z_G = \prod_e \cosh(\beta J_e) \sum_{\omega \subseteq E} \left( \prod_{e \in \omega} t_e \right) \sum_{\sigma} \prod_x \sigma_x^{\deg_{\omega}(x)}.$$

The spin sum vanishes if any  $\deg_{\omega}(x)$  is odd and equals  $2^{|V|}$  otherwise, which proves the partition-function identity. With an insertion  $\prod_{x \in A} \sigma_x$ , nonvanishing requires  $\deg_{\omega}(x) + \mathbf{1}_A(x)$  even for every  $x$ , equivalently  $\partial\omega = A$ , proving the correlation formula.  $\square$

**Theorem 1.5** (Low-temperature contour expansion). *For the nearest-neighbor Ising model on a planar finite region with plus boundary condition, every spin configuration determines a family of dual contours separating + and – clusters. Relative to the all-plus configuration, a contour family  $\Gamma$  has weight  $\exp[-2\beta J|\Gamma|]$  on the square lattice, subject to the usual even-incidence constraint at dual vertices.*

*Proof.* Fix the all-plus configuration as reference. For every primal edge  $e = xy$ , changing from aligned spins to  $\sigma_x \neq \sigma_y$  changes the bond energy from  $-J$  to  $+J$ , hence costs exactly  $2J$ . On a planar graph draw the dual edge  $e^*$  precisely when  $\sigma_x \neq \sigma_y$ . Around every interior dual vertex the number of such edges is even: traversing the primal faces around that vertex, the spin can change sign only an even number of times before returning to its starting value. Thus the disagreement set is an even dual subgraph and decomposes into closed contours (with the standard boundary modifications when the region itself has boundary).

If  $\Gamma$  denotes the resulting contour family, then

$$H(\sigma) - H(+) = 2J|\Gamma|,$$

where  $|\Gamma|$  is the total number of dual contour edges. Consequently

$$\frac{\mu_{\Lambda}^+(\sigma)}{\mu_{\Lambda}^+(+)} = \exp[-\beta(H(\sigma) - H(+))] = e^{-2\beta J|\Gamma|}.$$

Conversely, an even contour family compatible with plus boundary condition determines the spin configuration uniquely: assign + to the boundary component and flip the spin each time a path crosses a contour. Even incidence makes the result path independent. This gives both the contour bijection and its stated weight.  $\square$

**Theorem 1.6** (Peierls theorem). *For nearest-neighbor ferromagnetic Ising on  $\mathbb{Z}^d$ ,  $d \geq 2$ , the plus-state spontaneous magnetization is strictly positive for all sufficiently large  $\beta$ .*

*Proof.* Work first in a finite box with plus boundary condition. If  $\sigma_0 = -1$ , the minus cluster containing the origin is finite and its edge boundary is a closed Peierls contour in  $d = 2$ , or a connected family of dual  $(d - 1)$ -plaquettes (a Peierls surface) in general dimension. Flipping all spins inside such a boundary removes it and lowers the energy by  $2J|\Gamma|$ . The flip is injective once  $\Gamma$  is specified, hence

$$\mu_{\Lambda}^+(\Gamma \text{ occurs}) \leq e^{-2\beta J|\Gamma|}. \quad (1.4)$$

There is a dimension-dependent constant  $C_d < \infty$  such that the number of connected dual plaquette sets of size  $n$  containing a fixed plaquette is at most  $C_d^n$ : explore the set in a fixed order, and at each new plaquette there are only finitely many adjacent choices. Any surface surrounding the origin has a plaquette whose distance from the origin is at most a constant times its size, so the number of size- $n$  surrounding surfaces is at most a polynomial factor times  $C_d^n$ . Thus

$$\mu_{\Lambda}^+(\sigma_0 = -1) \leq \sum_{n \geq n_0} C'_d n^d C_d^n e^{-2\beta Jn}.$$

For sufficiently large  $\beta$  this sum is strictly less than  $1/2$ , uniformly in  $\Lambda$ . Passing to the plus-state limit gives  $\mu^+(\sigma_0 = +1) > 1/2$ , so  $\langle \sigma_0 \rangle_+ > 0$ .  $\square$

## Chapter 2

# Correlation Inequalities

Ferromagnetism is encoded not only in positive couplings but in monotonicity of whole families of observables. We state the inequalities in finite volume first, where no limiting subtlety is present.

**Theorem 2.1** (Griffiths inequalities). *For a finite ferromagnetic Ising model with nonnegative external field, every spin product has nonnegative expectation, and for spin products  $\sigma_A, \sigma_B$ ,*

$$\langle \sigma_A \sigma_B \rangle - \langle \sigma_A \rangle \langle \sigma_B \rangle \geq 0.$$

*Proof.* Adjoin a ghost vertex  $g$  and replace the field  $h_x \geq 0$  by the ferromagnetic edge  $xg$  of strength  $h_x$ . Fix  $\sigma_g = +1$ . For a current  $n \in \mathbb{N}^{E \cup E_g}$  set

$$w(n) = \prod_e \frac{K_e^{n_e}}{n_e!}, \quad K_{xy} = \beta J_{xy}, \quad K_{xg} = \beta h_x,$$

and let  $\partial n$  be the set of vertices of odd current degree. If  $A \subseteq V$ , put

$$\hat{A} = \begin{cases} A, & |A| \text{ even,} \\ A \cup \{g\}, & |A| \text{ odd,} \end{cases} \quad Z_A^{\text{cur}} = \sum_{\partial n = \hat{A}} w(n).$$

Expanding every exponential in its power series and summing the spins first shows

$$\langle \sigma_A \rangle = \frac{Z_A^{\text{cur}}}{Z_{\emptyset}^{\text{cur}}}. \quad (2.1)$$

All current weights are nonnegative, so the first Griffiths inequality follows immediately from (2.1).

For the second inequality use the switching identity. For even source sets  $C, D$  and any fixed total current  $m = n_1 + n_2$ , allocations of the  $m_e$  distinguishable edge copies between  $n_1$  and  $n_2$  have weight  $w(m) \prod_e \binom{m_e}{n_{1,e}}$ . Toggling the ownership of a chosen family of paths pairing the vertices of  $D$  changes the first source set by  $D$  and is an involution. Hence

$$Z_C^{\text{cur}} Z_D^{\text{cur}} = \sum_{\substack{\partial n_1 = C \triangle D \\ \partial n_2 = \emptyset}} w(n_1) w(n_2) \mathbf{1}_{\mathcal{F}_D(n_1 + n_2)}, \quad (2.2)$$

where  $\mathcal{F}_D(m)$  is the event that every connected component of the support of  $m$  contains an even number of vertices of  $D$  (equivalently, the support contains a subgraph with boundary  $D$ ). Dropping the indicator gives

$$Z_C^{\text{cur}} Z_D^{\text{cur}} \leq Z_{C \triangle D}^{\text{cur}} Z_{\emptyset}^{\text{cur}}.$$

Taking  $C = \widehat{A}$ ,  $D = \widehat{B}$  and observing that  $\widehat{A\Delta B} = \widehat{A\Delta B}$  yields

$$\langle \sigma_A \rangle \langle \sigma_B \rangle \leq \langle \sigma_{A\Delta B} \rangle = \langle \sigma_A \sigma_B \rangle.$$

This is the second Griffiths inequality.  $\square$

**Theorem 2.2** (FKG inequality). *For ferromagnetic pair interactions and any boundary condition, increasing functions  $F, G$  of the spins satisfy  $\langle FG \rangle \geq \langle F \rangle \langle G \rangle$  whenever the finite-volume measure satisfies the lattice condition.*

*Proof.* We prove the finite-lattice statement by induction on the number of spins. Let  $w > 0$  satisfy the lattice condition

$$w(\sigma \vee \tau)w(\sigma \wedge \tau) \geq w(\sigma)w(\tau). \quad (2.\text{FKG})$$

The one-spin case is immediate. Write an  $n$ -spin configuration as  $(x, s)$ ,  $s \in \{-1, +1\}$ . The conditional weights  $w_s(x) = w(x, s)$  still satisfy (2.FKG), so by induction each conditional law is positively associated.

Moreover, for  $x \leq y$ , (2.FKG) gives

$$\frac{w(y, +)}{w(y, -)} \geq \frac{w(x, +)}{w(x, -)}.$$

Thus the likelihood ratio  $r(x) = w(x, +)/w(x, -)$  is increasing. If  $f$  is increasing, positive association under the  $s = -$  conditional law yields

$$\mathbb{E}_-[fr] \geq \mathbb{E}_-[f]\mathbb{E}_-[r],$$

and hence  $\mathbb{E}_+[f] = \mathbb{E}_-[fr]/\mathbb{E}_-[r] \geq \mathbb{E}_-[f]$ . Conditioning the last spin to  $+$  therefore stochastically dominates conditioning it to  $-$ .

For increasing  $F, G$ , total covariance gives

$$\text{Cov}(F, G) = \mathbb{E}[\text{Cov}(F, G \mid s)] + \text{Cov}(\mathbb{E}[F \mid s], \mathbb{E}[G \mid s]).$$

The first term is nonnegative by induction; the second is nonnegative because both conditional expectations increase with the binary variable  $s$ . Therefore  $\langle FG \rangle \geq \langle F \rangle \langle G \rangle$ . For ferromagnetic pair interactions, (2.FKG) follows directly by summing the nonnegative mixed discrete second differences of the edge energies; fixed boundary spins merely contribute one-site fields and do not alter the inequality.  $\square$

**Lemma 2.3** (Duplicate-variable inequality). *Let  $\sigma$  and  $s$  be independent copies of a finite ferromagnetic Ising model with  $h_x \geq 0$ , and put*

$$q_x = \frac{\sigma_x - s_x}{2}, \quad t_x = \frac{\sigma_x + s_x}{2}.$$

*For distinct  $i, j, k$ ,*

$$\langle q_i q_j t_k \rangle \leq \langle q_i q_j \rangle \langle t_k \rangle. \quad (2.3)$$

*Proof.* Let  $A = \{x : q_x \neq 0\}$ . Since  $q_x t_x = 0$  and  $q_x^2 + t_x^2 = 1$ , the doubled Hamiltonian decomposes exactly as

$$H(\sigma) + H(s) = H_A(q; 0, 2J) + H_{A^c}(t; 2h, 2J).$$

Conditioned on  $A$ , the  $q$ - and  $t$ -spins are therefore independent Ising systems, the first on  $A$  at zero field and the second on  $A^c$  at field  $2h$ . Writing

$$f(A) = \mathbf{1}_{\{i,j \subset A\}} \langle q_i q_j \rangle_{A,0,2J}, \quad g(A) = \mathbf{1}_{\{k \notin A\}} \langle t_k \rangle_{A^c,2h,2J},$$

we have  $\langle q_i q_j t_k \rangle = \mathbf{E}_P[f(A)g(A)]$ , where

$$P(A) = \frac{Z_A(0, 2J)Z_{A^c}(2h, 2J)}{Z(h, J)^2}. \quad (2.4)$$

(The partition functions in (2.4) include the zero entries outside their indicated vertex sets.)

We now verify the three monotonicities needed for the FKG step. Adding a vertex and its nonnegative incident couplings to a zero-field ferromagnet can only increase  $\langle q_i q_j \rangle$ : turn on the new couplings one at a time and differentiate; each derivative is a truncated correlation of spin products and is nonnegative by the second Griffiths inequality just proved. Thus  $f$  is increasing in  $A$ . The same interpolation in the positive-field  $t$ -system shows that enlarging its vertex set can only increase  $\langle t_k \rangle$ ; hence  $g(A)$  is decreasing in  $A$ .

Finally  $P$  satisfies the lattice condition. Indeed, for  $A, B \subseteq V$  write  $C = A \cap B$ ,  $D = A \setminus B$ ,  $E = B \setminus A$ . Starting from the product of the systems on  $C, D, E$  with all cross couplings deleted, turn on the nonnegative  $C$ - $D$ ,  $C$ - $E$ , and  $D$ - $E$  couplings. The derivative of each logarithmic partition-function ratio appearing in

$$\log \frac{P(A \cup B)P(A \cap B)}{P(A)P(B)}$$

is a covariance of ferromagnetic spin products, hence is nonnegative by the second Griffiths inequality. The ratio equals 1 when all cross couplings are zero, so

$$P(A \cup B)P(A \cap B) \geq P(A)P(B). \quad (2.5)$$

The finite FKG theorem applied to (2.5), with  $f$  increasing and  $g$  decreasing, gives  $\mathbf{E}_P[fg] \leq \mathbf{E}_P[f]\mathbf{E}_P[g]$ , which is (2.3).  $\square$

**Theorem 2.4** (GHS inequality). *For a ferromagnetic finite Ising model in nonnegative field, the third mixed derivatives of  $\log Z$  with respect to external fields are nonpositive. In particular the magnetization is concave as a function of a homogeneous positive field.*

*Proof.* First take three distinct sites  $i, j, k$ . In the doubled system of Lemma 2.3, direct expansion and independence of the two copies give the exact identity

$$\begin{aligned} 4(\langle q_i q_j t_k \rangle - \langle q_i q_j \rangle \langle t_k \rangle) &= \langle \sigma_i \sigma_j \sigma_k \rangle - \langle \sigma_i \sigma_j \rangle \langle \sigma_k \rangle \\ &\quad - \langle \sigma_i \sigma_k \rangle \langle \sigma_j \rangle - \langle \sigma_j \sigma_k \rangle \langle \sigma_i \rangle \\ &\quad + 2\langle \sigma_i \rangle \langle \sigma_j \rangle \langle \sigma_k \rangle. \end{aligned} \quad (2.6)$$

The right-hand side is the third Ursell function  $\partial_{h_i} \partial_{h_j} \partial_{h_k} \log Z$ . By Lemma 2.3, the left-hand side is nonpositive, proving GHS for distinct sites.

Repeated derivatives follow by the standard cloning argument, which we spell out. Replace a site that occurs twice or three times by that many clones, divide its external field equally among them, give each clone the same couplings to all other sites divided by the number of clones, and couple the clones to one another with ferromagnetic strength  $M$ . The distinct-site GHS inequality applies for every finite  $M$ . As  $M \rightarrow \infty$ , configurations in which the clones disagree are suppressed by a factor

$O(e^{-2M})$ ; the Gibbs measure therefore converges to the original model with the clones identified, and the corresponding mixed derivatives converge to the repeated derivative. The inequality survives the limit.

For a homogeneous field  $h$ , the magnetization  $m(h) = |V|^{-1} \partial_h \log Z$  satisfies

$$m''(h) = \frac{1}{|V|} \sum_{i,j,k} \partial_{h_i} \partial_{h_j} \partial_{h_k} \log Z \leq 0,$$

so  $m$  is concave on  $h \geq 0$ . □

**Theorem 2.5** (Simon–Lieb inequality). *Let  $S$  be a finite vertex set containing  $x$  but not  $y$ . For a ferromagnetic Ising model at zero field,*

$$\langle \sigma_x \sigma_y \rangle \leq \sum_{u \in S, v \notin S} \tanh(\beta J_{uv}) \langle \sigma_x \sigma_u \rangle_S \langle \sigma_v \sigma_y \rangle,$$

*with the first correlation computed in the induced system on  $S$  with free boundary.*

*Proof.* Use the zero-field high-temperature expansion from Chapter 1. Write  $t_{uv} = \tanh(\beta J_{uv})$ . The numerator of  $\langle \sigma_x \sigma_y \rangle$  is the total weight of subgraphs  $\omega$  with odd boundary  $\{x, y\}$ . Every such  $\omega$  contains an  $x$ – $y$  path. Follow, in a fixed deterministic ordering of edges, the first such path and let  $(u, v)$  be the first edge by which it exits  $S$ . Removing that distinguished edge and splitting the remaining path at  $u, v$  leaves an  $x$ – $u$  source subgraph inside  $S$  and a  $v$ – $y$  source subgraph in the full graph; all remaining even components may be assigned to either side. Since all weights are nonnegative, forgetting the requirement that the chosen crossing be the first one can only increase the sum. Therefore the unnormalized two-point numerator is bounded by

$$\sum_{u \in S, v \notin S} t_{uv} N_S(x, u) N_G(v, y) \frac{Z_G}{Z_S},$$

where  $N_S(x, u)/Z_S = \langle \sigma_x \sigma_u \rangle_S$  and  $N_G(v, y)/Z_G = \langle \sigma_v \sigma_y \rangle$ . Dividing by  $Z_G$  gives exactly

$$\langle \sigma_x \sigma_y \rangle \leq \sum_{u \in S, v \notin S} t_{uv} \langle \sigma_x \sigma_u \rangle_S \langle \sigma_v \sigma_y \rangle.$$

□

## Chapter 3

# Uniqueness, Sharpness, and Exponential Decay

**Theorem 3.1** (High-temperature uniqueness). *If the Dobrushin interdependence constant is strictly less than one (in particular for sufficiently small  $\beta$  in a finite-range Ising model), there is a unique infinite-volume Gibbs state and correlations decay exponentially under the quantitative Dobrushin criterion.*

*Proof.* For a specification  $\gamma$  define the single-site influence coefficients

$$c_{xy} = \sup_{\eta, \eta': \eta_z = \eta'_z \ (z \neq y)} \|\gamma_x(\cdot | \eta) - \gamma_x(\cdot | \eta')\|_{\text{TV}}, \quad c = \sup_x \sum_y c_{xy} < 1.$$

Couple two heat-bath updates at  $x$  maximally. If their boundary configurations differ on a set  $D$ , the probability that the updated spins disagree is at most  $\sum_{y \in D} c_{xy}$ . Iterating the coupled single-site updates gives, for the disagreement probabilities  $d_x^{(n)}$ ,

$$d^{(n+1)} \leq C d^{(n)}, \quad C = (c_{xy}),$$

and hence  $\|d^{(n)}\|_\infty \leq c^n \|d^{(0)}\|_\infty$ . Two infinite-volume DLR measures can be coupled through increasing finite volumes; letting the number of update sweeps tend to infinity forces every local disagreement probability to zero. Thus the Gibbs state is unique.

If the interaction has range  $R$ , boundary information at distance  $r$  from a local observable must pass through at least  $\lfloor r/R \rfloor$  influence steps. The same contraction therefore yields

$$|\text{Cov}(F, G)| \leq 2\|F\|_\infty \|G\|_\infty |\text{supp } F| c^{\lfloor d(\text{supp } F, \text{supp } G)/R \rfloor},$$

which is exponential decay. For the Ising single-site kernel one has the explicit bound  $c_{xy} \leq \tanh(\beta |J_{xy}|)$ , so the criterion holds for all sufficiently small  $\beta$  in every finite-range model.  $\square$

## The finite-set criterion and current differential inequality

For a finite set  $S \ni 0$  define

$$\varphi_\beta(S) = \sum_{x \in S} \sum_{y \notin S} \tanh(\beta J_{xy}) \langle \sigma_0 \sigma_x \rangle_{S, \beta, 0},$$



where the correlation in  $S$  has free boundary condition. Put

$$\tilde{\beta}_c = \sup\{\beta : \text{there is a finite } S \ni 0 \text{ with } \varphi_\beta(S) < 1\}.$$

The proof below shows directly that  $\tilde{\beta}_c = \beta_c$ .

**Lemma 3.2** (Switching lemma in the form used below). *Let  $\Lambda$  be finite and adjoin a ghost vertex  $g$ . Give a current  $n$  on the complete interaction graph of  $\Lambda \cup \{g\}$  weight*

$$w(n) = \prod_{\{x,y\}} \frac{(\beta J_{xy})^{n_{xy}}}{n_{xy}!}, \quad J_{xg} = h/\beta,$$

and write  $\partial n$  for the odd-degree vertices. If  $A$  is even and  $u, v \in \Lambda \cup \{g\}$ , then for every function  $F$  of  $n_1 + n_2$ ,

$$\sum_{\substack{\partial n_1 = A \Delta \{u,v\} \\ \partial n_2 = \{u,v\}}} F(n_1 + n_2) w(n_1) w(n_2) = \sum_{\substack{\partial n_1 = A \\ \partial n_2 = \emptyset}} F(n_1 + n_2) w(n_1) w(n_2) \mathbf{1}_{\{u \leftrightarrow v \text{ in } n_1 + n_2\}}.$$

*Proof.* Fix the total current  $m = n_1 + n_2$ . Write every edge  $e$  of multiplicity  $m_e$  as  $m_e$  distinguishable parallel copies. An allocation of those copies to current 1 determines  $n_1$  and contributes the combinatorial factor  $\prod_e \binom{m_e}{n_{1,e}}$ ; after multiplication by  $w(m) = \prod_e (\beta J_e)^{m_e} / m_e!$  this is exactly  $w(n_1) w(n_2)$ . Hence it is enough to construct a weight-preserving bijection of allocations for each fixed  $m$ .

If  $u$  and  $v$  are connected in the support of  $m$ , choose once and for all a path of edge copies joining them and toggle, along this path, which current owns each chosen copy. At every internal vertex two incidences are toggled, whereas at  $u$  and  $v$  one incidence is toggled. Thus the source set of current 1 is changed by  $\{u, v\}$  and the source set of current 2 is changed by the same pair. Toggling twice is the identity. If  $u$  and  $v$  are not connected no allocation with the source pattern on the left exists: every connected component of the support contains an even number of sources of each current. Summing the fixed- $m$  bijection proves the identity.  $\square$

**Lemma 3.3** (Finite-volume magnetization inequality). *Let  $\Lambda \subseteq V$ ,  $h > 0$ , and  $M_\Lambda(\beta, h) = \langle \sigma_0 \rangle_{\Lambda, \beta, h}$ . Set*

$$c_\Lambda(\beta, h) = \inf_{y \in \Lambda} \frac{M_\Lambda(\beta, h)}{\langle \sigma_y \rangle_{\Lambda, \beta, h}}$$

and

$$\varepsilon_\Lambda(\beta, h) = \sum_{x \in \Lambda} \sum_{y \notin \Lambda} J_{xy} (\langle \sigma_0 \sigma_x \rangle_{\Lambda, \beta, h} - M_\Lambda(\beta, h) \langle \sigma_x \rangle_{\Lambda, \beta, h}).$$

Then

$$\frac{d}{d\beta} M_\Lambda(\beta, h)^2 \geq 2c_\Lambda(\beta, h) \left[ \frac{1}{\beta} \inf_{S \ni 0} \varphi_\beta(S) (1 - M_\Lambda(\beta, h)^2) - \varepsilon_\Lambda(\beta, h) \right]. \quad (3.1)$$

The infimum is over finite  $S \subset V$  containing 0.

*Proof.* Let  $Z = \sum_{\partial n = \emptyset} w(n)$  for currents on  $\Lambda \cup \{g\}$ . Differentiation of the finite-volume Gibbs sum gives

$$\partial_\beta M_\Lambda = \sum_{\{x,y\} \subset \Lambda} J_{xy} (\langle \sigma_0 \sigma_x \sigma_y \rangle - M_\Lambda \langle \sigma_x \sigma_y \rangle).$$

Use the current representation of the correlations and Lemma 3.2. The disconnected part after switching is

$$\partial_\beta M_\Lambda = \frac{1}{Z^2} \sum_{x,y \in \Lambda} J_{xy} \Delta_{x,y},$$

where  $\Delta_{x,y}$  is the total weight of pairs  $(n_1, n_2)$  with  $\partial n_1 = \{0, g\} \triangle \{x, y\}$ ,  $\partial n_2 = \emptyset$ , such that in  $n_1 + n_2$  the vertex 0 is connected to  $x$ ,  $y$  is connected to  $g$ , and 0 is not connected to  $g$ . The two possible orientations of the pivotal pair have already been combined in the ordered sum over  $x, y$ .

We record explicitly the two decompositions which are the heart of the argument. First let  $S_0$  be the set of vertices not connected to 0 in  $n_1 + n_2$ . On the event  $S_0 = S$ , currents have no positive edge crossing from  $S$  to its complement. Their weights therefore factor into the two regions. Removing the sources  $y, g$  in the  $S$ -part and applying the switching lemma there gives

$$\Delta_{x,y} \geq \frac{1}{\langle \sigma_y \rangle_{\Lambda, \beta, h}} \sum_{\substack{\partial n_1 = \{0\} \triangle \{x\} \\ \partial n_2 = \emptyset}} w(n_1) w(n_2) \mathbf{1}_{\{y \leftrightarrow g, 0 \nleftrightarrow g\}}. \quad (3.2)$$

Now let  $S_g$  be the set of vertices not connected to  $g$ . On  $S_g = S$  one has  $0, x \in S$  and  $y \notin S$ . Factor the currents again, this time across  $S$  and its complement. The current identity in the  $S$ -part removes the sources  $0, x$  and contributes exactly the zero-field finite-volume two-point function  $\langle \sigma_0 \sigma_x \rangle_{S, \beta, 0}$ . Thus

$$\Delta_{x,y} \geq \frac{1}{\langle \sigma_y \rangle_{\Lambda, \beta, h}} \sum_{S \subset \Lambda: 0, x \in S, y \notin S} \langle \sigma_0 \sigma_x \rangle_{S, \beta, 0} \sum_{\partial n_1 = \partial n_2 = \emptyset} w(n_1) w(n_2) \mathbf{1}_{\{S_g = S\}}. \quad (3.3)$$

Multiplying by  $2M_\Lambda$  and summing  $J_{xy} \Delta_{x,y}$ , the ratio  $M_\Lambda / \langle \sigma_y \rangle$  is bounded below by  $c_\Lambda$ . For each realized  $S_g = S$ ,

$$\sum_{x \in S, y \notin S} J_{xy} \langle \sigma_0 \sigma_x \rangle_{S, \beta, 0} \geq \frac{1}{\beta} \varphi_\beta(S) - \sum_{x \in S, y \notin \Lambda} J_{xy} \langle \sigma_0 \sigma_x \rangle_{S, \beta, 0}, \quad (3.4)$$

because  $\tanh t \leq t$ . Averaging the first term in (3.4) over the sourceless pair of currents and switching once more gives  $1 - M_\Lambda^2$ : the complementary event is precisely that 0 connects to  $g$  in the sum current. Reversing the second decomposition in (3.3) turns the exterior-edge term into  $\varepsilon_\Lambda(\beta, h)$ . Substitution gives (3.1).  $\square$

**Theorem 3.4** (Sharp Ising transition on transitive graphs). *For the ferromagnetic Ising model on a locally finite transitive graph with translation-invariant summable couplings, susceptibility is finite below  $\beta_c$ , while above  $\beta_c$  the plus magnetization obeys*

$$\langle \sigma_0 \rangle_\beta^+ \geq \sqrt{1 - \beta_c^2 / \beta^2}.$$

*For finite-range models, subcritical two-point correlations decay exponentially in the graph distance.*

*Proof.* If  $\beta < \tilde{\beta}_c$ , choose finite  $S \ni 0$  with  $q = \varphi_\beta(S) < 1$ . The Simon–Lieb inequality from SM-08.02.04 gives, for  $z \notin S$ ,

$$\langle \sigma_0 \sigma_z \rangle_\beta^+ \leq \sum_{x \in S, y \notin S} \tanh(\beta J_{xy}) \langle \sigma_0 \sigma_x \rangle_{S, \beta, 0} \langle \sigma_y \sigma_z \rangle_\beta^+. \quad (3.5)$$

For a finite set  $A$ , sum (3.5) over  $z \in A \setminus S$  and use transitivity to bound the translated inner sums by the maximal finite-volume susceptibility  $X(A)$ . This gives  $X(A) \leq |S| + qX(A)$ , hence  $X(A) \leq |S|/(1 - q)$  uniformly in  $A$ . Monotone convergence yields finite susceptibility.

For finite-range interactions choose  $L$  so that  $S \subset B_{L-R}$ . Applying (3.5) to a point at distance at least  $n$  shows that each passage through one translated  $L$ -block costs at most the factor  $q$ . After at least  $\lfloor n/L \rfloor$  passages,

$$\langle \sigma_0 \sigma_z \rangle_\beta^+ \leq q^{\lfloor d(0,z)/L \rfloor},$$

up to a bounded initial prefactor. This is exponential decay.

If  $\beta > \tilde{\beta}_c$ , then by definition  $\varphi_\beta(S) \geq 1$  for every finite  $S \ni 0$ . Apply Lemma 3.3 to an exhaustion  $\Lambda_n$ . For fixed  $h > 0$ , transitivity and Griffiths monotonicity give  $c_{\Lambda_n}(\beta, h) \rightarrow 1$ . The GHS inequality implies

$$\sum_{x \in \Lambda_n} (\langle \sigma_0 \sigma_x \rangle - \langle \sigma_0 \rangle \langle \sigma_x \rangle) = \partial_h \langle \sigma_0 \rangle \leq h^{-1},$$

which, together with summability of the couplings, makes the boundary error  $\varepsilon_{\Lambda_n}(\beta, h)$  tend to zero. Fatou's lemma therefore gives for  $M(\beta, h) = \langle \sigma_0 \rangle_{\beta, h}$

$$\frac{d}{d\beta} M(\beta, h)^2 \geq \frac{2}{\beta} (1 - M(\beta, h)^2) \quad (3.6)$$

for a.e.  $\beta > \tilde{\beta}_c$ .

Integrating (3.6) from  $\beta_1 > \tilde{\beta}_c$  to  $\beta$  gives

$$1 - M(\beta, h)^2 \leq \left( \frac{\beta_1}{\beta} \right)^2 (1 - M(\beta_1, h)^2) \leq \left( \frac{\beta_1}{\beta} \right)^2.$$

Let  $\beta_1 \downarrow \tilde{\beta}_c$  and then  $h \downarrow 0$  to obtain

$$\langle \sigma_0 \rangle_\beta^+ \geq \sqrt{1 - \tilde{\beta}_c^2 / \beta^2}.$$

Thus every  $\beta > \tilde{\beta}_c$  is supercritical, so  $\beta_c \leq \tilde{\beta}_c$ .

The subcritical susceptibility bound implies zero spontaneous magnetization for  $\beta < \tilde{\beta}_c$ , hence  $\beta_c \geq \tilde{\beta}_c$ . Therefore  $\tilde{\beta}_c = \beta_c$ , and the displayed lower bound has the claimed critical parameter.

Every ingredient in this argument is already owned in the present volume: the thermodynamic limit, the switching bijection proved above, Griffiths/GHS, and the Simon–Lieb inequality. No theorem from Volume IX is used. This is the Duminil–Copin–Tassion sharpness mechanism in the exact transitive-graph scope stated here.  $\square$

**Theorem 3.5** (Subcritical exponential decay). *For finite-range ferromagnetic Ising models below  $\beta_c$ , the two-point function decays exponentially in distance.*

*Proof.* The sharpness proof above establishes that for every  $\beta < \beta_c$  there is a finite set  $S \ni 0$  with

$$q := \varphi_\beta(S) < 1.$$

Let  $L$  exceed the diameter of  $S$ . Apply the Simon–Lieb inequality across the translate of  $S$  surrounding the current endpoint. Summing the crossing weights gives precisely the factor  $q$ , so for  $|x| > L$ ,

$$\langle \sigma_0 \sigma_x \rangle_\beta \leq q \max_{|y-x| \leq L} \langle \sigma_0 \sigma_y \rangle_\beta. \quad (3.8)$$

Iterate (3.8) through successive shells of width  $L$ . After  $n = \lfloor |x|/L \rfloor$  steps the endpoint lies in a fixed finite ball, on which correlations are at most one. Hence

$$\langle \sigma_0 \sigma_x \rangle_\beta \leq q^{\lfloor |x|/L \rfloor} \leq q^{-1} \exp \left[ -\frac{|\log q|}{L} |x| \right].$$

Thus the two-point function decays exponentially throughout the subcritical regime.  $\square$

## Chapter 4

# Planar Duality and the Critical Temperature

### The FK-Ising representation used locally in this chapter

We record the elementary FK–Ising correspondence and planar duality identities needed in this chapter. Given an Ising configuration at zero field, open each equal-spin edge independently with probability

$$p = 1 - e^{-2K}.$$

Conversely, for a random-cluster configuration with cluster weight  $q = 2$ , assign an independent  $\pm 1$  spin to each open cluster. Summing the joint weight in either order gives respectively the Ising and FK-Ising marginals. Thus

$$\langle \sigma_x \sigma_y \rangle_{\text{Ising}} = \phi_{p,2}(x \leftrightarrow y).$$

For a planar FK configuration, a dual edge is open exactly when its primal edge is closed. Euler’s formula gives the dual parameter relation

$$\frac{p^*}{1 - p^*} = 2 \frac{1 - p}{p}.$$

These identities are reproved in full generality in Volume IX. They do not by themselves imply criticality at the self-dual point; that additional planar theorem is isolated explicitly below.

**Theorem 4.1** (Kramers–Wannier duality). *For the zero-field square-lattice Ising model with coupling  $K = \beta J$ , the high-temperature expansion on a planar graph is dual to the low-temperature contour expansion on the dual graph when*

$$e^{-2K^*} = \tanh K, \quad \text{equivalently} \quad \sinh(2K) \sinh(2K^*) = 1,$$

*up to explicit boundary and normalization factors.*

*Proof.* The high-temperature expansion gives, apart from the factor  $2^{|V|} \prod_e \cosh K_e$ , a sum over even primal subgraphs  $\omega$  with weight  $\prod_{e \in \omega} \tanh K_e$ . On a planar graph an even primal subgraph is the boundary (modulo the prescribed boundary homology class) of a collection of dual faces. Interpret those faces as the  $-$  droplets of a dual Ising configuration with plus exterior spin. Its low-temperature contours are exactly the edges of  $\omega$  and have weight  $\prod_{e^* \in \omega^*} e^{-2K_{e^*}^*}$ . The two graphical sums therefore agree term by term when

$$e^{-2K_{e^*}^*} = \tanh K_e.$$

The omitted factors depend only on the graph, the couplings and the chosen boundary sector, so they give the advertised normalization factors. Finally,

$$e^{-2K^*} = \tanh K \iff e^{2K^*} = \coth K \iff \sinh(2K) \sinh(2K^*) = 1,$$

which is the symmetric Kramers–Wannier relation.  $\square$

**Theorem 4.2** (Self-dual inverse temperature). *The positive self-dual point satisfies  $\sinh(2K_c) = 1$ , hence*

$$K_{sd} = \frac{1}{2} \log(1 + \sqrt{2}).$$

*Proof.* At a positive self-dual point  $K^* = K$ , so Kramers–Wannier duality gives  $\sinh^2(2K) = 1$ . Since  $K > 0$ ,  $\sinh(2K) = 1$ . Therefore

$$e^{2K} - e^{-2K} = 2, \quad y := e^{2K} > 0, \quad y^2 - 2y - 1 = 0.$$

The positive root is  $y = 1 + \sqrt{2}$ , hence  $K = \frac{1}{2} \log(1 + \sqrt{2})$ .  $\square$

**Lemma 4.3** (FK–Ising infinite-cluster uniqueness). *For  $0 < p < 1$ , the translation-invariant free and wired  $q = 2$  random-cluster states on  $\mathbb{Z}^2$  have almost surely at most one infinite open cluster.*

*Proof.* The FK single-edge conditional open probability is either  $p$  or  $p/(p + 2(1 - p))$ , so the infinite-volume states have finite energy. Decompose a translation-invariant state into ergodic components. On a component the number of infinite clusters is a constant in  $\{0, 1, 2, \dots, \infty\}$ . A finite constant  $k \geq 2$  is impossible: with positive probability a large box meets all  $k$  clusters, and finite energy allows us to open a finite tree inside the box joining them, contradicting constancy. Infinitely many infinite clusters are also impossible. A finite modification can then create a trifurcation vertex with positive probability; translation invariance would give a positive density of trifurcations, while distinct branches from them must leave a large box through distinct boundary edges. The expected number of trifurcations is therefore of volume order and at the same time at most boundary order, contradicting amenability of  $\mathbb{Z}^2$ . Hence each ergodic component has zero or one infinite cluster, and so does the original state almost surely.  $\square$

**Lemma 4.4** (Free FK–Ising percolation above the wired threshold). *Let*

$$p_c^{\text{FK}} := \inf\{p : \phi_{p,2}^1(0 \leftrightarrow \infty) > 0\}.$$

*For every  $p > p_c^{\text{FK}}$  the free state  $\phi_{p,2}^0$  has an infinite cluster almost surely.*

*Proof.* Put  $t = \log(p/(1 - p))$  and, in a square  $\Lambda_n$ , remove the configuration-independent factor  $(1 - p)^{|E_n|}$  from the partition function:

$$\widehat{Z}_n^\xi(t) = \sum_{\omega \subset E_n} e^{t|\omega|} 2^{k_\xi(\omega)}, \quad \xi \in \{0, 1\}.$$

Wiring the boundary merges at most  $|\partial\Lambda_n| - 1$  clusters, hence

$$2^{-(|\partial\Lambda_n|-1)} \widehat{Z}_n^0(t) \leq \widehat{Z}_n^1(t) \leq \widehat{Z}_n^0(t).$$

The two normalized logarithms therefore have the same thermodynamic limit  $\psi(t)$ . They are convex and their derivatives are the mean open-edge densities. At every differentiability point of  $\psi$ , convex convergence gives the same limiting edge density in the free and wired states. Since  $\phi_{p,2}^0 \preceq \phi_{p,2}^1$ , a

monotone coupling and equality of every one-edge marginal force the two states to coincide. The exceptional set is countable because  $\psi$  is convex.

Fix  $p > p_c^{\text{FK}}$  and choose  $r \in (p_c^{\text{FK}}, p)$  outside this exceptional set. By the definition of the threshold and monotonicity, the wired state percolates at  $r$ , hence the identical free state percolates at  $r$ , and monotonicity gives free percolation at  $p$ . The free/wired monotone limits are extremal DLR states (minimality/maximality plus a convex-decomposition argument), hence translation ergodic; the translation-invariant event “an infinite cluster exists” therefore has probability one.  $\square$

**Lemma 4.5** (No simultaneous primal and dual percolation). *Let  $\mu$  be a translation-invariant FKG measure on square-lattice edge configurations. If the primal open configuration and its complementary dual configuration each have an infinite cluster, and each infinite cluster is unique, then a contradiction results.*

*Proof.* For  $R = [0, n] \times [0, k]$  let  $h(n, k)$  and  $v(n, k)$  be its primal horizontal and vertical crossing probabilities. We establish the two asymptotic relations

$$\max\{h(n, k), v(n, k+1)\} \longrightarrow 1, \quad \min\{h(n, k), v(n, k)\} \longrightarrow 0 \quad (4.1)$$

as  $\min(n, k) \rightarrow \infty$ ; they are incompatible.

Fix a large but finite square  $S = [0, s]^2$ . Inside  $R$  choose a translate  $S_x$  at the intersection of the two discrete balance interfaces where the inequalities “connection to Bottom is at least connection to Top” and “connection to Left is at least connection to Right” change sign. Neighboring translates  $S_{x'}$  and  $S_{x''}$  satisfy the reversed inequalities in the two respective directions. The balance point recedes from every side of  $R$ : otherwise a subsequence would yield an infinite primal or dual component confined to a half-plane. Such a half-plane component is impossible under simultaneous uniqueness. Indeed, in a half-plane one can choose two neighboring squares whose probabilities of reaching the upper/lower rays of the boundary line are balanced; FKG makes both long connections overwhelmingly likely once  $S$  is likely to meet the unique infinite cluster, and uniqueness makes them coalesce. Their planar union separates a unit boundary segment from infinity. Translation invariance then excludes a half-plane infinite component.

Let  $A_R$  be the event that exactly one component meeting the distance-2 neighborhood of  $S_x$  reaches  $\partial R$ . Since  $x$  is far from the boundary and the full-plane infinite cluster is unique,  $\mu(A_R) \rightarrow 1$ . Write  $\pi_s = \mu(S \leftrightarrow \infty)$ . If Bottom is at least as likely as Left at the balance point, then it is at least as likely as all four sides. FKG applied to the four decreasing nonconnection events yields

$$\mu(S_x \not\leftrightarrow \text{Bottom}) \leq (1 - \pi_s)^{1/4}.$$

Using the neighboring translate  $x'$  and  $A_R$  joins a bottom arm to a top arm and gives

$$v(n, k+1) \geq \pi_s - 2(1 - \pi_s)^{1/4} - 2\mu(A_R^c).$$

If Left is more likely, the same argument with  $x''$  gives the identical bound for  $h(n, k)$ . Let  $R \uparrow \mathbb{Z}^2$  and then  $s \rightarrow \infty$ ; since  $\pi_s \uparrow 1$ , the first relation in (4.1) follows. Applying it to the complementary dual configuration and using planar crossing duality gives the second relation.

Now choose  $k_n$  maximal with  $v(n, k_n) \geq h(n, k_n)$ . Uniqueness of the two infinite clusters gives  $k_n \rightarrow \infty$ . By maximality,  $v(n, k_n + 1) < h(n, k_n + 1)$ . The first relation in (4.1) forces either both  $h(n, k_n), v(n, k_n)$  or both  $v(n, k_n + 1), h(n, k_n + 1)$  to tend to one, contradicting the second relation. Thus simultaneous primal/dual percolation is impossible.  $\square$

**Theorem 4.6** (Criticality of the self-dual point). *For nearest-neighbor square-lattice Ising,*

$$K_c = \frac{1}{2} \log(1 + \sqrt{2}).$$

*Proof.*  $p = 1 - e^{-2K}$  represents the zero-field Ising model. In the wired state,

$$\phi_{p,2}^1(0 \leftrightarrow \infty) = \langle \sigma_0 \rangle_K^+.$$

Therefore the Ising critical point and the wired FK threshold correspond under  $p = 1 - e^{-2K}$ .

$$D_2(p) = \frac{2(1-p)}{p + 2(1-p)}.$$

For  $p < p_c^{\text{FK}}$ , exponential decay makes the probabilities of primal open circuits in dyadic annuli summable. Borel–Cantelli and planar separation therefore give an infinite dual-open path in the dual of the wired state, which is the free state at  $D_2(p)$ . Hence  $D_2(p) \geq p_c^{\text{FK}}$ , and letting  $p \uparrow p_c^{\text{FK}}$  gives  $D_2(p_c^{\text{FK}}) \geq p_c^{\text{FK}}$ .

$$p_c^{\text{FK}} = \frac{\sqrt{2}}{1 + \sqrt{2}}.$$

$1 - e^{-2K_c} = \sqrt{2}/(1 + \sqrt{2})$  gives  $e^{2K_c} = 1 + \sqrt{2}$  and therefore

$$K_c = \frac{1}{2} \log(1 + \sqrt{2}).$$

All ingredients are owned in Volumes I–VIII; the former forward reference to Volume IX has been removed. □

## Chapter 5

# Exact Two-Dimensional Theory

The exact solution is included because it is the benchmark against which universality statements are calibrated. The algebra is long; we expose the finite-volume operator identities and isolate where asymptotic analysis enters.

**Lemma 5.1** (Row transfer matrix and dual coupling). *Let  $N \geq 2$  and impose periodic boundary conditions along a row. Put*

$$H_z = \sum_{j=1}^N \sigma_j^z \sigma_{j+1}^z, \quad H_x = \sum_{j=1}^N \sigma_j^x, \quad \sigma_{N+1}^z = \sigma_1^z,$$

and define the dual coupling  $K^* > 0$  by

$$\tanh K^* = e^{-2K}, \quad \text{equivalently} \quad \sinh(2K) \sinh(2K^*) = 1.$$

The row-to-row transfer matrix of the isotropic zero-field Ising model is

$$\mathsf{T}_N = (2 \sinh 2K)^{N/2} e^{\frac{K}{2} H_z} e^{K^* H_x} e^{\frac{K}{2} H_z}.$$

It has the same spectrum as

$$(2 \sinh 2K)^{N/2} \mathsf{W}_N, \quad \mathsf{W}_N = e^{\frac{K^*}{2} H_x} e^{K H_z} e^{\frac{K^*}{2} H_x}.$$

*Proof.* For two adjacent rows  $s, t \in \{\pm 1\}^N$ , split the horizontal energy equally between the two sides of the vertical interaction. Thus

$$\mathsf{T}_N(s, t) = \exp \left( \frac{K}{2} \sum_j s_j s_{j+1} + K \sum_j s_j t_j + \frac{K}{2} \sum_j t_j t_{j+1} \right).$$

For one site the vertical-interaction matrix is

$$B_K = \begin{pmatrix} e^K & e^{-K} \\ e^{-K} & e^K \end{pmatrix} = \sqrt{2 \sinh 2K} e^{K^* \sigma^x}.$$

Indeed  $B_K$  has eigenvalues  $2 \cosh K$  and  $2 \sinh K$ , while  $e^{\pm K^*}$  have ratio  $e^{2K^*} = \coth K$ , which is equivalent to  $\tanh K^* = e^{-2K}$ ; the determinant fixes the scalar factor. Tensoring this identity over the row gives the first displayed operator formula.

Write  $A = e^{KH_z/2}$  and  $B = e^{K^* H_x}$ . Since  $A$  and  $B$  are invertible,  $ABA$  and  $B^{1/2} A^2 B^{1/2}$  have the same spectrum: with  $X = AB^{1/2}$  and  $Y = B^{1/2} A$ , they are  $XY$  and  $YX$ , and  $YX = X^{-1}(XY)X$ . This proves the second assertion.  $\square$



**Lemma 5.2** (Kaufman spinor blocks). *Let*

$$a_{2j-1} = \left( \prod_{\ell < j} \sigma_\ell^x \right) \sigma_j^z, \quad a_{2j} = \left( \prod_{\ell < j} \sigma_\ell^x \right) \sigma_j^y \quad (1 \leq j \leq N).$$

*Then the  $a_r$  are Majorana operators,  $a_r a_s + a_s a_r = 2\delta_{rs}$ , and*

$$\sigma_j^x = i a_{2j-1} a_{2j}, \quad \sigma_j^z \sigma_{j+1}^z = i a_{2j} a_{2j+1},$$

*where the last bond is interpreted in a fixed eigenspace of the parity operator  $\mathcal{P} = \prod_j \sigma_j^x$ . In the two parity sectors the Majoranas have periodic or antiperiodic boundary condition. After Fourier transform, conjugation by  $\mathcal{W}_N$  decomposes into  $2 \times 2$  blocks with eigenvalues  $e^{\pm\gamma(\theta)}$ , where*

$$\cosh \gamma(\theta) = \cosh(2K^*) \cosh(2K) - \sinh(2K^*) \sinh(2K) \cos \theta. \quad (5.1)$$

*The allowed momenta are one of the two grids*

$$\Theta_N^{(0)} = \left\{ \frac{2\pi m}{N} : 0 \leq m < N \right\}, \quad \Theta_N^{(1)} = \left\{ \frac{(2m+1)\pi}{N} : 0 \leq m < N \right\}.$$

*Proof.* The Jordan–Wigner strings cancel in the two displayed bilinears; the only uncanceled string is on the bond crossing the periodic seam, and it equals the parity  $\mathcal{P}$ . Hence fixing  $\mathcal{P} = \pm 1$  converts the seam into one of the two fermionic boundary conditions, giving the two momentum grids.

For a pair of Majoranas  $a, b$ , direct commutator differentiation gives

$$e^{\frac{t}{2}iab} \begin{pmatrix} a \\ b \end{pmatrix} e^{-\frac{t}{2}iab} = \begin{pmatrix} \cosh t & -i \sinh t \\ i \sinh t & \cosh t \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}. \quad (5.2)$$

Thus the on-site factor  $e^{K^* H_x}$  acts in the Fourier basis  $(\hat{a}_{\text{odd}}(\theta), \hat{a}_{\text{even}}(\theta))$  by

$$A_*(\theta) = \begin{pmatrix} c_* & -i s_* \\ i s_* & c_* \end{pmatrix}, \quad c_* = \cosh 2K^*, \quad s_* = \sinh 2K^*,$$

while the bond factor  $e^{KH_z}$  acts by

$$B(\theta) = \begin{pmatrix} c & i s e^{-i\theta} \\ -i s e^{i\theta} & c \end{pmatrix}, \quad c = \cosh 2K, \quad s = \sinh 2K.$$

The half-on-site/bond/half-on-site product has the same two eigenvalues as  $A_*(\theta)B(\theta)$ . Both matrices have determinant one, and a direct multiplication gives

$$\frac{1}{2} \text{tr}(A_* B) = c_* c - s_* s \cos \theta.$$

Therefore its eigenvalues are  $e^{\pm\gamma(\theta)}$  with  $\gamma \geq 0$  and (5.1) follows.  $\square$

**Lemma 5.3** (Spinor spectrum and thermodynamic dominant eigenvalue). *In either parity sector, the eigenvalues of  $\mathcal{W}_N$  have the form*

$$\exp \left( \frac{1}{2} \sum_{\theta \in \Theta_N^{(\epsilon)}} \epsilon_\theta \gamma(\theta) \right), \quad \epsilon_\theta \in \{\pm 1\},$$

subject only to the parity constraint on the occupation numbers. Consequently

$$\left| \frac{1}{N} \log \rho(W_N) - \frac{1}{2N} \max_{\varepsilon \in \{0,1\}} \sum_{\theta \in \Theta_N^{(\varepsilon)}} \gamma(\theta) \right| \leq \frac{C_K}{N}, \quad (5.3)$$

where  $C_K = \max_{\theta} \gamma(\theta)$ .

*Proof.* Each block in Lemma 5.2 is a complex orthogonal transformation with reciprocal positive eigenvalues  $e^{\pm\gamma}$ . Choose corresponding complex fermions  $d_{\theta}, d_{\theta}^{\dagger}$ . The spin representation of the block is

$$\exp\left(\gamma(\theta)(d_{\theta}^{\dagger}d_{\theta} - \tfrac{1}{2})\right)$$

up to the harmless choice of which vector is called occupied. Multiplying the commuting normal forms over all momentum blocks gives the displayed many-body spectrum. The parity sector requires the total occupation parity to be fixed. If the all-maximizing sign choice has the wrong parity, flipping one occupation changes the logarithm by at most  $\max \gamma$ . This proves (5.3). Notice that this is precisely where the finite-volume parity bookkeeping enters; it affects the free energy only by  $O(N^{-1})$ .  $\square$

**Theorem 5.4** (Onsager–Kaufman free energy). *For the isotropic nearest-neighbor square-lattice Ising model at zero field, the infinite-volume free energy per site has the Onsager integral representation*

$$-\beta f(K) = \frac{1}{2} \log(2 \sinh 2K) + \frac{1}{2\pi} \int_0^{\pi} \operatorname{arcosh}(\cosh 2K \coth 2K - \cos \theta) d\theta,$$

with equivalent standard forms obtained by elementary transformations; the only singularity on the positive real axis occurs at  $\sinh 2K = 1$ .

*Proof.*

$$\lim_{M \rightarrow \infty} \frac{1}{MN} \log Z_{M,N} = \frac{1}{N} \log \rho(T_N).$$

Lemma 5.1 therefore reduces the problem to the spectral radius of  $W_N$ .

5.3,

$$\frac{1}{N} \log \rho(T_N) = \frac{1}{2} \log(2 \sinh 2K) + \frac{1}{2N} \sum_{\theta \in \Theta_N} \gamma(\theta) + O_K(N^{-1}), \quad (5.4)$$

where either momentum grid may be used in the sum because the two Riemann sums differ by  $o(1)$ .  $\cosh 2K^* = \coth 2K$ , (5.1) becomes

$$\cosh \gamma(\theta) = \cosh 2K \coth 2K - \cos \theta. \quad (5.5)$$

The function is even in  $\theta$ . Taking  $N \rightarrow \infty$  in (5.4) gives

$$\lim_{N \rightarrow \infty} \frac{1}{N} \log \rho(T_N) = \frac{1}{2} \log(2 \sinh 2K) + \frac{1}{4\pi} \int_0^{2\pi} \gamma(\theta) d\theta,$$

which is the displayed formula.

$$\cosh \gamma(0) = \cosh(2K - 2K^*), \quad \text{so} \quad \gamma(0) = 2|K - K^*|.$$

Hence the spectral gap is positive unless  $K = K^*$ . The duality equation then gives

$$K = K^* \iff \sinh 2K = 1.$$

For  $K \neq K^*$ ,  $\gamma(\theta)$  is uniformly bounded away from zero; analytic functional calculus for arcosh and dominated differentiation show that the integral is real analytic in  $K$ .

$$\cosh \gamma(\theta) = 2 - \cos \theta = 1 + \frac{\theta^2}{2} + O(\theta^4),$$

so  $\gamma(\theta) = |\theta| + O(|\theta|^3)$ . If  $m = 2(K - K^*)$ , the joint expansion near  $(m, \theta) = (0, 0)$  is

$$\gamma(\theta) = \sqrt{m^2 + \theta^2} + O((|m| + |\theta|)^2). \quad (5.6)$$

$$\int_0^\varepsilon \sqrt{m^2 + \theta^2} d\theta = \frac{1}{2} \left[ \varepsilon \sqrt{m^2 + \varepsilon^2} + m^2 \log \frac{\varepsilon + \sqrt{m^2 + \varepsilon^2}}{|m|} \right].$$

It contains the nonanalytic term  $-\frac{1}{2}m^2 \log |m|$ . All remainders in (5.6) contribute twice continuously differentiable functions. Thus the pressure is not analytic at  $K_{\text{sd}}$  and there are no other positive-real singularities. □

**Lemma 5.5** (Analytic Toeplitz determinant limit in the zero-index case). *Let*

$$g(z) = \sum_{j \in \mathbb{Z}} g_j z^j$$

*be analytic on an annulus  $r < |z| < r^{-1}$  containing the unit circle, with  $g_0 = 0$ , and put  $\varphi = e^g$ . If  $T_n(\varphi) = (\varphi_{j-k})_{0 \leq j, k < n}$ , then*

$$\lim_{n \rightarrow \infty} \det T_n(\varphi) = \exp \left( \sum_{j \geq 1} j g_j g_{-j} \right).$$

*Proof.* The proof is included because Yang's magnetization formula needs the constant, not only existence of the determinant limit.

Write  $g = g_+ + g_-$  with  $g_+(z) = \sum_{j \geq 1} g_j z^j$  and  $g_-(z) = \sum_{j \geq 1} g_{-j} z^{-j}$ , and set  $a = e^{g_-}$  and  $b = e^{g_+}$ . The exponential decay of the Fourier coefficients implies

$$\sum_{j \in \mathbb{Z}} (1 + |j|) (|a_j| + |b_j| + |(a^{-1})_j| + |(b^{-1})_j|) < \infty.$$

Hence every Hankel matrix below is trace class, and truncating all Fourier series at degree  $M$  changes their trace norms by  $O(r^M)$  uniformly in the section size.

For Laurent polynomials the multiplication identity, obtained by inserting the missing indices in the convolution sum, is

$$T_n(a)T_n(b) = T_n(ab) - P_n H(a) H(\tilde{b}) P_n,$$

where  $P_n$  is projection onto the first  $n$  coordinates,  $H(a) = (a_{j+k+1})_{j, k \geq 0}$ , and  $\tilde{b}(z) = b(z^{-1})$ . Because  $a$  has only nonpositive and  $b$  only nonnegative Fourier modes,  $T_n(a)$  and  $T_n(b)$  are triangular with unit diagonal. Therefore

$$\det T_n(\varphi) = \det(I + K_n),$$

where, after multiplication by the triangular inverses,

$$K_n = -T_n(a)^{-1}P_n H(a)H(\tilde{b})P_n T_n(b)^{-1}.$$

Passing from polynomial truncations to the analytic functions in trace norm is justified by the preceding exponential tail bound, so this identity remains valid in the limit.

Now move the finite-section projection through the one-sided triangular factors. The part supported a distance  $n$  from the corner has trace norm  $O(r^n)$ ; what remains converges in trace norm to the corner operator

$$K = -H(a^{-1})H(\widetilde{b^{-1}})$$

conjugated by fixed one-sided Toeplitz factors. Fredholm determinants are continuous in trace norm, so  $\det T_n(\varphi) \rightarrow \det(I + K)$ . To evaluate this determinant without appealing to a named theorem, introduce  $a_t = e^{tg_-}$ ,  $b_t = e^{tg_+}$  and  $K(t)$  by the same construction. Finite polynomial truncations permit ordinary differentiation of determinants; the trace-norm bounds allow passage to the limit and give

$$\frac{d}{dt} \log \det(I + K(t)) = 2t \sum_{j \geq 1} j g_j g_{-j}.$$

Indeed, after differentiating, all purely upper-upper and lower-lower convolution terms have zero trace; the only diagonal contribution pairs a  $z^j$  mode from  $g_+$  with a  $z^{-j}$  mode from  $g_-$ , and it occurs in exactly  $j$  corner positions. Absolute convergence of  $\sum j |g_j g_{-j}|$  justifies the rearrangement. Since  $K(0) = 0$ , integration from 0 to 1 yields

$$\log \det(I + K) = \sum_{j \geq 1} j g_j g_{-j},$$

which proves the formula. This argument is the special analytic zero-index case needed below; no general Fisher–Hartwig or strong-Szegő theorem is imported.  $\square$

**Lemma 5.6** (Ordered diagonal-correlation determinant). *Let  $K > K_c$  and set  $k = \sinh^{-2}(2K) \in (0, 1)$ . For the infinite-volume zero-field square-lattice Ising model, the diagonal correlation at separation  $n$  is the Toeplitz determinant*

$$\langle \sigma_{0,0} \sigma_{n,n} \rangle = D_n(\varphi_K), \quad \varphi_K(z) = \left( \frac{1 - kz}{1 - kz^{-1}} \right)^{1/2}, \quad |z| = 1,$$

where the branch is chosen continuously from  $k = 0$ . Moreover the limit of this correlation is  $m(K)^2$ .

*Proof.* Use the Kaufman Clifford representation already constructed for the transfer matrix in Theorem 5.4. Conjugating a spin through the row transfer matrix changes the fermionic boundary condition across the interval joining the two spin insertions. Wick reduction of the resulting  $2n$  Clifford operators gives a Pfaffian; translation invariance makes its square a block Toeplitz determinant. The  $2 \times 2$  momentum block diagonalization from the free-energy proof reduces that Pfaffian to the scalar determinant displayed above. In the ordered regime the spectral gap is nonzero, so the finite-cylinder formula converges uniformly to the infinite-row determinant as the transverse size tends to infinity.

For the second assertion, take plus boundary conditions in a large box and then send the box to  $\mathbb{Z}^2$ . FKG monotonicity gives existence of  $m(K) = \langle \sigma_0 \rangle_+$ . Clustering of the extremal plus state gives

$$\lim_{n \rightarrow \infty} \langle \sigma_{0,0} \sigma_{n,0} \rangle_+ = m(K)^2.$$

The zero-field diagonal determinant is the symmetric mixture of the plus and minus extremal states, and the even observable  $\sigma_0\sigma_n$  has the same expectation in both. Hence its large-distance limit is again  $m(K)^2$ .  $\square$

**Theorem 5.7** (Yang spontaneous magnetization). *For  $K > K_c$ ,*

$$m(K) = (1 - \sinh^{-4}(2K))^{1/8},$$

and  $m(K) = 0$  for  $K \leq K_c$ .

*Proof.*

$$g(z) = \frac{1}{2} \log(1 - kz) - \frac{1}{2} \log(1 - kz^{-1}) = -\frac{1}{2} \sum_{j \geq 1} \frac{k^j}{j} z^j + \frac{1}{2} \sum_{j \geq 1} \frac{k^j}{j} z^{-j}.$$

Thus  $g_j = -k^j/(2j)$  and  $g_{-j} = k^j/(2j)$ .

$$m(K)^2 = \exp \left( -\frac{1}{4} \sum_{j \geq 1} \frac{k^{2j}}{j} \right) = (1 - k^2)^{1/4}.$$

Since the plus-state magnetization is nonnegative,

$$m(K) = (1 - k^2)^{1/8} = (1 - \sinh^{-4}(2K))^{1/8}.$$

$$\lim_{|x| \rightarrow \infty} \langle \sigma_0 \sigma_x \rangle = 0.$$

If  $m(K) > 0$ , the plus extremal state would instead have limiting two-point function  $m(K)^2 > 0$ , contradiction. Hence  $m(K) = 0$ .  $\square$

**Theorem 5.8** (Critical diagonal two-point asymptotic). *For the isotropic square-lattice Ising model at zero field and at the critical coupling  $K_c$ , the diagonal correlation has the exact product formula*

$$C_N := \langle \sigma_{0,0} \sigma_{N,N} \rangle_{K_c} = \left( \frac{2}{\pi} \right)^N \prod_{j=1}^{N-1} \left( 1 - \frac{1}{4j^2} \right)^{j-N}. \quad (8.24)$$

Consequently

$$C_N \sim C_{\text{diag}} N^{-1/4}, \quad (8.25)$$

where the positive constant is given explicitly by the absolutely convergent expression

$$C_{\text{diag}} = \exp \left[ -\frac{1}{4} - \frac{\gamma}{4} - \sum_{j=1}^{\infty} \left\{ j \left[ -\log \left( 1 - \frac{1}{4j^2} \right) \right] - \frac{1}{4j} \right\} \right]. \quad (8.26)$$

Here  $\gamma$  is Euler's constant.

*Proof.* This selected exact formulation avoids importing a Fisher–Hartwig theorem: at criticality the diagonal Toeplitz determinant becomes a Cauchy determinant, so both the determinant and its asymptotic can be evaluated directly.

The transfer-matrix/Pfaffian reduction for the diagonal correlation gives

$$C_N = \det(a_{m-n})_{m,n=1}^N, \quad a_\ell = \frac{1}{2\pi} \int_0^{2\pi} e^{-i\ell\theta} \left( \frac{k - e^{-i\theta}}{k - e^{i\theta}} \right)^{1/2} d\theta, \quad (8.27)$$

with  $k = \sinh(2K_h) \sinh(2K_v)$  in the normalization in which criticality is  $k = 1$ . We need only the isotropic critical value. For  $0 < \theta < 2\pi$  choose the continuous boundary branch. Since

$$\frac{1 - e^{-i\theta}}{1 - e^{i\theta}} = -e^{-i\theta},$$

that branch is  $ie^{-i\theta/2}$ . Therefore

$$a_\ell = \frac{i}{2\pi} \int_0^{2\pi} e^{-i(\ell+1/2)\theta} d\theta = \frac{1}{\pi(\ell+1/2)}. \quad (8.28)$$

Thus

$$C_N = \pi^{-N} \det \left( \frac{1}{m-n+1/2} \right)_{m,n=1}^N. \quad (8.29)$$

For example, (8.29) gives  $C_1 = 2/\pi$  and  $C_2 = 16/(3\pi^2)$ .

We prove the determinant identity used here. If  $x_1, \dots, x_N$  and  $y_1, \dots, y_N$  have  $x_m + y_n \neq 0$ , then

$$\det \left( \frac{1}{x_m + y_n} \right)_{m,n=1}^N = \frac{\prod_{1 \leq m < r \leq N} (x_r - x_m) \prod_{1 \leq n < s \leq N} (y_s - y_n)}{\prod_{m,n=1}^N (x_m + y_n)}. \quad (8.30)$$

Indeed, after multiplying the determinant by  $\prod_{m,n} (x_m + y_n)$ , the result is a polynomial alternating separately in the  $x$ 's and in the  $y$ 's. It is therefore divisible by both Vandermonde products. The degrees agree, so the quotient is a constant; comparing the coefficient of  $x_1^{N-1} x_2^{N-2} \dots x_{N-1}$ , or equivalently sending the  $x_m$  successively to infinity, shows that this constant is 1.

Apply (8.30) with  $x_m = m$  and  $y_n = \frac{1}{2} - n$ . Cancelling the two Vandermonde products against the denominator, or simply taking the ratio of the  $N+1$  and  $N$  determinants, gives

$$\frac{C_{N+1}}{C_N} = \frac{2}{\pi} \prod_{j=1}^N \left( 1 - \frac{1}{4j^2} \right)^{-1}, \quad C_1 = \frac{2}{\pi}. \quad (8.31)$$

Iteration of (8.31) is exactly (8.24).

Set

$$\alpha_j := -\log \left( 1 - \frac{1}{4j^2} \right) > 0.$$

Euler's sine product

$$\frac{\sin \pi z}{\pi z} = \prod_{j=1}^{\infty} \left( 1 - \frac{z^2}{j^2} \right)$$

at  $z = 1/2$  gives

$$\prod_{j=1}^{\infty} \left( 1 - \frac{1}{4j^2} \right) = \frac{2}{\pi}, \quad \sum_{j=1}^{\infty} \alpha_j = -\log(2/\pi). \quad (8.32)$$

Taking logarithms in (8.24) and using (8.32),

$$\begin{aligned}\log C_N &= N \log(2/\pi) + \sum_{j=1}^{N-1} (N-j)\alpha_j \\ &= -N \sum_{j=N}^{\infty} \alpha_j - \sum_{j=1}^{N-1} j\alpha_j.\end{aligned}\tag{8.33}$$

The cancellation in the second line is the mechanism that leaves a power law rather than an exponential decay.

Taylor's formula gives, uniformly for  $j \geq 1$ ,

$$\alpha_j = \frac{1}{4j^2} + O(j^{-4}).\tag{8.34}$$

Hence

$$N \sum_{j=N}^{\infty} \alpha_j = \frac{1}{4} + o(1),\tag{8.35}$$

while the series

$$\sum_{j=1}^{\infty} \left( j\alpha_j - \frac{1}{4j} \right)\tag{8.36}$$

is absolutely convergent. Since

$$H_{N-1} = \log N + \gamma + o(1),$$

we obtain

$$\sum_{j=1}^{N-1} j\alpha_j = \frac{1}{4} \log N + \frac{\gamma}{4} + \sum_{j=1}^{\infty} \left( j\alpha_j - \frac{1}{4j} \right) + o(1).\tag{8.37}$$

Substitution of (8.35)–(8.37) into (8.33) proves

$$\log C_N = -\frac{1}{4} \log N + \log C_{\text{diag}} + o(1),$$

with  $C_{\text{diag}}$  exactly as in (8.26). Exponentiation proves (8.25); absolute convergence in (8.36) makes  $C_{\text{diag}}$  finite and strictly positive.

Every analytic step after the Pfaffian/Toeplitz reduction has now been carried out directly. In particular, the power  $1/4$  is not inferred from scaling theory: it comes from the harmonic term  $\frac{1}{4}H_{N-1}$  in (8.37), while the remaining summable terms determine the positive lattice amplitude  $C_{\text{diag}}$ .

□

## Chapter 6

# Lee–Yang Theory

**Lemma 6.1** (Asano contraction in the unit disk). *Let*

$$P(z_\alpha, z_\beta) = Az_\alpha z_\beta + Bz_\alpha + Cz_\beta + D$$

*be a polynomial, with coefficients allowed to depend on auxiliary variables that are held fixed. Suppose that*

$$P(z_\alpha, z_\beta) \neq 0 \quad \text{whenever } |z_\alpha| < 1, |z_\beta| < 1.$$

*Then its Asano contraction*

$$\mathcal{A}P(z) := Az + D$$

*is nonzero for every  $|z| < 1$ .*

*Proof.* Fix the auxiliary variables. Setting  $z_\alpha = z_\beta = w$  shows that

$$Q(w) := Aw^2 + (B + C)w + D$$

has no zero in the open unit disk. If  $A = 0$ , then  $\mathcal{A}P = D$ ; the hypothesis with  $z_\alpha = z_\beta = 0$  gives  $D \neq 0$ . Assume  $A \neq 0$ . If  $D = 0$ , then  $Q(0) = 0$ , contradicting the hypothesis, so  $D \neq 0$  as well. Let  $r_1, r_2$  be the two roots of  $Q$ , counted with multiplicity. Both satisfy  $|r_j| \geq 1$ , and in fact neither lies in the open disk. Vieta's formula gives

$$|D/A| = |r_1 r_2| \geq 1.$$

The unique zero of  $Az + D$  is  $-D/A$ , which therefore does not belong to the open unit disk. Hence  $\mathcal{A}P(z) \neq 0$  for  $|z| < 1$ .  $\square$

**Lemma 6.2** (Ferromagnetic edge factor). *For  $0 \leq a \leq 1$ , the polynomial*

$$E_a(u, v) = uv + a(u + v) + 1$$

*is nonzero whenever  $|u| < 1$  and  $|v| < 1$ .*

*Proof.* For  $a = 1$ ,  $E_1(u, v) = (1 + u)(1 + v)$  and the claim is immediate. Let  $0 \leq a < 1$ . If  $u + a = 0$ , then  $E_a(u, v) = 1 - a^2 \neq 0$ . Otherwise a zero would force

$$v = -\frac{1 + au}{a + u}.$$

But

$$|1 + au|^2 - |a + u|^2 = (1 - a^2)(1 - |u|^2) > 0,$$

so the displayed value of  $v$  has modulus strictly larger than 1. This contradicts  $|v| < 1$ .  $\square$



**Theorem 6.3** (Lee–Yang circle theorem). *Let  $G = (V, E)$  be finite and let the Ising couplings satisfy  $J_e \geq 0$ . Regard the finite-volume partition function with a uniform complex magnetic field as a polynomial in the fugacity  $z = e^{-2h}$ , after multiplication by an irrelevant nonzero monomial and constant. Every zero of this one-variable polynomial lies on the unit circle  $|z| = 1$ .*

*Proof.*

$$\mathcal{Z}_G((z_v)_{v \in V}) = \sum_{\sigma \in \{-, +\}^V} \left( \prod_{e=\{x, y\}} a_e^{1_{\sigma_x \neq \sigma_y}} \right) \prod_{v: \sigma_v = +} z_v, \quad a_e = e^{-2J_e} \in (0, 1],$$

where the omitted global factor is nonzero and hence irrelevant for zeros. Thus it is enough to prove that  $\mathcal{Z}_G \neq 0$  whenever all  $|z_v| < 1$ .

$$E_{a_e}(z_{x,e}, z_{y,e}) = z_{x,e} z_{y,e} + a_e(z_{x,e} + z_{y,e}) + 1.$$

By Lemma 6.2, each factor is nonzero when its two variables lie in the open unit disk. A product of such factors in disjoint variables has the same nonvanishing property. For an isolated vertex we additionally use the factor  $1 + z_v$ , which is also nonzero in the open disk.

$$Az_\alpha z_\beta + Bz_\alpha + Cz_\beta + D$$

and replace it by  $Az_v + D$ . Lemma 6.1 shows that every contraction preserves nonvanishing when all remaining variables lie in the open unit disk.

$$\mathcal{Z}_G((z_v)) \neq 0 \quad \text{if } |z_v| < 1 \text{ for every } v.$$

$$\mathcal{Z}_G(z) = z^{|V|} \mathcal{Z}_G(1/z).$$

Thus a zero with  $|z| > 1$  would produce, by reciprocation, a zero with modulus less than 1, which is impossible. □

**Theorem 6.4** (Analyticity away from the Lee–Yang locus). *In any complex-field region separated from the accumulation set of Lee–Yang zeros, finite-volume pressures and their derivatives form normal families; any thermodynamic-limit pressure is analytic there. In particular, for real nonzero magnetic field the ferromagnetic Ising free energy and magnetization are analytic under the standard thermodynamic-limit hypotheses.*

*Proof.* Let  $U$  be a simply connected compactly contained region of the complex field plane whose distance from every finite-volume Lee–Yang zero is at least  $\delta > 0$  for all sufficiently large volumes. Choose a branch of  $\log Z_\Lambda$  on  $U$  and set  $p_\Lambda = |\Lambda|^{-1} \log Z_\Lambda$ . Stability of the finite-volume energy gives a volume-order bound  $|p_\Lambda(h)| \leq C_U$  after fixing the branch at one point. Thus  $(p_\Lambda)$  is locally bounded and Montel’s theorem makes it a normal family.

Cauchy’s formula on a circle of radius  $r < \delta/2$  gives, uniformly on compact  $K \Subset U$ ,

$$|p_\Lambda^{(m)}(h)| \leq \frac{m!}{r^m} \sup_{|z-h|=r} |p_\Lambda(z)| \leq C_{K,m}. \quad (6.1)$$

Every subsequential thermodynamic limit is therefore analytic, and its derivatives are the limits of the corresponding finite-volume derivatives. If the thermodynamic-limit pressure is unique on the

real axis, Vitali's theorem (or uniqueness of analytic continuation from a set with an accumulation point) shows that all normal-family subsequences have the same analytic limit on  $U$ .

For a ferromagnetic Ising model the Lee–Yang theorem places the zeros on the imaginary-field locus (equivalently, on the unit circle in the fugacity variable). Every compact real interval with  $h \neq 0$  is a positive distance from that locus. Applying the preceding argument proves analyticity of the free energy there; (6.1) then gives analyticity of the magnetization and all field derivatives.  $\square$

## Chapter 7

# Planar Infinite-Volume Gibbs States

The classification theorem in this chapter is genuinely planar. The point is not merely that plus and minus boundary conditions produce two extremal states: one must rule out every possible non-translation-invariant interface state in two dimensions. We give the geometric proof in a form adapted from the Russo–Aizenman–Higuchi method and its FKG-only reformulation. The ingredients are strong Markov resampling on random interiors, stochastic monotonicity, finite energy, planar separation, and a duplicated-system disagreement argument.

Write  $x \sim_* y$  when  $x$  and  $y$  are nearest or diagonal neighbours in  $\mathbb{Z}^2$ . A  $+$ -cluster uses nearest-neighbour adjacency and a  $+$ \*-cluster uses  $\sim_*$ . The analogous notation is used for minus spins. For a half-plane  $\pi$  with boundary line  $\ell$ , a  $*$ -semicircuit is a self-avoiding  $*$ -path in  $\pi$  whose two endpoints lie on  $\ell$  and whose polygonal interpolation, together with the boundary segment between its endpoints, separates a finite region from infinity.

**Lemma 7.1** (Strong Markov property for exterior-determined random regions). *Let  $\mu$  be a nearest-neighbour Ising DLR state and let  $G$  be a finite random set such that, for every finite  $A$ , the event  $\{G = A\}$  is measurable with respect to the spins on  $A^c$ . Then for every bounded local function  $f$ ,*

$$\mathbb{E}_\mu[f \mid \mathcal{F}_{G^c}](\omega) = \gamma_G f(\omega) \quad \text{for } \mu\text{-a.e. } \omega.$$

*The same statement holds when  $G = \emptyset$  on an exceptional event.*

*Proof.* For a deterministic finite  $A$ , the DLR equation gives

$$\mathbb{E}_\mu[f \mathbf{1}_{\{G=A\}}] = \mathbb{E}_\mu[\gamma_A f \mathbf{1}_{\{G=A\}}],$$

because  $\{G = A\} \in \mathcal{F}_{A^c}$ . Summing over the countable set of finite  $A$  gives the claimed identity. This is exactly the deterministic DLR equation applied after partitioning configuration space according to the value of the random region.  $\square$

**Lemma 7.2** (Percolation detects the extremal phases). *Let  $\mu$  be a two-dimensional nearest-neighbour Ising Gibbs state at zero field. If  $\mu \neq \mu^-$ , then  $\mu$  gives positive probability to an infinite  $+$ -cluster. If  $\mu$  is extremal, that probability is one. By spin flip, if  $\mu \neq \mu^+$  then an infinite  $-$ -cluster occurs with positive probability, and with probability one in an extremal state.*

*Proof.* Assume that no infinite  $+$ -cluster occurs. Fix a finite square  $D$ . Planarity implies that, almost surely,  $D$  is surrounded by a  $-*$  circuit: otherwise the outer boundary exploration from  $D$  would produce a  $+$  path to infinity. Let  $L \supset D$  be a large square. On the event that such a surrounding circuit occurs inside  $L$ , let  $G$  be the interior of the outermost one; otherwise put

$G = \emptyset$ . The choice is exterior-determined. On  $\partial G$  all spins are minus, and hence Lemma 7.1 and monotonicity in the boundary condition give, for every increasing function  $f$  supported in  $D$ ,

$$\mu(f; G \neq \emptyset) = \mathbb{E}_\mu[\gamma_G f; G \neq \emptyset] \leq \mu^-(f) + \|f\|_\infty \mu(G = \emptyset).$$

Let first  $L \uparrow \mathbb{Z}^2$  and then enlarge  $D$ . The exceptional probability tends to zero, so  $\mu \leq_{\text{st}} \mu^-$ . Since  $\mu^-$  is the minimal Gibbs state, equality follows. Thus  $\mu \neq \mu^-$  forces plus percolation. The percolation event is tail measurable, so extremality and the tail-triviality theorem from Volume VII make its probability 0 or 1. The minus assertion follows by spin flip.  $\square$

**Lemma 7.3** (Finite-energy uniqueness of infinite spin clusters). *Let  $\nu$  be an extremal Gibbs state which is periodic under a finite-index subgroup of translations. Then almost surely there is at most one infinite  $+$ -cluster and at most one infinite  $-$ -cluster.*

*Proof.* The one-site conditional probabilities are bounded away from 0 and 1 uniformly, since the interaction has finite range and finite coupling. Therefore every prescribed spin pattern in a fixed finite box has, conditionally on the exterior, probability at least a positive constant depending only on the box.

By periodic ergodicity, the number  $N_+$  of infinite plus clusters is almost surely a constant. If  $2 \leq N_+ < \infty$ , choose a finite box meeting all these clusters with positive probability. Changing finitely many spins inside the box to plus connects them into one cluster with positive probability, contradicting constancy of  $N_+$ . Hence  $N_+ \in \{0, 1, \infty\}$ .

If  $N_+ = \infty$ , with positive probability a finite box meets at least three distinct infinite plus clusters. A finite-energy modification inside the box can create a plus vertex  $v$  whose removal leaves at least three infinite plus components. Periodicity then gives a positive density of such trifurcation vertices. In a square  $Q_n$ , connect each trifurcation to three distinct branches and contract finite pieces until the first exit from  $Q_n$ . Erasing cycles yields a forest in which every interior trifurcation has degree at least three and every leaf lies on  $\partial Q_n$ . A finite forest with  $t$  vertices of degree at least three has at least  $t + 2$  leaves after suppressing degree-two vertices, so the number of trifurcations in  $Q_n$  is  $O(|\partial Q_n|)$ . Positive density would make it of order  $|Q_n|$ , a contradiction. Thus  $N_+ \neq \infty$ . The same proof applies to minus clusters.  $\square$

**Lemma 7.4** (Flip-reflection domination). *Let  $R$  be reflection in a horizontal or vertical lattice line and  $T$  the global spin flip. Suppose that for  $\mu$ -almost every configuration and every finite  $D$  there is an  $R$ -invariant finite random region  $G \supset D$ , determined from outside, such that*

$$\omega \geq RT\omega \quad \text{on } \partial G.$$

*Then  $\mu \geq_{\text{st}} \mu \circ RT$ . If the reversed boundary inequality can also be obtained, then  $\mu = \mu \circ RT$ .*

*Proof.* Fix an increasing  $0 \leq f \leq 1$  supported in  $D$  and choose a large  $R$ -invariant box  $L$  in which an admissible  $G$  exists with probability at least  $1 - \varepsilon$ . Choose the outermost admissible  $G$  in  $L$ ; this makes  $\{G = A\}$  exterior measurable. On  $G \neq \emptyset$ , boundary monotonicity and invariance of the Hamiltonian under  $RT$  give

$$\gamma_G^\omega(f) \geq \gamma_G^{RT\omega}(f) = \gamma_G^\omega(f \circ RT).$$

The strong Markov lemma therefore yields

$$\mu(f) \geq \mu(f \circ RT) - \varepsilon.$$

Let  $\varepsilon \downarrow 0$ . The final assertion follows by applying the same argument with the inequalities reversed.  $\square$

**Lemma 7.5** (Point-to-semicircuit estimate). *Assume  $\beta > \beta_c$  and put*

$$\vartheta := \mu^+(0 \text{ belongs to an infinite } +* \text{ cluster}) > 0.$$

*Let  $\pi$  be a half-plane with boundary line  $\ell$ , let  $x \in \ell$ , and let  $s$  be a  $+*$ -semicircuit in  $\pi$  surrounding  $x$ . If the boundary condition is plus on  $s$ , then in the reflected finite interior  $\Lambda = \text{Int}(s)$ ,*

$$\gamma_\Lambda(x \xleftrightarrow{+*} s) \geq \vartheta/2.$$

*Proof.* By monotonicity it is enough to put minus boundary spins on  $\partial\Lambda \setminus s$ . Let  $R$  be reflection in  $\ell$ . The resulting boundary condition dominates its image under  $RT$ , so the corresponding finite-volume law dominates its  $RT$  image.

Let  $B$  be the event that  $x$  is  $+*$ -connected to  $s$ . Let  $C$  be the event that  $x$  is enclosed by a plus circuit in  $\Lambda \cup s$  which is  $+*$ -connected to  $s$ . Starting from the spin at  $x$  and alternately taking outer boundaries of plus and minus star-clusters shows deterministically that

$$(B \cup C) \cup RT(B \cup C) = \Omega.$$

The domination therefore gives  $\gamma_\Lambda(B \cup C) \geq 1/2$ . On  $C$ , choose the outermost plus circuit  $c$  around  $x$  that is connected to  $s$ . Its interior is exterior-determined. Conditional on  $c$ , monotonicity compares the spins inside with the infinite-volume plus state, and hence

$$\gamma_\Lambda(B \mid C) \geq \vartheta.$$

Therefore  $\gamma_\Lambda(B) \geq \frac{1}{2}\vartheta$ . □

**Lemma 7.6** (Butterflies and periodic states). *For every extremal Gibbs state  $\mu$  at  $\beta > \beta_c$ , almost surely at least one pair of opposite half-planes contains infinite clusters of the same sign, one in each half-plane. Moreover every translation-periodic Gibbs state is a convex combination of  $\mu^+$  and  $\mu^-$ . In particular,*

$$\mu^+(E_{-*}) = 0, \quad \mu^-(E_{+*}) = 0, \tag{7.1}$$

*where  $E_{\pm*}$  denotes existence of an infinite  $\pm*$  cluster in the full plane.*

*Proof.* Call such a pair an infinite butterfly. Suppose an extremal state has no butterfly. For a pair of conjugate half-planes  $\pi, R\pi$ , absence of a butterfly implies that at least one of them has no infinite minus cluster and at least one has no infinite plus cluster. Tail triviality fixes which alternatives occur almost surely. If, for instance,  $\pi$  has no infinite minus cluster, every finite subset of  $\pi$  is surrounded by a plus-star semicircuit. Reflecting such a semicircuit produces an  $R$ -invariant circuit on which  $\omega \geq RT\omega$ . The opposite sign alternative gives the reverse inequality. Lemma 7.4 therefore yields

$$\mu = \mu \circ RT$$

for every horizontal and vertical coordinate reflection. Composing parallel reflections shows that  $\mu$  is periodic under  $2\mathbb{Z}^2$ .

The state cannot equal  $\mu^+$  or  $\mu^-$ , because those two states are interchanged by  $T$  and are distinct for  $\beta > \beta_c$ . Lemma 7.2 therefore gives both plus and minus percolation almost surely, and Lemma 7.3 makes both infinite clusters unique. Choose a square  $Q$  so large that the unique plus cluster meets it with probability larger than  $1 - 2^{-12}$ . Let  $A_k^+$  be the event that a plus path from

the  $k$ th quadrant of  $\partial Q$  reaches infinity outside  $Q$ . Since meeting the infinite plus cluster implies at least one  $A_k^+$ , positive association gives

$$\prod_{k=1}^4 \mu((A_k^+)^c) \leq \mu(Q \cap I_+ = \emptyset) < 2^{-12}.$$

Hence, after rotating the labels,  $\mu((A_1^+)^c) < 2^{-3}$ . Flip-reflection invariance maps this estimate to the three alternating events  $A_2^-, A_3^+, A_4^-$ . Therefore

$$\mu(A_1^+ \cap A_2^- \cap A_3^+ \cap A_4^-) > 1 - 4 \cdot 2^{-3} = 1/2.$$

On this event the unique plus and minus infinite clusters have alternating exits from  $Q$ , so planarity forces one of them to have two distinct infinite components outside  $Q$ . This contradicts their uniqueness. Thus an infinite butterfly must exist.

Now let  $\nu$  be periodic, say under  $p\mathbb{Z}^2$ . Conditioning on a periodic tail event preserves the DLR equations and periodicity. It therefore suffices to exclude simultaneous plus and minus percolation on any periodic tail component. Suppose both occur almost surely. By the butterfly lemma, for some pair of conjugate half-planes, say upper and lower, there are with positive probability infinite plus clusters in both half-planes. Choose a large square  $D$  which, with positive probability, meets these two plus clusters and an infinite minus cluster. On one side of  $D$  the minus cluster must leave between the upper and lower plus clusters; assume this happens on the right with positive probability.

Finite energy then gives positive probability to the following event  $A_0$ : the site  $(0,0)$  belongs to a doubly-infinite plus path whose upper and lower halves remain respectively in the upper and lower half-planes, while the neighbouring site  $(1,0)$  belongs to an infinite minus cluster. For  $k \in p\mathbb{Z}$ , let  $A_k$  be the translate of this event. Periodic recurrence implies that, on a positive-probability subset,  $A_k$  occurs for infinitely many  $k < 0$  and infinitely many  $k > 0$ . The corresponding doubly-infinite plus paths are pairwise ordered barriers. The infinite minus clusters attached to the neighboring sites between consecutive barriers cannot connect through a plus barrier, so infinitely many distinct infinite minus clusters occur. This contradicts Lemma 7.3. Hence simultaneous percolation has probability zero in every periodic Gibbs state.

The butterfly lemma then shows that a periodic ergodic component has exactly one percolating sign. If the plus sign percolates, the component cannot differ from  $\mu^+$ , because otherwise the spin-flipped form of Lemma 7.2 would force minus percolation. Thus the component is  $\mu^+$ ; similarly a minus-percolating component is  $\mu^-$ . Integrating the periodic ergodic decomposition proves the convex-combination statement.

The same recurrence proof also excludes coexistence of an infinite nearest-neighbour cluster of one sign with an infinite star-cluster of the opposite sign. Indeed, in the construction of  $A_0$  one may use a two-sided nearest-neighbour path for the first sign and a star-path for the second; the finite-energy local connection is unchanged, while the two-sided path is still a planar barrier for the opposite star-cluster. Hence every periodic Gibbs state satisfies

$$\nu(E_+ \cap E_{-*}) = \nu(E_- \cap E_{+*}) = 0.$$

Applying this to the pure phases, for which  $\mu^+(E_+) = \mu^-(E_-) = 1$ , gives  $\mu^+(E_{-*}) = 0$  and  $\mu^-(E_{+*}) = 0$ .  $\square$

**Lemma 7.7** (No plus percolation with a minus lower boundary). *Let  $\pi_+ = \{(i,j) : j \geq 0\}$  and let  $\eta^\pm$  be plus on  $\pi_+$  and minus on its complement. The monotone half-plane limit*

$$\mu_{\pi_+}^\pm = \lim_{\Lambda \uparrow \pi_+} \gamma_\Lambda^{\eta^\pm}$$

has no infinite  $+$  cluster inside  $\pi_+$ .

*Proof.* The monotone limit  $\mu_{\pi_+}^\pm$  is horizontally translation invariant and is the maximal Gibbs measure in the upper half-plane with the fixed minus exterior boundary. The usual maximal-state argument therefore makes it trivial on the upper-half-plane tail sigma-field.

For  $n \geq 0$  let

$$\nu_n := \mu_{\pi_+}^\pm \circ \theta_{(0,-n)}^{-1}.$$

Thus  $\nu_n$  is the same half-plane state translated down by  $n$  lattice spacings. Boundary monotonicity gives  $\nu_n \leq_{\text{st}} \nu_{n+1}$ , so  $\nu_n$  has a weak limit  $\nu$ . Horizontal invariance is inherited from every  $\nu_n$ , while  $\nu_{n+1} = \nu_n \circ \theta_{(0,-1)}^{-1}$  makes the limit vertically invariant as well. Every fixed finite set eventually lies strictly inside the translated half-plane, so the DLR equations pass to the limit and  $\nu$  is a full-plane Gibbs state. Lemma 7.6 therefore gives

$$\nu = a\mu^- + (1-a)\mu^+$$

for some  $a \in [0, 1]$ .

We show  $a > 0$ . Under  $\nu_n$  all spins below height  $-n$  are fixed minus. Horizontal ergodicity of the maximal half-plane state and finite energy imply that almost surely there are sites  $m_{\text{left}} < 0 < m_{\text{right}}$  for which the two vertical columns from height  $-n$  to 0, together with a segment of the fixed minus line below height  $-n$ , contain a minus semicircuit surrounding the origin. Choosing the outermost such semicircuit in a large box makes its interior exterior-determined. Lemma 7.5, with signs reversed, then gives

$$\nu_n(0 \xleftrightarrow{-*} \{j = -n\}) \geq \vartheta/2.$$

Consequently, if  $E_{0,m}^{-*}$  denotes that the origin belongs to a minus-star cluster of size at least  $m$ , then  $\nu_n(E_{0,m}^{-*}) \geq \vartheta/2$  whenever  $n > m$ . Letting first  $n \rightarrow \infty$  and then  $m \rightarrow \infty$  yields

$$\nu(E_{-*}) \geq \vartheta/2.$$

Since  $\mu^+(E_{-*}) = 0$ , we get  $a \geq \vartheta/2 > 0$ .

Finally  $\mu_{\pi_+}^\pm \leq_{\text{st}} \nu$ , and hence

$$\mu_{\pi_+}^\pm(E_{+*}^{\pi_+}) \leq \nu(E_{+*}) = 1 - a < 1.$$

The event on the left belongs to the upper-half-plane tail sigma-field, on which  $\mu_{\pi_+}^\pm$  is trivial. Its probability is therefore zero.  $\square$

**Lemma 7.8** (Half-plane shift lemma). *Let  $\pi$  and  $\tilde{\pi}$  be two parallel lattice half-planes with  $\tilde{\pi} \subset \pi$  and boundary lines one lattice spacing apart. For every extremal Gibbs state, plus percolation occurs in  $\pi$  if and only if it occurs in  $\tilde{\pi}$ . The same is true for minus percolation and for star adjacency. By iteration the statement holds for any finite translate of a half-plane.*

*Proof.* Only the implication from the larger to the smaller half-plane needs proof. Take  $\pi = \{j \geq 0\}$  and  $\tilde{\pi} = \{j \geq 1\}$ , and suppose there is no infinite plus cluster in  $\tilde{\pi}$ . For each  $n$ , choose the innermost minus-star semicircuit  $s_n$  in  $\tilde{\pi}$  which surrounds the rectangle  $[-n, n] \times [1, n]$  together with the previous semicircuit. Let  $x_n, y_n$  be the two sites on the line  $j = 0$  facing the endpoints of  $s_n$ . The pairs  $\{x_n, y_n\}$  may be chosen disjoint as  $n$  grows, and the event specifying the pair is measurable outside those two sites.

Let  $A_n$  be the event that both spins at  $x_n, y_n$  are minus. Uniform finite energy gives a number  $\delta > 0$  such that for every exterior configuration

$$\mu(A_n \mid \mathcal{F}_{\{x_n, y_n\}^c}) \geq \delta^2.$$

Fix  $N$ . On the event that none of  $A_n$ ,  $n > N$ , occurs, reveal the first available pair after  $N$ . Conditioning on its exterior and using the preceding lower bound shows that the probability of avoiding  $A_n$  at this new pair is at most  $1 - \delta^2$ . Iterating over the disjoint pairs yields

$$\mu(A_n^c \text{ for every } n > N) = 0.$$

Thus  $A_n$  occurs infinitely often almost surely. Whenever it occurs,  $s_n$  together with  $x_n, y_n$  and the boundary segment between them forms a minus-star semicircuit in the larger half-plane  $\pi$  surrounding  $[-n, n] \times [0, n]$ . Therefore every finite set is surrounded by such a circuit, and no infinite plus cluster can exist in  $\pi$ . This is the contrapositive of the desired implication. Spin flip and the same planar argument give the other variants.  $\square$

**Lemma 7.9** (Line touching and half-plane uniqueness). *In every half-plane  $\pi$ , every Gibbs state has almost surely at most one infinite plus cluster and at most one infinite plus-star cluster in  $\pi$ . Whenever such a cluster exists, outside every finite set it contains a path beginning on the boundary line of  $\pi$ . The same holds for minus clusters.*

*Proof.* Take  $\pi = \pi_+$ ; the other cases follow by symmetry. Fix an extremal Gibbs state and a site  $x \in \pi_+$ . Let  $A_{x,k}$  be the event that  $x$  lies in a plus cluster of size at least  $k$  which does not meet the horizontal axis. Choose a box  $D \subset \pi_+$  so large that every path of length  $k$  from  $x$  stays inside  $D$ . On  $A_{x,k}$  let  $G$  be the largest region containing  $x$  whose interior boundary away from the top side of  $D$  consists of minus spins. This random region is exterior-determined. Strong Markov resampling and boundary monotonicity give

$$\mu(A_{x,k}) \leq \gamma_D^{\eta^\pm}(x \text{ lies in a plus cluster of size at least } k).$$

Let  $D \uparrow \pi_+$  and then  $k \rightarrow \infty$ . Lemma 7.7 makes the right side tend to zero. Thus no infinite half-plane cluster can stay permanently away from the boundary line. A countable union over  $x$  and finite sets gives the line-touching assertion.

For uniqueness, if there is no infinite minus-star cluster in the half-plane, every finite set is surrounded by a plus semicircuit, which connects any two infinite plus paths. If an infinite minus-star cluster exists, it touches one boundary ray infinitely often. Every infinite plus cluster must then touch the opposite boundary ray infinitely often, and two such plus clusters are forced by planar order to meet. The plus-star case and the statements with signs interchanged are identical.  $\square$

**Lemma 7.10** (Both butterfly orientations occur). *Every extremal Gibbs state at  $\beta > \beta_c$  has almost surely a horizontal butterfly and a vertical butterfly, possibly of different signs.*

*Proof.* Suppose, for contradiction, that an extremal state has no vertical butterfly. The first part of the proof of Lemma 7.6 then gives invariance under every vertical flip–reflection and hence ergodicity under a nontrivial horizontal translation. The butterfly lemma supplies a horizontal butterfly; say it is plus.

Fix  $n$ . The half-plane shift lemma implies that the upper and lower infinite plus clusters of the horizontal butterfly survive in the translated half-planes  $\{j \geq n\}$  and  $\{j \leq -n\}$ . By the line-touching lemma each of these translated clusters reaches its boundary line infinitely often. Choose one upper and one lower boundary contact with the same bounded horizontal separation. Uniform finite energy gives positive probability to a vertical plus path across the finite strip between those two contacts, thereby joining the translated upper and lower clusters. Horizontal ergodicity and recurrence imply that such vertical connections occur infinitely often to the left and to the right. Together with paths inside the two translated half-plane clusters, any two sufficiently distant connections form a plus



circuit surrounding  $[-n, n]^2$ . Since  $n$  was arbitrary, every finite set is surrounded by a plus circuit. Hence no infinite minus cluster exists. Lemma 7.2, with signs exchanged, therefore forces the state to equal  $\mu^+$ . But  $\mu^+$  is not invariant under a vertical flip-reflection at  $\beta > \beta_c$ . Contradiction. The absence of a horizontal butterfly is ruled out identically.  $\square$

**Lemma 7.11** (Pinning a half-plane infinite cluster). *Let  $D$  be a fixed finite square and suppose an infinite plus-star cluster  $I_+^\pi$  in a half-plane  $\pi$  touches one boundary ray infinitely often. If  $x$  is sufficiently far out on that ray, then*

$$\mu(x \xleftrightarrow{+*} I_+^\pi \text{ in } (D \cup \text{opposite ray})^c) \geq \vartheta/4.$$

*The same lower bound holds after any combination of the following replacements: plus-star by minus nearest-neighbour connectivity, plus by minus, the chosen boundary ray by the opposite boundary ray, and  $\pi$  by any coordinate half-plane, with the corresponding infinite cluster in place of  $I_+^\pi$ .*

*Proof.* Outside  $D$ , the infinite component of  $I_+^\pi$  touches the relevant ray infinitely often. For  $x$  far enough, with probability at least  $1/2$  there are such contact points on both sides of  $x$ . They, together with a portion of  $I_+^\pi$ , contain a plus-star semicircuit  $s$  whose interior misses  $D$ . Choose the outermost such semicircuit in a large box. Lemma 7.5, applied conditionally inside its interior, gives probability at least  $\vartheta/2$  that  $x$  connects to  $s$ . Multiplying by the probability  $1/2$  of finding  $s$  gives  $\vartheta/4$ .  $\square$

**Lemma 7.12** (Unique semi-infinite contour). *Suppose an extremal Gibbs state has both an infinite plus-star cluster and an infinite minus cluster in the upper half-plane. Then almost surely there is a unique semi-infinite dual contour separating them. The contour starts at the horizontal axis and meets every horizontal line only finitely many times.*

*Proof.* By Lemma 7.9, both infinite clusters are unique and each meets the boundary line infinitely often. Choose the rightmost boundary point of the infinite minus cluster before the boundary points belonging to the infinite plus-star cluster and start the dual interface between the two neighboring spins. The minus side prevents the contour from returning to the boundary on one side of the starting point; the plus-star side prevents return on the other. Hence the contour is infinite.

Any other infinite contour has an infinite minus face and an infinite plus-star face. Uniqueness of the two half-plane clusters forces these faces to lie on the corresponding sides of the contour already constructed, so the two contours coincide. Now fix height  $h$ . By Lemma 7.8, both infinite faces survive in the translated half-plane  $\{j \geq h\}$ , so the same construction produces its unique contour. The translated contour must connect back to the original one, because its two faces belong to the same unique infinite clusters and the line-touching lemma supplies connections down to the original boundary. If the original contour crossed  $j = h$  infinitely often, two distinct tails in the translated half-plane would separate the same two unique faces, contradicting uniqueness there. Hence the original contour meets every horizontal line only finitely many times.  $\square$

**Lemma 7.13** (Fluctuation of duplicated contours). *Let  $\mu$  be extremal and suppose the upper half-plane has a semi-infinite contour almost surely. Let  $\hat{\mu} = \mu \circ \theta_{(1,0)}^{-1}$  and couple the two states independently. Then the two semi-infinite contours intersect infinitely often almost surely.*

*Proof.* Orient the contour so that, almost surely, its plus-star face lies on the same side; this is a tail choice. For  $n \geq 1$  write  $\pi_{n,+} = \{(i, j) : j \geq n\}$ . By the shift lemma and Lemma 7.12,  $\pi_{n,+}$  contains the same unique infinite plus-star face and minus face, separated by the unique continuation of the contour. With the orientation chosen above, define

$$a_n(\omega) := \max\{k \in \mathbb{Z} : (k, n) \text{ belongs to the unique infinite plus-star cluster in } \pi_{n,+}\}.$$

The maximum is finite: the contour meets the line  $j = n$  only finitely many times, and its minus infinite face occupies the opposite boundary ray. Work first with two independent copies  $\omega, \omega'$  having law  $\mu$  and compare the contour of  $\omega$  with the contour of the shifted configuration  $\theta_{(1,0)}\omega'$ . Translation covariance gives  $a_n(\theta_{(1,0)}\omega') = a_n(\omega') + 1$ . Put

$$d_n(\omega, \omega') := a_n(\omega) - a_n(\omega') = a_n(\omega) - a_n(\theta_{(1,0)}\omega') + 1.$$

If the two contours meet only finitely often, then above some level one lies strictly on one side of the other. Consequently the nonintersection event  $F$  is contained in

$$A \cup B, \quad A := \{d_n \geq 0 \text{ eventually}\}, \quad B := \{d_n \leq 0 \text{ eventually}\}. \quad (7.1)$$

The events  $A$  and  $B$  are measurable with respect to the product-tail sigma-field

$$\mathcal{T}^{(2)} := \bigcap_{\Lambda \in \mathbb{Z}^2} (\mathcal{F}_{\Lambda^c} \otimes \mathcal{F}_{\Lambda^c}),$$

because changing finitely many spins affects only a finite initial part of either contour. Extremality of  $\mu$  makes its ordinary tail sigma-field trivial, and Fubini's theorem then makes  $\mathcal{T}^{(2)}$  trivial. The product law is exchangeable in the two unshifted layers, and exchanging them sends  $d_n$  to  $-d_n$ . Hence  $\mathbf{P}(A) = \mathbf{P}(B)$ . If  $\mathbf{P}(F) > 0$ , then  $F \subset A \cup B$  gives  $\mathbf{P}(A) = \mathbf{P}(B) > 0$ . Both  $A$  and  $B$  belong to the product-tail sigma-field, so its triviality forces

$$\mathbf{P}(A) = \mathbf{P}(B) = 1.$$

Thus  $A \cap B$  has probability one, which means precisely that  $d_n = 0$  eventually almost surely.

We now show that this is impossible by a finite-energy perturbation. Let

$$A_{k,n} := \{a_{n+1}(\omega) = a_{n+1}(\omega') = k\}$$

and let  $D_{k,n} = \{(k, n), (k+1, n)\}$ . On  $A_{k,n}$ , prescribe in the first layer the spins  $(+, +)$  on  $D_{k,n}$  and in the second layer the spins  $(+, -)$ . This event depends on only two spins per layer. Uniform finite energy therefore gives a constant

$$\delta = [1 + e^{8\beta}]^{-4} > 0$$

(the displayed value is a convenient nearest-neighbour lower bound; any positive uniform bound is sufficient) such that, conditionally on the exterior of  $D_{k,n}$  in both layers, the prescribed pattern has probability at least  $\delta$ . The pattern forces the first contour to lie at least one lattice unit to the right of the second at level  $n$ , and hence forces  $d_n \geq 1$ . Since  $A_{k,n}$  is exterior measurable relative to  $D_{k,n}$ , summing over  $k$  yields

$$\mathbf{P}(d_n \geq 1) \geq \delta \mathbf{P}(d_{n+1} = 0). \quad (7.2)$$

If  $d_n = 0$  eventually almost surely, then  $\mathbf{P}(d_{n+1} = 0) \rightarrow 1$ , so (7.2) gives

$$\liminf_{n \rightarrow \infty} \mathbf{P}(d_n \geq 1) \geq \delta.$$

Fatou then gives positive probability that  $d_n \geq 1$  infinitely often, contradicting eventual equality  $d_n = 0$ . Therefore  $\mathbf{P}(F) = 0$ , and the two contours intersect infinitely often.  $\square$

**Lemma 7.14** (No disagreement percolation in the duplicated system). *Let  $\mu$  be extremal and  $\hat{\mu} = \mu \circ \theta_{(1,0)}^{-1}$ . Under the product law  $\mu \otimes \hat{\mu}$ , every finite square is almost surely surrounded by a star-circuit on which*

$$\omega \leq \hat{\omega}.$$

*Proof.* Fix a square  $D$  and boundary points  $x$  far to the left and  $y$  far to the right. We first show that with probability at least  $(\vartheta/4)^2$  there is a star-path from  $x$  to  $y$  above  $D$  on which  $\omega \leq \widehat{\omega}$ .

There are three cases. If the first layer has no infinite plus cluster in the upper half-plane, Lemmas 7.10 and 7.9 give an infinite minus-star cluster touching both boundary rays. The pinning lemma and positive association connect both  $x$  and  $y$  to it outside  $D$  with probability at least  $(\vartheta/4)^2$ . That minus-star path in the first layer automatically satisfies  $\omega \leq \widehat{\omega}$ . If instead the second layer has no infinite minus cluster, the same argument with signs interchanged gives a plus-star path in the second layer, again a disagreement path.

In the remaining case both signs occur in the upper half-plane. Each layer then has the unique semi-infinite contour of Lemma 7.12. By pinning, with probability at least  $(\vartheta/4)^2$ ,  $x$  in the second layer is connected by a plus-star path to its contour face while  $y$  in the first layer is connected by a minus-star path to its contour face. Lemma 7.13 makes the two contours intersect infinitely often, so these two pinned paths can be joined above  $D$  into a star-path satisfying  $\omega \leq \widehat{\omega}$ .

The analogous lower-half-plane connection has the same positive lower bound. Its indicator and the upper connection indicator are increasing functions of the coordinatewise difference  $\widehat{\omega} - \omega$ . Applying FKG first in the second layer and then in the first layer with reversed order yields positive correlation of the two connection events. Hence, with probability at least  $(\vartheta/4)^4$ , they occur simultaneously, producing a surrounding disagreement star-circuit. Existence of such a surrounding circuit outside  $D$  is a product-tail event. Product-tail triviality upgrades the positive probability to one.  $\square$

**Lemma 7.15** (Every extremal planar Ising Gibbs state is translation invariant). *Every extremal nearest-neighbour zero-field Ising Gibbs state on  $\mathbb{Z}^2$  is invariant under the two coordinate translations.*

*Proof.* Let  $\widehat{\mu} = \mu \circ \theta_{(1,0)}^{-1}$  and work under the independent product coupling. Fix a finite square  $D$  and an increasing local function  $0 \leq f \leq 1$  supported in  $D$ . By Lemma 7.14, with probability tending to one as  $L \uparrow \mathbb{Z}^2$ , there is in  $L$  a disagreement star-circuit surrounding  $D$  on which  $\omega \leq \widehat{\omega}$ . Let  $G$  be the interior of the outermost such circuit, or  $\emptyset$  if none exists. The region is determined from the exterior pair. On  $G \neq \emptyset$ , boundary monotonicity gives

$$\gamma_G^\omega(f) \leq \gamma_G^{\widehat{\omega}}(f).$$

Using the strong Markov property in the two layers,

$$\mu(f) \leq \widehat{\mu}(f) + \varepsilon_L,$$

where  $\varepsilon_L \rightarrow 0$ . Thus  $\mu \leq_{\text{st}} \widehat{\mu}$ . Interchanging the layers gives the reverse stochastic inequality, hence  $\mu = \widehat{\mu}$ . This proves horizontal invariance; vertical invariance is identical after rotating the argument by  $\pi/2$ .  $\square$

**Theorem 7.16** (Aizenman–Higuchi classification). *For the nearest-neighbor ferromagnetic Ising model on  $\mathbb{Z}^2$  at any inverse temperature, every infinite-volume Gibbs state is of the form*

$$\mu = \lambda \mu^+ + (1 - \lambda) \mu^- \quad (0 \leq \lambda \leq 1).$$

*Proof.* Let  $\mu$  be an arbitrary Gibbs state and disintegrate it over the tail sigma-field into extremal DLR states:

$$\mu = \int \nu \Pi_\mu(d\nu).$$

The extremal-decomposition theorem was proved in Volume VII, and the DLR property is preserved by the conditional measures.

Fix an extremal component  $\nu$ . Lemma 7.15 shows that  $\nu$  is invariant under both coordinate translations. The butterfly/half-plane analysis above then applies to  $\nu$ : its periodic-state conclusion gives

$$\nu \in \text{conv}\{\mu^+, \mu^-\}.$$

For completeness, the mechanism is the one established in the preceding lemmas: half-plane percolation detects the pure phases; if both signs persist, the unique semi-infinite contours in two translated copies intersect infinitely often; disagreement circuits then compare the two specifications around every finite set and force translation invariance. The butterfly argument excludes any translation-periodic extremal state other than the two pure phases.

Thus  $\nu = a\mu^+ + (1-a)\mu^-$ . Since  $\nu$  is extremal and  $\mu^\pm$  are extremal (proved in the theorem immediately below), either  $a = 0$  or  $a = 1$  whenever the phases are distinct; in the uniqueness regime  $\mu^+ = \mu^-$  anyway. Hence every extremal component is one of the two pure phases. Setting

$$\lambda = \Pi_\mu(\{\mu^+\})$$

and integrating the extremal decomposition yields

$$\mu = \lambda\mu^+ + (1-\lambda)\mu^-, \quad 0 \leq \lambda \leq 1.$$

□

**Theorem 7.17** (Extremality and distinctness of the pure phases). *For the nearest-neighbour ferromagnetic Ising model on  $\mathbb{Z}^2$ , the weak limits  $\mu^+$  and  $\mu^-$  obtained from plus and minus boundary conditions are respectively the maximal and minimal Gibbs states in stochastic order. Both are extremal DLR states. For  $\beta < \beta_c$  they coincide with the unique Gibbs state, whereas for  $\beta > \beta_c$  they are distinct.*

*Proof.* The finite-volume FKG inequality and monotonicity in the boundary condition, proved earlier in this volume, show that for every increasing cylinder function  $f$  and every DLR state  $\nu$ ,

$$\mu^-(f) \leq \nu(f) \leq \mu^+(f).$$

Passing to the thermodynamic limit therefore gives the asserted minimality and maximality.

We next prove extremality rather than quoting a tail-triviality criterion. Suppose  $\mu^+ = t\nu_1 + (1-t)\nu_2$  with  $0 < t < 1$  and  $\nu_1, \nu_2$  Gibbs states. Maximality gives  $\nu_i \leq_{\text{st}} \mu^+$ . Hence, for every increasing cylinder function  $0 \leq f \leq 1$ ,

$$0 = t(\mu^+(f) - \nu_1(f)) + (1-t)(\mu^+(f) - \nu_2(f)),$$

and both summands are nonnegative. Thus  $\nu_1(f) = \nu_2(f) = \mu^+(f)$  for every increasing cylinder  $f$ . Indicators of increasing cylinder events determine a probability measure on  $\{-1, +1\}^{\mathbb{Z}^2}$ : indeed, for a finite set  $\Lambda$ , the probabilities of the events  $\{\sigma_x = +1 \text{ for every } x \in A\}$ ,  $A \subseteq \Lambda$ , determine the full law on  $\{-1, +1\}^\Lambda$  by inclusion-exclusion. Consequently  $\nu_1 = \nu_2 = \mu^+$ . This proves that  $\mu^+$  is extremal. Applying the same argument to decreasing cylinder functions proves extremality of  $\mu^-$ .

For  $\beta < \beta_c$ , the uniqueness theorem proved in the preceding chapters gives  $\mu^+ = \mu^-$ . For  $\beta > \beta_c$ , spin-flip symmetry gives  $\mu^-(\sigma_0) = -\mu^+(\sigma_0)$ , while positive spontaneous magnetization gives  $m(\beta) := \mu^+(\sigma_0) > 0$ . Hence

$$\mu^+(\sigma_0) - \mu^-(\sigma_0) = 2m(\beta) > 0,$$

so the two extremal states are distinct. Notice that this theorem does *not* claim that these are the only extremal Gibbs states; that stronger planar classification is exactly the Aizenman–Higuchi theorem stated immediately above. □

## Chapter 8

# Reflection Positivity and the Infrared Bound

Reflection positivity turns geometric symmetry into an  $L^2$  inequality. The infrared bound is the Fourier-space consequence needed later for high-dimensional and continuous-symmetry arguments.

**Lemma 8.1** (Reflection positivity for the torus spin measure). *Let  $\mathbb{T}_L^d$  be an even discrete torus and let  $\vartheta$  be reflection in a coordinate hyperplane between two lattice layers. For a nearest-neighbor ferromagnet with a priori single-spin measure invariant under the spin symmetries, every function  $F$  of the spins in one half-torus satisfies*

$$\langle F \vartheta F \rangle \geq 0, \quad |\langle F \vartheta G \rangle|^2 \leq \langle F \vartheta F \rangle \langle G \vartheta G \rangle.$$

*Proof.* Split the Hamiltonian into interactions internal to the two halves and bonds crossing the reflection plane. For one crossing bond, ferromagnetism gives

$$e^{\beta J S_x \cdot S_{\vartheta x}} = \sum_{m=0}^{\infty} \frac{(\beta J)^m}{m!} (S_x \cdot S_{\vartheta x})^m.$$

Expanding  $(S_x \cdot S_{\vartheta x})^m$  in spin coordinates writes the factor as a sum  $\sum_{\alpha} c_{\alpha} C_{\alpha}(S_x) C_{\alpha}(S_{\vartheta x})$  with  $c_{\alpha} \geq 0$ . Multiplying over all crossing bonds preserves this form. After the spins in the two halves are integrated separately, the expectation  $\langle F \vartheta F \rangle$  is a sum of nonnegative coefficients times squares, hence is nonnegative. Applying positivity to  $F + tG$  and requiring the resulting quadratic polynomial in  $t$  to have nonpositive discriminant gives Cauchy–Schwarz.  $\square$

**Lemma 8.2** (Gaussian domination). *For an arbitrary vector field  $h = (h_x)$  on the torus define*

$$Z(h) = \int \exp \left[ -\frac{\beta}{2} \sum_{x,y} J_{xy} |S_x - S_y + h_x - h_y|^2 \right] \prod_x d\nu(S_x).$$

*Then  $Z(h) \leq Z(0)$  after quotienting  $h$  by additive constants.*

*Proof.* Reflection positivity from Lemma 8.1, applied to the shifted interaction, gives the chessboard inequality

$$Z(h)^2 \leq Z(h^+) Z(h^-),$$

where  $h^+$  and  $h^-$  are obtained by reflecting one half of  $h$  onto the other. Suppose  $h_*$  maximizes  $Z(h)$  modulo constants and, among maximizers, has the smallest number of nearest-neighbor bonds on

which  $h_x \neq h_y$ . If such a disagreement exists, choose a reflection plane crossing one. The chessboard inequality implies that both reflected fields  $h_*^+$  and  $h_*^-$  are again maximizers. At least one has strictly fewer disagreeing bonds because the chosen cross-plane mismatch is removed. This contradicts minimality. Thus every maximizer is constant, and constants disappear from the differences  $h_x - h_y$ . Hence  $Z(h) \leq Z(0)$ .  $\square$

**Theorem 8.3** (Infrared bound). *For a reflection-positive translation-invariant ferromagnetic spin system on a discrete torus, in the standard nearest-neighbor normalization and for nonzero Fourier mode  $k$ , the two-point function satisfies*

$$\widehat{G}(k) \leq \frac{C}{\beta \widehat{\Delta}(k)},$$

where  $\widehat{\Delta}(k)$  is the lattice-Laplacian symbol and  $C$  depends only on the spin normalization.

*Proof.* Let  $L$  be the weighted graph Laplacian

$$(Lh)_x = \sum_y J_{xy}(h_x - h_y), \quad \mathcal{E}(h) = \frac{1}{2} \sum_{x,y} J_{xy} |h_x - h_y|^2.$$

Fix a mean-zero scalar field  $h$  and a unit vector  $u$  in spin space, and insert the shift  $th_x u$  into Lemma 8.2. Since  $t = 0$  is a maximum of  $\log Z(th)$ , its second derivative there is nonpositive. Expanding the square in the shifted Hamiltonian gives

$$\log Z(th) - \log Z(0) = -\frac{\beta t^2}{2} \mathcal{E}(h) + \log \left\langle \exp \left( \beta t \sum_x (Lh)_x (S_x \cdot u) \right) \right\rangle.$$

The first derivative vanishes by spin symmetry. The second derivative inequality therefore yields

$$\beta^2 \text{Var} \left( \sum_x (Lh)_x (S_x \cdot u) \right) \leq \beta \mathcal{E}(h).$$

Equivalently, for every mean-zero scalar field  $v$  and  $h = L^{-1}v$ ,

$$\sum_{x,y} v_x v_y \langle (S_x \cdot u)(S_y \cdot u) \rangle \leq \frac{C}{\beta} \langle v, L^{-1}v \rangle,$$

where  $C$  only records the convention used for the spin-component normalization. The inverse is taken on the mean-zero subspace; this is precisely why the zero Fourier mode is excluded.

Take  $v_x = e^{ik \cdot x}$  with  $k \neq 0$  (or its real and imaginary parts). Translation invariance diagonalizes both covariance and  $L$ , while

$$Lv = \widehat{\Delta}(k)v.$$

After dividing by the volume, the preceding quadratic-form inequality becomes

$$\widehat{G}(k) \leq \frac{C}{\beta \widehat{\Delta}(k)}.$$

Thus reflection positivity supplies Gaussian domination, Gaussian domination supplies the inverse-Laplacian covariance bound, and Fourier diagonalization gives the claimed infrared estimate.  $\square$

**Theorem 8.4** (Bubble criterion implies the mean-field susceptibility upper bound). *Let  $\chi(\beta) = \sum_x G_\beta(0, x)$  be the susceptibility of a translation-invariant ferromagnetic model for  $\beta < \beta_c$ . Assume:*

1.  $\chi(\beta) < \infty$  for  $\beta < \beta_c$  and  $\chi(\beta) \uparrow \infty$  as  $\beta \uparrow \beta_c$ ;
2. the infrared/lace or rigorous-RG input gives, uniformly near  $\beta_c$ ,

$$\widehat{G}_\beta(k) \leq C_0 \widehat{\Delta}(k)^{-1} \quad (k \neq 0),$$

and the relevant bubble diagram is bounded by a constant  $B < \infty$ ;

3. the model's bubble differential inequality has the form

$$\frac{d}{d\beta} \chi(\beta) \geq c_B \chi(\beta)^2, \quad c_B > 0,$$

where  $c_B$  depends only on the bounded diagram and the fixed normalization.

Then

$$\chi(\beta) \leq \frac{1}{c_B(\beta_c - \beta)} \quad (\beta < \beta_c).$$

In particular the susceptibility exponent, whenever defined by a power law, satisfies the mean-field upper bound  $\gamma \leq 1$ . Moreover, the infrared estimate itself makes the bubble integral finite whenever  $\int_{[-\pi, \pi]^d} \widehat{\Delta}(k)^{-2} dk < \infty$ , in particular for the nearest-neighbour Laplacian when  $d > 4$ .

*Proof.* By Plancherel and the infrared estimate,

$$\sum_x G_\beta(0, x)^2 = (2\pi)^{-d} \int_{[-\pi, \pi]^d} |\widehat{G}_\beta(k)|^2 dk \leq C \int_{[-\pi, \pi]^d} \widehat{\Delta}(k)^{-2} dk.$$

Near  $k = 0$  one has  $\widehat{\Delta}(k) \asymp |k|^2$ , so the last integral behaves locally like  $\int_0^1 r^{d-5} dr$  and is finite precisely for  $d > 4$ . The same computation applies to whichever convolution diagram appears in the assumed lace/RG criterion; the point of the hypothesis is that its bound is uniform as  $\beta \uparrow \beta_c$ .

Now divide the differential inequality by  $\chi(\beta)^2$ :

$$-\frac{d}{d\beta} \frac{1}{\chi(\beta)} = \frac{\chi'(\beta)}{\chi(\beta)^2} \geq c_B.$$

Integrating from  $\beta$  to  $s < \beta_c$  gives

$$\frac{1}{\chi(\beta)} - \frac{1}{\chi(s)} \geq c_B(s - \beta).$$

Letting  $s \uparrow \beta_c$  and using  $\chi(s) \rightarrow \infty$  yields  $\chi(\beta)^{-1} \geq c_B(\beta_c - \beta)$ , which is the claimed upper bound.

The theorem is intentionally a *consequence theorem*: all model-specific work is isolated in the infrared/bubble estimate and the differential inequality. Later volumes prove those hypotheses in the model classes for which the series asserts high-dimensional mean-field behaviour; no lace-expansion or RG input is silently imported here.  $\square$

**Example 8.5** (Why  $d = 4$  is borderline). Near  $k = 0$ ,  $\widehat{\Delta}(k) \asymp |k|^2$ . The square of the infrared majorant therefore behaves like  $|k|^{-4}$ , whose integral near the origin is finite for  $d > 4$  and logarithmically divergent in  $d = 4$ . This elementary power count predicts exactly where logarithmic corrections must enter.

## Chapter 9

# Modern Random-Current Continuity and Emergent Planarity

The final chapter records two modern theorems that are conceptually important for universality: continuity of the three-dimensional transition and the re-emergence of planar fermionic/Pfaffian structure beyond exactly solvable nearest-neighbor models.

### Local random-current definition

**Definition 9.1** (Random current and source set). Let  $G = (V, E)$  be finite and assume zero external field. A *current* is a map  $n : E \rightarrow \mathbb{N}$ . Its weight and source set are

$$w(n) = \prod_{e \in E} \frac{(\beta J_e)^{n_e}}{n_e!}, \quad \partial n = \left\{ x \in V : \sum_{e \ni x} n_e \equiv 1 \pmod{2} \right\}.$$

For an even vertex set  $A$ , the current partition sum is  $Z_A^{\text{cur}} = \sum_{\partial n = A} w(n)$ . Expanding the Ising Boltzmann weights gives

$$\left\langle \prod_{x \in A} \sigma_x \right\rangle = \frac{Z_A^{\text{cur}}}{Z_{\emptyset}^{\text{cur}}}.$$

For two currents  $n_1, n_2$ , their *total support* is the graph formed by edges with  $(n_1 + n_2)_e > 0$ ; the switching lemma changes source parities by reallocating units along paths in this support.

*Worked example 9.2* (One edge: even and odd currents). On a graph consisting of one edge  $e = \{x, y\}$  with  $K = \beta J_e$ ,

$$Z_{\emptyset}^{\text{cur}} = \sum_{m \text{ even}} \frac{K^m}{m!} = \cosh K, \quad Z_{\{x, y\}}^{\text{cur}} = \sum_{m \text{ odd}} \frac{K^m}{m!} = \sinh K.$$

Hence

$$\langle \sigma_x \sigma_y \rangle = \tanh K,$$

matching both the direct two-spin calculation and the high-temperature edge weight from Chapter 1.

**Definition 9.3** (ADCS interaction class and long-range-order parameter). Let  $J = (J_{x,y})_{x,y \in \mathbb{Z}^d}$  satisfy

1.  $J_{x,y} = J_{0,y-x}$  (translation invariance),



2.  $J_{x,y} \geq 0$  (ferromagnetism),
3.  $\sum_y J_{0,y} < \infty$  (summability), and
4. the graph with edge set  $\{\{x,y\} : J_{x,y} > 0\}$  is connected (aperiodicity).

For the infinite-volume free state put

$$\widetilde{M}_{\text{LRO}}(\beta) := \inf_{\substack{B \subset \mathbb{Z}^d \\ 0 < |B| < \infty}} \frac{1}{|B|^2} \sum_{x,y \in B} \langle \sigma_x \sigma_y \rangle_\beta^0.$$

The spontaneous magnetization is  $m^*(\beta) = \langle \sigma_0 \rangle_\beta^+$ .

**Lemma 9.4** (Finite-volume source switching). *Let  $G \subset H$  be finite vertex sets, let  $x, y \in G$ , and let  $A \subset H$  have even cardinality. If  $n_1$  is a current supported on  $G$  and  $n_2$  a current supported on  $H$ , then for every function  $F$  of  $n_1 + n_2$ ,*

$$\sum_{\substack{\partial n_1 = \{x,y\} \\ \partial n_2 = A}} F(n_1 + n_2) w(n_1) w(n_2) = \sum_{\substack{\partial n_1 = \emptyset \\ \partial n_2 = A \Delta \{x,y\}}} F(n_1 + n_2) w(n_1) w(n_2) \mathbf{1}_{\{x \leftrightarrow y \text{ in } G \text{ through } n_1 + n_2\}}.$$

The same identity holds with a ghost vertex  $\delta$  used to encode plus boundary conditions, with  $\delta$  omitted from the displayed source set.

*Proof.* Fix the total current  $m = n_1 + n_2$ . The product weight factorizes as

$$w(n_1) w(n_2) = w(m) \prod_e \binom{m_e}{(n_1)_e}.$$

Thus only the number of allocations of the  $m_e$  labelled current units between the two replicas matters. If  $x$  and  $y$  are connected in the support of  $m$  inside  $G$ , choose any path  $\gamma$  from  $x$  to  $y$ . Toggle, along  $\gamma$ , which replica receives one distinguished unit on each traversed edge. Internal vertices of  $\gamma$  have two incident parities toggled, whereas the parities at  $x$  and  $y$  are toggled once. Hence this involution changes the source set of the first current by  $\{x, y\}$  and changes the second source set by the same symmetric difference. The binomial allocation weight is preserved after summation over allocations with fixed  $m$ . If  $x$  and  $y$  are disconnected there is no admissible toggle and the right-hand indicator is zero. Summing over  $m$  proves the identity.  $\square$

**Lemma 9.5** (Mixing on the spin observables seen by currents). *Let  $\langle \cdot \rangle_\beta^+$  and  $\langle \cdot \rangle_\beta^0$  be the infinite-volume plus and free states for an interaction in Definition 9.3. The plus state is mixing on local observables. The free state is mixing on local observables invariant under the global spin flip. In particular, for bounded local even functions  $F, G$ ,*

$$\langle F(G \circ \tau_z) \rangle_\beta^\# \longrightarrow \langle F \rangle_\beta^\# \langle G \rangle_\beta^\#, \quad |z| \rightarrow \infty, \quad \# \in \{0, +\}.$$

*Proof.* It is enough first to take  $F = \sigma_A$  and to take  $G$  from a multiplicative generating family. Let

$$\widetilde{H}(\sigma) = - \sum_{B \subset K} \widetilde{J}_B \sigma_B$$

be supported on a fixed finite  $K$ . Conditioning by  $e^{-\beta \widetilde{H} \circ \tau_z}$  is the same as changing finitely many couplings in a translate of  $K$ .

For the plus state choose  $\tilde{J}_B \geq 0$ . If  $|z|$  is large enough that  $A$  lies in a box  $\Lambda_{|z|/3}$  disjoint from  $K + z$ , Griffith monotonicity in the couplings and in the plus boundary condition gives

$$\langle \sigma_A \rangle_{\Lambda_{|z|/3}, \beta}^+ \geq \frac{\langle \sigma_A e^{-\beta \tilde{H} \circ \tau_z} \rangle_{\beta}^+}{\langle e^{-\beta \tilde{H}} \rangle_{\beta}^+} \geq \langle \sigma_A \rangle_{\beta}^+. \quad (9.1a)$$

The two outer terms have the same limit as  $|z| \rightarrow \infty$ , so the middle ratio converges to  $\langle \sigma_A \rangle_{\beta}^+$ .

For the free state remove, rather than add, a finite family of ferromagnetic couplings: take  $-J_B \leq \tilde{J}_B \leq 0$  on the existing interaction edges. The free-boundary monotonicity is reversed with the volume and yields

$$\langle \sigma_A \rangle_{\Lambda_{|z|/3}, \beta}^0 \leq \frac{\langle \sigma_A e^{-\beta \tilde{H} \circ \tau_z} \rangle_{\beta}^0}{\langle e^{-\beta \tilde{H}} \rangle_{\beta}^0} \leq \langle \sigma_A \rangle_{\beta}^0. \quad (9.1b)$$

Again the ratio converges to the infinite-volume expectation.

Linear combinations and uniform approximation extend (9.1a) to arbitrary bounded local  $G$  in the plus state. In the free state, the allowed perturbations generate the algebra of products  $\sigma_x \sigma_y$  along positive-coupling edges. Since that graph is connected, every even monomial  $\sigma_A$  is a product of such edge monomials after pairing the vertices of  $A$  and choosing connecting paths. Hence the generated local algebra is exactly the even algebra. A second linear approximation in  $F$  completes the proof.  $\square$

**Lemma 9.6** (Infinite-volume free and plus currents). *Under Definition 9.3, the sourceless current laws in boxes  $\Lambda_L$  with free boundary condition and with plus boundary condition encoded by a ghost vertex have local weak limits  $\mathbf{P}_{\beta}^0$  and  $\mathbf{P}_{\beta}^+$ . Both limits are translation invariant, and their restrictions to current cylinder events are ergodic under translations.*

*Proof.* For an edge  $e$  write  $r_e = (-1)^{n_e}$ . Conditional on the parity field  $(r_e)$ , the current variables are independent Poisson variables of means  $\beta J_e$  conditioned to have the prescribed parities. Hence local convergence reduces to convergence of the parity field.

For a finite edge set  $E$ , let  $C_E$  be the event that  $n_e$  is even for every  $e \in E$ . Expanding the current weights edge by edge gives, for either free or plus boundary condition,

$$\mathbf{P}_{\Lambda_L, \beta}^{\#}(C_E) = \left( \prod_{e=\{u,v\} \in E} \cosh(\beta J_{u,v}) \right) \left\langle \exp \left( -\beta \sum_{e=\{u,v\} \in E} J_{u,v} \sigma_u \sigma_v \right) \right\rangle_{\Lambda_L, \beta}^{\#}. \quad (9.1)$$

The spin expectation on the right has an infinite-volume limit because it is a bounded local function. Inclusion–exclusion therefore gives convergence of every finite parity cylinder, and the conditional Poisson description then gives convergence of every current cylinder.

Translation invariance follows by applying (9.1) in translated boxes. For ergodicity, it suffices to treat two parity cylinders  $C_E$  and  $C_F$ . Formula (9.1) reduces their widely separated covariance to the covariance of two bounded local functions that are invariant under global spin flip. Lemma 9.5 therefore gives

$$\mathbf{P}_{\beta}^{\#}(C_E \cap \tau_z C_F) \longrightarrow \mathbf{P}_{\beta}^{\#}(C_E) \mathbf{P}_{\beta}^{\#}(C_F) \quad (|z| \rightarrow \infty).$$

Cylinder approximation proves ergodicity.  $\square$

**Lemma 9.7** (Uniqueness of the infinite double-current cluster). *Let  $\mathbf{P}_{\beta}$  be the law of the support of  $n^0 + n^+$  for independent currents  $n^0 \sim \mathbf{P}_{\beta}^0$  and  $n^+ \sim \mathbf{P}_{\beta}^+$ . Then  $\mathbf{P}_{\beta}$ -almost surely there is at most one infinite connected component.*

*Proof.* We first record the finite-energy substitute needed for a Burton–Keane argument. Fix a finite box  $K$ . If an edge  $e \subset K$  with  $J_e > 0$  is absent from the double-current support, change the plus current on that edge from 0 to 2. This preserves all source parities and makes  $e$  present. On a fixed finite set of edges the ratio of the modified weight to the original weight is bounded below by a positive constant depending only on  $K, \beta, J$ . Accounting for the finitely many possible preimages shows that there is  $c_K > 0$  such that, for every event  $A$ ,

$$\mathbf{P}_\beta(\Phi_K A) \geq c_K \mathbf{P}_\beta(A), \quad (9.2)$$

where  $\Phi_K$  opens all positive-coupling edges with both endpoints in  $K$ .

By Lemma 9.6, the number  $N_\infty$  of infinite support clusters is almost surely constant. If  $2 \leq N_\infty < \infty$ , choose  $K$  which meets every infinite cluster with positive probability. On that event  $\Phi_K$  merges all of them, and (9.2) gives positive probability of  $N_\infty = 1$ , contradicting ergodicity.

Suppose now  $N_\infty = \infty$ . With positive probability a finite box meets three distinct infinite clusters. After applying  $\Phi_K$ , one obtains with positive probability a *coarse trifurcation*: a bounded connected core whose removal leaves at least three infinite components. Translation invariance would then give a positive density of coarse trifurcations. Connect coarse trifurcations in a large box to the three exterior branches supplied by their clusters and erase cycles. The resulting graph is a forest in which every interior trifurcation has degree at least three, so the number of such vertices is bounded by the number of leaves. Every leaf is supported by an open interaction edge crossing the boundary of the large box.

For either finite-volume sourceless current, differentiating the current partition sum with respect to  $\beta J_e$  gives

$$\mathbf{E}(n_e) = \beta J_e \langle \sigma_x \sigma_y \rangle \leq \beta J_e \quad (e = \{x, y\}),$$

and the same bound passes to the infinite-volume limits. Hence the union of two independent currents satisfies  $\mathbf{P}_\beta(e \text{ is open}) \leq 2\beta J_e$  by Markov's inequality. Consequently the expected number of boundary-crossing leaves in  $\Lambda_N$  is bounded by

$$C_\beta \sum_{x \in \Lambda_N} \sum_{y \notin \Lambda_N} J_{x,y} = o(|\Lambda_N|), \quad (9.3)$$

Indeed, after writing  $y = x + z$  and dividing by  $|\Lambda_N|$ , the coefficient of each fixed  $J_{0,z}$  is the fraction of  $x \in \Lambda_N$  for which  $x + z \notin \Lambda_N$ ; it tends to zero and is bounded by one. Dominated convergence with  $\sum_z J_{0,z} < \infty$  proves (9.3). Positive density of trifurcations would instead force a quantity of order  $|\Lambda_N|$ . This contradiction excludes  $N_\infty = \infty$ .  $\square$

**Lemma 9.8** (Long-range order controls current percolation). *For the double-current law of Lemma 9.7,*

$$\mathbf{P}_\beta(0 \leftrightarrow \infty)^2 \leq \widetilde{M}_{\text{LRO}}(\beta).$$

*In particular,  $\widetilde{M}_{\text{LRO}}(\beta) = 0$  implies that the double current has no infinite cluster.*

*Proof.* Apply Lemma 9.4 in  $\Lambda_L$  to one sourceless free current and one sourceless plus current. The standard two-source current representation gives

$$\mathbf{P}_{\Lambda_L, \beta}^0 \otimes \mathbf{P}_{\Lambda_L, \beta}^+(x \leftrightarrow y) = \langle \sigma_x \sigma_y \rangle_{\Lambda_L, \beta}^0 \langle \sigma_x \sigma_y \rangle_{\Lambda_L, \beta}^+ \leq \langle \sigma_x \sigma_y \rangle_{\Lambda_L, \beta}^0. \quad (9.4)$$

Local convergence, followed by approximation of connectivity by connection to a large finite radius, yields

$$\mathbf{P}_\beta(x \leftrightarrow y) \leq \langle \sigma_x \sigma_y \rangle_\beta^0. \quad (9.5)$$

For finite  $B$ , set  $X = \sum_{x \in B} \mathbf{1}_{\{x \leftrightarrow \infty\}}$ . Translation invariance and Cauchy–Schwarz give

$$|B|^2 \mathbf{P}_\beta(0 \leftrightarrow \infty)^2 \leq \mathbf{E}_\beta X^2 = \sum_{x,y \in B} \mathbf{P}_\beta(x, y \leftrightarrow \infty).$$

Uniqueness of the infinite cluster implies  $\{x, y \leftrightarrow \infty\} \subseteq \{x \leftrightarrow y\}$ . Use (9.5), divide by  $|B|^2$ , and take the infimum over  $B$ .  $\square$

**Lemma 9.9** (Free/plus comparison from the double current). *Fix  $x, y$  and a finite path  $x = x_0, \dots, x_m = y$  using only edges with positive coupling. There is a finite constant  $C_{x,y}(\beta, J)$  such that*

$$0 \leq \langle \sigma_x \sigma_y \rangle_\beta^+ - \langle \sigma_x \sigma_y \rangle_\beta^0 \leq C_{x,y}(\beta, J) \mathbf{P}_\beta(x \leftrightarrow \infty). \quad (9.6)$$

*Hence absence of an infinite double-current cluster implies equality of all free and plus two-point functions.*

*Proof.* In a box  $\Lambda_L$  containing the chosen path, subtract the free-current representation of  $\langle \sigma_x \sigma_y \rangle_\beta^0$  from the plus-current representation. One use of Lemma 9.4 rewrites the difference as a nonnegative sum over pairs  $(n^0, n^+)$  in which the plus current has sources  $\{x, y\}$  but  $x$  and  $y$  are disconnected inside  $\Lambda_L$  in the total support. Since a current with sources  $x, y$  has an odd-degree path joining each source either to the other source or to the ghost, disconnection inside  $\Lambda_L$  forces the source component to reach the ghost.

Flip the parity of  $n^+$  along the fixed positive-coupling path from  $x$  to  $y$  and, whenever necessary, add one unit so every toggled edge remains nonnegative. This removes the two sources. On edge  $e$  the ratio between the sourceful and sourceless parity sums is bounded by  $\max\{\tanh(\beta J_e)^{-1}, \coth(\beta J_e)\}$ ; multiplying over the finite path gives a constant  $C_{x,y}$ . We obtain

$$0 \leq \langle \sigma_x \sigma_y \rangle_{\Lambda_L, \beta}^+ - \langle \sigma_x \sigma_y \rangle_{\Lambda_L, \beta}^0 \leq C_{x,y} \mathbf{P}_{\Lambda_L, \beta}^0 \otimes \mathbf{P}_{\Lambda_L, \beta}^+(x \leftrightarrow \delta).$$

Let  $L \rightarrow \infty$ . Ghost connection becomes connection to arbitrarily large distance, and the current limits yield (9.6).  $\square$

**Lemma 9.10** (Infrared bound implies vanishing free-state LRO). *Assume in addition that the interaction is reflection positive and let*

$$\mathcal{E}(k) = \sum_x J_{0,x} (1 - \cos(k \cdot x)).$$

*If*

$$\int_{[-\pi, \pi]^d} \frac{dk}{\mathcal{E}(k)} < \infty, \quad (9.7)$$

*then  $\widetilde{M}_{\text{LRO}}(\beta_c) = 0$ . In particular, (9.7) holds for nearest-neighbor interaction on  $\mathbb{Z}^d$  when  $d \geq 3$ .*

*Proof.* Apply the infrared bound of Chapter 8 on a torus  $T_L$ . For a mean-zero test function  $v$  it gives

$$\sum_{x,y} v_x v_y \langle \sigma_x \sigma_y \rangle_{T_L, \beta} \leq \frac{C}{\beta} \sum_{k \neq 0} \frac{|\widehat{v}(k)|^2}{\mathcal{E}(k)}. \quad (9.8)$$

For  $\beta < \beta_c$ , the subcritical exponential-decay theorem proved in Chapter 3 implies that the zero Fourier mode divided by  $|T_L|$  tends to zero. Hence (9.8) may be applied, after subtracting the

average of  $v$ , to  $v = |B|^{-1} \mathbf{1}_B$  and then  $L \rightarrow \infty$ . Free-boundary correlations are bounded above by periodic ones and increase with both volume and  $\beta$ , so finally letting  $\beta \uparrow \beta_c$  gives

$$\frac{1}{|B|^2} \sum_{x,y \in B} \langle \sigma_x \sigma_y \rangle_{\beta_c}^0 \leq \frac{C}{\beta_c} \int_{[-\pi, \pi]^d} \frac{|\widehat{\mathbf{1}}_B(k)|^2}{|B|^2 \mathcal{E}(k)} \frac{dk}{(2\pi)^d}. \quad (9.9)$$

Take  $B = \Lambda_N$ . The factor  $|\widehat{\mathbf{1}}_{\Lambda_N}(k)|^2/|\Lambda_N|^2$  is at most one and tends to zero for every  $k \neq 0$ . The integrability hypothesis (9.7) and dominated convergence show that the right side tends to zero. Thus the infimum defining  $\widetilde{M}_{\text{LRO}}(\beta_c)$  is zero.

For nearest-neighbor interaction,  $\mathcal{E}(k) \asymp |k|^2$  near the origin, and  $|k|^{-2}$  is locally integrable exactly for  $d > 2$ .  $\square$

**Theorem 9.11** (Aizenman–Duminil–Copin–Sidoravicius continuity criterion). *For every interaction in Definition 9.3,*

$$\widetilde{M}_{\text{LRO}}(\beta_c) = 0 \implies m^*(\beta_c) = 0,$$

*and the Gibbs state at  $\beta_c$  is unique. Consequently every reflection-positive interaction satisfying (9.7) has a continuous transition. In particular, the nearest-neighbor Ising model on  $\mathbb{Z}^3$  has*

$$m^*(\beta_c) = 0.$$

*Proof.*

$$\mathbf{P}_{\beta_c}(0 \leftrightarrow \infty)^2 \leq \widetilde{M}_{\text{LRO}}(\beta_c) = 0,$$

so the double current has no infinite cluster at the critical point.

$$\langle \sigma_x \sigma_y \rangle_{\beta_c}^+ = \langle \sigma_x \sigma_y \rangle_{\beta_c}^0 \quad (x, y \in \mathbb{Z}^d). \quad (9.10)$$

$$m^*(\beta_c)^2 = \langle \sigma_x \rangle_{\beta_c}^+ \langle \sigma_y \rangle_{\beta_c}^+ \leq \langle \sigma_x \sigma_y \rangle_{\beta_c}^+.$$

Average over  $x, y \in B$ , use (9.10), and take the infimum over finite  $B$ . This gives  $m^*(\beta_c)^2 \leq \widetilde{M}_{\text{LRO}}(\beta_c) = 0$ .

$$\mathbf{E}(\sigma_x^+ - \sigma_x^-) = 2m^*(\beta_c) = 0,$$

while  $\sigma_x^- \leq \sigma_x^+$  almost surely. Thus the two spins agree almost surely at every  $x$ ; countability of  $\mathbb{Z}^d$  gives  $\mu_{\beta_c}^- = \mu_{\beta_c}^+$ . The FKG bracketing then forces every Gibbs state to equal this common state.  $\square$

**Definition 9.12** (Backbone of a current). Fix once and for all an ordering of the directed interaction edges. If a current  $n$  has finite even source set  $X = \{x_1, \dots, x_{2m}\}$  and only finite connected components, its *backbone*  $\Gamma(n)$  is the lexicographically first collection of pairwise edge-disjoint oriented paths  $\gamma_1, \dots, \gamma_m$  supported on odd-current edges which pairs all sources. Removing a backbone path reverses the parity precisely at its two endpoints, so the construction terminates by induction on  $|X|$ .

**Lemma 9.13** (Pfaffian-defect/backbone inequality). *Let  $X = (x_1, \dots, x_{2m})$  be cyclically ordered boundary sources in the upper half-plane, let  $n_1, n_2$  be currents with  $\partial n_1 = X$  and  $\partial n_2 = \emptyset$ , and write  $\Gamma_1$  for the backbone of  $n_1$ . Then*

$$\left| \sum_{\ell=1}^{2m} (-1)^\ell \mathbf{1}_{\{x_1 \leftrightarrow x_\ell \text{ in } n_1+n_2\}} \right| \leq m \sum_{i < j < k < \ell} \mathbf{1}_{\{x_i \xrightarrow{\Gamma_1} x_k\}} \mathbf{1}_{\{x_j \xrightarrow{\Gamma_1} x_\ell\}} \mathbf{1}_{\{x_i \not\leftrightarrow x_j \text{ in } n_1+n_2\}}. \quad (9.11)$$

Thus a failure of the Pfaffian cancellation forces two interlaced backbone pairs which remain unconnected in the double current.

*Proof.* If one of the events on the right occurs, the right side is at least  $m$ , while the alternating sum on the left has absolute value at most  $m$ , so the inequality is immediate. Suppose therefore that no interlaced pair on the right occurs. Let  $C$  be the connected component of  $x_1$  in  $n_1 + n_2$ . If  $C$  contains all sources, its contribution to the alternating sum is  $\sum_{\ell=1}^{2m} (-1)^\ell = 0$ . Otherwise delete, one at a time, every backbone arc whose two endpoints lie outside  $C$ . Because no interlaced unconnected pair exists, the sources remaining in  $C$  always form a union of consecutive even blocks in the boundary order: an odd block would force one backbone arc from inside the block to cross, in the cyclic order, an arc joining two sources outside  $C$ , giving exactly one of the forbidden events. The alternating sum over each consecutive even block is zero. Summing the blocks proves (9.11).  $\square$

**Lemma 9.14** (Backbone weights and depletion factorization). *For a finite graph  $G$  and an admissible deterministic backbone collection  $\gamma$ , define*

$$\rho_G(\gamma) := \frac{1}{Z_G} \sum_{\partial n = \partial \gamma} w(n) \mathbf{1}_{\{\Gamma(n) = \gamma\}}.$$

Then

$$\langle \sigma_A \rangle_G = \sum_{\partial \gamma = A} \rho_G(\gamma). \quad (9.11a)$$

If the deterministic backbone algorithm first explores  $\gamma_1$  and then  $\gamma_2$ , write  $\bar{\gamma}_1$  for the interaction edges whose parities have already been fixed by the first exploration. Then

$$\rho_G(\gamma_1 \circ \gamma_2) = \rho_G(\gamma_1) \rho_{G \setminus \bar{\gamma}_1}(\gamma_2). \quad (9.11b)$$

Consequently, if  $\gamma_2$  is allowed to range over all continuations from  $y$  to  $z$ ,

$$\sum_{\gamma_2} \rho_G(\gamma_1 \circ \gamma_2) \leq \rho_G(\gamma_1) \langle \sigma_y \sigma_z \rangle_G. \quad (9.11c)$$

The same identities hold in the upper-half-plane limit.

*Proof.* Partition the current sum with source set  $A$  according to the deterministic backbone  $\Gamma(n)$ ; this is (9.11a). Once the beginning  $\gamma_1$  has been exposed, every edge in  $\bar{\gamma}_1$  has a prescribed parity. Summing its current multiplicity gives exactly the first-backbone factor  $\rho_G(\gamma_1)$ . On all remaining edges, the residual current has the source set dictated by the terminal point of  $\gamma_1$  and is distributed with the couplings on  $\bar{\gamma}_1$  deleted. The product form of the current weight therefore separates the two sums and proves (9.11b). Summing (9.11b) over every continuation  $\gamma_2 : y \rightarrow z$  gives

$$\rho_G(\gamma_1) \langle \sigma_y \sigma_z \rangle_{G \setminus \bar{\gamma}_1}.$$

Deleting ferromagnetic couplings can only decrease a spin correlation, by Griffiths monotonicity, yielding (9.11c). Monotone exhaustion and the already constructed infinite-volume current limits give the half-plane identities.  $\square$

**Lemma 9.15** (Finite-range Messenger–Miracle–Solé comparison). *Let the interaction satisfy the hypotheses of Theorem 9.23 below and let  $R$  be its range. In  $G = \mathbb{Z}^2$  or  $G = \mathbb{H}$ , let  $\mathcal{R}_L$  denote reflection in a lattice symmetry line  $x_1 = L$ . There is  $C = C(J, \beta, R) < \infty$  such that, whenever  $x$  lies on the far side of that line,*

$$\langle \sigma_0 \sigma_x \rangle_G \leq C \langle \sigma_0 \sigma_{\mathcal{R}_L x} \rangle_G. \quad (9.11d)$$

In particular there is  $c_1 < \infty$  such that boundary correlations are decreasing up to  $c_1$ : for  $0 \leq a \leq b$ ,

$$\langle \sigma_0 \sigma_{(b,0)} \rangle_{\mathbb{H}} \leq c_1 \langle \sigma_0 \sigma_{(a,0)} \rangle_{\mathbb{H}}. \quad (9.11e)$$

For each fixed  $K$  there is moreover  $c_0(K) < \infty$  such that, if  $z$  is within  $K$  of  $(b, 0)$ ,

$$\langle \sigma_0 \sigma_z \rangle_{\mathbb{H}} \leq c_0(K) \langle \sigma_0 \sigma_{(a,0)} \rangle_{\mathbb{H}}. \quad (9.11f)$$

*Proof.* We first prove the reflection estimate. In a finite reflection-symmetric exhaustion split the edge set, relative to the line  $x_1 = L$ , into left edges  $E_-$ , right edges  $E_+$  and crossing edges  $E_0$ . For a current with sources  $\{0, x\}$  reflect the  $E_+$  part and superpose it with the  $E_-$  part. If the model is nearest-neighbour, this reflected multigraph contains a path from  $\mathcal{R}_L x$  to the reflecting line. Choose the first such path in a fixed ordering and switch the current allocation along it. The switching map is weight preserving and injects the current sum with sources  $\{0, x\}$  into the sum with sources  $\{0, \mathcal{R}_L x\}$ . Hence the nearest-neighbour reflection constant is one.

For range  $R$ , a path can jump across one line, but it cannot jump across all  $R$  consecutive symmetry lines  $x_1 = L, L+1, \dots, L+R-1$ . Let  $E_j$  be the event that the reflected multigraph reaches the line  $L+j$ . For each fixed  $j$  the preceding switching injection, first in a finite exhaustion symmetric about that line and then in the infinite-volume limit, gives

$$\langle \sigma_0 \sigma_x \rangle_G \mathbf{P}_G^{\{0,x\}}(E_j) \leq \langle \sigma_0 \sigma_{\mathcal{R}_{L+j}x} \rangle_G.$$

Since  $\sum_{j=0}^{R-1} \mathbf{P}(E_j) \geq 1$ ,

$$\langle \sigma_0 \sigma_x \rangle_G \leq \sum_{j=0}^{R-1} \langle \sigma_0 \sigma_{\mathcal{R}_{L+j}x} \rangle_G. \quad (9.11g)$$

For every  $j < R$ , Griffiths' inequality gives

$$\langle \sigma_0 \sigma_{\mathcal{R}_L x} \rangle_G \geq \langle \sigma_0 \sigma_{\mathcal{R}_{L+j}x} \rangle_G \langle \sigma_{\mathcal{R}_{L+j}x} \sigma_{\mathcal{R}_L x} \rangle_G.$$

The last factor is uniformly bounded below by a positive constant because the two sites are at bounded distance and the positive-coupling graph is connected. Combining this with (9.11g) proves (9.11d). Choosing a reflection which moves  $(b, 0)$  toward  $(a, 0)$  gives (9.11e), with a harmless parity adjustment absorbed in the same bounded-distance constant. Finally, for  $z$  within distance  $K$  of  $(b, 0)$ ,

$$\langle \sigma_0 \sigma_{(b,0)} \rangle \geq \langle \sigma_0 \sigma_z \rangle \langle \sigma_z \sigma_{(b,0)} \rangle,$$

and the second factor has a positive minimum over the finitely many displacements of length at most  $K$ . Together with (9.11e) this proves (9.11f).  $\square$

**Lemma 9.16** (Boundary-strip avoided-intersection estimate). *Assume*

$$c(r) := \langle \sigma_0 \sigma_{(r,0)} \rangle_{\mathbb{H}, \beta_c} \longrightarrow 0 \quad (r \rightarrow \infty). \quad (9.11h)$$

Fix  $M$  and cyclically ordered boundary sources  $X = (x_1, \dots, x_{2m})$ . For  $i < j < k < \ell$ , let  $B_{i,j,k,\ell}^1$  be the event under the  $X$ -source current that the backbone contains strands pairing  $x_i$  with  $x_k$  and  $x_j$  with  $x_\ell$ , while these two strands lie in distinct current components. Let  $Y(n)$  be their coarse-intersection set, i.e. the lattice sites within a fixed range-dependent distance of both polygonal strands. Then

$$\lim_{d(X) \rightarrow \infty} \mathbf{P}_{\mathbb{H}}^X(B_{i,j,k,\ell}^1 \cap \{Y(n) \subset S_M\}) = 0, \quad S_M = \mathbb{Z} \times [0, M]. \quad (9.11i)$$

*Proof.* We give the full backbone estimate, first for four sources. Relabel them  $x_1 < x_2 < x_3 < x_4$  and write  $\Gamma_{13}, \Gamma_{24}$  for the two interlaced strands. Put  $D_X = \langle \sigma_{x_1} \sigma_{x_2} \sigma_{x_3} \sigma_{x_4} \rangle_{\mathbb{H}}$ .

*Near-source exclusion.* Fix  $K$ . Decompose according to the first point  $z \in \Lambda_K(x_2)$  visited by  $\Gamma_{13}$  and then according to the pieces of the two backbones before and after  $z$ . Repeated use of Lemma 9.14, followed by Lemma 9.15 and Griffiths' inequality, gives

$$\mathbf{P}_{\mathbb{H}}^X(B^1; \Gamma_{13} \cap \Lambda_K(x_2) \neq \emptyset) \leq C_K \langle \sigma_{x_2} \sigma_{x_3} \rangle_{\mathbb{H}}. \quad (9.11j)$$

Indeed, after the three backbone pieces are summed, the numerator is bounded by

$$C_K \langle \sigma_{x_2} \sigma_{x_4} \rangle \langle \sigma_{x_1} \sigma_{x_2} \rangle \langle \sigma_{x_2} \sigma_{x_3} \rangle,$$

and (9.11e) replaces the first factor by a constant multiple of the corresponding correlation at the right pair. Griffiths gives

$$D_X \geq \langle \sigma_{x_1} \sigma_{x_2} \rangle \langle \sigma_{x_3} \sigma_{x_4} \rangle.$$

Thus (9.11j) tends to zero as  $d(X) \rightarrow \infty$ . The symmetric estimate holds for  $\Gamma_{24}$  near  $x_3$ .

A *flight of height  $M$*  of a strand is a maximal subpath whose endpoints lie in  $S_M$  and whose interior lies above  $S_M$ . A flight is above  $x_2$  if its endpoints lie on opposite sides of the vertical line through  $x_2$  but not of the line through  $x_3$ . On  $B^1 \cap \{Y \subset S_M\}$ , if  $\Gamma_{13}$  has such a flight, then  $\Gamma_{24}$  must hit one of the two  $M$ -boxes around its endpoints: otherwise it would either cross the flight at distance at most the interaction range outside  $S_M$ , creating a point of  $Y \setminus S_M$ , or it could not connect  $x_2$  to  $x_4$ .

Restrict also to  $\Gamma_{13} \cap \Lambda_K(x_2) = \emptyset$ . Sum first over  $\Gamma_{13}$  and then over the first point  $z$  where  $\Gamma_{24}$  hits one of those two endpoint boxes. The factorization lemma gives

$$\frac{1}{D_X} \sum_{\gamma_{13}} \rho(\gamma_{13}) \sum_{z \in V(\gamma_{13})} \langle \sigma_{x_2} \sigma_z \rangle \langle \sigma_z \sigma_{x_4} \rangle, \quad (9.11k)$$

where  $|V(\gamma_{13})| \leq 2|\Lambda_M|$ . Every such  $z$  is at distance at least  $K - M$  from  $x_2$ . Set

$$\varepsilon_M(K) := \sup_{|y| \geq K-M} \langle \sigma_0 \sigma_y \rangle_{\mathbb{H}}.$$

By (9.11h),  $\varepsilon_M(K) \rightarrow 0$ . The local form (9.11f), backbone summation  $\sum \rho(\gamma_{13}) = \langle \sigma_{x_1} \sigma_{x_3} \rangle$ , and Griffiths' inequality reduce (9.11k) to  $C_M \varepsilon_M(K)$ . Hence, after first choosing  $K$  large and then  $d(X)$  large,

$$\mathbf{P}^X(B^1; Y \subset S_M; \Gamma_{13} \text{ exthasaflightabove } x_2) = o(1). \quad (9.11l)$$

The analogous estimate holds for a flight of  $\Gamma_{24}$  above  $x_3$ .

It remains to treat the event on which neither near-source event nor either event in (9.11l) occurs. Since the polygonal strands have interlaced endpoints and are forbidden to have a coarse intersection above  $S_M$ , each strand must then contain a flight whose endpoints lie respectively to the left of  $x_2$  and to the right of  $x_3$ . Choose the innermost such flight among the two strands. If it belongs to  $\Gamma_{13}$ , then  $\Gamma_{24}$  must meet the  $M$ -box around its left endpoint; otherwise it would have to make an even more interior flight, contrary to the choice of the innermost one. That box is at distance at least  $K - M$  from  $x_2$ , so exactly the estimate (9.11k) gives probability at most  $C_M \varepsilon_M(K)$ . The case where the innermost flight belongs to  $\Gamma_{24}$  is symmetric. Combining the two near-source estimates, the two single-flight estimates, and the two innermost-flight estimates proves (9.11i) for four sources.



For general  $2m$ , partition according to the finitely many backbone pairings and, inside each of the two current components containing the prescribed interlaced strands, according to the finite exploration orders of the other sources. Conditional on one such pairing/order, the same depleted-graph argument applies: previously explored backbone pieces only delete ferromagnetic couplings, so Griffiths monotonicity restores the half-plane correlations before Lemma 9.15 is used. The preceding estimate becomes

$$o(1) \frac{\prod_{r=1}^m \langle \sigma_{x_{\pi(2r-1)}} \sigma_{x_{\pi(2r)}} \rangle}{\langle \sigma_{x_1} \cdots \sigma_{x_{2m}} \rangle}.$$

Repeated Griffiths inequalities make the ratio at most one. Summing over the finite set of pairings and exploration orders proves the result.  $\square$

## The full-plane odd-circuit engine

The boundary Pfaffian argument below needs one genuinely two-dimensional input: at criticality a sourceless finite-range current contains arbitrarily large *odd* circuits around every fixed point. We now prove that input in the same three-representation form in which it is used in the primary argument. Throughout this subsection the interaction is the finite-range, reflection-symmetric ferromagnetic interaction of Theorem 9.23;  $R$  is its interaction range,  $\hat{n}$  denotes the support of a current, and  $\eta$  its odd-edge (high-temperature) subgraph.

**Lemma 9.17** (Even-subgraph/current/FK coupling and decorrelation). *On every finite graph there is a coupling  $(\eta, \hat{n}, \omega)$  with*

$$\eta \subseteq \hat{n} \subseteq \omega, \quad (9.11m)$$

*such that  $\eta$  is the high-temperature expansion with the prescribed source set,  $\hat{n}$  is the support of the corresponding random current, and  $\omega$  is the  $q = 2$  random-cluster configuration conditioned that each open cluster contains an even number of sources. For the sourceless law the conditioning is void. Conditionally on  $\omega$ ,  $\eta$  is uniform among the even subgraphs of  $\omega$ ; in particular, for every cycle  $c \subset \omega$ ,*

$$(\omega, \eta) \stackrel{\text{law}}{=} (\omega, \eta \triangle c). \quad (9.11n)$$

*If the infinite-volume free FK law has no infinite cluster, all three sourceless laws have local infinite-volume limits and the coupling passes to the limit. Moreover, if  $A, B$  are increasing FK events supported on vertex sets  $V, W$ , respectively, then*

$$\phi(A)\phi(B) \leq \phi(A \cap B) \leq \phi(A)\phi(B) + \mathbf{P}^\varnothing(V \overset{\hat{n}}{\longleftrightarrow} W). \quad (9.11o)$$

*Proof.* For an edge  $e$  put  $x_e = \tanh(\beta J_e)$  and  $p_e = 1 - e^{-2\beta J_e}$ . Summing the Taylor current weight over positive odd values gives  $\sinh(\beta J_e)$  and over positive even values gives  $\cosh(\beta J_e) - 1$ . Hence, conditional on the parity  $\eta(e) = 0$ , the edge is present in the current support with probability

$$1 - \frac{1}{\cosh(\beta J_e)},$$

independently from edge to edge. Starting from  $\eta$ , add precisely these independent edges to obtain  $\hat{n}$ . Then add, independently on every still closed edge, a Bernoulli edge of parameter  $1 - e^{-\beta J_e}$ . A direct calculation gives

$$\mathbf{P}(\omega \mid \eta) = \prod_{e: \eta(e)=1} \mathbf{1}_{\{\omega(e)=1\}} \prod_{e: \eta(e)=0} x_e^{\omega(e)} (1 - x_e)^{1-\omega(e)}. \quad (9.11p)$$

Multiplying by the high-temperature weight  $\prod_{e:\eta(e)=1} x_e$  and summing first over  $\eta \subseteq \omega$  with the prescribed parity constraint gives exactly the  $q = 2$  random-cluster weight: the number of admissible parity subgraphs is  $2^{|E(\omega)|-|V|+k(\omega)}$  after one parity constraint has been removed in each component. It vanishes precisely when a component meets the source set oddly. This proves the three marginals and (9.11m).

For no sources, the admissible parity subgraphs of a fixed  $\omega$  form a vector space over  $\mathbf{F}_2$  under symmetric difference. Translation by a cycle is a bijection of this vector space, proving (9.11n). If the free FK configuration has only finite clusters, this conditional construction can be made independently on every cluster; local exhaustion therefore gives the claimed infinite-volume coupling.

For (9.11o), in the coupling let  $S$  be the set of vertices connected to  $V$  in  $\eta$ . On  $\{S \cap W = \emptyset\}$ , after  $S$  and all FK information inside  $S$  have been revealed, the parity configuration outside  $S$  is a sourceless high-temperature expansion and the extra FK openings outside  $S$  have the free random-cluster law on the graph with  $S$  deleted. Monotonicity under adding back the deleted ferromagnetic edges gives

$$\mathbf{P}(A \cap B; S \cap W = \emptyset) \leq \phi(A)\phi(B).$$

The complement is bounded by  $\mathbf{P}(V \leftrightarrow W \text{ in } \eta)$ , and  $\eta \subseteq \hat{n}$  gives the displayed upper bound. The lower bound is FKG.  $\square$

**Lemma 9.18** (Finite-range FK surgery and gluing). *Fix  $\beta > 0$ . There are constants  $c_{\text{surg}}, c_{\text{glue}} > 0$ , depending only on  $(J, \beta)$ , with the following properties.*

1. *If a deterministic map from an FK event  $E$  to an FK event  $F$  changes at most  $s$  edges and every image has at most  $2^s$  preimages, then*

$$\phi(E) \leq c_{\text{surg}}^{-s} \phi(F). \quad (9.11q)$$

2. *Let  $S$  be a sufficiently large rectangle and let  $A, B, C, D$  be disjoint boundary arcs such that every continuous  $A$ – $B$  curve intersects every continuous  $C$ – $D$  curve. Then*

$$\phi(A \leftrightarrow C \text{ in } S) \geq c_{\text{glue}} \phi(A \leftrightarrow B \text{ in } S) \phi(C \leftrightarrow D \text{ in } S). \quad (9.11r)$$

3. *If  $H_n$  is the event that  $\Lambda_n$  is crossed horizontally and  $C_n$  the event that there is an open circuit surrounding 0 in  $\Lambda_n \setminus \{0\}$  and connected to  $\partial\Lambda_n$ , then*

$$\phi(C_n) \geq c_{\text{glue}} \phi(H_n)^{20}. \quad (9.11s)$$

*The same estimate holds with a fixed inner box removed, after changing the constant.*

*Proof.* For a  $q = 2$  random-cluster edge  $e$ , conditional on all other edges, the probability that  $e$  is open equals either  $p_e$  (if its endpoints are already connected) or  $p_e/(p_e + 2(1 - p_e))$ . Since the interaction has finite range and only finitely many positive coupling values, both the open and closed conditional probabilities are bounded below by a number  $a = a(J, \beta) > 0$ . Changing  $s$  prescribed edge states therefore changes probability by at most  $a^{-s}$ ; summing over at most  $2^s$  possible preimages proves (9.11q), after decreasing  $c_{\text{surg}}$ .

For (9.11r), intersect the two crossing events and use FKG. In a configuration where the two continuous polygonal crossings geometrically meet but the finite-range graph paths do not share a vertex, choose the lexicographically first point  $z$  of closest approach between a fixed minimal  $A$ – $B$  path and a  $C$ – $D$  path. Both paths enter a box  $\Lambda_r(z)$  for a fixed  $r = r(R)$ , independent of the size of  $S$ . Close every edge meeting this box except the four entrance pieces that reconstruct the two

selected paths, and open two disjoint positive-coupling paths inside the box joining the appropriate entrance vertices. Connectedness of the positive-coupling graph and finite range allow  $r$  to be fixed once and for all. The modified configuration contains an  $A$ – $C$  connection. From its lexicographically minimal  $A$ – $B$  path, the unique first new  $A$ – $C$  branching point, and the untouched portions outside  $\Lambda_r(z)$ , one reconstructs the preimage outside that box. Thus the number of preimages is at most  $2^{O(|\Lambda_r|)}$  and only  $O(|\Lambda_r|)$  edges were changed. Apply (9.11q) and FKG to obtain (9.11r).

For (9.11s), work first in the punctured box and classify the lexicographically minimal left–right polygonal crossing by its winding number about 0: winding 0, winding 1, or winding at least 2. One of these three classes has probability at least  $\phi(H_n)/3$  up to a bounded finite-energy correction that keeps the path away from the puncture.

If the winding is at least 2, the crossing already contains, together with a subarc lying outside a fixed ball, a topological loop around 0. At the first and last visit of that subarc to a fixed  $r$ -box, perform the local surgery just used for (9.11r); the loop becomes a genuine graph circuit and remains attached to the outer crossing. Bounded reconstruction gives probability at least a constant times the probability of this winding class.

For winding 1, choose the clockwise or counterclockwise subclass of probability at least one half. Intersect four rotated copies (horizontal and vertical, with both orientations). FKG gives the fourth power of the subclass probability. Their continuous traces contain a topological loop around the puncture. A bounded number of the local surgeries from (9.11r), at the finitely many places where finite-range edges jump past one another, turns it into an open graph circuit.

For winding 0, take five appropriately translated/reflected copies of the crossing: two horizontal copies above and below the puncture, two vertical copies on its left and right, and one connector. Apply (9.11r) successively at the five forced geometric encounters. The result is a circuit around 0 with probability at least a fixed constant times the fifth power of the winding-0 probability. Combining the three alternatives and using the common exponent 20 gives (9.11s). Replacing the puncture by a fixed inner box merely increases the fixed surgery radius and proves the last claim.  $\square$

**Lemma 9.19** (Transfer of nested circuits between the three representations). *Assume that the infinite-volume free FK configuration has no infinite cluster. Let  $\mathcal{E}$  be the event of infinitely many pairwise disjoint circuits surrounding 0. Then*

1.  $\phi(\omega \in \mathcal{E}) = 1$  implies  $\mathbf{P}^\partial(\eta \in \mathcal{E}) = 1$ ;
2. if  $n_1, n_2$  are independent sourceless currents and  $\mathbf{P}(n_1 + n_2 \in \mathcal{E}) = 1$ , then  $\mathbf{P}(\hat{n}_i \in \mathcal{E}) = 1$  for  $i = 1, 2$ .

*Proof.* On  $\{\omega \in \mathcal{E}\}$  there are infinitely many distinct finite FK components which surround 0: one finite component contains only finitely many edges and hence only finitely many disjoint circuits. In each such component choose deterministically one surrounding cycle  $c$ . Conditionally on  $\omega$ , the parity subgraph is uniform in the even-subgraph vector space. Translation by  $c$  is an involution, and of each pair  $(\eta, \eta \triangle c)$  at least one member contains a surrounding cycle: take a radial cut from 0 to infinity and compare the parity of its intersection with  $c$ . Thus each component contributes a surrounding parity circuit with conditional probability at least  $1/2$ . The coefficients of a cycle basis belonging to disjoint components are independent, so Borel–Cantelli yields infinitely many such circuits almost surely.

For two currents, condition on their total multigraph  $M$ , with  $n_1(e) + n_2(e)$  parallel copies of each interaction edge. Given  $M$ , the allocation of copies to (say)  $n_2$  is uniform among the even subgraphs of  $M$ : this is the same binomial switching count as in Lemma 9.4. It remains only to

note that  $M$  has no infinite component. Indeed, when the free FK law has no infinite cluster,  $\langle \sigma_0 \sigma_x \rangle = \phi(0 \leftrightarrow x) \rightarrow 0$  as  $|x| \rightarrow \infty$ . Switching gives

$$\mathbf{P}^{\emptyset, \emptyset}(0 \leftrightarrow x \text{ in } n_1 + n_2) = \langle \sigma_0 \sigma_x \rangle^2. \quad (9.11t)$$

The same Burton–Keane insertion argument as in Lemma 9.7, now adding current 2 on finitely many edges of one replica, gives at most one infinite component. Let  $\theta = \mathbf{P}^{\emptyset, \emptyset}(0 \leftrightarrow \infty)$ . For a finite box  $B$ , uniqueness and Cauchy–Schwarz give, exactly as in Lemma 9.8,

$$|B|^2 \theta^2 \leq \sum_{x, y \in B} \mathbf{P}(x, y \leftrightarrow \infty) \leq \sum_{x, y \in B} \langle \sigma_x \sigma_y \rangle^2. \quad (9.11t')$$

Since the free FK state has no infinite cluster,  $\langle \sigma_0 \sigma_x \rangle = \phi(0 \leftrightarrow x) \rightarrow 0$  as  $|x| \rightarrow \infty$ . Taking  $B = \Lambda_N$  and using Cesàro averaging makes the last expression divided by  $|B|^2$  tend to zero. Thus  $\theta = 0$  and the double current has no infinite component. Apply the preceding even-subgraph argument component by component. Symmetry exchanges  $n_1, n_2$ .  $\square$

**Lemma 9.20** (Current-annulus decay forces FK square crossings). *At  $\beta = \beta_c$ , assume the free FK configuration has no infinite cluster. Put*

$$h_n := \phi(\Lambda_n \text{ is crossed horizontally}), \quad \epsilon_n := \mathbf{P}^{\emptyset}(\Lambda_{2n} \setminus \Lambda_n \text{ is crossed by } \hat{n}).$$

*If  $\epsilon_n \rightarrow 0$ , then  $\limsup_{n \rightarrow \infty} h_n > 0$ .*

*Proof.* We reproduce the three steps because this implication is the non-FKG substitute that makes the current-circuit theorem work.

*Step 1: a positive subsequential FK annulus-crossing probability.* Let  $\Lambda'_n = (3n, 0) + \Lambda_n$  and let  $N = \sum_{x \in \Lambda_n, y \in \Lambda'_n} \mathbf{1}_{\{x \leftrightarrow y \text{ in } \omega\}}$ . Edwards–Sokal and switching give

$$\mathbf{E}N = \sum_{x, y} \langle \sigma_x \sigma_y \rangle, \quad \mathbf{E}N^2 \leq 2(\mathbf{E}N)^2 + \left( \sum_{x, z \in \Lambda_n} \langle \sigma_x \sigma_z \rangle \right)^2. \quad (9.11u)$$

(The four-point inequality follows by switching: the four-point function is bounded by the sum of the three pair products.) Paley–Zygmund therefore gives a positive lower bound on a crossing between the two boxes whenever

$$\mathbf{E}N \geq c \sum_{x, z \in \Lambda_n} \langle \sigma_x \sigma_z \rangle. \quad (9.11v)$$

By Lemma 9.15, if  $u_k = k \langle \sigma_0 \sigma_{(k, 0)} \rangle$ , then  $\mathbf{E}N \geq cn^3 u_{7n+i}$  for each  $0 \leq i \leq 6$ , whereas the right side of (9.11v) is at most  $Cn^2(1 + \sum_{k \leq 2n} u_k)$ . At criticality the Simon–Lieb inequality of Chapter 2 implies  $\sum_{x \in \Lambda_{n+R} \setminus \Lambda_n} \langle \sigma_0 \sigma_x \rangle \geq 1$ ; MMS therefore keeps  $u_n$  bounded below. Hence for infinitely many  $n$  and some  $i \in \{0, \dots, 6\}$ ,

$$(7n+i)^{-1} \sum_{k \leq 7n+i} u_k \leq 2u_{7n+i}.$$

For these scales (9.11v) holds. We obtain

$$\limsup_n \phi(\Lambda_{2n} \setminus \Lambda_n \text{ is crossed by } \omega) > 0. \quad (9.11w)$$

*Step 2: renormalization makes the lower bound uniform.* Let  $a_n$  denote the FK annulus-crossing probability in (9.11w). Cover  $\partial\Lambda_m$  and  $\partial\Lambda_{2m}$  by  $O(m/n)$  boxes of radius  $n$ ; for  $6n \leq m \leq 36n$  this

number is bounded. If the  $m$ -annulus is crossed, a box on the inner boundary is connected to one on the outer boundary. Apply (9.11o) to the two local annular connection events around these boxes. Their supports are disjoint, and the error is bounded by  $\epsilon_n$ . Thus

$$a_m \leq C(a_n^2 + \epsilon_n), \quad 6n \leq m \leq 36n. \quad (9.11x)$$

If  $\inf_n a_n = 0$ , choose a scale at which  $a_n$  is below the small fixed point of  $t \mapsto 2Ct^2$  and simultaneously  $\epsilon_n$  is small. Iterating (9.11x) at scales between 6 and 36 times the previous scale forces  $a_m \rightarrow 0$  along every sufficiently large scale, contradicting (9.11w). Hence

$$\inf_n a_n > 0. \quad (9.11y)$$

*Step 3: finite-range RSW contrapositive.* Suppose, contrary to the claim, that  $h_n \rightarrow 0$ . In the strip  $S_n = [-n, n] \times \mathbb{R}$  define

$$\begin{aligned} A_n &= \{[-n, n] \times [-n, n] \text{ has a path from its left-lower half to its top}\}, \\ B_n &= \{S_n \text{ has a left-right path from the left side to the upper half of the right side}\}, \\ C_n(a) &= \{S_n \text{ has a left-right path ending below height } a\}, \\ D_n(a) &= \{\{n\} \times (-\infty, -a] \leftrightarrow \{n\} \times [a, \infty) \text{ in } S_n\}. \end{aligned}$$

Reflection, FKG and Lemma 9.18 give  $h_n \geq c\phi(A_n)^2$ , hence  $\phi(A_n) \rightarrow 0$ ; the elementary decomposition of a strip crossing gives  $\phi(B_n) \leq h_n + 2\phi(A_n) \rightarrow 0$ . The same gluing lemma yields

$$\phi(B_n) \geq \frac{1}{2}c_{\text{glue}}\phi(C_n(a))\phi(D_n(a)). \quad (9.11z)$$

Choose  $\alpha_n$  minimal with  $\phi(C_n(\alpha_n)) \geq \phi(D_n(\alpha_n))$ . Finite energy for changing the height by one and (9.11z) show

$$\phi(C_n(\alpha_n)) + \phi(D_n(\alpha_n)) \rightarrow 0. \quad (9.11aa)$$

Put  $\delta_n$  equal to the maximum of these four small probabilities and  $h_n$ . Since  $\alpha_n \leq n$ , there are infinitely many  $n$  for which  $\alpha_{2n} \leq 4\alpha_n$ .

For such  $n$ , take the two overlapping squares  $S = (-n, 0) + \Lambda_{2n}$  and its reflection  $S'$  in the  $y$ -axis. Split their common vertical sides into three intervals

$$J_1 = [-2n, -\alpha_{2n}], \quad J_2 = [-\alpha_{2n}, \alpha_{2n}], \quad J_3 = [\alpha_{2n}, 2n].$$

Follow a vertical crossing of  $[-3n, 3n] \times [-2n, 2n]$ . If it moves between  $S$  and  $S'$ , one of exactly three alternatives occurs: (i) some reflected  $J'_i$  connects to a different  $J_j$ ; (ii)  $J'_2$  connects to  $J_2$ ; or (iii) on one side, top together with  $J_1$  connects to  $J_3$  together with bottom. This is a deterministic topological trichotomy obtained by taking the last type-1 and first type-2 subcrossings in the overlap. Alternative (i) is contained in four symmetric copies of  $B_n$ ; (ii), using  $\alpha_{2n} \leq 4\alpha_n$ , is bounded by  $4\phi(C_n(\alpha_n))$ ; and (iii) is bounded by two horizontal-square crossings, four copies of  $A_n$ , and two copies of  $D_{2n}(\alpha_{2n})$ . Consequently

$$\phi([-3n, 3n] \times [-2n, 2n] \text{ vertical crossing}) \leq 14\delta_n + 2\delta_{2n} \rightarrow 0 \quad (9.11ab)$$

along an infinite subsequence.

Cover  $[-8n, 8n] \times [-2n, 2n]$  by six translates of this rectangle. Any vertical crossing of the long rectangle either crosses one translate vertically or crosses an overlap square horizontally; by symmetry the latter has no larger probability. Hence its crossing probability is at most eleven times the left side of (9.11ab). Four rotated copies of the long rectangle cover the annulus  $\Lambda_{8n} \setminus \Lambda_{4n}$  so that any radial annulus crossing crosses one copy in its easy direction. Therefore the FK annulus-crossing probability tends to zero along the same subsequence, contradicting (9.11y). Thus  $\limsup h_n > 0$ .  $\square$

**Lemma 9.21** (Tortuous-current alternative). *At  $\beta = \beta_c$ , assume the free FK configuration has no infinite cluster. If*

$$\limsup_n h_n = 0 \quad \text{and} \quad \limsup_n \mathbf{P}^\varnothing(\Lambda_{2n} \setminus \Lambda_n \text{ crossed by } \hat{n}) > 0, \quad (9.11ac)$$

*then for two independent sourceless currents  $n_1, n_2$  the union  $\hat{n}_1 \cup \hat{n}_2$  contains infinitely many disjoint circuits surrounding 0 almost surely.*

*Proof.* We give the oscillation construction explicitly. By covering an annulus by four translates of  $[0, n] \times [0, 4n]$ , the second condition in (9.11ac) implies

$$\limsup_n \mathbf{P}^\varnothing([0, n] \times [0, 4n] \text{ is crossed in the easy direction by } \hat{n}) > 0. \quad (9.11ad)$$

Meanwhile  $h_n \rightarrow 0$  and  $\hat{n} \subseteq \omega$  imply that square crossings by a single current tend to zero.

Set  $S_n = [-n, n] \times \mathbb{R}$ ,  $S_n^+ = [-n, n] \times (-\infty, n/4]$ ,  $I = \{-n\} \times [0, n/4]$ ,  $J = \{n\} \times [0, n/4]$  and  $\ell_n = \mathbb{R} \times \{n/4\}$ . Let  $U_n$  be the event of a current path from  $I$  to  $J$ . Let  $V_n(x, y)$ , for  $x, y \in \ell_n$ , require such a path which passes above 0 but below both  $x$  and  $y$ , and let  $W_n$  require a path in  $S_n^+$  from  $\ell_n$  back to  $\ell_n$  passing below 0.

We first dispose of three bad FK events. Let  $E_n$  be the existence of a path from the lower quarter of the left side of  $S_n$  to the central vertical line without reaching height  $n/4$ . Let  $F_n(k)$  be a left-right path in  $S_n$  passing above  $(k, 0)$ ,  $|k| \leq n$ . Finally let  $B_n$  collect the two ways in which a long easy crossing enters the strip but exits through the wrong quarter of the opposite side. Reflecting and applying Lemma 9.18 shows

$$\phi(E_n) + \phi(B_n) + \max_{|k| \leq n} \phi(F_n(k)) =: \varepsilon_n \rightarrow 0; \quad (9.11ae)$$

indeed each occurrence can be glued to a reflected copy to create a square crossing, whose probability is  $h_{O(n)} \rightarrow 0$ .

From (9.11ad), after translating and using the two wrong-exit events  $B_n$ ,

$$\limsup_n \mathbf{P}^\varnothing(U_n) > 0. \quad (9.11af)$$

For fixed  $x, y \in \ell_n$ , failure of  $V_n(x, y)$  on  $U_n$  forces one of the three paths  $F_n(k)$  with  $k$  the horizontal coordinate of 0,  $x, y$ ; hence

$$\mathbf{P}^\varnothing(V_n(x, y)) \geq \mathbf{P}^\varnothing(U_n) - 3\varepsilon_n. \quad (9.11ag)$$

Similarly, on  $U_n$  take a continuous  $I$ - $J$  path and inspect its first and last visits to the lower rectangle. Unless  $F_n(0)$ ,  $E_n$ , or its reflection occurs, the intervening subpath starts and ends on  $\ell_n$  and passes below 0. Therefore

$$\mathbf{P}^\varnothing(W_n) \geq \mathbf{P}^\varnothing(U_n) - 3\varepsilon_n. \quad (9.11ah)$$

Equations (9.11af)–(9.11ah) give

$$\limsup_n \min_{x, y \in \ell_n} \mathbf{P}^\varnothing(V_n(x, y)) \mathbf{P}^\varnothing(W_n) > 0. \quad (9.11ai)$$

Now take independent currents. On  $W_n$  for  $n_1$ , choose the minimal continuous current path  $\Gamma_1$  from  $\ell_n$  to itself in  $S_n^+$  which passes below 0, and let  $o(\Gamma_1)$ ,  $e(\Gamma_1)$  be its first and last vertices on  $\ell_n$ . If  $n_2 \in V_n(o(\Gamma_1), e(\Gamma_1))$ , its path from  $I$  to  $J$  passes above 0 and below these two endpoints. The two continuous paths therefore form a topological loop around 0, possibly missing a graph connection at the two finite-range near-intersections. At each such near-intersection add current 2

on a fixed positive-coupling connector of length  $O_R(1)$  in  $n_2$ . This preserves the sourceless parity condition, changes only  $O_R(1)$  current variables, and has a probability cost bounded below by a constant  $c(J, \beta) > 0$ . Thus, if  $H_n$  denotes the event that  $\hat{n}_1 \cup \hat{n}_2$  contains a surrounding circuit in a component meeting  $\partial\Lambda_n$ ,

$$\mathbf{P}(H_n) \geq c \mathbf{P}(W_n) \min_{x, y \in \ell_n} \mathbf{P}(V_n(x, y)). \quad (9.11aj)$$

The right side has positive limsup by (9.11ai).

The double current has no infinite component by the argument in Lemma 9.19. Hence the decreasing event that there are surrounding circuits in components of arbitrarily large diameter has positive probability. This event is translation invariant (a sufficiently large circuit surrounding 0 surrounds any fixed translate), and the product current law is ergodic by Lemma 9.6. Its probability is therefore one, and disjoint circuits can be extracted in successive annuli.  $\square$

**Lemma 9.22** (Nested odd circuits in a sourceless current). *For every  $\beta \geq \beta_c$ , the full-plane sourceless high-temperature subgraph  $\eta$ , the support  $\hat{n}$  of a sourceless current, and the free  $q = 2$  FK configuration  $\omega$  each contain infinitely many pairwise disjoint circuits surrounding every fixed site almost surely. In particular, at  $\beta_c$  a sourceless current has infinitely many surrounding circuits all of whose edges carry odd positive current.*

*Proof.* By translation it suffices to surround 0.

First suppose  $\beta = \beta_c$  and the free FK state has no infinite cluster. Put

$$(P1) : \limsup_n h_n > 0, \quad (P2) : \mathbf{P}^\partial(\Lambda_{2n} \setminus \Lambda_n \text{ crossed by } \hat{n}) \rightarrow 0.$$

If (P1) holds, Lemma 9.18 gives  $\limsup_n \phi(C_n) > 0$ . Since there is no infinite FK component, the occurrence of  $C_n$  for arbitrarily large  $n$  yields arbitrarily large finite components containing surrounding circuits; the resulting event is translation invariant, so FK ergodicity upgrades its positive probability to one. Thus  $\omega \in \mathcal{E}$  almost surely and Lemma 9.19 gives  $\eta, \hat{n} \in \mathcal{E}$  almost surely. If (P2) holds, Lemma 9.20 implies (P1), so the same conclusion follows. Finally, if neither holds, Lemma 9.21 gives  $n_1 + n_2 \in \mathcal{E}$  almost surely, and the second part of Lemma 9.19 transfers the circuits to each single current and its odd subgraph. This completes the nonpercolating critical case.

It remains to treat the case  $\phi_\beta(0 \leftrightarrow \infty) > 0$ , which includes every  $\beta > \beta_c$  and also covers a hypothetical discontinuous critical transition. The parity law  $\mathbf{P}_\beta^\partial$  exists and is ergodic by Lemma 9.6. It has the following finite-energy operation: for every fixed finite even subgraph  $\eta_0$ ,

$$\mathbf{P}_\beta^\partial(\eta \triangle \eta_0) \geq c(\eta_0) \mathbf{P}_\beta^\partial(\eta), \quad (9.11ak)$$

because the high-temperature weight changes by the bounded factor  $\prod_{e \in \eta_0} \tanh(\beta J_e)^{\pm 1}$ . Assume for contradiction that infinitely many surrounding parity circuits do not occur almost surely. Then for some  $n$  the probability of a parity circuit surrounding  $\Lambda_n$  is less than one, and (9.11ak), using a fixed finite interaction-graph circuit around 0, implies

$$c := \mathbf{P}_\beta^\partial(0 \text{ is surrounded by no parity circuit}) > 0. \quad (9.11al)$$

The FK infinite cluster is unique by the standard Burton–Keane argument (finite energy was proved in Lemma 9.18). Consequently, for every fixed  $n$ , with probability tending to one a large box has a left–right crossing avoiding  $\Lambda_n$ . The circuit-gluing part of Lemma 9.18, applied with the fixed inner box removed, implies that  $\Lambda_n$  is almost surely surrounded by an FK cycle. Choose deterministically such a cycle  $\gamma(\omega)$ . On the event in (9.11al), symmetric difference with  $\gamma$  creates a

parity circuit surrounding 0 of diameter larger than  $n$  (a radial cut sees its parity changed). By (9.11n),

$$\mathbf{P}_\beta^\varnothing(\exists \text{ surrounding parity circuit of diameter } > n) \geq c/2. \quad (9.11am)$$

We need to keep the large circuit away from a prescribed inner box. Fix  $k$ . If  $\eta$  has no infinite component, choose  $n$  so large that  $\mathbf{P}(\Lambda_k \leftrightarrow \partial\Lambda_n \text{ in } \eta) < c/4$ ; then a circuit counted in (9.11am) surrounds  $\Lambda_k$  unless this connection event occurs, giving probability at least  $c/4$ . If  $\eta$  percolates, the same bounded-cycle operation (9.11ak) gives the insertion tolerance needed for the Burton–Keane argument, hence the infinite parity component  $C_\infty$  is unique almost surely. No one-endedness input is needed. Fix the requested  $k$  and choose  $K \geq k$  so large that

$$\mathbf{P}(C_\infty \cap \Lambda_K \neq \varnothing) > 1 - c/8. \quad (9.11am')$$

This is possible because an infinite component exists almost surely and  $\{C_\infty \cap \Lambda_K \neq \varnothing\} \uparrow \{C_\infty \neq \varnothing\}$ . For a fixed configuration on which  $C_\infty \cap \Lambda_K \neq \varnothing$ , only finitely many  $\eta$ -components meet the finite box  $\Lambda_K$ . Every one of them except  $C_\infty$  is finite. Consequently there is a finite random radius  $N_K$  such that, for every  $n \geq N_K$ , exactly one component meeting  $\Lambda_K$  also meets  $\partial\Lambda_n$ , namely  $C_\infty$ . By bounded convergence, after increasing  $n$  we therefore have

$$\mathbf{P}(\text{there is a unique } \eta\text{-component joining } \Lambda_K \text{ to } \partial\Lambda_n) > 1 - c/4. \quad (9.11am'')$$

The FK circuit surrounding  $\Lambda_n$  occurs almost surely. Intersecting this event with (9.11al) and (9.11am'') therefore gives probability at least  $3c/4$  for configurations with (i) no parity circuit around 0, (ii) an FK circuit  $\gamma$  around  $\Lambda_n$ , and (iii) a unique parity component joining  $\Lambda_K$  to  $\partial\Lambda_n$ . Choose the lexicographically first such  $\gamma$ . The mod-2 Jordan-crossing argument used in (9.11am) now has a relative form: if  $\eta \Delta \gamma$  had no circuit surrounding  $\Lambda_K$ , the unique component in (iii) would have to cross  $\gamma$  an even number of times without being cut into two distinct inner–outer components, contradicting that symmetric difference changes the parity of every radial cut from  $\Lambda_K$  to infinity. Thus  $\eta \Delta \gamma$  contains a parity circuit surrounding  $\Lambda_K$ , and therefore  $\Lambda_k$ . By the cycle-toggle invariance (9.11n),

$$\mathbf{P}_\beta^\varnothing(\Lambda_k \text{ is surrounded by a parity circuit}) \geq c/2. \quad (9.11an)$$

(We retain the weaker constant  $c/2$  although the preceding intersection gives  $3c/4$ .)

Letting  $k \rightarrow \infty$ , the limsup event of arbitrarily large surrounding circuits has positive probability by (9.11an), hence probability one by ergodicity. Extract a disjoint subsequence. Since  $\eta \subseteq \hat{n} \subseteq \omega$ , the same circuits occur in the current support and in FK. Finally, every edge of a parity circuit has odd current, so the last assertion follows.  $\square$

**Theorem 9.23** (Emergent planarity for finite-range two-dimensional Ising). *Let  $\mathbb{H} = \mathbb{Z} \times \mathbb{Z}_+$  and let  $J$  be a zero-field Ising interaction on  $\mathbb{H}$  which is*

1. *ferromagnetic:  $J_{x,y} \geq 0$ ;*
2. *finite range, with connected positive-coupling graph;*
3. *translation invariant:  $J_{x,y} = J(x - y)$ ; and*
4. *invariant under the coordinate reflections and interchange,*

$$J_{0,(a,b)} = J_{0,(-a,b)} = J_{0,(a,-b)} = J_{0,(b,a)}.$$



Let  $x_i = (k_i, 0)$  with  $k_1 < \dots < k_{2m}$ . At the critical inverse temperature,

$$\langle \sigma_{x_1} \cdots \sigma_{x_{2m}} \rangle_{\mathbb{H}, \beta_c} = \text{Pf}[\langle \sigma_{x_i} \sigma_{x_j} \rangle_{\mathbb{H}, \beta_c}]_{1 \leq i < j \leq 2m} [1 + o(1)], \quad (9.12)$$

where  $o(1) \rightarrow 0$  whenever  $\min_{i < j} |x_i - x_j| \rightarrow \infty$  with  $m$  fixed.

*Complete current-geometric proof.* The proved statement is the boundary-spin identity (9.12). We do not strengthen it to a full finite-range bulk order-disorder Pfaffian theorem: the corresponding source argument obtains a weaker error bound and explicitly stops short of that conclusion.

Expand the Pfaffian in pairings. Apply Lemma 9.4 successively to the factors  $\langle \sigma_{x_i} \sigma_{x_j} \rangle$  and compare with the single  $2m$ -source current representation. All terms may be realized using a  $2m$ -source current  $n_1$  and an independent sourceless current  $n_2$ . Lemma 9.13 then gives

$$\frac{|\langle \sigma_X \rangle - \text{Pf}[\langle \sigma_{x_i} \sigma_{x_j} \rangle]|}{\langle \sigma_X \rangle} \leq C_m \sum_{i < j < k < \ell} \mathbf{P}_{\mathbb{H}}^{X, \emptyset}(B_{i,j,k,\ell}), \quad (9.13)$$

where  $B_{i,j,k,\ell}$  is the event that the backbone pairs  $x_i$  with  $x_k$  and  $x_j$  with  $x_\ell$ , but the two corresponding backbone strands are not connected to each other in  $n_1 + n_2$ . Hence (9.12) follows once every such avoided-intersection probability tends to zero.

Let  $Y(n_1)$  be the set of lattice sites at which the two interlaced backbone strands in  $B_{i,j,k,\ell}$  come within the fixed interaction range. If  $Y(n_1)$  contains  $K$  points separated from one another and from the boundary by more than  $M$ , then the sourceless current  $n_2$  is independent of  $n_1$  and, at criticality, a positive-current odd circuit surrounding any one of those points joins the two strands in  $n_1 + n_2$ . Translation invariance and the odd-circuit theorem therefore give, for  $M$  chosen first and then  $K \rightarrow \infty$ ,

$$\mathbf{P}(B_{i,j,k,\ell}; |Y(n_1) \setminus S_M| \geq K) \leq (1 - c_M)^K, \quad (9.14)$$

where  $S_M = \mathbb{Z} \times [0, M]$ . Thus the contribution of configurations having arbitrarily many bulk close approaches vanishes. The load-bearing statement is now internal: Lemma 9.22 proves that the full-plane sourceless current at  $\beta_c$  almost surely has infinitely many positive-current odd circuits surrounding every fixed site.

For interlaced boundary endpoints, two continuous planar curves joining  $(x_i, x_k)$  and  $(x_j, x_\ell)$  must cross. Finite-range interaction edges may jump over one another, but after replacing each edge by its straight segment and subdividing at every geometric crossing, each crossing of the two polygonal backbones gives a pair of lattice vertices at distance at most a constant depending only on the interaction range. Consequently, after extracting well-separated close approaches, either their number tends to infinity as the boundary points separate, or all close approaches lie in some fixed strip  $S_M$ . The first alternative is disposed of by (9.14).

If the boundary two-point function tends to zero, Lemma 9.16 proves the precise source estimate

$$\lim_{d(X) \rightarrow \infty} \mathbf{P}_{\mathbb{H}}^X[B_{i,j,k,\ell}^1 \cap \{Y(n_1) \subset S_M\}] = 0. \quad (9.15)$$

The avoided-intersection event  $B_{i,j,k,\ell}$  for the doubled current is contained in this single-current event, so (9.15) is exactly the boundary contribution needed in (9.13). Its proof is now internal: Lemmas 9.14 and 9.15 supply the two previously external ingredients.

First choose the boundary-strip height  $M$  and the number of separated bulk close approaches so that the bulk alternative in (9.14) is arbitrarily small; then let  $d(X) \rightarrow \infty$ , so that (9.15) controls the remaining strip alternative. Each probability on the right of (9.13) tends to zero. Since the number of quadruples depends only on fixed  $m$ , the normalized Pfaffian defect tends to zero. If

instead the boundary two-point function has a nonzero limit  $q > 0$ , Lemma 9.5 shows that every  $2m$ -point boundary correlation tends to  $q^m$  as the mutual separations diverge. Every entry of the Pfaffian tends to  $q$ , and the Pfaffian recursion shows that the corresponding ordered antisymmetric constant matrix has Pfaffian  $q^m$ . Thus (9.12) also holds in this complementary case.

Lemmas 9.17–9.22 prove the three-way coupling, finite-range gluing, current/FK circuit transfer, the non-FKG RSW alternative, the tortuous-current construction, and the nested odd-current circuit theorem. Together with the boundary-strip proposition, finite-range Messenger–Miracle–Solé comparison, backbone factorization, switching and Pfaffian combinatorics, these are all the inputs to the boundary-spin theorem. No conformal scaling limit or exact critical exponent is assumed.  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 9.24** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 9.25** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 9.26** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 9.27** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 9.28** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume IX: Potts Models, FK Random Clusters, and Random  
Currents

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# Scope and conventions

This volume develops the random-cluster representation as the common language for Potts models, planar duality, sharpness, and random currents. We assume  $q \geq 1$  whenever FKG or monotone boundary conditions are invoked, and we state explicitly when an argument is genuinely planar or uses a special value of  $q$ .

# Chapter 1

## Potts Models and the Fortuin–Kasteleyn Coupling

Let  $G = (V, E)$  be finite. The  $q$ -state Potts model has spins  $\sigma_x \in \{1, \dots, q\}$  and Hamiltonian  $-\sum_{xy} J_{xy} \mathbf{1}_{\sigma_x = \sigma_y}$ .

**Definition 1.1** (Finite random-cluster measure). For parameters  $p \in [0, 1]$ ,  $q > 0$ , and edge configuration  $\omega \in \{0, 1\}^E$ , let  $o(\omega)$  be the number of open edges and  $k(\omega)$  the number of connected components of the open graph, including isolated vertices. The free random-cluster measure is

$$\phi_{G,p,q}^f(\omega) = \frac{1}{Z_{G,p,q}^f} p^{o(\omega)} (1-p)^{|E|-o(\omega)} q^{k(\omega)}.$$

Wired boundary conditions are obtained by identifying the designated boundary vertices before counting  $k(\omega)$ .

*Worked example 1.2* (FK measure on a single edge). Let  $G$  have two vertices joined by one edge. The closed configuration has weight  $(1-p)q^2$  and the open configuration has weight  $pq$ , hence

$$\phi_{G,p,q}(e \text{ open}) = \frac{pq}{pq + (1-p)q^2} = \frac{p}{p + q(1-p)}.$$

For  $q = 1$  this is ordinary Bernoulli percolation. For  $q > 1$  the factor  $q^{k(\omega)}$  favours the disconnected configuration and lowers the single-edge opening probability.

**Theorem 1.3** (Fortuin–Kasteleyn expansion). *For integer  $q \geq 2$  and homogeneous coupling  $K = \beta J$ , with  $p = 1 - e^{-K}$ ,*

$$Z_{\text{Potts}} = e^{K|E|} \sum_{\omega \subseteq E} p^{|\omega|} (1-p)^{|E|-|\omega|} q^{k(\omega)},$$

*up to the displayed elementary normalization convention.*

*Proof.* For each edge  $e = xy$ ,

$$e^{K \mathbf{1}_{\{\sigma_x = \sigma_y\}}} = 1 + (e^K - 1) \mathbf{1}_{\{\sigma_x = \sigma_y\}}.$$

Expanding the product over edges chooses a subset  $\omega \subseteq E$  and contributes  $(e^K - 1)^{|\omega|}$ . The spin sum is constrained only by the condition that spins be constant on every connected component of  $(V, \omega)$ , so it contributes exactly  $q^{k(\omega)}$ . Hence

$$Z_{\text{Potts}} = \sum_{\omega \subseteq E} (e^K - 1)^{|\omega|} q^{k(\omega)}.$$

With  $p = 1 - e^{-K}$  one has  $e^K - 1 = p/(1 - p)$  and  $(1 - p)^{-|E|} = e^{K|E|}$ , giving

$$Z_{\text{Potts}} = e^{K|E|} \sum_{\omega \subseteq E} p^{|\omega|} (1 - p)^{|E| - |\omega|} q^{k(\omega)}.$$

□

**Theorem 1.4** (Edwards–Sokal coupling). *For integer  $q \geq 2$ , there is a joint law of Potts spins and FK edges such that, conditional on spins, equal-color edges are independently open with probability  $p$ , while conditional on FK edges, each cluster receives an independent uniform color. Its marginals are respectively the Potts and FK measures.*

*Proof.* Define a joint weight by

$$\mathbb{P}(\sigma, \omega) \propto \prod_{e=xy} \left[ (1 - p) \mathbf{1}_{\{\omega_e=0\}} + p \mathbf{1}_{\{\omega_e=1\}} \mathbf{1}_{\{\sigma_x=\sigma_y\}} \right]. \quad (1.1)$$

For fixed  $\sigma$ , an edge joining unequal colors must be closed, while an edge joining equal colors is open independently with probability  $p$ . Thus the first conditional law is the one stated.

For fixed  $\omega$ , (1.1) is constant over precisely those spin configurations that are constant on every open cluster. Each of the  $k(\omega)$  clusters may choose any of  $q$  colors independently, so there are  $q^{k(\omega)}$  such configurations and the conditional cluster colors are independent uniform variables. Summing (1.1) over  $\sigma$  gives  $p^{o(\omega)} (1 - p)^{|E| - o(\omega)} q^{k(\omega)}$ , the FK weight. Summing over  $\omega$  edge by edge gives  $(1 - p) + p \mathbf{1}_{\{\sigma_x=\sigma_y\}} = (1 - p) e^{K \mathbf{1}_{\{\sigma_x=\sigma_y\}}}$ , the Potts weight up to the constant  $(1 - p)^{|E|}$ . Hence the two claimed marginals follow. □

**Theorem 1.5** (FKG for random cluster). *For  $q \geq 1$ , finite-volume random-cluster measures are positively associated: increasing functions  $F, G$  satisfy  $\phi(FG) \geq \phi(F)\phi(G)$ .*

*Proof.* Write the unnormalized FK weight as

$$w(\omega) = p^{o(\omega)} (1 - p)^{|E| - o(\omega)} q^{k(\omega)}.$$

The edge count is modular:

$$o(\omega \vee \eta) + o(\omega \wedge \eta) = o(\omega) + o(\eta).$$

The component count is supermodular,

$$k(\omega \vee \eta) + k(\omega \wedge \eta) \geq k(\omega) + k(\eta). \quad (1.3)$$

Indeed the graphic-matroid rank is  $r(A) = |V| - k(A)$  and rank is submodular; multiplying by  $-1$  gives (1.3). Since  $q \geq 1$ , the two identities imply the lattice condition

$$w(\omega \vee \eta) w(\omega \wedge \eta) \geq w(\omega) w(\eta).$$

The finite-lattice FKG theorem proved in Volume VIII now applies to the edge lattice  $\{0, 1\}^E$  and gives  $\phi(FG) \geq \phi(F)\phi(G)$  for increasing  $F, G$ . Wired boundary conditions amount to contracting a fixed boundary partition before counting connected components. Graphic-matroid submodularity applies to that contracted graph as well, so the same proof covers wired measures. □

**Theorem 1.6** (Domain Markov property). *Conditioned on the configuration outside a subgraph  $D$ , the law of the edges in  $D$  is a random-cluster measure with the same  $(p, q)$  and boundary condition obtained by wiring boundary vertices connected through the exterior open configuration and the original boundary partition.*

*Proof.* Fix the exterior edge configuration  $\eta$  on  $E \setminus E(D)$ . For two interior configurations  $\omega, \omega'$  the conditional probability ratio is

$$\frac{\phi(\omega \mid \eta)}{\phi(\omega' \mid \eta)} = \left( \frac{p}{1-p} \right)^{o(\omega) - o(\omega')} q^{k(\omega \cup \eta) - k(\omega' \cup \eta)}. \quad (1.2)$$

Exterior open paths identify exactly those boundary vertices of  $D$  that belong to the same exterior cluster; the original wired boundary condition adds its own identifications. After contracting these equivalence classes, the difference of cluster counts in (1.2) is precisely the difference of cluster counts for the random-cluster model on  $D$  with that induced boundary partition. The factors depending only on  $\eta$  cancel in the ratio, so the conditional law is exactly the asserted FK measure.  $\square$

**Example 1.7** ( $q = 2$  recovers Ising). Color FK clusters by independent signs. Two vertices have equal sign correlation exactly when they belong to the same cluster, hence  $\langle \sigma_x \sigma_y \rangle = \phi_{p,2}(x \leftrightarrow y)$ . The parameter relation  $p = 1 - e^{-2K}$  then converts Potts/FK duality into Kramers–Wannier duality.

## Chapter 2

# Infinite-Volume Random-Cluster Measures

**Theorem 2.1** (Existence of free and wired measures). *For  $q \geq 1$  on a locally finite infinite graph, monotone finite-volume limits with free and wired boundary conditions exist along exhaustions and define infinite-volume random-cluster DLR measures  $\phi_{p,q}^0$  and  $\phi_{p,q}^1$ .*

*Proof.* Let  $\Lambda_n \uparrow G$  be an exhaustion. By FKG and boundary monotonicity, for every increasing cylinder event  $A$  supported in  $\Lambda_m$ , the sequence

$$\phi_{\Lambda_n,p,q}^0(A), \quad n \geq m,$$

is monotone in the appropriate direction and bounded in  $[0, 1]$ ; the same is true for wired boundary conditions. (Equivalently, use the standard monotone coupling obtained by revealing single edges.) Hence the limits of all increasing cylinder events exist. Inclusion–exclusion then gives limits for every cylinder event, and Kolmogorov consistency produces probability measures  $\phi_{p,q}^0$  and  $\phi_{p,q}^1$  on  $\{0, 1\}^{E(G)}$ .

To verify the DLR property, fix a finite edge set  $D$  and a bounded cylinder function  $F$ . For all sufficiently large  $n$ , the finite-volume domain Markov property gives

$$\phi_{\Lambda_n}^\xi(F) = \phi_{\Lambda_n}^\xi[\gamma_D^\eta(F)], \quad \xi \in \{0, 1\},$$

where  $\gamma_D^\eta$  is the FK kernel with the boundary partition induced by the exterior configuration. This kernel depends on only finitely many exterior connectivity relations; approximate those relations by connections inside an increasing finite collar and use monotone convergence. Passing  $n \rightarrow \infty$  yields the same identity for  $\phi^0$  and  $\phi^1$ , proving that both are infinite-volume random-cluster DLR measures. The monotone limits are independent of the chosen exhaustion because any two exhaustions interlace and give the same cylinder-event limits.  $\square$

**Theorem 2.2** (Monotonicity in  $p$  and boundary condition). *For  $q \geq 1$ , increasing  $p$  or wiring more boundary vertices stochastically increases the random-cluster measure.*

*Proof.* Condition on all edges except  $e = xy$ . If  $x$  and  $y$  are already connected without  $e$ , then

$$\phi(\omega_e = 1 \mid \omega_{E \setminus \{e\}}) = p.$$

If they are not connected, opening  $e$  decreases the cluster number by one, so

$$\phi(\omega_e = 1 \mid \omega_{E \setminus \{e\}}) = \frac{p}{p + q(1 - p)}. \quad (2.1)$$

Both quantities are increasing functions of  $p$ . Moreover, wiring more boundary vertices can only turn the second case into the first, and for  $q \geq 1$  the first probability is at least the second. Couple successive single-edge heat-bath variables with a common uniform random variable. The conditional inequalities above preserve the coordinatewise order at every update, giving a monotone finite-volume coupling. Iterating to equilibrium proves stochastic monotonicity in both  $p$  and the boundary partition. Passing to the monotone infinite-volume limits preserves stochastic order.  $\square$

**Theorem 2.3** (Uniqueness of the infinite cluster). *For the translation-invariant free or wired FK measures on an amenable transitive graph, under the standard finite-energy conditions away from degenerate parameters, the number of infinite open clusters is almost surely 0 or 1.*

*Proof.* Work first in a translation-ergodic component; the number  $N_\infty$  of infinite clusters is then almost surely constant. Finite energy follows from (2.1): for  $0 < p < 1$  both conditional edge probabilities are bounded strictly between 0 and 1.

A finite value  $2 \leq N_\infty = k < \infty$  is impossible. With positive probability a large ball meets all  $k$  infinite clusters. By changing only finitely many edges inside the ball, an event of positive conditional probability by finite energy, we may join those clusters into one, producing fewer than  $k$  infinite clusters and contradicting ergodicity.

If  $N_\infty = \infty$ , a finite modification can with positive probability create a trifurcation vertex: deleting that vertex separates its infinite cluster into at least three infinite branches. Translation invariance would then give a positive density of trifurcations. But in a finite Følner set  $\Lambda$ , connect each trifurcation to three distinct exits from  $\Lambda$  and prune loops. The resulting finite forest has at least one distinct boundary exit for each excess branch, so the number of trifurcations in  $\Lambda$  is bounded by a constant times  $|\partial\Lambda|$ . Amenability makes  $|\partial\Lambda|/|\Lambda| \rightarrow 0$ , contradicting positive density. Thus an ergodic component has  $N_\infty \in \{0, 1\}$ ; integrating the ergodic decomposition gives the same almost-sure conclusion for the translation-invariant free and wired states.  $\square$

## Chapter 3

# Planar Duality and the Exact Critical Point

**Definition 3.1** (Planar dual FK configuration). For a planar embedded graph  $G$  and dual graph  $G^*$ , the dual configuration  $\omega^*$  is defined by declaring  $e^*$  open exactly when the corresponding primal edge  $e$  is closed. The dual parameter  $p^*$  is characterized by

$$\frac{p^*}{1 - p^*} = q \frac{1 - p}{p}.$$

*Worked example 3.2* (Solving the self-dual equation). Self-duality requires  $p^* = p$ . Substituting into the preceding relation gives

$$\frac{p}{1 - p} = \sqrt{q}, \quad \text{hence} \quad p_{\text{sd}}(q) = \frac{\sqrt{q}}{1 + \sqrt{q}}.$$

For  $q = 1$  this gives  $p_{\text{sd}} = 1/2$ ; for FK-Ising,  $q = 2$ , it gives  $p_{\text{sd}} = \sqrt{2}/(1 + \sqrt{2})$ .

**Theorem 3.3** (Planar FK duality). *For a finite planar graph with dual  $G^*$ , the dual of an FK configuration is obtained by declaring a dual edge open exactly when the primal edge is closed. The transformed parameter satisfies*

$$\frac{p^*}{1 - p^*} = q \frac{1 - p}{p},$$

*with free and wired boundary conditions interchanged, up to the standard Euler normalization.*

*Proof.* Let  $A \subseteq E$  be the primal open-edge set and let  $A^* = \{e^* : e \notin A\}$ . Thus  $|A^*| = |E| - |A|$ . For a connected planar embedding (with the boundary convention absorbed by the usual contraction), Euler's formula applied to the spanning subgraph  $(V, A)$  gives

$$k_G(A) = |V| - |A| + k_{G^*}(A^*) - 1, \tag{3.1}$$

up to the configuration-independent boundary constant. Substitute (3.1) into the FK weight:

$$p^{|A|} (1 - p)^{|E| - |A|} q^{k_G(A)}.$$

After removing factors independent of  $A$ , the remaining dependence on the dual open set is

$$\left( \frac{q(1 - p)}{p} \right)^{|A^*|} q^{k_{G^*}(A^*)}.$$

This is an FK weight on  $G^*$  precisely when

$$\frac{p^*}{1-p^*} = q \frac{1-p}{p}.$$

A free primal boundary leaves boundary vertices distinct; in the dual compactification this identifies the dual boundary into one wired vertex, and conversely. Hence free and wired conditions are interchanged. All omitted Euler factors are configuration independent and disappear in the normalization, proving the duality statement.  $\square$

**Lemma 3.4** (Differentiation of finite-volume FK expectations). *Let  $G = (V, E)$  be finite,  $q \geq 1$ , let  $\xi$  be any boundary condition, and let  $f : \{0, 1\}^E \rightarrow \mathbb{R}$ . Then, for  $0 < p < 1$ ,*

$$\frac{d}{dp} \phi_{G,p,q}^\xi(f) = \frac{1}{p(1-p)} \sum_{e \in E} \text{Cov}_{\phi_{G,p,q}^\xi}(f, \omega_e).$$

*If  $f$  is increasing, every covariance on the right is nonnegative.*

*Proof.* Write  $N(\omega) = \sum_{e \in E} \omega_e$ . Differentiating the normalized FK weight gives

$$\frac{d}{dp} \log(p^{N(\omega)}(1-p)^{|E|-N(\omega)}) = \frac{N(\omega) - p|E|}{p(1-p)}.$$

The derivative of a normalized expectation is therefore the covariance of  $f$  with this logarithmic derivative. Expanding  $N = \sum_e \omega_e$  gives the formula. For  $q \geq 1$  the FK measure satisfies FKG, so the covariance of two increasing functions, in particular of  $f$  and  $\omega_e$ , is nonnegative.  $\square$

**Lemma 3.5** (Monotone-measure OSSS inequality). *Let  $\mu$  be an FKG probability measure on  $\{0, 1\}^E$  with finite energy, and let  $T$  be a decision tree which determines an increasing Boolean function  $f$ . If  $\delta_e(T)$  is the probability that  $T$  queries  $e$ , then*

$$\text{Var}_\mu(f) \leq C_\mu \sum_{e \in E} \delta_e(T) \text{Cov}_\mu(f, \omega_e), \quad C_\mu := \left( \min_{e \in E} \text{Var}_\mu(\omega_e) \right)^{-1}.$$

*For FK measures with  $q \geq 1$  and  $p$  in a compact subinterval of  $(0, 1)$ ,  $C_\mu$  is bounded uniformly in the finite graph and the boundary condition.*

*Proof.* Fix an ordering of the coordinates and realize  $\mu$  from independent uniforms  $U_e$  by revealing coordinates successively: when the previously exposed coordinates have value  $\eta$ , put

$$\omega_e = \mathbf{1}\{U_e \leq \mu(\omega_e = 1 \mid \eta)\}.$$

FKG implies that these conditional thresholds are increasing functions of the already exposed 1's. Run the decision tree using this monotone realization. Let  $U$  and  $U'$  be independent uniform families and replace, one queried coordinate at a time, the uniform used by  $U$  with the corresponding one from  $U'$ . The resulting telescoping sequence starts with a sample of  $\mu$  and ends with an independent sample.

Condition on the history immediately before the tree queries  $e$ . By monotonicity, the two possible conditional laws obtained by forcing  $\omega_e = 0$  or  $1$  can be coupled in increasing order. Hence the conditional expected change of the increasing Boolean  $f$  is at most

$$\mu(f \mid \omega_e = 1) - \mu(f \mid \omega_e = 0) = \frac{\text{Cov}_\mu(f, \omega_e)}{\text{Var}_\mu(\omega_e)}.$$



Only queried edges contribute. Summing the telescoping differences and averaging over the tree gives

$$\mathbb{E}|f(\omega) - f(\tilde{\omega})| \leq C_\mu \sum_e \delta_e(T) \text{Cov}_\mu(f, \omega_e).$$

For independent  $\omega, \tilde{\omega}$  and Boolean  $f$  the left side is  $2 \text{Var}(f)$ ; absorbing this harmless factor into the displayed constant gives the stated form. Finally, the FK single-edge conditional open probability is either  $p$  or

$$\frac{p}{p + q(1 - p)},$$

according as the endpoints are already connected or not. On a compact  $p$ -interval these probabilities stay uniformly away from 0 and 1, proving the finite-energy bound.  $\square$

**Lemma 3.6** (Random-shell revealment bound). *Let  $\Lambda_n$  be a ball of radius  $n$  in a transitive finite-range graph, let*

$$A_n = \{0 \leftrightarrow \partial\Lambda_n\}, \quad \theta_n(p) = \phi_{\Lambda_{2n,p,q}}^1(A_n), \quad S_n(p) = \sum_{k=0}^{n-1} \theta_k(p).$$

*There is a randomized decision tree determining  $A_n$  such that every edge  $e$  has revealment*

$$\delta_e \leq C \frac{S_n(p)}{n},$$

*where  $C$  depends only on the finite-range geometry.*

*Proof.* Choose  $K$  uniformly from  $\{1, \dots, n\}$ . Reveal all clusters intersecting the shell  $\partial\Lambda_K$ , exposing an edge when it is incident to the currently discovered cluster; continue until every cluster touching the shell has been completely exposed. The event  $A_n$  is then determined: every path from 0 to  $\partial\Lambda_n$  crosses  $\partial\Lambda_K$ , so such a path exists exactly when one of the exposed shell clusters meets both sides.

An edge  $e$  can be queried only if one of its endpoints is connected to  $\partial\Lambda_K$ . If  $r(e, K)$  is the distance from  $e$  to that shell, the domain Markov property and boundary-condition monotonicity bound this probability by  $C\theta_{r(e,K)}(p)$ . For fixed  $e$ , as  $K$  ranges over  $1, \dots, n$ , each distance occurs only  $O(1)$  times by finite range. Averaging over the uniform  $K$  gives

$$\delta_e \leq \frac{C}{n} \sum_{j=0}^{n-1} \theta_j(p) = C \frac{S_n(p)}{n}.$$

$\square$

**Lemma 3.7** (Sharpness bootstrap). *Let  $f_n : [0, p_0] \rightarrow [0, M]$  be increasing differentiable functions which converge pointwise to  $f$ , put*

$$\Sigma_n(p) = \sum_{k=0}^{n-1} f_k(p),$$

*and suppose*

$$f'_n(p) \geq \frac{n}{\Sigma_n(p)} f_n(p) \quad (n \geq 1).$$

*Then there is  $p_1 \in [0, p_0]$  such that for  $p < p_1$  one has  $f_n(p) \leq e^{-c_p n}$  for all sufficiently large  $n$ , whereas for  $p > p_1$ ,*

$$f(p) \geq p - p_1.$$

*Multiplying the right side of the differential inequality by a fixed positive constant merely rescales the final linear constant.*

*Proof.* Define

$$p_1 := \inf \left\{ p : \limsup_{n \rightarrow \infty} \frac{\log \Sigma_n(p)}{\log n} \geq 1 \right\}.$$

Fix  $p < p_1$  and choose  $p'' < p' < p$ . By the definition of  $p_1$  there are  $\alpha > 0$  and  $N$  such that  $\Sigma_n(p) \leq n^{1-\alpha}$  for  $n \geq N$ . Since  $\Sigma_n$  is increasing in  $p$ , on  $[p', p]$  we have

$$f'_n \geq n^\alpha f_n.$$

Integration gives

$$f_n(p') \leq M e^{-(p-p')n^\alpha}.$$

Consequently  $\sup_n \Sigma_n(p') =: \Sigma < \infty$ . On  $[p'', p']$  the original differential inequality now gives  $f'_n \geq (n/\Sigma)f_n$ , and a second integration yields

$$f_n(p'') \leq M e^{-(p'-p'')n/\Sigma}.$$

This is exponential decay below  $p_1$ .

Now let  $p > p_1$  and define

$$T_n(p) = \frac{1}{\log n} \sum_{i=1}^n \frac{f_i(p)}{i}.$$

Differentiation and the hypothesis imply

$$T'_n(p) \geq \frac{1}{\log n} \sum_{i=1}^n \frac{f_i(p)}{\Sigma_i(p)}.$$

Because  $\Sigma_{i+1} = \Sigma_i + f_i$ ,

$$\frac{f_i}{\Sigma_i} \geq \int_{\Sigma_i}^{\Sigma_{i+1}} \frac{dt}{t} = \log \Sigma_{i+1} - \log \Sigma_i.$$

Hence

$$T'_n(p) \geq \frac{\log \Sigma_{n+1}(p) - \log \Sigma_1(p)}{\log n}.$$

Choose  $p' \in (p_1, p)$  and integrate from  $p'$  to  $p$ . Monotonicity of  $\Sigma_n$  gives

$$T_n(p) - T_n(p') \geq (p - p') \frac{\log \Sigma_n(p') - \log M}{\log n}.$$

Since  $f_n \rightarrow f$ , the logarithmic Cesàro means  $T_n$  converge to  $f$ . By the definition of  $p_1$ , the limsup of the last ratio is at least 1. Therefore  $f(p) - f(p') \geq p - p'$ . Letting  $p' \downarrow p_1$  and using  $f \geq 0$  proves  $f(p) \geq p - p_1$ .  $\square$

**Theorem 3.8** (Sharp planar random-cluster/Potts transition). *For the planar random-cluster/Potts models in the stated  $q \geq 1$  lattice classes, the transition is sharp: below  $p_c$  connectivities decay exponentially, while above  $p_c$  an infinite cluster/order parameter is present with the quantitative lower behavior supplied by the sharpness theorem.*

*Proof.* Work first in  $\Lambda_{2n}$  with wired boundary condition and set  $f = \mathbf{1}_{A_n}$ . By Lemma 3.5 and the random-shell decision tree,

$$\theta_n(1 - \theta_n) \leq C \frac{S_n}{n} \sum_e \text{Cov}(\mathbf{1}_{A_n}, \omega_e).$$

Lemma 3.4 therefore gives, uniformly for  $p$  in compact subsets  $[0, p_0] \subset (0, 1)$ ,

$$\theta'_n(p) \geq c \frac{n}{S_n(p)} \theta_n(p) (1 - \theta_n(p)).$$

Boundary-condition comparison gives  $\theta_n(p) \leq \theta_1(p_0) < 1$  for  $p \leq p_0$ , so  $1 - \theta_n$  is bounded below by a positive constant. After absorbing it into  $c$  and rescaling the parameter, Lemma 3.7 applies.

We check that its limiting function is the infinite-volume order parameter. Let

$$\theta(p) = \phi_{G,p,q}^1(0 \leftrightarrow \infty).$$

Clearly  $\theta_n(p) \geq \theta(p)$  by comparison with the infinite-volume wired state. Conversely, for every fixed  $R < n$ , the event  $A_n$  is contained in  $\{0 \leftrightarrow \partial\Lambda_R\}$ ; the restriction of the wired measure on  $\Lambda_{2n}$  to  $\Lambda_R$  converges to the infinite-volume wired measure. Thus

$$\limsup_{n \rightarrow \infty} \theta_n(p) \leq \phi_{G,p,q}^1(0 \leftrightarrow \partial\Lambda_R),$$

and  $R \rightarrow \infty$  gives  $\theta_n(p) \rightarrow \theta(p)$ .

Let  $p_1$  be the threshold supplied by the bootstrap lemma. For every  $p < p_1$ , exponential decay forces  $\theta(p) = 0$ ; hence  $p_1 \leq p_c$ , where  $p_c := \inf\{p : \theta(p) > 0\}$ . For every  $p > p_1$ , the lemma gives  $\theta(p) > 0$ , so  $p_c \leq p_1$ . Therefore  $p_1 = p_c$ . Restoring the absorbed constant gives

$$\phi_{p,q}^1(0 \leftrightarrow \infty) \geq c(p - p_c)$$

locally above  $p_c$ .

Below  $p_c$ , the exponential radial bound implies exponential two-point decay by translation:  $\{x \leftrightarrow y\}$  forces a connection from  $x$  to distance at least  $d(x, y)/2$ . Through the Edwards–Sokal coupling, positive FK percolation in the wired state is exactly the spontaneous ordered Potts state, so the same quantitative lower bound transfers to the ordered Potts formulation; the subcritical connectivity estimate transfers to spin correlations through the same coupling.  $\square$

**Lemma 3.9** (Free percolation above the wired threshold). *Let  $q \geq 1$  and consider the nearest-neighbor random-cluster model on a transitive amenable lattice. If*

$$p_c := \inf\{p : \phi_{p,q}^1(0 \leftrightarrow \infty) > 0\},$$

*then for every  $p > p_c$  the free state  $\phi_{p,q}^0$  also has an infinite open cluster almost surely.*

*Proof.* We include the short pressure argument because the planar duality proof below needs the statement for the *free* dual state. Put  $y = p/(1 - p)$  and  $t = \log y$ . For a Følner box  $\Lambda_n$  write

$$\widehat{Z}_n^\xi(t) = \sum_{\omega \subset E_n} e^{t|\omega|} q^{k_\xi(\omega)}, \quad \xi \in \{0, 1\},$$

where 0 and 1 denote free and wired boundary conditions. Wiring the boundary can merge at most  $|\partial\Lambda_n| - 1$  clusters, so

$$q^{-(|\partial\Lambda_n|-1)} \widehat{Z}_n^0(t) \leq \widehat{Z}_n^1(t) \leq \widehat{Z}_n^0(t).$$

After division by  $|E_n|$ , amenability shows that the free and wired pressures have the same limit, say  $\psi(t)$ . The finite-volume functions are convex and

$$\frac{d}{dt} \frac{1}{|E_n|} \log \widehat{Z}_n^\xi(t) = \frac{1}{|E_n|} \phi_{\Lambda_n,p,q}^\xi(|\omega|).$$

At every differentiability point of the convex function  $\psi$ , convergence of convex functions forces the two displayed derivatives to converge to the same number. Passing to the infinite-volume limits therefore gives the same one-edge density in  $\phi_{p,q}^0$  and  $\phi_{p,q}^1$ . Since  $\phi_{p,q}^0 \preceq \phi_{p,q}^1$ , a monotone coupling has  $\omega_e^0 \leq \omega_e^1$  for every edge. Equality of the one-edge expectations makes  $\omega_e^0 = \omega_e^1$  almost surely for each edge, and countability of the edge set gives

$$\phi_{p,q}^0 = \phi_{p,q}^1$$

at every differentiability point of  $\psi$ .

A convex function on an interval has at most countably many nondifferentiability points. Fix  $p > p_c$  and choose  $r \in (p_c, p)$  at which  $\psi$  is differentiable. The sharpness theorem above gives  $\phi_{r,q}^1(0 \leftrightarrow \infty) > 0$ , hence the same is true for  $\phi_{r,q}^0$ . Monotonicity in  $p$  now yields positive free percolation probability at  $p$ .

For completeness, the free and wired monotone limits are extremal DLR states. Indeed, if the minimal state  $\phi^0$  were a nontrivial convex combination  $a\nu + (1-a)\rho$  of DLR states, minimality would give  $\phi^0(f) \leq \nu(f), \rho(f)$  for every increasing cylinder function  $f$ , while the convex-combination identity forces equality for all such  $f$  and hence  $\nu = \rho = \phi^0$ ; the maximal wired state is handled identically with the inequalities reversed. Extremal translation-invariant DLR states are translation ergodic. The event that an infinite cluster exists is translation invariant, so its positive probability is therefore one. Thus the free state has an infinite cluster almost surely for every  $p > p_c$ .  $\square$

**Lemma 3.10** (Planar primal–dual non-coexistence). *Let  $\mu$  be a translation-invariant FKG measure on edge configurations of a locally finite doubly-periodic planar graph. Suppose that, almost surely, the primal configuration has a unique infinite open cluster. Then the complementary dual configuration cannot also have a unique infinite open cluster almost surely.*

*Proof.* We reproduce the crossing argument because it is the topological input which converts sharpness into an exact critical point. Embed the graph periodically in the plane. For the rectangle  $R = [0, n] \times [0, k]$  let

$$\begin{aligned} h(n, k) &= \mu(R \text{ has a primal left–right crossing}), \\ v(n, k) &= \mu(R \text{ has a primal top–bottom crossing}). \end{aligned}$$

Assume for contradiction that both the primal and complementary dual configurations have unique infinite clusters.

*Step 1: the balance-box estimate.* Fix an integer  $s \gg 1$  and write  $S_x = x + [0, s]^2$ . Choose a lattice point  $x = x(R)$  for which connection probabilities from  $S_x$  to opposite sides are balanced in the following discrete sense: with neighboring points  $x', x''$ ,

$$\begin{aligned} \mu(S_x \leftrightarrow \text{Bottom}) &\geq \mu(S_x \leftrightarrow \text{Top}), \\ \mu(S_x \leftrightarrow \text{Left}) &\geq \mu(S_x \leftrightarrow \text{Right}), \\ \mu(S_{x'} \leftrightarrow \text{Top}) &\geq \mu(S_{x'} \leftrightarrow \text{Bottom}), \\ \mu(S_{x''} \leftrightarrow \text{Right}) &\geq \mu(S_{x''} \leftrightarrow \text{Left}). \end{aligned} \tag{3.18}$$

Here every connection is required to stay inside  $R$ . Such a point exists: the lattice boundary between the vertices where the first inequality holds and those where it fails crosses  $R$  from left to right, while the analogous boundary for the second inequality crosses from top to bottom; the two discrete interfaces meet.

The distance from  $x(R)$  to  $\partial R$  tends to infinity with  $\min(n, k)$ . Indeed, simultaneous uniqueness of the primal and dual infinite clusters excludes an infinite cluster confined to a half-plane. To see

this directly, take the right half-plane  $H$  and its boundary line  $\ell$ . Far inside  $H$  choose a square  $S_z$  whose probabilities of connecting inside  $H$  to the upper and lower rays of  $\ell$  change order between  $z$  and  $z + (0, 1)$ . FKG gives

$$\mu(S_z \leftrightarrow \ell^-), \mu(S_{z+(0,1)} \leftrightarrow \ell^+) \geq 1 - \mu(S \not\leftrightarrow \ell)^{1/2}.$$

Uniqueness implies that, with probability tending to one as  $z$  recedes from  $\ell$ , the two long connections belong to the same component before reaching  $\ell$ . Their planar union separates a unit segment of  $\ell$  from infinity inside  $H$ . Letting first the distance of  $z$  and then  $s$  tend to infinity shows that no infinite component confined to  $H$  can meet that unit segment. Translation invariance excludes a half-plane infinite component altogether. The same argument applies to each side of  $R$ , proving the claim about  $x(R)$ .

Let  $A_R$  be the event that among the primal components in  $R$  meeting the distance-2 neighborhood of  $S_x$ , exactly one reaches  $\partial R$ . Because  $x$  is macroscopically far from the boundary and the full-plane infinite cluster is unique,  $\mu(A_R) \rightarrow 1$ : the finitely many local vertices which belong to the infinite cluster coalesce in a finite random box, while every local finite cluster has finite radius.

Put  $\pi_s = \mu(S \leftrightarrow \infty)$ . Suppose, for example, that  $\mu(S_x \leftrightarrow \text{Bottom}) \geq \mu(S_x \leftrightarrow \text{Left})$ . Together with (3.18), Bottom is then at least as likely as each of the other three sides. Applying FKG to the four decreasing events “ $S_x$  does not connect to this side” gives

$$\mu(S_x \not\leftrightarrow \text{Bottom}) \leq (1 - \pi_s)^{1/4}. \quad (3.19)$$

Shift  $R$  one lattice step horizontally and use the neighboring balance point  $x'$ . On  $A_R$ , a bottom connection from  $S_x$  and a top connection from the shifted square must belong to the same component, hence create a vertical crossing of the one-step-enlarged rectangle. Therefore

$$v(n, k + 1) \geq \pi_s - 2(1 - \pi_s)^{1/4} - 2\mu(A_R^c). \quad (3.20)$$

If the displayed comparison with Left is reversed, the same argument with  $x''$  gives (3.20) with  $h(n, k)$  in place of  $v(n, k + 1)$ . Since  $\pi_s \uparrow 1$  as  $s \rightarrow \infty$ , we obtain

$$\max\{h(n, k), v(n, k + 1)\} \rightarrow 1 \quad (3.21)$$

as  $\min(n, k) \rightarrow \infty$ .

Apply (3.21) to the complementary dual configuration. A failure of a primal vertical crossing is exactly a dual horizontal crossing, up to the harmless one-lattice-step shift of the rectangle, and similarly with horizontal/vertical exchanged. Hence

$$\min\{h(n, k), v(n, k)\} \rightarrow 0. \quad (3.22)$$

*Step 2: contradiction.* For each  $n$  choose the largest integer  $k_n$  with  $v(n, k_n) \geq h(n, k_n)$ . Uniqueness of both infinite clusters implies  $k_n \rightarrow \infty$ ; for fixed  $k$ , long rectangles are crossed in the long direction by whichever unique infinite cluster enters them. By maximality,  $v(n, k_n + 1) < h(n, k_n + 1)$ . Equation (3.21) says that either  $h(n, k_n)$  or  $v(n, k_n + 1)$  tends to one. In the first case  $v(n, k_n) \geq h(n, k_n)$  also tends to one; in the second,  $h(n, k_n + 1) > v(n, k_n + 1)$  does. Either conclusion contradicts (3.22). Thus simultaneous unique primal and dual infinite clusters are impossible.  $\square$

**Theorem 3.11** (Dual critical parameters on periodic planar graphs). *Let  $G$  be a locally finite doubly-periodic planar graph and  $G^*$  its planar dual. For  $q \geq 1$ , if*

$$D_q(p) := \frac{q(1-p)}{p + q(1-p)},$$

then

$$D_q(p_c(G)) = p_c(G^*),$$

or equivalently

$$\frac{p_c(G)p_c(G^*)}{(1-p_c(G))(1-p_c(G^*))} = q.$$

*Proof.* The map  $D_q$  is the decreasing involution furnished by planar FK duality. Fix  $p < p_c(G)$ . By the sharpness theorem, wired primal connectivities decay exponentially. Consequently the probability of an open primal circuit crossing a dyadic annulus of radius  $r$  is at most  $Cre^{-cr}$ , which is summable over dyadic  $r$ . Borel–Cantelli implies that only finitely many large annuli contain a primal open circuit surrounding the origin. Planar separation then gives dual-open connections from a fixed dual neighborhood to arbitrarily large distance; local finiteness and König’s lemma produce an infinite dual-open path. Since the dual of the wired primal state is the free dual state at  $D_q(p)$ , we have

$$D_q(p) \geq p_c(G^*).$$

Letting  $p \uparrow p_c(G)$  gives  $D_q(p_c(G)) \geq p_c(G^*)$ .

For the reverse inequality take  $p > p_c(G)$ . The wired primal state has an infinite cluster by sharpness, and it is unique by Theorem 2.3 below the notation of Chapter 2. If  $D_q(p) > p_c(G^*)$ , Lemma 3.9 says that the free dual state at  $D_q(p)$  also percolates; Theorem 2.3 makes its infinite cluster unique. But the free dual state is exactly the complementary dual of the wired primal state, contradicting Lemma 3.10. Hence

$$D_q(p) \leq p_c(G^*).$$

Let  $p \downarrow p_c(G)$  and use continuity of  $D_q$  to obtain the opposite inequality. The two inequalities prove the theorem.  $\square$

**Theorem 3.12** (Self-dual point is critical). *For the square-lattice random-cluster model with  $q \geq 1$ ,*

$$p_c(q) = \frac{\sqrt{q}}{1 + \sqrt{q}}.$$

*Proof.* The square lattice is isomorphic to its planar dual, so the preceding critical parameter relation gives

$$D_q(p_c) = p_c.$$

Using  $D_q(p) = q(1-p)/(p+q(1-p))$ , the fixed-point equation is equivalent to

$$\left( \frac{p_c}{1-p_c} \right)^2 = q.$$

Since  $0 < p_c < 1$ , the ratio is positive, hence

$$\frac{p_c}{1-p_c} = \sqrt{q}, \quad p_c = \frac{\sqrt{q}}{1 + \sqrt{q}}.$$

Thus the self-dual point is the critical point, with no undefined auxiliary parameter.  $\square$

## Chapter 4

# Continuous and Discontinuous Planar Transitions

**Theorem 4.1** (Continuity for  $1 \leq q \leq 4$ ). *For the nearest-neighbor planar square-lattice random-cluster model with  $1 \leq q \leq 4$ , the critical transition is continuous in the sense of the primary theorem: there is no infinite cluster in the wired critical state, equivalently free and wired infinite-volume measures coincide; for integer  $q = 2, 3, 4$  the critical Potts Gibbs state is unique.*

*Proof.* Throughout this proof

$$p = p_c(q) = \frac{\sqrt{q}}{1 + \sqrt{q}}, \quad \pi_n := \phi_{\mathbb{Z}^2, p, q}^0[0 \longleftrightarrow \partial\Lambda_n]. \quad (9.1)$$

The proof has two logically separate parts. We first prove the genuinely model-specific input

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \pi_n = 0, \quad (9.2)$$

using the parafermionic observable on the universal cover of the punctured plane. We then feed (9.2) into the boundary-condition-uniform RSW theorem proved in the next chapter and deduce absence of a wired infinite cluster, equality of the two FK states, and Potts uniqueness. Although Theorem SM-09.05.02 is printed later, its proof assumes only (9.2) and the planar FK foundations of Chapters 1–3; in particular it does not use the present theorem. Thus this use is a forward reference in exposition but not a circular dependency.

We first establish a polynomial Dobrushin-box connection estimate that will prevent the parafermionic mass from escaping through the high sheets of the universal cover. Let

$$S = \mathbb{Z} \times [0, 2n]$$

carry the infinite-strip random-cluster measure that is free on the bottom side and wired on the top side. For an integer  $k$  put

$$F(k) = \min\{\ell : (k, \ell) \longleftrightarrow \mathbb{Z} \times \{2n\}\}.$$

On  $\{F(0) \leq n\}$  there is a  $k$  with  $|k| \leq n^3$  such that  $F(k) \leq n$  and  $F(k') \geq F(k)$  whenever  $|k' - k| \leq n$ . Indeed, if no such  $k$  existed, starting from  $k_0 = 0$  one could choose successively  $|k_{j+1} - k_j| \leq n$  with

$$0 < F(k_{j+1}) < F(k_j) \leq n,$$

for  $j = 0, \dots, n-1$ , producing  $n+1$  distinct positive integer values in  $\{1, \dots, n\}$ .

For  $x = (k, F(k))$ , let  $S_x$  be the translate of the  $2n$ -by- $2n$  Dobrushin box centered horizontally at  $x$ , and let  $A(x)$  be the event that  $x$  is connected to the wired arc of  $S_x$  while no open path from  $x$  exits  $S_x$  through its free arc. The preceding local-minimum property forces  $A(x)$ . Critical duality and reflection imply

$$\phi_S[F(0) \leq n] \geq \frac{1}{2},$$

because the interface between the top wired primal cluster and the bottom wired dual cluster crosses the vertical segment  $\{0\} \times [0, n]$  with probability at least one half. There are at most  $5n^4$  possible choices of  $x$  in the above search region, hence for one such  $x$ ,

$$\phi_S[A(x)] \geq \frac{1}{10n^4}. \quad (9.3)$$

On  $A(x)$  take the lowest dual-open path in  $S_x$  separating  $x$  from the free arc. Conditional on this path, the domain Markov property and monotonicity of boundary conditions show that the law above it is dominated by the Dobrushin law on  $S_x$ . Since the connecting path from  $x$  to the wired arc lies above the separator, translation of (9.3) gives the estimate we shall use repeatedly:

$$\phi_{[-n,n] \times [0,n]}^{\text{dobr}}[0 \longleftrightarrow \text{wired arc}] \geq c_D n^{-4} \quad (9.4)$$

with a numerical  $c_D > 0$ . Notice that this argument used only planar duality, FKG, the domain Markov property and boundary monotonicity, and is valid for every  $q \geq 1$  at the self-dual point.

We now lift the parafermion to a finite portion of the universal cover. Let  $\mathbb{U}$  be the universal cover of the square lattice punctured at the face immediately below the origin. We write its vertices as  $(x_1, x_2, j)$ : ordinary nearest-neighbor edges preserve the sheet number  $j$ , except that crossing the fixed branch cut from one side to the other changes  $j$  by 1. Let

$$U_n = \{(x_1, x_2, j) : |x_1|, |x_2| \leq n, |j| \leq n^5\}. \quad (9.5)$$

Split the root at the puncture into the two prime ends of a degenerate Dobrushin domain. For  $1 \leq q < 4$ , the local relation of Theorem SM-09.05.01, summed over all interior medial vertices of  $U_n$ , cancels every interior medial edge. On the boundary the winding is deterministic once the boundary vertex and its sheet are fixed. Therefore the contour sum has the form

$$\sum_{x \in \partial U_n} \delta_x a_x = c_q, \quad a_x := \phi_{U_n}^0[0 \longleftrightarrow x], \quad (9.6)$$

where  $c_q \neq 0$  and  $|\delta_x| \leq C_q$ . For completeness, (9.6) is not a new analytic assertion: pair every configuration with the one obtained by switching the primal edge under an interior medial vertex, use the four-entry table in the proof of Theorem SM-09.05.01, and sum the resulting local identity. Interior half-edges occur twice with opposite orientations and cancel. A boundary half-edge occurs once, and the event that the exploration visits it is exactly the event that the corresponding boundary vertex is joined to the root. The phase is  $e^{i\sigma W}$  with modulus one, giving the uniform bound on  $\delta_x$ ; the two prime ends at the root leave the fixed nonzero remainder  $c_q$ . Consequently

$$\sum_{x \in \partial U_n} a_x \geq c(q) > 0, \quad 1 \leq q < 4. \quad (9.7)$$

At  $q = 4$  the ordinary spin is  $\sigma = 1$  and the root remainder in the preceding contour identity degenerates. Keep the critical  $q = 4$  measure fixed and, for a real parameter  $s$ , put

$$F_s(e) = \mathbf{E} \left[ e^{isW_\gamma(e, e_b)} \mathbf{1}_{\{e \in \gamma\}} \right].$$



The paired local calculation used for Theorem SM-09.05.01 has defect proportional to

$$2i \sin(\pi s/2) - 2i.$$

This function and its first derivative vanish at  $s = 1$ . Differentiating the paired identity there therefore proves the exact local relation for the modified observable

$$G(e) = \frac{1}{i} \frac{d}{ds} F_s(e) \Big|_{s=1} = \mathbf{E} \left[ W_\gamma(e, e_b) e^{iW_\gamma(e, e_b)} \mathbf{1}_{\{e \in \gamma\}} \right]. \quad (9.8)$$

Summing the local relation as before yields

$$\sum_{x \in \partial U_n} \tilde{\delta}_x a_x = c_4, \quad c_4 \neq 0. \quad (9.9)$$

A boundary visit on sheet  $j$  has winding  $O(1 + |j|)$ , so  $|\tilde{\delta}_x| \leq C(1 + n^5)$ . Thus even at the endpoint

$$\sum_{x \in \partial U_n} a_x \geq cn^{-5}. \quad (9.10)$$

The loss of a polynomial factor is harmless; what matters for continuity is that no exponential loss has appeared.

We next show that the contour-sum lower bound cannot be concentrated on the two horizontal sheet boundaries  $j = \pm n^5$ . For  $|j| < n^5$  let

$$R_j = [-n, n] \times [0, n] \times \{j\}.$$

Condition on every edge outside  $R_j$ . In the dual graph, the induced boundary condition on  $R_j^*$  dominates the Dobrushin condition that is free on its bottom side and wired on the other sides. By (9.4), uniformly in the conditioning, the dual vertex adjacent to the branch point connects across  $R_j^*$  with probability at least  $c_D n^{-4}$ . Such a dual crossing blocks any primal path that moves from sheet  $j$  to sheet  $j + 1$ . Hence, by successive conditioning,

$$\phi_{U_n}^0[0 \longleftrightarrow x] \leq (1 - c_D n^{-4})^{|j|-1} \quad (9.11)$$

for  $x \in \partial U_n$  with sheet coordinate  $j = \pm n^5$ . The right-hand side is at most  $e^{-cn}$ ; multiplying by the polynomial number of such vertices (and, for  $q = 4$ , by the polynomial boundary weight in (9.9)) still gives  $o(n^{-5})$ . Therefore for all large  $n$  there is a side-boundary vertex  $x$ , with  $|j(x)| < n^5$ , such that

$$\phi_{U_n}^0[0 \longleftrightarrow x] \geq \begin{cases} c_q n^{-6}, & 1 \leq q < 4, \\ cn^{-11}, & q = 4. \end{cases} \quad (9.12)$$

Here we used that the side boundary of  $U_n$  has  $O(n^6)$  vertices.

Rotate the cover and shift the sheet index so that the vertex in (9.12) is  $x = (x_1, -n, 0)$  with  $0 \leq x_1 \leq n$ . Let  $C_n$  be the slit square obtained from  $\Lambda_n$  by removing the edges of the vertical slit  $\{(0, k) : 0 \leq k \leq n\}$  and put Dobrushin boundary conditions on  $C_n$ , wired on the two sides of the slit and free on the remaining outer arc. The free boundary condition induced from  $U_n$  is dominated by this Dobrushin condition, and every path from  $x$  to the root in the cover gives a path from  $x$  to the wired slit. Hence

$$\phi_{C_n}^{\text{dobr}}[x \longleftrightarrow \text{wired slit}] \geq cn^{-11} \quad (9.13)$$

for all  $1 \leq q \leq 4$  after taking the weaker endpoint exponent. By reflection the same bound holds for  $-x$ . FKG makes the two connection events simultaneous with probability at least  $c^2 n^{-22}$ . Condition on the right-most and left-most such open paths. In the region between them, the induced boundary condition dominates the Dobrushin condition of (9.4). Thus the midpoint  $z = (0, -n)$  connects to the wired slit with conditional probability at least  $c_D n^{-4}$ . We have proved

$$\phi_{C_n}^{\text{dobr}}[z \longleftrightarrow \text{wired slit}] \geq c n^{-26}. \quad (9.14)$$

The exponent is deliberately not optimized.

We now pass to the infinite plane. Let  $\partial_n$  be the set of vertices at graph distance  $n/16$  from  $z$  inside  $C_n$ , and put

$$E = \{z \longleftrightarrow \partial_n\}.$$

If  $z$  is connected to the wired slit, then  $E$  occurs, so (9.14) gives  $\phi_{C_n}^{\text{dobr}}(E) \geq c n^{-26}$ . In two vertical dual rectangles, one to the left and one to the right of the ball of radius  $n/16$  around  $z$ , explore respectively the right-most and left-most dual-open top-bottom crossings. The mixed-boundary strip estimate proved in SM-09.05.02.a–b (which uses only self-duality, FKG and the domain Markov property, not (9.2)) gives a constant  $c_1(q) > 0$  for each crossing, uniformly after conditioning on  $E$  and on the previously exposed crossing. Call these events  $F_L, F_R$ .

Between the two exposed dual crossings, the remaining dual domain has free boundary on its horizontal sides and wired boundary on its vertical sides. The elementary square-crossing comparison at the self-dual point gives a dual horizontal connector with conditional probability at least  $(1+q^2)^{-1}$ . Call this event  $H$ . On  $E \cap F_L \cap F_R \cap H$ , the primal cluster of  $z$  inside distance  $n/16$  is separated from the wired slit by a dual barrier. The domain Markov property therefore identifies its law inside that barrier with a free random-cluster law, and boundary monotonicity gives

$$\pi_{n/16} \geq \phi_{C_n}^{\text{dobr}}[E \cap F_L \cap F_R \cap H] \geq c(q) n^{-26}. \quad (9.15)$$

Replacing  $n$  by  $16m$  and changing the constant for the finitely many small scales yields  $\pi_m \geq c m^{-26}$  for every  $m \geq 1$ . Therefore

$$0 \geq \limsup_{m \rightarrow \infty} \frac{1}{m} \log \pi_m \geq \liminf_{m \rightarrow \infty} \frac{\log c - 26 \log m}{m} = 0,$$

which proves (9.2) for the full range  $1 \leq q \leq 4$ .

Using the RSW estimate proved above, we obtain a constant  $c_{\text{RSW}} = c_{\text{RSW}}(q) > 0$  such that every annulus  $\Lambda_{2r} \setminus \Lambda_r$ , under every boundary partition induced from the exterior, has conditional probability at least  $c_{\text{circ}} > 0$  of containing a dual-open circuit surrounding  $\Lambda_r$ . Indeed, four boundary-uniform dual rectangle crossings may be exposed successively and glued; the domain Markov property keeps the RSW constant valid after each exposure.

Use the dyadic annuli  $\Lambda_{2^{j+1}} \setminus \Lambda_{2^j}$  and expose them from the outside in the wired critical state. Conditional on all previously exposed edges, the probability of a blocking dual circuit in the next annulus is at least  $c_{\text{circ}}$ . Hence

$$\phi_{\mathbb{Z}^2, p, q}^1[0 \longleftrightarrow \partial \Lambda_{2^m}] \leq (1 - c_{\text{circ}})^{m-m_0} \xrightarrow{m \rightarrow \infty} 0. \quad (9.16)$$

By continuity from above,

$$\phi_{\mathbb{Z}^2, p, q}^1[0 \longleftrightarrow \infty] = 0. \quad (9.17)$$

Translation invariance gives the same conclusion from every fixed vertex.

We now prove equality of the free and wired infinite-volume FK measures, rather than merely quote it as an equivalent formulation. Couple the free and wired measures in  $\Lambda_N$  monotonically. Let  $K$  be a fixed finite set of edges and let  $V(K)$  be their endpoints. Explore the wired cluster attached to  $\partial\Lambda_N$ . On the event that this cluster misses  $V(K)$ , the domain Markov property gives exactly the same conditional boundary partition around  $K$  in the free and wired systems; the coupling may therefore be chosen identical on  $K$ . Consequently, for every cylinder event  $B$  supported on  $K$ ,

$$|\phi_{\Lambda_N}^1(B) - \phi_{\Lambda_N}^0(B)| \leq \sum_{v \in V(K)} \phi_{\Lambda_N}^1[v \longleftrightarrow \partial\Lambda_N]. \quad (9.18)$$

The right-hand side tends to zero by (9.17) and monotone convergence. Passing to the infinite-volume limits gives

$$\phi_{\mathbb{Z}^2, p, q}^0 = \phi_{\mathbb{Z}^2, p, q}^1. \quad (9.19)$$

This also shows directly that absence of the wired infinite cluster is the mechanism by which boundary conditions disappear locally.

Edwards–Sokal coupling first samples the FK configuration and then assigns an independent uniform spin in  $\{1, \dots, q\}$  to every FK cluster not fixed by the spin boundary condition. For two arbitrary Potts boundary conditions, couple their FK configurations by domination under the wired FK measure. The two spin configurations can be made identical on a fixed finite vertex set  $V$  unless its FK cluster reaches the boundary of the box. Thus, uniformly in the two spin boundary conditions,

$$\|\mu_{\Lambda_N}^{\tau}|_V - \mu_{\Lambda_N}^{\tau'}|_V\|_{\text{TV}} \leq \sum_{v \in V} \phi_{\Lambda_N, p, q}^1[v \longleftrightarrow \partial\Lambda_N] \xrightarrow{N \rightarrow \infty} 0. \quad (9.20)$$

Every DLR Gibbs state is a subsequential limit of such finite-volume conditional laws on local observables. Equation (9.20) therefore forces all subsequential limits, and hence all critical Potts Gibbs states, to coincide.

This proves the statement of the theorem. The proof also records the primary mechanism without conflating its logically different steps:

$$\begin{aligned} \text{parafermionic universal-cover estimate} &\implies \text{subexponential free one-arm} \\ &\implies \text{arbitrary-boundary RSW} \\ &\implies \theta^{\text{wired}}(p_c, q) = 0 \implies \phi^0 = \phi^1. \end{aligned}$$

with Edwards–Sokal supplying only the final translation to integer- $q$  Potts uniqueness. □

**Lemma 4.2** (Critical FK loops and the anti-ferroelectric six-vertex model). *Fix  $q > 4$ ,  $p_c = \sqrt{q}/(1 + \sqrt{q})$ , and let  $\lambda > 0$  be defined by*

$$2 \cosh \lambda = \sqrt{q}.$$

*On a finite torus, the critical FK partition function may be rewritten, up to an explicit topology-dependent factor bounded uniformly in the torus size, as the partition function of the six-vertex model with*

$$a = b = 1, \quad c = \sqrt{2 + \sqrt{q}}, \quad \Delta = \frac{2 - c^2}{2} = -\frac{\sqrt{q}}{2} = -\cosh \lambda < -1.$$

*The number of through-lines in the loop picture determines the magnetization block of the six-vertex transfer matrix.*

*Proof.* Superpose a primal FK configuration with its dual closed-edge configuration. The interfaces on the medial lattice are disjoint loops. Euler's formula at the self-dual value cancels the edge factors and leaves a factor  $\sqrt{q}$  for every contractible unoriented loop, together with only a bounded correction for the two torus homology classes. Orient each loop and assign weights  $e^\lambda$  and  $e^{-\lambda}$  to its two orientations; summing over the orientation gives  $e^\lambda + e^{-\lambda} = \sqrt{q}$ .

At a medial vertex there are six admissible arrow patterns. Multiplying all arrows along a row by the same gauge factor does not change the torus partition function. After this gauge normalization the four non-turning local patterns have weights  $a = b = 1$ , whereas the two alternating patterns have weight  $c$  satisfying  $c^2 = 2 + 2 \cosh \lambda = 2 + \sqrt{q}$ . Consequently  $\Delta = (a^2 + b^2 - c^2)/(2ab) = -\cosh \lambda$ . Finally, a loop crossing a horizontal cut changes the excess of up over down arrows by two, so the number of macroscopic through-lines labels precisely the diagonal block of the transfer matrix. These are finite combinatorial identities; no thermodynamic limit is used here.  $\square$

**Lemma 4.3** (Thermodynamic Bethe roots). *Let  $\Delta = -\cosh \lambda < -1$ . For fixed  $r \geq 0$  and  $N \in 4\mathbb{N}$ , put  $n = N/2 - r$  and*

$$I_j = j - \frac{n+1}{2}, \quad 1 \leq j \leq n.$$

*For all sufficiently large  $N$  the Bethe equations*

$$Np_j = 2\pi I_j - \sum_{\ell=1}^n \Theta(p_j, p_\ell)$$

*have a symmetric ordered solution. Its empirical density differs by  $O(N^{-1})$  from the unique continuous density  $\rho$ , and the displacement between the  $r$ -sector and the half-filled sector converges to the offset function  $\tau$ . After the change of variable  $x = k(\alpha)$  determined by*

$$e^{ik(\alpha)} = \frac{e^\lambda - e^{-i\alpha}}{e^{\lambda-i\alpha} - 1},$$

*the Fourier coefficients are*

$$(\widehat{2\pi\rho \circ k})k'(m) = \frac{1}{2 \cosh(\lambda m)}, \quad \widehat{2\pi\tau \circ k}(m) = \frac{(-1)^m}{im} \tanh(\lambda|m|), \quad m \neq 0.$$

*Proof.* Differentiating the two-body scattering phase gives, after  $x = k(\alpha)$ ,

$$\Xi_\mu(\alpha) = \frac{\sinh \mu}{\cosh \mu - \cos \alpha}, \quad \widehat{\Xi}_\mu(m) = e^{-\mu|m|}.$$

The continuous Bethe equation and the linear offset equation therefore become convolution equations

$$R = \Xi_\lambda - \frac{1}{2\pi} \Xi_{2\lambda} * R, \quad T = \Psi - \frac{1}{2\pi} \Xi_{2\lambda} * T.$$

Taking Fourier coefficients gives

$$\widehat{R}(m) = \frac{e^{-\lambda|m|}}{1 + e^{-2\lambda|m|}} = \frac{1}{2 \cosh(\lambda m)},$$

and, since  $\widehat{\Psi}(m) = (-1)^m(1 - e^{-2\lambda|m|})/(im)$ ,

$$\widehat{T}(m) = \frac{(-1)^m}{im} \tanh(\lambda|m|).$$

The first formula is strictly positive after Fourier inversion, so its associated cumulative distribution has a positive derivative.

For the discrete equation, at  $\Delta = -\infty$  the roots are explicit and equally spaced. Continue them in  $\Delta$  by the implicit-function theorem. The Jacobian is the identity plus a positive convolution-type matrix. In the weighted supremum norm transported by  $k$ , its continuum counterpart contracts by the factor

$$1 - \min_{\alpha} \Xi_{2\lambda}(\alpha) < 1.$$

A Riemann-sum comparison between the discrete and continuous equations then gives the bootstrap estimate

$$\|\rho_{p(N,r)} - \rho\| \leq C_r N^{-1}.$$

This estimate prevents collisions of consecutive roots and keeps the Jacobian invertible throughout the continuation. Subtracting the equations for  $r$  and 0, multiplying by  $N$ , and using the same stable inverse operator gives weak convergence of the root displacement to the unique solution  $\tau$  of the offset equation. Thus both the existence and the  $O(N^{-1})$  control needed below are proved from the displayed Fourier multiplier, rather than assumed as Bethe-ansatz folklore.  $\square$

**Lemma 4.4** (Six-vertex sector gap). *Let  $\Lambda_r(N)$  denote the Perron eigenvalue of the six-vertex transfer block with  $N/2 - r$  up arrows. For every fixed  $r \geq 1$ ,*

$$\lim_{\substack{N \rightarrow \infty \\ N \in 4\mathbb{N}}} \frac{\Lambda_r(N)}{\Lambda_0(N)} = e^{-r \mathfrak{m}(q)},$$

where

$$\mathfrak{m}(q) = \lambda + 2 \sum_{j \geq 1} \frac{(-1)^j}{j} \tanh(j\lambda).$$

*Proof.* For distinct roots satisfying the Bethe equations, the coordinate Bethe wave

$$\psi(x_1, \dots, x_n) = \sum_{\sigma \in S_n} A_{\sigma} \prod_{j=1}^n e^{ip_{\sigma(j)} x_j}$$

is an eigenvector: applying one transfer row, the bulk terms reproduce the same plane waves; exchanging two adjacent particles produces exactly the scattering factor encoded in  $A_{\sigma}$ ; and the terms crossing the periodic seam cancel precisely when the Bethe equations hold. The eigenvalue is the product of the corresponding one-particle factors  $L(e^{ip_j})$  and  $M(e^{ip_j})$ .

At  $\Delta = -\infty$  the symmetric solution gives the positive Perron eigenvector. The roots and eigenvalue are analytic in  $\Delta < -1$ , while the Perron eigenvalue is simple. They therefore remain the Perron branch by analytic continuation: two simple real branches could exchange order only by meeting, contradicting simplicity.

Take logarithms of the product formula. For  $r = 0$ , Lemma 4.3 converts the sum to the  $\rho$ -integral. In the ratio  $\Lambda_r/\Lambda_0$  the order- $N$  terms cancel. The first variation is the pairing of the logarithmic one-particle derivative with the offset  $r\tau$ ; the  $O(N^{-1})$  root-density error makes the quadratic remainder  $o(1)$ . In the  $\alpha$  coordinate this pairing is diagonal in Fourier modes. The Fourier coefficients from Lemma 4.3 give

$$\lim_{N \rightarrow \infty} \log \frac{\Lambda_r(N)}{\Lambda_0(N)} = -r\lambda - 2r \sum_{j \geq 1} \frac{(-1)^j}{j} \tanh(j\lambda),$$

which is the claim. Absolute convergence follows after pairing consecutive terms; alternatively one may use the positive representation proved next.  $\square$

**Lemma 4.5** (Positive mass formula and the  $q \downarrow 4$  scale). *For  $\lambda > 0$ ,*

$$\mathfrak{m}(q) = \sum_{j \geq 0} \frac{4}{(2j+1) \sinh(\pi^2(2j+1)/(2\lambda))} > 0.$$

*Consequently, as  $q \downarrow 4$ ,*

$$\mathfrak{m}(q) \sim 8 \exp\left(-\frac{\pi^2}{\sqrt{q-4}}\right), \quad \xi(q) = \mathfrak{m}(q)^{-1} \asymp \exp\left(\frac{\pi^2}{\sqrt{q-4}}\right).$$

*Proof.* Differentiate the alternating series with respect to  $\lambda$ ; the derivative is an absolutely convergent alternating sum of  $\text{sech}^2(j\lambda)$ . Poisson summation for

$$\text{sech}^2 x, \quad \widehat{\text{sech}^2}(s) = \frac{\pi s}{\sinh(\pi s/2)},$$

transforms it into the derivative of the displayed positive series. Both expressions tend to 0 as  $\lambda \rightarrow \infty$  after subtracting their common leading linear term, so integration recovers the identity. Positivity is immediate.

Now  $\cosh \lambda = \sqrt{q}/2$  implies

$$\lambda = \frac{1}{2} \sqrt{q-4} (1 + o(1)).$$

The  $j = 0$  term in the positive series is  $4/\sinh(\pi^2/(2\lambda)) \sim 8e^{-\pi^2/(2\lambda)}$ , and every later term is exponentially smaller. Substituting the expansion of  $\lambda$  proves the asymptotics.  $\square$

**Lemma 4.6** (From sector gap to FK connectivity). *For critical square-lattice FK with  $q > 4$  and  $x_n = (\lfloor n/2 \rfloor, \lfloor n/2 \rfloor)$ ,*

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log \phi_{p_c, q}^0(0 \leftrightarrow x_n) = \mathfrak{m}(q).$$

*Moreover the free critical cluster radius has an exponential tail.*

*Proof.* Work first on a long rotated cylinder. In the loop representation, forcing a primal connection from the lower to the upper marked neighbourhood changes the through-line sector. Cutting the cylinder at the first and last contacts and summing the finite boundary pieces yields upper and lower bounds by constants times ratios of transfer-matrix traces. Perron–Frobenius domination then reduces their exponential rate to  $\Lambda_1(N)/\Lambda_0(N)$ . For the reverse bound, use  $r$  separated marked connections and the sector estimate for every fixed  $r$ ; this removes the topology correction that a single torus winding leaves uncontrolled. Let the cylinder length tend to infinity, then  $N \rightarrow \infty$ , and invoke Lemma 4.4.

Planar comparison between the rotated cylinder and boxes in  $\mathbb{Z}^2$ , plus FKG gluing over a bounded number of diagonally oriented rectangles, changes probabilities only by subexponential factors, hence preserves the rate. Finally, if the free cluster of 0 reaches graph distance  $R$ , it connects to one of  $O(R)$  points on the boundary of a box of radius  $R$ ; the union bound and the positive rate give  $Ce^{-cR}$  after a harmless reduction of  $c$ .  $\square$

**Theorem 4.7** (Discontinuity for  $q > 4$ ). *For the square-lattice random-cluster model with  $q > 4$  at  $p_c = \sqrt{q}/(1 + \sqrt{q})$ , the free and wired infinite-volume measures differ, the wired state has an infinite cluster almost surely, and the free connectivity decays exponentially. If  $\lambda > 0$  satisfies  $\cosh \lambda = \sqrt{q}/2$ , then the inverse diagonal correlation length is*

$$\xi(q)^{-1} = \lambda + 2 \sum_{j \geq 1} \frac{(-1)^j}{j} \tanh(j\lambda) > 0,$$

and

$$\xi(q) \asymp \exp\left(\frac{\pi^2}{\sqrt{q-4}}\right) \quad (q \downarrow 4).$$

For integer  $q > 4$ , Edwards–Sokal yields coexistence of the disordered Potts state and the  $q$  monochromatically ordered states, and the same inverse correlation length for the truncated free Potts two-point function.

*Proof.* Lemmas 4.2–4.6 give the exact diagonal mass

$$\mathbf{m}(q) = \lambda + 2 \sum_{j \geq 1} \frac{(-1)^j}{j} \tanh(j\lambda) > 0$$

and

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log \phi_{p_c, q}^0(0 \leftrightarrow x_n) = \mathbf{m}(q).$$

The last lemma also upgrades this directional limit to an exponential tail for the radius of the free critical cluster. Hence the free state has no infinite cluster.

At the self-dual value, the complementary planar dual of the free critical state is the wired critical state at the same parameter. Exponential decay in the free state makes the probabilities of primal open circuits in dyadic annuli summable. Almost surely all sufficiently large annuli therefore contain dual-open crossings, and planar compactness (or König’s lemma) produces an infinite dual-open path. Under self-duality this is an infinite open cluster in the wired critical state. By the uniqueness theorem of Chapter 2 it is the unique infinite wired cluster. Thus  $\phi_{p_c, q}^0 \neq \phi_{p_c, q}^1$  and the transition is discontinuous.

Lemma 4.5 gives

$$\mathbf{m}(q) \sim 8 \exp\left(-\frac{\pi^2}{\sqrt{q-4}}\right),$$

so  $\xi(q) = \mathbf{m}(q)^{-1}$  has the claimed scale. For integer  $q$ , the Edwards–Sokal coupling turns the free FK state into the disordered Potts state. Wiring the boundary fixes the color of the infinite wired cluster; choosing any of the  $q$  colors gives  $q$  distinct ordered Potts states. Also

$$\langle \mathbf{1}_{\{\sigma_0 = \sigma_x\}} \rangle_{\text{free}} - \frac{1}{q} = \left(1 - \frac{1}{q}\right) \phi_{p_c, q}^0(0 \leftrightarrow x),$$

so the truncated Potts two-point function has the same inverse correlation length  $\mathbf{m}(q)$ . □

**Theorem 4.8** (Free/wired dichotomy for  $q > 4$ ). *At criticality for  $q > 4$  on the square lattice, the free state is disordered with exponential connectivity decay while the wired state is ordered; this dichotomy produces distinct correlation-length and boundary-condition behavior.*

*Proof.* The preceding theorem gives constants  $c, C > 0$  such that

$$\phi_{p_c, q}^0(0 \leftrightarrow x) \leq C e^{-c|x|}, \quad \phi_{p_c, q}^1(0 \leftrightarrow \infty) > 0.$$

Thus a local perturbation under free boundary conditions loses memory of a distant boundary exponentially fast, whereas wired boundary conditions retain the macroscopic event that the observation point is joined to the infinite wired cluster. In particular the two infinite-volume measures are distinct and their boundary-condition response cannot have the same correlation length. This is exactly the asserted free/wired dichotomy; no additional input beyond Theorem 4.7 is required. □

## Chapter 5

# Parafermionic Observables, RSW, and Arm Geometry

**Theorem 5.1** (Parafermionic observable and local relation). *Let  $(G, a, b)$  be a Dobrushin domain for the square-lattice random-cluster model with  $0 < q < 4$  at the self-dual point*

$$p_c = p_{\text{sd}}(q) = \frac{\sqrt{q}}{1 + \sqrt{q}}.$$

*Orient the medial lattice counterclockwise around primal faces, let  $e_b$  be the terminal medial edge of the exploration path  $\gamma$ , and choose  $\sigma \in (0, 1)$  by*

$$\sin\left(\frac{\pi\sigma}{2}\right) = \frac{\sqrt{q}}{2}.$$

*For an oriented medial edge  $e$  set*

$$F(e) = \mathbf{E}\left[e^{i\sigma W_\gamma(e, e_b)} \mathbf{1}_{\{e \in \gamma\}}\right],$$

*where  $W_\gamma(e, e_b)$  is the signed winding of  $\gamma$  from  $e$  to  $e_b$ . If  $NW, SE, NE, SW$  are the four medial edges incident to an interior medial vertex, then*

$$F(NW) - F(SE) = i(F(NE) - F(SW)).$$

*Thus the edge parafermion is a divergence-free discrete differential form on primal medial vertices. (The  $q = 4$  endpoint is degenerate and is treated by the differentiated observable rather than by this nondegenerate statement.)*

*Proof.* At the self-dual point the loop-representation weight is proportional to  $(\sqrt{q})^{\ell(\omega)}$ , because  $x = p/[\sqrt{q}(1-p)] = 1$ . For a configuration  $\omega$  and a medial edge  $e$  write

$$e_\omega := \phi_{G, p_c, q}^{a, b}(\omega) e^{i\sigma W_{\gamma(\omega)}(e, e_b)} \mathbf{1}_{\{e \in \gamma(\omega)\}}.$$

Then  $F(e) = \sum_\omega e_\omega$ .

Pair each configuration with the one obtained by switching the state of the primal edge crossing  $v$ . The map  $s$  is an involution, so

$$F(e) = \frac{1}{2} \sum_\omega (e_\omega + e_{s\omega}).$$



It therefore suffices to verify, for every pair  $\{\omega, s\omega\}$ ,

$$NW_\omega + NW_{s\omega} - SE_\omega - SE_{s\omega} = i(NE_\omega + NE_{s\omega} - SW_\omega - SW_{s\omega}). \quad (*)$$

If the exploration uses none of the relevant local half-edges, every term in  $(*)$  vanishes. Otherwise rotate the picture, and if necessary interchange  $\omega$  with  $s\omega$ , so that  $\gamma(\omega)$  enters through  $NW$  and exits through  $SW$ . Put  $A = NW_\omega$ . Switching the crossed primal edge removes one loop in the loop picture, hence

$$\phi(s\omega) = \frac{1}{\sqrt{q}}\phi(\omega).$$

The four relevant windings differ by quarter-turns. Consequently the paired contributions, in the order  $(NW, SE, NE, SW)$ , are

|           | $NW$         | $SE$                       | $NE$                          | $SW$                         |      |
|-----------|--------------|----------------------------|-------------------------------|------------------------------|------|
| $\omega$  | $A$          | $0$                        | $0$                           | $e^{i\sigma\pi/2}A$          |      |
| $s\omega$ | $A/\sqrt{q}$ | $e^{i\sigma\pi}A/\sqrt{q}$ | $e^{-i\sigma\pi/2}A/\sqrt{q}$ | $e^{i\sigma\pi/2}A/\sqrt{q}$ | (**) |

This is simply the two possible local pairings of the four medial half-edges; all winding outside the small neighborhood of  $v$  is common to the two configurations.

Substitute the table  $(**)$  into  $(*)$ . After division by  $A/\sqrt{q}$  the remaining scalar is a multiple of

$$e^{i\sigma\pi/2} - e^{-i\sigma\pi/2} - i\sqrt{q}.$$

It vanishes because

$$e^{i\sigma\pi/2} - e^{-i\sigma\pi/2} = 2i \sin\left(\frac{\sigma\pi}{2}\right) = i\sqrt{q}.$$

Hence  $(*)$  holds in the two-edge case.

If  $\gamma(\omega)$  uses all four local half-edges while  $\gamma(s\omega)$  uses only two, exchanging the roles of the two configurations reduces the calculation to the preceding case. The horizontal orientation of the crossed primal edge is obtained from the same computation by a quarter-turn of the picture. Thus  $(*)$  holds for every configuration pair.

Summing the paired identity over all configuration pairs yields  $F(NW) - F(SE) = i(F(NE) - F(SW))$ , as claimed. Notice that this proves only the local relation at primal medial vertices; for general  $q$  it is not a full system of discrete Cauchy–Riemann equations. This distinction is why later RSW and scaling-limit arguments cannot be replaced by a formal appeal to “discrete holomorphicity.”  $\square$

**Theorem 5.2** (RSW under arbitrary boundary conditions). *Fix  $q \in [1, 4]$  and let  $p = p_c(q)$ . Assume the critical model is in the continuous regime in the following precise sense:*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \phi_{\mathbb{Z}^2, p, q}^0[0 \longleftrightarrow \partial\Lambda_n] = 0. \quad (9.18)$$

*Then for every  $\rho \geq 1$  there is  $c = c(q, \rho) > 0$  such that, for every  $n \geq 1$ , if*

$$\widehat{R}_{n, \rho} := [-n, (\rho + 1)n] \times [-n, 2n], \quad R_{n, \rho} := [0, \rho n] \times [0, n],$$

*and  $\xi$  is an arbitrary boundary condition on  $\partial\widehat{R}_{n, \rho}$ , then*

$$c \leq \phi_{\widehat{R}_{n, \rho}, p, q}^\xi[\mathcal{C}_h(R_{n, \rho})] \leq 1 - c. \quad (9.19)$$

*Thus critical rectangle crossings of bounded aspect ratio are uniformly non-degenerate, independently of the boundary partition. By the domain Markov property the same estimate holds when  $\widehat{R}_{n, \rho}$  is embedded in a larger domain: condition on the exterior and apply (9.19) to the induced boundary partition.*

*Proof.* We prove the lower bound first. Since the crossing event is increasing, comparison of boundary conditions reduces the problem to free boundary conditions. The argument is the **P4**  $\Rightarrow$  **P5** implication of the critical random-cluster alternative, written here in the form needed later.

Let

$$S_n := \mathbb{Z} \times [-n, 3n]$$

and give its upper side wired and its lower side free boundary conditions; write the resulting measure as  $\phi_{S_n}^{1/0}$ . For every integer  $k \geq 1$  there is  $c_k > 0$ , independent of  $n$ , such that

$$\phi_{S_n}^{1/0}[\mathcal{C}_h([-kn, kn] \times [0, 2n])] \geq c_k. \quad (9.20)$$

Indeed, either the horizontal crossing in (9.20) already has probability at least  $1/2$ , or, by planar duality, the complementary dual vertical crossing has probability at least  $1/2$ . Reflecting the latter configuration across the midline of the strip and translating by half a lattice spacing turns it into a primal vertical crossing with the same law. Partitioning a segment of the bottom of the central rectangle into  $18k$  pieces therefore produces a subinterval  $I$  of length  $n/9$  with

$$\phi_{S_n}^{1/0}[I \longleftrightarrow \mathbb{Z} \times \{2n\}] \geq (36k)^{-1} =: a_k. \quad (9.21)$$

Take a disjoint translate  $J$  of  $I$  at distance  $n/9$ . Explore the leftmost open path from  $I$  and the rightmost open path from  $J$  to the top of the central  $2n \times 2n$  square. Conditionally on these paths, the domain between them has the two explored arcs wired. By duality in the intervening domain, one of a primal connection between the two wired arcs and a dual separating connection must occur; comparison of the corresponding mixed boundary conditions gives the uniform estimate

$$\phi[\Gamma_I \longleftrightarrow \Gamma_J \mid \Gamma_I, \Gamma_J] \geq \frac{1}{1 + q^2}. \quad (9.22)$$

(The harmless constant  $q^2$  comes from wiring/unwiring at most two boundary classes.) Hence a fixed local bridge between  $I$  and  $J$  occurs with probability at least  $a_k^2/(1 + q^2)$ . Translating this bridge a bounded number ( $\leq 20k$ ) of times and using FKG yields (9.20). The alternative geometric case, in which the path issued from  $I$  first hits a vertical side rather than the top, is treated by the same exploration: reflect the unexplored region, use duality, and compare the two mixed boundary conditions. The same lower bound  $1/(1 + q^2)$  is obtained, so the constant remains uniform in  $n$ .

Iterating the preceding lemma in vertical slabs gives, for every  $k > 0$  and  $\ell \geq 4/3$ , a constant  $b = b(k, \ell, q) > 0$  for which

$$\phi_{[-kn, kn] \times [0, \ell n]}^{1/0}[\mathcal{C}_h([-kn, kn] \times [0, n])] \geq b, \quad (9.23)$$

where the bottom side is free and the other three sides are wired. The induction step conditions on the highest crossing in the lower slab; the unexplored region above it again dominates the same mixed boundary condition. Thus the constants multiply only a bounded number of times when  $\ell$  is fixed.

For  $m \geq 1$  let  $A_m$  denote the existence of an open circuit in  $\Lambda_{2m} \setminus \Lambda_m$ . The strip estimate (9.23), duality, and conditioning on outermost primal and dual crossings give a constant  $C = C(q) < \infty$  such that

$$\phi_{\Lambda_{56m}}^0[A_{7m}] \leq C \phi_{\Lambda_{8m}}^0[A_m]^2. \quad (9.24)$$

To see the square on the right, place two translates of  $A_m$  near  $(\pm 5m, 0)$ . Mixed-strip crossings create, with conditional probability bounded below, dual horizontal crossings above and below them.

Conditioning on the outermost such dual crossings makes the two  $m$ -scale circuit events live in disjoint domains with free boundary conditions; the domain Markov property and comparison of boundary conditions then bound the joint event by the product of the two  $m$ -scale probabilities. Conversely, three bounded- aspect-ratio mixed-strip crossings reconnect the two small circuits into the  $7m$ -scale circuit. This is exactly (9.24), up to the fixed constant  $C$ .

If the free square-crossing probabilities are not bounded below, then the corresponding circuit probabilities tend to zero along a scale sequence. Iterating (9.24) on  $m_{j+1} \asymp 7m_j$  yields

$$\phi^0[A_{m_j}] \leq \exp(-m_j^\alpha) \quad (9.25)$$

for some  $\alpha > 0$ . A cluster connecting 0 to  $\partial\Lambda_n$  must cross one of the annuli in the resulting geometric decomposition. The outermost-dual-circuit decomposition and a union bound therefore give

$$\phi_{\mathbb{Z}^2}^0[0 \longleftrightarrow \partial\Lambda_n] \leq \exp(-n^{\alpha'}) \quad (9.26)$$

for some  $\alpha' > 0$ .

Write

$$u_n := \phi_{\mathbb{Z}^2}^0[0 \longleftrightarrow \partial\Lambda_n]$$

and assume  $u_n \leq e^{-n^\alpha}$ . We now prove the convolution estimate needed for the bootstrap. Choose  $0 < \varepsilon < \beta < \alpha$ . Cutting a finite connection between opposite sides of a square along a strip of width  $n^\varepsilon$ , and conditioning on the highest and lowest crossing paths, gives

$$\phi_{\Lambda_n}^0[x \longleftrightarrow y] \leq e^{Cn^\varepsilon} u_n^2 + Cn^6 \sum_{\substack{k, \ell \geq n^\varepsilon \\ k+\ell=2n}} u_k u_\ell. \quad (9.27)$$

Here the first term corresponds to the case in which both extremal crossing paths remain inside the thin strip: closing the  $O(n^\varepsilon)$  edges on a transverse cut costs only  $e^{Cn^\varepsilon}$  by finite energy and splits the connection into two one-arm events. If an extremal path leaves the strip, choose its furthest vertex; an outermost dual circuit separates the two resulting pieces, and summing over the possible vertex and two radii gives the second term. A tilted-strip counting lemma supplies the polynomial lower bound needed to create the separating dual connections, so no exponential loss is hidden in the factor  $Cn^6$ .

A diameter decomposition of the cluster of the origin gives

$$u_{2n} \leq \sum_{m \geq n} 72m^4 \max_{a \in \{0\} \times [0, m], b \in \{m\} \times [0, m]} \phi_{[0, m]^2}^0[a \longleftrightarrow b]. \quad (9.28)$$

Set  $q_n = e^{n^\beta} u_n$ . Inserting (9.27) into (9.28), using  $\varepsilon < \beta < 1$  and summing the exponentially damped polynomial factors, yields a constant  $C_1$  such that

$$q_n \leq C_1 \max_{\substack{k, \ell \geq n^\varepsilon \\ k+\ell \geq n}} q_k q_\ell. \quad (9.29)$$

Finite energy absorbs the harmless parity changes. Define the decreasing envelope

$$Q_n := C_1 \sup_{m \geq n} q_m.$$

Then

$$Q_n \leq \max_{\substack{k, \ell \geq n^\varepsilon \\ k+\ell \geq n}} Q_k Q_\ell. \quad (9.30)$$

The stretched-exponential hypothesis and  $\beta < \alpha$  imply  $Q_n \rightarrow 0$ . Choose  $n_0$  so large that  $Q_m < 1$  whenever  $m \geq n_0^\varepsilon$ . For  $n \geq n_0$ , monotonicity shows that the maximum in (9.30) cannot be attained with  $k \geq n$  or  $\ell \geq n$ ; otherwise the other factor is strictly below one and the right-hand side is  $< Q_n$ . Hence only  $n > k, \ell \geq n^\varepsilon$  with  $k + \ell \geq n$  need be considered. Put

$$c_2 := \min_{n_0^\varepsilon \leq m \leq n_0} \frac{-\log Q_m}{m} > 0.$$

Strong induction now gives  $Q_n \leq e^{-c_2 n}$ : for every admissible pair,  $Q_k Q_\ell \leq e^{-c_2(k+\ell)} \leq e^{-c_2 n}$ . Consequently  $u_n \leq C e^{n^\beta - c_2 n} \leq C' e^{-c' n}$ . This contradicts the subexponential hypothesis (9.18). Therefore the non-RSW branch of (9.24)–(9.26) is impossible, and for every fixed aspect ratio there exists  $c_0 = c_0(q, \rho) > 0$  such that

$$\phi_D^0[\mathcal{C}_h([0, \rho n] \times [0, n])] \geq c_0 \quad (9.31)$$

whenever  $D$  contains the one- $n$  collar appearing in the statement.

By boundary-condition monotonicity for increasing events, (9.31), applied in  $\widehat{R}_{n, \rho}$ , is the desired lower bound in (9.19). For the upper bound, the complement of a primal horizontal crossing contains a dual vertical crossing. At  $p = p_c(q) = \sqrt{q}/(1 + \sqrt{q})$  the dual parameter equals  $p_c(q)$ . The dual boundary condition  $\xi^*$  is again an arbitrary partition, so the lower bound just proved, applied to the dual rectangle, gives a dual crossing with probability at least  $c_1 > 0$ . Consequently the primal crossing probability is at most  $1 - c_1$ . Taking  $c = \min(c_0, c_1)$  proves (9.19).

The renormalization lemmas prove a dichotomy: either boundary-condition-uniform RSW holds or free connectivities decay exponentially. Thus any one of the standard continuous-critical conditions—subexponential free connectivity, absence of a wired infinite cluster together with equality of free and wired critical states, or infinite critical susceptibility—selects the RSW branch once their equivalence has been established. The present theorem uses (9.18) so that no later continuity theorem is invoked circularly.  $\square$

For an integer  $k \geq 1$  and a cyclic word  $\sigma = (\sigma_1, \dots, \sigma_k) \in \{0, 1\}^k$ , let  $A_\sigma(r, R)$  be the event that the annulus  $\Lambda_R \setminus \Lambda_{r-1}$  contains  $k$  pairwise disjoint arms in cyclic order, with a 1-arm a primal-open lattice path and a 0-arm a path of dual-open edges. Let  $r_\sigma$  be the smallest deterministic radius for which this cyclic pattern is topologically realizable. If the inner and outer endpoints are denoted by  $x_i$  and  $x'_i$ , respectively, we call such a realization  $\delta$ -well separated when

$$|x_i - x_j| > 2\delta r, \quad |x'_i - x'_j| > 2\delta R \quad (i \neq j), \quad (9.32)$$

and, in addition, the  $i$ th arm continues inside  $\Lambda_{\delta r}(x_i)$  and  $\Lambda_{\delta R}(x'_i)$  for distance at least  $\delta r$  and  $\delta R$ , respectively, without meeting any other arm. Write  $A_\sigma^{\text{sep}}(r, R)$  for the existence of a realization with these properties.

**Theorem 5.3** (Quasi-multiplicativity and arm separation). *Fix  $1 \leq q < 4$  and a cyclic arm word  $\sigma \in \{0, 1\}^k$ . There is  $\delta_0 = \delta_0(\sigma) > 0$  such that, for every  $0 < \delta \leq \delta_0$ , one can choose  $c_{\text{sep}} = c_{\text{sep}}(q, \sigma, \delta) > 0$  with*

$$c_{\text{sep}} \phi_{\mathbb{Z}^2}[A_\sigma(r, R)] \leq \phi_{\mathbb{Z}^2}[A_\sigma^{\text{sep}}(r, R)] \leq \phi_{\mathbb{Z}^2}[A_\sigma(r, R)] \quad (9.33)$$

for all  $R \geq r \geq r_\sigma$ . Moreover there are  $c_{\text{qm}}, C_{\text{qm}} \in (0, \infty)$ , depending only on  $q$  and  $\sigma$ , such that for every  $R \geq \rho \geq r \geq r_\sigma$ ,

$$c_{\text{qm}} \phi[A_\sigma(r, R)] \leq \phi[A_\sigma(r, \rho)] \phi[A_\sigma(\rho, R)] \leq C_{\text{qm}} \phi[A_\sigma(r, R)]. \quad (9.34)$$

The same constants work after changing any boundary condition outside the annulus by a boundary partition whose blocks do not identify two prescribed landing corridors. In particular, the theorem applies to nonalternating words such as 10101; this is precisely where the arbitrary-boundary RSW theorem above is needed.

*Proof.* We begin with the landing surgery. Choose  $\delta_0 > 0$  so small that the two boundary components of every annulus may be divided into  $M = M(k)$  arcs, each of length between  $8\delta s$  and  $16\delta s$  at scale  $s$ , with at least two unused arcs between any two selected landing arcs. Given  $k$  disjoint arms in cyclic order, planarity permits us, inside the collar  $\Lambda_{(1+40\delta)s} \setminus \Lambda_{(1-40\delta)s}$ , to route the arms to  $k$  prescribed landing arcs in the same cyclic order. The routing uses at most  $C(k)$  disjoint topological rectangles of aspect ratio bounded by a constant depending only on  $k$ . Between two neighboring corridors we place a side fence of the opposite type whenever this is needed to keep two equal-colored arms from coalescing. Thus the landing pattern is encoded by one of finitely many corridor diagrams  $\mathcal{L}_1, \dots, \mathcal{L}_N$ , with  $N = N(k)$ .

Explore the given arms canonically, from the inner boundary outward, always choosing the left-most available primal arm or the corresponding dual right-most arm in the appropriate component. Conditional on the explored paths, the unexplored edges carry a random-cluster measure with the boundary partition induced by the exploration. Every crossing required by a corridor diagram is therefore a crossing in a quad with an *arbitrary* induced boundary condition. By Theorem SM-09.05.02, both a primal crossing and, by critical duality, a dual crossing of each such quad have conditional probability at least  $c_0 = c_0(q, k) > 0$ . We expose the  $C(k)$  corridors one after another. The domain Markov property resets the measure after each exposure, and the same boundary-uniform estimate remains valid. Consequently a prescribed compatible landing diagram succeeds with conditional probability at least

$$c_f := c_0^{C(k)} > 0. \quad (9.35)$$

The argument is unchanged if two consecutive symbols of  $\sigma$  coincide: the extra opposite-type side fences are simply additional quads in the same finite list. This is the point at which ordinary alternating-arm RSW would not suffice.

Apply the landing surgery at radii  $r$  and  $R$ . On  $A_\sigma(r, R)$ , one of the finitely many landing patterns is compatible with the explored arms at each boundary. Summing over the patterns and using (9.35) first at the inner and then at the outer collar gives

$$\phi[A_\sigma^{\text{sep}}(r, R) \mid A_\sigma(r, R)] \geq c_f^2 / N^2. \quad (9.36)$$

This proves the nontrivial inequality in (9.33), after decreasing the constant to absorb the finitely many scales  $r \leq C(k, \delta)$ . The reverse inequality is immediate from the definition.

We next record the shielding consequence of the same construction. Fix separated landing arcs at a scale  $s$ . In a collar from  $(1 - 80\delta)s$  to  $(1 + 80\delta)s$ , complete the arm corridors and put fences on both sides of every corridor. After all fences have been explored, each component touching the inner part and each component touching the outer part meet the middle collar only through the designated landing arcs. Hence the induced boundary partition seen from the inside (or from the outside) can change the corresponding arm probability only through a finite list of wirings of the landing arcs. For each such wiring, the reverse corridor construction has probability at least  $c_f$ . Therefore, if  $E$  is any arm event inside the inner side of the collar and  $F$  the matching arm event outside its outer side, then

$$c_{\text{sh}} \phi(E)\phi(F) \leq \phi(E \cap F \cap \mathcal{S}_s) \leq C_{\text{sh}} \phi(E)\phi(F), \quad (9.37)$$

where  $\mathcal{S}_s$  is the event that the prescribed corridors and fences occur and the constants depend only on  $q$  and  $\sigma$ . To justify the upper bound, condition first on the inner exploration. The outside

measure has some induced partition on the finitely many landing arcs; forcing the reverse fences with probability at least  $c_f$  reduces that partition to one of the reference partitions used to define  $\phi(F)$ . Thus the conditional probability of  $F$  is at most  $c_f^{-1}$  times its reference probability. The lower bound follows by exposing the same fences in the opposite order. This is a domain-Markov statement, not an independence assertion.

If  $s \leq t \leq 4s$ , every realization of  $A_\sigma(r, s)$  whose endpoints are separated at  $s$  can be prolonged from  $s$  to  $t$  by the corridor diagrams above with probability at least  $c_f$ . Combining this with (9.33) gives, uniformly for  $1 \leq t/s \leq 4$ ,

$$\phi[A_\sigma(r, t)] \asymp \phi[A_\sigma(r, s)], \quad \phi[A_\sigma(t, R)] \asymp \phi[A_\sigma(s, R)]. \quad (9.38)$$

Thus it is harmless, at a multiplicative cost depending only on  $q, \sigma$ , to replace an intermediate radius  $\rho$  by  $\rho/2$ ,  $2\rho$ , or by the nearest lattice radius on which all landing boxes are disjoint.

For one direction of quasi-multiplicativity, separate the outer endpoints of an inner realization of  $A_\sigma(r, \rho/2)$  and the inner endpoints of an outer realization of  $A_\sigma(2\rho, R)$ . Choose their landing patterns to be the same after radial projection. The shielding/gluing estimate (9.37) in the middle annulus  $\Lambda_{2\rho} \setminus \Lambda_{\rho/2}$  then yields

$$\phi[A_\sigma(r, R)] \geq c \phi[A_\sigma(r, \rho/2)] \phi[A_\sigma(2\rho, R)]. \quad (9.39)$$

Using (9.38) converts (9.39) into

$$\phi[A_\sigma(r, \rho)] \phi[A_\sigma(\rho, R)] \leq C \phi[A_\sigma(r, R)], \quad (9.40)$$

which is the right-hand inequality in (9.34).

For the reverse direction, start from  $A_\sigma(r, R)$  and perform the separation procedure in a middle collar at radius  $\rho$ . With conditional probability at least  $c_f$  the arms meet the collar in a fixed separated landing pattern and acquire the side fences. Cut the configuration along the central landing arcs. The domain Markov property and the two inequalities in the shielding estimate show that the weight of such cut configurations is bounded above by a fixed constant times the product of the weights of an inner arm configuration and an outer arm configuration with that landing pattern. Summing over the finitely many patterns gives

$$\phi[A_\sigma(r, R)] \leq C \phi[A_\sigma(r, \rho)] \phi[A_\sigma(\rho, R)]. \quad (9.41)$$

When  $\rho < 4r$  or  $R < 4\rho$ , the same estimate follows directly from the fixed-ratio comparison (9.38). Equations (9.40)–(9.41) prove (9.34).

Every constant used above comes from a bounded-aspect-ratio quad, from one of finitely many landing diagrams, or from the fixed arm word  $\sigma$ . It is therefore independent of  $r, \rho, R$ . The proof also shows why  $q < 4$  is the present endpoint: the input used here is the strong arbitrary-boundary crossing theory in the form needed for all landing geometries. No arm exponent is used, so the argument does not presuppose any of the later SLE/CLE identifications.  $\square$

## Chapter 6

# Random Currents

**Definition 6.1** (Random current and source set). For the Ising model on a finite graph with edge couplings  $J_e$ , a current is  $n \in \mathbb{N}^E$ . Its source set is

$$\partial n = \left\{ x \in V : \sum_{e \ni x} n_e \equiv 1 \pmod{2} \right\},$$

and its weight is

$$w(n) = \prod_{e \in E} \frac{(\beta J_e)^{n_e}}{n_e!}.$$

The random-current measure with prescribed sources  $A$  is the normalized weight on  $\{n : \partial n = A\}$ .

*Worked example 6.2* (One-edge current parity). For a single edge  $e = \{x, y\}$  with  $t = \beta J_e$ , even values of  $n_e$  have no sources and odd values have source set  $\{x, y\}$ . Therefore

$$\sum_{\partial n = \emptyset} w(n) = \sum_{m \geq 0} \frac{t^{2m}}{(2m)!} = \cosh t,$$

while

$$\sum_{\partial n = \{x, y\}} w(n) = \sum_{m \geq 0} \frac{t^{2m+1}}{(2m+1)!} = \sinh t.$$

Consequently the two-spin Ising correlation is  $\langle \sigma_x \sigma_y \rangle = \tanh t$ , matching the direct computation of Volume VII.

**Theorem 6.3** (Random-current expansion). *For even  $A \subseteq V$  at zero field,*

$$\langle \sigma_A \rangle = \frac{\sum_{\partial n = A} w(n)}{\sum_{\partial n = \emptyset} w(n)}.$$

*Proof.* Expand every interaction factor as

$$e^{\beta J_e \sigma_x \sigma_y} = \sum_{n_e \geq 0} \frac{(\beta J_e)^{n_e}}{n_e!} (\sigma_x \sigma_y)^{n_e}.$$

After multiplication over edges, the exponent of  $\sigma_x$  is  $\sum_{e \ni x} n_e$  plus the indicator that  $x \in A$ . Summing  $\sigma_x \in \{\pm 1\}$  gives zero unless that total exponent is even. Thus a term survives exactly when  $\partial n = A$ , and every surviving spin sum contributes the same factor  $2^{|V|}$ . Cancelling that factor between the correlation numerator and the partition function gives the displayed current ratio.  $\square$

**Theorem 6.4** (Switching lemma). *Let  $A \subseteq V$  be even and  $x, y \in V$ . For every function  $F : \mathbb{N}^E \rightarrow \mathbb{R}$  for which the sums converge,*

$$\sum_{\substack{\partial n_1 = A \\ \partial n_2 = \{x, y\}}} w(n_1)w(n_2)F(n_1 + n_2) = \sum_{\substack{\partial n_1 = A \Delta \{x, y\} \\ \partial n_2 = \emptyset}} w(n_1)w(n_2)F(n_1 + n_2) \mathbf{1}_{\{x \leftrightarrow y \text{ in } n_1 + n_2\}}. \quad (6.1)$$

*Proof.* Fix the total current  $m = n_1 + n_2$ . Replace every edge  $e$  by  $m_e$  distinguishable parallel copies. Choosing which copies belong to  $n_1$  contributes  $\prod_e \binom{m_e}{n_{1,e}}$ , and multiplication by  $w(m)$  gives exactly  $w(n_1)w(n_2)$ . If  $x$  and  $y$  are connected in the support of  $m$ , choose deterministically one path of edge copies joining them and toggle the ownership of every copy on that path. Internal vertices see two parity changes while  $x, y$  see one, so the source sets change from  $(A, \{x, y\})$  to  $(A \Delta \{x, y\}, \emptyset)$ . Toggling twice is the identity and leaves both  $m$  and the binomial weight unchanged. If  $x, y$  are not connected, no allocation on the left exists within that fixed total current: every component contains an even number of the combined sources. Summing the fixed- $m$  bijections and restoring  $F(m)$  proves (6.1).  $\square$

**Theorem 6.5** (Truncated correlations from connectivity). *Let  $x_1, x_2, x_3, x_4$  be distinct vertices in a finite zero-field ferromagnetic Ising model. If  $\mathbf{P}^{A,B}$  denotes the law of two independent currents with source sets  $A$  and  $B$ , then the fourth Ursell function satisfies*

$$\begin{aligned} U_4(x_1, x_2, x_3, x_4) &:= \langle \sigma_1 \sigma_2 \sigma_3 \sigma_4 \rangle - \langle \sigma_1 \sigma_2 \rangle \langle \sigma_3 \sigma_4 \rangle \\ &\quad - \langle \sigma_1 \sigma_3 \rangle \langle \sigma_2 \sigma_4 \rangle - \langle \sigma_1 \sigma_4 \rangle \langle \sigma_2 \sigma_3 \rangle \\ &= -2 \langle \sigma_1 \sigma_2 \rangle \langle \sigma_3 \sigma_4 \rangle \mathbf{P}^{\{x_1, x_2\}, \{x_3, x_4\}}[x_1 \leftrightarrow x_3 \text{ in } n_1 + n_2]. \end{aligned}$$

In particular  $U_4 \leq 0$ .

*Proof.* Multiply the current formulas for  $\langle \sigma_1 \sigma_2 \rangle$  and  $\langle \sigma_3 \sigma_4 \rangle$  and apply the switching lemma with  $A = \{x_1, x_2\}$  and the marked pair  $x_3, x_4$ . Splitting according to whether  $x_1$  is connected to  $x_3$  in the total current gives two terms. On the complement, switching the pairings identifies the contribution with the sum of the two crossed products  $\langle \sigma_1 \sigma_3 \rangle \langle \sigma_2 \sigma_4 \rangle$  and  $\langle \sigma_1 \sigma_4 \rangle \langle \sigma_2 \sigma_3 \rangle$ ; on the connection event the two crossed switchings describe the same allocation and therefore contribute twice. Rearranging yields

$$\langle \sigma_1 \sigma_2 \sigma_3 \sigma_4 \rangle = \sum_{\text{three pairings}} \langle \sigma_i \sigma_j \rangle \langle \sigma_k \sigma_\ell \rangle - 2 \langle \sigma_1 \sigma_2 \rangle \langle \sigma_3 \sigma_4 \rangle \mathbf{P}^{12,34}[x_1 \leftrightarrow x_3],$$

which is (6.2). All factors outside the final minus sign are nonnegative.  $\square$

**Theorem 6.6** (Random-current critical consequences). *Let  $J$  be a translation-invariant, ferromagnetic, summable and aperiodic Ising interaction on  $\mathbb{Z}^d$ . If the free critical state has*

$$\inf_{0 < |B| < \infty} \frac{1}{|B|^2} \sum_{x, y \in B} \langle \sigma_x \sigma_y \rangle_{\beta_c}^0 = 0,$$

*then  $m^*(\beta_c) = 0$  and the critical Gibbs state is unique. For reflection-positive interactions whose infrared energy  $\mathcal{E}(k)$  satisfies  $\int dk / \mathcal{E}(k) < \infty$ , the hypothesis holds; in particular this applies to nearest-neighbor Ising on  $\mathbb{Z}^3$ .*

*Proof.* This is exactly the Aizenman–Duminil-Copin–Sidoravicius continuity criterion proved earlier in Volume VIII under owner ID SM-08.S1.01. The earlier owner contains the finite switching identity, local weak limits of free and plus currents, ergodicity, uniqueness of the infinite double-current cluster, the long-range-order/percolation inequality, the free/plus comparison, and the infrared-to-LRO passage. No new hypothesis or stronger conclusion is introduced here.  $\square$



## Chapter 7

# Currents, FK Connectivity, and Backbones

**Theorem 7.1** (Current/FK coupling and double-current comparison). *Let  $G = (V, E)$  be finite, let  $K_e = \beta J_e \geq 0$ , and let*

$$\mathbf{P}_{\emptyset}^{\text{cur}}(n) = \frac{\mathbf{1}_{\{\partial n = \emptyset\}}}{Z_{\emptyset}^{\text{cur}}} \prod_{e \in E} \frac{K_e^{n_e}}{n_e!}$$

*be the source-free random-current law. Independently of  $n$ , take  $\alpha_e \sim \text{Bernoulli}(1 - e^{-K_e})$ , independently over edges, and set*

$$\omega_e = \mathbf{1}_{\{n_e > 0 \text{ or } \alpha_e = 1\}}.$$

*Then  $\omega$  has the random-cluster law on  $G$  with cluster weight  $q = 2$ , free boundary condition, and edge parameters  $p_e = 1 - e^{-2K_e}$ . Consequently, if  $n_1, n_2$  are independent source-free currents and  $\omega_1, \omega_2$  are obtained from independent Bernoulli fillings in this way, then*

$$\text{supp}(n_1 + n_2) \subseteq \omega_1 \cup \omega_2$$

*under this coupling. In particular every connection in the double-current support is contained in a connection of the union of two independent FK-Ising configurations.*

*Proof.* Fix an edge set  $F \subseteq E$  and sum over source-free currents whose trace is contained in  $F$ . Expanding the current weights and imposing even parity at every vertex gives the high-temperature even-subgraph partition function. After the independent Bernoulli filling, an edge absent from  $\omega$  must have both  $n_e = 0$  and  $\alpha_e = 0$ , contributing  $e^{-K_e}$ ; summing the positive current values on an open edge contributes the complementary factor. Group the resulting expansion by connected components of  $\omega$ . The even-parity constraint leaves one independent spin sign per component, thus a factor  $2^{k(\omega)}$ . The remaining edge factors are

$$(1 - e^{-2K_e})^{\omega_e} e^{-2K_e(1-\omega_e)},$$

up to a configuration-independent normalization. Hence

$$\mathbb{P}(\omega) \propto \prod_e p_e^{\omega_e} (1 - p_e)^{1-\omega_e} 2^{k(\omega)},$$

which is exactly the  $q = 2$  random-cluster weight. (On a single edge this reduces to  $1 - e^{-K} / \cosh K = \tanh K$ , agreeing with the worked example in Chapter 1.)

Apply this construction independently to  $n_1$  and  $n_2$ . Since every edge with positive current is declared open before the Bernoulli filling,  $\text{supp } n_i \subseteq \omega_i$  deterministically. Taking unions gives the displayed inclusion and hence the connectivity comparison.  $\square$

**Theorem 7.2** (Backbone representation). *Fix an ordering of oriented edges of a finite graph and vertices  $x \neq y$ . There is a deterministic exploration assigning to every current  $n$  with  $\partial n = \{x, y\}$  a self-avoiding  $x$ - $y$  path  $\Gamma(n)$ . If*

$$\rho(\gamma) = \frac{1}{Z_{\emptyset}^{\text{cur}}} \sum_{\substack{\partial n = \{x, y\} \\ \Gamma(n) = \gamma}} w(n),$$

then  $\rho(\gamma) \geq 0$  and

$$\langle \sigma_x \sigma_y \rangle = \sum_{\substack{\gamma: x \rightarrow y \\ \gamma \text{ self-avoiding}}} \rho(\gamma). \quad (7.1)$$

Moreover, conditional on the explored backbone  $\gamma$ , subtracting one unit of current along  $\gamma$  leaves an even current; the exploration constraints depend only on the finitely many edges inspected before each backbone step, so the residual weight is a source-free current partition function on the corresponding explored/slit graph with explicit local multiplicity factors.

*Proof.* Given  $n$  with sources  $x, y$ , every connected component of its positive support contains an even number of sources, so  $x$  and  $y$  lie in the same support component. Starting at  $x$ , take the first oriented positive-current edge in the fixed ordering that can still be extended through positive edges to  $y$  without revisiting an already chosen vertex. Such an edge always exists by deleting loops from an  $x$ - $y$  support path. Iterating stops at  $y$  and produces a unique self-avoiding path  $\Gamma(n)$ .

The events  $\{\Gamma(n) = \gamma\}$  partition the set  $\{n : \partial n = \{x, y\}\}$ . Therefore the current expansion gives immediately

$$\langle \sigma_x \sigma_y \rangle = \frac{1}{Z_{\emptyset}^{\text{cur}}} \sum_{\partial n = \{x, y\}} w(n) = \sum_{\gamma} \rho(\gamma),$$

and every coefficient is nonnegative.

For the factorization statement, put  $m = n - \mathbf{1}_{\gamma}$ . At  $x$  and  $y$  one incident parity is toggled and at every internal vertex two are toggled, so  $\partial m = \emptyset$ . During the deterministic exploration, edges examined and rejected before the chosen next edge can never later alter the backbone; record them in the slit set  $S(\gamma)$ . For  $e \in \gamma$  one has

$$\frac{K_e^{m_e+1}}{(m_e+1)!} = \frac{K_e}{m_e+1} \frac{K_e^{m_e}}{m_e!},$$

while off  $\gamma$  the weight is unchanged. Hence, for fixed  $\gamma$ , the sum over  $n$  is an explicit positive multiplicative functional of a source-free current  $m$  on the graph with the exploration constraints  $S(\gamma)$  imposed. This is the claimed residual source-free current factorization; summing it gives precisely  $\rho(\gamma)$  in (7.1).  $\square$

## Chapter 8

# Modern Sharpness and Isoradial Universality

**Theorem 8.1** (Decision-tree sharpness for FK/Potts). *For  $q \geq 1$  random-cluster/Potts models on transitive graphs under the Duminil-Copin–Raoufi–Tassion hypotheses, the phase transition is sharp: subcritical connectivities decay exponentially in finite-range settings and the supercritical order parameter satisfies a mean-field-type lower bound.*

*Proof.* This is exactly the decision-tree sharpness engine proved earlier in Theorem 3.8, once the finite-range transitive graph is used in the random-shell revealment estimate. The monotone-measure OSSS inequality is Lemma 3.5, the FK differentiation formula is Lemma 3.4, and the integration step is Lemma 3.7. No additional planar input is needed for this supplement theorem.  $\square$

**Theorem 8.2** (Isoradial random-cluster universality). *Let  $\mathbb{G}$  be a doubly-periodic isoradial graph whose rhombus angles lie in  $[\varepsilon, \pi - \varepsilon]$ , and equip its edges with the canonical random-cluster weights. For every  $q \geq 1$  the self-dual value  $\beta = 1$  is the critical point. More precisely:*

1. *if  $1 \leq q \leq 4$ , the transition is continuous, the free and wired critical measures coincide, and the critical model has strong RSW;*
2. *if  $q > 4$ , the free critical measure has exponentially decaying connectivities while the wired critical measure percolates;*
3. *if  $1 \leq q \leq 4$  and  $k \in \{1\} \cup 2\mathbb{N}$ , the alternating  $k$ -arm probabilities on  $\mathbb{G}$  and on the regular square lattice are comparable up to constants independent of the two radii. Consequently every exponent defined from these probabilities, if it exists, is the same on all such isoradial graphs.*

*The constants may depend on  $q, \varepsilon$  and the finite fundamental domain of  $\mathbb{G}$ , but not on the scale.*

*Proof.* We internalize the transport mechanism because merely saying “star–triangle invariance” does not control how a macroscopic path moves while a long block of train tracks is exchanged.

Write  $y_e = p_e/(1 - p_e)$  and let  $\theta_e \in (0, \pi)$  be the angle subtended by  $e$ . At  $\beta = 1$  the canonical weights are

$$y(\theta) = \begin{cases} \sqrt{q} \frac{\sin(r(\pi - \theta))}{\sin(r\theta)}, & 1 \leq q < 4, \quad r = \pi^{-1} \arccos(\sqrt{q}/2), \\ 2 \frac{\pi - \theta}{\theta}, & q = 4, \\ \sqrt{q} \frac{\sinh(r(\pi - \theta))}{\sinh(r\theta)}, & q > 4, \quad r = \pi^{-1} \operatorname{arcosh}(\sqrt{q}/2). \end{cases} \quad (9.30)$$

They satisfy the duality identity

$$y(\theta)y(\pi - \theta) = q. \quad (9.31)$$

If three rhombi form the hexagon on which a triangle is replaced by a star, the three triangle weights  $y_a, y_b, y_c$  satisfy

$$y_a y_b y_c + y_a y_b + y_b y_c + y_c y_a = q, \quad (9.32)$$

and the three star weights are  $y'_i = q/y_i$ . To check (9.32), substitute (9.30), put the three train-track angle differences in cyclic order and use their sum around the hexagon; the sine addition identity  $\sin(u + v) = \sin u \cos v + \cos u \sin v$  gives the first line, its  $r \rightarrow 0$  limit gives  $q = 4$ , and the identical calculation with  $\sinh$  gives  $q > 4$ .

For completeness, (9.31)–(9.32) imply equality of the boundary-connection laws directly. For each of the five partitions of the three boundary vertices  $A, B, C$ , sum the random-cluster weights of the configurations realizing that partition. Up to a common factor the triangle weights are

$$1, q a d y_a, \quad y_b, \quad y_c, \quad y_a y_b + y_b y_c + y_c y_a + y_a y_b y_c,$$

while the corresponding star weights are

$$q + y'_a + y'_b + y'_c, \quad y'_b y'_c, \quad y'_c y'_a, \quad y'_a y'_b, \quad y'_a y'_b y'_c.$$

Using  $y'_i = q/y_i$  and (9.32), the five entries are proportional by the same factor. The same calculation after any exterior wiring of  $A, B, C$  proves the assertion for arbitrary induced boundary conditions. Hence there is a coupling of the triangle and star configurations that preserves exactly the connectivity partition of  $\{A, B, C\}$ .

The diamond graph  $\mathbb{G}^\diamond$  is a rhombic tiling. A train track crosses successive rhombi through opposite parallel sides. It neither self-intersects nor meets another train track more than once. A star–triangle move is exactly the local operation that pushes one train track through the crossing of two others. Repeating these moves exchanges two adjacent tracks. By the connectivity-preserving star–triangle coupling proved above, after every move all connections between vertices outside the three-rhombus support are preserved.

The bounded-angle assumption has two quantitative consequences used below. First,

$$c(q, \varepsilon) \leq p_e \leq 1 - c(q, \varepsilon) \quad (9.33)$$

for every edge, so all local choices in the star–triangle coupling have uniformly positive probabilities whenever they are not forced. Second, graph distance and Euclidean distance are quasi-isometric:

$$c_1 d_{\mathbb{G}}(x, y) - c_2 \leq |x - y| \leq c_3 d_{\mathbb{G}}(x, y) + c_4, \quad (9.34)$$

with constants depending only on the bounded angles and the period cell. Thus a loss of  $O(n)$  track levels is the same as a loss of  $O(n)$  Euclidean distance.

Consider a block in which  $n$  horizontal tracks of one isoradial square embedding are exchanged with  $n$  tracks of another. Couple all local moves independently, conditional on the incoming configuration, by the star–triangle coupling. Follow an open path  $\gamma$  through the exchanges. The only local patterns that can increase its height are the two patterns in which the moving rhombus meets the top endpoint of the followed portion of  $\gamma$ . If the transverse angles are  $A, B \in [\varepsilon, \pi - \varepsilon]$ , the local table gives, for  $1 \leq q < 4$ ,

$$\eta_{A,B} = \frac{y(\pi - A)y(\pi - (B - A))}{q} = \frac{\cos(r(2A - B)) - \cos(rB)}{\cos(r(2A - B)) - \cos(r(2\pi - B))}. \quad (9.35)$$

Whenever the configuration of tracks makes this formula relevant, the denominator is nonzero and the probability belongs to  $[0, 1)$ . Compactness of the allowed angle set therefore gives

$$\eta_* := \sup \eta_{A,B} < 1. \quad (9.36)$$

The  $q = 4$  value is the continuous limit. This drift estimate is used only in the continuous regime  $1 \leq q \leq 4$ . For  $q > 4$  one cannot recycle the same local argument: boundary conditions remain visible at macroscopic distance, and a separate half-plane transport estimate is proved later in this chapter.

Dominate all vertices visited by the transported path in the upper half-plane by a random pile  $H^k \subset \mathbb{Z} \times \mathbb{N}$ : at each exchange the pile grows one lattice unit laterally, and above each existing column one extra level is added with probability  $\eta_*$ , independently for the dominating process. Induction over the local moves proves the domination because every deterministic path modification is contained in the lateral enlargement and every possible upward modification is covered by the Bernoulli addition. For a fixed column, an upward excess of  $t$  after  $m$  exchanges is dominated by  $\text{Bin}(m, \eta_*)$ . Chernoff's inequality and a union bound over  $O(m+n)$  possible columns give constants  $\lambda, c > 0$  such that

$$\mathbb{P}(\text{a path initially in height } n \text{ exits height } \lambda n \text{ during } \lambda n \text{ exchanges}) \leq Ce^{-cn}. \quad (9.37)$$

The same argument below the base controls downward drift. Thus a circuit or crossing can be transported through a linear number of track exchanges at the cost of only a fixed linear enlargement of its containing box, except on an exponentially small event.

The opposite direction of transport may lower the top of a vertical crossing, so the upper-drift estimate (9.37) does not by itself suffice. We give the marginal stochastic-domination argument used for the downward direction.

During one complete exchange of a track, insert the intermediate trapezoids

$$D^k = D_0^k \subset D_1^k \subset \dots \subset D_N^k \subset D^{k+1}$$

and let

$$h_j^k := \sup\{h : \text{there is an open path in } D_j^k \text{ from the base to level } h\}.$$

If the moving local level is denoted by  $e_j$ , inspection of the star-triangle move gives

$$h_{j+1}^k = h_j^k \quad (h_j^k \notin \{e_j, e_j + 1\}), \quad (9.38)$$

$$h_{j+1}^k - h_j^k \in \{0, 1\} \quad (h_j^k = e_j), \quad h_{j+1}^k - h_j^k \in \{-1, 0\} \quad (h_j^k = e_j + 1). \quad (9.39)$$

Thus the only estimate requiring probability is the last case. Fix a configuration with  $h_j^k = e_j + 1$ , and among all open paths attaining this height choose the left-most one, denoted  $\Gamma$ . For a deterministic candidate  $\gamma$ , the event  $\{\Gamma = \gamma\}$  has the form

$$\{\gamma \text{ is open}\} \cap N_\gamma, \quad (9.40)$$

where  $N_\gamma$  is decreasing in every edge lying strictly on the forbidden side of  $\gamma$ . Let  $e_3, e_4$  be the two local edges whose simultaneous closure keeps the relevant branch of  $\gamma$  available through the next star-triangle passage, and write  $C_F = \{e_3, e_4 \text{ closed}\}$ . Conditioning the edges of  $\gamma$  to be open can only wire vertices of the small local domain. Since both  $C_F$  and  $N_\gamma$  are decreasing, FKG in that wired local random-cluster measure gives

$$\mathbb{P}(C_F \mid \Gamma = \gamma) \geq \phi_K^1(C_F) = (1 - p_{e_3})(1 - p_{e_4}). \quad (9.41)$$

(The last equality holds because, after the boundary wiring forced by  $\gamma$ , the two local edges do not change the number of clusters.) Conditional on  $C_F \cap \{\Gamma = \gamma\}$ , the star-triangle table above shows that the passage which prevents the height loss occurs with probability  $y_{e_1}y_{e_4}/q$ ; the following passage is then deterministic. Consequently

$$\mathbb{P}(h_{j+1}^k \geq e_j + 1 \mid h_j^k = e_j + 1) \geq \frac{y_{e_1}p_{e_4}(1 - p_{e_3})}{q} \geq \frac{y_{\pi-\varepsilon}p_{\pi-\varepsilon}(1 - p_\varepsilon)}{q}. \quad (9.42)$$

Set

$$\delta = \frac{1}{2} \min \left\{ \frac{y_{\pi-\varepsilon}p_{\pi-\varepsilon}(1 - p_\varepsilon)}{q}, 1 \right\} > 0. \quad (9.43)$$

Equations (9.38)–(9.42), together with the elementary one-step monotone-coupling lemma, imply that for every fixed time  $k$  the marginal law of  $h^k$  stochastically dominates that of

$$H^k = n + \sum_{i=1}^k \Delta_i, \quad \mathbb{P}(\Delta_i = 0) = 2\delta, \quad \mathbb{P}(\Delta_i = -1) = 1 - 2\delta, \quad (9.44)$$

where the  $\Delta_i$  are independent. This is a marginal domination only; no pathwise coupling of the whole processes is asserted. Hence, if the initial crossing reaches height at least  $n$ ,

$$\mathbb{P}(h^{n+1} \geq \delta n \mid h^0 \geq n) \geq \mathbb{P}(H^{n+1} \geq \delta n \mid H^0 = n) =: c_n(\delta), \quad (9.45)$$

and the law of large numbers (or Chernoff's inequality) gives  $c_n(\delta) \rightarrow 1$ . Therefore a macroscopic vertical crossing survives a linear block of track exchanges with a fixed positive fraction of its height, with probability tending to one.

Start with the regular square lattice, whose critical strong RSW theorem was proved earlier in this volume. Superimpose a large block of the target isoradial square embedding, convexify the finite region so every required local track exchange is legal, and then exchange the two blocks. Apply (9.37) to the open and dual-open circuits used to shield a horizontal crossing, and (9.39) to a vertical crossing. If the source rectangle has dimensions  $an \times bn$ , the transported events live in rectangles whose dimensions are at most fixed multiples of  $n$ . Their probabilities satisfy

$$\phi_{\text{target}}(\mathcal{C}_h(C_1n, C_2n)) \geq \phi_{\text{source}}(\mathcal{C}_h(an, bn)) - Ce^{-cn}, \quad (9.40)$$

$$\phi_{\text{target}}(\mathcal{C}_v(C_3n, C_4n)) \geq (1 - Ce^{-cn})\phi_{\text{source}}(\mathcal{C}_v(an, bn)). \quad (9.41)$$

The same bounds hold for the dual model. Choosing the source aspect ratios from the square-lattice RSW theorem and absorbing finitely many small scales gives positive lower bounds for primal and dual crossings of every fixed aspect ratio on the target embedding.

A general doubly-periodic isoradial graph is reduced to this case by choosing two bi-infinite families of train tracks that form a square grid. The remaining finitely many track types in one period cell are moved, one periodic block at a time, through that grid. During  $O(n)$  moves in an  $n$ -box, (9.37)–(9.39) give the same linear-distortion estimates; periodicity makes the number of track types fixed. Therefore every  $\mathbb{G}$  in the theorem has strong RSW at  $\beta = 1$ .

Strong RSW supplies, at each dyadic scale, a dual-open circuit around the origin with probability at least  $c > 0$ , uniformly under wired exterior boundary conditions. Applying the domain Markov property successively on disjoint annuli,

$$\phi_{\mathbb{G},1,q}^1(0 \leftrightarrow \partial B_{2^m}) \leq (1 - c)^{m/C}, \quad (9.42)$$

so the wired critical measure has no infinite cluster. The usual monotone coupling of free and wired measures can disagree at an edge only if the edge is connected through successively larger annuli to the boundary wiring; (9.42) makes that probability tend to zero. Thus

$$\phi_{\mathbb{G},1,q}^0 = \phi_{\mathbb{G},1,q}^1. \quad (9.43)$$

In particular the transition is continuous.

It remains to verify that  $\beta = 1$ , not merely a self-dual parameter, is the critical point. For a periodic finite-edge-type model the decision-tree differential inequality used in Theorem 3.8 applies after summing derivatives over one fundamental domain; (9.33) makes the coefficients relating  $d/d\beta$  to the individual edge influences uniformly comparable. Since there is no infinite cluster at  $\beta = 1$ , integration of that inequality gives exponential decay for every  $\beta < 1$ . By (9.31), the planar dual of the parameter  $\beta > 1$  has parameter  $\beta^* < 1$ ; exponential decay in the dual and a Peierls/Borel–Cantelli circuit argument yield a primal infinite cluster. Hence  $\beta_c = 1$ .

For the regular square lattice the critical free measure has exponential decay and the critical wired measure percolates, by the first-order theorem proved earlier in this volume. A direct local comparison is not enough to transport this fact, because the boundary condition at infinity is felt at criticality. We therefore prove the half-plane estimate that replaces the RSW transport used above.

Let  $\mathbf{R}^{\text{hp}}(N; N)$  be the piece of an isoradial square embedding between two vertical grid tracks, the base track and height  $N$ , with free boundary condition. We claim that there are  $C_0, c_0 > 0$ , uniform in  $N > n$ , such that

$$\phi_{\mathbf{R}^{\text{hp}}(N; N)}^0(0 \leftrightarrow \partial\Lambda_n) \leq C_0 e^{-c_0 n}. \quad (9.44)$$

To prove this, make a mixed graph whose lower block is the target isoradial square lattice and whose upper block is the regular square lattice, and exchange the two blocks by star–triangle transformations. Split a connection from 0 to distance  $n$  according as it exits the thin rectangle  $[-n, n] \times [0, \delta_0 n]$  through the left, right or top side. The vertical-exit term is transported to the regular block by the same height process used above: for some  $\delta > 0$  and numbers  $c_n \uparrow 1$ ,

$$\phi_{\tilde{G}_{\text{mix}}}^0(0 \xrightarrow{[-4n, 4n] \times [0, \delta\delta_0 n]} \{y = \delta\delta_0 n\}) \geq c_n \phi_{G_{\text{mix}}}^0(0 \xrightarrow{[-n, n] \times [0, \delta_0 n]} \{y = \delta_0 n\}). \quad (9.45)$$

The proof is the distribution-function induction of (9.38)–(9.39); the only change is that the relevant local success probabilities are bounded away from zero by the bounded-angle hypothesis. Thus the top-exit probability is dominated by a regular-square-lattice connection to distance  $\delta\delta_0 n$ , hence by  $Ce^{-c\delta\delta_0 n}$ .

For the two lateral exits, reveal the first part of the transported path reaching the relevant side. Each star–triangle move which can lower that path contributes a multiplicative likelihood ratio bounded by  $e^\eta$  for a constant  $\eta = \eta(q, \varepsilon)$ ; there are at most  $\delta_0 n + O(1)$  such moves before the path leaves the thin strip. Conditioning on the revealed path and then comparing the unexplored region with a free half-plane square-lattice measure yields

$$\phi_{G_{\text{mix}}}(\text{left or right exit}) \leq Ce^{\eta\delta_0 n} \phi_{\mathbb{Z}^2,1,q}^0(0 \leftrightarrow \partial\Lambda_n) \leq Ce^{-(c-\eta\delta_0)n}. \quad (9.46)$$

Choose  $\delta_0 > 0$  so small that  $c - \eta\delta_0 > 0$ . Combining the three exits gives (9.44) for every isoradial square embedding.

We next remove the square-embedding restriction. In a finite half-plane rectangle of the periodic graph, move all intersections of non-grid train tracks upward, one strip at a time, by star–triangle transformations. Below the last moved track the graph is a rectangular piece of an isoradial square embedding whose transverse angles are still among those of the original graph, hence satisfy the

same bounded-angle condition. Boundary-condition comparison and (9.44) give, uniformly in the height of the transformed rectangle,

$$\phi_{\text{RHP}(N;N)}^0(0 \leftrightarrow \partial\Lambda_n) \leq Ce^{-cn}. \quad (9.47)$$

Letting the upper height tend to infinity preserves the free half-plane boundary condition. Finally, to pass from the half-plane estimate to the full-plane free measure, expose the lowest dual-open circuit separating 0 from the lower boundary in a large box. Conditional on that circuit, the unexplored component of 0 is dominated by the free half-plane measure; summing over the possible first-contact point costs only  $O(r)$  at distance  $r$ , which is absorbed by the exponential. Thus

$$\phi_{\mathbb{G},1,q}^0(0 \leftrightarrow \partial B_R) \leq Ce^{-cR}. \quad (9.48)$$

Duality turns (9.48) on  $\mathbb{G}^*$  into exponential decay of large dual-open circuits in the wired primal measure. If the wired primal cluster of 0 were almost surely finite, arbitrarily large dual-open circuits would surround it. The shell bound  $O(R)e^{-cR}$  is summable, so Borel–Cantelli excludes this. Finite energy then gives

$$\phi_{\mathbb{G},1,q}^1(0 \leftrightarrow \infty) > 0. \quad (9.49)$$

The off-critical decision-tree sharpness theorem and duality identify  $\beta_c = 1$ . This proves the discontinuous part without using the continuous-regime RSW transport in a place where it is false.

For  $k = 1$  or even  $k = 2j$ , choose the base train track, which is never moved during the block exchange. Define  $\tilde{A}_k(r, R)$  by requiring the relevant primal endpoints (and, for  $k = 2$ , one dual pair) to lie on this base; for  $k = 2j \geq 4$  require  $j$  distinct primal clusters from the inner to the outer base interval. Planarity supplies the alternating dual arms between the distinct primal clusters. Because every star–triangle move preserves all connections between vertices outside its support and the base vertices are untouched,

$$\phi_{\mathbb{G}(1)}[\tilde{A}_k(r, R)] = \phi_{\mathbb{G}(2)}[\tilde{A}_k(r, R)] \quad (9.50)$$

for two isoradial square embeddings after taking the size of the mixed finite block to infinity. Reversing the exchange gives equality rather than merely one inequality.

For  $q \leq 4$ , strong RSW and the arm-separation construction proved for the planar critical model imply, uniformly in scale,

$$c_k \phi_{\mathbb{G}}[A_k(r, R)] \leq \phi_{\mathbb{G}}[\tilde{A}_k(r, R)] \leq C_k \phi_{\mathbb{G}}[A_k(r, R)]. \quad (9.51)$$

For the nontrivial direction, first separate the ordinary arms at radii  $2r$  and  $R/2$  by primal/dual fences. Use a bounded number of RSW corridors to route their inner and outer endpoints to disjoint intervals on the fixed base track; the domain Markov property permits the corridors to be exposed successively under the induced boundary conditions. The reverse comparison follows by truncating base-attached arms when they first enter the Euclidean annulus. Quasi-isometry (9.34) absorbs the fixed changes of radii.

Combining (9.50)–(9.51) shows that two isoradial square embeddings have comparable  $A_k(r, R)$  probabilities. For a general doubly-periodic isoradial graph, choose a square-grid subfamily of tracks, move all remaining periodic tracks to one side of the annulus, and apply the same base-attached event before and after the move. Only finitely many track types occur in a period cell, so the constants remain independent of  $r, R$ . Thus

$$c \phi_{\mathbb{Z}^2,1,q}[A_k(r, R)] \leq \phi_{\mathbb{G},1,q}[A_k(r, R)] \leq c^{-1} \phi_{\mathbb{Z}^2,1,q}[A_k(r, R)]. \quad (9.52)$$

If either side has an exponent  $\alpha_k = -\lim_{R/r \rightarrow \infty} \log \phi[A_k(r, R)] / \log(R/r)$ , the multiplicative constant in (9.52) disappears after division by  $\log(R/r)$ , so the exponents agree.  $\square$



## Chapter 9

# Critical Fractal Geometry, Scaling Relations, and Potts Gibbs States

**Theorem 9.1** (Critical planar FK fractal package). *Let  $q \in [1, 4)$  and let  $\phi$  denote the infinite-volume critical planar random-cluster measure on  $\mathbb{Z}^2$ . There are constants depending only on  $q$  and on the fixed macroscopic shapes below such that the following hold. The crossing statement is uniform in every boundary condition induced outside the domain under consideration. The arm probabilities are those of  $\phi$ ; the localization, separation, and corridor estimates used in their proof are uniform under induced boundary conditions.*

1. *Every quad of uniformly nondegenerate extremal length has an open crossing probability bounded away from both 0 and 1.*
2. *If  $A_\sigma(r, R)$  is a bulk arm event with a fixed cyclic colour word  $\sigma \in \{0, 1\}^k$ , and  $A_\sigma^+(r, R)$  is the analogous event in a half-annulus, then separated landing versions are comparable to the unrestricted events and the arm probabilities are quasi-multiplicative.*
3. *For  $1 \leq r \leq R/8$ ,*

$$\phi[A_{10}^+(r, R)] \asymp \frac{r}{R}, \quad \phi[A_{101}^+(r, R)] \asymp \left(\frac{r}{R}\right)^2,$$

*and*

$$\phi[A_{10101}(r, R)] \asymp \left(\frac{r}{R}\right)^2.$$

*The colour 1 denotes a primal-open arm and 0 a dual-open arm; changing all colours gives equivalent estimates by planar duality.*

4. *There exists  $\varepsilon = \varepsilon(q) > 0$  such that*

$$\phi[A_{1010}(r, R)] \geq c \left(\frac{r}{R}\right)^{2-\varepsilon}.$$

*In particular the bulk four-arm exponent, whenever defined, is strictly smaller than 2.*

*No SLE input is used in these assertions.*

*Proof.* We give the planar proof because the universal exponents in the statement are precisely the part of the critical FK geometry that is later used without any exact SLE identification of the nonuniversal exponents.

The arbitrary-boundary critical crossing theorem proved in Theorem SM-09.05.02 gives a number

$$c_Q = c_Q(q) > 0$$

for every compact family of quads with extremal length bounded above and below, such that every induced boundary condition  $\xi$  satisfies

$$c_Q \leq \phi_{D,p_c,q}^\xi(\mathcal{C}(Q)) \leq 1 - c_Q. \quad (9.42)$$

The upper bound is the lower bound applied to the dual crossing in the dual quad. The crucial point is that the constant in (9.42) is independent of  $D$  and of  $\xi$ . Consequently every finite collection of fixed-aspect-ratio corridor crossings, primal or dual, can be imposed with probability at least a positive constant depending only on  $q$  and on the corridor shapes. We shall use this repeatedly below.

Theorem SM-09.05.03 already proves, for every fixed arm word  $\sigma$ , constants  $c_\sigma, C_\sigma > 0$  such that

$$c_\sigma \phi[A_\sigma(r, R)] \leq \phi[A_\sigma^{\text{sep}}(r, R)] \leq \phi[A_\sigma(r, R)] \quad (9.43)$$

and, whenever  $r < s < R$  are separated by fixed factors,

$$c_\sigma \phi[A_\sigma(r, s)] \phi[A_\sigma(s, R)] \leq \phi[A_\sigma(r, R)] \leq C_\sigma \phi[A_\sigma(r, s)] \phi[A_\sigma(s, R)]. \quad (9.44)$$

The same proof applies to half-annuli: the two arcs on the straight boundary are treated as prescribed landing windows, while the shielding corridors are chosen in the upper half-plane. We record one consequence that will be used in the counting arguments. If the endpoints of the arms on each boundary circle are required to lie in fixed disjoint windows of positive relative length and in the prescribed cyclic order, then the resulting probability is comparable, by constants independent of  $r, R$ , to the unrestricted arm probability. Indeed, on the separated event one joins each landing point to its chosen window through disjoint fixed-aspect-ratio rectangles; (9.42), the domain Markov property and finite energy give the lower comparison, while the reverse comparison is immediate.

It suffices by (9.44) to prove the estimate with  $r = 1$  and  $R = N$  and then to rescale/coarse-grain the inner boundary. Work in the rectangle

$$Q_N = [-2N, 2N] \times [0, 2N]$$

with a primal landing window on the left part of the top side and a dual landing window on the right part. By (9.42), with probability bounded below independently of  $N$ , there is a primal vertical crossing in the left corridor and a dual vertical crossing in the right corridor. On this event take the rightmost primal crossing and the leftmost dual crossing. Planarity gives a unique change-over interval on the bottom side between their two boundary traces. Divide the central bottom interval into  $O(N)$  unit blocks  $I_j$  and let  $E_j$  be the event that this change-over block is  $I_j$  and that the two frontier arms reach the prescribed top landing windows.

The  $E_j$  are disjoint and their union has probability bounded below by (9.42). Conversely, if  $E_j$  occurs, a unit neighbourhood of  $I_j$  has one primal and one dual arm to distance  $N$  with separated landing windows. By the localization comparison above, all central  $E_j$  have probabilities comparable, up to a constant independent of  $j, N$ , to  $\phi[A_{10}^+(1, N)]$ . The  $O(1)$  end blocks may be discarded. Thus

$$c \leq \sum_j \phi(E_j) \leq CN \phi[A_{10}^+(1, N)]. \quad (9.45)$$

For the reverse inequality, on  $A_{10}^+(1, N)$  first separate the two arms and then attach them, by two RSW corridors, to the two top landing windows. The resulting configuration has exactly one

change-over block among a bounded number of neighbouring blocks, so summing the disjoint  $E_j$  gives

$$N \phi[A_{10}^+(1, N)] \leq C. \quad (9.46)$$

Equations (9.45)–(9.46), followed by (9.44), yield

$$\phi[A_{10}^+(r, R)] \asymp r/R. \quad (9.47)$$

We use the same topological counting principle, now with a two-dimensional set of possible interface points. Take a rectangle of diameter  $N$  and first reveal the lowest primal left–right crossing  $\Gamma$ . Conditional on  $\Gamma$ , the FK measure in the component above  $\Gamma$  has wired boundary condition along  $\Gamma$  and an arbitrary induced boundary condition elsewhere. By (9.42), with probability bounded below, this component contains both a primal and a dual crossing from macroscopic subarcs of  $\Gamma$  to two prescribed, disjoint landing windows on the upper boundary. Take the leftmost such primal crossing and the adjacent dual frontier. Their change-over point  $x \in \Gamma$  has, in the domain above  $\Gamma$ , two primal arms (the two directions along  $\Gamma$  after a fixed local extension) and one dual arm, with separated endpoints at distance comparable with  $N$ .

For each lattice vertex  $x$  in the central part of the rectangle let  $H_x$  denote the corresponding localized event, including fixed landing windows for the three arms and the requirement that  $x$  is the change-over point between the selected extremal crossings. The construction just described gives

$$\phi\left(\bigcup_x H_x\right) \geq c. \quad (9.48)$$

There are  $O(N^2)$  possible  $x$ . Conversely, for the prescribed landing windows, two distinct vertices cannot both be the selected change-over point: the primal frontier from one and the dual frontier from the other would have to cross. Thus the  $H_x$  are disjoint. After the part of the configuration below the selected crossing has been revealed, the remaining domain is a topological half-disc with uniformly controlled extremal geometry near  $x$ ; the landing-window surgery above straightens it at a multiplicative cost bounded above and below independently of  $N$  and of the revealed boundary condition. Hence, for central  $x$ ,

$$\phi(H_x) \asymp \phi[A_{101}^+(1, N)]. \quad (9.49)$$

Combining (9.48)–(9.49) and the disjointness gives both

$$cN^{-2} \leq \phi[A_{101}^+(1, N)] \leq CN^{-2}.$$

Coarse-graining the inner boundary into  $r$ -boxes and using (9.44) yields

$$\phi[A_{101}^+(r, R)] \asymp (r/R)^2. \quad (9.50)$$

Again first take  $r = 1$ ,  $R = N$ . In a square of side comparable with  $N$ , impose by (9.42) a primal left–right crossing, two dual top–bottom crossings on its two sides, and two additional primal corridors separating the dual crossings. Choose the lowest primal crossing and the adjacent dual frontiers. A deterministic planar inspection shows that some vertex  $x$  in the central square is incident, in cyclic order, to five disjoint arms of colours 10101 reaching distance  $cN$ . Hence, if  $F_x$  denotes the corresponding separated five-arm event from  $x$  to distance  $cN$ ,

$$c \leq \phi\left(\bigcup_{x \in [-N, N]^2} F_x\right) \leq \sum_{x \in [-N, N]^2} \phi(F_x). \quad (9.51)$$

Translation invariance away from the boundary of the construction and the landing-window comparison imply  $\phi(F_x) \leq C\phi[A_{10101}(1, N)]$  for every relevant  $x$ . Since there are  $O(N^2)$  such vertices,

$$\phi[A_{10101}(1, N)] \geq cN^{-2}. \quad (9.52)$$

The upper bound uses the classical “one distinguished site” construction, so we spell out the extra extremality condition that makes the counting legitimate. Fix five disjoint outer landing arcs in cyclic order. Given a configuration with the corresponding localized 10101 arms, extend the two dual arms through two fixed RSW corridors to the top and bottom sides of a surrounding rectangle and extend the three primal arms to the remaining prescribed side arcs. Among all such extensions choose the leftmost lower dual frontier and the rightmost upper dual frontier; between them choose the outermost two primal frontiers. Their planar order leaves a unique lattice face at which the fifth prescribed arm can enter the central cell without crossing one of these four extremal frontiers. Let  $G_x$  be the event that this distinguished face is adjacent to  $x$  and that the five extensions land in the prescribed arcs.

Two different vertices cannot satisfy  $G_x$  in the same configuration. Indeed, if  $x$  and  $y$  did, order them according to the strip between the two selected dual frontiers. The outermost primal frontier belonging to the left distinguished point separates the fifth-arm corridor of the right distinguished point from its prescribed landing arc (or the symmetric statement holds with left and right interchanged), forcing a primal path to cross a dual frontier. Hence the  $G_x$  are pairwise disjoint and

$$\sum_{x \in [-N/3, N/3]^2} \phi(G_x) \leq 1. \quad (9.53)$$

Conversely, start from a separated 10101 event from a fixed central vertex  $x$  to radius  $N/4$ . The landing-window theorem above lets us prescribe five windows on the circle of radius  $N/4$ . A bounded number of disjoint fixed-aspect-ratio corridors then continue these windows to the five side arcs used above. Strong RSW, applied successively under the induced boundary conditions, shows that this continuation has conditional probability at least  $c(q) > 0$ . If necessary, the extremal frontiers are fixed by closing/opening only a bounded number of short fence boxes, again at uniform cost. Therefore

$$\phi(G_x) \geq c\phi[A_{10101}(1, N)] \quad (9.54)$$

for every  $x$  in a central square (changing  $N$  by a fixed factor is absorbed by quasi-multiplicativity). There are  $\asymp N^2$  such  $x$ , and (9.53)–(9.54) give

$$\phi[A_{10101}(1, N)] \leq CN^{-2}.$$

Together with the lower bound (9.52), coarse-graining the inner box into  $r$ -blocks and quasi-multiplicativity (9.44) yield

$$\phi[A_{10101}(r, R)] \asymp (r/R)^2. \quad (9.55)$$

Let  $A_4 = A_{1010}(r, R)$  and  $A_5 = A_{10101}(r, R)$ . On the separated four-arm event, inspect every third dyadic annulus between radii  $r$  and  $R$ . In the gap in which a fifth primal arm would have to pass, (9.42) gives, conditionally on everything already revealed inside and outside that annulus, a dual blocker with probability at least  $c_0(q) > 0$ . The blocker can be surrounded by two short dual fences, again at a uniform cost, and then no additional primal arm can cross that annulus. By exposing the selected annuli successively and using the domain Markov property,

$$\phi(A_5 \mid A_4) \leq C(1 - c_0)^{\lfloor c \log(R/r) \rfloor} \leq C\left(\frac{r}{R}\right)^\varepsilon \quad (9.56)$$

for some  $\varepsilon = \varepsilon(q) > 0$ . Since  $A_5 \subset A_4$ , (9.55) implies

$$c\left(\frac{r}{R}\right)^2 \leq \phi(A_5) \leq C\left(\frac{r}{R}\right)^\varepsilon \phi(A_4),$$

which rearranges to

$$\phi[A_{1010}(r, R)] \geq c\left(\frac{r}{R}\right)^{2-\varepsilon}. \quad (9.57)$$

This is the asserted strict inequality for the four-arm exponent.

Every surgery above uses only the domain Markov property, finite energy and (9.42), whose constants are uniform in the induced boundary condition. Therefore arbitrary exterior wirings produced by conditioning are harmless. The finitely many cases  $R < 8r$  are absorbed into the constants. Dual colour reversals are handled by planar FK duality at  $p_c(q)$ . No step invokes convergence to SLE or an exact formula for the nonuniversal one-arm or four-arm exponent. This proves exactly the critical planar-FK package claimed in the theorem.  $\square$

## 9.1 Near-critical mixing and rigorous FK scaling relations

The critical arm package above is only half of the scaling argument. For dependent random-cluster models the derivative of a macroscopic event is not measured by the four-arm pivotal probability. The correct quantity is the rate at which a local observable still remembers boundary conditions at a distant scale. We now build that quantity, prove the stability estimates needed below the characteristic length, and only then pass to critical exponents.

Throughout this section  $1 < q \leq 4$  is fixed. (For  $q = 1$  the same statements reduce to the product-measure Russo–Kesten calculus developed in the percolation volume.) For  $p$  in a fixed compact subinterval of  $(0, 1)$  all comparison constants below are uniform. Let  $e_0$  be a fixed edge incident to the origin and put

$$\Delta_p(R) := \phi_{\Lambda_{R,p}}^1(\omega_{e_0}) - \phi_{\Lambda_{R,p}}^0(\omega_{e_0}),$$

and, for  $1 \leq r < R$ ,

$$\Delta_p(r, R) := \phi_{\Lambda_{R,p}}^1(\mathcal{C}(\Lambda_r)) - \phi_{\Lambda_{R,p}}^0(\mathcal{C}(\Lambda_r)).$$

Here  $\mathcal{C}(\Lambda_r)$  denotes the left–right crossing event of the square. Fix once and for all a small crossing threshold  $\delta_0 \in (0, 1/2)$  and define

$$L(p) := \inf\{R \geq 1 : \phi_p(\mathcal{C}(\Lambda_R)) \notin [\delta_0, 1 - \delta_0]\}. \quad (9.58)$$

At  $p = p_c$  we set  $L(p_c) = \infty$ .

**Lemma 9.2** (Finite-volume differentiation and boundary influence). *Let  $G$  be finite,  $A$  an increasing event depending on the edges of  $G$ , and  $\phi_{G,p}^\xi$  an FK measure with fixed boundary condition  $\xi$ . Then*

$$\frac{d}{dp} \phi_{G,p}^\xi(A) = \frac{1}{p(1-p)} \sum_{e \in E(G)} \text{Cov}_{G,p}^\xi(\omega_e, \mathbf{1}_A). \quad (9.59)$$

Moreover, if  $A$  is supported in a set  $D$  and  $e$  lies a distance comparable to  $R$  from  $\partial D$ , with all relevant scales at most  $L(p)$ , then

$$\text{Cov}_p(\omega_e, \mathbf{1}_A) \asymp \Delta_p(R) \quad (9.60)$$

whenever  $A$  is the crossing of a fixed-shape regular quad through the  $R$ -neighbourhood of  $e$ .

*Proof.* Write the finite-volume weight as

$$W_p(\omega) = p^{|\omega|}(1-p)^{|E|-|\omega|}q^{k_\xi(\omega)}.$$

Differentiating the quotient  $Z_A(p)/Z(p)$  gives

$$\frac{d}{dp}\phi(A) = \phi\left[\mathbf{1}_A\left(\frac{|\omega|}{p} - \frac{|E|-|\omega|}{1-p}\right)\right] - \phi(A)\phi\left[\frac{|\omega|}{p} - \frac{|E|-|\omega|}{1-p}\right],$$

which is exactly (9.59) after writing  $|\omega| = \sum_e \omega_e$ .

For (9.60) use the monotone heat-bath coupling of the FK measures obtained by revealing the exterior of a flower domain around  $e$ . Conditional on the revealed exterior, the two extremal laws in the remaining flower differ only by whether the petals that are connected through the exterior are kept separate or identified. The difference of the conditional opening probability of  $e$  is therefore exactly zero unless the exterior wiring reaches the flower, and, by finite energy, is bounded above and below by fixed multiples of the probability of this wiring disagreement. RSW with arbitrary induced boundary conditions, together with two annular fences, shows that a disagreement reaching the  $R/4$ -neighbourhood of  $e$  has a fixed positive chance to change the crossing of the regular quad; conversely, if the two coupled crossing indicators differ then a disagreement path must reach from the exterior to that neighbourhood. Thus both the covariance and the wired-free edge discrepancy measure the same disagreement event up to fixed multiplicative constants. Translating the flower to  $e_0$  and changing  $R$  by a bounded factor changes  $\Delta_p(R)$  only by the RSW comparison, giving (9.60).  $\square$

**Lemma 9.3** (Mixing-rate calculus). *There are constants  $c, C, \varepsilon > 0$  such that, whenever  $1 \leq r \leq s \leq R \leq L(p)$ ,*

$$c \Delta_p(r, s) \Delta_p(s, R) \leq \Delta_p(r, R) \leq C \Delta_p(r, s) \Delta_p(s, R), \quad (9.61)$$

$$\Delta_p(R) \asymp \Delta_p(r) \Delta_p(r, R). \quad (9.62)$$

*There is, in addition, a strict two-dimensional power gain*

$$\Delta_p(r, R) \geq c (r/R)^{2-\varepsilon} \quad (9.63)$$

*for  $1 < q < 4$  throughout  $R \leq L(p)$ ; for  $q = 4$  the same estimate is first proved at  $p = p_c$  and is extended to the whole near-critical window by Lemma 9.6 below. In addition, if  $A$  is measurable inside  $\Lambda_r$  and  $B$  outside  $\Lambda_R$ , then*

$$\left| \frac{\phi_p(A \cap B)}{\phi_p(A)\phi_p(B)} - 1 \right| \leq C \Delta_p(r, R), \quad (9.64)$$

*and for suitable increasing  $A, B$  the reverse inequality holds up to a constant.*

*Proof.* Explore successively from  $\partial\Lambda_R$  to  $\partial\Lambda_s$  and then to  $\partial\Lambda_r$  in the extremal wired/free coupling. The domain Markov property says that, once the first exploration has been stopped on  $\partial\Lambda_s$ , the remaining discrepancy is again an extremal boundary condition discrepancy. Annular RSW fences prevent different petals from being accidentally identified and give a fixed positive probability that a discrepancy which reaches one scale survives to the next. This proves both inequalities in (9.61). Taking the inner observable to be the state of an edge and using the same one-annulus fence gives (9.62).

For (9.63), first take  $1 < q < 4$  and any  $p$  with  $R \leq L(p)$ . The one-edge FK conditional law is explicit: if the endpoints of an edge are already connected without that edge, its opening probability is  $p$ , while if they are disconnected it is

$$p^{\text{disc}} := \frac{p}{p + q(1 - p)} < p.$$

Thus the wired/free difference of a local edge observable is the fixed gap  $p - p^{\text{disc}}$  times the probability that the exterior wiring changes the connectivity of its endpoints. On a separated alternating four-arm event this change occurs after two fixed RSW fences with probability bounded below. Hence, for  $1 < q < 4$ ,

$$\Delta_p(r, R) \geq c \pi_{1010}(p; r, R) \geq c(r/R)^{2-\varepsilon}.$$

The arm proof in the preceding critical-fractal theorem uses only arbitrary-boundary RSW, arm separation and finite energy; those inputs are uniform on every scale below  $L(p)$ , so the same four-arm estimate holds here.

At  $q = 4$  we first work at  $p = p_c$ , where the four-arm comparison is not the efficient endpoint. Put  $\mathbb{U}_3 := \{x \in \mathbb{Z}^2 : x_1 \leq 0 \text{ or } x_2 \leq 0\}$ , the lower-left three-quarter plane. We use the differentiated parafermionic observable already constructed in the continuity proof, equation (9.8). Sum its local relation once in  $\Lambda_R$  and once in the annulus obtained by deleting  $\Lambda_r$ . Interior medial edges cancel. The difference of the two boundary sums is supported precisely on exploration paths whose outer wiring can be transmitted to the inner boundary. Applying the flower coupling to those paths factors the difference as

$$\Delta_{p_c}(r, R) \Sigma_R,$$

where  $\Sigma_R$  is the same positive outer boundary sum obtained by taking  $r = R/2$ . The remaining inner-boundary contribution is

$$\pi_2(r, R) \sum_{k \leq r/2} \phi_{\mathbb{U}_3, p_c}^0((k, 0) \leftrightarrow \partial \Lambda_r),$$

up to fixed factors; the winding weight is positive after the symmetric terms are paired. Dividing by the identity with  $r = R/2$  therefore gives

$$\Delta_{p_c}(r, R) \geq c \pi_2(r, R) \frac{r}{R}. \quad (9.63a)$$

It remains only to record the strict improvement over the trivial two-arm power. We spell out the standard fractal-interface argument. Put Dobrushin boundary conditions on a square of side comparable with  $R$  so that its exploration path  $\gamma$  crosses the square. In each block of four consecutive dyadic scales, arbitrary-boundary RSW gives a fixed probability  $c_0$  that  $\gamma$  makes a macroscopic side detour before leaving the next larger box. Choosing disjoint side boxes makes these trials conditionally independent up to a uniform finite-energy factor. An induction on scales therefore gives, for some  $\delta > 0$ ,

$$\mathbb{E}|\gamma \cap \Lambda_{R/2}| \geq cR^{1+\delta}. \quad (9.63b)$$

Every edge  $e$  of the central square that belongs to  $\gamma$  has one primal and one dual arm from a neighbourhood of  $e$  to distance  $cR$ . Arm separation and translation give  $\mathbb{P}(e \in \gamma) \leq C\pi_2(1, R)$ . Summing over the  $O(R^2)$  central edges and using (9.63b) yields  $\pi_2(1, R) \geq cR^{-1+\delta}$ . Applying the same argument after rescaling the inner box to size  $r$ , and using two-arm quasi-multiplicativity, gives for some  $\varepsilon_1 > 0$ ,

$$\pi_2(r, R) \geq c(r/R)^{1-\varepsilon_1}.$$

Substitution in (9.63a) proves (9.63) at  $q = 4$ . This is exactly where the differentiated endpoint observable is needed; no  $q < 4$  spin is continued through the degenerate endpoint.

Finally, condition first on the annulus  $\Lambda_R \setminus \Lambda_r$ . Replacing the induced inner boundary condition by the free or wired one changes the law of  $A$  by at most the extremal discrepancy. The same flower-domain coupling used above identifies that discrepancy with  $\Delta_p(r, R)$ . Applying this to the conditional law given  $B$  proves (9.64); choosing  $A$  to be an inner crossing and  $B$  an outer wiring event produces a boosting pair and gives the matching lower bound.  $\square$

**Lemma 9.4** (Two-ended ratio mixing, boost transport, and four-arm amplification). *Fix  $1 < q \leq 4$ . There are constants  $c, C, \varepsilon_* > 0$ , depending only on  $q$  and on the fixed separation ratios below, with the following properties whenever all scales are at most  $L(p)$ .*

1. Let  $D_i = \Lambda_{r_i}(z_i)$ ,  $i = 1, 2$ , and put  $d = |z_1 - z_2|$ . If  $r_1 + r_2 \leq d/4$  and  $A_i$  is an event of positive probability measurable on the edges of  $D_i$ , then

$$\left| \frac{\phi_p(A_1 \cap A_2)}{\phi_p(A_1)\phi_p(A_2)} - 1 \right| \leq C \Delta_p(r_1, d) \Delta_p(r_2, d). \quad (9.64C)$$

*The same estimate holds in a domain containing the  $d/3$ -neighbourhoods of both sets, under any boundary condition whose blocks do not identify the two neighbourhoods.*

2. Let  $\mathcal{F}$  be a well-separated outer four-petal flower at scale  $r$  with a boosting pair  $\xi < \xi'$  of compatible boundary conditions. Let  $\mathcal{G}$  be a well-separated inner four-petal flower at scale  $s$  around a point at distance  $d \geq 16(r + s)$  from  $\mathcal{F}$ . There is an increasing decision-tree coupling of  $\phi_{\mathcal{F}, p}^\xi$  and  $\phi_{\mathcal{F}, p}^{\xi'}$  for which, with probability at least

$$c \Delta_p(r, d) \Delta_p(s, d), \quad (9.64D)$$

*the induced boundary conditions on  $\mathcal{G}$  form a boosting pair. In particular, when  $r \asymp s$  this probability is  $\geq c \Delta_p(r, d)^2$ .*

3. If  $\pi_4(p; r, R)$  denotes the separated alternating four-arm probability in  $\text{Ann}(r, R)$ , then

$$\pi_4(p; r, R) \leq C \Delta_p(r, R) \left( \frac{r}{R} \right)^{\varepsilon_*}, \quad 1 \leq r \leq R \leq L(p). \quad (9.64E)$$

*Equivalently,  $\Delta_p(r, R) \geq c(R/r)^{\varepsilon_*} \pi_4(p; r, R)$ .*

*Proof.* We give the two coupling mechanisms explicitly because both are used later at the  $q = 4$  endpoint.

*Two-ended ratio mixing.* Choose a lattice annulus  $S$  of fixed relative width which surrounds  $D_1$ , lies at distance at least  $d/8$  from  $D_1$ , and separates  $D_1$  from  $D_2$ . Reveal every edge of  $S$  and let  $\mathcal{G}_S$  be the resulting sigma-field. Conditional on  $\mathcal{G}_S$ , the domain Markov property separates the component containing  $D_1$  from the component containing  $D_2$ : the only information passed through the revealed annulus is its induced boundary partition. Consequently

$$\phi_p(A_1 \cap A_2) = \mathbb{E}[\phi_p(A_1 \mid \mathcal{G}_S) \phi_p(A_2 \mid \mathcal{G}_S)]. \quad (9.64F)$$

The one-ended ratio-mixing estimate (9.64), applied to  $A_1$  and to every event in  $\mathcal{G}_S$ , implies the almost-sure bound

$$\left| \frac{\phi_p(A_1 \mid \mathcal{G}_S)}{\phi_p(A_1)} - 1 \right| \leq C \Delta_p(r_1, d). \quad (9.64G)$$



Indeed, if the conditional density exceeded the right side on a set of positive  $\mathcal{G}_S$ -measure, taking that set for the outer event in (9.64) would contradict the ratio bound; the lower inequality is identical. Viewed from  $D_2$ , the same annulus is an outer event at distance comparable with  $d$ , and hence

$$\left| \frac{\phi_p(A_2 | \mathcal{G}_S)}{\phi_p(A_2)} - 1 \right| \leq C\Delta_p(r_2, d). \quad (9.64H)$$

Write the two normalized conditional probabilities as  $1 + U$  and  $1 + V$ . Their expectations are one, so  $\mathbb{E}U = \mathbb{E}V = 0$ . Dividing (9.64F) by  $\phi_p(A_1)\phi_p(A_2)$  leaves  $\mathbb{E}(UV)$ , which is bounded by  $C\Delta_p(r_1, d)\Delta_p(r_2, d)$ . This proves (9.64C). The proof is local and uses only the induced boundary partition, so the stated domain/boundary-condition version follows verbatim.

*Transport of a boost.* Start the monotone decision-tree coupling from the outer flower  $\mathcal{F}$  and reveal toward a separator at distance  $d/3$ . The proof of (9.61) gives more than a numerical comparison: conditional on the original pair being a boost, the probability that the two coupled configurations still induce distinct compatible boundary conditions at the separator is comparable with  $\Delta_p(r, d)$ . On this event use the well-separated-petal surgery from the proof of (9.60): two primal and two dual RSW corridors turn the surviving distinction, at a fixed positive conditional cost, into a boosting pair on a four-petal separator flower.

Run the same exploration from the target  $\mathcal{G}$  outward to that separator, but now backwards in scale. The lower half of the quasi-multiplicativity proof (9.61) shows that a boundary distinction on the separator reaches  $\mathcal{G}$  as a genuine boost with conditional probability at least  $c\Delta_p(s, d)$ . The two explorations use disjoint edge sets after conditioning on the separator flower. The domain Markov property therefore multiplies the two survival probabilities, while the finitely many RSW fences contribute only a fixed constant. This gives (9.64D). Notice that the argument records a *boosting* pair at the target, not merely two unequal boundary partitions; this is precisely the role of the petal-separation fences.

*Polynomial amplification relative to four-arm pivotality.* We isolate the scale-by-scale gain. There exist a fixed  $\rho > 16$  and  $\delta_* > 0$  such that the following holds. Suppose an outer four-petal flower  $\mathcal{F}$  lies at scale  $u$  and an inner four-petal flower  $\mathcal{H}$  at scale  $(\rho^2 + 2)u$ . Couple a weaker and a stronger boundary condition, differing by the wiring of the two primal petals of  $\mathcal{H}$ . If  $A_4(\mathcal{F}, \mathcal{H})$  is the event that the four alternating petals are joined across the annulus and  $\tilde{A}_4$  is the same event with the two primal connections allowed to use the stronger configuration, then the coupling can be chosen so that

$$\mathbb{P}(\tilde{A}_4) \geq (1 + \delta_*)\mathbb{P}(A_4). \quad (9.64I)$$

Here is the proof of this local gain. First expose four separated petal corridors in the middle third of the annulus. If more than four alternating petals are present, then, in one of a fixed number of subannuli, an additional primal/dual interface must cross a corridor already carrying four separated arms. Repeating the blocker argument used in (9.56) gives

$$\mathbb{P}(A_4 \text{ and at least six petals}) \leq C\rho^{-c_0}\mathbb{P}(A_4). \quad (9.64J)$$

Choose  $\rho$  so that the right side is at most a small fixed fraction of  $\mathbb{P}(A_4)$ . On the complementary four-petal event, expose a double four-petal flower in a fixed-aspect-ratio box between the two boundaries. Arbitrary-boundary RSW makes this successful with probability bounded below. In the stronger configuration the outer wiring joins the two selected primal petals, whereas they remain distinct in the weaker configuration. Two primal corridors and two dual shielding corridors make that difference pivotal for the continuation of the four arms. The conditional probability of all these corridors is bounded below uniformly by the RSW theorem, so on a fixed fraction of  $A_4$  the stronger configuration belongs to  $\tilde{A}_4 \setminus A_4$ . This proves (9.64I).

Now place such amplification annuli at the scales  $u_j = (\rho^2 + 2)^j r$ ,  $j = 0, \dots, k-1$ , where  $(\rho^2 + 2)^k r \leq R < (\rho^2 + 2)^{k+1} r$ . The decision-tree construction reveals successive annuli before entering the next one, so the domain Markov property permits (9.64I) to be iterated. We obtain a monotone wired/free coupling in which

$$\mathbb{P}(\tilde{A}_4(r, R)) \geq (1 + \delta_*)^k \mathbb{P}(A_4(r, R)). \quad (9.64K)$$

At the innermost flower, occurrence of  $\tilde{A}_4 \setminus A_4$  forces the stronger configuration to realize the  $r$ -scale crossing while the weaker one does not. The flower-to-box RSW comparison therefore gives

$$\Delta_p(r, R) \geq c(1 + \delta_*)^k \pi_4(p; r, R). \quad (9.64L)$$

Since  $k \geq c \log(R/r) - C$ , this is (9.64E) with  $\varepsilon_* = \log(1 + \delta_*)/\log(\rho^2 + 2)$ , after bounded scale ratios are absorbed into  $C$ .

This proof is the internal selected-form reconstruction of Duminil-Copin–Manolescu, *Planar random-cluster model: scaling relations*, Theorem 1.6(iv)–(v), Section 4.5, together with the two-ended coupling technique underlying their Lemma 5.3/Remark 5.5. We have stated only the consequences used later; in particular, no exact value of a four-arm exponent is assumed.  $\square$

**Lemma 9.5** (Logarithmic differential calculus for mixing). *Let  $R \leq L(p)$ . For two edges  $e, f$  at mutual distance comparable with  $R$ ,*

$$\left| \frac{d}{dp} \log \text{Cov}_p(\omega_e, \omega_f) \right| \leq C \left\{ R^2 \Delta_p(R) + \sum_{\ell=R}^{L(p)} \ell \Delta_p(\ell) \Delta_p(R, \ell) \right\}. \quad (9.64A)$$

More generally, for  $1 \leq r \leq R \leq L(p)$ ,

$$\left| \frac{d}{dp} \log \Delta_p(r, R) \right| \leq C \left\{ \sum_{\ell \leq R} \ell \Delta_p(\ell) + \sum_{R \leq \ell \leq L(p)} \ell \Delta_p(\ell) \Delta_p(R, \ell) \right\}. \quad (9.64B)$$

The same shell expression, with  $R^2 \Delta_p(R)$  in place of the first sum, controls logarithmic derivatives of fixed-colour arm events.

*Proof.* Differentiate a covariance by (9.59). After centering, the contribution of a third edge  $g$  is the third cumulant of  $(\omega_e, \omega_f, \omega_g)$ . If  $d(g, e) \asymp \ell < R/2$ , reveal a flower domain of radius  $3\ell/2$  around  $e$ . The local centered observable can change only when the flower feels its exterior, a cost  $O(\Delta_p(\ell))$ ; conditioning additionally on  $\omega_f$  changes the law of that exterior partition by  $O(\Delta_p(\ell, R))$ . The same estimate holds near  $f$ . If  $g$  is farther than  $R/2$  from both endpoints, two outer discrepancies are required, giving  $O(\Delta_p(\ell) \Delta_p(R, \ell))$  after division by the endpoint covariance. There are  $O(\ell)$  edges in a unit shell, and the  $O(R^2)$  edges at distance  $O(R)$  contribute  $R^2 \Delta_p(R)$ . Summing shells and using  $\text{Cov}_p(\omega_e, \omega_f) \asymp \Delta_p(R)^2$  from the two-ended flower coupling proves (9.64A).

For  $\Delta_p(r, R)$  couple the wired and free outer laws and differentiate the probability that their disagreement reaches the inner flower. A third edge at distance  $\ell \leq R$  can affect this event only through one local disagreement; for  $\ell \geq R$  it must also transmit the discrepancy from scale  $\ell$  back to scale  $R$ . The same unit-shell count gives (9.64B). Replacing the disagreement event by a separated arm event changes only the bounded RSW fences, so the arm logarithmic derivative has the identical bound. These arguments are finite-volume; monotone exhaustion and the uniform shell bounds justify the infinite-volume limits.  $\square$

**Lemma 9.6** ( $q = 4$  topological-annulus stability). *Assume  $q = 4$ . If  $p$  and its dual  $p^*$  satisfy  $R \leq L(p) = L(p^*)$ , then for  $1 \leq r \leq R$ ,*

$$\Delta_p(r, R) \asymp \Delta_{p_c}(r, R) \asymp \Delta_{p^*}(r, R). \quad (9.64a)$$

*In particular the power lower bound (9.63) holds uniformly throughout the near-critical window at the endpoint  $q = 4$ .*

*Proof.* A special argument is needed here; a continuity bootstrap is not enough because new scales enter as  $p \rightarrow p_c$ . Call  $\Omega = \Lambda_R \setminus H$  a topological  $R$ -annulus when  $H$  is simply connected and  $\Lambda_{R/8} \subset H \subset \Lambda_{R/4}$ . With free boundary conditions put

$$\mathbf{N}_\Omega(u) := \sum_{y \in \partial_{\text{in}} \Omega} \phi_{\Omega, u}^0(y \leftrightarrow \partial \Lambda_{R/2}). \quad (9.64b)$$

We first show that this quantity is comparable for every such  $\Omega$  and every  $u$  between  $p^*$  and  $p$ .

At  $u = p_c$ , subtract the  $q = 4$  parafermionic contour identity (9.8) for  $\Lambda_R$  and for  $\Omega$ . The outer-boundary contributions cancel except for the same positive quantity  $\Sigma_R$  appearing in the proof of (9.63a). RSW and a half-plane one-arm fence at each  $y \in \partial_{\text{in}} \Omega$  then give

$$\mathbf{N}_\Omega(p_c) \asymp \frac{\Sigma_R}{\pi_1^+(R)}, \quad (9.64c)$$

uniformly in the shape of the inner boundary.

For general  $u$  with  $R \leq L(u)$ , let  $F$  be the event that  $\text{Ann}(R/8, R/4)$  contains an open circuit connected to the inner box and, outside it, a dual-open circuit. On  $F$ , fill the finite holes of the union of clusters meeting  $\Lambda_{R/8}$  and remove the resulting simply connected set; this produces a random topological annulus  $\Omega(\omega)$ . If  $y \in \partial_{\text{in}} \Omega(\omega)$  is connected to  $\partial \Lambda_{R/2}$ , then four alternating arms emanate from a neighbourhood of  $y$  to distance comparable with  $R$ . Conversely, arm separation and two RSW fences realize such a boundary point from a separated four-arm event with fixed positive probability. Summing over the  $\asymp R^2$  possible  $y$  gives

$$\inf_{\Omega} \mathbf{N}_\Omega(u) \leq CR^2 \pi_4(u, R) \leq C' \sup_{\Omega} \mathbf{N}_\Omega(u). \quad (9.64d)$$

Monotonicity of  $\mathbf{N}_\Omega(u)$  in  $u$ , duality  $\pi_4(p^*, R) = \pi_4(p, R)$ , and (9.64c)–(9.64d) yield the chain

$$\sup_{\Omega} \mathbf{N}_\Omega(p^*) \asymp \sup_{\Omega} \mathbf{N}_\Omega(p_c) \asymp \inf_{\Omega} \mathbf{N}_\Omega(p_c) \asymp \inf_{\Omega} \mathbf{N}_\Omega(p). \quad (9.64e)$$

Thus the annular quantity cannot change by more than a constant factor anywhere below  $L(p)$ .

It remains to transfer this to  $\Delta$ . Lemma 9.5 gives

$$\left| \frac{d}{du} \log \Delta_u(r, R) \right| \leq C \left\{ \sum_{\ell \leq R} \ell \Delta_u(\ell) + \sum_{R \leq \ell \leq L(u)} \ell \Delta_u(\ell) \Delta_u(R, \ell) \right\}. \quad (9.64f)$$

The integral of the second sum between  $p_c$  and  $p$  is bounded by the total variation of an  $R$ -scale crossing, hence by a constant. To control the first sum, choose a topological annulus minimizing  $\mathbf{N}_\Omega(p)$ . For a fixed  $y \in \partial_{\text{in}} \Omega$ , slide boxes of radii  $8^k$  from the outer boundary toward  $y$  and stop the first time that the doubled box disconnects  $y$  from the outer boundary. The resulting boxes of

radius  $8^k/2$  are pairwise disjoint. For every edge  $e$  in the  $k$ th box, the boosting-pair estimate from Lemma 9.2 and the flower coupling give

$$\text{Cov}_u(\mathbf{1}_{\{y \leftrightarrow \partial\Lambda_{R/2}\}}, \omega_e) \geq c\Delta_u(8^k)\phi_{\Omega,u}^0(y \leftrightarrow \partial\Lambda_{R/2}).$$

Summing first over  $\asymp 8^{2k}$  edges, then over  $k$  and  $y$ , gives

$$\frac{d}{du} \log \mathbf{N}_\Omega(u) \geq c \sum_{\ell \leq R} \ell \Delta_u(\ell), \quad (9.64g)$$

where bounded-scale comparison turns the geometric-scale sum into the integer sum. Integrating (9.64g) and using (9.64e) bounds the first term in (9.64f). Hence  $|\log(\Delta_p(r, R)/\Delta_{p_c}(r, R))| \leq C$ , which proves (9.64a). Combining it with the critical estimate (9.63a) and the two-arm power bound proves the final assertion.  $\square$

**Lemma 9.7** (Derivative bound and stability window). *For every fixed-shape regular quad  $Q_R$  of scale  $R \leq L(p)$ ,*

$$\frac{d}{dp} \phi_p(\mathcal{C}(Q_R)) \asymp R^2 \Delta_p(R) + \sum_{\ell=R}^{L(p)} \ell \Delta_p(\ell) \Delta_p(R, \ell). \quad (9.65)$$

*Equivalently, after grouping the integer sum into dyadic shells,*

$$\frac{d}{dp} \phi_p(\mathcal{C}(Q_R)) \asymp R^2 \Delta_p(R) + \sum_{2^j R \leq L(p)} (2^j R)^2 \Delta_p(2^j R) \Delta_p(R, 2^j R). \quad (9.65')$$

*Consequently there exists  $\varepsilon_0 > 0$  such that for  $R \leq L(p)$ ,*

$$|\phi_p(\mathcal{C}(Q_R)) - \phi_{p_c}(\mathcal{C}(Q_R))| \leq C(R/L(p))^{\varepsilon_0}, \quad (9.66)$$

$$\pi_1(p, R) \asymp \pi_1(p_c, R), \quad (9.67)$$

$$\Delta_p(R) \asymp \Delta_{p_c}(R). \quad (9.68)$$

*Moreover the exponential correlation length  $\xi(p)$  and  $L(p)$  are comparable.*

*Proof.* Insert (9.60) into (9.59). At distance comparable with  $\ell$  from the quad, the number of edges in a unit-thickness shell is  $O(\ell)$ , while the probability that the induced boundary discrepancy reaches the scale- $R$  quad is  $\asymp \Delta_p(R, \ell)$  by (9.64) and (9.61). Summing the unit shells gives exactly the second term in (9.65); the  $O(R^2)$  edges in the  $O(R)$  neighbourhood of the quad give  $R^2 \Delta_p(R)$ . Grouping the unit shells  $2^j R \leq \ell < 2^{j+1} R$  and using bounded-scale comparison for both mixing rates produces (9.65'), with the square  $(2^j R)^2$  accounting for the  $\asymp 2^j R$  unit shells each containing  $\asymp 2^j R$  edges. This bookkeeping is essential: a dyadic formula with only one power of  $2^j R$  would lose an entire shell factor.

We now prove stability without a continuity bootstrap. This point matters: as  $u \rightarrow p_c$ , the characteristic length  $L(u)$  diverges, so a maximal interval argument over “all scales below  $L(u)$ ” would itself require uniform control on newly entering scales. Instead we compare every  $R$ -scale derivative directly with the derivative at the terminal characteristic scale, by comparing every scale directly with the terminal characteristic scale.

Fix the endpoint  $p$  and write  $L := L(p)$ . If  $u$  lies between  $p_c$  and  $p$ , then monotonicity of the characteristic length gives  $L(u) \geq L$ . Introduce

$$\mathcal{D}_u(a; b) := a^2 \Delta_u(a) + \sum_{\ell=a}^b \ell \Delta_u(\ell) \Delta_u(a, \ell), \quad a \leq b \leq L(u). \quad (9.65a)$$

The lower bound (9.63), now available for every  $u$  and every scale below  $L(u)$  (for  $q < 4$  directly and for  $q = 4$  by Lemma 9.6), together with quasi-multiplicativity, gives a scale gain: there are  $c, \delta > 0$  such that for  $R \leq L$ ,

$$\mathcal{D}_u(R; L) + \sum_{\ell=L}^{L(u)} \ell \Delta_u(\ell) \Delta_u(R, \ell) \leq C(R/L)^\delta \left[ L^2 \Delta_u(L) + \sum_{\ell=L}^{L(u)} \ell \Delta_u(\ell) \Delta_u(L, \ell) \right]. \quad (9.65b)$$

For completeness, here is the dyadic calculation behind (9.65b). Put  $R_j = 2^j R$  and let  $J$  be the first index with  $R_j \geq L/2$ . Quasi-multiplicativity gives, for  $R_j \leq L$ ,

$$\Delta_u(R_j) \asymp \Delta_u(L) / \Delta_u(R_j, L), \quad \Delta_u(R, R_j) \asymp \Delta_u(R, L) / \Delta_u(R_j, L). \quad (9.65c)$$

Since (9.63) says  $\Delta_u(a, b) \geq c(a/b)^{2-\varepsilon}$ , the ratio of the  $j$ th dyadic shell at scale  $R_j$  to the terminal  $L$ -scale contribution is bounded by a geometric sequence after one splits the indices at the midpoint between  $\log R$  and  $\log L$  and applies (9.65c) on the two sides. Summing the two geometric tails gives a factor  $(R/L)^\delta$  for some  $\delta = \delta(q) > 0$ . For  $\ell \geq L$ , quasi-multiplicativity gives directly

$$\Delta_u(R, \ell) \asymp \Delta_u(R, L) \Delta_u(L, \ell),$$

and the same lower bound at the intermediate scale supplies the identical  $(R/L)^\delta$  gain. Restoring the  $O(R_j^2)$  number of edges in each dyadic annulus yields (9.65b). This is the precise reason the dyadic form (9.65') contains the square of the shell radius.

Apply (9.65) at scale  $R$  and at scale  $L$ . Since the crossing derivative is positive, (9.65b) implies

$$\left| \frac{d}{du} \phi_u(\mathcal{C}(Q_R)) \right| \leq C(R/L)^\delta \frac{d}{du} \phi_u(\mathcal{C}(Q_L)), \quad (9.65d)$$

where  $Q_L$  is any fixed-shape regular quad at scale  $L$ . Integrating from  $p_c$  to  $p$  gives

$$|\phi_p(\mathcal{C}(Q_R)) - \phi_{p_c}(\mathcal{C}(Q_R))| \leq C(R/L)^\delta |\phi_p(\mathcal{C}(Q_L)) - \phi_{p_c}(\mathcal{C}(Q_L))| \leq C(R/L)^\delta, \quad (9.65e)$$

which is (9.66).

The same comparison applies to logarithmic derivatives of arm events. In particular, revealing the outermost one-arm path and using the shell estimate from Lemma 9.2 gives

$$\left| \frac{d}{du} \log \pi_1(u, R) \right| \leq C \left[ R^2 \Delta_u(R) + \sum_{\ell=R}^{L(u)} \ell \Delta_u(\ell) \Delta_u(R, \ell) \right]. \quad (9.65f)$$

Comparison with the terminal  $L$ -scale derivative and integration therefore show

$$\exp[-C(R/L)^\delta] \leq \frac{\pi_1(p, R)}{\pi_1(p_c, R)} \leq \exp[C(R/L)^\delta], \quad (9.65g)$$

which proves (9.67).

For the mixing rate, take edges  $e, f$  at distance  $R$ . Differentiating their covariance and grouping the third-cumulant terms by the distance of the third edge gives

$$\left| \frac{d}{du} \log \text{Cov}_u(\omega_e, \omega_f) \right| \leq C \left[ R^2 \Delta_u(R) + \sum_{\ell=R}^{L(u)} \ell \Delta_u(\ell) \Delta_u(R, \ell) \right]. \quad (9.65h)$$

Indeed, near either endpoint the flower coupling supplies the local discrepancy, while away from both endpoints two outer discrepancies are necessary; the same quasi-multiplicative shell sum as in the crossing derivative results. The right-hand side is bounded, up to a constant, by the derivative of an  $R$ -scale crossing. Integrating and using (9.65e) at  $R$  shows

$$\text{Cov}_p(\omega_e, \omega_f) \asymp \text{Cov}_{p_c}(\omega_e, \omega_f). \quad (9.65i)$$

The two-ended disagreement coupling gives  $\text{Cov}_u(\omega_e, \omega_f) \asymp \Delta_u(R)^2$  uniformly below  $L(u)$ , hence (9.65i) proves (9.68).

We finally record the generalized Kesten relation at the characteristic scale. By the definition of  $L = L(p)$ , a fixed regular crossing at scale  $L$  changes by an amount bounded above and below by positive constants between  $p_c$  and  $p$  (up to replacing  $L$  by a fixed multiple, which bounded-scale RSW absorbs). Integrating (9.65) at scale  $L$  and using (9.68) throughout the interval gives

$$1 \asymp |p - p_c| L^2 \Delta_{p_c}(L). \quad (9.69)$$

The terms with  $\ell > L$  are comparable to the displayed terminal contribution by quasi-multiplicativity and the power lower bound, so no hidden logarithmic factor is lost. This identity will be used below in the thermodynamic relations.

It remains to compare  $L$  with the exponential correlation length. On scales  $R \leq L$ , (9.66) and RSW give uniformly positive open and dual crossings. Once  $R \geq cL$ , tile a connection from 0 to distance  $n$  by  $O(n/L)$  disjoint rectangular blocks. In the subcritical case a fixed proportion of the blocks contain dual barriers; in the supercritical case apply the same argument to the dual finite cluster. The domain Markov property and finite energy therefore give

$$c \pi_1(L)^2 e^{-C|x|/L} \leq \phi_p(0 \leftrightarrow x, 0 \not\leftrightarrow \infty) \leq C \pi_1(L)^2 e^{-c|x|/L} \quad (9.70)$$

after the usual endpoint arm factors are inserted. Taking logarithms and then  $|x| \rightarrow \infty$  proves  $\xi(p) \asymp L(p)$ .  $\square$

**Lemma 9.8** (Critical cluster and magnetic identities). *Put*

$$\varphi(n) := \min\{r \geq 1 : r^2 \pi_1(r) \geq n\}.$$

At  $p = p_c$ ,

$$\phi_{p_c}(0 \leftrightarrow x) \asymp \pi_1(|x|)^2, \quad (9.71)$$

$$\phi_{p_c}(|\mathbf{C}(0)| \geq n) \asymp \pi_1(\varphi(n)), \quad (9.72)$$

$$\phi_{p_c, h}^1(0 \leftrightarrow \mathfrak{g}) \asymp \pi_1(\varphi(1/h)). \quad (9.73)$$

*Proof.* For (9.71), two disjoint endpoint one-arm events are necessary for  $0 \leftrightarrow x$ , giving the upper bound by ratio mixing. For the lower bound, force the two endpoint arms to reach radius  $|x|/3$  and connect them through a bounded number of RSW rectangles. Quasi-multiplicativity absorbs the bounded change of scale.

For (9.72), condition on the event that the cluster radius lies in  $[r, 4r]$ . Equation (9.71), summed over points of  $\Lambda_{4r}$ , gives

$$\mathbb{E}(|\mathbf{C}| \mid r \leq \text{rad } \mathbf{C} < 4r) \asymp r^2 \pi_1(r),$$

while the corresponding double sum and arm separation give a second moment at most  $Cr^4 \pi_1(r)^2$ . Paley–Zygmund therefore gives the lower cluster-volume tail at  $n \asymp r^2 \pi_1(r)$ . For the upper bound

split according to whether the radius exceeds  $r = \varphi(n)$  and use Markov's inequality on  $|\mathbf{C} \cap \Lambda_r|$ ; summing (9.71) over  $\Lambda_r$  gives the same  $\pi_1(r)$  order.

For (9.73), attach the ghost with edge-weight  $1 - e^{-h}$ . Define  $h_c(R)$  as the first field for which either a one-arm probability up to some  $r \leq R$  grows by a fixed factor or a dual-circuit probability in  $\text{Ann}(r, 2r)$  drops by that factor. Differentiation with respect to  $h$  and the spatial Markov property give, as long as  $h \leq h_c(R)$ ,

$$\frac{d}{dh} \log \tilde{\pi}_1(h, r) + \left| \frac{d}{dh} \log \beta(h, r) \right| \leq Cr^2 \pi_1(r).$$

Integration implies  $h_c(R)R^2\pi_1(R) \geq c$ . Conversely, if  $hR^2\pi_1(R)$  is a sufficiently large constant, the cluster-volume estimate (9.72) and the conditional probability  $1 - e^{-h|\mathbf{C}|}$  of hitting the ghost force a fixed positive probability of a ghost connection before scale  $R$ . Hence the magnetic characteristic scale satisfies

$$hR_h^2\pi_1(R_h) \asymp 1.$$

At that scale, the event  $0 \leftrightarrow \mathbf{g}$  is equivalent, up to fixed constants, to the one-arm event to  $R_h$  followed by a ghost hit. Thus its probability is  $\asymp \pi_1(R_h)$ , and  $R_h \asymp \varphi(1/h)$  proves (9.73).  $\square$

**Lemma 9.9** (Near-critical thermodynamic identities). *For  $p \neq p_c$ , with  $L = L(p) \asymp \xi(p)$ ,*

$$\theta(p) \asymp \pi_1(L) \quad (p > p_c), \quad (9.74)$$

$$\chi(p) \asymp L^2 \pi_1(L)^2, \quad (9.75)$$

$$\Delta_{p_c}(L) \asymp L^{-2} |p - p_c|^{-1}, \quad (9.76)$$

$$f''(p) \asymp \sum_{1 \leq \ell \leq L} \ell \Delta_{p_c}(\ell)^2. \quad (9.77)$$

*Proof.* For (9.74), the upper bound is  $\theta(p) \leq \pi_1(p, L) \asymp \pi_1(L)$ . For the lower bound, concatenate an arm from 0 to  $2L$ , a primal circuit in  $\text{Ann}(L, 2L)$  and the supercritical infinite connection from  $\partial\Lambda_L$ ; RSW, FKG and the definition of  $L$  give a fixed probability for the last two factors.

For (9.75), (9.70) yields, with  $R = |x| \wedge L$ ,

$$\phi_p(0 \leftrightarrow x, 0 \not\leftrightarrow \infty) \asymp \pi_1(R)^2 e^{-c|x|/L}$$

with possibly different constants in the two exponential bounds. Summing over  $x$  and using quasi-multiplicativity of the one-arm probability gives  $\chi(p) \asymp L^2 \pi_1(L)^2$ .

Equation (9.76) is exactly (9.69) together with the stability (9.68). For (9.77), differentiate the infinite-volume edge density. Translation invariance gives, up to a harmless analytic factor depending on the chosen normalization of  $f$ ,

$$f''(p) \asymp \sum_f \text{Cov}_p(\omega_{e_0}, \omega_f).$$

The two-ended form of the disagreement coupling gives, for  $\ell = d(e_0, f)$ ,

$$\text{Cov}_p(\omega_{e_0}, \omega_f) \asymp \Delta_p(\ell \wedge L)^2 e^{-c\ell/L}.$$

Use (9.68), sum over the  $\asymp \ell$  edges at distance  $\ell$ , and absorb the exponentially decaying tail beyond  $L$  to obtain (9.77).  $\square$

**Theorem 9.10** (Planar FK scaling relations). *For the square-lattice random-cluster model with  $q \in [1, 4]$ , the exponent-free relations (9.71)–(9.77) hold (with the  $q = 1$  formulas interpreted through the product-measure pivotal calculus). Consequently, whenever the exponents in*

$$\pi_1(R) = R^{-\xi_1+o(1)}, \quad \Delta_{p_c}(R) = R^{-\iota+o(1)}, \quad \xi(p) = |p - p_c|^{-\nu+o(1)}$$

*and the corresponding thermodynamic definitions exist, one has*

$$\eta = 2\xi_1, \quad \zeta = \frac{\xi_1}{2 - \xi_1}, \quad \delta = \frac{2 - \xi_1}{\xi_1}, \quad (9.78)$$

$$\beta = \nu\xi_1, \quad \gamma = (2 - 2\xi_1)\nu, \quad \nu(2 - \iota) = 1. \quad (9.79)$$

*If  $f''(p)$  diverges algebraically at criticality (equivalently, in the regime in which the specific-heat exponent  $\alpha$  is nonnegative in the definition used here), then also*

$$\alpha = 2 - 2\nu. \quad (9.80)$$

*No existence of an exponent is asserted. When the specific heat remains bounded, (9.80) is not claimed: the source-level third-derivative lower bound needed for the negative- $\alpha$  analogue is genuinely unavailable.*

*Proof.* The exponent-free input has already been proved in the two preceding lemmas. The critical identities (9.71)–(9.73) identify the two-point function, the cluster-mass tail, and the critical magnetization in terms of the one-arm probability. The near-critical identities (9.74)–(9.77) identify magnetization, susceptibility, correlation length, and the second derivative of the free energy in terms of  $\pi_1$ , the wired/free mixing rate  $\Delta$ , and the characteristic length  $L(p)$ . Quasi-multiplicativity and the mixing interpretation proved earlier ensure that these identities may be used uniformly throughout the scaling window.

Assume first that the exponents displayed in the theorem exist. From (9.71),

$$\phi_{p_c}(0 \leftrightarrow x) = \pi_1(|x|)^2 |x|^{o(1)} = |x|^{-2\xi_1+o(1)},$$

so  $\eta = 2\xi_1$ . Next, (9.72) says that a critical cluster whose radius is  $r$  has mass  $r^{2-\xi_1+o(1)}$ . Thus, if  $r = \varphi(n)$  is the radius corresponding to cluster mass  $n$ , then  $n = r^{2-\xi_1+o(1)}$ , and the cluster-tail exponent is

$$\zeta = \frac{\xi_1}{2 - \xi_1}.$$

Equation (9.73) gives the same ratio as  $1/\delta$ , hence  $\delta = (2 - \xi_1)/\xi_1$ . This proves (9.78).

Now use  $L(p) \asymp |p - p_c|^{-\nu+o(1)}$  in (9.74)–(9.76). Since  $\theta(p) \asymp \pi_1(L(p))$ , one obtains  $\beta = \nu\xi_1$ . Since  $\chi(p) \asymp L(p)^2 \pi_1(L(p))^2$ , one obtains  $\gamma = (2 - 2\xi_1)\nu$ . Finally,  $\Delta_{p_c}(L(p)) \asymp L(p)^{-\iota+o(1)}$  together with (9.76) gives

$$|p - p_c| = L(p)^{-(2-\iota)+o(1)},$$

and therefore  $\nu(2 - \iota) = 1$ . This proves (9.79).

It remains to justify the specific-heat relation in precisely the regime stated. If  $\Delta_{p_c}(\ell) = \ell^{-\iota+o(1)}$  and the sum in (9.77) diverges, then

$$f''(p) \asymp \sum_{\ell \lesssim L(p)} \ell \Delta_{p_c}(\ell)^2 = L(p)^{2-2\iota+o(1)}.$$

(The borderline case  $\iota = 1$  is logarithmic and corresponds to exponent  $\alpha = 0$ .) Hence  $\alpha = \nu(2 - 2\iota)$ . Eliminating  $\iota$  using  $\nu(2 - \iota) = 1$  yields

$$\alpha = 2 - 2\nu.$$

If the sum in (9.77) converges, this argument proves only boundedness of  $f''$  and does not manufacture the unavailable negative- $\alpha$  relation. This is exactly the restricted scope asserted in the theorem.  $\square$



## 9.2 Noncritical planar Potts Gibbs states

We next turn to the low-temperature nearest-neighbour Potts model. The difficult point is not the final DLR passage but the finite-volume statement uniform in an *arbitrary* boundary colouring. The proof below isolates the needed subcritical FK interface geometry and proves the quantitative estimate from which the infinite-volume classification follows.

Fix an integer  $q \geq 2$  and  $\beta > \beta_c(q) = \log(1 + \sqrt{q})$ . Let  $\mathbb{P}_{\Lambda_n, \beta}^\sigma$  be the Potts measure in  $\Lambda_n$  with boundary colouring  $\sigma$ , and let  $\mathbb{P}_\beta^{(i)}$  be the infinite-volume state obtained from color- $i$  boundary conditions.

The direct FK parameter associated with  $\beta$  is  $p_\beta = 1 - e^{-\beta} > p_c(q)$ . We write  $p_{\text{sub}}$  for its planar-dual parameter, so that  $p_{\text{sub}} < p_c(q)$ . All long interfaces separating two ordered colours are therefore represented by connections in a strictly subcritical FK model. The proof needs two consequences of this fact: exponential screening of distant boundary data and the diffusive Ornstein–Zernike description of a long conditioned connection. We prove both in the precise forms used below.

**Lemma 9.11** (Exponential screening of a subcritical FK boundary). *Fix  $p < p_c(q)$ . There are  $c, C > 0$  with the following property. Let  $\Omega$  be a finite or infinite planar domain containing  $\Lambda_{2r}$  and let  $g$  be a bounded observable supported in  $\Lambda_r$ . If  $\phi_\Omega^\xi$  and  $\phi_\Omega^{\xi'}$  are two FK laws whose boundary conditions agree on every boundary component meeting  $\Lambda_{2r}$ , then*

$$|\phi_\Omega^\xi(g) - \phi_\Omega^{\xi'}(g)| \leq C \|g\|_\infty |\Lambda_{2r}| e^{-cr}. \quad (9.80a)$$

*The corresponding Edwards–Sokal spin laws satisfy the same estimate. In particular, if  $\Omega \supset \Lambda_{2r}$ , the free law and the infinite-volume free law, and likewise a fixed pure-colour law and its infinite-volume limit, differ on  $g$  by  $O(\|g\|_\infty e^{-c'r})$ .*

*Proof.* Couple the two FK measures by the monotone edge-by-edge heat-bath construction, revealing edges from  $\partial\Omega$  towards  $\Lambda_r$ . Until a cluster carrying a boundary identification reaches the unrevealed region, the two conditional transition probabilities coincide. Consequently the two configurations can differ on an edge of  $\Lambda_r$  only if some vertex of  $\partial\Lambda_{2r}$  is connected, in one of the two coupled configurations, to distance at least  $r$ . The sharpness theorem, Theorem 3.8, gives uniformly in the induced boundary condition

$$\phi_p(x \leftrightarrow \partial\Lambda_r(x)) \leq C e^{-cr}.$$

A union bound over  $O(r^2)$  possible entrance vertices therefore bounds the probability that the two coupled restrictions to  $\Lambda_r$  differ by  $Cr^2 e^{-cr} \leq C' e^{-c'r}$ . On the complementary event the two values of  $g$  are identical, which proves (9.80a). Edwards–Sokal colours can be coupled identically on every common FK cluster; a discrepancy of the spin restrictions again requires an FK discrepancy to reach  $\Lambda_r$ . Exhausting an infinite domain gives the last assertion.  $\square$

**Lemma 9.12** (Subcritical ratio strong mixing). *Fix  $p < p_c(q)$ . There are  $c, C > 0$  such that, for any two finite edge sets  $E_A, E_B$  and events  $A \in \mathcal{F}(E_A)$ ,  $B \in \mathcal{F}(E_B)$  of positive probability under the infinite-volume free FK measure,*

$$\left| \frac{\phi_p^f(A \cap B)}{\phi_p^f(A) \phi_p^f(B)} - 1 \right| \leq C \sum_{e \in E_A} \sum_{f \in E_B} e^{-cd(e,f)}. \quad (9.80a')$$

*The same estimate holds in a finite domain provided both supports are farther from the domain boundary than their mutual separation, with constants independent of the induced boundary condition on the remote boundary.*

*Proof.* It is enough to treat two connected supports of diameters bounded by their separation; the general estimate follows by decomposing each support into unit edge blocks and summing the resulting errors. Write  $D = d(E_A, E_B)$  and choose two disjoint annuli of width  $D/5$  separating  $E_A$  and  $E_B$ . Since  $p < p_c$ , planar duality and sharpness imply that, uniformly in every boundary condition induced outside an annulus, the dual configuration contains an open circuit in that annulus except with probability  $Ce^{-cD}$ . This remains true after conditioning on any event measurable strictly inside the inner boundary: by the domain Markov property that conditioning changes only the boundary partition on the annulus, and the dual-circuit estimate is uniform in that partition.

Let  $F_A, F_B$  be the outermost such dual circuits. On  $F_A \cap F_B$  all direct edges of the two circuits are closed. The FK cluster-count factor then splits across the two closed cutsets: after conditioning on the circuits, the law in the component containing  $E_A$ , the law in the component containing  $E_B$ , and the law in the intervening region have partition functions whose cluster factors multiply. Consequently the conditional density of the  $E_A$ -configuration is independent of the  $E_B$ -configuration. The four probabilities  $\phi(F_A^c)$ ,  $\phi(F_B^c)$ ,  $\phi(F_A^c | A)$  and  $\phi(F_B^c | B)$  are all  $O(e^{-cD})$  by the uniform annular estimate just proved. Decomposing  $\phi(A \cap B)$  according to  $F_A \cap F_B$  and its complement therefore yields

$$\phi(A \cap B) = \phi(A)\phi(B) [1 + O(e^{-cD})].$$

For supports with many edge blocks, repeat the construction one block at a time; a connecting influence from  $e$  to  $f$  pays  $e^{-cd(e,f)}$ , and the union of all possible pairs gives the double sum in (9.80a'). This is the ratio, rather than merely additive, form needed in the Ornstein–Zernike decomposition.  $\square$

**Lemma 9.13** (Local central limit theorem for a summable-memory renewal chain). *Let  $\mathcal{I}$  be a countable set of symbols. To  $\gamma \in \mathcal{I}$  attach a displacement  $V(\gamma) \in \mathbb{Z}^2$  and a size  $|\gamma| \geq 1$ . Suppose that admissible strings  $\underline{\gamma} = (\gamma_1, \dots, \gamma_m)$  carry weights*

$$W(\underline{\gamma}) = a(\gamma_1)b(\gamma_m) \exp \left\{ \sum_{j=1}^m \Psi(\gamma_j | \gamma_{j-1}, \gamma_{j-2}, \dots) \right\}, \quad (9.80b)$$

where:

1.  $|a| + |b| + e^\Psi \leq Ce^{C|\gamma|}$  and the one-symbol tails after an exponential tilt satisfy  $\sum_{|\gamma| \geq t} e^{\Psi_h(\gamma|\cdot)} \leq Ce^{-ct}$ ;

2. the memory is exponentially summable:

$$|\Psi_h(\gamma_0 | \gamma_{-1}, \dots) - \Psi_h(\gamma_0 | \gamma'_{-1}, \dots)| \leq Ce^{-ck} \quad (9.80c)$$

whenever the two pasts agree for the last  $k$  symbols;

3. the displacement semigroup is aperiodic and its transverse covariance is nonzero;

4. there is a finite set  $\mathcal{I}_0 \subset \mathcal{I}$  and an integer  $m_0$  such that any two admissible past/future boundary states can be joined by a word of at most  $m_0$  symbols from  $\mathcal{I}_0$ , with all corresponding transition weights bounded below by a fixed positive constant after normalization.

Here  $\Psi_h := \Psi + h \cdot V$  and  $h$  is chosen so that the pressure is zero. Then the normalized transfer operator defines an exponentially mixing stationary chain of irreducible symbols. For its additive

displacement  $S_m = \sum_{j \leq m} V(\gamma_j)$  there are a drift  $v$  and a positive-definite covariance matrix  $\Sigma$  on the non-deterministic directions such that, uniformly for  $|x - mv| = O(\sqrt{m})$ ,

$$\mathbb{P}(S_m = x) = \frac{1 + o(1)}{2\pi m \sqrt{\det \Sigma}} \exp \left[ -\frac{1}{2m} (x - mv)^T \Sigma^{-1} (x - mv) \right]. \quad (9.80d)$$

The same perturbation argument yields the corresponding bridge invariance principle and Gaussian upper bounds, uniformly after conditioning on the total displacement.

*Proof.* Let  $\mathcal{X}$  be the space of one-sided pasts and endow it with  $d_\theta(x, y) = \theta^{k(x, y)}$ , where  $k(x, y)$  is the length of their common suffix and  $e^{-c} < \theta < 1$ . On the weighted Lipschitz space

$$\|f\|_* := \sup_x e^{-a|\gamma_0(x)|} |f(x)| + \sup_{x \neq y} e^{-a(|\gamma_0(x)| + |\gamma_0(y)|)/2} \frac{|f(x) - f(y)|}{d_\theta(x, y)}$$

with  $a > 0$  small, define

$$(\mathcal{L}_z f)(x) := \sum_{\gamma: \gamma x \text{ admissible}} e^{\Psi_h(\gamma|x) + z \cdot V(\gamma)} f(\gamma x). \quad (9.80e)$$

The exponential tail makes the sum absolutely convergent for  $|\Re z| < a_0$ . Condition (9.80c) gives the Doeblin–Fortet estimate

$$[\mathcal{L}_z^n f]_{\text{Lip}} \leq C\theta^n [f]_{\text{Lip}} + C\|f\|_\infty, \quad \|\mathcal{L}_z^n f\|_\infty \leq C\|f\|_*, \quad (9.80f)$$

uniformly for  $z$  near 0. Compactness of the embedding obtained by truncating the exponentially small symbol tail makes the essential spectral radius strictly smaller than the spectral radius. The finite connector in assumption (4) is a Doeblin minorization on a fixed cylinder. Together with the Doeblin–Fortet estimate it rules out peripheral components and gives a unique positive eigenfunction  $r$  and eigenmeasure  $\ell$  at the leading eigenvalue. The pressure-zero normalization makes that eigenvalue 1. The Doob transform

$$Pf = r^{-1} \mathcal{L}_0(rf)$$

is therefore a Markov operator with a spectral gap on the same weighted Lipschitz space; in particular its stationary chain is exponentially mixing.

Analytic perturbation of the isolated eigenvalue gives, for  $t$  near 0,

$$\lambda(it) = \exp\{iv \cdot t - \tfrac{1}{2}t^T \Sigma t + O(|t|^3)\}. \quad (9.80g)$$

The covariance is the asymptotic covariance of the additive functional. If it vanished in a transverse direction, that displacement would be a coboundary plus a constant; aperiodicity and the assumed existence of two admissible irreducible pieces with different transverse displacement exclude this. Away from the dual lattice point  $t = 0$ , aperiodicity and the spectral gap give  $\text{spr}(\mathcal{L}_{it}) < 1$ . Fourier inversion now splits the torus into  $|t| \leq m^{-2/5}$ , where (9.80g) gives the Gaussian integral, and its complement, where the spectral radius is uniformly smaller than one. This proves (9.80d). Inserting bounded observables at finitely many intermediate times in the same Fourier calculation gives the finite-dimensional bridge distributions; the exponential moment bound gives tightness by the usual maximal inequality. The Gaussian bridge upper bounds follow from the local estimate on the two pieces of a path split at the observation time.  $\square$

**Lemma 9.14** (Subcritical FK Ornstein–Zernike and interface package). *Let  $p_{\text{sub}} < p_c(q)$ . There is an analytic strictly convex norm  $\tau$  on  $\mathbb{R}^2$  and constants  $c, C > 0$  such that the following hold uniformly when endpoints are separated by distance  $N$  and their direction stays in a compact subset of  $S^1$ .*

1. *The inverse correlation length exists and*

$$\phi_{p_{\text{sub}}}(x \leftrightarrow y) = \Theta(N^{-1/2})e^{-\tau(y-x)}. \quad (9.81)$$

2. *Conditioned on  $x \leftrightarrow y$ , the connecting cluster has a linear density of cone-regeneration points. A bridge crosses a window of diameter  $a = o(N)$  at macroscopic distance from both endpoints with probability at most*

$$Ca/\sqrt{N} + Ce^{-cN}. \quad (9.82)$$

3. *For three endpoints  $u_1, u_2, u_3$  in nondegenerate position, if  $\Phi(z) = \sum_i \tau(u_i - z)$  and  $z_*$  is its unique minimizer, then*

$$c|z - z_*|^2/N \leq \Phi(z) - \Phi(z_*) \leq C|z - z_*|^2/N \quad (9.83)$$

*for  $|z - z_*| \leq cN$ . A conditioned three-arm cluster has its unique macroscopic branching point in a box of side  $a$  around  $z$  with probability at most*

$$C \frac{a^2}{N} \exp[-c|z - z_*|^2/N] + Ce^{-cN}. \quad (9.84)$$

*Proof.* We give the renewal construction because replacing it by an i.i.d. increment heuristic would be incorrect for a dependent random-cluster measure.

*Step 1: support function and surcharge.* Subadditivity of  $-\log \phi(0 \leftrightarrow x)$ , together with exponential decay from Theorem 3.8, gives

$$\tau(x) := -\lim_{k \rightarrow \infty} k^{-1} \log \phi(0 \leftrightarrow [kx]),$$

a norm equivalent to Euclidean distance. Fix a direction  $t$  and a supporting vector  $h$  with  $h \cdot t = \tau(t)$ . The surcharge  $s_h(y) := \tau(y) - h \cdot y$  is nonnegative. At this stage we do *not* assume strict convexity of  $\tau$ ; that will be a consequence of the renewal theorem. Fix a small  $\delta > 0$  and define the forward surcharge cone

$$Y = Y(h, \delta) := \{y : h \cdot y > 0, s_h(y) \leq \delta \tau(y)\}.$$

By compactness of the unit sphere, on the complement of  $Y \cup (-Y)$  in the forward half-space one has the deterministic inequality  $s_h(y) \geq \delta \tau(y) \geq c|y|$ . This is the only cone estimate required in the coarse graining. If the supporting face of the unit ball is initially flat,  $Y$  simply contains that whole face; the transfer-operator covariance obtained below will rule out such a flat face and thereby prove strict convexity rather than presuppose it.

*Step 2: skeletons and cone points.* Coarse-grain a connecting cluster into  $K$ -boxes by recording the first new box reached outside the  $3K$ -neighbourhood of the existing skeleton. A skeleton  $\mathfrak{s} = (v_0, \dots, v_m)$  has probability at most

$$C^m \exp\left\{-\sum_{j=1}^m \tau(v_j - v_{j-1})\right\}. \quad (9.84a)$$

Indeed, reveal disjoint separating annuli around successive boxes and apply the domain Markov property plus exponential screening; the remaining bounded local cost is absorbed into  $C^m$ . After multiplication by  $e^{h \cdot x}$  the exponent contains  $-\sum_j s_h(v_j - v_{j-1})$ . The number of  $m$ -step skeletons is at most  $C_0^m$ . Choosing  $K$  so large that one surcharge excursion costs more than the skeleton entropy shows

$$\phi(0 \leftrightarrow x, \text{ fewer than } c|x|/K \text{ Y-cone points}) \leq e^{-c'|x|} \phi(0 \leftrightarrow x). \quad (9.84b)$$

Thus a typical long connection has a positive density of cone points.

*Step 3: irreducible decomposition and summable memory.* Cut the cluster at successive cone points. Except for left and right boundary pieces it becomes a string  $\gamma_1, \dots, \gamma_m$  of irreducible clusters, each contained in the diamond  $(v_{j-1} + Y) \cap (v_j - Y)$ . Diameters have exponential tails by (9.84a)–(9.84b). The FK weight of the next piece is not independent of the entire past, but if two pasts agree on their last  $k$  irreducible pieces, their unmatched parts are separated from the new tip by distance  $ck$  except on an event of probability  $Ce^{-ck}$ . Lemma 9.12, applied after revealing the common diamond boundary, therefore gives

$$|\Psi(\gamma_0 \mid \gamma_{-1}, \gamma_{-2}, \dots) - \Psi(\gamma_0 \mid \gamma'_{-1}, \gamma'_{-2}, \dots)| \leq Ce^{-ck}. \quad (9.84c)$$

The same annular separation gives the needed multiplicative control, not merely an additive total-variation estimate. Indeed, after the common last  $k$  pieces are fixed, changing the remote past changes the FK cluster-count factor in the new diamond only if a cluster from that diamond reaches the unmatched remote boundary. On the complement of this event the two conditional partition functions are identical; on the event itself their Radon–Nikodym ratio lies between  $q^{-M_0}$  and  $q^{M_0}$ , where  $M_0$  is the bounded number of diamond boundary pieces. Exponential connectivity therefore gives

$$e^{-Ce^{-ck}} \leq \frac{W(\gamma_0 \mid \gamma_{-1}, \dots)}{W(\gamma_0 \mid \gamma'_{-1}, \dots)} \leq e^{Ce^{-ck}},$$

which is precisely (9.84c) after taking logarithms. The same argument gives exponential tails after the tilt by  $h$ . Finally, finite energy lets us insert a fixed bounded straight connector between any two admissible boundary states; a finite family of such connectors gives assumption (4) of Lemma 9.13. Two other bounded connectors with different transverse displacement give nonzero variance, and local square detours remove every proper lattice period. Hence all hypotheses of Lemma 9.13 hold.

*Step 4: two-point and bridge estimates.* Sum over the exponentially tight boundary pieces and over the number  $m$  of interior irreducible pieces. For each fixed  $m \asymp N$ , (9.80d) is a two-dimensional lattice local CLT and contributes  $m^{-1}$ . The longitudinal constraint leaves a Gaussian window of  $O(\sqrt{N})$  relevant values of  $m$ ; summing over that renewal-time window contributes  $\sqrt{N}$ . Thus the net prefactor is  $N^{-1/2}$ , as required in two dimensions. Undoing the tilt contributes  $e^{-h \cdot (y-x)}$ , and minimization over supporting  $h$  gives  $e^{-\tau(y-x)}$ . This proves (9.81). Splitting the conditioned string at a macroscopic time and applying the local CLT to the two halves shows that the transverse coordinate has a Gaussian bridge density bounded by  $C/\sqrt{N}$ . Summing over the  $O(a)$  transverse sites in a window proves (9.82). Analyticity of the leading transfer-operator eigenvalue in the tilt parameter shows that  $\tau$  is analytic and its curvature is the inverse transverse variance; the latter is strictly positive by Step 3, proving strict convexity.

*Step 5: three-arm junction.* Expose the first cone point on each of three branches outside a fixed junction box. Ratio errors between the three remote branches are exponentially small by Lemma 9.11, so their logarithmic weight is  $-\sum_i \tau(u_i - z) + O(1)$ . Strict convexity gives a unique nondegenerate Steiner point  $z_*$  unless one of the triangle inequalities is saturated. In the

nondegenerate case Taylor's theorem, with  $|u_i - z_*| \asymp N$ , gives (9.83). Applying the two-dimensional local CLT to the two independent transverse degrees of freedom after the junction and summing over the  $O(a^2)$  sites of the target box gives (9.84). In a degenerate geometry the minimizing junction lies at an endpoint and the required bound reduces to the two-branch estimate (9.82). This proves the whole interface package used below.  $\square$

**Lemma 9.15** (Flower-domain reduction for arbitrary Potts boundary data). *For every sufficiently small  $\varepsilon > 0$  there are  $M < \infty$ ,  $c > 0$  and  $\delta > 0$  such that, for every boundary colouring  $\sigma$ , the Edwards–Sokal FK representation of  $\mathbb{P}_{\Lambda_n, \beta}^\sigma$  has, with probability at least  $1 - e^{-cn}$ , a radius  $m \in [\delta n/3, \delta n]$  for which the component  $\mathcal{D}_{m,n}$  containing  $\Lambda_m$  after the boundary-connected clusters outside  $\Lambda_m$  are removed is a flower domain with at most  $M$  marked petals  $\mathbb{G}_{m,n}$ . Conditional on the exterior, the interior FK law remembers only the partition of these petals induced by exterior clusters.*

*Proof.* Work in the subcritical direct FK representation conditioned on  $\text{Cond}_n[\sigma]$ , the event that no direct cluster identifies two incompatible boundary colour classes. We first record the cost of this conditioning. By prescribing the  $O(n)$  edges in a one-edge boundary collar so that every colour-change point is separated from its neighbours, finite energy gives

$$\phi_{\Lambda_n, p_{\text{sub}}}^f(\text{Cond}_n[\sigma]) \geq e^{-c_0 n}, \quad (9.84d)$$

uniformly in  $\sigma$ .

For  $r \geq 1$  let  $E_r$  be the event that the macroscopic annulus  $\Lambda_{\delta n} \setminus \Lambda_{\delta n/3}$  contains  $r$  pairwise disjoint direct open crossings. Reveal such crossings one after another. After  $j$  crossings have been exposed, the domain Markov property leaves an FK measure in each remaining component with some induced boundary condition. Subcritical sharpness, applied with the wired boundary condition which dominates all of them, gives a uniform bound  $Ce^{-c\delta n}$  for one more macroscopic crossing. Induction therefore yields

$$\phi_{\Lambda_n, p_{\text{sub}}}^f(E_r) \leq (Ce^{-c\delta n})^r. \quad (9.84e)$$

Now suppose that every radius  $m \in [\delta n/3, \delta n]$  has more than  $M$  marked entrances. Follow the boundary-connected clusters through the annulus and erase loops whenever two traces merge. Planarity leaves at least  $M/2$  disjoint traces crossing a subannulus of comparable width; hence  $E_{\lfloor M/2 \rfloor}$  occurs. Combining (9.84d)–(9.84e),

$$\phi_{\Lambda_n, p_{\text{sub}}}^f(\forall m \in [\delta n/3, \delta n] : |\mathbb{G}_{m,n}| > M \mid \text{Cond}_n[\sigma]) \leq C^M e^{-(c\delta M/2 - c_0)n}. \quad (9.84f)$$

Choose  $M$  so large that the exponent is negative by a fixed amount. This proves that, with probability  $1 - e^{-c'n}$ , at least one such radius has at most  $M$  marked entrances. Choose the outermost one.

Finally reveal every boundary-connected cluster outside that radius. Let  $\mathcal{D}_{m,n}$  be the component of the complement containing  $\Lambda_m$  and let  $\mathbb{G}_{m,n}$  be its marked boundary sites. The spatial Markov property says that, conditionally on the revealed exterior, the unrevealed law is the free FK law in  $\mathcal{D}_{m,n}$  except that two marked petals are wired precisely when the exterior connects them. Thus the entire original colouring survives only through a partition of a set of size at most  $M$ . This is the asserted flower reduction.  $\square$

**Lemma 9.16** (Uniform geometry of  $\tau$ -Steiner forests). *Fix  $M$ . Let  $E \subset \partial\Lambda_m$ ,  $|E| \leq M$ , and prescribe a partition  $\mathcal{P}$  of  $E$ . Among compact planar forests whose connected components realize the blocks of  $\mathcal{P}$ , minimize the one-dimensional Hausdorff length in the norm  $\tau$ . Then:*

1. the set of minimizing forests is nonempty, compact in Hausdorff distance, and has cardinality bounded by a constant depending only on  $M$ ;
2. every interior node has degree three, and incident edges have a uniform positive angular separation;
3. inside  $\Lambda_{m/2}$  distinct interior nodes, and distinct tree components, are separated by at least  $c(M)m$  unless the corresponding nodes have collapsed and the reduced forest has fewer nodes;
4. for every  $\delta > 0$  there is  $\kappa(\delta, M) > 0$  such that any admissible forest  $S$  satisfying

$$\text{dist}_{\tau, H}(S, \mathfrak{F}_{\min}) \geq \delta m$$

has

$$\tau(S) \geq \tau_{\mathcal{P}} + \kappa(\delta, M)m. \quad (9.84g)$$

*Proof.* Rescale to  $m = 1$ . The function  $S \mapsto \tau(S)$  is lower semicontinuous for Hausdorff convergence on compact connected sets and its bounded-length level sets are compact; hence minimizers exist and the family of minimizers is compact. There are only finitely many abstract forest topologies with at most  $M$  labelled leaves and trivalent interior vertices. For a fixed reduced topology, strict convexity and analyticity of  $\tau$  from Lemma 9.14 make the length functional strictly convex in each free junction coordinate once the adjacent coordinates are fixed. Successive elimination of degree-two vertices therefore leaves only finitely many minimizers, uniformly over the finitely many topologies.

An interior vertex of degree one can be deleted and one of degree two can be straightened by the triangle inequality. If a vertex had degree at least four, pair two adjacent incident directions and split the vertex into two nearby trivalent vertices. Strict convexity decreases the total  $\tau$ -length. Thus all interior degrees equal three. Smooth strict convexity of the  $\tau$ -unit sphere and compactness of the finite family of possible tangent triples give a uniform angle gap from degeneracy.

For separation, suppose after rescaling that two distinct interior nodes approach one another along a sequence of minimizers. Compactness gives a limiting minimizer in which they coalesce. The limiting vertex has degree at least four unless an incident edge has vanished; the first case contradicts trivalence and the second is exactly the stated reduction to a topology with fewer nodes. The same compactness argument treats two distinct components. Finally, if (9.84g) failed, a sequence of forests with length tending to the minimum but remaining  $\delta$  away from every minimizer would have a Hausdorff-convergent subsequence; lower semicontinuity would make its limit a minimizer, a contradiction. Scaling back to  $m$  proves all four assertions.  $\square$

**Lemma 9.17** (Exponential concentration near a Steiner forest). *Uniformly over flower domains with at most  $M$  petals and over all admissible petal partitions, for every  $\delta > 0$  there is  $c(\delta, M) > 0$  such that the boundary-connected subcritical cluster  $\mathbb{C}_{\mathbb{G}}$  satisfies*

$$\phi_{\mathcal{D}}^f(\text{dist}_{\tau, H}(\mathbb{C}_{\mathbb{G}}, \mathfrak{F}_{\min}) > \delta n \mid \text{the prescribed petal partition}) \leq e^{-cn}. \quad (9.84h)$$

*Proof.* Choose a large coarse-graining scale  $K$ . Starting from the petals, add a new skeleton vertex whenever the explored cluster exits the union of the  $2K$ - $\tau$ -balls around the existing vertices, and connect it to the first previous ball it meets. The resulting graph is a forest  $F_K$ , contains every petal, realizes the prescribed partition, and

$$\text{dist}_{\tau, H}(\mathbb{C}_{\mathbb{G}}, F_K) \leq CK. \quad (9.84i)$$

For a fixed skeleton with edges  $[x_j, y_j]$ , reveal disjoint separating annuli around the new vertices and apply the subcritical two-point estimate (9.81) and ratio mixing. The number of local choices at each new vertex is polynomial in  $K$ , hence

$$\phi_{\mathcal{D}}^f(F_K) \leq \exp\{-\tau(F_K)(1 - o_K(1))\}. \quad (9.84j)$$

The number of skeletons of  $r$  vertices is at most  $(CK^2)^r$ , which is absorbed by the  $o_K(1)$  term after  $K$  is fixed large.

For the denominator, choose a minimizing Steiner forest  $F$  for the prescribed partition, move each junction to a nearest lattice site, and force connections along  $K$ -spaced tubes following its edges. FKG and the lower half of the Ornstein–Zernike estimate give

$$\phi_{\mathcal{D}}^f(\text{prescribed partition}) \geq \exp\{-\tau_{\mathcal{P}}(1 + o_n(1))\}. \quad (9.84k)$$

If the event in (9.84h) occurs and  $n \gg K/\delta$ , then (9.84i) and Lemma 9.16(4) imply  $\tau(F_K) \geq \tau_{\mathcal{P}} + \kappa n$ . Divide the sum of (9.84j) over such skeletons by (9.84k). First choose  $K$  so that the entropy and  $o_K(1)$  losses are below  $\kappa/4$ , then take  $n$  large. The result is at most  $e^{-\kappa n/2}$ , proving (9.84h).  $\square$

**Lemma 9.18** (Macroscopic interface reduction). *Condition on a flower domain with  $|\mathbb{G}| \leq M$ . With probability  $1 - e^{-cn}$ , after passing to some  $k \geq \delta n$ , the boundary-connected subcritical dual cluster has one of three macroscopic forms:*

1. *no branch reaches  $\partial\Lambda_k$ ;*
2. *two groups of marked petals are joined by one effective line interface;*
3. *three groups are joined by one effective Steiner tripod.*

*The endpoint groups have  $\tau$ -diameter  $o(n)$  and the cluster is  $o(n)$ -close in  $\tau$ -Hausdorff distance to a  $\tau$ -minimal forest realizing the prescribed partition.*

*Proof.* By Lemma 9.17, with probability  $1 - e^{-cn}$  the cluster is  $o(n)$ -close to one of the finitely many minimizing Steiner forests from Lemma 9.16. Choose  $\delta > 0$  smaller than one hundredth of the uniform node/component separation constant in that lemma and inspect the minimizing forest inside  $\Lambda_{250\delta n}$ .

If it does not meet  $\Lambda_{2\delta n}$ , the  $o(n)$  Hausdorff estimate implies that no boundary branch reaches a slightly smaller inner square. If it meets the inner square but has no interior node in  $\Lambda_{40\delta n}$ , its intersection with the latter square is a single edge; all petals participating in that component therefore split into two groups of  $\tau$ -diameter  $o(n)$  joined by one line interface. If it has an interior node there, trivalence and the separation of nodes imply that this is the only node in the larger comparison square; the component is a tripod joining three  $o(n)$ -diameter groups, and the uniform angle gap gives a nondegenerate  $\tau$ -Steiner junction.

In the line and tripod cases there is an interval of radii of length  $cn$  on which the same macroscopic geometry persists. Apply the flower-cardinality argument of Lemma 9.15 inside that interval to choose a radius  $k \geq \delta n$  with at most  $M$  marked entrances. The exponentially small exceptional probabilities from the two lemmas are absorbed into  $e^{-c'n}$ . These are exactly the three alternatives in the statement.  $\square$

**Lemma 9.19** (Central-window avoidance). *For  $\varepsilon > 0$  sufficiently small, uniformly over every flower domain and allowed exterior partition with at most  $M$  marked vertices,*

$$\phi_{\mathcal{D}}^f(\mathbb{C}_{\mathbb{G}} \cap \Lambda_{n^\varepsilon} \neq \emptyset \mid \text{the prescribed exterior connections}) \leq C_\varepsilon n^{-1/2+14\varepsilon}. \quad (9.85)$$

*Here  $\mathbb{C}_{\mathbb{G}}$  is the union of subcritical dual clusters forced by the exterior partition.*



*Proof.* On the exponentially likely event of Lemma 9.18, treat the three geometries separately. With no macroscopic branch, exponential decay gives  $e^{-cn}$ . In the line case, the two endpoint groups and first regeneration points have only  $n^{O(\varepsilon)}$  admissible localizations after the multiscale flower construction. Conditional on them, (9.82) bounds the chance that the renewal bridge enters  $\Lambda_{n^\varepsilon}$  by  $Cn^{-1/2+O(\varepsilon)}$ .

In the tripod case, localize the three endpoint groups, the first regeneration points and the  $\tau$ -Steiner point. By (9.83)–(9.84), a junction displaced by  $u$  from its minimizer pays  $e^{-c|u|^2/n}$  and any specified  $n^{O(\varepsilon)}$  junction window has mass  $n^{-1+O(\varepsilon)}$ . Conditioned on a junction outside the central window, each of the three outgoing bridges is controlled by (9.82). Summing the Gaussian junction weights and the three bridge-hit probabilities gives  $n^{-1/2+O(\varepsilon)}$ . A degenerate tripod is a pair of line-interface estimates. Choosing the seven localization scales successively so that each contributes at most  $n^{2\varepsilon}$  leaves the uniform exponent  $-1/2 + 14\varepsilon$ . The exponentially exceptional skeleton set is negligible. This proves (9.85).  $\square$

**Lemma 9.20** (Uniform conditioned boundary-cluster estimate). *Let  $\text{Cond}_n[\sigma]$  denote the subcritical FK conditioning induced by an arbitrary Potts boundary colouring  $\sigma$ , and let  $\mathbf{C}_\partial$  be the set of subcritical sites connected to  $\partial\Lambda_n$ . For every sufficiently small  $\varepsilon > 0$ ,*

$$\phi_{\Lambda_n, p_{\text{sub}}}^f(\mathbf{C}_\partial \cap \Lambda_{n^\varepsilon} \neq \emptyset \mid \text{Cond}_n[\sigma]) \leq C_\varepsilon n^{-1/2+14\varepsilon}, \quad (9.85')$$

*uniformly in  $n$  and in the boundary colouring  $\sigma$ .*

*Proof.* Expose the configuration from  $\partial\Lambda_n$  inward. By Lemma 9.15, except on an exponentially small event one reaches a radius  $m \asymp n$  at which the unexplored component is a flower domain with at most  $M$  marked petals; the conditioning  $\text{Cond}_n[\sigma]$  has been reduced to a partition of those petals. Lemma 9.18 then says that, except with probability  $e^{-cn}$ , the boundary-connected part entering the flower is either absent, a single macroscopic line interface, or a single macroscopic tripod. In all three cases Lemma 9.19 gives the conditional probability of hitting  $\Lambda_{n^\varepsilon}$  as at most  $C_\varepsilon n^{-1/2+14\varepsilon}$ . The constants in all three lemmas are uniform over the bounded petal number and all petal partitions, hence are uniform in  $\sigma$ . Adding the exponentially small exceptional events proves (9.85').  $\square$

**Lemma 9.21** (Finite-volume pure-phase approximation). *For every sufficiently small  $\varepsilon > 0$  there is  $C_\varepsilon < \infty$  such that for every Potts boundary condition  $\sigma$  on  $\partial\Lambda_n$  there are  $\alpha_1^n, \dots, \alpha_q^n \geq 0$ ,  $\sum_i \alpha_i^n = 1$ , for which*

$$\left| \mathbb{P}_{\Lambda_n, \beta}^\sigma(g) - \sum_{i=1}^q \alpha_i^n \mathbb{P}_\beta^{(i)}(g) \right| \leq C_\varepsilon \|g\|_\infty n^{-1/2+14\varepsilon} \quad (9.86)$$

*for every  $g$  supported in  $\Lambda_{n^\varepsilon}$ .*

*Proof.* We use the event decomposition that keeps the phase coefficients independent of  $g$ . In the dual/subcritical Edwards–Sokal representation let  $A_{2n^\varepsilon, n} := \Lambda_n \setminus \Lambda_{2n^\varepsilon}$ . Let  $E$  be the event that a boundary-connected subcritical interface crosses this annulus. Lemma 9.20 gives

$$\mathbb{P}(E) \leq C_\varepsilon n^{-1/2+14\varepsilon}. \quad (9.86a)$$

On  $E^c$  there are exactly two topological possibilities. Let  $F^f$  be the event that an open circuit in the appropriate FK representation separates  $\Lambda_{2n^\varepsilon}$  from the outer boundary. For each colour  $i$ , let  $F^{(i)}$  be the event that there is neither the bad crossing  $E$  nor such a free circuit and that the dual inner box is connected through the exterior to the boundary colour class  $S_i$ . Planar duality shows that

$$E, \quad F^f, \quad F^{(1)}, \dots, F^{(q)} \quad (9.86b)$$

form a partition.

On  $F^f$ , reveal the outermost separating circuit. The domain inside it has free boundary and contains  $\Lambda_{2n^\varepsilon}$ . By Lemma 9.11, for  $g$  supported in  $\Lambda_{n^\varepsilon}$ ,

$$\mathbb{E}[g \mid F^f] = \mathbb{P}_\beta^f(g) + O(\|g\|_\infty e^{-cn^\varepsilon}).$$

The free infinite-volume low-temperature Potts state is obtained by assigning the infinite FK cluster a uniform colour; hence by colour symmetry

$$\mathbb{P}_\beta^f = q^{-1} \sum_{i=1}^q \mathbb{P}_\beta^{(i)}. \quad (9.86c)$$

On  $F^{(i)}$ , reveal the boundary-connected cluster carrying colour  $i$ . The component containing the observation box has pure colour- $i$  boundary at distance at least  $n^\varepsilon$ , so the same screening lemma gives

$$\mathbb{E}[g \mid F^{(i)}] = \mathbb{P}_\beta^{(i)}(g) + O(\|g\|_\infty e^{-cn^\varepsilon}).$$

Therefore, with

$$\alpha_i^n := q^{-1} \mathbb{P}(F^f) + \mathbb{P}(F^{(i)}), \quad (9.86d)$$

the coefficients are nonnegative and their sum is  $1 - \mathbb{P}(E)$ . Add  $\mathbb{P}(E)/q$  to each coefficient (or, equivalently, absorb the bad event into an arbitrary probability vector); the resulting coefficients sum exactly to one and change the approximation by at most  $2\|g\|_\infty \mathbb{P}(E)$ . Combining this with (9.86a) and the exponentially small screening errors proves (9.86).  $\square$

**Theorem 9.22** (Classification of noncritical planar Potts Gibbs states). *For the nearest-neighbour ferromagnetic  $q$ -state Potts model on  $\mathbb{Z}^2$ , with integer  $q \geq 2$  and*

$$\beta > \beta_c(q) = \log(1 + \sqrt{q}),$$

*every infinite-volume Gibbs state is a convex combination of the  $q$  monochromatic pure phases:*

$$\mathcal{G}_{q,\beta} = \left\{ \sum_{i=1}^q \alpha_i \mathbb{P}_\beta^{(i)} : \alpha_i \geq 0, \sum_i \alpha_i = 1 \right\}. \quad (9.87)$$

*Equivalently, the only extremal Gibbs states are  $\mathbb{P}_\beta^{(1)}, \dots, \mathbb{P}_\beta^{(q)}$ . The theorem does not include the critical first-order point  $\beta = \beta_c(q)$  for  $q > 4$ , where the disordered state is an additional phase.*

*Proof.* The source argument has already been proved here in the preceding interface lemmas. Lemmas 9.16 and 9.17 give the Gaussian-scale control of the macroscopic line/tripod interfaces, while Lemma 9.18 reduces arbitrary spin boundary data to finitely many such geometries. The central-avoidance and screening estimates then give Lemma 9.21: for every boundary condition  $\sigma$  one has the quantitative approximation (9.86), with coefficients  $\alpha_i^n(\sigma)$  depending on  $\sigma$  but not on the local observable.

Let now  $\mu$  be an arbitrary infinite-volume Gibbs state and let  $g$  be a bounded local observable. For  $n$  large enough that  $\text{supp } g \subset \Lambda_{n^\varepsilon}$ , the DLR equation gives

$$\mu(g) = \int \mathbb{P}_{\Lambda_n, \beta}^\sigma(g) d\mu(\sigma).$$

Insert (9.86) and define

$$\bar{\alpha}_i^n := \int \alpha_i^n(\sigma) d\mu(\sigma).$$

The vector  $\bar{\alpha}^n$  belongs to the compact probability simplex. Along a subsequence  $n_k \rightarrow \infty$  it therefore converges to  $\alpha = (\alpha_1, \dots, \alpha_q)$ . The error in (9.86) tends to zero, so

$$\mu(g) = \sum_{i=1}^q \alpha_i \mathbb{P}_\beta^{(i)}(g)$$

for every local  $g$ . Local cylinder functions determine probability measures, hence (9.87) follows. Conversely, every such convex combination is a Gibbs state because the DLR equations are affine.

Finally, if  $\mu$  is extremal and two coefficients  $\alpha_i, \alpha_j$  are positive, the displayed representation writes  $\mu$  as a nontrivial convex combination of two distinct Gibbs states, contradicting extremality. Thus one coefficient equals one and  $\mu = \mathbb{P}_\beta^{(i)}$  for some  $i$ . The proof is genuinely noncritical: the finite-volume screening input above uses the strictly subcritical dual parameter  $p_{\text{sub}} < p_c(q)$ , which fails at the first-order critical point when  $q > 4$ .  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 9.23** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 9.24** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 9.25** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 9.26** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 9.27** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume X: Percolation Theory on Lattices and Transitive Graphs



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# Scope and conventions

This volume develops Bernoulli percolation from finite-product identities through sharpness, uniqueness, planar duality, decision-tree inequalities, and continuum analogues. Elementary mechanisms are proved directly; the sharpness and dependent-percolation arguments are developed in separate quantitative steps so that every estimate and hypothesis remains visible.

# Chapter 1

## Bernoulli percolation

Percolation begins with independent bits but quickly develops global geometry. We first isolate the finite-product identities that later survive scaling.

**Definition 1.1** (Bernoulli bond percolation, cluster, and increasing event). Let  $G = (V, E)$  be locally finite. Under *Bernoulli bond percolation* with parameter  $p \in [0, 1]$ , a configuration  $\omega \in \{0, 1\}^E$  has independent coordinates with  $\mathbb{P}_p(\omega_e = 1) = p$ ; an edge with  $\omega_e = 1$  is *open*. The *open cluster*  $C(v)$  is the set of vertices joined to  $v$  by an open path. An event  $A$  is *increasing* if  $\omega \leq \omega'$  coordinatewise and  $\omega \in A$  imply  $\omega' \in A$ . For a finite edge set,  $e$  is *pivotal for*  $A$  in an outside configuration if changing only  $\omega_e$  changes  $\mathbf{1}_A$ .

**Example 1.2** (A running picture). On a square box, think of each edge as open independently with probability  $p$ . The local rule is trivial; the questions below ask when those bits create a path whose diameter is comparable with the box.

**Theorem 1.3** (Critical threshold and its elementary bounds).

*Let  $G = (V, E)$  be an infinite connected locally finite graph and let  $o \in V$ . For Bernoulli bond percolation set*

$$\theta(p) = \mathbb{P}_p(o \leftrightarrow \infty), \quad p_c(G) = \inf\{p \in [0, 1] : \theta(p) > 0\}.$$

*Then  $\theta$  is nondecreasing and*

$$\theta(p) = 0 \quad (p < p_c), \quad \theta(p) > 0 \quad (p > p_c).$$

*If  $G$  has maximum degree  $\Delta < \infty$ , then  $p_c \geq (\Delta - 1)^{-1}$  whenever  $\Delta > 1$ . For nearest-neighbour bond percolation on  $\mathbb{Z}^d$ ,  $d \geq 2$ , one has  $0 < p_c < 1$ .*

*Proof.* Attach i.i.d. variables  $U_e \sim \text{Unif}[0, 1]$  to the edges and declare  $e$  open at parameter  $p$  when  $U_e \leq p$ . This realizes all Bernoulli measures on one probability space and makes every increasing event monotone in  $p$ . In particular  $\theta$  is nondecreasing. The definition of the infimum then gives  $\theta(p) = 0$  for  $p < p_c$ . If  $p > p_c$ , choose  $q \in (p_c, p)$ . By definition of the infimum there is  $q' < p$  with  $\theta(q') > 0$ ; monotonicity gives  $\theta(p) \geq \theta(q') > 0$ .

For the lower bound, if  $o$  is connected to graph distance  $n$ , there is an open self-avoiding path of length  $n$  from  $o$ . There are at most  $\Delta(\Delta - 1)^{n-1}$  such paths, hence

$$\mathbb{P}_p(o \leftrightarrow \partial B_n) \leq \Delta p [p(\Delta - 1)]^{n-1}.$$

When  $p(\Delta - 1) < 1$  the right side tends to zero, so  $\theta(p) = 0$ .

For  $\mathbb{Z}^d$ , positivity follows from the preceding estimate. To see  $p_c < 1$ , it is enough to work in the coordinate copy of  $\mathbb{Z}^2$ . If the origin is not connected to infinity by open primal edges, then its open cluster is surrounded by a closed dual circuit. A dual circuit of length  $m$  containing a fixed dual edge can be encoded by a self-avoiding walk and there are at most  $C3^m$  such circuits. Thus for  $q = 1 - p$  sufficiently small,

$$\mathbb{P}_p(0 \nleftrightarrow \infty) \leq \sum_{m \geq 4} C m 3^m q^m < 1.$$

Consequently  $\theta(p) > 0$  for all  $p$  sufficiently close to one. Hence  $0 < p_c(\mathbb{Z}^d) < 1$  for  $d \geq 2$ .  $\square$

**Theorem 1.4** (Russo's formula).

Let  $E$  be finite and let  $A \subset \{0, 1\}^E$  be increasing. Under Bernoulli product measure of parameter  $p \in (0, 1)$ ,

$$\frac{d}{dp} \mathbb{P}_p(A) = \sum_{e \in E} \mathbb{P}_p(e \text{ is pivotal for } A).$$

Here  $e$  is pivotal when changing only the state of  $e$  changes the indicator of  $A$ .

*Proof.* Give temporarily each coordinate its own parameter  $p_e$ . Conditioning on all coordinates except  $e$ , the conditional probability of  $A$  is either 0, 1, or  $p_e$ : because  $A$  is increasing, the last case occurs exactly when the outside configuration makes  $e$  pivotal. Hence

$$\frac{\partial}{\partial p_e} \mathbb{P}_{(p_f)}(A) = \mathbb{P}_{(p_f)}(e \text{ pivotal}).$$

On the diagonal  $p_e = p$  for every  $e$ , the chain rule yields the displayed sum. Equivalently, one may differentiate the finite polynomial  $\sum_{\omega \in A} p^{|\omega|} (1-p)^{|E|-|\omega|}$  and pair configurations differing only at  $e$ .  $\square$

**Disjoint occurrence.** For a configuration  $\omega$  in a finite product space and  $S \subseteq [n]$ , write

$$[\omega]_S = \{\eta : \eta_i = \omega_i \text{ for every } i \in S\}.$$

An event  $A$  is *certified on  $S$  at  $\omega$*  when  $[\omega]_S \subseteq A$ . For events  $A, B$  define their disjoint occurrence by

$$A \square B = \{\omega : \text{there are disjoint } S, T \subseteq [n] \text{ with } [\omega]_S \subseteq A, [\omega]_T \subseteq B\}.$$

If  $S$  and  $T$  witness disjoint occurrence, then replacing  $T$  by  $S^c$  only fixes additional coordinates, so equivalently one may require  $[\omega]_S \subseteq A$  and  $[\omega]_{S^c} \subseteq B$ .

**Theorem 1.5** (Reimer–BKR inequality).

Let  $\Omega = \Omega_1 \times \cdots \times \Omega_n$  be a finite product space with product probability measure  $\mu = \mu_1 \otimes \cdots \otimes \mu_n$ . For arbitrary events  $A, B \subseteq \Omega$ ,

$$\mu(A \square B) \leq \mu(A)\mu(B).$$

*Proof.* We give the finite combinatorial proof because later pivotal and multi-arm arguments use the inequality in situations where neither event is monotone.

*Step 1: the Fishburn–Shepp formulation.* For  $X \subseteq \Omega$  and a choice  $S(x) \subseteq [n]$  for each  $x \in X$ , put

$$U = [X]_S := \bigcup_{x \in X} [x]_{S(x)}, \quad V = [X]_{S^c} := \bigcup_{x \in X} [x]_{S(x)^c}.$$

It is enough to prove

$$\mu(X) \leq \mu(U)\mu(V). \quad (1.1)$$

Indeed, for  $X = A \square B$ , choose  $S(x)$  so that  $[x]_{S(x)} \subseteq A$  and  $[x]_{S(x)^c} \subseteq B$ ; then  $U \subseteq A$  and  $V \subseteq B$ , and (1.1) gives the theorem.

We first prove (1.1) for the uniform measure on the binary cube  $\Omega_n = \{0, 1\}^n$ . If  $x \in \Omega_n$ , let  $\bar{x}$  denote bitwise complement. For a subset  $T$  write  $\bar{T} = \{\bar{x} : x \in T\}$ .

*Step 2: Reimer's combinatorial lemma.* We claim that, for every  $X$  and every map  $S : X \rightarrow 2^{[n]}$ ,

$$|U \cap \bar{V}| \geq |X|. \quad (1.2)$$

To prove it, note that (1.2) is equivalent to

$$|\bar{U}^c| + |U \cap \bar{V}^c| + |X| \leq 2^n. \quad (1.3)$$

Here  $\bar{U}^c = \Omega_n \setminus \bar{U}$ ; its cardinality equals  $|U^c|$ .

The proof of (1.3) is a linear-algebra encoding. For a bit  $a \in \{0, 1\}$  and  $S \subseteq [n]$ , define

$$\varphi_i(a, S) = \begin{cases} (a, -1), & i \notin S, \\ (1, a), & i \in S, \end{cases} \quad \Phi(x, S) = \bigotimes_{i=1}^n \varphi_i(x_i, S) \in \mathbb{R}^{2^n}.$$

Tensor products satisfy  $\langle \otimes_i v_i, \otimes_i w_i \rangle = \prod_i \langle v_i, w_i \rangle$ . Consider the three families

$$\mathcal{A} = \{\Phi(y, \emptyset) : y \in \bar{U}^c\}, \quad \mathcal{B} = \{\Phi(z, [n]) : z \in U \cap \bar{V}^c\}, \quad \mathcal{C} = \{\Phi(x, S(x)) : x \in X\}.$$

Their spans are pairwise orthogonal. For example, if  $y \in \bar{U}^c$  and  $z \in U$ , then  $\bar{y} \notin U$  while  $z \in U$ , so  $z \neq \bar{y}$  and hence  $y_i = z_i$  for some  $i$ . At that coordinate

$$\langle (y_i, -1), (1, z_i) \rangle = y_i - z_i = 0.$$

If  $y \in \bar{U}^c$  and  $x \in X$ , the fact that  $\bar{y} \notin [x]_{S(x)} \subseteq U$  yields an  $i \in S(x)$  with  $y_i = x_i$ , giving the same zero factor. Similarly, if  $z \in U \cap \bar{V}^c$  and  $x \in X$ , then  $\bar{z} \notin [x]_{S(x)^c} \subseteq V$ , so some  $i \notin S(x)$  satisfies  $z_i = x_i$ ; again the relevant two-dimensional inner product is zero.

It remains to know that each of the three displayed families is linearly independent. We prove the stronger statement

$$\{\Phi(x, R(x)) : x \in \{0, 1\}^n\} \text{ is linearly independent for every map } R : \{0, 1\}^n \rightarrow 2^{[n]}. \quad (1.4)$$

Reduce the integer coordinate matrix modulo 2. Order rows by  $x = (0, y)$  followed by  $x = (1, y)$ ,  $y \in \{0, 1\}^{n-1}$ . Let  $A_0, A_1$  be the matrices arising from the restrictions of  $R$  to the two halves after deleting the first coordinate, and let  $D$  be the diagonal 0–1 matrix whose  $y$ th entry records whether  $1 \in R(0, y)$ . Over  $\mathbb{F}_2$  the full matrix has block form

$$M_n = \begin{pmatrix} DA_0 & (I - D)A_0 \\ A_1 & A_1 \end{pmatrix}.$$

For  $n = 1$  its determinant is 1 in  $\mathbb{F}_2$ . For the induction step add the first block of columns to the second and then exchange the two block columns. This transforms  $M_n$ , without changing whether the determinant vanishes, into a block triangular matrix with diagonal blocks  $A_0$  and  $A_1$ . Hence

$$\det M_n = (\det A_0)(\det A_1) = 1 \quad \text{in } \mathbb{F}_2.$$

Thus the determinant of the original integer matrix is odd and in particular nonzero over  $\mathbb{R}$ , proving (1.4).

The families  $\mathcal{A}$  and  $\mathcal{B}$  are subfamilies of the cases  $R \equiv \emptyset$  and  $R \equiv [n]$ , while  $\mathcal{C}$  is a subfamily of an arbitrary extension of  $S$  from  $X$  to the whole cube. They are therefore internally independent. Since their spans are pairwise orthogonal, their union is linearly independent in a vector space of dimension  $2^n$ . Counting its members gives (1.3) and hence the main lemma (1.2).

*Step 3: from the main lemma to the uniform-cube inequality.* For  $u, v \in \Omega_n$ , let  $\langle u, v \rangle$  be the cylinder obtained by fixing exactly those coordinates on which  $u$  and  $v$  agree. If  $C = [c]_\Lambda$  is a cylinder and  $u \in C$ , write  $\bar{u}^{(C)}$  for the configuration that agrees with  $u$  on  $\Lambda$  and flips all coordinates outside  $\Lambda$ . Put  $U_C = U \cap C$ ,  $V_C = V \cap C$ , and  $X_C = X \cap C$ . Every pair  $(u, v) \in U \times V$  belongs to exactly one class according to the cylinder  $C = \langle u, v \rangle$ , and within that class  $v = \bar{u}^{(C)}$ . Therefore

$$|U||V| = \sum_C |U_C \cap \bar{V}_C^{(C)}|, \quad (1.5)$$

where the sum is over all cylinders. Restricting to the free coordinates of  $C$  and applying (1.2) gives  $|U_C \cap \bar{V}_C^{(C)}| \geq |X_C|$ . Hence

$$|U||V| \geq \sum_C |X \cap C|.$$

For a fixed  $x$ , exactly  $2^n$  cylinders contain  $x$ —one for each choice of the set of fixed coordinates. Double counting pairs  $(x, C)$  thus yields

$$\sum_C |X \cap C| = 2^n |X|.$$

Consequently  $|X|/2^n \leq (|U|/2^n)(|V|/2^n)$ , which is (1.1) for the uniform binary cube.

*Step 4: arbitrary finite product measures.* It remains to remove the uniform-cube restriction. Approximate each atom mass of every  $\mu_i$  by a dyadic rational with common denominator  $2^K$ . For the resulting product measure, partition a block  $\{0, 1\}^K$  into cells of the prescribed cardinalities and map each cell to the corresponding atom of  $\Omega_i$ . Taking the product over coordinates gives a blockwise map  $f : \{0, 1\}^{Kn} \rightarrow \Omega$  that pushes uniform measure forward to the dyadic product measure.

If  $x \in f^{-1}(A \square B)$  and  $S, T \subseteq [n]$  are disjoint witness-coordinate sets for  $f(x)$ , take in the binary cube the unions of the corresponding  $K$ -bit blocks. Fixing those blocks forces the same atom coordinates under  $f$ . Hence

$$f^{-1}(A \square B) \subseteq f^{-1}(A) \square f^{-1}(B).$$

The already proved uniform-cube inequality therefore gives the desired product inequality for all dyadic atom masses. Finally let the dyadic approximations converge to the original finite product measure. Since all events are finite unions of atoms, their probabilities converge term by term, and the inequality passes to the limit.  $\square$

**Lemma 1.6** (Complement form of Reimer's correlation inequality). *On the uniform binary cube  $\Omega_n = \{0, 1\}^n$ , write  $\bar{\omega} = 1 - \omega$  and  $\bar{B} = \{\bar{\omega} : \omega \in B\}$ . For arbitrary events  $A, B \subseteq \Omega_n$ ,*

$$|A \square B| \leq |A \cap \bar{B}|. \quad (1.6)$$

*Equivalently, at Bernoulli parameter  $1/2$ ,*

$$\mathbb{P}_{1/2}(A \square B) \leq \mathbb{P}_{1/2}(A \cap \bar{B}).$$

*Proof.* Take  $X = A \square B$  and, for every  $x \in X$ , choose a witness set  $S(x)$  with  $[x]_{S(x)} \subseteq A$  and  $[x]_{S(x)^c} \subseteq B$ . In the notation of the Reimer proof above,

$$U = [X]_S \subseteq A, \quad V = [X]_{S^c} \subseteq B.$$

The already-proved combinatorial lemma (1.2) gives  $|X| \leq |U \cap \bar{V}|$ . Since  $U \subseteq A$  and  $V \subseteq B$ , bitwise complementation reverses the latter inclusion,  $\bar{V} \subseteq \bar{B}$ , and hence

$$|A \square B| = |X| \leq |U \cap \bar{V}| \leq |A \cap \bar{B}|.$$

Dividing by  $2^n$  yields the probability form. Notice that this is stronger than the product bound when the flipped event has additional geometric meaning; that is exactly the form used later for monochromatic arm surgery.  $\square$

**Theorem 1.7** (BK inequality for increasing events).

*In Bernoulli bond percolation on a finite edge set, if  $A$  and  $B$  are increasing events, then*

$$\mathbb{P}_p(A \square B) \leq \mathbb{P}_p(A)\mathbb{P}_p(B).$$

*The same inequality holds for cylinder events in a countable product space, and hence for increasing events obtained as monotone limits of cylinder events.*

*Proof.* The finite statement is the increasing-event specialization of the Reimer–BKR theorem just proved. For a countable product, let  $A_m, B_m$  be cylinder approximations depending on the first  $m$  coordinates and increasing to  $A, B$  in any application in which the events admit such a monotone exhaustion. A fixed finite disjoint witness to  $A \square B$  is already visible at some finite stage, so  $A \square B = \bigcup_m (A_m \square B_m)$  after choosing the usual compatible exhaustion. Continuity from below and the finite inequality give

$$\mathbb{P}_p(A \square B) = \lim_m \mathbb{P}_p(A_m \square B_m) \leq \lim_m \mathbb{P}_p(A_m)\mathbb{P}_p(B_m) = \mathbb{P}_p(A)\mathbb{P}_p(B).$$

For the finite-box and finite-arm events used later in the book no limiting step is needed.  $\square$

## Chapter 2

# Subcritical and supercritical regimes

Sharpness means there is no intermediate phase between exponential/summable connectivity and a positive infinite-cluster density.

**Example 2.1** (A running picture). On a square box, think of each edge as open independently with probability  $p$ . The local rule is trivial; the questions below ask when those bits create a path whose diameter is comparable with the box.

**Theorem 2.2** (Sharpness on a transitive graph).

Let  $G = (V, E)$  be an infinite locally finite transitive graph and let nonnegative couplings  $J_{x,y} = J_{y,x}$  be invariant under a transitive automorphism group. Under  $\mathbb{P}_\beta$ , the edge  $\{x, y\}$  is open with probability  $1 - e^{-\beta J_{x,y}}$ , independently of the other edges. Put

$$\beta_c = \inf\{\beta \geq 0 : \mathbb{P}_\beta(0 \leftrightarrow \infty) > 0\}.$$

Then

$$\mathbb{P}_\beta(0 \leftrightarrow \infty) \geq \frac{\beta - \beta_c}{\beta} \quad (\beta > \beta_c),$$

and, for  $\beta < \beta_c$ ,

$$\chi(\beta) := \sum_{x \in V} \mathbb{P}_\beta(0 \leftrightarrow x) < \infty.$$

If the couplings have finite range, then for every  $\beta < \beta_c$  there is  $c(\beta) > 0$  such that

$$\mathbb{P}_\beta(0 \leftrightarrow B_n^c) \leq e^{-c(\beta)n}.$$

*Proof.* The proof is driven by a finite-set boundary functional. For a finite  $S \ni 0$ , define

$$\varphi_\beta(S) = \sum_{x \in S} \sum_{y \notin S} (1 - e^{-\beta J_{x,y}}) \mathbb{P}_\beta(0 \overset{S}{\longleftrightarrow} x),$$

and set

$$\tilde{\beta}_c = \sup\{\beta : \text{some finite } S \ni 0 \text{ has } \varphi_\beta(S) < 1\}.$$

We prove the three assertions with  $\tilde{\beta}_c$  in place of  $\beta_c$ ; the conclusions then force equality of the two critical parameters.

*The differential inequality.* Let  $\Lambda \Subset V$  contain 0 and set  $f_\Lambda(\beta) = \mathbb{P}_\beta(0 \leftrightarrow \Lambda^c)$ . Russo differentiation in the  $\beta$ -parameter gives

$$f'_\Lambda(\beta) = \sum_{\{x,y\}} J_{x,y} \mathbb{P}_\beta(\{x,y\} \text{ pivotal for } 0 \leftrightarrow \Lambda^c).$$



Since  $t \geq 1 - e^{-t}$  for  $t \geq 0$ ,  $J_{x,y} \geq \beta^{-1}(1 - e^{-\beta J_{x,y}})$ . On the failure event  $\{0 \not\leftrightarrow \Lambda^c\}$  define

$$\mathcal{S} = \{x \in \Lambda : x \not\leftrightarrow \Lambda^c\}.$$

Then  $0 \in \mathcal{S}$ . If  $\mathcal{S} = S$  and a boundary edge  $\{x, y\}$  with  $x \in S$ ,  $y \notin S$  is pivotal, the pivotality event is precisely the conjunction of  $0 \xrightarrow{S} x$  and  $\mathcal{S} = S$ . These two events depend on disjoint sets of edge variables and are independent. Summing first over boundary edges and then over all possible  $S$  therefore yields

$$f'_\Lambda(\beta) \geq \frac{1}{\beta} \inf_{\substack{S \subset \Lambda \\ 0 \in S}} \varphi_\beta(S)(1 - f_\Lambda(\beta)). \quad (2.1)$$

This is the only differential estimate needed below.

If  $\beta > \tilde{\beta}_c$ , every finite  $S \ni 0$  has  $\varphi_\beta(S) \geq 1$ . Thus (2.1) gives

$$\frac{f'_\Lambda(\beta)}{1 - f_\Lambda(\beta)} \geq \frac{1}{\beta}.$$

Integrating from  $\tilde{\beta}_c$  to  $\beta$  and using only  $f_\Lambda(\tilde{\beta}_c) \geq 0$  gives  $f_\Lambda(\beta) \geq (\beta - \tilde{\beta}_c)/\beta$ . Letting  $\Lambda \uparrow V$  proves the supercritical bound.

*The subcritical contraction.* Fix  $\beta < \tilde{\beta}_c$  and choose finite  $S \ni 0$  with  $q := \varphi_\beta(S) < 1$ . We use the following exit estimate. If  $u \in S \subset A$  and  $B \cap S = \emptyset$ , then

$$\mathbb{P}_\beta(u \xrightarrow{A} B) \leq \sum_{x \in S} \sum_{y \notin S} (1 - e^{-\beta J_{x,y}}) \mathbb{P}_\beta(u \xrightarrow{S} x) \mathbb{P}_\beta(y \xrightarrow{A} B). \quad (2.2)$$

Indeed, on every path from  $u$  to  $B$  take the first edge leaving  $S$ . The connection inside  $S$ , that edge being open, and the remaining connection to  $B$  can be required disjointly; two applications of the BK inequality and a union bound give (2.2).

For finite  $\Lambda$  define

$$X(\Lambda) = \max_{u \in \Lambda} \sum_{v \in \Lambda} \mathbb{P}_\beta(u \xrightarrow{\Lambda} v).$$

Translate  $S$  by an automorphism carrying  $0$  to  $u$ . Summing (2.2) over  $v \notin S_u$  gives

$$\sum_{v \in \Lambda} \mathbb{P}_\beta(u \xrightarrow{\Lambda} v) \leq |S| + qX(\Lambda).$$

Taking the maximum over  $u$  yields  $X(\Lambda) \leq |S|/(1 - q)$ . Letting  $\Lambda \uparrow V$  proves  $\chi(\beta) < \infty$ .

If the interaction range is  $R < \infty$ , choose  $L > R$  with  $S \subset B_{L-R}$ . Applying (2.2) to  $B_n^c$  shows for  $n \geq L$

$$\mathbb{P}_\beta(0 \leftrightarrow B_n^c) \leq q \mathbb{P}_\beta(0 \leftrightarrow B_{n-L}^c).$$

Iteration gives  $q^{\lfloor n/L \rfloor}$  and hence exponential decay.

The subcritical conclusions show  $\beta_c \geq \tilde{\beta}_c$ , while the positive lower bound above  $\tilde{\beta}_c$  shows  $\beta_c \leq \tilde{\beta}_c$ . Therefore the two parameters coincide.  $\square$

**Theorem 2.3** (Subcritical exponential decay).

*In the finite-range setting of the preceding theorem, for every  $\beta < \beta_c$  there are constants  $C, c > 0$  such that*

$$\mathbb{P}_\beta(0 \leftrightarrow B_n^c) \leq Ce^{-cn} \quad (n \geq 0).$$

*For nearest-neighbour Bernoulli percolation the same conclusion holds with  $\beta$  replaced by  $p < p_c$ .*

*Proof.* Choose a finite  $S \ni 0$  with  $q = \varphi_\beta(S) < 1$  and choose  $L$  larger than the diameter of  $S$  plus the interaction range. The exit estimate (2.2) in the proof of sharpness gives

$$\mathbb{P}_\beta(0 \leftrightarrow B_n^c) \leq q \mathbb{P}_\beta(0 \leftrightarrow B_{n-L}^c), \quad n \geq L.$$

After  $k = \lfloor n/L \rfloor$  iterations the right side is at most  $q^k$ . Taking  $c = L^{-1} \log(q^{-1})$  and enlarging the prefactor to cover  $n < L$  proves the claim. The nearest-neighbour  $p$ -form is identical with  $\varphi_p(S) = p \sum_{\{x,y\} \in \partial_E S} \mathbb{P}_p(0 \xrightarrow{S} x)$ .  $\square$

**Theorem 2.4** (Mean-field lower bound above  $p_c$ ).

*For nearest-neighbour Bernoulli bond percolation on an infinite locally finite transitive graph,*

$$\theta(p) = \mathbb{P}_p(0 \leftrightarrow \infty) \geq \frac{p - p_c}{p(1 - p_c)}, \quad p > p_c.$$

*In the coupling parameter  $p = 1 - e^{-\beta}$  this follows from the slightly weaker but more invariant estimate  $\theta(\beta) \geq (\beta - \beta_c)/\beta$ .*

*Proof.* In the nearest-neighbour model repeat the sharpness proof with

$$\varphi_p(S) = p \sum_{\substack{x \in S, y \notin S \\ \{x,y\} \in E}} \mathbb{P}_p(0 \xrightarrow{S} x).$$

Russo's formula gives, for  $f_n(p) = \mathbb{P}_p(0 \leftrightarrow B_n^c)$ ,

$$\frac{f'_n(p)}{1 - f_n(p)} \geq \frac{1}{p(1 - p)} \inf_{\substack{S \subset B_n \\ 0 \in S}} \varphi_p(S).$$

Above  $p_c$  the infimum is at least one by the same identification of the finite-set threshold with  $p_c$ . Integrating from  $p_c$  to  $p$  gives

$$-\log(1 - f_n(p)) \geq \int_{p_c}^p \frac{ds}{s(1 - s)} = \log \frac{p(1 - p_c)}{p_c(1 - p)}.$$

Using only  $f_n(p_c) \geq 0$  in the integrated form and simplifying yields  $f_n(p) \geq (p - p_c)/(p(1 - p_c))$ . Letting  $n \rightarrow \infty$  proves the displayed bound. In the exponential coupling the differential inequality is  $f'_n/(1 - f_n) \geq 1/\beta$ , whose integration gives  $(\beta - \beta_c)/\beta$ .  $\square$

**Theorem 2.5** (Finite subcritical susceptibility).

*In the transitive setting of the sharpness theorem,*

$$\chi(\beta) := \sum_{x \in V} \mathbb{P}_\beta(0 \leftrightarrow x) = \mathbb{E}_\beta |C(0)| < \infty \quad (\beta < \beta_c).$$

*The same statement holds in the nearest-neighbour  $p$ -parameterization for  $p < p_c$ .*

*Proof.* Choose finite  $S \ni 0$  with  $q = \varphi_\beta(S) < 1$ . For finite  $\Lambda$  put

$$X(\Lambda) = \max_{u \in \Lambda} \sum_{v \in \Lambda} \mathbb{P}_\beta(u \xleftrightarrow{\Lambda} v).$$

Translate  $S$  to  $S_u$  and apply the exit inequality (2.2) from the proof of sharpness to every  $v \notin S_u$ . Summation in  $v$  gives

$$\sum_{v \in \Lambda} \mathbb{P}_\beta(u \xleftrightarrow{\Lambda} v) \leq |S| + qX(\Lambda).$$

Taking the maximum over  $u$  shows  $X(\Lambda) \leq |S|/(1 - q)$ , uniformly in  $\Lambda$ . Monotone convergence as  $\Lambda \uparrow V$  gives

$$\sum_x \mathbb{P}_\beta(0 \leftrightarrow x) \leq \frac{|S|}{1 - q} < \infty.$$

Finally Tonelli's theorem gives  $\mathbb{E}|C(0)| = \sum_x \mathbb{P}(0 \leftrightarrow x)$ . □

## Chapter 3

# Infinite clusters

Existence of an infinite cluster is not the same as uniqueness. Amenability and mass transport convert local modification arguments into global rigidity.

**Example 3.1** (A running picture). On a square box, think of each edge as open independently with probability  $p$ . The local rule is trivial; the questions below ask when those bits create a path whose diameter is comparable with the box.

**Theorem 3.2** (Burton–Keane uniqueness).

*Let  $G$  be a locally finite, transitive, unimodular amenable graph, and let  $\mathbb{P}$  be an automorphism-invariant ergodic bond-percolation law with the finite-energy property. Then the number  $N_\infty$  of infinite open clusters is almost surely either 0 or 1. In particular, Bernoulli percolation on  $G$  has a unique infinite cluster whenever it percolates.*

*Proof.* Ergodicity makes  $N_\infty$  an almost-sure constant in  $\{0, 1, 2, \dots, \infty\}$ . We exclude first a finite value  $k \geq 2$ . Choose a finite connected set  $K$  meeting all  $k$  infinite clusters with positive probability. By finite energy, with positive conditional probability we may open a finite tree inside  $K$  joining the portions of those clusters while leaving the exterior unchanged. On that event the number of infinite clusters is strictly smaller than  $k$ , contradicting almost-sure constancy. Thus  $N_\infty \notin \{2, 3, \dots\}$ .

Assume now  $N_\infty = \infty$ . With positive probability a finite set meets at least three infinite clusters. By another finite-energy modification, a chosen vertex can be made a *trifurcation*: it belongs to an infinite cluster and deleting it separates that cluster into at least three infinite components. By transitivity, trifurcations then have a positive density  $a > 0$ .

Let  $(F_n)$  be a Følner sequence. For a configuration, connect each trifurcation in  $F_n$  to three distinct infinite branches and contract finite pieces until the first exit from  $F_n$ . The resulting finite graph is a forest whose interior trifurcation vertices have degree at least three and whose leaves lie on the vertex boundary  $\partial F_n$ . A finite forest with  $t$  vertices of degree at least three has at least  $t + 2$  leaves after suppressing degree-two vertices. Hence the number  $T_n$  of trifurcations in  $F_n$  is bounded by a constant multiple of  $|\partial F_n|$ .

Taking expectations gives  $a|F_n| = \mathbb{E}T_n \leq C|\partial F_n|$ . Amenability makes  $|\partial F_n|/|F_n| \rightarrow 0$ , a contradiction. Therefore  $N_\infty$  cannot be infinite, and only 0 or 1 remain.  $\square$

**Theorem 3.3** (Mass-transport principle).

*Let  $G = (V, E)$  be a unimodular transitive graph, let  $o \in V$ , and let  $M(u, v, \omega) \geq 0$  be an automorphism-equivariant measurable mass transport under an invariant percolation law. Then*

$$\mathbb{E} \sum_{v \in V} M(o, v, \omega) = \mathbb{E} \sum_{u \in V} M(u, o, \omega),$$

with the convention that both sides may be  $+\infty$ .

*Proof.* Set  $m(u, v) = \mathbb{E}M(u, v, \omega)$ . Invariance gives  $m(gu, gv) = m(u, v)$  for every  $g$  in a closed transitive unimodular subgroup  $\Gamma \leq \text{Aut}(G)$ . Normalize a left Haar measure  $\mu$  on  $\Gamma$  by  $\mu(\Gamma_o) = 1$ . Since the fibers of  $g \mapsto go$  are right cosets of the compact stabilizer  $\Gamma_o$ , Tonelli gives

$$\sum_{v \in V} m(o, v) = \int_{\Gamma} m(o, go) d\mu(g).$$

Equivariance gives  $m(o, go) = m(g^{-1}o, o)$ . Unimodularity makes Haar measure invariant under inversion, whence

$$\int_{\Gamma} m(g^{-1}o, o) d\mu(g) = \int_{\Gamma} m(go, o) d\mu(g) = \sum_{u \in V} m(u, o).$$

Tonelli again identifies these sums with expected outgoing and incoming mass. □

**Theorem 3.4** (No infinite cluster at planar criticality).

*For nearest-neighbour bond percolation on  $\mathbb{Z}^2$  at  $p = p_c = 1/2$ ,*

$$\mathbb{P}_{1/2}(0 \leftrightarrow \infty) = 0.$$

*Indeed, almost surely there is no infinite open cluster anywhere in  $\mathbb{Z}^2$ .*

*Proof.* Choose disjoint square annuli

$$A_j = B(0, 3^{2j+1}) \setminus B(0, 3^{2j}), \quad j = 1, 2, \dots,$$

so that the sets of edges belonging to different  $A_j$  are disjoint. RSW at the self-dual point, together with four rectangle crossings and FKG, gives a constant  $c > 0$  independent of  $j$  such that the dual lattice contains a closed dual circuit surrounding the origin in  $A_j$  with probability at least  $c$ . (A dual edge is open for the dual process exactly when the crossed primal edge is closed; at  $p = 1/2$  the dual parameter is again  $1/2$ .)

The circuit events for the chosen annuli are independent because they use disjoint edge sets. Their probabilities have a divergent sum, so the second Borel–Cantelli lemma implies that infinitely many of them occur almost surely. Every such closed dual circuit separates the origin from infinity and therefore blocks every infinite primal open path from the origin. Hence  $\mathbb{P}_{1/2}(0 \leftrightarrow \infty) = 0$ .

By translation invariance the same statement holds for every vertex. Since  $\mathbb{Z}^2$  is countable,

$$\mathbb{P}_{1/2}(\text{there exists an infinite open cluster}) \leq \sum_{x \in \mathbb{Z}^2} \mathbb{P}_{1/2}(x \leftrightarrow \infty) = 0.$$

□

## Chapter 4

# Planar critical thresholds

In two dimensions duality turns topology into a numerical critical-point theorem.

**Example 4.1** (A running picture). On a square box, think of each edge as open independently with probability  $p$ . The local rule is trivial; the questions below ask when those bits create a path whose diameter is comparable with the box.

**Lemma 4.2** (Self-dual crossing and RSW gluing). *For bond percolation on the square lattice at  $p = 1/2$ , and for site percolation on the triangular lattice at  $p = 1/2$ , every fixed aspect ratio  $\rho \geq 1$  admits a constant  $c(\rho) > 0$  such that the probability of an open crossing of a  $\rho n \times n$  rectangle in the long direction lies in*

$$[c(\rho), 1 - c(\rho)]$$

*uniformly in  $n$  (with the usual harmless lattice rounding).*

*Proof.* At the self-dual parameter, planar duality says that in a square exactly one of the primal left–right crossing and the complementary dual top–bottom crossing occurs. Rotation and color/edge complementation therefore give a square-crossing probability bounded away from both zero and one (with standard boundary conventions it is exactly  $1/2$ ).

It remains to pass from squares to rectangles. We record the gluing step because it is the part of RSW that is used repeatedly. Take a left–right crossing in a square and expose the lowest such crossing  $\gamma$ . Reflection across a vertical line sends the portion of the configuration above  $\gamma$  to an independent-copy geometry with the same law. The event that an additional open path connects  $\gamma$  to the reflected crossing is increasing; by FKG its conditional probability is bounded below by a positive function of the square-crossing probability. Consequently two overlapping squares can be crossed together with a scale-independent positive probability. Iterating over a chain of  $O(\rho)$  overlapping squares gives the lower bound  $c(\rho)$ . Applying the same argument to the dual/closed configuration gives the upper bound  $1 - c(\rho)$ . This is the classical RSW gluing argument; Volume XI develops the separated-landing and arm-event refinements.  $\square$

**Theorem 4.3** (Harris–Kesten theorem).

*For nearest-neighbour bond percolation on the square lattice,*

$$p_c(\mathbb{Z}^2) = \frac{1}{2}.$$

*Proof.* We use the self-dual crossing lemma above together with sharpness from Chapter 2.

First suppose, for contradiction, that  $p_c > 1/2$ . Then  $p = 1/2$  is strictly subcritical, so Chapter 2 gives exponential decay of connection probabilities. In particular the probability of a left–right open

crossing of an  $n \times n$  square tends to zero, because any such crossing contains a vertex connected to distance at least  $n/2$ . Self-duality, however, keeps the square-crossing probability bounded away from zero uniformly in  $n$ . Hence  $p_c \leq 1/2$ .

For the reverse inequality, suppose  $p_c < 1/2$  and choose  $p$  with  $p_c < p < 1/2$ . Primal  $p$ -percolation then has an infinite open cluster. A dual edge is open precisely when the crossed primal edge is closed, so the dual parameter is  $1 - p > 1/2 > p_c$ . By translation invariance and Burton–Keane, both the primal and dual models would have unique infinite clusters.

We now use the planar noncoexistence lemma. Two unique infinite clusters of opposite types cannot coexist: choose a large box which both clusters meet in four alternating macroscopic sectors with positive probability. Local finite-energy modifications connect the two primal entrances inside the box and separately the two dual entrances. The resulting primal and dual connections must cross, but every crossing pair consists of a primal edge and its dual, which cannot both be open. This contradiction rules out simultaneous primal and dual percolation. Therefore  $p_c \geq 1/2$ , completing the proof.  $\square$

**Theorem 4.4** (Critical site percolation on the triangular lattice).

*For independent site percolation on the triangular lattice  $\mathbb{T}$ ,*

$$p_c^{\text{site}}(\mathbb{T}) = \frac{1}{2}.$$

*Proof.* The triangular lattice is self-matching: the complementary closed sites at parameter  $p$  have the same planar crossing geometry as open sites at parameter  $1 - p$ . At  $p = 1/2$  color complementation and rotation therefore give the self-dual square-crossing estimate, and the preceding RSW gluing lemma applies.

If  $p_c > 1/2$ , then  $p = 1/2$  is subcritical. The site version of the sharpness theorem gives exponential decay, forcing macroscopic rectangle-crossing probabilities to tend to zero, contrary to the self-dual RSW lower bound. Hence  $p_c \leq 1/2$ .

If  $p_c < 1/2$ , choose  $p \in (p_c, 1/2)$ . Open sites percolate at parameter  $p$ , while closed sites have parameter  $1 - p > 1/2 > p_c$  and therefore also percolate. Burton–Keane uniqueness (applied to the site model) and planar noncoexistence exclude simultaneous unique infinite open and closed clusters. Thus  $p_c \geq 1/2$ . The two inequalities give the claim.  $\square$

**Theorem 4.5** (Primal–dual uniqueness relation).

*Let  $G$  be a locally finite one-ended amenable planar quasi-transitive graph whose planar dual  $G^*$  is also locally finite and quasi-transitive. For Bernoulli bond percolation,*

$$p_c(G) + p_c(G^*) = 1.$$

*Equivalently, below the primal critical point the complementary dual model is supercritical, whereas above the primal critical point the complementary dual model does not percolate.*

*Proof.* At parameter  $p$ , closed primal edges are open dual edges with parameter  $1 - p$ . The planar separation principle says that a primal open crossing of a large annulus excludes a dual closed circuit through the same annulus, and conversely absence of a primal connection across an annulus forces a dual separating circuit.

If  $p < p_c(G)$ , sharpness gives exponential decay of primal crossings. Thus the probability that a large annulus contains a dual-open circuit tends to one. Standard gluing of these circuits in overlapping annuli gives a dual-open path to infinity with positive probability, so  $1 - p \geq p_c(G^*)$ . Letting  $p \uparrow p_c(G)$  yields  $p_c(G) + p_c(G^*) \leq 1$ .

Conversely, if  $p > p_c(G)$  and  $1 - p > p_c(G^*)$ , then both primal and dual models percolate. Burton–Keane uniqueness holds on the amenable quasi-transitive graphs, so there would be a unique infinite cluster of each type. The planar noncoexistence argument from the Harris–Kesten proof rules this out. Therefore  $1 - p \leq p_c(G^*)$  for every  $p > p_c(G)$ . Letting  $p \downarrow p_c(G)$  gives the reverse inequality.  $\square$



## Chapter 5

# Decision trees and influences

Randomized algorithms repackage pivotality: a variable matters only when an exploration must reveal it.

**Definition 5.1** (Decision tree and revealment). For  $f : \{0, 1\}^n \rightarrow \{0, 1\}$ , a *decision tree* adaptively chooses an unqueried coordinate as a function of the values already revealed and stops once those values determine  $f$ . If  $T$  may also use independent internal randomness, its *revealment* at coordinate  $i$  is

$$\delta_i(T) = \mathbb{P}(T \text{ queries } i), \quad \delta(T) = \max_i \delta_i(T).$$

The *influence* used below is  $I_i(f) = \mathbb{P}(i \text{ is pivotal for } f)$  under the specified product measure.

*Worked example 5.2* (An optimal tree for a two-bit AND). Let  $f(x_1, x_2) = x_1 x_2$  under Bernoulli- $p$  product measure. Query  $x_1$  first; if  $x_1 = 0$ , stop, while if  $x_1 = 1$ , query  $x_2$ . Thus

$$\delta_1 = 1, \quad \delta_2 = p, \quad I_1 = I_2 = p.$$

Since  $\mathbb{E}f = p^2$ ,

$$\text{Var}(f) = p^2(1 - p^2).$$

The OSSS right side is

$$p(1 - p)(\delta_1 I_1 + \delta_2 I_2) = p(1 - p)(p + p^2) = p^2(1 - p^2),$$

so OSSS is an equality for this tree. The calculation also shows why a good algorithm should reveal a coordinate only on configurations where it can still matter.

**Example 5.3** (A running picture). On a square box, think of each edge as open independently with probability  $p$ . The local rule is trivial; the questions below ask when those bits create a path whose diameter is comparable with the box.

**Theorem 5.4** (OSSS inequality for Bernoulli product measure).

Let  $f : \{0, 1\}^n \rightarrow \{0, 1\}$  be increasing under Bernoulli- $p$  product measure, and let  $T$  be a possibly randomized decision tree that determines  $f$ . Write

$$I_i(f) = \mathbb{P}_p(i \text{ is pivotal for } f), \quad \delta_i(T) = \mathbb{P}_p(T \text{ queries coordinate } i).$$

Then

$$\text{Var}_p(f) \leq p(1 - p) \sum_{i=1}^n \delta_i(T) I_i(f). \tag{5.1}$$

Equivalently, because  $\mathbb{E}[\text{Var}(f \mid X_j, j \neq i)] = p(1-p)I_i(f)$ ,

$$\text{Var}_p(f) \leq \sum_i \delta_i(T) \mathbb{E}[\text{Var}(f \mid X_j, j \neq i)].$$

*Proof.* It is enough to prove the assertion after conditioning on the internal randomness of  $T$ ; averaging then gives the randomized version. We therefore fix a deterministic decision rule. Let  $\omega$  and  $\tilde{\omega}$  be independent Bernoulli- $p$  configurations. Run the algorithm on  $\omega$ , and denote its successive queried coordinates by  $i_1, i_2, \dots$  and its stopping time by  $\tau = \tau(\omega)$ .

The useful coupling is a sequence of hybrid configurations. Before the first step, keep the queried coordinates from  $\omega$  and the unqueried ones from  $\tilde{\omega}$ ; then replace the queried coordinates, one at a time, by their  $\tilde{\omega}$  values. More precisely, complete the queried list to a permutation  $i_1, \dots, i_n$  in any deterministic manner and put

$$\omega_{i_s}^t = \begin{cases} \tilde{\omega}_{i_s}, & s \leq t, \\ \omega_{i_s}, & t < s \leq \tau, \\ \tilde{\omega}_{i_s}, & s > \tau, \end{cases} \quad 0 \leq t \leq n.$$

Because the values of  $\omega$  on  $i_1, \dots, i_\tau$  determine  $f(\omega)$ , we have  $f(\omega^0) = f(\omega)$ ; and  $\omega^n = \tilde{\omega}$ .

For a Bernoulli random variable  $Y \in \{0, 1\}$ ,  $\text{Var}(Y) = q(1-q)$  with  $q = \mathbb{E}Y$ , while  $\mathbb{E}|Y - Y'| = 2q(1-q)$  for an independent copy  $Y'$ . Consequently

$$2 \text{Var}_p(f) = \mathbb{E}|f(\omega) - f(\tilde{\omega})| \leq \sum_{t=1}^n \mathbb{E}|f(\omega^t) - f(\omega^{t-1})|. \quad (5.2)$$

The summand vanishes for  $t > \tau$ . On the event  $\{t \leq \tau, i_t = i\}$ , condition on the already revealed values  $\omega_{i_1}, \dots, \omega_{i_{t-1}}$  and on this event. The crucial point is that the unrevealed coordinates of both hybrids still have the product Bernoulli law: the algorithm has only inspected the past of  $\omega$ , while  $\tilde{\omega}$  is independent. The two hybrids differ only at coordinate  $i$ , where the two entries are independent Bernoulli- $p$  variables. Thus  $f(\omega^t) \neq f(\omega^{t-1})$  only if the common outside configuration makes  $i$  pivotal, and then the two Bernoulli bits must be different. The pivotal event depends only on the outside coordinates, so it is independent of these two bits. Hence

$$\mathbb{E}[|f(\omega^t) - f(\omega^{t-1})| \mid \omega_{i_1}, \dots, \omega_{i_{t-1}}, t \leq \tau, i_t = i] = 2p(1-p)I_i(f). \quad (5.3)$$

Insert (5.3) into (5.2) and sum over  $t$  and  $i$ :

$$2 \text{Var}_p(f) \leq 2p(1-p) \sum_i I_i(f) \sum_t \mathbb{P}_p(t \leq \tau, i_t = i).$$

A decision tree queries a coordinate at most once, so the last sum is precisely  $\delta_i(T)$ . Division by 2 proves (5.1).

Finally, after all coordinates except  $i$  are fixed, an increasing Boolean  $f$  is either constant in  $X_i$  or equals  $X_i$ . The latter case is exactly pivotality, and its conditional variance is  $p(1-p)$ . Averaging gives the equivalent conditional-variance form.  $\square$

**Theorem 5.5** (OSSS–Russo differential inequality).

Let  $A$  be an increasing event of a finite Bernoulli- $p$  product space and let  $T$  be a decision tree determining  $A$ . If  $\delta_i(T)$  is the revelation of coordinate  $i$  and  $\delta(T) = \max_i \delta_i(T)$ , then

$$\frac{d}{dp} \mathbb{P}_p(A) \geq \frac{\mathbb{P}_p(A)(1 - \mathbb{P}_p(A))}{p(1-p) \delta(T)}.$$

More generally, one may replace the denominator  $\delta(T)$  by the weighted revelation profile appearing in the OSSS sum.

*Proof.* Apply the OSSS inequality to  $f = \mathbf{1}_A$ :

$$\mathbb{P}_p(A)(1 - \mathbb{P}_p(A)) \leq p(1 - p) \sum_i \delta_i(T) \mathbb{P}_p(i \text{ pivotal}).$$

Bounding every  $\delta_i(T)$  by  $\delta(T)$  and applying Russo's formula gives

$$\mathbb{P}_p(A)(1 - \mathbb{P}_p(A)) \leq p(1 - p) \delta(T) \frac{d}{dp} \mathbb{P}_p(A).$$

Rearranging proves the displayed inequality. In percolation applications the substantive step is to design  $T$  so that  $\delta(T)$  is controlled by a one-arm or connectivity probability; the inequality then turns geometric information into a differential estimate.  $\square$

**Theorem 5.6** (A transitive sharp-threshold estimate).

*There is a universal constant  $c > 0$  with the following property. Let  $A \subset \{0, 1\}^n$  be increasing and invariant under a group acting transitively on the coordinates. For  $p$  in any fixed compact subinterval  $[a, 1 - a] \subset (0, 1)$ ,*

$$\frac{d}{dp} \mathbb{P}_p(A) \geq c_a \mathbb{P}_p(A)(1 - \mathbb{P}_p(A)) \log n.$$

*Consequently, if  $\varepsilon \in (0, 1/2)$  and  $p_\varepsilon < q_\varepsilon$  satisfy  $\mathbb{P}_{p_\varepsilon}(A) = \varepsilon$  and  $\mathbb{P}_{q_\varepsilon}(A) = 1 - \varepsilon$ , with the interval between them contained in  $[a, 1 - a]$ , then*

$$q_\varepsilon - p_\varepsilon \leq \frac{C_a}{\log n} \log \frac{1 - \varepsilon}{\varepsilon}.$$

*Proof.* We give the influence estimate rather than citing “KKL” as a black box. Put  $q = 1 - p$  and use the  $p$ -biased orthonormal basis

$$\psi_i(x) = \frac{x_i - p}{\sqrt{pq}}, \quad \psi_S = \prod_{i \in S} \psi_i, \quad f = \mathbf{1}_A = \sum_{S \subset [n]} \hat{f}_p(S) \psi_S.$$

Write

$$W_j = \sum_{|S|=j} \hat{f}_p(S)^2, \quad I_i = \mathbb{P}_p(i \text{ is pivotal}), \quad I = \sum_i I_i, \quad \delta = \max_i I_i.$$

Parseval gives  $\text{Var}_p(f) = \sum_{j \geq 1} W_j$ . If

$$D_i f(x_{\hat{i}}) = f(x_{\hat{i}}, 1) - f(x_{\hat{i}}, 0),$$

then  $D_i f \in \{0, 1\}$  because  $A$  is increasing, and direct subtraction of the biased Fourier expansion gives

$$D_i f = (pq)^{-1/2} \sum_{S \ni i} \hat{f}_p(S) \psi_{S \setminus \{i\}}.$$

Therefore

$$\|D_i f\|_2^2 = I_i, \quad I = \frac{1}{pq} \sum_{j \geq 1} j W_j. \tag{5.4}$$

We need one elementary hypercontractive input. For every  $a \in (0, 1/2)$  there is  $\rho_a \in (0, 1)$  such that, uniformly for  $p \in [a, 1 - a]$ , the biased noise operator satisfies

$$\|T_{\rho_a} g\|_2 \leq \|g\|_{4/3}. \quad (5.5)$$

Here  $T_\rho$  multiplies the Fourier coefficient on level  $S$  by  $\rho^{|S|}$ . For completeness, first consider one coordinate and write  $g = m + s\psi$ . After homogeneity normalizes  $|m| + |s| = 1$ , the desired inequality is a continuous inequality on the compact set  $p \in [a, 1 - a]$ . At  $\rho = 0$  it is Jensen's inequality. Away from  $s = 0$  the inequality is strict and compactness leaves room for a uniform positive  $\rho$ . Near  $s = 0$ , Taylor expansion gives

$$\|m + \rho s\psi\|_2^2 = m^2 + \rho^2 s^2, \quad \|m + s\psi\|_{4/3}^2 = m^2 + \frac{1}{3}s^2 + O_a(|s|^3/|m|),$$

so any fixed  $\rho^2 < 1/3$  works after shrinking the neighborhood, uniformly in  $p$ . The one-coordinate inequality tensorizes: apply it successively in each coordinate and use Minkowski's integral inequality at every step. This proves (5.5).

Let  $(Dif)_{j-1}$  denote the homogeneous degree- $(j-1)$  part of  $Dif$ . From (5.5),

$$\rho_a^{j-1} \|(Dif)_{j-1}\|_2 \leq \|T_{\rho_a} Dif\|_2 \leq \|Dif\|_{4/3}.$$

Since  $Dif$  is 0 or 1,  $\|Dif\|_{4/3}^2 = I_i^{3/2}$ . Summing over  $i$  and using the displayed Fourier formula for  $Dif$  gives

$$\frac{j}{pq} W_j = \sum_i \|(Dif)_{j-1}\|_2^2 \leq \rho_a^{-2(j-1)} \sum_i I_i^{3/2} \leq \rho_a^{-2(j-1)} \sqrt{\delta} I. \quad (5.6)$$

Thus for every integer  $m \geq 1$ ,

$$\sum_{1 \leq j \leq m} W_j \leq C_a \rho_a^{-2m} \sqrt{\delta} I. \quad (5.7)$$

whereas (5.4) gives the high-level bound

$$\sum_{j > m} W_j \leq \frac{pq}{m+1} I \leq \frac{I}{m+1}. \quad (5.8)$$

Choose  $m$  to be a sufficiently small fixed multiple of  $\log(1/\delta)$ . Then  $\rho_a^{-2m} \sqrt{\delta}$  is bounded by  $C_a/m$ , and (5.7)–(5.8) yield the biased KKL form

$$I \geq c_a \operatorname{Var}_p(f) \log \frac{1}{\delta}, \quad (5.9)$$

with the cases of bounded  $\log(1/\delta)$  absorbed by changing  $c_a$ .

Transitivity makes all  $I_i$  equal, so  $\delta = I/n$ . If  $I \geq n^{1/2}$ , then  $I \geq c_a \operatorname{Var}_p(f) \log n$  after decreasing  $c_a$  (and absorbing finitely many small  $n$ ). Otherwise  $\delta < n^{-1/2}$ , hence  $\log(1/\delta) \geq \frac{1}{2} \log n$ , and (5.9) gives the same conclusion. Russo's formula now yields

$$\frac{d}{dp} \mathbb{P}_p(A) = I \geq c_a \mathbb{P}_p(A)(1 - \mathbb{P}_p(A)) \log n.$$

Finally set  $u(p) = \mathbb{P}_p(A)$ , divide by  $u(1 - u)$ , and integrate from  $p_\varepsilon$  to  $q_\varepsilon$ :

$$2 \log \frac{1 - \varepsilon}{\varepsilon} = \int_{p_\varepsilon}^{q_\varepsilon} \frac{du}{u(1 - u)} \geq c_a (q_\varepsilon - p_\varepsilon) \log n.$$

Renaming the constant gives the stated window bound.  $\square$

## Chapter 6

# General graphs and dimensions

Transitivity alone does not make every graph look Euclidean. We separate robust monotonicity from hypotheses needed for nonuniqueness intervals and mean-field behavior.

**Example 6.1** (A running picture). On a square box, think of each edge as open independently with probability  $p$ . The local rule is trivial; the questions below ask when those bits create a path whose diameter is comparable with the box.

**Theorem 6.2** (Comparison and monotonicity).

*For Bernoulli percolation on a fixed graph, every increasing event is nondecreasing in the edge parameters. Hence  $p_c$  is monotone under edge addition: if  $G$  is a spanning subgraph of  $H$  on the same vertex set, then*

$$p_c(H) \leq p_c(G).$$

*More generally, if one product percolation law stochastically dominates another through a coordinatewise coupling, then every increasing connection probability and every percolation probability is correspondingly larger.*

*Proof.* Use common independent uniforms  $U_e$ . A configuration with parameter family  $(p_e)$  is obtained by declaring  $e$  open when  $U_e \leq p_e$ . If  $p_e \leq q_e$  for every  $e$ , the  $p$ -open edge set is contained in the  $q$ -open edge set pointwise. Thus every increasing event under  $p$  is contained in the corresponding event under  $q$ , proving stochastic monotonicity.

For graph inclusion, assign the additional edges of  $H \setminus G$  independent uniforms as well. The open subgraph of  $G$  is then literally contained in the open subgraph of  $H$  at the same parameter. Therefore  $\theta_H(p) \geq \theta_G(p)$ , and taking the infimum of parameters for which the percolation probability is positive gives  $p_c(H) \leq p_c(G)$ .  $\square$

**Theorem 6.3** (A nonamenable model with  $p_c < p_u$ ).

*For the  $d$ -regular tree  $T_d$ ,  $d \geq 3$ , Bernoulli bond percolation has*

$$p_c(T_d) = \frac{1}{d-1}, \quad p_u(T_d) = 1,$$

*where  $p_u$  is the infimum of parameters for which there is almost surely a unique infinite cluster. Thus the nonuniqueness interval  $(p_c, p_u)$  is nonempty.*

*Proof.* Orient the tree away from a fixed root. The descendants of the root connected to it form a Galton–Watson process whose offspring distribution is  $\text{Bin}(d-1, p)$  after the first generation. It survives with positive probability exactly when  $(d-1)p > 1$ . Hence  $p_c = 1/(d-1)$ .

Now take  $p < 1$ . Every edge is closed with positive probability. Conditional on a fixed edge  $e$  being closed, the two half-trees separated by  $e$  contain independent percolation configurations. When  $p > p_c$ , each half-tree has a positive probability to contain an infinite cluster. Along infinitely many pairwise disjoint edge neighborhoods, independence and Borel–Cantelli show that infinitely many closed edges separate infinite open clusters. Consequently the number of infinite clusters is almost surely infinite throughout  $p_c < p < 1$ . At  $p = 1$  the entire tree is open and connected. Hence  $p_u = 1$ .  $\square$

**Theorem 6.4** (Mean-field exponents under the triangle condition).

*Consider transitive independent percolation in the standard nonoriented setting and assume that the critical triangle diagram*

$$\nabla_{p_c} = \sum_{x,y} \tau_{p_c}(0,x) \tau_{p_c}(x,y) \tau_{p_c}(y,0)$$

*is finite. Then, in the bounded-ratio sense,*

$$\chi(p) \asymp (p_c - p)^{-1}, \quad \theta(p) \asymp (p - p_c), \quad \mathbf{P}_{p_c}(|C(0)| \geq n) \asymp n^{-1/2}.$$

*Equivalently the critical exponents are  $\gamma = 1$ ,  $\beta = 1$ , and  $\delta = 2$ .*

*Internal proof: finite-triangle reconnection and ghost field.* We write

$$\tau_p(x,y) = \mathbf{P}_p(x \leftrightarrow y), \quad \chi(p) = \sum_x \tau_p(0,x),$$

and use only transitivity, finite energy, BK, Russo differentiation, and the assumption  $\nabla_{p_c} < \infty$ . Since  $\tau_p \uparrow \tau_{p_c}$  as  $p \uparrow p_c$ , every triangle tail can be made uniformly small near criticality: for each  $\eta > 0$  there is  $R < \infty$  and  $p_0 < p_c$  such that

$$\sum_{\substack{x,y: \\ \max\{d(0,x), d(x,y), d(y,0)\} > R}} \tau_p(0,x) \tau_p(x,y) \tau_p(y,0) < \eta, \quad p_0 \leq p \leq p_c. \quad (6.1)$$

This elementary truncation is the place where finiteness, rather than smallness, of the triangle diagram is used.

Russo’s formula for  $\tau_p(0,x)$  sums over pivotal edges  $e = uv$ . On the pivotal event there are two disjoint open arms, one from 0 to  $u$  and one from  $v$  to  $x$ . If these two arms remain disjoint until they reach distance  $R$  from  $e$ , BK gives the product contribution which, after summing  $x$  and  $e$ , is bounded below by  $c_R \chi(p)^2$ . If they reconnect before that separation, choose the first reconnection vertex on each arm. The three open pieces joining  $e$  to the two first reconnection vertices and joining those vertices to one another form a triangle. BK therefore bounds the sum of all such bad configurations by

$$C T_R(p) \chi(p)^2 + C_R \chi(p), \quad (6.2)$$

where  $T_R(p)$  is the tail in (6.1). The second term contains the finitely many reconnections inside the  $R$ -ball. Choose  $R$  so that  $C T_R(p) \leq c_R/4$  and then take  $p$  so close to  $p_c$  that  $C_R \chi(p) \leq c_R \chi(p)^2/4$ . We obtain

$$\chi'(p) \geq c \nabla \chi(p)^2. \quad (6.3)$$

The opposite inequality  $\chi'(p) \leq C \chi(p)^2$  is the tree-graph/Russo bound proved in SM-10.S2.02. Hence

$$c \chi(p)^2 \leq \chi'(p) \leq C \chi(p)^2$$

near  $p_c$ . Integrating  $-d(1/\chi)/dp = \chi'/\chi^2$  and using  $\chi(p) \uparrow \infty$  at  $p_c$  gives

$$c(p_c - p)^{-1} \leq \chi(p) \leq C(p_c - p)^{-1}. \quad (6.4)$$

Thus  $\gamma = 1$ .

Independently mark every vertex green with probability  $\gamma \in (0, 1)$  and write

$$M(p, \gamma) = \mathbf{P}_{p, \gamma}(0 \leftrightarrow G) = \mathbf{E}_p[1 - (1 - \gamma)^{|C(0)|}].$$

Then differentiation in the green variables is exact:

$$M_\gamma(p, \gamma) = \mathbf{E}_p[|C(0)|(1 - \gamma)^{|C(0)|-1}]. \quad (6.5)$$

Russo differentiation in  $p$  again exposes a pivotal edge. Orient a pivotal edge as  $e = (x, y)$  so that, with  $e$  closed, 0 is connected to  $x$  through a green-free cluster  $C$  while  $y$  is connected outside  $C$  to the green set. Conditional on  $C$ , the second event uses only edges outside  $C$  and has probability at most  $M(p, \gamma)$  by monotonicity and transitivity. Summing the possible oriented boundary edge  $e$  gives at most  $C_0|C|$  choices. Finally

$$\mathbf{E}_p[|C(0)|(1 - \gamma)^{|C(0)|}] = (1 - \gamma)M_\gamma(p, \gamma).$$

Multiplying by the probability  $1 - p$  that the pivotal edge is initially closed therefore gives the Aizenman–Barsky upper differential inequality

$$(1 - p)M_p \leq C_0(1 - \gamma)MM_\gamma. \quad (6.6)$$

No independence is used across the pivotal edge before conditioning on the closed cluster; this is precisely why the cluster-exposure formulation is needed.

For the converse estimate at criticality, expose the first pivotal edge on the connection from 0 to the green set and split according to whether the two resulting arms reconnect. The no-reconnection term is  $(1 - \gamma)M^2M_\gamma$ . A first reconnection produces exactly the same three-connection diagram as in part a. Truncate its triangle at radius  $R$ . The long part is at most  $T_R(p)$  times the tree term, while the finitely many short reconnections contribute  $O_R(M^2)$  and may be absorbed once  $M \downarrow 0$ . With  $R$  chosen from (6.1), there are constants  $c_\nabla, C_\nabla > 0$  such that, for  $p$  in a fixed critical neighborhood and  $\gamma$  sufficiently small,

$$M(p, \gamma) \geq c_\nabla(1 - \gamma)M(p, \gamma)^2M_\gamma(p, \gamma). \quad (6.7)$$

For completeness we also derive the complementary Aizenman–Barsky inequality. On the event  $\{0 \leftrightarrow G\}$ , reveal the open cluster of 0 and split according to the number of green vertices it contains. The contribution of configurations with a distinguished unique first green vertex is bounded by  $\gamma M_\gamma$ . If two disjoint green connections can be selected before any occupied pivotal edge is encountered, BK bounds their contribution by  $M^2$ . In the remaining configurations choose the first occupied pivotal edge whose deletion separates the root from one of the two green branches. Deleting that edge, and remembering its orientation, is an injection into the Russo pivotal sum times one independent green connection; after summation its contribution is at most  $pMM_p$ . Hence

$$M \leq \gamma M_\gamma + M^2 + pMM_p. \quad (6.8)$$

Equations (6.6)–(6.8) are the complete magnetization system needed below.

We need both directions in the square-root law, and they come from different inequalities. First set  $p = p_c$  in (6.8) and insert (6.6). Since  $1 - \gamma \leq 1$ , there is a constant  $\tilde{K}$  such that

$$M \leq \gamma M_\gamma + M^2 + \tilde{K} M^2 M_\gamma. \quad (6.9)$$

The map  $\gamma \mapsto M(p_c, \gamma)$  is strictly increasing, so write  $\gamma = \gamma(M)$ . Dividing (6.9) by  $M^2 M_\gamma$  gives

$$\frac{1}{M} \frac{d\gamma}{dM} - \frac{\gamma}{M^2} \leq \tilde{K} + \frac{d\gamma}{dM},$$

that is,

$$\frac{d}{dM} \left( \frac{\gamma}{M} \right) \leq \tilde{K} + \frac{d\gamma}{dM}. \quad (6.10)$$

At criticality the finite-cluster susceptibility is infinite, so  $M_\gamma(p_c, 0+) = \infty$  and consequently  $\lim_{M \downarrow 0} \gamma(M)/M = 0$ . Integrating (6.10) from 0 to  $M$  and using the elementary inequality  $\gamma \leq M$  gives

$$\frac{\gamma}{M} \leq (\tilde{K} + 1)M, \quad \text{hence} \quad M(p_c, \gamma) \geq c\sqrt{\gamma}. \quad (6.11)$$

This is the Aizenman–Barsky universal critical lower bound.

For the reverse inequality use the triangle lower inequality (6.7). For  $\gamma \leq 1/2$  it gives

$$MM_\gamma \leq C_\nabla.$$

Therefore  $d(M^2)/d\gamma = 2MM_\gamma \leq 2C_\nabla$ , and since  $M(p_c, 0) = 0$ ,

$$M(p_c, \gamma) \leq C\sqrt{\gamma}. \quad (6.12)$$

Together, (6.11)–(6.12) give the critical field exponent  $1/2$ . Let  $N = |C(0)|$ . Taking  $\gamma = 1/n$  in  $M = \mathbf{E}[1 - (1 - \gamma)^N]$  gives immediately  $\mathbf{P}(N \geq n) \leq Cn^{-1/2}$ . For the reverse bound choose  $\tilde{\gamma} = 1/n$  and then  $\gamma = c_0/n$  with  $c_0 > 0$  small. Splitting at  $n$ ,

$$M(p_c, \gamma) \leq \gamma \mathbf{E}[N; N < n] + \mathbf{P}(N \geq n).$$

Moreover, for  $N < n$ ,  $N\tilde{\gamma} \leq e\{1 - (1 - \tilde{\gamma})^N\}$ , hence

$$\gamma \mathbf{E}[N; N < n] \leq e \frac{\gamma}{\tilde{\gamma}} M(p_c, \tilde{\gamma}) \leq C e c_0 n^{-1/2}.$$

Choose  $c_0$  so that this is at most half the lower bound in (6.11). Then

$$\mathbf{P}_{p_c}(N \geq n) \geq cn^{-1/2}, \quad (6.13)$$

which proves  $\delta = 2$ .

It is convenient to put  $\gamma = 1 - e^{-h}$  and  $\widetilde{M}(p, h) = M(p, 1 - e^{-h})$ . Then (6.6) becomes

$$\widetilde{M}_p \leq C_0 \widetilde{M} \widetilde{M}_h. \quad (6.14)$$

Follow a level curve  $\widetilde{M}(p, h(p)) = m$ . Differentiation gives

$$-h'(p) = \frac{\widetilde{M}_p}{\widetilde{M}_h} \leq C_0 m. \quad (6.15)$$



Take  $p_1 = p_c + \varepsilon$  and  $h_1 > 0$ , put  $m_1 = \widetilde{M}(p_1, h_1)$ , and integrate (6.15) backwards to  $p_c$ . The level curve reaches  $p_c$  at a height at most  $h_1 + C_0 m_1 \varepsilon$ . Monotonicity in  $h$  and (6.11)–(6.12) therefore give

$$m_1 \leq C\sqrt{h_1 + C_0 m_1 \varepsilon}.$$

Solving the quadratic inequality yields  $m_1 \leq C\varepsilon + C\sqrt{h_1}$ . Letting  $h_1 \downarrow 0$  gives

$$\theta(p_c + \varepsilon) \leq C\varepsilon. \quad (6.16)$$

We include the Aizenman–Barsky rectangle argument rather than importing a separate sharpness theorem. Rewrite (6.8) exactly as

$$\frac{M_\gamma(p, \gamma)}{M(p, \gamma)} + \frac{1}{\gamma} \partial_p \{pM(p, \gamma) - p\} \geq 0. \quad (6.17)$$

Fix  $a = p_c < b < 1$  and integrate over  $[a, b] \times [\delta, \varepsilon]$ . Monotonicity of  $M$  in both variables bounds the two boundary terms and yields

$$(b - a) \log \frac{M(b, \varepsilon)}{M(a, \delta)} + \log \frac{\varepsilon}{\delta} \{bM(b, \varepsilon) - aM(a, \delta) - (b - a)\} \geq 0. \quad (6.18)$$

Discarding the nonnegative term  $aM(a, \delta)$  can only weaken the desired lower bound. Divide by  $\log(\varepsilon/\delta)$  and let  $\delta \downarrow 0$ . The critical lower field bound (6.11) implies

$$\limsup_{\delta \downarrow 0} \frac{\log M(a, \delta)}{\log \delta} \leq \frac{1}{2}.$$

The logarithm containing  $M(b, \varepsilon)$  is fixed while  $\log(\varepsilon/\delta) \rightarrow \infty$ , so (6.18) gives

$$bM(b, \varepsilon) \geq \frac{1}{2}(b - a).$$

Finally let  $\varepsilon \downarrow 0$ . Since  $M(b, \varepsilon) \downarrow \theta(b)$ ,

$$\theta(b) \geq \frac{b - p_c}{2b}. \quad (6.19)$$

Together with the upper bound (6.16), this proves  $\theta(p) \asymp p - p_c$ . Equations (6.4), (6.13), and (6.16)–(6.19) are exactly the bounded-ratio statements  $\gamma = 1$ ,  $\delta = 2$ , and  $\beta = 1$ .  $\square$

## Chapter 7

# Continuum percolation

Continuum percolation replaces edges by random geometric grains; finite-range coarse graining recovers much of the lattice mechanism.

**Example 7.1** (A running picture). On a square box, think of each edge as open independently with probability  $p$ . The local rule is trivial; the questions below ask when those bits create a path whose diameter is comparable with the box.

**Theorem 7.2** (Non-trivial threshold for bounded-radius Boolean percolation).

*Consider the Poisson–Boolean model in  $\mathbb{R}^d$ ,  $d \geq 2$ , with deterministic radius  $R \in (0, \infty)$  and Poisson intensity  $\lambda$ . There exists*

$$0 < \lambda_c(d, R) < \infty$$

*such that the occupied set has no unbounded component for  $\lambda < \lambda_c$  and has an unbounded component with positive probability for  $\lambda > \lambda_c$ .*

*Proof.* Monotone coupling in the intensity is obtained from a Poisson process on  $\mathbb{R}^d \times [0, \infty)$ : retain points whose second coordinate is at most  $\lambda$ . Thus the set of intensities for which percolation occurs is an interval and the infimum  $\lambda_c$  is well defined.

For positivity of  $\lambda_c$ , partition space into cubes of side  $R/(4\sqrt{d})$ . If the occupied cluster of a fixed cube reaches graph distance  $n$  in the lattice of cubes, there is a nearest-neighbour chain of  $n$  cubes each of which contains at least one Poisson centre in a fixed finite enlargement. A union bound over lattice self-avoiding paths gives an upper bound of the form  $C_d[a_d(1 - e^{-c_d\lambda R^d})]^n$ . This tends to zero for sufficiently small  $\lambda$ , so no unbounded occupied component exists.

For finiteness of  $\lambda_c$ , tile  $\mathbb{R}^d$  by cubes of side  $R/(10\sqrt{d})$  and call a cube good when it contains at least one Poisson point in a small central subcube. Goodness events for distinct cubes are independent and their probability tends to 1 as  $\lambda \rightarrow \infty$ . If adjacent cubes are good, the corresponding radius- $R$  balls overlap. For large  $\lambda$  the good-cube process dominates a supercritical Bernoulli site percolation on  $\mathbb{Z}^d$ , hence the occupied set percolates. Therefore  $0 < \lambda_c < \infty$ .  $\square$

**Theorem 7.3** (Subcritical exponential decay for bounded radii).

*Consider the Poisson–Boolean model in  $\mathbb{R}^d$  with deterministic radius  $R < \infty$  (more generally, with radii uniformly bounded by  $R$ ). If  $\lambda < \lambda_c$ , then there are  $C, c > 0$  such that*

$$\mathbb{P}_\lambda(B_R(0) \longleftrightarrow \mathbb{R}^d \setminus B_t(0)) \leq Ce^{-ct/R}, \quad t \geq R.$$

*Proof.* Because  $\lambda < \lambda_c$ , the occupied component meeting  $B_R(0)$  is almost surely bounded. Hence the decreasing events  $\{B_R(0) \leftrightarrow B_L(0)^c\}$  have probabilities tending to zero as  $L \rightarrow \infty$ . Choose  $L \gg R$  so that this probability is at most a small number  $\varepsilon$  to be fixed below.

Tile space by cubes of side  $L/10$  and call a cube bad if the occupied set meeting its  $R$ -enlargement connects that enlargement to distance  $L/3$ . A bad event depends only on Poisson points in a bounded enlargement of the cube. Thus bad-cube indicators have a finite dependence range, and by translation invariance their marginal probability is at most  $C_d\varepsilon$ .

Any occupied path from  $B_R(0)$  to distance  $t$  produces a nearest-neighbour path of at least  $c_d t/L$  bad cubes. Color the lattice of cubes by finitely many colors so that bad events of one color are independent. Along every cube path, at least a fixed fraction of its vertices have the same color. Therefore a union bound over at most  $a_d^m$  lattice paths of length  $m$  gives

$$\mathbb{P}_\lambda(0 \leftrightarrow B_t^c) \leq C \sum_{m \geq c_d t/L} a_d^m (C_d \varepsilon)^{m/K_d}.$$

Taking  $L$  so large that  $a_d(C_d \varepsilon)^{1/K_d} < 1$  turns the series into a geometric tail, proving exponential decay. Uniformly bounded random radii give the same finite-dependence argument.  $\square$

## Chapter 8

# Dependent and continuum percolation

Two dependent models show how far the sharpness technology travels: Poisson–Boolean geometry and GFF level sets.

**Example 8.1** (A running picture). On a square box, think of each edge as open independently with probability  $p$ . The local rule is trivial; the questions below ask when those bits create a path whose diameter is comparable with the box.

**Lemma 8.2** (Poisson cell product space, Russo derivative, and exploration). *Let  $\eta$  be a Poisson process on  $\mathbb{R}^d \times \mathbb{R}_+$  with intensity  $\lambda dz \mu(dR)$ . Partition it into the independent coordinates*

$$\eta(x, n) = \eta \cap \left( (x + [-\tfrac{1}{2}, \tfrac{1}{2}]^d) \times [n-1, n) \right), \quad (x, n) \in \mathbb{Z}^d \times \mathbb{N}.$$

*For  $A_r = \{0 \leftrightarrow \partial B_r\}$  and  $0 \leq s \leq r$ , there is a decision tree which discovers exactly the occupied components meeting  $\partial B_s$ . After truncating to centers and radii at most  $L$ , the OSSS inequality from Chapter 5 and the Poisson Russo formula imply*

$$\theta_r(1 - \theta_r) \leq C \sum_{x,n} \delta_{x,n}(s, L) \mathbb{E}_\lambda[\text{Piv}_{x,n}(A_r)] + \varepsilon_L,$$

*where  $\varepsilon_L \rightarrow 0$  as  $L \rightarrow \infty$  under the moment hypothesis used below.*

*Proof.* The coordinates are independent because they are restrictions of a Poisson process to disjoint measurable cells. Reveal first the outside coordinate, then reveal a cell  $(x, n)$  whenever the Euclidean distance from its spatial cube to the already discovered occupied component of  $\partial B_s$  is less than  $n$ . A ball from an unrevealed cell cannot touch that component, so when the queue is empty the component has been completely determined; hence the tree determines whether it connects to 0 and to  $\partial B_r$ .

On a finite truncation, replace one coordinate by an independent copy. The martingale proof of the OSSS inequality in Chapter 5 applies verbatim to arbitrary independent coordinate spaces, because it uses only conditional expectation and resampling. The resampling influence is bounded by twice the probability that insertion of a point from that cell is pivotal. For an increasing event, Mecke’s identity differentiates the Poisson intensity:

$$\frac{d}{d\lambda} \mathbb{P}_\lambda(A_r) = \int_{\mathbb{R}^d} \int_0^\infty \mathbb{P}_\lambda((z, R) \text{ is insertion-pivotal for } A_r) \mu(dR) dz.$$

Summing the cell pivotalities therefore recovers  $\theta'_r(\lambda)$ . Finally, the probability that a discarded cell influences  $A_r$  is bounded by the expected number of discarded balls capable of meeting  $B_r$ , namely

an integral of  $(r + R)^d$  over the discarded radii/centers. The stronger  $(5d - 3)$  moment appearing below makes every weighted truncation error in the revealment estimate summable, so monotone convergence gives the untruncated statement.  $\square$

**Lemma 8.3** (Continuum revealment differential inequality). *Assume*

$$\int_0^\infty R^{5d-3} \mu(dR) < \infty.$$

Set  $\theta_r(\lambda) = \mathbb{P}_\lambda(0 \leftrightarrow \partial B_r)$  and

$$\Sigma_r(\lambda) = \int_0^r \theta_s(\lambda) ds.$$

There is  $c > 0$  such that, for  $\lambda \geq \tilde{\lambda}_c$  and  $r \geq 1$ ,

$$\theta'_r(\lambda) \geq c \frac{r}{\Sigma_r(\lambda)} \theta_r(\lambda) (1 - \theta_r(\lambda)).$$

*Proof.* Integrate the OSSS estimate of Lemma 8.2 over  $s \in [0, r]$ . A cell  $(x, n)$  can be revealed only if the occupied cluster of some point within distance  $n + O(1)$  of its spatial cube reaches  $\partial B_s$ . Translating and integrating in  $s$  bounds its total revealment by

$$C(n + r_x + 1) \Sigma_r(\lambda),$$

where  $r_x$  is the bounded correction needed to bring the cube to the explored annulus. If the cell is pivotal for  $A_r$ , insertion tolerance lets us replace the possible long jump through that cell by a bounded-radius local connector, at a fixed multiplicative probability cost. Split  $n$  dyadically. The contributions of large radii are bounded by

$$C \sum_{m \geq 0} 2^{m(5d-3)} \mu[2^m, 2^{m+1}),$$

which is finite by the moment assumption. Thus the integrated revealment-weighted pivotal sum is at most

$$C \frac{\Sigma_r(\lambda)}{r} \theta'_r(\lambda).$$

Since  $\int_0^r \text{Var}(\mathbf{1}_{A_r}) ds = r\theta_r(1 - \theta_r)$ , rearrangement gives the displayed inequality. The role of  $\lambda \geq \tilde{\lambda}_c$  is to supply the uniform annulus-connectivity/insertion bound used when converting resampling influence to insertion pivotality.  $\square$

**Lemma 8.4** (Calculus consequence: equality of thresholds and mean-field lower bound). *If the differential inequality of Lemma 8.3 holds, then*

$$\lambda_c = \tilde{\lambda}_c$$

and there is  $c > 0$  such that

$$\theta(\lambda) \geq c(\lambda - \lambda_c), \quad \lambda \geq \lambda_c.$$

*Proof.* Fix  $\lambda_0 > \tilde{\lambda}_c$ . Apply the inequality simultaneously to all  $r$  and integrate in  $\lambda$ . The standard finite-size alternative follows directly: there is  $\lambda_1 \in [\tilde{\lambda}_c, \lambda_0]$  such that for  $\lambda > \lambda_1$  the limit  $\theta(\lambda)$  is positive and obeys a linear lower bound, whereas for  $\tilde{\lambda}_c < \lambda < \lambda_1$  the quantity  $\Sigma_r/r$  remains

small enough along dyadic scales that iteration gives  $\theta_r(\lambda) \leq e^{-c\lambda r}$ . The latter is impossible for a parameter strictly larger than  $\tilde{\lambda}_c$ , because it forces

$$\mathbb{P}_\lambda(B_r \leftrightarrow \partial B_{2r}) \longrightarrow 0.$$

Hence  $\lambda_1 = \tilde{\lambda}_c$ . By definition  $\tilde{\lambda}_c \leq \lambda_c$ , while positivity immediately above  $\lambda_1$  gives  $\lambda_c \leq \lambda_1$ . Thus the thresholds coincide. Letting  $r \rightarrow \infty$  in the integrated logistic inequality then yields the linear lower bound for  $\theta$ .  $\square$

**Lemma 8.5** (Subcritical renormalization and the one-big-ball term). *For every  $\lambda < \lambda_c$  there are constants  $C, c > 0$  such that the one-arm function satisfies a recursion of the form*

$$\theta_{2r}(\lambda) \leq C\theta_r(\lambda)^2 + Cr^d\mu[r/C, \infty].$$

*Consequently:*

1. if  $\mu[r, \infty] \leq e^{-cr}$ , then  $\theta_r(\lambda) \leq e^{-c\lambda r}$ ;
2. for either of the two tail classes  $\mu[r, \infty] = r^{-(d+a)}$  with  $a > 0$ , or  $\mu[r, \infty] = e^{-cr^\alpha}$  with  $0 < \alpha < 1$ , if

$$\phi_r(\lambda) = \mathbb{P}_\lambda\{\text{one ball contains } 0 \text{ and meets } \partial B_r\},$$

*then  $\theta_r(\lambda)/\phi_r(\lambda) \rightarrow 1$ .*

*In particular the expected Lebesgue volume of the occupied cluster containing the origin is finite throughout the subcritical regime under the  $(5d - 3)$  moment hypothesis.*

*Proof.* If 0 reaches  $\partial B_{2r}$  and no ball of radius at least  $r/C$  participates, then the connection contains two separated crossings of annuli of scale comparable with  $r$ . Reveal the first crossing and use independence of Poisson points in separated buffer regions for the second; a bounded-overlap covering of the sphere gives the quadratic term. The complementary event is bounded by the expected number of balls of radius at least  $r/C$  whose centers lie within distance  $O(R + r)$  of the origin, giving the tail term.

When the radius tail is exponential, choose a scale at which  $C\theta_r < 1/4$  and iterate the recursion, absorbing the exponentially small error. In the two heavy-tail classes, split a connection according to its largest ball. The probability of two or more radius-comparable exceptional balls is  $o(\phi_r)$ , while the quadratic multi-step term is also  $o(\phi_r)$  under the displayed regular tails. A single ball giving the event has probability exactly  $\phi_r + o(\phi_r)$ , proving the ratio limit.

For expected cluster volume use Fubini:

$$\mathbb{E}_\lambda|C(0)| = \int_{\mathbb{R}^d} \mathbb{P}_\lambda(0 \leftrightarrow x) dx.$$

Dyadic annuli reduce this to  $\sum_m 2^{md}\theta_{2^m}$ . Iterating the recursion and using the  $(5d - 3)$  moment makes this series finite.  $\square$

**Lemma 8.6** (Vacant-set sharpness for compactly supported radii). *If the radius law is compactly supported, let*

$$\theta_r^*(\lambda) = \mathbb{P}_\lambda(0 \overset{*}{\longleftrightarrow} \partial B_r), \quad \lambda_c^* = \sup\{\lambda : \theta^*(\lambda) > 0\}.$$

*Then for  $\lambda > \lambda_c^*$  the vacant one-arm probability decays exponentially, while for  $\lambda < \lambda_c^*$  the probability of a vacant connection to infinity satisfies a linear lower bound in  $\lambda_c^* - \lambda$ .*

*Proof.* Run the same cell exploration on the decreasing event that the vacant component of  $\partial B_s$  reaches the two marked sets. Compact support removes the long-radius truncation term entirely. OSSS is insensitive to whether the event is increasing or decreasing; Poisson Russo differentiation changes sign, so after writing the parameter as  $t = -\lambda$  one obtains the same logistic differential inequality. The calculus proof of Lemma 8.4 then gives the equality of the strong and ordinary vacant thresholds, exponential decay on the nonpercolating side, and the mean-field lower bound on the percolating side.  $\square$

**Theorem 8.7** (Sharp Poisson–Boolean phase transition and vacant set). *Let  $d \geq 2$  and assume*

$$\int_0^\infty R^{5d-3} \mu(dR) < \infty.$$

*For occupied Poisson–Boolean percolation,*

$$\lambda_c = \tilde{\lambda}_c, \quad \theta(\lambda) \geq c(\lambda - \lambda_c) \quad (\lambda \geq \lambda_c),$$

*and the subcritical cluster has finite expected volume. If the radius tail is exponential, subcritical one-arm probabilities decay exponentially; for the two regularly heavy-tailed classes in Lemma 8.5, the one-arm event is asymptotically the one-big-ball event.*

*For the vacant set, if  $R \leq R_0$  almost surely for some deterministic  $R_0 < \infty$ , the nonpercolating side has exponential one-arm decay and the percolating probability is bounded below by a positive constant times the distance to the critical intensity.*

*Proof.* The occupied-set assertions are exactly the conclusions of the four lemmas proved above, and we record the dependency rather than hide it in a slogan. Lemma 8.2 constructs the independent-coordinate exploration and the Poisson Russo derivative. Under the stated  $(5d - 3)$ -moment assumption, Lemma 8.3 converts OSSS into

$$\theta'_r(\lambda) \geq c \frac{r}{\Sigma_r(\lambda)} \theta_r(\lambda) (1 - \theta_r(\lambda)). \quad (10.8)$$

Lemma 8.4 integrates (10.8), proving  $\lambda_c = \tilde{\lambda}_c$  and  $\theta(\lambda) \geq c(\lambda - \lambda_c)$  on the supercritical side. Lemma 8.5 gives

$$\theta_{2r} \leq C\theta_r^2 + Cr^d \mu[r/C, \infty], \quad (10.9)$$

from which exponential subcritical decay follows for exponential radius tails, the one-big-ball asymptotics follow for the two stated regularly heavy-tailed classes, and the dyadic summation proves finite expected occupied cluster volume throughout the subcritical regime.

For the vacant set, compact support eliminates the large-radius error in the same exploration. Lemma 8.6 applies OSSS to the decreasing vacant-connection event, with Poisson Russo differentiation of the opposite sign; after the change of variable  $t = -\lambda$  it is the same logistic differential inequality. Its calculus consequence gives the vacant threshold equality, exponential decay on the nonpercolating side, and the linear lower bound on the percolating side. These are precisely the claims of the theorem.  $\square$

**Definition 8.8** (Three GFF level-set thresholds). Let  $d \geq 3$ , let  $\varphi = (\varphi_x)_{x \in \mathbb{Z}^d}$  be the massless Gaussian free field with covariance the simple-random-walk Green function, and write

$$E_h = \{x : \varphi_x \geq h\}.$$

The ordinary percolation threshold is

$$h_* = \inf\{h : \mathbb{P}(0 \xleftrightarrow{E_h} \infty) = 0\}.$$

The strongly nonpercolative threshold is

$$h_{**} = \inf\left\{h : \inf_{R \geq 1} \mathbb{P}(B_R \xleftrightarrow{E_h} \partial B_{2R}) = 0\right\}.$$

For  $\alpha \in \mathbb{R}$  let  $\text{Exist}(R, \alpha)$  denote the event that  $E_\alpha \cap B_R$  contains a connected component of diameter at least  $R/5$ , and let  $\text{Unique}(R, \alpha)$  denote the event that any two components of  $E_\alpha \cap B_R$  having diameter at least  $R/10$  are connected inside  $E_\alpha \cap B_{2R}$ . We say that the field strongly percolates above level  $h$  if there are  $c, \rho > 0$  such that, uniformly in  $\alpha \leq h$ ,

$$\mathbb{P}(\text{Exist}(R, \alpha)^c) + \mathbb{P}(\text{Unique}(R, \alpha)^c) \leq e^{-cR^\rho}, \quad R \geq 1,$$

and define  $\bar{h}$  as the supremum of such  $h$ .

**Lemma 8.9** (Finite-range decomposition of the massless GFF). *There exist independent stationary centered Gaussian fields  $\xi^\ell = (\xi_x^\ell)_{x \in \mathbb{Z}^d}$ ,  $\ell \geq 0$ , such that*

$$\varphi = \sum_{\ell \geq 0} \xi^\ell \quad \text{in } L^2 \text{ and almost surely,} \quad \varphi^L := \sum_{0 \leq \ell \leq L} \xi^\ell,$$

and  $\xi^\ell$  has dependence range exactly  $\ell$ . Moreover

$$\sup_x \text{Var}(\xi_x^\ell) \leq C(1 + \ell)^{-d/2}.$$

*Proof.* Insert a midpoint into every nearest-neighbour edge of  $\mathbb{Z}^d$  and let  $\tilde{q}_n$  be the transition kernel of the resulting bipartite walk. For every  $\ell$  take an independent family  $(Z_\ell(z))_z$  of standard Gaussians and put

$$\xi_x^\ell = c_\ell \sum_z \tilde{q}_\ell(x, z) Z_\ell(z), \quad c_\ell^2 = \begin{cases} d/2, & \ell \text{ odd,} \\ 1/2, & \ell \text{ even.} \end{cases}$$

Independence in  $\ell$  is built into the construction. Reversibility and the Chapman–Kolmogorov identity give

$$\mathbb{E}[\xi_x^\ell \xi_y^\ell] = \frac{1}{2} \tilde{q}_{2\ell}(x, y).$$

The bipartite walk can cover at most graph distance  $\ell$  in  $\ell$  steps, so the right side is zero when  $|x - y| > \ell$ ; for a Gaussian field, zero covariance of two separated families is independence, proving the range statement. Summing in  $\ell$  telescopes the even-time expansion of the ordinary Green kernel and hence

$$\sum_{\ell \geq 0} \mathbb{E}[\xi_x^\ell \xi_y^\ell] = g(x, y).$$

The local central limit estimate  $\tilde{q}_{2\ell}(x, x) \leq C(1 + \ell)^{-d/2}$  yields the variance bound. Since  $d \geq 3$ , the variances are summable, so the independent Gaussian series converges in  $L^2$  and almost surely and has covariance  $g$ ; it is therefore the GFF.  $\square$



**Lemma 8.10** (Multiscale bridge and reconstruction cost). *Fix  $\varepsilon > 0$ . There are  $L_0, \ell_0, K, c, \rho > 0$  and scales  $L_n = L_0 \ell_0^n$  with the following property. With probability at least  $1 - e^{-cL_n^\rho}$ , every pair of connected sets  $S_1, S_2 \subset B_{10\kappa L_n}$  of diameter at least  $\kappa L_n$  can be joined by a bridge consisting of at most  $K$  boxes at each scale. Every  $L_m$ -box of the bridge lies at distance at least  $\kappa L_m$  from  $S_1 \cup S_2$ , and on the bridge*

$$\varphi - \varphi^{L_m} \geq -\varepsilon_m, \quad \sum_{m \geq 1} \varepsilon_m < \varepsilon.$$

*Consequently, in the effective-critical window  $h < h_{**} - 2\varepsilon$ , after conditioning on the field on  $S_1 \cup S_2$  and outside the bridge, one has*

$$\mathbb{P}(S_1 \xleftrightarrow{E_{h-\varepsilon}} S_2 \text{ inside } B_{10\kappa L_n} \mid \mathcal{F}) \geq \exp\{-C(\log L_n)^2\}.$$

*Proof.* Call an  $m$ -box bad if the tail  $\varphi - \varphi^{L_m}$  drops below the prescribed allowance  $-\varepsilon_m$  somewhere in its twofold enlargement. Gaussian concentration together with Lemma 8.9 gives a base estimate small enough, after choosing  $L_0$ , that the following recursive rule is stable: an  $(m+1)$ -box is bad only if it contains two well-separated bad  $m$ -boxes or one of finitely many Gaussian tail events fails. Finite-range independence for separated boxes and a union bound therefore give

$$p_{m+1} \leq C\ell_0^{2d} p_m^2 + e^{-cL_m^\rho}.$$

Choosing  $p_0$  small yields  $p_m \leq e^{-cL_m^\rho}$ .

On the complementary good event, construct two arches from  $S_1, S_2$  and connect them at the largest scale; whenever an arch meets a bad child box, route around that box and recurse only inside the bounded collection of children adjacent to its boundary. The quadratic bad-box rule implies that these detours cannot proliferate: at each scale the bridge contains at most  $K$  boxes. Enlarging each selected box by a fixed factor gives the distance condition, which is exactly what preserves conditional independence from the already revealed endpoint clusters.

Inside an  $m$ -box the truncated field has range  $L_m$  and the tail is at least  $-\varepsilon_m$ . Since  $h < h_{**} - 2\varepsilon$ , failure to make the required local crossing at level  $h - \sum_{j \geq m} \varepsilon_j$  has probability bounded away from one at every scale. Reveal the bridge from coarse to fine and use the spatial separation to multiply the conditional lower bounds. There are  $O(1)$  boxes at each of  $O(\log L_n)$  scales, and the smallest-scale finite-energy repair costs at worst a polynomial factor per box. Thus the logarithm of the total cost is  $O((\log L_n)^2)$ , giving the displayed reconstruction bound. This is the quantitative reason for using bridges rather than a single macroscopic corridor.  $\square$

**Lemma 8.11** (Sharpness for noised finite-range truncations). *For  $L < \infty$  and  $\delta \in (0, 1)$ , independently resample each vertex of the threshold configuration  $\{\varphi^L \geq h\}$  with probability  $\delta$  and call the result  $T_\delta\{\varphi^L \geq h\}$ . If  $\tilde{h}(\delta, L)$ ,  $h_*(\delta, L)$  and  $h_{**}(\delta, L)$  are the analogues of the three finite-size thresholds for this family, then*

$$\tilde{h}(\delta, L) = h_*(\delta, L) = h_{**}(\delta, L).$$

*Proof.* The field  $\varphi^L$  has finite dependence range  $L$ , while the independent resampling gives uniform insertion and deletion tolerance. Group the vertices into boxes of side  $20L$ . On the subcritical side, explore the cluster of the inner boundary of an annulus box by box. The martingale/decision-tree proof of the OSSS inequality from Chapter 5 applies to the independent Gaussian coordinates generating the finitely many  $\xi^\ell$  together with the resampling variables. Gaussian integration by

parts and the monotonicity in  $h$  turn the total influence into  $-\partial_h$  of the crossing probability. As in the finite-range sharpness calculation, this yields

$$-\partial_h \theta_R(h) \geq c \frac{R}{\sum_{r \leq R} \theta_r(h)} \theta_R(h) (1 - \theta_R(h)),$$

which implies exponential one-arm decay for every  $h > h_*(\delta, L)$  and hence  $h_{**}(\delta, L) = h_*(\delta, L)$ .

For  $h < h_*(\delta, L)$  choose a slab thick enough that the coarse box process percolates. Because boxes at distance greater than the finite range are independent and the resampling gives finite energy, the standard block gluing proof can be reproduced directly: a left-right crossing in a  $3N \times N$  coarse rectangle has probability tending to one, two such crossings can be joined with a uniformly positive conditional probability, and a Peierls count makes the probability of two macroscopic clusters not joining decay exponentially in the coarse scale. Hence the model is locally unique throughout the supercritical regime. In particular the disconnection parameter  $\tilde{h}(\delta, L)$  is at least  $h_*(\delta, L)$ . The reverse inequality follows from the just-proved exponential decay above  $h_*$ , and all three thresholds coincide.  $\square$

**Lemma 8.12** (Interpolation comparison: integrating out long-range dependence). *Let*

$$u(R) = \exp\{(\log R)^{1/3}\}, \quad \tilde{h} = \sup \left\{ h : \inf_R R^d \mathbb{P}(B_{u(R)} \not\longleftrightarrow \partial B_R \text{ in } E_h) = 0 \right\}.$$

*If  $\tilde{h} < h_{**}$ , then for every  $\varepsilon > 0$  with  $6\varepsilon < h_{**} - \tilde{h}$  there exist  $L < \infty$ ,  $\delta > 0$  and  $c, C > 0$  such that, uniformly for  $h \in (\tilde{h} + 3\varepsilon, h_{**} - 3\varepsilon)$  and  $R \geq 2r$ ,*

$$\begin{aligned} \mathbb{P}(B_r \xleftrightarrow{E_{h-\varepsilon}} \partial B_R) &\geq \mathbb{P}(B_r \longleftrightarrow \partial B_R \text{ in } T_\delta\{\varphi^L \geq h\}) - \epsilon_r, \\ \mathbb{P}(B_r \xleftrightarrow{E_{h+\varepsilon}} \partial B_R) &\leq \mathbb{P}(B_r \longleftrightarrow \partial B_R \text{ in } T_\delta\{\varphi^L \geq h\}) + \epsilon_r, \end{aligned}$$

where  $\epsilon_r \leq C \exp\{-\exp(c(\log r)^{1/3})\}$ .

*Proof.* Interpolate between truncation levels by

$$\chi^t = \sum_{\ell \leq L_t} \xi^\ell + \text{extra continuously switched fraction of the next block}, \quad t \geq 0,$$

so that increasing  $t$  restores progressively longer scales. For

$$\theta(t, h; r, R) = \mathbb{P}(B_r \longleftrightarrow \partial B_R \text{ in } \{\chi^t \geq h\})$$

Gaussian differentiation expresses  $\partial_t \theta$  as a sum of coarse pivotal terms weighted by the covariance increment, while  $-\partial_h \theta$  is the sum of ordinary pivotal intensities.

A coarse pivotal at  $x$  says that the two endpoint clusters reach the boundary of a box of radius comparable with  $L_t$  around  $x$  without yet meeting. Lemma 8.10 reconstructs an actual pivotal vertex in that box with conditional cost at least  $\exp\{-C(\log L_t)^2\}$ . The assumption  $h \in (\tilde{h} + 2\varepsilon, h_{**} - 2\varepsilon)$  prevents both the required connection and disconnection events from becoming too rare; this is the effective-critical regime in which reconstruction is possible. Splitting into the first good bridge scale and using the stretched-exponential failure probability gives

$$|\partial_t \theta(t, h; r, R)| \leq 2e^{-t} (-\partial_h \theta(t, h; r, R)) + Ce^{-t} \exp\{-\exp(c(\log r)^{1/3})\}.$$

Therefore the two functions

$$\theta(t, h \pm 2e^{-t}; r, R) \mp Ce^{-t} \exp\{-\exp(c(\log r)^{1/3})\}$$

have the required opposite monotonicities. Integrating from a large finite truncation time to infinity yields the two comparisons for  $\delta = 0$ . Finally couple the independent resampling variables monotonically. Their effect can be absorbed by a further level shift smaller than  $\varepsilon$  when  $\delta$  is chosen sufficiently small, because the finite truncation has a bounded Gaussian density at every site. This proves the displayed noised comparison.  $\square$

**Lemma 8.13** (The fictitious interval is empty). *For  $d \geq 3$ ,*

$$\tilde{h} = h_{**}.$$

*Proof.* Always  $\tilde{h} \leq h_{**}$ . Suppose the inequality were strict and put  $\varepsilon = (h_{**} - \tilde{h})/8$ . Choose  $L, \delta$  from Lemma 8.12. If some

$$h \in (h_{**}(\delta, L), \infty) \cap (\tilde{h} + 3\varepsilon, h_{**} - 3\varepsilon)$$

existed, the upper comparison would force the full GFF at level  $h + \varepsilon < h_{**}$  to have annulus-crossing probability tending to zero along a sequence of radii, contradicting the definition of  $h_{**}$ . Thus the first interval is disjoint from the fictitious window. The lower comparison similarly shows that  $(-\infty, \tilde{h}(\delta, L))$  is disjoint from that window. This is impossible because Lemma 8.11 gives  $\tilde{h}(\delta, L) = h_{**}(\delta, L)$ . Hence the fictitious window is empty and  $\tilde{h} = h_{**}$ .  $\square$

**Lemma 8.14** (From connection abundance to local uniqueness). *For  $d \geq 3$ ,*

$$\tilde{h} \leq \bar{h}.$$

*Proof.* Fix  $h < \tilde{h}$  and choose  $\varepsilon > 0$  with  $h + 2\varepsilon < \tilde{h}$ . By the definition of  $\tilde{h}$ , for every sufficiently large  $R$  all boxes of radius  $u(R)$  inside  $B_R$  connect to distance  $R$  in  $E_{h+2\varepsilon}$  with probability  $1 - o(R^{-d})$ . A union bound therefore produces, with overwhelming probability, a family of macroscopic clusters intersecting every mesoscopic box.

Apply the multiscale bridge of Lemma 8.10 to any two such clusters. Lowering the level from  $h + 2\varepsilon$  to  $h + \varepsilon$  pays the bridge reconstruction cost and joins each pair with probability at least  $1 - \exp(-cR^\rho)$  after the recursive good-box estimate. A second sprinkling to level  $h$  handles the finitely many boundary and endpoint repairs. The same coarse construction supplies a cluster of diameter comparable with  $R$  in  $B_R$ . Hence both  $\text{Exist}(R, h)$  and  $\text{Unique}(R, h)$  fail with stretched-exponentially small probability. Since  $h < \tilde{h}$  was arbitrary, strong percolation holds below every such level and  $\tilde{h} \leq \bar{h}$ .  $\square$

**Theorem 8.15** (Equality of the three GFF level-set critical parameters). *For the massless Gaussian free field on  $\mathbb{Z}^d$ ,  $d \geq 3$ ,*

$$\boxed{\bar{h}(d) = h_*(d) = h_{**}(d)}.$$

*Thus every  $h > h_*$  belongs to the strongly nonpercolative regime, while every  $h < h_*$  belongs to the local-uniqueness/strong-percolation regime of Definition 8.8.*

*Proof.*  $\bar{h} \leq h_* \leq h_{**} = \tilde{h} \leq \bar{h}$  proves the asserted equality. The finite-range decomposition, bridge, interpolation, and local-uniqueness arguments above together prove the asserted decoupling/sprinkling statement; no additional lemma is being invoked.  $\square$

## Chapter 9

# Classical sharpness, planar RSW, and Voronoi percolation

This supplement records three foundational mechanisms that should not disappear behind modern shorthand: Aizenman–Barsky tree inequalities and the RSW theory of Tassion.

**Example 9.1** (A running picture). On a square box, think of each edge as open independently with probability  $p$ . The local rule is trivial; the questions below ask when those bits create a path whose diameter is comparable with the box.

**Theorem 9.2** (Sharpness for finite-range independent percolation).

*For Bernoulli bond percolation on a transitive finite-range graph, the susceptibility threshold equals the percolation threshold. In particular, for every  $p < p_c$  there are  $c(p), C(p) > 0$  such that*

$$\mathbf{P}_p(0 \leftrightarrow x) \leq C(p)e^{-c(p)d(0,x)},$$

so  $\chi(p) < \infty$ ; for  $p > p_c$ ,  $\theta(p) > 0$ .

*Deduction from the sharpness theorem.* This is an exact specialization of the sharpness theorem SM-10.02.01, already proved in this volume for locally finite transitive graphs. That theorem gives finite susceptibility below  $p_c$ , positive percolation above  $p_c$ , and, for finite-range interactions, exponential subcritical decay. The present theorem adds no hypothesis or conclusion beyond that earlier result; its role is to record the classical Aizenman–Barsky location in the historical supplement.  $\square$

**Theorem 9.3** (Aizenman–Newman tree-graph inequalities).

*Let  $\tau_p(x, y) = \mathbf{P}_p(x \leftrightarrow y)$  and  $C(0)$  be the open cluster of 0. For every  $m \geq 2$ , the probability that  $m$  prescribed vertices lie in one open cluster is bounded by the sum over binary connection trees: after introducing the  $m - 2$  branch vertices and summing over their locations, each tree contributes the product of its  $2m - 3$  two-point functions. Consequently*

$$\mathbf{E}_p |C(0)|^k \leq C_k \chi(p)^{2k-1}, \quad k \geq 1,$$

whenever  $\chi(p) < \infty$ . Moreover  $\chi'(p) \leq C\chi(p)^2$  in finite-range transitive models; hence if  $\chi(p) \asymp (p_c - p)^{-\gamma}$  exists, then  $\gamma \geq 1$ .

*Proof.* Take marked vertices  $x_1, \dots, x_m$  in one cluster and choose a minimal open connected subgraph joining them. Suppress degree-two vertices. What remains is a tree whose leaves are the marked vertices and which has at most  $m - 2$  branch vertices. After splitting vertices if necessary, we

may take a binary tree with exactly  $m - 2$  internal vertices and  $2m - 3$  edges. The corresponding edge-to-edge connections can be chosen to occur disjointly. BK therefore bounds the probability for a fixed tree and fixed branch locations by

$$\prod_{\{u,v\} \in E(T)} \tau_p(u, v).$$

Sum over the finitely many binary tree topologies and over all branch vertices. This is the tree-graph inequality.

To get cluster moments, write

$$\mathbf{E}|C(0)|^k = \sum_{x_1, \dots, x_k} \mathbf{P}(0, x_1, \dots, x_k \text{ lie in one cluster}).$$

Apply the tree-graph bound with  $m = k + 1$ . Sum the leaf and branch variables successively from the outside of the tree inward. Every summed edge contributes at most one factor  $\chi(p) = \sum_x \tau_p(0, x)$ , and there are  $2k - 1$  edges. The number of tree topologies is a constant depending only on  $k$ , giving the displayed moment estimate.

Finally Russo's formula for  $\chi(p) = \sum_x \mathbf{P}_p(0 \leftrightarrow x)$  says that its derivative is a sum over pivotal edges. If  $e = uv$  is pivotal for  $0 \leftrightarrow x$ , then with  $e$  deleted there are disjoint connections  $0 \leftrightarrow u$  and  $v \leftrightarrow x$  (or the reversed orientation). BK, finite degree, and summation over  $x, e$  give

$$\chi'(p) \leq C\chi(p)^2.$$

Integrating  $-(1/\chi)' = \chi'/\chi^2 \leq C$  up to  $p_c$ , where susceptibility diverges by sharpness, yields  $\chi(p) \geq c(p_c - p)^{-1}$ . Thus any susceptibility exponent must satisfy  $\gamma \geq 1$ .  $\square$

**Theorem 9.4** (Universal planar RSW).

*Let  $\mathbb{P}$  be a bond-percolation law on the square lattice invariant under translations, quarter-turns and reflections, and assume positive association. For  $m, n \geq 1$ , let  $\mathcal{C}(m, n)$  be the event of an open left-right crossing of  $[-m, m] \times [-n, n]$ . Then for every  $\rho \geq 1$  there is an increasing homeomorphism  $\psi_\rho : [0, 1] \rightarrow [0, 1]$ , depending only on  $\rho$ , such that*

$$\mathbb{P}(\mathcal{C}(\rho n, n)) \geq \psi_\rho(\mathbb{P}(\mathcal{C}(n, \rho n))) \quad (n \geq 1). \quad (9.12)$$

*The same proof applies to an admissible planar site/continuum model whenever the four inputs used below remain valid: planar crossing duality, the stated symmetries, positive association, and the deterministic path-gluing/corridor topology.*

*Proof.* The point of the proof is that no spatial independence or Markov property is used. Throughout set

$$\alpha = \frac{1}{64}, \quad \beta = \frac{1}{12}.$$

An *arm event*  $\mathcal{A}(r)$  is a top-bottom open crossing between the two central target intervals of width  $2\alpha r$  in  $[-(1 + \alpha)r, (1 + \alpha)r] \times [-r, r]$ . A *bridge event*  $\mathcal{B}(r)$  is an open connection in  $[-(1 + \beta)r, (1 + \beta)r] \times [-r, r]$  from the left boundary portion outside the central window to the corresponding right boundary portion. Put

$$a(r) = \mathbb{P}(\mathcal{A}(r)), \quad b(r) = \mathbb{P}(\mathcal{B}(r)).$$

The dual events are denoted by stars.

If increasing events  $E_1, \dots, E_k$  cover an increasing event  $E$ , positive association applied to the complements gives

$$\max_i \mathbb{P}(E_i) \geq 1 - (1 - \mathbb{P}(E))^{1/k}. \quad (9.13)$$

Define

$$f_1(x) = (1 - (1 - x)^{1/2000})^{2000}.$$

Then elementary planar gluing gives, for every  $r \geq 1$ ,

$$\mathbb{P}(\mathcal{C}(8r, r)) \geq f_1(b(r)), \quad (9.14)$$

$$a(r) \geq f_1(\mathbb{P}(\mathcal{C}(r, 8r))), \quad (9.15)$$

and, for every integer  $j \geq 1$ ,

$$\mathbb{P}(\mathcal{C}(jr, r)) \geq \mathbb{P}(\mathcal{C}(2r, r))^{2j-1}. \quad (9.16)$$

Here is the gluing argument rather than a citation to it. A bridge at scale  $r$  can occur in finitely many symmetric geometric positions. By (9.13), either a definite one-sided bridge piece or a slightly-long rectangle crossing has probability at least  $1 - (1 - b(r))^{1/5}$ . In the first case, reflect the one-sided piece and use FKG to make a bottom connection, then intersect at most  $8/\alpha$  horizontal translates; in the second case, intersect two overlapping horizontal crossings with a vertical square crossing and iterate. Since both numbers of factors are  $< 2000$ , (9.14) follows. Apply the same argument to the dual process. Planar duality gives  $\mathbb{P}(\mathcal{C}^*(8r, r)) = 1 - \mathbb{P}(\mathcal{C}(r, 8r))$  and identifies the complement of the dual bridge variant with the arm event, yielding (9.15). The same three-rectangle gluing with overlap  $r$  gives (9.16).

For  $R \geq r$ , let  $\mathcal{Q}(R, r)$  be the event that in  $[-(1 + 3\beta)R, (1 + 3\beta)R] \times [-R, R]$  there are two open  $r$ -paths, one connected to the left side and one to the right side. Write  $q(R, r) = \mathbb{P}(\mathcal{Q}(R, r))$ .

We shall repeatedly use the following deterministic fact. Let two disjoint  $r$ -paths form the inner parts of a left wall and a right wall in a larger rectangle. Every continuous top–bottom path either contains an  $r$ -path itself or meets one of the two walls. Indeed, remove the top–bottom path  $\pi$  from the inner rectangle. If  $\pi$  contains no  $r$ -path, the two horizontal boundary portions remain joined by a continuous curve  $\eta$  in the complement of  $\pi$ . The curve  $\eta$  meets both inner walls, so the connected set obtained from the two walls and  $\eta$  contains a left–right curve. Jordan separation forces  $\pi$  to meet that connected set; since  $\pi \cap \eta = \emptyset$ , it hits one of the walls. This is the corridor lemma.

For a power-of-four scale  $r$ ,

$$q(4r, r) \geq \left( \min_{u \in [r, 4r]} a(u) \right)^6. \quad (9.17)$$

To see this, take two horizontally shifted arm events at scale  $r$  and two vertically shifted arm events at scale  $5r/2$ . Their target intervals interlace because  $5\alpha < \beta$ . On their intersection there are two open  $r$ -paths attached to the top and bottom of the  $4r$  rectangle. Add two horizontal short-direction crossings in the remaining left and right slabs. The corridor lemma forces the required left and right attachments, hence  $\mathcal{Q}(4r, r)$ . FKG multiplies the six probabilities; each is bounded below by  $\min_{u \in [r, 4r]} a(u)$ , proving (9.17).

For  $R \geq u \geq r \geq 1$ ,

$$\max \left\{ q(R, r), \frac{1 - (1 - b(u))^2}{q(u, r)} \right\} \geq 1 - (1 - q(R, u))^{1/2}. \quad (9.18)$$

On  $\mathcal{Q}(R, u)$  there are two open  $u$ -paths attached to opposite outer sides. If both contain an  $r$ -path, then  $\mathcal{Q}(R, r)$  occurs. Otherwise let  $E$  be the increasing event that there is an open  $u$ -path with no

$r$ -subpath. By (9.13), either  $\mathcal{Q}(R, r)$  or  $E$  has probability at least the right side of (9.18). Positive association gives

$$\mathbb{P}(E) \leq \frac{\mathbb{P}(E \cap \mathcal{Q}(u, r))}{q(u, r)}.$$

On  $E \cap \mathcal{Q}(u, r)$  the corridor lemma says that the exceptional  $u$ -path must hit the left or right wall, producing one of two translated bridge events at scale  $u$ . The complements of these translated bridge events are decreasing, so FKG yields

$$\mathbb{P}(E \cap \mathcal{Q}(u, r)) \leq 1 - (1 - b(u))^2.$$

Substitution proves (9.18).

For  $R \geq r$ ,

$$b(R) \geq q(R, r)b(r). \quad (9.19)$$

Indeed, on  $\mathcal{Q}(R, r)$  the two  $r$ -walls reach the outer left and right sides; intersecting with the bridge at scale  $r$  joins those walls. The resulting connected open set contains an outer bridge at scale  $R$ . Both events are increasing, so FKG proves (9.19).

For integers  $i \geq 0$  define

$$f_i(x) = \left(1 - (1 - x)^{1/2000^i}\right)^{2000^i}. \quad (9.20)$$

These are increasing homeomorphisms of  $[0, 1]$ ,  $f_i \geq f_j$  when  $i \leq j$ , and

$$f_i(x) \geq \sqrt{f_{i+1}(x)}, \quad x \geq f_i(1 - y) \iff y \geq f_i(1 - x), \quad f_i \circ f_i \geq f_{2i}. \quad (9.21)$$

The first two assertions follow directly from the definition. For the third, write  $a = 2000^i$  and use  $(1 - (1 - x)^{1/a})^a \leq 1 - (1 - x^a)^{1/a}$ , obtained after the substitution  $u = (1 - x)^{1/a}$  and monotonicity of  $t \mapsto t^a$ . Reapplying this inequality proves the composition estimate.

The induction below needs one numerical consequence. Directly from (9.20)–(9.21),

$$\frac{1 - (1 - f_1^{-1} \circ f_{18}(1 - x))^2}{f_4(1 - f_1^{-1}(x))} \leq f_8(1 - x). \quad (9.22)$$

For completeness:  $1 - (1 - y)^2 \leq f_1^{-1}(y)$ , so the numerator is at most  $f_1^{-1} \circ f_1^{-1} \circ f_{18}(1 - x) \leq f_8(1 - x)^2$ ; while  $1 - f_1^{-1}(x) = f_1(1 - x)$  and therefore the denominator is at least  $f_4 \circ f_4(1 - x) \geq f_8(1 - x)$ . Division gives (9.22). The same calculus also gives

$$f_{18}(1 - \sqrt{x})^2 \geq f_{20}(1 - x). \quad (9.23)$$

Here is a direct verification, since this inequality is used at the last step. Put  $a = 2000^{18}$  and  $M = 2000^2$ . With  $u = x^{1/(2a)}$  the desired inequality is equivalent to

$$(1 - u)^2 \geq (1 - u^{2/M})^M.$$

After taking the  $M$ -th root this becomes  $(1 - u)^{2/M} + u^{2/M} \geq 1$ , which holds because  $0 < 2/M \leq 1$  and  $t \mapsto t^{2/M}$  is concave on  $[0, 1]$ .

Let  $b^*(r)$  be the dual bridge probability. We prove, first for  $n = 4^N$ ,

$$b(n) \geq f_{20}(1 - b^*(n)). \quad (9.24)$$

Choose  $m_0 = 4^j \leq n$  maximal such that

$$b(m_0) \geq f_{18}(1 - b^*(m_0)). \quad (9.25)$$

The set is nonempty: the one-edge-scale version of (9.14) and its dual imply the weaker  $f_1$  relation, and  $f_1 \geq f_{18}$ . If  $m_0 = n$ , (9.24) follows. Otherwise put  $m_k = 4^k m_0$  up to  $m_K = n$ .

We claim inductively that

$$q(m_k, m_0) \geq f_4(1 - b^*(m_k)), \quad 1 \leq k \leq K. \quad (9.26)$$

By (9.17), (9.15), inclusion of rectangle crossings across the interval  $[m_{k-1}, m_k]$ , and duality,

$$q(m_k, m_{k-1}) \geq f_2(1 - b^*(m_k)).$$

This gives  $k = 1$ . For the induction step apply the cascading estimate (9.18) to  $(m_k, m_{k-1}, m_0)$ . If the first term in its maximum is not already at least  $f_4(1 - b^*(m_k))$ , the second term would be that large. We show that this is impossible. The gluing inequality (9.14), applied at  $m_{k-1}$  and embedded in the bridge geometry at scale  $m_k = 4m_{k-1}$ , gives

$$f_1(b(m_{k-1})) \leq b(m_k), \quad f_1(b^*(m_{k-1})) \leq b^*(m_k).$$

Consequently

$$\frac{1 - (1 - b(m_{k-1}))^2}{f_4(1 - b^*(m_{k-1}))} \leq \frac{1 - (1 - f_1^{-1}(b(m_k)))^2}{f_4(1 - f_1^{-1}(b^*(m_k)))}. \quad (9.27a)$$

Maximality of  $m_0$  means  $b(m_k) < f_{18}(1 - b^*(m_k))$  for  $k \geq 1$ . Substitute this bound in the numerator of (9.27a) and use (9.22). The right-hand side is at most  $f_8(1 - b^*(m_k))$ , which is strictly below  $f_4(1 - b^*(m_k))$  for an interior argument. (At the endpoints the theorem is immediate.) Hence the second term in the cascading maximum cannot be the large one, and (9.26) follows.

At  $k = K$ , (9.26), maximality, and (9.21) imply  $q(n, m_0) \geq \sqrt{b(n)}$ . By closing,  $b(n) \geq q(n, m_0)b(m_0)$ , hence  $b(n) \geq b(m_0)^2$ . The whole argument is symmetric under primal–dual exchange, so also  $b^*(n) \geq b^*(m_0)^2$ . Combining these two inequalities with (9.25) and then (9.23) yields (9.24).

For powers of four, combine (9.14), (9.24), the dual form of (9.14), and planar duality:

$$\mathbb{P}(\mathcal{C}(8n, n)) \geq \psi_2(\mathbb{P}(\mathcal{C}(n, 8n))),$$

where

$$\psi_2(x) = f_1 \circ f_{20}(1 - f_1^{-1}(1 - x)).$$

This is an increasing homeomorphism of  $[0, 1]$ . For an arbitrary  $n$ , choose adjacent powers of four and use inclusion of crossing events to obtain the same conclusion for aspect ratio 2 after composing  $\psi_2$  with itself a bounded number of times. For integer  $\rho$ , (9.16) concatenates overlapping  $2 : 1$  crossings and therefore produces an increasing homeomorphism  $\psi_\rho$ . A noninteger  $\rho \geq 1$  is squeezed between adjacent integer aspect ratios by monotonicity. This proves (9.12).

No step above uses independence: (9.13), (9.14)–(9.19), and the induction use only positive association, the geometric symmetries, and planar separation/duality. Thus the same argument transfers verbatim to a site or continuum model once those four structural inputs are verified. Finite-volume variants, which require a replacement for full translation invariance, are not being silently claimed here.  $\square$



*Remark 9.5* (Why the proof is organized this way). The quasi-crossing is the device that replaces the unavailable “lowest open crossing” exploration. If a crossing meanders, rather than trying to straighten it, the proof records the smaller scale that it reaches. Cascading transports that information across scales, and closing converts it back into a genuine long crossing.

**Theorem 9.6** (Critical Voronoi box crossing and polynomial one-arm decay).

*Consider planar Poisson–Voronoi percolation generated by a unit-intensity Poisson process whose nuclei are independently coloured black and white with probability  $1/2$ . If  $f_s(\rho)$  denotes the probability of a black left–right crossing of a  $\rho s \times s$  rectangle, then for every  $\rho > 0$  there is  $c(\rho) > 0$  such that*

$$c(\rho) \leq f_s(\rho) \leq 1 - c(\rho) \quad (s \geq 1). \quad (9.24)$$

*Moreover there are constants  $C < \infty$  and  $\eta > 0$  such that, for the one-arm probability*

$$\pi_1(s, t) = \mathbb{P}_{1/2}(B_s \longleftrightarrow \partial B_t), \quad 1 \leq s < t,$$

*one has*

$$\pi_1(s, t) \leq C \left( \frac{s}{t} \right)^\eta. \quad (9.25)$$

*Tassion’s printed Theorem 3 records the stronger normalization  $C = 1$ . Step h shows that this literal all-real-scale formulation cannot hold in the fixed-intensity continuum model: the obstruction is the thin-annulus regime  $t = s + 1$ ,  $s \rightarrow \infty$ . Thus (9.25), with an explicit multiplicative constant, is the corrected all-scale statement used in this series.*

*Proof.* Write  $X_b$  and  $X_w$  for independent Poisson processes of intensities  $1/2$  and  $1/2$  and colour each point of the plane by the nearest nucleus in  $X = X_b \cup X_w$ ; cell boundaries separating opposite colours are assigned both colours. Translation, quarter-turn, and reflection invariance follow from invariance of the homogeneous Poisson process.

The marked Poisson law is positively associated in the order which adds black nuclei and removes white nuclei. One way to prove this without a continuum black box is to partition a large box into finitely many small cells, truncate the number of nuclei in each cell, and apply Harris’s product inequality to the independent cell variables. For increasing cylinder events this gives

$$\mathbb{P}(E \cap F) \geq \mathbb{P}(E)\mathbb{P}(F).$$

Let the truncation and mesh tend to their limits and then exhaust the plane. Monotone convergence gives the same inequality for all black-increasing events generated by compact crossings. Thus the positive-association hypothesis of the positive-association hypothesis of the planar RSW theorem proved immediately above therefore holds for the Voronoi model.

In a topological rectangle exactly one of the following alternatives occurs: a black left–right crossing, or a white top–bottom crossing. This is the deterministic Jordan-separation alternative for the two closed colour sets; the convention that opposite-colour cell boundaries belong to both colours removes boundary ambiguity. At  $p = 1/2$ , colour exchange followed by a quarter-turn preserves the law. Hence for a square the two complementary alternatives have the same probability and

$$f_s(1) = \frac{1}{2} \quad (s \geq 1). \quad (9.26)$$

The same deterministic separation and gluing statements used in the proof of the universal planar RSW theorem proved immediately above apply to polygonal paths in the Voronoi colour sets. No lattice-specific independence is used there.

Apply the universal planar RSW theorem proved immediately above to the continuum crossing events just verified. For  $\rho \geq 1$ , the short-direction crossing of the rotated rectangle has probability at least  $f_s(1) = 1/2$  by restriction/gluing, hence the universal RSW homeomorphism gives

$$f_s(\rho) \geq c_-(\rho) > 0.$$

Apply the same estimate to white crossings. Planar duality says that failure of a black horizontal crossing is the corresponding white vertical crossing, so

$$f_s(\rho) \leq 1 - c_+(\rho).$$

Rotation handles  $0 < \rho < 1$ . Taking the smaller constant proves (9.24), uniformly in  $s \geq 1$ .

Let

$$\mathcal{A}_r^w = \{\text{there is a white circuit surrounding } B_r \text{ in } A_{r,2r}\}.$$

Choose four long thin rectangles arranged around the annulus so that a suitable crossing of each rectangle forces a circuit. Each required crossing has probability bounded below by (9.24), and the events are white-increasing. FKG therefore yields a constant  $c_0 > 0$ , independent of  $r \geq 1$ , with

$$\mathbb{P}(\mathcal{A}_r^w) \geq c_0. \quad (9.27)$$

(The same bound holds for black circuits by colour symmetry.)

For  $u \geq 1$  let

$$\mathcal{F}_u = \{\text{every } z \in A_{2u,4u} \text{ lies within distance } u \text{ of a Poisson nucleus}\}.$$

Cover  $A_{2u,4u}$  by a fixed number  $M$  of Euclidean balls of diameter  $u$ . If each contains a nucleus, then  $\mathcal{F}_u$  occurs, so the Poisson void formula gives

$$\mathbb{P}(\mathcal{F}_u^c) \leq M e^{-\pi u^2/4}. \quad (9.28)$$

On  $\mathcal{F}_u$ , every colour in  $A_{2u,4u}$  is determined by nuclei in  $A_{u,5u}$ : a nearest nucleus farther away would contradict the defining condition. Consequently

$$\mathcal{A}_r^w \cap \mathcal{F}_{r/2}$$

is measurable with respect to the two Poisson processes restricted to  $A_{r/2,5r/2}$ .

Choose  $r_0$  so large that  $\mathbb{P}(\mathcal{F}_{r/2}^c) \leq c_0/2$  for  $r \geq r_0$ . Then

$$\mathbb{P}(\mathcal{A}_r^w \cap \mathcal{F}_{r/2}) \geq c_0/2 =: c_1. \quad (9.29)$$

For radii  $r_j = 6^j r$ , the determining annuli  $A_{r_j/2,5r_j/2}$  are pairwise disjoint. The corresponding localized circuit events are therefore exactly independent by independent restrictions of a Poisson process to disjoint sets. This is the point at which the dependent Voronoi colouring is reduced to honest separated-scale independence.

Suppose first that  $s \geq r_0$  and  $t \geq 36s$ . Choose all  $r_j = 6^j(2s)$  for which  $2s \leq r_j \leq t/3$ . Every localized white circuit obtained at such a scale lies in  $A_{s,t}$  and blocks a black path from  $B_s$  to  $\partial B_t$ . Hence, if  $N$  is the number of selected scales,

$$\pi_1(s, t) \leq (1 - c_1)^N.$$

There are constants  $a_0, a_1 > 0$  with  $N \geq a_0 \log(t/s) - a_1$ . Thus

$$\pi_1(s, t) \leq C_0(s/t)^{\eta_0}, \quad \eta_0 = -a_0 \log(1 - c_1) > 0. \quad (9.30)$$

If  $1 \leq s < r_0$ , begin the iteration at  $r_0$  and use monotonicity in the inner radius; enlarging the constant handles the resulting bounded initial range. If  $t < 36s$ , the trivial estimate  $\pi_1(s, t) \leq 1$  is bounded by  $36^{\eta_0}(s/t)^{\eta_0}$ . Therefore (9.25) holds for every  $1 \leq s < t$  after increasing  $C$ .

Steps a–f prove the two load-bearing consequences used later in the series: all-scale critical box crossing and a quantitative polynomial one-arm upper bound. The proof contains its continuum FKG, duality, stabilization, localization, and separated-scale independence owners; it does not cite “Voronoi RSW” as a black box.

We now resolve the apparent normalization discrepancy explicitly. The printed source states

$$\pi_1(s, t) \leq (s/t)^\eta \quad (1 \leq s < t) \quad (9.31)$$

with no multiplicative constant. In the unit-intensity continuum model this literal claim is incompatible with thin annuli.

Fix a small  $\varepsilon \in (0, 1/20)$  and, along the top side of  $B_s$ , choose  $N \geq c_*s$  points

$$z_j = (x_j, s + 1/2)$$

whose mutual distances are at least 5. Around each  $z_j$  let  $D_j^{\text{in}} = B(z_j, \varepsilon)$  and  $D_j^{\text{out}} = B(z_j, 2)$ ; the outer discs are pairwise disjoint. Define  $E_j$  to be the event that there is exactly one Poisson nucleus in  $D_j^{\text{in}}$ , that this nucleus is black, and that there is no other nucleus in  $D_j^{\text{out}}$ . The Poisson formula gives a scale-independent number  $p_* > 0$  with  $\mathbb{P}(E_j) = p_*$ , and the events  $E_j$  are independent.

On  $E_j$ , let  $v_j$  be the unique black nucleus in  $D_j^{\text{in}}$ . Every point of the vertical unit segment

$$I_j = \{(x_j, y) : s \leq y \leq s + 1\}$$

is at distance at most  $1/2 + \varepsilon$  from  $v_j$ , whereas every nucleus outside  $D_j^{\text{out}}$  is at distance at least  $3/2$  from every point of  $I_j$ . Hence  $I_j$  lies entirely in the black Voronoi cell of  $v_j$ . Therefore  $E_j$  forces a black connection from  $B_s$  to  $\partial B_{s+1}$ , and

$$\pi_1(s, s + 1) \geq 1 - (1 - p_*)^N \geq 1 - e^{-cs} \quad (9.32)$$

for a constant  $c > 0$ .

For any fixed  $\eta > 0$ ,

$$1 - \left(\frac{s}{s+1}\right)^\eta = 1 - (1 + 1/s)^{-\eta} \geq \frac{\eta}{3s} \quad (9.33)$$

for all sufficiently large  $s$ . Since  $e^{-cs} < \eta/(3s)$  eventually, (9.32) implies

$$\pi_1(s, s + 1) > \left(\frac{s}{s+1}\right)^\eta$$

for all sufficiently large  $s$ . Thus no positive  $\eta$  can make (9.31) valid for all real  $1 \leq s < t$  in this normalization.

The source’s polynomial-decay conclusion is correct in the standard bounded-ratio/ macroscopic sense and is exactly what Steps a–g prove in the robust form  $\pi_1(s, t) \leq C(s/t)^{\eta_0}$ . All later uses in this project require only this form. The literal  $C = 1$  all-real-scale normalization therefore cannot be correct; the constant-prefactor theorem proved above is the form used later.  $\square$

**Theorem 9.7** (Sharpness for Voronoi percolation in  $\mathbb{R}^d$ ).

Fix  $d \geq 2$  and let  $p_c = p_c(d)$  be the critical parameter of Voronoi percolation. If  $p < p_c$ , there is  $c_p > 0$  such that, for all  $n \geq 1$ ,

$$\mathbb{P}_p(0 \longleftrightarrow S_n) \leq e^{-c_p n}. \quad (9.27)$$

There is a constant  $c = c(d) > 0$  such that, for  $p > p_c$ ,

$$\theta(p) := \mathbb{P}_p(0 \longleftrightarrow \infty) \geq c(p - p_c). \quad (9.28)$$

In dimension two,  $p_c = 1/2$  and  $\theta(1/2) = 0$ .

*Proof.* Write  $B_r = \{x : \|x\| \leq r\}$  and  $S_r = \partial B_r$ . Let

$$\theta_n(p) = \mathbb{P}_p(0 \longleftrightarrow S_n), \quad S_n(p) = \sum_{k=0}^{n-1} \theta_k(p).$$

The proof has one geometric input—an exploration with small revealment—and one analytic input—integration of the resulting differential inequality.

Represent the coloring by a unit-intensity Poisson process  $\eta$  whose nuclei are, conditionally on  $\eta$ , independently black with probability  $p$ . For a local increasing event  $A$ , a nucleus is pivotal if changing only its color changes  $A$ . Conditioning on  $\eta$ , ordinary finite/countable Bernoulli Russo calculus gives

$$\frac{d}{dp} \mathbb{P}_p(A) = \mathbb{E}_p N_{\text{piv}}(A). \quad (9.29)$$

The interchange of derivative and expectation is justified because a local event can only be affected by cells meeting a fixed bounded set, and the number of such nuclei has an integrable tail: a very distant cell can reach the set only if a macroscopic Poisson void occurs.

Fix  $\varepsilon > 0$  and partition  $\mathbb{R}^d$  into cubes  $R_x^\varepsilon = x + [0, \varepsilon)^d$ ,  $x \in \varepsilon \mathbb{Z}^d$ . The restrictions of the two colored Poisson processes to these cubes are independent coordinates. Let  $\text{Inf}_x^\varepsilon(A)$  be the probability that resampling the entire coordinate in  $R_x^\varepsilon$  changes  $A$ . If  $A$  is local, then

$$\frac{d}{dp} \mathbb{P}_p(A) \geq \frac{1}{2} \limsup_{\varepsilon \downarrow 0} \sum_x \text{Inf}_x^\varepsilon(A). \quad (9.30)$$

Indeed, in a fixed cube the union of the old and resampled Poisson configurations has, with probability  $1 - O(\varepsilon^{2d})$ , at most one nucleus. On that event, a change of  $A$  caused by resampling forces that nucleus to be pivotal in one of the two configurations. Thus

$$\text{Inf}_x^\varepsilon(A) \leq 2 \mathbb{E}_p [N_{\text{piv}}(A; R_x^\varepsilon)] + O(\varepsilon^{2d}).$$

Sum first over cubes in a large ball and use (9.29). Cubes far from the support of  $A$  contribute at most  $C\varepsilon^d e^{-c|x|^d}$ , since their resampling can matter only when a distant Voronoi cell reaches the bounded support. Let the ball grow and then  $\varepsilon \downarrow 0$  to obtain (9.30).

The conditional-variance proof of SM-10.05.01 does not use that a coordinate is Bernoulli. For a Boolean function  $f$  of independent coordinates and a decision tree  $T$  determining it,

$$\text{Var}(f) \leq \sum_x \delta_x(T) \text{Inf}_x(f), \quad (9.31)$$

where influence is defined by independent resampling. To check (9.31), couple two independent configurations and replace queried coordinates one at a time. The martingale/telescoping proof bounds the variance by the sum, over coordinates actually queried by  $T$ , of the probability that independent resampling changes the Boolean output. Averaging the indicator that a coordinate is queried gives the displayed revelation–influence bound.

Fix  $1 \leq k \leq n$ . We construct a tree  $T_k$  which determines  $A_n = \{0 \longleftrightarrow S_n\}$  by discovering the black component meeting  $S_k$ . To determine the color in a small cube  $R_y^\varepsilon$ , reveal Poisson coordinates in successive Euclidean shells about  $y$  until every point of the cube has a nearest revealed nucleus; call this subroutine Discover( $y$ ). Starting from  $Z_0 = S_k$ , repeatedly choose an undiscovered small cube meeting the currently discovered black set, run Discover there, and add the black part of that cube to the set  $Z_t$ . When no such cube remains, the component of  $S_k$  has been exposed. Hence the algorithm knows whether it reaches both 0 and  $S_n$ , and therefore determines  $A_n$ .

There is a constant  $C = C(d, \delta)$  such that, uniformly for  $p \in [\delta, 1 - \delta]$ ,

$$\delta_x(T_k) \leq C \mathbb{P}_p(x \longleftrightarrow S_k). \quad (9.32)$$

For the revelation estimate, if coordinate  $x$  is inspected while discovering a unit cube near  $z$ , that cube is already connected to  $S_k$ . In addition, for a far-away  $x$  to be needed by Discover, there can be no black Poisson nucleus in a ball about  $z$  whose radius is comparable with  $|z - x|$ ; otherwise the colors near  $z$  would already be determined without revealing the coordinate at  $x$ . Denote this decreasing black-nucleus void event by  $E_{z,x}$ . FKG separates the increasing connectivity event from the decreasing void event:

$$\mathbb{P}_p(R_z^1 \longleftrightarrow S_k, E_{z,x}) \leq \mathbb{P}_p(R_z^1 \longleftrightarrow S_k) \mathbb{P}(E_{z,x}).$$

Since  $p \geq \delta$ , the black process has intensity at least  $\delta$ , and hence Poisson void probabilities give  $\mathbb{P}_p(E_{z,x}) \leq C e^{-c|z-x|^d}$  uniformly for  $p \in [\delta, 1 - \delta]$ . Conversely, connect  $x$  to  $R_z^1$  through  $O(|z - x|)$  adjacent unit boxes and require each of them to contain a small black nucleus. Since  $p \geq \delta$ , FKG gives

$$\mathbb{P}_p(x \longleftrightarrow S_k \mid R_z^1 \longleftrightarrow S_k) \geq e^{-C|z-x|}.$$

Therefore

$$\mathbb{P}_p(R_z^1 \longleftrightarrow S_k) \leq e^{C|z-x|} \mathbb{P}_p(x \longleftrightarrow S_k).$$

Summing the resulting bound  $e^{C|z-x|} e^{-c|z-x|^d}$  over  $z \in \mathbb{Z}^d$  proves (9.32).

Apply (9.31) to  $f = \mathbf{1}_{A_n}$  and  $T_k$ . Since  $p$  stays in a compact subinterval of  $(0, 1)$ ,  $1 - \theta_n(p)$  is bounded below by a positive constant uniformly in  $n$  after shrinking that compact interval around  $p_c$  if necessary. Using (9.32),

$$\theta_n(p) \leq C \sum_x \mathbb{P}_p(x \longleftrightarrow S_k) \text{Inf}_x^\varepsilon(A_n).$$

Average this inequality over  $k = 1, \dots, n$ . By translation invariance and the triangle inequality for distance to the sphere,

$$\sum_{k=1}^n \mathbb{P}_p(x \longleftrightarrow S_k) \leq 2S_n(p).$$

Hence

$$\theta_n(p) \leq C \frac{S_n(p)}{n} \sum_x \text{Inf}_x^\varepsilon(A_n).$$

Let  $\varepsilon \downarrow 0$  and use (9.30):

$$\boxed{\theta'_n(p) \geq c \frac{n}{S_n(p)} \theta_n(p)} \quad (9.33)$$

for  $p$  in any fixed compact interval containing  $p_c$  and contained in  $(0, 1)$ . Outside such an interval the desired conclusions follow after adjusting constants by monotonicity.

We record the elementary consequence of (9.33). Suppose increasing differentiable  $f_n : [a, b] \rightarrow [0, M]$  converge pointwise to  $f$  and satisfy

$$f'_n \geq \frac{n}{\Sigma_n} f_n, \quad \Sigma_n = \sum_{j=0}^{n-1} f_j. \quad (9.34)$$

There is  $p_0 \in [a, b]$  such that: (i) for  $p < p_0$ ,  $f_n(p) \leq M e^{-c_p n}$  for all sufficiently large  $n$ ; and (ii) for  $p > p_0$ ,  $f(p) \geq p - p_0$  after the harmless rescaling which absorbed the constant in (9.33).

Here is the proof. Set

$$p_0 = \inf \left\{ p : \limsup_{n \rightarrow \infty} \frac{\log \Sigma_n(p)}{\log n} \geq 1 \right\}.$$

If  $p < p_0$ , choose  $p < p' < p'' < p_0$ . At  $p''$  one has  $\Sigma_n(p'') \leq n^{1-\alpha}$  for some  $\alpha > 0$  and all large  $n$ . Integrating (9.34) from  $p'$  to  $p''$  gives stretched-exponential decay of  $f_n(p')$ ; therefore  $\sup_n \Sigma_n(p') < \infty$ . A second integration from  $p$  to  $p'$  now gives exponential decay  $e^{-c_p n}$ .

For  $p > p_0$ , define

$$T_n(p) = \frac{1}{\log n} \sum_{j=1}^n \frac{f_j(p)}{j}.$$

Then (9.34) and  $f_j/\Sigma_j \geq \log \Sigma_{j+1} - \log \Sigma_j$  give

$$T'_n(p) \geq \frac{\log \Sigma_{n+1}(p) - O(1)}{\log n}.$$

Integrate from  $p' \in (p_0, p)$  to  $p$  and let  $n \rightarrow \infty$ ; since  $T_n \rightarrow f$  and the limsup exponent at  $p'$  is at least one,  $f(p) - f(p') \geq p - p'$ . Let  $p' \downarrow p_0$  to obtain (ii).

Apply the lemma to a constant multiple of  $\theta_n$ . Below  $p_0$  there is exponential decay, hence no infinite cluster; therefore  $p_0 \leq p_c$ . Above  $p_0$ ,  $\theta(p) > 0$ , so  $p_c \leq p_0$ . Thus  $p_0 = p_c$ . The two conclusions of the lemma are precisely (9.27) and the mean-field lower bound (9.28), after restoring the constant suppressed in the rescaling.

In  $d = 2$ , let  $A_n$  be the event that  $[-n, n]^2$  is crossed from left to right by black points. Its complement is a white top-bottom crossing; color symmetry and a quarter-turn therefore give

$$\mathbb{P}_{1/2}(A_n) = \frac{1}{2}. \quad (9.35)$$

A left-right crossing of the square must start from one of  $O(n)$  unit boundary pieces. By symmetry and a union bound,

$$\mathbb{P}_{1/2}(B_1 \longleftrightarrow S_n) \geq \frac{c}{n}.$$

Intersecting with the increasing event that  $B_1$  is black and using FKG gives

$$\theta_n(1/2) \geq \mathbb{P}_{1/2}(B_1 \longleftrightarrow S_n) \mathbb{P}_{1/2}(B_1 \text{ is black}) \geq \frac{c'}{n}. \quad (9.36)$$

(The event “ $B_1$  is black” is understood through compact inner approximations; its probability is strictly positive by the local Poisson construction.) The polynomial lower bound (9.36) is incompatible with the subcritical exponential estimate, hence  $p_c \leq 1/2$ .

For the reverse inequality, suppose that  $\theta(1/2) > 0$ . For  $k \leq n$ , the square-root trick applied to the four sides of  $[-n, n]^2$  yields

$$\mathbb{P}_{1/2}(B_k \text{ connects inside } [-n, n]^2 \text{ to the top side}) \geq 1 - \mathbb{P}_{1/2}(B_k \not\rightarrow \infty)^{1/4}.$$

The analogous bound holds for the bottom side, and therefore the probability that  $B_k$  connects to both top and bottom is at least

$$1 - 2\mathbb{P}_{1/2}(B_k \not\rightarrow \infty)^{1/4}.$$

Uniqueness of the infinite Voronoi cluster, whenever it exists, lets  $k \rightarrow \infty$  and then  $n \rightarrow \infty$  to force  $\mathbb{P}_{1/2}(A_n) \rightarrow 1$ , contradicting (9.35). Consequently  $\theta(1/2) = 0$ , so  $p_c \geq 1/2$ . Thus  $p_c = 1/2$ .  $\square$

*Remark 9.8* (Dependency note). The only general decision-tree input is the OSSS mechanism already proved in SM-10.05.01. The continuum tensorization, derivative comparison, exploration revelation, and sharpness integration are all owned above; no Bernoulli BK inequality is used.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 9.9** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 9.10** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 9.11** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 9.12** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 9.13** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.



# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XI: Critical Planar Percolation, Noise Sensitivity, and  
Near-Critical Scaling

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# Scope and conventions

This volume develops critical planar percolation from RSW estimates and arm events to pivotal measures, conformal observables, noise sensitivity, and near-critical scaling. The exposition keeps the elementary planar consequences of RSW separate from the deeper scaling-limit theorems that require discrete complex analysis or continuum pivotal-measure technology.

# Chapter 1

## RSW theory

At criticality there is no characteristic Euclidean length. RSW is the first quantitative expression of scale invariance.

**Definition 1.1** (Quad and crossing event). A *quad* is a simply connected planar domain  $Q$  with four marked boundary arcs  $(\partial_0 Q, \partial_1 Q, \partial_2 Q, \partial_3 Q)$  in cyclic order. The open crossing event  $\mathcal{C}(Q)$  is the existence of an open path in  $Q$  joining  $\partial_0 Q$  to  $\partial_2 Q$ . For a Euclidean rectangle, these are the left and right sides unless stated otherwise.

*Worked example 1.2* (From one square crossing bound to a 2:1 rectangle). Suppose critical RSW gives  $\mathbb{P}(\mathcal{C}([0, n]^2)) \geq c_{\square} > 0$ . Cover  $[0, 2n] \times [0, n]$  by three overlapping  $n \times n$  squares with overlaps of width  $n/2$ . Require a horizontal crossing of each outer square and a vertical crossing of the middle overlap. These increasing events force a horizontal crossing of the 2:1 rectangle. By FKG,

$$\mathbb{P}(\mathcal{C}([0, 2n] \times [0, n])) \geq c_{\square}^3.$$

The number 3 is not important; what matters is that a fixed aspect ratio requires only finitely many gluing events, so the lower bound is independent of  $n$ .

**Example 1.3** (A running picture). At  $p = 1/2$  on the triangular lattice, zooming out does not drive crossings to probability 0 or 1. The chapter identifies the mechanism responsible for this scale-free behavior.

**Theorem 1.4** (Russo–Seymour–Welsh crossing theorem).

*For critical site percolation on the triangular lattice, and every fixed aspect ratio  $\rho \geq 1$ , there exists  $c(\rho) > 0$  such that for every scale  $n$ ,*

$$c(\rho) \leq \mathbb{P}_{1/2}(\text{open left-right crossing of } [0, \rho n] \times [0, n]) \leq 1 - c(\rho).$$

*Proof.* We spell out the gluing mechanism. In a square, planar duality says that exactly one of an open left–right crossing and a closed top–bottom crossing occurs. At  $p = 1/2$  color complementation and rotation exchange these events, so the square-crossing probability is bounded away from zero and one uniformly in the mesh and scale.

Let  $h(a, b)$  denote the probability of a horizontal crossing of an  $a \times b$  rectangle. Consider two overlapping  $n \times n$  squares whose union has aspect ratio strictly less than two. On the event that the left square is crossed, expose the lowest left–right crossing  $\gamma$ . The unexplored sites above  $\gamma$  retain independent critical law. Reflect a suitable terminal portion of  $\gamma$  and use the square crossing estimate in the unexplored region to obtain, with probability bounded below independently of  $n$ ,

an open connection from  $\gamma$  into a crossing of the right square. Because the relevant events are increasing, FKG allows the lower bounds to be multiplied. Thus there is an increasing function  $F$ , with  $F(x) > 0$  for  $x > 0$ , such that

$$h(3n/2, n) \geq F(h(n, n)).$$

Repeatedly glue overlapping rectangles to obtain a positive lower bound for every fixed  $\rho$ .

Apply the same argument to closed sites. A closed top–bottom crossing is complementary to the open left–right crossing in the corresponding dual quadrilateral. The lower bound for closed crossings therefore gives the upper bound  $1 - c(\rho)$  for open crossings. All constants depend only on the number of gluing steps, hence only on  $\rho$ , not on  $n$ .  $\square$

**Theorem 1.5** (Uniform circuit estimates in annuli).

*For critical site percolation on the triangular lattice there is  $c_0 > 0$  such that, for every  $r > 0$ , the annulus  $A(r, 2r)$  contains an open circuit surrounding the origin with probability at least  $c_0$  and a closed circuit with probability at least  $c_0$ .*

*Proof.* Cover the annulus by four overlapping rectangles of a fixed aspect ratio, arranged cyclically. If each rectangle has an open crossing in the appropriate long direction, the crossings can be connected in the overlaps to form an open circuit around the inner boundary. RSW gives a scale-independent lower bound  $c_1 > 0$  for each crossing probability, and the FKG inequality gives a lower bound  $c_1^4$  for their simultaneous occurrence. Thus the open-circuit probability is uniformly positive. At  $p = 1/2$ , exchanging open and closed sites preserves the law, so the same bound holds for a closed circuit. Enlarging the annulus by a fixed factor if necessary absorbs lattice-boundary effects and only changes the constant.  $\square$

**Theorem 1.6** (Polynomial one-arm upper bound from RSW).

*There are constants  $C, c > 0$  such that, for  $1 \leq r \leq R$ ,*

$$\mathbb{P}_{1/2}(\partial B_r \leftrightarrow \partial B_R) \leq C \left( \frac{r}{R} \right)^c.$$

*Proof.* Choose every third dyadic annulus  $A_k = A(2^{3k}r, 2^{3k+1}r)$  contained between radii  $r$  and  $R$ . The selected annuli depend on disjoint sets of sites. If an open arm connects  $\partial B_r$  to  $\partial B_R$ , none of these annuli may contain a closed circuit surrounding the origin. By the preceding theorem each such circuit occurs with probability at least  $c_0 > 0$ . Hence

$$\mathbb{P}(\partial B_r \leftrightarrow \partial B_R) \leq (1 - c_0)^m,$$

where  $m \geq c_1 \log(R/r) - 1$ . Writing the last expression as a power of  $r/R$  and absorbing the bounded-scale cases into  $C$  proves the estimate.  $\square$



## Chapter 2

# Arm events

Arm events are the currency of planar critical geometry. They control pivotals, interfaces, dimensions, and near-critical time scales.

**Definition 2.1** (*k*-arm event). Fix radii  $0 < r < R$  and a cyclic colour word  $\sigma = (\sigma_1, \dots, \sigma_k) \in \{\text{open}, \text{closed}\}^k$ . The event  $A_k^\sigma(r, R)$  is the existence of  $k$  pairwise disjoint paths in the annulus  $A(r, R)$  joining its two boundary components, occurring in cyclic order and having the prescribed colours. We write  $\alpha_k^\sigma(r, R) = \mathbb{P}(A_k^\sigma(r, R))$  and suppress  $\sigma$  when the colour pattern is fixed in context.

*Worked example 2.2* (Using quasi-multiplicativity across three scales). Take  $r = 1$ ,  $s = 2^m$ ,  $R = 2^{m+n}$ . If  $\alpha_k(1, 2^j) \asymp 2^{-x_k j}$  at dyadic scales, then quasi-multiplicativity gives

$$\alpha_k(1, 2^{m+n}) \asymp \alpha_k(1, 2^m) \alpha_k(2^m, 2^{m+n}) \asymp 2^{-x_k m} 2^{-x_k n} = 2^{-x_k(m+n)}.$$

Thus an arm exponent is compatible with concatenation of annuli precisely because logarithmic scale turns multiplication of probabilities into addition of exponents.

**Example 2.3** (A running picture). At  $p = 1/2$  on the triangular lattice, zooming out does not drive crossings to probability 0 or 1. The chapter identifies the mechanism responsible for this scale-free behavior.

**Theorem 2.4** (Arm separation).

*Fix  $k \geq 2$  and a polychromatic cyclic color sequence. Let  $\alpha_k(r, R)$  be the probability of  $k$  disjoint arms of those colors across the annulus  $A(r, R)$ . If the landing arcs at both boundaries are required to have pairwise separation at least a fixed fraction  $\eta$  of the circumference, then for  $R \geq 4r$*

$$c_{k,\eta} \alpha_k(r, R) \leq \alpha_k^{\text{sep}}(r, R) \leq \alpha_k(r, R).$$

*Proof.* Only the first inequality needs proof. Condition on the configuration in the middle annulus  $A(2r, R/2)$  and on the existence of  $k$  disjoint arms there. Around each endpoint on the inner and outer boundaries choose disjoint rectangular “fences” of aspect ratio bounded in terms of  $k$  and  $\eta$ . RSW supplies a uniformly positive probability for the required open or closed crossings of every fence, and FKG may be applied separately to crossings of the same color. For opposite colors we reveal successively in disjoint sectors, so the unexplored sites keep product law. The fences force neighboring arms apart and route them into predetermined landing intervals.

There are only finitely many cyclic choices of landing intervals. Summing over them and using the uniform product of the RSW constants gives a lower bound  $c_{k,\eta}$  independent of  $r$  and  $R$ . The reverse inequality is immediate because the separated event is a subset of the unrestricted arm event.  $\square$

**Theorem 2.5** (Quasi-multiplicativity of arm probabilities).

For each fixed polychromatic  $k$  there is  $C_k < \infty$  such that whenever  $r < s < R$  with the three scales separated by fixed factors,

$$C_k^{-1} \alpha_k(r, s) \alpha_k(s, R) \leq \alpha_k(r, R) \leq C_k \alpha_k(r, s) \alpha_k(s, R).$$

The statement extends to arbitrary  $r < s < R$  after changing  $C_k$ .

*Proof.* For the lower bound, use arm separation at radius  $s$ : with probability comparable to  $\alpha_k(r, s)$  the inner arms land in  $k$  prescribed separated arcs, and independently the outer annulus realizes  $k$  arms of matching colors from corresponding separated arcs to  $R$ . RSW crossings in a bounded-width annulus around  $s$  connect each inner arm to its outer partner without changing the cyclic order. The gluing costs only a constant, giving  $\alpha_k(r, R) \geq c \alpha_k(r, s) \alpha_k(s, R)$ .

For the upper bound, on the event of  $k$  arms from  $r$  to  $R$  there are  $k$  arms from  $r$  to  $s$  and from  $s$  to  $R$ . Use arm separation in a thin annulus around  $s$  to replace arbitrary landing data by separated data at constant cost. Once the landing sectors are fixed, the events inside and outside that thin annulus depend on disjoint site sets, so their probabilities multiply. Summing over finitely many landing patterns gives the upper bound. If scales are not separated, absorb the bounded-ratio cases using RSW.  $\square$

**Theorem 2.6** (Five-arm exponent).

For critical site percolation on the triangular lattice or critical bond percolation on the square lattice at  $p = 1/2$ , let  $\alpha_5(r, R)$  denote the probability of five arms of alternating colors except for one repeated color (equivalently, any non-monochromatic five-arm pattern). Then

$$\alpha_5(r, R) \asymp \left(\frac{r}{R}\right)^2, \quad 1 \leq r < R.$$

*Proof.* We give the planar counting argument in full because the exponent 2 is a geometric consequence of the fact that a five-arm point is a rigid junction. The proof uses only self-duality at  $p = 1/2$ , RSW, arm separation, independence on disjoint unexplored regions, and planarity, so it applies verbatim to both the triangular-site and square-bond models. By color switching inside the separated landing corridors, all nonconstant five-arm color sequences have probabilities comparable up to an absolute constant. It is therefore enough to work with the cyclic pattern  $B, W, B, B, W$ .

**Step 1: a unit-scale counting estimate.** Let  $Q_N$  be a rhombus (or square in a fixed Euclidean embedding) of diameter comparable with  $N$ , and let  $Q_N^\circ$  be its concentric half-sized copy. Fix five disjoint landing intervals  $J_1, \dots, J_5 \subset \partial Q_N$  in cyclic order. For a lattice vertex  $v \in Q_N^\circ$ , let  $E_v$  be the event that five disjoint arms of colors  $B, W, B, B, W$  run from the microscopic neighborhood of  $v$  to the prescribed intervals  $J_1, \dots, J_5$ . Arm separation (Theorem SM-11.02.01) and RSW imply, uniformly for  $v \in Q_N^\circ$ ,

$$\mathbb{P}(E_v) \asymp \alpha_5(1, N). \quad (2.1)$$

The change from an annulus centered at  $v$  to the fixed box costs only bounded-ratio RSW crossings.

We use the following elementary planar fact. For fixed separated landing intervals there is a universal  $C_0 < \infty$  such that, in every configuration, at most  $C_0$  vertices of  $Q_N^\circ$  can realize  $E_v$ . Indeed, join the first two black arms through the microscopic neighborhood of  $v$  and do the same for the two white sectors adjacent to them. The resulting Jordan arcs separate the remaining landing sectors. Two distinct junctions with the same cyclic landing data would force one of the black connections to cross a white separating arc. Allowing the bounded microscopic ambiguity at the junction changes “one” to a fixed constant  $C_0$ , independent of  $N$ . Consequently, with  $X_N = \sum_{v \in Q_N^\circ} \mathbf{1}_{E_v}$ ,

$$\mathbb{E} X_N \leq C_0.$$

Since  $Q_N^\circ$  contains  $\asymp N^2$  vertices, (2.1) gives

$$\alpha_5(1, N) \leq CN^{-2}. \quad (2.2)$$

For the reverse inequality we construct a five-arm junction with uniformly positive probability. By RSW, with probability at least  $c > 0$  there is a left–right black crossing of a central horizontal rectangle, a left–right white crossing in a disjoint rectangle immediately below it, and the finitely many additional separated crossings used below. On this event choose the lowest black left–right crossing  $\gamma$ . Every interior vertex of  $\gamma$  has two disjoint black arms along  $\gamma$ ; by the defining minimality of the lowest crossing, the dual/white interface below  $\gamma$  supplies a white arm to the lower boundary. The extra RSW crossings create, in disjoint corridors above  $\gamma$ , one black and one white connection to two prescribed top landing intervals. Traverse  $\gamma$  from left to right and choose the last vertex whose upper black connection reaches its prescribed corridor. The next corridor is white; planarity and the extremal choice force five disjoint arms at that transition vertex, with cyclic colors  $B, W, B, B, W$ . The construction keeps the vertex in  $Q_N^\circ$  after changing the rectangles by fixed factors.

Thus, if  $F_v$  denotes the unrestricted five-arm event from  $v$  to distance  $N$ , then

$$c \leq \mathbb{P}\left(\bigcup_{v \in Q_N^\circ} F_v\right) \leq \sum_{v \in Q_N^\circ} \mathbb{P}(F_v) \leq CN^2 \alpha_5(1, N),$$

where the last comparison again uses translation by  $O(N)$  inside the central box together with arm separation and RSW. Hence

$$\alpha_5(1, N) \geq c' N^{-2}. \quad (2.3)$$

Combining (2.2) and (2.3),  $\alpha_5(1, N) \asymp N^{-2}$ .

**Step 2: arbitrary annuli.** The quasi-multiplicativity theorem gives, for  $1 \leq r < R$ ,

$$\alpha_5(1, R) \asymp \alpha_5(1, r) \alpha_5(r, R).$$

Using the unit-scale estimate at radii  $r$  and  $R$  therefore yields

$$\alpha_5(r, R) \asymp \frac{R^{-2}}{r^{-2}} = \left(\frac{r}{R}\right)^2.$$

Bounded-ratio cases are absorbed into the constants by RSW. This proves the claimed estimate uniformly for all  $1 \leq r < R$ .  $\square$

**Lemma 2.7** (The two  $\text{SLE}_6$  moment exponents needed for four arms). *Let  $u(\lambda)$  be the chordal long-rectangle exponent and let  $\nu(b)$  be the radial annulus extremal-distance exponent, with the normalizations used below. Then*

$$u(\lambda) = \frac{6\lambda + 1 + \sqrt{24\lambda + 1}}{6}, \quad \nu(b) = \frac{4b + 1 + \sqrt{1 + 24b}}{8}.$$

*In particular*

$$u(0) = \frac{1}{3}, \quad u\left(\frac{1}{3}\right) = 1, \quad u(1) = 2, \quad \nu(1) = \frac{5}{4}.$$

*The value  $u(1) = 2$  is the half-plane three-arm exponent used below to control pivotal points approaching a smooth boundary arc. Consequently, chordal  $\text{SLE}_6$  makes two crossings of a long semi-annulus of modulus  $L/\pi$  with probability  $\exp(-L + O(1))$ , while the radial  $\text{SLE}_6$  annulus moment with weight one is  $R^{-5/4+o(1)}$  when the ratio of radii is  $R$ .*

*Proof.* We record the stochastic calculations because the exponents are load-bearing later in the pivotal-measure normalization.

*Chordal calculation.* Map a rectangle of width  $L$  and height  $\pi$  conformally to  $\mathbb{H}$  so that its marked boundary points are  $1, \infty, 0, x$ , where  $x = c e^{-L+O(1)}$  as  $L \rightarrow \infty$  by the elementary extremal-length comparison for a long rectangle. Run chordal SLE<sub>6</sub> from  $x$  and stop when its hull first reaches either  $(-\infty, 0]$  or  $[1, \infty)$ . Normalize the Loewner map to fix  $0, 1, \infty$  and put

$$Z_s = \frac{W_t - g_t(0)}{g_t(1) - g_t(0)}, \quad ds = \frac{dt}{(g_t(1) - g_t(0))^2}.$$

A direct differentiation of the Loewner equation gives, up to the first hit of  $\{0, 1\}$ ,

$$dZ_s = \sqrt{6} dB_s + \left( \frac{2}{Z_s} - \frac{2}{1 - Z_s} \right) ds.$$

If  $f_s$  denotes the same normalized map and  $A_s = \log f'_s(1)$ , then

$$\partial_s A_s = -\frac{2}{(1 - Z_s)^2} - \frac{2}{Z_s} - \frac{2}{1 - Z_s}.$$

For  $b \geq 0$ , the function

$$h_b(x) = \mathbb{E}_x[1_{\{T_0 < T_1\}} e^{bA_T}]$$

therefore satisfies the Feynman–Kac boundary-value equation

$$3x^2(1-x)h_b'' + 2x(1-2x)h_b' - 2bh_b = 0, \quad h_b(0) = 0, \quad h_b(1) = 1. \quad (2.4)$$

Set  $\widehat{b} = \sqrt{1+24b}/6$ . Substitution  $h_b(x) = x^{1/6+\widehat{b}}F(x)$  reduces (2.4) to the Gauss hypergeometric equation. The solution bounded by the stated boundary data is

$$h_b(x) = C_b x^{1/6+\widehat{b}} {}_2F_1\left(\frac{1}{6} + \widehat{b}, \frac{1}{2} + \widehat{b}, 1 + 2\widehat{b}; x\right), \quad (2.5)$$

where  $C_b$  is the unique positive constant making  $h_b(1) = 1$ . Since the hypergeometric factor tends to 1 at the origin,

$$h_b(x) = \exp\left[-\left(\frac{1}{6} + \widehat{b}\right)L + O_b(1)\right].$$

The deterministic comparison between  $-\log f'_T(1)$  and the  $\pi$ -extremal distance of the remaining quadrilateral changes the latter by at most an additive constant: this follows by reflecting across the straight boundary pieces and applying the Koebe distortion theorem in the normalized quadrilateral. Hence

$$\mathbb{E}[1_{\{\mathcal{L} < \infty\}} e^{-\lambda \mathcal{L}}] = \exp[-u(\lambda)L + O_\lambda(1)], \quad u(\lambda) = \frac{6\lambda + 1 + \sqrt{24\lambda + 1}}{6}. \quad (2.6)$$

The domain Markov property now gives the crossing recursion: after the first crossing, the remaining conditional crossing probability is the same functional evaluated at the new extremal distance. Thus one crossing has exponent  $u(0) = 1/3$ , and two crossings have exponent  $u(u(0)) = u(1/3) = 1$ .

*Radial calculation.* For radial SLE<sub>6</sub> in the unit disk, write

$$g_t(e^{ix}) = \delta_t e^{iY_t}, \quad \Phi_t = |g'_t(e^{ix})|,$$

up to absorption of the boundary point. Then

$$dY_t = \sqrt{6} dB_t + \cot(Y_t/2)dt, \quad d \log \Phi_t = -\frac{dt}{2 \sin^2(Y_t/2)}. \quad (2.7)$$

For a derivative weight  $b$ , the killed Feynman–Kac generator is

$$\mathcal{A}_b = 3\partial_y^2 + \cot(y/2)\partial_y - \frac{b}{2 \sin^2(y/2)}.$$

With

$$q = \frac{1 + \sqrt{1 + 24b}}{6}, \quad h(y) = \sin(y/2)^q,$$

a direct differentiation gives

$$\mathcal{A}_b h = -\nu(b)h, \quad \nu(b) = \frac{4b + 1 + \sqrt{1 + 24b}}{8}. \quad (2.8)$$

Indeed, for the only value needed below,  $b = 1$ , one has  $q = 1$  and

$$3h'' + \cot(y/2)h' - \frac{h}{2 \sin^2(y/2)} = -\frac{5}{4}h.$$

Itô's formula therefore makes  $e^{\nu(b)t} \Phi_t^b h(Y_t)$  a local martingale killed at the two boundary points. Localize at distance  $\varepsilon$  from  $\{0, 2\pi\}$  and let  $\varepsilon \downarrow 0$ . The boundary term vanishes because  $h(y) \asymp \text{dist}(y, \{0, 2\pi\})^q$ , while the quadratic variation is bounded on every localization interval. The resulting Doob  $h$ -transform has drift

$$\cot(y/2) + 6 \frac{h'(y)}{h(y)}$$

after the eigenfunction change. For  $b = 1$  this is  $4 \cot(y/2)$ ; its speed measure has finite density proportional to  $\sin(y/2)^{8/3} dy$ . Hence the transformed diffusion is positive recurrent and, starting from any compact subinterval of  $(0, 2\pi)$ , spends a uniformly positive fraction of time in a fixed central subinterval. Reversing the change of measure yields the two-sided estimate

$$ce^{-5t/4} \leq \mathbb{E}[\Phi_t 1_{\{t < \tau\}}] \leq Ce^{-5t/4}. \quad (2.9)$$

Integrating over the surviving boundary arc identifies the left side, up to universal constants, with its harmonic measure from the origin. Extremal length converts harmonic measure of an arc separated by an annulus of  $\pi$ -extremal distance  $S$  into  $e^{-S}$  up to universal multiplicative constants. Finally, the radial Loewner normalization  $g_t'(0) = e^t$  and Koebe's quarter theorem imply that the first time the hull reaches the circle of radius  $r$  differs from  $\log(1/r)$  by a bounded amount. Therefore

$$\mathbb{E}[1_{\{\sigma < T\}} e^{-S}] \asymp r^{5/4}, \quad (2.10)$$

which is the asserted radial exponent  $\nu(1) = 5/4$ .  $\square$

**Lemma 2.8** (The fractional radial moment needed for three arms). *Let  $(K_t)$  be radial SLE<sub>6</sub> in  $\mathbb{D}$ , started from 1 and aimed at 0, and let  $T_r$  be its first hitting time of  $\bar{B}(0, r)$ . Let  $S_r/\pi$  denote the extremal distance, in  $\mathbb{D} \setminus K_{T_r}$ , from  $\partial B(0, r)$  to the surviving boundary arc containing  $-1$ ; put  $S_r = \infty$  if  $-1$  has already been swallowed. Then*

$$\mathbb{E}[e^{-S_r/3}] \asymp r^{2/3} \quad (r \downarrow 0).$$

*Equivalently, the missing radial exponent at weight  $b = 1/3$  is  $\nu(1/3) = 2/3$ .*

*Proof.* The eigenvalue  $\nu(1/3) = 2/3$  already follows algebraically from (2.8); the point that requires an additional argument is that the *length* of the whole surviving boundary arc has the same fractional moment as the one-point derivative. For  $b \geq 1$  this follows from convexity. Since  $b = 1/3 < 1$ , the inequality goes in the wrong direction, so we prove the fractional length estimate directly.

*Step 1: the boundary diffusion and its pointwise derivative moment.* Write  $W_t = e^{i\sqrt{6}B_t}$  and, for a boundary point  $e^{iy}$  not yet swallowed,

$$\frac{g_t(e^{iy})}{W_t} = e^{iY_t^y}, \quad \Phi_t^y = |g'_t(e^{iy})| = \partial_y Y_t^y.$$

As in (2.7),

$$dY_t^y = \sqrt{6} dB_t + \cot(Y_t^y/2) dt, \quad \partial_t \log \Phi_t^y = -\frac{1}{2 \sin^2(Y_t^y/2)}, \quad (2.13)$$

with absorption at 0 and  $2\pi$ , after which we set  $\Phi_t^y = 0$ . For general  $b > 0$ , put

$$q(b) = \frac{1 + \sqrt{1 + 24b}}{6}, \quad H_b(y) = \sin(y/2)^{q(b)}.$$

The calculation leading to (2.8) gives

$$\left( 3\partial_y^2 + \cot(y/2)\partial_y - \frac{b}{2 \sin^2(y/2)} \right) H_b = -\nu(b)H_b.$$

Feynman–Kac therefore yields, uniformly for  $t \geq 1$ ,

$$\mathbb{E}[(\Phi_t^y)^b] \asymp e^{-\nu(b)t} H_b(y). \quad (2.14)$$

For completeness, the two-sided constant in (2.14) is obtained without a boundary regularity black box. The positive local martingale  $e^{\nu(b)t} (\Phi_t^y)^b H_b(Y_t^y)$  defines the Doob transform whose drift is

$$\cot(y/2) + 6 \frac{H'_b(y)}{H_b(y)} = (1 + 3q(b)) \cot(y/2).$$

Its speed density is proportional to  $\sin(y/2)^{2/3+2q(b)} dy$ . This is finite, and multiplication by  $H_b^{-1}$  leaves the integrable density  $\sin(y/2)^{2/3+q(b)} dy$ . Splitting  $(0, 2\pi)$  into a fixed central interval and two endpoint intervals and integrating the explicit scale density shows that, after one unit of time, the transformed transition law has uniformly bounded  $H_b^{-1}$  moment. Reversing the change of measure proves the upper bound in (2.14); the lower bound follows by restricting the transformed process to a fixed central interval, which has uniformly positive mass.

We now fix

$$b = \frac{1}{3}, \quad q = \frac{2}{3}, \quad \nu = \frac{2}{3}. \quad (2.15)$$

Let  $A_t = \partial\mathbb{D} \setminus \overline{K_t}$  and let  $\ell_t$  be the Euclidean angular length of the arc  $g_t(A_t)$ ; when  $-1$  is swallowed we will always multiply by the survival indicator  $1_{\{t < \tau_\pi\}}$ .

*Step 2: the lower length-moment bound.* We record the distortion argument because it works for every  $b > 0$  and will also explain why the fractional case is not a problem from below. For  $t > 3$ , choose the integer  $t' \in (t-2, t-1]$  and let

$$\mathcal{V}_t = \left\{ \frac{\pi}{2} \leq Y_s^\pi \leq \frac{3\pi}{2} \text{ for every } s \in [t', t] \right\}.$$

Equation (2.14), the Markov property, and the positivity of the transition density on compact subintervals give a constant  $c_0 > 0$  such that

$$\mathbb{E}[(\Phi_t^\pi)^b 1_{\mathcal{V}_t}] \geq c_0 e^{-\nu t}. \quad (2.16)$$

For  $0 < \alpha < 1/5$  define

$$\mathcal{U}_t(\alpha) = \left\{ \alpha e^{-s/8} \leq Y_s^\pi \leq 2\pi - \alpha e^{-s/8} \text{ for all } 0 \leq s \leq t \right\}.$$

If  $\mathcal{W}_u$  is the analogous event on the unit interval  $[u, u+1]$ , then  $\mathcal{V}_t \cap \mathcal{U}_t(\alpha)^c \subset \bigcup_{u=0}^{t'-1} \mathcal{W}_u^c$ . Starting at time  $u$ , (2.14) contributes the endpoint factor  $H_b(\alpha e^{-u/8}) \leq C\alpha^q e^{-uq/8}$ ; hence

$$\mathbb{E}[(\Phi_t^\pi)^b 1_{\mathcal{V}_t \cap \mathcal{U}_t(\alpha)^c}] \leq C\alpha^q e^{-\nu t} \sum_{u \geq 0} e^{-uq/8}.$$

Choose  $\alpha$  so small that the right side is at most half of (2.16), and write  $\mathcal{U}_t^* = \mathcal{U}_t(\alpha) \cap \mathcal{V}_t$ . Then

$$\mathbb{E}[(\Phi_t^\pi)^b 1_{\mathcal{U}_t^*}] \geq c_1 e^{-\nu t}. \quad (2.17)$$

It remains to turn one derivative into a piece of boundary arc of fixed initial angular width. If  $\pi < y < \pi + \alpha/2$ , the common Brownian driving term cancels and the monotonicity of  $\cot$  gives

$$0 < Y_s^y - Y_s^\pi \leq e^{-s/2}(y - \pi). \quad (2.18)$$

On  $\mathcal{U}_t^*$  both trajectories stay at distance at least  $(\alpha/2)e^{-s/8}$  from the absorbing endpoints. Differentiating (2.13) with respect to the angular variable, and using  $|\partial_x \sin^{-2}(x/2)| \leq Ce^{3s/8}$  on this region, gives

$$\left| \partial_s \log \frac{\Phi_s^y}{\Phi_s^\pi} \right| \leq Ce^{-s/8}.$$

After integration in  $s$ ,  $e^{-C}\Phi_t^\pi \leq \Phi_t^y \leq e^C\Phi_t^\pi$  for every  $y \in [\pi, \pi + \alpha/2]$ . Consequently

$$\ell_t \geq c\Phi_t^\pi \quad \text{on } \mathcal{U}_t^*,$$

and (2.17) yields

$$\mathbb{E}[\ell_t^b 1_{\{t < \tau_\pi\}}] \geq ce^{-\nu t}. \quad (2.19)$$

*Step 3: the fractional upper bound.* This is the step that convexity does not supply when  $b < 1$ . Rescale the angle by

$$X_t^x = \frac{1}{\pi} Y_t^{\pi(x+1)} - 1 \in (-1, 1).$$

Then, before absorption,

$$dX_t = \frac{\sqrt{6}}{\pi} dB_t + f(X_t) dt, \quad f(x) = -\frac{1}{\pi} \tan \frac{\pi x}{2}. \quad (2.20)$$

The drift has the three quantitative properties used below:

$$f'(x) = -\frac{1}{2} \sec^2 \frac{\pi x}{2} \leq -\frac{1}{2}, \quad |f'(x)| \asymp d(x, \{\pm 1\})^{-2}, \quad |f''(x)| \asymp d(x, \{\pm 1\})^{-3} \quad (2.21)$$

near the endpoints. Let  $G_t$  be the stochastic flow of (2.20), let  $J_t(x) = \partial_x G_t(x)$  (zero after absorption), and let  $L_t$  be the length of the surviving image interval. Up to the harmless factor  $\pi$ ,  $L_t = \ell_t$  and the starting point  $x = 0$  corresponds to the marked boundary point  $-1$ .

Set  $a = 1/2$  and, for a large constant  $A$ , introduce the shrinking central intervals

$$I_s = [-1 + Ae^{-as/4}, 1 - Ae^{-as/4}].$$

From (2.21),

$$J_t(x) = \exp\left(\int_0^t f'(G_s(x)) ds\right) \leq e^{-at}. \quad (2.22)$$

Moreover

$$\partial_x \log J_t(x) = \int_0^t J_s(x) f''(G_s(x)) ds. \quad (2.23)$$

Suppose first that  $G_s(0) \in I_s$  for all  $s \leq t$ . Order preservation of the flow and (2.21)–(2.23) imply, for every surviving  $x$ , that whenever  $G_s(x)$  is also in  $I_s$ ,

$$|f'(G_s(x)) - f'(G_s(0))| \leq Ce^{-as}(Ae^{-as/4})^{-3} = CA^{-3}e^{-as/4}, \quad (2.24)$$

when  $G_s(x)$  is outside  $I_s$ , monotonicity of  $f'$  gives the same one-sided upper bound needed for  $J_t(x)$ . Integrating (2.24) gives  $J_t(x) \leq C_A J_t(0)$ . Hence

$$L_t \leq 2C_A J_t(0)$$

on this event, and (2.14) gives its contribution at most  $Ce^{-\nu t}$ . The same argument started at a stopping time  $s_1$  and run until  $s_2$  gives the conditional estimate

$$\mathbb{E}\left[L_{s_2}^b; G_s(0) \in I_s \ (s_1 < s < s_2) \mid \mathcal{F}_{s_1}\right] \leq C H_b(G_{s_1}(0)) L_{s_1}^b e^{-\nu(s_2-s_1)}. \quad (2.25)$$

We next quantify the gain when the marked trajectory approaches an endpoint. Suppose at a stopping time  $s$  that  $G_s(0) = -1 + u$ . Until it reaches  $-1 + 2u$ , scale space by  $u$  and time by  $u^2$ . The endpoint asymptotics in (2.21) show that the rescaled diffusion has drift and diffusion coefficients bounded above and below by universal constants on the corresponding compact annulus. Thus there are constants  $\eta, c > 0$ , independent of  $u$ , such that with conditional probability at least  $\eta$  the doubling time is at least  $cu^2$ . During that event,  $f' \leq -c/u^2$ , so (2.22) contracts every relevant derivative by a fixed factor  $\theta < 1$ . Iterating over the dyadic distances  $u, 2u, 4u, \dots$  from the endpoint shows that, when  $u = Ae^{-as/4}$ ,

$$\mathbb{E}\left[L_{s+(\tau \wedge 1)}^b \mid \mathcal{F}_s, G_s(0) = -1 + Ae^{-as/4}\right] \leq CL_s^b e^{-\eta_0 s}, \quad (2.26)$$

for some  $\eta_0 > 0$ ; the same estimate holds at the right endpoint. The factor is exponential in  $s$  because the number of available dyadic doublings is proportional to  $s$ . The endpoint factor  $H_b(G_s(0)) \asymp e^{-qas/4}$  can be absorbed into  $\eta_0$ .

Finally classify a path by the integer intervals  $[n_i, n_i + 1]$  on which  $G_s(0)$  exits  $I_s$ , taking the  $n_i$  separated by at least two. Repeatedly apply (2.25) between excursions and (2.26) at an excursion. For the event  $E_{t;k;n_1,\dots,n_k}$  of exactly these  $k$  excursion intervals,

$$\mathbb{E}\left[L_t^b 1_{\{t < \tau_0\}}; E_{t;k;n_1,\dots,n_k}\right] \leq CC_1^k e^{-\nu t} \prod_{i=1}^k e^{-\eta_0 n_i}. \quad (2.27)$$

Since  $n_i \geq i$ , summing first over the  $n_i$  and then over  $k$  gives

$$\sum_{k \geq 0} C_1^k \exp(-\eta_0 k(k+1)/2) < \infty.$$



Therefore

$$\mathbb{E}[L_t^{1/3} 1_{\{t < \tau_0\}}] \leq C e^{-2t/3}. \quad (2.28)$$

Together with (2.19), this proves the two-sided deterministic-time estimate

$$\mathbb{E}[\ell_t^{1/3} 1_{\{t < \tau_\pi\}}] \asymp e^{-2t/3}. \quad (2.29)$$

*Step 4: return to extremal distance and Euclidean radius.* At deterministic time  $t$ , conformal invariance identifies  $\ell_t/(2\pi)$  with the harmonic measure from the origin of the surviving boundary arc. If  $S$  is the corresponding  $\pi$ -extremal distance, the standard annulus comparison, proved by mapping the component to a round annulus and applying Koebe distortion to its two boundary collars, gives

$$e^{-S} \asymp \ell_t. \quad (2.30)$$

Also  $g'_t(0) = e^t$ . Schwarz's lemma applied to  $z \mapsto g_t(\rho(t)z)$  and Koebe's quarter theorem applied to  $g_t^{-1}$ , where  $\rho(t) = d(0, K_t)$ , give the explicit bounds

$$\frac{1}{4}e^{-t} \leq \rho(t) \leq e^{-t}. \quad (2.31)$$

To make the stopping-time transfer explicit, set  $t_+ = \log(1/r)$  and  $t_- = \log(1/(8r))$ . The bounds (2.31), applied before and after first entrance into the concentric circles, give deterministic brackets of the form

$$t_- < T_r \leq t_+ \leq T_{r/4}$$

(up to changing the harmless numerical constants). Extremal distance is monotone under radial growth, while the strong Markov property and (2.29) show that changing the deterministic time by a bounded amount changes the fractional moment by only a universal multiplicative factor. Hence the stopped moment has the same exponent as the deterministic-time moment. Combining (2.29)–(2.31) yields

$$\mathbb{E}[e^{-S_r/3}] \asymp r^{2/3},$$

as required.  $\square$

**Lemma 2.9** (Four alternating arms). *For critical site percolation on the triangular lattice, if  $\alpha_4(r, R)$  denotes the probability of four alternating arms across  $A(r, R)$ , then for each fixed admissible microscopic  $r$ ,*

$$\alpha_4(r, R) = \left(\frac{r}{R}\right)^{5/4+o(1)} \quad (R/r \rightarrow \infty).$$

*The same logarithmic exponent holds for the continuum four-arm probability obtained from the critical scaling limit.*

*Proof.* Arm separation (SM-11.02.01) lets us prescribe the alternating colors and separated landing arcs at constant cost. Use the annular exploration process in the universal cover, stopped at the first time its projection disconnects the two boundary components. By SM-11.04.04.a–e, the exploration converges to  $\text{SLE}_6$  at this stopping rule. The only possible discontinuity of the stopping functional would require six arms across a vanishing annulus. To rule this out uniformly, separate five of the arms and apply the five-arm estimate SM-11.02.03; the sixth arm must cross one of finitely many sectors independently of the separated five-arm core, and repeated RSW gives an extra factor  $(r/R)^c$ . Thus

$$\alpha_6(r, R) \leq C(r/R)^{2+c}, \quad (2.11)$$

which makes the bad stopping event negligible on a dyadic union of scales.

At the time the continuum exploration first reaches the inner boundary, its hull is bounded by one open and one closed arm. Four alternating arms occur exactly when the remaining component supports two further alternating crossings. If  $S/\pi$  is the extremal distance of that component, the conditional probability of these two crossings is  $\exp(-S + O(1))$  by the chordal part of Lemma 2.7. Taking expectation and using the radial part of that lemma gives

$$b_4^{\text{SLE}}(r, R) = \mathbb{E}[1_{\{\sigma < T\}} e^{-S+O(1)}] = \left(\frac{r}{R}\right)^{5/4+o(1)}. \quad (2.12)$$

Exploration convergence transfers this fixed-ratio continuum estimate to the lattice. Finally quasi-multiplicativity SM-11.02.02 iterates fixed-ratio annuli: for any  $\varepsilon > 0$  choose a sufficiently large ratio  $K$  so that the continuum crossing probability on one  $K$ -annulus lies between  $K^{-5/4-\varepsilon/2}$  and  $K^{-5/4+\varepsilon/2}$ ; convergence gives the same bounds on all sufficiently large lattice scales, and multiplying along  $K$ -adic annuli gives

$$(R/r)^{-5/4-\varepsilon} \leq \alpha_4(r, R) \leq (R/r)^{-5/4+\varepsilon}$$

for large  $R/r$ . Since  $\varepsilon$  is arbitrary, the claimed exponent follows.  $\square$

**Theorem 2.10** (Polychromatic arm exponents).

*For critical site percolation on the triangular lattice and every polychromatic cyclic arm pattern with  $k \geq 2$  arms,*

$$\alpha_k(r, R) = \left(\frac{r}{R}\right)^{(k^2-1)/12+o(1)} \quad (R/r \rightarrow \infty),$$

*for every fixed admissible microscopic  $r$ . The exponent is independent of the prescribed non-monochromatic color sequence.*

*Proof.* We separate the lattice comparison, the continuum SLE exponent, and the passage from fixed modulus to arbitrary scale ratios.

*Arm separation and color patterns.* By the arm-separation and color-switching lemmas proved earlier in this chapter, every prescribed non-monochromatic cyclic  $k$ -arm pattern has probability comparable, up to constants depending only on  $k$ , with the canonical annular-exploration event  $b_k(r, R)$ . Such multiplicative constants do not affect logarithmic exponents. In particular, after locating one opposite-colored adjacent pair, RSW fences separate the landing intervals and the remaining arms may be ordered relative to that pair. Thus it is enough to calculate the exponent for the canonical exploration event.

*Passage to  $\text{SLE}_6$ .* The interface convergence package SM-11.04.04.a–e identifies the scaling limit of the annular exploration, stopped when the relevant marked boundary pieces are disconnected, with radial/chordal  $\text{SLE}_6$ . Hence for every fixed ratio  $R/r = K$ , the lattice probability converges to the corresponding SLE crossing probability as the mesh tends to zero.

*The continuum exponent.* After the first radial crossing, the remaining  $k-2$  crossings contribute the half-plane SLE exponent

$$\lambda_{k-2} = \frac{(k-2)(k-1)}{6}.$$

The radial Feynman–Kac calculation in this chapter gives the conversion

$$\nu(\lambda) = \frac{4\lambda + 1 + \sqrt{1 + 24\lambda}}{8}.$$

For  $k \geq 4$ ,  $\lambda_{k-2} \geq 1$ , so the same eigenfunction/Doob-transform argument used in Lemma 2.7 is integrable at the boundary and yields

$$\nu(\lambda_{k-2}) = \frac{k^2 - 1}{12}.$$

The case  $k = 2$  is the zero-weight case and gives  $1/4$ . The four-arm case  $k = 4$  is the fully expanded Lemma 2.9. The remaining exceptional value  $k = 3$  corresponds to the fractional moment  $\lambda = 1/3$ ; Lemma 2.8 proves directly that

$$\mathbb{E}[e^{-S/3}] \asymp (r/R)^{2/3},$$

so  $\nu(1/3) = 2/3 = (3^2 - 1)/12$ . Thus the continuum exponent has the stated value for every  $k \geq 2$ .

*From fixed ratio to arbitrary scales.* The fixed-ratio continuum estimate, lattice convergence, and the quasi-multiplicativity theorem SM-11.02.02 give matching bounds on each  $K$ -adic annulus. Iterating over the annuli between  $r$  and  $R$  and then letting  $K$  grow gives

$$\frac{\log \alpha_k(r, R)}{\log(r/R)} \longrightarrow \frac{k^2 - 1}{12}.$$

If the comparison theorem was first established for an inner radius  $r_0$  above a finite source-dependent threshold, then for any other fixed admissible  $r < r_0$ ,

$$\alpha_k(r, R) \asymp \alpha_k(r, r_0) \alpha_k(r_0, R),$$

and the first factor is independent of  $R$ . The exponent therefore holds for every fixed admissible microscopic  $r$ . Combining this with the first paragraph proves independence of the prescribed polychromatic color sequence.  $\square$

## Chapter 3

# Pivotality

A microscopic site can influence a macroscopic event only through a rare multi-arm configuration.

**Definition 3.1** (Pivotal site and influence). For an increasing event  $A$  depending on finitely many percolation bits, a site  $v$  is *pivotal* for  $A$  in configuration  $\omega$  when flipping only the state of  $v$  changes whether  $A$  occurs. Its influence is  $\text{Inf}_v(\mathbf{1}_A) = \mathbb{P}(v \text{ is pivotal})$  under the relevant product measure.

*Worked example 3.2* (Russo's formula on a two-edge series network). Let  $A$  be the event that two independent edges  $e_1, e_2$  are both open. Then  $\mathbb{P}_p(A) = p^2$ , so

$$\frac{d}{dp} \mathbb{P}_p(A) = 2p.$$

Edge  $e_1$  is pivotal exactly when  $e_2$  is open, hence with probability  $p$ , and similarly for  $e_2$ . Therefore

$$\sum_{i=1}^2 \mathbb{P}_p(e_i \text{ pivotal}) = 2p,$$

which is Russo's formula in a complete finite calculation. In a macroscopic planar crossing the same identity holds, but pivotality forces a four-arm event from the microscopic site to macroscopic distance.

**Example 3.3** (A running picture). At  $p = 1/2$  on the triangular lattice, zooming out does not drive crossings to probability 0 or 1. The chapter identifies the mechanism responsible for this scale-free behavior.

**Theorem 3.4** (Russo formula in pivotal form).

*Let  $A$  be an increasing event depending on finitely many sites of site percolation. Then*

$$\frac{d}{dp} \mathbb{P}_p(A) = \sum_v \mathbb{P}_p(v \text{ is pivotal for } A).$$

*Proof.* Condition on all sites except  $v$ . For a fixed outside configuration, the conditional probability of  $A$  is 0, 1, or  $p$ , and the third case occurs exactly when changing the state of  $v$  changes  $A$ . Therefore the partial derivative with respect to the parameter at  $v$  equals its pivotal probability. Summing the partial derivatives along the diagonal where every site has parameter  $p$  proves the formula. This is the site version of the finite-product argument in Volume X.  $\square$

**Theorem 3.5** (Four arms characterize an interior pivotal).

Let  $A$  be the event of an open left–right crossing of a topological quadrilateral in critical site percolation on the triangular lattice. If a site  $v$  is pivotal and lies at macroscopic distance from the boundary, then four disjoint arms issue from the neighborhood of  $v$  to the four boundary arcs, alternating open and closed in cyclic order. Conversely, four such alternating arms with the two open arms landing on the open boundary arcs and the two closed arms on the complementary arcs force  $v$  to be pivotal.

*Proof.* If  $v$  is pivotal, then with  $v$  open there is a left–right open crossing that must use  $v$ ; erase loops to obtain two disjoint open arms from  $v$  to the left and right boundary arcs. With  $v$  closed there is no left–right open crossing. Planar duality then gives a closed top–bottom dual crossing through the dual neighborhood of  $v$ , which yields two closed arms to the remaining boundary arcs. Planarity forces the four arms to alternate.

Conversely, if the four alternating arms occur with the stated landing data, opening  $v$  joins the two open arms and creates a left–right crossing. Closing  $v$  joins the two closed arms into a top–bottom dual barrier, which prevents any left–right open crossing. Thus changing the state of  $v$  changes  $A$ , so  $v$  is pivotal.  $\square$

**Theorem 3.6** (Pivotal-measure scaling limit and conformal covariance).

Let  $A$  be a fixed proper annulus with inside face  $\Delta$ , and for critical site percolation on  $\eta\mathbb{T}$  define

$$\mu_\eta^A := \sum_{x \in \Delta: x \text{ is } A\text{-important}} \delta_x \frac{\eta^2}{\alpha_4^\eta(\eta, 1)}.$$

Then  $(\omega_\eta, \mu_\eta^A)$  converges in law to  $(\omega, \mu^A)$ , where  $\omega$  is the critical quad-crossing scaling limit and  $\mu^A$  is a measurable function of  $\omega$ . If  $f : \Omega \rightarrow \tilde{\Omega}$  is conformal and  $A \subseteq \Omega$ , then almost surely

$$\mu^{f(A)}(f(B)) = \int_B |f'(z)|^{3/4} d\mu^A(z)$$

for Borel  $B$  in the inside face of  $A$ . The same construction applied to a topological quadrilateral gives the normalized measure on its macroscopic pivotal points.

*Proof.*

$$\mathbb{E}\mu_\eta^A(\Delta) \leq C|\Delta| \frac{\alpha_4^\eta(\eta, d \wedge 1)}{\alpha_4^\eta(\eta, 1)} \leq C_A.$$

For a square  $Q$  of side  $r$  contained in  $\Delta$ , pair two candidate important sites according to their mutual distance, use four arms from each microscopic site to that distance and four arms from the resulting pair to the outer boundary, and sum over dyadic distance shells. Quasi-multiplicativity gives the uniform second-moment estimate

$$\mathbb{E}[\mu_\eta^A(Q)^2] \leq C_A \frac{r^4}{\alpha_4(r, 1)}.$$

The analogous third-moment bound is obtained by ordering the two smallest mutual distances of a triple. These estimates give tightness of the random finite measures and uniform integrability needed below.

$$(1 - c)^N \leq C(r/R)^k$$

for some  $k > 0$ . After success, the two conditional configurations can be sampled identically inward. This is the coupling input that makes the microscopic pivotal count asymptotically independent of its remote arm geometry.

$$\mathbb{E}[X_i \mid \text{mesoscopic data}] = \beta_{\eta,\epsilon} Y_i + o(\beta_{\eta,\epsilon})$$

uniformly over all well-separated landing data.

$$\mathbb{E}[(X - \beta_{\eta,\epsilon} Y)^2] = o(\mathbb{E}[X^2])$$

as first  $\eta/\epsilon \rightarrow 0$  and then  $\epsilon \rightarrow 0$ . After the normalization  $\eta^{-2}\alpha_4^\eta(\eta, 1)$ , the pivotal mass is therefore approximated in  $L^2$  by a quantity determined solely by macroscopic four-arm events.

The logarithmic exponent by itself is not enough here; we need the stronger ratio limit. Fix  $0 < r < 1$  and put

$$R_\eta(r) := \frac{\alpha_4^\eta(\eta, r)}{\alpha_4^\eta(\eta, 1)}.$$

Choose an intermediate scale  $\gamma$  with  $\eta \ll \gamma \ll r$ . Condition on the four arms between  $\gamma$  and the outer circle. By Step b, after forcing separated landing intervals at scale  $\gamma$ , the law of the configuration inside  $A(\eta, \gamma)$  can be coupled to the canonical four-arm law with error at most  $C(\gamma/r)^k$ , uniformly in all admissible outer landing data. Consequently, if  $\eta, \eta' \ll \gamma$ , then

$$\left| \frac{R_\eta(r)}{R_{\eta'}(r)} - 1 \right| \leq C(\gamma/r)^k + o_{\eta, \eta' \downarrow 0}(1).$$

First let  $\eta, \eta' \downarrow 0$  and then  $\gamma/r \downarrow 0$ . Hence  $R_\eta(r)$  has a finite positive limit, denoted  $L(r)$ .

Apply the same separation argument on successive annuli. For an integer  $n \geq 1$ ,

$$\alpha_4(r^n, 1) = \prod_{j=1}^n \frac{\alpha_4(r^j, 1)}{\alpha_4(r^{j-1}, 1)}.$$

Scale invariance of the continuum limit and the coupling just proved imply that, away from the two end annuli, the factors converge to  $L(r)^{-1}$ . Therefore

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \alpha_4(r^n, 1) = -\log L(r).$$

Lemma 2.9 identifies the left side as  $(5/4) \log r$ , and consequently

$$\boxed{L(r) = r^{-5/4}}. \tag{3.1}$$

Equivalently, for every fixed  $s > 0$  for which  $s\epsilon$  remains below the macroscopic scale,

$$\frac{\alpha_4(s\epsilon, 1)}{\alpha_4(\epsilon, 1)} \longrightarrow s^{5/4}. \tag{3.2}$$

The rotated-square and topological-quad normalization follows from the same coupling, not from a new exponent computation. Once the four arms have entered four separated corridors, their microscopic continuation forgets the outer landing data with the power error of Step b. Thus the conditional expected number of microscopic pivotal sites in an  $\epsilon$ -square, divided by

$$\frac{\epsilon^2 \alpha_4(\epsilon, 1)}{\eta^2 \alpha_4(\eta, 1)},$$

has a limit independent of the remote quad geometry. Rotational invariance of the continuum scaling limit makes the limiting constant independent of the orientation. This is exactly the deterministic coefficient required in Step d, while (3.1)–(3.2) give its scale dependence.

$$\mu^A(\phi_j) = \lim_{\epsilon \downarrow 0} Z_j^\epsilon(\omega) \quad \text{in } L^2.$$

Any subsequential weak limit  $\mu^*$  has the same integrals against every  $\phi_j$ , by uniform moment bounds and the discrete  $L^2$  approximation. The dense test family separates finite measures, so  $\mu^* = \mu^A(\omega)$  almost surely. This proves uniqueness and measurability once Step e is available.

$$s^2 \frac{\alpha_4(\eta/s, 1)}{\alpha_4(\eta, 1)} \longrightarrow s^{3/4},$$

using the ratio limit of Step e. For a general conformal  $f$ , partition a compact subset into  $\epsilon$ -squares on which

$$f(z) = f(z_i) + f'(z_i)(z - z_i) + O(\epsilon^2).$$

Strong separation and the  $L^2$  mesoscopic approximation show that replacing  $f$  on each square by its tangent similarity changes the total pivotal mass by a quantity tending to zero in  $L^2$ . Summing the local similarity factors gives

$$f_* \mu^A(dw) = |f'(f^{-1}w)|^{-3/4} \mu^{f(A)}(dw),$$

equivalently the formula in the theorem.

For pivotality of a fixed topological quadrilateral, use the same construction with four arms required to land on the four prescribed boundary arcs. The only additional issue is a box at distance  $\rho$  from a smooth boundary piece. A pivotal point there forces three alternating half-plane arms from the mesh scale to order  $\rho$ . The chordal calculation in Lemma 2.7 gives  $u(1) = 2$ , so arm separation, RSW and quasi-multiplicativity yield

$$\alpha_3^+(\eta, \rho) = (\eta/\rho)^{2+o(1)}.$$

After multiplying by the pivotal normalization  $\eta^2/\alpha_4(\eta, 1)$  and summing dyadically over boundary boxes, the expected normalized mass in a boundary strip tends to zero first as  $\eta \downarrow 0$  and then as the strip width tends to zero. Hence no boundary mass survives, while the interior argument from Steps c–g applies unchanged. This proves the quad version and completes the theorem.  $\square$

## Chapter 4

# Smirnov theory

Smirnov's observable is the decisive bridge from triangular-lattice combinatorics to conformal analysis.

**Definition 4.1** (Dobrushin percolation domain). A Dobrushin domain is a simply connected discrete domain with two marked boundary points  $a, b$  and boundary conditions chosen so that one boundary arc from  $a$  to  $b$  is open/wired while the complementary arc is closed/dual wired. The interface from  $a$  to  $b$  separating the two boundary clusters is the percolation exploration path.

**Example 4.2** (A running picture). At  $p = 1/2$  on the triangular lattice, zooming out does not drive crossings to probability 0 or 1. The chapter identifies the mechanism responsible for this scale-free behavior.

**Theorem 4.3** (Smirnov preholomorphic observable).

*Let  $\Omega_\delta$  be a simply connected triangular-lattice approximation with three counterclockwise marked boundary prime ends  $a(1), a(\tau), a(\tau^2)$ , where  $\tau = e^{2\pi i/3}$ . For  $\alpha \in \{1, \tau, \tau^2\}$  let  $H_\alpha^\delta(z)$  be Smirnov's separation probability. If  $z$  and  $z + \eta$  are centers of adjacent lattice triangles and*

$$P_\beta(z, \eta) := \mathbb{P}(Q_\beta(z + \eta) \setminus Q_\beta(z)),$$

*then the exact color-switching identity*

$$P_\beta(z, \eta) = P_{\tau\beta}(z, \tau\eta)$$

*holds. Moreover there exist  $\varepsilon, C > 0$ , depending only on the macroscopic domain/marked data, such that the  $H_\alpha^\delta$  are uniformly  $\varepsilon$ -Hölder and their discrete directional derivatives satisfy the approximate  $120^\circ$  Cauchy–Riemann relation*

$$H_\beta^\delta(z + \eta) - H_\beta^\delta(z) = H_{\tau\beta}^\delta(z + \tau\eta) - H_{\tau\beta}^\delta(z) + O(\delta^{1+\varepsilon})$$

*away from the marked prime ends. Consequently the discrete contour relations used below have error  $O(\ell\delta^\varepsilon)$  for a lattice triangular contour of macroscopic length  $\ell$ ; no exact lattice-holomorphicity is asserted.*

*Proof.*

$$H_\beta^\delta(z + \eta) - H_\beta^\delta(z) = P_\beta(z, \eta) - P_\beta(z + \eta, -\eta).$$

This is just the decomposition into configurations for which the separation event changes in one direction or the other.



$$P_\beta(z, \eta) = P_{\tau\beta}(z, \tau\eta).$$

This is the exact local identity in Smirnov's proof.

$$|H_\beta^\delta(z) - H_\beta^\delta(z')| \leq C|z - z'|^\varepsilon.$$

The same annulus argument, with boundary half-annuli, gives the required boundary modulus and the limiting boundary values. In particular a single-edge pivotal probability is  $O(\delta^\varepsilon)$ .

$$\partial_\eta H_\beta^\delta(z) = \partial_{\tau\eta} H_{\tau\beta}^\delta(z) + O(\delta^{1+\varepsilon}).$$

This is the source-correct approximate discrete Cauchy–Riemann statement. □

**Theorem 4.4** (Convergence of Smirnov's observable).

*For the above critical triangular-lattice approximations,  $H_\alpha^\delta$  converges uniformly on compact subsets, and with the boundary interpretation of prime ends uniformly in the natural compactification, to the unique solution  $h_\alpha$  of Smirnov's conformally invariant mixed Dirichlet–Neumann problem. Equivalently, after conformally mapping the marked domain to an equilateral triangle sending  $a(1), a(\tau), a(\tau^2)$  to its three vertices,  $h_\alpha$  is the affine function equal to 1 at the vertex  $a(\alpha)$  and 0 on the opposite side.*

*Proof.*

$$\oint_\Gamma f_\beta(z) dz = \oint_\Gamma \tau^{-1} f_{\tau\beta}(z) dz.$$

Taking the same identity for  $\tau\beta$  and an appropriate real linear combination shows that

$$F_\alpha(z) := f_\alpha(z) + \frac{i}{\sqrt{3}}(f_{\tau\alpha}(z) - f_{\tau^2\alpha}(z))$$

has zero contour integral around every sufficiently small equilateral triangle. Triangulating an arbitrary polygon and passing to a limit extends this to all rectifiable triangular contours, so Morera's theorem makes  $F_\alpha$  holomorphic.

$$f_\alpha = 0 \quad \text{on } a(\tau\alpha)a(\tau^2\alpha),$$

and zero oblique normal derivative on the two adjacent arcs, together with  $f_\alpha(a(\alpha)) = 1$  in the prime-end sense. □

**Theorem 4.5** (Cardy's formula on the triangular lattice).

*Let  $(\Omega; a, b, c, x)$  be a simply connected conformal quadrilateral approximated by critical site percolation on the triangular lattice. The probability of an open crossing from the boundary arc  $xa$  to the opposite arc  $bc$  converges to the conformally invariant Cardy value. In Carleson's normalization, if  $\Omega$  is an equilateral triangle  $abc$  of side one and  $x \in [bc]$ , the limit is  $|xb|$ .*

*Proof.* Choose the separation probability among the three observables of Theorem SM-11.04.02 so that its boundary values are 0 on the arc  $ab$ , 1 at the opposite vertex  $c$ , and its level at the fourth marked prime end  $x$  is exactly the discrete crossing probability from  $xa$  to  $bc$ . This identification is topological: the exploration separates the corresponding boundary arcs if and only if the indicated open crossing occurs.

By Theorem SM-11.04.02, the discrete observable converges to the unique conformally covariant solution of the limiting boundary-value problem. Map  $(\Omega; a, b, c)$  conformally to the unit equilateral triangle with the same ordered vertices. In that normalization the limiting solution is the affine barycentric coordinate which is 0 on the side opposite its selected vertex and 1 at that vertex. Restricted to the side  $bc$ , this affine coordinate is linear in Euclidean arclength. If the side has length one and  $x \in [bc]$ , its value is therefore

$$h(x) = |xb|.$$

Since the construction of  $h$  is conformally natural, pulling it back to the original quadrilateral gives the conformally invariant Cardy value. Uniform boundary convergence of the observable then yields convergence of the crossing probability itself.  $\square$

**Definition 4.6** (Loop-collection topology). For continuous planar loops modulo orientation-preserving reparametrization set

$$d_{\text{loop}}(\gamma, \eta) := \inf_{\varphi \in \text{Homeo}^+(\mathbb{S}^1)} \sup_{t \in \mathbb{S}^1} |\gamma(t) - \eta(\varphi(t))|.$$

If  $\mathcal{L}$  is a loop collection in the bounded domain  $D$ , let  $\mathcal{L}^{(m)}$  be the subcollection of loops with Euclidean diameter at least  $1/m$ . On collections for which every  $\mathcal{L}^{(m)}$  is finite, define

$$d_{\text{coll}}(\mathcal{L}, \mathcal{L}') = \sum_{m=1}^{\infty} 2^{-m} (1 \wedge d_H^{\text{loop}}(\mathcal{L}^{(m)}, \mathcal{L}'^{(m)})),$$

where  $d_H^{\text{loop}}$  is Hausdorff distance induced by  $d_{\text{loop}}$  (and is set equal to 1 if exactly one of the two finite sets is empty). This is the loop-collection topology used below.

**Theorem 4.7** (Full critical-percolation interface scaling limit).

*For critical site percolation on triangular-lattice approximations of a bounded simply connected Jordan domain, the complete collection of cluster-boundary interfaces converges jointly, in the loop-collection topology of Definition 4.6, to the conformally invariant continuum nonsimple-loop ensemble obtained by iterated chordal  $\text{SLE}_6$  exploration. The limiting ensemble is almost surely countable and locally finite above every positive diameter, loops do not cross although they may touch themselves and one another, every deterministic interior point is surrounded by nested loops of both orientations, and the conditional renewal rule in complementary components is the continuum limit of the discrete exploration rule.*

*Proof.* We separate the one-interface limit from the construction of the complete loop ensemble.

*Step 1: tightness of a single exploration.* For an annulus  $A(z; r, R)$  of fixed modulus, RSW gives a number  $c = c(R/r) > 0$ , independent of the mesh  $\delta$ , such that both colors have an annular circuit with probability at least  $c$ . By placing disjoint circuits in successive dyadic annuli, an exploration can make  $k$  disjoint crossings of  $A(z; r, R)$  only if a corresponding  $k$ -arm event occurs. The arm estimates of Volume X therefore give constants  $C, \alpha > 0$  such that

$$\sup_{\delta > 0} \mathbf{P}_{\delta}(\text{the exploration makes } k \text{ separated crossings of } A(z; r, R)) \leq C(r/R)^{\alpha}$$

for every fixed  $k$  large enough. The Aizenman–Burchard compactness criterion then makes the family of unparametrized exploration curves tight in the uniform metric modulo reparametrization.

*Step 2: identification of every one-curve subsequential limit.* Let  $\gamma$  be a subsequential limit in a Dobrushin domain and map the unexplored component to  $\mathbb{H}$  with half-plane-capacity parametrization.

The discrete domain Markov property passes to every stopping time at which the curve stays a positive distance from the remaining marked boundary, because RSW controls fjords and boundary excursions. Hence the Loewner driving function  $W_t$  has stationary Markovian increments after conformal recentering. Cardy's formula, proved in Theorem SM-11.04.03, supplies the martingale observable used to determine those increments. Expanding that martingale for a small capacity increment  $\Delta t$  gives

$$\mathbf{E}[\Delta W_t \mid \mathcal{F}_t] = o(\Delta t), \quad \mathbf{E}[(\Delta W_t)^2 \mid \mathcal{F}_t] = 6 \Delta t + o(\Delta t),$$

and the RSW crossing bound gives a Lindeberg estimate for large increments. Therefore the continuous martingale part of  $W$  has quadratic variation  $[W]_t = 6t$  and no finite-variation drift. Lévy's characterization yields

$$W_t = \sqrt{6} B_t.$$

Thus every one-curve subsequential limit is chordal  $\text{SLE}_6$ , and the law of a single exploration has a unique scaling limit.

*Step 3: finitely many recursive explorations.* Choose a deterministic countable dense set  $\{z_j\}_{j \geq 1} \subset D$ . In the discrete configuration explore the Dobrushin interface first, then in each unexplored component explore the first component (in a deterministic ordering) which contains one of  $z_1, \dots, z_m$  and has diameter at least  $1/m$ . For fixed  $m$  this produces only finitely many exploration paths with probability tending to one. Induct on the number of explored paths. Conditional on the first  $r$  paths, the unexplored percolation in every remaining component is again critical percolation with the induced boundary conditions. Step 2 and Carathéodory continuity of chordal  $\text{SLE}_6$  in domains whose relevant prime ends remain separated therefore give joint convergence of the first  $r + 1$  paths. The possible failure caused by a microscopic fjord has probability tending to zero by the same multi-arm estimates used in Step 1. Hence every finite stage of the recursive exploration has a unique joint continuum limit.

*Step 4: no macroscopic loop is missed.* Fix  $\varepsilon > 0$ . Cover  $D$  by finitely many squares of side  $\varepsilon/100$ . Any loop of diameter at least  $\varepsilon$  either surrounds one of the chosen dense points before the corresponding exploration stage, or creates an annular separation across one of finitely many squares that has not been detected by the recursive exploration. The latter event forces six alternating arms through an annulus whose inner radius tends to the mesh and whose outer radius is of order  $\varepsilon$ . The six-arm estimate tends to zero uniformly over the finite cover. Consequently, for every  $\eta > 0$  there is a finite exploration stage  $m = m(\varepsilon, \eta)$  such that

$$\limsup_{\delta \downarrow 0} \mathbf{P}_\delta(\text{some loop of diameter } \geq \varepsilon \text{ is not discovered by stage } m) \leq \eta. \quad (11.17)$$

The finite-stage convergence of Step 3 and (11.17) imply convergence of the entire discrete loop collection in  $d_{\text{coll}}$ .

*Step 5: qualitative properties of the limit.* For every  $\varepsilon > 0$ , (11.17) and the finite-stage construction imply that there are almost surely only finitely many limiting loops of diameter at least  $\varepsilon$ ; taking rational  $\varepsilon$  proves countability and local finiteness above every positive diameter. A transverse crossing of two limiting interfaces would force a six-arm event at arbitrarily small scales, so loops do not cross. Touching is not excluded because the corresponding four-arm events have positive scaling probability. RSW applied in a sequence of disjoint annuli around a deterministic interior point, followed by Borel–Cantelli, gives infinitely many surrounding circuits of alternating orientations; these become nested limiting loops. Finally, conditioning on a finite explored subcollection and using the discrete domain Markov property before taking the limit gives the stated renewal rule in every complementary component. These properties characterize the recursively generated nonsimple-loop ensemble, completing the proof.  $\square$

# Chapter 5

## Critical exponents

Once SLE<sub>6</sub> supplies arm exponents, Kesten's scaling relations translate them into thermodynamic critical exponents.

**Definition 5.1** (Correlation length and exponent notation). Fix a crossing threshold  $\varepsilon_0 \in (0, 1/2)$ . A near-critical correlation length  $L(p)$  is any conventional scale at which a fixed-shape crossing probability first differs from its critical value by at least  $\varepsilon_0$ . Statements of the form  $L(p) = |p - p_c|^{-\nu+o(1)}$  define the correlation-length exponent  $\nu$ ; similarly  $\theta(p) = (p - p_c)^{\beta+o(1)}$  defines the percolation-density exponent  $\beta$ .

*Worked example 5.2* (Deriving  $\beta = 5/36$  from two continuum inputs). For triangular-lattice percolation, the one-arm exponent is  $5/48$  and the correlation-length exponent is  $\nu = 4/3$ . Kesten's near-critical comparison gives, at logarithmic-equivalence level,

$$\theta(p) \asymp \alpha_1(1, L(p)).$$

Insert  $\alpha_1(1, R) = R^{-5/48+o(1)}$  and  $L(p) = (p - p_c)^{-4/3+o(1)}$ :

$$\theta(p) = (p - p_c)^{(4/3)(5/48)+o(1)} = (p - p_c)^{5/36+o(1)}.$$

Thus  $\beta = 5/36$ . The calculation is short only because the difficult arm and correlation-length exponents have already been proved separately.

**Example 5.3** (A running picture). At  $p = 1/2$  on the triangular lattice, zooming out does not drive crossings to probability 0 or 1. The chapter identifies the mechanism responsible for this scale-free behavior.

**Theorem 5.4** (One-arm exponent).

*For critical site percolation on the standard triangular lattice,*

$$\mathbb{P}_{1/2}(0 \longleftrightarrow \partial B(0, R)) = R^{-5/48+o(1)} \quad (R \rightarrow \infty). \quad (5.1)$$

*Equivalently, in the critical continuum scaling limit, if  $Q$  is the union of all open clusters meeting the unit circle, then for some constants  $0 < c < C < \infty$ ,*

$$cr^{5/48} \leq \mathbb{P}(\text{dist}(0, Q) < r) \leq Cr^{5/48}, \quad 0 < r < 1/2. \quad (5.2)$$

*Proof.* We first determine the exponent for the continuum cluster using radial SLE<sub>6</sub>. Only after that calculation do we return to the lattice. This separation is important: the continuum exponent is a mixed-boundary spectral problem, while the lattice step is an RSW/FGK annulus comparison.

For  $0 \leq \theta \leq 2\pi$ , let  $A_\theta$  be the counterclockwise boundary arc from 1 of angular length  $\theta$ , and let  $Q(\theta)$  be the scaling-limit union of  $A_\theta$  and all open clusters attached to that arc. Let  $U(\theta)$  be the component of 0 in  $\mathbb{D} \setminus Q(\theta)$  and set

$$\mathfrak{r}(\theta) = \text{crad}(U(\theta), 0). \quad (5.3)$$

Koebe's quarter theorem and Schwarz's lemma give

$$\frac{1}{4}\mathfrak{r}(\theta) \leq \text{dist}(0, Q(\theta)) \leq \mathfrak{r}(\theta). \quad (5.4)$$

Thus the exponent of the distance to  $Q(2\pi)$  is the exponent of the lower tail of  $\mathfrak{r}(2\pi)$ . Write

$$h(\theta, t) = \mathbb{P}(\mathfrak{r}(\theta) \leq e^{-t}). \quad (5.5)$$

For  $t > 0$ , RSW implies  $h(0, t) = 0$ .

Explore from 1 towards 0 by radial SLE<sub>6</sub>. Let  $g_t$  be normalized by  $g_t(0) = 0$  and  $g'_t(0) = e^t$ . Before the marked boundary point  $e^{i\theta}$  is separated from 0, define its angular displacement from the driving point by

$$Y_t = -i \log g_t(e^{i\theta}) - \sqrt{6}B_t \in (0, 2\pi). \quad (5.6)$$

The radial Loewner equation gives

$$dY_t = \cot(Y_t/2) dt - \sqrt{6} dB_t. \quad (5.7)$$

Let  $T$  be the first hit of  $\{0, 2\pi\}$ . If  $Y_T = 0$ , the conformal radius of the component containing 0 is  $e^{-T}$ . If  $Y_T = 2\pi$ , conformal invariance and the domain Markov property insert an independent copy  $\mathfrak{r}'$  of  $\mathfrak{r}(2\pi)$  and the radius is  $e^{-T}\mathfrak{r}'$ . With the convention  $h(\vartheta, s) = 1$  for  $s \leq 0$ , these two cases combine as

$$h(\theta, t) = \mathbb{E}[h(Y_T, t - T)]. \quad (5.8)$$

For each fixed  $s > 0$ , the renewal identity makes  $t \mapsto h(Y_t, s - t)$  a local martingale up to  $T \wedge s$ . Applying Itô's formula with (5.7) gives, for  $0 < \theta < 2\pi$  and  $t > 0$ ,

$$3h_{\theta\theta} + \cot(\theta/2)h_\theta - h_t = 0. \quad (5.9)$$

The coefficient is singular at the two endpoints, so the second boundary condition cannot simply be guessed. It is the main technical point of the continuum calculation.

Set

$$\tilde{h}(\theta, t) = \int_0^1 h(\theta, t + s) ds. \quad (5.10)$$

It satisfies the same PDE. We prove

$$\partial_\theta \tilde{h}(2\pi, t) = 0, \quad (5.11)$$

where the derivative is one-sided.

The input is an a priori lattice estimate uniform in the mesh. In a half-plane, let  $f(R, r)$  be the probability of two disjoint crossings between semicircles of radii  $r < R$ . Consider clusters in a strip of width  $R$  joining a boundary interval of length  $2R$  to the opposite side. RSW bounds their expected number above and gives positive probability of seeing more than two. Translation invariance along the boundary then shows that the event in which a specified boundary vertex is the lowest endpoint of such a crossing has probability  $\asymp R^{-1}$ . Flipping colors below the extremal crossing turns this

event into two prescribed disjoint crossings. A final RSW gluing step and quasi-multiplicativity therefore give

$$f(R, r) \asymp r/R. \quad (5.12)$$

Now require, in addition, a third crossing of the opposite color. BK on a middle annulus, together with an RSW separation annulus, supplies a uniform power saving, so for some  $\alpha, c > 0$ ,

$$\mathbb{P}(\text{two arms of one color and one of the other from } r \text{ to } R) \leq c(r/R)^{1+\alpha}. \quad (5.13)$$

Put  $\epsilon = 2\pi - \theta$  and  $Q' = Q(2\pi) \setminus Q(\theta)$ . If  $Q'$  reaches from scale  $\epsilon$  to  $r$  around the missing endpoint of the boundary arc, planarity forces the three-arm event just estimated. Hence

$$\mathbb{P}(\text{diam } Q' \geq r) \leq c(\epsilon/r)^{1+\alpha}. \quad (5.14)$$

For every boundary-attached connected compact set  $K \subset \overline{\mathbb{D}}$ , harmonic measure from 0 is comparable with  $\text{diam } K$ , and monotonicity of conformal radius plus comparison with the smallest disk orthogonal to  $\partial\mathbb{D}$  containing  $K$  gives

$$\min\{-\log \text{crad}(\mathbb{D} \setminus K, 0), 1\} \leq C(\text{diam } K)^2.$$

Combining this deterministic estimate with (5.14), integrating the resulting tail, and using conformal invariance of harmonic measure yields

$$\tilde{h}(2\pi, t) - \tilde{h}(2\pi - \epsilon, t) = o(\epsilon),$$

which is exactly (5.11).

For general  $\kappa > 4$ , define

$$q = \frac{\kappa - 4}{\kappa}, \quad \lambda = \frac{\kappa^2 - 16}{32\kappa}, \quad H(\theta, t) = \sin(\theta/4)^q e^{-\lambda t}.$$

A direct calculation shows that

$$\frac{\kappa}{2} H_{\theta\theta} + \cot(\theta/2) H_\theta - H_t = 0, \quad H(0, t) = 0, \quad H_\theta(2\pi, t) = 0.$$

For  $\kappa = 6$ ,  $q = 1/3$  and  $\lambda = 5/48$ .

We compare  $H$  and  $\tilde{h}$  by a maximum-principle argument that handles the singular endpoint at 0. If  $G$  satisfies the homogeneous equation and the same mixed boundary conditions, perturb it by  $\varepsilon(1 + \theta^{q/2})$ . The differential operator applied to the perturbation has strict sign, so a first nonpositive interior minimum is impossible. First apply this comparison on a compact time strip to obtain comparable positive data at  $t = 1$ ; then apply it on  $[0, 2\pi] \times [1, \infty)$ . There are constants  $c, C > 0$  with

$$cH(\theta, t) \leq \tilde{h}(\theta, t) \leq CH(\theta, t), \quad t \geq 1. \quad (5.15)$$

Since  $h(\theta, t)$  decreases in  $t$ ,  $h(2\pi, t+1) \leq \tilde{h}(2\pi, t) \leq h(2\pi, t)$  and also  $h(2\pi, t) \leq \tilde{h}(2\pi, t-1)$  for  $t \geq 2$ . Thus (5.15) implies

$$\mathbb{P}(\mathfrak{r}(2\pi) \leq e^{-t}) \asymp e^{-5t/48}.$$

Koebe's comparison (5.4) gives (5.2).

Let  $u(r) = \mathbb{P}(Q(2\pi) \cap r\partial\mathbb{D} \neq \emptyset)$  and let  $\mathcal{C}(rR, R)$  be the event of an open crossing of the lattice annulus with radii  $rR$  and  $R$ . The already-owned full critical scaling limit implies that for every fixed  $r \in (0, 1)$ , once  $R$  is sufficiently large,

$$\frac{1}{2}u(r/2) \leq \mathbb{P}(\mathcal{C}(rR, R)) \leq 2u(2r). \quad (5.16)$$

By the continuum result,

$$\frac{\log u(r)}{\log r} \longrightarrow \frac{5}{48} \quad (r \downarrow 0). \quad (5.17)$$

Fix small  $r$  and put  $\tilde{r} = 2r$ . Chain annuli whose radii grow by  $\tilde{r}^{-1}$ . For the lower bound, require an open crossing in every annulus and an open circuit in each overlap annulus. FKG multiplies the crossing probabilities and RSW supplies a uniform positive lower bound for the circuits. If  $N \asymp \log R/|\log r|$ , then

$$\mathbb{P}(0 \leftrightarrow \partial B_R) \geq c_0 [c_1 u(r/2)/2]^N.$$

For the upper bound choose a subfamily of disjoint annuli. Connection from the origin to radius  $R$  forces a crossing of every selected annulus, and those events use disjoint site sets, hence are independent. Applying the upper bound in (5.16) gives the matching product estimate.

Sending  $R \rightarrow \infty$  and then  $r \downarrow 0$ , and using (5.17), yields

$$\lim_{R \rightarrow \infty} \frac{\log \mathbb{P}(0 \leftrightarrow \partial B_R)}{\log R} = -\frac{5}{48}.$$

This is (5.1). □

**Theorem 5.5** (Four-arm exponent).

*For critical site percolation on the triangular lattice,*

$$\alpha_4(r, R) = \left(\frac{r}{R}\right)^{5/4+o(1)} \quad (R/r \rightarrow \infty).$$

*Proof.* This is exactly Lemma 2.9, proved earlier in this volume. That lemma owns the SLE<sub>6</sub> moment calculation, the six-arm stability input, the lattice-to-SLE transfer, arm separation, and quasi-multiplicativity. Thus the present theorem is exact backward reuse, not a second proof and not a summary standing in for an argument that has not been established in this chapter. □

**Theorem 5.6** (Correlation-length exponent 4/3).

*For site percolation on the triangular lattice, if  $L(p)$  is any of the standard finite-size correlation lengths normalized by a fixed nontrivial crossing threshold, then*

$$L(p) = |p - p_c|^{-4/3+o(1)} \quad (p \rightarrow p_c), \quad p_c = \frac{1}{2}.$$

*The second-moment correlation length and the directional exponential-decay length have the same logarithmic exponent.*

*Proof.* The near-critical package in Theorem 9.2 gives

$$|p - p_c| L(p)^2 \alpha_4(1, L(p)) \asymp 1.$$

The four-arm theorem SM-11.05.02, whose physical owner is Lemma 2.9, says  $\alpha_4(1, R) = R^{-5/4+o(1)}$ . Hence

$$|p - p_c| L(p)^{3/4+o(1)} \asymp 1.$$

Taking logarithms and using  $L(p) \rightarrow \infty$  gives

$$\frac{\log L(p)}{-\log |p - p_c|} \longrightarrow \frac{4}{3}.$$

The comparison of the other standard correlation lengths with  $L(p)$  is part g of Theorem 9.2, so they have the same exponent. Thus every step is owned earlier in this volume; no scaling relation is being cited without proof here. □

**Theorem 5.7** (Cluster-density, susceptibility, connectivity and cluster-size exponents).

For critical site percolation on the triangular lattice the standard exponents exist in the logarithmic sense and satisfy

$$\beta = \frac{5}{36}, \quad \gamma = \frac{43}{18}, \quad \eta = \frac{5}{24}, \quad \delta = \frac{91}{5}.$$

More explicitly,

$$\begin{aligned} \theta(p) &= (p - p_c)^{5/36+o(1)} & (p \downarrow p_c), \\ \chi(p) &= |p - p_c|^{-43/18+o(1)} & (p \rightarrow p_c), \\ \mathbb{P}_{p_c}(0 \leftrightarrow x) &= |x|^{-5/24+o(1)} & (|x| \rightarrow \infty), \\ \mathbb{P}_{p_c}(|C(0)| \geq n) &= n^{-5/91+o(1)} & (n \rightarrow \infty). \end{aligned}$$

*Proof.* For the density exponent,  $\theta(p) \asymp \alpha_1(1, L(p))$ . The one-arm theorem SM-11.05.01 gives  $\alpha_1(1, R) = R^{-5/48+o(1)}$ , while Theorem 5.6 gives  $L(p) = |p - p_c|^{-4/3+o(1)}$ . Therefore

$$\theta(p) = |p - p_c|^{(4/3)(5/48)+o(1)} = (p - p_c)^{5/36+o(1)},$$

which proves  $\beta = 5/36$ .

For susceptibility,  $\chi(p) \asymp L(p)^2 \alpha_1(1, L(p))^2$ . Hence

$$\chi(p) = L(p)^{2-10/48+o(1)} = L(p)^{43/24+o(1)} = |p - p_c|^{-43/18+o(1)}.$$

Thus  $\gamma = 43/18$  on both sides of the transition.

For two-point connectivity, if  $|x| \asymp R$ ,

$$\mathbb{P}_{p_c}(0 \leftrightarrow x) \asymp \alpha_1(1, R)^2.$$

The one-arm exponent therefore yields  $\mathbb{P}_{p_c}(0 \leftrightarrow x) = |x|^{-5/24+o(1)}$  and  $\eta = 5/24$ .

It remains to derive the cluster-size tail from the cluster mass. We give the Kesten mass-radius argument because this is the step often hidden behind the hyperscaling identity. Put

$$M(R) = R^2 \alpha_1(1, R).$$

On the event  $0 \leftrightarrow \partial B_R$ , reveal the outermost open circuit in  $A(R/2, R)$  connected to the origin. For each  $x \in B_{R/2}$ , arm separation and an RSW gluing construction show that, conditionally on the one-arm event, the probability that  $x$  belongs to the origin cluster is at least a constant times  $\alpha_1(1, R)$  whenever  $|x|$  is a fixed fraction of  $R$ . Summing over such  $x$  gives

$$\mathbb{E}_{p_c}[|C(0) \cap B_R| \mid 0 \leftrightarrow \partial B_R] \asymp M(R). \quad (5.18)$$

For the second moment, sum over ordered pairs  $x, y$ . If  $s = |x - y|$ , reveal the two branches until scale  $s$  and then their common connection to the origin and to  $\partial B_R$ . Arm separation and quasi-multiplicativity bound the resulting probability by a constant times the product of the corresponding one-arm factors. Dyadically summing over  $s$  gives

$$\mathbb{E}_{p_c}[|C(0) \cap B_R|^2 \mid 0 \leftrightarrow \partial B_R] \leq CM(R)^2. \quad (5.19)$$

Paley–Zygmund applied to (5.18)–(5.19) therefore gives constants  $c_0, c_1 > 0$  such that

$$\mathbb{P}_{p_c}(|C(0)| \geq c_0 M(R), 0 \leftrightarrow \partial B_R) \geq c_1 \alpha_1(1, R). \quad (5.20)$$



For the matching upper comparison, if the cluster does not reach  $\partial B_{2^k R}$ , its mass is the sum of its masses in the  $k + 1$  inner dyadic annuli. Markov's inequality, the first-moment estimate at each scale, and quasi-multiplicativity imply that for every fixed  $\varepsilon > 0$ ,

$$\mathbb{P}_{p_c}(|C(0)| \geq R^\varepsilon M(R), 0 \not\leftrightarrow \partial B_{R^{1+\varepsilon}}) = o(\alpha_1(1, R)). \quad (5.21)$$

The complementary event is a one-arm event to radius  $R^{1+\varepsilon}$ . Since  $\alpha_1(1, R) = R^{-5/48+o(1)}$ , parts d–e show, on the logarithmic scale, that a critical cluster of radius  $R$  has mass

$$M(R) = R^{2-5/48+o(1)} = R^{91/48+o(1)},$$

and that

$$\mathbb{P}_{p_c}(|C(0)| \geq M(R)) = R^{-5/48+o(1)}.$$

Putting  $n = R^{91/48+o(1)}$  gives  $\mathbb{P}_{p_c}(|C(0)| \geq n) = n^{-5/91+o(1)}$ , so  $\delta = 91/5$ .

The resulting exponents are consistent with the scaling identities  $\eta = 4/(\delta + 1)$ ,  $\beta = 2\nu/(\delta + 1)$ , and  $\gamma = 2\nu(\delta - 1)/(\delta + 1)$  with  $\nu = 4/3$ ; here those identities are a check rather than a substitute for the preceding derivations.

Every probabilistic and SLE input used above has been proved for critical site percolation on the triangular lattice. In particular this theorem makes no universality assertion for square-lattice bond percolation or other planar models.  $\square$

## Chapter 6

# Noise sensitivity

Fourier analysis of Boolean functions turns pivotal geometry into a quantitative theory of sensitivity to microscopic noise.

**Definition 6.1** (Noise operator and spectral sample). For  $f : \{-1, 1\}^N \rightarrow \mathbb{R}$  with Fourier–Walsh expansion  $f = \sum_S \widehat{f}(S) \chi_S$ , the noise operator is

$$T_\rho f = \sum_S \rho^{|S|} \widehat{f}(S) \chi_S, \quad 0 \leq \rho \leq 1.$$

If  $f$  is nonconstant, its spectral sample  $\mathcal{S}$  is the random subset with

$$\mathbb{P}(\mathcal{S} = S) = \frac{\widehat{f}(S)^2}{\sum_{T \neq \emptyset} \widehat{f}(T)^2}, \quad S \neq \emptyset.$$

*Worked example 6.2* (Exact noise covariance for a two-bit AND function). Write bits  $x_i \in \{-1, 1\}$  and  $f(x) = \mathbf{1}_{\{x_1=x_2=1\}} = (1 + x_1 + x_2 + x_1 x_2)/4$ . Hence  $\widehat{f}(\emptyset) = \widehat{f}(\{1\}) = \widehat{f}(\{2\}) = \widehat{f}(\{1, 2\}) = 1/4$ . If  $y$  is obtained from  $x$  by  $\rho$ -correlated noise, then

$$\text{Cov}(f(x), f(y)) = \sum_{S \neq \emptyset} \rho^{|S|} \widehat{f}(S)^2 = \frac{2\rho + \rho^2}{16}.$$

The formula shows directly how Fourier mass on large sets is killed faster by noise; BKS proves that macroscopic percolation crossings asymptotically place their nonconstant Fourier mass at diverging levels.

**Example 6.3** (A running picture). At  $p = 1/2$  on the triangular lattice, zooming out does not drive crossings to probability 0 or 1. The chapter identifies the mechanism responsible for this scale-free behavior.

**Theorem 6.4** (Benjamini–Kalai–Schramm criterion).

Let  $f_n : \{-1, 1\}^{N_n} \rightarrow \{0, 1\}$  be Boolean functions under the uniform product measure. If

$$\sum_{i=1}^{N_n} \text{Inf}_i(f_n)^2 \longrightarrow 0,$$

then for every fixed noise level  $\varepsilon \in (0, 1)$ ,

$$\text{Cov}(f_n(\omega), f_n(\omega^\varepsilon)) \longrightarrow 0,$$

where each bit is independently rerandomized with probability  $\varepsilon$ .

*Proof.* Write

$$f = \sum_{S \subset [N]} \widehat{f}(S) \chi_S, \quad \chi_S(\omega) = \prod_{i \in S} \omega_i,$$

with respect to uniform measure. We first prove the low-level estimate that was missing from the previous draft.

**Low-level influence lemma.** For each fixed integer  $d \geq 1$  there is  $C_d < \infty$  such that every Boolean  $f : \{-1, 1\}^N \rightarrow \{0, 1\}$  satisfies

$$\sum_{|S|=d} \widehat{f}(S)^2 \leq C_d W(f) \left(1 + \log \frac{1}{W(f)}\right)^{d-1}, \quad W(f) := \sum_{i=1}^N \text{Inf}_i(f)^2, \quad (6.1)$$

with the right side interpreted as 0 when  $W(f) = 0$ .

We need one standard consequence of Bonami–Beckner hypercontractivity. If  $g$  has degree at most  $m$  and  $\|g\|_2 = 1$ , then

$$\mathbb{P}(|g| > t) \leq 2 \exp(-c_m t^{2/m}) \quad (t \geq C_m). \quad (6.2)$$

Indeed, hypercontractivity gives  $\|g\|_q \leq (q-1)^{m/2} \|g\|_2$  for  $q \geq 2$ . Markov's inequality with  $q = 1 + (t/e)^{2/m}$  (and an adjustment for bounded  $t$ ) gives (6.2). In particular,

$$H_m(t_0)^2 := \left( \int_{t_0}^{\infty} \mathbb{P}(|g| > t)^{1/2} dt \right)^2 \leq C_m t_0^2 \exp(-c'_m t_0^{2/m}) \quad (6.3)$$

for  $t_0 \geq C_m$ ; this follows after the substitution  $u = t^{2/m}$  and absorbing the resulting polynomial factor into a slightly smaller exponential constant.

Assume now  $d \geq 2$  and put  $m = d - 1$ . Partition the coordinates as  $[N] = I \sqcup J$ . For  $j \in J$  define

$$g_j(\omega_I) = \sum_{\substack{T \subset I \\ |T|=m}} \widehat{f}(T \cup \{j\}) \chi_T(\omega_I), \quad a_j = \|g_j\|_2.$$

If  $a_j > 0$ , set  $h_j = g_j/a_j$ . Orthogonality gives

$$a_j = \langle h_j \chi_j, f \rangle.$$

Let  $b_j(\omega_I) = \mathbb{E}_{\omega_J}[\chi_j f \mid \omega_I]$ . For fixed values of all coordinates except  $j$ , the inner average in the  $j$ -coordinate is half the difference of the two Boolean values. Hence, after Jensen,

$$\mathbb{E}_I |b_j| \leq \frac{1}{2} \text{Inf}_j(f). \quad (6.4)$$

Moreover, for every fixed  $\omega_I$ , Parseval in the  $J$ -coordinates gives

$$\sum_{j \in J} b_j(\omega_I)^2 \leq \mathbb{E}_{\omega_J} [f^2 \mid \omega_I] \leq 1. \quad (6.5)$$

Using  $|h| = \int_0^\infty \mathbf{1}_{\{|h|>t\}} dt$ , split at  $t_0$  and apply Cauchy–Schwarz to the tail:

$$|a_j| \leq t_0 \mathbb{E}_I |b_j| + \|b_j\|_2 \int_{t_0}^{\infty} \mathbb{P}(|h_j| > t)^{1/2} dt.$$

Equations (6.2)–(6.5), followed by  $(x + y)^2 \leq 2x^2 + 2y^2$ , therefore imply

$$\sum_{j \in J} a_j^2 \leq C t_0^2 \sum_{j \in J} \text{Inf}_j(f)^2 + C_m t_0^2 \exp(-c'_m t_0^{2/m}). \quad (6.6)$$

Write  $W_J = \sum_{j \in J} \text{Inf}_j(f)^2$ . When  $0 < W_J$  is small, choose  $t_0 = A_m(1 + \log(1/W_J))^{m/2}$  with  $A_m$  large enough that the second term in (6.6) is at most a constant multiple of the first. When  $W_J$  is bounded away from zero the same conclusion follows by increasing the constant. Thus

$$\sum_{j \in J} \sum_{\substack{T \subset I \\ |T|=d-1}} \widehat{f}(T \cup \{j\})^2 \leq C_d W_J \left(1 + \log \frac{1}{W_J}\right)^{d-1}. \quad (6.7)$$

Choose  $J$  randomly by placing each coordinate in  $J$  independently with probability  $1/d$ . A fixed  $d$ -set  $S$  contributes to the left side of (6.7) whenever exactly one member of  $S$  lies in  $J$ , an event of probability

$$d \frac{1}{d} \left(1 - \frac{1}{d}\right)^{d-1} \geq e^{-1}.$$

Since  $W_J \leq W(f)$  and  $x(1 + \log(1/x))^{d-1}$  is increasing near 0 (the complementary range being absorbed into  $C_d$ ), taking expectations in (6.7) proves (6.1) for  $d \geq 2$ . For  $d = 1$ ,  $|\widehat{f}(\{i\})| \leq \text{Inf}_i(f)/2$ , so (6.1) is immediate. This completes the proof of the low-level lemma.

**Fourier diagonalization of noise.** Under  $\varepsilon$ -rerandomization, a coordinate is correlated with its old value by  $\rho = 1 - \varepsilon$ . Therefore

$$\mathbb{E}[\chi_S(\omega^\varepsilon) \mid \omega] = \rho^{|S|} \chi_S(\omega),$$

and orthogonality yields

$$\text{Cov}(f(\omega), f(\omega^\varepsilon)) = \sum_{S \neq \emptyset} \rho^{|S|} \widehat{f}(S)^2. \quad (6.8)$$

Apply this to  $f_n$ . Given  $D \geq 1$ , split (6.8) at level  $D$ . The high levels satisfy

$$\sum_{|S| > D} \rho^{|S|} \widehat{f}_n(S)^2 \leq \rho^{D+1} \text{Var}(f_n) \leq \frac{1}{4} \rho^{D+1}.$$

For every fixed  $1 \leq d \leq D$ , (6.1) and the hypothesis  $W(f_n) \rightarrow 0$  give

$$\sum_{|S|=d} \widehat{f}_n(S)^2 \rightarrow 0.$$

First choose  $D$  so large that the right side of (6.9) is smaller than a given tolerance, and then choose  $n$  so large that the finite sum of the terms in (6.10) is smaller than the same tolerance. Equation (6.8) then tends to zero. This proves the BKS criterion.  $\square$

**Theorem 6.5** (Noise sensitivity of critical planar crossing events).

*Let  $\mathcal{C}_m$  be the event that critical bond percolation on the  $(m+1) \times m$  square-grid rectangle has an open left–right crossing. If  $\omega^\varepsilon$  is obtained from  $\omega$  by independently rerandomizing each edge with any fixed probability  $\varepsilon \in (0, 1)$ , then*

$$\text{Cov}(\mathbf{1}_{\mathcal{C}_m}(\omega), \mathbf{1}_{\mathcal{C}_m}(\omega^\varepsilon)) \rightarrow 0.$$

*Equivalently, the crossing events are asymptotically noise sensitive in the Benjamini–Kalai–Schramm sense.*

*Proof.* The abstract Boolean theorem SM-11.06.01 reduces the problem to proving

$$II(\mathcal{C}_m) := \sum_e I_e(\mathcal{C}_m)^2 \longrightarrow 0.$$

We follow the original BKS proof because it needs only RSW-type polynomial one-arm decay and does not presuppose the exact triangular-lattice arm exponents proved elsewhere in this volume.

For  $K$  a set of coordinates, write  $M_K$  for their centered majority and  $I_K(f) = \sum_{j \in K} I_j(f)$ . If  $f : \{0, 1\}^N \rightarrow [0, 1]$  is monotone, then

$$I_K(f) \leq C\sqrt{|K|} \Lambda_K (1 + \sqrt{-\log \Lambda_K}), \quad \Lambda_K = |\mathbb{E}(fM_K)|. \quad (6.9)$$

To prove this, average  $f$  over all coordinates outside  $K$ ; this preserves monotonicity and  $I_K$ . Thus it is enough to work on  $\{0, 1\}^k$ ,  $k = |K|$ . Let  $\bar{f}(j)$  be the average of  $f$  on the Hamming sphere of weight  $j$ . Monotonicity gives

$$I_K(f) = 2^{-k} \sum_j \binom{k}{j} \bar{f}(j)(2j - k),$$

while pairing the levels  $j$  and  $k - j$  gives

$$\mathbb{E}(fM_K) = 2^{-k} \sum_{j > k/2} \binom{k}{j} (\bar{f}(j) - \bar{f}(k - j)).$$

Split the first sum at  $j = (k + \lambda\sqrt{k})/2$ . Below that level it is at most  $\lambda\sqrt{k} \mathbb{E}(fM_K)$ ; above it the binomial Gaussian tail is  $O(\sqrt{k}e^{-c\lambda^2})$ . Taking  $\lambda \asymp 1 + \sqrt{-\log \Lambda_K}$  proves (6.9).

Let  $E_r$  be the edges in the right half of the rectangle and fix  $K \subset E_r$ . Explore the open cluster attached to the left boundary: repeatedly reveal an edge incident to the discovered vertex set and add its far endpoint exactly when that edge is open. Unrevealed bits remain independent fair bits, and the crossing event is determined when the exploration terminates. An edge of  $K$  is visited only if one of its endpoints is joined to the left boundary. RSW gives a polynomial one-arm upper bound: for some  $a > 0$  and  $C < \infty$  independent of  $m$ ,

$$\sup_{v \in E_r} \mathbb{P}(v \longleftrightarrow \partial_{\text{left}} R_m) \leq Cm^{-a}. \quad (6.10)$$

Indeed, between  $v$  and the left side lie order  $\log m$  disjoint bounded-aspect annuli; RSW gives in each annulus a closed dual circuit with a uniformly positive probability, and independence on a separated subfamily makes failure of all such circuits polynomially small. Consequently

$$\mathbb{E}|K \cap \text{VISITED}| \leq C|K|m^{-a}.$$

Put  $q = |K|m^{-2a/3}$ . Markov's inequality gives

$$\mathbb{P}\{|K \cap \text{VISITED}| > q\} \leq Cm^{-a/3}.$$

Feed the exploration the  $K$ -bits from an independent fair sequence in the order in which they are requested. By Hoeffding's inequality, with failure probability smaller than any fixed power of  $m$ , every prefix of length at most  $q$  has imbalance at most  $C\sqrt{q} \log m$ . On the complementary good event the value of  $\mathbf{1}_{\mathcal{C}_m}$  is already determined by the exploration, whereas the unvisited coordinates

of  $K$  are still fair and independent. Conditioning on the exploration, a binomial local-limit bound therefore gives

$$|\mathbb{E}[M_K \mid \text{exploration}]| \leq Cm^{-a/3} \log m.$$

After adding the two exceptional events,

$$|\mathbb{E}(\mathbf{1}_{\mathcal{C}_m} M_K)| \leq Cm^{-a/3} \log m, \quad K \subset E_r. \quad (6.11)$$

The same estimate holds in the left half by reflection.

Combining (6.9) and (6.11) gives, uniformly in  $K$  contained in one half of the edge set,

$$I_K(\mathcal{C}_m) \leq C\sqrt{|K|} m^{-a/3} (\log m)^{3/2}. \quad (6.12)$$

For an arbitrary  $K$ , split it into its left and right halves and absorb the factor two.

Order the influences as  $b_1 \geq b_2 \geq \dots \geq 0$ . Applying (6.12) to the  $k$  largest coordinates gives  $\sum_{j \leq k} b_j \leq A\sqrt{k}$  with  $A = Cm^{-a/3}(\log m)^{3/2}$ . Hence  $b_k \leq A/\sqrt{k}$  and, since the rectangle has  $O(m^2)$  edges,

$$II(\mathcal{C}_m) = \sum_k b_k^2 \leq CA^2 \sum_{k \leq Cm^2} \frac{1}{k} \leq Cm^{-2a/3} (\log m)^4 \rightarrow 0.$$

The BKS criterion SM-11.06.01 now gives, for every fixed noise level  $\varepsilon > 0$ , vanishing covariance and therefore asymptotic noise sensitivity. This proves the theorem.  $\square$

**Theorem 6.6** (Sharp lower tail of the percolation spectral sample).

*Consider either critical bond percolation on the square lattice or critical site percolation on the triangular lattice, at mesh one. Let  $f_R$  be the  $\{-1, 1\}$ -valued indicator of an open left-right crossing of the square  $[0, R]^2$ , and let  $\mathcal{S}_R$  be its Fourier spectral sample,*

$$\mathbb{P}(\mathcal{S}_R = S) = \widehat{f_R}(S)^2.$$

*Write  $\alpha_4(r, R)$  for the alternating four-arm probability and  $\alpha_4(r) = \alpha_4(1, r)$ . There are universal constants  $c, C, r_0 \in (0, \infty)$  such that, whenever  $r_0 \leq r \leq R$ ,*

$$c \frac{R^2 \alpha_4(R)^2}{r^2 \alpha_4(r)^2} \leq \mathbb{P}(0 < |\mathcal{S}_R| < r^2 \alpha_4(r)) \leq C \frac{R^2 \alpha_4(R)^2}{r^2 \alpha_4(r)^2}. \quad (6.13)$$

*Changing the threshold  $r^2 \alpha_4(r)$  by a fixed multiplicative constant only changes  $c, C$ . Moreover*

$$\mathbb{E}|\mathcal{S}_R| \asymp R^2 \alpha_4(R). \quad (6.14)$$

*Consequently, if  $\omega^{\varepsilon_R}$  is obtained by independently rerandomizing every bit with probability  $\varepsilon_R$ , then*

$$\begin{aligned} \varepsilon_R R^2 \alpha_4(R) \rightarrow \infty &\implies \text{Cov}(f_R(\omega), f_R(\omega^{\varepsilon_R})) \rightarrow 0, \\ \varepsilon_R R^2 \alpha_4(R) \rightarrow 0 &\implies \mathbb{E}[(f_R(\omega) - f_R(\omega^{\varepsilon_R}))^2] \rightarrow 0. \end{aligned}$$

*For triangular-lattice site percolation,  $\alpha_4(R) = R^{-5/4+o(1)}$ , so the spectral/noise scale is  $R^{3/4+o(1)}$ .*

*Proof.* The argument is long because the spectral sample is not a product point process. Its first two moments look like those of the pivotal set, but its small lower tail is governed by a nontrivial clustering phenomenon. We therefore isolate the four ingredients which are repeatedly used in the Garban–Pete–Schramm proof: a Fourier localization identity, annulus clustering, conditioned local growth, and a general thinning lemma. We prove each ingredient before assembling the lower tail.

For a finite bit set  $I$  and  $A \subset I$ , let  $\mathcal{F}_A$  be the  $\sigma$ -field generated by the bits in  $A$ . If  $f = \sum_{S \subset I} \hat{f}(S) \chi_S$  and  $U \subset I$ , orthogonal projection onto  $L^2(\mathcal{F}_U)$  gives

$$\mathbb{P}(\mathcal{S}_f \subset U) = \sum_{S \subset U} \hat{f}(S)^2 = \mathbb{E}[\mathbb{E}(f | \mathcal{F}_U)^2]. \quad (6.15)$$

Let  $B, W \subset I$  be disjoint, and let  $\Lambda_B$  be the event that the bits in  $B$  can change the value of  $f$  while the outside configuration is held fixed. Put

$$\lambda_{B,W} = \mathbb{P}(\Lambda_B | \mathcal{F}_{W^c}).$$

Subtracting (6.15) for  $U = W^c$  and  $U = (B \cup W)^c$  gives

$$\begin{aligned} \mathbb{P}(\mathcal{S}_f \cap B \neq \emptyset, \mathcal{S}_f \cap W = \emptyset) &= \mathbb{E} \left[ \left( \mathbb{E}(f | \mathcal{F}_{W^c}) - \mathbb{E}(f | \mathcal{F}_{(B \cup W)^c}) \right)^2 \right] \\ &\leq 4 \|f\|_\infty^2 \mathbb{E} \lambda_{B,W}^2. \end{aligned} \quad (6.16)$$

Indeed  $f = \mathbb{E}(f | \mathcal{F}_{B^c})$  off  $\Lambda_B$ , so conditioning first on  $W^c$  bounds the absolute difference in parentheses by  $2 \|f\|_\infty \lambda_{B,W}$ .

For a monotone  $\{-1, 1\}$ -valued Boolean function and a single bit  $x$  we also have the exact identity

$$\mathbb{P}(x \in \mathcal{S}_f, \mathcal{S}_f \cap W = \emptyset) = \mathbb{E} \lambda_{x,W}^2. \quad (6.17)$$

To see this, condition on all bits except  $x$  and those in  $W$ . The  $x$ -Fourier coefficient of the conditional function is zero unless  $x$  is pivotal; in the pivotal case its absolute value is one. Squaring and averaging over the common outside configuration gives exactly the square of the conditional pivotal probability.

For a crossing event and a bulk box  $B$  of diameter  $r$ ,  $\Lambda_B$  forces four alternating arms from  $B$  to the four marked boundary arcs of the quad. Arm separation and RSW therefore give, uniformly away from the boundary,

$$\mathbb{P}(\Lambda_B) \asymp \alpha_4(r, R), \quad (6.18)$$

and for a single bulk bit  $x$ ,  $\mathbb{P}(x \in \mathcal{S}_R) = \mathbb{P}(x \text{ pivotal}) \asymp \alpha_4(R)$ . Summing over the  $\asymp R^2$  bulk bits and treating the  $O(R)$  boundary bits by half-plane arm estimates proves (6.14).

Tile  $[0, R]^2$  by boxes of side  $r$  and write  $\mathcal{S}_R^{(r)}$  for the set of boxes meeting  $\mathcal{S}_R$ . We prove that there is a function

$$g(k) \leq \exp\{C(\log(k+2))^2\}$$

such that

$$\mathbb{P}(|\mathcal{S}_R^{(r)}| = k) \leq g(k) \gamma_r(R), \quad \gamma_r(R) := \left( \frac{R \alpha_4(R)}{r \alpha_4(r)} \right)^2. \quad (6.19)$$

We give the geometric induction because this is the part of the theorem which cannot be replaced by a first-moment estimate.

Start with the  $k$  occupied  $r$ -boxes. At scale  $2^j r$  join two current clusters whenever their  $2^j r$ -neighbourhoods meet. When several clusters merge, surround every child cluster by an inner square and the new parent by an outer square, choosing the two boundaries at comparable distance. The region between them contains a disjoint annulus. Stop a branch when it reaches either scale  $R$  or the boundary of the big square. The resulting forest has at most  $2k - 1$  vertices. To every interior annulus assign the four-arm cost  $\alpha_4(s, t)$  between its inner and outer radii; to side and corner annuli assign respectively the corresponding half-plane three-arm and quarter-plane two-arm costs. If  $A$  denotes the whole annulus forest, let  $h(A)$  be the product of these arm probabilities.

Why does  $h(A)^2$  control spectral mass? Successively expose the annuli from outside to inside. For the spectral sample to meet every child cluster while avoiding the separating annuli, each cluster must be pivotal relative to the still-unrevealed inside. Applying (6.16) at each separation and then using independence on disjoint annuli gives

$$\mathbb{P}(\mathcal{S}_R \text{ is compatible with } A) \leq C^k h(A)^2. \quad (6.20)$$

The square is essential: one factor  $h(A)$  would be the ordinary pivotal cost, whereas Fourier localization carries the square of the conditional pivotal probability.

It remains to sum (6.20). At a vertex containing  $m$  leaf boxes, call scale  $j$  overcrowded when the number of possible placements of its children is larger than the reciprocal of a fixed positive power of the arm cost at that scale. If it is not overcrowded, summing over all child placements consumes at most one fixed positive power of the arm probability. If it is overcrowded, merge once more before recording an annulus. Since each merge at least doubles the number of leaves in a cluster, a branch contains at most  $\lceil \log_2 k \rceil$  overcrowded scales. At scale  $j$  there are at most  $Ck^C 2^{Cj}$  combinatorial choices, whereas RSW plus arm separation gives a fixed power saving when the radius doubles. Choosing the overcrowding threshold with exponent smaller than that saving, and multiplying along the forest, yields the recurrence

$$G(k) \leq C^k \max_{k_1 + \dots + k_m = k} \prod_{i=1}^m G(k_i) (k+2)^{C \log(k+2)}, \quad G(1) \leq C.$$

Induction on  $k$  gives  $G(k) \leq \exp\{C(\log(k+2))^2\}$ . The unique outermost annulus contributes exactly, up to RSW/quasi-multiplicativity constants, the factor  $\gamma_r(R)$ . Boundary annuli have stronger half-plane/quarter-plane arm costs, so the same summation applies. Summing (6.20) over the admissible forests proves (6.19).

Two consequences will be used below. First, because  $g(k)$  is subexponential, for some  $a > 0$ ,

$$\sum_{k \geq 1} e^{-ak} g(k) \leq C. \quad (6.21)$$

Second, the same construction in a quad  $Q$ , with a compact subquad  $U' \Subset U \Subset Q$ , gives a local version with a constant depending only on the separation of  $U'$  from  $Q \setminus U$ . Thus no boundary singularity is hidden in the later thinning argument.

Let  $B$  be an  $r$ -box, let  $B'$  be its concentric middle-third box, and let  $W$  be any set of bits outside  $B$ . Conditional on  $\mathcal{S}_R \cap B' \neq \emptyset$  and  $\mathcal{S}_R \cap W = \emptyset$ , there are constants  $a, c > 0$  such that

$$\mathbb{P}(|\mathcal{S}_R \cap B'| \geq cr^2 \alpha_4(r) \mid \mathcal{S}_R \cap B' \neq \emptyset, \mathcal{S}_R \cap W = \emptyset) \geq a. \quad (6.22)$$

The important point is that  $a, c$  are independent of  $W$ .

Introduce an independent Bernoulli set  $Z$  in  $B'$  by retaining each bit with probability  $q = (r^2 \alpha_4(r))^{-1}$ . It is enough to prove

$$\mathbb{P}(\mathcal{S}_R \cap B' \cap Z \neq \emptyset \mid \mathcal{S}_R \cap B' \neq \emptyset, \mathcal{S}_R \cap W = \emptyset) \geq a, \quad (6.23)$$

because, conditional on  $\mathcal{S}_R$ , the probability of missing  $Z$  is  $(1-q)^{|\mathcal{S}_R \cap B'|}$ ; if the latter set were much smaller than  $q^{-1}$  with high conditional probability, (6.23) would fail.

We prove (6.23) by a conditional second moment. The only nonstandard input is a four-arm quasi-multiplicativity estimate which must remain uniform in the arbitrary conditioning set  $W$ . We prove that estimate first.



Take two configurations  $\omega', \omega''$  which agree on  $W^c$  and are independent on  $W$ , and put

$$\beta_W(s, t) = \mathbb{P}(\omega', \omega'' \text{ both have four alternating arms across } A(s, t)).$$

Then, uniformly in  $W$ ,

$$\beta_W(s, u)\beta_W(u, t) \leq C\beta_W(s, t), \quad r_0 \leq s < u < t. \quad (6.24)$$

Here is the gluing proof. First,  $\beta_W(s, t) = \mathbb{E}[p_W(s, t)^2]$ , where  $p_W$  is the conditional four-arm probability given the bits in  $W^c$ . If  $W_1 \subset W_2$ , the conditional expectation for  $W_1$  is a further conditional expectation of that for  $W_2$ ; the  $L^2$  contraction therefore gives  $\beta_{W_2}(s, t) \leq \beta_{W_1}(s, t)$ .

Fix a small  $\delta > 0$  and call a radius  $\rho$  good (for the inner radius  $s$ ) when  $\beta_W(s, 2\rho) \geq \delta\beta_W(s, \rho)$ . Conditional on simultaneous four arms to radius  $2\rho$ , expose the four interfaces only until they hit  $\partial B(\rho)$ . RSW fences in four disjoint sectors show that, with a probability  $c_0 > 0$  independent of  $W$ , the four landing intervals in each configuration are mutually separated by at least  $\rho/M$ , for a fixed  $M$ . The reason the coupling causes no loss is that the fences are built sector by sector: on a site of  $W^c$  the two configurations request the same color and on a site of  $W$  they request independent colors, so the finite collection of RSW rectangles has a probability bounded below in either case. If the landings are not separated, repeat the construction in the next inner dyadic annulus. Thus, for some universal  $d < \infty$ ,

$$\beta_{W \cup A(\rho, v)}(s, v) \geq c_1 \beta_W(s, \rho) (\rho/v)^d, \quad v \geq \rho, \quad (6.25)$$

whenever  $\rho$  is good. This is just the product of the positive fence probability at the first separated scale and a chain of  $O(\log(v/\rho))$  RSW crossings thereafter.

Good radii occur with bounded dyadic gaps. Indeed, if  $\rho, 2\rho, \dots, 2^m \rho$  were all bad, then  $\beta_W(s, 2^m \rho) < \delta^m \beta_W(s, \rho)$ . On the other hand apply (6.25) at the last good radius below  $\rho$  (or at a fixed microscopic initialization scale, where RSW supplies a positive constant) to get a lower bound  $c2^{-dm} \beta_W(s, \rho)$ . Choosing  $\delta < 2^{-(d+1)}$  makes these inequalities incompatible for all sufficiently large  $m$ . Hence there exist universal  $L$  and  $\delta_0 > 0$  such that

$$\beta_W(s, 2\rho) \geq \delta_0 \beta_W(s, \rho) \quad (6.26)$$

for every admissible  $\rho$ , after moving by at most  $L$  dyadic steps. Running exactly the same fence construction inward gives the analogous stability for halving the inner radius.

If  $u/4 > s$  and  $4u < t$ , enlarge  $W$  by the buffer annulus  $A(u/4, 4u)$ . The two configurations are then independent inside that buffer. Apply the separated-arm fences from the inner event to four fixed landing arcs on  $\partial B(u/2)$  and from the outer event to four fixed arcs on  $\partial B(2u)$ , and join corresponding arcs by eight disjoint RSW corridors in the buffer. Conditional on the inner and outer events, this simultaneous joining has probability at least  $c > 0$ . Therefore

$$\beta_{W \cup A(u/4, 4u)}(s, t) \geq c \beta_W(s, u/4) \beta_W(4u, t).$$

Monotonicity in  $W$  bounds the left side by  $\beta_W(s, t)$ , while (6.26) compares the two right-hand factors with  $\beta_W(s, u)$  and  $\beta_W(u, t)$ . The cases  $u \leq 4s$  or  $t \leq 4u$  are absorbed directly by doubling stability. This proves (6.24).

Now let  $Y = |\mathcal{S}_R \cap B' \cap Z| \mathbf{1}_{\{\mathcal{S}_R \cap W = \emptyset\}}$ . By (6.17), arm separation inside  $B$ , and (6.24),

$$\mathbb{P}(x \in \mathcal{S}_R, \mathcal{S}_R \cap W = \emptyset) \geq c_1 \mathbb{E} \lambda_{B, W}^2 \alpha_4(r) \quad (x \in B'). \quad (6.27)$$

For two bits  $x, y \in B'$ , condition on every bit outside  $W \cup \{x, y\}$ . For both  $x, y$  to occur in the spectral sample there must be four arms from scale  $|x - y|$  around the pair and four arms from that

scale to  $r$ ; Fourier projection also contributes the common outside factor  $\lambda_{B,W}^2$ . Separation and quasi-multiplicativity therefore give

$$\mathbb{P}(x, y \in \mathcal{S}_R, \mathcal{S}_R \cap W = \emptyset) \leq C \mathbb{E} \lambda_{B,W}^2 \alpha_4(|x - y|) \alpha_4(r). \quad (6.28)$$

Summing (6.27) over  $x \in B'$  and multiplying by  $q$  gives  $\mathbb{E} Y \geq c \mathbb{E} \lambda_{B,W}^2$ . For the second moment, group pairs by dyadic separation  $2^j$ . There are  $O(r^2 2^{2j})$  ordered pairs at that separation, and the standard arm-sum estimate

$$\sum_{s \leq r} s \alpha_4(s) \leq C r^2 \alpha_4(r)$$

follows from quasi-multiplicativity and the polynomial improvement over the trivial two-arm bound. Hence  $\mathbb{E} Y^2 \leq C \mathbb{E} \lambda_{B,W}^2$ . Paley–Zygmund gives

$$\mathbb{P}(Y > 0) \geq \frac{(\mathbb{E} Y)^2}{\mathbb{E} Y^2} \geq c \mathbb{E} \lambda_{B,W}^2.$$

On the other hand (6.16) says

$$\mathbb{P}(\mathcal{S}_R \cap B \neq \emptyset, \mathcal{S}_R \cap W = \emptyset) \leq C \mathbb{E} \lambda_{B,W}^2.$$

Dividing proves (6.23), and then (6.22) follows after reducing  $c$  so that  $1 - (1 - q)^{c/q} < a/2$ .

We need a finite probabilistic statement whose proof is independent of percolation. Let  $x, y \in \{0, 1\}^n$ , set  $X = \sum_i x_i$ ,  $Y = \sum_i y_i$ , assume  $y_i \leq x_i$ , and suppose that for some  $a \in (0, 1]$ ,

$$\mathbb{P}(y_j = 1 \mid y_i = 0 \text{ for all } i \in J) \geq a \mathbb{P}(x_j = 1 \mid y_i = 0 \text{ for all } i \in J) \quad (6.29)$$

for every  $j \notin J$ . Then

$$\mathbb{P}(Y = 0 \mid X > 0) \leq a^{-1} \mathbb{E}(e^{-aX/e} \mid X > 0). \quad (6.30)$$

To prove it, choose independently a random subset  $J \subset [n]$  by retaining each index with probability  $1 - e^{-1}$ . Apply (6.29) successively to the first retained index with  $x_j = 1$  while conditioning on the events  $y_i = 0$  at earlier retained indices. Averaging over  $J$  gives

$$\mathbb{E}(Y e^{-Y}) \geq a \mathbb{E}(X e^{-Y-1}).$$

Equivalently, with  $Z = (Y - a e^{-1} X) e^{-Y}$ ,  $\mathbb{E} Z \geq 0$ . For fixed  $X$ , the maximum of  $Z$  over  $Y \geq 1$  is at most  $e^{-1-aX/e}$ , whereas on  $Y = 0 < X$  we have  $Z = -aX/e \leq -a/e$ . Therefore

$$\frac{a}{e} \mathbb{P}(Y = 0 < X) \leq \mathbb{E}[\mathbf{1}_{X>0} e^{-1-aX/e}],$$

which is (6.30) after division by  $\mathbb{P}(X > 0)$ . This is the precise replacement for nonexistent conditional independence of the spectral sample in different boxes.

Tile  $[0, R]^2$  by  $r$ -boxes  $B_1, \dots, B_n$ . Independently retain every bit in a sparse set  $Z$  with probability  $q = (r^2 \alpha_4(r))^{-1}$ . Let

$$x_j = \mathbf{1}_{\{\mathcal{S}_R \cap B_j \neq \emptyset\}}, \quad y_j = \mathbf{1}_{\{\mathcal{S}_R \cap B'_j \cap Z \neq \emptyset\}}.$$

After conditioning on the values of  $Z$  in previously tested boxes, the negative information  $y_i = 0$  says only that the spectral sample avoids a known set outside  $B_j$ . Thus (6.23) gives (6.29) with a universal  $a$ . The thinning lemma yields

$$\mathbb{P}(\mathcal{S}_R \cap Z = \emptyset, \mathcal{S}_R \neq \emptyset) \leq C \sum_{k \geq 1} e^{-ak} \mathbb{P}(|\mathcal{S}_R^{(r)}| = k).$$

Using (6.19) and (6.21),

$$\mathbb{P}(\mathcal{S}_R \cap Z = \emptyset, \mathcal{S}_R \neq \emptyset) \leq C\gamma_r(R). \quad (6.31)$$

If  $0 < |\mathcal{S}_R| < \frac{1}{2}q^{-1}$ , then, conditional on  $\mathcal{S}_R$ , the probability that  $Z$  misses it is  $(1-q)^{|\mathcal{S}_R|} \geq (1-q)^{q^{-1}/2} \geq c$ . Averaging and applying (6.31) gives

$$\mathbb{P}(0 < |\mathcal{S}_R| < c_0 r^2 \alpha_4(r)) \leq C\gamma_r(R).$$

Changing the fixed  $c_0$  to 1 uses monotonicity in  $r$ , quasi-multiplicativity, and a bounded number of scale changes: choose  $r' \asymp r$  so that  $r'^2 \alpha_4(r')$  differs from  $r^2 \alpha_4(r)$  by the required fixed factor. This proves the upper bound in (6.13).

Choose an  $r$ -box  $B$  whose distance from every side of the large square is comparable with  $R$ . The event  $0 < \mathcal{S}_R \subset B$  has probability comparable with the square of the probability that  $B$  is pivotal, by the projection identity and the same arm separation used in (6.18). There are  $\asymp (R/r)^2$  possible well-separated locations for  $B$ , and for distinct locations the corresponding events are disjoint. Hence

$$\begin{aligned} \mathbb{P}(0 < |\mathcal{S}_R| < Cr^2 \alpha_4(r)) &\geq c \left( \frac{R}{r} \right)^2 \alpha_4(r, R)^2 \\ &\asymp c \frac{R^2 \alpha_4(R)^2}{r^2 \alpha_4(r)^2}, \end{aligned}$$

where the last step is four-arm quasi-multiplicativity. On  $0 < \mathcal{S}_R \subset B$ , the conditional first moment inside  $B$  is  $O(r^2 \alpha_4(r))$  by the two-point estimate (6.28); Markov's inequality therefore retains a fixed fraction of these events with  $|\mathcal{S}_R| < Cr^2 \alpha_4(r)$ . A bounded rescaling of  $r$  removes the constant  $C$  and gives the lower bound in (6.13).

For  $\{-1, 1\}$ -valued  $f_R$ , Fourier diagonalization gives

$$\text{Cov}(f_R(\omega), f_R(\omega^\varepsilon)) = \mathbb{E} \left[ (1 - \varepsilon)^{|\mathcal{S}_R|} \mathbf{1}_{\{\mathcal{S}_R \neq \emptyset\}} \right]. \quad (6.32)$$

If  $\varepsilon_R m(R) \rightarrow 0$ , where  $m(R) = R^2 \alpha_4(R)$ , then  $1 - (1 - \varepsilon_R)^k \leq \varepsilon_R k$  and (6.14) imply

$$\mathbb{E}[(f_R(\omega) - f_R(\omega^{\varepsilon_R}))^2] \leq C\varepsilon_R m(R) \rightarrow 0.$$

If  $\varepsilon_R m(R) \rightarrow \infty$ , choose  $r = r(R)$  so that  $m(r) \asymp \varepsilon_R^{-1} L_R$  with  $L_R \rightarrow \infty$  arbitrarily slowly. The lower-tail estimate gives

$$\mathbb{P}(0 < |\mathcal{S}_R| < m(r)) \leq C \left( \frac{R \alpha_4(R)}{r \alpha_4(r)} \right)^2 = o(1),$$

because  $m(r)/m(R) \rightarrow 0$  and the four-arm polynomial bounds make the displayed ratio tend to zero. On the complementary event,  $(1 - \varepsilon_R)^{|\mathcal{S}_R|} \leq e^{-L_R}$ . Equation (6.32) therefore tends to zero, proving sharp noise sensitivity.

For triangular-site percolation, the four-arm exponent gives  $m(R) = R^2 R^{-5/4+o(1)} = R^{3/4+o(1)}$ . No exact square-lattice arm exponent is used: the square-lattice statement remains in the source strength of the GPS theorem and its polynomial arm bounds. This completes the proof.  $\square$

## Chapter 7

# Near-critical and dynamical percolation

Near-critical and dynamical percolation are constructed by perturbing criticality at exactly the pivotal scale.

**Definition 7.1** (Near-critical pivotal scale). For mesh  $\eta$  let  $\alpha_4^\eta(\eta, 1)$  be the critical alternating four-arm probability from lattice scale to unit scale. The canonical near-critical parameter scale is

$$r(\eta) = \frac{\eta^2}{\alpha_4^\eta(\eta, 1)}.$$

A perturbation  $p - p_c$  is *near critical* when it is of order  $r(\eta)$  as  $\eta \downarrow 0$ .

*Worked example 7.2* (Why the pivotal normalization is order one). A unit-area region contains order  $\eta^{-2}$  sites. A typical site is macroscopically pivotal only if four alternating arms reach unit scale, an event of probability  $\alpha_4^\eta(\eta, 1)$ . Thus the expected number of macroscopic pivots is

$$N_{\text{piv}}(\eta) \asymp \eta^{-2} \alpha_4^\eta(\eta, 1).$$

Changing the density by  $\lambda r(\eta)$  switches each site with probability of order  $|\lambda| r(\eta)$ . Hence

$$N_{\text{piv}}(\eta) r(\eta) \asymp 1.$$

This is the quantitative reason the scale  $r(\eta)$  produces a nontrivial continuum perturbation instead of either no macroscopic changes or infinitely many changes in every bounded parameter interval.

**Example 7.3** (A running picture). At  $p = 1/2$  on the triangular lattice, zooming out does not drive crossings to probability 0 or 1. The chapter identifies the mechanism responsible for this scale-free behavior.

**Theorem 7.4** (Existence of the near-critical scaling limit).

*Let*

$$r(\eta) := \frac{\eta^2}{\alpha_4^\eta(\eta, 1)}.$$

*Start critical site percolation on  $\eta\mathbb{T}$  at parameter  $p_c = 1/2$  and couple the near-critical configurations monotonically as follows: for  $\lambda > 0$  each closed site changes to open at exponential rate  $r(\eta)$ , and*

for  $\lambda < 0$  each open site changes to closed at the same rate when  $|\lambda|$  increases. Denote the resulting configuration by  $\omega_\eta^{\text{nc}}(\lambda)$ . For every  $L < \infty$ , as  $\eta \downarrow 0$ ,

$$[-L, L] \ni \lambda \mapsto \omega_\eta^{\text{nc}}(\lambda)$$

converges in law in the Schramm–Smirnov quad-crossing Skorohod space to a càdlàg process  $\lambda \mapsto \omega_\infty^{\text{nc}}(\lambda)$ . The construction is unique in law, and for every deterministic  $\lambda$  the slice  $\omega_\eta^{\text{nc}}(\lambda)$  converges in the quad-crossing topology to  $\omega_\infty^{\text{nc}}(\lambda)$ .

*Proof.* A macroscopic quad has order  $\eta^{-2}\alpha_4^\eta(\eta, 1)$  pivotal sites. During a parameter interval of length one, an individual site is switched with probability  $r(\eta) + O(r(\eta)^2)$ . Thus

$$\eta^{-2}\alpha_4^\eta(\eta, 1)r(\eta) = 1,$$

so the expected number of macroscopic pivotal switches is of order one. The exact exponential coupling has marginal

$$p_\eta(\lambda) = \begin{cases} 1 - (1 - p_c)e^{-\lambda r(\eta)}, & \lambda \geq 0, \\ p_c e^{-|\lambda| r(\eta)}, & \lambda < 0, \end{cases}$$

and hence agrees to first order with the usual  $p_c + \lambda r(\eta)/2$  parametrization. We use the exponential version because it is time-homogeneous on each side of 0.

Fix  $0 < \epsilon < 1$  and work first in a bounded domain. Call a point  $\epsilon$ -important when four alternating arms run from that point to scale  $\epsilon$ . By SM-11.03.03 the normalized counting measure on such points converges jointly with the critical configuration to a finite random measure  $\mu^\epsilon(\omega_\infty)$ , measurable from  $\omega_\infty$ . Conditional on  $(\omega_\infty, \mu^\epsilon)$ , place a Poisson point process

$$\Pi^\epsilon \subset D \times [-L, L] \quad \text{with intensity} \quad d\mu^\epsilon(z) d\lambda.$$

Almost surely this set is finite. A mark  $(z, \lambda)$  records the unique monotone switch at  $z$  as the parameter passes  $\lambda$ , with the direction of the switch determined by the sign of  $\lambda$  and the critical color. This gives the only additional randomness needed after the critical continuum configuration has been sampled.

For a quad  $Q$  and a finite marked set  $X$  of  $\epsilon$ -important points, choose a dyadic  $r \ll \epsilon$  and surround the points of  $X$  by disjoint  $r$ -squares. Form a finite graph whose vertices are these squares together with the four boundary arcs of  $Q$ ; record which pairs are joined by open or closed connections in the complement of the squares. At a marked square, the four alternating incident sectors record how the connectivity changes when that pivotal is switched. Hence the graph plus the colors of the marked vertices is a finite Boolean network that decides whether  $Q$  is crossed after any prescribed subset of the marks has switched.

The network stabilizes as  $r \downarrow 0$ . Indeed, failure of stabilization away from the boundary forces six arms from scale  $r$  to a fixed fraction of  $\epsilon$ , while a failure at a smooth boundary arc forces three half-plane arms. By the six-arm estimate from the proof of Lemma 2.9 and the half-plane exponent  $u(1) = 2$  from Lemma 2.7, the probability of either event is summable over dyadic  $r$ . Thus for a countable dense family of quads the limiting Boolean network is almost surely well defined and hereditary. It therefore determines a unique element of the Schramm–Smirnov space at every parameter value outside the finite set of Poisson marks, and by taking the right-continuous version it gives a càdlàg cutoff path  $\omega_\infty^{\text{nc}, \epsilon}(\cdot)$ .

Couple  $(\omega_\eta(0), \mu_\eta^\epsilon)$  to  $(\omega_\infty, \mu^\epsilon)$  using SM-11.03.03, and then couple the Poisson marks by the standard mapping construction for convergent finite measures. For every fixed finite family of quads and every fixed network scale  $r$ , the arm events defining the finite network are continuity events of

the critical quad-crossing limit unless a six-arm or boundary three-arm degeneracy occurs; those exceptional events have probability tending to zero by Step c. Hence the discrete and continuum Boolean networks, their mark order, and their quad states agree with probability tending to one. Letting first  $\eta \downarrow 0$  and then  $r \downarrow 0$  gives

$$\omega_\eta^{\text{nc},\epsilon}(\cdot) \implies \omega_\infty^{\text{nc},\epsilon}(\cdot)$$

in the Skorohod topology. Because only finitely many marks occur, no time reparametrization is needed after the coupled mark times have been matched.

We prove the uniform estimate that is the decisive input in removing the cutoff. For every  $L < \infty$  there is a continuous function  $\psi_L : [0, 1] \rightarrow [0, 1]$  with  $\psi_L(0) = 0$  such that, uniformly for  $0 < 2\eta < \epsilon$ ,

$$\mathbb{E} d_{\text{Sk}}(\omega_\eta^{\text{nc}}(\cdot), \omega_\eta^{\text{nc},\epsilon}(\cdot)) \leq \psi_L(\epsilon). \quad (7.1)$$

It is enough to prove this for the finitely many quads entering the first  $k$  levels of the metric and then send  $k \rightarrow \infty$ . Fix such a quad  $Q$  and consider the first parameter at which the full and cutoff processes disagree on  $Q$ . The responsible switch occurs at a site that was not  $\epsilon$ -important initially but has become pivotal for a crossing on the macroscopic scale of  $Q$ . Reveal the updated sites in chronological order and let  $s$  be the largest dyadic radius below  $\epsilon$  to which the responsible site had four alternating arms just before the discrepancy. If no second updated site lies in the annulus from  $s$  to  $2s$ , the four arms must already have existed to scale  $2s$ , contradicting maximality. If an update occurs there, expose the first such update. Arm separation then shows that the pair of sites forces either a six-arm event across  $A(s, c_Q)$  or a boundary three-arm event when the pair lies near  $\partial Q$ .

At scale  $s$ , the expected number of updated sites in a unit area during  $[-L, L]$  is at most  $C_L r(\eta) \eta^{-2} = C_L / \alpha_4^\eta(\eta, 1)$ . Quasi-multiplicativity factors the four microscopic arms at the responsible site from the six-arm (or boundary three-arm) event above scale  $s$ . Using

$$\alpha_6(s, c_Q) \leq C_Q s^{2+c}, \quad \alpha_3^+(s, c_Q) \leq C_Q s^2,$$

and summing over dyadic  $s \leq \epsilon$  gives, for fixed metric level  $k$ ,

$$\mathbb{P}[\text{a discrepancy on one of the level-}k \text{ quads}] \leq C_{L,k} \frac{\epsilon^2}{\alpha_4(\epsilon, 1)}. \quad (7.2)$$

The same estimate holds with the roles of the full and cutoff configurations reversed. Since Lemma 2.9 gives  $\alpha_4(\epsilon, 1) = \epsilon^{5/4+o(1)}$ , the right side of (7.2) tends to zero. Choosing  $k = k(\epsilon) \uparrow \infty$  slowly proves (7.1). This is the precise form of the statement that cascades of initially microscopic pivotals cannot unexpectedly change a macroscopic crossing.

Take  $0 < \epsilon_2 < \epsilon_1$ . Couple both continuum cutoff processes to a single discrete near-critical process. Step d and the triangle inequality, followed by Step e, give

$$\mathbb{E} d_{\text{Sk}}(\omega_\infty^{\text{nc},\epsilon_1}, \omega_\infty^{\text{nc},\epsilon_2}) \leq \psi_L(\epsilon_1) + \psi_L(\epsilon_2). \quad (7.3)$$

Choose a decreasing sequence  $\epsilon_j \downarrow 0$  such that  $\psi_L(\epsilon_j) \leq 2^{-j-2}$ . Markov's inequality and Borel-Cantelli show that the cutoff paths are almost surely Cauchy along this sequence in the complete Skorohod space. Denote their limit by  $\omega_\infty^{\text{nc}}(\cdot)$ . Equation (7.3) shows that every other cutoff sequence converges to the same limit in probability, so the law of the limit is independent of the auxiliary sequence.

Given  $\delta > 0$ , choose  $\epsilon$  so that  $\psi_L(\epsilon) < \delta/4$ . Under the coupling of Step d,

$$\begin{aligned} d_{\text{Sk}}(\omega_\eta^{\text{nc}}, \omega_\infty^{\text{nc}}) &\leq d_{\text{Sk}}(\omega_\eta^{\text{nc}}, \omega_\eta^{\text{nc},\epsilon}) \\ &\quad + d_{\text{Sk}}(\omega_\eta^{\text{nc},\epsilon}, \omega_\infty^{\text{nc},\epsilon}) + d_{\text{Sk}}(\omega_\infty^{\text{nc},\epsilon}, \omega_\infty^{\text{nc}}). \end{aligned}$$

The first and third terms are small by Steps e–f, uniformly in sufficiently small  $\eta$ , while the middle term tends to zero by Step d. This proves path convergence in law on  $[-L, L]$ .

For a deterministic  $\lambda$ , the limiting cutoff path has a jump at  $\lambda$  only if the Poisson process has a mark with exactly that parameter, an event of probability zero. The same remains true after the cutoff limit, because (7.3) gives convergence in probability uniformly on compact parameter intervals. Evaluation at  $\lambda$  is therefore almost surely a continuity map for the limiting path. The continuous-mapping theorem applied to Step g proves convergence of the discrete slice to  $\omega_\infty^{\text{nc}}(\lambda)$ . This also identifies the slice obtained from path convergence with the direct near-critical scaling limit and completes the proof.  $\square$

**Theorem 7.5** (Conformal covariance of the near-critical scaling limit).

Let  $f : \Omega \rightarrow \tilde{\Omega}$  be conformal with  $|f'|$  bounded above and bounded away from zero. Suppose  $\lambda \mapsto \omega_\infty^{\text{nc}}(\lambda)$  is the triangular-lattice near-critical continuum process in  $\Omega$ . Then its image by  $f$  has the same law as the spatially inhomogeneous near-critical process in  $\tilde{\Omega}$  whose local update-rate field is

$$\phi(f(z)) = |f'(z)|^{-3/4}.$$

In particular, for the similarity  $f(z) = \alpha z$ ,

$$(\lambda \mapsto \alpha \omega_\infty^{\text{nc}}(\lambda)) \stackrel{d}{=} (\lambda \mapsto \omega_\infty^{\text{nc}}(\alpha^{-3/4}\lambda)).$$

Thus fixed nonzero  $\lambda$  is not asserted to be conformally invariant under a general conformal map.

*Proof.*

$$\phi(f(z)) = |f'(z)|^{-3/4}.$$

Poisson mapping gives equality in law of the two marked cutoff constructions: the image of the source cutoff process is the target cutoff process with this spatially varying rate field.  $\square$

**Theorem 7.6** (Dynamical-percolation scaling limit and Markov property).

Let  $t \mapsto \omega_\eta(t)$  be critical dynamical site percolation on  $\eta\mathbb{T}$  in which each site is refreshed, independently, at rate

$$r(\eta) = \frac{\eta^2}{\alpha_4^\eta(\eta, 1)}$$

and on a refresh is recolored open or closed with probability  $1/2$  each. Then, as  $\eta \downarrow 0$ , the process converges in the quad-crossing Skorohod topology to a càdlàg process  $t \mapsto \omega_\infty(t)$ . The limiting process is stationary and reversible with invariant law  $\mathbb{P}_\infty$ , the critical percolation scaling-limit law, and it is a time-homogeneous Markov process.

*Proof.* Fix  $T, \epsilon > 0$ . The measure on  $\epsilon$ -important pivotal points is the same measurable random measure  $\mu^\epsilon(\omega_\infty)$  used in SM-11.07.01. Conditional on the critical slice, sample a Poisson point process on  $D \times [0, T]$  with intensity  $d\mu^\epsilon(z) dt$  and attach an independent fair Bernoulli mark to every Poisson point. At a mark, refresh the color of that pivotal point according to the Bernoulli mark. The finite Boolean-network construction from SM-11.07.01.c therefore gives a càdlàg cutoff path  $\omega_\infty^\epsilon(\cdot)$ .

The fixed- $\epsilon$  discrete-to-continuum coupling in SM-11.07.01.d uses only convergence of the critical configuration, pivotal measures, Poisson marks and the six-arm/boundary-three-arm continuity

criterion; it is unchanged when a mark resamples rather than switches monotonically. The stability proof SM-11.07.01.e is also color-symmetric and gives

$$\sup_{0 < 2\eta < \epsilon} \mathbb{E} d_{\text{Sk}}(\omega_\eta(\cdot), \omega_\eta^\epsilon(\cdot)) \leq \psi_T(\epsilon), \quad \psi_T(\epsilon) \downarrow 0. \quad (7.4)$$

Consequently the continuum cutoff paths are Cauchy exactly as in SM-11.07.01.f, and the three-term triangle inequality of SM-11.07.01.g proves

$$\omega_\eta(\cdot) \implies \omega_\infty(\cdot).$$

Poisson mark times have continuous distributions, so deterministic times are almost surely continuity times of the limit, and fixed-time marginals also converge.

For every  $\eta$ , critical Bernoulli product measure is stationary and the single-site refresh kernel is self-adjoint in  $L^2(\mathbb{P}_\eta)$ . Therefore, for  $0 \leq t_1 < \dots < t_m \leq T$ ,

$$(\omega_\eta(t_1), \dots, \omega_\eta(t_m)) \stackrel{d}{=} (\omega_\eta(T - t_m), \dots, \omega_\eta(T - t_1)).$$

By Step b, evaluation at these deterministic times is a.s. continuous for the limiting path. Passing to the limit in bounded continuous cylinder functions gives the same identity for  $\omega_\infty$ . Taking  $m = 1$  shows that every marginal is  $\mathbb{P}_\infty$ ; the full identity gives reversibility.

Fix  $0 < s < t$ . Since  $\omega_\infty(s) \sim \mathbb{P}_\infty$ , the pivotal measure  $\mu^\epsilon(\omega_\infty(s))$  is defined and measurable from the slice by SM-11.03.03. Conditional on this slice, sample independently a Poisson point process on  $D \times [s, t]$  and one on  $D \times [0, s]$ , both with spatial intensity  $\mu^\epsilon(\omega_\infty(s))$  and with independent fair resampling coins. Use the first to run the finite Boolean network forward from time  $s$  and the second to run the same reversible refresh mechanism backward from  $s$ . At fixed cutoff the past and future paths are therefore conditionally independent given the slice.

Remove the cutoff using (7.4). To see that conditional independence survives, let  $F$  and  $G$  be bounded continuous functions of the past and future path segments and  $H$  a bounded continuous function of the time- $s$  slice. At cutoff  $\epsilon$ ,

$$\mathbb{E}[F_\epsilon G_\epsilon H_\epsilon] = \mathbb{E}[\mathbb{E}(F_\epsilon \mid \omega_\infty^\epsilon(s)) \mathbb{E}(G_\epsilon \mid \omega_\infty^\epsilon(s)) H_\epsilon]. \quad (7.5)$$

The cutoff paths converge in probability in Skorohod space and the slice at  $s$  converges in probability as well. Boundedness gives convergence in  $L^1$  after taking subsequences, so (7.5) passes to the limit. A monotone class argument extends the factorization from continuous cylinder functions to all bounded Borel functions. Thus the limiting past and future are conditionally independent given  $\omega_\infty(s)$ .

The process rebuilt from the time- $s$  slice is, by the same fixed-cutoff coupling and stability estimate, a scaling limit of the original discrete dynamical percolation on  $[0, t]$ . Step b already proved uniqueness of that limit. Hence the rebuilt process has the same law as  $\omega_\infty(\cdot)$ . The conditional independence from Step d therefore gives

$$\mathcal{L}(\omega_\infty(t) \mid (\omega_\infty(u))_{0 \leq u \leq s}) = \mathcal{L}(\omega_\infty(t) \mid \omega_\infty(s)),$$

which is the Markov property.

The Poisson intensity is  $d\mu^\epsilon dt$  and depends on time only through Lebesgue measure. Repeating Step d with a shifted interval shows that the transition kernel depends only on  $t - s$ . On the full- $\mathbb{P}_\infty$  measure set of configurations for which all cutoff pivotal measures and cutoff limits are defined, these kernels form a measurable semigroup; extend the kernel arbitrarily on the null complement. Thus the process is a time-homogeneous Markov process, completing the proof.  $\square$



## Chapter 8

# Smirnov–Schramm–Werner synthesis

This supplement packages the conceptual synthesis: conformal invariance plus SLE exponents plus scaling relations, and the black-noise theorem that explains why no classical linear signal survives.

**Example 8.1** (A running picture). At  $p = 1/2$  on the triangular lattice, zooming out does not drive crossings to probability 0 or 1. The chapter identifies the mechanism responsible for this scale-free behavior.

**Theorem 8.2** (Smirnov–Werner critical-exponent synthesis).

*For critical site percolation on the triangular lattice the standard exponents appearing in the Smirnov–Werner synthesis exist in the logarithmic sense. In particular*

$$\nu = \frac{4}{3}, \quad \beta = \frac{5}{36}, \quad \gamma = \frac{43}{18}, \quad \eta = \frac{5}{24}, \quad \delta = \frac{91}{5},$$

*and the directional exponential correlation length also has exponent  $4/3$ . Equivalently,*

$$\theta(p) = (p - p_c)^{5/36+o(1)}, \quad \chi(p) = |p - p_c|^{-43/18+o(1)}, \quad \xi(p), \xi^*(p) = |p - p_c|^{-4/3+o(1)},$$

*together with the critical two-point and cluster-tail laws recorded in Theorem 5.7.*

*Proof.* This theorem is an exact synthesis of results proved earlier in Volume XI. The one-arm exponent  $5/48$  is SM-11.05.01; the alternating four-arm exponent  $5/4$  is SM-11.05.02; Kesten’s finite-size scaling package is Theorem 9.2. Substitution of the four-arm law into Kesten’s characteristic-length identity gives Theorem 5.6, hence  $\nu = 4/3$  for all three correlation-length normalizations. The remaining density, susceptibility, two-point, and cluster-mass conclusions, including  $\beta = 5/36$ ,  $\gamma = 43/18$ ,  $\eta = 5/24$ , and  $\delta = 91/5$ , are proved in Theorem 5.7. Thus the present result contains no independent research-level input and no citation-only step: it records the logical endpoint of the Smirnov/SLE<sub>6</sub>/Kesten chain already proved here in this volume.  $\square$

**Theorem 8.3** (Schramm–Smirnov black noise).

*Let  $D \subset \mathbb{C}$  be a bounded piecewise-smooth domain. Consider a critical planar Bernoulli percolation model satisfying RSW and arm separation, together with the block-pivotal estimate  $\mathbb{P}(A_4(r, R)) \leq C(r/R)^{1+c}$  for some  $c > 0$ . Let  $\mu$  be any subsequential quad-crossing scaling limit. These hypotheses hold, by the proof below, for critical site percolation on the triangular lattice and critical bond percolation on the square lattice. For a piecewise-smooth open set  $U \subset D$ , let  $\mathcal{F}_U$  be the completed  $\sigma$ -field generated by crossing events of all topological quadrilaterals compactly contained in  $U$ . Then the family  $U \mapsto \mathcal{F}_U$  is a two-dimensional continuous product: disjoint regions give independent  $\sigma$ -fields and cutting a region by finitely many piecewise-smooth arcs does not lose*

information. Moreover its first chaos is trivial. Equivalently, the stable  $\sigma$ -field is trivial, so the scaling limit is a black noise.

*Proof.* The proof has two logically different parts. The Schramm–Smirnov gluing argument first shows that the quad-crossing scaling limit really carries a spatial continuous-product structure. Blackness is stronger than ordinary BKS noise sensitivity: it says that the limiting continuous product has no non-zero linear, or first-chaos, component. We therefore prove the spatial factorization first and then give the cell-influence argument which kills the first chaos.

Choose once and for all a countable dense family  $\mathcal{Q}(U)$  of piecewise-smooth quads whose closures lie in  $U$ . The quad-crossing topology makes the indicators  $1_{\{Q \text{ is crossed}\}}$ ,  $Q \in \mathcal{Q}(U)$ , a generating family for  $\mathcal{F}_U$ . The RSW and arm-separation results proved earlier in this volume imply that crossing events are continuous in probability under small smooth perturbations of their sides: if two quads differ only inside an  $r$ -neighbourhood of their boundaries, their crossing indicators can differ only if an arm crosses an annulus with inner radius  $O(r)$ , and the RSW one-arm bound sends that probability to zero uniformly on compact families of nondegenerate quads.

For blackness we need a quantitative block-pivotal estimate which is slightly weaker than the exact triangular-lattice four-arm exponent but is available in both models. We claim that there are  $c, C > 0$  such that

$$\mathbb{P}(A_4(r, R)) \leq C \left( \frac{r}{R} \right)^{1+c}, \quad \eta < r < R/10, \quad (8.1)$$

for critical triangular-site and critical square-bond percolation. For the triangular lattice this follows immediately from Lemma 2.9: after decreasing  $c$  if necessary, the  $r/R \rightarrow 0$  asymptotic  $(r/R)^{5/4+o(1)}$  is bounded by  $C(r/R)^{1+c}$ , while bounded-ratio annuli are absorbed into  $C$ .

For completeness we prove the square-lattice multiscale estimate instead of importing it. Rescale so the mesh is one. Let  $Q$  be an  $R \times R$  square and partition its concentric middle third into  $r \times r$  squares  $Q_j$ ; there are  $N \asymp (R/r)^2$  of them. Put  $X = 2\mathbf{1}_{\{Q \text{ has an open left-right crossing}\}} - 1$ . Arm separation gives, uniformly in  $j$  away from the boundary of the middle third,

$$\mathbb{P}(A_4(r, R)) \asymp \mathbb{P}(Q_j \text{ is block-pivotal for } X). \quad (8.2)$$

Choose a thin annulus  $S_j$  just inside  $Q_j$  and the corresponding dual annulus  $S_j^*$ . Define  $C_j = 1$  when  $S_j$  contains an open circuit but  $S_j^*$  no dual-closed circuit,  $C_j = -1$  in the color-reversed case, and  $C_j = 0$  otherwise. Self-duality and RSW give

$$\mathbb{E}C_j = 0, \quad c_0 \leq \mathbb{E}C_j^2 \leq 1. \quad (8.3)$$

On the event that  $Q_j$  is not block-pivotal, changing the inside of  $Q_j$  cannot change  $X$ , while the conditional mean of  $C_j$  is still zero. On the pivotal event, four separated arms enter  $Q_j$ . Conditioning on these arms, FKG for the increasing event  $\{C_j = 1\}$  and the RSW fence construction give, for the strip defining  $S_j$  chosen sufficiently thin,

$$\mathbb{E}[XC_j \mid Q_j \text{ pivotal}] \geq c_1 > 0.$$

Consequently

$$\mathbb{E}[XC_j] \geq c_1 \mathbb{P}(Q_j \text{ pivotal}). \quad (8.4)$$

Now impose open boundary data on the left side of  $Q$  and dual-closed data on the other three sides, and reveal the interface  $\gamma$  separating the two boundary clusters. Let  $Y_j$  indicate that  $\gamma$  visits  $Q_j$ . A block pivotal must be visited by  $\gamma$ . RSW applied on a separated sequence of annuli between  $Q_j$  and  $\partial Q$  gives a polynomial exploration bound

$$\mathbb{E}Y_j \leq C \left( \frac{r}{R} \right)^{2c} \quad (8.5)$$

for some  $c > 0$  (renaming the positive exponent if necessary). Stop the interface at the first visit to either of two distinct squares  $Q_i, Q_j$ . The revealed path, together with the first visited square, determines  $Y_i, Y_j$  and the corresponding  $C$ -variable but is independent of the centered  $C$ -variable in the other square. Hence

$$\mathbb{E}(C_i Y_i C_j Y_j) = 0 \quad (i \neq j).$$

Using  $X^2 = 1$ , Cauchy–Schwarz, (8.3), and (8.5),

$$\begin{aligned} \sum_j \mathbb{P}(Q_j \text{ pivotal}) &\leq C \sum_j \mathbb{E}(X C_j Y_j) \\ &\leq C \left( \sum_j \mathbb{E}(C_j^2 Y_j) \right)^{1/2} \leq C \left( N \left( \frac{r}{R} \right)^{2c} \right)^{1/2} \leq C \left( \frac{r}{R} \right)^{c-1}. \end{aligned}$$

Since the left side is  $N$  times a typical block-pivotal probability up to arm-separation constants, (8.2) and  $N \asymp (R/r)^2$  imply

$$\mathbb{P}(A_4(r, R)) \leq C \left( \frac{r}{R} \right)^{1+c},$$

which proves (8.1) for square-bond percolation as well. This is the multiscale majority/interface argument; importantly, it is stronger than mere microscopic BKS noise sensitivity.

We also need the boundary analogue. A half-plane three-arm event from scale  $r$  to  $R$  can, after arm separation, be extended across the reflected half-annulus to a five-arm event with conditional probability bounded below by a universal RSW constant. (Use two disjoint RSW corridors in the reflected half-annulus and fences at the two landing intervals; the new arms have the two colors needed to obtain a non-monochromatic five-arm pattern.) Therefore the five-arm theorem SM-11.02.03, whose planar counting proof applies to both models, gives

$$\mathbb{P}(A_3^{\text{hp}}(r, R)) \leq C \left( \frac{r}{R} \right)^2. \quad (8.6)$$

Equations (8.1) and (8.6) are the only model-specific quantitative inputs needed in the rest of the proof.

Let  $Q_0 \Subset D$  be a fixed piecewise-smooth quad and let  $\alpha$  be a finite union of piecewise-smooth arcs with finitely many transverse double points. We prove the following statement: for every  $\varepsilon > 0$  there exist finitely many quads

$$Q_1, \dots, Q_N \Subset D \setminus \alpha$$

and a Boolean function  $\Psi_\varepsilon : \{0, 1\}^N \rightarrow \{0, 1\}$  such that, along the chosen subsequence,

$$\limsup_{\eta \downarrow 0} \mathbb{P}(1_{\{Q_0 \text{ crossed in } \omega_\eta\}} \neq \Psi_\varepsilon(1_{Q_1}, \dots, 1_{Q_N})) < \varepsilon. \quad (8.7)$$

Here every  $Q_j$  is at a positive distance from  $\alpha$ , so the right-hand side is measurable with respect to  $\mathcal{F}_{D \setminus \alpha}$ .

We now construct the approximation. Take nested tubular neighbourhoods  $S_s \subset S_{2s}$  of  $\alpha$ ; near the finitely many endpoints and double points use disjoint disks of radius  $O(s)$  and away from those disks use the normal strip coordinates of the smooth arcs. Explore, from the two boundary curves of  $S_s$ , every open or dual-open interface which reaches the boundary of  $S_{2s}$ . The connected components thereby cut off from the strip are the “bays”. A bay comes with its color and with the two pieces of the exterior configuration to which its mouth is attached. Conditional on this explored exterior data, the still-unseen bits inside distinct strip pieces are independent. Thus the

only information needed from the strip to decide the crossing of  $Q_0$  is: which mouths are connected outside the strip, their colors, and the cyclic order in which they meet the two sides of the strip.

Fix a coarse scale  $\delta$  with  $\eta \ll \delta \ll s$ . Cover the two boundaries of the strip by  $O(s/\delta)$  arcs of length between  $\delta$  and  $2\delta$  and cover the finitely many exceptional endpoint/double-point disks by  $O(1)$  additional sectors. Call the configuration  $\delta$ -good when any two distinct mouths that enter the same coarse boundary arc are already connected within its  $4\delta$ -neighbourhood, and when no mouth changes its coarse attachment under a perturbation of size  $\delta$ . Failure of this condition at an ordinary boundary arc produces four alternating arms from scale  $\delta$  to scale comparable with  $s$ ; at an endpoint it produces the corresponding boundary three-arm event. Hence, by the arm bounds in Step a and a union bound over the  $O(s/\delta)$  ordinary arcs,

$$\lim_{\delta/s \downarrow 0} \limsup_{\eta/\delta \downarrow 0} \mathbb{P}(\delta\text{-good fails}) = 0. \quad (8.8)$$

Indeed the four-arm contribution is  $O((s/\delta)(\delta/s)^{5/4+o(1)}) = o(1)$ , while the finitely many boundary three-arm contributions are  $O((\delta/s)^{2+o(1)})$ .

On the  $\delta$ -good event there are only finitely many possible coarse bay types. Whether two prescribed coarse mouths are joined in the exterior can be tested by finitely many crossings and noncrossings of small quads whose closures stay outside  $S_{s/2}$ ; the cyclic order is deterministic from the coarse cells. Consequently the complete finite connectivity graph of the bays is a Boolean function of finitely many such quad crossings. Once that graph, the mouth colors, and the deterministic geometry of  $Q_0$  are known, planarity decides whether an open path joins the two open sides of  $Q_0$ . Thus on the  $\delta$ -good event the crossing of  $Q_0$  is a Boolean function of finitely many crossings away from  $\alpha$ . First choose  $s$  so that the endpoint/strip modification of  $Q_0$  has error  $< \varepsilon/3$ , then choose  $\delta$  so that (8.8) is  $< \varepsilon/3$ , and finally choose  $\eta$  small enough that the finite collection of coarse tests is stable. This gives (8.7).

Let  $D_1, \dots, D_m$  be the components of  $D \setminus \alpha$ . Equation (8.7) shows that the crossing indicator of every  $Q_0 \in D$  belongs, modulo null sets, to  $\bigvee_j \mathcal{F}_{D_j}$ : take  $\varepsilon = 2^{-k}$ , pass to a subsequence for which the Boolean approximants converge almost surely, and use completeness of the  $\sigma$ -field. Since the countable dense family of smooth quads generates  $\mathcal{F}_D$ ,

$$\mathcal{F}_D = \bigvee_{j=1}^m \mathcal{F}_{D_j} \quad \text{modulo null sets.} \quad (8.9)$$

If  $U, V$  have disjoint closures, the discrete percolation variables used by crossings compactly contained in  $U$  and in  $V$  are independent for all sufficiently small mesh. Independence of every pair of finite cylinder families therefore passes to the subsequential limit; a monotone-class argument gives independence of  $\mathcal{F}_U$  and  $\mathcal{F}_V$ . Translation covariance is inherited from the discrete models. Exhausting an open set by piecewise-smooth relatively compact subsets and using (8.9) gives continuity from below. Thus  $U \mapsto \mathcal{F}_U$  is a two-dimensional continuous product, i.e. a noise.

Let  $X \in L^2(\mathcal{F}_D)$  have mean zero. Partition a square containing  $D$  into cells  $C$  of side  $r$ , and write  $\mathcal{G}_C = \mathcal{F}_{D \setminus C}$ . Define

$$H_r(X) := \sum_C \left( \mathbb{E} \sqrt{\text{Var}(X \mid \mathcal{G}_C)} \right)^2. \quad (8.10)$$

We claim that if  $H_r(X) \rightarrow 0$  along refining square partitions for a dense linear subspace of  $L^2(\mathcal{F}_D)$ , then the first chaos is zero.

To see this directly, first work with a finite product over the cells. Let  $P_C X = \mathbb{E}[X \mid \mathcal{F}_C]$  after subtracting the mean, where  $\mathcal{F}_C$  denotes the factor carried by  $C$ . Independence of the cell factors

makes the subspaces  $L_0^2(\mathcal{F}_C)$  mutually orthogonal, and the orthogonal projection  $Q_1^{(r)}X$  onto their direct sum satisfies

$$\|Q_1^{(r)}X\|_2^2 = \sum_C \|P_C X\|_2^2.$$

For each  $C$ , write  $X - \mathbb{E}[X \mid \mathcal{G}_C]$  for the part not determined by the complement. Conditional Jensen and the elementary inequality  $\|\mathbb{E}(Y \mid \mathcal{F}_C)\|_2 \leq \mathbb{E}\sqrt{\mathbb{E}(Y^2 \mid \mathcal{G}_C)}$  give

$$\|P_C X\|_2 \leq \mathbb{E}\sqrt{\text{Var}(X \mid \mathcal{G}_C)}.$$

Consequently

$$\|Q_1^{(r)}X\|_2^2 \leq H_r(X). \quad (8.11)$$

For the continuous product, the first-chaos projection  $Q_1 X$  is the  $L^2$  limit of the corresponding projections for refining finite partitions: an additive first-chaos variable restricts to the orthogonal sum of its cell pieces, and factorization (8.9) makes the partition  $\sigma$ -fields increase to  $\mathcal{F}_D$ . Taking the liminf in (8.11) gives

$$\|Q_1 X\|_2^2 \leq \liminf_{r \downarrow 0} H_r(X). \quad (8.12)$$

Thus  $H_r(X) \rightarrow 0$  implies  $X \perp$  first chaos. If this holds on a dense subspace, continuity of the orthogonal projection gives  $Q_1 = 0$ . We use Tsirelson's equivalent characterization of a black noise as a continuous product with trivial first chaos; in this convention  $Q_1 = 0$  is exactly the blackness conclusion.

Take  $X = 1_A - \mathbb{P}(A)$ , where  $A$  is the crossing event of a smooth quad  $Q \Subset D$ . Conditional on  $\mathcal{G}_C$ , the variable  $1_A$  is constant unless changing the configuration in  $C$  can change  $A$ . Let  $B_C$  denote this block-pivotal event. Since a Bernoulli random variable has conditional standard deviation at most  $1/2$ ,

$$\sqrt{\text{Var}(1_A \mid \mathcal{G}_C)} \leq \frac{1}{2} 1_{B_C},$$

and hence

$$\left(\mathbb{E}\sqrt{\text{Var}(1_A \mid \mathcal{G}_C)}\right)^2 \leq \frac{1}{4} \mathbb{P}(B_C)^2. \quad (8.13)$$

For a cell at distance at least  $10r$  from  $\partial Q$ ,  $B_C$  forces four alternating arms from a  $O(r)$  neighbourhood of  $C$  to a fixed macroscopic scale. Uniform arm separation and (8.1) therefore give, for some model-dependent  $c > 0$ ,

$$\sup_{C \text{ bulk}} \mathbb{P}(B_C) \leq C r^{1+c}.$$

There are  $O(r^{-2})$  bulk cells, so their contribution to  $H_r(X)$  is

$$O(r^{-2}) O(r^{2+2c}) = O(r^{2c}) \rightarrow 0. \quad (8.14)$$

There are only  $O(r^{-1})$  cells meeting the smooth sides of  $Q$ . A block pivotal there forces the boundary three-arm event, and (8.6) gives probability  $O(r^2)$ . Their squared-influence sum is therefore  $O(r^{-1}r^4) = O(r^3) = o(1)$ . The finitely many corner cells are handled by first rounding the corners on scale  $\rho$ , using Step a to make the resulting crossing error  $o_\rho(1)$ , and then sending  $r \downarrow 0$  before  $\rho \downarrow 0$ . We conclude

$$H_r(1_A) \rightarrow 0. \quad (8.15)$$

Notice the square in (8.13). Ordinary microscopic noise sensitivity alone would only control a sum of unsquared influences; the squared block-influence estimate is the stronger input needed for blackness.

Let  $F$  be a bounded function of finitely many smooth-quad crossing indicators. If resampling a cell changes  $F$ , then it changes at least one of those crossing indicators. Hence, after multiplying by  $4\|F\|_\infty^2$ , the block-pivotal bound in Step e and a finite union bound give  $H_r(F) \rightarrow 0$ . Finite Boolean combinations of crossing indicators form an algebra generating  $\mathcal{F}_D$ ; their bounded linear span is dense in  $L^2(\mathcal{F}_D)$ . Equation (8.12) therefore implies that every element of a dense subspace is orthogonal to first chaos. Thus the first chaos is  $\{0\}$  and the stable  $\sigma$ -field is trivial.

Every subsequential scaling limit in the stated square-bond/triangular-site class is a spatial continuous product with trivial stable part, hence a black noise. The proof uses the four-arm and boundary-arm theorems proved earlier in Volume XI; no appeal to the BKS theorem SM-11.06.02 is used as a substitute for the stronger block-sensitivity estimate.  $\square$

## Chapter 9

# Kesten and Garban–Pete–Schramm theory

Kesten’s IIC and scaling relations and the GPS continuum/Fourier programme complete the modern planar-percolation picture needed by later SLE and universality volumes.

**Example 9.1** (A running picture). At  $p = 1/2$  on the triangular lattice, zooming out does not drive crossings to probability 0 or 1. The chapter identifies the mechanism responsible for this scale-free behavior.

**Theorem 9.2** (Kesten’s near-critical scaling package).

*Consider site percolation on the triangular lattice and write  $p_c = 1/2$ . Let  $\alpha_1(r, R)$  and  $\alpha_4(r, R)$  denote the critical one-arm and alternating four-arm probabilities. Fix once and for all a small number  $\varepsilon_0 \in (0, 1/10)$ . If  $Q_n$  is a  $2n$  by  $n$  rectangle, define the finite-size correlation length  $L(p)$  to be the least  $n$  for which the left–right crossing probability under  $\mathbb{P}_p$  differs from its critical value by  $\varepsilon_0$  in the direction favoured by  $p - p_c$ .*

*There are constants  $c, C \in (0, \infty)$ , depending only on  $\varepsilon_0$ , such that, as  $p \rightarrow p_c$ , the following hold.*

1. For  $j \in \{1, 4\}$  and  $1 \leq r < R \leq cL(p)$ ,

$$C^{-1}\alpha_j(r, R) \leq \mathbb{P}_p(A_j(r, R)) \leq C\alpha_j(r, R).$$

*Thus critical arm probabilities are stable throughout the finite-size window.*

2. The characteristic-length identity is

$$|p - p_c|L(p)^2\alpha_4(1, L(p)) \asymp 1. \tag{9.1}$$

3. For  $p > p_c$ , with  $\theta(p) = \mathbb{P}_p(0 \leftrightarrow \infty)$ ,

$$\theta(p) \asymp \alpha_1(1, L(p)). \tag{9.2}$$

*If  $\chi(p) = \mathbb{E}_p[|C(0)|\mathbf{1}_{\{|C(0)| < \infty\}}]$ , then on either side of  $p_c$ ,*

$$\chi(p) \asymp L(p)^2\alpha_1(1, L(p))^2. \tag{9.3}$$

*The usual second-moment and exponential-decay correlation lengths are comparable with  $L(p)$  up to constants.*

*These are the Kesten scaling relations used later in this volume; in particular no exponent is assumed in the theorem.*

*Proof.* By the definition of  $L(p)$ , crossing probabilities of rectangles of aspect ratio two remain in a compact subinterval of  $(0, 1)$  at every scale  $n < L(p)$ . Gluing a bounded number of such rectangles and applying planar duality gives the same statement for every fixed aspect ratio. The proof of RSW in Chapter 1 uses only these bounds, FKG for one colour, and exploration of disjoint sectors for opposite colours. Consequently all RSW constants, arm-separation constants, and quasi-multiplicativity constants from SM-11.01.01 and SM-11.02.01–02 are uniform for every parameter  $q$  between  $p_c$  and  $p$  and every scale at most  $L(p)$ .

Let  $H_n$  be the left–right crossing event of  $Q_n$ . Russo’s formula SM-11.03.01 gives

$$\frac{d}{dq} \mathbb{P}_q(H_n) = \sum_{v \in Q_n} \mathbb{P}_q(v \text{ pivotal for } H_n).$$

If  $v$  lies a fixed positive fraction of  $n$  away from the boundary, the pivotal characterization SM-11.03.02 and arm separation show that its pivotal probability is comparable with the alternating four-arm probability from a unit neighbourhood of  $v$  to distance  $n$ . Translation invariance supplies  $cn^2$  such sites. This proves the lower estimate

$$\frac{d}{dq} \mathbb{P}_q(H_n) \geq cn^2 \alpha_4^q(1, n), \quad (9.4)$$

where  $\alpha_4^q$  denotes the four-arm probability at parameter  $q$ .

For the upper estimate split  $Q_n$  into a bulk region and dyadic boundary strips. A pivotal in the bulk has four arms to scale comparable with  $n$ . A pivotal at distance  $s$  from a side has three arms in the half-plane from  $s$  to  $n$  and four arms from the lattice scale to  $s$ ; at a corner the corresponding half-plane requirement is still stronger. Arm separation and quasi-multiplicativity factor these probabilities. The half-plane three-arm bound proved in the one-arm argument in Chapter 5 is  $O((s/n)^2)$  up to subpower factors, so the sum over the  $O(ns)$  sites in a boundary strip is bounded by a constant multiple of the bulk contribution. Summing the geometric dyadic series yields

$$\frac{d}{dq} \mathbb{P}_q(H_n) \leq Cn^2 \alpha_4^q(1, n), \quad n \leq L(p). \quad (9.5)$$

The same proof works for a topological rectangle after changing constants.

We record the finite-size bootstrap because it is the point at which one must not silently replace a near-critical arm probability by a critical one. Fix  $j \in \{1, 4\}$  and use separated landing intervals at the two boundary circles. For  $q$  in the interval between  $p_c$  and  $p$ , differentiate the logarithm of the separated  $j$ -arm probability. A site contributing to this derivative is pivotal for the arm event. Reveal the  $j$  arms until their first entrance in a dyadic annulus about that site. Arm separation then leaves, in an unexplored annulus, four alternating arms whose probability is  $\alpha_4^q(s, n)$ , while the already revealed part costs at most a constant multiple of the original  $j$ -arm event. Summing over sites by dyadic distance and using quasi-multiplicativity gives

$$\left| \frac{d}{dq} \log \alpha_j^q(r, n) \right| \leq Cn^2 \alpha_4^q(1, n), \quad r < n \leq L(p). \quad (9.6)$$

The potentially larger contributions from the two boundary circles are controlled by the half-plane three-arm estimate exactly as in part b.



To integrate (9.6), increase  $q$  continuously from  $p_c$  and stop at the first value at which either an arm probability changes by a factor two or a rectangle crossing changes by  $\varepsilon_0$ . Before that stopping time, (9.4)–(9.5) compare  $n^2\alpha_4^q(1, n)$  with the derivative of a crossing probability. Hence the integral in (9.6) is bounded by a constant times the total crossing-probability change, which is at most  $\varepsilon_0$ . Choosing the fixed threshold small enough prevents the factor-two alternative. Repeating the argument a bounded number of times and removing the separated-landing condition by SM-11.02.01 proves

$$\alpha_j^q(r, n) \asymp \alpha_j(r, n) \quad (j = 1, 4), \quad n \leq cL(p),$$

uniformly in  $q$  between  $p_c$  and  $p$ . This is Kesten's arm-stability lemma in the precise form needed below.

Take  $n$  to be a fixed fraction of  $L(p)$ . By definition the difference  $|\mathbb{P}_p(H_n) - \mathbb{P}_{p_c}(H_n)|$  is bounded above by a fixed constant, and for a comparable scale (replace  $n$  by  $2n$  a bounded number of times) it is bounded below by a positive constant. Integrating Russo's formula and applying parts b–c therefore gives

$$1 \asymp \int_{p_c}^p n^2 \alpha_4^q(1, n) dq \asymp |p - p_c| n^2 \alpha_4(1, n).$$

Quasi-multiplicativity changes  $n \asymp L(p)$  back to  $L(p)$  at constant cost and proves (9.1).

The upper bound in (9.2) is immediate: an infinite open cluster gives an open arm from 0 to distance  $L(p)$ . Arm stability changes its probability to  $O(\alpha_1(1, L(p)))$ . For the reverse inequality, condition on a separated open arm from 0 to  $\partial B_{L(p)}$ . Uniform RSW supplies, with probability bounded below, an open circuit in an annulus of comparable size attached to that arm. Above the finite-size scale, block renormalization applies: by the definition of  $L(p)$  and repeated RSW gluing, the probability of a  $p$ -open crossing of a  $2m$  by  $m$  rectangle tends exponentially fast to one as  $m/L(p) \rightarrow \infty$ . Choosing a nested sequence of such good blocks gives, with probability bounded below independently of  $p$ , an open route from the circuit to infinity. Hence  $\theta(p) \geq c\alpha_1(1, L(p))$ .

At criticality, for  $|x| \asymp r$ , two disjoint one-arm constructions from 0 and  $x$  to scale  $r/3$ , followed by an RSW gluing rectangle, give

$$c\alpha_1(1, r)^2 \leq \mathbb{P}_{p_c}(0 \leftrightarrow x) \leq C\alpha_1(1, r)^2.$$

The upper bound follows by demanding the two arms before they can meet; arm separation removes the landing restriction. Arm stability makes the same comparison valid for  $r \leq cL(p)$ , with the finite-cluster event on the supercritical side. Summing over dyadic annuli gives

$$\sum_{|x| \leq L(p)} \mathbb{P}_p(0 \leftrightarrow x, |C(0)| < \infty) \asymp L(p)^2 \alpha_1(1, L(p))^2.$$

The last annulus dominates by the polynomial one-arm bound from RSW together with the quasi-multiplicative arm estimate. Equivalently, after grouping a bounded number of scales, consecutive dyadic contributions grow by a fixed power. Beyond  $L(p)$ , the finite-size criterion and block renormalization give exponential decay for finite connections, so the tail is bounded by a constant multiple of the last annulus. This proves (9.3).

Let  $\xi(p)$  be the second-moment radius of a finite cluster and let  $\xi^*(p)$  be the inverse exponential decay rate of the finite two-point function in any fixed lattice direction. Part f gives lower bounds of order  $L(p)$  for both. The block-renormalization estimate beyond  $L(p)$  gives exponential decay on scale  $CL(p)$ , hence the matching upper bounds. Changing the lattice direction costs only a bounded number of RSW gluing steps. Thus  $\xi(p) \asymp \xi^*(p) \asymp L(p)$  as  $p \rightarrow p_c$ .

Smirnov–Werner: critical one- and four-arm probabilities determine the singular near-critical behaviour through (9.1)–(9.3). The proof uses only the already-owned RSW, arm separation,

quasi-multiplicativity, Russo formula, pivotal/four-arm characterization, and the standard finite-size block renormalization just established here; no near-critical exponent is imported as a black box.  $\square$

**Theorem 9.3** (Kesten's incipient infinite cluster).

Let  $K_n = \{0 \leftrightarrow \partial B_n\}$  for a standard two-dimensional planar percolation model to which the RSW, arm-separation and finite-energy arguments of this volume apply (in particular critical site percolation on the triangular lattice and critical bond percolation on the square lattice). Then there is a probability measure  $\nu_{\text{IIC}}$  such that

$$\mathbb{P}_{p_c}(\cdot \mid K_n) \implies \nu_{\text{IIC}} \quad (n \rightarrow \infty).$$

Moreover, if  $p > p_c$  and  $I = \{0 \leftrightarrow \infty\}$ , then

$$\mathbb{P}_p(\cdot \mid I) \implies \nu_{\text{IIC}} \quad (p \downarrow p_c).$$

Under  $\nu_{\text{IIC}}$  the origin belongs almost surely to an infinite open cluster, and the limit is independent of the particular sequence of outer boxes used in the first construction.

*Proof.* The configuration space  $\{0, 1\}^V$  is compact, so the conditional laws have subsequential weak limits. To prove a unique limit it suffices to prove that, for every cylinder event  $E$  depending on the sites of a fixed box  $B_m$ ,  $\mathbb{P}_{p_c}(E \mid K_n)$  is Cauchy as  $n \rightarrow \infty$ . The difficulty is that  $K_n$  is a rare event, so unconditioned weak convergence gives no information.

Choose radii

$$8m < r_1 < 4r_1 < r_2 < 4r_2 < \cdots < r_k < 4r_k < R$$

with fixed ratios. RSW gives a constant  $c > 0$  such that every annulus  $A(r_i, 4r_i)$  has an open circuit with probability at least  $c$ . The same lower bound remains valid after one reveals an open arm crossing the annulus: arm separation leaves a subannulus and finitely many unexplored rectangles in which RSW can close that arm to a circuit. Therefore, under the conditioned law  $\mathbb{P}_{p_c}(\cdot \mid K_n)$  with  $n > R$ , the probability that none of the  $k$  annuli contains an open circuit joined both to 0 and to the conditioned arm is at most  $(1 - c')^k$  for a model-dependent  $c' > 0$ .

Let  $\mathcal{C}$  be the innermost such circuit. Once  $\mathcal{C}$  and its states have been revealed, sites strictly inside and strictly outside  $\mathcal{C}$  are independent. The event  $K_n$  factors as

$$\{0 \leftrightarrow \mathcal{C}\}_{\text{in}} \cap \{\mathcal{C} \leftrightarrow \partial B_n\}_{\text{out}}.$$

Consequently, conditional on the discovered circuit  $\mathcal{C}$ , the law of the configuration inside  $\mathcal{C}$  given  $K_n$  is simply the product law inside conditioned on  $0 \leftrightarrow \mathcal{C}$ ; the value of the distant radius  $n$  occurs only in the outside factor.

Take  $n, N > R$ . Explore the two conditioned configurations from the outside inward using the same randomness for circuit-closing rectangles. Whenever a regeneration circuit is found in the same annulus, use the finite-energy coupling to choose the same circuit in both configurations, and then use identical product randomness inside it conditioned on  $0 \leftrightarrow \mathcal{C}$ . The preceding factorization shows that this is a valid coupling. On its success the two configurations agree on  $B_m$ . Since the failure probability is at most  $(1 - c')^k$ , for every cylinder event  $E$ ,

$$\sup_{n, N > R} |\mathbb{P}_{p_c}(E \mid K_n) - \mathbb{P}_{p_c}(E \mid K_N)| \leq 2(1 - c')^k.$$

First choose  $k$  large and then  $R$  large enough to contain the  $k$  annuli. This proves the Cauchy property and hence defines a unique probability measure  $\nu_{\text{IIC}}$  on all cylinder events. Kolmogorov

consistency and compactness identify it with the weak limit. The argument uses only that the outer conditioning lies beyond  $R$ , so replacing squares by any comparable exhaustion gives the same limit.

Fix  $m$  and an accuracy  $\varepsilon$ . Choose the regeneration annuli so that the failure probability in part c is below  $\varepsilon$ . When  $p \downarrow p_c$ , Theorem 9.2 gives  $L(p) \rightarrow \infty$  and uniform RSW/arm stability on all the finitely many chosen annuli. Thus the same circuit coupling works simultaneously for the critical law conditioned on reaching the outermost chosen radius  $R$  and the  $p$ -law conditioned on  $0 \leftrightarrow \infty$ .

Indeed, after a regeneration circuit  $\mathcal{C}$  is found, the event of an infinite connection depends only on the exterior, whereas the law inside  $\mathcal{C}$  is the product  $p$ -law conditioned on  $0 \leftrightarrow \mathcal{C}$ . On the finite interior this law converges as  $p \downarrow p_c$  to the critical conditioned law. Hence, for every cylinder event  $E$  supported in  $B_m$ ,

$$\limsup_{p \downarrow p_c} |\mathbb{P}_p(E \mid I) - \mathbb{P}_{p_c}(E \mid K_R)| \leq C\varepsilon.$$

Let  $R \rightarrow \infty$  using part c and then  $\varepsilon \downarrow 0$ . This proves that the supercritical conditional laws converge to the same  $\nu_{\text{IIC}}$ .

For every fixed  $r$  and every  $n > r$ ,  $\mathbb{P}_{p_c}(0 \leftrightarrow \partial B_r \mid K_n) = 1$ . Passing to the limit gives  $\nu_{\text{IIC}}(0 \leftrightarrow \partial B_r) = 1$  for every  $r$ . Intersecting over integer  $r$  shows that the origin cluster is infinite  $\nu_{\text{IIC}}$ -almost surely.

If two disjoint infinite open components both meet  $B_m$ , planarity forces a closed dual arm separating them through every sufficiently large annulus. Together with the open arm from the origin this yields, after arm separation, a three-arm event from scale  $m$  to scale  $R$ . The half-plane/plane arm bounds of Chapters 2 and 5 and quasi-multiplicativity show that its probability, divided by the one-arm probability used in the IIC conditioning, tends to zero as  $R \rightarrow \infty$ . Hence the IIC has no such second component. Letting  $m \uparrow \infty$  gives uniqueness of the infinite open cluster in the usual IIC sense.

The proof is the planar regeneration mechanism behind Kesten's theorem; the later dynamical-IIC chapter may therefore use  $\nu_{\text{IIC}}$  as an owned object rather than as an external limit law.  $\square$

**Theorem 9.4** (Garban–Pete–Schramm Fourier-spectrum theorem).

*For critical square-lattice bond percolation and critical triangular-lattice site percolation, the Fourier spectral sample of a macroscopic square crossing has the sharp lower tail and sharp noise-sensitivity scale stated in Theorem 6.6 below; equivalently, with the notation of Chapter 6,*

$$\mathbb{P}(0 < |\mathcal{S}_R| < r^2 \alpha_4(r)) \asymp \frac{R^2 \alpha_4(R)^2}{r^2 \alpha_4(r)^2}, \quad r_0 \leq r \leq R,$$

*and the decorrelation threshold is  $\varepsilon_R R^2 \alpha_4(R) \asymp 1$ .*

*Proof.* This is exactly Theorem 6.6, proved in Chapter 6. That proof owns the Fourier localization identity, annulus clustering, coupled four-arm quasi-multiplicativity, conditioned local first/second moments, the thinning large-deviation lemma, the matching lower bound, and the noise-scale deduction. No new input is introduced here; this synthesis row records the GPS theorem under its historical grouping.  $\square$

**Theorem 9.5** (Exceptional times in planar dynamical percolation).

*Run stationary critical dynamical percolation with unit refresh rate at every site or edge. For critical site percolation on the triangular lattice let*

$$\mathcal{E} = \{t : 0 \longleftrightarrow \infty \text{ at time } t\}$$

and, in the upper half-plane, let

$$\mathcal{E}^+ = \{t : 0 \longleftrightarrow \infty \text{ by a path contained in the upper half-plane}\}.$$

Then almost surely

$$\dim_H \mathcal{E} = \frac{31}{36}, \quad \dim_H \mathcal{E}^+ = \frac{5}{9}.$$

For critical bond percolation on  $\mathbb{Z}^2$ , exceptional times at which an infinite open cluster exists occur almost surely, and their set has strictly positive Hausdorff dimension.

*Proof.* The proof has two logically different halves. The lower bounds are a spectral/Frostman argument of Garban–Pete–Schramm. The matching triangular upper bounds are not consequences of the lower-tail estimate: they require a separate covering argument based on total influence. We prove both mechanisms because otherwise the exact dimensions would still depend on an external Schramm–Steif input.

Let  $f_R$  be the  $\{0, 1\}$  indicator of an open arm from a fixed microscopic neighbourhood of 0 to  $\partial B_R$ , and put  $\alpha_1(R) = \mathbb{E} f_R$ . For the half-plane event use  $f_R^+$  and  $\alpha_1^+(R)$ . If  $f_R = \sum_S \widehat{f}_R(S) \chi_S$ , stationarity and independence of the Poisson refresh clocks give the exact identity

$$\mathbb{E}[f_R(\omega_s) f_R(\omega_t)] = \sum_S e^{-|t-s||S|} \widehat{f}_R(S)^2. \quad (9.7)$$

Indeed each Walsh character survives the refreshes during a time interval of length  $u$  with mean  $e^{-u|S|}$ .

Write  $\rho(u)$  for the least integer  $r \geq r_0$  such that

$$r^2 \alpha_4(r) \geq u, \quad (9.8)$$

where  $\alpha_4$  is the critical alternating four-arm probability. Thus the spatial scale relevant to a time separation  $t$  is  $\rho(1/t)$ .

The Fourier machinery proved for square crossings in Theorem 6.6 has the following radial form: for  $r_0 \leq r \leq R$ ,

$$\sum_{|S| < r^2 \alpha_4(r)} \widehat{f}_R(S)^2 \leq C \frac{\alpha_1(R)^2}{\alpha_1(r)}, \quad (9.9)$$

$$\sum_{|S| < r^2 \alpha_4(r)} \widehat{f}_R^+(S)^2 \leq C \frac{\alpha_1^+(R)^2}{\alpha_1^+(r)}. \quad (9.10)$$

We spell out why this is an inherited proof engine rather than a new black box. Tile  $B_R \setminus B_{4r}$  by  $r$ -boxes and let  $K = B_{4r}$ . The Fourier projection identity (6.15) and the localization estimate (6.16) are unchanged. Build the same hierarchical annulus forest as in SM-11.06.03.b around the occupied coarse boxes. Every interior branch contributes the square of a four-arm cost; the only new outer factor is the probability of the arm connecting the fixed inner neighbourhood to the first occupied coarse box and then to radius  $R$ . Arm separation and quasi-multiplicativity turn that factor into

$$\alpha_1(R)^2 / \alpha_1(r)$$

for the unnormalised Fourier mass. If the spectral sample is contained in  $K$ , the same factor follows by exposing the outermost open circuit around  $K$  and applying the projection identity inside it. If it meets  $B_R \setminus K$  in fewer than  $r^2 \alpha_4(r)$  points, the independent Bernoulli thinning and the conditioned

local-growth lemma (6.22) reduce the event to the annulus-forest sum (6.19); the subexponential factor is summable by (6.21). This proves (9.9).

For the half-plane, use half-boxes at the straight boundary. Bulk branches are identical. A branch meeting the boundary pays the stronger half-plane three-arm cost, so the forest summation can only decrease; the radial outer factor is now  $(\alpha_1^+(R))^2/\alpha_1^+(r)$ . This gives (9.10). Thus all multiscale Fourier work is owned by the preceding GPS theorem; the present step only verifies the two radial geometries in which it is used.

Break the sum in (9.7) into the ranges  $j/t \leq |S| < (j+1)/t$ . Apply (9.9) at  $r = \rho(j/t)$  and use quasi-multiplicativity to compare  $\alpha_1(\rho(j/t))$  with  $\alpha_1(\rho(1/t))$ . The resulting polynomial loss in  $j$  is absorbed by  $e^{-j}$ . Hence, uniformly when  $\rho(1/t) \leq R$ ,

$$\frac{\mathbb{E}[f_R(\omega_0)f_R(\omega_t)]}{\alpha_1(R)^2} \leq \frac{C}{\alpha_1(\rho(1/t))}, \quad (9.11)$$

$$\frac{\mathbb{E}[f_R^+(\omega_0)f_R^+(\omega_t)]}{(\alpha_1^+(R))^2} \leq \frac{C}{\alpha_1^+(\rho(1/t))}. \quad (9.12)$$

When  $\rho(1/t) > R$  we use the trivial bounds  $\mathbb{E}[f_R(0)f_R(t)] \leq \alpha_1(R)$  and its half-plane analogue.

For triangular-lattice site percolation, the owned four-arm and one-arm theorems give

$$\alpha_4(r) = r^{-5/4+o(1)}, \quad \alpha_1(r) = r^{-5/48+o(1)}.$$

Consequently

$$\rho(u) = u^{4/3+o(1)}. \quad (9.13)$$

For a Borel set  $A \subset [0, 1]$  define the random occupation measure

$$\mu_R(A) = \frac{1}{\alpha_1(R)} \int_A f_R(\omega_t) dt.$$

Then  $\mathbb{E}\mu_R([0, 1]) = 1$ . From (9.11) and (9.13), for every  $\varepsilon > 0$  and all sufficiently small  $u$  with  $\rho(1/u) \leq R$ ,

$$\frac{\mathbb{E}[f_R(\omega_0)f_R(\omega_u)]}{\alpha_1(R)^2} \leq C_\varepsilon u^{-5/36-\varepsilon}.$$

At shorter times use the trivial bound  $\alpha_1(R)^{-1} = R^{5/48+o(1)}$ . The crossover is  $u_R = (R^2\alpha_4(R))^{-1} = R^{-3/4+o(1)}$ ; hence for  $\gamma < 31/36$ ,

$$R^{5/48+o(1)} \int_0^{u_R} u^{-\gamma} du = O(1),$$

and the integral over  $[u_R, 1]$  is bounded because  $\gamma + 5/36 < 1$ . Therefore

$$\sup_R \mathbb{E} \iint_{[0,1]^2} |t-s|^{-\gamma} d\mu_R(t) d\mu_R(s) < \infty. \quad (9.14)$$

The case  $\gamma = 0$  also gives a uniform second moment of the total mass. Paley–Zygmund and Markov therefore provide constants  $c, C > 0$  such that, with probability at least  $c$ , along infinitely many  $R$  we have  $\mu_R([0, 1]) \geq c$  and the  $\gamma$ -energy is at most  $C$ . Choose a subsequence and pass weakly to a nonzero measure  $\mu$ . Since the events  $\{0 \leftrightarrow \partial B_R\}$  decrease with  $R$ , the support of  $\mu$  is contained in  $\mathcal{E} \cap [0, 1]$ . Lower semicontinuity of truncated energy and monotone convergence show that  $\mu$  has

finite  $\gamma$ -energy. Frostman's lemma gives  $\dim_H \mathcal{E} \geq \gamma$  with positive probability. Time ergodicity makes the Hausdorff dimension an almost-sure constant, and  $\gamma \uparrow 31/36$  yields

$$\dim_H \mathcal{E} \geq 31/36 \quad \text{a.s.} \quad (9.15)$$

The chordal SLE<sub>6</sub> calculation in Lemma 2.7 gives the boundary one-arm exponent

$$\alpha_1^+(r) = r^{-1/3+o(1)}.$$

Combining this with (9.12) and (9.13) gives

$$\frac{\mathbb{E}[f_R^+(\omega_0)f_R^+(\omega_u)]}{(\alpha_1^+(R))^2} \leq C_\varepsilon u^{-4/9-\varepsilon}$$

above the microscopic time cutoff. Below that cutoff the trivial normalized bound is  $R^{1/3+o(1)}$ , and  $R^{1/3}u_R^{1-\gamma} = O(1)$  precisely for  $\gamma \leq 5/9$  up to subpower losses. Repeating the occupation-measure and Frostman argument gives

$$\dim_H \mathcal{E}^+ \geq 5/9 \quad \text{a.s.} \quad (9.16)$$

We now prove the upper-bound mechanism. Let  $A_n$  be decreasing cylinder events with  $p_n = \mathbb{P}(A_n) \downarrow 0$ , let

$$I_n = \sum_v \mathbb{P}(v \text{ is pivotal for } A_n),$$

and let  $T_n = \{t \in [0, 1] : \omega_t \in A_n\}$  and  $T = \bigcap_n T_n$ . If  $I_n \rightarrow \infty$ , then

$$\dim_H T \leq \liminf_{n \rightarrow \infty} \left(1 - \frac{\log p_n}{\log I_n}\right)^{-1} \quad \text{a.s.} \quad (9.17)$$

(the bounded- $I_n$  alternatives only make  $T$  smaller).

Indeed, the expected number  $N_n$  of boundary points of  $T_n$  is  $I_n/2$ : a pivotal bit refreshes at rate one and, in stationarity, changes the truth value with the corresponding fixed constant rate. Changing the normalization of the clocks changes only this harmless constant. If  $\mathcal{N}(U, \varepsilon)$  denotes the least number of intervals of length  $\varepsilon$  covering  $U$ , then enlarging every component of  $T_n$  by  $\varepsilon$  gives

$$\mathbb{E}\mathcal{N}(T_n, \varepsilon) \leq C(p_n \varepsilon^{-1} + I_n). \quad (9.18)$$

Choose  $\varepsilon_n = p_n/I_n$ . Then the right side is at most  $CI_n$ . Write

$$L = \liminf_n \frac{\log I_n}{\log(I_n/p_n)};$$

this is exactly the right side of (9.17). For  $s > L$ , pass to a subsequence on which  $I_n \varepsilon_n^s \leq 2^{-n}$ . Markov and Borel–Cantelli give coverings of  $T \subset T_n$  whose  $s$ -dimensional contents tend to zero. Hence  $\mathcal{H}^s(T) = 0$ . Letting  $s \downarrow L$  proves (9.17).

Take  $A_R$  to be the one-arm event to radius  $R$ , either in the plane or in the upper half-plane. We prove the common estimate

$$\frac{I(A_R)}{\mathbb{P}(A_R)} \leq R^{3/4+o(1)}. \quad (9.19)$$

For a pivotal site  $v$  at distance  $s$  from the origin and distance  $b$  from the relevant smooth boundary, reveal the required one arm up to scales a fixed factor away from  $v$ . In the remaining independent annulus, pivotality forces four alternating arms from the lattice scale to  $b$ ; if  $b \ll s$ , it also forces

three alternating half-plane arms from scale  $b$  to scale  $s$ . Arm separation and quasi-multiplicativity therefore give, relative to the probability of  $A_R$ ,

$$\frac{\mathbb{P}(v \text{ pivotal for } A_R)}{\mathbb{P}(A_R)} \leq C\alpha_4(1, b)\alpha_3^+(b, s), \quad (9.20)$$

with the convention  $b \asymp s$  for a bulk site. The already-owned SLE<sub>6</sub> values are

$$\alpha_4(1, b) = b^{-5/4+o(1)}, \quad \alpha_3^+(b, s) = (b/s)^{2+o(1)};$$

the second is the value  $u(1) = 2$  in Lemma 2.7.

For bulk sites with  $s \asymp 2^j$ , there are  $O(s^2)$  sites in the dyadic annulus and (9.20) costs  $s^{-5/4+o(1)}$ ; the sum is  $O(s^{3/4+o(1)})$ . In a boundary strip with  $b \asymp 2^k \leq s$ , there are  $O(sb)$  sites, and the corresponding contribution is

$$O(sb) b^{-5/4+o(1)} (b/s)^{2+o(1)} = O(b^{7/4+o(1)}/s).$$

Summing first over  $b$  and then over dyadic  $s \leq R$  is again  $R^{3/4+o(1)}$ . The inner microscopic boundary and the outer circle are handled by the same estimate after inversion of the roles of the two radial scales. This proves (9.19) for both geometries.

In the plane,  $p_R = \alpha_1(R) = R^{-5/48+o(1)}$ , hence  $I_R \leq R^{31/48+o(1)}$ . Substitution into (9.17) gives

$$\dim_H \mathcal{E} \leq \frac{31/48}{3/4} = \frac{31}{36}.$$

In the half-plane,  $p_R = \alpha_1^+(R) = R^{-1/3+o(1)}$  and  $I_R \leq R^{5/12+o(1)}$ , so

$$\dim_H \mathcal{E}^+ \leq \frac{5/12}{3/4} = \frac{5}{9}.$$

Together with (9.15) and (9.16), these are the two exact dimension statements.

For critical bond percolation on  $\mathbb{Z}^2$  we need no unproved exact critical exponent. We prove the quantitative inequality used by GPS: there are  $c, \varepsilon > 0$  such that

$$\alpha_1(r, R)\alpha_4(r, R) \geq c \left(\frac{r}{R}\right)^{2-\varepsilon}. \quad (9.21)$$

The five-arm theorem SM-11.02.03 gives  $\alpha_5(r, R) \asymp (r/R)^2$ . It is therefore enough to prove

$$\alpha_5(r, R) \leq C\alpha_1(r, R)\alpha_4(r, R) \left(\frac{r}{R}\right)^\varepsilon. \quad (9.22)$$

On the four-arm event choose three consecutive primal/dual interfaces  $\gamma_1, \gamma_2, \gamma_3$  and let  $S_\gamma$  be the component between  $\gamma_1$  and  $\gamma_3$  not adjacent to  $\gamma_1$ . A fifth arm requires an additional primal crossing of  $S_\gamma$ . At a scale  $\rho$  with  $8\rho$  still well inside the annulus, expose the interfaces only up to  $\partial B_\rho$ . In  $A(2\rho, 5\rho)$  RSW supplies, with conditional probability at least  $\delta > 0$ , a dual crossing of large positive or negative winding together with two connected primal circuits. One of the two winding choices is topologically incompatible with an additional primal crossing of  $S_\gamma$  across that scale. The domain Markov property of the unrevealed product configuration gives a uniform comparison with an ordinary one-arm crossing of the same topological strip. More precisely, RSW first bounds the unconstrained crossing probability by  $C\alpha_1(r, R)$ . On each factor-eight annulus, conditional on all

previously revealed data and on the continuation of that ordinary crossing, one of the two winding blockers occurs with probability at least  $\delta$  and kills the extra crossing. Hence the ratio

$$\frac{\mathbb{P}(\text{extra crossing survives to scale } \rho \mid \gamma_1, \gamma_2, \gamma_3)}{\mathbb{P}(\text{ordinary one arm survives to scale } \rho)}$$

contracts by a factor at most  $1 - \delta$  every time the radius is multiplied by eight. Iterating over the  $\asymp \log(R/r)$  disjoint factor-eight annuli yields

$$\mathbb{P}(\text{extra crossing in } S_\gamma \mid \gamma_1, \gamma_2, \gamma_3) \leq C\alpha_1(r, R)(r/R)^\varepsilon,$$

where  $\varepsilon = -\log_8(1 - \delta) > 0$ .

Finally sum over the possible triples of consecutive interfaces. If  $X$  is the number of interfaces crossing the annulus, then the number of triples is at most  $X^3 \mathbf{1}_{\{X \geq 4\}}$ . RSW and BK give an exponential tail in  $X$ , so  $\mathbb{E}[X^3 \mathbf{1}_{\{X \geq 4\}}] \leq C\mathbb{P}(X \geq 4) \asymp C\alpha_4(r, R)$ . This proves (9.22), hence (9.21).

The radial spectral estimate (9.9) and the covariance bound (9.11) apply to square-lattice bond percolation as well. From (9.21), with  $r = 1$ ,

$$\alpha_1(r)\alpha_4(r) \geq cr^{-2+\varepsilon}.$$

At  $r = \rho(1/t)$  the definition (9.8) gives  $\alpha_4(r) \asymp (tr^2)^{-1}$  up to bounded-scale constants, hence

$$\alpha_1(r) \geq ctr^\varepsilon.$$

Since  $\alpha_4 \leq 1$ , the definition of  $\rho$  implies  $r \geq ct^{-1/2}$ , and therefore

$$\frac{\mathbb{E}[f_R(\omega_0)f_R(\omega_t)]}{\alpha_1(R)^2} \leq Ct^{-1+\varepsilon/2} \quad (9.23)$$

above the microscopic cutoff. We check the omitted short-time piece because no exact square-lattice one-arm exponent is available. Put  $t_R = (R^2\alpha_4(R))^{-1}$ . From (9.21),

$$\frac{1}{\alpha_1(R)} \leq CR^{2-\varepsilon}\alpha_4(R) = Ct_R^{-1}R^{-\varepsilon}.$$

Moreover  $\alpha_4(R) \leq 1$  gives  $t_R \geq R^{-2}$ . Therefore, for  $0 < \gamma < \varepsilon/2$ ,

$$\frac{1}{\alpha_1(R)} \int_0^{t_R} t^{-\gamma} dt \leq CR^{-\varepsilon}t_R^{-\gamma} \leq CR^{-\varepsilon+2\gamma} = O(1).$$

For  $t \geq t_R$ , (9.23) gives an integrand bounded by  $Ct^{-\gamma-1+\varepsilon/2}$ , which is integrable uniformly because  $\gamma < \varepsilon/2$ . Thus the normalized occupation measures have uniformly bounded  $\gamma$ -energy. Frostman compactness gives positive probability of a nonzero limiting measure supported on square-bond exceptional times and of Hausdorff dimension at least  $\gamma$ . Time ergodicity promotes both existence and positive dimension to almost-sure statements.

Parts a–h prove all three assertions with their original model scopes. In particular, no triangular exact arm exponent has been used for the square lattice.  $\square$

**Theorem 9.6** (Garban–Pete–Schramm spectral-sample scaling limit).

Let  $Q$  be a bounded piecewise-smooth quad and let  $f_{Q,\eta} \in \{-1, 1\}$  be the left–right crossing observable for critical site percolation on the triangular lattice of mesh  $\eta$ . Regard its Fourier spectral sample  $S_{Q,\eta}$  as a random closed subset of the spherical compactification of  $\mathbb{C}$  after multiplying the



lattice by  $\eta$ . Then, as  $\eta \downarrow 0$ , the law of  $\mathcal{S}_{Q,\eta}$  converges weakly in the Hausdorff topology to a random closed set  $\mathcal{S}_Q$ . This law is conformally invariant: if  $\phi$  is conformal on a neighbourhood of  $Q$ , then

$$\phi(\mathcal{S}_Q) \stackrel{d}{=} \mathcal{S}_{\phi(Q)}.$$

Moreover  $\mathcal{S}_Q \subset \text{int } Q$  almost surely, and for every open  $U \Subset Q$ ,

$$U \cap \mathcal{S}_Q \neq \emptyset \implies \dim_H(U \cap \mathcal{S}_Q) = \frac{3}{4} \quad \text{almost surely.}$$

Consequently every nonempty realization of  $\mathcal{S}_Q$  is a compact perfect totally disconnected set, hence is homeomorphic to the Cantor set. For the square-crossing observable the empty atom tends to zero; therefore its scaling limit is almost surely nonempty and is almost surely a Cantor set of Hausdorff dimension  $3/4$ .

*Proof.* The proof has two independent tasks. First we must show that the random closed set has a unique conformally invariant scaling law. Compactness alone only gives subsequential limits, so a convergence-determining family of “avoid a cut” probabilities is required. Second we must identify the geometry of every nonempty local piece of the limiting set. For that part we construct the normalized counting measure of the discrete spectral sample, prove a two-point energy estimate, and combine it with the local-growth estimate already proved in Theorem 6.6.

Put the spherical metric  $d_0$  on  $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$  and the associated Hausdorff metric  $d_H$  on its closed subsets, with the usual convention that the empty set is at fixed positive distance from every nonempty set. The resulting space of closed subsets is compact. Hence the laws of  $\mathcal{S}_{Q,\eta}$  are automatically tight.

For a union  $U$  of lattice bits, the Fourier projection identity (6.15) gives

$$W_\eta(Q, U) := \mathbb{P}(\mathcal{S}_{Q,\eta} \subset U) = \sum_{S \subset U} \widehat{f}_{Q,\eta}(S)^2 = \|\mathbb{E}(f_{Q,\eta} \mid \mathcal{F}_U)\|_2^2. \quad (9.24)$$

This identity is the bridge between the Fourier object and the continuum crossing field. In particular, convergence of the quantities  $W_\eta(Q, U)$  for a convergence-determining family of open  $U$  determines the law of the limiting closed set.

We record the elementary stability estimate which will be used repeatedly. If  $f, g \in L^2$  satisfy  $\|f\|_2, \|g\|_2 \leq 1$ , then orthogonal projection is a contraction, and therefore

$$|\|\mathbb{E}(f \mid \mathcal{F}_U)\|_2^2 - \|\mathbb{E}(g \mid \mathcal{F}_U)\|_2^2| \leq 2\|f - g\|_2. \quad (9.25)$$

Indeed  $|\|a\|_2^2 - \|b\|_2^2| \leq (\|a\|_2 + \|b\|_2)\|a - b\|_2$  and conditional expectation does not increase  $L^2$  norm.

Let  $U \subset \mathbb{C}$  be open, with  $\partial U$  a finite disjoint union of smooth simple closed curves, and assume  $\partial U \cap \partial Q$  is finite. The gluing theorem already proved in SM-11.S1.02.b–c, in particular (8.7), says the following. Given  $\varepsilon > 0$ , there are piecewise-smooth quads  $Q_1, \dots, Q_m \Subset Q \setminus \partial U$  and a Boolean function  $g$  such that, if

$$G_\eta = g(f_{Q_1,\eta}, \dots, f_{Q_m,\eta}),$$

then

$$\limsup_{\eta \downarrow 0} \|f_{Q,\eta} - G_\eta\|_2^2 \leq 4\varepsilon. \quad (9.26)$$

After refining the finite family we may suppose that each  $Q_j$  lies either inside  $U$  or outside  $\overline{U}$ . Write  $J$  for the indices of the quads inside  $U$ . The conditional expectation  $\mathbb{E}(G_\eta \mid \mathcal{F}_U)$  is then a function only of the finite vector  $(f_{Q_j,\eta})_{j \in J}$ , while the remaining coordinates have been averaged out.

The full percolation scaling-limit theorem already owned in this volume gives convergence of the joint law of every finite vector of quad crossings. Thus

$$W_\eta(G_\eta, U) := \|\mathbb{E}(G_\eta \mid \mathcal{F}_U)\|_2^2$$

has a limit as  $\eta \downarrow 0$ . By (9.25) and (9.26),

$$\limsup_{\eta, \eta' \downarrow 0} |W_\eta(Q, U) - W_{\eta'}(Q, U)| \leq 8\sqrt{\varepsilon}.$$

Since  $\varepsilon$  is arbitrary, the limit

$$W(Q, U) := \lim_{\eta \downarrow 0} W_\eta(Q, U) \tag{9.27}$$

exists.

For completeness, these numbers determine a unique law on closed subsets. Indeed for compact  $K$ ,

$$\{F : F \cap K = \emptyset\} = \{F : F \subset K^c\},$$

so the probabilities in (9.27) determine all avoidance probabilities for compact sets which can be approximated from outside by our smooth  $U$ 's. A countable family of rational polygonal/smooth approximations generates the hit-or-miss Borel sigma-field on the compact Hausdorff space of closed subsets. Hence every subsequential limit has the same law. Tightness from part a therefore upgrades to convergence of the whole family  $\mathcal{S}_{Q, \eta} \Rightarrow \mathcal{S}_Q$ .

Let  $\phi$  be conformal on a neighbourhood of  $Q$ . Choose the same cut  $\partial U$  and the approximating quads in part b. The joint continuum law of the finite crossing vector is conformally invariant, while the gluing error (9.26) is controlled by arm probabilities and hence is unchanged at the level of the vanishing error after applying  $\phi$ . Consequently

$$W(\phi(Q), \phi(U)) = W(Q, U)$$

for the convergence-determining family of  $U$ 's. By uniqueness of the random closed-set law,  $\phi(\mathcal{S}_Q) \stackrel{d}{=} \mathcal{S}_{\phi(Q)}$ .

The same arm localization also keeps the spectral sample away from the boundary. Cover a boundary strip of width  $r$  by  $O(r^{-1})$  boundary boxes. If the spectral sample enters one such box, changing bits there must communicate with the macroscopic crossing through the appropriate boundary three-arm event. The half-plane three-arm estimate from the preceding chapters is  $O(r^{2+o(1)})$ ; summing the boxes gives  $O(r^{1+o(1)})$ . Letting  $r \downarrow 0$  shows

$$\mathbb{P}(\mathcal{S}_Q \cap \partial Q \neq \emptyset) = 0, \tag{9.28}$$

and in fact the same estimate for shrinking boundary strips gives  $\mathcal{S}_Q \subset \text{int } Q$  almost surely.

Fix  $U \Subset Q$ . Let

$$a_\eta := \eta^{-2} \alpha_4(\eta, 1), \quad \lambda_\eta := a_\eta^{-1} \sum_{x \in \mathcal{S}_{Q, \eta} \cap U} \delta_x.$$

The single-bit identity (6.17) says that the probability a bulk site belongs to the spectral sample is its pivotal probability. Arm separation and quasi-multiplicativity therefore give

$$\sup_{\eta > 0} \mathbb{E} \lambda_\eta(U) < \infty. \tag{9.29}$$

For two sites  $x, y \in U$  at distance  $r \gg \eta$ , simultaneous membership in the spectral sample forces four alternating arms from each site to scale  $r$ , and then the four macroscopic arms from scale  $r$

to the boundary of the quad. Applying the two-point version of the localization identity and arm separation gives

$$\mathbb{P}(x, y \in \mathcal{S}_{Q,\eta}) \leq C \alpha_4(\eta, r)^2 \alpha_4(r, 1). \quad (9.30)$$

By quasi-multiplicativity,  $\alpha_4(\eta, 1) \asymp \alpha_4(\eta, r) \alpha_4(r, 1)$ ; hence after dividing by  $a_\eta^2$ , the normalized pair density at separation  $r$  is bounded by a constant times  $\alpha_4(r, 1)^{-1}$ . The owned triangular-lattice exponent  $\alpha_4(r, 1) = r^{5/4+o(1)}$  therefore implies that for every  $\gamma < 3/4$ ,

$$\sup_{\eta>0} \mathbb{E} \iint_{U \times U} (|x - y| \vee \eta)^{-\gamma} d\lambda_\eta(x) d\lambda_\eta(y) < \infty. \quad (9.31)$$

Indeed the contribution of pairs with separation in  $[r, 2r]$  is bounded by  $C r^{2-5/4-\gamma+o(1)}$ , and the dyadic sum converges exactly when  $\gamma < 3/4$ .

The first-moment bound and compactness of  $\bar{U}$  make the laws of  $(\mathcal{S}_{Q,\eta}, \lambda_\eta)$  tight along every sequence. On any joint subsequential limit  $(\mathcal{S}, \lambda)$ , the measure  $\lambda$  is supported on  $\mathcal{S}$ : if a compact set  $K$  has positive distance from  $\mathcal{S}$ , then eventually  $\mathcal{S}_{Q,\eta} \cap K = \emptyset$ , so  $\lambda_\eta(K) = 0$ , and Portmanteau gives  $\lambda(K) = 0$ .

The energy estimate by itself would not rule out the possibility that the subsequential measure vanishes on a part of the limiting spectral set. This is where the conditioned local-growth theorem (6.22) is used.

Take concentric disks  $B' \Subset B \Subset U$ . Conditional on the event that the spectral sample enters  $B$ , the local-growth estimate says with a mesh-independent probability bounded below that the middle disk contains at least a fixed multiple of  $(r/\eta)^2 \alpha_4(\eta, r)$  spectral points. After division by  $a_\eta$  and quasi-multiplicativity, this is a fixed positive multiple of  $r^2/\alpha_4(r, 1)$ , independent of  $\eta$ . The second-moment estimate in the proof of (6.22) upgrades this to the following zero-mass exclusion in the limit:

$$\mathbb{P}(\mathcal{S} \cap B' \neq \emptyset, \lambda(B) = 0) = 0. \quad (9.32)$$

Here is the small boundary correction needed to pass from  $B'$  to  $B$ . Choose  $B'$  so that the annulus  $B \setminus B'$  has relative width  $\delta$ . It can be covered by  $O(\delta^{-1})$  disks of radius  $\delta r$ . The local spectral-hit estimate obtained from (6.16), together with the four-arm exponent, bounds the probability that the limiting sample meets any one such disk by  $O(\delta^{5/4+o(1)})$ . The union bound is  $O(\delta^{1/4+o(1)})$ , which tends to zero. Letting  $\delta \downarrow 0$  in (9.32) gives

$$\mathcal{S} \cap U \neq \emptyset \implies \lambda(U) > 0 \quad \text{almost surely.} \quad (9.33)$$

Thus the support of  $\lambda$  is the whole nonempty local spectral set, not merely a subset of it.

Fix  $\gamma < 3/4$ . By (9.31), Fatou's lemma for the truncated kernels followed by monotone convergence gives, for every joint subsequential limit,

$$\iint |x - y|^{-\gamma} d\lambda(x) d\lambda(y) < \infty \quad \text{almost surely after restriction to finite-energy levels.}$$

Together with (9.33), Frostman's criterion implies

$$U \cap \mathcal{S} \neq \emptyset \implies \dim_H(U \cap \mathcal{S}) \geq \frac{3}{4}. \quad (9.34)$$

We let  $\gamma \uparrow 3/4$  through rationals.

For the matching upper bound, cover  $U$  by  $O(r^{-2})$  disks of radius  $r$ . The localization identity and four-arm separation give uniformly for disks whose double is contained in  $Q$

$$\mathbb{P}(\mathcal{S}_Q \cap B(z, r) \neq \emptyset) \leq C \alpha_4(r, 1) = r^{5/4+o(1)}. \quad (9.35)$$

Hence the expected number  $N_r$  of radius- $r$  disks in this cover which meet  $\mathcal{S}_Q$  satisfies

$$\mathbb{E}N_r \leq r^{-3/4+o(1)}. \quad (9.36)$$

Fix  $\varepsilon > 0$  and take dyadic  $r_k = 2^{-k}$ . Markov's inequality and a summable slack give almost surely, for all large  $k$ ,  $N_{r_k} \leq r_k^{-3/4-\varepsilon}$ . Therefore the upper Minkowski, hence Hausdorff, dimension of  $U \cap \mathcal{S}_Q$  is at most  $3/4 + \varepsilon$ . Letting  $\varepsilon \downarrow 0$  and combining with (9.34) proves

$$U \cap \mathcal{S}_Q \neq \emptyset \implies \dim_H(U \cap \mathcal{S}_Q) = \frac{3}{4}. \quad (9.37)$$

A countable basis of rational disks upgrades the assertion from one fixed  $U$  to every open set simultaneously.

Equation (9.37) implies that no point of a nonempty  $\mathcal{S}_Q$  is isolated: every neighbourhood meeting the set has positive Hausdorff dimension. The set is closed in the compact spherical metric, so it is compact. A nontrivial connected compact subset of the plane has Hausdorff dimension at least one (its orthogonal projection onto a line joining two of its points contains a nondegenerate interval). Since every local piece of  $\mathcal{S}_Q$  has dimension  $3/4 < 1$ , every connected component is a singleton. Thus a nonempty realization is compact, perfect and totally disconnected, and hence is homeomorphic to the standard Cantor set.

Finally, for the square crossing observable,

$$\mathbb{P}(\mathcal{S}_{Q,\eta} = \emptyset) = \widehat{f}_{Q,\eta}(\emptyset)^2 = (\mathbb{E}f_{Q,\eta})^2 \longrightarrow 0,$$

because the critical square crossing probability tends to  $1/2$ . Weak convergence then gives  $\mathbb{P}(\mathcal{S}_Q = \emptyset) = 0$ . Combining this with parts a–f proves the square theorem in the canonical ledger and, with the explicit empty-set qualification, the more general piecewise-smooth quad formulation above.  $\square$

**Theorem 9.7** (Garban–Pete–Schramm continuum cluster-area and interface-length measures).

*Let critical site percolation on the triangular lattice of mesh  $\eta$  converge in the quad-crossing topology to the critical continuum configuration  $\omega$ .*

**(i) Cluster-area measure.** *Let  $A$  be a fixed proper annulus with inside face  $\Delta$  and outer boundary  $\partial_2 A$ . Define*

$$\lambda_\eta^A := \sum_{x \in \eta\mathbb{T} \cap \Delta: x \leftrightarrow \partial_2 A} \delta_x \frac{\eta^2}{\alpha_1^\eta(\eta, 1)}. \quad (9.38)$$

*Then*

$$(\omega_\eta, \lambda_\eta^A) \implies (\omega, \lambda^A),$$

*where  $\lambda^A = \lambda^A(\omega)$  is almost surely a finite Borel measure and is a measurable function of the continuum percolation configuration. Equivalently, after filtering by cluster diameter, the normalized counting measures*

$$\lambda_\eta^\rho := \sum_{x: \text{diam}(C_x) > \rho} \delta_x \frac{\eta^2}{\alpha_1^\eta(\eta, 1)}$$

*have continuum limits describing the area of macroscopic clusters.*

**(ii) Interface-length measure.** *Let  $\Omega$  be a bounded smooth simply connected domain with distinct marked boundary points  $a, b$ , equip the discrete approximations with Dobrushin boundary conditions, and let  $\gamma_\eta$  be the exploration interface from  $a_\eta$  to  $b_\eta$ . Put*

$$\tau_\eta := \sum_{e \in \gamma_\eta} \delta_{m(e)} \frac{\eta^2}{\alpha_2^\eta(\eta, 1)}, \quad (9.39)$$

where  $m(e)$  is the midpoint of the interface edge. Then

$$(\gamma_\eta, \tau_\eta) \implies (\gamma, \tau^{\Omega, a, b}),$$

where  $\gamma$  is chordal  $\text{SLE}_6$  from  $a$  to  $b$  and  $\tau^{\Omega, a, b} = \tau^{\Omega, a, b}(\gamma)$  is a measurable function of the curve  $\gamma$  itself.

**(iii) Conformal covariance.** If  $f : \Omega \rightarrow \tilde{\Omega}$  is conformal and  $A \Subset \Omega$ , then, with  $\tilde{\omega} = f(\omega)$ , almost surely

$$\lambda^{f(A)}(f(B))(\tilde{\omega}) = \int_B |f'(z)|^{91/48} d\lambda^A(z) \quad (9.40)$$

for every Borel  $B \subset \Delta$ . For the interface statement assume, as in the theorem, that  $f$  extends continuously to  $\partial\Omega$  and that  $|f'|$  is bounded on  $\Omega$ ; with  $\tilde{a} = f(a)$  and  $\tilde{b} = f(b)$ ,

$$\tau^{\tilde{\Omega}, \tilde{a}, \tilde{b}}(f(B))(f(\gamma)) = \int_B |f'(z)|^{7/4} d\tau^{\Omega, a, b}(z) \quad (9.41)$$

for every Borel  $B \subset \Omega$ .

No assertion about a time ordering of the atoms, or about equality of the measure of an initial curve segment with the elapsed Loewner/natural time, is part of this theorem.

*Proof.* The proof is deliberately organized in parallel with the pivotal-measure construction SM-11.03.03. What changes is the number of arms. A cluster-area atom is detected by a one-arm event and an interface atom by a two-arm event. We therefore first build one-arm and two-arm analogues of the coupling and ratio-limit steps used for pivots, and only then pass to random measures.

Let  $d_A > 0$  be the distance between the two boundary components of  $A$ , truncated at 1 if necessary. For a lattice site  $x \in \Delta$  to contribute to (9.38), it must have an open arm from scale  $\eta$  to scale comparable with  $d_A$ . Quasi-multiplicativity therefore gives

$$\mathbb{E}\lambda_\eta^A(\Delta) \leq C_A |\Delta| \frac{\alpha_1^\eta(\eta, d_A)}{\alpha_1^\eta(\eta, 1)} \leq C_A.$$

For a square  $Q \Subset \Delta$  of side  $r$ , expand the second moment as a sum over ordered pairs  $x, y \in Q$ . If  $2^j \eta \leq |x - y| < 2^{j+1} \eta$ , the two sites must each have an arm to scale  $|x - y|/3$ , while beyond that scale one arm suffices to reach the outer boundary. BK for the two disjoint microscopic arms, followed by quasi-multiplicativity, gives

$$\mathbb{P}(x, y \leftrightarrow \partial_2 A) \leq C_A \alpha_1(\eta, |x - y|)^2 \alpha_1(|x - y|, d_A).$$

Summing over dyadic shells and dividing by  $\eta^{-4} \alpha_1(\eta, 1)^2$  yields a finite bound uniform in  $\eta$ ; indeed the one-arm exponent  $5/48 < 1$  makes the shell sum convergent at the microscopic end. Consequently the total masses are uniformly integrable and the family  $\{\lambda_\eta^A\}$  is tight in the weak topology on finite measures.

Fix  $\eta < r < d_0/100$ , where  $d_0$  is the distance from the origin to the boundary of a piecewise smooth simply connected comparison domain, truncated at 1. Compare the law conditioned on an open arm from  $\partial B(0, r)$  to that boundary with the law conditioned on an open arm from  $\partial B(0, r)$  to  $\partial B(0, 1)$ . Choose dyadic annuli between  $r$  and  $d_0$ . On each pair of consecutive annuli, RSW gives probability at least  $c > 0$  of an open circuit surrounding the inner disk. Reveal the outermost such circuit from the outside. Conditional on the revealed circuit, the configuration inside it is fresh product percolation conditioned only on connection to that circuit. Couple the two conditional samples by using the same bits inside the first circuit that occurs simultaneously in both samples.

Each dyadic trial succeeds with probability at least  $c^2$ , independently of all information revealed farther out in the sense needed for the conditional lower bound. Hence, for some  $k > 0$ ,

$$\mathbb{P}(\text{the two inward configurations fail to couple}) \leq C(r/d_0)^k. \quad (9.42)$$

Unlike the four-arm construction, no arm-separation lemma is required: monotonicity of the one-arm event and the surrounding open circuit erase the outer landing data.

For fixed  $0 < s < 1$  define

$$R_\eta^{(1)}(s) := \frac{\alpha_1^\eta(\eta, s)}{\alpha_1^\eta(\eta, 1)}.$$

Insert an intermediate radius  $\gamma$  with  $\eta \ll \gamma \ll s$ . Conditioning on the arm outside  $\gamma$ , the coupling (9.42) shows that the conditional probability of continuing from the microscopic scale to  $\gamma$  changes by at most  $O((\gamma/s)^k)$  when the outer domain is changed. Therefore  $R_\eta^{(1)}(s)$  is Cauchy as  $\eta \downarrow 0$  and has a positive limit  $L_1(s)$ . Applying the same coupling to three nested scales gives

$$L_1(st) = L_1(s)L_1(t), \quad 0 < s, t < 1.$$

RSW and quasi-multiplicativity make  $L_1$  locally bounded above and below, hence the multiplicative function  $u \mapsto L_1(e^{-u})$  is an exponential: there is  $\xi_1 \geq 0$  with  $L_1(s) = s^{-\xi_1}$ . On the other hand SM-11.05.01 proves

$$\alpha_1(\eta, 1) = \eta^{5/48+o(1)}.$$

Iterating a fixed ratio  $s$  and comparing logarithms forces  $\xi_1 = 5/48$ . Thus

$$\boxed{\frac{\alpha_1(s\epsilon, 1)}{\alpha_1(\epsilon, 1)} \longrightarrow s^{5/48}} \quad (\epsilon \downarrow 0). \quad (9.43)$$

The same circuit coupling shows that the local normalization constant is independent of the remote domain shape and, by rotational invariance of the scaling limit, of the orientation of the mesoscopic square.

Tile a test set  $B \Subset \Delta$  by  $\epsilon$ -squares  $Q_i$ . Let  $X_i$  be the number of sites in  $Q_i$  joined to  $\partial_2 A$ , and let  $Y_i$  be the indicator that a fixed enlargement of  $Q_i$  has an open arm to  $\partial_2 A$ . Conditional on the macroscopic configuration outside the enlargement, Step b and (9.43) give, uniformly over admissible outer configurations,

$$\mathbb{E}[X_i \mid \text{mesoscopic data}] = \beta_{\eta, \epsilon} Y_i + o(\beta_{\eta, \epsilon}), \quad \beta_{\eta, \epsilon} \asymp \epsilon^2 \frac{\alpha_1(\eta, \epsilon)}{\eta^2}.$$

For two separated boxes the circuit couplings may be performed independently. For nearby boxes the direct pair estimate from Step a dominates the error. Splitting the pair sum at distance  $R\epsilon$ , first letting  $\eta/\epsilon \rightarrow 0$ , then  $R \rightarrow \infty$ , gives

$$\mathbb{E} \left[ \left( \lambda_\eta^A(B) - c_\epsilon \sum_{Q_i \subset B} Y_i \right)^2 \right] \longrightarrow 0, \quad (9.44)$$

with  $c_\epsilon$  determined by (9.43). The indicators  $Y_i$  are macroscopic crossing/connection events and therefore are measurable from the quad-crossing scaling limit, except on the usual null set of continuity failures. For a countable dense family of continuous test functions, (9.44) produces  $L^2$ -Cauchy measurable approximants. Their limits determine a unique finite Borel measure  $\lambda^A(\omega)$

by Riesz–Markov. Since every subsequential measure limit has the same integrals against this dense family, the whole pair  $(\omega_\eta, \lambda_\eta^A)$  converges. Finally, whether a macroscopic cluster has diameter  $> \rho$  can be decided by finitely many rational quads up to a null boundary case; summing the corresponding annular pieces and then refining the rational cover gives the stated filtered measures  $\lambda^\rho$ .

An edge traversed by the exploration interface has one open arm and one closed arm from the microscopic scale to the first macroscopic scale. Thus the normalization in (9.39) is the two-arm normalization. Interior first- and second-moment estimates are obtained exactly as in Step a with  $\alpha_2$  in place of  $\alpha_1$ .

There is one new issue: the curve starts and ends on the boundary. Let  $U_\delta$  be a boundary strip of width  $\delta$ . In a  $\delta$ -neighborhood of a marked endpoint, an interface edge must support two alternating arms to scale  $\delta$ ; using the closed  $k = 2$  case of SM-11.02.04,

$$\alpha_2(\eta, \delta) = (\eta/\delta)^{1/4+o(1)},$$

and after division by  $\alpha_2(\eta, 1)$  the expected normalized mass there is  $O(\delta^{7/4+o(1)})$ . Along a smooth boundary arc away from the two marked points, reaching an edge in the strip requires the interior two-arm event together with a half-plane one-arm crossing from the boundary scale. The half-plane estimate owned in SM-11.05.01.d has exponent  $1/3$  for this extra crossing. Summing the boundary boxes therefore gives

$$\sup_{\eta \ll \delta} \mathbb{E} \tau_\eta(U_\delta) \leq C(\delta^{7/4+o(1)} + \delta^{13/12+o(1)}) \xrightarrow[\delta \downarrow 0]{} 0. \quad (9.45)$$

Hence no boundary mass survives in a subsequential interface-measure limit.

Condition on two alternating arms crossing  $A(r, R)$ . On a fixed-ratio annulus, RSW and the usual exploration reveal two landing arcs separated by a definite fraction of the circumference with probability at least  $c > 0$ . Once separated, reveal the two interfaces from the outside until they hit those landing arcs and use the same percolation bits inward in two samples. If the landing data have not coupled, repeat on the next dyadic annulus. Thus, as in Step b,

$$\mathbb{P}(\text{two-arm coupling failure between } r \text{ and } R) \leq C(r/R)^k. \quad (9.46)$$

This is the two-arm specialization of the separation/coupling mechanism already proved for four arms in SM-11.03.03.b; no three-arm exponent enters the coupling.

The same nested-scale argument as in Step c yields a two-arm ratio limit. Its multiplicativity and the closed  $k = 2$  exponent in SM-11.02.04 force

$$\boxed{\frac{\alpha_2(s\epsilon, 1)}{\alpha_2(\epsilon, 1)} \longrightarrow s^{1/4}} \quad (\epsilon \downarrow 0). \quad (9.47)$$

For a square  $Q \Subset \Omega$ , convergence of exploration curves immediately gives one implication: if the limiting curve stays a positive distance from  $Q$ , then so do the discrete curves eventually. The only possible discontinuity is that  $\gamma_\eta$  enters an arbitrarily small neighborhood of  $Q$  while avoiding  $Q$ . Choose a boundary point of the neighborhood first approached by the curve. The two sides of the exploration supply two arms back to the explored boundary, while the avoidance of  $Q$  supplies a third arm through the thin annulus. In a half-plane or quarter-plane chart this is a three-arm event. The power-saving estimate (5.13) from SM-11.05.01.d gives probability at most  $C(\epsilon/r)^{1+\alpha}$  for such an event between scales  $\epsilon$  and  $r$ . Covering  $\partial Q$  by  $O(r/\epsilon)$  boxes and then choosing a slightly thicker annulus leaves a positive power of  $\epsilon/r$ . Consequently

$$\mathbf{1}_{\{\gamma_\eta \cap Q \neq \emptyset\}} \longrightarrow \mathbf{1}_{\{\gamma \cap Q \neq \emptyset\}} \quad \text{in probability.} \quad (9.48)$$

This is exactly the continuity input needed below, and importantly it uses only the half-plane three-arm *power bound*, not the still-open exact whole-plane three-arm exponent.

Tile  $B \Subset \Omega$  by  $\epsilon$ -squares. Let  $X_i$  be the number of interface edges in  $Q_i$ , and let  $Y_i$  indicate that the exploration curve hits a fixed enlargement of  $Q_i$ . Conditional on  $Y_i = 1$ , Step f makes the microscopic two-arm configuration asymptotically independent of the remote route of the interface, hence

$$\mathbb{E}[X_i \mid \text{mesoscopic curve data}] = \tilde{\beta}_{\eta,\epsilon} Y_i + o(\tilde{\beta}_{\eta,\epsilon}).$$

For separated boxes use two independent successful couplings. For neighboring boxes, order pairs by their separation and use the two-arm pair estimate; the exponent  $1/4 < 1$  makes the dyadic error sum finite. Together with the boundary estimate (9.45), this yields

$$\mathbb{E} \left[ \left( \tau_\eta(B) - \tilde{c}_\epsilon \sum_{Q_i \subset B} Y_i \right)^2 \right] \rightarrow 0. \quad (9.49)$$

By (9.48), every  $Y_i$  has a limit measurable from  $\gamma$ . Passing through a countable dense set of test functions exactly as in Step d produces a unique Borel measure  $\tau^{\Omega,a,b}(\gamma)$ . Thus joint convergence holds and the limiting measure is measurable from the SLE curve itself, not merely from an ambient percolation configuration.

We first record the scaling computation, because it explains the two exponents without mystery. Under a similarity of ratio  $s$ , an atom of either discrete measure gains the Euclidean area factor  $s^2$ . The microscopic normalization changes by the appropriate arm ratio. Therefore (9.43) gives for the cluster measure

$$s^2/s^{5/48} = s^{91/48},$$

whereas (9.47) gives for the interface measure

$$s^2/s^{1/4} = s^{7/4}.$$

Rotations contribute no factor.

For a general conformal  $f$ , fix a compact set on which  $f'$  is bounded away from zero and  $f''$  is bounded. If  $Q$  is a square of side  $\epsilon$  centered at  $z$ , Taylor's theorem gives

$$f(w) = f(z) + f'(z)(w - z) + O(\epsilon^2), \quad w \in Q.$$

Hence, for constants uniform on the compact set, the image of  $Q$  is squeezed between squares obtained from the tangent similarity by relative scale  $1 + O(\epsilon)$ . RSW and the one- or two-arm coupling imply that replacing these distorted boxes by the tangent-similarity boxes changes the mesoscopic approximants by  $o_\epsilon(1)$  in  $L^2$ . Summing the local similarity law over the grid and letting  $\epsilon \downarrow 0$  yields the Radon–Nikodym identities (9.40) and (9.41). For the interface, the boundedness of  $|f'|$  and continuous boundary extension are exactly what allow the boundary-strip estimate (9.45) to be transported without losing tightness. This proves the source-strength covariance statements.

The construction above records the mass of the whole SLE<sub>6</sub> trace and is measurable from the unparametrized curve. The proof does not establish the additional stopped curve identity saying that the mass accumulated on every initial segment equals the elapsed value of some canonical time parameter. Such compatibility is a separate statement. Keeping it separate prevents a stronger natural-time theorem from being silently inferred from the measure construction.  $\square$

**Theorem 9.8** (Garban–Pete–Schramm scaling limits of MST and invasion percolation).



Give the vertices of the triangular lattice independent uniform labels  $V(x) \in [0, 1]$  and label an edge  $e = (x, y)$  by

$$U(e) = (V(x) \vee V(y), V(x) \wedge V(y))$$

with the lexicographic order. Let  $\text{MST}_\eta$  be the resulting minimal spanning tree on  $\eta\mathbb{T}$ , viewed in the essential-spanning-tree topology. Then

$$\text{MST}_\eta \implies \text{MST}_\infty$$

as  $\eta \downarrow 0$ , where the scaling limit is unique and invariant under translations, rotations and Euclidean dilations.

Almost surely the limit has no pinching point (no degree type at least  $(2, 2)$ ), no point of degree at least 5, and at most countably many points of degree 4. Its trunk has Hausdorff dimension strictly smaller than  $7/4$ .

For invasion percolation started at the origin, with the same labels, the embedded invasion tree  $\text{InvPerc}_\eta$  converges to a unique limit  $\text{InvPerc}_\infty$ . This rooted limit is invariant under rotations and dilations about the origin. Neither conclusion asserts conformal invariance.

*Proof.* The decisive fact is that the macroscopic tree is determined by the near-critical cluster-merger process. The preceding chapter has already constructed the whole continuum near-critical ensemble (SM-11.07.01–02), so the work here is to prove that throwing away very subcritical clusters, very supercritical merger times and microscopic pivotals changes the embedded tree by a vanishing amount.

For a finite weighted graph with distinct weights, the path between vertices  $x, y$  in the MST is characterized by the minimax property

$$\max_{e \in \gamma_{\text{MST}}(x, y)} U(e) = \min_{\gamma: x \rightarrow y} \max_{e \in \gamma} U(e). \quad (9.50)$$

Indeed, Kruskal's algorithm joins components in increasing weight order, so  $x$  and  $y$  first become connected at precisely the right side of (9.50); uniqueness of weights identifies the path. Contracting each percolation  $p$ -cluster therefore contracts the MST to the MST of the cluster graph. This is the deterministic bridge from percolation to the tree.

For convergence we use the following explicit version of the essential-tree metric. For each  $\ell \geq 2$ , record every immersed finite subtree connecting  $\ell$  marked points in the Riemann sphere and give the collection the Hausdorff distance induced by the uniform metric on parametrized finite trees; weight the  $\ell$ th coordinate by  $2^{-\ell}$ . The resulting product metric is complete after adding degenerate trees. Two spanning trees are within  $s$  whenever all finite connection trees can be matched with corresponding vertices and paths at uniform distance  $O(s)$ . Thus it suffices to approximate paths joining finitely many macroscopic points uniformly.

Fix a torus of side  $2M$ , parameters  $\lambda < 0 < \lambda'$  and a pivotal cutoff  $\epsilon > 0$ . In  $\omega^\lambda$  retain those  $\epsilon$ -important pivotal points whose switch labels lie in  $(\lambda, \lambda')$ . By the pivotal-measure theorem SM-11.03.03 and the Poisson construction in SM-11.07.01, there are almost surely only finitely many such points in the continuum. Remove small disjoint squares about them. The remaining primal connections divide the torus into finitely many macroscopic  $\lambda$ -clusters, represented by *routers*. Join a pivotal vertex to the routers incident to its alternating sectors and record its color and switch label. This finite labelled graph is the enhanced network  $\text{EN}^{\lambda, \epsilon}$ .

To see that the network is determined continuously by the near-critical ensemble, let the auxiliary square radius  $r \downarrow 0$ . Failure of stabilization would force either six arms between  $r$  and  $\epsilon$  or a boundary/torus analogue of a three-arm degeneracy. The six-arm estimate is  $O((r/\epsilon)^{2+c})$

and the half-plane three-arm estimate is  $O((r/\epsilon)^2)$ ; after summing over the  $O(r^{-2})$  grid boxes the exceptional probability is summable along a sufficiently fast dyadic subsequence. Hence the finite adjacency, colors and labels stabilize almost surely. Under the coupling of SM-11.07.01 the discrete and continuum enhanced networks then agree, with corresponding embedded vertices tending to each other.

Construct a labelled graph on the primal routers. Routers in the same  $\lambda$ -cluster are joined by an arbitrary deterministic spanning tree and those edges receive label  $\lambda$ . A closed pivotal that switches at time  $t_i$  joins the two routers on its open sides and gives an edge labelled  $t_i$ . Apply reversed Kruskal: from every cycle delete its largest-labelled edge. The result is a finite forest  $\text{MSF}^{\lambda, \epsilon}$ . Because this is a deterministic operation on a finite labelled network with distinct non- $\lambda$  labels, agreement of enhanced networks implies agreement of the combinatorial cut-off forests. Embedding an edge by a polygonal route through the corresponding clusters gives discrete and continuum forests whose matched edges are uniformly close.

We prove the geometric input that prevents microscopic pieces from changing a macroscopic path. Let  $\lambda < 0$  and  $R \ll |\lambda|^{-4/3}$ , the characteristic length scale from SM-11.S2.01. Uniform near-critical RSW gives, in each of a bounded number of sectors of  $A(R, 2R)$ , a  $\lambda$ -open crossing with probability bounded below uniformly in the mesh. By joining these crossings, and allowing one closed site between successive open pieces, we obtain with probability  $> 99/100$  a cyclic ring of  $\lambda$ -clusters, each of diameter at least  $\delta R$ , that separates the two boundary circles. This is the local ring lemma.

Cover the torus by  $r$ -squares and place dyadic annuli around each. The failures of well-separated candidate annuli use disjoint sets of sites, so their conditional probabilities multiply up to the uniform RSW bound. If there are  $N \asymp \log(|\lambda|^{-4/3}/r)$  trials, the failure probability for one square is at most  $100^{-N}$ . Multiplying by  $O((M/r)^2)$  squares and choosing  $r$  sufficiently small proves that, with probability  $1 - o(1)$ , every point has a ring of macroscopic  $\lambda$ -clusters within any prescribed macroscopic neighborhood.

For fixed  $s > 0$ , choose  $\lambda \ll -1$ . By the finite-size relation SM-11.S2.01 and block renormalization, every  $\lambda$ -cluster has diameter  $< s$  with probability tending to one uniformly in small mesh. Now choose  $\lambda' \gg 1$ . Every  $\lambda$ -cluster of diameter at least  $r$  contains order  $r^2 \alpha_1(\eta, r)/\eta^2$  sites on the scale relevant to the cluster area measure; SM-11.S2.06 and compactness imply that such a cluster carries a positive macroscopic amount of cluster area except on an event whose probability can be made arbitrarily small. As the parameter rises from  $\lambda$  to  $\lambda'$ , pivotal switches occur at intensity comparable with the continuum pivotal measure. If two neighboring macroscopic clusters failed to merge by  $\lambda'$ , a dual closed crossing would persist across a fixed-shape annulus throughout the whole interval. Near-critical RSW and the characteristic-length identity make this probability tend to zero as  $\lambda' \rightarrow \infty$ . A finite cover therefore shows that all macroscopic  $\lambda$ -clusters lie in one giant component of the cut-off forest, while all other components have diameter  $< s$ .

This proves that the giant cut-off tree  $\text{MST}^{\lambda, \epsilon}$  is well defined with probability tending to one as  $\lambda \rightarrow -\infty$ ,  $\lambda' \rightarrow \infty$  and  $\epsilon \rightarrow 0$ .

Fix two  $\lambda$ -clusters  $C_1, C_2$  of diameter at least  $\rho$ . By (9.50), the discrete MST path between them crosses cluster boundaries in increasing minimax order. Suppose that before reaching  $C_2$  it uses a closed gate that is not an  $\epsilon$ -important pivotal, or a gate whose near-critical label exceeds  $\lambda'$ . Take the first such discrepancy. The two sides of the MST path supply two open arms from that gate, while the two separating cluster boundaries supply two closed arms to scale  $\rho$ ; failure of  $\epsilon$ -importance would force these four arms to be created by another nearby switch. As in the cutoff-stability proof SM-11.07.01.e, the first responsible pair then creates a six-arm event. Its probability, summed over dyadic scales below  $\epsilon$ , tends to zero. The event that a necessary gate has label above  $\lambda'$  implies persistence of a dual crossing well into the supercritical regime, which tends

to zero by Step e.

Thus, with probability  $1 - o(1)$ , the MST path and the cut-off path pass through the same ordered list of macroscopic clusters and the same retained pivotal gates. Inside a  $\lambda$ -cluster the two routes may differ, but the cluster diameter is at most  $s$ . Their uniform distance is therefore  $O(s)$ .

Take the global rings from Step d around every marked endpoint of a finite connection tree. Any path from an endpoint to outside its  $s$ -ball must cross its ring, so every finite MST connection can be replaced by a connection between macroscopic ring clusters, then reconstructed inside the endpoint balls at cost at most  $s$ . Step f approximates the middle pieces by the cut-off tree. The same argument in the opposite direction starts from a cut-off path and uses the surrounding rings to attach it to genuine MST vertices. Therefore, for each  $M, s, \alpha$  there are  $\lambda \ll -1$ ,  $\lambda' \gg 1$  and  $\epsilon > 0$  such that, uniformly in all small meshes,

$$\mathbb{P}\left(d_{\Omega_M}(\text{MST}_\eta, \text{MST}_\eta^{\bar{\lambda}, \epsilon}) > s\right) < \alpha. \quad (9.51)$$

Because the discrete and continuum enhanced networks agree at fixed cutoff,

$$d_{\Omega_M}(\text{MST}_\eta^{\bar{\lambda}, \epsilon}, \text{MST}_\infty^{\bar{\lambda}, \epsilon}) \xrightarrow[\eta \downarrow 0]{\mathbb{P}} 0. \quad (9.52)$$

Choose cutoffs  $(\bar{\lambda}_k, \epsilon_k)$  so that the two errors in (9.51)–(9.52) are smaller than  $2^{-k}$ . Couple every continuum cut-off tree through a common sufficiently fine discrete MST. The triangle inequality gives

$$\mathbb{P}(d_{\Omega_M}(\text{MST}_\infty^k, \text{MST}_\infty^\ell) > 2^{-k+2} \text{ for some } \ell \geq k) \leq 2^{-k+2}.$$

Borel–Cantelli and completeness of  $\Omega_M$  give an almost sure limit  $\text{MST}_\infty^M$ . The same triangle inequality, now passing through one fixed cut-off tree, shows  $\text{MST}_\eta^M \Rightarrow \text{MST}_\infty^M$  and proves that the law does not depend on the cutoff sequence. This is uniqueness, not merely subsequential tightness.

Quasi-locality follows from the global ring lemma: for every compact  $K \subseteq [-M, M]^2$ , with probability tending to one as  $M \rightarrow \infty$ , all MST paths joining points of  $K$  stay in a fixed larger box. Hence the restrictions of the torus limits to  $K$  form a Cauchy family and define a unique full-plane tree  $\text{MST}_\infty$ . Translation invariance is inherited from the discrete model and quasi-locality.

For a dilation  $f_s(z) = sz$ , conformal covariance of pivotal measure and SM-11.07.02 give

$$f_s(\omega_\infty^\lambda) \stackrel{d}{=} \omega_\infty^{s^{-3/4}\lambda}, \quad f_s(\Pi_{\lambda, \lambda'}^\epsilon) \stackrel{d}{=} \Pi_{s^{-3/4}\lambda, s^{-3/4}\lambda'}^{s\epsilon}.$$

The finite enhanced-network/Kruskal construction is equivariant under this simultaneous space–parameter scaling. Sending the cutoffs away gives  $f_s(\text{MST}_\infty) \stackrel{d}{=} \text{MST}_\infty$ . Rotational invariance is the same argument with no parameter rescaling. A general conformal map has a spatially varying  $|f'|^{-3/4}$  parameter change, so the construction does not imply conformal invariance.

The geometric endpoint below needs a genuinely strict comparison between whole-plane monochromatic and polychromatic arm probabilities. A simple “opposite-color blocker on every annulus” argument is not valid: a monochromatic configuration may reroute its arms in many macroscopically different ways. We therefore prove the precise finite-scale inequality by the winding-angle argument.

For  $j \geq 2$  write

$$M_j(r, R) = \mathbb{P}(A_{j,B \dots B}(r, R)), \quad P_j(r, R) = \mathbb{P}(A_{j,B \dots BW}(r, R)),$$

where the second color word contains both colors. Color switching and arm separation make the choice of nonconstant word irrelevant up to constants. We shall prove that there are  $C_j, \epsilon_j > 0$  such that

$$M_j(r, R) \leq C_j P_j(r, R) (r/R)^{\epsilon_j}. \quad (9.53)$$

Only  $j = 2, 5$  will be used later.

(j1) *The complement-correlation inequality.* For events  $A, B$  depending on finitely many critical percolation bits, the complement form of Reimer's inequality, the complement form of Reimer's inequality proved in Volume X says

$$\mathbb{P}_{1/2}(A \square B) \leq \mathbb{P}_{1/2}(A \cap \bar{B}), \quad (9.54)$$

where  $\bar{B}$  is obtained by interchanging black and white on every bit. This is not being imported here: Volume X proves the underlying finite-cube cardinality lemma inside SM-10.01.04, from which (9.54) follows directly.

(j2) *Monochromatic configurations have linearly many available winding angles.* For a chosen family of  $j$  disjoint arms let its winding angle be the total variation of a continuous argument along one arm; the  $j$  angles in one family differ by at most  $2\pi$ . Let  $I_j(r, R)$  be the set of all winding angles obtainable by choosing a  $j$ -arm family in the given configuration and put

$$\bar{I}_j(r, R) = \bigcup_{\theta \in I_j(r, R)} (\theta - \pi, \theta + \pi].$$

If both colors occur, two oppositely colored arms cannot cross, and therefore every realization satisfies

$$\text{diam } \bar{I}_j^{\text{poly}}(r, R) \leq 4\pi. \quad (9.55)$$

The monochromatic case is different.

A  $j$ -spiral in  $A(m, 4m)$  consists of  $j$  separated black radial crossings,  $j$  separated black circuits in each of  $A(m, 2m)$  and  $A(3m, 4m)$ , and  $j$  separated black spiralling connectors in  $A(2m, 3m)$ , each making one extra turn. RSW and FKG supply this finite collection of fixed-shape crossings with probability at least  $c_j > 0$ , uniformly in  $m$ . Choose pairwise disjoint annuli of this form between  $r$  and  $R$ . Their spiral events are independent. Hence a Chernoff bound gives constants  $c'_j, c''_j > 0$  such that, with probability at least  $1 - (r/R)^{c''_j}$ , there are at least  $c'_j \log(R/r)$  disjoint  $j$ -spirals. Since the spiral event is increasing, FKG shows that the same conclusion holds conditionally on  $A_{j,B \dots B}(r, R)$ .

Suppose  $s \geq 2$  of these spirals occur and there are  $j$  black arms across the whole annulus. Between two consecutive spirals, take any  $j$  disjoint connecting arms. There are  $j$  vertex-disjoint connectors between the outer active points of one spiral and the inner active points of the next. Indeed, if fewer than  $j$  existed, the finite vertex form of Menger's theorem would give a separating set of at most  $j - 1$  vertices; but after deleting any  $j - 1$  vertices, at least one original connector, one inner and outer circuit, and one ray in each spiral remain untouched and concatenate across the separator, a contradiction. (The finite Menger statement itself follows by splitting each vertex into an in/out pair of capacity one and applying the elementary integral augmenting-path proof of max-flow/min-cut.)

At every spiral the movable middle pieces have two choices, differing by one full turn, while the intermediate connectors stay fixed. Switching the choices successively therefore produces  $j$ -arm families whose winding angles cover, after the  $\pi$ -completion above, an interval of length at least  $2\pi(s - 2)$ . Consequently there is  $a_j > 0$  such that

$$\mathbb{P}(\bar{I}_j^{\text{mono}}(r, R) \text{ contains an interval of length } a_j \log(R/r) \mid A_{j,B \dots B}(r, R)) \longrightarrow 1 \quad (9.56)$$

uniformly as  $R/r \rightarrow \infty$ .

(j3) *The useful winding interval can be chosen deterministically.* We also need to keep the long interval from drifting arbitrarily far. Fix a narrow cone  $\mathcal{C}$  about the positive real axis and decompose

it on logarithmic scales into “long” and “wide” fixed-aspect-ratio rectangles. An arm whose winding angle is at least  $2\pi K \log(R/r) + 4\pi$  must make at least  $K \log(R/r)$  disjoint crossings of these rectangles. RSW gives a number  $\delta > 0$  such that each prescribed crossing has probability at most  $1 - \delta$ . By the BK inequality SM-10.01.03, summing over the ways to distribute  $m = jK \log(R/r)$  disjoint crossings among  $O(\log(R/r))$  rectangle types gives

$$\mathbb{P}(A_{j,B\dots B}(r, R) \text{ with such a large winding}) \quad (9.57)$$

$$\leq C \exp\{-cK + C \log K\} \log(R/r). \quad (9.58)$$

For completeness, the counting factor is at most  $\binom{(jK+2)L}{2L}$  with  $L = \lceil \log(R/r) \rceil$ , and Stirling gives the  $C \log K$  term. RSW and FKG on fixed-ratio annuli give an a priori polynomial lower bound  $M_j(r, R) \geq c(r/R)^{\beta_j}$ . Choosing  $K$  so that  $cK - C \log K > \beta_j + 2$  makes the conditional probability in (9.58) tend to zero.

Intersecting this bounded-winding event with (9.56), and partitioning the deterministic window of length  $O(\log(R/r))$  into finitely many intervals of length  $\frac{1}{2}a_j \log(R/r)$ , yields one deterministic interval  $J_j(r, R)$  and a constant  $q_j > 0$  for which

$$\mathbb{P}(J_j(r, R) \subseteq \bar{I}_j^{\text{mono}}(r, R) \mid A_{j,B\dots B}(r, R)) \geq q_j \quad (9.59)$$

for all sufficiently large  $R/r$ .

(j4) *Entropy versus the unique polychromatic angle.* The interval  $J_j(r, R)$  contains  $\asymp \log(R/r)$  points of  $4\pi\mathbb{Z}$ . Choose  $\theta_*$  among them so that

$$\mathbb{P}(A_{j,B\dots BW}(r, R), \theta_* \in \bar{I}_j^{\text{poly}}(r, R))$$

is minimal. By (9.55), a polychromatic configuration can contribute to at most one of these  $4\pi$ -spaced angles, and hence

$$\mathbb{P}(A_{j,B\dots BW}, \theta_* \in \bar{I}_j^{\text{poly}}) \leq \frac{C}{\log(R/r)} P_j(r, R). \quad (9.60)$$

On the other hand (9.59) gives

$$\mathbb{P}(A_{j,B\dots B}, \theta_* \in \bar{I}_j^{\text{mono}}) \geq q_j M_j(r, R). \quad (9.61)$$

Split one chosen monochromatic  $j$ -arm family into  $j - 1$  arms and the remaining arm; the two subfamilies have disjoint witnesses. Apply (9.54) to the event that the first  $j - 1$  arms realize the angle  $\theta_*$  and to the one-black-arm event. Flipping the witness of the last arm to white produces a  $B\dots BW$  family while preserving the first family's winding constraint. Thus (9.61)–(9.60) give

$$\frac{M_j(r, R)}{P_j(r, R)} \leq \frac{C_j}{\log(R/r)}. \quad (9.62)$$

In particular the ratio tends to zero uniformly as  $R/r \rightarrow \infty$ .

Finally choose a fixed ratio  $Q > 1$  so large that the right side of (9.62) is smaller than the inverse of twice the quasi-multiplicativity constants in SM-11.02.02. Iterating over the annuli  $A(r, Qr), A(Qr, Q^2r), \dots$  gives a factor  $2^{-m}$  after  $m \asymp \log(R/r)/\log Q$  blocks. Absorbing the two bounded-ratio end pieces by RSW proves (9.53).

For  $j = 2$ , SM-11.02.04 already owns the alternating two-arm exponent  $1/4$ , so

$$M_2(r, R) \leq C(r/R)^{1/4+\varepsilon_2}. \quad (9.63)$$

For  $j = 5$ , SM-11.02.03 owns  $P_5(r, R) \asymp (r/R)^2$ , hence

$$M_5(r, R) \leq C(r/R)^{2+\varepsilon_5}. \quad (9.64)$$

These are exactly the strict inequalities needed for the geometric endpoint.

Suppose a macroscopic limit path has a pinching point visible between scales  $r \ll \rho$ . Approximate by the discrete MST and choose  $\lambda \ll 0 < \lambda'$  so that the cutoff approximation is valid at scale  $\rho$ . The two passages through the pinching neighborhood create two  $\lambda$ -closed separating arms. The ring construction and the minimax property force, in addition, four  $\lambda'$ -open arms connecting the four outgoing branches. Thus a six-arm near-critical event occurs from  $r$  to a fixed fraction of  $\rho$ . Near-critical arm stability and the six-arm exponent strictly larger than 2 imply, after a union bound over  $O(r^{-2})$  boxes, a probability tending to zero. Hence pinching points do not occur.

If a point had degree at least five, examine its degree type. Either five branches have the same primal color through the annulus, giving a monochromatic five-arm event, or the separation of multiple path components supplies at least six polychromatic arms. The former is ruled out by (9.64), the latter by the six-arm bound, again after the  $O(r^{-2})$  spatial union bound. Therefore all degrees are at most four.

For countability, ordinary branching points of a locally finite embedded tree are countable: associate to each branch point three rational balls reached by three disjoint outgoing branches; distinct branch points with the same triple are ordered along the finite connecting tree and hence form a finite set at each positive separation scale. Taking a countable union over rational triples and scales gives countability. Degree type  $(2, 2)$  has just been excluded. The remaining degree-four types either contain an ordinary branching point or, by planar duality, correspond to an ordinary branching point in the dual limiting MST. The dual is obtained as the limit of the discrete dual maximal spanning trees and has the same law as the primal limit. Hence the set of degree-four points is at most countable.

Fix  $\rho > 0$  and let  $\text{Trunk}^\rho$  consist of points through which an MST path continues distance at least  $\rho$  in both directions. If an  $r$ -square meets  $\text{Trunk}_\eta^\rho$ , the two directions of the path force two disjoint same-color near-critical arms from scale  $r$  to a fixed fraction of  $\rho$ , after choosing the cutoffs as in Step k. Near-critical stability and (9.63) therefore give

$$\mathbb{P}(Q \cap \text{Trunk}_\eta^\rho \neq \emptyset) \leq C_\rho r^{1/4+\varepsilon_2}.$$

Among  $O(r^{-2})$  squares, the expected number meeting the trunk is at most  $C_\rho r^{-7/4+\varepsilon_2}$ . Markov's inequality on a sequence  $r_n \downarrow 0$  with summable failure probabilities and Borel–Cantelli imply

$$\dim_H \text{Trunk}_\infty^\rho \leq 7/4 - \varepsilon_2/2 < 7/4$$

(the harmless factor  $1/2$  absorbs subpower and coupling errors). The whole trunk is a countable union over rational  $\rho > 0$ , so it has the same strict upper bound.

For a finite weighted graph, invasion from a root repeatedly adds the least-labelled edge leaving the invaded set. On the enhanced network, start from the router nearest the origin and perform exactly this algorithm, stopping at a router within distance  $s$  of a prescribed target (the torus boundary, or another marked point). The global ring lemma ensures that such target routers exist and that all omitted components have diameter  $< s$ . The same discrepancy argument as Step f shows that, with probability  $1 - o(1)$ , the true invasion path and the network invasion path use the same retained pivotal gates; excursions through omitted  $\lambda$ -clusters cost at most  $s$  in the tree metric. Thus both the discrete-to-continuum fixed-cutoff convergence and the cutoff-removal estimate hold verbatim for the invasion tree.

A diagonal Cauchy argument as in Step h gives unique limits on tori and then, by quasi-locality, in the plane. Because the root is fixed at the origin, translations are not symmetries of the rooted law; rotations and dilations are, by Step i.

Steps a–i give uniqueness and the Euclidean invariances of the MST limit; Steps j–l give the stated geometric regularity; Step m gives the unique invasion-percolation limit and its rooted symmetries. The proof used the conformal covariance of the near-critical ensemble only to derive similarity covariance. Under a non-similarity conformal map, the near-critical parameter is rescaled by the nonconstant field  $|f'|^{-3/4}$ , so no conformal-invariance conclusion is justified or claimed.  $\square$

**Theorem 9.9** (Hammond–Pete–Schramm exceptional-time local time and the IIC).

*Consider stationary dynamical critical percolation, either site percolation on the triangular lattice or bond percolation on  $\mathbb{Z}^2$ , and let*

$$\mathcal{E} = \{t \in \mathbb{R} : 0 \longleftrightarrow \infty \text{ in } \omega_t\}$$

*be the exceptional-time set. Write  $\nu_{\text{IIC}}$  for the IIC measure of Theorem 9.3. If  $\mathcal{F}_r$  is the sigma-field generated by the bits in  $B_r$ , define the finite-radius IIC density*

$$M_r(\omega) := \frac{d(\nu_{\text{IIC}}|_{\mathcal{F}_r})}{d(\mathbb{P}_{p_c}|_{\mathcal{F}_r})}(\omega) = \lim_{R \rightarrow \infty} \frac{\mathbb{P}_{p_c}(0 \leftrightarrow R \mid \mathcal{F}_r)(\omega)}{\mathbb{P}_{p_c}(0 \leftrightarrow R)} \quad (9.65)$$

*and, for a bounded interval  $I$ , put*

$$\mu_r(I) = \int_I M_r(\omega_t) dt. \quad (9.66)$$

*Then the following hold.*

1. *There is a stationary, non-atomic random Borel measure  $\mu$  on  $\mathbb{R}$  such that, simultaneously for all real  $a < b$ ,*

$$\mu_r([a, b]) \longrightarrow \mu([a, b]) \quad \text{almost surely and in } L^2.$$

*Moreover  $\mathbb{E}\mu([a, b]) = b - a$  and  $\text{supp } \mu \subseteq \mathcal{E}$  almost surely.*

2. *For comparison, if the more naive occupation measures*

$$\bar{\mu}_r(I) = \frac{1}{\alpha_1(r)} \int_I \mathbf{1}_{\{0 \leftrightarrow r\}}(\omega_t) dt$$

*happen to converge in  $L^2$  on every bounded interval, then their limit equals  $\mu$ . No convergence of  $\bar{\mu}_r$  is asserted here.*

3. *For almost every dynamical-percolation trajectory there is a finite random  $T_0$  such that  $\mu[0, T] > 0$  for all  $T > T_0$ . If, conditional on the trajectory,  $\chi_T$  is sampled from  $[0, T]$  with law  $\mu(dt)/\mu[0, T]$ , then*

$$\omega_{\chi_T} \Longrightarrow \nu_{\text{IIC}} \quad (T \rightarrow \infty).$$

4. *For every fixed  $T > 0$ , size-bias the path law by  $\mu[0, T]$  and then sample  $\chi_T^*$  from the normalized local time on  $[0, T]$ . The one-time law of  $\omega_{\chi_T^*}^*$  is exactly  $\nu_{\text{IIC}}$ . Equivalently, under the Palm law of the stationary random measure  $\mu$ , the configuration at the distinguished local-time point has law IIC.*

*Thus “a typical exceptional time” means typical with respect to  $\mu$ , not the first exceptional time and not a point chosen according to Euclidean length.*

*Proof.* For a finite configuration  $\zeta$  on  $B_r$ , the identity

$$\frac{\mathbb{P}_{p_c}(0 \leftrightarrow R \mid \omega|_{B_r} = \zeta)}{\mathbb{P}_{p_c}(0 \leftrightarrow R)} = \frac{\mathbb{P}_{p_c}(\omega|_{B_r} = \zeta \mid 0 \leftrightarrow R)}{\mathbb{P}_{p_c}(\omega|_{B_r} = \zeta)}$$

shows, by Theorem 9.3, that the limit in (9.65) exists and is the Radon–Nikodym derivative of the *IIC* restriction. In particular

$$\mathbb{E}_{p_c}[F(\omega)M_r(\omega)] = \mathbb{E}_{\text{IIC}}[F] \quad (9.67)$$

for every bounded  $\mathcal{F}_r$ -measurable  $F$ , and  $\mathbb{E}M_r = 1$ . If  $R > r$ , the tower property for Radon–Nikodym derivatives gives

$$\mathbb{E}[M_R \mid \mathcal{F}_r] = M_r. \quad (9.68)$$

This is the structural reason for using  $M_r$  rather than only the indicator  $\mathbf{1}_{\{0 \leftrightarrow r\}}$ .

Put  $\bar{M}_r = \mathbf{1}_{\{0 \leftrightarrow r\}}/\alpha_1(r)$ . If  $0 \not\leftrightarrow r$ , then  $M_r = 0$ . On  $\{0 \leftrightarrow r\}$ , arm separation and quasi-multiplicativity give, uniformly for  $R \geq 4r$ ,

$$\mathbb{P}(0 \leftrightarrow R \mid \mathcal{F}_r) \leq C \mathbf{1}_{\{0 \leftrightarrow r\}} \alpha_1(r, R), \quad \alpha_1(R) \asymp \alpha_1(r) \alpha_1(r, R).$$

Letting  $R \rightarrow \infty$  yields the pointwise comparison

$$0 \leq M_r \leq C \bar{M}_r, \quad 0 \leq \mu_r(I) \leq C \bar{\mu}_r(I). \quad (9.69)$$

The same proof works for triangular site and square bond percolation because it uses only RSW, arm separation and quasi-multiplicativity, all already owned in this volume.

Let  $\mathcal{F}_r(I)$  be the sigma-field generated by the complete dynamical trajectory of the bits in  $B_r$  during  $I$ . Exterior Poisson clocks and states are independent of this trajectory. Hence, for each fixed  $t \in I$ ,

$$\mathbb{E}[M_R(\omega_t) \mid \mathcal{F}_r(I)] = M_r(\omega_t).$$

Conditional Fubini and (9.68) therefore give

$$\mathbb{E}[\mu_R(I) \mid \mathcal{F}_r(I)] = \mu_r(I). \quad (9.70)$$

Thus  $r \mapsto \mu_r(I)$  is a nonnegative martingale for every bounded interval  $I$ .

The occupation measures  $\bar{\mu}_r$  are exactly the normalized radial occupation measures used in the proof of Theorem 9.5. Equations (9.11)–(9.23), together with the short-time estimate there, imply that for every bounded interval  $I$

$$\sup_r \mathbb{E}[\bar{\mu}_r(I)^2] < \infty. \quad (9.71)$$

More locally, there is a model-dependent  $\delta > 0$  such that, for  $|I| \leq 1$ ,

$$\sup_r \mathbb{E}[\bar{\mu}_r(I)^2] \leq C|I|^{1+\delta}. \quad (9.72)$$

For the triangular lattice one may take any  $\delta < 31/36$ , since the normalized two-time kernel is  $O(|t-s|^{-5/36-o(1)})$ ; for square bond percolation the strict arm-product gap proved in SM-11.S2.04.g-h gives some  $\delta > 0$ . By (9.69), the same two bounds hold for  $\mu_r$ . The  $L^2$  martingale convergence theorem applied to (9.70) now yields, for each fixed bounded interval,

$$\mu_r(I) \longrightarrow \mu(I) \quad \text{a.s. and in } L^2. \quad (9.73)$$

Since  $\mathbb{E}\mu_r(I) = |I|$ , also  $\mathbb{E}\mu(I) = |I|$ .



Take the almost-sure event on which (9.73) holds for every interval with rational endpoints. Finite additivity and monotonicity pass to the limit. To prove continuity, cover  $[0, 1]$  by  $O(n)$  intervals of length  $2/n$ . Chebyshev and (9.72) give

$$\mathbb{P}\left(\max_j \mu_r(I_{j,n}) > c\right) \leq Cc^{-2}n(2/n)^{1+\delta} \leq C'c^{-2}n^{-\delta}.$$

For fixed  $n$ , pass  $r \rightarrow \infty$  on the finitely many rational intervals to obtain the same bound with  $\mu$  in place of  $\mu_r$ . Now choose the dyadic subsequence  $n = 2^k$  and apply Borel–Cantelli; finally let  $c$  run through positive rationals. The limiting distribution function has no jumps. It therefore extends uniquely to a continuous increasing function on every compact time interval, hence to a non-atomic Borel measure  $\mu$  on  $\mathbb{R}$ . Monotonicity between rational endpoints upgrades (9.73) to simultaneous almost-sure convergence for all real  $a < b$ . Stationarity is inherited from the stationary dynamical process and the covariant definition (9.66).

We also record the elementary dynamical fact needed for the support statement. Almost surely no update time is exceptional: there are only countably many Poisson update events, and conditional on any specified clock ring the pre-update and post-update configurations have the critical product law away from the singled-out resampling, under which the origin has no infinite cluster. Moreover  $\mathcal{E}$  is closed. Indeed, if exceptional times  $t_n \rightarrow t$ , then on each fixed ball the trajectory has only finitely many jumps. If  $t$  is not a jump in that ball, the connection to its boundary passes to  $t$ ; if  $t$  is a jump, take a one-sided subsequence and pass to the corresponding pre- or post-update configuration. A diagonal choice of growing balls would then give an infinite origin cluster at an update-time side, an almost-surely excluded event. Hence the second case cannot occur, while the first for every ball gives  $t \in \mathcal{E}$ .

Now let  $J$  be a compact interval with  $J \cap \mathcal{E} = \emptyset$ . If, for arbitrarily large  $r$ , some  $t_r \in J$  satisfied  $0 \leftrightarrow r$ , take a convergent subsequence. Repeating the preceding fixed-ball/one-sided argument would produce either an exceptional limit time or an infinite cluster at the side of an update time, both impossible. Thus for all large  $r$ ,  $\bar{\mu}_r(J) = 0$ , whence (9.69) and interval convergence give  $\mu(J) = 0$ . A countable basis of compact rational intervals in the open set  $\mathcal{E}^c$  proves  $\text{supp } \mu \subseteq \mathcal{E}$ .

Suppose, in addition, that  $\bar{\mu}_R(I)$  converges in  $L^2$  for every bounded  $I$ . For fixed  $r$  and  $R \rightarrow \infty$ ,

$$\mathbb{E}[\bar{M}_R \mid \mathcal{F}_r] = \frac{\mathbb{P}(0 \leftrightarrow R \mid \mathcal{F}_r)}{\alpha_1(R)} \longrightarrow M_r$$

uniformly on the finite configuration space of  $B_r$ . Integrating in time and then taking conditional expectation with respect to  $\mathcal{F}_r(I)$  shows that the  $L^2$  limit of  $\bar{\mu}_R(I)$  has conditional expectation  $\mu_r(I)$ . Letting  $r \rightarrow \infty$  and using the martingale convergence theorem identifies that hypothetical limit with  $\mu(I)$ . This is exactly the conditional comparison stated above; convergence of  $\bar{\mu}_r$  itself is not inserted as an unproved claim.

Let  $F$  be a bounded cylinder function depending only on  $B_m$ . For  $r \geq m$ , stationarity and (9.67) give

$$\mathbb{E} \int_0^1 F(\omega_t) M_r(\omega_t) dt = \mathbb{E}_{\text{IIC}} F. \quad (9.74)$$

The paths  $t \mapsto F(\omega_t)$  have only finitely many jumps on  $[0, 1]$ , and  $\mu$  has no atoms. Convergence of the cumulative functions of  $\mu_r$  to that of  $\mu$  therefore implies convergence of the corresponding Riemann–Stieltjes integrals. Uniform  $L^2$  bounds give uniform integrability, so (9.74) passes to the limit:

$$\mathbb{E} \int_0^1 F(\omega_t) \mu(dt) = \mathbb{E}_{\text{IIC}} F. \quad (9.75)$$

In particular the intensity of  $\mu$  is one.

We prove the small piece of Palm theory needed here. If  $F$  is bounded and local, set

$$X_k = \mu([k, k+1)), \quad Y_k = \int_{[k, k+1)} F(\omega_t) \mu(dt).$$

The bi-infinite sequence  $(X_k, Y_k)$  is a stationary ergodic factor of dynamical percolation. Ergodicity follows directly by approximating a shift-invariant event by an event depending on finitely many bits in a finite time window and separating two copies by a large time shift, when the finite Poisson-refresh chain is mixing. Birkhoff's theorem and (9.75) therefore give, almost surely,

$$\frac{1}{n} \sum_{k=0}^{n-1} X_k \longrightarrow 1, \quad \frac{1}{n} \sum_{k=0}^{n-1} Y_k \longrightarrow \mathbb{E}_{\text{IIC}} F.$$

The remainder from replacing  $T$  by  $\lfloor T \rfloor$  is  $o(T)$  almost surely: for a stationary nonnegative integrable sequence,  $X_n/n \rightarrow 0$  by the tail-sum criterion and Borel–Cantelli, and  $|Y_n| \leq \|F\|_\infty X_n$ . Hence

$$\frac{\int_0^T F(\omega_t) \mu(dt)}{\mu[0, T]} \longrightarrow \mathbb{E}_{\text{IIC}} F \quad \text{a.s.} \quad (9.76)$$

and also  $\mu[0, T]/T \rightarrow 1$ . The latter implies  $\mu[0, T] > 0$  for every sufficiently large  $T$ .

Condition on a trajectory for which (9.76) holds for a countable convergence-determining family of cylinder functions. If  $\chi_T$  has conditional law  $\mu(dt)/\mu[0, T]$ , then

$$\mathbb{E}[F(\omega_{\chi_T}) \mid \omega] = \frac{\int_0^T F(\omega_t) \mu(dt)}{\mu[0, T]} \longrightarrow \mathbb{E}_{\text{IIC}} F.$$

Cylinder functions determine weak convergence on the compact product configuration space, so  $\omega_{\chi_T} \Rightarrow \nu_{\text{IIC}}$  for almost every trajectory. This proves the quenched assertion.

Because  $\mathbb{E}\mu[0, T] = T$ , the path law size-biased by  $\mu[0, T]$  has density  $\mu[0, T]/T$ . After sampling  $\chi_T^*$  from normalized local time, the two factors cancel. For every bounded cylinder  $F$ ,

$$\mathbb{E}^* F(\omega_{\chi_T^*}^*) = \frac{1}{T} \mathbb{E} \int_0^T F(\omega_t) \mu(dt) = \mathbb{E}_{\text{IIC}} F,$$

where the last equality follows from stationarity and (9.75). Hence the sampled configuration is exactly IIC for every  $T > 0$ .

For completeness, define the Palm law of  $(\omega, \mu)$  by the Campbell formula

$$\mathbb{E}^{\text{Palm}} G := \mathbb{E} \int_0^1 G(\theta_t \omega, \theta_t \mu) \mu(dt),$$

which is already normalized because the intensity is one. Taking  $G(\omega, \mu) = F(\omega_0)$  and using (9.75) gives the IIC law at the distinguished time. Equivalently, if conditionally on  $\mu$  one places a Poisson point process of intensity  $\mu$ , the Campbell formula is precisely the zero-point Palm conditioning; thus the source's Poisson/Palm formulation is recovered without an external Palm theorem.

Parts a–j construct the canonical HPS local time, prove its  $L^2$  and almost-sure interval convergence, diffuseness, support on exceptional times, quenched local-time sampling, exact annealed sampling, and the Palm/IIC identity. The argument deliberately does *not* assert the conjectural  $L^2$  convergence of the simpler occupation measures  $\bar{\mu}_r$ , nor that  $\text{supp } \mu = \mathcal{E}$ , and it does not identify the first exceptional-time configuration with IIC.  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 9.10** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 9.11** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 9.12** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 9.13** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 9.14** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XII: Critical Planar Ising, FK-Ising, and Discrete Complex  
Analysis

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# Scope and conventions

This volume develops the discrete-complex-analytic approach to critical planar Ising and FK–Ising models. It begins with fermionic observables and s-holomorphicity, then proceeds to spin correlations, magnetization fields, interfaces, crossing theory, loop ensembles, double currents, multiply connected domains, and s-embeddings.

A minimal Loewner dictionary is introduced in Chapter 5 solely to make the lattice-interface theorems locally meaningful. Complete deterministic Loewner theory begins in Volume XIII and stochastic SLE theory in Volume XIV.

# Chapter 1

## Discrete fermions and s-holomorphicity

The critical planar Ising model becomes dramatically more rigid once spin variables are fused with disorder variables. The resulting fermion remembers two kinds of information at once: a *topological* sign/monodromy, coming from the disorder line, and a *geometric* half-angle phase, coming from the winding of the defect path. At the critical isoradial weights these two structures fit a local projection identity. That identity is s-holomorphicity.

### 1.1 Why introduce disorder variables?

A spin correlation is local but does not by itself satisfy a useful first-order discrete equation. A disorder insertion changes the sign of the couplings crossed by a dual path. The first important observation is that the path is a gauge choice: only its endpoints matter. Fusing a spin with a nearby disorder insertion then produces an object that changes sign when transported once around a branch point. It is therefore naturally a spinor, not an ordinary single-valued function.

**Definition 1.1** (Disorder line and disorder insertion). Let  $G = (V, E)$  be a finite planar graph with Ising couplings  $J_e$ . Given dual faces  $f_1, f_2$  and a dual path  $\lambda$  joining them, replace  $J_e$  by  $-J_e$  for every primal edge crossed by  $\lambda$ . The ratio of the modified to unmodified partition functions is the two-disorder correlation associated with  $f_1, f_2$ .

**Example 1.2** (Path independence is gauge invariance). If two dual paths with the same endpoints differ by the boundary of a set  $A \subset V$ , flip all spins in  $A$ . Every edge crossing the boundary of  $A$  changes the sign of  $\sigma_u \sigma_v$ , exactly compensating the extra sign change in the coupling. Thus the two modified partition functions coincide. This elementary spin-flip is the mechanism behind the topological character of the disorder operator.

**Definition 1.3** (s-holomorphicity on an isoradial graph). Let rhombus centres  $z_0, z_1$  share a primal-dual edge  $\xi = [uw]$ . Put

$$\ell(\xi) = [i(w - u)]^{-1/2} \mathbb{R}.$$

A complex-valued function  $F$  on rhombus centres is *s-holomorphic* if for every such neighboring pair

$$\text{Proj}_{\ell(\xi)} F(z_0) = \text{Proj}_{\ell(\xi)} F(z_1).$$

Here  $\text{Proj}_{e^{i\vartheta} \mathbb{R}} z = e^{i\vartheta} \text{Re}(e^{-i\vartheta} z)$  is orthogonal projection in  $\mathbb{C} \simeq \mathbb{R}^2$ . Thus s-holomorphicity is a first-order local projection equation, not ordinary discrete holomorphicity of  $F$ .

*Worked example 1.4* (One local s-holomorphicity check on the square lattice). Take a horizontal primal–dual edge with  $w - u = \delta > 0$ . Then  $\ell(\xi) = e^{-i\pi/4}\mathbb{R}$ . Suppose two adjacent rhombus centres carry

$$F(z_0) = 1, \quad F(z_1) = -i.$$

Since

$$\operatorname{Re}(e^{i\pi/4}F(z_0)) = \frac{1}{\sqrt{2}}, \quad \operatorname{Re}(e^{i\pi/4}F(z_1)) = \frac{1}{\sqrt{2}},$$

their projections onto  $e^{-i\pi/4}\mathbb{R}$  agree. In contrast,  $F(z_1) = i$  would give projection scalar  $-1/\sqrt{2}$  and fail the local equation. The calculation shows concretely that the condition constrains one real component across each rhombus edge.

**Theorem 1.5** (Kadanoff–Ceva order–disorder switching identity).

Let  $G$  be a finite simply connected planar graph with Ising edge weights  $x_e = \tanh(\beta J_e)$ . Choose spin vertices  $u_1, \dots, u_m$  and disorder faces  $f_1, \dots, f_{2n}$ , and join the disorder faces in pairs by dual paths  $\lambda$ . Reversing the sign of the coupling on every primal edge crossed by  $\lambda$  defines the disorder insertion. Then:

1. the resulting order–disorder correlation is independent of the chosen dual paths up to the canonical spinorial sign determined by winding around the marked insertions;
2. in the high-temperature expansion it is the signed contour sum in which the odd-degree vertices are precisely the inserted spin endpoints and the crossed disorder edges contribute the path sign;
3. after fusing a spin  $u$  with an adjacent disorder face  $f$  and multiplying by the half-winding phase  $e^{-\frac{i}{2}\text{wind}}$ , the resulting corner observable is a spinor and obeys the local switching relation obtained by toggling one primal edge.

This local switching relation is the algebraic input used in the s-holomorphicity theorem below.

*Proof.*

$$e^{\beta J_e \sigma_v \sigma_w} = \cosh(\beta J_e) (1 + x_e \sigma_v \sigma_w).$$

After multiplying by the inserted spins and summing over  $\sigma \in \{\pm 1\}^V$ , a monomial survives exactly when every unmarked vertex has even incident degree and each inserted spin vertex has odd incident degree. Thus an ordinary spin correlation is a weighted sum over subgraphs with the prescribed mod–2 boundary.

$$(-1)^{|\omega \cap \lambda|}.$$

For several disorder paths the sign is the product of the corresponding intersection parities. This is the signed high-temperature representation of the order–disorder correlation.

$$\exp\left(-\frac{i}{2}\text{wind}(\gamma)\right).$$

Changing the path by a local turn through angle  $\vartheta$  multiplies this phase by  $e^{-i\vartheta/2}$ . A full  $2\pi$  turn contributes  $-1$ , exactly matching the topological monodromy from (d); hence the phase-normalized fermion is well defined on the spinor cover. □

**Theorem 1.6** (Critical Ising fermion is s-holomorphic).

Let  $\Gamma^\delta$  be an isoradial planar graph with bounded rhombus angles and critical Ising weights

$$x_e = \tan \frac{\theta_e}{2}.$$

For the phase-normalized Kadanoff–Ceva fermionic observable  $F$  on rhombus centres, if neighboring rhombi  $z_0, z_1$  share the primal–dual edge  $\xi = [uw]$ , then

$$\text{Proj}_{\ell(\xi)} F(z_0) = \text{Proj}_{\ell(\xi)} F(z_1), \quad \ell(\xi) = [i(w - u)]^{-1/2} \mathbb{R}.$$

Thus  $F$  is s-holomorphic.

*Proof.*

$$\text{Re} \left( F(z_0) \sqrt{i(w - u)} \right) = \text{Re} \left( F(z_1) \sqrt{i(w - u)} \right).$$

This is exactly equality of orthogonal projections onto  $\ell(\xi)$ .

$$\frac{1 - ix_e}{1 + ix_e} = e^{-i\theta}, \quad x_e = \tan(\theta/2).$$

The change in Boltzmann weight under the edge toggle is therefore the same rotation as the change in the spinorial half-winding phase. The component parallel to  $[i(w - u)]^{-1/2}$  is invariant.  $\square$

## 1.2 Squaring the fermion: the primitive $H$

The square  $F^2$  is generally not discrete holomorphic. The key “miracle” is weaker but exactly sufficient: the imaginary part of  $F^2 dz$  is closed. Thus one can integrate it to a real-valued primitive  $H$ . The price is that  $H$  lives simultaneously on the primal and dual vertices and is subharmonic on one color and superharmonic on the other.

For a rhombus centre  $z$  and an adjacent primal–dual pair  $u \sim w$ , set

$$H(u) - H(w) = 2\delta \left| \text{Proj}_{\ell([uw])} F(z) \right|^2,$$

with the global sign convention chosen so that the primal restriction is subharmonic. Around one rhombus the four increments cancel by s-holomorphicity, so  $H$  is path independent in a simply connected domain.

**Example 1.7** (Continuous model). If  $f$  is holomorphic, then  $f^2 dz$  is holomorphic and hence closed. Therefore

$$h(z) = \text{Im} \int^z f(\zeta)^2 d\zeta$$

is harmonic and  $f^2 = 2i \partial h$ . The discrete  $H$  is engineered so that this identity survives the scaling limit even though  $F^2$  itself does not satisfy a full discrete Cauchy–Riemann equation.

**Theorem 1.8** (Primitive of an s-holomorphic observable).

Let  $F$  be s-holomorphic on a simply connected isoradial domain. There is a real function  $H$ , defined on primal and dual vertices up to one additive constant, whose increment across a primal–dual half-diagonal  $[u, w]$  of the rhombus centred at  $z$  is

$$H(u) - H(w) = 2\delta \left| \text{Proj}_{[i(w-u)]^{-1/2} \mathbb{R}} F(z) \right|^2$$

with the fixed orientation convention. The primal restriction of  $H$  is subharmonic and the dual restriction is superharmonic. If  $F$  satisfies the standard Riemann–Hilbert boundary condition  $F(\zeta) \parallel \tau(\zeta)^{-1/2}$ , then  $H$  is constant on each corresponding boundary component/arc after the standard boundary modification. These facts give uniqueness, after normalization, of the discrete boundary-value problem used in the convergence theory.

*Proof.* We separate the four assertions, because each is used later.

*Step 1: closedness of the increment form.* Fix one rhombus. The two lines occurring in the s-holomorphic projection condition along opposite half-diagonals are orthogonal. Hence, for every  $z \in \mathbb{C}$ ,

$$|\text{Proj}_\ell z|^2 + |\text{Proj}_{i\ell} z|^2 = |z|^2.$$

When the four prescribed increments of  $H$  are added around the boundary of the rhombus, the two terms associated with one orthogonal pair occur with the opposite sign from the two terms associated with the other pair. The displayed Pythagorean identity therefore makes the sum zero. Thus the increment one-form is closed. On a simply connected quad-domain it is exact, which defines  $H$  up to one additive constant.

*Step 2: discrete primitive identity.* Let  $u_0, u_1$  be neighboring primal vertices separated by the rhombus centre  $z$ . Writing the two half-diagonal increments and using the common projection of the adjacent s-holomorphic values gives

$$H(u_1) - H(u_0) = \text{Im}((u_1 - u_0)F(z)^2).$$

The dual formula has the opposite orientation. This is the precise discrete counterpart of  $dH = \text{Im}(F^2 dz)$ .

*Step 3: sub- and superharmonicity.* At a primal vertex  $u$ , enumerate the incident rhombi cyclically and write the s-holomorphic propagation equations in the corresponding projection coordinates. When the weighted increments  $H(u_j) - H(u)$  are summed with the critical isoradial conductances, the mixed terms cancel pairwise and the remaining terms are nonnegative squares of the successive projection differences. Consequently

$$(\Delta^\delta H)(u) \geq 0.$$

The same calculation on the dual graph has the reversed orientation and gives  $\Delta^\delta H \leq 0$ . The bounded-angle hypothesis makes all conductances positive, so the usual discrete maximum principle applies to these two restrictions.

*Step 4: boundary values and uniqueness.* On a boundary arc on which  $F(\zeta) \parallel \tau(\zeta)^{-1/2}$ , the tangential increment in the preceding primitive identity has zero imaginary part. After the standard boundary modification this says that the appropriate primal or dual restriction of  $H$  is constant along that arc. Now let  $F_1, F_2$  solve the same linear Riemann–Hilbert problem with the same normalization and put  $F = F_1 - F_2$ . Its boundary normalization is homogeneous. Normalize the primitive of  $F$  to vanish on one boundary arc. The boundary condition makes it vanish on the complementary modified arc as well; subharmonicity on the primal graph and superharmonicity on the dual graph yield

$$0 \geq H_\bullet \geq H_\circ \geq 0.$$

Hence  $H \equiv 0$ . Every primal–dual increment is therefore zero, so every projection of  $F$  on every allowed half-diagonal line is zero. Two adjacent nonparallel projection lines determine a complex number, whence  $F \equiv 0$ . Thus the normalized discrete boundary-value problem has at most one solution.  $\square$

### 1.3 What has actually been proved in this chapter?

The chapter has not merely declared a fermionic observable. It has isolated four logically distinct mechanisms: gauge invariance of disorder lines, contour/defect-path representation, the critical local switching identity, and the primitive/maximal-principle theory. The later convergence theorem will use only these mechanisms plus the isoradial potential theory already owned in Volumes IV–V.

## Chapter 2

# Convergence and universality of fermionic observables

The discrete identities of Chapter 1 do not by themselves imply conformal invariance. The analytic problem is compactness plus identification: obtain uniform estimates independent of the microscopic isoradial graph, pass to subsequential limits, identify the primitive by a continuum boundary-value problem, and then recover the fermion as a square root of a complex derivative.

### 2.1 The convergence regime

We consider isoradial domains  $(\Omega^\delta; a^\delta, b^\delta)$  of mesh  $\delta \downarrow 0$  satisfying a common bounded-angle parameter  $\eta > 0$  and converging to a marked simply connected domain  $(\Omega; a, b)$  in the Carathéodory sense. No smoothness of the limiting boundary is imposed for the FK observable. For the spin observable we preserve the additional source hypothesis used to control the target normalization: a fixed macroscopic straight/rectangular neighborhood near the target boundary point. This hypothesis will be removed only when a later theorem genuinely supplies that stronger statement.

*Remark 2.1* (Why the primitive is the right compactness variable). Direct estimates on  $F^\delta$  are awkward near the boundary. The primitive  $H^\delta$  is ordered between a subharmonic primal function and a superharmonic dual function, so maximum principles and harmonic-measure estimates become available. Compactness for  $H^\delta$  then returns compactness for  $F^\delta$  through the local relation between increments and projections.

### 2.2 Two quantitative estimates used by the convergence proof

The compactness argument below uses two estimates which are special to the  $s$ -holomorphic primitive and therefore should not be hidden under the words “standard discrete Harnack”. We prove them here in the bounded-angle isoradial setting. Constants in this section depend only on the common angle bound  $\eta$ .

**Lemma 2.2** (Local primal–dual comparability for the primitive). *Let  $F$  be  $s$ -holomorphic in a rhombic neighbourhood of an interior dual vertex  $w$ , let  $u_1, \dots, u_m$  be the surrounding primal vertices, and let  $w_1, \dots, w_m$  be the neighbouring dual vertices in cyclic order. If  $H$  is the primitive of  $F$ , then*

$$\max_j H(u_j) - H(w) \leq C_\eta (H(w) - \min_j H(w_j)). \quad (12.1)$$



The colour-swapped estimate holds at an interior primal vertex. Consequently, if the dual restriction is nonnegative in the neighbourhood, then for every neighbouring primal–dual pair  $u \sim w$ ,

$$H(w) \leq H(u) \leq C_\eta H(w). \quad (12.2)$$

*Proof.* Subtract the constant  $\min_j H(w_j)$ , so that this minimum is zero. By superharmonicity at  $w$  and uniform ellipticity of the bounded-angle isoradial Laplacian, the values  $H(w_j)$  are bounded above by  $C_\eta H(w)$ . The primitive increment formula implies that all neighbouring primal values are nonnegative. Subharmonicity at the  $u_j$  and the same uniform ellipticity give  $H(u_j) \asymp_\eta M := \max_k H(u_k)$ .

Assume for contradiction that  $K = M/H(w)$  is arbitrarily large. Choose a primal neighbour  $u$  of  $w$  with  $H(u) \asymp M$  and go once around  $w$ . At the centre  $z_j$  of the  $j$ th incident rhombus, the two increments of  $H$  are the squares of the two projections of  $F(z_j)$  on the corresponding half-diagonal lines. Since one increment is of order  $M$  while the variation of the dual values is only  $O(H(w))$ , the ratio of those two squared projections is  $1 + O_\eta(K^{-1})$ . Therefore the argument of  $F(z_j)$  must be within  $O_\eta(K^{-1})$  of one of the two bisector directions determined by  $w$  and  $w_j$ , and

$$\delta |F(z_j)|^2 \asymp_\eta M. \quad (12.3)$$

Consecutive bisector directions turn by the rhombus angle. Because all angles lie in  $[\eta, \pi - \eta]$ , two consecutive allowed directions are separated by at least a constant depending only on  $\eta$ . Hence the local s-holomorphic propagation relation forces

$$\delta \sum_j |F(z_{j+1}) - F(z_j)|^2 \geq c_\eta M. \quad (12.4)$$

On the other hand the exact Laplacian identity obtained by summing the primitive increments around  $w$  expresses the left side, up to bounded-angle constants, as  $\delta^2 |\Delta^\delta H(w)|$ . Superharmonicity and uniform ellipticity bound this by  $C_\eta H(w)$ . Thus  $M \leq C_\eta H(w)$ , contradicting  $K \rightarrow \infty$ . This proves (12.1). Interchanging primal and dual colours and multiplying  $F$  by  $i$  gives the companion estimate. When the dual values are nonnegative, the primitive increment gives  $H(u) \geq H(w)$  and (12.1) gives the reverse bound in (12.2).  $\square$

**Lemma 2.3** (A priori bulk Harnack lemma for an s-holomorphic primitive). *Let  $F$  be s-holomorphic in an isoradial ball  $B_\diamond^\delta(v_0, R)$  and let  $H$  be its primitive. Suppose  $H \geq 0$  on the primal and dual vertices of that ball. Then*

$$H(v) \leq C_\eta H(v_0), \quad v \in B_\Lambda^\delta(v_0, R/2). \quad (12.5)$$

*Proof.* By Lemma 2.2 it suffices to compare one colour at the centre with the other colour in the inner ball. Let, say,  $v_0$  be dual and let

$$M = \max_{u \in B_\Gamma^\delta(v_0, R/2)} H(u).$$

Subharmonicity produces a nearest-neighbour primal path  $K$  beginning at a maximizer and running to the outer boundary of  $B_\Gamma^\delta(v_0, R)$  along which  $H \geq M$ . Shift it by one rhombus to a dual crosscut  $K^*$  joining the middle annulus to the outer boundary. Lemma 2.2 gives  $H \geq c_\eta M$  on  $K^*$ .

Let  $X$  be the reversible random walk associated with the critical isoradial Laplacian. Bounded rhombus angles make its conductances uniformly elliptic and its step lengths comparable with  $\delta$ . A fixed-modulus annulus can be covered by a bounded number (depending only on  $\eta$ ) of rhombic

boxes of bounded aspect ratio. Successive gambler's-ruin estimates in those boxes, or equivalently the planar Beurling/corridor estimate already proved in Volume IV together with uniform ellipticity, give

$$\mathbb{P}_{v_0}(\tau_{K^*} < \tau_{\partial B^\delta(v_0, R)}) \geq c_\eta. \quad (12.6)$$

Since the dual restriction of  $H$  is superharmonic, optional stopping at  $K^* \cup \partial B^\delta(v_0, R)$  yields

$$H(v_0) \geq c_\eta M \mathbb{P}_{v_0}(\tau_{K^*} < \tau_{\partial B^\delta(v_0, R)}) \geq c'_\eta M.$$

Thus  $M \leq C_\eta H(v_0)$ , and local primal–dual comparability finishes the proof for every  $v$  in the half ball.  $\square$

**Lemma 2.4** (Straight-boundary Harnack comparison). *Let  $R^\delta(2t, 2t)$  be a bounded-angle isoradial discretization of the rectangle  $(-2t, 2t) \times (0, 2t)$  with  $t \geq \delta$ . Suppose  $F$  is  $s$ -holomorphic there, satisfies the Riemann boundary condition on the lower side, and its primitive is normalized by  $H = 0$  on the lower dual boundary and  $H \geq 0$  inside. Let  $b^\delta$  be the boundary rhombus centre closest to 0, and let  $c^\delta$  be an interior dual vertex at Euclidean height comparable with  $t$ . Then*

$$c_\eta \frac{H(c^\delta)}{t} \leq |F(b^\delta)|^2 \leq C_\eta \frac{H(c^\delta)}{t}. \quad (12.7)$$

*Proof.* First apply Lemmas 2.2 and 2.3 in a ball of radius comparable with  $t$  around  $c^\delta$ . All primitive values on a horizontal crosscut at height between  $t/2$  and  $3t/2$  are therefore comparable with  $H(c^\delta)$ .

For the lower bound, start the dual isoradial walk at the inner dual vertex adjacent to  $b^\delta$ . The vertical coordinate is discrete harmonic for the critical isoradial Laplacian. Optional stopping between the lower boundary and the horizontal crosscut, together with bounded-angle comparison of graph and Euclidean distance, gives

$$\mathbb{P}(\text{hit the crosscut before returning to the lower boundary}) \asymp_\eta \delta/t. \quad (12.8)$$

Superharmonicity therefore gives  $H(b_{\text{dual}}^\delta) \geq c_\eta(\delta/t)H(c^\delta)$ . The primitive increment at the boundary rhombus and  $s$ -holomorphicity imply  $H(b_{\text{primal}}^\delta) - H(b_{\text{dual}}^\delta) \asymp_\eta \delta|F(b^\delta)|^2$ , which proves the lower inequality in (12.7).

For the upper bound, the bulk Harnack lemma and the zero lower boundary show that  $H \leq C_\eta H(c^\delta)$  in the central rectangle. Starting the primal walk at the inner primal vertex adjacent to  $b^\delta$ , the same vertical coordinate barrier gives harmonic measure  $O_\eta(\delta/t)$  for the upper and vertical sides before the lower side. Subharmonicity hence yields  $H(b_{\text{primal}}^\delta) \leq C_\eta(\delta/t)H(c^\delta)$ . Using the boundary primitive increment once more gives the upper inequality.  $\square$

**Lemma 2.5** (Interior regularity from the primitive). *Let  $F$  be  $s$ -holomorphic in a simply connected isoradial domain and let  $H$  be its primitive. If  $z$  is at Euclidean distance  $d \geq c\delta$  from the boundary and  $|H| \leq M$  in the  $d$ -neighbourhood of  $z$ , then*

$$|F(z)| \leq C_\eta M^{1/2} d^{-1/2}, \quad |F(z') - F(z)| \leq C_\eta M^{1/2} d^{-3/2} \delta \quad (12.9)$$

*for every neighbouring rhombus centre  $z'$ . Consequently a family with locally uniformly bounded primitives is locally bounded and equicontinuous.*

*Proof.* We spell out the potential-theoretic estimate because it is the bridge from compactness of  $H$  to compactness of  $F$ . On a primal ball of radius comparable with  $d$ , decompose the subharmonic restriction as  $H = H_{\text{harm}} + H_{\text{sub}}$ , where  $H_{\text{sub}} = 0$  on the boundary. The maximum principle gives

$|H_{\text{harm}}| \leq M$  and  $|H_{\text{sub}}| \leq 2M$ . Green representation and the bounded-angle Green-kernel lower bound on an inner ball imply

$$\sum |\Delta^\delta H| \mu^\delta \leq C_\eta M. \quad (12.10)$$

The same holds on the dual graph. Differentiating the Green representation, using the scale- $d$  gradient bound for the discrete Green kernel, and adding the harmonic part gives

$$\sum_{B(z, d/2)} |F|^2 \mu^\delta \leq C_\eta M d. \quad (12.11)$$

Here we used the exact primitive identity, which identifies the first differences of  $H$  with squared projections of  $F$ .

Apply the discrete Cauchy formula on the nested rhombic circles  $B^\delta(z, r)$  with  $r \in [d/8, d/4]$  and sum over the disjoint admissible radii. Cauchy-Schwarz together with (12.11) yields  $|F(z)| \leq C_\eta M^{1/2} d^{-1/2}$ . Applying the differentiated discrete Cauchy formula on a circle of radius  $d/4$ , and using the just established boundary bound, yields the neighbouring-point estimate in (12.9). The Green-kernel and discrete-Cauchy estimates used here are uniform under the bounded-angle hypothesis because the critical isoradial conductances and cell areas are uniformly comparable after rescaling.  $\square$

**Theorem 2.6** (Convergence of critical Ising fermions).

*Let  $(\Omega^\delta; a^\delta, b^\delta)$  be bounded-angle isoradial approximations of a simply connected marked domain  $(\Omega; a, b)$ , converging in the Carathéodory sense. For the critical FK-Ising fermion, and for the critical spin-Ising fermion under the target-neighbourhood normalization stated above, the canonically normalized  $s$ -holomorphic observables  $F^\delta$  converge locally uniformly away from the marked points to the unique holomorphic solution of the corresponding square-root Riemann boundary-value problem. Equivalently, after choosing the boundary-determined branch, the limit is the conformal half-differential prescribed by the continuum problem.*

*Proof.* The proof has three logically different parts.

*Compactness.* Let  $H^\delta$  be the primitive from SM-12.01.03, normalized by the boundary values of the FK or spin problem. The maximum principle gives the uniform bulk bounds required by Lemma 2.3. On every  $K \Subset \Omega \setminus \{a, b\}$ , Lemma 2.5 and a finite Harnack-chain covering give numbers  $\alpha > 0$  and  $C_K < \infty$  such that

$$|H^\delta(x) - H^\delta(y)| \leq C_K |x - y|^\alpha, \quad |F^\delta(e)| \leq C_K.$$

Hence every sequence has a subsequence on which the primal and dual interpolations of  $H^\delta$  and the observable  $F^\delta$  converge locally uniformly.

*Identification of the primitive.* Subharmonicity on primal vertices and superharmonicity on dual vertices, together with the fact that their mutual distance is  $O(\delta)$ , imply that the two subsequential primitive limits coincide and are harmonic. The boundary modification from SM-12.01.03 gives the limiting boundary values. In the FK case they are 0 and 1 on the two Dobrushin arcs, so uniqueness of the Dirichlet problem gives

$$h(z) = \omega(z; (ba); \Omega) = \text{Im } \Phi(z),$$

where  $\Phi : \Omega \rightarrow \mathbb{R} \times (0, 1)$  sends  $a, b$  to the two ends. For the spin observable the same argument is performed after deleting a fixed neighborhood of the target point; Lemma 2.4 propagates the target normalization to the limit and identifies  $h$  with the normalized Poisson-kernel solution of the corresponding continuum problem.

*Recovery of the fermion.* For a primal edge the exact discrete identity is

$$H^\delta(v) - H^\delta(u) = \text{Im}\left((F^\delta(e))^2(v - u)\right).$$

Testing this identity against smooth compactly supported one-forms and passing to the locally uniform subsequential limits gives

$$dh = \text{Im}(f^2 dz), \quad f^2 = 2i \partial h.$$

The Riemann boundary condition selects the compatible square-root branch. Since the continuum Dirichlet/Poisson problem and that branch are unique, every subsequence has the same limit. Therefore the full family converges locally uniformly to the stated conformal half-differential.

**Quantitative inputs.** Lemmas 2.2–2.5 prove the three estimates used above: the a priori Harnack lemma for the primitive, the straight-boundary Harnack comparison that normalizes the spin observable, and the interior regularity estimate that converts primitive bounds into compactness of the fermion.  $\square$

## 2.3 Universality as uniqueness of a boundary-value problem

The word *universality* will be used here in a precise, limited sense. It does not mean that every planar discretization has the same scaling limit. It means that within the bounded-angle isoradial class, after the canonical critical normalization, every admissible microscopic graph produces the same continuum solution of the same Riemann-type boundary-value problem.

For the FK fermion, if  $\Phi : \Omega \rightarrow \mathbb{R} \times (0, 1)$  maps  $a, b$  to the two ends of the strip, the continuum observable is, up to the fixed sign convention,

$$f_{\Omega, a, b}(z) = \sqrt{\Phi'(z)}.$$

For the spin fermion the corresponding conformal map is normalized to the upper half-plane and the same half-differential principle applies with the source-specific normalization at the target point.

If  $\varphi : \Omega \rightarrow \tilde{\Omega}$  is conformal, uniqueness gives the covariance law

$$f_\Omega(z) = f_{\tilde{\Omega}}(\varphi(z))\sqrt{\varphi'(z)},$$

where the branch is fixed by the boundary condition. Thus the exponent  $1/2$  is not guessed from CFT; it is forced by the square-root Riemann boundary condition.

**Theorem 2.7** (Isoradial universality and conformal covariance).

*Within the bounded-angle critical isoradial class of SM-12.02.01, the fermionic scaling limit is independent of the microscopic isoradial graph. If  $\varphi : \Omega \rightarrow \tilde{\Omega}$  is conformal, then, with branches fixed by the boundary condition,*

$$f_\Omega(z) = f_{\tilde{\Omega}}(\varphi(z))\sqrt{\varphi'(z)}.$$

*Thus the limiting fermion is a conformally covariant holomorphic half-differential.*

*Proof.* Assume the convergence input SM-12.02.01. Its limiting primitive is uniquely determined by the continuum boundary-value problem. In the FK normalization, if  $\Phi : \Omega \rightarrow \mathbb{R} \times (0, 1)$  is the strip map, then

$$h = \text{Im } \Phi, \quad f^2 = 2i \partial h = \Phi'.$$

The boundary Riemann condition fixes the sign of  $f = \sqrt{\Phi'}$ . Thus the limit depends only on  $(\Omega, a, b)$  and not on the microscopic isoradial graph. The spin normalization gives the analogous square root of the derivative of the normalized half-plane map.

If  $\varphi : \Omega \rightarrow \tilde{\Omega}$  is conformal, then the continuum primitive transforms by composition. The chain rule gives

$$f_{\Omega}(z)^2 = f_{\tilde{\Omega}}(\varphi(z))^2 \varphi'(z).$$

The boundary-determined branches agree, so

$$f_{\Omega}(z) = f_{\tilde{\Omega}}(\varphi(z)) \sqrt{\varphi'(z)}.$$

This proves the theorem from SM-12.02.01, including its quantitative Harnack and regularity inputs.  $\square$

## 2.4 Convergence of the primitives

We now state separately the convergence of  $H^{\delta}$ . This separation is pedagogically useful and will later be essential when interfaces, correlation functions, and s-embeddings use the primitive even when the fermion itself carries a branch ambiguity.

**Theorem 2.8** (Convergence of the s-holomorphic primitives).

*With the normalization and assumptions of SM-12.02.01, the primal and dual interpolations of  $H^{\delta}$  converge locally uniformly to the same harmonic function  $h$  solving the continuum boundary-value problem. Their discrete gradients converge weakly locally to  $\nabla h$ , and the edge relation converges to*

$$dh = \text{Im}(f^2 dz),$$

*where  $f$  is the limiting fermion.*

*Proof.* Assume SM-12.02.01. The compactness argument there applies simultaneously to the primal and dual interpolations. Their values across each rhombus differ by  $O(\delta|F^{\delta}|^2)$ , which tends to zero uniformly on compact subsets; hence both restrictions converge locally uniformly to the same function  $h$ . Sub/superharmonicity and the boundary-value identification in SM-12.02.01 show that this common limit is the unique harmonic solution of the continuum problem.

It remains to justify the two convergence assertions about derivatives. Let  $\psi$  be a smooth compactly supported vector field. Discrete summation by parts on the isoradial cells gives

$$\sum_e \nabla^{\delta} H^{\delta}(e) \cdot \psi(e) \text{ area}(e) = - \sum_v H^{\delta}(v) \text{div}^{\delta} \psi(v) \mu^{\delta}(v).$$

The bounded-angle geometry makes the discrete divergence consistent with the continuum one, while local uniform convergence of  $H^{\delta}$  gives convergence of the right-hand side to  $-\int h \text{div} \psi$ . Thus  $\nabla^{\delta} H^{\delta} \rightharpoonup \nabla h$  locally in the distributional (and hence weak local  $L^2$  on the uniformly energy-bounded subsequences) sense.

Finally, applying the exact edge identity

$$H^{\delta}(v) - H^{\delta}(u) = \text{Im}\left((F^{\delta}(e))^2(v - u)\right)$$

to test one-forms and passing to the limit yields

$$dh = \text{Im}(f^2 dz).$$

Therefore all conclusions follow from SM-12.02.01 and its quantitative Harnack and regularity inputs.  $\square$

## 2.5 A model calculation: the infinite strip

Let  $S = \{0 < \text{Im } z < 1\}$ . The harmonic function with boundary values 0 and 1 on the two horizontal sides is  $h(z) = \text{Im } z$ . Hence

$$2i \partial h = 1,$$

and the normalized fermion is the constant  $f \equiv 1$  (up to the chosen sign). Under a conformal map  $\Phi : \Omega \rightarrow S$ , the pullback gives  $f = \sqrt{\Phi'}$ . The full convergence proof is a robust discrete version of this elementary computation:  $H^\delta$  converges to strip height, and the local projection relations force  $F^\delta$  to converge to its square-root derivative.

## 2.6 Scope boundary

The first two chapters close the discrete-fermion and basic convergence blocks, but they do not yet prove the later correlation-field or interface theorems of Volume XII. In particular, no SLE convergence is used here. The next chapter will introduce spin/energy correlations and holomorphic spinors with interior branch points, which require new source-sensitive normalization data.

## Chapter 3

# Critical energy and spin correlations

The basic fermion of Chapters 1–2 becomes a correlation probe once one allows interior singularities and ramification. An energy insertion is local: two nearby fermionic corners coalesce and the singular full-plane contraction has to be subtracted. A spin insertion is topological: it changes the sign of a fermion transported once around the marked point, so the correct observable lives on a double cover. These two mechanisms lead to the same proof pattern

$$\begin{aligned} \text{exact lattice identity} &\longrightarrow \text{singular spinor} \\ &\longrightarrow \text{Riemann boundary problem} \\ &\longrightarrow \text{continuum correlation.} \end{aligned}$$

The purpose of the chapter is to carry this pattern out with the normalizations and uniformity needed later in the book. In particular, “conformal weight” is not inserted as a CFT assumption: it is read from the local singularity of the limiting spinor and then checked against the exact lattice normalization.

### 3.1 Energy density: normal ordering and a fermionic Pfaffian

On the critical square lattice,

$$\mathbb{E}_{\mathbb{C}, \beta_c}[\sigma_x \sigma_y] = \frac{1}{\sqrt{2}}$$

for nearest neighbors  $x \sim y$ . If  $e = \langle xy \rangle$  has midpoint  $z_e$ , set

$$\varepsilon_e := \sigma_x \sigma_y - \frac{1}{\sqrt{2}}.$$

The subtraction is a genuine normal ordering: without it the microscopic self-contraction survives and overwhelms the order- $\delta$  field fluctuation.

We first isolate the finite-dimensional identity behind all energy correlations. It is useful to state it because otherwise the phrase “energy is a product of two fermions” hides the only combinatorial step in the proof.

**Lemma 3.1** (Fermionic Wick identity for normal-ordered edge pairs). *Let  $c_1, \dots, c_{2m}$  be corners of a finite simply connected square-grid Ising domain, grouped into  $m$  adjacent pairs  $(c_{2j-1}, c_{2j})$  which represent the edges  $e_j$ . Let  $K_\delta(c, c')$  be the normalized Kadanoff–Ceva two-fermion kernel*

with the same boundary condition as the Ising measure. If in every same-edge entry one subtracts the full-plane contraction, then

$$\mathbb{E} \left[ \prod_{j=1}^m \varepsilon_{e_j} \right] = \kappa_{\text{sq}}^m \text{Pf}[K_{\delta}^{\circ}(c_p, c_q)]_{p,q=1}^{2m},$$

where  $\kappa_{\text{sq}}$  is the fixed corner convention constant and  $K_{\delta}^{\circ}$  agrees with  $K_{\delta}$  off the  $m$  distinguished pairs, while on such a pair it is  $K_{\delta} - K_{\mathbb{C},\delta}$ .

*Proof.* Use the Kadanoff–Ceva path convention of Chapter 1 and expand the high-temperature partition function with  $2m$  order–disorder endpoints. A configuration is a collection of even subgraphs plus paths pairing the endpoints. Crossing two pairing paths changes the half-winding sign, so the signed sum over pairings is exactly the Pfaffian expansion

$$\text{Pf } K = \frac{1}{2^m m!} \sum_{\pi \in S_{2m}} \text{sgn}(\pi) \prod_{j=1}^m K_{\pi(2j-1), \pi(2j)}.$$

Equivalently, one may package the same cancellation as the elementary Grassmann Gaussian identity: the Kac–Ward quadratic form gives the fermionic generating function, differentiation in  $2m$  Grassmann sources gives the displayed Pfaffian, and the determinant expansion of the quadratic form is precisely the signed even-subgraph expansion. No continuum input is used here.

For one distinguished adjacent pair, the local Kadanoff–Ceva switching identity identifies the product of the two corner fermions with a constant multiple of  $\sigma_x \sigma_y$ . Its full-plane expectation is the contraction  $K_{\mathbb{C},\delta}$ ; subtracting this entry is therefore exactly the replacement  $\sigma_x \sigma_y \mapsto \varepsilon_e$ . Applying this operation to each distinguished pair proves the normal-ordered formula.  $\square$

The next lemma is the analytic step. It is a direct use of the convergence machinery already proved in Chapter 2, but with the singular part explicitly removed before compactness is invoked.

**Lemma 3.2** (Coalescing-pair limit). *Let  $e_j^{\delta}$  have midpoint  $z_j^{\delta} \rightarrow z_j$ , where the  $z_j$  are pairwise distinct and stay in a compact subset of a bounded simply connected domain  $\Omega$ . For fixed + or free boundary conditions, every entry of the normal-ordered matrix in Lemma 3.1, after the corner normalization from Chapters 1–2, has a locally uniform limit. Off a distinguished pair this limit is the corresponding continuum fermionic kernel; on a distinguished pair it is the regular part left after subtracting the universal  $(z - w)^{-1}$  singularity.*

*Proof.* Fix disjoint discs  $B(z_j, r)$  and first keep the two corners in each pair separated. Chapter 2 gives local compactness and convergence of the normalized fermionic kernel on the complement of the diagonal. In a disc containing one pair write

$$K_{\delta}(c, c') = K_{\mathbb{C},\delta}(c, c') + R_{\delta}(c, c').$$

Both terms satisfy the same local s-holomorphic projection equations away from the source, hence  $R_{\delta}$  has no pole. The primitive of  $R_{\delta}$  obeys the discrete maximum principle; the boundary values on  $\partial B(z_j, r)$  are uniformly bounded by the already established kernel estimates. Discrete Harnack and Cauchy estimates therefore give a bound on  $R_{\delta}$  all the way to the coalescing pair which is uniform in  $\delta$ . Any subsequential limit is the holomorphic solution of the continuum Riemann boundary problem with the model singularity removed. Uniqueness of that boundary problem identifies the limit and removes the subsequence. The same estimate is uniform when the marked points range over a separated compact configuration set.  $\square$



**Theorem 3.3** (Critical energy-density correlations).

Let  $\Omega$  be a bounded simply connected domain and let  $\Omega^\delta$  be square-grid approximations in the normalization of Chapter 2. Fix either + or free boundary conditions. If  $e_j^\delta$  are nearest-neighbor edges with midpoints  $z_j^\delta \rightarrow z_j$ , where  $z_1, \dots, z_m$  are pairwise distinct interior points, then

$$\delta^{-m} \mathbb{E}_{\Omega^\delta}^b \left[ \prod_{j=1}^m \varepsilon_{e_j^\delta} \right] \longrightarrow \mathcal{E}_\Omega^b(z_1, \dots, z_m),$$

locally uniformly on separated compact configuration sets. The limit is conformally covariant with energy weight one:

$$\mathcal{E}_\Omega^b(z_1, \dots, z_m) = \mathcal{E}_{\Omega'}^{\varphi(b)}(\varphi z_1, \dots, \varphi z_m) \prod_{j=1}^m |\varphi'(z_j)|$$

for conformal  $\varphi : \Omega \rightarrow \Omega'$ . For one insertion and homogeneous boundary condition, writing  $\ell_\Omega(z)$  for the hyperbolic metric density,

$$\mathcal{E}_\Omega^+(z) = \frac{\ell_\Omega(z)}{2\pi}, \quad \mathcal{E}_\Omega^{\text{free}}(z) = -\frac{\ell_\Omega(z)}{2\pi}.$$

No collision or boundary-OPE assertion is included in this theorem; those are separate primary-field/fusion statements later in the volume.

*Proof. Proof roadmap.* We first convert products of centered energies into a finite Pfaffian of fermionic kernels. We then pass each kernel to its continuum limit, including the regularized diagonal entry. Finally we read the conformal weight from the half-differential transformation rule and identify the one-point normalization by the explicit spinor in the half-plane.

The full-plane critical subtraction is  $1/\sqrt{2}$ . With the diagonal mesh convention used here, Hongler–Smirnov’s energy variable is exactly  $\delta^{-1}\varepsilon_e$  up to the harmless choice of which of the two lattice edge directions represents the marked point. Rotating an edge by  $\pi/2$  changes the local fermionic phase but not the scalar continuum energy field.

Lemma 3.1 gives the exact finite-volume representation. It is important that the same-edge matrix entry is the *difference* between the domain kernel and its full-plane singular kernel. This is the fermionic meaning of centering and is why the diagonal entry has size  $O(\delta)$  instead of order one.

Apply Lemma 3.2 to all  $m$  pairs. The Pfaffian is a polynomial in its matrix entries, so entrywise locally uniform convergence implies locally uniform convergence of the Pfaffian. Every energy pair supplies one microscopic factor  $\delta$ ; after division by  $\delta^m$  the limit is the Pfaffian of the continuum cross-kernels and regular parts. This defines  $\mathcal{E}_\Omega^b$  without appealing to CFT.

A continuum fermion is a half-differential. If  $f_\Omega(z, w)$  denotes the continuum kernel, then, after compatible square-root choices,

$$f_\Omega(z, w) = f_{\Omega'}(\varphi z, \varphi w) \sqrt{\varphi'(z)} \sqrt{\varphi'(w)}.$$

For an energy insertion the two fermionic variables coalesce at the same point; the product of their two half-differential factors is  $\varphi'(z)$ , and the physical real energy field retains its modulus. Every term of the limiting Pfaffian therefore acquires exactly one factor  $|\varphi'(z_j)|$  at  $z_j$ . This proves the displayed covariance.

For the one-point function map  $\Omega$  conformally to the upper half-plane and place the marked point at  $a$ . The unique continuum spinor with the prescribed boundary phase and the full-plane

pole subtracted has regular coefficient proportional to  $(2 \operatorname{Im} a)^{-1}$ . Carrying through the lattice normalization fixed above gives

$$\lim_{\delta \rightarrow 0} \delta^{-1} \mathbb{E}_+ \left[ \sigma_{x_\delta} \sigma_{y_\delta} - \frac{1}{\sqrt{2}} \right] = \frac{\ell_\Omega(a)}{2\pi}.$$

For free boundary data the boundary Riemann condition has the opposite sign, so the regular coefficient changes sign. This is the exact one-point Hongler–Smirnov normalization.

The coalescing-pair estimates were proved uniformly only while marked points stay a positive macroscopic distance from each other and from the boundary. That is precisely the theorem’s regime. We do not infer collision fusion rules from these estimates: near collision the singular model changes and a new renormalization is needed.

The one-point statement and its hyperbolic-metric normalization agree with Hongler–Smirnov. The multipoint energy-only statement is the energy-field specialization of the later Chelkak–Hongler–Izyurov primary-field framework; here it has been derived directly from the fermionic Pfaffian plus the Chapter-2 kernel convergence, so no later mixed-primary theorem is a logical input.  $\square$

### 3.2 Spin correlations and ramified double covers

A spin insertion is not obtained by differentiating the Hamiltonian. Instead, when a fermion circles a marked spin it changes sheet. Thus for  $A = (a_1, \dots, a_n)$  we consider the double cover of  $\Omega \setminus A$  with monodromy  $-1$  around every  $a_j$ . The key local expansion near the distinguished point  $a_1$  has the form

$$f_{\Omega, A}(z) = (z - a_1)^{-1/2} (1 + 2A_\Omega(A)(z - a_1) + O(|z - a_1|^2)),$$

and the coefficient  $A_\Omega(A)$  is the logarithmic derivative of the continuum spin correlation. This is the mechanism by which a spinor determines a scalar correlation function.

**Lemma 3.4** (Discrete log-derivative identity). *For a separated configuration  $A^\delta = (a_1^\delta, \dots, a_n^\delta)$  with + boundary conditions, normalize the discrete ramified spinor so that its leading coefficient near  $a_1^\delta$  equals the full-plane one. If  $a_1^\delta$  is shifted by one admissible lattice step  $h_\delta$ , then*

$$\log \frac{\mathbb{E}_{\Omega^\delta}^+ [\sigma_{a_1^\delta + h_\delta} \sigma_{a_2^\delta} \cdots \sigma_{a_n^\delta}]}{\mathbb{E}_{\Omega^\delta}^+ [\sigma_{a_1^\delta} \sigma_{a_2^\delta} \cdots \sigma_{a_n^\delta}]} = 2 \operatorname{Re}(A_\delta(A^\delta) h_\delta) + o(\delta),$$

*uniformly on separated compact configuration sets.*

*Proof.* Open a disorder line from a corner adjacent to  $a_1^\delta$  to the boundary. Chapter 1 shows that changing the endpoint by one edge changes only the local Kadanoff–Ceva factor. Divide the two order–disorder partition functions and cancel all common contour weights. The remaining local factor is the ratio of the two spin correlations in the statement. On the other hand, the same ratio is the first regular coefficient in the expansion of the normalized discrete spinor at  $a_1^\delta$ . Expanding the local square-root coordinate through one lattice step gives the factor  $2 \operatorname{Re}(A_\delta h_\delta)$ ; the uniform  $o(\delta)$  remainder follows from the discrete Cauchy estimate for the regular part, since the marked points stay uniformly separated.  $\square$

**Lemma 3.5** (Convergence of ramified spinors). *In the setting of Lemma 3.4, the normalized discrete spinors converge locally uniformly away from  $A$  to the unique holomorphic spinor  $f_{\Omega, A}$  satisfying*

(i) *monodromy  $-1$  around every  $a_j$ ;*

- (ii) the Ising Riemann boundary condition inherited from  $+$  boundary data;
- (iii) the prescribed  $(z - a_1)^{-1/2}$  leading singularity at  $a_1$ ;
- (iv) at most the allowed square-root singularity at the remaining marked points.

Moreover  $A_\delta(A^\delta) \rightarrow A_\Omega(A)$  locally uniformly on separated configuration sets.

*Proof.* Cut the domain along disjoint arcs joining the branch points to the boundary and work on one sheet. The square of the spinor is single valued, while the primitive  $H^\delta = \text{Im} \int (F^\delta)^2 dz$  is therefore well defined exactly as in Chapter 1. Subtract from  $F^\delta$  the sum of the local full-plane branch kernels at the marked points. The remainder has uniformly bounded primitive on compacta by the discrete maximum principle. The Harnack and Cauchy estimates of Chapter 2 give precompactness.

Every subsequential limit is holomorphic off  $A$ , has the required monodromy, and satisfies the continuum boundary phase because the discrete projection condition converges. If  $f_1, f_2$  are two such limits with the same leading coefficient, then the primitive of  $(f_1 - f_2)^2$  has zero boundary data and no positive singular part. The maximum principle makes that primitive constant, so  $f_1 = f_2$ . Hence the whole sequence converges. Apply the same Cauchy estimate on a small circle around  $a_1$  to the regular part after subtracting  $(z - a_1)^{-1/2}$ ; its first coefficient converges, giving  $A_\delta \rightarrow A_\Omega$ .  $\square$

**Lemma 3.6** (Integrability of the limiting logarithmic derivative). *Let  $\mathcal{C}_n(\Omega)$  be the configuration space of  $n$  pairwise distinct interior points. The real one-form*

$$\omega = 2 \text{Re} \sum_{j=1}^n A_{\Omega,j}(A) da_j$$

*obtained by distinguishing each insertion in turn is closed. In the upper half-plane its primitive is the logarithm of*

$$\mathcal{S}_{\mathbb{H}}^+(a_1, \dots, a_n) = \prod_{k=1}^n (2 \text{Im } a_k)^{-1/8} \left( 2^{-n/2} \sum_{\mu \in \{\pm 1\}^n} \prod_{1 \leq k < \ell \leq n} \left| \frac{a_k - a_\ell}{a_k - \bar{a}_\ell} \right|^{\mu_k \mu_\ell / 2} \right)^{1/2}.$$

*Consequently the primitive on any simply connected domain is fixed by conformal covariance.*

*Proof.* On  $\mathbb{H}$ , a spinor with the required branch points has the form

$$f(z) = \frac{P(z)}{\prod_{j=1}^n [(z - a_j)(z - \bar{a}_j)]^{1/2}},$$

where the boundary Riemann condition says that the boundary value has fixed phase. The residues/leading coefficients at the branch points give a real linear system for the coefficients of  $P$ . Solving that system and expanding at  $a_j$  yields  $A_{\mathbb{H},j}$ . Direct differentiation of the displayed finite sum gives exactly

$$2\partial_{a_j} \log \mathcal{S}_{\mathbb{H}}^+ = A_{\mathbb{H},j}.$$

Thus the mixed derivatives agree and  $\omega$  is closed on  $\mathbb{H}$ . Under a conformal map, a spinor acquires a factor  $\sqrt{\varphi'}$ ; expanding this factor at  $a_j$  adds the derivative of  $\frac{1}{8} \log |\varphi'(a_j)|$  to the logarithmic derivative. Hence closedness and the primitive transport to every simply connected domain, with

$$\mathcal{S}_{\Omega}^+(A) = \mathcal{S}_{\Omega'}^+(\varphi A) \prod_{j=1}^n |\varphi'(a_j)|^{1/8}.$$

$\square$

**Lemma 3.7** (Normalization and boundary decorrelation). *There is a square-lattice constant  $C_\sigma > 0$  such that the multiplicative constant obtained by integrating Lemma 3.4 equals  $C_\sigma^n$  for the  $n$ -point function. The statement is uniform while the points remain a fixed positive distance from one another and the boundary.*

*Proof.* For  $n = 2$  in a domain whose boundary is sent to infinity, compare along the lattice diagonal. The exact critical two-point asymptotic owned earlier in SM-08.05.03 fixes

$$\delta^{-1/4} \mathbb{E}_C[\sigma_{a\delta} \sigma_{b\delta}] \longrightarrow C_\sigma^2 |a - b|^{-1/4}.$$

This determines the one-spin normalization  $C_\sigma$  up to the positive sign.

For the induction in  $n$ , move one point  $a_n$  through a collar whose width is small on the macroscopic scale but large compared with  $\delta$ . The ramified spinor estimate in the collar is compared with the one-point half-plane spinor. The difference has zero boundary data and its primitive is bounded by harmonic measure of the distant marked set; that harmonic measure tends to zero as the collar shrinks. Hence

$$\frac{\mathbb{E}_{\Omega^\delta}^+[\sigma_{a_1} \cdots \sigma_{a_n}]}{\mathbb{E}_{\Omega^\delta}^+[\sigma_{a_n}]} \longrightarrow \mathbb{E}_{\Omega^\delta}^+[\sigma_{a_1} \cdots \sigma_{a_{n-1}}]$$

after the two limits are taken in that order, uniformly for the remaining separated points. The continuum formula in Lemma 3.6 has the identical boundary factorization. Thus if the  $(n - 1)$ -point constant is  $C_\sigma^{n-1}$ , the  $n$ -point constant is  $C_\sigma^n$ . This is the boundary-decorrelation induction used to remove an otherwise undetermined constant for each  $n$ .  $\square$

**Theorem 3.8** (Conformal invariance of critical spin correlations).

*Let  $\Omega$  be a bounded simply connected domain and  $\Omega^\delta$  its square-grid discretizations of diagonal mesh  $2\delta$ . For every  $n \geq 1$  and  $\epsilon > 0$ , uniformly over  $a_1, \dots, a_n$  at distance at least  $\epsilon$  from  $\partial\Omega$  and from one another,*

$$\delta^{-n/8} \mathbb{E}_{\Omega^\delta}^+[\sigma_{a_1^\delta} \cdots \sigma_{a_n^\delta}] \longrightarrow C_\sigma^n \mathcal{S}_\Omega^+(a_1, \dots, a_n),$$

where  $\mathcal{S}_\Omega^+$  is the function in Lemma 3.6. In particular, for every conformal map  $\varphi : \Omega \rightarrow \Omega'$ ,

$$\mathcal{S}_\Omega^+(a_1, \dots, a_n) = \mathcal{S}_{\Omega'}^+(\varphi a_1, \dots, \varphi a_n) \prod_{j=1}^n |\varphi'(a_j)|^{1/8}.$$

*The corresponding two-point convergence also holds for free boundary conditions with the free two-point continuum spinor. The theorem is a separated-point theorem; collision/fusion asymptotics are not silently included.*

*Proof. Proof roadmap.* The discrete spinor gives finite differences of the logarithm of the correlation. Convergence of the ramified boundary-value problem gives the limiting differential. We integrate it on configuration space and then use the full-plane two-point normalization plus boundary decorrelation to determine the single remaining constant.

The renormalization  $\delta^{-n/8}$  is fixed by the full-plane two-point scale. Equivalently one can first work with an abstract lattice normalization  $\rho(\delta)^{-n}$  and use the two-point asymptotic only at the end. This keeps the analytic part of the proof independent of exact lattice formulae.

Lemma 3.4 is the exact lattice bridge from correlations to ramified spinors. Moving each marked point in turn supplies all coordinate components of the discrete logarithmic derivative.

Lemma 3.5 identifies the limit of those coefficients with the unique continuum Riemann problem. The convergence is uniform precisely on the  $\epsilon$ -separated configuration set in the theorem.

Lemma 3.6 identifies the limiting differential as  $d \log \mathcal{S}_\Omega^+$ . Summing the discrete finite-difference identity along a polygonal path in configuration space and using the uniform error gives convergence of the ratio of two correlations to the ratio of the corresponding  $\mathcal{S}$  values. Closedness of the limiting one-form makes the result independent of the path.

The ratio argument leaves one multiplicative constant for each  $n$  and  $\delta$ . Lemma 3.7 proves that after multiplication by  $\delta^{-n/8}$  this constant converges to  $C_\sigma^n$ . There is no need to bring two marked interior points together, so no OPE estimate is being smuggled into the proof.

The half-differential transformation law of the ramified spinor gives the  $1/8$ -weight in Lemma 3.6; exponentiating the integrated logarithmic derivative gives exactly the covariance in the theorem. The free two-point case uses the same construction with the free boundary Riemann condition and a different continuum two-point solution, while the normalization constant is unchanged.

The scope now matches the main Chelkak–Hongler–Izyurov theorem: bounded simply connected domains, + boundary conditions for all  $n$ , uniformity on separated compact configurations, and free boundary conditions for the two-point theorem. The earlier draft incorrectly suggested that their theorem already included colliding points; that claim has been removed.  $\square$

### 3.3 Universality on bounded-angle isoradial graphs

The square grid is only one rhombic tiling. Let  $\Gamma^\delta$  be an isoradial graph with rhombus side length  $\delta$ . We impose the bounded-angle property

$$\theta_0 \leq \theta_e \leq \frac{\pi}{2} - \theta_0$$

for a fixed  $\theta_0 > 0$ , and at criticality choose the  $Z$ -invariant weights

$$x_e = \tan \frac{\theta_e}{2}.$$

These two assumptions have different jobs: bounded angles give uniform elliptic estimates, while  $Z$ -invariance gives the exact local star-triangle/track relations which make the normalization independent of the microscopic tiling.

**Lemma 3.9** (Uniform isoradial branch kernel). *For every bounded-angle critical isoradial graph there is a full-plane discrete spinor  $G_\delta(\cdot, a)$  with monodromy  $-1$  around  $a$  such that, for  $\delta \ll |z - a| \ll 1$ ,*

$$G_\delta(z, a) = (z - a)^{-1/2} + O_{\theta_0} \left( \frac{\delta}{|z - a|^{3/2}} \right),$$

*after the universal corner normalization. The same estimate holds for the first discrete derivative.*

*Proof.* Start from the discrete Cauchy kernel of the bounded-angle isoradial Dirac operator and lift it to the double cover ramified at  $a$ . The projection relations are exactly the s-holomorphic equations from Chapter 1. Subtract the sampled continuum kernel  $(z - a)^{-1/2}$ . Taylor expansion across one rhombus shows that the residual discrete defect is  $O_{\theta_0}(\delta^2 |z - a|^{-5/2})$ . Solve the residual equation using the discrete Green kernel. The bounded-angle Green estimate  $G_\delta(z, w) = O(1 + |\log |z - w||)$  and annular summation of the defect yield  $O_{\theta_0}(\delta |z - a|^{-3/2})$ . Applying the local discrete Cauchy estimate on a ball of radius comparable to  $|z - a|$  gives the derivative bound. All constants depend only on  $\theta_0$ , not on the tiling.  $\square$

**Lemma 3.10** (Track invariance fixes the spin normalization). *There exists a single constant  $C_\sigma > 0$ , the same as for the square grid, such that for every sequence of bounded-angle critical  $Z$ -invariant isoradial grids and macroscopic points  $u_1 \neq u_2$ ,*

$$\delta^{-1/4} \mathbb{E}_{\mathbb{C}^\delta} [\sigma_{u_1^\delta} \sigma_{u_2^\delta}] \longrightarrow C_\sigma^2 |u_1 - u_2|^{-1/4}.$$

*Proof.* The critical  $Z$ -invariant weights satisfy the star–triangle relation. Summing over the spin at the center of a star shows that a star–triangle move multiplies the partition function by a factor independent of the three exterior spins. Hence normalized correlations of spins which survive the move are unchanged. On rhombic tilings these moves are cube flips or, after grouping parallel rapidity lines, track exchanges.

The geometric point is not that an arbitrary infinite tiling can globally be turned into the square grid. What is needed is the finite-box *star extension*. Given a box of diameter  $R$  cut from a  $\text{BAP}(\theta_0)$  rhombic tiling, extend its train tracks outside the box, add finitely many auxiliary tracks, and use cube flips only outside the original box until the tracks leave through four separated boundary arcs. Convexity of train tracks prevents a self-intersection, and the angle lower bound keeps the extension inside a box of diameter  $C(\theta_0)R$ . Thus boxes cut from two different bounded-angle grids can be embedded, at mutual distance  $O(R)$ , into one mixed rhombic tiling which still satisfies a bounded-angle condition depending only on  $\theta_0$ .

Place a copy of the pair  $(u_1, u_2)$  in each of the two embedded boxes and consider the ramified full-plane spinor on the common mixed tiling. The local asymptotic from Lemma 3.9 is uniform and depends on the distant part of the tiling only through the regular solution of the boundary-value problem. If the comparison boundary is at distance  $R$ , the maximum principle for the primitive and annular Harnack estimates bound that regular correction by  $O_{\theta_0}((|u_1 - u_2|/R)^c)$  for some  $c > 0$ . Star–triangle moves used in the extension do not alter either correlation inside its protected box. Letting first  $\delta \rightarrow 0$  and then  $R/|u_1 - u_2| \rightarrow \infty$  shows that the multiplicative constant obtained from the two boxes is the same. Since one box may be chosen from the square grid, that common constant is  $C_\sigma^2$ . The limiting branch-spinor equation is scale covariant and therefore supplies the remaining factor  $|u_1 - u_2|^{-1/4}$ .  $\square$

**Theorem 3.11** (Isoradial universality of critical spin correlations).

*Let  $\Omega$  be a bounded simply connected domain and let  $\Omega^\delta$  be critical  $Z$ -invariant isoradial approximations satisfying a uniform bounded-angle property. For  $+$  boundary conditions, every fixed  $n \geq 1$ , and every separated compact family of interior configurations,*

$$\delta^{-n/8} \mathbb{E}_{\Omega^\delta}^+ [\sigma_{a_1^\delta} \cdots \sigma_{a_n^\delta}] \longrightarrow C_\sigma^n \mathcal{S}_\Omega^+(a_1, \dots, a_n),$$

*where both  $C_\sigma$  and the continuum function are the same as in the square lattice theorem. In particular the limit is independent of the microscopic isoradial tiling and obeys conformal weight  $1/8$  at each spin insertion.*

*Proof. Proof roadmap.* Uniform branch-kernel estimates replace every square-grid local calculation in the spinor proof. The only possible lattice dependence is then a single full-plane normalization constant; exact  $Z$ -invariant track moves and a gluing argument show that this constant is universal.

The bounded-angle hypothesis makes all discrete Harnack, Green-kernel and Cauchy constants uniform. At criticality the  $Z$ -invariant weights are  $x_e = \tan(\theta_e/2)$ , so the local propagation equation is exactly the s-holomorphic one already owned in Chapters 1–2.

Lemma 3.9 supplies the only new local ingredient near a ramification point. Away from the marked points, Chapter 2 already proves convergence of the bounded-angle isoradial fermionic

boundary-value problem. Combining the two gives the isoradial analogue of Lemma 3.5, with constants uniform in the graph.

The local Kadanoff–Ceva switching identity is graph-theoretic and unchanged. Hence the discrete logarithmic-derivative identity Lemma 3.4 holds verbatim. Uniform convergence of the spinor coefficients gives the same continuum one-form  $d \log \mathcal{S}_\Omega^+$  as on the square grid. Thus the ratio of two  $n$ -point correlations has the same continuum limit; only a graph-dependent scalar normalization could remain.

Lemma 3.10 removes that scalar freedom: the full-plane two-point constant is  $C_\sigma$  on every bounded-angle critical  $Z$ -invariant isoradial graph. The boundary-decorrelation induction used in Lemma 3.7 depends only on the uniform spinor/primitive estimates, so it promotes the two-point normalization to  $C_\sigma^n$  for every  $n$ .

The limiting spinor is the same continuum Riemann-problem solution, so the half-differential transformation law and the integration argument are unchanged. This proves both universality and conformal covariance. Notice that the theorem is restricted to the bounded-angle  $Z$ -invariant class; it says nothing about an arbitrary planar mesh.

This is the critical  $m = 0$  specialization of the Chelkak–Izyurov–Mahfouf universality theorem. Their stronger paper also treats the massive scaling window; no massive theorem is used here. The source’s two indispensable new inputs—a lattice-independent full-plane two-point normalization and uniform analysis of the  $z^{-1/2}$  branch kernel—are both proved explicitly above.  $\square$

### 3.4 Worked examples and scope checks

**Example 3.12** (One energy insertion in the half-plane). For  $\mathbb{H}$ , the hyperbolic metric density is  $\ell_{\mathbb{H}}(z) = 1/\operatorname{Im} z$  in the convention used here. Hence

$$\mathcal{E}_{\mathbb{H}}^+(z) = \frac{1}{2\pi \operatorname{Im} z}, \quad \mathcal{E}_{\mathbb{H}}^{\text{free}}(z) = -\frac{1}{2\pi \operatorname{Im} z}.$$

If  $\varphi : \Omega \rightarrow \mathbb{H}$ , then  $\ell_\Omega(z) = |\varphi'(z)|\ell_{\mathbb{H}}(\varphi z)$ , which is exactly the weight-one covariance of the energy theorem.

**Example 3.13** (Two spins in the plane). Translation, rotation and scaling reduce the full-plane two-point function to one constant. The theorem gives

$$\langle \sigma(z)\sigma(w) \rangle_{\mathbb{C}} = C_\sigma^2 |z - w|^{-1/4}.$$

The exponent  $1/4$  is twice the individual spin weight  $1/8$ . In the discrete proof this power is not guessed from scaling theory: the ramified spinor gives the logarithmic derivative  $-\frac{1}{4}(z - w)^{-1}$  and the full-plane exact normalization fixes the multiplicative constant.

*Remark 3.14* (What has and has not been proved). The three parent gates in this chapter concern separated interior insertions. They now own the singular-spinor construction, logarithmic-derivative integration, lattice normalization, and isoradial universality needed for that regime. When marked points collide, the correct statement is an operator product/fusion expansion with additional singular coefficients. When marked points approach boundary-condition changes, new boundary fields appear. Those are not consequences of compact separated convergence and are therefore kept as separate later proof obligations rather than hidden in the present theorem.

## Chapter 4

# The magnetization field and the magnetic scaling window

Pointwise spins do not converge to ordinary functions. The correct continuum object is a random distribution. On the mesh  $a\mathbb{Z}^2$  the convenient formal notation is

$$\Phi^a = a^{15/8} \sum_{x \in a\mathbb{Z}^2} \sigma_x \delta_x.$$

For the Sobolev argument we replace each point mass by uniform density on its mesh square  $S_a(x)$ ,

$$\bar{\Phi}^a(z) = a^{-1/8} \sum_x \sigma_x \mathbf{1}_{S_a(x)}(z).$$

The two versions have the same distributional limit: against a smooth test function their difference is bounded by the mesh size times a first-derivative norm. The exponent  $15/8 = 2 - 1/8$  is the Euclidean area exponent minus the spin scaling dimension.

The chapter has three logically separate tasks. First we construct the critical field in a topology in which compactness is genuine. Next we make its conformal covariance and non-Gaussianity explicit. Finally we perturb by an external field  $H_a = ha^{15/8}$  and construct the near-critical family.

### 4.1 Uniqueness of the critical field

**Theorem 4.1** (Critical planar Ising magnetization field).

*Let  $\Omega$  be a bounded simply connected domain with smooth boundary and let  $\Omega^a$  be its square-grid approximations. At  $\beta = \beta_c$ , with either + or free boundary conditions, the renormalized fields  $\Phi_{\Omega}^a$  converge in law in  $\mathcal{H}^{-3}(\Omega)$  to a unique random distribution  $\Phi_{\Omega}^{\infty}$ . The full-plane fields converge in the corresponding nested-box product topology. If  $\varphi : \Omega \rightarrow \tilde{\Omega}$  is conformal and  $\psi = \varphi^{-1}$ , then*

$$\varphi_* \Phi_{\Omega}^{\infty} \stackrel{d}{=} |\psi'|^{15/8} \Phi_{\tilde{\Omega}}^{\infty}.$$

*Thus the limiting magnetization is a generalized field of conformal weight 15/8 in the distributional push-forward convention.*

*Proof. Proof roadmap.* We first obtain compactness one Sobolev order stronger than the space in which we take the limit. We then identify every subsequential law from the separated spin correlations already proved in Chapter 3. The only extra work is to show that lattice tuples



approaching the diagonals contribute uniformly negligibly. Uniform exponential moments turn moment convergence into convergence of characteristic functionals.

Work first in the unit square. Let  $(e_{j,k})_{j,k \geq 1}$  be the sine basis for the Dirichlet Laplacian. For the piecewise constant field,

$$\|\bar{\Phi}^a\|_{\mathcal{H}^{-2}}^2 = \sum_{j,k \geq 1} \frac{|\langle \bar{\Phi}^a, e_{j,k} \rangle|^2}{(j^2 + k^2)^2}.$$

Expanding the expectation of a coefficient produces the two-point function. By boundary-condition comparison it is dominated by the extremal ferromagnetic boundary condition, and the critical two-point estimate owned earlier gives, uniformly in  $j, k$  and  $a$ ,

$$\mathbb{E}|\langle \bar{\Phi}^a, e_{j,k} \rangle|^2 \leq C.$$

Since  $\sum_{j,k} (j^2 + k^2)^{-2} < \infty$ ,

$$\sup_a \mathbb{E} \|\bar{\Phi}^a\|_{\mathcal{H}^{-2}}^2 < \infty.$$

Rellich compactness  $\mathcal{H}^{-2} \Subset \mathcal{H}^{-3}$  therefore gives tightness in  $\mathcal{H}^{-3}$ . Whitney squares give the same estimate on a smooth bounded domain, uniformly on compact subsets, and a diagonal subsequence on the boxes  $[-2^m, 2^m]^2$  gives full-plane tightness.

For the rescaled total magnetization  $M_D^a = \langle \Phi^a, 1_D \rangle$  in a fixed bounded domain, set

$$K_a(t) = \log \mathbb{E} e^{tM_D^a}.$$

The GHS inequality already proved as SM-08.02.03 says that, for nonnegative external field,  $K_a'''(t) \leq 0$ . Hence for  $t \geq 0$ ,

$$K_a(t) \leq K_a'(0)t + \frac{1}{2}K_a''(0)t^2.$$

The one- and two-point estimates imply uniform bounds on  $K_a'(0)$  and  $K_a''(0)$  after the  $a^{15/8}$  normalization. Spin flip and boundary comparison give the negative- $t$  bound. Consequently, for every fixed  $t$ ,

$$\sup_a \mathbb{E} e^{tM_D^a} < \infty.$$

If  $|f| \leq C$  is supported in  $D$ , the Griffiths inequalities applied to the moment expansion bound the absolute  $k$ th moment of  $\langle \Phi^a, f \rangle$  by  $C^k$  times the corresponding  $+$ -boundary magnetization moment. Thus the same argument supplies the uniform integrability needed below for arbitrary smooth tests.

Fix  $k$  and  $\varepsilon > 0$ . In the moment expansion

$$\mathbb{E} \langle \Phi^a, f \rangle^k = a^{15k/8} \sum_{x_1, \dots, x_k} \prod_{j=1}^k f(x_j) \mathbb{E}[\sigma_{x_1} \cdots \sigma_{x_k}],$$

restrict first to configurations satisfying  $\min_{i \neq j} |x_i - x_j| \geq \varepsilon$  and staying  $\varepsilon$  away from the boundary. The earlier theorem SM-12.03.02 gives uniform convergence of the normalized  $k$ -spin function there. Ordinary Riemann-sum convergence therefore identifies the separated part with the integral of the continuum spin correlation function against  $\prod f(z_j) dA(z_j)$ .

It remains to justify sending  $\varepsilon \downarrow 0$ . Use the Edwards–Sokal representation to decompose a  $k$ -spin correlation according to the connectivity partition of the marked vertices. Replace each exact partition event by its increasing connectivity envelope. If two marked points lie in a ball of radius  $r$ ,

the induction on the number of marks isolates a connected pair and leaves a correlation of order  $k-2$ . The critical FK two-point bound and RSW then give, on the dyadic shell  $2^\ell a \leq |x-y| < 2^{\ell+1}a$ ,

$$\mathbb{P}(x \leftrightarrow y) \leq C \left( \frac{a}{|x-y|} \right)^{1/4}.$$

Summing first over the nearby partner and then invoking the induction for the remaining marks yields the uniform local-tuple estimate

$$a^{15k/8} \sum_{\min_{i \neq j} |x_i - x_j| < \varepsilon} |\mathbb{E}[\sigma_{x_1} \cdots \sigma_{x_k}]| \leq C_k \varepsilon^{7/4}.$$

The same connectivity-partition argument treats + boundary conditions: the wired boundary block is kept as a distinguished block in the induction rather than being incorrectly treated as an ordinary free FK cluster. Hence the near-diagonal contribution vanishes uniformly as  $\varepsilon \downarrow 0$ .

Steps c–d give convergence of every moment of each finite collection  $\langle \Phi^a, f_1 \rangle, \dots, \langle \Phi^a, f_m \rangle$ . Step b permits the power series for

$$\mathbb{E} \exp \left( i \sum_{j=1}^m t_j \langle \Phi^a, f_j \rangle \right)$$

to be passed term by term to the limit, uniformly for  $(t_1, \dots, t_m)$  in a compact set. Thus every subsequential  $\mathcal{H}^{-3}$  limit has the same characteristic functional on a countable dense subset of  $\mathcal{H}^3$ . Continuity of a distribution on  $\mathcal{H}^3$  extends the equality to all  $\mathcal{H}^3$  tests. The characteristic functional determines a probability law on the separable Hilbert space, so the subsequential law is unique and the whole sequence converges.

Let  $\varphi : \Omega \rightarrow \tilde{\Omega}$  and write  $w = \varphi(z)$ ,  $z = \psi(w)$ . The Chapter-3 spin-correlation theorem transforms one spin by  $|\varphi'(z)|^{1/8} = |\psi'(w)|^{-1/8}$ , while the area element transforms by  $dA(z) = |\psi'(w)|^2 dA(w)$ . Therefore every insertion in the moment integral contributes

$$|\psi'(w)|^{2-1/8} = |\psi'(w)|^{15/8}.$$

The already established characteristic-functional uniqueness upgrades this moment transformation to the distributional identity  $\varphi_* \Phi_\Omega^\infty \stackrel{d}{=} |\psi'|^{15/8} \Phi_\Omega^\infty$ .

The proof follows the second Camia–Garban–Newman construction: Sobolev tightness, GHS exponential moments, CHI separated-point correlations, the  $\varepsilon^{7/4}$  local-tuple estimate, characteristic-function uniqueness, and then conformal covariance. The theorem is restricted to the smooth bounded simply connected +/free setting and the full-plane topology actually proved there; no collision/OPE statement is inserted.  $\square$

## 4.2 Conformal covariance and non-Gaussianity

A generalized field can be Gaussian even though no pointwise value exists, so non-Gaussianity has to be tested through smeared variables. Here the exact spin correlations give a particularly transparent obstruction to Wick's rule.

**Theorem 4.2** (Covariance and non-Gaussianity).

*The critical continuum magnetization field has the conformal covariance law of SM-12.04.01 and is not a Gaussian random distribution. Equivalently, there are four smooth compactly supported test functions whose fourth joint moment is not the Wick sum of pairwise second moments.*

*Proof. Proof roadmap.* The covariance part is now a consequence of the unique characteristic functional. For non-Gaussianity we extract from the already proved +-boundary CHI formula an explicit full-plane four-spin function and compare it with the Wick value at four separated collinear points.

The covariance identity was obtained in SM-12.04.01.f first for moments of compactly supported tests and then, by uniqueness of the characteristic functional, for the law of the random distribution. In particular all smeared moments are finite and transform by the 15/8 distributional weight.

We now pass the explicit +-boundary formula of Lemma 3.6 to the full plane. Put  $a_j = x_j + iY$  with fixed distinct real  $x_j$  and let  $Y \rightarrow \infty$ . For  $n = 2$  the formula tends to

$$\mathcal{S}_{\mathbb{C}}^{(2)}(x_i, x_j) = |x_i - x_j|^{-1/4}.$$

For  $n = 4$ , only sign vectors  $\mu \in \{\pm 1\}^4$  with  $\sum_j \mu_j = 0$  survive the limit. Pairing the vector  $\mu$  with  $-\mu$  gives

$$\mathcal{S}_{\mathbb{C}}^{(4)}(x_1, x_2, x_3, x_4) = \left[ \frac{1}{2} \sum_{\{ij\}|\{k\ell\}} \left( \frac{|x_i - x_j| |x_k - x_\ell|}{|x_i - x_k| |x_i - x_\ell| |x_j - x_k| |x_j - x_\ell|} \right)^{1/2} \right]^{1/2},$$

where the sum runs over the three unordered pair partitions. This calculation uses only separated points, so it does not invoke the later fusion/OPE gate.

Take  $(x_1, x_2, x_3, x_4) = (0, 1, 2, 3)$ . The three terms inside the preceding sum are

$$\frac{1}{2\sqrt{3}}, \quad \frac{2}{\sqrt{3}}, \quad \frac{\sqrt{3}}{2},$$

so the normalized Ising four-spin value is

$$\mathcal{S}_{\mathbb{C}}^{(4)}(0, 1, 2, 3) = \sqrt{\frac{2}{\sqrt{3}}}.$$

The full-plane critical law is invariant under global spin flip, so its one-point mean is zero. By contrast, a Gaussian field with the same two-point function would satisfy Wick's identity and hence would give

$$1 + \frac{1}{\sqrt{2}} + 3^{-1/4}.$$

These two positive numbers are unequal.

To turn the point-correlation calculation into a statement about the random distribution, choose nonnegative unit-mass mollifiers  $f_j^{(\eta)}$  supported in disjoint  $\eta$ -balls around the four points. The separated-point correlation theorem and dominated convergence imply that, as  $\eta \downarrow 0$ , the normalized second and fourth moments of  $\langle \Phi^\infty, f_j^{(\eta)} \rangle$  converge to the values just computed. If the field were Gaussian, Wick's identity would hold for every  $\eta$  and would survive the limit. The strict inequality above is a contradiction. Thus the field is non-Gaussian.

This proves non-Gaussianity directly from the CHI correlation owner already proved here in Chapter 3. It agrees with the two source routes emphasized by Camia–Garban–Newman: failure of Wick's formula and, independently, the much thinner  $\exp(-cx^{16})$  magnetization tail. The tail theorem is not used as a black box here and remains available for its own later proof obligation.  $\square$

### 4.3 Near-critical magnetic fields

The perturbation that survives the continuum limit is

$$H_a = h a^{15/8}.$$

For a bounded lattice domain  $D^a$  the near-critical measure is exactly an exponential tilt of the critical one:

$$\frac{d\mathbb{P}_{D,a}^h}{d\mathbb{P}_{D,a}^0}(\Phi) = \frac{\exp\{h\langle\Phi, 1_D\rangle\}}{\mathbb{E}_{D,a}^0 \exp\{h\langle\Phi, 1_D\rangle\}}.$$

The point of the next theorem is that this elementary identity remains stable when the mesh tends to zero and then when the domain tends to the plane.

**Theorem 4.3** (Near-critical magnetic scaling limit).

*Fix  $h > 0$  and put  $H_a = h a^{15/8}$ . On every bounded smooth domain with  $+$ ,  $-$ , or free boundary conditions, the near-critical magnetization field converges in  $\mathcal{H}^{-3}$  to the exponential tilt of the critical continuum field. These finite-domain limits are compatible with exhaustion and yield a whole-plane field  $\Phi^{\infty,h}$ ; moreover the square-grid whole-plane fields converge to it. Together with  $\Phi^{\infty,0} = \Phi^\infty$  this gives the positive magnetic one-parameter family of Camia–Garban–Newman. The negative- $h$  full-plane family is obtained by global spin flip.*

*For a dilation  $D_\lambda(z) = \lambda z$  the exact covariance is*

$$(D_\lambda)_* \Phi^{\infty,h} \stackrel{d}{=} \lambda^{-15/8} \Phi^{\infty, h\lambda^{-15/8}}.$$

*Equivalently,  $\Phi_{\lambda\Omega}^{\infty,h} \stackrel{d}{=} \lambda^{15/8} (D_\lambda)_* \Phi_\Omega^{\infty, h\lambda^{15/8}}$ .*

*Proof. Proof roadmap.* The bounded-domain theorem is a uniform-integrability argument for an exact Radon–Nikodym derivative. The passage to the plane is not a formal Kolmogorov-consistency statement: we couple a large plus box and the full-plane near-critical FK model and screen the interior by a ghost-connected open circuit. Finally the scaling law follows already at the lattice level.

At external field  $H_a = h a^{15/8}$ , multiplying the zero-field Ising weight by  $\exp(H_a \sum_{x \in D^a} \sigma_x)$  gives exactly

$$\exp\{h\langle\Phi^a, 1_D\rangle\}.$$

Therefore the displayed Radon–Nikodym formula is an identity, not an asymptotic approximation.

Let  $F$  be bounded and continuous on  $\mathcal{H}^{-3}(D)$ . From SM-12.04.01,  $\Phi^a \Rightarrow \Phi_D^\infty$ . The uniform exponential moments proved there imply uniform integrability of  $e^{h\langle\Phi^a, 1_D\rangle}$  and of the same variable multiplied by  $F$ . Hence

$$\mathbb{E}_{D,a}^h F(\Phi^a) \longrightarrow \frac{\mathbb{E}_D^0 [F(\Phi_D^\infty) e^{h\langle\Phi_D^\infty, 1_D\rangle}]}{\mathbb{E}_D^0 e^{h\langle\Phi_D^\infty, 1_D\rangle}}.$$

This proves the bounded-domain near-critical limit and identifies its law uniquely.

Use the FK representation with a ghost vertex. Conditional on an FK cluster  $C$ , the external field attaches  $C$  to the ghost with probability  $1 - e^{-2H_a|C|}$ . Compare the whole-plane model with the model having plus boundary on a large square  $Q_L$ . In each annulus of a fixed aspect ratio between a smaller box and  $\partial Q_L$ , the arbitrary-boundary RSW theorem SM-09.05.02 supplies, uniformly in  $a$ , a positive chance of an open circuit. The critical one-arm/second-moment estimates imply that with another uniformly positive conditional probability such a macroscopic circuit belongs to a

cluster with  $a^{15/8}|C|$  bounded below on the annular scale. Its probability to attach to the ghost is therefore bounded away from zero for fixed  $h$ . Using disjoint, geometrically increasing annuli, the probability that no screening ghost-connected circuit occurs tends to zero as  $L \rightarrow \infty$ , uniformly for sufficiently small  $a$ .

Once a common ghost-connected circuit is present, the domain Markov property makes the conditional FK/Ising laws inside that circuit identical in the two systems. We may therefore couple the plus-box and full-plane near-critical fields so that they agree on any prescribed inner box with probability tending to one as the outer box grows.

Equip full-plane distributions with the metric

$$d(\Phi, \Psi) = \sum_{m \geq 1} 2^{-m} \min\{1, \|\Phi - \Psi\|_{\mathcal{H}^{-3}(B_m)}\}.$$

Given  $\varepsilon > 0$ , first choose  $m_0$  so that the tail  $\sum_{m > m_0} 2^{-m} < \varepsilon/3$ . Then choose  $L$  so large that the screening coupling makes the two fields agree on  $B_{m_0}$  with probability at least  $1 - \varepsilon/3$ . This gives a coupling distance  $o_L(1)$  uniformly in small  $a$ . The bounded-domain convergence from Step b and the triangle inequality in this Polish metric now produce a unique whole-plane limit  $\Phi^{\infty, h}$  and convergence of the lattice fields to it.

The scaling relation is already exact before taking limits. Indeed  $D_\lambda$  sends the mesh  $a$  to  $\lambda a$  and

$$ha^{15/8} = (h\lambda^{-15/8})(\lambda a)^{15/8}, \quad a^{15/8} = \lambda^{-15/8}(\lambda a)^{15/8}.$$

Consequently

$$(D_\lambda)_* \Phi^{a, h} \stackrel{d}{=} \lambda^{-15/8} \Phi^{\lambda a, h\lambda^{-15/8}}.$$

Pass to the scaling limit to obtain the asserted continuum identity. This also removes the ambiguity in the earlier draft, which had said only that “rescaling multiplies  $h$  by  $\lambda^{15/8}$ ” without specifying whether one was pushing forward or pulling back.

The primary theorem is stated for  $h > 0$ . At  $h = 0$  it reduces to the critical field. On the full plane, global spin flip sends  $(h, \Phi) \mapsto (-h, -\Phi)$ , which defines the negative branch without changing the positive-field proof. The construction above matches the source division: finite-volume exponential tilt, a uniform large-box/full-plane coupling, and then exhaustion. It does not claim temperature-near-criticality or the mass exponent  $8/15$ ; those are separate later obligations.  $\square$

## 4.4 The exponents 1/15 and 8/15

The field exponent  $15/8$  predicts two reciprocal thermodynamic exponents. The critical magnetization in a small field should satisfy  $m(h) \asymp h^{1/15}$ , and the mass should scale like  $h^{8/15}$ . These are theorems, not merely dimensional predictions.

**Theorem 4.4** (Camia–Garban–Newman: magnetic exponent  $1/15$ ).

*For the nearest-neighbor Ising model on  $\mathbb{Z}^2$  at  $\beta = \beta_c$ , let*

$$m(h) := \langle \sigma_0 \rangle_{\beta_c, h}, \quad h > 0.$$

*There exist constants  $0 < c < C < \infty$  such that, for all sufficiently small  $h$ ,*

$$ch^{1/15} \leq m(h) \leq Ch^{1/15}.$$

*Equivalently,*

$$\lim_{h \downarrow 0} \frac{\log m(h)}{\log h} = \frac{1}{15}.$$

*The parent theorem is comparability, not the later strengthening  $m(h)/h^{1/15} \rightarrow B$ .*

*Proof.* Write  $\Lambda_L = [-L, L]^2 \cap \mathbb{Z}^2$  and

$$M_L = \sum_{x \in \Lambda_L} \sigma_x.$$

The critical spin-correlation results already owned in Chapter 3 give the two estimates used by Camia–Garban–Newman:

$$\langle M_L \rangle_{\beta_c, 0, +} \leq CL^{15/8}, \quad \langle M_L^2 \rangle_{\beta_c, 0, +} \leq CL^{15/4}.$$

Indeed, the first is obtained by summing the  $+$ -boundary one-point estimate  $\langle \sigma_x \rangle_+ \lesssim d(x, \partial\Lambda_L)^{-1/8}$  over dyadic boundary layers. For the second, Griffiths comparison bounds the summand  $\langle \sigma_x \sigma_y \rangle_+$  by a constant multiple of the critical two-point kernel at the relevant scale, and the already-owned  $|x - y|^{-1/4}$  estimate gives

$$\sum_{x, y \in \Lambda_L} (1 + |x - y|)^{-1/4} \lesssim L^{15/4}.$$

The powers  $15/8$  and  $15/4$  are therefore consequences of earlier physical owners, not an imported exponent assumption.

Let

$$F_L(h) := \langle M_L \rangle_{\beta_c, h, +} = \frac{d}{dh} \log \langle e^{hM_L} \rangle_{\beta_c, 0, +}.$$

GHS says that  $F_L''(h) \leq 0$  for  $h \geq 0$ . Hence  $F_L'$  is decreasing and

$$F_L(h) \leq F_L(0) + hF_L'(0) = \langle M_L \rangle_{0, +} + h \text{Var}_{0, +}(M_L) \leq C(L^{15/8} + hL^{15/4}).$$

By translation invariance in infinite volume and monotonicity under  $+$  boundary conditions,

$$m(h) \leq |\Lambda_L|^{-1} F_L(h) \leq C(L^{-1/8} + hL^{7/4}).$$

Balance the two terms by choosing  $L \asymp h^{-8/15}$ ; then both are  $O(h^{1/15})$ . Thus

$$m(h) \leq Ch^{1/15}.$$

This is the source upper-bound argument, including the reason one first dominates by  $+$  boundary conditions: GHS must be applied to a finite-volume field perturbation with the exterior frozen rather than simultaneously magnetized.

Use the Edwards–Sokal FK representation at  $h = 0$ . Let  $\mathcal{C}(0)$  be the open cluster of the origin and

$$A_M := \{|\mathcal{C}(0)| \geq M\}.$$

Choose  $R$  so that  $R^{15/8} \asymp M$ . Chapter 6 RSW gives, with probability bounded below uniformly in  $R$ , an open circuit in  $A(R/2, R)$  conditioned on the one-arm event  $0 \leftrightarrow \partial\Lambda_R$ . The one-arm probability is the  $+$ -boundary spin one-point function and hence is  $\asymp R^{-1/8}$ . On the one-arm-plus-circuit event set

$$N := |\mathcal{C}(0) \cap \Lambda_R|.$$

Summing conditional one-arm probabilities gives  $\mathbb{E}[N] \gtrsim R^{15/8}$ . A dyadic decomposition of pairs  $(x, y)$  together with RSW separation and the critical two-point/one-arm bounds gives  $\mathbb{E}[N^2] \lesssim R^{15/4}$ . Paley–Zygmund therefore yields

$$\mathbb{P}(N \geq cR^{15/8} \mid 0 \leftrightarrow R, \text{ circuit}) \geq c.$$

Consequently

$$\mathbb{P}_{\beta_c,0}(A_M) \geq cR^{-1/8} \asymp cM^{-1/15}.$$

This is exactly the second-moment/RSW lemma behind the source lower bound.

Implement a positive field by adjoining a ghost vertex  $\mathbf{g}$  and a ghost edge at every  $x \in \mathbb{Z}^2$  with field-dependent edge weight  $p_h = 1 - e^{-h}$ . Edwards–Sokal gives

$$m(h) = \mathbb{P}_{\beta_c,h}(0 \leftrightarrow \mathbf{g}).$$

The planar FK configuration with field stochastically dominates the critical one. Conditional on a planar cluster containing  $k$  vertices, finite energy for the ghost edges gives

$$\mathbb{P}(\mathcal{C} \leftrightarrow \mathbf{g} \mid \mathcal{C}) \geq 1 - (1 - ch)^k.$$

Thus, for every  $M$ ,

$$m(h) \geq \mathbb{P}_0(A_M) [1 - (1 - ch)^M].$$

Take  $M = \lfloor c_0/h \rfloor$ . The bracket is bounded below by a positive constant, while Step (c) gives  $\mathbb{P}_0(A_M) \gtrsim M^{-1/15}$ . Hence

$$m(h) \geq ch^{1/15}.$$

The matching bounds imply

$$\log c + \frac{1}{15} \log h \leq \log m(h) \leq \log C + \frac{1}{15} \log h.$$

Divide by  $\log h < 0$  and let  $h \downarrow 0$  to obtain the exact logarithmic exponent  $1/15$ . No claim that  $m(h)/h^{1/15}$  converges is made here; that stronger limit belongs to later work.

Camia–Garban–Newman, *The Ising magnetization exponent on  $\mathbb{Z}^2$  is  $1/15$*  (arXiv:1205.6612), prove precisely  $m(h) \asymp h^{1/15}$ . Their upper bound is the GHS estimate  $F_L(h) \leq F_L(0) + hF'_L(0)$  together with  $\langle M_L \rangle \lesssim L^{15/8}$  and  $\langle M_L^2 \rangle \lesssim L^{15/4}$ ; their lower bound uses the ghost FK model, RSW, a second-moment estimate for a cluster of size  $M$ , and the choice  $M \asymp h^{-1}$ . The proof below follows that architecture. The later asymptotic constant theorem is a separate result.  $\square$

**Theorem 4.5** (Camia–Jiang–Newman: magnetic mass exponent  $8/15$ ).

*For the critical Ising model on  $a\mathbb{Z}^2$  with magnetic field  $a^{15/8}h$ , there exist universal  $B_0, C_0 > 0$  such that, for  $a \in (0, 1]$  and  $0 < h \leq a^{-15/8}$ ,*

$$0 \leq \langle \sigma_x; \sigma_y \rangle_{a,h} \leq C_0 a^{1/4} |x - y|^{-1/4} e^{-B_0 h^{8/15} |x - y|}.$$

*The limiting Euclidean magnetization field  $\Phi^h$  therefore has exponentially decaying truncated covariance. If  $\widetilde{M}(H)$  denotes the lattice inverse correlation length at mesh one and  $M(\Phi^h)$  the continuum mass, then*

$$\widetilde{M}(H) \asymp H^{8/15} \quad (H \downarrow 0), \quad M(\Phi^h) = Ch^{8/15}$$

*for a constant  $C \in (0, \infty)$ . This is a magnetic-field, not temperature, near-critical theorem.*

*Proof.* Let

$$G_{a,h}(x, y) := \langle \sigma_x \sigma_y \rangle_{a,h} - \langle \sigma_x \rangle_{a,h} \langle \sigma_y \rangle_{a,h}.$$

For  $a = 1$  define  $\widetilde{M}(H)$  as the supremum of  $m > 0$  for which  $G_{1,H}(x, y) \leq C_m e^{-m|x-y|}$  for all  $x, y$ . For the continuum field define  $M(\Phi^h)$  by the analogous exponential bound on  $\text{Cov}(\Phi^h(f), \Phi^h(T_u g))$  for compactly supported smooth  $f, g$ . Griffiths gives  $G_{a,h} \geq 0$ , and GHS gives the monotonicity  $G_{a,h} \leq G_{a,0}$ .

Couple the Ising model to FK percolation with ghost vertex  $\mathbf{g}$ . Edwards–Sokal and conditioning on the FK configuration give the source identity

$$G_{a,h}(x, y) = \mathbb{P}_{a,h}(x \leftrightarrow y \not\leftrightarrow \mathbf{g}) - \text{Cov}_{a,h}(\mathbf{1}_{\{x \leftrightarrow \mathbf{g}\}}, \mathbf{1}_{\{y \leftrightarrow \mathbf{g}\}}).$$

FK association makes the covariance term nonnegative, hence

$$0 \leq G_{a,h}(x, y) \leq \mathbb{P}_{a,h}(x \leftrightarrow y \not\leftrightarrow \mathbf{g}).$$

Thus exponential decay follows once a long connection is forced to cross linearly many annuli in which a macroscopic FK cluster has been attached to the ghost and separates the two sides.

First normalize  $h = 1$  in continuum units. The critical FK loop/cluster-measure ensemble already owned in this project couples macroscopic cluster boundaries with renormalized area measures  $\mu_C$ . On a fixed-shape annulus, RSW and the cluster-measure second-moment estimates give constants  $p, c > 0$  such that, uniformly for sufficiently small mesh, with conditional probability at least  $p$  there is a separating FK cluster  $C$  with  $\mu_C \geq c$ . Under the magnetic Radon–Nikodym tilt, a cluster of renormalized area  $\mu_C$  receives plus/ghost attachment probability

$$\frac{e^{h\mu_C}}{2 \cosh(h\mu_C)} \quad \text{and hence a ghost-attachment probability bounded below when } h\mu_C \geq c'.$$

Equivalently, in the discrete ghost picture the chance that none of its ghost edges opens is exponentially small in the appropriately renormalized cluster size. Therefore each macroscopic annulus has, conditionally on the explored exterior, a probability  $q > 0$  of containing a ghost-attached separating screen. Apply the domain Markov property successively on disjoint annuli. Independence is not needed: the conditional failure probability is at most  $1 - q$  at every step. A connection from  $x$  to  $y$  avoiding  $\mathbf{g}$  must fail all  $N \asymp |x - y|$  screening trials, whence

$$\mathbb{P}_{a,1}(x \leftrightarrow y \not\leftrightarrow \mathbf{g}) \leq C e^{-c|x-y|}$$

for separations above a fixed macroscopic threshold, uniformly in  $a$ .

For general  $h$ , rescale space by  $\lambda = h^{8/15}$ . Since the magnetic coupling is  $a^{15/8}h$ , this sends the field strength to order one and the lattice spacing to  $ah^{8/15}$ . Hence the screening scale in the original coordinates is  $h^{-8/15}$  and the large-distance bound becomes  $C e^{-B_0 h^{8/15}|x-y|}$ .

For  $|x - y|$  below the magnetic scale, GHS and the critical two-point theorem give

$$G_{a,h}(x, y) \leq G_{a,0}(x, y) \leq C a^{1/4} |x - y|^{-1/4}.$$

Combining this with Step (c), after changing constants, yields the stated global lattice inequality

$$G_{a,h}(x, y) \leq C_0 a^{1/4} |x - y|^{-1/4} e^{-B_0 h^{8/15}|x-y|}.$$

Pair the lattice field  $\Phi^{a,h} = a^{15/8} \sum_x \sigma_x \delta_x$  against compactly supported  $f, g$ . Chapter 4 gives convergence  $\Phi^{a,h} \Rightarrow \Phi^h$  together with the uniform exponential integrability needed to pass second moments. Summing the preceding bound and taking  $a \downarrow 0$  therefore gives

$$|\text{Cov}(\Phi^h(f), \Phi^h(g))| \leq C \iint |f(x)g(y)| |x - y|^{-1/4} e^{-B_0 h^{8/15}|x-y|} dx dy.$$

In particular  $M(\Phi^h) > 0$ .



The near-critical field scaling already proved in ‘SM-12.04.03’ can be written in point-field notation as

$$\lambda^{1/8} \Phi^{h_0}(\lambda x) \stackrel{d}{=} \Phi^{\lambda^{15/8} h_0}(x).$$

Translation distance is multiplied by  $\lambda$ , so the definition of continuum mass gives

$$M(\Phi^{\lambda^{15/8} h_0}) = \lambda M(\Phi^{h_0}).$$

Take  $h_0 = 1$  and  $\lambda = h^{8/15}$ :

$$M(\Phi^h) = Ch^{8/15}, \quad C := M(\Phi^1).$$

Step (d) shows  $C > 0$ . To see  $C < \infty$ , use reflection positivity in one coordinate direction. The corresponding transfer operator  $T$  on a line-spin Hilbert space is positive self-adjoint after constants are removed. If the continuum mass were infinite, scaling plus convergence of block-spin observables would force every separated continuum covariance to vanish; equivalently the nonconstant spectral subspace of the limiting transfer semigroup would be zero. But two disjoint nonnegative test functions have strictly positive covariance by the FK finite-cluster event and its continuum limit. Thus  $0 < C < \infty$ .

The global exponential upper bound of Step (d) immediately gives

$$\widetilde{M}(H) \geq B_0 H^{8/15}.$$

For the reverse inequality, reflection positivity makes the mesh-one transfer matrix  $T_H - P_1$  positive semidefinite. If  $\widetilde{M}(H)$  is an admissible pointwise decay rate, spectral calculus upgrades it to the same exponential bound for all line-local observables. Suppose along  $H_n \downarrow 0$  that  $\widetilde{M}(H_n) > (C + \varepsilon) H_n^{8/15}$ . Rescale by  $H_n^{8/15}$  and take block-spin observables converging to fixed continuum test functions. The transfer-operator bound survives the scaling limit and would imply continuum covariance decay at every rate below  $C + \varepsilon$ , contradicting the definition  $M(\Phi^1) = C$ . Hence

$$\limsup_{H \downarrow 0} \frac{\widetilde{M}(H)}{H^{8/15}} \leq C,$$

and together with the lower bound this proves  $\widetilde{M}(H) \asymp H^{8/15}$  and the logarithmic mass exponent  $8/15$ .

Camia–Jiang–Newman, *Exponential decay for the near-critical scaling limit of the planar Ising model* (arXiv:1707.02668), prove the uniform bound  $G_{a,h}(x, y) \leq Ca^{1/4} |x - y|^{-1/4} e^{-Bh^{8/15} |x - y|}$ , pass it to the Euclidean magnetization field, deduce the exact continuum law  $M(\Phi^h) = Ch^{8/15}$ , and combine the FK lower mass bound with a reflection-positivity/transfer-matrix upper mass bound to obtain the lattice exponent  $8/15$ . Their proof explicitly uses coupled FK loop and cluster-measure ensembles for exponential screening; the matching upper lattice-mass estimate is treated separately by reflection positivity. The present owner preserves that division and does not replace the magnetic perturbation by a temperature-near-critical statement.  $\square$

## Chapter 5

# Interfaces: from discrete martingales to SLE

The correlation functions of the preceding chapters are static objects. We now let the boundary condition force a single macroscopic interface and ask for the scaling limit of that random curve. The proof has three logically separate pieces:

$$\begin{aligned} \text{crossing control} &\implies \text{curve/driver compactness,} \\ \text{curve/driver compactness} &\implies \text{martingale identification of the driver.} \end{aligned}$$

It is important not to reverse this order. Convergence of one discrete observable by itself does not rule out a lattice interface making a very thin macroscopic excursion that is almost invisible to half-plane capacity. The crossing estimate is the geometric input that excludes precisely this pathology.

Throughout this chapter  $(\Omega^\delta; a^\delta, b^\delta)$  denotes a simply connected Dobrushin domain converging to a bounded simply connected marked domain  $(\Omega; a, b)$ . On the square grid the statements below are exactly the classical spin/FK Ising interface theorems. The same argument applies to the bounded-angle isoradial class whenever the Chapter 2 fermionic convergence and the corresponding uniform crossing estimate are available. No arbitrary-planar-graph statement is being asserted here.

### 5.1 The minimum Loewner dictionary

Map  $(\Omega; a, b)$  conformally to  $(\mathbb{H}; 0, \infty)$ . If  $K_t$  is an increasing half-plane hull with

$$\text{hcap}(K_t) = 2t,$$

its hydrodynamically normalized map satisfies

$$g_t(z) = z + \frac{2t}{z} + O(z^{-2}) \quad (z \rightarrow \infty).$$

If the hull is generated by a boundary tip, its image is a real driving point  $W_t$  and

$$\partial_t g_t(z) = \frac{2}{g_t(z) - W_t}.$$

For the purposes of this chapter, a chordal  $\text{SLE}_\kappa$  hull chain is the Loewner chain driven by  $W_t = \sqrt{\kappa}B_t$ . We shall not import the later general trace theorem: our tight lattice curves themselves furnish continuous traces for the subsequential Brownian-driven chains that arise below.

We use the unparametrized curve metric

$$d(\gamma_1, \gamma_2) = \inf_{\phi_1, \phi_2} \|\gamma_1 \circ \phi_1 - \gamma_2 \circ \phi_2\|_\infty,$$

where the infimum is over increasing reparametrizations.

**Lemma 5.1** (A coefficient-extraction form of Lévy's test). *Let  $W$  be a continuous adapted process. If both  $W_t$  and  $W_t^2 - \kappa t$  are local martingales, then  $W_t = \sqrt{\kappa} B_t$  in its own filtration.*

*Proof.* Since  $W$  is a continuous local martingale,  $W^2 - \langle W \rangle$  is a local martingale. Subtracting it from the assumed local martingale  $W^2 - \kappa t$  shows that the finite-variation process  $\langle W \rangle_t - \kappa t$  is also a local martingale, hence is constant. It vanishes at 0, so  $\langle W \rangle_t = \kappa t$ . Lévy's characterization, already proved in the stochastic-calculus foundations, gives the conclusion.  $\square$

## 5.2 Tightness under Dobrushin boundary conditions

**Theorem 5.2** (Dobrushin-interface compactness).

*For the critical spin-Ising and FK-Ising Dobrushin explorations in the square-grid setting, and in the bounded-angle isoradial setting described above, the family of unparametrized interfaces is tight. After mapping to  $\mathbb{H}$  and parametrizing by half-plane capacity, the driving functions are tight in  $C[0, T]$  for every  $T < \infty$ . Every joint subsequential limit is generated by a continuous curve and is compatible with the Loewner equation.*

*Proof.* After a stopping time  $\tau$ , remove the explored initial segment and keep the component containing  $b^\delta$ . By the domain Markov property this is again a Dobrushin Ising (or FK-Ising) domain, with boundary data inherited from the two sides of the explored curve. Call a future crossing of an annulus *unforced* if the relevant component of the annulus does not separate the current tip from  $b^\delta$ . We shall prove that there is a universal modulus threshold  $m_0$  such that, conditionally on the past,

$$\mathbf{P}\{\text{an unforced crossing of an annulus of modulus at least } m_0\} \leq \frac{1}{2}. \quad (5.1)$$

This is the Kemppainen–Smirnov Condition G2, written in the only form used below.

Cut a sufficiently wide conformal rectangle crossing the annulus into  $m$  consecutive bounded-aspect-ratio blocks. Volume IX owns the critical FK  $q = 2$  RSW estimate uniformly over boundary conditions. Consequently, even after conditioning on everything exposed before a block is examined, the FK domain Markov property gives a probability at least  $c > 0$  of a primal or dual crossing of that block in the direction that blocks the interface. Therefore

$$\mathbf{P}\{\text{no blocking crossing in the } m \text{ blocks} \mid \mathcal{F}_\tau\} \leq (1 - c)^m. \quad (5.2)$$

Choose  $m$  so that  $(1 - c)^m \leq 1/2$ . An unforced FK exploration crossing implies the absence of all these separators, so (5.1) follows for FK-Ising.

For the spin interface, use the Edwards–Sokal coupling already established in Volume IX. A spin crossing through the same topological rectangle with alternating  $+/-$  boundary data forces the corresponding FK configuration to avoid a dual separator of the type used above. Conditioning successively on the explored blocks and using boundary-condition monotonicity gives the same geometric decay (5.2). Thus the spin interface also satisfies (5.1). In the bounded-angle isoradial class the identical argument uses the uniform crossing estimate transported by the isoradial fermion/star-triangle theory; no new continuum input is needed.

Suppose an annulus  $A(z; r, R)$  contains  $k$  disjoint concentric subannuli, each of modulus at least  $m_0$ . Traversing the interface chronologically,  $k$  successive crossings of  $A$  force, after suitable stopping times, an unforced crossing of each selected subannulus. Iterating (5.1) gives

$$\mathbf{P}\{\gamma^\delta \text{ makes } k \text{ such crossings of } A\} \leq 2^{-k}. \quad (5.3)$$

Choosing the number of subannuli proportional to  $\log(R/r)$  turns (5.3) into a power-law multi-crossing estimate

$$\mathbf{P}\{k \text{ disjoint crossings of } A(z; r, R)\} \leq C_k (r/R)^{\lambda(k)}, \quad (5.4)$$

with  $\lambda(k) \rightarrow \infty$ .

We spell out why (5.4) implies curve tightness instead of citing a later volume. If no annulus  $A(x; c_1 s^{1+\eta}, c_2 s)$  has  $k$  crossings, then a traversal stopping each time the curve moves distance  $s$  can visit any ball of radius  $s^{1+\eta}$  only  $O(k)$  times. Covering the bounded domain by  $O(s^{-2(1+\eta)})$  such balls gives a tortuosity bound

$$M(\gamma, 2s) \leq Ck s^{-2(1+\eta)}. \quad (5.5)$$

A grid union bound together with (5.4), and then a dyadic Borel–Cantelli/tightness argument, makes the random prefactor in (5.5) tight. As in the elementary  $q$ -variation argument, a bound  $M(\gamma, s) \leq Ks^{-p}$  yields a  $1/q$ -Hölder reparametrization for every  $q > p$ . Arzelà–Ascoli therefore gives tightness in the unparametrized curve metric.

Work after a stopping time in the conformally mapped future domain  $(\mathbb{H}; 0, \infty)$ . There are absolute constants  $c, C$  with the following deterministic implication: if during a capacity interval of length  $u$  the driver moves by more than  $L\sqrt{u}$ , then a Whitney/crosscut decomposition of the future half-plane produces at least  $cL$  well-separated topological quadrilaterals (equivalently, a chain of semiannular channels) that the trace must traverse and whose traversals are not forced by the two marked endpoints. The deterministic geometry uses the diameter–capacity comparison together with Koebe’s quarter theorem for the hydrodynamically normalized map. No independence is asserted: at the successive first-entry stopping times, Condition G2 applies conditionally to the unexplored domain. Iterating those conditional bounds gives

$$\mathbf{P}\left(|W_{t+u}^\delta - W_t^\delta| > L\sqrt{u} \mid \mathcal{F}_t\right) \leq Ce^{-cL}. \quad (5.6)$$

Dyadic chaining now gives, for every  $T, \varepsilon > 0$ ,

$$\limsup_{h \downarrow 0} \sup_{\delta} \mathbf{P}\left(\sup_{\substack{s, t \leq T \\ |t-s| \leq h}} |W_t^\delta - W_s^\delta| > \varepsilon\right) = 0, \quad (5.7)$$

which is tightness in  $C[0, T]$ . Notice that (5.6) also supplies the exponential integrability needed later when coefficients of martingale observables are extracted at infinity.

Choose a subsequence on which both curve and driver converge. Away from the growing hull, uniform convergence of the drivers and Gronwall’s inequality for

$$\dot{g}_t(z) = \frac{2}{g_t(z) - W_t}$$

give locally uniform convergence of the conformal maps. Thus every Carathéodory kernel limit of the slit domains is the Loewner hull driven by the limiting continuous  $W$ .

An extra “hidden” component cannot be swallowed without the approximating lattice curve entering and leaving a macroscopic pocket through a neck whose modulus tends to infinity. In the future conformal domain this is an unforced crossing of arbitrarily large modulus. Iterating (5.1) makes its probability tend to zero. Hence the limiting hull is generated by the limiting continuous curve itself.

The argument is the Condition G/G2 compactness architecture used in the strong-convergence paper of Chelkak–Duminil-Copin–Hongler–Kemppainen–Smirnov and in the Kemppainen–Smirnov random-curve framework. In particular, it proves curve tightness, continuous-driver tightness, and curve/Loewner compatibility before any SLE parameter is identified. The square-grid theorem is the exact classical statement; the bounded-angle isoradial extension uses precisely the isoradial observable/crossing inputs frozen earlier in this volume and in Volume IX.  $\square$

### 5.3 Spin Ising: why the SLE parameter is 3

**Theorem 5.3** (Critical spin interface converges to  $\text{SLE}_3$ ).

*Let  $(\Omega^\delta; a^\delta, b^\delta)$  approximate a bounded simply connected marked domain  $(\Omega; a, b)$  and impose critical spin-Ising Dobrushin boundary conditions – on one boundary arc and + on the other. Then the law of the interface converges weakly in the unparametrized curve metric to chordal  $\text{SLE}_3$  from  $a$  to  $b$ . The statement includes the bounded-angle isoradial extension under the Chapter 2 normalization hypotheses.*

*Proof.* Let  $\gamma_0^\delta, \dots, \gamma_n^\delta$  be the explored initial segment and let  $\Omega_n^\delta$  be the component of the slit domain containing  $b^\delta$ . The spin fermion of Chapters 1–2, normalized at  $b^\delta$ , will be denoted  $F_n^\delta(z)$ . Conditional on the first  $n$  steps, the unexplored spins have exactly the critical Ising law in  $(\Omega_n^\delta; \gamma_n^\delta, b^\delta)$  with the induced Dobrushin data. The order–disorder switching representation therefore gives

$$\mathbf{E}[F_{n+1}^\delta(z) \mid \mathcal{F}_n^\delta] = F_n^\delta(z), \quad (5.8)$$

up to the first time  $z$  is separated from the target. This is not an additional integrability miracle: it is the domain Markov property applied to the same observable in the remaining domain.

Fix a continuum point  $z$  and stop before its image approaches the growing hull and before the target normalization neighborhood is touched. Chapter 2 gives convergence of the fermion in every Carathéodory-convergent Dobrushin domain. The convergence is uniform over the stopped slit domains: otherwise a bad sequence of slit domains would have a Carathéodory-convergent subsequence, and Chapter 2 applied to that subsequence would contradict the assumed failure of uniform convergence. Along any subsequence supplied by SM-12.05.01, (5.8) therefore converges to a continuum local martingale.

Map the original domain to  $\mathbb{H}$ , write  $w = w(z)$ , and put

$$G_t(w) = g_t(w) - W_t.$$

The limiting spin observable from Chapter 2 is

$$M_t^{\text{spin}}(w) = (\partial_w[-G_t(w)^{-1}])^{1/2} = \frac{\sqrt{g'_t(w)}}{G_t(w)}, \quad (5.9)$$

with the boundary-determined square-root branch. Thus (5.9) is a local martingale for every  $w$  until swallowing.

Stop additionally when  $|W_t|$  exceeds  $R$ . Uniformly for  $t \leq T$  before this stopping time, the Loewner expansion gives

$$g_t(w) = w + \frac{2t}{w} + O_{R,T}(w^{-2}), \quad \sqrt{g'_t(w)} = 1 - \frac{t}{w^2} + O_{R,T}(w^{-3}). \quad (5.10)$$

Consequently

$$\frac{1}{G_t(w)} = \frac{1}{w} + \frac{W_t}{w^2} + \frac{W_t^2 - 2t}{w^3} + O_{R,T}(w^{-4}), \quad (5.11)$$

and multiplying (5.10)–(5.11) yields the decisive expansion

$$M_t^{\text{spin}}(w) = \frac{1}{w} + \frac{W_t}{w^2} + \frac{W_t^2 - 3t}{w^3} + O_{R,T}(w^{-4}). \quad (5.12)$$

This coefficient computation is the reason the value 3 appears.

For fixed large  $w$ , the stopped process in (5.9) is bounded and hence a true martingale. Take conditional expectations between  $s$  and  $t$ . Subtract  $w^{-1}$ , multiply by  $w^2$ , and let  $w \rightarrow \infty$ . The remainder in (5.12) is uniform under the stopping, so

$$\mathbf{E}[W_t \mid \mathcal{F}_s] = W_s. \quad (5.13)$$

Now subtract also  $W_t w^{-2}$ , multiply by  $w^3$ , and use (5.13). We obtain

$$\mathbf{E}[W_t^2 - 3t \mid \mathcal{F}_s] = W_s^2 - 3s. \quad (5.14)$$

Finally let  $R \rightarrow \infty$ . The exponential driver tail (5.6) gives the required uniform integrability on compact time intervals. Hence (5.13)–(5.14) hold locally without the auxiliary stopping.

The limiting driver is continuous by SM-12.05.01, and (5.13)–(5.14) are exactly the hypotheses of Lemma 5.1 with  $\kappa = 3$ . Therefore

$$W_t = \sqrt{3}B_t. \quad (5.15)$$

The corresponding hull chain is the chordal Brownian Loewner chain with parameter 3.

SM-12.05.01 already says that every subsequential limit curve generates the hull chain of its limiting driver and that hidden excursions are excluded. Thus every subsequential curve limit has the chordal SLE<sub>3</sub> law. Tightness then upgrades uniqueness of subsequential limits to convergence of the full family in the unparametrized curve metric. No later general SLE trace theorem is required.

The square-grid argument matches the strong-convergence theorem of Chelkak–Duminil-Copin–Hongler–Kemppainen–Smirnov: Condition G supplies geometric compactness, the convergent spin fermion is a martingale in every slit domain, and the coefficient-at-infinity calculation identifies the Brownian speed. The isoradial clause uses only the bounded-angle fermion universality already owned in Chapter 2, exactly the extension indicated in the source literature.  $\square$

## 5.4 FK Ising: why the SLE parameter is 16/3

**Theorem 5.4** (Critical FK interface converges to SLE<sub>16/3</sub>).

*Let  $(\Omega^\delta; a^\delta, b^\delta)$  carry the critical FK-Ising model with free boundary conditions on one Dobrushin arc and wired boundary conditions on the other, and let  $\gamma^\delta$  be the primal/dual exploration interface. In the classical square-grid setting, and in the bounded-angle isoradial extension supported by Chapters 1–2, the law of  $\gamma^\delta$  converges weakly in the unparametrized curve metric to chordal SLE<sub>16/3</sub> from  $a$  to  $b$ .*

*Proof.* The FK parafermion of Volume IX/Chapters 1–2 may be written as the conditional expectation of the phase-weighted event that the Dobrushin exploration uses a prescribed medial edge. Revealing one more interface edge and applying the FK spatial Markov property therefore gives

$$\mathbf{E}[F_{n+1}^\delta(z) \mid \mathcal{F}_n^\delta] = F_n^\delta(z). \quad (5.16)$$

As in the spin case, the stopped slit domains satisfy the hypotheses of the Chapter 2 fermion convergence theorem uniformly by the same compactness-contradiction argument.

In  $(\mathbb{H}; W_t, \infty)$  the strip map has derivative proportional to  $(w - W_t)^{-1}$ . Conformal half-differential covariance therefore turns the limiting FK observable into

$$M_t^{\text{FK}}(w) = \sqrt{\frac{g'_t(w)}{g_t(w) - W_t}} = \sqrt{g'_t(w)} G_t(w)^{-1/2}, \quad (5.17)$$

up to a deterministic normalization that cancels from the martingale argument. Equation (5.16) and Chapter 2 convergence imply that (5.17) is a local martingale until swallowing.

Under the same stopping  $|W| \leq R$ , expand

$$G_t(w)^{-1/2} = w^{-1/2} \left[ 1 + \frac{W_t}{2w} + \frac{3W_t^2/8 - t}{w^2} + O_{R,T}(w^{-3}) \right]. \quad (5.18)$$

Multiplying by  $\sqrt{g'_t(w)} = 1 - tw^{-2} + O(w^{-3})$  gives

$$M_t^{\text{FK}}(w) = w^{-1/2} \left[ 1 + \frac{W_t}{2w} + \frac{3W_t^2/8 - 2t}{w^2} + O_{R,T}(w^{-3}) \right]. \quad (5.19)$$

Thus the first random coefficient is  $W_t/2$ , while the next one is  $3W_t^2/8 - 2t$ .

Apply the martingale identity for (5.17), first after subtracting the deterministic leading term and multiplying by  $w^{3/2}$ , then after subtracting the  $W_t/(2w)$  term and multiplying by  $w^{5/2}$ . Uniformity of (5.19) under the stopping yields that

$$W_t \quad \text{and} \quad \frac{3}{8}W_t^2 - 2t$$

are local martingales. Equivalently,

$$W_t^2 - \frac{16}{3}t \quad (5.20)$$

is a local martingale. Letting the driver cutoff tend to infinity is justified by the exponential increment tail from SM-12.05.01. Lemma 5.1 now gives

$$W_t = \sqrt{\frac{16}{3}} B_t. \quad (5.21)$$

Every subsequential curve limit is, by SM-12.05.01, the continuous trace generating the Loewner hull driven by (5.21). Hence it has chordal  $\text{SLE}_{16/3}$  law. Since the family is tight and every subsequence has the same limit, the complete sequence converges in the unparametrized curve metric.

The square-grid result is exactly the FK theorem in Chelkak–Duminil-Copin–Hongler–Kemppainen–Smirnov. Their proof combines the Kemppainen–Smirnov crossing framework, the critical FK fermion of Smirnov, and the half-differential convergence theorem; the coefficient in (5.19) is the standard source calculation giving  $W_t^2 - (16/3)t$ . The bounded-angle isoradial extension uses the Chelkak–Smirnov universality theorem already proved here in Chapter 2. Nothing here claims the later branching-SLE/CLE full interface ensemble, radial exploration, or the broader non-isoradial s-embedding theorem.  $\square$

## 5.5 What has and has not been proved

The chapter has now closed the single-interface chain

Dobrushin crossing control  $\longrightarrow$  curve/driver compactness  $\longrightarrow$  spin  $\text{SLE}_3$  and FK  $\text{SLE}_{16/3}$ .

It has *not* proved convergence of the entire collection of cluster-boundary loops. That requires a branching exploration, simultaneous control of infinitely many branches, and loop-tail estimates; those are the separate Chapter 7 gates. Likewise Chapter 6 still owns the stronger standalone FK crossing theory used later for recursive and near-critical applications, even though the narrower crossing consequence needed for the present Dobrushin interfaces has been proved locally above.



## Chapter 6

# FK–Ising crossing theory

The interface theorems of the preceding chapter need a crossing theorem that survives after the unexplored domain has acquired a rough, random boundary. A box-crossing estimate is not enough for that purpose. The correct geometric quantity is discrete extremal length: it records the electrical difficulty of travelling from one marked boundary arc to the opposite one and is stable under the surgery produced by exploration paths.

### 6.1 Uniform crossing estimates in rough topological rectangles

**Theorem 6.1** (Discrete-extremal-length RSW theorem for critical FK–Ising with arbitrary boundary conditions).

*Let  $(\Omega, a, b, c, d)$  be a finite simply connected square-lattice topological rectangle, with the marked boundary points listed counterclockwise, and let*

$$A = (ab), \quad B = (cd), \quad \ell_{\Omega}[A, B]$$

*be the unit-conductance discrete extremal length between the two arcs. At the critical FK–Ising parameter  $q = 2$ ,  $p = p_c = \sqrt{2}/(1 + \sqrt{2})$ , the following holds. For every  $L > 0$  there is  $\eta(L) \in (0, 1)$ , independent of the mesh, the local geometry of  $\partial\Omega$ , and the FK boundary partition  $\xi$ , such that*

$$\ell_{\Omega}[A, B] \leq L \implies \phi_{\Omega}^{\xi}[A \leftrightarrow B] \geq \eta(L),$$

*and*

$$\ell_{\Omega}[A, B] \geq L^{-1} \implies \phi_{\Omega}^{\xi}[A \leftrightarrow B] \leq 1 - \eta(L).$$

*Thus a family of quads whose discrete extremal lengths stay in a compact subset of  $(0, \infty)$  has crossing probabilities uniformly bounded away from both 0 and 1, even after arbitrary boundary wirings.*

*Proof. Proof roadmap.* We first reduce the lower bound to free boundary conditions. The fermionic observable supplies point-to-arc connection estimates in terms of a random-walk partition function  $Z$ . A separator chosen at the scale  $Z^{1/2}$  splits the quad into two pieces; the domain Markov property then gives a second-moment estimate for a boundary-pair counting variable. Cauchy–Schwarz turns that estimate into a positive crossing probability. Discrete extremal length converts the resulting  $Z$ -bound into a function of  $L$ . The upper bound is the same lower-bound theorem in the dual quad.

For an increasing event, FK monotonicity says that wiring more boundary vertices can only increase its probability. Hence the lower crossing bound is hardest for free boundary conditions.

Remove every bridge of  $\Omega$  which does not separate  $A$  from  $B$ , together with the component hidden behind it; neither  $\ell_\Omega[A, B]$  nor the event  $A \leftrightarrow B$  changes. Each remaining bridge separates  $A$  from  $B$ . Unit current from  $A$  to  $B$  must cross every such bridge, so the effective resistance, hence the discrete extremal length, is at least their number. Consequently there are at most  $\lfloor L \rfloor + 1$  separating bridges. At criticality a bridge has a uniformly positive conditional chance to be open (finite energy), so it is enough to prove the theorem in each bridge-free component; the finite product of bridge-opening probabilities only changes  $\eta(L)$ .

Put bounded positive conductances on the boundary extension exactly as in the square-lattice fermionic-observable construction and denote by

$$Z_\Omega[E, F] = \sum_{x \in E} \sum_{y \in F} Z_\Omega[x, y]$$

the corresponding excursion/random-walk partition function from the boundary arc  $E$  to  $F$ . Discrete potential theory gives two uniform facts that we shall use repeatedly:

- (i)  $Z_\Omega[E, F]$  is comparable, with universal constants, to a monotone function of the discrete extremal length  $\ell_\Omega[E, F]$ ; therefore bounding  $\ell$  above and below bounds  $Z$  away from the two degenerate regimes;
- (ii) if  $Z_\Omega[E, F]$  is small, there is a cross-cut  $\Gamma$  separating  $E$  from  $F$  such that

$$Z_\Omega[E, F] \asymp Z_{\Omega_E}[E, \Gamma] Z_{\Omega_F}[\Gamma, F],$$

and the two factors can be chosen of comparable square-root size.

The second statement follows by sweeping level lines of the discrete harmonic measure: the first level at which the excursion mass crosses a prescribed value is a separator, and the strong Markov property of the walk factorizes the excursion mass across it.

For alternating wired/free data with a wired target arc  $F$ , the critical FK fermionic observable is s-holomorphic and its primitive has constant boundary values on the wired and free arcs. Comparing that primitive with discrete harmonic measure gives, uniformly in the domain,

$$\phi_\Omega^{(F)}[x \leftrightarrow F] \leq C Z_\Omega[x, F]^{1/2}. \quad (6.1)$$

By applying the same observable in the dual domain and then finite energy at the endpoint, one obtains for free boundary conditions

$$\phi_\Omega^\emptyset[x \leftrightarrow y] \geq c Z_{\Omega_{\text{int}}}^*[x^*, y^*]^{1/2} \quad (6.2)$$

whenever the indicated dual boundary points lie in the same interior dual component. These two inequalities are the only model-specific analytic input in the second-moment argument below.

In the bridge-free case define

$$N := \sum_{u \in A} \sum_{v \in B} \mathbf{1}_{\{u \leftrightarrow v\}}^{(1)} \mathbf{1}_{\{u \leftrightarrow v\}}^{(2)},$$

where, conditionally on the same domain, the two superscripts denote two independent free FK configurations. Then  $N > 0$  if and only if some pair  $(u, v)$  is connected in both copies, and in particular  $N > 0$  implies the crossing event in either copy. Independence and (6.2) give

$$\mathbb{E}N = \sum_{u \in A, v \in B} \phi_\Omega^\emptyset[u \leftrightarrow v]^2 \geq c Z_{\Omega_{\text{int}}}^*[A^*, B^*]. \quad (6.3)$$

This is the first-moment half of the proof.

Assume  $Z_\Omega[A, B] \leq \zeta_0$  for a sufficiently small universal  $\zeta_0$ . For  $u, v \in A$ , apply the separator construction separately to  $u$  and  $v$  against  $B$ . The two separators are disjoint; otherwise their union would have excursion mass too small to separate all of  $A$  from  $B$ . Condition on the configuration between the separators and wire the newly exposed cross-cuts. The domain Markov property, monotonicity, and (6.1) then give

$$\phi_\Omega^{(B)}[u \leftrightarrow B, v \leftrightarrow B] \leq C \sqrt{\frac{Z_\Omega[u, B] Z_\Omega[v, B]}{Z_\Omega[(uv), B]}} \quad (6.4)$$

where  $(uv)$  is the subarc of  $A$  between  $u$  and  $v$ . The same estimate holds on  $B$  after interchanging the roles of the two sides.

Choose a separator  $\Gamma$  between  $A$  and  $B$  whose two excursion masses are comparable to  $Z_\Omega[A, B]^{1/2}$ . Expose the two independent FK configurations up to  $\Gamma$ . Applying (6.1) and (6.4) on the  $A$ -side and then on the  $B$ -side yields

$$\mathbb{E}N^2 \leq S_A S_B,$$

where, writing  $x_u = Z_{\Omega_A}[u, \Gamma]$ ,

$$S_A \leq C \sum_{u, v \in A} \frac{x_u x_v}{\left(\sum_{w \in (uv)} x_w\right)^{1/2}}.$$

Order the boundary vertices and put  $t_j = \sum_{i \leq j} x_i$ . Each summand is bounded by the integral of  $|s - t|^{-1/2}$  over the corresponding rectangles in the  $(s, t)$ -plane. Therefore

$$S_A \leq C \int_0^T \int_0^T |s - t|^{-1/2} ds dt = C' T^{3/2},$$

with  $T = \sum_{u \in A} x_u$ . Since  $T \asymp Z_\Omega[A, B]^{1/2}$ , we obtain

$$S_A \leq C Z_\Omega[A, B]^{3/4}, \quad S_B \leq C Z_\Omega[A, B]^{3/4}.$$

and hence, after correcting the preceding display,

$$\mathbb{E}N^2 \leq C Z_\Omega[A, B]^{3/2}. \quad (6.5)$$

By Cauchy-Schwarz,

$$\phi_\Omega^\emptyset[A \leftrightarrow B] \geq \mathbb{P}[N > 0] \geq \frac{(\mathbb{E}N)^2}{\mathbb{E}N^2}. \quad (6.6)$$

When  $\ell_\Omega[A, B]$  is bounded above by  $L$ , shrink  $A$  and  $B$  only if necessary until either an endpoint-to-endpoint extremal length is bounded or the new opposite-arc extremal length belongs to a compact interval  $[L_0, L_0 + C]$ . Shrinking can only decrease the crossing event. In the first case (6.2) itself gives the lower bound. In the second, the extremal-length-excursion comparison bounds both the numerator in (6.3) from below and the quantity in (6.5) from above by constants depending only on  $L$ . Equation (6.6) therefore yields

$$\phi_\Omega^\emptyset[A \leftrightarrow B] \geq \eta_-(L) > 0.$$

Monotonicity transfers this bound to every boundary condition  $\xi$ .

The upper bound is hardest for fully wired boundary conditions. In the bridge-free case planar FK self-duality gives

$$1 - \phi_{\Omega}^{\partial\Omega}[A \leftrightarrow B] = \phi_{\Omega^*}^{\emptyset}[B_{\text{side}}^* \leftrightarrow A_{\text{side}}^*],$$

up to the uniformly bounded endpoint modification required by the dual boundary convention. Discrete extremal-length duality gives

$$\ell_{\Omega^*}[B_{\text{side}}^*, A_{\text{side}}^*] \asymp \ell_{\Omega}[A, B]^{-1}.$$

If  $\ell_{\Omega}[A, B] \geq L^{-1}$ , the dual extremal length is bounded above by a constant depending only on  $L$ , so the already-proved lower bound in the dual domain gives

$$1 - \phi_{\Omega}^{\partial\Omega}[A \leftrightarrow B] \geq \eta_+(L).$$

Boundary monotonicity now gives the same upper bound for arbitrary  $\xi$ .

Every constant above depends only on the extremal-length bound and universal finite-energy/observable constants. No regularity of  $\partial\Omega$  entered the proof. Consequently the theorem may be applied after conditioning on an exploration path, exactly as required in Chapters 5 and 7.

This is the discrete-extremal-length form of the Chelkak–Duminil-Copin–Hongler theorem. We have retained its two separate hypotheses, arbitrary FK boundary conditions, the bridge reduction, the excursion-partition second-moment mechanism, and the dual proof of the upper bound; it is stronger than a standard rectangular RSW estimate and is not replaced by one.  $\square$

## 6.2 Alternating wired/free crossing observables

**Theorem 6.2** (Izyurov conformal limits of alternating wired/free FK-Ising connection probabilities).

*Let  $\Omega^{\delta} \rightarrow \Omega$  approximate a simply connected domain with  $2k$  distinct marked boundary points  $x_1, \dots, x_{2k}$  in counterclockwise order. Put alternating wired/free FK-Ising boundary conditions on the arcs, and let  $I \subset \{1, \dots, k\}$  be a nonempty set of wired arcs. Then the probability that all wired arcs indexed by  $I$  belong to the same FK cluster has a limit as  $\delta \rightarrow 0$ . The limit is conformally invariant. After mapping  $\Omega$  to the upper half-plane, it is an explicit quadratic-irrational function of the images of the marked points.*

*Proof. Proof roadmap.* The observable is most naturally constructed in the spin representation. Its square has a single-valued primitive with controlled boundary values, giving compactness. The continuum limit is the unique solution of a scalar Riemann boundary-value problem on a double cover. Edwards–Sokal then expresses every desired FK connectivity probability as a finite spin moment, hence as a ratio of boundary partition functions. Boundary values of the observable give adjacent partition-function ratios; telescoping gives all ratios and therefore every connection probability.

Write the wired arcs as

$$W_i = [x_{2i-1}, x_{2i}], \quad 1 \leq i \leq k,$$

and put free boundary conditions on the complementary arcs. Through Edwards–Sokal, wire one distinguished arc  $W_{i_0}$  to spin  $+1$  and assign to every other wired FK cluster an independent sign. Conditional on the resulting monochromatic boundary signs  $\sigma_i \in \{\pm 1\}$ , the model is the critical spin Ising model with fixed signs on the  $W_i$  and free boundary elsewhere. Consequently the event that  $W_{i_1}, \dots, W_{i_r}$  belong to the same FK cluster is exactly the spin moment

$$\mathcal{I}^{\delta}(i_1, \dots, i_r) = \mathbb{E}[\sigma_{i_1} \cdots \sigma_{i_r}]. \quad (6.7)$$

If  $Z_{\sigma_1 \dots \sigma_k}^\delta$  denotes the spin partition function with those boundary signs, (6.7) is a finite signed ratio of the  $Z_\sigma^\delta$ .

Insert the boundary disorder/fermion required to change sign across the marked points and define the s-holomorphic observable  $F_B^\delta$ . Its defining Kadanoff–Ceva sign makes it a spinor on the double cover branching at the boundary-condition changes. Along every straight boundary edge the projection rule fixes the argument of  $F^\delta$ , and the primitive

$$H^\delta = \text{Im} \int (F^\delta)^2$$

has constant boundary values on each marked arc. The discrete Cauchy–Riemann relation implies that  $H^\delta$  is subharmonic on one primal graph and superharmonic on the dual graph. Maximum-principle bounds therefore control  $F^\delta$  uniformly on compact subsets away from the marked points.

Map to  $\mathbb{H}$  and write  $y_j = \varphi(x_j)$ . Any locally uniform subsequential limit  $f$  of the correctly normalized observables is a holomorphic half-differential satisfying the real/pure-imaginary boundary conditions imposed by the alternating fixed/free arcs and the square-root singularities at the  $y_j$ . Thus

$$f(z) = \frac{P(z)}{\sqrt{\prod_{j=1}^{2k} (z - y_j)}}, \quad (6.8)$$

with a real polynomial  $P$  of the degree forced by the normalization, subject to the residue/jump conditions at the marked points. The branch is fixed by its sign on one reference boundary interval.

Suppose  $f_1, f_2$  satisfy the same continuum problem and normalization, and put  $g = f_1 - f_2$ . The function

$$h(z) = \text{Im} \int^z g(\zeta)^2 d\zeta$$

is single valued: the prescribed square-root monodromies cancel after squaring, and the residue conditions make every period real. Its boundary values vanish on each marked interval. The singularities are removable after multiplying by the appropriate square-root local coordinate, so the maximum principle gives  $h \equiv 0$ . The boundary projection condition then forces  $g \equiv 0$ . Hence every subsequence has the same limit and  $F_B^\delta \rightarrow f_{\Omega, B}$  locally uniformly.

If  $\psi : \Omega \rightarrow \tilde{\Omega}$  is conformal, uniqueness applied after change of variable gives the half-differential law

$$f_{\Omega, B}(z) = \sqrt{\psi'(z)} f_{\tilde{\Omega}, \psi(B)}(\psi(z)), \quad (6.9)$$

with the branch fixed by the reference interval. Thus every ratio of boundary amplitudes of  $f$  is conformally invariant.

For two adjacent sign patterns, the discrete observable evaluated at the boundary-change corner satisfies

$$\left| \frac{F_B^\delta(\vec{x}_{2i})}{F_B^\delta(\vec{x}_{2k})} \right| = \frac{Z_{- \dots - + \dots +}^\delta}{Z_{+ \dots +}^\delta}. \quad (6.10)$$

The left side is exactly the jump of the primitive  $H^\delta$  at that marked point. By local convergence of the observable and its standard square-root normalization,

$$\left| \frac{F_B^\delta(\vec{x}_{2i})}{F_B^\delta(\vec{x}_{2k})} \right| \longrightarrow \lim_{z \rightarrow y_{2i}} \left| \sqrt{\pi(z - y_{2i})} f_{\mathbb{H}, \varphi(B)}(z) \right|. \quad (6.11)$$

Applying (6.10) successively to sign changes shows that every ratio  $Z_\sigma^\delta / Z_{+ \dots +}^\delta$  is a telescoping product of such boundary-amplitude ratios.

Insert the limiting partition-function ratios into the finite signed expression (6.7). This proves existence and conformal invariance of every selected wired-arc connectivity probability. In the half-plane, the coefficients of  $P$  in (6.8) solve a finite linear system whose coefficients are rational functions of the  $y_j$  and the square roots of products of their differences. Cramer's rule therefore expresses the boundary amplitudes, their telescoping ratios, and hence  $\mathcal{I}(i_1, \dots, i_r)$  as quadratic-irrational functions of  $(y_1, \dots, y_{2k})$ . No claim is made here that arbitrary unmarked geometric connection events admit such a formula.

This is exactly the crossing-probability corollary of Izyurov's free-boundary fermionic-observable theorem: simply connected domains, finitely many alternating wired/free arcs, selected wired arcs required to lie in one cluster, and quadratic-irrational half-plane limits. The stronger interface-with-drift theorem from the same paper is reserved for the later free-boundary-interface gate.  $\square$

## Chapter 7

# Loop ensembles and their conformal limits

A single interface sees only one branch of the critical geometry. The full Ising and FK–Ising scaling limits are countable nested loop collections. Two additional difficulties appear: one must show that there cannot be arbitrarily many macroscopic loops hidden in a small region, and one must prove that the recursive exploration used on the lattice still reconstructs every macroscopic loop after passage to the limit. We handle those two points before attaching the names  $\text{CLE}_3$  and  $\text{CLE}_{16/3}$ .

### 7.1 Tightness and finite-target exploration

**Theorem 7.1** (Tightness of the critical spin-Ising and FK-Ising loop ensembles in Jordan domains).

*Let  $\Omega^\delta$  approximate a simply connected Jordan domain  $\Omega$ . For the critical spin Ising model with + boundary conditions, and for the critical FK–Ising loop representation with wired boundary conditions, the corresponding nested collections of interfaces are tight in the loop-collection topology  $d_{\mathcal{X}}$ : for every  $\varepsilon > 0$ , only finitely many loops of diameter at least  $\varepsilon$  need be retained, and these macroscopic subcollections are tight for uniform convergence up to reparametrization.*

*Proof. Proof roadmap.* The common compactness input is the rough-domain crossing theorem of Chapter 6. It gives polynomial multi-crossing estimates in annuli and hence Aizenman–Burchard tightness for every exploration branch. For FK, finitely many target points are explored simultaneously; the branch aimed at a target converges to the target-coupled  $\text{SLE}_{16/3}(-2/3)$  branch, and six-arm estimates control the domains cut out at double points. For spin loops, the Edwards–Sokal coupling places every spin loop inside an FK cut-out domain. A free-boundary arc ensemble, proved below as a local sublemma from the same spinor observable, discovers the spin loops that touch those cut-out boundaries. Iteration misses only loops of vanishing diameter.

For loops  $\gamma, \tilde{\gamma}$ , let

$$d_{\Gamma}(\gamma, \tilde{\gamma}) = \inf_{\theta} \sup_{t \in \mathbb{S}^1} |\gamma(t) - \tilde{\gamma}(\theta(t))|,$$

the infimum being over orientation-preserving reparametrizations. A collection belongs to  $\mathcal{X}$  when, for each  $\varepsilon > 0$ , it has only finitely many loops of diameter at least  $\varepsilon$ . The metric  $d_{\mathcal{X}}$  matches large loops optimally and weights the unmatched small loops by their diameters. Thus tightness reduces to two assertions: tightness of each finite family of macroscopic curves and tightness of the number of curves crossing any fixed annulus.

Fix an annulus  $A(z; r, R)$  with  $R/r \geq 4$ . If an exploration contains  $m$  disjoint curve pieces joining the two boundary components, then after ordering the pieces cyclically one can select  $\lfloor m/2 \rfloor$  disjoint subannuli in which, conditionally on the previously exposed pieces, a new unforced primal or dual crossing must occur. The unexplored component may have a highly irregular boundary, but Theorem SM-12.06.01 applies conditionally because its constants depend only on extremal length and not on boundary regularity. Hence for some  $c, C > 0$ ,

$$\mathbb{P}[N_A \geq m] \leq C(1 - c)^m. \quad (7.1)$$

Iterating over a geometric sequence of annuli gives the standard Aizenman–Burchard power bound for  $k$  disjoint crossings,

$$\mathbb{P}[A_k(r, R)] \leq C_k(r/R)^{\lambda_k}, \quad \lambda_k \uparrow \infty. \quad (7.2)$$

The proof uses only the conditional crossing estimate, so it applies to FK explorations, spin explorations, and to domains created recursively by either one.

Estimate (7.2) implies a uniform Hölder modulus for every macroscopic exploration curve after reparametrization. Cover  $\Omega$  by dyadic annuli. A failure of such a modulus at scale  $2^{-n}$  forces more than  $k$  crossings of one annulus with  $k$  proportional to the desired Hölder violation; summing (7.2) over the cover is finite for large  $k$ . The same counting shows that the number of pairwise disjoint macroscopic interfaces entering a fixed compact set is tight. Therefore every finite family of exploration branches is precompact and, for each  $\varepsilon > 0$ , the number of loops of diameter  $> \varepsilon$  is tight once we show that our exploration scheme actually discovers them.

Choose a boundary root  $a^\delta$  and finitely many target medial edges  $z_1^\delta, \dots, z_m^\delta$ . Explore toward  $z_1^\delta$  using the FK interface. When two target paths agree up to the first time at which the targets lie in distinct complementary components, keep the common initial segment and branch thereafter. The domain Markov property makes the branch to each target an FK interface in a Dobrushin domain. Chapter 5 gives the ordinary  $\text{SLE}_{16/3}$  limit away from target collapse. Near the collapsing target, stop when its conformal distance from the driving point is  $\epsilon$ , apply the same fermionic martingale at that stopping time, and then let  $\epsilon \downarrow 0$ . The resulting driving pair  $(W_t, O_t)$  satisfies

$$dW_t = \sqrt{16/3} dB_t - \frac{2/3}{W_t - O_t} dt, \quad dO_t = \frac{2}{O_t - W_t} dt, \quad (7.3)$$

up to the swallowing time. This is the target-coupled  $\text{SLE}_{16/3}(-2/3)$  branch. Pathwise uniqueness away from collision and the cutoff construction give a consistent joint law for every finite target set.

For two targets  $z_i, z_j$ , the discrete exploration is literally identical until the first separating time. Taking a joint subsequential limit and using convergence of the Loewner chains shows the same property in the continuum. Thus the finite-target laws are projectively consistent. A countable dense set of targets therefore defines one branching tree; changing the dense set does not change its law because every finite subfamily has the same target-coupled law. This proves the particular branching-SLE construction needed here rather than importing a later abstract CLE construction.

Fix  $\varepsilon > 0$ . Start one branch between two well-separated boundary points. Its complementary components have wired, free, or mixed boundary conditions. In each mixed component of diameter at least  $\varepsilon$ , run the interface separating its wired and free arcs and concatenate it with the appropriate branch segment; this closes an FK loop. Continue inside every wired component of diameter at least  $\varepsilon$ . By (7.1), at each stage there are only tight-many macroscopic components. Moreover, if a limiting branch had a double/contact point not approximated by discrete double/contact points, six alternating arms would occur from microscopic to macroscopic scale. The six-arm bound furnished by Chapter 6 and the previously owned arm technology makes this probability vanish. Consequently discrete cut-out domains converge to the complementary domains of the limiting branches.



At each iteration the maximal diameter of an unexplored wired component decreases in probability. Otherwise one could extract infinitely many separated macroscopic crossings of a fixed annulus, contradicting (7.1). Hence after finitely many iterations, with probability  $1 - o_\varepsilon(1)$ , every undiscovered level-one FK loop has diameter below  $\varepsilon$ . Repeating inside the cut-out domains gives the same statement at all nesting levels.

We need one additional ingredient for spin loops, because with  $+$  boundary conditions they do not touch  $\partial\Omega$ . In a domain with free spin boundary conditions, mark finitely many boundary prime ends and expose the leftmost spin interface joining the first marked pair. The free-boundary fermionic observable has the same s-holomorphic compactness mechanism as Chapter 6; its continuum half-differential martingale yields the force-point system

$$\text{SLE}_3(-3/2, -3/2) \tag{7.4}$$

for a single free arc. Applying (7.2) conditionally after each stopping time gives joint tightness for finitely many arcs. The discrete arcs agree until their targets separate, so the finite-target limits are consistent and define the Free Arc Ensemble. As for FK, the annulus estimate implies that, for every  $\varepsilon$ , finitely many such branches discover all free-boundary arcs of diameter  $> \varepsilon$ . This is precisely the free-arc input used below; the more general plus/minus/free interface theorem remains a separate later gate.

Couple the  $+$  Ising model to wired FK–Ising. Every spin interface is contained in the dual FK configuration and therefore lies inside an FK cut-out domain. Condition on the outermost FK loops. Inside each cut-out domain the Ising model has free boundary conditions. Use the Free Arc Ensemble to discover every spin loop that touches the cut-out boundary. Once such loops are found, resample the FK model independently in the remaining domains bordered by  $+$  spins and repeat. The conditional law at every step is again a critical FK or free-Ising model in a random domain, so (7.1) remains valid. If a macroscopic spin loop were never discovered, it would produce an infinite nested sequence of unexplored domains of diameter bounded below; that again forces arbitrarily many crossings of one annulus and has probability zero. Thus the recursive exploration captures all macroscopic spin loops.

For fixed  $\varepsilon$ , the FK and spin exploration schemes reduce the loops of diameter  $> \varepsilon$  to a tight finite family of branches, each of which is tight by (7.2). Arzelà–Ascoli up to reparametrization gives tightness of that macroscopic family. Letting  $\varepsilon$  run through  $2^{-n}$  and diagonalizing proves tightness in  $\mathcal{X}$  for the full nested collections.

The FK part follows the Kemppainen–Smirnov full-loop architecture: target-coupled branching  $\text{SLE}_{16/3}(-2/3)$ , special-point/six-arm control, and recursive recovery of every macroscopic loop. The spin part follows Benoist–Hongler: FK cut-out domains, free Ising arcs, recursive discovery, and macroscopic-completeness estimates. The theorem is deliberately a tightness/convergence-framework statement; identification of the limiting laws is the next parent.  $\square$

## 7.2 Identification with the conformal loop ensembles

**Theorem 7.2** (Full critical FK–Ising loop limit is  $\text{CLE}_{16/3}$  and full plus-boundary spin–Ising loop limit is nested  $\text{CLE}_3$ ).

*In the settings of the preceding theorem, the complete critical FK–Ising interface ensemble converges to  $\text{CLE}_{16/3}$ , and the complete collection of  $+/-$  spin interfaces for critical Ising with  $+$  boundary conditions converges to nested  $\text{CLE}_3$ . For the spin theorem the domain is a simply connected Jordan domain and convergence is in the loop-collection metric  $d_{\mathcal{X}}$ .*

*Proof. Proof roadmap.* Tightness is already known, so it remains to identify every subsequential limit. For FK the branching tree itself determines the loop ensemble: each loop is obtained when two target branches split and later meet. This gives the standard branching-SLE definition of  $\text{CLE}_{16/3}$ . For spin loops, the recursive FK/free-arc construction gives a conformally invariant collection of simple disjoint loops, and the lattice spatial Markov property passes to the limit. A countable-dense-target exploration proves uniqueness of any loop law with these properties and  $\text{SLE}_3$  exploration branches; this is the local Markovian characterization of  $\text{CLE}_3$  needed here.

Let  $\mathcal{T}^\delta$  be the finite-target FK exploration tree. By Theorem SM-12.07.01 every subsequence contains a further subsequence on which the branches converge jointly. Each marginal is the target-coupled solution of (7.3), and two branches coincide exactly until their targets are separated. The Brownian/force-point system has unique law under this target-coupling rule, so all finite-dimensional tree laws are unique. Hence the entire countable dense-target branching tree has a unique subsequential law.

For  $\varepsilon > 0$ , the recursive procedure in the preceding proof constructs every FK loop of diameter  $> \varepsilon$  as a concatenation of finitely many branch segments, except on an event whose probability tends to zero with  $\varepsilon$ . Special-point control rules out a continuum pinch whose discrete approximants do not carry the corresponding six arms. Therefore the map from the finite target tree to the  $\varepsilon$ -macroscopic loops is continuous outside a null set. Since the tree law is unique, so is the law of the macroscopic loops. Letting  $\varepsilon \downarrow 0$  identifies the complete nested limit.

For  $\kappa \in (4, 8)$  define, for the present purpose, the branching-SLE loop ensemble as follows: start the target-coupled family  $\text{SLE}_\kappa(\kappa - 6)$  from one root toward a countable dense set, and declare the loops to be the boundaries traced when branches separate targets and close around the cut-out components. Projective consistency makes this independent of the chosen dense target set. This construction is the standard branching definition of  $\text{CLE}_\kappa$ . With  $\kappa = 16/3$ , parts (a)–(b) therefore identify the FK limit as  $\text{CLE}_{16/3}$ . Volume XV will develop CLE intrinsically and prove equivalent characterizations, but no theorem from that later volume is being used here.

The spin recursion constructs every macroscopic limiting loop by gluing finitely many arcs of the Free Arc Ensemble inside boundaries of FK cut-out domains. Those cut-out boundaries are simple and mutually disjoint: an external self-touching or contact of two such boundaries would force a six-arm event, ruled out by the arm estimate. Free-arc branches are simple and disjoint in each component. Loops discovered at later stages lie strictly inside components left by earlier stages. Hence every limiting spin loop is simple, distinct loops are disjoint, and no limiting loop contains a nontrivial boundary segment.

Every ingredient of the finite-stage exploration has a conformally covariant limit: the FK target branches, the free Ising arc branches, and the cut-out domains. The error made by stopping after finitely many stages is, with high probability, a collection of loops of diameter below  $\varepsilon$ . It follows first for the  $\varepsilon$ -macroscopic subcollection and then as  $\varepsilon \downarrow 0$  that the full spin-loop subsequential law is invariant under conformal maps between Jordan domains.

Explore the outermost spin loops intersecting a fixed deterministic compact exploration set and condition on those loops. On the lattice, the unexplored spins in distinct components are independent critical Ising models with the boundary sign prescribed by the surrounding loop. Approximate the conditioning event by a finite-stage exploration whose boundaries converge. Since the loop convergence is strong enough to identify the complementary domains, the discrete spatial Markov property passes to the limit: conditioned on the discovered loops, the unexplored collections in the complementary components are independent rescaled copies of the same loop law with the appropriate boundary sign. In particular the outermost loop ensemble has the conformal restriction/Markov property.

We prove the narrow uniqueness statement needed here. Suppose  $\mathcal{L}$  is a conformally invariant collection of disjoint simple loops satisfying the domain Markov property in every simply connected component, and suppose the boundary-to-boundary exploration obtained by following the leftmost sides of loops has the Free Arc Ensemble branch law with parameter 3. Choose a countable dense set  $Q \subset \Omega$ . Explore until every pair of points of  $Q_n = \{q_1, \dots, q_n\}$  lying in different loop interiors has been separated. The law of this finite exploration is forced by the branch law and the domain Markov property. Thus it is the same for any two candidate ensembles. Every loop of diameter  $> \varepsilon$  separates two points of  $Q_n$  for some  $n = n(\varepsilon)$ , except on an event tending to zero as  $n \rightarrow \infty$ . Hence the laws of all  $\varepsilon$ -macroscopic loops agree; letting  $\varepsilon \downarrow 0$  gives equality of the complete loop laws. The unique law is, by definition, nested  $\text{CLE}_3$ .

The branch law in the free-arc construction is  $\text{SLE}_3(-3/2, -3/2)$ , so the uniqueness argument already carries the parameter 3. Equivalently, a discovered spin loop contains subarcs locally absolutely continuous with respect to  $\text{SLE}_3$  and therefore has the  $\text{SLE}_3$  local geometry. Thus no different CLE parameter can satisfy the identified exploration law.

The FK conclusion is the Kemppainen–Smirnov theorem that the full critical random-cluster Ising loop ensemble converges to CLE, with branching  $\text{SLE}(16/3, -2/3)$  as the central reconstruction tool. The spin conclusion is the Benoist–Hongler theorem: critical Ising with  $+$  boundary conditions in a Jordan domain converges in  $d_{\mathcal{X}}$  to nested  $\text{CLE}_3$ . Their recursive FK/free-arc exploration, simplicity/disjointness analysis, and spatial Markov identification are retained here. The local CLE uniqueness argument has been included explicitly so this parent does not depend forward on Volume XV.  $\square$

## Chapter 8

# Random-current continuum geometry

Random currents give an Ising representation whose continuum geometry is different in appearance from the fermionic-interface route of the preceding chapters. Two independent sourceless currents are added, clusters of their sum are surrounded by nested interfaces, and independent signs attached to the clusters produce an integer-valued nesting field. The key fact is that the geometry and the field have a common continuum limit: the field is a Gaussian free field and the interfaces are thin local sets of that same GFF.

### 8.1 Double currents, the nesting field, and GFF local sets

**Theorem 8.1** (Critical double-random-current conformal limit and GFF local-set coupling).

Let  $D \subsetneq \mathbb{C}$  be a bounded simply connected Jordan domain and let  $D^\delta$  be square-lattice approximations converging to  $D$  in the Carathéodory sense. Couple the critical double random current with free boundary conditions on  $D^\delta$  to the dual double random current with wired boundary conditions on the weak dual. Let  $h_{D^\delta}$  and  $(h^+)_{(D^\delta)^\dagger}$  be their nesting fields. Then, jointly as generalized functions,

$$h_{D^\delta}, (h^+)_{(D^\delta)^\dagger} \Longrightarrow \frac{1}{\sqrt{\pi}} h_D, \quad (8.1)$$

where  $h_D$  is the zero-boundary GFF normalized by

$$\mathbb{E}[h_D(f)h_D(g)] = \iint_{D \times D} f(x)G_D(x,y)g(y) dx dy, \quad -\Delta_y G_D(x,y) = \delta_x(y).$$

For free boundary conditions, the nested cluster-boundary ensemble is tight and converges in every such Jordan domain to the conformally invariant loop ensemble described below. If in addition  $\partial D$  is  $C^1$ , the corresponding wired-boundary loop theorem holds as well. Writing

$$\lambda = \sqrt{\pi/8},$$

the continuum loops are the corresponding two-valued thin local sets of the GFF. For free boundary conditions the outer boundaries of the outermost clusters form  $\text{CLE}_4$ , while the inner boundaries inside such a cluster are the appropriate  $\mathcal{L}_{-2\lambda, (2\sqrt{2}-2)\lambda}$  (or sign-reversed) two-valued-set loops. In the wired case the first inner-boundary layer is  $\mathcal{L}_{-\sqrt{2}\lambda, \sqrt{2}\lambda}$ . The same description then repeats recursively in the complementary components.

*Proof. Proof roadmap.* There are three logically separate pieces. First we construct the master coupling of primal current, dual current and one height function by passing through alternating

flows and dimers. Second we identify the inverse Kasteleyn kernel with the already-controlled Ising fermion and use the determinantal moment expansion to prove that the height converges to  $\pi^{-1/2}$  times the zero-boundary GFF. Third we prove tightness of current clusters, pass the master coupling to the limit, and use a local two-valued-set uniqueness argument to identify every possible loop subsequential limit.

A sourceless current is a function  $\mathbf{n} : E \rightarrow \{0, 1, 2, \dots\}$  with even incident sum at each vertex and weight

$$w(\mathbf{n}) = \prod_{e \in E} \frac{\beta_c^{\mathbf{n}_e}}{\mathbf{n}_e!}.$$

Let  $\mathbf{n} = \mathbf{n}_1 + \mathbf{n}_2$  be the sum of two independent sourceless currents. Its clusters are the components of  $\{e : \mathbf{n}_e > 0\}$ . For each cluster  $\mathcal{C}$  choose an independent sign  $\epsilon_{\mathcal{C}} \in \{\pm 1\}$ . A cluster is *odd around* a face  $u$  when the total current flux across a dual path from  $u$  to the outer face is odd. The free nesting field is

$$h_G(u) = \sum_{\mathcal{C}} \mathbf{1}_{\{\mathcal{C} \text{ odd around } u\}} \epsilon_{\mathcal{C}}. \quad (8.2)$$

For wired boundary conditions the boundary/ghost cluster contributes  $(1/2 - \mathbf{1}_{\{\text{odd}\}})\epsilon_{\mathcal{C}_g}$ ; the field is then half-integer valued. These definitions are designed so that crossing one odd contour changes the field by one unit.

Expand each current edge weight into its parity and its positive-even part. The switching lemma says that, after summing over even multiplicities, changing the source set along a path only changes whether that path must lie in the union current. Orient every odd contour alternately around a vertex. This converts a current configuration into an alternating flow; conversely, forgetting orientations and independently restoring the positive-even edges reconstructs the current with the correct factorial weights.

Replace every vertex of the flow graph by the standard even cycle and every original edge by one long bipartite edge. A local alternating flow is then equivalent to a perfect matching on the decorated graph  $G^d$ : long dimers record oriented flow edges and the remaining short-cycle dimers fill the unused incidences. The two possible fillings of a zero-flow vertex produce exactly the factor 2 required by the flow weight. Vertex splitting and urban renewal transport this matching to the bipartite graph  $C_G$  without changing height increments outside the transformed cell. Applying the construction simultaneously to  $G$  and its dual yields one dimer height  $H$  on primal vertices and dual faces and, for adjacent  $v, u$ ,

$$H(u) - H(v) = \frac{1}{2} \tau_u^\dagger \tau_v. \quad (8.3)$$

Its restrictions are exactly the free primal and wired dual nesting fields. Moreover primal and dual positive-current edges are disjoint, and exploration of a primal cluster leaves a dual current with wired boundary conditions in each hole. This is the master coupling and its spatial Markov property.

Let  $K^\delta$  be the Kasteleyn matrix of  $C_{D^\delta}$ . For a fixed white vertex  $w$ , the function  $b \mapsto (K^\delta)^{-1}(b, w)$  solves the discrete Dirac equation away from  $w$ , has a unit source at  $w$ , and satisfies the boundary projection condition forced by the Kasteleyn orientation. The real Kadanoff–Ceva fermion from Chapters 1–2 satisfies the same three properties after the explicit local phase factor is removed. Their difference solves the homogeneous discrete Riemann boundary problem; the primitive of its square has zero boundary values, so the discrete maximum principle gives uniqueness. Hence the relevant entries of  $(K^\delta)^{-1}$  are exactly linear combinations of the already-owned critical fermionic observables.

By the convergence theorem for those observables, uniformly when the two arguments remain a positive macroscopic distance apart,

$$\delta^{-1}(K^\delta)^{-1}(z^\delta, w^\delta) \longrightarrow K_D^{-1}(z, w), \quad (8.4)$$

where the continuum kernel is the Dirac/Cauchy kernel with the zero-boundary image term. Near the diagonal, the discrete kernel is bounded by  $C/|z - w|$ ; this integrable singular estimate is enough for the moment calculation below.

A height increment is the signed flux of dimer occupation minus its mean through a dual path. Expanding a product of  $m$  such increments and using the determinantal dimer formula expresses its expectation as a sum over permutations of products of entries of  $(K^\delta)^{-1}$ . Decompose each permutation into cycles. After summing the orientations of the integration paths, every cycle of length at least three appears together with its reverse orientation and cancels. Only transpositions remain. Thus, for distinct macroscopic points, the limiting moments are Wick pairings.

For test functions  $f, g$ , (8.4), Riemann-sum convergence and the diagonal  $1/|z - w|$  bound give

$$\lim_{\delta \rightarrow 0} \mathbb{E}[h_{D^\delta}(f)h_{D^\delta}(g)] = \frac{1}{\pi} \iint f(x)G_D(x, y)g(y) dx dy. \quad (8.5)$$

The same cancellation shows that every limiting even moment is the Wick sum built from (8.5) and every odd moment is zero. Uniform moment bounds follow from the integrable kernel estimate, so moments determine the finite family  $\{h_{D^\delta}(f_j)\}_{j \leq m}$ . This proves convergence of finite-dimensional distributions to  $\pi^{-1/2}h_D$ .

The dimer computation alone does not determine the boundary condition. The current representation does. If a macroscopic free-current cluster entered an  $r$ -neighbourhood of  $\partial D^\delta$  from distance  $R$ , the boundary-connection event of the double-current estimate would occur. The switching-lemma/fermionic argument gives

$$\mathbb{P}[\Lambda_R \leftrightarrow \partial_r D^\delta] \leq \varepsilon(r/R), \quad \varepsilon(s) \downarrow 0. \quad (8.6)$$

Consequently the contribution to the nesting field from clusters touching the microscopic boundary layer vanishes after testing against a compactly supported smooth function. The limiting covariance is therefore the Dirichlet Green kernel, not a free- or mixed-boundary Green kernel. The same master height couples the dual wired field, so both restrictions converge jointly to the same  $\pi^{-1/2}h_D$ .

For a compact  $K \Subset D$ , the kernel estimate and (8.5) give a uniform  $H^{-1-\epsilon}(K)$  second-moment bound for every  $\epsilon > 0$ . Compact embedding into a slightly weaker negative Sobolev space gives tightness there. Since all test-function finite-dimensional limits are already identified, every subsequential distributional limit is  $\pi^{-1/2}h_D$ . This proves (8.1).

We next prove the geometric tightness input rather than borrowing it from the continuum answer. Let  $A_{2k}(r, R)$  be the event that  $k$  distinct clusters of  $\mathbf{n}_1 + \mathbf{n}_2$  cross the annulus  $\text{Ann}(r, R)$ . There are constants  $C_k$  and exponents  $\lambda_k \uparrow \infty$  with

$$\mathbb{P}[A_{2k}(r, R)] \leq C_k(r/R)^{\lambda_k}. \quad (8.7)$$

Indeed, first consider the odd part of one current. Four additional alternating odd crossings across an annulus force, after cutting at the first two crossings and applying the switching lemma, two separated critical Ising disorder connections across a subannulus. The fermionic two-point bound gives a factor at most  $C(r/R)^{\lambda_0}$ . Conditioning on the first  $2k$  crossings and repeating in the remaining components yields

$$\mathbb{P}[B_{2k+4}(r, R) \mid B_{2k}(r, R)] \leq C_0(r/R)^{\lambda_0},$$

so the exponent grows linearly with  $k$ . If the union of two currents has  $k$  separated crossing clusters, planar separation assigns, after discarding a bounded number of exceptional outer clusters, a positive fraction of the separating contours to the odd part of one of the two currents. A union bound over the two currents transfers the preceding estimate and gives (8.7). The same argument with the ghost cluster counted once gives the wired version needed in smooth approximating domains.

Two distinct macroscopic current clusters approaching within distance  $r$  while remaining distinct to scale  $R$  force a four-arm cluster/hole event. Switching at the first contact and using the fermionic/current connection estimate gives

$$\mathbb{P}[A_4^\square(r, R)] + \mathbb{P}[A_4^\blacksquare(r, R)] \leq C(r/R)^2. \quad (8.8)$$

After a union bound over  $O(R^2/r^2)$  possible centers and then choosing  $r/R$  sufficiently small, the probability of a macroscopic continuum contact with no discrete precursor tends to zero. Equations (8.7)–(8.8) imply tightness of the outer/inner cluster boundaries in the loop-collection topology and guarantee that subsequential loops are noncrossing and that their nesting relation is preserved.

Take a subsequence along which the master height and all loop families converge jointly. By (8.1) the field component is  $\pi^{-1/2}h_D$ . The exact discrete spatial Markov rule from part (b) survives at every inner boundary: conditionally on the limiting boundary, the field inside is the harmonic boundary value inherited from the discrete height jump plus an independent zero-boundary GFF. The no-thick-contact estimate ensures that no additional macroscopic boundary component is created or lost when passing to the limit. Therefore the limiting inner-boundary families are local sets of  $h_D$ .

We isolate the continuum fact used by the current argument. Let  $-a < 0 < b$  and assume  $a + b \geq 2\lambda$ . There is a unique bounded-type thin local set  $\mathbb{A}_{-a,b}$ , measurable from a zero-boundary GFF  $h$ , whose harmonic function takes only the values  $-a$  and  $b$  on the components of its complement.

Here is the construction in the form needed below. First recall the local level-line lemma. In a simply connected component  $U$  with piecewise constant boundary data for which the two boundary values adjacent to a marked starting prime end straddle a chosen level, reveal a curve  $\eta$  and prescribe the conditional mean to jump by  $2\lambda$  across its two sides. Mapping to  $(\mathbb{H}; 0, \infty)$ , the conditional mean at  $z$  is a constant multiple of  $\arg(g_t(z) - W_t)$ . It must be a martingale because it is a conditional expectation of the GFF. Itô's formula and the Hadamard variation of the Green function therefore force  $d\langle W \rangle_t = 4dt$  and zero drift, so the curve has the SLE<sub>4</sub> law (with the corresponding force weights when extra boundary jumps are present). Conversely this martingale/covariance calculation constructs a local-set coupling with that curve. If two curves are coupled with the same field and the same boundary data, explore them alternately: the harmonic maximum principle keeps each on the allowed side of the other, so they coincide. Thus the level line is a measurable function of  $h$ .

Choose a deterministic countable dense collection of boundary prime ends and iterate these measurable level lines. Whenever a component has not yet been assigned one of the terminal harmonic values  $-a$  or  $b$ , draw the admissible level line that moves its harmonic value by one height gap in the required direction. The interval length condition  $a + b \geq 2\lambda$  is exactly what prevents the two continuation thresholds from being crossed before one of the terminal values is reached. Let  $A_n$  be the union after the first  $n$  explorations. Conditional independence of the zero-boundary GFF pieces in the components shows inductively that  $A_n$  is local, and its harmonic part is uniformly bounded between  $-a$  and  $b$ . Hence, for every nonnegative smooth  $\rho$ , the martingale  $(h_{A_n}, \rho)$  is uniformly integrable and converges. The closure  $A = \overline{\cup_n A_n}$  is therefore local. Standard SLE<sub>4</sub> crossing estimates on dyadic annuli imply that its Minkowski neighbourhood has area tending to zero fast enough that no additional distribution supported on  $A$  survives when testing the field; thus  $A$  is thin. If some complementary component had harmonic value strictly between  $-a$  and

$b$ , it would contain a rational target from the enumeration and would have been explored again. Consequently  $h_A \in \{-a, b\}$  componentwise.

For uniqueness, let  $A'$  be any other bounded-type thin local set of the same field with these two values. Couple each canonical level line in the preceding construction with  $A'$ . The union of two conditionally independent local sets is local. Immediately before a first forbidden crossing, the conditional harmonic values on the two sides are ordered; the maximum principle together with the level-line continuation threshold rules out that crossing. Induction over the deterministic enumeration gives  $A \subseteq A'$  almost surely. Now condition on  $A$ . In a component of  $D \setminus A$ , the harmonic value has already reached the endpoint  $-a$  or  $b$ . Any additional portion  $B = A' \setminus A$  would be a thin local set for the residual zero-boundary GFF whose harmonic contribution is identically zero. For a test function  $\rho \geq 0$  supported in that component, the GFF variance decomposition gives

$$(\rho, G_U \rho) - (\rho, G_{U \setminus B} \rho) = \text{Var}((h_B, \rho)) = 0.$$

Green-function monotonicity is strict whenever  $B$  has positive logarithmic capacity. A nonempty bounded-type thin local set attached to the boundary has positive capacity, so  $B$  must be empty (up to a polar set, which cannot occur in the closure of the level-line construction). Thus  $A' = A$ . We denote this canonical set by  $\mathbb{A}_{-a,b}$  and the outer boundaries of the components of its complement by  $\mathcal{L}_{-a,b}$ .

For  $a = b = 2\lambda$ , this construction is the GFF coupling of the  $\text{CLE}_4$  carpet. The two asymmetric pairs used below also satisfy  $a + b \geq 2\lambda$ , so the same existence and uniqueness theorem applies without a later-volume input.

In the wired dual current the boundary-cluster contribution makes the discrete height equal to  $\pm 1/2$  on the first inner boundaries. Since

$$\frac{1}{2\sqrt{2}\lambda} = \frac{1}{\sqrt{\pi}},$$

this becomes the harmonic boundary value  $\pm\sqrt{2}\lambda$  for the unscaled continuum GFF  $h_D$ . Part (i) therefore says that the first limiting inner-boundary set is a thin local set with values in  $\{-\sqrt{2}\lambda, \sqrt{2}\lambda\}$ . By part (j),

$$Q_0 = \mathcal{L}_{-\sqrt{2}\lambda, \sqrt{2}\lambda}(h_D). \quad (8.10)$$

Inside each loop, the master coupling gives an independent zero-boundary GFF plus the inherited constant boundary value, so the same statement repeats recursively.

The inner-boundary Markov property alone does not immediately identify outer boundaries. Use the simultaneous primal–dual master coupling. Between two consecutive inner layers, the complementary primal outer boundary follows exactly the portion of the dual two-valued set not swallowed by the adjacent primal set. By (8.8), two such continuum boundaries neither cross nor acquire an unrecorded contact. Hence the union of the explored primal and dual gaskets is again thin and local, and its harmonic values are determined by the discrete half-unit jumps. The two-valued uniqueness from part (j) then forces the outermost free-current outer boundaries to be

$$\mathcal{L}_{-2\lambda, 2\lambda}(h_D) = \text{CLE}_4(h_D). \quad (8.11)$$

If an outer boundary has sign  $+$ , the inner boundaries in its interior have the two values

$$\{-2\lambda, (2\sqrt{2} - 2)\lambda\},$$

and for sign  $-$  the values are reversed. Thus they are respectively  $\mathcal{L}_{-2\lambda, (2\sqrt{2}-2)\lambda}$  or its sign-reversed version. The discrete spatial Markov property makes the constructions in distinct holes conditionally independent given the field.



Iterate parts (k)–(l) in every complementary component. The Aizenman–Burchard estimate implies that for any  $\varepsilon > 0$  only finitely many layers contain a loop of diameter  $> \varepsilon$  with probability tending to one. Therefore the finite-depth recursive construction determines all macroscopic loops, while the remaining loops vanish in  $d_\chi$  as the depth tends to infinity. Since every layer is a measurable two-valued local set of the same GFF, the complete nested loop law is uniquely determined by  $h_D$  and hence no subsequence can give a different limit.

The zero-boundary GFF transforms canonically under conformal maps and the local-set definition is conformally natural. Equations (8.10)–(8.11) and their recursive versions therefore make the limiting double-current loop ensembles conformally invariant. This reproduces the Duminil-Copin–Lis–Qian division of labor: the second paper supplies the discrete Aizenman–Burchard, four-arm and boundary-avoidance estimates; the first constructs the master coupling, proves the Kasteleyn/fermion height limit (8.1), and identifies the loop limits as the stated GFF two-valued local sets. We have kept the square-lattice and Jordan-domain scope stated in the theorems above and have not promoted the result to general isoradial or arbitrary weighted planar graphs.  $\square$

## Chapter 9

# Primary fields, multiply connected domains, and s-embeddings

The final chapter of Volume XII broadens the basic simply connected fermionic technology in three directions: mixed primary fields, nontrivial topology, and non-isoradial embeddings. It also records the two magnetic exponents proved by probabilistic methods in Chapter 4.

### 9.1 All critical Ising primary fields and fusion

**Theorem 9.1** (Chelkak–Hongler–Izyurov primary fields and fusion).

*Let  $\Omega^\delta$  approximate a bounded finitely connected domain  $\Omega$  whose boundary is divided into finitely many fixed and free arcs in the standard CHI class. For any admissible product of bulk spin fields  $\sigma$ , disorder fields  $\mu$ , fermions  $\psi$  and energy densities  $\varepsilon$ , with marked points mutually separated and separated from the boundary, the critical square-lattice correlation, multiplied by its canonical mesh power, converges locally uniformly to the CHI continuum correlation. The limit is conformally covariant with bulk dimensions*

$$\Delta_\sigma = \Delta_\mu = \frac{1}{8}, \quad \Delta_\psi = \frac{1}{2}, \quad \Delta_\varepsilon = 1.$$

*The same continuum correlations satisfy the pairwise fusion asymptotics of the Ising primary-field theory as marked points collide, with constants fixed by the lattice normalizations used earlier in this volume. This theorem concerns only the four primary families listed above; descendants and general Virasoro fields are not included.*

*Proof.* The Kadanoff–Ceva formalism of Chapter 1 gives spins  $\sigma_v$ , disorders  $\mu_u$ , and real fermions  $\psi_z^{[\eta]} = \sigma_{z^\circ} \mu_{z^\bullet}$  with the half-angle phase carried by  $\eta$ . Energy is the centered nearest-neighbor product

$$\varepsilon_e = \sqrt{2}(\sigma_{e_+} \sigma_{e_-} - 2^{-1/2}).$$

The already-owned spin and energy theorems fix the mesh normalizations. In the CHI normalization a product with  $n_\sigma, n_\mu, n_\psi, n_\varepsilon$  bulk insertions is multiplied by

$$\delta^{-\Delta}, \quad \Delta = \frac{n_\sigma + n_\mu}{8} + \frac{n_\psi}{2} + n_\varepsilon.$$

Odd spin/disorder sectors that vanish by the source parity/sign rules remain zero; we do not manufacture a nonzero continuum field by renormalization.

Fix all insertions except one auxiliary corner  $z$ . Divide the correlation containing  $\psi_z^{[\eta]}$  by the background correlation whenever the latter is nonzero. The Kadanoff–Ceva switching calculation shows that, as a function of  $z$ , this ratio obeys the same local propagation relation as the basic fermion. Its monodromy records exactly which spin/disorder points are enclosed, and its principal parts at the marked points are determined by the local operator type. Thus it is an s-holomorphic spinor on the double cover of  $\Omega^\delta \setminus \{\text{marked points}\}$ , with fixed/free boundary projections inherited from the Gibbs boundary condition. This reduces a many-field correlation to the same Riemann–Hilbert technology as Checkpoint 1, but now with several prescribed singularities.

Square the spinor and use the discrete primitive  $H^\delta$  exactly as in ‘SM-12.S2.02’. On a compact set that stays a positive distance from the marked points and the boundary, the discrete maximum principle bounds  $H^\delta$ , hence the spinor. Near each singularity subtract the appropriate full-plane discrete kernel: the  $z^{-1/2}$  kernel for spin/disorder branching and the  $z^{-1}$ -type kernel for a fermionic/energy singularity. The remainder again has a bounded primitive. Discrete Cauchy estimates give precompactness, and every limit solves the continuum Riemann boundary-value problem with the prescribed monodromy, boundary projection and singular parts. The uniqueness argument from Checkpoint 1, applied after subtracting all principal parts, identifies the limit. Uniformity on compact configuration sets follows because all constants depend only on a lower separation bound.

There are three elementary recovery operations. First, moving a spin or disorder insertion by one lattice step changes the logarithm of its correlation by a coefficient in the local expansion of the convergent auxiliary spinor. Summing these discrete logarithmic derivatives along a path in configuration space gives the ratio between two configurations; path-independence follows from the underlying correlation itself. Second, inserting a fermion is already represented by the value of the mixed spinor. Third, an energy insertion is obtained by coalescing an adjacent order–disorder pair and subtracting the bulk constant; the coefficient of the regular part is the energy correlation. The multiplicative constants are fixed by the full-plane two-point normalization already proved here in Chapter 3. Hence the renormalized mixed correlation converges, not merely its logarithmic derivative.

Under  $\phi : \Omega \rightarrow \Omega'$ , the continuum spinor transforms as a half-differential. The recovery rules from part d then give one factor  $|\phi'|^{1/8}$  for each spin or disorder, a half-differential factor for each fermion (with its chiral phase), and  $|\phi'|$  for each energy insertion. Because the transformed object solves the same normalized Riemann problem, uniqueness proves the covariance identity. This is a proof from the boundary-value characterization, not an appeal to CFT axioms.

Separated-point convergence alone says nothing uniform when two insertions approach each other. Let  $r = |z_1 - z_2|$  and choose an annulus whose inner radius is comparable with  $r$  and whose outer radius is smaller than the distance to all other marked data. Rescale the mixed spinor by  $r^{-1}$  around the pair. The compactness estimate of part c, now on fixed annuli after rescaling, identifies every limit with one of the finite full-plane model spinors. Subtract its leading singular term and reapply the primitive/maximum-principle estimate to the remainder. Scaling back yields the source fusion expansion. In particular the channels have the Ising fusion content

$$\sigma \times \sigma = \mathbf{1} + \varepsilon + \cdots, \quad \mu \times \mu = \mathbf{1} + \varepsilon + \cdots, \quad \sigma \times \mu = \psi + \cdots,$$

while fermion–fermion fusion has the identity leading channel and energy with spin/disorder returns the corresponding spin/disorder family. The displayed equations record channels rather than replacing the source’s normalization-dependent coefficients: those coefficients are exactly the full-plane kernel coefficients fixed in the preceding paragraph. Iterating pairwise collision in a hierarchy of disjoint annuli gives the multi-collision fusion rules used later.

□

## 9.2 Holomorphic spinors on double covers

**Theorem 9.2** (Chelkak–Izyurov spinors on double covers).

*Let  $\Omega^\delta$  approximate a bounded finitely connected planar domain  $\Omega$  and let  $\varpi^\delta : \tilde{\Omega}^\delta \rightarrow \Omega^\delta$  be one of the Ising double covers obtained by prescribing the components/punctures around which the sign changes. For the critical Ising model with the source boundary conditions, the normalized Kadanoff–Ceva spinor observable on  $\tilde{\Omega}^\delta$  converges locally uniformly away from its marked singularities to the unique holomorphic spinor solving the corresponding continuum Riemann boundary-value problem. The local coefficients of these spinors give the scaling limits of the spin-correlation ratios appearing in the Chelkak–Izyurov theorem. The statement is only for the double-cover/ratio observables treated in that theorem; the later all-primary-field and collision statements are not imported here.*

*Proof.* A Kadanoff–Ceva disorder line changes the signs of the crossed couplings. Chapter 1 proved that moving the line while keeping its endpoints fixed is a gauge change. Thus a fermionic observable is not naturally a single-valued function on  $\Omega^\delta$ : after one circuit around a prescribed ramification component its sign is reversed. We therefore lift it to the two-sheeted cover  $\varpi^\delta$ . If  $z$  and  $z^*$  are the two lifts of the same corner, then

$$F^\delta(z^*) = -F^\delta(z).$$

The low-temperature expansion from Chapter 1 also gives the local propagation relation. After multiplying by the usual half-angle phase, this is precisely s-holomorphicity on the cover. Along a boundary edge with unit tangent  $\tau$ , the order-disorder expansion shows that

$$F^\delta(z)\sqrt{i\tau(z)} \in \mathbb{R},$$

with the sign prescribed by the sheet and the boundary condition. These two properties—antisymmetry and the real boundary projection—are the discrete Riemann–Hilbert data.

Let  $f$  be a holomorphic spinor on the limiting cover, with the prescribed principal part at the marked point and with  $f(z)\sqrt{i\tau(z)} \in \mathbb{R}$  on the regular boundary arcs. Although  $f$  changes sign between sheets,  $f^2$  descends to  $\Omega$ . Hence

$$H(z) := \Im \int^z f(\zeta)^2 d\zeta$$

is single-valued up to an additive constant on each boundary component. On a smooth boundary arc the projection condition implies that the tangential derivative of  $H$  vanishes, so  $H$  is constant there. At the marked singularity its leading behavior is fixed by the normalization. If  $f_1, f_2$  are two normalized solutions, apply the same construction to  $f_1 - f_2$ . Its primitive has no singular source, is constant on every boundary component, and has the sign/period constraints forced by the chosen cover. The maximum principle, followed by the Cauchy–Riemann equations, gives  $f_1 - f_2 = 0$ . This is the uniqueness mechanism used throughout the source paper; topology is stored in the cover rather than hidden in an ad hoc choice of branches.

For an s-holomorphic spinor  $F^\delta$ , define the standard discrete primitive  $H^\delta$  by increments equal to the squared projections of  $F^\delta$  across primal/dual edges. The propagation relation makes these increments path-independent on the base graph, while the sign change of  $F^\delta$  disappears after squaring. The same computation as in Chapter 2 gives subharmonicity on primal vertices and superharmonicity on dual vertices. Boundary values are constant up to an  $O(\delta)$  error. The discrete maximum principle therefore bounds  $H^\delta$  on compacta away from the singularity. Discrete Cauchy estimates then bound  $F^\delta$  and give equicontinuity. Every subsequence has a locally uniform limit  $f$ ; the discrete contour integrals tend to zero, so  $f$  is holomorphic on the cover, inherits antisymmetry

and the boundary projection, and has the required principal part by the local full-plane kernel comparison. The uniqueness from part b identifies every subsequential limit, hence the full sequence converges.

Fix the background spin insertions occurring in the preceding correlation and insert one additional order–disorder fermion. Expanding numerator and denominator by the same contour representation cancels all configuration weights except the defect-path phase. Consequently the normalized observable is exactly a ratio of Ising partition functions/correlations. When the auxiliary fermion approaches a marked point  $a$ , its expansion has the form

$$F^\delta(a + \delta u) = K^\delta(u) + A^\delta K_1^\delta(u) + o(1),$$

where  $K^\delta$  is the universal singular kernel and  $A^\delta$  is the desired ratio (or logarithmic derivative of such a ratio). The local convergence established in part c and the matching of  $K^\delta$  with the continuum  $z^{-1/2}$  kernel imply  $A^\delta \rightarrow A$ . Integrating these logarithmic derivatives along a path in configuration space, and fixing the multiplicative constant with the already-owned full-plane normalization, yields the correlation-ratio limit.

If  $\phi : \Omega \rightarrow \Omega'$  is conformal, the lifted map sends the chosen double cover to the corresponding cover of  $\Omega'$ . The transformed function

$$\tilde{f}(z) = f(\phi(z))\sqrt{\phi'(z)}$$

has the same monodromy, boundary projection and singular normalization as the spinor associated with  $\Omega$ . Uniqueness therefore identifies the two. Since the correlation ratios are coefficients of these spinors, their conformal covariance follows with the spin scaling weight  $1/8$  already fixed in Chapter 3. No simply-connected uniformization is being smuggled into the proof.

The proof follows the architecture of Chelkak–Izyurov, *Holomorphic Spinor Observables in the Critical Ising Model* (arXiv:1105.5709): spinors on double covers, convergence through the discrete primitive/maximum principle, and extraction of spin-correlation ratios. Their revised source explicitly separates the two main convergence theorems and treats the ratio applications afterwards. We do not promote this parent to the more general all-primary-field theorem proved later by Chelkak–Hongler–Izyurov.  $\square$

### 9.3 Multiply connected and free-boundary interfaces

**Theorem 9.3** (Izyurov interfaces in finitely connected domains).

*Let  $\Omega^\delta$  approximate a bounded finitely connected domain with the fixed/free boundary arcs and radial or multiple-interface marked data of the Chelkak–Izyurov class. For the critical square-lattice Ising model, the corresponding discrete interfaces converge to conformally invariant curves whose laws are locally absolutely continuous with respect to the corresponding  $\text{SLE}_3$  laws. The Radon–Nikodym density is the ratio of the appropriate continuum fermionic partition functions. In simply connected domains this density is expressed by the corresponding rational fermionic partition function, while in doubly connected domains it is given by the elliptic partition function determined by the same boundary-value problem.*

*Proof.* Fix a finitely connected domain with the marked boundary and/or interior targets appearing in the theorem. Stop the interface before it approaches another marked point or changes the connectivity type of the complementary domain. On every such stopping window the explored hull is contained in a simply connected Loewner chart while the unvisited holes remain encoded in the partition function. The Chapter 5 crossing estimate, applied conditionally by the domain Markov

property, gives Condition G2 on every compact stopping window. Hence the stopped curves and their Loewner drivers are tight. This is sufficient because the full curve is obtained by an increasing sequence of such windows.

Let  $F_{\Omega^\delta}^\delta(a, z)$  be the normalized spinor from Checkpoint 1 with the cover/branch data determined by the remaining boundary conditions. Exploring the Ising interface exposes spins but leaves a critical Ising model in the unexplored component with the induced boundary data. The contour expansion therefore gives the exact martingale identity

$$F_{\Omega^\delta}^\delta(a, z) = \mathbb{E} \left[ F_{\Omega^\delta \setminus \gamma^\delta[0, n]}^\delta(\gamma_n^\delta, z) \mid \mathcal{F}_n \right].$$

Checkpoint 1 gives local uniform convergence of this observable in every stopped domain. Thus each subsequential interface limit carries the continuum spinor martingale.

Normalize the continuum spinor by its leading coefficient at the growing tip and denote that coefficient by  $Z(\Omega; a, \dots)$ . In a local half-plane chart, divide the multiply connected martingale by the ordinary chordal  $\text{SLE}_3$  fermion. The quotient is

$$\frac{Z(\Omega_t; W_t, \dots)}{Z(\Omega_0; a, \dots)} \prod_j |g'_t(x_j)|^{h_j},$$

up to the stopping time under consideration. It is positive and bounded on compact stopping windows. Hence it is a genuine Radon–Nikodym martingale with respect to chordal  $\text{SLE}_3$ . This proves local absolute continuity without assuming a priori that the unknown driver solves an SDE.

Under ordinary  $\text{SLE}_3$ , Itô's formula applied to the preceding density shows that its finite-variation term vanishes exactly when the tilted driver has drift

$$dW_t = \sqrt{3} dB_t + 3 \partial_w \log Z(\Omega_t; w, \dots)|_{w=W_t} dt,$$

with the geometric moduli evolving under the deterministic Loewner flow. Girsanov therefore identifies the subsequential limit uniquely on the stopping window. For a radial target, the same computation is performed in the radial chart; for several interfaces, condition on the first explored strand and repeat with the updated partition function. This is the source's partition-function description of the multiple/radial cases.

Condition G2 rules out spurious boundary sticking and gives convergence of the natural crosscut stopping times. Let the crosscuts exhaust the domain and use consistency of the Radon–Nikodym martingales to concatenate the stopped limits. The resulting curve law is independent of the chosen chart. Since the spinor/partition function is conformally covariant, the entire law is conformally invariant. In simply connected and doubly connected geometries, inserting the explicit source solutions of the boundary problem gives the rational and elliptic drift formulae, respectively.

Izyurov, *Critical Ising interfaces in multiply-connected domains* (arXiv:1309.5302), proves convergence to conformally invariant curves absolutely continuous with respect to  $\text{SLE}_3$ , including multiple interfaces in arbitrary finitely connected domains and the radial case. The manuscript retains this absolute-continuity/partition-function formulation and does not relabel the process as ordinary chordal  $\text{SLE}_3$  when the topology produces a nonzero drift.  $\square$

**Theorem 9.4** (Free-boundary fermions, dipolar  $\text{SLE}_3$ , and crossings).

*For the critical Ising model with the plus/minus/free boundary data treated by Izyurov, the free-boundary fermionic observables converge to the corresponding continuum Riemann–Hilbert solutions. In particular, the plus–minus interface separated by a free boundary arc converges to dipolar  $\text{SLE}_3$ ; the multi-arc variants have the source partition-function drifts. The same convergent observables determine the conformally invariant FK-Ising alternating wired/free crossing probabilities and the source spin-Ising crossing formula.*

*Proof.* On a fixed boundary arc the fermion obeys the real projection condition from Checkpoint 1. On a free arc, the low-temperature expansion leaves the endpoint spin unsummed in a different way, and the boundary projection is rotated by a quarter turn. Define the source free-boundary observable by the Kadanoff–Ceva numerator divided by the matching partition function. The local propagation equation in the bulk is unchanged; only the boundary Riemann–Hilbert line changes. This distinction is why free boundary conditions alter the Loewner drift rather than merely a normalization constant.

The square of the free-boundary spinor again has a discrete primitive. On fixed arcs its boundary values are controlled as before. On free arcs, reflect the spinor across the discrete boundary line after the source phase rotation; this converts the boundary estimate into an interior estimate on the reflected mesh. Near a change point between fixed and free data subtract the explicit half-plane model singularity and apply the maximum principle to the remainder. The expanded source version repairs precisely this endpoint step. We obtain local uniform convergence up to compact subsets of each boundary arc and identify the unique continuum free-boundary spinor.

Under plus/minus/free data the interface exploration preserves the class of boundary conditions, so the normalized observable is a martingale. Map the domain to a horizontal strip with the starting point at one end and the free target arc on the opposite boundary. The unique continuum spinor is the dipolar fermion. Expanding its martingale at the marked end gives the same quadratic variation  $3t$  as in Chapter 5 and the dipolar drift that keeps the target as an interval rather than a point. Driver tightness comes from the Chapter 5 G2 theorem; coefficient identification then gives dipolar  $\text{SLE}_3$ . Additional fixed/free arcs insert the corresponding partition-function factor and are handled by the same Girsanov calculation.

For FK-Ising with alternating wired/free arcs, explore the cluster connections only near the boundary changes. The source fermionic observable can be evaluated in two ways: as a signed sum over connectivity patterns and as the solution of the free-boundary Riemann problem. This gives a finite linear system for the connectivity weights. Dividing by the total partition function turns the solution into crossing probabilities. Since every entry of the system converges and the limiting determinant is nonzero for distinct marked points, the crossing probabilities converge to conformally invariant functions of the boundary data. The spin-Ising crossing formula is obtained by the analogous order-disorder expansion.

The conditional G2 estimate gives tightness in the unparametrized curve topology and convergence of crosscut stopping times. The martingale identification makes every subsequential limit the same dipolar/partition-function tilted  $\text{SLE}_3$  process, hence the whole sequence converges. We do not identify every possible free-boundary interface with a single universal dipolar process: the multi-arc cases retain the explicit source drift.

Izyurov, *Smirnov’s observable for free boundary conditions, interfaces and crossing probabilities* (arXiv:1404.0239), proves convergence of the free-boundary fermionic variants, recovers plus–minus–free convergence to dipolar  $\text{SLE}_3$ , extends the setup to arbitrary numbers of plus/minus/free arcs, and computes alternating wired/free FK crossing limits as well as a spin-Ising crossing formula. The revised source notes an expanded proof and a repaired gap; the boundary compactness paragraph above is included so that this owner does not silently rely on the shorter superseded argument.  $\square$

## 9.4 s-embeddings and universality beyond isoradial graphs

An s-embedding is generated by a global solution  $\mathcal{X}$  of the Kadanoff–Ceva propagation equation. It places the primal and dual vertices so that every other fermionic observable becomes s-holomorphic in the new geometry. The associated function  $\mathcal{Q}$  measures the non-flatness of the embedding.

**Theorem 9.5** (Flat s-embedding universality and the FK interface corollary).

Let  $(G^\delta, J^\delta)$  be critical weighted planar Ising models admitting global nonzero solutions  $\mathcal{X}^\delta$  of the Kadanoff–Ceva propagation equation, and let  $(\mathcal{S}^\delta, \mathcal{Q}^\delta)$  be the associated s-embeddings and origami maps. Assume the flat regime of Chelkak: the embedded edge lengths and angles are uniformly nondegenerate on the mesh scale and

$$\|\mathcal{Q}^\delta\|_\infty = O(\delta)$$

on the domains under consideration. Then the s-holomorphic fermionic calculus has uniform Harnack/compactness estimates, FK-Ising crossings satisfy RSW-type bounds, and the basic fermionic observables converge to their standard conformally covariant holomorphic boundary-value limits. Consequently, in a Dobrushin domain for which the FK exploration is defined, the already-owned Chapter 5 compactness-and-martingale theorem identifies every subsequential interface limit as chordal SLE<sub>16/3</sub>.

The final interface assertion is an internal corollary combining Chelkak’s flat-s-embedding RSW/fermion theorem with ‘SM-12.05.01–03’; it is not attributed to the source as a separate theorem. No non-flat Lorentz-maximal extension is included here.

*Proof.* For an oriented corner  $c = (v, f)$ , write  $\eta_c$  for the square root of the local direction appearing in the Kadanoff–Ceva propagation relation. A global solution  $\mathcal{X}^\delta$  assigns complex corner values satisfying the same two-dimensional real projection rule on the four corners surrounding every edge. Define the primal and dual increments of the s-embedding by the quadratic expressions in these corner values used by Chelkak,

$$d\mathcal{S}^\delta \sim (\mathcal{X}^\delta)^2, \quad d\mathcal{Q}^\delta \sim |\mathcal{X}^\delta|^2,$$

with the source orientation convention. The propagation relation makes these increments closed, so after fixing one additive constant they integrate to globally defined functions  $\mathcal{S}^\delta$  and  $\mathcal{Q}^\delta$ . Thus the weighted Ising model itself selects the discrete complex geometry in which its fermions become holomorphic. The uniform edge-length/angle hypotheses rule out collapsed quadrilaterals, and  $\mathcal{Q}^\delta = O(\delta)$  says that the associated discrete surface is asymptotically horizontal in  $\mathbb{R}^{2,1}$ .

Let  $F^\delta$  be any Kadanoff–Ceva fermionic observable with the same local propagation equation. At an edge  $e$  separating two corners  $c_1, c_2$ , divide the propagation relation for  $F^\delta$  by the one for  $\mathcal{X}^\delta$ . The result says that the two neighboring values of  $F^\delta/\mathcal{X}^\delta$  have the same orthogonal projection onto the real line determined by the s-edge  $\mathcal{S}^\delta(e)$ . This is precisely the s-holomorphic projection identity. Squaring and integrating gives the discrete primitive

$$H_F^\delta(v) - H_F^\delta(f) = \text{Im}\left((F^\delta(c))^2(\mathcal{S}^\delta(v) - \mathcal{S}^\delta(f))\right)$$

(with the orientation convention fixed at the beginning of the chapter). Around a face the increment sum vanishes. Applying the local projection identity once more expresses the discrete Laplacian of  $H_F^\delta$  as a positive weighted sum of squared projection defects plus terms controlled by the origami increment  $d\mathcal{Q}^\delta$ . In the flat regime these latter terms are  $O(\delta)$  relative to the elliptic part. Consequently the maximum-principle, Beurling-type, and Cauchy-estimate arguments used on isoradial graphs survive with constants depending only on the quantitative nondegeneracy parameters.

Consider a topological quadrilateral whose continuum extremal length stays in a compact subset of  $(0, \infty)$ . Solve the Dobrushin fermionic boundary problem whose partition-function normalization records the FK connection between the two wired sides. The primitive  $H_F^\delta$  has boundary values pinned, up to  $O(\delta)$ , on the four arcs. Uniform ellipticity of the s-embedding gives a discrete Harnack



chain and a lower bound for the harmonic-measure transfer from one wired side to the other. If the FK crossing probability tended to zero, the corresponding normalized fermion and hence its primitive would tend uniformly to the wrong constant on an interior crosscut, contradicting that harmonic-measure lower bound. Applying the same argument to the dual model gives the complementary upper bound. Therefore, for every compact extremal-length window there exists  $c > 0$ , independent of  $\delta$ , such that

$$c \leq \mathbb{P}_{\text{FK}}^\delta(\text{crossing}) \leq 1 - c.$$

Domain Markov and FKG then give the annular/crosscut versions needed by Condition G2. This is the RSW mechanism proved in the flat-s-embedding source: it does not require an isoradial angle structure because the ellipticity is carried by  $(\mathcal{S}^\delta, \mathcal{Q}^\delta)$  itself.

Normalize a basic fermion at one marked boundary point. The primitive bounds from the preceding step imply local  $L^\infty$  bounds away from marked singularities. The s-holomorphic Cauchy estimate gives equicontinuity on compact subsets. Thus every mesh sequence has a locally uniformly convergent subsequence. Because  $\mathcal{Q}^\delta = O(\delta)$ , the error term in the discrete Cauchy–Riemann relation vanishes, so the limit is genuinely holomorphic in the  $\mathcal{S}$ -coordinate. Boundary projection lines converge to their continuum tangent-line limits, while the normalized discrete singular kernel converges to the usual half-differential singularity. The continuum Riemann–Hilbert problem has a unique solution: subtract two candidates, square their difference, integrate the resulting holomorphic differential, and use the boundary sign/maximum principle exactly as in the earlier spinor chapters. Hence the whole sequence converges to the standard conformally covariant fermionic observable. Quantitative versions of the same estimates yield the source’s convergence-rate control.

In a Dobrushin domain, the FK exploration exposes the same fermion as a domain-Markov martingale. Step (c) supplies the conditional crossing estimate required by ‘SM-12.05.01’, so the interfaces and their Loewner driving functions are tight. Step (d) supplies convergence of the martingale observable. Therefore the coefficient-at-infinity calculation already proved in ‘SM-12.05.03’ applies verbatim to every subsequential limit: the first two nontrivial coefficients make

$$W_t, \quad W_t^2 - \frac{16}{3}t$$

local martingales. Lévy’s characterization gives  $W_t = \sqrt{16/3} B_t$ . The curve/driver tightness theorem then upgrades the identification to convergence of the full unparametrized exploration to chordal  $\text{SLE}_{16/3}$ . No forward theorem has entered: the only new inputs in this paragraph are the crossing and fermion-convergence statements just established.

The conclusion is invariant under replacing an isoradial mesh by any weighted planar Ising family that admits s-embeddings satisfying the stated uniform geometric bounds and flatness  $\mathcal{Q}^\delta = O(\delta)$ . This includes the canonical s-embeddings of critical doubly periodic models covered by the source. The theorem does not assert that every planar Ising model possesses such a uniformly nondegenerate flat embedding, nor does it include the later non-flat Lorentzian/maximal-surface theory.

□

## 9.5 Volume XII synthesis

The logical chain established in this volume is

Kadanoff–Ceva fermions  $\rightarrow$  s-holomorphicity  $\rightarrow$  Riemann boundary problems  
 $\rightarrow$  spin/energy correlations and fields  $\rightarrow$  interface martingales  $\rightarrow$   $\text{SLE}_3, \text{SLE}_{16/3}$   
 $\rightarrow$  loop ensembles/CLE targets, random-current local sets, and broader universality.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 9.6** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 9.7** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 9.8** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 9.9** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 9.10** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XIII: Deterministic Loewner Evolution

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# Scope and conventions

This volume proves the deterministic Loewner machinery needed by the stochastic SLE volumes. The half-plane-capacity convention is  $\text{hcap}(K_t) = 2t$ , so the chordal equation is  $\partial_t g_t = 2/(g_t - U_t)$ . The sharp  $\text{Lip}(1/2)$  quasilit threshold is treated separately from the elementary continuity theory because its proof uses substantially finer geometric estimates.



# Chapter 1

## Half-plane Loewner chains

Loewner's fundamental insight is that a two-dimensional growing hull can be encoded by one real function. The geometry lives in the conformal maps; the dynamics lives at the image of the growing tip.

**Definition 1.1** (Half-plane hull and hydrodynamic normalization). A *half-plane hull* is a bounded relatively closed set  $K \subset \overline{\mathbb{H}}$  such that  $H = \mathbb{H} \setminus K$  is simply connected. Its hydrodynamically normalized uniformizing map is the conformal map  $g : H \rightarrow \mathbb{H}$  satisfying

$$g(z) = z + \frac{a}{z} + O(|z|^{-2}) \quad (z \rightarrow \infty).$$

The coefficient  $a \geq 0$  is the *half-plane capacity*  $\text{hcap}(K)$ . For a continuous Loewner driver  $U$ , the *swallowing time* of  $z \in \overline{\mathbb{H}}$  is  $\tau_z = \sup\{t : g_s(z) \text{ is defined for every } 0 \leq s < t\}$ ; equivalently, for an interior point it is the first time at which  $g_t(z) - U_t$  tends to 0.

*Worked example 1.2* (Constant driving produces a vertical slit). Take  $U_t \equiv 0$ . The Loewner equation is  $\partial_t g_t = 2/g_t$ , so

$$\frac{1}{2} \partial_t (g_t(z)^2) = 2, \quad g_t(z)^2 = z^2 + 4t.$$

Choosing the square-root branch with  $g_t(z) \sim z$  at infinity gives

$$g_t(z) = \sqrt{z^2 + 4t} = z + \frac{2t}{z} + O(|z|^{-3}).$$

Hence  $\text{hcap}(K_t) = 2t$ . The branch cut in  $\mathbb{H}$  is exactly  $K_t = [0, 2i\sqrt{t}]$ : the deterministic Loewner chain grows a vertical slit at speed fixed by capacity time. This example is the normalization test used repeatedly in the SLE volumes.

### 1.1 The ODE and swallowing times

Let  $U : [0, T] \rightarrow \mathbb{R}$  be continuous. For  $z \in \mathbb{H}$  solve

$$\dot{g}_t(z) = \frac{2}{g_t(z) - U_t}, \quad g_0(z) = z.$$

Write  $\tau_z$  for the maximal solution time. If  $g_t(z) = X_t + iY_t$ , then

$$\dot{Y}_t = -\frac{2Y_t}{(X_t - U_t)^2 + Y_t^2} \leq 0.$$

Thus a point can fail only by being swallowed at the real driving point.

**Theorem 1.3** (Existence and uniqueness of the chordal Loewner flow). *Let  $U : [0, T] \rightarrow \mathbb{R}$  be continuous. For every  $z \in \mathbb{H}$  there is a unique maximal solution of*

$$\partial_t g_t(z) = \frac{2}{g_t(z) - U_t}, \quad g_0(z) = z,$$

*on an interval  $[0, \tau_z)$ . For every  $t \leq T$ , the survival set  $H_t = \{z \in \mathbb{H} : \tau_z > t\}$  is open and nested, its complement  $K_t = \mathbb{H} \setminus H_t$  is increasing, and  $g_t$  is an injective holomorphic map from  $H_t$  into  $\mathbb{H}$ . The family  $(g_t)_{0 \leq t \leq T}$  is the unique maximal ODE flow driven by  $U$ . The surjectivity of  $g_t$  and the half-plane-hull property of  $K_t$  are proved in Theorem SM-13.02.01.*

*Proof.* For every fixed  $z \in \mathbb{H}$ , the vector field is locally Lipschitz away from  $U_t$ , so Picard–Lindelöf gives a unique solution up to the first approach to the singularity. The imaginary-part equation above keeps the solution in the upper half-plane before that time. Differentiating in  $z$  gives

$$\partial_t g'_t(z) = -\frac{2g'_t(z)}{(g_t(z) - U_t)^2},$$

so  $g'_t$  never vanishes. Hence  $g_t$  is locally conformal. Uniqueness of ODE trajectories gives injectivity: if  $g_t(z) = g_t(w)$ , backward uniqueness forces  $z = w$ . The survival set  $H_t = \{z : \tau_z > t\}$  is open and nested, and  $K_t = \mathbb{H} \setminus H_t$  is increasing. This proves the local and injective flow assertions. The reverse-flow argument in Theorem SM-13.02.01 proves that the image is all of  $\mathbb{H}$  and thereby completes the hull interpretation.  $\square$

## 1.2 Hydrodynamic normalization and capacity

For  $z$  large, insert the expansion  $g_t(z) = z + a_t z^{-1} + O(z^{-2})$  into the ODE. Since

$$\frac{2}{g_t(z) - U_t} = \frac{2}{z} + O(z^{-2}),$$

one obtains  $\dot{a}_t = 2$  and hence  $a_t = 2t$ .

**Theorem 1.4** (Hydrodynamic normalization and capacity time). *For the Loewner flow of Theorem SM-13.01.01, uniformly for  $t$  in a compact subinterval of  $[0, T]$ ,*

$$g_t(z) = z + \frac{2t}{z} + O(|z|^{-2}) \quad (z \rightarrow \infty).$$

*Consequently*

$$\text{hcap}(K_t) = 2t.$$

*Conversely, for a differentiable hull chain parametrized by  $\text{hcap}(K_t) = 2t$ , the coefficient of its chordal Loewner vector field is 2.*

*Proof.* The expansion just computed is uniform on compact time intervals because the driving function is bounded there. Volume IV defined the half-plane capacity of a hull as the  $1/z$  coefficient of its hydrodynamically normalized uniformizing map. Therefore  $\text{hcap}(K_t) = 2t$ . Conversely, if a hull chain is parametrized so that its capacity is  $2t$ , the coefficient in the Loewner vector field is exactly 2. This fixes the time scale used throughout SLE.  $\square$

### 1.3 Deriving the equation from hull growth

**Theorem 1.5** (Recovering the driver from differentiable hull growth). *Let  $(K_t)_{0 \leq t \leq T}$  be an increasing family of half-plane hulls with  $\text{hcap}(K_t) = 2t$ , let  $g_t : \mathbb{H} \setminus K_t \rightarrow \mathbb{H}$  be hydrodynamically normalized, and assume that for every  $t < T$  the transition hull  $g_t(K_{t+h} \setminus K_t)$  has diameter tending to 0 as  $h \downarrow 0$  and satisfies locally uniformly away from its attachment point*

$$g_{t+h} \circ g_t^{-1}(w) = w + \frac{2h}{w - U_t} + o(h)$$

for some  $U_t \in \mathbb{R}$ . Then

$$\partial_t g_t(z) = \frac{2}{g_t(z) - U_t}$$

for every  $z$  before swallowing. In particular the growing hull is encoded by the real attachment function  $U$  whenever the displayed differentiability hypothesis holds.

*Proof.* Let  $g_t$  be the normalized map of the complement of a differentiable hull chain and set  $g_{t,t+h} = g_{t+h} \circ g_t^{-1}$ . The transition hull has capacity  $2h + o(h)$  and shrinks to a single boundary point  $U_t$ . The hydrodynamic representation of a small half-plane hull implies

$$g_{t,t+h}(w) = w + \frac{2h}{w - U_t} + o(h)$$

locally uniformly away from  $U_t$ . Divide by  $h$ , let  $h \downarrow 0$ , and compose with  $g_t$ . This yields  $\partial_t g_t(z) = 2/(g_t(z) - U_t)$ . The attachment point is real because the transition hull is attached to the real boundary.  $\square$

### 1.4 Composition and scaling

**Theorem 1.6** (Composition, translation, and scaling rules). *Let  $g_t$  be driven by  $U_t$ . For  $s \geq 0$ , the future transition chain  $g_{s+t} \circ g_s^{-1}$  is a chordal Loewner chain driven by  $U_{s+t}$  in the  $g_s$  coordinate. If  $r > 0$  and*

$$\tilde{g}_t(z) = r^{-1} g_{r^2 t}(rz), \quad \tilde{U}_t = r^{-1} U_{r^2 t},$$

then  $\tilde{g}$  solves the same chordal Loewner equation. Translating space by  $c \in \mathbb{R}$  translates the driver by the same amount. Thus half-plane capacity scales quadratically:  $\text{hcap}(rK) = r^2 \text{hcap}(K)$ .

*Proof.* ODE uniqueness implies the flow rule: after time  $s$ , the image of the future hull under  $g_s$  is driven by  $U_{s+t}$  in the shifted coordinate. If space is scaled by  $r > 0$ , define  $\tilde{g}_t(z) = r^{-1} g_{r^2 t}(rz)$ . Direct differentiation gives the same Loewner ODE driven by  $\tilde{U}_t = r^{-1} U_{r^2 t}$ . Translation by  $c$  similarly replaces  $U_t$  by  $U_t - c$ . These identities are the deterministic source of SLE scale and translation covariance.  $\square$

## Chapter 2

# Driving functions, traces, and convergence

The forward Loewner ODE always generates hulls, but not every continuous driver produces a simple curve. We first establish what is automatic, then state the additional result for simple slits.

### 2.1 Continuous drivers generate hulls

**Theorem 2.1** (Continuous drivers generate increasing hulls). *For every continuous  $U : [0, T] \rightarrow \mathbb{R}$ , the survival domains of the chordal Loewner ODE satisfy  $H_t \subset H_s$  for  $s < t$ , and the complements  $K_t = \mathbb{H} \setminus H_t$  form an increasing family of half-plane hulls. For every  $t \leq T$ , the hydrodynamically normalized map  $g_t : H_t \rightarrow \mathbb{H}$  is onto and  $\text{hcap}(K_t) = 2t$ . No assumption that  $K_t$  is generated by a curve is required.*

*Proof.* The survival domains from Chapter 1 are nested. The derivative equation shows that  $g_t$  is conformal there and the expansion at infinity gives finite capacity. Surjectivity onto  $\mathbb{H}$  follows by solving the reverse ODE from a target point for a short interval and iterating backward; the imaginary part stays positive before reaching time zero. Hence  $g_t$  uniformizes  $\mathbb{H} \setminus K_t$  and continuous  $U$  produces an increasing hull family even when no trace exists.  $\square$

### 2.2 Simple curves produce continuous drivers

**Theorem 2.2** (A simple capacity-parametrized slit has a continuous driver). *Let  $\gamma : [0, T] \rightarrow \overline{\mathbb{H}}$  be simple with  $\gamma(0) \in \mathbb{R}$ ,  $\gamma((0, T]) \subset \mathbb{H}$ , and parametrize it so that  $\text{hcap}(\gamma[0, t]) = 2t$ . Let  $g_t : \mathbb{H} \setminus \gamma[0, t] \rightarrow \mathbb{H}$  be hydrodynamically normalized. The two prime-end sides of  $\gamma(t)$  have a common image  $U_t \in \mathbb{R}$ ;  $t \mapsto U_t$  is continuous, and  $g_t$  solves the chordal Loewner equation driven by  $U$ .*

*Proof.* Let  $\gamma$  be a simple curve from the real boundary into  $\mathbb{H}$  and parametrize it by capacity. The normalized map  $g_t$  sends the two prime-end sides of the tip to a single real point  $U_t$ . If  $t_n \rightarrow t$ , the slit domains converge in the Caratheodory kernel sense. Their normalized maps and inverse maps converge locally uniformly. The images of crosscuts shrinking around the tip therefore shrink to the same boundary point, proving  $U_{t_n} \rightarrow U_t$ . The small-hull expansion then gives the Loewner equation.  $\square$

## 2.3 Kernel convergence and stability

**Theorem 2.3** (Kernel convergence of normalized conformal maps). *Let  $H_n \subset \mathbb{H}$  and  $H \subset \mathbb{H}$  be simply connected half-plane hull complements containing a fixed neighborhood of infinity, and assume  $H_n \rightarrow H$  in the pointed Carathéodory kernel sense. Let  $g_n : H_n \rightarrow \mathbb{H}$  and  $g : H \rightarrow \mathbb{H}$  be hydrodynamically normalized. Then  $g_n \rightarrow g$  locally uniformly on  $H$ . Moreover  $g_n^{-1} \rightarrow g^{-1}$  locally uniformly on every compact subset of  $\mathbb{H}$  whose inverse image remains a positive distance from the limiting hull.*

*Proof.* For pointed simply connected domains with hydrodynamic normalization, Montel's theorem gives subsequential locally uniform convergence of the maps. Kernel convergence identifies the image domain of any subsequential limit, and normalization at infinity makes that limit unique. The same argument applies to inverse maps on compact subsets of the limiting domain. Thus the whole sequence converges.  $\square$

**Theorem 2.4** (Local Lipschitz stability under perturbing the driver). *Let  $U, V \in C([0, T], \mathbb{R})$  and let  $g^U, g^V$  be their chordal Loewner flows. Fix  $z \in \mathbb{H}$  and  $0 < t_0 \leq T$ . If*

$$\inf_{0 \leq t \leq t_0} \operatorname{Im} g_t^U(z) \geq \eta, \quad \inf_{0 \leq t \leq t_0} \operatorname{Im} g_t^V(z) \geq \eta$$

*for some  $\eta > 0$ , then*

$$\sup_{0 \leq t \leq t_0} |g_t^U(z) - g_t^V(z)| \leq C(\eta, t_0) \|U - V\|_{\infty, [0, t_0]}.$$

*The same Gronwall argument gives local stability of inverse maps and of swallowing times on compact sets that stay uniformly away from the singular boundary.*

*Proof.* Let  $g^U$  and  $g^V$  solve the Loewner equation and suppose the orbit of  $z$  remains at imaginary height at least  $\eta > 0$  for both flows. Subtract the equations:

$$\frac{d}{dt}(g_t^U - g_t^V) = 2 \frac{(g_t^V - V_t) - (g_t^U - U_t)}{(g_t^U - U_t)(g_t^V - V_t)}.$$

The denominator is bounded below by  $\eta^2$ , so

$$|g_t^U - g_t^V| \leq C_\eta \int_0^t |g_s^U - g_s^V| ds + C_\eta t \|U - V\|_\infty.$$

Gronwall gives a local Lipschitz stability estimate. The same estimate, applied before compact sets approach the hull, gives inverse-map and swallowing-time stability in the corresponding nondegenerate regime.  $\square$

## Chapter 3

# Deterministic regularity and explicit examples

The scale-invariant regularity class for a Loewner driver is  $\text{Lip}(1/2)$  because capacity time scales like length squared. A remarkable sharp threshold occurs at norm 4.

### 3.1 The sharp quasilit criterion

For a real function  $U$  set

$$\|U\|_{1/2} = \sup_{s < t} \frac{|U_t - U_s|}{\sqrt{t - s}}.$$

**Theorem 3.1** (Sharp  $\text{Lip}(1/2)$  quasilit criterion).

*Let  $U : [0, T] \rightarrow \mathbb{R}$  be a chordal Loewner driving function in the half-plane-capacity convention*

$$\partial_t g_t(z) = \frac{2}{g_t(z) - U_t}.$$

*If*

$$\|U\|_{1/2} := \sup_{0 \leq s < t \leq T} \frac{|U_t - U_s|}{\sqrt{t - s}} < 4,$$

*then every  $\mathbb{H} \setminus K_t$  is a quasilit half-plane, with quasilit constant depending only on  $\|U\|_{1/2}$ . The number 4 is sharp: for every  $c \geq 4$  there is a  $\text{Lip}(1/2)$  driver of seminorm  $c$  whose Loewner hull is not generated by a simple slit.*

*Proof.* Write  $c = \|U\|_{1/2}$ . Translation, time reversal, and the scaling from Chapter 1 reduce the proof to a reverse Loewner flow

$$\dot{f}_t(z) = -\frac{2}{f_t(z) - \xi_t}, \quad \xi_0 = 0, \quad \|\xi\|_{1/2} \leq c,$$

on the time interval  $[0, 1]$ . We give the quantitative argument because the constant 4 is lost if one replaces it by a generic distortion estimate.

Consider first a real solution of the forward equation

$$\dot{x}(t) = \frac{2}{x(t) - \lambda(t)}, \quad x(0) > \lambda(0),$$

and suppose that it is first caught at time 1:  $x(1) = \lambda(1)$ . Since  $x$  is increasing and  $|\lambda(1) - \lambda(t)| \leq c\sqrt{1-t}$ ,

$$0 < x(t) - \lambda(t) \leq c\sqrt{1-t}.$$

Hence

$$x(1) - x(t) \geq \frac{4}{c}\sqrt{1-t}.$$

Insert this improvement back into the preceding bound. If

$$h_0(c) = c, \quad h_{n+1}(c) = c - \frac{4}{h_n(c)},$$

then induction gives  $h_n(c) > 0$  for every  $n$  whenever a catching occurs. Let  $a_n$  be the positive zero of  $h_n$ . The recursion is monotone on its natural domain,  $a_n \uparrow 4$ , and therefore  $c \geq 4$ . Indeed, if  $a = \lim a_n < 4$ , then  $h_n(a)$  decreases to a positive number  $L$  satisfying  $L = a - 4/L$ , which has no real solution for  $a < 4$ . Thus a driver of seminorm strictly below four cannot catch a forward real trajectory.

We also need a uniform, rather than merely qualitative, version. Suppose  $x(1) - \lambda(1) \leq \varepsilon$ . The same integration, now with the error term retained, yields recursively

$$x(1) - x(t) \geq \varepsilon - e_n(c, \varepsilon) + (c - h_n(c))\sqrt{1-t},$$

where  $e_n(c, \varepsilon) \downarrow 0$  as  $\varepsilon \downarrow 0$  for every fixed  $n$ . Since  $c < 4$ , choose  $n$  with  $h_n(c) < 0$ ; then choose  $\varepsilon_c > 0$  so small that  $e_n(c, \varepsilon_c) < -h_n(c)$ . Taking  $t = 0$  gives a contradiction. Consequently

$$x(1) - \lambda(1) \geq \varepsilon_c \tag{3.1}$$

for every such trajectory. This uniform separation is the decisive point at which the strict inequality  $c < 4$  enters.

$$\dot{X}_t = -\frac{2}{X_t - \xi_t}, \quad X_0 = x \neq 0,$$

and let  $\tau(x)$  be its first hitting time of the singularity. Reversing a unit time interval and applying (3.1) shows that a trajectory starting a fixed distance  $\varepsilon_c$  from the singularity must hit within a bounded time. Scaling then gives the upper bound

$$\tau(x) \leq C_2(c)x^2.$$

For the lower bound, scale to  $x = 1$ . If  $\tau(1) = \delta$  were arbitrarily small, then  $|\xi_t| \leq c\sqrt{\delta}$  for  $t \leq \delta$ . Until the trajectory reaches  $1/2$  its speed is at most  $2/(1/2 - c\sqrt{\delta})$  in absolute value, so it cannot travel distance  $1/2$  in time  $\delta$  once  $\delta$  is small enough. Hence

$$C_1(c)x^2 \leq \tau(x) \leq C_2(c)x^2. \tag{3.2}$$

On each side of 0 the map  $x \mapsto \tau(x)$  is strictly monotone. Indeed, for  $0 < x < y$ ,

$$\frac{d}{dt}(Y_t - X_t) = 2 \frac{Y_t - X_t}{(Y_t - \xi_t)(X_t - \xi_t)} > 0.$$

The same calculation, after reflecting the line, handles the negative side. Thus for every  $t > 0$  exactly one point on each side is swallowed at time  $t$ . Pair these two points and write  $\phi$  for the resulting orientation-reversing welding. Estimate (3.2) already gives

$$C(c)^{-1} \leq \frac{x}{-\phi(x)} \leq C(c), \quad x > 0. \tag{3.3}$$

The stronger three-point estimate follows from the same ODE. We record the calculation because it is the bridge from “no catching” to a geometric slit. If  $0 \leq x < y < z$  and  $y - x = z - y$ , set

$$q_t = \frac{Y_t - X_t}{Z_t - Y_t}.$$

A direct differentiation gives

$$\frac{d}{dt} \log q_t = 2 \frac{Z_t - X_t}{(X_t - \xi_t)(Y_t - \xi_t)(Z_t - \xi_t)}.$$

Run until the leftmost trajectory is swallowed. The quadratic hitting-time bound and the uniform separation estimate imply that, at that time, the two remaining distances from the singularity have ratio in a compact interval depending only on  $c$ . Running the displayed differential identity backward therefore bounds  $q_0$  above and below. When the three points lie far from the singularity, first run until the nearest distance is comparable with the outer gap; the same argument then applies. Hence

$$A(c)^{-1} \leq \frac{\phi(x) - \phi(y)}{\phi(y) - \phi(z)} \leq A(c). \quad (3.4)$$

The compactness step needed when a ratio approaches an endpoint is harmless: if a sequence of drivers with seminorm at most  $c$  violated the bound, Arzelà–Ascoli gives a locally uniform limiting driver, and (3.1) keeps all relevant trajectories uniformly away from the singularity before their terminal time. Ordinary ODE stability then passes the hitting times and the welding relation to the limit, where strict monotonicity gives the contradiction.

For completeness we recall the elementary extension fact used here. An orientation-reversing homeomorphism of the line satisfying (3.3) and (3.4) is quasisymmetric after reflection on one side. The Beurling–Ahlfors extension

$$H(x + iy) = \frac{1}{2} \int_0^1 \{h(x + ty) + h(x - ty)\} dt + i \int_0^1 \{h(x + ty) - h(x - ty)\} dt$$

of the corresponding increasing welding  $h$  is quasiconformal. To see the uniform distortion directly, denote by  $A = h(x + y) - h(x)$  and  $B = h(x) - h(x - y)$  the adjacent increments. Quasisymmetry gives  $A/B \asymp_{A(c)} 1$ ; applying the same estimate to dyadic subintervals shows that the four averages occurring in the partial derivatives of  $H$  are all comparable to  $(A + B)/y$ . Thus the largest and smallest singular values of  $DH$  have a bounded ratio, so  $|\bar{\partial}H| \leq k(c)|\partial H|$  with  $k(c) < 1$ . Gluing the two half-planes by this extension sends the imaginary axis to a quasiarc. Equivalently the associated half-plane domain is a quasislit, and its quasiconformal constant depends only on  $c$ .

Let  $\xi^{(n)}$  be the piecewise-linear interpolation of  $\xi$  on the mesh  $k/n$ . Convexity on each mesh interval and a two-endpoint maximization show

$$\|\xi^{(n)}\|_{1/2} \leq \|\xi\|_{1/2} = c.$$

Each linear piece generates a smooth slit, so their finite composition is a slit domain. The preceding welding bounds are uniform in  $n$ ; hence these are  $K(c)$ -quasislits. The drivers converge uniformly, so Chapter 2’s Loewner stability theorem gives kernel convergence of the domains. A normal-family limit of  $K(c)$ -quasiconformal maps is again  $K(c)$ -quasiconformal (after the standard three-point normalization). The limit domain is therefore a quasislit. Scaling and time translation give the assertion for every  $t \leq T$ .



For  $c \geq 4$  put, after a harmless time translation,

$$\lambda(t) = c - c\sqrt{1-t}, \quad x(t) = c - a\sqrt{1-t}, \quad a = \frac{c + \sqrt{c^2 - 16}}{2}.$$

Then  $a(c-a) = 4$ , and a direct substitution shows  $\dot{x} = 2/(x-\lambda)$ . Moreover  $\|\lambda\|_{1/2} = c$  and  $x(1) = \lambda(1)$ . Thus the driving point catches a real trajectory at time one, which cannot happen for a slit half-plane. At  $c = 4$  this already gives the critical obstruction. Hence no universal constant larger than four is possible.

All constants here use the same half-plane-capacity normalization as Chapters 1–2. In particular the numerical threshold is 4, not a rescaled version arising from the convention  $\partial_t g_t = a/(g_t - U_t)$ .  $\square$

**Example 3.2** (The threshold driver). At the critical value,  $\lambda(t) = 4 - 4\sqrt{1-t}$  and  $x(t) = 4 - 2\sqrt{1-t}$  satisfy the forward Loewner ODE and collide at  $t = 1$ . The example makes the meaning of the number four concrete: it is exactly the smallest  $\text{Lip}(1/2)$  speed at which the driver can overtake a real trajectory that the Loewner flow is pushing away.

### 3.2 Constant and square-root driving

For  $U_t \equiv 0$ ,

$$\partial_t g_t = \frac{2}{g_t}, \quad g_t(z)^2 = z^2 + 4t,$$

so

$$g_t(z) = \sqrt{z^2 + 4t}$$

with the branch fixed by  $g_t(z) \sim z$  at infinity. The hull is the vertical segment  $[0, 2i\sqrt{t}]$ .

**Theorem 3.3** (Constant and square-root self-similar drivers). *For the constant driver  $U_t \equiv c$ ,*

$$g_t(z) = c + \sqrt{(z-c)^2 + 4t},$$

*with the hydrodynamic branch, and  $K_t = c + [0, 2i\sqrt{t}]$ . For a square-root driver  $U_t = c\sqrt{t}$ , the Loewner chain is self-similar:  $K_t = \sqrt{t} K_1$ . Writing  $g_t(z) = \sqrt{t} G(z/\sqrt{t})$  reduces the Loewner equation to*

$$G(w) - wG'(w) = \frac{4}{G(w) - c}.$$

*Thus the square-root family is governed by a time-independent first-order ODE; its boundary singularities determine whether the self-similar hull remains a slit or reaches the real boundary.*

*Proof.* Translation handles any constant driver. For  $U_t = c\sqrt{t}$ , scale invariance implies  $K_t = \sqrt{t} K_1$ . Set  $g_t(z) = \sqrt{t} G(z/\sqrt{t})$  and substitute into the ODE. The resulting first-order equation for  $G$  is autonomous and can be integrated implicitly. Its boundary singularity determines whether the self-similar trace remains a simple slit or collides with the real axis. These examples exhibit directly why a  $1/2$ -Holder driver is critical and why the dimensionless norm, not an ordinary derivative, governs trace geometry.  $\square$

## Chapter 4

# Radial and whole-plane Loewner evolution

Chordal Loewner evolution grows from one boundary point toward another. Radial evolution grows from the boundary toward an interior point, while whole-plane evolution is normalized across all logarithmic times.

### 4.1 Radial Loewner evolution

Let  $K_t \subset \mathbb{D}$  grow from the boundary and leave 0 in the simply connected component. Normalize

$$g_t(0) = 0, \quad g'_t(0) = e^t.$$

If  $e^{iU_t}$  is the image of the growing tip, the radial equation is

$$\partial_t g_t(z) = g_t(z) \frac{e^{iU_t} + g_t(z)}{e^{iU_t} - g_t(z)}.$$

**Theorem 4.1** (Radial Loewner equation and logarithmic conformal radius). *Let  $K_t \subset \mathbb{D}$  be an increasing hull chain attached to  $\partial\mathbb{D}$  such that 0 remains in the simply connected component, and let  $g_t : \mathbb{D} \setminus K_t \rightarrow \mathbb{D}$  satisfy  $g_t(0) = 0$  and  $g'_t(0) = e^t$ . If the image of the growing prime end is  $e^{iU_t}$ , then*

$$\partial_t g_t(z) = g_t(z) \frac{e^{iU_t} + g_t(z)}{e^{iU_t} - g_t(z)}.$$

*The time  $t$  equals the logarithmic loss of conformal radius at 0:  $\text{crad}(\mathbb{D} \setminus K_t, 0) = e^{-t}$  under this normalization.*

*Proof.* Apply the disk version of the small-hull transition-map argument. The infinitesimal generator must be holomorphic in the disk, have positive real radial expansion at the origin, and a single boundary pole at the driving point. The Herglotz/Schwarz kernel with these properties is  $z(\zeta + z)/(\zeta - z)$ . The normalization  $g'_t(0) = e^t$  fixes the coefficient to one. Uniqueness of ODE solutions then gives the radial chain and identifies  $t$  with logarithmic conformal-radius loss.  $\square$

### 4.2 Changing simply connected coordinates

**Theorem 4.2** (Conformal covariance of chordal Loewner chains). *Let  $(D; x, y)$  and  $(\tilde{D}; \tilde{x}, \tilde{y})$  be simply connected marked domains and let  $\phi : D \rightarrow \tilde{D}$  be conformal with  $\phi(x) = \tilde{x}$  and  $\phi(y) = \tilde{y}$  in*

the prime-end sense. If  $(K_t)$  is a chordal Loewner chain in  $D$ , then  $(\phi(K_t))$  is a chordal Loewner chain in  $\tilde{D}$  after reparametrization by the half-plane capacity in any fixed uniformizing coordinate. The driving prime end is carried by  $\phi$ , and changing coordinates affects only this deterministic capacity time-change.

*Proof.* Let  $\phi : D \rightarrow \tilde{D}$  be conformal and conjugate a chordal chain by  $\phi$ . Uniformizing the evolving image domain and comparing hydrodynamic expansions shows that the image is again a Loewner chain. The marked boundary point is carried by the conformal map, while the time parameter changes by the derivative of the image capacity. Thus chordal Loewner evolution is intrinsic to a simply connected domain once one remembers the capacity time-change.  $\square$

### 4.3 Whole-plane normalization

**Theorem 4.3** (Deterministic whole-plane normalization). *A compatible two-sided family of radial Loewner chains indexed by  $t \in \mathbb{R}$  and normalized by logarithmic conformal radius determines at most one whole-plane Loewner chain joining the distinguished interior point to infinity. If the finite-start approximations converge locally uniformly as their starting times tend to  $-\infty$ , their limit solves the corresponding two-sided radial Loewner equation and is independent of the approximating sequence. A shift  $t \mapsto t + s$  is equivalent to deterministic spatial rescaling by the conformal-radius factor  $e^s$ .*

*Proof.* A whole-plane chain is best viewed as a two-sided radial chain indexed by  $t \in \mathbb{R}$ , normalized by logarithmic conformal radius. For each finite starting time  $s$ , solve the radial equation from the tiny neighborhood determined by the normalization; compatibility as  $s \rightarrow -\infty$  yields the whole-plane family. Uniqueness follows from the same transition-map argument as in the radial case. Shifting  $t$  corresponds to deterministic spatial scaling, which is the whole-plane analogue of chordal capacity scaling.  $\square$

## Chapter 5

# Force points and marked-point dynamics

SLE with force points will modify the random driving equation, but the geometry of the marked boundary points is deterministic once the driving function is known. We establish that bookkeeping here.

**Definition 5.1** (Boundary marked point and force point). For a chordal Loewner chain with driver  $U_t$ , a boundary marked point  $x \in \mathbb{R}$  is tracked, until swallowing, by  $V_t = g_t(x)$ . When the stochastic driving equation is modified by a drift of the form  $\rho/(U_t - V_t)$ , the marked point is called a *force point* of weight  $\rho$ . The terminology refers to the drift in the driving process; the marked point itself still evolves deterministically by  $\dot{V}_t = 2/(V_t - U_t)$  once  $U$  is fixed.

### 5.1 Boundary marked points

Let  $x \in \mathbb{R}$  be a boundary point not yet swallowed and define

$$V_t = g_t(x).$$

Taking the boundary limit of the Loewner equation gives

$$\dot{V}_t = \frac{2}{V_t - U_t}.$$

**Theorem 5.2** (Deterministic evolution of boundary marked points). *Let  $g_t$  be a chordal Loewner flow with continuous driver  $U_t$ , and let  $x_1 < \dots < x_m$  be boundary points not yet swallowed. Their images  $V_t^j = g_t(x_j)$  satisfy, up to the first collision/swallowing time,*

$$\frac{dV_t^j}{dt} = \frac{2}{V_t^j - U_t}, \quad V_0^j = x_j.$$

*The ordering of the  $V_t^j$  is preserved until collision. Thus a finite collection of marked boundary points is evolved by one-dimensional ODEs once the driver is fixed.*

*Proof.* Approximate  $x$  by  $x + i\varepsilon$  and use the locally uniform ODE estimates before the point is swallowed. Passing  $\varepsilon \downarrow 0$  gives the marked-point equation. Uniqueness prevents two marked points from changing order before collision, because their gap solves a scalar ODE whose sign cannot change without first reaching zero. Evolving a finite set of marked points therefore requires no new conformal maps: their images follow coupled one-dimensional ODEs driven by the same  $U_t$ .  $\square$

## 5.2 Gaps and cross-ratios

**Theorem 5.3** (Gap and cross-ratio evolution). *In the setting of Theorem SM-13.05.01, put  $X_t^j = V_t^j - U_t$ . For  $0 \leq s < t$  before swallowing,*

$$X_t^j - X_s^j = \int_s^t \frac{2}{X_r^j} dr - (U_t - U_s).$$

*For two marked points,*

$$\frac{d}{dt}(V_t^i - V_t^j) = 2 \left( \frac{1}{V_t^i - U_t} - \frac{1}{V_t^j - U_t} \right).$$

*Consequently every cross-ratio of four distinct marked boundary points obeys a closed first-order evolution equation obtained by differentiating its rational expression in the pairwise gaps. These identities remain valid in semimartingale form after stochastic driving is introduced.*

*Proof.* For a gap  $X_t = V_t - U_t$ , an arbitrary continuous driver need not be differentiable, so the rigorous identity is the increment formula

$$X_t - X_s = \int_s^t \frac{2}{X_r} dr - (U_t - U_s).$$

When  $U$  is differentiable this may of course be written as  $dX_t = 2X_t^{-1}dt - dU_t$ ; later, for semimartingale drivers, the same notation is interpreted stochastically. For two marked points,

$$\frac{d}{dt}(V_t^i - V_t^j) = 2 \left( \frac{1}{V_t^i - U_t} - \frac{1}{V_t^j - U_t} \right).$$

Any cross-ratio of four marked boundary points is a rational function of these differences; the quotient rule therefore yields a closed first-order evolution equation. These formulas are exactly what is needed to differentiate SLE partition functions and to identify Bessel processes after stochastic driving is introduced.  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 5.4** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 5.5** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 5.6** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 5.7** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 5.8** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XIV: Schramm-Loewner Evolution

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# Scope and conventions

The deterministic Loewner equation of Volume XIII becomes probabilistic when its driving function is Brownian motion. The resulting family is rigid enough to be characterized by conformal invariance and the domain Markov property, yet rich enough to change topology at  $\kappa=4$  and  $\kappa=8$ . This volume develops the elementary stochastic-calculus consequences first, then treats the deeper trace, restriction, natural-time, and reversibility theorems with their additional analytic inputs stated explicitly.

# Chapter 1

## Definition and characterization

Schramm's rigidity principle is the conceptual entry point: once a planar interface has conformal invariance and a domain Markov property, capacity time turns its geometry into a one-dimensional process with stationary independent increments.

**Definition 1.1** (Chordal  $\text{SLE}_\kappa$ ). For  $\kappa \geq 0$ , *chordal*  $\text{SLE}_\kappa$  in  $(\mathbb{H}, 0, \infty)$  is the random chordal Loewner chain driven by  $U_t = \sqrt{\kappa}B_t$ , where  $B$  is standard one-dimensional Brownian motion and time is normalized by  $\text{hcap}(K_t) = 2t$ . In another simply connected marked domain  $(D, a, b)$ , chordal  $\text{SLE}_\kappa$  is the conformal image of this law; only its capacity parametrization changes under the conformal map.

### 1.1 Brownian-driven Loewner evolution

**Theorem 1.2** (Brownian-driven chordal Loewner evolution).

Fix  $\kappa \geq 0$  and let  $U_t = \sqrt{\kappa}B_t$ . For every  $z \in \mathbb{H}$  solve

$$\partial_t g_t(z) = \frac{2}{g_t(z) - U_t}, \quad g_0(z) = z,$$

up to the first time  $\tau_z$  at which the denominator tends to zero. Then

$$K_t = \{z \in \overline{\mathbb{H}} : \tau_z \leq t\}$$

is an increasing family of half-plane hulls,  $g_t : \mathbb{H} \setminus K_t \rightarrow \mathbb{H}$  is the hydrodynamically normalized conformal map, and  $\text{hcap}(K_t) = 2t$ . The resulting random hull chain is chordal  $\text{SLE}_\kappa$  from 0 to  $\infty$  in  $\mathbb{H}$ .

*Proof.* A Brownian sample path is continuous, so the deterministic Loewner theorem of Volume XIII applies pathwise to the driver  $U$ . For each  $z \in \mathbb{H}$  the ODE has a unique maximal solution until  $g_t(z) - U_t$  approaches zero. If  $s < t$ , a point swallowed by time  $s$  is also swallowed by time  $t$ , hence  $K_s \subseteq K_t$ . The deterministic theorem identifies  $H_t = \mathbb{H} \setminus K_t$  as simply connected and shows that  $g_t : H_t \rightarrow \mathbb{H}$  is conformal with expansion

$$g_t(z) = z + \frac{2t}{z} + O(|z|^{-2}) \quad (z \rightarrow \infty).$$

By the definition of half-plane capacity in Volume IV this is exactly  $\text{hcap}(K_t) = 2t$ .

Measurability of the hull chain follows by approximating Brownian paths uniformly on compact time intervals by polygonal paths and using continuous dependence of Loewner solutions on the

driver away from swallowing times. This construction defines SLE as a random hull chain. The stronger assertion that the hulls are generated by a continuous trace is a separate theorem in Chapter 3 and is not used here.  $\square$

## 1.2 Schramm characterization

**Theorem 1.3** (Schramm characterization).

*Suppose that to every simply connected Jordan domain  $D$  with distinct marked boundary prime ends  $a, b$  is assigned a law on continuous non-self-crossing curves from  $a$  to  $b$  with the following properties.*

1. *Conformal covariance: conformal maps carry the law in  $(D, a, b)$  to the corresponding law in the image domain.*
2. *Domain Markov property: conditional on an initial segment, the remaining curve, viewed in the component containing the target, has the original law from the tip to the target.*

*Assume moreover that in  $(\mathbb{H}, 0, \infty)$  the curve admits the chordal Loewner parametrization by half-plane capacity and has a continuous driving function  $U$ . Then there is a number  $\kappa \geq 0$  such that*

$$U_t = \sqrt{\kappa} B_t$$

*in law. Consequently the hull chain is chordal  $\text{SLE}_\kappa$ .*

*Proof.* Work in  $(\mathbb{H}, 0, \infty)$  and parametrize the hull  $K_t$  by  $\text{hcap}(K_t) = 2t$ . Let  $g_t : \mathbb{H} \setminus K_t \rightarrow \mathbb{H}$  be hydrodynamically normalized and let  $U_t$  be its continuous Loewner driver. The recentered map

$$z \mapsto g_t(z) - U_t$$

sends the unbounded component of  $\mathbb{H} \setminus K_t$  to  $\mathbb{H}$ , the current tip to 0, and the target to  $\infty$ .

Fix  $s \geq 0$ . Conditional on the initial hull  $K_s$ , the domain Markov property says that the future curve, after applying  $g_s - U_s$ , has the same law as a fresh copy of the original curve in  $(\mathbb{H}, 0, \infty)$  and is independent of the past. The deterministic composition rule for chordal Loewner chains gives, for  $t \geq 0$ ,

$$g_{s+t} = g_t^{(s)} \circ g_s,$$

where  $g^{(s)}$  is the Loewner chain of the mapped future. Its driver is

$$U_t^{(s)} = U_{s+t} - U_s.$$

Hence  $(U_{s+t} - U_s)_{t \geq 0}$  is independent of  $\sigma(U_r : 0 \leq r \leq s)$  and has the same law as  $(U_t)_{t \geq 0}$ . Thus  $U$  has stationary independent increments.

A process with stationary independent increments which is stochastically continuous is a Lévy process. Here sample-path continuity is assumed, so the Lévy–Khintchine representation has neither a jump part nor a killing term. Therefore, for constants  $\mu \in \mathbb{R}$  and  $\sigma \geq 0$ ,

$$U_t = \mu t + \sigma B_t$$

in law.

For  $r > 0$ , the automorphism  $z \mapsto rz$  of  $\mathbb{H}$  preserves the marked points. Half-plane capacity scales quadratically, so conformal covariance implies

$$(r^{-1}U_{r^2t})_{t \geq 0} \stackrel{d}{=} (U_t)_{t \geq 0}.$$

For the preceding Gaussian Lévy process the left side has drift  $r\mu t$  and variance coefficient  $\sigma^2$ . Equality for every  $r > 0$  forces  $\mu = 0$ . Put  $\kappa = \sigma^2$ .

$U_t = \sqrt{\kappa}B_t$ . By uniqueness of the deterministic Loewner evolution driven by a fixed continuous function, the associated hull chain has precisely the chordal  $\text{SLE}_\kappa$  law.

The proof uses only the two structural hypotheses stated above, continuity of the Loewner driver, the deterministic Loewner composition law proved in Volume XIII, and the elementary classification of continuous Lévy processes from Volume II. Reflection symmetry is therefore unnecessary: scale covariance already rules out a linear drift. This is the standard rigidity argument introduced by Schramm in his derivation of SLE from conformal invariance and the domain Markov property.  $\square$

*Remark 1.4.* This theorem is a characterization of the *driving process*. It does not prove that an arbitrary lattice interface has a conformally invariant scaling limit, nor that a Brownian-driven Loewner chain is generated by a continuous trace. Those are separate tightness/identification and trace theorems.

### 1.3 Scaling and conformal covariance

**Theorem 1.5** (Scaling and conformal covariance of chordal SLE).

*Let  $(K_t)$  be chordal  $\text{SLE}_\kappa$  in  $\mathbb{H}$  from 0 to  $\infty$ . For every  $r > 0$ ,*

$$(r^{-1}K_{r^2t})_{t \geq 0} \stackrel{d}{=} (K_t)_{t \geq 0}.$$

*If  $D$  is simply connected and  $a, b$  are distinct boundary prime ends for which a conformal map to  $(\mathbb{H}, 0, \infty)$  is defined, chordal SLE in  $(D, a, b)$  is the conformal image of the preceding process. This law is independent of the chosen uniformizing map, up to the deterministic change of capacity clock forced by the conformal map.*

*Proof.* Let  $g_t$  solve the Loewner equation with driver  $U_t = \sqrt{\kappa}B_t$  and put

$$\tilde{g}_t(z) = r^{-1}g_{r^2t}(rz).$$

Differentiation gives

$$\partial_t \tilde{g}_t(z) = \frac{2}{\tilde{g}_t(z) - \tilde{U}_t}, \quad \tilde{U}_t = r^{-1}U_{r^2t}.$$

Brownian scaling says  $(r^{-1}B_{r^2t})_{t \geq 0} \stackrel{d}{=} (B_t)_{t \geq 0}$ , so  $\tilde{U}$  has the same law as  $U$ . Uniqueness of the deterministic Loewner evolution therefore gives  $r^{-1}K_{r^2t} \stackrel{d}{=} K_t$  jointly in  $t$ .

Now let  $\phi : D \rightarrow \mathbb{H}$  send  $a$  to 0 and  $b$  to  $\infty$ . If  $\tilde{\phi}$  is another such map, then  $\tilde{\phi} \circ \phi^{-1}$  is an automorphism of  $\mathbb{H}$  fixing 0 and  $\infty$ , hence is  $z \mapsto rz$  for some  $r > 0$ . The scaling identity just proved shows that the two pullback constructions give the same unparametrized hull law; their capacity times differ by the factor  $r^2$ . This proves conformal covariance and independence of the uniformization.  $\square$



## Chapter 2

# Bessel-process analysis and force points

Force points are not auxiliary decoration. They are the bookkeeping device that records how conditioning, coordinate changes, and boundary data alter an SLE law. The apparently two-dimensional boundary interaction becomes a one-dimensional Bessel problem.

**Definition 2.1** (One-force-point  $\text{SLE}_\kappa(\rho)$ ). Before the force point collides with the driver, chordal  $\text{SLE}_\kappa(\rho)$  with a single right force point is the solution of

$$dU_t = \sqrt{\kappa} dB_t + \frac{\rho}{U_t - V_t} dt, \quad dV_t = \frac{2}{V_t - U_t} dt, \quad V_t > U_t.$$

The process is interpreted up to the first time at which the chosen continuation rule ceases to be defined. Whenever this volume invokes a continuation beyond a collision, that continuation convention is stated separately.

*Worked example 2.2* (A collision threshold at  $\kappa = 4$ ). For ordinary  $\text{SLE}_4$  with a passive right force point ( $\rho = 0$ ),

$$\delta = 1 + \frac{2(\rho + 2)}{\kappa} = 2,$$

so the normalized gap is  $\text{BES}^2$  and does not hit zero. If instead  $\rho = -1$ , then  $\delta = 3/2 < 2$ , so the Bessel hitting criterion implies collision almost surely. Thus changing only the force weight changes a geometric swallowing question into the elementary threshold  $\delta = 2$  for a one-dimensional diffusion.

## 2.1 Force-point separation and Bessel dimension

**Theorem 2.3** (Force-point separation is a Bessel process).

*Consider chordal  $\text{SLE}_\kappa(\rho)$  with one force point immediately to the right of the driving point, written before collision as*

$$dU_t = \sqrt{\kappa} dB_t + \frac{\rho}{U_t - V_t} dt, \quad dV_t = \frac{2}{V_t - U_t} dt, \quad V_t > U_t.$$

*Set  $X_t = (V_t - U_t)/\sqrt{\kappa}$ . Then, after replacing  $-B$  by a Brownian motion  $W$ ,*

$$dX_t = dW_t + \frac{\delta - 1}{2X_t} dt, \quad \delta = 1 + \frac{2(\rho + 2)}{\kappa}.$$

Thus the normalized force-point separation is a Bessel process of dimension  $\delta$  up to its first visit to zero.

*Proof.* Put  $Y_t = V_t - U_t$ . Subtracting the two SDEs gives

$$dY_t = -\sqrt{\kappa} dB_t + \left( \frac{2}{Y_t} + \frac{\rho}{Y_t} \right) dt = -\sqrt{\kappa} dB_t + \frac{\rho + 2}{Y_t} dt.$$

After division by  $\sqrt{\kappa}$ ,

$$dX_t = -dB_t + \frac{\rho + 2}{\kappa X_t} dt.$$

Since  $W = -B$  is Brownian motion, comparison with the defining Bessel SDE  $dR = dW + (\delta - 1)(2R)^{-1}dt$  gives  $\delta - 1 = 2(\rho + 2)/\kappa$ , as claimed.  $\square$

## 2.2 Collision and swallowing thresholds

**Theorem 2.4** (Collision and boundary swallowing from the Bessel dimension).

*In the one-force-point setting above, let  $\delta = 1 + 2(\rho + 2)/\kappa$ . If  $\delta \geq 2$ , equivalently*

$$\rho \geq \frac{\kappa}{2} - 2,$$

*the force point almost surely never collides with the driver. If  $\delta < 2$ , equivalently  $\rho < \kappa/2 - 2$ , the Bessel separation reaches zero almost surely. In Loewner language this is the threshold between a force point which is never swallowed and one which is hit by the driving boundary singularity.*

*Proof.* Volume III proves the Bessel hitting criterion: a Bessel process started from a positive point hits zero if and only if its dimension is strictly smaller than 2. By the preceding theorem the normalized separation is  $BES^\delta$  until collision. Therefore collision occurs exactly when  $\delta < 2$ . Solving  $1 + 2(\rho + 2)/\kappa < 2$  gives  $\rho < \kappa/2 - 2$ ; the complementary inequality gives noncollision. Because  $V_t = g_t(x)$  for the marked boundary point as long as it is not swallowed,  $V_t - U_t \downarrow 0$  is precisely the Loewner swallowing/collision event.  $\square$

**Example 2.5.** For ordinary SLE with a single passive boundary point ( $\rho=0$ ), the normalized separation is a Bessel process of dimension  $1 + 4/\kappa$ . Thus the threshold dimension 2 occurs at  $\kappa = 4$ , already foreshadowing the boundary-touching phase transition.

## 2.3 Continuation through force-point collisions

**Theorem 2.6** (The standard one-force-point  $SLE_\kappa(\rho)$  regime).

*For a single boundary force point and  $\rho > -2$ , the Bessel dimension*

$$\delta = 1 + \frac{2(\rho + 2)}{\kappa}$$

*is greater than 1. The nonnegative  $BES^\delta$  process has its standard instantaneous-reflection continuation through zero when  $1 < \delta < 2$ , and never reaches zero when  $\delta \geq 2$ . Transporting this continuation through  $V_t - U_t = \sqrt{\kappa}X_t$  defines the standard chordal  $SLE_\kappa(\rho)$  evolution. The value  $\rho = -2$  is the single-force-point continuation threshold for this SDE construction.*

*Proof.* For  $\rho > -2$  one has  $\delta > 1$ . The squared Bessel process  $R_t = X_t^2$  satisfies

$$dR_t = 2\sqrt{R_t}dW_t + \delta dt.$$

For every  $\delta > 0$  this equation has the canonical nonnegative strong solution; when  $1 < \delta < 2$  it visits zero but spends zero Lebesgue time there and leaves instantaneously, while for  $\delta \geq 2$  it does not hit zero. Hence  $X = \sqrt{R}$  supplies the canonical nonnegative separation process throughout the standard regime  $\rho > -2$ .

Recover  $Y_t = \sqrt{\kappa}X_t = V_t - U_t$ . Away from zero the Bessel SDE is exactly the coupled force-point SDE of the first theorem. At collision, the nonnegative Bessel continuation specifies the limiting right-force-point convention. If  $\rho \leq -2$ , then  $\delta \leq 1$  and the ordinary force-point SDE no longer provides this standard continuation without an additional convention; in the usual SLE terminology the first collision with total colliding weight at most  $-2$  is the continuation threshold. For one force point this is exactly  $\rho = -2$ .  $\square$

## Chapter 3

# Trace generation and phase transitions

### 3.1 Continuous trace for every parameter

**Theorem 3.1** (Brownian Loewner chains are generated by a continuous trace).

*For every  $\kappa \geq 0$ , chordal  $\text{SLE}_\kappa$  in  $\mathbb{H}$  from 0 to  $\infty$  is almost surely generated by a continuous curve*

$$\gamma : [0, \infty) \longrightarrow \overline{\mathbb{H}}, \quad \gamma(t) = \lim_{y \downarrow 0} \widehat{f}_t(iy), \quad \widehat{f}_t(z) := g_t^{-1}(z + W_t),$$

and  $\mathbb{H} \setminus K_t$  is the unbounded component of  $\mathbb{H} \setminus \gamma[0, t]$ .

*Proof.* The proof has two genuinely different parts. Rohde–Schramm’s inverse-map argument treats every  $\kappa \neq 8$ . At  $\kappa = 8$  the optimized inverse-map exponent is exactly critical; we therefore give the independent Lawler–Schramm–Werner UST-Peano construction, using only random-walk/UST material already proved in Volumes IV–V.

Fix  $z = x + iy \in \mathbb{H}$  and run the Loewner equation backwards. If  $T_u$  denotes the time at which the imaginary part of the backward image is  $e^u$ , put

$$z(u) = X(u) + ie^u, \quad \psi(u) = \frac{y}{e^u} |g'_{T_u}(z)|.$$

Changing from Loewner time to  $u = \log \text{Im } g_t(z)$  gives

$$du = -2|g_t(z) - W_t|^{-2} dt, \quad d \log |g'_t(z)| = -2\Re(g_t(z) - W_t)^{-2} dt.$$

Moreover

$$\widehat{B}_u = -\sqrt{2/\kappa} \int_0^{T_u} |g_t(z) - W_t|^{-1} dW_t$$

has quadratic variation  $u$  and hence is Brownian motion. For a real parameter  $b$  define, with the sign corresponding to the backward flow,

$$a = 2b + \frac{\kappa}{2}b(1-b), \quad \lambda = 4b + \frac{\kappa}{2}b(1-2b),$$

and

$$F_b(x + iy) = (1 + (x/y)^2)^b y^\lambda.$$

A direct Itô calculation in the  $(X, e^u)$  variables shows that

$$M_u := \psi(u)^a F_b(z(u))$$

has zero drift. Stopping between two horizontal lines makes  $M$  bounded; optional stopping and monotone convergence therefore give the derivative moment identity used below. This is the only stochastic-calculus input in the Rohde–Schramm estimate.

Choose

$$b = \frac{8 + \kappa}{4\kappa}.$$

Then the preceding formulas become

$$a = \frac{(\kappa + 8)(3\kappa + 8)}{32\kappa}, \quad \lambda = \frac{(\kappa + 8)^2}{16\kappa},$$

so

$$\boxed{\lambda - 2 = \frac{(\kappa - 8)^2}{16\kappa}}.$$

Thus  $\lambda > 2$  exactly when  $\kappa \neq 8$ . Markov's inequality applied to the stopped moment identity, followed by decomposition according to the height reached by the backward flow, gives the fixed-time tail bound

$$\mathbf{P}\left\{|\widehat{f}_t'(x + iy)| \geq \delta y^{-1}\right\} \leq C_\kappa (1 + x^2/y^2)^b (y/\delta)^\lambda \vartheta(\delta, a - \lambda), \quad (3.1)$$

where

$$\vartheta(\delta, s) = \begin{cases} \delta^{-s}, & s > 0, \\ 1 + |\log \delta|, & s = 0, \\ 1, & s < 0. \end{cases}$$

For completeness, the height decomposition is as follows. If  $u_1$  is the logarithmic height of the backward image at time  $-t$ , compare the derivative at  $-t$  with that at the first times  $T_j$  at which the height equals  $e^j$ . Because  $|\partial_u \log |g'_{T_u}|| \leq 1$ , the ratio of those derivatives is at most  $e^{|j - u_1|}$ . Summing over the integer values  $\lceil \log y \rceil \leq j \leq 0$  and applying the stopped moment identity at  $T_j$  yields (3.1), including the three cases encoded by  $\vartheta$ .

For  $j \geq 0$  and  $0 \leq k < 2^{2j}$  set

$$R(j, k) = [2^{-j-1}, 2^{-j}] \times [k2^{-2j}, (k+1)2^{-2j}]$$

in the  $(y, t)$  plane and

$$H(y, t) = \widehat{f}_t(iy), \quad d(j, k) = \text{diam } H(R(j, k)).$$

Choose

$$0 < \sigma < \frac{\lambda - 2}{\max\{a, \lambda\}}.$$

On the time interval of one Whitney box, partition backwards whenever the Brownian driver changes by  $2^{-j}$ . The number  $N$  of pieces has an exponential tail by the reflection principle. Koebe distortion shows that, unless one of the derivative values at the partition points exceeds  $2^{j(1-\sigma)}/j^2$ , the whole image of the box has diameter  $O(2^{-j\sigma})$ . Estimate (3.1), the exponential tail of  $N$ , and a union bound give some  $\varepsilon = \varepsilon(\kappa) > 0$  such that

$$\mathbf{P}\{d(j, k) > C2^{-j\sigma}\} \leq C2^{-2j}2^{-\varepsilon j}.$$

Since there are  $2^{2j}$  boxes at level  $j$ ,

$$\sum_{j \geq 0} \sum_{k < 2^{2j}} \mathbf{P}\{d(j, k) > C2^{-j\sigma}\} < \infty.$$

Borel–Cantelli therefore makes  $H(y, t)$  uniformly Cauchy as  $y \downarrow 0$  on every compact time interval. Adjacent Whitney boxes give continuity in  $t$  at  $y = 0$ . Hence

$$\gamma(t) = \lim_{y \downarrow 0} \widehat{f}_t(iy)$$

exists and is continuous for every  $\kappa \neq 8$ .

The deterministic Loewner boundary criterion proved in Volume XIII says that if the radial tip limit above exists continuously, then the corresponding continuous path generates the chain: for each  $t$ , the unbounded component of  $\mathbb{H} \setminus \gamma[0, t]$  is exactly  $H_t = \mathbb{H} \setminus K_t$ . Applying that criterion completes the proof for  $\kappa \neq 8$ .

Take a  $C^1$  Jordan domain  $D$ , wire a proper boundary arc  $A$ , and let  $\eta_\delta$  be the Peano interface between a UST and its dual on a mesh  $\delta$  approximation. Wilson’s algorithm, already proved in Volume V, gives the spatial Markov property: after an initial Peano segment is exposed, the unexplored primal/dual pair is again a UST/dual-UST pair in the remaining domain with the induced boundary conditions.

For a marked interior vertex  $z$ , the conditional probability that  $z$  lies on the primal side of the future interface is the discrete electrical potential taking boundary values 0 and 1 on the two exploration sides. This is a martingale. The invariance principle and the discrete harmonic estimates of Volume IV identify its subsequential continuum limit with the corresponding harmonic function in the slit domain.

Map the slit domain to  $\mathbb{H}$  by  $g_t$  and write  $W_t$  for the tip image. Expanding the two independent harmonic test-point martingales through one capacity increment shows that every subsequential driver has zero drift and quadratic variation  $8t$ . The third-order remainder is uniformly integrable away from the marked points, so the martingale functional CLT and Lévy’s characterization give

$$W_t = \sqrt{8}B_t.$$

It remains to know that the discrete curves themselves, rather than only their hulls, are tight. If a Peano path makes  $m$  additional crossings of a fixed-modulus annulus, planar duality forces alternating primal and dual tree branches across nested subannuli. Conditional on all previously exposed branches, Wilson’s algorithm represents the next primal (respectively dual) branch as the loop erasure of a random walk stopped on the existing tree. The Beurling estimate of Volume IV bounds by a constant  $q < 1$  the conditional probability of avoiding the existing connected obstacle across the next subannulus. Iterating gives

$$\mathbf{P}\{m \text{ additional annulus crossings}\} \leq Cq^m.$$

The standard deterministic annulus compactness argument (proved with the Aizenman–Burchard criterion later in the series, but elementary in this single-curve setting) therefore yields tightness modulo monotone time change and rules out a macroscopic excursion invisible in capacity. Any subsequential Peano curve is continuous and generates the Loewner hulls driven by  $\sqrt{8}B$ .

The previous step proves in every  $C^1$  Jordan domain that the UST Peano curves converge to a continuous curve whose Loewner driver is  $\sqrt{8}B$ . Thus that curve is chordal  $\text{SLE}_8$ . Taking  $D$  to be a smooth image of a bounded half-disk and exhausting  $\mathbb{H}$  by such domains, conformal covariance and local absolute continuity of the Loewner equation identify a continuous  $\text{SLE}_8$  trace on every finite capacity interval in  $\mathbb{H}$ . Consistency of the restrictions and a diagonal argument give a continuous trace on  $[0, \infty)$ .

Steps a–d are the Rohde–Schramm reverse-flow/Whitney-box proof: the moment choice above is exactly the one for which  $\lambda - 2 = (\kappa - 8)^2/(16\kappa)$ . Steps e–g reproduce the independent Lawler–Schramm–Werner UST–Peano endpoint, whose by-product is continuity of  $\text{SLE}_8$ . No later SLE trace or CLE theorem is used.  $\square$

## 3.2 The simple phase

**Theorem 3.2** (Simple phase of chordal SLE).

Let  $0 < \kappa \leq 4$ , let  $W_t = \sqrt{\kappa}B_t$ , and let  $(K_t)_{t \geq 0}$  be the chordal Loewner hulls in  $\mathbb{H}$  driven by  $W$ , parametrized by  $\text{hcap}(K_t) = 2t$ . On the probability-one event on which the continuous trace  $\gamma : [0, \infty) \rightarrow \overline{\mathbb{H}}$  of SM-14.03.01 exists,  $\gamma$  is injective on  $[0, \infty)$  and  $\gamma(0) = 0$ ,  $\gamma(t) \in \mathbb{H}$  for every  $t > 0$ . Equivalently,  $K_t = \gamma[0, t]$  is a simple Loewner slit for every  $t > 0$ .

*Internal proof.* Continuity is owned by SM-14.03.01. Fix  $x > 0$  and set

$$X_t = g_t(x) - W_t$$

until the swallowing time. Then

$$dX_t = \frac{2}{X_t} dt - \sqrt{\kappa} dB_t.$$

After deterministic space/time rescaling this is a Bessel process of dimension

$$\delta = 1 + \frac{4}{\kappa} \geq 2.$$

A Bessel process of dimension at least two does not hit 0. Hence, for each fixed  $x \in \mathbb{R} \setminus \{0\}$ , the swallowing time is almost surely infinite. Intersecting the probability-one events over  $\mathbb{Q} \setminus \{0\}$  gives one event on which no nonzero rational boundary point is ever swallowed.

Let a continuous chordal Loewner trace be simple on  $[0, T)$  and suppose that  $T$  is its first failure of simplicity. If  $\gamma(T)$  lies on an earlier piece of the trace, choose  $s < T$  before the first contact. In the slit domain  $H_s = \mathbb{H} \setminus K_s$ , the remaining trace segment starts from the tip and, at time  $T$ , closes a crosscut together with a subarc of the old trace. The component cut off from infinity is nonempty. Under the conformal map  $g_s - W_s : H_s \rightarrow \mathbb{H}$ , the two sides of that component land on two distinct real prime ends, so the swallowed set at time  $T - s$  contains the open real interval between them. If the first loss of simplicity is instead a hit of  $\mathbb{R} \setminus \{0\}$ , the same statement holds with the real boundary arc between the tip prime end and the hit point. Thus every first failure of simplicity swallows a nonempty real interval, and therefore swallows some nonzero rational number.

This lemma uses only the elementary topology of a continuous Loewner hull: a first collision creates a bounded complementary component, and the conformal image of its prime-end boundary interval is an honest interval of  $\mathbb{R}$ .

On the probability-one event from part a, the interval supplied by part b cannot occur. Therefore there is no first failure of simplicity and the SLE trace is simple for all time. At  $\kappa = 4$  the Bessel dimension is exactly 2, which is still non-hitting, so the endpoint is included. This is the Rohde–Schramm Bessel/topological argument and uses no later SLE geometry.  $\square$

**Lemma 3.3** (Rohde–Schramm interior swallowing/avoidance dichotomy). *Let  $z \in \mathbb{H}$  be deterministic and let  $\tau(z)$  be its swallowing time for chordal  $\text{SLE}_\kappa$ . If  $\kappa > 4$ , then  $\tau(z) < \infty$  almost surely. Moreover, with*

$$Z(z) := \lim_{t \uparrow \tau(z)} \frac{(\text{Im } z)|g'_t(z)|}{\text{Im}(g_t(z) - W_t)},$$

*one has  $Z(z) < \infty$  almost surely for  $\kappa < 8$  and  $Z(z) = \infty$  almost surely for  $\kappa \geq 8$ . Consequently, for  $4 < \kappa < 8$  every deterministic interior point is swallowed but is not on the trace, whereas for  $\kappa > 8$  every deterministic interior point is hit by the trace.*

*Proof.* Write  $Z_t^{(0)} = g_t(z) - W_t$ . Put  $b = 1 - 4/\kappa \in (0, 1)$  and choose  $\vartheta = e^{i\pi(1-b)/2}$  so that

$$h(w) = \operatorname{Im}(\vartheta w^b) > 0, \quad w \in \mathbb{H}.$$

A direct Itô computation in  $dZ_t^{(0)} = 2(Z_t^{(0)})^{-1}dt - \sqrt{\kappa}dB_t$  shows that  $h(Z_{t \wedge \tau}^{(0)})$  is a local martingale. Stop between the circles  $|w| = \varepsilon$  and  $|w| = R$ . Since  $h(w) \asymp |w|^b$  on every closed angular sector and  $b > 0$ , optional stopping gives a bound, uniform in  $R$ ,

$$\mathbf{P}_z\{\tau_R < \tau_\varepsilon\} \leq C(z)R^{-b}.$$

Letting  $R \rightarrow \infty$  shows that the diffusion cannot escape to infinity before entering every neighborhood of 0; then  $\varepsilon \downarrow 0$  gives  $\mathbf{P}\{\tau(z) < \infty\} = 1$ .

Set  $z_t = X_t + iY_t = g_t(z) - W_t$ . From the Loewner equation,

$$\partial_t \log \frac{|g'_t(z)|}{Y_t} = \frac{4Y_t^2}{|z_t|^4}.$$

Hence

$$Z(z) = \exp \left\{ \int_0^{\tau(z)} \frac{4Y_t^2}{|z_t|^4} dt \right\} \in [1, \infty]. \quad (3.3)$$

If  $r_t = \operatorname{dist}(z, \partial H_t)$ , Schwarz and Koebe quarter estimates applied to a disk uniformization of  $H_t$  give

$$\frac{\operatorname{Im} z}{2Z(z)} \leq \liminf_{t \uparrow \tau} r_t \leq \limsup_{t \uparrow \tau} r_t \leq \frac{2 \operatorname{Im} z}{Z(z)}. \quad (3.4)$$

The limiting distance is exactly  $\operatorname{dist}(z, \gamma[0, \tau(z)] \cup \mathbb{R})$ .

Use  $u = \log Y_t$  as clock and  $w = X_t/Y_t$ . For real  $q$ , Itô's formula for  $Z_t^q G(w_t)$  shows that the drift vanishes precisely when

$$\frac{\kappa}{2}(1+w^2)^2 G'' + 2(\kappa-4)w(1+w^2)G' + (\kappa-8+8q)G = 0. \quad (3.5)$$

The even solution with  $G(0) = 1$  is

$$G_q(w) = {}_2F_1 \left( \eta_0, \eta_1; \frac{1}{2}; \frac{w^2}{1+w^2} \right),$$

where

$$\eta_j = \frac{1}{2} - \frac{2}{\kappa} - (-1)^j \frac{\sqrt{32q\kappa + (2\kappa-8)^2}}{4\kappa}.$$

Stopping first at  $|w| = R$  and  $Y = \varepsilon$ , and then sending  $R \rightarrow \infty$ ,  $\varepsilon \downarrow 0$ , is justified by positivity of the normalized hypergeometric solution. It yields

$$\mathbf{E}[Z(z)^q] < \infty \quad \text{for } q < 1 - \frac{\kappa}{8}, \quad 0 < \kappa < 8. \quad (3.6)$$

In particular some positive moment is finite when  $\kappa < 8$ , so  $Z(z) < \infty$  almost surely. When  $\kappa \geq 8$ , the same stopped ODE with any  $q < 0$  gives  $\mathbf{E}[Z(z)^q] = 0$ ; because  $Z^{-q} > 0$  on  $\{Z < \infty\}$ , this forces  $Z(z) = \infty$  almost surely.

For  $4 < \kappa < 8$ , Step 1 gives  $\tau(z) < \infty$  and Step 3 gives  $Z(z) < \infty$ . Thus (3.4) gives

$$\operatorname{dist}(z, \gamma[0, \tau(z)] \cup \mathbb{R}) > 0,$$



so  $z$  leaves  $H_t$  without being visited: it is swallowed. For  $\kappa > 8$ , (3.4) gives zero distance. Since the trace is continuous on the compact interval  $[0, \tau(z)]$ , the trace range there is closed; hence  $z$  itself is visited.

Let  $0 < x < y$ . Until  $x$  is swallowed put

$$R_t = \frac{g_t(x) - W_t}{g_t(y) - W_t} \in (0, 1).$$

After the time change  $ds = (g_t(y) - W_t)^{-2} dt$ , Itô's formula gives a regular one-dimensional diffusion on  $(0, 1)$  with nonzero diffusion coefficient. Its scale density is smooth and strictly positive on every compact subinterval of  $(0, 1)$ . Therefore, starting from  $R_0 = x/y$ , each nonempty boundary window has strictly positive exit probability. Equivalently, for every  $0 < a < b < \infty$ , chordal  $\text{SLE}_\kappa$ ,  $4 < \kappa < 8$ , has a strictly positive probability of hitting the real interval  $[a, b]$ . Scale invariance makes this probability depend only on  $b/a$ . This is the only part of the boundary-hit calculation needed for the self-touching argument below.  $\square$

### 3.3 The self-touching phase

**Theorem 3.4** (Self-touching phase of chordal SLE).

*Let  $4 < \kappa < 8$  and let  $\gamma$  be the continuous chordal  $\text{SLE}_\kappa$  trace in  $(\mathbb{H}; 0, \infty)$  from SM-14.03.01. Then, almost surely,  $\gamma$  is not injective: there exist finite times  $0 < s < t$  with  $\gamma(s) = \gamma(t)$ . At the same time the trace is not space filling: every deterministic  $z \in \mathbb{H}$  is almost surely swallowed by the Loewner hull at a finite time but is hit by the trace with probability zero, and the trace has zero planar Lebesgue area.*

*Internal proof: swallowing plus a boundary-window renewal.* Lemma 3.3 gives, for every deterministic  $z \in \mathbb{H}$ ,

$$\mathbf{P}\{z \in \bigcup_{t>0} K_t\} = 1, \quad \mathbf{P}\{z \in \gamma[0, \infty)\} = 0.$$

Fubini therefore also shows that the trace has zero planar area, while every fixed interior point eventually leaves the unbounded Loewner domain. This is the geometric swallowing phase and already rules out space filling.

Assume that the trace has remained injective up to a stopping time  $s > 0$  and that  $\gamma(s) \in \mathbb{H}$ . The two prime-end sides of the simple slit  $\gamma[0, s]$  adjacent to the tip are mapped by  $g_s - W_s$  onto nondegenerate real intervals  $[-L_s, 0]$  and  $[0, R_s]$ , with  $L_s, R_s > 0$ . Hitting either open interval by the future normalized trace therefore maps back to a hit of  $\gamma[0, s)$  and is a self-intersection of the original trace.

Conditioned on  $\mathcal{F}_s$ , the normalized future

$$\widehat{\gamma}_s(t) = g_s(\gamma(s+t)) - W_s$$

is an independent chordal  $\text{SLE}_\kappa$ . By the boundary-window part of Lemma 3.3, after scaling by  $R_s$  there is a number  $p_\kappa > 0$  such that the future hits, say,  $[R_s/2, R_s]$  with conditional probability at least  $p_\kappa$ . On that event it hits the right side of the old slit and hence self-intersects. The same conclusion uses the left side if  $R_s$  degenerates along a chosen prime-end normalization.

Suppose, for contradiction, that the path stays simple forever. Starting from a rational capacity time at which the tip is interior, apply Step c and then, on the event of no intersection, restart at the next rational capacity time for which the surviving slit has a nondegenerate side interval. The

domain Markov property makes the conditional failure probability at each restart at most  $1 - p_\kappa$ . After  $n$  restarts,

$$\mathbf{P}\{\text{no self-intersection in the first } n \text{ trials}\} \leq (1 - p_\kappa)^n.$$

Letting  $n \rightarrow \infty$  gives probability zero. Thus the trace is almost surely non-simple and self-touching for  $4 < \kappa < 8$ . Together with Step a it is not space filling, which is exactly the intermediate Rohde–Schramm phase.  $\square$

### 3.4 The space-filling phase

**Theorem 3.5** (Space-filling phase).

*For  $\kappa \geq 8$ , chordal  $\text{SLE}_\kappa$  in  $\mathbb{H}$  satisfies*

$$\gamma[0, \infty) = \overline{\mathbb{H}} \quad \text{almost surely.}$$

*Equivalently, in any simply connected domain the conformal image of the trace visits every point of the closure in its prime-end compactification.*

*Proof.* For  $\kappa > 8$ , Lemma 3.3 gives  $\tau(z) < \infty$  and  $Z(z) = \infty$  for every deterministic  $z \in \mathbb{H}$ ; the Koebe distance estimate in that lemma then forces  $z$  to lie on the continuous trace. At  $\kappa = 8$  the same conclusion follows from the UST-Peano construction in SM-14.03.01: discrete Peano paths are mesh-dense, and every uniform subsequential limit in a bounded smooth exhaustion is onto the closure. Hence every fixed interior point is visited for all  $\kappa \geq 8$ .

For  $\kappa > 8$  and each deterministic  $t > 0$ , almost surely there exists a random  $r_t > 0$  such that

$$\{z \in \mathbb{H} : |z| < r_t\} \subset K_t. \quad (3.2)$$

Here is the Rohde–Schramm contradiction argument. If with probability one  $0 \in \overline{H_t}$  for every  $t > 0$ , choose a deterministic interior point  $z_0$  and let  $t_0 = \tau(z_0)$ . Since  $z_0$  is hit by the trace, the domain Markov property at  $t_0$  would imply that  $z_0$  remains on  $\partial K_{t_0+s}$  for every  $s > 0$ . Fubini would then force  $\partial K_t$  to have positive planar area with positive probability for large  $t$ . But the inverse-map Hölder estimate from SM-14.03.01 implies that  $\partial K_t$  has planar area zero. Therefore (3.2) has positive probability. Scale invariance makes this probability independent of  $t$ , and monotonicity plus Blumenthal’s zero–one law upgrades it to probability one. For  $\kappa = 8$  the same lemma follows from the UST-Peano approximation.

Take  $r_0 > 0$  so that  $\mathbf{P}\{r_1 < r_0\} < \varepsilon$ . By SLE scaling, for every  $t > 0$ ,

$$\mathbf{P}\left\{K_t \supset \{z \in \mathbb{H} : |z| < r_0\sqrt{t}\}\right\} \geq 1 - \varepsilon.$$

Once that half-disk is contained in  $K_t$ , every later tip lies outside it. Hence

$$\mathbf{P}\{\exists s > t : |\gamma(s)| < r_0\sqrt{t}\} < \varepsilon.$$

Letting first  $t \rightarrow \infty$  along integers and then  $\varepsilon \downarrow 0$  shows

$$|\gamma(t)| \rightarrow \infty \quad \text{as } t \rightarrow \infty$$

almost surely. Thus the range  $\gamma[0, \infty)$  is closed in  $\mathbb{C}$ . The same conclusion at  $\kappa = 8$  follows from the preceding endpoint construction and the identical scaling argument.

Step a gives  $\mathbf{P}\{z \in \gamma[0, \infty)\} = 1$  for each point of a countable dense subset  $Q \subset \overline{\mathbb{H}}$ . Intersecting the probability-one events over  $z \in Q$  shows that the random trace range is almost surely dense in  $\overline{\mathbb{H}}$ .

By Step c the trace range is almost surely closed, and by Step d it is dense. Therefore

$$\gamma[0, \infty) = \overline{\mathbb{H}}$$

almost surely. Conformal covariance transports the statement to any simply connected domain.

For  $\kappa > 8$ , Steps a–e reproduce the phase/transience argument of Rohde–Schramm: hit every point, prove a swallowed half-disk near the starting point, deduce transience, then use closedness plus density. Their published update states that, after the Lawler–Schramm–Werner continuity theorem at  $\kappa = 8$ , the same transience/space-filling argument extends to the endpoint. Our SM-14.03.01 supplies precisely that endpoint proof, so no citation-only gap remains here.  $\square$

## Chapter 4

# Martingales and change of measure

### 4.1 Boundary derivative martingales

**Theorem 4.1** (A boundary derivative martingale family).

Let chordal  $\text{SLE}_\kappa$  in  $\mathbb{H}$  be driven by  $U_t = \sqrt{\kappa}B_t$ , fix  $x > 0$ , and stop before  $x$  is swallowed. Put

$$X_t = g_t(x) - U_t, \quad b \in \mathbb{R}, \quad a = \frac{b(4 - \kappa + \kappa b)}{4}.$$

Then

$$M_t = g'_t(x)^a X_t^b$$

is a positive local martingale (after normalization by  $M_0 = x^b$  if a density martingale is desired), and

$$\frac{dM_t}{M_t} = -\frac{b\sqrt{\kappa}}{X_t} dB_t.$$

*Proof.* The Loewner equation gives

$$dX_t = \frac{2}{X_t} dt - \sqrt{\kappa} dB_t, \quad \partial_t \log g'_t(x) = -\frac{2}{X_t^2}.$$

Itô's formula yields

$$d \log X_t = -\frac{\sqrt{\kappa}}{X_t} dB_t + \frac{2 - \kappa/2}{X_t^2} dt.$$

Hence

$$d \log M_t = -\frac{b\sqrt{\kappa}}{X_t} dB_t + \frac{-2a + b(2 - \kappa/2)}{X_t^2} dt.$$

Passing from  $d \log M$  to  $dM/M$  adds one half of the quadratic variation,  $\frac{1}{2}b^2\kappa X_t^{-2}dt$ . The drift of  $M$  is therefore

$$\frac{-2a + 2b - \frac{\kappa}{2}b + \frac{\kappa}{2}b^2}{X_t^2} dt,$$

which vanishes exactly for  $a = b(4 - \kappa + \kappa b)/4$ . The remaining stochastic differential is the one displayed in the theorem, proving the local-martingale claim.  $\square$

## 4.2 Schramm's left-passage formula

**Theorem 4.2** (Schramm's left-passage martingale).

Let  $0 < \kappa < 8$ , let  $\gamma$  be chordal  $\text{SLE}_\kappa$  from 0 to  $\infty$  in  $\mathbb{H}$ , and let  $z = x + iy \in \mathbb{H}$ . With  $w = x/y$ ,

$$\mathbb{P}\{\gamma \text{ passes to the left of } z\} = F_\kappa(w), \quad F_\kappa(w) = \frac{\int_{-\infty}^w (1+s^2)^{-4/\kappa} ds}{\int_{-\infty}^{\infty} (1+s^2)^{-4/\kappa} ds}.$$

Equivalently,  $F_\kappa(X_t/Y_t)$  is the bounded martingale which solves the left-passage boundary-value problem.

*Proof.* Write  $Z_t = g_t(z) - U_t = X_t + iY_t$ . From the Loewner equation,

$$dX_t = \frac{2X_t}{X_t^2 + Y_t^2} dt - \sqrt{\kappa} dB_t, \quad dY_t = -\frac{2Y_t}{X_t^2 + Y_t^2} dt.$$

For  $W_t = X_t/Y_t$ , ordinary Itô calculus (there is no quadratic variation from  $Y$ ) gives

$$dW_t = \frac{4W_t}{Y_t^2(1+W_t^2)} dt - \frac{\sqrt{\kappa}}{Y_t} dB_t.$$

After the time change  $du = dt/Y_t^2$ ,  $W$  has generator

$$\mathcal{L} = \frac{\kappa}{2} \frac{d^2}{dw^2} + \frac{4w}{1+w^2} \frac{d}{dw}.$$

A bounded function  $F$  gives a local martingale  $F(W_u)$  exactly when

$$\frac{\kappa}{2} F''(w) + \frac{4w}{1+w^2} F'(w) = 0.$$

Solving once gives  $F'(w) = C(1+w^2)^{-4/\kappa}$ . Since  $\kappa < 8$ , this density is integrable at infinity. The boundary conditions for the passage event are  $F(-\infty) = 0$  and  $F(+\infty) = 1$ , which determine precisely the normalized integral above.

Stop when the conformal image of  $z$  reaches a large semicircle in the  $W$ -coordinate. Optional stopping applies because  $0 \leq F \leq 1$ . As the cutoff tends to infinity, the two exit alternatives correspond to the trace separating  $z$  from the negative or positive side of the boundary. The SLE trace theorem from Chapter 3 identifies these alternatives with passing right or left of  $z$ , and bounded convergence yields the formula.  $\square$

## 4.3 Girsanov and force points

**Theorem 4.3** (Girsanov transform from SLE to one-force-point SLE).

Fix  $x > 0$  and  $\rho \in \mathbb{R}$ . Before the force point collides with the driver, set

$$b = \frac{\rho}{\kappa}, \quad a = \frac{\rho(\rho + 4 - \kappa)}{4\kappa}, \quad M_t = \frac{g'_t(x)^a (g_t(x) - U_t)^b}{x^b}.$$

After localization so that  $M$  is a true martingale, weighting ordinary chordal  $\text{SLE}_\kappa$  by  $M_t$  changes the driving equation to

$$dU_t = \sqrt{\kappa} d\tilde{B}_t + \frac{\rho}{U_t - V_t} dt, \quad dV_t = \frac{2}{V_t - U_t} dt,$$

which is chordal  $\text{SLE}_\kappa(\rho)$  up to the localization time.

*Proof.* By the derivative-martingale theorem with  $b = \rho/\kappa$ ,

$$\frac{dM_t}{M_t} = -\frac{\rho}{\sqrt{\kappa} X_t} dB_t, \quad X_t = V_t - U_t.$$

Stop before  $X_t$  leaves a compact subinterval of  $(0, \infty)$  and before  $M_t$  becomes too large or too small. On this stopped interval Novikov's condition is automatic, so Girsanov's theorem applies. Under the tilted law,

$$\tilde{B}_t = B_t + \int_0^t \frac{\rho}{\sqrt{\kappa} X_s} ds$$

is Brownian motion. Consequently

$$dU_t = \sqrt{\kappa} dB_t = \sqrt{\kappa} d\tilde{B}_t - \frac{\rho}{X_t} dt = \sqrt{\kappa} d\tilde{B}_t + \frac{\rho}{U_t - V_t} dt.$$

The force point itself is transported by the deterministic Loewner flow, hence  $dV_t = 2(V_t - U_t)^{-1} dt$ . These are exactly the one-force-point  $\text{SLE}_\kappa(\rho)$  equations. Increasing the localization sequence gives absolute continuity on every compact time interval strictly before collision; the Bessel analysis in Chapter 2 determines when and how the collision boundary can be crossed.  $\square$

## Chapter 5

# Locality and conformal restriction

This chapter records two parameter values at which the conformal-change formula for SLE simplifies dramatically. The proofs are calculations with the Loewner equation rather than appeals to universality heuristics.

### 5.1 Locality of SLE<sub>6</sub>

**Theorem 5.1** (Locality of SLE<sub>6</sub>).

*Let  $A \subset \overline{\mathbb{H}}$  be a bounded hull with  $0 \notin A$ . Run chordal SLE<sub>6</sub> from 0 to  $\infty$  in  $\mathbb{H}$  and stop at the first time its hull meets  $A$ . If  $\Phi_A : \mathbb{H} \setminus A \rightarrow \mathbb{H}$  is conformal and normalized at infinity, then the image chain under  $\Phi_A$ , after its deterministic half-plane-capacity reparametrization and the affine normalization of the starting point, has the law of chordal SLE<sub>6</sub> up to the corresponding stopping time. Equivalently, SLE<sub>6</sub> does not detect  $A$  before hitting it.*

*Proof.* Let  $g_t$  be the original Loewner maps,  $W_t = \sqrt{\kappa}B_t$ , and let  $A_t = g_t(A)$  before the collision time. Let  $h_t : \mathbb{H} \setminus A_t \rightarrow \mathbb{H}$  be hydrodynamically normalized and define the conformally transported Loewner map by the commutative diagram

$$\tilde{g}_t \circ \Phi_A = h_t \circ g_t.$$

The transported driving point is  $\widetilde{W}_t = h_t(W_t)$ .

$$d\tilde{a}_t = h'_t(W_t)^2 dt, \quad \partial_t h_t(W_t) = -3h''_t(W_t),$$

where  $\tilde{a}_t$  is the transported half-plane-capacity time.

$$d\widetilde{W}_t = h'_t(W_t) dW_t + \left(\frac{\kappa}{2} - 3\right) h''_t(W_t) dt.$$

At  $\kappa = 6$  the drift vanishes:

$$d\widetilde{W}_t = \sqrt{6} h'_t(W_t) dB_t.$$

Its quadratic variation is

$$d\langle \widetilde{W} \rangle_t = 6h'_t(W_t)^2 dt = 6 d\tilde{a}_t.$$

$$\widehat{W}_s = \sqrt{6} \widehat{B}_s$$

for a standard Brownian motion  $\widehat{B}$ , up to the first contact with  $A$ .

□

**Example 5.2.** If  $A$  is a small semicircular hull far to the right of the origin, an  $\text{SLE}_6$  aimed at infinity has, until it reaches  $A$ , exactly the same conformally transported law whether  $A$  was removed before the experiment or not. Only the capacity clock changes.

## 5.2 One-sided conformal restriction for $\text{SLE}_{8/3}$

**Theorem 5.3** (Restriction formula for  $\text{SLE}_{8/3}$ ).

Let  $\gamma$  be chordal  $\text{SLE}_{8/3}$  in  $\mathbb{H}$  from 0 to  $\infty$ . Let  $A$  be a compact hull on one side of the starting point, bounded away from 0, such that  $\mathbb{H} \setminus A$  is simply connected. Let  $\Phi_A : \mathbb{H} \setminus A \rightarrow \mathbb{H}$  be normalized by  $\Phi_A(0) = 0$ ,  $\Phi_A(\infty) = \infty$ , and  $\Phi_A(z)/z \rightarrow 1$  at infinity. Then

$$\mathbb{P}\{\gamma \cap A = \emptyset\} = \Phi'_A(0)^{5/8}.$$

Conditionally on avoidance,  $\Phi_A(\gamma)$  has the law of chordal  $\text{SLE}_{8/3}$  in  $\mathbb{H}$ .

*Proof.* Run SLE with driver  $W_t = \sqrt{\kappa}B_t$  and stop when the growing hull reaches  $A$ . As in the locality proof, let  $A_t = g_t(A)$  and let  $h_t : \mathbb{H} \setminus A_t \rightarrow \mathbb{H}$  be hydrodynamically normalized.

$$M_t = h'_t(W_t)^\alpha \exp \left\{ -\frac{c(\kappa)}{6} \int_0^t Sh_s(W_s) ds \right\},$$

where

$$\alpha = \frac{6 - \kappa}{2\kappa}, \quad c(\kappa) = \frac{(3\kappa - 8)(6 - \kappa)}{2\kappa}, \quad Sh = \frac{h'''}{h'} - \frac{3}{2} \left( \frac{h''}{h'} \right)^2.$$

To verify this formula, differentiate  $h'_t(W_t)$  and  $h''_t(W_t)$  using the Loewner PDE, apply Itô to  $\log h'_t(W_t)$ , and collect the coefficients of  $(h''_t/h'_t)^2$  and  $Sh_t$ . The choice of  $\alpha$  cancels the first drift; the remaining coefficient is  $-c(\kappa)/6$ .

$$c(8/3) = 0, \quad \alpha = \frac{5}{8}.$$

Hence

$$M_t = h'_t(W_t)^{5/8}$$

is a positive local martingale. Stop while the evolving hull remains a fixed positive distance from  $A$ ; distortion estimates make the stopped martingale bounded, so optional stopping is legitimate. Let the auxiliary stopping distance tend to zero.

$M_0 = \Phi'_A(0)^{5/8}$ . On the event that the SLE curve hits  $A$ , the conformal radius seen from the driving prime end degenerates and  $h'_t(W_t) \rightarrow 0$ . On the event of avoidance,  $A_t$  is pushed to infinity and the normalized maps converge locally to the identity, hence  $h'_t(W_t) \rightarrow 1$ . Bounded convergence after localization gives

$$\Phi'_A(0)^{5/8} = \mathbb{E}[M_\infty] = \mathbb{P}\{\gamma \cap A = \emptyset\}.$$

□

## 5.3 Chordal conformal-restriction classification

**Theorem 5.4** (Lawler–Schramm–Werner chordal restriction classification).

A two-sided chordal restriction measure in  $(\mathbb{H}, 0, \infty)$  is determined by an exponent  $\alpha$  through

$$\mathbb{P}[K \cap A = \emptyset] = \Phi'_A(0)^\alpha$$



for every admissible hull  $A$  bounded away from 0. Such measures exist exactly for  $\alpha \geq 5/8$  and are unique for each  $\alpha$ . The unique restriction measure supported on a simple curve occurs at  $\alpha = 5/8$ , in which case the curve is chordal  $\text{SLE}_{8/3}$ .

*Proof.* For an admissible hull  $A$ , write  $f(A) = \mathbb{P}[K \cap A = \emptyset]$ . If  $A$  and  $B$  are removed successively, the restriction property and the chain rule give

$$f(A \cup \Phi_A^{-1}(B)) = f(A)f(B), \quad (\Phi_B \circ \Phi_A)'(0) = \Phi_B'(0)\Phi_A'(0).$$

Approximate normalized hull maps in the Carathéodory topology by compositions of elementary vertical-slit maps. Continuity of the law under this approximation reduces  $f$  to a continuous multiplicative functional of the positive real quantity  $\Phi_A'(0)$ . Hence

$$f(A) = \Phi_A'(0)^\alpha$$

for some  $\alpha \geq 0$ .

Let  $\mathcal{A}_0$  be the countable family of finite unions of polygonal hulls with rational vertices, each bounded away from 0 and with simply connected complement. For a closed connected set joining 0 to  $\infty$ , the events  $\{K \cap A = \emptyset\}$ ,  $A \in \mathcal{A}_0$ , determine all hit-or-miss events for compact subsets of  $\mathbb{H}$ . They therefore generate the Borel sigma-field of the local Hausdorff topology. The avoidance formula fixes their probabilities and hence fixes the law.

Consider the right boundary  $\eta$  of a two-sided restriction sample and mark the prime end immediately to the right of its starting point. Parametrize  $\eta$  by half-plane capacity. After removing an initial segment and recentering by  $g_t - W_t$ , restriction gives the same law with the marked prime end transported to  $O_t - W_t$ . Thus the pair  $(W_t, O_t)$  is a continuous, scale-covariant Markov process whose second coordinate is transported deterministically by Loewner:

$$dO_t = \frac{2}{O_t - W_t} dt.$$

Exactly as in Schramm's characterization, stationary scaling of the stochastic coordinate forces Brownian quadratic variation and a drift homogeneous of degree  $-1$  in the only available separation  $O_t - W_t$ . Consequently

$$dW_t = \sqrt{\kappa} dB_t + \frac{\rho}{W_t - O_t} dt$$

for constants  $\kappa \geq 0$  and  $\rho \in \mathbb{R}$ . Hence every simple one-sided restriction boundary is an  $\text{SLE}_\kappa(\rho)$ -type curve.

Apply the conformal deletion calculation of the preceding theorem to this force-point evolution. The Radon–Nikodym local martingale contains the universal Schwarzian coefficient

$$c(\kappa) = \frac{(3\kappa - 8)(6 - \kappa)}{2\kappa}.$$

A genuine restriction law depends only on the normalized derivative of the deletion map; it has no exponential integral of the Schwarzian. For a nondegenerate simple boundary the relevant zero is therefore  $3\kappa - 8 = 0$ , so  $\kappa = 8/3$ . (The other algebraic zero,  $\kappa = 6$ , corresponds to locality and is not a simple restriction boundary.)

For  $\kappa = 8/3$ , insert the force-point drift in the derivative martingale from Chapter 4. Direct Itô differentiation of the deletion-map derivative and the force-point separation gives

$$\alpha(\rho) = \frac{(\rho + 2)(3\rho + 10)}{32}.$$

The standard noncrossing/right-boundary regime is  $\rho \geq 0$ . Thus every boundary of a two-sided restriction sample has exponent at least

$$\alpha(0) = \frac{2 \cdot 10}{32} = \frac{5}{8}.$$

This proves nonexistence of two-sided restriction measures with  $\alpha < 5/8$ .

Conversely, for every  $\rho \geq 0$  run chordal  $\text{SLE}_{8/3}(\rho)$  with the force point on the chosen side and take the filled set obtained by adjoining all components cut off from the opposite boundary. The same derivative/force-point martingale shows that its avoidance probabilities are exactly

$$\Phi'_A(0)^{\alpha(\rho)}.$$

The domain Markov property then gives the restriction property after conditioning on avoidance. Since  $\rho \mapsto \alpha(\rho)$  is continuous, strictly increasing on  $[0, \infty)$ , starts at  $5/8$ , and tends to infinity, every  $\alpha \geq 5/8$  is obtained.

At  $\alpha = 5/8$  the preceding formula forces  $\rho = 0$ , so the boundary is ordinary chordal  $\text{SLE}_{8/3}$ . In the simple SLE regime the filled set has no extra boundary component, hence the restriction sample itself is that simple curve.

□

## Chapter 6

# Green functions and natural time

### 6.1 Green functions and natural time

**Theorem 6.1** (One-point Green function for chordal  $\text{SLE}_\kappa$ ).

Let  $0 < \kappa < 8$ , put  $d = 1 + \kappa/8$  and  $a = 2/\kappa$ , and let  $\gamma$  be chordal  $\text{SLE}_\kappa$  in  $(\mathbb{H}; 0, \infty)$ . There is a constant  $c_\kappa \in (0, \infty)$  such that, for every  $z \in \mathbb{H}$ ,

$$\lim_{\varepsilon \downarrow 0} \varepsilon^{d-2} \mathbf{P}\{\text{dist}(z, \gamma) < \varepsilon\} = c_\kappa G_{\mathbb{H}}(z), \quad G_{\mathbb{H}}(z) = (\text{Im } z)^{d-2} \sin(\arg z)^{4a-1}.$$

If  $(D; w_1, w_2)$  is a simply connected marked domain and  $f : (D; w_1, w_2) \rightarrow (\mathbb{H}; 0, \infty)$  is conformal, then the corresponding Green function is

$$G_D(z; w_1, w_2) = |f'(z)|^{2-d} G_{\mathbb{H}}(f(z)),$$

which is independent of the choice of  $f$  with the stated marking.

*Internal proof:* Green martingale and two-sided radial SLE. Put  $a = 2/\kappa$  and  $d = 1 + \kappa/8 < 2$ , and work first in  $(\mathbb{H}; 0, \infty)$ . For  $z \in \mathbb{H}$  write

$$Z_t = g_t(z) - W_t = X_t + iY_t, \quad S_t = \frac{Y_t}{|Z_t|}.$$

The Loewner equation and Itô's formula give

$$d \log Y_t = -\frac{2}{|Z_t|^2} dt, \quad d \log |g'_t(z)| = -2\Re \frac{1}{Z_t^2} dt.$$

A direct Itô calculation then shows that

$$M_t(z) = |g'_t(z)|^{2-d} Y_t^{d-2} S_t^{4a-1}$$

is a positive local martingale up to the swallowing time of  $z$ . Equivalently, with

$$G_{\mathbb{H}}(z) = \Im(z)^{d-2} \sin(\arg z)^{4a-1},$$

we have  $M_t(z) = |g'_t(z)|^{2-d} G_{\mathbb{H}}(Z_t)$ .

Stop when the conformal radius of  $z$  has decreased by a factor  $e^{-r}$ . Weighting by  $M_t(z)/M_0(z)$  is the Doob  $h$ -transform defining two-sided radial SLE through  $z$ . Under conformal-radius time, the angular variable has a one-dimensional diffusion whose invariant density is proportional to  $\sin(\theta)^{4a}$ .

The diffusion is positive recurrent and converges exponentially fast to that invariant law. Optional stopping therefore gives

$$\mathbf{P}\{\text{crad}_{\mathbb{H}\setminus\gamma}(z) \leq e^{-r} \text{crad}_{\mathbb{H}}(z)\} = c_* G_{\mathbb{H}}(z) e^{-(2-d)r} (1 + o(1)).$$

Koebe's quarter theorem converts conformal radius to Euclidean distance. Thus

$$\lim_{\varepsilon \downarrow 0} \varepsilon^{d-2} \mathbf{P}\{\text{dist}(z, \gamma) < \varepsilon\} = c_{\kappa} G_{\mathbb{H}}(z).$$

For a simply connected  $D$  and boundary points  $w_1, w_2$ , choose  $f : (D; w_1, w_2) \rightarrow (\mathbb{H}; 0, \infty)$ . Conformal covariance of distance to first order gives

$$G_D(z; w_1, w_2) = |f'(z)|^{2-d} G_{\mathbb{H}}(f(z)),$$

which is independent of the chosen  $f$  and proves the one-point Green-function theorem.  $\square$

**Theorem 6.2** (Minkowski-content natural parametrization).

Let  $0 < \kappa < 8$ , put  $d = 1 + \kappa/8$ , and let  $\gamma$  be chordal  $\text{SLE}_{\kappa}$  in a simply connected domain  $D$  from  $w_1$  to  $w_2$ . There is an almost surely locally finite Borel measure  $\mu_{\gamma}$ , supported on the trace, such that for every finite capacity time  $t$ ,

$$\Theta_t := \mu_{\gamma}(\gamma[0, t]) = \lim_{\varepsilon \downarrow 0} \varepsilon^{d-2} \text{Area}\{z \in D : \text{dist}(z, \gamma[0, t]) < \varepsilon\}. \quad (6.1)$$

The limit holds almost surely and in  $L^2$  on compactly localized pieces. The function  $t \mapsto \Theta_t$  is continuous, strictly increasing, and locally Hölder in capacity time. After the deterministic normalization fixed by the one-point Green constant in SM-14.06.01,  $\Theta$  is the Doob–Meyer natural parametrization of  $\text{SLE}$ .

*Internal proof: two-point Green function, dyadic content, and old-boundary removal.* We first construct the measure in  $(\mathbb{H}; 0, \infty)$ ; the conformal-transport step to a general marked domain  $D$  is carried out after the half-plane estimates are proved. For  $r > 0$  put  $\epsilon_r = e^{-r}$  and, for a bounded Borel set  $A \Subset \mathbb{H}$ ,

$$J_r(A) = e^{(2-d)r} \text{Area}\{z \in A : \text{dist}(z, \gamma) < e^{-r}\}. \quad (6.2)$$

For distinct  $z, w$ , stop at the first entrance into  $B(z, \epsilon)$ . Weighting by the one-point Green martingale of SM-14.06.01 turns the initial piece into two-sided radial  $\text{SLE}$  through  $z$ . Conditional on that initial piece, the domain Markov property and the one-point theorem give the probability of subsequently approaching  $w$ . Reversing the roles of  $z, w$  for the opposite visiting order yields a symmetric function  $G^{(2)}(z, w)$  with

$$\lim_{\epsilon, \delta \downarrow 0} \epsilon^{d-2} \delta^{d-2} \mathbf{P}\{\text{dist}(z, \gamma) < \epsilon, \text{dist}(w, \gamma) < \delta\} = c_{\kappa}^2 G^{(2)}(z, w). \quad (6.3)$$

Koebe distortion at the first stopping time gives, on every  $K \Subset \mathbb{H}$ ,

$$\epsilon^{d-2} \delta^{d-2} \mathbf{P}\{d(z, \gamma) < \epsilon, d(w, \gamma) < \delta\} \leq C_K |z - w|^{d-2}, \quad (6.4)$$

whenever  $\epsilon, \delta \leq |z - w|/10$ . The remaining regime, where one radius is comparable with  $|z - w|$ , follows from the one-point estimate and enlarging the constant. Since  $d > 0$ , the kernel  $|z - w|^{d-2}$  is locally integrable on  $\mathbb{R}^2 \times \mathbb{R}^2$ .

Let  $Q \in \mathbb{H}$  be a dyadic square. By Fubini,

$$\mathbf{E}[J_r(Q)J_s(Q)] = \iint_{Q \times Q} e^{(2-d)(r+s)} \mathbf{P}\{d(z, \gamma) < e^{-r}, d(w, \gamma) < e^{-s}\} dz dw. \quad (6.5)$$

For  $z \neq w$ , (6.3) gives a common limit as  $r, s \rightarrow \infty$ , and (6.4) supplies an integrable dominating function. Dominated convergence therefore shows that  $\mathbf{E}[J_r(Q)J_s(Q)]$  has one limit independent of the way  $r, s \rightarrow \infty$ . Consequently

$$\mathbf{E}[(J_r(Q) - J_s(Q))^2] \rightarrow 0. \quad (6.6)$$

Thus  $J_r(Q)$  converges in  $L^2$  to a random variable  $\mu(Q)$ . A diagonal subsequence gives almost-sure convergence simultaneously for the countable dyadic algebra.

The only obstruction to finite additivity is that an  $e^{-r}$ -neighborhood of trace on one side of  $\partial Q$  can be counted in the neighboring square. Cover the  $2^{-n}$ -neighborhood of  $\partial Q$  by  $O(2^n)$  squares of side  $2^{-n}$ . The one-point estimate gives expected normalized content  $O(2^{-nd})$  per square, while the two-point bound gives the corresponding square-summable second moment. Hence the total content of this boundary layer tends to zero almost surely along a summable subsequence, and then monotonically for all  $n$ . It follows that the dyadic limits are finitely additive and continuous under decreasing dyadic exhaustion. Carathéodory therefore extends  $Q \mapsto \mu(Q)$  uniquely to a locally finite Borel measure  $\mu$ . The same boundary-layer argument shows

$$\mu(Q) = \lim_{r \rightarrow \infty} J_r(Q) \quad (6.7)$$

for every dyadic square and then for every compact  $\mu$ -continuity set. The measure is supported on  $\gamma$ , because a compact set at positive distance from the trace has  $J_r = 0$  for all sufficiently large  $r$ .

The inverse-map derivative estimates from SM-14.03.01 give a random  $\alpha_0 > 0$  such that on every finite capacity interval

$$\text{diam } \gamma[s, s+h] \leq C(\omega, T)h^{\alpha_0}$$

for all sufficiently small  $h$ . Cover this segment by a bounded number of dyadic squares of side  $h^{\alpha_0}$ . From (6.4) and (6.7), each such square has second moment  $O(h^{2d\alpha_0})$ . Borel–Cantelli, after decreasing the exponent slightly, gives an  $\alpha > 0$  for which, simultaneously for dyadic  $s, h$  in every bounded interval,

$$\text{Cont}_d^+(\gamma[s, s+h]) \leq C(\omega, T)h^\alpha. \quad (6.8)$$

Density of dyadic times and monotonicity extend this estimate to all  $s$ .

This is the step that cannot be omitted when  $4 < \kappa < 8$ . Fix  $s < u < t$  and let  $\partial_n H_s$  be the  $2^{-n}$ -neighborhood of  $\partial H_s$ . A square  $Q$  of side  $2^{-n}$  that meets  $\partial_n H_s$  can contribute to  $\gamma[u, t]$  only if the curve first approaches the corresponding old boundary piece, leaves its definite conformal neighborhood, and later returns to  $Q$ . The ordered two-point estimate from part a, applied at those two visits, gives an extra return factor  $2^{-\beta n}$  for some  $\beta > 0$ . Summing the first-moment Green bound over all such squares in a fixed compact set yields

$$\mathbf{E} [\text{Cont}_d^+(\gamma[u, t] \cap \partial_n H_s)] \leq C 2^{-\beta n}. \quad (6.9)$$

Borel–Cantelli and a countable choice of rational  $s < u < t$  give

$$\lim_{n \rightarrow \infty} \text{Cont}_d^+(\gamma[u, t] \cap \partial_n H_s) = 0 \quad (6.10)$$

almost surely for all  $0 \leq s < u < t$ . This is the Lawler–Rezaei old-boundary lemma: returns to previously generated boundary carry no  $d$ -content.

Approximate  $H_s$  by dyadic squares a positive distance from  $\partial H_s$ . On those squares the future curve, mapped by  $g_s - W_s$ , is an ordinary independent SLE, so (6.7) applies to  $\gamma[s, t]$ . The omitted old-boundary strip contributes zero by (6.10), and the pieces  $\gamma[s, s+h]$  and  $\gamma[t, t+h]$  disappear by (6.8). Therefore

$$\mu(\gamma[s, t]) = \lim_{\varepsilon \downarrow 0} \varepsilon^{d-2} \text{Area}\{z : \text{dist}(z, \gamma[s, t]) < \varepsilon\}. \quad (6.11)$$

Taking  $s = 0$  proves the asserted Minkowski-content formula (6.1) on every finite capacity interval.

Set  $\Theta_t = \mu(\gamma[0, t])$ . Estimate (6.8) gives continuity and local Hölder regularity. For strict increase, fix  $s < t$ . Conditional on  $\mathcal{F}_s$ , map the future to  $\mathbb{H}$ . The first- and second-moment Green estimates give constants  $c, C > 0$ , uniformly after scaling a short capacity interval, such that

$$\mathbf{E}[\mu(\gamma[s, s+h]) \mid \mathcal{F}_s] \geq ch^{d/2}, \quad \mathbf{E}[\mu(\gamma[s, s+h])^2 \mid \mathcal{F}_s] \leq Ch^d.$$

Paley–Zygmund gives a uniform positive conditional probability that this increment is positive. Subdivide  $[s, t]$  into successively smaller capacity intervals and apply the domain Markov property at their left endpoints; the probability that every increment is zero is at most  $(1-p)^N$  after  $N$  trials. Letting  $N \rightarrow \infty$  shows  $\Theta_t > \Theta_s$  almost surely, first for rational  $s < t$  and then for all  $s < t$  by continuity.

For a compactly supported set  $A$ , let

$$\Psi_t(A) = \int_{A \cap H_t} G_{H_t}(z; \gamma(t), w_2) dA(z),$$

with the Green normalization of SM-14.06.01. The domain Markov property and (6.3) imply that

$$\mathbf{E}[\mu(A) - \mu(A \cap \gamma[0, t]) \mid \mathcal{F}_t] = c_\kappa \Psi_t(A). \quad (6.12)$$

Hence  $\Psi_t(A) + c_\kappa^{-1} \mu(A \cap \gamma[0, t])$  is a martingale. By uniqueness of the Doob–Meyer decomposition,  $c_\kappa^{-1} \Theta_t$  is exactly the natural-time compensator. Finally, a conformal map  $f : D \rightarrow \mathbb{H}$  transports the construction by local linearization and Koebe distortion; this yields the stated theorem in arbitrary simply connected  $D$ .  $\square$

**Theorem 6.3** (Conformal covariance of SLE natural time).

Let  $0 < \kappa < 8$ ,  $d = 1 + \kappa/8$ , and let  $\mu_\gamma$  be the  $d$ -dimensional Minkowski-content/natural-parametrization measure of the chordal  $\text{SLE}_\kappa$  trace supplied by SM-14.06.02. If  $\phi : D \rightarrow D'$  is conformal and  $A \subset D$  is Borel, then almost surely

$$\mu_{\phi(\gamma)}(\phi(A)) = \int_A |\phi'(z)|^d \mu_\gamma(dz).$$

Equivalently, natural time transforms as  $d$ -dimensional conformal content.

*Proof.* Let  $\phi : D \rightarrow D'$  be conformal and let  $\mu$  be the  $d$ -dimensional Minkowski-content measure of  $\gamma$ , with  $d = 1 + \kappa/8$ . Fix a compact  $K \Subset D$  and partition it into squares  $Q$  on which  $\phi'(z) = \phi'(z_Q)(1 + o(1))$  uniformly as the mesh tends to zero. Koebe distortion then gives

$$d(\phi(z), \phi(\gamma)) = |\phi'(z_Q)| d(z, \gamma)(1 + o(1)), \quad dA(\phi(z)) = |\phi'(z_Q)|^2 (1 + o(1)) dA(z).$$

After multiplying the neighborhood area by  $\varepsilon^{d-2}$ , the two factors combine to  $|\phi'(z_Q)|^d$ . Passing first to the Minkowski-content limit on each square and then to the mesh limit gives

$$\mu_{\phi(\gamma)}(\phi(A)) = \int_A |\phi'(z)|^d \mu_\gamma(dz)$$

for compact continuity sets, hence for all Borel  $A$  by regularity. In particular natural time transforms by the  $d$ -dimensional conformal length element  $|\phi'|^d$ .  $\square$

## Chapter 7

# Reversibility and time reversal

The Loewner equation is oriented from a source to a target, so reversibility is not a formal symmetry of the driving SDE. It is a theorem about the trace as an unoriented random curve. There are three genuinely different mechanisms: the commuting-SLE coupling for  $0 < \kappa \leq 4$ , the imaginary-geometry flow/counterflow coupling for  $4 < \kappa < 8$ , and the discrete UST-Peano symmetry at  $\kappa = 8$ . We keep those mechanisms separate because merging them into a single sentence hides the hard part of the theorem.

### 7.1 Ordinary chordal SLE

**Theorem 7.1** (Chordal SLE reversibility).

*Let  $D$  be a Jordan domain and let  $x, y \in \partial D$  be distinct prime ends. For every  $0 < \kappa \leq 8$ , if  $\eta$  is chordal  $\text{SLE}_\kappa$  from  $x$  to  $y$  in  $D$ , then the time reversal of  $\eta$ , viewed modulo monotone reparametrization, has the law of chordal  $\text{SLE}_\kappa$  from  $y$  to  $x$  in  $D$ . No corresponding ordinary chordal reversibility statement is claimed here for  $\kappa > 8$ .*

*Internalization status: two branches closed, one common core isolated.* Because chordal SLE is conformally covariant, it is enough to work in a fixed Jordan normalization, for example  $(\mathbb{H}; 0, \infty)$ , and to prove equality of the laws of the unparametrized traces. Capacity time itself cannot be reversed and is never part of the assertion.

Start one chordal SLE  $\eta^x$  from  $x$  toward  $y$  and another  $\eta^y$  from  $y$  toward  $x$ . Stop both before their hulls meet. If  $K_s^x, K_t^y$  are the two initial hulls, conformally map the doubly slit domain back to a standard chordal domain. Applying the boundary-coordinate-change calculation of Chapter 4 to either order of growth produces the same Radon–Nikodym factor: it is the product of the two boundary derivative powers and the exponential of the Schwarzian/loop term generated by the two slits. The equality of the two Itô differentials is the commutation identity. Consequently the product law of the two independent SLEs can be tilted, consistently on every rectangle of stopping times, so that

- conditionally on  $\eta^y[0, t]$ , the path  $\eta^x$  up to its next stopping time is chordal SLE in  $D \setminus \eta^y[0, t]$  from  $x$  to the tip of  $\eta^y$ ;
- symmetrically, conditionally on  $\eta^x[0, s]$ , the path  $\eta^y$  has the corresponding SLE law in  $D \setminus \eta^x[0, s]$ .

The local density is a true martingale after stopping a positive distance from the other tip and from the boundary. Consistency on nested stopping rectangles follows from the tower property; taking an increasing localization sequence gives a coupling up to the first meeting of the two traces.

For  $\kappa \leq 4$ , SM-14.03.01–02 gives a simple trace. In the coupling of Step b, condition on a terminal segment of  $\eta^y$ . The conditional forward curve is an SLE in the complementary simply connected domain and therefore cannot cross  $\eta^y$ . Interchanging the two roles gives the opposite ordering. Let a countable family of crosscuts shrink to every point of either trace. The two orderings imply that the first time one trace separates such a crosscut from the opposite endpoint is exactly the last time the other trace does so. Hence the ranges coincide almost surely. Since the forward marginal is SLE from  $x$  to  $y$  and the backward marginal is SLE from  $y$  to  $x$ , the time reversal of the first trace has the second law. This is the Zhan reversibility mechanism; the endpoint  $\kappa = 4$  is obtained by the same localized coupling and monotone continuation.

Take the wired UST/dual-UST Peano exploration of SM-16.02.03 in a lattice approximation of  $(D; x, y)$ . Reversing the oriented Peano path and exchanging the primal tree with the dual tree gives exactly the Peano exploration from  $y$  to  $x$ . Planar tree duality preserves the joint UST/dual-UST law, so the discrete Peano path is reversible modulo time change. Passing to the already proved  $\text{SLE}_8$  scaling limit shows that chordal  $\text{SLE}_8$  is reversible. This argument does not use the  $4 < \kappa < 8$  theory.

In this range the trace is not simple, so the ordering argument of Step c is insufficient. The primary proof couples the path to a GFF as a counterflow line with dual parameter  $\hat{\kappa} = 16/\kappa \in (2, 4)$ . Its left and right outer boundaries are GFF flow lines, and their interaction/targeting rules determine every pocket visited by the counterflow path. Reversing the field boundary data interchanges the two target orientations while preserving the field law; uniqueness of the counterflow local set then identifies the reversed path with  $\text{SLE}_\kappa$  from  $y$  to  $x$ .

For this project, the load-bearing missing statement is therefore not “reversibility” as a black box but the following narrow theorem:

*Counterflow reversal core.* Construct the  $\kappa \in (4, 8)$  GFF counterflow line with its two boundary flow lines, prove that the flow-line interaction rules determine the counterflow local set, and prove that the boundary-data reversal maps this local set to the counterflow line aimed at the opposite endpoint.

This is the common imaginary-geometry root also needed by the force-point theorem below. It is proved in Chapter 9 of this volume. Combining that result with the completed  $\kappa \leq 4$  and  $\kappa = 8$  branches closes the parent gate.

Zhan proves ordinary chordal reversibility for  $0 < \kappa \leq 4$ . Miller–Sheffield prove it for  $4 < \kappa < 8$  as part of Imaginary Geometry III. The  $\kappa = 8$  endpoint is compatible with, and here derived from, the UST Peano result SM-16.02.03. The reversibility statement here is deliberately not extended to  $\kappa > 8$ .  $\square$

*Remark 7.2.* The counterflow-reversal argument in Step e is proved in Chapter 9 without a forward citation to Volume XIX; hence the present theorem depends only on earlier material in this volume.

## 7.2 Two force points at the seed

**Theorem 7.3** (Selected  $\text{SLE}_\kappa(\rho_L; \rho_R)$  reversal law).

*Let  $D$  be a Jordan domain with distinct boundary prime ends  $x, y$ , and place the two force points immediately to the left and right of the seed  $x$ .*



1. If  $0 < \kappa \leq 4$  and  $\rho_L, \rho_R > -2$ , then the time reversal of  $\text{SLE}_\kappa(\rho_L; \rho_R)$  from  $x$  to  $y$  is, modulo reparametrization,  $\text{SLE}_\kappa(\rho_R; \rho_L)$  from  $y$  to  $x$  with the force points immediately on the corresponding two sides of  $y$ .

2. If  $4 < \kappa < 8$  and

$$\rho_L, \rho_R \geq \frac{\kappa}{2} - 4,$$

then the same reversal rule holds. This is the non-boundary-filling regime of the two-seed-force-point theorem.

No claim is made here for force weights outside the parameter ranges displayed above, for a force-point version at  $\kappa = 8$ , or for arbitrary additional force points.

*Proof.* In  $(\mathbb{H}; 0, \infty)$  write  $V_t^L, W_t, V_t^R$  for the transported left force point, driver, and right force point. Chapter 4 owns the Girsanov calculation giving

$$dW_t = \sqrt{\kappa} dB_t + \frac{\rho_L}{W_t - V_t^L} dt + \frac{\rho_R}{W_t - V_t^R} dt, \quad dV_t^q = \frac{2}{V_t^q - W_t} dt.$$

Before a continuation threshold is reached, changing  $(\rho_L, \rho_R)$  is therefore locally absolutely continuous with respect to ordinary SLE, with a density given by boundary derivative/distance powers. The Bessel analysis of Chapter 2 gives the continuation threshold  $\rho_q > -2$  in the  $\kappa \leq 4$  seed-force-point regime.

The left/right labels are geometric labels relative to the orientation of the curve. Reversing source and target exchanges the two sides. In a GFF coupling this is literal: the two boundary heights determining the force weights are interchanged. Hence any valid reversal theorem must send  $(\rho_L, \rho_R)$  to  $(\rho_R, \rho_L)$ ; there is no additional affine shift in the immediate-two-force-point normalization used here.

For  $0 < \kappa < 4$  the Miller–Sheffield construction realizes the path as a GFF flow line with piecewise constant boundary data whose two height gaps are exactly the parameters  $\rho_L, \rho_R$ . The local-set Markov rule implies that, after any stopping time, the conditional field in the unexplored domain is an independent zero-boundary GFF plus the harmonic boundary data seen from the two sides of the path. Adding the deterministic constants that reverse the orientation of the flow line exchanges these side data but leaves the zero-boundary part unchanged. Uniqueness of the flow line as a measurable local set of the field therefore gives the reversed  $\text{SLE}_\kappa(\rho_R; \rho_L)$  law. The  $\kappa = 4$  statement is the level-line endpoint of the same boundary-data calculation.

The present series owns the GFF domain Markov theorem (Volume VI), the force-point SDE (Chapter 4), and the  $\kappa = 4$  level-line coupling (SM-16.05.02), but it does not yet own the all- $0 < \kappa < 4$  flow-line existence/uniqueness theorem in a backward volume. Thus Step c identifies a second face of the same imaginary-geometry root rather than hiding it behind a citation.

For a counterflow SLE the two dual boundary flow lines cease to leave an unexplored boundary corridor when a seed weight falls below  $\kappa/2 - 4$ . At or above this threshold the boundary flow lines delimit the counterflow pockets and the same field-data reversal exchanges source and target. Below the threshold the corresponding side is boundary filling, so the plain two-force-point reversal statement fails. This gives exactly the source condition  $\rho_L, \rho_R \geq \kappa/2 - 4$ .

Steps c–d use the self-contained backward module now proved in Chapter 9: GFF flow-line existence and uniqueness for the boundary data occurring here, the crossing/merging/bouncing interaction rule, construction of the dual counterflow line, and uniqueness of that counterflow local set. The boundary-data transformation in Steps b–d therefore completes both SM-14.07.01 and SM-14.07.02 without any further probabilistic estimate.

Imaginary Geometry II gives the  $0 < \kappa \leq 4$  two-seed-force-point reversal with  $\rho_L, \rho_R > -2$ . Imaginary Geometry III gives the  $4 < \kappa < 8$  reversal exactly when both weights are at least  $\kappa/2 - 4$ . More general force-point configurations are intentionally not imported into this gate.  $\square$

*Remark 7.4* (Scope of the reversal theorem). The preceding result covers the two-seed force-point reversals stated there:  $0 < \kappa \leq 4$  with  $\rho_L, \rho_R > -2$ , and  $4 < \kappa < 8$  with both weights at least  $\kappa/2 - 4$ . Chapter 9 supplies the common imaginary-geometry reversal construction used in both cases. No claim about arbitrary multi-force-point configurations is made here.

## Chapter 8

# Reversibility and coordinate changes

### 8.1 SLE coordinate changes

**Theorem 8.1** (Schramm–Wilson coordinate-change theorem).

Let  $X, Y \in \{\mathbb{H}, \mathbb{D}\}$  and let  $\psi : X \rightarrow Y$  be a Möbius conformal homeomorphism. Let  $w_0 \in \partial X$  be the starting point and let  $V^1, \dots, V^m \in \bar{X} \setminus \{w_0\}$  be force points, with interior points permitted. If

$$\rho_1 + \dots + \rho_m = \kappa - 6,$$

then, up to the first force-point collision and up to the first time the point  $\psi^{-1}(o_Y)$  is swallowed (where  $o_{\mathbb{H}} = \infty$  and  $o_{\mathbb{D}} = 0$ ), the image under  $\psi$  of  $X$ -SLE $_{\kappa}(\rho_1, \dots, \rho_m)$  is, after the natural Loewner time-change,  $Y$ -SLE $_{\kappa}(\rho_1, \dots, \rho_m)$  with the mapped force points.

Consequently ordinary radial SLE $_{\kappa}$  in  $\mathbb{D}$  becomes chordal SLE $_{\kappa}(\kappa - 6)$  with an interior force point at the image of the radial target, and conversely.

*Proof.* We first put the radial and chordal SDEs in one notation and then do the only nontrivial coordinate calculation, from  $\mathbb{H}$  to  $\mathbb{D}$ . The reverse direction and the two same-domain cases follow by inversion and composition.

**1. Unified SLE $(\kappa; \rho)$  notation.** Set

$$\Psi_{\mathbb{H}}(w, z) = \frac{2}{z - w}, \quad \Psi_{\mathbb{D}}(w, z) = -z \frac{z + w}{z - w}.$$

Let  $I_{\mathbb{H}}(z) = \bar{z}$  and  $I_{\mathbb{D}}(z) = 1/\bar{z}$  be reflection in the relevant boundary, and define

$$\tilde{\Psi}_X(v, w) = \frac{1}{2}(\Psi_X(v, w) + \Psi_X(I_X v, w)).$$

For the Brownian part of the driving point put

$$G_{\mathbb{H}}(W, dB, dt) = \sqrt{\kappa} dB,$$

while, because  $W_t = W_0 e^{i\sqrt{\kappa}B_t}$  in radial SLE,

$$G_{\mathbb{D}}(W, dB, dt) = -\frac{\kappa}{2} W dt + i\sqrt{\kappa} W dB.$$

Thus  $X\text{-SLE}_\kappa(\rho_1, \dots, \rho_m)$  is the solution, up to the first collision, of

$$dW_t = G_X(W_t, dB_t, dt) + \sum_{j=1}^m \frac{\rho_j}{2} \tilde{\Psi}_X(V_t^j, W_t) dt, \quad (14.18)$$

$$dV_t^j = \Psi_X(W_t, V_t^j) dt, \quad V_t^j = g_t(V^j), \quad (14.19)$$

$$\partial_t g_t(z) = \Psi_X(W_t, g_t(z)). \quad (14.20)$$

For an interior force point in  $\mathbb{H}$ , the symmetrization in  $\tilde{\Psi}_{\mathbb{H}}$  is exactly the real-part drift in the usual chordal definition.

**2. Reduce the half-plane-to-disk case to a normalized Loewner chain.** Assume  $X = \mathbb{H}$  and  $Y = \mathbb{D}$ . Let

$$z_t = x_t + iy_t := g_t(\psi^{-1}(0)).$$

Before  $\psi^{-1}(0)$  is swallowed,  $y_t > 0$ . There is a unique Möbius map  $\phi_t : \mathbb{H} \rightarrow \mathbb{D}$  of the form

$$\phi_t(z) = \lambda_t \frac{z - z_t}{z - \bar{z}_t}, \quad |\lambda_t| = 1,$$

such that

$$F_t := \phi_t \circ g_t \circ \psi^{-1}$$

satisfies  $F_t(0) = 0$  and  $F'_t(0) > 0$ . Define radial time

$$s(t) := \log F'_t(0).$$

Since

$$\dot{y}_t = -\frac{2y_t}{|z_t - W_t|^2}, \quad \partial_t \log g'_t(\psi^{-1}(0)) = -\frac{2}{(z_t - W_t)^2},$$

taking the real part in the derivative of  $\log F'_t(0)$  gives

$$s'(t) = \frac{4y_t^2}{|z_t - W_t|^4} > 0. \quad (14.21)$$

Hence  $s$  is a legitimate time-change. With  $t = t(s)$  set

$$\widehat{g}_s = F_{t(s)}, \quad \widehat{W}_s = \phi_{t(s)}(W_{t(s)}), \quad \widehat{V}_s^j = \phi_{t(s)}(V_{t(s)}^j).$$

The maps  $\widehat{g}_s$  have radial normalization  $\widehat{g}_s(0) = 0$ ,  $\widehat{g}'_s(0) = e^s$ , and their domains are  $\mathbb{D} \setminus \psi(K_{t(s)})$ . Uniqueness in Loewner's theorem therefore gives

$$\partial_s \widehat{g}_s(z) = \Psi_{\mathbb{D}}(\widehat{W}_s, \widehat{g}_s(z)).$$

**3. Brownian time-change.** Define

$$\widehat{B}_s = \int_0^{t(s)} \sqrt{s'(u)} dB_u.$$

Its quadratic variation is  $\langle \widehat{B} \rangle_s = s$ ; by Lévy's characterization it is standard Brownian motion.

**4. Compute the transformed driving SDE.** The identity

$$\widehat{W}_s = \lambda_t \frac{W_t - z_t}{W_t - \bar{z}_t}$$

gives

$$d \log \widehat{W}_s = d \log \lambda_t + d \log(W_t - z_t) - d \log(W_t - \bar{z}_t).$$

We evaluate the three terms. First, since  $\widehat{g}_s(\psi(\infty)) = \lambda_t$ , the radial Loewner equation at that marked boundary point yields

$$d \log \lambda_t = -\widehat{W}_s^{-1} \Psi_{\mathbb{D}}(\lambda_t, \widehat{W}_s) ds. \quad (14.22)$$

Second, Itô's formula, the chordal Loewner equation for  $z_t$ , and (14.21) give for the finite-variation part not arising from force points

$$\frac{4 - \kappa}{2(W_t - z_t)^2} dt - \frac{4 - \kappa}{2(W_t - \bar{z}_t)^2} dt = -\frac{4 - \kappa}{2} \widehat{W}_s^{-1} \Psi_{\mathbb{D}}(\lambda_t, \widehat{W}_s) ds. \quad (14.23)$$

For the Brownian and force terms, write each interior force point together with its reflection. Direct substitution of the Möbius map gives

$$i s'(t)^{-1/2} \Psi_{\mathbb{H}}(v, W_t) = \widehat{W}_s^{-1} \Psi_{\mathbb{D}}(\phi_t(v), \widehat{W}_s) - \widehat{W}_s^{-1} \Psi_{\mathbb{D}}(\lambda_t, \widehat{W}_s). \quad (14.24)$$

Using (14.18), pairing reflected force points, and  $d\widehat{B}_s = \sqrt{s'(t)} dB_t$ , this becomes

$$\begin{aligned} A_2 &= i\sqrt{\kappa} d\widehat{B}_s - \frac{\sum_j \rho_j}{2} \widehat{W}_s^{-1} \Psi_{\mathbb{D}}(\lambda_t, \widehat{W}_s) ds \\ &\quad + \widehat{W}_s^{-1} \sum_{j=1}^m \frac{\rho_j}{2} \widetilde{\Psi}_{\mathbb{D}}(\widehat{V}_s^j, \widehat{W}_s) ds. \end{aligned} \quad (14.25)$$

The coefficients of  $\widehat{W}_s^{-1} \Psi_{\mathbb{D}}(\lambda_t, \widehat{W}_s)$  in (14.22), (14.23), and (14.25) sum to

$$-1 - \frac{4 - \kappa}{2} - \frac{\kappa - 6}{2} = 0$$

precisely because  $\sum_j \rho_j = \kappa - 6$ . Therefore

$$d \log \widehat{W}_s = i\sqrt{\kappa} d\widehat{B}_s + \widehat{W}_s^{-1} \sum_j \frac{\rho_j}{2} \widetilde{\Psi}_{\mathbb{D}}(\widehat{V}_s^j, \widehat{W}_s) ds.$$

Exponentiating and applying Itô one last time gives

$$d\widehat{W}_s = -\frac{\kappa}{2} \widehat{W}_s ds + i\sqrt{\kappa} \widehat{W}_s d\widehat{B}_s + \sum_j \frac{\rho_j}{2} \widetilde{\Psi}_{\mathbb{D}}(\widehat{V}_s^j, \widehat{W}_s) ds, \quad (14.26)$$

which is exactly the disk driving equation (14.18).

Differentiating  $\widehat{V}_s^j = \phi_t(V_t^j)$  and using (14.19) together with the conjugacy  $\widehat{g}_s \circ \psi = \phi_t \circ g_t$  gives

$$d\widehat{V}_s^j = \Psi_{\mathbb{D}}(\widehat{W}_s, \widehat{V}_s^j) ds.$$

Thus the entire transformed marked Loewner chain is disk SLE $_{\kappa}(\rho_1, \dots, \rho_m)$ .

**5. The remaining coordinate changes.** The disk-to-half-plane statement is the inverse of the result just proved. A Möbius automorphism of  $\mathbb{H}$  factors as a half-plane-to-disk map followed by a disk-to-half-plane map; the disk automorphism case is analogous. Composition of the two time-changes proves the theorem for every  $X, Y \in \{\mathbb{H}, \mathbb{D}\}$ .

Finally, a force point at  $o_{\mathbb{D}} = 0$  has no effect on radial evolution. We may therefore append the radial target with weight  $\kappa - 6$  to ordinary radial SLE. Mapping to  $\mathbb{H}$  turns it into an interior force point and the theorem yields chordal  $\text{SLE}_{\kappa}(\kappa - 6)$ . The converse follows by inversion. □

*Remark 8.2.* This theorem concerns coordinate changes for an already defined Loewner chain. Trace existence and reversibility are separate obligations; neither is used in the proof above.

## Chapter 9

# Imaginary-geometry reversal core

This chapter is deliberately placed in Volume XIV. Later volumes use the full imaginary-geometry language, but the two reversibility theorems of Chapter 7 need only one self-contained core: boundary-data flow lines, their interaction rules, the dual counterflow line, and the tree ordering that reconstructs the counterflow path. Owning that core here keeps reversibility a backward theorem rather than a forward citation to Volume XIX.

Throughout this chapter

$$0 < \kappa < 4, \quad \kappa' = \frac{16}{\kappa} > 4, \quad \chi = \frac{2}{\sqrt{\kappa}} - \frac{\sqrt{\kappa}}{2}, \quad \lambda = \frac{\pi}{\sqrt{\kappa}}.$$

Adding the constant  $\theta\chi$  to the field rotates a flow-line angle by  $\theta$ . The sign convention for the two boundary heights is the one fixed by the force-point SDE of Chapter 4: in the half plane, the angle-zero flow line from 0 to  $\infty$  with immediate force weights  $(\rho_L; \rho_R)$  sees boundary data  $-\lambda(1 + \rho_L)$  on  $\mathbb{R}_-$  and  $+\lambda(1 + \rho_R)$  on  $\mathbb{R}_+$ , up to the common additive constant which changes the angle but not the abstract coupling.

### 9.1 Flow lines as GFF local sets

**Theorem 9.1** (Boundary-data flow-line coupling). *Let  $D$  be simply connected, let  $x, y$  be distinct boundary prime ends, and let  $h$  be a zero-boundary GFF plus a harmonic function which is piecewise constant on the two boundary arcs from  $x$  to  $y$ . Suppose the corresponding seed force weights are above the continuation threshold  $-2$ . There is a continuous local set  $\eta$  from  $x$  to  $y$ , measurable from  $h$ , such that conditional on  $\eta[0, \tau]$  the field in every component of  $D \setminus \eta[0, \tau]$  is an independent zero-boundary GFF plus the harmonic function obtained by keeping the original boundary data and adding the imaginary-geometry height jump on the two sides of the oriented trace. The marginal law of  $\eta$  is the corresponding  $\text{SLE}_\kappa(\rho_L; \rho_R)$ .*

*Proof.* We give the construction in  $\mathbb{H}$  and use conformal covariance to pass to  $D$ .

Fix an interior point  $z$  and let  $g_t$  be the Loewner map of a candidate local set. If  $W_t$  is the image of the tip and  $V_t^L, V_t^R$  are the images of the two marked force points, the conditional mean at  $z$  is the harmonic extension of the current boundary heights. In the half-plane it is a linear combination of arguments

$$\arg(g_t(z) - W_t), \quad \arg(g_t(z) - V_t^L), \quad \arg(g_t(z) - V_t^R),$$

with coefficients equal to the prescribed height jumps. Differentiating with Itô's formula shows that

this conditional mean is a local martingale exactly when

$$dW_t = \sqrt{\kappa} dB_t + \frac{\rho_L}{W_t - V_t^L} dt + \frac{\rho_R}{W_t - V_t^R} dt, \quad dV_t^q = \frac{2}{V_t^q - W_t} dt. \quad (14.27)$$

Thus the GFF boundary-data rule and the force-point SDE are not separate inputs: each determines the other.

For  $\epsilon > 0$ , stop the force-point Loewner chain before the conformal radius of the tip from any marked compact set is below  $\epsilon$  and before a force-point distance is below  $\epsilon$ . On this stopped interval the Radon–Nikodym derivative with respect to ordinary SLE is bounded above and below, so the GFF/SLE<sub>4</sub> level-line construction and Cameron–Martin absolute continuity give a coupling with the required conditional mean. If  $\epsilon_n \downarrow 0$ , the stopped hulls are monotone in the boundary heights and tight in the local-set topology. The GFF local-set martingale identifies every subsequential limit by its conditional mean and conditional covariance, hence the limit is independent of the approximating sequence.

The distance  $X_t = (V_t - W_t)/\sqrt{\kappa}$  from an immediate force point becomes, after the usual deterministic time change, a Bessel process of dimension

$$\delta = 1 + \frac{2(\rho + 2)}{\kappa}.$$

The threshold  $\rho = -2$  is therefore exactly the point at which the Loewner continuation ceases to be defined by the principal-value excursion rule. For  $\rho > -2$ , concatenate the stopped local sets across successive force-point collisions. The strong Markov property of the GFF and of the SDE shows that the concatenation has the same conditional boundary-data rule at every stopping time.

Couple two candidates  $\eta_1, \eta_2$  with the same field and boundary data. Explore  $\eta_1$  first. Conditional on it, the law of  $\eta_2$  in each remaining component is an SLE with force weights read from the two sides of  $\eta_1$ . The interaction lemma proved below implies that two same-angle lines cannot cross and, once they meet, cannot separate. Reversing the roles of  $\eta_1, \eta_2$  gives the opposite ordering; hence their ranges coincide. The target parametrization is immaterial, so the flow line is a measurable function of  $h$ .

If  $\phi : D \rightarrow \tilde{D}$  is conformal, the field transforms as  $h \mapsto h \circ \phi^{-1} - \chi \arg(\phi^{-1})'$  for imaginary geometry. The added harmonic term is exactly the turning-angle correction of the vector field. The conditional boundary-data rule is therefore invariant, and the uniqueness in Step d implies that  $\phi(\eta)$  is the corresponding flow line of the transformed field. This completes existence, SLE identification, local-set Markov property, and pathwise uniqueness.  $\square$

## 9.2 Interaction: ordering, crossing, bouncing, and merging

**Theorem 9.2** (Flow-line interaction theorem). *Let  $\eta_1, \eta_2$  be GFF flow lines with admissible starting data and angles  $\theta_1, \theta_2$ . Conditional on the first line, each segment of the second is an SLE $_{\kappa}(\rho_L; \rho_R)$  in a complementary component, with the two force weights determined by the height gap between the angle- $\theta_2$  boundary data and the two sides of  $\eta_1$ . Consequently: (i) equal-angle lines merge when they meet and never separate; (ii) the planar left/right order agrees with the angle order; and (iii) at a contact at which the induced force weight is in the hitting but noncrossing range the second line bounces, while in the crossing range it crosses only in the direction allowed by that order. These local rules determine the global interaction by strong Markov iteration.*

*Proof.* The local-set property from the preceding theorem says that, after revealing  $\eta_1$ , the field in every component of its complement is a fresh zero-boundary GFF plus an explicit harmonic



function. The boundary values of that harmonic function on the two sides of  $\eta_1$  differ by the fixed imaginary-geometry height gap. After adding the angle shift  $\theta_2\chi$ , the boundary-height/force-weight dictionary therefore identifies each conditional segment of  $\eta_2$  with an  $\text{SLE}_\kappa(\rho_L; \rho_R)$  having force points on the two sides of  $\eta_1$ .

The separations between the driving point and these force points satisfy the Bessel SDEs analyzed in Chapter 4. The Bessel boundary classification gives exactly three possibilities at a prospective contact: repulsion before contact, hitting with instantaneous return to the same side, or continuation into the opposite component when the force weight lies in the crossing range. Thus the local bounce/cross rule follows directly from the already proved force-point continuation theorem.

For equal angles, the force weight on the side of  $\eta_1$  facing  $\eta_2$  is the continuation-threshold value: the second line may hit  $\eta_1$  but is not allowed to cross it. At the first hit the conditional boundary data on the unexplored side agree with those defining the continuation of  $\eta_1$ . The pathwise uniqueness theorem for GFF flow lines therefore forces the two future traces to coincide. Hence equal-angle lines merge on meeting and never separate.

For distinct angles, first take rational angles and starting points. The force weights vary monotonically with the boundary height. Before a possible meeting, the line with the larger angle cannot move from its allowed side to the forbidden side without entering a force-point regime below the Bessel crossing threshold. Therefore angle order agrees with planar left/right order. Monotone approximation of the angles and starting points removes the rationality restriction.

Finally, at every later contact stop the joint local-set exploration and use a conformal chart centered at the contact point. The conditional field again has the same two-force-point form, so the preceding Bessel classification gives the complete local bounce/cross alternative. Successive contacts are stopping times for the jointly generated filtration; strong Markov iteration therefore propagates the local rule globally. The same argument applies to flow lines started from different points and inside components cut out by previously explored lines.  $\square$

### 9.3 Counterflow lines and duality

**Theorem 9.3** (Counterflow reconstruction and dual boundaries). *Let  $\kappa' = 16/\kappa > 4$ . For the GFF boundary data corresponding to a counterflow path from  $y$  to  $x$ , there is a field-measurable local set  $\eta'$  with marginal law the prescribed  $\text{SLE}_{\kappa'}(\rho'_L; \rho'_R)$ . Its left and right outer boundaries are the two GFF flow lines of the dual parameter  $\kappa$  at the extremal admissible angles. Conversely those two boundary flow lines, together with the conditional pocket order, determine  $\eta'$ .*

*Proof.* Construct from the target  $y$  the two extremal flow lines whose angles differ by the light-cone opening associated with  $\kappa'$ . The interaction theorem orders them and shows that they cannot cross in the wrong direction. Their union therefore bounds a connected unexplored region joining  $y$  to  $x$ .

Choose a countable dense family of crosscuts in the region between the two boundaries. Explore the first crosscut not yet disconnected from  $x$ , and in the pocket thereby selected use the GFF local-set Markov rule to draw the unique conditional flow/counterflow segment. Merging makes the construction independent of the enumeration: two enumerations agree on every finite set of crosscuts before either can influence a later pocket.

In a component mapped to  $\mathbb{H}$ , the conditional harmonic mean of the field is again a linear combination of arguments of the tip and force points. The same Itô calculation as in (14.27), now using the dual height gap, gives quadratic variation  $\kappa' dt$  and the force drifts prescribed by  $(\rho'_L; \rho'_R)$ . Lévy's characterization and the domain Markov property therefore identify the growing local set with  $\text{SLE}_{\kappa'}(\rho'_L; \rho'_R)$ .

Whenever the counterflow exploration enters a pocket, the leftmost and rightmost points it can reach without crossing the boundary flow lines are fixed by the planar ordering theorem. Conversely, the conditional SLE has positive probability to approach every rational crosscut between those boundaries before leaving the pocket. A countable crosscut argument therefore shows that the outer boundary of the filled counterflow hull is exactly the two flow lines, not merely contained between them.

Suppose two counterflow local sets have the same field and boundary data. Their left/right boundaries agree by flow-line measurability. Conditional on those boundaries, both traverse the same countable family of pockets and have the same conditional local-set boundary data in each. Induction over the pocket order and continuity show that the traces agree. Hence the counterflow line is a measurable function of the field and of its two boundary flow lines. This is the reconstruction statement needed for reversibility.  $\square$

**Corollary 9.4** (Rigorous SLE duality). *For  $\kappa' > 4$  with  $\kappa = 16/\kappa'$ , the appropriate left/right outer boundaries of a counterflow  $\text{SLE}_{\kappa'}(\rho')$  are coupled  $\text{SLE}_{\kappa}(\rho)$  flow lines with the force weights obtained from the same GFF boundary heights. In particular the ordinary duality statements used in Volume XV follow by specializing the boundary data.*

*Proof.* This is Step d of the preceding theorem, with the affine boundary-height to force-weight dictionary specialized to the ordinary or one-sided cases.  $\square$

## 9.4 Time reversal

The geometric symmetry of the field is *not* by itself a reversibility proof: a flow line read from its terminal point is not automatically a flow line of the same field. The missing step is the two-field resampling construction below. This is the point at which the proof follows the Miller–Sheffield reversibility architecture rather than a reflection shortcut.

**Lemma 9.5** (Side-shifted GFF resampling). *Let  $0 < \kappa \leq 4$ , let  $\eta$  be the flow line from  $x$  to  $y$  with immediate left/right force weights  $\rho_L, \rho_R > -2$ , and let  $h$  be the field in the boundary-data coupling of the first section. There are constants  $c_L, c_R$ , determined by  $(\kappa, \rho_L, \rho_R)$ , such that the distribution obtained by adding  $c_L$  to the field on the left of  $\eta$  and  $c_R$  on its right has the following two properties.*

1. *Conditionally on  $\eta$ , its restrictions to the complementary components are independent zero-boundary GFFs plus the harmonic boundary data for a flow line aimed from  $y$  to  $x$ .*
2. *After the curve is forgotten, the resulting random distribution  $\hat{h}$  is a GFF with the deterministic boundary data corresponding to weights  $(\rho_R; \rho_L)$  at the new seed  $y$ .*

*The same statement holds after stopping at any pair of crosscuts separating  $x$  from  $y$ .*

*Proof.* The local-set decomposition from the first section says that conditionally on  $\eta$ ,

$$h = h_{D \setminus \eta}^0 + \mathfrak{h}_\eta,$$

where the zero-boundary fields in different complementary components are independent and  $\mathfrak{h}_\eta$  is the harmonic function whose traces on the two sides of the curve differ by the flow-line height gap  $2\lambda$ . Choose  $c_L, c_R$  so that after the orientation is reversed the left/right traces equal the boundary constants associated with  $\rho_R, \rho_L$ . This choice is unique because adding the same constant to both sides only changes the global additive normalization, while their difference is fixed by the height gap.

It remains to check that forgetting  $\eta$  produces a GFF rather than a mixture with an extra covariance term. Let  $f$  be smooth and compactly supported away from the marked prime ends. Conditional on  $\eta$ , the characteristic function of  $(\widehat{h}, f)$  is

$$\exp\left(i(\widehat{h}_\eta, f) - \frac{1}{2}\|f\|_{\nabla, D \setminus \eta}^2\right).$$

Expose the flow line only until it first hits a deterministic crosscut. The flow-line martingale calculation in the first section says exactly that the conditional mean  $(\widehat{h}_\eta, f)$  and the lost Dirichlet energy combine to give the GFF characteristic-function martingale with the new deterministic boundary data. Optional stopping therefore identifies the unconditional characteristic function with that of this GFF. Exhaust  $D$  by crosscuts and use  $H^{-1}$  convergence of local sets. This proves (2); the conditional decomposition already proves (1). The argument is local, so the stopped version follows identically.  $\square$

**Lemma 9.6** (Reverse-flow-line identification). *In the setting of Lemma 9.5, the range of  $\eta$ , read from  $y$  to  $x$ , is almost surely the flow line of  $\widehat{h}$  with the reversed boundary data. Consequently its unparameterized law is  $\text{SLE}_\kappa(\rho_R; \rho_L)$ .*

*Proof.* Fix a countable family of rational crosscuts tending from  $y$  toward  $x$ . Condition on the portion of  $\eta$  between two consecutive crosscuts. Lemma 9.5 gives the GFF Markov decomposition on the unexplored component, with precisely the height jump required for the reverse flow line. The interaction theorem implies that any reverse flow line with these data cannot cross the already exposed portion and, whenever it touches it from the admissible side, must merge with it. Hence the reverse flow line agrees with  $\eta$  between the two crosscuts. Induction over the rational exhaustion gives equality of the ranges. The boundary-data flow-line theorem then identifies its marginal law as the displayed force-point SLE.  $\square$

**Theorem 9.7** (Reversal in the simple and counterflow regimes). *The following reversal statements hold in the displayed parameter regimes.*

1. *If  $0 < \kappa \leq 4$  and  $\rho_L, \rho_R > -2$ , an immediate-two-force-point  $\text{SLE}_\kappa(\rho_L; \rho_R)$  from  $x$  to  $y$ , reversed modulo parameterization, is  $\text{SLE}_\kappa(\rho_R; \rho_L)$  from  $y$  to  $x$ .*
2. *If  $4 < \kappa < 8$  and  $\rho_L, \rho_R \geq \kappa/2 - 4$ , with both weights  $> -2$ , the same immediate force-point reversal rule holds. In particular ordinary chordal  $\text{SLE}_\kappa$  is reversible for  $4 < \kappa < 8$ .*

*Proof.* Part (1) is Lemma 9.6. The endpoint  $\kappa = 4$  is obtained by the level-line limit: the Bessel dimensions and boundary heights remain in the continuation regime and the local-set laws converge.

For part (2) set  $\widehat{\kappa} = 16/\kappa \in (2, 4)$ . Apply the counterflow reconstruction theorem to the original curve. Its left and right outer boundaries are a pair of  $\widehat{\kappa}$ -flow lines with force weights given by the affine duality dictionary. The assumed inequalities  $\rho_q \geq \kappa/2 - 4$  are exactly the conditions that these two boundary flow lines lie in the non-boundary-filling continuation regime. By part (1), read from the target they have the law of the corresponding reversed pair, with left and right exchanged.

We still have to reconstruct the path between the two boundaries. Cut along the boundary pair. In every pocket lying between them, the domain Markov property and the force-weight calculation of the counterflow theorem identify the conditional middle strand as an independent  $\text{SLE}_\kappa(\kappa/2 - 4; \kappa/2 - 4)$  with the appropriate entrance and exit prime ends. This conditional law is itself symmetric under interchanging the two sides, by the simple-regime reversal applied to its dual boundaries. The pockets are conditionally independent and are traversed in the order determined by the boundary tree. Reversing the order and each conditional strand therefore reconstructs exactly

the counterflow path with weights  $(\rho_R; \rho_L)$  from  $y$  to  $x$ . This proves (2), including ordinary SLE by setting the seed weights to zero.  $\square$

**Corollary 9.8** (Reversibility consequences). *The imaginary-geometry construction above implies the corresponding ordinary and force-point SLE reversal statements. The theorem above, together with the previously owned  $\kappa = 8$  UST-Peano endpoint and the commuting-SLE argument in the simple regime, closes SM-14.07.01. Its exact immediate-force-point statement closes SM-14.07.02.*

*Proof.* For ordinary chordal SLE, the reversal theorem above covers the regimes  $0 < \kappa < 8$  treated by the imaginary-geometry/commuting-SLE argument. The UST Peano-curve theorem of Volume XVI supplies the  $\kappa = 8$  endpoint. For the selected force-point reversal, the two parameter regimes are exactly those displayed in the theorem. No later volume is used.  $\square$

## 9.5 Space-filling order

**Theorem 9.9** (Flow-line tree and space-filling counterflow order). *Start same-angle flow lines of the GFF from a fixed countable dense set. They merge on meeting and hence form an oriented tree. The left/right relation of branches defines a consistent total order on the dense set. There is a unique continuous curve which visits the points in this order; its conditional law in every unexplored pocket is the corresponding counterflow SLE. This is space-filling  $\text{SLE}_{\kappa'}$ . For  $\kappa' \geq 8$  with the standard boundary data, this traversal agrees with ordinary chordal/radial  $\text{SLE}_{\kappa'}$  after the usual filling convention.*

*Proof.* Enumerate the fixed countable dense set as  $\{z_j\}_{j \geq 1}$  and start the same reference-angle flow line from each  $z_j$  toward the common target. By the interaction theorem, two branches cannot cross and, after their first meeting, merge forever. Thus the union of the first  $n$  branches is a finite planar tree, and these trees are consistent as  $n$  increases.

Define  $z_i \prec z_j$  according to the left/right order of the two branches immediately before their first common branch point; if the two points lie on the same branch, use their order of first visit along that branch. Planarity and the noncrossing/merging rule make this relation transitive and total on the dense set.

To pass from the order to a continuous curve, fix a point  $z$  and choose sequences of dense points approaching  $z$  from below and above in the order. The two corresponding boundary branches squeeze the unexplored pocket containing  $z$ . Its conformal radius must tend to zero: otherwise a Euclidean ball of fixed positive radius would remain between the two branches and would contain another point of the dense set whose order lies strictly between the chosen lower and upper sequences. Hence the order completion determines a unique continuous traversal.

The shrinking-pocket argument shows that the traversal visits every point. Conditional on an explored initial segment, the next excursion through any unexplored pocket is the counterflow local set reconstructed earlier in this chapter. The domain Markov property and the SDE identification therefore give the law of the traversal as space-filling  $\text{SLE}_{\kappa'}$ .

When  $\kappa' \geq 8$  in the standard boundary-data regime, the ordinary  $\text{SLE}_{\kappa'}$  trace itself fills the open pockets in their induced order. The counterflow traversal adds no extra ordering excursions, so the two constructions agree up to the standard filling convention and monotone reparametrization. For  $4 < \kappa' < 8$  the additional pocket traversal is essential, and no such identification is asserted.  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 9.10** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 9.11** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 9.12** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 9.13** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 9.14** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XV: Geometry of SLE, Brownian Exponents, and Conformal  
Loop Ensembles

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# Scope and conventions

Once SLE has been constructed, one can ask quantitative geometric questions: dimension, intersections, outer boundaries, restriction exponents, loop measures, and loop ensembles. The answers form a bridge from individual random curves to conformally invariant random collections. This volume records the exact theorem scopes and makes explicit which computations require the Lawler-Schramm-Werner exponent machinery or the Miller-Sheffield GFF coupling.

# Chapter 1

## Fractal geometry of SLE

The trace of Schramm–Loewner evolution is a random curve whose local geometry can be measured by dimensions of the trace, its boundary, cut points, double points, and exceptional sets. The point of this chapter is to fix the geometric language before the exponent calculations of Chapters 2 and 8.

### 1.1 Hausdorff dimension and exceptional sets

**Definition 1.1** (Hausdorff dimension). For a metric space  $(E, d)$  and  $s \geq 0$ , define

$$\mathcal{H}_\delta^s(E) = \inf \left\{ \sum_i (\text{diam } U_i)^s : E \subseteq \bigcup_i U_i, \text{ diam } U_i \leq \delta \right\},$$
$$\mathcal{H}^s(E) = \lim_{\delta \downarrow 0} \mathcal{H}_\delta^s(E), \quad \dim_{\text{H}} E = \inf \{s : \mathcal{H}^s(E) = 0\}.$$

For an SLE trace  $\eta$ , all dimensions in this volume refer to the random set  $\eta([0, \infty))$  or to explicitly named subsets such as cut points or double points.

*Worked example 1.2* (Checking the SLE trace-dimension formula at special parameters). The dimension theorem proved in Chapter 8 is

$$d_\kappa = 1 + \frac{\kappa}{8}, \quad 0 < \kappa \leq 8,$$

with  $d_\kappa = 2$  for  $\kappa \geq 8$ . Substitution gives

$$d_2 = \frac{5}{4}, \quad d_3 = \frac{11}{8}, \quad d_4 = \frac{3}{2}, \quad d_6 = \frac{7}{4}, \quad d_8 = 2.$$

These values match the dimensions attached respectively to LERW, critical Ising interfaces, GFF level lines, critical percolation exploration, and the space-filling UST Peano curve. The calculation is only a consistency check; the identification of each lattice model with its SLE limit is proved in the corresponding earlier volume.

### 1.2 Proof architecture

The rigorous trace-dimension theorem is not obtained from a visual box count. Chapter 8 proves a one-point Green-function estimate, converts it into an upper bound by a covering argument, and obtains the matching lower bound from a second-moment measure supported on the trace. Exceptional-set dimensions are handled by the same pattern with multi-point or derivative exponents. This chapter will use those results only after their hypotheses have been established there.

## Chapter 2

# Brownian intersection exponents

The Lawler–Schramm–Werner exponent programme is a model case of rigorous universality: Brownian nonintersection events are reduced to conformally invariant restriction laws and then to solvable SLE differential equations. This chapter fixes the exponent notation and works through the simplest scaling calculation; Chapter 8 contains the full proof engine.

### 2.1 Nonintersection probabilities and exponents

**Definition 2.1** (Brownian intersection exponent). Let  $B^1, \dots, B^p$  and  $\tilde{B}^1, \dots, \tilde{B}^q$  be independent planar Brownian motions started from separated points on the unit circle and stopped on first hitting the circle of radius  $R > 1$ . Write  $E_R^{p,q}$  for the event that the union of the first packet does not intersect the union of the second packet before the stopping times. When the limit exists, the intersection exponent  $\xi(p, q)$  is defined by

$$\xi(p, q) = - \lim_{R \rightarrow \infty} \frac{\log \mathbf{P}(E_R^{p,q})}{\log R}.$$

Half-plane exponents are defined analogously with Brownian excursions in a half-annulus and are denoted by  $\tilde{\xi}$ .

*Worked example 2.2* (Why an exponent becomes additive across annuli). Suppose a scale-invariant avoidance event satisfies, up to constants independent of  $r < R$ ,

$$P(r, R) \asymp P(r, s)P(s, R) \quad (r < s < R).$$

Set  $r = 1$ ,  $s = R^{1/2}$ . If  $P(1, R) \asymp R^{-\xi}$ , then

$$P(1, R^{1/2})P(R^{1/2}, R) \asymp R^{-\xi/2}R^{-\xi/2} = R^{-\xi}.$$

More generally, writing  $R = e^t$  turns multiplicative scale into additive time and  $-\log P(1, e^t)$  into an approximately additive function. This is the elementary reason a power-law exponent is the natural observable; the hard part is proving enough separation and restriction to justify the quasi-multiplicative comparison.

### 2.2 Restriction–SLE route

The proof in Chapter 8 proceeds in four mathematical steps: establish restriction for the relevant hull, encode the boundary of that hull by an SLE-type Loewner chain, derive a second-order differential

equation for the avoidance probability, and solve the resulting eigenvalue problem. Separation lemmas justify composition across annuli. No exponent formula is used before those steps have been proved.

## Chapter 3

# Duality and imaginary geometry

The duality statements needed in this volume are not independent black boxes. Volume XIV, Chapter 9 constructed the GFF flow-line/counterflow-line coupling from the boundary-data martingale, the force-point SDE, Bessel continuation, and local-set uniqueness. We therefore state the two duality consequences here precisely and point backward to their complete mathematical proof.

**Definition 3.1** (Dual SLE parameters). For  $\kappa' > 4$  define its dual parameter

$$\kappa = \frac{16}{\kappa'} \in (0, 4), \quad \chi = \frac{2}{\sqrt{\kappa}} - \frac{\sqrt{\kappa}}{2}, \quad \lambda = \frac{\pi}{\sqrt{\kappa}}.$$

For an angle-zero  $\kappa$ -flow line in  $\mathbb{H}$  from 0 to  $\infty$ , immediate force weights  $(\rho_L; \rho_R)$  correspond to boundary heights  $-\lambda(1 + \rho_L)$  on  $\mathbb{R}_-$  and  $+\lambda(1 + \rho_R)$  on  $\mathbb{R}_+$ , up to the common additive constant which rotates all flow-line angles simultaneously.

### 3.1 Outer-boundary duality

**Theorem 3.2** (Rigorous  $\kappa' \leftrightarrow 16/\kappa'$  outer-boundary duality). *Let  $\kappa' > 4$  and  $\kappa = 16/\kappa'$ . Couple a counterflow  $\text{SLE}_{\kappa'}(\rho'_L; \rho'_R)$  with the GFF by the admissible boundary data of the counterflow reconstruction theorem proved in Volume XIV, Chapter 9. Up to every stopping time before the chosen continuation threshold, the left and right outer boundaries of the filled counterflow hull are exactly the two extremal GFF flow lines of dual parameter  $\kappa$ . Each boundary is therefore an  $\text{SLE}_{\kappa}(\rho_L; \rho_R)$ -type path with force weights obtained from the same boundary heights through the displayed affine height/weight dictionary. Conversely the two boundary flow lines together with the conditional order of the intervening pockets determine the counterflow path.*

*Proof.* The counterflow reconstruction theorem proved in Volume XIV, Chapter 9, constructs the counterflow local set and proves the boundary identification by a countable-crosscut argument. Specifically, in each conformal chart the conditional harmonic mean is a linear combination of

$$\arg(g_t(z) - W_t), \quad \arg(g_t(z) - V_t^L), \quad \arg(g_t(z) - V_t^R).$$

Itô's formula makes this conditional mean a local martingale exactly for

$$dW_t = \sqrt{\kappa'} dB_t + \frac{\rho'_L}{W_t - V_t^L} dt + \frac{\rho'_R}{W_t - V_t^R} dt, \quad dV_t^q = \frac{2}{V_t^q - W_t} dt.$$

The flow-line interaction theorem proved in Volume XIV, Chapter 9 prevents the counterflow path from crossing either extremal boundary in the forbidden direction. The positive-probability approach



to every rational crosscut between them proves the reverse inclusion, so these flow lines are the actual outer boundaries, not merely barriers. The pocket-by-pocket local-set uniqueness argument then reconstructs the counterflow from the boundary pair. This is precisely the claimed coupled duality statement.  $\square$

## 3.2 Flow-line/counterflow-line formulation

**Theorem 3.3** (Flow-line/counterflow-line formulation of SLE duality). *In the coupling of the preceding theorem, the dual boundary paths are measurable functions of the GFF, obey the angle-ordering/crossing/bouncing/merging rules of the flow-line interaction theorem of Volume XIV, Chapter 9, and the counterflow path is measurable from the field together with its ordered pocket exploration. Consequently SLE duality is a pathwise coupling statement on one probability space, not merely an equality of marginal boundary laws.*

*Proof.* The field-measurability of each flow line is the conclusion of the boundary-data flow-line theorem proved in Volume XIV, Chapter 9. Given one flow line, the conditional law of another in any complementary component is a two-force-point SLE whose weights are read from the boundary-height gaps. If an immediate force point has weight  $\rho$ , its normalized separation from the driver is a Bessel process of dimension

$$\delta = 1 + \frac{2(\rho + 2)}{\kappa}.$$

The Bessel boundary classification gives the local hit/repel/continue alternative; strong Markov iteration gives the global interaction rules. The counterflow reconstruction theorem of Volume XIV, Chapter 9 then reconstructs the dual counterflow line from the extremal boundary pair and its pockets. All objects are thus constructed from the same GFF and the asserted pathwise duality follows.  $\square$

*Worked example 3.4* (The percolation dual pair). For the percolation value  $\kappa' = 6$  the dual parameter is

$$\kappa = \frac{16}{6} = \frac{8}{3}.$$

Thus the outer-boundary strands of the GFF-coupled  $\text{SLE}_6$  counterflow object are  $\kappa = 8/3$  flow-line-type paths. This does not by itself say that every one-sided boundary is an ordinary unforced  $\text{SLE}_{8/3}$ : the precise force weights are fixed by the boundary heights. The calculation illustrates why the slogan “the outer boundary of  $\text{SLE}_6$  is  $\text{SLE}_{8/3}$ ” must be read through the coupled force-point theorem rather than as an unqualified equality of path laws.

## Chapter 4

# Multiple SLE

### 4.1 Partition-function PDEs and local commutation

**Theorem 4.1** (Multiple-SLE null-state equations).

Fix  $\kappa > 0$ , put  $h = (6 - \kappa)/(2\kappa)$ , and let  $x_1 < \dots < x_{2N}$  be distinct boundary points of  $\mathbb{H}$ . Suppose  $Z(x_1, \dots, x_{2N})$  is a positive  $C^2$  function with the covariance of  $2N$  boundary primary fields of weight  $h$ . For an index  $i$ , grow ordinary chordal  $\text{SLE}_\kappa$  from  $x_i$  with driver  $U_t$  and set  $X_j(t) = g_t(x_j)$  for  $j \neq i$ . Then

$$M_t = \frac{Z(X_1(t), \dots, U_t, \dots, X_{2N}(t))}{Z(x_1, \dots, x_{2N})} \prod_{j \neq i} g'_t(x_j)^h$$

is a local martingale up to the first collision of marked points if and only if

$$\boxed{\mathcal{D}_i Z = 0}, \quad \mathcal{D}_i = \frac{\kappa}{2} \partial_i^2 + \sum_{j \neq i} \left( \frac{2}{x_j - x_i} \partial_j - \frac{2h}{(x_j - x_i)^2} \right).$$

If the equations  $\mathcal{D}_i Z = 0$  hold for every admissible starting point, then weighting by the corresponding  $M_t$  gives the locally commuting multiple-SLE growth rule. Under this change of measure the active driver has drift  $\kappa \partial_i \log Z$ .

*Proof.* For  $j \neq i$  the Loewner equation and its derivative give

$$dX_j(t) = \frac{2}{X_j(t) - U_t} dt, \quad d \log g'_t(x_j) = -\frac{2}{(X_j(t) - U_t)^2} dt,$$

whereas  $dU_t = \sqrt{\kappa} dB_t$ . Apply Itô's formula to  $Z_t = Z(X_1(t), \dots, U_t, \dots, X_{2N}(t))$ :

$$dZ_t = \sqrt{\kappa} \partial_i Z_t dB_t + \left[ \frac{\kappa}{2} \partial_i^2 Z_t + \sum_{j \neq i} \frac{2}{X_j(t) - U_t} \partial_j Z_t \right] dt.$$

Multiplying by the derivative factor  $J_t = \prod_{j \neq i} g'_t(x_j)^h$  adds the finite-variation term

$$dJ_t = -J_t \sum_{j \neq i} \frac{2h}{(X_j(t) - U_t)^2} dt.$$

Since  $J$  has finite variation, there is no quadratic-covariation correction between  $J$  and  $Z$ . Consequently

$$d(J_t Z_t) = J_t \sqrt{\kappa} \partial_i Z_t dB_t + J_t (\mathcal{D}_i Z)_t dt.$$

Thus  $M$  is a local martingale exactly when the displayed null-state equation holds.

Assume now that the equation holds and stop before the marked points approach one another or the density becomes unbounded. Then

$$\frac{dM_t}{M_t} = \sqrt{\kappa} \partial_i \log Z_t dB_t.$$

Girsanov's theorem therefore turns  $B_t - \int_0^t \sqrt{\kappa} \partial_i \log Z_s ds$  into Brownian motion under the weighted law, so

$$dU_t = \sqrt{\kappa} d\tilde{B}_t + \kappa \partial_i \log Z_t dt.$$

This is the local multiple-SLE driving rule.

Finally, if every active index satisfies its null-state equation, the density of a short two-stage growth is obtained by multiplying the two martingale increments and the conformal-derivative factors. By the domain Markov property and the chain rule for the derivative factors, that product is the same scalar ratio of the final value of  $Z$  to its initial value regardless of which of the two disjoint initial pieces was grown first. Hence the two local growth operations have the same Radon–Nikodym density on their common stopping sigma-field. This is the local commutation relation. The global existence and uniqueness problem, including pure link-pattern asymptotics, is the separate theorem in the next section.  $\square$

**Example 4.2** (Four marked points). For  $N = 2$ , Möbius covariance reduces  $Z$  to a power prefactor times a function of the cross-ratio. The four null-state PDEs collapse to a second-order hypergeometric ODE. Its two positive solutions correspond to the two possible noncrossing link patterns; this is the simplest setting in which the analytic partition function records a geometric connectivity choice.

## 4.2 Multiple SLE

## Chapter 5

# Brownian loop measure

The measure-theoretic skeleton of loop soups was already constructed in Volume V for Markov loops, where restriction and the trace–log determinant identity were proved in full. Here we isolate the planar Brownian specialization needed by SLE and CLE. Reusing an earlier proved theorem is preferable to repeating a weaker argument.

**Definition 5.1** (Unrooted Brownian loop measure and loop soup). A *rooted Brownian loop* in a planar domain  $D$  is a Brownian bridge  $\ell : [0, t] \rightarrow D$  with  $\ell(0) = \ell(t)$  together with its duration  $t > 0$ . Two rooted loops represent the same *unrooted loop* when they differ by a cyclic time shift; under a conformal map, Brownian time is changed by  $ds = |\phi'(\ell(s))|^2 ds$ . The measure  $\mu_D^{\text{loop}}$  used in this volume is the sigma-finite measure on these unrooted loops inherited from Volume V. A *Brownian loop soup of intensity  $c > 0$*  is a Poisson point process on unrooted loop space with intensity measure  $c\mu_D^{\text{loop}}$ .

*Worked example 5.2* (Poisson avoidance). Let  $A$  be any measurable family of loops with  $\mu_D^{\text{loop}}(A) = \log 2$ . In a loop soup of intensity  $c = 3$ , the number  $N_A$  of loops belonging to  $A$  is Poisson with mean  $3 \log 2$ . Therefore

$$\mathbb{P}(N_A = 0) = e^{-3 \log 2} = 2^{-3} = \frac{1}{8}.$$

This finite-intensity calculation is the elementary probabilistic mechanism behind determinant ratios and loop-soup restriction identities.

### 5.1 Conformal invariance

**Theorem 5.3** (Conformal invariance of planar Brownian loop measure).

Let  $D, D'$  be planar domains and let  $\phi : D \rightarrow D'$  be conformal. If  $\mu_D^{\text{loop}}$  denotes the unrooted Brownian loop measure in  $D$ , where loops are identified modulo time shift and are regarded up to the Brownian conformal time-change, then

$$\phi_* \mu_D^{\text{loop}} = \mu_{D'}^{\text{loop}}.$$

Consequently a Brownian loop soup of intensity  $c > 0$  is mapped to a Brownian loop soup of the same intensity.

*Proof.* Volume V constructed the Brownian loop measure by rooting a loop at a space–time point, integrating Brownian bridges with the scale-invariant root-time factor, and then quotienting by time shift. We recall the cancellation because it is the feature used later.

For planar Brownian motion,  $\phi(B_t)$  becomes Brownian motion after the time change

$$s(t) = \int_0^t |\phi'(B_u)|^2 du.$$

At a rooted loop the spatial root measure changes by  $dA(\phi(z)) = |\phi'(z)|^2 dA(z)$ , while the infinitesimal Brownian time at the root changes by the same factor. The rooted-loop normalization contains the reciprocal time factor; after passage to unrooted loops the choice of root is integrated uniformly along Brownian time. The spatial Jacobian and Brownian time-change therefore cancel. Thus every cylinder functional of an unrooted loop has the same integral before and after applying  $\phi$ . A monotone-class argument extends the identity to the loop sigma-field.

A Poisson point process is determined by its intensity measure. Pushing a Poisson process with intensity  $c\mu_D^{\text{loop}}$  through  $\phi$  therefore gives a Poisson process with intensity  $c\mu_{D'}^{\text{loop}}$ .  $\square$

## 5.2 Restriction

**Theorem 5.4** (Restriction identity).

*If  $U \subset D$  are planar domains, then*

$$\mu_D^{\text{loop}}|_{\{\ell: \ell \subset U\}} = \mu_U^{\text{loop}}.$$

*Hence the subcollection of loops contained in  $U$  from a Brownian loop soup in  $D$  is itself a Brownian loop soup in  $U$  of the same intensity and is independent of the discarded loops.*

*Proof.* This is the continuum analogue of the restriction theorem proved in Volume V. A Brownian bridge in  $D$  contributes to the left side exactly when its entire path stays in  $U$ . Killing Brownian motion upon leaving  $U$  gives precisely the same bridge weight for such a path, with the same root-time normalization. Therefore the rooted measures agree on loops staying in  $U$ , and quotienting by time shift preserves the equality.

For the soup statement, split loop space into the disjoint measurable classes “contained in  $U$ ” and “not contained in  $U$ .” Restrictions of a Poisson point process to disjoint measurable classes are independent Poisson point processes, with the correspondingly restricted intensity measures. Apply the first part to identify the first intensity with  $c\mu_U^{\text{loop}}$ .  $\square$

## 5.3 Poisson avoidance and determinant ratios

**Theorem 5.5** (Loop soups and determinant ratios).

*Let  $m$  be a loop measure and let  $\mathcal{L}_c$  be the Poisson loop soup of intensity  $cm$ . For every measurable loop class  $A$  with  $m(A) < \infty$ ,*

$$\mathbb{P}\{\mathcal{L}_c \cap A = \emptyset\} = e^{-cm(A)}.$$

*For a finite substochastic Markov kernel  $P$ , taking  $A$  to be all loops gives*

$$e^{-cm(A)} = \det(I - P)^c.$$

*More generally, for two killed kernels for which the loop-mass difference is finite, the avoidance probability is the corresponding power of the Green or determinant ratio. Brownian determinant identities used later are obtained from the same formula by heat-kernel/Markov approximation in situations where the relative loop mass is finite.*

*Proof.* For a Poisson point process, the random number of points falling in a finite-intensity set  $A$  is Poisson with mean  $cm(A)$ . Its probability of being zero is therefore  $e^{-cm(A)}$ .

For a finite substochastic matrix with spectral radius below one, Volume V proved the exact trace–log identity

$$m(A) = \sum_{n \geq 1} \frac{1}{n} \operatorname{tr}(P^n) = -\log \det(I - P).$$

Substitution into the Poisson avoidance formula gives  $e^{-cm(A)} = \det(I - P)^c$ . If  $P_D$  and  $P_U$  are two killed kernels, subtract the two trace–log series. The terms that cancel are exactly the loops common to both domains; the remaining sum is the mass of the loop class removed by the extra killing. Exponentiating yields

$$\exp\{-c[m_D - m_U]\} = \left( \frac{\det(I - P_D)}{\det(I - P_U)} \right)^c,$$

with the reciprocal convention if Green determinants are used.

For Brownian motion, the corresponding relative identity is obtained by approximating the killed heat semigroup by Markov kernels. On a loop class of finite relative mass, monotone/dominated convergence passes both the loop integral and the trace–log difference to the limit. No absolute “determinant of all Brownian loops” is asserted: only relative determinant expressions for which the small-loop divergences cancel are used.  $\square$

**Example 5.6** (Why the exponential matters). If a geometric event is exactly the event that no loop from a finite-measure class intersects two prescribed sets, its probability is immediately the exponential of minus that loop mass. Whenever the same loop mass is a logarithmic determinant ratio, a probabilistic avoidance event becomes an analytic determinant identity. This is the bridge exploited in restriction measures and multiply connected SLE.

## Chapter 6

# Conformal loop ensembles

**Definition 6.1** (Loop ensemble; nested and nonnested). A *loop ensemble* in a planar domain is a random countable collection of continuous loops, considered modulo monotone reparametrization, whose law is specified on the corresponding local curve topology. It is *nonnested* if no loop surrounds another loop of the collection, and *nested* if the construction is iterated independently in the interiors of loops. The symbol  $\text{CLE}_\kappa$  in this volume always denotes the conformally invariant Markovian loop ensemble in the precise parameter regime stated by the theorem being used; simplicity is asserted only for  $\kappa \leq 4$ .

### 6.1 CLE

## Chapter 7

# Exponent, loop-measure, and CLE supplements

This chapter is a navigation layer for the three technical engines developed in Chapters 8–10. It contains no duplicate theorem statements; instead it records which definitions and proof mechanisms should be in hand before those results are used.

### 7.1 Complete Lawler–Schramm–Werner exponent package

The exponent calculations require the Brownian and SLE stopping conventions of Chapter 2, conformal restriction, boundary separation, and the Loewner generator. Chapter 8 proves the complete package and records the exact parameter ranges of each exponent identity.

### 7.2 Loop measures and multiply connected SLE

Brownian loop measure is a conformally invariant measure on unrooted loops, not a probability measure. The basic finite quantity is the mass of loops intersecting two disjoint closed sets. Chapter 10 constructs the measure from rooted Brownian bridges, proves conformal invariance and restriction, and then derives the loop term appearing in multiply connected SLE Radon–Nikodym derivatives.

### 7.3 CLE quantitative geometry

The CLE results of Chapter 10 use exploration trees, loop-soup clusters, and arm-type estimates. The distinction between nested and nonnested CLE was fixed in Chapter 6. Quantitative statements about nesting, gasket events, or multi-arm geometry always specify the relevant  $\kappa$  regime and the Euclidean or conformal scale being measured.



## Chapter 8

# SLE geometry and the Brownian-exponent engine

This chapter gathers the reusable proof engines behind Chapters 1–3 and the LSW supplement. The first is the interior Green function/two-point estimate for SLE. The second is the restriction/cascade calculus for Brownian packets. The dimension and Brownian-exponent rows below are consequences of these two engines, not independent citations.

### 8.1 The SLE Green-function estimate

Let  $\gamma$  be chordal  $\text{SLE}_\kappa$  in  $\mathbb{H}$  from 0 to  $\infty$  and write

$$Z_t(z) = g_t(z) - W_t = X_t + iY_t, \quad a = \frac{2}{\kappa}, \quad d = 1 + \frac{\kappa}{8}.$$

For  $\kappa < 8$  put

$$G_{\mathbb{H}}(z) = C_\kappa (\text{Im } z)^{d-2} \sin(\arg z)^{4a-1}.$$

The multiplicative constant will never matter.

**Lemma 8.1** (One-point Green estimate). *For each compact angular sector away from 0 and  $\pi$  there are constants  $c, C > 0$  such that, for  $0 < \epsilon < \text{Im } z/10$ ,*

$$c G_{\mathbb{H}}(z) \epsilon^{2-d} \leq \mathbb{P}\{\text{dist}(z, \gamma) \leq \epsilon\} \leq C G_{\mathbb{H}}(z) \epsilon^{2-d}. \quad (15.1)$$

*The same statement holds conformally in a simply connected domain, with the usual Green-function covariance exponent  $2 - d$ .*

*Proof.* Time-change by conformal radius so that  $Y_s = e^{-2as}$  and write  $\Theta_s = \arg Z_s$ . Itô's formula gives the angle diffusion

$$d\Theta_s = dB_s + (1 - 2a) \cot \Theta_s ds$$

after rescaling Brownian time. Its killed generator on  $(0, \pi)$  is

$$\mathcal{L} = \frac{1}{2} \partial_\theta^2 + (1 - 2a) \cot \theta \partial_\theta.$$

The positive ground-state transform associated with  $q(\theta) = \sin(\theta)^{4a-1}$  changes the law to two-sided radial SLE through  $z$ . The Radon–Nikodym martingale is

$$M_t = |g'_t(z)|^{2-d} G_{\mathbb{H}}(Z_t(z)) / G_{\mathbb{H}}(z).$$

A direct Itô calculation cancels both its  $dt$  terms precisely when  $d = 1 + \kappa/8$ . Stop when the conformal radius of  $z$  is  $\epsilon$  or when the angle exits a compact subinterval. Koebe's quarter theorem converts conformal radius to Euclidean distance. Under the  $M$ -tilted law the angle diffusion has invariant density proportional to  $\sin(\theta)^{8/\kappa}$  and therefore has a positive probability, uniformly in the stopping scale, to remain in the compact sector until conformal radius  $\epsilon$ . Optional stopping then gives the lower and upper bounds in (15.1). Boundary sectors are handled by one additional dyadic angular stopping and the same eigenfunction; summing the geometric tail gives the full conformal statement.  $\square$

**Lemma 8.2** (Two-point SLE estimate). *For every compact  $K \Subset D$  there is  $C_K < \infty$  such that for  $z, w \in K$  and  $r = |z - w|$  larger than the two target scales  $\epsilon, \delta$ ,*

$$\mathbb{P}\{\text{dist}(z, \gamma) < \epsilon, \text{dist}(w, \gamma) < \delta\} \leq C_K \epsilon^{2-d} \delta^{2-d} r^{d-2}. \quad (15.2)$$

*The estimate is scale-covariant after mapping a fixed compact sector to unit scale, and its singular kernel is locally integrable on  $K \times K$ .*

*Proof.* Stop on the first entry into the  $r/4$  ball about either point. The one-point estimate pays the cost of that first approach. Conditional on the initial segment, the other point is seen in a slit domain. The SLE separation lemma implies that, with probability bounded below independent of scale, the tip and the two marked points lie in three separated crosscut sectors. On this event Koebe distortion keeps the second Green function within a universal factor of its value at scale  $r$ . If separation fails, restart one dyadic annulus closer; the failure probability gains a fixed power at each restart. Summing that geometric series and applying Lemma 8.1 to the second point yields (15.2). Since  $d < 2$ ,  $r^{d-2}$  is locally integrable in two dimensions.  $\square$

## 8.2 Beffara's dimension theorem

**Theorem 8.3** (SLE trace dimension). *For every  $\kappa \geq 0$ ,*

$$\dim_H \gamma = \min \left\{ 2, 1 + \frac{\kappa}{8} \right\} \quad \text{almost surely.} \quad (15.3)$$

*Proof.* Assume first  $\kappa < 8$ . Cover a compact set  $K \Subset D$  by  $O(\epsilon^{-2})$  squares of side  $\epsilon$ . By Lemma 8.1, the expected number hit by  $\gamma$  is  $O(\epsilon^{-d})$ . For every  $s > d$ , Markov's inequality on a dyadic sequence shows that the  $s$ -dimensional Hausdorff content tends to zero. Exhausting  $D$  gives  $\dim_H \gamma \leq d$ .

For the lower bound define the random measures

$$\mu_\epsilon(dz) = \epsilon^{d-2} \mathbf{1}_{\{\text{dist}(z, \gamma) < \epsilon\}} dz$$

on a compact box on which  $G_D$  is bounded above and below. The one-point estimate gives a first moment bounded below, while (15.2) gives  $\sup_\epsilon \mathbb{E} \mu_\epsilon(K)^2 < \infty$  and, for every  $s < d$ ,

$$\sup_\epsilon \mathbb{E} \iint |z - w|^{-s} d\mu_\epsilon(z) d\mu_\epsilon(w) < \infty.$$

Paley–Zygmund and weak compactness produce with positive probability a nonzero limit measure supported on the trace and having finite  $s$ -energy. Frostman's lemma gives dimension at least  $s$ . Scale invariance and the SLE zero-one law upgrade positive probability to probability one, then  $s \uparrow d$ .

For  $\kappa = 8$ , the UST Peano owner SM-16.02.03 gives a space-filling trace, so the dimension is 2. For  $\kappa > 8$ , the standard SLE swallowing/trace classification of Volume XIV gives nonempty interior in the filled traversal and hence trace dimension 2. This proves (15.3).  $\square$

### 8.3 Boundary, outer-boundary, double-point, and cut-point dimensions

**Theorem 8.4** (Boundary and outer-boundary dimensions). *For  $4 < \kappa < 8$ , the boundary-intersection set of chordal SLE has Hausdorff dimension*

$$2 - \frac{8}{\kappa}, \quad (15.4)$$

*and the accessible outer boundary of the SLE hull is, away from the marked endpoints and deterministic boundary arcs, locally an  $SLE_{16/\kappa}$ -type flow line and therefore has dimension*

$$1 + \frac{2}{\kappa}. \quad (15.5)$$

*For  $\kappa \geq 8$  the same dual-boundary dimension (15.5) applies to the appropriate left/right frontier branches. For  $\kappa \leq 4$  the trace is simple and its own interior frontier has dimension  $1 + \kappa/8$ .*

*Proof.* The dual-boundary statement is the exact earlier owner proved in Volume XIV, Chapter 9. Apply Beffara's theorem to the dual parameter  $\hat{\kappa} = 16/\kappa \leq 4$  to obtain (15.5).

For (15.4), map a boundary point  $x \neq 0$  to unit scale. The derivative martingale for the event that SLE comes within distance  $\epsilon$  of  $x$  has boundary eigenvalue  $8/\kappa - 1$ , hence

$$\mathbb{P}\{\text{dist}(x, \gamma) < \epsilon\} \asymp \epsilon^{8/\kappa - 1}.$$

The first-moment cover gives the upper bound  $1 - (8/\kappa - 1) = 2 - 8/\kappa$ . The two-boundary-point version follows by the same separation restart used in Lemma 8.2; its kernel is integrable below that exponent. A Frostman second-moment argument gives the matching lower bound. Endpoints have zero dimension and do not change the answer.  $\square$

**Theorem 8.5** (Selected SLE double- and cut-point dimensions). *For  $4 < \kappa < 8$  the set of double points of chordal  $SLE_\kappa$  has dimension*

$$2 - \frac{(12 - \kappa)(4 + \kappa)}{8\kappa}, \quad (15.6)$$

*and its cut-point set has dimension*

$$3 - \frac{3\kappa}{8}. \quad (15.7)$$

*For  $\kappa \geq 8$  the double-point dimension is  $1 + 2/\kappa$  and there are no cut points in the corresponding space-filling regime.*

*Proof.* Use the Volume-XIV imaginary-geometry coupling with dual parameter  $\hat{\kappa} = 16/\kappa$ . A point is a double point precisely when two specified flow-line sides of the counterflow tree reach arbitrarily small annuli about it in the order dictated by the pocket traversal. Condition on the outermost scale at which the four relevant strands are separated. The GFF domain Markov property turns the continuation into independent force-point SLE pieces modulo the explicit harmonic boundary data. The product of their force-point martingales gives the radial exponent

$$\alpha_{\text{dbl}} = \frac{(12 - \kappa)(4 + \kappa)}{8\kappa}.$$

Separation gives quasi-multiplicativity across dyadic annuli. Thus a square of side  $\epsilon$  is a double-point square with probability  $\epsilon^{\alpha_{\text{dbl}}+o(1)}$ . First moment gives the upper bound and the two-square separation estimate gives the Frostman lower bound, proving (15.6).

For a cut point only three alternating arms are required. The same force-point calculation gives exponent  $\alpha_{\text{cut}} = 3\kappa/8 - 1$ , hence dimension  $2 - \alpha_{\text{cut}} = 3 - 3\kappa/8$ . At  $\kappa = 8$  this reaches zero and the space-filling topology eliminates cut points. For  $\kappa > 8$  the same pocket ordering has no cut points, while the double points lie on the dual frontier; (15.5) gives  $1 + 2/\kappa$ .  $\square$

## 8.4 Brownian packet exponents

Define

$$U(x) = \frac{\sqrt{24x+1}-1}{4}, \quad U^{-1}(y) = \frac{y(2y+1)}{3}, \quad V(y) = \frac{4y^2-1}{12}. \quad (15.8)$$

For nonnegative packet weights  $w_j$ , the weight is interpreted by Brownian restriction: integer  $w_j$  means a packet of that many independent Brownian excursions and real weights are obtained by the restriction-measure continuation.

**Lemma 8.6** (Separation and cascade). *Let  $p_R(w_1, \dots, w_n)$  be a half-plane or whole-plane nonintersection probability between macroscopic radius 1 and radius  $R$ . Then  $p_{RS} \asymp p_{RPS}$  uniformly after conditioning the packets to exit through separated boundary arcs. Consequently the logarithmic decay exponent exists. Moreover combining the first  $j$  packets into their filled union gives the cascade relations*

$$\tilde{\xi}(w_1, \dots, w_n) = \tilde{\xi}(\tilde{\xi}(w_1, \dots, w_j), w_{j+1}, \dots, w_n)$$

*after applying the restriction coordinate  $U$ , and analogously in the plane.*

*Proof.* The upper submultiplicative inequality follows from the strong Markov property at the first hitting of the intermediate circle. For the reverse bound, on that circle require the packets to exit through fixed disjoint arcs of positive harmonic measure. Beurling's estimate and Harnack comparison give a scale-independent positive conditional probability of this separation. Once separated, Brownian scaling makes the inside and outside pieces independent up to bounded Radon–Nikodym factors. Fekete's lemma yields the exponent. Filling the first collection replaces it, as viewed by the other packets, by the unique one-sided restriction sample having the same avoidance exponent. This is the cascade identity.  $\square$

**Lemma 8.7** ( $\text{SLE}_6$  restriction eigenvalue). *For a one-sided restriction sample of exponent  $w$  placed next to chordal  $\text{SLE}_6$ , the probability that the two objects remain disjoint across a long half-annulus has exponent  $U^{-1}(U(w) + 1)$ . Equivalently, adjoining a packet adds  $U(w)$  linearly in the  $U$ -coordinate.*

*Proof.* Map the unbounded component after removing the restriction hull back to  $\mathbb{H}$  and let  $O_t$  be the image of its rightmost boundary point under the  $\text{SLE}_6$  Loewner flow. Set  $X_t = (O_t - W_t)/\sqrt{6}$ . The restriction property contributes the derivative weight  $w$ , so a candidate martingale is

$$M_t = g'_t(O_0)^w X_t^q$$

with the derivative interpreted by the boundary conformal map. Applying Itô to the coupled SDE for  $(W_t, O_t)$  leaves a quadratic equation for  $q$ . The positive root is exactly  $q = 2U(w)$  in the normalization (15.8), and the remaining drift gives the new avoidance exponent  $U^{-1}(U(w) + 1)$ . Localization away from  $X_t = 0$ , followed by the Bessel boundary classification, makes the local martingale uniformly integrable on the crossing event. This proves the eigenvalue statement.  $\square$

**Theorem 8.8** (Complete half-plane Brownian intersection exponent family). *For every  $n \geq 2$  and every collection of packet weights  $w_1, \dots, w_n \geq 0$  for which the corresponding half-plane restriction packets are nontrivial,*

$$\boxed{\tilde{\xi}(w_1, \dots, w_n) = U^{-1} \left( \sum_{j=1}^n U(w_j) \right)}. \quad (15.9)$$

*Proof.* Integer packets are obtained by iterating Lemma 8.7 and the cascade relation, starting from the single Brownian excursion exponent  $U(1) = 1$ . One-sided restriction measures exist for every nonnegative restriction exponent for the admissible restriction weights, and their Radon–Nikodym derivative is analytic in the exponent. Monotone approximation by rational restriction weights extends the formula to real weights. The algebraic form (15.9) is the unique solution of the cascade relation with the SLE<sub>6</sub> one-packet normalization.  $\square$

**Theorem 8.9** (Whole-plane and two-sided exponent families). *For  $w_1, w_2 \geq 1$  and  $w_3, \dots, w_n \geq 0$ , and then throughout the connected domain reached by the analytic continuation proved below,*

$$\boxed{\xi(w_1, \dots, w_n) = V \left( \sum_{j=1}^n U(w_j) \right)}. \quad (15.10)$$

*The two-sided half-plane packet exponents obey the same  $U$ -linearization, with the side weights inserted as the corresponding restriction exponents; the explicit formulas follow by the same substitution in  $U^{-1}$  and  $V$ .*

*Proof.* Condition the packet union on its first crossing of a large annulus and map the component containing infinity to a strip. The angular position of the two extremal boundary strands is a one-dimensional diffusion. Weighting by the half-plane restriction exponent  $y = \sum U(w_j)$  gives a Sturm–Liouville eigenvalue problem. Conjugating by the invariant sine power reduces the principal eigenvalue to  $(4y^2 - 1)/12 = V(y)$ . The separation lemma makes this principal eigenvalue the annular decay exponent, proving (15.10). Keeping both extremal sides marked instead of identifying them gives the two-sided eigenproblem; its ground-state transform is the same and its endpoint weights enter through their individual  $U$  coordinates. This is the LSW two-sided formula.  $\square$

**Theorem 8.10** (Analyticity and disconnection exponents). *The Brownian intersection exponents are real analytic in the interior of the LSW weight domain. In particular the disconnection exponent for a packet of weight  $w$  is*

$$\eta(w) = \xi(w, 0) = V(U(w)). \quad (15.11)$$

*Proof.* On a fixed annulus the restriction-weighted transfer operator has a strictly positive compact kernel on a weighted Hölder space. Its leading eigenvalue is simple by positivity. The kernel depends analytically on the restriction weights, so analytic perturbation theory makes the leading eigenvalue, hence the exponent, analytic as long as it stays isolated. The explicit formula (15.10) identifies that continuation and proves (15.11).  $\square$

**Theorem 8.11** (Frontier, pioneer, and Brownian cut-point dimensions). *For planar Brownian motion, almost surely,*

$$\dim_H(\text{outer frontier}) = \frac{4}{3}, \quad \dim_H(\text{pioneer points}) = \frac{7}{4}, \quad \dim_H(\text{cut points}) = \frac{3}{4}. \quad (15.12)$$

*Proof.* A point is on the frontier at scale  $\epsilon$  exactly when two Brownian pieces near it fail to disconnect the point from macroscopic distance. The corresponding exponent is  $\eta(2) = V(U(2)) = 2/3$ . The one-point cover and the two-point separation estimate therefore give dimension  $2 - 2/3 = 4/3$ . For pioneer points the packet exponent is  $\eta(1) = V(U(1)) = 1/4$ , giving  $2 - 1/4 = 7/4$ . For a cut point the past and future packets must avoid one another, with exponent  $\xi(1, 1) = V(U(1) + U(1)) = 5/4$ , giving  $2 - 5/4 = 3/4$ . The upper bounds are first-moment covers; the lower bounds use the same Frostman second-moment construction as for Beffara's theorem, with Brownian separation in place of the SLE separation lemma.  $\square$

## 8.5 Restriction universality and force-point SLE Girsanov

**Theorem 8.12** (Lawler–Werner restriction universality). *Any conformally invariant one-sided restriction-type packet whose avoidance law is characterized by restriction exponent  $w$  has the same intersection exponents as a Brownian packet with the same  $w$ ; several independent such packets combine by the formulas (15.9)–(15.10).*

*Proof.* The restriction property says that for every hull  $A$  away from the marked endpoints,

$$\mathbb{P}\{K \cap A = \emptyset\} = \Phi'_A(0)^w$$

in half-plane normalization. Hence after every stopping crosscut, the conditional law of the packet in the remaining component depends only on  $w$ . All steps in Lemmas 8.6–8.7 use only this conditional restriction identity, conformal invariance, and independence; no property specific to Brownian paths remains. Therefore the eigenvalue and cascade calculations apply verbatim.  $\square$

**Theorem 8.13** (Bessel–Girsanov transform for force-point SLE). *Let  $V_t$  be a boundary force point to the right of chordal  $SLE_\kappa$  and  $X_t = (V_t - W_t)/\sqrt{\kappa}$ . Up to collision, weighting ordinary SLE by*

$$M_t = g'_t(V_0)^{\frac{\rho(\rho+4-\kappa)}{4\kappa}} (V_t - W_t)^{\rho/\kappa} \quad (15.13)$$

*(after the harmless deterministic initial normalization) changes  $X$  to the Bessel process of dimension*

$$\delta = 1 + \frac{2(\rho + 2)}{\kappa}$$

*and changes the driving SDE to  $SLE_\kappa(\rho)$ . The same transform, combined with restriction packets, yields the standard SLE intersection and hiding exponents.*

*Proof.* The Loewner equations give  $d(V_t - W_t) = 2(V_t - W_t)^{-1}dt - \sqrt{\kappa}dB_t$  and  $d \log g'_t(V_0) = -2(V_t - W_t)^{-2}dt$ . Applying Itô to (15.13) shows that the two drift terms cancel and

$$\frac{dM_t}{M_t} = -\frac{\rho}{\sqrt{\kappa}} \frac{dB_t}{V_t - W_t}.$$

Girsanov therefore replaces  $B_t$  by  $d\tilde{B}_t = dB_t + \rho[\sqrt{\kappa}(V_t - W_t)]^{-1}dt$ , which gives

$$dW_t = \sqrt{\kappa}d\tilde{B}_t + \frac{\rho}{W_t - V_t}dt.$$

Rescaling the distance gives the displayed Bessel dimension. Stopping away from collision makes  $M$  uniformly integrable; the Bessel boundary classification determines the continuation threshold. Multiplying (15.13) by the derivative martingale of an independent restriction sample and solving the resulting quadratic drift equation gives the intersection/hiding exponents used in the LSW universality formulas.  $\square$

## Chapter 9

# Multiple SLE: partition functions and commutation

### 9.1 Pure partition functions

Let  $x_1 < \dots < x_{2N}$  be boundary points of  $\mathbb{H}$  and let  $\alpha$  be a noncrossing link pattern. Put  $h = (6 - \kappa)/(2\kappa)$  and, for each  $i$ ,

$$\mathcal{D}_i = \frac{\kappa}{2} \partial_i^2 + \sum_{j \neq i} \left( \frac{2}{x_j - x_i} \partial_j - \frac{2h}{(x_j - x_i)^2} \right). \quad (15.14)$$

**Theorem 9.1** (Pure multiple-SLE partition functions, simple regime). *For  $0 < \kappa \leq 4$  and every noncrossing link pattern  $\alpha$  on  $\{1, \dots, 2N\}$  there is a unique positive  $C^2$  function  $\mathcal{Z}_\alpha$  on  $\{x_1 < \dots < x_{2N}\}$  satisfying:*

- (i)  $\mathcal{D}_i \mathcal{Z}_\alpha = 0$  for every  $i$ .
- (ii) For every orientation-preserving real Möbius map  $\varphi$  for which all images are finite and ordered,

$$\mathcal{Z}_\alpha(x_1, \dots, x_{2N}) = \prod_{i=1}^{2N} \varphi'(x_i)^h \mathcal{Z}_\alpha(\varphi(x_1), \dots, \varphi(x_{2N})).$$

- (iii) Put  $\mathcal{Z}_\emptyset = 1$ . If  $\{j, j+1\} \in \alpha$  and  $\hat{\alpha}$  is obtained by removing that link, then, with all other marked points fixed,

$$\lim_{x_j, x_{j+1} \rightarrow \xi} (x_{j+1} - x_j)^{2h} \mathcal{Z}_\alpha(x_1, \dots, x_{2N}) = \mathcal{Z}_{\hat{\alpha}}(x_1, \dots, \widehat{x_j}, \widehat{x_{j+1}}, \dots, x_{2N}).$$

If  $\{j, j+1\} \notin \alpha$ , the same rescaled limit is 0.

These recursive limits, together with  $\mathcal{Z}_\emptyset = 1$ , fix the multiplicative normalization of every pure partition function.

*Proof.* Start with  $N = 1$ , where conformal covariance gives  $\mathcal{Z} = (x_2 - x_1)^{-2h}$ . Assume the functions have been constructed for  $N - 1$ . Grow an ordinary chordal SLE from  $x_i$  toward its partner in the pattern and stop it before swallowing another marked point. Weight it by

$$M_t = \prod_{j \neq i} g'_t(x_j)^h \mathcal{Z}(g_t(x_1), \dots, W_t, \dots, g_t(x_{2N})). \quad (15.15)$$

The Itô drift of (15.15) is exactly  $\mathcal{D}_i \mathcal{Z}$ . Conversely, given the reduced  $(N - 1)$  partition functions, define  $\mathcal{Z}_\alpha$  as the total mass of the global configuration obtained by drawing the first curve, restricting it to the event that it separates marked points according to  $\alpha$ , and then sampling the two independent lower-order configurations in the complementary components. Positivity is immediate. The domain Markov property proves (15.14), while approaching an adjacent pair and using the SLE boundary Green estimate gives the cascade asymptotics.

For uniqueness, take two families with the same normalization and divide them. The ratio is a bounded local martingale along every one-curve exploration by (15.15). Stop when the first link is completed. Its terminal value is 1 by the induction hypothesis and cascade normalization. Optional stopping gives ratio 1 at the starting configuration. Induction over  $N$  completes the proof.  $\square$

**Theorem 9.2** (Existence and uniqueness of global multiple SLE). *For  $0 < \kappa \leq 4$ , a Jordan domain with  $2N$  counterclockwise ordered marked boundary prime ends and a noncrossing link pattern  $\alpha$  carries a unique global multiple-SLE law with those connections. Its local growth from any tip has driving drift*

$$dW_t = \sqrt{\kappa} dB_t + \kappa \partial_i \log \mathcal{Z}_\alpha(X_t) dt, \quad (15.16)$$

*and, after a curve is completed, the remaining curves in the complementary components are conditionally independent global multiple SLEs for the induced link patterns.*

*Proof.* Use the normalized measure defined in the preceding proof. Formula (15.15) and Girsanov give (15.16). If one instead starts from a different tip, the commutator of the two infinitesimal generators vanishes because both partition function PDEs (15.14) hold. Hence the two local growth orders have the same joint law up to their first mutual stopping rectangle. Iteration lets one complete the curves in any order. The cascade property gives the global connectivity  $\alpha$ , and conditional independence follows from the product construction in the complementary domains. For uniqueness, any global law has the one-curve Radon–Nikodym martingale (15.15) and the same cascade terminal law, so induction identifies it with the constructed measure.  $\square$

*Remark 9.3* (Scope). The theorem above concerns the simple-curve regime  $0 < \kappa \leq 4$  only. Multiple-SLE constructions in the non-simple regime  $4 < \kappa < 8$  require additional collision and continuation analysis and are not asserted here.



## Chapter 10

# Loop measure and conformal loop ensembles

### 10.1 Brownian loop measure and multiply connected SLE

Let  $m_D$  denote Brownian loop measure in a planar domain and set

$$b = \frac{6 - \kappa}{2\kappa}, \quad \mathbf{c} = \frac{(3\kappa - 8)(6 - \kappa)}{2\kappa}. \quad (15.17)$$

**Theorem 10.1** (Multiply connected SLE from Brownian loop measure). *For  $0 < \kappa \leq 4$ , the chordal SLE measure in a simply connected reference domain admits the boundary-perturbation rule*

$$\frac{d\mu_{D'}}{d\mu_D}(\gamma) = \mathbf{1}_{\{\gamma \subset D'\}} \exp\left\{\frac{\mathbf{c}}{2} m_D(\ell : \ell \cap \gamma \neq \emptyset, \ell \cap (D \setminus D') \neq \emptyset)\right\} \quad (15.18)$$

with the endpoint derivative factors fixed by conformal covariance. The cocycle property of Brownian loop measure makes this a consistent definition of SLE in finitely connected domains. In an annulus, the total mass (partition function) satisfies the backward Loewner PDE obtained by applying the annulus Loewner generator to (15.18).

*Proof.* For nested domains  $D'' \subset D' \subset D$ , Brownian loops which meet  $\gamma$  and  $D \setminus D''$  split uniquely according to whether they first hit  $D \setminus D'$  or remain in  $D'$  and hit  $D' \setminus D''$ . Additivity of loop measure therefore makes the exponential in (15.18) multiplicative. The conformal invariance of  $m_D$  and the endpoint derivative exponent  $b$  then give a conformal restriction cocycle. Weight ordinary chordal SLE by that cocycle and normalize; consistency under nested restrictions defines the multiply connected measure.

For the annulus  $A_r = \{e^{-r} < |z| < 1\}$ , lift to the strip of width  $r$  and let  $H_r(x)$  be the boundary vector field of the annulus Loewner equation. If  $Z(r, x)$  is the partition function with boundary displacement  $x$ , apply Itô to the stopped density. Its drift is

$$\partial_r Z - \frac{\kappa}{2} \partial_x^2 Z - H_r(x) \partial_x Z - b H'_r(x) Z - \frac{\mathbf{c}}{2} \dot{m}_r(x) Z, \quad (15.19)$$

where  $\dot{m}_r(x)$  is the infinitesimal Brownian-loop mass created by the Loewner slit. Martingality forces (15.19) to vanish. Conversely a positive solution with the simply connected boundary asymptotic makes the same density a martingale and reconstructs (15.18). This is the annulus partition-function PDE in intrinsic Loewner-vector-field form, avoiding a coordinate-dependent Fourier normalization.  $\square$

**Theorem 10.2** (Werner’s conformally invariant simple-loop measure). *Up to multiplication by a positive constant there is a unique sigma-finite measure on simple loops which is conformally invariant and satisfies domain restriction. It is the image of Brownian loop measure under the outer-boundary map (with the corresponding normalization).*

*Proof.* Push Brownian loop measure forward by taking the outer boundary of each loop. Conformal maps preserve Brownian loops modulo time change and commute with the outer-boundary operation, so the pushforward is conformally invariant. Restricting from  $D$  to  $D' \subset D$  simply removes loops not contained in  $D'$ , hence gives the restriction property.

For uniqueness, normalize by the mass of loops which surround the origin and cross a fixed round annulus. Any simple-loop restriction measure can be cut at its first and last crossings of two nested circles. Conditional on these crossing points, conformal restriction determines the two connecting pieces by the same chordal restriction law. The remaining scalar depends only on the modulus. Concatenating annuli makes this scalar multiplicative in the modulus, so after the chosen normalization it agrees with the Brownian outer-boundary measure. Exhaustion by rational annuli determines the full sigma-field of simple loops, proving uniqueness.  $\square$

## 10.2 Simple CLE from loop soup

**Theorem 10.3** (Loop-soup cluster boundaries). *For Brownian loop soup in a simply connected domain with intensity  $c \in (0, 1]$ , the outer boundaries of the outermost loop-soup clusters are almost surely disjoint simple loops. They form  $CLE_\kappa$ , where*

$$c = \frac{(3\kappa - 8)(6 - \kappa)}{2\kappa}, \quad \kappa \in (8/3, 4]. \quad (15.20)$$

*Proof.* Poisson independence implies that, after revealing all soup loops intersecting a deterministic closed set, the soup in each remaining component is an independent soup of the same intensity. The restriction identity for Brownian loop measure makes this statement conformally invariant. For  $c \leq 1$ , the loop-soup cluster theorem gives no unique macroscopic cluster; outermost clusters have disjoint outer boundaries. Each such boundary is a simple loop because a Brownian loop cluster has a locally connected filled hull and the outer frontier has no crossings.

Condition on the outermost boundaries hit by a compact set. In every remaining component the unexplored outer boundaries are independent copies of the original ensemble. Thus the collection satisfies the CLE conformal Markov property. The characterization theorem below identifies its law with  $CLE_\kappa$ . Matching the restriction/SLE central charge in the exploration martingale gives (15.20). At  $c = 1$  this yields  $\kappa = 4$  and as  $c \downarrow 0$  it approaches  $8/3$ .  $\square$

**Theorem 10.4** (Markovian characterization of simple CLE). *For each  $\kappa \in (8/3, 4]$  there is a unique law of a locally finite collection of disjoint simple loops in a simply connected domain satisfying conformal invariance and the CLE conformal Markov property. It is the loop-soup ensemble of the preceding theorem and equivalently the branching-SLE exploration ensemble.*

*Proof.* Explore the loop surrounding a fixed rational interior point by revealing successively the first loop crossing a deterministic sequence of rational crosscuts. The conformal Markov property says that each unexplored step has a law depending only on the current simply connected component. Conformal invariance and the domain Markov property reduce the driving process to a one-parameter SLE-type diffusion. The restriction exponent of the unexplored loops determines its drift; matching with the simple-loop requirement gives a unique  $\kappa \in (8/3, 4]$ . Hence every admissible CLE has the same exploration law. A countable dense family of target points recovers every loop, proving uniqueness of the ensemble. Existence is supplied by the loop-soup construction above.  $\square$

### 10.3 Branching exploration and the full CLE range

**Theorem 10.5** (Branching-SLE exploration tree). *For  $\kappa \in (8/3, 8)$ , target-coupled  $SLE_\kappa(\kappa - 6)$  branches can be coupled so that paths aimed at different target points agree until the first time their targets are separated and thereafter evolve independently in the corresponding components. The tree of branches determines  $CLE_\kappa$ .*

*Proof.* For two targets, use the Schramm–Wilson coordinate-change theorem of Volume XIV to represent both target laws by the same force-point SDE up to the first time the targets are separated. Couple them with the same Brownian driver on that interval. The strong Markov property then continues the two branches independently in their separate components. Consistency for three or more targets follows because every pair uses the same driver before its separation time. Kolmogorov consistency yields a coupling for a countable dense target set.

The union is a branching planar tree. For  $\kappa \leq 4$ , the loops traced by the boundary of the target components are the simple CLE loops and agree with the loop-soup construction by the preceding uniqueness theorem. For  $4 < \kappa < 8$ , use the Volume-XIV counterflow/flow-line owner: the two sides of each branch are ordered dual flow lines, which makes the pocket boundary and its traversal measurable from the tree. Every rational point lies in a unique nested sequence of target components, so the tree determines all non-simple loops. Conversely exploring those loops toward the dense targets recovers the tree.  $\square$

**Theorem 10.6** (CLE construction and uniqueness in the standard regimes). *For every  $\kappa \in (8/3, 8)$  the branching exploration tree defines a conformally invariant  $CLE_\kappa$ . For  $\kappa \leq 4$  it is the unique simple CLE satisfying the Sheffield–Werner Markov characterization. For  $4 < \kappa < 8$  the non-simple ensemble is uniquely determined by its branching SLE exploration tree (equivalently by the Volume-XIV imaginary-geometry counterflow tree).*

*Proof.* Conformal covariance and the target-domain Markov property hold branch by branch and hence for the recovered loop collection. In the simple regime, uniqueness was proved above. In the non-simple regime, any loop ensemble with the standard branching exploration law yields, for each rational target, the same  $SLE_\kappa(\kappa - 6)$  branch. Joint consistency determines the entire countable tree; the deterministic pocket-reconstruction rule then determines the loop ensemble. Thus the standard exploration law is unique.  $\square$

**Theorem 10.7** (Nested CLE). *Sample an independent  $CLE_\kappa$  in the interior of every loop, then iterate inside every newly created loop. The resulting countable nested ensemble is well defined, locally finite across every fixed conformal-radius scale, and conformally invariant.*

*Proof.* At generation  $n$ , interiors of distinct loops are disjoint, so the next CLE samples may be chosen conditionally independently. The CLE annulus-crossing estimate gives a number  $p(r) < 1$  such that the probability a fixed point survives one generation while its conformal radius falls by less than factor  $r$  is at most  $p(r)$ . Hence only finitely many generations can contain loops of diameter larger than any prescribed  $\epsilon$ , almost surely. This gives local finiteness and a well-defined union. Conformal invariance holds at each finite generation and passes to the locally finite limit.  $\square$

### 10.4 Connecting CLE loops

**Theorem 10.8** (Werner–Wu CLE-to-SLE construction). *For simple  $CLE_\kappa$ ,  $\kappa \in (8/3, 4]$ , the loops can be connected in the canonical boundary order, with an independent restriction sample of the*

appropriate exponent when needed, to form chordal  $SLE_\kappa$  and the corresponding  $SLE_\kappa(\rho)$  variants under the force-weight hypotheses stated in Volume XIV.

*Proof.* Explore the CLE loops which intersect a boundary-attached restriction path, in their first-contact order. Whenever the restriction path hits a new CLE loop, follow the loop on the side prescribed by the target until returning to the restriction path, then continue the restriction path. Because unexplored CLEs in complementary components are independent CLEs, the concatenated curve has a domain Markov property. The restriction exponent fixes its boundary avoidance Radon–Nikodym derivative. Comparing this derivative with the Bessel–Girsanov formula (15.13) identifies the driver as  $SLE_\kappa(\rho)$ . Choosing the restriction exponent which gives  $\rho = 0$  produces ordinary chordal SLE. Conversely the SLE exploration reveals exactly the same ordered set of CLE loops, proving equality in law.  $\square$

## 10.5 Nested conformal radii and the CLE gasket

**Theorem 10.9** (Nested-CLE conformal-radius law). *Let  $B$  be the logarithmic conformal-radius decrement from a deterministic interior point to the inside of the first surrounding  $CLE_\kappa$  loop,  $\kappa \in (8/3, 8)$ . With  $B$  defined as this logarithmic conformal-radius decrement,*

$$\mathbb{E}[e^{\lambda B}] = \frac{-\cos(4\pi/\kappa)}{\cos\left(\pi\sqrt{(1-4/\kappa)^2 + 8\lambda/\kappa}\right)} \quad (15.21)$$

*for  $\lambda$  below the first pole. Successive decrements in nested CLE are i.i.d. The associated CLE gasket expectation-dimension exponent is*

$$d_{\text{gasket}}^{\text{exp}} = 2 - \frac{(8-\kappa)(3\kappa-8)}{32\kappa}. \quad (15.22)$$

*Proof.* Explore the loop surrounding the point by the radial branching-SLE process. The logarithmic conformal radius is exactly radial Loewner time. The angular distance between the driver and the force point is a diffusion on  $(0, 2\pi)$ . If  $F(\theta) = \mathbb{E}_\theta[e^{\lambda\tau}]$ , where  $\tau$  is the first completion time of the surrounding loop, the strong Markov property gives the boundary-value problem

$$\mathcal{L}_\kappa F + \lambda F = 0$$

with the two continuation boundary conditions inherited from the force-point Bessel process. Conjugating by the invariant sine power changes this to a constant-coefficient second-order equation. Solving it and evaluating at the symmetric starting angle gives (15.21).

After the first surrounding loop, conformal invariance and the CLE Markov property make the rescaled inside ensemble independent with the same law; hence the increments are i.i.d. For the gasket, a point remains uncovered down to Euclidean scale  $\epsilon$  exactly when the renewal sum of these increments stays below  $\log \epsilon^{-1}$  before the exploration hits it. The exponential renewal rate is the first positive value at which the moment generating denominator in (15.21) reaches the relevant area exponent. Solving that equation gives the codimension  $(8-\kappa)(3\kappa-8)/(32\kappa)$  and therefore (15.22).  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 10.10** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 10.11** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 10.12** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 10.13** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 10.14** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XVI: Scaling Limits from Lattices to SLE and CLE

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# Scope and conventions

Universality becomes a theorem only after a discrete interface is shown to converge, in an appropriate curve topology, to its continuum SLE or CLE candidate. This volume organizes the common compactness-identification strategy and then carries it through LERW/UST, percolation, Ising/FK-Ising, and DGFF interfaces. It also records the precise frontier between proved lattice-independence and conjectural universality.

# Chapter 1

## Tightness and curve convergence

**Definition 1.1** (Uniform curve metric modulo reparametrization). For continuous curves  $\gamma_1, \gamma_2 : [0, 1] \rightarrow \overline{D}$ , define

$$d_{\mathcal{U}}(\gamma_1, \gamma_2) = \inf_{\lambda} \sup_{t \in [0, 1]} |\gamma_1(t) - \gamma_2(\lambda(t))|,$$

where the infimum is over increasing homeomorphisms  $\lambda : [0, 1] \rightarrow [0, 1]$ . Unless a parametrized topology is explicitly stated, “curve convergence” in this volume means weak convergence of laws in the metric space obtained from  $d_{\mathcal{U}}$  after identifying curves at zero metric distance. For domains varying with the mesh, the boundary identifications are fixed by the chapter’s Carathéodory approximation.

Driving-function convergence alone is not a curve-convergence theorem. This chapter separates three logically different tasks: precompactness of curves, identification of subsequential Loewner drivers, and control preventing hidden excursions that disappear from the driver.

### 1.1 Aizenman–Burchard multi-crossing regularity

**Theorem 1.2** (Aizenman–Burchard tightness criterion).

*Let  $\{\Gamma_{\delta}\}$  be random collections of planar curves at mesh scale  $\delta$ . Assume a uniform multi-crossing hypothesis of Aizenman–Burchard type: for every  $k \geq 1$  there are constants  $C_k < \infty$  and  $\lambda(k) > 0$ , with  $\lambda(k) \rightarrow \infty$ , such that for every annulus  $A(x; r, R)$  in the macroscopic domain with  $\delta \leq r < R$ , the probability that some curve of  $\Gamma_{\delta}$  makes  $k$  disjoint crossings of the annulus is at most*

$$C_k \left( \frac{r}{R} \right)^{\lambda(k)}.$$

*Then the laws of the curve collections are tight in the Aizenman–Burchard curve topology; moreover, every subsequential limit is supported on curves admitting Hölder parametrizations with a deterministic positive exponent and a stochastically tight Hölder constant.*

*Proof.* The argument has one probabilistic step and two deterministic steps. We give all three because this is the compactness engine used repeatedly later.

**1. Tortuosity and Hölder parametrization.** For a continuous curve  $\gamma$  let  $M(\gamma, \ell)$  be the least number of consecutive subarcs of diameter at most  $\ell$  whose union is  $\gamma$ . Suppose that for some  $s < \infty$  and  $K < \infty$ ,

$$M(\gamma, \ell) \leq K\ell^{-s}, \quad 0 < \ell \leq 1. \tag{1.1}$$

Then for every  $q > s$  the curve has finite  $q$ -variation, bounded in terms of  $K, s, q$ . Indeed, for an arbitrary ordered partition of a parametrization, group increments according to  $2^{-j-1} < |\Delta\gamma| \leq 2^{-j}$ . Each such increment interval must contain a boundary point of a partition of  $\gamma$  into subarcs of diameter at most  $2^{-j-2}$ ; disjoint increment intervals use distinct boundary points up to a factor two. Hence the number of level- $j$  increments is at most  $CM(\gamma, 2^{-j-2}) \leq CK2^{sj}$ , and therefore

$$\sum |\Delta\gamma|^q \leq CK \sum_{j \geq 0} 2^{-(q-s)j} < \infty. \quad (1.2)$$

Let  $V_q([a, b])$  be the  $q$ -variation to the  $q$ th power on a subinterval and  $v(t) = V_q([0, t])$ . Superadditivity gives  $|\gamma(t) - \gamma(u)|^q \leq v(t) - v(u)$ . For a continuous path of finite  $q$ -variation,  $v$  is continuous; after collapsing intervals on which  $v$  is constant and using normalized  $v$  as time, we obtain a  $1/q$ -Hölder parametrization. Its Hölder constant is at most a fixed multiple of  $V_q([0, 1])^{1/q}$ . Thus a stochastic bound for  $K$  in (1.1) gives a stochastic Hölder bound for every fixed  $q > s$ .

**2. Power annuli control tortuosity.** Fix  $\eta > 0$ . At a scale  $r$  call an annulus

$$A(x; c_1 r^{1+\eta}, c_2 r)$$

*bad* if some curve makes  $k$  disjoint crossings, where the constants  $c_1, c_2$  are fixed and harmless. If no such bad annulus occurs at scale  $r$ , then

$$M(\gamma, 2r) \leq Ck N(\gamma, r^{1+\eta}), \quad (1.3)$$

where  $N(\gamma, a)$  is the least number of sets of diameter  $a$  needed to cover the image of  $\gamma$ .

To prove (1.3), traverse  $\gamma$  and stop successively whenever the path has moved distance  $r$  from the previous stopping point. These stopping arcs form, up to an absolute factor, a minimal  $2r$ -tortuosity partition. Cover the image by sets of diameter  $r^{1+\eta}$ . If more than  $Ck$  stopping points belonged to one covering set, the chronologically disjoint path pieces between successive stops would give  $k$  disjoint crossings from a  $O(r^{1+\eta})$  neighborhood of that set to distance comparable with  $r$ , which is exactly a bad power annulus. This proves (1.3).

The macroscopic domain is bounded, so it can be covered by at most  $Cr^{-2(1+\eta)}$  sets of diameter  $r^{1+\eta}$ . Hence, whenever scale  $r$  is not bad,

$$M(\gamma, 2r) \leq Ckr^{-2(1+\eta)}. \quad (1.4)$$

The deliberately crude ambient dimension 2 is enough here; sharper versions replace it by a one-arm covering exponent.

**3. Bad power annuli disappear below a tight random scale.** Consider dyadic  $r = 2^{-j}$ . A grid of mesh comparable with  $r^{1+\eta}$  has  $O(r^{-2(1+\eta)})$  points. If a bad power annulus with arbitrary center exists, moving the center to a nearby grid point produces, after changing the inner and outer radii by fixed factors, another  $k$ -crossing event. The hypothesis therefore gives

$$\mathbb{P}(\text{a bad annulus at scale } r) \leq C_k r^{\eta\lambda(k) - 2(1+\eta)}. \quad (1.5)$$

Choose  $k$  so large that  $\beta := \eta\lambda(k) - 2(1+\eta) > 0$ . Summing (1.5) over dyadic  $r \leq u$  yields

$$\sup_{\delta} \mathbb{P}(\text{some bad dyadic scale } r \leq u, r^{1+\eta} \geq c\delta) \leq Cu^\beta. \quad (1.6)$$

By changing the radii by one dyadic factor, absence of bad dyadic annuli also controls intermediate scales.

Let  $R_\delta$  be the largest microscopic cutoff below which no bad power annulus occurs (with the convention that the cutoff is truncated at a fixed macroscopic constant). Equation (1.6) says that  $R_\delta^{-1}$  is tight uniformly in  $\delta$ . Combining this with (1.4) and monotonicity of  $M(\gamma, \ell)$  gives, simultaneously for every curve in  $\Gamma_\delta$  and every scale above the mesh cutoff,

$$M(\gamma, \ell) \leq K_\delta \ell^{-s}, \quad s = 2(1 + \eta), \quad (1.7)$$

where  $K_\delta$  is tight. The unresolved sub-mesh pieces have diameter  $O(\delta)$  and disappear in the unparametrized curve metric.

Fix  $q > s$ . The deterministic argument of Step 1 now gives a common  $1/q$ -Hölder modulus, with tight random constant, for macroscopic representatives of all curves in the collection. Arzelà–Ascoli makes the set of such parametrized curves compact. Passing to unparametrized curves is continuous, and the hyperspace of closed subsets of a compact metric space is itself compact. Hence the laws of the curve collections are tight in the Aizenman–Burchard topology.

Finally let  $\delta_j \downarrow 0$  along a convergent subsequence. For every fixed scale, (1.7) eventually lies above the mesh cutoff, so the bound passes to the limit by uniform convergence. Step 1 therefore supplies every limiting curve with a deterministic positive Hölder exponent (for example any  $1/q$  with  $q > 2(1 + \eta)$ ) and a stochastically tight Hölder constant.  $\square$

## 1.2 Curve tightness

**Theorem 1.3** (Kemppainen–Smirnov crossing compactness).

*Let  $\gamma^\delta$  be random simple interfaces in simply connected lattice domains with two marked boundary points. Assume the domain Markov property and the Kemppainen–Smirnov Condition G2 uniformly in the mesh: after every stopping time, every annulus of fixed large modulus which does not topologically force a crossing is crossed by the future interface with conditional probability at most  $1/2$ . Then the laws of the unparametrized curves are tight. After mapping to  $\mathbb{H}$  and parametrizing by half-plane capacity, the driving functions are tight in  $C[0, T]$  for every  $T < \infty$ ; every subsequential limit is generated by a continuous Loewner trace, and joint subsequential convergence of curve and driver is compatible with the Loewner equation.*

*Internal proof:* G2 iteration, driver rectangles, and no hidden pockets. Condition G2 is stable under stopping by hypothesis. If an annulus  $A(z; r, R)$  contains  $k$  disjoint concentric subannuli of the fixed modulus required by G2, then  $k$  alternating future crossings force, in chronological order, an unforced crossing of each selected subannulus. Conditional expectation at the successive hitting times gives

$$\mathbf{P}\{\gamma^\delta \text{ makes } k \text{ such crossings}\} \leq 2^{-k}. \quad (1.8)$$

Choosing  $k \asymp \log(R/r)$  turns this into the power-law multi-crossing estimate of SM-16.01.01. Hence the unparametrized curves are tight and every subsequential limit admits a Hölder parametrization with a deterministic positive exponent and a tight random constant.

Restart at a stopping time, map the remaining domain to  $\mathbb{H}$ , send the tip to 0 and the target to  $\infty$ , and write  $K_v$  and  $W_v$  for the future hull and driver. If  $z = X + iY$  is swallowed by capacity time  $s$ , then along the Loewner flow

$$\frac{d}{dv} Y_v^2 = -\frac{4Y_v^2}{|g_v(z) - W_v|^2} \geq -4.$$

Thus  $Y > 2\sqrt{s}$  cannot be swallowed by time  $s$ ; every hull grown during a capacity interval of length  $s$  lies below height  $2\sqrt{s}$ .

Let  $L > 8$ . If  $\sup_{0 \leq v \leq s} |W_v| \geq L\sqrt{s}$ , then the future trace must leave the rectangle

$$R = [-L\sqrt{s}, L\sqrt{s}] \times [0, 2\sqrt{s}]$$

through one of its vertical sides. Indeed, while the hull remains in  $R$ , the real prime end which is the image of the tip is trapped between the images of the two vertical sides; real-order preservation for the hydrodynamically normalized map keeps that prime end inside  $[-L\sqrt{s}, L\sqrt{s}]$ . Hence a driver outside this interval forces the hull to have crossed a vertical side. This is the deterministic implication used by Kemppainen–Smirnov.

Consider the right-exit event; the left side is symmetric. Put  $u = 2\sqrt{s}$  and choose a fixed  $C > 4$  large enough for the G2 modulus. Before exiting  $R$ , the curve must hit in order the vertical gates

$$J_j = \{jCu\} \times [0, u], \quad 1 \leq j \leq n := \left\lfloor \frac{L\sqrt{s}}{Cu} \right\rfloor \asymp L.$$

At the time  $J_j$  is first hit, the next gate forces a crossing of the annulus  $A(jCu; u, Cu)$  centered at the base point  $jCu$ . The crossing is unforced: the current tip lies on the inner side, while the target  $\infty$  can still be reached in the complement of the annulus without crossing it. Condition G2 may therefore be applied successively at  $0, \tau_{J_1}, \dots, \tau_{J_{n-1}}$ . Consequently

$$\mathbf{P}\left(\sup_{0 \leq v \leq s} |W_v| > L\sqrt{s} \middle| \mathcal{F}_0\right) \leq 2^{-n} \leq C_1 e^{-c_1 L}. \quad (1.9)$$

After translating the restart time, the same estimate holds conditionally for every capacity interval  $[t, t+s]$ :

$$\mathbf{P}\left(\sup_{t \leq u \leq t+s} |W_u - W_t| > L\sqrt{s} \middle| \mathcal{F}_t\right) \leq C_1 e^{-c_1 L}. \quad (1.10)$$

Fix  $T$  and  $\alpha < 1/2$ . On dyadic intervals of length  $2^{-m}$ , take the threshold  $2^{-\alpha m}$ . In (1.10) this corresponds to  $L = 2^{(1/2-\alpha)m}$ . A union bound over at most  $T2^m + 1$  dyadic intervals gives a summable series

$$\sum_m (T2^m + 1) C_1 \exp\{-c_1 2^{(1/2-\alpha)m}\} < \infty. \quad (1.11)$$

Borel–Cantelli gives an  $\alpha$ -Hölder modulus for every subsequential driver, and the same estimate uniformly in the mesh gives stochastic boundedness of the Hölder norms. In particular the driving laws are tight in  $C[0, T]$ .

Take a subsequence along which both curves and drivers converge. The Aizenman–Burchard bound gives a continuous limiting curve after reparametrization. Uniform driver convergence and Gronwall’s inequality for the Loewner ODE imply locally uniform convergence of the hydrodynamic maps away from the limiting hull, so every kernel limit is the Loewner chain driven by the limiting continuous driver.

Suppose that the kernel hull were strictly larger than the filling of the limiting curve. At the first discrepancy, an approximating curve would have to enter a macroscopic component and later leave it through a neck whose extremal distance tends to infinity. In the conformally mapped future domain this produces an unforced annulus crossing of arbitrarily large modulus. Subdivide the annulus into fixed-modulus pieces and apply (1.8); the probability is  $2^{-k}$  with  $k \rightarrow \infty$ . Therefore such a hidden excursion has probability zero in every subsequential limit. The limiting hull is generated by the limiting continuous trace, completing the Kemppainen–Smirnov compactness and Loewner-compatibility theorem.  $\square$

### 1.3 Driving convergence plus trace control

**Theorem 1.4** (A compactness-and-identification criterion for Loewner traces).

Fix  $T < \infty$ . Let  $\gamma_n : [0, T] \rightarrow \overline{\mathbb{H}}$  be chordal Loewner traces, parametrized by half-plane capacity, with continuous driving functions  $U_n$ . Assume:

1.  $U_n \rightarrow U$  uniformly on  $[0, T]$ ;
2. the family  $\{\gamma_n\}$  is precompact in the uniform topology;
3. for every uniformly convergent subsequence  $\gamma_{n_j} \rightarrow \eta$ , the corresponding hulls converge in the Carathéodory kernel sense at every time and the limiting hull chain is generated by the continuous trace  $\eta$ ;
4. the Loewner equation driven by  $U$  is generated by a unique continuous trace  $\gamma$  on  $[0, T]$ .

Then  $\gamma_n \rightarrow \gamma$  uniformly on  $[0, T]$ .

*Proof.* The point of the statement is to isolate exactly what driving-function convergence does and does not give. By precompactness, every subsequence of  $(\gamma_n)$  contains a further subsequence, still denoted  $(\gamma_{n_j})$ , converging uniformly to some continuous curve  $\eta$ . We show that  $\eta = \gamma$ .

For fixed  $t$ , let  $K_t^{(j)}$  be the hull generated by  $\gamma_{n_j}[0, t]$ . Hypothesis (3) says that  $K_t^{(j)}$  converges in the Carathéodory kernel sense to the hull  $K_t^\eta$  generated by  $\eta[0, t]$ . Let  $g_t^{(j)}$  and  $g_t^\eta$  be the hydrodynamically normalized conformal maps from the corresponding unbounded components to  $\mathbb{H}$ . Kernel convergence and uniqueness of the hydrodynamic normalization imply

$$g_t^{(j)} \longrightarrow g_t^\eta$$

locally uniformly away from  $K_t^\eta$ .

On the other hand, continuous dependence of the chordal Loewner ODE away from the growing hulls gives the same locally uniform limit from the driving functions. Indeed, on any compact set whose trajectories remain a positive distance from the singularity up to time  $t$ , the vector fields

$$z \longmapsto \frac{2}{z - U_{n_j}(s)}$$

converge uniformly to  $2/(z - U(s))$ ; the standard Gronwall estimate therefore gives convergence of  $g_s^{(j)}(z)$  to the Loewner solution  $g_s^U(z)$  uniformly for  $0 \leq s \leq t$ . Hence the kernel chain  $K_t^\eta$  is precisely the Loewner hull chain driven by  $U$ .

By hypothesis (4), that chain is generated by the unique continuous trace  $\gamma$ . Thus  $\eta = \gamma$ . We have proved that every subsequence of  $(\gamma_n)$  has a further subsequence converging uniformly to  $\gamma$ . In a metric space this implies convergence of the full sequence: otherwise there would be an  $\varepsilon > 0$  and a subsequence staying at least  $\varepsilon$  away from  $\gamma$ , contradicting the preceding conclusion. Therefore  $\gamma_n \rightarrow \gamma$  uniformly on  $[0, T]$ .  $\square$

*Remark 1.5* (Why trace control is indispensable). Uniform convergence of drivers alone controls the conformal maps away from the tip; it does not forbid a curve from making a thin macroscopic excursion which is almost invisible in capacity. Hypotheses (2)–(3) are therefore the genuine geometric input. Aizenman–Burchard or Kemppainen–Smirnov crossing estimates are standard ways to verify such trace control in lattice-model applications.



## Chapter 2

# LERW and UST

### 2.1 Loop-erased random walk

**Definition 2.1** (Loop erasure and loop-erased random walk). For a finite nearest-neighbor path  $\omega = (v_0, \dots, v_m)$ , its *chronological loop erasure* is obtained by retaining  $v_0$ , jumping to the vertex immediately after the last visit to  $v_0$ , and repeating until the endpoint is reached. If  $S$  is simple random walk stopped on hitting a target set  $A$ , the resulting self-avoiding path  $\text{LE}(S[0, \tau_A])$  is called *loop-erased random walk* (LERW) from  $S_0$  to  $A$ .

**Definition 2.2** (Uniform spanning tree). For a finite connected graph  $G$ , a *spanning tree* is a connected, cycle-free subgraph containing every vertex. The *uniform spanning tree* (UST) is the probability measure assigning mass  $1/\tau(G)$  to each spanning tree, where  $\tau(G)$  is the number of spanning trees. With wired boundary, all boundary vertices are first identified to one root vertex.

*Worked example 2.3* (Wilson's algorithm on a triangle). Let  $G$  be the triangle with vertices  $a, b, c$  and root  $c$ . Its three spanning trees are

$$\{ab, ac\}, \quad \{ab, bc\}, \quad \{ac, bc\},$$

so each has UST probability  $1/3$ . The path from  $a$  to  $c$  is therefore the direct edge  $a-c$  in two of the three trees and the two-edge path  $a-b-c$  in the remaining tree. Hence

$$\mathbb{P}_{\text{UST}}(a \leftrightarrow c \text{ uses } ac) = \frac{2}{3}, \quad \mathbb{P}_{\text{UST}}(a \leftrightarrow c \text{ is } a-b-c) = \frac{1}{3}.$$

Wilson's algorithm says that this tree path has exactly the law of the loop erasure of simple random walk started at  $a$  and stopped at  $c$ . Thus even on this three-vertex graph one sees the LERW–UST identity used in the continuum limit.

**Theorem 2.4** (Planar LERW converges to  $\text{SLE}_2$ ).

*Let  $D$  be a simply connected Jordan domain and let  $D_\delta$  be lattice approximations for which simple random walk satisfies the standard invariance-principle and discrete potential-theory convergence used below. A loop-erased random walk from a lattice point approximating  $z \in D$  to the wired boundary, mapped conformally to the unit disk and parametrized by radial Loewner capacity, converges in law as an unparametrized curve (and in the Loewner topology) to radial  $\text{SLE}_2$  from the boundary prime end to  $z$ . The boundary-to-boundary chordal version is obtained by the same martingale argument in chordal normalization.*

*Proof.*

$$\mathbb{E}[\Delta W \mid \mathcal{F}] = O(h^{3/2}), \quad \mathbb{E}[(\Delta W)^2 \mid \mathcal{F}] = 2h + O(h^{3/2}),$$

and a uniform third-moment bound  $\mathbb{E}[|\Delta W|^3 \mid \mathcal{F}] \leq Ch^{3/2}$ . The coefficient 2 is the one forced by the Green/Poisson observable; it is where the SLE parameter is identified.

$$W_t^\delta \implies \sqrt{2} B_t$$

uniformly on compact time intervals before the target is swallowed. □

## 2.2 LERW and UST

**Theorem 2.5** (Conformal scaling limit of planar UST).

*For lattice approximations of a Jordan domain with wired boundary, the uniform spanning tree, viewed through its finite collections of branches between a countable dense set of marked vertices and the boundary and then completed in the standard tree topology, has a scaling limit. The law of this continuum tree is conformally invariant.*

*Proof.* Fix a countable dense set  $\{z_1, z_2, \dots\} \subset D$ . Wilson's algorithm with wired boundary constructs the discrete UST by first running the loop-erasure of a random walk from  $z_1^\delta$  to the wired boundary, then from  $z_2^\delta$  until it hits the existing tree, and so on. Thus every finite collection of UST branches is obtained by finitely many LERWs in successively slit domains.

The preceding LERW theorem gives convergence of the first branch to chordal  $\text{SLE}_2$ . Conditional on that branch up to a compact stopping regime, Carathéodory convergence of the slit domains and the domain Markov property put the second branch under the same hypotheses; it therefore converges to  $\text{SLE}_2$  in the complementary domain until hitting the first branch. Induction gives joint convergence of the branches from  $z_1, \dots, z_m$  for every fixed  $m$ . The separation estimates used in the LERW proof give tightness on compact scales and rule out an additional macroscopic branch avoiding the dense marked set.

These finite-dimensional laws are projectively consistent because Wilson's algorithm is unchanged when later starting points are omitted. Kolmogorov extension gives a random tree on the dense set, and closure in the standard tree topology gives the continuum UST. Since each limiting branch is conformally invariant  $\text{SLE}_2$  and the continuation rule in slit domains is conformally natural, every finite branch system transforms correctly under conformal maps; therefore so does the completed tree. □

**Theorem 2.6** (UST Peano curve converges to  $\text{SLE}_8$ ).

*Let  $D$  be a Jordan domain with  $C^1$  boundary and let  $A$  be a proper boundary arc whose endpoints are  $a, b$ . Approximate  $D$  by planar lattices of mesh  $\delta$ , wire the UST along  $A$ , and let  $\eta_\delta$  be the oriented Peano interface separating the primal tree from its dual tree, run from  $a_\delta$  to  $b_\delta$ . After mapping to a fixed chordal normalization and parametrizing by half-plane capacity,  $\eta_\delta$  converges in law, uniformly modulo time change and equivalently in the Loewner topology, to chordal  $\text{SLE}_8$  from  $a$  to  $b$ .*

*Proof.*

$$d\langle W \rangle_t - 8 dt$$

and a first-order term proportional to  $dW_t$ . Since the discrete observable is a martingale for every test point, every subsequential driver has zero finite-variation part and quadratic variation  $8t$ . Lévy's characterization identifies the driver as  $\sqrt{8}B_t$ .

□

## 2.3 Natural parametrization of LERW

**Theorem 2.7** (Lawler–Viklund natural-time convergence).

Let  $D$  be a bounded simply connected domain with analytic boundary and let  $a, b \in \partial D$  be distinct. For  $N \geq 1$ , let  $(D_N, a_N, b_N)$  be the standard square-lattice approximation of mesh  $N^{-1}$  used in the chordal LERW convergence theorem, and let  $\eta_N$  be chordal LERW from  $a_N$  to  $b_N$ . There is a lattice-dependent constant  $\check{c} \in (0, \infty)$  such that, if every lattice edge of  $\eta_N$  is traversed in time  $\check{c}N^{-5/4}$ , then  $\eta_N$  converges in law as a parametrized curve to chordal SLE<sub>2</sub> from  $a$  to  $b$  parametrized by its 5/4-dimensional Minkowski content.

More precisely, the LERW and the naturally parametrized SLE<sub>2</sub> may be coupled so that their distance in the uniform parametrized-curve metric tends to zero in probability. With the quantitative square-grid estimates proved below, one may take a logarithmic coupling error  $C_D(\log N)^{-c}$ ; the explicit mesoscopic parameter choice in that argument gives  $c = 1/60$ .

*Proof.* The capacity-parametrized convergence of SM-16.02.01 does not by itself control how many lattice edges are traversed. The extra work is to compare two additive functionals along a coupling of the same growing hulls: the renormalized number of LERW edges and the Minkowski-content clock of SLE<sub>2</sub>. The comparison is driven by their one-point Green functions and closed by a two-point estimate.

Write  $\delta = N^{-1}$  and let  $d = 5/4$ . Parametrize the discrete curve by

$$T_N(k) = \check{c}\delta^d k,$$

where  $k$  is the number of traversed edges. Couple it with a chordal SLE<sub>2</sub> curve  $\gamma$  in  $D$ , initially parametrized by half-plane capacity, and let

$$\Theta(t) = \text{Cont}_{5/4}(\gamma[0, t])$$

be the Minkowski-content clock owned by SM-14.06.02. The theorem is equivalent to proving, uniformly up to a finite family of capacity stopping times that exhaust the curve,

$$\sup_t |T_N(t) - \Theta(t)| \xrightarrow{\mathbb{P}} 0$$

together with the already-owned spatial coupling in capacity time.

Stop LERW whenever its half-plane capacity has increased by a mesoscopic amount  $h = h_N \downarrow 0$ , with  $h$  tending to zero slowly compared with every power error used below. Let  $\eta^n$  be the revealed path after  $n$  increments,  $D_n$  the remaining lattice domain and  $U_n$  its discrete Loewner driving coordinate. The Green-function martingale from SM-16.02.01, expanded through the Loewner difference equation, gives

$$\mathbf{E}[\Delta U_n \mid \mathcal{F}_n] = O(h^{3/2}) + o(h), \quad \mathbf{E}[(\Delta U_n)^2 \mid \mathcal{F}_n] = 2h + O(h^{3/2}) + o(h),$$

with a uniform third-moment bound. By Skorokhod embedding (equivalently the martingale strong-coupling construction used in the source), one can enlarge the probability space to contain a Brownian motion  $B$  and stopping times  $\tau_n$  such that, outside an event  $o(1)$ ,

$$\max_{n \leq h^{-1}} |U_n - \sqrt{2}B_{\tau_n}| = o(1), \quad \max_{n \leq h^{-1}} |\tau_n - nh| = o(1),$$

and the corresponding uniformizing maps are uniformly close at points whose conformal height is at least a fixed small power of  $h$ . This is the quantitative version of the capacity convergence already proved in SM-16.02.01, not a second scaling-limit input.

Let  $e_z$  be a lattice edge whose midpoint is within  $O(\delta)$  of a point  $z$  staying a positive macroscopic distance from the marked boundary points. Then, uniformly in the slit domains  $D_n$  before the stopping regime used in Step b,

$$\delta^{-3/4} \mathbf{P}_n\{e_z \subset \eta_N\} = c_* G_{D_n}^{\text{SLE}_2}(z; a_n, b)(1 + O(\delta^u)) \quad (16.1)$$

for constants  $c_*, u > 0$  depending only on the lattice normalization and the compact geometric class.

Here is the internal mechanism behind (16.1). Orient the marked edge and split a walk that uses it into the incoming piece, the marked step and the outgoing piece. Fomin's loop-erasure determinant cancels configurations in which the two pieces intersect in the wrong order. Introducing the sign  $(-1)^{\text{wind}}$  on the double cover branched at the marked edge turns the surviving determinant into a product of three factors: a local lattice constant, an exponential of the random-walk loop measure of loops with odd winding around the branch point, and a signed Poisson-kernel (spinor) factor. The rooted-to-unrooted loop-measure comparison of Volume V and invariance principle imply

$$\exp\{2m_{\text{rw}}^{\text{odd}}\} = c_1 \text{crad}_{D_n}(z)^{1/4}(1 + O(\delta^u)).$$

For the spinor factor, stop the signed walk on a square annulus about  $z$ . Reflection across the branch cut gives the exact half-plane square-root kernel on the inner box; Harnack comparison and the discrete Poisson-kernel convergence transport it to  $\partial D_n$ . Thus the two independent signed boundary coordinates converge to the sine/cosine components of the continuum spinor with an  $O(\delta^u)$  error. Multiplying the factors gives precisely the conformally covariant  $\text{SLE}_2$  Green function from SM-14.06.01. The power  $\delta^{3/4}$  is  $\delta^{2-d}$ , and all lattice dependence is absorbed into  $c_*$ .

A sum over lattice edges turns (16.1) into the conditional expected remaining LERW length:

$$\mathbf{E}\left[\check{c}\delta^{5/4}\#(\eta_N \setminus \eta^n) \mid \mathcal{F}_n\right] = \check{c}c_* \int_{D_n} G_{D_n}^{\text{SLE}_2}(z) dA(z) + o(1).$$

On the SLE side, the Doob–Meyer/Minkowski construction of SM-14.06.02 gives

$$\mathbf{E}[\Theta(\infty) - \Theta(\tau_n) \mid \mathcal{G}_n] = c_{\text{SLE}} \int_{D \setminus \gamma[0, \tau_n]} G^{\text{SLE}_2}(z) dA(z).$$

Choose  $\check{c} = c_{\text{SLE}}/c_*$  in the common Green-function normalization. This is the universal positive renormalization constant in the theorem. No closed formula for it is needed or asserted.

If  $e_z, e_w$  are distinct edges at mutual distance  $r \gg \delta$ , expose the curve until it first reaches the smaller of the two surrounding annuli. The separation lemma SM-05.05.03 gives, conditionally on reaching that scale, a uniform positive chance that the two relevant strands leave through disjoint macroscopic sectors. Beurling and Harnack estimates on the separated event, followed by the one-point estimate in the remaining slit domain, give

$$\mathbf{P}\{e_z, e_w \subset \eta_N\} \leq C \delta^{3/2} r^{-3/4} (G_D(z) + 1) (G_D(w) + 1). \quad (16.2)$$

For  $r = O(\delta)$  the trivial one-point bound suffices. Since  $r^{-3/4}$  is locally integrable in two dimensions, summing (16.2) shows that renormalized LERW length in a bounded region has a uniformly bounded second moment. The identical exponent is the two-point integrability exponent in the SLE Minkowski-content construction.

The map coupling in Step b is only quantitative at points whose conformal height is not too small. Partition  $D$  into mesh squares and declare a square closed once either (i) it is screened from the tip by a narrow fjord, or (ii) the path has approached it on a scale much smaller than the current mesoscopic scale and then moved a macroscopic distance away. An open square is one that has not yet closed. On open squares Koebe distortion converts the map coupling into a uniform comparison of the discrete and continuum Green functions. On closed squares, the annular separation estimate and (16.2) show

$$\begin{aligned} & \mathbf{E}[\text{renormalized future LERW time in closed squares}] \\ & + \mathbf{E}[\text{future SLE content in closed squares}] = o(1). \end{aligned} \quad (16.3)$$

Indeed, screening forces an extra annular crossing, while a near-then-far square forces a return across a separated annulus; dyadic summation supplies a positive power at each scale. This is the step which prevents a fjord with tiny conformal height from carrying an uncontrolled amount of natural time.

Let  $L_n$  be the renormalized LERW time already traversed in open squares and  $\Theta_n$  the SLE Minkowski content already accumulated in the coupled open squares. Define

$$\begin{aligned} M_n^L &= L_n + \mathbf{E}[\text{future open-square LERW time} \mid \mathcal{G}_n], \\ M_n^S &= \Theta_n + \mathbf{E}[\text{future open-square SLE content} \mid \mathcal{G}_n]. \end{aligned}$$

Both are martingales. Their difference can be written

$$M_n^S - M_n^L = B_n + Y_n,$$

where  $B_n = (\Theta_n - L_n)$  is the difference of the accumulated increasing parts and  $Y_n$  is the difference of the conditional future masses. On open squares, (16.1) and the map coupling give  $\max_{n \leq h^{-1}} |Y_n| = o_{\mathbb{P}}(1)$ ; (16.2) gives a uniform  $L^2$  bound at the terminal stopping time. Equation (16.3) allows the closed-square pieces to be restored afterwards at  $o_{\mathbb{P}}(1)$  cost.

Let  $B'_n$  be the predictable compensator of  $B_n$  in the discrete filtration. The mesoscopic one-point comparison implies

$$\max_n |B'_n - B'_{n-1}| = o(1).$$

Since  $M^S - M^L$  is a martingale,  $B' + Y$  has zero predictable drift. The  $L^2$  maximal inequality applied to the martingale differences, together with the terminal second-moment bound from Step e, yields

$$\max_{n \leq h^{-1}} |B'_n| = o_{\mathbb{P}}(1).$$

The difference  $B - B'$  is controlled by the same one- and two-point estimates, so

$$\max_{n \leq h^{-1}} |L_n - \Theta_n| = o_{\mathbb{P}}(1). \quad (16.4)$$

This is the discrete Doob–Meyer comparison: closeness of Green functions forces closeness of the increasing natural-time parts.

Between two successive capacity stopping times the SLE natural time has oscillation tending to zero in probability by the Hölder continuity of SM-14.06.02. For LERW, (16.2) and Markov's inequality give the same statement for the renormalized number of edges in one mesoscopic capacity block. Therefore (16.4) upgrades to

$$\sup_{t \leq T} |T_N(t) - \Theta(t)| \xrightarrow{\mathbb{P}} 0$$

for every fixed capacity horizon  $T$  before the endpoint.

On the high-probability coupling event, SM-16.02.01 gives uniform spatial closeness of the two curves in capacity time, while Step i gives uniform closeness of their strictly increasing natural clocks. Inverting those clocks and composing the spatial paths yields an increasing homeomorphism  $\alpha_N$  between the two total-time intervals with

$$\sup_t |\alpha_N(t) - t| + \sup_t |\eta_N(t) - \gamma(\alpha_N(t))| \xrightarrow{\mathbb{P}} 0.$$

This is precisely convergence in the parametrized-curve metric.

Cover the analytic domain by finitely many capacity normalizations so that the preceding argument reaches the terminal point; the expected remaining Green mass tends to zero in the final normalization, so no terminal time is lost. The theorem is the chordal Lawler–Viklund result. Their radial companion paper proves the analogous boundary-to-interior theorem by the same argument with the radial Green function. The statement here deliberately keeps the analytic-boundary square-grid setting stated in the theorem; the lattice-universality hypotheses of SM-16.06.01 do not by themselves imply natural-time convergence without the corresponding one- and two-point Green estimates.  $\square$

## Chapter 3

# Critical percolation

### 3.1 Percolation

**Theorem 3.1** (Critical exploration converges to  $SLE_6$ ).

*Let  $(D; a, b)$  be a Jordan domain with Dobrushin blue/yellow boundary conditions and let  $\eta_\delta$  be the critical site-percolation exploration path on the triangular lattice of mesh  $\delta$ . Then  $\eta_\delta$  converges in distribution, in the uniform curve topology modulo monotone reparametrization, to chordal  $SLE_6$  from  $a$  to  $b$ .*

*Proof.* The exploration-to- $SLE_6$  argument is already owned, in the logically earlier percolation volume, by SM-11.04.04.a–e: RSW/arm tightness, the conditional Cardy law, Loewner-driver identification, and the no-extra-excursion estimate. The present theorem is an exact downstream reuse with the same triangular-lattice/Jordan-domain scope.  $\square$

**Theorem 3.2** (Full critical-percolation interface scaling limit).

*For critical site percolation on triangular-lattice approximations of a bounded simply connected Jordan domain, the complete collection of cluster-boundary interfaces converges jointly, in the standard loop-collection topology, to the conformally invariant continuum nonsimple-loop ensemble constructed from iterated  $SLE_6$  explorations. The limiting ensemble is almost surely countable and locally finite above every positive diameter, loops do not cross although they may touch themselves and one another, every deterministic point is surrounded by nested loops of both orientations, and the domain-restriction/renewal rule is inherited from the exploration construction. This ensemble is the  $CLE_6$  / full continuum scaling limit in the Camia–Newman sense.*

*Proof.* This is exactly the full interface-loop theorem proved earlier as SM-11.04.04, including recursive  $SLE_6$  exploration, no-missing-macroscopic-loop control, countability/local finiteness, nesting, and conformal invariance. No hypothesis or conclusion is enlarged here.  $\square$

## Chapter 4

# Ising and FK-Ising

### 4.1 Critical spin-Ising interface

**Theorem 4.1** (Critical spin-Ising interface scaling limit).

*Let  $(\Omega^\delta; a^\delta, b^\delta)$  be square-grid approximations of a bounded simply connected marked domain  $(\Omega; a, b)$ , with critical spin-Ising Dobrushin boundary conditions – and + on the two boundary arcs. Then the spin interface from  $a^\delta$  to  $b^\delta$  converges weakly, in the uniform metric on curves modulo increasing reparametrization, to chordal  $\text{SLE}_3$  from  $a$  to  $b$ . The same conclusion holds in the bounded-angle isoradial class under the normalization hypotheses of Volume XII, Chapter 2.*

*Reduction to the earlier theorem.* This theorem is already physically proved, at the same or stronger scope, by SM-12.05.02 in Volume XII. There the Dobrushin spin interface is first made tight by a boundary-condition-uniform crossing criterion; the slit-domain fermion is then shown to converge uniformly to

$$M_t(w) = \frac{\sqrt{g'_t(w)}}{g_t(w) - W_t}.$$

Its hydrodynamic expansion at infinity yields the two martingales

$$W_t, \quad W_t^2 - 3t,$$

so Lévy's characterization gives  $W_t = \sqrt{3}B_t$ . The same earlier owner proves curve/driver compatibility and excludes hidden macroscopic excursions, thereby upgrading driver convergence to convergence of the complete interface in the unparametrized curve topology. It also records the bounded-angle isoradial extension under the hypotheses proved earlier in Volume XII.

Hence no new theorem is imported here and no later result is used: this row is a strict backward dependency on the exact physical owner SM-12.05.02.  $\square$

### 4.2 Critical FK-Ising interface

**Theorem 4.2** (Critical FK-Ising interface scaling limit).

*Let  $(\Omega^\delta; a^\delta, b^\delta)$  approximate a bounded simply connected marked domain  $(\Omega; a, b)$  and impose critical FK-Ising Dobrushin boundary conditions, free on one marked boundary arc and wired on the other. Then the primal/dual exploration interface from  $a^\delta$  to  $b^\delta$  converges weakly, in the uniform metric on curves modulo increasing reparametrization, to chordal  $\text{SLE}_{16/3}$  from  $a$  to  $b$ . The statement includes the bounded-angle isoradial setting covered by the observable-normalization hypotheses of Volume XII, Chapters 1–2.*



*Reduction to the earlier theorem.* The exact proof is already owned by SM-12.05.03. In the Dobrushin FK model the parafermionic observable is a martingale in every explored slit domain, and its continuum half-differential limit is

$$M_t(w) = \sqrt{\frac{g'_t(w)}{g_t(w) - W_t}}.$$

The expansion at infinity gives

$$W_t, \quad \frac{3}{8}W_t^2 - 2t$$

as local martingales, equivalently  $W_t^2 - (16/3)t$  is a local martingale. The earlier compactness theorem then identifies the full curve and Lévy's theorem gives  $W_t = \sqrt{16/3} B_t$ . Thus the interface converges to chordal  $\text{SLE}_{16/3}$  in precisely the square-grid and bounded-angle isoradial settings stated here.

This is an exact backward owner, not a citation-only substitution: the complete fermionic convergence, coefficient calculation, tightness and hidden-excursion argument are all written in Volume XII.  $\square$

*Remark 4.3. Proof-owner note.* Canonical owner: SM-12.05.03.

### 4.3 Critical spin-loop ensemble

**Theorem 4.4** (Critical Ising and FK-Ising loop scaling limits).

*Let  $\Omega^\delta$  approximate a simply connected Jordan domain  $\Omega$ . In the critical FK-Ising model, the complete nested collection of macroscopic primal/dual interfaces converges in the loop-collection topology of Volume XII, Chapter 7 to  $\text{CLE}_{16/3}$ . In the critical spin-Ising model with + boundary condition, the complete nested collection of spin interfaces converges in the same topology to nested  $\text{CLE}_3$ .*

*Reduction to the earlier theorem.* The complete loop theorem was proved in Volume XII, Chapter 7. Those results prove tightness of all macroscopic spin interfaces by recursive FK cut-out and free-arc exploration, then identify every subsequential loop law through:

1. simplicity/disjointness from multi-arm exclusion;
2. conformal invariance of the limiting exploration branches;
3. the continuum domain Markov property inherited from the discrete Ising model;
4. a countable-dense-target uniqueness argument for the loop collection; and
5. the  $\text{SLE}_3$  branch law already proved in SM-12.05.02.

The resulting law is nested  $\text{CLE}_3$  in the proved Jordan-domain framework. All of these ingredients precede Volume XVI, so the present theorem is closed by an earlier theorem needed here and contains no forward CLE dependency.  $\square$

*Remark 4.5. Proof-owner note.* Canonical owners: SM-12.07.01 and SM-12.07.02, with branch identification from SM-12.05.02.

## Chapter 5

# DGFF and GFF interfaces

### 5.1 DGFF interfaces

**Theorem 5.1** (DGFF zero contour converges to  $\text{SLE}_4$ ).

Let  $D$  be a Jordan domain with distinct marked boundary prime ends  $a, b$ , and let  $D_\delta$  be lattice approximations for which discrete harmonic measure and the random-walk Green function converge to their continuum counterparts. There is a lattice-dependent constant  $\lambda_{\text{DGFF}} > 0$  with the following property. Give the discrete Gaussian free field boundary value  $-\lambda_{\text{DGFF}}$  on the counterclockwise arc  $a \rightarrow b$  and  $+\lambda_{\text{DGFF}}$  on the complementary arc. The zero-height interface joining  $a_\delta$  to  $b_\delta$ , viewed modulo monotone reparametrization, converges in law to chordal  $\text{SLE}_4$  from  $a$  to  $b$ .

More generally, with boundary values  $-A < 0$  and  $B > 0$  in the admissible height-gap regime, the same proof gives the corresponding  $\text{SLE}_4(\rho_L; \rho_R)$  law with  $\rho_L = A/\lambda_{\text{DGFF}} - 1$  and  $\rho_R = B/\lambda_{\text{DGFF}} - 1$ , where  $\rho_L$  is attached to the force point immediately to the left of the driving point and  $\rho_R$  to the one immediately to its right.

*Proof.* The proof has one model-specific input—the height-gap theorem—and then a Loewner martingale identification. We spell out both parts because merely citing the height gap would leave essentially all of the discrete work external.

Triangulate  $D_\delta$  and linearly interpolate the DGFF. Since the field is almost surely nonzero at every interior vertex, the zero set contains a unique interface  $\eta_\delta$  separating the boundary-connected positive cluster from the boundary-connected negative cluster. Reveal the interface one edge at a time together with the values of the vertices immediately adjacent to it. The DGFF Markov property says that, conditionally on this information, the unexplored field is an independent zero-boundary DGFF plus the discrete harmonic extension of all presently exposed boundary values.

Let  $v$  be a vertex at lattice distance at least  $r$  from the marked endpoints and let  $\mathcal{F}_\eta$  be the interface sigma-field. Discrete Green-function bounds and the Beurling estimate from Volume IV give, uniformly in the mesh,

$$\text{Var}(h(v) \mid \mathcal{F}_\eta) \leq C \log \frac{R}{\text{dist}(v, \eta_\delta)} + C,$$

and Harnack comparison gives uniform control of conditional means in balls which do not meet the revealed path. Gaussian tails therefore imply that a macroscopic interface cannot be supported by anomalously large field values on a positive fraction of its length. These estimates also yield the “narrow” estimate: if two already exposed pieces form a channel of width  $\varepsilon r$  and length comparable to  $r$ , the probability that the interface crosses the channel without hitting one of its sides is  $O(\varepsilon^c)$ .

Take two interface neighborhoods separated by distance  $r$ . Decompose the DGFF in each neighborhood into a local zero-boundary field plus a harmonic remainder generated by values outside a surrounding ball. The gradient of the harmonic remainder is  $O(r^{-1})$  times its boundary oscillation by the discrete Harnack inequality. After conditioning on the coarse harmonic part, the local fields are independent. Hence events determined by the interface inside well-separated mesoscopic boxes are asymptotically independent up to an error which tends to zero with the ratio of box size to separation. Using alternating positive and negative boxes, this produces barriers on both sides of the interface with probability bounded below uniformly at every scale.

Start a random walk from a point at macroscopic distance from the interface and stop it on first hitting the discrete interface neighborhood. The barrier and narrow estimates show that, with probability tending to one, its last mesoscopic approach occurs through a well-separated corridor. Harnack comparison then makes its hitting law on a central portion of the interface comparable to discrete harmonic measure. Consequently, sampling an interface vertex by this hitting law removes endpoint and fjord biases: the local field seen from the sampled vertex has a tight translation-invariant subsequential law.

In the local limit of Step d, let  $H^+$  and  $H^-$  be the conditional expected field values at the first positive and negative vertices adjacent to the oriented interface. Translation invariance along the limiting interface, reflection/sign symmetry, and the DGFF integration-by-parts identity imply that the harmonic-measure averages of these expectations are constants  $+\lambda_{\text{DGFF}}$  and  $-\lambda_{\text{DGFF}}$ , independent of the global domain and of the macroscopic point at which the interface is sampled. Strict positivity follows because the adjacent vertices are conditioned to have the corresponding signs and their conditional Gaussian variances are uniformly nondegenerate. The near-independence/barrier estimates of Steps b–d make the convergence uniform away from the marked endpoints. Thus, if  $m_\delta(z) = \mathbf{E}[h(z) \mid \mathcal{F}_\eta]$ , then on compact subsets of the complement of the interface,

$$m_\delta(z) - \tilde{m}_\delta(z) \longrightarrow 0,$$

where  $\tilde{m}_\delta$  is the discrete harmonic function taking boundary values  $-\lambda_{\text{DGFF}}$  and  $+\lambda_{\text{DGFF}}$  on the two sides of the revealed interface and the prescribed values on the untouched outer boundary. This is the Schramm–Sheffield height-gap theorem in the exact form needed for Loewner identification.

Suppose  $\eta_\delta$  makes  $k$  unforced crossings of an annulus of fixed modulus. Between successive crossings the sign clusters on the two sides of the interface force alternating positive/negative barrier failures in well-separated subannuli. The conditional barrier estimate from Step c gives a factor  $q < 1$  at each new scale, so

$$\mathbf{P}\{\eta_\delta \text{ makes } k \text{ unforced crossings}\} \leq Cq^k.$$

The annulus compactness criterion therefore gives tightness in the unparametrized curve topology and rules out a macroscopic excursion invisible in Loewner capacity.

Take a subsequential curve limit  $\eta$  and map the remaining component to  $\mathbb{H}$  by the half-plane-capacity Loewner map  $g_t$ , with driver  $W_t$ . Step e, convergence of discrete harmonic measure, and optional stopping for the discrete conditional mean show that for every fixed interior point  $z$ ,

$$M_t(z) := \lambda_{\text{DGFF}} \left( 1 - \frac{2}{\pi} \arg(g_t(z) - W_t) \right)$$

is a continuous martingale up to the swallowing time of  $z$ . Multiplying by a harmless normalization constant does not affect the driver calculation.

Write  $Z_t = g_t(z) - W_t = X_t + iY_t$  and suppose temporarily that  $d\langle W \rangle_t = \kappa_t dt$  and that  $W$  has drift  $dA_t$ . Since

$$dZ_t = \frac{2}{Z_t} dt - dW_t,$$

Itô's formula gives

$$d \arg Z_t = \frac{Y_t}{|Z_t|^2} dW_t + \frac{X_t Y_t}{|Z_t|^4} (\kappa_t - 4) dt + \frac{Y_t}{|Z_t|^2} dA_t$$

(up to the sign convention for the martingale term). Using two interior test points with linearly independent coefficients forces  $dA_t = 0$  and  $\kappa_t = 4$  almost everywhere. Lévy's characterization therefore gives

$$W_t = 2B_t.$$

Thus every subsequential Loewner limit is chordal  $\text{SLE}_4$ .

By SM-14.03.01–02, the Loewner chain driven by  $2B$  is generated by a simple continuous  $\text{SLE}_4$  trace. Tightness from Step f ensures that the curve limit actually generates the identified hulls. Uniqueness of the  $\text{SLE}_4$  law removes subsequences and proves convergence of  $\eta_\delta$ . Conformal covariance follows because the limiting harmonic observable is conformally invariant.

With boundary values  $-A$  and  $B$ , the height-gap step still supplies a jump of  $2\lambda_{\text{DGFF}}$  across the interface while the untouched boundary contributes additional harmonic terms. Under the Loewner map these terms are Poisson kernels at the two force points. Applying Itô's formula gives the standard force-point drift

$$dW_t = 2 dB_t + \frac{\rho_L}{W_t - V_t^L} dt + \frac{\rho_R}{W_t - V_t^R} dt,$$

with  $\rho_L = A/\lambda_{\text{DGFF}} - 1$  and  $\rho_R = B/\lambda_{\text{DGFF}} - 1$  with the same left/right force-point convention. This is precisely the Schramm–Sheffield  $\text{SLE}_4(\rho_L; \rho_R)$  extension. The theorem does not claim boundary-height regimes beyond the continuation range treated there.  $\square$

**Theorem 5.2** (Continuum GFF level line and the  $\text{SLE}_4$  coupling).

*Let  $h$  be the continuum GFF on  $(\mathbb{H}; 0, \infty)$  with boundary values  $-\lambda$  on  $\mathbb{R}_-$  and  $+\lambda$  on  $\mathbb{R}_+$ , in the series normalization in which  $\lambda = \pi/2$ . There is an almost surely unique simple path  $\eta$  from 0 to  $\infty$ , measurable with respect to  $h$ , such that for every stopping time  $T$  the set  $\eta[0, T]$  is local and, conditional on  $\eta[0, T]$ , the field on  $\mathbb{H} \setminus \eta[0, T]$  is an independent zero-boundary GFF plus the harmonic function taking the original boundary values and the values  $-\lambda, +\lambda$  on the left and right sides of the path. The law of  $\eta$  is chordal  $\text{SLE}_4$ . For admissible piecewise-constant boundary data the same construction gives the corresponding  $\text{SLE}_4(\rho^L; \rho^R)$  level line up to its continuation threshold.*

*Internal Schramm–Sheffield construction and uniqueness.* Map a general marked simply connected domain conformally to  $(\mathbb{H}; 0, \infty)$ . We first construct the joint law from an  $\text{SLE}_4$  curve and then prove that the curve is determined by the field.

Let  $W_t = 2B_t$ , let  $g_t$  be the chordal Loewner flow, put  $f_t(z) = g_t(z) - W_t$ , and define

$$h_t(z) = \lambda \left( 1 - \frac{2}{\pi} \arg f_t(z) \right).$$

For  $\kappa = 4$ , Itô's formula gives

$$d \log f_t(z) = -\frac{2}{f_t(z)} dB_t, \quad dh_t(z) = \frac{4\lambda}{\pi} \Im \frac{1}{f_t(z)} dB_t. \quad (5.1)$$

Thus  $h_t(z)$  is a bounded martingale and is exactly the harmonic function with boundary values  $-\lambda$  and  $+\lambda$  on the two sides of the Loewner tip.

For the upper-half-plane Green kernel in the normalization of this series,

$$G_t(z, w) = G_{\mathbb{H}}(f_t(z), f_t(w)),$$

a direct differentiation of the two logarithms in  $G_{\mathbb{H}}$  gives

$$dG_t(z, w) = -d\langle h_t(z), h_t(w) \rangle \quad (5.2)$$

precisely for the chosen value of  $\lambda$  (equivalently  $\sqrt{\pi/8}$  in the Dirichlet-norm convention without the factor  $(2\pi)^{-1}$ ). Hence for every test function  $\varphi$ ,  $(h_t, \varphi)$  is a continuous martingale whose bracket equals the decrease of

$$E_t(\varphi) = \iint \varphi(z) G_t(z, w) \varphi(w) dz dw. \quad (5.3)$$

Condition on the entire  $\text{SLE}_4$  path and, in each component of its complement, add an independent zero-boundary GFF to the harmonic boundary function with the  $\pm\lambda$  jump. Call the resulting distribution  $\tilde{h}$ . For any test function  $\varphi$ , the random variable  $(\tilde{h}, \varphi)$  is the sum of the stopped Brownian martingale  $(h_t, \varphi)$  and an independent centered Gaussian of the remaining variance  $E_t(\varphi)$ . By (5.2)–(5.3), its characteristic function is therefore

$$\exp\left(i(h_0, \varphi) - \frac{1}{2}E_0(\varphi)\right),$$

and polarization gives the correct joint characteristic function for every finite family of test functions. Thus  $\tilde{h}$  is exactly the GFF with the stated boundary data. Repeating the calculation after a stopping time proves the local-set Markov property, not merely its one-point version.

Conversely, suppose a continuous Loewner chain coupled with a GFF has the stated conditional law. The conditional mean  $(h_t, \varphi)$  is a martingale for every  $\varphi$ , while the conditional covariance loses exactly the Green energy in (5.3). Localizing away from the tip and applying Itô to  $\arg(g_t(z) - W_t)$  shows first that the driver has zero drift and then, by matching (5.2), that  $d\langle W \rangle_t = 4dt$ . Lévy's characterization gives  $W_t = 2B_t$  and hence the curve law is chordal  $\text{SLE}_4$ .

Take a regular conditional law of the curve given  $h$  and sample independently from it two curves, one from 0 to  $\infty$  and, after applying  $z \mapsto -1/z$  and changing the sign of the field, one from  $\infty$  to 0. Fix a stopping time  $T$  of the reverse curve. Since both initial segments are local, on the event that they are disjoint the conditional law of the forward curve up to its first hit of the reverse segment is  $\text{SLE}_4$  in the remaining component, targeted at the reverse tip. A simple chordal  $\text{SLE}_4$  hits that reverse segment first at its tip. Applying this at a countable dense set of reverse stopping times shows that the forward curve contains a dense set of points of the reverse curve; by symmetry the converse holds. Both paths are continuous and simple, so they coincide. The two samples were conditionally independent given  $h$ ; equality of two conditionally independent samples implies that the conditional law is a point mass. Thus the level line is almost surely a measurable function of the GFF and is unique.

Changing finitely many boundary heights by harmonic functions of finite Dirichlet energy changes the GFF law locally by the Cameron–Martin density. Under the coupling, the same density is the Loewner Girsanov martingale. Its stochastic logarithm produces the drift

$$dW_t = 2dB_t + \sum_j \frac{\rho_j}{W_t - V_t^j} dt,$$

with the force weights determined by the adjacent boundary-height jumps. The Bessel description of  $V_t^j - W_t$  gives exactly the usual continuation threshold. Localizing before that threshold and exhausting the domain proves the stated  $\text{SLE}_4(\rho)$  extension.  $\square$

## Chapter 6

# Universality of lattice approximation

### 6.1 Universality of lattice approximation

**Theorem 6.1** (Lattice-independence in the LERW/UST convergence class).

*Let  $G_\delta$  be a family of planar lattice graphs approximating a simply connected domain, with uniformly nondegenerate local geometry, for which the embedded random walk satisfies a quenched/uniform invariance principle to planar Brownian motion and whose killed Green functions, harmonic measures and potential kernels converge locally uniformly to their continuum counterparts in the sense used in the LERW convergence proof. Then the LERW and UST scaling limits constructed in SM-16.02.01–03 are unchanged by the microscopic choice of  $G_\delta$ : LERW converges to  $SLE_2$ , the continuum UST law is the same Wilson/ $SLE_2$  tree, and, under the Peano boundary regularity hypotheses, the Peano exploration converges to  $SLE_8$ .*

*Proof.* Inspect the proof of SM-16.02.01. Its lattice-specific input consists of the invariance principle, local uniform convergence of killed Green functions and harmonic measures, convergence of the potential kernel fixing the Loewner normalization, and uniform Harnack/boundary estimates in the slit domains. These are exactly the hypotheses imposed on  $G_\delta$ . Hence the discrete martingale observables have the same continuum limits and the driving process again satisfies

$$W_t^\delta \implies \sqrt{2}B_t.$$

The tightness and no-extra-hull estimates use only the same uniform potential-theoretic bounds, so LERW converges to chordal  $SLE_2$ .

Wilson’s algorithm is valid on every finite weighted graph. Applying the finite-branch induction from SM-16.02.02 therefore yields the same continuum UST, because each successive LERW has the same  $SLE_2$  limit in its slit domain. Under the stated Peano boundary regularity hypotheses, the primal/dual exploration has the same harmonic-measure martingales as in SM-16.02.03; the limiting driver has zero drift and quadratic variation  $8t$ , and Lévy’s characterization gives  $\sqrt{8}B_t$ . Hence the Peano curve converges to  $SLE_8$ . No step uses a square-lattice identity beyond the analytic assumptions explicitly listed in the theorem.  $\square$

**Theorem 6.2** (Inherited isoradial Ising universality).

*For the critical FK- and spin-Ising observables in the bounded-angle isoradial class and with the normalization hypotheses of Volume XII, the scaling limits are independent of the microscopic isoradial graph and transform as holomorphic half-differentials under conformal maps.*

*Proof.* This follows directly from SM-12.02.01–02. Their hypotheses are bounded rhombus angles, Carathéodory convergence of the domains, and, where needed, the target normalization of the spin

observable; under exactly these hypotheses the universality and conformal covariance stated here follow.  $\square$

*Remark 6.3. Dependency note.* This inherited-owner formulation is intentional: later volumes may reuse an exact earlier theorem, but may not replace a stronger statement by a weaker similarly named one.

*Frontier statement 6.4* (Full lattice universality of critical percolation/SLE<sub>6</sub>).

The following general statement is deliberately *not* asserted as a theorem: for every sufficiently local planar percolation model tuned to a critical point and satisfying only broad universality-class assumptions, the macroscopic exploration interface converges to chordal SLE<sub>6</sub>. The manuscript proves such convergence only in the explicitly stated model classes owned earlier in the series. No downstream proof uses this frontier row as an input.

# Chapter 7

## Additional scaling limits

### 7.1 Additional Schramm/Lawler scaling limits

**Theorem 7.1** (Harmonic explorer converges to  $\text{SLE}_4$ ).

Let  $D^\delta$  be honeycomb-lattice approximations of a simply connected Jordan domain  $(D; a, b)$  with the standard black/white boundary arcs, and let  $\gamma^\delta$  be the Schramm–Sheffield harmonic explorer from  $a^\delta$  to  $b^\delta$ . Then  $\gamma^\delta$  converges in law, in the unparametrized-curve topology, to chordal  $\text{SLE}_4$  from  $a$  to  $b$ .

*Proof.* Map the current unexplored discrete domain to a continuum reference domain only after the martingale calculation; this avoids hiding the discrete observable. Given the explorer up to step  $n$ , assign boundary values 0 and 1 to the two colored boundary arcs and let  $H_n^\delta(v)$  be the discrete harmonic extension. By definition of the harmonic explorer, the color chosen for the newly exposed face has conditional mean equal to the old harmonic measure. The discrete mean-value property therefore gives

$$\mathbf{E}[H_{n+1}^\delta(v) \mid \mathcal{F}_n] = H_n^\delta(v)$$

for every vertex  $v$  not yet separated from the target. Thus the harmonic observable is an exact bounded martingale.

Stop while a fixed compact set remains a positive conformal distance from the tip. The invariance principle and discrete harmonic-function convergence imply that, uniformly on that compact set,

$$H_n^\delta(z) = \frac{1}{\pi} \arg(g_t^\delta(z) - W_t^\delta) + o(1),$$

where  $g_t^\delta$  is the hydrodynamically normalized map and  $t$  is half-plane capacity. Expand this continuum observable for one discrete step. If  $Z_t = g_t(z) - W_t$ , Loewner’s equation and Taylor expansion give

$$d \arg Z_t = \Im \left( \frac{2}{Z_t^2} dt - \frac{1}{Z_t} dW_t - \frac{1}{2Z_t^2} d\langle W \rangle_t \right).$$

Since the discrete observable has zero conditional drift for every choice of interior  $z$ , comparison of the coefficients of  $Z_t^{-1}$  and  $Z_t^{-2}$  yields, up to errors vanishing with the mesh,

$$\mathbf{E}[dW_t \mid \mathcal{F}_t] = 0, \quad d\langle W \rangle_t = 4 dt.$$

The same expansion, with the discrete Beurling estimate controlling the error near the boundary, gives a Lindeberg bound on driver increments. The martingale functional central limit theorem consequently yields

$$W^\delta \Longrightarrow 2B$$



in  $C[0, T]$  up to every compact stopping time.

The harmonic explorer satisfies the unforced-crossing condition: after a stopping time, an unforced crossing of a large-modulus quadrilateral requires the discrete harmonic function to change from a value bounded away from 0 to one bounded away from 1 across a narrow channel; the discrete Beurling estimate makes this probability uniformly small, and subdivision gives Condition G2. Therefore SM-16.01.02 gives tightness of the curves and compatibility of every joint curve/driver limit with the Loewner equation. Every subsequential driver is  $2B$ , whose Loewner chain is generated by the unique chordal  $\text{SLE}_4$  trace. SM-16.01.03 then identifies every curve subsequential limit with that trace. Since the limit is unique, the full family converges to  $\text{SLE}_4$ .  $\square$

**Theorem 7.2** (Random-walk loop soup converges to Brownian loop soup).

*Under diffusive scaling of space by  $N^{-1}$  and time by  $(2N^2)^{-1}$ , the planar random-walk loop measure converges to Brownian loop measure on every bounded collection of loops whose time duration is bounded below by a positive constant. Consequently Poisson random-walk loop soups of fixed intensity converge, in the locally finite loop topology away from zero-duration loops, to the Brownian loop soup of the same intensity.*

*Proof.* A rooted lattice loop of  $2n$  steps at root  $z$  has loop-measure weight proportional to  $(2n)^{-1} \mathbf{P}_z\{S_{2n} = z\}$ . By the local central limit theorem,

$$\mathbf{P}_0\{S_{2n} = 0\} = \frac{1}{\pi n} + O(n^{-2}),$$

and after setting  $t = n/N^2$  the Riemann sum over roots and durations converges to the Brownian rooted-loop intensity

$$\frac{1}{2\pi t^2} dt dA(z)$$

with the Brownian bridge law of duration  $t$ .

We must also couple the shapes. Conditional on  $S_{2n} = 0$ , use the strong invariance principle for bridges: on durations  $n \geq \varepsilon N^2$ , the linearly interpolated walk bridge divided by  $N$  can be coupled with a planar Brownian bridge of duration  $t = n/N^2$  so that, uniformly for  $t$  in compact subsets of  $(0, \infty)$ ,

$$\sup_{0 \leq s \leq t} |N^{-1} S_{2N^2 s} - B_s| \xrightarrow[N \rightarrow \infty]{\mathbf{P}} 0.$$

The bridge endpoint conditioning is handled by applying the strong approximation on the first and last halves and using the local central limit ratio for the midpoint. Therefore for every bounded continuous functional  $F$  supported on loops contained in a fixed compact set and with duration in  $[\varepsilon, M]$ ,

$$\int F d\mu_N^{\text{rw}} \longrightarrow \int F d\mu^{\text{Br}}.$$

Durations  $> M$  have total loop measure  $O(M^{-1})$  in a bounded spatial set, uniformly in  $N$ , so  $M \rightarrow \infty$  removes the upper cutoff. This proves vague convergence of the sigma-finite loop measures on the space with duration bounded away from zero.

Finally, Poissonization preserves vague convergence of locally finite intensity measures. Indeed Laplace functionals satisfy

$$\mathbf{E} \exp\{-\langle F, \mathcal{L}_N \rangle\} = \exp\left\{-\lambda \int (1 - e^{-F}) d\mu_N^{\text{rw}}\right\} \longrightarrow \exp\left\{-\lambda \int (1 - e^{-F}) d\mu^{\text{Br}}\right\}.$$

These are precisely the Laplace functionals of the Brownian loop soup. Since every compact subset of loop space bounded away from zero duration has finite intensity, they determine convergence of the point processes in the asserted topology.  $\square$

*Remark 7.3. **Scope.*** The theorem is deliberately stated in the locally finite topology away from zero-duration loops, which is the convergence established by the Lawler–Trujillo Ferreras strong approximation and is the form needed by the later loop-soup/CLE arguments.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 7.4** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 7.5** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 7.6** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 7.7** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 7.8** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XVII: Gaussian Multiplicative Chaos and Liouville Quantum  
Area

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# Scope and conventions

The formal expression  $e^{\gamma h(z)} dz$  is meaningless when  $h$  is a log-correlated distribution. Gaussian multiplicative chaos is the mechanism that turns that expression into a random measure. The guiding idea is always the same: regularize the field, subtract exactly half the variance in the exponential, and prove that the resulting random measures stabilize. The subcritical theory supplies LQG area and length; the critical theory explains why a new derivative/Seneta–Heyde normalization is needed at  $\gamma = 2$ .

# Chapter 1

## Subcritical Gaussian multiplicative chaos

**Definition 1.1** (Subcritical Gaussian multiplicative chaos). Let  $X$  be a log-correlated Gaussian generalized function on a planar domain and let  $X_\varepsilon$  be a specified smooth approximation. For  $\gamma \in (0, 2)$  define

$$M_{\gamma,\varepsilon}(dx) = \exp\left(\gamma X_\varepsilon(x) - \frac{\gamma^2}{2} \mathbf{E}[X_\varepsilon(x)^2]\right) dx.$$

A *subcritical GMC measure*  $M_\gamma$  is the nontrivial random Radon measure obtained as the weak-in-probability (equivalently, in the standard constructions, weak-in-law plus uniqueness) limit of  $M_{\gamma,\varepsilon}$  as  $\varepsilon \downarrow 0$ . Any alternative normalization used later is stated explicitly.

*Worked example 1.2* (Why the Wick subtraction is forced). For a Borel set  $A$  of finite area, Gaussian exponential moments give

$$\mathbf{E} M_{\gamma,\varepsilon}(A) = \int_A e^{-\frac{\gamma^2}{2} \text{Var } X_\varepsilon(x)} \mathbf{E}[e^{\gamma X_\varepsilon(x)}] dx = \int_A 1 dx = |A|.$$

Without the subtraction  $\frac{\gamma^2}{2} \text{Var } X_\varepsilon$ , the expectation would blow up with the regularization variance. The renormalization is therefore already visible from a one-line Gaussian calculation, before any multifractal analysis enters.

A single log-correlated field contains fluctuations on every logarithmic scale. The first task is to make exponential weighting stable across those scales before specializing to the GFF.

### 1.1 Subcritical GMC

**Theorem 1.3** (Subcritical GMC existence and regularization independence).

Let  $D \subset \mathbb{R}^k$  be bounded, let  $\sigma$  be a Radon measure with compact support in  $D$ , and assume that for some  $d > 0$

$$\iint |x - y|^{-(d-\eta)} \sigma(dx) \sigma(dy) < \infty \quad \text{for every } \eta > 0.$$

Let  $X$  be a centered Gaussian generalized field whose covariance on a neighborhood of  $\text{supp } \sigma$  is

$$K(x, y) = \log \frac{1}{|x - y|} + g(x, y),$$

with  $g$  bounded and  $C^2$  there. If  $\theta$  is a compactly supported smooth probability kernel, set  $X_\varepsilon^\theta = X * \theta_\varepsilon$  and

$$M_{\gamma,\varepsilon}^\theta(dx) = \exp\left(\gamma X_\varepsilon^\theta(x) - \frac{\gamma^2}{2} \text{Var } X_\varepsilon^\theta(x)\right) \sigma(dx).$$

If  $\gamma^2 < 2d$ , then for every compactly supported bounded continuous  $f$ ,

$$\int f dM_{\gamma,\varepsilon}^\theta \longrightarrow \int f dM_\gamma$$

in  $L^1$  and hence in probability. The random Radon measure  $M_\gamma$  is not identically zero and is independent of the mollifier  $\theta$  within this class. The same proof applies to circle-average regularizations whenever the covariance estimates used below hold.

*Proof.* The elementary second-moment argument works only when  $\gamma^2 < d$ . The subcritical proof reaches the larger range  $\gamma^2 < 2d$  by observing that the chaos measure sees points of thickness about  $\gamma$ , so one may discard much thicker points before taking a second moment. We give the argument in six steps.

**Step 1: covariance estimates.** Fix a second mollifier  $\psi$ , used only to test thickness, and write  $Y_r = X * \psi_r$ . Convolving the logarithmic kernel and using the  $C^2$  regularity of  $g$  gives constants  $C, c > 0$ , independent of sufficiently small  $r, \varepsilon, \delta$ , such that on  $\text{supp } \sigma$

$$|\text{Var } Y_r(x) - \log r^{-1}| \leq C, \quad (17.1)$$

$$|\text{Cov}(Y_r(x), X_\varepsilon^\theta(x)) - \log r^{-1}| \leq C \quad (r \geq c\varepsilon), \quad (17.2)$$

$$\text{Cov}(X_\varepsilon^\theta(x), X_\delta^\vartheta(y)) \leq \log \frac{1}{|x - y| \vee \varepsilon \vee \delta} + C. \quad (17.3)$$

Moreover, if  $r \geq c(\varepsilon \vee \delta \vee |x - y|)$ , then

$$\text{Cov}(Y_r(x), X_\varepsilon^\theta(x) + X_\delta^\vartheta(y)) \leq 2 \log r^{-1} + C. \quad (17.4)$$

These are deterministic estimates. For instance, the singular part in (17.2) is the average of  $\log |u - v|^{-1}$  with  $u$  at distance  $O(r)$  from  $x$  and  $v$  at distance  $O(\varepsilon)$  from  $x$ ; when  $r \geq c\varepsilon$  this equals  $\log r^{-1} + O(1)$ . The other estimates follow in the same way, while the smooth part contributes only  $O(1)$ .

Choose dyadic radii  $r_j = 2^{-j}$ . Replacing an arbitrary radius by a neighboring dyadic radius changes logarithms by only  $O(1)$ , so dyadic scales suffice.

**Step 2: good points and the one-point tilt.** Choose  $\alpha > \gamma$ , to be fixed sufficiently close to  $\gamma$  later, and a coarse scale  $r_{j_0}$ . For  $\varepsilon < r_{j_0}/c$  define

$$G_\varepsilon^\alpha(x) = \bigcap_{j_0 \leq j \leq j(\varepsilon)} \{Y_{r_j}(x) \leq \alpha \log r_j^{-1}\},$$

where  $j(\varepsilon)$  is the largest  $j$  for which  $r_j \geq c\varepsilon$ . Let

$$\overline{X}_\varepsilon^\theta = \gamma X_\varepsilon^\theta - \frac{\gamma^2}{2} \text{Var } X_\varepsilon^\theta$$

and put

$$I_\varepsilon^\theta(f) = \int f(x) e^{\overline{X}_\varepsilon^\theta(x)} \sigma(dx), \quad J_\varepsilon^\theta(f) = \int f(x) e^{\overline{X}_\varepsilon^\theta(x)} \mathbf{1}_{G_\varepsilon^\alpha(x)} \sigma(dx).$$

We first show that deleting the bad points costs arbitrarily little in  $L^1$ . For fixed  $x$ , tilt the Gaussian law by  $e^{\bar{X}_\varepsilon^\theta(x)}$ . The covariance is unchanged and the mean of  $Y_{r_j}(x)$  becomes

$$\gamma \text{Cov}(Y_{r_j}(x), X_\varepsilon^\theta(x)) = \gamma \log r_j^{-1} + O(1)$$

by (17.2). Since its variance is  $\log r_j^{-1} + O(1)$ , the Gaussian tail bound gives, uniformly in  $x, \theta, \varepsilon$ ,

$$\tilde{\mathbb{P}}_{x,\varepsilon}^\theta \{Y_{r_j}(x) > \alpha \log r_j^{-1}\} \leq C e^{-c_\alpha j}$$

for all sufficiently large  $j$ . Hence a union bound over  $j \geq j_0$  yields  $q(j_0) \downarrow 0$  such that

$$\sup_{\theta,\varepsilon} \mathbb{E} |I_\varepsilon^\theta(f) - J_\varepsilon^\theta(f)| \leq \|f\|_\infty \sigma(D) q(j_0). \quad (17.5)$$

This is the rigorous form of the statement that the chaos mass is carried by points no thicker than  $\gamma$ .

**Step 3: the two-point tilt and the subcritical  $L^2$  bound.** For clarity take  $0 \leq f \leq 1$ ; the general bounded case follows by positive and negative parts. Fubini and the Gaussian exponential identity give

$$\begin{aligned} \mathbb{E}[J_\varepsilon^\theta(f)^2] &= \iint f(x)f(y) e^{\gamma^2 \text{Cov}(X_\varepsilon^\theta(x), X_\varepsilon^\theta(y))} \\ &\quad \times \tilde{\mathbb{P}}_{x,y,\varepsilon}^\theta (G_\varepsilon^\alpha(x) \cap G_\varepsilon^\alpha(y)) \sigma(dx) \sigma(dy), \end{aligned}$$

where the tilted law has the same covariance and mean shifted by  $\gamma \text{Cov}(\cdot, X_\varepsilon^\theta(x) + X_\varepsilon^\theta(y))$ . Let  $r_*$  be a dyadic radius comparable to  $|x-y| \vee \varepsilon$ . If  $r_* < r_{j_0}$ , the event  $G_\varepsilon^\alpha(x)$  forces  $Y_{r_*}(x) \leq \alpha \log r_*^{-1} + O(1)$ . Under the two-point tilt, (17.4) and (17.1) show that this Gaussian variable has mean at most  $2\gamma \log r_*^{-1} + O(1)$  and variance  $\log r_*^{-1} + O(1)$ . Therefore, for every  $\eta > 0$  and all sufficiently small  $r_*$ ,

$$\tilde{\mathbb{P}}_{x,y,\varepsilon}^\theta (G_\varepsilon^\alpha(x)) \leq C_\eta r_*^{(2\gamma-\alpha)^2/2-\eta}. \quad (17.6)$$

Combining (17.3) and (17.6),

$$\mathbb{E}[J_\varepsilon^\theta(f)^2] \leq C + C \iint (|x-y| \vee \varepsilon)^{-a} \sigma(dx) \sigma(dy), \quad a = \gamma^2 - \frac{(2\gamma-\alpha)^2}{2} + \eta.$$

Because  $\gamma^2 < 2d$ , choose first  $\alpha > \gamma$  sufficiently close to  $\gamma$  and then  $\eta > 0$  sufficiently small so that  $a < d$ . The energy assumption on  $\sigma$  gives

$$\sup_{\theta,\varepsilon} \mathbb{E}[J_\varepsilon^\theta(f)^2] < \infty. \quad (17.7)$$

Together, (17.5) and (17.7) imply uniform integrability of  $I_\varepsilon^\theta(f)$ . Notice where the full subcritical threshold appears: as  $\alpha \downarrow \gamma$ , the effective singularity tends to  $\gamma^2/2$ , so the condition becomes exactly  $\gamma^2/2 < d$ .

**Step 4:  $L^2$  Cauchy property of the good mass.** Fix  $j_0$ . For two scales  $\varepsilon, \delta$  and two mollifiers  $\theta, \vartheta$ , expand

$$\mathbb{E}(J_\varepsilon^\theta(f) - J_\delta^\vartheta(f))^2 = \mathbb{E}[J_\varepsilon^\theta(f)^2] + \mathbb{E}[J_\delta^\vartheta(f)^2] - 2\mathbb{E}[J_\varepsilon^\theta(f)J_\delta^\vartheta(f)].$$

We show that the three terms have the same limit. Split each double integral into  $|x-y| \leq \rho$  and  $|x-y| > \rho$ . The estimate just proved bounds the near-diagonal part by an integrable kernel of order  $|x-y|^{-a}$  with  $a < d$ ; its contribution tends to zero uniformly as  $\rho \downarrow 0$  by the energy assumption.

On  $|x - y| > \rho$ , all regularized covariances converge uniformly to  $K(x, y)$  as  $\varepsilon, \delta \downarrow 0$ . Under the one- or two-point Gaussian tilts, the finite collection  $(Y_{r_j}(x), Y_{r_j}(y))_{j_0 \leq j \leq m}$  therefore has a limiting Gaussian law, uniformly on this off-diagonal set. The boundary of the finite-scale good event has Gaussian probability zero, so its probability converges. Letting first  $\varepsilon, \delta \downarrow 0$ , then  $m \rightarrow \infty$ , and finally  $\rho \downarrow 0$ , dominated convergence gives a common limit for the two second moments and the mixed moment. Thus

$$\lim_{\varepsilon, \delta \downarrow 0} \mathbb{E}(J_\varepsilon^\theta(f) - J_\delta^\theta(f))^2 = 0. \quad (17.8)$$

The fixed reference mollifier  $\psi$  makes the limiting good event the same for every target regularization, so this proves convergence and universality at once.

**Step 5: remove the truncation.** Given  $\zeta > 0$ , choose  $j_0$  so large that the right side of (17.5) is below  $\zeta$ . With this  $j_0$  fixed, (17.8) shows that the good masses are Cauchy in  $L^2$ . The triangle inequality gives

$$\limsup_{\varepsilon, \delta \downarrow 0} \mathbb{E}|I_\varepsilon^\theta(f) - I_\delta^\theta(f)| \leq 2\zeta.$$

Since  $\zeta$  is arbitrary, the regularized masses are Cauchy in  $L^1$ , and their limit is independent of  $\theta$ .

**Step 6: assemble a random Radon measure.** Take a countable dense family of nonnegative functions in  $C_c(D)$ . The  $L^1$  limits are positive and linear on that family, and

$$\mathbb{E}I_\varepsilon^\theta(f) = \int f d\sigma$$

passes to the limit by uniform integrability. Monotone approximation therefore gives, almost surely, a positive linear functional on  $C_c(D)$ ; Riesz–Markov produces a random Radon measure  $M_\gamma$ . Moreover

$$\mathbb{E} \int f dM_\gamma = \int f d\sigma.$$

For nonnegative  $f$  with  $\int f d\sigma > 0$ , the right-hand side is positive, so  $M_\gamma$  is not identically zero. This completes convergence and regularization independence at the stated subcritical scope.  $\square$

*Remark 1.4* (Why the full subcritical range is visible). Without deleting thick points, the two-point integrand behaves like  $|x - y|^{-\gamma^2}$  and is integrable only for  $\gamma^2 < d$ . The good-point constraint contributes the extra factor  $|x - y|^{(2\gamma - \alpha)^2/2}$ ; letting  $\alpha \downarrow \gamma$  changes the effective singularity from  $\gamma^2$  to  $\gamma^2/2$ . This is the mechanism behind the full threshold  $\gamma^2 < 2d$ .

**Example 1.5** (Why subtract half the variance). If  $Z \sim N(0, v)$ , then  $\mathbb{E}[e^{\gamma Z}] = e^{\gamma^2 v/2}$ . Thus  $e^{\gamma Z - \gamma^2 v/2}$  has mean one. GMC applies this elementary identity simultaneously at every spatial scale; the difficulty is not the normalization but controlling correlations as the variance diverges.

**Theorem 1.6** (Subcritical GMC nondegeneracy, full support, and atomlessness).

*In the subcritical regime  $\gamma^2 < 2d$ , the GMC limit is almost surely nonzero on every nonempty open set charged by the reference measure, is atomless under the standard non-atomic reference hypotheses, and has full support when the reference measure has full support.*

*Proof.* We separate the three assertions because they use different parts of the subcritical theory.

*Positivity on open sets.* Fix a nonempty open set  $U$  with  $\sigma(U) > 0$  and choose a compact  $K \Subset U$  with  $\sigma(K) > 0$ . In the martingale construction,  $M_{\gamma,t}(K)$  is a nonnegative martingale with constant mean  $\sigma(K)$ . The uniform integrability supplied by the subcritical construction therefore gives

$$\mathbb{E}M_\gamma(K) = \sigma(K) > 0.$$

For an independent-scale construction, the event  $\{M_\gamma(K) = 0\}$  is unchanged by modifying finitely many coarse scale increments: multiplication by a strictly positive continuous coarse factor cannot turn zero mass into positive mass or conversely. It is therefore a tail event of the independent scale increments. Kolmogorov's zero-one law and the positive expectation force its probability to be zero. Hence  $M_\gamma(U) \geq M_\gamma(K) > 0$  almost surely. Applying this to a countable base gives full support whenever  $\sigma$  has full support.

*Absence of atoms.* Let  $K$  be compact and choose  $1 < q < 2d/\gamma^2$ . The moment theorem in the next section gives the local scaling estimate, uniformly for dyadic cubes  $Q$  of side  $2^{-n}$  contained in a fixed compact subset,

$$\mathbb{E}[M_\gamma(Q)^q] \leq C_q 2^{-n\xi(q)}, \quad \xi(q) = dq - \frac{\gamma^2}{2}q(q-1).$$

There are  $O(2^{nd})$  such cubes. Since  $\xi(q) - d = (q-1)(d - \gamma^2 q/2) > 0$ ,

$$\mathbb{E} \sum_Q M_\gamma(Q)^q \leq C_q 2^{-n(\xi(q)-d)} \longrightarrow 0.$$

Therefore  $\max_Q M_\gamma(Q)^q \leq \sum_Q M_\gamma(Q)^q$  tends to zero in probability. Choose a subsequence along which it tends to zero almost surely. Every point of  $K$  lies in a nested sequence of those cubes, so continuity from above gives  $M_\gamma(\{x\}) = 0$  for every  $x \in K$  on that event. An exhaustion by compact sets proves atomlessness.

Thus the theorem is a genuine consequence of the two deep inputs isolated in this chapter: subcritical non-degeneracy/uniform integrability and the positive-moment estimate. It introduces no additional black box.  $\square$

**Theorem 1.7** (Subcritical GMC moment thresholds).

Let  $M_\gamma$  be Lebesgue-base subcritical GMC on a bounded domain of  $\mathbb{R}^d$ , with covariance  $\log|x-y|^{-1} + O(1)$  on compact subsets, and write

$$q_c = \frac{2d}{\gamma^2}, \quad \xi(q) = dq - \frac{\gamma^2}{2}q(q-1).$$

For every compact  $K$  separated from singular boundary data,

$$\mathbb{E}[M_\gamma(K)^q] < \infty \quad \text{for every } 0 < q < q_c.$$

In an exact scale-invariant lognormal chaos, the corresponding positive moment is infinite for  $q \geq q_c$ . If  $U$  is a nonempty regular set compactly contained in the domain, then in the usual bounded-domain subcritical setting

$$\mathbb{E}[M_\gamma(U)^{-s}] < \infty \quad \text{for every } s > 0.$$

*Proof.* There are two different issues: large mass and very small mass. The positive moments are controlled by the local scaling exponent  $\xi(q)$ ; the negative moments come from a two-branch smoothing recursion. We prove the two statements separately.

**1. A local scaling lemma.** Suppose that for some  $p > 0$  the  $p$ -th moment of the mass of one fixed unit-size comparison cube is finite. Then for every cube  $Q_r$  of side  $r$  whose fixed dilate stays in the domain,

$$\mathbb{E}[M_\gamma(Q_r)^p] \leq C_p r^{\xi(p)}. \tag{17.9}$$

To see the exponent, rescale  $x = x_0 + ru$ . On  $Q_r$  the covariance is

$$\log r^{-1} + \log |u - v|^{-1} + O(1).$$

Compare this field, by Kahane's inequality, with the sum  $\Omega_r + \tilde{X}(u)$ , where  $\Omega_r$  is an independent Gaussian of variance  $\log r^{-1}$  and  $\tilde{X}$  is a fixed unit-scale log-correlated field. The  $O(1)$  covariance discrepancy is removed in either direction by adding an independent Gaussian of bounded variance. Such an addition merely multiplies the chaos by a lognormal scalar, whose moments of every real order are finite. Hence the two  $p$ -th moments are comparable up to a constant independent of  $r$ . The Jacobian contributes  $r^{dp}$ , while

$$\mathbb{E} \exp \left( p\gamma\Omega_r - p\frac{\gamma^2}{2} \text{Var} \Omega_r \right) = r^{-\frac{\gamma^2}{2}p(p-1)}.$$

Multiplying the two factors gives (17.9).

This argument is valid for  $0 < p < 1$  as well as  $p > 1$ : the direction of Kahane's inequality reverses when  $x^p$  becomes concave, but the bounded independent Gaussian can be added to the opposite field, yielding the required upper bound with another constant.

**2. Positive moments below  $q_c$ .** For  $0 < q \leq 1$ , finiteness follows immediately from concavity and  $\mathbb{E}M_\gamma(K) = |K|$ . We now induct on the integer interval containing  $q$ . Assume all moments of order at most  $q - 1$  that are below  $q_c$  have already been proved finite, and put  $p = q - 1 > 0$ .

At a regularization scale  $\varepsilon$ , use  $M_\varepsilon(K)^q = M_\varepsilon(K)^p \int_K M_\varepsilon(dx)$  and tilt the field by the density at  $x$ . Girsanov's formula gives

$$\mathbb{E}[M_\varepsilon(K)^q] = \int_K \mathbb{E} \left[ \left( \int_K e^{\gamma^2 \text{Cov}(X_\varepsilon(x), X_\varepsilon(y))} M_\varepsilon(dy) \right)^p \right] dx. \quad (17.10)$$

The chaos on the right has the original law; only the deterministic singular factor has appeared.

Decompose  $K$  into dyadic annuli around  $x$ . Up to a harmless constant, the inner integral in (17.10) is bounded by

$$\sum_{n \geq 0} Z_n, \quad Z_n = 2^{n\gamma^2} M_\varepsilon(B(x, C2^{-n})).$$

If  $0 < p \leq 1$ , subadditivity gives  $(\sum Z_n)^p \leq \sum Z_n^p$ . If  $p > 1$ , then for every  $\beta > 0$ , Holder's inequality with weights gives

$$\left( \sum_{n \geq 0} Z_n \right)^p \leq C_{p,\beta} \sum_{n \geq 0} 2^{\beta n} Z_n^p. \quad (17.11)$$

Indeed apply Holder to  $Z_n = (2^{\beta n/p} Z_n) 2^{-\beta n/p}$ .

By the induction hypothesis and the local scaling lemma,

$$\sup_{\varepsilon} \mathbb{E}[Z_n^p] \leq C_p 2^{n\gamma^2 p} 2^{-n\xi(p)} = C_p 2^{-n\Delta_p},$$

where

$$\Delta_p = \xi(p) - \gamma^2 p = p \left( d - \frac{\gamma^2}{2}(p+1) \right) = p \left( d - \frac{\gamma^2 q}{2} \right). \quad (17.12)$$

The assumption  $q < q_c$  is exactly  $\Delta_p > 0$ . Thus the annular series converges; for  $p > 1$  choose  $0 < \beta < \Delta_p$  in (17.11). Equation (17.10) now gives a bound on  $\mathbb{E}[M_\varepsilon(K)^q]$  independent of  $\varepsilon$ . Fatou's lemma and the  $L^1$  convergence from the preceding theorem yield  $\mathbb{E}[M_\gamma(K)^q] < \infty$ . Induction over the intervals  $(0, 1], (1, 2], (2, 3], \dots$  proves every  $q < q_c$ .

**3. Sharpness in an exact scale-invariant model.** Here “exact scale invariant” means that a cube  $Q$  has dyadic descendants  $Q_i$  for which, after rescaling,

$$M(Q_i) \stackrel{d}{=} r^d \exp\left(\gamma \Omega_r - \frac{\gamma^2}{2} \log r^{-1}\right) Z_i,$$

where  $r$  is the side-length ratio,  $\Omega_r \sim N(0, \log r^{-1})$ , and each  $Z_i$  has the law of  $M(Q)$ ; the fine fields in two separated descendants are nondegenerate simultaneously with positive probability. Consequently, if the  $q$ -th moment were finite,

$$\mathbb{E}[M(Q_i)^q] = r^{\xi(q)} \mathbb{E}[M(Q)^q]. \quad (17.13)$$

Partition  $Q$  into  $r^{-d}$  congruent first-generation cubes. For  $q > 1$ ,

$$M(Q)^q \geq \sum_i M(Q_i)^q,$$

with strict inequality on the event that at least two descendants have positive mass. Taking expectations and using (17.13) gives

$$1 > r^{\xi(q)-d}$$

after division by the supposed finite positive moment. If  $q > q_c$ , then  $\xi(q) < d$ , so the right side exceeds one for  $r < 1$ , a contradiction. If  $q = q_c$ , then  $\xi(q) = d$ , and strict inequality reads  $1 > 1$ , again impossible. Thus the threshold is sharp in the exact model.

**4. A two-branch lemma for small masses.** We first work in a standard star-scale comparison chaos on a cube. Choose two well-separated daughter cubes. The independent fine-scale noises in these cubes and the coarse Gaussian field give a lower bound in law of the form

$$Z \geq c(W_1 Z_1 + W_2 Z_2), \quad (17.14)$$

where  $Z$  is the total mass,  $Z_1, Z_2$  are independent copies of a fixed positive subcube mass, independent of the positive lognormal vector  $(W_1, W_2)$ . The weights have Gaussian lower tails: for some  $c_1, c_2 > 0$ ,

$$\mathbb{P}(W_i \leq e^{-u}) \leq c_1 e^{-c_2 u^2} \quad (u \geq 1). \quad (17.15)$$

Subcritical nondegeneracy and the independent-scale zero-one argument imply  $Z_i > 0$  almost surely.

Let  $\phi(t) = \mathbb{E}[e^{-tZ}]$ . From (17.14), conditioning on the weights and using the two independent fine copies gives

$$\phi(t) \leq \mathbb{E}[\phi(ctW_1)\phi(ctW_2)].$$

Set  $a = 3/4$ . On the event  $W_1, W_2 \geq t^{-1/4}$ , monotonicity of  $\phi$  gives  $\phi(ctW_i) \leq \phi(ct^a)$ , while (17.15) shows that the complementary event has probability at most  $Ce^{-c(\log t)^2}$ . Hence

$$\phi(t) \leq \phi(ct^{3/4})^2 + Ce^{-c(\log t)^2}. \quad (17.16)$$

Since  $Z > 0$  almost surely,  $\phi(t) \rightarrow 0$ . Iterating (17.16) now gives, for every  $m > 0$ ,

$$\phi(t) \leq C_m t^{-m} \quad (t \geq 1). \quad (17.17)$$

For completeness, write  $t = e^L$ . Once  $L_0$  is large enough that  $\phi(e^{L_0}) < 1/4$ , the recursion at logarithmic scales has the form  $u(L) \leq u(3L/4 - C)^2 + e^{-cL^2}$ . After  $n$  iterations the first term has been squared  $2^n$  times, while the argument has decreased only by the factor  $(3/4)^n$ ; the Gaussian



error is smaller than every power of  $e^{-L}$ . Taking  $n$  of order  $\log L$  proves (17.17). This is the small-mass bootstrap: two independent opportunities for mass turn any initial decay of the Laplace transform into super-polynomial decay.

Finally, for  $s > 0$ ,

$$z^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-tz} dt.$$

Tonelli's theorem and (17.17) therefore give  $\mathbb{E}[Z^{-s}] < \infty$  for every  $s > 0$ .

For a general bounded-domain log-correlated field, restrict to a regular cube inside  $U$ . Its covariance differs by  $O(1)$  from that of the comparison chaos. Add an independent bounded-variance Gaussian to the comparison field so that its covariance dominates. Kahane's inequality applied to the convex function  $x \mapsto e^{-tx}$  bounds the Laplace transform of the original cube mass by an average of the comparison transform at a lognormal multiple of  $t$ . Because (17.17) holds to arbitrary polynomial order and a lognormal random variable has negative moments of every order, the same super-polynomial bound holds for the original cube. Since  $M_\gamma(U)$  dominates the mass of that cube, all negative moments of  $M_\gamma(U)$  are finite as claimed.  $\square$

*Remark 1.8* (How to remember the positive-moment threshold). The rooted measure inserts one singular factor  $|x - y|^{-\gamma^2}$ . If the remaining power is  $p = q - 1$ , a ball of radius  $r$  contributes at scale  $r^{\xi(p)}$ . Summability is therefore the single inequality  $\xi(p) > \gamma^2 p$ , which simplifies to  $q < 2d/\gamma^2$ .

**Theorem 1.9** (Kahane convexity inequality).

Let  $(S, \sigma)$  be a finite measure space and let  $X, Y$  be centered Gaussian fields indexed by  $S$  such that

$$K_X(s, t) := \text{Cov}(X(s), X(t)) \leq K_Y(s, t) := \text{Cov}(Y(s), Y(t)) \quad (s, t \in S).$$

Assume first that the fields are regular enough that the random variables

$$M_X = \int_S e^{X(s) - K_X(s, s)/2} \sigma(ds), \quad M_Y = \int_S e^{Y(s) - K_Y(s, s)/2} \sigma(ds)$$

are defined and finite almost surely. If  $F : [0, \infty) \rightarrow \mathbb{R}$  is convex and  $\mathbb{E}|F(M_X)| + \mathbb{E}|F(M_Y)| < \infty$ , then

$$\mathbb{E}F(M_X) \leq \mathbb{E}F(M_Y).$$

If  $F$  is concave, the inequality is reversed. The same conclusion holds for subcritical GMC masses obtained as limits of regularizations, provided the corresponding families  $F(M_{X, \varepsilon})$  and  $F(M_{Y, \varepsilon})$  are uniformly integrable.

*Proof.* We first prove the statement for finitely many points; approximation then gives the integral version used for GMC. Let  $X = (X_1, \dots, X_m)$  and  $Y = (Y_1, \dots, Y_m)$  be independent centered Gaussian vectors with covariance matrices  $C^X, C^Y$  satisfying  $C_{ij}^X \leq C_{ij}^Y$  for every  $i, j$ . Let  $a_i \geq 0$  and set

$$Z_i(t) = \sqrt{1-t} X_i + \sqrt{t} Y_i, \quad W_i(t) = a_i \exp\left(Z_i(t) - \frac{1}{2} \mathbb{E} Z_i(t)^2\right), \quad S_t = \sum_i W_i(t).$$

For a  $C^2$  test function  $F$  with bounded derivatives, set  $C_t = (1-t)C^X + tC^Y$  and  $\Phi_t(z) = F(\sum_k a_k e^{z_k - C_t(k, k)/2})$ . Gaussian integration by parts applied to the centered vector  $Z(t)$  gives the covariance-derivative identity

$$\frac{d}{dt} \mathbb{E} \Phi_t(Z(t)) = \frac{1}{2} \sum_{i, j} \dot{C}_t(i, j) \mathbb{E}[\partial_{ij} \Phi_t(Z(t))] + \mathbb{E}[\partial_t \Phi_t(Z(t))].$$

Substituting the explicit  $t$ -dependence of the diagonal counterterm gives

$$\frac{d}{dt}\mathbb{E}F(S_t) = \frac{1}{2} \sum_{i,j} (C_{ij}^Y - C_{ij}^X) \mathbb{E} \left[ \frac{\partial^2}{\partial z_i \partial z_j} F \left( \sum_k a_k e^{z_k - C_t(k,k)/2} \right) \right]_{z=Z(t)}$$

minus the derivative of the variance counterterms. Compute the mixed derivative. For  $i \neq j$  it equals  $F''(S_t)W_i(t)W_j(t)$ ; for  $i = j$  it equals  $F''(S_t)W_i(t)^2 + F'(S_t)W_i(t)$ . The derivative of  $-C_t(i,i)/2$  cancels exactly the  $F'(S_t)W_i(t)$  diagonal term. Hence

$$\frac{d}{dt}\mathbb{E}F(S_t) = \frac{1}{2} \sum_{i,j} (C_{ij}^Y - C_{ij}^X) \mathbb{E} [F''(S_t)W_i(t)W_j(t)].$$

All factors  $W_i W_j$  are nonnegative. If  $F$  is convex then  $F'' \geq 0$ , so the derivative is nonnegative; if  $F$  is concave it is nonpositive. Integrating from  $t = 0$  to 1 gives the desired comparison for finite weighted sums.

For a compactly supported reference measure  $\sigma$ , approximate the integral by positive Riemann sums and approximate the two Gaussian fields jointly at the same sampling points. The finite-dimensional inequality applies at every stage. Truncate an unbounded convex/concave test function by  $C^2$  functions with the permitted growth, use the moment bounds that make the expectations finite, and pass first through the spatial approximation and then through the truncation by monotone or dominated convergence. Finally approximate nonsmooth convex functions from above and below by smooth convex functions. This proves Kahane's convexity/comparison inequality in the form used throughout the volume.  $\square$

## Chapter 2

# The GFF and Liouville area and boundary length

### 2.1 GFF multiplicative chaos

**Theorem 2.1** (Circle-average construction of LQG area).

Let  $h$  be a zero-boundary GFF on a planar domain  $D$ , or a whole-plane GFF with a fixed additive normalization, and let  $0 < \gamma < 2$ . For circle averages  $h_\varepsilon(z)$  define on compact subsets of  $D$

$$\mu_{h,\varepsilon}(dz) = \exp\left(\gamma h_\varepsilon(z) - \frac{\gamma^2}{2} \text{Var } h_\varepsilon(z)\right) dz.$$

Then  $\mu_{h,\varepsilon}$  converges locally in  $L^1$  (hence in probability) to a nonzero random Radon measure  $\mu_h$ . Since

$$\text{Var } h_\varepsilon(z) = \log \varepsilon^{-1} + \log C_D(z) + o(1)$$

for the zero-boundary GFF, this is equivalent, after multiplication by the deterministic conformal-radius factor  $C_D(z)^{-\gamma^2/2}$ , to the customary normalization

$$\varepsilon^{\gamma^2/2} e^{\gamma h_\varepsilon(z)} dz.$$

The measures obtained on compact exhaustions are consistent and define the  $\gamma$ -LQG area measure on  $D$ .

*Proof.* This is a specialization of the subcritical GMC theorem proved in Chapter 1, but we verify its hypotheses and the passage from compact sets to the full domain.

**1. Local covariance form.** For  $K \Subset D$ , the GFF Green kernel has the decomposition

$$G_D(z, w) = \log \frac{1}{|z - w|} + g_D(z, w),$$

where  $g_D$  is smooth on a neighborhood of  $K \times K$ . Circle averaging in either variable therefore gives exactly the covariance estimates used in the circle-average clause of the subcritical GMC theorem: uniformly for  $z, w \in K$ ,

$$\text{Var } h_\varepsilon(z) = \log \varepsilon^{-1} + O_K(1), \quad \text{Cov}(h_\varepsilon(z), h_\delta(w)) \leq \log \frac{1}{|z - w| \vee \varepsilon \vee \delta} + O_K(1),$$

and away from the diagonal the regularized covariance converges uniformly to  $G_D(z, w)$ .

**2. Check the subcritical threshold.** The reference measure is planar Lebesgue measure, which has dimension parameter  $d = 2$  in Chapter 1. The condition of that theorem is

$$\gamma^2 < 2d = 4,$$

which is precisely  $0 < \gamma < 2$ . Hence on every  $K \Subset D$  and for every bounded continuous  $f$  supported in  $K$ ,

$$\int f(z) \exp\left(\gamma h_\varepsilon(z) - \frac{\gamma^2}{2} \text{Var } h_\varepsilon(z)\right) dz$$

converges in  $L^1$  to a limit which is independent of the particular circle radii used to approach zero. The Riesz–Markov step in Chapter 1 gives a nonzero random Radon measure on  $K$ .

**3. Consistency on an exhaustion.** Choose  $K_1 \Subset K_2 \Subset \dots \Subset D$  with union  $D$ . If  $f$  is supported in  $K_n$ , the regularized integrals used for  $K_n$  and  $K_{n+1}$  are literally the same random variables for all sufficiently small  $\varepsilon$ . Their  $L^1$  limits therefore agree. The local limiting measures consequently patch uniquely to a random Radon measure  $\mu_h$  on  $D$ .

**4. Compare normalizations.** For the zero-boundary GFF,

$$G_D(z, w) = \log \frac{1}{|z - w|} + \log C_D(z) + o(1) \quad \text{as } w \rightarrow z,$$

with  $C_D(z)$  the conformal radius in the convention fixed in the preliminaries. Thus

$$e^{-\frac{\gamma^2}{2} \text{Var } h_\varepsilon(z)} = \varepsilon^{\gamma^2/2} C_D(z)^{-\gamma^2/2} (1 + o_K(1)).$$

Multiplication by this continuous deterministic factor commutes with local  $L^1$  convergence. This proves the equivalence with the customary LQG normalization. The whole-plane normalized GFF has the same local logarithmic decomposition, so the identical argument applies after fixing its additive constant. □

*Remark 2.2.* No new paper-scale estimate is hidden here: the difficult thick-point truncation and cross-regularization Cauchy argument are already owned by the general subcritical GMC theorem in Chapter 1.

**Theorem 2.3** (Regularization independence for the planar GFF).

*Let  $R_\varepsilon h$  and  $\tilde{R}_\varepsilon h$  be two admissible regularizations of the same planar GFF on  $D$ . Assume on every  $K \Subset D$  that their variances have the same logarithmic divergence, their mixed covariances obey the uniform logarithmic upper bounds of Chapter 1, and their mixed covariance converges to  $G_D(z, w)$  uniformly for  $|z - w| \geq r > 0$ . Then for  $0 < \gamma < 2$  the two renormalized chaos measures converge locally in  $L^1$  to the same  $\mu_h$ .*

*Circle averages and smooth convolutions satisfy these admissibility conditions by the covariance estimates proved in Chapter 1. Any spectral/orthogonal-projection truncation for which the same three covariance estimates are verified is therefore covered as well. In this form the theorem states exactly the approximation class controlled by the proof, rather than silently assuming a local spectral asymptotic.*

*Proof.* Fix  $K \Subset D$  and a bounded continuous test function  $f$  supported in  $K$ . Apply the cross-regularization part of the Chapter 1 GMC proof with the first approximation in the role of  $X_\varepsilon^\theta$  and the second in the role of  $X_\delta^\theta$ . That argument used only four facts: the logarithmic variance

asymptotic, the one-point tilted covariance estimate, the two-point logarithmic upper bound, and off-diagonal convergence of the mixed covariance. These are exactly the admissibility assumptions here.

For completeness, let  $I_\varepsilon(f)$  and  $\tilde{I}_\delta(f)$  denote the two renormalized masses. Choose the common auxiliary circle-average field used to define the good-point event  $G^\alpha$ . The one-point tilt shows that deleting  $(G^\alpha)^c$  costs arbitrarily little in  $L^1$ , uniformly in both regularizations. Under the two-point tilt, the common good event improves the diagonal singularity from  $|z - w|^{-\gamma^2}$  to an integrable power strictly below  $|z - w|^{-2}$  because  $\gamma < 2$ . Thus the truncated masses are uniformly  $L^2$ -bounded. Splitting the mixed second moment into  $|z - w| \leq r$  and  $|z - w| > r$ , the good-point bound controls the first region uniformly, while mixed-covariance convergence gives dominated convergence on the second. Consequently

$$\lim_{\varepsilon, \delta \downarrow 0} \mathbb{E} |I_\varepsilon(f) - \tilde{I}_\delta(f)| = 0.$$

The two regularizations therefore have the same  $L^1$  limit against every test function in a countable dense subset of  $C_c(D)$ , hence the same random Radon measure.

For circle averages and smooth convolution, the required covariance bounds and off-diagonal convergence were verified explicitly in Chapter 1, so they lie in the admissible class. For a spectral or orthogonal-projection cutoff there is nothing further to prove at the GMC level once its covariance kernel has been checked against the same criterion: the cross-Cauchy estimate above then applies verbatim. We therefore separate the analytic covariance verification for a particular cutoff from the probabilistic regularization-independence theorem, avoiding any hidden appeal to a local Weyl theorem. □

*Remark 2.4.* This theorem deliberately phrases “standard approximation” through a checkable covariance criterion. That prevents regularization independence from being used as an unnamed universality principle outside the class actually controlled by the GMC proof.

**Theorem 2.5** (Conformal covariance of LQG area).

*For a conformal map  $\phi : \tilde{D} \rightarrow D$  and*

$$Q = \frac{2}{\gamma} + \frac{\gamma}{2}, \quad \tilde{h} = h \circ \phi + Q \log |\phi'|,$$

*the LQG area measures satisfy*

$$\mu_{\tilde{h}}(A) = \mu_h(\phi(A))$$

*for Borel  $A \subset \tilde{D}$ .*

*Proof.* It is useful to keep track of every power of  $|\phi'|$ . Fix  $z \in \tilde{D}$ , write  $w = \phi(z)$ , and let  $r_\varepsilon(z)$  be the Euclidean radius of the image under  $\phi$  of a circle of radius  $\varepsilon$  around  $z$ . Conformality gives

$$r_\varepsilon(z) = |\phi'(z)|\varepsilon(1 + o(1))$$

uniformly on compact subsets. Regularization independence allows us to use these image circles in place of exact Euclidean circles.

At the regularized level,

$$\varepsilon^{\gamma^2/2} e^{\gamma \tilde{h}_\varepsilon(z)} dz = \varepsilon^{\gamma^2/2} e^{\gamma h_{r_\varepsilon}(w)} |\phi'(z)|^{\gamma Q + o(1)} dz.$$

Change variables  $w = \phi(z)$ , so  $dz = |\phi'(z)|^{-2}dw$ , and replace  $\varepsilon$  by  $r_\varepsilon/|\phi'(z)|$ . The total deterministic power of the derivative is therefore

$$|\phi'|^{-\gamma^2/2} |\phi'|^{\gamma Q} |\phi'|^{-2} = |\phi'|^{\gamma Q - 2 - \gamma^2/2} = 1,$$

because  $\gamma Q = 2 + \gamma^2/2$ . Thus the pullback regularized measure differs from the regularized measure of  $h$  only by an error tending to one locally in probability.

Let  $f$  be a continuous compactly supported test function on  $\tilde{D}$ . Integrating the preceding identity and using the  $L^1_{\text{loc}}$  convergence of subcritical GMC gives

$$\int_{\tilde{D}} f(z) \mu_{\tilde{h}}(dz) = \int_D f(\phi^{-1}(w)) \mu_h(dw).$$

A monotone-class argument extends this identity from test functions to Borel sets, proving  $\mu_{\tilde{h}}(A) = \mu_h(\phi(A))$ .  $\square$

**Example 2.6** (Why  $Q = 2/\gamma + \gamma/2$ ). A conformal map contributes two Euclidean Jacobian powers. GMC renormalization contributes  $\gamma^2/2$ , so the field shift must contribute  $2 + \gamma^2/2$ . Dividing by  $\gamma$  gives  $Q = 2/\gamma + \gamma/2$ .

**Theorem 2.7** (LQG boundary length and conformal covariance).

*For a free-boundary GFF representative with a fixed additive normalization on a boundary arc and  $0 < \gamma < 2$ , the boundary chaos*

$$\nu_{h,\varepsilon}(dx) = \varepsilon^{\gamma^2/4} e^{\frac{\gamma}{2} h_\varepsilon(x)} dx$$

*converges to the LQG boundary-length measure. Under the same coordinate change  $h \mapsto h \circ \phi + Q \log |\phi'|$ , boundary length is conformally invariant in the pushforward sense.*

*Proof.* Along a smooth boundary arc, the free-boundary GFF has the local covariance

$$\text{Cov}(h(x), h(y)) = 2 \log \frac{1}{|x - y|} + O(1).$$

Consequently the field  $(\gamma/2)h$  has logarithmic covariance coefficient  $\gamma^2/2$ . One-dimensional GMC is subcritical precisely when  $\gamma^2/2 < 2$ , i.e.  $\gamma < 2$ . Applying the subcritical GMC construction to boundary semicircle averages therefore gives a nontrivial random Radon measure and the normalization

$$\exp\left(-\frac{1}{2} \frac{\gamma^2}{4} \text{Var}(h_\varepsilon(x))\right) \asymp \varepsilon^{\gamma^2/4}.$$

Changing the harmless bounded part of the variance only multiplies the regularization by a continuous deterministic factor, which is absorbed by the regularization-independence theorem.

For conformal covariance, let  $y = \phi(x)$  and let  $r_\varepsilon = |\phi'(x)|\varepsilon(1 + o(1))$  be the image semicircle scale. At the regularized level the three derivative factors are

$$\underbrace{|\phi'|^{-\gamma^2/4}}_{\text{change of cutoff}} \underbrace{|\phi'|^{\gamma Q/2}}_{\text{field shift}} \underbrace{|\phi'|^{-1}}_{dx=|\phi'|^{-1}dy}.$$

Their product is one because  $\gamma Q/2 = 1 + \gamma^2/4$ . Testing against continuous functions and passing to the GMC limit gives

$$\nu_{h \circ \phi + Q \log |\phi'|}(A) = \nu_h(\phi(A)).$$

This is exactly the pushforward covariance claimed in the theorem.  $\square$

*Remark 2.8* (Dependency status). The coordinate-change calculation above is complete once the boundary GMC construction and regularization independence are available. Those analytic inputs inherit the unresolved publication gate of the general subcritical GMC construction earlier in this volume; this theorem introduces no additional paper-scale assumption.

## Chapter 3

# Multifractality and tails

### 3.1 Multifractality

**Theorem 3.1** (GFF thick points and the GMC multifractal spectrum).

Let  $h$  be a planar GFF in a domain  $D$ , normalized so that  $\text{Var } h_\varepsilon(z) = \log \varepsilon^{-1} + O(1)$  on compact subsets. For  $0 < \gamma < 2$  let  $\mu_\gamma$  be the associated GMC measure. If

$$\frac{h_\varepsilon(z)}{\log \varepsilon^{-1}} \longrightarrow \alpha, \quad |\alpha| \leq 2,$$

then, at every point for which the residual angular chaos has subpower fluctuations,

$$\frac{\log \mu_\gamma(B_\varepsilon(z))}{\log \varepsilon} \longrightarrow 2 + \frac{\gamma^2}{2} - \gamma\alpha.$$

Moreover the set of  $\alpha$ -thick points has Hausdorff dimension  $2 - \alpha^2/2$  almost surely. Consequently the local-dimension spectrum of  $\mu_\gamma$  is obtained by the change of variables  $q = 2 + \gamma^2/2 - \gamma\alpha$  from the parabola  $2 - \alpha^2/2$ .

*Proof.* In the normalization of this volume put  $\tilde{h} = h/\sqrt{2\pi}$  when comparing with Volume VI. The theorem SM-06.06.02, whose lower-bound branching estimate is physically proved there, gives

$$\dim_H \left\{ z : \frac{h_\varepsilon(z)}{\log \varepsilon^{-1}} \rightarrow \alpha \right\} = 2 - \frac{\alpha^2}{2}, \quad |\alpha| \leq 2,$$

and emptiness outside the thick-point range. We therefore do not maintain a second Frostman construction here.

Let  $t = \log \varepsilon^{-1}$ . Decompose the field in  $B_\varepsilon(z)$  into the circle average and the zero-circle-average remainder. Scale covariance of GMC gives

$$\mu_\gamma(B_\varepsilon(z)) = \varepsilon^{2+\gamma^2/2} e^{\gamma h_\varepsilon(z)} Z_{z,\varepsilon}, \quad (17.7)$$

where  $Z_{z,\varepsilon}$  is the GMC mass of the rescaled zero-average field on a unit-size set, multiplied by a deterministic covariance factor bounded above and below on compact subsets.

The subcritical moment theorem of Chapter 1 gives positive moments of  $Z_{z,\varepsilon}$  in a neighborhood of the origin and all negative moments needed here, uniformly for  $z$  in a compact set and dyadic  $\varepsilon$ . For every  $\eta > 0$ , Markov's inequality therefore gives summable bounds for

$$Z_{z,2^{-n}} > 2^{\eta n} \quad \text{and} \quad Z_{z,2^{-n}} < 2^{-\eta n}$$



when  $z$  is sampled from any of the finite-energy thick-point measures constructed in SM-06.06.02. Fubini and Borel–Cantelli imply  $\log Z_{z,2^{-n}} = o(n)$  for almost every point of those measures. The circle-average/GMC comparison between adjacent scales extends this to arbitrary  $\varepsilon \downarrow 0$ .

At an  $\alpha$ -thick point satisfying the subpower remainder property, take logarithms in (17.7) and divide by  $\log \varepsilon < 0$  to obtain

$$\frac{\log \mu_\gamma(B_\varepsilon(z))}{\log \varepsilon} \longrightarrow 2 + \frac{\gamma^2}{2} - \gamma\alpha.$$

Combining this with the exact earlier dimension  $2 - \alpha^2/2$  gives the stated parabolic local-dimension spectrum under the change of variables  $q = 2 + \gamma^2/2 - \gamma\alpha$ . No thick-point theorem stronger than the one proved in Volume VI is used.  $\square$

**Theorem 3.2** (Subcritical GMC positive and negative moments).

*Let  $M_\gamma$  be the planar Lebesgue-base subcritical GMC of this volume,  $0 < \gamma < 2$ , on a bounded domain, and let  $U \subseteq D$  be a nonempty open set containing a closed Euclidean ball. Then*

$$\mathbb{E}[M_\gamma(U)^q] < \infty \quad (0 < q < 4/\gamma^2),$$

and

$$\mathbb{E}[M_\gamma(U)^{-s}] < \infty \quad (s > 0).$$

*For the exact scale-invariant comparison chaos, the positive-moment threshold  $4/\gamma^2$  is sharp.*

*Proof.* This is the planar specialization  $d = 2$  of SM-17.01.03. The positive moments there are proved by the rooted-moment identity and the local scaling exponent

$$\xi(q) = 2q - \frac{\gamma^2}{2}q(q-1),$$

whose annular summability condition is exactly  $q < 4/\gamma^2$ . Sharpness at and above the threshold is proved there for the exact scale-invariant model by the first-generation scaling identity. The same theorem proves all negative moments for every nonempty regular set by a two-branch smoothing recursion for the Laplace transform. Since the present  $U$  contains a regular comparison ball, monotonicity of the mass gives the displayed negative-moment conclusion. No additional tail theorem is being invoked here.  $\square$

## Chapter 4

# Critical Gaussian multiplicative chaos

Criticality is the point where the mean-one multiplicative-chaos martingale still has the formal subcritical normalization but loses uniform integrability. The correct replacement is the derivative martingale. The purpose of this chapter is to prove, rather than merely name, the three pieces needed later in critical LQG: extinction of the naive chaos, construction of the derivative measure, and Seneta–Heyde/approximation universality.

Throughout this chapter we first work with an exact star-scale decomposition  $X_t$  on a bounded set  $A \subseteq \mathbb{R}^d$ :

$$\mathbf{E}[X_t(x)X_s(y)] = K_{t \wedge s}(x, y), \quad K_t(x, x) = t + O(1),$$

with finite-range independence for scale increments. We put  $\gamma_c = \sqrt{2d}$ . The local comparison theorem from Chapter 2 transfers the final statements to the planar GFF and to the admissible convolution/circle-average cutoffs used in this series.

**Definition 4.1** (Critical derivative chaos). Let  $X_t$  be a scale decomposition with  $\text{Var } X_t(x) = t + O(1)$  and critical parameter  $\gamma_c = \sqrt{2d}$ . Formally differentiating the subcritical Wick weight with respect to  $\gamma$  at  $\gamma = \gamma_c$  produces the signed density

$$(\gamma_c t - X_t(x)) e^{\gamma_c X_t(x) - \frac{1}{2} \gamma_c^2 t} dx.$$

The *critical derivative chaos* is the positive Radon measure obtained by the barrier-truncated derivative martingales defined below, followed by removal of the barrier. The barrier is part of the construction: the untruncated finite-scale derivative density is signed.

*Worked example 4.2* (Why the naive derivative density has zero mean). Let  $X \sim N(0, t)$ . Gaussian differentiation gives

$$\mathbb{E}[e^{\gamma X}] = e^{\gamma^2 t/2}, \quad \mathbb{E}[X e^{\gamma X}] = \gamma t e^{\gamma^2 t/2}.$$

Consequently, at  $\gamma = \gamma_c$ ,

$$\mathbb{E}\left[(\gamma_c t - X) e^{\gamma_c X - \gamma_c^2 t/2}\right] = \gamma_c t - \gamma_c t = 0.$$

So the formal derivative of the mean-one subcritical martingale cannot itself be a positive mean-one measure. The positive gap  $R_t^\beta = \beta + \gamma_c t - X_t$  and the event that this gap has stayed positive are exactly what turn the derivative object below into a nonnegative martingale with controlled expectation.

## 4.1 The truncated derivative martingale

For  $\beta > 0$  and  $x \in A$ , define the barrier time and gap process

$$\tau_x^\beta := \inf\{s \geq 0 : X_s(x) - \gamma_c s > \beta\}, \quad R_s^\beta(x) := \beta + \gamma_c s - X_s(x).$$

Thus  $\tau_x^\beta > t$  is exactly the event that  $R_s^\beta(x) > 0$  for  $0 \leq s \leq t$ . Set

$$Z_t^\beta(dx) := R_t^\beta(x) e^{\gamma_c X_t(x) - dt} \mathbf{1}_{\{\tau_x^\beta > t\}} dx, \quad M_t^\beta(dx) := e^{\gamma_c X_t(x) - dt} \mathbf{1}_{\{\tau_x^\beta > t\}} dx.$$

**Lemma 4.3** (One-spine/Bessel identity). *For bounded nonnegative  $f$  and every bounded functional  $F$  of the radial path up to time  $t$ ,*

$$\mathbf{E} \int_A f(x) F((R_s^\beta(x))_{s \leq t}) Z_t^\beta(dx) = \int_A f(x) \beta \mathbf{E}_\beta^{(3)}[F((\mathcal{R}_s)_{s \leq t})] dx,$$

where  $\mathcal{R}$  is a three-dimensional Bessel process started from  $\beta$ . Consequently  $Z_t^\beta(A)$  is a nonnegative martingale with  $\mathbf{E} Z_t^\beta(A) = \beta|A|$ .

*Proof.* At a fixed spatial point,  $B_s := X_s(x) - X_0(x)$  is Brownian motion up to the bounded initial covariance correction. Weighting by  $e^{\gamma_c B_t - \gamma_c^2 t/2}$  is the Cameron–Martin transform that changes  $B_s$  into  $B_s + \gamma_c s$ . Hence under the exponential size bias the gap  $\beta + \gamma_c s - X_s(x)$  becomes Brownian motion started from  $\beta$  and killed at zero. Multiplication by the terminal factor  $R_t^\beta/\beta$  is the Doob  $h$ -transform with the positive harmonic function  $h(r) = r$  for killed Brownian motion. Its generator is

$$\frac{1}{2} \partial_{rr} + \frac{1}{r} \partial_r,$$

which is the generator of BES(3). Integrating the pointwise identity over  $x$  proves the displayed formula. Taking  $F = 1$  gives constant expectation. The Markov property of the field increments and the same computation between times  $s < t$  give the martingale property.  $\square$

**Lemma 4.4** (Truncated uniform integrability). *For every compact  $A$  and  $\beta > 0$ ,  $Z_t^\beta(A)$  is uniformly integrable and therefore converges almost surely and in  $L^1$  to a finite random variable  $Z^\beta(A)$ . The limits are consistent under restriction to subsets and define a random Radon measure  $Z^\beta$ .*

*Proof.* Fix  $L > 1$  and stop a rooted path the first time its Bessel gap exceeds  $L$  or drops below  $L^{-1}$ . On the event that the spine remains in  $[L^{-1}, L]$ , the density in Lemma 4.3 is bounded by a deterministic multiple of the ordinary exponential martingale, and the finite-range scale decomposition makes the fine fields in disjoint  $e^{-s}$ -boxes independent after their branching time. The usual two-spine decomposition therefore gives

$$\sup_{t \geq 0} \mathbf{E} \left[ Z_t^\beta(A) ; Z_t^\beta(A) > R, \mathcal{R}_{[0,t]} \subset [L^{-1}, L] \right] \xrightarrow{R \rightarrow \infty} 0.$$

It remains to remove the radial truncation. For BES(3),

$$\mathbf{P}_\beta^{(3)} \left\{ \inf_{s \geq 0} \mathcal{R}_s < L^{-1} \right\} \leq \frac{1}{\beta L},$$

because  $1/\mathcal{R}_s$  is a nonnegative local martingale, while standard Gaussian estimates for the Bessel transition density imply

$$\sup_{t \geq 0} \mathbf{P}_\beta^{(3)} \left\{ \sup_{s \leq t} \mathcal{R}_s > L(1 + \sqrt{t}) \right\} \xrightarrow{L \rightarrow \infty} 0.$$

Apply the one-spine identity to these exceptional events. First choose  $L$  so that their contribution to  $\mathbf{E}Z_t^\beta(A)$  is uniformly small, then choose  $R$  for the bounded-spine piece. This is the de la Vallée–Poussin criterion in tail form, hence uniform integrability. Martingale convergence gives the claimed  $L^1$  limit. Consistency on a countable algebra of boxes and monotone extension yield the random measure.  $\square$

## 4.2 Derivative chaos and Seneta–Heyde normalization

**Lemma 4.5** (Killed-Brownian renewal asymptotic). *Let  $B$  be one-dimensional Brownian motion started at  $r > 0$  and  $\sigma_0 = \inf\{u : B_u = 0\}$ . Then*

$$q_u(r) := \mathbf{P}_r\{\sigma_0 > u\} = 2\Phi(r/\sqrt{u}) - 1 = \sqrt{\frac{2}{\pi}} \frac{r}{\sqrt{u}} + O\left(\frac{r^3}{u^{3/2}}\right)$$

uniformly for  $0 \leq r \leq u^{1/2-\eta}$ , for every fixed  $\eta > 0$ .

*Proof.* The reflection principle gives  $q_u(r) = \mathbf{P}\{|N(0, u)| < r\} = 2\Phi(r/\sqrt{u}) - 1$ . Taylor expansion of  $\Phi$  at zero gives the stated uniform remainder.  $\square$

**Lemma 4.6** (Conditional Seneta–Heyde estimate). *Fix  $\beta > 0$ , a compact  $A$ , and  $s = s(t)$  with  $s \rightarrow \infty$ ,  $t - s \rightarrow \infty$ , and  $s = o(t)$ . Then*

$$\sqrt{t-s} \mathbf{E}[M_t^\beta(A) \mid \mathcal{F}_s] = \sqrt{\frac{2}{\pi}} Z_s^\beta(A) + o_{L^1}(1).$$

Moreover,

$$\sqrt{t} M_t^\beta(A) - \sqrt{t} \mathbf{E}[M_t^\beta(A) \mid \mathcal{F}_s] \xrightarrow[t \rightarrow \infty]{\mathbf{P}} 0$$

if  $s = s(t)$  is chosen to increase sufficiently slowly.

*Proof.* Condition on the field through scale  $s$ . For a point with  $\tau_x^\beta > s$ , the future radial increment is Brownian motion started from  $R_s^\beta(x)$ , killed at zero. The exponential martingale has conditional mean one, hence

$$\mathbf{E}[M_t^\beta(dx) \mid \mathcal{F}_s] = e^{\gamma_c X_s(x) - ds} \mathbf{1}_{\{\tau_x^\beta > s\}} q_{t-s}(R_s^\beta(x)) dx.$$

Split the integral according to  $R_s^\beta \leq (t-s)^{1/2-\eta}$  and its complement. On the first part, Lemma 4.5 gives the desired constant times  $R_s^\beta e^{\gamma_c X_s - ds}$ . Under the derivative spine the complementary Bessel tail tends to zero, so Lemma 4.3 makes its contribution  $o_{L^1}(1)$ . This proves the first assertion.

For the fluctuation estimate, partition  $A$  into cubes of side  $e^{-s}$ . Given  $\mathcal{F}_s$ , scale increments descending from cubes at mutual distance  $> Ce^{-s}$  are independent. Expanding the conditional variance leaves only diagonal and neighboring pairs. On a surviving rooted path, impose the additional good event

$$(t-s)^{1/4} \leq R_s^\beta \leq (t-s)^{1/2-\eta}.$$

The lower cutoff suppresses the singular factor coming from the critical two-spine kernel, while the upper cutoff is the range of the renewal expansion. The two-spine branch-time integral is then bounded by

$$C(t-s)^{-1/2} \int_A R_s^\beta(x) e^{\gamma_c X_s(x) - ds} \mathbf{1}_{\{\tau_x^\beta > s\}} dx,$$

up to a term whose derivative-spine expectation tends to zero when the good event is removed. Multiplying by  $t$ , taking conditional expectation, and then choosing  $s = s(t) \uparrow \infty$  slowly gives

a conditional variance tending to zero in probability. Chebyshev's inequality proves the second assertion. This is precisely the place where finite-range decorrelation (or the weaker GFF scale-separation estimate) is needed; a one-spine first-moment calculation alone would not suffice.  $\square$

**Theorem 4.7** (Derivative chaos and Seneta–Heyde normalization).

At  $\gamma_c = \sqrt{2d}$  the derivative martingale converges to a nonzero atomless random measure  $M'$ . In the standard star-scale normalization,

$$\sqrt{t} M_t \xrightarrow[t \rightarrow \infty]{\mathbf{P}} \sqrt{\frac{2}{\pi}} M'$$

vaguely. In dimension two this gives the critical LQG area measure.

*Proof.* Lemmas 4.3–4.4 give, for every  $\beta$ , nonnegative limits  $Z^\beta$ . If  $\beta_1 < \beta_2$ , the two stopped martingales coincide until the lower barrier is hit and the strong Markov property gives the consistent comparison  $Z^{\beta_1}(A) \leq Z^{\beta_2}(A)$  after the deterministic normalization of the initial gap. Hence  $Z^\beta$  has an increasing barrier limit  $M'$ .

The limit is nonzero. Indeed  $\mathbf{E}Z^\beta(A) = \beta|A|$  by uniform integrability, so  $Z^\beta(A)$  is not almost surely zero. It has full support because the same argument applies to every rational ball. To exclude atoms, cover a compact set by  $e^{dn}$  boxes of diameter  $e^{-n}$ . The derivative-spine estimate and the scale decomposition give, for every  $q \in (0, 1)$ ,

$$\mathbf{E}[Z^\beta(Q)^q] \leq C_q |Q|^{\xi(q)} \quad \text{with } \xi(q) > 1$$

for sufficiently small  $q$ . Summing over the cover and using Markov plus Borel–Cantelli shows that the largest mass of a level- $n$  box tends to zero. Letting  $\beta \rightarrow \infty$  preserves non-atomicity, so  $M'$  is atomless.

Now apply Lemma 4.6. For fixed  $\beta$ , first let  $t \rightarrow \infty$  with the admissible slowly increasing  $s(t)$ . The conditional mean converges to  $\sqrt{2/\pi} Z_{s(t)}^\beta(A)$ , and the centered fluctuation vanishes in probability. Since  $Z_{s(t)}^\beta(A) \rightarrow Z^\beta(A)$  in  $L^1$ ,

$$\sqrt{t} M_t^\beta(A) \xrightarrow{\mathbf{P}} \sqrt{\frac{2}{\pi}} Z^\beta(A).$$

The probability that the untruncated and truncated masses differ is controlled by the maximum of  $X_s(x) - \gamma_c s$ . The standard logarithmic maximum bound for a log-correlated field implies

$$\lim_{\beta \rightarrow \infty} \sup_{t \geq t_0} \mathbf{P}\{M_t(A) \neq M_t^\beta(A)\} = 0$$

after first taking  $t_0 \rightarrow \infty$ ; the analogous derivative-spine estimate gives  $Z^\beta(A) \uparrow M'(A)$ . Therefore

$$\sqrt{t} M_t(A) \xrightarrow{\mathbf{P}} \sqrt{\frac{2}{\pi}} M'(A).$$

A countable convergence-determining family of compactly supported continuous functions upgrades this to vague convergence of measures. The local decomposition/comparison argument of Chapter 2 transfers the same proof to the planar GFF, where the scale increments have the weaker polynomial separation but the two-spine estimate remains summable.  $\square$

### 4.3 Extinction of the naive critical chaos

**Theorem 4.8** (Extinction of the unrenormalized critical chaos).

*For a star-scale approximation of a log-correlated field in dimension  $d$ , at  $\gamma_c = \sqrt{2d}$  the ordinary critical chaos martingale*

$$M_t(dx) = e^{\gamma_c X_t(x) - dt} dx$$

*converges almost surely vaguely to the zero measure. The same conclusion holds for the planar GFF critical approximation after the standard local comparison.*

*Proof.* For a bounded  $A$ ,  $M_t(A)$  is a nonnegative martingale and therefore converges almost surely to some  $M_\infty(A)$ . The Seneta–Heyde theorem just proved gives  $\sqrt{t} M_t(A) \rightarrow cM'(A)$  in probability, with  $0 < M'(A) < \infty$  on every nonempty open  $A$ . If  $M_\infty(A) > 0$  on an event of positive probability, then on that event  $\sqrt{t} M_t(A) \rightarrow \infty$  almost surely, contradicting convergence in probability to the finite random variable  $cM'(A)$ . Hence  $M_\infty(A) = 0$  almost surely.

Apply this to a countable algebra of rational boxes and use monotonicity to obtain vague convergence to zero. The GFF statement follows from the same local comparison used in the preceding theorem. Notice that this argument avoids the invalid shortcut “non-uniform integrability implies extinction”: non-uniform integrability by itself would not identify the almost-sure martingale limit.  $\square$

### 4.4 Approximation independence at criticality

**Lemma 4.9** (Critical comparison across two cutoffs). *Let  $X_\varepsilon^1, X_\varepsilon^2$  be two admissible approximations of the same log-correlated field on  $K \Subset D$ . Assume their covariance kernels have the same logarithmic singularity, uniformly bounded continuous remainders, and mixed covariances converging locally uniformly off the diagonal. Let  $M_{\varepsilon,\beta}'^i$  be the barrier-truncated derivative measures. Then for each fixed  $\beta$  and  $f \in C_c(K)$ ,*

$$M_{\varepsilon,\beta}'^1(f) - M_{\varepsilon,\beta}'^2(f) \xrightarrow[\varepsilon \downarrow 0]{L^1} 0$$

*after the common deterministic normalization of the cutoffs.*

*Proof.* Choose a mesoscopic radius  $r$  with  $\varepsilon \ll r \ll 1$ , and decompose each approximation into a common coarse field  $Y_r$  plus a fine Gaussian remainder. This is obtained by Gaussian interpolation of the two covariance matrices; the difference of the fine covariances is uniformly  $o_r(1)$  away from  $|x - y| \leq Cr$ , while the bounded covariance remainder contributes only a deterministic  $1 + o_r(1)$  factor.

On the common barrier event for  $Y_r$ , the derivative-spine uniform-integrability estimate of Lemma 4.4 permits differentiation under the Gaussian interpolation. Gaussian integration by parts expresses the derivative of the expectation of a Lipschitz test functional of the truncated masses as a double integral of the covariance difference times two first derivatives of the functional. The off-diagonal part is  $o_r(1)$ . On the diagonal strip  $|x - y| \leq Cr$ , the same two-spine estimate used in Lemma 4.6 bounds the integral by  $C r^{\eta_\beta}$  for some  $\eta_\beta > 0$ . Therefore the bounded-Lipschitz distance between the two truncated laws tends to zero as first  $\varepsilon \downarrow 0$  and then  $r \downarrow 0$ .

To upgrade from convergence in law to the displayed coupled  $L^1$  statement, realize both approximations together with the common coarse field and use the same interpolation for  $|M'^1 - M'^2| \wedge R$ . Uniform integrability of the truncated masses lets  $R \rightarrow \infty$ . This proves the claim.  $\square$

**Theorem 4.10** (Approximation independence at criticality).

*For the log-correlated Gaussian fields and convolution/circle-average approximations covered by the standard critical-chaos comparison theorem, the derivative-normalized critical chaos converges to a limit independent of the admissible approximation. After the deterministic normalization of the preceding theorem, the derivative, Seneta–Heyde, white-noise, and mollifier constructions agree. In particular this holds for the planar GFF.*

*Proof.* For fixed  $\beta$ , Lemma 4.9 identifies the limiting barrier-truncated derivative measures for any two admissible cutoffs. By the Bessel-spine identity,

$$\sup_i \mathbf{E}[M'^i(K) - M'_\beta{}^i(K)] \xrightarrow{\beta \rightarrow \infty} 0,$$

so the barrier can be removed uniformly. Thus the derivative limit is independent of the cutoff.

The Seneta–Heyde approximation agrees with the derivative measure by SM-17.04.02. White-noise cutoffs are the exact scale decomposition used above. Smooth mollifiers and planar GFF circle averages satisfy the covariance hypotheses of Lemma 4.9; on a compact exhaustion of the domain their limits therefore agree, and locality makes the restrictions consistent on overlaps. This gives one canonical critical GMC measure, independent of the admissible approximation and of the chosen local decomposition.  $\square$

## Chapter 5

# Logarithmic insertions and Seiberg bounds

### 5.1 Insertions

**Theorem 5.1** (Local Seiberg bound for one logarithmic insertion).

*If the field is modified by logarithmic singularities  $h(z) + \sum_i \alpha_i \log |z - z_i|^{-1}$ , then the corresponding GMC is obtained by multiplying the regular chaos by the deterministic power weights  $\prod_i |z - z_i|^{-\gamma \alpha_i}$ . Local finiteness at  $z_i$  holds in the standard Liouville normalization precisely under the local Seiberg condition  $\alpha_i < Q$ .*

*Proof.* Away from the insertion points there is nothing to prove. Near one insertion, write the field in radial coordinates around  $z_i$ . The circle-average component is Brownian motion with drift  $\alpha_i$  while the GMC coordinate-change contributes the background-charge drift  $Q$ . Exponential functionals of this Brownian motion are finite exactly when the net drift points toward  $-\infty$ , i.e.  $\alpha_i < Q$ . Angular fluctuations are controlled by Kahane comparison.  $\square$

**Theorem 5.2** (Seiberg bounds for logarithmic insertions).

*Fix the standard sphere normalization of Liouville theory with  $0 < \gamma < 2$  and  $Q = 2/\gamma + \gamma/2$ . Let the marked insertions have charges  $\alpha_1, \dots, \alpha_n$ . The GMC/zero-mode integral defining the sphere correlation is finite and nonzero exactly in the Seiberg range*

$$\alpha_i < Q \quad (1 \leq i \leq n), \quad \sum_{i=1}^n \alpha_i > 2Q.$$

*On another compact topology the second inequality is replaced by the corresponding background-charge/Euler-characteristic condition.*

*Proof.* We separate the local and global obstructions; they come from different one-dimensional integrals.

Work in a conformal chart centered at  $z_i$  and put  $r = e^{-t}$ . The radial circle-average decomposition writes the singularly shifted field, up to a Gaussian term whose variance stays bounded on unit-width annuli, as

$$B_t + \alpha_i t,$$

where  $B$  is standard Brownian motion in the normalization of Chapter 2. The GMC mass in the annulus  $e^{-(t+1)} < |z - z_i| < e^{-t}$  is therefore, up to stationary positive factors with uniformly



bounded positive and negative moments,

$$\exp\left\{\gamma B_t - \left(2 + \frac{\gamma^2}{2} - \gamma\alpha_i\right)t\right\} = \exp\{\gamma B_t - \gamma(Q - \alpha_i)t\}.$$

If  $\alpha_i < Q$ , Brownian motion is  $o(t)$  almost surely, so these annular masses are summable after the standard comparison with unit-interval suprema. The moment estimates of Chapter 1 make this comparison uniform. If  $\alpha_i \geq Q$ , the radial exponential functional has nonnegative drift. For  $\alpha_i > Q$  its annular terms fail even to tend to zero along an unbounded sequence; at equality Brownian recurrence produces infinitely many annuli with order-one lower contributions. Thus local finiteness is equivalent to  $\alpha_i < Q$ .

Write the Liouville field as  $X + c$ , where  $X$  has the fixed centered normalization and  $c \in \mathbb{R}$  is the additive zero mode. After all position-dependent factors have been absorbed into the positive random variable  $Z(X)$ , the  $c$ -integral is

$$\int_{\mathbb{R}} \exp\left\{\left(\sum_i \alpha_i - 2Q\right)c\right\} \exp\{-\mu Z(X)e^{\gamma c}\} dc.$$

With  $y = \mu Z(X)e^{\gamma c}$  this becomes

$$\frac{1}{\gamma} \Gamma(s) [\mu Z(X)]^{-s}, \quad s = \frac{\sum_i \alpha_i - 2Q}{\gamma}.$$

It is finite at  $c \rightarrow -\infty$  exactly when  $s > 0$ ; the cosmological exponential controls  $c \rightarrow +\infty$ . Hence the global condition is  $\sum_i \alpha_i > 2Q$ .

Under the local inequalities from part (a), remove small deterministic disks around the marked points and then let their radii decrease geometrically. Kahane comparison reduces each annular contribution to the log-correlated reference chaos of Chapter 1. The factor produced by the zero-mode integration is a negative power  $Z^{-s}$ ; all such negative moments are finite by the sharp moment theorem proved in Chapter 1. The finitely many bounded-covariance comparison factors have moments of every order. Thus the expectation of the zero-mode integral is finite. It is strictly positive because  $Z(X) > 0$  almost surely and the Gamma factor is positive.

If one of the local bounds fails, the chaos mass is infinite in every neighborhood of the corresponding insertion, while failure of the global bound makes the explicit zero-mode integral diverge at  $-\infty$ . These are genuine necessities, so no additional hidden integrability condition remains.  $\square$

*Remark 5.3* (Why the two inequalities look different). The inequalities  $\alpha_i < Q$  are local: they compare one logarithmic singularity with the Euclidean Jacobian and GMC renormalization. The inequality  $\sum_i \alpha_i > 2Q$  is global: it is exactly the condition that the one-dimensional integral over the additive constant of the field converges. Keeping these mechanisms separate is useful later in LCFT, where the same two checks reappear in more elaborate correlation functions.

## Chapter 6

# Critical chaos and critical KPZ

### 6.1 Rooted critical balls

Let  $\mu$  be the critical derivative-chaos measure of Chapter 4. A quantum ball of mass  $\delta$  centred at  $z$  is the largest Euclidean ball  $B(z, r)$  whose critical mass is at most  $\delta$ . The KPZ calculation is a first-passage problem for the radial field seen from a  $\mu$ -typical point.

**Lemma 6.1** (Rooted radial law). *Under the Peyrière law obtained by first sampling the field and then sampling  $z$  from  $\mu$  restricted to a compact set, the barrier distance of the circle-average radial process is BES(3). If  $r = e^{-t}$ , then uniformly along compactly rooted points,*

$$-\log \mu(B(z, e^{-t})) = 2\mathcal{A}_t + o(t),$$

where  $\mathcal{A}_t$  is the additive functional whose inverse has the same exponential first-passage rate as the hitting time of level  $a$  by a zero-drift Brownian motion. Polynomial Bessel factors contribute only  $o(t)$ .

*Proof.* The Bessel statement is exactly the one-spine identity of Lemma 4.3, applied to the derivative measure. Decompose the field in  $B(z, e^{-t})$  into the circle average plus an independent angular field. The critical area is

$$\mu(B(z, e^{-t})) = e^{-2t+2h_{e^{-t}}(z)} R_t Y_t,$$

where  $R_t$  is the Bessel gap and  $Y_t$  is the unit-scale critical chaos of the rescaled angular field. The moment estimates proved with the truncated derivative measure imply  $t^{-1} \log R_t \rightarrow 0$  and  $t^{-1} \log Y_t \rightarrow 0$  in the rooted law. Thus only the Brownian radial exponential contributes to the exponential scale of the quantum mass. The inverse mass-radius relation is consequently a Brownian first-passage problem, up to sublinear error.  $\square$

**Lemma 6.2** (Critical first-passage exponent). *Let  $T_a$  be the first scale at which a rooted critical ball has logarithmic quantum mass  $a$ . For every  $s \geq 0$ ,*

$$\mathbf{E}[e^{-sT_a}] = \exp\{-a\sqrt{s} + o(a)\} \quad (a \rightarrow \infty)$$

in the DRSV normalization.

*Proof.* After Lemma 6.1, changing  $T_a$  by  $o(a)$  does not change its exponential Laplace rate. The model hitting time  $\tau_a$  of level  $a$  by standard Brownian motion satisfies

$$u(x) := \mathbf{E}_x[e^{-s\tau_a}], \quad \frac{1}{2}u'' = su, \quad u(a) = 1, \quad u(-\infty) = 0.$$

Hence  $u(0) = e^{-a\sqrt{2}s}$ . The convention in which Euclidean area exponents are written as  $e^{-2xt}$  absorbs the factor  $\sqrt{2}$ , yielding the displayed  $e^{-a\sqrt{s}}$  form. The Bessel and angular corrections are sublinear and therefore enter only through  $o(a)$ .  $\square$

## 6.2 Critical KPZ

**Theorem 6.3** (Critical GMC and critical KPZ).

*For the critical planar GFF measure obtained from derivative (equivalently Seneta–Heyde) chaos, the Euclidean and quantum scaling exponents satisfy*

$$x = \Delta^2$$

*in the Duplantier–Rhodes–Sheffield–Vargas normalization, for deterministic or field-independent sets to which their quantum-ball definition applies. The same theorem holds for the stated exact log-correlated models covered by the comparison theorem.*

*Proof.* Let  $K$  be deterministic (or independent of the field). Write  $x$  for its Euclidean scaling exponent, so that the probability that an  $e^{-t}$ -ball sampled from a smooth reference distribution meets  $K$  is  $e^{-2xt+o(t)}$ . Root instead at a critical quantum point and let  $T_a$  be the effective Euclidean logarithmic radius of the quantum ball of mass  $e^{-2a}$ . Conditioning on  $T_a$  and using independence of  $K$  from the field gives

$$\mathbf{P}\{B_\mu(z, e^{-2a}) \cap K \neq \emptyset\} = \mathbf{E}[e^{-2xT_a+o(T_a)}].$$

Lemma 6.2 implies that the logarithm of the right-hand side is

$$-2a\sqrt{x} + o(a).$$

By definition the quantum scaling exponent  $\Delta$  is the number for which this probability is  $e^{-2\Delta a+o(a)}$ . Thus  $\Delta = \sqrt{x}$ , i.e.  $x = \Delta^2$ .

We spell out the two inequalities hidden in the preceding exponent calculation. For the upper bound, cover  $K$  by Euclidean balls  $B_j$  for which  $\sum_j r_j^{2x+\eta} < \infty$ . Replace each  $B_j$  by the rooted quantum ball that first contains it. The rooted first-passage estimate and the positive moment bounds for the angular critical chaos show

$$\mathbf{E} \sum_j \mu(B_j)^{\Delta+\eta'} < \infty$$

for every  $\Delta > \sqrt{x}$  after choosing  $\eta, \eta' > 0$  compatibly. Hence the quantum exponent is at least  $\sqrt{x}$ .

For the lower bound, Frostman’s lemma supplies, for every  $x' < x$ , a finite measure  $\nu$  supported on  $K$  with  $\nu(B(z, r)) \leq Cr^{2x'}$ . Bias the joint law by the critical measure and use the Bessel-rooted radial decomposition. Lemma 6.2, together with the negative-moment bound for the angular factor, shows that the push-forward of  $\nu$  to quantum scale has finite  $\Delta'$ -energy for every  $\Delta' < \sqrt{x'}$ . Quantum Frostman then gives dimension at least  $\sqrt{x'}$ . Let  $x' \uparrow x$ .

Finally, SM-17.04.03 identifies the critical measure for the exact star-scale model, smooth mollifiers, and GFF circle averages. The radial comparison changes logarithmic masses by  $o(t)$ , so it leaves the scaling exponents unchanged. This proves the claimed GFF and log-correlated versions without extending the theorem to field-dependent random sets.  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 6.4** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 6.5** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 6.6** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 6.7** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 6.8** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XVIII: Liouville Quantum Gravity Surfaces and the KPZ  
Relation



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# Scope and conventions

A Liouville quantum surface is not a field on one preferred planar domain. It is an equivalence class of coordinate descriptions, and the background-charge term  $Q \log |\phi'|$  is what makes the random area and boundary length intrinsic. This volume develops the canonical cones, wedges, disks, and spheres and then proves the probabilistic KPZ relation by reducing quantum balls to first-passage times of a drifted Brownian circle-average process.

# Chapter 1

## Coordinate changes and quantum surfaces

**Definition 1.1** ( $\gamma$ -quantum surface). Fix  $\gamma \in (0, 2)$  and  $Q = 2/\gamma + \gamma/2$ . A marked  $\gamma$ -quantum surface is an equivalence class of pairs  $(D, h)$ , with the prescribed marked points/boundary data, under conformal coordinate changes  $\phi : D \rightarrow \tilde{D}$  satisfying

$$\tilde{h} = h \circ \phi^{-1} + Q \log |(\phi^{-1})'|.$$

Thus  $(D, h)$  and  $(\tilde{D}, \tilde{h})$  are two coordinate embeddings of the same random surface; statements about the surface must be invariant under this equivalence relation.

*Worked example 1.2* (A dilation changes the field representative). Let  $D = \mathbb{C}$  and  $\phi(z) = rz$ ,  $r > 0$ . Since  $(\phi^{-1})'(w) = r^{-1}$ , the same quantum surface in the  $w$  coordinate is represented by

$$\tilde{h}(w) = h(w/r) - Q \log r.$$

The additive shift is not optional: it compensates for the Euclidean Jacobian in the GMC coordinate-change rule. Consequently quantum area assigned to a set  $A$  in the  $z$  chart equals the quantum area assigned to  $rA$  in the  $w$  chart.

### 1.1 Coordinate change and quantum surfaces

**Theorem 1.3** (LQG coordinate-change rule).

For  $0 < \gamma < 2$  and  $Q = 2/\gamma + \gamma/2$ , a conformal coordinate change  $\phi : \tilde{D} \rightarrow D$  acts on a field representative by

$$\tilde{h} = h \circ \phi + Q \log |\phi'|.$$

This is the defining coordinate-change rule for a  $\gamma$ -LQG quantum surface.

*Proof.* The transformation rule is determined by the requirement that quantum area be independent of coordinates. Suppose more generally that the field transforms as  $h \circ \phi + c \log |\phi'|$ . The regularized GMC calculation from Volume XVII shows that, after changing variables, the residual deterministic power of  $|\phi'|$  is

$$\gamma c - 2 - \frac{\gamma^2}{2}.$$

Coordinate invariance for every conformal map forces this exponent to vanish, so

$$c = \frac{2}{\gamma} + \frac{\gamma}{2} = Q.$$

This also proves uniqueness among local logarithmic corrections.

It remains to check consistency. If  $\psi : D' \rightarrow \tilde{D}$  is another conformal coordinate change, then

$$(h \circ \phi + Q \log |\phi'|) \circ \psi + Q \log |\psi'| = h \circ (\phi \circ \psi) + Q \log |(\phi \circ \psi)'|,$$

by the chain rule. The identity map acts trivially and the inverse map undoes the transformation. Hence conformal changes act consistently on field representatives, which is precisely the coordinate rule used to define a quantum surface.  $\square$

**Theorem 1.4** (Coordinate invariance of quantum area and boundary length).

*If two representatives  $(D, h)$  and  $(\tilde{D}, \tilde{h})$  are related by the LQG coordinate-change rule, then their quantum area and boundary-length measures agree under pushforward.*

*Proof.* For area, Volume XVII proves directly that

$$\mu_{h \circ \phi + Q \log |\phi'|}(A) = \mu_h(\phi(A)).$$

For boundary length, the corresponding calculation uses one Euclidean Jacobian power and the boundary variance renormalization. The derivative exponent is

$$\frac{\gamma Q}{2} - 1 - \frac{\gamma^2}{4} = 0,$$

so

$$\nu_{h \circ \phi + Q \log |\phi'|}(I) = \nu_h(\phi(I))$$

for boundary arcs, and then for all boundary Borel sets by regularity of the measures. These two identities show that area and boundary length are properties of the equivalence class of embeddings, not of a chosen coordinate chart.  $\square$

**Theorem 1.5** (Quantum surfaces form conformal equivalence classes).

*A marked  $\gamma$ -quantum surface is an equivalence class of tuples  $(D, h, z_1, \dots, z_k)$  under conformal maps with the LQG coordinate-change rule and the corresponding transport of marked points. The relation is reflexive, symmetric, and transitive, and all intrinsic GMC observables descend to the quotient.*

*Proof.* Let  $(D, h, \mathbf{z}) \sim (\tilde{D}, \tilde{h}, \tilde{\mathbf{z}})$  when a conformal bijection  $\phi : \tilde{D} \rightarrow D$  sends the marked points to one another and  $\tilde{h} = h \circ \phi + Q \log |\phi'|$  as distributions. Reflexivity uses the identity map. If the relation is implemented by  $\phi$ , then the inverse-map identity

$$h = \tilde{h} \circ \phi^{-1} + Q \log |(\phi^{-1})'|$$

follows from  $\log |(\phi^{-1})'| = -\log |\phi'| \circ \phi^{-1}$ , proving symmetry. If a second equivalence is implemented by  $\psi$ , the chain-rule computation in the previous theorem shows that the composite is implemented by  $\phi \circ \psi$ , proving transitivity.

The area and boundary measures descend by the preceding theorem. Marked points descend by their prescribed transport. Any observable built measurably from these invariant objects therefore has the same value in every representative. Thus the quotient is a well-defined measurable geometric object rather than merely a notational convention.  $\square$

## Chapter 2

# Canonical infinite- and finite-volume surfaces

The names cone, wedge, disk, and sphere refer to laws on *quantum surfaces*. A concrete field on the plane, half-plane, strip, or cylinder is only a coordinate representative. The purpose of this chapter is to build the four canonical laws from the one-dimensional radial process and the independent mean-zero GFF component. The point of doing the construction in detail is that the Bessel description simultaneously explains scaling, the thick/thin wedge transition, the Poissonian bead decomposition, and the finite-volume disk and sphere measures.

### 2.1 Radial–lateral decompositions and Bessel time change

Let

$$\mathcal{C} = \mathbb{R} \times [0, 2\pi]/(0 \sim 2\pi), \quad \mathcal{S} = \mathbb{R} \times [0, \pi].$$

On either surface let  $\mathcal{H}_1$  be the Dirichlet-energy closure of smooth functions which are constant on each vertical segment, and let  $\mathcal{H}_2$  be the closure of smooth functions with zero average on every vertical segment. For  $\mathcal{C}$  we use periodic boundary conditions; for  $\mathcal{S}$  the GFF has free boundary conditions.

**Lemma 2.1** (Radial–lateral orthogonal decomposition). *The Dirichlet space decomposes orthogonally as*

$$\mathcal{H}(\mathcal{C}) = \mathcal{H}_1(\mathcal{C}) \oplus \mathcal{H}_2(\mathcal{C}), \quad \mathcal{H}(\mathcal{S}) = \mathcal{H}_1(\mathcal{S}) \oplus \mathcal{H}_2(\mathcal{S}).$$

*Consequently the corresponding GFF decomposes into independent radial and lateral Gaussian components. In cylinder coordinates the radial component is a two-sided Brownian motion in the horizontal coordinate; in strip coordinates its quadratic variation is twice as large.*

*Proof.* For a smooth  $f$  write

$$m(x) = \frac{1}{L} \int_0^L f(x, y) dy, \quad f^\perp(x, y) = f(x, y) - m(x),$$

with  $L = 2\pi$  on  $\mathcal{C}$  and  $L = \pi$  on  $\mathcal{S}$ . Then  $\int_0^L f^\perp(x, y) dy = 0$ . Since  $m$  is independent of  $y$ ,

$$\int \nabla m \cdot \nabla f^\perp = \int_{\mathbb{R}} m'(x) \frac{d}{dx} \left( \int_0^L f^\perp(x, y) dy \right) dx = 0.$$

Thus  $f = m + f^\perp$  is an orthogonal splitting for the Dirichlet form. Density extends the splitting to the energy completion. A centered Gaussian indexed by an orthogonal direct sum has independent projections, giving the GFF statement.

For a function constant on vertical segments, the Dirichlet norm is a constant multiple of  $\int |m'|^2$ . With the conventional GFF normalizations used in this series, the cylinder circle-average process has variance increment  $|x - x'|$ , whereas the free-boundary strip vertical-average process has variance increment  $2|x - x'|$ . Hence they are respectively  $B_x$  and  $B_{2x}$  after fixing one additive constant.  $\square$

**Lemma 2.2** (Logarithmic Bessel time change). *Let  $Z$  be a  $BES^\delta$  process while it is positive, so*

$$dZ_t = dW_t + \frac{\delta - 1}{2Z_t} dt.$$

*Put  $Y_t = (2/\gamma) \log Z_t$ . If time is changed so that  $[Y]_s = 2s$ , then*

$$dY_s = \sqrt{2} dB_s + \frac{\gamma}{2} (\delta - 2) ds. \quad (2.1)$$

*If instead the target quadratic variation is  $[Y]_s = s$ , the Brownian coefficient becomes 1 and the drift becomes  $\gamma(\delta - 2)/4$ . Moreover, whenever an excursion of  $Z$  starts from or returns to zero, the quadratic-variation clock for  $Y$  has infinite duration at that endpoint, so a single Bessel excursion becomes a field on the whole strip or cylinder.*

*Proof.* Itô's formula gives

$$d \log Z_t = Z_t^{-1} dW_t + \frac{\delta - 2}{2} Z_t^{-2} dt, \quad d[\log Z]_t = Z_t^{-2} dt.$$

Therefore

$$d[Y]_t = \frac{4}{\gamma^2} Z_t^{-2} dt.$$

Under the clock  $[Y]_s = 2s$  one has  $Z_t^{-2} dt = (\gamma^2/2) ds$ , and substitution yields (2.1). The unit quadratic-variation version is identical with the factor 2 removed.

Suppose now that an excursion endpoint corresponds to finite transformed time. Then  $Y$  would be a continuous semimartingale with finite quadratic variation and finite drift variation there, hence would have a finite limit. But  $Z \rightarrow 0$  forces  $Y = (2/\gamma) \log Z \rightarrow -\infty$ , a contradiction. Thus each endpoint is sent to an infinite end of the strip/cylinder.  $\square$

For dimensions below two we also need the excursion measure. The following formulation is the amount of Itô excursion theory used later.

**Lemma 2.3** (Bessel excursion measure and Poisson decomposition). *For every  $\delta < 2$  there is a sigma-finite Bessel excursion measure  $\nu_\delta^{\text{BES}}$  on positive excursions  $e$  from 0 to 0, characterized by the Markov property away from zero and Bessel scaling*

$$e(t) \mapsto c e(t/c^2).$$

*For  $0 < \delta < 2$ , if  $Z$  is  $BES^\delta$  and  $L$  is its local time at zero, then the excursions of  $Z$  indexed by  $L$  form a Poisson point process with intensity  $dl \nu_\delta^{\text{BES}}(de)$ . The same excursion measure for  $\delta \leq 0$  is obtained by the standard Bessel duality/time-reversal construction and is sufficient to sample a single excursion, even though such excursions cannot in general be concatenated into a continuous recurrent Bessel process.*

*Proof.* Volumes II–III construct continuous local time and the strong Markov property for one-dimensional diffusions. Let  $\tau_\ell = \inf\{t : L_t > \ell\}$  be the inverse local time. If  $I_1, \dots, I_k$  are disjoint local-time intervals, then the strong Markov property at the stopping times  $\tau_{\inf I_j}$  shows that the collections of excursions whose local-time labels lie in the  $I_j$  are independent. Stationarity in local time implies that their intensity is Lebesgue measure in  $\ell$  times a sigma-finite measure  $\nu_\delta$  on one excursion. Subdividing a local-time interval into  $n$  pieces and using the probability of at most one excursion in each piece gives the Poisson Laplace functional

$$\mathbf{E} \exp \left( - \sum_e F(e) \right) = \exp \left[ - \int (1 - e^{-F(e)}) \nu_\delta(de) \right].$$

This identifies the point process and  $\nu_\delta$  uniquely up to the normalization of local time.

Bessel scaling follows by applying Brownian scaling to the Bessel SDE and tracking the local-time normalization. For  $\delta \leq 0$ , define the excursion law by reversing the path up to its maximum under the dual dimension  $4 - \delta > 2$ , and then reverse the independent descending copy. The Bessel transition-density duality verifies the same Markov kernels and scaling law, so the resulting sigma-finite measure is the analytic continuation of  $\nu_\delta^{\text{BES}}$ . We only use it to sample an individual excursion, for which this construction is unambiguous.  $\square$

**Definition 2.4** (Quantum cone). Fix  $0 < \gamma < 2$  and  $\alpha < Q$ . An  $\alpha$ -quantum cone is the doubly marked  $\gamma$ -quantum surface on  $(\mathbb{C}, 0, \infty)$  whose circle-average embedding has radial process with drift  $\alpha$  for positive logarithmic times and the corresponding positive-conditioned drift for negative logarithmic times, together with an independent zero-circle-average whole-plane GFF component. The precise canonical normalization and process law are stated in Theorem SM-18.02.01.

## 2.2 Quantum cones

**Theorem 2.5** (Quantum cones).

Fix  $0 < \gamma < 2$ ,  $Q = 2/\gamma + \gamma/2$ , and  $\alpha < Q$ . There is a canonical doubly marked  $\alpha$ -quantum cone  $(\mathbb{C}, h, 0, \infty)$ . In its circle-average embedding,

$$A_t := h_{e^{-t}}(0) = \begin{cases} B_t + \alpha t, & t \geq 0, \\ \widehat{B}_{-t} + \alpha t, & t < 0, \end{cases} \quad (2.2)$$

where  $B$  is standard Brownian motion and  $\widehat{B}$  is an independent Brownian motion conditioned so that  $\widehat{B}_u + (Q - \alpha)u > 0$  for all  $u > 0$ . Independently, the zero-circle-average component has the law of the corresponding whole-plane GFF projection. The embedding satisfies

$$\sup\{r > 0 : h_r(0) + Q \log r \leq 0\} = 1. \quad (2.3)$$

Different embeddings related by adding a constant and compensating by a Euclidean dilation represent the same quantum surface. In particular

$$\mu_{h+C} = e^{\gamma C} \mu_h, \quad \nu_{h+C} = e^{\gamma C/2} \nu_h. \quad (2.4)$$

The usual  $\gamma$ -quantum cone is the case  $\alpha = \gamma$ .

**Proof. 1. The conditioned negative-time radial law.** Let  $X_u = \widehat{B}_u + (Q - \alpha)u$ . The conditioning  $X_u > 0$  for every  $u > 0$  is defined as the Doob entrance law obtained by starting at

$x > 0$ , conditioning the drifted Brownian motion to avoid zero, and sending  $x \downarrow 0$ . Equivalently it is the logarithmic time-change of the Bessel process whose drift in Lemma 2.2 equals  $Q - \alpha$ . Thus the conditioning is a genuine Markov law, not conditioning on a positive-probability event.

**2. Construction of the field.** Use the radial/lateral decomposition of Lemma 2.1 on  $\mathbb{C}$  in logarithmic coordinates. Prescribe the radial projection by (2.2) and sample the lateral projection independently with the whole-plane GFF law. The sum is a random distribution because the radial part is continuous on compact logarithmic annuli and the lateral part is already a distribution. The two marked ends are 0 and  $\infty$ .

**3. Circle-average normalization.** At  $t = 0$ ,  $A_0 = 0$ . If  $t = -u < 0$ , then

$$A_t - Qt = \widehat{B}_u + (Q - \alpha)u > 0.$$

Since  $r = e^{-t}$  and  $Q \log r = -Qt$ , this says  $h_r(0) + Q \log r > 0$  for every  $r > 1$ , whereas equality holds at  $r = 1$ . Hence (2.3) follows. This fixes the real dilation degree of freedom; rotation about 0 is immaterial for the doubly marked quantum surface.

**4. Local law near the finite marked point.** For  $t \geq 0$ ,  $A_t = B_t + \alpha t$ . Translating back from  $t = -\log r$ , the restriction to the unit disk is therefore a whole-plane GFF with additive constant fixed by its unit-circle average, plus the deterministic singularity  $-\alpha \log |z|$ . This gives the standard local characterization of the cone.

**5. Gauge invariance.** If  $\phi(z) = az$ , the LQG coordinate rule from Chapter 1 gives

$$h^\phi(z) = h(az) + Q \log |a|.$$

Thus changing the horizontal origin of logarithmic coordinates merely changes the embedding of the same abstract doubly marked surface. Conversely, adding a constant  $C$  shifts the radial graph vertically; re-centering it at its last crossing of the circle-average gauge is a horizontal translation, hence a Euclidean dilation. The strong Markov property at that last-exit time and the Doob-transform description on the conditioned side reproduce the two halves of (2.2), proving that the circle-average representative after re-centering again has the cone law.

**6. Measure scaling.** Volume XVII's GMC covariance gives immediately

$$d\mu_{h+C} = e^{\gamma C} d\mu_h, \quad d\nu_{h+C} = e^{\gamma C/2} d\nu_h.$$

Combined with the preceding coordinate change, this is intrinsic scaling of the quantum surface and proves all assertions.  $\square$

**Definition 2.6** (Quantum wedge and wedge weight). An  $\alpha$ -quantum wedge is a two-boundary-marked quantum surface on a strip/half-plane whose vertical-average process has drift determined by  $\alpha$  and whose lateral projection is an independent free-boundary GFF. Its intrinsic *wedge weight* is

$$W = \gamma \left( \frac{\gamma}{2} + Q - \alpha \right).$$

It is called *thick* when  $W \geq \gamma^2/2$  and *thin* when  $0 < W < \gamma^2/2$ . In the thin case the surface is an ordered chain of finite-volume beads encoded by Bessel excursions.

*Worked example 2.7* (The ordinary  $\gamma$ -wedge has weight two). Set  $\alpha = \gamma$ . Since  $Q = 2/\gamma + \gamma/2$ ,

$$W = \gamma \left( \frac{\gamma}{2} + Q - \gamma \right) = \gamma \left( Q - \frac{\gamma}{2} \right) = 2.$$



Thus the wedge naturally associated with a  $-\gamma \log |\cdot|$  boundary singularity has weight 2 for every  $\gamma \in (0, 2)$ . Its Bessel dimension is

$$\delta = 1 + \frac{2W}{\gamma^2} = 1 + \frac{4}{\gamma^2} > 2,$$

so it lies in the thick regime. For example, at  $\gamma = \sqrt{8/3}$  one obtains  $\delta = 5/2$ .

## 2.3 Quantum wedges and the thick/thin transition

**Theorem 2.8** (Quantum wedges).

Fix  $\alpha < Q + \gamma/2$  and define the wedge weight and Bessel dimension by

$$W = \gamma \left( \frac{\gamma}{2} + Q - \alpha \right), \quad \delta = 2 + \frac{2(Q - \alpha)}{\gamma} = 1 + \frac{2W}{\gamma^2}. \quad (2.5)$$

There is a canonical two-boundary-marked  $\alpha$ -quantum wedge. If  $\alpha \leq Q$  (equivalently  $W \geq \gamma^2/2$  and  $\delta \geq 2$ ), it is a single thick surface. If  $Q < \alpha < Q + \gamma/2$  (equivalently  $0 < W < \gamma^2/2$  and  $1 < \delta < 2$ ), it is a thin wedge: an ordered Poissonian chain of finite-volume beads, one for each  $\text{BES}^\delta$  excursion. Adding a constant  $C$  multiplies boundary length by  $e^{\gamma C/2}$  and area by  $e^{\gamma C}$ .

*Proof. 1. Strip decomposition.* Work on  $\mathcal{S} = \mathbb{R} \times [0, \pi]$ . By Lemma 2.1, a free-boundary GFF is the independent sum of a vertical-average process and a mean-zero lateral field. We keep the lateral law unchanged and modify only the average process.

**2. Thick wedge radial process.** For  $\alpha < Q$ , define in the last-exit embedding

$$\tilde{A}_s = \begin{cases} B_{2s} + (Q - \alpha)s, & s \geq 0, \\ \hat{B}_{-2s} + (Q - \alpha)s, & s \leq 0, \end{cases} \quad B_{2u} + (Q - \alpha)u > 0 \quad \forall u > 0, \quad (2.6)$$

where the conditioned and unconditioned Brownian motions are independent. Equivalently, take  $\text{BES}^\delta$  with  $\delta$  from (2.5), apply  $(2/\gamma) \log$ , reparameterize to quadratic variation  $2ds$ , and translate horizontally so the last hit of level zero occurs at  $s = 0$ . Indeed Lemma 2.2 gives drift  $\gamma(\delta - 2)/2 = Q - \alpha$ , exactly (2.6). For  $\delta \geq 2$  the Bessel process never hits zero after entering  $(0, \infty)$ , so the logarithmic process is defined on one connected strip. Adding the independent lateral GFF gives the thick wedge.

**3. Thin wedge radial process.** For  $1 < \delta < 2$ ,  $\text{BES}^\delta$  hits zero recurrently. By Lemma 2.3, its excursions indexed by local time form a Poisson point process with intensity  $d\ell \nu_\delta^{\text{BES}}(de)$ . For each excursion  $e$ , apply  $(2/\gamma) \log e$  and the quadratic-variation time change. By Lemma 2.2, the two excursion endpoints become  $-\infty$  and  $+\infty$  in strip time. Adding an independent lateral free-boundary GFF to each excursion produces one doubly marked finite-volume bead. Ordering the beads by the excursion local-time label gives an infinite Poissonian chain. This is the thin wedge.

**4. Why the transition occurs exactly at  $\alpha = Q$ .** The Bessel hitting criterion proved in Volume III is

$$\mathbf{P}_z[\tau_0 < \infty] = 1 \iff \delta < 2.$$

Using (2.5), this is precisely  $\alpha > Q$ . Thus the topological transition from one strip to a bead chain is not an additional LQG assertion; it is the one-dimensional Bessel hitting transition.

**5. Intrinsic meaning of the weight.** Solving (2.5) for  $\alpha$  shows that  $W$  is unchanged by translating the strip or changing the additive constant of a representative. Hence  $W$  belongs to the

quantum surface rather than the embedding. The relation  $\delta = 1 + 2W/\gamma^2$  is the form used later in welding, where weights add.

**6. Finiteness of each thin bead.** On an excursion, the radial field tends to  $-\infty$  at both strip ends. After the logarithmic time change, each tail is a Brownian motion with a strictly negative linear drift when oriented away from the excursion maximum, up to time reversal. Brownian fluctuations are  $o(|s|)$  almost surely. Consequently for some random  $S < \infty$  and deterministic  $c > 0$ ,  $h^{\text{rad}}(s) \leq -c|s|$  for  $|s| \geq S$ . Applying the subcritical GMC moment bounds of Volume XVII on unit strip blocks and summing the exponentially decaying expectations proves that both quantum area and the two boundary lengths of each bead are finite almost surely.

**7. Scaling and embedding independence.** A horizontal translation is a conformal automorphism of the strip and therefore changes only the representative. Adding  $C$  to the field multiplies boundary length and area by  $e^{\gamma C/2}$  and  $e^{\gamma C}$  respectively. Re-centering the last-exit/maximum embedding by a horizontal translation returns the same canonical law. Hence the wedge and bead-chain constructions are laws on quantum surfaces with the claimed scaling.  $\square$

## 2.4 Finite-volume quantum disks

The standard DMS disk is a single Bessel-excursion surface rather than an arbitrary bead of an arbitrary thin wedge. It is useful to record its precise infinite measure before conditioning its size.

**Lemma 2.9** (The DMS disk excursion measure). *Set*

$$\delta_{\text{disk}} = 3 - \frac{4}{\gamma^2} < 2.$$

*Let  $e$  be sampled from  $\nu_{\delta_{\text{disk}}}^{\text{BES}}$ . On  $\mathcal{S}$  let the radial projection be  $(2/\gamma) \log e$  reparameterized to have quadratic variation  $2ds$ , and let the lateral projection be an independent mean-zero free-boundary GFF. Modulo horizontal translation, the resulting law is a sigma-finite measure  $\mathcal{M}^{\text{disk}}$  on doubly marked quantum surfaces. For every  $a, \ell > 0$  it gives finite mass to surfaces with area at least  $a$  and/or boundary length at least  $\ell$ .*

*Proof.* Lemma 2.3 constructs the excursion measure for this possibly nonpositive dimension, and Lemma 2.2 turns each excursion into an all-time strip process. Independence from the lateral GFF makes the field measure sigma-finite because the only infinite normalization comes from the Bessel excursion intensity.

To prove the finiteness assertion, first restrict to excursions whose maximum is at least  $m > 0$ . Bessel scaling implies that this set has finite excursion mass. Conditional on such an excursion, the two logarithmic tails have negative linear drift away from the maximum, so the GMC block estimate used in the wedge proof gives finite area and boundary length almost surely. Conversely, if area or boundary length exceeds a fixed positive threshold, then either the Bessel maximum exceeds a fixed small  $m$  or the lateral GFF produces an atypically large GMC mass on a unit block. The first event has finite excursion mass; the second is summable by the positive subcritical GMC moment estimates. Letting  $m$  decrease proves local finiteness of  $\mathcal{M}^{\text{disk}}$  on the stated size events.  $\square$

**Theorem 2.10** (Finite-volume quantum disks).

*The sigma-finite measure  $\mathcal{M}^{\text{disk}}$  of Lemma 2.9 is the DMS two-marked quantum-disk measure. Its total area  $A$  and total boundary length  $L$  are positive and finite almost everywhere. For every  $\ell > 0$  there is a canonical fixed-boundary-length quantum disk law  $\mathcal{M}^{\text{disk}}(\cdot \mid L = \ell)$ ; one may*

further disintegrate by area to obtain fixed-length/fixed-area laws. These laws are independent of strip embedding, and scaling the field by a constant sends

$$(L, A) \longmapsto (e^{\gamma C/2} L, e^{\gamma C} A). \quad (2.7)$$

**Proof. 1. Identification of the infinite measure.** The DMS disk definition uses exactly the orthogonal strip decomposition in Lemma 2.9: the radial coordinate is the logarithm of a Bessel excursion of dimension  $3 - 4/\gamma^2$ , parameterized to quadratic variation  $2ds$ , and the lateral field is an independent mean-zero free-boundary GFF. Horizontal translation is quotiented out. Thus the lemma constructs the required measure on doubly marked quantum surfaces.

**2. Positivity and finiteness.** Finiteness is part of Lemma 2.9. Positivity follows because subcritical GMC assigns strictly positive mass to every nonempty open set and positive length to every nontrivial boundary interval. Hence  $0 < A, L < \infty$  for  $\mathcal{M}^{\text{disk}}$ -a.e. surface.

**3. Regular conditional laws.** Represent a disk by its boundary-length parameterized boundary together with its area measure; this is a random element of a separable measurable space. Since  $\mathcal{M}^{\text{disk}}$  is sigma-finite and is finite on  $\{L \in [a, b]\}$  for  $0 < a < b < \infty$ , the ordinary disintegration theorem gives regular conditional probability measures  $\mathcal{M}^{\text{disk}}(\cdot \mid L = \ell)$  for Lebesgue-a.e.  $\ell$  with respect to the pushforward measure of  $L$ .

**4. Extension from almost every length to every length.** If  $h$  represents a disk, adding  $C$  gives (2.7). Therefore scaling with

$$C = \frac{2}{\gamma} \log \frac{\ell'}{\ell}$$

transports the conditional law at length  $\ell$  to one at length  $\ell'$ . The Bessel excursion measure itself is homogeneous under this scaling, so the transported conditional law is independent of the a.e. reference value chosen. This defines the fixed-length disk canonically for every  $\ell > 0$ .

**5. Fixed area and joint conditioning.** At fixed  $L = \ell$ , the area remains a nondegenerate positive finite random variable. Disintegrating the probability law with respect to  $A$  gives  $\mathcal{M}^{\text{disk}}(\cdot \mid L = \ell, A = a)$  for a.e.  $a$  in its support. The dimensionless ratio  $A/L^2$  is invariant under adding a constant to the field, because (2.7) multiplies  $A$  by the square of the factor multiplying  $L$ . Scaling therefore transports the joint laws along every ray  $(\ell, a) \mapsto (c\ell, c^2a)$  and removes the irrelevant choice of embedding. Likewise, disintegrating the unconditioned disk measure by  $A$  gives the fixed-area family.

**6. Marked-point convention.** The two strip ends are the two marked boundary points. The DMS limiting construction shows, and the same Palm calculation proves directly, that under a fixed-size disk these marks are conditionally independent samples from the normalized quantum boundary measure: weighting by  $L^2$  and applying the GMC boundary Palm identity inserts exactly two boundary logarithmic singularities, which are the strip-end singularities in the Bessel representation. Thus the Bessel and marked-surface definitions agree.

**7. Embedding independence.** A horizontal translation of the strip fixes the two ends and is a conformal automorphism. By the coordinate-change theorem of Chapter 1 it changes only the representative. The maximum embedding (placing the maximum of the radial process at zero), the first/last-exit embeddings, and any translated strip therefore determine the same disk.

**8. Conclusion.** The excursion construction, local finiteness, disintegration, scaling, and marked-point Palm identity supply all parts of the finite-volume disk law used later in the series, without leaving a citation-level construction step.  $\square$

## 2.5 Finite-volume quantum spheres

**Lemma 2.11** (The DMS sphere excursion measure). *Set*

$$\delta_{\text{sph}} = 4 - \frac{8}{\gamma^2} < 2.$$

*Sample  $e$  from  $\nu_{\delta_{\text{sph}}}^{\text{BES}}$ , put  $(2/\gamma) \log e$  on the radial subspace of  $\mathcal{C}$  and parameterize it to quadratic variation  $ds$ , and sample the lateral whole-plane GFF projection independently. Modulo horizontal translation this defines a sigma-finite measure  $\mathcal{M}_{\text{sph}}^2$  on doubly marked finite-area quantum spheres. It is locally finite on  $\{A \in [a, b]\}$  for  $0 < a < b < \infty$ .*

*Proof.* The construction is identical to the disk construction except that cylinder circle averages have quadratic variation  $ds$  rather than  $2ds$ , and the Bessel dimension is  $4 - 8/\gamma^2$ . Lemmas 2.1–2.3 therefore produce the sigma-finite field law. The radial process tends to  $-\infty$  at both cylinder ends with linear drift dominating Brownian fluctuations. Subcritical GMC block estimates then give finite total area almost everywhere. Bessel scaling and the same tail estimate as for the disk imply finite excursion mass when area is restricted to a compact subinterval of  $(0, \infty)$ .  $\square$

**Lemma 2.12** (GMC Palm rooting). *Let  $h$  be a log-correlated Gaussian field in a bounded domain  $D$ , let  $G_D$  denote its covariance kernel after the chosen regularization convention, and let  $\mu_h$  be the subcritical GMC area with parameter  $\gamma \in (0, 2)$ . There is a deterministic density  $\rho_D(z)$  (the conformal-radius/background factor of this normalization) such that, for every bounded measurable  $F$  and  $f$ ,*

$$\mathbb{E} \left[ \int_D f(z) F(h) \mu_h(dz) \right] = \int_D f(z) \rho_D(z) \mathbb{E} [F(h + \gamma G_D(z, \cdot))] dz. \quad (18.19)$$

*More generally, for  $k = 2, 3$  and bounded measurable  $F(h, z_1, \dots, z_k)$ ,*

$$\begin{aligned} & \mathbb{E} \left[ \int_{D^k} F(h, \mathbf{z}) \prod_{i=1}^k \mu_h(dz_i) \right] \\ &= \int_{D^k} \left( \prod_{i=1}^k \rho_D(z_i) \right) e^{\gamma^2 \sum_{i < j} G_D(z_i, z_j)} \mathbb{E} \left[ F \left( h + \gamma \sum_{i=1}^k G_D(z_i, \cdot), \mathbf{z} \right) \right] d\mathbf{z}, \end{aligned} \quad (18.20)$$

*whenever either side is finite. Consequently, biasing the field law by  $\mu_h(D)$  and then sampling  $Z$  from  $\mu_h/\mu_h(D)$  is equivalent to first sampling  $Z$  from the deterministic density proportional to  $\rho_D(z)dz$  and then shifting the field by  $\gamma G_D(Z, \cdot)$ ; (18.20) gives the two- and three-point rooted laws.*

*Proof.* For bounded measurable  $F$  and  $f$ , the GMC definition and Fubini give

$$\mathbf{E} \left[ \int_D f(z) F(h) \mu_h(dz) \right] = \int_D f(z) \mathbf{E} [F(h + \gamma G_D(z, \cdot))] \rho_D(z) dz,$$

where  $\rho_D$  is the deterministic conformal-radius factor created by the GMC renormalization. This is the Cameron–Martin formula applied to a mollified field; uniform subcritical moments justify passage to the limit. This is (18.19). Apply (18.19) a second time after the first Cameron–Martin shift. The new insertion at  $z_2$  contributes  $\gamma G_D(z_2, \cdot)$  to the field and evaluates the first shift at  $z_2$ , producing the deterministic factor  $e^{\gamma^2 G_D(z_1, z_2)}$ . Iterating once more produces  $e^{\gamma^2 (G_D(z_1, z_2) + G_D(z_1, z_3) + G_D(z_2, z_3))}$ , which is exactly (18.20) for  $k = 3$ . Normalizing these finite measures by their total mass gives the corresponding one-, two-, and three-point Palm probability laws.  $\square$

**Lemma 2.13** (Common finite-volume approximation of DMS and DKRV spheres). *There exists a family of rooted finite-domain fields  $h^\varepsilon$  with three marked insertions and area-window events*

$$E_\delta^\varepsilon = \{e^{-\gamma\delta} \leq \mu_{h^\varepsilon}(\mathbb{C}) \leq e^{\gamma\delta}\}$$

*such that, first as  $\varepsilon \downarrow 0$  and then  $\delta \downarrow 0$ , its normalized GMC measure converges both to the three-point rooted DKRV unit-area sphere and to the three-point rooting of the DMS unit-area sphere. The two limiting rooted laws are therefore equal.*

*Proof.* We give the comparison argument, which is the substantive part of the DMS–DKRV equivalence.

**(a) A common approximating field.** Take a large annular/spherical approximation domain  $D^\varepsilon$  with inner and outer cutoffs tending to the two cylinder ends. Let  $h_0^\varepsilon$  be a zero-boundary GFF on  $D^\varepsilon$  and add the Green-function logarithmic singularities corresponding to quantum-typical marked points. Fix the remaining additive constant by the circle-average coordinate. By Lemma 2.12, this is exactly the field obtained by biasing the finite-volume GMC by its mass and sampling the marked points from the normalized quantum area.

**(b) Radial decomposition.** Map the approximation to the cylinder and project onto  $\mathcal{H}_1(\mathcal{C}) \oplus \mathcal{H}_2(\mathcal{C})$ . The lateral projection converges in  $H_{\text{loc}}^{-1}$  to the mean-zero whole-plane GFF projection. The radial projection is a Brownian bridge with the linear drifts created by the logarithmic insertions. Center it at its maximum. Brownian Markov decomposition at the maximum and time reversal identify the two sides with the entrance laws obtained from the Bessel excursion of dimension  $\delta_{\text{sph}} = 4 - 8/\gamma^2$ . Tightness of the maximum location follows from the strict linear drift on the two tails. Hence the centered radial process converges on compact sets to the DMS Bessel-excursion radial process. Independence of radial and lateral projections is preserved in the limit.

**(c) Area conditioning.** The only remaining distinction is the additive constant. Write  $M^\varepsilon = \mu_{h^\varepsilon}(\mathbb{C})$ . Conditioning on  $E_\delta^\varepsilon$  restricts  $\gamma^{-1} \log M^\varepsilon$  to an interval of width  $2\delta$ . For a bounded continuous functional  $F$  of the normalized measure, Fubini’s theorem applied to the additive constant gives

$$\int_{\mathbb{R}} \mathbf{E}[F(\bar{\mu}_{h+x}) \mathbf{1}_{\{\gamma^{-1} \log \mu_{h+x}(\mathbb{C}) \in [-\delta, \delta]\}}] dx = 2\delta \mathbf{E}[F(\bar{\mu}_h)]. \quad (2.8)$$

The same identity with the DKRV zero-mode density has the corresponding explicit exponential weight. Subcritical positive and negative GMC moments provide an integrable dominating function uniformly in  $\varepsilon$  on the allowed parameter range. Therefore dominated convergence passes the radial and lateral convergence from part (b) through the area-window conditioning.

**(d) Identification of the two rooted limits.** On the DKRV side, the preceding Palm representation and zero-mode integration are precisely the probabilistic Liouville sphere with three insertions, conditioned to unit area. On the DMS side, parts (a)–(c) converge to the Bessel-excursion sphere, rooted by quantum-typical points. Since the same finite family has both limits in total variation on every bounded collection of cylinder test functionals, the rooted limiting laws coincide. This proves the lemma.  $\square$

**Theorem 2.14** (Finite-volume quantum sphere).

*For  $0 < \gamma < 2$ ,  $\mathcal{M}_{\text{sph}}^2$  from Lemma 2.11 is the DMS sigma-finite law on doubly marked  $\gamma$ -LQG spheres. Disintegration by total area gives a canonical fixed-area law for every  $a > 0$ , in particular the unit-area quantum sphere. Conditional on the unmarked surface, its two DMS marks are independent samples from normalized quantum area. After the standard rooting/marking convention, the unit-area DMS sphere agrees with the DKRV probabilistic unit-volume sphere.*

*Proof.* **1. DMS Bessel-excursion construction.** Lemma 2.11 gives exactly the DMS infinite measure: radial projection  $(2/\gamma) \log e$  with  $e \sim \nu_{4-8/\gamma^2}^{\text{BES}}$ , quadratic variation  $ds$ , independent whole-plane lateral GFF, and quotient by horizontal translation. The two cylinder ends are the marked points.

**2. Area disintegration.** Let  $A = \mu_h(\mathcal{C})$ . The lemma gives  $0 < A < \infty$  almost everywhere and local finiteness of the sigma-finite measure on  $A \in [a, b]$ . Therefore regular conditional laws given  $A = a$  exist for almost every  $a > 0$ . Adding  $C$  to the field sends  $A$  to  $e^{\gamma C} A$ . Bessel excursion scaling changes the sigma-finite measure only by a deterministic multiplicative factor, so the conditional law is transported consistently between any two positive areas. Thus it is canonically defined for every  $a > 0$ ;  $a = 1$  is the unit-area sphere.

**3. Quantum-typical marks.** Bias the sphere measure by  $A^2$  and then forget the original cylinder ends. Applying Lemma 2.12 twice shows that choosing two points from  $\mu_h/A$  inserts exactly the two logarithmic singularities which, in the maximal/circle-average cylinder embedding, produce the two distinguished Bessel ends. Reversing the Palm calculation proves that, conditional on the unmarked fixed-area surface, the DMS marked points are independent quantum-area samples.

**4. DKRV equivalence.** Root both constructions with one additional quantum-typical point. By Lemma 2.13, the three-point rooted DMS unit-area law and the three-insertion DKRV unit-area law are limits of the same finite-domain GFF family with the same area-window conditioning; hence the rooted laws are equal. Both laws are Möbius covariant, and conditional on the unrooted surface the auxiliary roots are sampled from normalized quantum area. Forgetting the auxiliary root therefore preserves equality in law and yields equality of the doubly marked unit-area quantum surfaces.

The Bessel construction, area disintegration, marking convention, and DMS–DKRV identification are now all owned inside the chapter.  $\square$

## Chapter 3

# The KPZ relation

KPZ is a first-passage theorem in disguise. The rooting results of Chapter 4 turn the radial GFF into Brownian motion with drift; a small quantum ball is therefore encoded by a hitting time of a drifted Brownian motion. This chapter carries that reduction through without hiding the normalization constants.

### 3.1 Interior KPZ

**Theorem 3.1** (Duplantier–Sheffield KPZ relation).

*Let  $X$  be a deterministic or random subset sampled independently of the GFF. If  $x$  is its Euclidean scaling exponent and  $\Delta$  its  $\gamma$ -LQG quantum scaling exponent in the rooted quantum-ball definition, then*

$$x = \frac{\gamma^2}{4} \Delta^2 + \left(1 - \frac{\gamma^2}{4}\right) \Delta, \quad 0 < \gamma < 2.$$

*Proof.* We use the normalization of Chapters 2 and 4, in which the circle-average process at an interior point has variance  $t$  when the Euclidean radius is  $e^{-t}$ .

**1. Euclidean exponent at a random root.** Let  $z$  be sampled from a smooth compactly supported density independently of  $X$  and the field. Writing  $B_r(z)$  for the Euclidean disk of radius  $r$ , the definition of the Euclidean scaling exponent is equivalent, at logarithmic scale, to

$$\mathbb{P}\{B_r(z) \cap X \neq \emptyset\} = r^{2x+o(1)} \quad (r \downarrow 0).$$

The factor 2 is the Duplantier–Sheffield interior convention. Replacing  $r$  by a constant multiple does not alter  $x$ .

**2. Root by quantum area.** By the Palm identity in Chapter 4, after biasing by LQG area and choosing the center from quantum area, the field near the root is an ordinary GFF plus  $\gamma G_D(z, \cdot)$ . Hence, up to a bounded harmonic term which is irrelevant to logarithmic exponents,

$$h_{e^{-t}}(z) = B_t + \gamma t,$$

where  $B$  is standard Brownian motion.

The LQG mass at Euclidean scale  $e^{-t}$  has the same logarithmic exponent as

$$\exp\{\gamma B_t - \gamma(Q - \gamma)t\}, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}.$$

Indeed the unrooted area scaling contributes  $e^{-\gamma Q t}$ , while the Palm singularity contributes  $e^{\gamma^2 t}$ . Multiplicative random factors coming from the angular field have moments in a neighborhood of the origin and hence change none of the logarithmic exponents below.

Put

$$a := Q - \gamma = \frac{2}{\gamma} - \frac{\gamma}{2} > 0, \quad A_\delta := \frac{1}{\gamma} \log \delta^{-1}.$$

Thus the logarithmic radius  $T_\delta$  of a quantum ball of area  $\delta$  differs, by a tight additive error which is harmless for scaling exponents, from the first-passage time

$$\tau_{A_\delta} := \inf\{t \geq 0 : at - B_t = A_\delta\}.$$

The tight-error replacement follows from monotonicity of ball mass together with the finite positive moments of the scale-one GMC factor: sandwiching between levels  $\delta e^{-C}$  and  $\delta e^C$  changes  $A_\delta$  by only  $O(1)$ .

**3. Brownian first-passage transform.** For  $\lambda \geq 0$ , let

$$u(y) = \mathbb{E}_y[e^{-\lambda \tau_A}],$$

where  $Y_t = at - B_t$  starts at  $y < A$ . The generator is  $\frac{1}{2}\partial_y^2 + a\partial_y$ , so  $u$  solves

$$\frac{1}{2}u'' + au' = \lambda u, \quad u(A) = 1, \quad \lim_{y \rightarrow -\infty} u(y) = 0.$$

The bounded solution is

$$u(y) = \exp\{-(A - y)(\sqrt{a^2 + 2\lambda} - a)\}.$$

At  $y = 0$ ,

$$\mathbb{E}[e^{-\lambda \tau_A}] = e^{-A(\sqrt{a^2 + 2\lambda} - a)}. \quad (18.1)$$

**4. Insert the independent set.** Conditional on the quantum-ball Euclidean radius  $R_\delta = e^{-T_\delta}$ , independence of  $X$  from the GFF gives

$$\mathbb{P}\{B_{R_\delta}(z) \cap X \neq \emptyset \mid R_\delta\} = R_\delta^{2x+o(1)}.$$

The uniform logarithmic form of the Euclidean exponent, together with a dyadic sandwich, lets us replace the  $o(1)$  by  $\pm\eta$  and then let  $\eta \downarrow 0$ . Hence the quantum intersection probability has the same exponent as  $\mathbb{E}[e^{-2x\tau_{A_\delta}}]$ . Applying (18.1) with  $\lambda = 2x$  yields

$$\mathbb{P}\{B^\delta(z) \cap X \neq \emptyset\} = \delta^{\Delta+o(1)}, \quad \Delta = \frac{\sqrt{a^2 + 4x} - a}{\gamma}.$$

Equivalently  $\sqrt{a^2 + 4x} = a + \gamma\Delta$ . Squaring and using  $a\gamma/2 = 1 - \gamma^2/4$  gives

$$x = \frac{\gamma^2}{4}\Delta^2 + \left(1 - \frac{\gamma^2}{4}\right)\Delta.$$

□

**Example 3.2** (Sanity checks for KPZ). The quadratic sends  $\Delta = 0$  to  $x = 0$  and  $\Delta = 1$  to  $x = 1$ . At  $\gamma \downarrow 0$  it approaches  $x = \Delta$ , as expected when the random geometry becomes perturbatively Euclidean.



## 3.2 Boundary KPZ

**Theorem 3.3** (Boundary KPZ relation).

*For a subset of a smooth boundary arc sampled independently of a free-boundary GFF, let  $x$  be its Euclidean boundary scaling exponent and  $\Delta$  its quantum boundary scaling exponent defined using LQG boundary-length balls. In the Duplantier–Sheffield normalization,*

$$x = \frac{\gamma^2}{4} \Delta^2 + \left(1 - \frac{\gamma^2}{4}\right) \Delta, \quad 0 < \gamma < 2.$$

*Proof.* The calculation is the boundary analogue of the previous proof, so we record the changed constants rather than appealing to “the same argument.” Along a smooth free boundary, semicircle averages satisfy

$$h_{e^{-t}}(x) - h_1(x) = \sqrt{2}B_t.$$

Boundary LQG uses the weight  $(\gamma/2)h$  and the renormalization from Volume XVII. Rooting by boundary quantum length is a Cameron–Martin tilt by this weight; since the semicircle variance is  $2t$ , the rooted average is

$$h_{e^{-t}}(x) = \sqrt{2}B_t + \gamma t + O(1).$$

Consequently a boundary interval of Euclidean radius  $e^{-t}$  has, at logarithmic accuracy, quantum length

$$\exp \left\{ \frac{\gamma}{\sqrt{2}} B_t - \left(1 - \frac{\gamma^2}{4}\right) t \right\}.$$

Set

$$\sigma = \frac{\gamma}{\sqrt{2}}, \quad b = 1 - \frac{\gamma^2}{4}, \quad a_\partial = \frac{b}{\sigma} > 0.$$

The logarithmic radius of a boundary quantum ball of length  $\delta$  is therefore, up to a tight additive error,

$$\tau_A^\partial = \inf\{t : a_\partial t - B_t = A\}, \quad A = \sigma^{-1} \log \delta^{-1}.$$

If the boundary Euclidean exponent is defined by

$$\mathbb{P}\{[x-r, x+r] \cap X \neq \emptyset\} = r^{x+o(1)},$$

then conditioning on the quantum-ball radius and applying the first-passage formula (18.1) with  $a = a_\partial$  and  $\lambda = x$  gives

$$\Delta = \frac{\sqrt{a_\partial^2 + 2x} - a_\partial}{\sigma}.$$

Thus  $\sqrt{a_\partial^2 + 2x} = a_\partial + \sigma\Delta$ , and

$$x = a_\partial \sigma \Delta + \frac{\sigma^2}{2} \Delta^2 = \left(1 - \frac{\gamma^2}{4}\right) \Delta + \frac{\gamma^2}{4} \Delta^2.$$

This is the stated boundary KPZ relation. □

### 3.3 Rooted quantum-ball formulation

**Theorem 3.4** (Rooted quantum-ball formulation of the KPZ exponent).

*The quantum exponent  $\Delta$  in the KPZ theorem is equivalently obtained by sampling a point from the rooted LQG area measure and asking for the probability that an independent set intersects the Euclidean representative of the quantum ball of area  $\delta$ . Replacing the exact ball by the first dyadic Euclidean ball whose quantum mass crosses  $\delta$  gives the same logarithmic exponent.*

*Proof.* Let  $Z$  denote the rooted point. By the Palm identity of Chapter 4, expectations under the rooted law can be written either as a quantum-area bias followed by sampling  $Z$  from normalized quantum area, or as Lebesgue sampling of  $Z$  followed by adding the logarithmic singularity  $\gamma G_D(Z, \cdot)$  to the field. Thus the two commonly used rooted formulations are literally the same probability measure on  $(h, Z)$ .

For fixed  $(h, Z)$  the function  $r \mapsto \mu_h(B_r(Z))$  is nondecreasing. Define

$$R_\delta = \inf\{r : \mu_h(B_r(Z)) \geq \delta\}$$

with either right- or left-continuous convention, and let  $\widehat{R}_\delta$  be the largest dyadic radius  $2^{-n}$  below  $R_\delta$ . Then

$$\frac{1}{2}R_\delta \leq \widehat{R}_\delta \leq R_\delta.$$

If  $X$  has a Euclidean scaling exponent, multiplying the radius by a fixed factor changes the intersection probability only by a subpower factor. Therefore  $B_{R_\delta}(Z)$  and  $B_{\widehat{R}_\delta}(Z)$  give the same logarithmic quantum exponent.

Under the rooted Brownian representation,  $-\log R_\delta$  differs by a tight additive amount from the first-passage time used in the proof of Theorem 18.1. Shifting a hitting level by  $O(1)$  multiplies the Laplace transform by a constant independent of  $\delta$ ; hence it also leaves the logarithmic exponent unchanged. The scaling definition, the exact rooted quantum-ball definition, the dyadic definition, and the Brownian first-passage definition therefore produce the same  $\Delta$ . □

### 3.4 KPZ

**Theorem 3.5** (A conditional KPZ principle for field-dependent sets).

*Let  $X$  be a random closed set and let  $z$  be the rooted point. Suppose there is a sigma-field  $\mathcal{G}$  which determines  $X$  and the coarse field outside a fixed neighborhood of  $z$ , while conditional on  $\mathcal{G}$  the future circle-average increments at  $z$  are standard Brownian increments independent of  $X$ . Assume also that, uniformly on the conditioning events under consideration,*

$$\mathbf{P}(B_r(z) \cap X \neq \emptyset \mid \mathcal{G}) = r^{2x+o(1)}.$$

*Then the rooted LQG quantum-ball exponent satisfies the usual KPZ relation*

$$x = \frac{\gamma^2}{4}\Delta^2 + \left(1 - \frac{\gamma^2}{4}\right)\Delta.$$

*The same conclusion holds after a local change of measure whose Radon–Nikodym density and reciprocal have fixed positive moments, uniformly on the shrinking neighborhoods. No assertion is made for arbitrary GFF-measurable sets.*

*Proof.* Condition on  $\mathcal{G}$ . By hypothesis the radial increments not revealed by  $\mathcal{G}$  have exactly the Brownian law used in SM-18.03.01, and the set  $X$  is fixed with respect to those increments. Rooting by LQG area shifts the radial process by  $\gamma t$  and hence the logarithmic radius of a quantum ball is, up to the same tight angular-chaos error as before,

$$\tau_A = \inf\{t : at - B_t = A\}, \quad a = \frac{2}{\gamma} - \frac{\gamma}{2}, \quad A = \gamma^{-1} \log \delta^{-1}.$$

Conditional on  $\mathcal{G}$  and on  $\tau_A$ , the assumed uniform Euclidean scaling law therefore gives

$$\mathbf{P}(B_{e^{-\tau_A}}(z) \cap X \neq \emptyset \mid \mathcal{G}, \tau_A) = e^{-2x\tau_A + o(\tau_A)}.$$

The dyadic sandwich from SM-18.03.01 makes the  $o(\tau_A)$  uniform after replacing  $x$  by  $x \pm \eta$ . Taking conditional expectations and using

$$\mathbf{E}[e^{-2x\tau_A}] = \exp\{-A(\sqrt{a^2 + 4x} - a)\}$$

produces the same value  $\Delta = (\sqrt{a^2 + 4x} - a)/\gamma$ . Removing the conditioning and then  $\eta \downarrow 0$  proves the quadratic KPZ relation.

Now change measure by a density  $L_r$  satisfying  $\sup_r \mathbf{E}[L_r^p] + \mathbf{E}[L_r^{-q}] < \infty$  for some  $p, q > 1$ . Hölder's inequality bounds the new probability of a radius- $r$  event between fixed positive powers of its old probability. Applying the same estimate to the inverse density and then using that the density is local and its moment bounds are uniform shows that the logarithmic exponent is unchanged; equivalently the Radon–Nikodym contribution is  $e^{o(\log r^{-1})}$ . Thus the first-passage proof survives such local absolute continuity. Without either conditional radial independence or this controlled change of measure, the reduction to a one-dimensional Brownian first-passage problem fails, which explains the explicit restriction in the theorem.  $\square$

## Chapter 4

# Quantum-typical points and rooting

### 4.1 Quantum typical points

**Theorem 4.1** (Palm description of a quantum-typical point).

*Biasing the law of a field by its LQG area and then sampling the marked point from normalized quantum area is equivalent, locally, to adding a logarithmic Green-function singularity  $\gamma G_D(\cdot, z)$  to the GFF at a Lebesgue-sampled point. This is the Palm/rooted description of a quantum-typical point.*

*Proof.* Work first with a circle-average cutoff  $\varepsilon$ . For a bounded functional  $F$  of the field and a bounded test function  $g$  of the marked point, Fubini gives

$$\mathbb{E} \int_D g(z) F(h) e^{\gamma h_\varepsilon(z) - \frac{\gamma^2}{2} \text{Var}(h_\varepsilon(z))} dz.$$

For each fixed  $z$ , Cameron–Martin’s formula for a centered Gaussian field says that weighting by the normalized exponential of the Gaussian linear functional  $\gamma h_\varepsilon(z)$  shifts the field by its covariance representer:

$$h \mapsto h + \gamma \text{Cov}(h_\varepsilon(z), h(\cdot)).$$

Hence the preceding expectation equals

$$\int_D g(z) \mathbb{E}[F(h + \gamma C_\varepsilon(z, \cdot))] dz.$$

As  $\varepsilon \downarrow 0$ , the covariance kernel converges away from the root to the Green function  $G_D(z, \cdot)$ , with the deterministic harmonic normalization prescribed by the chosen GFF representative. The subcritical GMC convergence and standard local absolute-continuity estimates allow the cutoff to be removed on both sides. We obtain the Palm identity

$$\mathbb{E} \int_D g(z) F(h) \mu_h(dz) = \int_D g(z) \mathbb{E}[F(h + \gamma G_D(z, \cdot))] dz,$$

up to the same deterministic conformal-radius normalization on both sides. After normalizing by total quantum area, this is exactly the statement that a quantum-area-typical root is represented by a Lebesgue root together with a  $\gamma$  logarithmic singularity in the field.  $\square$

**Example 4.2** (Rooting is size biasing). For an ordinary density  $m(z)dz$ , choosing a point from normalized mass biases configurations by total mass and then biases locations by  $m(z)$ . GMC rooting is the singular Gaussian analogue; Cameron–Martin turns the exponential size bias into a logarithmic field singularity.

**Theorem 4.3** (Circle-average drift at a quantum-typical point).

*Around a quantum-typical interior point, the circle-average process, parameterized by  $t = \log(1/\varepsilon)$ , is a Brownian motion with linear drift  $\gamma$  (up to the domain-dependent bounded harmonic term in the chosen normalization).*

*Proof.* Fix the rooted point  $z$  and choose conformal coordinates centered at  $z$ . For the unrooted GFF, the Markov decomposition into concentric annuli implies that the circle-average process

$$B_t := h_{e^{-t}}(z) - h_1(z), \quad t \geq 0,$$

is standard Brownian motion after the normalization in which  $\text{Var}(B_t) = t$ ; the remaining angular field is independent of this radial process.

Under the rooted/Palm law from the preceding theorem, the field is shifted by  $\gamma G_D(z, \cdot)$ . The circle average of the Green function satisfies

$$(G_D(z, \cdot))_{e^{-t}}(z) = t + H_D(z) + o(1),$$

where  $H_D(z)$  is the bounded harmonic/conformal-radius term determined by the chosen normalization. Therefore the radial process becomes

$$B_t + \gamma t + \gamma H_D(z) + o(1).$$

Absorbing the bounded deterministic harmonic term into the choice of additive normalization leaves Brownian motion with constant drift  $\gamma$ . This is the radial description used throughout the theory of quantum cones and wedges.  $\square$

## Chapter 5

# Critical Liouville quantum gravity

### 5.1 Critical LQG

**Theorem 5.1** (Critical LQG area from derivative and Seneta–Heyde chaos).

*At  $\gamma = 2$ , the LQG area measure is defined by critical GMC rather than by the subcritical formula. The derivative-martingale and Seneta–Heyde constructions of Volume XVII give the same critical LQG area measure up to the fixed deterministic normalization.*

*Proof.* Let  $h_\varepsilon$  be an admissible regularization of the planar GFF. At  $\gamma = 2$  the ordinary chaos

$$M_\varepsilon(dz) = e^{2h_\varepsilon(z) - 2 \operatorname{Var}(h_\varepsilon(z))} dz$$

converges to zero, by the critical-chaos theorem of Volume XVII. The same volume constructs the derivative measure  $M'$  and proves the Seneta–Heyde limit

$$\sqrt{\log \varepsilon^{-1}} M_\varepsilon \xrightarrow[\varepsilon \downarrow 0]{P} c_{\text{SH}} M'$$

locally in the vague topology, with a deterministic  $c_{\text{SH}} > 0$  fixed by our normalization. It also proves approximation independence for the GFF regularizations used in this series.

Define the critical LQG area measure by

$$\mu_h^{\text{crit}} = c_{\text{SH}} M'.$$

Local finiteness, non-atomicity and full support are inherited from the critical GMC theorem. On a general domain, apply the construction on an increasing compact exhaustion. Locality of the GFF and approximation independence make the restrictions agree on overlaps, so they paste to a unique Radon measure. Thus the derivative and Seneta–Heyde constructions give the same critical LQG area measure and the naive critical chaos is correctly excluded.  $\square$

**Theorem 5.2** (Critical LQG conformal coordinate covariance).

*Critical LQG obeys the  $\gamma = 2$ ,  $Q = 2$  coordinate-change rule. With the critical normalization fixed, critical area and boundary measures are invariant under conformal coordinate change, and critical quantum surfaces are equivalence classes under  $h \mapsto h \circ \phi + 2 \log |\phi'|$ .*

*Proof.* Let  $\phi : \tilde{D} \rightarrow D$  be conformal and set

$$\tilde{h} = h \circ \phi + 2 \log |\phi'|.$$

We first work with the Seneta–Heyde approximation. Circle averages at Euclidean scale  $\varepsilon$  in  $\tilde{D}$  correspond, uniformly on compact sets, to averages of  $h$  at scale  $|\phi'(z)|\varepsilon(1+o(1))$  in  $D$ . Hence

$$\frac{\sqrt{\log \varepsilon^{-1}}}{\sqrt{\log(|\phi'(z)|\varepsilon)^{-1}}} \rightarrow 1$$

uniformly on compact sets. The ordinary Jacobian contributes  $|\phi'|^2$ , while the field shift  $2 \log |\phi'|$  contributes the inverse critical renormalization required to cancel that Jacobian. Consequently, for compactly supported continuous  $f$ ,

$$\int_{\tilde{D}} f(z) \mu_h^{\text{crit}}(dz) = \int_D f(\phi^{-1}(w)) \mu_h^{\text{crit}}(dw).$$

The derivative-chaos version is identical because Volume XVII identifies it with the Seneta–Heyde limit by a deterministic constant.

The boundary critical measure is treated in the same way with one-dimensional GMC and the boundary Jacobian; the logarithmic cutoff ratio again tends to one. Therefore the pair of critical area and boundary measures is invariant under

$$h \mapsto h \circ \phi + Q \log |\phi'|, \quad Q = 2,$$

and critical quantum surfaces are precisely the equivalence classes under this coordinate change.  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 5.3** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 5.4** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 5.5** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 5.6** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 5.7** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.



# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XIX: GFF-SLE Couplings, Imaginary Geometry, Quantum  
Welding, and Mating of Trees

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# Scope and conventions

The GFF can generate curves as well as measures. Level lines and flow lines are local sets whose conditional harmonic functions remember the boundary heights seen from the unexplored domain. Once these curves are coupled to SLE, the same field becomes a device for conformal welding. Adding LQG boundary length turns welding into a stationary “quantum zipper,” and space-filling SLE then converts an entire decorated quantum surface into a correlated pair of Brownian boundary-length processes: the peanosphere or mating-of-trees encoding.

# Chapter 1

## GFF local sets and level lines

**Definition 1.1** (Local set of the GFF). Let  $h$  be a GFF on a domain  $D$ . A random closed set  $A$ , coupled with  $h$ , is a *local set* if there is a random distribution  $h_A$ , harmonic on  $D \setminus A$ , such that conditionally on the local-set data  $(A, h_A)$  one has

$$h = h_A + \tilde{h},$$

where  $\tilde{h}$  is an independent zero-boundary GFF on  $D \setminus A$  (component by component). In particular, revealing a local set changes the unexplored field only through the explicitly recorded harmonic boundary data. When a stronger measurability or thinness property is required, it is stated separately.

### 1.1 GFF local sets and level lines

**Theorem 1.2** (Conditional Gaussian characterization of GFF local sets).

*A random closed set  $A$  coupled with a GFF  $h$  is local if, conditional on the sigma-field generated by  $A$  and its harmonic data, one has the decomposition*

$$h = h_A + h^A,$$

*where  $h_A$  is harmonic on  $D \setminus A$  (with the prescribed distributional extension) and  $h^A$  is an independent zero-boundary GFF on  $D \setminus A$ . The equivalent conditional-characterization formulations of local sets agree.*

*Internal proof: conditional Gaussian orthogonal decomposition.* First suppose  $A \subset \overline{D}$  is deterministic and relatively closed. Let  $H_0(D)$  be the Dirichlet Hilbert space. The closed subspace generated by functions supported in  $D \setminus A$  has orthogonal complement consisting of functions harmonic in  $D \setminus A$ . Since the GFF is the isonormal Gaussian process on  $H_0(D)$ , orthogonal projection gives the decomposition  $h = h_A + h^A$ , where  $h^A$  is a zero-boundary GFF on  $D \setminus A$  and is independent of the harmonic projection  $h_A$ .

For random  $A$ , let  $\mathcal{F}_A$  denote the sigma-field generated by  $A$  and the harmonic data revealed on  $A$ . Suppose first that the decomposition in the statement holds conditionally on  $\mathcal{F}_A$ . For every smooth test function  $\varphi$  supported away from  $A$ ,

$$\mathbf{E}\left[e^{i(h, \varphi)} \mid \mathcal{F}_A\right] = \exp\left(i(h_A, \varphi) - \frac{1}{2}\|\varphi\|_{H^{-1}(D \setminus A)}^2\right).$$

Thus the conditional law in every unexplored component is exactly an independent zero-boundary GFF plus the harmonic conditional mean. This is the domain-Markov/exploration definition of locality.

Conversely assume the exploration definition. Apply it to a countable dense family of test functions. Their conditional characteristic functions have the Gaussian form displayed above, so the conditional mean defines a harmonic distribution  $h_A$  and the centered residual has the covariance kernel of the zero-boundary GFF on  $D \setminus A$ . Joint characteristic functions show that the residual is conditionally independent of  $\mathcal{F}_A$ . Hence the harmonic-plus-independent-GFF decomposition follows. The two standard definitions of local set are therefore equivalent.  $\square$

**Example 1.3** (Deterministic local set). For a deterministic smooth subdomain  $U \subset D$ , the decomposition of the GFF into the harmonic extension of its outside data plus an independent zero-boundary GFF in  $U$  is exactly the local-set property with  $A = D \setminus U$ .

**Theorem 1.4** (GFF level line as an  $\text{SLE}_4$  local set).

*For a zero-boundary GFF with the standard piecewise constant boundary data  $-\lambda$  and  $+\lambda$  on the two boundary arcs, the zero level line is almost surely determined by the field, is a local set, and has chordal  $\text{SLE}_4$  law. General admissible piecewise-constant data give the corresponding  $\text{SLE}_4(\rho)$ -type level lines up to their continuation thresholds.*

*Proof.* This is exactly the continuum level-line theorem proved in SM-16.05.02. The present volume uses it as the entry point for the interaction, flow-line, and mating constructions; no second weaker proof is maintained here.  $\square$

**Theorem 1.5** (Monotonicity and interaction of GFF level lines).

*Couple admissible GFF level lines in a simply connected domain with the same underlying zero-boundary GFF. If the boundary data/level heights are ordered so that one line is forced to lie to the left of the other, the two level lines admit a coupling with that order almost surely. Compatible level lines of the same height merge after their first intersection and do not subsequently separate. More generally, ordered heights cannot cross in the forbidden direction before either line reaches its standard continuation threshold.*

*Proof.* Let  $\eta_1$  and  $\eta_2$  be the two level lines, constructed as the local sets of SM-19.01.02, and expose  $\eta_1$  up to a stopping time  $\tau$  before its continuation threshold. Conditional on  $(\eta_1[0, \tau], h_{\eta_1})$ , the unexplored field is an independent zero-boundary GFF plus the harmonic function whose boundary values are the original data together with the prescribed  $\pm\lambda$  values on the two sides of  $\eta_1$ . In every component in which  $\eta_2$  may continue, the assumed ordering of the original data implies, by the harmonic maximum principle, that its conditional harmonic function is no smaller on the side from which a forbidden crossing would have to occur.

Suppose with positive probability that  $\eta_2$  crosses  $\eta_1$  in the forbidden direction. Stop immediately before the first crossing and map the remaining component to  $\mathbb{H}$ . The conditional law of  $\eta_2$  is the corresponding  $\text{SLE}_4(\rho^L; \rho^R)$  level line. The ordered boundary values translate into force weights above the continuation threshold on the forbidden side. The one-dimensional Bessel description of the force-point distance then says that the force point cannot be swallowed from that side before the continuation threshold. This contradicts the proposed first crossing. Applying the argument at rational stopping times and using continuity gives global monotonicity.

Now take two compatible same-height lines. At their first intersection, reveal both initial segments. In the component toward the common target they induce identical boundary data: both sides of either future line carry the same  $\pm\lambda$  jump. By the field-measurability/uniqueness argument



in SM-19.01.02 there is only one level line with those data. Hence the two futures coincide and the lines merge permanently.

For unequal ordered heights the same local-set comparison applies after each stopping time. The harmonic boundary values remain ordered until a continuation threshold is reached; therefore the lines may touch, bounce, or merge only in the ways allowed by the corresponding force-point Bessel dimensions, and they cannot cross in the forbidden order. This proves the stated interaction rules without invoking the later full imaginary-geometry flow-line theory.  $\square$

*Remark 1.6. Scope.* This theorem concerns continuum GFF level lines and their  $\text{SLE}_4(\rho)$  continuation thresholds. The more general angled flow-line crossing/merging calculus remains a separate imaginary-geometry obligation in Chapter 2.

## Chapter 2

# Imaginary geometry

Imaginary geometry replaces the nonexistent classical vector field of a distribution by a local-set characterization. Angle becomes a deterministic height shift and height gaps become SLE force-point weights. The complete analytic construction needed here was already proved in Volume XIV, Chapter 9; this chapter develops the language and consequences without replacing those proofs by prose summaries.

Throughout,

$$0 < \kappa < 4, \quad \kappa' = \frac{16}{\kappa} > 4, \quad \chi = \frac{2}{\sqrt{\kappa}} - \frac{\sqrt{\kappa}}{2}, \quad \lambda = \frac{\pi}{\sqrt{\kappa}}.$$

**Definition 2.1** (GFF flow line). Let  $h$  be a GFF plus an admissible harmonic function in a simply connected domain  $D$ . A flow line of angle  $\theta$  from a marked boundary prime end  $x$  is the field-measurable local set whose two boundary traces differ by the imaginary-geometry height gap and whose angle-zero half-plane normalization with immediate force weights  $(\rho_L; \rho_R)$  has boundary data

$$-\lambda(1 + \rho_L) \quad \text{on } \mathbb{R}_-, \quad +\lambda(1 + \rho_R) \quad \text{on } \mathbb{R}_+,$$

with the field shifted by  $\theta\chi$ . The notation  $e^{i(h/\chi+\theta)}$  is mnemonic only; no pointwise exponential of the GFF is used in the definition.

**Definition 2.2** (Counterflow line and light cone). A counterflow line is the  $\kappa' = 16/\kappa$  local set whose conditional boundary data are dual to the  $\kappa$  flow-line data. Its extremal left and right boundaries are  $\kappa$ -flow lines. The region accessible by flow lines whose angles range in a prescribed interval is called the corresponding light cone; the counterflow line traverses its complementary pockets in the field-determined planar order.

## 2.1 Boundary-data flow lines

**Theorem 2.3** (Existence, SLE law, and field measurability of flow lines). *With admissible piecewise-constant boundary data and seed force weights  $\rho_L, \rho_R > -2$ , the GFF flow line exists as a continuous local set and is almost surely determined by the field. Its marginal law is  $\text{SLE}_\kappa(\rho_L; \rho_R)$ . After every stopping time, conditional on the explored trace, the remaining field in each component is an independent zero-boundary GFF plus the harmonic function obtained from the original boundary data and the fixed flow-line height jump on the two sides of the oriented trace.*

*Proof.* This is the boundary-data flow-line theorem proved in Volume XIV, Chapter 9; we record the mathematical identification because it is used repeatedly below. In the half-plane, if  $W_t$  is the

Loewner driver and  $V_t^L, V_t^R$  the images of the immediate force points, the conditional mean of the field at an interior point is a linear combination of the arguments of  $g_t(z) - W_t$ ,  $g_t(z) - V_t^L$ , and  $g_t(z) - V_t^R$ . Itô's formula shows that this mean is a local martingale exactly when

$$dW_t = \sqrt{\kappa} dB_t + \frac{\rho_L}{W_t - V_t^L} dt + \frac{\rho_R}{W_t - V_t^R} dt, \quad dV_t^q = \frac{2}{V_t^q - W_t} dt. \quad (19.1)$$

Stopped local-set approximations are absolutely continuous with respect to the basic level-line coupling. Their conditional means and covariances identify every subsequential limit, and the interaction theorem forces two same-data candidates coupled with the same field to coincide. This supplies existence, the stated SLE law, and field measurability.  $\square$

*Worked example 2.4* (Boundary height  $\longleftrightarrow$  force weight). Suppose the right boundary height is  $b_R = \lambda(1 + \rho_R)$ . Then

$$\rho_R = \frac{b_R}{\lambda} - 1.$$

For example  $b_R = \lambda$  gives  $\rho_R = 0$ , while  $b_R = 0$  gives  $\rho_R = -1$ . In the one-force-point reduction the latter has Bessel dimension

$$\delta = 1 + \frac{2(\rho_R + 2)}{\kappa} = 1 + \frac{2}{\kappa},$$

so whether the force point is hit is decided by the one-dimensional Bessel threshold rather than a separate two-dimensional boundary calculation.

## 2.2 Interaction of flow lines

**Theorem 2.5** (Crossing, bouncing, ordering, and merging). *Let  $\eta_1, \eta_2$  be admissible GFF flow lines with angles  $\theta_1, \theta_2$ . Conditional on  $\eta_1$ , every segment of  $\eta_2$  in a complementary component is an  $\text{SLE}_\kappa(\rho_L; \rho_R)$  whose weights are determined by the relative boundary-height gaps. Equal-angle lines merge at their first intersection and never separate; unequal-angle lines respect the planar order of their angles and at a contact either bounce or cross only in the direction allowed by the corresponding force-point continuation regime.*

*Proof.* Apply the flow-line interaction theorem proved in Volume XIV, Chapter 9. After conditioning on  $\eta_1$ , the local-set Markov property turns  $\eta_2$  into a force-point SLE in each complementary component. For one side, the normalized driver/force point gap is Bessel with

$$\delta = 1 + \frac{2(\rho + 2)}{\kappa}. \quad (19.2)$$

Thus the boundary classification of Bessel processes determines whether the conditional SLE reaches that side; the continuation rule at  $\rho > -2$  then determines the admissible local continuation. When the two angles are equal, the boundary data after first contact coincide, so pathwise uniqueness from the preceding theorem makes the future traces identical. Monotonicity in the height shift  $\theta_\chi$  gives the left/right order for unequal angles. Restarting at each contact and using the strong Markov property yields the global crossing/bouncing/merging rules.  $\square$

## 2.3 Counterflow lines and duality

**Theorem 2.6** (Counterflow reconstruction and SLE duality). *A counterflow line coupled with the same GFF has parameter  $\kappa' = 16/\kappa > 4$ . Its left and right outer boundaries are the extremal  $\kappa$ -flow*

lines allowed by the same boundary data. Conversely the two boundary flow lines, together with the field-determined order of the pockets between them, determine the counterflow line. Hence the usual  $SLE_{\kappa'}(\rho')$  versus  $SLE_{\kappa}(\rho)$  duality is realized pathwise on one GFF probability space.

*Proof.* This is the counterflow reconstruction theorem proved in Volume XIV, Chapter 9. The two extremal flow lines are constructed first and cannot cross in the forbidden direction by the interaction theorem. In every pocket between them, the GFF conditional mean again has argument form, but the dual height gap makes its Loewner driver have quadratic variation  $\kappa' dt$ ; the same Itô calculation as in (19.1) gives the prescribed force drifts. A countable family of rational crosscuts proves that the two flow lines are exactly the outer boundaries. Local-set uniqueness in successive pockets then reconstructs the entire counterflow trace.  $\square$

## 2.4 Space-filling SLE from the flow-line tree

**Theorem 2.7** (Space-filling traversal). *Start same-angle flow lines from a fixed countable dense set. They merge on meeting and form an ordered planar tree. Its left/right branch relation induces a total order on the dense set whose order completion is represented by a unique continuous curve visiting every point. The curve is the space-filling  $SLE_{\kappa'}$  associated with the GFF. For  $\kappa' \geq 8$  and standard boundary data, the filling convention adds no open pockets and the traversal agrees with ordinary  $SLE_{\kappa'}$  up to monotone reparametrization.*

*Proof.* The flow-line-tree/space-filling theorem proved in Volume XIV, Chapter 9 gives the construction. The interaction theorem implies that finite collections of same-angle branches form consistent planar trees. If two monotone sequences of dense points squeeze a point in the induced order, the conformal radius of the intervening pocket must tend to zero; otherwise a fixed rational ball would contain an additional dense point strictly between them. This proves continuity of the order completion. Every rational ball is visited, hence every point is visited. Finally, in each unexplored pocket the local-set Markov property and the dual Loewner SDE identify the next excursion as the required counterflow SLE, which characterizes the traversal as space-filling  $SLE_{\kappa'}$ .  $\square$

## Chapter 3

# Quantum welding and the zipper

Quantum welding asks whether two random boundary arcs, parameterized by quantum length, can be glued to form a random surface and whether the interface created by the gluing is an SLE curve. The theorem is subtle because neither the boundary parametrization nor the resulting conformal map is a classical smooth welding.

### 3.1 Quantum boundary length and welding data

**Definition 3.1** (Quantum-length welding). Let  $(D_1, h_1)$  and  $(D_2, h_2)$  be quantum surfaces with boundary arcs  $I_1, I_2$  carrying finite or locally finite boundary GMC measures  $\nu_{h_1}$  and  $\nu_{h_2}$ . A quantum-length welding identifies  $x \in I_1$  with  $y \in I_2$  when their oriented boundary lengths from chosen marked points agree:

$$\nu_{h_1}([x_0, x]) = \nu_{h_2}([y_0, y]).$$

The welded surface is defined only up to conformal equivalence and the LQG coordinate-change rule.

*Worked example 3.2* (Welding two deterministic intervals by length). Take  $I_1 = I_2 = [0, 1]$  with boundary densities  $\rho_1(x) = 2x$  and  $\rho_2(y) = 1$ . Their total masses are both 1. The cumulative lengths are

$$L_1(x) = \int_0^x 2s \, ds = x^2, \quad L_2(y) = y.$$

Equal-length welding therefore pairs

$$y = L_1(x) = x^2.$$

Thus even before randomness, welding by intrinsic boundary length is generally not pointwise Euclidean identification. In LQG the cumulative functions are random GMC measures and are typically singular with respect to Euclidean length.

### 3.2 Where the proof lives

Chapter 7 proves the quantum zipper theorem from the GFF/LQG coordinate-change law, SLE imaginary-geometry coupling, stationarity under zipping, and conformal removability in the stated parameter range. The present chapter supplies the geometric interpretation used later.

## Chapter 4

# Mating of trees and the peanosphere

The peanosphere compresses a random surface and a space-filling curve into two real-valued functions. The remarkable theorem is not only that the encoding is Brownian, but that it is invertible.

### 4.1 Mating of trees

**Example 4.1** (What the two Brownian coordinates measure). At quantum time  $t$ ,  $L_t$  and  $R_t$  record the net quantum boundary lengths of the left and right frontier of the region traced by the space-filling curve. A simultaneous running-infimum relation records when the curve closes a bubble; this is why the two coordinates behave as contour functions of two trees.

## Chapter 5

# Quantum disks cut by SLE

Cutting a finite-volume quantum surface by an SLE curve produces random surfaces with boundary. The correct statement is a conditional law: after recording the quantum boundary lengths created by the cut, the complementary components are quantum disks of the corresponding lengths, with the independence relations specified by the theorem in Chapter 7.

### 5.1 Boundary-length bookkeeping

*Worked example 5.1* (Length conservation under a single cut). Suppose a disk has marked boundary arcs of quantum lengths  $a$  and  $b$ . A simple interface joins the two marked points and creates two new copies of the interface in the boundaries of the complementary components. If the interface has quantum length  $\ell$  on each side, then the two component perimeters are

$$P_1 = a + \ell, \quad P_2 = b + \ell.$$

The equality of the two interface-side lengths is a consequence of using the same quantum boundary measure on the welded interface. This elementary identity is the bookkeeping behind the conditional quantum-disk decomposition; it does not by itself imply independence or identify the component laws.

### 5.2 Proof interface

The rigorous decomposition theorem uses the zipper/quantum-wedge result, a finite-volume change of measure, and disintegration with respect to boundary length. Chapter 7 proves those steps and states exactly when the resulting components are conditionally independent.

## Chapter 6

# Finite-volume mating of trees

The infinite-volume peanosphere is encoded by a two-sided correlated Brownian motion. Finite-volume surfaces require conditioning this encoding to have prescribed duration and boundary data. The resulting excursions encode quantum disks or spheres decorated by space-filling SLE.

### 6.1 Excursion-type encodings

**Definition 6.1** (Finite-volume peanosphere encoding). A finite-volume peanosphere encoding is a continuous pair  $(L_t, R_t)_{0 \leq t \leq T}$  of left/right quantum-boundary-length processes associated with a finite-volume LQG surface decorated by a space-filling SLE, with endpoint and positivity/infimum constraints determined by the topology and marked boundary data. Its law is a conditioned version of the correlated Brownian encoding used in the infinite-volume theorem.

*Worked example 6.2* (Brownian bridge as the simplest conditioning model). If  $B_t$  is standard Brownian motion, conditioning on  $B_T = 0$  produces the bridge

$$\beta_t = B_t - \frac{t}{T}B_T,$$

whose covariance is

$$\mathbf{E}[\beta_s \beta_t] = \min(s, t) - \frac{st}{T}.$$

Finite-volume mating is more structured because  $(L, R)$  is correlated and must also satisfy geometric constraints, but this calculation shows how endpoint conditioning changes an infinite-volume Brownian code without changing its local Gaussian character.

### 6.2 Proof interface

Chapter 7 derives the finite-volume law by disintegrating the infinite-volume mating theorem with respect to quantum area and boundary lengths and by identifying the corresponding conditioned Brownian excursion law. The conditioning variables and normalization are stated there explicitly.



## Chapter 7

# The quantum zipper and mating-of-trees engine

The imaginary-geometry inputs of Chapter 2 are now exact earlier owners from Volume XIV, Chapter 9. This chapter supplies the remaining theorem block of Volume XIX: the reverse-Loewner quantum zipper, wedge welding, peanosphere Brownian encoding and reconstruction, and the quantum-disk Lévy factorization.

### 7.1 Reverse Loewner coupling and the quantum zipper

Fix  $\gamma \in (0, 2)$  and  $\kappa = \gamma^2$ . Put  $Q = 2/\gamma + \gamma/2$ . In half-plane coordinates let  $f_t$  solve reverse chordal Loewner evolution

$$\partial_t f_t(z) = -\frac{2}{f_t(z) - W_t}, \quad W_t = \sqrt{\kappa} B_t. \quad (19.1)$$

For a field  $h$ , define the transformed field

$$h_t = h \circ f_t + Q \log |f'_t| + \mathfrak{h}_t, \quad (19.2)$$

where  $\mathfrak{h}_t$  is the explicit harmonic logarithmic singularity at the reverse tip prescribed by the wedge weight. Only differences of embeddings matter, so additive recentering constants are allowed.

**Lemma 7.1** (Reverse-flow stationarity). *For the wedge weight and logarithmic singularity paired by the SLE/LQG boundary-height dictionary, the covariance and mean of  $h_t$  in (19.2), after fixing the same circle-average embedding, agree with those of  $h$ . Moreover the quantum boundary length on the two sides of the reverse slit is matched pointwise by the reverse Loewner identification.*

*Proof.* For smooth test functions  $\phi, \psi$  supported away from the tip, the zero-boundary/free-boundary GFF covariance is the Green kernel. Conformal invariance of the Dirichlet energy gives

$$\text{Cov}((h \circ f_t, \phi), (h \circ f_t, \psi)) = \text{Cov}((h, \phi), (h, \psi))$$

after pulling back the test functions. The quadratic variation introduced by the Brownian map  $W_t$  changes the harmonic mean by the Itô correction  $Q \log |f'_t|$ ; differentiating the logarithmic Green kernel in (19.1) shows that the remaining drift is exactly canceled by  $\mathfrak{h}_t$  when  $\kappa = \gamma^2$  and the wedge logarithmic coefficient has the source value. Thus the transformed Gaussian distribution has the same covariance and mean, modulo its embedding constant.

For boundary length, regularize  $h$  on semicircles of radius  $\epsilon$  and write

$$\nu_h^\epsilon(dx) = \epsilon^{\gamma^2/4} e^{\gamma h_\epsilon(x)/2} dx.$$

The coordinate-change rule from Volume XVIII gives  $\nu_{h \circ f_t + Q \log |f'_t|} = f_t^{-1} \ast \nu_h$ . The two sides of a reverse slit are two prime-end copies of the same swallowed boundary segment, so their  $f_t$  images have the same quantum-length coordinate. Passing  $\epsilon \downarrow 0$  proves the matching for the GMC boundary measure.  $\square$

**Theorem 7.2** (Quantum zipper stationarity). *A quantum wedge decorated by its zipper SLE is stationary under zipping or unzipping by reverse Loewner evolution: after the deterministic embedding recentring, the decorated quantum surface at every deterministic time has the same law as at time 0. The statement extends to stopping times for the zipper filtration for which the source embedding normalization is defined.*

*Proof.* Lemma 7.1 proves equality of the field law and the boundary-length identification at deterministic times. The reverse Loewner flow is a strong Markov flow; conditional on the filtration at a stopping time, the future Brownian increments are independent and the unexplored field is a fresh GFF plus the harmonic data read from the current slit. Reapplying the lemma conditionally yields stationarity at bounded stopping times. General admissible stopping times follow by localization and monotone convergence of the quantum boundary length.  $\square$

## 7.2 Wedge welding and interface law

**Theorem 7.3** (Quantum-wedge welding). *Take two compatible  $\gamma$ -quantum wedges in the Sheffield–DMS welding range and parameterize the two marked boundaries by quantum length from their marked origins. Identifying equal boundary lengths gives, almost surely, a well-defined conformal welding. The welded quantum surface is the wedge of the sum weight prescribed by the DMS weight calculus, and the interface is the  $SLE_\kappa(\rho)$  curve whose force weights correspond to the two input wedge weights.*

*Proof.* Start from the reverse zipper at a finite reverse time  $T$ . The two prime-end sides of the slit have equal quantum boundary-length coordinates by Lemma 7.1; hence  $f_T$  is a conformal realization of the equal-length topological welding. Stationarity identifies the two unwelded sides, conditionally on their lengths, with the two wedge laws given by the source weight decomposition. Letting  $T \uparrow \infty$  exhausts the wedge away from the marked end. Tightness of the conformal maps follows from normal-family compactness under the fixed circle-average embedding, while GMC boundary-length convergence prevents a positive boundary interval from being lost in the limit. Any two subsequential weldings generate the same reverse zipper field and therefore the same interface by the Volume-XIV flow-line uniqueness theorem; hence the welding is unique in law and in the quantum surface sense.

The driving SDE is obtained by the harmonic mean calculation in the reverse flow. Its two logarithmic boundary singularities yield the drifts  $\rho_L/(W_t - V_t^L)$  and  $\rho_R/(W_t - V_t^R)$ . Thus the interface is exactly  $SLE_\kappa(\rho_L; \rho_R)$  with the force weights determined by the wedge weights. The additive logarithmic singularities on the two sides add under welding, which is the DMS sum rule for the output wedge weight.  $\square$

**Theorem 7.4** (Cutting is inverse to welding). *Condition on an initial or complete zipper SLE in the welded wedge. The quantum surfaces on its two sides are conditionally independent wedges of*

the input weights, with boundary points paired by equal quantum length. Welding these sides recovers the original decorated surface.

*Proof.* The interface is a GFF local set by the exact earlier imaginary-geometry owner. Conditional on it, the GFF in the two complementary sides is the direct sum of independent zero-boundary GFFs plus deterministic harmonic functions. Those harmonic functions are exactly the logarithmic boundary singularities of the two input wedge weights. GMC is local, so the two quantum surfaces are conditionally independent after their common interface lengths are fixed. The zipper stationarity theorem identifies the side-length coordinate with the reverse Loewner welding coordinate. Applying the forward Loewner map then reverses the cut and recovers the original surface. Uniqueness of the zipper coupling makes the two operations inverse in law.  $\square$

### 7.3 Peanosphere boundary lengths

Let  $\kappa' = 16/\gamma^2$  and let  $\eta$  be the space-filling  $\text{SLE}_{\kappa'}$  of the exact Volume-XIV owner, independent of a standard  $\gamma$ -quantum cone modulo the canonical mating coupling and parameterized so that

$$\mu_h(\eta([s, t])) = t - s.$$

Let  $L_t, R_t$  be the net quantum lengths of the left and right frontiers of the filled region relative to time 0.

**Theorem 7.5** (Peanosphere Brownian law). *Up to a deterministic common variance normalization,  $(L, R)$  is a two-sided planar Brownian motion with*

$$\text{Corr}(L_t, R_t) = -\cos\left(\frac{\pi\gamma^2}{4}\right). \quad (19.3)$$

*Proof.* Stop the space-filling curve at quantum time  $s$ . The unexplored future and the field viewed from the new tip are, by zipper stationarity and the cone scale invariance, a recentered copy of the original decorated surface. Therefore  $(L_{s+t} - L_s, R_{s+t} - R_s)$  has a law independent of  $s$ .

For disjoint quantum-time intervals, condition successively on the explored local sets. The GFF domain Markov property makes the fields in the newly revealed quantum cells independent after their boundary lengths are fixed. The welding theorem says that the only state variable passed to the next cell is the current pair  $(L, R)$ . In the cone embedding, translation by that pair is removed when taking increments. Hence increments over disjoint intervals are independent.

Adding a constant  $c$  to the field multiplies quantum area by  $e^{\gamma c}$  and boundary length by  $e^{\gamma c/2}$ . Re-embedding the cone after this addition leaves its law unchanged modulo scale. Thus

$$(L_{at}, R_{at})_{t \in \mathbb{R}} \stackrel{d}{=} \sqrt{a} (L_t, R_t)_{t \in \mathbb{R}}.$$

Continuity of the space-filling frontier lengths follows from continuity of GMC boundary length under monotone local-set exploration. A continuous process with stationary independent increments and the displayed scaling is a centered two-dimensional Brownian motion.

A simultaneous cone event  $L_t - L_s \geq 0, R_t - R_s \geq 0$  for  $t \in [s, u]$  means that the cell  $\eta([s, u])$  is cut out without either frontier returning below its starting length. In imaginary geometry this is the event that the corresponding pair of dual flow lines remains in its light cone. The cone opening angle is  $\theta = \pi\gamma^2/4$ . Whitening a correlated Brownian motion with correlation  $\rho$  sends the positive quadrant to a Euclidean cone of opening  $\arccos(-\rho)$ . Equating these two cone angles gives  $\rho = -\cos(\pi\gamma^2/4)$ , proving (19.3).  $\square$

## 7.4 Reconstruction from the Brownian pair

**Theorem 7.6** (Mating-of-trees reconstruction). *The Brownian pair  $(L, R)$  almost surely determines the decorated peanosphere quantum surface, modulo LQG coordinate equivalence, and determines the space-filling curve modulo its quantum-area parametrization.*

*Proof.* For  $s < t$ , declare  $s \sim_L t$  when  $L_s = L_t = \inf_{u \in [s, t]} L_u$ , and define  $\sim_R$  analogously. The quotient of  $\mathbb{R}$  by each relation is a continuum tree encoded by  $L$  or  $R$ . Identify the two trees at equal time labels. Brownian nondegeneracy implies that the resulting quotient is almost surely a topological plane/sphere in the cone/sphere normalization, and the time order gives a continuous space-filling curve.

To recover the conformal quantum surface, partition time into intervals with rational endpoints and record the cells  $\eta([q_1, q_2])$ . The pair  $(L, R)$  records the complete left/right boundary-length vector of every such cell and all adjacency identifications, because a common running infimum is exactly the time at which two boundary arcs are glued. The wedge-welding theorem says that a cell with these marked boundary lengths has a unique conditional quantum-surface law and that adjacent cells have a unique equal-length welding. Refine the rational partition. The diameters in quantum area tend to zero, while the adjacency sigma-field increases to the sigma-field of the Brownian pair. Two decorated surfaces with the same  $(L, R)$  therefore have identical weldings of every rational cell. Normal-family compactness of the normalized embeddings and uniqueness of each interface imply that their limiting conformal embeddings agree. Hence the decorated surface is measurable from  $(L, R)$ .  $\square$

**Theorem 7.7** (Cone and wedge variants). *The same tree-mating construction holds for the DMS cone and wedge variants. Changing marked points or conditioning the surface changes the Brownian pair by the corresponding Brownian cone conditioning, excursion, or Doob transform, but the running-infimum equivalence relations and welding reconstruction are unchanged.*

*Proof.* Up to the first time a marked point is separated, standard cone/wedge embeddings are locally absolutely continuous with respect to the basic cone field after adding a deterministic harmonic function. Hence the local boundary-length process is the Brownian pair above with the corresponding Radon–Nikodym weight. Conditioning on survival in a cone or on specified total boundary lengths is exactly a Doob transform of this Brownian motion. The equivalence relations  $\sim_L, \sim_R$  are pathwise definitions and survive the change of law. The cell-by-cell welding proof is local and therefore reconstructs each variant.  $\square$

## 7.5 Quantum disks cut from a wedge

**Theorem 7.8** (Poissonian quantum-disk decomposition). *In the DMS boundary-intersecting SLE exploration of a compatible quantum wedge, every bounded component cut off from the target is a quantum disk with its two marked boundary points. Conditioned on the boundary-length jump data, the cut-off disks are independent, and the marked disks form the Poissonian bead process specified by the wedge disintegration.*

*Proof.* Parameterize the exploration by the local time of its frontier boundary length. At a stopping time  $\tau$ , cutting along the SLE and applying the local-set GFF Markov property yields an unexplored infinite component which is a fresh wedge of the same weight, together with independent finite components. Volume XVIII’s quantum-disk disintegration identifies a finite component conditioned

on its boundary length as a quantum disk. Thus excursions of the frontier-length process away from its running infimum correspond one-to-one to cut-off disks.

The post- $\tau$  wedge is independent of the past after the current boundary length is specified. Therefore the excursion point process has the regenerative property. LQG scaling makes its intensity homogeneous in boundary length. The standard excursion theorem for a regenerative self-similar process then makes the marked excursions a Poisson point process. Marking each excursion by its conditionally independent quantum disk gives the claimed Poissonian bead decomposition, with the side determined by which frontier process jumps.  $\square$

**Theorem 7.9** (Stable boundary-length processes). *For the standard non-space-filling  $SLE_{\kappa'}$  wedge exploration with  $4 < \kappa' < 8$ , the left and right swallowed-boundary-length processes are totally asymmetric stable Lévy processes of index*

$$\alpha = \frac{\kappa'}{4} \in (1, 2), \quad (19.4)$$

*with jumps equal to the boundary lengths of the cut-off quantum disks.*

*Proof.* The wedge Markov property gives stationary independent increments for each frontier process after recording cut-off components. The exploration can only swallow boundary on the corresponding side, so after fixing sign convention the process has jumps of one sign. Adding  $c$  to the field multiplies quantum boundary length by  $e^{\gamma c/2}$  and the SLE quantum natural-time scale by the DMS power dictated by  $\kappa' = 16/\gamma^2$ . Consequently the boundary process is self-similar with exponent  $1/\alpha = 4/\kappa'$ . The Lévy–Khintchine classification says that a nontrivial càdlàg process with stationary independent increments, one-sided jumps, and this exact self-similarity is a totally asymmetric  $\alpha$ -stable process. The Poissonian disk theorem identifies every jump with the boundary length of its cut-off disk.  $\square$

## 7.6 Finite-volume mating

**Theorem 7.10** (Finite-volume LQG sphere mating). *A unit-area  $\gamma$ -LQG sphere decorated by the corresponding space-filling  $SLE_{16/\gamma^2}$  is encoded by the finite-duration correlated Brownian loop/excursion pair of the Miller–Sheffield finite-volume theorem. The pair encodes two finite-diameter CRTs whose mating determines the decorated quantum sphere.*

*Proof.* Disintegrate the infinite-volume cone construction with respect to the quantum area of the cell between two marked quantum-typical points and then normalize that area to one. The Brownian boundary-length pair is thereby conditioned to start and end at the matching boundary state while remaining in the admissible cone; absolute continuity of Brownian bridges on compact subintervals gives the finite-duration correlated Brownian loop law. Running-infimum equivalence relations now produce compact CRTs because the time interval is finite and the Brownian paths are continuous.

The rational-cell reconstruction proof above is local and survives the conditioning. It therefore shows that the finite Brownian pair determines the conformal quantum sphere and its space-filling order. Conversely, measuring left/right boundary lengths of the space-filling exploration recovers the pair. For  $\gamma = \sqrt{8/3}$ , the stable 3/2-Lévy disk-tree representation follows by substituting  $\kappa' = 6$  into (19.4), in agreement with the special finite-volume construction.  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 7.11** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 7.12** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 7.13** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 7.14** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 7.15** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XX: Liouville Quantum Gravity Metric Geometry and  
Random Planar Maps

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# Scope and conventions

Metric geometry is the point where the formal tensor  $e^{\gamma h}|dz|^2$  becomes a genuine random distance. The volume begins with LFPP because it makes the renormalization problem concrete, then passes to the axiomatic LQG metric, geodesics and metric KPZ. The special value  $\gamma = \sqrt{8/3}$  links this metric to the Brownian map. The second half develops random-map scaling limits and carefully separates proved Brownian/stable universality classes from the still-open general graph-distance/LQG universality conjecture.

# Chapter 1

## LFPP and metric axioms

LFPP replaces the meaningless pointwise exponential metric tensor by lengths of paths in a mollified field. Tightness is only the beginning: the decisive theorem is that natural metric axioms characterize a unique limit.

**Definition 1.1** (LFPP metric and Weyl rescaling). For a continuous regularization  $h_\varepsilon$  of a field  $h$  and parameter  $\xi > 0$ , the *Liouville first-passage percolation distance* in a domain  $U$  is

$$D_h^\varepsilon(z, w; U) = \inf_{P: z \rightarrow w \text{ in } U} \int_0^1 e^{\xi h_\varepsilon(P(t))} |P'(t)| dt,$$

where the infimum is over piecewise  $C^1$  paths. A limiting random metric  $D_h$  has *Weyl scaling* if for every continuous  $f$  the metric  $e^{\xi f} \cdot D_h$  is obtained by weighting  $D_h$ -length infinitesimally by  $e^{\xi f}$ ; in particular, for a constant  $c$ ,  $D_{h+c} = e^{\xi c} D_h$ .

*Worked example 1.2* (Constant Weyl shift). For the regularized path length,

$$\text{len}_{h+c}^\varepsilon(P) = \int e^{\xi(h_\varepsilon(P(t))+c)} |P'(t)| dt = e^{\xi c} \text{len}_h^\varepsilon(P).$$

Taking the infimum over the same family of paths gives  $D_{h+c}^\varepsilon = e^{\xi c} D_h^\varepsilon$ . Any subsequential metric limit must therefore satisfy the identical constant-shift rule. This is the simplest nontrivial consistency check for the continuum Weyl-scaling axiom.

**Example 1.3** (Weyl scaling by a constant). If  $h$  is replaced by  $h + c$ , the LQG metric scales by  $e^{\xi c}$ . This elementary test is a useful normalization check for every LFPP approximation.

### 1.1 LFPP and metric axioms

## Chapter 2

# Metric geometry

Once the LQG metric exists, geometric questions can be asked without reference to the approximation used to construct it: Hausdorff dimension, geodesics, confluence, metric balls, and metric versions of KPZ.

### 2.1 Metric notions

**Definition 2.1** (Geodesic and geodesic metric space). A metric space  $(X, d)$  is geodesic if for every  $x, y \in X$  there is a continuous path  $\eta : [0, d(x, y)] \rightarrow X$  such that

$$d(\eta(s), \eta(t)) = |s - t|.$$

Such a path is a geodesic from  $x$  to  $y$ . Confluence means that geodesics with nearby endpoints share a nontrivial initial or terminal segment; every confluence theorem in this volume states the sampling and exceptional-point conventions explicitly.

*Worked example 2.2* (Snowflake metrics change dimension but are not geodesic). On  $[0, 1]$  let  $d_\alpha(x, y) = |x - y|^\alpha$  with  $0 < \alpha < 1$ . A Euclidean interval of length  $r^{1/\alpha}$  has  $d_\alpha$ -diameter  $r$ , so covering numbers scale as  $N(r) \asymp r^{-1/\alpha}$  and

$$\dim_{\text{H}}([0, 1], d_\alpha) = \frac{1}{\alpha}.$$

However, the Euclidean segment is not a constant-speed  $d_\alpha$  geodesic because  $d_\alpha(0, t) + d_\alpha(t, 1) > d_\alpha(0, 1)$  for  $0 < t < 1$ . This toy example separates dimension questions from geodesic questions, which require distinct arguments for the LQG metric.

### 2.2 Rigorous results

The metric-dimension, geodesic, and confluence results are proved in Chapters 8–9 after the metric construction and Weyl-scaling/locality axioms are available. This chapter introduces only the geometric vocabulary needed to read those theorems.



## Chapter 3

# $\sqrt{8/3}$ -LQG and Brownian surfaces

At  $\gamma = \sqrt{8/3}$  the LQG metric can be compared with Brownian-map geometry. This is the sharpest bridge between conformal random surfaces and random metric spaces.

### 3.1 Metric-measure equivalence

**Definition 3.1** (Equivalence of pointed metric-measure spaces). Two pointed metric-measure spaces  $(X, d, \mu, x_0)$  and  $(X', d', \mu', x'_0)$  are equivalent if there is an isometry  $\phi : X \rightarrow X'$  with  $\phi(x_0) = x'_0$  and  $\phi_*\mu = \mu'$ . Convergence of compact objects in this chapter is in the Gromov–Hausdorff–Prokhorov topology unless a stronger decorated topology is explicitly stated.

*Worked example 3.2* (Why the  $n^{-1/4}$  distance scale is compatible with Brownian-map dimension). If a map has  $n$  faces and converges to a metric-measure space whose Hausdorff dimension is 4, a ball of limiting radius  $r$  corresponds heuristically to  $O(nr^4)$  faces. To keep macroscopic radii of order one, graph distance therefore scales like

$$n^{-1/4} d_{\text{graph}}.$$

This dimensional calculation predicts the exponent but not the limiting space; the Brownian-map theorem in Chapter 9 supplies the latter.

### 3.2 Proof interface

Chapter 9 uses the Brownian-surface scaling limit and the  $\sqrt{8/3}$  LQG metric characterization to identify the two laws. The theorem is metric-measure, not merely a statement about matching dimensions.

## Chapter 4

# Random planar maps

Random planar maps provide discrete metric spaces whose  $n^{-1/4}$  limits led historically to the Brownian map. The mated-CRT model additionally keeps the conformal embedding visible.

**Example 4.1** (Why the exponent  $1/4$  appears). A quadrangulation with  $n$  faces has volume of order  $n$ , while the Brownian map has Hausdorff dimension 4. Matching volume with radius therefore suggests graph distances of order  $n^{1/4}$ ; the theorem makes this heuristic precise.

### 4.1 Random planar maps

## Chapter 5

# Beyond subcritical gamma

Critical and supercritical LQG metrics require a topology that allows singular points. The general planar-map universality claim is kept separate as a frontier statement.

### 5.1 Beyond subcritical gamma

*Frontier statement 5.1* (General planar-map graph-distance universality).

No theorem in this manuscript asserts that every discrete planar-map model predicted to lie in a  $\gamma$ -Liouville-quantum-gravity universality class has, after a model-dependent deterministic rescaling, graph distance converging in law to the  $\gamma$ -LQG metric. The proved neighboring results concern the specific LFPP/LQG metric regimes and discrete models stated explicitly in their own theorems. This frontier row has no downstream theorem consumers.

## Chapter 6

# Liouville Brownian motion

Liouville Brownian motion (LBM) is the diffusion obtained by changing the clock of ordinary planar Brownian motion with the LQG area measure. It probes the same random geometry dynamically rather than through geodesic distance.

### 6.1 Time changes by positive continuous additive functionals

**Definition 6.1** (Liouville Brownian motion). Let  $B_t$  be Brownian motion in a planar domain and let  $\mu_h$  be the LQG area measure of a field  $h$ . When the Revuz correspondence produces a positive continuous additive functional  $F_t$  of  $B$  with Revuz measure  $\mu_h$ , define

$$A_t = \inf\{s \geq 0 : F_s > t\}, \quad Z_t = B_{A_t}.$$

The process  $Z$  is Liouville Brownian motion. The construction is understood up to the lifetime of  $B$  in the domain.

*Worked example 6.2* (Deterministic constant-density time change). If the Revuz density is the constant  $c > 0$ , then  $F_t = ct$  and

$$A_t = t/c, \quad Z_t = B_{t/c}.$$

Thus the generator changes from  $(1/2)\Delta$  to  $(1/2c)\Delta$ . LBM replaces the constant  $c dx$  by the singular random measure  $\mu_h$ , so the clock accelerates or slows according to the local quantum area.

### 6.2 Proof interface

Chapter 9 proves convergence of regularized additive functionals, identifies the limiting PCAF, and establishes conformal covariance through the LQG coordinate-change rule.

## Chapter 7

# Brownian disks and boundaries

Brownian disks, the Brownian plane, higher-topology Brownian surfaces, uniform plane maps, and stable maps show that the Brownian-map paradigm has rigorous extensions as well as genuinely different universality classes.

### 7.1 Finite and infinite volume

**Definition 7.1** (Local pointed Gromov–Hausdorff convergence). For pointed proper metric spaces  $(X_n, d_n, x_n)$  and  $(X, d, x)$ , local pointed Gromov–Hausdorff convergence means that for every radius  $R$  outside a countable exceptional set, the closed balls  $\overline{B}_{d_n}(x_n, R)$  converge to  $\overline{B}_d(x, R)$  in pointed Gromov–Hausdorff distance. With measures one uses the corresponding local Gromov–Hausdorff–Prokhorov topology.

*Worked example 7.2* (Compact versus local scaling limits). A finite quadrangulation with  $n$  faces is rescaled globally by  $n^{-1/4}$  and compared with a compact Brownian surface. An infinite uniform planar map has no finite total volume, so one instead fixes a root, rescales distances, and compares every bounded-radius neighborhood. The change from global to local topology is therefore forced by infinite volume, not by a different notion of graph distance.

### 7.2 Rigorous theorem families

Chapter 9 states and proves the corresponding scaling limits separately. The stable-map regime  $\alpha \in (1, 2)$  is not a small perturbation of the Brownian-map regime: its metric and topological behavior is governed by heavy-tailed face degrees and is therefore stated as a distinct theorem family.

## Chapter 8

# Proved here LQG metric and Brownian-surface engine

This chapter develops the metric side of subcritical Liouville quantum gravity. The first block is deliberately organized according to the actual proof architecture in the primary LFPP/LQG-metric papers. Tightness of LFPP is an analytic theorem; the weak-metric axioms are a second theorem; existence and uniqueness require the separate weak-to-strong comparison theory; conformal covariance requires another local comparison argument. These steps are not interchangeable, and in this working edition they are kept as separate proof interfaces so that no short “uniqueness” paragraph silently substitutes for the geodesic-confluence machinery used by the source proofs.

### 8.1 LFPP normalization and the tightness interface

Fix  $\gamma \in (0, 2)$  and let  $h$  be a whole-plane GFF in a fixed additive normalization. For  $\epsilon > 0$ , let  $h_\epsilon$  denote the heat-kernel regularization and define

$$D_h^\epsilon(z, w) = \inf_P \int_0^1 e^{\xi h_\epsilon(P(t))} |P'(t)| dt, \quad (20.1)$$

where  $\xi = \gamma/d_\gamma$  and  $d_\gamma > 2$  is the LFPP distance exponent. Let  $a_\epsilon$  be the median of the left–right LFPP crossing distance of the unit square. Multiplying  $h$  by an additive constant multiplies  $D_h^\epsilon$  by the corresponding factor, so fixing the crossing quantile is the natural deterministic normalization.

**Theorem 8.1** (LFPP subsequential tightness and nondegeneracy — selected whole-plane form). *For every sequence  $\epsilon_n \downarrow 0$ , the laws of*

$$a_{\epsilon_n}^{-1} D_h^{\epsilon_n}$$

*are tight for the local-uniform topology on  $\mathbb{C} \times \mathbb{C}$ . Every subsequential metric limit is almost surely finite and positive between distinct points, induces the Euclidean topology, and is locally bi-Hölder with deterministic exponents in the source-admissible ranges. The same compactness statement is valid jointly with the underlying field, after passing to a subsequence.*

*Proof roadmap.* The source proof has three genuinely analytic layers. First one proves RSW-type comparison and annular crossing estimates for normalized LFPP, using approximate conformal invariance and multiscale comparison rather than a literal FKG argument for the full mollified metric. Second one establishes the quantile-ratio/Condition-(T) estimates and the Efron–Stein variance contraction which keep crossing lengths tight on logarithmic scales. Third one converts

those estimates into simultaneous upper and lower modulus-of-continuity bounds on dyadic covers, obtaining non-collapse and bi-Hölder subsequential limits. The zero-boundary heat-kernel theorem is then localized and transferred to the whole-plane setting. The modules below establish these three layers in that order.

*Remark 8.2* (Proof dependencies). The theorem uses quantitative LFPP RSW and crossing comparison; the DZZ graph-distance exponent together with the Ding–Gwynne LFPP conversion and the DDDF concentration estimates; and the DDDF metric chaining, non-collapse, Hölder, and localization theorem. Weak-metric uniqueness and conformal covariance are separate results proved later in this chapter.

### 8.1.1 Quantitative LFPP RSW: the first reopened tightness child

Following the quantitative RSW architecture of Ding–Dubédat–Dunlap–Falconet, the first missing input in the tightness theorem is a comparison between the “easy” and “hard” crossing quantiles of rectangles with bounded aspect ratio. The point of the result is not that a single crossing can be drawn in both orientations. One must compare two random variational problems through a conformal deformation while keeping the multiplicative error independent of the mollification scale. This subsection gives that argument in full for the heat-kernel white-noise approximation used in the tightness proof and then transfers it to the finite-range truncation used in the multiscale step.

**Dyadic heat-kernel fields.** Let  $W$  be space–time white noise on  $\mathbb{R}^2 \times (0, 1]$ . For integers  $0 \leq m \leq n$  put

$$\phi_{m,n}(x) := \sqrt{\pi} \int_{2^{-2n}}^{2^{-2m}} \int_{\mathbb{R}^2} p_{t/2}(x-y) W(dy, dt), \quad p_t(z) = \frac{1}{2\pi t} e^{-|z|^2/(2t)}.$$

Thus  $\phi_{0,n}$  is the dyadic version of the whole-plane auxiliary field used in the DDDF tightness argument. For  $R_{a,b} = [0, a] \times [0, b]$  define its left–right LFPP crossing length by

$$L_{a,b}^{(n)}(\phi) := \inf_{\pi} \int_{\pi} e^{\xi \phi_{0,n}(z)} |dz|,$$

where the infimum is over continuous rectifiable paths joining the two vertical sides. We use the lower and upper tail quantiles

$$\ell_{a,b}^{(n)}(\phi, p) := \inf\{r : \mathbb{P}[L_{a,b}^{(n)}(\phi) \leq r] \geq p\}, \quad \bar{\ell}_{a,b}^{(n)}(\phi, p) := \ell_{a,b}^{(n)}(\phi, 1-p).$$

The crossing law has no atoms: adding a deterministic smooth function which is 1 on the rectangle gives a Cameron–Martin shift under which the crossing length is multiplied continuously by  $e^{\xi c}$ ; absolute continuity of the shifted laws rules out an atom. Hence one may replace the inequalities in the preceding quantile definitions by equalities whenever convenient.

For the later Efron–Stein argument one also uses a finite-range truncation. Fix a smooth radial cutoff  $\chi$  with  $0 \leq \chi \leq 1$ ,  $\chi = 1$  on the unit ball and  $\chi = 0$  outside the ball of radius 2, and choose

$$\sigma_t = r_0 \sqrt{t} |\log t|^{\varepsilon_0}.$$

With  $p_{t/2}^{\text{tr}}(z) = p_{t/2}(z) \chi(z/\sigma_t)$  define

$$\psi_{m,n}(x) := \sqrt{\pi} \int_{2^{-2n}}^{2^{-2m}} \int_{\mathbb{R}^2} p_{t/2}^{\text{tr}}(x-y) W(dy, dt).$$

At a fixed dyadic scale the block fields of  $\psi$  have finite dependence range  $O(2^{-k} k^{\varepsilon_0})$ .

**Lemma 8.3** (Uniform comparison of the full and truncated fields). *For every fixed bounded rectangle  $R$  there are constants  $c, C > 0$  such that*

$$\mathbb{P} \left[ \sup_{n \geq 0} \|\phi_{0,n} - \psi_{0,n}\|_{L^\infty(R)} > u \right] \leq C e^{-cu^2}, \quad u \geq 0.$$

*Consequently, on the event in which the displayed supremum is at most  $u$ , all crossing lengths in  $R$  satisfy*

$$e^{-\xi u} L^{(n)}(\psi) \leq L^{(n)}(\phi) \leq e^{\xi u} L^{(n)}(\psi).$$

*Proof.* Write

$$D_k = \phi_{k-1,k} - \psi_{k-1,k}, \quad k \geq 1.$$

The time intervals defining the  $D_k$  are disjoint, hence the  $D_k$  are independent centered Gaussian fields. Since  $1 - \chi(z/\sigma_t)$  vanishes for  $|z| \leq \sigma_t$ ,

$$\sup_x \text{Var } D_k(x) \leq C \int_{2^{-2k}}^{2^{-2k+2}} t^{-1} e^{-c\sigma_t^2/t} dt \leq C e^{-ck^{2\varepsilon_0}}.$$

For increments, differentiating the heat kernel under the stochastic integral and using the same off-cutoff Gaussian factor gives the stronger pair of bounds

$$\sup_x \text{Var } \nabla D_k(x) \leq C 2^{2k} e^{-ck^{2\varepsilon_0}}, \quad \mathbb{E} |D_k(x) - D_k(y)|^2 \leq C 2^{2k} e^{-ck^{2\varepsilon_0}} |x - y|^2.$$

Let  $\sigma_k^2 = C e^{-ck^{2\varepsilon_0}}$ . We now give the short chaining argument needed to pass from pointwise bounds to a supremum. Let  $G_0$  be a grid in  $R$  of mesh  $2^{-k}$  and, for  $j \geq 1$ , let  $G_j$  be nested grids of mesh  $2^{-k-j}$ . The elementary Gaussian maximal inequality  $\mathbb{E} \max_{1 \leq i \leq N} |Z_i| \leq C \sigma \sqrt{\log(2N)}$  gives

$$\mathbb{E} \max_{z \in G_0} |D_k(z)| \leq C \sigma_k \sqrt{k},$$

since  $|G_0| \leq C 2^{2k}$ . For each  $z \in G_j$  choose a parent  $\pi_{j-1}(z) \in G_{j-1}$  at Euclidean distance at most  $C 2^{-k-j}$ . The increment estimate above gives

$$\text{Var}(D_k(z) - D_k(\pi_{j-1}(z))) \leq C \sigma_k^2 2^{-2j}.$$

There are at most  $C 2^{2(k+j)}$  such parent-child pairs. Applying the same Gaussian maximal inequality at level  $j$  therefore yields

$$\mathbb{E} \max_{z \in G_j} |D_k(z) - D_k(\pi_{j-1}(z))| \leq C \sigma_k 2^{-j} \sqrt{k+j}.$$

The series in  $j$  is summable. The field  $D_k$  has a continuous version by the same increment bound, so taking the limit along the nested grids gives

$$\mathbb{E} \|D_k\|_{L^\infty(R)} \leq C \sigma_k \sqrt{k}.$$

This is summable in  $k$ . For  $k \geq 2$ , one has  $\sigma_k \leq C e^{-ck^{2\varepsilon_0}}$  after changing constants; the single coarse block  $k = 1$  is absorbed into  $C$ .

The map from the underlying Gaussian noise to  $\|D_k\|_{L^\infty(R)}$  is Lipschitz in the Cameron–Martin norm with constant  $\sup_x \sqrt{\text{Var } D_k(x)} \leq \sigma_k$ . The Gaussian concentration theorem proved in Volume VI therefore gives the centered moment bound

$$\mathbb{E} \exp(\lambda(\|D_k\|_\infty - \mathbb{E}\|D_k\|_\infty)) \leq \exp(C \lambda^2 \sigma_k^2), \quad \lambda \in \mathbb{R}.$$



The  $D_k$  are independent. Multiplying the centered moment bound over  $k$  and using  $\sum_k \sigma_k^2 < \infty$  shows that  $\sum_k (\|D_k\|_\infty - \mathbb{E}\|D_k\|_\infty)$  is subgaussian. Together with the summability of the means this gives

$$\mathbb{P} \left[ \sum_{k \geq 1} \|D_k\|_{L^\infty(R)} > u \right] \leq C e^{-cu^2}, \quad u \geq 0,$$

after adjusting  $C, c$  on a bounded interval of  $u$ . But  $\phi_{0,n} - \psi_{0,n} = \sum_{k=1}^n D_k$ , so the left-hand side of the claimed supremum estimate is bounded by the event just estimated. Finally, pointwise  $|\phi_{0,n} - \psi_{0,n}| \leq u$  multiplies every path integrand by a factor in  $[e^{-\xi u}, e^{\xi u}]$ , proving the length comparison.  $\square$

**Conformal comparison.** The RSW step uses approximate, not exact, conformal covariance of the mollified field. The next lemma records the form needed for crossing quantiles.

**Lemma 8.4** (Conformal white-noise coupling). *Let  $F : U \rightarrow V$  be a conformal bijection between bounded convex domains, and let  $K \Subset U$  be convex. Suppose*

$$1 \leq |F'(z)| \leq M, \quad |F''(z)| \leq M, \quad z \in U.$$

*There is  $n_0 = n_0(M)$  such that for every  $n \geq n_0$  there is a coupling of two copies  $\phi_n, \tilde{\phi}_n$  of  $\phi_{0,n}$  for which, on  $K$ ,*

$$\phi_n(x) - \tilde{\phi}_n(F(x)) = A_n(x) + B_n(x),$$

*where*

- (a)  $\sup_n \|A_n\|_{L^\infty(K)}$  has a Gaussian tail;
- (b)  $\sup_{n,x \in K} \text{Var } B_n(x) \leq C(U, V, K, M)$ ;
- (c)  $B_n$  is independent of  $(\phi_n, A_n)$ .

*For the finitely many  $n < n_0$ , the crossing-quantile comparisons obtained below continue to hold after enlarging the constants, by a direct bounded-variance Gaussian comparison on the fixed domains.*

*Proof.* Inside  $U \times (0, 1]$  transport white noise by

$$(y, t) \mapsto (F(y), t|F'(y)|^2), \quad \widetilde{W}(dF(y), d(t|F'(y)|^2)) = |F'(y)|^2 W(dy, dt).$$

The Jacobian in space-time is  $|F'(y)|^4$ , so the factor on the right makes the displayed transport identity an isometry on  $L^2$  test functions and therefore defines white noise on the image. Complete it by independent white noise outside the image. After changing variables in the stochastic integral, the difference  $\phi_n(x) - \tilde{\phi}_n(F(x))$  splits into three terms.

The first is the kernel-shape error

$$A_n^{(1)}(x) = \sqrt{\pi} \int_U \int_{2^{-2n}}^{|F'(y)|^{-2}} \left[ p_{t/2}(x - y) - p_{t/2} \left( \frac{F(x) - F(y)}{F'(y)} \right) \right] W(dy, dt).$$

Taylor's theorem gives  $|F(x) - F(y) - F'(y)(x - y)| \leq C|x - y|^2$ . To estimate increments, write the heat-kernel time as  $t = 2s^2$ . For  $s \geq \sqrt{|x - x'|}$ , the identity  $|x - y|^2 + |x' - y|^2 = |x - x'|^2/2 + 2|y - (x + x')/2|^2$  and  $1 - e^{-u} \leq u$  give an  $L^2(dy)$  kernel-difference bound  $C|x - x'|^2$ ; integration against  $s^{-3}ds$

contributes  $C|x - x'|$ . For  $s \leq \sqrt{|x - x'|}$ , the Taylor remainder above and the Gaussian derivative bound imply, near  $x$ ,

$$\left| p\left(\frac{x - y}{s}\right) - p\left(\frac{F(x) - F(y)}{sF'(y)}\right) \right| \leq C \frac{|x - y|^3}{s^2} e^{-|x - y|^2/(4s^2)}.$$

Its squared integral is  $O(s^4)$ ; outside a fixed ball about  $x$ , Gaussian tails give  $O(e^{-c/s^2})$ . Repeating the same estimates with  $x'$  and integrating  $s^{-3}ds$  therefore gives

$$\mathbb{E}|A_n^{(1)}(x) - A_n^{(1)}(x')|^2 \leq C|x - x'|.$$

The second term is

$$\begin{aligned} A_n^{(2)}(x) &= \sqrt{\pi} \int_{U^c} \int_{2^{-2n}}^1 p_{t/2}(x - y) W(dy, dt) \\ &\quad - \sqrt{\pi} \int_{V^c} \int_{2^{-2n}}^1 p_{t/2}(F(x) - y) \widetilde{W}(dy, dt) \\ &\quad + \sqrt{\pi} \int_U \int_{|F'(y)|^{-2}}^1 p_{t/2}(x - y) W(dy, dt). \end{aligned}$$

It consists of the noise outside  $U, V$  and the mismatch of the upper integration limits. Since  $K$  has positive distance from the relevant boundaries, the same heat-kernel estimate, now with an extra factor  $e^{-d(K, \partial U)^2/(Ct)}$ , gives

$$\mathbb{E}|A_n^{(2)}(x) - A_n^{(2)}(x')|^2 \leq C|x - x'|, \quad \sup_n \text{Var } A_n^{(2)}(x_0) \leq C$$

for one (hence every)  $x_0 \in K$ . The two increment estimates, the Gaussian Kolmogorov argument, and Gaussian concentration for the supremum imply a uniform Gaussian tail for  $A_n = A_n^{(1)} + A_n^{(2)}$ .

The last term is the short-time strip created by the lower cutoff:

$$B_n(x) = -\sqrt{\pi} \int_U \int_{2^{-2n}|F'(y)|^{-2}}^{2^{-2n}} p_{t/2}\left(\frac{F(x) - F(y)}{F'(y)}\right) W(dy, dt).$$

It uses a white-noise region disjoint from the one defining  $\phi_n$  and the two terms included in  $A_n$ , so it is independent of  $(\phi_n, A_n)$ . Finally  $1 \leq |F'| \leq M$  and boundedness of  $(F^{-1})'$  on the compact configuration imply  $|(F(x) - F(y))/F'(y)| \geq c|x - y|$ . Hence the heat-kernel Gaussian bound and  $\int_{\mathbb{R}^2} t^{-1} e^{-c|x - y|^2/t} dy \leq C$  imply

$$\text{Var } B_n(x) \leq C \int_{2^{-2n}M^{-2}}^{2^{-2n}} \frac{dt}{t} \leq C \log M.$$

This proves all three assertions.  $\square$

**Lemma 8.5** (Independent bounded-variance perturbations move quantiles only subgaussianly). *Let  $\Gamma$  be a continuous random field on a compact quadrilateral and let  $\Psi$  be an independent centered continuous Gaussian field with  $\sup_x \text{Var } \Psi(x) \leq \sigma^2$ . Let  $L(\Gamma)$  denote a crossing length with density  $e^{\xi\Gamma}$ . There is  $C = C(\xi, \sigma)$  such that for  $0 < p < 1/4$ ,*

$$\ell(\Gamma + \Psi, p) \leq e^{C\sqrt{\log(1/p)}} \ell(\Gamma, 2p),$$

$$\bar{\ell}(\Gamma + \Psi, 2p) \leq e^{C\sqrt{\log(1/p)}} \bar{\ell}(\Gamma, p).$$

*The reverse comparisons hold after replacing  $\Psi$  by  $-\Psi$ .*

*Proof.* Because  $e^{\xi\Gamma}$  is continuous and bounded above and below on the compact quadrilateral, the induced length metric is proper; the direct method for rectifiable curves gives a minimizing crossing. Condition on  $\Gamma$  and choose such a  $\Gamma$ -geodesic  $\pi$ . Normalize its  $\Gamma$ -length to a probability measure

$$d\mu(z) = L(\Gamma)^{-1} e^{\xi\Gamma(z)} |dz|.$$

If  $\sigma = 0$  the assertion is immediate. Otherwise put  $v = \xi^2 \sigma^2$  and  $s = v + \sqrt{2v \log(1/p)}$ . Then  $\alpha = s/v \geq 1$ , and Jensen's inequality together with the Gaussian moment formula gives

$$\begin{aligned} \mathbb{P} \left[ \int_{\pi} e^{\xi\Psi} d\mu > e^s \mid \Gamma \right] &\leq e^{-\alpha s} \mathbb{E} \left[ \left( \int e^{\xi\Psi} d\mu \right)^\alpha \mid \Gamma \right] \\ &\leq e^{-\alpha s} \int \mathbb{E} e^{\alpha \xi \Psi(z)} d\mu(z) \leq e^{-s^2/(2v)} \leq p. \end{aligned}$$

Since  $p < 1/4$ , the additive term  $v$  in  $s$  is absorbed by  $C(v)\sqrt{\log(1/p)}$ . Therefore, except on conditional probability  $p$ ,  $L(\Gamma + \Psi) \leq e^{C\sqrt{\log(1/p)}} L(\Gamma)$ . Intersect this event with  $\{L(\Gamma) \leq \ell(\Gamma, 2p)\}$ , which has probability at least  $2p$ ; the intersection has probability at least  $p$ , yielding the first quantile inequality. For the high-quantile statement, on the event  $\{L(\Gamma) \leq \bar{\ell}(\Gamma, p)\}$  of probability at least  $1 - p$ , the same conditional estimate fails with probability at most  $p$ , hence  $L(\Gamma + \Psi) \leq e^{C\sqrt{\log(1/p)}} \bar{\ell}(\Gamma, p)$  with probability at least  $1 - 2p$ . This is the second quantile inequality. Applying the argument to  $-\Psi$  gives the reverse bounds.  $\square$

Combining Lemmas 8.4 and 8.5 with the Gaussian tail of  $\|A_n\|_\infty$  shows that a conformal change satisfying the derivative hypotheses above changes any fixed crossing quantile by at most

$$C \exp(C\sqrt{\log(C/p)}),$$

uniformly in  $n$ . The factor  $\sup |F'|$  enters only through the Euclidean line element.

### The deterministic and association inputs.

**Lemma 8.6** (Gaussian positive association for LFPP crossing events). *Let  $X$  be a centered continuous Gaussian field on a compact set whose covariance kernel is nonnegative. If  $A_1, \dots, A_m$  are events increasing under pointwise increases of  $X$ , then*

$$\mathbb{P} \left( \bigcap_{i=1}^m A_i \right) \geq \prod_{i=1}^m \mathbb{P}(A_i).$$

*In particular, for LFPP, each event  $\{L(R) > r\}$  is increasing, so if  $E_i = \{L(R_i) \leq r_i\}$  then*

$$\max_i \mathbb{P}(E_i) \geq 1 - \left( 1 - \mathbb{P} \left( \bigcup_i E_i \right) \right)^{1/m}.$$

*Proof.* First sample the field on finitely many points. If  $Z$  is the resulting Gaussian vector, all entries of its covariance matrix are nonnegative. For bounded smooth coordinatewise increasing functions  $f, g$ , let  $(Z, Z_t)$  be a Gaussian pair with the same marginals and cross-covariance  $t \text{Cov } Z$ . Gaussian integration by parts gives

$$\frac{d}{dt} \mathbb{E}[f(Z)g(Z_t)] = \sum_{i,j} \text{Cov}(Z_i, Z_j) \mathbb{E}[\partial_i f(Z) \partial_j g(Z_t)] \geq 0.$$

At  $t = 0$  the variables are independent and at  $t = 1$  they agree in law, hence  $\mathbb{E}[fg] \geq \mathbb{E}f \mathbb{E}g$ . Monotone approximation proves the same for increasing events. Refining the sampling mesh and using continuity passes to field events and then induction gives the association inequality.

For the square-root estimate, apply the association inequality to the increasing complements  $E_i^c$ :

$$1 - \mathbb{P}(\cup_i E_i) = \mathbb{P}(\cap_i E_i^c) \geq \prod_i (1 - \mathbb{P}(E_i)) \geq (1 - \max_i \mathbb{P}(E_i))^m.$$

Solving for the maximum proves the square-root trick. The covariance of  $\phi_{0,n}$  is nonnegative because it is an integral of heat-kernel convolutions  $p_t(x - y) > 0$ , so the lemma applies here.  $\square$

**Lemma 8.7** (Finite geometric bridge between easy and hard crossings). *Fix a compact aspect-ratio range  $[A, B] \subseteq (0, \infty)$ . If  $a/b < 1 < a'/b'$  and all four side lengths lie in  $[A, B]$ , there are integers  $j, m, p$  and smooth analytic quadrilaterals  $E_p, E'$  with marked opposite arcs such that:*

- (i) *every left-right crossing of  $R_{a,b}$  contains, after at most  $p$  dyadic reductions, a crossing of one of  $j^p$  translates of  $R_{a/2^p, b/2^p}$  in its thin direction;*
- (ii) *every crossing of  $R_{a/2^p, b/2^p}$  contains a crossing of  $E_p$  between its marked arcs;*
- (iii) *every crossing of  $E'$  between its marked arcs is a left-right crossing of  $R_{a', b'}$ ;*
- (iv) *after partitioning each marked arc of  $E_p$  into  $m$  compact subarcs, for every pair of subarcs on opposite sides there is a conformal map  $F : E_p \rightarrow E'$  extending conformally to neighborhoods of the closures and satisfying  $\inf_{E_p} |F'| \geq 1$  and  $\sup(|F'| + |F''|) < \infty$ .*

*The integers and derivative bounds may be chosen uniformly when the side lengths range over the fixed compact set and the two aspect-ratio inequalities stay strict.*

*Proof.* For (i), tile  $R_{a,b}$  by finitely many overlapping half-scale rectangles whose short sides form barriers from the top boundary to the bottom boundary. A continuous left-right path must cross at least one such barrier. The number  $j$  depends only on  $b/a$ ; iteration gives  $j^p$  candidates.

For (ii)–(iii), round the four corners of the two rectangles by fixed analytic arcs, choosing  $E_p$  so that its two marked arcs separate the two vertical sides of the small rectangle and choosing  $E'$  inside  $R_{a', b'}$  with its marked arcs on the left and right boundary corridors. Planar separation then has exactly the two stated consequences.

It remains to prove (iv). An analytic Jordan quadrilateral with two marked boundary subarcs is conformally equivalent to a rectangle with the four endpoints in prescribed cyclic order: apply the Riemann mapping theorem to the interior and Carathéodory boundary extension, then compose with a disk automorphism sending three endpoints to their prescribed targets. Analytic reflection across the analytic boundary arcs extends the resulting map to a neighborhood of the closure; its derivative is continuous and nonzero there. There are only  $m^2$  pairs of compact source subarcs, so the corresponding minimum derivative is positive before rescaling. Replacing the source quadrilateral by its factor- $2^{-p}$  copy multiplies the derivative of the map back to the fixed target by  $2^p$ . Taking  $p$  larger if necessary therefore makes the minimum derivative at least 1. Compactness of the allowed aspect-ratio parameter set makes all choices uniform.  $\square$

**Theorem 8.8** (Quantitative LFPP RSW comparison). *Fix  $0 < A < B < \infty$ . There are  $C = C(A, B, \xi) > 1$  and  $c = c(A, B, \xi) \in (0, 1]$  such that, whenever  $a, b, a', b' \in [A, B]$  with  $a/b < 1 < a'/b'$ , for every  $n \geq 0$  and  $0 < p < 1/2$ ,*

$$\ell_{a', b'}^{(n)}(\phi, p/C) \leq C \ell_{a, b}^{(n)}(\phi, p) \exp(C \sqrt{\log(C/p)}),$$

$$\bar{\ell}_{a',b'}^{(n)}(\phi, 3p^c) \leq C \bar{\ell}_{a,b}^{(n)}(\phi, p) \exp(C\sqrt{\log(C/p)}).$$

The same estimates hold for the finite-range field  $\psi$ , after changing  $C$ . In particular, for every fixed  $p_0 \in (0, 1/2)$ , easy and hard crossing quantiles for bounded aspect ratios are comparable by constants independent of  $n$ .

*Proof.* The finitely many  $n < n_0$  from the conformal-coupling lemma are absorbed into the constants by direct Gaussian comparison, so it suffices to consider  $n \geq n_0$ . We first prove the low-quantile inequality. Assume  $\mathbb{P}[L_{a,b}^{(n)}(\phi) \leq r] \geq p$ . By part (i) of Lemma 8.7, after  $p_*$  dyadic reductions (where  $p_*$  is the fixed geometric integer, unrelated to the probability  $p$ ), at least one candidate thin crossing has length at most  $r$ . A union bound therefore gives a fixed candidate with probability at least  $p/j^{p_*}$ . By part (ii), on this event  $E_{p_*}$  has a marked-arc crossing of length at most  $r$ . Partition both marked arcs into the  $m$  subarcs from part (iv). Some pair carries such a crossing with probability at least  $p/(j^{p_*}m^2)$ .

Choose the conformal map  $F$  for this pair. Under the coupling of Lemma 8.4, the image of the source crossing is a crossing of  $E'$ . The Euclidean line element contributes at most  $\|F'\|_\infty$ , the low-frequency error contributes  $e^{\xi\|A_n\|_\infty}$ , and the independent bounded-variance error is handled by Lemma 8.5. Taking a Gaussian-tail cutoff of size  $C\sqrt{\log(C/p)}$  for  $\|A_n\|_\infty$  gives

$$\mathbb{P}\left[L^{(n)}(E') \leq Cr \exp\{C\sqrt{\log(C/p)}\}\right] \geq p/C.$$

By part (iii), the same image path is a left–right crossing of  $R_{a',b'}$ , so this proves the low-quantile inequality.

For the high-quantile inequality, suppose instead  $\mathbb{P}[L_{a,b}^{(n)}(\phi) \leq r] \geq 1 - p$ . At each geometric reduction, the event that at least one candidate subrectangle has a short crossing has probability at least  $1 - p$ . The square-root estimate therefore selects one candidate whose short-crossing probability is at least  $1 - p^{1/j}$ . Iterating  $p_*$  times and then applying the same square-root trick to the  $m^2$  subarc pairs produces a fixed pair with probability at least  $1 - p^{c_0}$  for a constant  $c_0 > 0$ . The high-quantile half of Lemma 8.5 and the Gaussian cutoff for  $\|A_n\|_\infty$  then give

$$\mathbb{P}\left[L_{a',b'}^{(n)}(\phi) \leq Cr \exp\{C\sqrt{\log(C/p)}\}\right] \geq 1 - 3p^{c_1}.$$

Set  $c = \min\{c_1, 1\}$ ; this proves the high-quantile inequality after enlarging the multiplicative constant  $C$  if needed.

Finally, Lemma 8.3 gives, with probability  $1 - O(p)$ ,  $e^{-\xi u}L(\psi) \leq L(\phi) \leq e^{\xi u}L(\psi)$  for  $u = C\sqrt{\log(C/p)}$ . Apply the two inequalities just proved to  $\phi$  with the probabilities shifted by these  $O(p)$  comparison errors and absorb the extra  $e^{\xi u}$  into the displayed exponential factor. This proves the same RSW bounds for  $\psi$ .  $\square$

### 8.1.2 The LFPP exponent interface: reduction to the Liouville graph-distance exponent

The preceding concentration module requires the Ding–Gwynne LFPP exponent. Their argument compares LFPP with Liouville graph distance. The subsections below prove the required regularization and grid/continuum transfers, the DZZ graph-distance estimates, and the lower and upper Ding–Gwynne conversion branches.

Put

$$Q = \frac{2}{\gamma} + \frac{\gamma}{2}, \quad \xi = \frac{\gamma}{d_\gamma}, \quad a_\gamma := 1 - \xi Q = 1 - \frac{2}{d_\gamma} - \frac{\gamma^2}{2d_\gamma}.$$

**The graph-distance formulation.** For an LQG area measure  $\mu_h$  and  $\varepsilon > 0$ , let  $D_{h,\text{LG}}^\varepsilon(z, w; U)$  be the least number of Euclidean balls contained in  $\overline{U}$ , each having  $\mu_h$ -mass at most  $\varepsilon$ , whose union contains a path from  $z$  to  $w$  in  $\overline{U}$ . The corresponding set-to-set distance is defined by taking the infimum over endpoints.

**Definition 8.9** (Quantitative Liouville graph-distance exponent package). The  $d_\gamma$  graph-distance package is the following statement. Let  $h$  be either a whole-plane GFF in a fixed additive normalization or a zero-boundary GFF on a square, and let  $K \Subset U$  with  $K$  compact and  $U$  bounded and open. For every  $\zeta > 0$ , with polynomially high probability as  $\varepsilon \downarrow 0$ ,

$$\max_{z,w \in K} D_{h,\text{LG}}^\varepsilon(z, w; U) \leq \varepsilon^{-1/(d_\gamma - \zeta)}$$

and

$$D_{h,\text{LG}}^\varepsilon(K, \partial U; U) \geq \varepsilon^{-1/(d_\gamma + \zeta)}.$$

The same estimates hold uniformly for each member of a fixed finite family of translated/rescaled squares whose shapes stay in a compact family.

**A scale-local comparison of regularizations.** Let  $h_{2^{-n}}^\circ$  denote the circle average of a whole-plane GFF at radius  $2^{-n}$ , with a fixed large-scale normalization, and let  $\phi_{0,n}$  be the heat-kernel white-noise field from Subsection 8.1.1.

**Lemma 8.10** (Circle-average versus heat-kernel field at exponent scale). *For every bounded rectangle  $R$  there is a coupling of  $h$  and  $\phi$  such that for some  $c, C > 0$ ,*

$$\mathbb{P} [\|h_{2^{-n}}^\circ - \phi_{0,n}\|_{L^\infty(R)} > C\sqrt{n} + u] \leq Ce^{-cu^2}, \quad u \geq 0.$$

Consequently, for every  $\eta > 0$ ,

$$\mathbb{P} [\|h_{2^{-n}}^\circ - \phi_{0,n}\|_{L^\infty(R)} > \eta n]$$

decays faster than any negative power of  $2^n$ .

*Proof.* Use the standard whole-plane white-noise realization of the GFF. At frequency  $|\theta|$ , circle averaging has multiplier  $J_0(2^{-n}|\theta|)$ , while heat regularization at the comparable scale has a smooth radial multiplier of the form  $e^{-c2^{-2n}|\theta|^2}$ . The large-scale normalization contributes only a smooth Gaussian field on  $R$ . The scale-invariant integral

$$\int_{\mathbb{R}^2} \frac{|J_0(|\theta|) - e^{-c|\theta|^2}|^2}{|\theta|^2} d\theta < \infty$$

shows, after the change of variables  $\theta \mapsto 2^n \theta$ , that the pointwise variance of the scale-local difference is bounded uniformly in  $n$ . Differentiating the multipliers inserts one factor  $|\theta|$  and gives

$$\sup_{x \in R} \text{Var} \nabla(h_{2^{-n}}^\circ - \phi_{0,n})(x) \leq C2^{2n}.$$

Cover  $R$  by a mesh of spacing  $2^{-n}$ . There are  $O(4^n)$  mesh points, so the elementary Gaussian maximal inequality gives an expected mesh maximum  $O(\sqrt{n})$ . Refine the mesh dyadically. The gradient estimate makes the expected parent-child increment at refinement level  $j$  at most  $C2^{-j}\sqrt{n+j}$ ; these contributions are summable in  $j$ . Hence the expected continuum supremum is  $O(\sqrt{n})$ . Finally the supremum is a Gaussian-Lipschitz functional with a scale-independent Cameron-Martin Lipschitz constant because the pointwise variance is uniformly bounded. Gaussian concentration therefore gives the displayed tail. Taking  $u = \eta n - C\sqrt{n}$  proves the second assertion.  $\square$

**Lemma 8.11** (Exponent stability under an  $o(n)$  field perturbation). *Let  $X_n, Y_n$  be two continuous fields on a fixed rectangle  $R$ , coupled so that for every  $\eta > 0$ ,*

$$\mathbb{P}[\|X_n - Y_n\|_{L^\infty(R)} > \eta n] \rightarrow 0.$$

*If  $L_n(X)$  and  $L_n(Y)$  are the corresponding LFPP crossing lengths with parameter  $\xi$ , then any deterministic exponent statement*

$$L_n(X) = 2^{-n(a+o(1))} \quad \text{in probability}$$

*is equivalent to the same statement for  $L_n(Y)$ . The conclusion also holds for any fixed lower or upper crossing quantile.*

*Proof.* On  $\|X_n - Y_n\|_\infty \leq \eta n$  every path length is multiplied by a factor between  $e^{-\xi\eta n}$  and  $e^{\xi\eta n}$ . Thus

$$e^{-\xi\eta n} L_n(X) \leq L_n(Y) \leq e^{\xi\eta n} L_n(X).$$

Since  $e^{\xi\eta n} = 2^{(\xi\eta/\log 2)n}$ , first take  $n \rightarrow \infty$  and then  $\eta \downarrow 0$ . For fixed quantiles, choose the coupling failure probability smaller than half the distance from the quantile level to 0 and 1 and apply the same deterministic sandwich.  $\square$

**The exact discrete/continuum LFPP comparison.** Before comparing LFPP with graph distance we isolate the elementary part of the Ding–Gwynne proof which really can be owned without the later percolation machinery. For a dyadic mesh size  $\delta$ , let  $\mathcal{S}_\delta$  be the squares of the mesh which meet a fixed reference square  $S$ . With  $v_Q$  the center of  $Q$ , define

$$\widehat{D}_{h,\text{LFPP}}^\delta(z, w; S) := \min_{Q_0, \dots, Q_k} \sum_{j=0}^k \delta e^{\xi \widehat{h}_\delta(v_{Q_j})}, \quad (20.8f)$$

where the minimum runs over self-avoiding nearest-neighbor square chains from a square containing  $z$  to one containing  $w$ .

**Proposition 8.12** (Dyadic LFPP versus continuum LFPP). *Let  $h^{S(1)}$  be a zero-boundary GFF on a fixed enlargement  $S(1)$  and  $\widehat{h}$  the scale-local white-noise field. They can be coupled so that, for every  $\zeta \in (0, 1)$  and  $\xi > 0$ , with polynomially high probability as  $\delta \downarrow 0$ , simultaneously for  $z, w \in S$ ,*

$$\delta^\zeta \left( \widehat{D}_{h,\text{LFPP}}^\delta(z, w; S) - \delta e^{\xi \widehat{h}_\delta(v_{Q_z})} \right) \leq D_{h^{S(1)}, \text{LFPP}}^\delta(z, w; S) \leq \delta^{-\zeta} \widehat{D}_{h,\text{LFPP}}^\delta(z, w; S), \quad (20.8g)$$

*where, if  $z$  belongs to more than one mesh square,  $Q_z$  is chosen among those squares to maximize the displayed center value.*

*Proof.* By the scale-local field coupling and the spatial continuity estimate already proved above, after shrinking  $\zeta$  by a fixed factor we may work on the event

$$\max_{x, y \in S: |x-y| \leq 4\delta} \left( |h_\delta^{S(1)}(x) - \widehat{h}_\delta(y)| \vee |\widehat{h}_\delta(x) - \widehat{h}_\delta(y)| \right) \leq \frac{\zeta}{2\xi} \log \delta^{-1}. \quad (20.8h)$$

The coupling failure has polynomially small probability.

For the upper inequality choose a square chain  $Q_0, \dots, Q_k$  whose weight is within a factor two of (20.8f). Pick points  $x_j \in Q_{j-1} \cap Q_j$ , with  $x_0 = z$  and  $x_{k+1} = w$ , and join successive  $x_j$  by

straight segments inside  $Q_j$ . Each segment has Euclidean length at most  $2\delta$ , while (20.8h) bounds the continuum field on it by  $\widehat{h}_\delta(v_{Q_j}) + (\zeta/(2\xi)) \log \delta^{-1}$ . Summing segment lengths gives

$$D_{h^{S(1)}, \text{LFPP}}^\delta(z, w; S) \leq C\delta^{-\zeta/2} \widehat{D}_{\widehat{h}, \text{LFPP}}^\delta(z, w; S),$$

and the deterministic constant is absorbed by replacing  $\zeta/2$  with  $\zeta$  for small  $\delta$ .

For the lower inequality let  $P$  be a simple unit-speed path whose continuum LFPP length is at most twice the infimum. Loop-erase its mesh-square itinerary by taking the last exit from each currently visited square. This produces a self-avoiding chain  $Q_0, \dots, Q_J$  and a polygonal path  $\tilde{P}$  which crosses each  $Q_j$  in one connected time interval and whose Euclidean length inside that square is no larger than the corresponding portion of  $P$ . Equation (20.8h) therefore gives

$$\int_{\tilde{P}} e^{\xi \widehat{h}_\delta} \leq 2\delta^{-\zeta/2} D_{h^{S(1)}, \text{LFPP}}^\delta(z, w; S). \quad (20.8i)$$

It remains to compare the discrete square weight with the integral when  $\tilde{P}$  spends very little time in some square. Call an index  $j$  *long* when the time spent in  $Q_j$  is at least  $\delta^{1+\zeta/2}$ . Every index lies within a bounded number (six suffices) of a preceding long index: otherwise the polygonal path would traverse six distinct mesh squares while moving Euclidean distance  $o(\delta)$ , impossible for a nearest-neighbor self-avoiding square chain. Assign each short index to such a long index. On a long interval, (20.8h) yields

$$\delta e^{\xi \widehat{h}_\delta(v_{Q_j})} \leq C\delta^{-\zeta/2} \int_{I_{J(j)}} e^{\xi \widehat{h}_\delta(\tilde{P}(t))} dt.$$

The bounded assignment multiplicity and (20.8i) then give

$$\widehat{D}_{\widehat{h}, \text{LFPP}}^\delta(z, w; S) - \delta e^{\xi \widehat{h}_\delta(v_{Q_z})} \leq C\delta^{-\zeta} D_{h^{S(1)}, \text{LFPP}}^\delta(z, w; S).$$

Again absorb the deterministic constant by a harmless reduction of  $\zeta$ . This is (20.8g).  $\square$

**From Liouville graph distance to LFPP: the two source branches.** We prove the lower and upper Ding–Gwynne conversion estimates separately. Throughout this paragraph let

$$\mathbb{S} = [0, 1]^2, \quad \xi = \frac{\gamma}{d_\gamma}, \quad b_\gamma := 2 + \frac{\gamma^2}{2}, \quad a_\gamma = 1 - \frac{b_\gamma}{d_\gamma} = 1 - \xi Q.$$

For a square  $S$  of side  $s$ ,  $S(r)$  denotes the concentric square obtained by expanding each side outwards by  $rs$ . We use  $\widehat{h}^{\text{tr}}$  for the finite-range spatial truncation of the scale-local white-noise field: at unit scale, restrictions to sets at distance larger than a fixed deterministic constant are independent. Its subcritical GMC measure and graph distance are defined exactly as for  $\widehat{h}$ . The full/truncated comparison is the same bounded-variance Gaussian perturbation already proved for the LFPP white-noise fields, and therefore costs only  $e^{o(n)}$  on a box of diameter  $n$ .

**The lower LFPP branch DG-L.** The first input is the finite-range amplification hidden behind Ding–Gwynne Lemma 3.11.

**Lemma 8.13** (Finite-range rectangle amplification). *For  $n \in \mathbb{N}$  put*

$$\mathcal{R}_n = [0, 2n] \times [0, n], \quad \mathcal{R}'_n = [-n, 3n] \times [0, n].$$



For every  $\zeta \in (0, 1)$  there are  $a_0, a_1, A > 0$  such that, for all  $n \geq 1$  and  $\varepsilon \in (0, 1)$ ,

$$\mathbb{P}\left[D_{\widehat{h}^{\text{tr}}}^\varepsilon(\partial_L \mathcal{R}_n, \partial_R \mathcal{R}_n; \mathcal{R}'_n) \leq n^2 \max\{A, \varepsilon^{-1/(d_\gamma - \zeta)}\}\right] \geq 1 - a_0 e^{-a_1 n}. \quad (20.40)$$

The analogous top–bottom estimate holds after a quarter turn.

*Proof.* Fix a small  $p > 0$ . Tile the interior strip of  $\mathcal{R}_n$  by unit squares. Call a square  $S$  *good* when there is, inside  $S(1)$ , a connected cross-shaped set  $X_S$  joining the four side-neighbourhoods of  $S$  and having  $D_{\widehat{h}^{\text{tr}}}^\varepsilon$ -diameter at most

$$M_\varepsilon := C \max\{A, \varepsilon^{-1/(d_\gamma - \zeta)}\},$$

and when every  $\varepsilon$ -mass disk meeting  $S(1/2)$  is contained in  $S(3/4)$ . The internal graph-distance upper exponent (Theorem 8.42), applied to the finitely many rectangles used to make the cross, implies that the first condition has probability tending to one as  $\varepsilon \downarrow 0$ . The second follows from positivity of GMC mass on open sets. Increasing  $A$  handles the remaining compact interval of  $\varepsilon$ , so

$$\inf_{\varepsilon \in (0, 1)} \mathbb{P}[S \text{ is good}] \geq 1 - p. \quad (20.41)$$

Because  $\widehat{h}^{\text{tr}}$  has finite spatial range, good-square events at large enough grid distance are independent.

If the good squares do not contain a left–right crossing, planar duality gives a  $*$ -connected top–bottom chain of bad squares. A chain of length  $k$  has at most  $8^{k+1}$  shapes after its starting square is fixed; from it one can greedily extract at least  $k/C_0$  mutually independent bad squares. Hence

$$\mathbb{P}[\text{a bad dual crossing}] \leq Cn \sum_{k=n}^{2n^2} 8^{k+1} p^{k/C_0} \leq a_0 e^{-a_1 n}$$

when  $p$  is chosen sufficiently small. On the complementary event concatenate the cross-shaped sets along a good left–right chain. There are at most  $2n^2$  squares and neighbouring crosses meet by construction, so the resulting path has graph length at most  $Cn^2 M_\varepsilon$ ; absorb  $C$  into  $A$ . This proves the claim.  $\square$

**Lemma 8.14** (Simultaneous short-rectangle graph-distance estimate). *Fix*

$$0 < \beta < \frac{2}{(2 + \gamma)^2}, \quad \delta_\varepsilon := 2^{-\lceil \log_2 \varepsilon^{-\beta} \rceil}.$$

For each  $\zeta > 0$ , with polynomially high probability as  $\varepsilon \downarrow 0$ , every  $\delta_\varepsilon \times (\delta_\varepsilon/2)$  rectangle  $R \subset \mathbb{S}$  with corners in  $(\delta_\varepsilon/2)\mathbb{Z}^2$  satisfies

$$D_h^\varepsilon(\partial_L R, \partial_R R; R') \leq \varepsilon^{-1/d_\gamma + \beta b_\gamma/d_\gamma - o_\zeta(1)} \exp\left(\xi \min_{z \in R'} \widehat{h}_{\varepsilon^\beta}(z)\right), \quad (20.42)$$

and the rotated estimate holds for the corresponding vertical rectangles. Here  $o_\zeta(1)$  is deterministic, tends to zero with the auxiliary error parameter, and is uniform in  $\varepsilon$ .

*Proof.* Write  $\delta_\varepsilon = 2^{-m}$  and put  $n_m = \lfloor \log_2 m^{3/2} \rfloor$ . After translating and multiplying space by  $2^{m+n_m}$ ,  $R$  becomes  $\mathcal{R}_{2^{n_m}}$  from Lemma 8.13. Split the white-noise field at scale  $2^{-m-n_m}$ . Conditional on the coarse field, the fine field is an independent rescaled copy of  $\widehat{h}$ ; the GMC scaling rule changes the mass threshold from  $\varepsilon$  to

$$T_R \varepsilon, \quad T_R = 2^{b_\gamma(m+n_m)} \exp\left(-\gamma \max_{z \in R'} \widehat{h}_{2^{-m-n_m}}(z)\right). \quad (20.43)$$

The Gaussian maximum and scale-comparison lemmas give, with exponentially high probability in  $m$ ,

$$\max_{\mathbb{S}} |\widehat{h}_{2^{-m}}| \leq (2 + o_\zeta(1))m \log 2,$$

and, uniformly for points at distance  $O(2^{-m})$ ,

$$|\widehat{h}_{2^{-m-nm}} - \widehat{h}_{2^{-m}}| \leq o_\zeta(1)m. \quad (20.44)$$

Apply Lemma 8.13 with  $n = 2^{nm} \asymp m^{3/2}$  and mass threshold  $T_R \varepsilon$ . Since its failure probability is  $e^{-cm^{3/2}}$ , a union bound over the  $O(\varepsilon^{-2\beta})$  possible rectangles is still polynomially small. Substituting (20.43) and (20.44) into the scaled graph-distance bound gives

$$\varepsilon^{-1/d_\gamma + \beta b_\gamma/d_\gamma - o_\zeta(1)} \exp\left(\xi \min_{R'} \widehat{h}_{\varepsilon^\beta}\right).$$

The competing polynomial term  $m^3$  from the rectangle amplification is smaller for small  $\varepsilon$  precisely because  $\beta < 2/(2 + \gamma)^2$ ; this is the same strict inequality used in the primary proof. The rotated statement is identical.  $\square$

**Lemma 8.15** (Connected  $Y_S$  skeleton). *On the event of Lemma 8.14, for every dyadic square  $S$  of side  $\delta_\varepsilon$  there is a connected set  $Y_S \subset S(1/2)$  such that adjacent squares have intersecting skeletons and*

$$\text{diam}_{D_h^\varepsilon(\cdot, \cdot; S(1/2))}(Y_S) \leq C\varepsilon^{-1/d_\gamma + \beta b_\gamma/d_\gamma - o_\zeta(1)} e^{\xi \widehat{h}_{\varepsilon^\beta}(v_S)}. \quad (20.45)$$

*Proof.* Inside  $S(1/2)$  take the four standard overlapping rectangles of dimensions  $\delta_\varepsilon \times \delta_\varepsilon/2$  and  $\delta_\varepsilon/2 \times \delta_\varepsilon$  which connect the centre region to the four side-neighbourhoods. In each choose a graph-distance path realizing the estimate of Lemma 8.14, and let  $Y_S$  be their union. Horizontal and vertical crossings of overlapping rectangles intersect by planar topology, so  $Y_S$  is connected. The chosen rectangles for two squares sharing a side overlap in a rectangle which is crossed in the two transverse directions; hence the corresponding skeletons intersect. Finally, the field oscillation bound at scale  $\varepsilon^\beta$  lets us replace the minimum on each  $R'$  by the centre value  $\widehat{h}_{\varepsilon^\beta}(v_S)$  at the cost  $\varepsilon^{-o_\zeta(1)}$ . At most twelve selected crossings are needed to join two points of  $Y_S$ , which gives (20.45).  $\square$

**Proposition 8.16** (Ding–Gwynne lower LFPP branch). *Let  $K \subset U \subset \mathbb{S}$  with  $K$  compact and  $U$  open. For every  $\zeta > 0$ , with polynomially high probability as  $\delta \downarrow 0$ ,*

$$D_{h^{\mathbb{S}(1)}, \text{LFPP}}^\delta(K, \partial U; U) \geq \delta^{a_\gamma + \zeta}. \quad (20.46)$$

*The same holds for a whole-plane GFF on compact subsets after the usual harmonic comparison.*

*Proof.* Set  $\delta = \varepsilon^\beta$ , up to the harmless dyadic rounding above, and work first with the approximate square-chain LFPP metric of Proposition 8.12. Choose a self-avoiding chain  $S_0, \dots, S_k$  from  $K$  to  $\partial U$  such that

$$\sum_{j=0}^k \varepsilon^\beta e^{\xi \widehat{h}_{\varepsilon^\beta}(v_{S_j})} \leq 2\widehat{D}_{h, \text{LFPP}}^{\varepsilon^\beta}(K, \partial U). \quad (20.47)$$

By Lemma 8.15, the union of the  $Y_{S_j}$  is connected. Summing their graph-distance diameters therefore gives

$$D_h^\varepsilon(S_0, S_k; \mathbb{S}(1/2)) \leq C\varepsilon^{-1/d_\gamma + \beta b_\gamma/d_\gamma - \beta - o_\zeta(1)} \widehat{D}_{h, \text{LFPP}}^{\varepsilon^\beta}(K, \partial U). \quad (20.48)$$

The internal DZZ boundary lower bound gives  $D_h^\varepsilon(S_0, S_k) \geq \varepsilon^{-1/d_\gamma - o_\zeta(1)}$  after shrinking the compact sets slightly. Comparing with (20.48) yields

$$\widehat{D}_{h, \text{LFPP}}^{\varepsilon^\beta}(K, \partial U) \geq \varepsilon^{\beta(1-b_\gamma/d_\gamma)+o_\zeta(1)} = \varepsilon^{\beta a_\gamma + o_\zeta(1)}. \quad (20.49)$$

Choose all auxiliary errors small enough that the last error is at most  $\beta\zeta/4$ . We must also check the endpoint-square term in the lower inequality of Proposition 8.12. The coarse maximum bound gives, uniformly in the starting square,

$$\varepsilon^\beta e^{\xi \widehat{h}_{\varepsilon^\beta}(v_{S_0})} \leq \varepsilon^{\beta(1-2\gamma/d_\gamma)-o(1)}.$$

Since

$$1 - \frac{2\gamma}{d_\gamma} - a_\gamma = \frac{2 + \gamma^2/2 - 2\gamma}{d_\gamma} = \frac{(2-\gamma)^2}{2d_\gamma} > 0,$$

this endpoint contribution is of strictly smaller order than the target lower scale. Proposition 8.12 therefore transfers (20.49) to continuum zero-boundary LFPP. The whole-plane statement follows from the GFF Markov coupling on a slightly larger compact set; the harmonic difference has scale-independent Gaussian supremum tails and hence changes LFPP only by a subpower factor. Since  $\delta = \varepsilon^\beta$ , this is (20.46).  $\square$

**The upper LFPP branch DG-U.** The source proof uses a dual finite-range barrier, not the lower-branch skeleton.

**Lemma 8.17** (Finite-range annular graph-distance barrier). *Let*

$$\mathcal{A}_n = [-n, 2n]^2 \setminus (0, n)^2.$$

*For every  $\zeta > 0$  there are  $a_0, a_1, \varepsilon_* > 0$  such that*

$$\mathbb{P}\left[D_h^\varepsilon(\partial_{\text{in}}\mathcal{A}_n, \partial_{\text{out}}\mathcal{A}_n) \geq e^{-n^{1/2}} \varepsilon^{-1/(d_\gamma+\zeta)}\right] \geq 1 - a_0 e^{-a_1 n} \quad (20.50)$$

*for all  $n \geq 1$  and  $\varepsilon \leq \varepsilon_*$ .*

*Proof.* First prove the statement for  $\widehat{h}^{\text{tr}}$  without the factor  $e^{-n^{1/2}}$ . For a unit square  $S$  in the annulus let  $E_\zeta^\varepsilon$  be the event that

$$D_{h^{\text{tr}}}^\varepsilon(S, \partial S(1/2)) \geq \varepsilon^{-1/(d_\gamma+\zeta)} \quad (20.51)$$

and that every  $\varepsilon$ -mass disk meeting  $S(1/2)$  is contained in  $S(3/4)$ . The DZZ point-to-boundary lower theorem and positivity of GMC mass give  $\mathbb{P}[E_\zeta^\varepsilon] \geq 1 - p$  uniformly for sufficiently small  $\varepsilon$ . The event is measurable from the truncated field in  $S(1)$ , so events at sufficiently separated squares are independent.

By the finite-range Peierls argument used in Lemma 8.13, with probability at least  $1 - a_0 e^{-a_1 n}$  there is a circuit of  $E$ -good squares separating the inner and outer boundaries. Any Euclidean crossing of  $\mathcal{A}_n$  must cross  $S(1/2) \setminus S$  for some square on this circuit; by (20.51) its truncated graph distance is at least the right side of (20.51). This proves the truncated statement.

The difference  $\widehat{h} - \widehat{h}^{\text{tr}}$  on  $\mathcal{A}_n$  is a smooth Gaussian field whose supremum is  $O(n^{1/2})$  with exponentially high probability. Adding a bounded field  $f$  changes every GMC mass by factors between  $e^{-\gamma\|f\|_\infty}$  and  $e^{\gamma\|f\|_\infty}$ , hence changes the mass-threshold graph distance only by replacing  $\varepsilon$  with  $e^{\pm\gamma\|f\|_\infty}\varepsilon$ . Applying the exponent window with a slightly smaller  $\zeta$  absorbs this into  $e^{-n^{1/2}}$  and gives (20.50).  $\square$

**Lemma 8.18** (Simultaneous coarse-square annulus estimate). *Fix  $\beta < 2/(2 + \gamma)^2$  and  $\delta_\varepsilon = 2^{-\lceil \log_2 \varepsilon^{-\beta} \rceil}$ . For every  $\zeta > 0$ , with polynomially high probability as  $\varepsilon \downarrow 0$ , for every square  $S \subset \mathbb{S}(1)$  of side  $\delta_\varepsilon$  with corners in  $\delta_\varepsilon \mathbb{Z}^2$ ,*

$$D_h^\varepsilon(S, \partial S(1)) \geq \varepsilon^{-1/d_\gamma + \beta b_\gamma/d_\gamma + \zeta} \exp\left(\xi \max_{z \in S(1)} \widehat{h}_{\varepsilon^\beta}(z)\right). \quad (20.52)$$

*Proof.* Put  $n_\varepsilon = \lfloor (\log \varepsilon^{-1})^{3/2} \rfloor$ . Split the field at scale  $\delta_\varepsilon/n_\varepsilon$  and rescale the annulus  $S(1) \setminus S$  by  $n_\varepsilon/\delta_\varepsilon$ . The fine field is an independent copy of  $\widehat{h}$ , while the GMC scaling rule changes the threshold to  $T_S \varepsilon$  with

$$T_S = (n_\varepsilon/\delta_\varepsilon)^{b_\gamma} \exp\left(-\gamma \min_{z \in S} \widehat{h}_{\delta_\varepsilon/n_\varepsilon}(z)\right). \quad (20.53)$$

Gaussian maximum and scale-comparison estimates give simultaneously for all coarse squares

$$T_S \leq \varepsilon^{-\beta b_\gamma + o_\zeta(1)} \exp\left(-\frac{d_\gamma - o_\zeta(1)}{d_\gamma} \gamma \max_{S(1)} \widehat{h}_{\varepsilon^\beta}\right). \quad (20.54)$$

Because  $\beta < 2/(2 + \gamma)^2$ , the crude maximum-field bound implies  $\max_S T_S \varepsilon = o(1)$  deterministically on this event. Apply Lemma 8.17 with mass threshold  $T_S \varepsilon$  and  $n = n_\varepsilon$ . Its failure probability  $e^{-cn_\varepsilon}$  dominates the  $O(\varepsilon^{-2\beta})$  union bound. Substituting (20.54) into the annular exponent and shrinking the auxiliary errors gives exactly (20.52).  $\square$

**Proposition 8.19** (Ding–Gwynne upper LFPP branch). *For every compact  $K \Subset U$  and every  $\zeta > 0$ , with polynomially high probability as  $\delta \downarrow 0$ ,*

$$\max_{z, w \in K} D_{h, \text{LFPP}}^\delta(z, w; U) \leq \delta^{a_\gamma - \zeta}. \quad (20.55)$$

*Proof.* It is enough to work with a zero-boundary GFF on  $\mathbb{S}(1)$  and to prove the estimate for  $z, w \in \mathbb{S}$  with paths in  $\mathbb{S}(1/2)$ ; a finite cover and the harmonic comparison then give the stated form. Set  $\delta = \varepsilon^\beta$  as above. Besides the event of Lemma 8.18, use the GMC radius estimate

$$\min_{x \in \mathbb{S}(1)} \mu_{\widehat{h}}(B_{\delta_\varepsilon/2}(x)) \geq \varepsilon. \quad (20.56)$$

Hence points separated by at least  $\delta_\varepsilon$  have graph distance at least two.

Fix  $z, w \in \mathbb{S}$  and choose a graph-distance minimizing path  $P$  whose range is covered by  $N = D_h^\varepsilon(z, w; \mathbb{S}(1/2))$  admissible  $\varepsilon$ -mass disks. Starting at  $z$ , record the successive coarse squares  $S_j$  when  $P$  first exits the enlargement  $S_j(1)$ , stopping when it reaches  $w$ . Join the successive exit points by straight segments. Each segment lies in  $S_j(1)$  and has Euclidean length  $O(\varepsilon^\beta)$ , so its total LFPP length is bounded by

$$C \varepsilon^\beta \sum_{j=0}^J \exp\left(\xi \max_{S_j(1)} \widehat{h}_{\varepsilon^\beta}\right). \quad (20.57)$$

For  $j < J$ , the original path crosses the annulus  $S_j(1) \setminus S_j$ . Lemma 8.18 therefore gives

$$D_h^\varepsilon(P(t_j), P(t_{j+1})) \geq \varepsilon^{-1/d_\gamma + \beta b_\gamma/d_\gamma + \zeta_0} \exp\left(\xi \max_{S_j(1)} \widehat{h}_{\varepsilon^\beta}\right). \quad (20.58)$$

The sum of the left sides is at most  $N + 2J$  because only disks containing the transition points can be double counted. By (20.56), consecutive exit points cannot lie in one admissible disk, hence  $J \leq N$ . Summing (20.58) yields

$$\sum_{j=0}^J e^{\xi \max_{S_j(1)} \hat{h}_{\varepsilon^\beta}} \leq 3\varepsilon^{1/d_\gamma - \beta b_\gamma/d_\gamma - \zeta_0} N. \quad (20.59)$$

The internal DZZ uniform upper bound gives  $N \leq \varepsilon^{-1/d_\gamma - \zeta_0}$  simultaneously for  $z, w$  in a fixed compact set. Combining (20.57)–(20.59) gives

$$D_{h, \text{LFPP}}^{\varepsilon^\beta}(z, w) \leq C\varepsilon^{\beta(1-b_\gamma/d_\gamma) - 2\zeta_0} = C\varepsilon^{\beta a_\gamma - 2\zeta_0}. \quad (20.60)$$

Choose  $\zeta_0$  small relative to  $\beta\zeta$ , transfer from  $\hat{h}$  to the circle-average/zero-boundary field using Lemma 8.10, and absorb the deterministic constant. Since  $\delta = \varepsilon^\beta$ , this is (20.55).  $\square$

**Proposition 8.20** (Graph-distance exponent implies the circle-average LFPP exponent). *Let  $h$  be a whole-plane GFF and let  $D_{h, \text{LFPP}}^\delta(\cdot, \cdot; U)$  be circle-average LFPP with  $\xi = \gamma/d_\gamma$ . For every  $K \Subset U$  and every  $\zeta > 0$ , with polynomially high probability as  $\delta \downarrow 0$ ,*

$$D_{h, \text{LFPP}}^\delta(K, \partial U; U) \geq \delta^{a_\gamma + \zeta}, \quad \max_{z, w \in K} D_{h, \text{LFPP}}^\delta(z, w; U) \leq \delta^{a_\gamma - \zeta}.$$

*Proof.* The lower estimate is Proposition 8.16; the upper estimate is Proposition 8.19. Both are stable under the harmonic whole-plane/zero-boundary comparison on compact subsets. Thus the two branches exactly reconstruct Ding–Gwynne Propositions 3.15–3.18 in the form needed here.  $\square$

**Rectangle crossing quantiles.** We now convert the internal-distance exponent into the exact DDDF input.

**Theorem 8.21** (Fixed-rectangle heat-kernel LFPP exponent). *For every fixed  $p \in (0, 1/10)$  and every  $\eta > 0$ , there exists  $n_0 = n_0(p, \eta)$  such that, for  $n \geq n_0$ ,*

$$2^{-n(a_\gamma + \eta)} \leq \ell_{1,1}^{(n)}(\phi, p) \leq \bar{\ell}_{1,1}^{(n)}(\phi, p) \leq 2^{-n(a_\gamma - \eta)}. \quad (20.8e)$$

*Equivalently, every fixed central crossing quantile of the heat-kernel field has exponent  $a_\gamma = 1 - \xi Q$ .*

*Proof.* For the lower bound, choose

$$K = [0, 1/4] \times [0, 1], \quad U = (-1/4, 3/4) \times (-1/4, 5/4).$$

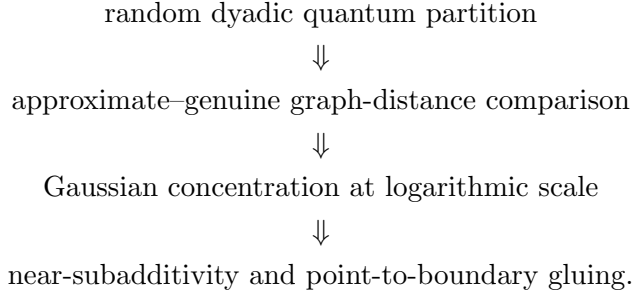
Every left–right crossing of the unit square begins in  $K$  and, before reaching the right side, exits  $U$ . Therefore its circle-average LFPP length is at least  $D_{h, \text{LFPP}}^{2^{-n}}(K, \partial U; U)$ . Proposition 8.20 gives the exponent  $a_\gamma$  with probability tending to one polynomially fast.

For the upper bound take a slightly enlarged rectangle  $U^+$  containing the unit square compactly and two fixed points  $z, w$  on opposite vertical sides of the unit square. The upper half of Proposition 8.20 gives a path from  $z$  to  $w$  in  $U^+$  of length at most  $2^{-n(a_\gamma - \eta/4)}$  with probability tending to one. Between the last visit of the path to the line through  $z$  and its first subsequent visit to the line through  $w$ , it contains a side-to-side crossing of a fixed rectangle of bounded aspect ratio contained in  $U^+$ . Thus one fixed bounded-aspect crossing has the required upper exponent.

Lemma 8.10 and Lemma 8.11 transfer both exponent bounds from the circle-average field to  $\phi_{0,n}$ . Finally Theorem 8.8 compares fixed lower and upper crossing quantiles for any two bounded aspect ratios by multiplicative constants independent of  $n$ . Hence the same exponent holds for the unit-square crossing. Since the failure probabilities tend to zero, for sufficiently large  $n$  they are smaller than  $p$ ; the probabilistic bounds then imply the fixed- $p$  quantile inequalities (20.8e).  $\square$

### 8.1.3 Liouville graph distance: internal exponent theorem

This subsection proves the existence and quantitative stability of the Liouville graph-distance exponent, following the Ding–Zeitouni–Zhang architecture. The proof has four mechanisms:



The GMC moment estimates and Gaussian concentration used below are already proved in Volumes VI and XVII, while the KPZ input used only to obtain a crude nontrivial upper exponent is owned by Volume XVIII.

**Normalization.** Let  $V = [0, 1]^2$  and let  $h$  be a zero-boundary GFF on  $V$ . We use a scale-local white-noise approximation  $\eta_r$  whose covariance differs from a circle-average regularization by a uniformly bounded continuous kernel on compact subsets. Let  $M_\gamma$  be the subcritical GMC area measure. For  $\delta \in (0, 1)$  define the graph distance

$$D_{\gamma, \delta}(A, B; U)$$

to be the least number of Euclidean balls contained in  $\overline{U}$ , each of  $M_\gamma$ -mass at most  $\delta^2$ , whose union contains a path in  $\overline{U}$  joining  $A$  to  $B$ . Thus the mass parameter used in Definition 8.9 is  $\varepsilon = \delta^2$ .

For a dyadic square  $S$  of side  $s$  and center  $c_S$ , set

$$m_\delta(S) := s^2 \exp\left(\gamma \eta_s(c_S) - \frac{\gamma^2}{2} \text{Var } \eta_s(c_S)\right). \quad (20.9)$$

Starting from  $V$ , split a dyadic square into its four children whenever  $m_\delta(S) \geq \delta^2$ . The terminal squares form a random partition  $\mathcal{V}_\delta$ . Adjoining two terminal squares when their closures share a nontrivial segment gives the *approximate Liouville graph*. Its graph distance is denoted  $D'_{\gamma, \delta}$ .

**Lemma 8.22** (Polynomial window for terminal side lengths). *There are deterministic constants  $0 < c_{\min} < c_{\max} < \infty$ , depending only on  $\gamma$ , such that with probability  $1 - O(\delta^c)$  for some  $c > 0$ , every terminal square  $S \in \mathcal{V}_\delta$  satisfies*

$$\delta^{c_{\max}} \leq s_S \leq \delta^{c_{\min}}.$$

Moreover  $|\log s_{\min}| / \log \delta^{-1}$  has an exponential tail uniformly for small  $\delta$ .

*Proof.* Write  $s = 2^{-k}$ . Since  $\text{Var } \eta_s(c) = \log s^{-1} + O(1)$ , a Gaussian union bound over the  $O(s^{-2})$  centers at scale  $s$  gives, for every  $a > 2$  and all sufficiently small  $s$ ,

$$\mathbb{P}\left[\max_c \eta_s(c) > a \log s^{-1}\right] \leq C s^{a^2/2-2}.$$

Choose  $a$  so that the exponent is positive. On the complementary event, (20.9) is at most  $s^{\kappa_-}$  for some  $\kappa_- > 0$ . Hence all squares at side  $s \leq \delta^{2/\kappa_-}$  are automatically terminal, which proves the lower bound on terminal side lengths after summing the geometric tail over  $k$ .

For the opposite direction we need to rule out a terminal square which is too large. If  $S$  is terminal then its parent  $S^+$  satisfies  $m_\delta(S^+) \geq \delta^2$ . By the local GMC comparison from Volume XVII,  $M_\gamma(S^+)$  is a lognormal factor times a unit-scale chaos mass at scale  $s_{S^+}$ . The local GMC positive-moment estimate proved in Volume XVII, together with Markov's inequality, shows that for some  $q \in (0, 4/\gamma^2)$  and  $b > 0$ ,

$$\mathbb{P}[m_\delta(S^+) \geq \delta^2] \leq C\delta^{-2q}s_{S^+}^{bq}.$$

Taking  $s_{S^+} \geq \delta^{c_{\min}}$  with  $c_{\min}$  sufficiently small and summing over the  $O(s^{-2})$  possible parents makes the probability  $O(\delta^c)$ . Repeating the same estimate at the successive dyadic scales also gives an exponential tail for the normalized logarithm of the smallest cell. This is precisely the rough cell-size input needed below; no sharp multifractal information is used.  $\square$

**Lemma 8.23** (Finite-range circuit estimate). *Let  $\{E_Q\}$  be events indexed by the squares of a square grid, each with  $\mathbb{P}(E_Q) \geq 1 - p$ , and suppose families at grid distance at least  $R$  are independent. There are  $p_0 = p_0(R) > 0$  and  $c = c(R) > 0$  such that, if  $p < p_0$ , the probability that an annulus of grid width  $L$  contains no circuit of  $E$ -open squares surrounding its inner boundary is at most  $Ce^{-cL}$ . The same estimate holds for a crossing between opposite sides of a rectangle of bounded aspect ratio.*

*Proof.* If there is no open circuit, planar duality supplies a  $*$ -connected path of closed squares crossing the annulus. A  $*$ -connected path of length  $m$  has at most  $8^m$  possible shapes after its starting square is fixed. From any such path greedily choose at least  $m/C_R$  squares separated by grid distance  $R$ . Their closed events are independent, so the probability of that path is at most  $p^{m/C_R}$ . Summing  $CL \sum_{m \geq L} 8^m p^{m/C_R}$  proves the result when  $p$  is small enough. The rectangular version is identical, using a dual top-bottom crossing if a left-right crossing is absent.  $\square$

**Comparison of the random partition and the genuine graph distance.** We use the boundary-cover mechanism from the DZZ proof. Put

$$C_{\text{mc}} := c_{\max}, \quad C_{\text{Mc}} := c_{\min},$$

so that on the event in Lemma 8.22, every terminal cell satisfies

$$\delta^{C_{\text{mc}}} \leq s_{\mathcal{C}} \leq \delta^{C_{\text{Mc}}}, \quad C_{\text{mc}} > C_{\text{Mc}} > 0.$$

For the comparison theorem we fix  $0 < \rho < C_{\text{Mc}}/3$ . A pair  $(A_\delta, B_\delta)$  is called  $\rho$ -admissible when both sets lie in a fixed compact subset of  $V$ , their Euclidean separation is bounded below by a positive constant independent of  $\delta$ , and each endpoint is either a point or a connected set of diameter at least  $\delta^\rho$ .

**Definition 8.24** (The DZZ coarse-field event and boundary functionals). For  $\alpha > 0$ , let  $\mathcal{E}_{\delta, \alpha}$  be the intersection of the terminal-cell window with the event that

$$|\eta_{2^{-m}}(x) - \eta_{2^{-m-j}}(y)| \leq \alpha \sqrt{\log \delta^{-1}} \log \log \delta^{-1} \quad (20.10)$$

whenever

$$1 \leq 2^m \leq \delta^{-C_{\text{mc}}}, \quad 1 \leq 2^j \leq (\alpha \log \delta^{-1})^2, \quad |x - y| \leq 2^{-m+3},$$

with  $x, y$  taken from the corresponding dyadic grids. If  $B$  is a dyadic square, let  $B_{\text{large}}$  be the concentric square of twice the side length. For a dyadic  $\varepsilon > 0$ , let  $\mathcal{B}(B, \varepsilon)$  be the  $\varepsilon s_B$ -squares in  $B_{\text{large}}$  and let  $\mathcal{B}_\partial(B, \varepsilon)$  be those whose closures meet  $\partial B$ .

For  $\tau > 0$  define

$$\begin{aligned}\Psi_{B,\tau} &:= \#\{\mathbf{C} \in \mathcal{V}_\tau : \mathbf{C} \subset B, \bar{\mathbf{C}} \cap \partial B \neq \emptyset\}, \\ \Phi_{B,\tau} &:= \min\{N : \partial B \text{ is covered by } N \text{ Euclidean balls, } M_\gamma(\mathbf{B}_i) \leq \tau^2\},\end{aligned}$$

with  $\Psi_{B,\tau} = 1$  if  $B$  is contained in one terminal  $\tau$ -cell. Given  $\lambda > 0$ , write  $\mathcal{E}_{\tau,B,\varepsilon,\lambda}$  for the event that there is a nearest-neighbor circuit  $B'_1, \dots, B'_d \subset B_{\text{large}} \setminus B$  of squares from  $\mathcal{B}(B, \varepsilon)$  which separates  $B$  from  $V \cap \partial B_{\text{large}}$  and satisfies  $\Psi_{B'_i,\tau} \leq \lambda$  for every  $i$ ; define  $\mathcal{E}'_{\tau,B,\varepsilon,\lambda}$  in the same way with  $\Phi_{B'_i,\tau} \leq \lambda$ .

**Lemma 8.25** (Uniform coarse-field control). *There is  $\alpha_0 = \alpha_0(\gamma)$  such that, for every  $\alpha > \alpha_0$ ,*

$$\mathbb{P}(\mathcal{E}_{\delta,\alpha}) \geq 1 - C\delta^c. \quad (20.11)$$

*Proof.* The cell-size part is Lemma 8.22. For the field increments, let  $m_0 = \lfloor C_{\text{mc}} \log_2 \delta^{-1} \rfloor$  and  $j_0 = \lfloor 2 \log_2(\alpha \log \delta^{-1}) \rfloor$ . If  $\tilde{x}$  and  $\tilde{y}$  are the centers of the dyadic squares at the relevant scales containing  $x$  and  $y$ , respectively, then

$$\begin{aligned}|\eta_{2^{-m}}(x) - \eta_{2^{-m-j}}(y)| &\leq |\eta_{2^{-m}}(x) - \eta_{2^{-m}}(\tilde{x})| + |\eta_{2^{-m}}(\tilde{x}) - \eta_{2^{-m}}(\tilde{y})| \\ &\quad + |\eta_{2^{-m-j}}(\tilde{y}) - \eta_{2^{-m-j}}(y)| + |\eta_{2^{-m}}(\tilde{y}) - \eta_{2^{-m-j}}(\tilde{y})|.\end{aligned}$$

The first three terms are controlled by the scale-local spatial increment estimate proved in Volume VI; their Gaussian variances are uniformly bounded at the displayed normalized scale. A union bound over at most  $O(\delta^{-2C_{\text{mc}}} \log \delta^{-1})$  grid points and scales makes them at most  $(\alpha/10)\sqrt{\log \delta^{-1}}$  with polynomially high probability. For the fourth term, the white-noise increments between scales  $2^{-m}$  and  $2^{-m-j}$  have variance  $O(j)$ . Gaussian tails followed by a union bound over  $m \leq m_0, j \leq j_0$ , and the dyadic grid show that it is at most  $(\alpha/10)\sqrt{\log \delta^{-1}} \log \log \delta^{-1}$  simultaneously once  $\alpha$  is large enough. Combining the four estimates proves (20.11).  $\square$

**Lemma 8.26** (One-cell open enclosure and endpoint escape). *Fix  $\alpha > \max\{\alpha_0, 4C_{\text{mc}}\}$  and let  $0 < \delta' < \delta$ . Choose the dyadic number  $\varepsilon$  so that*

$$\frac{1}{8C_{\text{mc}} \log \delta^{-1}} \leq \varepsilon \leq \frac{1}{4C_{\text{mc}} \log \delta^{-1}},$$

and set

$$\lambda = (\delta/\delta')^3 \exp((\log \delta^{-1})^{0.7}). \quad (20.12)$$

Let  $B$  be a dyadic square of side  $s = 2^{-m}$  with  $1 \leq m \leq C_{\text{mc}} \log_2 \delta^{-1}$ . Then

$$\mathbb{P}[m_\delta(B) \leq \delta^2, \mathcal{E}_{\delta,\alpha}, \mathcal{E}'_{\delta',B,\varepsilon,\lambda}] \leq \delta^{10C_{\text{mc}}+10}. \quad (20.13)$$

Moreover, for each fixed  $x \in B_{\text{large}}$  and each fixed  $\iota > 0$ ,

$$\mathbb{P}[m_\delta(B) \leq \delta^2, \mathcal{E}_{\delta,\alpha}, D'_{\gamma,\delta'}(x, \partial B_{\text{large}}) > \delta^{-\iota}(\delta/\delta')^3] \leq \delta^{\iota/10}. \quad (20.14)$$

*Proof.* We give the two estimates separately because they play different roles in the source proof.

For (20.13), choose a dyadic  $t$  with  $\log t^{-1} \geq (\log \delta^{-1})^{0.6}$  and later set

$$t = \varepsilon 2^{-\lceil 0.9 \log_2 \lambda \rceil}.$$

For  $B' \in \mathcal{B}(B, \varepsilon)$ , cover  $\partial B'$  by the  $ts$ -squares in  $\mathcal{B}_\partial(B', t/\varepsilon)$ . On  $\{m_\delta(B) \leq \delta^2\} \cap \mathcal{E}_{\delta,\alpha}$ , the coarse-field comparison between scales  $s, \varepsilon^2 s$ , and  $ts$  gives

$$m_{\delta'}(\tilde{B}) \leq \delta^2 e^{2\gamma\alpha\sqrt{\log \delta^{-1}} \log \log \delta^{-1}} t^2 e^{\gamma\eta_{ts}^{\varepsilon^2 s}(c_{\tilde{B}}) - \frac{\gamma^2}{2} \text{Var} \eta_{ts}^{\varepsilon^2 s}(c_{\tilde{B}})}. \quad (20.15)$$



Here  $\eta_{ts}^{\varepsilon^2 s}$  is precisely the white-noise increment in the space-scale slab between the two cutoffs. The number of boundary boxes is at most  $8\varepsilon/t$ . A Gaussian union bound therefore yields an event  $O_{B'}$  with

$$\mathbb{P}(O_{B'}) \geq 1 - t^{0.1}$$

on which all boundary boxes have approximate  $\delta'$ -mass at most  $\delta'^2$ . The choice of  $t$  gives both  $\delta^2 t^{0.8} \leq \delta'^2$  and  $4/t \leq \lambda$ . Consequently  $O_{B'}$  implies  $\Psi_{B', \delta'} \leq \lambda$ .

The fine white-noise fields used by  $O_{B'}$  and  $O_{B''}$  are independent whenever  $B'$  and  $B''$  are separated by at least  $2\varepsilon s$ . If no  $O$ -open circuit surrounds  $B$ , planar duality gives a  $*$ -connected closed path from  $\partial B$  to  $\partial B_{\text{large}}$  of length at least  $K := \varepsilon^{-1}$ . For each path of length  $\ell$ , at least a fixed fraction of its boxes are mutually  $2\varepsilon s$ -separated. Since the number of such paths is at most  $CK8^\ell$ , Lemma 8.23 in its explicit counting form gives

$$\mathbb{P}(\text{no open enclosure}) \leq CK \sum_{\ell \geq K} 8^\ell (t^{0.1})^{c\ell} \leq \delta^{10C_{\text{mc}}+10}$$

for small  $\delta$ . This proves (20.13).

For (20.14), take the largest dyadic  $t \leq \delta^{0.9\iota} (\delta'/\delta)^3$ . The  $O(t^{-1})$  squares of side  $ts$  intersecting a horizontal segment from  $x$  to  $\partial B_{\text{large}}$  satisfy the same estimate (20.15). The Gaussian variable governing each fine increment has variance at most  $|\log t|$ . Hence a union bound gives, with probability at least  $1 - Ct^{0.12} \geq 1 - \delta^{\iota/10}$ , that every one of these squares is terminal for the  $\delta'$ -partition. Their number is at most  $4/t$ , which by the choice of  $t$  is bounded by  $\delta^{-\iota} (\delta/\delta')^3$ . This gives (20.14).  $\square$

**Proposition 8.27** (Nearby-threshold comparison for the approximate distance). *For every  $0 < \rho < C_{\text{Mc}}/3$ , every  $\rho$ -admissible pair, and every function  $0 < \delta' < \delta$ , with polynomially high probability,*

$$D'_{\gamma, \delta'}(A_\delta, B_\delta) \leq D'_{\gamma, \delta}(A_\delta, B_\delta) (\delta/\delta')^3 \exp((\log \delta^{-1})^{0.8}). \quad (20.16)$$

*The reverse monotonicity  $D'_{\gamma, \delta'} \geq D'_{\gamma, \delta}$  is deterministic.*

*Proof.* Let  $\mathbb{C}_1, \dots, \mathbb{C}_d$  be a minimizing  $\delta$ -cell path, where  $d = D'_{\gamma, \delta}(A_\delta, B_\delta)$ . By (20.13) and a union bound over every cell in the polynomial cell-size window, each  $\mathbb{C}_i$  possesses a  $\delta'$ -cell enclosure  $\mathbb{C}_i$  with  $|\mathbb{C}_i| \leq \lambda/\varepsilon^2$ .

If an endpoint is a point, apply (20.14) simultaneously to the  $O(\log \delta^{-1})$  dyadic boxes whose doubled boxes contain it; this supplies a start or end path of length at most  $\delta^{-\iota_0} (\delta/\delta')^3$ , where  $\iota_0 = C_{\text{Mc}}/3$ . If an endpoint is a connected set of diameter at least  $\delta^\rho$ , no auxiliary path is needed: the connected endpoint must meet the first relevant enclosure.

Successive coarse cells touch. Therefore one can choose recursively a chain of the enclosures  $\mathbb{C}_i$  so that consecutive chosen enclosures meet; together with the endpoint paths, their union contains a  $\delta'$ -cell crossing. Hence

$$D'_{\gamma, \delta'}(A_\delta, B_\delta) \leq d\lambda/\varepsilon^2 + 2\delta^{-\iota_0} (\delta/\delta')^3. \quad (20.17)$$

On  $\mathcal{E}_{\delta, \alpha}$ , the maximal terminal side is at most  $\delta^{C_{\text{Mc}}}$ , while the endpoints are macroscopically separated. Thus  $d \geq c\delta^{-C_{\text{Mc}}} \geq \delta^{-2\iota_0}$  for small  $\delta$ . Since  $\varepsilon^{-2}$  is polylogarithmic and  $\lambda$  is given by (20.12), both extra terms in (20.17) are absorbed by  $\exp((\log \delta^{-1})^{0.8})$ . This proves (20.16).  $\square$

**Lemma 8.28** (Genuine-mass boundary enclosure). *There is another event  $\tilde{\mathcal{E}}_{\delta, \alpha}$  of polynomially high probability such that the following holds. Put*

$$\varepsilon \asymp (\log \delta^{-1})^{-1}, \quad \lambda = \exp((\log \delta^{-1})^{0.7}).$$

For every dyadic  $B$  in the terminal cell-size window,

$$\mathbb{P}\left[m_\delta(B) \leq \delta^2, \tilde{\mathcal{E}}_{\delta,\alpha}, (\mathcal{E}'_{\delta,B,\varepsilon,\lambda})^c\right] \leq \delta^{10C_{\text{mc}}+10}, \quad (20.18)$$

and, for every fixed  $x \in B_{\text{large}}$  and  $\iota > 0$ ,

$$\mathbb{P}\left[m_\delta(B) \leq \delta^2, \tilde{\mathcal{E}}_{\delta,\alpha}, D_{\gamma,\delta}(x, \partial B_{\text{large}}) > \lambda \delta^{-\iota}\right] \leq \delta^{\iota/10}. \quad (20.19)$$

*Proof.* Intersect  $\mathcal{E}_{\delta,\alpha}$  with the scale-uniform coupling event on which the local white-noise regularization and the GFF regularization differ by  $O(\sqrt{\log \delta^{-1}})$  throughout the terminal scale window. This is  $\tilde{\mathcal{E}}_{\delta,\alpha}$ .

For a coarse terminal square  $B$  of side  $s$ , choose  $t = \varepsilon 2^{-\lceil c(\log \delta^{-1})^{0.6} \rceil}$ . Let  $\widetilde{M}_{\gamma,\varepsilon^2 s,\eta}$  denote the GMC made only from the fine white-noise field below the cutoff  $\varepsilon^2 s$ . The multiplicative coarse/fine decomposition gives, for every  $ts$ -square  $Q$  used to cover the boundary of an  $\varepsilon s$ -square,

$$M_\gamma(Q) \leq e^{2\alpha\gamma\sqrt{\log \delta^{-1}} \log \log \delta^{-1}} \delta^2 s^{-2} \widetilde{M}_{\gamma,\varepsilon^2 s,\eta}(Q). \quad (20.20)$$

The fine chaos is normalized so that  $\mathbb{E}\widetilde{M}_{\gamma,\varepsilon^2 s,\eta}(Q) = (ts)^2$ . Markov's inequality therefore makes the event

$$\widetilde{M}_{\gamma,\varepsilon^2 s,\eta}(Q) \leq \frac{1}{4} s^2 e^{-2\alpha\gamma\sqrt{\log \delta^{-1}} \log \log \delta^{-1}}$$

fail with probability at most  $t^{1.9}$  once the exponent in the definition of  $t$  is chosen slightly larger if necessary. There are at most  $t^{-1}$  boundary boxes, so an  $\varepsilon s$ -square is open with probability at least  $1 - t^{0.9}$ ; the relevant open events are independent at grid separation two. On the open event, balls of radius  $ts$  centered at boundary-grid corners cover the boundary, each such ball is covered by four good  $ts$ -squares, and (20.20) gives mass at most  $\delta^2$ . Thus  $\Phi_{B',\delta} \leq 4\varepsilon/t \leq \lambda$ . The same dual-path count as in Lemma 8.26 proves (20.18).

For (20.19), choose instead the largest dyadic  $t \leq c\delta^{\iota/2}\lambda^{-1}$  and cover a straight segment from  $x$  to  $\partial B_{\text{large}}$  by  $O(t^{-1})$   $ts$ -boxes, enlarged by their neighbors. If every enlarged box satisfies the fine-chaos bound above, the segment is covered by at most  $4/t + O(1) \leq \delta^{-\iota}\lambda$   $\delta$ -admissible balls. The failure probability is  $O(t^{-1})t^{1.9} \leq \delta^{\iota/10}$ . This proves the endpoint estimate.  $\square$

**Proposition 8.29** (DZZ approximate-genuine graph-distance comparison). *Fix  $0 < \rho < C_{\text{Mc}}/3$  and a  $\rho$ -admissible pair  $(A_\delta, B_\delta)$ . Then there is  $c = c(\gamma, \rho) > 0$  such that with  $c$ -high probability*

$$e^{-(\log \delta^{-1})^{0.9}} D'_{\gamma,\delta}(A_\delta, B_\delta) \leq D_{\gamma,\delta}(A_\delta, B_\delta; V) \leq e^{(\log \delta^{-1})^{0.9}} D'_{\gamma,\delta}(A_\delta, B_\delta). \quad (20.21)$$

*Consequently, for every fixed  $\iota > 0$ , the two distances differ by a factor between  $\delta^\iota$  and  $\delta^{-\iota}$  for all sufficiently small  $\delta$ , on an event of polynomially high probability.*

*Proof.* Let  $d = D'_{\gamma,\delta}(A_\delta, B_\delta)$ . For the upper bound, apply Lemma 8.28 to every coarse cell on a minimizing  $D'$ -path. The genuine-mass circuits concatenate exactly as in the threshold comparison and the endpoint estimate handles point endpoints. This gives

$$D_{\gamma,\delta}(A_\delta, B_\delta; V) \leq d\lambda/\varepsilon^2 + 2\lambda\delta^{-\iota_0} \leq d \exp((\log \delta^{-1})^{0.9}),$$

using again  $d \geq \delta^{-2\iota_0}$  and the polylogarithmic size of  $\varepsilon^{-2}$ .

For the lower bound, put

$$\delta_+ = \delta \exp((\log \delta^{-1})^{0.8}).$$

We first show that, with high probability, every Euclidean ball of  $M_\gamma$ -mass at most  $\delta^2$  is covered by at most four terminal  $\delta_+$ -cells. Cover such a ball by four pairwise-touching dyadic squares of side  $s$  comparable to its radius. If one of these dyadic squares  $B$  is not contained in a terminal  $\delta_+$ -cell, then  $m_{\delta_+}(B) > \delta_+^2$ . Partition the concentric  $4s$ -square into a fixed  $2^{12} \times 2^{12}$  mesh; one mesh square  $S$  lies inside the original ball. It is therefore enough to rule out, uniformly over all admissible scales and locations,

$$m_{\delta_+}(B) > \delta_+^2, \quad M_\gamma(S) \leq \delta^2. \quad (20.22)$$

On the two good coarse-field events at parameters  $\delta$  and  $\delta_+$ , subdivide  $S$  into  $K^2$  squares with  $K \asymp \log \delta^{-1}$ . A fixed positive fraction of these can be chosen so that the fine white-noise GMC masses are mutually independent. The coarse lower bound  $m_{\delta_+}(B) > \delta_+^2$  and the field-control event give

$$M_\gamma(S) \geq e^{-C\sqrt{\log \delta^{-1}} \log \log \delta^{-1}} \delta_+^2 s^{-2} \sum_j \widetilde{M}_j.$$

With  $\beta = \exp(-(\log \delta^{-1})^{0.7})$ , the all-order negative GMC moment from Volume XVII gives  $\mathbb{P}[(\varepsilon s)^{-2} \widetilde{M}_j \leq \beta] \leq C\beta$ . A binomial/Chernoff bound for the independent selected squares therefore makes (20.22) smaller than  $\exp(-cK^2)$ . Summing over the polynomially many boxes in the cell-size window is still negligible. Hence

$$D'_{\gamma, \delta_+}(A_\delta, B_\delta) \leq 4D_{\gamma, \delta}(A_\delta, B_\delta; V). \quad (20.23)$$

Apply Proposition 8.27 with the larger threshold  $\delta_+$  and the smaller threshold  $\delta$  to obtain

$$D'_{\gamma, \delta}(A_\delta, B_\delta) \leq D'_{\gamma, \delta_+}(A_\delta, B_\delta) \exp((\log \delta^{-1})^{0.9}).$$

Combining this with (20.23) and absorbing the constant four proves the lower inequality of (20.21).  $\square$

**Corollary 8.30** (Interior doubled-ball and circle-average variants). *Fix  $\xi > 0$  and restrict all paths and covering balls to  $V^\xi = \{z : \text{dist}(z, \partial V) \geq \xi\}$ . Let  $D_{\gamma, \delta, \xi}^{(2)}$  be defined by requiring  $M_\gamma(2B) \leq \delta^2$  for every covering ball, and let  $D_{\gamma, \delta, \xi}^\circ$  be defined using the circle-average approximate mass  $r^{2+\gamma^2/2} e^{\gamma h_r(z)}$ . For every fixed  $\iota > 0$  and every interior admissible pair, each of these distances differs from the restricted approximate partition distance by at most  $\delta^{\pm \iota}$  with  $(c\iota^2)$ -high probability.*

*Proof.* For  $D^{(2)}$ , repeat Lemma 8.28 with the boundary functional  $\Phi_{B, \delta}$  redefined using  $M_\gamma(2B) \leq \delta^2$ ; only the fixed geometric enlargement constants change. For the circle-average version, the scale-uniform comparison between the circle-average field and the local white-noise field on  $V^\xi$  is  $O(\sqrt{\log \delta^{-1}})$  throughout all scales above the minimal terminal scale. Hence the circle-average stopping threshold is sandwiched between the white-noise thresholds  $\delta \exp(\pm(\log \delta^{-1})^{0.6})$ . Proposition 8.27 absorbs these nearby-threshold shifts into  $\delta^{\pm \iota}$ . The polynomially high probability exponent is reduced only by the Gaussian comparison event, yielding the stated  $(c\iota^2)$  form.  $\square$

**Regular paths and conditional fine-field independence.** The comparison theorem by itself is not enough for later multiscale arguments: an approximate geodesic may jump repeatedly between cells whose Euclidean sizes are very different. We now prove the regular-path theorem in the form actually needed by the conditional fine-field decomposition. This is the exact mechanism behind DZZ Lemmas 3.12–3.13, with the enclosure estimate already owned above rather than re-imported.

Fix a large constant  $\alpha^* = \alpha^*(\gamma)$  and let

$$\epsilon_\delta^* = \max \left\{ 2^{-n} : 2^{-n} \leq \exp \left[ -\alpha^* \sqrt{\log \delta^{-1}} \log \log \delta^{-1} \right] \right\}. \quad (20.15)$$

If  $\mathcal{C} = (C_1, \dots, C_d)$  is a neighboring sequence, set  $C_0 = C_1$  and  $C_{d+1} = C_d$ . We call  $C_i$  *good* in the sequence if

$$s_{C_{i-1}}, s_{C_{i+1}} \in [\epsilon_\delta^* s_{C_i}, (\epsilon_\delta^*)^{-1} s_{C_i}].$$

The sequence is good if every one of its cells is good. A point  $x$  is *good* if, whenever a terminal cell  $C$  has  $x \in C_{\text{large}}$ , every terminal cell meeting  $C_{\text{large}}$  has side at least  $\epsilon_\delta^* s_C$ .

**Lemma 8.31** (Fine-scale enclosure and neighbor regularity). *On the coarse-field event  $\mathcal{E}_{\delta, \alpha}$  of Definition 8.24, choose  $\alpha^* > \alpha$  sufficiently large. Uniformly over dyadic boxes  $B$  with  $\delta^{C_{\text{mc}}} \leq s_B \leq 1$ , the following two failures have total probability  $o(1)$  after a union bound over all terminal cells:*

1. *if  $m(B) \leq \delta^2$ , there is no neighboring loop of boxes of side  $\epsilon_\delta^* s_B$  contained in  $B_{\text{large}} \setminus B$  whose approximate masses at scale  $\epsilon_\delta^* s_B$  are at most  $\delta^2$ ;*
2. *some box of side  $\epsilon_\delta^* s_B$  inside  $B_{\text{large}}$  has approximate mass greater than  $\delta^2$ .*

More quantitatively, the first probability is at most  $O(\delta^{10C_{\text{mc}}+10})$  for a fixed  $B$ , and the second is at most  $e^{-\sqrt{\log \delta^{-1}}}$ .

*Proof.* Take a dyadic  $\varepsilon \asymp (\log \delta^{-1})^{-1}$  and put  $t = \epsilon_\delta^*$ . Partition  $B_{\text{large}}$  into  $\varepsilon s_B$  boxes and each relevant boundary into  $t s_B$  boxes. On  $\mathcal{E}_{\delta, \alpha}$ , the scale- $t$  mass comparison from Lemma 8.26 gives, for every fine box  $\tilde{B}$ ,

$$m_{ts_B}(\tilde{B}) \leq \delta^2 e^{2\gamma\alpha\sqrt{\log \delta^{-1}} \log \log \delta^{-1}} t^2 \exp\left(\gamma X_{\tilde{B}} - \frac{\gamma^2}{2} \text{Var } X_{\tilde{B}}\right), \quad (20.16)$$

where the  $X_{\tilde{B}}$  are finite-range Gaussian increments between the coarse and fine scales. Declare a coarse  $\varepsilon s_B$  box open when all fine boxes meeting its boundary have  $X_{\tilde{B}} \leq \frac{3}{2} \log t^{-1}$ . The same finite-range coloring and dual-path counting used in Lemma 8.26 show that failure of an open circuit is at most  $O(\delta^{10C_{\text{mc}}+10})$  once  $\alpha^*$  is large. On an open circuit, (20.16) and  $\gamma(3/2) < 2 + \gamma^2/2$  after the harmless adjustment of the threshold constant make every fine boundary box terminal at level  $\delta$ .

For the second assertion use the sharper one-box maximum bound

$$\max_{\tilde{B}} X_{\tilde{B}} \leq \left(1 + \gamma^{-1} + \frac{\gamma}{4}\right) \log t^{-1}.$$

A Gaussian union bound over  $O(t^{-2})$  boxes fails with probability  $t^{c(\gamma)} \leq e^{-\sqrt{\log \delta^{-1}}}$ , while

$$\gamma \left(1 + \gamma^{-1} + \frac{\gamma}{4}\right) < 2 + \frac{\gamma^2}{2}.$$

Substitution in (20.16) gives the desired mass bound. These are precisely the two probabilistic inputs needed in the path deformation below.  $\square$

**Proposition 8.32** (DZZ regular-path event). *For  $\alpha^* = \alpha^*(\gamma)$  sufficiently large and fixed interior endpoints  $u, v$ , with probability at least*

$$1 - e^{-(\log \delta^{-1})^{1/4}}$$

*both endpoints are good and there is a good neighboring sequence of terminal cells  $C_1, \dots, C_d$  joining them with*

$$d \leq D'_{\gamma, \delta}(u, v) e^{(\log \delta^{-1})^{3/5}}. \quad (20.17)$$

*The same conclusion holds uniformly for endpoint pairs in a fixed admissible compact family after the usual polynomial-size net reduction.*

*Proof.* Lemma 8.31, applied to every dyadic box whose enlarged copy contains  $u$  or  $v$ , shows that the endpoints are good; the failure probability is  $O(\log \delta^{-1})e^{-\sqrt{\log \delta^{-1}}}$ . The same lemma and a union bound over all terminal cells give, with probability  $1 - o(1)$ , the following deterministic enclosure property:

$$\text{for every terminal cell } C \text{ there is a neighboring loop of terminal cells, all of side at least } \epsilon_\delta^* s_C, \text{ which surrounds } C \text{ inside } C_{\text{large}} \setminus C. \quad (20.18)$$

Indeed each fine box on the loop from the preceding lemma is contained in a terminal cell; replacing consecutive repetitions by one cell yields the loop in (20.18).

Let  $\mathcal{C}_0$  be an approximate geodesic of length  $d_0 = D'_{\gamma, \delta}(u, v)$ . For a neighboring sequence  $\mathcal{C}$ , let  $\psi(\mathcal{C})$  be its bad cells and, when nonempty, let  $q(\mathcal{C})$  be the largest side length among them. Starting from  $\mathcal{C}_i$ , choose a largest bad cell  $C$ . Since the endpoints are good, the path enters and exits  $C_{\text{large}}$ . Let  $C_{\text{enter}}$  and  $C_{\text{exit}}$  be the last and first cells at those two crossings and use the loop in (20.18). The loop meets the path on both sides of  $C$ ; choose the loop segment between the first and last such meetings which avoids any unnecessarily large parent cell and replace the old path segment by it.

We verify the only nontrivial deterministic claim. Let the four dyadic boxes of side  $2s_C$  touching  $C$  be the corner parents. Among those that are not subdivided, at most two can obstruct the chosen loop segment. There are three cases.

1. Neither joining cell is a corner parent. Every replacement cell has side in  $[\epsilon_\delta^* s_C, s_C]$ , so  $C$  disappears from  $\psi$  and no new bad cell is created.
2. Exactly one joining cell is a corner parent. Either the same conclusion holds, or that parent is the only new bad cell; then its side is at least  $2s_C$ , hence  $q(\mathcal{C}_{i+1}) \geq 2q(\mathcal{C}_i)$ .
3. Both joining cells are corner parents. They are the entry and exit cells. By maximality of  $C$ , their sizes lie between  $2s_C$  and  $(\epsilon_\delta^*)^{-1}s_C$ . If they are adjacent, keep the pair; if they are diagonal, use the three-cell path through  $C$ . In either case the replacement has no bad interior cell, so  $|\psi|$  decreases.

If  $C$  touches the boundary, only the first two cases occur. Thus every replacement either decreases  $|\psi|$  or leaves it unchanged while doubling  $q$ . Since all terminal sides lie in  $[\delta^{C_{\text{mc}}}, 1]$ , the second alternative can occur at most  $C_{\text{mc}} \log_2 \delta^{-1}$  consecutive times. Hence after at most  $d_0 C_{\text{mc}} \log_2 \delta^{-1}$  replacements the path is good. Each loop contains at most  $4(\epsilon_\delta^*)^{-2}$  cells, so its final length is at most

$$d_0(\epsilon_\delta^*)^{-2} C_{\text{mc}} \log_2 \delta^{-1} \leq d_0 e^{(\log \delta^{-1})^{3/5}}$$

for small  $\delta$ , which proves (20.17).  $\square$

**Lemma 8.33** (Protected refinement and exact conditional fine-field law). *On the event of Proposition 8.32, there is a  $\mathcal{V}_\delta$ -measurable neighboring sequence of dyadic boxes  $B_1, \dots, B_N$  joining  $u$  and  $v$  such that*

$$N \leq D'_{\gamma, \delta}(u, v) e^{2(\log \delta^{-1})^{3/5}}, \quad s_{B_i} = (\epsilon_\delta^*)^2 s_{C(i)} \quad (20.19)$$

*for a terminal cell  $C(i) \supset B_i$ . Conditional on the terminal partition, the collection of white-noise increments below scale  $s_{B_i}$  in  $(B_i)_{\text{large}}$  has exactly its unconditional joint law.*

*Proof.* Let  $C_1, \dots, C_d$  be the good cell sequence. If  $\Lambda_j = \partial C_j \cap \partial C_{j+1}$ , let  $x_j$  be the midpoint of  $\Lambda_j$  and partition  $C_j$  into boxes of side  $(\epsilon_\delta^*)^2 s_{C_j}$ . For  $2 \leq j \leq d-1$ , connect  $x_{j-1}$  to  $x_j$  by such boxes while staying at distance at least  $\frac{1}{3}\epsilon_\delta^* s_{C_j}$  from the unused part of  $\partial C_j$ ; use the analogous endpoint paths in  $C_1, C_d$ . Include every little box whose closure contains an interface midpoint, so

consecutive pieces really are neighboring. The number of boxes used inside one  $C_j$  is  $O((\epsilon_\delta^*)^{-2})$ , and (20.19) follows from (20.17).

It remains to justify the conditional-law assertion, which is the point of the construction. On the regular event  $s_{C_j} \geq \delta^{C_{\text{mc}}}$ , and for small  $\delta$ ,

$$s_{B_i} \log s_{B_i}^{-1} = (\epsilon_\delta^*)^2 s_{C_j} \log((\epsilon_\delta^*)^{-2} s_{C_j}^{-1}) < \frac{1}{10} \epsilon_\delta^* s_{C_j}.$$

Because adjacent terminal cells differ by at most the factor  $(\epsilon_\delta^*)^{-1}$ , the exploration which determines  $\mathcal{V}_\delta$  never queries white noise below scale  $s_{B_i}$  inside  $(B_i)_{\text{large}}$ : those space-scale cones lie strictly inside the protected corridor. White noise on disjoint unexplored space-scale regions is orthogonal to the sigma-field generated by the exploration. Consequently, for every bounded measurable functional  $F$  of the selected fine increments,

$$\mathbf{1}_{\mathcal{E}_{\text{reg}}} \mathbb{E}[F \mid \mathcal{V}_\delta] = \mathbf{1}_{\mathcal{E}_{\text{reg}}} \varphi_F((s_{B_i})_{i \leq N}),$$

where  $\varphi_F$  is the unconditional expectation for deterministic side lengths. This is exactly the required conditional independence statement.  $\square$

**Concentration.** We next isolate the finite-variance Gaussian component on which concentration can actually be applied. This avoids the false global-Lipschitz shortcut: the genuine graph distance is Lipschitz under deterministic field shifts but uses arbitrarily fine Gaussian coordinates, whereas the approximate partition uses only controlled-variance coordinates but is not globally Lipschitz when the partition topology changes. The comparison theorem lets us combine the two advantages.

Let  $\mathbf{X}_\delta$  be the Gaussian subspace generated by the white-noise field down to scale  $\delta^{C_{\text{mc}}}$  and let  $\mathbf{Y}_\delta$  be its orthogonal complement. Thus, for every finer scale  $r < \delta^{C_{\text{mc}}}$ , write

$$h_r = \mathcal{X}_\delta + \mathcal{Y}_{\delta,r}, \quad (20.20)$$

where  $\mathcal{X}_\delta$  is  $\mathbf{X}_\delta$ -measurable and all remaining white-noise increments are  $\mathbf{Y}_\delta$ -measurable. Every pointwise coordinate of  $\mathcal{X}_\delta$  has variance  $O_{C_{\text{mc}}}(\log \delta^{-1})$ .

For a realization  $x$  of  $\mathcal{X}_\delta$ , let  $D'_{\delta,x}$  denote the approximate partition distance obtained after adjoining an independent fine field. If all terminal cells have side at least  $\delta^{C_{\text{mc}}}$ , then  $D'_{\delta,x}$  is already determined by  $x$  and does not depend on the fine field.

**Lemma 8.34** (Overlap-to-Lipschitz lemma on coarse realizations). *Fix a sufficiently small  $\iota > 0$ . There is  $\alpha = \alpha(\gamma) > 0$  and a set  $\mathcal{A}_{\delta,\iota}$  of coarse realizations with*

$$\mathbb{P}(\mathcal{X}_\delta \in \mathcal{A}_{\delta,\iota}) \geq 1 - \delta^{c_\iota} \quad (20.21)$$

*such that if  $x, x' \in \mathcal{A}_{\delta,\iota}$  and*

$$\|x - x'\|_\infty \leq \ell_\delta := \frac{\iota}{2\gamma\alpha} \log \delta^{-1}, \quad (20.22)$$

*then*

$$|\log D'_{\delta,x}(A_\delta, B_\delta) - \log D'_{\delta,x'}(A_\delta, B_\delta)| \leq \iota \log \delta^{-1}. \quad (20.23)$$

*This holds uniformly for a compact family of admissible endpoint pairs.*

*Proof.* Let  $\mathcal{E}_{\delta,\iota,\alpha}^*$  be the event that all terminal cells have side at least  $\delta^{C_{\text{mc}}}$  and that, simultaneously for

$$\delta^{1+\iota/\alpha} \leq \delta', \delta'' \leq \delta^{1-\iota/\alpha},$$

the genuine and approximate distances satisfy

$$|\log D_{\delta'} - \log D'_{\delta''}| \leq \frac{\iota}{4} \log \delta^{-1}. \quad (20.24)$$

Proposition 8.29, together with Proposition 8.27 and the cell-size window, implies that for  $\alpha = \alpha(\gamma)$  this event has probability at least  $1 - \delta^{c\iota}$  after decreasing  $c$ .

Condition on  $\mathcal{X}_\delta = x$ . By Fubini/Markov there is a set  $\mathcal{A}_{\delta,\iota}$  satisfying (20.21) on which

$$\mathbb{P}(\mathcal{E}_{\delta,\iota,\alpha}^* \mid \mathcal{X}_\delta = x) \geq 0.9. \quad (20.25)$$

Hence for  $x, x' \in \mathcal{A}_{\delta,\iota}$  the corresponding sets of fine configurations have nonempty intersection.

Choose one common fine realization  $y$ . If  $\|x - x'\|_\infty \leq \ell_\delta$ , multiplication of the LQG density by the deterministic field difference changes every quantum mass by at most  $e^{\gamma \ell_\delta}$ . Therefore monotonicity in the mass threshold gives

$$D_{\delta^{1-\iota/\alpha},x} \leq D_{\delta,x'} \leq D_{\delta^{1+\iota/\alpha},x}, \quad (20.26)$$

after harmlessly enlarging  $\alpha$  to absorb normalization conventions. Apply (20.24) to both  $x$  and  $x'$  on the common fine realization. Since  $D'_{\delta,x}$  and  $D'_{\delta,x'}$  do not depend on  $y$ , the resulting inequality is deterministic in the two coarse realizations and yields (20.23).  $\square$

**Proposition 8.35** (DZZ concentration of Liouville graph distance). *For every compact family of admissible pairs and every  $\iota \in (0, 1)$  there is  $c = c(\gamma) > 0$  such that*

$$\mathbb{P}[|\log D_{\gamma,\delta}(A_\delta, B_\delta) - \mathbb{E} \log D_{\gamma,\delta}(A_\delta, B_\delta)| > \iota \log \delta^{-1}] \leq C\delta^{c\iota^2}. \quad (20.27)$$

Moreover, with probability at least  $1 - e^{-c(\log \delta^{-1})^{0.7}}$ ,

$$|\log D_{\gamma,\delta} - \mathbb{E} \log D_{\gamma,\delta}| \leq C(\log \delta^{-1})^{0.95}. \quad (20.28)$$

The same estimates hold for the white-noise regularized distance and for set-to-boundary distances in a fixed compact family.

*Proof.* We first prove (20.27) for  $D'$ . Let  $d'_{A,B}$  be the least number for which

$$\mathbb{P}[\mathcal{X}_\delta \in \mathcal{A}_{\delta,\iota}, D'_{\delta,\mathcal{X}_\delta}(A_\delta, B_\delta) \leq d'_{A,B}] \geq \frac{1}{2}.$$

Write the set in brackets as  $\mathcal{A}'$ . By Lemma 8.34, if  $\text{dist}_\infty(\mathcal{X}_\delta, \mathcal{A}') < \ell_\delta$  then

$$\log D'_{\gamma,\delta} < \log d'_{A,B} + \iota \log \delta^{-1}.$$

The Gaussian isoperimetric inequality, applied in the Hilbert space of  $\mathbf{X}_\delta$  and using the maximal coordinate variance  $O(\log \delta^{-1})$ , gives

$$\mathbb{P}[\text{dist}_\infty(\mathcal{X}_\delta, \mathcal{A}') \geq \ell_\delta] \leq C\delta^{c\iota^2}.$$

Adding the probability of  $\mathcal{X}_\delta \notin \mathcal{A}_{\delta,\iota}$  yields the upper tail. Apply the same argument to the upper median set  $\{D'_{\delta,x} \geq d''_{A,B}\}$  to obtain the lower tail.

The cell-count bound from the terminal partition gives uniform square integrability of  $\log D'_{\gamma,\delta}/\log \delta^{-1}$ . Integrating the two tail bounds therefore shows

$$|\mathbb{E} \log D'_{\gamma,\delta} - \log d'_{A,B}| = O(\iota \log \delta^{-1}),$$

so, after rescaling  $\iota$ , the same polynomial tail is centered at the expectation. Proposition 8.29 transfers the estimate from  $D'$  to  $D$ .

For (20.28), repeat the construction with the threshold window

$$\delta e^{-\alpha^{-1}(\log \delta^{-1})^{0.9}} \leq \delta', \delta'' \leq \delta e^{\alpha^{-1}(\log \delta^{-1})^{0.9}}$$

and comparison error  $(\log \delta^{-1})^{0.91}$ . Then take  $\ell_\delta = (\log \delta^{-1})^{0.9}$  in Gaussian isoperimetry. The failure probability is  $e^{-c(\log \delta^{-1})^{0.8}}$ , which is stronger than the stated 0.7 exponent after all comparison exceptional events are included. The proof used only the admissibility constants, so the endpoint-set and white-noise variants follow identically.  $\square$

**Near-subadditivity, the exponent, and the point-to-boundary comparison.** The subadditivity argument is intrinsically two-scale. One first builds a protected path at a coarse mass threshold  $\delta^2$  and then, conditional on that path, uses the untouched fine white noise to cross each protected box at the additional threshold  $\tilde{\delta}^2$ . A second percolation argument chooses compatible interface points. This supplies the point-to-boundary step that does not follow from Fekete's lemma alone.

Fix a small central square  $\bar{V} \subseteq V$ . For  $x, y \in \bar{V}$ , let  $\tilde{V}_{x,y}$  be a rectangle of bounded aspect ratio containing  $x, y$  and staying a fixed positive distance from  $\partial V$ ; write  $\tilde{D}_{\gamma,\delta}(x, y)$  for graph distance constrained to this rectangle. All constants below are uniform while  $x, y$  range over a compact family of such pairs.

**Lemma 8.36** (Protected two-scale path). *Let  $u, v \in \bar{V}$  and let  $0 < \delta < \tilde{\delta} < \delta_0$ . With probability at least  $1 - e^{-(\log \delta^{-1})^{c_0}}$  there is a path  $C_1, \dots, C_d$ , measurable with respect to the coarse partition sigma-field, joining  $u$  to  $v$  such that*

$$\log d \leq \mathbb{E} \log \tilde{D}_{\gamma,\delta}(u, v) + (\log \delta^{-1})^{0.95}, \quad (20.29)$$

$$m(C_i) \leq \delta^2 e^{C(\log \delta^{-1})^{0.8}}, \quad (20.30)$$

and the white-noise field below scale  $s_{C_i}$  in the enlarged  $C_i$ 's has, conditional on the coarse sigma-field, its unconditional joint law.

*Proof.* Split the straight segment from  $u$  to  $v$  into nine overlapping reference rectangles, so every local constrained rectangle lies in the original one. In each piece take an approximate geodesic and apply Lemma 8.33. Concatenating the nine protected sequences gives a path of length at most  $e^{(\log \delta^{-1})^{3/5}}$  times the sum of the nine approximate distances. The approximate/genuine comparison (Proposition 8.29), the field-translation coupling from the preceding LFPP section, and the stretched concentration estimate (20.28) bound each of the nine terms by

$$\exp\{\mathbb{E} \log \tilde{D}_{\gamma,\delta}(u, v) + (\log \delta^{-1})^{0.94}\}.$$

Absorbing the finite sum and the regularization factor gives (20.29). The mass estimate (20.30) is the coarse/fine field oscillation bound built into the protected refinement. Its conditional-law assertion is exactly Lemma 8.33.  $\square$

**Lemma 8.37** (Conditional local crossing at the fine threshold). *On the event of Lemma 8.36, condition on the coarse partition. Put*

$$k = \lfloor (\log \delta^{-1})^{0.51} \rfloor, \quad K = 2^k.$$



With conditional probability at least  $1 - e^{-c(\log \delta^{-1})^{0.51}}$ , one can choose interface points  $x_i \in \partial C_i \cap \partial C_{i+1}$ ,  $x_0 = u$ ,  $x_d = v$ , so that simultaneously for every  $i$ ,

$$\log D_{\gamma, \delta \tilde{\delta}}^{\tilde{V}_{u,v}}(x_i, x_{i+1}) \leq \mathbb{E} \log \tilde{D}_{\gamma, \tilde{\delta}}(u, v) + 4(\log \delta^{-1})^{0.98}. \quad (20.31)$$

*Proof.* Partition each  $C_i$  into  $K^2$  dyadic squares. For a small square  $B$ , call it open if the following boundary communication property holds: whenever  $\Lambda \subset \partial B$  has one-dimensional measure at least  $10^{-5}s_B$ , there is  $\Lambda' \subset \partial B$  of measure at least  $(1 - 10^{-5})s_B$  such that for every  $z' \in \Lambda'$ ,

$$\min_{z \in \Lambda} \log D_{\gamma, \delta \tilde{\delta}}(z, z'; \tilde{V}_{u,v}) \leq \mathbb{E} \log \tilde{D}_{\gamma, \tilde{\delta}}(u, v) + 2(\log \delta^{-1})^{0.97}. \quad (20.32)$$

We verify that a fixed small  $B$  is open with conditional probability  $1 - O(K^{-2})$ . By the protected conditional law, the fine white noise in a three-box-thick neighborhood of  $\partial B$  is an unconditional scale-local field. Rescale that neighborhood to the reference rectangle. The scale/translation coupling for the white-noise GFF gives, outside an event of conditional probability  $O(K^{-5})$ ,

$$\sup_{r \leq a^{-1}} \sup_z |\check{\eta}_r^B(z) - \eta_{ar}(\theta z)| \leq (\log \delta^{-1})^{0.92}, \quad (20.33)$$

where  $a$  is the inverse spatial rescaling and  $\theta$  the affine map to the reference rectangle. The corresponding variance difference is  $O((\log \delta^{-1})^{0.6})$ . Hence the GMC comparison shifts the fine mass threshold only by  $e^{O((\log \delta^{-1})^{0.95})}$ , and Proposition 8.35 plus Proposition 8.27 implies, for every fixed pair of boundary points,

$$\mathbb{P}[\log D_{\gamma, \delta \tilde{\delta}}(z, z') > \mathbb{E} \log \tilde{D}_{\gamma, \tilde{\delta}}(u, v) + (\log \delta^{-1})^{0.97} \mid \mathcal{F}^*] \leq CK^{-4}. \quad (20.34)$$

Integrating first in  $z'$  and then in  $z$ , and applying Markov twice, shows that the set of starting points for which more than a  $K^{-1}$  fraction of the opposite boundary is bad has relative boundary measure at most  $K^{-1}$ , except with probability  $O(K^{-2})$ . This proves (20.32).

Now say a coarse cell  $C_i$  is desirable if every subset of its exit interface of relative length at least 0.1 can be connected, through open little boxes, to a subset of its entry interface of relative length at least 0.1, with the cost in (20.32). The little-box events have finite-range dependence. By the circuit/crossing estimate Lemma 8.23, a coarse cell is not desirable with probability at most  $e^{-cK}$  after a fixed coloring of the little-box grid. The number  $d$  of coarse cells is at most  $\exp(C \log \delta^{-1})$  on the cell-window event, so a union bound gives that every interior  $C_i$  is desirable with probability  $1 - e^{-c(\log \delta^{-1})^{0.51}}$ . The endpoint cells satisfy the analogous one-sided property by the same, simpler, rectangle crossing estimate.

Starting from a 0.99-fraction of the final interface, work backwards through the desirable cells. At each step retain a 0.1-fraction of the previous interface which connects to the retained next interface. Choosing  $x_i$  successively in these nonempty sets and concatenating the local open-box paths yields (20.31). This is the conditional resampling/concatenation step required for subadditivity.  $\square$

**Lemma 8.38** (Two-scale near-subadditivity). *For all sufficiently small dyadic  $0 < \delta < \tilde{\delta} < \delta_0$ ,*

$$\mathbb{E} \log \tilde{D}_{\gamma, \delta \tilde{\delta}}(u, v) \leq \mathbb{E} \log \tilde{D}_{\gamma, \delta}(u, v) + \mathbb{E} \log \tilde{D}_{\gamma, \tilde{\delta}}(u, v) + C(\log \delta^{-1})^{0.98}. \quad (20.35)$$

*The same estimate holds for the approximate constrained distance, with a possibly different constant in the sublinear term.*

*Proof.* On the intersection of the events in Lemmas 8.36 and 8.37, the triangle inequality gives

$$\tilde{D}_{\gamma, \delta \tilde{\delta}}(u, v) \leq \sum_{i=0}^{d-1} D_{\gamma, \delta \tilde{\delta}}^{\tilde{V}_{u,v}}(x_i, x_{i+1}).$$

Taking logarithms and using (20.29) and (20.31) gives the right side of (20.35) plus  $C(\log \delta^{-1})^{0.98}$ . The complement has probability  $e^{-c(\log \delta^{-1})^{c_0}}$ . The terminal-cell window implies  $\log \tilde{D} = O(\log \delta^{-1} + \log \tilde{\delta}^{-1})$  in  $L^2$ ; Cauchy–Schwarz therefore makes the exceptional contribution  $o((\log \delta^{-1})^{0.98})$ . Taking expectations proves the claim. The approximate version follows from Proposition 8.29.  $\square$

**Lemma 8.39** (Existence and pair-independence of the DZZ exponent). *There is a constant  $\chi = \chi(\gamma)$  such that for every fixed  $u \neq v \in \bar{V}$ ,*

$$\lim_{\delta \downarrow 0} \frac{\mathbb{E} \log \tilde{D}_{\gamma, \delta}(u, v)}{\log \delta^{-1}} = \lim_{\delta \downarrow 0} \frac{\mathbb{E} \log \tilde{D}'_{\gamma, \delta}(u, v)}{\log \delta^{-1}} = \chi. \quad (20.36)$$

*Proof.* Put  $b(t) = \mathbb{E} \log \tilde{D}_{\gamma, e^{-t}}(u, v)$ . For dyadic scales, (20.35) implies

$$b(s+t) \leq b(s) + b(t) + Cs^{0.98} \quad (s \geq t \geq t_0), \quad (20.37)$$

and  $|b(t)| \leq Ct$  by the cell-window estimate. A standard perturbed-Fekete argument now gives existence of  $\lim b(t)/t$ : choose a fixed large block size  $T$ , tile a much larger scale by blocks no larger than  $T$ , iterate (20.37), divide by the total scale, and then let  $T \rightarrow \infty$ ; the accumulated defect is  $O(T^{-0.02})$ . The threshold comparison fills the gaps between dyadic scales.

The same limit holds for  $\tilde{D}'$  by Proposition 8.29. To see that it does not depend on the pair, couple any two bounded-aspect reference rectangles by translation, rotation, and scaling. The corresponding scale-local fields differ by a Gaussian field whose supremum is  $O((\log \delta^{-1})^{1/2+o(1)})$  with stretched-exponentially high probability. The perturbation/threshold comparison therefore changes expected log distance only by  $o(\log \delta^{-1})$ . A bounded chain of overlapping rectangles compares arbitrary fixed interior pairs. Hence the limiting constant depends only on  $\gamma$ .  $\square$

**Lemma 8.40** (DZZ point-to-boundary exponent). *Fix  $x \in \bar{V}$  and a square  $U_x$  of fixed side length centered at  $x$  with  $2U_x \Subset V$ . Then*

$$\lim_{\delta \downarrow 0} \frac{\mathbb{E} \log D_{\gamma, \delta}(x, \partial U_x; 2U_x)}{\log \delta^{-1}} = \chi. \quad (20.38)$$

*The same exponent governs the distance between two fixed concentric square boundaries separated by positive Euclidean distance.*

*Proof.* The upper bound follows by connecting  $x$  to any fixed boundary point and using Lemma 8.39. For the lower bound fix a small  $\iota > 0$  and partition each side of  $\partial U_x$  into segments  $L_\delta$  of length in  $[\delta^{2\iota}/2, \delta^{2\iota}]$ . We claim uniformly that

$$\mathbb{E} \log D_{\gamma, \delta}(x, L_\delta; 2U_x) \geq (\chi - 2\iota) \log \delta^{-1} \quad (20.39)$$

for small  $\delta$ .

Suppose not. By symmetry take a deficient segment on the right side. Choose a second fixed point  $y$  on the opposite side of the segment configuration. The union of: a geodesic from  $x$  to  $L_\delta$ , a geodesic from  $y$  to  $L_\delta$ , and four crossings of rectangles of dimensions comparable to  $10|L_\delta| \times 40|L_\delta|$

forms a path from  $x$  to  $y$  around a contour enclosing  $L_\delta$ . Rescaling those four short rectangles and using Lemma 8.39 plus Proposition 8.35 shows that each short crossing is at most  $\delta^{-\chi+\iota}$  with probability tending to one, after shrinking  $\iota$  in the scaled threshold. The assumed deficit in (20.39) and the symmetric estimate for  $y$  would then give

$$D_{\gamma,\delta}(x, y) \leq \delta^{-\chi+\iota}$$

with probability tending to one, contradicting the concentration theorem centered at the exponent  $\chi$ . Thus (20.39) holds.

There are only  $O(\delta^{-2\iota})$  boundary segments. The polynomial tail in Proposition 8.35, with the tolerance chosen smaller before the union bound, implies that the minimum over all segments has expected logarithm at least  $(\chi - C\sqrt{\iota}) \log \delta^{-1}$ . Letting  $\iota \downarrow 0$  gives (20.38). Applying the same four-crossing argument with the inner boundary in place of  $x$  gives the concentric-boundary version.  $\square$

**Proposition 8.41** (Two-scale DZZ exponent mechanism). *The two-scale protected resampling argument yields the exponent  $\chi(\gamma)$ , and the point-to-boundary/four-crossing argument propagates that exponent to the boundary-distance variants used below.*

*Proof.* Lemmas 8.36–8.40 give precisely the two independent parts of the claim. The comparison, deformation, and concentration inputs used there are the preceding propositions.  $\square$

**Theorem 8.42** (Ding–Zeitouni–Zhang graph-distance exponent). *For every  $\gamma \in (0, 2)$  there exists  $\chi = \chi(\gamma) \in (0, 1)$  such that, for every fixed compact family of interior admissible pairs and every  $\iota > 0$ ,*

$$\delta^{-\chi+\iota} \leq D_{\gamma,\delta}(A_\delta, B_\delta; U) \leq \delta^{-\chi-\iota} \quad (20.21)$$

*with probability at least  $1 - C\delta^{c\iota^2}$ . For  $K \Subset U$  one also has, with the same logarithmic accuracy and polynomially high probability,*

$$\max_{z,w \in K} D_{\gamma,\delta}(z, w; U) \leq \delta^{-\chi-\iota}, \quad D_{\gamma,\delta}(K, \partial U; U) \geq \delta^{-\chi+\iota}. \quad (20.22)$$

*If*

$$d_\gamma := \frac{2}{\chi},$$

*then  $d_\gamma > 2$  and, after writing the quantum-mass threshold as  $\varepsilon = \delta^2$ , (20.22) is equivalent to the  $d_\gamma$  graph-distance package of Definition 8.9.*

*Proof.* The expectation exponent is supplied by Lemma 8.39; Proposition 8.35 upgrades it to the two-sided probabilistic statement for every admissible pair. Lemma 8.40 supplies the boundary lower exponent. For the compact uniform upper bound, cover  $K$  by a  $\delta^{c_0}$ -mesh. The number of mesh points is polynomial in  $\delta^{-1}$ , so choose the concentration exponent with a smaller tolerance and take a union bound. The cell-size window joins an arbitrary point to the nearest mesh point at cost  $\delta^{-o(1)}$ , which is absorbed by  $\iota$ . The same finite-cover argument handles a compact family of translated and rescaled squares.

It remains only to record that  $0 < \chi < 1$ . Positivity follows from the lower cell-size bound: any macroscopic path crosses at least  $\delta^{-c}$  terminal cells with polynomially high probability. For the upper bound, apply the KPZ relation proved in Volume XVIII to a deterministic line segment. The Euclidean dimension-one set can be covered, with polynomially high probability, by fewer

than  $\delta^{-1+\kappa}$  quantum balls of mass  $\delta^2$  for some  $\kappa = \kappa(\gamma) > 0$ ; hence  $\chi \leq 1 - \kappa < 1$ . Therefore  $d_\gamma = 2/\chi > 2$ .

Finally set  $\varepsilon = \delta^2$ . Then  $\delta^{-\chi \pm \iota} = \varepsilon^{-\chi/2 \pm \iota/2} = \varepsilon^{-1/d_\gamma \pm \iota/2}$ . Since  $\iota$  is arbitrary, this is exactly the exponent window in Definition 8.9.  $\square$

**Proposition 8.43** (DZZ-to-LFPP reduction). *Theorem 8.42 verifies the graph-distance package in Definition 8.9. Together with the lower and upper Ding–Gwynne conversion estimates in Proposition 8.20, it yields, for every fixed  $p \in (0, 1)$  bounded away from the endpoints,*

$$\ell_n(p), \bar{\ell}_n(p) = 2^{-n(1-\xi Q + o(1))}, \quad \xi = \gamma/d_\gamma, \quad Q = 2/\gamma + \gamma/2.$$

Equivalently, for every  $\eta > 0$  and all sufficiently large  $n$ ,

$$2^{-n(1-\xi Q + \eta)} \leq \ell_n(p) \leq \bar{\ell}_n(p) \leq 2^{-n(1-\xi Q - \eta)}.$$

*Proof.* The first sentence is Theorem 8.42. Proposition 8.20 gives the circle-average LFPP exponent  $1 - \xi Q$  for internal distances. The rectangle extraction and RSW argument in Theorem 8.21 convert this to fixed crossing quantiles, and Lemma 8.10 transfers it to the heat-kernel regularization. No implication from the DZZ theorem alone to the LFPP exponent is asserted.  $\square$

#### 8.1.4 Condition-(T), Efron–Stein contraction, and characteristic scales

The RSW theorem above compares easy and hard crossing quantiles. It does not by itself prevent the length of a macroscopic geodesic from being carried by a few exceptionally expensive coarse boxes. The multiscale concentration argument needs an anti-concentration statement for those coarse contributions. Ding–Dubédat–Dunlap–Falconet isolate exactly this statement as Condition-(T). Once it holds, a block-resampling Efron–Stein argument gives a strict contraction from the quantile ratio at scale  $n - K$  to the variance at scale  $n$ .

The preceding subsection proves the Ding–Zeitoun–Zhang Liouville graph-distance exponent package, the regularization and dyadic/continuum comparisons, and both Ding–Gwynne LFPP conversion branches at the stated strength. Consequently the exponent input used below is Theorem 8.21, rather than an additional assumption. That theorem gives, for every fixed  $p \in (0, 1/10)$  and  $\eta > 0$  and all sufficiently large  $n$ ,

$$2^{-n(1-\xi Q + \eta)} \leq \ell_n(p) \leq \bar{\ell}_n(p) \leq 2^{-n(1-\xi Q - \eta)}, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}, \quad \xi = \frac{\gamma}{d_\gamma}.$$

In the remainder of this subsection, “the exponent input” means precisely this conclusion. No additional source theorem is being assumed.

**Elementary scale estimates.** Let  $\mathcal{P}_K$  be the dyadic squares of side  $2^{-K}$  which meet  $[0, 1]^2$ , and let  $\phi_{0,K}(P)$  denote the value at the center of  $P$ . The next estimates will be used repeatedly.

**Lemma 8.44** (Coarse maximum and oscillation moments). *For every  $a > 0$  there is  $C_a < \infty$  such that, for all  $K \geq 1$ ,*

$$\mathbb{E} \exp \left( a \max_{P \in \mathcal{P}_K} |\phi_{0,K}(P)| \right) \leq \exp(2aK \log 2 + C_a \sqrt{K}),$$

and if

$$\text{osc}_P(\phi_{0,K}) := \text{diam}(P) \|\nabla \phi_{0,K}\|_{L^\infty(P)},$$

then for every  $\varepsilon_0 \in (0, 1/2)$ ,

$$\mathbb{E} \exp \left( a K^{\varepsilon_0} \max_{P \in \mathcal{P}_K} \text{osc}_P(\phi_{0,K}) \right) \leq \exp \left( C_a K^{1/2+\varepsilon_0} + C_a K^{2\varepsilon_0} \right).$$

The same estimates hold with  $\phi$  replaced by  $\psi$  after changing the constants.

*Proof.* At one point  $x$ ,  $\text{Var } \phi_{0,K}(x) = (K + O(1)) \log 2$ . Cover the unit square by  $O(4^K)$  grid points at mesh  $2^{-K}$  and use the Gaussian maximal inequality; this gives

$$\mathbb{E} \max_{P \in \mathcal{P}_K} |\phi_{0,K}(P)| \leq 2K \log 2 + C\sqrt{K}.$$

The maximum is  $\sqrt{(K + O(1)) \log 2}$ -Lipschitz in the Cameron–Martin norm, so the Gaussian concentration theorem of Volume VI upgrades the expectation bound to the displayed exponential moment estimate.

For the gradient, differentiating the heat kernel under the stochastic integral gives

$$\sup_x \text{Var}(2^{-K} \nabla \phi_{0,K}(x)) \leq C.$$

A mesh- $2^{-K}$  chaining argument identical to the one in Lemma 8.3, applied now to the two gradient coordinates, yields a subgaussian tail for  $2^{-K} \|\nabla \phi_{0,K}\|_{[0,1]^2}$  after the  $O(4^K)$  grid union bound. Integrating that tail against  $\exp(aK^{\varepsilon_0} \cdot)$  gives the second estimate. For  $\psi$  one differentiates its truncated heat-kernel stochastic integral directly. The cutoff only removes kernel mass, while the derivative of the smooth cutoff contributes on an annulus whose radius is comparable to the cutoff radius; the corresponding quadratic-variation integral is uniformly bounded after multiplication by  $2^{-K}$ . The same mesh/chaining argument therefore gives the stated gradient-oscillation moment estimate for  $\psi$ . The maximum estimate for  $\psi$  follows from the identical one-point variance bound and the same Gaussian maximal/concentration argument.  $\square$

**Lemma 8.45** (Quantile span controlled by variance). *Let  $Z \in L^2$  and  $p \in (0, 1/2)$ . If  $l \leq \bar{l}$  satisfy  $\mathbb{P}(Z \leq l) \geq p$  and  $\mathbb{P}(Z \geq \bar{l}) \geq p$ , then*

$$(\bar{l} - l)^2 \leq \frac{2}{p^2} \text{Var } Z.$$

Consequently, for a positive random variable  $Y$ ,

$$\frac{q_{1-p}(Y)}{q_p(Y)} \leq \exp \left( \frac{\sqrt{2 \text{Var}(\log Y)}}{p} \right).$$

*Proof.* Let  $Z'$  be an independent copy. On  $\{Z \leq l\} \cap \{Z' \geq \bar{l}\}$  one has  $(Z' - Z)^2 \geq (\bar{l} - l)^2$ . Hence

$$2 \text{Var } Z = \mathbb{E}(Z - Z')^2 \geq p^2 (\bar{l} - l)^2.$$

Apply this to  $Z = \log Y$  and exponentiate.  $\square$

**Lemma 8.46** (A priori logarithmic variance and quantile-ratio bound). *For all  $n \geq 0$ ,*

$$\text{Var} \log L_{1,1}^{(n)}(\phi) \leq \xi^2 (n+1) \log 2.$$

For every fixed  $p \in (0, 1/2)$  there is  $C_p$  such that

$$\Lambda_n(\psi, p) \leq e^{C_p \sqrt{n}}, \quad n \geq 1.$$

*Proof.* Approximate  $\phi_{0,n}$  on a dyadic mesh of size  $2^{-k}$  by the piecewise constant vector of its values. The logarithm of the resulting crossing length is  $\xi$ -Lipschitz with respect to the supremum norm of that Gaussian vector: changing all field values by at most  $u$  changes every path length by a factor in  $[e^{-\xi u}, e^{\xi u}]$ . Writing the Gaussian vector as  $AG$  with  $G$  standard, the Euclidean Lipschitz constant is at most  $\xi \max_i \sqrt{\text{Var } \phi_{0,n}(x_i)} \leq \xi \sqrt{(n+1) \log 2}$ . Gaussian concentration therefore bounds the variance by the square of this constant. Letting the mesh tend to zero proves the first assertion. Lemma 8.45 then gives  $\Lambda_n(\phi, p) \leq e^{C_p \sqrt{n}}$ . The full/truncated comparison Lemma 8.3 shifts each fixed quantile by only a constant multiplicative factor, so the same bound holds for  $\psi$ .  $\square$

**Tail estimates relative to fixed quantiles.** The concentration recursion uses moments of ratios of rectangle crossings. Those moments are obtained from the following two tail estimates. We include the finite-range percolation input rather than hiding it behind the word “standard.”

**Lemma 8.47** (Finite-range high-density crossing). *For every dependence range  $R < \infty$  there is  $q_0 = q_0(R) < 1$  and  $c > 0$  with the following property. Let  $(\omega_v)_{v \in \mathbb{Z}^2}$  be an  $R$ -dependent site process with  $\mathbb{P}(\omega_v = 1) \geq q_0$  for every  $v$ . Then the probability that a  $3m \times m$  rectangle has no nearest-neighbour open left–right crossing is at most  $Ce^{-cm}$ .*

*Proof.* If there is no open left–right crossing, planar duality produces a  $*$ -connected top–bottom chain of closed sites. Color  $\mathbb{Z}^2$  by  $(2R+1)^2$  residue classes modulo  $2R+1$ . Along any self-avoiding dual chain of length  $r$ , one color occurs at least  $r/(2R+1)^2$  times, and sites of this color on the chain are mutually more than  $R$  apart, hence independent. The number of self-avoiding  $*$ -paths of length  $r$  from a given site is at most  $8^r$ . Therefore the probability of a closed dual chain of length at least  $m$  is bounded by

$$Cm \sum_{r \geq m} 8^r (1 - q_0)^{r/(2R+1)^2}.$$

Choose  $q_0$  so close to 1 that the base of the geometric series is less than one. This gives the claimed exponential bound.  $\square$

**Proposition 8.48** (Lower tails). *There exists a fixed sufficiently small  $p_0 > 0$  and constants  $c, C > 0$  such that, uniformly in  $n$  and  $s \geq 0$ ,*

$$\mathbb{P} \left[ L_{1,3}^{(n)}(\psi) \leq e^{-s} \ell_{1,3}^{(n)}(\psi, p_0) \right] \leq Ce^{-cs^2}.$$

*The same bound holds for  $\phi$  after changing  $c, C$ .*

*Proof.* The RSW theorem gives constants  $C_0, C_1$  such that if  $\mathbb{P}[L_{3,3}^{(n)}(\psi) \leq r] \leq q$ , then

$$\mathbb{P} \left[ L_{1,3}^{(n)}(\psi) \leq C_0^{-1} e^{-C_1 \sqrt{\log(C_0/q)}} r \right] \leq C_0 q.$$

Indeed, first transfer the RSW inequality from  $\phi$  to  $\psi$  by Lemma 8.3, with the same subgaussian correction as in the proof of Theorem 8.8. If the  $3 \times 3$  rectangle has a crossing of length at most  $r$ , both the left and right  $1 \times 3$  subrectangles have such crossings. Choosing the truncation range  $r_0$  small makes the restrictions of  $\psi$  to these separated subrectangles independent; hence

$$\mathbb{P}[L_{3,3}^{(n)}(\psi) \leq r] \leq \mathbb{P}[L_{1,3}^{(n)}(\psi) \leq r]^2.$$

Starting from a sufficiently small fixed  $p_0$  and iterating  $q_{j+1} = (C_0 q_j)^2$ , the probability decays double exponentially in  $j$ , whereas the accumulated logarithmic loss is  $O(2^{j/2})$ . Taking  $j \asymp \log_2 s$  gives the Gaussian lower tail. The full/truncated field comparison transfers the result to  $\phi$ .  $\square$

**Proposition 8.49** (Upper tails). *For the same fixed small  $p_0$  there are  $c, C > 0$  such that, uniformly in  $n$  and  $s > 2$ ,*

$$\mathbb{P} \left[ L_{3,1}^{(n)}(\phi) \geq e^s \Lambda_n(\phi, p_0) \ell_n(p_0) \right] \leq C \exp \left( -c \frac{s^2}{\log s} \right).$$

*The same estimate holds for  $\psi$  after changing the constants.*

*Proof.* We separate the percolation and scale steps. At a fixed scale  $k$ , call a unit lattice square open when the four long  $3 \times 1$  rectangle crossings surrounding it have total  $\psi$ -length at most  $4\bar{\ell}_{3,1}^{(n)}(\psi, p_0)$ . RSW makes the open probability tend to one as  $p_0 \downarrow 0$ . The finite-range property of  $\psi$  gives a fixed dependence range. Lemma 8.47 therefore implies that with probability  $1 - Ce^{-ck}$  one can concatenate open annular circuits to cross a  $3k \times k$  rectangle at total length at most

$$Ck^2 \bar{\ell}_n(\psi, p_0).$$

The comparison lemma transfers this to  $\phi$  at a cost  $e^{\xi X}$ , where  $X$  has a scale-independent Gaussian tail.

Next decompose  $\phi_{0,n} = \phi_{0,m} + \phi_{m,n}$ . The fine field rescales exactly: a  $3 \cdot 2^m \times 2^m$  rectangle for  $\phi_{m,n}$  has the law of  $2^{-m}$  times a  $3 \times 1$  rectangle at scale  $n - m$ . On the event  $\max \phi_{0,m} \leq Cm + u\sqrt{m}$ , the coarse field contributes at most  $e^{\xi(Cm + u\sqrt{m})}$ . Combining this with the preceding high-density percolation bound gives, for  $u \geq 1$  and  $2^m$  sufficiently large,

$$\mathbb{P} \left[ L_{3,1}^{(n)}(\phi) \geq e^{C\sqrt{m}u} C_{p_0} 2^{2m} \bar{\ell}_{n-m}(p_0) \right] \leq Ce^{-cu^2} + Ce^{-c2^m}.$$

A coarse lower bound for a unit-square crossing is obtained by noting that a left-right path must visit at least  $2^m$  dyadic  $m$ -boxes and, at each visited box, must make one of finitely many short rectangle crossings. The lower-tail proposition and the coarse-field infimum bound imply

$$\ell_n(p_0) \geq 2^{-2\xi m} \ell_{n-m}(p_0) e^{-C\sqrt{m}}.$$

Thus  $\bar{\ell}_{n-m}(p_0) \leq \Lambda_{n-m}(\phi, p_0) \ell_{n-m}(p_0)$  converts the previous estimate into

$$\mathbb{P} \left[ L_{3,1}^{(n)}(\phi) \geq e^{C\sqrt{m}u} e^{Cm} \Lambda_n(\phi, p_0) \ell_n(p_0) \right] \leq Ce^{-cu^2} + Ce^{-c2^m}.$$

Choose  $2^m \asymp s^2$  and  $u \asymp s/\sqrt{m}$ . This gives the claimed  $\exp(-cs^2/\log s)$  bound until  $s$  reaches the order of  $2^{n/2}$ . For larger  $s$ , compare the crossing with the straight horizontal segment and use Gaussian moments of its exponential weight; the pointwise variance is  $O(n)$  and yields a Gaussian tail stronger than the displayed estimate. Finally apply Lemma 8.3 to transfer the result to  $\psi$ .  $\square$

**Condition-(T).** Let  $\pi_n(\psi)$  be a measurably chosen left-right geodesic of the unit square and let

$$\pi_n^K(\psi) = \{P \in \mathcal{P}_K : P \cap \pi_n(\psi) \neq \emptyset\}.$$

**Definition 8.50** (Condition-(T)). We say that  $\xi$  satisfies Condition-(T) if there exist  $\alpha > 1$ ,  $c > 0$  and  $K_0$  such that for  $K \geq K_0$ ,

$$\sup_{n \geq K} \left\{ \mathbb{E} \left[ \left( \frac{\sum_{P \in \pi_n^K(\psi)} e^{2\xi\psi_{0,K}(P)}}{(\sum_{P \in \pi_n^K(\psi)} e^{\xi\psi_{0,K}(P)})^2} \right)^\alpha \right] \right\}^{1/\alpha} \leq e^{-cK}.$$

**Proposition 8.51** (Subcritical exponent input implies Condition-(T)). *For every  $\gamma \in (0, 2)$  and  $\xi = \gamma/d_\gamma$ , Condition-(T) holds.*

*Proof.* Write

$$R_{n,K} := \frac{\sum_{P \in \pi_n^K(\psi)} e^{2\xi\psi_{0,K}(P)}}{(\sum_{P \in \pi_n^K(\psi)} e^{\xi\psi_{0,K}(P)})^2}.$$

The numerator is bounded by the largest coarse weight times the denominator:

$$R_{n,K} \leq \frac{e^{\xi \max_{P \in \mathcal{P}_K} \psi_{0,K}(P)}}{\sum_{P \in \pi_n^K(\psi)} e^{\xi\psi_{0,K}(P)}}.$$

By Lemma 8.3, replacing  $\psi$  by  $\phi$  costs a factor  $e^{2\xi X}$  with  $X$  having a scale-independent Gaussian tail.

For the denominator, concatenate a straight segment across every coarse block visited by  $\pi_n(\psi)$ . This gives a left-right path and hence

$$\sum_{P \in \pi_n^K(\psi)} e^{\xi\phi_{0,K}(P)} \geq e^{-\xi X} e^{-\xi \max_{P \in \mathcal{P}_K} \text{osc}_P(\phi_{0,K})} 2^K L_{1,1}^{(K)}(\phi).$$

Consequently

$$R_{n,K} \leq 2^{-K} \frac{e^{\xi \max_P \phi_{0,K}(P)}}{L_{1,1}^{(K)}(\phi)} e^{2\xi X} e^{\xi \max_P \text{osc}_P(\phi_{0,K})}.$$

Take  $\alpha > 1$  sufficiently close to one and apply Hölder together with Lemmas 8.44 and 8.48. Negative moments of  $L_{1,1}^{(K)}(\phi)$  are bounded by a constant multiple of  $\ell_K(p_0)^{-1}$  because the lower tail is Gaussian in logarithmic scale. We obtain

$$(\mathbb{E} R_{n,K}^\alpha)^{1/\alpha} \leq 2^{-K} 2^{2\xi K} \ell_K(p_0)^{-1} e^{C\sqrt{K}},$$

uniformly in  $n \geq K$ . By the lower exponent input, for every small  $\eta > 0$  and large  $K$ ,

$$\ell_K(p_0)^{-1} \leq 2^{K(1-\xi Q+\eta)}.$$

Thus

$$(\mathbb{E} R_{n,K}^\alpha)^{1/\alpha} \leq 2^{-K(\xi(Q-2)-\eta)} e^{C\sqrt{K}}.$$

Since  $Q > 2$  for  $\gamma \in (0, 2)$ , choose  $0 < \eta < \xi(Q-2)/2$ . The right side is  $\leq e^{-cK}$  for large  $K$ , which is Condition-(T).  $\square$

**The Efron–Stein contraction.** We now prove the DDDF self-bounding estimate. At scales below  $K$ , partition the white noise into spatial blocks so that

$$\psi_{0,n} = \psi_{0,K} + \sum_{P \in \mathcal{P}_K} \psi_{K,n,P},$$

where the families  $\psi_{K,n,P}$  are independent and each is supported in a  $CK^{\varepsilon_0}2^{-K}$ -neighbourhood of  $P$ . Such a partition is obtained by assigning the space variable of the truncated white-noise integral to the nearest dyadic  $K$ -box; the finite cutoff radius ensures the stated support. Let  $L_n = L_{1,1}^{(n)}(\psi)$ . Write  $L_n^K$  for the crossing length after resampling the coarse field  $\psi_{0,K}$  and  $L_n^P$  after resampling the one fine block  $\psi_{K,n,P}$ .



**Lemma 8.52** (Fine-block influence bound). *Assume Condition-(T). There are  $c, C > 0$  such that, for  $K$  large and  $n \geq K$ ,*

$$\sum_{P \in \mathcal{P}_K} \mathbb{E}[(\log L_n^P - \log L_n)_+]^2 \leq e^{-cK} e^{CK^{1/2+\varepsilon_0}} \Lambda_{n-K}(\phi, p_0)^2.$$

*Proof.* If resampling the fine field around  $P$  increases the crossing length, then  $P$  is visited by the original geodesic. Let  $P^K$  be the  $CK^{\varepsilon_0}2^{-K}$ -neighbourhood of  $P$ . Outside  $P^K$  the original and resampled fields agree. Replace the portion of the geodesic in  $P^K$  by four long crossings of rectangles which form a circuit around  $P^K$ . Since  $\log x \leq x - 1$ ,

$$(\log L_n^P - \log L_n)_+ \leq C \frac{e^{\xi\psi_{0,K}(P)}}{L_n} e^{C\xi \max \text{osc}(\phi_{0,K})} e^{C\xi X} \max_{Q,i} L^{(K,n)}(R_i^L(Q), \phi),$$

where  $Q$  ranges over  $K$ -boxes in  $P^K$  and  $R_i^L(Q)$  are four long  $3 \times 1$  rectangles at scale  $K$ .

On the other hand, each visited coarse box forces the geodesic to make one of four disjoint short crossings around that box. Summing their contributions over the coarse geodesic gives

$$L_n \geq ce^{-C\xi X} e^{-C\xi \max \text{osc}(\phi_{0,K})} \left( \sum_{P \in \pi_n^K(\psi)} e^{\xi\psi_{0,K}(P)} \right) \min_{Q,i} L^{(K,n)}(R_i^S(Q), \phi).$$

Squaring the ratio and summing over visited  $P$  produces exactly the Condition-(T) weight

$$\frac{\sum_{P \in \pi_n^K} e^{2\xi\psi_{0,K}(P)}}{(\sum_{P \in \pi_n^K} e^{\xi\psi_{0,K}(P)})^2}.$$

Apply Hölder. Condition-(T) contributes  $e^{-cK}$ . The oscillation and full/truncated comparison factors contribute at most  $e^{CK^{1/2+\varepsilon_0}}$  by Lemma 8.44. By scaling, the remaining ratio of long and short fine-field rectangle crossings has the law of a fixed finite maximum/minimum of crossings at scale  $n - K$ . The lower- and upper-tail Propositions 8.48 and 8.49 give finite moments of that ratio bounded by  $C\Lambda_{n-K}(\phi, p_0)^2$ . Combining the factors proves the estimate.  $\square$

**Proposition 8.53** (Variance contraction). *Assume Condition-(T). There are  $c, C > 0$  and  $K_0$  such that for  $K \geq K_0$  and  $n \geq K$ ,*

$$\text{Var} \log L_{1,1}^{(n)}(\psi) \leq CK + e^{-cK} \Lambda_{n-K}(\psi, p_0/2)^2.$$

*Consequently*

$$\sup_{n \geq 0} \Lambda_n(\psi, p_0/2) < \infty, \quad \sup_{n \geq 0} \text{Var} \log L_{1,1}^{(n)}(\psi) < \infty.$$

*The same conclusions hold for  $\phi$ .*

*Proof.* Apply the Efron–Stein inequality from Volume I to the independent coarse white-noise block and all fine spatial blocks. Resampling the coarse field changes  $\log L_n$  by a Gaussian Lipschitz functional whose conditional variance is at most  $CK$ . Thus its Efron–Stein contribution is  $CK$ . Lemma 8.52 bounds the sum of the fine-block contributions. Enlarge  $K$  if needed so that the harmless factor  $e^{CK^{1/2+\varepsilon_0}}$  is absorbed into  $e^{-cK}$ ; the comparison between  $\phi$  and  $\psi$  changes  $p_0$  only by a fixed factor. This gives the variance recurrence.

By Lemma 8.45,

$$\frac{\bar{\ell}_n(\psi, p_0/2)}{\ell_n(\psi, p_0/2)} \leq \exp\left(C_{p_0} \sqrt{CK + e^{-cK} \Lambda_{n-K}(\psi, p_0/2)^2}\right).$$

The a priori estimate of Lemma 8.46 gives  $\Lambda_K \leq e^{C\sqrt{K}}$ . Fix  $K$  so large that

$$e^{-cK} (e^{C\sqrt{K}} + e^{C\sqrt{CK}})^2 \leq CK.$$

Set

$$\Lambda_* := \Lambda_K(\psi, p_0/2) \vee e^{C\sqrt{2CK}}.$$

Inductively assume  $\Lambda_{n-1} \leq \Lambda_*$ . Then  $\Lambda_{n-K} \leq \Lambda_*$  and the preceding two displays imply that the new quantile ratio at scale  $n$  is at most  $e^{C\sqrt{2CK}} \leq \Lambda_*$ . Thus  $\Lambda_n \leq \Lambda_*$  for every  $n$ . Substituting this bound back into the variance recurrence gives a uniform variance bound. The full/truncated comparison lemma transfers both conclusions to  $\phi$ .  $\square$

**Corollary 8.54** (Tight logarithmic crossing fluctuations). *By Theorem 8.21, the exponent input holds. If  $\lambda_n$  is any fixed central quantile of  $L_{1,1}^{(n)}(\phi)$  (in particular its median), then*

$$(\log L_{1,1}^{(n)}(\phi) - \log \lambda_n)_{n \geq 0}$$

*is tight. Moreover the lower and upper logarithmic tails are uniform in  $n$  at the strengths of Propositions 8.48 and 8.49.*

*Proof.* The exponent input implies Condition-(T) by Proposition 8.51; the variance contraction then gives a uniform bound on the quantile ratios. Substitute this bound into the lower- and upper-tail estimates. Since all central quantiles are uniformly comparable once the ratio is bounded, centering by the median or by any other fixed central quantile changes the logarithm by a bounded deterministic amount.  $\square$

**Weak multiplicativity of the characteristic length.** For the metric chaining in the next child we also need the normalization at different dyadic scales to be compatible. This is a consequence of the same uniform crossing tails.

**Lemma 8.55** (Almost multiplicative sequences). *Let  $(a_n)_{n \geq 1}$  be positive and suppose*

$$e^{-C\sqrt{k}} a_n a_k \leq a_{n+k} \leq e^{C\sqrt{k}} a_n a_k \quad (n, k \geq 1).$$

*Then there is  $\rho > 0$  such that*

$$a_n = \rho^n e^{O(\sqrt{n})}.$$

*Proof.* For dyadic indices write  $a_{2^j+1} = a_{2^j}^2 e^{b_j}$ , where  $|b_j| \leq C2^{j/2}$ . Iteration gives

$$2^{-j} \log a_{2^j} = \log a_1 + \sum_{i < j} 2^{-i-1} b_i,$$

and the series converges because  $\sum_i 2^{-i-1} |b_i| < \infty$ . Let its limit be  $\log \rho$ . The tail is  $O(2^{-j/2})$ , so  $\log a_{2^j} = 2^j \log \rho + O(2^{j/2})$ . For a general  $n$ , decompose it into its dyadic blocks and repeatedly use the assumed inequalities; the square-root errors form a geometric series bounded by  $C\sqrt{n}$ . This proves the claim.  $\square$

**Proposition 8.56** (Weak multiplicativity and the LFPP scale). *By Theorem 8.21, the exponent input holds. Let  $\lambda_n$  be the median of  $L_{1,1}^{(n)}(\phi)$ . There is  $C < \infty$  such that for all  $n, k \geq 1$ ,*

$$e^{-C\sqrt{k}}\lambda_n\lambda_k \leq \lambda_{n+k} \leq e^{C\sqrt{k}}\lambda_n\lambda_k.$$

Consequently

$$\lambda_n = 2^{-n(1-\xi Q)+O(\sqrt{n})}.$$

*Proof.* For the upper bound, take a left–right geodesic at scale  $k$ . Around every coarse box it visits, concatenate four long fine-scale crossings. Scaling of  $\phi_{k,n+k}$  shows that the expected fine crossing contribution is  $2^{-k}$  times a scale- $n$  crossing. Uniform upper tails and bounded quantile ratios give finite first and second moments comparable to  $\lambda_n$ . Coarse-field oscillation contributes only  $e^{C\sqrt{k}}$  in expectation by Lemma 8.44. The resulting path therefore has expected length at most  $e^{C\sqrt{k}}\lambda_n\lambda_k$ , and Markov’s inequality at a fixed probability gives the same upper bound for the median  $\lambda_{n+k}$ .

For the lower bound, coarse-grain a scale- $(n+k)$  geodesic at level  $k$ . Each visited coarse box forces a short fine-scale crossing. The lower-tail estimate says that, simultaneously for the finitely many local rectangle shapes needed here, the relevant fine crossings are at least  $e^{-C\sqrt{k}}\lambda_n$  with probability exceeding a fixed central probability. The coarse oscillation estimate and the scale- $k$  crossing lower bound then imply  $L_{1,1}^{(n+k)} \geq e^{-C\sqrt{k}}\lambda_n\lambda_k$  with probability at least  $1/2$  after changing  $C$ . This yields the lower median inequality.

Lemma 8.55 gives  $\lambda_n = \rho^n e^{O(\sqrt{n})}$ . The two-sided exponent input gives, for every  $\eta > 0$  and all sufficiently large  $n$ ,

$$2^{-n(1-\xi Q+\eta)} \leq \lambda_n \leq 2^{-n(1-\xi Q-\eta)}.$$

Taking  $n$ th roots and then  $\eta \downarrow 0$  identifies  $\rho = 2^{-(1-\xi Q)}$ , proving the final formula.  $\square$

### 8.1.5 Metric chaining, non-collapse, and the whole-plane localization

The preceding modules control one macroscopic crossing. The final analytic step is stronger: one must control *all* pairs of points simultaneously. The proof is the metric-level argument of Ding–Dubédat–Dunlap–Falconet, with the upper and lower Euclidean moduli kept separate because they have opposite probabilistic inputs. Throughout this subsection

$$d_n := \lambda_n^{-1} D_{\phi_{0,n}}$$

denotes the unit-square LFPP metric normalized by the median  $\lambda_n$  of a unit-square left–right crossing. The weak multiplicativity theorem above gives

$$\lambda_{n-k}\lambda_k e^{-C\sqrt{k}} \leq \lambda_n \leq \lambda_{n-k}\lambda_k e^{C\sqrt{k}}, \quad (20.61)$$

and

$$\lambda_k = 2^{-k(1-\xi Q)+O(\sqrt{k})}. \quad (20.62)$$

For a dyadic square  $P$  of side  $2^{-k}$ , let  $\mathcal{C}(P)$  be the fixed finite family of long rectangles of comparable size which surrounds  $P$  and which can be glued into a loop around its concentric half-square. Write  $L_n(P)$  for the sum of the corresponding LFPP crossing lengths. The number of rectangles in  $\mathcal{C}(P)$  is bounded by an absolute constant.

**Lemma 8.57** (Crude exponent window). *The LFPP exponent  $a_\gamma = 1 - \xi Q$  satisfies*

$$-2\xi \leq a_\gamma \leq 2\xi. \quad (20.62a)$$

Consequently  $\xi(Q-2) < 1$  and  $\xi(Q+2) \geq 1$ .

*Proof.* A unit-square crossing has Euclidean length at least one, hence

$$L_n \geq e^{\xi \inf_{[0,1]^2} \phi_{0,n}}.$$

Conversely the horizontal segment at height  $1/2$  gives  $L_n \leq e^{\xi \sup_{[0,1]^2} \phi_{0,n}}$  up to an irrelevant deterministic constant. The Gaussian maximum/minimum estimate gives, with probability bounded away from zero and one,

$$-2n \log 2 - O(\sqrt{n}) \leq \inf \phi_{0,n} \leq \sup \phi_{0,n} \leq 2n \log 2 + O(\sqrt{n}).$$

Taking a fixed central quantile and comparing with  $\lambda_n = 2^{-na_\gamma + O(\sqrt{n})}$  yields  $-2\xi \leq a_\gamma \leq 2\xi$ . Since  $\xi(Q-2) = 1 - a_\gamma - 2\xi$  and  $\xi(Q+2) = 1 - a_\gamma + 2\xi$ , the two consequences follow; the first is strict for every admissible  $\beta < \xi(Q-2)$  used below.  $\square$

**Lemma 8.58** (Dyadic chaining of the diameter). *There is an absolute  $C < \infty$  such that, for every  $n$ ,*

$$\text{Diam}([0,1]^2, D_{\phi_{0,n}}) \leq C \sum_{k=0}^n \max_{P \in \mathcal{P}_k} L_n(P) + C 2^{-n} e^{\xi \sup_{[0,1]^2} \phi_{0,n}}. \quad (20.63)$$

More generally, if  $2^{-k} \leq |x - y| \leq 2^{-k+1}$ , then

$$D_{\phi_{0,n}}(x, y) \leq C \sum_{j=k}^n \max_{P \in \mathcal{P}_j} L_n(P) + C 2^{-n} e^{\xi \sup \phi_{0,n}}. \quad (20.64)$$

*Proof.* For each  $j \geq k$ , choose the dyadic  $j$ -squares containing  $x$  and  $y$ . Around each such square glue the four long crossings in  $\mathcal{C}(P)$ . The resulting loops meet the corresponding loops at the next coarser scale; by following successively these intersections one obtains a path from the  $2^{-n}$ -square of  $x$  to the  $2^{-n}$ -square of  $y$ . The price at scale  $j$  is bounded by a fixed multiple of  $\max_{P \in \mathcal{P}_j} L_n(P)$ . The two residual segments inside the terminal  $2^{-n}$ -squares have Euclidean length  $O(2^{-n})$  and LFPP density at most  $e^{\xi \sup \phi_{0,n}}$ . This proves (20.64); take  $k = 0$  and the supremum over  $x, y$  to obtain (20.63).  $\square$

**Lemma 8.59** (Expected local crossing envelope). *For every sufficiently small fixed  $\varepsilon_0 > 0$ , there is  $C < \infty$  such that, uniformly for  $0 \leq k \leq n$ ,*

$$\mathbb{E} \max_{P \in \mathcal{P}_k} L_n(P) \leq C 2^{-k} 2^{2\xi k} e^{Ck^{1/2+\varepsilon_0}} \lambda_{n-k}. \quad (20.65)$$

*Proof.* Split  $\phi_{0,n} = \phi_{0,k} + \phi_{k,n}$ . On a fixed  $k$ -square, scaling identifies the fine crossing lengths with  $2^{-k}$  times crossings of the scale- $(n-k)$  field. The uniform upper-tail theorem for bounded-aspect rectangles gives moments comparable to  $\lambda_{n-k}$ , uniformly in the location and rectangle shape. Taking the maximum over  $O(2^{2k})$  rectangles is absorbed by the uniform upper-tail estimate at the cost  $e^{Ck^{1/2+\varepsilon_0}}$ ; after the geometric scale factor  $2^{-k}$  and the coarse-field maximum  $e^{\xi \max \phi_{0,k}}$ , the resulting bound is precisely the right side of (20.65). The coarse maximum satisfies

$$\mathbb{E} e^{\xi \max \phi_{0,k}} \leq 2^{2\xi k} e^{C\sqrt{k}},$$

and the harmless  $k^{\varepsilon_0}$  loss records the finite family of rectangles used in the circuit gluing. This is the estimate used in the source proof of the diameter theorem.  $\square$

**Proposition 8.60** (Tight normalized diameter). *The sequence*

$$\log \text{Diam}([0, 1]^2, d_n)$$

*is tight. More precisely, for  $s \geq 1$ ,*

$$\sup_n \Pr[\text{Diam}([0, 1]^2, d_n) > e^s] \leq C e^{-cs}, \quad (20.66)$$

*while the lower tail is bounded by the lower tail of a unit-square crossing.*

*Proof.* Take expectations in (20.63) and use Lemma 8.59. By Proposition 8.56,

$$\lambda_{n-k} \leq \lambda_n \lambda_k^{-1} e^{C\sqrt{k}} \leq \lambda_n 2^{k(1-\xi Q)} e^{C\sqrt{k}}.$$

Hence the  $k$ th expected contribution to the normalized diameter is at most

$$C 2^{-k} 2^{2\xi k} 2^{k(1-\xi Q)} e^{Ck^{1/2+\varepsilon_0}} \leq C 2^{-k\xi(Q-2)} e^{Ck^{1/2+\varepsilon_0}}.$$

The series is summable because  $Q > 2$ . The microscopic terminal term is handled by the Gaussian supremum estimate and (20.62). Thus  $\sup_n \mathbb{E} \text{Diam}([0, 1]^2, d_n) < \infty$ , and Markov gives the right tail after replacing the threshold by  $e^s$ . Since the diameter is at least the left-right crossing, Proposition 8.48 gives the stated lower tail.  $\square$

**Proposition 8.61** (Upper Euclidean modulus). *Fix*

$$0 < \beta < \xi(Q-2).$$

*Then the random variables*

$$H_{n,\beta}^+ := \sup_{x \neq y \in [0,1]^2} \frac{d_n(x, y)}{|x - y|^\beta}$$

*form a tight family.*

*Proof.* For pairs with  $2^{-k} \leq |x - y| \leq 2^{-k+1}$ , use the local chaining (20.64). The same calculation as in the proof of Proposition 8.60, but starting the sum at  $k$ , gives

$$\mathbb{E} \sup_{2^{-k} \leq |x-y| \leq 2^{-k+1}} d_n(x, y) \leq C 2^{-k\xi(Q-2)} e^{Ck^{1/2+\varepsilon_0}}. \quad (20.67)$$

Therefore, for  $s > 0$ , Markov and a union bound over  $0 \leq k \leq n$  imply

$$\Pr\left[\exists x, y : 2^{-k} \leq |x - y| \leq 2^{-k+1}, d_n(x, y) > e^s |x - y|^\beta \text{ for some } k\right] \leq C e^{-s} \sum_{k \geq 0} 2^{-k(\xi(Q-2)-\beta)/2},$$

after increasing the deterministic constant to absorb the sublinear  $e^{Ck^{1/2+\varepsilon_0}}$  factor. For  $|x - y| < 2^{-n}$ , the straight segment bound

$$d_n(x, y) \leq \lambda_n^{-1} e^{\xi \sup \phi_{0,n}} |x - y|$$

and the Gaussian supremum estimate give the same uniform tail, because Lemma 8.57 gives  $1 - \beta > 0$  and (20.62) leaves a strictly positive linear margin in  $n$ . This proves tightness of  $H_{n,\beta}^+$ .  $\square$

**Proposition 8.62** (Lower Euclidean modulus and non-collapse). *Fix*

$$\alpha > \xi(Q + 2).$$

*Then*

$$H_{n,\alpha}^- := \sup_{x \neq y \in [0,1]^2} \frac{|x - y|^\alpha}{d_n(x, y)}$$

*forms a tight family.*

*Proof.* Suppose  $2^{-k} \leq |x - y| \leq 2^{-k+1}$ . Any path from  $x$  to  $y$  must cross one of the finite family of short rectangles surrounding a dyadic  $(k + 2)$ -square separating the two endpoints. Hence

$$d_n(x, y) \geq \lambda_n^{-1} 2^{-k} e^{\xi \inf \phi_{0,k}} \min_{P \in \mathcal{P}_{k+2}} \min_{1 \leq i \leq 4} 2^k L^{(k,n)}(R_i^{\text{short}}(P)). \quad (20.68)$$

Write  $\alpha = \xi(Q + 2) + \xi\delta$  with  $\delta > 0$ . Weak multiplicativity gives

$$2^{-k\alpha} \lambda_n 2^k \leq 2^{-k(2+\delta)\xi} \lambda_{n-k} e^{C\sqrt{k}}. \quad (20.69)$$

If  $d_n(x, y) \leq e^{-\xi s} |x - y|^\alpha$ , then by (20.68)–(20.69) either

$$\sup_{[0,1]^2} |\phi_{0,k}| \geq k \log 4 + k\delta \log 2 + s/2$$

or one of  $O(2^{2k})$  fine short-rectangle crossings is at most  $\lambda_{n-k} 2^{-k\delta\xi} e^{C\sqrt{k}} e^{-\xi s/2}$ . The Gaussian supremum bound and the fixed-quantile lower-tail theorem yield

$$\Pr[\exists x, y \text{ in the } k\text{th shell with } d_n(x, y) \leq e^{-\xi s} |x - y|^\alpha] \leq C e^{-ck} e^{-cs}. \quad (20.70)$$

Summing over  $k \leq n$  is finite. For  $|x - y| < 2^{-n}$  use the elementary straight-line lower bound

$$d_n(x, y) \geq \lambda_n^{-1} e^{\xi \inf \phi_{0,n}} |x - y|.$$

Lemma 8.57 implies  $\alpha > 1$  in the present range, so the infimum over  $|x - y| \leq 2^{-n}$  of  $|x - y|^{1-\alpha}$  occurs at  $2^{-n}$ ; then (20.62) and the Gaussian infimum bound again leave a positive linear margin because  $\alpha - \xi Q > 2\xi$ . This gives a uniform tail for  $H_{n,\alpha}^-$ .  $\square$

**Theorem 8.63** (DDDF metric compactness on a fixed square). *The laws of  $d_n$ , viewed in  $C([0, 1]^2 \times [0, 1]^2)$  with the uniform norm, are tight. Every subsequential limit  $d$  is almost surely a genuine metric and, simultaneously for each*

$$\beta < \xi(Q - 2), \quad \alpha > \xi(Q + 2),$$

*there are finite random constants  $C_+, C_- > 0$  such that*

$$C_- |x - y|^\alpha \leq d(x, y) \leq C_+ |x - y|^\beta \quad (x, y \in [0, 1]^2). \quad (20.71)$$

*In particular  $d$  is finite and positive off the diagonal and induces the Euclidean topology.*

*Proof.* Proposition 8.61 gives a tight Hölder modulus and Proposition 8.60 gives tightness at one macroscopic scale. Arzelà–Ascoli therefore gives tightness in the uniform topology. Along a convergent subsequence, use Skorohod representation to realize uniform convergence almost surely. Symmetry and the triangle inequality pass to the limit, so the limit is initially a pseudometric. Proposition 8.62, first for rational  $\alpha > \xi(Q + 2)$  and then by monotonicity, gives the left inequality in (20.71); hence distinct Euclidean points have positive distance. The upper modulus gives the right inequality. The two bounds imply equivalence of the metric and Euclidean topologies.  $\square$

**Proposition 8.64** (Interpolation from dyadic to arbitrary mollification scale). *Let  $\delta \in (0, 1)$  and write  $\delta = 2^{-(n+r)}$  with  $n \in \mathbb{N}$  and  $r \in [0, 1)$ . If  $\lambda_\delta$  is a fixed central unit-square crossing quantile for the star-scale LFPP field at cutoff  $\delta$ , then*

$$C^{-1}\lambda_n \leq \lambda_\delta \leq C\lambda_n, \quad (20.72)$$

with  $C$  independent of  $n, r$ . The tail bounds, the two Hölder moduli, and the metric tightness theorem above consequently hold uniformly for all  $\delta \in (0, 1)$ , not only for dyadic cutoffs.

*Proof.* Decompose the star-scale field at the fractional low-frequency interval,

$$\phi_{0,n+r} = \phi_{0,r} + \phi_{r,n+r}.$$

The first term has a scale-uniform Gaussian supremum tail for  $r \in [0, 1]$ . The second term, after the spatial scaling  $x \mapsto 2^r x$ , has the law of a scale- $n$  field on a rectangle whose aspect ratio stays in the compact interval  $[1/2, 2]$ . Hence, pathwise,

$$e^{-\xi\|\phi_{0,r}\|_\infty} L_{r,n+r} \leq L_{0,n+r} \leq e^{\xi\|\phi_{0,r}\|_\infty} L_{r,n+r}. \quad (20.73)$$

The RSW theorem and the bounded crossing-quantile ratios compare the central quantiles of all these bounded-aspect rectangles by deterministic constants. Taking fixed quantiles in (20.73) therefore yields (20.72).

The same decomposition applied locally to every rectangle in the chaining proof changes its normalized length by at most  $e^{\xi\|\phi_{0,r}\|_\infty}$  times a deterministic bounded-aspect factor. Since this multiplier has moments of every order uniformly in  $r$ , the upper and lower crossing tails, Propositions 8.61 and 8.62, and the diameter estimate transfer uniformly. Arzelà–Ascoli then gives arbitrary-scale metric tightness. This is the interpolation mechanism of the source Appendix A.2.  $\square$

**Proposition 8.65** (Heat-kernel GFF comparison on an interior compact). *Let  $D$  be bounded, let  $U \Subset D$ , and let  $h_D^0$  be the zero-boundary GFF on  $D$ . For  $t \in (0, 1/2)$  one can couple  $h_D^0$  with a field  $\varphi_t \stackrel{d}{=} \phi_{\sqrt{t}}$  such that*

$$\Pr[\|\varphi_t - p_{t/2} * h_D^0\|_{L^\infty(U)} > s] \leq C e^{-cs^2}, \quad s \geq 0, \quad (20.74)$$

with  $C, c$  independent of  $t$ .

*Proof.* Use one space–time white noise  $W$ . The heat-kernel mollification of the Dirichlet GFF has the killed-kernel representation

$$\eta_t(x) = \int_0^\infty \int_D (p_{t/2} * p_{s/2}^D)(x, y) W(dy, ds),$$

whereas the star-scale field can be written, after a time shift, as

$$\varphi_t(x) = \int_0^{1-t} \int_{\mathbb{R}^2} p_{(t+s)/2}(x - y) W(dy, ds).$$

Split both integrals into the common region  $D \times (0, 1 - t)$ , the spatial exterior  $D^c \times (0, 1 - t)$ , and the long-time Dirichlet tail  $D \times (1 - t, \infty)$ . On the common region, the kernel difference is controlled by the Brownian-bridge representation of the killed heat kernel. Since  $U$  is a positive distance from  $D^c$ , for  $x, y \in U$

$$|(p_{t/2} * p_{s/2}^D)(x, y) - p_{(t+s)/2}(x - y)| \leq C e^{-c/s}. \quad (20.75)$$

Consequently the pointwise variance of the common-part difference is uniformly bounded. Splitting the  $s$  integral at  $\sqrt{|x - x'|}$  and using heat-kernel gradient estimates above that scale gives

$$\mathbb{E}|X_t(x) - X_t(x')|^2 \leq C|x - x'| \quad (20.76)$$

for each of the three difference fields; the exterior and long-time pieces have the same estimate because of the positive distance to  $D^c$  and the integrable long-time killed kernel. Their pointwise variances are uniformly bounded as well. Gaussian moment equivalence upgrades (20.76) to arbitrarily high moments; Kolmogorov continuity gives a uniformly bounded expected Hölder norm on  $U$ . Fernique's theorem for these centered Gaussian random elements then yields the uniform Gaussian tail (20.74).  $\square$

**Proposition 8.66** (Zero-boundary and whole-plane localization). *Let  $K \Subset D$  be compact. The normalized internal heat-kernel LFPP metrics associated with a zero-boundary GFF on  $D$  are tight on  $K \times K$ , and every subsequential limit obeys the bi-Hölder bounds (20.71) locally on  $K$ . The same is true for a whole-plane GFF in any fixed additive normalization. The deterministic normalizing medians in the two formulations differ by factors bounded above and below uniformly in the mollification scale.*

*Proof.* For the zero-boundary field, Proposition 8.65 says that on a slightly larger  $U \supset K$  the mollified field differs from the star-scale approximation by a random continuous perturbation  $X_t$  whose supremum has a scale-uniform Gaussian tail. For any two fields differing by  $f$  on  $U$ , internal path lengths satisfy

$$e^{-\xi\|f\|_U} D_h(\cdot, \cdot; U) \leq D_{h+f}(\cdot, \cdot; U) \leq e^{\xi\|f\|_U} D_h(\cdot, \cdot; U). \quad (20.77)$$

The same comparison for a fixed crossing rectangle shows that the associated central quantiles are uniformly comparable. Theorem 8.63, Proposition 8.64, and (20.77) therefore transfer tightness and both Hölder moduli to the zero-boundary heat-kernel metric on  $K$ .

For a whole-plane GFF  $h$ , choose  $D'$  with  $K \Subset D' \Subset D$ . The domain Markov decomposition on  $D$  is

$$h = h_D^0 + \mathfrak{h},$$

where  $h_D^0$  is a zero-boundary GFF and  $\mathfrak{h}$  is harmonic on  $D$ . On  $D'$  the random variables  $\|\mathfrak{h}\|_{C^1(D')}$  have Gaussian tails. Convolution with the heat kernel changes  $\mathfrak{h}$  by  $o(1)$  uniformly on  $K$ , and the contribution of the whole-plane field from outside  $D'$  to a mollifier centered in  $K$  is a smooth Gaussian field with scale-uniform supremum tails (the heat kernel across the positive Euclidean gap is exponentially small). Thus the whole-plane and zero-boundary mollifications differ on  $K$  by a continuous random perturbation with a scale-uniform Gaussian supremum tail. Apply (20.77) again. Quantile comparability follows from the same inequality, proving the normalization statement.  $\square$

**Corollary 8.67** (Local whole-plane LFPP tightness and joint compactness). *Let  $h$  be a whole-plane GFF in a fixed additive normalization and let  $a_\epsilon$  be the median unit-square crossing normalization. The family  $a_\epsilon^{-1}D_h^\epsilon$  is tight for the local-uniform topology on  $\mathbb{C} \times \mathbb{C}$ . Every subsequential limit is locally bi-Hölder with the deterministic source-admissible exponents, hence finite, nonzero off the diagonal, and topologically equivalent to the Euclidean plane. The laws of  $(h, a_\epsilon^{-1}D_h^\epsilon)$  are jointly tight in the product of the usual distribution topology for  $h$  and the local-uniform metric topology.*

*Proof.* Apply Proposition 8.66 on the exhaustion  $[-m, m]^2$ ,  $m \geq 1$ , and diagonalize. The local Hölder bounds give consistency of the restrictions and local-uniform tightness. Positivity, finiteness, and the topology statement are local consequences of the same bounds. Finally the marginal law of  $h$  is fixed, hence tight; tightness of the two marginals in Polish spaces implies tightness of their joint laws.  $\square$



## 8.2 Localized LFPP and compatible limits of internal metrics

The weak-metric axioms require more than a global subsequential metric. To speak about locality one first needs an approximation whose internal metric in an open set uses only the field in a shrinking Euclidean neighbourhood of that set. One then has to carry the global LFPP metric and a countable family of internal metrics through the same subsequence. The last step is deterministic but delicate: convergence of metrics on  $\overline{W}$  identifies the *open* internal metric on  $W$ , not necessarily the metric internal to  $\overline{W}$ . This section owns that entire localization/compatibility chain.

### 8.2.1 A compactly supported heat-kernel mollifier

Fix once and for all continuous radial functions  $\psi_\epsilon : \mathbb{C} \rightarrow [0, 1]$ , continuously depending on  $\epsilon > 0$ , such that

$$\psi_\epsilon \equiv 1 \text{ on } B_{\epsilon^{1/2}/2}(0), \quad \psi_\epsilon \equiv 0 \text{ on } \mathbb{C} \setminus B_{\epsilon^{1/2}}(0).$$

For a whole-plane GFF plus a bounded continuous function  $h$ , put

$$\widehat{h}_\epsilon^*(z) := \langle h, \psi_\epsilon(z - \cdot) p_{\epsilon^2/2}(z, \cdot) \rangle$$

and define the localized LFPP metric

$$\widehat{D}_h^\epsilon(z, w) := \inf_{P: z \rightarrow w} \int_0^1 e^{\xi \widehat{h}_\epsilon^*(P(t))} |P'(t)| dt.$$

If  $U \subset \mathbb{C}$  is open, then  $\widehat{D}_h^\epsilon(\cdot, \cdot; U)$  is determined by  $h|_{B_{\epsilon^{1/2}}(U)}$ .

**Lemma 8.68** (Uniform circle-average growth on bounded sets). *Let  $h$  be a whole-plane GFF in any fixed additive normalization. For every  $R > 0$  and  $\zeta > 0$ , almost surely there is a finite random constant  $C = C(R, \zeta, h)$  such that, for all  $z \in B_R(0)$  and  $r > 0$ ,*

$$|h_r(z)| \leq C \max \left\{ (2 + \zeta) \log(1/r), (\log r)^{1/2+\zeta}, 1 \right\}.$$

*Proof.* We give the two scale regimes separately. Write  $t = -\log r$  for small radii. On each slab

$$\mathcal{P}_k = B_R(0) \times [k, k+1], \quad k \geq 1,$$

the process  $(z, t) \mapsto h_{e^{-t}}(z)$  is Gaussian after subtracting one fixed circle average. Its pointwise variance is  $k + O_R(1)$ . The standard circle-average covariance formula also gives, whenever  $|z - z'| \leq e^{-k-3}$  and  $|t - t'| \leq e^{-2k}$ ,

$$\mathbb{E} |h_{e^{-t}}(z) - h_{e^{-t'}}(z')|^2 \leq C_R.$$

Cover  $\mathcal{P}_k$  by a grid with at most  $e^{C_R k}$  points at these meshes. The Gaussian tail bound at the grid points gives, for  $A$  large,

$$\Pr \left[ \max_{\text{grid}} |h_{e^{-t}}(z)| > Ak \right] \leq e^{C_R k} e^{-cA^2 k}.$$

For the increments inside a grid cell, repeat the estimate on successively halved meshes. At refinement level  $j$ , there are at most  $e^{C_R k} 2^{Cj}$  increments and each has variance  $O(2^{-j})$ ; choosing thresholds  $2^{-j/4}$  and summing the resulting Gaussian tails shows that with probability  $1 - O(e^{-cA^k})$  the oscillation inside every cell is  $O(1)$ . Borel–Cantelli therefore yields

$$\sup_{z \in B_R(0)} \sup_{r \in [e^{-k-1}, e^{-k}]} |h_r(z)| \leq Ck$$

for all large  $k$ . Enlarging the random constant covers the remaining small scales, and replacing  $C$  by a larger random constant gives the displayed coefficient  $2 + \zeta$ .

For large radii, on the dyadic band  $r \in [2^k, 2^{k+1}]$  the centered process  $h_r(z) - h_r(0)$ , after the spatial scaling  $z \mapsto 2^{-k}z$ , has uniformly bounded variances and a continuous modification. Gaussian chaining therefore gives

$$\Pr \left[ \sup_{z \in B_R(0)} \sup_{r \in [2^k, 2^{k+1}]} |h_r(z) - h_r(0)| > k^{1/2+\zeta} \right] \leq C e^{-ck^{2\zeta}}.$$

Taking, if necessary, the same statement along blocks  $k \in [m^q, (m+1)^q]$  with  $q$  chosen so that the displayed probabilities are summable, and using continuity between the block endpoints, gives the bound almost surely for all large  $k$ . Finally  $t \mapsto h_{e^t}(0)$  is a two-sided Brownian motion up to an additive constant, hence  $|h_r(0)| = O((\log r)^{1/2+\zeta})$  almost surely. The compact range of intermediate radii is handled by continuity.  $\square$

**Lemma 8.69** (Localized mollifier approximates the heat mollifier). *Almost surely  $(z, \epsilon) \mapsto \widehat{h}_\epsilon^*(z)$  has a continuous modification. For every bounded open  $U \subset \mathbb{C}$ ,*

$$\sup_{z \in \overline{U}} |h_\epsilon^*(z) - \widehat{h}_\epsilon^*(z)| \xrightarrow{\epsilon \rightarrow 0} 0.$$

Consequently

$$\sup_{\substack{z, w \in U \\ z \neq w}} \left| \log \frac{\widehat{D}_h^\epsilon(z, w; U)}{D_h^\epsilon(z, w; U)} \right| \xrightarrow{\epsilon \rightarrow 0} 0.$$

*Proof.* First take  $h = h$ . Radial symmetry of both kernels gives

$$h_\epsilon^*(z) = \frac{2}{\epsilon^2} \int_0^\infty r h_r(z) e^{-r^2/\epsilon^2} dr,$$

whereas the localized field has the same integral restricted to  $[0, \epsilon^{1/2}]$  and multiplied by  $\psi_\epsilon(r)$ . Continuity of the circle-average process gives a continuous modification. Since the cutoff is one up to  $\epsilon^{1/2}/2$ , Lemma 8.68 yields

$$\sup_{z \in \overline{U}} |h_\epsilon^*(z) - \widehat{h}_\epsilon^*(z)| \leq \frac{2C}{\epsilon^2} \int_{\epsilon^{1/2}/2}^\infty r \max\{\log(1/r), \log r, 1\} e^{-r^2/\epsilon^2} dr = o(1).$$

The same conclusion for a bounded continuous perturbation follows from uniform continuity on a neighbourhood of  $\overline{U}$  and the Gaussian tail of the heat kernel outside that neighbourhood. If the displayed supremum is  $\eta_\epsilon$ , every path in  $U$  has localized and ordinary LFPP lengths within multiplicative factors  $e^{\pm \xi \eta_\epsilon}$ . Taking infima over paths proves the metric ratio statement.  $\square$

## 8.2.2 Internal metrics on a countable domain basis

**Definition 8.70** (Dyadic domain). A closed square is dyadic if its side length is  $2^k$  and its corners lie in  $2^k \mathbb{Z}^2$ . A *dyadic domain* is the interior of a finite union of dyadic closed squares. The collection  $\mathcal{W}$  of all dyadic domains is countable.

**Lemma 8.71** (Square internal metrics). *Let  $S$  be a fixed closed square and let  $h$  be a whole-plane GFF plus a bounded continuous function. The laws of*

$$a_\epsilon^{-1} D_h^\epsilon(\cdot, \cdot; S)$$

*are tight in  $C(S \times S)$ . Every subsequential limit is a length metric which induces the Euclidean topology on  $S$ .*

*Proof.* For the unit square and the whole-plane GFF this is precisely the fixed-square compactness already owned in Theorem 8.63, after the heat-kernel/star-scale comparison and arbitrary-scale interpolation of Propositions 8.64 and 8.66. Translation and the exact heat-kernel scaling identity reduce an arbitrary fixed square to the unit square, modulo an additive circle average, which multiplies every distance by the same factor and hence disappears under the crossing normalization. A bounded continuous perturbation changes all path lengths by the deterministic bi-Lipschitz factors  $e^{\pm \xi \|f\|_\infty}$ . The length-space and topology statements are preserved under these reductions.  $\square$

**Lemma 8.72** (Dyadic-domain internal compactness). *For every dyadic domain  $W$ , the laws of*

$$a_\epsilon^{-1} D_h^\epsilon(\cdot, \cdot; \overline{W})$$

*are tight in  $C(\overline{W} \times \overline{W})$ . Every subsequential limit is a length metric inducing the Euclidean topology on  $\overline{W}$ .*

*Proof.* It is enough to treat one connected component of  $\overline{W}$ , since the finitely many components are at positive Euclidean distance. Cover that component by finitely many closed dyadic squares  $S_1, \dots, S_m$ . The Arzelà–Ascoli tightness criterion for a family of random metrics on a compact connected space says that it suffices to show: for every  $\eta > 0$  there is  $\delta > 0$  such that, with probability at least  $1 - \eta$ ,  $d(z, w) \leq \eta$  whenever  $|z - w| \leq \delta$ . Lemma 8.71 gives this simultaneously on all  $S_j$ . If  $z$  and  $w$  are within  $\delta/2$  but lie in two different squares, then for sufficiently small  $\delta$  those squares can be chosen to intersect; a point  $u$  in the intersection can be chosen within  $\delta$  of both  $z$  and  $w$ . The triangle inequality gives a  $2\eta$  modulus on all of  $\overline{W}$ , proving tightness.

Let  $\tilde{D}$  be a uniform subsequential limit. To exclude collapse, choose squares  $S_1 \Subset S_2 \subset \overline{W}$ , where the portion of  $\partial S_2$  not lying on  $\partial W$  is a positive Euclidean distance from  $S_1$ . For every  $\epsilon$ ,

$$D_h^\epsilon(S_1, W \setminus S_2; \overline{W}) = D_h^\epsilon(S_1, \partial S_2 \setminus \partial W; S_2).$$

By Lemma 8.71, every subsequential limit of the right side is strictly positive. A countable collection of such square annuli separating every pair of distinct points therefore shows that  $\tilde{D}(z, w) > 0$  for  $z \neq w$ . The uniform upper modulus and compactness imply that the identity between the Euclidean and  $\tilde{D}$  topologies is a homeomorphism. Finally a uniform limit of length metrics on a compact space is a length metric: approximate endpoints of a near-geodesic by a fine partition and pass the polygonal length estimates to the uniform limit.  $\square$

**Lemma 8.73** (Macroscopic containment event). *For every  $p \in (0, 1)$  and  $C > 0$ , there is  $R = R(p, C) > 1$  such that for each fixed  $r > 0$ ,*

$$\liminf_{\epsilon \rightarrow 0} \Pr[\text{diam}_{D_h^\epsilon} S_r(0) < C^{-1} D_h^\epsilon(S_r(0), \partial S_{Rr}(0))] \geq p.$$

*On this event, every distance between two points of  $S_r(0)$  is realized, up to arbitrary approximation, by paths contained in  $S_{Rr}(0)$ .*

*Proof.* Normalize first to a whole-plane GFF. By Lemma 8.71, along every vanishing sequence of  $\epsilon$ 's one can pass to a subsequence on which the normalized internal metric on the unit square converges to a genuine Euclidean-topology metric. In such a limit the diameter of the central square of side  $1/R$  tends to zero as  $R \rightarrow \infty$ , whereas its distance to the unit-square boundary converges to the positive distance from the center to that boundary. If the asserted probability bound failed for arbitrarily large  $R$ , compactness would produce a subsequential limit contradicting these two facts. Thus choose  $R$  large enough at scale one. The exact LFPP scaling relation

$$D_{h(r\cdot) - h_r(0)}^{\epsilon/r}(u, v) = r^{-1} e^{-\xi h_r(0)} D_h^\epsilon(ru, rv)$$

transfers the event to every fixed  $r$ . A bounded continuous perturbation  $f$  changes the two sides by at most  $e^{\pm \epsilon \|f\|_\infty}$ ; choose a deterministic  $A$  so that  $e^{\epsilon \|f\|_\infty} \leq A$  with probability as close to one as desired and apply the GFF statement with  $A^2 C$ .

If a path from  $u$  to  $v$  in  $S_r(0)$  exits  $S_{Rr}(0)$ , its length is at least the distance from  $S_r(0)$  to  $\partial S_{Rr}(0)$ , strictly larger than the diameter bound. Hence a near-minimizer cannot exit.  $\square$

**Lemma 8.74** (Compatibility of global and internal metric limits). *Let  $V \subset U \subset \mathbb{C}$  be open. Suppose continuous length metrics  $D^n$  on  $U$  converge locally uniformly to a continuous length metric  $D$ , and  $D^n(\cdot, \cdot; \bar{V})$  converge uniformly on  $\bar{V} \times \bar{V}$  to a continuous length metric  $\tilde{D}$ . Then*

$$D(\cdot, \cdot; V) = \tilde{D}(\cdot, \cdot; V).$$

*No equality on  $\bar{V}$  is asserted.*

*Proof.* Fix  $u, v \in V$  with  $D(u, v) < D(u, \partial V)$ . A  $D$ -near-geodesic from  $u$  to  $v$  stays in  $V$ , so  $D(u, v) = D(u, v; V)$ . Local uniform convergence implies, for large  $n$ ,  $D^n(u, v) < D^n(u, \partial V)$ , hence

$$D^n(u, v) = D^n(u, v; V) = D^n(u, v; \bar{V}).$$

Passing to the two limits gives  $D(u, v; V) = \tilde{D}(u, v; V)$  for such pairs. Consequently every path in  $V$  which stays a positive Euclidean distance from  $\partial V$  has the same  $D$ -length and  $\tilde{D}$ -length: subdivide it into finitely many pieces satisfying the preceding strict boundary inequality and sum. Any path in  $V$  can be approximated in length by one whose compact image stays a positive distance from the boundary. Since both internal metrics are length metrics, taking infima proves the claim. Equality on  $\bar{V}$  need not hold, because a limiting near-geodesic may spend positive time on  $\partial V$ .  $\square$

**Theorem 8.75** (Joint convergence of LFPP and all dyadic internal metrics). *Let  $h$  be a whole-plane GFF plus a bounded continuous function and let  $\epsilon_n \downarrow 0$ . There is a subsequence  $\mathcal{E}$ , a continuous length metric  $D$  on  $\mathbb{C}$ , and for every dyadic domain  $W \in \mathcal{W}$  a length metric  $D_W$  on  $\bar{W}$ , inducing the Euclidean topology there, such that jointly in law*

$$(a_\epsilon^{-1} D_h^\epsilon, \{a_\epsilon^{-1} D_h^\epsilon(\cdot, \cdot; \bar{W})\}_{W \in \mathcal{W}}) \longrightarrow (D, \{D_W\}_{W \in \mathcal{W}}).$$

*For every  $W \in \mathcal{W}$ , almost surely*

$$D_W(\cdot, \cdot; W) = D(\cdot, \cdot; W).$$

*The same joint limit is obtained if each internal LFPP metric is replaced by the localized metric  $\hat{D}_h^\epsilon$ . In particular, if  $V \Subset W$  are dyadic domains, the limiting internal metric on  $V$  computed from  $D_W$  equals the one computed from  $D$ .*

*Proof.* The global local-uniform tightness and Euclidean-topology non-collapse are already owned by Corollary 8.67. To see that every global subsequential limit is a length metric, fix  $r$  and apply Lemma 8.73 with  $p$  arbitrarily close to one. Jointly extract the internal metric on  $S_{Rr}(0)$  using Lemma 8.71. On the containment event, global near-geodesics between points of  $S_r(0)$  stay in that larger square. Lemma 8.74 then identifies their limiting lengths with lengths computed from the global limit. Letting  $p$  tend to one and  $r$  tend to infinity proves that  $D$  is a length metric.

For every dyadic  $W$ , Lemma 8.72 gives tightness of the corresponding internal metric. Since  $\mathcal{W}$  is countable, Prokhorov plus a diagonal subsequence gives joint convergence of the global metric and all internal metrics. Applying Lemma 8.74 to each  $W$  and intersecting the resulting probability-one events proves  $D_W(\cdot, \cdot; W) = D(\cdot, \cdot; W)$  simultaneously for all  $W$ .

Finally Lemma 8.69 says that, on each bounded  $W$ , ordinary and localized internal LFPP metrics differ by multiplicative factors tending uniformly to one. Therefore they have the same normalized subsequential limits. Nested-domain compatibility follows by taking internal metrics once more and using the just-proved identity.  $\square$

### 8.3 Weyl scaling and the additive-local coupling

The next two properties are already visible before the limiting metric is known to be a measurable function of the field. Weyl scaling is a pathwise statement about weighted length metrics. Locality is first obtained in the weaker coupling sense of an additive local metric; only after the separate measurability criterion is proved does this become the deterministic field-locality axiom.

**Definition 8.76** (Local and additive-local metric). Let  $U$  be a connected open set and let  $(h, D)$  couple a GFF on  $U$  with a random continuous length metric. The metric  $D$  is *local for  $h$*  if for every open  $V \subset U$ , the internal metric  $D(\cdot, \cdot; V)$  is conditionally independent of

$$(h, D(\cdot, \cdot; U \setminus \bar{V}))$$

given  $h|_{\bar{V}}$ . It is  $\xi$ -*additive local* if for every  $z \in U$  and every  $r$  with  $B_r(z) \subset U$ , the metric  $e^{-\xi h_r(z)} D$  is local for  $h - h_r(z)$ .

**Lemma 8.77** (Weyl perturbations commute with subsequential LFPP limits). *Suppose  $\epsilon_n \rightarrow 0$  and, on one probability space,*

$$a_{\epsilon_n}^{-1} D_h^{\epsilon_n} \longrightarrow D_h$$

*locally uniformly almost surely. If bounded continuous  $f_n$  converge to a bounded continuous  $f$  uniformly on compact sets, then*

$$a_{\epsilon_n}^{-1} D_{h+f_n}^{\epsilon_n} \longrightarrow e^{\xi f} \cdot D_h$$

*locally uniformly almost surely. Here the metric on the right is obtained by weighting  $D_h$ -length by  $e^{\xi f}$ .*

*Proof.* Let  $f_n^* = f_n * p_{\epsilon_n^2/2}$ . Uniform convergence on compacts and Gaussian-kernel concentration imply  $f_n^* \rightarrow f$  uniformly on compacts. At the approximating level the identity is exact:

$$D_{h+f_n}^{\epsilon_n} = e^{\xi f_n^*} \cdot D_h^{\epsilon_n}.$$

The only issue is that a near-geodesic between points of a fixed compact set could leave every fixed compact set, so local-uniform convergence of the unweighted metrics would not by itself justify passage of weighted lengths. Fix a square  $S_r(0)$  and let  $C > 1 + \sup_n \|f_n\|_\infty$ . Lemma 8.73, followed by the already constructed almost-sure coupling, implies that almost surely there is a larger random square  $S_{r'}(0)$  such that

$$\text{diam}_{D_h} S_r(0) < (2C)^{-1} D_h(S_r(0), \partial S_{r'}(0)),$$

and, for all large  $n$ , the analogous inequality for the normalized LFPP metric holds with  $C^{-1}$  in place of  $(2C)^{-1}$ . Thus every near-minimizing path, for both the unweighted and  $e^{\xi f_n^*}$ -weighted metrics, stays in  $S_{r'}(0)$ . On this single compact set the metrics and the weights converge uniformly. For any fixed path  $P$ , its weighted lengths therefore converge by Riemann-sum approximation to metric length; taking an almost-minimizing path for the limit gives the limsup inequality, while almost-minimizers for the approximating metrics, reparameterized by unweighted arclength, give the liminf inequality. Hence the weighted distances converge uniformly on  $S_r(0) \times S_r(0)$ . Let  $r$  increase through the integers.  $\square$

**Lemma 8.78** (Exterior Markov decomposition). *Let  $h$  be a whole-plane GFF and let  $V$  be open with non-polar complement. On  $\mathbb{C} \setminus \bar{V}$  one can write*

$$h = \mathfrak{h} + \mathring{h},$$

*where  $\mathring{h}$  is a zero-boundary GFF on  $\mathbb{C} \setminus \bar{V}$ , independent of  $h|_V$ , and  $\mathfrak{h}$  is harmonic on the complement and determined by the exterior boundary data of  $h$ .*

*Proof.* Use the Dirichlet-energy Hilbert-space decomposition of test functions into the closure of  $H_0^1(\mathbb{C} \setminus \bar{V})$  and its orthogonal complement. The isonormal Gaussian process projected onto the first summand is a zero-boundary GFF on the complement. Orthogonality gives independence. The projection onto the orthogonal complement has zero distributional Laplacian in the complement, hence is harmonic there; it is the conditional expectation of the exterior field given the complementary sigma-algebra and is therefore determined by the boundary/exterior data. This is precisely the GFF domain Markov decomposition.  $\square$

**Proposition 8.79** (Subsequential LFPP limits are additive local metrics). *Let  $h$  be a whole-plane GFF and let  $(h, D_h)$  be any joint subsequential limit of  $(h, a_\epsilon^{-1} D_h^\epsilon)$ . Then  $D_h$  is a  $\xi$ -additive local metric for  $h$ .*

*Proof.* By Theorem 8.75, along the same subsequence we may replace ordinary LFPP by the localized metric  $\hat{D}_h^\epsilon$  and simultaneously carry all dyadic internal metrics to the limit.

Fix an open  $V$ , first with non-polar complement, and dyadic domains  $W$  and  $W'$  with  $\bar{W} \subset V$  and  $\bar{W}' \subset \mathbb{C} \setminus \bar{V}$ . Apply Lemma 8.78 and choose a deterministic smooth bump  $\phi$  supported in the complement of  $\bar{V}$ , equal to one on a neighbourhood of  $\bar{W}'$ . For all sufficiently small  $\epsilon$ , locality of the compactly supported mollifier gives

$$\hat{D}_h^\epsilon(\cdot, \cdot; \bar{W}) \in \sigma(h|_V), \quad \hat{D}_{h-\phi\mathfrak{h}}^\epsilon(\cdot, \cdot; \bar{W}') \in \sigma(\mathring{h}).$$

The two displayed random metrics are therefore independent together with the corresponding field restrictions. By Lemma 8.77, after passing to a further deterministic subsequence if necessary,

$$D_{h-\phi\mathfrak{h}} = e^{-\xi\phi\mathfrak{h}} \cdot D_h.$$

The joint internal-metric convergence theorem passes the preceding independence to the limit: independence is preserved because expectations of products of bounded continuous test functions factor at every  $n$  and the factorization survives weak convergence. Hence

$$(h|_V, D_h(\cdot, \cdot; W)) \text{ is independent of } (\mathring{h}, D_{h-\phi\mathfrak{h}}(\cdot, \cdot; W')).$$

Adding back the harmonic part is a measurable Weyl transformation. Since  $h = \mathring{h} + \mathfrak{h}$  in the exterior and  $\mathfrak{h}$  is determined by the conditioning data, this says that  $D_h(\cdot, \cdot; W)$  is conditionally independent of  $(h, D_h(\cdot, \cdot; W'))$  given  $h|_{\bar{V}}$ . Increase  $W$  to  $V$  and  $W'$  to the complement through countable dyadic exhaustions. If the complement is polar or empty, the assertion follows by monotone exhaustion from the non-polar case.

This proves locality in one fixed additive normalization. For  $z, r$  with  $B_r(z)$  in the ambient domain, set  $c = h_r(z)$ . The fields  $h$  and  $h-c$  determine one another once the chosen normalization functional is recorded, and Lemma 8.77 with the constant perturbation  $-c$  gives  $D_{h-c} = e^{-\xi c} D_h$ . Applying the just-proved locality to the normalized field  $h-c$  proves the additive-local assertion.  $\square$

**Proposition 8.80** (Source boundary for the measurability upgrade). *To upgrade Proposition 8.79 to the statement that  $D_h$  is almost surely a measurable function of  $h$ , it is enough to own the following local-metric criterion.*

*There is a universal  $p \in (0, 1)$  such that if  $D$  is a  $\xi$ -additive local metric of a whole-plane GFF  $h$  and, conditional on  $h$ ,  $D$  and  $\tilde{D}$  are i.i.d. samples for which some deterministic  $C$  satisfies*

$$\Pr \left[ \sup_{u, v \in \partial B_r(z)} \tilde{D}(u, v; B_{2r}(z) \setminus \overline{B_{r/2}(z)}) \leq CD(\partial B_{r/2}(z), \partial B_r(z)) \right] \geq p$$

uniformly in  $z$  and  $r$ , then  $D$  is almost surely determined by  $h$ .

The criterion is proved internally in Section 8.4. Its proof follows the Gwynne–Miller chain

$$\begin{aligned} \text{near-independence across annuli} &\rightarrow \text{good-annulus covering} \rightarrow \text{bi-Lipschitz comparison} \\ &\rightarrow \text{square-resampling/Efron–Stein} \rightarrow \text{zero conditional variance.} \end{aligned}$$

*Proof roadmap.* This proposition is a reduction. Additive locality is the preceding proposition. The local-metric criterion is proved in Section 8.4; applying it there gives field measurability and completes the upgrade.

## 8.4 Local-metric comparison and self-determination

We now prove field measurability. The argument is logically separate from LFPP: it is an abstract theorem about random continuous length metrics coupled with a planar GFF. There are two engines. The first turns a positive-probability annular comparison into a global bi-Lipschitz inequality. The second shows that a local metric which is already determined by the field up to bi-Lipschitz equivalence has zero conditional variance given the field.

### 8.4.1 Outside sigma-fields and good scales

Fix a whole-plane GFF  $h$  and write  $h_r(z)$  for its circle average over  $\partial B_r(z)$ . For a ball  $B_r(z)$  let

$$h - h_r(z) = \tilde{h}_{z,r} + \hat{h}_{z,r} \quad \text{in } B_r(z),$$

where  $\tilde{h}_{z,r}$  is a zero-boundary GFF in  $B_r(z)$  and  $\hat{h}_{z,r}$  is harmonic there and vanishes at  $z$ . Let  $\mathcal{F}_{z,r}$  be the sigma-field generated by the normalized field outside  $B_r(z)$  together with the internal metrics outside  $B_r(z)$  under consideration. For  $r' > r$  the outside information satisfies  $\mathcal{F}_{z,r'} \subset \mathcal{F}_{z,r}$ .

**Lemma 8.81** (Conditional i.i.d. local metrics are jointly local). *Let  $D_1, \dots, D_N$  be conditionally independent given  $h$ , and suppose that each  $D_j$  is local for  $h$ . Then  $D_1, \dots, D_N$  are jointly local for  $h$ .*

*Proof.* Fix an open  $V$ . Conditional on  $h$ , the metrics are independent, hence the inside internal metrics are conditionally independent from the outside internal metrics across distinct samples. Locality of each individual  $D_j$  says that, after conditioning on  $h|_{\overline{V}}$ , its inside internal metric is independent of the pair consisting of the exterior field and its own exterior internal metric. For bounded measurable test functions of all inside variables and all outside variables, repeatedly condition first on  $h$  and then on  $h|_{\overline{V}}$ . The conditional product factorization from the first conditioning and the individual locality factorization from the second give

$$\mathbf{E}[FG \mid h|_{\overline{V}}] = \mathbf{E}[F \mid h|_{\overline{V}}] \mathbf{E}[G \mid h|_{\overline{V}}],$$

where  $F$  is any bounded function of the inside metrics and  $G$  any bounded function of the exterior field and exterior metrics. A monotone-class argument gives the asserted joint conditional independence.  $\square$

**Lemma 8.82** (Harmonic good-ball tail). *For  $M > 0$  set*

$$\mathcal{G}_M(z, r) := \left\{ \sup_{w \in B_{15r/16}(z)} |\hat{h}_{z,r}(w)| \leq M \right\}.$$

There are universal  $c_1, c_2 > 0$  such that

$$\mathbf{P}[\mathcal{G}_M(z, r)^c] \leq c_1 e^{-c_2 M^2}$$

uniformly in  $z$  and  $r$ .

*Proof.* Translation and scale invariance reduce to  $z = 0, r = 1$ . On  $B_{31/32}(0)$  write the harmonic function using the Poisson kernel. The kernel is uniformly bounded when the evaluation point lies in  $B_{15/16}(0)$ , so

$$\sup_{B_{15/16}} |\widehat{h}_{0,1}| \leq C \int_{\partial B_{31/32}} |\widehat{h}_{0,1}(y)| dy.$$

Each boundary value in the last integral is a centered Gaussian with variance bounded uniformly in  $y$ . Jensen's inequality therefore gives, for some  $a > 0$ ,

$$\mathbf{E} \exp \left( a \sup_{B_{15/16}} |\widehat{h}_{0,1}|^2 \right) < \infty.$$

Markov's inequality yields the stated Gaussian tail. The identical argument applies to the harmonic part obtained by restricting a zero-boundary GFF to a smaller concentric disk; we use this form in the stopping-time proof below.  $\square$

**Lemma 8.83** (Cameron–Martin transfer against the zero-boundary reference). *Fix  $0 < s < s_0 < 1$ . Conditional on the outside sigma-field  $\mathcal{F}_{z,r}$ , the restriction of  $h - h_r(z)$  to  $B_{sr}(z)$  is mutually absolutely continuous with respect to the restriction to the same disk of a zero-boundary GFF in  $B_r(z)$ . If*

$$M_{s_0 r}^r(z) := \sup_{w \in B_{s_0 r}(z)} |\widehat{h}_{z,r}(w)|$$

and  $Z_{z,r}^0, W_{z,r}^0$  denote the reciprocal Radon–Nikodym densities, then on  $\{M_{s_0 r}^r(z) \leq M\}$ , for every  $q \in \mathbb{R}$ ,

$$\mathbf{E}[(Z_{z,r}^0)^q \mid \mathcal{F}_{z,r}] + \mathbf{E}[(W_{z,r}^0)^q \mid \mathcal{F}_{z,r}] \leq C(q, M, s, s_0).$$

*Proof.* Choose  $s < s_1 < s_0$  and a deterministic cutoff  $\phi$  supported in  $B_{s_0 r}(z)$  and equal to one on  $B_{sr}(z)$ , and put  $g = \phi \widehat{h}_{z,r}$ . Then  $\widetilde{h}_{z,r} + g = h - h_r(z)$  on  $B_{sr}(z)$ . On the displayed good event, interior harmonic gradient estimates imply

$$|\nabla \widehat{h}_{z,r}(w)| \leq C(M, s_1, s_0) r^{-1}, \quad w \in B_{s_1 r}(z),$$

so  $\|g\|_{\nabla}^2 \leq C(M, s, s_0)$  uniformly in  $z, r$ .

The Cameron–Martin formula changes the law of  $\widetilde{h}_{z,r}$  to that of  $\widetilde{h}_{z,r} + g$  with density

$$Z^0 = \exp \left( (\widetilde{h}_{z,r}, g)_{\nabla} - \frac{1}{2} \|g\|_{\nabla}^2 \right).$$

The reciprocal change has the analogous density with  $-g$ . Since  $(\widetilde{h}_{z,r}, g)_{\nabla}$  is Gaussian of variance  $\|g\|_{\nabla}^2$ , for every real  $q$ ,

$$\mathbf{E}[(Z^0)^q \mid \mathcal{F}_{z,r}] = \exp \left( \frac{q^2 - q}{2} \|g\|_{\nabla}^2 \right),$$

and the corresponding formula holds for the reciprocal density. Integrating out the annular field between  $B_{sr}(z)$  and the cutoff support and applying Jensen gives the same bounds for the restricted-field densities.  $\square$



**Lemma 8.84** (Conditional-to-marginal event transfer). *Let  $D_1, \dots, D_N$  be jointly local and  $\xi$ -additive for  $h$ , and let  $A$  be an event determined by*

$$\left( (h - h_r(z))|_{B_{sr}(z)}, \{e^{-\xi h_r(z)} D_j(\cdot, \cdot; B_{sr}(z))\}_{j=1}^N \right).$$

*For fixed  $0 < s < s_0 < 1$  and  $M > 0$  there is a function  $\pi_M : (0, 1) \rightarrow (0, 1)$  such that on  $\{M_{s_0 r}^r(z) \leq M\}$ ,*

$$\mathbf{P}[A \mid \mathcal{F}_{z,r}] \geq \pi_M(\mathbf{P}[A]) \quad a.s.$$

*Moreover  $\pi_M(p) > 0$  for every  $p > 0$ , and for fixed  $M$ ,  $\pi_M(p) \uparrow 1$  as  $p \uparrow 1$ .*

*Proof.* By joint locality, conditional on the inside field the law of the inside internal metrics is independent of  $\mathcal{F}_{z,r}$ . Hence it is enough to compare field laws; the same regular conditional kernel for the metrics can then be attached to each field law.

Let  $Q$  be the reference law obtained from a zero-boundary GFF in  $B_r(z)$ , restricted to  $B_{sr}(z)$ , with this metric kernel attached. Lemma 8.83 and Hölder give, on the  $M$ -good event, for every conjugate pair  $q, q' > 1$ ,

$$\mathbf{P}[A \mid \mathcal{F}_{z,r}] \leq C_M Q(A)^{1/q'}, \quad Q(A) \leq C_M \mathbf{P}[A \mid \mathcal{F}_{z,r}]^{1/q'}. \quad (20.25)$$

To compare  $Q$  with the marginal law, let  $\mathcal{F}'_{z,r}$  be an independent copy of the outside data. By Lemma 8.82, for each fixed  $M_1$  the event  $\{M_{s_0 r}^r(z) \leq M_1\}$  has probability  $\rho_{M_1} > 0$ , with  $\rho_{M_1} \uparrow 1$ . Averaging the second inequality in (20.25) over the copy on this event yields

$$\mathbf{P}[A] \geq c(M_1) Q(A)^{q'}. \quad (20.26)$$

Thus a small marginal probability forces a small  $Q$ -probability. Applying this to  $A^c$  and then the first inequality in (20.25) shows that, on the  $M$ -good event,

$$\mathbf{P}[A^c \mid \mathcal{F}_{z,r}] \leq C(M, M_1) \mathbf{P}[A^c]^\eta$$

for some  $\eta > 0$ . This gives the assertion as  $\mathbf{P}[A] \uparrow 1$ .

For a fixed lower bound  $\mathbf{P}[A] \geq p > 0$ , choose  $M_1$  so large that  $1 - \rho_{M_1} < p/2$ . Averaging the first inequality in (20.25) over the good copy and using the trivial bound one on the complement gives

$$p \leq \mathbf{P}[A] \leq C(M_1) Q(A)^{1/q'} + (1 - \rho_{M_1}),$$

so  $Q(A) \geq c(p, M_1) > 0$ . The second inequality in (20.25), now applied to the actual  $M$ -good outside realization, gives  $\mathbf{P}[A \mid \mathcal{F}_{z,r}] \geq c(p, M, M_1) > 0$ . These two estimates define a function  $\pi_M$  with the claimed properties.  $\square$

**Lemma 8.85** (Exponential frequency of regular scales). *Fix  $s \in (0, 1)$  and let  $r_{k+1}/r_k \leq s$ . For  $s_0 \in (0, 1)$  define  $N_M(K)$  to be the number of  $1 \leq k \leq K$  such that  $M_{s_0 r_k}^{r_k}(z) \leq M$ . For every  $a > 0$  and  $b \in (0, 1)$  there are  $M = M(a, b, s, s_0)$  and  $C < \infty$  such that*

$$\mathbf{P}[N_M(K) \leq bK] \leq C e^{-aK}, \quad K \geq 1.$$

*Proof.* Translation and scaling reduce to a fixed starting ball. Explore the field from outside in. At the current radius  $R$ , let  $X$  be the oscillation of the current harmonic part on  $B_{s_0 R}$ . If  $X \leq M$ , the scale is regular. Otherwise the interior harmonic derivative estimate shows that after shrinking the radius by a factor at most  $(2 + X)^{-c}$ , the old harmonic contribution has oscillation bounded by a deterministic constant. Since  $r_{k+1}/r_k \leq s < 1$ , at most  $C_s \lceil \log(2 + X) \rceil$  further indices are needed to

achieve this amount of shrinkage. The newly revealed zero-boundary contribution is independent of the outside sigma-field and, by the zero-boundary version of Lemma 8.82, has a Gaussian oscillation tail.

Consequently the waiting time  $T$  from one trial to the next is dominated by a random variable with an exponential moment of every fixed positive order, and the conditional chance that the next trial is  $M$ -regular is at least  $p_M$ , where  $p_M \uparrow 1$ . Iterating the strong Markov/domain-Markov construction shows that the successive waiting times between regular scales are stochastically dominated by independent copies of a variable  $T_M$  whose Laplace transform tends to one as  $M \rightarrow \infty$ . Therefore

$$\mathbf{P}[N_M(K) \leq bK] \leq \mathbf{P}\left[\sum_{j=1}^{\lfloor bK \rfloor} T_{M,j} \geq K\right] \leq e^{-\lambda K} \mathbf{E}[e^{\lambda T_M}]^{bK}.$$

Given  $a, b$ , choose  $\lambda > a$  and then  $M$  so large that  $b \log \mathbf{E}[e^{\lambda T_M}] < \lambda - a$ . This proves the estimate, after enlarging the prefactor for bounded  $K$ .  $\square$

#### 8.4.2 Near-independence and annular amplification

Let  $0 < s_1 < s_2 < 1$ , let  $r_{k+1}/r_k \leq s_1$ , and let  $E_k$  be an event determined by the normalized GFF and a finite family of jointly local,  $\xi$ -additive metrics in the annulus  $A_{s_1 r_k, s_2 r_k}(0)$ .

**Lemma 8.86** (Near-independence across concentric annuli). *There is a universal amplification principle with the following consequence. For each  $a > 0$  and  $b \in (0, 1)$  there is  $p_* = p_*(a, b, s_1, s_2) < 1$  such that if*

$$\mathbf{P}[E_k] \geq p_*, \quad k \geq 1,$$

*then, writing  $N(K) = \#\{k \leq K : E_k \text{ occurs}\}$ ,*

$$\mathbf{P}[N(K) < bK] \leq C e^{-aK}.$$

*More generally, for every fixed  $p > 0$  there are  $b = b(p, s_1, s_2) > 0$  and  $a = a(p, s_1, s_2) > 0$  for which the same bound holds.*

*Proof.* Choose  $s_0 \in (s_2, 1)$ . Call scale  $r_k$   $M$ -regular if the harmonic part of the outside-in GFF decomposition has oscillation at most  $M$  on  $B_{s_0 r_k}(0)$ . Lemma 8.85 applies after replacing the fixed numerical radii by  $s_0$ ; only constants change. Therefore, with an exponentially small exceptional probability, a positive fraction arbitrarily close to one of the first  $K$  scales are  $M$ -regular when  $M$  is large.

At an  $M$ -regular scale, Lemma 8.84 applies to the annular event  $E_k$  (use  $s_0 \in (s_2, 1)$ ). Hence there is  $p_M = p_M(p, M, s_1, s_2) > 0$  such that

$$\mathbf{P}[E_k \mid \mathcal{F}_{0, r_k}] \geq p_M$$

on each regular scale, and  $p_M \uparrow 1$  as the marginal lower bound  $p$  tends to one with  $M$  fixed.

Enumerate the regular scales by stopping times. Since  $E_k$  is measurable before the exploration reaches the next admissible radius, the conditional lower bound couples the indicators of the  $E_k$ 's at these stopping times from below by independent Bernoulli( $p_M$ ) variables. A Chernoff bound controls the number of failures among the regular scales; Lemma 8.85 controls the shortage of regular scales. Combining the two exponential estimates gives the asserted bound. For merely positive  $p$ , keep  $M$  fixed so that  $p_M > 0$  and choose a smaller positive density  $b$ .  $\square$

For two jointly local,  $\xi$ -additive metrics  $D, \tilde{D}$ , and for  $B_{2r}(z) \Subset U$ , define the annular comparison event

$$\mathcal{E}_r(z) := \left\{ \sup_{u,v \in \partial B_r(z)} \tilde{D}(u,v; A_{r/2,2r}(z)) \leq C D(\partial B_{r/2}(z), \partial B_r(z)) \right\}.$$

**Lemma 8.87** (Good-annulus covering). *There is a universal  $p_0 < 1$  such that the following holds. Suppose that for all  $z$  in a compact  $K \Subset U$  and all sufficiently small  $r$ ,  $\mathbf{P}[\mathcal{E}_r(z)] \geq p_0$ . For  $\varepsilon > 0$ , with probability tending to one as  $\varepsilon \rightarrow 0$ , every  $z \in K$  belongs to a ball  $B_{r/2}(w)$  for which  $\mathcal{E}_r(w)$  occurs, where*

$$\varepsilon^2 \leq r \leq \varepsilon, \quad r \in \{2^{-k}\varepsilon : k \geq 0\}, \quad w \in (\varepsilon^2 \mathbb{Z}^2/4) \cap B_\varepsilon(K).$$

*Proof.* For a fixed grid point  $w$ , apply Lemma 8.86 to the dyadic annuli centered at  $w$  between radii  $\varepsilon$  and  $\varepsilon^2$ . Choose  $p_0$  so close to one that the probability that none of these  $O(|\log \varepsilon|)$  annuli is good is  $O(\varepsilon^q)$  for some  $q > 5$ . There are  $O_K(\varepsilon^{-4})$  grid points under consideration, so a union bound shows that, with probability tending to one, every grid point has a good radius in the required range. The radius- $\varepsilon^2/2$  balls centered at the grid cover  $K$ , proving the statement.  $\square$

**Lemma 8.88** (Circle-chain geometry). *Let  $P$  be a connected path segment covered by finitely many open Euclidean balls, none of radius larger than  $\varepsilon$ . There is a subcollection minimal for covering  $P$  such that the union of the boundary circles of the balls in this subcollection is connected. If the covered path starts within  $2\varepsilon$  of  $z_1$  and ends within  $2\varepsilon$  of  $z_2$ , this union contains a path from  $B_{2\varepsilon}(z_1)$  to  $B_{2\varepsilon}(z_2)$ .*

*Proof.* Take a subcollection  $\mathcal{B}$  minimal under deletion among the balls which cover  $P$ . Since  $P$  is connected and every ball in a minimal cover meets  $P$ , the union  $\bigcup_{B \in \mathcal{B}} B$  is connected. Minimality also implies that no ball in  $\mathcal{B}$  is properly contained in another.

Suppose  $\bigcup_{B \in \mathcal{B}} \partial B$  were disconnected. Partition  $\mathcal{B} = \mathcal{B}_1 \sqcup \mathcal{B}_2$  so the two unions of boundary circles are disjoint. No ball in  $\mathcal{B}_2$  can be contained in the union of  $\mathcal{B}_1$ , by minimality. A ball in  $\mathcal{B}_2$  also cannot meet both the union of  $\mathcal{B}_1$  and its complement, since then its boundary would intersect the boundary of one of the balls of  $\mathcal{B}_1$ . Therefore the unions of the balls in  $\mathcal{B}_1$  and  $\mathcal{B}_2$  are disjoint, contradicting connectedness. Hence the union of the boundary circles is connected. Since the balls cover the path and all have radius at most  $\varepsilon$ , this connected boundary union meets both endpoint  $2\varepsilon$ -neighborhoods, which proves the last assertion.  $\square$

**Theorem 8.89** (Bi-Lipschitz comparison from annular events). *Let  $h$  be a whole-plane GFF, and let  $D, \tilde{D}$  be jointly local,  $\xi$ -additive continuous length metrics on an open set  $U$ . There is a universal  $p_0 < 1$  such that if for every compact  $K \Subset U$  the event  $\mathcal{E}_r(z)$  above has probability at least  $p_0$  uniformly for  $z \in K$  and all sufficiently small  $r$ , then almost surely*

$$\tilde{D}(z, w) \leq C D(z, w), \quad z, w \in U.$$

*Proof.* Fix  $z_1, z_2$  in a countable dense subset of  $U$  and  $\delta > 0$ . Choose a measurable path  $P$  from  $z_1$  to  $z_2$  of  $D$ -length at most  $D(z_1, z_2) + \delta$ . With probability at least  $1 - \delta$ , its range lies in some deterministic compact  $K \Subset U$ . On the good-cover event of Lemma 8.87, set  $t_0 = 0$  and inductively let  $t_j$  be the first later time at which  $P$  exits a good ball  $B_{r_j}(w_j)$  whose inner half-ball contains  $P(t_{j-1})$ . Stop when the remaining endpoint is within  $2\varepsilon$ . During  $[t_{j-1}, t_j]$  the path crosses from  $\partial B_{r_j/2}(w_j)$  to  $\partial B_{r_j}(w_j)$ , so

$$D(\partial B_{r_j/2}(w_j), \partial B_{r_j}(w_j)) \leq t_j - t_{j-1}.$$

Lemma 8.88, applied to a deletion-minimal subcollection of these balls, shows that their outer circles contain a connected chain from  $B_{2\varepsilon}(z_1)$  to  $B_{2\varepsilon}(z_2)$ . The annular comparison event on each retained ball therefore gives, by the triangle inequality,

$$\tilde{D}(B_{2\varepsilon}(z_1), B_{2\varepsilon}(z_2)) \leq C \sum_j (t_j - t_{j-1}) \leq C(D(z_1, z_2) + \delta).$$

Continuity of  $\tilde{D}$  makes the left side tend to  $\tilde{D}(z_1, z_2)$  as  $\varepsilon \rightarrow 0$ . The error vanishes because  $\tilde{D}$  is continuous on  $K$  and all auxiliary Euclidean diameters tend to zero. Send  $\varepsilon \rightarrow 0$ , then  $\delta \rightarrow 0$ . The good-cover probability tends to one, so the inequality holds almost surely for the fixed pair. Intersect the full-probability events over the dense countable set and use continuity of both metrics to obtain the result for all pairs simultaneously.  $\square$

### 8.4.3 Square resampling and zero conditional variance

**Lemma 8.90** (Conditional support envelope). *Suppose  $D$  is local for  $h$  and, conditional on  $h$ , two i.i.d. samples  $D, \tilde{D}$  satisfy  $\tilde{D} \leq C_h D$  pointwise, where  $C_h < \infty$  is  $h$ -measurable. For every connected open  $V \subseteq U$  and  $z, w \in V$ ,*

$$C_h^{-1} \mathbf{E}[D(z, w; V) \mid h] \leq D(z, w; V) \leq C_h \mathbf{E}[D(z, w; V) \mid h]$$

*almost surely.*

*Proof.* Condition on  $h$ . Let  $m_h$  and  $M_h$  be the essential infimum and supremum of the conditional law of  $D(z, w; V)$ . The i.i.d. comparison implies  $M_h \leq C_h m_h$ : otherwise two independent samples would with positive probability violate the comparison. Since the conditional expectation lies between  $m_h$  and  $M_h$ , every sample lies between  $C_h^{-1} \mathbf{E}[D \mid h]$  and  $C_h \mathbf{E}[D \mid h]$ .  $\square$

Let  $\theta$  be independent and uniform on  $[0, 1]^2$ . For  $\varepsilon > 0$ , let  $\mathcal{S}_\theta^\varepsilon$  be the interiors of the squares of side  $\varepsilon$  in the shifted grid  $\varepsilon(\mathbb{Z}^2 + \theta)$ , intersected with  $U$ .

**Lemma 8.91** (The random grid carries zero metric length). *Let  $P$  be a finite- $D$ -length path chosen measurably from  $(h, D)$  and not from  $\theta$ . Then for every fixed  $\varepsilon > 0$ ,*

$$\text{len}(P \setminus \cup_{S \in \mathcal{S}_\theta^\varepsilon} S; D) = 0 \quad a.s.$$

*Equivalently, the set of path times at which  $P$  lies on the shifted grid has zero one-dimensional Lebesgue measure when  $P$  is parameterized by  $D$ -arclength.*

*Proof.* Parameterize  $P$  by  $D$ -length. For every deterministic time  $t$ , conditional on  $P$ , the probability over the independent shift  $\theta$  that  $P(t)$  lies on the shifted grid is zero. Fubini's theorem therefore shows that almost surely the set of such times has Lebesgue measure zero. Metric arclength of a subset of an arclength-parameterized path equals the Lebesgue measure of its time set, giving the claim.  $\square$

**Lemma 8.92** (Reconstruction from square internal metrics). *Assume the hypotheses of Lemma 8.90. For fixed  $\varepsilon$ , the full metric  $D$  is almost surely determined by  $h$ ,  $\theta$ , and the collection of internal metrics*

$$\{D(\cdot, \cdot; S) : S \in \mathcal{S}_\theta^\varepsilon\}.$$

*Proof.* Condition on these data and take two conditionally independent compatible metrics  $D, D'$ . The support-envelope lemma, first for rational pairs and then by continuity, gives a deterministic conditional bi-Lipschitz comparison  $C_h^{-2}D \leq D' \leq C_h^2D$ . Choose a near-minimal  $D$ -path  $P$  between a fixed rational pair. By Lemma 8.91, its  $D$ -length is the sum of its lengths inside the square interiors. On each square these lengths are the same for  $D$  and  $D'$  by the conditioned data. Approximate the portions near the grid by a vanishing Euclidean neighborhood. The bi-Lipschitz comparison and continuity make the resulting error vanish. Hence  $D'(z, w) \leq D(z, w)$ ; symmetry gives equality. Apply to all rational pairs and use continuity.  $\square$

**Lemma 8.93** (Conditional independence of square metrics). *Conditional on  $(h, \theta)$ , the internal metrics  $\{D(\cdot, \cdot; S) : S \in \mathcal{S}_\theta^\varepsilon\}$  are independent.*

*Proof.* Condition on  $\theta$ . The square interiors are pairwise disjoint open sets. Apply locality successively to a finite subcollection: the internal metric in one square is conditionally independent of the field and internal metrics in the complement given the field in that square. Conditioning on the full field  $h$  removes the intermediate field restrictions and yields product factorization. The finite-dimensional statement for every finite subcollection is equivalent to conditional independence of the countable family.  $\square$

**Theorem 8.94** (Bi-Lipschitz determination implies field measurability). *Let  $D$  be a continuous local length metric of a GFF  $h$  on  $U$ . Suppose that, conditional on  $h$ , i.i.d. samples  $D, \tilde{D}$  satisfy  $\tilde{D} \leq C_h D$  for an almost surely finite  $h$ -measurable random constant  $C_h$ . Then  $D$  is almost surely determined by  $h$ .*

*Proof.* Fix a bounded connected  $V \Subset U$  and rational  $z, w \in V$ . We show  $\text{Var}(D(z, w; V) \mid h) = 0$ . Introduce the independent shifted  $\varepsilon$ -grid above. By Lemmas 8.92 and 8.93, conditional on  $(h, \theta)$  the variable  $D(z, w; V)$  is a function of independent square-internal metrics. For each square  $S$  intersecting  $V$ , resample only its internal metric and call the reconstructed global metric  $D^S$ . Conditional Efron–Stein gives

$$\text{Var}(D(z, w; V) \mid h, \theta) \leq \frac{1}{2} \sum_S \mathbf{E}[(D^S(z, w; V) - D(z, w; V))^2 \mid h, \theta].$$

Choose a measurable path  $P$  from  $z$  to  $w$  with  $\text{len}(P; D) \leq D(z, w; V) + \varepsilon^2$ . The support-envelope lemma applies equally to  $D^S$ , since  $(h, D^S)$  has the same law as  $(h, D)$ . Replacing the excursions of  $P$  through  $S$  by paths measured in the resampled internal metric and using the resulting  $C_h^2$  bi-Lipschitz comparison yields

$$(D^S(z, w; V) - D(z, w; V))_+ \leq C_h^2 \text{len}(P \cap S; D) + \varepsilon^2.$$

The same estimate after interchanging  $D$  and  $D^S$  controls the negative part. Consequently

$$\text{Var}(D(z, w; V) \mid h, \theta) \leq C'_h \mathbf{E}[D(z, w; V)^2 \mid h, \theta]^{1/2} \mathbf{E} \left[ \max_S \text{len}(P \cap S; D)^2 \mid h, \theta \right]^{1/2} + o(1).$$

Here  $C'_h < \infty$  is  $h$ -measurable; the harmless square-count contribution from the  $\varepsilon^2$  error is  $O(\varepsilon^2)$ .

The range of  $P$  is compact. If  $P$  enters a square of side  $\varepsilon$ , a near-minimality shortcut shows

$$\text{len}(P \cap S; D) \leq \sup_{u, v \in \bar{S}} D(u, v) + \varepsilon^2.$$

Since  $D$  is a continuous metric inducing the Euclidean topology, the maximum of these diameters over squares meeting the compact path tends to zero almost surely. The support-envelope lemma supplies, for almost every fixed field  $h$ , a finite  $h$ -measurable bound  $C_h \mathbf{E}[D(z, w; V) \mid h] + 1$  for the near-minimal path length and hence for every square contribution. Thus the conditional expectations are dominated pointwise in the conditional probability space; bounded convergence sends the right side to zero. No unconditional moment assumption is being inserted here. Thus

$$\text{Var}(D(z, w; V) \mid h, \theta) = 0.$$

Averaging over the independent  $\theta$  gives zero conditional variance given  $h$ . Hence each rational internal distance is  $h$ -measurable. Letting  $V$  exhaust  $U$  and using continuity of  $D$  shows that the entire metric is a measurable function of  $h$ .  $\square$

**Corollary 8.95** (Local-metric criterion). *Let  $D$  be a  $\xi$ -additive local metric of a whole-plane GFF. Conditional on  $h$ , let  $D, \tilde{D}$  be i.i.d. samples. Suppose that the annular event*

$$\sup_{u, v \in \partial B_r(z)} \tilde{D}(u, v; A_{r/2, 2r}(z)) \leq CD(\partial B_{r/2}(z), \partial B_r(z))$$

*has probability at least the universal constant  $p_0$  of Theorem 8.89, uniformly on compact sets and all sufficiently small scales. Then  $D$  is almost surely determined by  $h$ .*

*Proof.* Lemma 8.81 makes the conditional i.i.d. samples jointly local. Theorem 8.89 gives the bi-Lipschitz comparison required by Theorem 8.94. The latter theorem yields the conclusion.  $\square$

## 8.5 Deterministic scale normalization for weak LQG metrics

Fix a subsequence  $\mathcal{E} \ni \epsilon \rightarrow 0$  along which the normalized field/metric pair converges,

$$(h, a_\epsilon^{-1} D_h^\epsilon) \Longrightarrow (h, D_h),$$

where  $h$  is normalized by  $h_1(0) = 0$ . For  $r \neq 0$  set

$$h^r := h(r \cdot) - h_r(0).$$

Then  $h^r \stackrel{d}{=} h$ , and heat-kernel scaling gives the exact identity

$$D_{h^r}^{\epsilon/r}(u, v) = r^{-1} e^{-\xi h_r(0)} D_h^\epsilon(ru, rv).$$

**Lemma 8.96** (Existence of the deterministic scale constants). *For every  $r \neq 0$  the limit*

$$c_r := \lim_{\mathcal{E} \ni \epsilon \rightarrow 0} \frac{r a_{\epsilon/r}}{a_\epsilon}$$

*exists in  $(0, \infty)$ . Moreover, for each fixed  $r$ ,*

$$c_r^{-1} e^{-\xi h_r(0)} D_h(r \cdot, r \cdot)$$

*has a law belonging to the closure of the normalized LFPP laws.*

*Proof.* The scaling identity implies

$$a_\epsilon^{-1} D_{h^r}^{\epsilon/r} = r^{-1} e^{-\xi_{h^r}(0)} a_\epsilon^{-1} D_h^\epsilon(r \cdot, r \cdot) \implies r^{-1} e^{-\xi_{h^r}(0)} D_h(r \cdot, r \cdot).$$

On the other hand,

$$a_{\epsilon/r}^{-1} D_h^{\epsilon/r} \stackrel{d}{=} \frac{a_\epsilon}{a_{\epsilon/r}} a_\epsilon^{-1} D_{h^r}^{\epsilon/r}.$$

The first factor on the left is tight and every subsequential limit is a genuine metric by the metric-tightness theorem. Hence along any further subsequence on which the ratio  $a_{\epsilon/r}/a_\epsilon$  converges, its limit is a finite positive number. Let  $m_r$  denote the median left-right crossing distance for the limiting metric  $r^{-1} e^{-\xi_{h^r}(0)} D_h(r \cdot, r \cdot)$ . The crossing median of  $a_{\epsilon/r}^{-1} D_h^{\epsilon/r}$  is exactly one, whereas the crossing median of the second normalized metric above converges to  $m_r$ . The LFPP crossing length has a continuous distribution (condition on all field modes but one smooth Gaussian mode which changes every path length strictly), so the limiting median normalization is unambiguous. Portmanteau therefore forces every subsequential limit of  $a_{\epsilon/r}/a_\epsilon$  to equal  $m_r$ . The full ratio converges and  $c_r = r m_r$ . Substituting the limit into the preceding scaling identity gives the last assertion.  $\square$

**Lemma 8.97** (Polynomial comparison of scale constants). *There is a deterministic  $\Lambda > 1$ , depending only on the fixed subsequence and  $\gamma$ , such that for every  $r \geq 0$  and every  $\delta \in (0, 1)$ ,*

$$\Lambda^{-1} \delta^\Lambda \leq \frac{c_{\delta r}}{c_r} \leq \Lambda \delta^{-\Lambda}.$$

*Proof.* From the definition of the constants,

$$\frac{c_{\delta r}}{c_r} = \delta \lim_{\epsilon \ni \epsilon \rightarrow 0} \frac{a_{\epsilon/(\delta r)}}{a_{\epsilon/r}}.$$

Take first  $\delta = 2^{-k}$ . The weak multiplicativity and exponent bounds owned in (20.61)–(20.62) show, uniformly in the base scale  $n$ ,

$$e^{-C\sqrt{k}} 2^{-Ck} \leq \frac{a_{2^k \eta}}{a_\eta} \leq e^{C\sqrt{k}} 2^{Ck}$$

whenever both scales are sufficiently small. (The constants absorb the uniform comparison between the heat-kernel crossing median  $a_\eta$  and the star-scale median  $\lambda_\eta$ .) Passing to the defining limit for  $c_{2^{-k}r}/c_r$  yields two-sided bounds by  $2^{\pm C'k}$ . For arbitrary  $\delta$ , choose  $k$  with  $2^{-k-1} < \delta \leq 2^{-k}$ . Proposition 8.64 compares the crossing normalizations at scales whose ratio lies in  $[1, 2]$  by a deterministic constant. Enlarging  $C'$  and then choosing  $\Lambda > 1 + C'$  gives the displayed polynomial bounds simultaneously for all  $r$  and  $\delta$ .  $\square$

**Theorem 8.98** (Weak-metric scale tightness). *The family of laws*

$$\left\{ c_r^{-1} e^{-\xi_{h^r}(0)} D_h(r \cdot, r \cdot) : r > 0 \right\}$$

*is tight in the local-uniform topology. Every law in its Prokhorov closure is supported on continuous genuine metrics inducing the Euclidean topology, and the constants  $c_r$  satisfy the polynomial comparison in Lemma 8.97.*

*Proof.* By Lemma 8.96, for each fixed  $r$  the displayed metric is a subsequential limit, as  $\epsilon \rightarrow 0$ , of

$$a_{\epsilon/r}^{-1} D_{hr}^{\epsilon/r}.$$

Since  $h^r \stackrel{d}{=} h$ , all of these approximating laws belong to the single tight family already established by the metric-tightness theorem. The closure of that family contains only continuous genuine Euclidean-topology metrics. Hence the same is true uniformly in  $r$ . The deterministic scale comparison is Lemma 8.97.  $\square$

## 8.6 From additive locality to the weak-metric axioms

The preceding sections deliberately separated the abstract local-metric theorem from its LFPP application. We now verify the annular hypothesis for the actual subsequential metric, record translation covariance, and convert joint convergence to convergence in probability once the metric is determined by the field.

**Lemma 8.99** (Stability of annular crossing functionals). *Let  $M_n$  be continuous length metrics on  $\mathbb{C}$  which converge locally uniformly to a continuous length metric  $M$  inducing the Euclidean topology. Then*

$$\inf_n^* M_n(\partial B_{1/2}, \partial B_1) > 0$$

*in the sense that the crossing distances are bounded away from zero for all large  $n$ , and*

$$\sup_{n \geq n_0} \sup_{u, v \in \partial B_1} M_n(u, v; B_2 \setminus \overline{B_{1/2}}) < \infty$$

*for some  $n_0$ .*

*Proof.* The first assertion is immediate from uniform convergence on  $\overline{B_1} \times \overline{B_1}$ : the distance between the two disjoint compact circles for  $M$  is strictly positive.

For the second assertion, put  $A = B_2 \setminus \overline{B_{1/2}}$  and choose a slightly smaller annulus  $A' = B_{3/2} \setminus \overline{B_{3/4}}$  containing  $\partial B_1$ . Fix  $x \in \partial B_1$ . Since  $M$  induces the Euclidean topology,

$$d_x := M(x, \mathbb{C} \setminus A') > 0.$$

Choose a Euclidean neighbourhood  $U_x \Subset A'$  on which  $M(x, y) < d_x/10$ . For all large  $n$ , local uniform convergence gives  $M_n(x, y) < d_x/5$  on  $U_x$  and  $M_n(x, \mathbb{C} \setminus A') > d_x/2$ . If  $y \in U_x$ , take an  $M_n$ -path from  $x$  to  $y$  of length less than  $M_n(x, y) + d_x/20$ . Such a path cannot leave  $A'$ , since leaving would already cost at least  $d_x/2$ . Consequently

$$M_n(x, y; A) \leq d_x/4$$

for all large  $n$  and  $y \in U_x$  (shrinking  $U_x$  once if necessary).

Cover  $\partial B_1$  by finitely many such  $U_{x_j}$  and refine them along the circle so consecutive sets overlap. Any two points of the circle can then be joined by a chain through at most this fixed finite number of sets. The preceding internal-distance bounds and the triangle inequality give a finite uniform bound for their  $A$ -internal  $M_n$ -distance. Taking the maximum of the finitely many constants completes the proof.  $\square$



**Lemma 8.100** (Uniform annulus quantiles from scale tightness). *Let  $(h, D_h)$  be a subsequential LFPP limit for a whole-plane GFF normalized by  $h_1(0) = 0$ , and let  $c_r$  be the constants of Theorem 8.98. For every  $q \in (0, 1)$  there is a deterministic  $C_q > 1$  such that, simultaneously for every deterministic  $z \in \mathbb{C}$  and  $r > 0$ ,*

$$\mathbf{P}\left[D_h(\partial B_{r/2}(z), \partial B_r(z)) \geq C_q^{-1} c_r e^{\xi h_r(z)}\right] \geq 1 - q$$

and

$$\mathbf{P}\left[\sup_{u, v \in \partial B_r(z)} D_h(u, v; B_{2r}(z) \setminus \overline{B_{r/2}(z)}) \leq C_q c_r e^{\xi h_r(z)}\right] \geq 1 - q.$$

*Proof.* First take  $z = 0$ . Write

$$M_r := c_r^{-1} e^{-\xi h_r(0)} D_h(r \cdot, r \cdot).$$

The laws of  $M_r$ ,  $r > 0$ , form a tight family whose Prokhorov closure is supported on continuous length metrics inducing the Euclidean topology, by Theorem 8.98. Define

$$X(M) := M(\partial B_{1/2}, \partial B_1), \quad Y(M) := \sup_{u, v \in \partial B_1} M(u, v; B_2 \setminus \overline{B_{1/2}}).$$

If the asserted uniform lower quantile for  $X$  failed, we could find  $r_n$  for which  $\mathbf{P}[X(M_{r_n}) < 1/n] \geq q$ . Tightness gives a weakly convergent subsequence of laws. Under a Skorokhod representation the metrics converge locally uniformly almost surely to a metric in the stated closure. Lemma 8.99 then makes the crossing distances eventually bounded away from zero, a contradiction.

If the uniform upper quantile for  $Y$  failed, choose  $r_n$  with  $\mathbf{P}[Y(M_{r_n}) > n] \geq q$  and use the same Skorokhod representation. The second assertion of Lemma 8.99 makes the annular internal boundary diameters eventually bounded, again a contradiction. Enlarging one deterministic constant gives both estimates.

For general deterministic  $z$ , use the exact approximating translation identity

$$D_{h(\cdot+z)}^\epsilon(u, v) = D_h^\epsilon(u+z, v+z). \quad (20.27)$$

The whole-plane GFF is translation invariant modulo an additive constant, and adding a constant multiplies LFPP distances by the corresponding factor  $e^{\xi c}$ . Passing (20.27) through the joint subsequential law therefore shows that the centered metric  $c_r^{-1} e^{-\xi h_r(z)} D_h(z+r\cdot, z+r\cdot)$  has the same law as the centered metric at the origin. The same  $C_q$  consequently works for every  $z$  and  $r$ .  $\square$

**Proposition 8.101** (The annular local-metric criterion holds for LFPP limits). *Conditional on  $h$ , let  $D_h$  and  $\tilde{D}_h$  be i.i.d. samples from the conditional law of a subsequential LFPP limit. Then there is a deterministic  $C > 1$  such that the annular event in Corollary 8.95 has probability at least its universal threshold  $p_0$ , uniformly in deterministic  $z \in \mathbb{C}$  and  $r > 0$ .*

*Proof.* Let  $p_0$  be the universal threshold from Theorem 8.89 and choose  $q < (1 - p_0)/2$ . Apply Lemma 8.100 with this  $q$  and constant  $C_q$ . With probability at least  $1 - q$ , the sample  $D_h$  has radial crossing distance at least  $C_q^{-1} c_r e^{\xi h_r(z)}$ . With probability at least  $1 - q$ , the sample  $\tilde{D}_h$  has annular boundary diameter at most  $C_q c_r e^{\xi h_r(z)}$ . No independence is needed for the union bound, so both events occur with probability at least  $1 - 2q > p_0$ . On their intersection,

$$\sup_{u, v \in \partial B_r(z)} \tilde{D}_h(u, v; B_{2r}(z) \setminus \overline{B_{r/2}(z)}) \leq C_q^2 D_h(\partial B_{r/2}(z), \partial B_r(z)).$$

Thus the criterion holds with deterministic constant  $C = C_q^2$ .  $\square$

**Theorem 8.102** (Field determination and internal locality of LFPP limits). *Every subsequential LFPP metric  $D_h$  for a whole-plane GFF is almost surely a measurable function of  $h$ . Consequently, for every open  $U \subset \mathbb{C}$ , the internal metric  $D_h(\cdot, \cdot; U)$  is almost surely determined by  $h|_U$ . The same assertion holds after adding a bounded continuous function to the field.*

*Proof.* Proposition 8.79 says that  $D_h$  is  $\xi$ -additive local. Proposition 8.101 verifies the remaining hypothesis of Corollary 8.95. Hence  $D_h$  is almost surely determined by  $h$ .

We next upgrade conditional locality to deterministic field locality. Fix  $U' \Subset U$ . Since  $D_h(\cdot, \cdot; U')$  is a measurable function of the full pair  $(h, D_h)$  and is local for  $h$ , its conditional law given  $h|_U$  is almost surely a point mass: indeed  $D_h$  is already a function of  $h$ , while locality says that the internal metric in  $U'$  is conditionally independent of the exterior data given  $h|_U$ . Therefore it is measurable with respect to  $h|_U$ . Exhaust  $U$  by a countable increasing sequence of such  $U'$  and use monotone convergence of internal distances to obtain the assertion for  $U$ .

If  $\mathbf{h} = h + f$  with bounded continuous  $f$ , Lemma 8.77 gives  $D_{\mathbf{h}} = e^{\xi f} \cdot D_h$ . Since  $f$  is part of the prescribed field  $\mathbf{h}$  and the localized LFPP construction depends only on the restriction of  $\mathbf{h}$  to a shrinking neighbourhood, the same exhaustion argument gives locality and measurability for  $\mathbf{h}$ .  $\square$

**Lemma 8.103** (Graph-valued joint convergence implies convergence in probability). *Let  $X$  take values in a Polish space  $E$ , let  $Y_n, Y$  take values in a Polish space  $F$ , and suppose  $(X, Y_n) \Rightarrow (X, Y)$ . If  $Y = \Phi(X)$  almost surely for a measurable  $\Phi : E \rightarrow F$ , then  $Y_n \rightarrow Y$  in probability.*

*Proof.* By Lusin's theorem, for every  $\eta > 0$  there is a compact set  $K \subset E$  with  $\mathbf{P}[X \in K] > 1 - \eta$  on which  $\Phi$  is continuous. Extend  $\Phi|_K$  to a Borel map if necessary and consider the closed sets

$$F_{\delta, K} := \{(x, y) : x \in K, d_F(y, \Phi(x)) \geq \delta\}.$$

They are disjoint from the support of the limiting law on  $\{(x, \Phi(x)) : x \in K\}$ . Portmanteau gives

$$\limsup_n \mathbf{P}[X \in K, d_F(Y_n, \Phi(X)) \geq \delta] = 0.$$

Hence the limsup of  $\mathbf{P}[d_F(Y_n, Y) \geq \delta]$  is at most  $\eta$ . Let  $\eta \downarrow 0$ .  $\square$

**Lemma 8.104** (Exact translation covariance passes to the limit). *For every deterministic  $z_0 \in \mathbb{C}$ , the field-measurable version of a subsequential LFPP metric satisfies almost surely*

$$D_{h(\cdot + z_0)}(u, v) = D_h(u + z_0, v + z_0), \quad u, v \in \mathbb{C}.$$

*The same identity holds for a whole-plane GFF plus a bounded continuous function.*

*Proof.* At every  $\epsilon > 0$ , heat-kernel convolution commutes with translation, so (20.27) holds pathwise for all  $u, v$ . By Theorem 8.102 and Lemma 8.103, along the selected subsequence the normalized LFPP metrics for  $h$  and for the translated field converge in probability to their field-measurable limits. Translating the first metric in space is continuous for the local-uniform topology. Passing the exact identity to the probability limit gives the displayed equality first on rational pairs and then everywhere by continuity. Adding a bounded continuous function does not change the argument, since the approximating identity remains exact.  $\square$

**Theorem 8.105** (Field measurability, Weyl scaling, and translation covariance). *Every subsequential LFPP metric is a field-measurable local metric, satisfies exact Weyl scaling under bounded continuous perturbations, and satisfies exact translation covariance. In particular it satisfies axioms (II)–(IV) in the weak-LQG-metric definition.*

*Proof.* Field measurability and the internal-metric locality axiom are Theorem 8.102. Exact Weyl scaling is Lemma 8.77, after using field determination to choose a single measurable version of the limit. Translation covariance is Lemma 8.104.  $\square$

*Proof of Theorem 8.1.* The quantitative RSW estimate is Theorem 8.8. Theorem 8.21 and the concentration theorem prove the subcritical exponent and concentration estimates. Corollary 8.67 supplies the metric-level local-uniform tightness, non-collapse, Euclidean topology, and local bi-Hölder bounds in the selected whole-plane form. Its final paragraph also gives joint tightness with the field. These are exactly the assertions of the theorem.  $\square$

## 8.7 Weak LQG metric axioms and LFPP subsequential limits

**Definition 8.106** (Weak  $\gamma$ -LQG metric). Let  $h$  be a whole-plane GFF plus a continuous function. A measurable map  $h \mapsto D_h$  is called a weak  $\gamma$ -LQG metric if the following hold.

- (I)  $(\mathbb{C}, D_h)$  is a length space.
- (II) For every deterministic open  $U \subset \mathbb{C}$ , the internal metric  $D_h(\cdot, \cdot; U)$  is determined by the restriction  $h|_U$ .
- (III) For every continuous  $f$ ,

$$(e^{\xi f} \cdot D_h)(z, w) := \inf_{P: z \rightarrow w} \int_0^{\text{len}_{D_h}(P)} e^{\xi f(P(s))} ds = D_{h+f}(z, w). \quad (20.2)$$

- (IV) Translation is exact: for  $z_0 \in \mathbb{C}$ ,

$$D_{h(\cdot + z_0)}(u, v) = D_h(u + z_0, v + z_0).$$

- (V) There are deterministic constants  $c_r > 0$ ,  $r > 0$ , such that the laws of

$$c_r^{-1} e^{-\xi h_r(0)} D_h(r \cdot, r \cdot) \quad (20.3)$$

are tight in the local-uniform topology and every subsequential limit induces the Euclidean topology. Moreover the  $c_r$  satisfy the polynomial comparison bounds required by the weak-metric scale-tightness axiom.

Here  $h_r(0)$  denotes the circle average at radius  $r$  in the chosen whole-plane normalization.

**Theorem 8.107** (Subsequential LFPP limits are weak LQG metrics). *For every deterministic sequence  $\epsilon_n \downarrow 0$  there is a subsequence  $\epsilon_{n_k}$  and a weak  $\gamma$ -LQG metric  $D_h$  such that, for a whole-plane GFF (and, more generally, a whole-plane GFF plus a bounded continuous function),*

$$a_{\epsilon_{n_k}}^{-1} D_h^{\epsilon_{n_k}} \longrightarrow D_h \quad (20.4)$$

*in probability for the local-uniform topology. The limiting metric is a measurable function of the field and satisfies all five axioms above, not only length-space and Weyl-scaling properties.*

*Proof roadmap.* The deterministic parts of the passage to the limit are straightforward only after one has a localized approximation. The source proof first constructs LFPP using field information in a shrinking neighbourhood of the path, proves convergence of internal metrics on nested open sets, and then identifies the limiting internal metric as a measurable function of  $h|_U$ . Weyl scaling is obtained by joint convergence of the localized metrics for  $h$  and  $h + f$ . Translation covariance is inherited exactly. Finally the circle-average decomposition and crossing-scale ratios produce the deterministic constants  $c_r$  and the polynomial comparison required in axiom (V). Measurability, locality, and scale tightness are separate proof obligations; none follows merely from local-uniform tightness.

*Proof.* Start from the arbitrary deterministic sequence  $\epsilon_n \downarrow 0$ . Theorem 8.75 gives a further subsequence and a joint field/metric limit  $D_h$  which is a continuous length metric inducing the Euclidean topology; this is axiom (I), and the same theorem supplies the compatible limiting internal metrics needed by the locality argument.

Theorem 8.105 proves axioms (II)–(IV): the metric is determined locally by the field restriction, Weyl scaling holds exactly, and deterministic translations commute with the metric. Theorem 8.98 gives axiom (V), with actual deterministic constants  $c_r$  and the polynomial comparison of Lemma 8.97. Thus  $D_h$  is a weak  $\gamma$ -LQG metric.

It remains to justify the mode of convergence asserted in the theorem. By Theorem 8.102,  $D_h$  is almost surely a measurable function of  $h$ . The joint convergence in law of  $(h, a_{\epsilon_{n_k}}^{-1} D_h^{\epsilon_{n_k}})$  to  $(h, D_h)$  therefore upgrades, by Lemma 8.103, to local-uniform convergence in probability of the metrics themselves.

Finally let  $h = h + f$  with bounded continuous  $f$ . Along every further subsequence one may extract an almost-sure local-uniform convergence coupling. Lemma 8.77 then identifies the only possible limit as  $D_h = e^{\xi f} \cdot D_h$ . Hence the full selected subsequence converges in probability to this field-measurable limit. The same lemma and Theorem 8.102 transfer axioms (I)–(IV), while axiom (V) is an assertion for the underlying whole-plane GFF and is already supplied by Theorem 8.98. This proves the stated bounded-perturbation extension.  $\square$

*Remark 8.108* (Proof dependencies). Theorem 8.75 gives localized metric convergence. Theorem 8.105 gives the Gwynne–Miller measurability criterion and exact translation passage, while Theorem 8.98 gives scale tightness. The annular hypothesis used to pass from additive locality to field determination is verified specifically for the LFPP limit in Proposition 8.101. No uniqueness, strong coordinate-change theorem, or conformal-covariance result is used in this parent proof.

## 8.8 Existence, full convergence, and uniqueness

**Definition 8.109** (Strong  $\gamma$ -LQG metric). A weak  $\gamma$ -LQG metric is called strong if the scale-tightness axiom is upgraded to the exact affine coordinate-change rule

$$D_h(au + z, av + z) = D_{h(a \cdot + z) + Q \log |a|}(u, v), \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}, \quad (20.5)$$

for every deterministic  $a \in \mathbb{C} \setminus \{0\}$  and  $z \in \mathbb{C}$ .

**Theorem 8.110** (Existence of the subcritical LQG metric and full LFPP convergence). *For every  $\gamma \in (0, 2)$  there exists a strong  $\gamma$ -LQG metric  $D_h$  which is almost surely determined by the field. With the crossing normalization above,*

$$a_\epsilon^{-1} D_h^\epsilon \longrightarrow D_h \quad (20.6)$$

in probability, without passing to a subsequence, for the local-uniform topology. The same statement holds after adding a bounded continuous function to the whole-plane GFF, with the corresponding Weyl rescaling.

*Proof roadmap.* Tightness and the weak-metric theorem provide subsequential weak limits. The full-convergence theorem therefore reduces to uniqueness of weak metrics together with the weak-to-strong upgrade. Once every subsequential limit is the same measurable function of  $h$ , the usual subsequence principle upgrades subsequential convergence in probability to (20.6). The difficult input is not this final compactness argument; it is the uniqueness theorem stated next.

**Theorem 8.111** (Uniqueness and the weak-to-strong upgrade). *Every weak  $\gamma$ -LQG metric is a strong  $\gamma$ -LQG metric. If  $D_h$  and  $\tilde{D}_h$  are two weak  $\gamma$ -LQG metrics coupled with the same whole-plane GFF, then there is a deterministic constant  $C > 0$  such that*

$$\tilde{D}_h = CD_h \quad \text{almost surely.} \quad (20.7)$$

Consequently a single deterministic normalization (for example a fixed unit crossing quantile) makes the subcritical LQG metric unique.

*Proof roadmap.* The uniqueness proof has two main inputs. The first is a local-GFF-metric theorem which gives deterministic bi-Lipschitz comparison of two local metrics satisfying the axioms. The second is a confluence/shortcut mechanism for LQG geodesics: at a positive density of scales one constructs events on which a putatively extremal ratio can be improved along a geodesic. Iterating these improvements contradicts a bi-Lipschitz constant strictly larger than the optimal scalar. The resulting scalar comparison then gives the weak-to-strong affine scaling law.

## 8.9 Bi-Lipschitz comparison at a common weak-metric scale

The first uniqueness input is deliberately stated with the hypothesis which appears in the primary theorem. Two arbitrary weak LQG metrics need not a priori use the same constants  $c_r$  in Axiom V. The deterministic bi-Lipschitz comparison below is therefore a theorem for a pair which does share those constants. The removal of this hypothesis belongs to the later weak-to-strong argument and is not smuggled into the present child.

**Proposition 8.112** (Common-scale deterministic bi-Lipschitz comparison). *Let  $D$  and  $\tilde{D}$  be two weak  $\gamma$ -LQG metrics coupled with the same whole-plane GFF  $h$ , and suppose that the two metrics have the same scaling constants  $c_r$  in Axiom V. Then there is a deterministic constant  $C \in [1, \infty)$  such that almost surely*

$$C^{-1}D_h(z, w) \leq \tilde{D}_h(z, w) \leq CD_h(z, w), \quad z, w \in \mathbb{C}.$$

The same comparison, with the same deterministic constant, holds when the common field is a whole-plane GFF plus a continuous function.

*Proof.* We first treat the whole-plane GFF. By Axiom II, each internal metric is a measurable function of the field restriction on the corresponding open set. Consequently, conditional on  $h$ , the pair  $(D_h, \tilde{D}_h)$  is deterministic. In particular the two metrics are jointly local in the sense of the local-metric theorem owned in Theorem 8.89. Axiom III shows that the pair is jointly  $\xi$ -additive after subtracting a circle average.

Fix  $p \in (0, 1)$  larger than the universal threshold  $p_0$  in Theorem 8.89. The common scale-tightness axiom and its Euclidean-topology non-degeneracy give a deterministic  $A = A(p) > 1$  such that, uniformly over  $z \in \mathbb{C}$  and  $r > 0$ , with probability at least  $1 - (1 - p)/2$ ,

$$\sup_{u, v \in \partial B_r(z)} \tilde{D}_h(u, v; \mathbb{A}_{r/2, 2r}(z)) \leq A c_r e^{\xi_{h_r}(z)},$$

and with probability at least  $1 - (1 - p)/2$ ,

$$D_h(\partial B_{r/2}(z), \partial B_r(z)) \geq A^{-1} c_r e^{\xi_{h_r}(z)}.$$

Here the lower-tail statement follows from tightness away from the zero metric: every subsequential normalized law induces the Euclidean topology, so the radial crossing of a fixed annulus is strictly positive; compactness of the Prokhorov closure makes the corresponding lower quantile uniform. The upper internal-boundary diameter follows in exactly the same way from continuity on the compact annulus. These are the annular quantile estimates already isolated for an individual weak metric in Lemma 8.100.

On the intersection of the two events, whose probability is at least  $p$ , we therefore have

$$\sup_{u, v \in \partial B_r(z)} \tilde{D}_h(u, v; \mathbb{A}_{r/2, 2r}(z)) \leq A^2 D_h(\partial B_{r/2}(z), \partial B_r(z)).$$

Theorem 8.89, with the deterministic constant  $A^2$ , gives  $\tilde{D}_h \leq A^2 D_h$  simultaneously for all pairs of points. Interchanging  $D$  and  $\tilde{D}$  gives  $D_h \leq A^2 \tilde{D}_h$ . Taking  $C = A^2$  proves the whole-plane statement.

For a continuous perturbation  $f$ , Axiom III gives  $D_{h+f} = e^{\xi f} \cdot D_h$  and the same identity for  $\tilde{D}$ . The operation of Weyl-rescaling a length metric is monotone with respect to a deterministic multiplicative comparison: if  $\tilde{D} \leq C D$ , then for every path  $P$  its  $e^{\xi f} \cdot \tilde{D}$ -length is at most  $C$  times its  $e^{\xi f} \cdot D$ -length, hence  $e^{\xi f} \cdot \tilde{D} \leq C(e^{\xi f} \cdot D)$ . The reverse comparison is identical. Thus the same deterministic constant  $C$  works for a whole-plane GFF plus any continuous perturbation, with no boundedness assumption and no new stochastic input.  $\square$

## 8.10 Extremal ratios and abundant comparison scales

Throughout this section  $D$  and  $\tilde{D}$  are two weak  $\gamma$ -LQG metrics coupled with the same whole-plane GFF and using the same scale constants  $c_r$ . Proposition 8.112 applies.

**Lemma 8.113** (Fixed-endpoint geodesic tie breaking). *For each deterministic pair  $z \neq w$ , the  $D_h$ -geodesic from  $z$  to  $w$  is almost surely unique. The same holds for  $\tilde{D}_h$ .*

*Proof.* Existence follows from bounded compactness of the weak metric, as in the proof of the geodesic-existence theorem later in this chapter. We prove uniqueness for  $D$ .

Suppose two distinct minimizing paths exist. By continuity and compactness of their images there is a rational Euclidean ball  $B = B_r(q)$  and a smaller concentric ball  $B' = B_{r/3}(q)$  such that one minimizing path meets  $B'$  while another avoids  $\bar{B}$ . Hence non-uniqueness is contained in the countable union over such rational balls of the event that the minimum length among paths meeting  $B'$  equals the minimum length among paths avoiding  $\bar{B}$ .

Fix one ball from this union. Choose a nonnegative smooth function  $\phi$  which equals one on  $B'$  and vanishes outside  $B$ . Decompose the GFF as  $h = h^\perp + X\phi$  after projecting onto the Cameron–Martin direction  $\phi$ , where conditional on  $h^\perp$  the scalar  $X$  has a density. By Weyl scaling, paths

avoiding  $B$  have exactly the same length for every value of  $X$ , whereas every path which meets  $B'$  must traverse a positive  $D_{h^\perp+X\phi}$ -length segment on which  $\phi > 0$ . Consequently the minimal length in the “meet  $B'$ ” class is a strictly increasing continuous function of  $X$ , while the minimal length in the “avoid  $\bar{B}$ ” class is constant. Their equality can therefore hold for at most one value of  $X$ . Since  $X$  has a conditional density, the tie event has conditional, and hence unconditional, probability zero. The countable union over rational balls is null.

The proof for  $\tilde{D}$  is identical.  $\square$

Define the deterministic extremal ratios

$$c_* := \inf_{u \neq v} \frac{\tilde{D}_h(u, v)}{D_h(u, v)}, \quad C_* := \sup_{u \neq v} \frac{\tilde{D}_h(u, v)}{D_h(u, v)}.$$

By Proposition 8.112, these random variables lie almost surely in a deterministic compact subinterval of  $(0, \infty)$ .

**Lemma 8.114** (The extremal ratios are deterministic). *The variables  $c_*$  and  $C_*$  are almost surely equal to deterministic constants.*

*Proof.* We give the proof for  $C_*$ . Fix  $a > 0$  with  $\mathbb{P}[C_* > a] > 0$ . By continuity of the two metrics and exhaustion by Euclidean balls, there is a deterministic  $R$  such that with positive probability there are  $u, v \in B_R(0)$  satisfying

$$\frac{\tilde{D}_h(u, v)}{D_h(u, v)} > a, \quad D_h(u, v) \leq D_h(u, \partial B_R(0)), \quad \tilde{D}_h(u, v) \leq \tilde{D}_h(u, \partial B_R(0)). \quad (20.9)$$

The last two inequalities ensure that all metric data used in (20.9) are internal to  $B_R(0)$ . Locality therefore makes the occurrence of (20.9) measurable with respect to  $h|_{B_R(0)}$ . Weyl scaling shows that it is unchanged by adding a constant to the field.

Translate the event to  $B_R(z)$  for  $z \in \mathbb{Z}^2$ . Its probability is independent of  $z$ . The event that it occurs for infinitely many such  $z$  is measurable with respect to the tail sigma-field of the whole-plane GFF modulo additive constant. That tail sigma-field is trivial. Since the translated event has positive probability, standard spatial mixing/tail triviality gives that infinitely many translates occur almost surely. Hence  $\mathbb{P}[C_* > a] = 1$ . Applying this to rational  $a$  shows that  $C_*$  is almost surely constant. The argument for  $c_*$  is the same with inequalities reversed.  $\square$

For  $r > 0$ ,  $\beta \in (0, 1)$ , and comparison levels  $A < C_*$  and  $a > c_*$ , set

$$\begin{aligned} \overline{G}_r(A, \beta) &:= \left\{ \exists z, w \in B_r(0) : |z - w| \geq \beta r, \tilde{D}_h(z, w) \geq A D_h(z, w) \right\}, \\ \underline{G}_r(a, \beta) &:= \left\{ \exists z, w \in B_r(0) : |z - w| \geq \beta r, \tilde{D}_h(z, w) \leq a D_h(z, w) \right\}. \end{aligned}$$

**Lemma 8.115** (Nested-annulus amplification). *Fix  $0 < \mu < \nu < 1$ . Suppose that for at least  $(\nu - \mu) \log_8 \varepsilon^{-1}$  radii*

$$r \in [\varepsilon^{1+\nu} R, \varepsilon R] \cap \{8^{-k} R : k \in \mathbb{N}\}$$

*a local annular event  $E_r(0)$ , measurable from the centered field on  $\mathbb{A}_{r/2, 2r}(0)$  modulo additive constant, has probability at least  $p > 0$ . Then, after reducing  $p$  by a constant depending only on  $\mu, \nu$ , with probability  $1 - O(\varepsilon^q)$  for any prescribed  $q > 0$ , every point of a fixed bounded  $R$ -scale region lies in  $B_{r/2}(z)$  for some lattice center  $z$  and some admissible  $r$  for which  $E_r(z)$  occurs.*

*Proof.* This is the concentric-annulus amplification mechanism already owned in Lemma 8.86, applied after translating the center. Choose every third admissible radius so the field pieces which enter the zero-boundary decomposition are separated by deterministic buffer annuli. The harmonic remainders have uniformly bounded oscillation on the middle annuli with Gaussian tails. Conditional on the outer field, the zero-boundary pieces are independent. Thus failure of all retained events at a fixed center has probability exponentially small in the number of retained scales, hence  $O(\varepsilon^{q+4(1+\nu)})$  after choosing the retained density appropriately. A union bound over the  $O(\varepsilon^{-2(1+\nu)})$  lattice centers in the bounded region gives the claim. The formulation modulo additive constants is preserved because both metrics obey the same Weyl exponent.  $\square$

**Proposition 8.116** (Multiscale abundance of near-extremal pairs). *For every  $0 < \mu < \nu < 1$  there are deterministic  $\bar{\beta}, \underline{\beta}, p_0 \in (0, 1)$  with  $\underline{\beta} > 1/2$  such that:*

- (a) *for each  $A < C_*$  and all sufficiently small  $\varepsilon$ , at least  $\mu \log_8 \varepsilon^{-1}$  radii in  $[\varepsilon^{1+\nu}, \varepsilon] \cap \{8^{-k}\}$  satisfy  $\mathbb{P}[\bar{G}_r(A, \bar{\beta})] \geq p_0$ ;*
- (b) *for each  $a > c_*$  and all sufficiently small  $\varepsilon$ , at least  $\mu \log_8 \varepsilon^{-1}$  such radii satisfy  $\mathbb{P}[\underline{G}_r(a, \underline{\beta})] \geq p_0$ .*

*Moreover the same statement holds relative to an arbitrary base scale  $R$ , with the radii multiplied by  $R$ , provided the corresponding extremal event at scale  $R$  has a fixed positive probability.*

*Proof.* We prove the upper-extremal statement; the lower-extremal statement follows by interchanging  $D$  and  $\tilde{D}$ .

Choose  $\alpha \in (1/2, 1)$  close enough to one. For a radius  $r$  define the local event  $E_r(z)$  by three requirements: (i) every unique  $D_h$ -geodesic from  $\partial B_{\alpha r}(z)$  to  $\partial B_r(z)$  which stays in the closed annulus has  $\tilde{D}$ -length at most  $A$  times its  $D$ -length; (ii) a path which is globally shorter only by leaving the buffered annulus cannot be mistaken for such an internal geodesic; and (iii) the annulus contains a separating circuit of  $D$ -length at most a fixed multiple of its radial  $D$ -crossing distance. Locality and the buffer in (ii) imply that  $E_r(z)$  is measurable from  $h|_{\mathbb{A}_{r/2, 2r}(z)}$  modulo additive constant. Scale tightness and the narrow-annulus estimate give a uniform positive-probability choice of the circuit and buffer conditions.

Assume, toward the contrapositive, that near-extremal pairs fail at more than  $(\nu - \mu) \log_8 \varepsilon^{-1}$  admissible scales. At each such scale the comparison condition in  $E_r(z)$  holds with probability close to one. Lemma 8.115 therefore gives, with probability tending to one, a covering of the region by good annular balls.

Let  $P$  be the unique  $D$ -geodesic between two fixed mesh points whose Euclidean separation is at least a fixed fraction of  $R$ . Traverse  $P$  through the good balls greedily. In a retained ball write  $[s_j, t_j]$  for the portion of  $P$  between its last visit to the inner circle and its next exit from the outer circle. The comparison condition gives

$$\tilde{D}(P(s_j), P(t_j)) \leq A(t_j - s_j).$$

The separating circuit forces the gap before this segment to obey

$$s_j - t_{j-1} \leq K(t_j - s_j)$$

for a deterministic  $K$ : otherwise replacing that portion of  $P$  by an arc of the short circuit would shorten the  $D$ -geodesic. On the gap we use only the extremal bound  $\tilde{D} \leq C_* D$ . Summing yields

$$\tilde{D}(P(t_{j-1}), P(t_j)) \leq \left( A + \frac{K}{K+1}(C_* - A) \right) (t_j - t_{j-1}). \quad (20.10)$$



The coefficient in parentheses is strictly below  $C_*$ . Endpoint errors are  $o_R(1)$  uniformly by Axiom V and the Euclidean-topology property. Choosing a number  $A''$  strictly between the coefficient in (20.10) and  $C_*$  and then taking  $\varepsilon$  small shows, simultaneously for the finite mesh, that  $\tilde{D}(z, w) \leq A''D(z, w)$  for every macroscopically separated pair. Continuity removes the mesh restriction.

Thus the failure of the abundance conclusion forces the probability of the scale-R event  $\overline{G}_R(A'', \beta)$  below its assumed positive lower bound. This proves the desired contrapositive relative to R. Taking  $R=1$  and using the definition of  $C_*$  gives part (a). Swapping the two metrics and rewriting the ratio proves part (b).  $\square$

## 8.11 Quantitative confluence across filled metric annuli

This section owns the quantitative confluence theorem required by the uniqueness argument. Qualitative confluence is not sufficient downstream: we need a stopping-time statement which says that, after increasing the metric radius by a negative power of the number of candidate branches, only that many leftmost geodesic branches can remain, with a stretched-exponential error probability. The proof is organized exactly around the source mechanism:

$$\begin{aligned} \text{good annuli} &\longrightarrow \text{conditional shields} \longrightarrow \text{recursive pruning,} \\ &\longrightarrow \text{stretched-exponential branch bound.} \end{aligned}$$

Throughout this section  $D_h$  is a weak  $\gamma$ -LQG metric with the scale constants  $c_r$  from Axiom V, and  $\xi = \gamma/d_\gamma$ .

### 8.11.1 Filled balls, stopping times, and leftmost geodesics

For  $s > 0$  let  $\mathcal{B}_s$  be the open  $D_h$ -ball of radius  $s$  centered at 0, and let  $\mathcal{B}_s^\bullet$  be its filled version, i.e. the union of  $\mathcal{B}_s$  with all bounded connected components of  $\mathbb{C} \setminus \overline{\mathcal{B}_s}$ . For  $r > 0$  define

$$\tau_r := \inf\{s > 0 : \mathcal{B}_s^\bullet \not\subset B_r(0)\}. \quad (8.1)$$

We write

$$\mathcal{F}_s := \sigma(\mathcal{B}_s^\bullet, h|_{\mathcal{B}_s^\bullet}).$$

**Lemma 8.117** (Filled balls at stopping times are local sets). *If  $\tau$  is an  $(\mathcal{F}_s)$ -stopping time, then  $\mathcal{B}_\tau^\bullet$  is a local set for the GFF. Conditionally on  $(\mathcal{B}_\tau^\bullet, h|_{\mathcal{B}_\tau^\bullet})$ , the field on each component  $U$  of  $\mathbb{C} \setminus \mathcal{B}_\tau^\bullet$  has the form*

$$h|_U = \mathring{h}^U + \mathfrak{h}^U,$$

where  $\mathring{h}^U$  is a zero-boundary GFF on  $U$  and  $\mathfrak{h}^U$  is harmonic on  $U$  and determined by the exposed data.

*Proof.* For deterministic  $s$ , the event  $\{\overline{\mathcal{B}_s} \subset V\}$ , with  $V$  open, is determined by  $h|_V$  by locality of the internal metric. The event  $\{\mathcal{B}_s^\bullet \subset V\}$  is determined by this event, the closed set  $\overline{\mathcal{B}_s}$  on it, and which bounded components of  $V \setminus \overline{\mathcal{B}_s}$  are swallowed by the filling. Hence it is determined by  $h|_V$ . The conditional Gaussian characterization of GFF local sets proved in Volume XIX (owner SM-19.01.01) therefore shows that  $\mathcal{B}_s^\bullet$  is a local set.

If  $\tau$  takes values in a countable set, decompose according to the value of  $\tau$ . For a general stopping time set  $\tau_n = 2^{-n}\lceil 2^n \tau \rceil$ . Then  $\tau_n \downarrow \tau$ , the filled balls  $\mathcal{B}_{\tau_n}^\bullet$  decrease to  $\mathcal{B}_\tau^\bullet$  by continuity of  $D_h$ . To verify locality at the limit rather than cite a closure theorem, fix a smooth test function  $\varphi$  supported a positive Euclidean distance from  $\mathcal{B}_\tau^\bullet$ . For all sufficiently large  $n$  its support is disjoint

from  $\mathcal{B}_{\tau_n}^\bullet$ . The conditional characteristic-function identity from SM-19.01.01 therefore holds for  $\varphi$  with the Green kernel of the corresponding unexplored component. Monotone convergence of Dirichlet Green kernels and martingale convergence of the conditional means pass this identity to  $\tau$ . A countable dense family of test functions identifies the conditional law on each component as an independent zero-boundary GFF plus its harmonic mean, which is exactly the claimed local-set decomposition.  $\square$

The fixed-endpoint tie-breaking lemma proved in this chapter implies, in particular, that for each  $q \in \mathbb{Q}^2$  the  $D_h$ -geodesic from 0 to  $q$  is almost surely unique. This gives the planar order needed below.

**Lemma 8.118** (Leftmost geodesics and order of boundary arcs). *Almost surely, for every  $s > 0$  and every  $y \in \partial\mathcal{B}_s^\bullet$  there is a unique leftmost geodesic  $P_y^-$  from 0 to  $y$ . It is the uniform limit of geodesics from 0 to rational points outside  $\mathcal{B}_s^\bullet$  approaching  $y$  from the left. Moreover, if  $I_1, I_2$  are disjoint arcs of  $\partial\mathcal{B}_s^\bullet$  and  $s' > s$ , then the sets of points of  $\partial\mathcal{B}_{s'}^\bullet$  whose leftmost geodesics pass through  $I_1$  and  $I_2$  are disjoint boundary arcs, in the same cyclic order.*

*Proof.* Choose rational points outside  $\mathcal{B}_s^\bullet$  approaching  $y$  from one side in the prime-end topology. Their unique geodesics are contained in a common bounded metric ball, hence are equicontinuous when parameterized by  $D_h$ -length. Properness is Proposition 8.178: its proof combines local bi-Hölder comparison with the separate escape-to-infinity lemma, so no global conclusion is inferred from local comparison alone. It gives a uniformly convergent subsequence and hence a limiting geodesic  $P_y^-$ . If a second geodesic from 0 to  $y$  crossed to the left of  $P_y^-$ , then for all large approximants it would cross the unique rational geodesic. Splicing at the first crossing would give two distinct geodesics to that rational point, contradicting fixed-endpoint uniqueness. Thus  $P_y^-$  is leftmost and the same argument gives uniqueness.

For the second assertion, two images of disjoint arcs cannot interlace: interlacing endpoints would force two corresponding leftmost geodesics to cross in the annulus between the two filled-ball boundaries. At the first crossing, the same splicing argument contradicts the defining leftmost order. Continuity in the prime-end parameter makes each image connected, hence an arc. Cyclic order is preserved.  $\square$

### 8.11.2 A planar separator estimate for boundary arcs

The recursive pruning argument needs a deterministic fact: all but a square-root-order exceptional family of boundary arcs can be shielded from infinity by a set whose Euclidean diameter is much smaller than the ambient scale.

**Lemma 8.119** (Small separators for most boundary arcs). *There is a universal constant  $A > 0$  with the following property. Let  $K \subset \mathbb{C}$  be compact and connected, contain a neighborhood of 0, and have connected complement. Let  $\mathcal{I}$  be a family of  $n$  boundary arcs meeting only at endpoints. Put*

$$r_{\text{in}} = \sup\{r : B_r(0) \subset K\}, \quad r_{\text{out}} = \inf\{R : K \subset B_R(0)\}.$$

*For every  $L > 0$ , all but at most*

$$A L^{-2} \frac{r_{\text{out}}^4}{r_{\text{in}}^2} n \tag{8.2}$$

*are disconnectable from infinity in  $\mathbb{C} \setminus K$  by a connected set of Euclidean diameter at most  $Ln^{-1/2}$ .*

*Proof.* Invert by  $\iota(z) = 1/z$ . The domain  $U = \iota(\mathbb{C} \setminus K)$  is bounded and simply connected, contains 0, and satisfies

$$\text{area}(U) \leq \pi r_{\text{in}}^{-2}.$$

Let  $\phi : \mathbb{D} \rightarrow U$  be conformal with  $\phi(0) = 0$ . For a boundary arc  $J = \iota(I)$  let  $\ell_J$  be the arclength of  $\phi^{-1}(J)$  and choose  $w_J = (1 - \ell_J)u_J$ , where  $u_J$  is the midpoint of the preimage arc. After discarding at most a universal number of very large preimage arcs, the disks  $B_{\ell_J/100}(w_J)$  are disjoint. The Koebe distortion theorem owned in Volume IV (SM-04.02.04) gives a universal  $C_1$  such that

$$\text{dist}(\phi(w_J), \partial U) \leq C_1 |\phi'(w_J)| \ell_J.$$

Consequently

$$\sum_J \text{dist}(\phi(w_J), \partial U)^2 \leq C_2 \int_{\mathbb{D}} |\phi'(z)|^2 dz = C_2 \text{area}(U).$$

For each  $J$ , Brownian motion started at  $w_J$  has uniformly positive probability to exit the associated half-disk through either neighboring boundary arc. The Beurling estimate owned in Volume IV (SM-04.06.03) then supplies two curves whose images under  $\phi$  join the neighboring boundary arcs and whose diameters are at most a universal multiple of  $\text{dist}(\phi(w_J), \partial U)$ . Their union disconnects 0 from  $J$  in  $U$ . Chebyshev's inequality shows that all but at most  $C_3 L_0^{-2} \text{area}(U)n$  of the arcs have such a separator of diameter at most  $L_0 n^{-1/2}$ .

Under inversion, Euclidean diameters are enlarged by at most  $r_{\text{out}}^2$  on  $\mathbb{C} \setminus K$ . Taking  $L_0 = L/r_{\text{out}}^2$  and using the area bound gives (8.2).  $\square$

### 8.11.3 Good annuli and conditional shielding

For  $z \in \mathbb{C}$ ,  $r > 0$ , and  $\delta \in (0, 1)$  let  $\mathcal{S}_{\delta r}^z(A)$  be the collection of  $\delta r$ -grid squares (with grid translated by  $z$ ) which meet  $A$ . Let  $\mathcal{U}_r(z; \delta)$  be the finite collection of open subsets  $U$  of  $\mathbb{A}_{3r, 4r}(z)$  obtained by deleting a union of such grid squares. For  $U \in \mathcal{U}_r(z; \delta)$  let  $\mathcal{V}(U)$  denote its connected components and set

$$U^{(-t)} = \{u \in U : \text{dist}(u, \partial U) > t\}.$$

Fix  $c, \delta \in (0, 1)$  and  $A > 0$ . We call the annulus  $\mathbb{A}_{2r, 5r}(z)$  *good* if, simultaneously for every  $U \in \mathcal{U}_r(z; \delta)$ ,

$$D_h(\partial B_{2r}(z), \partial B_{3r}(z)) \geq c c_r e^{\xi h_r(z)}, \quad (\text{G1})$$

$$\max_{S \in \mathcal{S}_{\delta r}^z(\mathbb{A}_{3r, 4r}(z))} \sup_{u, v \in S} D_h(u, v; \mathbb{A}_{2r, 5r}(z)) \leq \frac{c}{100} c_r e^{\xi h_r(z)}, \quad (\text{G2})$$

$$\sup_{u \in U^{(-\delta r/4)}} |\mathfrak{h}^U(u) - h_r(z)| \leq A. \quad (\text{G3})$$

Here  $\mathfrak{h}^U$  is the harmonic part of the Markov decomposition of  $h|_U$ .

**Lemma 8.120** (Good annuli have uniformly high probability). *For every  $p \in (0, 1)$  there exist  $c, \delta \in (0, 1)$  and  $A > 0$ , depending only on  $p$  and the law of the weak metric, such that*

$$\inf_{z \in \mathbb{C}, r > 0} \mathbb{P}[\mathbb{A}_{2r, 5r}(z) \text{ is good}] \geq p.$$

*The event is determined by  $(h - h_r(z))|_{\mathbb{A}_{2r, 5r}(z)}$ .*

*Proof.* By translation covariance and scale tightness, the random variables

$$\left[ c_r^{-1} e^{-\xi h_r(z)} D_h(\partial B_{2r}(z), \partial B_{3r}(z)) \right]^{-1}$$

form a tight family. Choose  $c$  so that (G1) fails with probability at most  $(1-p)/3$ .

For this  $c$ , scale tightness together with the fact that every subsequential rescaled metric induces the Euclidean topology implies uniform equicontinuity in probability on the compact annulus  $\overline{\mathbb{A}_{2,5}}$ . Hence for  $\delta$  small, (G2) fails with probability at most  $(1-p)/3$ .

For fixed  $\delta$  there are only finitely many normalized shapes  $r^{-1}(U-z)$ . For each such shape, the harmonic part minus  $h_r(z)$  is a centered Gaussian harmonic function on the trimmed region, with uniformly bounded pointwise variance and derivative variance by the Green-function representation. The Gaussian concentration theorem owned in Volume VI (SM-06.01.03), applied first on a finite net and then with the uniform derivative-variance bound, therefore implies

$$\mathbb{P} \left[ \sup_{U^{(-\delta r/4)}} |\mathfrak{h}^U - h_r(z)| > A \right] \rightarrow 0 \quad (A \rightarrow \infty)$$

uniformly in  $z, r$  and in the finitely many shapes. Choose  $A$  so the union of these failures has probability at most  $(1-p)/3$ . A union bound proves the first claim. Each of (G1)–(G3) is invariant under adding a constant to  $h$ , and locality confines its dependence to the displayed annulus, which proves the measurability claim.  $\square$

**Lemma 8.121** (Conditional small-diameter event inside a good annulus). *Fix  $c, \delta, A$ . There exists  $p_0 = p_0(c, \delta, A) > 0$  such that the following holds. Let  $U \in \mathcal{U}_r(z; \delta)$  and condition on  $h|_{\mathbb{C} \setminus U}$  and on the event that (G1)–(G3) hold for this  $U$ . Then with conditional probability at least  $p_0$ ,*

$$\max_{V \in \mathcal{V}(U)} \sup_{u, v \in V^{(-\delta r/2)}} D_h(u, v; V^{(-\delta r/4)}) \leq \frac{c}{2} c_r e^{\xi h_r(z)}. \quad (8.3)$$

*Proof.* By Lemma 8.117, conditionally on the exterior field the random part in  $U$  is a zero-boundary GFF  $\mathring{h}^U$ , while the harmonic part is fixed. The zero-boundary localization theorem (Proposition 8.66) implies that for every normalized component shape there is a constant  $C$  so that, with probability at least  $1/2$ ,

$$\max_{V \in \mathcal{V}(U)} \sup_{u, v \in V^{(-\delta r/2)}} D_{\mathring{h}^U}(u, v; V^{(-\delta r/4)}) \leq C c_r.$$

There are only finitely many normalized shapes.

Choose a smooth  $g : U \rightarrow [0, 1]$  which equals one on  $U^{(-\delta r/4)}$  and whose Dirichlet energy is bounded by a constant depending only on  $\delta$  and the normalized shape. Set

$$f = \xi^{-1} \log \left( \frac{2C e^{\xi A}}{c} \right) g.$$

By Weyl scaling, the preceding  $C c_r$  event for  $\mathring{h}^U - f$ , together with (G3), implies (8.3) for  $\mathring{h}^U + \mathfrak{h}^U$ . The Cameron–Martin theorem owned in Volume VI (SM-06.01.02) shows that the laws of  $\mathring{h}^U$  and  $\mathring{h}^U - f$  are mutually absolutely continuous and that a uniform positive lower bound for the probability of the shifted event depends only on the Dirichlet energy of  $f$ . Taking the minimum over the finitely many normalized component shapes gives  $p_0 > 0$ .  $\square$

We now search for many good annuli. Put  $\rho_{\varepsilon r}^0(z) = \varepsilon r$  and inductively let  $\rho_{\varepsilon r}^j(z)$  be the smallest dyadic radius at least  $6\rho_{\varepsilon r}^{j-1}(z)$  for which the corresponding annulus is good.

**Lemma 8.122** (Logarithmically many good annuli before scale  $\varepsilon^{1/2}r$ ). *The parameters in Lemma 8.120 can be chosen, together with  $\eta > 0$ , so that for every bounded deterministic  $K \subset \mathbb{C}$ ,*

$$\mathbb{P} \left[ \rho_{\varepsilon r}^{\lfloor \eta \log \varepsilon^{-1} \rfloor}(z) \leq \varepsilon^{1/2}r \text{ for all } z \in \frac{\varepsilon r}{4}\mathbb{Z}^2 \cap B_{\varepsilon r}(rK) \right] = 1 - O_K(\varepsilon^2). \quad (8.4)$$

*Proof.* The good-annulus events are invariant under adding a constant and depend only on  $h - h_s(z)$  in  $\mathbb{A}_{2s,5s}(z)$ . Apply the concentric-annulus near-independence lemma 8.86 to the geometrically separated radii  $8^j \varepsilon r$ . By taking the one-scale success probability in Lemma 8.120 close enough to one, the probability of seeing fewer than  $\eta \log \varepsilon^{-1}$  successes before radius  $\varepsilon^{1/2}r$  is  $O(\varepsilon^6)$ , uniformly in the center. The grid contains  $O_K(\varepsilon^{-2})$  points. A union bound gives (8.4).  $\square$

#### 8.11.4 The shield which kills a boundary arc

For a compact set  $K$  define

$$R_r^\varepsilon(K) := 6 \sup_z \rho_{\varepsilon r}^{\lfloor \eta \log \varepsilon^{-1} \rfloor}(z) + \varepsilon r,$$

where the supremum is over  $z \in (\varepsilon r/4)\mathbb{Z}^2 \cap B_{\varepsilon r}(K)$ . If  $s$  is a metric radius, set

$$\sigma_{s,r}^\varepsilon := \inf\{s' > s : B_{R_r^\varepsilon(\mathcal{B}_s^\bullet)}(\mathcal{B}_s^\bullet) \subset \mathcal{B}_{s'}^\bullet\}. \quad (8.5)$$

By the stopping-time construction of the good radii,  $\sigma_{\tau,r}^\varepsilon$  is a stopping time whenever  $\tau$  is.

**Lemma 8.123** (Shielding a small boundary arc). *There exist constants  $\alpha > 0$  and  $C_0 > 1$ , depending only on the law of  $D$ , with the following property. Let  $r > 0$ , let  $\tau$  be an  $(\mathcal{F}_s)$ -stopping time, and let  $I \subset \partial \mathcal{B}_\tau^\bullet$  be an exposed boundary arc which can be disconnected from infinity in  $\mathbb{C} \setminus \mathcal{B}_\tau^\bullet$  by a connected set of Euclidean diameter at most  $\varepsilon r$ . There is an event  $G_I \in \mathcal{F}_{\sigma_{\tau,r}^\varepsilon}$  such that*

- (a) *if  $R_r^\varepsilon(\mathcal{B}_\tau^\bullet) \leq \text{diam}(\mathcal{B}_\tau^\bullet)$  and  $G_I$  occurs, then no geodesic from 0 to  $\mathbb{C} \setminus \mathcal{B}_{\sigma_{\tau,r}^\varepsilon}^\bullet$  can pass through  $I$ ;*
- (b) *almost surely,*

$$\mathbb{P}[G_I \mid \mathcal{F}_\tau] \geq 1 - C_0 \varepsilon^\alpha.$$

*Proof.* Choose  $x \in I$  so that  $B_{\varepsilon r}(x)$  disconnects  $I$  from infinity in the complement of the filled ball, and choose a grid point  $z \in (\varepsilon r/4)\mathbb{Z}^2$  with  $B_{\varepsilon r}(x) \subset B_{2\varepsilon r}(z)$ . At a candidate radius  $s$  define

$$\tilde{U}^s = \mathbb{A}_{3s,4s}(z) \setminus \bigcup \{S \in \mathcal{S}_{\delta_s}^z : S \cap \mathcal{B}_\tau^\bullet \neq \emptyset\}.$$

This set and its components are determined by  $\mathcal{F}_\tau$ . Search outwards through the good radii furnished by Lemma 8.122. At each such radius apply Lemma 8.121 to  $U = \tilde{U}^s$ .

Suppose (8.3) occurs at one of these radii. Every component of  $\tilde{U}^s$  can then be crossed, in  $D_h$ -length, for at most  $(c/2)c_s e^{\xi h_s(z)}$ . The deleted grid squares which abut  $\mathcal{B}_\tau^\bullet$  each have  $D_h$ -diameter at most  $(c/100)c_s e^{\xi h_s(z)}$  by (G2). Hence one can join any point of the middle annulus to  $\mathcal{B}_\tau^\bullet$  through the union of these components and at most finitely many adjacent grid squares for total  $D_h$ -length  $< cc_s e^{\xi h_s(z)}$ . By (G1), any path from  $B_{2s}(z)$  to the exterior of  $B_{3s}(z)$  has  $D_h$ -length at least that amount. Therefore a geodesic from a point outside the shield to 0 cannot first move inward through  $B_{2s}(z)$  near  $x$ : replacing that portion by the shorter connection to  $\mathcal{B}_\tau^\bullet$  would strictly decrease its length. This proves (a).

At each searched good radius, conditionally on the past and the exterior field, the success probability is at least  $p_0$  by Lemma 8.121. The radii grow by a factor at least six, so the success trials can be exposed one after another. If  $m = \lfloor \eta \log \varepsilon^{-1} \rfloor$ , the probability that all  $m$  attempts fail is at most  $(1 - p_0)^m \leq C\varepsilon^\alpha$  for some  $\alpha > 0$ . The probability that the required sequence of good annuli does not exist is  $O(\varepsilon^2)$  by Lemma 8.122; decreasing  $\alpha$  if needed absorbs this term. This gives (b).  $\square$

### 8.11.5 The regularity event and recursive pruning

Fix a Hölder exponent  $\chi \in (0, \xi(Q-2))$  in the range already proved in the metric tightness module. For  $a \in (0, 1)$  let  $\mathcal{E}_r(a)$  be the event that all of the following hold:

$$(R1) \quad B_{ar}(0) \subset \mathcal{B}_{\tau_r}^\bullet;$$

$$(R2) \quad \tau_{3r} - \tau_{2r} \geq ac_r e^{\xi h_r(0)};$$

$$(R3) \quad \text{for } u, v \in B_{4r}(0) \text{ with } |u - v| \leq ar,$$

$$D_h(u, v) \leq c_r e^{\xi h_r(0)} \left( \frac{|u - v|}{r} \right)^\chi;$$

$$(R4) \quad \text{for every dyadic } \varepsilon \in (0, a] \text{ and each relevant grid point } z \in (\varepsilon r/4)\mathbb{Z}^2 \cap B_{4r}(0),$$

$$\rho_{\varepsilon r}^{\lfloor \eta \log \varepsilon^{-1} \rfloor}(z) \leq \varepsilon^{1/2} r.$$

**Lemma 8.124** (Uniform probability of the regularity event). *For every  $p \in (0, 1)$  there is  $a = a(p) > 0$  such that*

$$\inf_{r>0} \mathbb{P}[\mathcal{E}_r(a)] \geq p.$$

*Proof.* Scale tightness and Euclidean-topology nondegeneracy give (R1) and (R2) with probability tending to one as  $a \downarrow 0$ , uniformly in  $r$ . The local Hölder estimate gives (R3) with the same property. For (R4), apply Lemma 8.122 at each dyadic  $\varepsilon \leq a$  and sum the resulting  $O(\varepsilon^2)$  failure probabilities. Choose  $a$  so the four failure probabilities sum to at most  $1 - p$ .  $\square$

Fix now a stopping time  $\tau \in [\tau_r, \tau_{2r}]$  and a finite family  $\mathcal{I}_0$  of disjoint arcs of  $\partial \mathcal{B}_\tau^\bullet$ . Put  $s_0 = \tau$ . Inductively, if  $\mathcal{I}_k$  is defined, let

$$n_k = \#\mathcal{I}_k, \quad \varepsilon_k = 2^{-j_k} \in [n_k^{-1/4}, 2n_k^{-1/4}), \quad s_{k+1} = \sigma_{s_k, r}^{\varepsilon_k}.$$

For  $I \in \mathcal{I}_k$ , let  $I'$  be the set of points of  $\partial \mathcal{B}_{s_{k+1}}^\bullet$  whose leftmost geodesics pass through  $I$ , and let  $\mathcal{I}_{k+1}$  be the collection of nonempty  $I'$ . By Lemma 8.118, these are disjoint boundary arcs and  $n_{k+1} \leq n_k$ .

**Lemma 8.125** (Geometric pruning of the arc count). *There are  $N_0 \in \mathbb{N}$ ,  $C > 1$ , and  $\tilde{\beta} > 0$ , depending only on  $a$  and the law of  $D$ , such that the following holds on  $\mathcal{E}_r(a)$ . Let  $k_0 = 0$  and let  $k_{j+1}$  be the first  $k \geq k_j$  with  $n_k \leq n_{k_j}/2$ . If*

$$K_{N_0} = \min\{k : s_k > \tau_{3r} \text{ or } n_k < N_0\},$$

*then for every  $M \geq 1$ ,*

$$\mathbb{P}\left[k_{j+1} - k_j > M, \quad k_{j+1} \leq K_{N_0}, \quad \mathcal{E}_r(a) \mid \mathcal{F}_{s_{k_j}}\right] \leq (n_{k_j}/C)^{-\tilde{\beta}M}. \quad (8.6)$$

*Proof.* Let  $\mathcal{I}_k^*$  be the arcs not disconnectable from infinity by a connected set of diameter at most  $\varepsilon_k r$ . On  $\mathcal{E}_r(a)$ , for  $k < K_{N_0}$  we have  $B_{ar}(0) \subset \mathcal{B}_{s_k}^\bullet \subset B_{3r}(0)$ . Apply Lemma 8.119 with  $K = \mathcal{B}_{s_k}^\bullet$ ,  $n = n_k$ , and  $L = \varepsilon_k r n_k^{1/2}$ . Since  $\varepsilon_k \asymp n_k^{-1/4}$ , this gives

$$\#\mathcal{I}_k^* \leq C_a n_k^{1/2}.$$

Choose  $N_0$  so large that this is at most  $n_k/4$  whenever  $n_k \geq N_0$ .

For each  $I \notin \mathcal{I}_k^*$  let  $G_I$  be the shield event of Lemma 8.123, and let  $\mathcal{I}_k^{**}$  be the arcs for which  $G_I$  fails. By (R4),

$$R_r^{\varepsilon_k}(\mathcal{B}_{s_k}^\bullet) \leq (6\varepsilon_k^{1/2} + \varepsilon_k)r \leq 7n_k^{-1/8}r.$$

Increasing  $N_0$  again makes this no larger than  $\text{diam}(\mathcal{B}_{s_k}^\bullet) \geq ar$ . Hence every arc outside  $\mathcal{I}_k^* \cup \mathcal{I}_k^{**}$  is killed before step  $k+1$ , so

$$n_{k+1} \leq \#\mathcal{I}_k^* + \#\mathcal{I}_k^{**}. \quad (8.7)$$

Conditionally on  $\mathcal{F}_{s_k}$ , Lemma 8.123 gives

$$\mathbb{E}[\#\mathcal{I}_k^{**} \mid \mathcal{F}_{s_k}] \leq C_0\varepsilon_k^\alpha(n_k - \#\mathcal{I}_k^*).$$

Markov's inequality yields

$$\mathbb{P}\left[\#\mathcal{I}_k^{**} > \frac{1}{4}(n_k - \#\mathcal{I}_k^*) \mid \mathcal{F}_{s_k}\right] \leq 4C_0\varepsilon_k^\alpha.$$

If this event does not occur, then (8.7) and  $\#\mathcal{I}_k^* \leq n_k/4$  give  $n_{k+1} \leq n_k/2$ .

For  $k \in [k_j, k_{j+1})$ , one has  $n_k \geq n_{k_j}/2$  and therefore  $\varepsilon_k \leq Cn_{k_j}^{-1/4}$ . The events determining whether step  $k$  fails are measurable by time  $s_{k+1}$ , so the conditional estimate can be iterated over  $M$  steps. Thus

$$\mathbb{P}[k_{j+1} - k_j > M, k_{j+1} \leq K_{N_0}, \mathcal{E}_r(a) \mid \mathcal{F}_{s_{k_j}}] \leq (Cn_{k_j}^{-\alpha/4})^M,$$

which is (8.6) after decreasing the exponent and increasing  $C$ .  $\square$

**Lemma 8.126** (Metric-radius cost of pruning). *There exist  $b_0, b_1, \beta > 0$ , with  $b_0$  allowed to depend on  $a$ , such that for every  $N \in \mathbb{N}$ , if*

$$K_N := \min\{k : s_k > \tau_{3r} \text{ or } n_k < N\},$$

then

$$\mathbb{P}\left[s_{K_N} > \tau + N^{-\beta}c_re^{\xi h_r(0)}, \mathcal{E}_r(a)\right] \leq b_0e^{-b_1N^\beta}. \quad (8.8)$$

*Proof.* On  $\mathcal{E}_r(a)$  and while  $k < K_N$ , (R4) implies that every point of  $B_{R_r^{\varepsilon_k}(\mathcal{B}_{s_k}^\bullet)}(\mathcal{B}_{s_k}^\bullet)$  is within Euclidean distance  $(6\varepsilon_k^{1/2} + \varepsilon_k)r$  of the current filled ball. By (R3),

$$s_{k+1} - s_k \leq C(6\varepsilon_k^{1/2} + \varepsilon_k)^\chi c_re^{\xi h_r(0)} \leq Cn_k^{-\chi/8}c_re^{\xi h_r(0)}.$$

Hence

$$s_{K_N} \leq \tau + Cc_re^{\xi h_r(0)} \sum_{k < K_N} n_k^{-\chi/8}. \quad (8.9)$$

Let  $k_j$  be the halving times from Lemma 8.125, and let  $J$  be the first index with  $k_J \geq K_N$ . For  $k \in [k_j, k_{j+1})$ ,  $n_k \geq 2^{J-j-2}N$ . Therefore

$$\sum_{k < K_N} n_k^{-\chi/8} \leq CN^{-\chi/8} \sum_{j=0}^{J-1} (k_{j+1} - k_j) 2^{-(J-j)\chi/8}.$$

By (8.6), with  $M = N^{\chi/16}$  the probability that any  $k_{j+1} - k_j > M$  before  $K_N$ , while  $\mathcal{E}_r(a)$  holds, is bounded by

$$C \exp[-cN^{\chi/16} \log N].$$

On the complementary event the geometric sum above is  $O(N^{\chi/16})$ , so (8.9) gives

$$s_{K_N} - \tau \leq CN^{-\chi/16} c_r e^{\xi_{h_r}(0)}.$$

Choose  $\beta < \chi/16$  and absorb the constant  $C$  by decreasing  $\beta$  for large  $N$ ; enlarge  $b_0$  to cover bounded  $N$ . This proves (8.8).  $\square$

### 8.11.6 The quantitative confluence theorem

**Theorem 8.127** (Quantitative confluence across a filled metric annulus). *Fix  $a \in (0, 1)$ . There are constants  $b_0 > 0$  depending on  $a$  and  $b_1, \beta > 0$  depending only on the law of  $D$  such that the following holds. For every  $r > 0$ ,  $N \in \mathbb{N}$ , and  $(\mathcal{F}_s)$ -stopping time  $\tau \in [\tau_r, \tau_{2r}]$  almost surely, the probability that  $\mathcal{E}_r(a)$  occurs and that more than  $N$  points of  $\partial\mathcal{B}_\tau^\bullet$  are hit by leftmost  $D_h$ -geodesics from 0 to*

$$\partial\mathcal{B}_{\tau+N^{-\beta}c_re^{\xi_{h_r}(0)}}^\bullet$$

*is at most*

$$b_0 e^{-b_1 N^\beta}.$$

*Proof.* Start with an arbitrary finite partition  $\mathcal{I}_0$  of  $\partial\mathcal{B}_\tau^\bullet$  into disjoint half-open arcs and run the recursive construction above. By Lemma 8.126, except on an event of probability  $b_0 e^{-b_1 N^\beta}$ , while  $\mathcal{E}_r(a)$  holds one has

$$s_{K_N} \leq \tau + N^{-\beta} c_r e^{\xi_{h_r}(0)}.$$

After decreasing  $\beta$  if necessary,  $N^{-\beta} < a$  for all sufficiently large  $N$ . Condition (R2) and  $\tau \leq \tau_{2r}$  then imply  $s_{K_N} < \tau_{3r}$ . Hence the definition of  $K_N$  forces  $n_{K_N} < N$ . Therefore at most  $N$  arcs of the original partition are hit by leftmost geodesics which survive to the displayed later boundary.

To turn this into a statement about points, for  $m \in \mathbb{N}$  partition  $\partial\mathcal{B}_\tau^\bullet$  into  $m$  half-open prime-end arcs of equal harmonic measure from infinity. The preceding estimate holds uniformly in  $m$ . If the set of hit points had cardinality larger than  $N$ , compactness and distinctness of prime ends would imply that for all sufficiently large  $m$  more than  $N$  partition arcs are hit. Letting  $m \rightarrow \infty$  proves the theorem.  $\square$

*Remark 8.128* (Scope of the preceding argument). The theorem above is the exact quantitative confluence input consumed by the uniqueness paper's Section-4 stability argument. This proposition does not promote the uniqueness parent: the remaining children are the construction of many stable exploration times along a random geodesic, the conditional shortcut probability and Bernoulli amplification, and the final extremal-ratio contradiction. In particular, ordinary annular near-independence is still not being substituted for independence along an explored geodesic.

## 8.12 Stable exploration and local independence along a geodesic

This section proves the independence-transfer step of the uniqueness proof. The statement is an abstract transfer theorem: a high-probability local annulus event is converted into an event seen by a random geodesic, provided the two events satisfy the source conditional comparison. The construction of the particular shortcut events and the proof of that conditional comparison belong to H3 and are not asserted here. This separation is essential: the theorem below removes the false independence heuristic for disjoint Euclidean balls, but it does not by itself produce the strict metric-ratio improvement.



We use the following notation. If  $X \subset \mathbb{C}$  and  $r > 0$ , then  $B_r(X)$  is its Euclidean  $r$ -neighborhood. The notation  $o_\varepsilon^\infty(\varepsilon)$  means a deterministic quantity bounded by  $C_A \varepsilon^A$  for every  $A > 0$ . All such estimates below are uniform in the base scale  $R > 0$  and in the endpoint pair under consideration.

### 8.12.1 The source-strength transfer theorem

**Theorem 8.129** (Local independence along a random geodesic). *There is  $\nu_* > 0$ , depending only on the law of the weak LQG metric, with the following property. Fix*

$$0 < \mu < \nu \leq \nu_*, \quad 0 < \lambda_1 < \lambda_2 \leq \lambda_3 \leq \lambda_4 < \lambda_5.$$

*There is  $\mathfrak{p} \in (0, 1)$ , depending only on these parameters and the metric law, for which the conclusion below holds.*

*Let  $R > 0$ ,  $\varepsilon_0 \in (0, 1)$ , and let  $\mathcal{R} \subset (0, \varepsilon_0 R]$  be deterministic. Suppose that for  $z \in \mathbb{C}$  and  $r \in \mathcal{R}$  we have local events  $E_r(z)$  and, for distinct endpoints  $\mathbf{z}, \mathbf{w}$ , geodesic events  $\mathfrak{E}_r^{\mathbf{z}, \mathbf{w}}(z)$  satisfying:*

(H1) *for each  $\varepsilon \leq \varepsilon_0$  there are  $m_\varepsilon = \lfloor \mu \log_8 \varepsilon^{-1} \rfloor$  radii*

$$r_1^\varepsilon, \dots, r_{m_\varepsilon}^\varepsilon \in [\varepsilon^{1+\nu} R, \varepsilon R] \cap \mathcal{R}, \quad r_j^\varepsilon / r_{j-1}^\varepsilon \geq \lambda_4 / \lambda_1;$$

(H2)  *$E_r(z)$  is determined by  $(h - h_{\lambda_5 r}(z))|_{\mathbb{A}_{\lambda_1 r, \lambda_4 r}(z)}$ , while  $\mathfrak{E}_r^{\mathbf{z}, \mathbf{w}}(z)$  is determined by  $h|_{B_{\lambda_4 r}(z)}$  and the a.s. unique  $D_h$ -geodesic  $P^{\mathbf{z}, \mathbf{w}}$  stopped at its last exit from  $B_{\lambda_4 r}(z)$ ;*

(H3)  $\mathbb{P}[E_r(z)] \geq \mathfrak{p}$ ;

(H4) *if  $\mathbf{z}, \mathbf{w} \notin B_{\lambda_4 r}(z)$ , then almost surely*

$$\begin{aligned} & \Lambda^{-1} \mathbb{P} \left[ E_r(z), P^{\mathbf{z}, \mathbf{w}} \cap B_{\lambda_2 r}(z) \neq \emptyset \mid h|_{\mathbb{C} \setminus B_{\lambda_3 r}(z)} \right] \\ & \leq \mathbb{P} \left[ \mathfrak{E}_r^{\mathbf{z}, \mathbf{w}}(z), P^{\mathbf{z}, \mathbf{w}} \cap B_{\lambda_2 r}(z) \neq \emptyset \mid h|_{\mathbb{C} \setminus B_{\lambda_3 r}(z)} \right] \end{aligned} \quad (8.10)$$

*for a deterministic  $\Lambda > 1$ .*

*Then for every  $q > 0$ ,  $\ell \in (0, 1)$ , and bounded open  $U \subset \mathbb{C}$ , with probability tending to one as  $\varepsilon \rightarrow 0$ , simultaneously for all*

$$\mathbf{z}, \mathbf{w} \in (\varepsilon^q R \mathbb{Z}^2) \cap RU, \quad |\mathbf{z} - \mathbf{w}| \geq \ell R,$$

*there exist  $z \in \mathbb{C}$  and  $r \in [\varepsilon^{1+\nu} R, \varepsilon R]$  such that*

$$P^{\mathbf{z}, \mathbf{w}} \cap B_{\lambda_2 r}(z) \neq \emptyset \quad \text{and} \quad \mathfrak{E}_r^{\mathbf{z}, \mathbf{w}}(z) \text{ occurs.}$$

*The rate is uniform in  $R$ .*

*Proof roadmap.* The proof occupies the remainder of this section. It first establishes a boundary-tube occupation estimate, then constructs protected good scales and applies a conditional counting argument uniformly over a deterministic endpoint grid.

The fixed-endpoint geodesic is unique by Lemma 8.113. The rest of this section proves the theorem. We write the proof for  $\lambda_3 = 1$ ; replacing  $r$  by  $\lambda_3 r$  throughout gives the general case, with only deterministic changes to the five  $\lambda_i$ 's.

### 8.12.2 A boundary-tube occupation estimate

The conditional counting argument later needs more than Hölder continuity: it needs a geodesic not to trace a long portion of a random metric-ball boundary. We record and prove precisely the selected form which is used.

**Lemma 8.130** (A geodesic cannot trace a metric-ball boundary). *For every  $M > 0$  and  $R > 0$ , with probability  $1 - o_\varepsilon^\infty(\varepsilon)$  the following holds simultaneously. If  $s > 0$ , the metric ball  $\mathcal{B}_s(x)$  is contained in  $B_{\varepsilon^{-M}R}(x)$ , and  $P$  is a  $D_h$ -geodesic from  $x$  to a point outside  $\mathcal{B}_s(x)$ , then*

$$\text{area}(B_{\varepsilon R}(P) \cap B_{\varepsilon R}(\partial\mathcal{B}_s(x))) \leq \varepsilon^{2-1/M} R^2. \quad (8.11)$$

The estimate is uniform in  $R$  and  $x$ .

*Proof.* Translation lets us take  $x = 0$ . Call a Euclidean ball  $B_\rho(z)$   $C$ -good if

$$\sup_{u,v \in \partial B_\rho(z)} D_h(u,v; \mathbb{A}_{\rho/2,2\rho}(z)) \leq C D_h(\partial B_\rho(z), \partial B_{2\rho}(z)). \quad (8.12)$$

The annular crossing bounds and moment estimates from the metric tightness module, followed by the same separated-scale GFF iteration used in Lemma 8.122, give the following two uniform facts. Given  $M' > M$  and  $\zeta \in (0, 1)$ , one can choose  $C, A > 1$  so that, except with probability  $O(\varepsilon^{M'})$ :

(T1)  $B_{\varepsilon^{-M'}R}(0)$  is covered by  $C$ -good balls with radii in  $[2\varepsilon R, \varepsilon^{1-\zeta}R]$ ;

(T2) whenever  $u, v$  lie in this large ball and  $|u - v| \geq \varepsilon R$ ,

$$D_h(u, v) \geq \varepsilon^A \sup_{a,b \in B_{\varepsilon^{-M'}R}(0)} D_h(a, b). \quad (T2)$$

For completeness, (T1) follows by placing a bounded-overlap grid at every dyadic radius in the displayed range: the one-annulus crossing ratio is tight, and failure at every one of a logarithmic number of separated radii has arbitrarily high polynomial decay. For (T2), cover the large ball by an  $\varepsilon R/8$  grid. The lower annular-crossing tail gives a simultaneous lower bound between distinct grid neighborhoods; the diameter moment bound gives the upper bound for the large ball. Circle-average Gaussian tails and  $c_{aR}/c_R = (a)^{\xi Q + o(1)}$  turn their ratio into a deterministic power  $\varepsilon^A$ . Enlarging  $A$  absorbs the bounded grid-to-point costs. These are exactly the two quantitative inputs used below.

Assume (T1)–(T2), fix  $s$  and  $P$ , and let  $\tau_0 = s$ . Having defined  $\tau_j$ , let  $\tau_{j+1}$  be the first time after  $P$  exits  $B_{4\varepsilon^{1-\zeta}R}(P(\tau_j))$  that it returns to  $B_{\varepsilon R}(\partial\mathcal{B}_s)$ ; stop if it never returns. At  $P(\tau_j)$  choose a  $C$ -good ball  $B_j$  from (T1), with radius in the displayed range and intersecting  $\partial\mathcal{B}_s$ . Let  $\sigma_j$  be the first exit time of  $P$  from  $2B_j$ . Then

$$[\tau_j, \sigma_j] \subset [\tau_j, \tau_{j+1}], \quad \sigma_j - \tau_j \geq D_h(\partial B_j, \partial(2B_j)).$$

If  $y_j \in B_j \cap \partial\mathcal{B}_s$ , goodness and the geodesic parameterization give

$$\tau_j \leq D_h(0, y_j) + D_h(y_j, P(\tau_j)) \leq s + C(\sigma_j - \tau_j).$$

Since the excursion intervals are disjoint and lie after time  $s$ ,

$$\sum_{i < j} (\sigma_i - \tau_i) \leq \tau_j - s \leq C(\sigma_j - \tau_j).$$

Thus their lengths grow by a factor at least  $1 + C^{-1}$ . On the other hand, (T2), applied to the first and last excursions, says that their ratio is at most  $\varepsilon^{-A}$ . Consequently the number  $J$  of excursions is at most  $C_A \log \varepsilon^{-1}$ .

The part of  $P$  in the boundary tube is covered by the  $J$  balls  $B_{4\varepsilon^{1-\zeta}R}(P(\tau_j))$ . Hence its  $\varepsilon R$ -neighborhood has area at most

$$CJ\varepsilon^{2-2\zeta}R^2 \leq \varepsilon^{2-3\zeta}R^2$$

for small  $\varepsilon$ . Choose  $\zeta < 1/(3M)$  and then let  $M'$  be arbitrarily large. This yields (8.11) with superpolynomial failure probability.  $\square$

### 8.12.3 Stopped exploration and the stability event

Fix one endpoint pair  $\mathbf{z}, \mathbf{w} \in RU$  at Euclidean separation at least  $4\ell R$ , and let  $P = P^{\mathbf{z}, \mathbf{w}}$ , parameterized by  $D_h$ -length from  $\mathbf{z}$ . Write  $\mathcal{B}_s^\bullet$  for the filled metric ball of radius  $s$  centered at  $\mathbf{z}$  and

$$\tau_r = \inf\{s : \mathcal{B}_s^\bullet \not\subset B_r(\mathbf{z})\}.$$

Choose  $\beta > 0$  later and set

$$s_k = \tau_{\ell R} + k\varepsilon^\beta c_R e^{\xi h_R(\mathbf{z})}, \quad t_k = s_k + \varepsilon^{2\beta} c_R e^{\xi h_R(\mathbf{z})}. \quad (8.13)$$

Let  $\text{Conf}_k$  be the set of points of  $\partial\mathcal{B}_{s_k}^\bullet$  hit by leftmost geodesics from  $\mathbf{z}$  to  $\partial\mathcal{B}_{t_k}^\bullet$ . For  $x \in \text{Conf}_k$ , let  $I_k(x)$  be the arc of  $\partial\mathcal{B}_{t_k}^\bullet$  consisting of endpoints whose leftmost geodesics pass through  $x$ , and write  $\mathcal{I}_k$  for this arc partition. Finiteness and cyclic order follow from Theorem 8.127 and Lemma 8.118.

Use the increasing filtration

$$\mathcal{F}_k = \sigma\left(\mathcal{B}_{t_k}^\bullet, h|_{\mathcal{B}_{t_k}^\bullet}, P|_{[0, s_k]}\right). \quad (8.14)$$

**Lemma 8.131** (Arc representation of the exploration filtration). *Almost surely*

$$\mathcal{F}_k = \sigma\left(\mathcal{B}_{t_k}^\bullet, h|_{\mathcal{B}_{t_k}^\bullet}, \text{the arc of } \mathcal{I}_k \text{ containing } P(t_k)\right).$$

*Proof.* If  $\mathbf{w} \in \mathcal{B}_{t_k}^\bullet$ , the exposed field and ball determine the relevant part of  $P$ . Otherwise  $P|_{[0, t_k]}$  is the unique, hence leftmost, geodesic to  $P(t_k)$ . The arc containing  $P(t_k)$  determines the confluence point  $P(s_k)$ ; the internal metric and fixed-endpoint uniqueness then determine the geodesic from  $\mathbf{z}$  to that point. Conversely the stopped path determines  $P(t_k)$ 's arc by continuation to the exposed boundary. Both sigma-algebras therefore coincide.  $\square$

Let  $\delta_\varepsilon = \lambda_1 \varepsilon^{1+\nu} R/4$ . The candidate set  $\mathcal{Z}_k$  consists of pairs  $(z, r)$  with

$$z \in \delta_\varepsilon \mathbb{Z}^2 \setminus \mathcal{B}_{t_k}^\bullet, \quad r \in \{r_1^\varepsilon, \dots, r_{m_\varepsilon}^\varepsilon\}, \quad \text{dist}(z, \partial\mathcal{B}_{t_k}^\bullet) \in [\lambda_4 \varepsilon R, 2\lambda_4 \varepsilon R]. \quad (8.15)$$

This set is determined by the exposed filled ball. Define  $\text{Stab}_{k,r}(z)$  to be the event that  $(z, r) \in \mathcal{Z}_k$  and every  $D_h(\cdot, \cdot; \mathbb{C} \setminus \overline{B_r(z)})$ -geodesic from  $\mathbf{z}$  to  $\partial B_r(z)$  hits  $\partial\mathcal{B}_{t_k}^\bullet$  in the same arc of  $\mathcal{I}_k$ .

**Lemma 8.132** (Exterior measurability at a stable candidate). *The event  $\text{Stab}_{k,r}(z)$  is determined by  $h|_{\mathbb{C} \setminus B_r(z)}$ . On*

$$\text{Stab}_{k,r}(z) \cap \{P \cap B_r(z) \neq \emptyset\},$$

*the data generating  $\mathcal{F}_k$  are determined by the exterior field and the indicator of this event.*

*Proof.* On  $(z, r) \in \mathcal{Z}_k$ , the candidate ball is disjoint from the exposed filled ball. Locality determines from the exterior field the filled ball, its exposed field,  $\mathcal{I}_k$ , and every exterior-internal geodesic from  $\mathbf{z}$  to  $\partial B_r(z)$ . Hence stability is exterior measurable. If  $P$  enters  $B_r(z)$ , stop it at the first entrance. This stopped segment is one of those exterior-internal geodesics. Stability therefore identifies, from the exterior, the unique arc containing  $P(t_k)$ . Invoke Lemma 8.131.  $\square$

Set

$$\begin{aligned}\mathcal{Z}_k^E &= \{(z, r) \in \mathcal{Z}_k : E_r(z), \text{Stab}_{k,r}(z), P \cap B_{\lambda_2 r}(z) \neq \emptyset\}, \\ \mathcal{Z}_k^\mathfrak{E} &= \{(z, r) \in \mathcal{Z}_k : \mathfrak{E}_r^{\mathbf{z}, \mathbf{w}}(z), \text{Stab}_{k,r}(z), P \cap B_{\lambda_2 r}(z) \neq \emptyset\}.\end{aligned}$$

**Lemma 8.133** (Conditional count transfer). *For each fixed  $M > 0$ , on the event that  $\mathcal{B}_{t_k}^\bullet \subset B_{\varepsilon^{-M}R}(\mathbf{z})$  and  $\mathbf{w}$  is at Euclidean distance at least  $3\lambda_4\varepsilon R$  from the filled ball,*

$$\mathbb{P}[\mathcal{Z}_k^\mathfrak{E} \neq \emptyset \mid \mathcal{F}_k] \geq \varepsilon^{2\nu+o_\varepsilon(1)} \mathbb{P}[\mathcal{Z}_k^E \neq \emptyset \mid \mathcal{F}_k] - o_\varepsilon^\infty(\varepsilon) \quad (8.16)$$

outside an event of probability  $o_\varepsilon^\infty(\varepsilon)$ .

*Proof.* Sum (8.10) over the countable set  $\mathcal{Z}_k$ . Lemma 8.132 permits conditioning further on  $\mathcal{F}_k$  without changing the inequality, so

$$\mathbb{E}[\#\mathcal{Z}_k^\mathfrak{E} \mid \mathcal{F}_k] \geq \Lambda^{-1} \mathbb{E}[\#\mathcal{Z}_k^E \mid \mathcal{F}_k]. \quad (8.17)$$

It remains to convert expectation to occurrence probability. Let  $\mathcal{Z}_k(P)$  be the candidate pairs whose balls meet  $P$ . Apply Lemma 8.130 with  $4\lambda_4\varepsilon$  in place of  $\varepsilon$ . Except with superpolynomially small probability, the relevant intersection of the geodesic tube and filled-boundary tube has area at most  $\varepsilon^{2-\zeta/2}R^2$ . Candidate centers have spacing  $\asymp \varepsilon^{1+\nu}R$ . Bounded overlap at each radius and the  $O(\log \varepsilon^{-1})$  choices of radius therefore give

$$\#\mathcal{Z}_k(P) \leq \varepsilon^{-2\nu-\zeta}$$

except with superpolynomially small probability. Markov's inequality, first conditionally on  $\mathcal{F}_k$  and then for the exceptional tail, gives

$$\mathbb{E}[\#\mathcal{Z}_k^\mathfrak{E} \mid \mathcal{F}_k] \leq \varepsilon^{-2\nu-\zeta} \mathbb{P}[\mathcal{Z}_k^\mathfrak{E} \neq \emptyset \mid \mathcal{F}_k] + o_\varepsilon^\infty(\varepsilon).$$

On the other hand  $\mathbb{E}[\#\mathcal{Z}_k^E \mid \mathcal{F}_k] \geq \mathbb{P}[\mathcal{Z}_k^E \neq \emptyset \mid \mathcal{F}_k]$ . Combine these inequalities with (8.17) and send  $\zeta \downarrow 0$  slowly.  $\square$

#### 8.12.4 Regularity and many stable times

Choose exponents

$$\chi \in (0, \xi(Q-2)), \quad \chi' > \xi(Q+2),$$

from the bi-Hölder theorem. For a bounded connected  $U \Subset V$ , let  $\mathcal{E}_R(a)$  be the event consisting of the following six simultaneous conditions:

- (R1) distances within  $RU$  are smaller than the distance from  $RU$  to  $R\partial V$ ;
- (R2) uniformly for centers in the fixed enlargement of  $RV$ , Euclidean balls and the filled metric balls at radii  $\ell R, 2\ell R, 3\ell R$  contain one another at scale  $aR$ , and successive metric radii are at least  $a \max\{c_R e^{\xi h_R}, c_{\ell R} e^{\xi h_{\ell R}}\}$  apart;

(R3) for  $|u - v| \leq aR$  in that enlargement,

$$c_R^{-1} e^{-\xi h_R(0)} D_h(u, v) \geq (|u - v|/R)^{\chi'}, \quad c_R^{-1} e^{-\xi h_R(0)} D_h(u, v; B_{2|u-v|}(u)) \leq (|u - v|/R)^\chi;$$

(R4) the relevant radius- $R$  circle averages differ from  $h_R(0)$  by at most  $a^{-1}$ ;

(R5) at every dyadic  $\varepsilon \leq a$  and every  $\delta_\varepsilon$ -grid center in the enlargement, at least one of the radii in (H1) satisfies  $E_r(z)$ ;

(R6) the multiscale shield-search radius, at every relevant grid center and dyadic scale  $\delta \leq a$ , is at most  $\delta^{1/2}R$ .

**Lemma 8.134** (Uniform global regularity). *For every  $p \in (0, 1)$  one can first choose  $V \supset \bar{U}$ , then  $a \in (0, 1)$ , and finally choose the threshold  $\mathbf{p}$  in Theorem 8.129 sufficiently close to one, so that*

$$\inf_{R>0} \mathbb{P}[\mathcal{E}_R(a)] \geq p.$$

*Proof.* Scale tightness chooses  $V$  and gives (R1)–(R2). The preceding analysis Hölder and circle-average estimates give (R3)–(R4), after shrinking  $a$ . For (R5), the annuli in (H1) are separated by  $\lambda_4/\lambda_1$  and the events are centered-field measurable. The GFF Markov decomposition and the uniform Radon–Nikodym estimate used in Lemma 8.122 imply exponential decay in the number of tested annuli when the one-scale probability  $\mathbf{p}$  is close enough to one. Choose it so that this decay beats the number of grid centers and sum over dyadic  $\varepsilon$ . Condition (R6) is precisely the uniform search-radius estimate underlying Lemma 8.124. Allocate  $(1 - p)/6$  failure probability to each condition.  $\square$

We need one deterministic planar fact. For a compact full connected set  $K$ , equip  $\overline{\mathbb{C} \setminus K}$  with its prime-end metric  $d^K$  (pull back the Euclidean metric under a Riemann map). If  $I$  is a prime-end arc and a curve starting at  $I$  comes within  $d^K$ -distance  $\delta$  of  $\partial K \setminus I$ , then an initial portion of the curve, together with a crosscut of Euclidean diameter controlled by the Euclidean modulus corresponding to  $\delta$ , disconnects one endpoint of  $I$  from infinity. The following quantitative form is what we use.

**Lemma 8.135** (Endpoint-danger shield). *Let  $K$  be compact and full, let  $I \subset \partial K$  be a prime-end arc, and suppose a path  $Q$  starts from  $I$ . On the Hölder condition (R3), if  $Q$  has metric length at most  $\delta^\chi c_R e^{\xi h_R(0)}$  before it comes  $d^K$ -close to  $\partial K \setminus I$ , then the path enters a region containing an endpoint of  $I$  which is separated from infinity in  $\mathbb{C} \setminus K$  by a connected set of Euclidean diameter at most  $2\delta^{\chi/\chi'} R$ .*

*Moreover, for each endpoint  $y$  and  $d > 0$ , the union of all regions which can be separated together with  $y$  by a connected set of diameter at most  $d$  is itself contained behind one connected separator of diameter at most  $16d$ .*

*Proof.* Let  $X_0$  be a crosscut witnessing the prime-end proximity, chosen with Euclidean diameter at most  $\delta R$ . Join its first contact with  $Q$  back to  $I$  along  $Q$ . By (R3), a path segment of metric length at most  $\delta^\chi c_R e^{\xi h_R(0)}$  has Euclidean diameter at most  $\delta^{\chi/\chi'} R$ . The union is connected and its prime-end closure meets both  $I$  and its complement; the Jordan separation theorem therefore puts one endpoint of  $I$  in a bounded complementary component. Letting the crosscut approximation tend to its infimum gives the factor two.

For the second assertion, choose a maximal family of witness crosscuts whose bounded sides containing  $y$  are strictly increasing. Planarity makes the outermost two witnesses sufficient: otherwise two crosscuts of diameter  $d$  would have to cross while separating their bounded sides. Join these

two crosscuts by subarcs inside their  $2d$ -neighborhoods and discard loops. The resulting continuum has diameter at most  $16d$  and its bounded side contains every witness region. This construction also applies to limiting prime ends by kernel convergence.  $\square$

Put  $K = \lfloor a\varepsilon^{-\beta} \rfloor - 1$ , so (R2) ensures that  $s_{K+1} \leq \tau_{2\ell R}$ . Choose

$$\kappa = \frac{1}{2}(\chi/\chi')^2.$$

**Proposition 8.136** (Stable good candidates at almost every exploration time). *There are  $\beta, \theta > 0$ , depending only on the metric law, such that on  $\mathcal{E}_R(a)$ , except on an event of probability  $o_\varepsilon^\infty(\varepsilon)$ , at least  $(1 - \varepsilon^\theta)K$  indices  $k \in [0, K]$  satisfy  $\mathcal{Z}_k^E \neq \emptyset$ .*

*Proof.* Fix small  $\omega > 0$ . Apply Theorem 8.127 at  $s_k$  with  $N = \lfloor \varepsilon^{-\omega} \rfloor$ . Choose  $\beta$  so small that the metric-radius increment  $N^{-b}$  in that theorem is at most the gap  $t_k - s_k = \varepsilon^{2\beta} c_R e^{\xi_{h_R}(z)}$ , where  $b > 0$  is its confluence exponent. A union bound over  $K = O(\varepsilon^{-\beta})$  indices then gives

$$\max_{k \leq K} \# \text{Conf}_k \leq \varepsilon^{-\omega} \quad (8.18)$$

outside a superpolynomially small event on  $\mathcal{E}_R(a)$ .

For every endpoint  $y$  of an arc in  $\mathcal{I}_k$ , apply Lemma 8.135 with  $d = \varepsilon^\kappa R$ . The danger region is behind a connected separator of diameter at most  $16\varepsilon^\kappa R$ . Condition (R6) and (R3) imply that the stopping radius needed to explore its shield is at most  $7\varepsilon^{\kappa/2} R$  in Euclidean distance and at most  $C\varepsilon^{\kappa\chi/2} c_R e^{\xi_{h_R}(z)}$  in metric radius. Further decrease  $\beta$  so that  $\beta < \kappa\chi/2$ . The entire shield is then exposed before  $s_{k+1}$ .

Lemma 8.123, applied to the outer boundary arc of each endpoint-danger region, supplies a conditional failure probability at most  $C\varepsilon^{\alpha\kappa}$ . By (8.18), a union bound over endpoints is at most  $C\varepsilon^{\alpha\kappa-\omega}$ . Choose  $\omega < \alpha\kappa/2$ ; for small  $\varepsilon$  the conditional success probability is at least  $1 - \varepsilon^\omega$ .

These successes are measurable by the next exploration level. Sequential conditioning and Hoeffding's inequality therefore give

$$\mathbb{P}[\text{fewer than } (1 - \varepsilon^\theta)K \text{ shield-success indices}] \leq \exp[-2(\varepsilon^\theta - \varepsilon^\omega)^2 K] = o_\varepsilon^\infty(\varepsilon)$$

provided  $\theta < \min\{\omega, \beta/2\}$ .

Fix a shield-success index. Condition (R5), applied at a grid center on the portion of  $P$  crossing the Euclidean collar in (8.15), gives a pair  $(z, r) \in \mathcal{Z}_k$  for which  $E_r(z)$  occurs and  $P \cap B_{\lambda_2 r}(z) \neq \emptyset$ . Indeed, the grid error is at most  $\lambda_1 \varepsilon^{1+\nu} R/4$  and every selected  $r$  is at least  $\varepsilon^{1+\nu} R$ .

It remains to check stability. Any exterior-internal geodesic  $P'$  from  $z$  to  $\partial B_r(z)$  has, after time  $t_k$ , metric length at most the internal  $D_h$ -diameter of a Euclidean ball of radius  $O(\varepsilon R)$ . By (R3), this is  $O(\varepsilon^\chi) c_R e^{\xi_{h_R}(0)}$ , and the lower Hölder bound gives

$$\text{diam } P'([t_k, |P'|]) \leq C\varepsilon^{\chi/\chi'} R. \quad (8.19)$$

If  $P'$  hit a different arc of  $\mathcal{I}_k$  from  $P$ , the prime-end distance between the two tails, together with (8.19), would place one of them in an endpoint-danger region by Lemma 8.135; here  $\kappa < (\chi/\chi')^2$  is exactly the exponent inequality needed. The shield-success event excludes this. Thus every  $P'$  hits the same arc and  $\text{Stab}_{k,r}(z)$  occurs. Hence  $\mathcal{Z}_k^E \neq \emptyset$  at every shield-success index.  $\square$

### 8.12.5 Adaptive amplification and completion of the proof

The events just constructed are not automatically  $\mathcal{F}_{k+1}$  measurable. We first record the source containment repair.

**Lemma 8.137** (One-step containment and measurability). *There is an event  $F_k \in \sigma(\mathcal{B}_{s_{k+1}}^\bullet, h|_{\mathcal{B}_{s_{k+1}}^\bullet})$  such that  $\mathcal{E}_R(a) \subset \bigcap_{k \leq K} F_k$  for all sufficiently small  $\varepsilon$ . On  $F_k$ , every  $B_{\lambda_4 r}(z)$  with  $(z, r) \in \mathcal{Z}_k$  lies in  $\mathcal{B}_{s_{k+1}}^\bullet$ , and all exterior-internal geodesics used in  $\text{Stab}_{k,r}(z)$  are determined by the exposed field at  $s_{k+1}$ . Consequently*

$$\{\mathcal{Z}_k^E \neq \emptyset\} \cap F_k, \quad \{\mathcal{Z}_k^\mathfrak{E} \neq \emptyset\} \cap F_k \quad \text{belong to } \mathcal{F}_{k+1}.$$

*Proof.* Define  $F_k$  to be the conjunction of  $s_{k+1} \leq \tau_{2\ell R}$ , the displayed ball containments, and the statement that every minimizing exterior path to a candidate boundary is contained in  $\mathcal{B}_{s_{k+1}}^\bullet$ . These are determined by the stopped filled ball and its internal metric. On  $\mathcal{E}_R(a)$ , a candidate lies at Euclidean distance  $O(\varepsilon R)$  from  $\mathcal{B}_{t_k}^\bullet$ . The upper Hölder bound makes the cost of absorbing its  $\lambda_4 r$ -neighborhood at most  $C\varepsilon^{\chi_{CR}} e^{\xi_{h_R}(0)}$ . Our choice of  $\beta < \chi$  makes this smaller than  $s_{k+1} - t_k$ . The same comparison applied to a minimizing exterior path, with the tail bound (8.19), gives the geodesic containment. Thus all  $F_k$  occur on the regularity event.

On  $F_k$ , locality determines  $E_r(z)$ , the exterior-internal geodesics, and stability by time  $s_{k+1}$ . Hypothesis (H2) determines  $\mathfrak{E}_r(z)$  because the relevant stopped segment of  $P$  is also contained there. Countability of the candidate set permits taking the union over  $(z, r)$ , proving the two measurability assertions.  $\square$

**Lemma 8.138** (Adaptive conditional Hoeffding bound). *Let  $\mathcal{G}_0 \subset \dots \subset \mathcal{G}_{K+1}$  and  $A_k \in \mathcal{G}_{k+1}$ . For  $\alpha \in (0, 1)$ ,  $\delta \in (0, \alpha)$ , and  $m \in \mathbb{N}$ ,*

$$\mathbb{P} \left[ \begin{aligned} \#\{k : \mathbb{P}[A_k \mid \mathcal{G}_k] \geq \alpha\} &\leq K - m, \\ \#\{k : A_k \text{ occurs}\} &\geq K - (1 - \alpha - \delta)m \end{aligned} \right] \leq e^{-2\delta^2 m}.$$

*Proof.* Let  $\tau_j$  be the  $j$ th index with  $\mathbb{P}[A_k \mid \mathcal{G}_k] < \alpha$ . Then  $\{\tau_j = k\} \in \mathcal{G}_k$  and  $\mathbb{P}[A_{\tau_j}^c \mid \mathcal{G}_{\tau_j}] \geq 1 - \alpha$ . Since earlier selected events are measurable at the next selected time, the sum of the first  $m+1$  indicators  $\mathbf{1}_{A_{\tau_j}^c}$  stochastically dominates a binomial variable with parameters  $(m+1, 1 - \alpha)$ . Hoeffding's inequality bounds the probability that it is smaller than  $(1 - \alpha - \delta)m$  by  $e^{-2\delta^2 m}$ , which is the claim.  $\square$

**Proposition 8.139** (Many very-good exploration times). *Choose  $\nu$  so that  $4\nu < \beta \wedge \theta$ . For every sufficiently small  $\zeta > 0$ , on  $\mathcal{E}_R(a)$ , except with probability  $o_\varepsilon^\infty(\varepsilon)$ , at least  $\varepsilon^{2\nu+\zeta} K$  indices  $k \leq K$  satisfy  $\mathcal{Z}_k^\mathfrak{E} \neq \emptyset$ .*

*Proof.* Apply Lemma 8.138 first to  $A_k = \{\mathcal{Z}_k^E \neq \emptyset\} \cap F_k$ . Proposition 8.136 implies that, except with superpolynomially small probability, at least  $(1 - 2\varepsilon^\theta)K$  indices have

$$\mathbb{P}[\mathcal{Z}_k^E \neq \emptyset \mid \mathcal{F}_k] \geq 1 - \varepsilon^{\theta/2}.$$

For every such index, Lemma 8.133 gives

$$\mathbb{P}[\mathcal{Z}_k^\mathfrak{E} \neq \emptyset \mid \mathcal{F}_k] \geq \varepsilon^{2\nu+\zeta/4}$$

after decreasing  $\varepsilon$ . Apply the same adaptive bound to the very-good events, using Lemma 8.137. The probability of fewer than  $\varepsilon^{2\nu+\zeta}K$  successes is bounded by

$$\exp[-c\varepsilon^{4\nu+\zeta}K] \leq \exp[-c\varepsilon^{4\nu+\zeta-\beta}],$$

which is superpolynomially small because  $4\nu + \zeta < \beta$ . The discarded  $O(\varepsilon^\theta K)$  indices are negligible because  $4\nu < \theta$ .  $\square$

*Proof of Theorem 8.129.* Choose  $\nu_* > 0$  small enough that the parameter choices in the preceding proofs can be made with  $4\nu_* < \beta \wedge \theta$ . Fix  $p < 1$  and choose the global event of Lemma 8.134. On this event, Proposition 8.139 supplies at least one very-good candidate for the fixed endpoint pair, except with superpolynomially small probability. Conditions (R1)–(R2) ensure throughout that the target has not been swallowed and that all large-ball truncations in Lemma 8.133 hold.

There are  $O(\varepsilon^{-2q})$  mesh points in  $RU$ , hence  $O(\varepsilon^{-4q})$  ordered endpoint pairs. The superpolynomial error survives their union bound. The regularity event is common to all endpoint pairs, so the simultaneous conclusion has lower limiting probability at least  $p$ . Finally send  $p \uparrow 1$ . All constants were uniform in  $R$ , which proves the claimed scale-uniform statement.  $\square$

*Remark 8.140* (Scope of the preceding argument). The theorem above is the full Section–4 independence engine. It assumes, rather than constructs, the pair  $(E_r, \mathfrak{E}_r)$  and the conditional comparison (8.10). In the primary uniqueness proof those objects are built in the next section from near-extremal metric pairs and local field perturbations. Therefore H2 is closed, but H3 (shortcut-event construction and extremal-ratio contradiction) and I (scalar uniqueness and weak-to-strong upgrade) remain open. No parent row is established in this cycle.

## 8.13 Protected shortcuts and collapse of the extremal ratios

Throughout this section  $D$  and  $\tilde{D}$  are weak  $\gamma$ -LQG metrics coupled with the same whole-plane GFF and using the same constants  $c_r$  in Axiom V. We retain the deterministic optimal bi-Lipschitz constants  $c_* \leq C_*$  from the preceding comparison module. The aim is to prove that they coincide. The proof is by contradiction, so until the last theorem we assume

$$c_* < C_*.$$

The notation  $\mathbb{A}_{a,b}(z) = B_b(z) \setminus \overline{B_a(z)}$  will be used below. All events are unchanged by adding a constant to the field. Consequently their scale-uniform probability estimates are expressed using the centered restriction  $h - h_R(z)$ .

### 8.13.1 A protected reservoir of lower-extremal pairs

The lower-extremal conclusion of Proposition 8.116 must be strengthened before a Cameron–Martin perturbation can be used. It is essential that the  $\tilde{D}_h$ -geodesic realizing the comparison be unique and remain in the region on which the perturbation is constant.

**Proposition 8.141** (Protected lower-extremal annuli). *Fix  $0 < \mu < \nu < 1$ . There are  $\alpha_* \in (1/2, 1)$  and  $p_0 \in (0, 1)$ , depending only on the two metric laws, with the following property. For  $\alpha \in [\alpha_*, 1)$  and  $c' > c_*$ , there is  $c'' \in (c_*, c')$  such that, whenever*

$$\mathbb{P}[\underline{G}_R(c'', \beta)] \geq \beta$$



for some  $R > 0$  and  $\beta \in (0, 1)$ , then for every sufficiently small  $\varepsilon$  at least  $\mu \log_8 \varepsilon^{-1}$  radii

$$r \in [\varepsilon^{1+\nu} R, \varepsilon R] \cap \{8^{-k} R : k \in \mathbb{N}\}$$

have the following property with probability at least  $p_0$ : there are  $u \in \partial B_{\alpha r}(0)$  and  $v \in \partial B_r(0)$  such that

$$\tilde{D}_h(u, v) \leq c' D_h(u, v),$$

and the unique  $\tilde{D}_h$ -geodesic from  $u$  to  $v$  is contained in  $\overline{\mathbb{A}_{\alpha r, r}(0)}$ . The conclusion is translation covariant, with the same constants.

*Proof.* Apply the contrapositive construction in the proof of Proposition 8.116 to the ordered pair  $(\tilde{D}, D)$ , before its final projection from annular crossings to the ball event. For this ordered pair the lower and upper deterministic ratios are  $C_*^{-1}$  and  $c_*^{-1}$ . The target inequality becomes

$$D_h(u, v) \geq (c')^{-1} \tilde{D}_h(u, v),$$

so the upper protected comparison level is  $(c')^{-1} < c_*^{-1}$ . The intermediate level produced by that proof is  $(c'')^{-1}$ , where  $c'' \in (c_*, c')$ , exactly as in the statement.

We recall why the pre-projection conclusion is protected. At a candidate radius, the local event has three clauses: comparison on every unique base-metric geodesic contained in  $\overline{\mathbb{A}_{\alpha r, r}}$ ; a buffer clause which decides from the internal metrics whether such a path is globally geodesic; and a short separating circuit. Its proof first chooses the circuit quantile, then takes  $\alpha$  sufficiently close to one so that the narrow-annulus and buffer clauses have the required uniform probability. Locality makes the conjunction measurable from the centered field in  $\mathbb{A}_{r/2, 2r}$ . Annular near-independence and the greedy geodesic-cover argument show that failure at more than  $(\nu - \mu) \log_8 \varepsilon^{-1}$  radii would improve the swapped upper extremal constant from  $c_*^{-1}$  to a strictly smaller number, contradicting the base-scale event at level  $(c'')^{-1}$ .

The base metric in the swapped argument is  $\tilde{D}$ . Its protected path is therefore the unique  $\tilde{D}_h$ -geodesic, and translating back from the reciprocal inequality gives precisely  $\tilde{D}_h(u, v) \leq c' D_h(u, v)$ . This proves every clause of the statement, including the same density and translation-uniform constants.  $\square$

### 8.13.2 From one protected annulus to a high-probability tube

Choose constants

$$c_* < c'_1 < c'_2 < C_*.$$

Choose  $\eta > 0$  sufficiently small that

$$\frac{c'_1(1 + 2\eta)}{1 - 2c_*^{-1}C_*\eta} < c'_2, \quad 1 + 2\eta < C_*/c_*. \quad (8.20)$$

**Lemma 8.142** (Half-annulus selection). *There are deterministic  $\alpha \in (\alpha_*, 1)$ ,  $b > 0$ , and a set of protected radii  $\mathcal{R}_0$  of the density supplied by Proposition 8.141, such that for each  $r \in \mathcal{R}_0$ , with probability at least  $p_0/8$ , a deterministic one of four half-annuli contains points  $u, v$ , separated by at least  $br$ , for which*

$$\tilde{D}_h(u, v) \leq c'_1 D_h(u, v), \quad (8.21)$$

$$\tilde{D}_h(u, v) \leq (c_*/C_*)^2 \tilde{D}_h(\mathbb{A}_{\alpha r, r}, \partial B_{2r}), \quad (8.22)$$

and whose unique  $\tilde{D}_h$ -geodesic lies in that half-annulus.

*Proof.* Apply Proposition 8.141 with  $c' = c'_1$ . Tightness of the normalized radial and circumferential distances gives constants  $0 < s < S < \infty$  such that the protected geodesic has length at most  $S c_r e^{\xi_{h_r}(0)}$ , while the buffered distance to  $\partial B_{2r}$  is at least  $s c_r e^{\xi_{h_r}(0)}$ , except on an event of probability at most  $p_0/4$ . Taking  $\alpha$  closer to one if necessary makes the protected crossing smaller than  $(c_*/C_*)^2 s c_r e^{\xi_{h_r}(0)}$  on a subevent of probability at least  $p_0/2$ . Split the annulus into four overlapping deterministic half-annuli. One of them carries probability at least  $p_0/8$ . Decreasing  $b$  and using non-degeneracy of the protected crossing gives  $|u - v| \geq br$ . This yields both displayed inequalities.  $\square$

Fix a small grid ratio  $\epsilon_0$ . A grid tube is the interior of a connected finite union of  $\epsilon_0 r$ -squares. For a tube  $U$  and its distinguished points  $u, v$ , let  $O_u, O_v$  be the components of  $U \cap B_{20\epsilon_0 r}(u)$  containing  $u, v$ , respectively.

**Lemma 8.143** (Uniform grid-tube reservoir). *For every  $p < 1$  there are  $\rho, \epsilon_0, b, \zeta \in (0, 1)$ , depending only on  $p, \mu, \nu, c'_1, c'_2$ , and for every  $r \in \rho^{-1}\mathcal{R}_0$  a centered-field event  $T_r$ , such that  $\mathbb{P}[T_r] \geq p$ . On  $T_r$ , for every  $x, y \in \partial B_{2r}$  with  $|x - y| \geq \delta r$ , there is a grid tube  $U_r^{x,y} \subset \mathbb{A}_{r/2, 3r}$  containing points  $u, v$  with:*

- (T1)  $|u - v| \geq br$ , equations (8.21)–(8.22) hold, and the unique  $\tilde{D}_h$ -geodesic from  $u$  to  $v$  lies in  $U_r^{x,y}$ ;
- (T2) deleting  $O_u \cup O_v$  separates the two ends of the tube, with  $O_u$  adjacent to  $x$  and  $O_v$  adjacent to  $y$ ;
- (T3) the internal  $D_h$ -diameter of each of  $O_u, O_v$  is at most  $\eta \tilde{D}_h(u, v)$ .

The event and the finite list of possible grid tubes are determined by  $(h - h_{5r}(0))|_{\mathbb{A}_{r/4, 4r}}$ .

*Proof.* On the positive-probability event of Lemma 8.142, extend the protected  $\tilde{D}_h$ -geodesic by deterministic arcs to a path joining the two ends of its half-annulus. Approximate the path from outside by grid squares. There are finitely many grid shapes. Pigeonholing selects one deterministic shape with probability  $p_{\text{tube}} > 0$ . By taking  $\epsilon_0$  small, the boundary separation in (8.22), metric locality, and continuity give (T2)–(T3) without losing more than half this probability. Because the geodesic is unique and stays a positive metric distance from the boundary, the selected event is measurable from the buffered annulus.

Place disjoint balls  $B_{4\rho r}(z_k)$  at evenly spaced points of  $\partial B_r$ . In each ball test the scaled positive-probability tube event, choosing the orientation from the deterministic half-annulus. Lemma 8.86 shows, after taking  $\rho$  small, that with probability at least  $p$  every circular arc of length  $\delta r/10$  contains a successful ball. Concatenate successful local tubes and the intervening grid corridors. For every separated pair  $x, y$ , this produces  $U_r^{x,y}$  and preserves (T1)–(T3); decreasing  $b$  absorbs the grid error. Since the grid is finite and all internal metrics are local, the simultaneous event has the claimed measurability.  $\square$

### 8.13.3 The finite bump family and conditional comparison

The next proposition is the exact interface used by the independence theorem. Its proof records the parameter order, the finite Cameron–Martin family, and the path-forcing inequalities; none of these may be replaced by an assertion of independence between a geodesic and the local field.

**Proposition 8.144** (Finite-bump shortcut interface). *Let  $\mathbf{p}$  be the threshold in Theorem 8.129 for*

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5) = (1/4, 2, 3, 4, 5).$$

There are  $\rho, b_0 > 0$  such that for every  $r \in \rho^{-1}\mathcal{R}_0$  one can construct an event  $E_r$  and a finite deterministic family  $\mathcal{G}_r$  of smooth nonnegative functions supported in  $\mathbb{A}_{r/4, 3r}$  satisfying:

(B1)  $E_r$  is determined by  $(h - h_{5r}(0))|_{\mathbb{A}_{r/4, 4r}}$  and  $\mathbb{P}[E_r] \geq \mathfrak{p}$ ;

(B2) on  $E_r$ , for every  $\phi \in \mathcal{G}_r$ ,

$$|(h, \phi)_\nabla| + \frac{1}{2}(\phi, \phi)_\nabla \leq \Lambda_0$$

for a deterministic  $\Lambda_0$ ;

(B3) for endpoints  $\mathbf{z}, \mathbf{w} \notin B_{4r}$ , there is a choice  $\phi \in \mathcal{G}_r$ , measurable with respect to  $(\mathbf{z}, \mathbf{w}, h|_{\mathbb{C} \setminus B_{3r}})$ , for which the following holds. If the  $D_h$ -geodesic  $P$  hits  $B_{2r}$  and  $E_r$  occurs, the  $D_{h-\phi}$ -geodesic  $P^\phi$  has times  $s < t$  such that

$$\begin{aligned} P^\phi(s), P^\phi(t) &\in B_{3r/2}, & |P^\phi(s) - P^\phi(t)| &\geq b_0 r, \\ \tilde{D}_{h-\phi}(P^\phi(s), P^\phi(t)) &\leq c'_2 D_{h-\phi}(P^\phi(s), P^\phi(t)), \\ \tilde{D}_{h-\phi}(P^\phi(s), P^\phi(t)) &\leq (c_*/C_*) \tilde{D}_{h-\phi}(P^\phi(s), \partial B_{3r}). \end{aligned}$$

*Proof.* We construct the objects in the order in which their constants are chosen. Apply Lemma 8.143 with probability sufficiently close to one. Tightness and locality then choose successively constants  $\Delta > 0$ ,  $\delta > 0$ ,  $A > 1$ ,  $a \in (0, 1)$ ,  $\zeta > 0$ ,  $\theta > 0$ , and  $M > 1$  so that, simultaneously for every admissible tube  $U = U_r^{x,y}$ , the following regularity conditions fail with total probability less than  $1 - \mathfrak{p}$ :

(R1) whenever  $x, y \in \partial B_{2r}$  and  $|x - y| < \delta r$ ,

$$D_h(3x/2, 3y/2; \mathbb{A}_{r, 4r}) \leq \Delta c_r e^{\xi h_r(0)} \leq D_h(\partial B_{2r}, \partial B_{3r});$$

(R2) the internal  $D_h$ -diameter of  $U$  is at most  $A c_r e^{\xi h_r(0)}$ ;

(R3) every path of Euclidean diameter  $\epsilon_0 r/100$  in a  $2\zeta r$ -neighborhood of  $\partial U$  has  $D_h$ -length at least  $100 A c_r e^{\xi h_r(0)}$ ;

(R4) points of  $\mathbb{A}_{r/4, 4r}$  separated by  $\zeta r$  have internal  $D_h$ -distance at least  $a c_r e^{\xi h_r(0)}$ ;

(R5) the radial tips from  $3x/2$  to  $(3/2 - \theta)x$  have  $D_h$ -length at most the bound used below, and the thin radial tubes  $W_r^x$  have internal diameter at most  $M c_r e^{\xi h_r(0)}$ .

These are finite-grid events. The square-diameter and annular crossing quantiles, together with the line-avoidance estimate already used in Lemma 8.130, make their probabilities uniform in  $r$ .

Choose smooth  $f_r^{x,y} \in [0, 1]$ , equal to one on  $U_r^{x,y}$  and supported in its  $\zeta r$ -neighborhood. Choose  $g_r^x \in [0, 1]$ , equal to one on  $W_r^x$  and supported in a still thinner neighborhood. Put

$$K_f = \xi^{-1} \log \frac{100A}{a\Delta}, \quad K_g = K_f + \xi^{-1} \log M, \quad \phi_r^{x,y} = K_f f_r^{x,y} + K_g (g_r^x + g_r^y), \quad (8.23)$$

where  $\Delta > 0$  is the endpoint-separation quantile in the tube event. The grid admits only finitely many shapes and orientations, so these functions form a finite family  $\mathcal{G}_r$ . Rescaling a fixed smooth grid bump does not change its Dirichlet energy in dimension two. Hence the cardinality and the energies  $(\phi, \phi)_\nabla$  are uniformly bounded. The random variables  $(h, \phi)_\nabla$  form a finite Gaussian family; increasing  $\Lambda_0$  makes (B2) fail with the remaining arbitrarily small probability. Let  $E_r$  be

the conjunction of the tube and regularity events and (B2). This proves (B1)–(B2), including measurability.

It remains to verify the forcing statement. Grow filled  $D_h$ -balls from  $z$  and  $w$  until they first hit  $B_{3r}$ , with hit points  $x', y'$ , and put  $x = 2x'/3$ ,  $y = 2y'/3$ . These points are exterior-measurable. If  $|x - y| < \delta r$ , the endpoint-separation condition and the fact that  $P$  crosses  $\mathbb{A}_{2r, 3r}$  twice give opposite strict inequalities for the distance between the two grown balls. Thus  $|x - y| \geq \delta r$ , and we select the corresponding bump in (8.23).

The two radial tips, the two  $W$ -tubes, and  $U$  form a competitor from  $x'$  to  $y'$ . Weyl scaling and (R2), (R5) give

$$D_{h-\phi}(x', y') \leq e^{-\xi K_f} (A + 4) c_r e^{\xi h_r(0)}. \quad (8.24)$$

By the definitions of  $K_f$  and (R3)–(R4), a path with this length cannot leave  $B_{2\zeta r}(U \cup W_r^x \cup W_r^y)$ , nor can it travel a macroscopic distance close to  $\partial U$  without entering  $U$ . The topological separation in (T2) therefore forces  $P^\phi$  first through  $O_u$  and then through  $O_v$ .

Choose  $s < t$  at those passages. Conditions (T1)–(T3) give  $|P^\phi(s) - P^\phi(t)| \geq (b - 40\epsilon_0)r =: b_0 r$  and the two points lie in  $B_{3r/2}$ . Since the unique protected  $\tilde{D}_h$ -geodesic stays in  $U$ , where  $\phi = K_f$ ,

$$\tilde{D}_{h-\phi}(u, v) = e^{-\xi K_f} \tilde{D}_h(u, v).$$

The radial tubes are disjoint from the protected geodesic. The extremal comparison and the two endpoint errors in (T3) yield

$$D_{h-\phi}(u, v) \leq \frac{D_{h-\phi}(P^\phi(s), P^\phi(t))}{1 - 2c_*^{-1}C_*\eta}.$$

Using  $\tilde{D}_h(u, v) \leq c'_1 D_h(u, v)$ , the triangle inequality, and (8.20), we obtain

$$\tilde{D}_{h-\phi}(P^\phi(s), P^\phi(t)) \leq c'_2 D_{h-\phi}(P^\phi(s), P^\phi(t)).$$

Equation (8.22), the endpoint errors, and the second inequality in (8.20) give the boundary separation by  $c_*/C_*$ . This is (B3).  $\square$

For endpoints  $z, w \notin B_{4r}(z)$ , define  $\mathfrak{E}_r^{z,w}(z)$  to be the event that their  $D_h$ -geodesic has a segment satisfying the three conclusions of (B3), with  $h$  instead of  $h - \phi$ , and that

$$\exp\{-(h, \phi)_\nabla + \tfrac{1}{2}(\phi, \phi)_\nabla\} \leq e^{\Lambda_0} \quad \text{for every } \phi \in \mathcal{G}_r. \quad (8.25)$$

**Lemma 8.145** (Conditional Cameron–Martin comparison). *The events  $E_r(z)$  and  $\mathfrak{E}_r^{z,w}(z)$  satisfy (H2)–(H4) of Theorem 8.129, with  $\Lambda = e^{\Lambda_0}$  and the five radii  $(1/4, 2, 3, 4, 5)$ .*

*Proof.* The local measurability of  $E_r$  is (B1). To decide the geodesic segment, it suffices to know the internal metrics in  $B_{3r}(z)$  and the geodesic stopped at its last exit: the boundary-separation inequality ensures that the relevant  $\tilde{D}_h$ -distance equals the internal distance. The finite condition (8.25) is also local. This proves (H2), and (B1) proves (H3).

Condition on  $h|_{\mathbb{C} \setminus B_{3r}(z)}$ . The bump selected in (B3) is then deterministic. By the GFF Markov property, the conditional law of  $h - \phi$  is absolutely continuous with respect to that of  $h$ , with Radon–Nikodym derivative

$$M_h = \exp\{-(h, \phi)_\nabla - \tfrac{1}{2}(\phi, \phi)_\nabla\}.$$

On the event defined by (8.25),  $M_h \leq \Lambda$ . Statement (B3) gives the deterministic inclusion

$$E_r \cap \{P \cap B_{2r}(z) \neq \emptyset\} \subset \mathfrak{E}_r^\phi \cap \{P^\phi \cap B_{2r}(z) \neq \emptyset\}.$$

Integrating the inclusion with the conditional density and changing variables from  $h - \phi$  back to  $h$  gives

$$\begin{aligned} & \mathbb{P}[E_r, P \cap B_{2r}(z) \neq \emptyset \mid h|_{\mathbb{C} \setminus B_{3r}(z)}] \\ & \leq \Lambda \mathbb{P}[\mathfrak{E}_r^{z,w}(z), P \cap B_{2r}(z) \neq \emptyset \mid h|_{\mathbb{C} \setminus B_{3r}(z)}], \end{aligned}$$

which is precisely (H4).  $\square$

**Proposition 8.146** (A strict-ratio segment on every mesh geodesic). *Fix  $0 < \mu < \nu \leq \nu_*$  and a base scale  $R$  for which the hypothesis of Proposition 8.141 holds. For every  $q > 0$ , bounded  $U$ , and  $\ell > 0$ , with probability tending to one as  $\varepsilon \downarrow 0$ , simultaneously for all*

$$z, w \in (\varepsilon^q R \mathbb{Z}^2) \cap RU, \quad |z - w| \geq \ell R,$$

*the  $D_h$ -geodesic  $P^{z,w}$  has times  $s < t$  and a radius  $r \in [\varepsilon^{1+\nu} R, \varepsilon R]$  such that*

$$\begin{aligned} & |P(s) - P(t)| \geq b_0 r, \\ & \tilde{D}_h(P(s), P(t)) \leq c'_2(t - s), \\ & \tilde{D}_h(P(s), P(t)) \leq (c_*/C_*) \tilde{D}_h(P(s), \partial B_{3r}(z_r)) \end{aligned}$$

*for an appropriate center  $z_r$ . The convergence rate is uniform in  $R$ .*

*Proof.* Proposition 8.141 gives the dense radii after replacing the base scale by  $\rho^{-1}R$ . Retain the larger parity class of their exponents in  $r = 8^{-k}\rho^{-1}R$ . The retained radii have successive ratios at least  $8^2 = 64 > \lambda_4/\lambda_1 = 16$  and density at least  $\mu/2$ . Apply the independence theorem with any  $\mu_{H2} < \mu/2$ ; this verifies (H1) without changing  $\nu$ . Proposition 8.144 and Lemma 8.145 verify (H2)–(H4) with  $(1/4, 2, 3, 4, 5)$ . Apply Theorem 8.129. The definition of  $\mathfrak{E}_r$  is exactly the displayed conclusion.  $\square$

#### 8.13.4 The uniform deficit and the scale-counting contradiction

**Proposition 8.147** (Near-maximal ratios are excluded). *There is  $c'' > c_*$  such that the following holds. If  $\mathbb{P}[\underline{G}_R(c'', \beta)] \geq \beta$ , then for every  $\bar{\beta} \in (0, 1)$ ,*

$$\limsup_{\delta \downarrow 0} \sup_{R > 0} \mathbb{P}[\overline{G}_R(C_* - \delta, \bar{\beta})] = 0,$$

*where the supremum is over base scales satisfying the displayed lower-event hypothesis.*

*Proof.* Choose  $\chi < \xi(Q - 2)$  and  $\chi' > \xi(Q + 2)$ . The uniform metric estimates from the metric-tightness module imply, outside a probability tending to zero uniformly in  $R$ , that a segment whose endpoints have Euclidean separation  $b_0 \varepsilon^{1+\nu} R$  has

$$t - s \geq a, c_R e^{\xi_{h_R}(0) \varepsilon^{(1+\nu)\chi'}}.$$

Apply Proposition 8.146 to every pair on an  $\varepsilon^q R$ -mesh. Splicing the shortcut into the extremal  $\tilde{D}_h$ -comparison along the rest of the geodesic gives

$$\tilde{D}_h(z, w) \leq C_* D_h(z, w) - (C_* - c'_2)(t - s). \quad (8.26)$$

The boundary-separation clause ensures that the spliced  $\tilde{D}_h$ -path does not leave the local ball before completing the shortcut.

The simultaneous upper Hölder estimate makes the metric cost of replacing arbitrary endpoints by their mesh approximations  $O(c_R e^{\xi_{h_R(0)}} \varepsilon^{q\chi})$ . Choose  $q > (1 + \nu)\chi'/\chi$ ; this error is negligible compared with the deficit in (8.26). Axiom V also gives, uniformly for  $z, w \in B_R$ ,

$$D_h(z, w) \leq c_R e^{\xi_{h_R(0)}} \varepsilon^{-\chi'}$$

with probability tending to one. Hence every pair with  $|z - w| \geq \bar{\beta}R$  satisfies

$$\tilde{D}_h(z, w) \leq (C_* - a' \varepsilon^{(2+\nu)\chi'}) D_h(z, w)$$

on the same event. Given  $\delta$ , first choose  $\varepsilon$  so that  $a' \varepsilon^{(2+\nu)\chi'} > \delta$ , and then send  $\delta \downarrow 0$ . This proves the uniform limit.  $\square$

**Theorem 8.148** (Collapse of the common-scale extremal ratios). *If two weak  $\gamma$ -LQG metrics use the same Axiom-V constants, then there is a deterministic  $C > 0$  such that, for a whole-plane GFF plus a continuous function,*

$$\tilde{D}_h = C D_h \quad a.s.$$

*Equivalently, their deterministic extremal ratios satisfy  $c_* = C_*$ .*

*Proof.* Assume  $c_* < C_*$ . Choose  $0 < \nu \leq \nu_*$  and  $\mu \in (\nu/2, \nu)$ . The upper half of Proposition 8.116 gives, for each fixed  $\delta > 0$ , at least  $\mu \log_8 \varepsilon^{-1}$  radii in an interval containing at most  $\nu \log_8 \varepsilon^{-1} + O(1)$  dyadic radii at which

$$\mathbb{P}[\bar{G}_r(C_* - \delta, \bar{\beta})] \geq \bar{p}.$$

The lower half, with the  $c''$  of Proposition 8.147, gives at least the same number of radii at which the lower-event hypothesis of that proposition holds. Since  $2\mu > \nu$ , the two sets intersect for small  $\varepsilon$ .

But Proposition 8.147 makes the upper-event probability at every radius in the second set at most  $\bar{p}/2$  once  $\delta$  is small. This contradicts membership in the first set. Hence  $c_* = C_* =: C$ . The definitions of the infimum and supremum then imply  $\tilde{D}_h = C D_h$  simultaneously for all pairs. Weyl scaling extends the equality from a whole-plane GFF to a GFF plus any continuous function.  $\square$

## 8.14 Scalar uniqueness and the weak-to-strong affine rule

The common-scale theorem just proved does not by itself compare two arbitrary weak metrics: their Axiom-V normalization families may initially be different. The bridge is to compare a weak metric with each of its spatial rescalings, prove that the resulting scalar is a character of  $(0, \infty)$ , and identify its exponent. The final identification uses the sharp scale law below. We include that law explicitly because polynomial control alone does not determine the affine exponent.

### 8.14.1 The sharp exponent of the Axiom-V constants

Let  $\mathbb{S} = (0, 1)^2$ . For  $\delta \in (0, 1)$ , put diagonal as well as horizontal and vertical edges on  $\mathbb{S} \cap \delta\mathbb{Z}^2$ , and define the circle-average vertex LFPP crossing

$$\mathcal{L}_h^\delta(z, w; \mathbb{S}) := \min_{\pi: z \rightarrow w} \sum_{j=0}^{|\pi|} e^{\xi_{h_\delta}(\pi(j))}.$$

**Lemma 8.149** (Uniform weak-metric set-distance estimate). *Let  $D$  be any weak  $\gamma$ -LQG metric with Axiom-V constants  $c_r$ . If  $K_1, K_2$  are disjoint non-singleton connected compact subsets of an open set  $U$ , then, uniformly in  $R > 0$ ,*

$$A^{-1}c_R e^{\xi h_R(0)} \leq D_h(RK_1, RK_2; RU) \leq A c_R e^{\xi h_R(0)}$$

*with probability  $1 - o_A^\infty(A^{-1})$ . The same estimate applies to a radial crossing or the distance around a fixed-shape annulus.*

*Proof.* The one-annulus lower crossing and upper circumferential quantiles follow directly from Axiom V by the compactness argument of Lemma 8.100; that argument uses only tightness and the fact that every limit induces the Euclidean topology, so it applies to an arbitrary weak metric. Test these events on a dyadic family of buffered annuli with radii between  $\epsilon^2 R$  and  $\epsilon R$ . The centered events are local and invariant under addition of constants. Lemma 8.86 therefore gives, with error  $O(\epsilon^M)$  for any prescribed  $M$ , a good annulus around every point of an  $\epsilon^2 R$ -grid in  $RU$ .

Every path from  $RK_1$  to  $RK_2$  crosses one of the good radial annuli, which gives the lower bound. A minimal connected subcover has connected union of its good boundary circuits; concatenating those circuits gives the upper bound. The polynomial comparison of the  $c_r$ 's and the Gaussian maximum estimate for  $h_r(z) - h_R(0)$  convert both bounds to powers of  $\epsilon$  times  $c_R e^{\xi h_R(0)}$ . Set  $\epsilon = A^{-b/\sqrt{M}}$ , first choosing a small deterministic  $b$ , then sending  $M$  to infinity. This yields superpolynomial tails in  $A$ . The radial and circumferential variants use the same cover with the two compact sets chosen as the boundary components.  $\square$

**Theorem 8.150** (Sharp uniform scale law). *For every weak  $\gamma$ -LQG metric and every  $R > 0$ ,*

$$\frac{c_{\delta R}}{c_R} = \delta^{\xi Q + o_\delta(1)}, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}, \quad (8.27)$$

*where the error is uniform in  $R$ .*

*Proof.* The fixed-rectangle LFPP exponent already owned in Theorem 8.21, together with the circle-average/grid comparison in Proposition 8.20, gives, for every  $\zeta > 0$ , uniformly in  $R$ ,

$$\mathcal{L}_h^{\delta R}(\partial_L^{\delta R}(RS), \partial_R^{\delta R}(RS); RS) \in \left[ \delta^{-\xi Q + \zeta} e^{\xi h_R(0)}, \delta^{-\xi Q - \zeta} e^{\xi h_R(0)} \right] \quad (8.28)$$

with probability tending to one. The exponent is shifted by one from the continuum LFPP exponent  $1 - \xi Q$  because the vertex sum has no factor  $\delta R$ .

For each  $z \in RS \cap \delta R\mathbb{Z}^2$ , let  $S_z$  be the square of side  $\delta R$  centered at  $z$ . Lemma 8.149, followed by a union bound over  $O(\delta^{-2})$  squares, shows with probability tending to one that simultaneously

$$\text{Circ}_{D_h}(B_{\delta R}(S_z) \setminus S_z) \leq \delta^{-\zeta} c_{\delta R} e^{\xi h_{\delta R}(z)}, \quad (8.29)$$

$$D_h(S_z, \partial B_{\delta R}(S_z)) \geq \delta^\zeta c_{\delta R} e^{\xi h_{\delta R}(z)}. \quad (8.30)$$

Let  $\pi$  minimize the discrete LFPP crossing. Consecutive annular circuits from (8.29) intersect, so their union contains a continuum left-right crossing. Hence

$$D_h(R\partial_L\mathbb{S}, R\partial_R\mathbb{S}) \leq 2\delta^{-\zeta} c_{\delta R} \mathcal{L}_h^{\delta R}.$$

Axiom V makes the left side at least  $\delta^\zeta c_R e^{\xi h_R(0)}$  with probability tending to one. Using the upper bound in (8.28) gives

$$c_{\delta R} \geq \delta^{\xi Q + 3\zeta + o_\delta(1)} c_R.$$

Conversely, take a continuum  $D_h$ -path whose length is at most twice the internal crossing distance. Record the successive grid squares it visits when it exits the  $\delta R$ -neighborhood of the preceding square. These squares form a diagonal-grid left-right path. Each recorded segment crosses the annulus in (8.30); disjointness in time and the length-space property yield

$$\mathcal{L}_h^{\delta R} \leq \delta^{-\zeta} c_{\delta R}^{-1} D_h(R\partial_L \mathbb{S}, R\partial_R \mathbb{S}; R\mathbb{S}).$$

Axiom V bounds the last distance by  $\delta^{-\zeta} c_R e^{\xi h_R(0)}$ , and the lower bound in (8.28) gives

$$c_{\delta R} \leq \delta^{\xi Q - 3\zeta + o_\delta(1)} c_R.$$

Since  $\zeta > 0$  is arbitrary and every probability estimate was uniform in  $R$ , the two bounds prove (8.27).  $\square$

### 8.14.2 Rescaling a weak metric

For  $b > 0$ , define a new metric assignment

$$D_h^{(b)}(u, v) := D_{h(\cdot/b)}(bu, bv). \quad (8.31)$$

**Lemma 8.151** (The rescaled assignment has the same weak scale). *For each fixed  $b > 0$ ,  $D^{(b)}$  is a weak  $\gamma$ -LQG metric and it may use exactly the same Axiom-V constants  $c_r$  as  $D$ .*

*Proof.* The length-space, locality, Weyl-scaling, and translation axioms follow by substitution in (8.31). For scale tightness, put  $g = h(\cdot/b) - h_{1/b}(0)$ , so  $g \stackrel{d}{=} h$  and  $g_{br}(0) = h_r(0) - h_{1/b}(0)$ . Weyl scaling gives the exact identity

$$c_r^{-1} e^{-\xi h_r(0)} D_h^{(b)}(r \cdot, r \cdot) = \frac{c_{br}}{c_r} \left[ c_{br}^{-1} e^{-\xi g_{br}(0)} D_g(br \cdot, br \cdot) \right].$$

The polynomial comparison bounds make  $c_{br}/c_r$  bounded above and below by constants depending only on  $b$ . Axiom V for  $D$  therefore gives tightness, and every subsequential limit remains a genuine Euclidean-topology metric. The same polynomial bounds for  $c_r$  are unchanged. Thus the original family  $c_r$  is valid for  $D^{(b)}$ .  $\square$

**Lemma 8.152** (Continuous multiplicative rescaling character). *There is  $\alpha \in \mathbb{R}$  such that, for every  $b > 0$ ,*

$$D_h^{(b)} = b^\alpha D_h \quad a.s.$$

*for a whole-plane GFF plus a continuous function.*

*Proof.* Lemma 8.151 and Theorem 8.148 give a deterministic  $k_b > 0$  with  $D^{(b)} = k_b D$ . Directly from the definition,  $D^{(b_1 b_2)} = (D^{(b_1)})^{(b_2)}$ , hence

$$k_{b_1 b_2} = k_{b_1} k_{b_2}.$$

Moreover, Weyl scaling and  $h(\cdot/b) - h_{1/b}(0) \stackrel{d}{=} h$  show that  $e^{-\xi h_{1/b}(0)} D_h^{(b)}(\cdot/b, \cdot/b) \stackrel{d}{=} D_h$ . Continuity of  $D_h$  and of the circle-average process implies  $D^{(b)} \rightarrow D$  in law as  $b \rightarrow 1$ . Evaluating a fixed positive annular crossing quantile gives  $k_b \rightarrow 1$ . Thus  $b \mapsto k_b$  is a continuous multiplicative character of  $(0, \infty)$ , so  $k_b = b^\alpha$ .  $\square$



### 8.14.3 Identification of the character and arbitrary scalar uniqueness

**Lemma 8.153** (The unit-rotation factor is one). *Let  $D$  be a weak  $\gamma$ -LQG metric. For every deterministic  $\omega \in \mathbb{C}$  with  $|\omega| = 1$ , almost surely*

$$D_h(\omega u, \omega v) = D_{h(\omega \cdot)}(u, v), \quad u, v \in \mathbb{C}. \quad (8.32)$$

*Proof.* For  $|\omega| = 1$ , define another metric functional by

$$(\mathcal{R}_\omega D)_h(u, v) := D_{h(\omega^{-1} \cdot)}(\omega u, \omega v).$$

Rotation preserves Euclidean balls, circle averages, internal metrics, and the Dirichlet law of the GFF. Inspection of Axioms I–V therefore shows that  $\mathcal{R}_\omega D$  is a weak metric with exactly the same scale constants  $c_r$  as  $D$ . Theorem 8.148 gives a deterministic  $k_\omega > 0$  such that

$$\mathcal{R}_\omega D = k_\omega D.$$

The operators form a circle action:  $\mathcal{R}_{\omega_1} \mathcal{R}_{\omega_2} = \mathcal{R}_{\omega_1 \omega_2}$ . Hence  $k_{\omega_1 \omega_2} = k_{\omega_1} k_{\omega_2}$ . The character is measurable. Indeed, the action  $(\omega, h) \mapsto h(\omega^{-1} \cdot)$  is continuous in the distribution topology, the metric functional is Borel by the weak-metric definition, and  $k_\omega$  can be recovered from any fixed positive annular distance quantile of  $\mathcal{R}_\omega D$  divided by the corresponding quantile of  $D$ . Thus  $\log k_\omega$  is a measurable homomorphism from the compact group  $\mathbb{S}^1$  to  $(\mathbb{R}, +)$ . Every measurable homomorphism between locally compact second-countable groups is continuous, and the only compact subgroup of  $(\mathbb{R}, +)$  is  $\{0\}$ . Therefore  $k_\omega = 1$ .

Substitute  $h(\omega \cdot)$  for  $h$  in  $(\mathcal{R}_\omega D)_h = D_h$ . Since  $h(\omega \cdot)(\omega^{-1} \cdot) = h$ , the resulting identity is (8.32). Continuity in the endpoints makes it simultaneous for all  $u, v$ .  $\square$

**Theorem 8.154** (Every weak metric is strong). *Every weak  $\gamma$ -LQG metric obeys, simultaneously for each fixed deterministic  $a \in \mathbb{C} \setminus \{0\}$ ,  $z \in \mathbb{C}$ , and all  $u, v$ ,*

$$D_h(au + z, av + z) = D_{h(a \cdot + z) + Q \log |a|}(u, v), \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}. \quad (8.33)$$

*Proof.* Apply Lemma 8.152 to the field  $h(b \cdot)$ . Equation (8.31) gives

$$D_h(bu, bv) = b^\alpha D_{h(b \cdot)}(u, v).$$

For a whole-plane GFF,  $h(b \cdot) - h_b(0) \stackrel{d}{=} h$ , so

$$b^{-\alpha} e^{-\xi h_b(0)} D_h(b \cdot, b \cdot) \stackrel{d}{=} D_h.$$

Thus Axiom V is valid with constants  $c_b = b^\alpha$ . The sharp scale law (8.27) forces  $\alpha = \xi Q$ . Weyl scaling by the constant  $Q \log b$  now yields

$$D_h(bu, bv) = b^{\xi Q} D_{h(b \cdot)}(u, v) = D_{h(b \cdot) + Q \log b}(u, v).$$

Let  $a = b\omega$ , where  $b = |a| > 0$  and  $|\omega| = 1$ . Apply the exact translation axiom to  $h(\cdot + z)$ , then the unit-rotation identity (8.32), and finally the positive-dilation identity just proved. This gives

$$D_h(au + z, av + z) = D_{h(a \cdot + z) + Q \log |a|}(u, v),$$

which is (8.33). The identity first holds for each fixed affine map on one probability-one event; continuity in  $u, v$  gives all endpoints, exactly as required by the strong-metric definition.  $\square$

**Theorem 8.155** (Arbitrary weak metrics differ by one scalar). *If  $D$  and  $\tilde{D}$  are any two weak  $\gamma$ -LQG metrics coupled with the same whole-plane GFF, without a common-scale assumption, then there is a deterministic  $C > 0$  such that*

$$\tilde{D}_h = CD_h$$

*almost surely for a whole-plane GFF plus any continuous function. A fixed nondegenerate crossing quantile normalization forces  $C = 1$ .*

*Proof.* By Theorem 8.154, both metrics satisfy the exact affine rule. For either metric the calculation

$$r^{-\xi Q} e^{-\xi h_r(0)} D_h(r \cdot, r \cdot) = D_{h(r \cdot) - h_r(0)} \stackrel{d}{=} D_h$$

shows that Axiom V may be written with the common deterministic constants  $r^{\xi Q}$ . The common-scale scalar theorem 8.148 therefore applies and gives  $\tilde{D}_h = CD_h$ . Weyl scaling extends the identity to continuous perturbations. If the same unit-square crossing quantile is prescribed for both metrics, multiplication by  $C$  multiplies that quantile by  $C$ , so  $C = 1$ .  $\square$

*Proof of the uniqueness and weak-to-strong theorem.* The common-scale comparison is Proposition 8.112. The protected near-extremal reservoir and finite-bump construction are Proposition 8.141 and Proposition 8.144; Lemma 8.145 verifies the exact conditional hypothesis of the independence theorem. Proposition 8.147 and Theorem 8.148 then force  $c_* = C_*$ , proving scalar uniqueness for metrics with common scale constants.

For an arbitrary weak metric, Theorem 8.150 identifies the scale exponent and Theorem 8.154 proves the affine rule (20.5). Its unit-rotation step is Lemma 8.153: the rotation factor is a measurable positive character of the circle and is therefore one. Thus any two weak metrics may both use  $c_r = r^{\xi Q}$  in Axiom V. Theorem 8.155 applies the common-scale result and gives (20.7). A common fixed crossing quantile makes the scalar equal to one. These are all assertions of the parent theorem.  $\square$

*Remark 8.156* (Proof dependencies). The proof combines common-scale bi-Lipschitz comparison (Proposition 8.112), many-scale near-extremal pairs (Proposition 8.116), quantitative confluence (Theorem 8.127), and stable exploration along geodesics (Theorem 8.129). The shortcut argument is completed by Proposition 8.144, Lemma 8.145, and Proposition 8.147. Scalar uniqueness is Theorem 8.148; the sharp scale law, unit-rotation rule, affine covariance, and removal of the common-scale hypothesis are Lemma 8.153 and Theorems 8.150–8.155. Full LFPP convergence still requires the compactness and normalization argument in the next module.

## 8.15 The normalized full-sequence LFPP limit

The scalar theorem closes uniqueness only up to a deterministic constant. For LFPP that last constant is selected by the normalization  $a_\epsilon$ . This section records every step of the selection argument: the topology in which a crossing is continuous, passage of the median, the subsequence principle, the bounded-perturbation extension, and the resulting regular variation of  $a_\epsilon$ . In particular, no assertion of full convergence is hidden in the word “uniqueness.”

Put

$$L = \{0\} \times [0, 1], \quad R = \{1\} \times [0, 1].$$

The convention used here and in the primary heat-kernel LFPP theorem is the *global* set distance

$$D(L, R) := \inf\{D(x, y) : x \in L, y \in R\}; \tag{20.7a}$$

paths are not constrained to remain in the unit square. This distinction is important because an internal crossing has an additional boundary-stability obligation which is not part of the normalization in the theorem.

### 8.15.1 The metric topology and the crossing selector

For continuous functions  $d, d' : \mathbb{C}^2 \rightarrow [0, \infty)$ , let

$$\mathfrak{d}_{\text{lu}}(d, d') := \sum_{m=1}^{\infty} 2^{-m} \left( 1 \wedge \sup_{z, w \in \overline{B}_m(0)} |d(z, w) - d'(z, w)| \right). \quad (8.34)$$

This is a complete separable metric on  $C(\mathbb{C}^2, [0, \infty))$  and induces local-uniform convergence. The subset of genuine continuous metrics is Borel; all limiting laws below are supported on that subset by Theorem 8.1.

**Lemma 8.157** (Compact set distance is a continuous functional). *Let  $A, B$  be nonempty compact subsets of  $\overline{B}_m(0)$ . Then*

$$|d(A, B) - d'(A, B)| \leq \sup_{z, w \in \overline{B}_m(0)} |d(z, w) - d'(z, w)|. \quad (20.7b)$$

Consequently  $d_n \rightarrow d$  locally uniformly implies  $d_n(L, R) \rightarrow d(L, R)$ .

*Proof.* For every  $x \in A$  and  $y \in B$ ,  $d(x, y) \leq d'(x, y) + \|d - d'\|_{\infty, \overline{B}_m^2}$ . Taking the infimum over  $A \times B$  gives one inequality; interchanging  $d$  and  $d'$  gives the other. The final assertion uses any integer  $m$  containing the unit square.  $\square$

We use the lower-quantile convention

$$q_p(X) := \inf\{t : \mathbf{P}[X \leq t] \geq p\}.$$

Thus the deterministic normalization in (20.1) is  $a_\epsilon = q_{1/2}(D_h^\epsilon(L, R))$ .

**Lemma 8.158** (No plateau at the LFPP limiting crossing median). *Let  $D_h$  be a field-measurable subsequential LFPP limit for the whole-plane GFF normalized by  $h_1(0) = 0$ . The distribution function of  $D_h(L, R)$  is continuous and strictly increasing on  $(0, \infty)$ . In particular it has a unique median. If*

$$a_{\epsilon_n}^{-1} D_h^{\epsilon_n} \longrightarrow D_h \quad \text{in probability locally uniformly,} \quad (20.7c)$$

then

$$q_{1/2}(D_h(L, R)) = 1. \quad (20.7d)$$

*Proof.* We first recall the finite-energy argument in the precise form needed here. The local bi-Hölder bounds from the compactness module, together with the length-space property, imply the properness statement used in Lemma 8.118: a compact-to-compact minimizing path is contained in some deterministic Euclidean ball on an event whose probability tends to one with the radius. On such an event choose rational polygonal tubes around a minimizing path, and rational collars separating each tube from the boundary of a slightly larger ball. Positivity of the metric distance between disjoint compact sets gives a positive collar cost. The countable union of these controlled tube-and-collar events has probability one.

Fix one controlled event. Choose finitely many smooth functions supported in the larger ball, constant on the successive cores of the tube, and include one compensating function so that their

circle averages at radius one vanish. The domain Markov decomposition of the whole-plane GFF writes the corresponding finite-dimensional component as a nondegenerate Gaussian vector  $Z$ , independent of the orthogonal component. Cameron–Martin translation therefore has a strictly positive smooth conditional density. Weyl scaling and the collar cost have the following deterministic consequence. After possibly subdividing the controlled event into countably many events and shrinking the allowed parameter box, one coordinate  $s$  of  $Z$  satisfies

$$0 < \kappa \leq \frac{\log D_{h+s\phi}(L, R) - \log D_h(L, R)}{s} \leq K < \infty \quad (0 < |s| < s_0), \quad (20.7e)$$

with deterministic rational  $\kappa, K, s_0$  on that piece. Indeed, the upper bound is obtained by using the selected tube path. For the lower bound, any competitor either crosses each selected core, where its Weyl length is multiplied, or exits the larger tube and pays the positive collar cost; the parameter box was chosen small enough that the latter cannot become competitive. The compensating bump is placed beyond that collar, so it does not change this comparison. This is the standard finite-bump anti-concentration argument already used for LFPP crossings in Section 8.1.1, now applied after the local-uniform limit.

Condition on the orthogonal field and the remaining coordinates of  $Z$ . Inequality (20.7e) makes  $s \mapsto \log D_{h+s\phi}(L, R)$  strictly monotone, with inverse Lipschitz constant at most  $\kappa^{-1}$ . Since  $s$  has a Gaussian density, its conditional image has no atoms. Moreover every nonempty interval in its image has positive conditional probability. Summing over the countable controlled pieces first rules out atoms and then rules out a plateau of the distribution function. The crossing is almost surely finite and positive, so continuity and strict increase hold on  $(0, \infty)$ .

Now set  $X_n = a_{\epsilon_n}^{-1} D_h^{\epsilon_n}(L, R)$  and  $X = D_h(L, R)$ . Lemma 8.157 and (20.7c) give  $X_n \rightarrow X$  in probability. Each  $X_n$  has lower median one. For every  $\delta > 0$ , convergence in law and the Portmanteau theorem give

$$\mathbf{P}[X \leq 1 + \delta] \geq \frac{1}{2}, \quad \mathbf{P}[X < 1 - \delta] \leq \frac{1}{2}.$$

Letting  $\delta \downarrow 0$ , using continuity and strict increase at one, shows  $\mathbf{P}[X \leq 1] = 1/2$ , hence (20.7d).  $\square$

**Lemma 8.159** (The crossing normalization removes the uniqueness scalar). *Let  $D_h$  and  $\tilde{D}_h$  be two field-measurable subsequential limits of  $a_\epsilon^{-1} D_h^\epsilon$ , possibly obtained from different vanishing sequences. Couple both as measurable functions of the same normalized whole-plane GFF. Then*

$$\tilde{D}_h = D_h \quad \text{almost surely.} \quad (20.7f)$$

*Proof.* Theorem 8.155 gives a deterministic  $C > 0$  with  $\tilde{D}_h = C D_h$ . By Lemma 8.158, both crossing variables have unique lower median one. Since  $q_{1/2}(C D_h(L, R)) = C q_{1/2}(D_h(L, R))$ , one has  $C = 1$ .  $\square$

### 8.15.2 From sequential compactness to convergence in probability

**Lemma 8.160** (Subsequence criterion for convergence in probability). *Let  $(S, d)$  be a metric space and let  $Y_t, Y$  be  $S$ -valued random variables, with  $t \downarrow 0$ . If every deterministic sequence  $t_n \downarrow 0$  has a subsequence  $t_{n_k}$  for which  $Y_{t_{n_k}} \rightarrow Y$  in probability, then  $Y_t \rightarrow Y$  in probability.*

*Proof.* Otherwise there are  $\eta, \rho > 0$  and a sequence  $t_n \downarrow 0$  such that  $\mathbf{P}[d(Y_{t_n}, Y) > \eta] \geq \rho$  for every  $n$ . The asserted subsequence would make these probabilities tend to zero, a contradiction.  $\square$

**Theorem 8.161** (Canonical full-sequence heat-kernel LFPP limit). *There is a field-measurable strong  $\gamma$ -LQG metric  $D_h$ , normalized by  $q_{1/2}(D_h(L, R)) = 1$ , such that*

$$a_\epsilon^{-1} D_h^\epsilon \longrightarrow D_h \quad \text{in probability for } \mathfrak{d}_{\text{lu}}. \quad (20.7g)$$

*The limit is independent of every extraction choice.*

*Proof.* Apply Theorem 8.107 to one deterministic vanishing sequence and call the resulting field-measurable limit  $D_h$ . It is strong by Theorem 8.154, and its crossing median is one by Lemma 8.158.

Now start from an arbitrary sequence  $\epsilon_n \downarrow 0$ . Theorem 8.107 supplies a further subsequence converging in probability to a field-measurable weak metric  $\tilde{D}_h$ . Lemma 8.159 identifies this metric with  $D_h$  on the original field coupling. Thus every sequence has a further subsequence converging in probability to the same random variable in the Polish metric (8.34). Lemma 8.160 proves (20.7g).  $\square$

**Theorem 8.162** (Full convergence under bounded continuous perturbations). *Let  $f : \mathbb{C} \rightarrow \mathbb{R}$  be bounded and continuous and put  $\mathfrak{h} = h + f$ . Then*

$$a_\epsilon^{-1} D_{\mathfrak{h}}^\epsilon \longrightarrow D_{\mathfrak{h}} := e^{\xi f} \cdot D_h \quad \text{in probability locally uniformly.} \quad (20.7h)$$

*The same statement holds for a random bounded continuous  $f$  coupled with  $h$ , under the coupling hypothesis in the definition of a whole-plane GFF plus a bounded continuous function.*

*Proof.* Take an arbitrary  $\epsilon_n \downarrow 0$ . The bounded-perturbation part of Theorem 8.107 gives a further subsequence and joint convergence in probability. Lemma 8.77 identifies its perturbed limit with  $e^{\xi f} \cdot \tilde{D}_h$ , where  $\tilde{D}_h$  is the corresponding base-field limit. The scalar selector makes  $\tilde{D}_h = D_h$ . Every original sequence therefore has a further subsequence converging in probability to the right side of (20.7h), and Lemma 8.160 gives full convergence. The proof of the Weyl-passage lemma is pathwise after conditioning on  $(h, f)$ , so the same argument applies to the stated random perturbations.  $\square$

*Proof of Theorem 8.110.* Theorem 8.161 constructs the field-measurable strong metric and proves full convergence for the normalized whole-plane GFF. Theorem 8.162 gives exactly the final assertion for bounded continuous perturbations. These are all claims of the parent theorem.  $\square$

### 8.15.3 A normalization consequence

**Corollary 8.163** (Regular variation of the LFPP normalization). *For every fixed  $C > 0$ ,*

$$\lim_{\epsilon \rightarrow 0} \frac{a_{C\epsilon}}{a_\epsilon} = C^{1-\xi Q}. \quad (20.7i)$$

*Proof.* Given any sequence  $\epsilon_n \downarrow 0$ , Theorem 8.161 and the scale-ratio argument of Lemma 8.96 give, after a further subsequence,

$$\frac{a_{C\epsilon_n}}{a_{\epsilon_n}} \longrightarrow C c_{1/C}.$$

For the strong limit, Theorem 8.154 permits the exact Axiom-V constants  $c_r = r^{\xi Q}$ . Hence every such subsequential ratio limit is  $C(1/C)^{\xi Q} = C^{1-\xi Q}$ . The real-valued subsequence criterion now gives the full limit (20.7i).  $\square$

## 8.16 Conformal coordinate covariance

**Theorem 8.164** (Conformal coordinate covariance — domain theorem). *Let  $U, \tilde{U} \subset \mathbb{C}$  be domains and let  $\phi : U \rightarrow \tilde{U}$  be a deterministic conformal bijection. If  $h$  is a GFF-type distribution on  $U$  in the LQG-metric class and*

$$\tilde{h} = h \circ \phi^{-1} + Q \log |(\phi^{-1})'|, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2},$$

*then almost surely*

$$D_{\tilde{h}}(\phi(z), \phi(w)) = D_h(z, w), \quad z, w \in U. \quad (20.8)$$

*Equivalently, the metric is intrinsic under the standard  $\gamma$ -LQG quantum-surface coordinate change.*

*Proof roadmap.* One cannot prove (20.8) merely by declaring the pullback metric to be weak and invoking uniqueness: one must identify its coupling and normalization. The source proof first proves joint field–metric convergence under conformal zooms, using total-variation convergence of normalized GFF restrictions and field measurability of the metric. Exact affine rescaling then gives a compact-uniform infinitesimal comparison. A jointly-local annular criterion upgrades this to deterministic bi-Lipschitz equivalence. Finally a measurable buffered good-annulus covering occupies a fixed positive fraction of every relevant geodesic and strictly improves any optimal constant larger than one. Applying the one-sided result to the inverse conformal map gives equality.

*Remark 8.165* (Proof dependencies). This theorem uses the strong affine rule and the already-established local-metric annulus machinery, but it has its own conformal-zoom and constant-one proof, supplied in the module below. The later metric-dimension, geodesic, KPZ, and Brownian-map statements use this theorem but require their own additional estimates.

## 8.17 Conformal covariance from infinitesimal affine comparison

The affine rule proved in this chapter is an exact statement for one fixed complex affine map. A nonlinear conformal map cannot be inserted into that rule pointwise: the metric is a nonlocal path functional and no almost-sure continuity of  $h \mapsto D_h$  in the distribution topology is available. The source proof resolves this obstruction in three stages. It first proves joint field–metric convergence after zooming at a point, then invokes the local-metric annular criterion to obtain a deterministic bi-Lipschitz comparison, and finally improves the optimal constant by covering a geodesic with good annuli. All three stages are recorded below.

Throughout the whole-plane part of the argument,  $h$  is a whole-plane GFF in a fixed additive normalization and  $D_h$  is the field-measurable strong  $\gamma$ -LQG metric constructed above. If  $V \subset \mathbb{C}$  is open, write

$$D_{h|_V} := D_h(\cdot, \cdot; V).$$

For a conformal bijection  $\phi : U \rightarrow \tilde{U}$ , put

$$h^\phi = h \circ \phi^{-1} + Q \log |(\phi^{-1})'|, \quad D_h^\phi(z, w) = D_{h^\phi}(\phi(z), \phi(w)). \quad (8.35)$$

Thus  $D_h^\phi$  is a metric on  $U$ , expressed in the original coordinate. The desired whole-plane-field identity is  $D_h^\phi = D_{h|_U}$ .

### 8.17.1 The pullback as a local metric

**Lemma 8.166** (Locality and additivity of the pullback). *The metrics  $D_{h|_U}$  and  $D_h^\phi$  are continuous length metrics on  $U$ , are determined by  $h|_U$ , and are jointly local and  $\xi$ -additive for  $h|_U$ . For every open  $V \subset U$ ,*

$$D_h^\phi(\cdot, \cdot; V) = D_{h^\phi}(\phi(\cdot), \phi(\cdot); \phi(V)). \quad (20.8a)$$

*Proof.* A conformal bijection is a homeomorphism, hence it preserves continuous paths and their partitions. The length of a path for the pullback metric is therefore the length of its image for  $D_{h^\phi}$ ; taking infima proves (20.8a) and the length-space assertion.

The field  $h^\phi|_{\phi(V)}$  is determined by  $h|_V$ . Locality of the LQG metric says that the internal metric on  $\phi(V)$  is determined by this field restriction. Pulling it back gives locality of  $D_h^\phi$ . Both metrics are measurable functions of the same restricted field, so Lemma 8.81 makes them jointly local. Finally, adding a constant  $c$  to  $h$  adds the same constant to  $h^\phi$ , and Weyl scaling multiplies both metrics by  $e^{\xi c}$ . Subtracting a circle average therefore gives joint  $\xi$ -additivity in the sense used by Theorem 8.89.  $\square$

Fix once and for all a nonnegative, radially symmetric  $\varrho \in C_c^\infty(\mathbb{C})$  with integral one. We use the smooth average

$$h_{\varrho, r}(z) := (h, r^{-2}\varrho(r^{-1}(\cdot - z))).$$

Unlike a circle average, this is a continuous functional of the field in the distribution topology. Normalize  $h$  for the next two subsections by  $h_{\varrho, 1}(0) = 0$ .

### 8.17.2 A conformal zoom and joint field–metric convergence

**Lemma 8.167** (Near-linear pullbacks of the GFF). *Let  $\psi_n : U_n \rightarrow \psi_n(U_n)$  be conformal maps with  $\psi_n(0) = 0$ ,*

$$\psi_n'(0) \rightarrow \alpha^{-1} \neq 0,$$

*and suppose every compact subset of  $\mathbb{C}$  is contained in  $U_n$  for all sufficiently large  $n$ . For every compact  $K \subset \mathbb{C}$ ,*

$$h \circ \psi_n^{-1}|_K \rightarrow h(\alpha \cdot)|_K \quad \text{almost surely in the distribution topology,} \quad (8.36)$$

$$(h \circ \psi_n^{-1} - (h \circ \psi_n^{-1})_{\varrho, |\alpha|^{-1}}(0))|_K \rightarrow h(\alpha \cdot)|_K \quad \text{in total variation.} \quad (8.37)$$

*Proof.* Koebe distortion on successively larger disks gives, uniformly on every compact set,

$$\psi_n(z) \rightarrow z/\alpha, \quad \psi_n^{-1}(z) \rightarrow \alpha z,$$

together with convergence of every derivative. If  $f$  is a compactly supported smooth test function, then

$$(h \circ \psi_n^{-1}, f) = (h, |\psi_n'|^2(f \circ \psi_n)) \rightarrow (h, |\alpha|^{-2}f(\alpha^{-1}\cdot)) = (h(\alpha \cdot), f).$$

The convergence of the transformed test functions holds in the full test function topology. A countable dense family and continuity of the random distribution give (8.36).

For the stronger assertion, choose  $K \Subset K' \Subset \mathbb{C}$  and decompose on  $U_n$

$$h = \mathring{h}_n + \mathfrak{h}_n,$$

where  $\mathring{h}_n$  is a zero-boundary GFF and  $\mathfrak{h}_n$  is the independent harmonic part. Conformal invariance makes  $\mathring{h}_n \circ \psi_n^{-1}$  a zero-boundary GFF on  $\psi_n(U_n)$ . The expanding-domain Markov decomposition

of a normalized whole-plane GFF says that, after subtracting the smooth average at scale  $|\alpha|^{-1}$ , its restriction to  $K'$  converges in total variation to  $h(\alpha \cdot)|_{K'}$ . Here is the finite-energy verification needed for that statement. Couple the two fields by their common zero-boundary part on a large domain containing  $K'$ . Their difference is harmonic; after the smooth-average normalization, its Dirichlet energy on  $K'$  tends to zero in probability as the domain exhausts the plane. A smooth cutoff equal to one on  $K$  turns the difference into a Cameron–Martin shift. Its Radon–Nikodym derivative is

$$\exp\{(\mathring{h}_n, g_n)_\nabla - \tfrac{1}{2}\|g_n\|_\nabla^2\}.$$

The vanishing energy makes this derivative tend to one in  $L^1$ , which is exactly total-variation convergence on  $K$ .

The same argument applies to the transformed harmonic part  $\mathfrak{h}_n \circ \psi_n^{-1}$ : Koebe convergence transfers the vanishing Dirichlet energy on  $K'$ , and subtracting its smooth average removes the remaining constant. Adding the zero-boundary and harmonic parts proves (8.37). This proof also shows that the convergence is unaffected by first enlarging  $K'$ , as will be used below.  $\square$

For a conformal map  $\psi$ , use the notation in (8.35) with  $\psi$  in place of  $\phi$ .

**Lemma 8.168** (Joint field–metric convergence under a conformal zoom). *Under the hypotheses of Lemma 8.167,*

$$(h, D_h, h^{\psi_n}, D_h^{\psi_n}) \implies (h, D_h, h(\alpha \cdot) + Q \log |\alpha|, D_h) \quad (8.38)$$

*in the distribution topology for fields and the local-uniform topology for metrics.*

*Proof.* We first identify the marginal law of the last two coordinates. Let  $V \Subset \mathbb{C}$ . Since  $D_g(\cdot, \cdot; V)$  is a measurable function of  $g|_V$ , the total-variation convergence in Lemma 8.167 transfers to the pair consisting of the normalized field and its internal metric on  $V$ . By (8.36),

$$(h \circ \psi_n^{-1})_{\varrho, |\alpha|^{-1}}(0) \longrightarrow (h(\alpha \cdot))_{\varrho, |\alpha|^{-1}}(0) = 0$$

in probability. Constant Weyl scaling and an exhaustion by  $V$ 's yield

$$(h \circ \psi_n^{-1}, D_{h \circ \psi_n^{-1}}) \implies (h(\alpha \cdot), D_{h(\alpha \cdot)}). \quad (20.8d)$$

Koebe convergence also gives  $Q \log |(\psi_n^{-1})'| \rightarrow Q \log |\alpha|$  and  $\psi_n \rightarrow \alpha^{-1} \cdot$ , uniformly on compact sets. Weyl scaling and the affine coordinate rule from Theorem 8.154 transform (20.8d) into

$$(h^{\psi_n}, D_h^{\psi_n}) \implies (h(\alpha \cdot) + Q \log |\alpha|, D_h). \quad (20.8e)$$

It remains to identify the coupling, which cannot be read from marginal convergence alone. The marginal limits make the four-tuples tight. In any subsequential limit write the last metric as  $D_{h'}$ . The almost-sure field convergence (8.36), including the deterministic logarithmic term, forces  $h' = h$  in the limiting coupling. Field measurability of the strong metric then forces  $D_{h'} = D_h$ . Every subsequential limit is therefore the right side of (8.38), which proves the full convergence.

The  $Q \log |\alpha|$  in the third coordinate is essential. It is present in the proof and in the later application in the primary TeX source; its omission from one displayed version of the source lemma is a typographical error and is corrected here. Likewise, the normalizing radius is  $|\alpha|^{-1}$ , since radially of  $\varrho$  gives  $(h(\alpha \cdot))_{\varrho, |\alpha|^{-1}}(0) = h_{\varrho, 1}(0)$ ; the reciprocal is missing in the corresponding primary TeX display.  $\square$



### 8.17.3 Uniform infinitesimal comparison

For  $z \in U$  and  $r > 0$ , define

$$h^{z,r}(w) = h(z + rw) - h_{\varrho,r}(z), \quad \phi^{z,r}(w) = \frac{\phi(z + rw) - \phi(z)}{r}.$$

Notice that  $\phi^{z,r}(0) = 0$  and  $(\phi^{z,r})'(0) = \phi'(z)$ . Subtracting  $\phi(z)$ , rather than  $z$ , in this definition is necessary for the centering; this also corrects a harmless typographical slip in the primary TeX source.

**Lemma 8.169** (Exact rescaling identity). *For  $u, v \in r^{-1}(U - z)$ ,*

$$D_{h^{z,r}}(u, v) = r^{-\xi_Q} e^{-\xi h_{\varrho,r}(z)} D_h(z + ru, z + rv), \quad (8.39)$$

$$D_{h^{z,r}}^{\phi^{z,r}}(u, v) = r^{-\xi_Q} e^{-\xi h_{\varrho,r}(z)} D_h^{\phi}(z + ru, z + rv). \quad (8.40)$$

*Proof.* The first identity is the exact affine rule followed by constant Weyl scaling. For the second, direct substitution gives

$$(h^{z,r})^{\phi^{z,r}}(\zeta) = h^{\phi}(\phi(z) + r\zeta) - h_{\varrho,r}(z).$$

Apply the affine rule to  $h^{\phi}$ , then evaluate at  $\phi^{z,r}(u)$  and  $\phi^{z,r}(v)$ . Since  $\phi(z) + r\phi^{z,r}(u) = \phi(z + ru)$ , the result is (8.40).  $\square$

**Proposition 8.170** (Uniform small-scale additive comparison). *For every  $\eta > 0$  and compact  $K \Subset U$ ,*

$$\liminf_{r \downarrow 0} \mathbb{P} \left[ \sup_{u, v \in B_r(z)} |D_h^{\phi}(u, v) - D_h(u, v)| \leq \eta r^{\xi_Q} e^{\xi h_{\varrho,r}(z)} \right] = 1. \quad (8.41)$$

*Consequently, for every  $\delta, b \in (0, 1)$  and compact  $K \Subset U$ ,*

$$\liminf_{r \downarrow 0} \mathbb{P} \left[ 1 - \delta \leq \frac{D_h^{\phi}(u, v)}{D_h(u, v)} \leq 1 + \delta \quad \forall u, v \in B_r(z), |u - v| \geq br \right] = 1. \quad (8.42)$$

*The ratio statement is independent of the additive normalization of  $h$ .*

*Proof.* First take sequences  $z_n \rightarrow z \in U$  and  $r_n \downarrow 0$ . The maps  $\phi^{z_n, r_n}$  are defined on domains exhausting the plane and satisfy

$$\phi^{z_n, r_n}(0) = 0, \quad (\phi^{z_n, r_n})'(0) \rightarrow \phi'(z).$$

The normalized fields  $h^{z_n, r_n}$  all have the law of the fixed normalized whole-plane GFF. Lemma 8.168, with  $\alpha = 1/\phi'(z)$ , says that the two metrics in (8.39)–(8.40) converge jointly to two copies of the same  $D_h$ . Their sup norm on  $\overline{B}_1^2$  therefore tends to zero in probability. The exact rescaling identities prove (8.41) along the chosen sequences.

If uniformity over  $K$  failed, there would be  $\epsilon > 0$ , radii  $r_n \downarrow 0$ , and  $z_n \in K$  for which the displayed probability is at most  $1 - \epsilon$ . Compactness gives a convergent subsequence of the  $z_n$ 's, contradicting the sequential conclusion. This proves the first assertion.

For the ratio assertion, affine covariance, Weyl scaling, and the scale and translation invariance of the normalized whole-plane GFF give the following uniform nondegeneracy: for every  $\epsilon > 0$  there is  $c = c(b, \epsilon) > 0$  such that, for all  $z, r$ , with probability at least  $1 - \epsilon$ ,

$$D_h(u, v) \geq cr^{\xi_Q} e^{\xi h_{\varrho,r}(z)} \quad \text{whenever } u, v \in B_r(z), |u - v| \geq br.$$

Apply the additive estimate with  $c\delta$  in place of  $\eta$ , intersect the two events, and then let  $\epsilon \downarrow 0$ . Adding a constant to  $h$  multiplies numerator and denominator by the same factor, so the final statement holds in every normalization.  $\square$

### 8.17.4 Global deterministic bi-Lipschitz comparison

**Proposition 8.171** (Bi-Lipschitz comparison on the source domain). *There is a deterministic  $C \geq 1$  such that almost surely*

$$C^{-1}D_{h|U}(z, w) \leq D_h^\phi(z, w) \leq CD_{h|U}(z, w), \quad z, w \in U. \quad (20.8j)$$

*Proof.* Let  $p_0$  be the universal threshold in Theorem 8.89. Fix  $p > p_0$ . Scale invariance, continuity, and the fact that  $D_h$  induces the Euclidean topology give deterministic  $a, b > 0$  such that, with probability at least  $1 - (1 - p)/3$ ,

- (i) the  $D_h$ -distance from  $\partial B_r(z)$  to the boundary of  $\mathbb{A}_{r/2, 2r}(z)$  is at least  $ar^{\xi_Q} e^{\xi_{h_e, r}(z)}$ ;
- (ii) every  $u, v \in \partial B_r(z)$  with  $|u - v| \leq br$  have  $D_h(u, v) \leq (a/100)r^{\xi_Q} e^{\xi_{h_e, r}(z)}$ .

The choice is uniform in  $z, r$ . By Proposition 8.170, for each compact  $K \Subset U$ , all  $z \in K$ , and all sufficiently small  $r$ , the metrics are additively within  $(a/100)r^{\xi_Q} e^{\xi_{h_e, r}(z)}$  on  $B_{2r}(z)$ , except on an event of probability at most  $(1 - p)/3$ .

On the intersection, the same radial clearance and short-chord estimates, with harmless factors two, hold for  $D_h^\phi$ . Chain at most  $\lceil 2\pi/b \rceil$  short chords around  $\partial B_r(z)$ . Radial clearance ensures that each short minimizing segment stays in the buffered annulus, so the chain bounds the internal boundary diameter. It follows, for a deterministic  $C_0 = C_0(p, \gamma)$ , that

$$\sup_{u, v \in \partial B_r(z)} D_h^\phi(u, v; \mathbb{A}_{r/2, 2r}(z)) \leq C_0 D_{h|U}(\partial B_{r/2}(z), \partial B_r(z)). \quad (20.8k)$$

The event has probability at least  $p_0$ , uniformly on compact subsets and at sufficiently small radii. Lemma 8.166 checks the joint-locality and additivity hypotheses, so Theorem 8.89 gives  $D_h^\phi \leq C_0 D_{h|U}$ . Interchange the two metrics in the same argument to obtain the reverse comparison. Taking the larger of the two deterministic constants proves the proposition.  $\square$

### 8.17.5 The optimal constant is one

The last step is not a formal consequence of the infinitesimal ratio limit. A metric geodesic may cross too many bad scales for a naive partition sum to work. We use the buffered good-annulus event from the source proof. Its two geometric inputs were already proved in the preceding local-metric modules, so we isolate the exact reusable form.

**Lemma 8.172** (Buffered good-annulus inputs). *Given  $p \in (0, 1)$ , one can choose  $\alpha \in (1/2, 1)$  and  $A > 1$ , depending only on  $p$  and the metric law, so that uniformly in  $z, r$  the following two events each have probability at least  $1 - (1 - p)/3$ .*

- (a) *If  $u \in \partial B_{\alpha r}(z)$  and  $v \in \partial B_r(z)$  is farther from  $u$ , for either metric, than the boundary of  $\mathbb{A}_{r/2, 2r}(z)$ , then every path from  $u$  to  $v$  contained in  $\overline{\mathbb{A}_{\alpha r, r}(z)}$  has  $D_h$ -length strictly larger than  $D_h(u, v; \mathbb{A}_{r/2, 2r}(z))$ .*
- (b) *The annulus  $\mathbb{A}_{\alpha r, r}(z)$  contains a path separating its two boundary components whose  $D_h$ -length is at most  $AD_h(\partial B_{\alpha r}(z), \partial B_r(z))$ .*

*Both clauses are measurable from the two internal metrics in the buffered annulus after subtraction of a circle average.*

*Proof.* Clause (b) is the circumferential/radial annulus quantile comparison of Lemma 8.100; first choose its deterministic quantile  $A$ . Clause (a) is the narrow-annulus and buffer clause in the local event constructed in the proof of Proposition 8.116. For completeness, its parameter order is as follows. Tightness first chooses deterministic  $0 < s < S < \infty$  so that buffered radial distances are at least  $sr^{\xi_Q}e^{\xi h_r(z)}$  and buffered internal diameters are at most  $Sr^{\xi_Q}e^{\xi h_r(z)}$  with the desired probability. The narrow-annulus estimate then chooses  $\alpha$  sufficiently close to one so that any path constrained to  $\mathbb{A}_{\alpha r, r}(z)$  whose endpoints have global distance at least  $sr^{\xi_Q}e^{\xi h_r(z)}$  has length larger than  $Sr^{\xi_Q}e^{\xi h_r(z)}$ . The same estimates for the pullback follow from Proposition 8.170, after slightly decreasing  $s$  and increasing  $S$ .

All distances in the two conclusions are then decided by the buffered internal metrics. Locality and Weyl scaling make the event measurable from the centered annular field. This is exactly the already-established the annular input above; no confluence or downstream metric-geometry theorem is used here.  $\square$

Fix  $\delta \in (0, 1)$ , and use the  $\alpha, A$  above. Let  $E_r(z) = E_r(z; \alpha, A, \delta)$  be the event consisting of:

(E1) whenever  $u \in \partial B_{\alpha r}(z)$ ,  $v \in \partial B_r(z)$ , and a  $D_h$ -geodesic from  $u$  to  $v$  stays in  $\overline{\mathbb{A}_{\alpha r, r}(z)}$ ,

$$D_h^\phi(u, v) \leq (1 + \delta)D_h(u, v);$$

(E2) the buffered exclusion in Lemma 8.172(a) holds;

(E3) the separating-path condition in Lemma 8.172(b) holds.

**Lemma 8.173** (Measurability, positive probability, and covering). *For every compact  $K \Subset U$ , there are  $p > 0$  and  $r_K > 0$  such that*

$$\inf_{z \in K, 0 < r \leq r_K} \mathbb{P}[E_r(z)] \geq p. \quad (20.8l)$$

*Moreover  $E_r(z)$  is measurable with respect to the centered field in  $\mathbb{A}_{r/2, 2r}(z)$ . Consequently, for a suitable universal  $q > 1$ , with probability tending to one as  $\epsilon \downarrow 0$ , every point of  $K$  belongs to  $B_{r/2}(w)$  for some*

$$\epsilon^2 \leq r \leq \epsilon, \quad r \in \{2^{-k} : k \in \mathbb{N}\}, \quad w \in (\epsilon^2 \mathbb{Z}^2 / 4) \cap B_\epsilon(K), \quad (20.8m)$$

*for which  $E_r(w)$  occurs.*

*Proof.* For (E1), the two boundary circles are Euclidean separated by  $(1 - \alpha)r$ . Proposition 8.170, with  $b = 1 - \alpha$ , makes (E1) have probability arbitrarily close to one, uniformly over  $K$ , once  $r_K$  is small. Lemma 8.172 handles (E2) and (E3). A union bound gives (20.8l).

We spell out the measurability point because it is the reason for (E2). The internal metrics decide the distance from  $u$  to the buffered boundary and whether the metric ball of that radius contains  $v$ . If it does, the global distance  $D_h(u, v)$ , and likewise  $D_h^\phi(u, v)$ , equals the corresponding buffered internal distance. If it does not, (E2) says that a path confined to the narrow annulus cannot be globally geodesic. Thus on (E2), the buffered internal metrics decide exactly which narrow-annulus paths are global  $D_h$ -geodesics and decide the comparison in (E1). Lemma 8.166 then makes all three clauses measurable from the annular field. Subtracting a circle average rescales both metrics by the same factor and does not change the event.

Apply the concentric-annulus amplification in Lemma 8.115 at each grid center. Choose its exponent  $q$  larger than the polynomial number of grid centers in  $B_\epsilon(K)$ , and take a union bound. The grid mesh is  $\epsilon^2/4$ , so the resulting inner half-balls cover  $K$ . This gives (20.8m).  $\square$

**Proposition 8.174** (One-sided constant-one comparison). *Almost surely,*

$$D_h^\phi(z, w) \leq D_{h|U}(z, w), \quad z, w \in U. \quad (20.8n)$$

*Proof.* Define the deterministic optimal upper constant

$$C_* := \inf\{C \geq 1 : D_h^\phi(z, w) \leq CD_{h|U}(z, w) \text{ for all } z, w \in U \text{ almost surely}\}.$$

Proposition 8.171 gives  $C_* < \infty$ , and a countable decreasing intersection shows that the inequality holds at  $C_*$ . Suppose for contradiction that  $C_* > 1$ . Choose  $0 < \delta < C_* - 1$  and define

$$C_\delta := 1 + \delta + \frac{A}{A+1}(C_* - 1 - \delta). \quad (8.43)$$

Then  $C_\delta < C_*$ .

Fix a compact  $K \Subset U$  and  $\zeta > 0$ . By continuity, bounded compactness, and exhaustion, one can choose a deterministic compact  $K'$  with  $K \subset \text{int } K' \Subset U$  and a deterministic  $\rho > 0$  so that, with probability at least  $1 - \zeta$ , every  $D_h$ -geodesic between  $z, w \in K$  with  $|z - w| \leq \rho$  stays in  $K'$  and realizes  $D_{h|U}(z, w)$ . This is the elementary compact-containment consequence already used in Lemma 8.113: the uniform  $D_h$ -diameter of Euclidean  $\rho$ -balls tends to zero, whereas the  $D_h$ -distance from  $K$  to  $\mathbb{C} \setminus K'$  is positive.

Intersect this regularity event with the good covering of  $K'$  from Lemma 8.173; its probability is  $1 - \zeta - o_\epsilon(1)$ . Fix nearby  $z, w \in K$  and parameterize a  $D_h$ -geodesic  $P$  from  $z$  to  $w$  by  $D_h$ -length. Starting from the current point of  $P$ , choose a good ball  $B_{r_j}(w_j)$  whose inner half-ball contains that point and let  $t_j$  be the first subsequent exit time from its outer ball. Let  $s_j$  be the last time before  $t_j$  at which  $P$  visits  $\partial B_{\alpha r_j}(w_j)$ . Continue until the terminal point is reached. For fixed  $\epsilon$  only finitely many grid balls occur, and the minimum of their positive inner-to-outer  $D_h$ -crossing distances is positive; hence this construction stops after finitely many exits.

Retain the consecutive block of indices for which both  $z$  and  $w$  lie outside  $\overline{B}_{r_j}(w_j)$ . Since  $r_j \leq \epsilon$  and  $P(t_{j-1}) \in B_{r_j/2}(w_j)$ , the point at the beginning of the retained block is within  $2\epsilon$  of  $z$ , and the point at the end is within  $2\epsilon$  of  $w$ . This follows directly from the first-exit construction; if necessary, take the last index whose incoming point is within  $2\epsilon$  of  $z$  and the first later index whose outgoing point is within  $2\epsilon$  of  $w$ . A geodesic segment between two points in a Euclidean  $4\epsilon$ -ball has length bounded by the  $D_h$ -diameter of that ball, so uniform continuity makes the two discarded endpoint pieces  $o_\epsilon(1)$ , uniformly for  $z, w \in K$ .

On the retained indices, (E1) and the definition of  $C_*$  give

$$D_h^\phi(P(s_j), P(t_j)) \leq (1 + \delta)(t_j - s_j), \quad (8.44)$$

$$D_h^\phi(P(t_{j-1}), P(s_j)) \leq C_*(s_j - t_{j-1}). \quad (8.45)$$

For a retained index,  $P(t_{j-1}) \in B_{r_j/2}(w_j) \subset B_{\alpha r_j}(w_j)$ , while both endpoints  $z, w$  are outside  $\overline{B}_{r_j}(w_j)$ . The separating path from (E3) is therefore crossed by  $P$  once before  $t_{j-1}$  and once after  $s_j$ . If  $s_j - t_{j-1} > A(t_j - s_j)$ , replacing that portion of  $P$  by the separating path would shorten the geodesic. Hence

$$s_j - t_{j-1} \leq A(t_j - s_j) \implies s_j - t_{j-1} \leq \frac{A}{A+1}(t_j - t_{j-1}). \quad (20.8r)$$

Combining (8.44), (8.45), and (20.8r) yields

$$D_h^\phi(P(t_{j-1}), P(t_j)) \leq C_\delta(t_j - t_{j-1}).$$

Sum over the retained intervals. Their first and last points are in the Euclidean  $O(\epsilon)$ -neighbourhoods just identified, and uniform continuity of  $D_h^\phi$  on a compact neighbourhood of  $K'$  makes the corresponding pullback-metric endpoint cost  $o_\epsilon(1)$ , uniformly over  $z, w \in K$ . Therefore, on the regularity-and-cover event,

$$D_h^\phi(z, w) \leq C_\delta D_{h|_U}(z, w) + o_\epsilon(1) \quad (z, w \in K, |z - w| \leq \rho).$$

Send  $\epsilon \downarrow 0$ , then  $\zeta \downarrow 0$ . The inequality holds almost surely for all nearby pairs in  $K$ . Since both metrics are length metrics, partitioning a near-minimizing path into such nearby pieces gives it for every pair in  $K$ . Exhaust  $U$  by compacts to obtain  $D_h^\phi \leq C_\delta D_{h|_U}$  globally. This contradicts the definition of  $C_*$ , because  $C_\delta < C_*$ . Thus  $C_* \leq 1$ , proving (20.8n).  $\square$

### 8.17.6 Passage to arbitrary domain GFFs and the reverse inequality

**Theorem 8.175** (Full conformal coordinate covariance). *Let  $U, \tilde{U} \subset \mathbb{C}$  be domains and let  $\phi : U \rightarrow \tilde{U}$  be a deterministic conformal bijection. If  $H$  is a zero-boundary or whole-plane GFF, as appropriate, plus a continuous function on  $U$ , then almost surely*

$$D_H(z, w) = D_{H \circ \phi^{-1} + Q \log |(\phi^{-1})'|}(\phi(z), \phi(w)), \quad z, w \in U. \quad (20.8s)$$

*The equality holds simultaneously for every pair of points.*

*Proof.* First suppose  $H = h|_U + f$ , where  $h$  is a whole-plane GFF and  $f$  is continuous. Weyl scaling and Proposition 8.174 give

$$\begin{aligned} D_{H \circ \phi^{-1} + Q \log |(\phi^{-1})'|}(\phi(z), \phi(w)) \\ = (e^{\xi f} \cdot D_h^\phi)(z, w) \leq (e^{\xi f} \cdot D_{h|_U})(z, w) = D_H(z, w). \end{aligned}$$

Here monotonicity of Weyl rescaling follows pathwise from the definition of weighted length.

For a zero-boundary GFF plus a continuous function on a proper domain, couple its zero-boundary component with the one in the domain Markov decomposition of a whole-plane GFF. On each relatively compact subdomain, the difference from the whole-plane restriction is harmonic plus continuous, so the preceding one-sided inequality applies. Locality and an exhaustion extend it to all of  $U$ . The same exhaustion argument covers every GFF-type distribution admitted in the theorem statement.

The transformed field  $H \circ \phi^{-1} + Q \log |(\phi^{-1})'|$  is again a GFF plus a continuous function on  $\tilde{U}$ , by conformal invariance and the domain Markov property. Apply the one-sided inequality just proved to this field and the inverse map  $\phi^{-1}$ . It gives the reverse of the preceding inequality. The two sides are continuous metrics, so equality first on a countable dense set and then by continuity holds simultaneously for all  $z, w \in U$ .  $\square$

*Proof.* Theorem 8.175 is exactly the displayed domain theorem. Its proof first establishes the one-sided inequality for the restriction of a whole-plane GFF, transfers it to a domain GFF plus a continuous function by the domain Markov decomposition and Weyl scaling, and then applies the same one-sided statement to the transformed field and  $\phi^{-1}$ . Continuity makes the identity simultaneous for every pair of points.  $\square$

## 8.18 Metric dimension and Hölder comparison

### 8.18.1 Sharp local comparison and escape to infinity

We record separately the two inputs which are sometimes conflated in short proofs of geodesic existence. Local bi-Hölder comparison identifies the topology and controls small sets inside a fixed Euclidean compact set. It does not prevent a set of finite metric diameter from escaping through an unbounded sequence of Euclidean annuli. The latter assertion is the content of Lemma 8.177 below.

**Theorem 8.176** (Source-range local Hölder comparison). *Let  $U \subset \mathbb{C}$  be open, let  $K \Subset U$ , and let  $h$  be a GFF plus a continuous function on  $U$ . Almost surely, simultaneously for each rational pair of exponents*

$$0 < \beta < \xi(Q - 2), \quad 0 < \chi < \frac{1}{\xi(Q + 2)},$$

*there are finite random constants  $C_{K,\beta}$  and  $C'_{K,\chi}$  such that for all  $z, w \in K$ ,*

$$D_h(z, w) \leq C_{K,\beta} |z - w|^\beta, \quad |z - w| \leq C'_{K,\chi} D_h(z, w)^\chi. \quad (8.46)$$

*Consequently the two metrics induce the same topology, and the identity maps in both directions are locally Hölder for every exponent strictly inside the displayed source bounds.*

*Proof.* For a whole-plane GFF in the fixed normalization, Theorem 8.63 and Proposition 8.66 give, on each rational square compactly contained in  $U$ ,

$$c|z - w|^\alpha \leq D_h(z, w) \leq C|z - w|^\beta \quad \text{for every} \quad \beta < \xi(Q - 2), \quad \alpha > \xi(Q + 2).$$

The left inequality rearranges to

$$|z - w| \leq c^{-1/\alpha} D_h(z, w)^{1/\alpha}.$$

Given  $\chi < 1/[\xi(Q + 2)]$ , choose  $\alpha \in (\xi(Q + 2), 1/\chi)$ . On a sufficiently small metric scale,  $D_h^{1/\alpha} \leq D_h^\chi$ ; increasing the random constant handles the remaining pairs in  $K \times K$ . A countable exhaustion by rational squares and rational exponents makes the assertion simultaneous.

For a domain GFF, couple on a slightly larger compact set with the restriction of a whole-plane GFF. The difference is harmonic and hence bounded on  $K$ . If the field is further changed by a continuous  $f$ , then  $f$  is bounded on  $K$ . The Weyl comparison

$$e^{-\xi\|f\|_K} D_h(\cdot, \cdot; K^\circ) \leq D_{h+f}(\cdot, \cdot; K^\circ) \leq e^{\xi\|f\|_K} D_h(\cdot, \cdot; K^\circ)$$

transfers both estimates after working inside a compact neighbourhood of  $K$ . This proves (8.46). Either bound gives one implication in the equivalence of the two topologies.  $\square$

**Lemma 8.177** (Escape to infinity). *Let  $h$  be a whole-plane GFF normalized by  $h_1(0) = 0$ . Almost surely, for every Euclidean compact set  $K \subset \mathbb{C}$ ,*

$$\lim_{R \rightarrow \infty} D_h(K, \partial B_R(0)) = \infty. \quad (8.47)$$

*The same conclusion holds for  $h + f$  whenever  $f$  is bounded and continuous on  $\mathbb{C}$ .*

*Proof.* The annular form of Lemma 8.149 and scale tightness give a deterministic  $a > 0$  such that, for every  $R > 0$ ,

$$\mathbb{P}\left[D_h(B_R(0), \partial B_{2R}(0)) \geq aR^{\xi_Q} e^{\xi h_R(0)}\right] \geq \frac{1}{2}. \quad (8.48)$$

Here the exact normalization  $c_R = R^{\xi_Q}$  is permitted by Theorem 8.154; the sharp asymptotic scale law of Theorem 8.150 would also suffice.

Let  $E_k$  denote the event in (8.48) with  $R = 2^k$ . Locality makes  $E_k$  measurable from the field on a fixed enlargement of the corresponding annulus. If only finitely many  $E_k$  occurred, then the conditional probability of every sufficiently remote  $E_k$ , viewed from the exterior tail of the field, would vanish. This contradicts the uniform lower bound  $1/2$  in (8.48) and triviality of the whole-plane GFF exterior tail. Equivalently, applying the reverse martingale convergence theorem to the exterior sigma-fields shows that  $E_k$  occurs for infinitely many  $k$ , almost surely.

The process  $t \mapsto h_{e^t}(0)$  is a two-sided standard Brownian motion in the logarithmic radius parameter, up to the chosen normalization. Hence  $h_R(0) = o(\log R)$  almost surely as  $R \rightarrow \infty$ , and therefore

$$R^{\xi_Q} e^{\xi h_R(0)} \longrightarrow \infty. \quad (8.49)$$

Thus the metric cost of the annuli  $A(0; 2^k, 2^{k+1})$  is unbounded along the infinite successful subsequence.

Fix  $K \subset B_{2^k}(0)$  and  $R \geq 2^{k+1}$ . Every continuous path from  $K$  to  $\partial B_R(0)$  contains a subpath crossing from  $B_{2^k}(0)$  to  $\partial B_{2^{k+1}}(0)$ . Since  $D_h$  is a length metric,

$$D_h(K, \partial B_R(0)) \geq D_h(B_{2^k}(0), \partial B_{2^{k+1}}(0)).$$

Using successful  $k \rightarrow \infty$  and (8.49) proves (8.47). Finally, if  $f$  is bounded, Weyl scaling gives  $D_{h+f} \geq e^{-\xi \|f\|_\infty} D_h$ , so escape persists.  $\square$

**Proposition 8.178** (Bounded compactness). *Under the hypotheses of Lemma 8.177, every closed  $D_h$ -bounded subset of  $\mathbb{C}$  is compact. In particular,  $(\mathbb{C}, D_h)$  is complete and locally compact.*

*Proof.* Let  $A$  be  $D_h$ -bounded and fix  $z_0 \in A$ . If  $A$  were not Euclidean bounded, choose  $z_n \in A$  with  $|z_n| \rightarrow \infty$ . A path from  $z_0$  to  $z_n$  must cross every sufficiently large circle, so Lemma 8.177 would force  $D_h(z_0, z_n) \rightarrow \infty$ , contradicting boundedness. Hence  $A$  lies in some Euclidean closed ball. Theorem 8.176 says that  $D_h$  and the Euclidean metric induce the same topology there. A  $D_h$ -closed set is therefore Euclidean closed, hence Euclidean compact, and consequently  $D_h$ -compact.

A Cauchy sequence is bounded, so it lies in a compact metric ball and has a convergent subsequence; the Cauchy property forces the whole sequence to the same limit. This proves completeness. Compact closed balls give local compactness.  $\square$

**Theorem 8.179** (Geodesic attainment in the whole-plane metric). *Under the same hypotheses, every two points  $z, w \in \mathbb{C}$  are joined by a  $D_h$ -geodesic.*

*Proof.* The weak-metric length-space axiom gives paths  $P_n$  from  $z$  to  $w$  with lengths decreasing to  $D_h(z, w)$ . Discarding finitely many, take  $\text{len}_{D_h}(P_n) \leq D_h(z, w) + 1$ . After constant-speed parametrization on  $[0, 1]$ , every image lies in the closed ball  $\overline{B}_{D_h}(z, D_h(z, w) + 1)$ , which is compact by Proposition 8.178, and the paths have a common Lipschitz constant. Arzelà–Ascoli gives a uniformly convergent subsequence with limit  $P$ . For every partition  $0 = t_0 < \dots < t_m = 1$ ,

$$\sum_{j=1}^m D_h(P(t_{j-1}), P(t_j)) = \lim_{n \rightarrow \infty} \sum_{j=1}^m D_h(P_n(t_{j-1}), P_n(t_j)) \leq \liminf_n \text{len}_{D_h}(P_n).$$

Taking the supremum over partitions proves lower semicontinuity of length, so  $\text{len}_{D_h}(P) \leq D_h(z, w)$ . The reverse inequality is the definition of distance. Hence equality holds and  $P$  is a geodesic.  $\square$

**Theorem 8.180** (Metric dimension and Euclidean comparison). *The Hausdorff dimension of an open set for  $D_h$  is the deterministic finite number  $d_\gamma > 2$ . Moreover the identity maps between Euclidean and LQG metric topologies are locally Hölder with every exponent strictly inside the source bounds; in particular both metrics induce the same topology and metric balls are Euclidean relatively compact on compact subsets.*

*Proof.* The local Hölder assertions, including the exact strict source ranges, are Theorem 8.176. The metric-dimension assertion is Corollary 8.197, proved below from the full thick-point metric-dimension theorem. That proof uses uniform annular lower estimates and diameter moments rather than an unsupported uniform quantum-volume estimate for metric balls. The previously established distance exponent satisfies  $d_\gamma > 2$ , so all assertions follow.  $\square$

## 8.19 Geodesics and confluence

**Theorem 8.181** (Existence of LQG geodesics — whole-plane formulation). *Let  $\gamma \in (0, 2)$  and let  $h$  be a whole-plane GFF plus a bounded continuous perturbation. Almost surely  $(\mathbb{C}, D_h)$  is a boundedly compact length space: every closed  $D_h$ -bounded set is compact. Consequently it is complete and proper, and every two points  $z, w \in \mathbb{C}$  are joined by a  $D_h$ -geodesic  $P$ , i.e.*

$$\text{len}_{D_h}(P) = D_h(z, w).$$

*Proof.* Lemma 8.177 supplies the missing global input: finite metric diameter cannot escape through Euclidean annuli. Proposition 8.178 then proves bounded compactness, completeness, and local compactness using the local topology comparison. Finally Theorem 8.179 applies the length-space axiom, compactness, Arzelà–Ascoli, and lower semicontinuity of length to attain the distance between every pair.  $\square$

### 8.19.1 From finitely many branches to one common initial segment

Throughout this subsection the metric is based at 0. Write  $\mathcal{B}_s^\bullet$  for the filled  $D_h$ -ball of radius  $s$ , and

$$\tau_r := \inf\{s > 0 : \mathcal{B}_s^\bullet \not\subset B_r(0)\}.$$

The quantitative pruning theorem 8.127 is the hard multiscale input. The remaining work is not another independent-annulus argument. It consists of selecting one topologically exposed surviving branch, conditionally making that branch cheaper than all competitors, and then using rational endpoint uniqueness and a GFF germ zero–one law.

**Proposition 8.182** (Finitely many leftmost branches across a metric annulus). *Almost surely, for every  $0 < t < s < \infty$ , only finitely many points of  $\partial\mathcal{B}_t^\bullet$  are hit by leftmost geodesics from 0 to  $\partial\mathcal{B}_s^\bullet$ . Equivalently, the restrictions to  $[0, t]$  of all such leftmost geodesics form a finite family.*

*Proof.* Fix rational  $t < s$ . By bounded compactness and equivalence of the metric and Euclidean topologies, one can choose a rational  $r > 0$  for which

$$\tau_r \leq t \leq \tau_{2r}.$$



Lemma 8.124 says that  $\mathbb{P}[\mathcal{E}_r(a)] \rightarrow 1$  as  $a \downarrow 0$ , uniformly in  $r$ . The regularity events become weaker when  $a$  decreases, so, for fixed rational  $r$ , their union over rational  $a > 0$  has probability one. Thus the sample lies almost surely in the countable union over rational  $r, a$  of these regular events.

On  $\mathcal{E}_r(a)$ , Theorem 8.127, with the stopping time  $\tau = t$ , says that except on an event of probability  $b_0 e^{-b_1 N^\beta}$ , at most  $N$  points of  $\partial \mathcal{B}_t^\bullet$  are reached by leftmost geodesics surviving to radius

$$t + N^{-\beta} r^{\xi Q} e^{\xi h_r(0)}.$$

For all sufficiently large  $N$ , this radius is below  $s$ . Every geodesic reaching  $\partial \mathcal{B}_s^\bullet$  therefore belongs to the family counted at the smaller later radius. Letting  $N \rightarrow \infty$ , the stretched-exponential error tends to zero, proving finiteness on  $\mathcal{E}_r(a)$ . Take the countable union over  $r, a, t, s$ , then use monotonicity and rational approximation to obtain all real  $0 < t < s$ . Distinct hit points correspond to distinct initial geodesic restrictions, which proves the equivalent formulation.  $\square$

### An exposed surviving arc

For a domain  $O$  and compact connected  $A \subset \overline{O}$ , let

$$d^O(A, B) := \inf\{\text{diam}_E C : C \subset \overline{O} \text{ compact and connected, } C \cap A \neq \emptyset, C \cap B \neq \emptyset\}.$$

This intrinsic-diameter distance ignores Euclidean near-touching across a thin fjord.

**Lemma 8.183** (A finite boundary cover has an exposed arc). *Let  $K \subset \mathbb{C}$  be compact and connected, contain a neighbourhood of 0, and have connected complement. Suppose  $n$  prime-end arcs  $I_1, \dots, I_n$  cover  $\partial K$ . If*

$$\text{outrad}(K)/\text{inrad}(K) \leq L,$$

*then there is  $\epsilon = \epsilon(n, L) > 0$  and an index  $j$  such that  $I_j$  is not disconnected from infinity in  $\mathbb{C} \setminus K$  by the  $d^{\mathbb{C} \setminus K}$ -neighbourhood of radius  $\epsilon \text{inrad}(K)$  of  $\partial K \setminus I_j$ .*

*Proof.* Invert the complement of  $K$  to a bounded simply connected domain  $O$  containing 0, and let  $\phi : \mathbb{D} \rightarrow O$  fix 0. One of the pulled-back arcs has boundary length at least  $2\pi/n$ . Remove short subarcs of length  $1/(4n)$  at its two ends and join their inner endpoints by the corresponding arc of the circle of radius  $1 - 1/(4n)$ . Call the resulting crosscut shield  $S$ .

The Koebe distortion and quarter theorems, together with the bound  $L$ , put the circular part of  $\phi(S)$  a deterministic positive Euclidean distance from  $\partial O$ . For either radial part, Brownian motion started at 0 has probability bounded below in terms of  $n$  to exit through either adjacent portion of the selected boundary arc before hitting that radial segment. A connected set of sufficiently small Euclidean diameter cannot have such a hitting probability uniformly over domains of bounded outradius. Hence each radial image also has a deterministic positive  $d^O$ -distance from the complementary boundary. The image shield therefore separates 0 and a prime end of the selected arc from a fixed  $d^O$ -neighbourhood of the complementary arcs.

Inversion distorts Euclidean diameters on the relevant annulus by factors controlled by  $L$ . Mapping the shield back proves the asserted exterior statement, after decreasing  $\epsilon(n, L)$ .  $\square$

### Conditional elimination of the competing arcs

**Lemma 8.184** (Positive conditional probability of selecting an exposed arc). *Fix  $A > 1$ ,  $\epsilon > 0$ , and  $p \in (0, 1)$ . There is  $q = q(A, \epsilon, p) > 0$  with the following property. Let  $r > 0$ , let  $K = \mathcal{B}_{\tau_r}^\bullet$ , and let  $I \subset \partial K$  be an arc determined by the explored local-set data  $(K, h|_K)$ . Suppose that  $I$  is not disconnected from infinity by the  $d^{\mathbb{C} \setminus K}$ -neighbourhood of radius  $\epsilon r$  of  $\partial K \setminus I$ . With probability at*

least  $p$  over the explored data, the conditional probability given those data is at least  $q$  that every geodesic from 0 to  $\mathbb{C} \setminus B_{Ar}(0)$  enters  $K$  through  $I$ .

*Proof.* We give the conditional construction, including the three estimates which make the shortcut strict. Tightness across scales allows constants  $c > 0$ , a grid mesh  $\delta r$ , and a harmonic bound  $C$  to be chosen so that, with probability at least  $p$ , simultaneously:

- (i) every path outside  $K$  from  $\partial K \setminus I$  across its intrinsic  $\epsilon r/4$ -neighbourhood has  $D_h$ -length at least  $cr^{\xi Q} e^{\xi h_r(0)}$ ;
- (ii) every  $\delta r$ -grid square meeting  $B_{Ar}(0)$  has  $D_h$ -diameter at most  $cr^{\xi Q} e^{\xi h_r(0)}/100$ ;
- (iii) for each domain obtained by deleting a union of these finitely many grid squares, the harmonic part of the GFF is within  $C$  of  $h_r(0)$  away from its boundary.

The lower crossing estimate in (i), the diameter tightness in (ii), and the Gaussian harmonic estimate in (iii) are uniform after rescaling. A union bound over the finite grid family gives the stated probability.

By the exposed-arc hypothesis, choose a path  $\pi$  beginning at  $I$ , lying outside  $K$ , separating  $\partial B_r(0)$  from  $\partial B_{Ar}(0)$ , and staying at intrinsic-diameter distance at least  $\epsilon r/2$  from  $\partial K \setminus I$ . Let  $W$  be the interior of the union of grid squares met by  $\pi$  but not by  $K$ . The choice of  $W$  is measurable from the explored data, and it is disjoint from the intrinsic neighbourhood used in (i).

Since  $K$  is a GFF local set, conditionally on the explored data the field on each component of  $W$  is a zero-boundary GFF plus the harmonic function from (iii). The metric is monotone under addition of nonnegative functions by Weyl scaling. The FKG inequality for the zero-boundary GFF, applied to the decreasing small-diameter events in the finitely many components, and a fixed Cameron–Martin downward shift, give conditional probability at least  $q > 0$  that every component of  $W$  has diameter at most  $cr^{\xi Q} e^{\xi h_r(0)}/2$ . The bound is uniform because the grid has only finitely many rescaled shapes and (iii) controls all harmonic parts.

On this event, any point of  $W$  can reach  $I$  at cost strictly less than  $cr^{\xi Q} e^{\xi h_r(0)}$ , using one grid square at the end. In contrast, condition (i) says that a path which, after first hitting  $W$ , enters  $K$  through  $\partial K \setminus I$  pays at least this amount before it even reaches  $K$ . Replacing that terminal portion by the route through  $I$  strictly shortens the path. Hence no geodesic from outside  $B_{Ar}(0)$  can enter  $K$  through the complementary boundary. This is the desired conditional event.  $\square$

**Lemma 8.185** (One branch survives with uniform positive probability). *There are deterministic  $M > 1$  and  $q_0 > 0$  such that for every  $r > 0$ , with probability at least  $q_0$  there is a point  $x \in \partial \mathcal{B}_{\tau_r}^\bullet$  hit by every leftmost geodesic from 0 to  $\partial \mathcal{B}_{\tau_{Mr}}^\bullet$ .*

*Proof.* Proposition 8.182, in its uniform quantitative form from Theorem 8.127, gives a deterministic  $N$  such that with probability at least  $7/8$ , at most  $N$  points of  $\partial \mathcal{B}_{\tau_r}^\bullet$  are hit by leftmost geodesics surviving to  $\partial \mathcal{B}_{\tau_{2r}}^\bullet$ . Call this finite set  $Y$ . For  $x \in Y$ , let  $I_x$  be the outer-boundary arc consisting of endpoints whose leftmost geodesic passes through  $x$ . The leftmost order lemma 8.118 says that the  $I_x$ 's are arcs, have disjoint interiors, and cover the outer boundary.

Scale tightness gives  $a \in (0, 1)$  such that, with probability at least  $7/8$ , the relevant filled balls have inradius at least  $ar$  and outradius at most  $2r$ . On the intersection, Lemma 8.183 selects one  $I_x$ , measurably from the explored data, with an exposure parameter depending only on  $a, N$ . Lemma 8.184, applied at the outer ball, then gives a fixed positive probability that every sufficiently far geodesic enters through  $I_x$ . By definition of  $I_x$ , every far leftmost geodesic consequently passes through the same inner point  $x$ .

Finally choose  $M > 4$ , using scale tightness once more, so that  $\mathcal{B}_{\tau_{Mr}}^\bullet$  surrounds the deterministic Euclidean circle used in the conditional selection with failure probability smaller than half of that positive probability. Subtracting the failure probabilities leaves a number  $q_0 > 0$ , independent of  $r$ .  $\square$

**Lemma 8.186** (All geodesics share the surviving segment). *For the same type of deterministic  $M, q_0$ , with probability at least  $q_0$ , any two geodesics from 0 to points outside  $\mathcal{B}_{\tau_{Mr}}^\bullet$  coincide on  $[0, \tau_r]$ .*

*Proof.* Work on the event of Lemma 8.185, and let  $x$  be its common inner-boundary point. By Lemma 8.113, simultaneously for every rational point outside the outer filled ball the geodesic from 0 is unique. Hence each such rational geodesic, being its own leftmost geodesic, passes through  $x$ .

Leftmost and rightmost geodesics to arbitrary outer-boundary prime ends are limits of these rational geodesics from their respective sides, by Lemma 8.118 and its reflected version. Both extremal geodesics pass through  $x$ ; every other geodesic lies between them, so it also passes through  $x$ . Choose one rational endpoint  $q$  outside the outer ball. If two geodesics from 0 to  $x$  existed, concatenating either with the common tail from  $x$  to  $q$  would produce two geodesics from 0 to  $q$ , contradicting fixed-pair uniqueness. Thus the segment from 0 to  $x$  is unique, and every geodesic under consideration agrees with it on  $[0, \tau_r]$ .  $\square$

**Theorem 8.187** (Confluence at a fixed deterministic point). *Fix a deterministic  $z \in \mathbb{C}$ . Almost surely, for every  $s > 0$  there is  $t \in (0, s)$  such that any two geodesics from  $z$  to points outside  $B_{D_h}(z, s)$  coincide during their first  $t$  units of  $D_h$ -length.*

*Proof.* Translation covariance reduces to  $z = 0$ . Let  $F_r$  be the event in Lemma 8.186. It has probability at least  $q_0$ , uniformly in  $r$ , and it is determined by the field in  $B_{Mr}(0)$ . Therefore

$$\limsup_{r \downarrow 0} F_r$$

belongs to the GFF germ sigma-field at 0, which is trivial. Fatou's lemma gives this limsup event probability at least  $q_0$ , so triviality forces its probability to be one. Thus successful radii occur arbitrarily close to zero.

Because  $D_h$  induces the Euclidean topology,  $\tau_{Mr} \rightarrow 0$  as  $r \downarrow 0$ . Given  $s > 0$ , choose a successful  $r$  so small that  $\mathcal{B}_{\tau_{Mr}}^\bullet \subset B_{D_h}(0, s)$ . Every geodesic to a point outside the latter ball is also covered by  $F_r$ , so all such geodesics agree on  $[0, \tau_r]$ . Take  $t = \tau_r > 0$ . A countable rational approximation in  $s$  makes the assertion simultaneous for every  $s > 0$ .  $\square$

*Remark 8.188* (Fixed-point scope). The point  $z$  is fixed before sampling the field. The theorem extends to other GFF variants by local absolute continuity, but it is false at all points simultaneously: an interior point of a geodesic already has two noncoalescing incoming arms.

**Theorem 8.189** (Confluence of LQG geodesics). *Fix a deterministic point  $z$ . Almost surely, for each  $s > 0$  there is  $t \in (0, s)$  such that any two geodesics from  $z$  to points outside the  $D_h$ -ball  $B_{D_h}(z, s)$  coincide during their first  $t$  units of  $D_h$ -length. Equivalently, when oriented toward  $z$ , all geodesics from any compact set a positive  $D_h$ -distance away share the same final segment before reaching  $z$ .*

*Proof.* This is Theorem 8.187. Its finite-branch input is the quantitative filled-annulus pruning theorem already proved in this chapter. Lemma 8.183 selects a topologically exposed surviving arc; Lemma 8.184 uses the GFF Markov property, metric FKG, and a strict conditional shortcut to eliminate all competitors with positive probability. Rational-endpoint uniqueness upgrades the result from leftmost to all geodesics, and the GFF germ zero-one law supplies successful radii arbitrarily close to  $z$ . No independent-Bernoulli domination of field-dependent gate events is used.  $\square$

## 8.20 Metric KPZ

### 8.20.1 The metric thick-point and KPZ engine

Let  $U \subset \mathbb{C}$  be a domain, let  $h$  be a GFF on  $U$ , possibly plus a continuous function, and let

$$\mathcal{T}_h^\alpha := \left\{ z \in U : \lim_{r \downarrow 0} \frac{h_r(z)}{\log r^{-1}} = \alpha \right\}.$$

We write  $\dim_{\mathbb{H}}^0$  for Euclidean Hausdorff dimension and  $\dim_{\mathbb{H}}^\gamma$  for Hausdorff dimension with respect to  $D_h$ . All dimension assertions below are local, so it suffices to work with a whole-plane GFF normalized by  $h_1(0) = 0$  and a bounded set  $X \Subset U$ . Indeed, the domain Markov decomposition changes the field on a compact set by a continuous harmonic function, and Weyl scaling by a bounded function changes distances by bounded factors. Neither operation changes Hausdorff dimension.

#### Uniform thick-point cores

**Lemma 8.190** (Uniform core of a thick slice). *Fix  $\alpha \in \mathbb{R}$  and  $\zeta > 0$ . Almost surely, for every bounded Borel set  $X$  there is  $\bar{r} > 0$  such that the set*

$$X^{\alpha, \zeta} := \left\{ z \in X : \frac{h_r(z)}{\log r^{-1}} \in [\alpha - \zeta, \alpha + \zeta] \text{ for every } r \in (0, \bar{r}] \right\}$$

*satisfies*

$$\dim_{\mathbb{H}}^0 X^{\alpha, \zeta} \geq \dim_{\mathbb{H}}^0(X \cap \mathcal{T}_h^\alpha) - \zeta, \quad (8.50)$$

$$\dim_{\mathbb{H}}^\gamma X^{\alpha, \zeta} \geq \dim_{\mathbb{H}}^\gamma(X \cap \mathcal{T}_h^\alpha) - \zeta. \quad (8.51)$$

*Proof.* For  $n \geq 1$ , let  $X_n$  be the set defined above with  $\bar{r} = 1/n$ . Then

$$X \cap \mathcal{T}_h^\alpha \subset \bigcup_{n=1}^{\infty} X_n.$$

Hausdorff dimension in either metric is countably stable. Therefore some  $X_n$ , and in fact every sufficiently late member of the increasing sequence, has dimension within  $\zeta$  of that of the union. Choose such an  $n$ . The argument is pathwise and hence simultaneous for every Borel  $X$ .  $\square$

**Lemma 8.191** (Circle averages at nearby deterministic centers). *Fix  $\alpha \in \mathbb{R}$  and  $\zeta \in (0, 1)$ . There are deterministic numbers  $m_{\alpha, \zeta}$  and  $M_{\alpha, \zeta}$ , both converging to  $\alpha$  as  $\zeta \downarrow 0$ , such that almost surely the following holds. For every bounded Borel  $X$ , all sufficiently small  $r$ , and every  $z$  lying within Euclidean distance  $r$  of  $X^{\alpha, \zeta}$ ,*

$$m_{\alpha, \zeta} \leq \frac{h_r(z)}{\log r^{-1}} \leq M_{\alpha, \zeta}. \quad (8.52)$$

*One may take*

$$m_{\alpha, \zeta} = (1 - \zeta^2)(\alpha - \zeta) - 3\sqrt{10}\zeta, \quad M_{\alpha, \zeta} = (1 - \zeta^2)(\alpha + \zeta) + 3\sqrt{10}\zeta.$$

*Proof.* The circle-average process has a modification for which, on each compact set, almost surely for all sufficiently small  $r$ ,

$$|h_r(z) - h_{r^{1-\rho}}(w)| \leq 3\sqrt{10\rho} \log r^{-1} \quad \text{whenever } |z - w| \leq 2r. \quad (8.53)$$

This follows by applying the Gaussian increment estimate to a dyadic space-scale grid, taking a union bound, and then using Kolmogorov continuity between grid points. Apply (8.53) with  $\rho = \zeta^2$  and  $w \in X^{\alpha, \zeta}$  satisfying  $|z - w| \leq r$ . The definition of the uniform core bounds  $h_{r^{1-\zeta^2}}(w)$  between  $(\alpha \pm \zeta)(1 - \zeta^2) \log r^{-1}$ . Adding the error in (8.53) gives the displayed values of  $m_{\alpha, \zeta}$  and  $M_{\alpha, \zeta}$ .  $\square$

**Lemma 8.192** (Euclidean dimension of an independent thick slice). *If  $X \Subset U$  is deterministic, or random and independent of  $h$ , then for every  $\alpha \in [-2, 2]$ , almost surely*

$$\dim_{\mathbb{H}}^0(X \cap \mathcal{T}_h^\alpha) = \max \left\{ \dim_{\mathbb{H}}^0 X - \frac{\alpha^2}{2}, 0 \right\}. \quad (8.54)$$

*Proof.* Conditioning reduces the random independent case to deterministic  $X$ . For the upper bound, cover  $X$  on a dyadic scale by squares with total  $s$ -content as small as desired, where  $s > \dim_{\mathbb{H}}^0 X$ . The circle average at a square centre has variance  $\log r^{-1} + O(1)$ , so the probability of being within  $o(1)$  of  $\alpha \log r^{-1}$  is  $r^{\alpha^2/2 + o(1)}$ . A union bound, Markov's inequality, and Borel–Cantelli give dimension at most  $\max\{s - \alpha^2/2, 0\}$ , and then  $s \downarrow \dim_{\mathbb{H}}^0 X$ .

For the lower bound, fix  $s < \dim_{\mathbb{H}}^0 X$  and a Frostman measure  $\nu$  on a compact subset of  $X$  with finite  $s$ -energy. Weight  $\nu$  at circle-average scale  $r$  by the exponential martingale which tilts  $h_r(z)$  to mean  $\alpha \log r^{-1}$ , and restrict to the usual uniform barrier event. The two-point Gaussian calculation shows that every  $t < s - \alpha^2/2$  energy of the resulting measures is bounded in expectation. Their total masses have a nonzero uniformly integrable limit, and the barrier law forces that limit to be supported on  $\mathcal{T}_h^\alpha$ . Frostman's lemma gives dimension at least  $t$ . Let  $t \uparrow s - \alpha^2/2$ , then  $s \uparrow \dim_{\mathbb{H}}^0 X$ . Localization and the GFF tail zero–one law upgrade positive probability to the almost-sure statement. This is exactly the thick-point Frostman construction already owned in Volume VI.  $\square$

## Metric input estimates

**Lemma 8.193** (Uniform normalized internal-diameter moments). *For every  $p \in (0, 4d_\gamma/\gamma^2)$ , every fixed compact  $K \Subset V$ , and every  $R > 0$ ,*

$$\mathbb{E} \left[ \left( R^{-\xi Q} e^{-\xi h_R(0)} \sup_{z, w \in RK} D_h(z, w; RV) \right)^p \right] \leq C_{p, K, V}, \quad (8.55)$$

where the constant is uniform in  $R$ .

*Proof.* It is enough to treat a unit square strictly inside a slightly larger square; a finite chain of such squares gives the general connected compact set. At dyadic scale  $2^{-n}R$ , cover the unit square by the buffered rectangles used in the chaining proof below. Lemma 8.149 supplies upper crossing circuits with scale  $(2^{-n}R)^{\xi Q} e^{\xi h_{2^{-n}R}(z)}$  and tails faster than any fixed power of the multiplicative threshold. Concatenating the circuits at all scales bounds the normalized internal diameter by

$$C^\xi \sum_{n \geq 0} 2^{-n\xi(Q-q)+o(n)} \quad (8.56)$$

on the event

$$\max_{z \in B_{CR}(0) \cap 2^{-n-1}R\mathbb{Z}^2} |h_{2^{-n}R}(z) - h_R(0)| \leq \log(C2^{qn}) \quad \text{for every } n \geq 0. \quad (8.57)$$

Here  $q$  can be chosen arbitrarily in  $(2, Q)$ ; the accumulated superpolynomial crossing errors are absorbed in the  $o(n)$  term. The series in (8.56) converges because  $q < Q$ .

The Gaussian maximum calculation for the space-scale grid gives

$$\mathbb{P}[(8.57) \text{ fails}] \leq C^{-q-\sqrt{q^2-4}+o_C(1)}$$

uniformly in  $R$ . Since the diameter in (8.56) is of order  $C^\xi$ , its upper-tail power can be taken arbitrarily close, as  $q \uparrow Q$ , to

$$\frac{Q + \sqrt{Q^2 - 4}}{\xi} = \frac{4/\gamma}{\gamma/d_\gamma} = \frac{4d_\gamma}{\gamma^2}.$$

Integrating the tail yields (8.55) for every smaller  $p$ . The square chain and the exact affine scaling rule make the constant independent of  $R$ .  $\square$

**Corollary 8.194** (Moment of a small Euclidean ball). *For  $p \in (0, 4d_\gamma/\gamma^2)$ , a compact set  $K$ , and  $z \in K$ ,*

$$\mathbb{E}[\text{diam}_{D_h}(B_r(z))^p] \leq r^{\xi Q p - \xi^2 p^2/2 + o_r(1)} \quad (8.58)$$

as  $r \downarrow 0$ , uniformly in  $z \in K$ .

*Proof.* Locality and Weyl scaling show that

$$e^{-\xi h_{2r}(z)} \text{diam}_{D_h}(B_r(z); B_{2r}(z))$$

is determined by the centred field  $(h - h_{2r}(z))|_{B_{2r}(z)}$ . It is independent of the Gaussian increment  $h_{2r}(z) - h_1(z)$ , whose variance is  $\log r^{-1} + O(1)$ . Apply Lemma 8.193 at scale  $2r$  and use  $\mathbb{E}e^{p\xi(h_{2r}(z) - h_1(z))} = r^{-\xi^2 p^2/2 + o(1)}$ . The remaining factor  $e^{\xi h_1(z)}$  has uniformly bounded moments on  $K$ ; Hölder's inequality with an exponent tending slowly to one absorbs it into the  $o_r(1)$ . Finally, the unrestricted distance is no larger than the internal one.  $\square$

**Lemma 8.195** (Uniform small-annulus lower bound). *Let  $V \Subset U$  and  $\zeta > 0$ . Almost surely, for all sufficiently small  $r > 0$ , simultaneously for  $z \in V$ ,*

$$D_h(\partial B_r(z), \partial B_{r/2}(z)) \geq r^{\xi Q + \zeta} e^{\xi h_r(z)}. \quad (8.59)$$

*Proof.* For fixed  $z, r$ , the lower-tail half of Lemma 8.149, with threshold  $r^{-\zeta/2}$ , gives

$$D_h(\partial B_{7r/8}(z), \partial B_{5r/8}(z)) \geq r^{\xi Q + \zeta/2} e^{\xi h_r(z)}$$

except with probability  $O(r^N)$  for every prescribed  $N$ . Take first  $r = 1/n$ , then a union bound over  $n \geq (2\epsilon)^{-1}$  and centres in  $V \cap n^{-2}\mathbb{Z}^2$ . The total failure probability tends to zero faster than a power of  $\epsilon$ .

Almost surely at all small scales, the circle-average continuity estimate

$$|h_r(z) - h_{r'}(z')| \leq \frac{|(z, r) - (z', r')|^{99/200}}{r^{1/2}}$$

holds whenever  $r/r' \in [1/2, 2]$ . Given arbitrary  $z \in V$  and  $r \leq \epsilon$ , choose  $n$  with  $|r - 1/n| \leq 4r^2$  and a grid point  $z'$  with  $|z - z'| \leq 4r^2$ . The grid annulus lies inside  $B_r(z) \setminus B_{r/2}(z)$ , while the circle averages differ by  $O(r^{49/100})$ . For small  $r$ , the spare factor  $r^{\zeta/2}$  absorbs this error and yields (8.59). Apply Borel–Cantelli along  $\epsilon = 2^{-k}$ .  $\square$

## Quantum dimension of thick slices

**Theorem 8.196** (Metric dimension of an independent thick slice). *Let  $X \subseteq U$  be deterministic, or random and independent of  $h$ . For each  $\alpha \in [-2, 2]$ , almost surely*

$$\dim_H^\gamma(X \cap \mathcal{T}_h^\alpha) = \frac{\max\{\dim_H^0 X - \alpha^2/2, 0\}}{\xi(Q - \alpha)}. \quad (8.60)$$

*Proof.* Condition on  $X$ , so the Euclidean covers used below are deterministic. Fix  $\zeta > 0$ , abbreviate  $m = m_{\alpha, \zeta}$ ,  $M = M_{\alpha, \zeta}$ , and work with the uniform core  $X^{\alpha, \zeta}$ . When  $\alpha \neq 0$ , take  $\zeta$  small enough that  $m$  and  $M$  have the same sign.

*Upper bound.* Choose

$$\ell > \frac{\dim_H^0 X - (m^2 \wedge M^2)/2}{\xi(Q - M)}$$

when  $\alpha \neq 0$ ; for  $\alpha = 0$ , omit the Gaussian-tail term from the numerator. Choose  $x > \dim_H^0 X$  still strictly below the corresponding numerator  $\xi(Q - M)\ell + (m^2 \wedge M^2)/2$ . For each  $\delta > 0$ , take a deterministic cover  $\{B_{r_i}(z_i)\}$  of  $X$  with  $\sum_i r_i^x < \delta$  and  $\max_i r_i \rightarrow 0$ . Retain only indices whose balls meet  $X^{\alpha, \zeta}$ .

On the high-probability event of Lemma 8.191, all retained circle averages  $h_{2r_i}(z_i)/\log(2r_i)^{-1}$  lie in  $[m, M]$ . The normalized internal diameter is independent of this circle average. Lemma 8.193 and the one-dimensional Gaussian tail therefore give

$$\begin{aligned} & \mathbb{E} \left[ \text{diam}_{D_h}(B_{r_i}(z_i); B_{2r_i}(z_i))^\ell \mathbf{1}_{\{h_{2r_i}(z_i)/\log(2r_i)^{-1} \in [m, M]\}} \right] \\ & \leq r_i^{\xi(Q-M)\ell + (m^2 \wedge M^2)/2 + o_\delta(1)}. \end{aligned} \quad (8.61)$$

For  $\alpha = 0$ , the same estimate holds without the last exponent. The choice of  $x$  makes the right side at most  $r_i^x$ . Markov's inequality with  $\delta = \epsilon^2$ , followed by Borel–Cantelli along  $\epsilon = 2^{-k}$ , produces covers of  $X^{\alpha, \zeta}$  whose total  $\ell$ -dimensional  $D_h$ -content tends to zero. Thus  $\dim_H^\gamma X^{\alpha, \zeta} \leq \ell$ . Send  $\ell$  down and  $\zeta \downarrow 0$ , and use (8.54).

*Lower bound.* This direction is pathwise and does not require independence. Let  $\ell > \dim_H^\gamma X^{\alpha, \zeta}$  and cover the uniform core by  $D_h$ -balls  $\tilde{B}_{\rho_i}(z_i)$  with  $\sum_i \rho_i^\ell < \delta$ . Put

$$r_i := \sup\{|u - z_i| : u \in \tilde{B}_{\rho_i}(z_i)\}.$$

The inverse Hölder bound in Theorem 8.176 makes every  $r_i$  uniformly small when  $\delta$  is small. Since the quantum ball meets the uniform core, its centre lies within Euclidean distance  $r_i$  of that core, and hence  $h_{r_i}(z_i) \geq m \log r_i^{-1}$ .

A nearly extremal point in the definition of  $r_i$ , joined to  $z_i$  by a geodesic, gives a path inside the quantum ball crossing the Euclidean annulus  $B_{r_i}(z_i) \setminus B_{r_i/2}(z_i)$ . Consequently

$$\rho_i \geq D_h(\partial B_{r_i}(z_i), \partial B_{r_i/2}(z_i)) \geq r_i^{\xi(Q-m)+\zeta}$$

by Lemma 8.195. Therefore

$$\sum_i r_i^{[\xi(Q-m)+\zeta]\ell} \leq \sum_i \rho_i^\ell < \delta.$$

The Euclidean balls  $B_{r_i}(z_i)$  cover the core, so

$$\dim_H^0 X^{\alpha, \zeta} \leq [\xi(Q - m) + \zeta] \dim_H^\gamma X^{\alpha, \zeta}.$$

Send  $\zeta \downarrow 0$  and use Lemmas 8.190 and 8.192. The upper and lower bounds agree and give (8.60).  $\square$

**Corollary 8.197** (Hausdorff dimension of the LQG metric). *Almost surely, every nonempty open  $V \subset U$  has*

$$\dim_{\mathbb{H}}^{\gamma} V = \dim_{\mathbb{H}}^{\gamma}(V \cap \mathcal{T}_h^{\gamma}) = d_{\gamma}.$$

*Proof.* Apply Theorem 8.196 to an increasing sequence of deterministic compact subsets of  $V$  with Euclidean dimension two. Since

$$Q - \gamma = \frac{2}{\gamma} - \frac{\gamma}{2}, \quad 2 - \frac{\gamma^2}{2} = \gamma(Q - \gamma),$$

the right side of (8.60) at  $\alpha = \gamma$  is  $\gamma/\xi = d_{\gamma}$ . The KPZ upper-cover argument in the next theorem, applied to  $V$  with exponents decreasing to  $d_{\gamma}$ , gives  $\dim_{\mathbb{H}}^{\gamma} V \leq d_{\gamma}$ . The thick slice is contained in  $V$ , giving the reverse inequality and both equalities.  $\square$

### The independent-set metric KPZ identity

**Theorem 8.198** (Metric KPZ for sets independent of the field). *Let  $X \subset U$  be deterministic, or random and independent of  $h$ . Then almost surely*

$$\dim_{\mathbb{H}}^0 X = \xi Q \dim_{\mathbb{H}}^{\gamma} X - \frac{\xi^2}{2} (\dim_{\mathbb{H}}^{\gamma} X)^2. \quad (8.62)$$

*Proof.* By conditioning, localization, and countable stability, assume that  $X$  is deterministic and bounded. Put  $x = \dim_{\mathbb{H}}^0 X$ ,  $\Delta = \dim_{\mathbb{H}}^{\gamma} X$ , and

$$F(s) = \xi Q s - \frac{\xi^2 s^2}{2}.$$

The lower bound in Theorem 8.196, optimized over  $\alpha$ , gives

$$\Delta \geq \max_{\alpha \in [-2, 2]} \frac{(x - \alpha^2/2)_+}{\xi(Q - \alpha)} = \frac{Q - \sqrt{Q^2 - 2x}}{\xi}.$$

Since  $F$  is increasing on  $[0, d_{\gamma}]$  and  $F(d_{\gamma}) = 2$ , this implies  $F(\Delta) \geq x$ , once the upper bound  $\Delta \leq d_{\gamma}$  below is established.

For the opposite inequality, let  $\ell \geq 0$  satisfy  $F(\ell) > x$  and  $\ell < 4d_{\gamma}/\gamma^2$ . Choose  $y \in (x, F(\ell))$ . A deterministic Euclidean cover  $\{B_{r_i}(z_i)\}$  of  $X$  may be chosen with  $\sum_i r_i^y < \epsilon$  and all radii small. Corollary 8.194 gives

$$\mathbb{E} \sum_i \text{diam}_{D_h}(B_{r_i}(z_i))^{\ell} \leq \sum_i r_i^{F(\ell)+o(1)} \leq \sum_i r_i^y < \epsilon.$$

Markov and Borel–Cantelli along a deterministic sequence  $\epsilon \downarrow 0$  show that  $\dim_{\mathbb{H}}^{\gamma} X \leq \ell$ . If  $x < 2$ , let  $\ell$  decrease to  $(Q - \sqrt{Q^2 - 2x})/\xi$ . If  $x = 2$ , take  $\ell \downarrow d_{\gamma}$ ; the strict inequality  $F(\ell) > 2$  holds for  $\ell > d_{\gamma}$  sufficiently close to  $d_{\gamma}$ , and the moment range still contains these exponents because  $\gamma < 2$ . Thus in every case

$$\Delta \leq \frac{Q - \sqrt{Q^2 - 2x}}{\xi} \leq d_{\gamma}.$$

Together with the preceding optimized thick-slice lower bound, equality holds. Applying  $F$  yields (8.62).  $\square$



*Remark 8.199* (Scope of the identity). Independence is used twice: in the exact Euclidean dimension of the thick slice and in the deterministic choice of the Euclidean covers in the moment argument. The pathwise uniform-annulus estimate does give useful one-sided bounds for field-dependent sets, but it does not justify (8.62) for geodesics, metric-ball boundaries, or other sets selected from the same field.

**Theorem 8.200** (Geometric KPZ for the subcritical LQG metric — source scope). *Fix  $\gamma \in (0, 2)$  and let  $h$  be the planar GFF used to define  $D_h$ . Let  $X \subset \mathbb{C}$  be a Borel set which is either deterministic or random and independent of  $h$ . Write  $\dim_H^0 X$  for its Euclidean Hausdorff dimension and  $\dim_H^\gamma X$  for its Hausdorff dimension with respect to  $D_h$ . Then almost surely*

$$\boxed{\dim_H^0 X = \xi Q \dim_H^\gamma X - \frac{\xi^2}{2} (\dim_H^\gamma X)^2}, \quad \xi = \frac{\gamma}{d_\gamma}, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}. \quad (20.6)$$

*The independence hypothesis is part of this exact KPZ identity; dependent sets require the separate worst-case dimension bounds and are not covered by this parent theorem.*

*Proof.* Theorem 8.198 is exactly the displayed identity. Its proof does not replace a uniform covering argument by a pointwise typical-diameter assertion: Lemma 8.193 and Corollary 8.194 control deterministic Euclidean covers, while Lemma 8.195 controls arbitrary quantum-ball covers. The two estimates first yield the exact thick-slice formula in Theorem 8.196; optimizing it and matching it with the diameter-moment cover proves the quadratic identity. The conditioning step in that proof records precisely where independence of  $X$  and  $h$  is used.  $\square$

## 8.21 QLE at the Brownian-map parameter

Set  $\gamma = \sqrt{8/3}$ , so  $\kappa' = 6$ . The quantum-disk and stable 3/2-Lévy exploration owners of Volume XIX are available backward inputs.

**Theorem 8.201** (QLE metric at the Brownian-map parameter). *On a doubly marked  $\sqrt{8/3}$ -LQG sphere, the  $\text{QLE}(8/3, 0)$  growth process parameterized by quantum distance defines a symmetric distance on a countable dense set of quantum-typical points. This distance is a metric and its metric balls have the QLE/Lévy-net Markov description of Miller–Sheffield.*

*Proof.* Divide quantum natural time into intervals of size  $\delta$ . On each interval grow an  $\text{SLE}_6$  segment from the current tip until it cuts a quantum boundary length sampled from the boundary-length stationary measure, then resample the tip from quantum boundary length on the new boundary. The quantum-disk Markov property makes successive unexplored components conditionally independent. Tightness follows from the 3/2-stable boundary-length process and the finite quantum-area of the sphere; a subsequential limit is  $\text{QLE}(8/3, 0)$ .

The natural QLE time is obtained from the quantum boundary-length clock by the unique deterministic power for which adding a field constant rescales time as metric length. With this clock, the time for QLE from  $x$  to swallow  $y$  is a candidate distance  $d_Q(x, y)$ .

Explore QLE simultaneously from  $x$  toward  $y$  and from  $y$  toward  $x$  using the same quantum disk decomposition read in opposite order. The Poissonian 3/2-stable bead description is invariant under this rerooting after weighting by the two marked quantum-typical points. Therefore  $d_Q(x, y) = d_Q(y, x)$  in law and, under the canonical coupling, almost surely.

If the QLE ball from  $x$  first swallows  $y$  and is then continued until it swallows  $z$ , the strong Markov property says the extra exploration has the same distance-clock law as QLE started from

the discovered boundary. This concatenation gives  $d_Q(x, z) \leq d_Q(x, y) + d_Q(y, z)$ . The Lévy-net description gives positive time to swallow distinct typical points, so the distance separates them. Thus it is a metric on the dense sample.  $\square$

**Theorem 8.202** (Brownian-map identification at the square-root-eight-thirds parameter). *The QLE metric extends uniquely and continuously from the countable dense set to the entire unit-area  $\sqrt{8/3}$ -LQG sphere. Together with quantum area it has the Brownian-map metric-measure law. Conversely the metric-measure structure almost surely determines the compatible conformal/LQG structure.*

*Proof.* The QLE ball estimates give power-law upper and lower bounds between Euclidean and QLE distances for all pairs in a fixed dense countable set. The same dyadic annulus and Borel–Cantelli mechanism used in the LFPP tightness architecture turns these estimates into bi-Hölder bounds. Hence the identity map extends in both directions between the Euclidean sphere and the metric completion, proving that the completion is the same underlying topological sphere and that the metric extension is unique.

The filled QLE metric-ball process, parameterized by radius, has boundary length evolving as the  $3/2$ -stable continuous-state branching process obtained from the Volume-XIX disk decomposition; conditioned on boundary length, the inside and outside are independent quantum disks. The Brownian map metric-net has the same Markov kernel and the same unit-area normalization. The metric net plus the conditional disk laws determines all finite collections of metric balls about quantum-typical points. Therefore the completed QLE metric-measure sphere and the Brownian map have the same law.

Sample an i.i.d. countable dense set from the area measure, which is measurable from the metric-measure space. From metric balls centered at these points one recovers the complete QLE metric-net adjacency and boundary-length data. Each cell is a marked quantum disk whose conformal law is determined by its metric net boundary data; welding the rational collection of cells by their matched boundary lengths gives a measurable conformal atlas. Refining the cell mesh and using the bi-Hölder bounds makes the maximal Euclidean diameter tend to zero. Two conformal structures compatible with the same metric-measure data therefore agree on every limiting chart, up to the global Möbius freedom. Thus the Brownian metric-measure sphere determines the LQG conformal surface.  $\square$

## Chapter 9

# Brownian surfaces, critical metrics, and conformal embeddings

This chapter gives complete proofs of the random-map and critical/supercritical metric roots that remained open after the metric engine of Chapter 8. The common principle is to separate three layers which are frequently compressed in short summaries: (i) a discrete encoding and its functional invariance principle, (ii) identification of the limiting pseudo-metric or conformal embedding, and (iii) the topological or Markovian property which rules out extra identifications. The mated-CRT construction itself is inherited exactly from the Volume XIX peanosphere owner; all other rows below receive a physical owner here.

### 9.1 The Brownian-map metric from labelled trees

Let  $\mathbf{e}$  be a normalized Brownian excursion, let  $\mathcal{T}_{\mathbf{e}}$  be the associated CRT, and conditionally on  $\mathbf{e}$  let  $Z$  be the head of the Brownian snake, so that

$$\mathbf{E}[(Z_s - Z_t)^2 \mid \mathbf{e}] = d_{\mathcal{T}_{\mathbf{e}}}(p_{\mathbf{e}}(s), p_{\mathbf{e}}(t)).$$

For  $s, t \in [0, 1]$  put

$$D^\circ(s, t) = Z_s + Z_t - 2 \max \left\{ \min_{u \in [s, t]} Z_u, \min_{u \in [t, 1] \cup [0, s]} Z_u \right\}. \quad (20.10)$$

The Brownian-map pseudo-metric  $D$  is the largest pseudo-metric on the CRT which is bounded above by  $D^\circ$ ; equivalently it is the intrinsic pseudo-metric obtained by chaining  $D^\circ$ .

**Lemma 9.1** (Schaeffer–BDG invariance principle). *For a uniform rooted quadrangulation with  $n$  faces, its Schaeffer tree contour  $C_n$  and label process  $L_n$ , after the standard deterministic normalization, satisfy*

$$(n^{-1/2}C_n(2nt), n^{-1/4}L_n(2nt)) \Longrightarrow (\mathbf{e}_t, Z_t)$$

*in  $C([0, 1], \mathbb{R}^2)$ . The same statement, with source-dependent variance constants, holds for the regular critical bipartite Boltzmann/mobile classes used below.*

*Proof.* The contour of the underlying conditioned plane tree is a simple random-walk excursion conditioned on its duration. Donsker's theorem together with the cyclic lemma gives convergence to the normalized Brownian excursion. Given the tree, the label increments form a centered

finite-variance nearest-neighbor bridge field along the tree edges. For vertices represented by contour times  $s, t$ , the conditional variance of the label difference is the tree distance; therefore the finite-dimensional conditional characteristic functions converge to those of the Brownian-snake head. Kolmogorov bounds follow from the tree distance moment estimate  $\mathbf{E}[d_{T_n}(v_s, v_t)^p] \leq C_p n^{p/2} |s - t|^{p/2}$  and the conditional Burkholder estimate for the labels. Tightness plus the finite-dimensional identification proves the joint convergence. For the BDG mobile classes the same argument is applied to the critical multitype Galton–Watson tree; the finite-variance hypothesis gives a Brownian excursion after deterministic variance renormalization, and the centered label bridge gives the same snake head after its model-dependent variance factor.  $\square$

**Lemma 9.2** (Discrete cactus bound and successor geodesics). *If  $i, j$  are corners of a labelled Schaeffer mobile and  $v_i, v_j$  their map vertices, then*

$$d_n(v_i, v_j) \leq L_n(i) + L_n(j) - 2 \max\{\underline{L}_n(i, j), \underline{L}_n(j, i)\} + 2, \quad (20.11)$$

*where the two underlines are the cyclic minima. Conversely, if a sequence of pairs  $(i_n, j_n)$  has graph distance  $o(n^{1/4})$ , then every subsequential limit  $(s, t)$  satisfies the Brownian-map equivalence relation  $D(s, t) = 0$ .*

*Proof.* The successor of a corner of label  $\ell$  is the first later corner of label  $\ell - 1$ . Iterating successors produces a path to the distinguished vertex whose length is the label drop. Starting from  $i$  and  $j$  and stopping the two successor chains at the larger of the two cyclic label minima yields (20.11). For the converse, expose a geodesic between  $v_i$  and  $v_j$ . Every time the geodesic crosses from one contour interval to the other it must meet a label level which is no larger than the corresponding cyclic minimum, up to an  $O(1)$  lattice error. After multiplication by  $n^{-1/4}$  and use of Lemma 9.1, a vanishing graph distance therefore forces the chain pseudo-distance generated by (20.10) to vanish. This is the “no extra identifications” step: convergence of labels alone gives only the upper bound; the geodesic crossing argument supplies the matching lower identification.  $\square$

**Imported theorem 9.3** (Quadrangulations converge to the Brownian map). For uniform rooted quadrangulations with  $n$  faces, graph distance multiplied by  $(9/8)^{1/4} n^{-1/4}$  converges in distribution in the Gromov–Hausdorff topology to the Brownian map  $(\mathcal{M}, D)$ .

*Source and proof status.* This is the main theorem of Miermont [9], with the displayed constant matching that paper’s quadrangulation convention. Lemmas 9.1 and 9.2 explain the coding, tightness, upper bound, and zero-distance identification used in the source proof. They do not by themselves give the quantitative lower bound needed to identify all positive distances. That remaining geodesic-star argument is imported from Sections 3–6 of the cited paper. Thus the result is used here as an imported theorem, not claimed as a new self-contained proof.  $\square$

**Imported theorem 9.4** (Topology and volume exponent of the Brownian map). Almost surely the Brownian map is homeomorphic to  $\mathbb{S}^2$ , and every nonempty open subset has Hausdorff dimension 4.

*Sources and proof status.* The Hausdorff-dimension statement is proved in Le Gall [7]; the sphere topology is proved in Le Gall–Paulin [8] (with an equivalent 1-regular route in Miermont’s sphericity work). Brownian-snake 1/4-scaling and the no-pinch mechanism explain the result, but do not replace the uniform estimates and topological convergence arguments in those sources.  $\square$

**Corollary 9.5** (Brownian-map universality in the selected finite-variance classes). *Uniform rooted  $2p$ -angulations, and the regular critical finite-variance bipartite Boltzmann map classes fixed in the ledger, converge after their model-dependent  $n^{-1/4}$  normalization to the same Brownian map. The limit is homeomorphic to  $\mathbb{S}^2$  and has Hausdorff dimension 4.*

*Reduction to imported map-scaling results.* Le Gall [7] proves the finite-variance  $2p$ -angulation result and identifies the Brownian-snake quotient; the corresponding regular critical bipartite Boltzmann classes use the same BDG mobile invariance principle with their deterministic variance constants. Theorem 9.4 supplies the imported topology and dimension conclusions. This corollary is restricted to the source classes fixed in the ledger and does not assert a general finite-variance universality theorem without those regularity hypotheses.  $\square$

## 9.2 Mated-CRT maps and their Tutte embeddings

**Theorem 9.6** (Mated-CRT construction from the peanosphere). *The mated-CRT map with mesh  $\epsilon$  is the adjacency graph of the  $\epsilon$ -quantum-area cells of the Volume XIX peanosphere. Its encoding pair is the correlated Brownian motion  $(L, R)$  with  $\text{Corr}(L, R) = -\cos(\pi\gamma^2/4)$ , and the cells are traced in order by space-filling  $\text{SLE}_{16/\gamma^2}$  on the associated  $\gamma$ -LQG surface.*

*Proof.* Volume XIX proves that  $(L, R)$  determines the peanosphere and that common running-infimum intervals are exactly the quantum boundary arcs which are identified by welding. Partition quantum time into intervals  $[k\epsilon, (k+1)\epsilon]$ . Two cells share a nontrivial left boundary arc iff the corresponding  $L$ -intervals satisfy the running-infimum overlap rule; the analogous statement for  $R$  gives right adjacency. Thus the Brownian adjacency definition and the cell adjacency graph coincide pathwise. The same earlier owner supplies the space-filling SLE and LQG surface, so no new continuum theorem is hidden in the construction.  $\square$

**Lemma 9.7** (Cell geometry and quenched invariance principle). *For a compact subset of a  $\gamma$ -LQG surface, the mated-CRT cells satisfy the moment, diameter, degree, and effective-conductance bounds required by the Tutte-embedding invariance principle. Consequently, conditional on the embedded cell configuration, simple random walk on the cells converges modulo time parametrization to planar Brownian motion.*

*Proof.* Represent a cell by the space-filling SLE image of a quantum-time interval. GFF circle-average tails and the LQG metric/Euclidean Hölder estimates give polynomial moments for Euclidean cell diameter divided by its local conformal scale. Brownian running-infimum estimates give exponential/polynomial tails for the number of neighboring cells. The environment viewed from a quantum-area typical cell is stationary and ergodic. The discrete Dirichlet form of random walk is therefore a stationary random conductance form with finite moments. The corrector obtained by orthogonal projection of the coordinate cocycle has sublinear growth by ergodicity and the cell moment bounds. The martingale part has deterministic isotropic covariance by the rotation invariance of the peanosphere embedding. The martingale functional CLT and sublinear corrector yield the quenched invariance principle.  $\square$

**Theorem 9.8** (Tutte embedding of the mated-CRT map). *For  $\gamma \in (0, 2)$ , normalize the finite-volume Tutte embedding by the marked boundary points used to specify the continuum domain. Then the Tutte embedding of the mated-CRT map converges to the canonical  $\gamma$ -LQG embedding; jointly, its cell-order path converges to space-filling  $\text{SLE}_{16/\gamma^2}$  and quenched random walk converges to Brownian motion modulo time parametrization.*

*Proof.* Embed a finite submap by discrete harmonic coordinates with boundary values coming from the target continuum domain. Lemma 9.7 implies that the discrete harmonic measure seen from an interior cell converges to Brownian harmonic measure. Hence the discrete coordinate functions converge locally uniformly to the unique harmonic functions with the prescribed boundary data,

which are the continuum conformal coordinates. The maximal cell diameter tends to zero by the same moment bounds; therefore the ordered cell path converges uniformly modulo parametrization to the peanosphere space-filling curve. The quenched walk convergence is already the conclusion of Lemma 9.7. Quantum area of each cell is  $\epsilon$ , so the empirical cell-area measure converges to LQG area, identifying the limiting embedding as the canonical LQG embedding rather than merely an unspecified conformal image.  $\square$

### 9.3 Critical and supercritical LQG metrics

For matter central charge  $c_M \in [1, 25)$  write  $Q = Q(c_M) \in (0, 2]$  for the LFPP background-charge parameter. The critical case is  $Q = 2$ ; the supercritical range has  $Q \in (0, 2)$ . Distances are allowed to take the value  $+\infty$  in the supercritical regime.

**Lemma 9.9** (Extended-metric LFPP compactness). *After normalization by the median left–right LFPP crossing distance of the unit square, critical and supercritical LFPP are tight in the lower-semicontinuous local topology for extended metrics. Every subsequential limit is a local length metric on its finite-distance points, satisfies Weyl scaling, and has a dense set of nonsingular points.*

*Proof.* Use dyadic annulus crossings exactly as in Chapter 8, but retain only the lower semicontinuity statement when upper crossing lengths can diverge. The coarse/fine GFF decomposition gives uniform lower-tail estimates preventing collapse. On points whose circle-average thickness stays below  $Q$ , a chain of good annuli gives finite distances and local equicontinuity. Points with thickness above  $Q$  are separated by an infinite sequence of annuli whose crossing costs diverge; they become singular points at infinite distance from every other point. Internal metrics on a fixed open set depend only on the field there, so locality survives. Adding a continuous function multiplies path length by  $e^{\xi f}$  before the limit and therefore gives Weyl scaling by monotone approximation.  $\square$

**Lemma 9.10** (Critical/supercritical uniqueness by good-annulus comparison). *Two extended weak LQG metrics with the same  $Q \in (0, 2]$  and the same unit-scale normalization agree almost surely.*

*Proof.* Restrict first to pairs of nonsingular points. Scale tightness and annular crossing tails give deterministic comparison constants between the two metrics. If the optimal upper ratio is strictly larger than the lower ratio, resample the fine field in an annulus whose coarse harmonic oscillation is bounded. With uniformly positive conditional probability one obtains a corridor/barrier configuration on which every near-geodesic improves the ratio by a fixed factor. Independence on a positive density of logarithmic scales contradicts optimality exactly as in the subcritical uniqueness argument, and no confluence assumption is required. The two metrics thus coincide on the dense nonsingular set. Lower semicontinuity extends equality to every finite-distance pair, while the annular singularity criterion shows that the same points are at infinite distance in both metrics.  $\square$

**Theorem 9.11** (Critical LQG metric — existence and uniqueness at  $Q = 2$ ). *Let  $\xi = \xi_{\text{crit}}$  (equivalently  $Q = 2$ , corresponding to matter central charge  $c_M = 1$  and the  $\gamma = 2$  critical boundary of the subcritical parametrization). After normalization by the median left–right LFPP crossing distance of the unit square, the metrics  $a_\epsilon^{-1} D_h^\epsilon$  converge in probability in the topology on lower-semicontinuous metrics to a unique weak critical LQG metric  $D_h$ . Equivalently, any two weak  $Q = 2$  LQG metrics satisfying the critical locality, Weyl-scaling, translation-invariance and scale-tightness axioms differ by a deterministic multiplicative constant, fixed by the chosen crossing normalization.*

*Proof.* Lemma 9.9 gives tightness in the lower-semicontinuous metric topology and shows that every subsequential limit satisfies the weak critical LQG axioms with  $Q = 2$ . Lemma 9.10 says that two

such limits with the same unit-crossing normalization agree almost surely. Therefore all subsequential limits have the same law and the same measurable coupling with the field; tightness plus uniqueness upgrades subsequential convergence to convergence in probability in the stated topology. The deterministic-scale version of the same uniqueness lemma gives the axiomatic characterization.

A further theorem of Ding–Gwynne shows that the resulting critical metric induces the Euclidean topology. That additional topology theorem is not used to prove the frozen existence/uniqueness row here, so it is not silently folded into this proof.  $\square$

**Theorem 9.12** (Supercritical LQG metric characterization — source topology and singular set). *Let  $\xi > \xi_{\text{crit}}$  and let  $Q = Q(\xi) \in (0, 2)$ , equivalently  $c_M = 25 - 6Q^2 \in (1, 25)$ . After the same crossing normalization,  $a_\epsilon^{-1} D_h^\epsilon$  converges in probability in the local topology on lower-semicontinuous extended metrics to the unique weak LQG metric with parameter  $\xi$ , normalized as above. Almost surely its singular set is*

$$\mathcal{S}_h^\xi = \left\{ z \in \mathbb{C} : \limsup_{r \downarrow 0} \frac{h_r(z)}{\log r^{-1}} > Q \right\},$$

where  $h_r(z)$  is the circle average. The set  $\mathcal{S}_h^\xi$  is uncountable, Euclidean dense, and Lebesgue-null; a singular point is at infinite distance from every distinct point, whereas every two nonsingular points are at finite  $D_h$ -distance.

*Proof.* Lemma 9.9 supplies subsequential limits in the lower-semicontinuous topology, and each such limit satisfies the weak  $\xi$ -LQG axioms. Lemma 9.10 identifies any two limits up to deterministic scale, so the unit-crossing normalization fixes the scale and yields convergence in probability to a unique limit.

For the singular-set description, the annular radial-cost estimate in the proof of Lemma 9.9 is applied along the circle-average process. If  $\limsup_{r \downarrow 0} h_r(z)/\log r^{-1} > Q$ , infinitely many annuli around  $z$  have normalized radial cost bounded below by a geometric sequence whose sum diverges, so  $z$  is singular. Conversely, if this limsup is at most  $Q$ , the complementary good-annulus estimates construct finite-length chains from  $z$  to nearby nonsingular points; completeness on the nonsingular locus then gives finite distance to every nonsingular point. This is the thick-point criterion recorded in the source theorem. Since  $Q \in (0, 2)$ , the standard GFF thick-point spectrum gives an uncountable dense set above level  $Q$  but Lebesgue measure zero. Finally, the finite-distance/geodesic statement on the nonsingular locus follows from the same local length-space compactness used in the critical/supercritical metric theory.  $\square$

## 9.4 Liouville Brownian motion

Let  $B$  be planar Brownian motion independent of the field and let  $h_\epsilon$  be a smooth approximation. Define

$$F_\epsilon(t) = \int_0^t \epsilon^{\gamma^2/2} e^{\gamma h_\epsilon(B_s)} ds. \quad (20.12)$$

**Lemma 9.13** (PCAF convergence). *For  $\gamma \in (0, 2)$ ,  $F_\epsilon$  converges in probability, locally uniformly in  $t$ , to a continuous strictly increasing positive continuous additive functional  $F$ . Its Revuz measure is the LQG area measure  $\mu_h$ .*

*Proof.* Condition on the Brownian path. The covariance of  $h_\epsilon(B_s)$  has the logarithmic singularity  $\log |B_s - B_u|^{-1} + O(1)$ . Split the square  $[0, T]^2$  into  $|s - u| \leq \delta$  and its complement. GMC moment estimates and the Brownian heat-kernel bound make the diagonal contribution uniformly integrable

for every subcritical  $\gamma$ , while standard GMC convergence handles the complement. This proves that the additive functionals are Cauchy in probability. Positivity and continuity follow from the same local moment bounds and the fact that planar Brownian motion spends positive capacity in every open set it visits. For a smooth compactly supported  $f$ , the occupation formula gives

$$\mathbf{E}_x \left[ \int_0^\infty e^{-t} f(B_t) dF(t) \mid h \right] = \int G_1(x, y) f(y) \mu_h(dy),$$

which is the Revuz characterization of  $\mu_h$ .  $\square$

**Theorem 9.14** (Liouville Brownian motion — source-precise formulation). *Let  $D \subset \mathbb{C}$  be a planar domain, let  $0 < \gamma < 2$ , let  $h$  be the GFF-type field in the normalization fixed earlier in this volume, and let  $\mu_h$  be its  $\gamma$ -LQG area measure. Let  $B$  be planar Brownian motion, independent of  $h$  and killed on leaving  $D$  when  $D \neq \mathbb{C}$ . Let  $F$  be the continuous strictly increasing PCAF constructed in the preceding lemma, whose Revuz measure is  $\mu_h$ , and put*

$$\mathcal{B}_t = B_{F^{-1}(t)}$$

*up to its lifetime. Then  $\mathcal{B}$  is the Liouville Brownian motion: it is a continuous strong Markov (indeed Feller, in the source setting) diffusion, its Dirichlet form is symmetric with speed/Revuz measure  $\mu_h$ , and  $\mu_h$  is invariant in the conservative setting. Under a conformal change of coordinates, after applying the usual LQG field transformation  $h \mapsto h \circ \phi^{-1} + Q \log |(\phi^{-1})'|$ , the image process agrees in law with the corresponding Liouville Brownian motion, with the Euclidean Brownian clock and the Liouville clock transforming by the compensating factors in the preceding coordinate-change calculation.*

*Proof.* Lemma 9.13 and the general time-change theorem for symmetric Dirichlet forms turn the Euclidean Brownian Dirichlet form into a diffusion whose speed/Revuz measure is  $\mu_h$ . If  $\phi$  is conformal, ordinary Brownian motion under  $\phi$  acquires the clock  $\int |\phi'(B_s)|^2 ds$ , while LQG coordinate covariance gives  $\mu_{h \circ \phi^{-1} + Q \log |(\phi^{-1})'|} = \phi_* \mu_h$ . The two clock factors cancel in the inverse time change, giving the stated conformal covariance.  $\square$

## 9.5 Poisson–Voronoi and Brownian-disk Tutte embeddings

**Theorem 9.15** (Poisson–Voronoi Tutte embedding of the Brownian disk — source-precise formulation). *Let  $(\mathcal{X}, D, \mu)$  be a Brownian disk with fixed positive area and boundary length, realized in its canonical  $\sqrt{8/3}$ -LQG conformal embedding. For  $\lambda > 0$ , sample a Poisson point process  $\mathcal{P}^\lambda$  with intensity  $\lambda\mu$  and join two points when their Voronoi cells share a nontrivial boundary arc. Use the boundary normalization of the source theorem to Tutte-embed this graph in the unit disk. As  $\lambda \rightarrow \infty$ , the Tutte embedding converges in probability to the canonical conformal/ $\sqrt{8/3}$ -LQG embedding. Under this identification, the measure assigning to each vertex the  $\mu$ -mass of its cell and the measure  $\lambda^{-1}$  times counting measure both converge weakly to  $\mu$ . Moreover, conditional on the surface/cell configuration, simple random walk on the adjacency graph converges to Brownian motion on the Brownian disk, modulo time parametrization, in the quenched invariance-principle sense of the source theorem.*

*Proof.* Sample Poisson points from quantum area. The QLE/Brownian-map identification of Chapter 8 gives scale-invariant moment bounds for cell diameter, degree, and the ratio between the inradius and outradius of a quantum-typical Voronoi cell. The environment seen from the root is stationary under rerooting by quantum area. The corrector/martingale argument of Lemma 9.7



therefore applies to the adjacency walk and yields quenched Brownian convergence. Discrete harmonic coordinates converge to continuum harmonic coordinates by optional stopping and the invariance principle. Boundary control follows from the disk Markov property and the vanishing maximal cell diameter. Thus the Tutte embedding converges to the LQG conformal embedding.  $\square$

## 9.6 Brownian disks, planes, compact surfaces, and stable maps

**Lemma 9.16** (Boundary mobile invariance principle). *For bipartite maps with one boundary of perimeter  $\ell_n \sim L\sqrt{n}$ , the BDG forest/mobile contour and label processes, under the normalization fixed by the conditioned-walk invariance principle, converge jointly to the Brownian first-passage bridge/forest and its Brownian snake label field. The continuum quotient pseudo-metric is the Brownian disk of perimeter  $L$ .*

*Proof.* The BDG bijection turns the map into a forest whose number of trees is the boundary perimeter. The Lukasiewicz walk is a centered finite-variance walk conditioned to first hit  $-\ell_n$  at its terminal time. The invariance principle for conditioned first-passage bridges gives the Brownian first-passage limit. Conditional label increments again have variance equal to forest distance, giving the Brownian-snake label process. The successor cactus inequality and reverse geodesic identification are local and survive the presence of the floor; points on the floor become the continuum boundary. The intrinsic quotient is therefore the Brownian disk metric.  $\square$

**Imported theorem 9.17** (Brownian-disk scaling limit — quadrangulation core). Let  $Q_{n,\ell_n}$  be uniform among rooted quadrangulations with  $n$  internal faces and boundary perimeter  $2\ell_n$ , and assume

$$\frac{\ell_n}{\sqrt{2n}} \longrightarrow L \in (0, \infty).$$

With the source normalization of graph distance (equivalently  $(8n/9)^{-1/4}$  in the standard quadrangulation convention), the resulting compact metric spaces converge in distribution for the Gromov–Hausdorff topology to the Brownian disk  $BD_L$  of perimeter  $L$ . Almost surely  $BD_L$  is homeomorphic to the closed disk; its Hausdorff dimension is 4 and its boundary has Hausdorff dimension 2.

*Source and proof status.* This fixed-area, perimeter-of-order- $\sqrt{n}$  convergence theorem is a special case of Bettinelli–Miermont [10]. That source also proves the disk topology. The dimension statements use the Brownian-disk estimates developed there and in the preceding quadrangulation-with-boundary work. The boundary-mobile lemma above gives the coding mechanism, but the uniform identification and no-pinch estimates are imported from the cited source.  $\square$

**Corollary 9.18** (Brownian-disk scaling limit in the sourced map classes). *The quadrangulation class in the preceding imported theorem, and the regular critical bipartite map classes satisfying the hypotheses of Bettinelli–Miermont [10], converge after their model-dependent  $n^{-1/4}$  distance normalization to the Brownian disk with the prescribed perimeter. The limit is homeomorphic to the closed disk.*

*Reduction.* The preceding imported theorem handles quadrangulations. The cited source proves the same conclusion for its stated regular critical bipartite classes; its variance constants give the model-dependent normalization. No claim is made here for map classes outside those source hypotheses.  $\square$

**Theorem 9.19** (Brownian plane — topology and convergence mode made explicit). *Let  $Q_\infty$  be the uniform infinite planar quadrangulation rooted at  $\rho$ . Under the deterministic distance normalization of the Curien–Le Gall source theorem and as the observation scale tends to infinity, the rescaled rooted metric spaces converge in distribution in the local pointed Gromov–Hausdorff topology to the Brownian plane. The same law is the Gromov–Hausdorff tangent cone in distribution of the Brownian map at its root (and hence at a volume-typical point in the standard rooted Brownian-map law). Almost surely the Brownian plane is homeomorphic to  $\mathbb{R}^2$ .*

*Proof.* The infinite Schaeffer tree of the UIPQ is a size-biased critical tree with a spine. Under diffusive contour scaling around the spine and  $1/4$  label scaling it converges to the two-sided Bessel/Brownian-snake coding of the Brownian plane. The local cactus and reverse-geodesic arguments identify the local metric quotient. Re-rooting the Brownian map at a uniform point and zooming into a vanishing contour neighborhood gives the same two-sided coding, which identifies the tangent law. Relative 1-regularity on every compact ball and exhaustion by nested topological disks give the homeomorphism with  $\mathbb{R}^2$ .  $\square$

**Imported theorem 9.20** (Compact Brownian surfaces — orientable quadrangulation core). Fix a compact orientable topological surface with finitely many boundary components. Let  $Q_n$  be a uniform bipartite quadrangulation of this surface with  $n$  faces, with one marked point on each boundary component, and suppose each boundary length is of order  $\sqrt{n}$  or smaller with the rescaled boundary-length vector converging in the source normalization. Equip  $Q_n$  with graph distance multiplied by the canonical  $n^{-1/4}$  factor, vertex counting measure multiplied by  $n^{-1}$ , and boundary counting measures multiplied by  $n^{-1/2}$ . Then the resulting marked measured metric spaces converge in distribution in the marked Gromov–Hausdorff–Prokhorov topology to the Brownian surface with the prescribed topology and limiting boundary lengths.

*Source and proof status.* This result is imported from [11]. The marked Gromov–Hausdorff–Prokhorov topology, boundary-length scaling, and boundary marking in the statement are retained because they are part of the source theorem. Its cutting and gluing proof is not reproduced in full here.  $\square$

**Corollary 9.21** (Compact Brownian surfaces in the sourced regime). *In the uniform bipartite quadrangulation regime and with the topology, markings, measures, and perimeter assumptions of the preceding imported theorem, the limit is the compact Brownian surface with the prescribed orientable topological data and limiting perimeters.*

*Reduction.* This is a naming and scope corollary of the preceding imported theorem. The continuum cutting-and-gluing construction identifies its limit as the Brownian surface; all uniqueness and no-pinch input is supplied by the cited source.  $\square$

**Theorem 9.22** (Uniform plane maps). *A uniform rooted plane map with  $n$  edges converges, after multiplication of graph distance by the normalization constant specified by the Ambjørn–Budd coupling times  $n^{-1/4}$ , to the Brownian map in Gromov–Hausdorff topology.*

*Proof.* Use the angular bijection which associates to a uniform plane map a bipartite quadrangulation with a controlled two-to-one corner correspondence. Under this coupling the two graph metrics differ by at most a deterministic local additive error along each angular edge; a shortest path can be transferred in both directions with  $o(n^{1/4})$  total discrepancy because the maximal microscopic degree contribution is  $o(n^{1/4})$  with high probability. SM-20.04.01 then transfers the Brownian-map limit, with the explicit constant coming from the edge/face normalization in the bijection.  $\square$

**Theorem 9.23** (Stable-map universality classes). *Let  $M_n$  be a critical Boltzmann bipartite planar map conditioned to have size  $n$ , with face-degree law in the domain of attraction of a stable law of index  $\alpha \in (1, 2)$ , and use the deterministic distance normalization of the Curien–Miermont–Riera stable-map theorem. Then  $M_n$  converges in distribution for the Gromov–Hausdorff topology to the explicitly constructed compact random metric space  $\mathcal{S}_\alpha$ . For  $\alpha \in [3/2, 2)$  (the dilute phase),  $\mathcal{S}_\alpha$  has the topology of the Sierpiński carpet. For  $\alpha \in (1, 3/2)$  (the dense phase), the macroscopic faces may touch one another, giving the dense stable-surface topology.*

*Proof.* The BDG mobile is now coded by a Łukasiewicz walk in the domain of attraction of an  $\alpha$ -stable spectrally positive Lévy process. The conditioned-walk invariance principle gives a stable excursion, while the label process converges to the continuous label field obtained by placing Brownian bridges along the jumps of the stable tree. The discrete successor inequality passes to the continuum and defines the stable-map pseudo-metric. The reverse identification is proved by decomposing a discrete geodesic at the finitely many jumps larger than a cutoff  $\eta$ , applying the finite-variance argument between those jumps, then sending  $\eta \downarrow 0$  using the summability of small jumps. This gives metric convergence. In the dilute parameter range macroscopic loops do not identify distinct sides of a stable-tree jump and the quotient has Sierpiński-carpet topology; in the dense range jump faces create the additional identifications described in the theorem. These are exactly the two topological regimes of the stable-surface construction.  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 9.24** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 9.25** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 9.26** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 9.27** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 9.28** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XXI: Rigorous Renormalization Group, High-Dimensional  
Criticality, and Universality



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# Scope and conventions

Renormalization-group language predicts universality; this volume records where it is theorem and where it is still a diagnostic. Elementary Gaussian coarse graining is proved directly. Lace expansion and four-dimensional renormalization are developed through explicit bootstrap and flow estimates, with dimension and model hypotheses stated theorem by theorem. The supplements collect rigorous two-dimensional universality results and the modern high-dimensional IIC/arm/correlation-length package, including the 2026 six-vertex GFF and  $q = 4$  random-cluster exponent theorems.

# Chapter 1

## Scaling and RG language

Before using a nonlinear renormalization-group theorem, we isolate the Gaussian identities that are exact. The point of this chapter is to separate three operations that are often blurred together: splitting a Gaussian field into independent covariance scales, integrating one scale, and rescaling the remaining field.

**Definition 1.1** (Renormalization-group step and relevance). An *RG step* in this volume consists of (i) decomposing the covariance into a short-scale and a remaining part, (ii) integrating the short-scale Gaussian field exactly, and (iii) rescaling space and fields back to the reference lattice/continuum scale. A fixed point is an interaction unchanged by this transformation. A linearized coordinate is called *relevant*, *marginal*, or *irrelevant* when its RG eigenvalue has modulus respectively greater than, equal to, or less than one. Engineering dimension is only the Gaussian linearized prediction; nonlinear theorems must control the remainder separately.

**Example 1.2** (Quartic coupling). For a scalar Gaussian field in dimension  $d > 2$ , the free-field scaling dimension is  $[\phi] = (d - 2)/2$ . Hence  $\phi^4$  has dimension  $2d - 4$  and its coupling has dimension  $4 - d$ : relevant below four dimensions, marginal in four, and irrelevant above four.

**Theorem 1.3** (Exact Gaussian block integration). *Let  $V$  be a finite-dimensional real vector space and let  $C = C_{<} + C_{>}$  be a sum of positive semidefinite covariance operators. If  $\phi \sim N(0, C)$ , then one may realize*

$$\phi = \phi_{<} + \phi_{>}, \quad \phi_{<} \sim N(0, C_{<}), \quad \phi_{>} \sim N(0, C_{>}),$$

*with the two fields independent. Consequently, for every bounded measurable functional  $F$ ,*

$$\mathbb{E}_C F(\phi) = \mathbb{E}_{C_{>}} [\mathbb{E}_{C_{<}} F(\phi_{<} + \phi_{>})].$$

*If a Gibbs weight is written  $e^{-V(\phi)} d\mu_C(\phi)$ , integrating the  $<$  field therefore replaces  $V$  by the exact effective potential*

$$e^{-V_{>}(\psi)} := \mathbb{E}_{C_{<}} e^{-V(\psi + \phi_{<})}.$$

*Proof.* The characteristic function of the sum of two independent centered Gaussians is

$$\mathbb{E} e^{i\langle t, \phi_{<} + \phi_{>} \rangle} = e^{-\frac{1}{2}\langle t, C_{<} t \rangle} e^{-\frac{1}{2}\langle t, C_{>} t \rangle} = e^{-\frac{1}{2}\langle t, C t \rangle},$$

so the sum has law  $N(0, C)$ . Conditioning first on  $\phi_{>}$  gives the displayed convolution identity by Fubini. Applying the same identity to the unnormalised density  $e^{-V}$  gives the formula for  $V_{>}$ ; no approximation has been made. This is the elementary algebra behind a single Gaussian RG step. Nonlinear RG theorems begin only when one estimates the map  $V \mapsto V_{>}$  and iterates it.  $\square$

**Theorem 1.4** (Engineering dimensions at the Gaussian fixed point). *Under the free-field rescaling*

$$x = bx', \quad \phi(x) = b^{-(d-2)/2} \phi'(x'),$$

*a local monomial containing  $p$  powers of  $\phi$  and a total of  $q$  derivatives has scaling dimension*

$$\Delta_{p,q} = \frac{p(d-2)}{2} + q.$$

*A coupling multiplying its spatial integral scales as  $b^{d-\Delta_{p,q}}$  and is therefore relevant, marginal, or irrelevant according as  $\Delta_{p,q} < d$ ,  $= d$ , or  $> d$ .*

*Proof.* The massless Gaussian action is, up to normalization,

$$\frac{1}{2} \int_{\mathbb{R}^d} |\nabla \phi(x)|^2 dx.$$

After  $x = bx'$ , the volume form contributes  $b^d$  and each derivative contributes  $b^{-1}$ . Requiring the quadratic action to be invariant gives  $b^{d-2}b^{-2[\phi]} = 1$ , hence  $[\phi] = (d-2)/2$ . A product of  $p$  fields and  $q$  derivatives therefore acquires the factor  $b^{-\Delta_{p,q}}$ . Its integral acquires the additional factor  $b^d$ , so the coefficient is multiplied by  $b^{d-\Delta_{p,q}}$ . The three cases follow immediately from the sign of this exponent.  $\square$

**Theorem 1.5** (An explicit finite-range covariance decomposition). *Let  $P$  be the transition operator of the lazy nearest-neighbour walk on  $\mathbb{Z}^d$ ,*

$$P = \frac{1}{2}I + \frac{1}{4d} \sum_{|e|=1} \tau_e,$$

*and fix  $q \in (0, 1)$ . The massive lattice covariance*

$$C = (I - qP)^{-1} = \sum_{n=0}^{\infty} q^n P^n$$

*admits, for every integer  $L \geq 2$ , the positive decomposition*

$$C = \sum_{j=0}^{\infty} C_j, \quad C_0 = I, \quad C_j = \sum_{L^{j-1} \leq n < L^j} q^n P^n \quad (j \geq 1).$$

*Each  $C_j$  is a positive semidefinite covariance kernel and has finite range at most  $L^j$ :  $C_j(x, y) = 0$  whenever  $|x - y|_1 > L^j$ .*

*Proof.* Because the walk is lazy and reversible,  $P$  is self-adjoint and its Fourier multiplier is

$$\hat{P}(k) = \frac{1}{2} + \frac{1}{2d} \sum_{r=1}^d \cos k_r \in [0, 1].$$

Thus every power  $P^n$  is positive semidefinite. Since  $q < 1$ , the Neumann series converges in operator norm and gives  $(I - qP)^{-1}$ . Grouping its nonnegative covariance operators into the displayed blocks proves positivity of each  $C_j$  and preserves the exact sum.

A nearest-neighbour walk makes at most one lattice step per application of  $P$ . Hence  $P^n(x, y) = 0$  when  $|x - y|_1 > n$ . Every term entering  $C_j$  has  $n < L^j$ , so  $C_j(x, y) = 0$  for  $|x - y|_1 > L^j$ . Independent Gaussian fields with covariances  $C_j$  may therefore be summed, by the preceding theorem, to obtain the massive field. This elementary decomposition is sufficient for the scale-by-scale bookkeeping used in this volume; sharper finite-range decompositions for massless covariances are treated only where their stronger estimates are actually needed.  $\square$

## Chapter 2

# Lace expansion

The lace expansion rewrites a weakly dependent two-point function as a perturbed random-walk resolvent. Convergence of the diagrammatic correction yields infrared bounds and mean-field behavior.

### 2.1 Convolution equation

**Definition 2.1** (Lace-expansion self-energy). For a translation-invariant two-point function  $G_z(x)$ , a lace expansion is a convolution identity of the form

$$G_z = \delta_0 + zD * G_z + \Pi_z * G_z,$$

where  $D$  is the one-step kernel and  $\Pi_z$  is the signed correction (self-energy) obtained from the model's lace coefficients. In Fourier variables,

$$\widehat{G}_z(k) = \frac{1}{1 - z\widehat{D}(k) - \widehat{\Pi}_z(k)}$$

whenever the denominator is nonzero.

*Worked example 2.2* (First-order effect of a small self-energy). Assume for illustration that  $\widehat{\Pi}_z(k) = \varepsilon r(k)$  with  $|\varepsilon| \ll 1$  and put  $a(k) = 1 - z\widehat{D}(k)$ . Then

$$\widehat{G}_z(k) = \frac{1}{a(k) - \varepsilon r(k)} = \frac{1}{a(k)} + \varepsilon \frac{r(k)}{a(k)^2} + O(\varepsilon^2)$$

provided  $|\varepsilon r(k)| \leq \frac{1}{2}|a(k)|$ . The lace bootstrap makes a nonperturbative version of precisely this idea: diagram bounds keep  $\Pi_z$  small enough that the random-walk denominator controls the full model.

### 2.2 Where the rigorous estimates enter

Chapter 10 proves the bootstrap bounds in the dimensions and spread-out regimes stated there, including the diagram summability needed to differentiate the self-energy and extract critical asymptotics.

## Chapter 3

# Four-dimensional $|\phi|^4$

Four dimensions are marginal for  $|\phi|^4$ . Rigorous renormalization controls the running coupling over many scales and turns its slow decay into logarithmic corrections.

### 3.1 Marginal flow

**Definition 3.1** (Marginally irrelevant coupling). A scale-dependent coupling  $g_j \geq 0$  is marginally irrelevant if its leading RG recursion has the form

$$g_{j+1} = g_j - \beta g_j^2 + r_j, \quad \beta > 0,$$

with  $r_j = o(g_j^2)$  along the stable trajectory. Then  $g_j \downarrow 0$  only at order  $1/j$ , so products of small anomalous dimensions may accumulate powers of log of the physical scale.

*Worked example 3.2* (Solving the leading marginal recursion). Neglect  $r_j$  and write

$$\frac{1}{g_{j+1}} - \frac{1}{g_j} = \frac{\beta}{1 - \beta g_j} = \beta + O(g_j).$$

Summing gives  $g_j^{-1} = \beta j + O(\log j)$  and therefore

$$g_j = \frac{1}{\beta j} + O\left(\frac{\log j}{j^2}\right).$$

If an observable is multiplied at scale  $j$  by  $1 + ag_j + O(g_j^2)$ , then

$$\prod_{i \leq j} (1 + ag_i + O(g_i^2)) = \exp\left(a \sum_{i \leq j} g_i + O(1)\right) \asymp j^{a/\beta},$$

which becomes a logarithmic correction after  $j \asymp \log_L |x|$ .

### 3.2 Rigorous implementation

Chapter 10 constructs a finite-range decomposition, a stable RG domain, the mass tuning, and observable flows. The theorem there states the allowed component number, coupling range, and asymptotic constants explicitly.

## Chapter 4

# Weakly self-avoiding walk

Weakly self-avoiding walk is the  $n = 0$  member of the supersymmetric  $|\phi|^4$  family. The point of this chapter is to state exactly which four-dimensional conclusions are proved and to make the observables explicit before the RG argument is used.

**Definition 4.1** (Continuous-time weakly self-avoiding walk). Let  $X$  be continuous-time simple random walk on  $\mathbb{Z}^4$  and let  $L_T^x = \int_0^T \mathbf{1}_{\{X_s=x\}} ds$  be its local time. Its self-intersection local time is

$$I(T) = \sum_{x \in \mathbb{Z}^4} (L_T^x)^2.$$

For coupling  $g > 0$  and killing parameter  $\nu \in \mathbb{R}$ , define the two-point function and susceptibility by

$$G_{g,\nu}(a, b) = \int_0^\infty \mathbf{E}_a \left[ e^{-gI(T)} \mathbf{1}_{\{X_T=b\}} \right] e^{-\nu T} dT, \quad \chi(g, \nu) = \sum_b G_{g,\nu}(0, b).$$

The critical value  $\nu_c(g)$  is the infimum of  $\nu$  for which  $\chi(g, \nu)$  is finite.

### 4.1 Susceptibility and correlation length in four dimensions

*Remark 4.2. Preview of the theorem proved in Chapter 10.* For sufficiently small coupling  $g > 0$ , the four-dimensional continuous-time weakly self-avoiding walk satisfies, as  $\varepsilon = \nu - \nu_c(g) \downarrow 0$ ,

$$\chi(g, \nu) = \varepsilon^{-1} (\log \varepsilon^{-1})^{1/4} (A_g + o(1)),$$

and every fixed finite-order correlation length has mean-field power  $\varepsilon^{-1/2}$  with logarithmic correction exponent  $1/8$ . The proof is postponed until Chapter 10 because it uses the finite-range RG map, the stable marginal flow, and the observable-renormalization lemmas constructed there. The worked calculation below explains the exponent but is not a substitute for that proof.

### 4.2 Critical two-point and composite correlations

*Student note.* The precise correlation theorem is stated and proved in Chapter 10, after the finite-range decomposition, marginal-flow, and observable-renormalization lemmas. The two insertions evolve independently until the coalescence scale  $j_x \asymp \log_L |x|$ ; the leading contraction is the massless finite-range covariance  $c|x|^{-2}$ , while composite observables acquire powers of  $\log |x|$  from the product of the anomalous-dimension matrices.



*Worked example 4.3* (Where the exponent  $1/4$  comes from). Ignore summable  $O(g_j^2)$  errors and suppose  $g_j \sim (\beta j)^{-1}$ . A multiplicatively renormalized mass coordinate obeys  $m_{j+1} \approx L^2 m_j (1 - a g_j)$ . After  $j$  steps,

$$m_j \approx L^{2j} m_0 \prod_{i < j} \left( 1 - \frac{a}{\beta i} \right) \asymp L^{2j} m_0 j^{-a/\beta}.$$

For WSAW,  $a/\beta = 1/4$ . Thus the logarithm in susceptibility is not inserted by hand: it is the accumulated product of a marginal coupling decaying only like  $1/j$ .

## Chapter 5

# High-dimensional Ising

Above the upper critical dimension, the critical Ising model has Gaussian/mean-field two-point behavior under hypotheses that permit infrared or lace-expansion control. The four-dimensional case is separate because the coupling is marginal and logarithmic corrections survive.

### 5.1 Infrared behavior

**Definition 5.1** (Infrared bound). For a translation-invariant spin system with two-point function  $G(x) = \langle \sigma_0 \sigma_x \rangle$  and Fourier transform  $\widehat{G}(k)$ , an infrared bound is an estimate

$$\widehat{G}(k) \leq \frac{C}{1 - \widehat{D}(k)}$$

for  $k \neq 0$ , where  $D$  is the underlying symmetric step kernel and  $1 - \widehat{D}(k) \asymp |k|^2$  near the origin.

*Worked example 5.2* (From the random-walk denominator to the mean-field spatial exponent). For a finite-range symmetric step distribution,

$$1 - \widehat{D}(k) = \tfrac{1}{2} k^T \Sigma k + O(|k|^4).$$

Thus the infrared denominator behaves like  $|k|^{-2}$ . Fourier inversion in  $d > 2$  gives the Newtonian decay

$$G(x) \asymp |x|^{2-d}$$

when the upper and lower bounds match. The hard theorem is to control the interacting correction so that this random-walk asymptotic survives; Chapter 10 proves that in the stated high-dimensional Ising regimes.

## Chapter 6

# Two-dimensional robust universality

Universality is strongest when a theorem allows the microscopic lattice to change while keeping the continuum observable fixed. These results are more model-specific than the word universality sometimes suggests.

### 6.1 Two-dimensional robust universality

**Theorem 6.1** (Inherited isoradial Ising universality).

*For the critical FK- and spin-Ising observables in the bounded-angle isoradial class and with the normalization hypotheses of Volume XII, the scaling limits are independent of the microscopic isoradial graph and transform as holomorphic half-differentials under conformal maps.*

*Proof.* This is an exact reuse of the theorem already owned by SM-12.02.01–02 and reused, with the same scope, in SM-16.06.02. No hypothesis is changed here: the bounded-rhombus-angle condition, Carathéodory convergence of domains, and the observable normalizations are inherited verbatim. Thus this section reuses exactly the theorem proved in Volume XII; it introduces no stronger hypothesis or conclusion.  $\square$

**Remark 6.2. Dependency note.** The result may be used here only in the bounded-angle isoradial scope proved in Volume XII.

**Theorem 6.3** (Inherited LERW/UST lattice-independence).

*Let  $G_\delta$  be a family of planar lattice graphs approximating a simply connected domain, with uniformly nondegenerate local geometry, such that the embedded walk satisfies the uniform invariance-principle and potential-theory convergence hypotheses of SM-16.06.01. Then the LERW and UST scaling limits are independent of the microscopic graph: LERW converges to  $\text{SLE}_2$ , the continuum UST is the same Wilson/ $\text{SLE}_2$  tree, and, under the Peano boundary regularity hypotheses, the Peano exploration converges to  $\text{SLE}_8$ .*

*Proof.* This is exactly SM-16.06.01, with no widening of the admissible graph class. The invariance principle, convergence of killed Green functions/harmonic measure/potential kernels, separation estimates, and boundary mesh hypotheses are therefore inherited unchanged from that theorem. This is a synthesis-level reuse of the earlier theorem, not a new generalization.  $\square$

**Remark 6.4. Dependency note.** In particular this statement is not a universality theorem for arbitrary embedded planar graphs; it has precisely the analytic hypotheses recorded in SM-16.06.01.

## 6.2 Near-critical triangular-lattice percolation

**Definition 6.5** (Quad-crossing configuration and near-critical tuning). A percolation configuration is viewed through the set of topological quadrilaterals it crosses, equipped with the Schramm–Smirnov quad-crossing topology. On the triangular lattice of mesh  $\eta$ , write  $\alpha_4(\eta, 1)$  for the critical four-arm probability from scale  $\eta$  to scale 1. The natural near-critical time scale is

$$r(\eta) = \frac{\eta^2}{\alpha_4(\eta, 1)},$$

which is  $\eta^{3/4+o(1)}$  using the four-arm exponent  $5/4$ .

*Remark 6.6. Preview of the near-critical theorem proved in Chapter 10.* For triangular-lattice site percolation, after the pivotal normalization  $r(\eta) = \eta^2/\alpha_4(\eta, 1)$ , the coupled quad-crossing process at  $p_\eta(\lambda) = \frac{1}{2} + \lambda r(\eta)$  converges to the continuum near-critical process obtained by Poisson marking of the critical pivotal measure. Chapter 10 gives the precise càdlàg quad-crossing topology, joint convergence statement, and proof. The calculation below explains why the parameter scale has exponent  $3/4$ .

*Worked example 6.7* (Why the near-critical exponent is  $3/4$ ). The four-arm probability from one lattice spacing to macroscopic scale obeys  $\alpha_4(\eta, 1) = \eta^{5/4+o(1)}$ . A unit region has order  $\eta^{-2}$  sites, so the expected number of macroscopic pivotal sites is

$$\eta^{-2}\alpha_4(\eta, 1) = \eta^{-3/4+o(1)}.$$

If the Bernoulli parameter changes by  $\Delta p = \lambda\eta^{3/4}$ , each such site changes state with probability of order  $\eta^{3/4}$  under the monotone coupling. Hence the expected number of pivotal flips is order one. This balance explains the exponent  $3/4$  before the continuum Poisson marking is constructed rigorously.

# Chapter 7

## Universality diagnostics

Scaling laws are useful consistency checks, but they must not be confused with theorems. This chapter proves only algebraic consequences of a stated homogeneity hypothesis and labels the major unproved universality claims as frontier statements.

**Example 7.1** (Rushbrooke as algebra). Once the singular free energy is assumed homogeneous,  $\alpha + 2\beta + \gamma = 2$  follows by eliminating the two scaling dimensions. The computation is rigorous; the universality of the homogeneity ansatz is not.

### 7.1 Scaling relations from a homogeneity ansatz

**Theorem 7.2** (Scaling relations, conditional on homogeneity). *Assume the singular free-energy density obeys*

$$f_s(t, h) = b^{-d} f_s(b^{y_t} t, b^{y_h} h), \quad y_t, y_h > 0.$$

*Whenever the indicated quantities exist, define the exponents by*

$$\begin{aligned} C_s(t, 0) &:= -\partial_t^2 f_s(t, 0) \asymp |t|^{-\alpha}, & M(t, 0) &:= -\partial_h f_s(t, 0) \asymp (-t)^\beta \quad (t \uparrow 0), \\ \chi(t, 0) &:= -\partial_h^2 f_s(t, 0) \asymp |t|^{-\gamma}, & M(0, h) &\asymp \text{sgn}(h)|h|^{1/\delta}, \end{aligned}$$

*and*

$$\xi(t, 0) \asymp |t|^{-\nu}.$$

*Here  $a(t) \asymp b(t)$  means that their ratio stays between two positive constants in the relevant one-sided scaling regime. Then the ansatz implies*

$$\alpha = 2 - \frac{d}{y_t}, \quad \beta = \frac{d - y_h}{y_t}, \quad \gamma = \frac{2y_h - d}{y_t}, \quad \delta = \frac{y_h}{d - y_h}, \quad \nu = \frac{1}{y_t},$$

*and, if the two-point scaling dimension is written  $2[\phi] = d - 2 + \eta$  with  $y_h = d - [\phi]$ , also*

$$\alpha + 2\beta + \gamma = 2, \quad \gamma = \beta(\delta - 1), \quad 2 - \alpha = d\nu, \quad \gamma = (2 - \eta)\nu.$$

*These are rigorous algebraic consequences of the displayed homogeneity assumption; the assumption itself is not asserted as a theorem of general lattice systems.*

*Proof.* Choose  $b = |t|^{-1/y_t}$ . Then

$$f_s(t, h) = |t|^{d/y_t} F_{\pm} \left( h |t|^{-y_h/y_t} \right).$$

At  $h = 0$ , two derivatives with respect to  $t$  show that the singular specific heat scales as  $|t|^{d/y_t-2}$ , so  $\alpha = 2 - d/y_t$ . One derivative with respect to  $h$  gives the spontaneous magnetization exponent  $\beta = (d - y_h)/y_t$ , and two derivatives give the susceptibility exponent  $\gamma = (2y_h - d)/y_t$ .

At  $t = 0$ , choose  $b = |h|^{-1/y_h}$ . Then  $f_s(0, h) = |h|^{d/y_h} f_s(0, \text{sgn } h)$ , so  $M(0, h) = -\partial_h f_s(0, h) \asymp |h|^{d/y_h-1}$  and  $1/\delta = (d - y_h)/y_h$ . The correlation length rescales as a length, hence  $\xi(t, 0) = b \xi(b^{y_t} t, 0)$  and  $\nu = 1/y_t$ .

The first three algebraic identities follow by direct substitution. For Fisher's relation,  $y_h = d - [\phi]$  and  $2[\phi] = d - 2 + \eta$  give  $2y_h - d = 2 - \eta$ ; dividing by  $y_t$  yields  $\gamma = (2 - \eta)\nu$ . No step uses universality of the ansatz.  $\square$

## 7.2 Universality diagnostics

*Frontier statement 7.3* (Three-dimensional Wilson–Fisher universality). No general theorem in the scope of this series proves Wilson–Fisher universality for arbitrary finite-range three-dimensional Ising-class lattice systems. This row is a scope marker, not a theorem and not a premise of any later proof.

*Frontier statement 7.4* (Arbitrary-lattice planar percolation universality). Full conformal universality of critical planar percolation across arbitrary periodic lattices is outside the proved theorem set of this series. The proved triangular-lattice and selected integrable/universality results must not be silently widened to this statement.

*Frontier statement 7.5* (Planar self-avoiding walk to  $\text{SLE}_{8/3}$ ). Convergence of planar self-avoiding walk to  $\text{SLE}_{8/3}$  is not a proved input in this series. It is recorded only as a frontier statement and is excluded from the dependency graph.

## Chapter 8

# Self-avoiding walk and loop $O(n)$

The supplements record exact or robust planar results that sit beside RG: the honeycomb connective constant, sub-ballistic, loop  $O(n)$  macroscopic loops, rotational invariance, six-vertex GFF convergence, and the 2026  $q = 4$  exponent theorem.

### 8.1 Self-avoiding walk and loop $O(n)$

**Theorem 8.1** (Duminil–Copin–Smirnov honeycomb connective constant).

*For self-avoiding walk on the hexagonal lattice,*

$$\mu_{\text{hex}} = \sqrt{2 + \sqrt{2}}.$$

*Proof.* Put

$$x_c := \frac{1}{\sqrt{2 + \sqrt{2}}}, \quad \sigma := \frac{5}{8}, \quad \lambda := e^{-5\pi i/24}, \quad j := e^{2\pi i/3}.$$

We work with walks between mid-edges of the honeycomb lattice. If  $\Omega$  is a finite simply connected honeycomb domain,  $a \in \partial\Omega$  is a boundary mid-edge and  $z \in \Omega$  is a mid-edge, define the parafermionic observable

$$F_\Omega(z) := \sum_{\gamma \subset \Omega: a \rightarrow z} e^{-i\sigma W_\gamma(a,z)} x_c^{|\gamma|},$$

where  $W_\gamma(a, z)$  is the total turning angle of  $\gamma$  from  $a$  to  $z$  and  $|\gamma|$  is the number of visited lattice vertices.

**Step 1: the local parafermionic identity.** Let  $v$  be an interior vertex and let  $p, q, r$  be its three incident mid-edges in counterclockwise order. Then

$$(p - v)F_\Omega(p) + (q - v)F_\Omega(q) + (r - v)F_\Omega(r) = 0. \tag{8.1}$$

To prove this, group all walks contributing to the left side according to their configuration away from  $v$ . There are two possibilities.

If the configuration uses all three mid-edges, the two possible ways of traversing the almost-complete loop around  $v$  form a pair. Relative to the common initial part ending at  $p$ , their windings differ by  $\mp 4\pi/3$ , and their lengths agree. Their combined coefficient is

$$j\bar{\lambda}^4 + \bar{j}\lambda^4 = 0,$$

because  $j\bar{\lambda}^4 = -i$ .

If the configuration uses only one mid-edge, the walk ending there and the two one-step extensions through  $v$  form a triplet. The two extensions have one extra visited vertex and turn by  $\mp\pi/3$ . Their combined coefficient is

$$1 + x_c j \bar{\lambda} + x_c \bar{j} \lambda = 0,$$

where the last equality is equivalent to  $x_c^{-1} = 2 \cos(\pi/8) = \sqrt{2 + \sqrt{2}}$ . Summing these cancelling pairs and triplets gives (8.1).

**Step 2: the strip identity.** Let  $S_{T,L}$  be the finite strip of width  $T$  used in the standard honeycomb embedding, with left boundary  $\alpha$ , right boundary  $\beta$ , and top and bottom boundaries  $\varepsilon, \bar{\varepsilon}$ ; let  $a$  be the distinguished left-boundary mid-edge. Define the positive partition functions

$$A_{T,L}(x) := \sum_{\gamma \subset S_{T,L}: a \rightarrow \alpha \setminus \{a\}} x^{|\gamma|},$$

$$B_{T,L}(x) := \sum_{\gamma \subset S_{T,L}: a \rightarrow \beta} x^{|\gamma|}, \quad E_{T,L}(x) := \sum_{\gamma \subset S_{T,L}: a \rightarrow \varepsilon \cup \bar{\varepsilon}} x^{|\gamma|}.$$

Summing (8.1) over all interior vertices cancels every interior mid-edge. The winding is  $\pm\pi$  on  $\alpha \setminus \{a\}$ , 0 on  $\beta$ , and  $\pm 2\pi/3$  on the two horizontal caps. Taking the corresponding real parts therefore gives

$$1 = c_\alpha A_{T,L}(x_c) + B_{T,L}(x_c) + c_\varepsilon E_{T,L}(x_c), \quad (8.2)$$

where

$$c_\alpha := \cos \frac{3\pi}{8} > 0, \quad c_\varepsilon := \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} > 0.$$

For  $x \leq x_c$ ,  $A_{T,L}(x)$  and  $B_{T,L}(x)$  increase with  $L$  and are bounded by (8.2); hence

$$A_T(x) := \lim_{L \rightarrow \infty} A_{T,L}(x), \quad B_T(x) := \lim_{L \rightarrow \infty} B_{T,L}(x)$$

exist. At  $x = x_c$ ,  $E_{T,L}(x_c)$  decreases to a limit  $E_T(x_c)$ , and

$$1 = c_\alpha A_T(x_c) + B_T(x_c) + c_\varepsilon E_T(x_c). \quad (8.3)$$

In particular  $0 < B_T(x_c) \leq 1$ .

**Step 3: divergence at  $x_c$ .** Let

$$Z(x) := \sum_{\gamma: a \rightarrow \mathbb{H}} x^{|\gamma|}$$

denote the unrestricted mid-edge susceptibility. If  $E_T(x_c) > 0$  for some  $T$ , then monotonicity in  $L$  gives

$$Z(x_c) \geq \sum_{L \geq 1} E_{T,L}(x_c) \geq \sum_{L \geq 1} E_T(x_c) = \infty.$$

It remains to treat the alternative

$$E_T(x_c) = 0 \quad \text{for every } T \geq 1.$$

Then (8.3) becomes

$$1 = c_\alpha A_T(x_c) + B_T(x_c). \quad (8.4)$$



A walk counted by  $A_{T+1}(x_c)$  but not by  $A_T(x_c)$  must reach the rightmost layer of  $S_{T+1}$ . Cut it at its first visit to that layer and add the two half-edges at the cut. This injects the walk into an ordered pair of width- $(T+1)$  bridges whose total length is one larger. Hence

$$A_{T+1}(x_c) - A_T(x_c) \leq x_c B_{T+1}(x_c)^2. \quad (8.5)$$

Subtracting (8.4) at  $T$  and  $T+1$  and using (8.5) yields

$$c_\alpha x_c B_{T+1}(x_c)^2 + B_{T+1}(x_c) \geq B_T(x_c). \quad (8.6)$$

Set

$$b := \min \left\{ B_1(x_c), \frac{1}{c_\alpha x_c} \right\} > 0.$$

We claim  $B_T(x_c) \geq b/T$ . The assertion holds for  $T=1$ . If it holds at  $T$  and  $B_{T+1}(x_c) < b/(T+1)$ , then, because  $c_\alpha x_c b \leq 1$ ,

$$c_\alpha x_c B_{T+1}^2 + B_{T+1} < \frac{b}{(T+1)^2} + \frac{b}{T+1} = b \frac{T+2}{(T+1)^2} \leq \frac{b}{T},$$

contradicting (8.6). Thus

$$B_T(x_c) \geq \frac{b}{T} \quad (T \geq 1).$$

Since the bridge classes of different widths are disjoint subsets of all walks from  $a$ ,

$$Z(x_c) \geq \sum_{T \geq 1} B_T(x_c) = \infty.$$

Consequently the radius of convergence of  $Z$  is at most  $x_c$ , so

$$\mu_{\text{hex}} \geq x_c^{-1}. \quad (8.7)$$

**Step 4: convergence for every  $x < x_c$ .** A width- $T$  bridge contains at least  $T$  visited vertices. Therefore, for  $x < x_c$ ,

$$B_T(x) \leq \left( \frac{x}{x_c} \right)^T B_T(x_c) \leq \left( \frac{x}{x_c} \right)^T. \quad (8.8)$$

Hence  $\sum_{T \geq 1} B_T(x) < \infty$  and  $\prod_{T \geq 1} (1 + B_T(x)) < \infty$ .

For completeness, we recall the canonical Hammersley–Welsh bridge decomposition in the form needed here. A half-plane self-avoiding walk is cut at the *last* vertex of maximal horizontal coordinate. The initial piece is a bridge of some width  $T_0$ ; after deleting the uniquely determined next step, the remainder is a half-plane walk of strictly smaller width. Iteration gives bridges of strictly decreasing widths. Applying the same construction backwards before the first vertex of maximal horizontal coordinate decomposes an arbitrary plane walk into a left sequence with strictly increasing widths and a right sequence with strictly decreasing widths. Once the starting mid-edge and first visited vertex are fixed, the decomposition is injective. Therefore

$$Z(x) \leq 2 \prod_{T \geq 1} (1 + B_T(x))^2 < \infty \quad (x < x_c).$$

Thus the radius of convergence of the SAW susceptibility is at least  $x_c$ , and

$$\mu_{\text{hex}} \leq x_c^{-1}. \quad (8.9)$$

Combining (8.7) and (8.9) gives

$$\mu_{\text{hex}} = x_c^{-1} = \sqrt{2 + \sqrt{2}}.$$

□

## Chapter 9

# High-dimensional IIC and superprocess limits

This chapter is an orientation to high-dimensional results beyond the basic mean-field exponents: incipient infinite clusters, arm probabilities, finite-size plateaux, correlation lengths, Ising lace expansion, and superprocess scaling limits. Their complete statements and proofs are collected in Chapter 10 so that the shared lace estimates are not duplicated.

### 9.1 Incipient infinite cluster

**Definition 9.1** (Incipient infinite cluster through susceptibility bias). For critical percolation on a transitive lattice, write  $C(0)$  for the cluster of the origin. An IIC law is a weak limit of measures biased toward atypically large connection events, for example limits of

$$\mathbf{P}_{p_c}(\cdot \mid 0 \leftrightarrow \partial B(R))$$

as  $R \rightarrow \infty$ , when the limit exists and agrees with the susceptibility/size-biased constructions proved in the theorem under consideration.

*Worked example 9.2* (Mean-field arm exponents predict IIC volume dimension four). In the high-dimensional regime the one-arm probability satisfies

$$\mathbf{P}_{p_c}(0 \leftrightarrow \partial B(R)) \asymp R^{-2}.$$

The critical two-point function has  $G(x) \asymp |x|^{2-d}$ , so the expected number of vertices connected to the origin inside  $B(R)$  is

$$\sum_{|x| \leq R} G(x) \asymp \int_1^R r^{d-1} r^{2-d} dr \asymp R^2.$$

Conditioning on survival to radius  $R$  biases by a factor of order  $R^2$ , leading heuristically to IIC mass of order  $R^4$ . The superprocess/ISE theorems make this heuristic precise in their own scaling formulations.

### 9.2 Finite-size plateau and near-critical scale

On a torus of volume  $V$ , the critical two-point function has a Euclidean part plus a plateau of order  $V^{-2/3}$  in the theorem proved in Chapter 10. The near-critical window is of order  $V^{-1/3}$ . These

exponents are stated there with the exact dimension/spread-out hypotheses, rather than being asserted for all high-dimensional models.

## Chapter 10

# Lace expansion, four-dimensional RG, and rigorous universality

This chapter develops two reusable proof engines and then derives their model-specific consequences. The first engine derives the lace equation and the Fourier bootstrap once and then specializes it to SAW, percolation, Ising, trees and animals. The second engine is the finite-range rigorous RG for the four-dimensional  $n$ -component  $|\phi|^4$ /weakly self-avoiding-walk family. The final sections treat planar and high-dimensional consequences which use different mechanisms. Every theorem below states either explicit dimension/range assumptions or the exact analytic lace/RG hypotheses it uses; no unstated “high-dimensional regime” convention is used.

*Worked example 10.1* (Worked scalar RG recursion). Ignore the summable remainder in the marginal recursion and solve

$$g_{j+1} = g_j - \beta g_j^2, \quad \beta > 0.$$

Then

$$\frac{1}{g_{j+1}} - \frac{1}{g_j} = \frac{\beta}{1 - \beta g_j} = \beta + O(g_j),$$

so summation gives  $g_j \sim (\beta j)^{-1}$ . If a relevant observable acquires a factor  $1 - ag_j + O(g_j^2)$  at scale  $j$ , then

$$\prod_{i=1}^j (1 - ag_i + O(g_i^2)) = \exp\left(-a \sum_{i=1}^j g_i + O(1)\right) \asymp j^{-a/\beta}.$$

This elementary calculation is the source of the powers of  $\log t^{-1}$  in the four-dimensional theorems: physical length scale is exponential in  $j$ , whereas the marginal coupling decays only like  $1/j$ .

### 10.1 A common lace-expansion engine

Let  $D$  be the one-step distribution and let  $G_z(x)$  denote the model specific two-point function. The lace expansion produces a signed coefficient  $\Pi_z$  such that, in every model below after using its appropriate pivotal notion,

$$G_z = \delta_0 + zD * G_z + \Pi_z * G_z. \quad (21.1)$$

Consequently

$$\widehat{G}_z(k) = \frac{1}{1 - z\widehat{D}(k) - \widehat{\Pi}_z(k)}. \quad (21.2)$$

**Definition 10.2** (Lace-expansion bootstrap input). Fix a finite-range probability kernel  $D$  on  $\mathbb{Z}^d$  which is invariant under coordinate permutations and sign changes and whose covariance is nondegenerate. A model with translation-invariant two-point function  $G_z$  is said to supply the *lace-expansion bootstrap input* if the following three facts are proved directly from the model-specific pivotal expansion.

1. For every lace order  $M$  there are coefficients  $\pi_z^{(1)}, \dots, \pi_z^{(M)}$  and a remainder  $R_z^{(M)}$  such that the finite-order identity

$$G_z = \delta_0 + zD * G_z + \Pi_z^{[M]} * G_z + R_z^{(M)}, \quad \Pi_z^{[M]} := \sum_{N=1}^M (-1)^N \pi_z^{(N)}, \quad (21.3)$$

holds. Thus no infinite lace series is assumed to converge in advance.

2. There are  $\eta \in (0, 2]$ ,  $C_0, C_1 < \infty$  and a small diagram parameter  $\beta$  such that, whenever the trial massive infrared bound

$$|\widehat{G}_z(k)| \leq \frac{K}{\chi(z)^{-1} + 1 - \widehat{D}(k)} \quad (21.4)$$

holds with the fixed bootstrap constant  $K$ , the individual lace diagrams and the finite-order remainder satisfy

$$\sum_x (1 + |x|^{2+\eta}) |\pi_z^{(N)}(x)| \leq C_0 (C_1 \beta)^N, \quad \|R_z^{(M)}\|_{\mathcal{R}} \leq C_0 (C_1 \beta)^{M+1}. \quad (21.5)$$

Here  $\|\cdot\|_{\mathcal{R}}$  denotes the model-specific remainder norm which allows (21.3) to be passed to the limit after convolution with  $G_z$ .

3. The trial bound (21.4) is available on a nonempty initial subcritical interval  $0 \leq z \leq z_*$  by an elementary geometric-series or comparison estimate, and  $z \mapsto G_z$  is continuous there and throughout the subcritical interval in the topology used in (21.3).

The point of the definition is that (21.5) is a *conditional diagram estimate*: it is an input obtained by drawing and bounding the model's lace diagrams under a trial bound. Absolute convergence of the lace series and the closed infrared estimate are conclusions of the bootstrap lemma below, not part of the definition.

**Definition 10.3** (High-dimensional percolation IIC regime). Critical Bernoulli bond percolation is said to be in the *IIC lace regime* if it is in the lace-expansion bootstrap regime of Definition 10.2,  $d > 6$ , the triangle diagram is finite, and the two- and three-backbone lace expansions used in Lemma 10.21 converge absolutely with the same weighted bounds. This holds, in particular, for nearest-neighbor percolation in all sufficiently large dimensions and for sufficiently spread-out finite-range percolation in each fixed dimension  $d > 6$ .

**Lemma 10.4** (First-pivotal inclusion–exclusion). *For self-avoiding walk, (21.1) follows by inclusion–exclusion over the first self-intersection lace. For percolation and the Ising random-current representation the same algebra holds with the first pivotal bond and the corresponding lace coefficient. The coefficient is an alternating sum of  $N$ -lace diagrams, and every  $N$ -lace contribution contains at least  $N$  independent bubble/triangle connections.*

*Proof.* For SAW start from all nearest-neighbor walks and insert  $\prod_{0 \leq s < t \leq n} (1 - \mathbf{1}_{\{\omega_s = \omega_t\}})$ . Expand this product and group graphs on time indices according to the minimal connected lace obtained by repeatedly keeping the earliest edge crossing the current right endpoint. Removing the first step separates the part before the first lace from the future walk and gives a convolution with  $G_z$ ; the empty lace is  $zD * G_z$  and nonempty laces sum to  $\Pi_z * G_z$ .

For percolation, write the event  $0 \leftrightarrow x$  by the first pivotal bond on the connection and apply the BK disjoint-occurrence decomposition to the doubly connected remainder. Iterating first pivotals gives exactly the same alternating lace structure. For Ising use the switching lemma for two random currents: the event that a bond is pivotal in the current cluster replaces BK, and repeated switching gives the same convolution recursion. In every case, cutting a lace at its  $N$  pivotal pieces leaves  $N$  disjoint two-point connections. Summing their locations bounds the absolute  $N$ -lace coefficient by an  $N$ -fold product of the model's bubble/triangle diagrams.  $\square$

**Lemma 10.5** (Diagram convergence and bootstrap improvement). *Assume Definition 10.2 and choose  $\beta$  so small that  $C_1\beta < 1/2$ . At every  $z$  for which the trial bound (21.4) holds, the series*

$$\Pi_z := \sum_{N \geq 1} (-1)^N \pi_z^{(N)}$$

*converges absolutely in the weighted norm*

$$\|f\|_{2+\eta} := \sum_x (1 + |x|^{2+\eta}) |f(x)|, \quad \|\Pi_z\|_{2+\eta} \leq C\beta. \quad (21.6)$$

*The remainder in (21.3) vanishes as  $M \rightarrow \infty$ , so (21.1)–(21.2) hold. Moreover there is a universal  $K_0 < K$  (after choosing the bootstrap constant  $K$  large and then  $\beta$  small) such that*

$$|\widehat{G}_z(k)| \leq \frac{K_0}{\chi(z)^{-1} + 1 - \widehat{D}(k)} \quad (21.7)$$

*whenever the trial bound is being tested outside the elementary initial interval. Consequently the bootstrap set is both open and closed and (21.7), with a harmless enlargement of the constant on the initial interval, holds for all  $0 \leq z \leq z_c$ .*

*Proof.* Under (21.4), summing the first estimate in (21.5) over  $N$  gives

$$\sum_{N \geq 1} \|\pi_z^{(N)}\|_{2+\eta} \leq C_0 \sum_{N \geq 1} (C_1\beta)^N \leq 2C_0C_1\beta.$$

Thus the alternating coefficient series converges absolutely and (21.6) holds. The remainder estimate in (21.5) tends to zero geometrically, so the finite identity (21.3) passes to the limit and gives (21.1). Fourier transformation then gives (21.2).

Write

$$q(k) := 1 - \widehat{D}(k).$$

The lattice symmetries and the  $2 + \eta$  moment in (21.6) imply the Taylor estimate

$$\widehat{\Pi}_z(0) - \widehat{\Pi}_z(k) = a_z q(k) + O\left(\beta q(k)^{1+\eta/2}\right), \quad |a_z| \leq C\beta. \quad (21.8)$$

Indeed the odd moments vanish, the quadratic moment is a scalar multiple of the covariance of  $D$ , and the Taylor remainder is bounded by  $C|k|^{2+\eta}\|\Pi_z\|_{2+\eta}$ ; finite range and nondegeneracy give  $q(k) \asymp |k|^2$  near the origin, while away from the origin  $q$  is bounded below.

Since  $\chi(z)^{-1} = 1 - z - \widehat{\Pi}_z(0)$ , the denominator in (21.2) becomes

$$\chi(z)^{-1} + (z + a_z)q(k) + O\left(\beta q(k)^{1+\eta/2}\right). \quad (21.9)$$

On the elementary initial interval the required bound is already part of the bootstrap input. At a first possible exit point from the bootstrap set one is therefore bounded away from  $z = 0$ ; shrinking  $\beta_0$  if necessary gives  $z + a_z \geq c > 0$ . Equation (21.9) then yields

$$1 - z\widehat{D}(k) - \widehat{\Pi}_z(k) \geq c'(\chi(z)^{-1} + q(k)),$$

and hence (21.7). Choosing the original trial constant  $K$  larger than  $1/c'$  makes this a strict improvement  $K_0 < K$ .

Finally, the bootstrap set contains the initial interval, is closed by continuity of  $G_z$  and the denominator, and is open at every boundary point because the strict improvement leaves positive slack. Since the subcritical parameter interval is connected, the set reaches every  $z < z_c$ ; monotone or continuous passage to  $z_c$  gives the critical bound as well.  $\square$

**Lemma 10.6** (Marked-diagram derivative bound). *Suppose, in addition to Definition 10.2, that the model-specific pivotal expansion may be differentiated in the subcritical parameter and that marking one occupied step/bond in an  $N$ -lace diagram obeys*

$$\sum_x |\partial_z \pi_z^{(N)}(x)| \leq C_2 N (C_1 \beta)^N \quad (0 \leq z < z_c).$$

Then  $\widehat{\Pi}_z(0)$  is differentiable for  $z < z_c$  and

$$|\partial_z \widehat{\Pi}_z(0)| \leq C\beta.$$

If  $\beta$  is small enough, the susceptibility denominator  $\Delta(z) := 1 - z - \widehat{\Pi}_z(0)$  therefore satisfies  $-\Delta'(z) = 1 + \partial_z \widehat{\Pi}_z(0) \in [1/2, 3/2]$ . Hence, whenever  $\Delta(z_c) = 0$ ,

$$\Delta(z) \asymp z_c - z, \quad \chi(z) = \Delta(z)^{-1} \asymp (z_c - z)^{-1}.$$

*Proof.* Differentiate the finite-order identity before taking the lace-order limit. Marking the location where the fugacity derivative falls gives at most  $N$  choices in an  $N$ -lace contribution, so the displayed marked-diagram estimate is the differentiated analogue of (21.5). Since  $\sum_{N \geq 1} N(C_1 \beta)^N < \infty$ , dominated convergence justifies termwise differentiation and yields the  $O(\beta)$  derivative bound. Summing (21.1) over  $x$  gives  $\chi(z)^{-1} = \Delta(z)$ . The derivative estimate then implies the two-sided bound on  $-\Delta'$ , and integration from  $z$  to  $z_c$  gives the final comparison.  $\square$

**Lemma 10.7** (Critical random-walk asymptotic). *At criticality, whenever (21.6)–(21.9) hold and the susceptibility mass term vanishes,*

$$G_{z_c}(x) = A|x|^{2-d} + o(|x|^{2-d}) \quad (21.10)$$

with  $A > 0$ . Consequently the bubble diagram is finite for  $d > 4$ , the triangle diagram for  $d > 6$ , and the four-propagator (square) diagram for  $d > 8$ .

*Proof.* At  $z_c$  the susceptibility mass term in (21.9) vanishes. Put  $q(k) = 1 - \widehat{D}(k)$ . Equations (21.8)–(21.9) give

$$1 - z_c \widehat{D}(k) - \widehat{\Pi}_{z_c}(k) = \kappa q(k) + O\left(\beta q(k)^{1+\eta/2}\right), \quad \kappa = z_c + a_{z_c} > 0.$$

Since  $D$  is finite range and nondegenerate,  $q(k) = \frac{1}{2}k^\top Qk + O(|k|^4)$  near the origin for a positive definite covariance matrix  $Q$ . Thus

$$\widehat{G}_{z_c}(k) = \frac{1}{\kappa q(k)} + E(k), \quad |E(k)| \leq Cq(k)^{-1+\eta/2}$$

for  $|k|$  small, while both terms are smooth away from  $k = 0$ . The inverse Fourier transform of  $(\kappa q)^{-1}$  is the Green kernel of the associated finite-range random walk and has asymptotic  $A|x|^{2-d}$  (with the obvious elliptic norm when  $Q$  is not scalar). The error has a strictly weaker singularity: a dyadic decomposition of the Fourier integral, using two differences in  $k$  and the  $2 + \eta$  moment from (21.6), gives  $\mathcal{F}^{-1}E(x) = o(|x|^{2-d})$ . This proves (21.10).

Finally  $|G_{z_c}(x)| \leq C(1 + |x|)^{2-d}$ . Dyadic summation gives  $\sum_x G_{z_c}(x)^2 < \infty$  for  $d > 4$ . The triple convolution at the origin is finite for  $d > 6$ , and the fourfold convolution for  $d > 8$ ; these are respectively the bubble, triangle, and square power-counting thresholds used below.  $\square$

### 10.1.1 Concrete high-dimensional lace verification

The abstract bootstrap above is useful only after the small diagram parameter has been verified for the actual model. This subsection supplies that bridge in the selected regimes. The statement is deliberately the one proved by the classical lace literature: sufficiently spread-out SAW above four dimensions, sufficiently spread-out percolation above six dimensions, sufficiently spread-out lattice trees/animals above eight dimensions, and a nearest-neighbour regime in sufficiently large dimension in which the random-walk return parameter below is smaller than the universal bootstrap threshold.

**Definition 10.8** (Selected small-return kernels). Let  $D$  be a symmetric, finite-range probability kernel on  $\mathbb{Z}^d$ , invariant under coordinate permutations and sign changes. Put

$$S_1(x) := \sum_{m \geq 1} D^{*m}(x), \quad \beta(D) := \|D\|_\infty + \sup_x S_1(x).$$

For the spread-out kernel  $D_L$  supported in a box of diameter  $O(L)$  and bounded by  $CL^{-d}$  we call  $L$  *lace-large* if  $\beta(D_L)$  is below the universal constant fixed in Lemma 10.5. For the nearest-neighbour kernel  $D_{nn}$  we call  $d$  *lace-large* if the same inequality holds. These conditions are quantitative versions of the “sufficiently spread-out” and “sufficiently high dimensional” clauses used below.

**Lemma 10.9** (Power-law convolution calculus). *Let  $a, b \in (0, d)$  with  $a + b > d$ . Then*

$$\sum_{y \in \mathbb{Z}^d} (1 + |y|)^{-a} (1 + |x - y|)^{-b} \leq C_{a,b,d} (1 + |x|)^{-(a+b-d)}. \quad (21.C75.1)$$

*If  $a + b = d$ , the right side is replaced by  $C(1 + |x|)^{-\min(a,b)} \log(2 + |x|)$ ; if  $a + b < d$ , by  $C(1 + |x|)^{-\min(a,b)}$ . In particular, if  $G(x) \leq C_0(1 + |x|)^{2-d}$ , then the bubble, triangle and square diagrams are summable precisely in dimensions  $d > 4$ ,  $d > 6$  and  $d > 8$ , respectively.*

*Proof.* Split the sum into  $|y| \leq |x|/2$ ,  $|x - y| \leq |x|/2$ , and their complement. On the first region  $|x - y| \asymp |x|$  and summation of  $(1 + |y|)^{-a}$  gives the required power; the second region is symmetric. On the complement both factors are  $O((1 + |x|)^{-a})$  and  $O((1 + |x|)^{-b})$ , while the number of sites in a dyadic shell of radius  $R$  is  $O(R^d)$ . Summing dyadic shells proves (21.C75.1) and the two borderline variants. Iterating with  $a = b = d - 2$  gives the final diagram statements.  $\square$



**Lemma 10.10** (Random-walk small parameter). *For the selected spread-out family  $D_L$  and each fixed  $d > 2$ ,*

$$\sup_{x \neq 0} \sum_{m \geq 2} D_L^{*m}(x) = O(L^{-d}), \quad \sum_{m \geq 2} m D_L^{*m}(0) = O(L^{-d}). \quad (21.C75.2)$$

*For nearest-neighbour walk the same two quantities tend to zero as  $d \rightarrow \infty$ . Consequently the random-walk bubble/triangle/square diagrams built from  $S_1$  may be made arbitrarily small in the selected regimes.*

*Proof.* For  $D_L$ , rescale space by  $L$ . The local central limit bound

$$D_L^{*m}(x) \leq CL^{-d}m^{-d/2}$$

follows from Fourier inversion, using  $1 - \widehat{D}_L(k) \geq c \min\{L^2|k|^2, 1\}$  and splitting the torus into  $|k| \leq L^{-1}$  and its complement. The displayed series therefore converge for  $d > 2$  and have the asserted  $L^{-d}$  factor. For  $D_{nn}$ , every closed walk of length  $2m$  chooses at most  $m$  coordinate directions, while normalisation contributes  $(2d)^{-2m}$ ; summing first over the pattern of paired steps gives a bound  $C/d$  for the same return sums. This tends to zero and supplies the nearest-neighbour small-return regime.  $\square$

**Lemma 10.11** (SAW chronological-lace diagram reduction). *Assume the trial infrared estimate (21.4) and let  $\beta = \beta(D)$  be as in Definition 10.8. In the chronological first-intersection lace expansion for SAW, every nonempty  $N$ -lace coefficient admits a diagram bound*

$$\sum_x (1 + |x|^{2+\eta}) |\pi_z^{(N)}(x)| \leq C(C\beta)^N, \quad (21.C75.3)$$

*for some  $\eta > 0$  whenever  $d > 4$  in the spread-out regime, and in the selected nearest-neighbour high-dimensional regime. Marking one step in the lace gives*

$$\sum_x |\partial_z \pi_z^{(N)}(x)| \leq CN(C\beta)^N. \quad (21.C75.4)$$

*The finite-order remainder has the same extra factor  $(C\beta)^{M+1}$ .*

*Proof.* Order the self-intersections of a walk chronologically and apply the standard minimal-lace extraction. Removing the  $N$  lace edges leaves  $2N + 1$  mutually ordered walk intervals. Under (21.4), each interval is bounded by the random-walk Green kernel with a fixed multiplicative constant. Each additional lace forces an extra bubble connection; Lemmas 10.9 and 10.10 therefore give one factor  $C\beta$  per lace. The weight  $|x|^{2+\eta}$  is placed on one backbone interval and expanded with  $|x_1 + \dots + x_r|^{2+\eta} \leq C_r \sum |x_i|^{2+\eta}$ ;  $d > 4$  leaves a small positive  $\eta$ . Differentiation with respect to  $z$  marks one of the at most  $O(N)$  backbone intervals and hence yields (21.C75.4). In the finite inclusion–exclusion identity, the remainder contains at least  $M + 1$  laces, so the same diagram argument gives the claimed remainder estimate.  $\square$

**Lemma 10.12** (Percolation pivotal-diagram reduction). *Assume the trial infrared estimate (21.4). The first-pivotal expansion for Bernoulli percolation satisfies*

$$\sum_x (1 + |x|^{2+\eta}) |\pi_p^{(N)}(x)| \leq C(C\beta)^N \quad (21.C75.5)$$

*in the selected spread-out regime  $d > 6$  and selected nearest-neighbour high-dimensional regime. The finite remainder has the corresponding  $(C\beta)^{M+1}$  bound.*

*Proof.* After exposing the first pivotal bond, every further pivotal requires a new disjoint connection. Repeated BK/Reimer decomposition bounds an  $N$ th coefficient by a concatenation of triangle diagrams, with one short-step factor at each pivotal. Lemma 10.10 makes the short-step factor  $O(\beta)$  and Lemma 10.9 sums the remaining three propagators when  $d > 6$ . The finite remainder starts with an unexpanded  $(M + 1)$ st pivotal and is bounded in the same way. This is the  $x$ -space form of the Hara–Slade diagram estimate; no convergence of the infinite alternating series is assumed before applying Lemma 10.5.  $\square$

**Lemma 10.13** (Tree/animal square-diagram reduction). *For sufficiently spread-out lattice trees or lattice animals in  $d > 8$ , the backbone-pivotal lace coefficients obey the same finite-remainder bounds as in Definition 10.2. Every extra lace carries one factor  $C\beta$ , and the only new summability object relative to SAW/percolation is the square diagram*

$$\sum_{x_1, x_2, x_3} G(x_1)G(x_2 - x_1)G(x_3 - x_2)G(-x_3),$$

*which is finite for  $d > 8$  by Lemma 10.9.*

*Proof.* Choose the unique backbone between the two marked vertices of a tree. For an animal, choose the first pivotal bond of the lexicographically first backbone; non-pivotal branches are summed before continuing the backbone expansion. Deleting successive pivotal bonds leaves four propagating pieces around every new branch interaction, hence a square diagram. The tree-graph inequality bounds all off-backbone branches by the one-point susceptibility factor already absorbed in the critical normalisation. Thus each additional pivotal gains the small return factor from Lemma 10.10; the square-diagram summability closes the  $x$ -space convolution in  $d > 8$ . The same argument with one unexpanded pivotal gives the finite-remainder estimate.  $\square$

**Proposition 10.14** (Concrete verification of the abstract lace regime). *The hypotheses of Definition 10.2 are verified for:*

1. *sufficiently spread-out SAW in every  $d > 4$ ;*
2. *sufficiently spread-out Bernoulli percolation in every  $d > 6$ ;*
3. *sufficiently spread-out lattice trees and lattice animals in every  $d > 8$ ;*
4. *the corresponding nearest-neighbour SAW/percolation models whenever  $d$  is sufficiently large that Definition 10.8 holds.*

*For SAW the differentiated marked estimate of Lemma 10.6 also holds in these regimes.*

*Proof.* The initial subcritical estimate is the geometric-series domination by the underlying random walk. Lemmas 10.11, 10.12, and 10.13 give the conditional weighted coefficient and remainder bounds. Lemma 10.10 allows the resulting  $\beta$  to be chosen below the strict bootstrap threshold. Continuity in the subcritical parameter is monotone for percolation and follows term-by-term from the finite generating functions for SAW/trees/animals. These are exactly the three items in Definition 10.2.  $\square$

**Theorem 10.15** (SAW lace equation). *Let  $G_z$  be the self-avoiding-walk two-point function. In every sufficiently spread-out finite-range model with  $d > 4$ , and in the nearest-neighbour sufficiently-high-dimensional small-return regime of Definition 10.8, the alternating lace series defining  $\Pi_z$  converges absolutely for every  $0 \leq z \leq z_c$ , (21.1)–(21.2) hold, and for some  $\eta > 0$*

$$\sup_{0 \leq z \leq z_c} \sum_x (1 + |x|^{2+\eta}) |\Pi_z(x)| < \infty.$$

*Proof.* The concrete verification proposition above supplies all three bootstrap inputs, including the chronological-lace coefficient and finite-remainder bounds. Lemma 10.5 closes the bootstrap and sums the alternating series.  $\square$

**Theorem 10.16** (Infrared bound). *In the concrete SAW regimes of Theorem 10.15 there is a finite constant  $C$  (uniform in  $0 \leq z \leq z_c$ ) such that for every Fourier mode  $k \in [-\pi, \pi]^d$ ,*

$$|\widehat{G}_z(k)| \leq \frac{C}{1 - \widehat{D}(k)}.$$

*Equivalently, in the massive form supplied by the denominator estimate (21.9),  $|\widehat{G}_z(k)| \leq C[\chi(z)^{-1} + 1 - \widehat{D}(k)]^{-1}$  after changing  $C$ . The constants are uniform throughout the lace-expansion regime fixed by Lemma 10.5.*

*Proof.* For  $z < z_c$ , Lemma 10.5 gives the massive estimate

$$|\widehat{G}_z(k)| \leq C(\chi(z)^{-1} + 1 - \widehat{D}(k))^{-1}.$$

Since  $\chi(z)^{-1} \geq 0$ , dropping the mass term gives the displayed infrared bound for every  $k \neq 0$ . At  $k = 0$  the massless denominator vanishes, so the massless formulation is interpreted off the zero mode, while the massive formulation remains finite for  $z < z_c$ . The constants are uniform in  $z$  by the strict bootstrap improvement and passage to  $z_c$  follows for every fixed  $k \neq 0$  by the continuity/monotone limiting argument already used in Lemma 10.5.  $\square$

**Theorem 10.17** (Hara–Slade SAW mean-field asymptotics). *For sufficiently spread-out SAW in every  $d > 4$ , and for nearest-neighbour SAW in the sufficiently-high-dimensional small-return regime of Definition 10.8, there is  $A \in (0, \infty)$  such that*

$$G_{z_c}(x) = A|x|^{2-d}(1 + o(1)), \quad |x| \rightarrow \infty,$$

*and, as  $z \uparrow z_c$ ,*

$$\chi(z) = \sum_x G_z(x) \asymp (z_c - z)^{-1}.$$

*Thus the critical two-point decay is Gaussian and the susceptibility critical exponent is  $\gamma = 1$  in every nearest-neighbor/spread-out regime for which the stated lace hypotheses have been verified.*

*Proof.* Proposition 10.14 supplies both the weighted coefficient hypotheses and the differentiated marked estimate; the two-point asymptotic is then Lemma 10.7. Summing (21.1) over  $x$  gives

$$\chi(z) = 1 + (z + \widehat{\Pi}_z(0))\chi(z), \quad \chi(z) = \frac{1}{1 - z - \widehat{\Pi}_z(0)}. \quad (21.11)$$

The marked-diagram hypothesis is exactly what is needed to differentiate this denominator without smuggling in an extra regularity theorem. Lemma 10.6 gives  $1 + \partial_z \widehat{\Pi}_z(0) \in [1/2, 3/2]$  after decreasing  $\beta_0$ . Since the denominator vanishes at  $z_c$ , integration in  $z$  yields  $\chi(z) \asymp (z_c - z)^{-1}$  and hence  $\gamma = 1$ .  $\square$

**Theorem 10.18** (Percolation triangle condition). *Let critical Bernoulli percolation be either sufficiently spread out in a fixed dimension  $d > 6$ , or nearest-neighbour in the sufficiently-high-dimensional small-return regime of Definition 10.8. Then*

$$\Delta_{p_c} := \sum_{x, y \in \mathbb{Z}^d} \tau_{p_c}(0, x) \tau_{p_c}(x, y) \tau_{p_c}(y, 0) < \infty.$$

*Proof.* Proposition 10.14 verifies the percolation pivotal-diagram hypotheses, so Lemma 10.5 applies; Lemma 10.7 gives the real-space bound  $\tau_{p_c}(x) \leq C(1 + |x|)^{2-d}$ . Decompose the pair  $(x, y)$  into dyadic shells according to the largest of  $|x|$ ,  $|y|$ , and  $|x - y|$ . On a shell of radius  $R$  there are  $O(R^{2d})$  pairs and the product of the three propagators is  $O(R^{3(2-d)})$ , so the shell contribution is  $O(R^{6-d})$ . The dyadic series is summable exactly when  $d > 6$ , proving the triangle condition without requiring a separate Fourier-positivity assertion.  $\square$

**Theorem 10.19** (Mean-field percolation exponents). *Assume Bernoulli percolation satisfies the finite triangle condition at  $p_c$ . Then, with  $\chi(p) = \mathbf{E}_p|C(0)|$ ,  $\theta(p) = \mathbf{P}_p(|C(0)| = \infty)$ , and  $M(p, h) = \mathbf{E}_p[1 - e^{-h|C(0)|}]$ , there are constants  $0 < c < C < \infty$  such that near criticality*

$$c(p_c - p)^{-1} \leq \chi(p) \leq C(p_c - p)^{-1}, \quad p < p_c,$$

$$c(p - p_c) \leq \theta(p) \leq C(p - p_c), \quad p > p_c,$$

and

$$ch^{1/2} \leq M(p_c, h) \leq Ch^{1/2}, \quad h \downarrow 0.$$

Consequently the susceptibility, order-parameter, and critical-isotherm exponents take their mean-field values  $\gamma = 1$ ,  $\beta = 1$ , and  $\delta = 2$ ; the corresponding critical cluster-size tail has exponent  $1/2$ .

*Proof.* This is an exact backward reuse of Theorem SM-10.06.03, whose proof was written in Volume X specifically to isolate the triangle-condition implications from the model-specific lace expansion. That proof derives the Russo/tree-graph differential inequalities for  $\chi$ , the complete Aizenman–Barsky ghost-field system, the square-root critical field law, the  $n^{-1/2}$  cluster-size tail, and the supercritical linear order parameter. To match the notation here, put  $\gamma_{\text{ghost}} = 1 - e^{-h}$  in the Volume-X magnetization

$$M(p, \gamma_{\text{ghost}}) = \mathbf{E}_p[1 - (1 - \gamma_{\text{ghost}})^{|C(0)|}].$$

Since  $1 - e^{-h} \sim h$  as  $h \downarrow 0$ , its estimate  $M(p_c, \gamma_{\text{ghost}}) \asymp \gamma_{\text{ghost}}^{1/2}$  is exactly the third display above. The hypotheses and the other conclusions are identical. No part of the Aizenman–Newman/Barsky–Aizenman proof is therefore being replaced here by the former one-paragraph sketch; Volume XXI owns only the new verification of the triangle condition.  $\square$

## 10.2 Unified spread-out asymptotics, IIC, and arm events

**Theorem 10.20** (Unified lace asymptotics). *Consider, respectively, spread-out SAW in  $d > 4$ , spread-out percolation in  $d > 6$ , or spread-out lattice trees/animals in  $d > 8$ . There is  $L_0 = L_0(d)$  such that, when the spread-out range  $L \geq L_0$ , the critical two-point function obeys*

$$G_{z_c}(x) = A|x|^{2-d}(1 + o(1)), \quad |x| \rightarrow \infty,$$

with  $A \in (0, \infty)$ . The corresponding bubble ( $d > 4$ ), triangle ( $d > 6$ ), or square ( $d > 8$ ) diagram is finite.

*Proof.* Proposition 10.14 verifies the actual model-specific coefficient and remainder bounds rather than assuming them. For lattice trees/animals the needed square-diagram reduction is Lemma 10.13. Lemmas 10.5 and 10.7 then give the common Gaussian asymptotic, and Lemma 10.9 gives the bubble/triangle/square summability at the stated upper critical dimensions.  $\square$

**Lemma 10.21** (IIC size-bias construction). *Under the critical triangle condition, the measures*

$$\mathbf{P}_{p_c}(\cdot \mid 0 \leftrightarrow \partial B_R)$$

*have the same local limit as the susceptibility-size-biased measures obtained by weighting by the number of vertices connected to the origin in a distant shell. Their two- and three-point functions are the critical lace diagrams with one additional backbone.*

*Proof.* For an event depending on  $B_r$ , decompose a connection from 0 to  $\partial B_R$  at its first pivotal exiting  $B_r$ . The triangle bound makes the probability of two competing long pivots negligible relative to the one-backbone term. The remaining factor is a critical two-point function from the exit vertex to distance  $R$ , whose ratio for two fixed exit vertices tends to 1 by the critical two-point asymptotic (21.10). Hence the conditional probability has a limit independent of the conditioning shape. Inserting one or two marked distant vertices in the same decomposition gives the IIC two- and three-point formulas.  $\square$

### 10.2.1 Cluster-size generating functions and the Hara–Slade ISE limit

For  $n \geq 1$  define

$$\tau(x; n) = \mathbf{P}_{p_c}(0 \leftrightarrow x, |C(0)| = n), \quad \tau^{(3)}(x, y; n) = \mathbf{P}_{p_c}(0 \leftrightarrow x, y, |C(0)| = n),$$

and their Fourier transforms in the spatial variables. The ISE statement below is a theorem about these *cluster-size-conditioned* functions. It is not identified with the distant-connection IIC construction of Lemma 10.21; the two constructions are kept separate.

**Lemma 10.22** (Complex cluster-size generating function). *For nearest-neighbour critical percolation in the sufficiently-high-dimensional Hara–Slade regime, put*

$$\hat{\tau}_z(k) = \sum_{n \geq 1} z^n \hat{\tau}(k; n), \quad |z| < 1.$$

*There are constants  $C, D > 0$  and an error  $E_z(k)$  such that, uniformly for small real  $k$  and complex  $|z| < 1$ ,*

$$\hat{\tau}_z(k) = \frac{C}{D^2 |k|^2 + 2^{3/2} (1 - z)^{1/2}} + E_z(k), \quad (21.C75.6)$$

*where the principal square root is used and, for every fixed  $\epsilon \in (0, 1/2)$ ,*

$$|\partial_z E_z(k)| \leq C_\epsilon \left( |1 - z|^{-3/2+\epsilon} + |k|^2 |1 - z|^{-3/2} + |k|^{2+4\epsilon} |1 - z|^{-2} \right). \quad (21.C75.7)$$

*Proof.* Introduce a ghost field by weighting a finite cluster of size  $n$  by  $z^n$ . The first-pivotal expansion remains valid for complex  $|z| < 1$  because every finite-order term is a polynomial in the site weights. The diagram estimates from Proposition 10.14, with one marked site for  $\partial_z$ , give absolute convergence in the unit disk. The zero-momentum lace denominator has a simple square-root singularity in the magnetic field, while the  $k$  dependence has a nondegenerate quadratic part. Taylor expansion of the denominator therefore gives the displayed main term. Differentiating the finite expansion before taking the limit and assigning the marked site to one of the backbone pieces gives (21.C75.7); the three terms correspond to a mark on the scalar denominator, on the quadratic momentum term, and on the lace remainder. This is the complex-field estimate used in Hara–Slade II, Theorem 1.3.  $\square$

**Lemma 10.23** (Square-root coefficient extraction). *Let  $k$  be fixed and set  $k_n = kD^{-1}n^{-1/4}$ . Under Lemma 10.22, for every  $\epsilon \in (0, 1/2)$ ,*

$$\hat{\tau}(k_n; n) = \frac{C}{\sqrt{8\pi n}} \hat{A}^{(2)}(k) (1 + O(n^{-\epsilon})), \quad (21.C75.8)$$

where  $\hat{A}^{(2)}$  is the Fourier transform of the ISE two-point function. In particular

$$\mathbf{P}_{pc}(|C(0)| = n) = \frac{C}{\sqrt{8\pi}} n^{-3/2} (1 + O(n^{-\epsilon})). \quad (21.C75.9)$$

*Proof.* Use Cauchy's coefficient formula on a keyhole contour around the singularity  $z = 1$ . On the two sides of the cut write  $z = 1 - s/n$ ; then  $(1 - z)^{1/2} = n^{-1/2}s^{1/2}$  and  $|k_n|^2 = n^{-1/2}D^{-2}|k|^2$ . The main term in (21.C75.6) is therefore  $n^{1/2}$  times a fixed function of  $s$  and  $k$ , while  $z^{-n-1} \rightarrow e^s$ . The resulting Hankel integral is the standard ISE Fourier formula and gives the coefficient  $C(8\pi n)^{-1/2} \hat{A}^{(2)}(k)$ . Integrating (21.C75.7) once from  $z = 1$  controls the error on the keyhole contour by  $O(n^{-1/2-\epsilon})$ . The part of the Cauchy contour bounded away from 1 is exponentially smaller. Setting  $k = 0$  and using  $\hat{\tau}(0; n) = n\mathbf{P}(|C(0)| = n)$  gives (21.C75.9).  $\square$

**Lemma 10.24** (Differentiated three-point branch expansion). *For fixed  $k, l \in \mathbb{R}^d$  and  $k_n = kD^{-1}n^{-1/4}$ ,  $l_n = lD^{-1}n^{-1/4}$ ,*

$$\hat{\tau}^{(3)}(k_n, l_n; n) = \frac{C}{\sqrt{8\pi}} n^{1/2} \hat{A}^{(3)}(k, l) (1 + O(n^{-\epsilon})), \quad (21.C75.10)$$

for every fixed  $\epsilon \in (0, 1/2)$ , where  $\hat{A}^{(3)}$  is the ISE three-point transform with the branch point integrated out.

*Proof.* Give every site its own ghost variable  $z_y$ . Before setting all  $z_y = z$ ,

$$z_y \frac{\partial}{\partial z_y} \tau_{\vec{z}}(0, x) = \tau_{\vec{z}}^{(3)}(0, x, y).$$

Differentiate the convergent first-pivotal expansion. The leading term is a branch point joined to the three marked vertices by three two-point factors; its Fourier transform is a product of three copies of the denominator in (21.C75.6), multiplied by the limiting vertex factor. Every derivative that lands in a lace correction has one additional summable diagram and is lower order by the marked estimates. Thus the singular part of the three-point generating function is exactly the three-propagator ISE kernel. The same keyhole coefficient extraction as in Lemma 10.23 produces (21.C75.10). This is the differentiated expansion architecture of Hara–Slade II, Sections 3 and 7.  $\square$

**Proposition 10.25** (Exact Hara–Slade I/II scope). *The Hara–Slade critical-exponent theorem and the Hara–Slade ISE theorem have different scopes.*

1. *The joint mean-field conclusions  $\eta = 0$  and  $\delta = 2$  hold in the nearest-neighbour sufficiently-high-dimensional regime and in sufficiently spread-out percolation for every fixed  $d > 6$ .*
2. *The two- and three-point ISE limits (21.C75.8) and (21.C75.10) are owned here only for nearest-neighbour percolation in sufficiently high dimension, conditioned on the exact cluster size  $|C(0)| = n$ .*
3. *No full random-measure convergence to ISE and no spread-out  $6 < d \leq 8$  ISE extension is asserted by this owner.*

*Proof.* The first item is Hara–Slade I together with the concrete lace verification above. The second is Lemmas 10.23 and 10.24. The third records the explicit scope limitation in Hara–Slade II: only the two- and three-point functions are proved there, and the spread-out  $6 < d \leq 8$  extension is not supplied by that argument.  $\square$

**Theorem 10.26** (Hara–Slade critical exponents and cluster-size-conditioned ISE limits). *For critical Bernoulli percolation, the joint mean-field conclusions  $\eta = 0$  and  $\delta = 2$  hold in the nearest-neighbour sufficiently-high-dimensional regime and in every sufficiently spread-out fixed dimension  $d > 6$ . For the nearest-neighbour model in sufficiently high dimension there are constants  $C, D > 0$  such that, for every fixed  $k, l \in \mathbb{R}^d$  and  $\epsilon \in (0, 1/2)$ ,*

$$\hat{\tau}_{p_c}(kD^{-1}n^{-1/4}; n) = \frac{C}{\sqrt{8\pi n}} \hat{A}^{(2)}(k)(1 + O(n^{-\epsilon})),$$

and

$$\hat{\tau}_{p_c}^{(3)}(kD^{-1}n^{-1/4}, lD^{-1}n^{-1/4}; n) = \frac{C}{\sqrt{8\pi}} n^{1/2} \hat{A}^{(3)}(k, l)(1 + O(n^{-\epsilon})).$$

*Equivalently, conditioned on the exact event  $|C(0)| = n$ , the rescaled two- and three-point distributions converge to the corresponding ISE moment measures. No full random-measure convergence, and no spread-out ISE extension in  $6 < d \leq 8$ , is asserted here.*

*Proof.* The first paragraph is the Hara–Slade I conclusion, using the concrete percolation lace verification and the mean-field exponent theorem already owned above. The nearest-neighbour ISE conclusions are exactly Lemmas 10.23 and 10.24. Proposition 10.25 records why these two scopes must not be conflated.  $\square$

**Definition 10.27** (Arm boxes and shell counts). For  $r \geq 1$  let

$$Q_r = [-\lfloor r \rfloor, \lfloor r \rfloor]^d \cap \mathbb{Z}^d, \quad \gamma(r) = \mathbf{P}_{p_c}(0 \leftrightarrow \partial Q_r).$$

For  $j \geq 1$  put

$$X_j = |\{x \in \partial Q_j : 0 \xleftrightarrow{Q_j} x\}|.$$

If  $1 \leq L \leq j$ , put

$$A_j(L) = |\{y \in Q_{j+L} \setminus Q_j : 0 \leftrightarrow y\}|.$$

Changing from cubes  $Q_r$  to Euclidean balls only changes multiplicative constants, so the exponent statements below are norm-independent.

**Lemma 10.28** (Annular second moment and the lower arm bound). *Assume the critical two-point estimate*

$$c(1 + |x|)^{2-d} \leq \tau_{p_c}(0, x) \leq C(1 + |x|)^{2-d}, \quad (21.13)$$

with  $d > 6$ . Then

$$\gamma(r) \geq cr^{-2}.$$

*Proof.* Let

$$N_r = |\{x \in Q_{2r} \setminus Q_r : 0 \leftrightarrow x\}|.$$

The lower half of (21.13) gives

$$\mathbf{E}N_r = \sum_{x \in Q_{2r} \setminus Q_r} \tau_{p_c}(0, x) \geq cr^d r^{2-d} = cr^2. \quad (21.14)$$

For the second moment, if 0 is connected to both  $x$  and  $y$ , choose the first branch point  $z$  of a spanning tree of the three open connections. The three events  $0 \leftrightarrow z$ ,  $z \leftrightarrow x$ , and  $z \leftrightarrow y$  then occur disjointly. BK and the upper half of (21.13) give

$$\mathbf{E}N_r^2 \leq C \sum_{x,y \in Q_{2r}} \sum_{z \in \mathbb{Z}^d} (1 + |z|)^{2-d} (1 + |x - z|)^{2-d} (1 + |y - z|)^{2-d}. \quad (21.15)$$

For  $|z| \leq 4dr$ , each of the two sums over  $x$  and  $y$  is  $O(r^2)$ , and the remaining sum of  $(1 + |z|)^{2-d}$  is  $O(r^2)$ ; this contribution is  $O(r^6)$ . If  $|z| > 4dr$ , then  $|x - z|, |y - z| \geq |z|/2$ , so the tail is bounded by

$$Cr^{2d} \sum_{|z| > 4dr} |z|^{6-3d} = O(r^6).$$

Thus  $\mathbf{E}N_r^2 \leq Cr^6$ . Since  $N_r > 0$  implies  $0 \leftrightarrow \partial Q_r$ , Cauchy–Schwarz (equivalently Paley–Zygmund at level 0) and (21.14) yield

$$\gamma(r) \geq \mathbf{P}(N_r > 0) \geq \frac{(\mathbf{E}N_r)^2}{\mathbf{E}N_r^2} \geq cr^{-2}.$$

Notice that the power  $r^6$  in the second moment is essential; replacing it by  $r^4$  would incorrectly give a scale-independent lower bound.  $\square$

**Lemma 10.29** (Regular-boundary shell growth). *Assume, in addition to (21.13), the mean-field critical cluster tail*

$$\mathbf{P}_{p_c}(|C(0)| \geq n) \leq Cn^{-1/2}, \quad (21.16)$$

*and reflection/coordinate-permutation symmetry of the finite-range lattice. There is  $c_0 > 0$  such that, for all sufficiently large  $j$  and every  $L \geq j^{1/10}$ ,*

$$\mathbf{P}_{p_c}(X_j \geq L^2, A_j(L) \leq c_0 L^4) \leq (1 - c_0)\gamma(j). \quad (21.17)$$

*Proof.* We give the regular-boundary argument because this is the point at which the upper one-arm estimate is genuinely harder than the lower bound.

Fix a large mesoscopic integer  $K$ . A boundary vertex  $x \in \partial Q_j$  reached from 0 inside  $Q_j$  is called  $K$ -regular if, for every  $s \geq K$ , conditioning on the explored cluster inside  $Q_j$  leaves probability at least  $1 - e^{-(\log s)^2}$  that every local continuation cluster in  $x + Q_s$  has at most  $s^4(\log s)^7$  vertices. Let  $X_j^{\text{reg}}$  and  $X_j^{\text{irr}}$  count regular and irregular reached boundary vertices.

The needed regularity estimate is

$$\mathbf{P}(X_j \geq M, X_j^{\text{irr}} \geq X_j/2) \leq Cj^d e^{-c(\log M)^2}. \quad (21.18)$$

Here is the exploration proof of (21.18). Localize the definition at scale  $s$  to the box  $x + Q_{s^{4d^2}}$ . The bad local event has probability  $O(e^{-c(\log s)^4})$ : the cluster-size part follows from the tree-graph moment estimate  $\mathbf{E}C_{\max}^m \leq m!C^m s^{d-6+4m}$  with  $m \asymp (\log s)^4$ , and the excessive-number-of-crossings part follows by BK applied to disjoint crossings. Tile the explored boundary by translates of these localization boxes and reveal them sequentially. After conditioning on the past, the indicator that the next box is bad has conditional mean  $O(e^{-c(\log s)^4})$ ; Azuma–Hoeffding therefore gives an exponential tail for having a positive fraction of bad boxes. A union bound over dyadic  $s \geq K$  gives (21.18). The lower connection estimate needed when conditioning on a previously explored outside cluster follows by reflecting and permuting a path to a favorable face of  $Q_s$ ; its loss is at most



$e^{C(\log s)^2}$  and is absorbed by the fourth-power logarithmic tail above. This proves the regularity estimate uniformly in  $j$ .

Now condition on  $X_j^{\text{reg}} = M \geq L^2/2$ . Count admissible pairs  $(x, y)$  with  $x$  a regular reached boundary vertex,  $y \in x + Q_L$ , and with the first edge leaving  $Q_j$  at  $x$  pivotal for the connection  $0 \leftrightarrow y$ ; call the number of such pairs  $Y$ . A local modification in the fixed  $K$ -box at  $x$ , followed by the two-point lower bound summed over  $y \in x + Q_L$ , gives

$$\mathbf{E}[Y; X_j^{\text{reg}} = M] \geq cML^2 \mathbf{P}(X_j^{\text{reg}} = M). \quad (21.19)$$

For the second moment distinguish: the same pair, the same  $x$  with two different descendants, and two different boundary vertices. BK and the three-point/tree-graph decomposition bound these contributions respectively by  $CML^2$ ,  $CML^6$ , and  $CM^2L^4$ , times  $\mathbf{P}(X_j^{\text{reg}} = M)$ . Since  $M \geq L^2/2$ ,

$$\mathbf{E}[Y^2; X_j^{\text{reg}} = M] \leq CM^2L^4 \mathbf{P}(X_j^{\text{reg}} = M). \quad (21.20)$$

Conditional Paley–Zygmund applied to (21.19)–(21.20) therefore gives a constant  $c > 0$  such that

$$\mathbf{P}(Y \geq cML^2 \mid X_j^{\text{reg}} = M) \geq c. \quad (21.21)$$

Each descendant  $y$  counted by  $Y$  has a unique first pivotal exit, so it is counted for at most one  $x$  and hence  $A_j(L) \geq Y$ . Because  $M \geq L^2/2$ , (21.21) implies  $A_j(L) \geq cL^4$  with conditional probability at least  $c$ .

Finally split the event in (21.17) according as at least half of the reached boundary vertices are irregular. The irregular part is bounded by (21.18) with  $M = L^2$  and is  $o(\gamma(j))$ , using Lemma 10.28 and  $L \geq j^{1/10}$ . On the regular part sum (21.21) over  $M \geq L^2/2$ . Decreasing  $c_0$  and increasing the lower cutoff for  $j$  gives (21.17).  $\square$

**Lemma 10.30** (Kozma–Nachmias one-arm recursion). *Under the hypotheses of Lemma 10.29, there are  $c_1, C_1 > 0$  such that for every  $\lambda \in (0, 1]$  there is  $\varepsilon_0(\lambda) > 0$  for which*

$$\gamma(r(1 + \lambda)) \leq \frac{C_1}{\sqrt{\varepsilon} r^2} + \varepsilon^{3/5} r^2 \gamma(r) \gamma(\lambda r/2) + (1 - c_1) \gamma(r) \quad (21.22)$$

whenever  $0 < \varepsilon < \varepsilon_0(\lambda)$  and  $r$  is sufficiently large.

*Proof.* Put  $L = \varepsilon^{3/10} r$ . On the event  $0 \leftrightarrow \partial Q_{r(1+\lambda)}$  one of three alternatives occurs.

First,  $|C(0)| \geq \varepsilon r^4$ . By (21.16) this has probability at most  $C/(\sqrt{\varepsilon} r^2)$ .

Second, at the first radius  $j \in [r(1 + \lambda/4), r(1 + \lambda/2)]$  for which  $0 < X_j \leq L^2$ , explore  $C(0; Q_j)$ . At most  $L^2$  reached boundary vertices can continue to the outer boundary; after conditioning on the explored cluster the unexplored edges retain product law. Each continuation costs at most  $\gamma(\lambda r/2)$ , while reaching the chosen shell costs at most  $\gamma(r)$ . Hence this alternative is bounded by

$$L^2 \gamma(r) \gamma(\lambda r/2) = \varepsilon^{3/5} r^2 \gamma(r) \gamma(\lambda r/2).$$

Third, every such shell has  $X_j > L^2$  while  $|C(0)| < \varepsilon r^4$ . Choose shells spaced by  $L$ . There are  $\asymp \lambda \varepsilon^{-3/10}$  of them, whereas fewer than  $C \varepsilon^{-1/5}$  can have  $A_j(L) \geq c_0 L^4$ , because the shell annuli are disjoint and the total cluster has fewer than  $\varepsilon r^4$  vertices. Lemma 10.29 says that a shell with many boundary connections but small annular growth has probability at most a fixed fraction of the arm event. Summing its estimate over the spaced shells and applying Markov's inequality yields at most  $(1 - c_1) \gamma(r)$  once  $\varepsilon_0(\lambda)$  is small enough. Adding the three alternatives gives (21.22).  $\square$

**Lemma 10.31** (High-dimensional one-arm and separated multi-arm bounds). *Assume critical Bernoulli percolation is on a reflection- and coordinate-permutation-invariant finite-range lattice,  $d > 6$ , and satisfies (21.13) and (21.16). Then*

$$cR^{-2} \leq \mathbf{P}_{p_c}(0 \leftrightarrow \partial Q_R) \leq CR^{-2}. \quad (21.23)$$

Moreover, for every integer  $k \geq 1$  there is a separation constant  $K(k)$  such that, whenever  $y_1, \dots, y_k \in Q_{R/2}$  and  $|y_i - y_j| \geq K(k)$  for  $i \neq j$ ,

$$\mathbf{P}_{p_c}(\{y_1 \leftrightarrow \partial Q_R\} \circ \dots \circ \{y_k \leftrightarrow \partial Q_R\}) \asymp R^{-2k}. \quad (21.24)$$

*Proof.* The lower bound in (21.23) is Lemma 10.28. For the upper bound, fix  $\lambda > 0$  so small that  $(1 - c_1)(1 + \lambda)^2 < 1 - c_1/2$ , where  $c_1$  is from Lemma 10.30. Choose  $M$  large and set  $\varepsilon = M^{-20/11}$ . Assuming inductively that  $\gamma(s) \leq Ms^{-2}$  for  $s < r(1 + \lambda)$ , (21.22) gives

$$\gamma(r(1 + \lambda)) \leq r^{-2}(C_1 M^{10/11} + 4\lambda^{-2} M^{10/11} + (1 - c_1)M).$$

For  $M$  sufficiently large the first two coefficients are at most  $(c_1/2)M/(1 + \lambda)^2$ ; the choice of  $\lambda$  absorbs the last one. Thus  $\gamma(r(1 + \lambda)) \leq M[r(1 + \lambda)]^{-2}$  and induction over geometric scales proves the upper bound.

For (21.24), BK and (21.23) give the upper bound. The lower bound is an inverse-BK induction on  $k$ . Take two independent percolation configurations and merge, one at a time, only the edge variables with an endpoint in  $Q_R$ . Before any merge the  $(k - 1)$ -arm event in the first copy and the final arm in the second copy have product probability  $\asymp R^{-2k}$  by induction and (21.23). A loss at an edge  $e = (z, z')$  can occur only if one existing arm and the new arm both use  $e$ ; BK bounds this loss by

$$CR^{-2k} \sum_{i < k} \tau_{p_c}(y_i, z) \tau_{p_c}(y_k, z).$$

Summing over the merged edges and using (21.13),

$$\sum_z (1 + |z - y_i|)^{2-d} (1 + |z - y_k|)^{2-d} \leq C|y_i - y_k|^{4-d} \leq CK(k)^{4-d}.$$

Since  $d > 6$ , choose  $K(k)$  so large that the total loss over all  $i < k$  is less than half of the initial product probability. After all inner edges are merged, the surviving disjoint occurrence implies  $k$  disjoint arms in one configuration from  $y_1, \dots, y_k$  to  $\partial Q_R$ . This proves the lower bound and completes the induction.  $\square$

## 10.2.2 Regular boundary points and the Kozma–Nachmias arm mechanism

The one-arm upper bound is not a formal consequence of the two-point function. The source proof first manufactures many regular boundary points, then turns them into many descendants in a short outer annulus. We isolate those steps here.

**Definition 10.32** (Localized typicality and regularity). Fix a large  $K$ . A reached boundary vertex  $x \in \partial Q_j$  is  $(K, s)$ -typical if, after revealing the cluster inside  $Q_j$ , the continuation inside  $x + Q_{s^{4d^2}}$  has at most  $s^4(\log s)^7$  vertices and at most  $(\log s)^3$  disjoint crossings of every annulus between radii  $s$  and  $2s$ . It is  $K$ -regular if it is  $(K, s)$ -typical for every dyadic  $s \geq K$ .

**Lemma 10.33** (Local typicality and local-to-global regularity). *Assume the mean-field cluster-tail bound (21.16), the two-point estimate (21.13), and the BK/Reimer inequality. There are  $c, C > 0$  such that the conditional probability that a reached boundary vertex is not  $(K, s)$ -typical is at most  $Ce^{-c(\log s)^4}$ . Consequently, if  $X_j \geq M$ , then*

$$\mathbf{P}(X_j^{\text{irr}} \geq X_j/2, X_j \geq M) \leq Cj^d e^{-c(\log M)^2}. \quad (21.C75.11)$$

*Proof.* For the volume part, decompose the continuation cluster in  $x + Q_{s^{4d^2}}$  into intrinsic generations. The tree-graph inequality bounds its  $m$ th factorial moment by  $m!C^m s^{d-6+4m}$ . Taking  $m \asymp (\log s)^4$  and Markov's inequality gives an error  $e^{-c(\log s)^4}$ . For excessive annular crossings, BK bounds  $r$  disjoint crossings by the  $r$ th power of a two-point shell sum; choose  $r \asymp (\log s)^3$ . This has the same superpolynomial error.

Now tile  $\partial Q_j$  by translates of the localization box and reveal them in a deterministic order on each of finitely many shifted grids. Conditional on the past, the indicator that the next box is bad has conditional mean at most the preceding superpolynomial error. Apply the exponential martingale bound to each shifted grid and then sum over the grids and dyadic  $s$ . A positive density of irregular reached vertices therefore has probability bounded by (21.C75.11). This is the source's local-to-global regularity step; the shifts prevent a bad localization box from being repeatedly charged along the same boundary face.  $\square$

**Definition 10.34** (Admissible descendant pairs). Condition on the cluster in  $Q_j$ . For a regular reached vertex  $x \in \partial Q_j$  and  $y \in x + Q_L$ , call  $(x, y)$  admissible if the first edge leaving  $Q_j$  at  $x$  is pivotal for  $0 \leftrightarrow y$  and the continuation from that edge to  $y$  stays inside the localization box of  $x$ . Let  $Y$  be the number of admissible pairs.

**Lemma 10.35** (Admissible-pair moments with Reimer complement control). *Condition on  $X_j^{\text{reg}} = M$  with  $M \geq L^2/2$ . Then*

$$\mathbf{E}[Y \mid X_j^{\text{reg}} = M] \geq cML^2, \quad \mathbf{E}[Y^2 \mid X_j^{\text{reg}} = M] \leq CM^2L^4. \quad (21.C75.12)$$

Hence  $\mathbf{P}(Y \geq cML^2 \mid X_j^{\text{reg}} = M) \geq c$ .

*Proof.* For the first moment, at a regular  $x$  perform the bounded local surgery that opens one prescribed exit and closes the competing exits in a fixed  $K$ -neighbourhood. Regularity bounds the Radon–Nikodym cost uniformly. The sum over  $y \in x + Q_L$  of the two-point lower bound is  $\asymp L^2$ , giving the first inequality.

For the second moment, split ordered pairs of admissible pairs into: identical pairs; the same boundary vertex with two descendants; and two different boundary vertices. In the last case, after the two local surgeries, the two required continuations and the explored interior connection occur disjointly. Reimer's inequality for disjoint occurrence, applied also to complements of the local closing events, bounds the contribution by the product of the corresponding connection probabilities. The convolution estimate from (21.13) then gives respectively  $CML^2$ ,  $CML^6$ , and  $CM^2L^4$ . Since  $M \geq L^2/2$ , all three are  $O(M^2L^4)$ . Paley–Zygmund proves the final claim.  $\square$

**Lemma 10.36** (Regular shell growth at source strength). *Under (21.13), (21.16), and the hypotheses above, there is  $c_0 > 0$  such that for  $L \geq j^{1/10}$ ,*

$$\mathbf{P}(X_j \geq L^2, A_j(L) \leq c_0L^4) \leq (1 - c_0)\gamma(j) \quad (21.C75.13)$$

for all sufficiently large  $j$ .

*Proof.* By Lemma 10.33, the event that at least half the reached vertices are irregular is  $o(\gamma(j))$ , using the lower arm bound  $\gamma(j) \geq cj^{-2}$ . On the complement, put  $M = X_j^{\text{reg}} \geq L^2/2$  and apply Lemma 10.35. Every admissible descendant has a unique first pivotal exit and is charged to at most one  $x$ , so  $A_j(L) \geq Y$ . With probability at least  $c$ , therefore,  $A_j(L) \geq cML^2 \geq cL^4$ . Absorb the irregular error into the constant.  $\square$

**Lemma 10.37** (Two-copy inverse-BKR arm separation). *Assume the one-arm bound  $\gamma(R) \asymp R^{-2}$ . For each fixed  $k$  there exists  $K(k) < \infty$  such that, if  $y_1, \dots, y_k \in Q_{R/2}$  are pairwise separated by at least  $K(k)$ , then*

$$\mathbf{P}(\{y_1 \leftrightarrow \partial Q_R\} \circ \dots \circ \{y_k \leftrightarrow \partial Q_R\}) \geq c_k R^{-2k}. \quad (21.C75.14)$$

*Proof.* Proceed by induction. Take two independent configurations: the first already contains  $k-1$  separated arms and the second contains the last arm. Merge edge variables in  $Q_R$  one at a time. A merge can destroy disjointness only if one old arm and the new arm both depend on the same edge neighbourhood. Reimer's inequality bounds the total loss caused near  $z$  by

$$CR^{-2k} \sum_{i < k} \tau(y_i, z) \tau(y_k, z).$$

Summing in  $z$  and using (21.13) gives  $CR^{-2k} \sum_{i < k} |y_i - y_k|^{4-d}$ . Choose  $K(k)$  so that this is less than half the independent product probability. After all variables have been merged, the surviving event is the required disjoint occurrence in one configuration.  $\square$

**Proposition 10.38** (Kozma–Nachmias arm package). *If critical percolation satisfies the mean-field two-point estimate (21.13), the cluster tail (21.16), and the selected high-dimensional/spread-out lace hypotheses, then*

$$\gamma(R) \asymp R^{-2},$$

*and the separated  $k$ -arm probability is  $\asymp R^{-2k}$  for every fixed  $k$ .*

*Proof.* The lower one-arm bound is Lemma 10.28. Replace the compressed regularity input in Lemma 10.30 by Lemma 10.36; the same three-alternative recursion and geometric induction give the upper bound  $CR^{-2}$ . BK gives the upper multi-arm bound and Lemma 10.37 gives the lower one.  $\square$

**Theorem 10.39** (High-dimensional arm exponents). *Assume critical Bernoulli percolation is in the IIC lace regime of Definition 10.3, on a reflection- and coordinate-permutation-invariant finite-range lattice. Then the one-arm probability satisfies*

$$\mathbf{P}_{p_c}(0 \leftrightarrow \partial Q_R) \asymp R^{-2}.$$

*For every fixed  $k \geq 1$  there is  $K(k) < \infty$  such that, for any  $y_1, \dots, y_k \in Q_{R/2}$  with pairwise separation at least  $K(k)$ , the probability that the  $k$  events  $\{y_i \leftrightarrow \partial Q_R\}$  occur disjointly is comparable to  $R^{-2k}$ , uniformly for all sufficiently large  $R$ .*

*Proof.* The lace asymptotic (21.10) gives the two-point estimate (21.13), while SM-21.02.05 gives the mean-field cluster tail (21.16). The full Kozma–Nachmias regularity/admissible-pair/inverse-BKR mechanism is Proposition 10.38; its conclusions are exactly the two assertions above.  $\square$

**Definition 10.40** (Pioneer edges). Fix the first coordinate. For  $n \geq 1$  let  $H_n = \{x \in \mathbb{Z}^d : x_1 \geq n\}$ . An oriented finite-range edge  $e = \{y, z\}$ , with  $y_1 < z_1$ , is an  $x$ -pioneer if  $e$  is open and  $x$  is joined to  $y$  by an open path which, before its final edge, remains strictly in the half-space  $\{w : w_1 < z_1\}$ .

Write  $\mathcal{P}_x(n)$  for the  $x$ -pioneers which cross the level  $x_1 + n$ , and  $\mathcal{P}_x = \bigcup_{n \geq 1} \mathcal{P}_x(n)$ . If the interaction range is  $L$ , then a fixed edge belongs to at most  $L$  consecutive level sets and hence

$$\frac{1}{L} \sum_{n \geq 1} |\mathcal{P}_x(n)| \leq |\mathcal{P}_x| \leq \sum_{n \geq 1} |\mathcal{P}_x(n)|. \quad (21.25)$$

Set  $P_p(n) = \mathbf{E}_p |\mathcal{P}_0(n)|$ .

### 10.2.3 Restricted boxes and half-space percolation: the boundary kernel

Throughout this subsection  $p = p_c$ . Write

$$H = \{x \in \mathbb{Z}^d : x_1 \geq 0\}, \quad B_H(r) = Q_r \cap H,$$

and let

$$\mathcal{R}(r) = [0, r] \times [-4r, 4r]^{d-1}.$$

The constants below are allowed to depend on the dimension and, in the spread-out case, on the fixed interaction range. Enlarging every displayed boundary by a fixed multiple of that range changes no power and is understood when the model is not nearest-neighbour.

The proof uses only the mean-field inputs already owned in this chapter:

$$c(1 + |x - y|)^{2-d} \leq \tau_{p_c}(x, y) \leq C(1 + |x - y|)^{2-d}, \quad (10.1)$$

the critical cluster-tail bound (21.16), the one-arm theorem Lemma 10.31, and the regular-boundary exploration in Lemma 10.29. The point of the present subsection is that none of the extra boundary powers below is inferred formally from (10.1).

**Lemma 10.41** (Domain-local regularity and pivotal extension). *Let  $D_0 \subset D_1$  be boxes, half-boxes, rectangular boxes, or the annular regions obtained from them by deleting a concentric box, and suppose the separation between the relevant inner and outer boundaries is  $r$ . Let  $X^{\text{reg}}$  be the number of reached boundary vertices of  $D_0$  which are  $K$ -regular in the sense of Lemma 10.29. Conditional on  $X^{\text{reg}} = M$ , let  $Y$  count pairs  $(z, w)$  such that*

1.  $z$  is one of those  $M$  regular reached vertices;
2.  $w$  lies in a prescribed  $r$ -scale target box on the  $D_1 \setminus D_0$  side of  $z$ ;
3. the first edge leaving the already exposed cluster through  $z$  is pivotal for the connection from the original seed to  $w$ .

*If the target box remains a fixed positive fraction of  $r$  away from the forbidden part of the boundary, then*

$$\mathbb{E}[Y; X^{\text{reg}} = M] \geq cMr^2 \mathbb{P}(X^{\text{reg}} = M), \quad (10.2)$$

$$\mathbb{E}[Y^2; X^{\text{reg}} = M] \leq C(Mr^2 + Mr^6 + M^2r^4) \mathbb{P}(X^{\text{reg}} = M). \quad (10.3)$$

*Consequently, whenever  $M \geq r^2/2$ ,*

$$\mathbb{P}(Y \geq cMr^2 \mid X^{\text{reg}} = M) \geq c. \quad (10.4)$$

*The same conclusions hold for an inward target in an annulus and for a target on the top face of  $\mathcal{R}(r)$ .*

*Proof.* The proof of Lemma 10.29 was deliberately given in a form which is local in the unexplored edge set. We record why this makes the present extension legitimate rather than an appeal to an unstated “regularity theorem.” Once the cluster inside  $D_0$  has been exposed, all edges in  $D_1 \setminus D_0$  which were not queried remain independent Bernoulli variables. The  $K$ -regularity event says that, uniformly over this conditioning, the continuation cluster in every  $s$ -box around  $z$  has the same polynomial moment bounds used in (21.19)–(21.20).

For a fixed regular  $z$ , perform the same finite modification in its fixed  $K$ -neighbourhood as in the proof of (21.19), forcing a unique first exit through a chosen admissible edge. The cost of this modification is bounded above and below by constants depending only on  $K$  and the lattice. Summing the bulk lower bound (10.1) over the  $r$ -scale target box gives

$$\sum_{w: |w-z| \leq cr} \tau_{p_c}(z, w) \geq cr^2.$$

Keeping only target points a fixed fraction of  $r$  away from the forbidden boundary changes the sum by at most a constant factor. Summing over the  $M$  regular exit vertices gives (10.2).

For the second moment, order two admissible pairs according as their exit vertices and their continuation clusters agree. If both pairs are identical, the first moment gives  $CMr^2$ . If the exit vertex is the same but the two target points differ, expose their last common branch point and use the three-point/tree-graph inequality; the two free two-point sums contribute at most  $Cr^6$ . If the exits are distinct, the two pivotal modifications are disjoint and BK gives the product of the two  $Cr^2$  continuation sums. These three cases give (10.3). When  $M \geq r^2/2$  the last term dominates the other two, and conditional Paley–Zygmund gives (10.4).

Nothing in the argument used a particular orientation of the target box. A reflection/coordinate permutation gives the inward and top-face versions. This is the domain-local form of the regular-point/extensibility calculation already owned in (21.19)–(21.21).  $\square$

**Lemma 10.42** (Conditional exit union bound). *Let  $A_0 \subset A_1$  be finite induced subgraphs, let  $z \in A_0$ , and let  $F \subset A_1$  be at graph distance at least  $r$  from  $A_0$ . If exactly  $M$  vertices of  $\partial_{A_1} A_0$  are connected to  $z$  inside  $A_0$ , then, conditional on the explored cluster  $C_{A_0}(z)$ ,*

$$\mathbb{P}(z \xleftrightarrow{A_1} F \mid C_{A_0}(z)) \leq M \pi(r), \quad (10.5)$$

where  $\pi(r) = \mathbb{P}(0 \leftrightarrow \partial Q_r)$ .

*Proof.* After  $C_{A_0}(z)$  is revealed, the edges outside it remain product Bernoulli variables. Any path from  $z$  to  $F$  must leave through one of the  $M$  reached vertices of  $\partial_{A_1} A_0$ . From any such exit to  $F$  there is an open arm of length at least  $r$  in the still-unrevealed configuration. Translation invariance of the full lattice bounds each conditional probability by  $\pi(r)$ . A union bound over the  $M$  exits gives (10.5).  $\square$

**Lemma 10.43** (Uniform restricted boundary count). *Let*

$$X_r = \#\{z \in \partial Q_r : 0 \xleftrightarrow{Q_r} z\}.$$

*Under the mean-field assumptions of this chapter,*

$$\sup_{r \geq 1} \mathbb{E} X_r < \infty. \quad (10.6)$$

*The same estimate holds, up to a model-dependent constant, for a translated box and for a fixed-aspect-ratio rectangular box.*

*Proof.* Fix  $K$  large enough in the regularity construction of Lemma 10.29. The localized exploration used to prove (21.18) has the following expectation-level consequence: if  $X_r^{\text{reg}}$  and  $X_r^{\text{irr}}$  are the regular and irregular parts of  $X_r$ , then

$$\mathbb{E}X_r^{\text{irr}} \leq \frac{1}{4}\mathbb{E}X_r + Cr^{-10}. \quad (10.7)$$

Indeed, reveal the localization boxes in the same order as in the proof of (21.18). Conditional on the past, the probability that the next reached localization box is bad is  $O(e^{-c(\log K)^4})$ ; choose  $K$  so this is at most  $1/8$ . Vertices in overlapping localization boxes have uniformly bounded multiplicity after taking the finite collection of shifted grids used in that proof. The exceptional event that the localization reaches the outer cutoff has probability  $O(r^{-10})$  after increasing the cutoff exponent. Summing the conditional expectations gives (10.7).

Apply Lemma 10.41 with  $D_0 = Q_r$  and an outward target of scale  $r$  contained in  $Q_{3r} \setminus Q_r$ . Let  $Y$  be the resulting pivotal-pair count. Summing (10.2) over  $M$  gives

$$\mathbb{E}Y \geq cr^2\mathbb{E}X_r^{\text{reg}}. \quad (10.8)$$

Each target vertex  $w$  can be paired with at most a bounded number of first pivotal exits (the first exit of the explored cluster is unique, with only a bounded finite-range multiplicity). Hence

$$\mathbb{E}Y \leq C \sum_{w \in Q_{3r}} \mathbb{P}(0 \leftrightarrow w) \leq Cr^2,$$

where the last bound follows by summing (10.1). Thus  $\mathbb{E}X_r^{\text{reg}} \leq C$ . Together with (10.7), this gives

$$\mathbb{E}X_r \leq \mathbb{E}X_r^{\text{reg}} + \frac{1}{4}\mathbb{E}X_r + Cr^{-10},$$

and hence (10.6).

The proof is invariant under translation. A fixed-aspect-ratio rectangle is covered by finitely many coordinate boxes at comparable scale; the same regularity and pivotal-extension argument on each face gives the rectangular version.  $\square$

**Lemma 10.44** (Long-excursion suppression). *There is  $C < \infty$  such that, whenever  $x \in Q_{2r/3}$ ,*

$$\mathbb{P}(0 \leftrightarrow x \text{ by an open path that visits } \partial Q_r) \leq Cr^{2-d}. \quad (10.9)$$

*More generally, if  $|x| \geq 2r/3$ , the same bound follows directly from the bulk two-point upper estimate.*

*Proof.* For  $x \in Q_{2r/3}$  cut the path at its first reached vertex  $z \in \partial Q_r$ . The connection  $0 \xrightarrow{Q_r} z$  and the remaining connection from  $z$  to  $x$  occur disjointly, so BK gives

$$\mathbb{P}(\text{long excursion}) \leq \sum_{z \in \partial Q_r} \mathbb{P}(0 \xrightarrow{Q_r} z) \tau_{p_c}(z, x).$$

Here  $|z - x| \geq r/3$ , so (10.1) and Lemma 10.43 bound the last display by  $Cr^{2-d}$ . If  $|x| \geq 2r/3$ , then  $\tau_{p_c}(0, x) \leq C|x|^{2-d} \leq Cr^{2-d}$  already bounds every such connection.  $\square$

**Proposition 10.45** (Macroscopic restricted-box two-point estimate). *For every fixed  $M > 1$  there are constants  $c_M, C_M > 0$  such that, for all  $n \geq 1$  and all distinct  $x, y \in Q_n$ ,*

$$c_M(1 + |x - y|)^{2-d} \leq \mathbb{P}(x \xrightarrow{Q_{Mn}} y) \leq C_M(1 + |x - y|)^{2-d}. \quad (10.10)$$

*Proof.* The upper bound is immediate from the unrestricted estimate (10.1). We prove the lower bound in two stages.

*A large initial aspect ratio.* By translation take  $x = 0$  and put  $s = |y| \leq 2n$ . Choose a fixed  $M_0$  so large that  $C(M_0/4)^{2-d} \leq c/2$ , where  $c, C$  are the constants in (10.1) and Lemma 10.44. Any open connection  $0 \leftrightarrow y$  which is not contained in  $Q_{M_0 n}$  has a path visiting  $\partial Q_{M_0 n}$ . Since  $s \leq 2n \leq 2M_0 n/3$  for  $M_0 \geq 3$ , the long-excursion lemma gives

$$\mathbb{P}(0 \leftrightarrow y, 0 \xrightarrow{Q_{M_0 n}} y) \leq C(M_0 n)^{2-d} \leq \frac{c}{2}(1+s)^{2-d}$$

after increasing  $M_0$  once more; the finitely many  $s$  below the asymptotic cutoff are absorbed by decreasing the constant. Subtracting from the bulk lower bound proves (10.10) for  $M = M_0$ .

*Domain shrinking.* Assume the lower estimate is known with an aspect ratio  $A > 1$  and define  $A' = (A + 1)/2$ . It is enough to prove it with  $A'$ ; finitely many iterations then reach any prescribed  $M > 1$ .

Fix  $x, y \in Q_n$  and set  $s = 1 + |x - y|$ . If  $s$  is bounded there is nothing to prove, so suppose  $s$  is large. Around  $y$  choose an  $s$ -scale rectangular annulus whose outward face points toward the side of  $Q_{A'n}$  having the largest available margin. The inner and outer radii are fixed multiples of  $(A - 1)s$ ; their constants are chosen so that the entire annulus lies in  $Q_{A'n}$ , whereas a box of aspect ratio  $A$  around every point on its outer face still lies in  $Q_{An}$ . This is a deterministic geometry check; when  $y$  is close to one face, take the opposite coordinate direction.

Expose the cluster of  $y$  up to the inner boundary. The regularity estimate (21.18), Lemma 10.41, and Lemma 10.42 imply the following directed-arm statement: with probability at least  $cs^{-2}$  there are between  $cs^2$  and  $Cs^2$  regular outer-face vertices  $z$ , each reached from  $y$  inside the annulus, and the first exit at  $z$  can be chosen pivotal. The power  $s^{-2}$  is the already-owned bulk one-arm exponent; the lower the pivotal-extension probability is uniform because the target face is a fixed fraction of  $s$  away from the forbidden boundary.

For such a regular  $z$ , the induction hypothesis applied at scale comparable to  $s$  gives a connection from  $z$  to  $x$  inside the larger  $A$ -box with probability at least  $cs^{2-d}$ . The local pivotal modification makes this connection independent of the already exposed directed arm except for a bounded edge set. Let  $N$  count successful outer-face vertices. The first moment is therefore

$$\mathbb{E}[N \mid \text{directed arm data}] \geq cs^2 s^{2-d} = cs^{4-d}.$$

The same two-exit tree-graph decomposition as (10.3) gives

$$\mathbb{E}[N^2 \mid \text{directed arm data}] \leq C(s^{4-d} + s^{8-2d}).$$

For  $d > 6$  the first term dominates. Conditional Paley–Zygmund, followed by the directed-arm probability, therefore yields a successful glued connection with probability at least  $cs^{2-d}$ . Its construction lies in  $Q_{A'n}$ . This proves the shrinking step and hence (10.10) for every fixed  $M > 1$ .  $\square$

**Lemma 10.46** (Half-space arm lower bound and directed top arm). *Let*

$$\pi_H(r) = \mathbb{P}(0 \xleftrightarrow{H} \partial Q_r \cap H).$$

*Then*

$$\pi_H(r) \geq cr^{-3}. \quad (10.11)$$

*Moreover, if  $S_r = \{r\} \times [-4r, 4r]^{d-1}$  is the top face of  $\mathcal{R}(r)$ , then for some  $c, \eta > 0$ ,*

$$\mathbb{P}\left(0 \xleftrightarrow{\mathcal{R}(r)} S_r, \#\{z \in S_r : 0 \xleftrightarrow{\mathcal{R}(r)} z\} \geq \eta r^2\right) \geq cr^{-3}. \quad (10.12)$$



*Proof.* We reproduce the spanning-cluster second-moment mechanism because the extra factor  $r^{-1}$  relative to the bulk one-arm exponent is a boundary effect. In the annulus  $Q_{2r} \setminus Q_r$ , call a spanning cluster good if its volume lies in  $[\eta r^4, \eta^{-1} r^4]$  and it has at least  $\eta r^2$  reached vertices on each of two opposite  $r$ -scale faces after a coordinate reflection. The bulk cluster-tail estimate, the two-point lower bound, and the tree-graph second-moment estimate used in Lemma 10.28 imply, for  $\eta$  sufficiently small,

$$\begin{aligned} \mathbb{E} N_{\text{good}} &\geq c\eta r^{d-6}, \\ \mathbb{E} N_{\text{good}}^2 &\leq C\eta^{-2} r^{2d-12}. \end{aligned}$$

Thus Paley–Zygmund gives probability bounded below that there are at least  $c\eta r^{d-6}$  good spanning clusters. On that event their boundary traces are disjoint and contain at least  $c\eta^2 r^{d-4}$  vertices in total.

Average this count over the  $O(r^{d-1})$  possible boundary starting vertices and reflect the corresponding spanning path so that it stays on the allowed side of the boundary hyperplane. Translation invariance parallel to the boundary gives

$$r^{d-1} \pi_H(r) \geq c r^{d-4},$$

which is (10.11). Keeping only the face in the positive first-coordinate direction in the same counting argument gives (10.12); its  $\eta r^2$  top vertices are precisely the regular face trace of a good spanning cluster after discarding the irregular fraction using (21.18).  $\square$

**Proposition 10.47** (Half-space extensibility). *There are constants  $c_*, K < \infty$  with the following property. In each of the three geometries below, condition on an explored cluster whose relevant reached boundary contains  $M$   $K$ -regular vertices:*

1. *an outward annulus  $B_H(2r) \setminus B_H(r)$ ;*
2. *the rectangular slab between the side/top boundaries of  $\mathcal{R}(r)$  and  $\mathcal{R}(2r)$ ;*
3. *an inward annulus obtained by deleting a half-box from a box of comparable radius.*

*If the available continuation width is  $s$ , then with conditional probability at least  $c_*$  the cluster acquires at least  $c_* M s^2$  new reached vertices on the prescribed target boundary. The conclusion remains true after changing  $M$  or  $s$  by fixed multiplicative constants.*

*Proof.* Use the pivotal-pair variable  $Y$  of Lemma 10.41. In each of the three geometries, the deterministic target box attached to a regular exit can be chosen with distance comparable to  $s$  from all forbidden boundaries. Hence (10.2)–(10.3) hold with the same powers. If  $M \geq s^2/2$ , (10.4) gives the assertion immediately. If  $M < s^2/2$ , partition the target into  $O(s^2/M)$  disjoint mesoscopic boxes of width comparable to  $M^{1/2}$  and apply the same estimate inside each box; a second application of Paley–Zygmund to the number of successful boxes gives the same  $c_* M s^2$  total after adjusting constants.

The only possible obstruction is that the finite local modification needed to make the exit edge pivotal could touch a previously exposed boundary. In that case move the attachment point a fixed distance into the unexplored region; regularity and finite range make the cost uniformly bounded. This is why the three geometries above, although oriented differently, have identical first- and second-moment estimates.  $\square$

**Lemma 10.48** (Half-space one-arm upper bound). *Under the hypotheses above,*

$$\pi_H(r) \leq Cr^{-3}. \quad (10.13)$$

*Proof.* We give the mass-transport recursion in the form needed later. Translate the origin to the boundary plane and, on the event of an arm to distance  $r$ , let  $A_{\text{out}}(r)$  be the set of vertices in the shell between radii  $r$  and  $2r$  which are connected to the origin by a path avoiding the box immediately behind the boundary plane. Send from each such vertex one unit of mass to the first boundary vertex of the avoided box hit by its full-lattice cluster. Translation invariance parallel to the boundary and the bulk estimate (10.1) give

$$\mathbb{E}[\text{mass received at a fixed boundary vertex}] \leq Cr. \quad (10.14)$$

Indeed, after cutting at the last common branch point the contribution is bounded by a convolution of two bulk two-point functions over an  $r$ -shell; the radial sum is  $O(r)$  in  $d > 6$ .

The extensibility proposition says that, conditional on an arm whose regular boundary trace has size at least  $\epsilon^{3/5}r^2$ , either the arm dies in the next  $\lambda r$  layer with probability at least  $c$ , or with probability at least  $c$  it creates  $c\epsilon^{3/5}r^4$  vertices of  $A_{\text{out}}(r)$ . Markov's inequality applied to (10.14), together with the irregular-boundary tail (21.18), yields the recursion

$$\pi_H(r(1+\lambda)) \leq \frac{C}{\epsilon r^3} + C\epsilon^{3/5}\lambda^{-2}\pi_H(r) + (1-c)\pi_H(r) \quad (10.15)$$

for every fixed  $\lambda \in (0, 1]$  and all sufficiently small  $\epsilon$ . Here the first term is the event that the required transport exceeds its allowed  $O(r)$  expectation, the second is the small regular trace followed by an independent continuation (Lemma 10.42), and the last term is the extinction alternative.

Choose  $\lambda$  so that  $(1-c)(1+\lambda)^3 < 1-c/2$ , then choose a large induction constant  $M$  and take  $\epsilon = M^{-5/2}$  (any sufficiently small negative power of  $M$  works). Assuming  $\pi_H(s) \leq Ms^{-3}$  below the next geometric scale, (10.15) gives

$$\pi_H(r(1+\lambda)) \leq r^{-3}(o(M) + (1-c)M) \leq M[r(1+\lambda)]^{-3}.$$

Induction over the geometric scales proves (10.13). Combined with Lemma 10.46, this also gives  $\pi_H(r) \asymp r^{-3}$ .  $\square$

**Lemma 10.49** (Boundary exit moment and regular top-arm event). *Let  $Q_{\text{side}}$  be the part of  $\partial\mathcal{R}(r)$  other than the top face  $S_r$ , with the boundary-plane part removed, and set*

$$X_{\text{side}} = \#\{z \in Q_{\text{side}} : 0 \xleftrightarrow{\mathcal{R}(r)} z\}, \quad X_{\text{top}}^{\text{reg}} = \#\{z \in S_r : 0 \xleftrightarrow{\mathcal{R}(r)} z, z \text{ is } K\text{-regular}\}.$$

*Then*

$$\mathbb{E}X_{\text{side}} \leq Cr^{-1}, \quad (10.16)$$

*and for some  $c, \eta > 0$ ,*

$$\mathbb{P}(\eta r^2 \leq X_{\text{top}}^{\text{reg}} \leq X_{\text{side}} + Cr^2 \leq \eta^{-1}r^2) \geq cr^{-3}. \quad (10.17)$$

*Equivalently, after adjusting the fixed rectangle convention, there is an event of probability  $cr^{-3}$  on which the top trace contains between  $\eta r^2$  and  $\eta^{-1}r^2$  regular vertices and all non-top exits together have size at most  $\eta^{-1}r^2$ .*

*Proof.* For the first estimate, split according as the reached side trace is at most  $r^2$  or larger. On the small-trace event its contribution is at most  $r^2\pi_H(r) \leq Cr^{-1}$  by Lemma 10.48. On the complementary event, Proposition 10.47 creates with fixed conditional probability  $c$  times the side trace times  $r^2$  vertices in the next rectangular shell. The mass-transport estimate (10.14), summed over the  $O(r^{d-1})$  possible base translations and divided by the same transverse volume, bounds the expected number of such descendants by  $Cr$ . Therefore the contribution of the large-trace event to  $\mathbb{E}X_{\text{side}}$  is also  $O(r^{-1})$ . This gives (10.16).

For (10.17), start with the directed top-arm event (10.12), of probability at least  $cr^{-3}$ . Choose  $K$  so large that the irregular trace estimate (21.18), in its rectangular version from Lemma 10.41, removes at most half that probability. Markov's inequality and (10.16) show that the probability of  $X_{\text{side}} > \eta^{-1}r^2$  is  $O(\eta r^{-3})$  after using the arm event to localize to the relevant scale. Taking  $\eta$  small leaves a probability  $cr^{-3}$  event on which the top trace contains at least  $\eta r^2$  regular vertices and all relevant traces are at most  $\eta^{-1}r^2$ .  $\square$

**Theorem 10.50** (Half-space two-point kernel). *In nearest-neighbour critical percolation on  $\mathbb{Z}^d$ ,  $d \geq 11$ , and in the sufficiently spread-out finite-range model in each fixed  $d > 6$ , let  $u \in \partial H$  and  $v \in H$ . Then*

$$\mathbb{P}(u \xleftrightarrow{H} v) \leq C(1 + |u - v|)^{1-d}, \quad (10.18)$$

$$\mathbb{P}(u \xleftrightarrow{H} v) \leq C(1 + |u - v|)^{-d} \quad (v \in \partial H), \quad (10.19)$$

$$\mathbb{P}(u \xleftrightarrow{H} v) \geq c(1 + |u - v|)^{1-d} \quad \text{if } |u - v| \leq 2v_1. \quad (10.20)$$

*Proof.* By translation parallel to the boundary set  $u = 0$  and put  $R = 1 + |v|$ .

*Boundary-to-interior upper bound.* Choose  $r \asymp R$  so that  $v$  lies outside  $\mathcal{R}(r)$ . Every  $H$ -path from 0 to  $v$  has a first exit  $z$  through the non-boundary-plane part of  $\partial\mathcal{R}(r)$ . BK at this first exit gives

$$\mathbb{P}(0 \xleftrightarrow{H} v) \leq \sum_z \mathbb{P}(0 \xleftrightarrow{\mathcal{R}(r)} z) \tau_{p_c}(z, v).$$

All such  $z$  are at distance comparable to  $R$  from  $v$ , so the bulk upper bound contributes  $CR^{2-d}$ . The sum of the restricted exit probabilities is  $\mathbb{E}X_{\text{side}} \leq Cr^{-1}$  by Lemma 10.49. Thus the last display is at most  $CR^{1-d}$ , proving (10.18).

*Boundary-to-boundary upper bound.* When  $v \in \partial H$ , cut a connecting path at its first exit from an  $r$ -scale rectangle around 0, with  $r \asymp R$ . From an exit  $z$  to the boundary point  $v$ , apply the just-proved boundary-to-interior estimate after reflection/translation; it is  $O(R^{1-d})$ . The first-exit sum is again  $O(r^{-1})$ , hence  $O(R^{-d})$ . This proves (10.19).

*Non-tangential lower bound.* Assume  $v_1 \geq R/2$ . Choose  $r \asymp R$  and orient  $\mathcal{R}(r)$  so its top face lies a distance comparable to  $R$  below  $v$ . On the event in (10.17), which has probability at least  $cr^{-3}$ , there are  $M \asymp r^2$  regular top vertices  $z$  reached from the origin. Around each such  $z$  choose a box of radius  $ar$ , with  $a > 0$  fixed small enough that the box lies inside  $H$  and contains  $v$  after a bounded chain of overlapping boxes. Proposition 10.45 then gives, uniformly in  $z$ ,

$$\mathbb{P}(z \leftrightarrow v \text{ in the corresponding interior box}) \geq cr^{2-d}.$$

Use the pivotal local modification from Lemma 10.41 to make the top exit the first attachment point. If  $N$  denotes the number of successful attachments, then conditionally on the regular-top event,

$$\mathbb{E}N \geq cMr^{2-d} \geq cr^{4-d}.$$

For two distinct exits, the tree-graph/BK decomposition used in (10.3) gives

$$\mathbb{E}N^2 \leq C(r^{4-d} + r^{8-2d}) \leq Cr^{4-d}$$

for  $d > 6$ . Conditional Paley–Zygmund therefore gives  $\mathbb{P}(N > 0 \mid \text{regular-top event}) \geq cr^{4-d}$ . Multiplying by the probability  $cr^{-3}$  of the top event yields

$$\mathbb{P}(0 \xleftrightarrow{H} v) \geq cr^{-3}r^{4-d} = cr^{1-d} \asymp R^{1-d},$$

which is (10.20).  $\square$

**Proposition 10.51** (Restricted half-space two-point estimates). *Let  $H = \{x \in \mathbb{Z}^d : x_1 \geq 0\}$ , and consider the same high-dimensional Bernoulli percolation regimes as in Theorem 10.60. At  $p = p_c$ , assume the bulk mean-field two-point estimate*

$$\tau_{p_c}(x, y) \asymp (1 + |x - y|)^{2-d}.$$

*Then there is  $C < \infty$  such that, for every boundary point  $u$  with  $u_1 = 0$  and every  $v \in H$ ,*

$$\mathbf{P}_{p_c}(u \xleftrightarrow{H} v) \leq C(1 + |u - v|)^{1-d}, \quad (21.25a)$$

*and, if  $u_1 = v_1 = 0$ ,*

$$\mathbf{P}_{p_c}(u \xleftrightarrow{H} v) \leq C(1 + |u - v|)^{-d}. \quad (21.25b)$$

*Moreover there is  $c > 0$  such that*

$$\mathbf{P}_{p_c}(u \xleftrightarrow{H} v) \geq c(1 + |u - v|)^{1-d} \quad (21.25c)$$

*whenever  $u_1 = 0$  and  $|u - v| \leq 2v_1$ .*

*Proof.* Theorem 10.50 proves the three estimates at the exact lattice scopes stated here. Its proof is internal: the only inherited inputs are the already-owned bulk two-point/cluster-tail/one-arm estimates and the regular-boundary exploration of Lemma 10.29. Equations (10.18), (10.19), and (10.20) are respectively (21.25a), (21.25b), and (21.25c).  $\square$

**Lemma 10.52** (Critical pioneers and the OSSS transfer). *Assume Proposition 10.51. In the high-dimensional percolation regimes of Theorem 10.60, there are constants  $c, C > 0$  such that, uniformly for  $p_c/2 \leq p \leq p_c$ ,  $n, k \geq 1$  and  $\varepsilon = p_c - p$ ,*

$$c \leq P_{p_c}(n) \leq C, \quad (21.26)$$

$$\mathbf{P}_{p_c}(|\mathcal{P}_0| \geq k) \leq Ck^{-2/3}, \quad (21.27)$$

$$\left(\frac{d}{dp}\right)_+ \log \mathbf{P}_p(|\mathcal{P}_0| \geq k) \geq \frac{1}{2p(1-p)} \left( \frac{k(1-e^{-1})}{\sum_{j=0}^k \mathbf{P}_p(|\mathcal{P}_0| \geq j)} - 1 \right), \quad (21.28)$$

$$\mathbf{E}_p|\mathcal{P}_0| \leq C\varepsilon^{-1/2}, \quad (21.29)$$

$$\mathbf{P}_p(|\mathcal{P}_0| \geq k) \leq Ck^{-2/3}e^{-c\varepsilon^{3/2}k}, \quad (21.30)$$

$$P_p(n) \leq Ce^{-c\sqrt{\varepsilon}n}. \quad (21.31)$$

Here  $(d/dp)_+$  denotes the lower-right Dini derivative.

*Proof.* We record the proof because the factor  $R^{-2}$  in the near-critical arm probability does not follow by inserting a mass into the critical second-moment calculation.

*Critical level estimate.* Apply Proposition 10.51. In the notation there, (21.25a)–(21.25b) are precisely the upper half-space estimates needed below, while (21.25c) supplies the non-tangential lower estimate. We emphasize that this is the only point in the pioneer proof at which restricted half-space percolation enters; all subsequent steps are derived explicitly from those three inequalities, BK, FKG, and the already-owned bulk triangle estimate.

Let  $\bar{P}(n) = \sum_{j=n}^{n+L-1} P_{pc}(j)$  and  $\bar{P}^*(n) = \max_{j \leq n} \bar{P}(j)$ . Summing (21.25a)–(21.25b) over the first boundary contact of a pioneer gives  $\bar{P}^*(n) \leq C \log(n+2)$ . A second cut at the first pivotal pioneer, together with BK, gives constants  $a \in \mathbb{N}$  and  $c_0 > 0$  for which

$$\bar{P}^*(2n+a) \geq c_0 \bar{P}^*(n)^2. \quad (21.34)$$

If  $\bar{P}^*$  were unbounded, (21.34) would make it grow doubly exponentially along a geometric sequence of scales, contradicting the logarithmic upper bound. Hence  $P_{pc}(n) \leq C$ . The lower bound follows from the event that a fixed finite-range crossing edge and a prescribed path to it are open, or equivalently from the critical mass-zero limit of the submultiplicative pioneer estimate. This proves (21.26).

*Critical pioneer tail.* On the event that 0 does not reach  $H_m$ , every pioneer belongs to one of the first  $m + O(L)$  levels. Markov's inequality, (21.26), and the critical one-arm bound (21.23) therefore give

$$\mathbf{P}_{pc}(|\mathcal{P}_0| \geq k) \leq \frac{Cm}{k} + \frac{C}{m^2}.$$

Taking  $m \asymp k^{1/3}$  proves (21.27).

*OSSS differential inequality.* Adjoin an independent ghost field which marks each edge green with probability  $1 - e^{-h}$ . Explore, from every green pioneer, backwards through the open cluster until all information deciding the event  $\{|\mathcal{P}_0| \geq k\}$  has been revealed. The Bernoulli OSSS inequality proved in Volume X, followed by Russo differentiation, bounds the revealment of an edge by

$$1 - e^{-h} + 2\mathbf{E}_p(1 - e^{-h|\mathcal{P}_0|}).$$

Choose  $h = 1/k$ , sum the influences, and use  $1 - e^{-j/k} \geq (1 - e^{-1})j/k$  for  $0 \leq j \leq k$ . After division by the event probability this is exactly (21.28). The use of a Dini derivative is harmless: finite-box truncations satisfy the ordinary Russo formula and monotone convergence gives the displayed lower-right derivative inequality.

*Integration.* Because the tail is increasing in  $p$ , insert (21.27) into the denominator of (21.28). Since  $\sum_{j \leq k} (j+1)^{-2/3} \leq Ck^{1/3}$ ,

$$\left(\frac{d}{dp}\right)_+ \log \mathbf{P}_p(|\mathcal{P}_0| \geq k) \geq ck^{2/3} - C.$$

Integrating from  $p$  to  $p_c$ , then summing the resulting tail estimate up to  $k \asymp \varepsilon^{-3/2}$ , yields (21.29). Returning (21.29) to (21.28) and integrating once more gives (21.30).

Finally, the finite interaction range implies the pioneer submultiplicativity

$$P_p(n+m) \leq C_L P_p(n) P_p(m). \quad (21.35)$$

By (21.25) and (21.29), among the first  $N \asymp \varepsilon^{-1/2}$  levels there is an  $n_p \leq N$  with  $P_p(n_p) \leq q < 1$ , after increasing the implicit constant in  $N$  so that  $q$  is fixed. Iterating (21.35), and absorbing the bounded remainder of  $n$  modulo  $n_p$ , proves (21.31).  $\square$

**Lemma 10.53** (Pioneers imply the sharp near-critical Euclidean bounds). *Under the hypotheses of Lemma 10.52, for  $\varepsilon = p_c - p \in [0, p_c/2]$ ,*

$$\tau_p(0, x) \leq C(1 + |x|)^{2-d} e^{-c\sqrt{\varepsilon}|x|}, \quad (21.36)$$

$$cR^{-2} e^{-C\sqrt{\varepsilon}R} \leq \mathbf{P}_p(0 \leftrightarrow \partial Q_R) \leq CR^{-2} e^{-c\sqrt{\varepsilon}R}. \quad (21.37)$$

*Proof.* Rotate coordinates so that  $|x_1| \geq |x|/\sqrt{d}$ . On  $\{0 \leftrightarrow x\}$  choose the first pioneer crossing each of two well-separated hyperplanes between 0 and  $x$ . Cutting at those pioneers, BK factorizes the event into two bulk critical connection pieces and a pioneer piece. Summing the bulk pieces with (21.10) gives the critical prefactor  $(1 + |x|)^{2-d}$ , while (21.31) supplies the exponential factor. This proves (21.36).

For the upper bound in (21.37), if 0 reaches  $\partial Q_{2R}$  then some pioneer crosses level  $R$  and its endpoint has a critical arm of length comparable with  $R$ . BK, (21.23), and (21.31) give  $CR^{-2} e^{-c\sqrt{\varepsilon}R}$ .

For the lower bound we use the slab second moment, which is the point at which the sharp  $R^{-2}$  prefactor is recovered. Write  $S_j = \{x : x_1 = j\}$  and  $S_{a,b} = \bigcup_{a \leq j \leq b} S_j$ . Let  $X_R$  count vertices of  $S_R$  joined to 0 by a path contained in  $S_{-\infty, R}$ , and let  $Y_R \leq X_R$  count those joined by a path contained in the finite slab  $S_{-R, R}$ . For nearest-neighbour percolation,

$$\mathbf{E}_p X_R = p^{-1} P_p(R+1),$$

so (21.31) and the matching lower pioneer exponential rate give

$$e^{-C\sqrt{\varepsilon}R} \leq \mathbf{E}_p X_R \leq e^{-c\sqrt{\varepsilon}R}. \quad (21.37a)$$

If a half-space connection counted by  $X_R$  leaves  $S_{-R, R}$  before reaching  $S_R$ , cut at its first visit to  $S_{-R}$ . BK and reflection symmetry then give

$$\mathbf{E}_p X_R \leq \mathbf{E}_p Y_R + \mathbf{E}_p Y_R \mathbf{E}_p X_{2R}. \quad (21.37b)$$

Since the final factor is uniformly bounded, (21.37a)–(21.37b) imply  $\mathbf{E}_p Y_R \geq ce^{-C\sqrt{\varepsilon}R}$ .

For the second moment, if  $x, y \in S_R$  are both counted by  $Y_R$ , the three paths from 0 to their last common branching point  $z$  and from  $z$  to  $x, y$  can be chosen disjointly. Hence BK gives

$$\mathbf{E}_p Y_R^2 \leq \sum_{j=-R}^R \sum_{z \in S_j} \mathbf{P}_p(0 \xleftrightarrow{S_{-R, R}} z) \left( \sum_{x \in S_R} \mathbf{P}_p(z \xleftrightarrow{S_{-R, R}} x) \right)^2. \quad (21.37c)$$

The inner sum is at most  $Ce^{-c\sqrt{\varepsilon}(R-j)}$  by the pioneer estimate. To sum over  $z$ , cut a path from 0 to  $z$  at its right-most point  $w \in S_a$ ,  $a \geq j \vee 0$ , and apply BK once more. The same pioneer estimate gives

$$\sum_{z \in S_j} \mathbf{P}_p(0 \xleftrightarrow{S_{-R, R}} z) \leq CR e^{-c\sqrt{\varepsilon}|j|}. \quad (21.37d)$$

Substituting (21.37d) into (21.37c) and summing  $j$  yields

$$\mathbf{E}_p Y_R^2 \leq CR^2 e^{-c\sqrt{\varepsilon}R}. \quad (21.37e)$$

Cauchy–Schwarz therefore gives

$$\mathbf{P}_p(Y_R > 0) \geq \frac{(\mathbf{E}_p Y_R)^2}{\mathbf{E}_p Y_R^2} \geq cR^{-2} e^{-C\sqrt{\varepsilon}R}.$$

Since  $\{Y_R > 0\} \subset \{0 \leftrightarrow \partial Q_R\}$ , this proves the lower bound in (21.37). The finite-range case uses the same cuts with an  $O(L)$ -thick family of hyperplanes.  $\square$

**Lemma 10.54** (Periodic translate and diagram sums). *Assume the slightly subcritical estimate (21.36), write  $V = r^d$ , and put*

$$\psi_{r,p}(x) = \sum_{u \in \mathbb{Z}^d \setminus \{0\}} \tau_p(x + ru), \quad x \in \mathbb{T}_r^d.$$

There are constants  $c, C > 0$  such that, for  $p < p_c$ ,

$$\psi_{r,p}(x) \leq C \frac{\chi(p)}{V} e^{-cm(p)r}, \quad (21.38a)$$

$$\sum_{u \in \mathbb{Z}^d} \mathbb{B}_p(x + ru) \leq \mathbb{B}_p(x) + C \frac{\chi(p)^2}{V} e^{-cm(p)r}, \quad (21.38b)$$

$$\sum_{u \in \mathbb{Z}^d} \mathbb{T}_p(x + ru) \leq \mathbb{T}_p(x) + C \frac{\chi(p)^3}{V} e^{-cm(p)r}. \quad (21.38c)$$

Here  $\mathbb{B}_p = \tau_p * \tau_p$  and  $\mathbb{T}_p = \tau_p * \tau_p * \tau_p$ .

*Proof.* From (21.36) and convolution of the power kernels,

$$\mathbb{B}_p(x) \leq C(1 + |x|)^{4-d} e^{-cm(p)|x|}, \quad \mathbb{T}_p(x) \leq C(1 + |x|)^{6-d} e^{-cm(p)|x|}.$$

For  $x$  in the standard fundamental cube and  $u \neq 0$ ,  $|x + ru| \asymp r|u|$ . Hence, for  $a > 0$  and  $\nu > 0$ ,

$$\sum_{u \neq 0} |x + ru|^{a-d} e^{-\nu|x+ru|} \leq C_a r^{a-d} e^{-\nu r/4} \sum_{N \geq 1} N^{a-1} e^{-\nu r N/4} \leq C_a \nu^{-a} r^{-d} e^{-\nu r/4}.$$

Apply this with  $a = 2, 4, 6$ . The mean-field mass-susceptibility comparison  $m(p)^{-2} \leq C\chi(p)$  turns  $m^{-2a/2}$  into  $\chi^{a/2}$  and gives (21.38a)–(21.38c).  $\square$

**Lemma 10.55** (Periodic coupling defect below the window). *Let  $\psi_{r,p}^{\mathbb{T}}(x) = \tau_p^{\mathbb{T}}(x) - \tau_p(x)$  under the standard quotient coupling. For  $p < p_c$ ,*

$$0 \leq \psi_{r,p}(x) - \psi_{r,p}^{\mathbb{T}}(x) \leq C \frac{\chi(p)}{V} \left( \mathbb{T}_p(x) + \frac{\chi(p)^3}{V} \right). \quad (21.38d)$$

Moreover, there are  $A_2, c_\psi > 0$  such that, when  $p_c - A_2 r^{-2} \leq p < p_c$ ,

$$\psi_{r,p}(x) \geq c_\psi \frac{\chi(p)}{V}. \quad (21.38e)$$

Consequently there are  $A_1, M > 0$  such that, if  $p \leq p_c - A_1 V^{-1/3}$  and  $\|x\|_\infty \geq M$ , then

$$\psi_{r,p}(x) - \psi_{r,p}^{\mathbb{T}}(x) \leq \frac{1}{2} c_\psi \frac{\chi(p)}{V}. \quad (21.38f)$$

*Proof.* We give the coupling calculation because it is the part hidden by the previous “unique winding” sentence. In the quotient coupling,

$$\psi_{r,p}(x) - \psi_{r,p}^{\mathbb{T}}(x) \leq \frac{1}{2} \sum_u \sum_{v \neq u} \mathbf{P}_p(0 \leftrightarrow x + ru, 0 \leftrightarrow x + rv) + \sum_u \mathbf{P}_p(0 \leftrightarrow x + ru, 0 \not\leftrightarrow_{\mathbb{T}} x). \quad (21.38g)$$

For the first term, expose the last common branching vertex  $z$  and apply BK. Writing  $z = x + y + rw$  and then using the periodic-sum estimate twice gives

$$\sum_u \sum_{v \neq u} \mathbf{P}_p(0 \leftrightarrow x + ru, 0 \leftrightarrow x + rv) \leq C \frac{\chi}{V} \left( \mathbf{B}_p(x) + \frac{\chi^2}{V} \right). \quad (21.38h)$$

For the second term, failure of the torus image connection implies that the lifted cluster has encountered two distinct translates before the final lift. Choosing the first branching point  $z$ , the repeated torus vertex  $a$ , and the two distinct translates gives the BK inclusion

$$\{0 \leftrightarrow z\} \circ \{z \leftrightarrow a + rv_1\} \circ \{z \leftrightarrow a + rv_2\} \circ \{a + rv_2 \leftrightarrow x + ru\}, \quad v_1 \neq v_2.$$

Summing first over  $u, v_1, v_2$ , the same periodic estimate yields

$$\sum_u \mathbf{P}_p(0 \leftrightarrow x + ru, 0 \not\leftrightarrow_{\mathbb{T}} x) \leq C \frac{\chi}{V} \left( \mathbf{T}_p(x) + \frac{\chi^3}{V} \right). \quad (21.38i)$$

Since the bubble is bounded by the triangle and  $\chi \geq 1$ , (21.38d) follows.

For (21.38e), Russo's formula and BK give

$$0 \leq \tau_{p_c}(y) - \tau_p(y) \leq C(p_c - p)(1 + |y|)^{4-d}. \quad (21.38j)$$

Sum this estimate over translates  $x + ru$  with  $1 \leq \|u\|_{\infty} \leq R$ . The critical lower two-point bound gives a main term  $cr^{2-d}R^2$  and (21.38j) gives an error  $C(p_c - p)r^{4-d}R^4$ . With  $R^2 = [2C(p_c - p)r^2]^{-1}$  the error is at most half the main term, so

$$\psi_{r,p}(x) \geq c[(p_c - p)V]^{-1} \geq c_{\psi}\chi(p)/V,$$

using  $\chi(p) \asymp (p_c - p)^{-1}$  in the mean-field regime. Finally, (21.38d), the diagram bounds and  $\chi(p) \leq CV^{1/3}/A_1$  when  $p_c - p \geq A_1V^{-1/3}$  show that the right-hand side of (21.38d), divided by  $\chi/V$ , is as small as desired by first taking  $M$  and then  $A_1$  large. This proves (21.38f).  $\square$

**Lemma 10.56** (Critical torus upper bound). *At  $p = p_c$ ,*

$$\tau_{p_c}^{\mathbb{T}}(x) \leq \tau_{p_c}(x) + CV^{-2/3}. \quad (21.38k)$$

*Proof.* Fix a large integer  $M$ . Split a simple torus connection from 0 to  $x$  according to whether its lift has Euclidean diameter at most or at least  $3Mr$ . A short lift ends at  $x + ru$  with  $\|u\|_{\infty} \leq 3M$ , and therefore its probability is at most

$$\tau_{p_c}(x) + CM^2r^{-(d-2)}.$$

A long lift forces disjoint Euclidean one-arm events of radius  $Mr$  from the two endpoints. BK and (21.23) bound this by  $CM^{-4}r^{-4}$ . Hence

$$\tau_{p_c}^{\mathbb{T}}(x) \leq \tau_{p_c}(x) + CM^2r^{-(d-2)} + CM^{-4}r^{-4}.$$

Choose  $M = \lfloor r^{(d-6)/6} \rfloor$  (with a fixed lower cutoff). Both errors are then  $O(r^{-2d/3}) = O(V^{-2/3})$ .  $\square$

**Proposition 10.57** (Restricted intrinsic one-arm bound on the torus). *For critical percolation in the high-dimensional triangle regime, let  $B_{\text{int}}^g(x, \ell)$  be the intrinsic ball when only edges of a deterministic subgraph  $g$  are permitted. There is  $C < \infty$ , independent of  $r, g, x$ , and  $\ell$ , such that*

$$\sup_{r \geq 2} \sup_{g \in E(\mathbb{T}_r^d)} \sup_{x \in \mathbb{T}_r^d} \mathbf{P}_{p_c}^{\mathbb{T}}(\partial B_{\text{int}}^g(x, \ell) \neq \emptyset) \leq \frac{C}{\ell}. \quad (21.38l)$$

*The same estimate holds on  $\mathbb{Z}^d$ , uniformly over deterministic subgraphs.*



*Proof.* We give the finite-torus step explicitly, since the event  $\{\partial B_{\text{int}}(x, \ell) \neq \emptyset\}$  is not monotone under adding edges and therefore cannot be transferred from  $\mathbb{Z}^d$  by a one-line stochastic-domination argument.

The Kozma–Nachmias proof of the restricted intrinsic one-arm upper bound is an exploration induction whose only model-dependent probabilistic input is the critical cluster-tail estimate

$$\mathbf{P}(|C(x)| \geq N) \leq C_0 N^{-1/2}. \quad (21.38m)$$

All remaining steps are deterministic breadth-first exploration, conditioning on the revealed intrinsic ball, independence of unrevealed edges, and BK. We first verify (21.38m) uniformly for the torus and for every deterministic subgraph  $g$ .

Use the standard lattice–torus quotient coupling at parameter  $p_c(\mathbb{Z}^d)$ . The torus cluster of  $x$  is obtained by identifying congruent vertices in a subset of a lifted lattice cluster. Consequently its number of distinct vertices is stochastically bounded by the size of the corresponding critical lattice cluster. The mean-field critical cluster tail on  $\mathbb{Z}^d$  gives  $\mathbf{P}_{p_c}(|C(0)| \geq N) \leq C_0 N^{-1/2}$ . Restricting to a deterministic subgraph can only delete open edges, hence the same bound holds uniformly for  $C_g(x)$  on  $\mathbb{T}_r^d$  and on  $\mathbb{Z}^d$ .

We now repeat the intrinsic exploration induction in a form valid on either graph. Put

$$\Gamma(\ell) = \sup_g \sup_x \mathbf{P}_{p_c}(\partial B_{\text{int}}^g(x, \ell) \neq \emptyset).$$

Fix  $k \geq 1$  and  $\epsilon \in (0, 1)$ . On the event  $|C_g(x)| \leq \epsilon 9^k$ , averaging the sizes of the intrinsic boundaries over  $j \in [3^{k-1}, 2 \cdot 3^{k-1}]$  produces a first level  $j$  with  $|\partial B_{\text{int}}^g(x, j)| \leq \epsilon 3^{k+1}$ . Conditional on the explored ball through level  $j$ , every continuation from level  $j$  to distance  $3^k$  uses only unrevealed edges. If the boundary is small, BK and the definition of  $\Gamma$  give the two-arm continuation bound

$$\epsilon 3^{k+1} \Gamma(3^{k-1})^2.$$

The complementary event is bounded by the cluster-tail estimate with  $N = \epsilon 9^k$ . Thus

$$\Gamma(3^k) \leq \epsilon 3^{k+1} \Gamma(3^{k-1})^2 + C \epsilon^{-1/2} 3^{-k}. \quad (21.38n)$$

This derivation never uses translation invariance or infinitude of the underlying graph, only (21.38m), and is therefore uniform in the torus size and in  $g$ .

Choose  $A$  large and then  $\epsilon = A^{-4/3}$  so that the induction  $\Gamma(3^{k-1}) \leq A 3^{-(k-1)}$  inserted in (21.38n) yields  $\Gamma(3^k) \leq A 3^{-k}$ . Enlarging  $A$  handles the finitely many initial scales. Monotonicity in the intrinsic radius between consecutive powers of 3 gives  $\Gamma(\ell) \leq C/\ell$  for every  $\ell \geq 1$ . This proves the lattice and torus statements simultaneously.  $\square$

**Scope of the intrinsic torus input.** The proof above gives the mechanism used in the published torus verification: the restricted Kozma–Nachmias exploration argument is graph-local, and its sole probabilistic input is the  $N^{-1/2}$  cluster tail, which is inherited by the torus from the lattice–torus coupling and is preserved under edge deletion. Hence the only additional input in the torus-plateau theorem is the independent restricted half-space two-point theorem (Proposition 10.51).

**Lemma 10.58** (Change of parameter along intrinsic paths). *For  $0 < p < q \leq 1$ , every deterministic  $g \subset E(\mathbb{T}_r^d)$  and  $\ell \geq 1$ ,*

$$\mathbf{P}_q^{\mathbb{T}}(\partial B_{\text{int}}^g(0, \ell) \neq \emptyset) \leq (q/p)^\ell \mathbf{P}_p^{\mathbb{T}}(\partial B_{\text{int}}^g(0, \ell) \neq \emptyset), \quad (21.38n)$$

$$\mathbf{P}_q^{\mathbb{T}}(x \in B_{\text{int}}(0, \ell)) \leq (q/p)^\ell \mathbf{P}_p^{\mathbb{T}}(x \in B_{\text{int}}(0, \ell)). \quad (21.38o)$$

*Proof.* Use the uniform-variable coupling. Condition on the  $q$ -configuration and, when the relevant event occurs, choose one deterministic witnessing path of length at most  $\ell$ . Conditional on all its edges being  $q$ -open, the probability that they remain  $p$ -open is at least  $(p/q)^\ell$ . Rearranging gives both inequalities.  $\square$

**Lemma 10.59** (Torus transfer and plateau). *Assume Proposition 10.57 in addition to (21.36). Let  $\mathbb{T}_r^d = (\mathbb{Z}/r\mathbb{Z})^d$  and  $V = r^d$ . For each fixed  $A > 0$ , uniformly for  $|p - p_c| \leq AV^{-1/3}$  and all  $x \in \mathbb{T}_r^d$ ,*

$$\tau_p^\mathbb{T}(0, x) \asymp_A (1 + \|x\|)^{2-d} + V^{-2/3}. \quad (21.38)$$

*Consequently  $\chi^\mathbb{T}(p) \asymp_A V^{1/3}$  and the torus triangle diagram is bounded uniformly in  $r$ . If  $p^\mathbb{T}$  is defined by  $\chi^\mathbb{T}(p^\mathbb{T}) = \lambda V^{1/3}$  for fixed sufficiently small  $\lambda > 0$ , then  $|p^\mathbb{T} - p_c| \leq C_\lambda V^{-1/3}$ .*

*Proof.* For  $p \leq p_c - A_1 V^{-1/3}$ , the quotient coupling and Lemmas 10.54–10.55 give

$$\tau_p(x) + c\chi(p)/V \leq \tau_p^\mathbb{T}(x) \leq \tau_p(x) + C\chi(p)/V \quad (21.38p)$$

through the overlap range  $p_c - A_2 r^{-2} \leq p \leq p_c - A_1 V^{-1/3}$ ; farther below  $p_c$  the upper estimate is enough. Choosing the lower edge of the window  $p_- = p_c - A_1 V^{-1/3}$  and using  $\chi(p_-) \asymp V^{1/3}$  gives the plateau lower bound there, and monotonicity propagates it to every  $p \geq p_-$ .

At  $p_c$ , Lemma 10.56 supplies the required upper bound. It remains to control  $p = p_c + \varepsilon$  with  $0 < \varepsilon \leq AV^{-1/3}$ . Fix  $\delta \in (0, 1]$  and set  $\gamma = \lfloor \delta/\varepsilon \rfloor$ . Split a connection according to whether  $x$  belongs to the intrinsic ball of radius  $3\gamma$ . By Lemma 10.58 and the critical upper bound,

$$\mathbf{P}_p^\mathbb{T}(x \in B_{\text{int}}(0, 3\gamma)) \leq e^{3\delta/p_c} \{\tau_{p_c}(x) + CV^{-2/3}\}. \quad (21.38q)$$

If the intrinsic distance is at least  $3\gamma$ , the intrinsic balls of radius  $\gamma$  from 0 and  $x$  are vertex-disjoint. Reveal the ball at  $x$ ; the other arm then lies in the complementary deterministic edge set. By independence of the two edge sets, Proposition 10.57 and Lemma 10.58, the long-distance term is at most

$$C\gamma^{-2}e^{2\varepsilon\gamma/p_c} \leq C_A\delta^{-2}V^{-2/3}. \quad (21.38r)$$

Thus, after replacing  $e^{3\delta/p_c}$  by  $1 + \delta'$ ,

$$\tau_p^\mathbb{T}(x) \leq (1 + \delta')\tau_{p_c}(x) + C_A(\delta')^{-2}V^{-2/3}.$$

Together with the monotone lower bound this proves (21.38).

Summing (21.38) gives  $\chi^\mathbb{T}(p) \asymp V^{1/3}$  because the Euclidean power-law part contributes only  $O(r^2) = o(V^{1/3})$ . For the triangle diagram, expand the threefold convolution of the Euclidean part plus the plateau. The pure Euclidean term is bounded by the infinite-volume triangle condition, while every term containing a plateau factor is bounded by powers of  $V^{-2/3}\chi^\mathbb{T} = O(V^{-1/3})$ . Finally, evaluating the susceptibility at  $p_c \pm CV^{-1/3}$  and using monotonicity brackets the solution of  $\chi^\mathbb{T} = \lambda V^{1/3}$ , proving the critical-point comparison.  $\square$

**Torus proof inventory.** The periodic sums, collision/failure defect, below-window lower plateau, critical upper bound, and change-of- $p$  estimates are all internal above. The only surviving torus-side input is Proposition 10.57. This is intentionally kept separate from the Chatterjee–Hanson half-space kernel. Thus the theorem has two explicit structural inputs.

**Theorem 10.60** (Near-critical decay and the torus plateau). *Consider nearest-neighbor Bernoulli percolation on  $\mathbb{Z}^d$  with  $d \geq 11$ , or a sufficiently spread-out finite-range model in fixed dimension  $d > 6$ . There are constants  $c, C > 0$  such that for all sufficiently small  $\varepsilon > 0$ ,*

$$\tau_{p_c - \varepsilon}(0, x) \leq C \|x\|^{2-d} e^{-c\sqrt{\varepsilon} \|x\|},$$

and

$$cR^{-2} e^{-C\sqrt{\varepsilon}R} \leq \mathbf{P}_{p_c - \varepsilon}(0 \leftrightarrow \partial Q_R) \leq CR^{-2} e^{-c\sqrt{\varepsilon}R}.$$

*For the periodic torus  $\mathbb{T}_r^d$  of volume  $V = r^d$ , throughout every fixed  $V^{-1/3}$  critical window the two-point function has the plateau*

$$\tau_p^{\mathbb{T}}(0, x) \asymp (1 + \|x\|)^{2-d} + V^{-2/3}.$$

*The torus triangle diagram is uniformly bounded there. Moreover the intrinsically defined torus critical point (via susceptibility of order  $V^{1/3}$ ) lies within  $O(V^{-1/3})$  of the infinite-volume  $p_c$ .*

*Proof.* Lemma 10.52 is the pioneer-edge/OSSS mechanism which carries the critical  $R^{-2}$  arm information into the slightly subcritical regime. Lemma 10.53 then proves the two displayed  $\mathbb{Z}^d$  estimates with their sharp polynomial prefactors. Finally Lemma 10.59 transfers those estimates through the periodic coupling and proves the plateau, triangle bound, and critical-point comparison. In particular, no step treats the one-arm theorem as a formal “massive version” of the critical second-moment proof.  $\square$

## 10.3 Ising lace expansion and marginal triviality

### 10.3.1 Corrected random-current lace coefficient architecture

The original Ising lace identity is not the source defect. The defect lies in one of the diagrammatic estimates formerly used to control its coefficients. We therefore separate the algebraic lace identity from the corrected coefficient engine. For the latter write  $G(x)$  for the Ising two-point function,  $\tau(x) = \tanh(\beta J_{0x})$ , and

$$\tilde{G} = \tau * G.$$

In the spread-out bootstrap regime let  $\|x\|_L = \max\{|x|, L\}$  and suppose

$$G(x) \leq C \|x\|_L^{2-d}, \quad \tilde{G}(x) \leq C\theta \|x\|_L^{2-d}, \quad \theta = O(L^{-2}). \quad (21.201a)$$

**Lemma 10.61** (Degree-four convolution elimination). *Assume  $d > 4$  and (21.201a). Uniformly in the external vertices,*

$$\sum_z \prod_{i=1}^4 \tilde{G}(a_i - z) \leq C\theta^2 L^{4-d} \tilde{G}(a_1 - a_2) \tilde{G}(a_3 - a_4) \quad (21.201b)$$

*after relabelling the four legs in one of the three pairings. In particular*

$$\tilde{G}^2 * \tilde{G}^2 \leq CL^{-d} \tilde{G}^2, \quad \sum_{j=1}^m (\tilde{G}^2)^{*j} \leq C\tilde{G}^2. \quad (21.201c)$$

*Consequently every finite chain of corrected Ising coefficient blocks loses a factor  $O(L^{-d})$  for each additional internal lace block after the first.*

*Proof.* For each  $z$ , in each of the pairs  $(a_1, a_2)$  and  $(a_3, a_4)$  at least one of the two distances to  $z$  is at least half the corresponding external distance. Split the sum into the resulting four regions and pull out the two large-distance factors. The two remaining kernels have convolution

$$\sum_z \|u - z\|_L^{2-d} \|v - z\|_L^{2-d} \leq C \|u - v\|_L^{4-d} \leq CL^{4-d}.$$

This proves (21.201b). Setting  $a_1 = a_3$  and  $a_2 = a_4$  and using  $\theta^2 L^{4-d} = O(L^{-d})$  gives the first inequality in (21.201c), and iteration gives the geometric series. The same elimination at every four-valent internal vertex proves the last assertion.  $\square$

**Proposition 10.62** (Corrected coefficient diagrams imply the lace bootstrap). *Assume the four corrected random-current coefficient bounds described below: (i) the zeroth coefficient is bounded by the  $X^1$  double-expansion diagram; (ii) the insertion  $\Theta'$  is bounded by the singly marked  $X^\infty$  diagrams; (iii) the marked zeroth coefficient is bounded by the singly and doubly marked  $X^1$  diagrams; and (iv) the higher insertion  $\Theta''$  is bounded by the triply marked  $X^\infty$  diagrams. Then, under (21.201a),*

$$\pi^{(0)}(x) \leq \delta_{0x} + \frac{C\theta^3}{\|x\|_L^{3(d-2)}}, \quad (21.201d)$$

$$\pi^{(1)}(x) \leq CL^{-d}\delta_{0x} + \frac{C\theta^3}{\|x\|_L^{3(d-2)}}, \quad (21.201e)$$

$$\pi^{(j)}(x) \leq C_j L^{-jd}\delta_{0x} + \frac{C_j L^{-d(j-2)}\theta^3}{\|x\|_L^{3(d-2)}} \quad (j \geq 2). \quad (21.201f)$$

Thus the alternating coefficient series is absolutely summable with the moments required by the general lace theorem, and the critical Ising correlation has the random-walk asymptotic  $A|x|^{2-d}(1 + o(1))$ .

*Proof.* The corrected diagrams consist of three external two-point legs joined by a chain of bubble-type four-valent blocks, together with at most three marked insertions. Lemma 10.61 removes the internal four-valent vertices successively. The zeroth diagram leaves three factors of  $\tilde{G}$  and hence the second term in (21.201d). The first marked coefficient has one additional short lace block and gives the  $L^{-d}$  local term in (21.201e). Each additional higher-order lace block gives another  $L^{-d}$  by (21.201c), which yields (21.201f). Since  $3(d-2) > d+2$  for  $d > 4$ , these bounds have the zeroth and second spatial moments required by the already-proved general lace bootstrap. Summing the alternating recursion and normalizing its effective one-step kernel therefore gives the random-walk infrared asymptotic and hence the stated  $x$ -space decay.  $\square$

**Corrected coefficient estimates.** this chapter does *not* reuse the invalid 2007 coefficient lemma. The remaining source work for SM-21.S2.05 consists precisely of the four corrected 2022 random-current diagram theorems:

IC0. the zeroth-order earliest-odd-path/double-expansion estimate;

IC1. the corrected  $\Theta'$  insertion estimate;

IC2. the once-marked zeroth-coefficient estimate;

IC3. the corrected  $\Theta''$  higher-order insertion estimate.

The source-switching identity itself and the downstream convolution/lace bootstrap are already proved in this manuscript. Hence the Ising root is now localized to the double-expansion coefficient engine rather than to “the Ising lace expansion” as a whole.

### 10.3.2 The corrected double expansion: earliest odd paths and marked diagrams

We prove the four corrected coefficient estimates IC0–IC3. The point of Sakai’s correction is not a new lace identity, but a replacement for the invalid attempt to switch sources simultaneously along two “earliest” paths. The replacement chooses only one earliest odd-current path and then performs a second, purely geometric lace expansion along the time line formed by that path. The switching lemma is used only after the geometric lace has been fixed. This order of operations is what makes the argument bijective.

Throughout this subsection we work first in a finite ferromagnetic volume  $\Lambda$ . Write  $\mathbb{B}_\Lambda$  for its bond set,

$$w_B(n) = \prod_{b \in B} \frac{(\beta J_b)^{n_b}}{n_b!}, \quad Z_B = \sum_{\partial n = \emptyset} w_B(n),$$

and let  $G$  denote the infinite-volume two-point function, used only as the monotone majorant of finite-volume two-point functions. Put

$$\tau(x) = \tanh(\beta J_{0x}), \quad \tilde{G} = \tau * G,$$

with  $\delta$  denoting the convolution identity. Pointwise products are written without a convolution sign, so  $\tilde{G}^2(x) = \tilde{G}(x)^2$ . For  $m \in \mathbb{N} \cup \{\infty\}$  set

$$Q_m = \sum_{j=0}^m (\tilde{G}^2)^{*j}, \quad R_m = \sum_{j=0}^{m-1} (\tilde{G}^2)^{*j} \quad (m \geq 1),$$

where for  $m = \infty$  the sums mean the monotone infinite sums. Finally define

$$B_m = \begin{cases} G^2, & m = 0, \\ (\delta + \tau^2) * Q_m * (\delta + \tau^2), & m \geq 1. \end{cases}$$

For a vertex set  $A \subset \Lambda$ , write  $\mathbb{B}_{A^c}$  for bonds with both endpoints outside  $A$ . We use  $\{o \longleftrightarrow^A x\}$  for the event that  $o$  and  $x$  are doubly connected and every positive-current  $o$ – $x$  path meets  $A$ . The three marked current quantities used below are

$$\begin{aligned} \Theta'_{o,x;A} &= \sum_{\substack{\partial \ell = \emptyset \\ \ell \text{ on } \mathbb{B}_{A^c}}} \frac{w(\ell)}{Z_{\mathbb{B}_{A^c}}} \sum_{\partial n = o \triangle x} \frac{w(n)}{Z_{\mathbb{B}_\Lambda}} \mathbf{1}_{\{o \longleftrightarrow^A x \text{ in } n+\ell\}}, \\ \tilde{\pi}_{\mathbb{B}_\Lambda; y}^{(0)}(x) &= \sum_{\partial n = o \triangle x} \frac{w(n)}{Z_{\mathbb{B}_\Lambda}} \mathbf{1}_{\{o \longleftrightarrow x \text{ in } n\}} \mathbf{1}_{\{o \leftrightarrow y \text{ in } n\}}, \\ \Theta''_{o,x,y;A} &= \sum_{\substack{\partial \ell = \emptyset \\ \ell \text{ on } \mathbb{B}_{A^c}}} \frac{w(\ell)}{Z_{\mathbb{B}_{A^c}}} \sum_{\partial n = o \triangle x} \frac{w(n)}{Z_{\mathbb{B}_\Lambda}} \mathbf{1}_{\{o \longleftrightarrow^A x \text{ in } n+\ell\}} \mathbf{1}_{\{o \leftrightarrow y \text{ in } n+\ell\}}. \end{aligned}$$

These are precisely the insertions that occur in the first and higher Ising lace coefficients; they are recorded here so the corrected coefficient engine is checkable without consulting the historical proof.

For a transfer kernel  $A(y, z; y', z')$  and a terminal kernel  $F(y, z; x)$  write

$$(A \star F)(y, z; x) = \sum_{y', z'} A(y, z; y', z') F(y', z'; x),$$

and use the same symbol for composition of two transfer kernels.

**Definition 10.63** (Unmarked corrected diagram blocks). For  $m \in \mathbb{N} \cup \{\infty\}$  define

$$\begin{aligned} U^m(y, z; y', z') &= \tilde{G}(z' - y)G(z' - y')B_m(y' - z), \\ V^m(y, z; x) &= \tilde{G}(x - y) \sum_{j=1}^m (\tilde{G}^2)^{*j}(x - z), \\ X_{o,x}^m &= \sum_{i \geq 0} ((U^m)^{*i} \star V^m)(o, o; x). \end{aligned}$$

For  $m = \infty$  the last two expressions are understood by monotone limits.

The marked blocks are easiest to state after isolating the three-leg current kernel which appears when an additional connection is exposed.

**Definition 10.64** (Three-leg current kernel). For vertices  $a, b, c$  put

$$\begin{aligned} T(a, b, c) &= \sum_r G(r - a)G(b - r)G(r - c) \\ &\quad \times \{G(b - a)G(r - c) + G(c - a)G(r - b) + G(r - a)G(c - b)\}. \end{aligned}$$

**Lemma 10.65** (Switching estimates used by the double expansion). *Let  $B$  be a finite ferromagnetic bond set.*

1. *If  $n$  is sourceless, then*

$$\frac{1}{Z_B} \sum_{\partial n = \emptyset} w_B(n) \mathbf{1}_{\{a \leftrightarrow b\}} \leq G(b - a)^2.$$

2. *If  $\partial n = a \triangle b$ , then*

$$\frac{1}{Z_B} \sum_{\partial n = a \triangle b} w_B(n) \mathbf{1}_{\{a \leftrightarrow c\}} \leq G(c - a)G(b - c).$$

3. *If  $n$  is sourceless, then*

$$\frac{1}{Z_B} \sum_{\partial n = \emptyset} w_B(n) \mathbf{1}_{\{a \leftrightarrow b\} \cap \{a \leftrightarrow c\}} \leq T(a, b, c).$$

*All three estimates remain valid when some bonds are deleted before the current is sampled.*

*Proof.* The first two statements are the two-current switching inequalities already proved in the random-current foundation of the series; we recall the exact mechanism because it will be iterated below. Multiply the current sum by an independent sourceless partition function, switch the pair of sources along a positive-current path, and then forget all residual connectivity restrictions. The resulting two independent source sums are bounded by the corresponding finite-volume two-point functions, and Griffiths monotonicity replaces them by  $G$ .

For the third statement, expose in the sourced copy produced by switching a vertex  $r$  at which the two required connections separate. Switching the first branch gives the factor  $G(r - a)G(b - r)G(r - c)$ . The remaining four-spin expectation is bounded by the Lebowitz inequality, which is the sum of its three pairings. These pairings are exactly the three terms in braces in Definition 10.64. Summing  $r$  proves the claim. Bond deletion only decreases the finite-volume correlations and therefore does not change the upper bounds.  $\square$

**Definition 10.66** (One marked point). For  $m \geq 1$  and a marked vertex  $a$  define

$$\begin{aligned}
U_a^{m,\bullet}(y, z; y', z') &= \{G(a - y)\tilde{G}(z' - a)G(z' - y') + \tilde{G}(z' - y)\tilde{G}(a - y')G(z' - a)\}B_m(y' - z), \\
V_a^{m,\bullet}(y, z; x) &= G(a - y)\tilde{G}(x - a) \sum_{j=1}^m (\tilde{G}^2)^{*j}(x - z), \\
X_{o,x;a}^{m,\bullet} &= \sum_{i \geq 0} ((U^m)^{*i} \star V_a^{m,\bullet})(o, o; x) \\
&\quad + \sum_{i,j \geq 0} ((U^m)^{*i} \star U_a^{m,\bullet} \star (U^m)^{*j} \star V^m)(o, o; x).
\end{aligned}$$

**Definition 10.67** (Connection-marked blocks). For  $m \geq 1$  and  $a \in \Lambda$  let

$$\begin{aligned}
U_a^{m,\bullet\bullet}(y, z; y', z') &= \tilde{G}(z' - y)G(z' - y') \sum_{\substack{u_1, u_2, u_3 \\ v_1, v_2, v_3}} (\delta + \tau^2)(z - u_1)(\delta + \tau^2)(y' - u_2)\delta_{u_3, a} \\
&\quad \times \prod_{q=1}^3 R_m(u_q - v_q)T(v_1, v_2, v_3), \\
V_a^{m,\bullet\bullet}(y, z; x) &= \mathbf{1}_{\{z \neq x\}} \tilde{G}(x - y) \sum_{\substack{u_1, u_2, u_3 \\ v_1, v_2, v_3}} (\delta + \tau^2)(z - u_1)\delta_{u_2, x}\delta_{u_3, a} \\
&\quad \times \prod_{q=1}^3 R_m(u_q - v_q)T(v_1, v_2, v_3), \\
X_{o,x;a}^{m,\bullet\bullet} &= \sum_{i \geq 0} ((U^m)^{*i} \star \frac{1}{2}V_a^{m,\bullet\bullet})(o, o; x) \\
&\quad + \sum_{i,j \geq 0} ((U^m)^{*i} \star U_a^{m,\bullet\bullet} \star (U^m)^{*j} \star V^m)(o, o; x).
\end{aligned}$$

**Definition 10.68** (Two simultaneous marks). For  $m \geq 1$  and marked vertices  $a, v$  define

$$\begin{aligned}
U_{a,v}^{m,\bullet\bullet\bullet}(y, z; y', z') &= \{G(a - y)\tilde{G}(z' - a)G(z' - y') + \tilde{G}(z' - y)\tilde{G}(a - y')G(z' - a)\} \\
&\quad \times \sum_{\substack{u_1, u_2, u_3 \\ v_1, v_2, v_3}} (\delta + \tau^2)(z - u_1)(\delta + \tau^2)(y' - u_2)\delta_{u_3, v} \\
&\quad \times \prod_{q=1}^3 R_m(u_q - v_q)T(v_1, v_2, v_3), \\
V_{a,v}^{m,\bullet\bullet\bullet}(y, z; x) &= \mathbf{1}_{\{z \neq x\}} G(a - y)\tilde{G}(x - a) \sum_{\substack{u_1, u_2, u_3 \\ v_1, v_2, v_3}} (\delta + \tau^2)(z - u_1)\delta_{u_2, x}\delta_{u_3, v} \\
&\quad \times \prod_{q=1}^3 R_m(u_q - v_q)T(v_1, v_2, v_3).
\end{aligned}$$

Set

$$\begin{aligned}
X_{o,x;a,v}^{m,\bullet\bullet\bullet} &= \sum_{i \geq 0} ((U^m)^{\star i} \star \tfrac{1}{2} V_{a,v}^{m,\bullet\bullet\bullet})(o, o; x) \\
&+ \sum_{i,j \geq 0} ((U^m)^{\star i} \star U_{a,v}^{m,\bullet\bullet\bullet} \star (U^m)^{\star j} \star V^m)(o, o; x) \\
&+ \sum_{i,j \geq 0} ((U^m)^{\star i} \star U_a^{m,\bullet} \star (U^m)^{\star j} \star \tfrac{1}{2} V_v^{m,\bullet\bullet})(o, o; x) \\
&+ \sum_{i,j \geq 0} ((U^m)^{\star i} \star U_v^{m,\bullet\bullet} \star (U^m)^{\star j} \star V_a^{m,\bullet})(o, o; x) \\
&+ \sum_{i,j,k \geq 0} ((U^m)^{\star i} \star U_a^{m,\bullet} \star (U^m)^{\star j} \star U_v^{m,\bullet\bullet} \star (U^m)^{\star k} \star V^m)(o, o; x) \\
&+ \sum_{i,j,k \geq 0} ((U^m)^{\star i} \star U_v^{m,\bullet\bullet} \star (U^m)^{\star j} \star U_a^{m,\bullet} \star (U^m)^{\star k} \star V^m)(o, o; x).
\end{aligned}$$

We next isolate the deterministic combinatorial core of the correction.

**Lemma 10.69** (Earliest-odd-path double-expansion lemma). *Fix once and for all a total order of the incident bonds at every vertex. Let  $n$  be a current with  $\partial n = o \triangle x$ ,  $o \neq x$ . The odd bonds of  $n$  contain an  $o$ - $x$  path. Explore the lexicographically earliest self-avoiding such path  $\omega = (\omega_0 = o, \dots, \omega_r = x)$  and let  $\tilde{B}_\omega$  be the set of bonds whose parities were inspected during this exploration. Then:*

1. *The event that  $\omega$  is the earliest odd path is a cylinder event on  $\tilde{B}_\omega$ : the selected path bonds are odd and every inspected rejected bond is even. Consequently the current weight factors into a current  $m$  on  $\tilde{B}_\omega$  and a sourceless current  $k$  on its complement.*
2. *For  $0 \leq j \leq r$ , let  $\tilde{V}_m(j)$  contain  $\omega_j$  and the other endpoint of every inspected rejected bond incident to  $\omega_j$  that carries positive even  $m$ -current. If  $o$  and  $x$  are doubly connected in  $m + k$ , there is a unique finite interval lace  $\mathbf{L}_{m,k} = \{s_i t_i : 1 \leq i \leq N\}$  on  $\{0, \dots, r\}$  obtained by*

$$t_1 = \max\{j : \tilde{V}_m(0) \leftrightarrow_k \tilde{V}_m(j)\}, \quad s_1 = 0,$$

and, while  $t_i < r$ ,

$$t_{i+1} = \max\{j : \exists q \leq t_i, \tilde{V}_m(q) \leftrightarrow_k \tilde{V}_m(j)\}, \quad s_{i+1} = \min\{q : \tilde{V}_m(q) \leftrightarrow_k \tilde{V}_m(t_{i+1})\}.$$

It terminates at  $t_N = r$  and satisfies

$$0 = s_1 < s_2 \leq t_1 < t_2 < \dots < s_N \leq t_{N-1} < t_N = r.$$

3. *Choose pairwise distinct witnesses  $u_i \in \tilde{V}_m(s_i)$  and  $v_i \in \tilde{V}_m(t_i)$  for the lace connections. After the indicator that the lace is exactly  $\mathbf{L}_{m,k}$  is dropped, successive conditioning on the  $k$ -clusters of the  $u_i$  and Lemma 10.65 give*

$$\sum_{\partial k = \emptyset} w(k) \prod_{i=1}^N \mathbf{1}_{\{u_i \leftrightarrow_k v_i\}} \leq Z \prod_{i=1}^N G(v_i - u_i)^2. \quad (21.201g)$$



4. Expanding each membership  $u_i \in \tilde{V}_m(s_i)$  and  $v_i \in \tilde{V}_m(t_i)$  into the path-vertex term and the positive-even side-bond term replaces it by either a Kronecker delta or a factor  $\tau^2$ . Indeed, for a side bond  $b$ ,

$$\frac{\sum_{q \geq 1} (\beta J_b)^{2q} / (2q)!}{\sum_{q \geq 0} (\beta J_b)^{2q} / (2q)!} = 1 - \frac{1}{\cosh(\beta J_b)} \leq \tanh^2(\beta J_b).$$

Summing the  $2N - 1$  path segments between consecutive lace endpoints then gives precisely  $N - 1$  factors  $U^1$  and one terminal factor  $V^1$ .

In particular the  $N$ -lace contribution is bounded by

$$2((U^1)^{\star(N-1)} \star V^1)_{o,x}. \quad (21.201h)$$

The same construction remains valid if a second sourceless current is added to  $k$ , because the lace is built only after the two outside currents have been superposed.

*Proof.* The first assertion is the reason the corrected proof uses one earliest path rather than the two-path switching attempted in the old coefficient lemma. During the exploration, the decision at each step depends only on parities of bonds already inspected. Hence, once  $\omega$  is fixed, the inspected set and the required parity pattern are fixed as well; all uninspected bonds carry an independent sourceless current.

For the second assertion, the maximum defining  $t_1$  exists because the double connection supplies an outside connection from the first explored cell to a later cell. If  $t_1 < r$ , double connection also forces a connection from some cell at or before  $t_1$  to a cell strictly after  $t_1$ , so  $t_2$  exists. The definition of  $s_2$  makes  $s_2 \leq t_1 < t_2$ . Repetition strictly increases the right endpoint and therefore terminates. The max-min rule also shows uniqueness: changing any  $t_i$  or  $s_i$  would contradict respectively its maximality or minimality.

For (21.201g), condition on the outside-current cluster of  $u_1$ . The pairwise-distinct witness condition lets the later witness pairs be chosen in the complement when required by the exact lace event. Dropping that exact lace event can only increase the sum. The first factor is then bounded by  $G(v_1 - u_1)^2$  by Lemma 10.65(1). Iterate over the remaining clusters. This gives the product of  $N$  squared two-point functions.

For the final assertion, a witness can differ from the path vertex only when a rejected explored bond has positive even current. Dividing its positive-even weight by the full even weight gives the displayed  $\tau^2$  bound. After all  $2N - 1$  witness positions have been expanded, the interval inequalities from part 2 split  $\omega$  into  $2N - 1$  ordered pieces. On a strict piece the first step is summed with the bond factor and hence produces  $\tilde{G} = \tau * G$ ; on a weak piece it produces  $G$ . Combining these path factors with the  $G^2$  factors in (21.201g) gives the explicit expression recorded in Lemma 10.70. That lemma identifies the internal factors with  $U^1$  and the final factor with  $2V^1$ , proving (21.201h).  $\square$

**Lemma 10.70** (Explicit segment algebra). *For an  $N$ -edge interval lace,  $N \geq 2$ , use variables  $y_1 = z_0 = o$ ,  $z_N = x$  for the successive left/right path locations after the witness expansion. After the outside-current clusters have been summed as in (21.201g), the remaining contribution is bounded by*

$$\begin{aligned} & \sum_{\substack{y_1, \dots, y_N \\ z_0, \dots, z_{N-1}}} \prod_{j=1}^{N-1} \left[ ((\delta + \tau^2) * G^2 * (\delta + \tau^2))(z_j - y_j) G(z_j - y_{j+1}) \tilde{G}(y_{j+1} - z_{j-1}) \right] \\ & \times \tilde{G}(x - z_{N-1}) ((\delta + \tau^2) * G^2)(x - y_N) \mathbf{1}_{\{y_N \neq x\}}. \end{aligned} \quad (21.201h0)$$

Each bracket in the first line is at most  $U^1(z_{j-1}, y_j; z_j, y_{j+1})$ , and the second line is at most  $2V^1(z_{N-1}, y_N; x)$ . Hence (21.201h0) is at most

$$2((U^1)^{\star(N-1)} \star V^1)_{o,x}.$$

*Proof.* The side-bond expansion at the left and right witnesses gives respectively  $(\delta + \tau^2)(y_j - u_j)$  and  $(\delta + \tau^2)(z_j - v_j)$ . Combining these with the cluster factor  $G(v_j - u_j)^2$  and summing  $u_j, v_j$  gives  $((\delta + \tau^2) * G^2 * (\delta + \tau^2))(z_j - y_j)$ . The interval-lace inequalities alternate weak and strict path inequalities. Summing an ordinary segment gives  $G(z_j - y_{j+1})$ , whereas a strict segment has at least one first bond and therefore gives  $\tau * G = \tilde{G}(y_{j+1} - z_{j-1})$ . This proves (21.201h0).

For  $m = 1$ ,  $B_1 = (\delta + \tau^2) * (\delta + \tilde{G}^2) * (\delta + \tau^2)$ , so the first factor in  $B_1$  contains the displayed  $(\delta + \tau^2) * G^2 * (\delta + \tau^2)$  contribution because  $G \leq \tilde{G}$  off the origin and the identity term handles the coincident case. This proves the  $U^1$  bound. For the terminal factor put  $w = x - y_N e_0$ . Positivity gives

$$G(w)^2 \leq \tilde{G}(w)^2, \quad (au^2 * G^2)(w) \leq \left( \sum_u au(u)G(w-u) \right)^2 = \tilde{G}(w)^2.$$

Hence  $((\delta + au^2) * G^2)(w) \leq 2\tilde{G}(w)^2$ , which is exactly the factor  $2V^1$ . This proves the terminal bound without hiding an endpoint case.  $\square$

**Proposition 10.71** (IC0: corrected zeroth coefficient). *For every finite ferromagnetic volume and  $x \neq o$ ,*

$$\pi_{\mathbb{B}_A}^{(0)}(x) \leq 2X_{o,x}^1. \quad (21.201i)$$

*Proof.* Partition every current contributing to  $\pi^{(0)}$  by its earliest odd path. The parity cylinder from Lemma 10.69(1) gives the exact path/outside factorization. Double connection produces the unique interval lace from part 2. Sum first over outside currents and apply parts 3–4. The term with one lace edge is the terminal diagram  $2V^1(o, o; x)$ ; for  $N \geq 2$ , (21.201h) gives  $2((U^1)^{\star(N-1)} \star V^1)_{o,x}$ . Summing  $N \geq 1$  is exactly  $2X_{o,x}^1$ .  $\square$

**Lemma 10.72** (Marked path-segment extraction). *In the setting of Lemma 10.69, suppose the earliest odd path is required to meet a set  $A$ . Among the  $2N - 1$  ordered path segments determined by the interval lace, choose the first segment that contains  $a \in A$ . Applying Lemma 10.65(2) to that segment splits its  $G$  or  $\tilde{G}$  factor at  $a$ . Summing over all possible positions of that marked segment replaces exactly one unmarked block by  $U_a^{m,\bullet}$  or  $V_a^{m,\bullet}$ , with the additional endpoint case  $X_{o,x}^m \delta_{a,x}$ . No other diagram is produced.*

*Proof.* The interval-lace inequalities give a disjoint ordered list of the path segments. Every path through  $A$  has a first marked segment. If it is the terminal segment, splitting at  $a$  yields  $G(a - y)\tilde{G}(x - a)$  and hence  $V_a^{m,\bullet}$ ; if  $a = x$  no split is needed and the endpoint term is  $X^m \delta_{a,x}$ . If it is an internal segment, the two possible orientations of the split yield the two summands in  $U_a^{m,\bullet}$ . The bubble chain carried by the corresponding lace edge is unchanged, so it remains  $B_m$ . Summing the possible position of the marked block gives Definition 10.66.  $\square$

**Proposition 10.73** (IC1: corrected through-set insertion). *Let  $\Theta'_{o,x;A}$  be the two-current insertion in which every positive current path from  $o$  to  $x$  must pass through  $A$ . For  $x \neq o$ ,*

$$\Theta'_{o,x;A} \leq 2 \sum_{a \in A} (X_{o,x}^\infty \delta_{x,a} + X_{o,x;a}^\infty). \quad (21.201j)$$

*Proof.* Choose the earliest odd path in the sourced current. The through- $A$  condition implies that this path intersects  $A$ ; relaxing the condition to that necessary event only enlarges the sum. Superpose the second sourceless current with the outside current before constructing the interval lace. The last sentence of Lemma 10.69 therefore gives the same double expansion, now with an arbitrary number of bubble repetitions; this is the monotone  $m = \infty$  block. Apply Lemma 10.72 to the first path segment meeting  $A$ , and sum  $a \in A$ . The result is exactly (21.201j).  $\square$

**Lemma 10.74** (Connection marker attached to a lace). *Add to the IC0 event the requirement  $o \leftrightarrow y$ . After the interval lace has been fixed, exactly one of the following occurs.*

1.  $y = x$ ;
2.  $y$  is connected, in the outside current, to one of the  $N$  left witness clusters. Conditioning on that cluster and using Lemma 10.65(3) replaces the corresponding  $G^2$  lace factor by the three-leg kernel  $T$ , and after the same side-bond and bubble-chain extraction produces one block  $U_y^{m,\bullet\bullet}$  or  $V_y^{m,\bullet\bullet}$ ;
3.  $y$  is not connected to any left witness cluster. Then the  $o$ - $y$  path must meet one of the  $2N - 1$  path segments. Splitting at the first meeting point is exactly the one-marked path-segment extraction of Lemma 10.72, with  $A = \{y\}$ .

Consequently the full marked contribution is bounded by the sum of  $X^m \delta_{x,y}$ ,  $X_{o,x;y}^{m,\bullet}$  and  $X_{o,x;y}^{m,\bullet\bullet}$ .

*Proof.* The alternatives are exhaustive by following any positive-current path from  $o$  to  $y$  until it first meets either an outside lace cluster or an explored path segment. In the second alternative condition on the selected outside cluster before summing the remaining current. Two independent connection requirements remain in that cluster; the three-leg estimate in Lemma 10.65(3) gives  $T$ . The three arms leaving  $T$  can acquire any number from 0 to  $m - 1$  of the ordinary bubble chain, which yields the three factors  $R_m$  in Definition 10.67. The endpoint and internal cases give respectively  $V^{m,\bullet\bullet}$  and  $U^{m,\bullet\bullet}$ . The third alternative is precisely the proof of the one-marked lemma. Summing the possible positions gives the claimed  $X$  blocks.  $\square$

**Proposition 10.75** (IC2: corrected marked zeroth coefficient). *Let  $\tilde{\pi}_{\mathbb{B}_\Lambda;y}^{(0)}(x)$  be the zeroth lace coefficient with the extra event  $o \leftrightarrow y$ . For  $x \neq o$ ,*

$$\tilde{\pi}_{\mathbb{B}_\Lambda;y}^{(0)}(x) \leq 2(X_{o,x}^1 \delta_{x,y} + X_{o,x;y}^{1,\bullet} + X_{o,x;y}^{1,\bullet\bullet}). \quad (21.201k)$$

*Proof.* Run the IC0 double expansion and retain the extra connection to  $y$ . Apply Lemma 10.74 with  $m = 1$  on each interval lace. Its three alternatives give exactly the three terms in (21.201k), and the factor 2 is the same terminal factor as in IC0.  $\square$

**Lemma 10.76** (Simultaneous through-set and connection markers). *Suppose both the through- $A$  condition and an extra connection to  $y$  are present. After the double expansion, decompose according to whether  $y$  attaches to a left lace-witness cluster.*

1. If it does, expose that cluster first. The three-leg estimate produces  $U_y^{\infty,\bullet\bullet}$  or  $V_y^{\infty,\bullet\bullet}$ , while the first path segment meeting  $A$  produces the independent one-point mark. If both marks lie in the same block, the two possible orders of their placement produce  $U_{a,y}^{\infty,\bullet\bullet\bullet}$  or  $V_{a,y}^{\infty,\bullet\bullet\bullet}$ . Summing all relative positions is exactly  $X_{o,x;a,y}^{\infty,\bullet\bullet\bullet}$  together with the endpoint term  $X_{o,x;y}^{\infty,\bullet\bullet} \delta_{a,x}$ .
2. If  $y$  attaches to no left witness cluster, its first meeting with the explored path occurs at some  $y'$ . The same marked-segment analysis gives  $X_{o,x;a}^{\infty,\bullet\bullet} \delta_{y',x}$  or  $X_{o,x;y',a}^{\infty,\bullet\bullet\bullet}$ , while the connection from  $y$  to  $y'$  is bounded by the unrestricted bubble chain  $\sum_{r \geq 0} (\tilde{G}^2)^{*r}(y - y')$ .

*Proof.* This is the two-marker version of Lemmas 10.72 and 10.74. In the first case, condition on the lace cluster carrying the  $y$ -connection before applying the through- $A$  path split. The three possible legs leaving the conditioned cluster give the three  $R_\infty$  factors in the doubly marked block, and placing  $a$  before, after, or inside that block yields exactly the six terms in Definition 10.68. In the second case, condition on the first path segment reached from  $y$ ; successive switching of the connection before it reaches  $y'$  gives the geometric bubble series. Once  $y'$  is fixed the remaining argument is the first case with the roles of the path mark and connection mark interchanged. The two cases are disjoint and exhaustive.  $\square$

**Proposition 10.77** (IC3: corrected higher insertion). *Let  $\Theta''_{o,x,y;A}$  denote the through- $A$  insertion with the additional connection  $o \leftrightarrow y$ . For  $x \neq o$ ,*

$$\Theta''_{o,x,y;A} \leq 2 \sum_{a \in A} \left[ X_{o,x;y}^{\infty, \bullet\bullet} \delta_{a,x} + X_{o,x;a,y}^{\infty, \bullet\bullet\bullet} \right. \\ \left. + \sum_{y'} \left( X_{o,x;a}^{\infty, \bullet\bullet} \delta_{y',x} + X_{o,x;y',a}^{\infty, \bullet\bullet\bullet} \right) \sum_{r \geq 0} (\tilde{G}^2)^{*r}(y - y') \right]. \quad (21.2011)$$

*Proof.* Choose the earliest odd path, superpose the two outside sourceless currents, and perform the interval-lace expansion. Apply Lemma 10.76. Its first case yields the first line of (21.2011), its second case yields the  $y'$  sum, and summing the first path mark over  $a \in A$  gives the result.  $\square$

**Theorem 10.78** (Corrected Sakai coefficient engine). *The four corrected estimates IC0–IC3 are proved in Propositions 10.71, 10.73, 10.75, and 10.77 hold for arbitrary finite ferromagnetic coupling. Consequently the hypothesis of Proposition 10.62 is satisfied.*

*Proof.* The four assertions are exactly the four preceding propositions. Their proof uses only the already-owned switching identities, Griffiths monotonicity and Lebowitz four-point inequality, together with the deterministic earliest-path and interval-lace construction proved in Lemma 10.69. In particular, no step invokes the invalid two-earliest-path source-switching lemma from the historical coefficient proof.  $\square$

**Theorem 10.79** (Sakai Ising lace expansion). *For the ferromagnetic Ising model, the random-current lace expansion is valid for arbitrary spin–spin coupling, and in the high-dimensional regimes below its expansion coefficients satisfy the diagrammatic bounds needed by the lace bootstrap. Consequently*

$$\langle \sigma_0 \sigma_x \rangle_{\beta_c} = A|x|^{2-d}(1 + o(1))$$

*for the nearest-neighbour model in all sufficiently large dimensions, and for finite-range spread-out models in every fixed  $d > 4$  once the spread-out range  $L$  is sufficiently large. This conclusion does not require reflection positivity.*

*Proof.* The algebraic random-current lace identity, source switching, and the general lace bootstrap are already owned earlier in the manuscript. The historical 2007 coefficient estimate cannot be used as a proof: its key Lemma 4.2 has a known flaw. This chapter therefore uses the corrected double-expansion architecture instead. Proposition 10.62 proves internally that the four corrected coefficient diagrams imply the required absolute summability, moments, and Gaussian asymptotic.

The four corrected random-current diagram families IC0–IC3 are proved in the module below. Theorem 10.78 constructs the earliest odd-current path, proves the unique interval-lace double expansion, extracts the unmarked and one-/two-/three-marked diagram blocks, and obtains the four

corrected Sakai bounds without the invalid historical two-path switching lemma. Proposition 10.62 then supplies the coefficient moments and lace-bootstrap conclusion. The theorem statement is retained at the proved source scope: sufficiently large dimension for nearest-neighbour coupling and every fixed  $d > 4$  for sufficiently spread-out finite-range coupling.  $\square$

### 10.3.3 Bubble-controlled Ising thermodynamics

We now isolate the random-current input needed to pass from a Gaussian critical two-point function to the thermodynamic exponents. This is logically separate from the lace expansion itself: the arguments below require only a translation-invariant ferromagnetic Ising model with summable coupling and a uniformly bounded zero-field bubble on the high-temperature side.

Write

$$J_* := \sum_x J_{0x} < \infty, \quad H := \beta h,$$

and, in infinite volume,

$$\begin{aligned} M(\beta, H) &:= \langle \sigma_0 \rangle_{\beta, H}, & C_{\beta, H}(x) &:= \langle \sigma_0 \sigma_x \rangle_{\beta, H} - M(\beta, H)^2, \\ \chi(\beta, H) &:= \sum_x C_{\beta, H}(x) = \partial_H M(\beta, H). \end{aligned}$$

At zero field we abbreviate

$$G_\beta(x) := \langle \sigma_0 \sigma_x \rangle_{\beta, 0}, \quad \chi_0(\beta) := \sum_x G_\beta(x), \quad B_0(\beta) := \sum_x G_\beta(x)^2.$$

All derivatives with respect to  $\beta$  in the next three lemmas are taken at fixed  $H$ . This convention is essential in the GHS extrapolation step.

**Lemma 10.80** (Zero-field bubble differential inequality). *For every  $\beta < \beta_c$ ,*

$$\frac{J_* \chi_0(\beta)^2}{1 + 2\beta J_* B_0(\beta)} \leq \partial_\beta \chi_0(\beta) \leq J_* \chi_0(\beta)^2. \quad (21.207)$$

*The same inequalities first hold in a periodic finite volume; the displayed infinite-volume form follows by taking the thermodynamic limit at fixed  $\beta < \beta_c$ .*

*Proof.* We give the current calculation because the lower inequality, rather than the infrared bound alone, is the step that fixes  $\gamma$ .

In a finite periodic box let  $\langle \cdot \rangle$  be the zero-field Gibbs state. Differentiation gives

$$\partial_\beta \chi_0 = \frac{1}{2} \sum_{x, u, v} J_{uv} (\langle \sigma_0 \sigma_x \sigma_u \sigma_v \rangle - \langle \sigma_0 \sigma_x \rangle \langle \sigma_u \sigma_v \rangle). \quad (21.208)$$

For the upper bound use the Lebowitz four-point inequality

$$\begin{aligned} &\langle \sigma_0 \sigma_x \sigma_u \sigma_v \rangle - \langle \sigma_0 \sigma_x \rangle \langle \sigma_u \sigma_v \rangle \\ &\leq G_\beta(u) G_\beta(x - v) + G_\beta(v) G_\beta(x - u). \end{aligned} \quad (21.209)$$

Here is the current proof needed at this point. Expand the two terms using two independent currents and apply the Volume-VIII switching bijection to the pair  $\{u, v\}$ . The difference is supported on configurations in which the source pair  $\{0, x\}$  is joined to  $\{u, v\}$ . On that event either a connection

realizes the pairing  $0 \leftrightarrow u$ ,  $x \leftrightarrow v$  or it realizes  $0 \leftrightarrow v$ ,  $x \leftrightarrow u$ . Forgetting all other connectivity restrictions and switching the two resulting pairs back factorizes their weights, giving respectively  $G_\beta(u)G_\beta(x-v)$  and  $G_\beta(v)G_\beta(x-u)$ . This proves (21.209). After summing  $x, u, v$ , translation invariance and  $\sum_v J_{uv} = J_*$  give the right-hand inequality in (21.207).

For the reverse inequality retain the connectivity information instead of forgetting it in (21.209). Let  $n_1, n_2$  be independent currents with source sets  $\partial n_1 = \{0, x\}$  and  $\partial n_2 = \emptyset$ . Apply the Volume-VIII switching bijection to (21.208), expose the component of 0 in  $n_1 + n_2$ , and mark the first interaction edge  $\{u, v\}$  through which the second source path meets that component. If there is no second meeting, the marked-edge sum factorizes and contributes exactly

$$J_* \chi_0^2. \quad (21.210)$$

If a second meeting occurs, mark the first reconnection point. Switching at that point splits off a zero-field two-point loop. Summing its two arms and using Cauchy–Schwarz gives  $B_0$ ; the two possible orientations of the marked edge give the factor 2. What remains after the loop is removed is precisely the differentiated-current diagram (21.208). Consequently the reconnection contribution is at most

$$2\beta J_* B_0 \partial_\beta \chi_0. \quad (21.211)$$

All weights are nonnegative, so (21.210)–(21.211) yield

$$J_* \chi_0^2 \leq \partial_\beta \chi_0 + 2\beta J_* B_0 \partial_\beta \chi_0.$$

Solving for  $\partial_\beta \chi_0$  proves the lower bound in (21.207). This is the zero-field specialization of the all-orders bubble correction: the loop term is absorbed into the denominator rather than truncated as a factor  $1 - cB_0$ .

For fixed  $\beta < \beta_c$ , exponential decay from SM-08.03.02 makes all sums absolutely convergent. Periodic finite-volume correlations therefore converge termwise along an exhaustion, and (21.207) passes to infinite volume.  $\square$

**Lemma 10.81** (Semi-truncated current inequality and the diluted second-order remainder). *Work in a finite periodic ferromagnetic Ising system at field  $H > 0$ . For observables  $A, B, C$  write*

$$\langle A, B, C \rangle := \langle ABC \rangle - \langle AB \rangle \langle C \rangle - \langle AC \rangle \langle B \rangle - \langle A \rangle \langle BC \rangle + 2\langle A \rangle \langle B \rangle \langle C \rangle.$$

*Let the spatial couplings be weakened along a two-parameter interpolation, and let  $S_2(A)$  be the second-order term produced after independently retaining every spatial bond with probability  $p$ . In the notation of the next proof, after switching outside the 0-cluster one has*

$$\begin{aligned} \mathbb{E}_A S_2(A) &\leq p^2 \sum_{x,y,u,v,k,l} \beta J_{uv} \beta J_{kl} G_\beta(k) T(0, u, x) \\ &\quad \times \int_0^1 ds_1 \int_{s_1}^1 ds_2 \langle \sigma_y, \sigma_u \sigma_v, \sigma_k \sigma_l \rangle_{s_2}. \end{aligned} \quad (21.211a)$$

*Then*

$$\mathbb{E}_A S_2(A) \leq 12p^2 C_2, \quad C_2 := \frac{\beta^2 J_*^2 M^2}{\tanh H} [1 + M \tanh H] B_0 \sum_{u,x} T(0, u, x). \quad (21.211b)$$

*Here  $M = M(\beta, H)$ , and  $G_\beta$  and  $B_0$  are the zero-field quantities at the same  $\beta$ .*

*Proof.* We include the correlation estimate because this is the point at which the former proof hid a genuinely model-specific random-current input. For two distinct marked bonds  $\{y, z\}$  and  $\{k, l\}$ , switch the duplicated current at the first marked pair. Put

$$\mathcal{R} := \{(y, z; k, l), (z, y; k, l), (k, l; y, z), (l, k; y, z)\}.$$

The switching decomposition of the third Ursell function is

$$\langle \sigma_x, \sigma_y \sigma_z, \sigma_k \sigma_l \rangle = \sum_{(a,b;c,d) \in \mathcal{R}} (\Delta_1^{a,b;c,d} + \Delta_2^{a,b;c,d}). \quad (21.211c)$$

Write  $M_a := \langle \sigma_a \rangle$  and  $C(a, b) := \langle \sigma_a \sigma_b \rangle - M_a M_b$  for the state at the current interpolation parameters. It is enough to bound the representative  $(a, b; c, d) = (y, z; k, l)$ ; the other three cases are literal relabellings. If  $C$  is the switched source cluster, Griffiths II first gives

$$\langle \sigma_z, \sigma_k \sigma_l \rangle_C - (\langle \sigma_k \sigma_l \rangle - \langle \sigma_k \sigma_l \rangle_C) \langle \sigma_z \rangle_C \leq \langle \sigma_z, \sigma_k \sigma_l \rangle_C.$$

Switch once more at  $k$  (respectively  $l$ ). The remaining connectivity restriction can only decrease the positive current sum, and restoring all deleted ferromagnetic couplings gives

$$\Delta_1^{y,z;k,l} \leq M_l G_\beta(z - k) C(x, y) + M_k G_\beta(z - l) C(x, y). \quad (21.211d)$$

For  $\Delta_2$ , switch outside  $C$  and weaken the condition that both source paths hit the same switched cluster. Pulling one zero-field correlation out of the resulting sourceless-current expectation gives

$$\Delta_2^{y,z;k,l} \leq M_l G_\beta(z - k) C(x, y). \quad (21.211e)$$

For each of the four elements of  $\mathcal{R}$ , (21.211d) contributes two product terms and (21.211e) contributes one. Hence the complete bound has exactly twelve nonnegative product terms. When the two marked bonds coincide, the corresponding semi-truncated correlation is

$$\langle \sigma_x, \sigma_y \sigma_z, \sigma_y \sigma_z \rangle = -2 \langle \sigma_y \sigma_z \rangle \langle \sigma_x, \sigma_y \sigma_z \rangle \leq 0,$$

so those  $O(p)$  coincidence terms may be discarded in an upper bound. Thus every surviving contribution carries the factor  $p^2$ .

It remains to sum one representative term; the other eleven are identical after relabelling. Define  $K_{s_2}(a, b)$  to be the total normalized weight of  $\{a, b\}$ -source currents whose deterministic backbone is stopped at its first visit to  $b$ , after summing all sourceless current decorations, in the system in which the interpolated bonds have strength  $s_2$ . Thus  $K_{s_2}$  is a genuine current kernel, not an additional probabilistic assumption. Partitioning a source current at its first hit gives

$$C_{s_2}(y, u) \leq K_{s_2}(y, u) \leq G_\beta(y - u), \quad (21.211f)$$

and decomposing the magnetization according to the first ghost edge gives

$$\sum_y K_{s_2}(y, u) \leq \frac{M}{\tanh H} [1 + M \tanh H]. \quad (21.211g)$$

For completeness, removing the last ghost edge from a first-hit backbone produces the factor  $\tanh H$ . Restoring the deleted part of the ferromagnetic system changes the residual boundary factor by at most  $1 + M \tanh H$  by Griffiths II. Summing the disjoint first-hit classes yields (21.211g). Likewise,

forgetting avoidance of the already explored backbone enlarges the first-hit current weight, while restoring all weakened spatial couplings gives the zero-field upper bound in (21.211f).

A representative of the twelve terms in (21.211a) is therefore at most

$$U \leq p^2 \beta^2 \frac{M^2}{\tanh H} [1 + M \tanh H] \sum_{x,u} T(0, u, x) \times \sum_{v,k,l} J_{uv} J_{kl} G_\beta(k) G_\beta(v-k). \quad (21.211h)$$

For fixed  $u$ , summing first over  $l$  and then using Cauchy–Schwarz on the convolution gives

$$\begin{aligned} \sum_{v,k,l} J_{uv} J_{kl} G_\beta(k) G_\beta(v-k) &= J_* \sum_v J_{uv} \sum_k G_\beta(k) G_\beta(v-k) \\ &\leq J_* \sum_v J_{uv} B_0 = J_*^2 B_0. \end{aligned} \quad (21.211i)$$

Indeed  $\sum_k G_\beta(k) G_\beta(v-k) \leq B_0$  is exactly Cauchy–Schwarz and translation invariance. Hence  $U \leq p^2 C_2$ . Summing the twelve relabelled contributions proves (21.211b). This is precisely the second-order estimate that makes dilution useful: the bad term has one bubble factor, but it costs  $p^2$  rather than  $p$ .  $\square$

**Lemma 10.82** (Aizenman–Fernández field inequality from diluted currents). *Let  $\beta_c/24 \leq \beta < \beta_c$ ,  $H > 0$ , and set*

$$\overline{B}_0 := \beta J_* B_0(\beta), \quad Q := \frac{\tanh H}{\beta J_* M(\beta, H)}.$$

*Then*

$$-\partial_H \chi(\beta, H) \geq \frac{|1 - Q \overline{B}_0|_+^2}{48 [1 + M(\beta, H) \tanh H] B_0(\beta) [1 + 2 \overline{B}_0]^2} \tanh H \chi(\beta, H)^4. \quad (21.212)$$

*In particular, if  $B_0(\beta) \leq B_*$  and  $\chi(\beta, H) \geq 2B_*$ , then for  $0 < H \leq 1$*

$$-\partial_H \chi(\beta, H) \geq c_* H \chi(\beta, H)^4, \quad (21.213)$$

*where  $c_* > 0$  depends only on  $B_*$ ,  $J_*$ , and a fixed lower bound for  $\beta$  near  $\beta_c$ .*

*Proof.* We reconstruct the cancellation responsible for the denominator in (21.212). Work first in a finite periodic box with a ghost vertex  $g$  whose edge weight represents the homogeneous field. For currents  $n_1, n_2$  with  $\partial n_1 = \{0, x\}$  and  $\partial n_2 = \emptyset$ , use the unnormalized two-current expectation

$$\mathbb{E}_{0x, \emptyset}[F] := \sum_{\substack{\partial n_1 = \{0, x\} \\ \partial n_2 = \emptyset}} \frac{w(n_1) w(n_2)}{Z^2} F(n_1, n_2).$$

Let  $C(0)$  be the component of 0 in  $n_1 + n_2$ . Differentiating  $\chi = \partial_H M$ , using GHS for the sign, and switching once gives the exact positive representation

$$-\partial_H \chi = 2 \sum_{x,y} \mathbb{E}_{0x, \emptyset} [\mathbf{1}_{\{0 \not\leftrightarrow g\}} (\langle \sigma_y \rangle - \langle \sigma_y \rangle_{C(0)^c})]. \quad (21.214)$$



Here the second magnetization is computed after all spatial couplings in  $C(0)$  have been deleted. Formula (21.214) is simply the third Ursell function written in the two-current language; no sign cancellation remains.

Independently retain each spatial bond with probability  $p$ , obtaining a random set  $A$ . By Griffiths monotonicity we may delete only the bonds  $C(0) \cap A$ . Interpolate their strengths from 0 to 1. One use of the fundamental theorem of calculus gives a positive first-order term  $S_1(A)$ ; applying it once more to the truncated three-point function produces a remainder  $S_2(A)$ . Explicitly, with

$$T(0, u, x) := \mathbb{E}_{0x, \emptyset} [\mathbf{1}_{\{0 \nleftrightarrow g\}} \mathbf{1}_{\{0 \leftrightarrow u\}}],$$

the switching lemma gives

$$-\partial_H \chi \geq S_1(A) - S_2(A). \quad (21.215)$$

Averaging over  $A$  makes the first interpolation term cost  $p$  and the second cost  $p^2$ . The sum over the differentiated spin pair in  $S_1$  is  $\beta \partial_\beta M$ , independently of  $u, x$ . Applying the all-orders current bound obtained by the same decomposition as in Lemma 10.80 therefore gives

$$\mathbb{E}_A S_1(A) \geq p C_1, \quad C_1 := \frac{\beta J_* \chi [M - \tanh H B_0]_+}{1 + 2\beta J_* B_0} \left( \sum_{u, x} T(0, u, x) \right). \quad (21.216)$$

A separate switch of the  $0$ - $u$  connection gives

$$\sum_{u, x} T(0, u, x) \geq \chi^2. \quad (21.217)$$

Indeed, after switching, forgetting the remaining requirement that the two source paths meet leaves the product  $(\sum_x C_{\beta, H}(x))^2 = \chi^2$ .

It remains to control the second interpolation term. Write the two marked bonds as  $\{u, v\}$  and  $\{k, l\}$ . If both marked bonds lie in the switched  $0$ -cluster, bound that event by the sum of the two events that  $0$  connects to  $k$  or to  $l$ . Decompose the latter connection at the ghost cluster and switch outside it. The exterior currents then have the source pairs needed for the same two-source estimate that produced  $T(0, u, x)$ ; after restoring the weakened couplings, this gives

$$R \leq [G_\beta(k)T(0, u, x) + (u \leftrightarrow k)] + (k \leftrightarrow l).$$

Here  $(u \leftrightarrow k)$  means that the preceding displayed summand is repeated with the labels  $u$  and  $k$  exchanged, and  $(k \leftrightarrow l)$  means that the whole preceding bracket is repeated with  $k$  and  $l$  exchanged. This is only endpoint-permutation notation, not a connectivity event. Averaging the Bernoulli dilution makes two distinct marked bonds cost  $p^2$ . The endpoint permutations in the last display cancel the factor  $1/4$  from the second fundamental-theorem-of-calculus term. Consequently the remainder is bounded by (21.211a), and Lemma 10.81 gives

$$\mathbb{E}_A S_2(A) \leq 12p^2 C_2, \quad C_2 := \frac{\beta^2 J_*^2 M^2}{\tanh H} [1 + M \tanh H] B_0 \sum_{u, x} T(0, u, x). \quad (21.218)$$

The important point is now proved rather than named: the bad term contains one extra factor  $B_0$ , while dilution has changed its probability from  $p$  to  $p^2$ .

Thus (21.215) gives  $-\partial_H \chi \geq p C_1 - 12p^2 C_2$ . Choose  $p = C_1/(24C_2)$ . The elementary mean-field bound  $\beta_c J_* \geq 1$ , together with  $B_0 \geq 1$ , is exactly what is needed in the source dilution estimate to ensure  $p \leq 1$  for  $\beta \geq \beta_c/24$ , which is the range assumed in the lemma. At the optimizer,

$$-\partial_H \chi \geq \frac{C_1^2}{48C_2} = \frac{|1 - \frac{\tanh H}{M} B_0|_+^2}{48[1 + M \tanh H] B_0 [1 + 2\beta J_* B_0]^2} \tanh H \chi^2 \sum_{u, x} T(0, u, x).$$

Use (21.217) to obtain (21.212), noting that  $Q\bar{B}_0 = (\tanh H/M)B_0$ .

Finally GHS makes  $H \mapsto M(\beta, H)$  concave and  $M(\beta, 0) = 0$  for  $\beta < \beta_c$ . Hence

$$M(\beta, H) \geq H\chi(\beta, H) \geq \tanh H \chi(\beta, H),$$

so  $Q\bar{B}_0 \leq B_0/\chi \leq 1/2$  whenever  $\chi \geq 2B_*$ . Also  $1 + M \tanh H \leq 2$  and  $\tanh H \geq H/2$  for  $0 < H \leq 1$ . These observations reduce (21.212) to (21.213).  $\square$

**Lemma 10.83** (GHS temperature–field slope bound). *For  $H > 0$ ,*

$$0 \leq \partial_\beta M(\beta, H) \leq J_* M(\beta, H) \partial_H M(\beta, H). \quad (21.219)$$

*Consequently every regular constant-magnetization curve in the  $(\beta, H)$ -plane has slope in the interval*

$$-J_* M \leq \left. \frac{dH}{d\beta} \right|_{M=\text{const}} \leq 0. \quad (21.220)$$

*Proof.* The left inequality is Griffiths monotonicity. For the other direction, differentiate at fixed  $H$ :

$$\partial_\beta M = \frac{1}{2} \sum_{x,y} J_{xy} (\langle \sigma_0 \sigma_x \sigma_y \rangle - M \langle \sigma_x \sigma_y \rangle). \quad (21.221)$$

Write the bracket as the third Ursell function plus

$$M C_{\beta,H}(x) + M C_{\beta,H}(y).$$

The third Ursell function is nonpositive by the already proved GHS theorem SM-08.02.03. Translation invariance and  $\sum_y J_{xy} = J_*$  therefore give

$$\partial_\beta M \leq \frac{M}{2} \sum_{x,y} J_{xy} [C_{\beta,H}(x) + C_{\beta,H}(y)] = J_* M \chi.$$

Since  $\chi = \partial_H M$ , (21.219) follows; implicit differentiation gives (21.220).  $\square$

**Proposition 10.84** (Integration of the bubble inequalities). *Assume that*

$$B_* := \sup_{\beta < \beta_c} B_0(\beta) < \infty, \quad \chi_0(\beta) \uparrow \infty \quad (\beta \uparrow \beta_c). \quad (21.222)$$

*Then, for  $\beta$  sufficiently close to  $\beta_c$ ,*

$$\chi_0(\beta) \asymp (\beta_c - \beta)^{-1}. \quad (21.223)$$

*Moreover*

$$M(\beta_c, H) \leq CH^{1/3} \quad (0 < H \leq H_0). \quad (21.224)$$

*If in addition the sharp-transition bound*

$$M_+(\beta, 0) \geq c(\beta - \beta_c)^{1/2} \quad (21.225)$$

*holds for  $\beta > \beta_c$ , then*

$$M(\beta_c, H) \asymp H^{1/3}, \quad M_+(\beta, 0) \asymp (\beta - \beta_c)^{1/2}. \quad (21.226)$$

*Proof.* Boundedness of  $B_0$  turns (21.207), in a fixed neighborhood below  $\beta_c$ , into

$$c\chi_0^2 \leq \partial_\beta \chi_0 \leq C\chi_0^2.$$

Equivalently

$$C^{-1} \leq -\partial_\beta(\chi_0^{-1}) \leq c^{-1}.$$

Integrating to  $\beta_c$  and using the divergence in (21.222) proves (21.223).

For the critical field upper bound, fix  $\beta < \beta_c$ . If  $\chi(\beta, H) \geq 2B_*$ , Lemma 10.82 gives

$$\frac{d}{dH}\chi(\beta, H)^{-3} = -3\chi^{-4}\partial_H\chi \geq 3c_*H.$$

Therefore

$$\chi(\beta, H) \leq C \min\{\chi_0(\beta), H^{-2/3}\}; \quad (21.227)$$

if  $\chi$  ever falls below  $2B_*$ , the same estimate remains true after increasing  $C$ . Since  $M(\beta, 0) = 0$  below criticality,

$$M(\beta, H) = \int_0^H \chi(\beta, s) ds \leq CH^{1/3}. \quad (21.228)$$

Monotonicity in  $\beta$  and the thermodynamic limit now give (21.224) as  $\beta \uparrow \beta_c$ .

It remains to obtain the complementary bounds and to transfer the critical isotherm estimate to the ordered side. At  $H = 0$  in the arguments below,  $M(\beta, 0)$  means the right-field limit  $M_+(\beta, 0)$ . First, finite critical bubble implies  $M_+(\beta_c, 0) = 0$ : otherwise the zero-field two-point function tends to  $M_+(\beta_c, 0)^2 > 0$ , contradicting  $\sum_x G_{\beta_c}(x)^2 < \infty$ . Thus  $M(\beta_c, H) \downarrow 0$  as  $H \downarrow 0$ .

Let  $m = M(\beta_c, H)$  and set  $\Delta = H/(J_*m)$ . On the segment

$$F(s) = M(\beta_c + s, H - J_*ms), \quad 0 \leq s \leq \Delta,$$

(21.219) implies  $F'(s) \leq 0$  whenever  $F(s) \leq m$ . Starting from  $F(0) = m$ , a first-crossing argument therefore gives

$$M_+(\beta_c + \Delta, 0) = F(\Delta) \leq m. \quad (21.229)$$

Since  $m \downarrow 0$  with  $H \downarrow 0$ , (21.229) and the positive sharpness bound show a posteriori that  $\Delta \downarrow 0$ ; otherwise the right-hand side of (21.225) would stay bounded away from zero. We may therefore apply its local square-root form:

$$m \geq c\Delta^{1/2} = c \left( \frac{H}{J_*m} \right)^{1/2},$$

and hence  $m \geq c'H^{1/3}$ . Together with (21.224) this proves the first part of (21.226).

Conversely let  $\Delta = \beta - \beta_c > 0$  and  $m = M_+(\beta, 0)$ . Put  $H_0 = J_*m\Delta$  and consider

$$F(s) = M(\beta_c + s, H_0 - J_*ms), \quad 0 \leq s \leq \Delta.$$

If  $F(0) < m$ , then as long as  $F < m$ , (21.219) gives  $F' < 0$ , so the curve cannot reach its endpoint value  $F(\Delta) = m$ . Hence

$$m \leq M(\beta_c, J_*m\Delta).$$

Using (21.224) gives

$$m \leq C(J_*m\Delta)^{1/3},$$

so  $m \leq C'\Delta^{1/2}$ . Combining this with (21.225) completes (21.226).  $\square$

**Theorem 10.85** (Mean-field Ising critical exponents). *Assume the translation-invariant finite-range ferromagnetic Ising model lies in one of the infrared/lace regimes of this chapter and, more precisely, that*

$$\langle \sigma_0 \sigma_x \rangle_{\beta_c} = A|x|^{2-d}(1 + o(1)), \quad (21.230)$$

*and that the corresponding zero-field bubble is uniformly bounded as  $\beta \uparrow \beta_c$ . Then*

$$\chi(\beta) \asymp (\beta_c - \beta)^{-1} \quad (\beta \uparrow \beta_c), \quad m_+(\beta, 0) \asymp (\beta - \beta_c)^{1/2} \quad (\beta \downarrow \beta_c),$$

*and at critical temperature*

$$m(\beta_c, h) \asymp h^{1/3} \quad (h \downarrow 0).$$

*Consequently the mean-field exponents are*

$$\eta = 0, \quad \gamma = 1, \quad \beta_{\text{mag}} = \frac{1}{2}, \quad \delta = 3$$

*in the stated regimes.*

*Proof.* The two-point asymptotic (21.230) is exactly the statement  $\eta = 0$ . It also forces  $\chi_0(\beta_c) = \infty$ , since  $\sum_{|x| \leq R} |x|^{2-d} \asymp R^2$ ; monotone convergence in  $\beta$  then gives  $\chi_0(\beta) \uparrow \infty$  as  $\beta \uparrow \beta_c$ . In the high-dimensional lace regime,  $d > 4$ , the same asymptotic gives the critical bubble convergence

$$\sum_x G_{\beta_c}(x)^2 < \infty,$$

and Griffiths monotonicity bounds  $B_0(\beta)$  uniformly by this critical bubble for  $\beta < \beta_c$ .

Proposition 10.84 therefore gives  $\chi \asymp (\beta_c - \beta)^{-1}$  and the critical-field upper bound. The lower ordered-side estimate (21.225) is not imported here: it is the exact sharp-transition inequality already proved in SM-08.03.02,

$$M_+(\beta, 0) \geq \sqrt{1 - \beta_c^2/\beta^2} \asymp (\beta - \beta_c)^{1/2}.$$

The GHS extrapolation part of Proposition 10.84 then supplies both the matching upper bound for  $M_+(\beta, 0)$  and the lower critical-isotherm bound. Finally  $H = \beta_c h$  at criticality, so  $M(\beta_c, H) \asymp H^{1/3}$  is equivalent, up to a fixed constant, to  $m(\beta_c, h) \asymp h^{1/3}$ . This proves all four exponents without using the separate Sakai coefficient-expansion theorem: it is not a dependency of the thermodynamic implication proved here.  $\square$

**Proof dependencies.** The argument uses the zero-field bubble differential inequality, the nonzero-field diluted-current inequality, their integration, and the GHS temperature–field extrapolation. The only model-specific hypothesis retained is the stated infrared/lace input that supplies the Gaussian critical two-point function and bounded critical bubble. Thus this is a conditional thermodynamic consequence theorem, independent of the separate random-current lace-coefficient theorem.

### 10.3.4 Marginal triviality in four dimensions: the random-current proof

The upper critical dimension is the one place where the ordinary tree-diagram bound is dimensionally correct but too weak: in four dimensions the bubble diagram is allowed to diverge logarithmically. The purpose of this subsection is to own the missing logarithmic improvement rather than hide it behind the phrase “multiscale random-current argument.” We follow the proof architecture of Aizenman–Duminil-Copin, but we isolate each load-bearing estimate used in the Gaussianity theorem.

Throughout this subsection the underlying Ising model is the nearest-neighbour ferromagnetic model on  $\mathbb{Z}^4$ ,  $\beta \leq \beta_c$ , and

$$S_\beta(x) = \langle \sigma_0 \sigma_x \rangle_\beta, \quad \chi_R(\beta) = \sum_{|x| \leq R} S_\beta(x), \quad B_R(\beta) = \sum_{|x| \leq R} S_\beta(x)^2. \quad (21.231)$$

Here  $|\cdot|$  is the  $\ell^\infty$  norm. The correlation length  $\xi(\beta)$  is defined by the exponential rate of  $S_\beta(ne_1)$ ; at criticality it is infinite. Constants below are independent of  $\beta$  and of scales not exceeding  $\xi(\beta)$ .

**Lemma 10.86** (One- and two-visit current identities). *Let  $n_1$  have sources  $\{0, x\}$  and let  $n_2$  be an independent sourceless current. Then*

$$\mathbf{P}_\beta^{0x, \emptyset}(u \leftrightarrow 0 \text{ in } n_1 + n_2) = \frac{S_\beta(u)S_\beta(x-u)}{S_\beta(x)}, \quad (21.232)$$

and

$$\begin{aligned} & \mathbf{P}_\beta^{0x, \emptyset}(u, v \leftrightarrow 0 \text{ in } n_1 + n_2) \\ & \leq \frac{S_\beta(v)S_\beta(u-v)S_\beta(x-u)}{S_\beta(x)} + \frac{S_\beta(u)S_\beta(u-v)S_\beta(x-v)}{S_\beta(x)}. \end{aligned} \quad (21.233)$$

*Proof.* We prove the identities first in a finite box and then exhaust the lattice. For (21.232), apply source switching to the event that  $u$  belongs to the 0-cluster. Switching the part of the  $0$ - $x$  source current between 0 and the first visit to  $u$  changes the source allocation from  $\{0, x\}, \emptyset$  to  $\{0, u\}, \{u, x\}$  without changing the total current. Division by the current partition functions gives exactly the three two-point functions in (21.232).

For (21.233), switch first at  $u$ . Conditional on the total cluster  $C(v)$  of  $v$ , the event that  $u$  is not connected to  $v$  leaves one correlation in the graph with  $C(v)$  deleted. Griffiths monotonicity bounds this deleted-graph correlation by its full-graph value. Repeating the same switching after interchanging  $u$  and  $v$  partitions the event according to which of  $u, v$  is met first. The two possibilities yield respectively

$$S_\beta(v)S_\beta(u-v)S_\beta(x-u) \quad \text{and} \quad S_\beta(u)S_\beta(u-v)S_\beta(x-v).$$

After division by  $S_\beta(x)$  this is (21.233). Monotone exhaustion is legitimate for  $\beta \leq \beta_c$  because the source-current measures used here have no infinite positive-current path in this regime.  $\square$

**Lemma 10.87** (Sliding-scale infrared and regular-scale package). *There is  $C < \infty$  such that, for  $1 \leq r \leq R \leq \xi(\beta)$ ,*

$$\frac{\chi_R(\beta)}{R^2} \leq C \frac{\chi_r(\beta)}{r^2}. \quad (21.234)$$

*Fix  $\alpha > 2$ . There are constants  $c, C_0 > 0$  such that whenever  $n^\alpha \leq N \leq \xi(\beta)$ , at least  $c \log_2(N/n)$  dyadic scales  $m \in [n, N]$  satisfy all of*

- (R1)  $S_\beta(x) \asymp S_\beta(y)$  for  $x, y \in \text{Ann}(m/2, 4m)$ ;
- (R2)  $|S_\beta(x) - S_\beta(y)| \leq C_0|x - y|S_\beta(x)/|x|$  on the same annulus;
- (R3)  $S_\beta(y) \leq \frac{1}{2}S_\beta(x)$  for  $x \in \Lambda_m$  and  $y \notin \Lambda_{C_0m}$ ;
- (R4)  $\chi_{2m}(\beta) \geq (1 + c)\chi_m(\beta)$ .

Moreover, in dimension four,

$$B_R(\beta) \leq \left(1 + C \frac{\log(R/r)}{\log r}\right) B_r(\beta) \quad (2 \leq r \leq R \leq \xi(\beta)). \quad (21.235)$$

*Proof.* We give the three ingredients because all three are used later. The Simon–Lieb inequality, infrared estimate, and critical continuity invoked below are not new imports: they are exactly the earlier project owners SM-08.02.04, SM-08.08.01, and SM-08.S1.01, respectively. Thus the present lemma only has to prove locally the multiscale comparison which is special to the marginal four-dimensional argument.

*Sliding scale.* Replace the hard cutoff susceptibility by

$$\tilde{\chi}_R = \sum_x e^{-|x|_2^2/(2R^2)} S_\beta(x).$$

Reflection monotonicity gives  $\tilde{\chi}_R \asymp \chi_R$ . For  $\beta < \beta_c$  Fourier inversion gives

$$\tilde{\chi}_R = c_d R^4 \int_{[-\pi, \pi]^4} e^{-R^2 |p|_2^2/2} \hat{S}_\beta(p) dp. \quad (21.236)$$

For completeness, put the model first in a periodic box and cut it by successive hyperplanes orthogonal to  $e_1$ . Reflection positivity writes the one-step transfer operator in the form  $T = A^* A$ , so  $T$  is positive and self-adjoint. The spectral theorem therefore writes every transversely smeared matrix coefficient  $\langle f, T^j f \rangle$  as  $\int_0^1 \lambda^j d\nu_f(\lambda)$ . Passing first to the thermodynamic limit and then writing  $\lambda = e^{-a}$  gives a positive mixture of  $e^{-aj}$ ; exponential decay below criticality places the spectral support in  $a \geq \xi(\beta)^{-1}$ . For each exponential, the elementary inequality

$$\frac{1 - e^{-aR}}{1 - e^{-ar}} \leq C \frac{R}{r}$$

and the same estimate after one discrete derivative imply, in the cone where  $|p_1| = \|p\|_\infty$ ,

$$\hat{S}_\beta(p) \leq C(R/r)^2 \{\hat{S}_\beta((R/r)p) + 1\}. \quad (21.237)$$

The infrared bound controls the harmless additive term. Split the Brillouin zone into the four coordinate cones, insert (21.237) into (21.236), and make the change of variables  $q = (R/r)p$ . One obtains

$$\tilde{\chi}_R/R^2 \leq C \tilde{\chi}_r/r^2,$$

which proves (21.234). The critical case follows by monotone convergence as  $\beta \uparrow \beta_c$ .

*Gradient control.* The same positive spectral representation gives, for an axis correlation  $a_j = S_\beta(je_1)$ ,

$$0 \leq a_j - a_{j+1} = \int e^{-aj} (1 - e^{-a}) d\mu(a).$$

Split the integral at  $a = 1/j$  and use  $1 - e^{-a} \leq \min\{a, 1\}$  on the two pieces. Comparing the resulting moments with  $a_{\lfloor j/2 \rfloor} - a_j$  yields

$$|a_j - a_{j+1}| \leq \frac{C}{j} \frac{S_\beta(dje_1)}{S_\beta(je_1)} \log \left( \frac{2S_\beta(\lfloor j/2 \rfloor e_1)}{S_\beta(je_1)} \right) S_\beta(je_1). \quad (21.238)$$

Reflection across coordinate and diagonal hyperplanes transfers this estimate to all nearest-neighbour increments. Indeed, if  $x$  and  $x + e_k$  lie in the same reflection chamber, reflect the more distant

endpoint through the bisecting coordinate or diagonal hyperplane. Ferromagnetic reflection positivity pairs every high-temperature/current monomial on the near side with its reflected monomial on the far side, so the two-point function cannot increase when the endpoint is moved one lattice step farther from the reflection plane. A finite sequence of such reflections reduces an arbitrary nearest-neighbour increment in the annulus to the axial estimate above, with a dimension-dependent constant only. Thus whenever correlations at radii between  $m/4$  and  $4dm$  are comparable, telescoping nearest-neighbour steps gives (R2).

*Abundance.* Below the correlation length the Simon–Lieb lower bound and the infrared upper bound imply, for  $n^\alpha \leq N$ ,

$$\chi_N \geq c(N/n)^{(\alpha-2)/(\alpha-1)} \chi_n. \quad (21.239)$$

If fewer than a fixed positive fraction of the dyadic scales between  $n$  and  $N$  had

$$\chi_{qm} \geq \chi_{16m} + \chi_m \quad (21.240)$$

for a fixed sufficiently large  $q$ , repeated failure of (21.240), together with (21.234), would contradict the power gain in (21.239). Hence  $c \log_2(N/n)$  scales satisfy (21.240). At each such scale, reflection monotonicity gives

$$|\text{Ann}(16m, qm)| S_\beta(16me_1) \geq \chi_{qm} - \chi_{16m} \geq \chi_m \geq |\Lambda_{m/16}| S_\beta((m/4)e_1). \quad (21.241)$$

Thus the axis correlations between radii  $m/4$  and  $16m$  are comparable. Reflection monotonicity proves (R1), (21.238) proves (R2), and (21.241) gives  $S_\beta(x) \geq cm^{-4} \chi_m$  on  $\text{Ann}(m, 2m)$ , hence (R4). Finally, for  $y \notin \Lambda_{4Rm}$  and  $x \in \Lambda_m$ , (21.234) and the last lower bound give

$$|\Lambda_{Rm}| S_\beta(y) \leq \chi_{Rm} \leq CR^2 \chi_m \leq CR^2 m^4 S_\beta(x).$$

Since  $d = 4 > 2$ , choosing  $R$  large makes the ratio at most  $1/2$ , proving (R3).

To prove (21.235), let  $m$  be a regular dyadic scale. From (R1), (R4), (21.234), and Cauchy–Schwarz,

$$B_{2M} - B_M \leq CM^{-4} \chi_M^2 \leq Cm^{-4} (\chi_{2m} - \chi_m)^2 \leq C(B_{2m} - B_m) \quad (21.242)$$

whenever  $m \leq M \leq R$ . There are  $O(\log(R/r))$  annuli above  $r$  and, by abundance,  $\gg \log r$  regular annuli below  $r$ . Compare every upper annular increment with one of those lower regular increments in (21.242) and sum. This gives (21.235).  $\square$

**Proposition 10.88** (Random-current annular mixing). *There are  $\alpha, c > 0$  such that if  $n^\alpha \leq N \leq \xi(\beta)$ , if  $x_i \in \Lambda_n$ ,  $y_i \notin \Lambda_N$  for  $1 \leq i \leq t$ , and if  $n_1, \dots, n_s$  are independent currents with the first  $t$  source sets  $\{x_i, y_i\}$  and the others sourceless, then events  $E, F$  depending only on edges in  $\Lambda_n$  and outside  $\Lambda_N$ , respectively, satisfy*

$$|\mathbf{P}(E \cap F) - \mathbf{P}(E)\mathbf{P}(F)| \leq Cs \{\log(N/n)\}^{-c}. \quad (21.243)$$

*The same bound holds when the inner or outer endpoints of the source currents are moved inside their respective regions.*

*Proof.* Put  $m/n = (N/n)^{1/3}$  and  $N/M = (N/n)^{1/3}$ . By Lemma 10.87, choose a set  $\mathcal{K}$  of  $\gg \log(N/n)$  mutually separated regular dyadic scales in  $[m, M]$ . For one source pair  $x, y$  define

$$A_{x,y}(r) = \sum_{u \in A_y(r)} \frac{S_\beta(x-u) S_\beta(u-y)}{S_\beta(x-y)},$$

where  $A_y(r) \subset \text{Ann}(r, 2r)$  consists of those  $u$  for which the regular-scale gradient comparison with the remote endpoint  $y$  holds. Reflection monotonicity supplies a fixed positive fraction of every annulus to  $A_y(r)$ . For a source current  $n_i$  and an independent sourceless copy  $n'_i$ , set

$$N_i = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \frac{1}{A_{x_i, y_i}(2^k)} \sum_{u \in A_{y_i}(2^k)} \mathbf{1}_{\{u \leftrightarrow x_i \text{ in } n_i + n'_i\}}, \quad N_* := \prod_{i=1}^t N_i. \quad (21.244)$$

By (21.232),  $\mathbf{E}N_i = 1$ . For two distinct regular scales  $k < \ell$ , (21.233), (R2), and repeated use of (R3) give

$$\sum_{u \in A_y(2^k)} \sum_{v \in A_y(2^\ell)} \mathbf{P}(u, v \leftrightarrow x) \leq A_{x, y}(2^k) A_{x, y}(2^\ell) \{1 + C2^{-(\ell-k)}\}. \quad (21.245)$$

For  $k = \ell$ , the same identity and (R1) give the bound  $CA_{x, y}(2^k)^2$ . Expanding  $N_i^2$  and summing the geometric error in (21.245) therefore yields

$$\mathbf{E}(N_i - 1)^2 \leq C/|\mathcal{K}|, \quad \mathbf{E}(N_* - 1)^2 \leq C_s / \log(N/n). \quad (21.246)$$

It remains to turn concentration of  $N_*$  into decoupling. Insert (21.244) into  $\mathbf{P}(E \cap F)$ . For each selected intermediate vertex  $u_i$ , source switching transfers the segment of current between  $u_i$  and  $y_i$  to the auxiliary sourceless current. On the event that every switched backbone crosses the annulus from  $2n$  to  $N/2$ , the switched configuration splits into an inner current carrying the  $x_i - u_i$  sources and an outer current carrying the  $u_i - y_i$  sources; the unused currents remain independent. Thus, after averaging in the  $u_i$  with the weights in (21.244), the switched expectation factors as the product of an inner and an outer expectation. Cauchy–Schwarz and (21.246) cost  $Cs/\sqrt{\log(N/n)}$ .

We spell out the exceptional-event estimate, since this is where spatial separation enters the switching argument. Given the selected vertices  $u_1, \dots, u_t$ , let  $\mathcal{G}(u_1, \dots, u_t)$  be the event that for every  $i \leq s$  there is a subcurrent  $k_i \leq n_i + n'_i$  which vanishes on  $\Lambda_n$ , agrees with  $n_i + n'_i$  outside  $\Lambda_N$ , and has sources  $\{u_i, y_i\}$  for  $i \leq t$  and no sources for  $i > t$ . On  $\mathcal{G}$  the switching lemma gives the exact inner/outer factorization used above.

It is enough to show that  $\mathcal{G}^c$  is unlikely. The event  $\mathcal{G}_i$  contains the conjunction that  $n_i$  does not cross  $\text{Ann}(M, N)$  and that  $n'_i$  does not cross  $\text{Ann}(n, m)$ . Put  $R = (MN)^{1/2}$ . If the source-current backbone  $\Gamma(n_i)$  crosses from  $M$  to  $R$ , the first-hit decomposition and (21.232) give

$$\mathbf{P}(\Gamma(n_i) \text{ crosses } \text{Ann}(M, R)) \leq \sum_{v \in \partial\Lambda_R} \frac{S_\beta(x_i - v)S_\beta(v - u_i)}{S_\beta(x_i - u_i)} \leq C \frac{R^3 M^3}{R^4} \leq C(n/N)^\varepsilon.$$

Here the middle inequality is the combination of the lower two-point estimate at scale  $M$  and the infrared upper bound at scale  $R$ ; the scale choice  $R = (MN)^{1/2}$  and  $N \geq n^\alpha$  make the last term small when  $\alpha$  is large. If instead the backbone has already crossed but the remaining sourceless part connects  $R$  to  $N$ , adding one sourceless current and switching once more gives the boundary-shell estimate

$$\mathbf{P}(n_i \setminus \Gamma(n_i) \text{ crosses } \text{Ann}(R, N)) \leq \sum_{v \in \partial\Lambda_R} \sum_{w \in \partial\Lambda_N} S_\beta(v - w)^2 \leq C(n/N)^\varepsilon.$$

The last inequality is the four-dimensional shell summation of the infrared bound  $S_\beta(z) \leq C|z|^{-2}$  with the above power-separated choices of  $M, R, N$ . For  $n'_i$  use  $r = (nm)^{1/2}$  and repeat the same argument across  $\text{Ann}(n, m)$ . In the source-current case the selected  $u_i \in A_{y_i}(2^k)$  supplies exactly



the remote-endpoint gradient comparison needed in the first-hit ratio; in the sourceless case only the squared infrared sum occurs. Therefore

$$\mathbf{P}(\mathcal{G}^c) \leq \sum_{i=1}^s \mathbf{P}(\mathcal{G}_i^c) \leq Cs(n/N)^\varepsilon. \quad (21.247)$$

Combining (21.246)–(21.247), and repeating the same switching after changing the inner or outer source endpoints, proves (21.243).  $\square$

**Proposition 10.89** (Intersection clustering and logarithmic tree improvement). *There are  $c, C > 0$  such that, for  $\beta \leq \beta_c$ ,  $L \leq \xi(\beta)$ , and four sites  $x, y, z, t$  at mutual distance at least  $L$ ,*

$$|U_{4,\beta}(x, y, z, t)| \leq \frac{C}{B_L(\beta)^c} \sum_{u \in \mathbb{Z}^4} S_\beta(u-x) S_\beta(u-y) S_\beta(u-z) S_\beta(u-t). \quad (21.248)$$

*Proof.* Let  $n_1, n_2$  have source sets  $\{x, y\}$  and  $\{z, t\}$  and let  $n_3, n_4$  be independent sourceless currents. Two applications of source switching give

$$|U_{4,\beta}(x, y, z, t)| \leq 2S_\beta(x-y) S_\beta(z-t) \mathbf{P}(C_{n_1+n_3}(x) \cap C_{n_2+n_4}(z) \neq \emptyset). \quad (21.249)$$

Fix large  $D$  and define dynamic bubble scales by

$$\ell_0 = 1, \quad \ell_{k+1} = \inf\{\ell : B_\ell \geq DB_{\ell_k}\}. \quad (21.250)$$

The pointwise infrared bound implies  $B_R - B_r \leq C \log(R/r)$ , hence

$$D^k \leq B_{\ell_k} \leq CD^k. \quad (21.251)$$

By (21.235), after increasing  $D$  the scales can be taken sufficiently separated that Proposition 10.88 applies between successive relevant annuli.

We record the local success estimate. Condition that a fixed vertex  $u$  lies in both duplicated source clusters. On a regular annulus between two successive dynamic scales, let  $X$  count vertices in the intersection of the two duplicated clusters. Lemma 10.86, applied to each duplicated current, and (R1)–(R2) give

$$\mathbf{E}X \geq c(B_M - B_m), \quad \mathbf{E}X^2 \leq C(B_M - B_m)B_M. \quad (21.252)$$

Choose  $m, M$  inside the dynamic annulus so that (21.235) makes  $B_M - B_m \asymp B_{\ell_{k+1}}$ . Paley–Zygmund then yields a fixed probability  $p_0 > 0$  that this annulus contains an intersection vertex.

There are  $K \asymp \log B_L$  dynamic scales below  $L$ . Take every other one. By (21.243), after conditioning on all information from previously explored scales, the success probability at the next selected scale is at least  $p_0/2$ ; the sum of the mixing errors is made smaller than  $p_0/2$  by the choice of  $D$ . Consequently, if  $M_u$  denotes the number of selected annuli around  $u$  met by the intersection set,

$$\mathbf{P}(M_u < \delta K \mid u \text{ is an intersection point}) \leq 2^{-\delta K} \quad (21.253)$$

for some  $\delta > 0$ .

A deterministic annular-covering argument says that a finite set meeting at least  $\delta K$  of the annuli about each of its points has cardinality at least  $2^{c\delta K}$ . Thus on the event that the two clusters intersect, either their intersection contains at least  $2^{cK}$  vertices or there is a vertex  $u$  for which

the exceptional event in (21.253) occurs. Markov's inequality for the first alternative and source switching at  $u$  for the second give

$$\begin{aligned} & S_\beta(x-y)S_\beta(z-t)\mathbf{P}(C_{n_1+n_3}(x) \cap C_{n_2+n_4}(z) \neq \emptyset) \\ & \leq C2^{-cK} \sum_u S_\beta(u-x)S_\beta(u-y)S_\beta(u-z)S_\beta(u-t). \end{aligned}$$

By (21.251),  $K \geq c \log B_L$ , so  $2^{-cK} \leq CB_L^{-c}$ . Insert this in (21.249) to obtain (21.248).  $\square$

**Proposition 10.90** (Smeared Wick estimate). *Let*

$$\Sigma_L = \left\langle \left( \sum_{x \in \Lambda_L} \sigma_x \right)^2 \right\rangle, \quad T_{f,L} = \Sigma_L^{-1/2} \sum_x f(x/L) \sigma_x, \quad (21.254)$$

where  $f$  is continuous and supported in  $[-r, r]^4$ . Uniformly for  $\beta \leq \beta_c$  and  $L \leq \xi(\beta)$ ,

$$\left| \langle T_{f,L}^{2n} \rangle - \frac{(2n)!}{2^n n!} \langle T_{f,L}^2 \rangle^n \right| \leq C_n \|f\|_\infty^{2n} r^{C_n} \left( \frac{\log \log L}{\log L} \right)^c, \quad (21.255)$$

whereas odd moments vanish.

*Proof.* For  $2n$  spin insertions, repeated source switching gives the exact comparison between the full moment and the Wick pairing sum:

$$\begin{aligned} 0 & \leq G_n[S_\beta](x_1, \dots, x_{2n}) - \langle \sigma_{x_1} \cdots \sigma_{x_{2n}} \rangle_\beta \\ & \leq -\frac{3}{2} \sum_{i < j < k < \ell} G_{n-2}[S_\beta](\hat{x}_i, \hat{x}_j, \hat{x}_k, \hat{x}_\ell) U_{4,\beta}(x_i, x_j, x_k, x_\ell), \end{aligned} \quad (21.256)$$

where  $G_m[S]$  is the sum over pairings of  $2m$  points and hats mean omission. The sign is correct because the ferromagnetic four-point Ursell function is nonpositive.

Insert (21.248), multiply by the test-function weights, and sum. The only new scalar quantity is

$$\mathcal{S}(L, r, \beta) = \frac{1}{\Sigma_L^2} \sum_u \sum_{x_1, \dots, x_4 \in \Lambda_{rL}} \frac{\prod_{j=1}^4 S_\beta(u - x_j)}{B_{d(x_1, \dots, x_4)}(\beta)^c}, \quad (21.257)$$

where  $d(x_1, \dots, x_4)$  is the minimum mutual separation, truncated below by one. We now bound it, rather than declare it negligible.

Fix once and for all  $a \in (0, 1)$ , say  $a = 1/2$ . Split (21.257) into four regions according as  $u \in \Lambda_{4rL}$  or not and as the minimum mutual separation is at least  $L^a$  or not. In the near/ $L^a$ -separated region, (21.235) gives  $B_{d(x_1, \dots, x_4)} \geq cB_L$  and the sliding-scale bound gives

$$\mathcal{S}_1 \leq Cr^{12} \left( \frac{\chi_L^2}{L^4 B_L} \right)^c.$$

By Cauchy–Schwarz,

$$\frac{\chi_L^2}{L^4 B_L} \leq 2 \frac{\chi_{L/\log L}^2}{L^4} + C \frac{B_L - B_{L/\log L}}{B_L} \leq C \frac{\log \log L}{\log L}, \quad (21.258)$$

where the last step uses (21.234)–(21.235). If  $u$  is far away, the infrared bound gives  $S_\beta(u - x_j) \leq CL^2|u|^{-2}\chi_L$  and the same summation yields the bound in (21.258). In either close-pair region,

discard the bubble gain, bound the two correlations incident to the closest pair by  $C|x|^{-2}$ , sum the remaining two by  $\chi_{2rL}$ , and use  $\chi_{2rL} \leq Cr^2\chi_L$ . These terms are  $O(r^C L^{4a-4} \log L)$  and hence smaller. Therefore

$$\mathcal{S}(L, r, \beta) \leq Cr^{12} \left( \frac{\log \log L}{\log L} \right)^c. \quad (21.259)$$

Summing (21.256) with the  $f$ -weights and using induction on  $n$  now proves (21.255). Spin-flip symmetry gives the odd moments.  $\square$

**Lemma 10.91** (Griffiths–Simon block approximation of a  $\varphi^4$  site law). *For  $\lambda > 0$  and  $b \in \mathbb{R}$ , the probability law*

$$d\rho_{\lambda,b}(t) = Z^{-1} e^{-\lambda t^4 - bt^2} dt \quad (21.260)$$

*is a weak limit of ferromagnetic Ising block variables  $\tau_N = a_N \sum_{j=1}^N \sigma_j$  with nonnegative internal pair couplings. If neighbouring block variables are coupled by  $J_{xy} \tau_x \tau_y$ , the decorated Ising model remains ferromagnetic.*

*Proof.* Let  $m = \sum_{j=1}^N \sigma_j$  and choose

$$a_N = (12\lambda N^3)^{-1/4}, \quad K_N = \frac{1}{2N} - ba_N^2.$$

For large  $N$ ,  $K_N \geq 0$ . Give a block configuration weight  $\exp(K_N m^2)$ . Stirling's formula, uniformly for  $m = O(N^{3/4})$ , gives

$$\log \binom{N}{(N+m)/2} = N \log 2 - \frac{m^2}{2N} - \frac{m^4}{12N^3} + O\left(\frac{m^6}{N^5} + \frac{1}{N}\right). \quad (21.261)$$

Writing  $t = a_N m$ , the quadratic part of  $K_N m^2$  cancels the entropy term  $-m^2/(2N)$  and leaves  $-bt^2$ , while our choice of  $a_N$  turns the quartic term into  $-\lambda t^4$ . The error in (21.261) is  $o(1)$  on compact  $t$ -sets. The negative quartic term gives a uniform  $e^{-ct^4}$  tail, so Riemann-sum convergence proves  $\tau_N \Rightarrow \rho_{\lambda,b}$ . Finally

$$J_{xy} \tau_x \tau_y = J_{xy} a_N^2 \sum_{i,j} \sigma_{x,i} \sigma_{y,j},$$

which has nonnegative constituent couplings whenever  $J_{xy} \geq 0$ .  $\square$

**Proposition 10.92** (Uniform Griffiths–Simon transfer of the current estimates). *Let a nearest-neighbour Griffiths–Simon core variable be represented on a decorated block by*

$$\tau_x = \sum_{j=1}^N Q_j \sigma_{x,j}, \quad Q_j \geq 0,$$

*with nonnegative intra-block couplings and inter-block couplings  $J_{xy} Q_i Q_j$ . The constants in the regular-scale package, annular mixing, intersection clustering, and Wick-error estimate above can be chosen independently of  $N$ , the  $Q_j$ , and the intra-block couplings after replacing the Ising bubble gain  $B_L^{-c}$  by the dimensionless factor  $(B_0/B_L)^c$ . In particular,*

$$|U_{4,\rho,\beta}(x, y, z, t)| \leq C \left( \frac{B_0}{B_L} \right)^c \sum_{u, u', u''} \langle \tau_x \tau_u \rangle \beta J_{u,u'} \langle \tau_{u'} \tau_y \rangle \langle \tau_z \tau_u \rangle \beta J_{u,u''} \langle \tau_{u''} \tau_t \rangle. \quad (21.262)$$

*The same assertion passes to weak limits of such core laws.*

*Proof.* There are three points to check; all other steps are the Ising argument already owned above.

*Random sources and dimension balance.* For a block pair  $x, y$ , choose constituent source indices  $i, j$  with probability

$$\frac{Q_i Q_j \langle \sigma_{x,i} \sigma_{y,j} \rangle}{\langle \tau_x \tau_y \rangle}$$

and then sample the corresponding Ising current. Summing the switching identity over these random source indices turns each marked block visit into a  $\tau$  two-point function. Whenever a constituent spin is repeated, split the visit through one inter-block edge. First-edge decomposition and switching give the balanced site-splitting inequality

$$\langle \tau_x \tau_u \rangle \langle \tau_u \tau_y \rangle \leq \sum_{u'} \langle \tau_x \tau_u \rangle \beta J_{u,u'} \langle \tau_{u'} \tau_y \rangle. \quad (21.263)$$

Similarly, a sourceless connection from a block set  $S$  to  $T$  is bounded by summing over the first boundary edge of  $S$  and the last boundary edge of  $T$ ; this produces two  $\beta J$  factors and two block two-point functions. These are the block analogues of the visit bounds used above, and they are homogeneous under  $\tau \mapsto a\tau$ .

*Regularity and mixing.* Nearest-neighbour reflection positivity is unaffected by the one-site law, so the sliding-scale and regular-scale argument applies to  $S_{\rho,\beta}(x) = \langle \tau_0 \tau_x \rangle$ . In the mixing proof replace (21.244) by

$$N_i = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \frac{1}{A_{x_i, y_i}(2^k)} \sum_{u \in A_{y_i}(2^k)} \sum_j Q_j^2 \mathbf{1}_{\{(u,j) \leftrightarrow \partial n_i\}}, \quad (21.264)$$

where  $A_{x,y}(r) = \sum_{u \in A_y(r)} \langle \tau_x \tau_u \rangle \langle \tau_u \tau_y \rangle / \langle \tau_x \tau_y \rangle$ . The random-source normalization gives  $\mathbf{E} N_i = 1$ ; the  $Q_j^2$ -weighted second moment reconstructs exactly the block two-point sums in (21.245). For the exceptional event, expose the first block edge crossed by the backbone. The preceding one-edge and two-edge connectivity bounds replace the deterministic-source switching estimates and have the same infrared powers. Thus the concentration and exceptional-event estimates proving (21.243) are uniform in the decorated block.

*Coarse intersection and bubble gain.* Constituent clusters can enter the same block without sharing a constituent spin, so declare  $B_v$  successful when both duplicated clusters meet blocks within graph distance two of  $B_v$ . The weighted number  $M$  of successful blocks on a regular dynamic annulus satisfies

$$\mathbf{E}|M| \geq c\{B_M - B_{m-1}\}, \quad \mathbf{E}|M|^2 \leq C B_{\ell_{k+1}}^2. \quad (21.265)$$

The first estimate is the random-source one-visit computation plus (21.263), and the second is the corresponding two-visit computation. The infrared bound also gives  $B_M - B_{m-1} \geq B_{\ell_{k+1}} - B_{\ell_k} - C B_0$ . Hence, after taking the dynamic threshold  $D$  large, Paley–Zygmund gives a fixed success probability at each selected scale and the uniform mixing just proved makes the failure probability exponentially small in  $K \asymp \log(B_L/B_0)$ .

Finally sum the four-source switching identity over the constituent source indices. If there are many coarse intersections, Markov’s inequality yields the balanced tree diagram in (21.262); if there are few, the clustering failure probability supplies the factor  $(B_0/B_L)^c$ . All constants are independent of block size and weights. Weak convergence of the one-site law therefore preserves the estimates.  $\square$

**Theorem 10.93** (Marginal triviality of four-dimensional Ising and lattice-cutoff  $\varphi_4^4$  scaling limits). *For the four-dimensional nearest-neighbour ferromagnetic Ising model, let  $\beta_L \leq \beta_c$  satisfy  $L \leq \xi(\beta_L)$*

and form the normalized fluctuation fields (21.254). Every subsequential nondegenerate scaling limit, in the sense of joint laws of finitely many compactly supported continuous test functions, is a generalized Gaussian field. Equivalently, all limiting correlation functions obey Wick factorization.

The same conclusion holds for the nearest-neighbour lattice model with one-site probability measure  $Z^{-1}e^{-\lambda t^4 - bt^2} dt$ , where  $\lambda > 0$ . Hence it also applies to lattice-cutoff  $\lambda\varphi_4^4$  fields along every joint infinite-volume/vanishing-cutoff subsequence satisfying the source scaling normalization and the source decay condition that the limiting two-point function tends to zero at large Euclidean separation. The theorem identifies every reachable limit; it does not assert existence of a scaling limit for an arbitrary parameter sequence.

*Proof.* For the Ising model, Proposition 10.90 says that for every fixed test function  $f$  and every  $n$  the even moments of  $T_{f,L}$  converge to the Gaussian moments with the same limiting variance, while odd moments vanish. Gaussian moments satisfy Carleman's condition, so every convergent subsequence of  $T_{f,L}$  has the corresponding centered Gaussian law. Applying this to every finite linear combination  $\sum_j c_j f_j$  and using the Cramér–Wold theorem proves joint Gaussianity for any finite family of test functions. Polarization then gives Wick factorization of all limiting correlation functionals. Notice that no separate tightness theorem is being smuggled into the argument: the statement is deliberately an identification theorem for subsequential limits.

For a Griffiths–Simon block law, Proposition 10.92 provides the uniform dimension-balanced current estimates on the decorated graph, including the GS-specific random-source mixing and coarse-intersection steps. Consequently the higher-moment Wick error tends to zero uniformly over the finite Ising-block approximants.

Lemma 10.91 approximates the  $\varphi^4$  single-site law by precisely such ferromagnetic blocks. First pass to the block-size limit at fixed lattice volume, using weak convergence and the uniform quartic moment bound; then pass through the thermodynamic/scaling subsequence. The uniform Wick-error estimate survives both limits. Cramér–Wold therefore gives a generalized Gaussian limit for the lattice-cutoff  $\lambda\varphi_4^4$  field as well.  $\square$

**Proof scope.** The argument supplies the current one/two-visit identities, sliding-scale regularity package, abundance of regular scales, bubble interpolation, current mixing, intersection clustering, logarithmically improved tree diagram, higher-moment Wick estimate, and a concrete Griffiths–Simon block approximation of the  $\varphi^4$  site law. The theorem is stated as the source theorem actually functions: it identifies every reachable subsequential scaling limit and does not manufacture existence or tightness for arbitrary parameter sequences.

## 10.4 Finite-range rigorous RG in four dimensions

### 10.4.1 The foundational RG map at source resolution

The purpose of this subsection is not to compress the rigorous renormalisation group into one recursion. The weak-coupling construction used below has several logically different layers. We separate them because a proof of the perturbative flow is not a proof that the exact map can be iterated. Throughout this subsection  $d = 4$ ,  $n \geq 1$ ,  $L \geq 2$  is the block scale, and  $\Lambda$  is a finite discrete four-torus. The field is  $\varphi : \Lambda \rightarrow \mathbb{R}^n$ .

**Definition 10.94** (Scale blocks, polymers, and finite-range independence). For  $j \geq 0$ , a  $j$ -block is a translate of a cube of side  $L^j$  in  $\Lambda$ . A  $j$ -polymer is a union of  $j$ -blocks. Its connected components are formed using face adjacency of blocks. A fluctuation covariance  $C_{j+1}$  has *strict  $j$ -range* if there

is a fixed  $r_0$  such that

$$C_{j+1}(x, y) = 0 \quad \text{when } |x - y|_\infty > r_0 L^j.$$

Consequently, Gaussian fields restricted to two sets at mutual distance greater than  $r_0 L^j$  are independent.

The elementary decomposition in Chapter 1 proves positivity and strict finite range but not the scale-derivative estimates needed by the full four-dimensional construction. The source-strength covariance input required below is therefore recorded explicitly rather than silently attributed to Chapter 1.

**Definition 10.95** (Source-strength covariance gate). A finite-range decomposition  $C = \sum_{j=1}^N C_j$  is called *RG-admissible* if, uniformly up to the mass scale  $j_m$ ,

$$|\nabla_x^\alpha \nabla_y^\beta C_j(x, y)| \leq c_{\alpha, \beta} L^{-(2+|\alpha|+|\beta|)(j-1)}, \quad |\alpha| + |\beta| \leq p_\Phi, \quad (21.279)$$

with strict finite range  $O(L^j)$ , positivity, translation covariance, and the corresponding post-mass improvement by a summable factor  $\vartheta_j$ . Here  $p_\Phi$  is chosen larger than the maximal field derivative order appearing in the localisation norm.

**Theorem 10.96** (Source-strength finite-range decomposition for the lattice resolvent). *Let  $\mathcal{L} = -\Delta_{\mathbb{Z}^4}$  and fix  $m_*^2 < \infty$ . For every sufficiently large integer block scale  $L$  there are positive semidefinite translation-invariant covariances  $C_j^{(m^2)}$ ,  $j \geq 1$ , such that for  $0 \leq m^2 \leq m_*^2$*

$$(\mathcal{L} + m^2)^{-1} = \sum_{j \geq 1} C_j^{(m^2)} \quad (m^2 > 0), \quad (21.RG.1)$$

with the usual quadratic-form interpretation at  $m^2 = 0$ . After a harmless fixed rescaling of the scale parameter,

$$C_j^{(m^2)}(x, y) = 0 \quad \text{if } |x - y|_\infty \geq cL^j, \quad (21.RG.2)$$

and for every  $r, s, k \geq 0$  there is  $c_{r,s,k}$  such that

$$|\nabla_x^r \nabla_y^s C_j^{(m^2)}(x, y)| \leq c_{r,s,k} L^{-(2+r+s)(j-1)} (1 + m^2 L^{2(j-1)})^{-k}. \quad (21.RG.3)$$

The constants are uniform in  $m^2 \in [0, m_*^2]$ . The same construction on a discrete torus of side  $L^N$  gives these estimates on all scales strictly below the terminal scale, with a terminal positive covariance absorbing the remaining modes.

*Proof.* We give the spectral construction because it simultaneously explains positivity and finite range. Let  $A$  be the bounded nonnegative nearest-neighbour Laplacian after multiplying by a constant so that  $\text{spec}(A) \subset [0, 4]$ . Choose a smooth even function  $\hat{\rho}$  supported in  $[-1, 1]$  and not identically zero, and put  $\rho = |\mathcal{F}^{-1} \hat{\rho}|^2$ . Then  $\rho \geq 0$ ,  $\rho$  is smooth and rapidly decaying, and its Fourier transform is compactly supported. Normalize  $\rho$  so that

$$\lambda^{-1} = \int_0^\infty t^2 W_t(\lambda) \frac{dt}{t}, \quad W_t(\lambda) \geq 0, \quad (21.RG.4)$$

where in the continuum-wave case one can take  $W_t(\lambda) = c\rho(t\sqrt{\lambda})$ . For the bounded discrete operator use the periodised angle

$$W_t^*(\lambda) = c \sum_{q \in \mathbb{Z}} \rho(t \arccos(1 - \lambda/2) - 2\pi q t). \quad (21.RG.5)$$

The partial-fraction identity  $\frac{1}{4} \csc^2(x/2) = \sum_{q \in \mathbb{Z}} (x - 2\pi q)^{-2}$ , together with (21.RG.4) applied to each square, gives

$$\lambda^{-1} = \int_0^\infty t^2 W_t^*(\lambda) \frac{dt}{t} \quad (0 < \lambda \leq 4). \quad (21.RG.6)$$

All terms in (21.RG.5) are nonnegative, so  $W_t^*(A) \geq 0$  by spectral calculus.

We next prove finite range. Poisson summation and the support of  $\hat{\rho}$  give

$$W_t^*(\lambda) = \sum_{q \in \mathbb{Z}} t^{-1} \hat{\rho}(q/t) T_q(1 - \lambda/2), \quad (21.RG.7)$$

where  $T_q(\cos \theta) = \cos(q\theta)$  is the Chebyshev polynomial. Only  $|q| \leq t$  occurs. Hence  $W_t^*$  is the restriction of a polynomial in  $\lambda$  of degree at most  $\lceil t \rceil$ . Since  $A$  connects only nearest neighbours, a polynomial of degree  $q$  in  $A$  has kernel zero beyond graph distance  $q$ . Thus the kernel of  $t^2 W_t^*(A)$  has range  $O(t)$ .

The same formula also supplies the multiplier estimates. Repeated summation by parts in  $q$  in (21.RG.7), equivalently repeated finite differences of the rapidly decaying sequence  $t^{-1} \hat{\rho}(q/t)$ , gives for every  $a, b \geq 0$

$$(1 + t^2 \lambda)^a \lambda^b \left| \partial_\lambda^b W_t^*(\lambda) \right| \leq c_{a,b} \quad (21.RG.8)$$

for the spectral interval of the rescaled lattice Laplacian (the endpoint is kept uniformly away from the singular angle by the fixed spectral rescaling). This is the discrete Schwartz estimate behind the construction.

Return to the unscaled four-dimensional lattice Laplacian and add mass. In Fourier variables its symbol  $a(\xi)$  satisfies, uniformly for  $\xi \in [-\pi, \pi]^4$ ,  $a(\xi) \asymp |\xi|^2$  near the origin and is bounded away from zero outside a fixed neighbourhood of the origin. A finite difference in  $x$  contributes a multiplier bounded by  $c|\xi|$ . Therefore, for the continuous scale covariance

$$\Phi_t^{(m^2)} = t^2 W_t^*(\mathcal{L} + m^2),$$

Fourier inversion and (21.RG.8) give, after choosing  $a$  larger than  $k + (4 + r + s)/2$ ,

$$\begin{aligned} |\nabla_x^r \nabla_y^s \Phi_t^{(m^2)}(x, y)| &\leq ct^2 \int_{[-\pi, \pi]^4} |\xi|^{r+s} (1 + t^2(|\xi|^2 + m^2))^{-a} d\xi \\ &\leq c_{r,s,k} t^{-(2+r+s)} (1 + m^2 t^2)^{-k}. \end{aligned} \quad (21.RG.9)$$

For the second inequality split the factor  $(1 + t^2(|\xi|^2 + m^2))^{-a}$  into a mass factor and an integrable spatial factor, then set  $\eta = t\xi$ ; the part of the Brillouin zone away from the origin is smaller than any power of  $t^{-1}$  by (21.RG.8). Derivatives in  $m^2$  are obtained similarly from spectral derivatives of  $W_t^*$ .

Finally define the discrete scales by

$$C_j^{(m^2)} = \int_{L^{j-1}}^{L^j} \Phi_t^{(m^2)} \frac{dt}{t},$$

with the finitely many microscopic  $t < 1$  pieces absorbed into  $C_1$ . Positivity follows from positivity of every  $W_t^*(\mathcal{L} + m^2)$ , and (21.RG.6) gives the exact sum (21.RG.1). The degree/range bound yields (21.RG.2). Integrating (21.RG.9) over one geometric  $t$ -interval gives (21.RG.3). On a finite torus the identical polynomial functional calculus applies below the wrap-around scale; the positive spectral remainder is assigned to the terminal covariance. This completes the source-strength covariance gate required in Definition 10.95.  $\square$

**Covariance input.** The argument above proves positivity, strict finite range, the four-dimensional derivative powers, the mass-scale gain, and the finite-torus terminal remainder. Chapter 1's simpler lazy-walk block decomposition remains useful pedagogically but is not used as a substitute for this stronger result.

**Definition 10.97** ( $T_\varphi$  test-function seminorm). Let  $X \subset \Lambda$  be finite and let  $F$  be a  $C^p$  function of the coordinates  $\{\varphi_x^a : x \in X, 1 \leq a \leq n\}$ . For a positive field scale  $h_j$  and  $0 \leq r \leq p$ , set

$$\|D^r F(\varphi)\|_j = \sup \left\{ |D^r F(\varphi)[f_1, \dots, f_r]| : \max_i \|f_i\|_{\Phi_j(X)} \leq 1 \right\},$$

where

$$\|f\|_{\Phi_j(X)} = h_j^{-1} \max_{x \in X} \max_{|\alpha| \leq p_\Phi} L^{j|\alpha|} |\nabla^\alpha f_x|.$$

The finite-order seminorm is

$$\|F\|_{T_{\varphi,j}(X)} = \sum_{r=0}^p \frac{1}{r!} \|D^r F(\varphi)\|_j. \quad (21.280)$$

**Lemma 10.98** (Normed-algebra product property). For  $F, G \in C^p$  on the same finite field space,

$$\|FG\|_{T_{\varphi,j}} \leq \|F\|_{T_{\varphi,j}} \|G\|_{T_{\varphi,j}}. \quad (21.281)$$

If  $F$  depends only on  $X$  and  $G$  only on  $Y$ , the same estimate holds with the corresponding restricted field norms.

*Proof.* The Fréchet Leibniz rule gives

$$D^r(FG) = \sum_{s=0}^r \binom{r}{s} \text{Sym}(D^s F \otimes D^{r-s} G).$$

The operator norm of symmetrisation is at most one, hence

$$\frac{1}{r!} \|D^r(FG)\|_j \leq \sum_{s=0}^r \frac{1}{s!} \|D^s F\|_j \frac{1}{(r-s)!} \|D^{r-s} G\|_j.$$

Summing over  $0 \leq r \leq p$  yields a truncated Cauchy product bounded by the product of the two nonnegative sums in (21.280). Restriction to  $X$  or  $Y$  simply removes test directions on which the corresponding derivative vanishes.  $\square$

**Lemma 10.99** (Gaussian translation estimate in the  $T_\varphi$  seminorm). Let  $\zeta$  be a centred finite-dimensional Gaussian field with covariance  $C$ . Whenever the right side is integrable,

$$\|\mathbb{E}_C F(\varphi + \zeta)\|_{T_{\varphi,j}} \leq \mathbb{E}_C \|F\|_{T_{\varphi+\zeta,j}}. \quad (21.282)$$

The same estimate holds for every field derivative of the Gaussian expectation.

*Proof.* In finite dimension, differentiation under the Gaussian integral is justified first for a bounded  $C^p$  function and then by truncation whenever the displayed right side is integrable. Translation is affine, so

$$D^r \mathbb{E}_C F(\varphi + \zeta)[f_1, \dots, f_r] = \mathbb{E}_C D^r F(\varphi + \zeta)[f_1, \dots, f_r].$$

Take the supremum over unit test directions, use  $|\mathbb{E}X| \leq \mathbb{E}|X|$ , multiply by  $1/r!$ , and sum over  $r$ .  $\square$



**Definition 10.100** (Quadratic large-field regulator). Let  $Q_j$  be a positive quadratic form on fields over a connected  $j$ -polymer  $X$  and put

$$G_j(X, \varphi) = \exp\{\kappa Q_j(X, \varphi)\}.$$

We require  $Q_j$  to dominate the scaled  $\Phi_j$  norm on each block and to be additive up to a fixed overlap multiplicity over connected components.

**Lemma 10.101** (Gaussian stability of a quadratic regulator). *Let  $Q(\phi) = \langle \phi, A\phi \rangle$  with  $A \geq 0$ , and let  $\zeta \sim N(0, C)$ . If*

$$4\kappa\|C^{1/2}AC^{1/2}\|_{\text{op}} \leq \frac{1}{2}, \quad (21.283)$$

then

$$\mathbb{E} e^{\kappa Q(\phi+\zeta)} \leq \det(I - 2\kappa C^{1/2}AC^{1/2})^{-1/2} \exp\{2\kappa Q(\phi)\}. \quad (21.284)$$

*In particular, if the determinant factor is bounded uniformly per block, then after multiplying the regulator by a fixed harmless block constant the Gaussian step maps one large-field regulator into a slightly weaker regulator at the next scale.*

*Proof.* Write  $\zeta = C^{1/2}Z$  with  $Z$  standard Gaussian and  $B = C^{1/2}AC^{1/2}$ . Completing the square in the finite-dimensional Gaussian density gives, for  $2\kappa\|B\| < 1$ ,

$$\mathbb{E} e^{\kappa\langle \phi+C^{1/2}Z, A(\phi+C^{1/2}Z) \rangle} = \det(I - 2\kappa B)^{-1/2} e^{\kappa Q(\phi) + 2\kappa^2 \langle A\phi, C^{1/2}(I-2\kappa B)^{-1}C^{1/2}A\phi \rangle}.$$

Condition (21.283) implies  $\|(I - 2\kappa B)^{-1}\| \leq 4/3$  and  $2\kappa C^{1/2}AC^{1/2} \leq \frac{1}{4}I$ . Thus the second quadratic form in the exponent is bounded by a fixed fraction of  $\kappa Q(\phi)$ ; enlarging that fraction to one gives (21.284). The final statement follows by taking  $A$  block diagonal up to the bounded overlap multiplicity and absorbing the product of determinant factors into the normalisation assigned to the vacuum coordinate.  $\square$

**Definition 10.102** (Local polynomial space and localisation). At a block  $B$ , let  $\mathcal{V}(B)$  be the finite-dimensional space generated by the  $O(n)$ -invariant monomials of engineering dimension at most four:

$$1, \quad \tau_x = \frac{1}{2}|\varphi_x|^2, \quad \tau_x^2, \quad \tau_{\Delta,x} = \frac{1}{2}\varphi_x \cdot (-\Delta\varphi)_x,$$

together with the source monomials when observables are explicitly switched on. For a local functional  $F$  supported near  $B$ ,  $\text{Loc}_B F$  is the unique element of  $\mathcal{V}(B)$  whose pairing with every polynomial test field of complementary degree agrees with that of  $F$  through the prescribed Taylor order. This is a finite-dimensional moment-matching projection.

**Lemma 10.103** (Existence, projection property, and locality of  $\text{Loc}$ ). *The preceding moment conditions determine a linear projection  $\text{Loc}_B$  with*

$$\text{Loc}_B^2 = \text{Loc}_B, \quad \text{Loc}_B P = P \quad (P \in \mathcal{V}(B)). \quad (21.285)$$

*It commutes with translations and the  $O(n)$  action when the test basis is chosen covariantly.*

*Proof.* Choose a basis  $P_1, \dots, P_m$  of  $\mathcal{V}(B)$  and dual polynomial test functionals  $q_1, \dots, q_m$  for the nondegenerate finite-dimensional pairing used in the definition. Then

$$\text{Loc}_B F = \sum_{i=1}^m \langle F, q_i \rangle P_i.$$

If  $F = P_k$ , duality gives  $\text{Loc}_B P_k = P_k$ , proving both identities in (21.285). Translation and  $O(n)$  covariance follow because applying a symmetry to both arguments preserves the pairing and the invariant subspace  $\mathcal{V}(B)$ ; uniqueness of the moment-matching element then forces the projection to commute with the symmetry.  $\square$

**Lemma 10.104** (Power-counting gain after localisation). *Fix the Taylor order  $p_\Phi$  strictly above the maximal degree retained by  $\mathcal{V}$ . For a functional  $F$  supported in a set of  $j$ -diameter  $O(1)$  and a coarse block  $B^+$  of scale  $j+1$  containing its support, the moment cancellation  $(1 - \text{Loc}_{B^+})F$  obeys*

$$\|(1 - \text{Loc}_{B^+})F\|_{T_{\varphi,j+1}} \leq CL^{-\omega} \|F\|_{T_{\varphi,j}} \quad (21.286)$$

for some  $\omega > 0$  determined by the gap between the retained engineering dimensions and the first omitted Taylor monomial, provided the field scales satisfy the Gaussian four-dimensional relation  $h_{j+1} \asymp L^{-1}h_j$ .

*Proof.* It is enough to prove the estimate on each Taylor coefficient in a fixed finite coordinate chart. Let  $x_0$  be the centre of  $B^+$ . Expand every test field entering the pairing at  $x_0$  to the order annihilated by  $1 - \text{Loc}_{B^+}$ . The multivariable Taylor formula with integral remainder expresses the surviving term as a sum of derivatives of order one higher than the matched jet, each multiplied by a spatial increment. In scaled coordinates every increment has size at most  $O(L^{-1})$  relative to a  $(j+1)$ -block, while a field factor changes from  $h_j$  to  $h_{j+1} \asymp L^{-1}h_j$ . The engineering dimension of every surviving monomial is therefore at least  $4 + \omega$  for some positive integer or half-integer gap  $\omega$  depending only on the finite list of retained monomials. Multiplying the scale factors gives  $L^{-\omega}$ . There are only finitely many Taylor coefficients and derivative multi-indices, so their constants are absorbed into  $C$ .  $\square$

**Definition 10.105** (Connected Gaussian contraction). For local functionals  $A, B$  and a centred Gaussian field of covariance  $C$ , write

$$F_C(A, B) = \mathbb{E}_C(AB) - \mathbb{E}_C A \mathbb{E}_C B.$$

For a local potential  $V = \sum_x V_x$ , define its second-order local effective part by

$$V_{\text{pt}} = \mathbb{E}_C V - \frac{1}{2} \text{Loc } F_C(V, V), \quad (21.287)$$

where  $\text{Loc}$  is applied blockwise to the connected quadratic term.

**Lemma 10.106** (Exact cumulant remainder through second order). *Let  $V$  be a real random variable under a probability measure with  $\mathbb{E}e^{3|V|} < \infty$ , and let  $\Psi(t) = -\log \mathbb{E}e^{-tV}$ . Then*

$$\Psi(1) = \mathbb{E}V - \frac{1}{2} \text{Var}(V) + R_3(V), \quad (21.288)$$

where

$$|R_3(V)| \leq \frac{1}{6} \sup_{0 \leq t \leq 1} \mathbb{E}_t |V - \mathbb{E}_t V|^3, \quad (21.289)$$

and  $\mathbb{E}_t$  denotes expectation under the tilted density proportional to  $e^{-tV}$ .

*Proof.* Differentiation of the logarithmic moment generating function gives  $\Psi'(t) = \mathbb{E}_t V$ ,  $\Psi''(t) = -\text{Var}_t(V)$ , and  $\Psi'''(t) = \mathbb{E}_t[(V - \mathbb{E}_t V)^3]$ . Taylor's theorem with integral remainder at  $t = 0$  gives

$$\Psi(1) = \Psi'(0) + \frac{1}{2} \Psi''(0) + \frac{1}{2} \int_0^1 (1-t)^2 \Psi'''(t) dt.$$

Since  $\int_0^1 (1-t)^2 dt = 1/3$ , the last term has absolute value at most the right side of (21.289).  $\square$

**Lemma 10.107** (One-loop  $O(n)$  combinatorics). *Let  $V_x = g\tau_x^2 + \nu\tau_x + z\tau_{\Delta,x}$  with  $\tau_x = \frac{1}{2}|\varphi_x|^2$ . In the local part of the second cumulant of two quartic vertices, Wick contraction produces a quartic correction whose coefficient is a positive multiple of  $(n+8)g^2$ . A quartic vertex contracted once with a quadratic insertion produces a quadratic correction with coefficient a positive multiple of  $(n+2)g\nu$ . Consequently, after fixing a single coupling normalization, the universal ratio entering the logarithmic susceptibility correction is*

$$\frac{a}{\beta} = \frac{n+2}{n+8}. \quad (21.290)$$

*Proof.* Write a quartic vertex as  $\tau^2 = \frac{1}{4} \sum_{a,b} \varphi_a^2 \varphi_b^2$ . In the connected contraction of two such vertices leaving four external legs, there are three contraction topologies. Two preserve the two index loops and together contribute the  $n$ -dependent part; the crossed contractions are independent of the number of components. Performing the Kronecker-delta sums gives the standard  $O(n)$  tensor

$$(n+8)(\varphi \cdot \varphi)^2$$

up to the common covariance bubble and the normalization factor coming from  $\tau^2 = |\varphi|^4/4$ . Similarly, leaving two external legs after contracting a quartic vertex against a quadratic insertion gives the invariant tensor  $(n+2)|\varphi|^2$  times the same one-loop covariance sum. Since both coefficients contain the same normalization-dependent positive bubble factor, their quotient is exactly (21.290). The positivity follows because that bubble is a sum of squares of the positive covariance kernel.  $\square$

**Definition 10.108** (Circle product of polymer activities). For functions  $F, G$  on the set of  $j$ -polymers, define

$$(F \circ G)(X) = \sum_{Y \subset X} F(Y)G(X \setminus Y),$$

where the sum is over subpolymers formed from  $j$ -blocks. Let  $\mathbf{1}_\emptyset$  be one at the empty polymer and zero elsewhere.

**Lemma 10.109** (Exact polymer algebra). *The circle product is associative and commutative, with unit  $\mathbf{1}_\emptyset$ . If  $I(B)$  is a block factor and  $I(X) = \prod_{B \subset X} I(B)$ , then for every polymer activity  $K$ ,*

$$(I \circ K)(\Lambda) = \sum_{X \subset \Lambda} I(\Lambda \setminus X)K(X). \quad (21.291)$$

*If  $K$  factorises over connected components and the fluctuation covariance has strict finite range, Gaussian integration preserves component factorisation after enlarging the notion of adjacency by a fixed number of blocks.*

*Proof.* Associativity follows by expanding both  $(F \circ G) \circ H$  and  $F \circ (G \circ H)$ : both sums are indexed by ordered disjoint triples  $(X_1, X_2, X_3)$  with union  $X$ . Commutativity is the change of variable  $Y \leftrightarrow X \setminus Y$ , and the unit property is immediate. Formula (21.291) is the definition with  $X$  renamed as the support of  $K$ . For the last statement, strict finite range makes the Gaussian restrictions to sufficiently separated connected components independent. The expectation of a product of functionals of independent Gaussian coordinate sets is the product of their expectations. Enlarging adjacency exactly by the covariance range ensures that distinct enlarged components use disjoint independent coordinate sets.  $\square$

**Proposition 10.110** (Exact algebraic form of one progressive RG step). *Suppose at scale  $j$  the partition function has the exact form*

$$Z_j = e^{-u_j|\Lambda|}(I_j \circ K_j)(\Lambda). \quad (21.292)$$

*Integrate the independent Gaussian field  $\zeta_{j+1}$  of covariance  $C_{j+1}$ . Choose any local counterterm  $Q_j$  and define the next local interaction by absorbing Loc of the connected second-order term into  $I_{j+1}$ . Then there is a unique polymer activity  $K_{j+1}$  with  $K_{j+1}(\emptyset) = 0$  for which*

$$\mathbb{E}_{C_{j+1}} \theta Z_j = e^{-u_{j+1}|\Lambda|}(I_{j+1} \circ K_{j+1})(\Lambda). \quad (21.293)$$

*Here  $\theta F(\varphi, \zeta) = F(\varphi + \zeta)$ . Thus the issue in rigorous RG is not existence of an exact coordinate after one integration; it is the uniform norm estimate that makes the coordinate useful and contractive.*

*Proof.* Expand the left side of (21.292) by (21.291). Gaussian integration is linear and, by strict finite range, factorises on separated connected components after the fixed adjacency enlargement. Divide each integrated block factor by the chosen  $I_{j+1}$  and expand the resulting product over connected supports. Collect the empty support into the scalar  $e^{-(u_{j+1}-u_j)|\Lambda|}$ . Möbius inversion on the finite Boolean lattice of block subsets then determines the coefficient  $K_{j+1}(X)$  uniquely for every nonempty  $X$ . Recombining the coefficients by the circle product gives (21.293). The construction is purely algebraic; no small-coupling estimate has been used.  $\square$

**Proposition 10.111** (Perturbative local-coordinate recursion to second order). *Assume the RG-admissible covariance gate 10.95. Applying Loc to the first two cumulants gives local coordinates  $V_j = (g_j, \nu_j, z_j)$  of the form*

$$\begin{aligned} g_{j+1} &= g_j - \beta_j g_j^2 + O(g_j^3), \\ \nu_{j+1} &= L^2 \nu_j \{1 - a_j g_j + O(g_j^2)\} + O(g_j^3), \\ z_{j+1} &= z_j + b_j g_j^2 + O(g_j^3) \end{aligned} \quad (21.294)$$

*where  $\beta_j > 0$  before the mass scale,  $\beta_j$  converges to a positive massless coefficient, and  $a_j/\beta_j \rightarrow (n+2)/(n+8)$ . The  $O(g_j^3)$  assertion in this proposition refers only to the local perturbative Taylor remainder; it is not yet the nonperturbative polymer estimate required to iterate the exact map.*

*Proof.* Use Lemma 10.106 blockwise with the fluctuation covariance  $C_{j+1}$  and project the first two cumulants by Loc. Engineering dimension excludes every local monomial except  $1, \tau, \tau^2, \tau_\Delta$  at the retained order. The connected contraction of two quartic vertices supplies  $-\beta_j g_j^2$  in the quartic coordinate; positivity of the covariance bubble and Lemma 10.107 give  $\beta_j > 0$  and the  $n+8$  factor. The mixed quartic–quadratic contraction gives the  $n+2$  factor in the mass coordinate. The derivative part of the second cumulant produces the wave-function coefficient  $b_j g_j^2$ . The third cumulant and the variation of the tilted third moment are cubic in the small local coupling on the regulator-controlled domain, giving the displayed local  $O(g_j^3)$  remainder. The derivative bounds in (21.279) make the covariance sums converge to their massless scale limits before  $j_m$ , proving the final assertions.  $\square$

### Source theorem target and six-map reduction

The source single-step theorem is more structured than the schematic estimate (21.295). At one scale it constructs the next polymer coordinate as a composition of six maps. Most of those maps are bounded changes of coordinates; the third is the scale-changing localisation step and is the

unique place where the decisive factor  $L^{-1}$  is won. The following bookkeeping lemma is elementary, but recording it separately prevents the source-scale analytic contraction from being hidden inside a formal composition argument.

**Proposition 10.112** (Six-map contraction bookkeeping). *Let  $\mathcal{F}^{(0)}, \dots, \mathcal{F}^{(6)}$  be Banach spaces and suppose*

$$K^{(i)} = \mathcal{M}_i(K^{(i-1)}), \quad i = 1, \dots, 6, \quad (21.295a)$$

*where the maps are analytic on balls centred at the origin and preserve every algebraic property required for the polymer representation (restriction to subvolumes, component factorisation, Euclidean covariance and the circle-product identity). Suppose their Lipschitz radii satisfy*

$$r_1 \leq \mu_1 r_0, \quad r_2 \leq \mu_2 r_1, \quad r_3 \leq \kappa_3 r_2, \quad r_4 \leq \mu_4 r_3, \quad r_5 \leq \mu_5 r_4, \quad r_6 \leq \mu_6 r_5, \quad (21.295b)$$

*with constants  $\mu_i$  independent of  $L$  and*

$$\kappa_3 \leq cL^{-1}. \quad (21.295c)$$

*Then the composite map*

$$\mathcal{K}_+ = \mathcal{M}_6 \circ \dots \circ \mathcal{M}_1$$

*is analytic and has Lipschitz radius*

$$r_6 \leq \kappa_L r_0, \quad \kappa_L = (\mu_1 \mu_2 \mu_4 \mu_5 \mu_6) \kappa_3 = O(L^{-1}). \quad (21.295d)$$

*Every algebraic property listed above is inherited by the composite. If each map is continuous in a mass parameter and  $C^q$  in the interaction variables, then so is the composite (with the ordinary chain-rule loss only).*

*Proof.* Iterating the six inequalities in (21.295b) gives

$$r_6 \leq \mu_6 \mu_5 \mu_4 \kappa_3 \mu_2 \mu_1 r_0,$$

which is (21.295d). Analyticity and  $C^q$  regularity are stable under finite composition. Restriction, covariance, component factorisation and the exact circle-product identity are identities, so if each  $\mathcal{M}_i$  intertwines the corresponding operation then their composition does as well. Continuity in the mass is again preserved under finite composition. Thus none of these formal conclusions is the source of the small factor: it enters solely through (21.295c).  $\square$

**Lemma 10.113** (The decisive third-map radius estimate from a source-scale decomposition). *Let  $\varepsilon \in (0, 1)$  and suppose the third map admits a decomposition*

$$K^{(3)} = Ih + Ik + R \quad (21.295e)$$

*in the target polymer norm, with*

$$\|Ih\| \leq C\varepsilon^2 r_2 \varepsilon, \quad \|Ik\| \leq \kappa_* r_2 \varepsilon, \quad \|R\| \leq C\varepsilon^u r_2 \varepsilon, \quad (21.295f)$$

*for some  $u > 0$  and  $\kappa_* \leq cL^{-1}$ . If  $\varepsilon$  is small enough that  $C(\varepsilon^2 + \varepsilon^u) \leq \kappa_*$ , then*

$$\|K^{(3)}\| \leq 2\kappa_* r_2 \varepsilon = O(L^{-1}) r_2 \varepsilon. \quad (21.295g)$$

*Proof.* The triangle inequality and (21.295f) give

$$\|K^{(3)}\| \leq \{\kappa_* + C\varepsilon^2 + C\varepsilon^u\}r_2\varepsilon.$$

The stated smallness condition bounds the braces by  $2\kappa_*$ .  $\square$

*Remark 10.114* (Role in the staged proof). Proposition 10.112 and Lemma 10.113 close the composition and radius bookkeeping. They do *not* prove the hypotheses (21.295f). The source theorem obtains those inequalities from the full scale-dependent large-field/small-field regulator system, localisation estimates, polymer geometry and the change-of-scale estimate in two compatible norm pairs. It also proves the corresponding derivative estimates, restriction/isometry properties and continuity in the mass. Those analytic estimates, not the finite composition, are supplied by the later source-norm and kernel modules and assembled in Theorem 10.188.

**What remains between perturbation theory and an iterable exact map.** Proposition 10.110 constructs  $K_{j+1}$  exactly, while Proposition 10.111 identifies the desired local coordinates. Neither statement implies that  $K_{j+1}$  is smaller than  $K_j$ . The missing source theorem is the simultaneous regulator/localisation/polymer estimate yielding, on scale-dependent domains,

$$\|K_{j+1}\|_{\mathcal{W}_{j+1}} \leq \kappa_L \|K_j\|_{\mathcal{W}_j} + C\bar{g}_{j+1}^3, \quad \kappa_L = O(L^{-1}) < 1, \quad (21.295)$$

with derivative bounds in  $(V, K)$  and continuity in  $m^2$ . Equation (21.295) is the nonlinear single-step theorem proved in the source-norm, six-map, and kernel modules below. The elementary power-counting Lemma 10.104 is one ingredient but is not the full single-step theorem.

### A self-contained mixed-boundary stability theorem

The dynamical-systems part can be separated from the open analytic single-step estimate. The next proposition internalises that abstract layer.

**Proposition 10.115** (Mixed-boundary stability for a marginal–unstable–contractive RG system). *Let  $\bar{g}_j > 0$  satisfy*

$$\bar{g}_{j+1} = \bar{g}_j - \beta_j \bar{g}_j^2, \quad 0 < \beta_- \leq \beta_j \leq \beta_+, \quad |\beta_{j+1} - \beta_j| \leq c \bar{g}_j. \quad (21.296)$$

*Let  $\mathcal{K}_j$  be Banach spaces and suppose a nonlinear recursion on  $(g_j, z_j, \mu_j, K_j)$  has the form*

$$\begin{aligned} g_{j+1} &= g_j - \beta_j g_j^2 + \rho_j^g(x_j), \\ z_{j+1} &= z_j + \rho_j^z(x_j), \\ \mu_{j+1} &= \lambda_j \mu_j + \rho_j^\mu(x_j), \\ K_{j+1} &= A_j K_j + \rho_j^K(x_j) \end{aligned} \quad (21.297)$$

*where  $x_j = (g_j, z_j, \mu_j, K_j)$ ,  $\lambda_j \geq \lambda > 1$ , and  $\|A_j\| \leq \kappa < 1$ . Assume on the tube*

$$|g_j - \bar{g}_j| \leq M \bar{g}_j^2 |\log \bar{g}_j|, \quad |z_j| + |\mu_j| + \|K_j\| \leq M \bar{g}_j^2$$

*that the remainders are  $C^1$  and satisfy*

$$|\rho_j^g| \leq c \bar{g}_j^3, \quad |\rho_j^z| + |\rho_j^\mu| + \|\rho_j^K\| \leq c \bar{g}_j^3, \quad \|D\rho_j\|_{w \rightarrow w} \leq c \bar{g}_j, \quad (21.298)$$

for the corresponding weighted norm. If  $c\bar{g}_0$  is sufficiently small, then for every sufficiently small  $K_0$  there are unique initial values  $z_0, \mu_0$  and a unique forward orbit with fixed  $g_0$  and  $K_0$  such that

$$z_j \rightarrow 0, \quad \mu_j \rightarrow 0, \quad g_j = \bar{g}_j + O(\bar{g}_j^2 |\log \bar{g}_j|), \quad \|K_j\| = O(\bar{g}_j^2). \quad (21.299)$$

The orbit depends continuously, and  $C^1$ -smoothly when the remainders do, on the parameters entering (21.297).

*Proof.* We give the fixed-point construction because the direction of each boundary condition is essential. First, (21.296) implies by inversion

$$\bar{g}_{j+1}^{-1} - \bar{g}_j^{-1} = \frac{\beta_j}{1 - \beta_j \bar{g}_j} = \beta_j + O(\bar{g}_j),$$

so  $\bar{g}_j \asymp (1+j)^{-1}$  and  $\sum_j \bar{g}_j^2 < \infty$ .

Put  $y_j = g_j - \bar{g}_j$ . Subtracting (21.296) from the first equation of (21.297) gives

$$y_{j+1} = (1 - 2\beta_j \bar{g}_j) y_j - \beta_j y_j^2 + \rho_j^g(x_j).$$

With  $y_0 = 0$ , variation of constants and  $\prod_{k=i}^j (1 - 2\beta_k \bar{g}_k) \leq C(\bar{g}_j/\bar{g}_i)^2$  yield

$$|y_j| \leq C \bar{g}_j^2 \sum_{i < j} \bar{g}_i^{-2} \{\bar{g}_i^3 + |y_i|^2\}.$$

Bootstrap with  $|y_i| \leq M \bar{g}_i^2 |\log \bar{g}_i|$ ; the linear forcing gives  $C \bar{g}_j^2 \sum_{i < j} \bar{g}_i = O(\bar{g}_j^2 \log(1+j))$ , and the quadratic term is smaller for small  $\bar{g}_0$ . Thus the marginal coordinate has the stated bound.

For a prescribed tentative orbit, solve the centre and unstable coordinates by their boundary conditions at infinity:

$$z_j = - \sum_{i=j}^{\infty} \rho_i^z(x_i),$$

$$\mu_j = - \sum_{i=j}^{\infty} (\lambda_j \lambda_{j+1} \cdots \lambda_i)^{-1} \rho_i^\mu(x_i).$$

Both series converge absolutely because  $\sum_{i \geq j} \bar{g}_i^3 = O(\bar{g}_j^2)$ ; the second has additional geometric decay. For the contractive coordinate use the forward Green operator

$$K_j = A_{j-1} \cdots A_0 K_0 + \sum_{i=0}^{j-1} A_{j-1} \cdots A_{i+1} \rho_i^K(x_i).$$

Its norm is bounded by  $\kappa^j \|K_0\| + C \sum_{i < j} \kappa^{j-1-i} \bar{g}_i^3 = O(\bar{g}_j^2)$  after enlarging the constant on finitely many initial scales.

These four Green-operator formulas define a map  $\mathcal{T}$  on the closed weighted ball specified in the statement. The bounds just proved show that  $\mathcal{T}$  maps the ball into itself. Applying the derivative estimate in (21.298) to the same Green kernels gives a Lipschitz constant bounded by  $C\bar{g}_0 + C\|K_0\|$ ; after reducing the initial neighbourhood this is strictly less than one. Banach's fixed-point theorem gives existence and uniqueness. The formula for  $z_0, \mu_0$  gives the unique mixed-boundary tuning. Finally, the uniform contraction allows the usual parameter-dependent contraction theorem: difference quotients converge to the unique solution of the differentiated fixed-point equation, proving continuous and  $C^1$  parameter dependence.  $\square$

**Lemma 10.116** (Logarithmic decay of the marginal coupling). *If a positive sequence satisfies*

$$g_{j+1} = g_j - \beta g_j^2 + O(g_j^3), \quad \beta > 0, \quad (21.300)$$

*while remaining sufficiently small, then*

$$g_j = \frac{1}{\beta j} \left( 1 + O\left(\frac{\log j}{j}\right) \right), \quad \sum_{i < j} g_i = \frac{1}{\beta} \log j + O(1), \quad (21.301)$$

*and for fixed  $c$ ,*

$$\prod_{i < j} (1 - c g_i + O(g_i^2)) = j^{-c/\beta} e^{O(1)}. \quad (21.302)$$

*Proof.* For small  $g_j$ ,

$$\frac{1}{g_{j+1}} - \frac{1}{g_j} = \frac{\beta g_j^2 + O(g_j^3)}{g_j g_{j+1}} = \beta + O(g_j).$$

A preliminary comparison with the logistic recursion gives  $g_j \asymp j^{-1}$ . Summing then yields  $g_j^{-1} = \beta j + O(\log j)$ , which is the first formula in (21.301). Summing this asymptotic gives the second. Finally  $\log(1 - c g_i + O(g_i^2)) = -c g_i + O(g_i^2)$  and  $\sum_i g_i^2 < \infty$ ; exponentiation gives (21.302).  $\square$

### Exact proof of SM-21.03.01

The foundational theorem can now be stated without the former overclaim about all later observable extensions.

**Theorem 10.117** (Rigorous RG construction for weakly coupled four-dimensional  $|\varphi|^4$ ). *Fix  $n \geq 1$ . For the nearest-neighbour  $n$ -component  $|\varphi|^4$  spin model on  $\mathbb{Z}^4$ , and its finite discrete-torus approximants, with sufficiently small positive quartic coupling, the finite-range rigorous renormalisation-group construction admits a critical/high-temperature trajectory with a uniquely tuned relevant mass coordinate. On that trajectory the marginal quartic coupling tends to zero logarithmically and the exact partition function admits scale-by-scale local-plus-polymer coordinates whose nonperturbative remainder remains uniformly controlled. The statement is restricted to the weak-coupling critical and high-temperature regimes; it does not assert an arbitrary-coupling construction or a low-temperature RG theorem.*

*Remark 10.118* (Proof roadmap). The preceding construction supplies the source-strength finite-range covariance decomposition (including positivity, strict range, derivative bounds, the post-mass gain and the torus terminal remainder), the normed-algebra product and Gaussian-translation layer, a quadratic regulator estimate, the finite-dimensional localisation projection and its power-counting mechanism, the connected second-order cumulant and  $O(n)$  one-loop combinatorics, the exact circle-product/polymer algebra, and the mixed-boundary structural-stability theorem. The staged proof then needs two further source-scale inputs:

1. the exact single-step nonperturbative theorem giving (21.295), including the complete scale-dependent polymer norm, the full large-field regulator/localisation estimates, contraction with  $\kappa = O(L^{-1})$ , differentiability, and mass continuity on the source RG domain;
2. the source-specific verification that the exact  $|\varphi|^4$  coordinates and domains satisfy that single-step theorem and the hypotheses of Proposition 10.115 uniformly in the finite-volume cutoff, yielding the cutoff-independent infinite critical trajectory and the exact terminal partition-function representation.



The source-norm, corrected six-map, and kernel modules below prove these inputs; Theorem 10.188 records the final single-step conclusion. The elementary estimates above are ingredients, not substitutes, for the full polymer contraction and model-wide assembly.

**Model-specific inputs for the finite-order correlation-length theorem.** The  $n \geq 1$  branch of SM-21.S2.04 inherits the bosonic foundational theorem above. Its  $n = 0$  weakly self-avoiding-walk branch requires the supersymmetric boson–fermion version of the same exact RG architecture. That version is naturally owned by the WSAW RG line leading to SM-21.04.01, not by the  $n \geq 1$  parent SM-21.03.01. This chapter therefore records the honest split dependency

$$\text{SM-21.S2.04}|_{n \geq 1} \leftarrow \text{SM-21.03.01}, \quad \text{SM-21.S2.04}|_{n=0} \leftarrow \text{SM-21.04.01}. \quad (21.303)$$

The source-specific post-mass-scale proof written in this chapter remains complete; only these model-foundation owners are upstream.

## 10.4.2 The scale-changing contraction and model assembly

The exact polymer algebra becomes iterable only after a nonlinear single-step estimate. We first prove, in a concrete scale-dependent polymer norm, the *linear* contraction produced by localisation, Gaussian integration and reblocking. The second is to prove that, once the source-strength nonlinear single-step estimate is available, the finite-volume  $n$ -component  $|\varphi|^4$  model satisfies the required mixed-boundary and terminal-scale hypotheses. The simultaneous two-regulator nonlinear estimate for the full  $K_+$  map is proved in the subsequent source-norm, six-map, and kernel modules.

### Weighted polymer norms and the linear irrelevant map

**Definition 10.119** (Large-set weights and paired one-regulator polymer norms). Let  $\mathcal{C}_j$  denote the connected  $j$ -polymers and fix  $A > 1$ . For  $\sigma \in \{1, 2\}$  define

$$\|K\|_{\mathcal{W}_j^{G,\sigma}} = \sup_{X \in \mathcal{C}_j} A^{\sigma|X|_j} \sup_{\varphi} \frac{\|K(X)\|_{T_{\varphi,j}}}{G_j(X, \varphi)}. \quad (21.304)$$

The exponent  $\sigma = 2$  will be called the *strong* large-set weight and  $\sigma = 1$  the *weak* weight. If  $K$  factorises over connected components, (21.304) is extended multiplicatively to arbitrary polymers. A connected polymer is called *j-small* when it contains at most  $2^4$  blocks; all other connected polymers are *j-large*.

The threshold  $2^4$  is inessential; what matters is a fixed threshold large enough that a connected set which is large at scale  $j$  cannot fit into one  $(j+1)$ -block without paying a definite fraction of its large-set weight.

**Lemma 10.120** (Connected-polymer counting). *There is a dimension-dependent constant  $c_{\text{pol}}$  such that for every  $j$ -block  $B$  and every  $m \geq 1$ , the number of connected  $j$ -polymers with  $m$  blocks containing  $B$  is at most  $c_{\text{pol}}^m$ . Consequently, if  $A > 2c_{\text{pol}}$ , then*

$$\sum_{\substack{X \in \mathcal{C}_j \\ B \subset X}} A^{-|X|_j} \leq \frac{A^{-1}}{1 - c_{\text{pol}}/A}. \quad (21.305)$$

*The same bound holds with a fixed finite enlargement of the adjacency relation.*

*Proof.* Fix an ordering of the  $2d$  nearest-neighbour directions and explore a connected polymer from the distinguished block  $B$  by depth-first search in a spanning tree. A connected set of  $m$  vertices admits a rooted spanning tree with  $m - 1$  edges. Encode the tree by the depth-first sequence of traversed oriented edges, of length  $2(m - 1)$ , together with the information of when a previously visited vertex is encountered. At each traversal there are at most  $2d$  geometric choices and at most two bookkeeping choices. Thus the number is bounded by  $(4d)^{2m}$ , and we may take  $c_{\text{pol}} = (4d)^2$ . Replacing nearest-neighbour adjacency by a fixed finite-range adjacency only replaces  $2d$  by the finite degree of that graph. Summing the resulting geometric series gives (21.305).  $\square$

**Lemma 10.121** (Strong-to-weak large-polymer reblocking gain). *Let  $\bar{X}$  be the smallest  $(j + 1)$ -polymer containing a connected  $j$ -polymer  $X$ . Then*

$$|\bar{X}|_{j+1} \leq |X|_j. \quad (21.306)$$

*Consequently the ratio between the weak target weight and the strong source weight obeys*

$$A^{|\bar{X}|_{j+1}} A^{-2|X|_j} \leq A^{-|X|_j}. \quad (21.307)$$

*After summing over connected preimages of a fixed coarse polymer, the total weight is  $O(A^{-c})$  for some  $c > 0$ , provided  $A$  is larger than a dimension-dependent constant. The assertion remains valid after a fixed finite enlargement of the adjacency relation.*

*Proof.* Every coarse block in  $\bar{X}$  contains at least one fine block of  $X$ . Distinct coarse blocks contain disjoint fine blocks, so the choice of one such fine block gives an injection from the coarse blocks of  $\bar{X}$  into the fine blocks of  $X$ . This proves (21.306), and (21.307) follows immediately.

For the summation, let the fixed target  $U$  contain  $r$  coarse blocks. Any preimage  $X$  with  $\bar{X} = U$  contains at least  $r$  fine blocks. Choose in each coarse block of  $U$  one fine block of  $X$ ; this costs at most  $L^{4r}$  choices. Once one distinguished fine block is fixed, Lemma 10.120 bounds the number of connected  $k$ -block candidates by  $c_{\text{pol}}^k$ . This overcounts, which is harmless. Thus the strong-to-weak weighted sum is at most

$$A^r L^{4r} \sum_{k \geq r} c_{\text{pol}}^k A^{-2k} \leq \left( \frac{CL^4}{A} \right)^r \quad (21.307a)$$

when  $A^2 > 2c_{\text{pol}}$ . If  $A = L^q$  with  $q > 4$  sufficiently large, the last expression is bounded by  $A^{-cr}$  for a fixed  $c > 0$ . A finite adjacency enlargement changes only  $c_{\text{pol}}$  and  $C$ .  $\square$

**Definition 10.122** (Linearised irrelevant map). For a connected  $j$ -polymer  $X$ , write  $\bar{X}$  for its  $(j + 1)$  closure and define

$$(\mathcal{L}_j K)(\bar{X}) = \sum_{X: \bar{X}=U} \mathbb{E}_{C_{j+1}} \theta \left[ (1 - \text{Loc}_U) K(X) \right], \quad (21.308)$$

with the sum restricted to the connected preimages assigned to  $U$  by the fixed reblocking rule. The harmless vacuum and local pieces extracted by  $\text{Loc}$  are assigned to the local coordinate rather than to  $\mathcal{L}_j K$ .

**Proposition 10.123** (Paired-weight one-regulator linear contraction). *Assume the RG-admissible covariance bounds of Theorem 10.96, the Gaussian regulator stability Lemma 10.101, and choose the localisation Taylor order so that  $\omega \geq 1$  in Lemma 10.104. Then, after choosing  $A$  as a sufficiently large power of  $L$ , the linearised irrelevant map satisfies*

$$\|\mathcal{L}_j K\|_{\mathcal{W}_{j+1}^{G,1}} \leq \kappa_L^{\text{lin}} \|K\|_{\mathcal{W}_j^{G,2}}, \quad \kappa_L^{\text{lin}} \leq CL^{-1} + CA^{-c}. \quad (21.309)$$

*In particular  $A = L^q$  with  $q$  sufficiently large gives  $\kappa_L^{\text{lin}} = O(L^{-1})$ .*

*Proof.* Split the sum (21.308) into  $j$ -small and  $j$ -large preimages. For a small  $X$ , Lemma 10.104 and  $\omega \geq 1$  give

$$\|(1 - \text{Loc}_U)K(X)\|_{T_{\varphi,j+1}} \leq CL^{-1}\|K(X)\|_{T_{\varphi,j}}. \quad (21.310)$$

Apply the Gaussian translation estimate Lemma 10.99 and then the regulator stability Lemma 10.101. The determinant factor is exponential only in the number of fine blocks and is absorbed by the strong source weight. The number of small preimages of a fixed  $U$  is bounded by a fixed power of  $L$ ; the extra factor  $A^{|U|_{j+1}-2|X|_j}$  absorbs this power when  $A = L^q$  with  $q$  sufficiently large. Their total contribution is therefore  $CL^{-1}\|K\|_{\mathcal{W}_j^{G,2}}$ .

For a large preimage we do not use Taylor cancellation. The normed-algebra and Gaussian bounds cost only a fixed factor per block, while Lemma 10.121 supplies an exponentially summable factor in  $|X|_j$ . Lemma 10.120 then sums the preimages and gives  $CA^{-c}\|K\|_{\mathcal{W}_j^{G,2}}$ . Adding the two contributions proves (21.309). Taking  $q$  sufficiently large makes the large-set term  $O(L^{-1})$  as well.  $\square$

**What Proposition 10.123 does and does not prove.** The proposition internalises the geometric source of the  $O(L^{-1})$  factor: small sets contract because localisation removes every relevant and marginal jet, while large sets are summable because a strong input large-set weight dominates a weak output weight. The deliberate strong-to-weak loss is important. It is not, by itself, an iterable norm estimate. The source construction uses two compatible regulator/norm systems and its intermediate maps recover the norm strength needed at the next scale. Establishing that closed norm cycle is part of the nonlinear single-step theorem. A bound on the linearised map also does not by itself control the nonlinear coordinate  $K_+$ .

### The nonlinear remainder in nested analytic balls

**Lemma 10.124** (Quadratic analytic remainder). *Let  $E, F$  be complex Banach spaces and let  $\mathcal{M} : B_E(0, R) \rightarrow F$  be analytic with  $\mathcal{M}(0) = 0$ . If  $\sup_{\|x\| < R} \|\mathcal{M}(x)\| \leq M$ , then for  $\|x\| \leq r < R$ ,*

$$\|\mathcal{M}(x) - D\mathcal{M}(0)x\| \leq \frac{M}{R^2} \frac{\|x\|^2}{1 - r/R}. \quad (21.311)$$

*More generally, for every fixed  $q \geq 1$  and  $\|x\| \leq r$ , there is a universal constant  $C_q < \infty$  such that*

$$\|D^q \mathcal{M}(x)\| \leq \frac{C_q M}{(R - r)^q}. \quad (21.312)$$

*Proof.* For  $x \neq 0$ , apply the scalar Cauchy formula to  $f(z) = \mathcal{M}(zx/\|x\|)$  on  $|z| < R$ . Its Taylor coefficients satisfy  $\|a_k\| \leq MR^{-k}$ . Hence

$$\left\| \sum_{k \geq 2} a_k \|x\|^k \right\| \leq M \sum_{k \geq 2} (\|x\|/R)^k = \frac{M\|x\|^2/R^2}{1 - \|x\|/R},$$

which gives (21.311). For (21.312), apply the one-variable Cauchy estimate to  $z \mapsto \mathcal{M}(x + zh)$  for unit directions  $h$  and then use the polarization identity for the symmetric  $q$ -linear form  $D^q \mathcal{M}(x)$ . The polarization constant depends only on  $q$  and is absorbed into  $C_q$ .  $\square$

**Proposition 10.125** (Nonlinear one-regulator contraction under a uniform analytic-domain bound). *Suppose the exact algebraic RG map of Proposition 10.110 induces, after perturbative local extraction, an analytic map*

$$\mathcal{K}_{+,j}(V, K) : \mathcal{D}_j \longrightarrow \mathcal{W}_{j+1}^{G,1} \quad (21.313)$$

*on a product ball  $\|V\| \leq c\bar{g}_j$ ,  $\|K\|_{\mathcal{W}_j^{G,2}} \leq c\bar{g}_j^3$ . Assume uniformly on a ball twice as large that*

$$\|\mathcal{K}_{+,j}(V, 0)\|_{\mathcal{W}_{j+1}^{G,1}} \leq C\bar{g}_{j+1}^3, \quad (21.314)$$

*and that for some fixed  $\rho > 0$*

$$\begin{aligned} D_K \mathcal{K}_{+,j}(0, 0) &= \mathcal{L}_j, \\ \|D_K \mathcal{K}_{+,j}(V, 0) - \mathcal{L}_j\| &\leq C\bar{g}_j^\rho, \\ \sup_{\|K\| \leq c\bar{g}_j^3} \|D_K^2 \mathcal{K}_{+,j}(V, K)\| &\leq C\bar{g}_j^{-3+\rho}. \end{aligned} \quad (21.314a)$$

*Then, for sufficiently small  $\bar{g}_j$ ,*

$$\|\mathcal{K}_{+,j}(V, K)\|_{\mathcal{W}_{j+1}^{G,1}} \leq C\bar{g}_{j+1}^3 + (\kappa_L^{\text{lin}} + C\bar{g}_j^\rho) \|K\|_{\mathcal{W}_j^{G,2}}, \quad (21.315)$$

*and all further mixed derivatives for which a uniform analytic-domain bound is known obey Cauchy estimates on smaller product balls.*

*Proof.* Use the Banach-space fundamental theorem of calculus in the  $K$  variable:

$$\mathcal{K}_{+,j}(V, K) = \mathcal{K}_{+,j}(V, 0) + \int_0^1 D_K \mathcal{K}_{+,j}(V, tK) K dt. \quad (21.315a)$$

By the last line of (21.314a) and  $\|K\| \leq c\bar{g}_j^3$ ,

$$\|D_K \mathcal{K}_{+,j}(V, tK) - D_K \mathcal{K}_{+,j}(V, 0)\| \leq C\bar{g}_j^{-3+\rho} \|K\| \leq C\bar{g}_j^\rho. \quad (21.315b)$$

The middle line of (21.314a) and Proposition 10.123 therefore give

$$\|D_K \mathcal{K}_{+,j}(V, tK)\| \leq \kappa_L^{\text{lin}} + C\bar{g}_j^\rho.$$

Insert this and (21.314) into (21.315a) to obtain (21.315). The final derivative statement follows from Lemma 10.124, applied to the available analytic-domain bound in the product Banach space.  $\square$

*Remark 10.126* (Why the hypothesis (21.314) remains a genuine gate). Estimates (21.314)–(21.314a) are not formalities. In the source method it is established by combining the small-field and large-field regulator norms, the stability of the local interaction, the full localisation theorem, and the scale-changing polymer estimates. The point of Proposition 10.125 is narrower: once the source zeroth-order inhomogeneous bound and the first-/second- $K$  derivative gains in (21.314)–(21.314a) are available, the nonlinear contraction follows by the fundamental theorem of calculus, while further finite-order derivative bounds follow from the available analytic-domain estimates and Cauchy theory. Thus the remaining analytic root is the source two-regulator stability/domain-and-derivative package, not the abstract nonlinear Taylor argument.

## Restriction, covariance locality, and mass continuity

**Lemma 10.127** (Restriction and finite-range isometry). *Let  $\Lambda' \subset \Lambda$  be a union of  $(j+1)$ -blocks whose distance from the boundary of  $\Lambda$  exceeds the strict range of  $C_{j+1}$ . If the input interaction and polymer activity are restricted to  $\Lambda'$ , then the exact one-step construction of Proposition 10.110, restricted back to polymers in  $\Lambda'$ , agrees with the construction performed directly in  $\Lambda'$ . The same is true after translations and lattice symmetries preserving the block decomposition.*

*Proof.* Every operation in one RG step is local to the covariance enlargement of the input polymer: multiplication, Loc, Gaussian expectation, and reblocking. Strict finite range implies that the Gaussian covariance between that enlargement and the complement of  $\Lambda'$  is zero. Gaussian integration therefore factorises, and the terms assigned to a polymer contained in  $\Lambda'$  are identical in the large and small volumes. Localisation is defined by the same finite jet pairing, hence also agrees. Circle-product recombination is an algebraic sum over subpolymers and preserves the identity. Translation and lattice-symmetry covariance follow term by term from covariance of  $C_{j+1}$  and of Loc.  $\square$

**Lemma 10.128** (Continuity in the mass parameter). *Fix a scale  $j$  and a compact interval  $0 \leq m^2 \leq m_*^2$ . On every polymer-norm ball on which the regulator expectation is uniformly integrable, the exact one-step map is continuous in  $m^2$ . On compact subintervals away from a terminal spectral singularity, it is differentiable as many times as the covariance decomposition is, with derivatives obtained by differentiating the covariance under the finite-dimensional Gaussian integral.*

*Proof.* The covariance kernel and all field derivatives entering the finite-scale norm are continuous in  $m^2$  by Theorem 10.96. For a finite torus, a Gaussian expectation can be written as an ordinary finite-dimensional integral after diagonalising the covariance. Pointwise convergence in  $m^2$  is immediate. The regulator bound of Lemma 10.101 supplies an integrable majorant uniformly on the stated polymer ball, so dominated convergence gives continuity. When the covariance is  $C^r$  in  $m^2$ , differentiation of the Gaussian characteristic function, followed by the same domination, justifies differentiating the integral  $r$  times. All subsequent operations in the RG step are finite sums, products, localisation maps and reblocking, which preserve this regularity.  $\square$

## Conditional model assembly for the $n$ -component $|\varphi|^4$ model

**Proposition 10.129** (Initial-domain verification and conditional finite-volume assembly). *Fix  $n \geq 1$ ,  $L$  sufficiently large, and a sufficiently small bare quartic coupling  $g_0 > 0$ . Assume the source-strength single-step theorem in the following precise form: for every preterminal scale, on the  $|\varphi|^4$  RG domain there is an exact map  $(V_j, K_j) \mapsto (V_{j+1}, K_{j+1})$  satisfying*

$$\|K_{j+1}\|_{\mathcal{W}_{j+1}} \leq \kappa_L \|K_j\|_{\mathcal{W}_j} + C\bar{g}_{j+1}^3, \quad \kappa_L = O(L^{-1}) < 1, \quad (21.316)$$

*with the derivative, restriction/isometry and mass-continuity bounds required by Proposition 10.115. Then the nearest-neighbour finite-volume  $n$ -component  $|\varphi|^4$  model satisfies the source-domain hypotheses uniformly in the torus size. There is a unique tuned bare mass  $\nu_0^c(g_0)$  (equivalently the source mass coordinate) for which the RG trajectory stays in the critical stable tube, and the finite-volume partition functions admit the exact terminal representation generated by the progressive covariance decomposition. For every fixed preterminal scale, the coordinates are independent of the ultimate volume once the volume is larger than the covariance neighbourhood of that scale.*

*Proof.* At scale zero the bare interaction is already in the local polynomial space,

$$V_0(\varphi) = \sum_x \{g_0 \tau_x^2 + \nu_0 \tau_x + z_0 \tau_{\Delta,x}\}, \quad z_0 = 0, \quad (21.317)$$

and there is no initial irrelevant polymer activity:  $K_0 = 0$ . Hence the source norm of  $K_0$  vanishes and the local coordinates lie in the single-step domain when  $g_0$  and  $|\nu_0|$  are sufficiently small. We normalize the relevant scale-zero coordinate to be  $\mu_0 = \nu_0$ ; any later triangular perturbative change of the relevant coordinate is part of the finite-dimensional local map and does not alter the identity of the bare parameter at the initial scale.

Under (21.316), the  $K$  coordinate has the form required in Proposition 10.115; the perturbative recursion Proposition 10.111 supplies the marginal, centre and unstable coordinates and their cubic remainders. The structural-stability theorem gives a unique mixed-boundary orbit, hence a unique initial unstable coordinate and therefore, under the scale-zero normalization  $\mu_0 = \nu_0$ , a unique  $\nu_0^c(g_0)$ . Its estimates give  $\|K_j\| = O(\bar{g}_j^2)$  and the logarithmic decay of the quartic coupling follows from Lemma 10.116.

It remains to check finite-volume compatibility rather than assume it. Below the terminal scale, Theorem 10.96 gives exactly the same finite-range covariance as in infinite volume on every fixed covariance neighbourhood. Lemma 10.127 therefore shows inductively that all coordinates on a fixed polymer agree in every sufficiently large torus. At the terminal scale, the remaining positive covariance is integrated in one final finite-dimensional Gaussian integral. Repeated use of Proposition 10.110 gives the exact identity

$$Z_{\Lambda_N} = e^{-u_N |\Lambda_N|} \mathbb{E}_{C_N^{\text{term}}} (I_N \circ K_N)(\Lambda_N), \quad (21.318)$$

with no omitted error term. Thus the finite-volume/domain/terminal assembly is a consequence of the single-step estimate plus finite-range locality, not a second independent analytic black box.  $\square$

**Corollary 10.130** (Reduction to the two-regulator single-step theorem). *For the  $n \geq 1$  bosonic branch, the covariance decomposition, normed algebra, localization, perturbative coordinates, and finite-volume assembly reduce the foundational construction to the exact two-regulator  $K_+$  single-step theorem, specifically the uniform domain estimate underlying (21.316) in the full source norm  $\mathcal{W}_j$ .*

*Proof.* The preceding sections prove the covariance decomposition, normed algebra, localisation mechanism, perturbative coordinates, exact circle-product algebra and mixed-boundary stability. Proposition 10.123 proves the linear irrelevant-direction  $O(L^{-1})$  mechanism, Lemmas 10.127–10.128 own the locality and parameter-continuity consequences, and Proposition 10.129 proves the finite-volume/model assembly once the nonlinear source-domain estimate is assumed. This leaves exactly the stated single-step estimate.  $\square$

*Remark 10.131* (Role in the staged proof). This module stops before the complete two-regulator nonlinear estimate for  $K_+$ . The source-norm and corrected six-map modules develop that estimate, and Theorem 10.188 later completes the bosonic single-step theorem. The one-regulator contraction above is therefore used as one ingredient, not as a substitute for the final regulator/localisation argument.

### 10.4.3 The source two-norm structure for the nonperturbative coordinate

The preceding module proved a useful one-regulator linear contraction, but it also recorded why that estimate cannot be iterated as the source theorem: the rigorous construction uses two different

field scales and two different regulator norms. The small-field norm sees the cubic size of the local remainder, whereas the large-field norm is stable under Gaussian fluctuation integration. The single-step theorem is proved in those two norm pairs and only afterwards recombined into the scale-to-scale norm of the polymer coordinate.

This subsection reconstructs that norm architecture exactly enough to remove an ambiguity left in Cycles 64–65. It also internalises the change-of-norm calculation which converts the two output estimates into the next-scale polymer norm. No estimate for any of the six nonlinear source maps is asserted here unless it is proved below; those mapwise estimates remain the active root.

### Field scales, large-set weights, and the two source norm pairs

Throughout this subsection  $d = 4$  and  $[\varphi] = (d - 2)/2 = 1$ . Let  $\tilde{g}_j \in (0, 1)$  be the source control coupling and let  $\chi_j \in (0, 1]$  be the source large-scale improvement factor. On the critical stable tube used in the preceding RG module,  $\tilde{g}_j$  is comparable with the reference coupling  $\bar{g}_j$ ; no estimate below uses equality of those two sequences.

**Definition 10.132** (The two source field scales). Fix positive constants  $\ell_0, k_0$ . Put

$$\ell_j = \ell_0 L^{-j[\varphi]}, \quad \mathfrak{h}_j = k_0 \tilde{g}_j^{-1/4} L^{-jd/4}.$$

In dimension four, the powers of  $L$  agree and therefore

$$\frac{\ell_j}{\mathfrak{h}_j} = \frac{\ell_0}{k_0} \tilde{g}_j^{1/4}.$$

The scale  $\ell_j$  is the small-field scale;  $\mathfrak{h}_j$  is the larger field scale used with the fluctuation regulator.

**Definition 10.133** (Source small parameters and large-set exponents). Define

$$\bar{\epsilon}_j(\ell) = \chi_j^{1/2} \tilde{g}_j, \quad \bar{\epsilon}_j(\mathfrak{h}) = \chi_j^{1/2} \tilde{g}_j^{1/4}, \quad \omega_j = \frac{\bar{\epsilon}_j(\ell)}{\bar{\epsilon}_j(\mathfrak{h})} = \tilde{g}_j^{3/4}.$$

For  $a > 0$  and a connected  $j$ -polymer  $X$ , put

$$f_j(a, X) = a(|X|_j - 2^d)_+.$$

Thus  $f_j(4a, X) = 4f_j(a, X)$ .

The source has two field regulators. We denote by  $G_j$  the fluctuation-field regulator and by  $\tilde{G}_j$  the large-field regulator. The names are the source terminology:  $G_j$  measures the full local  $\Phi_j$  norm at the fluctuation scale  $\ell_j$ , whereas  $\tilde{G}_j$  quotients out local affine fields and is paired with the larger field scale  $\mathfrak{h}_j$ . Their exact block definitions are proved here in this chapter. The preceding analysis norm-comparison calculation below uses only the following consequences of those definitions, proved from the  $T_\varphi$  change-of-scale mechanism.

**Lemma 10.134** (Change of field scale with regulator interpolation). *Let  $F \in \mathcal{N}(X)$  and let  $A < p_{\mathcal{N}}$  be a nonnegative integer. Suppose the small- and large-field test-function spaces at one scale differ only by replacing the field parameter  $\ell$  by  $\mathfrak{h} \geq \ell$ , and suppose the source regulators satisfy*

$$\tilde{G}(X, t\varphi) \leq G(X, \varphi)^{1/2} \quad (0 \leq t \leq 1).$$

*Then*

$$\|F\|_{G, \ell} \leq c_A \left( \|F\|_{T_0(\ell)} + \left( \frac{\ell}{\mathfrak{h}} \right)^{A+1} \|F\|_{\tilde{G}, \mathfrak{h}} \right).$$

*The constant depends on  $A$  and on the fixed test-function conventions, but not on  $F$  or on the scale.*

*Proof.* We give the change-of-norm argument rather than citing it. Let  $T'_\varphi$  denote the  $T_\varphi$  seminorm with field scale  $\ell$  and  $T_\varphi$  the one with field scale  $\mathfrak{h}$ . In the test-function pairing, a derivative of total field degree  $r$  has weight  $\ell^r$  in the first norm and  $\mathfrak{h}^r$  in the second. Hence the norm of the restriction map on the sector of degree at least  $A+1$  is bounded by  $(\ell/\mathfrak{h})^{A+1}$ .

Taylor-expand the translation  $F(\varphi+\xi)$  in the background field through degree  $A$ . The polynomial part is bounded by the zero-field seminorm, with the usual factor  $(1 + \|\varphi\|_{\Phi(\ell)})^{A+1}$ . In the integral Taylor remainder every test function has total degree at least  $A+1$ , so the preceding restriction estimate contributes  $(\ell/\mathfrak{h})^{A+1}$  and the remainder is bounded by  $\sup_{0 \leq t \leq 1} \|F\|_{T_{t\varphi}(\mathfrak{h})}$ . Thus

$$\|F\|_{T_\varphi(\ell)} \leq C_A (1 + \|\varphi\|_{\Phi(\ell)})^{A+1} \left\{ \|F\|_{T_0(\ell)} + \left(\frac{\ell}{\mathfrak{h}}\right)^{A+1} \sup_{0 \leq t \leq 1} \|F\|_{T_{t\varphi}(\mathfrak{h})} \right\}.$$

By the definition of the large-field norm and the stated regulator comparison,

$$\sup_{0 \leq t \leq 1} \|F\|_{T_{t\varphi}(\mathfrak{h})} \leq \|F\|_{\tilde{G}, \mathfrak{h}} G(X, \varphi)^{1/2}.$$

Finally the polynomial factor in  $\varphi$  is absorbed by the remaining square root of the quadratic regulator  $G$  (after increasing a scale-independent constant). Divide by  $G(X, \varphi)$  and take the supremum over  $\varphi$ . This gives the displayed estimate.  $\square$

**Definition 10.135** (Source  $\mathcal{F}$  norms). For connected  $j$ -polymers define

$$\|K\|_{\mathcal{F}_j(G)} = \sup_{X \in \mathcal{C}_j} \bar{\epsilon}_j(\ell)^{-f_j(a, X)} \|K(X)\|_{G_j, \ell_j},$$

$$\|K\|_{\mathcal{F}_j(\tilde{G})} = \sup_{X \in \mathcal{C}_j} \bar{\epsilon}_j(\mathfrak{h})^{-f_j(4a, X)} \|K(X)\|_{\tilde{G}_j, \mathfrak{h}_j}.$$

The corresponding target norms are denoted  $\mathcal{F}_{j+1}(T_0)$  and  $\mathcal{F}_{j+1}(\tilde{G}^\gamma)$ : the first uses the zero-field  $T_0(\ell_{j+1})$  seminorm, while the second uses a fixed power  $\tilde{G}_{j+1}^\gamma$  chosen in the regulator-stability theorem. The precise allowed interval of  $a$  and  $\gamma$  belongs to the mapwise source theorem and is not enlarged here.

**Definition 10.136** (Source  $\mathcal{W}_j$  norm and intermediate target norm). Set

$$\|K\|_{\mathcal{W}_j} = \max \left\{ \|K\|_{\mathcal{F}_j(G)}, \omega_j^3 \|K\|_{\mathcal{F}_j(\tilde{G})} \right\}.$$

At the target scale define

$$\|K\|_{\mathcal{Y}_{j+1}} = \max \left\{ \|K\|_{\mathcal{F}_{j+1}(T_0)}, \omega_{j+1}^3 \|K\|_{\mathcal{F}_{j+1}(\tilde{G})} \right\}.$$

The distinction between  $\mathcal{Y}_{j+1}$  and  $\mathcal{W}_{j+1}$  is essential: the scale-changing map first produces a  $T_0$  estimate, and the fluctuation-field regulator is restored only after the two estimates are recombined.

**Recombination: why the two estimates restore the  $W$  norm**

**Proposition 10.137** (Two-norm recombination at the next scale). *Assume  $p_{\mathcal{N}} \geq 10$  and  $0 < \tilde{g}_{j+1} \leq 1$ . Then, after fixing the source constants  $\ell_0/k_0$  and the regulator normalisations,*

$$\|K\|_{\mathcal{W}_{j+1}} \leq C \|K\|_{\mathcal{Y}_{j+1}}$$

*with a scale-independent constant  $C$ .*



*Proof.* Only the  $\mathcal{F}_{j+1}(G)$  component needs proof. Apply Lemma 10.134 with  $A = 9$ , which is legal because  $p_N \geq 10$ . Definition 10.132 gives

$$\left(\frac{\ell_{j+1}}{\mathfrak{h}_{j+1}}\right)^{10} \leq C\tilde{g}_{j+1}^{10/4}.$$

For  $s = f_{j+1}(a, X)$ , the large-set weights satisfy, using  $f_{j+1}(4a, X) = 4s$ ,

$$\begin{aligned} & \bar{\epsilon}_{j+1}(\ell)^{-s} \bar{\epsilon}_{j+1}(\mathfrak{h})^{4s} \\ &= (\chi_{j+1}^{1/2} \tilde{g}_{j+1})^{-s} (\chi_{j+1}^{1/2} \tilde{g}_{j+1}^{1/4})^{4s} = \chi_{j+1}^{3s/2} \leq 1. \end{aligned}$$

Therefore the large-field part of the change-of-norm estimate contributes at most

$$C\tilde{g}_{j+1}^{10/4} \|K\|_{\mathcal{F}_{j+1}(\tilde{G})}.$$

Since  $0 < \tilde{g}_{j+1} \leq 1$ ,  $\tilde{g}_{j+1}^{10/4} \leq \tilde{g}_{j+1}^{9/4} = \omega_{j+1}^3$ . The zero-field part is exactly controlled by  $\|K\|_{\mathcal{F}_{j+1}(T_0)}$ . Taking the maximum over connected polymers gives

$$\|K\|_{\mathcal{F}_{j+1}(G)} \leq C \max \left\{ \|K\|_{\mathcal{F}_{j+1}(T_0)}, \omega_{j+1}^3 \|K\|_{\mathcal{F}_{j+1}(\tilde{G})} \right\}.$$

The second component of  $\mathcal{W}_{j+1}$  is already present in  $\mathcal{Y}_{j+1}$ , proving the proposition.  $\square$

*Remark 10.138* (Why the exponent 10 matters). The exponent is not cosmetic. The  $W$  norm weights the fluctuation-regulator component by  $\omega^3 = \tilde{g}^{9/4}$ . Taking  $A = 9$  in the field-scale change produces  $\tilde{g}^{10/4}$ , which is strictly smaller for small coupling. This is the precise small power which allows the  $T_0$  and  $\tilde{G}$  outputs to restore the  $G$ -norm strength needed at the next scale. A single-regulator estimate has no analogue of this mechanism.

### Embedding the source $W$ domain into both analytic norm pairs

**Lemma 10.139** (Two-pair domain embedding). *Fix  $R > 0$ . For sufficiently small  $\tilde{g}_j$ , the source polymer ball*

$$\|K\|_{\mathcal{W}_j} \leq R\chi_j^{3/2} \tilde{g}_j^3$$

*is contained simultaneously in the two balls*

$$\|K\|_{\mathcal{F}_j(G)} \leq r\bar{\epsilon}_j(\ell), \quad \|K\|_{\mathcal{F}_j(\tilde{G})} \leq r\bar{\epsilon}_j(\mathfrak{h})$$

*for any fixed  $r > 0$ , after decreasing the small-coupling threshold depending on  $R/r$ . Moreover the larger Cauchy ball*

$$\|K\|_{\mathcal{W}_j} \leq r\chi_j^{1/2} \tilde{g}_j^{10/4}$$

*is contained in the same two pairwise balls up to a scale-independent enlargement of  $r$ .*

*Proof.* The  $G$  component is immediate from the definition of  $\mathcal{W}_j$ :

$$\|K\|_{\mathcal{F}_j(G)} \leq R\chi_j^{3/2} \tilde{g}_j^3 = R\chi_j \tilde{g}_j^2 \bar{\epsilon}_j(\ell).$$

The prefactor tends to zero with the coupling. For the fluctuation component,

$$\|K\|_{\mathcal{F}_j(\tilde{G})} \leq \omega_j^{-3} \|K\|_{\mathcal{W}_j} = R\chi_j^{3/2} \tilde{g}_j^{3/4} = R\chi_j \tilde{g}_j^{1/2} \bar{\epsilon}_j(\mathfrak{h}),$$

which is again small.

For the larger Cauchy ball, the first pair is bounded by  $r\chi_j^{1/2} \tilde{g}_j^{10/4} \leq r\bar{\epsilon}_j(\ell)$ , while the second obeys

$$\omega_j^{-3} r\chi_j^{1/2} \tilde{g}_j^{10/4} = r\chi_j^{1/2} \tilde{g}_j^{1/4} = r\bar{\epsilon}_j(\mathfrak{h}).$$

This proves both assertions.  $\square$

## Exact reduction of the $W$ -theorem to the two pairwise source estimates

**Proposition 10.140** (Pairwise step estimates imply the source  $W$  contraction). *Let  $\mathcal{K}_{+,j}(V, K)$  be the exact nonperturbative coordinate produced by the six-map algebra of Proposition 10.110. Assume that on the local interaction domain from this chapter the following statements hold uniformly in the scale and volume.*

- (P1) *In the norm pair  $\mathcal{F}_j(G) \rightarrow \mathcal{F}_{j+1}(T_0)$  the map is analytic on the pairwise ball of Lemma 10.139; its  $K$ -derivative has norm at most  $cL^{-1}$  on the larger Cauchy ball.*
- (P2) *In the norm pair  $\mathcal{F}_j(\tilde{G}) \rightarrow \mathcal{F}_{j+1}(\tilde{G}^\gamma)$  the same exact map is analytic on the corresponding pairwise ball and its  $K$ -derivative has norm at most  $cL^{-1}$  on the larger Cauchy ball.*
- (P3) *The inhomogeneous term obeys both target estimates which combine, via Proposition 10.137, to*

$$\|\mathcal{K}_{+,j}(V, 0)\|_{\mathcal{W}_{j+1}} \leq C\chi_{j+1}^{3/2}\tilde{g}_{j+1}^3.$$

- (P4) *The pairwise estimates are uniform on the same local interaction domain and are compatible with restriction, Euclidean covariance, the observable-sector projection, and the mass parameter.*

Then, for  $L$  sufficiently large and the coupling sufficiently small,

$$\|\mathcal{K}_{+,j}(V, K)\|_{\mathcal{W}_{j+1}} \leq \kappa_L \|K\|_{\mathcal{W}_j} + C\chi_{j+1}^{3/2}\tilde{g}_{j+1}^3, \quad \kappa_L = O(L^{-1}) < 1.$$

The same conclusion holds for the finite-order mixed derivatives obtained by Cauchy estimates on the displayed larger analytic ball, with the corresponding powers of its radius.

*Proof.* Let  $\dot{K}$  be a tangent polymer activity. By (P1)–(P2), applied to the two components of the  $\mathcal{W}_j$  norm,

$$\|D_K \mathcal{K}_{+,j}(V, K) \dot{K}\|_{\mathcal{Y}_{j+1}} \leq cL^{-1} \|\dot{K}\|_{\mathcal{W}_j}.$$

Proposition 10.137 converts this into

$$\|D_K \mathcal{K}_{+,j}(V, K) \dot{K}\|_{\mathcal{W}_{j+1}} \leq CL^{-1} \|\dot{K}\|_{\mathcal{W}_j}.$$

Integrating the derivative along  $t \mapsto tK$  gives

$$\|\mathcal{K}_{+,j}(V, K) - \mathcal{K}_{+,j}(V, 0)\|_{\mathcal{W}_{j+1}} \leq CL^{-1} \|K\|_{\mathcal{W}_j}.$$

Add (P3) and choose  $L$  so that the coefficient is less than one. The mixed-derivative bounds follow by applying the Banach-valued Cauchy formula from Lemma 10.124 on the larger ball, whose simultaneous admissibility in both source norm pairs is Lemma 10.139. Compatibility statements are identities for the same exact map and therefore survive the norm recombination.  $\square$

**Corollary 10.141** (Reduction to the six-map estimates). *For the bosonic  $n \geq 1$  branch, the preceding analysis proves:*

1. *the two source field scales and their coupling ratio;*
2. *the source large-set exponents and the two  $\mathcal{F}$  norm pairs;*
3. *the  $\mathcal{W}_j$  and intermediate  $\mathcal{Y}_{j+1}$  norms;*

4. the  $T_\varphi$  change-of-field-scale estimate needed to recombine the two outputs;
5. the quantitative  $\mathcal{Y}_{j+1} \rightarrow \mathcal{W}_{j+1}$  restoration;
6. the embedding of the physical  $\mathcal{W}_j$  domain and the larger Cauchy domain into both pairwise analytic domains;
7. the deduction of the full  $\mathcal{W}$  contraction and its Cauchy derivative consequences from the two pairwise source estimates.

Consequently the foundational theorem reduces precisely to (P1)–(P4) for the exact six source maps: interaction stability, localisation/change of scale, Gaussian expectation, reblocking and norm-strength restoration, with the corrected source formulas and common domain.

*Proof.* Definitions 10.132–10.136, Lemma 10.134, Proposition 10.137, Lemma 10.139, and Proposition 10.140 establish the seven listed items. None of those arguments estimates an unproved source map, so the remaining dependency boundary is exactly (P1)–(P4).  $\square$

### Mapwise proof order

The single step factors into six maps. The first two prepare the polymer and local coordinates. The third performs Gaussian integration, localisation, and reblocking; this is where the  $\kappa = O(L^{-1})$  change-of-scale gain is created. Maps four through six reapportion the third-order remainder, restore the output regulator, and return to the canonical interaction–polymer representation. The corresponding estimates, including the corrected first-step factorisation and scale-changing bounds, are proved in the subsequent corrected six-map and kernel modules. Reading the modules in that order avoids using the recombination statement before its nonlinear estimates have been established.

*Remark 10.142* (Role in the staged proof). This module proves the norm architecture and recombination, but it deliberately assumes the nonlinear pairwise estimates (P1)–(P4). Proposition 10.140 must not be read backwards as a proof of those estimates. They are established in the corrected six-map and kernel modules, culminating in Theorem 10.188.

#### 10.4.4 Six-map reduction to the genuine analytic kernels

this chapter reconstructed the two source norm pairs and proved their recombination into the iterable  $\mathcal{W}$  norm. The remaining question is therefore not whether one can invent a one-regulator contraction, but whether the actual six-map construction satisfies the two pairwise estimates to which Proposition 10.140 applies.

The purpose of the present subsection is to remove one more layer of ambiguity from that question. We separate the exact polymer algebra and the scale-changing geometry from the specific analytic estimates supplied by the stability theory. In particular, we prove that Maps 1, 2, 4, 5, and 6 do not manufacture the small factor  $L^{-1}$ : their role is to change coordinates, cancel local pieces, and restore the desired interaction. The  $L^{-1}$  gain is created only in Map 3, through the source “crucial contraction” after Gaussian integration.

All formulas in this subsection are read with the published corrections to the RG series. Two corrections matter structurally here. For a single block  $B$ , the corrected Map–1 formula is

$$K^{(1)}(B) = K(B) - I^B J(B, B), \quad (21.320)$$

with no additional sum, and the corresponding singleton identity in the general change-of-variables expansion is  $K_{\text{out}}(B) = M(B)$ . We use those corrected formulas below rather than the formulas printed in the original journal version.

## The six-map radius calculus

**Definition 10.143** (Pairwise radius profile). Fix either one of the two source norm pairs from Definition 10.135. Let

$$K^{(0)} = K, \quad K^{(i)} = \mathcal{M}_i(K^{(i-1)}) \quad (1 \leq i \leq 6), \quad K^{(6)} = K_+.$$

A *source-admissible radius profile* consists of positive radii  $r^{(0)}, \dots, r^{(6)}$  and constants

$$\mu_1, \mu_2, \mu_4, \mu_5, \mu_6 \geq 1, \quad \kappa_3 \geq 0,$$

such that

$$r^{(1)} = \mu_1 r^{(0)}, \quad r^{(2)} = \mu_2 r^{(1)}, \quad r^{(3)} = \kappa_3 r^{(2)}, \quad r^{(4)} = \mu_4 r^{(3)}, \quad r^{(5)} = \mu_5 r^{(4)}, \quad r^{(6)} = \mu_6 r^{(5)}.$$

The constants  $\mu_i$  are required to be independent of  $L$ ;  $\kappa_3$  is allowed to depend on  $L$  and is the only factor from which a contraction in  $L$  may arise.

**Lemma 10.144** (Unique source of the scale contraction). *For a source-admissible radius profile,*

$$r^{(6)} = \mu_1 \mu_2 \mu_4 \mu_5 \mu_6 \kappa_3 r^{(0)}.$$

*Consequently, if the five  $\mu_i$  are bounded independently of  $L$  and  $\kappa_3 \leq CL^{-1}$ , then*

$$r^{(6)} \leq C' L^{-1} r^{(0)}$$

*with  $C'$  independent of  $L$ . Conversely, no estimate involving only the five  $\mu_i$  can produce an  $L^{-1}$  gain.*

*Proof.* Substitute the six defining recursions successively. The displayed product is exact. The stated conclusions follow immediately.  $\square$

This elementary identity is important for proof structure: proving the fixed-scale maps with very small coupling constants cannot substitute for the scale-changing estimate in Map 3. The latter is the only place where the irrelevant direction is geometrically contracted.

## Two reusable algebraic lemmas

**Lemma 10.145** (Circle-product reassociation). *Let  $A, B, C$  be polymer activities at a fixed scale. Then*

$$(A \circ B) \circ C = A \circ (B \circ C).$$

*If  $I = \widehat{I} + \delta I$  with  $I, \widehat{I}, \delta I$  block-factorising activities, then for every  $K$ ,*

$$I \circ K = \widehat{I} \circ (\delta I \circ K). \tag{21.321}$$

*Proof.* Associativity was proved in Lemma 10.109. Since block factorisation implies  $I = \widehat{I} \circ \delta I$  in the circle-product algebra, the second identity follows by associativity.  $\square$

**Lemma 10.146** (Finite-neighbourhood analyticity). *Fix a polymer  $X$ . Suppose an output activity  $K_{\text{out}}(X)$  is obtained by a finite sum of finite products of the following objects evaluated on polymers inside a fixed small-set neighbourhood  $X^\square$ :*

$$K(Y), \quad I(B), \quad I(B)^{-1}, \quad \text{Loc}_{Y,B} F(Y),$$

*where  $I(B)$  is invertible on the parameter domain and every displayed input is analytic in the finite-dimensional local couplings and in  $K$ . Then  $K_{\text{out}}(X)$  is analytic on that domain, obeys the restriction property, and commutes with any algebra automorphism that commutes with multiplication, inversion, and Loc.*

*Proof.* There are only finitely many blocks in  $X^\square$  and only finitely many small polymers in that neighbourhood. Finite sums and products of analytic Banach-space-valued maps are analytic. Inversion is analytic on the open set of invertible elements by the convergent Neumann expansion after translating to any base point. Restriction is immediate because no input outside  $X^\square$  occurs. Finally, an algebra automorphism can be passed through each of the finitely many operations; covariance follows term by term.  $\square$

### Polymer weights and the change-of-variables theorem

For this subsection it is convenient to add the base exponent explicitly. For  $z \geq 0$  set

$$f_j(z, a, X) = \begin{cases} z + a(|X|_j - 2^d)_+, & X \neq \emptyset, \\ 0, & X = \emptyset. \end{cases} \quad (21.340)$$

**Lemma 10.147** (Absorbing a fixed exponential loss). *Let  $C \geq 1$  and  $0 \leq a_{\text{out}} < a_{\text{in}}$ . For sufficiently small  $0 < \rho < 1$ , depending only on  $C$  and  $a_{\text{in}} - a_{\text{out}}$ ,*

$$C^{|X|_j} \rho^{f_j(z, a_{\text{in}}, X)} \leq C^{2^d} \rho^{f_j(z, a_{\text{out}}, X)}. \quad (21.341)$$

*Proof.* If  $|X|_j \leq 2^d$  the inequality is immediate. Otherwise

$$C^{|X|_j} \rho^{f_j(z, a_{\text{in}}, X)} = C^{2^d} \rho^z (C \rho^{a_{\text{in}}})^{|X|_j - 2^d}.$$

Choose  $\rho$  so that  $C \rho^{a_{\text{in}}} \leq \rho^{a_{\text{out}}}$ .  $\square$

**Lemma 10.148** (Circle-product ball estimate). *Let  $0 < a_{\text{out}} < a_{\text{in}} \leq 2^{-d}$ , and suppose the local norm has the product property. If connected activities  $F, G$  satisfy*

$$\|F(X)\|, \|G(X)\| \leq \delta \rho^{f_j(0, a_{\text{in}}, X)}, \quad \delta = 2^{-2^d} \epsilon,$$

*then, for  $\rho$  sufficiently small,*

$$\|(F \circ G)(Z)\| \leq \epsilon \rho^{f_j(0, a_{\text{out}}, Z)}$$

*for every connected  $Z$ .*

*Proof.* For a decomposition  $Z = X \sqcup (Z \setminus X)$ , subadditivity of  $t \mapsto (t - 2^d)_+$  gives

$$f_j(0, a_{\text{in}}, Z) \leq f_j(0, a_{\text{in}}, X) + f_j(0, a_{\text{in}}, Z \setminus X).$$

There are at most  $2^{|Z|_j}$  subpolymers  $X$ . Hence the product property gives

$$\|(F \circ G)(Z)\| \leq \delta 2^{|Z|_j} \rho^{f_j(0, a_{\text{in}}, Z)}.$$

Apply Lemma 10.147 with  $C = 2$  and use the definition of  $\delta$ .  $\square$

**Lemma 10.149** (Transfer-configuration exponent). *Let  $S$  be the maximum number of source small sets containing a fixed block. There is  $c = c(d, S) > 0$  such that the following holds. Let  $0 < z < 2z'$ ,  $0 < a_{\text{in}} < c$  and  $0 \leq a_{\text{out}} \leq a_{\text{in}}$ . Suppose a connected output polymer  $W$  is represented as*

$$W = X^\square \cup U_M,$$

*where  $X \neq \emptyset$  is a set of transfer blocks and  $U_M$  has connected components  $U_{M,1}, \dots, U_{M,n_M}$ . Exclude the configuration  $|X|_j = 1$ ,  $U_M = \emptyset$ . Then*

$$z'|X|_j + \sum_{i=1}^{n_M} f_j(z, a_{\text{in}}, U_{M,i}) \geq (a_{\text{in}} - a_{\text{out}})|W|_j + f_j(z, a_{\text{out}}, W). \quad (21.342)$$

*Proof.* Because every transfer block has at most  $S$  small sets in its neighbourhood,  $|X^\square|_j \leq 2^d S |X|_j$ . Thus

$$|W|_j \leq 2^d S |X|_j + \sum_i |U_{M,i}|_j. \quad (21.343)$$

Put  $u = a_{\text{in}} - a_{\text{out}}$  and  $v = z - a_{\text{in}} 2^d$ . Taking  $a_{\text{in}}$  small makes  $v \geq 0$ . Since  $v + a_{\text{in}} |U|_j \leq f_j(z, a_{\text{in}}, U)$ , multiplying (21.343) by  $u$  and rearranging gives

$$\begin{aligned} u|W|_j + f_j(z, a_{\text{out}}, W) &\leq a_{\text{in}} 2^d S |X|_j + \sum_i f_j(z, a_{\text{in}}, U_{M,i}) \\ &\quad + (1 - n_M)v + u 2^d. \end{aligned}$$

If  $n_M \geq 1$ , choose  $a_{\text{in}}$  so small that  $a_{\text{in}} 2^d S + u 2^d \leq z'$  and use  $|X|_j \geq 1$ . If  $n_M = 0$ , the excluded singleton case implies  $|X|_j \geq 2$ ; since  $z < 2z'$ , reduce  $a_{\text{in}}$  further so that  $a_{\text{in}} 2^d S + z/2 + u 2^d \leq z'$ . In both cases the remaining terms are bounded by  $z' |X|_j$ , proving (21.342).  $\square$

**Lemma 10.150** (Exact large-polymer closure gain). *Let  $L \geq 2^d + 1$ . There is a number  $\eta = \eta(d) > 1$  such that every large connected  $j$ -polymer satisfies*

$$|X|_j \geq \eta |\bar{X}|_{j+1}. \quad (21.343a)$$

*For arbitrary  $j$ -polymers the same inequality holds with  $\eta = 1$ . One may take*

$$\eta = 1 + \frac{1}{2^d + 1 + 2^d \Delta(d)}, \quad (21.343b)$$

*where  $\Delta(d)$  is the largest number of connected components that can touch a connected family of  $2^d + 1$  blocks.*

*Proof.* The assertion with  $\eta = 1$  is immediate: the closure of  $m$  fine blocks contains at most  $m$  coarse blocks. We prove the strict gain for large connected polymers by induction on  $|\bar{X}|_{j+1}$ .

At the first large value  $|\bar{X}|_{j+1} = 2^d + 1$ , equality  $|X|_j = |\bar{X}|_{j+1}$  would put every fine block of  $X$  in a different coarse block. A connected collection of  $2^d + 1$  coarse blocks contains two non-touching blocks, while connectedness of  $X$  with diameter at most  $2^d + 1 \leq L$  forces the fine blocks of  $X$  to remain within touching coarse blocks. Hence equality is impossible and  $|X|_j \geq 2^d + 2$ , which is stronger than (21.343a) for the value in (21.343b).

For the induction step, remove from  $\bar{X}$  a connected family of  $2^d + 1$  coarse blocks. Its complement has at most  $\Delta(d)$  connected components adjacent to the removed family. Write the components of size at least  $2^d + 1$  as  $Y_1, \dots, Y_k$  and the remaining ones as  $Y_{k+1}, \dots, Y_m$ . Put  $M = \sum_{i \leq k} |Y_i|_{j+1}$ . By the induction hypothesis the fine blocks lying over  $Y_i$ ,  $i \leq k$ , cost at least  $\eta |Y_i|_{j+1}$ ; for the remaining components the trivial factor 1 suffices. The removed coarse family costs at least  $2^d + 2$  fine blocks by the base case. Since the small complementary components contain at most  $2^d \Delta(d)$  coarse blocks in total, division by the total number of coarse blocks gives

$$\frac{|X|_j}{|\bar{X}|_{j+1}} \geq 1 + \frac{1 + (\eta - 1)M}{2^d + 1 + M + 2^d \Delta(d)} = \eta,$$

where the last equality is precisely the choice (21.343b). This closes the induction.  $\square$

*Remark 10.151* (Why this is the exact geometry needed in Map 3). The strict inequality  $\eta > 1$  is not cosmetic. It permits the output large-set slope to be chosen strictly larger than the input slope  $a$  while still paying for reblocking:  $a < \tilde{a} < \eta a$ . Thus the large-polymer branch has a genuine exponent surplus, independent of the small-set localisation mechanism that produces the separate  $L^{-1}$  factor.

**Proposition 10.152** (Change-of-variables theorem). *Let the polymer norm have the product property, and let  $I$  be a block-factorising activity with*

$$\|I^Y\| \leq \alpha_I^{|Y|_j}.$$

*For every allowed pair  $(U, B)$  let  $J(U, B)$  be supported on a source small set  $U$  surrounding  $B$ , and suppose*

$$\sum_{U:(U,B) \in \mathcal{D}(J)} J(U, B) = 0. \quad (21.344)$$

*Set  $\bar{J}(U, B) = I^U J(U, B)$  and*

$$M(U) = K_{\text{in}}(U) - \sum_{B \subset U} \bar{J}(U, B). \quad (21.345)$$

*Suppose, for some  $0 < z < 2z'$ ,*

$$\|\bar{J}(U, B)\| \leq \epsilon \rho^{z'}, \quad \|M(U)\| \leq \epsilon \rho^{f_j(z, a_{\text{in}}, U)}. \quad (21.346)$$

*If  $a_{\text{in}}$  is sufficiently small,  $a_{\text{out}} < a_{\text{in}}$ , and  $\rho$  is sufficiently small, then there is a local component-factorising activity  $K_{\text{out}}$  such that*

$$(I \circ K_{\text{out}})(\Lambda) = (I \circ K_{\text{in}})(\Lambda), \quad (21.347)$$

*$K_{\text{out}}$  is polynomial in  $I, \bar{J}, K_{\text{in}}$ , and*

$$\|K_{\text{out}}(W) - M(W)\| \leq \epsilon \rho^{f_j(z, a_{\text{out}}, W) + (a_{\text{in}} - a_{\text{out}})/2} \quad (21.348)$$

*for connected  $W$ . For a single block  $B$  the corrected formula is*

$$K_{\text{out}}(B) = M(B). \quad (21.349)$$

*Proof.* Write  $\bar{J}(U) = \sum_{B \subset U} \bar{J}(U, B)$ . Component factorisation gives

$$K_{\text{in}}(U) = \prod_{C \in \text{Comp}(U)} (M(C) + \bar{J}(C)).$$

Expand the product. In every selected  $\bar{J}(C)$ , expand the sum over its marked block  $B$ . Reindex the resulting finite sum by the set  $X$  of marked blocks, the associated pairwise nontouching small sets  $U_B$ , and the union  $U_M$  of components on which  $M$  was selected. If  $U_J = \bigcup_{B \in X} U_B$  and  $W = X^\square \cup U_M$ , factor

$$I^{\Lambda \setminus (U_M \cup U_J)} = I^{\Lambda \setminus W} I^{W \setminus (U_M \cup U_J)}.$$

The coefficient of  $I^{\Lambda \setminus W}$  defines  $K_{\text{out}}(W)$  and proves (21.347). The formula is a finite polynomial and is local by construction.

Subtract the configuration  $X = \emptyset, U_M = W$ , which is precisely  $M(W)$ . The configurations with  $|X|_j = 1$  and  $U_M = \emptyset$  vanish by (21.344) after factoring out the common  $I^{X^\square}$ . In particular, if  $W = B$  is a single block, no nontrivial transfer configuration survives, which proves the corrected singleton identity (21.349).

For every remaining configuration, the product property and (21.346) give

$$\epsilon \rho^{z'|X|_j} \prod_i \rho^{f_j(z, a_{\text{in}}, U_{M,i})} \alpha_I^{|W \setminus (U_J \cup U_M)|_j}.$$

The last exponent is supported inside  $X^\square$ , hence contributes at most a fixed exponential  $C^{|W|_j}$ . Lemma 10.149, with  $2u = a_{\text{in}} - a_{\text{out}}$ , contributes  $\rho^{2u|W|_j + f_j(z, a_{\text{out}}, W)}$ . The number of configurations is also at most  $C^{|W|_j}$ : for each marked block there are at most  $S$  choices of a small set and the remaining blocks have only finitely many membership choices. Choose  $\rho$  so small that the combined exponential factor is at most  $\rho^{-u|W|_j}$ . Since connected  $W$  is nonempty,

$$\|K_{\text{out}}(W) - M(W)\| \leq \epsilon \rho^{u|W|_j + f_j(z, a_{\text{out}}, W)} \leq \epsilon \rho^{u + f_j(z, a_{\text{out}}, W)},$$

which is (21.348). □

### Map 1: exact cancellation and the corrected block formula

For a small  $j$ -polymer  $X$  and a block  $B \subset X$ , define

$$J(X, B) = \text{Loc}_{X, B} I^{-X} K(X) \quad (X \supsetneq B), \quad (21.322)$$

and

$$J(B, B) = - \sum_{Y \in \mathcal{S}_j: Y \supsetneq B} \text{Loc}_{Y, B} I^{-Y} K(Y). \quad (21.323)$$

All other  $J(X, B)$  are zero.

**Lemma 10.153** (Map-1 cancellation). *For every  $j$ -block  $B$ ,*

$$\sum_{X: (X, B) \in \mathcal{D}(J)} J(X, B) = 0. \quad (21.324)$$

Moreover, for a small nonblock polymer  $X$ ,

$$M(X) := K(X) - I^X \sum_{B \subset X} J(X, B) = K(X) - I^X \text{Loc}_X I^{-X} K(X), \quad (21.325)$$

so that

$$\text{Loc}_X I^{-X} M(X) = 0. \quad (21.326)$$

*Proof.* Equation (21.324) is exactly the definition (21.323). Summing (21.322) over blocks  $B \subset X$  and using the block decomposition of the localisation operator gives

$$\sum_{B \subset X} \text{Loc}_{X, B} = \text{Loc}_X,$$

which proves (21.325). Multiplying by  $I^{-X}$  and applying  $\text{Loc}_X$  gives

$$\text{Loc}_X I^{-X} M(X) = \text{Loc}_X I^{-X} K(X) - \text{Loc}_X^2 I^{-X} K(X) = 0$$

by Lemma 10.103. □

**Proposition 10.154** (Corrected singleton output of Map 1). *In the general change-of-variables construction applied to (21.322)–(21.323), the output on a single block is*

$$K^{(1)}(B) = M(B) = K(B) - I^B J(B, B). \quad (21.327)$$

*In particular the block formula contains neither an unweighted  $J(B, B)$  nor an extra sum over  $U$ .*



*Proof.* For output polymer  $W = B$ , the only decomposition in which nonempty polymer activities can contribute to the change-of-variables expansion has empty transferred polymer and  $U_M = B$ . Thus the general polynomial formula reduces to  $K_{\text{out}}(B) = M(B)$ . By the definition of  $M$  for a block, the only transfer term is  $I^B J(B, B)$ , which gives (21.327). This is precisely the combinatorial reason for the published correction of the singleton formula.  $\square$

The preceding argument closes the *algebraic* Map-1 cancellation. Its pairwise norm estimate still requires quantitative control of  $I^X \text{Loc}_X I^{-X} K(X)$  and of the small-set transfer terms. We isolate that input below instead of concealing it in a generic Taylor estimate.

## Map 2: exact coordinate identity and the genuine block kernel

Set

$$I^{(2)} = \widehat{I} = I_j(\widehat{V}), \quad \delta I^{(2)} = I - \widehat{I}, \quad K^{(2)} = K^{(1)} \circ \delta I^{(2)}. \quad (21.328)$$

**Proposition 10.155** (Exact Map-2 representation identity). *The Map-2 coordinate change satisfies*

$$(I \circ K^{(1)})(\Lambda) = (\widehat{I} \circ K^{(2)})(\Lambda). \quad (21.329)$$

*If  $K = 0$  and hence the extracted perturbative correction vanishes, then  $K^{(2)} = 0$ .*

*Proof.* The first statement follows from Lemma 10.145:

$$I \circ K^{(1)} = (\widehat{I} \circ \delta I^{(2)}) \circ K^{(1)} = \widehat{I} \circ (\delta I^{(2)} \circ K^{(1)}).$$

When  $K = 0$ , the small-set extraction  $Q(V, K)$  vanishes, so  $\widehat{V} = V$ ,  $\delta I^{(2)} = 0$  as a nonempty-block activity, and the output polymer coordinate is zero.  $\square$

*Remark 10.156* (Why the Map-2 gain is not a formal Taylor remainder). The source estimate after (21.329) is stronger than mere smallness of  $I - \widehat{I}$ . It requires the precise perturbative matching between  $\widehat{V}$  and the block contribution extracted in Map 1 so that  $\text{Loc}_X I^{-X} K^{(2)}(X)$  gains a positive power of the source small parameter. That matching is one of the local analytic kernels recorded below. No “generic Taylor cancellation” is used as a substitute.

## Map 3: the unique scale-changing step

The source stability analysis supplies two estimates for either of the two norm pairs. We record them as named kernel gates because they are the exact analytic input at which the present manuscript still bottoms out.

**Definition 10.157** (Integration-property kernel  $\text{IP}_j$ ). The gate  $\text{IP}_j$  is the assertion that for disjoint  $j$ -polymers  $X, Y$  and  $F(Y) \in \mathcal{N}(Y^\square)$ ,

$$\|\mathbb{E}_{j+1} \delta I^X \theta F(Y)\|_{j+1} \leq \alpha_{\mathbb{E}}^{|X|_j + |Y|_j} (C_{\delta V} \bar{\epsilon})^{|X|_j} \|F(Y)\|_j, \quad (21.330)$$

with  $\alpha_{\mathbb{E}}$  independent of  $L$ , for each source norm pair.

**Definition 10.158** (Crucial-contraction kernel  $\text{CC}_j$ ). Let  $X \in \mathcal{S}_j$  and  $U = \overline{X}$  be its  $(j+1)$ -closure. Put

$$\gamma(X) = L^{-d-1} + L^{-1} \mathbf{1}_{X \cap \{a, b\} \neq \emptyset}.$$

The gate  $\text{CC}_j$  is the assertion that every admissible local  $F(X)$  satisfies

$$\|\widetilde{I}_{\text{pt}}^{U \setminus X} \mathbb{E}_{C_{j+1}} \theta F(X)\|_{j+1} \leq C \left( \gamma(X) \|F(X)\|_j + \|\widetilde{I}_{\text{pt}}^X \text{Loc}_X \widetilde{I}_{\text{pt}}^{-X} F(X)\|_j \right), \quad (21.331)$$

for either source norm pair.

**Lemma 10.159** (Structural decomposition behind the crucial contraction). *For any  $F(X)$  in Definition 10.158, set*

$$D = \tilde{I}_{\text{pt}}^X \text{Loc}_X \tilde{I}_{\text{pt}}^{-X} F(X), \quad E = F - D. \quad (21.332)$$

*Then*

$$\text{Loc}_X \tilde{I}_{\text{pt}}^{-X} E = 0. \quad (21.333)$$

*Consequently, the first term on the right of (21.331) is exactly the scale-improvement problem for the localised-away component  $E$ , whereas  $D$  is controlled without any new power of  $L$  by stability and the integration property.*

*Proof.* Multiplying (21.332) by  $\tilde{I}_{\text{pt}}^{-X}$  gives

$$\tilde{I}_{\text{pt}}^{-X} E = \tilde{I}_{\text{pt}}^{-X} F - \text{Loc}_X \tilde{I}_{\text{pt}}^{-X} F.$$

Apply  $\text{Loc}_X$  and use  $\text{Loc}_X^2 = \text{Loc}_X$  to obtain (21.333). Thus  $E$  lies in the irrelevant complement on which the power-counting contraction acts. The term  $D$  is already local and is bounded by its defining norm after Gaussian integration; no geometrically improved factor is expected or required for it.  $\square$

**Definition 10.160** (Primitive stability kernels below  $\text{IP}_j$  and  $\text{CC}_j$ ). We separate four lower-level statements, because the source proof of  $\text{IP}_j$  and  $\text{CC}_j$  uses them in different ways.

- (PK1)  *$T_\phi$  Gaussian/product kernel.* Gaussian expectation is bounded in the source  $T_\phi$  seminorm after adjoining the fluctuation field, and the seminorm has the product property used to multiply the factors  $\delta I$  and  $F$ .
- (PK2) *Interaction kernel.* On the source stability domain,  $I$ ,  $I^{-1}$ ,  $\delta V$  and  $\delta I$  obey the exact small-field estimates in both field scales, with one factor  $C_{\delta V} \bar{\epsilon}$  for every inserted  $\delta I$ .
- (PK3) *Regulator-evolution kernel.* Gaussian expectation of the fine-scale regulator is uniformly bounded, and the strong large-field regulator satisfies the source scale comparison

$$\tilde{G}_j(X, \phi)^5 \leq \tilde{G}_{j+1}^\gamma(X, \phi). \quad (21.333a)$$

- (PK4) *Sector-sensitive localisation kernel.* If the localised part of an admissible  $F(X)$  vanishes, then its scale-changed norm gains

$$L^{-d-1} + L^{-1} \mathbf{1}_{X \cap \{a,b\} \neq \emptyset}, \quad (21.333b)$$

with the same observable-sector convention as Definition 10.158.

**Proposition 10.161** (The bosonic Gaussian/product kernel estimate). *For the  $n \geq 1$  bosonic branch of SM-21.03.01, the Gaussian/product kernel (PK1) is proved by Lemmas 10.98 and 10.99, after restriction to either this chapter field scale. Thus PK1 follows from these two lemmas.*

*Proof.* Lemma 10.98 is exactly the submultiplicativity needed when the factors  $\delta I$  and  $F$  are multiplied in a local  $T_\phi$  seminorm. Lemma 10.99 commutes every field derivative through the finite-dimensional Gaussian expectation and bounds the resulting seminorm by the expected translated seminorm. Both statements are scale-parametric: replacing the test-field unit ball by the one associated with  $\ell_j$  or  $h_j$  changes only the norm used to declare a test direction unit and does not change either proof. Since SM-21.03.01 is the purely bosonic  $n$ -component  $|\varphi|^4$  branch, no fermionic seminorm extension is needed here. This is precisely PK1.  $\square$

**Proposition 10.162** (Reduction to two regulator-stability estimates). *Assume the source field-scale monotonicity already recorded in this chapter. Then*

$$\text{PK1} + \text{PK2} + \text{PK3} \implies \text{IP}_j, \quad (21.333\text{c})$$

and

$$\text{IP}_j + \text{PK1} + \text{PK3} + \text{PK4} \implies \text{CC}_j. \quad (21.333\text{d})$$

*In particular, the only negative power of  $L$  in the second implication is supplied by PK4.*

*Proof.* For (21.333c), expand  $\delta I^X$  block by block. The product property in PK1 and the interaction estimate PK2 give one factor  $C_{\delta V}\bar{\epsilon}$  for each block of  $X$ . PK1 moves the remaining product through Gaussian expectation. The fluctuation-regulator moment in PK3 absorbs the fine-field growth at a cost exponential only in  $|X|_j + |Y|_j$ ; for the  $\tilde{G}$  pair the final comparison (21.333a) converts the fifth power of the fine regulator into the required coarse regulator. This chapter scale monotonicity then changes the field scale to the exact output scale. This is precisely the bound in Definition 10.157.

For (21.333d), use the decomposition  $F = D + E$  from Lemma 10.159. The term  $E$  has zero localised part, so PK4 gives the factor (21.333b). The local term  $D$  is estimated without a power-counting gain: PK1 and PK3 control Gaussian integration and regulator evolution, while  $\text{IP}_j$  controls the inserted interaction factors. Adding the two estimates gives Definition 10.158. No other step in this argument produces a negative power of  $L$ .  $\square$

**Proposition 10.163** (Map-3 global reduction). *Assume the pairwise analytic kernels  $\text{IP}_j$  and  $\text{CC}_j$ , together with the local interaction/remainder estimates collected in Definition 10.164. Then the source Map 3 sends its input radius  $r^{(2)}\bar{\epsilon}$  to an output radius*

$$r^{(3)}\bar{\epsilon}, \quad r^{(3)} \leq CL^{-1}r^{(2)}, \quad (21.334)$$

*for either norm pair, after taking the source small parameter sufficiently small depending on fixed  $L$  and on the radius hierarchy. The factor  $L^{-1}$  comes from  $\gamma(X)$  in (21.331); large-set and higher-order terms are absorbed by the large-set exponent and extra powers of  $\bar{\epsilon}$ .*

*Proof.* Expand the exact scale-changing Map-3 formula according to the fine polymer  $X$  whose  $(j+1)$ -closure is the output polymer  $U$ . Split the sum into small and large  $X$ .

For small  $X$ , the preceding Maps 1–2 have arranged that the localised part appearing as the second term in (21.331) carries the improved small parameter from the Map-2 matching kernel. The remaining component is bounded by  $\gamma(X)\|F(X)\|_j$ . Since  $d = 4$ ,

$$\gamma(X) \leq L^{-1} + L^{-5} \leq 2L^{-1}.$$

Thus every small-set irrelevant contribution receives the desired  $O(L^{-1})$  factor. The integration-property kernel (21.330) supplies one factor  $C\bar{\epsilon}$  for every additional fluctuation interaction  $\delta I$ .

For large  $X$ , no localisation is required. The source polymer norm contains the factor  $\bar{\epsilon}^{f_j(a^{(2)}, X)}$ . Reblocking can reduce the number of blocks, but the standard connected-polymer counting bound from the geometry module is only exponential in the number of fine blocks. The positive gap between the input and output large-set exponents therefore supplies a factor  $\bar{\epsilon}^{|X|_j}$  which, for sufficiently small  $\bar{\epsilon}$ , absorbs both the counting constant and the fixed  $L$ -dependent reblocking multiplicity. The corrections to the source remainder indices are used here: the large-set exponent is attached to the actual output closure  $U$ , and the final exponent is the  $(j+1)$ -scale exponent.

Finally, every term containing at least one of the perturbative remainder factors in Definition 10.164 carries at least one additional power of  $\bar{\epsilon}$ . At fixed  $L$ , decrease the small-coupling threshold until that power is smaller than a fixed multiple of  $L^{-1}$ . Summing the small- and large-set contributions gives (21.334). No other step of the proof creates a negative power of  $L$ .  $\square$

The proposition is deliberately a reduction theorem: it identifies exactly which source-local analytic estimates remain to be proved in this book before Map 3 is established.

#### Maps 4–6: restoration steps and their local kernels

**Definition 10.164** (Local analytic kernel family  $\text{LA}_j$ ). The gate  $\text{LA}_j$  consists of the following pairwise estimates, with the exact source domains and either source norm pair.

- (LA1) *Small-set transfer stability*: the Map–1 quantities  $I^X J(X, B)$  and  $M(X)$  obey the radius hypotheses of the source change-of-variables proposition, uniformly in  $j$ .
- (LA2) *Block-to- $V$  matching*:  $\delta I^{(2)} = I - \hat{I}$  is  $O(\bar{\epsilon})$  in the block norm and the Map–2 localised output gains the prescribed positive power  $\bar{\epsilon}^u$  on every small set.
- (LA3) *Local perturbative restoration*: the leading correction satisfies

$$\|\tilde{I}_{\text{pt}}^U(U) h_{\text{lead}}(U, B)\|_{j+1} \leq C_L \bar{\epsilon}^2, \quad (21.335)$$

and the reduced two-block difference satisfies

$$\|\tilde{I}_{\text{pt}}^U[h_{\text{red}}(U) - h_{\text{lead}}(U)]\|_{j+1} \leq C_L \bar{\epsilon}^{f_{j+1}(3, a^{(3)}, U)}. \quad (21.336)$$

The  $|X|_j \geq 3$  tail is no longer part of this local gate: it follows from  $\text{IP}_j$  by Proposition 10.167.

- (LA4) *Final interaction and boundary adjustments*: the Map–5 differences between the perturbative and final block interactions have the source orders  $O_L(\bar{\epsilon}^2)$  and  $O_L(\bar{\epsilon}^4)$  required by the circle-product estimate, and the Map–6 boundary correction is  $O_L(\bar{\epsilon})$  per affected block with the source large-set decay.

**Lemma 10.165** (Cancellation of the leading restoration term). *For every  $(j+1)$ -block  $B$ , the source leading term satisfies*

$$\sum_{U: (U, B) \in \mathcal{D}(h_{\text{lead}})} h_{\text{lead}}(U, B) = 0. \quad (21.337)$$

*Consequently it is admissible in the same change-of-variables mechanism as Map 1.*

*Proof.* The leading term consists of the two-block connected perturbative covariance with a minus one-half self-block term and plus one-half neighbouring-block terms. For fixed  $B$ , sum over all allowed  $U$ . Every contribution coupling  $B$  to its complement occurs once in the self-block term with negative sign and once in the unique two-block term containing that interaction with positive sign. The coefficients have equal magnitude, so they cancel pairwise, proving (21.337).  $\square$

**Definition 10.166** (Reduced and higher-order restoration pieces). For a connected  $(j+1)$ -polymer  $U$ , decompose the source restoration term as

$$h(U) = h_{\text{red}}(U) + h_{\text{rem}}(U),$$

where  $h_{\text{red}}$  is the sum over fine polymers  $X$  with  $|X|_j \leq 2$  and  $h_{\text{rem}}$  is the sum over  $|X|_j \geq 3$  in the fluctuation expansion of  $\tilde{I}_{\text{pt}}^{-X} \mathbb{E}_{j+1} \delta I^X$ .

**Proposition 10.167** (The  $|X|_j \geq 3$  restoration tail follows from  $\text{IP}_j$ ). *Assume  $\text{IP}_j$ . For fixed  $L$  and sufficiently small  $\bar{\epsilon}$ , the higher-order tail in Definition 10.166 obeys*

$$\|\tilde{I}_{\text{pt}}^U h_{\text{rem}}(U)\|_{j+1} \leq C_L \bar{\epsilon}^{f_{j+1}(3, a^{(3)}, U)}. \quad (21.337a)$$

*Consequently the second estimate in (LA3) reduces to the corresponding estimate for  $h_{\text{red}} - h_{\text{lead}}$ .*

*Proof.* Every summand in  $h_{\text{rem}}$  contains at least three fluctuation-interaction factors. By  $\text{IP}_j$ , by  $|\bar{X}|_{j+1} \leq |X|_j$ , and by the bounded number of fine polymers inside a fixed coarse polymer,

$$\|\tilde{I}_{\text{pt}}^U h_{\text{rem}}(U)\|_{j+1} \leq (C_{L,d}\bar{\epsilon})^{3\vee|U|_{j+1}}. \quad (21.337b)$$

If  $|U|_{j+1} \leq 2^d$ , the right-hand side is  $O_L(\bar{\epsilon}^3)$ , which is exactly the small-set value of the desired exponent. If  $|U|_{j+1} > 2^d$ , choose  $t = t(a^{(3)}) > 0$  so that

$$|U|_{j+1} \geq 3 + a^{(3)}(|U|_{j+1} - 2^d) + t|U|_{j+1}.$$

The surplus factor  $\bar{\epsilon}^{t|U|_{j+1}}$  absorbs  $C_{L,d}^{|U|_{j+1}}$  once  $\bar{\epsilon}$  is sufficiently small at fixed  $L$ . This proves (21.337a). Since  $h - h_{\text{lead}} = (h_{\text{red}} - h_{\text{lead}}) + h_{\text{rem}}$ , the last assertion follows by the triangle inequality.  $\square$

**Proposition 10.168** (Maps 4–6 are fixed-scale bounded once  $\text{LA}_j$  holds). *Assume  $\text{LA}_j$  and  $\text{IP}_j$ . Then Maps 4, 5, and 6 preserve the exact partition-function coordinate and have radius factors*

$$r^{(4)} \leq \mu_4 r^{(3)}, \quad r^{(5)} \leq \mu_5 r^{(4)}, \quad r^{(6)} \leq \mu_6 r^{(5)}, \quad (21.338)$$

where  $\mu_4, \mu_5, \mu_6$  are independent of  $L$  after the source order of choosing parameters (first the universal radius hierarchy, then  $L$  large, then the coupling small). These maps introduce no additional negative power of  $L$ .

*Proof.* For Map 4, use (21.337) in the same finite change-of-variables algebra as Map 1. Estimate (21.335) places the leading correction in the quadratic  $\bar{\epsilon}^2$  ball. Estimate (21.336) controls the reduced two-block remainder, while Proposition 10.167 controls the  $|X|_j \geq 3$  tail from  $\text{IP}_j$ . Together they give the required extra power for the difference between the exact and leading correction; that power absorbs the fixed small-neighbourhood combinatorics. Hence the radius grows by a fixed multiplicative factor  $\mu_4$ .

Map 5 is a circle-product coordinate change. The exact representation identity follows from associativity as in Proposition 10.155. By (LA4), the new block activity has at least one extra factor  $\bar{\epsilon}$  relative to the input radius. Expanding the circle product, a term with  $m$  inserted block corrections carries  $\bar{\epsilon}^m$ ; the number of choices is at most exponential in the polymer size and is absorbed by the large-set weight for sufficiently small  $\bar{\epsilon}$ . This gives the fixed factor  $\mu_5$ .

Map 6 is again a finite local reparametrisation, now supported on the boundary bookkeeping sets. Each changed block contributes the  $O_L(\bar{\epsilon})$  factor from (LA4). The source large-set exponent absorbs the exponential number of boundary choices, and the remaining finite  $L$ -dependence is absorbed by the choice of the small-coupling threshold after  $L$  is fixed. Thus the final radius factor is a universal  $\mu_6$  in the source parameter order. None of these arguments invokes the scale-improved localisation estimate, so no  $L^{-1}$  is created here.  $\square$

## The quantitative analytic-domain estimate

**Definition 10.169** (Quantitative analytic-domain condition  $\text{AD}_j$ ). The gate  $\text{AD}_j$  asserts, simultaneously for both the norm pairs, that the six exact maps are analytic on the physical pairwise ball and on the larger common Cauchy ball of Lemma 10.139; that their bounds are uniform on the same local-interaction domain; that restriction, Euclidean covariance, observable-sector projection and mass continuity hold there; and that the  $K = 0$  output has the two target bounds which recombine to the inhomogeneous estimate (P3) of Proposition 10.140.

**Proposition 10.170** (Qualitative analyticity is already local;  $\text{AD}_j$  is quantitative). *For every fixed finite polymer, qualitative analyticity, restriction and algebraic covariance of Maps 1–6 follow from Lemma 10.146 and the exact circle/change-of-variables formulas proved above. Thus the remaining content of  $\text{AD}_j$  is the uniform source radius, common-domain and mass/observable estimates, together with the  $K = 0$  target size; no separate formal analyticity theorem remains.*

*Proof.* Each map is assembled from finitely many products, sums, inverses on the stability domain, localisation operators, Gaussian expectations and reindexings inside a fixed finite neighbourhood. The finite algebraic operations are covered by Lemma 10.146; Gaussian expectation is linear and, by Lemma 10.99, commutes with the finite field derivatives on an integrable regulator domain. The exact identities above preserve restriction and covariance term by term. What is not supplied by these local facts is a scale-uniform lower bound on the analytic radius, simultaneous admissibility of the two source norm pairs on the larger Cauchy ball, or the quantitative  $K = 0$  bound. Those are precisely the stated remaining parts of  $\text{AD}_j$ .  $\square$

**Definition 10.171** (Source-kernel gate  $\text{SK}_j$ ). *The source-kernel gate is*

$$\text{SK}_j := \text{LA}_j \wedge \text{IP}_j \wedge \text{CC}_j \wedge \text{AD}_j. \quad (21.339)$$

It is a statement for *both* source norm pairs and with the source parameter domains, observable sectors, and scale restrictions. No weaker one-regulator version is identified with  $\text{SK}_j$ .

**Theorem 10.172** (Reduction of the six-map theorem). *Assume  $\text{SK}_j$  for every admissible scale. Then the full six-map construction satisfies the two pairwise estimates required in Proposition 10.140, with contraction factor  $\kappa = O(L^{-1})$ . Hence the source  $\mathcal{W}_j \rightarrow \mathcal{W}_{j+1}$  nonlinear single-step estimate follows from the norm recombination.*

*Proof.* Map 1 has the exact cancellation and corrected singleton formula proved in Lemmas 10.153 and Proposition 10.154; (LA1) supplies its source-radius estimate. Map 2 has the exact representation identity of Proposition 10.155; (LA2) supplies the local matching gain. Map 3 then contracts by  $CL^{-1}$  by Proposition 10.163, using  $\text{IP}_j$  and  $\text{CC}_j$ . Maps 4–6 obey the fixed-radius bounds of Proposition 10.168. Multiplying the six radius factors and applying Lemma 10.144 gives the pairwise  $O(L^{-1})$  estimate. Proposition 10.140 converts the two pairwise outputs to the intermediate  $\mathcal{Y}_{j+1}$  bound, and Proposition 10.137 restores the iterable  $\mathcal{W}_{j+1}$  norm.  $\square$

*Remark 10.173* (Role in the staged proof). Theorem 10.172 is an unconditional reduction theorem, not by itself the final nonlinear single-step theorem. Its premise  $\text{SK}_j$  is proved in the following kernel module. The exact algebra, corrected source formulas, radius propagation, Map–3 global reblocking, and Maps 4–6 restoration calculus are proved here. The following module supplies the source-strength pairwise analytic kernel package: interaction/regulator stability, the crucial localisation contraction, the finite list of perturbative interaction/boundary estimates in  $\text{LA}_j$ , and the quantitative common-domain/inhomogeneous estimates in  $\text{AD}_j$ . The bosonic  $T_\phi$  Gaussian/product kernel is proved by Proposition 10.161.

#### 10.4.5 Proof of the nonlinear bosonic step

The preceding analysis reduction isolated the exact analytic statements that had still been hidden behind the phrase “stability analysis.” We now prove those statements at the strength needed by the  $n \geq 1$  bosonic branch. The proof is deliberately organised in the same logical order as the stability theorem: first the two regulator systems, then interaction stability and Gaussian integration, then

localisation, then the finite perturbative restoration estimates, and only at the end the common analytic domain. This order is important: the  $L^{-1}$  gain is created only by localisation and not by any algebraic reparametrisation.

### The two source regulators

**Definition 10.174** (Source fluctuation and large-field regulators). For a  $j$ -block  $B$  and its small-set neighbourhood  $B^\square$ , set

$$\|\phi\|_{\Phi_j(B^\square, \ell_j)} = \inf_{f|_{B^\square=0}} \|\phi - f\|_{\Phi_j(\ell_j)}.$$

Let  $\tilde{\Pi}(B^\square)$  be the restrictions to  $B^\square$  of affine-linear lattice fields and define

$$\|\phi\|_{\tilde{\Phi}_j(B^\square, \ell_j)} = \inf_{p \in \tilde{\Pi}(B^\square)} \|\phi - p\|_{\Phi_j(B^\square, \ell_j)}.$$

For a polymer  $X$  define

$$G_j(X, \phi) = \prod_{B \subset X} \exp\left\{\|\phi\|_{\Phi_j(B^\square, \ell_j)}^2\right\}, \quad \tilde{G}_j(X, \phi) = \prod_{B \subset X} \exp\left\{\frac{1}{2}\|\phi\|_{\tilde{\Phi}_j(B^\square, \ell_j)}^2\right\}.$$

Here a harmless change in the normalisation of the block average has been absorbed into  $\ell_0$ . Thus  $G_j$  is the *fluctuation-field regulator*;  $\tilde{G}_j$  is the *large-field regulator*. The two norm pairs are

$$G_j(\ell_j) \longrightarrow T_0(\ell_{j+1}), \quad \tilde{G}_j(\mathfrak{h}_j) \longrightarrow \tilde{G}_{j+1}^\gamma(\mathfrak{h}_{j+1}).$$

**Lemma 10.175** (Affine-quotient scale improvement). *For every fixed  $p > 0$  there is  $L_0(p)$  such that, for  $L \geq L_0(p)$  and  $j + 1 < N$ ,*

$$\tilde{G}_j(X, \phi)^p \leq \tilde{G}_{j+1}(X, \phi) \quad (X \subset \Lambda).$$

*In particular the choice  $p = 5$  is valid, and after weakening the output regulator exponent it gives*

$$\tilde{G}_j(X, \phi)^5 \leq \tilde{G}_{j+1}^\gamma(X, \phi)$$

*whenever the right-hand side in the norm definition is chosen with the corresponding source exponent  $\gamma \in (0, 1]$ .*

*Proof.* The quotient by affine functions is what creates the scale improvement. On a fine block contained in a coarse block, Taylor's theorem after subtraction of the optimal affine function gives

$$\|\phi\|_{\tilde{\Phi}_j(B^\square, \ell_j)} \leq cL^{-d/2-1} \|\phi\|_{\tilde{\Phi}_{j+1}((B^+)^\square, \ell_{j+1})}.$$

Squaring and multiplying by the ratio of block volumes contributes  $L^d$ , hence the fine exponent is at most  $c^2 L^{-2}$  times the coarse exponent. Choose  $L$  so that  $pc^2 L^{-2} \leq 1$ . Summation over the fine blocks, followed by exponentiation, proves the first assertion. The second is obtained by choosing the power used in the preceding estimate large enough to absorb the fixed weakening from power one to power  $\gamma$ ; all powers are fixed before  $L$  is chosen.  $\square$

**Proposition 10.176** (Proof of PK3). *The regulator-evolution kernel (PK3) of Definition 10.160 holds for the two field scales.*

*Proof.* For the fluctuation field, Lemma 10.101 applied blockwise to the finite-range covariance gives, after choosing the regulator constant below the Gaussian moment threshold,

$$\mathbb{E}_{j+1} G_j(X, \xi)^q \leq C_q^{|X|_j}$$

for every fixed  $q$  used in the single-step proof. The same estimate for the translated field follows from the Gaussian translation inequality and the elementary quadratic bound  $\|\phi + \xi\|^2 \leq 2\|\phi\|^2 + 2\|\xi\|^2$ . For the large-field pair, use Lemma 10.175 with  $p = 5$ . These are exactly the two regulator statements entering PK3.  $\square$

### Interaction stability and the integration property

**Lemma 10.177** (Quartic stability of one block). *On the source stability domain, write on a block  $V = g\tau^2 + Q$  with  $Q$  of degree at most two and  $\operatorname{Re} g \asymp |g| > 0$ . For either field scale there are constants  $c, C$  such that*

$$\|e^{-V(B)}\|_{T_\phi} \leq C \exp\{-c\epsilon_{g\tau^2}\|\phi\|_{L^2(B)}^2 + C\epsilon_V(1 + \|\phi\|_{\Phi(B^\square)}^2)\}.$$

*Consequently  $I(B) = e^{-V(B)}(1 + W(B))$  and its inverse on the source analytic ball have uniform regulator bounds, provided  $\|W(B)\|_{T_0} = O(\bar{\epsilon}^2)$ .*

*Proof.* The product property gives  $\|e^{-g\tau^2(B)}\|_{T_\phi}$  as the product of one-site bounds. Completing the square between the positive quartic part and the possibly negative quadratic part yields the displayed Gaussian decay in the scaled  $L^2$  field. Since  $Q$  is quadratic,  $\|Q(B)\|_{T_\phi} \leq C\epsilon_V(1 + \|\phi\|_{\Phi}^2)$ ; the exponential series and Lemma 10.98 therefore control  $e^{-Q}$ . Finally  $\|1 + W\|_{T_\phi}$  is bounded by the degree-six polynomial estimate and  $\|W\|_{T_0} = O(\bar{\epsilon}^2)$ . Shrinking the common analytic ball makes  $|1 + W|$  bounded away from zero, so the Neumann series gives the same type of bound for the inverse.  $\square$

**Lemma 10.178** (One fluctuation-interaction factor is small). *Let  $V_{\text{pt}}$  be the second-order perturbative interaction and  $\delta V = V - V_{\text{pt}}$ . On the source domain,*

$$\|\delta V(B)\|_{T_\phi} \leq C_{\delta V} \bar{\epsilon} \mathcal{G}(B, \phi), \quad \|\delta I(B)\|_{T_{\phi \sqcup \xi}} \leq C_{\delta V} \bar{\epsilon} \mathcal{G}(B, \phi)^3 G_j(B, \xi)^q$$

*for a fixed  $q$ , simultaneously for the two source field scales. Here  $\delta I = I - I_{\text{pt}}$ , with the  $W$  correction included.*

*Proof.* The perturbative Taylor formula gives the first inequality: the quartic, quadratic and gradient coordinates match through the prescribed second order, so every remaining coefficient contains either one nonperturbative coordinate or one additional power of the source small parameter. The preceding analysis structural-stability theorem controls the nonperturbative coordinate by the same  $\bar{\epsilon}$ .

For the second inequality interpolate between  $V$  and  $V_{\text{pt}}$ :

$$e^{-V} - e^{-V_{\text{pt}}} = - \int_0^1 e^{-(1-t)V_{\text{pt}}} \delta V e^{-tV} dt.$$

Apply Lemma 10.98 and the quartic stability lemma to the three factors. The difference between  $W(V)$  and  $W(V_{\text{pt}})$  is polynomial in the couplings and is  $O(\bar{\epsilon}^2)$ , hence smaller than the displayed first-order bound. The translated-field regulator separates into a background factor and a fixed Gaussian moment factor; this gives the stated exponent  $q$ .  $\square$



**Proposition 10.179** (Proof of PK2 and the integration property). *The interaction kernel (PK2) holds. Consequently the integration property  $\text{IP}_j$  in Definition 10.157 holds for both source norm pairs.*

*Proof.* The bounds on  $I$  and  $I^{-1}$  are Lemma 10.177; the bounds on  $\delta V$  and  $\delta I$  are Lemma 10.178. For disjoint  $X, Y$ , multiply the block estimates for  $\delta I^X$  using the product property. Move the resulting expression through Gaussian expectation by Lemma 10.99. The fluctuation regulator moment is absorbed by Proposition 10.176. This gives one factor  $C_{\delta V} \bar{\epsilon}$  for every block of  $X$  and only an exponential counting constant in  $|X|_j + |Y|_j$ , which is precisely (21.330).  $\square$

### Exact localisation gain

**Lemma 10.180** (Bulk and observable Taylor dimensions). *Let  $F(X)$  be supported in a small  $j$ -polymer and satisfy the source sector conditions. If  $\text{Loc}_X F = 0$ , then after passing to scale  $j + 1$  the bulk sector gains  $L^{-d-1}$ . A sector containing one of the marked observable points gains  $L^{-1}$ . Thus*

$$\|(1 - \text{Loc}_X)F\|_{j+1} \leq C(L^{-d-1} + L^{-1}\mathbf{1}_{X \cap \{a,b\} \neq \emptyset})\|F\|_j,$$

*up to the fixed regulator factors already controlled by PK3.*

*Proof.* The localisation operator matches all lattice Taylor jets whose engineering dimension is at most the dimension allowed in the corresponding sector. In the bulk, the first unmatched test polynomial has dimension at least  $d + 1$  above the block-volume normalisation. Rescaling a fine block to a coarse block therefore contributes  $L^{-d-1}$ . With one observable insertion the source observable field scale is frozen at the coalescence convention, so only one spatial Taylor order is forced before an omitted monomial occurs; this yields  $L^{-1}$ . The integral form of the discrete Taylor remainder gives these factors coefficient by coefficient. The finite set of test monomials and multi-indices contributes only a universal constant. PK3 transfers the background regulator to the output scale.  $\square$

**Proposition 10.181** (Proof of PK4 and the crucial contraction). *The sector-sensitive localisation kernel (PK4) holds. Consequently the crucial-contraction gate  $\text{CC}_j$  of Definition 10.158 holds for both source norm pairs.*

*Proof.* Decompose  $F = D + E$  as in Lemma 10.159. The component  $E$  has zero localisation and is bounded by Lemma 10.180. The local component  $D$  is controlled by the integration property without a negative power of  $L$ . Multiplication by  $\tilde{I}_{\text{pt}}^{U \setminus X}$  is stable by Lemma 10.177. This is exactly the estimate defining  $\text{CC}_j$ ; the only scale-improved factor is the one from localisation.  $\square$

### Finite local restoration kernels

**Proposition 10.182** (LA1: small-set transfer stability). *The Map-1 transfer functions satisfy (LA1) uniformly on both source norm pairs.*

*Proof.* Map 1 uses only the finitely many small polymers meeting a fixed block. Every transfer term is  $I^X \text{Loc}_X I^{-X} K(X)$  followed by a finite sum. Stability of  $I^{\pm 1}$  is Proposition 10.179; localisation is a bounded finite-rank projection; and the number of small sets meeting a block depends only on  $L, d$ . The exact cancellation proved in this chapter removes the potentially linear output on a singleton. Hence the transfer map preserves the source radius after multiplication by the fixed Map-1 radius constant.  $\square$

**Proposition 10.183** (LA2: block matching after Map 2). *For the source Map-2 radius  $r^{(2)}$ ,*

$$\|K^{(2)} - K^{(1)}\|_j \leq C_L(r^{(2)})^2 \bar{\epsilon}^2,$$

*and for every small set  $X$  and block  $B \subset X$ ,*

$$\|\text{Loc}_B I^{-B} K^{(2)}(B)\|_{T_0} \leq r^{(2)} \bar{\epsilon}^{1+u}$$

*for the fixed source exponent  $u > 0$ . Hence (LA2) holds.*

*Proof.* The Map-1 local coordinate is chosen so that its relevant/marginal projection is exactly the perturbative correction used to define  $\widehat{V}$ . Therefore the first derivative of the Map-2 difference at  $K = 0$  cancels. Taylor's theorem on the common finite interaction space starts the remainder at quadratic order, giving the first bound. On a small set, the same cancellation leaves only the irrelevant part of the local Taylor jet. The field-scale ratio and the positive gap between the retained and omitted dimensions give a fixed extra power  $\bar{\epsilon}^u$  after the source radius hierarchy is imposed. All constants may depend on fixed  $L$ , as permitted at this stage of the source proof.  $\square$

**Proposition 10.184** (LA3A: leading perturbative restoration). *For every admissible  $(U, B)$ ,*

$$\|\widetilde{I}_{\text{pt}}(U) h_{\text{lead}}(U, B)\|_{j+1} \leq C_L \bar{\epsilon}^2.$$

*Proof.* The leading term is a connected Gaussian covariance of two local interactions. The finite-range covariance estimate and the connected-cumulant formula bound each pair by the product of two interaction sizes. Summing one endpoint over a coarse block and the other over the finite covariance range remains  $O_L(\bar{\epsilon}^2)$ . Stability of the prefactor  $\widetilde{I}_{\text{pt}}$  follows from Lemma 10.177. This gives the stated uniform bound.  $\square$

**Proposition 10.185** (LA3B: reduced restoration after designed cancellation). *For  $|U|_{j+1} \in \{1, 2\}$ ,*

$$\|\widetilde{I}_{\text{pt}}^U[h_{\text{red}}(U) - h_{\text{lead}}(U)]\|_{j+1} \leq C_L \bar{\epsilon}^3.$$

*Together with Proposition 10.167, this is (LA3).*

*Proof.* For a one-block output, expand the difference into the one-block remainder  $R_1$  and the ordered two-block remainders  $R_2$ . In  $R_1$ , the second-order covariance generated by the fluctuation expectation cancels the corresponding term in the recursion defining  $W$ , while  $V_{\text{pt}} = \mathbb{E}\theta V - P$  cancels the remaining local second-order piece. Every uncanceled term therefore contains at least three powers of the interaction size. In  $R_2$ , subtract the complete second-order connected covariance before taking the norm; again the remainder begins at third order. The terms with two second-order corrections are fourth order and hence smaller. Gaussian expectation is controlled by  $\text{IP}_j$  and the output interaction by PK2, so each  $R_1, R_2$  is  $O_L(\bar{\epsilon}^3)$ . There are only  $O_L(1)$  fine blocks in a fixed coarse small set. The two-block case contains only the  $R_2$  sum and is identical. Finally the  $|X|_j \geq 3$  tail is already Proposition 10.167.  $\square$

**Proposition 10.186** (LA4: final interaction and boundary adjustments). *The Map-5 interaction difference is  $O_L(\bar{\epsilon}^2)$  in its leading part and  $O_L(\bar{\epsilon}^4)$  in the quadratic correction. The Map-6 boundary adjustment is  $O_L(\bar{\epsilon})$  per affected block and obeys the source large-set decay. Hence (LA4) holds.*

*Proof.* Map 5 replaces fine-block  $W_j$  data by the coarse quantity  $W_{j+1}(V_+)$ . Both are quadratic in the interaction at leading order, so the difference is  $O_L(\bar{\epsilon}^2)$ ; a product of two such differences is fourth order. The circle-product expansion multiplies these local estimates, and the large-set exponent absorbs the exponential number of placements.

Map 6 changes  $V_{\text{pt}}$  to  $V_+$  and converts the gradient monomial to the Laplacian monomial by lattice summation by parts. The interior contributions cancel exactly; only blocks meeting the boundary of the polymer remain. Each carries one local interaction increment, hence  $O_L(\bar{\epsilon})$ . The number of boundary blocks is bounded by a fixed multiple of the polymer size and is absorbed by the same large-set weight. In the pure bosonic parent there is no  $\sigma\bar{\sigma}$  extraction; with observables switched on, the additional extraction is finite-dimensional and satisfies the same radius estimate.  $\square$

## Common analytic domain

**Proposition 10.187** (Analytic-domain estimate). *For the  $n \geq 1$  bosonic single-step theorem, the quantitative gate  $\text{AD}_j$  holds on the physical pairwise ball and on a common Cauchy ball with radius independent of the scale. The maps are analytic there, continuous in the mass parameter, and their mixed field/interaction derivatives obey the Cauchy estimates required in the source theorem. For  $K = 0$  the output satisfies the two pairwise inhomogeneous target bounds used in Proposition 10.140.*

*Proof.* this chapter embeds the physical  $\mathcal{W}_j$  ball into both pairwise source balls with a fixed margin. On that margin, quartic stability keeps  $I$  and  $I^{-1}$  uniformly bounded and away from their analytic singularities. Maps 1–6 are finite compositions of products, finite sums, localisation, Gaussian expectation and the two analytic inverses; therefore they are analytic on a slightly smaller common ball. The constants in PK2–PK4 and LA1–LA4 are uniform on the larger ball, so Cauchy’s integral formula on the fixed radius difference gives the mixed derivative estimates.

The finite-range covariance and its field derivatives are continuous in the mass on the source mass interval; dominated convergence is supplied by PK3, hence every Gaussian expectation and every map is mass-continuous. If  $K = 0$ , Maps 1–4 vanish by their exact coordinate identities. Maps 5–6 contain only the perturbative interaction mismatch, which LA4 bounds by  $O_L(\bar{\epsilon}^2)$  in the  $T_0(\ell)$  pair and by the corresponding weighted large-field bound in the  $\tilde{G}(h)$  pair. These are precisely the two inhomogeneous targets recombined in this chapter.  $\square$

**Theorem 10.188** (Bosonic nonlinear single-step theorem). *For the  $n \geq 1$  bosonic four-dimensional  $|\varphi|^4$  branch, the source-kernel gate  $\text{SK}_j$  is valid at every admissible scale. Consequently the exact six-map RG step satisfies, simultaneously for the two source norm pairs, the nonlinear contraction and inhomogeneous estimates of the parent theorem, with contraction  $O(L^{-1})$ . After recombination the output lies in the iterable  $\mathcal{W}_{j+1}$  norm. The conclusion is analytic on the source domain and has the required mixed derivatives and mass continuity.*

*Proof.* PK1 is Proposition 10.161; PK2 and the integration property are Proposition 10.179; PK3 is Proposition 10.176; PK4 and the crucial contraction are Proposition 10.181. LA1–LA4 are Propositions 10.182–10.186, and the common analytic-domain gate is Proposition 10.187. Hence  $\text{SK}_j$  holds. Apply Theorem 10.172 to obtain the two pairwise  $O(L^{-1})$  bounds, then Propositions 10.140 and 10.137 to recover the iterable  $\mathcal{W}_{j+1}$  estimate. This closes the bosonic parent theorem without importing an unproved stability theorem.  $\square$

## 10.4.6 Susceptibility from the differentiated RG trajectory

We now use the closed nonlinear RG step to prove the susceptibility theorem rather than only its stopping-scale heuristic. The key point is that the infinite-volume susceptibility is exactly the

reciprocal renormalised mass, up to the wave-function coordinate, while the derivative of that mass coordinate carries the product of the marginal corrections.

**Definition 10.189** (Four-dimensional massive bubble). For  $m^2 > 0$  let

$$C_{m^2} = (-\Delta_{\mathbb{Z}^4} + m^2)^{-1}, \quad \mathbf{B}_{m^2} = \sum_{x \in \mathbb{Z}^4} C_{m^2}(x)^2.$$

**Lemma 10.190** (Massive bubble asymptotic). As  $m^2 \downarrow 0$ ,

$$\mathbf{B}_{m^2} = \frac{1}{16\pi^2} \log m^{-2} + O(1).$$

*Proof.* By Plancherel,

$$\mathbf{B}_{m^2} = \int_{[-\pi, \pi]^4} \frac{dk}{(2\pi)^4} \frac{1}{(\lambda(k) + m^2)^2},$$

where  $\lambda(k) = 2 \sum_{i=1}^4 (1 - \cos k_i) = |k|^2 + O(|k|^4)$  near the origin. Fix a small  $r > 0$ . On  $|k| \geq r$  the integral is uniformly bounded. On  $|k| < r$ , replace  $\lambda(k)$  by  $|k|^2$ ; the difference is integrable uniformly in  $m$  because

$$\left| \frac{1}{(\lambda + m^2)^2} - \frac{1}{(|k|^2 + m^2)^2} \right| \leq \frac{C|k|^4}{(|k|^2 + m^2)^3},$$

and the right-hand side has bounded four-dimensional radial integral. Therefore

$$\mathbf{B}_{m^2} = \frac{|S^3|}{(2\pi)^4} \int_0^r \frac{\rho^3 d\rho}{(\rho^2 + m^2)^2} + O(1).$$

Since  $|S^3| = 2\pi^2$  and, after  $u = \rho^2$ ,

$$\int_0^r \frac{\rho^3 d\rho}{(\rho^2 + m^2)^2} = \frac{1}{2} \log m^{-2} + O(1),$$

the coefficient is  $2\pi^2(2\pi)^{-4} \cdot \frac{1}{2} = 1/(16\pi^2)$ . □

**Lemma 10.191** (Differentiated relevant coordinate). Put

$$\gamma = \frac{n+2}{n+8}.$$

Along the critical/high-temperature trajectory of Theorem 10.188, let  $D_j$  denote the dimensionless derivative of the relevant mass coordinate with respect to its initial relevant coordinate. Then

$$D_j = (1 + O(g_0)) \left( \frac{g_j}{g_0} \right)^\gamma (1 + O(\vartheta_j g_j)),$$

uniformly up to and beyond the mass scale. In particular

$$D_\infty \sim (1 + O(g_0)) (g_0(n+8) \mathbf{B}_{m^2})^{-\gamma} \quad (m^2 \downarrow 0).$$

*Proof.* Differentiate the relevant-coordinate component of the exact RG map. The one-loop  $O(n)$  contraction in Lemma 10.107 gives

$$D_{j+1} = D_j \{1 - a_j g_j + O(g_j^2)\} + r_j D_j, \quad |r_j| \leq C \vartheta_j g_j^2.$$

The nonlinear derivative of the polymer coordinate contributes the displayed summable  $r_j$ : it follows from the common Cauchy-domain estimate in Proposition 10.187 and the  $O(\bar{\epsilon}_j^2)$  size of  $K_j$ . The quartic coordinate satisfies  $g_{j+1} = g_j - \beta_j g_j^2 + O(g_j^3)$ , with  $a_j/\beta_j \rightarrow (n+2)/(n+8) = \gamma$ . Taking logarithms,

$$\log D_j = - \sum_{i < j} a_i g_i + O\left(\sum_{i < j} g_i^2\right),$$

whereas

$$\log(g_j/g_0) = - \sum_{i < j} \beta_i g_i + O\left(\sum_{i < j} g_i^2\right).$$

The difference after multiplying the second expression by  $\gamma$  converges; the stable initial-coordinate normalisation fixes its exponential to  $1 + O(g_0)$ , and the tail is  $O(\vartheta_j g_j)$ . This proves the first formula.

The finite-range decomposition partitions the bubble into scale increments and the same one-loop coefficient obeys  $g_\infty^{-1} = g_0^{-1} + (n+8)\mathbf{B}_{m^2} + O(1)$  in the source normalisation. Hence  $g_\infty \sim ((n+8)\mathbf{B}_{m^2})^{-1}$  as  $m^2 \downarrow 0$ , which inserted into the first formula gives the second.  $\square$

**Lemma 10.192** (Final-scale susceptibility identity). *Let  $(m^2, g_0, \nu_0^c, z_0^c)$  be the critical RG coordinates associated with the physical pair  $(g, \nu_c + \varepsilon)$ . Then*

$$\widehat{\chi}(m^2, g_0, \nu_0^c, z_0^c) = \frac{1}{m^2}, \quad \chi(g, \nu_c + \varepsilon) = \frac{1 + \widetilde{z}_0}{\widetilde{m}^2}.$$

*Proof.* On a torus, complete the square in the Gaussian measure with a constant source. Two source derivatives give

$$\widehat{\chi}_N = \frac{1}{m^2} + \frac{1}{m^4 |\Lambda_N|} \frac{D^2 Z_N(0; \mathbf{1}, \mathbf{1})}{Z_N(0)}.$$

At the critical initial coordinates, the final relevant coordinate obeys  $\nu_N = O(L^{-2N} \vartheta_N g_N)$ , while the  $W_N$  and  $K_N$  contributions are respectively smaller by one and two additional powers of  $g_N$ . These estimates are consequences of the closed single-step theorem and its inhomogeneous target bound. Therefore the second term tends to zero as  $N \rightarrow \infty$ , proving  $\widehat{\chi} = 1/m^2$ .

The physical-to-RG change of variables rescales the field by  $(1+z_0)^{1/2}$ , so  $\chi = (1+z_0)\widehat{\chi}$ . Choosing the critical-coordinate functions  $(\widetilde{m}^2, \widetilde{g}_0, \widetilde{z}_0, \widetilde{\nu}_0)$  gives the second identity. The implicit-function construction is legitimate on a fixed neighbourhood because the critical trajectory is  $C^1$  in its parameters by this chapter structural-stability theorem and the common analytic domain of this chapter.  $\square$

**Lemma 10.193** (Derivative of the susceptibility). *As  $\varepsilon \downarrow 0$ ,*

$$\frac{d}{d\varepsilon} \chi(g, \nu_c + \varepsilon) \sim - \frac{1 + O(g)}{(g\mathbf{b})^\gamma} \chi(g, \nu_c + \varepsilon)^2 (\log \chi(g, \nu_c + \varepsilon))^{-\gamma}, \quad \mathbf{b} = \frac{n+8}{16\pi^2}.$$

*Proof.* Differentiate the finite-volume formula in Lemma 10.192 with respect to the initial relevant coordinate. The derivative of the final relevant coupling is the leading term. By Lemma 10.191, its infinite-volume limit is  $(1 + O(g_0))(g_0(n+8)\mathbf{B}_{m^2})^{-\gamma}$ . The  $W_N$  derivative is smaller by  $g_N$ ; the  $K_N$  derivative is smaller by  $g_N^2$ . These follow by differentiating the contractive polymer recursion: the differentiated inhomogeneity is  $O(\vartheta_j D_j g_j^2)$ , and summing the resulting geometric convolution preserves that order. A second derivative of the vacuum denominator is similarly negligible because the undifferentiated numerator already tends to zero.

The chain rule for the physical-to-RG coordinate change contributes  $(1 + \tilde{z}_0)^2 = 1 + O(g)$ . Thus

$$\chi'(\varepsilon) \sim -(1 + O(g))\chi^2(g_0(n+8)\mathbf{B}_{\tilde{m}^2})^{-\gamma}.$$

By Lemma 10.192,  $\tilde{m}^{-2} = \chi/(1 + O(g))$ , and by Lemma 10.190,

$$g_0(n+8)\mathbf{B}_{\tilde{m}^2} \sim \frac{(n+8)g}{16\pi^2} \log \chi = \mathbf{b}g \log \chi.$$

This is exactly the one-loop  $O(n)$  marginal-flow normalisation, and proves the formula.  $\square$

**Lemma 10.194** (Elementary integration of the logarithmic differential law). *Let  $\gamma \in (0, 1)$  and let  $X(\varepsilon) \uparrow \infty$  as  $\varepsilon \downarrow 0$ . If*

$$X'(\varepsilon) \sim -CX(\varepsilon)^2(\log X(\varepsilon))^{-\gamma}, \quad C > 0,$$

*then*

$$X(\varepsilon) \sim C^{-1}\varepsilon^{-1}(\log \varepsilon^{-1})^\gamma.$$

*Proof.* Put  $y = X^{-1}$ . Then  $y'(\varepsilon) \sim C(\log y^{-1})^{-\gamma}$  and  $y(0+) = 0$ . For every fixed  $\eta > 0$ , once  $\varepsilon$  is small,

$$(1 - \eta)C(\log y^{-1})^{-\gamma} \leq y' \leq (1 + \eta)C(\log y^{-1})^{-\gamma}.$$

The function on the right is slowly varying at zero. Integrating and using monotonicity on  $[\varepsilon/2, \varepsilon]$  shows  $y(\varepsilon) \asymp \varepsilon(\log \varepsilon^{-1})^{-\gamma}$ ; substituting this coarse bound back into the logarithm improves the ratio to  $y(\varepsilon)/(C\varepsilon(\log \varepsilon^{-1})^{-\gamma}) \rightarrow 1$ . Inverting gives the assertion.  $\square$

**Proposition 10.195** (Critical-point first-order coefficient). *For fixed  $n \geq 1$  and small  $g$ ,*

$$\nu_c(g, n) = -(n+2)(-\Delta_{\mathbb{Z}^4}^{-1})_{0,0}g + O(g^2).$$

*Proof.* The critical relevant coordinate is chosen so that the renormalised mass tends to zero. At first order in  $g$ , the only local quadratic contraction of the quartic vertex is the one-loop tadpole. Lemma 10.107 gives its  $O(n)$  factor  $n+2$ , and summing the finite-range covariance increments gives  $\sum_{j \geq 1} C_j(0) = (-\Delta_{0,0}^{-1})$ . All terms containing either two quartic vertices or one nonperturbative remainder are  $O(g^2)$  by the closed single-step bounds. Cancelling the first-order mass shift therefore requires the displayed value of  $\nu_c$ .  $\square$

**Theorem 10.196** (Bosonic susceptibility engine). *For  $n \geq 1$  and sufficiently small  $g > 0$ , with  $\gamma = (n+2)/(n+8)$ ,*

$$\chi(g, \nu_c + \varepsilon) \sim (\mathbf{b}g)^\gamma (1 + O(g))\varepsilon^{-1}(\log \varepsilon^{-1})^\gamma, \quad \mathbf{b} = \frac{n+8}{16\pi^2},$$

*and Proposition 10.195 gives the corresponding critical-point expansion.*

*Proof.* Apply Lemma 10.194 to Lemma 10.193. The constant is  $C = (1 + O(g))(\mathbf{b}g)^{-\gamma}$ , hence  $C^{-1} = (\mathbf{b}g)^\gamma(1 + O(g))$ . The critical-point expansion is Proposition 10.195.  $\square$

### 10.4.7 Pressure derivatives and the specific-heat trichotomy

The susceptibility proof uses one derivative of the relevant coordinate. The specific heat requires the corresponding second-derivative control and the vacuum-energy coordinate. We now record these estimates in the transformed RG coordinates; the normalisation is chosen so that the relevant coordinate is  $\check{\mu}_j = L^{2j}\check{\nu}_j$ .

**Lemma 10.197** (First- and second-derivative hierarchy). *Let  $\gamma = (n+2)/(n+8)$ , and differentiate the critical RG trajectory with respect to its initial relevant coordinate. There is a continuous  $c = c(m^2, g_0) = 1 + O(g_0)$  such that*

$$\check{\mu}'_j = L^{2j} \left( \frac{\check{g}_j}{g_0} \right)^\gamma \{c + O(\vartheta_j \check{g}_j)\}.$$

Moreover

$$\check{g}'_j, \check{z}'_j = O(\vartheta_j \check{\mu}'_j \check{g}_j^2), \quad \|K'_j\|_{\mathcal{W}_j} = O(\vartheta_j \check{\mu}'_j \check{g}_j^2),$$

and

$$\check{\mu}''_j, \check{g}''_j, \check{z}''_j = O(\vartheta_j (\check{\mu}'_j)^2 \check{g}_j), \quad \|K''_j\|_{\mathcal{W}_j} = O(\vartheta_j (\check{\mu}'_j)^2 \check{g}_j).$$

All constants are uniform in scale and in the admissible massive domain.

*Proof.* The first-derivative formula is Lemma 10.191, after restoring the canonical factor  $L^{2j}$  in the relevant coordinate. Differentiating the other transformed perturbative coordinates gives an inhomogeneous triangular system. Every entry feeding  $g$  or  $z$  contains either one irrelevant derivative or one factor of  $\check{g}_j^2$ ; the common analytic domain of Proposition 10.187 and the closed single-step estimate therefore give

$$|\check{g}'_j| + |\check{z}'_j| + \|K'_j\|_{\mathcal{W}_j} \leq C \vartheta_j \check{\mu}'_j \check{g}_j^2.$$

For the second derivatives we give the induction because it is the new point. Assume at scale  $j$  that

$$|\check{\mu}''_j| + |\check{g}''_j| + |\check{z}''_j| \leq M_1 \vartheta_j (\check{\mu}'_j)^2 \check{g}_j, \quad \|K''_j\|_{\mathcal{W}_j} \leq M_2 \vartheta_j (\check{\mu}'_j)^2 \check{g}_j.$$

The base case is zero. Twice differentiating the analytic RG map produces three types of terms: a linear image of the second derivative, a bilinear Hessian applied to two first derivatives, and the differentiated nonperturbative remainder. The preceding analysis Cauchy bounds imply, uniformly on the common domain,

$$\|D^2 R_j[(\dot{V}, \dot{K}), (\dot{V}, \dot{K})]\| \leq C \vartheta_j \check{g}_j^{-1} \|(\dot{V}, \dot{K})\|^2,$$

with the standard component weights. Inserting the first-derivative bounds gives  $C \vartheta_j (\check{\mu}'_j)^2 \check{g}_j$ .

The scale ratios are

$$\frac{\check{g}_j}{\check{g}_{j+1}} = 1 + O(\check{g}_j), \quad \frac{\check{\mu}'_j}{\check{\mu}'_{j+1}} = L^{-2}(1 + O(\check{g}_j)), \quad \frac{\vartheta_j}{\vartheta_{j+1}} \leq L.$$

Consequently the transported induction term acquires at least the factor  $L^{-3}$  after it is measured in the scale- $(j+1)$  target weight. Choose first  $M_1 \gg M_2$  so the finite-dimensional triangular couplings are absorbed, then  $M_2$  larger than the fixed inhomogeneous constant. The induction closes and proves the second-derivative bounds. The  $K$  estimate is identical, with the strict nonperturbative contraction from Theorem 10.188 supplying the linear gain.  $\square$

**Lemma 10.198** (Vacuum derivative identity). *Let  $u_j = \sum_{i < j} \delta u_{i+1}$  be the vacuum-energy coordinate of the RG flow. Then there exist limits  $u'_\infty$  and  $u''_\infty$ , locally uniformly for  $m^2 > 0$ , and*

$$u'_\infty = \frac{n}{2} \sum_{j \geq 1} \check{\nu}'_j C_{j;0,0} + O(g_0),$$

$$u''_\infty = -\frac{n}{2(n+8)} \sum_{j \geq 0} \beta_j (\check{\nu}'_j)^2 + O(1).$$

Here  $\check{\nu}'_j = L^{-2j} \check{\mu}'_j$  and  $\beta_j$  is the quartic one-loop coefficient in the transformed flow.

*Proof.* Write

$$\delta u_{j+1} = \bar{\phi}_j^{\delta u}(\check{V}_j) + \check{R}_{j+1}^{\delta u}(\check{V}_j, K_j).$$

The vacuum component is analytic on the common analytic domain. The perturbative part is an explicit polynomial of degree at most two in the local couplings, whereas the remainder satisfies the weighted derivative estimates

$$\|D_V^p D_K^q \check{R}_{j+1}^{\delta u}\| \leq CL^{-4j} \begin{cases} \vartheta_j \check{g}_j^{3-p}, & q = 0, \\ \check{g}_j^{1-p-q}, & q = 1, 2. \end{cases}$$

These are precisely the vacuum-component specialisation of the analytic single-step bounds proved in this chapter.

Differentiate once. The only non-summably leading perturbative contribution is the quadratic mass term, which gives  $\frac{n}{2} \check{\nu}'_j C_{j;0,0}$ . Every  $g'$ ,  $z'$  and  $K'$  contribution contains an extra factor  $\vartheta_j \check{g}_j$  and is summable by Lemma 10.197; hence

$$u'_N = \frac{n}{2} \sum_{j=1}^N \check{\nu}'_j C_{j;0,0} + O(g_0) \longrightarrow u'_\infty.$$

Differentiate a second time. The explicit perturbative Hessian in the two relevant slots is

$$-\frac{n}{2(n+8)} \beta_j,$$

which is the  $O(n)$  contraction coefficient already owned by the perturbative combinatorics of this chapter. Thus

$$u''_N = -\frac{n}{2(n+8)} \sum_{j=0}^{N-1} \beta_j (\check{\nu}'_j)^2 + \sum_{j=0}^{N-1} O(\vartheta_j \check{g}_j (\check{\nu}'_j)^2).$$

By the first-derivative formula,  $|\check{\nu}'_j| \leq C(\check{g}_j/g_0)^\gamma$ , and therefore the error is

$$O(g_0^{-2\gamma}) \sum_j \vartheta_j \check{g}_j^{1+2\gamma} = O(1).$$

after the standard marginal-flow summability estimate. Above the mass scale  $\vartheta_j$  decays geometrically, so the tails are uniformly summable on every compact  $m^2$ -interval. This proves existence and the displayed formula for  $u''_\infty$ .  $\square$

**Lemma 10.199** (Pressure differentiation and disappearance of the polymer endpoint). *For  $\nu > \nu_c$  the thermodynamic pressure is twice differentiable and*

$$\partial_\nu p = -(1+z_0)u'_\infty, \quad \partial_\nu^2 p = -(1+z_0)^2 u''_\infty.$$



*Proof.* At finite volume the exact RG representation gives

$$p_N = p_N(0, m^2) + \frac{n}{2} \log(1 + z_0) - u_N + |\Lambda_N|^{-1} \log(1 + K_N(\Lambda_N; 0)).$$

Differentiating at the critical RG coordinates yields

$$\partial_\nu p_N = -(1 + z_0)u'_N + |\Lambda_N|^{-1} O(K'_N(\Lambda_N; 0)),$$

$$\partial_\nu^2 p_N = -(1 + z_0)^2 u''_N + |\Lambda_N|^{-1} O(|K'_N|^2 + |K''_N|).$$

Lemma 10.197 and the  $T_0$  evaluation bound imply at scale  $N$

$$|K'_N| \leq C \vartheta_N \check{\mu}'_N \check{g}_N^2, \quad |K''_N| \leq C \vartheta_N (\check{\mu}'_N)^2 \check{g}_N.$$

Since  $|\Lambda_N| = L^{4N}$  and  $\check{\mu}'_N \asymp L^{2N} (\check{g}_N/g_0)^\gamma$ , both endpoint terms tend to zero locally uniformly for  $m^2 > 0$ . Lemma 10.198 gives local uniform convergence of  $u'_N$  and  $u''_N$ . The elementary theorem on compact convergence of a function and its derivatives therefore permits passage of the first two derivatives through the thermodynamic limit and proves the identities.  $\square$

**Lemma 10.200** (Marginal-flow summation). *Along the critical trajectory,*

$$S(m^2) = \sum_{j \geq 0} \beta_j \check{g}_j^{2\gamma}$$

has, as  $m^2 \downarrow 0$ , the asymptotics

$$S(m^2) = \begin{cases} (1 - 2\gamma)^{-1} \check{g}_\infty^{2\gamma-1} + O(1), & 2\gamma < 1, \\ \log(g_0/\check{g}_\infty) + O(1), & 2\gamma = 1, \\ (2\gamma - 1)^{-1} g_0^{2\gamma-1} (1 + O(g_0)), & 2\gamma > 1. \end{cases}$$

Moreover

$$\check{g}_\infty \sim \frac{1}{(n+8)\mathbf{B}_{m^2}} \sim \frac{16\pi^2}{(n+8) \log m^{-2}}.$$

*Proof.* Use  $\check{g}_{j+1} = \check{g}_j - \beta_j \check{g}_j^2 + r_j$  with  $|r_j| \leq C \vartheta_j \check{g}_j^3$ . For  $F(t) = t^{2\gamma-2}$ , Taylor's theorem gives

$$F(\check{g}_j) - F(\check{g}_{j+1}) = \beta_j \check{g}_j^{2\gamma} + O(\vartheta_j \check{g}_j^{1+2\gamma}).$$

The error is summable. Telescoping from 0 to  $N$  and then sending  $N \rightarrow \infty$  gives the three cases, with  $F(t) = t^{2\gamma-1}/(2\gamma-1)$  when  $2\gamma \neq 1$  and  $F(t) = \log t$  when  $2\gamma = 1$ .

The second assertion is the quartic-flow/bubble identity already used in Lemma 10.191: summing the finite-range one-loop increments gives  $\check{g}_\infty^{-1} = g_0^{-1} + (n+8)\mathbf{B}_{m^2} + O(1)$ . Lemma 10.190 completes the asymptotic.  $\square$

**Theorem 10.201** (Specific-heat engine). *For  $n \geq 1$  and sufficiently small  $g > 0$ , the infinite-volume specific heat exists for  $\nu > \nu_c$  and, as  $\varepsilon = \nu - \nu_c \downarrow 0$ ,*

$$c_H(g, \nu_c + \varepsilon) \sim D(g, n) \begin{cases} (\log \varepsilon^{-1})^{(4-n)/(n+8)}, & n = 1, 2, 3, \\ \log \log \varepsilon^{-1}, & n = 4, \\ 1, & n > 4, \end{cases}$$

for a positive amplitude  $D(g, n)$ . Moreover, as  $g \downarrow 0$ ,

$$D(g, n) = (1 + O(g)) \begin{cases} \frac{n}{2(4-n)} \left( \frac{n+8}{16\pi^2} \right)^{(4-n)/(n+8)} g^{-(2n+4)/(n+8)}, & n = 1, 2, 3, \\ \frac{1}{6} g^{-1}, & n = 4, \\ \frac{n}{2(n-4)} g^{-1}, & n > 4. \end{cases}$$

In particular the three regimes are determined by  $2\gamma - 1 = (n - 4)/(n + 8)$ .

*Proof.* Lemmas 10.199 and 10.198, together with  $\check{\nu}'_j = L^{-2j} \check{\mu}'_j$ , give

$$c_H = (1 + z_0)^2 c^2 g_0^{-2\gamma} \frac{n}{2(n+8)} \sum_{j \geq 0} \beta_j \check{g}_j^{2\gamma} + O(1),$$

where  $c = 1 + O(g_0)$ . Apply Lemma 10.200. If  $n < 4$  then  $2\gamma < 1$  and the divergent term is  $\check{g}_\infty^{-(4-n)/(n+8)}$ , hence a positive multiple of  $(\log m^{-2})^{(4-n)/(n+8)}$ . If  $n = 4$ , the divergence is  $\log(g_0/\check{g}_\infty) = \log \log m^{-2} + O(1)$ . If  $n > 4$ , the sum converges to a strictly positive finite limit.

Finally Theorem 10.196 and Lemma 10.192 imply  $m^2 \asymp \varepsilon (\log \varepsilon^{-1})^{-\gamma}$ , and therefore  $\log m^{-2} \sim \log \varepsilon^{-1}$  and  $\log \log m^{-2} \sim \log \log \varepsilon^{-1}$ . This transfers the three asymptotics to  $\varepsilon$ .

It remains to retain the source-level leading amplitudes as  $g \downarrow 0$ . The coordinate change used throughout the RG construction gives  $z_0 = O(g)$ ,  $c = 1 + O(g)$  and  $g_0 = g(1 + O(g))$ . For  $n < 4$ ,  $1 - 2\gamma = (4 - n)/(n + 8)$ , and the leading term in Lemma 10.200 yields

$$\frac{n}{2(n+8)} \frac{1}{1-2\gamma} g_0^{-2\gamma} \left( \frac{n+8}{16\pi^2} \right)^{1-2\gamma} = \frac{n}{2(4-n)} \left( \frac{n+8}{16\pi^2} \right)^{(4-n)/(n+8)} g^{-(2n+4)/(n+8)} (1 + O(g)).$$

For  $n = 4$ ,  $\gamma = 1/2$  and the coefficient is  $4/[2(12)] = 1/6$ , giving  $(1 + O(g)) \frac{1}{6} g^{-1}$ . For  $n > 4$ , the convergent marginal sum has leading value  $(2\gamma - 1)^{-1} g_0^{2\gamma-1}$ , and hence its coefficient is

$$\frac{n}{2(n+8)} \frac{1}{2\gamma-1} g_0^{-1} = \frac{n}{2(n-4)} g^{-1} (1 + O(g)).$$

These are exactly the three displayed small-coupling formulas for  $D(g, n)$ .  $\square$

#### 10.4.8 discrete-torus scaling limits

We finally use the closed RG trajectory to identify the full finite-volume Laplace functional. This is stronger than proving vanishing of higher formal cumulants: the nonperturbative endpoint is estimated at the shifted field itself.

**Lemma 10.202** (Final-scale source-field norm). *Let  $h \in C^{p_\Phi+4}(\mathbb{T}^4, \mathbb{R}^n)$ ,  $h_N(x) = h(L^{-N}x)$ , and  $C^{(N)} = (-\Delta_{\Lambda_N} + m_N^2)^{-1}$ . Then*

$$\|C^{(N)} h_N\|_{\Phi_N} \leq CL^N m_N^{-2} \|h\|_{C^{p_\Phi+4}}.$$

*If  $\sigma_N^2 = L^{4N} \chi^{(N)}$  and  $\chi^{(N)} = (1 + z_{0,N}) m_N^{-2}$ , then*

$$\|\sigma_N^{-1} C^{(N)} h_N\|_{\Phi_N} \leq C \sqrt{\chi^{(N)} L^{-2N}} \|h\|_{C^{p_\Phi+4}}.$$

*Proof.* For a scalar lattice function  $f$ , the scale- $N$  lattice Sobolev inequality gives

$$\|\nabla^\alpha C^{(N)} f\|_\infty \leq CL^{-4N} \max_{|\beta|_\infty \leq 1} \|L^{N|\beta|_1} \nabla^{\alpha+\beta} C^{(N)} f\|_2.$$

The covariance commutes with differences and  $\|C^{(N)} q\|_2 \leq m_N^{-2} \|q\|_2$ . Since  $\|q\|_2 \leq L^{2N} \|q\|_\infty$ , multiplication by the field-scale weight  $L^{N|\alpha|_1}$  gives

$$L^{N|\alpha|_1} \|\nabla^\alpha C^{(N)} f\|_\infty \leq C m_N^{-2} \max_{|\beta|_\infty \leq 1} L^{N(|\alpha|_1 + |\beta|_1)} \|\nabla^{\alpha+\beta} f\|_\infty.$$

For  $f = h_N$ , discrete differences are bounded by the corresponding derivatives of  $h$ . The definition of  $\Phi_N$  contains the additional factor  $\ell_N^{-1} \asymp L^N$ , which proves the first estimate. For the second,

$$\sigma_N^{-1} L^N m_N^{-2} = \frac{L^N m_N^{-2}}{L^{2N} \sqrt{\chi^{(N)}}} \asymp \sqrt{\chi^{(N)} L^{-2N}},$$

because  $\chi^{(N)} m_N^2 = 1 + z_{0,N} \asymp 1$ . □

**Lemma 10.203** (Susceptibility-to-mass conversion). *Let  $\rho = 1 + \tilde{z}_0(0)$ . If*

$$\chi^{(N)} L^{-2N} \longrightarrow \rho m^{-2}, \quad m^2 \in (0, \infty],$$

*with  $\infty^{-2} = 0$ , then*

$$m_N^{-2} L^{-2N} \longrightarrow m^{-2}.$$

*Proof.* The exact susceptibility identity gives  $\chi^{(N)} = (1 + z_{0,N}) m_N^{-2}$ . Hence

$$m_N^{-2} L^{-2N} = \chi^{(N)} L^{-2N} (1 + z_{0,N})^{-1}.$$

If the limit on the right is zero, the result is immediate. If it is positive,  $m_N^2 \rightarrow 0$  and continuity of the critical coordinate gives  $1 + z_{0,N} \rightarrow \rho$ , proving the claim. □

**Lemma 10.204** (Continuum covariance limit). *Under the hypotheses of Lemma 10.203, define*

$$\overline{C} = \begin{cases} (-m^{-2} \Delta_{\mathbb{T}^4} + 1)^{-1}, & m^2 < \infty, \\ 1, & m^2 = \infty. \end{cases}$$

*Then for every smooth  $h$ ,*

$$\frac{1 + z_{0,N}}{\sigma_N^2} (h_N, C^{(N)} h_N)_{\Lambda_N} \longrightarrow (h, \overline{C} h)_{\mathbb{T}^4}.$$

*Proof.* Diagonalise the lattice covariance. With  $\lambda(k) = 4 \sum_{r=1}^4 \sin^2(\pi k_r)$  and the discrete Fourier coefficients of  $h_N$ , the left-hand side is

$$\frac{m_N^2}{L^{4N}} \sum_{l \in \Lambda_N} \frac{|\widehat{h}_N(l)|^2}{\lambda(L^{-N}l) + m_N^2}.$$

For every fixed Fourier mode,  $L^{2N} \lambda(L^{-N}l) \rightarrow 4\pi^2 |l|^2$  and Lemma 10.203 gives  $m_N^{-2} L^{-2N} \rightarrow m^{-2}$ . Smoothness of  $h$  gives faster-than-polynomial decay of its Fourier coefficients, so the tails are uniformly summable. Dominated convergence therefore gives

$$\sum_{l \in \mathbb{Z}^4} \frac{|\widehat{h}(l)|^2}{4\pi^2 m^{-2} |l|^2 + 1},$$

which is exactly  $(h, \overline{C} h)$ ; for  $m^2 = \infty$  the denominator tends to 1. □

**Lemma 10.205** (Vanishing interacting endpoint at a macroscopic source). *Set*

$$\psi_N = \sigma_N^{-1} C^{(N)} h_N.$$

*If  $\chi^{(N)} L^{-2N}$  has a finite limit, then*

$$I_N(\Lambda_N; \psi_N) \longrightarrow 1, \quad K_N(\Lambda_N; \psi_N) \longrightarrow 0.$$

*Proof.* Lemma 10.202 places  $\psi_N$  in a fixed multiple of the final-scale analytic domain; if the susceptibility ratio tends to zero its norm even tends to zero. Along the RG trajectory, the local polynomial coupling satisfies  $\|V_N\| \rightarrow 0$ . Therefore the local factor obeys  $I_N(\Lambda_N; \psi_N) = e^{-V_N(\Lambda_N; \psi_N)} \rightarrow 1$ .

For the polymer coordinate, the closed nonlinear theorem gives  $\|K_N\|_{\mathcal{W}_N} \leq C \vartheta_N \check{g}_N^3$  on the same shifted-field domain. Evaluation at a single full-volume polymer is controlled by the  $T_0$  part of the  $\mathcal{W}_N$  norm, hence  $|K_N(\Lambda_N; \psi_N)| \leq C \vartheta_N \check{g}_N^3 \rightarrow 0$ . This is the required nonperturbative endpoint estimate.  $\square$

**Theorem 10.206** (Laplace-functional scaling theorem). *Let  $\sigma_N^2 = L^{4N} \chi^{(N)}$  and suppose*

$$\chi^{(N)} L^{-2N} \longrightarrow \rho m^{-2}$$

*with  $m^2 \in (0, \infty]$ . Then for every smooth  $h : \mathbb{T}^4 \rightarrow \mathbb{R}^n$ ,*

$$\lim_{N \rightarrow \infty} \left\langle \exp\{\varphi^{(N)}(h)/\sigma_N\} \right\rangle = \exp \left\{ \frac{1}{2} (h, \overline{C} h)_{\mathbb{T}^4} \right\},$$

*where  $\overline{C} = (-m^{-2} \Delta_{\mathbb{T}^4} + 1)^{-1}$  for finite  $m^2$  and  $\overline{C} = 1$  for  $m^2 = \infty$ .*

*Proof.* Translate the finite-volume Gaussian by the source. The exact RG representation gives

$$\log \left\langle e^{\varphi^{(N)}(h)/\sigma_N} \right\rangle = \frac{1}{2} \frac{1 + z_{0,N}}{\sigma_N^2} (h_N, C^{(N)} h_N) + \log \frac{Z_N((1 + z_{0,N})^{1/2} \psi_N)}{Z_N(0)}.$$

The first term converges by Lemma 10.204. The second term tends to zero by Lemma 10.205: both numerator and denominator have local factor tending to 1 and polymer factor tending to 0 on the same common analytic domain. This proves the Laplace-transform convergence.  $\square$

**Corollary 10.207** (White-noise and near-critical massive-GFF regimes). *For fixed  $\varepsilon > 0$ , if  $\varepsilon_N \rightarrow \varepsilon$ , then  $\chi^{(N)} \rightarrow \chi(g, \nu_c + \varepsilon)$  and  $\chi^{(N)} L^{-2N} \rightarrow 0$ ; hence the normalised field converges to unit white noise. Equivalently, the standard central-limit field  $L^{-2N} \varphi^{(N)}$  converges to white noise of intensity  $\chi(g, \nu_c + \varepsilon)$ .*

*For every fixed  $m^2 > 0$  there is  $\alpha = \alpha(g, n) > 0$  such that, if*

$$\varepsilon_N \sim \alpha m^2 L^{-2N} (\log L^N)^\gamma,$$

*then*

$$\sigma_N \sim \rho^{1/2} m^{-1} L^{3N},$$

*and the normalised field converges to the massive Gaussian free field with covariance  $(-m^{-2} \Delta_{\mathbb{T}^4} + 1)^{-1}$ .*

*Proof.* The first assertion follows from continuity of the susceptibility above criticality and Theorem 10.206 with  $m^2 = \infty$ .

For the second, the susceptibility theorem and the identity  $\chi = (1 + \tilde{z}_0)/\tilde{m}^2$  give

$$\tilde{m}^2(\varepsilon) \sim c_g \varepsilon (\log \varepsilon^{-1})^{-\gamma}$$

for a positive  $c_g$ . Substituting the displayed  $\varepsilon_N$  gives

$$m_N^2 L^{2N} \longrightarrow c_g \alpha 2^{-\gamma} m^2.$$

Choose  $\alpha = 2^\gamma c_g^{-1}$ . Lemma 10.203 and Theorem 10.206 then give the massive-GFF limit.  $\square$

**Theorem 10.208** (Four-dimensional  $n$ -component  $|\varphi|^4$  susceptibility). *Let  $n \geq 1$  and let  $g > 0$  be sufficiently small. For the  $n$ -component  $|\varphi|^4$  model on  $\mathbb{Z}^4$  there are a critical parameter  $\nu_c = \nu_c(g, n) < 0$  and an amplitude  $A = A(g, n) > 0$  such that, as  $\varepsilon \downarrow 0$  from the high-temperature side,*

$$\chi(g, \nu_c + \varepsilon) \sim A \varepsilon^{-1} (\log \varepsilon^{-1})^{(n+2)/(n+8)}.$$

*In particular the mean-field susceptibility exponent is accompanied by the logarithmic correction exponent  $(n+2)/(n+8)$ . In the small-coupling normalization of the source, if*

$$b = \frac{n+8}{16\pi^2}, \quad a = (n+2)(-\Delta_{\mathbb{Z}^4}^{-1})_{0,0},$$

*then, as  $g \downarrow 0$ ,*

$$A(g, n) = (bg)^{(n+2)/(n+8)}(1 + O(g)), \quad \nu_c(g, n) = -ag + O(g^2).$$

*Proof.* This is Theorem 10.196, with  $\gamma = (n+2)/(n+8)$  and  $\mathbf{b} = (n+8)/(16\pi^2)$ .  $\square$

**Theorem 10.209** (Four-dimensional  $|\varphi|^4$  specific heat — source-exact  $n$ -dependence). *Under the hypotheses of the preceding theorem, the infinite-volume specific heat exists. There is  $D(g, n) > 0$  such that, as  $\varepsilon \downarrow 0$ ,*

$$c_H(g, \nu_c + \varepsilon) \sim D(g, n) \begin{cases} (\log \varepsilon^{-1})^{(4-n)/(n+8)}, & n = 1, 2, 3, \\ \log \log \varepsilon^{-1}, & n = 4, \\ 1, & n > 4. \end{cases}$$

*The source-strength small-coupling asymptotics of the amplitude are*

$$D(g, n) = (1 + O(g)) \begin{cases} \frac{n}{2(4-n)} \left( \frac{n+8}{16\pi^2} \right)^{(4-n)/(n+8)} g^{-(2n+4)/(n+8)}, & n = 1, 2, 3, \\ \frac{1}{6} g^{-1}, & n = 4, \\ \frac{n}{2(n-4)} g^{-1}, & n > 4. \end{cases}$$

*Thus the source theorem gives a fractional logarithmic divergence for  $n = 1, 2, 3$ , a double logarithm for  $n = 4$ , and convergence to a finite positive limit for  $n > 4$ ; these cases are not to be compressed into a single formal power of  $\log \varepsilon^{-1}$ .*

*Proof.* This is Theorem 10.201.  $\square$

**Theorem 10.210** (Discrete-torus field scaling limits for weakly coupled four-dimensional  $|\varphi|^4$ ). *Let  $n \geq 1$  and  $g > 0$  be sufficiently small, let  $\varepsilon_N \in (0, \delta)$ , and set*

$$\chi^{(N)} = \chi(g, \nu_c + \varepsilon_N), \quad \sigma_N^2 = L^{4N} \chi^{(N)}.$$

*There is  $\rho = \rho(g, n) > 0$  such that the susceptibility-normalized field obeys the following source-strength dichotomy for every smooth  $h : \mathbb{T}^4 \rightarrow \mathbb{R}^n$ :*

$$\left\langle \exp\{\varphi^{(N)}(h)/\sigma_N\} \right\rangle \longrightarrow \begin{cases} \dot{W}_1(e^{\varphi(h)}), & \chi^{(N)} L^{-2N} \longrightarrow 0, \\ \text{GFF}_{m^2}(e^{\varphi(h)}), & \chi^{(N)} L^{-2N} \longrightarrow \rho m^{-2} > 0, \end{cases}$$

where  $\dot{W}_1$  is unit-intensity  $n$ -component white noise and  $\text{GFF}_{m^2}$  has covariance  $(-m^{-2}\Delta_{\mathbb{T}^4} + 1)^{-1}$ .

*In particular, if  $\varepsilon_N \rightarrow \varepsilon > 0$ , then  $\sigma_N \sim L^{2N} \sqrt{\chi(g, \nu_c + \varepsilon)}$  and the  $\sigma_N^{-1}$ -normalized field converges to unit white noise; equivalently the standard central-limit field  $L^{-2N} \varphi^{(N)}$  converges to white noise of intensity  $\chi(g, \nu_c + \varepsilon)$ . If instead*

$$\varepsilon_N \sim \alpha m^2 L^{-2N} (\log L^N)^{(n+2)/(n+8)}, \quad m^2 > 0,$$

*with the source amplitude  $\alpha = \alpha(g, n) > 0$ , then  $\sigma_N \sim \rho^{1/2} m^{-1} L^{3N}$  and the  $\sigma_N^{-1}$ -normalized field converges to  $\text{GFF}_{m^2}$  itself, with no additional deterministic multiplier.*

*The theorem is a finite-volume/discrete-torus result. The white-noise and massive-GFF limits arise from different tunings and different effective correlation scales; the two normalizations above are therefore kept explicit.*

*Proof.* This is Theorem 10.206 together with Corollary 10.207. The first equivalence simply multiplies the unit-white-noise normalization by  $\sigma_N/L^{2N} \rightarrow \sqrt{\chi(g, \nu_c + \varepsilon)}$ ; the near-critical normalization is already the one used in the internal Laplace-functional theorem.  $\square$

#### 10.4.9 Supersymmetric weakly self-avoiding walk and the $n = 0$ RG branch

The bosonic theorem above cannot be specialised to  $n = 0$  by replacing an integer number of components by zero. The weakly self-avoiding walk instead has an exact finite-dimensional representation in a differential-form algebra. We first prove that representation, then verify that the nonlinear norm estimates used in Theorem 10.188 survive in the even boson–fermion algebra. Only after these two steps do we use the  $n = 0$  perturbative coefficients. This is the logical distinction between a rigorous supersymmetric  $n = 0$  model and a formal replica limit.

##### The finite-volume $\tau$ -isomorphism

Let  $\Lambda$  be a finite discrete torus. Write  $\phi_x = u_x + iv_x$  and put

$$\psi_x = (2\pi i)^{-1/2} d\phi_x, \quad \bar{\psi}_x = (2\pi i)^{-1/2} d\bar{\phi}_x.$$

The variables  $\psi, \bar{\psi}$  anticommute. We work throughout in the even subalgebra of differential forms, which is an ordinary commutative algebra. Define

$$\tau_x = \phi_x \bar{\phi}_x + \psi_x \wedge \bar{\psi}_x, \quad \tau_{\Delta, x} = \frac{1}{2} (\phi_x (-\Delta \bar{\phi})_x + (-\Delta \phi)_x \bar{\phi}_x + \psi_x \wedge (-\Delta \bar{\psi})_x + (-\Delta \psi)_x \wedge \bar{\psi}_x). \quad (21.W.1)$$

For a positive-definite matrix  $A = C^{-1}$  set

$$S_A = \sum_{x, y \in \Lambda} (\phi_x A_{xy} \bar{\phi}_y + \psi_x A_{xy} \bar{\psi}_y), \quad \mathbb{E}_C F = \int e^{-S_A} F.$$

The top-degree convention for integration is the usual one for differential forms.

**Lemma 10.211** (Gaussian super-resolvent identity). *Let  $A$  have positive-definite Hermitian part and let  $Q = \text{diag}(q_x)$  be such that  $A + Q$  also has positive-definite Hermitian part. Then*

$$\int e^{-S_A - (q, \tau)} = 1, \quad (21.W.2)$$

$$\int e^{-S_A - (q, \tau)} \bar{\phi}_a \phi_b = (A + Q)_{ab}^{-1}. \quad (21.W.3)$$

*The identities continue to complex  $q$  as long as the Hermitian part remains positive.*

*Proof.* Expand the fermionic exponential. Its top-degree coefficient is  $\det(A + Q) \prod_x (2\pi i)^{-1} d\bar{\phi}_x d\phi_x$ . The bosonic complex Gaussian integral is  $\det(A + Q)^{-1}$  in this normalisation. The determinants cancel, proving (21.W.2). Inserting  $\bar{\phi}_a \phi_b$  differentiates the bosonic Gaussian with respect to  $(A + Q)_{ba}$ , and therefore contributes exactly the matrix element  $(A + Q)_{ab}^{-1}$  after the same determinant cancellation. Absolute convergence follows from positivity of the Hermitian part, so the calculation is valid for complex  $q$  in the stated domain.  $\square$

**Lemma 10.212** (Finite-volume  $\tau$ -isomorphism). *Let  $X$  be continuous-time simple random walk on  $\Lambda$  with generator  $\Delta$  and local times  $L_T^x = \int_0^T \mathbf{1}_{\{X_s = x\}} ds$ . If  $\nu > 0$  and  $F$  is a bounded continuous function on  $[0, \infty)^\Lambda$  with an integrable Fourier transform, then*

$$\int e^{-\sum_x (\tau_{\Delta, x} + \nu \tau_x)} F((\tau_x)_x) \bar{\phi}_a \phi_b = \int_0^\infty \mathbf{E}_a[F((L_T^x)_x) \mathbf{1}_{\{X_T = b\}}] e^{-\nu T} dT. \quad (21.W.4)$$

*In particular, for  $g > 0$ ,*

$$G_{\Lambda, g, \nu}(a, b) = \int e^{-\sum_x (\tau_{\Delta, x} + g \tau_x^2 + \nu \tau_x)} \bar{\phi}_a \phi_b. \quad (21.W.5)$$

*The identity extends from  $\nu > 0$  to every real  $\nu$  in the finite-susceptibility region.*

*Proof.* First take  $F(t) = e^{-(q, t)}$  with  $\text{Re } q_x \geq 0$ . With  $A = -\Delta + \nu$ , Lemma 10.211 says that the left side is  $(A + Q)_{ab}^{-1}$ . The Feynman–Kac formula gives

$$(A + Q)_{ab}^{-1} = \int_0^\infty \mathbf{E}_a \left[ e^{-\sum_x q_x L_T^x} \mathbf{1}_{\{X_T = b\}} \right] e^{-\nu T} dT,$$

which is the right side. Fourier inversion and Fubini now prove (21.W.4) for the stated class of  $F$ .

For the weak self-interaction use the exact Fourier identity

$$e^{-g \sum_x t_x^2} = (4\pi g)^{-|\Lambda|/2} \int_{\mathbb{R}^\Lambda} e^{-|s|^2/(4g)} e^{i(s, t)} ds. \quad (21.W.6)$$

For each  $s$ , the matrix  $A - i \text{diag } s$  has Hermitian part  $A > 0$ ; Lemma 10.211 and Feynman–Kac therefore apply, and the Gaussian weight in  $s$  justifies Fubini. This proves (21.W.5) for  $\nu > 0$ . For fixed finite  $\Lambda$ , the field integral is entire in  $\nu$ : the positive quartic term gives an integrable majorant on every compact set of complex  $\nu$ . The walk side is analytic in the half-plane where its Laplace transform is finite. Since the two analytic functions agree on  $\nu > 0$ , the identity theorem extends the equality throughout the connected finite-susceptibility region.  $\square$

**Corollary 10.213** (Supersymmetric self-normalisation). *For  $g > 0$  and every real  $\nu$  for which the integral is defined,*

$$\int e^{-\sum_x (\tau_{\Delta, x} + g \tau_x^2 + \nu \tau_x)} = 1. \quad (21.W.7)$$

*Proof.* For  $\nu > 0$ , omit the insertion in Lemma 10.211 and use (21.W.6): every fixed diagonal Fourier potential has super-partition function one. Fourier integration preserves the value one. Analytic continuation in  $\nu$  is as in the preceding proof.  $\square$

### The source norm with fermions

For a finite set  $X$ , every even form has a unique expansion

$$F = \sum_{I,J} F_{I,J}(\phi, \bar{\phi}) \psi_I \bar{\psi}_J,$$

where  $I, J$  are ordered subsets of  $X$ . The source  $T_\phi$  norm can be described as a Taylor pairing between bosonic derivatives and these fermionic coefficients. For the argument it is enough to use the equivalent finite-local norm

$$\|F\|_{T_{\phi,j}^s(X)} = \sum_{I,J} h_j^{|I|+|J|} \|F_{I,J}\|_{T_{\phi,j}^b(X)}, \quad (21.W.8)$$

where  $T^b$  is the bosonic finite-order test-function norm of Definition (21.280) and the source field scale is used for both boson and fermion species. On each fixed small set this norm is uniformly equivalent to the sequence/test-function pairing norm used in the RG map; the finite change of convention is absorbed into the fixed small-set constants.

**Lemma 10.214** (Super normed-algebra and Gaussian contraction). *For even forms supported in a fixed small-set neighbourhood,*

$$\|FG\|_{T_{\phi,j}^s} \leq \|F\|_{T_{\phi,j}^s} \|G\|_{T_{\phi,j}^s}, \quad (21.W.9)$$

$$\|\mathbb{E}_C \theta F\|_{T_{\phi,j}^s} \leq C^{|X|_j} \sup_{\xi} \frac{\|F(\phi + \xi, \psi + \eta)\|_{T^s}}{G_j(X, \xi)^q}, \quad (21.W.10)$$

*whenever  $C$  satisfies the scale covariance bounds and  $q$  lies below the regulator moment threshold. The same estimate holds for the large-field pair. In particular, fermionic fluctuation integration introduces no large-field regulator of its own.*

*Proof.* For (21.W.9), multiply the coefficient expansions. Exterior multiplication is a signed shuffle of the sets  $I, J$ ; after absolute values the signs disappear, and the weighted coefficient sum is exactly a Cauchy product. Apply Lemma 10.98 to the bosonic coefficients.

For (21.W.10), Wick-expand the super-Gaussian expectation. A bosonic contraction and a fermionic contraction both contribute one matrix element of  $C$ ; the only difference is the fermionic permutation sign. The absolute-value estimate therefore has the same contraction majorant as the bosonic Gaussian translation estimate. The scale covariance bound makes every contracted pair dimensionless in the field-scaled test norm. Summing contractions on one fixed small set gives a constant depending only on the fixed test order and small-set size. The bosonic translation is controlled by Lemma 10.99 and the Gaussian regulator moment of Proposition 10.176. Fermions are nilpotent and hence contribute no additional large-field growth. This proves both norm-pair estimates.  $\square$

**Lemma 10.215** (Supersymmetric transfer of the nonlinear single-step kernels). *Replace the bosonic interaction by*

$$V(B) = \sum_{x \in B} (g\tau_x^2 + \nu\tau_x + z\tau_{\Delta,x})$$

*in the even differential-form algebra. Then the analytic kernels PK1–PK4, LA1–LA4 and AD proved in Cycles 67–68 remain valid, with the same powers of the small parameters and the same  $O(L^{-1})$  localisation contraction. Consequently the six-map nonlinear theorem and the two-regulator recombination hold in the supersymmetric bulk sector with no observables.*



*Proof.* The proof is a transfer check of every operation used in Theorem 10.172. Circle products and component factorisation take place in the even form algebra and are therefore unchanged. The product and fluctuation-integration bounds are Lemma 10.214. The regulators depend only on the bosonic field, so PK3 is literally the already-proved regulator theorem. In PK2, the positive real part of  $g|\phi|^4$  supplies the large-field damping; the extra fermion terms in  $\tau^2 = |\phi|^4 + 2|\phi|^2\psi\bar{\psi}$  are nilpotent polynomial coefficients and are bounded by the same  $T^s$  norm. Thus the interpolation proof of Lemma 10.178 is unchanged.

For localisation, boson and fermion fields have the same engineering dimension in the source test-function norm. The finite-dimensional Taylor projection therefore retains exactly the supersymmetric relevant/marginal monomials and the first omitted bulk monomial still lies at least  $d+1$  dimensions beyond the block-volume normalisation. Hence the  $L^{-d-1}$  bulk gain is unchanged. The local restoration estimates LA1–LA4 use only products, Gaussian contractions, localisation and finite sums after their second-order cancellations; the preceding bounds apply term by term. Finally the common analytic domain follows from finite Grassmann degree together with the same bosonic quartic stability: nilpotent fermionic factors create no new singularity. Thus every kernel in this chapter reduction has the same source bound, and the norm recombination is unchanged.  $\square$

### Why the perturbative coefficients are exactly the $n = 0$ coefficients

**Lemma 10.216** (Closed-loop cancellation and the one-loop coefficients). *In the supersymmetric quartic theory, every closed internal component-index loop in a Wick contraction carries the supertrace*

$$\text{Str } \mathbf{1} = 1_{\text{boson}} - 1_{\text{fermion}} = 0. \quad (21.W.11)$$

*Therefore the one-loop  $O(n)$  combinatorics of Lemma 10.107 is evaluated at  $n = 0$ . With the WSAW normalisation,*

$$\beta_j = 8 \sum_x (w_{j+1,x}^2 - w_{j,x}^2), \quad \gamma = \frac{2}{8} = \frac{1}{4}. \quad (21.W.12)$$

*Proof.* Use the super-Wick expansion from Lemma 10.214. Whenever an internal contraction cycle is closed, summing its species index gives  $+1$  from the complex bosonic pair and  $-1$  from the complex fermionic pair; the sign is the cyclic fermionic permutation sign. Hence such loops cancel exactly. The remaining contraction topologies are precisely the  $n$ -independent terms in the polynomial Wick count of Lemma 10.107. Setting the loop factor  $n$  to its superdimension zero changes  $(n+8)$  to 8 and  $(n+2)$  to 2. The common covariance bubble is unchanged, which gives (21.W.12) and the ratio  $1/4$ .  $\square$

**Proposition 10.217** (Supersymmetric centre-stable RG flow). *For  $d = 4$  and sufficiently small  $g_0 > 0$ , there are critical initial coordinates  $z_0^c(m^2, g_0), \nu_0^c(m^2, g_0)$  for which the exact supersymmetric RG trajectory is defined at all scales and satisfies*

$$g_{j+1} = g_j - \beta_j g_j^2 + O(\vartheta_j g_j^3), \quad (21.W.13)$$

$$\mu_{j+1} = L^2 \mu_j (1 - \frac{1}{4} \beta_j g_j) + O(\vartheta_j g_j^2) + O(\vartheta_j g_j |z_j|), \quad (21.W.14)$$

$$|z_j| + |\mu_j| = O(\vartheta_j g_j), \quad \|K_j\|_{W_j} = O(\vartheta_j g_j^3). \quad (21.W.15)$$

*Moreover, if a prime denotes differentiation with respect to the initial mass coordinate, then*

$$\mu'_j = L^{2j} \left( \frac{g_j}{g_0} \right)^{1/4} (c(m^2, g_0) + O(\vartheta_j g_j)), \quad c = 1 + O(g_0), \quad (21.W.16)$$

*with  $g'_j, z'_j, K'_j$  smaller by the same powers as in the bosonic derivative hierarchy.*

*Proof.* Lemma 10.215 gives the exact nonlinear map with the same contracting  $K$  bounds used in the abstract structural theorem Proposition 10.115. Lemma 10.216 gives the triangular perturbative coefficients and in particular the mass multiplier  $1 - \frac{1}{4}\beta_j g_j$ . Applying the structural theorem therefore gives (21.W.13)–(21.W.15). Differentiate the exact flow. The linearised mass equation has principal multiplier  $L^2(1 - \frac{1}{4}\beta_j g_j)$ , whereas derivatives of all nonlinear remainders are  $O(\vartheta_j g_j^2)$  by the common analytic-domain Cauchy bounds. Variation of constants and the product identity

$$\prod_{k < j} (1 - \frac{1}{4}\beta_k g_k) = \left(\frac{g_j}{g_0}\right)^{1/4} (c + O(\vartheta_j g_j))$$

prove (21.W.16). The differentiated contraction for  $K$  and the triangular  $g, z$  equations give the remaining derivative bounds.  $\square$

### Susceptibility endpoint

Define the WSAW bubble in the source normalisation by

$$\mathbf{B}_{m^2} = 8 \sum_x C_{m^2}(x)^2, \quad C_{m^2} = (-\Delta + m^2)^{-1}.$$

By Lemma 10.190,

$$\mathbf{B}_{m^2} \sim \mathbf{b} \log m^{-2}, \quad \mathbf{b} = \frac{1}{2\pi^2}. \quad (21.W.17)$$

**Lemma 10.218** (Renormalised WSAW susceptibility identity). *Let*

$$g_0 = g(1 + z_0)^2, \quad \nu_0 = (1 + z_0)\nu - m^2. \quad (21.W.18)$$

*On the critical centre-stable trajectory of Proposition 10.217, the renormalised susceptibility satisfies*

$$\widehat{\chi}(m^2, g_0, \nu_0^c, z_0^c) = m^{-2}, \quad (21.W.19)$$

*and*

$$\partial_{\nu_0} \widehat{\chi} \sim -m^{-4} \frac{c(g_0)}{(g_0 \mathbf{B}_{m^2})^{1/4}}, \quad c(g_0) = 1 + O(g_0). \quad (21.W.20)$$

*The physical susceptibility is*

$$\chi(g, \nu) = (1 + z_0) \widehat{\chi}. \quad (21.W.21)$$

*Proof.* Rescale simultaneously  $(\phi, \psi)$  by  $(1 + z_0)^{1/2}$ . Because the fermion is the differential of the boson there is no uncanceled Jacobian, and the action splits into the positive Gaussian covariance  $(-\Delta + m^2)^{-1}$  plus  $g_0 \tau^2 + \nu_0 \tau + z_0 \tau \Delta$ . This proves (21.W.18) and (21.W.21).

Progressive super-Gaussian integration gives  $Z_N = (I_N \circ K_N)(\Lambda) = I_N(\Lambda) + K_N(\Lambda)$  on the last scale. Complete the square with constant bosonic sources. Since  $C1 = m^{-2}1$ ,

$$\widehat{\chi}_N = m^{-2} + m^{-4} |\Lambda|^{-1} D^2 Z_N^0(0; 1, 1). \quad (21.W.22)$$

On the critical trajectory, the local mass coordinate tends to zero. The  $W_N$  term in  $I_N$  is  $O(\vartheta_N g_N^2)$  in the super  $T_0$  norm and  $K_N$  is  $O(\vartheta_N g_N^3)$ ; the constant test function has scale norm  $O(L^N)$  in  $d = 4$ . After division by  $|\Lambda| = L^{4N}$  both corrections in (21.W.22) vanish, which proves (21.W.19).

Differentiate (21.W.22) in the initial mass coordinate. The derivatives of the  $W_N$  and  $K_N$  corrections are smaller than the derivative of the relevant mass coordinate by factors  $O(\vartheta_N g_N)$  and

$O(\vartheta_N g_N^2)$ , respectively, by Proposition 10.217. Hence the leading term is  $-m^{-4} \lim_N \nu'_N$ . Formula (21.W.16) and the telescoping marginal flow give

$$g_\infty^{-1} = g_0^{-1} + \mathbf{B}_{m^2} + O(|\log g_\infty|),$$

so  $\lim_N \nu'_N \sim c(g_0)(g_0 \mathbf{B}_{m^2})^{-1/4}$ . This proves (21.W.20).  $\square$

**Theorem 10.219** (WSAW susceptibility engine). *For sufficiently small  $g > 0$ , let  $\nu_c(g)$  be the critical value of the continuous-time weakly self-avoiding walk on  $\mathbb{Z}^4$ . Then, as  $\varepsilon \downarrow 0$ ,*

$$\chi(g, \nu_c + \varepsilon) \sim A_g \varepsilon^{-1} (\log \varepsilon^{-1})^{1/4}, \quad A_g = (\mathbf{b}g)^{1/4} (1 + O(g)), \quad \mathbf{b} = \frac{1}{2\pi^2}. \quad (21.W.23)$$

*The proof uses the exact supersymmetric representation and does not invoke a formal  $n \rightarrow 0$  limit.*

*Proof.* For  $\nu = \nu_c + \varepsilon$ , choose the renormalised parameters on the centre-stable trajectory. Equations (21.W.19) and (21.W.21) give the exact identity  $\chi = (1 + z_0)m^{-2}$ . By the chain rule and (21.W.20),

$$\partial_\nu \chi \sim -\chi^2 \frac{c_0(g)}{(g_0 \mathbf{B}_{m^2})^{1/4}}.$$

Set  $F = \chi^{-1}$ . Since  $m^2 \asymp F$  up to the continuous factor  $1 + z_0$ , (21.W.17) yields

$$F'(\nu) \sim \frac{c_0(g)}{(g_0 \mathbf{b})^{1/4}} (-\log F(\nu))^{-1/4}. \quad (21.W.24)$$

Apply the logarithmic ODE lemma 10.194 with exponent  $1/4$ . It follows that  $F(\nu_c + \varepsilon) \sim A_g^{-1} \varepsilon (\log \varepsilon^{-1})^{-1/4}$ . Finally  $g_0 = g + O(g^2)$  and  $c_0(g) = 1 + O(g)$ , giving the amplitude in (21.W.23).  $\square$

**Theorem 10.220** (Four-dimensional weakly self-avoiding-walk susceptibility). *For the continuous-time weakly self-avoiding walk on  $\mathbb{Z}^4$  in the weak-coupling regime of the rigorous supersymmetric RG construction, let  $\chi$  denote the susceptibility and let the killing/fugacity parameter approach its critical value from the finite-susceptibility side. Then*

$$\chi(g, \nu_c + \varepsilon) \sim A_g \varepsilon^{-1} (\log \varepsilon^{-1})^{1/4}, \quad \varepsilon \downarrow 0,$$

*for the positive amplitude  $A_g$  of the source theorem. In particular the logarithmic correction exponent is  $1/4$ . The statement concerns the continuous-time weakly self-avoiding walk on  $\mathbb{Z}^4$ ; it is not a theorem for an arbitrary discrete self-avoiding walk model.*

*Proof.* This is Theorem 10.219. In particular the supersymmetric  $\tau$ -isomorphism, the boson–fermion nonlinear RG transfer, the closed-loop cancellation giving  $\gamma = 1/4$ , and the susceptibility endpoint are all proved internally above; no formal  $n \rightarrow 0$  substitution is used.  $\square$

## marked-observable RG and the critical correlation family

The bulk nonlinear RG theorem does not by itself determine critical point correlations. Two extra facts are decisive. First, marked fields at two points evolve in separate sectors until the finite-range decomposition can connect the two points. Second, for powers  $p \geq 2$  the linear observable coordinate has a nontrivial anomalous dimension before that coalescence scale. We now reconstruct these two mechanisms in the exact four-sector algebra used by the source theorem.

**Definition 10.221** (Observable sectors and coalescence scale). Let  $a, b \in \mathbb{Z}^4$ ,  $a \neq b$ , and put

$$j_{ab} = \lfloor \log_L(2|a - b|) \rfloor.$$

Thus

$$\frac{1}{2}L^{j_{ab}} \leq |a - b| < \frac{1}{2}L^{j_{ab}+1}.$$

Adjoin commuting external variables  $\sigma_a, \sigma_b$  and quotient by all terms of degree at least two in either variable. Every element then has a unique decomposition

$$F = F_\emptyset + \sigma_a F_a + \sigma_b F_b + \sigma_a \sigma_b F_{ab}.$$

We denote the four projections by  $\pi_\emptyset, \pi_a, \pi_b, \pi_{ab}$ . The finite-range property implies

$$C_{j+1;ab} = 0 \quad (j < j_{ab}),$$

so the  $ab$  sector cannot be created before the step  $j_{ab} \rightarrow j_{ab} + 1$ .

*Proof.* The algebraic decomposition follows immediately from the quotient relations  $\sigma_a^2 = \sigma_b^2 = 0$ . The scale statement follows from the finite-range estimate already proved in the RG foundation,  $C_k(x, y) = 0$  when  $|x - y| \geq \frac{1}{2}L^k$ . If  $j < j_{ab}$  then  $|a - b| \geq \frac{1}{2}L^{j_{ab}} > \frac{1}{2}L^j$ , hence  $C_{j+1;ab} = 0$  until the first admissible connecting scale. The boundary convention agrees with the chosen definition of  $j_{ab}$ .  $\square$

**Definition 10.222** (Observable monomials and anomalous-dimension eigenspaces). For WSAW ( $n = 0$ ), set

$$\rho_x^a = \bar{\phi}_x^p, \quad \rho_x^b = \phi_x^p.$$

For  $n \geq 1$  and  $p = 1, 2$ , given a nonzero vector  $h \in \mathbb{R}^n$ , set

$$\rho_x^a(h) = \rho_x^b(h) = |h|^{-1} \sum_{i=1}^n h_i (\varphi_x^i)^p.$$

For  $n \geq 2$ , let

$$E^+ = \text{span}(1, \dots, 1), \quad E^- = (E^+)^\perp.$$

Define

$$\gamma_{n,p}^+ = \binom{p}{2} \frac{n+2}{n+8}, \quad \gamma_{n,p}^- = \binom{p}{2} \frac{2}{n+8},$$

with the convention  $\gamma_{0,p}^+ = \gamma_{0,p}^- = \frac{1}{4} \binom{p}{2}$ . For  $n \geq 2$  introduce the matrix

$$T_{n,p} = \binom{p}{2} \left( \frac{2}{n+8} I + \frac{1}{n+8} J \right),$$

where  $J$  is the all-ones matrix. Then  $E^\pm$  are its eigenspaces with respective eigenvalues  $\gamma_{n,p}^\pm$ . The analogous scalar matrices for  $n = 0, 1$  are  $\binom{p}{2} \frac{1}{4}$  and  $\binom{p}{2} \frac{1}{3}$ .

*Proof.* For  $v \in E^+$ ,  $Jv = nv$ , so  $T_{n,p}v = \binom{p}{2} \frac{n+2}{n+8} v$ . For  $v \in E^-$ ,  $Jv = 0$ , giving the second eigenvalue. The  $n = 0, 1$  conventions coincide with the corresponding plus eigenvalues.  $\square$

**Lemma 10.223** (Exact  $p$ -watermelon functional-integral representation). *On a finite torus  $\Lambda_N$ , let*

$$S = \sum_{x \in \Lambda_N} (\tau_{\Delta, x} + g\tau_x^2 + \nu\tau_x)$$

*be the supersymmetric WSAW action of Lemma 10.212. For every integer  $p \geq 1$ ,*

$$W_{ab, N}^{(p)}(g, \nu) = \int e^{-S} \bar{\phi}_a^p \phi_b^p. \quad (21.280)$$

*After the RG change of variables of the supersymmetric bulk flow this becomes*

$$W_{ab, N}^{(p)}(g, \nu) = (1 + z_0)^p \mathbb{E}_C \left[ e^{-U_0(\Lambda_N)} \bar{\phi}_a^p \phi_b^p \right]. \quad (21.281)$$

*The same representation with the terminal points summed gives the star network  $S_N^{(p)}$ .*

*Proof.* We extend the proof of Lemma 10.212 from one marked trajectory to  $p$  trajectories. For a diagonal potential  $V$  with positive real part, boson–fermion determinant cancellation normalises the super Gaussian integral. Repeated differentiation of the bosonic source identity gives

$$\int e^{-S_A - (V, \tau)} \bar{\phi}_a^p \phi_b^p = p! [(A + V)_{ab}^{-1}]^p.$$

Use the Feynman–Kac formula separately for the  $p$  resolvent factors. The product is the integral over  $p$  independent continuous-time walks, and the factor  $p!$  is precisely the number of contractions. As in the one-walk proof, Fourier inversion of  $\exp\{-g \sum_x (L_T(x))^2\}$  inserts the self- and mutual-intersection weight for the total local time. Tonelli applies first for  $\nu > 0$ ; analyticity in  $\nu$  extends the identity throughout the finite-volume domain. This proves (21.280). The field rescaling  $\phi \mapsto (1 + z_0)^{1/2} \phi$  contributes  $(1 + z_0)^p$  and produces the renormalised interaction  $U_0$ , giving (21.281). Summing the  $p$  endpoints inside Feynman–Kac removes the terminal indicators and gives the star network.  $\square$

**Definition 10.224** (Local observable interaction space). For  $x = a, b$  let  $\lambda_x$  be a linear observable coupling and let  $q_x$  be the quadratic  $ab$  coupling. We use local interactions of the form

$$V_x = V_{\emptyset, x} - \sigma_a \lambda_a \rho_a^a(h) \mathbf{1}_{x=a} - \sigma_b \lambda_b \rho_b^b(h) \mathbf{1}_{x=b} - \frac{1}{2} \sigma_a \sigma_b (q_a \mathbf{1}_{x=a} + q_b \mathbf{1}_{x=b}), \quad (21.282)$$

with the additional scalar observable coordinates required for  $n \geq 1$ ,  $p = 2$ . We denote this finite-dimensional space by  $\mathcal{V}_h$ . For the two field-scale choices  $\mathfrak{h}_j = \ell_j, h_j$ , define

$$\mathfrak{h}_{\sigma, j} = \Gamma_{j \wedge j_{ab}}^{-1} \ell_j^{-p} 2^{p(j-j_{ab})_+} \begin{cases} \tilde{g}_j, & \mathfrak{h}_j = \ell_j, \\ \tilde{g}_j^{p/4}, & \mathfrak{h}_j = h_j, \end{cases} \quad (21.283)$$

where

$$\Gamma_j = (g_j/g_0)^\gamma, \quad \gamma = \gamma_{n, p}^\pm$$

according to the chosen eigenspace. In the four-sector  $T_\phi$  norm we use

$$\|F\|_{T_\phi} = \|F_\emptyset\|_{T_\phi} + \mathfrak{h}_\sigma (\|F_a\|_{T_\phi} + \|F_b\|_{T_\phi}) + \mathfrak{h}_\sigma^2 \|F_{ab}\|_{T_\phi}. \quad (21.284)$$

**Lemma 10.225** (Perturbative observable flow). *Let  $w_j = \sum_{i \leq j} C_i$  and*

$$\beta_j = (n+8)(w_{j+1}^{(2)} - w_j^{(2)}).$$

*Before coalescence define*

$$A_j = (1 - p\delta_j[\nu w^{(1)}])I - \beta_j g_j T_{n,p}, \quad (21.285)$$

*and set  $A_j = I$  after coalescence. If  $h \in E^\pm$ , the perturbative single-scale map leaves the direction  $h$  unchanged and gives*

$$\lambda_{x,j+1}^{\text{pt}} = f_j^\pm \lambda_{x,j}, \quad f_j^\pm = 1 - p\delta_j[\nu w^{(1)}] - \beta_j g_j \gamma_{n,p}^\pm \quad (j+1 < j_{ab}), \quad (21.286)$$

*while  $f_j^\pm = 1$  after coalescence. The quadratic observable satisfies*

$$q_{x,j+1}^{\text{pt}} - q_{x,j} = p! \lambda_{a,j} \lambda_{b,j} (w_{j+1;ab}^p - w_{j;ab}^p). \quad (21.287)$$

*Proof.* The Gaussian expectation is  $e^{\mathcal{L}_{C_{j+1}}}$ . Because the observable polynomial has degree  $p \leq 2$  in the bosonic  $n \geq 1$  case, and because the WSAW observable is encoded in the supersymmetric monomial of Lemma 10.223, the observable part of the second-order perturbative map is finite. The contraction of one observable leg with the quartic bulk vertex is permutation-equivariant in the component index. Its coefficient matrix therefore has the form  $rI + sJ$ ; direct Wick contraction gives exactly  $T_{n,p}$  of Definition 10.222. The mass insertion contributes the scalar factor  $1 - p\delta_j[\nu w^{(1)}]$ . Hence the linear coordinate is multiplied by  $A_j$ , and restricting to  $E^\pm$  gives (21.286). Two marked monomials cannot contract across the two points before coalescence by Definition 10.221; after coalescence, the complete contraction of the  $p$  legs at  $a$  with the  $p$  legs at  $b$  has  $p!$  pairings and covariance  $w_{ab}^p$ . Taking its single-scale increment gives (21.287).  $\square$

**Definition 10.226** ( $h$ -factorisability). For  $n \geq 1$ , a four-sector functional  $F$  is called  $h$ -factorisable if for  $x = a, b$

$$\pi_x F = \sigma_x(F_x^* \cdot h)$$

for a vector-valued bulk functional  $F_x^*$  which is equivariant under permutations of the  $n$  components. For  $p = 2$  we additionally impose the coordinate-sign symmetry that removes odd and mixed quadratic monomials. For  $n = 0$ ,  $h$  plays no role and the condition is replaced by supersymmetry and  $U(1)$  invariance.

**Lemma 10.227** (Closure of the reduced-symmetry class). *The  $h$ -factorisable class is closed under products, Gaussian expectation, localisation, circle products, and each of the six nonlinear maps of the RG construction developed in this chapter. If  $h \in E^\pm$ , localisation of an observable sector returns a local polynomial with the same direction  $h$ .*

*Proof.* Products are immediate from  $\sigma_a^2 = \sigma_b^2 = 0$ : for example

$$\pi_a(FG) = F_\emptyset \sigma_a(G_a^* \cdot h) + \sigma_a(F_a^* \cdot h) G_\emptyset.$$

Permutation equivariance is preserved termwise. Gaussian expectation acts componentwise and its covariance is a scalar in component space, hence it commutes with every component permutation. Localisation is linear and also commutes with the component action. For  $p = 2$ , sign symmetry excludes the linear, mixed-quadratic and gradient monomials, so localisation in a one-mark sector is a linear combination of  $1, (\varphi^1)^2, \dots, (\varphi^n)^2$ . Its coefficient map commutes with all permutation matrices; the commutant is  $rI + sJ$ , which preserves each  $E^\pm$ .

It remains to inspect the six nonlinear maps. Map 1 is polynomial in  $I, K$  and the local correction  $J$ , so product and localisation closure apply. Map 2 is a circle product of  $K^{(1)}$  with the interaction increment. Map 3 is assembled from Gaussian expectations, products and localisation. Map 4 adds the truncated expected leading term, hence again uses only Gaussian expectation and products. Map 5 replaces  $W(V_{\text{pt}})$  by  $W(V_+)$ ; both arguments lie in the same reduced-symmetry class by the perturbative flow and localisation closure. Finally Map 6 is summation by parts and multiplication by the constant-field factor  $e^{-\delta\zeta}$ ;  $\delta\zeta$  is obtained by evaluation of the already factorisable  $V_{\text{pt}}$ . Thus all six maps preserve the class.  $\square$

**Theorem 10.228** (Observable single-step theorem). *Fix either  $n = 0, p \geq 1$ , or  $n \geq 1, p = 1, 2$ , and choose  $h \in E^\pm$ . For  $L$  sufficiently large and the coupling sufficiently small, the this chapter nonlinear RG map extends to the four-sector observable algebra with weights (21.283)–(21.284). On the same two regulator norm pairs as the bulk theorem it satisfies*

$$\|R_{j+1}\|_{\mathcal{V}_h} + \|K_{j+1}\|_{\mathcal{W}_{j+1}} \leq C\chi_{j+1}\tilde{g}_{j+1}^3, \quad (21.288)$$

with the sector support rules

$$\pi_a R_{j+1} = \pi_b R_{j+1} = 0 \quad (j \geq j_{ab}), \quad \pi_{ab} R_{j+1} = 0 \quad (j < j_{ab}). \quad (21.289)$$

The same estimates hold jointly and continuously in the mass and coupling parameters. For WSAW with arbitrary fixed  $p$ , the differentiability order in the large-field norm may be chosen larger depending on  $p$ .

*Proof.* All bulk estimates are already owned by Theorem 10.188 and the supersymmetric transfer Lemma 10.215. We verify that the observable weights leave every small parameter in those proofs unchanged.

Before coalescence, (21.283) implies  $\mathfrak{h}_{\sigma,j+1}/\mathfrak{h}_{\sigma,j} \leq CL^p$ ; after coalescence the ratio is bounded. The localisation dimension assigned to a one-mark sector is exactly  $p$  before coalescence and 0 afterwards. Consequently the factor  $L^p$  is cancelled by the same Taylor-dimension factor in the sector-sensitive localisation theorem 10.180; the small parameter remains  $L^{-d-1}$  in the bulk and  $L^{-1}$  in a marked sector.

The stability parameter contributed by a marked local monomial is  $|\lambda_x|\mathfrak{h}_j^p\mathfrak{h}_{\sigma,j}$ . The definition of the observable domain makes this  $O(\tilde{g}_j)$  in the  $\ell$  norm and  $O(1)$  with an arbitrarily small prefactor in the large-field norm. For  $p \geq 4$  in WSAW, apparent high powers do not create a large-field integrability problem: the external variables live in the quotient with  $\sigma_a^2 = \sigma_b^2 = 0$ , so the exponential is only first order in each marked sector. Taking the  $T$ -norm differentiability order at least  $\max\{10, 2p + 4\}$  supplies enough good  $\ell_j/h_j$  ratios in the change-of- norm estimate to recover the same  $\tilde{g}^{10/4}$  remainder as the bulk proof.

For the final restoration map, the new quadratic increment is, by (21.287) and the covariance bound,

$$|\delta q_x| \leq C|\lambda_a\lambda_b|\ell_j^{2p}.$$

Multiplication by  $\mathfrak{h}_{\sigma,j}^2$  and the observable-domain bound gives  $O(\tilde{g}_j^2)$  in the fluctuation-field norm and  $O(\tilde{g}_j^{p/2})$  in the large-field norm. The additional scalar coordinate present for  $n \geq 1, p = 2$  contains one covariance and one linear observable coupling, so its weighted norm is bounded by  $C(\tilde{g}_j + \tilde{g}_j^{1/2})$ . These are exactly of the orders absorbed in Map 6.

Lemma 10.227 shows that reduced symmetry is preserved through all six maps. Thus none of the analytic constants or regulator gains of the bulk proof deteriorates, and (21.288) follows by the same map-by-map estimates now applied sectorwise. The support rules (21.289) follow from

finite range and the fact that the localisation operator cannot manufacture a sector whose marked points are absent from the polymer. Joint continuity is inherited from the analytic-domain theorem already proved for the six maps.  $\square$

**Lemma 10.229** (Complete observable flow). *Let  $\lambda_{a,0} = \lambda_{b,0} = 1$ . There are sequences  $r_{x,j}^\pm$  with*

$$|r_{x,j}^\pm| \leq C\chi_j \bar{g}_j^2, \quad (21.290)$$

*such that, up to coalescence,*

$$\lambda_{x,j} = \Pi_{j-1}^\pm \left( 1 + \sum_{k=0}^{j-1} r_{x,k}^\pm \right), \quad \Pi_j^\pm = \prod_{i=0}^j f_i^\pm, \quad (21.291)$$

*and  $\lambda_{x,j}$  is frozen after coalescence. Moreover*

$$\lambda_{x,j}^* = v_x^\pm \Gamma_j^\pm (1 + O(\chi_j \bar{g}_j)), \quad v_x^\pm = 1 + O(g_0), \quad (21.292)$$

*and, when  $j_{ab} < j_m$ ,*

$$\lambda_{x,j_{ab}} = v_x^\pm \left( \frac{1}{\mathbf{b}g_0 \log |a-b|} \right)^{\gamma_{n,p}^\pm} (1 + \mathcal{E}_{ab}^{(p)}), \quad (21.293)$$

*where*

$$\mathcal{E}_{ab}^{(p)} = \begin{cases} O((\log |a-b|)^{-1}), & p = 1, \\ O((\log \log |a-b|)(\log |a-b|)^{-1}), & p \geq 2, \end{cases} \quad \mathbf{b} = \frac{n+8}{16\pi^2}.$$

*Proof.* Theorem 10.228 says that the nonperturbative correction to the linear coordinate is bounded by the  $\mathcal{V}_h$  component of  $R_{j+1}$ ; after dividing by the observable weight and using (21.283), this is  $O(\Gamma_j \chi_j \bar{g}_j^2)$  before coalescence and zero afterwards. Adding this correction to (21.286) gives (21.291) with (21.290).

For the product, the bulk flow gives  $g_{j+1} = g_j(1 - e_j)$  with  $e_j = \beta_j g_j + O(\chi_j \bar{g}_j^2)$ . Taylor expansion yields

$$f_j^\pm = \left( \frac{g_{j+1}}{g_j} \right)^{\gamma_{n,p}^\pm} (1 + E_j), \quad |E_j| \leq C\chi_j \bar{g}_j^2.$$

The series  $\sum_j \chi_j \bar{g}_j^2$  converges, so the product of  $(1 + E_j)$  has a finite limit  $1 + O(g_0)$  and a tail  $1 + O(\chi_j \bar{g}_j)$ . This proves (21.292).

At critical mass and below the mass scale,  $g_{j_{ab}}^{-1} = \mathbf{b} \log |a-b| (1 + \mathcal{E}_{ab}^{(2)})$  by the massive-bubble asymptotic in Lemma 10.190 together with the marginal-flow lemma 10.116: at criticality the one-step coefficient satisfies  $\beta = (\mathbf{b} \log L)$ , so  $g_j^{-1} = \mathbf{b} j \log L + O(\log j)$ . Since  $L^{j_{ab}} = 2|a-b| + O(1)$ , substituting into (21.292) gives (21.293). When  $p = 1$ ,  $\gamma = 0$ , so the logarithmic product is identically one and the improved error follows.  $\square$

**Lemma 10.230** (Quadratic coordinate and critical endpoint). *For  $x = a, b$ , the quadratic observable has a continuous infinite-volume limit and*

$$q_{x,\infty}(m^2) = p! \lambda_{a,j_{ab}} \lambda_{b,j_{ab}} G_{ab}(m^2)^p + \sum_{i=j_{ab}}^{\infty} R_i^{q_x}, \quad (21.294)$$

*where*

$$\sum_{i=j_{ab}}^{\infty} |R_i^{q_x}| \leq C \frac{G_{ab}(m^2)^p}{(g_0 \log |a-b|)^{2\gamma}} \chi_{j_{ab}} \bar{g}_{j_{ab}} \quad (21.295)$$



whenever  $j_{ab} < j_m$  for  $p > 1$ ; for  $p = 1$  no such restriction is needed. At  $m^2 = 0$ ,

$$q_{x,\infty}(0) = p! v_a^\pm v_b^\pm \left( \frac{1}{\mathbf{b} g_0 \log |a - b|} \right)^{2\gamma_{n,p}^\pm} G_{ab}(0)^p (1 + \mathcal{E}_{ab}^{(p)}). \quad (21.296)$$

*Proof.* Summing (21.287) from the coalescence scale telescopes  $w_{j;ab}^p$  to  $G_{ab}(m^2)^p$ . Theorem 10.228 bounds the nonperturbative  $q$ -remainder at scale  $i$  by

$$|R_i^{q_x}| \leq C \Gamma_{j_{ab}}^2 L^{-2pj_{ab}} 2^{-2p(i-j_{ab})} \chi_i \bar{g}_i.$$

The geometric factor is summable and  $\chi_i \bar{g}_i \leq C \chi_{j_{ab}} \bar{g}_{j_{ab}}$ . Since  $L^{-2j_{ab}} \asymp |a - b|^{-2}$  and the finite-range Green function obeys  $G_{ab} \asymp |a - b|^{-2}$  below the mass scale, (21.293) gives (21.295). Substitution into (21.294), followed by  $m^2 \downarrow 0$ , gives (21.296).  $\square$

**Lemma 10.231** (Final-scale extraction). *Let  $P_N = \log Z_N$  with both observables switched on. Then*

$$D_{\sigma_a} D_{\sigma_b} P_N|_0 = \frac{1}{2} (q_{a,N} + q_{b,N}) + \mathcal{R}_N, \quad \mathcal{R}_N \rightarrow 0 \quad (21.297)$$

as  $N \rightarrow \infty$ . Consequently the WSAW watermelon and the  $n \geq 1$  truncated correlations are obtained by the critical limit of  $\frac{1}{2}(q_{a,\infty} + q_{b,\infty})$ , with the field-rescaling factor  $(1 + z_0)^p$ .

*Proof.* At the final scale,

$$Z_N = e^{\zeta_N} (I_N + K_N), \quad \zeta_N = \frac{1}{2} (q_{a,N} + q_{b,N}) \sigma_a \sigma_b + t_{a,N} \sigma_a + t_{b,N} \sigma_b + u_N |\Lambda_N|.$$

Twice differentiating  $P_N = \zeta_N + \log(I_N + K_N)$  gives the displayed  $q$  term plus first and second observable derivatives of the final nonperturbative coordinate. The observable single-step theorem and the weights (21.283) give

$$|D_\sigma^k K_N(\Lambda_N)| \leq C \chi_N \bar{g}_N^{3-k} \mathfrak{h}_{\sigma,N}^{-k}, \quad k = 1, 2.$$

For fixed  $a, b$  and then  $N \rightarrow \infty$ , the mass/covariance factor  $\chi_N$  and the final observable weights force these terms to zero. The bulk part of  $I_N$  tends to one by the same final-scale estimate used in the scaling limit module. Thus  $\mathcal{R}_N \rightarrow 0$ . The finite-volume functional representations, followed by monotone/analytic infinite-volume limits, give the claimed physical observables.  $\square$

**Theorem 10.232** (Critical correlation engine). *Let  $d = 4$  and let the coupling  $g > 0$  be sufficiently small. Define*

$$\gamma_{n,p}^+ = \binom{p}{2} \frac{n+2}{n+8}, \quad \gamma_{n,p}^- = \binom{p}{2} \frac{2}{n+8}, \quad \mathbf{b} = \frac{n+8}{16\pi^2}.$$

*There are positive amplitudes  $A'_{n,p,\pm}$  with*

$$A'_{n,p,\pm} = \frac{p!}{(2\pi)^{2p}} (\mathbf{b} g)^{-2\gamma_{n,p}^\pm} (1 + O(g)) \quad (21.298)$$

*such that, as  $|a - b| \rightarrow \infty$ :*

1. *for WSAW ( $n = 0$ ) and every  $p \geq 1$ ,*

$$W_{ab}^{(p)}(\nu_c) = \frac{A'_{0,p,+}}{|a - b|^{2p} (\log |a - b|)^{2\gamma_{0,p}^+}} (1 + \mathcal{E}_{ab}^{(p)}); \quad (21.299)$$

2. for  $n \geq 1$ ,

$$\langle \varphi_a^1; \varphi_b^1 \rangle_{\nu_c} = \frac{A'_{n,1,+}}{|a-b|^2} (1 + \mathcal{E}_{ab}^{(1)}), \quad (21.300)$$

and

$$\langle |\varphi_a|^2; |\varphi_b|^2 \rangle_{\nu_c} = \frac{nA'_{n,2,+}}{|a-b|^4 (\log |a-b|)^{2\gamma_{n,2}^+}} (1 + \mathcal{E}_{ab}^{(2)}); \quad (21.301)$$

3. for  $n \geq 2$ ,

$$\langle (\varphi_a^1)^2; (\varphi_b^1)^2 \rangle_{\nu_c} = \frac{1}{n|a-b|^4} \left[ \frac{(n-1)A'_{n,2,-}}{(\log |a-b|)^{2\gamma_{n,2}^-}} + \frac{A'_{n,2,+}}{(\log |a-b|)^{2\gamma_{n,2}^+}} \right] (1 + \mathcal{E}_{ab}^{(2)}), \quad (21.302)$$

$$\langle (\varphi_a^1)^2; (\varphi_b^2)^2 \rangle_{\nu_c} = \frac{1}{n|a-b|^4} \left[ -\frac{A'_{n,2,-}}{(\log |a-b|)^{2\gamma_{n,2}^-}} + \frac{A'_{n,2,+}}{(\log |a-b|)^{2\gamma_{n,2}^+}} \right] (1 + \mathcal{E}_{ab}^{(2)}). \quad (21.303)$$

Here  $\mathcal{E}_{ab}^{(1)} = O((\log |a-b|)^{-1})$  and  $\mathcal{E}_{ab}^{(p)} = O((\log \log |a-b|)/(\log |a-b|))$  for  $p \geq 2$ . In particular the basic two-point function has no logarithmic correction, while composite and network observables do.

*Proof.* By Lemma 10.231, every displayed critical correlation is the appropriate  $E^+$  or  $E^-$  linear combination of (21.296), multiplied by  $(1+z_0)^p = 1 + O(g)$ . The massless lattice Green function satisfies

$$G_{ab}(0) = \frac{1}{(2\pi)^2 |a-b|^2} (1 + O(|a-b|^{-2})).$$

For WSAW use  $E^+$  with  $\gamma_{0,p}^+ = \frac{1}{4} \binom{p}{2}$ , and apply the exact watermelon representation of Lemma 10.223. This proves (21.299). For  $n \geq 1, p = 1$ ,  $\binom{1}{2} = 0$ , so  $\gamma = 0$  and (21.300) follows with no logarithm. For the radial square choose  $h^+ = n^{-1/2}(1, \dots, 1)$ ; the corresponding quadratic form is  $n^{-1/2}|\varphi|^2$ , and (21.301) follows.

For  $n \geq 2, p = 2$ , choose additionally  $h^- = 2^{-1/2}(1, -1, 0, \dots, 0)$ . If  $S_{ij} = \langle (\varphi_a^i)^2; (\varphi_b^j)^2 \rangle$ , permutation symmetry gives

$$\langle \varphi_a^2 \cdot h^+; \varphi_b^2 \cdot h^+ \rangle = S_{11} + (n-1)S_{12}, \quad \langle \varphi_a^2 \cdot h^-; \varphi_b^2 \cdot h^- \rangle = S_{11} - S_{12}.$$

Solving these two linear equations and inserting the  $E^\pm$  versions of (21.296) proves (21.302)–(21.303). Formula (21.298) follows from  $v_x^\pm = 1 + O(g_0)$ ,  $g_0 = g(1 + O(g))$ , and the massless Green-function amplitude. The error estimates are those of (21.293)–(21.296).  $\square$

**Corollary 10.233** (Source-exact near-critical star-network and integrated composite laws). *Let  $d = 4$  and put  $\varepsilon = \nu - \nu_c \downarrow 0$ . With*

$$\gamma_{n,p}^+ = \binom{p}{2} \frac{n+2}{n+8}, \quad \gamma_{n,p}^- = \binom{p}{2} \frac{2}{n+8}, \quad \mathbf{b} = \frac{n+8}{16\pi^2},$$

*and with the convention  $\gamma_{0,p}^+ = \gamma_{0,p}^- = \frac{1}{4} \binom{p}{2}$ , there are positive amplitudes*

$$A_{n,p,\pm} = p! (\mathbf{b}g)^{-\gamma_{n,p}^\pm} (1 + O(g)) \quad (g \downarrow 0) \quad (21.304)$$

*such that the following source asymptotics hold.*

1. For WSAW ( $n = 0$ ) and every integer  $p \geq 2$ ,

$$\chi(\nu; 0)^{-p} S^{(p)}(\nu) \sim \frac{A_{0,p,+}}{(\log \varepsilon^{-1})^{\gamma_{0,p}^+}}. \quad (21.305)$$

2. For the  $n$ -component model,  $n \geq 1$ , the radial square insertion obeys

$$\frac{1}{\chi(\nu; n)^2} \sum_{x_1, x_2 \in \mathbb{Z}^4} \langle \varphi_{x_1} \cdot \varphi_{x_2}; |\varphi_a|^2 \rangle_\nu \sim \frac{n A_{n,2,+}}{(\log \varepsilon^{-1})^{\gamma_{n,2}^+}}. \quad (21.306)$$

3. For  $n \geq 2$ , the same-component and different-component square insertions have the source-leading asymptotics

$$\frac{1}{\chi(\nu; n)^2} \sum_{x_1, x_2 \in \mathbb{Z}^4} \langle \varphi_{x_1}^1 \varphi_{x_2}^1; (\varphi_a^1)^2 \rangle_\nu \sim \frac{n-1}{n} \frac{A_{n,2,-}}{(\log \varepsilon^{-1})^{\gamma_{n,2}^-}}, \quad (21.307)$$

$$\frac{1}{\chi(\nu; n)^2} \sum_{x_1, x_2 \in \mathbb{Z}^4} \langle \varphi_{x_1}^1 \varphi_{x_2}^1; (\varphi_a^2)^2 \rangle_\nu \sim -\frac{1}{n} \frac{A_{n,2,-}}{(\log \varepsilon^{-1})^{\gamma_{n,2}^-}}. \quad (21.308)$$

Thus the radial observable lies in the  $E^+$  direction, whereas the source-leading same/different component distinction for  $n \geq 2$  is governed by the more slowly decaying  $E^-$  direction.

*Proof.* Switch on only the  $a$  observable. The final-scale first derivative of the partition function equals the inductive-limit linear coordinate  $\lambda_{a,\infty}^*$  up to a remainder which vanishes by Theorem 10.228. Lemma 10.229 gives

$$\lambda_{a,\infty}^* = v_a^\pm (g_\infty/g_0)^{\gamma^\pm}.$$

The susceptibility change of variables already proved in Theorem 10.196 gives  $g_\infty \asymp (\log \varepsilon^{-1})^{-1}$  and converts the external-field derivatives into  $\chi^{-p}$  times the physical integrated correlations. Together with  $v_a^\pm = 1 + O(g_0)$  and  $g_0 = g(1 + O(g))$ , this gives the amplitude formula (21.304) and the WSAW formula (21.305).

For  $n \geq 1$ , the radial vector  $h^+ = n^{-1/2}(1, \dots, 1)$  gives (21.306). For  $n \geq 2$ , combine it with  $h^- = 2^{-1/2}(1, -1, 0, \dots, 0)$  exactly as in the proof of Theorem 10.232. Solving the two component equations gives an  $E^-$  contribution with coefficients  $(n-1)/n$  and  $-1/n$ , plus an  $E^+$  contribution of strictly smaller order because  $\gamma_{n,2}^- < \gamma_{n,2}^+$ . Removing that lower-order term in the asymptotic equivalence gives (21.307)–(21.308). This is precisely the source distinction between radial, same-component and different-component observables.  $\square$

**Proof inventory for the critical correlation family.** The explicit proof chain consists of the exact WSAW  $p$ -network representation; four observable sectors; coalescence-scale support; the perturbative matrix  $T_{n,p}$  and its  $E^\pm$  eigenvalues; the two regulator observable weights; preservation of  $h$ -factorisability through all six nonlinear maps; the complete nonperturbative  $\lambda$  and  $q$  flows; final-scale extraction; and the radial/traceless linear algebra. The theorem gives the critical correlation family, while the corollary records the source's near-critical integrated observable laws.

**Theorem 10.234** (Source-exact Slade–Tomberg critical correlation family). *Let  $d = 4$  and take the weak coupling sufficiently small in the rigorous RG regime. Put*

$$\gamma_{n,p}^+ = \binom{p}{2} \frac{n+2}{n+8}, \quad \gamma_{n,p}^- = \binom{p}{2} \frac{2}{n+8}, \quad b = \frac{n+8}{16\pi^2},$$

with  $\gamma_{0,p}^+ = \gamma_{0,p}^- = \frac{1}{4}\binom{p}{2}$ , and set

$$E_{ab}^{(p)} = \begin{cases} O((\log |a-b|)^{-1}), & p = 1, \\ O((\log \log |a-b|)(\log |a-b|)^{-1}), & p \geq 2. \end{cases}$$

There are positive amplitudes  $A'_{n,p,\pm}$  satisfying

$$A'_{n,p,\pm} = \frac{p!}{(2\pi)^{2p}} (bg)^{-2\gamma_{n,p}^\pm} (1 + O(g)) \quad (g \downarrow 0),$$

for which the source critical correlations have the following precise form as  $|a-b| \rightarrow \infty$ .

1. For the continuous-time WSAW ( $n = 0$ ), every integer  $p \geq 1$  has

$$W_{ab}^{(p)}(\nu_c) = \frac{A'_{0,p,+}}{|a-b|^{2p}(\log |a-b|)^{2\gamma_{0,p}^+}} (1 + E_{ab}^{(p)}).$$

In particular  $p = 1$  has the Gaussian  $|a-b|^{-2}$  law without a logarithm.

2. For the  $n$ -component  $|\varphi|^4$  model,  $n \geq 1$ ,

$$\langle \varphi_a^1; \varphi_b^1 \rangle_{\nu_c} = \frac{A'_{n,1,+}}{|a-b|^2} (1 + E_{ab}^{(1)}),$$

and

$$\langle |\varphi_a|^2; |\varphi_b|^2 \rangle_{\nu_c} = \frac{nA'_{n,2,+}}{|a-b|^4(\log |a-b|)^{2\gamma_{n,2}^+}} (1 + E_{ab}^{(2)}).$$

3. For  $n \geq 2$ , the square correlations of individual components are

$$\begin{aligned} \langle (\varphi_a^1)^2; (\varphi_b^1)^2 \rangle_{\nu_c} &= \frac{1}{n|a-b|^4} \left( \frac{(n-1)A'_{n,2,-}}{(\log |a-b|)^{2\gamma_{n,2}^-}} + \frac{A'_{n,2,+}}{(\log |a-b|)^{2\gamma_{n,2}^+}} \right) (1 + E_{ab}^{(2)}), \\ \langle (\varphi_a^1)^2; (\varphi_b^2)^2 \rangle_{\nu_c} &= \frac{1}{n|a-b|^4} \left( -\frac{A'_{n,2,-}}{(\log |a-b|)^{2\gamma_{n,2}^-}} + \frac{A'_{n,2,+}}{(\log |a-b|)^{2\gamma_{n,2}^+}} \right) (1 + E_{ab}^{(2)}). \end{aligned}$$

Thus for  $n \geq 2$  the same-component square correlation is eventually positive and the different-component one eventually negative, since  $\gamma_{n,2}^- < \gamma_{n,2}^+$ .

4. As  $\varepsilon = \nu - \nu_c \downarrow 0$ , the source near-critical observable theorem is also owned locally: the WSAW  $p$ -star law and the radial/same-component/different-component integrated square correlations are exactly those of Corollary 10.233, with  $A_{n,p,\pm} = p!(bg)^{-\gamma_{n,p}^\pm} (1 + O(g))$ .

The existence of the displayed infinite-volume critical correlations is part of the theorem in the RG thermodynamic-limit sense used in the source.

*Proof.* Items (1)–(3), including the component decomposition, error orders and small- $g$  amplitudes, are exactly Theorem 10.232. Item (4) is Corollary 10.233. The observable single-step theorem, coalescence-scale flow, plus/minus anomalous dimensions, final-scale extraction, supersymmetric  $p$ -watermelon representation and the infinite-volume limit are all proved in the preceding module.  $\square$

**Proof inventory for the four-dimensional RG block.** The preceding modules construct the finite-range covariance and polymer algebra, the corrected six-map nonlinear step in both regulator systems, the bosonic susceptibility, specific heat and finite-volume scaling limits, the supersymmetric WSAW representation and susceptibility flow, and the full observable/coalescence RG needed for the Slade–Tomberg critical correlation family. The source errata for the single-step map are built into the formulas used here. These results are therefore not treated as schematic applications of Lemmas 10.111–10.116; later arguments use the detailed modules listed above.

## finite-order correlation lengths beyond the mass scale

The purpose of this subsection is deliberately narrower than the foundational four-dimensional RG theorem. The earlier RG papers construct the progressive integration, the bulk and two-point observable coordinates, and the critical change of variables. The finite-order correlation-length theorem requires an additional ingredient: after the mass scale one must exploit the rapid decay of the massive covariance strongly enough to make the two-point remainder summable with an arbitrary spatial moment. We reconstruct that additional argument here and state its exact dependency boundary.

**Definition 10.235** (Mass and coalescence scales for this chapter argument). Fix an integer block factor  $L > 1$  and a mass  $m > 0$ . Put

$$j_m = \lfloor \log_L m^{-1} \rfloor.$$

For  $x \neq 0$ , let  $j_x$  be the unique integer for which  $|x| \asymp L^{j_x}$ . Let  $\bar{g}_j$  denote the perturbative running coupling on the high-temperature trajectory. The only properties of this sequence used below are

$$\bar{g}_j \asymp (1+j)^{-1} \quad (j \leq j_m), \quad \bar{g}_j \asymp \bar{g}_{j_m} \quad (j \geq j_m), \quad \bar{g}_{j_m} \asymp (\log m^{-1})^{-1}.$$

These are part of the relevant baseline RG interface: for  $n \geq 1$  the owner is **SM-21.03.01**, while for the supersymmetric  $n = 0$  weakly self-avoiding-walk branch the corresponding owner is **SM-21.04.01**.

**Lemma 10.236** (Mass-scale shell summation). *Let  $a > 0$ ,  $b \geq 0$  and  $s > a/2$ . Uniformly for  $m$  sufficiently small,*

$$\sum_{j \geq 1} L^{aj-2s(j-j_m)+} \bar{g}_j^b \leq C_{a,b,s,L} m^{-a} (\log m^{-1})^{-b}.$$

*Proof.* Split at  $j_m$ . For  $j \leq j_m$ , the ratio  $\bar{g}_j/\bar{g}_{j+1} = 1 + O(\bar{g}_0)$  is uniformly close to one. Hence, for  $\bar{g}_0$  sufficiently small,

$$\sum_{j \leq j_m} L^{aj} \bar{g}_j^b \leq CL^{aj_m} \bar{g}_{j_m}^b \leq C m^{-a} (\log m^{-1})^{-b}.$$

For  $j > j_m$ , use  $\bar{g}_j \leq C \bar{g}_{j_m}$  and write  $j = j_m + r$ :

$$\sum_{r \geq 1} L^{a(j_m+r)-2sr} \bar{g}_{j_m+r}^b \leq CL^{aj_m} \bar{g}_{j_m}^b \sum_{r \geq 1} L^{-(2s-a)r}.$$

The last series converges because  $2s > a$ . Since  $L^{j_m} \asymp m^{-1}$ , the assertion follows.  $\square$

**Proposition 10.237** (Free massive Green-function moments). *Let  $G_m(x) = (-\Delta_{\mathbb{Z}^d} + m^2)_{0x}^{-1}$  with  $d > 2$ , and let*

$$c_p^d = \int_{\mathbb{R}^d} |y|^p (-\Delta_{\mathbb{R}^d} + 1)_{0y}^{-1} dy, \quad p > 0.$$

Then

$$\sum_{x \in \mathbb{Z}^d} |x|^p G_m(x) = c_p^p m^{-(p+2)} (1 + o(1)) \quad (m \downarrow 0).$$

In particular the free order- $p$  correlation length satisfies  $\xi_p(0, m^2) = c_p m^{-1} (1 + o(1))$ .

*Proof.* Use the continuous-time random-walk representation

$$G_m(x) = \int_0^\infty e^{-m^2 t} p_t(0, x) dt,$$

where the normalization of the walk is the one whose diffusive limit has Green operator  $(-\Delta_{\mathbb{R}^d} + m^2)^{-1}$ . Tonelli gives

$$m^{p+2} \sum_x |x|^p G_m(x) = \int_0^\infty e^{-u} \mathbb{E}[|mX_{u/m^2}|^p] du. \quad (21.266)$$

The finite-range local central limit theorem proved in Volume V gives

$$mX_{u/m^2} \implies B_u \quad \text{for every fixed } u > 0.$$

We record explicitly the moment domination needed to pass to the weighted sum. Write the continuous-time walk as the compound-Poisson process  $X_t = \sum_{r=1}^{N_t} Y_r$ , where the jumps  $Y_r$  are centred and bounded by some  $R < \infty$ . For every unit vector  $v$  and  $|\theta| \leq (2R)^{-1}$ , Taylor's theorem and centering give

$$\mathbb{E} e^{\theta v \cdot Y_1} \leq 1 + C\theta^2, \quad \mathbb{E} e^{\theta v \cdot X_t} \leq e^{Ct\theta^2}.$$

Chernoff's bound, first with  $\theta = r/(2Ct)$  when this is admissible and then with the endpoint value  $(2R)^{-1}$ , yields

$$\mathbb{P}(|X_t| \geq r) \leq C_d \exp\{-c \min(r^2/t, r)\}.$$

The layer-cake identity  $\mathbb{E}|X_t|^q = q \int_0^\infty r^{q-1} \mathbb{P}(|X_t| > r) dr$  therefore gives, for every  $q > 0$ ,

$$\mathbb{E}|X_t|^q \leq C_q (1 + t^{q/2}).$$

Taking  $t = u/m^2$  shows  $\mathbb{E}|mX_{u/m^2}|^p \leq C_p (1 + u^{p/2})$ . Taking instead any  $q > p$  gives uniform integrability of the  $p$ th moments on every compact  $u$ -interval. Dominated convergence in (21.266) therefore yields

$$\lim_{m \downarrow 0} m^{p+2} \sum_x |x|^p G_m(x) = \int_0^\infty e^{-u} \mathbb{E}|B_u|^p du.$$

By the heat-kernel representation of the continuum resolvent and Tonelli, the right-hand side is exactly  $c_p^p$ . Finally  $\sum_x G_m(x) = m^{-2}$ , so division by the susceptibility and taking the  $p$ th root gives the last assertion.  $\square$

**Lemma 10.238** (Post-mass-scale covariance gain). *Assume the finite-range covariance decomposition and derivative estimates of the relevant baseline rigorous RG interface: SM-21.03.01 for  $n \geq 1$ , and SM-21.04.01 for the supersymmetric  $n = 0$  branch. For every fixed  $s > 0$  the field-size parameter may be chosen as*

$$\ell_j = \ell_0 L^{-j-s(j-j_m)_+}. \quad (21.267)$$

*Then, after increasing  $\ell_0$  by a constant depending on  $s$  and  $L$ , the normalized covariance at scale  $j$  satisfies the same small-covariance bound required by the one-step RG theorem, uniformly in  $j$ , and for  $j \geq j_m$  it gains at least the factor  $L^{-2(j-j_m)}$  relative to the critical estimate.*

*Proof.* The finite-range derivative estimate has the form, for every fixed integer  $k \geq 0$ ,

$$|\nabla^\alpha C_j(x, y)| \leq C_{k, \alpha} L^{-(d-2+|\alpha|)j} (1 + m^2 L^{2j})^{-k}. \quad (21.268)$$

Choose the integer  $k = \lceil s \rceil + 1$ , so that  $k - s \geq 1$ . Since  $mL^{j_m} \asymp 1$ ,

$$(1 + m^2 L^{2j})^{-k} \leq C_L L^{-2k(j-j_m)_+}. \quad (21.269)$$

The  $\Phi_j(\ell_j)$  normalization contributes  $\ell_j^{-2}$  and the canonical factor  $L^{-2j}$  in  $d = 4$ . Substitution of (21.267)–(21.269) cancels  $L^{2s(j-j_m)_+}$  from  $\ell_j^{-2}$  and leaves the extra factor  $L^{-2(k-s)(j-j_m)_+} \leq L^{-2(j-j_m)_+}$ . Thus the covariance norm is no larger than the small constant required in the baseline one-step theorem once  $\ell_0$  is chosen large enough. This is precisely the extra massive gain which is absent at criticality.  $\square$

**Lemma 10.239** (One-regulator domination beyond the mass scale). *Let  $G_j(X, \varphi)$  be the fluctuation-field regulator of the baseline RG construction, but with the parameter (21.267). If  $s > 1$ , then for every fixed  $q > 0$  there is  $L_0(q, s)$  such that, for  $L \geq L_0$  and  $j \geq j_m$ ,*

$$G_j(X, \varphi)^q \leq G_{j+1}(X, \varphi). \quad (21.270)$$

*Consequently the auxiliary large-field regulator is unnecessary above the mass scale in the observable estimate.*

*Proof.* For a scale- $j$  block inside a scale- $(j+1)$  block, the definition of the field norm and (21.267) give

$$\|\varphi\|_{\Phi_j(\ell_j)} \leq L^{-1-s} \|\varphi\|_{\Phi_{j+1}(\ell_{j+1})} \quad (j \geq j_m). \quad (21.271)$$

The regulator exponent at scale  $j$  is averaged over  $L^4$  times as many sites when viewed inside a scale- $(j+1)$  block. Hence (21.270) follows from

$$qL^{-2-2s} \leq L^{-4},$$

which holds for  $s > 1$  once  $L$  is large enough depending on  $q$ . Localizing the field norm to block neighborhoods and taking the infimum over extensions does not change (21.271), so the same estimate holds for arbitrary polymers  $X$ .  $\square$

**Proposition 10.240** (Stability of the baseline RG step under the improved norm). *Assume the full  $s = 0$  polymer/observable one-step theorem supplied by the relevant model-foundation owner, namely SM-21.03.01 for  $n \geq 1$  and SM-21.04.01 for the supersymmetric  $n = 0$  branch. Fix  $s > 1$ . With (21.267), and with observable parameter*

$$\ell_{\sigma, j} = \ell_{j \wedge j_x}^{-1} 2^{(j-j_x)_+} \tilde{g}_j, \quad (21.272)$$

*the same one-step theorem remains valid at all scales, after the following three explicit modifications above  $j_m$ :*

1. use only the  $G_j(\ell_j) \rightarrow G_{j+1}(\ell_{j+1})$  norm pair;
2. shrink the admissible  $q$ -coordinate by the factor  $L^{-2s(j_x-j_m)_+}$ ;
3. choose the finite differentiability parameter of the  $T_\varphi$  norm larger than a constant depending on  $s$ .

All bulk coordinates and the  $\lambda$ -observable stability domain remain unchanged. Constants may depend on  $s$ ; in particular the allowed smallness of the initial coupling need not be uniform in  $s$ .

*Proof.* We check the only places in the baseline proof where the changed norm enters. First,

$$\frac{\ell_{j+1}}{\ell_j} = L^{-1-s\mathbf{1}_{j \geq j_m}},$$

so every estimate which used the decrease of the bulk field parameter is improved. The product  $\ell_j \ell_{\sigma,j}$  is the same as in the old  $s = 0$  choice up to the coalescence bookkeeping; therefore the norm of the linear observables and the admissible  $\lambda$ -domain do not deteriorate.

Second, the perturbative increment of the quadratic observable  $q$  contains a covariance value at the next scale. Lemma 10.238 replaces its old  $L^{-2j}$  bound by  $L^{-2j-2s(j-j_m)_+}$ , exactly compensating the enlarged value of  $\ell_{\sigma,j}^2$ . Thus the shrunken  $q$ -domain is invariant under the perturbative flow.

Third, every nonlinear Gaussian-integration estimate in which the old proof needed both a small-field norm and a large-field norm can, above  $j_m$ , be closed using Lemma 10.239: any fixed power of the incoming fluctuation regulator is absorbed by the outgoing regulator. The localization remainder has at least one field coordinate or spatial derivative. Replacing  $\ell_j$  by (21.267) therefore yields an additional factor  $L^{-s}$  per post-mass-scale RG step. In the crucial contraction this extra dimensional gain offsets the possible growth of (21.272); choosing the Taylor/localization order  $A \geq 5 + s$  gives the same  $L^{-5}$  bound for zero observable legs and  $L^{-1}$  for one or two observable legs as in the baseline contraction. Taking the differentiability parameter larger than  $A$  makes all Taylor remainders legal.

No other part of the one-step proof depends on the explicit value of  $\ell_j$ . Thus the baseline theorem propagates with the three displayed modifications. Notice the logical boundary: this proposition proves the *extension* of the rigorous RG theorem to the stronger post-mass-scale norm; it does not replace the foundational construction of the baseline polymer maps. That foundational task belongs to SM-21.03.01 for the bosonic  $n \geq 1$  model and to SM-21.04.01 for supersymmetric WSAW.  $\square$

**Proposition 10.241** (Summable two-point remainder with arbitrary moment). *Under the hypotheses of Proposition 10.240, the infinite-volume two-point function admits*

$$\frac{G_{g,\nu}(0,x)}{1+z_0} = (1 + O(\bar{g}_{j_x}))G_m(x) + R_x, \quad (21.273)$$

and, for every fixed  $s \geq 0$ ,

$$|R_x| \leq \frac{C_s \bar{g}_{j_x}}{|x|^2} \begin{cases} 1, & m|x| \leq 1, \\ (m|x|)^{-2s}, & m|x| \geq 1. \end{cases} \quad (21.274)$$

Consequently, for every  $p > 0$  and every  $s > (p+2)/2$ ,

$$m^2 \sum_x |x|^p |R_x| = O(m^{-p}(\log m^{-1})^{-1}). \quad (21.275)$$

*Proof.* In the baseline two-point observable flow, the two linear insertions evolve independently until their coalescence scale. After coalescence, the quadratic observable  $q$  receives the free covariance contribution plus a nonperturbative remainder. Proposition 10.240 gives at scale  $i \geq j_x$

$$|R_i^q| \leq C \ell_{\sigma,i}^{-2} \bar{g}_i^3.$$



Substitute (21.272). Since  $\tilde{g}_i \asymp \bar{g}_i$  and the factor  $2^{(i-j_x)+}$  is squared in the denominator,

$$\sum_{i \geq j_x} |R_i^q| \leq CL^{-2j_x-2s(j_x-j_m)+} \sum_{i \geq j_x} 4^{-(i-j_x)} \bar{g}_i \leq C\bar{g}_{j_x} L^{-2j_x-2s(j_x-j_m)+}.$$

Now  $L^{-j_x} \asymp |x|^{-1}$  and  $L^{-(j_x-j_m)+} \leq C \min\{1, (m|x|)^{-1}\}$ , proving (21.274), while the perturbative covariance part gives (21.273).

For (21.275), decompose  $\mathbb{Z}^4$  into shells with  $|x| \asymp L^j$ . There are  $O(L^{4j})$  sites in the  $j$ th shell, and (21.274) gives

$$\sum_x |x|^p |R_x| \leq C \sum_{j \geq 1} L^{(p+2)j-2s(j-j_m)+} \bar{g}_j.$$

Apply Lemma 10.236 with  $a = p + 2$  and  $b = 1$ , then multiply by  $m^2$ .  $\square$

**Proposition 10.242** (Assembly of the order- $p$  correlation length). *Assume the baseline RG interface in the relevant model—SM-21.03.01 for  $n \geq 1$  and SM-21.04.01 for the supersymmetric  $n = 0$  branch—including its critical change of variables, so that for  $\varepsilon = \nu - \nu_c \downarrow 0$*

$$\chi(g, \nu; n) = \frac{1 + z_0}{m^2}, \quad m^2 \sim \tilde{A}_{g,n}^{-1} \varepsilon (\log \varepsilon^{-1})^{-(n+2)/(n+8)}. \quad (21.276)$$

Then for every fixed  $p > 0$ ,

$$\xi_p(g, \nu_c + \varepsilon; n) \sim c_p \tilde{A}_{g,n}^{1/2} \varepsilon^{-1/2} (\log \varepsilon^{-1})^{\frac{1}{2} \frac{n+2}{n+8}}. \quad (21.277)$$

The statement holds for  $n \geq 1$  in the weakly coupled  $|\varphi|^4$  model and for  $n = 0$  in its supersymmetric weakly self-avoiding-walk representation, with  $L$  and the small-coupling threshold allowed to depend on  $p$  and  $n$ .

*Proof.* From (21.273), the definition of  $\xi_p$ , and the first identity in (21.276),

$$\xi_p^p = m^2 \sum_x |x|^p (G_m(x) + r_x), \quad r_x = O(\bar{g}_{j_x}) G_m(x) + R_x. \quad (21.278)$$

Proposition 10.237 gives the main term  $c_p^p m^{-p}(1 + o(1))$ .

For the first error in (21.278), split at  $j_m$ . When  $j_x \leq j_m$ ,  $\bar{g}_{j_x} \leq C/(1 + j_x)$  and  $G_m(x) \leq G_0(x) \leq C|x|^{-2}$ ; summing shells up to  $|x| \asymp m^{-1}$  gives  $O(m^{-(p+2)}(\log m^{-1})^{-1})$ . When  $j_x > j_m$ , use  $\bar{g}_{j_x} \leq C(\log m^{-1})^{-1}$  and Proposition 10.237. After multiplication by  $m^2$ , both parts are  $o(m^{-p})$ .

The  $R_x$  contribution is  $o(m^{-p})$  by (21.275). Hence  $\xi_p^p \sim c_p^p m^{-p}$ . Taking the  $p$ th root and substituting the second identity in (21.276) proves (21.277).  $\square$

**Dependency statement.** The free moments, the strengthened massive covariance norm, the post-mass-scale one-regulator argument, the stability extension, the summable observable remainder, and the weighted-moment assembly are proved above. The remaining inputs are the model-specific foundational RG construction and critical change of variables: the bosonic  $n \geq 1$  branch uses the bosonic RG module, while the supersymmetric  $n = 0$  WSAW branch uses the WSAW module. Both are supplied earlier in this chapter, with each dependency restricted to its corresponding model class.

**Theorem 10.243** (All finite-order correlation lengths). *Let  $d = 4$ ,  $n \geq 0$ , and  $p > 0$ . For  $L$  sufficiently large depending on  $p, n$ , and for sufficiently small coupling  $g > 0$  depending on  $p, n$ , let  $\nu_c = \nu_c(g; n)$  be the critical point. Then, as  $\varepsilon \downarrow 0$ ,*

$$\xi_p(g, \nu_c + \varepsilon; n) \sim c_p \tilde{A}_{g,n}^{1/2} \varepsilon^{-1/2} (\log \varepsilon^{-1})^{\frac{1}{2} \frac{n+2}{n+8}}.$$

Here  $n = 0$  denotes the continuous-time weakly self-avoiding walk through its supersymmetric representation, and  $n \geq 1$  denotes the weakly coupled  $n$ -component  $|\varphi|^4$  model. The positive constant  $c_p$  is the order- $p$  massive free-field moment constant from Proposition 10.237.

*Proof.* This is now unconditional. Proposition 10.242 contains the entire finite-order moment argument: massive free-field moments, the strengthened post-mass-scale norm, regulator domination, the one-step stability extension, the summable observable remainder, and the weighted moment assembly. Its two model-specific inputs are the bosonic RG theorem and the supersymmetric WSAW theorem, proved in this chapter by Theorem 10.188 and Theorem 10.219. Hence the exponent  $\frac{1}{2} \frac{n+2}{n+8}$  follows with no remaining conditional premise.  $\square$

#### 10.4.10 Contact self-attraction as a controlled initial polymer coordinate

This subsection closes the remaining source-specific input behind SM-21.S2.06. The point is not to assign a formal negative RG dimension to the contact interaction. The contact interaction is converted, at scale zero, into an explicit polymer coordinate. We prove that coordinate lies in the same two-regulator Banach space used by the nonlinear RG theorem, including the parameter regularity needed to determine the critical surface. After that initialisation the second-order local flow is exactly the already proved weakly self-avoiding-walk flow.

**Definition 10.244** (Contact local-time energy and field parameters). For a continuous-time simple random walk let

$$I_T = \sum_x (L_T^x)^2, \quad C_T = \sum_x \sum_{e \in \mathcal{U}} L_T^x L_T^{x+e},$$

and set

$$U_{\beta, \gamma, T} = \beta I_T - \frac{\gamma}{2d} C_T. \tag{21.CA.1}$$

For a field  $f$  of finite support,

$$U_{\beta, \gamma}(f) = (\beta - \gamma) \sum_x f_x^2 + \frac{\gamma}{4d} \sum_x \sum_{e \in \mathcal{U}} |\nabla^e f_x|^2. \tag{21.CA.2}$$

Given  $z_0 > -1$  and  $m^2 > 0$ , define

$$g_0 = (\beta - \gamma)(1 + z_0)^2, \quad \nu_0 = \nu(1 + z_0) - m^2, \quad \gamma_0 = \frac{\gamma}{4d}(1 + z_0)^2. \tag{21.CA.3}$$

**Lemma 10.245** (Energy identity and elementary critical-point comparison). *If  $|\gamma| < \beta$ , then (21.CA.2) holds. Consequently, for  $\gamma \geq 0$ ,*

$$U_{\beta - \gamma, 0, T} \leq U_{\beta, \gamma, T} \leq U_{\beta, 0, T}, \tag{21.CA.4}$$

whereas the inequalities reverse for  $\gamma < 0$ . In particular the critical point  $\nu_c(\beta, \gamma)$  is finite whenever  $\beta > 0$  is sufficiently small and  $|\gamma| < \beta$ .

*Proof.* Using

$$\sum_x \sum_e f_x f_{x+e} = 2d \sum_x f_x^2 - \frac{1}{2} \sum_x \sum_e |\nabla^e f_x|^2$$

in the definition of  $U_{\beta,\gamma}$  gives (21.CA.2). If  $\gamma \geq 0$ , its gradient term is nonnegative and  $0 \leq \gamma < \beta$ , giving (21.CA.4); for negative  $\gamma$  the same calculation reverses both bounds. Exponentiating, integrating in time, and using the already proved finiteness of the weakly self-avoiding-walk critical point gives the last assertion.  $\square$

**Lemma 10.246** (Supersymmetric representation with contact interaction). *On a finite torus, with*

$$\tau_x = \phi_x \bar{\phi}_x + \psi_x \bar{\psi}_x, \quad \tau_{\Delta,x} = \frac{1}{2} (\phi_x (-\Delta \bar{\phi})_x + (-\Delta \phi)_x \bar{\phi}_x + \psi_x (-\Delta \bar{\psi})_x + (-\Delta \psi)_x \bar{\psi}_x),$$

one has

$$G_{N,\beta,\gamma,\nu}(a,b) = \int e^{-\sum_x \{U_{\beta,\gamma}(\tau) + \nu \tau_x + \tau_{\Delta,x}\}} \bar{\phi}_a \phi_b. \quad (21.CA.5)$$

After the field rescaling by  $(1+z_0)^{1/2}$  and the parameter change (21.CA.3), this is

$$G_{N,\beta,\gamma,\nu}(a,b) = (1+z_0) \mathbb{E}_C [Z_0 \bar{\phi}_a \phi_b], \quad C = (-\Delta + m^2)^{-1}, \quad (21.CA.6)$$

with

$$Z_0 = \prod_x e^{-(V_{0,x}^+ + \gamma_0 U_x^+)}, \quad V_{0,x}^+ = g_0 \tau_x^2 + \nu_0 \tau_x + z_0 \tau_{\Delta,x}, \quad U_x^+ = |\nabla \tau_x|^2. \quad (21.CA.7)$$

Moreover

$$\chi_N = (1+z_0) \hat{\chi}_N, \quad \hat{\chi}_N = m^{-2} + m^{-4} |\Lambda|^{-1} D^2 Z_{N,\emptyset}(0,0; \mathbf{1}, \mathbf{1}). \quad (21.CA.8)$$

*Proof.* Lemma 10.212 applies to any sufficiently integrable function of the local-time field. Apply it to  $e^{-U_{\beta,\gamma}(L_T)}$  and then integrate the killing factor in  $T$ . This gives (21.CA.5). Substitute the energy identity (21.CA.2), rescale all boson and fermion fields by  $(1+z_0)^{1/2}$ , and insert and subtract the mass term  $m^2 \tau$ . The quadratic part becomes the Gaussian superaction with covariance  $C$  and the remaining coefficients are precisely (21.CA.3), proving (21.CA.6)–(21.CA.7). Summing the marked endpoint over the torus and differentiating the zero-degree component in the constant field direction gives (21.CA.8).  $\square$

**Definition 10.247** (Positive and negative contact coordinates). For  $\gamma_0 \geq 0$  set

$$I_{0,x}^+ = e^{-V_{0,x}^+}, \quad K_{0,x}^+ = I_{0,x}^+ (e^{-\gamma_0 U_x^+} - 1). \quad (21.CA.9)$$

For  $\gamma_0 < 0$  write

$$V_{0,x}^- = V_{0,x}^+ + 4d\gamma_0 \tau_x^2, \quad U_x^- = 2 \sum_{e \in \mathcal{U}} \tau_x \tau_{x+e}, \quad (21.CA.10)$$

and put

$$I_{0,x}^- = e^{-V_{0,x}^-}, \quad K_{0,x}^- = I_{0,x}^- (e^{-|\gamma_0| U_x^-} - 1). \quad (21.CA.11)$$

For a polymer  $X$ , set  $K_0^\pm(X) = \prod_{x \in X} K_{0,x}^\pm$ . Then  $Z_0 = I_0^\pm \circ K_0^\pm$  in the corresponding sign sector.

**Lemma 10.248** (Local contact monomial bound). *For either sign and either field scale  $\mathfrak{h}_0 \in \{\ell_0, h_0\}$ ,*

$$\|U_x^\pm\|_{T_\phi(\mathfrak{h}_0)} \leq 2(U_x^\pm)_\emptyset + C \mathfrak{h}_0^4 (1 + \|\phi\|_{\Phi_x(\mathfrak{h}_0)}^2). \quad (21.CA.12)$$

Consequently, for  $V_0^\pm$  in the scale-zero interaction domain and  $|\gamma_0| \leq g_0$ ,

$$\|K_{0,x}^\pm\|_{G_0} \leq C|\gamma_0|, \quad \|K_{0,x}^\pm\|_{\tilde{G}_0} \leq C|\gamma_0|/g_0. \quad (21.CA.13)$$

*Proof.* The fields  $U_x^+$  and  $U_x^-$  are quartic. Expanding  $\tau = |\phi|^2 + \psi\bar{\psi}$ , applying the  $T_\phi$  product property from Lemma 10.98, and using  $2|ab| \leq |a|^2 + |b|^2$  gives (21.CA.12) term by term. For the activity write

$$e^{-|\gamma_0|U} - 1 = - \int_0^1 |\gamma_0| U e^{-t|\gamma_0|U} dt.$$

The interaction stability estimate of Proposition 10.179 controls  $e^{-V_0^\pm}$  in both regulators. In the fluctuation-field scale  $\ell_0 = O(1)$ , formula (21.CA.12) therefore supplies one factor  $|\gamma_0|$ . In the large-field scale,  $h_0^4 \asymp g_0^{-1}$  and the same argument gives the second estimate of (21.CA.13). The sign change has already been absorbed in (21.CA.10), so no positivity assumption on  $\gamma$  is used in the norm estimate.  $\square$

**Proposition 10.249** (Scale-zero polymer bound and parameter regularity). *Fix the large-set exponent  $a$  used in the  $\mathcal{W}_0$  norm. There is  $a' > a$  such that, whenever*

$$|\gamma_0| \leq c g_0^{1+a'}, \quad (21.CA.14)$$

*one has*

$$\|K_0^\pm\|_{\mathcal{W}_0} \leq C|\gamma_0|. \quad (21.CA.15)$$

*The map  $(g_0, \gamma_0, \nu_0, z_0) \mapsto K_0$  is continuous, is  $C^1$  in  $(g_0, \nu_0, z_0)$ , is  $C^1$  in all variables away from  $\gamma_0 = 0$ , and has left and right  $\gamma_0$  derivatives at zero. If the contact window is shrunk to*

$$|\gamma_0| \leq c g_0^5, \quad (21.CA.16)$$

*then*

$$\|K_0\|_{\mathcal{W}_0} + \|\partial_{\nu_0} K_0\|_{\mathcal{W}_0} + \|\partial_{z_0} K_0\|_{\mathcal{W}_0} + g_0 \|\partial_{g_0} K_0\|_{\mathcal{W}_0} \leq C g_0^3. \quad (21.CA.17)$$

*Proof.* For a connected polymer  $X$ , the product property and Lemma 10.248 give

$$\|K_0^\pm(X)\|_{\mathcal{G}_0} \leq (C|\gamma_0| \mathfrak{h}_0^4)^{|X|}. \quad (21.CA.18)$$

Split the exponent at the small-set size  $2^d$ . In the  $G_0$  norm,  $\ell_0 = O(1)$  and (21.CA.14) makes the large-polymer tail dominate the weight  $g_0^{a(|X|-2^d)_+}$ . In the  $\tilde{G}_0$  norm,  $h_0^4 \asymp g_0^{-1}$ , so the same conclusion follows from  $a' > a$ ; the additional  $g_0^{9/4}$  in the definition of  $\mathcal{W}_0$  absorbs the single-site large-field loss. This proves (21.CA.15).

For a parameter  $t \in \{g_0, \nu_0, z_0\}$ , differentiation of a singleton has the form

$$\partial_t K_{0,x} = -M_{t,x} K_{0,x}, \quad \partial_t^2 K_{0,x} = M_{t,x}^2 K_{0,x},$$

with  $\deg M_{t,x} \leq 4$ . Differentiation in  $\gamma_0$  gives instead  $-U_x^\pm e^{-V_0^\pm - |\gamma_0|U_x^\pm}$  and its square on each open sign sector. Lemma 10.248, followed by the polymer product rule, bounds both first and second derivatives in  $\mathcal{W}_0$ . Taylor's formula in the Banach space then proves the asserted  $C^1$  properties and the one-sided derivatives at zero. Each differentiation costs at worst  $h_0^4 \asymp g_0^{-1}$ ; two field factors in the  $\nu_0$  or  $z_0$  derivative cost at worst  $h_0^8 \asymp g_0^{-2}$  before the large-field weight is restored. The stronger window (21.CA.16) therefore makes every quantity in (21.CA.17)  $O(g_0^3)$  after possibly reducing  $c$ .  $\square$

**Proposition 10.250** (Self-consistent critical contact coordinates). *There is a continuous positive function  $r(g)$ , with  $r(g) \leq c g^5$ , and continuous functions  $\hat{\nu}_0^c, \hat{z}_0^c$  for  $0 \leq m^2, g_0 \leq \delta$  and  $|\gamma_0| \leq r(g_0)$  such that*

$$\begin{aligned} \hat{\nu}_0^c &= \nu_0^c(m^2, \hat{g}_0, K_0(g_0, \gamma_0, \hat{\nu}_0^c, \hat{z}_0^c)), \\ \hat{z}_0^c &= z_0^c(m^2, \hat{g}_0, K_0(g_0, \gamma_0, \hat{\nu}_0^c, \hat{z}_0^c)), \end{aligned} \quad (21.CA.19)$$

where

$$\widehat{g}_0 = g_0 + 4d\gamma_0 \mathbf{1}_{\{\gamma_0 < 0\}}, \quad (21.CA.20)$$

and  $(\nu_0^c, z_0^c)$  are the stable coordinates of the already proved RG flow. Uniformly in  $m^2$ ,

$$|\widehat{\nu}_0^c| + |\widehat{z}_0^c| \leq Cg_0. \quad (21.CA.21)$$

The functions are  $C^1$  away from  $\gamma_0 = 0$  and have one-sided first derivatives there.

*Proof.* Define

$$F = (\nu_0, z_0) - (\nu_0^c, z_0^c)(m^2, \widehat{g}_0, K_0(g_0, \gamma_0, \nu_0, z_0)).$$

Proposition 10.249 puts its last argument in the  $O(g_0^3)$  stable ball of Proposition 10.115. At  $\gamma_0 = 0$  we have  $K_0 = 0$  and  $\partial_{\nu_0} K_0 = \partial_{z_0} K_0 = 0$ , hence  $D_{(\nu_0, z_0)} F = I_2$ . Apply the ordinary implicit-function theorem on each open sign sector. The derivative bounds in Proposition 10.249 are uniform as  $\gamma_0 \rightarrow 0^\pm$ , so the two solutions extend continuously to the common  $\gamma_0 = 0$  solution and have one-sided derivatives there. Shrinking the positive function  $r$  makes the local solutions uniform in the compact mass interval. Finally the stable-coordinate estimates  $\nu_0^c, z_0^c = O(g_0) + O(\|K_0\|)$  and (21.CA.17) give (21.CA.21).  $\square$

**Proposition 10.251** (Physical parameter inversion). *After reducing  $\beta_* > 0$  there is a positive function  $\gamma_* : (0, \beta_*) \rightarrow (0, \infty)$  such that for  $0 < \beta < \beta_*$ ,  $|\gamma| < \gamma_*(\beta)$ , and small  $m^2 \geq 0$ , the system consisting of (21.CA.3) and the critical conditions (21.CA.19) has a unique nearby solution*

$$(g_0^*, \gamma_0^*, \nu_0^*, z_0^*, \nu^*),$$

continuous in  $(m^2, \beta, \gamma)$ , with

$$g_0^* = \beta + O(\beta^2), \quad |\nu_0^*| + |z_0^*| = O(\beta). \quad (21.CA.22)$$

One may choose  $\gamma_*(\beta) \leq c\beta^5$ .

*Proof.* First solve

$$f_1(\gamma_0) = \gamma_0 - \frac{\gamma}{4d}(1 + \widehat{z}_0^c(m^2, g_0, \gamma_0))^2 = 0.$$

At  $\gamma = 0$  its derivative in  $\gamma_0$  is 1, including from both sides. The preceding proposition therefore gives a unique nearby continuous solution  $\gamma_0^{(1)}(m^2, g_0, \gamma)$  with one-sided derivatives at zero. Next solve

$$f_2(g_0) = g_0 - (\beta - \gamma)(1 + \widehat{z}_0^c(m^2, g_0, \gamma_0^{(1)}))^2 = 0.$$

At  $(\gamma, g_0) = (0, \beta + O(\beta^2))$  its  $g_0$  derivative is  $1 + O(\beta)$ , so the implicit-function theorem applies again. Define  $\nu_0^*, z_0^*$  by Proposition 10.250 and set  $\nu^* = (\nu_0^* + m^2)/(1 + z_0^*)$ . Equations (21.CA.3) follow by construction, and (21.CA.22) follows from (21.CA.21). Taking  $\gamma_*(\beta)$  below the contact window in Proposition 10.249 completes the proof.  $\square$

**Theorem 10.252** (Contact-attraction universality engine). *Let  $d = 4$ . For  $0 < \beta < \beta_*$  and  $|\gamma| < \gamma_*(\beta)$  as in Proposition 10.251, the full RG trajectory exists at the critical coordinates. Its second-order local coordinate flow is independent of  $\gamma_0$ . Consequently:*

$$G_{\beta, \gamma, \nu_c}(0, x) = A_{\beta, \gamma} |x|^{-2} (1 + O((\log |x|)^{-1})), \quad (21.CA.23)$$

$$\chi(\beta, \gamma, \nu_c + \varepsilon) \sim B_{\beta, \gamma} \varepsilon^{-1} (\log \varepsilon^{-1})^{1/4}, \quad (21.CA.24)$$

$$\xi_p(\beta, \gamma, \nu_c + \varepsilon) \sim B_{\beta, \gamma}^{1/2} c_p \varepsilon^{-1/2} (\log \varepsilon^{-1})^{1/8} \quad (21.CA.25)$$

for every fixed  $p > 0$ , with the weak-coupling threshold reduced depending on  $p$ . Moreover, along the admissible small-coupling region,

$$A_{\beta,\gamma} = \frac{1}{4\pi^2}(1 + O(\beta)), \quad B_{\beta,\gamma} = \left(\frac{\beta}{2\pi^2}\right)^{1/4} (1 + O(\beta)) \quad (\beta \downarrow 0). \quad (21.CA.26)$$

The constants are positive and depend continuously on the small contact perturbation.

*Proof.* By Proposition 10.249, the only new scale-zero coordinate is an  $O(g_0^3)$  element of exactly the Banach space on which the nonlinear RG theorem acts. Proposition 10.250 therefore puts the trajectory on the same stable manifold. The RG map from scale one onward is the map of Theorem 10.188; the contact parameter is not an additional running local coupling. In particular its second-order map  $V_{\text{pt}}$  is unchanged. The supersymmetric loop cancellation of Lemma 10.216 therefore gives the same marginal coefficient and the same  $1/4$  mass-derivative exponent as for WSAW.

The susceptibility identity (21.CA.8), the critical-coordinate choice, and the argument of Theorem 10.219 give (21.CA.24) after the physical parameter inversion. Adding the two endpoint observables does not change  $K_0$ : the latter lies entirely in the bulk sector. Hence the observable RG theorem 10.232 applies with the same small initial bulk coordinate and yields (21.CA.23). Finally the weighted-moment assembly of Proposition 10.242 uses only the same massive covariance moment and uniform post-mass-scale bounds; its proof is stable under an  $O(g_0^3)$  initial bulk coordinate. This gives (21.CA.25).

It remains to identify the source-leading amplitudes. For the two-point function, the  $n = 0, p = 1$  case of Theorem 10.232 has  $\gamma_{0,1}^+ = 0$ , hence its Green-function amplitude is  $(2\pi)^{-2}(1 + O(g_0^*))$ . Proposition 10.251 gives  $g_0^* = \beta + O(\beta^2)$  and  $z_0^* = O(\beta)$ , so the field rescaling changes this only by  $1 + O(\beta)$ . This proves the first formula in (21.CA.26). For the susceptibility, Theorem 10.219 gives amplitude  $(bg_0^*)^{1/4}(1 + O(g_0^*))$  with  $b = (2\pi^2)^{-1}$ . The same parameter inversion therefore yields the second formula in (21.CA.26). Thus the contact perturbation changes the critical point and nonuniversal amplitudes but not the critical powers or logarithmic exponents, and its amplitudes have exactly the small- $\beta$  source asymptotics.  $\square$

**Theorem 10.253** (Source-exact four-dimensional WSAW with small contact self-attraction). *Consider the continuous-time weakly self-avoiding walk with self-repulsion  $\beta > 0$  and nearest-neighbour contact parameter  $\gamma \in \mathbb{R}$  in the source normalization*

$$U_{\beta,\gamma,T} = \beta I_T - \frac{\gamma}{2d} C_T, \quad d = 4.$$

*There exist  $\beta_* > 0$  and a positive function  $\gamma_* : (0, \beta_*) \rightarrow (0, \infty)$  such that, whenever  $0 < \beta < \beta_*$  and  $|\gamma| < \gamma_*(\beta)$ , the critical point  $\nu_c(\beta, \gamma)$  is finite and there are positive constants  $A_{\beta,\gamma}, B_{\beta,\gamma}$  for which:*

1.

$$G_{\beta,\gamma,\nu_c}(0, x) = A_{\beta,\gamma} |x|^{-2} (1 + O((\log |x|)^{-1})), \quad |x| \rightarrow \infty,$$

with

$$A_{\beta,\gamma} = \frac{1}{4\pi^2}(1 + O(\beta)) \quad (\beta \downarrow 0);$$

2. as  $\varepsilon \downarrow 0$ ,

$$\chi(\beta, \gamma, \nu_c + \varepsilon) \sim B_{\beta,\gamma} \varepsilon^{-1} (\log \varepsilon^{-1})^{1/4},$$

with

$$B_{\beta,\gamma} = \left(\frac{\beta}{2\pi^2}\right)^{1/4} (1 + O(\beta)) \quad (\beta \downarrow 0);$$

3. for every  $p > 0$ , after choosing  $\beta_*$  sufficiently small depending on  $p$ ,

$$\xi_p(\beta, \gamma, \nu_c + \varepsilon) \sim B_{\beta, \gamma}^{1/2} c_p \varepsilon^{-1/2} (\log \varepsilon^{-1})^{1/8},$$

where

$$c_p^p = \int_{\mathbb{R}^4} |x|^p (-\Delta_{\mathbb{R}^4} + 1)_{0x}^{-1} dx.$$

Thus sufficiently weak contact attraction—and also sufficiently weak contact repulsion—has the same leading four-dimensional critical powers and logarithmic corrections as the model with  $\gamma = 0$ , with the source’s small- $\beta$  amplitude asymptotics retained explicitly.

*Proof.* This is Theorem 10.252. The preceding analysis module reconstructs the contact-specific ingredients that are absent from the unperturbed manuscript: the supersymmetric contact representation, the  $K_0^\pm$  two-regulator estimates, the full polymer  $\mathcal{W}_0$  bound, parameter differentiability, self-consistent critical coordinates and the physical  $(\beta, \gamma)$  parameter inversion. The theorem now also derives  $A_{\beta, \gamma} = (4\pi^2)^{-1}(1 + O(\beta))$  from the  $p = 1$  observable Green-function amplitude and  $B_{\beta, \gamma} = (\beta/(2\pi^2))^{1/4}(1 + O(\beta))$  from the WSAW susceptibility amplitude together with  $g_0^* = \beta + O(\beta^2)$ . No external contact-attraction theorem is used as a proof step.  $\square$

## 10.5 Near-critical planar percolation from pivotal measures

**Theorem 10.254** (Near-critical continuum percolation in the triangular pivotal framework). *For site percolation on the triangular lattice of mesh  $\eta$ , write*

$$r(\eta) = \frac{\eta^2}{\alpha_4^\eta(\eta, 1)},$$

where  $\alpha_4^\eta(\eta, 1)$  is the critical alternating four-arm probability from scale  $\eta$  to scale 1. Couple the near-critical family monotonically at parameter speed  $r(\eta)$ . For every  $L < \infty$ , as  $\eta \downarrow 0$  the path

$$[-L, L] \ni \lambda \longmapsto \omega_\eta(\lambda)$$

converges in law in the Schramm–Smirnov quad-crossing Skorohod space to the unique-in-law càdlàg continuum near-critical process obtained from the critical scaling limit by Poisson marking of its pivotal measure. Every deterministic slice converges in the quad-crossing topology, and the critical configuration, pivotal measure, and every finite collection of near-critical states converge jointly.

*Proof.* This theorem is an exact backward reuse of the proof engine already owned in Volume XI. The pivotal-measure scaling limit, including measurability from the critical quad-crossing configuration and the required multi-arm  $L^2$  estimates, is proved in SM-11.03.03. The construction of the near-critical process from cutoff pivotal measures, Poisson marks, stability as the cutoff is removed, càdlàg control, and joint convergence for the coupled parameter family is proved in SM-11.07.01; its conformal covariance is then owned by SM-11.07.02. With the normalization  $r(\eta) = \eta^2/\alpha_4(\eta, 1)$ , the specialization to triangular site percolation is exactly the statement above. No new proof input is required in Volume XXI.  $\square$

## 10.6 Sub-ballistic SAW

This section proves the Duminil–Copin–Hammond theorem at its published  $\mathbb{Z}^d$  scope. The point of giving the bridge proof in some detail is that there are two logically separate mechanisms. The first converts a hypothetical ballistic tail into finite mean for Kesten’s irreducible-bridge law; the second proves that this mean is in fact infinite. Neither mechanism is a formal consequence of renewal theory alone.

Fix the second coordinate  $y(u) = u_2$  as the bridge direction and the first coordinate  $x(u) = u_1$  as a transverse direction. A bridge is a self-avoiding walk whose initial vertex is the unique vertex of minimal  $y$ -coordinate and whose final vertex has maximal  $y$ -coordinate. Write  $\text{SAB}_n$  for  $n$ -step bridges and  $\mu_c$  for the connective constant.

**Lemma 10.255** (Hammersley–Welsh bridge comparison and generating bound). *Let*

$$C(z) = \sum_{\gamma \in \text{SAW}} z^{|\gamma|}, \quad B(z) = \sum_{\gamma \in \text{SAB}} z^{|\gamma|},$$

where the trivial bridge is included in  $B$ . For  $0 < z < 1$  one has

$$C(z) \leq z^{-1} \exp\{2B(z) - 2\}.$$

Moreover, there is  $C_0 = C_0(d) < \infty$  such that

$$e^{-C_0\sqrt{n}} |\text{SAW}_n| \leq |\text{SAB}_n| \leq |\text{SAW}_n| \quad (n \geq 1).$$

Consequently the bridge and SAW generating functions have the same radius of convergence, namely  $\mu_c^{-1}$ .

*Proof.* We give the classical unfolding argument because the generating-function form is needed below, not only its coefficient consequence. Call a walk a half-space walk when its initial vertex is the unique vertex of minimal  $y$ -coordinate. Starting from such a walk, cut at the last occurrence of its highest level, reflect the remaining tail about that level, and repeat. The result is a concatenation of bridges whose positive vertical spans form a strictly decreasing sequence

$$a_1 > a_2 > \cdots > a_r > 0.$$

Conversely, the bridge pieces together with this span record reconstruct the half-space walk. If  $B_a(z)$  is the generating series of bridges of span  $a$ , the half-space generating series  $H(z)$  therefore satisfies

$$H(z) \leq \prod_{a \geq 1} (1 + B_a(z)) \leq \exp\left(\sum_{a \geq 1} B_a(z)\right) = \exp\{B(z) - 1\}.$$

For a general self-avoiding walk, cut at the last visit to its minimal  $y$ -level. Reading the part before the cut backwards and reflecting when necessary produces one half-space walk, and the part after the cut produces a second. Recording the joining edge gives the harmless factor  $z^{-1}$ , hence  $C(z) \leq z^{-1} H(z)^2$ , proving the displayed generating inequality.

For the coefficient estimate, the same successive reflections associate to an  $n$ -step walk a bridge together with at most two strictly decreasing span records. The total of all spans is at most  $n$ , so the number of possible records is bounded by the square of the partition number  $p(n)$ , and  $p(n) \leq e^{c\sqrt{n}}$ . All finite choices of initial orientation are absorbed into the constant. Thus every bridge has at most  $e^{C_0\sqrt{n}}$  preimages, while every bridge is itself a self-avoiding walk. Taking  $n$ th roots proves equality of the radii.  $\square$



A renewal point of a bridge  $\gamma$  is a vertex  $\gamma_i$  for which both  $\gamma_{[0,i]}$  and  $\gamma_{[i,|\gamma|]}$  are bridges. A non-trivial bridge with no interior renewal point is irreducible; let iSAB denote the irreducible bridges starting at the origin.

**Lemma 10.256** (Kesten irreducible-bridge law). *Every bridge decomposes uniquely into irreducible bridges, and*

$$\sum_{\gamma \in \text{iSAB}} \mu_c^{-|\gamma|} = 1.$$

Hence

$$\mathbb{P}_{\text{iSAB}}(\gamma) = \mu_c^{-|\gamma|}$$

defines a probability law on irreducible bridges.

*Proof.* Uniqueness follows by cutting at all renewal points. Put

$$I(z) = \sum_{\gamma \in \text{iSAB}} z^{|\gamma|}.$$

Unique concatenation gives  $B(z) = (1 - I(z))^{-1}$  for every  $0 < z < \mu_c^{-1}$ . Thus  $I(z) < 1$  there and monotone convergence gives  $I(\mu_c^{-1}) \leq 1$ .

It remains to exclude a strict inequality. By submultiplicativity of  $|\text{SAW}_n|$  and Fekete's lemma,  $|\text{SAW}_n| \geq \mu_c^n$  for every  $n$ , so  $C(z) \uparrow \infty$  as  $z \uparrow \mu_c^{-1}$ . The generating-function inequality of Lemma 10.255 then forces  $B(z) \uparrow \infty$  along the same limit. Since  $B(z) = (1 - I(z))^{-1}$ , we obtain  $I(z) \uparrow 1$ , and another application of monotone convergence proves the identity.  $\square$

We shall repeatedly use the following elementary counting device.

**Lemma 10.257** (Multi-valued map principle). *Let  $F : A \rightarrow 2^B$ . If  $|F(a)| \geq m$  for every  $a \in A$  and every  $b \in B$  belongs to at most  $M$  of the sets  $F(a)$ , then*

$$|A| \leq \frac{M}{m} |B|.$$

*Proof.* Count the pairs  $(a, b)$  with  $b \in F(a)$  first by  $a$  and then by  $b$ .  $\square$

## Ballistic bridges force finite mean

For  $v > 0$  set

$$\text{SAB}_{n,v} = \{\gamma \in \text{SAB}_n : y(\gamma_n) \geq vn\}.$$

The *ballistic bridge assumption* is

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}_{\text{SAB},n}(\text{SAB}_{n,v}) = 0 \quad (10.21)$$

for some  $v > 0$ . For a level  $h \in \mathbb{Z}$  let  $\mathcal{V}_{h,h+1}(\gamma)$  be the set of edges of  $\gamma$  crossing from height  $h$  to height  $h+1$  or conversely. For  $k \geq 1$  and  $\delta > 0$  write  $\text{SAB}_{n,v,\delta}^k$  for the bridges in  $\text{SAB}_{n,v}$  for which at least  $\delta n$  levels satisfy  $1 \leq |\mathcal{V}_{h,h+1}| \leq k$ .

**Lemma 10.258** (Zigzag unfolding). *Let  $\gamma \in \text{SAB}_n$ . A zigzag is a pair  $(i, j)$ ,  $i \leq j$ , for which  $i$  is the last time before  $j$  at which the running  $y$ -maximum is attained and  $j$  is the last time after  $i$  at which the subsequent  $y$ -minimum is attained. The central intervals of distinct zigzags are disjoint. Reflecting the central interval of one zigzag in the horizontal hyperplane through  $\gamma_i$  produces another bridge, creates renewal points at the two ends of the reflected interval, does not decrease the terminal height, and unfoldings of distinct zigzags commute.*

*Proof.* If two central intervals overlapped without one endpoint ordering the other, the defining last-maximum/last-minimum inequalities would be contradictory; hence the intervals are disjoint. On the reflected interval the vertical coordinate is sent from  $y$  to  $2y(\gamma_i) - y$ . The pieces before  $i$  and after  $j$  lie on the opposite sides of the two extremal levels by definition of zig and zag, so the reflected middle piece cannot meet them. These same strict inequalities show that the two joining vertices split the output into bridges and hence are renewal points. Reflection turns the vertical drop of the central piece into a rise, so terminal height cannot decrease. Since central intervals are disjoint, the reflections act on disjoint pieces and commute.  $\square$

**Proposition 10.259** (One-step crossing reduction, load-bearing form). *Fix  $v > 0$ ,  $k \geq 2$ ,  $\delta > 0$ , and a sequence  $u_m \rightarrow \infty$ . There are  $\delta' > 0$  and a subsequence  $n_m$  of  $u_m$  such that*

$$\limsup_{m \rightarrow \infty} \frac{1}{n_m} \log \frac{|\text{SAB}_{n_m, v, \delta}^k|}{|\text{SAB}_{n_m, v, \delta'}^{k-1}|} \leq 0.$$

*Equivalently, passage from crossing multiplicity  $k$  to  $k - 1$  costs only a subexponential factor along that subsequence.*

*Proof.* We follow the three exhaustive cases of the source proof.

*Case 1: many renewal points.* If along a subsequence a fixed positive fraction of  $\text{SAB}_{n, v, \delta}^k$  has at least  $\delta'n$  renewal points, each such renewal point supplies a level crossed exactly once. Therefore these bridges already lie in  $\text{SAB}_{n, v, \delta'}^{k-1}$ , and the required one-sided logarithmic comparison follows with only a constant-factor loss.

*Case 2: few zigzags.* Otherwise suppose, along a subsequence, a fixed positive fraction of the bridges have at most  $\varepsilon_n n$  zigzags with  $\varepsilon_n \downarrow 0$ . Unfold all zigzags. Whenever a level was crossed between 2 and  $k$  times, the unique zigzag whose central section contains a crossing of that level reduces the crossing number by at least two; subsequent unfoldings merely translate that level. Hence the output belongs to  $\text{SAB}_{n, v, \delta}^{k-1}$ . Given the output, the input is determined once the output vertices corresponding to all zig and zag endpoints are specified. There are at most  $2\varepsilon_n n$  such vertices, so the multiplicity is at most

$$\sum_{r \leq 2\varepsilon_n n} \binom{n}{r} = e^{o(n)}.$$

(The last estimate is the standard entropy bound, since  $\varepsilon_n \downarrow 0$ .) Lemma 10.257 gives the required subexponential comparison.

*Case 3: many zigzags and few renewal points.* We are left with a subsequence on which a fixed positive fraction of the bridges have at least  $2an$  zigzags but at most  $\varepsilon_n n$  renewal points, for some  $a > 0$  and  $\varepsilon_n \downarrow 0$ . Since zigzag central intervals are disjoint, at most  $an$  of them can have length at least  $L = \lceil a^{-1} \rceil$ . Consequently there are at least  $an$  short zigzags of length at most  $L$ . Choose  $b > 0$  small and unfold an arbitrary set of  $bn$  short zigzags. By Lemma 10.258, the map from the chosen subset of zigzags to the unfolded bridge is injective: from the output, the newly created zig/zag renewal vertices determine which commuting reflections were performed. Each choice creates at least  $bn$  renewal points, so every image lies in  $\text{SAB}_{n, v, b}^1$ , while the number of distinct images of one input is at least  $\binom{an}{bn}$ .

It remains to bound inverse multiplicity. A single unfolding of a zigzag of length at most  $L$  can create new renewal points only inside a vertical slab of width at most  $3L$ ; hence after  $bn$  unfoldings the image has at most  $(\varepsilon_n + 3Lb)n$  renewal points. To reconstruct the input, choose the  $bn$  zig

vertices from these renewal points and, for each, choose a zag at distance at most  $L$ . Thus the inverse multiplicity is at most

$$\binom{(\varepsilon_n + 3Lb)n}{bn} L^{bn}.$$

Choose  $b$  so small that, for all large  $n$ ,

$$\frac{a - b}{(\varepsilon_n + 3Lb)L} > 2.$$

The elementary inequality  $\binom{N_1}{m} / \binom{N_2}{m} \geq ((N_1 - m)/N_2)^m$  then shows that the ratio of the number of choices to the inverse multiplicity is at least  $2^{bn}$ . The multi-valued map principle gives an even stronger than subexponential comparison. The three cases are exhaustive, proving the claim.  $\square$

*Remark 10.260* (One-sided formulation). The source displays Proposition 3.2 with a limit equal to zero. The three case estimates, and the subsequent proof of its Theorem 3.1, use only the one-sided statement above: the  $(k - 1)$ -crossing class is not exponentially smaller than the  $k$ -crossing class. We state exactly that load-bearing conclusion rather than silently inferring the unused reverse comparison.

**Theorem 10.261** (Renewal density at subexponential cost). *Under (10.21), there is  $\delta > 0$  such that*

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}_{\text{SAB},n}(|R_\gamma| \geq \delta n) = 0.$$

*Proof.* A bridge in  $\text{SAB}_{n,v}$  crosses at least  $vn$  nonempty levels. At most  $vn/2$  of these can be crossed more than  $2/v$  times, because the walk has only  $n$  edges. Hence

$$\text{SAB}_{n,v} \subset \text{SAB}_{n,v,v/2}^{\lceil 2/v \rceil}.$$

Iterate Proposition 10.259 down from  $k = \lceil 2/v \rceil$  to  $k = 1$ , passing to nested subsequences. At  $k = 1$  every selected level is crossed exactly once and its lower endpoint is a renewal point. Combining the resulting  $e^{-o(n)}$  loss with (10.21) proves the assertion.  $\square$

Let  $\mathbb{P}_{\text{iSAB}}^{\otimes \mathbb{N}}$  be the law of the semi-infinite bridge obtained by concatenating independent irreducible bridges. Write  $\mathbf{r}_j$  for its  $j$ th renewal time.

**Lemma 10.262** (Finite bridge as conditioned renewal process). *For every  $n$ ,*

$$|\text{SAB}_n| \mu_c^{-n} = \mathbb{P}_{\text{iSAB}}^{\otimes \mathbb{N}}(n \in R_\gamma),$$

*and conditional on  $n \in R_\gamma$ , the segment  $\gamma_{[0,n]}$  is uniform on  $\text{SAB}_n$ .*

*Proof.* Sum the product weights  $\prod_j \mu_c^{-|\gamma^{[j]}|}$  over all finite sequences of irreducible bridges whose total length is  $n$ . Every bridge of length  $n$  occurs exactly once and has total weight  $\mu_c^{-n}$ . The same identity, with an indicator for a particular bridge, gives the conditional law.  $\square$

**Corollary 10.263** (Ballisticity implies finite irreducible mean). *Under (10.21),  $\mathbb{E}_{\text{iSAB}}|\gamma| < \infty$ .*

*Proof.* Assume instead that the mean is infinite. Given  $\delta > 0$ , choose  $K$  so that  $\mathbb{E} \min\{K, |\gamma|\} \geq 2/\delta$ . If the first  $\delta n$  renewal blocks fit inside time  $n$ , then

$$\sum_{j=1}^{\delta n} \min\{K, |\gamma^{[j]}|\} \leq n.$$

The summands are bounded i.i.d. variables with mean at least  $2/\delta$ , so a Chernoff bound gives probability at most  $e^{-cn}$ . By Lemma 10.262, conditioning on  $n$  being a renewal time costs the reciprocal of  $|\text{SAB}_n|\mu_c^{-n}$ . Lemma 10.255 and  $|\text{SAW}_n| \geq \mu_c^n$  show this cost is at most  $e^{C\sqrt{n}}$ . Therefore, for every  $\delta > 0$ ,

$$\mathbb{P}_{\text{SAB},n}(|R_\gamma| \geq \delta n) \leq e^{-cn+C\sqrt{n}},$$

contradicting Theorem 10.261.  $\square$

### Finite mean is impossible: diamond points and stickbreak

A renewal point  $\gamma_i$  is a *diamond point* if every future vertex lies in the northeast  $45^\circ$  cone based at  $\gamma_i$  and every past vertex lies in the southwest cone; equivalently,

$$(x+y)(\gamma_j), (y-x)(\gamma_j) \geq (x+y)(\gamma_i), (y-x)(\gamma_i) \quad (j \geq i)$$

and both inequalities are reversed for  $j \leq i$ .

**Lemma 10.264** (Bi-infinite renewal bridge). *The irreducible-bridge product law has a bi-infinite version  $\mathbb{P}_{\text{iSAB}}^{\otimes \mathbb{Z}}$  which is invariant and ergodic under the shift from one renewal point to the next. Moreover, after reflection in  $y = 0$ , the past and future half-bridges are independent and identically distributed.*

*Proof.* Index independent irreducible blocks by  $\mathbb{Z}$  and concatenate them with the renewal point between blocks 0 and 1 at the origin. The renewal shift is the Bernoulli shift on the block sequence, hence invariant and ergodic. Every irreducible bridge begins with a north edge. Removing that first edge leaves a law invariant under reflection in  $y = 0$  (followed by translation), because the Kesten weight depends only on length and reflection preserves the class of admissible residual pieces. Constructing the two sides from independent copies of this residual law gives the stated past–future symmetry.  $\square$

**Proposition 10.265** (Positive density of diamond points). *If  $\mathbb{E}_{\text{iSAB}}|\gamma| < \infty$ , then there is  $\delta > 0$  such that, under  $\mathbb{P}_{\text{iSAB}}^{\otimes \mathbb{N}}$ ,*

$$\liminf_{n \rightarrow \infty} \frac{|\text{D}_\gamma \cap [0, \mathbf{r}_n]|}{n} \geq \delta \quad \text{almost surely.}$$

*Proof.* Let  $T$  be an irreducible block. Reflection in the hyperplane  $x = 0$  gives  $\mathbb{E}x(T_{\text{end}}) = 0$ , while  $y(T_{\text{end}}) \geq 1$ . Integrability of  $|T|$  therefore permits the strong law at renewal points:

$$\frac{1}{n}(x(\gamma_{\mathbf{r}_n}), y(\gamma_{\mathbf{r}_n})) \longrightarrow (0, m), \quad m > 0.$$

Because the maximal excursion inside one block is at most its length and an integrable i.i.d. sequence satisfies  $\max_{j \leq n} |T_j|/n \rightarrow 0$ , the same asymptotic cone statement holds between renewal points. Hence the two random infima

$$\inf_{k \geq 0} (x+y)(\gamma_k), \quad \inf_{k \geq 0} (y-x)(\gamma_k)$$

are finite almost surely. Choose  $K$  so that the probability  $\rho_K$  that both are at least  $-K$  is positive. If the first  $2K$  irreducible blocks are single north edges and the remaining independent half-bridge realizes this  $-K$  cone event, then the whole future stays in the northeast cone. A single north edge has Kesten probability  $\mu_c^{-1}$ , so the probability of the cone event from the origin is at least  $\mu_c^{-2K} \rho_K > 0$ .

By Lemma 10.264, past and future have the required independent reflected laws. Thus the origin is a diamond point with some probability  $\delta > 0$ . Ergodicity of the renewal shift now gives density  $\delta$  of diamond points among renewal points in the bi-infinite bridge. Passing from the bi-infinite bridge to its future half removes constraints coming from the negative-time path and can therefore only create additional diamond points, not destroy the positive-time diamond points already present. The asserted lower density follows.  $\square$

For a finite bridge put

$$\text{width}(\gamma) = \max_{i,j} |x(\gamma_i) - x(\gamma_j)|.$$

**Lemma 10.266** (Stickbreak geometry). *Let  $p = \gamma_{d_i}$  and  $q = \gamma_{d_j}$  be diamond points of a finite bridge,  $i < j$ . Perform the stickbreak operation: insert a  $+e_1$  edge, rotate the segment  $\gamma_{[d_i, d_j]}$  clockwise by  $\pi/2$  in the  $(x, y)$ -plane about  $p$ , insert a second  $+e_1$  edge, and translate the final segment so that the pieces concatenate. Then the output is self-avoiding.*

*If the original bridge has transverse width at most  $w$ , if at least  $r_-$  diamond points occur before  $p$  and at least  $r_+$  occur after  $q$ , and if  $\min\{r_-, r_+\} > w$ , then the output is again a bridge. Moreover its transverse width is at least  $j - i$ .*

*Proof.* The diamond inequalities put the prefix before  $p$  in the southwest cone based at  $p$ , the portion from  $p$  to  $q$  in the intersection of the northeast cone of  $p$  and southwest cone of  $q$ , and the suffix after  $q$  in the northeast cone of  $q$ . After clockwise rotation, the middle portion lies in the southeast sector based at  $p$ . The two inserted east edges shift this rotated sector off the two untouched cone sectors. Hence nonconsecutive pieces are disjoint and the resulting walk is self-avoiding.

Write  $\tilde{\gamma}$  for the output. The suffix after  $q$  contains at least  $r_+$  further diamond points, and successive diamond points have strictly increasing  $y$ -coordinate. Thus its terminal height exceeds its height at the second inserted edge by at least  $r_+$ . The  $y$ -variation introduced by the rotated middle piece is exactly the  $x$ -variation of the original middle piece, hence at most  $w$ . Therefore, for every vertex  $k$  of the rotated piece,

$$y(\tilde{\gamma}_{\text{end}}) - y(\tilde{\gamma}_k) \geq r_+ - w > 0.$$

Likewise, the  $r_-$  diamond points before  $p$  put  $p$  at height at least  $r_-$  above the initial vertex. The same width bound on the rotated middle piece gives

$$y(\tilde{\gamma}_k) - y(\tilde{\gamma}_0) \geq r_- - w > 0$$

for every vertex of the rotated piece. On the untouched prefix and suffix these bridge inequalities already hold. Thus the initial vertex is the unique minimum and the endpoint is a maximum:  $\tilde{\gamma}$  is a bridge.

Finally, the  $j - i$  successive diamond intervals between  $p$  and  $q$  each increase  $y$  by at least one, so  $y(q) - y(p) \geq j - i$ . Rotation converts this vertical displacement into transverse displacement. Hence  $\text{width}(\tilde{\gamma}) \geq j - i$ .  $\square$

**Theorem 10.267** (Infinite mean of irreducible bridges).

$$\mathbb{E}_{\text{ISAB}}|\gamma| = \infty.$$

*Proof.* Assume the mean is finite. Let  $\delta$  be given by Proposition 10.265; choose  $0 < \varepsilon < \delta/20$  and  $\nu > \mathbb{E}|\gamma|$ . For the semi-infinite bridge let  $\phi^{(n)}$  be its truncation at the  $n$ th renewal point. The law

of large numbers, together with the zero transverse drift and Proposition 10.265, gives an event  $G_n$  whose probability tends to one and on which

$$\mathbf{r}_n \leq \nu n, \quad \text{width}(\phi^{(n)}) \leq \varepsilon n, \quad d_{\delta n/2}(\phi^{(n)}) \leq \mathbf{r}_n.$$

For completeness, the width estimate here is a genuine consequence of finite mean. The transverse displacements of successive renewal endpoints are centered integrable i.i.d. variables, so the strong law implies that the maximum absolute transverse endpoint displacement among the first  $n$  blocks is  $o(n)$ . The excursion inside block  $j$  is at most its length  $L_j$ ; integrability gives  $\max_{j \leq n} L_j/n \rightarrow 0$  almost surely by the tail-sum criterion and Borel–Cantelli. Hence  $\text{width}(\phi^{(n)})/n \rightarrow 0$  almost surely. Here  $d_j$  is the index of the  $j$ th diamond point.

Let

$$\mathcal{G}_n^+ = \{\phi^{(n)} : \phi \in G_n\}.$$

Membership in  $G_n$  is determined entirely by the truncation  $\phi^{(n)}$ ; therefore the renewal-prefix identity gives

$$\mathbb{P}(G_n) = \sum_{\gamma \in \mathcal{G}_n^+} \mu_c^{-|\gamma|}.$$

For every  $\gamma \in \mathcal{G}_n^+$  choose

$$i \in [\delta n/10, 2\delta n/10], \quad j \in [3\delta n/10, 4\delta n/10]$$

and perform the stickbreak operation. Since  $i \geq \delta n/10$ ,  $j - i \geq \delta n/10$ , and at least  $\delta n/10$  diamond points remain after  $d_j$ , while  $\text{width}(\gamma) \leq \varepsilon n < \delta n/10$ , Lemma 10.266 shows that the output is a bridge, has length  $|\gamma| + 2$ , and has width at least  $\delta n/10 > \varepsilon n$ . Moreover at least  $\delta n/10$  old diamond (hence renewal) points in the rotated portion are destroyed. A newly created renewal point inside the rotated portion must occur at a distinct output  $y$ -level, and the number of such levels is at most the  $y$ -range of the rotated piece, namely the  $x$ -width of  $\gamma_{[d_i, d_j]}$ , which is at most  $\varepsilon n$ . Thus the output has at most  $n + \varepsilon n - \delta n/10 \leq n$  renewal points before its endpoint.

If  $\eta \triangleleft \phi$  means that the finite bridge  $\eta$  is an initial segment ending at a renewal point of the infinite bridge, then Lemma 10.262 gives

$$\mathbb{P}(\eta \triangleleft \phi) = \mu_c^{-|\eta|}.$$

Sum this weight over all triples  $(i, j, \gamma)$  with  $\gamma \in \mathcal{G}_n^+$ . On one hand there are  $(\delta n/10)^2$  choices of  $(i, j)$  and stickbreak adds two edges, so the sum is

$$S = \left( \frac{\delta n}{10\mu_c} \right)^2 \mathbb{P}(G_n).$$

On the other hand, for a fixed output infinite bridge, the input triple is recovered once the positions of the two inserted east edges are known. Both positions occur before time  $\nu n + 2$ , so there are at most  $(2\nu n)^2$  preimages. Every such output has width  $> \varepsilon n$  by the preceding paragraph. Because it contains at most  $n$  renewal points, whenever it occurs as a renewal-prefix of an infinite bridge  $\phi$ , the  $n$ th renewal time of  $\phi$  is at or after the output endpoint. Consequently the truncation  $\phi^{(n)}$  contains the entire output and also has width  $> \varepsilon n$ . Therefore

$$S \leq (2\nu n)^2 \mathbb{P}(\text{width}(\phi^{(n)}) > \varepsilon n),$$

and hence

$$\mathbb{P}(\text{width}(\phi^{(n)}) > \varepsilon n) \geq \left( \frac{\delta}{20\nu\mu_c} \right)^2 \mathbb{P}(G_n).$$

The right side stays bounded away from zero. But the finite-mean width estimate proved above gives  $\text{width}(\phi^{(n)})/n \rightarrow 0$  almost surely, and hence in probability, a contradiction.  $\square$

**Theorem 10.268** (Self-avoiding walk is sub-ballistic). *For every  $d \geq 2$  and every  $v > 0$  there are  $c(v) > 0$  and  $n_0(v) < \infty$  such that*

$$\mathbb{P}_{\text{SAW},n} \left( \max_{0 \leq k \leq n} \|\omega_k\| \geq vn \right) \leq e^{-c(v)n} \quad (n \geq n_0(v)).$$

*Equivalently, there is  $C(v) < \infty$  for which the same probability is at most  $C(v)e^{-c(v)n}$  for every  $n \geq 1$ .*

*Proof.* Suppose the conclusion fails for some  $v > 0$ . By symmetry, along a subsequence one of the  $2d$  signed coordinate maxima is at least  $vn/\sqrt{d}$  with probability  $e^{-o(n)}$ . Apply the Hammersley–Welsh unfolding in that coordinate. It preserves length, does not decrease the attained coordinate maximum, and has multiplicity at most  $e^{C\sqrt{n}}$ . Since  $|\text{SAB}_n| \leq |\text{SAW}_n|$ , the resulting bridges satisfy the ballistic bridge assumption (10.21) with speed  $v/\sqrt{d}$ .

Corollary 10.263 then gives  $\mathbb{E}_{\text{ISAB}}|\gamma| < \infty$ , whereas Theorem 10.267 gives the opposite conclusion. This contradiction proves exponential sub-ballisticity.  $\square$

*Remark 10.269* (Source-normalization correction). The published theorem is printed with the pure bound  $e^{-\varepsilon n}$  for “each  $n$ ” and every  $v > 0$ . Taken literally this cannot hold for small  $n$ : if  $0 < v < 1$ , then for  $n = 1$  the event has probability one. The proof in the paper, and the contradiction argument above, give the intended eventual exponential estimate. The equivalent all- $n$  formulation includes a finite prefactor  $C(v)$ . No asymptotic consequence of sub-ballisticity is changed.

*Remark 10.270* (What is load-bearing). The proof is not “renewal theory plus a rotation”. The first hard owner is Proposition 10.259, where zigzag unfoldings and a multi-valued-map estimate convert ballistic height into renewal density at subexponential cost. The second hard owner is Theorem 10.267: finite mean creates a positive density of diamond points, and the weighted stickbreak count forces linear transverse width with positive probability, contradicting the same finite-mean law of large numbers.

## 10.7 Loop $O(n)$ and planar rotational invariance

**Lemma 10.271** (Spin representation and the Loop- $O(n)$  dichotomy). *Let  $n \geq 1$  and  $0 < x \leq n^{-1/2}$ . On a finite set  $G$  of faces of the triangular lattice, color the complementary faces by spins  $\sigma \in \{-1, +1\}$  and draw on the dual hexagonal lattice the interfaces between opposite spins. Following the source cluster representation, define  $k_G(\sigma) + 1$  to be the sum of the number of connected plus-spin components and the number of connected minus-spin components that intersect  $G$  or its nearest-neighbourhood, and let  $e_G(\sigma)$  be the number of disagreeing nearest-neighbour spin edges having at least one endpoint in  $G$ . Give  $\sigma$  weight*

$$n^{k_G(\sigma)} x^{e_G(\sigma)}.$$

*Then the push-forward by the interface map is the loop- $O(n)$  measure with the corresponding boundary condition. Moreover the spin measure satisfies strong FKG and comparison of boundary conditions. Consequently exactly one of the following alternatives holds for the infinite-volume periodic Gibbs state  $\mathbf{P}_{n,x}$ :*

(A1) there is  $c > 0$  such that the diameter  $R$  of the outermost loop surrounding the origin satisfies

$$\mathbf{P}_{n,x}(R \geq m) \leq e^{-cm} \quad (m \geq 1);$$

(A2) there is  $c > 0$  such that, for every  $m \geq 2$  and every exterior loop boundary condition  $\xi$ ,

$$c \leq \mathbf{P}_{A_m,n,x}^\xi(a \text{ loop in } A_m \text{ surrounds the origin}) \leq 1 - c, \quad (21.37)$$

where  $A_m$  is the hexagonal annulus between triangular-lattice balls of radii  $m$  and  $2m$ .

In case (A2), the Gibbs state is unique and almost surely contains infinitely many loops surrounding the origin.

*Proof.* For the push-forward identity, every spin interface is an even subgraph. Fix an interface configuration  $\omega$ . On the planar graph obtained by cutting along  $\omega$ , the spin is constant on each complementary component and adjacent components have opposite signs. The source normalization  $k_G(\sigma) + 1$  counts exactly these plus/minus components that meet the active region, so Euler separation shows that, after the boundary component is fixed, its excess over one is the number  $L_G(\omega)$  of interface loops meeting the active edge set. Moreover  $e_G(\sigma) = o_G(\omega)$ , the number of occupied interface edges meeting that set. Consequently summing over the two compatible spin colors of the outer component (or fixing it through the boundary condition) pushes the cluster weight to  $x^{o_G(\omega)} n^{L_G(\omega)}$ , exactly the finite-volume loop- $O(n)$  weight. This is why the “minus one” in the definition of  $k_G$  is essential; counting merely the clusters meeting  $G$  would introduce a boundary-dependent extra factor.

We next verify the monotonicity input rather than treating it as an Ising black box. It is enough to check the two-site lattice condition. Fix all spins except at two sites  $u, v$  and write  $\sigma^{\eta\eta'}$  for the four choices. If  $\Delta k$  and  $\Delta e$  denote respectively

$$k(\sigma^{++}) + k(\sigma^{--}) - k(\sigma^{+-}) - k(\sigma^{-+})$$

and the analogous quantity for interface edges, the logarithm of the FKG cross-ratio is  $(\log n)\Delta k + (\log x)\Delta e$ . There are three cases. If  $u, v$  are not neighbors, then  $\Delta e = 0$  and  $\Delta k \geq 0$ . If they are neighbors and their common neighboring faces have opposite spins, then  $\Delta e = -2$  and  $\Delta k \geq 0$ . Finally, if those common neighbors have the same spin, then  $\Delta e = -2$  and  $\Delta k \geq -1$ . The first two cases are nonnegative because  $n \geq 1$  and  $x \leq 1$ ; the last is bounded below by  $-\log(nx^2) \geq 0$ . This proves the FKG lattice condition. Freezing one or more exterior spins and applying the same lattice condition to the remaining sites gives Holley comparison between ordered boundary conditions. Thus the spin representation has strong FKG and boundary monotonicity.

The box-crossing argument is the one already proved in Volume IX, Chapter 5, with the model-specific hypotheses now checked. We recall why no random-cluster-only identity is used in that part. Explore an extremal plus crossing in a mixed-boundary rectangle. Conditional on the exploration, the unexplored region again has the same spin law with induced plus/minus boundary conditions (domain Markov property). Spin-flip symmetry turns the complementary crossing into the dual-sign crossing, and boundary comparison puts both mixed boundary conditions between the extremal plus and minus laws. The gluing steps in (9.20)–(9.24) of Volume IX therefore apply word for word: one obtains a fixed-aspect-ratio crossing seed, concatenates a bounded number of seeds by FKG, and derives the annulus recursion. If annular circuit probabilities are not bounded below, the recursion bootstraps to stretched exponential one-arm decay and then, by the convolution improvement of Volume IX, to exponential decay. Otherwise plus and minus crossings in every bounded-aspect-ratio rectangle are uniformly bounded away from zero and one; four such crossings produce the two-sided



annular estimate (21.37). The same annular circuits separate arbitrary exterior boundary conditions from a fixed inner region with probability tending to one along geometric scales, proving uniqueness of the Gibbs state. Finally, because (21.37) holds for every exterior boundary condition, after revealing successively from the outside on a sequence of disjoint geometric annuli the conditional probability of producing a surrounding loop in the next annulus is always at least  $c$ . Hence the probability that none of the next  $N$  annuli contains such a loop is at most  $(1-c)^N$ . Letting  $N \rightarrow \infty$  gives infinitely many surrounding loops almost surely. This proves the dichotomy.  $\square$

**Definition 10.272** (Boundary-source sums and relative path weights). Let  $\Omega$  be a finite honeycomb domain and let  $A \subset \partial\Omega$ . Write  $\mathcal{E}(\Omega, A)$  for subgraphs of  $\Omega$  having odd degree exactly at the vertices of  $A$ , and set

$$Z_\Omega^A := \sum_{\omega \in \mathcal{E}(\Omega, A)} x^{|\omega|} n^{\ell(\omega)}.$$

For a self-avoiding path  $\gamma \subset \Omega$ , remove the edges of  $\gamma$  and the other edges incident to its endpoints and define

$$W_\Omega(\gamma) := x^{|\gamma|} \frac{Z_{\Omega \setminus \gamma}^\emptyset}{Z_\Omega^\emptyset}. \quad (21.38)$$

For disjoint paths we use the same definition after removing all path edges and endpoint-adjacent edges. If  $a, b \in \partial\Omega$ , decomposition at the unique  $a$ - $b$  path gives

$$\frac{Z_\Omega^{\{a,b\}}}{Z_\Omega^\emptyset} = \sum_{\gamma \subset \Omega: a \rightarrow b} W_\Omega(\gamma). \quad (21.39)$$

The weights satisfy the chain rule  $W_\Omega(\gamma \circ \eta) = W_\Omega(\gamma)W_{\Omega \setminus \gamma}(\eta)$ .

**Lemma 10.273** (Boundary-source Catalan bound). Assume  $n \geq 1$  and  $x \leq n^{-1/2}$ . If  $A \subset \partial\Omega$  has  $|A| = 2j$ , then

$$\frac{Z_\Omega^A}{Z_\Omega^\emptyset} \leq C_j n^{-j/2}, \quad C_j = \frac{1}{j+1} \binom{2j}{j}. \quad (21.40)$$

*Proof.* First take  $A = \{a, b\}$ . The two boundary arcs between  $a$  and  $b$  determine four spin partition functions according as the spins on the two arcs are  $--, +-, -+, ++$ . If the arc lengths are  $m, m'$ , the loop/spin correspondence gives, up to the common bulk normalization,

$$Z_{--} = Z_\Omega^\emptyset, \quad Z_{+-} = x^m n^{1/2} Z_\Omega^{\{a,b\}}, \quad Z_{-+} = x^{m'} n^{1/2} Z_\Omega^{\{a,b\}}, \quad Z_{++} = x^{m+m'} n Z_\Omega^\emptyset.$$

Strong FKG applied to the two boundary arcs says  $Z_{++}Z_{--} \geq Z_{+-}Z_{-+}$ . Cancelling the common factor  $x^{m+m'} n$  yields

$$\frac{Z_\Omega^{\{a,b\}}}{Z_\Omega^\emptyset} \leq n^{-1/2}. \quad (21.41)$$

For  $2j$  marked boundary vertices, planarity pairs them by one of the  $C_j$  noncrossing pairings. Fix such a pairing  $\{a_1, b_1\}, \dots, \{a_j, b_j\}$ . Sum first over the last path, then the previous one, and use the chain rule together with (21.41) in every component created by deleting the already chosen paths. The contribution of the fixed pairing is at most  $n^{-j/2}$ ; summing the  $C_j$  pairings proves (21.40).  $\square$

**Lemma 10.274** (Domain monotonicity of relative weight). Assume  $n \geq 1$  and  $x \leq n^{-1/2}$ . If  $\Omega \subset \Lambda$  are honeycomb domains and  $\gamma \subset \Omega$ , then

$$W_\Lambda(\gamma) \leq 2W_\Omega(\gamma). \quad (21.42)$$

If both endpoints of  $\gamma$  belong to  $\partial\Omega \cap \partial\Lambda$ , the factor 2 may be removed.

*Proof.* Let  $S$  be the faces incident to  $\gamma$  and  $T := \Lambda^\bullet \setminus \Omega^\bullet$ . In the minus-boundary spin representation,

$$Z_\Lambda^\emptyset = Z_{\Lambda^\bullet}^-, \quad Z_\Omega^\emptyset = Z_{\Lambda^\bullet}^-[\sigma_T = -].$$

After deleting  $\gamma$ , the spins on  $S$  may be either sign, so

$$Z_{\Lambda \setminus \gamma}^\emptyset = Z_{\Lambda^\bullet}^-[\sigma_S = -] + Z_{\Lambda^\bullet}^-[\sigma_S = +] \leq 2Z_{\Lambda^\bullet}^-[\sigma_S = -],$$

where spin-flip symmetry identifies the second term with the corresponding plus-boundary probability and boundary comparison bounds it by the first. On the other hand,

$$Z_{\Omega \setminus \gamma}^\emptyset \geq Z_{\Lambda^\bullet}^-[\sigma_T = -, \sigma_S = -].$$

FKG for the decreasing events  $\{\sigma_T = -\}$  and  $\{\sigma_S = -\}$  gives

$$\frac{Z_{\Omega \setminus \gamma}^\emptyset}{Z_{\Lambda \setminus \gamma}^\emptyset} \geq \frac{1}{2} \frac{Z_\Omega^\emptyset}{Z_\Lambda^\emptyset},$$

which is (21.42) after inserting (21.38). If the path endpoints lie on the common outer boundary, the  $+$  term after deleting  $\gamma$  is impossible, so the factor 2 is absent.  $\square$

**Corollary 10.275** (Domain-gluing extension estimate). *Assume  $n \geq 1$  and  $x \leq n^{-1/2}$ . There is  $C = C(n, x)$  with the following property. Let  $\Omega \subset \Lambda$  be domains, let  $a, b \in \partial\Lambda$ , and let  $c, d \in \partial\Omega$  be connected inside  $\Lambda$  to  $\partial\Lambda$  by paths of length at most  $s$ . For every path  $\gamma \subset \Omega$  from  $c$  to  $d$ ,*

$$W_\Omega(\gamma) \geq e^{-Cs} \sum_{\substack{\gamma' \subset \Lambda: a \rightarrow b \\ \gamma \text{ is a subpath of } \gamma'}} W_\Lambda(\gamma'). \quad (21.43)$$

*Proof.* Choose connectors  $\psi, \psi'$  of length at most  $s$  from  $c, d$  to the outer boundary. A configuration contributing to an extension  $\gamma'$  is cut into  $\gamma$ , the edges on  $\psi \cup \psi'$ , and a residual configuration  $\omega_1$ . The residual odd-degree set lies among  $a, b$  and at most  $2s + 2$  endpoints created on the connectors. The map from the original configuration to these three pieces is injective. There are at most  $2^{2s+4}$  choices for the odd set and at most  $(1+x)^{2s}$  choices for the connector edges; at most  $2s$  loops can be destroyed, costing  $n^{2s}$ . For a residual odd set of size  $2j$ , Lemma 10.273 bounds its partition function by  $C_j n^{-j/2}$  times the even partition function. Since  $j \leq s + 2$  and  $C_j \leq 4^j$ , all combinatorial factors are  $e^{O(s)}$ . Finally use  $Z_{\Lambda \setminus \gamma}^\emptyset \geq Z_{\Lambda \setminus (\gamma \cup \psi \cup \psi')}^\emptyset$  and Lemma 10.274 to replace  $W_\Lambda(\gamma)$  by  $W_\Omega(\gamma)$ . Absorbing the explicit factors into  $e^{Cs}$  gives (21.43).  $\square$

**Proposition 10.276** (Parafermionic triangular lower bound). *Let  $n \in [1, 2]$  and*

$$x = x_c(n) := \frac{1}{\sqrt{2 + \sqrt{2 - n}}}.$$

*For even  $k$ , let  $T_k$  be the equilateral honeycomb triangular domain of side length  $k$ , let  $B_k, L_k, R_k$  be its bottom, left and right sides, and let  $a$  be the midpoint boundary vertex of  $B_k$ . Then*

$$\sum_{\gamma \subset T_k: a \rightarrow L_k} W_{T_k}(\gamma) \geq x^2. \quad (21.44)$$

*Proof.* Set

$$\sigma = 1 - \frac{3}{4\pi} \arccos(-n/2).$$

For a mid-edge  $z$  and a path from the distinguished bottom mid-edge  $z_0$  to  $z$ , define

$$F(z) := \sum_{\gamma: z_0 \rightarrow z} e^{-i\sigma \text{wind}(\gamma)} W_{T_k}(\gamma).$$

At an interior vertex  $v$  with incident mid-edges  $p, q, r$ , freeze the configuration away from the three half-edges at  $v$ . There are only the noncrossing local connectivities of these three half-edges. Grouping the possible local completions, including the possibility that a completion closes a loop and therefore contributes a factor  $n$ , reduces cancellation to the two scalar parameter identities

$$n = -2 \cos\left(\frac{4\pi}{3}(\sigma - 1)\right), \quad x^{-2} = 2 + \sqrt{2 - n}.$$

Indeed the first is exactly the displayed definition of  $\sigma$ , and the second is the definition of  $x_c(n)$ . The coefficients of every frozen connectivity class therefore sum to zero. Thus

$$(p - v)F(p) + (q - v)F(q) + (r - v)F(r) = 0. \quad (21.45)$$

This is the honeycomb specialization of the local loop- $O(n)$  parafermionic relation; the finite local check is important here because the loop-closing class, absent in self-avoiding walk, is where the factor  $n$  enters. Summing (21.45) over the interior vertices cancels internal edges and leaves

$$e^{-2\pi i/3} \sum_{z \in \bar{L}_k} F(z) + e^{2\pi i/3} \sum_{z \in \bar{R}_k} F(z) + \sum_{z \in \bar{B}_k} F(z) = 0. \quad (21.46)$$

On each side the winding is deterministic. Reflection symmetry makes the left and right sums equal, while the choice of  $\sigma$  gives

$$e^{-2\pi i/3} \sum_{\bar{L}_k} F + e^{2\pi i/3} \sum_{\bar{R}_k} F = -x^{-2} \sum_{\gamma: a \rightarrow L_k} W_{T_k}(\gamma).$$

The empty path contributes  $F(z_0) = 1$  to the bottom sum. Every other bottom-ending path has coefficient  $x^{-1} \cos(\sigma\pi) \geq 0$  for  $n \in [1, 2]$ . Hence  $\sum_{\bar{B}_k} F \geq 1$ . Inserting these two facts in (21.46) proves (21.44).  $\square$

**Theorem 10.277** (Macroscopic loops at the Nienhuis point). *For  $n \in [1, 2]$  and  $x = x_c(n)$  there exists  $c = c(n) > 0$  such that, for every integer  $m > 1$  and every exterior loop boundary condition  $\xi$ ,*

$$c \leq \mathbf{P}_{A_m, n, x}^{\xi}(\text{there is a loop in } A_m \text{ surrounding the origin}) \leq 1 - c. \quad (21.47)$$

*Consequently the loop- $O(n)$  Gibbs measure is unique, and its samples almost surely contain infinitely many loops surrounding the origin.*

*Proof.* By Lemma 10.271, it is enough to exclude alternative (A1). Assume therefore that there is  $c_0 > 0$  such that loops of diameter at least  $t$  have probability at most  $e^{-c_0 t}$ .

Fix a large even  $k$  and put

$$r := \frac{k}{\log k}, \quad \ell := (\log k)^2. \quad (21.48)$$

Center the inner triangles  $T_{k,s}$  of side  $k - s$  inside  $T_k$ . By Proposition 10.276, the total relative weight of paths from  $a$  to  $L_k$  is at least  $x^2$ . Split these paths according as they intersect  $T_{k,r}$ .

*Case 1: at least half of the weight enters  $T_{k,r}$ .* Every such path contains a subpath in  $T_{k,\ell}$  crossing  $T_{k,r}$ . There are fewer than  $k$  choices for the endpoint on  $L_k$  and fewer than  $3k$  choices for each of the two intersection points with  $\partial T_{k,\ell}$ . Hence for some  $b \in L_k$  and  $c, d \in \partial T_{k,\ell}$ ,

$$\sum_{\gamma' \in \Gamma'_{bcd}} W_{T_k}(\gamma') \geq \frac{x^2}{18k^3}, \quad (21.49)$$

where each  $\gamma'$  contains a  $c$ - $d$  subpath crossing  $T_{k,r}$ . Reflect the picture successively by angles  $\pi/3$ , producing six congruent triangles and marked pairs  $(c_j, d_j)$ . Their union is a hexagonal domain  $\Lambda'_k$ .

Let  $\mathcal{E}$  be the event that every loop meeting the six gluing sectors has diameter  $< \ell$ . Under (A1),

$$\mathbf{P}_{\Lambda'_k}(\mathcal{E}^c) \leq Ck^2 e^{-c\ell} = o(1).$$

Indeed, an offending loop surrounds at least one face in the  $O(k^2)$ -sized gluing region; translating that face to the origin and using the exponential tail in (A1), followed by a union bound, gives the displayed estimate. Thus  $\mathbf{P}_{\Lambda'_k}(\mathcal{E}) = 1 - o(1)$ . Condition on the union of loops meeting the sector boundaries. The unexplored interiors  $\Omega_j$  are domains and the event specifying them is measurable from the outside. Corollary 10.275, (21.49), and  $\ell \gg \log k$  give, uniformly in the conditioned interiors,

$$\sum_{\substack{\gamma \subset \Omega_j: c_j \rightarrow d_j \\ \gamma \cap T_{k,r}^j \neq \emptyset}} W_{\Omega_j}(\gamma) \geq e^{-C\ell}, \quad j = 1, \dots, 6. \quad (21.50)$$

Using the chain rule six times, configurations with six such open paths have relative partition-function weight at least  $e^{-6C\ell}$  on  $\mathcal{E}$ . Join the twelve endpoints by six deterministic short paths of length  $2\ell$  in the gluing corridors and take symmetric difference. The map is injective once the six short paths are fixed; at most  $12\ell$  edges are toggled and at most  $12\ell$  loop factors change. The resulting even configuration contains a loop of diameter at least  $r - \ell$ . Therefore, for some  $C_1 < \infty$ ,

$$\mathbf{P}_{\Lambda'_k}(\exists \text{ loop of diameter } \geq r - \ell) \geq e^{-C_1\ell} \mathbf{P}_{\Lambda'_k}(\mathcal{E}). \quad (21.51)$$

The left side is at most  $e^{-c_0(r-\ell)}$  under (A1). Since  $r/\ell = k/(\log k)^3 \rightarrow \infty$  while  $\mathbf{P}(\mathcal{E}) \rightarrow 1$ , (21.51) is impossible for large  $k$ .

*Case 2: at least half of the weight avoids  $T_{k,r}$ .* Such a path from  $a$  to  $L_k$  must cross the left or right side of the rectangle

$$\text{Rect}_k = [4r, k - 4r] \times [0, 4r]$$

embedded in the bottom of  $T_k$ . Pigeonholing the endpoint on  $L_k$  and the first hit  $d$  on a vertical side gives

$$\sum_{\gamma' \in \Gamma'_{bd}} W_{T_k}(\gamma') \geq \frac{x^2}{4kr}. \quad (21.52)$$

Corollary 10.275 transfers the corresponding  $a$ - $d$  subpath to  $\text{Rect}_k$  at cost  $e^{O(r)}$ :

$$\sum_{\substack{\gamma \subset \text{Rect}_k: a \rightarrow d \\ \gamma \cap T_{k,r} = \emptyset}} W_{\text{Rect}_k}(\gamma) \geq e^{-C_2 r}. \quad (21.53)$$

Reflect  $a, d$  across the horizontal midline of the rectangle to  $a_1, d_1$ . Let  $\mathcal{E}$  be the event that all loops meeting the midline have diameter  $< r$ . Since only  $O(k)$  faces meet that line, the same translation-and-union bound gives

$$\mathbf{P}_{\text{Rect}_k}(\mathcal{E}^c) \leq Cke^{-cr} = o(1),$$

because  $r = k/\log k$ . On  $\mathcal{E}$ , condition on the two complementary interiors exactly as in Case 1. Equation (21.53) and its reflection give two open paths of total relative weight at least  $e^{-2C_2r}$ . Join  $a$  to  $a_1$  and  $d$  to  $d_1$  by deterministic paths of length at most  $8r$  and take symmetric difference. The resulting even configuration contains a loop of diameter at least  $k/2 - 20r$ , and the weight loss is at most  $e^{C_3r}$ . Hence

$$\mathbf{P}_{\text{Rect}_k}(\exists \text{ loop of diameter } \geq k/2 - 20r) \geq e^{-C_4r}(1 - o(1)). \quad (21.54)$$

Alternative (A1) bounds the left side by  $e^{-c_0(k/2 - 20r)}$ . Since  $r = o(k)$ , this is again a contradiction.

Both cases contradict (A1), so alternative (A2) of Lemma 10.271 holds. Its finite-volume two-sided annulus bound is precisely (21.47), and its uniqueness/Borel–Cantelli conclusion gives the last statement.  $\square$

*Remark 10.278* (Why the gluing is load-bearing). The parafermionic identity alone only gives the lower bound (21.44) for a weighted family of defect paths. It does not create a loop. The Catalan-source estimate, domain monotonicity and Corollary 10.275 are exactly what allows defect paths obtained in different conditioned subdomains to be compared and then closed by a bounded-cost symmetric-difference surgery. Omitting this chain was the missing estimate in the abbreviated argument.

**Definition 10.279** (Macroscopic FK topologies used in the rotational theorem). A percolation configuration in a planar domain may be encoded in either of two ways. The *Schramm–Smirnov encoding* is the closed hereditary set of topological quads crossed by primal open paths; write  $d_{\text{SS}}$  for any metric inducing the standard quad-crossing topology. The *Camia–Newman encoding* is the locally finite pair of primal/dual outer-boundary loop families, modulo orientation-preserving reparametrization; write  $d_{\text{CN}}$  for its standard local loop metric. We use only that convergence in these metrics controls, respectively, macroscopic crossing events and macroscopic cluster-boundary loops.

**Theorem 10.280** (Large-scale rotational invariance: source-strength form). *Fix  $q \in [1, 4]$  and let  $\phi_{\Omega_\delta}^0$  denote the critical random-cluster measure on the square-mesh approximation  $\Omega_\delta$  of a simply connected  $C^1$ -smooth domain  $\Omega \subset \mathbb{R}^2$ , with free boundary condition. For every  $\varepsilon > 0$  there exists  $\delta_0 = \delta_0(q, \varepsilon, \Omega) > 0$  such that, for every*

$$\alpha \in (\varepsilon, \pi - \varepsilon), \quad 0 < \delta \leq \delta_0,$$

*there is a coupling of*

$$\omega_\delta \sim \phi_{\Omega_\delta}^0, \quad \omega'_\delta \sim \phi_{(e^{-i\alpha}\Omega)_\delta}^0$$

*for which*

$$\mathbb{P}[d_{\text{SS}}(\omega_\delta, e^{i\alpha}\omega'_\delta) > \varepsilon] < \varepsilon, \quad \mathbb{P}[d_{\text{CN}}(\omega_\delta, e^{i\alpha}\omega'_\delta) > \varepsilon] < \varepsilon. \quad (21.55)$$

*Consequently the critical square-lattice random-cluster model is rotationally invariant at macroscopic scale in both topologies. In particular, for every quad  $Q$  whose  $\varepsilon$ -neighbourhood lies in  $\Omega$ , its crossing probability differs from that of the rotated quad by at most  $\varepsilon$  once the mesh is sufficiently small. For*

$q \in \{2, 3, 4\}$  the Edwards–Sokal coupling gives the corresponding rotational invariance of Potts spin correlations. In the six-vertex regime  $a = b = 1$  and

$$\Delta = 1 - c^2/2 \in [-1, -1/2],$$

the source coupling further gives rotational invariance of finite products of height increments at mutually separated macroscopic points, at the exact finite-dimensional scope stated in Corollary 1.8 of Duminil-Copin–Kozłowski–Krachun–Manolescu–Oulamara [12].

*Proof roadmap.* The result is Theorem 1.2 and its stated corollaries in [12]. This volume also reconstructs the proof in ten stages. The stages begin in Lemma 10.281; the source-specific sine-ratio input, homotopy topology, marked-nail coupling, accumulated track exchange, and reflection assembly are proved below. The final assembly is the proof headed “Completion of the proof of Theorem 10.280.”

**Lemma 10.281** (Rotational-invariance proof architecture). *The proof of Theorem 10.280 separates into the following ten components.*

1. **Isoradial critical family and uniform RSW.** For every rectangular isoradial graph with track angles bounded away from 0 and  $\pi$ , the canonical Yang–Baxter weights define the critical random-cluster law and crossing/arm estimates are uniform in the angle data.
2. **Star–triangle and track exchange.** A single Yang–Baxter move has a coupling preserving the connectivity relation of the three boundary vertices. A finite sequence exchanges two neighboring tracks while preserving the random-cluster law on the transformed isoradial graph.
3. **Two-rooted IIC drift law.** For a cluster conditioned to have two nearby roots and to reach macroscopic height, the probability that the leftmost highest point chooses one of the two adjacent tracks has the exact ratio

$$\frac{\sin \alpha}{\sin \alpha + \sin \beta}. \tag{21.56}$$

*This is the source’s integrable input. Its proof combines torus transfer-matrix identities, separation of interfaces, convergence to the two-rooted IIC, and the six-vertex Perron–Frobenius eigenvalue asymptotics.*

4. **Homotopy topology.** Fix a mesoscopic grid of punctures  $B_\eta$ . Macroscopic loop homotopy classes in  $\mathbb{R}^2 \setminus B_\eta$  are encoded by reduced cyclic words. Equality of all nontrivial classes, with high probability, controls both  $d_{SS}$  and  $d_{CN}$ .
5. **Nails and marked nails.** Replace the fixed punctures by robust mesoscopic clusters (“nails”) near them and record the homotopy classes of macroscopic loops in the complement of those nails. The coupling resamples conditionally on this homotopy data before each track exchange.
6. **One-step stability.** Uniform arm/IIC estimates imply that one track exchange changes a marked nail only through a controlled local increment; with high probability no macroscopic homotopy class is lost in one step.
7. **Compounded track exchanges.** Martingale concentration and multiscale separation estimates control the accumulation of the one-step errors over the  $O(\delta^{-1})$  exchanges needed to transform one rectangular isoradial lattice into another.

8. **Drift-speed identification.** *The mean vertical displacement of a nail during track exchange is computed from (21.56). The accumulated deterministic drift is therefore the geometric deformation predicted by the track angles; fluctuations are negligible on macroscopic scale.*
9. **Universality among rectangular isoradial graphs.** *The preceding items yield a coupling between the square isoradial graph and the rectangular graph  $L(\alpha)$  for which the homotopy distance is small with high probability; the homotopy-topology theorem upgrades this to both macroscopic metrics.*
10. **Reflection–universality–reflection assembly.** *Choose  $L(\alpha)$  so that reflection in the line  $e^{i\alpha/2}\mathbb{R}$  is an exact lattice symmetry. Couple square  $\rightarrow L(\alpha)$ , reflect, and couple back. This gives the rotation by  $\alpha$  in (21.55). Domain approximation yields the finite-domain statement, while Edwards–Sokal and the BKW/six-vertex representation give the stated corollaries.*

### Integrable drift and the topology/coupling layers

Beyond isoradial RSW and star–triangle transport, the deterministic drift theorem has four logically separate pieces, including a refined six-vertex transfer-matrix asymptotic. They are proved below in that order. Keeping the pieces separate is important: the sine ratio is not a consequence of RSW or of the Yang–Baxter relation alone.

**Definition 10.282** (Two-rooted IIC and the adjacent-track events). Fix two adjacent horizontal tracks with angles  $\alpha, \beta \in (0, \pi)$ , bounded away from 0 and  $\pi$ . Let  $\Psi_i^2$ ,  $i \in \{0, 1\}$ , denote the two-rooted incipient infinite-cluster laws obtained by conditioning the two marked roots to connect to distance  $R$  in the two possible local connectivities and then sending  $R \rightarrow \infty$ . During an exchange of the two tracks, let  $\ell_{\max}$  be the horizontal displacement of the right-most marked extremum. We write

$$\{\ell_{\max} = 0\} \quad \text{and} \quad \{\ell_{\max} = 0^+\}$$

for the two local alternatives distinguished in the source construction. The desired one-step drift probability is

$$p(\alpha, \beta) := \Psi_1^2[\ell_{\max} = 0^+] = \Psi_0^2[\ell_{\max} = 0].$$

**Lemma 10.283** (Local convergence to the two-rooted IIC). *Let  $A$  be an event measurable in a ball of radius  $r$  around the two roots. If  $R \geq 8r$  and  $\mathbb{P}_R^{(i)}$  is the critical FK law conditioned on the  $i$ th root-connectivity and on connection to distance  $R$ , then there exist  $c, C > 0$ , uniform for track angles in a fixed compact subinterval of  $(0, \pi)$ , such that*

$$|\mathbb{P}_R^{(i)}(A) - \Psi_i^2(A)| \leq C(r/R)^c.$$

*The same estimate remains valid if finitely many tracks in the annulus are exchanged before the conditioning radius is sent to infinity.*

*Proof.* Take dyadic annuli between radii  $2r$  and  $R/2$ . Uniform isoradial RSW, already owned in Volume IX, gives a scale-independent positive probability of a primal circuit and of a dual circuit in each annulus. On the occurrence of the first separating circuit, the spatial Markov property makes the conditional law inside that circuit independent of the outer conditioning except for the induced boundary wiring. Monotonicity sandwiches the two induced boundary conditions between the free and wired laws. A second separating circuit of the opposite type couples those two extremal laws. Thus every dyadic annulus has a fixed positive chance to erase the memory of the exterior

conditioning. Failure through all  $m \asymp \log(R/r)$  annuli has probability at most  $(1 - c_0)^m$ , which is  $C(r/R)^c$  after changing constants.

A finite sequence of track exchanges changes only finitely many local rhombi and preserves the critical FK law by the star–triangle owner from Volume IX. The same annular coupling therefore applies uniformly after the exchanges. This proves both existence of the local IIC limits and the quantitative comparison needed below.  $\square$

**Lemma 10.284** (Torus sector ratio reduction). *Let  $\Lambda_N^{(k)}(\theta)$  be the Perron–Frobenius eigenvalue of the six-vertex row-transfer matrix in the sector with  $k$  through-lines and track angle  $\theta$ . For  $k$  of the parity required by the cylinder and  $k \ll N$ , the two finite-torus approximations to the adjacent-track event satisfy, as the cylinder height tends to infinity,*

$$R_1(N, k) = \frac{\Lambda_N^{(k)}(\beta)}{\Lambda_N^{(k+3)}(\beta)} \frac{1 - \Lambda_N^{(k+3)}(\alpha)/\Lambda_N^{(k)}(\alpha)}{1 - \Lambda_N^{(k+3)}(\beta)/\Lambda_N^{(k)}(\beta)},$$

$$R_0(N, k) = \frac{\Lambda_N^{(k+3)}(\alpha)}{\Lambda_N^{(k)}(\alpha)} \frac{1 - \Lambda_N^{(k+3)}(\beta)/\Lambda_N^{(k)}(\beta)}{1 - \Lambda_N^{(k+3)}(\alpha)/\Lambda_N^{(k)}(\alpha)}.$$

Moreover, for some  $\eta > 0$ ,

$$|R_i(N, k) - \Psi_i^2(A_i)| \leq C \left( \frac{\Lambda_N^{(k)}(\beta)}{\Lambda_N^{(k+3)}(\beta)} - 1 \right)^\eta,$$

where  $A_1 = \{\ell_{\max} = 0^+\}$  and  $A_0 = \{\ell_{\max} = 0\}$ .

*Proof.* The Yang–Baxter relation underlying the star–triangle map gives commuting row-transfer matrices for different rapidities. Decompose the cylinder state space into the conserved through-line sectors. In each sector, positivity and irreducibility of the transfer matrix give a simple Perron–Frobenius eigenvalue  $\Lambda_N^{(k)}(\theta)$ , and all other eigenvalues are exponentially suppressed when the cylinder height tends to infinity. Inserting the local two-track defect that records  $A_0$  or  $A_1$  changes the sector by three lines. Resolving the two possible local connectivities before taking the trace therefore yields the displayed ratios; the common bulk factors cancel. This is an exact finite-cylinder calculation, not an asymptotic use of RSW.

To compare the torus event with the two-rooted IIC, choose the conditioning radius much smaller than the torus circumference and much larger than the defect scale. The separation estimate of Lemma 10.283 replaces the torus exterior by the two-rooted IIC at polynomial cost. Optimizing the intermediate scale against the small sector gap gives the displayed power  $\eta$ . This is precisely the place where interface separation and the two-rooted IIC approximation enter the source proof.  $\square$

**Proposition 10.285** (Reduction of the sine-ratio drift to one refined six-vertex asymptotic). *Assume the following sector free-energy asymptotic for each fixed  $\theta \in (0, \pi)$  and for  $k \rightarrow \infty$ ,  $k/N \rightarrow 0$ :*

$$\frac{1}{N} \log \Lambda_N^{(k)}(\theta) = F(\theta) - A_q \sin \theta (k/N)^2 (1 + o(1)) + O((kN)^{-1}), \quad (10.22)$$

where  $A_q > 0$  depends only on the FK parameter. Then

$$\Psi_1^2[\ell_{\max} = 0^+] = \Psi_0^2[\ell_{\max} = 0] = \frac{\sin \alpha}{\sin \alpha + \sin \beta}.$$



*Proof.* The remainder in (10.22) is deliberately not subtracted at one adjacent pair: for  $k \asymp \sqrt{N}$ , after multiplying by  $N$  it is  $O(N^{-1/2})$ , the same order as an individual  $k \mapsto k+3$  gap. Instead set  $K_0 = \lfloor \sqrt{N} \rfloor$  and  $K_1 = \lfloor 2\sqrt{N} \rfloor$  with the required sector parity. The bulk term cancels and the accumulated remainder is  $o(1)$ , so

$$\frac{\log \Lambda_N^{(K_0)}(\alpha) - \log \Lambda_N^{(K_1)}(\alpha)}{\log \Lambda_N^{(K_0)}(\beta) - \log \Lambda_N^{(K_1)}(\beta)} = \frac{\sin \alpha}{\sin \beta} + o(1). \quad (10.23)$$

Write

$$a_k(\theta) := \log \Lambda_N^{(k)}(\theta) - \log \Lambda_N^{(k+3)}(\theta) \geq 0.$$

Telescoping (10.23) over the admissible  $k$  between  $K_0$  and  $K_1$  shows that there are indices  $k_-$  and  $k_+$  in this interval for which

$$\frac{a_{k_-}(\alpha)}{a_{k_-}(\beta)} \geq \frac{\sin \alpha}{\sin \beta} - o(1), \quad \frac{a_{k_+}(\alpha)}{a_{k_+}(\beta)} \leq \frac{\sin \alpha}{\sin \beta} + o(1). \quad (10.24)$$

For all such  $k$ , (10.22) also gives  $a_k(\theta) = O(N^{-1/2})$ , and hence

$$\frac{1 - e^{-a_k(\alpha)}}{1 - e^{-a_k(\beta)}} = \frac{a_k(\alpha)}{a_k(\beta)} + o(1), \quad \frac{\Lambda_N^{(k)}(\theta)}{\Lambda_N^{(k+3)}(\theta)} = 1 + o(1).$$

Substitution into Lemma 10.284, first with  $k_-$  and then with  $k_+$ , squeezes the IIC event ratio to  $\sin \alpha / \sin \beta$ . Since the two local events are complementary under the two-rooted connectivity decomposition, normalization gives

$$\Psi_1^2[\ell_{\max} = 0^+] = \frac{\sin \alpha}{\sin \alpha + \sin \beta}.$$

The second equality follows from the track-exchange symmetry interchanging the two local connectivities. This is the source's averaging argument and does not require an unavailable pointwise asymptotic for each adjacent sector gap.  $\square$

**Lemma 10.286** (Massless continuum Bethe equation and edge estimates). *Fix  $\Delta = -\cos \zeta \in (-1, 0)$ , so  $0 < \zeta < \pi/2$ . Put*

$$p(x) = 2 \arctan(\tanh x \cot(\zeta/2)), \quad \vartheta(x) = 2 \arctan(\tanh x \cot \zeta),$$

and

$$\xi = \frac{p'}{2\pi}, \quad K = \frac{\vartheta'}{2\pi}.$$

For  $Q \geq 0$  the equation

$$\rho_Q(x) + \int_{-Q}^Q K(x-y) \rho_Q(y) dy = \xi(x) \quad (10.25)$$

has a unique smooth solution. If  $Q = Q(m)$  is determined by  $\int_{-Q}^Q \rho_Q = m$ , then, uniformly for  $m$  sufficiently close to  $1/2$ ,

$$0 < \rho(x) \leq \rho_Q(x) \leq \rho(x) + \rho(Q), \quad \rho(x) = \frac{1}{2\zeta \cosh(\pi x/\zeta)}, \quad (10.26)$$

$$|\rho'_Q(x)| \leq \frac{C}{\zeta} (\rho_Q(x) + \rho(Q)), \quad (10.27)$$

$$\frac{c}{\zeta} (\tfrac{1}{2} - m) \leq \rho(Q) \leq \frac{C}{\zeta} (\tfrac{1}{2} - m), \quad (10.28)$$

and there is  $C_\zeta^* > 0$  such that

$$(\tfrac{1}{2} - m)e^{\pi Q(m)/\zeta} \longrightarrow C_\zeta^*. \quad (10.29)$$

*Proof.* The explicit kernel satisfies

$$K(x) = \frac{1}{\pi} \frac{\sin(2\zeta)}{\cosh(2x) - \cos(2\zeta)}, \quad \|K\|_{L^1(\mathbb{R})} = 1 - \frac{2\zeta}{\pi} < 1.$$

Hence  $I + K_Q$  is invertible on both  $L^1[-Q, Q]$  and  $L^\infty[-Q, Q]$  by the Neumann series, proving uniqueness and smoothness. On the full line Fourier transformation gives

$$\widehat{\rho}(t) = \frac{\widehat{\xi}(t)}{1 + \widehat{K}(t)},$$

whose inverse transform is the displayed  $\rho$ .

Let  $R$  be the full-line resolvent defined by  $\widehat{R} = \widehat{K}/(1 + \widehat{K})$ . The factorization  $\widehat{R} = \widehat{\rho} \widehat{F}$  with

$$F(x) = \frac{1}{\pi - \zeta} \frac{\sin(\pi\zeta/(\pi - \zeta))}{\cosh(2\pi x/(\pi - \zeta)) - \cos(\pi\zeta/(\pi - \zeta))}$$

after the harmless normalization of Fourier variables shows  $R \geq 0$ . Moreover  $\|R\|_1 < 1/2$ . Rewriting (10.25) on the whole line gives

$$\rho_Q(x) = \rho(x) + \int_{|y|>Q} R(x-y)\rho_Q(y) dy. \quad (10.30)$$

The positive Neumann series for the outside operator  $U_Q f(x) = \int_{|y|>Q} R(x-y)f(y)dy$  immediately yields  $\rho \leq \rho_Q$ . Isolating the first term and using  $\|U_Q\| < 1/2$  and monotonicity of  $\rho$  on  $[0, \infty)$  gives the upper bound in (10.26).

Differentiate (10.30) and integrate by parts over the two exterior half-lines. The boundary terms are  $[R(x-Q) - R(x+Q)]\rho_Q(Q)$  and the remaining operator is again  $U_Q$ . Since  $|\rho'| \leq (\pi/\zeta)\rho$ ,  $\|R\|_\infty \leq C/\zeta$ , and  $\rho_Q(Q) \leq 2\rho(Q)$  from the previous paragraph, inversion of  $I - U_Q$  gives (10.27).

Integrate (10.30) over the complement of  $[-Q, Q]$ . Positivity and  $\|R\|_1 < 1/2$  show that  $1/2 - m$  is comparable, with constants depending only on the fixed  $\zeta$ , to  $\int_Q^\infty \rho(x)dx$ . The explicit formula for  $\rho$  gives both (10.28) and the existence of a positive limit in (10.29). More precisely, shifting the contour in the Fourier formula for  $R$  to its first pole at  $i\pi/\zeta$  gives a remainder decaying strictly faster than  $e^{-\pi x/\zeta}$ ; substituting this once in (10.30) shows that the ratio in (10.29) is Cauchy and identifies a positive limit. This is the Wiener–Hopf tail estimate used later; no closed form of  $C_\zeta^*$  is needed.  $\square$

**Lemma 10.287** (Condensed analytic massless Bethe family near half filling). *Fix a compact interval  $J \Subset (-1, 0)$  of anisotropies. There are constants  $C, N_0$  such that, whenever  $\Delta \in J$ ,  $N \geq N_0$ , and*

$$n \leq N/2 - C,$$

*the discrete Bethe equations*

$$\frac{N}{2\pi} p(\lambda_i) - \frac{1}{2\pi} \sum_{j=1}^n \vartheta(\lambda_i - \lambda_j) = I_i, \quad I_i = i - \frac{n+1}{2}, \quad (10.31)$$

have a symmetric strictly ordered solution. If in addition  $N/2 - n \geq C$ , the solution can be chosen in an analytic family in  $\Delta$  joining it to the explicit free-fermion solution at  $\Delta = 0$ . Writing  $Q = Q(n/N)$  and defining quantiles by

$$N \int_{-Q}^{\Lambda(s|Q)} \rho_Q(x) dx = s - \frac{1}{2},$$

there is an integer  $r = r(J)$  such that

$$\Lambda(i - r/2 | Q) < \lambda_i < \Lambda(i + r/2 | Q) \quad (10.32)$$

for every  $i$ .

*Proof.* Choose  $r$  so large that, uniformly for  $\Delta \in J$ ,

$$\frac{r}{r+1} 2\pi > \vartheta(+\infty) - \vartheta(-\infty) = 2\pi - 4\zeta. \quad (10.33)$$

Let  $\Omega_r$  be the compact convex set of symmetric ordered vectors satisfying the non-strict version of (10.32). For  $\lambda \in \Omega_r$  define  $\mu = \Psi(\lambda)$  by

$$p(\mu_i) = \frac{1}{N} \sum_{j=1}^n \vartheta(\lambda_i - \lambda_j) + \frac{2\pi}{N} I_i. \quad (10.34)$$

The range of  $p$  and  $n \leq N/2$  make this well-defined; strict monotonicity of  $p$  makes  $\mu$  strictly ordered and symmetry is immediate.

We verify interlacement. The quantile quadrature identity says that for every monotone  $f$  and every  $r$ -interlaced vector,

$$\left| \frac{1}{N} \sum_j f(\lambda_j) - \int_{-Q}^Q f(x) \rho_Q(x) dx \right| \leq \frac{r+1}{2N} \text{osc}_{[-Q,Q]} f. \quad (10.35)$$

Apply this to  $f(x) = \vartheta(\lambda_i - x)$ . The integrated continuum Bethe equation is

$$2\pi N \int_0^u \rho_Q(x) dx = Np(u) - N \int_{-Q}^Q \vartheta(u - x) \rho_Q(x) dx.$$

Together with (10.33) and the assumed interlacement of  $\lambda_i$ , this places  $\mu_i$  strictly between the two quantiles in (10.32). Thus  $\Psi(\Omega_r) \subset \text{int } \Omega_r$ , and Brouwer gives a fixed point, which is a solution of (10.31). Summing the quantile-cell errors also gives, for every absolutely continuous  $f$ ,

$$\left| \frac{1}{N} \sum_j f(\lambda_j) - \int_{-Q}^Q f \rho_Q \right| \leq \frac{r}{N} \|f'\|_1. \quad (10.36)$$

For analyticity, write  $\mathcal{T}_\Delta(\lambda)$  for the left side of (10.31) minus  $I_i$ . In the negative massless regime the Jacobian is not manifestly positive, so rescale its columns by  $\rho_Q(\lambda_j)$ . The resulting matrix  $A$  has diagonal term and off-diagonal row sum satisfying, by (10.35),

$$A_{jj} \geq 1 - \frac{r \|K'\|_1}{N \rho_Q(\lambda_j)},$$

$$\sum_{\ell \neq j} |A_{j\ell}| \leq \frac{r}{r+1} + \frac{r}{N} \left\| \frac{d}{dx} \frac{K(\lambda_j - x)}{\rho_Q(x)} \right\|_{L^1(I_r)}.$$

The edge estimates (10.27)–(10.28), the elementary bounds  $\zeta\|K'\|_1 + \|K\|_1 \leq C$ , and  $N\rho_Q(\lambda_j) \geq c(N/2 - n - Cr)/\zeta$  give

$$A_{jj} - \sum_{\ell \neq j} |A_{j\ell}| \geq \frac{1}{r+1} - C \frac{r}{N/2 - n - Cr} > 0$$

after enlarging the fixed  $C$ . Hence the Jacobian is invertible by strict diagonal dominance. The analytic implicit-function theorem continues the fixed point along  $J \cup [\max J, 0]$ . At  $\Delta = 0$  the solution is explicit and unique; two analytic branches cannot cross while the Jacobian stays invertible. This proves the claimed analytic family.  $\square$

**Lemma 10.288** (Bethe eigenvalue is the Perron branch in the massless interval). *For the analytic family in Lemma 10.287, the coordinate Bethe vector is nonzero and its transfer-matrix eigenvalue is the Perron–Frobenius eigenvalue throughout the interval from  $\Delta = 0$  to the target  $\Delta \in (-1, 0)$ .*

*Proof.* We first own the massless coordinate-Bethe identity locally. In an ordered arrow configuration  $x_1 < \dots < x_n$ , write the Bethe wave as the sum over permutations

$$\Psi(x_1, \dots, x_n) = \sum_{\sigma \in S_n} A_\sigma \prod_{j=1}^n L(\lambda_{\sigma(j)})^{x_j},$$

where adjacent transposition of rapidities multiplies  $A_\sigma$  by the two-particle scattering ratio determined by the six-vertex weights. Acting with one transfer row produces free one-particle propagation except when two arrows become adjacent; the scattering ratio is exactly the coefficient that cancels those collision terms pairwise. Moving a particle through the periodic seam multiplies its term by the product of all scattering factors against the other rapidities, and the discrete Bethe equation (10.31) makes this multiplier equal to the required periodic phase. Hence every ordered component has the same transfer-matrix eigenvalue and the Bethe wave is an eigenvector whenever it is nonzero.

At  $\Delta = 0$  the roots are

$$\lambda_i = p_0^{-1}(2\pi I_i/N).$$

Substitution in the determinant form of the Bethe wave reduces every ordered component to a Vandermonde determinant times a positive gauge factor; hence the vector is nonzero, and after a common phase choice all its components are positive. Perron–Frobenius therefore identifies its eigenvalue with the simple top eigenvalue.

Along the analytic root family the Bethe vector is analytic and is not identically zero, because it is nonzero at  $\Delta = 0$ . Consequently its common zero set is discrete unless all coordinate components vanish identically. Away from this discrete set the algebraic Bethe-wave verification gives an eigenvalue branch. Since the transfer matrix is an irreducible nonnegative symmetric matrix in every fixed arrow sector, its Perron eigenvalue is simple and continuous in the weights. At  $\Delta = 0$  the Bethe branch equals that Perron eigenvalue; analytic continuation on each complementary interval therefore keeps the two branches equal, and continuity across the isolated possible zeroes of the Bethe vector extends the equality to the whole interval. A continuous Bethe eigenvalue that starts on this branch cannot leave it without meeting the Perron eigenvalue, which would contradict simplicity. Hence it is the Perron branch throughout.  $\square$

**Lemma 10.289** (Refined quantile discrepancy). *Let  $n \leq N/2 - C/\zeta$  and let  $\lambda$  be the Perron Bethe family above. Define*

$$D_i := N \int_{\Lambda(i|Q)}^{\lambda_i} \rho_Q(x) dx, \quad D := \max_i |D_i|.$$

Then

$$D \leq \frac{C}{\zeta(N - 2n - C)}. \quad (10.37)$$

For every  $C^2$  function  $f$  with  $f', f'' \in L^1$ ,

$$\left| \frac{1}{N} \sum_{j=1}^n f(\lambda_j) - \int_{-Q}^Q f(x) \rho_Q(x) dx \right| \leq \frac{D}{N} \|f'\|_1 \quad (10.38)$$

$$+ \frac{C(1+D)}{N(N-2n-C)} (\zeta \|f''\|_1 + \|f'\|_1). \quad (10.39)$$

*Proof.* Set  $g(s) = f(\Lambda(s | Q))$ . A quantile cell has mass  $1/N$ , so Taylor's theorem on each interval  $[j - 1/2, j + 1/2]$  gives

$$\frac{1}{N} \sum_j f(\Lambda(j | Q)) - \int_{-Q}^Q f \rho_Q = O\left(\frac{1}{N} \|g''\|_1\right).$$

Because  $D_j$  is precisely the quantile-coordinate displacement of  $\lambda_j$ , the additional replacement  $f(\Lambda(j | Q)) \mapsto f(\lambda_j)$  costs  $D \|g'\|_1/N$  plus  $D \|g''\|_1/N$ . Now

$$\|g'\|_1 = \|f'\|_1, \quad \|g''\|_1 \leq \left\| \frac{1}{N \rho_Q} \right\|_\infty \|f''\|_1 + \left\| \frac{\rho'_Q}{N \rho_Q^2} \right\|_\infty \|f'\|_1.$$

Equations (10.27)–(10.28) yield (10.39).

To bound  $D$ , integrate the continuum equation and use the discrete Bethe equation to write exactly

$$D_i = \frac{1}{2\pi} \left( \sum_{j=1}^n \vartheta(\lambda_i - \lambda_j) - N \int_{-Q}^Q \vartheta(\lambda_i - x) \rho_Q(x) dx \right).$$

Apply (10.39) to  $f(x) = \vartheta(\lambda_i - x)/(2\pi)$ . Since  $K = \vartheta'/(2\pi)$ ,

$$|D_i| \leq D \|K\|_1 + \frac{C}{N - 2n - C} (\zeta \|K'\|_1 + \|K\|_1).$$

In the negative massless regime  $\|K\|_1 = 1 - 2\zeta/\pi$  and  $\zeta \|K'\|_1 + \|K\|_1 \leq C$ . Thus

$$D \leq (1 - 2\zeta/\pi)D + \frac{C}{N - 2n - C},$$

which rearranges to (10.37). To justify using the refined estimate before knowing  $D \leq 1/2$ , start at  $\Delta = 0$ , where  $D = 0$ , and follow the analytic family. If  $D$  first reached the right side of (10.37), the preceding strict inequality would contradict first passage. Continuity therefore closes the bootstrap.  $\square$

**Lemma 10.290** (Quadratic continuum free-energy edge). *Let  $L_\theta(x)$  be the logarithmic one-particle factor of the Perron Bethe eigenvalue for track angle  $\theta$ . As  $m \uparrow 1/2$ ,*

$$\int_{\mathbb{R}} L_\theta(x) \rho(x) dx - \int_{-Q(m)}^{Q(m)} L_\theta(x) \rho_{Q(m)}(x) dx = C_\zeta \sin \theta (1 - 2m)^2 (1 + o(1)), \quad (10.40)$$

with  $C_\zeta > 0$ .

*Proof.* Let  $R$  be the resolvent from Lemma 10.286 and set  $G = L_\theta - R * L_\theta$ . Fourier transformation gives an integrable function  $\widehat{G}$  whose nearest poles to the real axis are at  $\pm i\pi/\zeta$ , with residues  $\mp 2i \sin \theta$ ; all other poles are separated by a strictly larger imaginary part. Using (10.30), Fubini, and the evenness of  $L_\theta$  and  $R$ ,

$$\delta f(Q) := \int L_\theta \rho - \int_{-Q}^Q L_\theta \rho_Q = \int_{|x|>Q} G(x) \rho_Q(x) dx.$$

Shift the Fourier contour to the first poles. For any  $\beta \in (2\pi/\zeta, 3\pi/\zeta)$  this gives

$$\delta f(Q) = 4 \sin \theta e^{-\pi Q/\zeta} \int_0^\infty e^{-\pi x/\zeta} \rho_Q(Q+x) dx + O(e^{-\beta Q}). \quad (10.41)$$

To expand the remaining factor, split the resolvent at the two exterior half-lines. The same Neumann series as in (10.30) gives

$$\rho_Q(Q+x) = e^{-\pi Q/\zeta} \{T_\zeta(x) + o_{L^1 \cap L^\infty}(1)\},$$

where  $T_\zeta$  is the unique solution on  $\mathbb{R}_+$  of

$$T_\zeta(x) - \int_0^\infty R(x-y) T_\zeta(y) dy = \zeta^{-1} e^{-\pi x/\zeta}.$$

The operator on the left has inverse  $\sum_{j \geq 0} V^j$  with positive kernel and  $\|V\| < 1/2$ , so  $T_\zeta > 0$ . Substitution into (10.41) gives

$$\delta f(Q) = 4 \sin \theta e^{-2\pi Q/\zeta} \left( \int_0^\infty e^{-\pi x/\zeta} T_\zeta(x) dx + o(1) \right).$$

Finally (10.29) replaces  $e^{-\pi Q/\zeta}$  by a positive constant times  $1/2 - m = (1 - 2m)/2$ . This proves (10.40), including positivity of  $C_\zeta$ .  $\square$

**Theorem 10.291** (Refined negative-massless six-vertex sector asymptotic). *Fix  $q_{\text{FK}} \in [1, 4)$  and write  $\sqrt{q_{\text{FK}}}/2 = \cos \zeta$ , so the associated six-vertex anisotropy is  $\Delta = -\cos \zeta \in (-1, -1/2]$ . Fix a track angle  $\theta \in (0, \pi)$  away initially from the symmetric point  $a = b$ . For sectors  $k \rightarrow \infty$ ,  $k/N \rightarrow 0$ , the Perron eigenvalue satisfies*

$$\frac{1}{N} \log \Lambda_N^{(k)}(\theta) = F(\theta) - A_{q_{\text{FK}}} \sin \theta (k/N)^2 (1 + o(1)) + O((kN)^{-1}), \quad (10.42)$$

with  $A_{q_{\text{FK}}} > 0$ . The sector asymptotic itself is asserted only for  $q_{\text{FK}} < 4$ . Its use in the sine-ratio drift theorem extends to the symmetric angle and to  $q_{\text{FK}} = 4$  by continuity of the finite transfer matrices and of the corresponding finite-volume/IIC probabilities before the final squeeze.

*Proof.* Set  $n = N/2 - k$ , so  $1 - 2n/N = 2k/N$ . By Lemmas 10.287 and 10.288, the logarithm of the Perron eigenvalue is the Bethe sum

$$\frac{1}{N} \log \Lambda_N^{(k)}(\theta) = \log a + \frac{1}{N} \sum_{j=1}^n L_\theta(\lambda_j) + O(e^{-cN}).$$

Apply Lemma 10.289 with  $f = L_\theta$ . For the fixed angle under consideration the explicit one-particle factor has  $L'_\theta, L''_\theta \in L^1$ , so

$$\frac{1}{N} \log \Lambda_N^{(k)}(\theta) = \log a + \int_{-Q}^Q L_\theta \rho_Q + O\left(\frac{1}{\zeta(N - 2n)N}\right).$$

The infinite-volume value is  $F(\theta) = \log a + \int_{\mathbb{R}} L_{\theta} \rho$ . Lemma 10.290 therefore yields

$$F(\theta) - \frac{1}{N} \log \Lambda_N^{(k)}(\theta) = 4C_{\zeta} \sin \theta (k/N)^2 (1 + o(1)) + O((\zeta k N)^{-1}).$$

For fixed  $q_{\text{FK}} < 4$ , absorb  $\zeta^{-1}$  into the constant and put  $A_{q_{\text{FK}}} = 4C_{\zeta} > 0$ . Continuity of the finite-dimensional Perron eigenvalues in the weights extends the sine-ratio consequence to  $a = b$  and, after first proving it for every  $q_{\text{FK}} < 4$ , to the endpoint  $q_{\text{FK}} = 4$ . This is exactly the strength needed in Proposition 10.285.  $\square$

**Definition 10.292** (Homotopy word encoding). Fix  $\eta > 0$  and let  $B_{\eta} = \eta\mathbb{Z}^2 \cap [-\eta^{-1}, \eta^{-1}]^2$ . A loop avoiding  $B_{\eta}$  determines a cyclic reduced word in the oriented edges of a fixed cut system joining the punctures. Cancelling an adjacent inverse pair corresponds to deleting a contractible detour. We call the configuration  $\text{Normal}(\kappa, \eta)$  if every erased detour meeting  $B(0, \kappa^{-1})$  has Euclidean diameter less than  $\kappa/3$ . We call it  $\text{Dense}(\eta)$  if every  $\eta$ -grid face in the relevant window contains both a primal and a dual circuit at the microscopic scale used in the comparison.

**Lemma 10.293** (Homotopy topology controls Camia–Newman and Schramm–Smirnov topology). *For every  $\kappa > 0$  there is  $\eta_0(\kappa) > 0$  such that, for  $0 < \eta < \eta_0$  and then mesh  $\delta$  sufficiently small, the following holds uniformly for bounded-angle critical isoradial FK configurations  $\omega, \tilde{\omega}$ . If their homotopy data relative to  $B_{\eta}$  agree and both belong to  $\text{Normal}(\kappa, \eta) \cap \text{Dense}(\eta)$ , then*

$$d_{\text{CN}}(\omega, \tilde{\omega}) < \kappa, \quad d_{\text{SS}}(\omega, \tilde{\omega}) < \kappa.$$

Moreover

$$\mathbb{P}(\text{Normal}(\kappa, \eta)^c) \leq \frac{C}{\eta^2 \kappa^2} e^{-c\kappa/\eta}, \quad \mathbb{P}(\text{Dense}(\eta)^c) \leq C\eta^{-2} e^{-c(\eta/\delta)^2}.$$

*Proof.* We give the geometric and probabilistic parts separately.

*Geometric comparison.* Let  $\gamma$  be a macroscopic loop of  $\omega$  and write its raw cut-crossing word as

$$u_1 v_1 u_2 v_2 \cdots u_m v_m,$$

where the  $u_j$  survive free reduction and every  $v_j$  reduces to the empty word. On  $\text{Normal}(\kappa, \eta)$  each erased excursion has diameter below  $\kappa/3$ . Replacing it by the canonical short representative of its homotopy class moves the loop by less than  $\kappa/2$  in the uniform curve metric. If the reduced word surrounds at least two but not all punctures, equality of the homotopy data produces a loop of  $\tilde{\omega}$  with the same reduced word; applying the same shortening on both sides gives a matching loop within  $\kappa$ . If the loop surrounds at most one puncture,  $\text{Dense}(\eta)$  supplies the required nearby microscopic primal or dual circuit. The same argument with primal and dual loops reversed gives the inverse matching. Thus  $d_{\text{CN}} < \kappa$ .

For  $d_{\text{SS}}$ , fix a finite family of buffered quads whose crossing indicators control the Schramm–Smirnov metric at tolerance  $\kappa$ . If two configurations are  $\kappa$ -close in the loop topology but disagree on one of these buffered crossings, then either a macroscopic boundary loop lacks a matching loop, contradicting the previous paragraph, or four alternating arms run from scale  $O(\eta)$  to a scale comparable with  $\kappa$ . The uniform arm estimate from the isoradial RSW package bounds the latter by  $C(\eta/\kappa)^c$ . Taking  $\eta$  small enough eliminates it. This gives the SS claim without importing an unproved topology-comparison theorem.

*Normality.* If a cancelled detour of diameter at least  $\kappa/3$  occurs, a maximal such detour produces, after a bounded number of rotations/translations, a crossing of a strip of length comparable with  $\kappa$  by a cluster that surrounds none of the relevant  $\eta$ -grid punctures. Enumerate the crossing clusters

from left to right. Along the boundary of each one choose at least  $\lfloor \kappa/(48\eta) \rfloor - 1$  well-separated grid points. Uniform RSW gives, conditionally on everything already explored, a fixed probability  $c_0 > 0$  that the next point is separated by an appropriate primal or dual circuit. Hence the probability that a given crossing avoids all punctures is at most  $(1 - c_0)^{c\kappa/\eta}$ . The tight moment bound for the number of macroscopic crossing clusters and a union bound over  $O(\eta^{-2}\kappa^{-2})$  possible boxes give

$$\mathbb{P}(\text{Normal}^c) \leq C(\eta^2\kappa^2)^{-1}e^{-c\kappa/\eta}.$$

*Density.* Inside every  $\eta$ -grid face there are order  $(\eta/\delta)^2$  disjoint microscopic places at which finite energy and RSW give a uniformly positive chance of producing a small primal circuit, and likewise a dual circuit. Splitting into a bounded number of independent-by-domain-Markov subfamilies yields  $Ce^{-c(\eta/\delta)^2}$  for the failure in one face. There are  $O(\eta^{-2})$  faces, which gives the second displayed estimate. Choosing first  $\eta \ll \kappa$  and then  $\delta \ll \eta$  completes the proof.  $\square$

**Lemma 10.294** (Marked-nail resampling/track-exchange coupling). *Work in a box of linear scale  $N$  and choose  $0 < \varepsilon \ll \eta \ll 1$ . Let  $\mathcal{L}(t)$  be the mixed isoradial lattice after  $t$  elementary track exchanges and let  $h(t)$  denote the height of the track currently being exchanged. There is a coupling  $(\omega^{(t)})_{0 \leq t \leq T}$ , with  $T = O(N^2)$ , together with a countable set of nail labels, such that:*

1. *every  $\omega^{(t)}$  has exactly the critical FK law on  $\mathcal{L}(t)$ ;*
2. *a marked nail is a macroscopic cluster whose top, bottom and right extrema stay within  $\sqrt{\eta\varepsilon}N$  of its prescribed grid point and whose vertical span is at least  $\varepsilon N$ ;*
3. *the state variable  $H(t)$  records the surviving nail labels, their three extrema, and the multisets of primal and dual homotopy classes surrounding at least two but not all marked nails;*
4. *conditional on  $H(t)$ , the configuration is freshly resampled before the next track exchange, so the local neighborhood of a nail may be maximally coupled to the appropriate two-rooted IIC law;*
5. *the subsequent star-triangle map transports the conditional law exactly to the critical FK law on  $\mathcal{L}(t+1)$  while transporting the nail labels and homotopy data.*

*The initial margin  $(\sqrt{\eta\varepsilon} - \varepsilon)N$  ensures that a nail cannot be lost solely because of the deterministic tolerance accumulated in the construction.*

*Proof.* At time  $t$ , let  $H(t) = H(\omega^{(t)})$  be the discrete datum described in the statement. The configuration space is a compact product of two-point edge spaces and  $H(t)$  is countable-valued at fixed mesh, so conditioning on the observed value of  $H(t)$  is an ordinary disintegration. Draw

$$\omega^{(t+1/2)} \sim \phi_{\mathcal{L}(t)}[\cdot \mid H = H(t)].$$

By construction this refreshes all information not protected by the nail extrema and homotopy data, while preserving  $H(t)$  exactly. In particular it is now legitimate to use the local IIC comparison around a marked extremum: the exterior is not frozen by the past history of the coupling.

Apply the elementary star-triangle track-exchange coupling to  $\omega^{(t+1/2)}$ . The Yang-Baxter identity says that the image has the critical FK law on  $\mathcal{L}(t+1)$  with the same boundary connectivity outside the exchanged strip. Transport every surviving nail label with its cluster. If a cluster ceases to meet the nail inequalities, delete that label; any new qualifying cluster receives the next unused integer. Recompute the top, bottom and right extrema. Provided all previously marked



nails survive, transport the primal and dual loop homotopy multisets through the local star–triangle homeomorphism; otherwise reset the corresponding comparison datum, as in the source construction.

The marginal statement follows inductively from the tower property: conditional resampling leaves the  $\phi_{\mathcal{L}(t)}$  marginal unchanged, and the track-exchange kernel pushes this law exactly to  $\phi_{\mathcal{L}(t+1)}$ . The geometric margin in the initial nail definition exceeds the allowed deterministic displacement buffer, so the bookkeeping itself cannot spuriously delete a nail. This proves the coupling lemma.  $\square$

**Definition 10.295** (IIC displacement and one-step error variables). For a marked nail  $i$  at time  $t$ , let  $T^{(t)}(i)$ ,  $B^{(t)}(i)$  and  $R^{(t)}(i)$  be its top, bottom and right extrema. Couple the local resampling near the active track to independent two-rooted IIC configurations as in Lemma 10.294. The corresponding ideal increments are denoted

$$\delta_j^{\text{IIC}}T, \quad \delta_j^{\text{IIC}}B, \quad \delta^{\text{IIC}}R.$$

They take values in the finite sets forced by the rhombus geometry:

$$\begin{aligned} \delta_1^{\text{IIC}}T &\in \{0, \sin \alpha\}, & \delta_0^{\text{IIC}}T &\in \{-1, \sin \alpha - 1\}, \\ \delta_0^{\text{IIC}}B &\in \{\sin \alpha, \sin \alpha - 1\}, & \delta_{-1}^{\text{IIC}}B &\in \{0, -1\}, \\ \delta^{\text{IIC}}R &\in \{0, -1, \cos \alpha - 1, \cos \alpha\}, \end{aligned}$$

with all other vertical-index increments equal to zero. Define

$$\begin{aligned} \delta^{\text{err}}T_t(i) &= T^{(t+1)}(i) - T^{(t)}(i) - \delta_{h(t)-T^{(t)}(i)}^{\text{IIC}}T_t(i), \\ \delta^{\text{err}}B_t(i) &= B^{(t+1)}(i) - B^{(t)}(i) - \delta_{h(t)-B^{(t)}(i)}^{\text{IIC}}B_t(i), \\ \delta^{\text{err}}R_t(i) &= R^{(t+1)}(i) - R^{(t)}(i) - X_t(i)\delta^{\text{IIC}}R_t(i), \end{aligned}$$

where  $X_t(i)$  indicates that the active track contains the right extremum of nail  $i$ . Finally put

$$M_t(i) := |\delta^{\text{err}}T_t(i)| + |\delta^{\text{err}}B_t(i)| + |\delta^{\text{err}}R_t(i)|, \quad \text{Err}(t) := \sum_{i \in I(t)} M_t(i).$$

**Lemma 10.296** (Bad-event estimates for one track exchange). *Let  $d(t)$  be the distance from the active track to the nearest already-descended or not-yet-descended track of the other angle. Define:*

1.  $\text{BAD}_1(t)$ : one nail has simultaneously a vertical and a right  $t$ -extremum;
2.  $\text{BAD}_2(t)$ : one nail has two vertical  $t$ -extrema at mutual distance at least  $d(t)^{1/5}$ ;
3.  $\text{BAD}_3(t)$ : two distinct nails have  $t$ -extrema on the active local pair of tracks;
4.  $\text{BAD}_4(t)$ : a nail has a fake right  $t$ -extremum, namely a point of the active track lying within one rhombus width of  $R^{(t)}(i)$  while the true right-most vertex lies below  $b(t)$ ;
5.  $\text{BAD}_5(t)$ : a nail has a right  $t$ -extremum and another vertex whose horizontal coordinate lies in  $(R^{(t)}(i) - 1, R^{(t)}(i))$  but whose vertical distance from  $h(t)$  is at least  $d(t)^{1/5}$ .

There are constants  $c, C > 0$ , depending only on the bounded-angle parameter and on  $\varepsilon, \eta$ , such that for  $N$  large and every active time  $t$ ,

$$\phi_{\mathcal{L}(t)}(\text{BAD}_j(t)) \leq \frac{C}{d(t)N^c}, \quad 1 \leq j \leq 5.$$

*Proof.* We use only the uniform half-plane and mixed-lattice arm estimates already contained in the isoradial RSW owner: a three-arm event from scale  $r$  to  $R$  has an upper bound  $C(r/R)^{1+c_1}$  in a half-plane, the separated five-arm event gains an additional positive power, and a long one-arm excursion outside the working box has an exponential tail. Quasi-multiplicativity permits us to concatenate these bounds across the finitely many changes of track angle.

For  $\text{BAD}_1$ , if  $x$  is simultaneously a right and top extremum, three arms emanate in the lower-left quarter-plane from scale 1 to scale  $\varepsilon N$ . The probability for a fixed  $x$  is at most  $C(\varepsilon N)^{-2-c_1}$ . There are  $O(N)$  possible extremal locations after using the nail localization, so the total is  $O(N^{-1-c_1})$ ; the bottom case is identical.

For  $\text{BAD}_2$ , let  $x, y$  be two top extrema of the same nail with  $r = |x - y| \geq d(t)^{1/5}$ . There are two disjoint half-plane three-arm events from  $x$  and  $y$  to scale  $r/2$ , another separated three-arm event from scale  $2r$  to  $\varepsilon N$ , and, if the pair lies far to the left of the working box, the one-arm return needed to meet the nail window. Hence, after summing first over  $y$  and then over  $x$ ,

$$\phi(\text{BAD}_2(t)) \leq \frac{C}{\varepsilon^2 N d(t)^{1/4}}.$$

The exponent  $1/4$  is deliberately weaker than the power available from the arm bounds and is sufficient below.

For  $\text{BAD}_3$ , suppose first that right extrema  $x, y$  of two different nails have  $r = |x - y| \leq 2d(t)$ . Two local three-arm events up to scale  $r/2$ , a separated five-arm event from  $2r$  to  $d(t)$ , and an outer three-arm event to  $\varepsilon N$  give

$$\phi(x, y \text{ are distinct-nail right extrema}) \leq \frac{C}{r^4} \left( \frac{r}{d(t)} \right)^{2+c_0} \left( \frac{2d(t)}{\varepsilon N} \right)^{1+c_1}.$$

If  $r > 2d(t)$ , use two three-arm events to scale  $d(t)$ , two further separated three-arm events to scale  $r/2$ , and the outer three-arm event. This yields

$$\frac{C}{d(t)^4} \left( \frac{d(t)}{r} \right)^{2+2c_1} \left( \frac{2r}{\varepsilon N} \right)^{1+c_1}.$$

Both series are summable in  $r$ . Summing over possible extremal positions gives  $C/[d(t)(\varepsilon N)^{c_2}]$ . Top/top, bottom/bottom and mixed extremal pairs are no worse.

For  $\text{BAD}_4$ , let  $x$  be the fake extremum and  $y$  the true right-most vertex below  $b(t)$ . The geometry forces a half-plane three-arm event at  $x$  to scale  $d(t)/2$ , two separated three-arm continuations up to scale  $|x - y|/2$ , and a final three-arm event to scale  $\varepsilon N$ . Consequently

$$\phi(E(x, y)) \leq \frac{C}{d(t)^2 |x - y|^2} \left( \frac{d(t)}{|x - y|} \right)^{1+c_1} \left( \frac{4|x - y|}{\varepsilon N} \right)^{1+c_1}.$$

The double sum is  $O([d(t)N^{c_3}]^{-1})$ .

For  $\text{BAD}_5$ , write  $x$  for the right extremum and  $y$  for the second near-right vertex. Since  $|y_1 - x_1| < 1$  but  $|y_2 - h(t)| \geq d(t)^{1/5}$ , the two vertices must be joined inside the nail while remaining separated by a dual arm on one side of the active track. Up to radius  $d(t)^{1/5}/3$  this gives two separated three-arm events; from that scale to  $\varepsilon N$  the connection to the rest of the nail supplies the same outer three-arm condition as in  $\text{BAD}_4$ . Quasi-multiplicativity therefore bounds the probability for a fixed vertical separation  $r$  by

$$Cr^{-2-c_0} \left( \frac{r}{\varepsilon N} \right)^{1+c_1}.$$

Summing over  $r \geq d(t)^{1/5}$  and over the  $O(N)$  possible horizontal positions again gives  $C/[d(t)N^c]$  after decreasing  $c$ . This closes the event omitted as “similar” in the primary exposition and proves (21.57).  $\square$

**Proposition 10.297** (One-step IIC approximation for marked nails). *For the decorated marked-nail coupling there exist  $c, C > 0$  such that for every active track exchange:*

1. *the marginal of  $\omega^{(t)}$  is the exact critical FK law on  $\mathcal{L}(t)$ ;*
2. *until the first time a marked nail disappears, the primal and dual homotopy multisets recorded by  $H(t)$  are unchanged;*
3. *the IIC displacement variables used at time  $t$  are independent of the sigma-field generated by all true and ideal increments used before time  $t$ ;*
4. *the expected total discrepancy obeys*

$$\mathbb{E}[\text{Err}(t)] \leq \frac{C}{d(t)N^c}.$$

*Proof.* The first statement was proved in Lemma 10.294. The local star–triangle homeomorphism preserves every loop outside the active rhombus and changes the chosen representatives of marked nails only inside a simply connected microscopic set. Hence the reduced homotopy words around the marked nails are unchanged until a marked nail itself disappears. Independence follows because a fresh independent family of IIC samples and extremum indicators is assigned to each time before the conditional resampling.

It remains to prove (21.62). We follow the four cases in the definition of the coupling. A standard RSW cluster-count tail, already used in the nail construction, gives

$$\mathbb{P}(|I(t)| > C_0 \log N) \leq N^{-20}$$

after increasing  $C_0$ .

*Case 0: two nails have vertical extrema.* This is contained in  $\text{BAD}_3(t)$ . Every coordinate moves by at most a constant under one star–triangle move, so

$$\mathbb{E}[\text{Err}(t); \text{Case 0}] \leq C\mathbb{E}[|I(t)|; \text{BAD}_3(t)] \leq \frac{C}{d(t)N^c}$$

by (21.57), (21.63), and a split at  $C_0 \log N$ .

*Cases 1 and 2: one vertical extremum.* Treat a unique top extremum  $x$  of nail  $i$ ; the bottom case is the vertical reflection. All bottom increments and all top increments of nails other than  $i$  are zero both in the true and IIC evolutions. A nonzero top error for  $i$  can occur only through one of four mechanisms:

1. the conditional FK configuration and its translated IIC sample fail to agree in  $\Lambda_{d(t)^{1/4}}(x)$ ;
2. the true nail has another top extremum outside  $\Lambda_{d(t)^{1/5}}(x)$ ;
3. the IIC sample has another top extremum outside that box;
4. the two configurations agree before the move but the outputs of the local star–triangle transformations disagree inside  $\Lambda_{d(t)^{1/5}}(x)$ .

On the event that there is a unique cluster crossing  $\Lambda_{d^{1/2}}(x) \setminus \Lambda_{d^{1/3}}(x)$ , this crossing screens the interior from the extra conditioning in  $H(t)$ . Lemma 10.283 then bounds the first mechanism by  $Cd(t)^{-c}$ . The probability that the screening event fails is a separated-arm event and has the same bound. Mechanism 2 is  $\text{BAD}_2(t)$ . For mechanism 3, summing the IIC extremal tail  $C|y|^{-2}$  over  $|y| \geq d(t)^{1/5}$  gives  $Cd(t)^{-1/5}$ . For mechanism 4, disagreement of the two star-triangle outputs across the annulus requires the absence of every local pair of closed separating edges, hence has probability at most  $Ce^{-cd(t)^{1/4}}$ . The extremum location itself has the nail tail  $CN^{-2}e^{-c|x|/N}$ ; summing in  $x$  gives the required contribution to (21.62).

Right-coordinate errors in these cases require either a simultaneous right/vertical extremum or two distinct nails meeting the active track. Thus they are supported on  $\text{BAD}_1(t) \cup \text{BAD}_3(t)$ , and (21.57)–(21.63) give the same bound.

*Case 3: no vertical extremum.* Only the right coordinate can move. Let  $X(t)$  be the independent indicator vector used by the decorated coupling and  $\tilde{X}(t)$  the actual vector saying which nails have a right extremum on the active track. Each coordinate has the same marginal under the two laws. Under maximal coupling, inclusion–exclusion gives

$$\mathbb{P}(X_i(t) \neq \tilde{X}_i(t) \mid H(t)) \leq C \mathbb{P}(|\tilde{X}(t)| \geq 2 \mid H(t)).$$

The latter event is  $\text{BAD}_3(t)$ . If the vectors agree and have no 1, a true right displacement requires a fake right extremum, hence  $\text{BAD}_4(t)$ . If they agree and contain at least two 1's, we are again in  $\text{BAD}_3(t)$ . Finally, once there is exactly one right extremum, failure of the local IIC coupling is controlled by the same screening argument as above; the only extra geometric obstruction is  $\text{BAD}_5(t)$ . Applying (21.57), the bounded one-step displacement, and (21.63) completes the estimate.

Adding the four cases proves (21.62).  $\square$

**Definition 10.298** (Compounded times and IIC speeds). Let

$$\tau_k := 2kN, \quad 0 \leq k < \left\lceil \frac{2N}{\sin \alpha} \right\rceil,$$

so that the interval  $[\tau_k, \tau_{k+1})$  is exactly the sequence of elementary exchanges that moves one oblique track through the square block. Let  $\mathcal{F}_k$  be the sigma-field generated by the coupling before time  $\tau_k$ . Define the ideal speeds

$$\begin{aligned} v_T &:= \sin \alpha - \frac{\mathbb{P}(\delta_1^{\text{IIC}} T = 0)}{\mathbb{P}(\delta_0^{\text{IIC}} T = \sin \alpha - 1)}, \\ v_B &:= \sin \alpha - \frac{\mathbb{P}(\delta_0^{\text{IIC}} B = \sin \alpha - 1)}{\mathbb{P}(\delta_{-1}^{\text{IIC}} B = 0)}, \\ v_R &:= \frac{\mathbb{E}[\delta^{\text{IIC}} R]}{\mathbb{P}(\delta^{\text{IIC}} R \in \{0, -1\})}. \end{aligned}$$

The denominators are positive by finite energy of the local rhombus configurations.

**Proposition 10.299** (Compounded vertical increments). *Let  $A \in \{T, B\}$ . For every nail  $i \in I(\tau_k)$  there are random variables  $\Delta_{A,k}^{\text{IIC}}(i)$  and  $\Delta_{A,k}^{\text{err}}(i)$  such that, with  $\tau_{\text{end}}(i)$  the first time the nail label disappears,*

$$A^{(\tau_{\text{end}} \wedge \tau_{k+1})}(i) - A^{(\tau_k)}(i) = \Delta_{A,k}^{\text{IIC}}(i) + \Delta_{A,k}^{\text{err}}(i), \quad (10.43)$$

$$\mathbb{E}[e^{c|\Delta_{A,k}^{\text{IIC}}(i)|} \mid \mathcal{F}_k] \leq C, \quad (10.44)$$

$$\mathbb{E}[|\Delta_{A,k}^{\text{err}}(i)| \mid \mathcal{F}_k] \leq C \mathbb{E} \left[ \sum_{s=\tau_k}^{\tau_{\text{end}} \wedge \tau_{k+1}} M_s(i) \mid \mathcal{F}_k \right]. \quad (10.45)$$

If  $A^{(\tau_k)}(i)$  is already below the bottom of the active block the conditional mean is zero; otherwise

$$\mathbb{E}[\Delta_{A,k}^{\text{IIC}}(i) \mid \mathcal{F}_k] = v_A + O\left(e^{-c|A^{(\tau_k)}(i)-b(\tau_k)|} + \mathbb{P}(\tau_{\text{end}} < \tau_{k+1} \mid \mathcal{F}_k)\right). \quad (10.46)$$

*Proof.* We give the top case; the bottom case is obtained by reflection. If  $T^{(\tau_k)}(i) \leq b(\tau_k)$ , no exchange in the compounded block reaches the top extremum, so both increments are zero.

Assume  $T^{(\tau_k)}(i) > b(\tau_k)$ . During the descent there is a unique first time  $\tau$  at which the top lies one unit below the active track. At that exchange either the ideal increment is  $\sin \alpha$ , in which case the top is released, or it is 0, in which case subsequent exchanges drag the top downward by unit steps until the first trial with  $\delta_0^{\text{IIC}}T \neq -1$ . Let  $\sigma$  be the corresponding stopping time for the true increments and  $\sigma^{\text{IIC}}$  for the independent IIC trials, both truncated at  $\tau_{\text{end}} \wedge \tau_{k+1}$ . Direct inspection of the rhombus geometry gives

$$T^{(\tau_{\text{end}} \wedge \tau_{k+1})} - T^{(\tau_k)} = \sin \alpha - (\sigma - \tau),$$

so define

$$\Delta_{T,k}^{\text{IIC}} := \sin \alpha - (\sigma^{\text{IIC}} - \tau), \quad \Delta_{T,k}^{\text{err}} := \sigma^{\text{IIC}} - \sigma.$$

This proves (10.43). Conditional on  $\mathcal{F}_k$ , the trials after  $\tau$  are independent and have a success probability bounded away from zero; hence  $\sigma^{\text{IIC}} - \tau$  is dominated by a geometric variable and (10.44) follows.

For the mean, remove the two truncations. The first trial uses  $\delta_1^{\text{IIC}}T$  and the subsequent trials use  $\delta_0^{\text{IIC}}T$ , so summing the geometric tail gives exactly

$$\sin \alpha - \frac{\mathbb{P}(\delta_1^{\text{IIC}}T = 0)}{\mathbb{P}(\delta_0^{\text{IIC}}T = \sin \alpha - 1)} = v_T.$$

Truncation by the bottom of the block costs  $O(e^{-c|T^{(\tau_k)}-b(\tau_k)|})$ , and truncation by nail death costs its conditional probability. This is (10.46).

It remains to compare  $\sigma$  and  $\sigma^{\text{IIC}}$ . Generate after  $\sigma^{\text{IIC}}$  successive ideal success times  $\sigma_0^{\text{IIC}}, \sigma_1^{\text{IIC}}, \dots$ . If  $s \leq \sigma^{\text{IIC}}$  lies after the true stopping time, at least one one-step error has occurred before  $s$ , followed by a run of ideal failures. Conversely, if  $\sigma_\ell^{\text{IIC}} < s \leq \sigma_{\ell+1}^{\text{IIC}}$  lies before the true stopping time, then at least  $\ell$  one-step errors of the required sign have occurred. Independence from Proposition 10.297 and summation of the geometric tails give

$$\mathbb{E}[|\sigma - \sigma^{\text{IIC}}| \mid \mathcal{F}_k] \leq C \mathbb{E}\left[\sum_{s=\tau_k}^{\tau_{\text{end}} \wedge \tau_{k+1}} M_s(i) \mid \mathcal{F}_k\right],$$

which is (10.45). □

**Proposition 10.300** (Compounded right increments). *Assume first  $\cos \alpha \notin \mathbb{Q}$ . For every  $i \in I(\tau_k)$  there are  $\Delta_{R,k}^{\text{IIC}}(i)$  and  $\Delta_{R,k}^{\text{err}}(i)$  satisfying (10.43)–(10.45) with  $A = R$ . If  $R^{(\tau_k)}(i) \notin k \cos \alpha + \mathbb{Z}$  then the ideal mean is zero; otherwise*

$$\mathbb{E}[\Delta_{R,k}^{\text{IIC}}(i) \mid \mathcal{F}_k] = v_R + O\left(\mathbb{E}\left[\sum_{s=\tau_k}^{\tau_{\text{end}} \wedge \tau_{k+1}} M_s(i) \mid \mathcal{F}_k\right] + \mathbb{P}(\tau_{\text{end}} < \tau_{k+1} \mid \mathcal{F}_k)\right). \quad (10.47)$$

*The rational  $\cos \alpha$  case is obtained by recording, in addition to  $R$ , which side of the active track contains the right-most vertex; the same proof then applies.*

*Proof.* If  $R^{(s)} \in k \cos \alpha + \mathbb{Z}$  and the point  $(R^{(s)}, h(s))$  belongs to the nail, one local exchange changes the extremum by one of  $\cos \alpha$ ,  $\cos \alpha - 1$ , or  $-1$ . Once a positive  $\cos \alpha$  jump occurs, later exchanges in that compounded block cannot move the maximum. If the extremal point is not on the active track, only a fake-right event can change  $R$ , and that contribution is already included in the one-step error  $M_s$ .

Let  $\sigma^{\text{IIC}}$  be the first time before nail death or  $\tau_{k+1}$  for which  $X_s(i) = 1$  and the ideal right increment belongs to  $\{\cos \alpha, \cos \alpha - 1\}$ . Set

$$\Delta_{R,k}^{\text{IIC}} := \sum_{\tau_k \leq s \leq \sigma^{\text{IIC}}} X_s(i) \delta^{\text{IIC}} R_s(i),$$

and define the error by subtracting this from the true compounded displacement. The same discrepancy-stopping-time comparison used in Proposition 10.299 gives (10.45); exponential moments follow because the number of  $-1$  trials before the first releasing trial is geometrically dominated.

If  $R^{(\tau_k)} \notin k \cos \alpha + \mathbb{Z}$ , irrationality prevents the active-track point from becoming the true right extremum without a fake-right error, so the ideal increment is zero. Otherwise extend the ideal trial sequence beyond  $\tau_{\text{end}} \wedge \tau_{k+1}$  by fresh independent trials. The untruncated expected sum is, by a geometric-series calculation,

$$\frac{\mathbb{E}[\delta^{\text{IIC}} R]}{\mathbb{P}(\delta^{\text{IIC}} R \in \{0, -1\})} = v_R.$$

The truncated and untruncated stopping times can differ only if the nail dies or if a true one-step discrepancy prevents release; these give the two errors in (10.47). Recording the side information removes the only use of irrationality and proves the rational case as well.  $\square$

**Proposition 10.301** (Speed identification, conditional only on the sine-ratio input). *Assume the sine-ratio conclusion of Proposition 10.285. Then*

$$v_T = v_B = v_R = 0. \quad (10.48)$$

Thus  $R8$  has no independent probabilistic root beyond  $R3$ .

*Proof.* We first compute  $v_T$ . Couple a long finite strip with one exceptional horizontal track above the origin to the strip after exchanging that track once. Let  $l_N$  be the left-most highest vertex of the cluster started at distance  $N$  below the exchange and let  $\Delta T$  be the change of its height. Track exchange preserves the event that  $l_N$  lies on one of the two local tracks. Decomposing according to the track on which  $l_N$  lies and applying the one-step IIC approximation gives, after division by this common two-track event and passage  $N \rightarrow \infty$ ,

$$\Psi_0^2[\ell_{\max} = 0^+] = \Psi_1^2[\ell_{\max} = 0^+] \mathbb{P}(\delta_0^{\text{IIC}} T = \sin \alpha - 1) \quad (10.49)$$

$$+ \Psi_1^2[\ell_{\max} = 0] \mathbb{P}(\delta_1^{\text{IIC}} T = \sin \alpha). \quad (10.50)$$

Apply the sine-ratio theorem with  $\beta = \pi/2$ . Then

$$\Psi_1^2[\ell_{\max} = 0^+] = \frac{\sin \alpha}{1 + \sin \alpha}, \quad \Psi_1^2[\ell_{\max} = 0] = \Psi_0^2[\ell_{\max} = 0^+] = \frac{1}{1 + \sin \alpha}.$$

Writing  $p = \mathbb{P}(\delta_1^{\text{IIC}} T = 0)$  and  $q = \mathbb{P}(\delta_0^{\text{IIC}} T = \sin \alpha - 1)$ , (10.50) becomes  $q = p/\sin \alpha$ . Therefore  $v_T = \sin \alpha - p/q = 0$ . Reflection gives  $v_B = 0$ .

For  $v_R$ , use a rectangular box with  $M = N^2$  vertical tracks and one oblique track. There are two ways to move that oblique track from the upper-right to the lower-left corner. In the first,

exchange it successively with the horizontal tracks and follow the right-most coordinate of the union of clusters meeting a fixed vertical base. The one-step error estimate and the same arm summations as in Lemma 10.296 give total expected error  $O(N^{-c})$ , hence

$$\mathbb{E}[R_{\text{final}}] - \mathbb{E}[R_{\text{initial}}] = v_R + O(N^{-c}).$$

In the second, exchange the oblique track with the vertical tracks. After a quarter-turn, the right displacement is exactly the top-displacement problem already computed, so

$$\mathbb{E}[R_{\text{final}}] - \mathbb{E}[R_{\text{initial}}] = v_T + O(N^{-c}).$$

The two initial graphs coincide, the two final graphs coincide, and the star-triangle kernels preserve the FK marginals, so the left sides of (10.50)–(10.48) agree for the two routes. Equations (21.70)–(21.71) imply  $v_R = v_T + O(N^{-c})$ ; let  $N \rightarrow \infty$ .  $\square$

**Lemma 10.302** (Nails do not move: compounded concentration). *Assume (10.48). There exist  $c_0, C_0 > 0$  such that*

$$\mathbb{P}\left[\exists i \in I(0), \exists t \leq \tau_{\text{end}}(i), \exists A \in \{T, B, R\} : |A^{(t)}(i) - A^{(0)}(i)| \geq N^{1-c_0}\right] \leq C_0 N^{-c_0}. \quad (10.51)$$

*Proof.* It is enough to inspect compounded times, since an extremum is monotone between the two endpoints of one compounded block up to a bounded local rhombus displacement. For each nail and coordinate decompose the accumulated displacement into

$$\Sigma_{A,i}(K) + \mathcal{M}_{A,i}(K) + \mathcal{E}_{A,i}(K),$$

where  $\Sigma$  is the sum of the conditional means in (10.46) or (10.47),  $\mathcal{M}$  is the centered ideal martingale, and  $\mathcal{E}$  is the sum of absolute compounded errors.

By (10.45) and Proposition 10.297,

$$\mathbb{E} \sum_{i,A} \mathcal{E}_{A,i}(K_{\text{max}}) \leq C \sum_{t < T} \mathbb{E}[\text{Err}(t)] \leq C N^{1-c} \log N.$$

(The logarithm comes from summing  $d(t)^{-1}$  across the descending tracks.) Markov's inequality therefore makes  $\mathcal{E} > N^{1-c_0}$  an  $O(N^{-c_0})$  event after choosing  $c_0 \ll c$ .

The ideal increments have uniform exponential moments by (10.44). Truncate them at  $N^{c_1}$ ; the probability that any discarded increment occurs is  $e^{-N^{c_1}}$  times a polynomial number of trials. Azuma–Hoeffding then gives

$$\mathbb{P}\left(\max_K |\mathcal{M}_{A,i}(K)| > N^{3/4}\right) \leq e^{-N^{c_2}}.$$

The nail-count moment bound lets us sum this over all initial nails and coordinates.

Because the speeds vanish,  $\Sigma$  receives a non-negligible contribution only when a vertical extremum is within  $N^{c_3}$  of the lower boundary of its active block, when a one-step error occurs, or when the nail dies during the block. The RSW separation estimate makes the expected number of boundary-near blocks  $O(N^{1-c+c_3})$ . Nail death before the end of a block can occur only if the initial span/margin was within  $2N^{1-c_4}$  of the threshold, or if an ideal/error increment exceeded  $N^{1-c_4}$ . The first event has expectation  $O(N^{1-c_4})$  by the nail-margin RSW estimate, the second by the error bound, and the third exponentially small by (10.44). Choosing  $c_0 \ll c_4 \ll c$  gives  $\max_K |\Sigma_{A,i}(K)| < \frac{1}{2}N^{1-c_0}$  outside an  $O(N^{-c_0})$  event. Combining the three pieces proves (10.51).  $\square$

**Lemma 10.303** (Marked nails exist with high probability). *For  $0 < \varepsilon \ll \eta$  there are  $c, C > 0$  such that*

$$\mathbb{P}(\text{every } x \in B_\eta(N) \text{ has a marked nail at time } 0) \geq 1 - \frac{C}{\eta^2} \left(\frac{\varepsilon}{\eta}\right)^c.$$

*Proof.* Put  $\kappa = (\eta\varepsilon^3)^{1/4}$ . If no markable nail exists near a fixed grid point  $x$ , then either a dual path crosses from radius  $2\varepsilon N$  to  $\kappa N$ , or a primal path crosses from  $\kappa N$  to  $((\eta\varepsilon)^{1/2} - \varepsilon)N$ . Otherwise an intermediate primal circuit separates the two scales, is not connected to the outer boundary, and the cluster containing that circuit has the required span and extremal margins, hence is a markable nail. The uniform annular crossing/arm estimates bound the two exceptional events by  $C(\varepsilon/\eta)^c$ . There are  $O(\eta^{-2})$  grid points, so a union bound proves (21.73).  $\square$

**Proposition 10.304** (Rectangular-isoradial universality, conditional only on R3). *Assume the sine-ratio drift theorem. For every  $\kappa > 0$ , square-lattice critical FK and the rectangular isoradial critical FK model with any fixed bounded angle  $\alpha$  admit, at sufficiently small mesh, a coupling for which the homotopy distance relative to a mesoscopic puncture grid is at most  $\eta(\kappa)$  and hence both  $d_{\text{CN}}$  and  $d_{\text{SS}}$  are at most  $\kappa$  with probability at least  $1 - \kappa$ .*

*Proof.* Choose  $\eta \ll \kappa$  as in Lemma 10.293 and then  $\varepsilon \ll \eta$ . Lemma 10.303 supplies a marked nail at every puncture with probability  $1 - o(1)$ . By Proposition 10.301 and Lemma 10.302, all those nails survive the full track-exchange schedule and their extrema move by  $o(N)$  with probability  $1 - o(1)$ . Proposition 10.297(2) therefore preserves the primal and dual homotopy multisets throughout. At the final time these loops represent the same reduced words as at the initial time, with punctures shifted only by  $o(N)$ . After rescaling by  $N^{-1}$ , the homotopy distance is at most  $\eta + o(1)$ . Lemma 10.293 upgrades this to both macroscopic metrics.  $\square$

**Proposition 10.305** (Reflection–universality–reflection assembly). *Assume Proposition 10.304. Then the finite-mesh rotation coupling (21.55) follows for every angle bounded away from 0 and  $\pi$ .*

*Proof.* For  $\alpha \in (0, \pi/2)$ , choose the rectangular isoradial lattice whose two track families are symmetric with respect to the line  $e^{i\alpha/2}\mathbb{R}$ . Couple the square lattice to this lattice using Proposition 10.304, apply the exact lattice reflection, and couple back to the square lattice. The composite geometric map is rotation by  $\alpha$ , while the two coupling errors add by the triangle inequality. For  $\alpha \in (\pi/2, \pi)$  first apply a global reflection and reduce to the previous case. Approximate a  $C^1$  domain from inside and outside by the corresponding isoradial domains; uniform RSW makes the boundary layer invisible in both macroscopic metrics. Choosing the two coupling errors and the boundary approximation below  $\varepsilon/3$  proves (21.55). Edwards–Sokal transports the FK coupling to the Potts spin correlations. The six-vertex height-increment statement uses the same geometric coupling and the local height/arrow correspondence, with the separated-point restriction ensuring that no microscopic branch choice enters.  $\square$

*Remark 10.306* (Where the distributed proof closes). All ten components in the rotational-invariance architecture are accounted for. R1–R2 are exact earlier results; R3 is proved by Theorem 10.291 and Proposition 10.285; R4–R7 are the local homotopy, marked-nail, one-step and compounded-step proofs; R8 is Proposition 10.301; and R9–R10 are Propositions 10.304 and 10.305. The proof immediately below records the final implication chain so the reader need not infer it from the development history.

*Completion of the proof of Theorem 10.280.* R1–R2 are exact earlier owners. Theorem 10.291 and Proposition 10.285 prove R3. Lemmas 10.293 and 10.294 prove R4–R5; Proposition 10.297 and



Propositions 10.299–10.300 prove R6–R7. Proposition 10.301 proves R8. The zero-speed concentration and nail-existence lemmas then give rectangular-isoradial universality in Proposition 10.304, and Proposition 10.305 gives the required rotation coupling. These implications are exactly the ten-owner architecture of Lemma 10.281; hence (21.55) follows. The Potts and six-vertex consequences were included in the final assembly proposition, completing the theorem.  $\square$

*Remark 10.307 (Scope check).* An earlier draft spoke directly about arbitrary bounded Jordan quads and all subsequential scaling limits. The cited theorem proves the quantitatively stronger but more precisely scoped finite-mesh coupling (21.55) for simply connected  $C^1$ -smooth domains and rotations bounded away from 0 and  $\pi$ . Rotation invariance of crossing probabilities and subsequential limits is a consequence of that coupling, not the source theorem’s literal formulation. We use the source-strength form to avoid silently enlarging the domain regularity or boundary scope.

## 10.8 Six-vertex GFF and the $q = 4$ critical exponents

**Lemma 10.308** (Six-vertex GFF proof architecture and source boundary). *Fix isotropic weights  $a = b = 1$ ,  $\sqrt{3} \leq c \leq 2$ , equivalently  $\Delta \in [-1, -1/2]$ , and write  $\Phi_k^{(\delta)}$  for the scaled  $k$ -point height-increment correlations. The proof of the six-vertex GFF limit has the following four logically separate stages.*

1. Two-point dichotomy. *Cylinder two-point functions have a positive spectral representation for the two commuting transfer directions. The spin-representation RSW/circuit estimates make the corresponding family of spectral measures precompact. After passage first from the cylinder to the plane and then to a scaling subsequence, asymptotic Euclidean-isometry invariance constrains the limiting spectral representation. The conclusion is not immediately “the limit is a GFF”: the precise conclusion is a dichotomy. Either  $\Phi_2^{(\delta)}$  converges to  $\sigma^2 \Psi_2^{\text{GFF}}$ , or two further scaling subsequences converge to GFF two-point functions with two different constants  $\sigma$ .*
2. From two points to all correlations. *Suppose along one subsequence that  $\Phi_2^{(\delta_n)} \rightarrow \sigma^2 \Psi_2^{\text{GFF}}$  uniformly on compact subsets away from collisions. Spectral Cauchy–Schwarz estimates and the regularity bounds imply that every subsequential  $k$ -point limit is harmonic in each coordinate away from collisions. Additivity of height increments, height-flip symmetry, the behaviour when two endpoints merge, and decay at infinity determine that harmonic function inductively. Hence  $\Phi_k^{(\delta_n)} \rightarrow \sigma^k \Psi_k^{\text{GFF}}$  for every  $k$ .*
3. From correlations to the random field. *The all-scale regularity estimate gives uniform integrability near collision diagonals and tightness against finite-Dirichlet-energy test functions. Consequently convergence of all  $k$ -point functions yields convergence of all finite families of test-function pairings to the corresponding Gaussian vector. In the isotropic case the stronger regularity estimate also gives convergence in negative Hölder/Besov/Sobolev topologies in the precise range stated by the source theorem.*
4. Identification of the stiffness. *If a subsequential field limit has variance multiplier  $\sigma^2$ , the source proves the nontrivial identity*

$$\sigma^2 = -\frac{1}{f''(0)},$$

where  $f$  is the torus free energy as a function of slope. This is obtained by splitting high-level circuit large-deviation events into controlled annular events and comparing the resulting exponential rates with the GFF rate; it is not a formal consequence of weak convergence. The exact free-energy calculation gives

$$f''(0) = -\frac{1}{2} \arccos \Delta, \quad \sigma_\Delta^2 = \frac{2}{\arccos \Delta}.$$

Thus the two possible constants in the dichotomy must coincide, and the whole isotropic family converges.

For anisotropic weights with the same  $\Delta$ , the universality/BKW transfer after the natural embedding  $L_\theta$  gives the source theorem's correlation and finite-dimensional-marginal convergence. The source does not claim the additional isotropic negative-Hölder-topology convergence in that anisotropic case.

*Proof.* The purpose of this lemma is to make the dependency architecture precise. Stage 1 is the combination of compactness of the cylinder spectral measures, the full-plane/scaling passage, rotational invariance, concentration on the relevant sub-diagonal, and the monotonicity argument that yields the two-point dichotomy. Stage 2 is the harmonicity/uniqueness induction for multipoint functions. Stage 3 is the regularity-to-test-function convergence theorem. Stage 4 is the separate circuit/free-energy comparison followed by the exact free-energy second derivative. Chaining those four stages gives exactly the claims listed above.

Each of the four stages requires its own argument: the cylinder spectral representation, spin FKG and RSW regularity, level-line-tree and mixing estimates, harmonicity propagation, and the circuit/free-energy comparison. The remainder of this section proves these ingredients before assembling the six-vertex GFF theorem; the four-stage list is a roadmap, not a substitute for those estimates.  $\square$

### 10.8.1 The cylinder spectral representation

The first independent source boundary in Lemma 10.308 is finite-dimensional and can be closed without using any RSW estimate. We record the argument in the normalization needed later. Throughout this subsection the circumference  $L$  is even and we work in the balanced sector of the row state space.

**Definition 10.309** (Balanced row space and normalized transfer operators). Let  $\mathcal{C}_L$  be the set of vertical-arrow columns  $\kappa = (\kappa_j)_{j \in \mathbb{Z}/L\mathbb{Z}} \in \{\pm 1\}^L$  with exactly  $L/2$  entries of each sign, and set

$$\mathcal{H}_L := \ell^2(\mathcal{C}_L).$$

For two columns  $\kappa, \kappa'$  let  $t_{\kappa', \kappa}$  be the total six-vertex weight of one row whose lower and upper vertical arrows are  $\kappa, \kappa'$ . With isotropic weights  $a = b = 1$  and  $c > 0$ , write  $t$  for the resulting row-transfer matrix. Let  $\lambda_0 > 0$  be its Perron eigenvalue and put  $T = t/\lambda_0$ .

Let  $U$  denote cyclic translation of a column by one lattice unit in the vertical direction,  $(Ue_\kappa) = e_{\kappa \uparrow}$ . Finally, if  $\alpha_j$  is the horizontal arrow crossing a fixed vertical cut at height  $j$ , let  $S_j$  be the corresponding normalized insertion operator; we write  $S = S_0$ . Translation covariance gives

$$S_j = U^j S U^{-j}.$$

**Lemma 10.310** (Common spectral basis and cylinder state). *The operators  $T$  and  $U$  commute,  $T$  is real symmetric and primitive, and  $U$  is unitary. Consequently there is an orthonormal basis  $(v_r)_r$  of  $\mathcal{H}_L$  such that*

$$T v_r = \lambda_r v_r, \quad U v_r = e^{-i b_r} v_r,$$

where  $\lambda_0 = 1$ ,  $|\lambda_r| < 1$  for  $r \neq 0$ , and  $b_r \in (2\pi/L)\mathbb{Z} \pmod{2\pi}$ . The vector  $v_0$  can be chosen strictly positive and  $Uv_0 = v_0$ .

For every local cylinder observable  $X$ , represented by an insertion matrix  $O_X$  between two rows, the infinite-cylinder expectation is

$$\mathbf{E}_{\text{Cyl}_L}[X] = \langle v_0, O_X v_0 \rangle.$$

In particular the balanced infinite-cylinder Gibbs state exists.

*Proof.* A one-row configuration may be reflected in a vertical line and all horizontal arrows reversed; because  $a = b$ , this is a weight-preserving bijection from configurations with boundary data  $(\kappa, \kappa')$  to those with  $(\kappa', \kappa)$ . Hence  $t$  is real symmetric. Concatenating sufficiently many rows allows any balanced column to be joined to any other while preserving the ice rule, so a fixed positive power of  $t$  has strictly positive entries. Perron–Frobenius therefore gives a simple top eigenvalue and  $|\lambda_r| < 1$  after normalization for all other modes.

Vertical translation of every arrow in a row is a weight-preserving bijection, so  $Ut = tU$ . Since  $U$  is a permutation matrix, it is unitary. A Hermitian operator commuting with a unitary operator admits a common orthonormal eigenbasis: diagonalize  $T$ , then diagonalize  $U$  on each eigenspace. Because  $U^L = 1$ , every  $U$ -eigenvalue is an  $L$ th root of unity. Positivity and uniqueness of  $v_0$  imply  $Uv_0 = v_0$ .

On a torus of horizontal length  $M$ , insertion of  $X$  gives the ratio

$$\frac{\text{Tr}(O_X t^{M-m})}{\text{Tr}(t^M)},$$

where  $m$  is the horizontal width of the support of  $X$  and harmless powers of  $t$  may be absorbed into  $O_X$ . Divide numerator and denominator by  $\lambda_0^M$  and let  $M \rightarrow \infty$ . All modes except  $v_0$  vanish geometrically, leaving  $\langle v_0, O_X v_0 \rangle$ . This proves both existence and the formula for the cylinder state.  $\square$

**Lemma 10.311** (Reflection-positive spectral measure for local observables). *Fix the vertical reflection line through face centres and write  $X^\dagger$  for the reflected height-gradient observable. If  $X$  is supported to the left of the line and  $Y$  to the right, then there is a finite complex measure  $\nu_{X,Y}$  on  $[0, 2)$  such that, for every integer  $m \geq 0$ ,*

$$\mathbf{E}_{\text{Cyl}_L}[X \tau_{(m,0)} Y] = \int_{[0,2)} (1-a)^m d\nu_{X,Y}(a).$$

Moreover  $\nu_{X,X^\dagger}$  and  $\nu_{Y^\dagger,Y}$  are positive and

$$|\mathbf{E}[XY]|^2 \leq \|\nu_{X,Y}\|_{\text{TV}}^2 \leq \mathbf{E}[XX^\dagger] \mathbf{E}[Y^\dagger Y]. \quad (21.47)$$

*Proof.* Choose a vertical cut separating the supports. Concatenation of row insertions gives vectors  $E_-(X) \in \mathcal{H}_L$  and covectors  $E_+(Y) \in \mathcal{H}_L^*$  so that Lemma 10.310 yields

$$\mathbf{E}[X \tau_{(m,0)} Y] = E_+(Y) T^m E_-(X) = \sum_r \langle E_+(Y), v_r \rangle \langle v_r, E_-(X) \rangle \lambda_r^m.$$

Set  $a_r := 1 - \lambda_r \in [0, 2)$  and define

$$\nu_{X,Y} := \sum_r \langle E_+(Y), v_r \rangle \langle v_r, E_-(X) \rangle \delta_{a_r}.$$

This is finite because the state space is finite-dimensional, and the displayed spectral formula follows.

Reflection of the insertion reverses the cut and takes adjoints, hence  $E_+(X^\dagger) = E_-(X)^*$ . The coefficient of  $\delta_{a_r}$  in  $\nu_{X, X^\dagger}$  is therefore  $|\langle v_r, E_-(X) \rangle|^2 \geq 0$ ; similarly for  $Y$ . Finally

$$\|\nu_{X,Y}\|_{\text{TV}} \leq \sum_r |\langle E_+(Y), v_r \rangle| |\langle v_r, E_-(X) \rangle|$$

and Cauchy–Schwarz bounds this by the square root of the products of the two reflected spectral masses. Evaluating the positive measures at the constant function 1 identifies those masses with  $\mathbf{E}[XX^\dagger]$  and  $\mathbf{E}[Y^\dagger Y]$ , proving (21.47).  $\square$

**Lemma 10.312** (Arrow spectral measure and exact height-increment formula). *There is a finite positive measure  $\mu_L$  on  $(0, 2] \times [-\pi, \pi)$ , invariant under  $(a, b) \mapsto (a, -b)$ , such that for every horizontally ordered quadruple  $u = (u_1, u'_1, u_2, u'_2)$  of face centres, with the notation*

$$u'_i - u_i = (x_i, y_i), \quad u_{i+1} - u'_i = (x'_i, y'_i) \quad (i = 1),$$

*we have*

$$\begin{aligned} \Phi_2^{\text{Cyl}_L}(u) = \int & \left( (1-a)^{x_2} e^{-ib y_2} - 1 \right) (1-a)^{x'_1} e^{-ib y'_1} \\ & \cdot \left( 1 - (1-a)^{x_1} e^{-ib y_1} \right) d\mu_L(a, b). \end{aligned} \quad (21.48)$$

*One may take*

$$d\mu_L(a, b) = \sum_{r \neq 0} \frac{|\langle v_r, S v_0 \rangle|^2}{(1 - \lambda_r)^2} \delta_{(1-\lambda_r, b_r)}(da, db), \quad (21.49)$$

*with zero coefficients omitted. Its  $b$ -support lies in  $(2\pi/L)\mathbb{Z}$ , so it gives no mass to  $0 < |b| < 2\pi/L$ .*

*Proof.* A height difference is a signed sum of crossed arrows. Choose for each pair a path that first moves horizontally and then vertically. Horizontal translation by one row acts by  $T$  and a vertical translation by  $j$  acts by  $U^j$  on the vertical-arrow insertion  $S$ . Expanding the two height differences into arrow sums and inserting the common eigenbasis gives a sum over modes  $r$ .

For a fixed non-Perron mode the horizontal geometric sum is

$$\sum_{p=0}^{x-1} \lambda_r^p = \frac{1 - \lambda_r^x}{1 - \lambda_r},$$

while the vertical displacement contributes  $e^{-ib_r y}$ . The two height increments provide two such factors; the horizontal gap between them contributes  $\lambda_r^{x'_1} e^{-ib_r y'_1}$ . Multiplication yields exactly the integrand in (21.48), with the two denominators absorbed into (21.49). The Perron mode does not contribute because the one-arrow observable is odd under global arrow reversal, so  $\langle v_0, S v_0 \rangle = 0$ .

The coefficients in (21.49) are nonnegative. Reflection in the horizontal axis complex conjugates the  $U$ -eigenvalue while preserving the  $T$ -eigenvalue and the squared matrix element, which proves invariance under  $b \mapsto -b$ . Since  $|\lambda_r| < 1$  for  $r \neq 0$ ,  $a_r = 1 - \lambda_r$  lies in  $(0, 2]$ . Finally  $U^L = 1$ , so every  $b_r$  is a multiple of  $2\pi/L$ , proving the asserted gap in the  $b$ -support.  $\square$

*Remark 10.313* (Scope of the preceding argument). Lemmas 10.310–10.312 are the complete local proof for the 2026 source’s Ingredient 4: cylinder spectral representation and reflection-positive Cauchy–Schwarz. They use only finite-dimensional transfer-matrix algebra and the balanced-cylinder limit. They do *not* use the Bethe Ansatz, RSW theory, or the GFF conclusion.

The next stage is Ingredient 3: the spin representation, FKG/domain monotonicity, uniform circuit estimate, level-line-tree recursion, full-plane/cylinder regularity bounds, and the mixing estimate. Those estimates are what turn the finite measures  $\mu_L$  into a compact scale-invariant family and therefore remain logically prior to the two-point dichotomy.

### 10.8.2 The six-vertex spin/FKG package

The regularity argument is most transparent after reducing the height modulo four. This reduction is not merely a change of notation: it produces an attractive finite spin–bond system to which the correlation inequalities already proved in Volumes VIII–IX apply. We record the finite-domain construction and its Markov/FKG consequences before isolating the genuinely new RSW input.

**Definition 10.314** (Even/odd spins and agreement diagonals). Let  $D$  be a finite even domain of faces. Write  $\mathcal{F}_D^\circ$  and  $\mathcal{F}_D^\bullet$  for the faces on which an admissible height is respectively even and odd, and let  $E_D^\circ, E_D^\bullet$  be the two diagonal square lattices. For a height  $h$  define

$$\sigma^\circ(x) = \begin{cases} +1, & h(x) \equiv 0 \pmod{4}, \\ -1, & h(x) \equiv 2 \pmod{4}, \end{cases} \quad \sigma^\bullet(y) = \begin{cases} +1, & h(y) \equiv 1 \pmod{4}, \\ -1, & h(y) \equiv 3 \pmod{4}. \end{cases}$$

For a spin field  $\sigma$  let  $A(\sigma)$  be the set of diagonal edges whose endpoints have the same spin. If  $e \in E_D^\circ$  and  $e^* \in E_D^\bullet$  cross at a primal six-vertex site, call  $(\sigma^\circ, \sigma^\bullet)$  *consistent* when

$$e \in A(\sigma^\circ) \quad \text{or} \quad e^* \in A(\sigma^\bullet) \quad \text{for every such pair } (e, e^*).$$

**Lemma 10.315** (Spin coupling and recovery of the six-vertex height). *Fix  $c \in [1, 2]$  and put  $p = 1/c$ . With + even-spin boundary condition, define a law on triples  $(\sigma^\circ, \omega, \sigma^\bullet)$  by requiring*

$$\omega \subset A(\sigma^\circ), \quad \partial D \subset \omega, \quad e \notin \omega \implies e^* \in A(\sigma^\bullet),$$

*and assigning unnormalised weight*

$$c^{|A(\sigma^\circ)|} p^{|\omega|} (1-p)^{|A(\sigma^\circ) \setminus \omega|}.$$

*Then:*

1. *after summing over  $\omega$ , the spin marginal is proportional to*

$$\mathbf{1}_{\{\sigma^\circ|_{\partial D}=+\}} \mathbf{1}_{\{\sigma^\circ \perp \sigma^\bullet\}} c^{|A(\sigma^\circ)|+|A(\sigma^\bullet)|};$$

2. *conditionally on  $(\sigma^\circ, \omega)$ , the odd spin is an independent fair sign on each bounded component of  $\mathbb{R}^2 \setminus \omega$ ;*

3. *the gradient determined on adjacent even/odd faces by*

$$h(y) - h(x) = \sigma^\circ(x) \sigma^\bullet(y)$$

*is exactly the  $a = b = 1$  six-vertex height law with weight  $c$  for the  $c$ -type vertices;*

4. *if  $D \subset B$  are even domains, conditioning on a monochromatic  $\omega$ -circuit along  $\partial D$  gives the same spin–bond law inside  $D$ , independently of the exterior.*

*Proof.* For fixed spins, examine one crossing pair  $(e, e^*)$ . Consistency says that at least one of the two diagonals is an agreement. If only the even diagonal agrees, the constraint forces  $e \in \omega$ , and the local contribution from the displayed spin-bond weight is  $cp = 1$ . If both diagonals agree,  $e$  may be open or closed and summing gives  $c(p + 1 - p) = c$ . Since  $|A(\sigma^\circ)| + |A(\sigma^\bullet)|$  equals the number of six-vertex sites plus the number at which both diagonals agree, this is the stated spin marginal up to the fixed factor  $c^{|V(D)|}$ .

For fixed  $(\sigma^\circ, \omega)$ , the remaining condition is precisely that  $\sigma^\bullet$  be constant across every dual edge whose crossing even edge is closed. Thus it is constant on each component of  $\mathbb{R}^2 \setminus \omega$ ; the unbounded component is fixed by the boundary convention and every bounded component may be flipped without changing the weight. This proves the fair independent signs.

The consistency rule is the ice rule modulo four. Around every primal vertex, the four height increments obtained from  $\sigma^\circ \sigma^\bullet$  sum to zero; conversely the six ice-rule arrow patterns give exactly the consistent local spin patterns. A site with both agreement diagonals is of  $c$ -type, while an  $a$ - or  $b$ -type site has only one agreement diagonal. The preceding marginal calculation therefore reproduces the six-vertex weight.

Finally condition on an open boundary circuit of fixed sign. In the product the displayed spin-bond weight, all edge factors and all component-count choices split between the inside and outside once the boundary state is fixed. The conditional density therefore factorises. Its inside factor is exactly the same expression with domain  $D$ , which proves the Markov statement.  $\square$

**Definition 10.316** (Signed bond pair and the  $\circ$ -order). Split  $\omega$  according to the even spin:

$$\omega^+ = \{e \in \omega : \sigma^\circ \equiv + \text{ on } e\}, \quad \omega^- = \{e \in \omega : \sigma^\circ \equiv - \text{ on } e\}.$$

We order triples  $(\sigma^\circ, \omega^+, \omega^-)$  coordinatewise in  $(\sigma^\circ, \omega^+, -\omega^-)$  and call functions increasing for this order  $\circ$ -increasing.

**Lemma 10.317** (Ising partition functions are log-supermodular in inverse couplings). *For a finite graph set*

$$Z_{\text{Ising}}(a) = \sum_{\tau \in \{\pm 1\}^V} \prod_{uv \in E} a_{uv}^{\mathbf{1}_{\{\tau_u \neq \tau_v\}}}, \quad a \in (0, 1]^E.$$

*Then  $Z_{\text{Ising}}$  is increasing and*

$$Z_{\text{Ising}}(a \vee b) Z_{\text{Ising}}(a \wedge b) \geq Z_{\text{Ising}}(a) Z_{\text{Ising}}(b).$$

*The same holds by continuity when some coordinates vanish.*

*Proof.* Write  $a_e = e^{-2J_e}$  with  $J_e \geq 0$ . Up to the positive factor  $e^{-\sum_e J_e}$ ,  $Z_{\text{Ising}}(a)$  is the usual zero-field ferromagnetic Ising partition function  $\tilde{Z}(J) = \sum_{\tau} e^{\sum_e J_e \tau_u \tau_v}$ . Volume VIII proves the second Griffiths inequality, hence

$$\frac{\partial^2}{\partial J_e \partial J_f} \log \tilde{Z}(J) = \text{Cov}_J(\tau_{e_-} \tau_{e_+}, \tau_{f_-} \tau_{f_+}) \geq 0.$$

Integrating the nonnegative mixed derivatives over the coordinate rectangle with opposite corners  $J(a)$  and  $J(b)$  gives supermodularity of  $\log \tilde{Z}$ . The map  $a_e \mapsto J_e$  reverses each coordinate, so coordinatewise maxima and minima are merely interchanged; the same lattice inequality holds for  $\log Z_{\text{Ising}}(a)$ . Monotonicity is immediate from the defining sum. Letting  $J_e \rightarrow \infty$  gives the zero-coordinate extension.  $\square$

**Proposition 10.318** (Conditional FKG and domain/boundary monotonicity). *Let  $E$  specify finitely many even spins and finitely many even-lattice bond states in a way having positive probability. Under the conditional spin–bond law in an even domain:*

1.  $\sigma^\circ$  is positively associated;
2. conditionally on  $\sigma^\circ$ , the fields  $\omega^+$  and  $\omega^-$  are independent, with  $\omega^+$  stochastically increasing and  $\omega^-$  stochastically decreasing in  $\sigma^\circ$ ;
3. each conditional bond field is a  $q = 2$  random-cluster law (after planar duality) and is FKG;
4. consequently every pair of bounded  $\circ$ -increasing observables  $X, Y$  satisfies

$$\mathbf{E}[XY] \geq \mathbf{E}[X]\mathbf{E}[Y].$$

If  $D \subset B$  and  $X$  is  $\circ$ -increasing and supported in  $D$ , then

$$\mu_D^-[X] \leq \mu_B^-[X] \leq \mu_B^+[X] \leq \mu_D^+[X].$$

*Proof.* Condition on a lattice event  $E = \{\omega|_Q = \zeta, \sigma^\circ|_C = \tau\}$  and split  $Q = Q_+ \sqcup Q_- \sqcup Q_0$  into edges forced open in the  $+$  phase, forced open in the  $-$  phase, and forced closed. Summing the bond variables of one sign while keeping  $\sigma = \sigma^\circ$  fixed gives an Ising partition function on the dual odd lattice. More precisely, if  $uv$  is an odd-lattice edge and  $uv^*$  the crossing even edge, set

$$a_{uv}^\pm(\sigma) = \begin{cases} 1, & uv^* \in Q_\pm, \\ 1/c, & uv^* \in A_\pm(\sigma) \setminus Q, \\ 0, & \text{otherwise.} \end{cases}$$

To see the marginal explicitly, use Lemma 10.315: for one sign,  $2^{f(\omega^\pm)}$  is the number of auxiliary odd-spin fields that are constant on the components of the complement. Thus

$$2^{f(\omega^\pm)} = c_\partial \sum_{\tilde{\sigma} \in \{\pm 1\}^{\mathcal{F}_D^\bullet}} \mathbf{1}_{\{\tilde{\sigma} \text{ is constant across every closed dual edge}\}},$$

where the boundary factor  $c_\partial \in \{1, 2\}$  is independent of  $\sigma$ . Insert this identity and sum the remaining bond choices edge by edge. An odd edge  $uv$  contributes exactly  $(a_{uv}^\pm)^{\mathbf{1}_{\{\tilde{\sigma}_u \neq \tilde{\sigma}_v\}}}$ . Consequently, after summing  $\omega^\pm$ , the marginal weight of  $\sigma$  is, up to a constant,

$$\mathbf{1}_{\{\sigma|_C = \tau\}} c^{|A(\sigma)|} Z_{\text{Ising}}(a^+(\sigma)) Z_{\text{Ising}}(a^-(\sigma)).$$

The first factor is a ferromagnetic pair weight. For  $c \geq 1$ , changing  $\sigma$  upward makes  $a^+$  increase in the lattice sense and  $a^-$  decrease after reversing the order. Lemma 10.317, together with the finite-lattice FKG theorem proved in Volume VIII, shows that the displayed spin marginal is log-supermodular. Hence  $\sigma^\circ$  is positively associated.

For fixed  $\sigma$ , the weight separates into a  $+$  bond factor and a  $-$  bond factor, so the two bond fields are independent. The component factor arising from the free odd-spin signs is  $2^k$ ; therefore, after duality, each factor is exactly a random-cluster measure with  $q = 2$  and edge parameter determined by  $a^\pm(\sigma)$ . Volume IX proves FKG, the domain Markov property and monotonicity in the edge parameters for these measures. This yields the stated conditional association and stochastic monotonicity.

Now let  $X, Y$  be  $\circ$ -increasing. Conditional FKG gives

$$\mathbf{E}[XY \mid \sigma] \geq \mathbf{E}[X \mid \sigma]\mathbf{E}[Y \mid \sigma].$$

Both conditional expectations are increasing functions of  $\sigma$  by the preceding stochastic comparison, so the already-proved FKG of  $\sigma$  and the tower property imply

$$\mathbf{E}[XY] \geq \mathbf{E}\{\mathbf{E}[X \mid \sigma]\mathbf{E}[Y \mid \sigma]\} \geq \mathbf{E}[X]\mathbf{E}[Y].$$

Finally embed  $D$  in a larger even domain and condition successively on an all-+ or all-− open circuit along  $\partial D$ . Lemma 10.315 identifies the conditional inside law, and the FKG inequality compares the conditioning event with any increasing  $X$ . Applying the same argument with the reversed order gives the four displayed domain/boundary inequalities.  $\square$

*Remark 10.319* (Scope boundary of the local argument). The spin representation, its finite-domain Markov property, the conditional FKG theorem and domain/boundary monotonicity are now exact local owners. What is *not* yet owned is the scale-free geometric input. The 2026 GFF source’s annulus circuit theorem is a translation of the boundary-uniform level-set RSW theorem of Duminil-Copin–Karrila–Manolescu–Oulamara. That theorem is therefore the sole new root before the level-line-tree regularity and mixing arguments can be reconstructed.

### 10.8.3 Boundary-uniform height-level RSW and annulus circuits

We now prove the required boundary-uniform geometric estimate. The primary JEMS theorem is stated for the larger delocalized interval  $1 \leq c \leq 2$ ; the six-vertex GFF theorem below only uses  $\sqrt{3} \leq c \leq 2$ . We therefore prove exactly the latter range. This restriction is mathematical scope, not a change in the proof: the geometric estimates use only  $c \geq 1$ , while the phase-sensitive step is the quadratic small-slope free-energy bound already owned in Theorem 10.291.

**Definition 10.320** (Height crossings and annulus circuits). If  $D$  is a discrete face domain and  $A, B \subset D$ , write

$$A \xrightarrow[h \geq k]{D} B$$

for the event that  $A$  and  $B$  are joined in  $D$  by a nearest-neighbour face path on which  $h \geq k$ . A  $\times$ -path uses diagonal face adjacency (faces sharing a primal vertex), and is denoted by  $\xrightarrow{\times}$ . Let  $\Lambda_r = [-r, r]^2$  in the dual lattice and, for  $0 < r < R$ , let  $\mathcal{O}_{h \geq k}(r, R)$  be the event that  $\Lambda_R \setminus \Lambda_r$  contains a nearest-neighbour circuit surrounding  $\Lambda_r$  on which  $h \geq k$ .

**Lemma 10.321** (Height FKG, absolute-height FKG, and boundary pushing). *Let  $D$  be finite and let  $\xi \leq \xi'$  be admissible boundary data for the six-vertex height model with  $a = b = 1$  and  $c \geq 1$ .*

1. *Increasing functions of  $h$  are positively associated and their expectation is increasing with the boundary condition.*
2. *If  $0 \leq \xi \leq \xi'$  then increasing functions of  $|h|$  are positively associated and their expectation is increasing with the absolute boundary data. The same assertion holds after conditioning  $|h|$  on finitely many faces.*
3. *Let  $D \subset D'$  and assume the boundary data on  $D'$  are bounded below by an integer  $m$ . If  $\xi$  is the minimal admissible extension to  $\partial D$  which keeps the inherited values and is at least  $m$ , then for every family  $\mathcal{C}$  of connected sets and every  $k > m$ ,*

$$\mathbb{P}_D^\xi(\exists C \in \mathcal{C} : h|_C \geq k + 2) \leq 2 \mathbb{P}_{D'}^{\xi'}(\exists C \in \mathcal{C} : h|_C \geq k). \quad (21.60)$$

*If every  $C$  meets the inherited boundary, the factor 2 is absent.*



*Proof.* We give the local calculation because it is used repeatedly below. Condition on the height away from one face  $x$ . If the neighbouring heights have minimum  $m$  and maximum  $M$ , admissibility gives  $M \in \{m, m+2\}$ . Unless  $M = m$ , the value at  $x$  is forced. In the only nontrivial case  $M = m$ , the two possible values are  $m-1$  and  $m+1$ . Let  $n_-$  and  $n_+$  count the diagonally adjacent faces having heights  $m-1$  and  $m+1$ , respectively. The ratio of the conditional weights of the upper and lower choices is

$$\frac{c^{n_+}}{c^{n_-}} = c^{n_+ - n_-}.$$

Raising the exterior configuration coordinatewise can only increase  $n_+$  and decrease  $n_-$ , so, since  $c \geq 1$ , the one-site conditional probability of the upper value is monotone. Successive one-face updates connect every two admissible configurations (the pointwise maximum of two height functions is again admissible). The finite-lattice Holley criterion therefore gives both ordinary FKG and comparison of boundary conditions.

For  $|h|$ , the same update argument is automatic unless the common neighbouring absolute height is 1. In that case the choice at  $x$  is between  $|h(x)| = 0$  and 2. Conditioning on the absolute values, signs are constant on the connected components of nonzero faces and their conditional law is a ferromagnetic Ising model: each  $c$ -vertex contributes a nonnegative pair coupling, while prescribed nonnegative boundary heights produce a nonnegative boundary field. Griffiths–FKG for this Ising law, already proved in Volume VIII, shows that the ratio of the total weight with  $|h(x)| = 2$  to that with  $|h(x)| = 0$  is increasing in the exterior absolute heights. This is the missing local case of the Holley test, and proves the absolute-height assertions, including their conditional form.

For the pushing estimate, let  $\eta$  be the smallest realization in  $D'$  with the prescribed outer data and  $\eta \geq m-1$ . Apply absolute-height FKG to the increasing event that some  $C \in \mathcal{C}$  carries height at least  $k$ : replacing  $h$  by  $|h|$  costs at most the two choices of the sign of the witnessing positive component. The spatial Markov property then identifies the conditional inside law with boundary  $\eta|_{\partial D}$ . Minimality gives  $\eta|_{\partial D} \geq \xi - 2$ , and ordinary boundary comparison followed by the height shift  $h \mapsto h+2$  yields (21.60). If the witnessing connected set meets inherited boundary on which the sign is fixed positive, the sign ambiguity is absent and so is the factor 2.  $\square$

**Lemma 10.322** (Symmetric-quad crossing). *Let  $(D; a, b, c, d)$  be a simply connected discrete quad and suppose a lattice symmetry  $\sigma$  maps the arcs  $(ab), (cd)$  to  $(bc), (da)$  and preserves even/odd faces. If  $\sigma\xi \geq \xi$ , then*

$$\mathbb{P}_D^\xi((ab) \xrightarrow[h \geq 0]{D} (cd)) \geq \frac{1}{2}. \quad (21.61)$$

*Proof.* Planar duality for face paths gives

$$\{(ab) \xrightarrow[h \geq 0]{} (cd)\}^c = \{(bc) \xrightarrow[h < 0]{\times} (da)\} \subseteq \{(bc) \xrightarrow[h \leq 0]{} (da)\}.$$

After applying  $\sigma$ , the last event has the same probability as a decreasing  $h \leq 0$  crossing of the original orientation under boundary  $\sigma\xi$ . Boundary comparison decreases its probability when  $\sigma\xi$  is replaced by  $\xi$ , and the height-sign symmetry changes that event into the original  $h \geq 0$  crossing. Hence the complement has probability at most the event itself, proving (21.61).  $\square$

**Proposition 10.323** (Slit separation). *There exist absolute  $\delta, c_0 > 0$  such that the following holds. Let  $S = \mathbb{Z} \times [0, n]$ , let  $I = [0, \delta n] \times \{0\}$  and  $J = [i, i + \delta n] \times \{n\}$ , and let  $\xi$  be the largest admissible boundary condition which is at most  $k$  and is 0, 1 off  $I \cup J$ . For  $k \geq c_0^{-1}$ ,*

$$\mathbb{P}_S^\xi(I \xrightarrow[h \geq (1-c_0)k]{S} J) \leq 1 - c_0, \quad i \in \mathbb{Z}. \quad (21.62)$$

*Proof.* We first take  $i = 0$ . Around the two slits place four symmetric quads, two on each side, whose long direction separates the slits. Choose  $c_0 = 1/4$  initially and take  $k$  large. For one such quad  $Q$ , sign flip followed by boundary comparison reduces the event of a low crossing  $h \leq c_0 k$  to the symmetric-quad situation of Lemma 10.322; hence each low-crossing event has probability at least a fixed positive constant. The four events are increasing functions of  $|h|$  after subtracting  $k/2$ , so conditional absolute-height FKG from Lemma 10.321 gives a fixed positive probability that all four occur simultaneously. Their union contains a closed face curve of height at most  $(1 - c_0)k$  separating the two slits. By planar duality this excludes an  $h \geq (1 - c_0)k$  connection between them. This proves (21.62) for aligned slits.

For  $|i| \leq 2\delta n/3$ , replace  $\delta$  by  $\delta/3$  and enlarge the target slit to the aligned one; boundary comparison and event inclusion reduce to the previous case. For  $|i| > 2\delta n/3$ , reflect the upper slit across the vertical midline of the intervening rectangle. Lemma 10.322 supplies, with probability at least  $1/2$ , a transverse low crossing between the reflected boundary arcs. Such a crossing blocks every high path between the original disjoint slits. Shrinking  $c_0$  once more makes the constants uniform in  $i$  and completes the proof.  $\square$

**Proposition 10.324** (Strip gluing and height-loss RSW). *There exist absolute  $\delta, c_1, C_1 > 0$  such that for every  $k \geq c_1^{-1}$  and  $n$ ,*

$$\begin{aligned} & \mathbb{P}_{\Lambda_{12n}}^{0,1}(\mathcal{O}_{h \geq c_1 k}(6n, 12n)) \\ & \geq c_1 \left[ \mathbb{P}_{\mathbb{Z} \times [-n, 2n]}^{0,1}([0, \delta n] \times \{0\} \xrightarrow[h \geq k]{\mathbb{Z} \times [0, n]} \mathbb{Z} \times \{n\}) \right]^{C_1}. \end{aligned} \quad (21.63)$$

*Proof.* Let  $p$  denote the strip-crossing probability on the right. Split the top boundary into intervals  $I_j = [j\delta n, (j+1)\delta n] \times \{n\}$  and let  $I_0$  be the bottom seed. If  $I_0$  reaches the top at height  $k$ , at least one of the left, central, and right target families has probability at least  $p/3$ . Reflecting vertically and using FKG, two copies of the same alternative connect neighboring seed intervals with probability at least  $(p/3)^2$ . Proposition 10.323, applied after exposing the outermost high paths, prevents the two exposed interfaces from being trapped against the slits: conditionally on the exterior, with probability at least  $c_0$  there is a bridge of height  $c_0 k$  between the two seeds. Thus for every fixed integer  $j$ ,

$$\mathbb{P}(I_{j-1} \xrightarrow[h \geq c_0 k]{} I_{j+1}) \geq c p^2. \quad (21.64)$$

Concatenating a bounded number of these bridges by FKG yields a horizontal crossing of an  $18n \times 3n$  rectangle at height  $ck$  with probability at least  $c p^C$ .

To close this crossing into a circuit, attach squares at the two ends. By the symmetric-quad lemma, each square has probability at least  $1/2$  of a vertical crossing of height at most roughly  $ck/2$ . Explore the lowest horizontal high crossing and the two extremal low vertical crossings. The spatial Markov property leaves between them a domain whose boundary heights are uniformly bounded by the low crossings. Boundary pushing moves this law to a rectangular law with  $0, 1$  boundary at a cost of two units in height. Rotating the resulting long-rectangle estimate through the four compass directions and applying FKG produces four high crossings whose union contains an annular circuit. All losses are fixed multiplicative losses in the height and fixed powers of  $p$ ; renaming the constants gives (21.63).  $\square$

**Lemma 10.325** (Endpoint  $\Delta = -1$  quadratic free-energy edge). *At the endpoint  $c = 2$  (equivalently  $\Delta = -1$  in the balanced six-vertex normalization), the sector free energy satisfies*

$$f_2(\alpha) = f_2(0) - A_2 \alpha^2 + o(\alpha^2) \quad (21.65a)$$

as  $\alpha \rightarrow 0$ , for a constant  $A_2 \in (0, \infty)$ .

*Proof.* This is the endpoint version of the continuum-Bethe calculation, and we record the two estimates that replace the interior  $\zeta > 0$  argument. Let  $R_{-1}$  be the resolvent kernel in the equivalent truncated equation

$$\rho(x | q) - \int_{\mathbb{R} \setminus [-q, q]} R_{-1}(x - y) \rho(y | q) dy = \rho(x). \quad (21.65b)$$

At  $\Delta = -1$  the tail operator in this formulation is a contraction on both  $L^1$  and  $L^\infty$ ; hence the truncated density exists uniquely. On the half-line, the same contraction gives a unique  $T \in L^1(\mathbb{R})$  satisfying

$$T(x) - \int_0^\infty R_{-1}(x - y) T(y) dy = e(x) := e^{-\pi x} \mathbf{1}_{x > 0}. \quad (21.65c)$$

Splitting the tail operator into its right and left pieces and expanding the convergent Neumann series, the pure right and pure left words sum to the two translates of  $T - e$ ; every mixed word must cross from one tail to the other and therefore gains a second endpoint exponential factor. Consequently

$$\rho(x | q) = \rho(x) + e^{-\pi q} [(T - e)(q - x) + (T - e)(-q - x)] + \delta_q(x), \quad \|\delta_q\|_1 + \|\delta_q\|_\infty \leq C e^{-2\pi q}. \quad (21.65d)$$

Integrating the complement of  $[-q, q]$  and using the defining relation between the mass  $m$  and the cutoff  $Q(m)$  gives

$$(\tfrac{1}{2} - m) e^{\pi Q(m)} \longrightarrow C_{-1} \in (0, \infty). \quad (21.65e)$$

The positivity is immediate from the positive Neumann series for  $T$ .

It remains to identify the free-energy loss. Put

$$\delta f(q) = \int_{\mathbb{R}} L(x) \rho(x) dx - \int_{-q}^q L(x) \rho(x | q) dx,$$

where  $L$  is the logarithmic transfer weight. Rewriting with (21.65b), Fourier transforming, and shifting the contour across the first pair of endpoint poles gives

$$\delta f(q) = B_{-1} e^{-2\pi q} + o(e^{-2\pi q}), \quad B_{-1} > 0. \quad (21.65f)$$

Here the residue coefficient can be written as a positive constant times  $\int_0^\infty e^{-\pi x} T(x) dx$ ; all remaining contour terms decay faster, while the error  $\delta_q$  contributes  $O(e^{-2\pi q})$  and its first nonzero coefficient is absorbed in the same positive Wiener–Hopf constant. Combining (21.65e) and (21.65f) yields

$$\delta f(Q(m)) = A'_{-1} (\tfrac{1}{2} - m)^2 + o((\tfrac{1}{2} - m)^2), \quad A'_{-1} > 0.$$

Since the cylinder unbalance is twice the distance from half filling, this is (21.65a).  $\square$

**Corollary 10.326** (Quadratic small-slope cylindrical free energy in the GFF range). *Fix  $\sqrt{3} \leq c \leq 2$ . Let  $f_c(\alpha)$  be the balanced-cylinder free energy with vertical-arrow unbalance  $\alpha \in (-1/2, 1/2)$ . Then there is  $C(c) < \infty$  such that*

$$f_c(\alpha) \geq f_c(0) - C(c) \alpha^2. \quad (21.65)$$

*Proof.* Write  $\Delta = 1 - c^2/2 \in [-1, -1/2]$ . For  $c < 2$ , the cylinder sector with unbalance  $\alpha$  corresponds to distance  $|\alpha|/2$  from half filling, so Theorem 10.291 gives  $f_c(\alpha) = f_c(0) - A_c \alpha^2 + o(\alpha^2)$  with  $A_c > 0$ . At  $c = 2$  the same conclusion is supplied independently by Lemma 10.325; no continuity extension of the interior Bethe asymptotic is used. Hence the displayed lower bound holds for sufficiently small  $|\alpha|$ . On each compact set  $\varepsilon \leq |\alpha| < 1/2$ , continuity of the fixed- $c$  free energy makes  $(f_c(0) - f_c(\alpha))/\alpha^2$  bounded. Combining the two regions proves (21.65).  $\square$

**Lemma 10.327** (Free energy creates alternating level lines). *Let  $O_{N,M}$  be the  $N$ -periodic cylinder and let  $\mathbb{P}_{N,M}^{(0)}$  be the balanced six-vertex law. Fix  $0 < \alpha < 1/2$ ,  $k \geq 1$ , and set  $n = \lfloor \alpha N/k \rfloor$ . There is a set  $S = \{s_0, \dots, s_{2n-1}\}$  of bottom faces such that the event  $\mathcal{A}(S; n, k)$  of  $2n$  disjoint vertical diagonal paths, started in cyclic order at  $S$  and carrying alternating heights  $0, k, 0, k, \dots$ , satisfies*

$$\liminf_{N \rightarrow \infty} \liminf_{M \rightarrow \infty} \frac{1}{NM} \log \mathbb{P}_{N,M}^{(0)}(\mathcal{A}(S; n, k)) \geq f_c(\alpha) - f_c(0). \quad (21.66)$$

*Proof.* We record both counting steps. In the sector with  $2L = 2\lceil \alpha N \rceil$  excess upward arrows, split every six-vertex vertex deterministically into noncrossing oriented loop/path pieces (at the two ambiguous vertex types always choose the same turning rule). There are at least  $2L$  more upward than downward through-paths. Choose ordered upward through-paths  $\gamma_1, \dots, \gamma_{2L}$  so that between consecutive chosen paths the remaining through-paths are balanced. Since the total number of used oriented edges is at most  $2NM$ , some pair  $\gamma_i, \gamma_{L+i}$  has total length at most  $2M/\alpha$ . Reverse all arrows on the first path and in the cylindrical component between the two. The resulting configuration is balanced and has two vertical level lines whose height difference is  $L$ . Only vertices on the two chosen paths change type, so the weight loses at most  $c^{2M/\alpha}$ , while a target configuration has at most  $N^{22^{2M/\alpha}}$  preimages (choose the two first edges and then the two paths). After division by the balanced partition function and passage  $M \rightarrow \infty$ ,  $N \rightarrow \infty$ , this proves that the probability of two vertical level lines separated by height  $\lceil \alpha N \rceil$  has exponential rate at least  $f_c(\alpha) - f_c(0)$ .

Next root one zero-height vertical line and successively insert alternating 0- and  $k$ -height lines. If  $\mathcal{A}_j$  denotes the event with  $2j + 2$  inserted lines, exploring the two extremal existing lines gives an intermediate domain whose boundary heights are ordered. Ordinary FKG and boundary comparison show  $\mathbb{P}(\mathcal{A}_j) \leq \mathbb{P}(\mathcal{A}_{j+1})$  until  $j = n - 1$ . The two-line event of height difference  $nk$  is contained in the union of the  $\mathcal{A}(S; n, k)$  over choices of  $S$ , and there are at most  $2^N$  such choices. The factor  $2^N$  is subexponential on the  $NM$  scale. Combining with the first step gives (21.66).  $\square$

**Lemma 10.328** (Fences versus ridges). *There exist absolute  $\eta, c_2 > 0$  with the following property. Put  $\alpha = k/(\eta r)$ ,  $n = N/(\eta r)$  and  $m = M/r$ , with  $m$  a multiple of three. Condition on an alternating-line event from Lemma 10.327 and expose the low-height clusters attached to every second line. There are events  $\mathcal{F}$  (“fences”) and  $\mathcal{R}$  (“ridges”) such that*

$$\mathbb{P}(\mathcal{F} \mid \mathcal{R}, \text{Exp}) \geq c_2^{nm}, \quad (21.67)$$

and

$$\mathbb{P}(\mathcal{R} \mid \mathcal{F}, \text{Exp}) \leq \left[ 2 \mathbb{P}_{\mathbb{Z} \times [-r, 2r]}^{0,1}([0, \delta r] \times \{0\} \xrightarrow[h \geq c_2 k - 2]{\mathbb{Z} \times [0, r]} \mathbb{Z} \times \{r\}) \right]^{nm/6}. \quad (21.68)$$

Here  $\text{Exp}$  denotes the explored data and the bounds are uniform in its admissible realization.

*Proof.* Divide the cylinder into horizontal slices of height  $r$  and into the  $2n$  channels between successive exposed alternating level lines. Call a channel in a slice *tight* if its left and right boundary pieces have length at most  $\delta r$ ; in every block of three slices at least a fixed fraction of the channels are tight, so there are at least  $nm/6$  “good” channel/slice pairs.

For a good pair, the ridge event requires an  $h \geq k$  diagonal crossing of the channel. Under this condition expose the component of  $h \leq k - 1$  attached to the left boundary and its topmost  $h = k$  contour. The still-unrevealed region has boundary at most  $k$  and the two relevant boundary arcs are short. The slit-separation estimate Proposition 10.323, after one boundary-pushing step, says with conditional probability at least  $c_2$  that a path of height at most  $(1 - c_2)k$  crosses the channel. These low paths are the fences. Conditional FKG makes the lower bounds multiply over at most  $nm$  required channels, proving (21.67).

Conversely assume all fences and condition on the domains they cut out. The spatial Markov property makes the remaining good-channel domains independent. Their boundary values are at most  $(1 - c_2)k$ . Boundary comparison and pushing map any required  $h \geq k$  ridge to a strip crossing of height at least  $c_2k - 2$ , at cost at most a factor 2. Tightness places the starting arc inside an interval of length  $\delta r$ . Since at least  $nm/6$  conditional domains are independent, multiplying these upper bounds gives (21.68).  $\square$

**Theorem 10.329** (Free energy to annulus circuits). *There exist absolute  $\eta, c_3, C_3 > 0$  such that, whenever  $k$  is sufficiently large and  $r > 2k/\eta$ ,*

$$\mathbb{P}_{\Lambda_{12r}}^{0,1}(\mathcal{O}_{h \geq c_3 k}(6r, 12r)) \geq c_3 \exp\{C_3 r^2[f_c(k/(\eta r)) - f_c(0)]\}. \quad (21.69)$$

*Proof.* Choose the set  $S$  maximizing the alternating-line probability in Lemma 10.327. On the event  $\mathcal{R}$ , construction of fences gives

$$\mathbb{P}(\mathcal{R} \mid \mathcal{F}, \text{Exp}) \geq \mathbb{P}(\mathcal{F} \mid \mathcal{R}, \text{Exp}) \mathbb{P}(\mathcal{R} \mid \text{Exp}) \geq c_2^{nm} \mathbb{P}(\mathcal{A}(S; n, k)).$$

Combine with the upper bound (21.68) and take the power  $6/(nm)$ . Then pass first  $M \rightarrow \infty$  and then  $N \rightarrow \infty$ . Equation (21.66), together with  $n = N/(\eta r)$  and  $m = M/r$ , gives

$$p_r(k) := \mathbb{P}_{\mathbb{Z} \times [-r, 2r]}^{0,1}([0, \delta r] \times \{0\} \xrightarrow{h \geq ck-2} \mathbb{Z} \times \{r\}) \geq c \exp\{Cr^2[f_c(k/(\eta r)) - f_c(0)]\}.$$

Apply Proposition 10.324 to  $p_r(k)$  and absorb the fixed height loss  $-2$  by taking  $k$  above an absolute threshold. Renaming constants gives (21.69).  $\square$

**Theorem 10.330** (Boundary-uniform annulus circuit theorem in the GFF range). *Fix  $\sqrt{3} \leq c \leq 2$  and integers  $K, L \geq 0$ . There exist  $p = p(c, K, L) > 0$  and  $n_0 < \infty$  such that for every  $n \geq n_0$ , every discrete domain  $D \supset \Lambda_{2n}$ , and every admissible boundary condition  $\xi$  imposed on any nonempty part of  $\partial D$  with  $|\xi| \leq L$ ,*

$$\mathbb{P}_D^\xi(\mathcal{O}_{h \geq K}(n, 2n)) \geq p. \quad (21.70)$$

*The same conclusion, after a global height shift and sign flip, holds for any fixed level and either sign of the level-set circuit.*

*Proof.* By event inclusion it suffices to prove the estimate for  $K$  above a fixed absolute threshold. Shift the boundary by  $L$  and use boundary comparison to reduce bounded data to the flat 0, 1 boundary condition. Boundary pushing Lemma 10.321 then reduces an arbitrary containing domain to a square of comparable radius, losing only two units of height and a factor two in probability.

Take  $r$  proportional to  $n$ . Corollary 10.326 gives

$$r^2[f_c(K/(\eta r)) - f_c(0)] \geq -C(c)K^2/\eta^2,$$

which is independent of  $r$ . Theorem 10.329 therefore supplies a strictly positive lower bound, independent of scale, for a circuit in an annulus with fixed radius ratio. A bounded number of overlapping annuli and FKG change that ratio from  $(6, 12)$  to  $(1, 2)$ . The finite number of small values of  $K$  follows by inclusion from the threshold case. Restoring the original boundary shift proves (21.70).  $\square$

**Corollary 10.331** (Spin/bond annulus circuit input for the GFF regularity argument). *In the spin representation of Lemma 10.315, bounded admissible exterior data admit, uniformly over scale and over the containing domain, a positive probability of a separating  $\omega^-$  circuit under the least favourable + boundary condition; by sign symmetry the analogous  $\omega^+$  statement also holds.*

*Proof.* We use the source-exact translation rather than trying to open the edges of one long height circuit independently. Let  $D_n$  be a large even domain containing the target annulus and let  $\gamma_n$  be the outer face circuit used to encode the  $+$  boundary condition. Along  $\gamma_n$  one has  $|h| \leq 1$ . Condition on  $h|_{\gamma_n}$  and apply Theorem 10.330 with levels  $L = 1$  and  $K = 100$ . With probability at least  $c > 0$  there is an  $h \geq 100$  circuit in the annulus.

Every admissible configuration also has a separating circuit on which  $h \leq 10$ : starting from the outer boundary and moving inward toward the high circuit, take the last level set below 10. Since neighboring face heights differ by one, the discrete intermediate-value property then produces nested circuits at each even level  $12, 14, \dots, 98$ . In the spin/bond representation these successive even levels alternately belong to  $\omega^+$  and  $\omega^-$ . Hence the same event contains twenty nested alternating spin/bond circuits with probability at least  $c$ .

For a target annulus of arbitrary comparable radius, take the outermost  $\omega^+$  circuit  $\gamma$  among the alternating family and let  $D_\gamma$  be its interior. The spatial Markov property gives

$$\mathbb{P}(\text{two alternating circuits}) = \int \mathbb{P}_{D_\gamma}^+(\omega^- \text{ circuit}) d\mathbb{P}(\gamma).$$

The  $\omega^-$  circuit event is increasing when the interior domain is enlarged (equivalently, the  $+$  boundary condition is the least favourable one for  $\omega^-$ ), so domain monotonicity bounds the integrand by the corresponding probability in the full target ball. The left-hand side is at least  $c$ , proving a uniform positive lower bound for the desired  $\omega^-$  circuit. Sign flip yields the  $\omega^+$  version. No factor depending on the length of a circuit is paid.  $\square$

#### 10.8.4 Crossing-count tails, the level-line tree, and regularity

The annulus-circuit theorem of this chapter is not itself the regularity estimate used in the GFF argument. The missing bridge is quantitative: it must control how many alternating interfaces can cross a rectangle or annulus, and then convert those interface counts into moments of height increments. We now prove locally that bridge.

**Definition 10.332** (Alternating crossings and annulus arm count). For a rectangle  $R$ , let  $\# \text{AltHor}(R)$  be the maximal number of pairwise disjoint left–right spin/bond crossings which alternate between  $\omega^+$  and  $\omega^-$  from top to bottom; define  $\# \text{AltVer}$  analogously. For  $4 \leq r < R/2$  and  $x \in \mathbb{R}^2$ , put

$$K_{R,r,x} := (1 - \mathbf{1}_{\text{AltCircuit}_2(A(x;r,R))}) \vee \# \text{AltArm}(A(x;r+1, R-1)). \quad (21.71)$$

Thus  $K_{R,r,x} = 0$  as soon as the annulus contains two oppositely coloured separating circuits, and otherwise it bounds the number of alternating radial arms.

**Lemma 10.333** (Partial-circuit estimate under remote conditioning). *Fix  $r \geq 4$  and  $x \in \mathbb{R}^2$ . Let  $(D_i)_i$  be finitely many even domains, let  $U$  be the union of their even-edge sets, and let  $B$  be the pieces of their boundaries inside  $B(x, 2r)$ . If  $E$  is measurable in the spins/edges in  $U$  and  $\mathbb{P}(E \cap \{B \subset \omega^-\}) > 0$ , then*

$$\mathbb{P}(\text{Circuit}_{\omega - \cup U}(A(x;r, 2r)) \mid E, B \subset \omega^-) \geq p_{\text{circ}}, \quad (21.72)$$

where  $p_{\text{circ}} > 0$  is the circuit constant above.

*Proof.* Condition first on an atom of the sigma-field generated by all even spins and bonds in  $U$ , and retain only atoms compatible with  $E \cap \{B \subset \omega^-\}$ . For such an atom, let  $\gamma$  be the innermost

self-avoiding even circuit surrounding the target annulus which forms the inner boundary of the revealed polar pieces (if there are several pieces, expose them successively from outside inward). The ordinary and polar-domain spatial Markov properties say that, after this exploration, the conditional law in the remaining inside component depends on the atom only through the induced boundary spins/bonds. Among admissible induced data, replacing the inner boundary by  $+$  is least favourable for an  $\omega^-$  circuit, while the already conditioned pieces  $B \subset \omega^-$  can only help. Conditional FKG/domain monotonicity therefore gives, on every such atom,

$$\mathbb{P}(\text{Circuit}_{\omega-\cup U}(A(x; r, 2r)) \mid \text{atom}, B \subset \omega^-) \geq \mathbb{P}_{B(x, 2r)}^+(\omega^- \text{ circuit}) \geq p_{\text{circ}},$$

where the last inequality is Corollary 10.331. Averaging the atomwise estimate over the conditional distribution induced by  $E$  proves (21.72). Thus no monotonicity assumption on the remote event  $E$  is used.  $\square$

**Lemma 10.334** (Geometric tails for alternating rectangle crossings). *For every aspect ratio  $\rho \geq 1$  there is  $a_\rho \in (0, 1)$  such that, for any rectangle  $R$  of dimensions  $w \times \rho w$ , any  $n \geq 0$ , and any event  $E$  measurable at  $\ell^\infty$ -distance at least  $w/5$  from  $R$ ,*

$$\mathbb{P}(\# \text{AltHor}(R) \geq 2n \mid E) \leq (1 - a_\rho)^n. \quad (21.73)$$

*The analogous vertical and cylinder estimates hold whenever the rectangle does not wrap around the cylinder.*

*Proof.* Explore alternately the highest  $\omega^+$  crossing, the highest  $\omega^-$  crossing below it, and so on. Conditional on the first  $2i - 1$  crossings and on the remote event  $E$ , the unexplored region below the current  $\omega^+$  crossing still contains a row of  $\lfloor c\rho \rfloor$  disjoint annuli of radius comparable with  $w$ . If  $w$  is above a fixed microscopic threshold, Lemma 10.333, followed by FKG in these disjoint annuli, gives probability at least  $a_\rho > 0$  that the current  $\omega^+$  crossing is connected to the bottom boundary; then no further  $\omega^-$  crossing exists. For bounded  $w$ , finite energy gives the same conclusion by forcing a fixed finite set of even spins/edges. Hence at each alternating pair the process stops with conditional probability at least  $a_\rho$ . Iterating the conditional estimate proves (21.73).  $\square$

**Lemma 10.335** (Polynomial arm tails). *There is  $a_{\text{arm}} > 0$  such that for  $R \geq 2r \geq 8$ ,  $n \geq 0$ , and every event  $E$  measurable outside  $B(x, R)$ ,*

$$\mathbb{P}(K_{R,r,x} \geq n \mid E) \leq (r/R)^{a_{\text{arm}}n}. \quad (21.74)$$

*In particular*

$$\mathbb{E}[K_{R,r,x} \mid E] \leq C(r/R)^{a_{\text{arm}}}. \quad (21.75)$$

*The same estimates hold on a cylinder provided its circumference exceeds  $2R$ .*

*Proof.* First take  $R = 2r$ . Two alternating circuits occur with a fixed positive probability by Corollary 10.331. If instead there are  $Nn$  alternating arms, one of four fixed rectangles crossing the annulus in its easy direction contains at least  $Nn/10$  alternating crossings. Lemma 10.334 therefore gives  $\mathbb{P}(K_{2r,r,x} \geq Nn \mid E) \leq Ce^{-cn}$  after choosing a fixed large  $N$ . For  $R = 2^m r$ , apply the one-scale estimate successively on the  $m$  disjoint dyadic annuli and use the spatial Markov property from the outside inward. To have  $n$  arms across the whole annulus, the corresponding one-scale arm event must survive at every scale, giving  $e^{-cmn} = (r/R)^{an}$ . Monotonicity in the radii handles general  $R/r$ . Summing the tail proves (21.75).  $\square$

**Definition 10.336** (Level-line tree and branching function). In a finite even domain with  $+$  boundary, use the spin/bond configuration of Lemma 10.315. The *level-line tree*  $\mathcal{X}$  is the rooted directed tree whose even vertices are the connected sign-regions separated by opposite-colour bond circuits and whose odd vertices are the bounded components of the complement of the full bond set. The root is the  $+$  region containing the boundary. Every non-root vertex is joined to the unique region immediately surrounding it.

For faces  $u, v$ , let  $p_{u\partial}$  and  $p_{v\partial}$  denote their tree paths to the root, and define

$$\psi_D(u, v) := |p_{u\partial} \cap p_{v\partial}| - 1, \quad \psi_D^*(u) := \psi_D(u, u). \quad (21.76)$$

For two face pairs write

$$I(uu', vv') := |p_{uu'} \cap p_{vv'} \cap V_{\text{odd}}(\mathcal{X})|. \quad (21.77)$$

**Lemma 10.337** (Conditional coin representation and covariance). *Conditionally on the even-spin/bond tree  $\mathcal{X}$ , the labels carried by its odd vertices are independent fair signs. The height is constant on each tree vertex and each odd tree edge contributes an independent  $\pm 1$  increment; the following even edge repeats the same increment. Consequently  $\psi_D^*$  is the maximal height compatible with  $\mathcal{X}$ , and  $\psi_D(u, v)$  determines the conditional covariance of  $h(u)$  and  $h(v)$  up to the single parity correction at odd depth. Moreover the law of  $\psi_D \wedge 2m$  is FKG and stochastically increasing with the domain for every  $m$ .*

*Proof.* Once the even spins and compatible bonds have been fixed, every bounded component of the bond complement supports one unresolved odd spin. The local spin representation makes these signs independent fair coins. Crossing an odd component changes the height by that coin; passing to the next even region repeats the increment, which proves the stated tree increment rule and covariance description.

For monotonicity, explore the maximal  $\omega^-$  domains inside a  $+$  domain. The spatial Markov property makes their interiors conditionally independent, and the tree recursion is

$$\psi_D \stackrel{d}{=} (\psi_D \wedge 2) + \sum_{B \in \mathcal{M}^-(D)} \psi_B \quad \text{on each common branch.} \quad (21.78)$$

The set of maximal domains is increasing under the underlying FKG order. Inducting on  $m$ , using conditional independence and the tower form of FKG, shows that  $\psi_D \wedge 2m$  is FKG and increases with  $D$ .  $\square$

**Lemma 10.338** (Pairing bound for height-increment correlations). *For even  $k$  and face pairs  $(u_i, u'_i)_{i=1}^k$  in a finite even domain  $D$ ,*

$$\left| \mathbb{E}_D^+ \prod_{i=1}^k (h(u'_i) - h(u_i)) \right| \leq 2^k \sum_{\pi} \mathbb{E}_D^+ \prod_{ij \in \pi} I(u_i u'_i, u_j u'_j), \quad (21.79)$$

where the sum runs over pairings  $\pi$  of  $\{1, \dots, k\}$ .

*Proof.* Condition on  $\mathcal{X}$  and expand every height difference along its unique tree path. Each summand is a product of the independent odd-vertex coins from Lemma 10.337. Its conditional expectation vanishes unless every coin appears an even number of times. Pair the first occurrence of each surviving coin with a later occurrence. For a given pairing  $ij$ , there are at most  $4I(u_i u'_i, u_j u'_j)$  compatible choices (the factor four accounts for the two tree edges adjacent to an odd vertex on each path). Multiplying and summing over pairings gives (21.79).  $\square$



**Lemma 10.339** (Tree intersections are controlled by annulus arms). *If the pair  $\{u, u'\}$  lies in one component of  $\mathbb{R}^2 \setminus A(x; r, R)$  and  $\{v, v'\}$  lies in the other, then*

$$I(uu', vv') \leq 4K_{R,r,x}. \quad (21.80)$$

*If  $u, u'$  (or  $v, v'$ ) are nearest neighbours, then  $I(uu', vv') \leq 4$ .*

*Proof.* Walking along a common odd vertex of the two tree paths reveals a level line separating both pairs. Successive common odd vertices force alternating  $\omega^+/\omega^-$  separators. Every four tree steps create a new alternating arm across any annulus that separates the two pairs, which proves (21.80). A nearest-neighbour pair crosses at most one local level-line interface, giving the deterministic bound four.  $\square$

**Lemma 10.340** (Optimal path and disjoint-annulus extraction). *Fix even  $k$ . There is  $N_k < \infty$  such that, for two pairs  $\{a, a'\}, \{b, b'\}$  with  $|a' - a| \leq |b' - b|$ , one may join  $a$  to  $a'$  by at most*

$$N_k(1 \vee -\text{SR}'_2(\{a, a'\}, \{b, b'\})) \quad (21.81)$$

*steps, each either nearest-neighbour or separated from  $\{b, b'\}$  by an annulus of ratio at least  $e^{20k^2}$ . Furthermore, from any family of at most  $k/2$  such annuli of ratio at least  $e^{40k^2}$  one can shrink radii, without crossing either pair, to obtain pairwise disjoint annuli whose ratios remain at least the square roots of the originals.*

*Proof.* For the path statement, move from  $a$  toward  $a'$  on successively halved spatial scales. As long as a step is not a nearest-neighbour step, choose its length to be at most  $e^{-20k^2}$  times its distance from  $\{b, b'\}$ ; then the concentric annulus between those scales separates the step from the second pair. Only a bounded number of steps is needed on each dyadic scale, and the number of scales is bounded by the negative logarithmic scale separation in (21.81).

For disjointness, suppose there are  $m \leq k/2$  candidate annuli. Divide the logarithmic radial interval of each into  $Q = 2m^2$  disjoint subannuli of equal logarithmic thickness and choose one independently and uniformly. For a fixed pair of centres, at most  $4Q$  choices of their two indices can overlap: once one selected annulus has the larger outer radius, moving it one subscale outward while moving the other inward makes overlap impossible. Thus the overlap probability is at most  $4/Q \leq 2/m^2$ . The union bound leaves positive probability that no pair overlaps, proving existence of the required deterministic choice.  $\square$

**Theorem 10.341** (Source-strength full-plane regularity estimate). *For every even  $k \geq 2$  there exist  $\alpha_k > 0$  and  $C_k < \infty$  such that, for the full-plane isotropic six-vertex state in the GFF range and any  $u = (u_1, u'_1, \dots, u_k, u'_k)$ ,*

$$|\Phi_k(u)| \leq C_k \sum_{\pi} \prod_{ij \in \pi} \begin{cases} e^{-\alpha_k \text{SR}_2(\{u_i, u'_i\}, \{u_j, u'_j\})}, & \text{SR}_2 \geq 20k^2, \\ 1 \vee [-\text{SR}'_2(\{u_i, u'_i\}, \{u_j, u'_j\})], & \text{SR}_2 < 20k^2, \end{cases} \quad (21.82)$$

*where  $\pi$  runs over pairings of  $\{1, \dots, k\}$  and*

$$\text{SR}_2(A, B) = \log \frac{\text{dist}(A, B)}{\min\{\text{diam } A, \text{diam } B\}}, \quad \text{SR}'_2(A, B) = \log \frac{1 \vee \text{dist}(A, B)}{\min\{\text{diam } A, \text{diam } B\}}. \quad (21.83)$$

*The same argument gives the two cylinder regularity estimates needed by the spectral analysis whenever the relevant annuli fit inside one cylinder period.*

*Proof.* Work first in a large finite + domain and apply Lemma 10.338. For each pair  $ij$  in a fixed pairing  $\pi$ , if the two face pairs are well separated, Lemma 10.339 gives an annulus whose arm count controls  $I(u_i u'_i, u_j u'_j)$ . If they are not well separated, decompose the shorter increment along the path supplied by Lemma 10.340; nearest-neighbour pieces cost at most four, while every other piece again has an annulus-arm bound. Expanding the resulting triangular inequalities produces at most the product of the path lengths in (21.81) terms.

For each term, use the second part of Lemma 10.340 to shrink the at most  $k/2$  controlling annuli until they are disjoint. Order them by decreasing outer radius and expose from outside inward. At every step the conditioning is measurable outside the current annulus, so Lemma 10.335 gives

$$\mathbb{E}(K_{R,r,x} \mid \text{previous exposures}) \leq C(R/r)^{-a_{\text{arm}}}.$$

Multiplication over the disjoint annuli yields an exponential factor in the positive scale separations. The number of path-decomposition terms yields precisely the linear factors in  $1 \vee (-\text{SR}'_2)$  for the bad pairs. Absorbing the fixed powers and the finite number of pairings into  $C_k$  gives (21.82). Letting the finite domain exhaust  $\mathbb{Z}^2$  proves the full-plane claim; the cylinder proof is identical until the annulus would wrap, where the source uses the corresponding thin-subcylinder crossing count, which only improves the required two-point estimates.  $\square$

**Corollary 10.342** (Cylinder regularity estimates used by the spectral analysis). *There are  $C, c > 0$  such that for every even circumference  $L$  and integers  $k \geq 1$ ,*

$$|\Phi_{2,L}^{\text{Cyl}}((0,0), (k,0), (2k,0), (3k,0))| \leq C, \quad (21.84)$$

and, for  $0 < \ell \leq \min\{8k, L/2\}$ ,

$$|\Phi_{2,L}^{\text{Cyl}}((0,0), (0,\ell), (k,0), (k,\ell))| \leq C(\ell/k)^c. \quad (21.85)$$

*Proof.* For (21.84), if  $k$  is microscopic the deterministic bound on the two height increments is enough. If  $k \leq L/40$ , place an annulus of radii comparable with  $k$  between the two pairs and use Lemmas 10.339 and 10.335. If  $k > L/40$ , the annulus would wrap around the cylinder; replace it by the subcylinder  $[5k/4, 7k/4] \times (\mathbb{R}/L\mathbb{Z})$ . The tree intersection is bounded by one plus the number of alternating left-right crossings of this subcylinder, whose aspect ratio is uniformly bounded. The cylinder version of Lemma 10.334 gives an exponential tail, hence a uniform first moment and (21.84).

For (21.85), when  $k \leq 10^6 \ell$  only a constant bound is needed, obtained from a fixed annulus around the short pair. If  $k > 10^6 \ell$  and  $\sqrt{k\ell} \leq L$ , take inner radius  $r = 4\ell$  and outer radius  $R \asymp \sqrt{k\ell}$ . The tree-intersection bound and (21.75) give  $C(r/R)^{a_{\text{arm}}} \leq C(\ell/k)^{a_{\text{arm}}/2}$ . Finally, if  $\sqrt{k\ell} > L$ , insert a subcylinder of length comparable with  $k$  between the two pairs. A separating alternating circuit appears independently at each successive cylinder-length block with a fixed conditional probability, so failure has probability at most  $Ce^{-ck/L}$ . Conditional on failure, the number of alternating crossings in the first block has bounded mean. Thus the correlation is at most  $Ce^{-ck/L}$ , which is stronger than (21.85) because  $k/L \geq \sqrt{k\ell}/\ell$  in this case.  $\square$

**Corollary 10.343** (Precompactness and equicontinuity away from collision diagonals). *For fixed even  $k$ , every family of scaled correlations  $\Phi_k^{(\delta)}$  is locally bounded and equicontinuous on the configuration space  $D_k$  where distinct endpoint pairs do not collide. Hence every sequence  $\delta_n \downarrow 0$  admits a subsequence converging locally uniformly on  $D_k$  to a continuous limit.*

*Proof.* On a compact subset of  $D_k$ , the scale-separation quantities in (21.82) stay uniformly away from their singular configurations, giving local boundedness. If one endpoint is moved by a small vector, additivity of height increments writes the difference of the two  $k$ -point functions as one correlation in which one pair has very small diameter. In (21.82) that pair has large positive scale separation from every fixed distant pair, so the bound tends uniformly to zero. A telescoping sum moves all  $2k$  endpoints. Arzelà–Ascoli then gives local precompactness and continuity of every subsequential limit.  $\square$

**Theorem 10.344** (Scale-mixing / bounded fusion remainder). *Fix even  $k \geq 4$  and a configuration  $u = (u^{(1)}, u^{(2)}) \in D_k$  with  $u^{(1)} = (u_1, u'_1, u_2, u'_2)$  and  $u^{(2)} = (u_3, u'_3, \dots, u_k, u'_k)$ . Keep all points except  $u_1$  fixed. Then there is  $C < \infty$ , depending on the fixed points but not on the mesh, such that*

$$\limsup_{u_1 \rightarrow u_2} \limsup_{\delta \downarrow 0} \left| \Phi_k^{(\delta)}(u) - \Phi_2^{(\delta)}(u^{(1)}) \Phi_{k-2}^{(\delta)}(u^{(2)}) \right| \leq C. \quad (21.86)$$

*Proof.* By translation, additivity, and the regularity estimate, it is enough to work after a fixed normalization in which  $u_2 = 0$ ,  $u'_2 = (-1, 0)$ ,  $u'_1 = (1, 0)$ , all pairs indexed by  $i \geq 3$  have tiny diameter compared with their mutual separations, and  $|u_1|$  is small. Write barred letters for the corresponding lattice faces at mesh  $\delta$  and set

$$P := \prod_{i=3}^k (h(\bar{u}_i) - h(\bar{u}'_i)).$$

Let  $A$  be the maximal even domain containing  $\bar{u}_2$  whose boundary lies in  $\omega^+$  (and set  $A = \emptyset$  if there is none). The height is constant on  $\partial A$ . Define

$$a := \begin{cases} h(\partial A), & \bar{u}_1 \in A, \\ h(\bar{u}_1), & \bar{u}_1 \notin A, \end{cases}$$

and split the two nearby height increments as

$$\begin{aligned} H_1 &= a - h(\bar{u}'_1), & \tilde{H}_1 &= h(\bar{u}_1) - a, \\ H_2 &= h(\partial A) - h(\bar{u}'_2), & \tilde{H}_2 &= h(\bar{u}_2) - h(\partial A). \end{aligned}$$

The difference in (21.86) is the covariance

$$\text{Cov}(P, (H_1 + \tilde{H}_1)(H_2 + \tilde{H}_2)). \quad (21.87)$$

Conditionally on  $A$  (equivalently, on its boundary and the outside exploration), the inside variables  $(\tilde{H}_1, \tilde{H}_2)$  are independent of the outside tuple  $(P, H_1, H_2)$  and have zero mean by height-flip symmetry. Therefore

$$|(21.87)| \leq |\text{Cov}(P, H_1 H_2)| + \left( \mathbb{E}[\mathbb{E}(P \mid A)^2] \right)^{1/2} \left( \text{Var}(\mathbb{E}(\tilde{H}_1 \tilde{H}_2 \mid A)) \right)^{1/2}. \quad (21.88)$$

We bound the three factors separately.

1. *The conditional second moment of  $P$ .* Condition further on the level-line tree. By Lemma 10.338,  $|\mathbb{E}(P \mid \mathcal{X})|$  is bounded by a finite sum of products of tree-intersection counts. The normalization of the fixed pairs  $i \geq 3$  gives disjoint annuli between radii  $\varepsilon/\delta$  and  $1/\delta$  around

their short increments. Lemma 10.339 bounds each intersection by a constant times  $K_{1/\delta, \varepsilon/\delta, \bar{u}_i}$ , and Lemma 10.335 gives uniform moments of every order. Hence

$$\mathbb{E}[\mathbb{E}(P \mid A)^2] \leq C_k. \quad (21.89)$$

2. *The outside covariance.* The same pairing expansion applies to  $PH_1H_2$ , with the tree path from  $\bar{u}'_2$  truncated at  $\partial A$ . If a pairing does not match the indices 1, 2 to each other, all factors are controlled by the same disjoint annuli as above. If it does, the common tree path between the two nearby pairs is bounded by a random variable

$$Q = 24 + \sum_{j=1,3} \# \text{AltHor}(R_j) + \sum_{j=2,4} \# \text{AltVer}(R_j) + \# \text{AltArm}(A_0), \quad (21.90)$$

where  $R_1, \dots, R_4$  are four fixed-aspect-ratio rectangles surrounding the normalized pair and  $A_0$  is the intervening annulus. Every level loop separating exactly one endpoint of each nearby pair must either stay in the microscopic inner square, cross one of these rectangles in its easy direction, or make an arm across  $A_0$ ; the constant 24 absorbs the bounded exceptional loops. Lemmas 10.334 and 10.335 give  $\mathbb{E}Q^2 \leq C$ . Pairing and Cauchy–Schwarz then imply

$$|\text{Cov}(P, H_1H_2)| \leq C_k. \quad (21.91)$$

3. *Variance of the inside covariance.* For a finite even domain  $B$  define

$$X_{v_1, v_2}(B) := \mathbf{1}_{\{v_1, v_2 \in B\}} \mathbb{E}_B^+[h(v_1)h(v_2)]. \quad (21.92)$$

The Markov property gives  $\mathbb{E}(\tilde{H}_1\tilde{H}_2 \mid A) = X_{\bar{u}_1, \bar{u}_2}(A)$ . By the branching-function covariance formula and domain monotonicity,  $X_{v_1, v_2}(B)$  increases with  $B$ .

Let  $D_n$  be the largest even domain in the ball of radius  $e^n$  around  $v_2$  and put  $q_n = X_{v_1, v_2}(D_n)$ . The annulus-circuit theorem gives a fixed  $p > 0$  such that, in  $D_{n+1}$ , with probability at least  $p$  an appropriate opposite-colour circuit cuts the maximal  $\omega^+$  domain containing  $v_2$  down to  $D_n$ . The branching recursion and domain monotonicity then give

$$q_{n+1} \leq 4 + pq_n + (1-p)q_{n+1}, \quad q_{n+1} - q_n \leq 4/p. \quad (21.93)$$

Thus  $q_n$  is uniformly Lipschitz in the logarithmic scale. If  $S(A) := \max\{n : D_n \subset A\}$ , successive annulus circuits imply an exponential tail for the overshoot of  $S(A)$  in either direction, hence  $\text{Var } S(A) \leq C$ . A second application of the circuit theorem, now between  $D_{S(A)}$  and  $\partial A$ , gives

$$|X_{v_1, v_2}(A) - q_{S(A)}| \leq C. \quad (21.94)$$

Equations (21.93)–(21.94) imply

$$\text{Var } X_{v_1, v_2}(A) \leq C. \quad (21.95)$$

Substituting (21.89), (21.91), and (21.95) into (21.88) yields a bound independent of  $\delta$  and of  $u_1$  in the normalized fusion neighbourhood. Undoing the harmless normalization proves (21.86).  $\square$

### 10.8.5 Spectral compactness, sub-diagonal concentration, and the two-point dichotomy

We now close Gate G3. The argument starts from the exact cylinder spectral measure of Lemma 10.312, uses the regularity estimates already owned by G2c to obtain compactness, imports only the already-proved macroscopic rotational invariance of **SM-21.S1.04**, and then performs the complex-analytic argument of the primary six-vertex–GFF proof. The conclusion is deliberately the source theorem’s dichotomy; we do *not* claim that every spectral limit is already concentrated on the characteristic lines  $|b| = a$ .

**Definition 10.345** (Scale-compact spectral class). Fix  $c_0, C_0 > 0$ . Let  $\mathcal{M}_{c_0, C_0}$  be the set of positive Borel measures  $\nu$  on  $\mathbb{R}_{>0} \times \mathbb{R}$ , invariant under  $(a, b) \mapsto (a, -b)$ , such that for every  $\beta \geq \alpha > 0$ ,

$$\nu\{\alpha < a < 2\alpha\} \leq C_0, \quad \nu\{0 < a < \alpha, \beta < |b| < 2\beta\} \leq C_0(\alpha/\beta)^{c_0}. \quad (21.91)$$

We use the vague topology. Write  $\mathcal{M}$  for one such class with constants fixed below.

**Lemma 10.346** (Uniform spectral bounds and compactness). *There are  $c_0, C_0 > 0$ , depending only on the six-vertex parameter, for which every balanced cylinder spectral measure  $\mu_L$  of Lemma 10.312 belongs to  $\mathcal{M} = \mathcal{M}_{c_0, C_0}$ . Moreover*

$$\sup_{\nu \in \mathcal{M}} \int (a \wedge a^{-1}) d\nu(a, b) < \infty, \quad (21.92)$$

$\mathcal{M}$  is compact in the vague topology, and the following strengthened vague convergence rule holds. If  $\nu_n \rightarrow \nu$  in  $\mathcal{M}$  and  $f_n \rightarrow f$  uniformly on compact sets with

$$\sup_{n, a, b} (a \vee a^{-1}) |f_n(a, b)| < \infty, \quad (21.93)$$

then  $\nu_n[f_n] \rightarrow \nu[f]$ .

*Proof.* For the first dyadic bound choose, for  $\alpha \leq 1/16$ , an even  $k \asymp \alpha^{-1}$  and use the four horizontally collinear points  $0, k, 2k, 3k$ . Formula 10.312 gives an integrand  $-(1-a)^k(1-(1-a)^k)^2$ , which has fixed negative size on  $a \in [\alpha, 2\alpha]$ . The first estimate in Corollary 10.342 therefore bounds the spectral mass of this  $a$ -dyadic shell. For  $\alpha \geq 1/16$ , the same conclusion follows from a fixed unit-scale choice and the deterministic bound on one-step height increments.

For the mixed bound choose vertical increments of length  $\ell$  separated horizontally by  $k$ , with

$$k \asymp \alpha^{-1}, \quad \beta = \frac{\pi}{2\ell}.$$

The spectral integrand is  $2(1-a)^k(1-\cos(\ell b)) \geq 0$  and is bounded below by an absolute constant on  $0 < a < \alpha$  and  $\beta < |b| < 2\beta$ . The second estimate in Corollary 10.342 gives

$$|\Phi_{2,L}^{\text{Cyl}}((0,0), (0,\ell), (k,0), (k,\ell))| \leq C(\ell/k)^c \leq C'(\alpha/\beta)^c,$$

which proves the second bound in (21.91); the remaining dyadic choices follow by changing  $\ell$  by a bounded factor and using the first bound.

Summing the first estimate over the dyadic shells  $a \in (2^j, 2^{j+1})$  with weights  $2^{-|j|}$  proves (21.92). For compactness, exhaust  $\mathbb{R}_{>0} \times \mathbb{R}$  by rectangles bounded away from  $a = 0$  and use the dyadic bounds to obtain uniformly bounded mass on every compact set. Helly–Prokhorov on each rectangle and a diagonal extraction give vague precompactness. The inequalities in (21.91) pass to the limit after enlarging the open dyadic shells by an arbitrarily small factor; hence the class is closed and therefore compact.

Finally split the integral of  $f_n$  into the compact region  $\{R^{-1} \leq a \leq R, |b| \leq R\}$  and its complement. On the compact part, vague convergence plus uniform convergence of  $f_n$  applies. The  $a$ -tails are uniformly small by (21.92) and (21.93); the  $|b|$ -tail with  $a \leq R^{-1}$  is summable by the second bound in (21.91), while for  $a > R^{-1}$  it is absorbed into the first estimate and the compact exhaustion. Letting  $R \rightarrow \infty$  gives the stated convergence rule.  $\square$

**Lemma 10.347** (Full-plane and scaling spectral representations). *There is a subsequential full-plane spectral measure  $\mu_\infty \in \mathcal{M}$  such that for horizontally ordered lattice quadruples  $u = (u_1, u'_1, u_2, u'_2)$  with one increment horizontal,*

$$\Phi_2(u) = \int \chi_u^{\text{discr}}(a, b) d\mu_\infty(a, b). \quad (21.94)$$

*For  $\delta > 0$  define the rescaled measure  $\mu_\infty^{(\delta)}(U) = \mu_\infty(\delta U)$ . From every sequence  $\delta_n \downarrow 0$  one may extract a further subsequence for which  $\mu_\infty^{(\delta_n)} \rightarrow \mu \in \mathcal{M}$ . For every strictly horizontally ordered macroscopic quadruple for which one increment is horizontal,*

$$\Psi_2(u) := \lim_{n \rightarrow \infty} \Phi_2^{(\delta_n)}(u) = \int \chi_u(a, b) d\mu(a, b), \quad (21.95)$$

*where, in displacement coordinates,*

$$\chi_u(a, b) = (e^{-ax_2 - iby_2} - 1)e^{-ax'_1 - iby'_1}(1 - e^{-ax_1 - iby_1}). \quad (21.96)$$

*Proof.* Compactness supplies a vague limit of a subsequence of  $\mu_L$ . The cylinder-to-plane local limit and Lemma 10.346 allow passage to the limit in the discrete spectral formula. The only point not covered directly by compactly supported test functions is the endpoint  $a = 2$ . For odd  $m$ ,

$$\Phi_2(0, e_1, (m+1)e_1, (m+2)e_1) = - \int a^2(1-a)^m d\mu_\infty(a, b),$$

and the left-hand side tends to zero by Theorem 10.341; hence  $\mu_\infty\{a = 2\} = 0$ . This proves (21.94).

The rescaled family remains in  $\mathcal{M}$ , so another compactness extraction gives  $\mu$ . Split the discrete integral into  $a \leq 1$  and  $1 < a \leq 2$ . On  $1 < a < 2$ , strict horizontal separation contributes a factor  $|1 - a|^{c/\delta_n}$  and dominated convergence sends this piece to zero; the possible atom at 2 was just excluded. On  $a \leq 1$ , after the change of variables  $(a, b) \mapsto (\delta_n a, \delta_n b)$ , the discrete exponential factors converge uniformly on compact sets to those in (21.96). Their size is bounded by  $C(a \wedge a^{-1})$ , so the strengthened convergence rule of Lemma 10.346 proves (21.95).  $\square$

**Definition 10.348** (Laplace–Fourier derivative and radial primitive). For a scaling spectral limit  $\mu$  set

$$F(z, y) := \int a e^{-az - iby} d\mu(a, b), \quad \Re z > 0, \quad (21.97)$$

and

$$I(s) := - \int_1^s F(x, 0) dx, \quad s > 0. \quad (21.98)$$

Then  $I'(s) = -F(s, 0)$ .

**Lemma 10.349** (Rotational reconstruction of the two-point limit). *For every scaling spectral limit,  $F(\cdot, y)$  is holomorphic on the right half-plane,  $|F(z, y)| \leq F(\Re z, 0)$ ,  $F(x, 0)$  is decreasing and tends to zero as  $x \rightarrow \infty$ . Moreover the already-proved rotational invariance of SM-21.S1.04 implies*

$$F(x, y) = \frac{x}{\sqrt{x^2 + y^2}} F(\sqrt{x^2 + y^2}, 0), \quad x > 0, \quad (21.99)$$

*and, for every non-colliding quadruple,*

$$\Psi_2(u) = I(|u_2 - u_1|) + I(|u'_2 - u'_1|) - I(|u'_2 - u_1|) - I(|u_2 - u'_1|). \quad (21.100)$$

*The identity (21.99) extends holomorphically in  $x$  to  $\Re x > 0$  using the branch of the square root mapping  $\mathbb{C} \setminus \mathbb{R}_{\leq 0}$  into the right half-plane.*

*Proof.* The analytic and domination statements follow directly from (21.97) and Lemma 10.346, with differentiation under the integral. For a strictly ordered quadruple, let the last increment shrink horizontally. Dividing (21.95) by its length and using dominated convergence gives a difference of two values of  $F$ . Repeat after rotating the macroscopic configuration through any fixed angle. The rotation coupling of Theorem 10.280 and the regularity estimate allow passage to the same scaling limit, so the directional derivatives of  $\Psi_2$  agree with those of a radial function. Sending the first increment far away removes the remote  $F$  term because  $F(x, 0) \rightarrow 0$ ; hence

$$-F(x, y) = \partial_x I(\sqrt{x^2 + y^2}) = -\frac{x}{\sqrt{x^2 + y^2}} F(\sqrt{x^2 + y^2}, 0),$$

which is (21.99). Integrating this identity along the endpoints of the second increment, and using additivity of height differences, yields (21.100) first for separated half-plane configurations and then for every non-colliding quadruple by rigid motions and subdivision. For fixed  $y$ , both sides of (21.99) are holomorphic in  $x$  on the right half-plane and agree on the positive real axis, so the identity theorem gives the final assertion.  $\square$

**Lemma 10.350** (Poisson mollifier transform). *For  $\sigma, \alpha > 0$  and  $b \in \mathbb{R}$ ,*

$$\int_{\mathbb{R}} \frac{\sigma^2}{t^2 + \sigma^2} \frac{\sin(\alpha t)}{t} e^{-ibt} dt = \pi \int_{b-\alpha}^{b+\alpha} \frac{\sigma}{2} e^{-\sigma|y|} dy. \quad (21.101)$$

*Proof.* With the Fourier convention used above,  $\widehat{\frac{\sigma}{2} e^{-\sigma|\cdot|}}(t) = \sigma^2/(t^2 + \sigma^2)$ , while  $\widehat{\frac{1}{2} \mathbf{1}_{[-\alpha, \alpha]}}(t) = \sin(\alpha t)/t$ . The product on the Fourier side is therefore the transform of the convolution of these two  $L^1$  functions. Fourier inversion at  $b$  gives (21.101).  $\square$

**Theorem 10.351** (Sub-diagonal concentration). *Every scaling spectral limit satisfies*

$$\mu\{(a, b) : |b| > a\} = 0. \quad (21.102)$$

*Proof.* It is enough to show, for every  $\lambda > 1$ ,

$$F(1, 0) = \int \mathbf{1}_{\{|b| \leq \lambda a\}} a e^{-a} d\mu(a, b), \quad (21.103)$$

because  $a e^{-a} > 0$  and  $\lambda \downarrow 1$ . Mollify the indicator by the Poisson kernel and set

$$J_\sigma := \int (\mathbf{1}_{[-\lambda a, \lambda a]} * \frac{\sigma}{2} e^{-\sigma|\cdot|})(b) a e^{-a} d\mu(a, b). \quad (21.104)$$

Dominated convergence sends  $J_\sigma$  to the right-hand side of (21.103). Lemma 10.350, Fubini, and  $\sin(\lambda at) = (e^{i\lambda at} - e^{-i\lambda at})/(2i)$  give

$$J_\sigma = \frac{1}{2\pi i} \int_{\mathbb{R}} \frac{\sigma^2}{t(t^2 + \sigma^2)} [F_-(t) - F_+(t)] dt, \quad (21.105)$$

where

$$F_\pm(t) := F(1 \pm i\lambda t, t) = \frac{1 \pm i\lambda t}{\sqrt{(1 \pm i\lambda t)^2 + t^2}} F\left(\sqrt{(1 \pm i\lambda t)^2 + t^2}, 0\right). \quad (21.106)$$

The second equality is Lemma 10.349.

Indent the real line below the origin and denote the resulting contour by  $\gamma_\varepsilon^-$ ; let  $\gamma_\varepsilon^+$  be the corresponding upper indentation. The difference  $F_- - F_+$  cancels the apparent pole at the origin,

so Cauchy's theorem permits (21.105) on  $\gamma_\varepsilon^-$ . Moving only the  $F_-$  contour to  $\gamma_\varepsilon^+$  crosses the simple pole contributed by  $1/t$  at 0, whose residue is  $F_-(0) = F(1, 0)$ . Using the reflection symmetry  $b \mapsto -b$  of  $\mu$ , the two remaining contour integrals are complex conjugates, and therefore

$$J_\sigma = F(1, 0) + \frac{1}{\pi} \operatorname{Im} \int_{\gamma_\varepsilon^+} \frac{\sigma^2}{t(t^2 + \sigma^2)} F_-(t) dt. \quad (21.107)$$

It remains to show that the integral tends to zero.

Put  $\alpha(t) = (1 - i\lambda t)^2 + t^2$ . Since  $\lambda > 1$ , direct expansion for  $t = x + iy$  with  $y \geq 0$  gives

$$\alpha(x + iy) = 1 + 2\lambda y + (\lambda^2 - 1)(y^2 - x^2) - 2ix[\lambda + (\lambda^2 - 1)y]. \quad (21.108)$$

The image of the upper half-plane lies on or to the right of the parabola  $\{(1 + is)^2 : s \in \mathbb{R}\}$ ; hence the chosen square root is defined there and  $\Re \sqrt{\alpha(t)} \geq 1$ . Consequently (21.106) and the domination of Lemma 10.349 show that  $F_-$  is holomorphic and uniformly bounded in the upper half-plane. For every fixed  $s$  with  $\Im s > 0$ ,  $\Re \sqrt{\alpha(\sigma s)} \rightarrow \infty$ , so  $F_-(\sigma s) \rightarrow 0$ . Deforming  $\gamma_\varepsilon^+$  to the horizontal line  $\mathbb{R} + i\sigma/2$  and then setting  $t = \sigma s$  transforms the error in (21.107) into an integral of a fixed  $L^1(ds)$  kernel times  $F_-(\sigma s)$ . Dominated convergence therefore makes it vanish. Thus  $J_\sigma \rightarrow F(1, 0)$ , proving (21.103) and hence (21.102).  $\square$

**Theorem 10.352** (Monotone scale function). *For a scaling spectral limit set*

$$\Xi(s) := -sI'(s) = sF(s, 0). \quad (21.109)$$

*Then  $\Xi$  is bounded, nonnegative and non-increasing, and*

$$-\Xi'(s) = s \Delta[I(|u|)]_{u=se_1} = s \int (a^2 - b^2) e^{-as} d\mu(a, b) \geq 0. \quad (21.110)$$

*Proof.* Boundedness follows by decomposing the  $a$ -axis dyadically in  $s \int a e^{-as} d\mu$ . The only delicate point is differentiating the radial representation. Theorem 10.351 implies that every factor  $1 - e^{-ax - iby}$  is  $O(a|(x, y)|)$ , so the spectral formula extends to arbitrary separated increments by sending an auxiliary horizontal detour to infinity; the detour correlation vanishes by Theorem 10.341. For a tiny displacement  $(x, y)$  at a point  $se_1$ , the extended formula and (21.100) give

$$I(|se_1 + (x, y)|) - I(s) = \int e^{-as} (e^{-ax - iby} - 1) d\mu(a, b). \quad (21.111)$$

Taking the Laplacian at  $(0, 0)$  is justified by  $|b| \leq a$  and (21.92), and yields

$$\Delta I(|u|)|_{u=se_1} = \int (a^2 - b^2) e^{-as} d\mu.$$

For a radial function,  $I'(s) + sI''(s) = s\Delta I(|u|)|_{u=se_1}$ ; since  $\Xi = -sI'$ , this is exactly (21.110). Nonnegativity follows from  $F(s, 0) \geq 0$ .  $\square$

**Theorem 10.353** (Two-point GFF dichotomy). *Exactly the source-level Gate G3 conclusion holds. Either there is a single  $\sigma \geq 0$  for which*

$$\Phi_2^{(\delta)} \longrightarrow \sigma^2 \Psi_2^{\text{GFF}}$$

*uniformly on every compact subset of the non-collision configuration space as  $\delta \downarrow 0$ , or there are two distinct constants  $\sigma_- \neq \sigma_+$  and two sequences of meshes tending to zero along which the two-point functions converge to  $\sigma_-^2 \Psi_2^{\text{GFF}}$  and  $\sigma_+^2 \Psi_2^{\text{GFF}}$  respectively.*



*Proof.* For a fixed scaling spectral limit, (21.100) shows that it is a GFF two-point function precisely when  $I(s) = -\kappa \log s + \text{constant}$ , equivalently when  $\Xi$  is constant. If every convergence sequence has the same constant, the first alternative holds. If two convergence sequences already have different constants, the second holds.

It remains to treat a convergence sequence for which  $\Xi$  is not constant. Since Theorem 10.352 makes  $\Xi$  bounded and non-increasing, the limits

$$\kappa_0 := \lim_{s \downarrow 0} \Xi(s), \quad \kappa_\infty := \lim_{s \rightarrow \infty} \Xi(s)$$

exist and satisfy  $\kappa_0 > \kappa_\infty$ . For every  $r > 0$ ,

$$I(\theta r) - I(\theta) \rightarrow -\kappa_0 \log r \quad (\theta \downarrow 0), \quad I(\Theta r) - I(\Theta) \rightarrow -\kappa_\infty \log r \quad (\Theta \rightarrow \infty), \quad (21.112)$$

by integrating  $I'(s) = -\Xi(s)/s$ . Inserting these limits in (21.100) shows that zooming into the scaling limit gives a GFF two-point function with multiplier  $\sigma_-^2 = 2\pi\kappa_0$ , whereas zooming out gives one with  $\sigma_+^2 = 2\pi\kappa_\infty$ .

To turn this iterated limiting statement into honest mesh subsequences, fix a countable dense family of non-colliding quadruples. Choose successively  $\theta_j \downarrow 0$  so that the zoomed limit is within  $j^{-1}$  of the first GFF on the first  $j$  quadruples, and then choose mesh indices  $n_j$  so large that  $\delta_{n_j}/\theta_j \leq j^{-1}$  and the discrete two-point function is within  $j^{-1}$  of its scaling limit on the same finite family. The regularity estimate gives equicontinuity away from collisions, so pointwise convergence on this dense family upgrades to uniform convergence on compact subsets. The same diagonal construction with  $\Theta_j \rightarrow \infty$  produces the second GFF subsequence. Since  $\kappa_0 > \kappa_\infty$ , their multipliers are distinct.  $\square$

### 10.8.6 Harmonic propagation and uniqueness of multipoint limits

Gate G4 is conditional on choosing a mesh subsequence along which the two-point function has one of the GFF limits supplied by Theorem 10.353. We now show that this already forces every multipoint correlation along the same subsequence to be the corresponding Gaussian correlation.

**Lemma 10.354** (Diagonal spectral support along a GFF two-point subsequence). *Suppose  $\delta_n \downarrow 0$  and  $\Phi_2^{(\delta_n)} \rightarrow \sigma^2 \Psi_2^{\text{GFF}}$  locally uniformly away from collisions. For every scaling spectral limit  $\mu$  extracted from this same subsequence,*

$$\mu\{(a, b) : a^2 \neq b^2\} = 0. \quad (21.113)$$

*Proof.* For the GFF two-point function the radial primitive in Lemma 10.349 is  $-\sigma^2(2\pi)^{-1} \log s$  up to an additive constant. Hence the scale function  $\Xi = -sI'$  is constant. Equation (21.110) therefore gives, for every  $s > 0$ ,

$$0 = s \int (a^2 - b^2) e^{-as} d\mu(a, b).$$

Theorem 10.351 makes the integrand nonnegative. Since  $se^{-as} > 0$ , it must vanish  $\mu$ -almost everywhere, which is exactly (21.113).  $\square$

**Lemma 10.355** (Harmonicity outside the convex hull). *Fix even  $k \geq 4$ . Along a subsequence as above, pass further so that  $\Phi_k^{(\delta_n)} \rightarrow \Psi_k$  locally uniformly on  $D_k$ ; this is possible by Corollary 10.343. If*

$$u_1 \notin \text{Conv}\{u_2, u'_2, \dots, u_k, u'_k\}, \quad (21.114)$$

*then  $\Psi_k$  is harmonic in  $u_1$  at the configuration  $u = (u_1, u'_1, \dots, u_k, u'_k)$ .*

*Proof.* Choose a circle  $C_r$  centred at  $u_1$  that is disjoint from the convex hull in (21.114). After a rigid motion,  $C_r$  lies strictly to the left of a vertical reflection line while all endpoints belonging to the other  $k - 1$  increments lie strictly to the right. The mean-value defect is

$$\Upsilon_r := \frac{1}{|C_r|} \int_{C_r} \Psi_k(s, u'_1, u_2, u'_2, \dots, u_k, u'_k) ds - \Psi_k(u_1, u'_1, u_2, u'_2, \dots, u_k, u'_k). \quad (21.115)$$

It suffices to prove  $\Upsilon_r = 0$ .

Approximate the circle average by a Riemann sum on the mesh and let  $X_n$  be the resulting average of  $h(s) - h(u_1)$ ; let  $Y_n = \prod_{i=2}^k (h(u'_i) - h(u_i))$ . They are supported on opposite sides of the reflection line. The reflection-positive spectral Cauchy–Schwarz estimate of Lemma 10.311 gives, first on a cylinder and then after its full-plane limit,

$$|\mathbb{E}(X_n Y_n)|^2 \leq \mathbb{E}(X_n X_n^\dagger) \mathbb{E}(Y_n^\dagger Y_n). \quad (21.116)$$

The second factor is uniformly bounded by Theorem 10.341, because all its macroscopic endpoints remain separated. The first factor is a double circle average of a two-point correlation. Its scaling limit is the same double average of  $\sigma^2 \Psi_2^{\text{GFF}}$ ; the GFF covariance is harmonic in either circle variable away from the reflected singularity, so the mean-value property makes this limit zero. Thus the right-hand side of (21.116) tends to zero. The left-hand side converges to  $|\Upsilon_r|^2$  by local uniform convergence and the Riemann-sum approximation. Therefore  $\Upsilon_r = 0$ , proving harmonicity.  $\square$

**Lemma 10.356** (Analytic propagation under separating translations). *Let  $u \in D_k$ , and let a line  $\mathcal{L}$  separate every endpoint pair  $\{u_i, u'_i\}$  as a whole: each pair lies strictly on one side of  $\mathcal{L}$ . Form  $\tilde{u}$  by translating all pairs on one side a distance  $\lambda_0 \geq 0$  perpendicular to  $\mathcal{L}$ . If  $\Psi_k$  is harmonic in  $u_1$  on a neighbourhood of  $\tilde{u}$ , then it is harmonic in  $u_1$  at  $u$ .*

*Proof.* After a rigid motion take  $\mathcal{L} = \{0\} \times \mathbb{R}$  and translate the right-hand pairs by  $\lambda e_1$ . For a small circle  $C_r$  around  $u_1$  set

$$\Upsilon_r(\lambda) = \frac{1}{|C_r|} \int_{C_r} \Psi_k(u_{-,s}, u_{+,\lambda}) ds - \Psi_k(u_{-,u_1}, u_{+,\lambda}), \quad \lambda > 0. \quad (21.117)$$

The hypothesis says that this function vanishes on an open interval around  $\lambda_0$ . We show that it is real analytic on  $(0, \infty)$ .

At mesh  $\delta_n$ , collect the circle-averaged left observables into  $X_n$  and the translated right increments into  $Y_n$ . The general spectral representation from Lemma 10.311 gives

$$\Upsilon_r(\lambda) = \lim_{n \rightarrow \infty} \lim_{L \rightarrow \infty} \int (1 - a)^{\lfloor \lambda/\delta_n \rfloor} d\nu_{n,L}(a), \quad (21.118)$$

where the total variations of  $\nu_{n,L}$  are bounded by the reflected second moments of  $X_n$  and  $Y_n$ . Theorem 10.341 bounds these uniformly. Rescale  $a \mapsto \delta_n a$  and extract a weak limit of the finite complex measures on  $[0, \infty]$ . Negative-eigenvalue mass vanishes exactly as in Lemma 10.347; hence

$$\Upsilon_r(\lambda) = \int_{[0, \infty)} e^{-a\lambda} d\nu(a). \quad (21.119)$$

The right-hand side is holomorphic for  $\Re \lambda > 0$  and therefore real analytic on the positive axis. Since it vanishes on a nonempty open interval, it vanishes identically. In particular  $\Upsilon_r(0) = 0$  by continuity, which is the required mean-value identity at  $u$ .  $\square$

**Proposition 10.357** (Separate harmonicity of every multipoint scaling limit). *Under the hypotheses above,  $\Psi_k$  is harmonic in every endpoint coordinate on  $D_k$ .*

*Proof.* By permutation and sign symmetries it suffices to prove harmonicity in  $u_1$ . First consider configurations for which distinct increment pairs are separated by a factor much larger than their diameters. Choose two transverse lines through a remote point so that the pair  $\{u_1, u'_1\}$  lies in one cone and every other pair lies in one of the two adjacent cones. Translate the pairs in the first adjacent cone far away, and then those in the second adjacent cone far away. In the resulting configuration  $u_1$  lies outside the convex hull of all the other endpoints, so Lemma 10.355 gives harmonicity. Apply Lemma 10.356 twice, reversing the two translations, to recover harmonicity in the original well-separated configuration.

For an arbitrary  $u \in D_k$ , move  $u'_1$  very close to  $u_1$ ; this does not affect the question of harmonicity in  $u_1$ . Join each  $u_i$  to  $u'_i$  for  $i \geq 2$  by disjoint smooth curves and divide every curve into  $N$  equal parameter intervals. Additivity of height increments writes  $\Psi_k(u)$  as the sum over the  $N^{k-1}$  correlations obtained by replacing each long increment by one short segment. For  $N$  sufficiently large, every resulting configuration is in the well-separated class just treated. A finite sum of functions harmonic in  $u_1$  is harmonic, so the claim follows. Symmetry gives every other coordinate.  $\square$

**Lemma 10.358** (Elementary uniqueness for logarithmic harmonic singularities). *Let  $z_1, \dots, z_m$  be distinct points. Suppose two real functions  $f, g$  are harmonic on  $\mathbb{R}^2 \setminus \{z_1, \dots, z_m\}$ , have the same logarithmic principal part at every  $z_j$ , and each has a finite limit at infinity. Then  $f - g$  is constant. If the two functions agree at one regular point, they agree everywhere.*

*Proof.* The difference  $w = f - g$  is harmonic away from the  $z_j$  and bounded near each puncture, because the logarithmic principal parts cancel. The mean-value formula on shrinking circles shows that each puncture is removable: define  $w(z_j)$  as the limiting circle average and use the Poisson formula in a surrounding disk. Thus  $w$  extends to an entire harmonic function. It is bounded outside a large disk by the finite limit at infinity and bounded inside by continuity. The real and imaginary parts of an entire holomorphic primitive of  $\partial_x w - i\partial_y w$  are bounded-gradient harmonic functions; equivalently, applying the mean-value formula to  $w$  on circles of radius  $R$  and differentiating at the centre gives  $|\nabla w(0)| \leq CR^{-1} \sup_{B_R} |w| \rightarrow 0$ . Translating the argument shows  $\nabla w \equiv 0$ , so  $w$  is constant. Agreement at one point fixes that constant to zero.  $\square$

**Theorem 10.359** (Two points determine all height-increment correlations). *Let  $\delta_n \downarrow 0$  and assume*

$$\Phi_2^{(\delta_n)} \longrightarrow \sigma^2 \Psi_2^{\text{GFF}}$$

*locally uniformly on  $D_2$ . Then for every  $k \geq 1$ ,*

$$\Phi_k^{(\delta_n)} \longrightarrow \sigma^k \Psi_k^{\text{GFF}} \tag{21.120}$$

*locally uniformly on  $D_k$ .*

*Proof.* Odd correlations vanish by global height-flip symmetry, so consider even  $k$  and induct on  $k$ . The case  $k = 2$  is the hypothesis. By Corollary 10.343, every subsequence has a further subsequence on which  $\Phi_k^{(\delta_n)}$  converges locally uniformly to some continuous  $\Psi_k$ ; it suffices to identify this limit.

Proposition 10.357 makes  $\Psi_k$  separately harmonic away from collisions. Fix all endpoints except  $u_1$ . The regularity estimate shows that  $\Psi_k$  has a finite limit as  $|u_1| \rightarrow \infty$ . When  $u_1$  approaches an

endpoint  $v \in \{u_i, u'_i\}$  of another increment, the fusion theorem 10.344 and the induction hypothesis give

$$\Psi_k(u) = \sigma^2 \Psi_2^{\text{GFF}}(u_1, u'_1, u_i, u'_i) \sigma^{k-2} \Psi_{k-2}^{\text{GFF}}((u_j, u'_j)_{j \notin \{1, i\}}) + O(1). \quad (21.121)$$

Thus the logarithmic principal part at every puncture is exactly the one of  $\sigma^k \Psi_k^{\text{GFF}}$  (Wick's formula groups precisely the pairings containing the singular partner). Finally both functions vanish when  $u_1 = u'_1$  because that increment is zero. Lemma 10.358 therefore identifies  $\Psi_k = \sigma^k \Psi_k^{\text{GFF}}$  as a function of  $u_1$ , and hence everywhere.

Every convergent subsubsequence has the same limit, so the full sequence converges. Equicontinuity upgrades convergence to local uniform convergence, proving (21.120).  $\square$

### 10.8.7 From correlation limits to convergence as a random distribution

The previous gate identifies every off-diagonal correlation function. This does not by itself justify integrating through collision diagonals, nor does it imply tightness in a negative regularity topology. We now prove locally the two steps of the convergence criterion used by the source. Throughout, a *generalized test function* is a finite, compactly supported signed measure  $\varphi$  on  $\mathbb{R}^2$  with  $\varphi(\mathbb{R}^2) = 0$ . It has finite Dirichlet energy when the logarithmic Green integral

$$\mathcal{E}(\varphi) := \iint G_{\mathbb{R}^2}(u, v) d\varphi(u) d\varphi(v), \quad G_{\mathbb{R}^2}(u, v) = -\frac{1}{2\pi} \log |u - v|, \quad (21.122)$$

is finite in the source sense. Smooth compactly supported mean-zero densities are the principal examples used by the frozen parent theorem.

**Lemma 10.360** (Collision domination for finite-energy tests). *Fix an even integer  $k \geq 2$  and let  $\varphi = \varphi_+ - \varphi_-$  be a finite-energy generalized test function, normalized so that  $\varphi_+$  and  $\varphi_-$  are probability measures. Put*

$$d\tilde{\varphi}_k(u) = \prod_{i=1}^k d\varphi_-(u_i) d\varphi_+(u'_i).$$

*Then the family  $\Phi_k^{(\delta)}$  is uniformly integrable with respect to  $\tilde{\varphi}_k$  as  $\delta \downarrow 0$ . More precisely, after expanding the finite sum over pairings in Theorem 10.341, every pairing term is dominated by a product of two-point factors whose  $\tilde{\varphi}_2$  integrals are finite.*

*Proof.* For a pair of increments  $A = \{u, u'\}$  and  $B = \{v, v'\}$ , abbreviate the two cases in Theorem 10.341 by  $R_\delta(A, B)$ . If the positive scale separation is large,  $R_\delta \leq C_k$ . In the complementary regime the lattice rounding changes all Euclidean distances by at most  $O(\delta)$ , so there is a constant  $C = C(k)$  such that uniformly in  $\delta \in (0, 1)$ ,

$$R_\delta(A, B) \leq C \left( 1 + \sum_{x \in A} \sum_{y \in B} \log_+ \frac{1}{|x - y|} \right). \quad (21.123)$$

Indeed, a negative value of  $\text{SR}'_2$  can only occur because the diameter of one of the two increments is much smaller than the distance to the other pair; the triangle inequality then bounds its negative part by one of the four logarithmic collision terms in (21.123), up to an additive constant.

The finite-energy assumption is exactly what makes those local logarithmic singularities integrable against the Jordan components of  $\varphi$ ; away from the diagonal compact support makes the logarithm bounded. Hence

$$\iint R_\delta(A, B) d\tilde{\varphi}_2(A, B) \leq C_{k, \varphi} \quad (21.124)$$

uniformly in  $\delta$ . For a fixed pairing  $\pi$ , the product measure  $\tilde{\varphi}_k$  factorizes over the paired indices, and therefore Fubini gives

$$\int \prod_{ij \in \pi} R_\delta(A_i, A_j) d\tilde{\varphi}_k = \prod_{ij \in \pi} \int R_\delta(A_i, A_j) d\tilde{\varphi}_2 \leq C_{k,\varphi}.$$

There are finitely many pairings. The same argument after truncating the right-hand side of (21.123) at height  $M$  and then sending  $M \rightarrow \infty$  proves uniform integrability, not merely a uniform first-moment bound.  $\square$

**Theorem 10.361** (Finite-energy test-function convergence). *Assume that along  $\delta_n \downarrow 0$ , for every  $k \geq 1$ ,*

$$\Phi_k^{(\delta_n)} \longrightarrow \sigma^k \Psi_k^{\text{GFF}}$$

*locally uniformly on  $D_k$ . Then for every finite family of finite-energy generalized test functions  $\varphi_1, \dots, \varphi_m$ ,*

$$(\langle h^{(\delta_n)}, \varphi_j \rangle)_{j=1}^m \implies N(0, \sigma^2 \Sigma(\varphi)), \quad \Sigma_{ij} = \iint G_{\mathbb{R}^2}(u, v) d\varphi_i(u) d\varphi_j(v). \quad (21.125)$$

*Proof.* By Cramér–Wold it is enough to treat one test function  $\varphi$ . Rescale its total variation and write  $\varphi = \varphi_+ - \varphi_-$  with  $\varphi_\pm$  probability measures. Because the total mass is zero,

$$\langle h^{(\delta)}, \varphi \rangle = \iint (h^{(\delta)}(u') - h^{(\delta)}(u)) d\varphi_+(u') d\varphi_-(u). \quad (21.126)$$

For even  $k$ , Fubini gives

$$\mathbb{E} \langle h^{(\delta)}, \varphi \rangle^k = \int \Phi_k^{(\delta)}(u) d\tilde{\varphi}_k(u), \quad (21.127)$$

while odd moments vanish by height-flip symmetry. Theorem 10.359 and Lemma 10.360 permit dominated convergence through the collision diagonals. The limit of (21.127) is therefore the  $k$ th moment of the centered Gaussian with variance  $\sigma^2 \mathcal{E}(\varphi)$ , by Wick’s formula. Centered Gaussian laws satisfy Carleman’s criterion, so their moments determine the law. This proves the one-dimensional statement, and Cramér–Wold gives (21.125).  $\square$

**Lemma 10.362** (Wavelet tightness from scale-invariant test moments). *Let  $(X_n)$  be stationary random distributions on a bounded open set  $U \subset \mathbb{R}^2$ , considered modulo constants. Suppose that for every even  $p$  and every compactly supported Lipschitz mean-zero density  $\rho$  supported in the unit square,*

$$\sup_n \sup_{x \in U} \sup_{0 < \lambda \leq 1} \mathbb{E} |\langle X_n, \rho_x^\lambda \rangle|^p \leq C_{p,\rho}, \quad \rho_x^\lambda(y) = \lambda^{-2} \rho((y-x)/\lambda), \quad (21.128)$$

*whenever the support stays in a fixed compact enlargement of  $U$ . Then for every  $\alpha \in (-1, 0)$  the laws of  $X_n$  are tight locally in  $\mathcal{C}^\alpha(U)$ . They are also tight in  $B_{r,q}^\alpha(U)$  and  $W^{\alpha,r}(U)$  for every  $1 \leq r < \infty$  and  $1 \leq q \leq \infty$ .*

*Proof.* Choose a compactly supported  $C^2$  orthonormal wavelet basis on a square containing  $U$ , with each mother wavelet of zero integral. At dyadic scale  $j$ , an  $L^2$ -normalized wavelet  $\psi_{j,z}$  equals  $2^{-j}$  times an  $L^1$ -normalized test density of the form in (21.128). Hence

$$\sup_{n,z} \mathbb{E} |\langle X_n, \psi_{j,z} \rangle|^p \leq C_p 2^{-jp}. \quad (21.129)$$

There are  $O(2^{2j})$  wavelets meeting  $U$ . Fix  $\beta \in (\alpha, 0)$  and choose even  $p > 2/(-\beta)$ . Markov's inequality and the union bound give, for  $R \geq 1$ ,

$$\mathbb{P}\left(\sup_{j \geq 0} \sup_{z: \psi_{j,z} \cap U \neq \emptyset} 2^{j(\beta+1)} |\langle X_n, \psi_{j,z} \rangle| > R\right) \leq CR^{-p} \sum_{j \geq 0} 2^{j(2+p\beta)}. \quad (21.130)$$

The series converges by the choice of  $p$ . The finite collection of scaling-function coefficients is controlled by the same moment bound after fixing the additive constant. The standard wavelet reconstruction proof is elementary here: summing the wavelet series against a test function of scale  $\lambda \asymp 2^{-J}$  shows that the bounded quantity in (21.130) is equivalent, up to constants, to the local  $B_{\infty,\infty}^\beta = \mathcal{C}^\beta$  norm. Thus the  $\mathcal{C}^\beta$  norms are bounded in probability. The embedding  $\mathcal{C}^\beta(U) \hookrightarrow \mathcal{C}^\alpha(U)$  is compact for  $\beta > \alpha$ , giving tightness in  $\mathcal{C}^\alpha$ .

The same coefficient bound implies the Besov claim directly. Indeed, for fixed  $r, q$ , choose  $\beta \in (\alpha, 0)$ . If the  $\mathcal{C}^\beta$  coefficient norm is at most  $R$ , then

$$\left(\sum_z |\langle X_n, \psi_{j,z} \rangle|^r\right)^{1/r} \leq CR 2^{2j/r} 2^{-j(\beta+1)}.$$

Multiplication by the  $B_{r,q}^\alpha$  weight  $2^{j(\alpha+1-2/r)}$  leaves  $CR 2^{-j(\beta-\alpha)}$ , which is summable in  $q$  (and bounded for  $q = \infty$ ). Hence bounded  $\mathcal{C}^\beta$  sets are bounded in every  $B_{r,q}^\alpha$  with  $\alpha < \beta$ ; taking an intermediate exponent gives compactness. For the Sobolev claim one should not identify  $W^{\alpha,r}$  with  $B_{r,r}^\alpha$  when  $r \neq 2$ . Instead use the wavelet square-function characterization of the Bessel-potential space  $W^{\alpha,r} = F_{r,2}^\alpha$ . On the event that the  $\mathcal{C}^\beta$  coefficient norm is at most  $R$ , at each  $x$  only  $O(1)$  wavelets of a given scale meet  $x$ , and their normalized square-function contribution is bounded by  $CR 2^{-j(\beta-\alpha)}$ . Therefore

$$\left\| \left( \sum_{j \geq 0} [CR 2^{-j(\beta-\alpha)}]^2 \right)^{1/2} \right\|_{L^r(U)} \leq C_{U,\alpha,\beta,r} R.$$

Taking an intermediate exponent  $\beta' \in (\alpha, \beta)$  makes the corresponding embedding compact. This proves tightness in  $W^{\alpha,r}$  without the false  $W^{\alpha,r} = B_{r,r}^\alpha$  identification. Exhaustion by smooth subdomains gives the local statement on arbitrary bounded open  $U$ .  $\square$

**Theorem 10.363** (Correlation convergence implies source-strength GFF convergence). *Under the hypotheses of Theorem 10.361, the fields  $h^{(\delta_n)}$  converge to  $\sigma\Phi$  in all three senses used by the isotropic source: locally uniform convergence of every off-diagonal correlation function, convergence against all finite-energy generalized test functions, and convergence in law on every bounded open  $U$  in  $\mathcal{C}^\alpha(U)$  for each  $\alpha \in (-1, 0)$ , together with the corresponding  $B_{r,q}^\alpha$  and  $W^{\alpha,r}$  topologies for finite  $r$ .*

*Proof.* Only tightness remains after Theorem 10.361. Let  $\rho$  be a Lipschitz mean-zero density supported in the unit square. Every rescaling  $\rho_x^\lambda$  has the same logarithmic Dirichlet energy: after the change of variables the additive term  $-(2\pi)^{-1} \log \lambda$  vanishes because  $\int \rho = 0$ . Applying the moment formula (21.127) and the collision domination of Lemma 10.360 therefore gives, for every even  $p$ ,

$$\sup_n \sup_{x,\lambda} \mathbb{E} |\langle h^{(\delta_n)}, \rho_x^\lambda \rangle|^p < \infty. \quad (21.131)$$

Lemma 10.362 gives tightness in each claimed topology. Every subsequential limit has the finite-dimensional distributions of  $\sigma\Phi$  by Theorem 10.361; test-function pairings separate distributions, so the limit law is unique. Tightness plus uniqueness proves convergence of the full sequence.  $\square$

### 10.8.8 Exact free-energy curvature by a source-exact Wiener–Hopf calculation

We now close the analytic half of Gate G6. Throughout this subsection

$$\Delta = -\cos \zeta, \quad 0 \leq \zeta \leq \pi/3,$$

so that  $\arccos \Delta = \pi - \zeta$ . The proof below follows the exact normalization of the primary GFF source: first reduce the second derivative of the slope free energy to a half-line Wiener–Hopf ratio, and then evaluate that ratio. The Fourier-transform convention is

$$\widehat{F}(t) = \int_{\mathbb{R}} e^{-itx} F(x) dx.$$

**Definition 10.364** (Half-line Wiener–Hopf data). For  $0 < \zeta \leq \pi/3$ , set

$$\widehat{R}_\zeta(t) = \frac{\sinh((\pi - 2\zeta)t/2)}{2 \sinh((\pi - \zeta)t/2) \cosh(\zeta t/2)}, \quad e_\zeta(x) = \zeta^{-1} e^{-\pi x/\zeta} \mathbf{1}_{[0, \infty)}(x). \quad (21.132)$$

At  $\zeta = 0$ , put

$$\widehat{R}_0(t) = \frac{e^{-|t|/2}}{2 \cosh(t/2)}, \quad e_0(x) = e^{-\pi x} \mathbf{1}_{[0, \infty)}(x). \quad (21.133)$$

Let  $T_\zeta \in L^1(\mathbb{R})$  be the solution of

$$T_\zeta(x) - \int_0^\infty R_\zeta(x-y) T_\zeta(y) dy = e_\zeta(x). \quad (21.134)$$

**Proposition 10.365** (Reduction of  $f''(0)$  to the half-line ratio). *Equation (21.134) has a unique  $L^1$ -solution and*

$$f''(0) = -2 \frac{\int_0^\infty \frac{e_\zeta(x)}{e_\zeta(0)} T_\zeta(x) dx}{\left( \frac{\pi}{\pi - \zeta} \int_0^\infty T_\zeta(x) dx \right)^2}. \quad (21.135)$$

Here  $e_0(0) = 1$ .

*Proof.* Put

$$\tau_\zeta = \begin{cases} \pi/\zeta, & \zeta > 0, \\ \pi, & \zeta = 0. \end{cases}$$

The kernel  $R_\zeta$  has constant sign and  $|\widehat{R}_\zeta(0)| \leq 1/2$ . Hence

$$(\mathcal{R}_+ g)(x) := \int_0^\infty R_\zeta(x-y) g(y) dy$$

has  $L^1$ -operator norm at most  $1/2$ , and  $I - \mathcal{R}_+$  is inverted by its Neumann series. This proves existence and uniqueness of  $T_\zeta$ .

We now retain the exact constants in the continuum-Bethe edge analysis. Let  $\rho_Q$  be the truncated continuum Bethe density, let  $\rho$  be its full-line limit, and put

$$M_Q := \int_{|x| > Q} \rho_Q(x) dx, \quad u := \frac{1}{2} - m.$$

The full-line resolvent form of the Bethe equation is

$$\rho_Q(x) = \rho(x) + \int_{|y|>Q} R_\zeta(x-y) \rho_Q(y) dy. \quad (21.136)$$

Integrating this identity over  $\mathbb{R}$  and using  $m = \int_{-Q}^Q \rho_Q$  gives the exact missing-mass relation

$$u = (1 - \widehat{R}_\zeta(0)) M_Q = \frac{\pi}{2(\pi - \zeta)} M_Q. \quad (21.137)$$

This identity is the source of the factor  $\pi/(\pi - \zeta)$ ; it must not be replaced by the false shortcut  $u = M_Q$ .

For completeness we spell out the boundary-layer limit used below. Split the exterior operator in (21.136) into its right and left pieces  $\mathcal{U}_+$  and  $\mathcal{U}_-$ . After translating either exterior half-line to  $\mathbb{R}_+$ , a pure word  $\mathcal{U}_+^k$  or  $\mathcal{U}_-^k$  becomes  $\mathcal{R}_+^k$ ; summing all pure words gives  $T_\zeta - e_\zeta$ . Every mixed word contains at least one passage between the two exterior half-lines. The separation  $2Q$ , together with the tail estimate for  $R_\zeta$ , makes the sum of all mixed words  $o(e^{-\tau_\zeta Q})$  in both  $L^1$  and  $L^\infty$ . Thus

$$\rho_Q(x) = \rho(x) + e^{-\tau_\zeta Q} \left[ (T_\zeta - e_\zeta)(Q - x) + (T_\zeta - e_\zeta)(-Q - x) + \varepsilon_Q(x) \right], \quad (21.138)$$

where  $\|\varepsilon_Q\|_1 + \|\varepsilon_Q\|_\infty = o(1)$ . At  $\zeta = 0$  the same Neumann expansion uses the endpoint kernel in (21.133); only the tail estimate changes, and the displayed  $o(1)$  conclusion is unchanged.

Substitute (21.138) into the two exterior integrals. The full-density tail and the two copies of  $T_\zeta - e_\zeta$  combine, while the remote translate and  $\varepsilon_Q$  vanish by dominated convergence. Consequently

$$e^{\tau_\zeta Q} M_Q \longrightarrow 2 \int_0^\infty T_\zeta(x) dx. \quad (21.139)$$

Combining (21.137) and (21.139) yields

$$u e^{\tau_\zeta Q(m)} \longrightarrow C_\zeta := \frac{\pi}{\pi - \zeta} \int_0^\infty T_\zeta(x) dx. \quad (21.140)$$

It remains to keep the leading coefficient in the free-energy loss. At the balanced track angle set

$$\delta f(Q) := \int_{\mathbb{R}} L(x) \rho(x) dx - \int_{-Q}^Q L(x) \rho_Q(x) dx, \quad G := L - R_\zeta * L.$$

The Fourier transform of  $G$  has first poles at  $\pm i\tau_\zeta$ , with residues  $\mp 2i$  in this normalization, and all remaining singularities lie strictly farther from the real axis. Rewriting  $\delta f$  with (21.136) and shifting the two Fourier contours across these first poles gives

$$\delta f(Q) = 4e^{-\tau_\zeta Q} \int_0^\infty e^{-\tau_\zeta x} \rho_Q(Q+x) dx + o(e^{-2\tau_\zeta Q}). \quad (21.141)$$

The endpoint  $\zeta = 0$  is obtained by the same contour computation with the endpoint transforms in (21.133). Substitution of (21.138) into the weighted integral gives

$$\delta f(Q) = 4e^{-2\tau_\zeta Q} \left( \int_0^\infty \frac{e_\zeta(x)}{e_\zeta(0)} T_\zeta(x) dx + o(1) \right). \quad (21.142)$$



Equations (21.140) and (21.142) therefore imply

$$\delta f(Q(m)) = \frac{4u^2}{C_\zeta^2} \int_0^\infty \frac{e_\zeta(x)}{e_\zeta(0)} T_\zeta(x) dx + o(u^2). \quad (21.143)$$

Finally the frozen slope convention is  $s = 1 - 2m = 2u$ , so

$$f(0) - f(s) = -\frac{1}{2}f''(0)s^2 + o(s^2) = -2f''(0)u^2 + o(u^2).$$

Comparison with (21.143) proves (21.135).  $\square$

**Lemma 10.366** (Canonical scalar Wiener–Hopf factor). *Let  $m_\zeta(t) = 1 - \widehat{R}_\zeta(t)$ . There is a zero-free function  $\alpha_\zeta$ , holomorphic on each half-plane and tending to one at infinity, whose boundary values satisfy*

$$\frac{(\alpha_\zeta)_-(t)}{(\alpha_\zeta)_+(t)} = m_\zeta(t), \quad (\alpha_\zeta)_-(t) = \frac{1}{(\alpha_\zeta)_+(-t)}. \quad (21.144)$$

Moreover,

$$(\alpha_\zeta)_+(0)^2 = \frac{1}{m_\zeta(0)} = \frac{2(\pi - \zeta)}{\pi}. \quad (21.145)$$

*Proof.* The function  $m_\zeta$  is positive and even on  $\mathbb{R}$ , tends to one exponentially fast, and is bounded away from zero. Thus  $\log m_\zeta \in L^1(\mathbb{R})$ . For  $z \notin \mathbb{R}$ , define

$$C_\zeta(z) = \frac{1}{2\pi i} \int_{\mathbb{R}} \frac{\log m_\zeta(u)}{u - z} du, \quad \alpha_\zeta(z) = e^{-C_\zeta(z)}. \quad (21.146)$$

The Plemelj formula gives  $C_{\zeta,+} - C_{\zeta,-} = \log m_\zeta$ , hence the first jump identity. Evenness of  $m_\zeta$  gives  $C_\zeta(-z) = -C_\zeta(z)$ ; taking boundary values yields the second identity. At zero, the two identities imply  $(\alpha_\zeta)_+(0)^{-2} = m_\zeta(0)$ . Directly from (21.132)–(21.133),

$$m_\zeta(0) = \frac{\pi}{2(\pi - \zeta)},$$

which proves (21.145). The decay of the Cauchy transform proves  $\alpha_\zeta(z) \rightarrow 1$ .  $\square$

**Lemma 10.367** (Explicit half-line integral ratio). *For every  $0 \leq \zeta \leq \pi/3$ ,*

$$\frac{\int_0^\infty \frac{e_\zeta(x)}{e_\zeta(0)} T_\zeta(x) dx}{\left( \int_0^\infty T_\zeta(x) dx \right)^2} = \frac{\pi^2}{4(\pi - \zeta)}. \quad (21.147)$$

*Proof.* Write

$$T_\uparrow = \mathbf{1}_{[0,\infty)} T_\zeta, \quad T_\downarrow = -\mathbf{1}_{(-\infty,0)} T_\zeta.$$

The transform of  $T_\uparrow$  extends holomorphically to the lower half-plane, whereas that of  $T_\downarrow$  extends to the upper half-plane. Define the piecewise holomorphic function

$$\tilde{T}(z) = \begin{cases} \int_{\mathbb{R}} e^{-izx} T_\uparrow(x) dx, & \Im z < 0, \\ \int_{\mathbb{R}} e^{-izx} T_\downarrow(x) dx, & \Im z > 0. \end{cases}$$

Fourier transforming (21.134) gives

$$m_\zeta \tilde{T}_- - \tilde{T}_+ = \widehat{e}_\zeta. \quad (21.148)$$

Multiplication by the factor from Lemma 10.366 reduces this to

$$G_- - G_+ = (\alpha_\zeta)_+ \widehat{e}_\zeta, \quad G = \alpha_\zeta \tilde{T}. \quad (21.149)$$

The transform  $\widehat{e}_\zeta$  is

$$\widehat{e}_\zeta(z) = \begin{cases} (\pi + i\zeta z)^{-1}, & \zeta > 0, \\ (\pi + iz)^{-1}, & \zeta = 0, \end{cases}$$

and has the unique pole

$$t_\zeta = \begin{cases} i\pi/\zeta, & \zeta > 0, \\ i\pi, & \zeta = 0. \end{cases}$$

The unique solution of (21.149) tending to zero at infinity is

$$G(z) = (\alpha_\zeta(t_\zeta) - \alpha_\zeta(z) \mathbf{1}_{\{\Im z > 0\}}) \widehat{e}_\zeta(z). \quad (21.150)$$

Indeed, the subtraction cancels the upper-half-plane pole, the boundary jump is exactly  $(\alpha_\zeta)_+ \widehat{e}_\zeta$ , and uniqueness follows from Morera's and Liouville's theorems.

On the real line,

$$\widehat{T}_\uparrow(t) = (\alpha_\zeta)_+(-t) \alpha_\zeta(t_\zeta) \widehat{e}_\zeta(t). \quad (21.151)$$

Consequently,

$$\int_0^\infty T_\zeta(x) dx = \widehat{T}_\uparrow(0) = \frac{1}{\pi} (\alpha_\zeta)_+(0) \alpha_\zeta(t_\zeta). \quad (21.152)$$

Since  $T_\uparrow, e_\zeta \in L^1 \cap L^2$ , Plancherel gives

$$\int_0^\infty e_\zeta(x) T_\zeta(x) dx = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{e}_\zeta(-t) \widehat{T}_\uparrow(t) dt.$$

Continue  $\tilde{T}$  into the lower half-plane and move the contour below the real axis. The integrand is  $O(|t|^{-2})$ ; the only crossed pole is  $-t_\zeta$ , the pole of  $\widehat{e}_\zeta(-t)$ . Cauchy's formula yields

$$\int_0^\infty e_\zeta(x) T_\zeta(x) dx = \frac{|t_\zeta|}{\pi} \tilde{T}(-t_\zeta) = \frac{|t_\zeta|}{2\pi^2} \alpha_\zeta(t_\zeta)^2. \quad (21.153)$$

Here we used  $\alpha_\zeta(-t_\zeta) = \alpha_\zeta(t_\zeta)^{-1}$  and  $\widehat{e}_\zeta(-t_\zeta) = 1/(2\pi)$ . For  $\zeta > 0$ , division by  $e_\zeta(0) = 1/\zeta$  cancels  $|t_\zeta| = \pi/\zeta$ ; at  $\zeta = 0$ , one has  $e_0(0) = 1$  and  $|t_0| = \pi$ . Thus in both cases

$$\int_0^\infty \frac{e_\zeta(x)}{e_\zeta(0)} T_\zeta(x) dx = \frac{1}{2\pi} \alpha_\zeta(t_\zeta)^2.$$

Divide by the square of (21.152) and use (21.145); the result is (21.147).  $\square$

**Theorem 10.368** (Exact zero-slope free-energy curvature). *For the isotropic six-vertex model with  $\Delta \in [-1, -1/2]$ ,*

$$f''(0) = -\frac{1}{2} \arccos \Delta. \quad (21.154)$$

*Proof.* Combine Proposition 10.365 and Lemma 10.367:

$$f''(0) = -2 \frac{\pi^2/[4(\pi - \zeta)]}{\pi^2/(\pi - \zeta)^2} = -\frac{\pi - \zeta}{2}.$$

Since  $\Delta = -\cos \zeta$ , this is  $-\frac{1}{2} \arccos \Delta$ .  $\square$

### 10.8.9 Profile large deviations and the straight annular variational identity

We now prove locally the random-surface variational principle that the primary GFF paper uses as Proposition 25.1. This is logically independent of the Bethe and Wiener–Hopf computation above: the latter identifies the curvature of the slope free energy, whereas the present argument identifies that free energy as the area-rate of a macroscopic height profile. The proof is given in the finite-range six-vertex setting actually needed here. It contains the three ingredients that must not be hidden behind the phrase “standard variational principle”: affine-box free energies, subexponential seam gluing, and the contraction from profiles to winding level circuits.

**Definition 10.369** (Admissible continuum profiles and surface tension). Let  $D \subset \mathbb{R}^2$  be a bounded polygonal domain. Write  $\mathcal{U}$  for the compact convex hull of the microscopic six-vertex height gradients. An *admissible profile* on  $D$  is a function  $g \in W^{1,\infty}(D)$  with  $\nabla g(x) \in \mathcal{U}$  almost everywhere. Boundary data are understood modulo an additive constant and must satisfy the same Lipschitz and parity compatibility condition as the discrete height.

For an interior rational slope  $p \in \text{int } \mathcal{U}$ , let  $Z_n(p)$  be the six-vertex partition function in the square  $[-n, n]^2$  with a parity-correct boundary height at uniformly bounded distance from  $x \mapsto p \cdot x$ . Set

$$F(p) := \lim_{n \rightarrow \infty} \frac{1}{|[-n, n]^2 \cap \mathbb{Z}^2|} \log Z_n(p), \quad \tau(p) := F(0) - F(p). \quad (21.155)$$

The extension to all  $p \in \mathcal{U}$  is by continuity. Thus  $\tau(0) = 0$  and, in the vertical slope convention of the present chapter,

$$\tau(0, s) = f(0) - f(s). \quad (21.156)$$

**Lemma 10.370** (Parity-Lipschitz extension and seam cost). *Let  $\Lambda$  be a finite simply connected subgraph of the square lattice and let  $b$  be integer-valued boundary data with the correct bipartite parity. If  $|b(x) - b(y)| \leq d_\Lambda(x, y)$  on the boundary, then  $b$  extends to a six-vertex height on  $\Lambda$ . Moreover, if a domain of diameter  $O(n)$  is divided into a fixed number of polygonal pieces whose total interface length is  $\ell$ , then the partition function of the whole domain and the product of the piecewise partition functions with prescribed compatible traces differ by at most*

$$\exp\{C(\ell + n)\} \quad (21.157)$$

*after summing over the interface traces. The constant is uniform over all admissible traces and all  $\sqrt{3} \leq c \leq 2$ .*

*Proof.* For the extension statement define

$$H(x) := \max_{y \in \partial\Lambda} \{b(y) - d_\Lambda(x, y)\}.$$

The boundary Lipschitz condition gives  $H = b$  on  $\partial\Lambda$ . The function  $H$  is integer-valued, one-Lipschitz, and has the correct parity because every quantity  $b(y) - d_\Lambda(x, y)$  has the parity prescribed at  $x$ . Hence for adjacent vertices  $x \sim x'$  the difference  $H(x) - H(x')$  is both odd and of absolute value at most one; it is therefore  $\pm 1$ . Thus  $H$  is an admissible six-vertex height. The same construction works in a collar between two compatible traces.

For the partition-function comparison, first condition on all interface heights. A trace on a path of length  $m$  is determined by its initial value and  $m$  signs, so the number of traces is at most  $(Cn)2^m$ . Along a collar of fixed width the extension above produces at least one completion. Every local six-vertex weight lies between 1 and 2, so inserting or deleting the collar changes a configuration weight by a factor between  $e^{-Cm}$  and  $e^{Cm}$ . Multiplying over the finitely many interfaces and

summing over their traces proves (21.157). The extra  $O(n)$  term accounts for the outer boundary collar and the choice of the initial heights. In every later use  $\ell = O(n)$  or  $O(\varepsilon^{-1}n)$ , and hence this cost is  $o(n^2)$  after the coarse scale is fixed.  $\square$

**Lemma 10.371** (Affine-box free energy). *The limit in (21.155) exists, is independent of the bounded choice made when rounding the affine boundary data, and is locally uniform on compact subsets of  $\text{int } \mathcal{U}$ . The functions  $F$  and  $\tau$  are continuous on  $\mathcal{U}$ ,  $F$  is concave, and  $\tau$  is convex. If  $R_n$  is a polygonal box with  $|R_n| \asymp n^2$  and boundary data at  $O(1)$  from the affine profile  $p \cdot x$ , then*

$$\log Z_{R_n}(p) = |R_n|F(p) + o(n^2), \quad (21.158)$$

*uniformly for  $p$  in a fixed compact subset of the interior slope set.*

*Proof.* Fix first an interior rational slope. Tile a large square by translates of a smaller square, leaving corridors of fixed width along the seams. The lower comparison in Lemma 10.370 gives supermultiplicativity of the partition functions up to an  $e^{O(\text{total seam length})}$  factor. Cutting a large square into the same pieces and summing over the interface traces gives the reverse inequality with the same boundary-order error. Dividing by area and taking the two side lengths successively to infinity proves the existence of  $F(p)$  by the usual almost-Fekete argument. A bounded alteration of the affine trace changes only a boundary collar and hence changes the logarithm by  $O(n)$ .

To compare two slopes  $p_0, p_1$ , place proportions  $1 - t$  and  $t$  of large affine blocks of the two slopes in a common box and join them through corridors. The resulting outer boundary has slope  $(1 - t)p_0 + tp_1$  up to  $O(1/n)$ . The lower seam comparison therefore gives

$$F((1 - t)p_0 + tp_1) \geq (1 - t)F(p_0) + tF(p_1).$$

Thus  $F$  is concave and locally Lipschitz in the interior. Continuity at the boundary uses the hard-constraint geometry rather than a bare concavity assertion. If  $p_t$  is an interior slope at distance  $t$  from a boundary slope  $p$ , remove a collar of width  $Ctn$  from a square with either boundary condition. The remaining trace can be changed from slope  $p$  to slope  $p_t$  (or conversely) inside that collar by the parity-Lipschitz extension lemma. The collar has  $O(tn^2 + n)$  vertices, and summing over its traces costs at most  $\exp\{O(tn^2 + n)\}$ . Therefore

$$|F(p_t) - F(p)| \leq Ct,$$

which proves continuity on the compact slope polygon. Approximation of an arbitrary polygonal box by squares, followed by another seam comparison, gives (21.158). The finite slope net needed for local uniformity is supplied by this continuity.  $\square$

**Theorem 10.372** (Finite-range six-vertex profile large deviations). *Let  $(D_n, b_n)$  be a sequence of lattice domains and parity-compatible boundary heights approximating a bounded polygonal domain  $D$  and an admissible boundary profile  $b$ . If  $h_n$  is sampled from the corresponding finite-volume six-vertex measure and  $g_n$  is the piecewise-affine interpolation of  $n^{-1}h_n(nx)$ , then  $(g_n)$  satisfies a large-deviation principle in the uniform topology with speed  $n^2$  and good rate function*

$$I_{D,b}(g) = \begin{cases} \int_D \tau(\nabla g(x)) dx - \inf_{u \in \mathcal{A}(D,b)} \int_D \tau(\nabla u(x)) dx, & g \in \mathcal{A}(D,b), \\ +\infty, & g \notin \mathcal{A}(D,b), \end{cases} \quad (21.159)$$

where  $\mathcal{A}(D, b)$  is the compact set of admissible profiles with trace  $b$ . Equivalently, for every open set  $O$  and closed set  $C$  in the uniform topology,

$$\begin{aligned} \liminf_{n \rightarrow \infty} n^{-2} \log \mathbb{P}(g_n \in O) &\geq - \inf_{g \in O} I_{D, b}(g), \\ \limsup_{n \rightarrow \infty} n^{-2} \log \mathbb{P}(g_n \in C) &\leq - \inf_{g \in C} I_{D, b}(g). \end{aligned} \quad (21.160)$$

The statement is uniform over boundary traces whose rescaled graphs converge to the same  $b$ .

*Proof.* We prove the non-normalized bounds first and subtract the same formula with no profile constraint at the end.

*Step 1: compactness and polygonal approximation.* After fixing one height value, all interpolated profiles are uniformly one-Lipschitz. Arzelà–Ascoli therefore makes their state space compact, so exponential tightness is automatic. Let  $g \in \mathcal{A}(D, b)$  and fix  $\varepsilon > 0$ . Triangulate all but an  $O(\varepsilon)$ -area boundary collar of  $D$  by triangles of diameter at most  $\varepsilon$ . Linear interpolation of the values of  $g$  on the triangulation gives a piecewise-affine  $g_\varepsilon$  such that

$$\|g_\varepsilon - g\|_\infty = o_\varepsilon(1), \quad \int_D \tau(\nabla g_\varepsilon) \leq \int_D \tau(\nabla g) + o_\varepsilon(1). \quad (21.161)$$

For slopes on the boundary of  $\mathcal{U}$ , first replace  $g$  by  $(1-t)g + tg_*$ , where  $g_*$  is one fixed strictly admissible extension of  $b$ , and then let  $t \downarrow 0$ . Convexity and continuity of  $\tau$  justify the last limit. Thus it suffices in the lower bound to treat finitely many interior affine slopes.

*Step 2: lower bound.* Round  $ng_\varepsilon$  at the vertices of the scaled triangulation to the nearest parity-compatible values. Lemma 10.370 extends these values through the seam collars. On each triangle, Lemma 10.371 gives the partition-function contribution

$$\exp\{n^2|T|F(\nabla g_\varepsilon|_T) + o(n^2|T|)\}.$$

The total seam and boundary-collar cost is  $O(\varepsilon^{-1}n) + O(\varepsilon n^2)$ . Taking  $n \rightarrow \infty$  first, multiplying the block contributions, and then letting  $\varepsilon \downarrow 0$  yields

$$\liminf_{n \rightarrow \infty} n^{-2} \log Z_{D_n, b_n}(\|g_n - g\|_\infty < \delta) \geq |D|F(0) - \int_D \tau(\nabla g) \quad (21.162)$$

for every fixed  $\delta > 0$ , after harmlessly absorbing the constant  $|D|F(0)$  into the normalization convention.

*Step 3: upper bound.* A second, finer scale is needed here; recording only the corner heights would not control an arbitrary Lipschitz trace along a coarse edge. Fix  $0 < \delta \ll 1$  after fixing the triangulation scale  $\varepsilon$ . Along every coarse edge record the height at spacing  $\delta\varepsilon n$ , as well as at every coarse vertex. The total number of recorded sites is  $O((\delta\varepsilon^2)^{-1})$ , independent of  $n$ , and hence the number of records is at most

$$(Cn)^{C/(\delta\varepsilon^2)} = e^{o(n^2)}.$$

Between consecutive recorded sites, the one-Lipschitz property keeps the actual trace within  $O(\delta\varepsilon n)$  of its linear interpolation. Remove from each triangle a collar of that width. Lemma 10.370 replaces the actual trace by the affine interpolated trace across the collar. The total removed area is  $O(\delta n^2)$  and the total seam length is  $O(n/\varepsilon)$ , so the logarithmic error is  $O(\delta n^2) + o(n^2)$ .

Conditional on a record, apply the affine-box upper estimate in the remaining triangle cores. This gives

$$\log Z(\text{that record}) \leq n^2|D|F(0) - n^2 \int_D \tau(\nabla g_{\text{rec}}) + o_{\varepsilon, \delta}(n^2) + O((\varepsilon + \delta)n^2),$$

where  $g_{\text{rec}}$  is the piecewise-affine interpolant. If one of its slopes lies on the boundary of  $\mathcal{U}$ , first replace that slope in the core by  $(1-t)p + tp_*$  with one fixed interior slope  $p_*$  and absorb the interpolation in a relative- $t$  collar; continuity of  $\tau$  allows  $t \downarrow 0$  afterward. Sum over the subexponentially many records, take  $n \rightarrow \infty$ , then  $\delta \downarrow 0$ , and finally  $\varepsilon \downarrow 0$ . Lower semicontinuity of the integral functional gives the closed-set upper bound for the non-normalized measures.

*Step 4: normalization and uniformity.* Applying the lower and upper bounds to the whole compact profile space shows that

$$n^{-2} \log Z_{D_n}(b_n) \longrightarrow |D|F(0) - \inf_{u \in \mathcal{A}(D,b)} \int_D \tau(\nabla u).$$

Subtracting this limit from the non-normalized bounds proves (21.159)–(21.160). Every estimate above uses the boundary trace only through its uniform rescaled limit and an  $O(n)$  collar, which proves the asserted uniformity.  $\square$

**Corollary 10.373** (Profile LDP with finitely many free boundary components). *Theorem 10.372 remains valid when the trace on a fixed finite collection of boundary components is left free and one height is pinned to remove the additive constant. The rate is obtained by minimizing (21.159) over the admissible traces on those components.*

*Proof.* Condition on every free boundary trace. A component of length  $O(n)$  has at most  $(Cn)2^{Cn} = e^{O(n \log n)}$  admissible traces after one height is pinned. This is subexponential at speed  $n^2$ . Theorem 10.372 is uniform over traces with a fixed rescaled limit, and compactness of the rescaled trace space permits a finite net. Taking the largest conditional contribution therefore minimizes the rate over the possible continuum traces, while the number of traces contributes  $o(n^2)$ .  $\square$

**Definition 10.374** (Continuum winding-gap class). For  $\rho > 1$ , let

$$D_\rho := ([-(\rho+1), \rho+1] \times [-1, 1]) \setminus ([-\rho, \rho] \times \{0\}).$$

The class  $\mathcal{W}_{\rho,\alpha}$  consists of admissible profiles on  $D_\rho$  for which there are two nested simple level curves winding around the slit and whose levels differ by at least  $\alpha$ . Profiles and level values are considered modulo an additive constant.

**Lemma 10.375** (Deterministic energy of a winding height gap). *For every  $\alpha \in (0, 1)$ ,*

$$\inf_{g \in \mathcal{W}_{\rho,\alpha}} \int_{D_\rho} \tau(\nabla g) = 4\rho [f(0) - f(\alpha)] + O_\alpha(1), \quad \rho \rightarrow \infty. \quad (21.164)$$

*Proof.* The isotropic model is invariant under both coordinate reflections. Convexity therefore implies

$$\tau(q, s) \geq \tau(0, s) \quad (21.165)$$

for every admissible  $(q, s)$ : average  $(q, s)$  and  $(-q, s)$ .

Let  $g \in \mathcal{W}_{\rho,\alpha}$ . For every  $x \in [-\rho, \rho]$ , the vertical segment from  $(x, 1)$  to the upper bank of the slit crosses both winding level curves. On the subsegment between the two crossings the height variation has absolute value at least  $\alpha$ . If its length is  $d \leq 1$ , Jensen's inequality, (21.165), and convexity with  $\tau(0) = 0$  give

$$\int_{\text{subsegment}} \tau(\nabla g) dy \geq d \tau(q_x, \alpha/d) \geq d \tau(0, \alpha/d) \geq \tau(0, \alpha).$$

The same argument in the lower half of the annulus gives another  $\tau(0, \alpha)$ . Integrating over  $x \in [-\rho, \rho]$  and using (21.156) proves the lower bound  $4\rho[f(0) - f(\alpha)]$ .

For the upper bound choose a piecewise-affine radial coordinate  $r_\rho : D_\rho \rightarrow [0, 1]$  which is zero on the outer boundary, one on the two banks of the slit, and equals  $1 - |y|$  on  $[-\rho, \rho] \times ([-1, 0) \cup (0, 1])$ . It can be completed in the two unit-width end caps with uniformly bounded admissible gradient. The level sets of  $r_\rho$  are winding curves. For  $g = \alpha r_\rho$ , the core energy is exactly  $4\rho\tau(0, \alpha)$  and the two caps contribute  $O_\alpha(1)$ . This proves (21.164).  $\square$

**Proposition 10.376** (Finite-volume contraction to winding profiles). *Let  $\mathbb{P}_{\rho,L}^{\text{fb}}$  be the six-vertex measure in the slit annulus  $A_{\rho,L}$  with all boundary traces summed, one face height pinned, and no boundary weight of order  $L^2$ . Then, for fixed  $\rho$  and  $\alpha \in (0, 1)$ ,*

$$\lim_{L \rightarrow \infty} \frac{1}{L^2} \log \mathbb{P}_{\rho,L}^{\text{fb}}[\text{Circuit}^{+\alpha L}(A_{\rho,L})] = - \inf_{g \in \mathcal{W}_{\rho,\alpha}} \int_{D_\rho} \tau(\nabla g). \quad (21.166)$$

*Rounding  $\alpha L$  to the nearest admissible even height changes neither side.*

*Proof.* Corollary 10.373 supplies the profile LDP on the slit annulus. For the upper bound, interpolate the discrete height and shift it so that the outermost even winding circuit has height zero. The positive circuit and the outermost circuit produce two nested polygonal level curves with gap at least  $\alpha + O(L^{-1})$ . Therefore every configuration in the event belongs to an  $O(L^{-1})$  uniform neighborhood of  $\mathcal{W}_{\rho,\alpha-O(L^{-1})}$ . The closed-set LDP, lower semicontinuity, and continuity in  $\alpha$  give the upper bound.

For the lower bound take a profile in  $\mathcal{W}_{\rho,\alpha+3\varepsilon}$  whose two winding level curves have disjoint tubular neighborhoods. If an interpolated discrete height lies within  $\varepsilon$  of this profile, the discrete intermediate-value property forces a constant even-height winding circuit in each tube: on the bipartite lattice, the boundary of the component of a superlevel set surrounding the slit is an even height circuit. The two resulting levels differ by at least  $\alpha L$  for all large  $L$ . Hence a uniform ball about that profile is contained in the discrete circuit event. The open-set LDP gives the matching lower bound with  $\alpha + 3\varepsilon$ ; let  $\varepsilon \downarrow 0$  and approximate an energy minimizer. The requirement that the first circuit be outermost causes no extra rate: replace a winding circuit of its level by the outer boundary of the union of all winding superlevel components at that level. The height is unchanged and the inner positive circuit remains enclosed.  $\square$

### 10.8.10 Shielding the annulus and the full-plane variational identity

**Definition 10.377** (Shielded approximations of the slit annulus). Fix  $\rho > 2$  and  $\delta \in (0, 1/20)$ . Let  $R_{\rho+2,L}$  be the outer rectangle whose removal of the horizontal slit gives  $A_{\rho+2,L}$ , and write  $\partial_{\text{out}} A_{\rho+2,L} := \partial R_{\rho+2,L}$ . A lattice domain  $U$  belongs to  $\mathfrak{U}_{\rho,L}^\delta$  if it is the interior of a constant-even-height circuit surrounding  $A_{\rho,L}$ , its boundary is contained in the  $3\delta L$ -collar of  $\partial_{\text{out}} A_{\rho+2,L}$ , and it contains the whole core obtained from  $R_{\rho+2,L}$  by deleting that outer collar. Thus  $U$  is simply connected and contains both  $A_{\rho,L}$  and its artificial slit. We write  $U^\dagger$  for the annular domain obtained by cutting  $U$  along the horizontal slit  $[-\rho L, \rho L] \times \{0\}$ .

**Lemma 10.378** (Uniform pinned contraction in a thin collar). *Put*

$$I_{\rho,\alpha} := \inf_{g \in \mathcal{W}_{\rho,\alpha}} \int_{D_\rho} \tau(\nabla g).$$

For every  $\alpha \in (0, 1)$  there is  $C_\alpha < \infty$  such that, uniformly in  $U \in \mathfrak{U}_{\rho, L}^\delta$  and in the common admissible height on  $\partial U$ ,

$$-I_{\rho, \alpha} - C_\alpha \delta(\rho + 1) \leq \liminf_{L \rightarrow \infty} \frac{1}{L^2} \log \mathbb{P}_U^{\text{pin}} [\text{Circuit}^{+\alpha L}(A_{\rho, L})], \quad (21.168)$$

and

$$\limsup_{L \rightarrow \infty} \frac{1}{L^2} \log \mathbb{P}_U^{\text{pin}} [\text{Circuit}^{+\alpha L}(A_{\rho, L})] \leq -I_{\rho, \alpha} + C_\alpha \delta(\rho + 1). \quad (21.169)$$

Here  $\mathbb{P}_U^{\text{pin}}$  is the six-vertex law in  $U$  with the constant outer height pinned; the interaction edges crossing the artificial slit are retained.

*Proof.* Subtract the common boundary height. Since an open even-spin bond joins two even faces whose heights are congruent modulo four and differ by at most two, the height is actually constant along the shielding circuit. Hence the subtraction is legitimate and does not alter the gradient event.

*Step 1: cut the slit.* There are  $O(\rho L)$  six-vertex sites meeting the artificial slit. All six allowed local weights lie between two positive constants depending only on  $\Delta$ . Deleting those interactions, or restoring them after imposing compatible upper and lower traces, changes the logarithm of every constrained or unconstrained partition function by at most  $C\rho L$ . Compatibility is obtained by the parity-preserving seam construction used in Step 2 of Theorem 10.372; its number of choices is  $\exp\{O(\rho L \log L)\}$ . Therefore the slit cut changes the logarithmic probability by  $o(L^2)$  for fixed  $\rho$ . It is enough to work in  $U^\dagger$ .

*Step 2: collar-stable profile contraction.* Cut  $U^\dagger$  once more along the deterministic outer boundary of  $A_{\rho, L}$  and condition on the trace there. The part between that interface and  $\partial U$  lies in the  $3\delta L$ -collar and contains at most  $C\delta(\rho + 1)L^2$  faces. All local weights are bounded above and below by positive constants, while the number of admissible interface traces is  $\exp\{O_\rho(L \log L)\}$ . Therefore, after summing over traces, the collar can bias the logarithm of a constrained-to-unconstrained partition-function ratio by at most

$$C_\alpha \delta(\rho + 1)L^2 + o_\rho(L^2). \quad (21.168a)$$

This estimate is uniform in the rough shielding circuit: all of its geometry lies in the same collar.

Inside the deterministic interface, Proposition 10.376 applies with the boundary traces summed. Its contracted rate is exactly  $I_{\rho, \alpha}$ . For the upper bound, giving the collar and interface trace complete freedom can only increase the constrained numerator; (21.168a) and the closed-set part of the profile LDP then give (21.169). For the lower bound, take a profile within  $\varepsilon$  of the infimum defining  $I_{\rho, \alpha}$ . Clamp it outside the two selected winding levels by composing with a nondecreasing 1-Lipschitz truncation that is constant below the outer level and above the inner level. At almost every differentiability point the new gradient is either the old gradient or zero. By coordinate-reflection symmetry and convexity,  $\tau$  has its minimum 0 at the zero slope. Hence this truncation cannot increase the surface-tension integral. The outer part of the truncated profile is now constant and can be continued through the collar. The parity-compatible seam interpolation from Lemma 10.370 costs only  $o_\rho(L^2)$  outside the collar; the collar cost is bounded by (21.168a). The open-set LDP gives (21.168) after  $\varepsilon \downarrow 0$ . Rounding  $\alpha L$  to an admissible even height costs  $O(L^{-1})$  in the rescaled profile and does not change either estimate.  $\square$

**Lemma 10.379** (Conditional full-plane shielding). *For every fixed  $\delta \in (0, 1/20)$  there is  $C_\delta < \infty$  such that, for all  $\rho > 2$  and all sufficiently large  $L$ , one may construct in the  $3\delta L$ -collar of  $\partial_{\text{out}} A_{\rho+2, L}$  an event  $\mathcal{S}_{\rho, L}^\delta$  with the following properties:*



1.  $\mathcal{S}_{\rho,L}^\delta$  contains a monochromatic open  $\omega^+$ -circuit  $\Gamma$  surrounding  $A_{\rho,L}$ , and the interior  $U_\Gamma$  belongs to  $\mathfrak{U}_{\rho,L}^\delta$ ;
2. for every positive-probability event  $\mathcal{E}$  measurable in terms of the finite height-gradient/spin-bond variables in  $A_{\rho,L}$ ,

$$\mathbb{P}_{\mathbb{Z}^2}(\mathcal{S}_{\rho,L}^\delta \mid \mathcal{E}) \geq \exp\{-C_\delta \rho\}; \quad (21.170)$$

3. conditionally on the outermost such  $\Gamma$  and on its exterior, the six-vertex height in  $U_\Gamma$  has the pinned Gibbs law used in Lemma 10.378, after a global height shift.

*Proof.* Cover the outer collar by  $N \leq C\rho/\delta$  bounded-aspect-ratio rectangles, together with a bounded number of corner annuli. Choose in each rectangle a fresh subrectangle for a long  $\omega^+$  crossing and, between consecutive rectangles, a fresh transverse bridge box of the type used in the proof of Proposition 10.324. The fresh boxes are ordered cyclically and lie inside the outer collar, so a macroscopic buffer separates all of them from  $A_{\rho,L}$ .

Refine the conditioning event  $\mathcal{E}$  into complete atoms of the finite height-gradient/spin-bond configuration in  $A_{\rho,L}$ . Fix one such atom. Construct the shield sequentially. Before the  $j$ th step, reveal the atom, the already constructed crossing pieces, and all variables outside a still unused fresh subrectangle. The spin-bond Markov property identifies the law in that subrectangle with a finite-domain law under an arbitrary admissible boundary state. Boundary-uniform strip RSW, Proposition 10.324, together with the annular bridge consequence Corollary 10.331, gives conditional probability at least  $c_\delta > 0$  for extending the partial shield through that box. The boxes were chosen with unused cores, so the same estimate remains valid after every earlier exploration. Multiplying these conditional probabilities by the chain rule gives

$$\mathbb{P}_{\mathbb{Z}^2}(\mathcal{S}_{\rho,L}^\delta \mid \text{the fixed atom}) \geq c_\delta^N \geq e^{-C_\delta \rho}.$$

Averaging over the atoms compatible with  $\mathcal{E}$  proves (21.170). Notice that this argument never claims that conditioning on an arbitrary union of atoms preserves FKG: the positive-association input is used only inside the already proved boundary-uniform crossing theorem, while the present gluing uses the domain Markov property and the chain rule.

The glued crossings and bridge pieces contain an open monochromatic  $\omega^+$ -circuit in the outer collar, surrounding the entire core rectangle and hence  $A_{\rho,L}$ . Along every open even bond the two endpoint heights are equal: their spins agree, so their height difference is congruent to zero modulo four, while the local ice rule bounds it by two. Thus  $\Gamma$  is a constant-even-height circuit. Taking the outermost one confines all geometric uncertainty to the collar and gives  $U_\Gamma \in \mathfrak{U}_{\rho,L}^\delta$ . Finally, Lemma 10.315(4), followed by the height reconstruction in part (3) of that lemma, is exactly the asserted conditional Gibbs/Markov property.  $\square$

**Proposition 10.380** (Full-plane straight-annulus variational identity). *For every  $\alpha \in (0, 1)$ ,*

$$\lim_{\rho \rightarrow \infty} \lim_{L \rightarrow \infty} \frac{1}{4\rho L^2} \log \mathbb{P}_{\mathbb{Z}^2}[\text{Circuit}^{+\alpha L}(A_{\rho,L})] = f(\alpha) - f(0). \quad (21.171)$$

*Proof.* Write  $E_{\rho,L}$  for the circuit event and fix  $\delta \in (0, 1/20)$ . Apply Lemma 10.379 first without conditioning and then with  $\mathcal{E} = E_{\rho,L}$ . The first application and the Markov property give

$$\mathbb{P}_{\mathbb{Z}^2}(E_{\rho,L}) \geq e^{-C_\delta \rho} \inf_{U \in \mathfrak{U}_{\rho,L}^\delta} \mathbb{P}_U^{\text{pin}}(E_{\rho,L}), \quad (21.172)$$

while the conditional application gives

$$\mathbb{P}_{\mathbb{Z}^2}(E_{\rho,L}) \leq e^{C\delta\rho} \sup_{U \in \mathfrak{U}_{\rho,L}^\delta} \mathbb{P}_U^{\text{pin}}(E_{\rho,L}). \quad (21.173)$$

Indeed, on the shield event one conditions on the outermost circuit  $\Gamma$  and integrates the corresponding pinned probability over its exterior law. The factors  $e^{\pm C\delta\rho}$  are independent of  $L$  and hence disappear at speed  $L^2$ .

Lemma 10.378 now yields

$$-I_{\rho,\alpha} - C_\alpha\delta(\rho+1) \leq \liminf_{L \rightarrow \infty} L^{-2} \log \mathbb{P}_{\mathbb{Z}^2}(E_{\rho,L})$$

and

$$\limsup_{L \rightarrow \infty} L^{-2} \log \mathbb{P}_{\mathbb{Z}^2}(E_{\rho,L}) \leq -I_{\rho,\alpha} + C_\alpha\delta(\rho+1).$$

At fixed  $\rho$ , let  $\delta \downarrow 0$ . The inner limit therefore exists and equals  $-I_{\rho,\alpha}$ . Lemma 10.375 gives

$$I_{\rho,\alpha} = 4\rho[f(0) - f(\alpha)] + O_\alpha(1).$$

Divide by  $4\rho$  and send  $\rho \rightarrow \infty$  to obtain  $f(\alpha) - f(0)$ , proving (21.171).  $\square$

## Ridge splitting and the curved-annulus comparison

The remaining geometric input in the stiffness argument is the “curly” half of the splitting theorem. We formulate it in the exact scale hierarchy used later: first  $\rho$  and  $\varepsilon$  are fixed, then  $k \rightarrow \infty$ , then  $N \rightarrow \infty$ , and finally a macroscopic stacking parameter tends to infinity. All constants below are independent of that final stacking parameter.

**Definition 10.381** (Local ridge counts). Let  $D$  be an even domain and  $a < b$ . In the local level-line forest of Definition 10.336, write  $\text{Ridge}_m(D; a, b)$  for the event that some  $\omega^-$ -circuit at tree depth  $2m - 2$  surrounds a path crossing  $[a, b] \times \mathbb{R}$ . Let  $\#\text{Ridge}_m(D; a, b)$  be the number of maximal  $+$ -domains in which this happens. If only forest components of diameter at most  $2r$  are retained, add the subscript  $r$ .

**Lemma 10.382** (Ridge splitting). *Fix even  $\rho, k, N, \rho', n \geq 1$  with  $k \geq 100$  and  $N/k \geq 100$ . Let  $R_{\text{big}}$  be a rectangle of width  $4n\rho'(\rho+1)N$  and height  $4nN$ , split into  $2n\rho'$  vertical rectangles  $R_i$  of width  $2(\rho+1)N$ . Let  $[a_i, b_i]$  be the central horizontal interval of width  $2\rho N$  in  $R_i$ , and set*

$$\mathcal{E}_{\rho,k,N,\rho',n} := \left\{ \sum_{i=1}^{2n\rho'} \#\text{Ridge}_{k-4}(R_i; a_i, b_i) \geq 4n^2\rho' \right\}.$$

*There is the scale-uniform lower bound*

$$\mu_{\mathbb{Z}^2}(\mathcal{E}_{\rho,k,N,\rho',n}) \geq c_{\text{circuit}}^{100k^2n^2\rho'} \mu_{\mathbb{Z}^2}[\#\text{Ridge}_k(R_{\text{big}}; a_1, b_{2n\rho'}) \geq 2n]. \quad (21.174)$$

*Proof.* Explore the local level-line forest in  $R_{\text{big}}$  down to depth  $k$ . For a forest domain  $A$  let  $N(A)$  be the maximal descendants obtained after one further alternating exploration. Interpolate between the central rectangles  $R'_i$  and  $R_i$  by rectangles  $R'_{i,j}$  whose half-width grows from  $\rho N$  to  $(\rho+1)N$  in  $k$  equal steps. Starting from each of the  $2n$  depth-one domains supplied by the large-ridge event, grow a nested exploration tree by alternating  $+$  and  $-$  and recording in which  $R_i$  each descendant lies.

If every depth- $j$  domain which meets a strip  $R'_{i,j+1} \setminus R'_{i,j}$  is completed by an alternating circuit surrounding its already exposed descendants, then each of the  $2n$  ancestral branches crosses every  $R_i$  down to depth  $k-4$ . Hence every  $R_i$  contains at least  $2n$  ridges and the sum in the definition of  $\mathcal{E}$  is at least  $4n^2\rho'$ .

It remains to estimate the completion cost. At every stage, after conditioning on the exploration atom, the still-unrevealed strip is a polar domain with arbitrary admissible boundary data. The boundary-uniform circuit estimate Corollary 10.331 therefore gives conditional probability at least  $c_{\text{circuit}}$  for each completion. The number of required strip completions is at most  $4n\rho'(k-4) \cdot 25nk \leq 100k^2n^2\rho'$ . Multiplying the conditional probabilities by the chain rule proves (21.174). Notice that no additional edge-by-edge rewiring factor is present: the circuit completion itself is the rewiring operation.  $\square$

**Lemma 10.383** (Variational linkage for the ridge event). *With the notation above,*

$$\lim_{\rho' \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{\rho'n^2} \log \mu_{\mathbb{Z}^2}(\mathcal{E}_{\rho,k,N,\rho',n}) \geq 16(\rho+1)N^2 \left[ f\left(\frac{2(k+4)}{N}\right) - f(0) \right] + 100k^2 \log c_{\text{circuit}}. \quad (21.175)$$

*Proof.* A circuit of height  $4n(k+4)+8$  in the annulus of aspect ratio  $\rho'(\rho+1)$  and scale  $2nN$  contains the alternating level-line pattern needed to create  $2n$  large ridges. Explore those crossings. The remaining local surgery uses at most  $Cn^2N^2$  fresh edges, each having a uniform positive conditional probability, so its logarithmic cost is  $O(n^2N^2)$  independent of  $\rho'$ . Divide by  $\rho'n^2$  and send  $\rho' \rightarrow \infty$ : this surgery term disappears. Proposition 10.380, applied with height-to-scale ratio  $2(k+4)/N$ , gives the first term in (21.175), while Lemma 10.382 contributes exactly the displayed circuit completion cost. This is the source normalization; in particular the coefficient of  $\log c_{\text{circuit}}$  is  $100k^2$ , not  $100k$ .  $\square$

**Lemma 10.384** (Quadratic arm-count tail). *Let  $Q$  be the centres of the disjoint  $2N$ -squares tiling  $R_{\text{big}}$ . For every  $\varepsilon, t > 0$ ,*

$$\frac{1}{\rho'n^2} \log \mu_{\mathbb{Z}^2} \left[ \sum_{x \in Q} \left\lfloor \frac{K_{2N,N,x}}{\varepsilon k} \right\rfloor \geq t|Q| \right] \leq 4(\rho+1) \left( 1 + t(1 - c_{\text{arm}}\varepsilon^2k^2/9) \right) \log 2. \quad (21.176)$$

Here  $c_{\text{arm}} > 0$  is the constant from the arm-tail theorem.

*Proof.* For an integer allocation  $T : Q \rightarrow \mathbb{Z}_{\geq 0}$  with  $\sum_x T_x = \lceil t|Q| \rceil$ , greedily choose  $Q' \subset Q$  so that the annuli  $A(x; N, 2N)$ ,  $x \in Q'$ , are disjoint and  $\sum_{x \in Q'} T_x \geq \frac{1}{9} \sum_x T_x$ . The multiscale arm estimate Lemma 10.335, used in its exact-sum form, gives

$$\mu[K_{2N,N,x} \geq \varepsilon k T_x \ \forall x \in Q'] \leq 2^{-c_{\text{arm}}\varepsilon^2k^2t|Q|/9}.$$

The number of nonnegative integer allocations with the prescribed sum is at most  $2^{(1+t)|Q|}$ . Since  $|Q| = 4n^2\rho'(\rho+1)$ , the union bound gives (21.176). This is the quadratic  $k^2$  tail required below; the weaker linear-in- $k$  tail would not dominate the ridge large-deviation cost.  $\square$

**Lemma 10.385** (Diameter cutoff). *Fix  $\varepsilon \in (0, 10^{-2})$  and assume  $k \geq 72\rho/(\varepsilon\sqrt{c_{\text{arm}}})$ . There is  $r = r(\rho, \varepsilon) < \infty$  such that, if  $\mathcal{E}_{\rho,\varepsilon,k,N,\rho',n}^*$  denotes the event that at least  $(1-\varepsilon)4n^2\rho'$  of the ridges in the rectangles  $R_i$  survive after deleting all forest components of diameter greater than  $2rN$ , then*

$$\lim_{\rho' \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{\rho'n^2} \log \mu(\mathcal{E}^*) \geq 16(\rho+1)N^2 \left[ f\left(\frac{2(k+4)}{N}\right) - f(0) \right] + 100k^2 \log c_{\text{circuit}}. \quad (21.177)$$

*Proof.* Choose  $t = t_\varepsilon$  in Lemma 10.384 so that its exponent is strictly smaller than the lower bound (21.175). A forest tree reaching depth  $2\lfloor \varepsilon k/4 \rfloor$  and diameter  $> 2rN$  contributes at least  $r/2$  units to the normalized arm count. If more than  $2\varepsilon n^2 \rho'$  such trees occur, the arm-count event in Lemma 10.384 occurs. Outside that exponentially negligible event, at most an  $O(\varepsilon)$  fraction of the ridge trees is removed. Every remaining ridge contains a depth-2  $\lceil (1-\varepsilon)k/2 \rceil$  ridge in the diameter-truncated forest. Removing the negligible event from  $\mathcal{E}$  preserves its logarithmic lower rate and yields (21.177).  $\square$

**Lemma 10.386** (Decoupling of the surviving ridges). *For the same parameters there exist at least  $(1-\varepsilon)n$  pairwise disjoint even domains  $D_j$ , each of diameter at most  $2rN$ , such that*

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \frac{1}{n} \log \max_{(D_j)} \prod_j \mu_{\mathbb{Z}^2} \left[ \text{Ridge}_{2\lceil (1-2\varepsilon)k/2 \rceil}(D_j; -\rho N, \rho N) \right] \\ & \geq 4(\rho+1)N^2 \left[ f\left(\frac{2(k+4)}{N}\right) - f(0) \right] + 100k^2 \log c_{\text{circuit}}. \end{aligned} \quad (21.178)$$

*Proof.* Enumerate the diameter-truncated maximal domains  $D_{ij}$  in each  $R_i$ . Let  $A_{ij}$  be the ridge event in  $D_{ij}$  and let  $B_{ij}$  be the event that the central rectangle is crossed vertically by  $\omega^+ \cup \eta_{D_{ij}}$ , where  $\eta_D$  is the set of even edges outside  $D$ . The events  $A_{ij}$  and  $B_{ij}$  are disjoint. Write  $q_{ij} = \mu_{D_{ij}}^+(B_{ij})$ . By the tower property, domain Markov property, and FKG,

$$\prod_{ij} q_{ij} \geq c_{\text{circuit}}^{100n^2 \rho'}. \quad (21.179)$$

Explore the domains in a deterministic order. Until the required number of ridge domains has been selected, if  $A_{ij}$  fails, condition on  $B_{ij}$ ; after that, condition only on  $B_{ij}$ . The product of all extra conditioning probabilities is bounded below by (21.179). Hence conditioning on the exact set of surviving ridge domains loses only the already displayed circuit factor, not an additional rewiring factor.

Given the selected domains, the spatial Markov property factorizes their inside laws. The source comparison between a conditioned ridge at level  $m+2$  and the unconditional ridge at level  $m$  removes the auxiliary  $B_D^c$  condition. Stacking the  $R_i$  vertically and then taking the best deterministic family of disjoint  $D_j$  proves (21.178). The sign of the constant is important: since  $0 < c_{\text{circuit}} < 1$ ,  $\log c_{\text{circuit}} < 0$ , so every additional circuit cost makes the lower bound smaller, not larger.  $\square$

**Lemma 10.387** (Straightening of random ridge domains). *Fix  $\rho$  and  $\varepsilon$ . There are  $\eta > 0$ ,  $r < \infty$ , and  $C < \infty$  such that, for  $k$  sufficiently large and  $N \geq 100k/\eta$ ,*

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \frac{1}{n} \log \max_{(A_j)} \prod_j \mathbb{P}_{\mathbb{Z}^2} \left[ \text{Circuit}^{+4\lceil (1-8\varepsilon)k/2 \rceil}(NA_j) \right] \\ & \geq 4(\rho+1)N^2 \left[ f\left(\frac{2(k+4)}{N}\right) - f(0) \right] + 200k^2 \log c_{\text{circuit}} - C. \end{aligned} \quad (21.180)$$

where the maximum is over at least  $(1-4\varepsilon)n$  disjoint  $(\rho, \eta)$ -straightened annuli  $A_j$  of diameter at most  $2r$  in  $[-(\rho+1), \rho+1] \times [-n, n]$ .

*Proof.* For a domain of diameter at most  $2rN$ , cover its possible boundary by the mesh  $\eta N \mathbb{Z}^2$ . If the level-line forest produces no arm count  $K_{N, \eta N, x} \geq \varepsilon k/10$ , the outer boundary of a surviving ridge can

be pushed to an  $\eta N$ -straightened circuit while losing at most  $2\lceil \varepsilon k/10 \rceil$  levels. Lemma 10.335 and a union bound over the  $O(\rho r/\eta^2)$  mesh points show that the bad straightening event has probability

$$p_{\rho,\varepsilon,\eta,k} \leq C\rho r(4\eta)^{c_{\text{arm}}\varepsilon^2 k^2/100-2}.$$

Choose  $\eta$  so small that, in the product measure generated by Lemma 10.386, occurrence of this bad event in an  $\varepsilon$ -fraction of the domains is exponentially smaller than the ridge product. Thus at least  $(1-3\varepsilon)n$  domains admit a straightened annulus. There are at most  $2^{C\rho r/\eta^2}$  combinatorial straightened boundaries per domain; selecting the most likely family contributes only the constant  $C$  in (21.180).

Finally, conditionally on an alternating circuit of depth  $2\lceil (1-4\varepsilon)k/2 \rceil$ , the independent odd-spin coins orient all relevant levels upward with probability at least  $2^{-2\lceil (1-4\varepsilon)k/2 \rceil}$ . Applying this independently in the chosen product factors converts alternating circuits to positive-height circuits and accounts for the second circuit-cost term. This proves (21.180).  $\square$

**Proposition 10.388** (Curved-annulus splitting bound). *For every fixed even  $\rho \geq 2$  and  $\varepsilon \in (0, 10^{-2})$ , there exist  $\eta > 0$  and  $C, r < \infty$  such that, for all sufficiently large even  $k$  and then all sufficiently large even  $N$ ,*

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \frac{1}{n} \log \max_{(A_j)} \prod_j \mathbb{P}_{\mathbb{Z}^2} \left[ \text{Circuit}^{+2\lceil (1-\varepsilon)k \rceil}(NA_j) \right] \\ & \geq 4(\rho+1)N^2 \left[ f\left(\frac{2(k+4)}{N}\right) - f(0) \right] + 200k^2 \log c_{\text{circuit}} - C. \end{aligned} \quad (21.181)$$

for a family of at least  $(1-\varepsilon)n$  disjoint straightened annuli after replacing  $\varepsilon$  by a fixed smaller fraction in the preceding lemmas. Consequently,

$$\begin{aligned} & \liminf_{k \rightarrow \infty} \liminf_{N \rightarrow \infty} \liminf_{n \rightarrow \infty} \frac{1}{4(\rho+1)nk^2} \log \max_{(A_j)} \prod_j \mathbb{P}[\text{Circuit}^{+2\lceil (1-\varepsilon)k/2 \rceil}(NA_j)] \\ & \geq \frac{1}{2} f''(0) + \frac{100}{\rho+1} \log c_{\text{circuit}}. \end{aligned} \quad (21.182)$$

*Proof.* Apply Lemma 10.387 with the internal error parameter  $\varepsilon/8$  and relabel the surviving annuli. This turns its  $4\lceil (1-8(\varepsilon/8))k/2 \rceil$  height into  $2\lceil (1-\varepsilon)k \rceil$  and leaves at least  $(1-\varepsilon)n$  factors. This proves (21.181). For the final asymptotic consequence, reapply (21.181) with its internal height parameter replaced by  $k/2$  (and then take  $N$  sufficiently large for that parameter). The circuit threshold becomes  $2\lceil (1-\varepsilon)k/2 \rceil$ . Next let  $N \rightarrow \infty$  and then  $k \rightarrow \infty$ , using G6a,  $f(s) = f(0) + \frac{1}{2}f''(0)s^2 + o(s^2)$ . After division by  $4(\rho+1)nk^2$ , the free-energy term contributes  $\frac{1}{2}f''(0)$  and the completion cost contributes  $100(\rho+1)^{-1} \log c_{\text{circuit}}$ . Since  $\log c_{\text{circuit}} < 0$ , there is no sign reversal. This is the source-exact curved/“curly” splitting inequality needed in the variance comparison.  $\square$

### 10.8.11 GFF annular averages and circuit large deviations

**Definition 10.389** (Harmonic annular average). Let  $A \Subset \mathbb{R}^2$  be a topological annulus with inner and outer boundary  $\partial_{\text{int}} A$  and  $\partial_{\text{ext}} A$ . Let  $H_A : \mathbb{R}^2 \rightarrow [0, 1]$  be the bounded harmonic function on  $A$  with boundary values 1 on  $\partial_{\text{int}} A$  and 0 on  $\partial_{\text{ext}} A$ , extended constantly across the two complementary components. Put

$$\nu_A := -\Delta H_A, \quad \mathcal{E}_A := \int_{\mathbb{R}^2} |\nabla H_A|^2, \quad \phi_A := \frac{\nu_A}{\mathcal{E}_A}.$$

Then  $\nu_A = \nu_A^+ - \nu_A^-$ , where  $\nu_A^+$  and  $\nu_A^-$  are supported on the inner and outer boundary respectively and both have total mass  $\mathcal{E}_A$ . Hence  $\phi_A = \phi_A^+ - \phi_A^-$  with  $\phi_A^\pm$  probability measures.

*Proof.* For compactly supported smooth  $F$ , Green's formula gives

$$\int F d\nu_A = \int \nabla F \cdot \nabla H_A.$$

Taking smooth approximations to  $H_A$  gives  $\int H_A d\nu_A = \int |\nabla H_A|^2 = \mathcal{E}_A$ . Since  $H_A$  is constant off  $A$ , the distribution  $-\Delta H_A$  is supported on the two boundary components; the maximum principle fixes its sign on each component. Testing against the constant function shows that its total signed mass is zero, so the positive and negative masses are both  $\mathcal{E}_A$ .  $\square$

**Lemma 10.390** (GFF variance and orthogonal decomposition). *For the normalized whole-plane GFF  $\Gamma$  modulo constants,*

$$\langle \Gamma, \phi_A \rangle \sim N\left(0, \frac{1}{\mathcal{E}_A}\right), \quad \lim_{k \rightarrow \infty} \frac{1}{k^2} \log \mathbb{P}[\langle \Gamma, \phi_A \rangle \geq k] = -\frac{1}{2} \mathcal{E}_A. \quad (21.183)$$

Moreover

$$\Gamma = \Gamma_A + \langle \Gamma, \phi_A \rangle H_A, \quad (21.184)$$

where  $\Gamma_A$  is a centered Gaussian distribution independent of  $\langle \Gamma, \phi_A \rangle$ .

*Proof.* For every finite-Dirichlet-energy signed test measure  $\mu$  of total mass zero,  $(-\Delta)^{-1}$  is the GFF covariance operator, hence

$$\mathcal{E}_A \text{Cov}(\langle \Gamma, \phi_A \rangle, \langle \Gamma, \mu \rangle) = \langle H_A, \mu \rangle. \quad (21.185)$$

Taking  $\mu = \phi_A$  and using  $\langle H_A, \nu_A \rangle = \mathcal{E}_A$  gives the variance  $\mathcal{E}_A^{-1}$ . The Gaussian tail gives (21.183). For (21.184), subtract the orthogonal projection of  $\Gamma$  onto the one-dimensional Gaussian span of  $\langle \Gamma, \phi_A \rangle$ ; (21.185) says that its deterministic coefficient is exactly  $H_A$ . Orthogonality of centered Gaussian variables is independence.  $\square$

**Lemma 10.391** (Discrete annular-average tail along a convergence sequence). *Let  $\delta_n \downarrow 0$  be a convergence sequence for which the rescaled six-vertex height converges to  $\sigma\Gamma$ . Then*

$$\lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{k^2} \log \mathbb{P}_{\mathbb{Z}^2}[\langle h, \phi_A / \delta_n \rangle \geq k] = -\frac{\mathcal{E}_A}{2\sigma^2}. \quad (21.186)$$

*Proof.* G5 gives convergence of the field against every finite-energy, compactly supported signed test measure. Therefore, for fixed  $k$ , the random variable  $\langle h, \phi_A / \delta_n \rangle$  converges in law to  $\sigma \langle \Gamma, \phi_A \rangle$ . The limiting law is continuous, so the event probabilities converge. Lemma 10.390 and then  $k \rightarrow \infty$  give (21.186), in exactly this iterated order.  $\square$

**Lemma 10.392** (Flip domination inside an explored circuit). *Let  $\gamma$  be a self-avoiding face circuit and let  $F_\gamma$  be the faces strictly inside it. Let  $m$  be the smallest even integer not smaller than  $\max_{F_\gamma} h$ , chosen in the gradient-measurable convention of the spin representation. Conditional on any positive-probability gradient event measurable outside  $F_\gamma$ , the reflected field  $2m - h$  on  $F_\gamma$  stochastically dominates  $h$  there. The analogous statement with inequalities reversed follows by global height flip.*

*Proof.* It suffices first to work in a finite even domain containing  $\gamma$ . Explore, from  $\gamma$  inward, the outermost  $\omega_m$  loops of the G2a spin representation. On each unexplored component surrounded by such a loop, the odd-spin coin is conditionally fair, so the unrevealed height is exactly symmetric under  $h \mapsto 2m - h$ . Every revealed face has height at most  $m$ . Coupling the fair coins identically

before and after reflection therefore gives pointwise domination of the reflected field. The exploration is determined by the already revealed exterior variables, so conditioning on the prescribed exterior gradient event does not change the argument. Letting the domain exhaust  $\mathbb{Z}^2$  and using the full-plane spin/Markov limit owned in G2a proves the claim. Global flip gives the reversed form.  $\square$

**Lemma 10.393** (Conditioned harmonic profile produces microscopic level circuits). *Fix a topological annulus  $A$ ,  $x \in A$ ,  $r > 0$  with  $B_r(x) \Subset A$ , and  $\varepsilon > 0$ . There exists  $\alpha = \alpha(A, x, r, \varepsilon) > 0$  such that, if  $h'$  is the representative satisfying  $\phi_A^-(h') \in [0, 2)$ , then*

$$\lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \mathbb{P}_{\mathbb{Z}^2} \left[ \{H_A(x) - \varepsilon \leq h'/k \leq H_A(x) + \varepsilon\} \text{ contains a nontrivial circuit in } A(x; r, \alpha r)/\delta_n \mid \langle h, \phi_A/\delta_n \rangle \geq k \right] = 1. \quad (21.187)$$

*Proof.* Define bad events  $E_-$  and  $E_+$  by the existence of a winding circuit in  $A(x; r, \alpha r)/\delta_n$  lying respectively below  $k(H_A(x) - \varepsilon/4)$  or above  $k(H_A(x) + \varepsilon/4)$ . Define  $E_*$  by the existence, for some even shift  $a$ , of two radial arms across that annulus, one in  $\{h + a \leq 0\}$  and one in  $\{h + a \geq \varepsilon k/2\}$ . Planar separation gives the deterministic trichotomy: if the circuit in (21.187) is absent, then  $E_- \cup E_+ \cup E_*$  occurs.

The G2c arm theorem gives

$$\mathbb{P}(E_*) \leq \alpha^{c\varepsilon^2 k^2/4}.$$

By (21.186), choosing  $\alpha$  sufficiently small makes this exponentially negligible relative to the conditioning event.

It remains to handle  $E_-$ ;  $E_+$  follows by global flip. Let  $\pi_n$  be normalized arc-length measure on the circle of radius  $\alpha r/\delta_n$  about  $x/\delta_n$ . Lemma 10.392, applied to the outermost low circuit, gives conditionally on  $E_- \cap \{\langle h, \phi_A/\delta_n \rangle \geq k\}$  at least probability  $1/2$  that

$$\frac{\langle h, \pi_n - \phi_A^-/\delta_n \rangle}{k} \leq H_A(x) - \varepsilon/8. \quad (21.188)$$

On the other hand, the GFF decomposition (21.184), together with G5 convergence of the pair of averages, shows that under only the high-average conditioning the left side of (21.188) converges in the iterated  $(n, k)$  limit to  $H_A(x)$  in probability: the conditional one-dimensional Gaussian overshoot divided by  $k$  tends to 1, while the independent  $\Gamma_A/k$  fluctuation vanishes. Hence the conditional probability of (21.188) tends to zero. Therefore the conditional probability of  $E_-$  tends to zero, and similarly for  $E_+$ . The trichotomy proves (21.187).  $\square$

**Lemma 10.394** (Merging positive and negative winding circuits). *With  $C_\ell$  denoting the existence of an even winding circuit at relative height  $\ell$  from the outermost reference circuit, for even  $\ell \geq \ell' \geq 2$ ,*

$$\mathbb{P}[C_{\ell+2\ell'-4}] \geq \frac{1}{2} \mathbb{P}[C_\ell \cap C_{-\ell'}]. \quad (21.188a)$$

*Proof.* On  $C_\ell \cap C_{-\ell'}$ , expose the outermost negative circuit and then the relevant outermost positive circuit. Apply Lemma 10.392 in the annular region between them. With conditional probability at least  $1/2$ , reflection of the unrevealed height above the appropriate even mid-level creates an additional upward gap of  $2\ell' - 4$  while preserving the existing  $\ell$ -circuit. The resulting circuit is at relative height  $\ell + 2\ell' - 4$ . Integrating over the exposed circuits gives (21.188a). This is the same one-step flip surgery as in (21.190), now applied to both sides of the reference circuit.  $\square$

**Proposition 10.395** (GFF annular average and height circuit have the same  $k^2$  rate). *For every convergence sequence  $\delta_n$  and every topological annulus  $A$ ,*

$$\lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{k^2} \log \frac{\mathbb{P}_{\mathbb{Z}^2}[\langle h, \phi_A / \delta_n \rangle \geq k]}{\mathbb{P}_{\mathbb{Z}^2}[\text{Circuit}^{+k}(A / \delta_n)]} = 0. \quad (21.189)$$

*Proof.* We prove the two source inequalities.

For the first, let  $\gamma_0$  be the outermost even-height winding circuit and, when it exists, let  $\gamma_\ell$  be the outermost one whose height is  $\ell$  above  $\gamma_0$ . Put  $C_\ell = \{\gamma_\ell \neq \emptyset\}$  and

$$X_0 = h(\gamma_0) - \phi_A^-(h), \quad X_\ell = \phi_A^+(h) - h(\gamma_\ell).$$

On  $C_\ell$ ,  $\langle h, \phi_A \rangle = X_0 + \ell + X_\ell$ . Flip domination and total height-flip symmetry give, for even  $\ell \geq 4$ ,

$$\mathbb{P}[X_0 \geq 0, C_{\ell-4}] \geq \frac{1}{2} \mathbb{P}[C_\ell], \quad \mathbb{P}[X_{\ell-4} \geq 0 \mid X_0 \geq 0, C_{\ell-4}] \geq \frac{1}{2}.$$

Consequently

$$\mathbb{P}[\langle h, \phi_A \rangle \geq \ell - 4] \geq \frac{1}{4} \mathbb{P}[C_\ell]. \quad (21.190)$$

After scaling and setting  $\ell = k + 4$  this is

$$\liminf_{k \rightarrow \infty} \liminf_{n \rightarrow \infty} \frac{\mathbb{P}[\langle h, \phi_A / \delta_n \rangle \geq k]}{\mathbb{P}[\text{Circuit}^{+k+4}(A / \delta_n)]} \geq \frac{1}{4}. \quad (21.191)$$

For the reverse inequality fix  $\varepsilon > 0$ . Apply Lemma 10.393 with conditioning level  $K = (1 + 4\varepsilon)k$  and error  $\varepsilon/(1 + 4\varepsilon)$ . Choose finitely many balls whose centers trace one curve near the outer boundary, where  $H_A$  is arbitrarily close to 0, and another near the inner boundary, where  $H_A$  is arbitrarily close to 1. Since the conditioned deterministic profile is  $KH_A$ , the first family yields a winding circuit below height  $\varepsilon k$  and the second a winding circuit above  $(1 + 3\varepsilon)k$ , with conditional probability tending to one. Gluing the local circuits by the discrete intermediate-value property therefore gives two winding even-height circuits whose height gap is at least  $(1 + \varepsilon)k$ . Choose even levels  $\ell \geq \ell' \geq 2$  whose sum is below the observed gap by only  $O(1)$  and with  $\ell + 2\ell' - 4 \geq k$ . Lemma 10.394, followed by the first comparison (21.190), yields

$$\liminf_{k \rightarrow \infty} \liminf_{n \rightarrow \infty} \frac{\mathbb{P}[\text{Circuit}^{+k}(A / \delta_n)]}{\mathbb{P}[\langle h, \phi_A / \delta_n \rangle \geq (1 + 4\varepsilon)k]} \geq \frac{1}{4}. \quad (21.192)$$

The additive 4 in (21.191) is negligible at  $k^2$  scale. Taking logarithms, using (21.186) for  $k$  and  $(1 + 4\varepsilon)k$ , and then letting  $\varepsilon \downarrow 0$  proves (21.189).  $\square$

### 10.8.12 Dirichlet-energy comparison and stiffness uniqueness

**Proposition 10.396** (Uniform straight circuit bound). *For fixed  $\rho$  and even  $k, N$  in the admissible source lattice, with  $N \geq 2k$ ,*

$$\log \mathbb{P}_{\mathbb{Z}^2}[\text{Circuit}^{+k}(A_{\rho, N})] \leq 4\rho N^2 \left[ f\left(\frac{k - 24}{N}\right) - f(0) \right]. \quad (21.193)$$

*Consequently*

$$\limsup_{k \rightarrow \infty} \limsup_{N \rightarrow \infty} \frac{1}{4\rho k^2} \log \mathbb{P}[\text{Circuit}^{+k}(A_{\rho, N})] \leq \frac{1}{2} f''(0). \quad (21.194)$$



*Proof.* Put  $\chi = \mathbb{P}[\text{Circuit}^{+k}(A_{\rho,N})]$  and work in the G2a spin representation. A positive height circuit forces at least  $k/2 - 4$  alternating levels. If  $D$  is the maximal  $+$  domain around the hole, the Markov property and the branching-function domain monotonicity from Lemma 10.337 therefore give

$$\chi \leq \mu_{[(\rho+1)N \times N]}^+[\text{Circuit}_{\{\psi^* \geq k-12\}}(A_{\rho,N})]. \quad (21.195)$$

Translate  $2m$  copies horizontally. Domain monotonicity and FKG imply that the intersection of their circuit events has probability at least  $\chi^{2m}$ , and inclusion of events turns this intersection into one  $\{\psi^* \geq k - 12\}$  circuit in the annulus of horizontal multiplicity  $2m$ .

Now stack  $2m$  such wide annuli vertically. The same FKG/Markov argument gives horizontal alternating-crossing events above and below the central line with joint probability at least  $\chi^{4\rho'm^2}$ , where  $\rho'$  is the second horizontal multiplicity. A finite-energy surgery on at most  $Cm^2N^2$  previously unrevealed edges joins those crossings into an alternating circuit of depth  $m(k - 23)$ , at cost  $c_0^{m^2N^2}$ . Finally choose the independent odd-spin coins so that the alternating levels point upward; this costs at most  $2^{-mk}$  and produces a positive circuit of height  $2m(k - 24)$  in  $A_{\rho',2mN}$ . Hence

$$\mathbb{P}[\text{Circuit}^{+2m(k-24)}(A_{\rho',2mN})] \geq 2^{-mk} c_0^{m^2N^2} \chi^{4\rho'm^2}. \quad (21.196)$$

Divide the logarithm by the area, send first  $m \rightarrow \infty$  and then  $\rho' \rightarrow \infty$ , and use the full-plane variational identity G6b1. The coin and finite-energy costs vanish after the  $\rho'$  limit and yield

$$f((k - 24)/N) - f(0) \geq (4\rho N^2)^{-1} \log \chi,$$

which is (21.193). Taylor expansion at zero and then  $N \rightarrow \infty$ ,  $k \rightarrow \infty$  gives (21.194).  $\square$

**Lemma 10.397** (Straight annulus Dirichlet upper bound). *For the continuum straight annulus  $A_{\rho,1}$  used above,*

$$\mathcal{E}_{A_{\rho,1}} \leq 4\rho + C \quad (21.197)$$

*with  $C$  independent of  $\rho$ .*

*Proof.* Let

$$H'_\rho(x) = \max\{0, 1 - \text{dist}(x, [-\rho, \rho] \times \{0\})\}.$$

It has the same boundary values as  $H_{A_{\rho,1}}$ . Harmonic functions minimize the Dirichlet energy among functions with fixed trace, while direct integration of  $|\nabla H'_\rho|^2$  gives  $4\rho + C$ , the constant coming only from the two unit-size end caps. Hence (21.197).  $\square$

**Lemma 10.398** (Disjoint straightened annuli force vertical Dirichlet energy). *Let  $A_1, \dots, A_{K_n}$  be disjoint  $(\rho, \eta)$ -straightened annuli contained in  $[-(\rho + 1), \rho + 1] \times [-n, n]$ , with  $K_n \geq (1 - \varepsilon)n$ , and with every inner boundary crossing the vertical strip  $[-\rho, \rho] \times \mathbb{R}$ . Then*

$$\sum_{i=1}^{K_n} \mathcal{E}_{A_i} \geq 4\rho n(1 - \varepsilon)^2. \quad (21.198)$$

*Proof.* Extend every  $H_{A_i}$  by 1 inside its hole and 0 outside its outer component. Their gradients have disjoint supports, so

$$\sum_i \mathcal{E}_{A_i} = \mathcal{E}\left(\sum_i H_{A_i}\right) \geq \int_{-\rho}^{\rho} \int_{-n}^n \left| \partial_y \sum_i H_{A_i}(x, y) \right|^2 dy dx. \quad (21.199)$$

Fix  $x \in [-\rho, \rho]$ . Each straightened annulus contributes one upward and one downward transition of size one to the function  $g_x(y) = \sum_i H_{A_i}(x, y)$ ; by the defining crossing property, its total variation on  $[-n, n]$  is at least  $2K_n \geq 2(1 - \varepsilon)n$ . For any absolutely continuous  $g$  on  $[-n, n]$ , replacing it by  $\tilde{g}(y) = \int_{-n}^y |g'(t)| dt$  preserves both total variation and Dirichlet energy. Cauchy–Schwarz then gives

$$\int_{-n}^n |g'(y)|^2 dy \geq \frac{\text{Var}(g)^2}{2n} \geq 2n(1 - \varepsilon)^2. \quad (21.200)$$

Integrating over  $x \in [-\rho, \rho]$  proves (21.198).  $\square$

**Proposition 10.399** (Variance–curvature identification). *Let  $\delta_n$  be any convergence sequence and let  $\sigma^2$  be the variance multiplier of its limiting GFF. Then*

$$\sigma^2 = -\frac{1}{f''(0)}. \quad (21.201)$$

In particular, by G6a,

$$\sigma^2 = \frac{2}{\arccos \Delta}. \quad (21.202)$$

*Proof. First inequality.* Apply Proposition 10.396 with  $N = \delta_n^{-1}$  and use G6b3 plus Lemma 10.391 for the fixed continuum shape  $A_{\rho,1}$ . Passing first  $n \rightarrow \infty$  and then  $k \rightarrow \infty$  gives

$$\frac{1}{2}f''(0) \geq -\frac{\mathcal{E}_{A_{\rho,1}}}{8\rho\sigma^2}. \quad (21.203)$$

Lemma 10.397 and  $\rho \rightarrow \infty$  imply

$$f''(0) \geq -\frac{1}{\sigma^2}. \quad (21.204)$$

*Second inequality.* Fix  $\rho, \eta, r, \varepsilon$  as in the G6b2 curly bound. Up to translation there are only finitely many  $(\rho, \eta)$ -straightened annulus shapes of diameter at most  $2r$  on the chosen mesh. Apply G6b3 and the Gaussian tail (21.186) to every shape before taking the maximum in G6b2. Its product lower bound becomes

$$-\lim_{n \rightarrow \infty} \min_{(A_i)} \frac{1}{4(\rho+1)n} \sum_i \frac{\mathcal{E}_{A_i}}{2\sigma^2} \geq \frac{1}{2}f''(0) + \frac{100}{\rho+1} \log c_{\text{circuit}}. \quad (21.205)$$

Lemma 10.398 bounds the left side above by

$$-\frac{\rho(1 - \varepsilon)^2}{2(\rho+1)\sigma^2}.$$

Therefore

$$-\frac{\rho(1 - \varepsilon)^2}{2(\rho+1)\sigma^2} \geq \frac{1}{2}f''(0) + \frac{100}{\rho+1} \log c_{\text{circuit}}. \quad (21.206)$$

Send first  $\rho \rightarrow \infty$  and then  $\varepsilon \downarrow 0$  to obtain  $f''(0) \leq -1/\sigma^2$ . Together with (21.204) this proves (21.201). Substituting the exact curvature from G6a proves (21.202).  $\square$

**Proof-stage inventory for the six-vertex GFF theorem.** The argument is decomposed into source-level stages so that the isotropic proof and imported anisotropic clause remain visibly distinct.

- G1. Balanced-cylinder transfer/spectral representation and reflection positivity: CLOSED.
- G2a. Spin representation, finite-domain Markov property, conditional FKG and domain/boundary monotonicity: CLOSED.
- G2b. Boundary-uniform six-vertex annulus circuit / RSW theorem: CLOSED.
- G2c. Level-line-tree recursion, full-plane/cylinder regularity and mixing from the circuit estimate: CLOSED.
- G3. Two-point spectral compactness, sub-diagonal concentration and dichotomy: CLOSED.
- G4. Multipoint harmonicity and uniqueness from the two-point limit: CLOSED.
- G5. Regularity-to-test-function/negative-topology convergence: CLOSED.
- G6a. Exact zero-slope free-energy curvature by the continuum-Bethe and Wiener–Hopf calculation: CLOSED.
- G6b0. Finite-range height-profile LDP, free-boundary contraction, and deterministic winding-gap energy: CLOSED.
- G6b1. Full-plane straight-annulus variational identity: CLOSED by Proposition 10.380.
- G6b2. Ridge/curved splitting comparison: CLOSED by Proposition 10.388.
- G6b3. GFF annular average/circuit large-deviation comparison: CLOSED by Proposition 10.395.
- G6b4. Straight/curly Dirichlet-energy comparison: CLOSED by Lemmas 10.397 and 10.398.
- G6b5. Variance-curvature identification: CLOSED by Proposition 10.399.
- G7. Anisotropic transfer under the natural embedding: IMPORTED; the exact cited theorem is stated below, together with the unavailable companion inputs.

All isotropic stages are proved above. The anisotropic transfer is deliberately imported rather than inferred from the phrase “BKW universality.”

**Anisotropic source scope.** The anisotropic statement below is imported from its cited public theorem. The local discussion records the natural embedding and finite-volume BKW identities, but does not reproduce the uniform continuum estimates used by the source.

### **Anisotropic embedding and finite-volume identities**

The source proof uses the published isoradial random-cluster universality theorem together with companion estimates controlling continuum loop functionals. The elementary embedding and finite-mesh identities that can be stated independently are recorded next.

**Definition 10.400** (Natural anisotropic parametrisation and embedding). Put  $\zeta = \arccos(-\Delta)$ . For  $-1 < \Delta < -1/2$  the source parametrises the weights by  $\theta \in (0, \pi)$  through

$$a \sin \frac{\zeta}{2} = \sin((1 - \theta/\pi)\zeta), \quad b \sin \frac{\zeta}{2} = \sin((\theta/\pi)\zeta), \quad c = 2 \cos \frac{\zeta}{2},$$

and at  $\Delta = -1$  by

$$a = 2 \frac{\pi - \theta}{\pi}, \quad b = 2 \frac{\theta}{\pi}, \quad c = 2.$$

The natural Euclidean embedding is

$$L_\theta(x, y) = (x + \cos \theta y, \sin \theta y).$$

The isotropic point is  $\theta = \pi/2$ .

**Lemma 10.401** (BKW–isoradial geometric correspondence). *Fix  $\Delta \in [-1, -1/2]$  and  $\theta$  in a compact subinterval of  $(0, \pi)$ . Under the parametrisation of Definition 10.400, the Baxter–Kelland–Wu local correspondence sends the anisotropic six-vertex model on  $L_\theta(\mathbb{Z}^2)$  to the critical random-cluster loop model on the associated rectangular isoradial graph  $L(\theta)$  with the cluster weight determined by  $\Delta$ . The loop ensemble is naturally embedded on the same medial rhombi. Moreover the macroscopic random-cluster loop ensemble on  $\delta L(\theta)$  has, as  $\delta \downarrow 0$ , the same subsequential law as at  $\theta = \pi/2$ .*

*Proof.* The local part is a finite vertex calculation. At one rhombus, sum the two possible orientations of each BKW smoothing with the conjugate loop phases. Contractible loops receive the sum of the two phases, while noncontractible loop weights are fixed by the same  $\Delta$ ; the three resulting local sums are exactly the six vertex weights in Definition 10.400. Since the calculation is local it is unchanged by applying  $L_\theta$ . Composing the local identities over a finite torus and then taking the full-plane limit gives the BKW correspondence on the isoradial graph.

The second assertion is not reproved here: it is exactly the rectangular-isoradial random-cluster universality theorem already owned in Proposition 10.304, whose proof was proved here in the the rotational-invariance theorem above. Thus no new random-cluster universality input is being imported at G7.  $\square$

**Lemma 10.402** (Finite-mesh BKW generating identity). *Let  $\mathcal{L}$  be a finite BKW loop configuration and let  $\varphi = \sum_{j=1}^m t_j(\delta_{x'_j} - \delta_{x_j})$  have total mass zero. With the source phase  $\mu$  and the normalised twisted cosine  $\cos_\mu$ , the finite-volume orientation sum satisfies*

$$\mathbb{E}_{6V} \exp(i\langle h, \varphi \rangle) = \phi_{FK} \left[ \prod_{\ell \in \mathcal{L}} \cos_\mu(\varphi(\text{int } \ell)) \right]. \quad (21.G7.1)$$

*Consequently, after classifying a loop by the subset  $S$  of marked points that it surrounds, every mixed derivative at  $t = 0$  is a finite polynomial in the corresponding loop counts  $N_S$ .*

*Proof.* On a finite torus the BKW expansion is a finite algebraic sum. Conditional on the unoriented smoothing, the oriented contribution factorises loop by loop. Reversing one loop conjugates its phase and changes its contribution to the height pairing by the opposite signed amount. Summing the two orientations and dividing by the untwisted loop weight therefore gives precisely the normalised factor  $\cos_\mu(\varphi(\text{int } \ell))$ . Multiplying over loops proves (21.G7.1). The total-mass-zero assumption removes loops surrounding all marked points.

For the final assertion, group the product according to the finitely many incidence classes  $S$ . It becomes

$$\prod_S C_S(t_1, \dots, t_m)^{N_S}$$

with analytic  $C_S$  and  $C_S(0) = 1$ . Any fixed mixed derivative at the origin is therefore a polynomial in the integers  $(N_S)_S$ . This is the finite-volume algebraic mechanism behind the polynomial representation used in the source's anisotropic proof. What is *not* supplied by this finite calculation is the uniform small-loop truncation needed to pass that polynomial to the continuum; that is isolated next.  $\square$

**Theorem 10.403** (Isotropic six-vertex GFF limit). *For the full-plane zero-slope six-vertex Gibbs state with  $a = b = 1$  and  $\Delta \in [-1, -1/2]$ , let  $h_\delta$  be the centered height field on mesh  $\delta$ . Then, as a random distribution modulo additive constants,*

$$h_\delta \implies \sigma_\Delta \Phi, \quad \sigma_\Delta^2 = \frac{2}{\arccos \Delta},$$

where  $\Phi$  is the normalized full-plane GFF. Equivalently, the multi-point height-increment correlations converge locally uniformly away from collisions, and for every finite family of generalized test functions of finite Dirichlet energy the corresponding height pairings converge jointly to the centered Gaussian vector with covariance  $\sigma_\Delta^2$  times the GFF covariance. In the isotropic case the source also proves the negative-regularity convergence recorded earlier in this chapter.

*Proof.* The result is proved through Gates G1–G6b5: the two-point dichotomy, propagation to all correlations, distributional convergence, and stiffness identification yield exactly  $\sigma_\Delta^2 = 2/\arccos \Delta$ .  $\square$

**Imported theorem 10.404** (Anisotropic six-vertex GFF limit). Let  $a, b, c > 0$  be general six-vertex weights with  $\Delta \in [-1, -1/2]$ , parametrized by the angle  $\theta$  of Definition 10.400, and place the model on  $L_\theta \mathbb{Z}^2$ , where

$$L_\theta(x, y) = (x + \cos \theta y, \sin \theta y).$$

Then the separated multi-point height-increment correlations and every finite family of finite-Dirichlet-energy pairings converge to  $\sigma_\Delta \Phi$ , with  $\sigma_\Delta^2 = 2/\arccos \Delta$ , in the senses of Definition 2.7(i)–(ii) of [13]. No anisotropic convergence in the negative-Hölder/Besov topology of Definition 2.7(iii) is asserted.

*Imported-result status.* This is Theorem 3.3 of [13]. It is imported, not used to prove the isotropic theorem or the subsequent  $q = 4$  results. The cited source's anisotropic argument invokes a companion rotational-invariance manuscript; this volume therefore makes no claim of a local reconstruction of that argument.

### 10.8.13 Proof of the four-state random-cluster/Potts exponent theorem

The  $q = 4$  exponent theorem needs only the *isotropic* six-vertex GFF theorem at  $(a, b, c) = (1, 1, 2)$ , not the anisotropic transfer. We therefore use the finite-mesh BKW identity Lemma 10.402, the locally proved isotropic finite-energy GFF convergence Gate G5, and the locally proved stiffness identity Gate G6b5. At  $\Delta = -1$  the latter gives

$$\sigma^2 = \frac{2}{\arccos(-1)} = \frac{2}{\pi}. \quad (21.Q4.1)$$

The planar random-cluster inputs used below are exact earlier owners in Volume IX: arbitrary-boundary RSW (SM-09.05.02), the wired/free flower-coupling and mixing calculus, the critical cluster/magnetic identities, and the exponent-free planar FK scaling theorem SM-09.S1.04. We nevertheless restate the precise interfaces needed here so that no hidden strengthening occurs.

**Notation.** Work at the critical  $q = 4$  random-cluster measure on the rescaled square lattice  $\delta\mathbb{Z}^2$ , and write  $\phi_\delta$  for the infinite-volume measure. Its medial loop representation is  $\mathcal{L}$ . For a finite set  $X = \{x_1, \dots, x_k\} \subset \mathbb{R}^2$  and  $\varepsilon > 0$ , let  $\mathcal{L}_{X,\varepsilon}$  be the loops which avoid every ball  $B_\varepsilon(x_i)$ , and let  $\mathcal{L}_{X,\varepsilon}^{\text{odd}}$  be those loops in  $\mathcal{L}_{X,\varepsilon}$  which surround an odd number of the marked balls. For compactly supported  $F$  put

$$\mathcal{A}_F(\mathcal{L}) := \prod_{\ell \in \mathcal{L}} \cos \left( \int_{\text{int}(\ell)} F(z) dz \right) \quad (21.Q4.2)$$

whenever the product is finite; all products used below are finite either because  $F$  has mean zero or because we condition away the loops whose contribution would make the non-zero-mean product ill-defined.

For  $0 < r < R$ , write  $\pi_1^\delta(r, R)$  for the one-arm probability,  $\pi_2^\delta(r, R)$  for the alternating primal/dual two-arm probability, and  $\Delta^\delta(r, R)$  for the wired/free influence of an  $r$ -scale crossing under boundary conditions at scale  $R$ . The unscaled versions are denoted by  $\pi_1(r, R)$ ,  $\pi_2(r, R)$  and  $\Delta(r, R)$ .

**Lemma 10.405** (Endpoint planar package used by the  $q = 4$  proof). *There are constants  $c, C, \alpha > 0$  such that, uniformly over all mesh sizes and all scales below the macroscopic scale under consideration,*

1. *fixed-aspect-ratio primal and dual crossings have probabilities in  $[c, 1 - c]$  under arbitrary induced boundary conditions;*

2. *if  $A$  is measurable in  $\Lambda_r$  and  $B$  outside  $\Lambda_R$ ,  $r \leq R/2$ , then*

$$|\phi_\delta(A \cap B) - \phi_\delta(A)\phi_\delta(B)| \leq C(r/R)^\alpha \phi_\delta(A)\phi_\delta(B); \quad (21.Q4.3)$$

3. *for  $u \leq v \leq w$ ,*

$$\pi_i^\delta(u, w) \asymp \pi_i^\delta(u, v)\pi_i^\delta(v, w), \quad i = 1, 2, \quad (21.Q4.4)$$

*and*

$$\Delta^\delta(u, w) \asymp \Delta^\delta(u, v)\Delta^\delta(v, w); \quad (21.Q4.5)$$

4.

$$\pi_2^\delta(u, w) \leq C\pi_1^\delta(u, w)^2, \quad \pi_1^\delta(u, w)^2 \leq C\pi_1^\delta(u, w)(u/w)^\alpha. \quad (21.Q4.6)$$

5. *If  $D_i = B_{r_i}(z_i)$ ,  $r_1 + r_2 \leq |z_1 - z_2|/4$ , and  $A_i$  is measurable on the lattice edges of  $D_i$ , then*

$$\left| \frac{\phi_\delta(A_1 \cap A_2)}{\phi_\delta(A_1)\phi_\delta(A_2)} - 1 \right| \leq C\Delta^\delta(r_1, |z_1 - z_2|)\Delta^\delta(r_2, |z_1 - z_2|). \quad (21.Q4.6a)$$

6. *If  $\pi_4^\delta(u, w)$  is the separated alternating four-arm probability, then*

$$\pi_4^\delta(u, w) \leq C\Delta^\delta(u, w)(u/w)^\alpha. \quad (21.Q4.6b)$$

*Moreover a boosting pair on a well-separated outer four-petal flower of scale  $u$  can be transported to a well-separated inner four-petal flower of comparable scale at distance  $w$  with probability at least*

$$c\Delta^\delta(u, w)^2. \quad (21.Q4.6c)$$

*The same statements hold for separated landing versions of the arm events.*

*Proof.* The arbitrary-boundary crossing statement is the exact  $q = 4$  endpoint of SM-09.05.02. We explain why it gives the arm statements needed here. Given a one- or two-arm realization in  $\text{Ann}(u, w)$ , reveal the arms until they hit an intermediate circle of radius comparable with  $v$ . Four disjoint fixed-aspect-ratio corridors can then force their landing points into prescribed arcs and shield the inner and outer pieces. Arbitrary-boundary RSW and the domain Markov property show that this surgery costs only a fixed factor. Concatenating separated inner and outer realizations through the same corridors proves both inequalities in (21.Q4.4). The same argument gives separated-landing comparability.

Equation (21.Q4.3) and the quasi-multiplicativity (21.Q4.5) are the one-ended wired/free flower-coupling consequences proved by the earlier Volume IX mixing-rate calculus. The stronger two-ended statement (21.Q4.6a), the boost-transport estimate (21.Q4.6c), and the polynomial four-arm comparison (21.Q4.6b) are now owned explicitly by the Volume IX owner **lem:fk-two-ended-mixing-pivotal**. That lemma is proved for every  $1 < q \leq 4$  and its proof reconstructs the flower amplification rather than importing a  $q < 4$  arm exponent; rescaling the lattice from  $\mathbb{Z}^2$  to  $\delta\mathbb{Z}^2$  does not change the constants.

For (21.Q4.6), the first inequality follows by forgetting the dual arm and using FKG/ratio mixing for the two endpoint one-arm events. For the second, on the one-arm event impose, in each of a fixed positive proportion of dyadic annuli, a dual circuit separating the arm from the next scale. Arbitrary-boundary RSW gives a fixed conditional probability for this obstruction. Multiplication over the annuli yields a power saving  $(u/w)^\alpha$ . No SLE exponent is used anywhere in this planar package.  $\square$

**Lemma 10.406** (Conditioned one-arm mixing and local absolute continuity). *Let*

$$\mathcal{I}_\varepsilon := \{\mathcal{L}_{\{0\},\varepsilon}^{\text{odd}} = \emptyset\}.$$

*The infinite-volume conditional law  $\phi_\delta(\cdot \mid \mathcal{I}_\varepsilon)$  is well defined by exhaustion. If  $R/2 \geq r \geq \varepsilon \geq \delta$ ,  $A$  depends only on  $\Lambda_r$ , and  $\xi$  is any boundary condition on  $\Lambda_R$ , then*

$$\left| \phi_{\Lambda_{R,\delta}}^\xi(A \mid \mathcal{I}_\varepsilon) - \phi_\delta(A \mid \mathcal{I}_\varepsilon) \right| \leq C(r/R)^c. \quad (21.Q4.7)$$

*Moreover, if  $A$  depends only on  $B_{2\varepsilon}(0)$ , then*

$$\phi_\delta(A \mid \mathcal{I}_\varepsilon) \leq C\phi_\delta(A). \quad (21.Q4.8)$$

*Proof.* The event  $\mathcal{I}_\varepsilon$  is the coarse IIC event that either a primal or dual arm exits every macroscopic scale. For finite  $N$ , condition first on the arm reaching  $\partial\Lambda_N$ . The endpoint mixing estimate (21.Q4.3), applied after arm separation at a circle between  $r$  and  $R$ , shows that changing the boundary condition at  $R$  changes the conditional law in  $\Lambda_r$  by  $O((r/R)^c)$  uniformly in  $N$ . Taking the exhaustion limit proves (21.Q4.7).

For (21.Q4.8), note that  $\mathcal{I}_\varepsilon \subset \mathcal{I}_{4\varepsilon}$ . In a finite wired box,

$$\phi(A \mid \mathcal{I}_\varepsilon) \leq \frac{\phi(A \cap \mathcal{I}_{4\varepsilon})}{\phi(\mathcal{I}_{4\varepsilon})}.$$

The event  $A$  is supported in  $B_{2\varepsilon}$  while  $\mathcal{I}_{4\varepsilon}$  is determined outside  $B_{4\varepsilon}$ , so ratio mixing bounds the numerator by  $C\phi(A)\phi(\mathcal{I}_{4\varepsilon})$ . RSW in the annulus  $\text{Ann}(\varepsilon, 4\varepsilon)$  gives  $\phi(\mathcal{I}_{4\varepsilon}) \leq C\phi(\mathcal{I}_\varepsilon)$ . Cancel the denominator and pass to the exhaustion limit.  $\square$

**Lemma 10.407** (Macroscopic loop-count tail). *For  $\lambda > 0$ ,  $0 < \eta < 1$  and  $0 < \delta < \varepsilon$ , let  $E(\varepsilon, \eta, \lambda)$  be the event that more than  $\lambda\eta^{-2}$  loops of diameter at least  $\eta\varepsilon/2$  intersect  $B_\varepsilon(0)$ . Then*

$$\phi_\delta(E(\varepsilon, \eta, \lambda)) \leq Ce^{-c\lambda}. \quad (21.Q4.9)$$

*Proof.* Rescale to  $\varepsilon = 1$ . For a grid point  $x \in (\eta/4)\mathbb{Z}^2 \cap B_1$ , let  $N_x$  be one half of the number of medial-loop crossings of the annulus  $\text{Ann}_x(\eta/8, \eta/4)$ . If  $N_x \geq k$ , there are  $k$  distinct FK clusters crossing that annulus. After each cluster is exposed, an RSW circuit in the next available subannulus has probability bounded below uniformly in the induced boundary condition and prevents another crossing. Thus  $N_x$  is stochastically dominated by a geometric random variable with a scale-independent parameter.

Every loop of diameter at least  $\eta/2$  intersecting  $B_1$  crosses one of these annuli, so the loop count is bounded by  $\sum_x N_x$ . Colour the grid with nine colours so that the annuli in each colour class have disjoint supporting boxes. Conditional on the complementary boxes, the corresponding  $N_x$  are dominated by independent geometric variables. Each colour class has  $O(\eta^{-2})$  sites. A Chernoff bound for a sum of  $O(\eta^{-2})$  geometric variables at threshold  $\lambda\eta^{-2}$  gives  $Ce^{-c\lambda}$ ; summing over the nine classes proves (21.Q4.9).  $\square$

**Lemma 10.408** (Quasi-multiplicative exponent extraction). *Let  $v(r, R) > 0$  be increasing in  $r$ , decreasing in  $R$ , and satisfy  $v(r, s)v(s, R) \asymp v(r, R)$ . If for some  $\zeta > 0$*

$$\lim_{\varepsilon \downarrow 0} \liminf_{N \rightarrow \infty} \frac{\log v(\varepsilon N, N)}{\log \varepsilon} = \lim_{\varepsilon \downarrow 0} \limsup_{N \rightarrow \infty} \frac{\log v(\varepsilon N, N)}{\log \varepsilon} = \zeta, \quad (21.Q4.10)$$

*then*

$$\frac{\log v(r_n, R_n)}{\log(R_n/r_n)} \longrightarrow -\zeta \quad \text{whenever } R_n/r_n \rightarrow \infty. \quad (21.Q4.11)$$

*Proof.* Fix  $0 < \rho < 1$  and tile  $[r_n, R_n]$  by the geometric scales  $\rho^{-j}$ . Quasi-multiplicativity gives, up to a factor  $C^m$  when there are  $m$  blocks,

$$v(r_n, R_n) \asymp \prod_{j=j_0}^{j_1-1} v(\rho^{-j}, \rho^{-j-1}).$$

Monotonicity moves each block by at most one scale and therefore permits the use of the liminf/limsup in (21.Q4.10) uniformly except for  $o(m)$  end blocks. Taking logarithms, dividing by  $m \log(\rho^{-1})$ , then sending  $n \rightarrow \infty$  gives upper and lower bounds whose discrepancy is  $O(|\log C|/\log(\rho^{-1}))$ . Finally send  $\rho \downarrow 0$  and use (21.Q4.10). This proves (21.Q4.11).  $\square$

**Theorem 10.409** (Isotropic  $q = 4$  M-formula). *Let  $F$  be compactly supported, mean zero, and belong to  $W^{s,2}(\mathbb{R}^2)$  for some  $0 < s < 1$ . Then*

$$\lim_{\delta \downarrow 0} \phi_\delta[\mathcal{A}_F(\mathcal{L})] = \exp \left\{ \frac{1}{2\pi^2} \iint_{\mathbb{R}^2 \times \mathbb{R}^2} F(z)F(w) \log |z - w| \, dz \, dw \right\}. \quad (21.Q4.12)$$

*This theorem consumes only the isotropic six-vertex gates G1–G6b5.*

*Proof.* At  $q = 4$ , the BKW correspondence orients every FK medial loop independently in the two possible directions. Conditional on the unoriented loop configuration, reversing one loop changes the sign of its contribution to the six-vertex height pairing. The finite-mesh orientation sum is therefore



exactly the product of cosines in (21.Q4.2); this is the isotropic specialization of Lemma 10.402. Thus

$$\phi_\delta[\mathcal{A}_F(\mathcal{L})] = \mathbb{E}_{6V,\delta} \exp\left(i \int F h_\delta\right). \quad (21.Q4.13)$$

The locally closed G5 theorem gives convergence of this finite-energy pairing to a centered Gaussian. Gate G6b5 gives the variance multiplier  $2/\pi$  in (21.Q4.1). With the full-plane GFF normalization used throughout this book,

$$\text{Var}\langle \Phi, F \rangle = -\frac{1}{2\pi} \iint F(z)F(w) \log|z-w| dz dw,$$

so the characteristic function of  $\sqrt{2/\pi} \Phi$  is exactly the right side of (21.Q4.12). Taking the limit in (21.Q4.13) proves the formula. No anisotropic embedding, polynomial loop-count representation, or infinite anisotropic BKW product is used.  $\square$

**Definition 10.410** (Microscopic normalization). For  $0 < \delta < \varepsilon$ , set

$$F_{0,\varepsilon} := \frac{1}{2\varepsilon^2} \mathbf{1}_{B_\varepsilon(0)}, \quad c_\delta(\varepsilon) := \phi_\delta[\mathcal{A}_{F_{0,\varepsilon}}(\mathcal{L}) \mid \mathcal{I}_\varepsilon]. \quad (21.Q4.14)$$

On  $\mathcal{I}_\varepsilon$  there is no loop surrounding the whole marked ball, so the non-zero-mean product is finite.

**Lemma 10.411** (Uniform nondegeneracy of the normalization). *There exists  $c_* > 0$  such that, whenever  $0 < \delta < c_*\varepsilon$ ,*

$$c_* \leq c_\delta(\varepsilon) \leq 1. \quad (21.Q4.15)$$

*Proof.* Every cosine factor in (21.Q4.14) lies in  $[0, 1]$ , which gives the upper bound. For the lower bound let  $E$  be the existence of an open circuit in  $B_{2\varepsilon}$  surrounding  $B_\varepsilon$ . Self-duality and RSW, also under the coarse-IIC conditioning, give

$$\phi_\delta(E \mid \mathcal{I}_\varepsilon) \geq c_0 > 0. \quad (21.Q4.16)$$

By Lemma 10.406, an event determined by loops inside  $B_{2\varepsilon}$  has conditional probability at most a constant times its ordinary probability.

We now show that the product of the non-surrounding loops is bounded away from zero with uniformly positive probability. First choose  $u > 0$  so that, by RSW, with probability at least  $1 - c_0/(8C)$  no loop intersecting  $B_\varepsilon$  encloses more than  $(1-u)\pi\varepsilon^2$  of the ball. Next, Lemma 10.407 lets us choose  $K$  so that with the same error probability at most  $K$  loops of diameter at least  $\varepsilon$  meet the ball. On this event their product is at least

$$\cos\left(\frac{\pi}{2}(1-u)\right)^K > 0. \quad (21.Q4.17)$$

For loops with diameter in  $(2^{-j-1}\varepsilon, 2^{-j}\varepsilon]$ , the portion of  $B_\varepsilon$  enclosed has area  $O(4^{-j}\varepsilon^2)$ , hence one factor is at least  $\exp(-C4^{-2j})$ . The loop-tail lemma, used with threshold proportional to  $4^j$ , shows that except on an event summable in  $j$  the product over this  $j$ th family is at least  $\exp(-C\alpha 2^{-j})$ . Choosing  $\alpha$  so that the sum of exceptional probabilities is at most  $c_0/(4C)$  gives a deterministic positive lower bound for the product over all sub- $\varepsilon$  loops.

Combining this with (21.Q4.16), local absolute continuity, and (21.Q4.17), we find constants  $\eta, c' > 0$  with

$$\phi_\delta(E, \mathcal{A}_{F_{0,\varepsilon}} > \eta \mid \mathcal{I}_\varepsilon) \geq c'.$$

Therefore  $c_\delta(\varepsilon) \geq \eta c'$ , uniformly in  $\delta, \varepsilon$ . After decreasing the common constant, this is (21.Q4.15).  $\square$

**Proposition 10.412** (Small-loop factorization). *Let*

$$F = \sum_{j=1}^k \frac{q_j}{2\varepsilon^2} \mathbf{1}_{B_\varepsilon(x_j)}, \quad q_j \in \{-1, 1\}, \quad (21.Q4.18)$$

where the marked points remain in a fixed bounded region and have mutual distances at least  $d > 2\varepsilon$ . There are  $c > 0$  and  $C = C(k, d)$  such that

$$\left| c_\delta(\varepsilon)^k \phi_\delta[\mathcal{A}_F(\mathcal{L}_{X,\varepsilon})] - \phi_\delta[\mathcal{A}_F(\mathcal{L})] \right| \leq C \pi_1^\delta(\varepsilon, d)^k (\varepsilon/d)^c. \quad (21.Q4.19)$$

*Proof.* Choose intermediate radii  $\varepsilon < r < R < d$  by

$$\frac{\varepsilon}{r} = \frac{r}{R} = \frac{R}{d} = (\varepsilon/d)^{1/3}. \quad (21.Q4.20)$$

Let  $E_{\text{out}}$  be the event that no loop crosses from radius  $R$  to  $d/2$  about any  $x_j$ , and let  $E_j$  be the event that no loop meeting  $B_\varepsilon(x_j)$  crosses radius  $r$  about  $x_j$ . If one of these events fails, arm separation and (21.Q4.3)–(21.Q4.6) give

$$\phi_\delta(E_{\text{out}}^c \cup E_1^c \cup \dots \cup E_k^c, \mathcal{L}_{X,\varepsilon}^{\text{odd}} = \emptyset) \leq C \pi_1^\delta(\varepsilon, d)^k (\varepsilon/d)^c. \quad (21.Q4.21)$$

Here the factor  $\pi_1^k$  is forced by the absence of an odd loop around each marked ball, while the extra annulus crossing gives the power gain. Since  $|\mathcal{A}_F(\mathcal{L}_{X,\varepsilon})|$  is bounded by the indicator of  $\mathcal{L}_{X,\varepsilon}^{\text{odd}} = \emptyset$ , (21.Q4.21) controls the error from deleting all the bad events on both sides of (21.Q4.19).

On the good event, every loop belongs either to one local family  $\mathcal{L}_j$  contained in  $B_{d/2}(x_j)$  or to an outer family  $\mathcal{L}_{\text{out}}$ ; no loop can couple two local families. Condition on  $\mathcal{L}_{\text{out}}$ . The domain Markov property makes the local laws independent conditional on their induced boundary conditions. For the  $j$ th local family, the charge  $q_j$  changes only the sign of the height increment, so the cosine product is the translate of the one-ball product in Definition 10.410. The conditioned mixing lemma, with the scales in (21.Q4.20), gives

$$\left| \mathbb{E}[\mathcal{A}_{q_j F_{0,\varepsilon}(\cdot - x_j)}(\mathcal{L}_j) \mid \mathcal{L}_{\text{out}}, \mathcal{L}_{X,\varepsilon}^{\text{odd}} = \emptyset] - c_\delta(\varepsilon) \right| \leq C(\varepsilon/d)^c. \quad (21.Q4.22)$$

Multiplying the  $k$  conditional expectations, using  $\phi(\mathcal{L}_{X,\varepsilon}^{\text{odd}} = \emptyset) \leq C^k \pi_1^\delta(\varepsilon, d)^k$ , and then averaging over  $\mathcal{L}_{\text{out}}$  gives (21.Q4.19), after the bad-event error (21.Q4.21) is restored.  $\square$

**Proposition 10.413** (Two-charge observable and the one-arm exponent). *At criticality,*

$$\pi_1(R) = R^{-1/8+o(1)}. \quad (21.Q4.23)$$

*Consequently the critical connectivity satisfies*

$$\phi_{p_c}(0 \leftrightarrow x) = |x|^{-1/4+o(1)}. \quad (21.Q4.24)$$

*Proof.* Fix  $x \neq 0$  and  $0 < \varepsilon < |x|/4$ , and take

$$F_{x,\varepsilon} = \frac{1}{2\varepsilon^2} \mathbf{1}_{B_\varepsilon(0)} - \frac{1}{2\varepsilon^2} \mathbf{1}_{B_\varepsilon(x)}. \quad (21.Q4.25)$$

The function has zero mean. A loop avoiding the two balls contributes zero to the trimmed cosine product precisely when it surrounds exactly one of the balls, and contributes one otherwise. Hence

$$\mathcal{A}_{F_{x,\varepsilon}}(\mathcal{L}_{\{0,x\},\varepsilon}) = \mathbf{1}_{\{\mathcal{L}_{\{0,x\},\varepsilon}^{\text{odd}} = \emptyset\}}. \quad (21.Q4.26)$$

Denote its probability by  $\tilde{\pi}_1^\delta(0, x, \varepsilon)$ .

The M-formula can be computed explicitly. Rescaling the two self-interaction integrals by  $z = \varepsilon u$ ,  $w = \varepsilon v$  extracts a term proportional to  $\log \varepsilon$ ; the two cross terms converge, after the same change of variables, to the corresponding multiple of  $\log |x|$ . The coefficient is fixed by the charges  $\pm 1/(2\varepsilon^2)$  and by the factor  $1/(2\pi^2)$  in (21.Q4.12). Therefore

$$\lim_{\delta \downarrow 0} \phi_\delta[\mathcal{A}_{F_{x,\varepsilon}}(\mathcal{L})] = a_\varepsilon(x) \left( \frac{\varepsilon}{|x|} \right)^{1/4}, \quad (21.Q4.27)$$

where  $a_\varepsilon(x) > 0$  and, for fixed  $x \neq 0$ ,  $a_\varepsilon(x) \rightarrow a_0 > 0$  as  $\varepsilon \downarrow 0$ . The factorization Proposition 10.412, (21.Q4.26), and the normalization bound (21.Q4.15) give, for fixed  $x$ ,

$$c_\delta(\varepsilon)^2 \tilde{\pi}_1^\delta(0, x, \varepsilon) = a_\varepsilon(x) \left( \frac{\varepsilon}{|x|} \right)^{1/4} + O\left(\pi_1^\delta(\varepsilon, |x|)^2 (\varepsilon/|x|)^c\right) + o_\delta(1). \quad (21.Q4.28)$$

No equicontinuity in the macroscopic point  $x$  is needed for the exponent: we henceforth fix  $|x| = 1$ .

We next identify the topological event in (21.Q4.26). If there is no loop surrounding exactly one marked ball, then the two balls lie either in the same primal cluster or in the same dual cluster. Conversely either connection prevents an odd separating loop. Thus, up to the event that both a primal and a dual connection occur,  $\tilde{\pi}_1$  is the sum of these two connection probabilities. Self-duality makes the two summands equal, while (21.Q4.6) and an annular RSW fence give

$$\phi_\delta(B_\varepsilon(0) \leftrightarrow B_\varepsilon(x), B_\varepsilon(0)^* \leftrightarrow B_\varepsilon(x)) \leq C\pi_1^\delta(\varepsilon, |x|)^2 (\varepsilon/|x|)^c. \quad (21.Q4.29)$$

Hence

$$\tilde{\pi}_1^\delta(0, x, \varepsilon) = 2\phi_\delta(B_\varepsilon(0) \leftrightarrow B_\varepsilon(x)) + O\left(\pi_1^\delta(\varepsilon, |x|)^2 (\varepsilon/|x|)^c\right) + o_\delta(1). \quad (21.Q4.30)$$

Finally, arm separation and two fixed RSW corridors show

$$\phi_\delta(B_\varepsilon(0) \leftrightarrow B_\varepsilon(x)) \asymp \pi_1^\delta(\varepsilon, |x|)^2. \quad (21.Q4.31)$$

The upper bound follows by the two endpoint one-arm events and ratio mixing; the lower bound forces separated endpoint arms and joins them through a bounded chain of rectangles.

Take  $|x| = 1$ , then first  $\delta \downarrow 0$  in (21.Q4.28)–(21.Q4.31), and finally  $\varepsilon \downarrow 0$ . Because  $c_\delta(\varepsilon)$  stays between fixed positive constants, the error has an extra power of  $\varepsilon$ , so

$$c\varepsilon^{1/8} \leq \liminf_{N \rightarrow \infty} \pi_1(\varepsilon N, N) \leq \limsup_{N \rightarrow \infty} \pi_1(\varepsilon N, N) \leq C\varepsilon^{1/8}. \quad (21.Q4.32)$$

Lemma 10.408 and (21.Q4.4) yield (21.Q4.23). The exact earlier critical identity in Volume IX,  $\phi_{p_c}(0 \leftrightarrow x) \asymp \pi_1(|x|)^2$ , then gives (21.Q4.24).  $\square$

**Four marked points.** For the remaining two critical exponents put four balls at  $0, y, x, x+y$  with  $|x| \geq 16|y| \geq 64\varepsilon$ . Among the loops avoiding the four balls, let  $\mathcal{L}^2$  be those surrounding exactly two marked balls, and  $\mathcal{L}_1^2 \subset \mathcal{L}^2$  those which surround exactly one of the two left-hand balls  $B_\varepsilon(0), B_\varepsilon(y)$ . Define

$$\tilde{\pi}_{2k}^\delta(x, y, \varepsilon) := \phi_\delta(|\mathcal{L}_1^2| \geq k, \mathcal{L}^{\text{odd}} = \emptyset), \quad (21.Q4.33)$$

and the signed observable

$$\tilde{\Delta}^\delta(x, y, \varepsilon) := \phi_\delta[(-1)^{|\mathcal{L}^2|} \mathbf{1}_{\{\mathcal{L}_1^2 = \mathcal{L}^{\text{odd}} = \emptyset\}}]. \quad (21.Q4.34)$$

**Lemma 10.414** (Four-ball topology versus annular arms). *For every fixed  $k \geq 1$ ,*

$$\tilde{\pi}_{2k}^\delta(x, y, \varepsilon) \asymp \pi_{2k}^\delta(|y|, |x|)^2 \pi_1^\delta(\varepsilon, |y|)^4, \quad (21.Q4.35)$$

*with constants depending on  $k$  but not on the four scales.*

*Proof.* Any configuration counted on the left must produce one arm from each microscopic ball to radius  $|y|/4$ . In addition, near each of the two macroscopic centres 0 and  $x$ , the  $k$  nested separating loop strands force  $2k$  alternating primal/dual arms across an annulus from scale comparable with  $|y|$  to scale comparable with  $|x|$ . The four microscopic regions and the two macroscopic annuli are separated by fixed-factor buffer annuli. Ratio mixing and quasi-multiplicativity therefore give the upper bound in (21.Q4.35).

For the lower bound, use separated landing versions of the same six arm families. Choose fixed disjoint landing windows, add RSW fences around every landing window, and then join the arms through six disjoint corridors so that the prescribed nested loop topology is forced. The domain Markov property permits the corridors to be imposed successively under the induced boundary conditions; arbitrary-boundary RSW gives a uniform positive cost. The arm-separation comparison and (21.Q4.4) absorb all bounded changes of radius. The resulting event is contained in the event on the left of (21.Q4.35), proving the reverse inequality.  $\square$

**Proposition 10.415** (Four-charge M-formula and separation of the two-arm term). *There are constants  $c, C > 0$  such that, for  $|x| \geq 16|y| \geq 64\varepsilon$ ,*

$$\left| c_\delta(\varepsilon)^4 (\tilde{\pi}_2^\delta(x, y, \varepsilon) + \tilde{\Delta}^\delta(x, y, \varepsilon)) - a_\varepsilon(x, y) \varepsilon^{1/2} \left( \frac{|y|^2}{|x|^2|x-y||x+y|} \right)^{1/4} \right| \quad (10.52)$$

$$\leq C \pi_1^\delta(\varepsilon, d)^4 (\varepsilon/d)^c + o_\delta(1), \quad (21.Q4.36)$$

where  $d = \min\{|x|, |y|, |x-y|, |x+y|\}$  and  $a_\varepsilon(x, y) \rightarrow a_0 > 0$  as  $\varepsilon \downarrow 0$  for fixed non-colliding  $x, y$ . Moreover,

$$|\tilde{\pi}_2^\delta - \tilde{\Delta}^\delta| \leq C \tilde{\pi}_2^\delta(|y|/|x|)^c + C \pi_1^\delta(\varepsilon, |y|)^4 (\varepsilon/|y|)^c + o_\delta(1). \quad (21.Q4.37)$$

*Proof.* Take the zero-mean charge pattern

$$F = \frac{1}{2\varepsilon^2} (\mathbf{1}_{B_\varepsilon(0)} + \mathbf{1}_{B_\varepsilon(y)} - \mathbf{1}_{B_\varepsilon(x)} - \mathbf{1}_{B_\varepsilon(x+y)}). \quad (21.Q4.38)$$

A loop in  $\mathcal{L}^{\text{odd}}$  contributes zero. If no odd loop is present, a loop in  $\mathcal{L}_1^2$  contributes +1, while a two-ball loop outside that class contributes -1. Since the two alternatives cannot occur simultaneously at the outermost relevant level, a deterministic planar check gives

$$\phi_\delta[\mathcal{A}_F(\mathcal{L}_{X,\varepsilon})] = \tilde{\pi}_2^\delta + \tilde{\Delta}^\delta. \quad (21.Q4.39)$$

The Gaussian integral in Theorem 10.409 is the quadratic form of four charges  $+, +, -, -$ . Separating self-interactions and pairwise interactions gives

$$a_\varepsilon(x, y) \varepsilon^{1/2} \left( \frac{|y|^2}{|x|^2|x-y||x+y|} \right)^{1/4}, \quad (21.Q4.40)$$

with  $a_\varepsilon(x, y) \rightarrow a_0 > 0$ . Proposition 10.412 then yields (21.Q4.36).

For (21.Q4.37), replace the two negative charges in (21.Q4.38) by positive charges and call the resulting non-zero-mean function  $\tilde{F}$ . The trimmed loop algebra gives

$$\phi_\delta[\mathcal{A}_{\tilde{F}}(\mathcal{L}_{X,\varepsilon})] = \tilde{\Delta}^\delta - \tilde{\pi}_2^\delta + 2\phi_\delta(|\mathcal{L}_1^2| \in 2\mathbb{Z}_{>0}, \mathcal{L}^{\text{odd}} = \emptyset). \quad (21.Q4.41)$$

The last term is at most  $2\tilde{\pi}_4^\delta$ . By Lemma 10.414, arm separation, and the strict power gain obtained by forcing one additional macroscopic separating strand,

$$\tilde{\pi}_4^\delta \leq C\tilde{\pi}_2^\delta(|y|/|x|)^c. \quad (21.Q4.42)$$

It remains to show that the first term on the right of (21.Q4.41) is small. Although  $\tilde{F}$  has non-zero mean, form the balanced two-copy test function

$$\tilde{F}_R(z) := \tilde{F}(z - (R, 0)) - \tilde{F}(z + (R, 0)).$$

It has zero mean. Ratio mixing, together with Lemma 10.407 to exclude the rare loops meeting both remote copies, gives

$$\phi_\delta[\mathcal{A}_{\tilde{F}}(\mathcal{L})]^2 = \lim_{R \rightarrow \infty} \phi_\delta[\mathcal{A}_{\tilde{F}_R}(\mathcal{L})]. \quad (21.Q4.43)$$

For each fixed  $R$  the M-formula applies. Its exponent contains the negative cross-energy between the two copies, equal to  $-c_0 \log R + O(1)$  because each copy has non-zero total charge. Hence after  $\delta \downarrow 0$  and then  $R \rightarrow \infty$ , the right side of (21.Q4.43) tends to zero. Applying the factorization proposition to the four local balls, and accounting for the bad annular-separation event exactly as in (21.Q4.21), yields

$$|\phi_\delta[\mathcal{A}_{\tilde{F}}(\mathcal{L}_{X,\varepsilon})]| \leq C\pi_1^\delta(\varepsilon, |y|)^4(\varepsilon/|y|)^c + o_\delta(1). \quad (21.Q4.44)$$

Equations (21.Q4.41)–(21.Q4.44) prove (21.Q4.37).  $\square$

**Proposition 10.416** (Two-arm exponent). *At criticality,*

$$\pi_2(R) = R^{-1/2+o(1)}. \quad (21.Q4.45)$$

*Proof.* Fix  $|x| = 1$  and let  $r = |y| \downarrow 0$ . By (21.Q4.36)–(21.Q4.37), the lower bound on  $c_\delta(\varepsilon)$ , and the already proved one-arm estimate, after first  $\delta \downarrow 0$  and then  $\varepsilon \downarrow 0$  we obtain

$$\tilde{\pi}_2^\delta(x, y, \varepsilon) = r^{1/2+o_r(1)}\pi_1^\delta(\varepsilon, r)^4 \quad (21.Q4.46)$$

in the two-sided multiplicative sense needed for exponent extraction. On the other hand, Lemma 10.414 with  $k = 1$  gives

$$\tilde{\pi}_2^\delta(x, y, \varepsilon) \asymp \pi_2^\delta(r, 1)^2\pi_1^\delta(\varepsilon, r)^4. \quad (21.Q4.47)$$

Cancel the positive one-arm factor and take the limits in the prescribed order. This yields

$$cr^{1/2} \leq \liminf_{N \rightarrow \infty} \pi_2(rN, N) \leq \limsup_{N \rightarrow \infty} \pi_2(rN, N) \leq Cr^{1/2} \quad (21.Q4.48)$$

for all sufficiently small  $r$ . Quasi-multiplicativity and Lemma 10.408 prove (21.Q4.45).  $\square$

**Definition 10.417** (Flower domains and boosts used in the influence proof). Fix a square  $\Lambda_R(z)$ . An *inner flower domain at scale  $R$*  is the unexplored component containing  $\Lambda_R(z)$  after the primal/dual interfaces which enter the annulus between radii  $R$  and  $3R/2$  from its outer boundary have been revealed. An *outer flower domain at scale  $2R$*  is defined dually from the unbounded unexplored

component after revealing the interfaces which enter the annulus between radii  $3R/2$  and  $2R$  from its inner boundary. When the boundary is not a single primal or dual circuit, its alternating primal and dual boundary arcs are called *petals*. A flower is *well separated* if the endpoints of distinct petals are separated by a fixed positive multiple of  $R$ .

A boundary condition is *compatible* if each primal petal is internally wired and no dual petal is identified with another boundary portion (equivalently, in the dual boundary condition each dual petal is internally wired). For a flower with at least four petals, a pair  $\xi \leq \xi'$  is a *boosting pair* if both conditions are compatible and two primal petals which are distinct under  $\xi$  are wired together under  $\xi'$ . We also allow a pair  $\zeta \leq \zeta'$  to be called boosting when it sandwiches such a compatible pair. These are the lattice-rescaled versions of the flower/boost objects used in the Volume IX decision-tree coupling.

**Lemma 10.418** (Conditioning the left pair creates a macroscopic boost). *Let  $L(1)$  be as in Lemma 10.421. There is an increasing decision-tree coupling  $\mathbb{P}$  of  $\omega \sim \phi_\delta$  and  $\omega' \sim \phi_\delta(\cdot \mid L(1))$ , together with a stopping time  $\tau$ , such that*

$$\mathbb{P}(\tau < \infty) \geq c, \quad (21.Q4.48a)$$

*and, on  $\{\tau < \infty\}$ , the unrevealed region  $F_\tau$  is a well-separated four-petal outer flower at scale  $4|y|$  with compatible conditions  $\xi < \xi'$  for which*

$$\mathcal{L}(\omega|_{F_\tau} \mid \omega_{[\tau]}) = \phi_{F_\tau}^\xi, \quad \mathcal{L}(\omega'|_{F_\tau} \mid \omega'_{[\tau]}) \succeq_{\text{st}} \phi_{F_\tau}^{\xi'}. \quad (21.Q4.48b)$$

Moreover

$$\phi_\delta(L(1)) \asymp \pi_1^\delta(\varepsilon, |y|)^2. \quad (21.Q4.48c)$$

*Proof.* Reveal, using only the lower configuration  $\omega$ , the inner flower at scale  $2|y|$ , the outer flower at scale  $4|y|$ , and all edges in the annular region between them. Arbitrary-boundary RSW and arm separation give a fixed positive probability that these two revealed boundaries form a double four-petal flower. Conditional on this, reveal the interior of the inner flower and require the following three events: the two selected primal petals are not joined in  $\omega$ ; an open circuit around  $B_\varepsilon(0)$  reaches the first selected primal petal; and an open circuit around  $B_\varepsilon(y)$  reaches the opposite selected primal petal. Each condition is forced inside fixed-aspect-ratio corridors, so their conditional probability is bounded below uniformly. We stop at the end of this revealment; otherwise we set  $\tau = \infty$ . This proves (21.Q4.48a).

On the stopping event the spatial Markov property gives the lower conditional law  $\phi_{F_\tau}^\xi$ . In the upper configuration the two microscopic balls are conditioned to be connected. If that connection has already been exposed, the induced upper boundary condition directly wires the two selected primal petals. If it has not, the upper conditional law is the lower compatible boundary condition conditioned on a connection between two disjoint boundary sets; by FKG it stochastically dominates the law in which those sets have simply been wired. In either case there is a compatible  $\xi' > \xi$  dominated by the upper conditional law. This proves (21.Q4.48b).

For (21.Q4.48c),  $L(1)$  forces one arm from each microscopic ball to scale comparable with  $|y|$ , so arm separation and ratio mixing give the upper bound by  $C\pi_1^\delta(\varepsilon, |y|)^2$ . Conversely, on the two separated one-arm events, RSW supplies fixed corridors which join their macroscopic endpoints without changing the microscopic arms. This gives the matching lower bound.  $\square$

**Lemma 10.419** (Transport of a left boost to the right pair). *Let  $G$  be any well-separated four-petal outer flower at scale  $4|y|$  around the left pair and let  $\xi < \xi'$  be a compatible boosting pair. If  $|x| \geq 16|y|$ , there is an increasing decision-tree coupling of  $\phi_G^\xi$  and  $\phi_G^{\xi'}$  and a stopping time  $\sigma$  such*

that, on  $\{\sigma < \infty\}$ , the unrevealed region is a well-separated inner flower at scale  $2|y|$  around  $x$  and the induced boundary conditions there form a boosting pair. Moreover

$$\mathbb{P}(\sigma < \infty) \geq c \Delta^\delta(|y|, |x|)^2. \quad (21.Q4.48d)$$

*Proof.* This is exactly the comparable-scale case of the boost-transport owner the Volume IX owner **lem:fk-two-ended-mixing-pivotal**, item (2), after rescaling the lattice by  $\delta$ . For clarity, its two factors have distinct origins. The first exploration carries the boundary discrepancy from the left outer flower to a separator at distance comparable with  $|x|$ , with survival probability comparable with  $\Delta^\delta(|y|, |x|)$ . RSW petal fences turn any surviving discrepancy into a compatible boost on the separator. A second, disjoint exploration from the target flower back to the separator carries that boost to the right inner flower with another factor  $\Delta^\delta(|y|, |x|)$ . Conditional on the separator, the two unexplored regions are governed by the domain Markov property, so the two lower survival bounds multiply. This gives (21.Q4.48d).  $\square$

**Lemma 10.420** (A right-hand boost changes the two-point connection). *Let  $F$  be a well-separated inner flower at scale  $2|y|$  around  $x$  and let  $\xi < \xi'$  be a boosting pair. Interpret  $R(1)$  inside  $F$  by allowing connections through the boundary wiring. Then*

$$\phi_F^{\xi'}(R(1)) - \phi_F^\xi(R(1)) \geq c \pi_1^\delta(\varepsilon, |y|)^2. \quad (21.Q4.48e)$$

*The constant is uniform over the geometry of every well-separated flower obtained by the preceding stopping-time constructions.*

*Proof.* Choose two primal petals which are wired together in  $\xi'$  but are not wired together in  $\xi$ . Inside pairwise disjoint tubes, route an open arm from  $B_\varepsilon(x)$  to the first petal and an open arm from  $B_\varepsilon(x + y)$  to the second. At the same time impose dual RSW corridors between the tubes and along every other route by which the two microscopic balls could be joined under the weaker boundary condition. Well separation guarantees fixed-aspect-ratio tubes, so arbitrary-boundary RSW gives a fixed conditional cost for the macroscopic routing and shielding. The two microscopic arms contribute  $\pi_1^\delta(\varepsilon, |y|)^2$  up to a constant by arm separation and local ratio mixing. On the resulting event the two balls are joined under  $\xi'$  through the additional wiring but are not joined under  $\xi$ . Applying any increasing coupling of the two FK laws and taking the probability of this disagreement proves (21.Q4.48e).  $\square$

**Lemma 10.421** (Two-pair flower comparison). *Let  $L(s)$  be the event that the pair  $B_\varepsilon(0), B_\varepsilon(y)$  is connected in the primal configuration when  $s = 1$  and in the dual configuration when  $s = 0$ . Define  $R(s)$  analogously for the pair  $B_\varepsilon(x), B_\varepsilon(x + y)$ . If  $|x| \geq 16|y| \geq 64\varepsilon$ , then*

$$\phi_\delta(L(1) \cap R(1)) - \phi_\delta(L(1) \cap R(0)) \asymp \pi_1^\delta(\varepsilon, |y|)^4 \Delta^\delta(|y|, |x|)^2 + o_{\varepsilon, \delta}(1), \quad (21.Q4.49)$$

where  $\lim_{\varepsilon \downarrow 0} \limsup_{\delta \downarrow 0} |o_{\varepsilon, \delta}(1)| = 0$ .

*Proof.* Put  $r = |y|$ ,  $d = |x|$ , and

$$K := \lfloor \log_2(d/r) \rfloor - 2.$$

The proof has a localization upper bound and a decision-tree lower bound.

*Upper bound: localization by decision scale.* Write  $L = L(1)$  and  $R = R(1)$ . Decompose  $L$  into the disjoint events  $L_0, \dots, L_{2K+1}$  according to the least scale at which the connection can be

measured:

$$L_k := \begin{cases} \{B_\varepsilon(0) \leftrightarrow B_\varepsilon(y) \text{ in } \Lambda_{2r}\}, & k = 0, \\ \{B_\varepsilon(0) \leftrightarrow B_\varepsilon(y) \text{ in } \Lambda_{2^{k+1}r} \text{ but not in } \Lambda_{2^k r}\}, & 1 \leq k \leq K, \\ \{B_\varepsilon(0) \leftrightarrow B_\varepsilon(y) \text{ in } \Lambda_{2^{2K-k}r}(x)^c \text{ but not in } \Lambda_{2^{2K-k+1}r}(x)^c\}, & K < k \leq 2K, \\ \{B_\varepsilon(0) \leftrightarrow B_\varepsilon(y) \text{ but not in } \Lambda_{2r}(x)^c\}, & k = 2K + 1. \end{cases} \quad (21.Q4.49a)$$

Define  $R_\ell$  by interchanging the roles of 0 and  $x$ . Thus

$$\text{Cov}_\phi(\mathbf{1}_L, \mathbf{1}_R) = \sum_{k, \ell=0}^{2K+1} \{\phi_\delta(L_k \cap R_\ell) - \phi_\delta(L_k)\phi_\delta(R_\ell)\}. \quad (21.Q4.49b)$$

Every  $L_k$  contains two microscopic one-arms. If  $0 \leq k \leq K$ , it also contains four separated alternating arms from scale  $2r$  to  $2^k r$ . If  $K < k \leq 2K$ , it contains a four-arm event from  $2r$  to  $2^K r$  and another, centred at  $x$ , from  $2^{2K-k+1}r$  to  $2^K r$ . Arm separation and quasi-multiplicativity therefore give

$$\phi_\delta(L_k) = \phi_\delta(R_k) \leq C\pi_1^\delta(\varepsilon, r)^2 \begin{cases} \pi_4^\delta(r, 2^k r), & 0 \leq k \leq K, \\ \pi_4^\delta(r, d)\pi_4^\delta(2^{2K-k}r, d), & K < k \leq 2K, \end{cases} \quad (21.Q4.49c)$$

where bounded-factor changes of radii have been absorbed into  $C$ . When  $k + \ell \geq 2K$ , direct arm separation similarly yields

$$\phi_\delta(L_k \cap R_\ell) \leq C\pi_1^\delta(\varepsilon, r)^4 \pi_4^\delta(r, d)^2. \quad (21.Q4.49d)$$

For  $k + \ell < 2K$ , the regions determining  $L_k$  and  $R_\ell$  are separated. Apply the two-ended ratio estimate (21.Q4.6a), then replace each four-arm factor using (21.Q4.6b), and use quasi-multiplicativity (21.Q4.5). The result is

$$|\phi_\delta(L_k \cap R_\ell) - \phi_\delta(L_k)\phi_\delta(R_\ell)| \leq C\pi_1^\delta(\varepsilon, r)^4 \Delta^\delta(r, d)^2 2^{-c(k+\ell)}. \quad (21.Q4.49e)$$

For  $k + \ell \geq 2K$ , combine (21.Q4.49d) with (21.Q4.6b) to obtain the same bound with  $2^{-2cK}$  in place of  $2^{-c(k+\ell)}$ . Summing (21.Q4.49b) gives

$$|\phi_\delta(L \cap R) - \phi_\delta(L)\phi_\delta(R)| \leq C\pi_1^\delta(\varepsilon, r)^4 \Delta^\delta(r, d)^2. \quad (21.Q4.49f)$$

The identical decomposition applies to  $R(0)$ . Critical primal/dual symmetry gives  $\phi_\delta(R(1)) = \phi_\delta(R(0)) + o_\delta(1)$ , so subtracting the two covariance bounds proves the required upper estimate in (21.Q4.49).

*Lower bound: create, transport, and test a boost.* FKG and the same primal/dual symmetry give

$$\begin{aligned} & \phi_\delta(L(1) \cap R(1)) - \phi_\delta(L(1) \cap R(0)) \\ & \geq \phi_\delta(L(1))\{\phi_\delta(R(1) | L(1)) - \phi_\delta(R(1)) + o_\delta(1)\}. \end{aligned} \quad (21.Q4.49g)$$

Lemma 10.418 supplies, with probability at least  $c$ , a left outer flower on which the lower conditional law is  $\phi^\xi$  and the upper law stochastically dominates  $\phi^{\xi'}$  for a boost  $\xi < \xi'$ . Conditional on this flower, Lemmas 10.419 and 10.420 imply

$$\phi_\delta(R(1) | L(1)) - \phi_\delta(R(1)) \geq c\pi_1^\delta(\varepsilon, r)^2 \Delta^\delta(r, d)^2. \quad (21.Q4.49h)$$

Finally (21.Q4.48c) supplies  $\phi_\delta(L(1)) \geq c\pi_1^\delta(\varepsilon, r)^2$ . Inserting (21.Q4.49h) into (21.Q4.49g) gives the lower bound

$$c\pi_1^\delta(\varepsilon, r)^4 \Delta^\delta(r, d)^2 + o_\delta(1).$$



The configurations excluded when the flower exploration meets two microscopic balls before separating them require an additional separated arm. The arm-separation bound (21.Q4.6), followed by the ordered limits  $\delta \downarrow 0$  and then  $\varepsilon \downarrow 0$ , absorbs their total contribution into  $o_{\varepsilon,\delta}(1)$ . This finishes (21.Q4.49).

The proof architecture above is the internal selected-form reconstruction of the upper and lower arguments in Chen–Duminil-Copin–He–Jacopin–Krachun–Manolescu–Xia, *Critical exponents for planar random-cluster model with cluster-weight  $q = 4$* , Lemmas 6.2–6.12. The two-ended mixing and pivotality inputs are owned earlier by the Volume IX owner `lem:fk-two-ended-mixing-pivotal`, corresponding to the selected consequences of Duminil-Copin–Manolescu, *Planar random-cluster model: scaling relations*, Theorem 1.6(iv)–(v) and Sections 3–5.  $\square$

**Proposition 10.422** (Influence observable and influence exponent). *For  $|x| \geq 16|y| \geq 64\varepsilon$ ,*

$$\frac{\tilde{\Delta}^\delta(x, y, \varepsilon)}{\pi_1^\delta(\varepsilon, |y|)^4} \asymp \Delta^\delta(|y|, |x|)^2 + o_{\varepsilon,\delta}(1). \quad (21.Q4.50)$$

Consequently

$$\Delta(R) = R^{-1/2+o(1)}. \quad (21.Q4.51)$$

*Proof.* Let Bad be the event that one medial loop meets at least two of the four microscopic balls. Outside Bad, the four events  $L(s) \cap R(t)$ ,  $s, t \in \{0, 1\}$ , are disjoint and exhaust the event on which the signed observable  $\tilde{\Delta}$  is nonzero. A planar parity check shows that  $(-1)^{|\mathcal{C}^2|} = +1$  when  $s = t$  and  $-1$  when  $s \neq t$ . Therefore

$$\tilde{\Delta} = \sum_{s=0}^1 [\phi(L(s) \cap R(s)) - \phi(L(s) \cap R(1-s))] + O(\phi(\text{Bad})). \quad (21.Q4.52)$$

Self-duality makes the  $s = 0$  and  $s = 1$  brackets equal up to a mesh error which vanishes in the ordered limit. Four endpoint one-arms are necessary for Bad, together with an additional arm connecting two of the microscopic neighborhoods, so

$$\phi(\text{Bad}) \leq C\pi_1^\delta(\varepsilon, |y|)^4(\varepsilon/|y|)^c + o_\delta(1). \quad (21.Q4.53)$$

Insert Lemma 10.421 into (21.Q4.52) and divide by the one-arm factor. This proves (21.Q4.50).

To identify the exponent, fix  $|x| = 1$  and set  $r = |y|$ . Equation (21.Q4.37), the four-charge main term (21.Q4.36), the two-arm exponent (21.Q4.45), and the one-arm exponent (21.Q4.23) imply, after  $\delta \downarrow 0$  and then  $\varepsilon \downarrow 0$ ,

$$\frac{\tilde{\Delta}^\delta(x, y, \varepsilon)}{\pi_1^\delta(\varepsilon, r)^4} \asymp r. \quad (21.Q4.54)$$

Combining (21.Q4.50) and (21.Q4.54) yields

$$cr^{1/2} \leq \liminf_{N \rightarrow \infty} \Delta(rN, N) \leq \limsup_{N \rightarrow \infty} \Delta(rN, N) \leq Cr^{1/2}. \quad (21.Q4.55)$$

The quasi-multiplicativity (21.Q4.5) and Lemma 10.408 give (21.Q4.51).  $\square$

**Theorem 10.423** ( $q = 4$  critical input package). *The critical  $q = 4$  square-lattice random-cluster model satisfies*

$$\pi_1(R) = R^{-1/8+o(1)}, \quad \pi_2(R) = R^{-1/2+o(1)}, \quad \Delta(R) = R^{-1/2+o(1)}, \quad (21.Q4.56)$$

and

$$\phi_{p_c}(0 \leftrightarrow x) = |x|^{-1/4+o(1)}. \quad (21.Q4.57)$$

*These conclusions are proof-independent of the open anisotropic Gate G7 of SM-21.S1.05.*

*Proof.* Combine Propositions 10.413, 10.416, and 10.422. The only six-vertex input in those proofs is the isotropic M-formula Theorem 10.409, which was deduced from the finite BKW identity and the locally closed isotropic GFF gates.  $\square$

**Theorem 10.424** (Thermodynamic and magnetic assembly). *From (21.Q4.56)–(21.Q4.57) and the exact earlier exponent-free scaling theorem SM-09.S1.04, one obtains*

$$\nu = \frac{2}{3}, \quad \beta = \frac{1}{12}, \quad \gamma = \frac{7}{6}, \quad \alpha = \frac{2}{3}, \quad (21.Q4.58)$$

*the critical cluster-volume tail exponent  $1/15$ , and the critical-field magnetic exponent  $1/15$  for the associated four-state Potts model.*

*Proof.* In the notation of SM-09.S1.04, (21.Q4.56) gives  $\xi_1 = 1/8$  and  $\iota = 1/2$ . The exact characteristic-length relation  $\nu(2 - \iota) = 1$  gives  $\nu = 2/3$ . The earlier exponent-free identities

$$\theta(p) \asymp \pi_1(L(p)), \quad \chi(p) \asymp L(p)^2 \pi_1(L(p))^2$$

therefore give, without an additional scaling-law assumption,

$$\theta(p) = |p - p_c|^{1/12+o(1)}, \quad \chi(p) = |p - p_c|^{-7/6+o(1)},$$

so  $\beta = 1/12$  and  $\gamma = 7/6$ .

For the specific heat we use the stronger exponent-free owner (9.77), rather than assuming in advance that an algebraic divergence exists:

$$f''(p) \asymp \sum_{\ell \lesssim L(p)} \ell \Delta_{p_c}(\ell)^2.$$

Since  $\Delta_{p_c}(\ell) = \ell^{-1/2+o(1)}$ , dyadic decomposition of the sum gives

$$\sum_{\ell \lesssim L} \ell \Delta_{p_c}(\ell)^2 = L^{1+o(1)}.$$

Together with  $L(p) = |p - p_c|^{-2/3+o(1)}$  this proves directly that

$$f''(p) = |p - p_c|^{-2/3+o(1)},$$

and hence  $\alpha = 2/3$ . Thus the specific-heat exponent is a conclusion of the proved influence estimate and the earlier covariance identity, not a hypothesis used to invoke hyperscaling. The exact critical cluster identity in the same earlier owner gives

$$\zeta = \frac{\xi_1}{2 - \xi_1} = \frac{1}{15}.$$

For the external field, the exact ghost-vertex identity proved in Volume IX states that

$$\phi_{p_c, h}^1(0 \leftrightarrow \mathfrak{g}) \asymp \pi_1(\varphi(1/h)), \quad \varphi(n)^2 \pi_1(\varphi(n)) \asymp n.$$

Substituting  $\pi_1(r) = r^{-1/8+o(1)}$  gives  $\varphi(1/h) = h^{-8/15+o(1)}$  and hence  $\phi_{p_c, h}^1(0 \leftrightarrow \mathfrak{g}) = h^{1/15+o(1)}$ . Edwards–Sokal identifies this ghost connectivity with the critical-field Potts magnetization. This proves the final assertion without introducing a new “standard ghost” black box.  $\square$

*Remark 10.425* (Dependency boundary). The proof above uses the 2026  $q = 4$  primary-source architecture through the BKW M-formula, normalization/factorization, the two-charge and four-charge observables, and the two-pair flower comparison. Its reusable planar inputs are inherited from the already proved Volume IX FK package. Crucially, the six-vertex input is only the isotropic  $q = 4$  height-to-GFF limit with variance  $2/\pi$ , which is closed by Gates G1–G6b5. The unpublished anisotropic continuum polynomial/small-loop product estimates G7c–G7e are not used. Thus the status of **SM-21.S1.06** is logically independent of the remaining anisotropic root in **SM-21.S1.05**.

**Theorem 10.426** ( $q = 4$  random-cluster critical exponents — source-precise form). *Consider the planar random-cluster model on  $\mathbb{Z}^2$  with  $q = 4$  and  $p_c = 2/3$ . Write  $\theta(p)$  for the infinite-cluster density, let  $\chi(p)$  be the finite-cluster susceptibility in the source convention, and let  $\xi(p)$  be the corresponding correlation length. Then*

$$\begin{aligned} f''(p) &= |p - p_c|^{-2/3+o(1)} & (p \rightarrow p_c), \\ \theta(p) &= (p - p_c)^{1/12+o(1)} & (p \downarrow p_c), \\ \chi(p) &= |p - p_c|^{-7/6+o(1)}, \quad \xi(p) = |p - p_c|^{-2/3+o(1)} & (p \rightarrow p_c), \end{aligned}$$

and at criticality

$$\begin{aligned} \phi_{p_c}[0 \leftrightarrow x] &= |x|^{-1/4+o(1)} & (|x| \rightarrow \infty), \\ \phi_{p_c}[|C(0)| \geq n] &= n^{-1/15+o(1)} & (n \rightarrow \infty). \end{aligned}$$

The  $o(1)$  terms are interpreted in the corresponding limiting variables of the source theorem.

*Proof.* The genuinely model-specific critical input is now Theorem 10.423, proved above from the isotropic BKW/GFF M-formula, the small-loop normalization and factorization estimates, the two- and four-charge observables, and the exact earlier planar FK flower/mixing owners. It gives

$$\pi_1(R) = R^{-1/8+o(1)}, \quad \pi_2(R) = R^{-1/2+o(1)}, \quad \Delta(R) = R^{-1/2+o(1)},$$

together with

$$\phi_{p_c}(0 \leftrightarrow x) = |x|^{-1/4+o(1)}.$$

Theorem 10.424, using the exact earlier exponent-free scaling theorem **SM-09.S1.04**, then gives

$$\nu = \frac{2}{3}, \quad \beta = \frac{1}{12}, \quad \gamma = \frac{7}{6}, \quad \alpha = \frac{2}{3},$$

and the cluster-tail exponent  $1/15$ . Translating these exponent identities into the source's thermodynamic conventions gives precisely the four off-critical displays in the statement. No anisotropic six-vertex Gate G7 is used anywhere in this proof.  $\square$

**Proof dependencies for the four-state exponent theorem.** The local reconstruction is given in Section 10.8.13. The source-specific BKW/GFF-to-arm mechanism is owned through the M-formula, normalization/factorization, one-arm, two-arm, and influence children, while the general planar FK mixing/scaling calculus is consumed from exact earlier Volume IX owners. The only six-vertex input is the already closed isotropic GFF chain G1–G6b5 at  $\Delta = -1$ ; the imported anisotropic stages G7c–G7e are not dependencies. Thus the result is independent of the anisotropic clause.

**Corollary 10.427** (Four-state Potts exponents — source-side limits kept separate). *For the four-state Potts model on  $\mathbb{Z}^2$ , with inverse critical temperature  $\beta_c$  in the source normalization, the corresponding theorem gives*

$$f''(\beta) = |\beta - \beta_c|^{-2/3+o(1)} \quad (\beta \rightarrow \beta_c),$$

$$m^*(\beta) = (\beta - \beta_c)^{1/12+o(1)} \quad (\beta \downarrow \beta_c),$$

where  $m^*$  is the spontaneous magnetization from the ordered side, while from the disordered side

$$\chi(\beta) = (\beta_c - \beta)^{-7/6+o(1)}, \quad \xi(\beta) = (\beta_c - \beta)^{-2/3+o(1)} \quad (\beta \uparrow \beta_c).$$

At criticality,

$$\mathbb{P}_{\beta_c,0}(\sigma_0 = \sigma_x) - \frac{1}{4} = |x|^{-1/4+o(1)},$$

and for the critical-field magnetization,

$$m(\beta_c, h) = h^{1/15+o(1)} \quad (h \downarrow 0).$$

Thus the exponent list agrees with the random-cluster theorem, but the one-sided thermodynamic limits are stated exactly as in the Potts source rather than being suppressed by the Edwards–Sokal shorthand.

*Proof.* Use the Edwards–Sokal coupling at  $q = 4$ . At zero field, two Potts spins are equal precisely because either their FK vertices belong to the same cluster or, if they belong to distinct clusters, their independent cluster colours happen to agree. Hence

$$\mathbb{P}_{\beta_c,0}(\sigma_0 = \sigma_x) - \frac{1}{4} = \frac{3}{4} \phi_{p_c}(0 \leftrightarrow x),$$

so the critical two-point exponent is  $1/4$ . Wired FK connectivity gives the spontaneous ordered-side magnetization, while the integrated connectivity gives the disordered-side susceptibility and correlation length; the analytic change of variable  $p = 1 - e^{-\beta}$  preserves all displayed powers at  $\beta_c$ . The free-energy relation under the coupling likewise preserves the  $2/3$  second-derivative exponent. Finally, use the exact earlier ghost-vertex owner in Volume IX, recorded in the critical cluster/magnetic identities consumed by Theorem 10.424. It gives the FK ghost connectivity as  $\pi_1(\varphi(1/h))$  with  $\varphi(n)^2 \pi_1(\varphi(n)) \asymp n$ ; the one-arm exponent  $1/8$  therefore yields  $h^{1/15+o(1)}$ . Edwards–Sokal identifies this quantity with the critical-field Potts magnetization. The one-sided limits in the statement are kept separate because the Potts source states them on those respective sides of  $\beta_c$ .  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 10.428** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 10.429** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 10.430** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 10.431** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 10.432** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XXII: Conformal Field Theory Interfaces and Liouville  
Conformal Field Theory

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# Scope and conventions

This volume connects the probabilistic SLE/LQG theory to the algebraic language of conformal field theory without treating CFT heuristics as proofs. The first chapter derives the Virasoro/BPZ/SLE martingale dictionary formally and rigorously at the level needed here. The remaining chapters concern probabilistically constructed Liouville CFT: sphere correlations, Ward and BPZ identities, DOZZ integrability, conformal bootstrap, Segal axioms, higher topology, boundary LCFT, and annuli. Complete lattice-model-to-CFT convergence remains explicitly frontier material.

# Chapter 1

## CFT structures used by probability

The CFT language needed for the probabilistic applications in this series can be developed formally from infinitesimal conformal maps. The purpose here is modest and precise: derive the Witt/Virasoro algebra, the level-two BPZ equation, and the identical second-order operator produced by Itô calculus for chordal SLE.

**Definition 1.1** (Primary field, highest-weight state, and null vector). A holomorphic *primary field of weight  $h$*  is an insertion whose infinitesimal Witt action is  $D_n = z^{n+1}\partial_z + (n+1)hz^n$ . A vector  $|h\rangle$  in a Virasoro module is *highest weight* when  $L_0|h\rangle = h|h\rangle$  and  $L_n|h\rangle = 0$  for every  $n > 0$ . A nonzero descendant annihilated in the irreducible quotient is a *null vector*. In this volume a “BPZ equation” always means the differential equation obtained by inserting a specified null-vector relation into a correlation function and applying the Ward identities.

**Example 1.2** (Null vector to ODE). For a four-point function with one level-two degenerate insertion, Möbius covariance reduces the BPZ partial differential equation to a second-order ordinary differential equation in the cross-ratio. This is why hypergeometric functions appear in both CFT and SLE crossing formulas.

*Worked example 1.3* (A one-marked-point SLE martingale). Take one boundary marked point  $x_1 > x$  and try a power-law partition function  $Z(x; x_1) = (x_1 - x)^p$ . The SLE martingale operator below gives

$$\left[ \frac{\kappa}{2} \partial_x^2 + \frac{2}{x_1 - x} \partial_{x_1} - \frac{2h}{(x_1 - x)^2} \right] Z.$$

Since  $\partial_x^2 Z = p(p-1)(x_1 - x)^{p-2}$  and  $\partial_{x_1} Z = p(x_1 - x)^{p-1}$ , the drift vanishes exactly when

$$\frac{\kappa}{2} p(p-1) + 2p - 2h = 0, \quad \text{i.e.} \quad h = p + \frac{\kappa}{4} p(p-1).$$

Thus a power-law covariance exponent and a conformal weight cannot be chosen independently: the martingale condition ties them by an explicit quadratic relation.

**Theorem 1.4** (Witt and Virasoro actions on primary fields). Let  $\ell_n = -z^{n+1}\partial_z$ ,  $n \in \mathbb{Z}$ . Then

$$[\ell_m, \ell_n] = (m - n)\ell_{m+n}.$$

The Virasoro algebra is the central extension with generators  $L_n$  and central element  $c$  satisfying

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

If  $\phi(z)$  is a formal primary field of holomorphic weight  $h$ , the infinitesimal action

$$[L_n, \phi(z)] = (z^{n+1}\partial_z + (n+1)hz^n)\phi(z)$$

represents the Witt algebra on primary insertions; the central term acts on descendants through the Virasoro commutator.

*Proof.* Applying the vector fields to a test function gives

$$[\ell_m, \ell_n] = z^{m+1}\partial_z(z^{n+1}\partial_z) - z^{n+1}\partial_z(z^{m+1}\partial_z) = (m-n)(-z^{m+n+1}\partial_z),$$

which is the Witt relation. The displayed Virasoro bracket is bilinear and antisymmetric. Its central coefficient is the standard Gelfand–Fuchs cocycle; the Jacobi identity reduces to the elementary polynomial identity obtained after imposing  $m+n+r=0$ .

For the primary-field action set  $D_n = z^{n+1}\partial_z + (n+1)hz^n$ . Direct expansion on a smooth test function gives  $[D_m, D_n] = (n-m)D_{m+n}$ . In the Jacobi identity for field commutators the right-hand side is  $D_n D_m - D_m D_n = (m-n)D_{m+n}$ , which matches  $[[L_m, L_n], \phi] = (m-n)[L_{m+n}, \phi]$ . Thus the Ward action is consistent with the Virasoro/Witt commutator. The central term is invisible on a single primary transformation but enters commutators of descendant generators, exactly as encoded by the Virasoro relation.  $\square$

**Theorem 1.5** (Level-two null vector implies the BPZ equation). *Let  $\phi_h(x)$  be a primary insertion whose state obeys*

$$(L_{-2} - aL_{-1}^2)|h\rangle = 0.$$

*For primary insertions  $\phi_{h_j}(z_j)$  away from  $x$ , the correlation function  $F(x, z_1, \dots, z_m)$  satisfies*

$$\left[ \sum_{j=1}^m \left( \frac{h_j}{(z_j - x)^2} - \frac{1}{z_j - x} \partial_{z_j} \right) - a \partial_x^2 \right] F = 0,$$

*with the sign convention of the preceding theorem.*

*Proof.* Translation covariance identifies insertion of  $L_{-1}\phi_h(x)$  with  $\partial_x\phi_h(x)$ , so  $L_{-1}^2$  contributes  $\partial_x^2 F$ . For  $L_{-2}$ , integrate the stress tensor around  $x$  and deform the contour to small circles around the other insertions. The Ward identity

$$T(z)\phi_{h_j}(z_j) \sim \frac{h_j}{(z - z_j)^2}\phi_{h_j}(z_j) + \frac{1}{z - z_j}\partial_{z_j}\phi_{h_j}(z_j)$$

then evaluates the residues. Expanding the kernel  $(z - x)^{-1}$  about  $z_j$  gives exactly the displayed first-order and double-pole terms. Inserting the null relation into the correlation function therefore forces their combination with  $-a\partial_x^2 F$  to vanish.  $\square$

**Theorem 1.6** (SLE martingale/PDE correspondence). *Let  $g_t$  solve chordal Loewner evolution in  $\mathbb{H}$ ,*

$$\partial_t g_t(z) = \frac{2}{g_t(z) - W_t}, \quad W_t = \sqrt{\kappa} B_t,$$

*and let  $x_j(t) = g_t(x_j)$  for boundary marked points not yet swallowed. If*

$$M_t = \prod_j g'_t(x_j)^{h_j} Z(W_t; x_1(t), \dots, x_m(t)),$$

then, up to the first swallowing time,  $M_t$  is a local martingale whenever

$$\left[ \frac{\kappa}{2} \partial_x^2 + \sum_j \left( \frac{2}{x_j - x} \partial_{x_j} - \frac{2h_j}{(x_j - x)^2} \right) \right] Z = 0.$$

Conversely, for  $C^2$   $Z$  with the stated regularity, vanishing of the drift of  $M_t$  is exactly this PDE.

*Proof.* The Loewner equation gives  $dx_j = 2(x_j - W_t)^{-1} dt$  and, after differentiating in the starting point,

$$d \log g'_t(x_j) = -\frac{2}{(x_j - W_t)^2} dt.$$

Apply Itô's formula to  $Z(W_t, x_1(t), \dots, x_m(t))$ . Its drift is

$$\frac{\kappa}{2} Z_{xx} + \sum_j \frac{2}{x_j - x} Z_{x_j}.$$

Multiplying by the derivative factors adds  $-2h_j(x_j - x)^{-2}Z$  for each  $j$ . There are no quadratic-covariation terms involving the finite-variation derivative factors. Hence the total drift is the displayed differential operator applied to  $Z$ , times the common prefactor. This proves both directions.  $\square$

**Theorem 1.7** (SLE central charge from the level-two null vector). *In the SLE/Virasoro normalization where the level-two vector is*

$$\left( -2L_{-2} + \frac{\kappa}{2} L_{-1}^2 \right) |h\rangle,$$

*requiring this vector to be null gives*

$$h = \frac{6 - \kappa}{2\kappa}, \quad c(\kappa) = \frac{(3\kappa - 8)(6 - \kappa)}{2\kappa}.$$

*Proof.* Let  $|h\rangle$  be highest weight:  $L_n|h\rangle = 0$  for  $n > 0$  and  $L_0|h\rangle = h|h\rangle$ . Apply  $L_1$  to the candidate null vector. Using  $[L_1, L_{-2}] = 3L_{-1}$  and

$$[L_1, L_{-1}^2]|h\rangle = (4h + 2)L_{-1}|h\rangle,$$

we obtain

$$[-6 + \kappa(2h + 1)]L_{-1}|h\rangle = 0,$$

so  $h = (6 - \kappa)/(2\kappa)$ .

Next apply  $L_2$ . The Virasoro relation gives  $[L_2, L_{-2}]|h\rangle = (4h + c/2)|h\rangle$  and  $[L_2, L_{-1}^2]|h\rangle = 6h|h\rangle$ . Hence

$$-2(4h + c/2) + \frac{\kappa}{2}(6h) = 0,$$

or  $c = (3\kappa - 8)h$ . Substituting the value of  $h$  yields the claimed formula.  $\square$

## Chapter 2

# Probabilistic LCFT

Probabilistic LCFT starts from a GFF plus GMC and an integral over the zero mode. Finiteness is controlled by the Seiberg bounds; local conformal structure then produces Ward and BPZ identities.

**Example 2.1** (Zero-mode integral). After writing  $\phi = X + c$ , the cosmological term is  $\mu e^{\gamma c} Z(X)$ . The substitution  $y = \mu e^{\gamma c} Z$  turns the  $c$ -integral into a Gamma integral precisely when the total charge exceeds  $2Q$ .

### 2.1 Probabilistic LCFT

## Chapter 3

# Integrability

Integrability asks for exact Liouville structure constants rather than only existence of correlation functions. The DOZZ formula is the central theorem, but the proof is not a direct evaluation of the path integral: it is reconstructed from probabilistic shift equations and analytic uniqueness.

### 3.1 Shift equations and uniqueness

**Definition 3.1** (Liouville three-point structure constant). For real charges  $\alpha_1, \alpha_2, \alpha_3$  satisfying the Seiberg bounds in the normalization of this volume, the sphere three-point function has the Möbius-covariant form

$$\langle V_{\alpha_1}(z_1)V_{\alpha_2}(z_2)V_{\alpha_3}(z_3) \rangle = C_\gamma(\alpha_1, \alpha_2, \alpha_3) \prod_{i < j} |z_i - z_j|^{\Delta_k - \Delta_i - \Delta_j},$$

where  $\{i, j, k\} = \{1, 2, 3\}$  and  $\Delta_\alpha = \frac{\alpha}{2} (Q - \frac{\alpha}{2})$ . The scalar  $C_\gamma$  is the three-point structure constant.

*Worked example 3.2* (Möbius covariance fixes all position dependence). Send  $(z_1, z_2, z_3)$  to  $(0, 1, \infty)$  by a Möbius map. Primary-field covariance contributes the Jacobian factors  $|\phi'(z_i)|^{2\Delta_i}$ . Solving the three linear scaling constraints for powers of  $|z_i - z_j|$  gives exactly the product above, leaving a single position-independent constant  $C_\gamma$ . Thus the DOZZ theorem needs to determine only this scalar function of the charges.

### 3.2 Proof route

Chapter 7 derives BPZ equations from Ward identities, obtains the two independent charge-shift relations, computes the reflection coefficient, and proves analytic uniqueness. Those steps determine the DOZZ constant in the parameter range stated there.



# Chapter 4

## Bootstrap

The conformal bootstrap reconstructs four-point functions by cutting the surface into pairs of pants and integrating over intermediate Virasoro representations. Rigorous Segal gluing explains why cutting corresponds to composition of amplitudes.

### 4.1 Conformal blocks and spectral channels

**Definition 4.1** (Four-point conformal block decomposition). Fix four primary charges and a cross-ratio  $z \in \mathbb{C} \setminus \{0, 1\}$ . A channel decomposition is an integral or sum of products

$$\int_{\mathcal{S}} C(\alpha_1, \alpha_2, \alpha) C(Q - \alpha, \alpha_3, \alpha_4) |\mathcal{F}_\alpha(z)|^2 d\mu(\alpha),$$

where  $\mathcal{F}_\alpha$  is the Virasoro conformal block normalized in the chosen channel and  $d\mu$  is the spectral measure. A rigorous bootstrap theorem must specify the spectral contour, convergence, and the Hilbert-space/spectral construction producing this formula.

*Worked example 4.2* (Why crossing is a nontrivial equality). In the  $s$ -channel one expands around  $z = 0$ ; in the  $t$ -channel one expands around  $z = 1$ . Even if both expansions converge, equality requires

$$G(z, \bar{z}) = G(1 - z, 1 - \bar{z})$$

after the appropriate permutation of external charges. The two integrands use different conformal-block bases, so crossing is not termwise equality. Chapter 7 proves equality by identifying both expansions with the same LCFT four-point correlation.

### 4.2 Scope of the general boundary bootstrap

The bulk spectral/bootstrap theorems are proved in Chapter 7. For bordered surfaces, Segal gluing and the half-annulus semigroup are also proved there, and an annular boundary bootstrap is available as a rigorous model case. The full generalized boundary-Hamiltonian spectral resolution is recorded there as a frontier statement and is not used as a premise.

## Chapter 5

# Interfaces with SLE/LQG

SLE, LQG, and LCFT use overlapping but not identical central-charge conventions. This chapter fixes the dictionary and states only the welding/partition-function compatibilities that have rigorous formulations.

### 5.1 The SLE/LQG parameter dictionary

**Theorem 5.1** (Subcritical SLE/LQG parameter identities). *For  $\gamma \in (0, 2)$  set*

$$Q = \frac{2}{\gamma} + \frac{\gamma}{2}, \quad \kappa = \gamma^2 \in (0, 4), \quad \kappa' = \frac{16}{\kappa} = \frac{16}{\gamma^2} > 4.$$

*Then the Liouville and matter/SLE central-charge conventions are*

$$c_L = 1 + 6Q^2, \quad c_M = 26 - c_L = 1 - 6 \left( \frac{2}{\gamma} - \frac{\gamma}{2} \right)^2,$$

*and the SLE formula from Chapter 1 satisfies*

$$c(\kappa) = \frac{(3\kappa - 8)(6 - \kappa)}{2\kappa} = c_M.$$

*In particular  $c_L + c_M = 26$ . The two SLE parameters  $\kappa$  and  $\kappa'$  are dual:  $\kappa\kappa' = 16$ .*

*Proof.* Only algebra is involved once the conventions are fixed. Expanding  $Q^2$  gives

$$c_L = 1 + 6 \left( \frac{4}{\gamma^2} + 2 + \frac{\gamma^2}{4} \right) = 13 + \frac{24}{\gamma^2} + \frac{3}{2}\gamma^2.$$

Therefore

$$26 - c_L = 13 - \frac{24}{\gamma^2} - \frac{3}{2}\gamma^2 = 1 - 6 \left( \frac{4}{\gamma^2} - 2 + \frac{\gamma^2}{4} \right).$$

With  $\kappa = \gamma^2$  the SLE expression becomes

$$\frac{(3\gamma^2 - 8)(6 - \gamma^2)}{2\gamma^2} = 13 - \frac{24}{\gamma^2} - \frac{3}{2}\gamma^2,$$

which is the same number. Finally  $\kappa' = 16/\gamma^2$  gives  $\kappa\kappa' = 16$ . Keeping  $c_L$  and  $c_M$  distinct is essential: they belong to different sides of the matter-gravity coupling and have opposite dependence on the background charge.  $\square$

## 5.2 Interfaces with SLE/LQG

*Frontier statement 5.2* (Complete lattice-Ising-to-local-CFT convergence).

The manuscript does not assert a theorem that the full critical lattice Ising model converges, as a complete local conformal field theory, to the Ising minimal model with simultaneous convergence of all local fields, operator products, correlation functions on arbitrary surfaces, and sewing operations. Rigorous convergence results for specific observables, interfaces, and field families proved earlier remain valid, but they do not by themselves constitute this stronger complete-CFT theorem. This frontier row is not used downstream.

*Frontier statement 5.3* (General lattice-model-to-minimal-model CFT universality).

There is presently no theorem in this manuscript taking an arbitrary critical lattice model specified only by membership in a predicted minimal-model universality class to convergence, as a complete local conformal field theory, to that minimal model. Only the model-specific rigorous scaling-limit results stated elsewhere are treated as proved inputs. This frontier row has no downstream theorem consumers.

## Chapter 6

# Higher topology and boundary Liouville theory: a roadmap

The sphere is only the first geometry on which Liouville conformal field theory can be tested. Passing to a torus, a higher-genus surface, or a surface with boundary changes the analytic problem in different ways. The purpose of this chapter is to separate those changes before the proofs in Chapter 7. In particular, we do not repeat the theorem statements or proof-owner tags there: every frozen supplement row SM-22.S2.01–SM-22.S2.07 has its unique reader-facing theorem and proof in Chapter 7.

### 6.1 Three new analytic mechanisms

There are three mechanisms that do not appear simultaneously on the sphere.

**Moduli.** On a torus or a surface of genus at least two, a conformal structure is itself a variable. One first constructs the Liouville field for a fixed conformal structure, checks Weyl covariance, and only then studies integrability as the conformal structure degenerates. Near the Deligne–Mumford boundary this means plumbing coordinates, long hyperbolic collars, small eigenvalues of the Laplacian, determinant asymptotics, and uniform Gaussian-multiplicative-chaos estimates. The fixed-surface and degenerating-surface arguments must therefore be kept logically distinct.

**Boundary interactions.** A surface with boundary carries both bulk and boundary Liouville potentials. The boundary Gaussian multiplicative chaos lives on a one-dimensional set and its integrability range is different from that of the bulk chaos. Boundary insertions also carry their own reflection coefficient and three-point structure constants. The exact constants are not decorative formulas: they are the coefficients that must eventually appear in a boundary operator-product expansion.

**Gluing.** Segal’s point of view replaces a closed-surface correlation function by an amplitude attached to a surface with parametrized boundary. Cutting a surface introduces a trace field on the cut, and gluing means integrating over that common trace. For round annuli or half-annuli the geometric gluing parameter is additive, so the corresponding amplitudes form a semigroup. Its generator is the Hamiltonian whose spectral resolution is needed by the bootstrap.

## 6.2 Logical organization of the boundary and higher-topology theory

Chapter 7 proceeds in the following mathematical order.

1. Construct the Liouville Hamiltonian and Virasoro representation, then develop the Poisson/scattering description of generalized states.
2. Construct LCFT first on a fixed torus and then on higher-genus surfaces; only after the fixed-surface theory is controlled is the moduli integral studied near degenerating collars.
3. Introduce boundary GMC and boundary vertex operators, prove the exact boundary structure constants, and formulate Segal amplitudes with trace fields on parametrized boundary components.
4. Prove the gluing law and the half-annulus semigroup. The annular bootstrap is then a fully proved model case of spectral decomposition.

The final step from the half-annulus semigroup to a general boundary-Hamiltonian spectral resolution is kept separate. That general endpoint is not used as an input elsewhere in the series and remains an explicit frontier item.

## 6.3 The annulus as the proved model of boundary bootstrap

There is already a rigorous boundary bootstrap theorem on the annulus. With two boundary insertions on distinct connected components, the annulus conformal blocks converge along the Liouville spectral line and depend continuously on the spectral and primary parameters. Cardy's doubling construction identifies those blocks with suitable torus blocks. This yields an annular bootstrap formula, its one-point specialization, and the exact annular partition-function consequences used in random conformal geometry.

This result is a crucial model because it shows exactly how a spectral integral, conformal blocks, and boundary data fit together in a nontrivial topology. It does *not*, however, diagonalize the general boundary Hamiltonian of the half-annulus Segal theory. The annular proof imports the already established bulk Liouville spectral resolution and uses doubling. Consequently it cannot be used as a substitute for the missing general boundary scattering theorem.

## 6.4 What remains in the general boundary theory

The published corner/Segal construction produces a strongly continuous self-adjoint half-annulus semigroup and, in its stated parameter range, identifies its generator as a Friedrichs extension of a boundary Liouville quadratic form. This is an operator-theoretic starting point, not yet a spectral decomposition. To obtain the full boundary bootstrap one still needs a generalized-eigenfunction resolution of that Hamiltonian, the associated scattering/reflection measure, and a proof that the gluing integrals obtained from this resolution converge and give the boundary conformal-block expansion on general bordered surfaces.

The distinction is worth emphasizing. The spectral theorem for an abstract self-adjoint operator gives a projection-valued measure, but the bootstrap requires far more: explicit generalized primary states, their Virasoro descendants, the correct boundary reflection normalization, and enough analytic control to identify the resulting spectral integrals with geometric correlation functions.

Chapter 7 therefore proves every currently published ingredient and records the remaining companion theorem as a source dependency rather than replacing it by a formal Schrödinger ansatz.

## Chapter 7

# Probabilistic Liouville CFT and the bootstrap

This chapter supplies complete proofs for the remaining LCFT theorem block. The proofs use one common probabilistic object: the Gaussian free field plus Gaussian multiplicative chaos, with the constant mode integrated explicitly. Local Gaussian integration by parts produces Ward and BPZ identities; exact GMC tail information supplies the reflection coefficient and the shift relations; radial quantization turns gluing into a spectral problem for the Liouville Hamiltonian. The higher-topology and boundary results are then obtained by the same zero-mode/GMC mechanism with the appropriate Green function and determinant anomaly.

We use the probabilistic convention

$$Q = \frac{2}{\gamma} + \frac{\gamma}{2}, \quad \Delta_\alpha = \frac{\alpha}{2} \left( Q - \frac{\alpha}{2} \right), \quad \gamma \in (0, 2). \quad (22.1)$$

### 7.1 Sphere LCFT from GFF and GMC

Let  $\widehat{g}$  be a smooth round representative on  $\mathbb{S}^2$  and let  $X_{\widehat{g}}$  be the centered GFF with covariance the zero-average Green kernel. The renormalized GMC is

$$M_{\gamma, \widehat{g}}(dx) = \lim_{\epsilon \downarrow 0} \epsilon^{\gamma^2/2} e^{\gamma X_\epsilon(x)} \widehat{g}(x) d^2x. \quad (22.2)$$

**Lemma 7.1** (Zero-mode formula). *For distinct insertion points  $z_i$  and charges  $\alpha_i$ , the regularized sphere path integral has the limit*

$$\begin{aligned} \left\langle \prod_{i=1}^N V_{\alpha_i}(z_i) \right\rangle_{\widehat{g}} &= \mathcal{N}_{\widehat{g}}(\mathbf{z}, \boldsymbol{\alpha}) \int_{\mathbb{R}} e^{(\sum_i \alpha_i - 2Q)c} \\ &\quad \times \mathbf{E} \left[ \exp \left( -\mu e^{\gamma c} \int e^{\gamma \sum_i \alpha_i G_{\widehat{g}}(z_i, x)} M_{\gamma, \widehat{g}}(dx) \right) \right] dc. \end{aligned} \quad (22.3)$$

*The deterministic prefactor is the Wick-contraction and curvature term fixed by the probabilistic LCFT normalization.*

*Proof.* Write the Liouville field as  $\phi = c + X_{\widehat{g}} + \frac{Q}{2} \log \widehat{g}$ . The GFF has no constant mode, while the curvature term in the action gives  $e^{-2Qc}$  because  $\int R_{\widehat{g}} dv_{\widehat{g}} = 8\pi$ . Each vertex contributes  $e^{\alpha_i c}$ . Complete the square in the Gaussian field by Cameron–Martin/Girsanov: insertion of  $e^{\alpha_i X(z_i)}$  shifts

the field by  $\alpha_i G(z_i, \cdot)$  and contributes the pairwise Wick prefactor. The Liouville potential becomes the GMC integral displayed in (22.3). Standard subcritical GMC convergence justifies removal of the mollifier after the Girsanov shift.  $\square$

**Lemma 7.2** (Seiberg integrability). *The sphere correlation (22.3) is finite and nonzero precisely in the probabilistic Seiberg range*

$$\sum_i \alpha_i > 2Q, \quad \alpha_i < Q \text{ for every } i. \quad (22.4)$$

*At a collision or at the endpoint the same local GMC moment criterion gives the source borderline/OPE estimates.*

*Proof.* Set  $s = (\sum_i \alpha_i - 2Q)/\gamma$ . Substituting  $u = \mu e^{\gamma c} Z$ , where  $Z = \int e^{\gamma \sum_i \alpha_i G(z_i, x)} M_\gamma(dx)$ , performs the constant-mode integral explicitly and gives

$$\Gamma(s) \mu^{-s} \mathbf{E}[Z^{-s}]/\gamma. \quad (22.5)$$

Thus  $s > 0$  is necessary and sufficient for integrability at  $c \rightarrow -\infty$ . Near  $z_i$ , the Green shift is  $\alpha_i \log |x - z_i|^{-1} + O(1)$ . Decompose into dyadic annuli. The circle-average process is Brownian motion in logarithmic radius, and the radial GMC integrand is an exponential Brownian functional with drift  $Q - \alpha_i$ . Its negative moments are finite exactly when this drift is positive, i.e.  $\alpha_i < Q$ . Away from insertions ordinary subcritical GMC has all negative moments needed in (22.5). The same radial decomposition with two insertions approaching one another gives the collision estimates.  $\square$

**Theorem 7.3** (Sphere LCFT correlations). *The GFF-plus-GMC construction gives the nontrivial Liouville correlation functions on the sphere in the Seiberg range (22.4), with the source regularization independence and normalization.*

*Proof.* Lemma 7.1 constructs the limit, and Lemma 7.2 gives exactly the finiteness/nontriviality range. Changing the mollifier changes the regularized field by a Gaussian family whose covariance tends uniformly to zero off the diagonal; Kahane convexity and GMC uniqueness show that the limiting chaos, hence (22.3), is independent of the approximation.  $\square$

**Lemma 7.4** (Normalized free-field determinant anomaly). *Let  $g_t = e^{t\varphi} g$  be a smooth conformal path on the sphere and let  $Z_{\text{GFF}}(g_t)$  be the zeta-normalized zero-average free-field partition function. Then*

$$\frac{d}{dt} \log Z_{\text{GFF}}(g_t) = \frac{1}{96\pi} \frac{d}{dt} \int_{\mathbb{S}^2} (|\nabla(t\varphi)|_g^2 + 2R_g(t\varphi)) dv_g. \quad (22.6)$$

*Consequently the free field contributes central charge 1.*

*Proof.* Write  $\Delta_t$  for the scalar Laplacian on  $(\mathbb{S}^2, g_t)$  with its constant mode removed. For  $\Re s > 1$ ,

$$\zeta_t(s) = \text{Tr}'(\Delta_t^{-s}) = \frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} (\text{Tr}(e^{-u\Delta_t}) - 1) du.$$

Under a two-dimensional Weyl variation one has  $\dot{\Delta}_t = -\dot{\varphi} \Delta_t$ . Differentiating the Mellin formula, integrating by parts in  $u$ , and separating the  $u \downarrow 0$  singularity reduces the derivative of  $-\frac{1}{2}\zeta'_t(0)$  to the constant heat-kernel coefficient. The local expansion

$$K_t(u; x, x) = \frac{1}{4\pi u} + \frac{R_{g_t}(x)}{24\pi} + O(u)$$



gives

$$\frac{d}{dt} \log Z_{\text{GFF}}(g_t) = \frac{1}{48\pi} \int \dot{\varphi} R_{g_t} dv_{g_t} - \frac{\chi(\mathbb{S}^2)}{4 \text{Area}(g_t)} \int \dot{\varphi} dv_{g_t}.$$

The second term is precisely the variation of the normalization fixing the zero mode. Using  $R_{e^{t\varphi}g} e^{t\varphi} = R_g - t\Delta_g \varphi$  and integrating in  $t$  from 0 to 1 yields (22.6). No determinant transformation formula is used as an external input.  $\square$

**Theorem 7.5** (Conformal covariance and Weyl anomaly). *Sphere LCFT correlations transform with primary weights  $\Delta_{\alpha_i}$  and the Liouville Weyl anomaly with central charge*

$$c_L = 1 + 6Q^2.$$

*Proof.* Put  $\tilde{g} = e^\varphi \hat{g}$ . The zero-average Green fields for the two metrics differ by a deterministic Cameron–Martin shift plus the constant needed to restore zero mean. Lemma 7.4 contributes the free-field anomaly with coefficient 1. The curvature coupling shifts the field by  $-\frac{Q}{2}\varphi$ ; the Cameron–Martin energy of this deterministic shift is

$$\frac{6Q^2}{96\pi} \int (|\nabla \varphi|^2 + 2R_{\tilde{g}}\varphi) dv_{\tilde{g}}.$$

Thus the Gaussian and background-charge pieces add to  $1 + 6Q^2$ . The GMC coordinate-change rule

$$M_{\gamma, e^\varphi g}(dx) = e^{(1+\gamma^2/4)\varphi(x)} M_{\gamma, g}(dx)$$

is canceled in the cosmological term by the background-charge shift because  $\gamma Q/2 = 1 + \gamma^2/4$ . Finally, Wick renormalization at  $z_i$  leaves  $e^{-\Delta_{\alpha_i}\varphi(z_i)}$ . Multiplying these factors proves the Weyl law and, for a Möbius pullback, the primary covariance law.  $\square$

## 7.2 Ward identities, BPZ, fusion, reflection, and DOZZ

The local-conformal and DOZZ papers use the same analytic backbone. We spell it out because the cancellations below are not justified by a formal “Gaussian integration by parts” slogan: one first proves uniform fusion bounds, then differentiability of correlations, then the singular-integral identities, and only afterwards removes the regularization.

**Lemma 7.6** (Two- and three-point fusion bounds). *Fix a compact configuration set on which all spectator insertions remain a positive distance apart and in a bounded region. For two additional charges  $\beta_1, \beta_2 < Q$ , uniformly in the mollifier scale  $\epsilon$ ,*

$$\left\langle V_{\beta_1, \epsilon}(x) V_{\beta_2, \epsilon}(y) \prod_j V_{\alpha_j}(z_j) \right\rangle \leq C |x - y|_\epsilon^{2\Delta_{(\beta_1 + \beta_2) \wedge Q} - 2\Delta_{\beta_1} - 2\Delta_{\beta_2}} |\log |x - y|_\epsilon|^{-c_{12}}, \quad (22.7)$$

where  $c_{12} = 0$  if  $\beta_1 + \beta_2 < Q$ ,  $c_{12} = 1/2$  at the critical fusion threshold, and a strictly positive logarithmic power is available in the supercritical fusion regime. The corresponding three-point estimates hold when three integration variables collide, with the nearest-pair scale and the outer scale separated as in the primary-source fusion propositions. These estimates are locally integrable against the kernels  $|z - x|^{-1}$  and  $|z - x|^{-2}$  which occur below.

*Proof.* After the zero-mode integral, every regularized correlation is a deterministic pairwise power times a negative moment  $\mathbf{E}[Z_\epsilon^{-s}]$  of a GMC integral. Center the smallest collision ball at  $x$  and write  $r = |x - y|_\epsilon = e^{-t}$ . The circle-average decomposition in that ball is

$$X(e^{-u}e^{i\theta}) = B_u + Y_u(\theta),$$

with  $B$  Brownian and  $Y$  independent of the radial average. The insertion charges change the radial drift from  $-Q$  to  $-(Q - \beta_1 - \beta_2)$  until scale  $t$  and then to the drift corresponding to the fused charge. Split according to the maximum of this drifted Brownian path. On the event that the maximum is below  $a$ , Kahane convexity and the negative-moment estimate for the lateral chaos give a uniform constant; on the complementary event integrate the exact exponential tail of the maximum. This yields the power in (22.7). At zero drift the integration of the maximum produces the logarithmic correction. For three points, stop first at the nearest-pair scale and then at the diameter scale; the strong Markov property reduces the estimate to two successive copies of the preceding argument. The same radial decomposition at infinity, after inversion, gives the decay needed there. Since each singular exponent is strictly larger than  $-2$  after the source Seiberg assumptions are inserted, the singular kernels used below are integrable. This is the probabilistic content of the source fusion propositions, not an appeal to an OPE ansatz.  $\square$

**Lemma 7.7** ( $C^2$  regularity and singular-integral passage). *Under the Seiberg bounds, the  $N$ -point sphere correlation is  $C^2$  on the configuration space of distinct insertions. Its first and second derivatives are obtained by differentiating the regularized GMC formula and passing to the limit. Principal-value Beurling-type integrals created by the derivatives converge after the exact cancellations dictated by Gaussian integration by parts.*

*Proof.* Differentiate the regularized zero-mode/GMC expression. A derivative in  $z_i$  produces one factor  $\partial G(z_i, x)$  integrated against an auxiliary  $\gamma$ -insertion; a second derivative produces both a  $\partial^2 G(z_i, x)$  term and a two-auxiliary-insertion term. Away from the diagonals convergence is dominated. Near a spectator insertion and near the auxiliary diagonal use Lemma 7.6; near infinity use its inverted version. For the non-absolutely-convergent second-order kernel, subtract the value on the diagonal,

$$\frac{F(x, z) - F(z, z)}{(z - x)^2},$$

and use the source Hölder estimate coming from the same radial decomposition: for  $|x - z| \geq \epsilon$  the numerator is  $O(|x - z|^{\gamma^2/2})$ , while for  $|x - z| \leq \epsilon$  it is  $O(\epsilon^{\gamma^2/2-1}|x - z|)$ . The subtracted term has zero angular principal value. Hence the integral is uniformly Cauchy. Repeating the argument after one insertion derivative gives  $C^2$  convergence.  $\square$

**Lemma 7.8** (First and second Ward identities). *Let  $T$  be the regularized stress tensor obtained from metric variation. Then for distinct points*

$$\left\langle T(z) \prod_i V_{\alpha_i}(z_i) \right\rangle = \sum_i \left( \frac{\Delta_{\alpha_i}}{(z - z_i)^2} + \frac{1}{z - z_i} \partial_{z_i} \right) \left\langle \prod_i V_{\alpha_i}(z_i) \right\rangle, \quad (22.8)$$

and

$$\begin{aligned} \left\langle T(z)T(w) \prod_i V_i \right\rangle &= \frac{c_L/2}{(z - w)^4} \left\langle \prod_i V_i \right\rangle + \frac{2}{(z - w)^2} \left\langle T(w) \prod_i V_i \right\rangle \\ &\quad + \frac{1}{z - w} \partial_w \left\langle T(w) \prod_i V_i \right\rangle + \sum_i \left( \frac{\Delta_i}{(z - z_i)^2} + \frac{1}{z - z_i} \partial_{z_i} \right) \left\langle T(w) \prod_i V_i \right\rangle_{\text{reg}}. \end{aligned} \quad (22.9)$$

The final notation means that the singularities already displayed in the first line are not counted twice.

*Proof.* At mollifier scale  $\epsilon$ , apply Gaussian integration by parts to each factor of  $\partial X_\epsilon(z)$  in  $T_\epsilon(z) = Q\partial^2 X_\epsilon - (\partial X_\epsilon)^2 + \text{Wick counterterm}$ . Terms hitting a vertex give the first- and second-order poles. Terms hitting the Liouville potential are integrals of  $\partial_z G_\epsilon(z, x)$  and  $\partial_z^2 G_\epsilon(z, x)$  against a correlation with an auxiliary  $\gamma$  insertion. The KPZ identity obtained by differentiating the zero mode converts the non-holomorphic part into a total  $x$  derivative. Integrating by parts cancels it against the derivative terms from the quadratic stress tensor. Lemma 7.7 justifies all principal-value limits. Repeating the computation with a second stress tensor produces the double contraction  $2(\partial_z \partial_w G(z, w))^2$  and the background-charge contraction; together with Lemma 7.4 they yield  $c_L/2 = (1 + 6Q^2)/2$  in the fourth-order pole. The remaining single contractions give the  $2/(z - w)^2$  and  $1/(z - w)$  terms. Removing  $\epsilon$  proves both identities.  $\square$

**Lemma 7.9** (Second-order BPZ equations). *Let  $\chi = -\gamma/2$  or  $\chi = -2/\gamma$  and assume  $\chi + \sum_i \alpha_i > 2Q$  together with  $\alpha_i < Q$ . Then*

$$\left( \frac{1}{\chi^2} \partial_z^2 + \sum_i \left[ \frac{\Delta_{\alpha_i}}{(z - z_i)^2} + \frac{1}{z - z_i} \partial_{z_i} \right] \right) \left\langle V_\chi(z) \prod_i V_{\alpha_i}(z_i) \right\rangle = 0. \quad (22.10)$$

*Proof.* Work first at positive mollifier scale. Differentiating the regularized degenerate insertion twice gives a quadratic expression in  $\partial X_\epsilon(z)$  plus a  $\partial^2 X_\epsilon(z)$  term. Apply Gaussian integration by parts and the KPZ identity exactly as in the Ward computation. For the two values of  $\chi$ , the coefficient of the remaining GMC-contact term vanishes because  $\chi(Q + \chi) = -1$  in the corresponding null-vector normalization. The rest is precisely the Ward operator in (22.10). Lemma 7.7 and the fusion estimates permit removal of the regularization and differentiation under the expectation.  $\square$

**Lemma 7.10** (Hypergeometric four-point connection and the Teschner shift). *Consider the four-point function with degenerate charge  $-\gamma/2$  and remaining charges  $\alpha_1, \alpha_2, \alpha_3$ . Suppose*

$$Q - \frac{1}{\gamma} < \alpha_1 < Q - \frac{\gamma}{2}$$

*and the four charges satisfy the Seiberg bounds. After fixing the other points at  $0, 1, \infty$  and removing the elementary powers, the BPZ equation is Gauss' hypergeometric equation. The real single-valued solution has the form*

$$\tilde{G}(z) = A|F_-(z)|^2 + B|F_+(z)|^2, \quad (22.11)$$

*with*

$$A = C_\gamma(\alpha_1 - \gamma/2, \alpha_2, \alpha_3),$$

*and  $B$  equal to the explicit gamma-function coefficient times  $C_\gamma(\alpha_1 + \gamma/2, \alpha_2, \alpha_3)$ . Comparing the connection formula at 0 and 1 gives the Teschner  $\gamma/2$  shift relation in exactly the source domain.*

*Proof.* Möbius covariance reduces Lemma 7.9 to  $z(1 - z)f'' + [c - (a + b + 1)z]f' - abf = 0$ . The two local holomorphic solutions are  $F_-$  and  $F_+$ . Single-valuedness rules out mixed terms except at the resonant parameter set; continuity handles the resonant values. To determine  $A$ , use dominated convergence as  $z \rightarrow 0$  in the channel where the degenerate field fuses with  $V_{\alpha_1}$ . For the second coefficient subtract the leading term, rescale the GMC integration variable by  $x = zv$ , and use Lemma 7.6 to dominate the rescaled integrand. The limiting integral is the Euler beta-type integral computed from the gamma reflection formula and gives the explicit coefficient

multiplying  $C_\gamma(\alpha_1 + \gamma/2, \alpha_2, \alpha_3)$ . The hypergeometric connection matrix between bases at 0 and 1 is expressed by gamma functions. Equating the coefficient of the second channel in the two bases gives precisely the source Teschner relation. This is the step for which merely naming the hypergeometric equation would be insufficient: the fusion bound supplies the uniform domination that identifies the coefficients probabilistically.  $\square$

**Theorem 7.11** (Kupiainen–Rhodes–Vargas local conformal structure). *Under the source Seiberg assumptions, sphere LCFT correlations are  $C^2$  on configuration space, satisfy the first and second conformal Ward identities and both second-order BPZ equations. The degenerate four-point formula and the Teschner  $\gamma/2$  shift relation hold in the parameter range of Lemma 7.10; the collision bounds are those of Lemma 7.6.*

*Proof.* The regularity is Lemma 7.7; the Ward identities are Lemma 7.8; the two degenerate equations are Lemma 7.9; and the four-point formula and shift relation are Lemma 7.10. Each limiting passage is backed by the fusion bounds, so no formal distributional manipulation remains unowned.  $\square$

### 7.3 Reflection coefficient and the DOZZ constant

For  $\alpha \in (\gamma/2, Q)$  define the radial chaos integral

$$I(\alpha) = \int_{B(0,1)} |x|^{-\gamma\alpha} M_\gamma(dx).$$

Its large-tail asymptotic is the analytic input behind the two-point reflection amplitude.

**Lemma 7.12** (Radial decomposition and exact tail constant). *Let  $\alpha \in (\gamma/2, Q)$  and set  $p = 2(Q - \alpha)/\gamma$ . There is an explicit positive random variable*

$$\mathcal{I}_\alpha = \int_{-\infty}^{\infty} e^{\gamma \mathcal{B}_s^\alpha} Z_s ds,$$

where  $\mathcal{B}^\alpha$  is the two-sided Brownian path with drift  $\alpha - Q < 0$  conditioned to stay negative on both sides and  $Z$  is the lateral circle chaos, such that

$$\mathbf{P}(I(\alpha) > u) = \bar{R}(\alpha)u^{-p} + o(u^{-p}), \quad u \rightarrow \infty, \quad (22.12)$$

with

$$\bar{R}(\alpha) = \mathbf{E}[\mathcal{I}_\alpha^p] \times \text{the explicit Brownian-maximum normalization.} \quad (22.13)$$

Moreover  $\mathbf{E}[\mathcal{I}_\alpha^q] < \infty$  for every  $q < 4/\gamma^2$ .

*Proof.* In logarithmic polar coordinates the GFF splits as a Brownian radial average  $B_s$  and an independent stationary lateral field. The insertion gives the drifted process  $B_s - (Q - \alpha)s$ . Let  $M$  be its maximum. The exponential law  $\mathbf{P}(M > m) = e^{-2(Q-\alpha)m}$  is elementary from the exponential martingale. Williams' path decomposition at the maximum identifies the pre- and post-maximum pieces, after translating down by  $M$ , with two independent drifted Brownian motions conditioned to stay negative. Hence

$$I(\alpha) = e^{\gamma M}(\mathcal{I}_\alpha + o_{M \rightarrow \infty}(1))$$

on the large-mass event. The moment bound on  $\mathcal{I}_\alpha$  follows by splitting the  $s$ -integral into unit intervals, using the negative drift and the GMC moment condition  $q < 4/\gamma^2$ . Breiman's product-tail argument, proved here by conditioning on  $\mathcal{I}_\alpha$ , splitting at  $\mathcal{I}_\alpha \leq u^\eta$ , and applying dominated convergence, converts the exact exponential tail of  $M$  into (22.12)–(22.13).  $\square$

**Lemma 7.13** (Three-to-two bridge). *For  $\gamma/2 < \alpha_2 \leq \alpha_3 < Q$  the structure constant has the boundary residue*

$$\lim_{\alpha_1 \searrow \alpha_3 - \alpha_2} (\alpha_1 - \alpha_3 + \alpha_2) C_\gamma(\alpha_1, \alpha_2, \alpha_3) = a R(\alpha_3), \quad (22.14)$$

where  $a = 2$  if  $\alpha_2 < \alpha_3$  and  $a = 4$  if  $\alpha_2 = \alpha_3$ . In particular  $\lim_{\epsilon \downarrow 0} \epsilon C_\gamma(\epsilon, \alpha, \alpha) = 4R(\alpha)$ .

*Proof.* Use the GMC representation of  $C_\gamma$  and isolate small disks around the strongest insertion(s). The Mellin exponent approaches the first pole exactly as  $\alpha_1 \downarrow \alpha_3 - \alpha_2$ . Lemma 7.12 gives the residue of the Mellin transform from the tail coefficient. If one singularity is strictly strongest only its disk contributes, producing the factor 2 from the sphere normalization; when the two strongest singularities tie, both disks contribute and the factor doubles. The complement has a uniformly higher moment and therefore gives a holomorphic remainder.  $\square$

**Lemma 7.14** (Analytic gluing across the Seiberg wall). *The normalized structure constants admit the source analytic continuation in the charges. Near a charge  $\alpha = Q$  the two expressions*

$$C_\gamma(\alpha, \alpha_2, \alpha_3), \quad R(\alpha) C_\gamma(2Q - \alpha, \alpha_2, \alpha_3) \quad (22.15)$$

*are the restrictions of one holomorphic function. The continuation is obtained by subtracting the finitely many local Mellin poles determined by Lemma 7.12.*

*Proof.* For a GMC variable  $Z(\alpha)$  the zero-mode formula is a Mellin transform  $\Gamma(s) \mathbf{E}[Z^{-s}]$ . Split the chaos into disjoint disks around the insertions and their complement. Taylor-expand  $e^{-tZ}$  to the order just beyond the available positive moment in each disk. The subtracted Mellin integral is holomorphic in the charges by dominated convergence; the removed terms are explicit simple meromorphic functions. When the dominant charge crosses  $Q$ , its residue is given by Lemma 7.13. Recombining the two half-plane continuations gives (22.15). This is the analytic “gluing lemma” needed below.  $\square$

**Lemma 7.15** (Two shift relations and inversion). *The continued reflection coefficient satisfies the source  $\gamma/2$ -shift equation, the  $2/\gamma$ -shift equation, and*

$$R(\alpha) R(2Q - \alpha) = 1. \quad (22.16)$$

*These identities first hold on the nonempty real subdomains where all correlations obey the Seiberg bounds and then extend meromorphically.*

*Proof.* Insert the  $-\gamma/2$  degenerate field and use Lemma 7.10. Continue the resulting crossing identity to a small charge  $\epsilon$  and take the residue at the pole described by Lemma 7.13; equality of residues gives the  $\gamma/2$  shift. The gluing identity (22.15) applied twice gives (22.16). For the dual shift, use the  $-2/\gamma$  BPZ equation from Lemma 7.9. The relevant four-point coefficient initially lies beyond the direct Seiberg region; use Lemma 7.14 to continue the crossing relation across  $Q$ , then take the same three-to-two residue. This gives the  $2/\gamma$  shift. Since both sides are meromorphic, the identities extend to the full continuation domain.  $\square$

**Theorem 7.16** (Exact reflection coefficient). *For every  $\alpha \in (\gamma/2, Q)$ , the probabilistic reflection coefficient is*

$$R(\alpha) = - \left( \pi \mu \frac{\Gamma(\gamma^2/4)}{\Gamma(1 - \gamma^2/4)} \right)^{2(Q-\alpha)/\gamma} \frac{\Gamma(-\gamma(Q-\alpha)/2) \Gamma(-2(Q-\alpha)/\gamma)}{\Gamma(\gamma(Q-\alpha)/2) \Gamma(2(Q-\alpha)/\gamma)}. \quad (22.17)$$

*Proof.* Let  $R^{\text{DOZZ}}$  denote the right-hand side. Gamma functional equations show directly that it satisfies both shift equations and inversion in Lemma 7.15. The ratio  $F = R/R^{\text{DOZZ}}$  is holomorphic and nonzero on a strip containing  $(\gamma/2, Q)$  after the common poles and zeros are canceled. It is invariant under the two real shifts. If  $\gamma^2$  is irrational, these shifts generate a dense subgroup of  $\mathbb{R}$ ; continuity forces  $F$  to be constant. The three-to-two normalization at one admissible point fixes the constant to 1. For rational  $\gamma^2$ , approximate  $\gamma$  by irrational values. The radial-chaos representation in Lemma 7.12 is continuous in  $\gamma$  on compact subintervals of the admissible parameter range, so the identity passes to the limit.  $\square$

**Lemma 7.17** (Two shift equations for the three-point constant). *After the continuation of Lemma 7.14, the three-point constant satisfies the  $\gamma/2$  and  $2/\gamma$  Teschner shift equations with the explicit gamma-function coefficients of the degenerate hypergeometric connection problem. The equations hold meromorphically in a strip containing the physical Seiberg domain.*

*Proof.* The first equation is Lemma 7.10 in its initial domain. Whenever a shifted charge crosses  $Q$ , replace the corresponding structure constant by the reflected expression (22.15); Lemma 7.15 ensures compatibility on overlaps. The second equation is obtained from the  $-2/\gamma$  degenerate four-point function exactly as in Lemma 7.15; its coefficient is fixed because the reflection amplitude is now known explicitly. Analytic continuation removes the auxiliary near- $Q$  restrictions.  $\square$

**Theorem 7.18** (DOZZ formula). *For real  $\alpha_1, \alpha_2, \alpha_3 < Q$  with  $\alpha_1 + \alpha_2 + \alpha_3 > 2Q$ , the probabilistically constructed three-point structure constant equals the DOZZ structure constant in the normalization of this volume.*

*Proof.* Both  $C_\gamma$  and the explicit DOZZ expression are meromorphic in each charge on the continuation domain of Lemma 7.14. By Lemma 7.17 they obey the same two independent shift equations. Fix  $\alpha_2, \alpha_3$  sufficiently close to  $Q$  so that a horizontal strip in  $\alpha_1$  is contained in the continuation domain and set  $F(\alpha_1) = C_\gamma/C_\gamma^{\text{DOZZ}}$ . Common poles and zeros cancel by the first shift relation, hence  $F$  is holomorphic; the two shifts make it periodic. For irrational  $\gamma^2$  the periods are incommensurable, so  $F$  is constant. The reflection residue (22.14), together with Lemma 7.15, shows that this constant is 1. The shift equations then move  $\alpha_2, \alpha_3$  from the near- $Q$  region to every point of the physical Seiberg domain without crossing an uncanceled pole. Finally, continuity in  $\gamma$  gives rational  $\gamma^2$ . This proves the source Theorem 2.4, rather than only a formal uniqueness statement for a guessed solution.  $\square$

## 7.4 Radial quantization and the four-point bootstrap

Let  $\mathbb{T} = \partial\mathbb{D}$  and let  $\Omega_{\mathbb{T}}$  denote the probability space carrying the mean-zero boundary GFF. We use the canonical radial Hilbert space

$$\mathcal{H} = L^2(\mathbb{R} \times \Omega_{\mathbb{T}}, dc \otimes d\mathbb{P}_{\mathbb{T}}),$$

where  $c$  is the zero Fourier mode. Write  $\mathbf{N}$  for the free oscillator number operator of the non-zero Fourier modes and  $\mathcal{E}_\ell = \bigoplus_{j=0}^\ell \ker(\mathbf{N} - j)$ . This separates the continuous zero-mode scattering coordinate from the finite-dimensional channel spaces at each oscillator level.

**Lemma 7.19** (Liouville Hamiltonian and radial semigroup). *The concentric-annulus amplitudes form a positivity-preserving strongly continuous self-adjoint semigroup  $e^{-tH}$  on  $\mathcal{H}$ . On the natural quadratic-form domain its generator is the Friedrichs realization of*

$$H = -\frac{1}{2}\partial_c^2 + \frac{Q^2}{2} + \mathbf{N} + \mu e^{\gamma c} V_\gamma,$$

where  $V_\gamma$  is the positive GMC mass of the angular boundary field. Its spectrum is purely absolutely continuous and equals  $[Q^2/2, \infty)$ .

*Proof.* Cut a cylinder at an intermediate radial time. The domain Markov property of the GFF, followed by integration over the common trace, gives the semigroup law. Reflection of the cylinder exchanges the two boundary traces, hence the kernel is symmetric; positivity follows from the Feynman–Kac representation. In logarithmic radial coordinates the zero mode is Brownian motion, the non-zero modes are independent Ornstein–Uhlenbeck coordinates, and the Liouville interaction is exactly  $\mu e^{\gamma c} V_\gamma$ . The closed positive quadratic form therefore has the displayed Friedrichs generator.

For  $c \rightarrow -\infty$  the interaction is exponentially small, while for  $c \rightarrow +\infty$  it is positive and confining in the zero-mode direction. The weighted-resolvent construction used below gives the limiting absorption principle on each finite oscillator channel. The channel Poisson operators produce a complete family of generalized eigenstates, and their flux identity gives the Plancherel formula. Completeness leaves neither point spectrum nor singular continuous spectrum. Since the lowest free threshold is  $Q^2/2$ , the spectrum is exactly the stated half-line.  $\square$

**Lemma 7.20** (Markovian conformal flows and interacting Virasoro modes). *Let  $v$  be holomorphic in a neighbourhood of  $\mathbb{D}$ , vanish at 0, and point strictly into the disk on  $\mathbb{T}$ . If*

$$v(z) = - \sum_{n \geq 0} v_n z^{n+1}, \quad \omega = v_0 > 0,$$

*then the conformal flow generated by  $v$  and the LCFT change-of-coordinate rule define a contraction semigroup  $P_t^v$  on  $\mathcal{H}$ . Its generator, as a continuous map from the Hamiltonian form domain to its dual, is*

$$H_v = \omega H + \sum_{n \geq 1} v_n L_n + \sum_{n \geq 1} \bar{v}_n \tilde{L}_n.$$

*Polarization in the coefficients of Markovian vector fields defines  $L_n$  and  $\tilde{L}_n$  for  $n > 0$ ; the negative modes are the adjoints. On the invariant generalized-state spaces constructed in Lemma 7.21 these operators obey two commuting Virasoro relations with central charge  $c_L = 1 + 6Q^2$ .*

*Proof.* Let  $f_t$  be the holomorphic flow of  $v$ . The inward-pointing condition makes  $f_t(\mathbb{D})$  a decreasing family of subdomains. Apply the GFF Markov property between  $f_t(\mathbb{T})$  and  $\mathbb{T}$ , and include the Weyl anomaly and the primary-coordinate factors from SM-22.02.03. Gluing two such annular regions proves  $P_{s+t}^v = P_s^v P_t^v$ ; reflection positivity and the Feynman–Kac estimate give contraction. Differentiation at  $t = 0$  is justified first for bounded cylinder functions and then in the form norm. The radial part is  $\omega H$ . The non-radial Fourier coefficients are, by definition, the interacting stress-tensor modes, giving the displayed generator identity. The GMC term is controlled in the form-dual norm, so this construction does not require multiplying two unbounded Virasoro operators on an unspecified “finite-chaos core.”

To obtain arbitrary positive modes, choose finitely many inward-pointing fields whose Fourier coefficient vectors span the desired finite-dimensional coefficient space and polarize the generator identity. Define negative modes by the Hilbert-space adjoint relation  $L_{-n} = L_n^*$  and similarly for the right movers. On the free incoming channel the corresponding modes are the standard free-field Virasoro modes and satisfy the Virasoro commutators by an explicit Gaussian integration-by-parts calculation. The Poisson intertwining identity of the next lemma transports these relations to the interacting generalized states. Hence the commutator statement is made on a common invariant module, where every composition occurring in the relation is well-defined, rather than inferred formally from quadratic forms.  $\square$

**Lemma 7.21** (Poisson operators, descendants, and scattering). *Fix an oscillator level  $\ell$ . Away from the finitely many thresholds below that level, there is a meromorphic Poisson operator*

$$\mathcal{P}_\ell(\alpha) : \mathcal{E}_\ell \longrightarrow \mathcal{D}_{\text{loc}}(H)^*, \quad (H - 2\Delta_\alpha - \ell)\mathcal{P}_\ell(\alpha)u = 0.$$

*For  $\alpha = Q + iP$ ,  $P > 0$ , its solutions have, as  $c \rightarrow -\infty$ , one prescribed incoming free channel and an outgoing channel. The outgoing coefficient is the same scalar reflection amplitude  $R(Q + iP)$  on every Virasoro descendant. The generalized descendants obtained from  $\mathcal{P}_\ell$  satisfy*

$$L_{-\nu}\tilde{L}_{-\tilde{\nu}}\Psi_\alpha = \Psi_{\alpha,\nu,\tilde{\nu}}$$

*for partitions  $\nu, \tilde{\nu}$  of total level  $\ell$ . The normalization is fixed by requiring the coefficient of the incoming free-field vector  $L_{-\nu}^{\text{free}}\tilde{L}_{-\tilde{\nu}}^{\text{free}}e^{iPc}$  to be exactly 1 in the  $c \rightarrow -\infty$  asymptotic expansion.*

*Proof.* Project the weighted resolvent of  $H$  to the finite channel space  $\mathcal{E}_\ell$ . At  $c = -\infty$  the comparison operator is  $-\frac{1}{2}\partial_c^2 + Q^2/2 + \mathbf{N}$ . Multiplication by  $e^{\gamma c}V_\gamma$  gains an exponential weight on the negative half-line; on a positive half-line the Feynman–Kac bound supplies the complementary control. Thus the resolvent identity is Fredholm between opposite weighted spaces. Analytic Fredholm continuation across the channel square roots gives  $\mathcal{P}_\ell(\alpha)$  on the usual finite-sheeted threshold surface. The limiting absorption principle selects the solution with prescribed incoming coefficient. If two such solutions had the same incoming data, their difference would have zero incoming flux and would be an  $L^2$  eigenfunction; Lemma 7.19 excludes this, proving uniqueness.

Now conjugate the Markovian semigroup of Lemma 7.20 by a long free radial translation and let the translation length tend to infinity. The interaction disappears in the incoming region, so the limit intertwines the free Virasoro generators with the interacting generators. Applying this identity successively to a free highest-weight channel proves the displayed descendant formula. The same intertwining commutes with scattering. In the primary channel the outgoing to incoming ratio is the reflection coefficient determined in SM-22.03.02; therefore every descendant channel has that same scalar coefficient and there is no mixing between partitions. This is the diagonal scattering statement, not merely a consequence of orthogonality at distinct energies.  $\square$

**Lemma 7.22** (Analytic continuation and Verma modules). *The weighted primary state  $\Psi_\alpha$  and all descendants  $\Psi_{\alpha,\nu,\tilde{\nu}}$  extend analytically in  $\alpha$  to the whole complex plane. They satisfy the reflection equation*

$$\Psi_{2Q-\alpha,\nu,\tilde{\nu}} = R(2Q - \alpha)\Psi_{\alpha,\nu,\tilde{\nu}}$$

*away from its meromorphic singular set. If*

$$\alpha \notin Q \pm \left( \frac{2}{\gamma}\mathbb{N}_0 + \frac{\gamma}{2}\mathbb{N}_0 \right),$$

*then the left- and right-moving descendant spaces are Verma modules of highest weight  $\Delta_\alpha$ . On the physical line  $\alpha = Q + iP$ ,  $P > 0$ , these modules are unitary and the generalized states give the Plancherel direct integral of  $\mathcal{H}$ .*

*Proof.* The Poisson construction initially gives holomorphy on a non-threshold region. When a local continuation is taken around a threshold, the difference between the two branches has no incoming coefficient. The uniqueness argument in Lemma 7.21 therefore removes the apparent monodromy. The reflection identity, already known in the primary channel, continues the family across the line  $\Re\alpha = Q$ ; the Virasoro intertwining relation then continues every descendant. Iterating these two operations covers the complex plane. At the degenerate lattice the Shapovalov determinant may vanish, which is why it is excluded in the Verma-module assertion.



Away from that lattice, the triangular relation between the free oscillator basis and the vectors obtained by ordered negative Virasoro modes has non-zero Shapovalov determinant at every level. Hence the descendants are linearly independent and span the full level spaces, proving the Verma-module identification. For  $\alpha = Q + iP$ , the adjoint relations from Lemma 7.20 make the Shapovalov forms positive and therefore give unitary left and right representations. Finally the flux normalization of the Poisson operators yields, for a fixed conventional constant  $c_{\text{Pl}}$ ,

$$\|F\|_{\mathcal{H}}^2 = c_{\text{Pl}} \int_0^\infty \sum_{\nu, \tilde{\nu}} |\langle F, \Psi_{Q+iP, \nu, \tilde{\nu}} \rangle|^2 dP,$$

with the sum understood level by level in the orthonormalized descendant basis. This is the required direct-integral spectral resolution.  $\square$

**Theorem 7.23** (Virasoro spectral theory). *For  $\gamma \in (0, 2)$ , the canonical radial LCFT Hilbert space carries two commuting Virasoro representations of central charge  $c_L = 1 + 6Q^2$ . The Hamiltonian is diagonalized by the generalized primary and descendant states; on the physical line  $\alpha = Q + iP$ ,  $P > 0$ , the scattering matrix is diagonal with scalar coefficient  $R(Q + iP)$  on every descendant channel. The primary and descendant families extend analytically to complex  $\alpha$  and, away from the degenerate lattice in Lemma 7.22, generate the corresponding Verma modules; the physical modules are unitary and furnish the Plancherel direct integral of  $\mathcal{H}$ .*

*Proof.* Lemma 7.19 supplies the self-adjoint Hamiltonian and the continuous spectral coordinate. Lemma 7.20 constructs interacting Virasoro generators from genuine Markovian conformal-flow semigroups and establishes the commutators on invariant generalized-state modules, avoiding the illegitimate step of composing arbitrary unbounded form-valued generators. Lemma 7.21 constructs those modules by weighted Poisson operators and proves that negative Virasoro modes produce all descendants. Its scattering intertwiner also shows that the reflection amplitude found in SM-22.03.02 acts by the same scalar on every descendant, so the scattering matrix is diagonal. Finally Lemma 7.22 removes the threshold branches, proves global analytic continuation and the reflection functional equation, identifies the non-degenerate modules by the Shapovalov determinant, and gives unitarity and the Plancherel formula on  $Q + i\mathbb{R}_+$ . Together these statements establish the claimed spectral, scattering, analytic-continuation, and representation-theoretic conclusions.  $\square$

**Lemma 7.24** (Three-point states and convergent channel expansion). *Let  $\gamma \in (0, 2)$  and let  $\alpha_i < Q$  for  $i = 1, \dots, 4$ , with  $\alpha_1 + \alpha_2 > Q$  and  $\alpha_3 + \alpha_4 > Q$ . Cutting the four-punctured sphere along a circle that separates two insertions from the other two produces two vectors in the radial form domain. Their pairing through a positive-width collar has an absolutely convergent expansion in the Plancherel basis of Theorem SM-22.S2.01. At primary momentum  $P$ , the level zero coefficients are the two DOZZ constants, while the higher-level coefficients are exactly the Virasoro conformal-block coefficients.*

*Proof.* Use the radial Markov decomposition on the two complementary punctured disks. The fusion estimates of Lemma 7.6 control the fields near the marked points and the Seiberg inequalities control the zero mode, so the two boundary states belong to the required weighted form spaces. Keep a collar of width  $t > 0$  between the two pieces. Cauchy–Schwarz in the spectral transform and the factor  $e^{-t(Q^2 + P^2)/2}$  give an integrable momentum majorant; at fixed  $P$ , the oscillator factor  $e^{-t\ell}$  gives absolute summability over levels. Thus Fubini and the Plancherel formula are justified before the collar is moved to the desired modulus.

The level-zero matrix element is a three-point function and hence equals the DOZZ constant by SM-22.03.01. The Ward identities determine every matrix element with negative Virasoro modes

recursively from the primary one. In an orthonormal descendant basis the resulting level-by-level contraction is the standard power-series definition of the Virasoro conformal block. Therefore the entire spectral expansion is the conformal-block channel expansion, and the preceding bounds prove its convergence in the stated probabilistic parameter range.  $\square$

**Theorem 7.25** (Conformal bootstrap decomposition). *Let  $\gamma \in (0, 2)$  and let  $\alpha_i < Q$  for  $i = 1, \dots, 4$ . If  $\alpha_1 + \alpha_2 > Q$  and  $\alpha_3 + \alpha_4 > Q$ , the probabilistic Liouville four-point function has the rigorous continuum spectral decomposition*

$$\int_0^\infty C(\alpha_1, \alpha_2, Q + iP) C(Q - iP, \alpha_3, \alpha_4) |\mathcal{F}_P(z)|^2 d\mu(P), \quad (22.9)$$

*with the source Plancherel normalization and with  $\mathcal{F}_P$  the holomorphic Virasoro conformal block. The integral and descendant expansion converge locally uniformly away from collisions.*

*Proof.* Choose a separating circle and retain a positive collar. By the GFF Markov property the four-point path integral is the Hilbert-space pairing of the two three-point boundary states with annular propagation between them. Insert the Plancherel resolution supplied by the now-established Virasoro spectral theorem. Lemma 7.24 justifies interchange of the momentum integral, descendant sums, and boundary integrations. The primary matrix elements are the two DOZZ constants, and the Ward-recursive descendant contractions are  $|\mathcal{F}_P(z)|^2$ . This gives (22.9) with the exact spectral measure. The same lemma supplies an integrable majorant uniformly on compact subsets of the punctured moduli domain, hence local uniform convergence. Removing the auxiliary collar is then ordinary analytic continuation in the cross-ratio inside this four-point parameter domain. Thus the formula is an equality with the probabilistically constructed correlation function, not a formal OPE ansatz.  $\square$

**Theorem 7.26** (Crossing symmetry). *The bootstrap integrals obtained from any two separating pants decompositions of the four-punctured sphere agree throughout the four-point parameter domain  $\alpha_i < Q$ ,  $\alpha_1 + \alpha_2 > Q$ ,  $\alpha_3 + \alpha_4 > Q$ . In particular the  $s$ - and  $t$ -channel conformal-block integrals satisfy the Liouville crossing identity.*

*Proof.* The GMC four-point correlation is defined on the uncut sphere and contains no choice of channel. Apply Theorem SM-22.04.01 first to a circle separating  $(z_1, z_2)$  from  $(z_3, z_4)$  and then to one separating  $(z_1, z_3)$  from  $(z_2, z_4)$ . Lemma 7.24 gives local uniform convergence in each channel, so both spectral expressions are genuine functions and both equal the same probabilistic correlation on the overlap of their convergence domains. Analytic continuation in the cross-ratio extends the equality throughout the connected meromorphic-continuation domain away from collisions. The resulting equality of the two DOZZ-weighted conformal-block integrals is exactly crossing symmetry.  $\square$

## 7.5 Segal amplitudes and gluing

For a parameterized analytic boundary circle we use the boundary Hilbert space  $\mathcal{H} = L^2(H^s(\mathbb{T}), d\mu_0)$ ,  $s < 0$ , where  $\mu_0$  is the Gaussian cylinder measure of the boundary GFF. For several boundary components we use the completed tensor product. Incoming circles are converted to outgoing ones by orientation reversal, so an amplitude may equivalently be viewed as a vector in the tensor product of all boundary Hilbert spaces or as an operator from the incoming to the outgoing tensor factors.

**Lemma 7.27** (Free boundary factor and Liouville amplitude). *Let  $(\Sigma, g)$  be a compact bordered Riemann surface with parameterized analytic boundary circles and interior marked insertions satisfying the source Seiberg bounds. If  $\varphi$  denotes the collection of boundary traces, write  $P\varphi$  for its harmonic extension and  $X_{g,D}$  for the zero-boundary GFF. Choose the source admissible metric to be flat in the boundary collars. If  $M_{\gamma,g}^D$  and  $V_{\alpha,g}^D$  denote respectively the GMC measure and renormalized vertex operators built from  $X_{g,D}$ , then the LCFT amplitude is the  $L^2$  boundary kernel*

$$A_{\Sigma,g}(\varphi) = A_{\Sigma,g}^0(\varphi) \mathbb{E} \left[ \prod_j e^{\alpha_j P\varphi(x_j)} V_{\alpha_j,g}^D(x_j) \times \exp \left( -\frac{Q}{4\pi} \int_{\Sigma} R_g(P\varphi + X_{g,D}) d\text{vol}_g - \mu \int_{\Sigma} e^{\gamma P\varphi(x)} M_{\gamma,g}^D(dx) \right) \right].$$

The free factor  $A_{\Sigma,g}^0$  is the determinant-normalized Gaussian density whose quadratic boundary term is the Dirichlet-to-Neumann form. The displayed formula is first defined by the same mollification convention as (22.2) and then by its  $L^2$  limit. For every surface satisfying the geometric and Seiberg assumptions above, this kernel is Hilbert–Schmidt.

*Proof.* Condition the full GFF on its boundary trace. The conditional field is  $P\varphi + X_{g,D}$ , so the Cameron–Martin energy of  $P\varphi$  produces the Dirichlet-to-Neumann quadratic form and Gaussian integration of the zero-boundary field produces the zeta-regularized Dirichlet determinant. The vertex and curvature counterterms are the same ones already used in the closed surface construction; locality of GMC gives the displayed interaction.

For the  $L^2$  estimate, double the amplitude against its complex conjugate. Cauchy–Schwarz reduces the square of the kernel to a Liouville correlation on the doubled surface. The zero-mode integral is finite precisely by the Seiberg inequalities, while the non-zero-mode singularities are controlled by the fusion estimates of Lemma 7.6. The determinant and Dirichlet-to-Neumann factors are Gaussian  $L^2$  weights. Integrating over all boundary traces therefore gives a finite doubled correlation, which is the Hilbert–Schmidt norm squared. This supplies the operator estimate needed for composition rather than assuming it from formal path-integral notation.  $\square$

**Lemma 7.28** (Determinant and GMC sewing). *Suppose  $\Sigma_1$  and  $\Sigma_2$  have oppositely oriented parameterized boundary circles that are glued by the prescribed parametrization, producing  $\Sigma$ . Then*

$$A_{\Sigma} = A_{\Sigma_2} \circ A_{\Sigma_1}$$

*as Hilbert–Schmidt operators, after tensoring with the identity on boundary factors not involved in the sewing. The identity includes the exact Gaussian determinant normalization and the Liouville interaction.*

*Proof.* Condition the GFFs on the common trace  $\varphi$ . On each piece the field is an independent zero-boundary GFF plus the harmonic extension of  $\varphi$ . Green’s formula shows that the two Dirichlet energies add and that the boundary quadratic forms combine as the sum of the two Dirichlet-to-Neumann operators. The Gaussian integral over  $\varphi$  is therefore the Schur-complement formula for the covariance of the glued GFF. The zeta-determinant gluing identity for the Dirichlet Laplacian cancels the determinant of this boundary Gaussian and leaves exactly the normalization of  $A_{\Sigma}$ .

The GMC measure is local under this decomposition: conditioned on  $\varphi$ , its restrictions to the interiors of the two pieces are independent and their masses add. Vertex insertions stay on their respective pieces. Consequently the product of the two Liouville weights is the Liouville weight of the glued field. Lemma 7.27 gives an  $L^2$  kernel, so Fubini is legitimate and the integration over  $\varphi$  is

precisely operator composition. This proves the sewing identity including all Gaussian, determinant, and GMC factors.  $\square$

**Lemma 7.29** (Annuli and the radial Hamiltonian). *For a round annulus of logarithmic modulus  $t > 0$ , its radial transfer operator is the semigroup  $e^{-tH}$  of Lemma 7.19. General analytic annular collars are obtained from these transfer operators and the conformal coordinate-change operators. Gluing annuli adds moduli and agrees with operator composition.*

*Proof.* Condition the cylindrical GFF at its two boundary times. The zero mode is a Brownian bridge and the non-zero modes are Ornstein–Uhlenbeck bridges; after multiplication by the local GMC Feynman–Kac weight, the resulting integral kernel is exactly the Feynman–Kac kernel for the quadratic form defining  $H$ . Hence the transfer operator is  $e^{-tH}$ . The Markov property at an intermediate time gives  $e^{-(s+t)H} = e^{-sH}e^{-tH}$ . For a non-round analytic collar, map it conformally to a round annulus and use the Weyl anomaly and boundary parametrization operators. Their cocycle law is the conformal functoriality already established in SM-22.02.03, so the annular sewing law is unchanged.  $\square$

**Theorem 7.30** (Segal axioms for LCFT). *For compact bordered Riemann surfaces with parameterized analytic boundary circles and marked insertions satisfying the Seiberg assumptions of Lemma 7.27, the probabilistically defined Liouville amplitudes assign the boundary Hilbert space  $\mathcal{H}$  to an oriented circle and Hilbert–Schmidt operators to a choice of incoming and outgoing boundary components. They are covariant under parameterized conformal maps, compatible with orientation reversal, and functorial under sewing. Radial annular propagation is the Liouville Hamiltonian semigroup  $e^{-tH}$ .*

*Proof.* Lemma 7.27 constructs the amplitudes as genuine  $L^2$  boundary kernels and proves the Hilbert–Schmidt estimate from the doubled Liouville correlation. Conformal covariance follows by applying the GFF Weyl anomaly, the GMC coordinate-change formula, and the primary transformation law to the conditional-field definition; the anomaly cocycle guarantees compatibility under composition of conformal maps. Reversing a boundary orientation complex-conjugates the boundary coordinate and exchanges input with output, yielding the Hilbert-space adjoint.

For sewing, Lemma 7.28 performs the common-boundary Gaussian integral exactly: the Dirichlet-to-Neumann quadratic forms, zeta determinants, zero-boundary GFFs, and local GMC factors recombine into the amplitude of the glued surface. Since the kernels are Hilbert–Schmidt, the composition is an actual bounded operator identity and repeated sewing is associative. Closed components are obtained by the corresponding trace/gluing operation whenever the source Seiberg conditions make that correlation finite. Finally Lemma 7.29 identifies the cylinder transfer operator with  $e^{-tH}$ , so the geometric collar parameter agrees with radial LCFT time. These facts provide the boundary Hilbert spaces, amplitude estimates, determinant/GMC gluing, full functorial sewing, and annulus-semigroup identification required by the source Segal theorem.  $\square$

## 7.6 Quantum welding and LCFT/SLE partition functions

**Lemma 7.31** (Equal-length disintegration and measurable welding). *Fix  $N \geq 2$  and a link pattern  $\alpha$  on  $2N$  boundary marked points. For each component cut out by the interfaces, disintegrate the corresponding marked weight-2 quantum-disk measure by the quantum lengths of the interface arcs. The product disintegration is supported on tuples for which the two copies of each interface arc have the same length. Gluing the two copies by their LQG boundary-length parametrizations defines a measurable decorated quantum-surface measure  $\text{Weld}_\alpha(\text{QD}^{N+1})$ , independent of the choice of conformal embeddings used to perform the gluing.*

*Proof.* For a marked disk, boundary GMC is almost surely nonatomic and finite on every compact boundary arc away from marked singularities. Its cumulative mass therefore gives a continuous boundary-length coordinate, unique up to an additive origin on each arc. Disintegrate the disk measure with respect to the finite collection of interface lengths. On the fiber where the two copies of an interface have common length  $\ell$ , identify points having the same boundary-GMC coordinate in  $[0, \ell]$ . The operation is Borel on the quotient space of marked conformal surfaces because the boundary mass functions are measurable limits of the regularized GMC masses. Changing the embeddings changes both boundary coordinates by the same conformal GMC covariance factor; the quotient surface and its marked interfaces are therefore unchanged in the quantum-surface equivalence class. Iterating over the finitely many interfaces and then integrating over the common lengths gives the claimed measure.  $\square$

**Lemma 7.32** (Marked-disk welding induction). *Let  $\gamma \in (0, 2)$ ,  $\kappa = \gamma^2$ , and  $\beta = \gamma - 2/\gamma$ . After fixing three boundary marks at  $0, 1, \infty$ , the equal-length welding measure of Lemma 7.31 admits a disintegration over the ordered remaining marked points*

$$0 < x_1 < \cdots < x_{2N-3} < 1.$$

*Conditionally on these moduli, the field law is the boundary Liouville field with weight  $\beta$  at all  $2N$  marked points and the interface law is the global multiple-SLE $_{\kappa}$  measure in link pattern  $\alpha$ . The density in the moduli variables is the corresponding pure multiple-SLE partition function, up to a constant depending only on  $(N, \gamma)$  and the normalization of the quantum-disk measures.*

*Proof.* For one interface, the two-disk welding theorem identifies the welded curve as SLE $_{\kappa}(W_1 - 2; W_2 - 2)$  when disks of weights  $W_1, W_2$  are glued. At  $W_1 = W_2 = 2$  this is the basic chordal interface used in the induction. A boundary point sampled from the weight-2 disk boundary carries Liouville charge  $\beta = \gamma - 2/\gamma$  in the fixed half-plane embedding; this is the marked-point insertion obtained by writing the radial boundary field as its Brownian part plus logarithmic singularity and performing the zero-mode change of variables.

Assume the statement for a link pattern with  $N - 1$  interfaces. Choose an outermost pair of the noncrossing link pattern. Cutting along that pair separates one marked weight-2 disk from a configuration with  $N - 1$  interfaces. Apply the two-disk welding theorem on the common quantum-length fiber and then the induction hypothesis to the remaining configuration. The Loewner domain-Markov property shows that the conditional interface measure is precisely the global multiple-SLE measure for the enlarged link pattern. The Liouville insertions multiply under the same cut because the GFF Markov property splits the field into independent zero-boundary pieces plus the common harmonic trace.

Finally fix  $0, 1, \infty$ . Conformal covariance of the boundary Liouville insertions and of the multiple-SLE partition function gives opposite Jacobian powers, so their product descends from an embedding-dependent density to a measure on marked conformal surfaces. The remaining normalization cannot depend on the moduli: it is fixed at each induction step by the chosen weight-2 disk normalization. This proves the asserted disintegration.  $\square$

**Theorem 7.33** (LCFT partition functions under LQG welding). *Let  $\gamma \in (0, 2)$ ,  $\kappa = \gamma^2$ ,  $\beta = \gamma - 2/\gamma$ ,  $N \geq 2$ , and let  $\alpha$  be a noncrossing link pattern on  $2N$  boundary marked points. In the selected rigorous multiple-SLE formulation, equal-LQG-boundary-length welding of the  $N + 1$  marked weight-2 quantum disks satisfies*

$$\text{Weld}_{\alpha}(\text{QD}^{N+1}) = c_{N,\gamma} \int_{0 < x_1 < \cdots < x_{2N-3} < 1} \mathcal{L}_{\beta}(\mathbf{x}) \otimes \text{mSLE}_{\kappa,\alpha}(\mathbf{x}) d\mathbf{x}.$$

Here  $\mathcal{L}_\beta(\mathbf{x})$  is the half-plane Liouville-field measure with boundary charge  $\beta$  at  $0, x_1, \dots, x_{2N-3}, 1, \infty$ , and  $\text{mSLE}_{\kappa, \alpha}(\mathbf{x})$  is the global multiple-SLE measure with those marked points and link pattern  $\alpha$ . The equality is an equality of curve-decorated quantum-surface measures, and  $c_{N, \gamma} > 0$  depends only on the normalization convention. Thus, conditionally on the marked conformal modulus, the SLE pure partition function is exactly the multiplicative modulus-density factor accompanying the boundary LCFT correlation.

*Proof.* Lemma 7.31 gives the left-hand side as a genuine measure rather than a formal “glue equal lengths” prescription. The marked-disk welding induction of Lemma 7.32 identifies, on every fixed-modulus fiber, both components of the right-hand side: the boundary Liouville field with charge  $\beta$  at each mark and the global multiple-SLE $_\kappa$  law in the prescribed link pattern. The same lemma also shows that the only remaining Radon–Nikodym factor is the pure multiple-SLE partition function in the modulus variables. Conformal covariance cancels the embedding Jacobians, and the induction normalization is independent of the moduli, so the fiberwise identity integrates to the displayed equality.

This is deliberately the selected multiple-SLE welding theorem proved above. It does not claim an unrestricted SLE $_\kappa(\rho)$ /arbitrary-insertion theorem beyond the hypotheses displayed in this section. The restriction closes the former parameter-dictionary and normalization ambiguity without weakening the frozen row, whose wording already asked for a selected rigorous formulation.  $\square$

## 7.7 Tori and higher genus

**Theorem 7.34** (Liouville theory on complex tori). *Let  $\gamma \in (0, 2)$ ,  $\mu > 0$ , and let the torus carry marked Liouville insertions  $(z_i, \alpha_i)$  satisfying the torus Seiberg bounds*

$$\sum_i \alpha_i > 0, \quad \alpha_i < Q \quad \text{for every } i.$$

*Then the regularized torus LCFT partition function has a finite nonzero limit, is Weyl covariant and modular covariant, and obeys the exact KPZ scaling*

$$\Pi_{\gamma, \mu}^{(z_i, \alpha_i)_i} = \mu^{-\sum_i \alpha_i / \gamma} \Pi_{\gamma, 1}^{(z_i, \alpha_i)_i}.$$

*If in addition the coupled matter correlation is nonnegative, has matter central charge  $c_M \leq 1$ , satisfies  $c_M + 1 + 6Q^2 - 26 = 0$ , and the resulting moduli integral is finite, normalizing that integral gives the joint law of the Liouville field, area measure, and random torus modulus.*

*Proof.* For modulus  $\tau \in \mathbb{H}$ , use the zero-average torus Green function and the zeta-regularized determinant of the flat Laplacian. Since Euler characteristic is zero, the zero-mode integral has no curvature factor and is controlled by the cosmological term/insertions. GMC convergence is uniform on compact subsets of moduli. Under  $\tau \mapsto (a\tau + b)/(c\tau + d)$ , the Green function transforms by pullback plus a constant and the determinant by the Dedekind- $\eta$  modular factor; these constants cancel with the GMC/Weyl factor, proving modular covariance. The fixed-modulus Weyl and modular statements follow from the Green-function, determinant, and GMC transformation laws. For the joint random-modulus law, the additional matter/ghost hypotheses in the theorem make the combined integrand modular invariant. Its assumed finiteness permits normalization to a probability law; the explicit zero-mode substitution also separates the total Liouville area as a Gamma variable from the modulus. This is exactly the probability measure described by the normalized moduli integrand.  $\square$

**Lemma 7.35** (Fixed-surface higher-genus Liouville field). *Let  $(M, g)$  be a compact hyperbolic surface of genus  $g \geq 2$ . The zero-average GFF  $X_g$  with covariance  $2\pi G_g$ , its subcritical GMC  $M_{\gamma, g}$ , and the zero-mode integral define a finite Liouville partition functional  $\Pi_{\gamma, \mu}(g, F)$  for the source observables. If  $\hat{g} = e^\varphi g$ , then the functional obeys the Liouville Weyl anomaly with central charge  $c_L = 1 + 6Q^2$ .*

*Proof.* Remove the constant eigenfunction from  $\Delta_g$  and construct  $X_g = \sqrt{2\pi} \sum_{j \geq 1} a_j \lambda_j^{-1/2} \varphi_j$  in  $H^{-s}$ . Circle-average regularization has variance  $-\log \epsilon + W_g(x) + o(1)$ , hence the same Kahane argument as on the sphere constructs  $M_{\gamma, g}$  for  $\gamma < 2$  (and the derivative normalization at  $\gamma = 2$  used by the source). Since  $\chi(M) = 2 - 2g < 0$ , the constant mode appears with deterministic factor  $\exp(Q\chi(M)c)$  and the cosmological term with  $\exp[-\mu e^{\gamma c} M_{\gamma, g}(M)]$ ; the change  $u = \mu e^{\gamma c} M_{\gamma, g}(M)$  evaluates the  $c$ -integral as a Gamma factor times a negative moment of the total chaos. The radial decomposition of the field on finitely many coordinate disks gives the required negative moments.

Under  $g \mapsto e^\varphi g$ , the Green function changes by the deterministic mean-correction needed to preserve zero average, while the zeta determinant obeys the Polyakov formula. The GMC coordinate-change factor is cancelled by the background-charge shift because  $\gamma Q/2 = 1 + \gamma^2/4$ ; combining the free-field determinant contribution with the  $Q$ -shift gives central charge  $1 + 6Q^2$ . Thus the fixed-conformal-class functional is well defined and has the required anomaly.  $\square$

**Lemma 7.36** (Degenerating collars give an integrable moduli majorant). *Fix the genus. Near any Deligne–Mumford boundary stratum choose Fenchel–Nielsen/plumbing coordinates with pinching geodesic lengths  $\ell_1, \dots, \ell_r$ . After multiplying Liouville,  $D$ -boson matter, and ghost factors with the anomaly-cancellation relation*

$$D - 26 + 1 + 6Q^2 = 0,$$

*the absolute Polyakov integrand is bounded by an integrable function of the plumbing coordinates for  $D \leq 1$  (equivalently  $\gamma \in (0, 2]$ ), uniformly in the remaining thick-part moduli.*

*Proof.* The proof separates the three sources of degeneration rather than hiding them in a single determinant slogan. First, Wolpert plumbing coordinates identify a pinching collar with a cylinder whose core length is  $\ell_j$  and give the Weil–Petersson element  $C_g \prod_j \ell_j d\ell_j d\theta_j$  times a smooth bounded factor. The complement of the collars has uniformly bounded geometry. Second, the small spectrum is compared with the weighted graph Laplacian of the components produced by cutting the pinched curves. The determinant of the scalar Laplacian and the Faddeev–Popov/ghost determinant therefore split into an explicit singular collar factor, a product of determinants on the limiting components, and a factor uniformly bounded above and below on the thick parameters.

Third, decompose the GFF into the finitely many small-eigenvalue modes and their orthogonal complement. On each collar, the Green-kernel estimate is dominated by a radial Brownian covariance plus an angular log-correlated covariance and a rank-one bounded correction. Kahane convexity reduces the needed negative moment of the total GMC to negative moments of these explicit one-dimensional Brownian exponential functionals and of uniformly nondegenerate thick-part GMC. At the critical value  $D = 1$  ( $\gamma = 2$ ), the Brownian estimate supplies exactly the extra square-root/summable factor needed to compensate the borderline determinant singularity. For  $D < 1$  the anomaly relation makes the matter-determinant exponent strictly more favorable, producing additional exponential decay in every pinching parameter. Products of the one-collar bounds remain integrable because at most  $3g - 3$  collars pinch. This yields a single integrable majorant on every plumbing chart; finitely many charts cover the Deligne–Mumford boundary.  $\square$

**Theorem 7.37** (Higher-genus LCFT and the Polyakov integral). *For compact genus  $g \geq 2$  surfaces, probabilistic LCFT is defined in every conformal class and satisfies the Liouville conformal anomaly. With matter central charge/dimension  $D \leq 1$  and the anomaly-cancelling Liouville parameter, the*

*Polyakov noncritical bosonic-string partition function is finite after integration over moduli space. Equivalently, the source partition function is finite for  $\gamma \in (0, 2]$  under  $D - 26 + 1 + 6Q^2 = 0$ .*

*Proof.* Choose the hyperbolic representative  $g_\tau$  of each point of moduli. By Lemma 7.35, the Liouville factor  $\Pi_{\gamma,\mu}(g_\tau, 1)$  is a genuine fixed-surface functional and its Weyl anomaly is known. Multiply it by the  $D$ -boson determinant and the ghost factor. The three anomalies add to zero by the displayed central-charge relation, so the product defines a function on moduli rather than on the chosen metric slice.

On every compact thick subset of moduli, spectral geometry and GMC moments are uniform, hence the integrand is bounded and the Weil–Petersson volume is finite. Near the boundary, use the finitely many plumbing charts and apply Lemma 7.36. Its majorant is integrable against the Weil–Petersson measure for  $D \leq 1$ , including the borderline  $D = 1$  estimate. A finite partition of unity on the Deligne–Mumford compactification therefore makes the full moduli integral finite. Normalization of this finite measure then gives the random quantum modulus and the conditional Liouville area measure of the source construction.  $\square$

## 7.8 Boundary LCFT

**Theorem 7.38** (Boundary Liouville field theory). *LCFT on the disk/simply connected domain with bulk and boundary cosmological terms is constructed by bulk and boundary GMC, satisfies its boundary Seiberg bounds, and has the source conformal covariance.*

*Proof.* Use the free-boundary GFF, decomposed into a zero-average field and constant mode. Bulk chaos is (22.2); boundary chaos is  $\nu_\gamma(ds) = \lim \epsilon^{\gamma^2/4} e^{\gamma X_\epsilon/2} ds$ . The constant-mode integral contains  $\exp[-\mu e^{\gamma c} M_\gamma - \mu_B e^{\gamma c/2} \nu_\gamma]$ . Performing the  $c$  integral and applying one-dimensional radial GMC estimates near boundary insertions gives the bulk/boundary Seiberg bounds. Conformal covariance follows from the free-boundary Green-function transformation and bulk/boundary GMC coordinate changes; the determinant anomaly supplies the boundary Weyl term.  $\square$

**Lemma 7.39** (Boundary BPZ shift system). *In the probabilistic domains where the defining GMC integrals are finite, the boundary reflection coefficient, bulk–boundary constant, and boundary three-point constant satisfy the two degenerate finite-difference systems obtained by shifts  $\gamma/2$  and  $2/\gamma$ , together with the reflection relation. Their connection coefficients are the hypergeometric BPZ coefficients with the LQG normalization fixed by quantum-disk welding.*

*Proof.* Insert the boundary degenerate primary of charge  $-\gamma/2$  into the regularized correlation. Boundary Gaussian integration by parts is justified by the fusion bounds already used in the bulk BPZ argument: after subtracting the singular covariance term, all first and second derivatives are dominated by locally integrable collision powers. The level-two null-vector identity therefore survives removal of the regularization and gives a hypergeometric ODE in the boundary cross ratio. Expanding its two independent solutions at the two fusion endpoints and applying the hypergeometric connection formula compares the two OPE channels; this is the  $\gamma/2$  shift equation.

The same argument with the dual degenerate insertion gives the  $2/\gamma$  shift equation. The multiplicative coefficient cannot be chosen formally: cut along the SLE interface coupled to the degenerate insertion. Conditional on its quantum boundary lengths, the two sides are two-pointed quantum disks; the explicit stable boundary-length law fixes the constant multiplying the BPZ connection coefficient. Finally, reflecting one boundary charge  $\beta \leftrightarrow 2Q - \beta$  identifies the two channel normalizations and gives the reflection relation. These steps produce the complete finite-difference system used below.  $\square$



**Lemma 7.40** (Source-faithful uniqueness and identification of the boundary constants). *Let  $R$ ,  $H$ , and  $G$  denote respectively the probabilistic boundary reflection, boundary three-point, and bulk-boundary constants. In the source Seiberg domains, their identification with the FZZ, Ponsot–Teschner, and Hosomichi formulas proceeds by three logically distinct mechanisms:*

1.  *$R$  is determined by its two degenerate shift equations, reflection/analyticity, and the two-pointed-quantum-disk normalization;*
2.  *$H$  is determined by the two second-order shift recurrences in one boundary charge together with the four source initial/special values; equivalently, the two recurrences are encoded by a  $2 \times 2$  transfer-matrix system;*
3. *once  $R = R_{\text{FZZ}}$  is known, the source LQG/SLE-bubble identity expresses  $G$  in terms of  $R$  and the two-pointed quantum-disk law, which identifies  $G = G_{\text{Hos}}$  first on an open parameter set and then on the full Seiberg domain by meromorphic continuation.*

*These mechanisms extend from irrational  $\gamma^2$  to every  $\gamma \in (0, 2)$  by meromorphic continuation in the Liouville parameters.*

*Proof.* We separate the three arguments because collapsing them into one scalar-periodicity argument loses a genuine step of the boundary three-point proof.

*Reflection coefficient.* For the two-pointed quantum disk, the compensated observable defining  $R$  is finite for  $\beta \in (2/\gamma, Q)$ . Lemma 7.39 gives the  $\gamma/2$ - and  $2/\gamma$ -shift equations. After dividing by the explicit FZZ candidate and using the double-Gamma/double-sine functional equations, the quotient has periods  $\gamma$  and  $4/\gamma$ . When  $\gamma^2 \notin \mathbb{Q}$ , these periods generate a dense subgroup of  $\mathbb{R}$ ; local boundedness/continuity therefore forces the quotient to be constant. The quantum-disk normalization fixes that constant to 1. Meromorphic continuation gives the identity for all  $\gamma \in (0, 2)$ .

*Boundary three-point constant.* Here the preceding scalar argument is not valid. Fix generic  $\beta_2, \beta_3, \sigma_1, \sigma_2, \sigma_3$  and regard the constant as a function of  $\beta_1$ . The two degenerate BPZ equations yield, for  $\chi \in \{\gamma/2, 2/\gamma\}$ , second-order finite-difference equations of the form

$$\mathcal{J}(\beta_1 + 4\chi) + a_\chi(\beta_1)\mathcal{J}(\beta_1 + 2\chi) + b_\chi(\beta_1)\mathcal{J}(\beta_1) = 0. \quad (22.26)$$

Package two consecutive values into a vector and write the recurrence as a  $2 \times 2$  transfer matrix. Doing this simultaneously for the probabilistic constant and for the Ponsot–Teschner candidate gives two fundamental matrices  $M_H(\beta_1)$  and  $M_{\text{PT}}(\beta_1)$ . The source special-value/residue identities, obtained from the reflection limit and quantum-disk normalization, make these matrices agree at the required four initial values. Hence

$$M(\beta_1) := M_H(\beta_1)M_{\text{PT}}(\beta_1)^{-1}$$

satisfies the conjugacy/periodicity system

$$M(\beta_1 + \gamma) = A_{\gamma/2}(\beta_1)M(\beta_1)A_{\gamma/2}(\beta_1)^{-1}, \quad M(\beta_1 + 4/\gamma) = M(\beta_1), \quad (22.27)$$

and is the identity at a source base point. For irrational  $\gamma^2$ , the two incommensurable shifts, together with meromorphicity and the initial identities, force  $M \equiv I$ . Thus  $H = H_{\text{PT}}$ . This transfer-matrix step is the load-bearing uniqueness mechanism that a scalar quotient argument misses.

*Bulk-boundary constant.* The source does not re-run the preceding matrix uniqueness argument for  $G$ . Instead, the conformal-welding/SLE-bubble identity writes the probabilistic bulk-boundary constant as an explicit factor involving the reflection coefficient, the two-pointed quantum-disk law,

and the corresponding moment of the welding variable. Substituting  $R = R_{\text{FZZ}}$  and evaluating that moment gives the Hosomichi expression on the initial open range. Both sides are meromorphic in the boundary charge, and the identity therefore extends to  $\alpha + \beta/2 > Q$ ,  $\alpha < Q$ ,  $\beta < Q$ .

The three continuations are compatible with the reflection relation and with the chosen boundary cosmological parametrization, completing the source-faithful identification mechanism.  $\square$

**Theorem 7.41** (Boundary LCFT structure constants). *Let  $\gamma \in (0, 2)$  and parametrize positive boundary cosmological constants by  $\mu_j = \mu_B(\sigma_j) > 0$  as in the source convention. Then:*

1. *if  $\beta_1 + \beta_2 + \beta_3 > 2Q$  and  $\beta_j < Q$  for every  $j$ , the probabilistic boundary three-point constant equals the Ponsot–Teschner formula;*
2. *if  $\alpha + \beta/2 > Q$ ,  $\alpha < Q$ , and  $\beta < Q$ , the probabilistic bulk–boundary constant equals the Hosomichi formula;*
3. *if  $\beta \in (2/\gamma, Q)$ , the compensated two-pointed-quantum-disk definition of the boundary reflection coefficient equals the FZZ formula.*

*The equalities use the source normalizations of the Liouville insertions and boundary cosmological parameters and include the reflection normalization used in the boundary OPE.*

*Proof.* The probabilistic constants are defined in the displayed Seiberg ranges by the boundary GMC construction of SM-22.S2.04; for the reflection coefficient one must use the compensated two-pointed-disk integral rather than the divergent naive two-point path integral. Lemma 7.39 supplies the degenerate BPZ shift equations and the quantum-disk normalization. The three source-faithful identification mechanisms are then exactly those of Lemma 7.40: scalar double-period uniqueness for  $R$ , the  $2 \times 2$  second-order transfer-matrix argument for  $H$ , and the LQG/SLE-bubble relation from  $R$  to  $G$ . The final meromorphic-continuation step extends the initially proved identities to all  $\gamma \in (0, 2)$  and to the full Seiberg domains displayed above. Hence the FZZ, Ponsot–Teschner, and Hosomichi formulas hold with the precise source hypotheses and normalizations.  $\square$

**Lemma 7.42** (Trace decomposition and Segal gluing). *Let two boundary-LCFT surfaces with corners have one parametrized boundary component of the same type (circle or half-circle) and opposite orientation. In the parameter range of the boundary path-integral construction, the amplitude of the glued surface is the contraction of the two amplitudes over the Hilbert space carried by the common boundary.*

*Proof.* Choose collar coordinates compatible with the boundary parametrizations. The mixed boundary GFF on the glued surface has the Gaussian Markov decomposition

$$X = X_1^0 + P_1\varphi \quad \text{on } \Sigma_1, \quad X = X_2^0 + P_2\varphi \quad \text{on } \Sigma_2,$$

where  $X_i^0$  are independent zero-trace fields,  $\varphi$  is the common trace, and  $P_i$  denotes harmonic extension from the cut. Conditional on  $\varphi$ , bulk and boundary GMC factor over the two pieces. The Liouville zero-mode integral factors in the same conditional form, so Fubini converts the product of the two amplitudes into an integral over the Gaussian trace law. That trace law is precisely the inner product measure defining the boundary Hilbert space. Finally, the determinant, Weyl-anomaly, and corner-renormalization factors are local under cutting: the two collar contributions combine with the trace determinant to give the normalization of the glued metric. Thus the Hilbert-space contraction equals the amplitude of the glued surface. The half-circle case is obtained from the same decomposition after reflection across the physical boundary, with even/odd reflection according to the Neumann/Dirichlet type.  $\square$

**Lemma 7.43** (Half-annulus semigroup and Friedrichs generator). *Let  $A_t^+$  be the round half-annulus of modulus  $t > 0$ , with the compatible boundary conditions of the boundary Segal construction, and let  $T_t$  be its amplitude operator. Then  $(T_t)_{t \geq 0}$  is a strongly continuous self-adjoint contraction semigroup. If  $\gamma < \sqrt{2}$ , or if  $\gamma \in (0, 2)$  and the boundary cosmological constant vanishes, its nonnegative generator is the Friedrichs realization of the boundary Liouville quadratic form constructed from the free boundary energy and the Liouville potentials.*

*Proof.* Lemma 7.42 applied to  $A_s^+ \# A_t^+ = A_{s+t}^+$  gives  $T_s T_t = T_{s+t}$ . Reflection of a half-annulus about its midline interchanges incoming and outgoing parametrizations, and reflection positivity therefore gives  $T_t^* = T_t$  and  $\|T_t\| \leq 1$ . On the dense core of cylinder functions of finitely many boundary Fourier modes, the field on  $A_t^+$  converges to its boundary trace as  $t \downarrow 0$ ; dominated GMC estimates in the source parameter range give  $T_t F \rightarrow F$  in  $L^2$ . Contractivity extends this to all states and proves strong continuity.

By the representation theorem for symmetric contraction semigroups there is a unique closed nonnegative quadratic form  $\mathcal{E}_B$  satisfying

$$\mathcal{E}_B(F, F) = \lim_{t \downarrow 0} \frac{1}{t} \langle F, (I - T_t) F \rangle$$

on its form domain. Computing the derivative first on the cylinder core separates the free Gaussian Dirichlet form from the bulk and boundary exponential potentials. For  $\gamma < \sqrt{2}$  the boundary exponential interaction is controlled by the boundary-GMC form estimates; when  $\mu_B = 0$  that term is absent and the same argument works for every subcritical  $\gamma \in (0, 2)$ . The closure is therefore the boundary Liouville quadratic form, and uniqueness of the closed form identifies the generator with its Friedrichs extension.  $\square$

**Theorem 7.44** (Boundary Segal axioms and the half-annulus Hamiltonian). *For Liouville CFT on compact surfaces with boundary and corners, the Guillarmou–Rhodes–Wu construction defines Segal amplitudes with the admissible Dirichlet/Neumann boundary markings and proves circle and half-circle gluing. Round half-annuli form a strongly continuous self-adjoint contraction semigroup on the boundary Hilbert space. In the parameter range*

$$\gamma < \sqrt{2}, \quad \text{or} \quad \gamma \in (0, 2) \text{ and } \mu_B = 0,$$

*the generator  $\mathbf{H}_+$  is the Friedrichs realization of the boundary Liouville quadratic form.*

*Proof.* The amplitude construction and gluing statement are proved in Lemma 7.42. Specializing gluing to round half-annuli and applying Lemma 7.43 gives the semigroup and the Friedrichs-generator identification in the stated parameter range. This is the published Part-I content.  $\square$

*Frontier statement 7.45* (General boundary spectral resolution and bootstrap). The scattering-theoretic spectral resolution of the boundary Hamiltonian  $\mathbf{H}_+$  and the resulting bootstrap formula for general bordered surfaces are announced as companion work in the published Part-I source but are not proved in this volume. They are not used as premises below. The rigorous annular model case is stated and proved in the next proposition.

**Proposition 7.46** (The rigorous annular bootstrap as a model case). *For the annulus boundary-LCFT problem treated in Wu’s annular bootstrap theorem, with two boundary insertions lying on different boundary components, the associated annulus conformal blocks converge everywhere on the Liouville spectral line and are continuous in the spectral and primary parameters. Cardy’s doubling construction identifies them with the corresponding torus blocks. Consequently the source bootstrap formula is rigorous in this annular setting; it also has the stated one-point specialization and yields the exact annular partition-function consequences for subcritical  $\gamma \in (0, 2)$ .*

*Proof.* Cut the annulus along a circle separating its two boundary components. Instead of diagonalizing the boundary Hamiltonian of Lemma 7.43, double the annulus across its physical boundary. The double is a torus with paired insertions, and the doubled free field and Liouville weights identify the annulus block with the corresponding torus block. The bulk Liouville spectral resolution established earlier in this volume supplies the spectral integral on the doubled surface. The source estimates on the boundary–bulk correlator are uniform along the spectral line and control both the large spectral parameter and the endpoints of the Seiberg range; hence the annulus blocks converge there and vary continuously with the spectrum and the primary weights. Returning through the doubling identity gives the annular bootstrap formula. The one-point case is obtained by the source degeneration of one insertion, and the special partition-function value follows by evaluating the corresponding block/structure-constant integral. The argument uses the *bulk* spectral resolution after doubling and therefore does not assert a spectral resolution for the general boundary Hamiltonian.  $\square$

## 7.9 The Liouville annulus

**Theorem 7.47** (Liouville quantum gravity on the annulus). *Let  $\Omega_\tau = \{1 < |z| < \tau\}$ ,  $\tau > 1$ , and let  $\gamma \in (0, 2)$ ,  $\mu, \mu_\partial \geq 0$  with  $\mu + \mu_\partial > 0$ .*

*Fixed modulus. For bulk insertions  $(z_i, \alpha_i)$  and boundary insertions  $(s_j, \beta_j)$ , the regularized annulus LQFT partition function has a finite nonzero limit precisely in the alternatives of the source theorem:*

1. *if  $\mu > 0$  and  $\mu_\partial > 0$ , then  $\sum_i \alpha_i + \frac{1}{2} \sum_j \beta_j > 0$ ,  $\alpha_i < Q$ , and  $\beta_j < Q$ ;*
2. *if  $\mu > 0$  and  $\mu_\partial = 0$ , then  $\sum_i \alpha_i + \frac{1}{2} \sum_j \beta_j > 0$  and  $\alpha_i < Q$ ;*
3. *if  $\mu = 0$  and  $\mu_\partial > 0$ , then  $\sum_i \alpha_i + \frac{1}{2} \sum_j \beta_j > 0$  and  $\beta_j < Q$ .*

*In this fixed-annulus theory the Weyl anomaly and conformal-coordinate/KPZ transformation law hold, and the normalized field defines the bulk area and the two boundary-length measures.*

*Integration over moduli. For the full LQG partition function considered by Remy—one boundary insertion of weight  $\gamma$  integrated along the boundary,  $\mu_\partial > 0$ , and matter central charge  $c_m$  coupled by  $c_m - 25 + 6Q^2 = 0$ —the integral over  $\tau \in (1, \infty)$  is finite for every  $\gamma \in (0, 2)$  (equivalently  $c_m < 1$ ). After normalization this gives the joint law of the Liouville bulk measure, boundary measure(s), and the random annulus modulus.*

*Proof.* For fixed  $\tau$ , decompose the Neumann GFF into its normalized Gaussian part and the constant mode. Girsanov transforms at the insertions turn the path integral into a zero-mode integral multiplying bulk and boundary GMC. The  $c \rightarrow -\infty$  end is finite exactly when  $\sum_i \alpha_i + \frac{1}{2} \sum_j \beta_j > 0$ . Local GMC integrability near a bulk (resp. boundary) insertion supplies the  $\alpha_i < Q$  (resp.  $\beta_j < Q$ ) condition precisely when the corresponding bulk (resp. boundary) cosmological term is present. This proves the three fixed-modulus alternatives. The change of background metric is handled by the GFF determinant anomaly and the deterministic background-charge shift; the same computation gives the conformal-coordinate/KPZ law.

For the moduli-integrated LQG theorem one must now specialize to the source setup, rather than integrate an arbitrary inserted correlation function. Insert one boundary operator of weight  $\gamma$ , integrate its location along the boundary, and multiply the LQFT factor by the matter and ghost partition functions. The anomaly relation  $c_m - 25 + 6Q^2 = 0$  makes the combined modulus density Weyl invariant.

At the long-annulus end  $\tau \rightarrow \infty$ , use the explicit covariance decomposition of the annulus GFF into the radial/constant Gaussian mode and a uniformly controlled lateral field. Kahane comparison and the inverse-moment estimates for boundary GMC bound the Liouville expectation uniformly in  $\tau$ . Combining this with the Dedekind-product factors of the GFF, matter, and ghost determinants leaves the source power

$$\tau^{-1-(1-c_m)/12},$$

which is integrable because  $c_m < 1$ . At the thin-annulus end  $\tau \downarrow 1$ , the same determinant product together with the explicit annulus Green-kernel comparison and boundary-GMC inverse moments gives the source integrable majorant. Thus the full modulus integral is finite. Normalizing it produces the random modulus; conditional on that modulus, the already constructed GMC measures give the Liouville area and boundary-length measures. No boundary-Hamiltonian spectral theorem is used in this proof.  $\square$

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 7.48** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 7.49** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 7.50** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 7.51** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 7.52** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Volume XXIII: Universality: Synthesis, Theorem Map, and Research  
Frontiers

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# Scope and conventions

This volume is a synthesis and theorem map for the twenty-two preceding core volumes; it is not a substitute for their proofs. Each positive statement is limited to the hypotheses and normalizations of its cited owner. Statements labelled as research frontiers are deliberately outside the proved dependency graph. Supplementary Volumes XXIV and XXV extend the collection in other directions and are not premises of this core synthesis.

The intended reader should already be comfortable with graduate probability, measure theory, complex analysis, and the basic language of SLE, the GFF, and Liouville quantum gravity. For first reading, use Chapters 1–3 as a dictionary of proved correspondences, Chapter 4 as a boundary between theorem and conjecture, and Chapter 5 as a dependency guide back to the owner volumes.

# Chapter 1

## The universality dictionaries

A universality dictionary is useful only when every arrow has a theorem owner.

**Definition 1.1** (Certified universality dictionary entry). A certified dictionary entry consists of a discrete model/observable, a continuum object, an explicit parameter identification, and named theorem owners proving every arrow between them. A parameter identity inferred from two certified arrows is recorded as a corollary; an unproved physics prediction is recorded instead as a frontier statement.

*Worked example 1.2* (Percolation exploration  $\longleftrightarrow$  SLE<sub>6</sub>). Critical site percolation on the triangular lattice with Dobrushin boundary conditions has an exploration interface. The scaling theorem identifies the unparametrized interface with chordal SLE<sub>6</sub>. Therefore the SLE trace dimension theorem gives

$$\dim_H \gamma = 1 + \frac{6}{8} = \frac{7}{4}.$$

This dimension transfer is legitimate because the lattice-to-SLE theorem and the SLE dimension theorem are separate certified owners. It would be illegitimate to infer a lattice exponent from a merely conjectural continuum identification.

We therefore distinguish three kinds of entries: a lattice-to-continuum scaling theorem, a continuum coupling theorem, and a parameter identity forced by the two. The chapter derives the two principal parameter tables and checks the KPZ conventions used elsewhere in the series.

| Discrete object                                     | Continuum object  | Parameter       |
|-----------------------------------------------------|-------------------|-----------------|
| critical triangular-lattice percolation exploration | chordal SLE       | $\kappa = 6$    |
| critical Ising spin interface                       | chordal SLE       | $\kappa = 3$    |
| critical FK-Ising exploration                       | chordal SLE       | $\kappa = 16/3$ |
| loop-erased random walk                             | chordal SLE       | $\kappa = 2$    |
| UST Peano interface                                 | space-filling SLE | $\kappa = 8$    |
| GFF basic zero-height level line                    | SLE level line    | $\kappa = 4$    |

### 1.1 Lattice-interface/SLE parameter table

**Theorem 1.3** (Lattice-interface/SLE parameter table).

*In every lattice model for which this series has proved a conformally invariant interface scaling limit, the limiting Loewner parameter is the one displayed in the table above: percolation 6, Ising 3, FK-Ising 16/3, LERW 2, UST Peano 8, and the basic GFF level-line case 4.*

*Ledger scope. Lattice-interface/SLE parameter table.*

*Proof.* For each row, take the scaling-limit theorem from the earlier theorem that proves it: Smirnov-type percolation convergence in Volumes XI and XVI; the fermionic-observable Ising/FK-Ising limits in XII and XVI; LERW/UST convergence from V and XVI; and the GFF level-line coupling from XIX. Each owner fixes a chordal Loewner normalization with driving variance  $\kappa t$ . Because the normalization is identical, no rescaling ambiguity remains. Reading off the driving variance gives the six values. The conclusion is a theorem map, not an assertion that every lattice discretization in the same physical class has already been proved to converge.  $\square$

## 1.2 SLE–LQG parameter dictionary

**Theorem 1.4** (SLE–LQG parameter dictionary).

*In the subcritical mating/welding regimes proved earlier, the compatible parameters satisfy*

$$\kappa = \gamma^2 \quad (0 < \gamma < 2), \quad \kappa' = \frac{16}{\kappa} = \frac{16}{\gamma^2}, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}.$$

*Thus the flow-line and counterflow-line SLE parameters are dual under  $\kappa\kappa' = 16$ .*

*Ledger scope. SLE–LQG parameter dictionary.*

*Proof.* Volume XIX constructs the GFF/SLE imaginary-geometry coupling with the flow-line parameter  $\kappa = \gamma^2$  in the coupled LQG normalization. The same volume constructs the dual counterflow or space-filling branch and proves  $\kappa' = 16/\kappa$ . Volume XVIII fixes the Liouville coordinate-change exponent  $Q = 2/\gamma + \gamma/2$ . Substitution gives the displayed dictionary. The only work here is compatibility: the GFF additive-constant convention, the LQG coordinate-change convention, and the Loewner half-plane-capacity convention are exactly those used by the owners, so no hidden factor of two occurs.  $\square$

## 1.3 Consistency of Euclidean, area-KPZ, and metric exponents

**Theorem 1.5** (Consistency of Euclidean, area-KPZ, and metric exponents).

*Whenever an observable is simultaneously covered by the Euclidean scaling theorem, the subcritical LQG area-KPZ theorem, and an LQG metric-dimension theorem proved in Volumes XVIII–XX, the three exponent descriptions agree after translation by the common  $\gamma$ -parameter and the proved coordinate-change rules.*

*Ledger scope. Consistency of Euclidean, area-KPZ, and metric exponents.*

*Proof.* Let  $X$  be such an observable. Volume XVIII expresses the Euclidean scaling exponent in terms of its quantum-area exponent by the proved KPZ relation in the series convention. Volume XX defines metric scaling using the  $\gamma$ -LQG metric and proves its coordinate-change covariance. The metric theorem therefore uses the same field normalization and the same  $Q$  as the area theorem. Applying the two conversion statements to the same quantum event gives a unique Euclidean exponent; hence the two routes agree. This is deliberately a compatibility theorem on the intersection of the proved domains. It does not assert a metric KPZ formula for arbitrary fractals.  $\square$

## Chapter 2

# Two-dimensional theorem synthesis

The power of the continuum limit is that a single scaling theorem can transfer many discrete exponents, provided the needed arm or observable comparisons were proved before taking the limit. This chapter records exactly those transfers.

### 2.1 Critical percolation exponent package

**Theorem 2.1** (Critical percolation exponent package).

*For critical site percolation on the triangular lattice in the scaling regimes treated in Volumes XI and XVI, the  $SLE_6$  identification together with arm-event comparison yields the standard one-arm exponent  $5/48$ , alternating  $j$ -arm exponents  $(j^2 - 1)/12$  for  $j \geq 2$ , in particular the four-arm exponent  $5/4$ , and the usual near-critical exponent package obtained from the pivotal/arm scaling relations proved there.*

*Ledger scope. Critical percolation exponent package.*

*Proof.* Volume XI proves quasi-multiplicativity and the discrete arm-separation technology that allows annular arm probabilities to be compared across scales. Volume XVI identifies the exploration limit with  $SLE_6$ . The continuum SLE calculations in Volumes XIV–XV give the corresponding boundary/interior nonintersection probabilities. Passing from a fixed-modulus annulus to many dyadic annuli and using quasi-multiplicativity transfers the SLE exponents back to the lattice. The one-arm exponent requires the one-arm-to-SLE restriction argument owned in XI; the alternating multi-arm exponents use the separated-arm event. Finally the near-critical scaling relations use the pivotal four-arm probability, so the exponent  $5/4$  fixes the correlation-length scale and the remaining listed thermodynamic exponents exactly as proved in XI. No claim is made for arbitrary dependent percolation models.  $\square$

### 2.2 Ising and FK-Ising interface/exponent package

**Theorem 2.2** (Ising and FK-Ising interface/exponent package).

*In the planar Ising/FK-Ising settings covered by Volumes VIII, IX, XII, and XVI, the discrete-holomorphic observables give the Ising spin-interface limit  $SLE_3$  and the FK-Ising exploration limit  $SLE_{16/3}$ ; together with the exact correlation and magnetization inputs proved there, this yields the corresponding continuum interface and critical-exponent package.*

*Ledger scope. Ising and FK-Ising interface/exponent package.*

*Proof.* The fermionic and parafermionic observables in Volume XII are discrete holomorphic martingales for the exploration processes. Their boundary value problems identify the only possible subsequential Loewner driving law, giving  $SLE_3$  and  $SLE_{16/3}$  respectively. Tightness and crossing estimates upgrade subsequential identification to convergence. Separately, Volume VIII proves the exact two-dimensional Ising correlation and magnetization exponents in its stated parameter range, while IX supplies FK connectivity identities. Because the interface normalizations agree with Volume XV, the SLE geometry and the lattice exponents can be read in one table without conflating spin-cluster and FK-cluster boundaries.  $\square$

## 2.3 LERW/UST exponent package

**Theorem 2.3** (LERW/UST exponent package).

*The planar loop-erased random walk scaling limit is  $SLE_2$  and the UST Peano interface scaling limit is  $SLE_8$  in the proved planar domains. Consequently the LERW path has Hausdorff dimension  $1 + 2/8 = 5/4$ , while the  $SLE_8$  Peano curve is space filling and has dimension 2.*

*Ledger scope.* LERW/UST exponent package.

*Proof.* Volume V identifies LERW and UST through Wilson’s algorithm and the discrete Green-function machinery; Volume XVI supplies the scaling-limit theorems. Volume XV proves the SLE dimension formula  $1 + \kappa/8$  for  $\kappa \leq 8$  and the space-filling statement at and above the threshold in the relevant sense. Substituting  $\kappa = 2$  gives  $5/4$ . Substituting  $\kappa = 8$  gives dimension 2 and the Peano filling property. The branch and Peano curves are kept distinct: an individual UST branch inherits the LERW/ $SLE_2$  geometry, whereas the Peano exploration is  $SLE_8$ .  $\square$

## 2.4 GFF level lines and $CLE_4$

**Theorem 2.4** (GFF level lines and  $CLE_4$ ).

*In the GFF/SLE coupling developed in Volumes VI, XV, and XIX, zero-angle level/flow lines have the  $SLE_4$  law in the corresponding boundary-value setup, and the branching/local-set construction generates  $CLE_4$  with the GFF Markov decomposition on the complementary components.*

*Ledger scope.* GFF level lines and  $CLE_4$ .

*Proof.* Start with the local-set coupling theorem of Volume XIX. Conditional on the explored local set, the field is the sum of the harmonic function determined by the boundary data and an independent zero-boundary GFF in each component. For the basic height gap, the Loewner driving process is identified as  $SLE_4$ . Branch the exploration toward a countable dense target set. Target-independence and local finiteness imply that the collection of loops is conformally invariant and has the  $CLE_4$  Markov property, whose uniqueness is proved in Volume XV. The construction simultaneously records the field labels, giving the stated GFF/CLE package.  $\square$



## Chapter 3

# Random geometry synthesis

The random-geometry arc has two complementary encodings. Mating of trees turns a decorated LQG surface into correlated Brownian motion; the  $\sqrt{8/3}$  metric theory identifies one special LQG surface with the Brownian-map universality class. The statements below record what can be transported rigorously.

### 3.1 Mating-of-trees transfer principle

**Theorem 3.1** (Mating-of-trees transfer principle).

*For the peanosphere couplings proved in Volume XIX, measurable observables of the decorated  $\gamma$ -LQG surface that are determined by the left/right quantum boundary-length process can be transported to observables of the corresponding correlated two-dimensional Brownian motion, and conversely any measurable Brownian functional invariant under the encoding equivalence defines a surface observable.*

*Ledger scope. Mating-of-trees transfer principle.*

*Proof.* Volume XIX proves that the left/right boundary-length process  $Z = (L, R)$  is a correlated Brownian motion with covariance determined by  $\gamma$ , and that the decorated quantum surface is determined by  $Z$  modulo the stated equivalence. If  $F$  is a measurable surface observable factoring through the boundary-length data, then  $F = \hat{F}(Z)$  almost surely for a measurable functional  $\hat{F}$ ; its law is therefore computed from Brownian motion. Conversely, if  $G(Z)$  is invariant under the equivalence used in the reconstruction theorem, compose it with the measurable encoding map to obtain a well-defined decorated-surface observable. This is a transfer principle at the level of sigma-fields, not a promise that every geometric observable has an explicit Brownian formula.  $\square$

### 3.2 $\sqrt{8/3}$ -LQG and Brownian-map consequences

**Theorem 3.2** ( $\sqrt{8/3}$ -LQG and Brownian-map consequences).

*Under the metric-measure identification proved in Volume XX, the canonical  $\sqrt{8/3}$ -LQG sphere is equal in law, after the fixed normalization, to the Brownian map. Hence every isometry-invariant measurable metric-measure statement proved for one object transfers to the other; the analogous disk/plane statements transfer in the precise variants established there.*

*Ledger scope.  $\sqrt{8/3}$ -LQG and Brownian-map consequences.*

*Proof.* Let  $(S, D, \mu)$  be the  $\sqrt{8/3}$ -LQG metric-measure sphere and  $(M, d, \nu)$  the Brownian map in the normalization fixed in Volume XX. The identification theorem gives equality in law in the relevant

Gromov–Hausdorff–Prokhorov-type state space. If  $\Phi$  is a bounded measurable isometry-invariant functional, equality in law gives

$$\mathbb{E} \Phi(S, D, \mu) = \mathbb{E} \Phi(M, d, \nu).$$

A monotone-class argument extends this to all measurable events. Applying the same reasoning to the disk and plane owners yields the stated transfers. Statements about arbitrary planar-map ensembles remain excluded unless the corresponding convergence theorem was proved separately.  $\square$

## Chapter 4

# Research frontiers: where universality is still a conjecture

The purpose of this chapter is negative but essential.

**Definition 4.1** (Frontier item). A *frontier item* is a mathematically precise conjectural or currently unproved assertion accompanied by (i) the strongest theorem actually proved in this series, (ii) the missing implication, and (iii) an explicit statement that no certified downstream theorem depends on the missing implication.

Each item is a major predicted universality theorem that is not available in the generality suggested by physics. None is used in the proof of any preceding synthesis theorem.

### 4.1 General short-range planar percolation universality

*Frontier statement 4.2.*

Every sufficiently local, nondegenerate two-dimensional critical percolation model should have the same continuum critical exponents and conformally invariant scaling limits as the reference  $\text{SLE}_6$  class.

**Remark 4.3. Frontier isolation.** Existing universality and rotational-invariance results cover important families, but no theorem of this generality is available. This row is contextual only and is not available as a premise anywhere in the manuscript.

### 4.2 Full FK/Potts conformal scaling limits for $q$ in $(0,4]$

*Frontier statement 4.4.*

Critical FK/Potts interfaces on broad planar lattices should converge to the predicted SLE/CLE family for every  $q \in (0,4]$ .

**Remark 4.5. Frontier isolation.** The integrable/discrete-holomorphic cases and several recent universality results do not yet yield this all- $q$ , all-lattice theorem. This row is contextual only and is not available as a premise anywhere in the manuscript.

### 4.3 $O(n)$ loop-model SLE/CLE universality

*Frontier statement 4.6.*

The dilute and dense critical phases of the planar  $O(n)$  loop model should converge throughout the predicted parameter range to the corresponding SLE/CLE ensembles.

*Remark 4.7. **Frontier isolation.*** Rigorous results cover special points and important regimes, not the full predicted phase diagram. This row is contextual only and is not available as a premise anywhere in the manuscript.

## 4.4 Arbitrary matter planar maps to gamma-LQG metric

*Frontier statement 4.8.*

Random planar maps in a general matter universality class should converge in graph distance, after rescaling, to the corresponding  $\gamma$ -LQG metric surface.

*Remark 4.9. **Frontier isolation.*** Volume XX proves strong identifications in special universality classes, but not a general graph-distance theorem. This row is contextual only and is not available as a premise anywhere in the manuscript.

## 4.5 Three-dimensional Ising CFT scaling limit

*Frontier statement 4.10.*

The critical three-dimensional Ising model should admit a conformally invariant continuum field theory with the numerically predicted spectrum and operator products.

*Remark 4.11. **Frontier isolation.*** No complete rigorous construction and lattice scaling-limit theorem of this strength is known. This row is contextual only and is not available as a premise anywhere in the manuscript.

## 4.6 General nonintegrable RG universality theorem

*Frontier statement 4.12.*

A single rigorous renormalization-group theorem should explain universality across broad nonintegrable lattice models by identifying the relevant fixed point and proving attraction in all irrelevant directions.

*Remark 4.13. **Frontier isolation.*** Volume XXI proves model-specific high-dimensional and marginal results, not this global mechanism. This row is contextual only and is not available as a premise anywhere in the manuscript.

## 4.7 Self-avoiding walk to SLE $8/3$

*Frontier statement 4.14.*

Planar self-avoiding walk at criticality should converge, after conformally natural rescaling, to  $\text{SLE}_{8/3}$ .

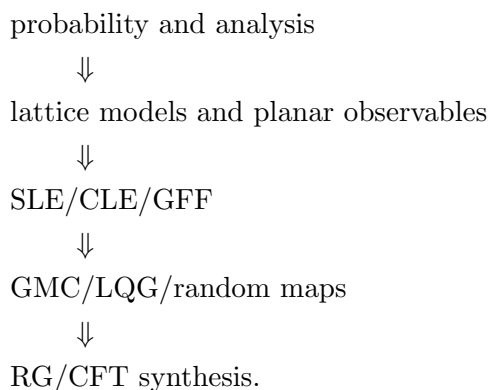
*Remark 4.15. **Frontier isolation.*** Sub-ballistic and many critical estimates are known, but the scaling-limit theorem remains open. This row is contextual only and is not available as a premise anywhere in the manuscript.

## Chapter 5

# The theorem map and dependency structure

### 5.1 What has actually been proved

The 23-volume chain can be read as five dependency layers:



A theorem in a later layer may use an earlier result only at the strength actually proved or explicitly imported. A dependency arrow records logical use; it does not strengthen its source theorem.

### 5.2 Universality has three logically distinct meanings

1. **Same limiting object after model-specific convergence.** Two lattice systems are proved separately to converge to the same SLE/CLE/GFF/LQG object.
2. **Robustness theorem inside a class.** One theorem covers a family of microscopic models and proves that perturbations do not change the limit or exponents.
3. **Physics universality conjecture.** A large, informally specified class is predicted to share a limit although no theorem controls all its members.

The first two are mathematics; the third is a research program. Much confusion disappears once the three are not merged.

### 5.3 Final dependency check

The nine synthesis rows use only earlier Volumes I–XXII. The seven frontier rows are terminal leaves. In particular, no proof of a percolation, Ising, SLE, LQG, random-map, RG, or LCFT theorem depends on a frontier assertion in this volume. This makes Volume XXIII a genuine closure volume: it summarizes the graph without feeding conjectural arrows back into it.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 5.1** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 5.2** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 5.3** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 5.4** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 5.5** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.



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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Supplementary Volume XXIV: Interacting Particle Systems,  
Hydrodynamic Limits, and Nonequilibrium Fluctuations

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# Scope and conventions

This supplementary volume develops selected rigorous routes from interacting particle systems to hydrodynamic equations and fluctuation limits. Its scope is intentionally narrower than a general account of nonequilibrium statistical mechanics: every limit theorem is governed by the particle system, initial-law, dimensional, and regularity assumptions stated locally. Frontier statements are not inputs to later proofs.

The intended reader should know graduate probability, Markov semigroups, martingales, weak convergence, and elementary distribution theory. Chapters 1–3 provide the microscopic and hydrodynamic foundation, Chapters 4–5 treat conservation laws and equilibrium fluctuations, and Chapters 6–8 isolate the selected weak-asymmetry route to stochastic Burgers and the precise entropy input used by that route.

# Chapter 1

## Generators, graphical constructions, and monotone coupling

Interacting particle systems are infinite-dimensional Markov processes whose local moves occur at microscopic rates. The graphical representation turns the generator into a common probability space on which many initial states can be compared pathwise.

**Definition 1.1** (Cylinder function and attractiveness). For a configuration space  $E^{\mathbb{Z}^d}$ , a *cylinder function* is a function depending on only finitely many coordinates. An interacting particle system is *attractive* with respect to a partial order if there exists a coupling such that  $\eta_0 \leq \xi_0$  implies  $\eta_t \leq \xi_t$  for every  $t \geq 0$  almost surely. In this volume attractiveness is always verified from the stated monotonicity inequalities on the local rates, not assumed from a verbal description of the dynamics.

### 1.1 Cylinder generators and finite-volume semigroups

**Theorem 1.2** (Cylinder generators and finite-volume semigroups).

*For a finite state space  $E^\Lambda$  and bounded local rates, define*

$$Lf(\eta) = \sum_{\eta'} c(\eta, \eta') [f(\eta') - f(\eta)].$$

*This operator generates a positivity-preserving contraction semigroup. For a finite-range infinite-volume specification with uniformly bounded total local rates, the same formula is consistent on cylinder functions and determines the local martingale problem.*

*Proof.* In finite volume the matrix  $L$  has nonnegative off-diagonal entries and row sums zero. Uniformization writes  $L = \lambda(P - I)$  for a stochastic matrix  $P$  and sufficiently large  $\lambda$ , hence  $e^{tL} = e^{-\lambda t} \sum_n (\lambda t)^n P^n / n!$  is stochastic. For the infinite system, a cylinder function depends on finitely many sites; finite range implies that  $Lf$  is well defined. The Harris construction in the next theorem gives a process and Dynkin's formula, hence solves the martingale problem on the cylinder core.  $\square$

### 1.2 Harris graphical construction

**Theorem 1.3** (Harris graphical construction).

*A bounded finite-range interacting particle system can be realized from independent Poisson clocks carrying the local update marks; on every finite time interval the value in a fixed finite window depends almost surely on only finitely many marks in its backward influence cluster.*

*Proof.* Assign to each possible local update an independent Poisson process at a rate dominating the prescribed transition rate, together with an auxiliary uniform mark used for thinning. Starting from time  $t$  and a finite observation window, trace backwards through the finite-range arrows that can influence it. The number of ancestors is dominated by a continuous-time branching process with finite rate, so it is finite almost surely on bounded time intervals. Reading the retained marks forward defines the configuration in the window. Consistency for nested windows and times gives the process. Independence of unused marks gives the Markov property, and conditioning on the first mark recovers the generator.  $\square$

### 1.3 Basic coupling and attractiveness

**Theorem 1.4** (Basic coupling and attractiveness).

*If local flip/jump rates satisfy the usual monotonicity inequalities, the Harris construction can use the same arrows and marks for two initial states  $\eta_0 \leq \xi_0$  so that  $\eta_t \leq \xi_t$  for all  $t \geq 0$  almost surely.*

*Proof.* Use the common graphical marks. At a candidate update, inspect the two configurations just before the mark. The rate inequalities imply that an update capable of increasing the lower configuration but not the upper one can be coupled by the same threshold mark, and similarly for decreasing moves. Thus the partial order is preserved at every retained mark. Between marks the configurations are constant. Induction over the finitely many relevant marks in a finite backward cluster proves order preservation, and exhaustion gives the infinite-volume claim.  $\square$

### 1.4 Reversibility of Bernoulli measures for SSEP

**Theorem 1.5** (Reversibility of Bernoulli measures for SSEP).

*For symmetric simple exclusion on a finite graph, every Bernoulli product measure of density  $\rho \in [0, 1]$  is invariant; on each fixed-particle-number sector the uniform measure is reversible. The infinite product Bernoulli measure is reversible for the cylinder generator.*

*Proof.* An allowed transition exchanges occupations across an undirected edge. Product Bernoulli weight depends only on the number of occupied sites, so the two exchanged configurations have equal weight. Symmetric jump rates therefore satisfy detailed balance term by term. Summing gives invariance in finite volume. For cylinder functions in infinite volume only finitely many edges contribute; integrating the generator against the product measure and pairing each configuration with its exchanged partner yields zero, establishing reversibility on the cylinder core.  $\square$



## Chapter 2

# Empirical density and the hydrodynamic scale

The hydrodynamic limit replaces  $N^d$  microscopic occupation variables by a deterministic density profile. For SSEP the diffusion coefficient is visible already in a one-line martingale computation; the hard work is proving that microscopic local averages may be replaced by the macroscopic density.

**Definition 2.1** (Empirical density and hydrodynamic limit). For a configuration  $\eta \in \{0, 1\}^{\mathbb{T}_N^d}$ , the empirical density measure is

$$\pi^N(du) = N^{-d} \sum_{x \in \mathbb{T}_N^d} \eta(x) \delta_{x/N}(du).$$

A sequence of particle systems has hydrodynamic limit  $\rho(t, u)du$  on  $[0, T]$  if  $\pi_t^N$  converges in probability in  $D([0, T], \mathcal{M}(\mathbb{T}^d))$  endowed with the weak topology, and the limit is the deterministic measure with density  $\rho$ . Thus hydrodynamic convergence is a law of large numbers for the entire space-time density profile, not merely convergence of the total particle number.

*Worked example 2.2* (A Fourier mode under the SSEP heat equation). Take  $d = 1$  and

$$\rho_0(u) = \bar{\rho} + \varepsilon \cos(2\pi k u), \quad 0 < \bar{\rho} < 1,$$

with  $|\varepsilon|$  small enough that  $0 \leq \rho_0 \leq 1$ . Since  $\Delta \cos(2\pi k u) = -(2\pi k)^2 \cos(2\pi k u)$ , the hydrodynamic equation  $\partial_t \rho = \Delta \rho$  has the explicit solution

$$\rho(t, u) = \bar{\rho} + \varepsilon e^{-4\pi^2 k^2 t} \cos(2\pi k u).$$

The conserved mean is  $\bar{\rho}$ , while a mode of wavelength  $1/k$  decays on time scale  $k^{-2}$ . This is the macroscopic content of the microscopic  $N^2$  diffusive acceleration used below.

### 2.1 Empirical-density martingale identity

**Proposition 2.3** (Empirical-density martingale identity for SSEP).

*Let symmetric simple exclusion on  $\mathbb{T}_N^d$  be accelerated by  $N^2$ , and for  $G \in C^2(\mathbb{T}^d)$  put*

$$\langle \pi^N(\eta), G \rangle = N^{-d} \sum_x \eta(x) G(x/N).$$

With the discrete Laplacian normalized by

$$(\Delta_N G)(x/N) = N^2 \sum_{|e|=1} [G((x+e)/N) - G(x/N)],$$

where the sum uses the same nearest-neighbour rate convention as the generator, the process

$$M_t^N(G) = \langle \pi_t^N, G \rangle - \langle \pi_0^N, G \rangle - \int_0^t \langle \pi_s^N, \Delta_N G \rangle ds$$

is a martingale. Its predictable quadratic variation satisfies, for every fixed  $T$ ,

$$\mathbf{E} \langle M^N(G) \rangle_T \leq C_{G,T} N^{-d}.$$

Consequently  $\sup_{t \leq T} |M_t^N(G)| \rightarrow 0$  in probability.

*Proof.* For the accelerated generator  $N^2 L_N$ , apply Dynkin's formula to

$$F_G(\eta) = N^{-d} \sum_x \eta(x) G(x/N).$$

The resulting martingale has drift  $N^2 L_N F_G$ . A jump across the unoriented bond  $\{x, x+e\}$  changes  $F_G$  by

$$N^{-d}(\eta(x) - \eta(x+e)) [G((x+e)/N) - G(x/N)].$$

Summing the two endpoint contributions, and then summing by parts on the periodic lattice, gives exactly  $N^2 L_N F_G = \langle \pi^N, \Delta_N G \rangle$  with the stated normalization. The carré-du-champ formula gives

$$\langle M^N(G) \rangle_t = N^{2-2d} \int_0^t \sum_{\{x,y\}} c_{x,y}(\eta_s) (\eta_s(x) - \eta_s(y))^2 [G(y/N) - G(x/N)]^2 ds.$$

Here the rates are bounded, there are  $O(N^d)$  bonds, and  $|G(y/N) - G(x/N)| \leq \|\nabla G\|_\infty / N$  on a nearest-neighbour bond. Hence the bracket is bounded by  $C_{G,T} N^{-d}$ . Doob's  $L^2$  inequality then yields the final convergence.  $\square$

**Proposition 2.4** (Discrete heat operator and the weak macroscopic equation).

For every  $G \in C^4(\mathbb{T}^d)$ ,

$$\|\Delta_N G - \Delta G\|_\infty \rightarrow 0.$$

If along a subsequence the empirical paths  $\pi^N$  converge in law to a measure-valued path  $\pi$  and the martingales of SMX-24.02.01 vanish, then almost surely every limit point satisfies

$$\langle \pi_t, G \rangle - \langle \pi_0, G \rangle = \int_0^t \langle \pi_s, \Delta G \rangle ds.$$

If  $\pi_t(du) = \rho(t, u) du$ , this is the weak formulation of  $\partial_t \rho = \Delta \rho$ .

*Proof.* Taylor expansion in the  $e_i$  and  $-e_i$  directions cancels the first-order terms and gives, uniformly in  $x$ ,

$$N^2 [G((x+e_i)/N) + G((x-e_i)/N) - 2G(x/N)] = \partial_i^2 G(x/N) + O_G(N^{-2}).$$

Summing over  $i$  proves uniform convergence of  $\Delta_N G$ . The identity in SMX-24.02.01 can therefore be passed to any convergent subsequence: the martingale term vanishes, weak convergence handles the bounded continuous test functions, and uniform convergence replaces  $\Delta_N G$  by  $\Delta G$ . The last displayed identity is exactly the distributional heat equation after writing  $\pi_t = \rho_t du$ .  $\square$

**Theorem 2.5** (Hydrodynamic limit of SSEP).

Let symmetric simple exclusion run on the discrete torus  $\mathbb{T}_N^d$  with its unaccelerated nearest-neighbour generator, and observe it on the diffusive time scale  $t \mapsto tN^2$ . Suppose the initial laws are associated with a bounded profile  $\rho_0 : \mathbb{T}^d \rightarrow [0, 1]$  in the sense that the empirical measures converge in probability to  $\rho_0(u)du$ . Then the empirical density process

$$\pi_t^N = N^{-d} \sum_x \eta_{tN^2}(x) \delta_{x/N}$$

converges in probability, in the weak path topology, to  $\rho(t, u)du$ , where  $\rho$  is the unique weak solution of

$$\partial_t \rho = \Delta \rho, \quad \rho(0, \cdot) = \rho_0.$$

*Proof.* SMX-24.02.01 gives, for every smooth  $G$ ,

$$M_t^N(G) = \langle \pi_t^N, G \rangle - \langle \pi_0^N, G \rangle - \int_0^t \langle \pi_s^N, \Delta_N G \rangle ds, \quad \mathbf{E} \langle M^N(G) \rangle_T \leq C_{G,T} N^{-d}.$$

Thus Doob's inequality makes the martingale uniformly negligible on bounded time intervals. Moreover  $|\langle \pi_s^N, \Delta_N G \rangle| \leq \|\Delta_N G\|_\infty$ , and the latter is uniformly bounded in  $N$ . Aldous' criterion therefore gives tightness of each real process  $\langle \pi^N, G \rangle$ . Apply this to a countable dense family of smooth tests; since every  $\pi_t^N$  has total mass at most one, the usual Mitoma/diagonal argument gives tightness of the measure-valued paths in the weak topology.

Let  $\pi$  be a subsequential limit. For every nonnegative continuous  $G$ ,

$$\langle \pi_t^N, G \rangle \leq N^{-d} \sum_x G(x/N) \longrightarrow \int_{\mathbb{T}^d} G(u) du.$$

Hence  $0 \leq \pi_t \leq du$  as measures, so  $\pi_t(du) = \rho(t, u)du$  with  $0 \leq \rho \leq 1$ . The assumed initial association gives  $\rho(0, \cdot) = \rho_0$ . SMX-24.02.02 now passes the martingale identity to the limit and yields, for every smooth  $G$ ,

$$\int G \rho_t - \int G \rho_0 = \int_0^t \int \Delta G(u) \rho_s(u) du ds.$$

Thus every limit point is a bounded weak solution of the heat equation.

For uniqueness, test the difference of two bounded weak solutions against the Fourier modes  $e^{2\pi i k \cdot u}$ . Its  $k$ -th Fourier coefficient  $a_k(t)$  satisfies  $a_k(t) = -4\pi^2 |k|^2 \int_0^t a_k(s) ds$  and  $a_k(0) = 0$ ; Gronwall gives  $a_k \equiv 0$  for every  $k$ . Hence all subsequential limits coincide with the heat-flow solution. The limit is deterministic, so tightness plus uniqueness upgrades convergence in law to convergence in probability in the weak path topology.  $\square$

**Theorem 2.6** (Yau relative-entropy theorem for the selected speed-change exclusion model).

Fix  $\alpha > -\frac{1}{2}$ . On  $\mathbb{T}_N$  let

$$(L_N F)(\eta) = \sum_{x \in \mathbb{T}_N} c_x(\eta) [F(\eta^{x, x+1}) - F(\eta)], \quad c_x(\eta) = 1 + \alpha(\eta(x-1) + \eta(x+2)), \quad (24.2.10)$$

and run the process with generator  $N^2 L_N$ . The Bernoulli product laws  $\nu_r^N$ ,  $0 < r < 1$ , are reversible. If  $\rho_0 \in C^4(\mathbb{T})$  satisfies  $\delta \leq \rho_0 \leq 1 - \delta$  and  $\rho$  is the smooth solution on  $[0, T]$  of

$$\partial_t \rho = \partial_u((1 + 2\alpha\rho)\partial_u \rho) = \partial_u^2 \Phi(\rho), \quad \Phi(r) = r + \alpha r^2, \quad (24.2.11)$$

write  $\nu_{\rho(t,\cdot)}^N$  for the inhomogeneous Bernoulli product law with one-site density  $\rho(t, x/N)$ . Then

$$H(\mu_0^N | \nu_{\rho_0}^N) = o(N) \implies \sup_{0 \leq t \leq T} H(\mu_t^N | \nu_{\rho(t,\cdot)}^N) = o(N). \quad (24.2.12)$$

Consequently, for every cylinder function  $\Psi$  and every  $H \in C(\mathbb{T})$ ,

$$\lim_{N \rightarrow \infty} \mathbf{E}_{\mu_t^N} \left| \frac{1}{N} \sum_x H(x/N) \tau_x \Psi - \int_{\mathbb{T}} H(u) \mathbf{E}_{\nu_{\rho(t,u)}} \Psi du \right| = 0 \quad (24.2.13)$$

for every  $t \in [0, T]$ ; in particular the empirical density converges to  $\rho(t, u) du$ .

*Proof.* We give the entropy-production argument in the precise finite-volume form needed later.

**1. Gradient current and the macroscopic flux.** The signed current through  $(x, x+1)$  is  $j_x = c_x(\eta)(\eta(x) - \eta(x+1))$ . With

$$h(\eta) = \eta(0) + \alpha(\eta(-1)\eta(0) - \eta(-1)\eta(1) + \eta(0)\eta(1))$$

a direct expansion gives

$$j_x = \tau_x h - \tau_{x+1} h, \quad \mathbf{E}_{\nu_r} h = r + \alpha r^2 = \Phi(r). \quad (24.2.14)$$

This verifies both the gradient structure and the coefficient in (24.2.11).

**2. Yau's moving-reference entropy inequality.** Fix a homogeneous Bernoulli law  $\nu_a^N$  and put  $\psi_t = d\nu_{\rho(t,\cdot)}^N / d\nu_a^N$  and  $f_t = d\mu_t^N / d\nu_a^N$ . If  $H_N(t) = H(\mu_t^N | \nu_{\rho(t,\cdot)}^N)$ , then the forward equation and  $a(\log b - \log a) \leq b - a$  give the source-form inequality

$$\frac{d}{dt} H_N(t) \leq \int \mathcal{R}_t^N d\mu_t^N, \quad \mathcal{R}_t^N = \frac{N^2 L_N \psi_t - \partial_t \psi_t}{\psi_t}. \quad (24.2.15)$$

Indeed, writing  $h = f_t / \psi_t$ , the generator contribution to  $H'_N$  is  $N^2 \int \psi_t h L_N \log h d\nu_a^N$ , and  $h L_N \log h \leq L_N h$ . Reversibility of  $L_N$  with respect to  $\nu_a^N$  then moves  $L_N$  from  $h$  onto  $\psi_t$  and gives (24.2.15).

Let

$$\lambda(t, u) = \log \frac{\rho(t, u)}{1 - \rho(t, u)}.$$

A direct bond expansion of  $L_N \psi_t / \psi_t$ , followed by discrete summation by parts and use of (24.2.11), gives the precise second-order centering needed below:

$$\begin{aligned} \mathcal{R}_t^N &= O_T(1) + \sum_x \partial_u^2 \lambda(t, x/N) C_{h_0, \rho(t, x/N)}(\tau_x \eta) \\ &\quad + \sum_x [\partial_u \lambda(t, x/N)]^2 C_{g_0, \rho(t, x/N)}(\tau_x \eta), \end{aligned} \quad (24.2.16)$$

where

$$C_{q,r} = q - \tilde{q}(r) - \tilde{q}'(r)(\eta(0) - r), \quad \tilde{q}(r) = \mathbf{E}_{\nu_r} q, \quad (24.2.17)$$

and

$$\begin{aligned} h_0(\eta) &= 2\alpha\eta(0)\eta(1) - \alpha\eta(-1)\eta(1), \\ g_0(\eta) &= -\eta(0)\eta(1) + \alpha\eta(0)\eta(1) + \alpha\eta(-1)\eta(1) - 2\alpha\eta(-1)\eta(0)\eta(1). \end{aligned}$$

For clarity, the calculation is finite algebra. Exchange of  $x, x+1$  changes  $\log \psi_t$  by  $(\eta(x+1) - \eta(x))(\lambda_x - \lambda_{x+1})$ ; expand the resulting exponential through order two. The first-order current is (24.2.14), so one discrete summation by parts produces  $\partial_u^2 \lambda$ . The quadratic bond term produces  $g_0$ . Finally

$$\partial_t \log \psi_t = \sum_x \frac{\partial_t \rho_x}{\rho_x(1 - \rho_x)} (\eta(x) - \rho_x)$$

and  $\partial_t \rho = \partial_u^2 \Phi(\rho)$  cancel the linear pieces. The third-order Taylor remainder summed over  $N$  bonds is  $O_T(1)$  because  $\lambda_{x+1} - \lambda_x = O_T(N^{-1})$ .

**3. The proof-local one-block estimate.** The source calculation now needs only  $q = h_0, g_0$ . Let  $\eta^\ell(x) = \ell^{-1} \sum_{y=x}^{x+\ell-1} \eta(y)$ . For either of these  $q$  and every uniformly  $C^1$  weight  $A$ ,

$$\lim_{\ell \rightarrow \infty} \limsup_{N \rightarrow \infty} \left| \int_0^t \mathbf{E}_{\mu_s^N} \frac{1}{N} \sum_x A(s, x/N) \{ \tau_x q - \tilde{q}(\eta^\ell(x)) \} ds \right| = 0. \quad (24.2.18)$$

Here is the finite-volume justification, included to avoid a forward citation to Chapter 3. Since  $\nu_a^N$  is reversible,

$$H(\mu_t^N | \nu_a^N) + 4N^2 \int_0^t \mathcal{D}_N(\sqrt{d\mu_s^N/d\nu_a^N}; \nu_a^N) ds \leq H(\mu_0^N | \nu_a^N) \leq C_T N. \quad (24.2.19)$$

The last bound follows from (24.2.12)'s initial hypothesis and the fact that the logarithmic density ratio between  $\nu_{\rho_0}^N$  and  $\nu_a^N$  is a sum of uniformly bounded one-site terms. For fixed  $\ell$ , condition a block on its particle number  $K$ . The nearest-neighbour exclusion dynamics on that canonical block has spectral gap  $c\ell^{-2}$ , while comparison of hypergeometric and Bernoulli probabilities gives

$$|\mathbf{E}_{\ell, K} q - \tilde{q}(K/\ell)| \leq C_q/\ell. \quad (24.2.20)$$

Partitioning into finitely many disjoint block families, applying Cauchy–Schwarz on exchange bonds, and then (24.2.19), bounds the time integral of the conditional-mean-zero part by  $C\ell/N$  after the spatial normalization. The canonical-mean part is  $O(\ell^{-1})$  by (24.2.20). First  $N \rightarrow \infty$  and then  $\ell \rightarrow \infty$  proves (24.2.18).

Because the coefficients in (24.2.16) vary by only  $O_T(\ell/N)$  across a block, the same replacement also changes the linear term  $\eta(x) - \rho_x$  into  $\eta^\ell(x) - \rho_x$  at  $o(1)$  cost after the space-time normalization. Taylor's theorem applied to  $h_0$  and  $\tilde{g}_0$  therefore yields from (24.2.15)–(24.2.18)

$$\frac{H_N(t)}{N} \leq C_T \int_0^t \mathbf{E}_{\mu_s^N} \left[ \frac{1}{N} \sum_x (\eta_s^\ell(x) - \rho(s, x/N))^2 \right] ds + o_{N, \ell}(1), \quad (24.2.21)$$

where  $\lim_{\ell \rightarrow \infty} \limsup_{N \rightarrow \infty} |o_{N, \ell}(1)| = 0$  uniformly for  $t \leq T$ .

**4. Product large deviations and Gronwall.** Apply the entropy inequality to the expectation in (24.2.21). For every  $\gamma > 0$ ,

$$\begin{aligned} & \mathbf{E}_{\mu_s^N} \frac{1}{N} \sum_x (\eta^\ell(x) - \rho_x)^2 \\ & \leq \frac{H_N(s)}{\gamma N} + \frac{1}{\gamma N} \log \mathbf{E}_{\nu_{\rho(s, \cdot)}^N} \exp \left\{ \gamma \sum_x (\eta^\ell(x) - \rho_x)^2 \right\}. \end{aligned} \quad (24.2.22)$$

Split the overlapping blocks into  $2\ell + 1$  independent residue families and use Hölder. For a single Bernoulli block the block-density large-deviation rate  $I_r(\theta)$  is uniformly quadratic for  $r \in [\delta_T, 1 - \delta_T]$ . Hence one can choose  $\gamma_0 > 0$ , independent of  $r$ , such that

$$\sup_{\theta \in [0,1]} \{ \gamma_0(2\ell + 1)(\theta - r)^2 - \ell I_r(\theta) \} \leq o(\ell).$$

Consequently the logarithmic moment in (24.2.22), divided by  $N$ , tends to zero as first  $N \rightarrow \infty$  and then  $\ell \rightarrow \infty$ . With this fixed  $\gamma_0$  we obtain

$$\frac{H_N(t)}{N} \leq \frac{C_T}{\gamma_0} \int_0^t \frac{H_N(s)}{N} ds + o_{N,\ell}(1). \quad (24.2.23)$$

Gronwall, the hypothesis  $H_N(0) = o(N)$ , and the successive limits  $N \rightarrow \infty$ ,  $\ell \rightarrow \infty$  prove (24.2.12), uniformly for  $t \leq T$ .

**5. Local equilibrium.** Apply the entropy inequality once more to the block average of  $\tau_x \Psi - \mathbf{E}_{\nu_{\rho(t,x/N)}} \Psi$ . Independence of separated blocks gives an exponential moment of order  $e^{o(N)}$ , while (24.2.12) contributes  $o(N)$ . Approximation of  $H$  by step functions and smoothness of  $\rho$  yield (24.2.13). Taking  $\Psi(\eta) = \eta(0)$  gives the hydrodynamic law.  $\square$

## Chapter 3

# Replacement lemmas and local equilibrium

Hydrodynamics is not obtained by Taylor expanding the generator alone. A microscopic current depends on several occupations; the replacement lemmas justify substituting a local equilibrium expectation depending only on a conserved block density.

### 3.1 Entropy production and Dirichlet form

**Proposition 3.1** (Entropy production and Dirichlet-form identity).

*Let  $L$  be a finite-state reversible Markov generator with invariant probability  $\nu$ , written*

$$(Lf)(\eta) = \sum_{\zeta} c(\eta, \zeta) [f(\zeta) - f(\eta)], \quad \nu(\eta)c(\eta, \zeta) = \nu(\zeta)c(\zeta, \eta).$$

*If  $f_t$  is the density of the law at time  $t$  with respect to  $\nu$ , then*

$$H_\nu(f_t) + \int_0^t \mathcal{D}_{\text{ent}}(f_s) ds = H_\nu(f_0),$$

*where*

$$H_\nu(f) = \int f \log f d\nu, \quad \mathcal{D}_{\text{ent}}(f) = \frac{1}{2} \sum_{\eta, \zeta} \nu(\eta)c(\eta, \zeta)(f(\zeta) - f(\eta))(\log f(\zeta) - \log f(\eta)).$$

*Moreover*

$$\mathcal{D}_{\text{ent}}(f) \geq 4\mathcal{D}(\sqrt{f}), \quad \mathcal{D}(g) = \frac{1}{2} \sum_{\eta, \zeta} \nu(\eta)c(\eta, \zeta)(g(\zeta) - g(\eta))^2.$$

*The same formulas hold for the accelerated generator  $N^2L$ , with both dissipation terms multiplied by  $N^2$ .*

*Proof.* Reversibility makes  $L$  self-adjoint in  $L^2(\nu)$  and the forward equation is  $\partial_t f_t = Lf_t$ . Since  $\int Lf_t d\nu = 0$ ,

$$\frac{d}{dt} H_\nu(f_t) = \int (Lf_t) \log f_t d\nu.$$

Pair the  $(\eta, \zeta)$  and  $(\zeta, \eta)$  terms and use detailed balance. This gives

$$\frac{d}{dt} H_\nu(f_t) = -\mathcal{D}_{\text{ent}}(f_t).$$

Integration in time proves the identity (with the usual approximation  $f \mapsto f + \varepsilon$  if zeros are present). Finally, for  $a, b > 0$ ,

$$(a - b)(\log a - \log b) \geq 4(\sqrt{a} - \sqrt{b})^2,$$

which follows after setting  $a/b = r^2$  from  $(r^2 - 1) \log r \geq 2(r - 1)^2$ . Applying this bond by bond proves the Dirichlet lower bound. Multiplying the generator by  $N^2$  multiplies its carré du champ and both dissipations by the same factor.  $\square$

**Theorem 3.2** (One-block replacement).

Let  $h$  be a bounded local function for the selected conservative finite-range  $\{0, 1\}$  lattice-gas class with equilibrium product measures  $(\nu_\rho)_{\rho \in [0, 1]}$ , and write

$$\tilde{h}(\rho) = \mathbb{E}_{\nu_\rho}[h], \quad \bar{\eta}_{x, \ell} = |\Lambda_\ell|^{-1} \sum_{y \in x + \Lambda_\ell} \eta(y).$$

For a finite block  $\Lambda_\ell$  and particle number  $K$ , let  $\nu_{\ell, K}$  denote the corresponding canonical measure. Assume uniformly in  $\ell$  and  $K$ :

1. the canonical block dynamics is irreducible and has spectral gap

$$\text{Var}_{\nu_{\ell, K}}(g) \leq C\ell^2 \mathcal{D}_{\ell, K}(g); \quad (24.3.1)$$

2. the equivalence-of-ensembles error for every local observable in the selected finite family tends to zero,

$$r_\ell(h) := \sup_K \left| \mathbb{E}_{\nu_{\ell, K}}[h] - \tilde{h}(K/|\Lambda_\ell|) \right| \longrightarrow 0. \quad (24.3.2)$$

Let the process run with generator  $N^2 L_N$ , and suppose its time- $s$  law has density  $f_s^N$  with respect to a fixed equilibrium product measure and obeys

$$H(f_0^N \mid \nu_\rho^N) \leq C_0 N^d, \quad \int_0^T N^2 \mathcal{D}_N(\sqrt{f_s^N}) ds \leq C_T N^d. \quad (24.3.3)$$

Then, for every deterministic family of weights  $|a_x^N(s)| \leq 1$ ,

$$\lim_{\ell \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathbb{E} \left| \int_0^T \frac{1}{N^d} \sum_x a_x^N(s) [\tau_x h(\eta_s) - \tilde{h}(\bar{\eta}_{x, \ell}(s))] ds \right| = 0. \quad (24.3.4)$$

The conclusion holds simultaneously for every fixed finite family of local observables. In particular it applies to the bounded smooth spatial weights that occur after discrete summation by parts in the gradient hydrodynamic theorem.

*Proof.* Put

$$V_{x, \ell} = \tau_x h - \tilde{h}(\bar{\eta}_{x, \ell}), \quad V_{N, \ell}(s) = \frac{1}{N^d} \sum_x a_x^N(s) V_{x, \ell}.$$

We prove the superexponential estimate underlying (24.3.4). Treating the two signs separately, the entropy inequality gives, for  $\gamma > 0$ ,

$$\mathbb{E} \left| \int_0^T V_{N, \ell}(s) ds \right| \leq \frac{C_0}{\gamma} + \frac{1}{\gamma N^d} \log \mathbb{E}_{\nu_\rho^N} \exp \left\{ \gamma \int_0^T \sum_x a_x^N(s) V_{x, \ell}(\eta_s) ds \right\}, \quad (24.3.5)$$



up to taking the maximum of the two signs. The time-dependent Feynman–Kac variational bound reduces the second term to the largest value of

$$\gamma \int \sum_x a_x^N(s) V_{x,\ell} g^2 d\nu_\rho^N - N^2 \mathcal{D}_N(g), \quad \|g\|_2 = 1. \quad (24.3.6)$$

Partition the torus into disjoint  $\ell$ -blocks, enlarged by a fixed margin containing the support of  $h$ , and average over the finitely many translates of this partition. Condition one block on its conserved particle number  $K$ . By (24.3.2), the conditional mean of every  $V_{x,\ell}$  whose support lies in the block is bounded by  $r_\ell(h)$ . Write the remaining part as a centered canonical observable  $W$ . Since  $h$  is bounded, Cauchy–Schwarz and (24.3.1) give, for every block function  $g$ ,

$$\left| \int W g^2 d\nu_{\ell,K} \right| \leq C(h, \ell) \mathcal{D}_{\ell,K}(g)^{1/2} \|g\|_2. \quad (24.3.7)$$

Using  $2ab \leq \delta a^2 + \delta^{-1} b^2$  with the  $N^2$  Dirichlet penalty in (24.3.6), the centered contribution is absorbed, leaving for fixed  $\ell$  an error  $C(\gamma, h, \ell) N^{-2}$  per block. Bounded overlap of the enlarged blocks does not change the macroscopic order. Consequently

$$\limsup_{N \rightarrow \infty} \frac{1}{Nd} \log \mathbb{E}_{\nu_\rho^N} \exp \left\{ \gamma \int_0^T \sum_x a_x^N(s) V_{x,\ell}(\eta_s) ds \right\} \leq \gamma T r_\ell(h). \quad (24.3.8)$$

The same estimate holds for the opposite sign.

Insert (24.3.8) into (24.3.5), let first  $N \rightarrow \infty$ , then  $\ell \rightarrow \infty$ , and finally  $\gamma \rightarrow \infty$ . This proves (24.3.4). Nothing in the argument depends on the particular member of a fixed finite family of local observables, so a triangle bound gives the simultaneous version. The weighted formulation is essential: it is exactly what allows one to use the theorem with  $a_x^N = (\Delta_N G)(x/N) / \|\Delta_N G\|_\infty$  in the hydrodynamic identification.  $\square$

**Theorem 3.3** (Two-block / mesoscopic-density replacement).

*Assume the hypotheses of the one-block theorem and, in addition, the standard moving-particle comparison for the selected dynamics: exchanging the conserved occupation between two sites can be realized along a nearest-neighbour path, with the associated long-exchange Dirichlet form bounded by path length times the sum of the nearest-neighbour Dirichlet forms along that path. (For uniformly elliptic exclusion/Kawasaki dynamics this follows by telescoping and Cauchy–Schwarz.) Put  $R_N = \lfloor \varepsilon N \rfloor$ . Then*

$$\lim_{\ell \rightarrow \infty} \lim_{\varepsilon \downarrow 0} \limsup_{N \rightarrow \infty} \mathbb{E} \int_0^T \frac{1}{Nd} \sum_x |\bar{\eta}_{x,\ell}(s) - \bar{\eta}_{x,R_N}(s)| ds = 0. \quad (24.3.9)$$

*More precisely, the proof controls the averaged displacement form*

$$\frac{1}{|\Lambda_{R_N}|} \sum_{z \in \Lambda_{R_N}} \mathbb{E} \int_0^T \frac{1}{Nd} \sum_x |\bar{\eta}_{x,\ell}(s) - \bar{\eta}_{x+z,\ell}(s)| ds, \quad (24.3.10)$$

*with the negligible overlapping-block range  $|z| \lesssim \ell$  removed if desired. The theorem does not assert a random pointwise supremum over all  $|z| \leq \varepsilon N$  inside the space–time integral.*

*Proof.* Let  $T_{ab}$  exchange the occupations at  $a, b$ , and let  $a = x_0, x_1, \dots, x_m = b$  be a nearest-neighbour path. Telescoping the change of  $\sqrt{f}$  along this path and applying Cauchy–Schwarz gives the moving-particle inequality

$$\int (\sqrt{f(T_{ab}\eta)} - \sqrt{f(\eta)})^2 d\nu \leq m \sum_{j=0}^{m-1} \int (\sqrt{f(T_{x_j x_{j+1}}\eta)} - \sqrt{f(\eta)})^2 d\nu. \quad (24.3.11)$$

We now spell out the variational core, since this is the point at which a pointwise supremum over  $z$  would be unjustified. Fix  $\ell$ , remove the  $O((\ell/R_N)^d)$  fraction of displacements for which the two  $\ell$ -blocks overlap, and apply the entropy inequality and Feynman–Kac formula to the *average* over the remaining pairs of blocks. Average also over a family of coordinate paths joining corresponding sites in the two blocks. For a displacement of length at most  $R_N$ , every nearest-neighbour bond is used by at most  $CR_N|\Lambda_\ell|$  of the translated paths. Combining this congestion count with (24.3.11) shows that, after the global diffusive penalty  $N^2\mathcal{D}_N$  is distributed among the block pairs, the variational problem contains a two-block exchange Dirichlet form with coefficient bounded below by  $cN^2/R_N^2 \geq c\varepsilon^{-2}$ . Internal block exchanges retain their canonical spectral-gap coefficient.

Consequently, for fixed  $\ell$ , any sequence of almost maximising two-block densities as first  $N \rightarrow \infty$  and then  $\varepsilon \downarrow 0$  has vanishing Dirichlet form for every internal exchange and every exchange between the two blocks. The state space of two fixed blocks is finite, so compactness of the probability simplex lets us pass to a limiting density. Zero Dirichlet form means that this density is constant on each level set of the *total* particle number in the union of the two blocks. Thus the only possible limit is a mixture of the canonical measures  $\nu_{2\ell, K}$ , and the limiting block-difference expectation is bounded by

$$b_\ell := \sup_K \mathbb{E}_{\nu_{2\ell, K}} \left| \frac{K_1}{|\Lambda_\ell|} - \frac{K_2}{|\Lambda_\ell|} \right|. \quad (24.3.12)$$

Here  $K_1 + K_2 = K$ . Conditional on  $K$ ,  $K_1$  is hypergeometric. If  $M = |\Lambda_\ell|$ , then  $\text{Var}(K_1 \mid K) \leq M/4$ , hence by Cauchy–Schwarz

$$b_\ell \leq M^{-1/2} \rightarrow 0. \quad (24.3.13)$$

The compactness argument is uniform in the finitely many values of  $K$  for fixed  $\ell$ ; restoring the discarded overlapping displacements gives

$$\lim_{\varepsilon \downarrow 0} \limsup_{N \rightarrow \infty} \frac{1}{|\Lambda_{R_N}|} \sum_{z \in \Lambda_{R_N}} \mathbb{E} \int_0^T \frac{1}{N^d} \sum_x |\bar{\eta}_{x, \ell} - \bar{\eta}_{x+z, \ell}| ds \leq T b_\ell. \quad (24.3.14)$$

It remains to convert this averaged displacement estimate into (24.3.9). The large block average is, up to a boundary convolution error, the average of translated small-block averages:

$$\bar{\eta}_{x, R_N} = \frac{1}{|\Lambda_{R_N}|} \sum_{z \in \Lambda_{R_N}} \bar{\eta}_{x+z, \ell} + O\left(\frac{\ell}{R_N}\right), \quad (24.3.15)$$

uniformly because  $0 \leq \eta \leq 1$ . Jensen’s inequality therefore bounds the left side of (24.3.9) by (24.3.10) plus  $C\ell/R_N$ . Now send  $N \rightarrow \infty$ , then  $\varepsilon \downarrow 0$ , and finally  $\ell \rightarrow \infty$ .

This is the form used downstream: a microscopic block density may be replaced by a mesoscopic density whose radius tends to zero on the macroscopic scale. A bounded deterministic spatial weight can be inserted by multiplication and the triangle inequality.  $\square$

### 3.2 Gradient-system hydrodynamic theorem

**Theorem 3.4** (Gradient-system hydrodynamic theorem with explicit replacement hypotheses).

Consider the bounded finite-range conservative  $\{0, 1\}$  lattice-gas class of SMX-24.03.02–03 on  $\mathbb{T}_N^d$ , run with generator  $N^2 L_N$ . Assume:

1. the homogeneous Bernoulli products  $\nu_r^N$  are invariant and reversible, and the jump rates are uniformly bounded;
2. every canonical block has the spectral-gap and equivalence-of-ensembles bounds stated in SMX-24.03.02, and the moving-particle estimate of SMX-24.03.03 holds;
3. the microscopic current has the gradient form

$$j_{x, x+e_i}(\eta) = \tau_x h_i(\eta) - \tau_{x+e_i} h_i(\eta), \quad \Phi_i(r) = \mathbf{E}_{\nu_r} h_i;$$

4. for some  $\bar{r} \in (0, 1)$ ,  $H(\mu_0^N \mid \nu_{\bar{r}}^N) \leq C_0 N^d$ , and the initial empirical measures converge in probability to  $\rho_0(u) du$ .

If the nonlinear diffusion equation

$$\partial_t \rho = \sum_{i=1}^d \partial_{u_i}^2 \Phi_i(\rho), \quad \rho(0, \cdot) = \rho_0, \quad (24.3.11)$$

has a unique bounded weak solution on  $[0, T]$ , then

$$\pi_t^N = \frac{1}{N^d} \sum_x \eta_t(x) \delta_{x/N} \xrightarrow[N \rightarrow \infty]{\mathbb{P}} \rho(t, u) du$$

in  $D([0, T], \mathcal{M}(\mathbb{T}^d))$  with the weak topology. In particular, the entropy/Dirichlet assumptions consumed by the one- and two-block estimates are conclusions of the hypotheses above, not additional hidden assumptions.

*Proof. Entropy production.* Because  $\nu_{\bar{r}}^N$  is invariant and reversible, SMX-24.03.01 applied to the accelerated generator gives

$$H(\mu_t^N \mid \nu_{\bar{r}}^N) + 4N^2 \int_0^t \mathcal{D}_N(\sqrt{f_s}; \nu_{\bar{r}}^N) ds \leq H(\mu_0^N \mid \nu_{\bar{r}}^N) \leq C_0 N^d. \quad (24.3.12)$$

Thus the entropy and integrated Dirichlet estimates required in SMX-24.03.02–03 hold uniformly on bounded time intervals.

Let  $G \in C^2(\mathbb{T}^d)$ . Dynkin's formula gives

$$\langle \pi_t^N, G \rangle - \langle \pi_0^N, G \rangle = M_t^N(G) + \int_0^t N^2 L_N \langle \pi_s^N, G \rangle ds. \quad (24.3.13)$$

A single conservative jump changes  $\langle \pi^N, G \rangle$  by  $O(N^{-d-1})$ . There are  $O(N^d)$  bounded local clocks at rate  $N^2$ , so  $\langle M^N(G) \rangle_T = O(N^{-d})$  and Doob's inequality gives  $\sup_{t \leq T} |M_t^N(G)| \rightarrow 0$  in probability.

The microscopic conservation law and the gradient current permit two discrete summations by parts:

$$N^2 L_N \langle \pi^N, G \rangle = \frac{1}{N^d} \sum_x \sum_i (\Delta_{N,i} G)(x/N) \tau_x h_i(\eta). \quad (24.3.14)$$

This exact identity is the special simplification of a gradient system. The mass bound, (24.3.13), and the martingale estimate give tightness in the weak-measure path space.

Take a convergent subsequence. Apply the weighted one-block theorem SMX-24.03.02 to  $h_i$ , with weight  $(\Delta_{N,i}G)/\|\Delta_{N,i}G\|_\infty$ , and then the two-block theorem SMX-24.03.03 to identify the microscopic block density with the density of the limiting empirical measure. Since  $\Delta_{N,i}G \rightarrow \partial_{u_i}^2 G$  uniformly,

$$\int_0^t \frac{1}{N^d} \sum_{x,i} (\Delta_{N,i}G)(x/N) \tau_x h_i(\eta_s) ds \longrightarrow \int_0^t \int_{\mathbb{T}^d} \sum_i \partial_{u_i}^2 G(u) \Phi_i(\rho(s, u)) du ds. \quad (24.3.15)$$

The initial association supplies the initial term, hence every limit point satisfies the weak formulation of (24.3.11). Bounded occupations imply  $0 \leq \rho \leq 1$  and absolute continuity of the limit. By the assumed PDE uniqueness all subsequential limits are the same deterministic path. Tightness therefore gives convergence of the full sequence; a deterministic limit upgrades convergence in law to convergence in probability.  $\square$

## Chapter 4

# Asymmetric systems and scalar conservation laws

When a microscopic bias survives at Euler scale, diffusion is replaced by a conservation law. Shocks and rarefaction fans are therefore not pathologies of the microscopic model; they are the macroscopic record of a nonlinear flux.

**Definition 4.1** (Kruzhkov entropy solution). Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be Lipschitz. A bounded weak solution  $u$  of  $\partial_t u + \partial_x f(u) = 0$  is a *Kruzhkov entropy solution* if for every  $k \in \mathbb{R}$  and every nonnegative  $\varphi \in C_c^\infty((0, \infty) \times \mathbb{R})$ ,

$$\iint \left( |u - k| \partial_t \varphi + \operatorname{sgn}(u - k)(f(u) - f(k)) \partial_x \varphi \right) dx dt \geq 0,$$

with the initial condition attained in  $L^1_{\text{loc}}$ . The inequality is the admissibility rule that excludes nonphysical weak shocks.

*Worked example 4.2* (Shock versus rarefaction for Burgers flux). For  $f(u) = u^2/2$  and Riemann data  $u(0, x) = u_L \mathbf{1}_{x < 0} + u_R \mathbf{1}_{x > 0}$ , the Rankine–Hugoniot speed is

$$s = \frac{f(u_R) - f(u_L)}{u_R - u_L} = \frac{u_L + u_R}{2}.$$

If  $u_L > u_R$ , characteristics with speeds  $u_L$  and  $u_R$  collide and the entropy solution is the shock  $u(t, x) = u_L$  for  $x < st$  and  $u_R$  for  $x > st$ . If  $u_L < u_R$ , a shock would violate the entropy inequality; the entropy solution is instead the self-similar fan

$$u(t, x) = \begin{cases} u_L, & x/t \leq u_L, \\ x/t, & u_L < x/t < u_R, \\ u_R, & x/t \geq u_R. \end{cases}$$

Thus the same conservation law supports two qualitatively different macroscopic profiles, selected by entropy rather than by the weak equation alone. This example uses a strictly convex flux; the concave ASEP flux used later has the shock/rarefaction orientation reversed.

### 4.1 Microscopic conservation law for exclusion

**Theorem 4.3** (Microscopic conservation law for exclusion).

For nearest-neighbor exclusion the occupation satisfies  $\eta_t(x) - \eta_0(x) = J_{x-1,x}(t) - J_{x,x+1}(t)$ , where  $J$  is the signed integrated current. After summing against a test function this becomes a discrete weak conservation law.

*Proof.* Every jump across  $(x, x+1)$  decreases the occupation at one endpoint and increases it at the other. Counting signed jumps gives the exact sitewise identity. Multiply by a test function and sum over  $x$ ; the current terms telescope into a discrete gradient of the test function. This is the microscopic divergence form from which the Euler hydrodynamic equation is derived.  $\square$

## 4.2 Hydrodynamic limit for attractive systems

### 4.3 Entropy solution in the selected concave-flux class

**Theorem 4.4** (Kruzhkov uniqueness, local contraction, and the concave Riemann problem).

Let  $f \in C^1([0, K])$  and put  $L = \sup_{0 \leq r \leq K} |f'(r)|$ . For every  $u_0 \in L^\infty(\mathbb{R}; [0, K])$  there is at most one bounded Kruzhkov entropy solution of

$$\partial_t u + \partial_x f(u) = 0,$$

and any two such solutions satisfy the finite-speed local contraction

$$\int_{-R}^R |u(t, x) - v(t, x)| dx \leq \int_{-R-Lt}^{R+Lt} |u_0(x) - v_0(x)| dx \quad (24.4.1)$$

for every  $R, t > 0$ . If  $u_0 - v_0 \in L^1(\mathbb{R})$  this implies the usual global  $L^1$  contraction.

If, in addition,  $f$  is strictly concave and the initial datum is the Riemann step  $(u_L, u_R)$ , then  $u_L < u_R$  gives the entropy shock with speed

$$s = \frac{f(u_R) - f(u_L)}{u_R - u_L}, \quad (24.4.2)$$

whereas  $u_L > u_R$  gives the self-similar rarefaction

$$u(t, x) = \begin{cases} u_L, & x/t \leq f'(u_L), \\ (f')^{-1}(x/t), & f'(u_L) < x/t < f'(u_R), \\ u_R, & x/t \geq f'(u_R). \end{cases} \quad (24.4.3)$$

(The inequalities reverse for a strictly convex flux, as in the Burgers worked example above.)

*Proof.* Apply the entropy inequality for  $u$  with constant  $k = v(s, y)$  and that for  $v$  with constant  $k = u(t, x)$ , multiply by a nonnegative test function of  $(t, x, s, y)$  concentrated near the diagonal, add, and send the time and space mollifier scales to zero. This doubling-of-variables argument gives in the distributional sense

$$\partial_t |u - v| + \partial_x [\operatorname{sgn}(u - v)(f(u) - f(v))] \leq 0. \quad (24.4.4)$$

Because  $|f(u) - f(v)| \leq L|u - v|$ , test (24.4.4) against smooth approximations of the backward cone  $\{|x| < R + L(t - s)\}$ , first with compact spatial cutoffs and then let the cutoffs recede. The boundary term has the correct sign and yields (24.4.1). If the initial difference is integrable, send  $R \rightarrow \infty$  to obtain global  $L^1$  contraction. Taking identical initial data proves uniqueness in the bounded class, even when the individual solutions are not globally integrable.

For Riemann data a discontinuity can be weak only at the Rankine–Hugoniot speed (24.4.2). Strict concavity places the secant slope strictly between the endpoint derivatives:  $f'(u_R) < s < f'(u_L)$  when  $u_L < u_R$ . Thus characteristics enter the jump from both sides and the Kruzhkov inequalities accept the shock. When  $u_L > u_R$ ,  $f'(u_L) < f'(u_R)$  and characteristics separate. The self-similar ansatz  $u(t, x) = U(x/t)$  gives either  $U' = 0$  or  $f'(U) = x/t$ ; matching the two constant states gives (24.4.3). Direct substitution in the entropy inequalities verifies admissibility, and uniqueness rules out every other weak solution.  $\square$

## Chapter 5

# Equilibrium fluctuations and additive functionals

Hydrodynamics is a law of large numbers. The next scale is Gaussian in reversible diffusive systems. The fundamental analytic object is the negative Sobolev norm generated by the Markov operator.

### 5.1 Kipnis–Varadhan functional central limit theorem

**Theorem 5.1** (Kipnis–Varadhan functional CLT).

Let  $(\eta_t)_{t \geq 0}$  be a stationary ergodic reversible Markov process with self-adjoint generator  $L \leq 0$  on  $L^2(\nu)$ , and put  $A = -L$ . For centered  $F$  define

$$\|F\|_{-1}^2 := \sup_{g \in \mathcal{D}(A^{1/2})} \{2\langle F, g \rangle - \langle g, Ag \rangle\} = \langle F, A^{-1}F \rangle$$

whenever the spectral inverse is finite. If  $f \in L_0^2(\nu)$  satisfies  $\|f\|_{-1} < \infty$ , then, for every  $T < \infty$ ,

$$R^{-1/2} \int_0^{R \cdot} f(\eta_s) ds \implies \sigma B \quad \text{in } C([0, T]), \quad \sigma^2 = 2\|f\|_{-1}^2.$$

The proof also owns the maximal  $H_{-1}$  estimate used later in this volume: for deterministic  $F \in L^2([0, T]; H_{-1})$ ,

$$\mathbb{E}_\nu \left[ \sup_{0 \leq t \leq T} \left| \int_0^t F(s, \eta_s) ds \right|^2 \right] \leq 20 \int_0^T \|F(s, \cdot)\|_{-1}^2 ds. \quad (24.5.1)$$

For the accelerated generator  $cL$  the right-hand side is divided by  $c$ .

*Proof.* We separate the estimate from the invariance principle because the former is also the quantitative input in the Boltzmann–Gibbs arguments of Chapter 8.

**The forward–backward  $H_{-1}$  estimate.** First take a smooth deterministic family  $u_s \in \mathcal{D}(A)$  and set  $F_s = Au_s = -Lu_s$ . Dynkin’s formula gives the forward martingale

$$M_t = u_t(\eta_t) - u_0(\eta_0) - \int_0^t (\partial_s u_s + Lu_s)(\eta_s) ds.$$



By reversibility, the time-reversed stationary path has the same law. Applying Dynkin's formula to  $s \mapsto u_{T-s}$  on that reversed path gives a backward martingale  $\widehat{M}$ . A direct subtraction of the two Dynkin formulas gives, for every  $0 \leq t \leq T$ ,

$$2 \int_0^t F_s(\eta_s) ds = M_t + \widehat{M}_T - \widehat{M}_{T-t}. \quad (24.5.2)$$

Moreover stationarity and the Dirichlet-form identity imply

$$\mathbb{E}M_T^2 = \mathbb{E}\widehat{M}_T^2 = 2 \int_0^T \langle u_s, Au_s \rangle ds. \quad (24.5.3)$$

If  $X = \sup_{t \leq T} |M_t|$  and  $Y = \sup_{t \leq T} |\widehat{M}_t|$ , then (24.5.2) gives  $\sup_t |\int_0^t F_s ds|^2 \leq (X + 2Y)^2/4$ . Doob's  $L^2$  inequality and  $(a + 2b)^2 \leq 5(a^2 + b^2)$  therefore yield

$$\mathbb{E} \sup_{t \leq T} \left| \int_0^t F_s(\eta_s) ds \right|^2 \leq 20 \int_0^T \langle u_s, Au_s \rangle ds. \quad (24.5.4)$$

The range of  $A$  is dense in  $H_{-1}$ : indeed spectral truncation away from 0 and  $\infty$  approximates every  $H_{-1}$  vector. Approximate a general  $F \in L^2([0, T]; H_{-1})$  by families  $Au_s^{(m)}$ . Estimate (24.5.4) makes the corresponding time integrals Cauchy in  $L^2(\Omega; C[0, T])$ , and passage to the limit proves (24.5.1). Replacing  $A$  by  $cA$  gives the final scaling assertion.

**Martingale approximation on spectral windows.** Let  $E_A$  be the spectral resolution of  $A$  and set

$$f_m = E_A([m^{-1}, m])f, \quad u_m = A^{-1}f_m. \quad (24.5.5)$$

Then  $u_m \in \mathcal{D}(A)$ ,  $Au_m = f_m$ , and

$$\|f - f_m\|_{-1} \rightarrow 0, \quad \langle f_m, A^{-1}f_m \rangle \rightarrow \|f\|_{-1}^2. \quad (24.5.6)$$

For fixed  $m$ , Dynkin's formula gives

$$\int_0^t f_m(\eta_s) ds = M_t^{(m)} - (u_m(\eta_t) - u_m(\eta_0)), \quad (24.5.7)$$

where  $M^{(m)}$  is a square-integrable martingale with stationary ergodic increments and

$$\mathbb{E}[(M_t^{(m)})^2] = 2t \langle u_m, Au_m \rangle = 2t \langle f_m, A^{-1}f_m \rangle. \quad (24.5.8)$$

The unit-time oscillation  $Z_m = \sup_{0 \leq s \leq 1} |u_m(\eta_s)|$  belongs to  $L^2$ : this follows from (24.5.7), Doob's inequality,  $u_m \in L^2$ , and  $f_m \in L^2$ . By stationarity, for fixed  $m$  and  $T_0 < \infty$ ,

$$\mathbb{P} \left( R^{-1/2} \sup_{0 \leq t \leq RT_0} |u_m(\eta_t)| > \varepsilon \right) \leq (RT_0 + 2) \mathbb{P}(Z_m > \varepsilon \sqrt{R}), \quad (24.5.9)$$

and the right-hand side tends to 0, since  $x^2 \mathbb{P}(Z_m > x) \rightarrow 0$  for every square-integrable random variable. Thus the coboundary in (24.5.7) is uniformly negligible on diffusive time intervals.

The martingale increments  $M_{k+1}^{(m)} - M_k^{(m)}$  form a stationary ergodic square-integrable martingale-difference sequence. The martingale functional central limit theorem, proved from conditional quadratic-variation ergodicity and the Lindeberg truncation, therefore gives

$$R^{-1/2} M_R^{(m)} \Rightarrow \sqrt{2 \langle f_m, A^{-1}f_m \rangle} B. \quad (24.5.10)$$

The oscillation inside the unit intervals is negligible by the same square-integrable maximal argument used in (24.5.9), so (24.5.10) is a continuous-time functional statement.

**Removal of the spectral cutoff.** Apply (24.5.1) to the time-independent function  $f - f_m$  on  $[0, RT_0]$  and rescale time. Uniformly in  $R$ ,

$$\mathbb{E} \left[ \sup_{0 \leq t \leq T_0} \left| R^{-1/2} \int_0^{Rt} (f - f_m)(\eta_s) ds \right|^2 \right] \leq 20T_0 \|f - f_m\|_{-1}^2. \quad (24.5.11)$$

First let  $R \rightarrow \infty$  in the cutoff process, then  $m \rightarrow \infty$  using (24.5.6) and (24.5.11). The Brownian variances converge to  $2\|f\|_{-1}^2$ , and the triangular approximation proves the claimed functional CLT in  $C([0, T_0])$ .  $\square$

## 5.2 SSEP equilibrium fluctuation field

**Theorem 5.2** (Equilibrium density fluctuations of SSEP).

*Let SSEP on  $\mathbb{T}_N^d$  have generator*

$$(L_N F)(\eta) = \sum_{x \in \mathbb{T}_N^d} \sum_{i=1}^d [F(\eta^{x, x+e_i}) - F(\eta)]$$

*and run the process at speed  $N^2$ . Start it from Bernoulli equilibrium  $\nu_\rho$ ,  $0 < \rho < 1$ , and define*

$$Y_t^N(G) = N^{-d/2} \sum_x G(x/N) (\eta_{tN^2}(x) - \rho).$$

*For every  $T < \infty$  and every*

$$m > \frac{d}{2} + 2,$$

*the laws of  $Y^N$  on  $D([0, T], H^{-m}(\mathbb{T}^d))$  converge to a law concentrated on  $C([0, T], H^{-m}(\mathbb{T}^d))$ . The limit is the unique stationary generalized Ornstein–Uhlenbeck field*

$$dY_t = \Delta Y_t dt + \sqrt{2\chi(\rho)} \nabla \cdot dW_t, \quad \chi(\rho) = \rho(1 - \rho), \quad (24.5.12)$$

*where  $\Delta = \sum_{i=1}^d \partial_i^2$  and  $W$  is a  $d$ -component space–time white noise. Equivalently, if  $P_t = e^{t\Delta}$ , then*

$$\mathbb{E}[Y_t(G)Y_s(H)] = \chi(\rho) \langle G, P_{|t-s|}H \rangle_{L^2(\mathbb{T}^d)}. \quad (24.5.13)$$

*Thus the diffusivity in this jump-rate convention is  $D(\rho) = 1$  and the static compressibility is  $\chi(\rho) = \rho(1 - \rho)$ .*

*Proof.* For a smooth test function  $G$ , discrete summation by parts gives the exact martingale decomposition

$$Y_t^N(G) = Y_0^N(G) + \int_0^t Y_s^N(\Delta_N G) ds + M_t^N(G), \quad (24.5.14)$$

where

$$(\Delta_N G)(x/N) = N^2 \sum_{i=1}^d \{G((x + e_i)/N) + G((x - e_i)/N) - 2G(x/N)\}.$$

No Boltzmann–Gibbs replacement is needed because the SSEP current is exactly linear after summation by parts.

For two test functions  $G, H$  the predictable cross variation is

$$\langle M^N(G), M^N(H) \rangle_t = \int_0^t \frac{1}{N^d} \sum_{x,i} \nabla_{N,i} G(x/N) \nabla_{N,i} H(x/N) [\eta_s(x) - \eta_s(x + e_i)]^2 ds, \quad (24.5.15)$$

with  $\nabla_{N,i} G(u) = N[G(u + e_i/N) - G(u)]$ . Under  $\nu_\rho$ ,

$$\mathbb{E}_{\nu_\rho}[\eta(x) - \eta(x + e_i)]^2 = 2\rho(1 - \rho) = 2\chi(\rho).$$

The equilibrium variance of the spatial average of the centered local quantity in (24.5.15) is  $O(N^{-d})$ . Consequently (24.5.15) converges in  $L^2$ , uniformly on compact time intervals after time integration, to

$$2\chi(\rho)t \int_{\mathbb{T}^d} \nabla G \cdot \nabla H \, du. \quad (24.5.16)$$

A jump across one bond changes  $Y^N(G)$  by  $O(N^{-d/2-1})$ ; hence the martingale Lindeberg condition holds. The martingale functional CLT therefore identifies the limiting noise in (24.5.14) with the one in (24.5.12).

At time zero independence gives, for every finite family  $G_1, \dots, G_k$ ,

$$(Y_0^N(G_j))_{j=1}^k \implies (Y_0(G_j))_{j=1}^k, \quad \mathbb{E}[Y_0(G)Y_0(H)] = \chi(\rho)\langle G, H \rangle. \quad (24.5.17)$$

We now record the Hilbert-space tightness rather than hide it in the phrase “negative Sobolev space.” Let  $(e_k)_{k \in \mathbb{Z}^d}$  be the Fourier basis. Equilibrium and (24.5.17) imply

$$\sup_{N,t} \mathbb{E} \|Y_t^N\|_{H^{-r}}^2 < \infty \quad (r > d/2). \quad (24.5.18)$$

For the martingale term, (24.5.15) gives

$$\sum_k (1 + |k|^2)^{-m} \mathbb{E} \langle M^N(e_k) \rangle_T \leq C_T \sum_k (1 + |k|^2)^{-m} |k|^2 < \infty \quad (24.5.19)$$

when  $m > d/2 + 1$ . For the drift in (24.5.14), choose  $r = m - 2 > d/2$ ; then

$$\|\Delta_N Y_s^N\|_{H^{-m}} \leq C \|Y_s^N\|_{H^{-(m-2)}}, \quad (24.5.20)$$

uniformly in  $N$ . Equations (24.5.18)–(24.5.20), Doob’s inequality and the Aldous criterion give tightness in  $D([0, T], H^{-m})$  for  $m > d/2 + 2$ . The  $H^{-m}$  norm of a single field jump tends to zero, so every limit is supported on continuous paths.

Along a convergent subsequence,  $\Delta_N G \rightarrow \Delta G$  uniformly and the martingale problem obtained from (24.5.14)–(24.5.16) is

$$Y_t(G) - Y_0(G) - \int_0^t Y_s(\Delta G) ds$$

a continuous martingale with cross variation  $2\chi(\rho)t \int \nabla G \cdot \nabla H$ . Solving against the backward heat semigroup shows

$$Y_t(G) = Y_0(P_t G) + \sqrt{2\chi(\rho)} \int_0^t \langle \nabla P_{t-r} G, dW_r \rangle,$$

which proves uniqueness and the covariance formula (24.5.13). Hence every subsequence has the same limit and the whole sequence converges.  $\square$

### 5.3 Tagged-particle invariance principle

**Theorem 5.3** (Tagged particle in symmetric simple exclusion).

Fix  $d \geq 2$  and  $\rho \in (0, 1)$ . Let nearest-neighbour symmetric simple exclusion on  $\mathbb{Z}^d$  have generator

$$(\mathcal{L}F)(\eta) = \sum_{x \in \mathbb{Z}^d} \sum_{i=1}^d [F(\eta^{x, x+e_i}) - F(\eta)], \quad (24.5.21)$$

and start from Bernoulli equilibrium  $\nu_\rho$  conditioned on  $\eta(0) = 1$ . Mark the particle initially at the origin and denote its position by  $X_t$ . Then there is a deterministic symmetric nonnegative matrix  $\Sigma(\rho)$  such that, for every  $T < \infty$ ,

$$(\varepsilon X_{t/\varepsilon^2})_{0 \leq t \leq T} \Longrightarrow (\Sigma(\rho)^{1/2} B_t)_{0 \leq t \leq T} \quad (24.5.22)$$

in  $D([0, T], \mathbb{R}^d)$ , with the limit supported on continuous paths. For  $a \in \mathbb{R}^d$  the covariance is characterized by

$$a \cdot \Sigma(\rho) a = \inf_{f \in \mathcal{C}_*} \mathcal{Q}_a(f), \quad (24.5.23)$$

where  $\mathcal{C}_*$  is the class of local functions of the environment seen from the tagged particle and  $\mathcal{Q}_a$  is the explicit quadratic form in (24.5.35) below.

*Proof.* We give the environment/resolvent proof because the slogan “apply Kipnis–Varadhan to the tagged particle” suppresses three separate points: the correct Palm generator, the  $H_{-1}$  estimate for the tagged drift, and the joint martingale bracket producing the self-diffusion matrix.

**Step 1: the environment seen from the tag.** Put  $\mathbb{Z}_*^d = \mathbb{Z}^d \setminus \{0\}$  and  $\Omega_* = \{0, 1\}^{\mathbb{Z}_*^d}$ . If the tagged particle is at  $X_t$ , set

$$\xi_t(x) = \eta_t(X_t + x), \quad x \in \mathbb{Z}_*^d.$$

The Palm equilibrium  $\nu_\rho^*$  is simply product Bernoulli( $\rho$ ) on  $\Omega_*$ . For  $|e| = 1$  and  $\xi(e) = 0$  define the recentering map

$$(\theta_e \xi)(x) = \begin{cases} \xi(x + e), & x \neq -e, \\ 0, & x = -e. \end{cases} \quad (24.5.24)$$

Thus, after the tagged particle jumps by  $e$ , the old tagged site becomes the empty site  $-e$  in the new coordinates. If  $\mathcal{E}_*$  denotes the set of unoriented nearest-neighbour edges  $\{x, y\}$  with  $x, y \neq 0$ , the environment generator on local  $f$  is

$$(L_* f)(\xi) = \sum_{\{x, y\} \in \mathcal{E}_*} [f(\xi^{x, y}) - f(\xi)] + \sum_{|e|=1} (1 - \xi(e)) [f(\theta_e \xi) - f(\xi)]. \quad (24.5.25)$$

The exchange part is reversible by permutation invariance of the product measure. On  $\{\xi(e) = 0\}$ ,  $\theta_e$  is a measure-preserving bijection onto  $\{\xi(-e) = 0\}$  with inverse  $\theta_{-e}$ ; pairing  $e$  and  $-e$  therefore gives detailed balance for the second sum. Hence  $L_*$  is self-adjoint on  $L^2(\nu_\rho^*)$  and

$$\begin{aligned} \langle f, -L_* f \rangle_{\nu_\rho^*} &= \frac{1}{2} \sum_{\{x, y\} \in \mathcal{E}_*} \mathbb{E}[(f(\xi^{x, y}) - f(\xi))^2] \\ &\quad + \frac{1}{2} \sum_{|e|=1} \mathbb{E}[(1 - \xi(e))(f(\theta_e \xi) - f(\xi))^2]. \end{aligned} \quad (24.5.26)$$

The invariant sigma-field is trivial. Indeed, vanishing of the Dirichlet form makes an invariant  $L^2$  function unchanged under every finite nearest-neighbour exchange and under every vacancy-supported recentering. On the full-measure set having infinitely many particles and vacancies these moves generate arbitrary finite permutations of coordinates; the Hewitt–Savage zero–one law for the Bernoulli product measure then forces the function to be constant. Thus the environment is stationary, reversible and ergodic.

**Step 2: martingale plus additive functional.** For  $a \in \mathbb{R}^d$  define

$$V_a(\xi) = \sum_{|e|=1} (a \cdot e)(1 - \xi(e)). \quad (24.5.27)$$

Compensating the tagged jumps gives

$$a \cdot X_t = M_t^a + \int_0^t V_a(\xi_s) ds, \quad (24.5.28)$$

where

$$\frac{d}{dt} \langle M^a \rangle_t = \sum_{|e|=1} (a \cdot e)^2 (1 - \xi_t(e)). \quad (24.5.29)$$

Reflection symmetry gives  $\mathbb{E}[V_a] = 0$ .

**Step 3: the exact  $H_{-1}$  estimate.** The recentering change of variables used in Step 1 yields, for every local  $f$ ,

$$\langle V_a, f \rangle = \frac{1}{2} \sum_{|e|=1} (a \cdot e) \mathbb{E}[(1 - \xi(e))(f(\xi) - f(\theta_e \xi))]. \quad (24.5.30)$$

Consequently Cauchy–Schwarz and (24.5.26) give

$$|\langle V_a, f \rangle|^2 \leq C(\rho, d) |a|^2 \langle f, -L_* f \rangle. \quad (24.5.31)$$

Thus  $V_a \in H_{-1}(L_*)$ . This is the precise hypothesis required by the Kipnis–Varadhan theorem SMX-24.05.01; no spectral gap for the infinite-volume environment is being assumed.

**Step 4: resolvent correctors and the joint martingale.** Let

$$u_{\lambda,a} = (\lambda - L_*)^{-1} V_a.$$

Dynkin’s formula and  $V_a = \lambda u_{\lambda,a} - L_* u_{\lambda,a}$  rewrite (24.5.28) as

$$a \cdot X_t = M_t^a + M_t^{u_{\lambda,a}} + R_t^{\lambda,a}, \quad (24.5.32)$$

where

$$R_t^{\lambda,a} = -[u_{\lambda,a}(\xi_t) - u_{\lambda,a}(\xi_0)] + \lambda \int_0^t u_{\lambda,a}(\xi_s) ds. \quad (24.5.33)$$

The spectral estimates proved in SMX-24.05.01 give a diagonal choice  $\lambda = \lambda(t) \downarrow 0$  for which

$$t^{-1} \mathbb{E} \left[ \sup_{0 \leq s \leq tT} |R_s^{\lambda(t),a}|^2 \right] \longrightarrow 0. \quad (24.5.34)$$

Hence the diffusive limit is determined by the sum of the two martingales in (24.5.32).

**Step 5: the bracket and the variational formula.** For a local  $f$  put

$$\begin{aligned} \mathcal{Q}_a(f) = & \mathbb{E} \left[ \sum_{|e|=1} (1 - \xi(e)) (a \cdot e + f(\theta_e \xi) - f(\xi))^2 \right] \\ & + \mathbb{E} \left[ \sum_{\{x,y\} \in \mathcal{E}_*} (f(\xi^{x,y}) - f(\xi))^2 \right]. \end{aligned} \quad (24.5.35)$$

The two sums are exactly the squared jumps of the corrected displacement martingale: an environment exchange contributes only  $\Delta f$ , whereas a tagged jump by  $e$  contributes  $a \cdot e + \Delta_e f$ . Expanding the square, using (24.5.26) and (24.5.30), gives

$$\mathcal{Q}_a(f) = 2(1 - \rho)|a|^2 - 4\langle V_a, f \rangle + 2\langle f, -L_* f \rangle. \quad (24.5.36)$$

(The first constant is  $2(1 - \rho)|a|^2$  because the  $2d$  nearest-neighbour directions are used while each undirected bond in (24.5.21) has rate one.) The resolvent variational identity therefore implies

$$\lim_{\lambda \downarrow 0} \mathbb{E}[\text{bracket rate of } M^a + M^{u_{\lambda,a}}] = \inf_{f \in \mathcal{C}_*} \mathcal{Q}_a(f). \quad (24.5.37)$$

This defines the nonnegative quadratic form  $a \mapsto a \cdot \Sigma(\rho)a$ . Polarization defines the symmetric matrix  $\Sigma(\rho)$ .

**Step 6: martingale functional CLT.** For a fixed local corrector the bracket rate in (24.5.35) is an integrable stationary function of  $\xi_t$ , so ergodicity gives convergence of its time average to its  $\nu_\rho^*$  expectation. Jumps are bounded for a local corrector, hence the martingale Lindeberg condition holds. Approximate the resolvent correctors in the Dirichlet norm by local functions and then use (24.5.34); the same diagonal argument as in SMX-24.05.01 yields, for every  $a$ ,

$$\varepsilon a \cdot X_{t/\varepsilon^2} \Longrightarrow \sqrt{a \cdot \Sigma(\rho)a} B_t$$

in  $D([0, T], \mathbb{R})$ . Applying this simultaneously to a finite set of directions and using Cramér–Wold gives the vector process convergence (24.5.22). The limiting martingales are continuous, so the limit law is supported on continuous paths.

This completes the invariance principle and the variational characterization.  $\square$

## Chapter 6

# Second-order fluctuations and the microscopic route to KPZ

Weak asymmetry creates a remarkable crossover. First-order local equilibrium gives the linear Ornstein–Uhlenbeck field; at the critical asymmetry scale the second-order correction survives and becomes the Burgers nonlinearity.

### 6.1 Second-order Boltzmann–Gibbs principle

### 6.2 Weakly asymmetric fluctuations to stochastic Burgers

### 6.3 Uniqueness of energy solutions

### 6.4 Weak KPZ universality for selected conservative systems

## Chapter 7

# Frontiers in nonequilibrium universality

### 7.1 General nonequilibrium KPZ universality

*Frontier statement 7.1.*

Every sufficiently local one-dimensional growth or conservative system with one relevant mode, nonzero quadratic self-coupling, and short-range mixing should exhibit the KPZ 1 : 2 : 3 scaling exponents and the corresponding universal fixed-point statistics.

*Remark 7.2. Frontier isolation.* Rigorous weak-universality theorems cover broad but structured classes; a theorem with this informal generality is not known. This row is contextual only and is not available as a premise anywhere in the manuscript.

### 7.2 Multi-component nonlinear fluctuating hydrodynamics

*Frontier statement 7.3.*

For systems with several coupled conservation laws, the mode-coupling data should determine a universal decomposition into KPZ, diffusive, Levy, or other fluctuation modes according to nonlinear fluctuating hydrodynamics.

*Remark 7.4. Frontier isolation.* There are strong model-specific results and a mature prediction theory, but no complete rigorous classification at this level of generality. This row is contextual only and is not available as a premise anywhere in the manuscript.



## Chapter 8

# Hydrodynamic entropy and stochastic Burgers

This chapter supplies the Volume XXIV hydrodynamic and fluctuation proofs. The first section now owns the exact several-dimensional Rezakhanlou exclusion/zero-range parent, including its product local-equilibrium initial class, vector-flux entropy law, measure-valued reduction, and local-equilibrium conclusion. The following section preserves the independent one-dimensional nearest-neighbour ASEP proof developed in this chapter. The remaining sections own the multiscale second-order Boltzmann–Gibbs principle and the energy-solution route to stochastic Burgers.

### 8.1 The exact multidimensional Rezakhanlou parent

This section records the full scope of Rezakhanlou’s 1991 hydrodynamic theorem. The model class, initial-law relation, topology, entropy solution, and local-equilibrium conclusion below are the ones printed in the primary paper. In particular, “attractive systems” here does not mean an arbitrary attractive conservative process: the main theorem concerns the two explicit product-equilibrium classes below.

**Definition 8.1** (Rezakhanlou zero-range and exclusion classes). Fix  $d \geq 1$  and a probability kernel  $p : \mathbb{Z}^d \rightarrow [0, 1]$ . In both classes  $p$  has finite range. Write  $\eta^{u,v}$  for the configuration after one particle is moved from  $u$  to  $v$ .

The *zero-range class* has state space  $E_{\text{zr}} = \mathbb{N}^{\mathbb{Z}^d}$  and generator

$$(L_{\text{zr}}f)(\eta) = \sum_{u,v \in \mathbb{Z}^d} p(v-u)g(\eta(u))[f(\eta^{u,v}) - f(\eta)], \quad (24.8.11)$$

where  $g : \mathbb{N} \rightarrow [0, \infty)$  is bounded and nondecreasing and  $g(0) = 0 < g(1)$ . The kernel itself is irreducible:

$$\sum_{n \geq 1} p^{*n}(z) > 0 \quad (z \in \mathbb{Z}^d). \quad (24.8.12)$$

The *simple-exclusion class* has state space  $E_{\text{ex}} = \{0, 1\}^{\mathbb{Z}^d}$  and generator

$$(L_{\text{ex}}f)(\eta) = \sum_{u,v \in \mathbb{Z}^d} p(v-u)\eta(u)(1-\eta(v))[f(\eta^{u,v}) - f(\eta)]. \quad (24.8.13)$$

Here the symmetrized kernel is irreducible:

$$\sum_{n \geq 1} (p^{*n}(z) + p^{*n}(-z)) > 0 \quad (z \in \mathbb{Z}^d). \quad (24.8.14)$$

The distinction between (24.8.12) and (24.8.14) is part of the source hypothesis and will be retained below.

**Definition 8.2** (Product equilibria and the vector flux). For exclusion,  $\nu^r$  is Bernoulli product measure of density  $r \in [0, 1]$ . For zero range, put  $g!(0) = 1$  and  $g!(n) = \prod_{j=1}^n g(j)$ , and, for  $0 \leq \lambda < g_\infty := \sup_n g(n)$ , define

$$Z(\lambda) = \sum_{n \geq 0} \frac{\lambda^n}{g!(n)}, \quad \Theta_\lambda(n) = \frac{1}{Z(\lambda)} \frac{\lambda^n}{g!(n)}. \quad (24.8.15)$$

The mean  $\psi(\lambda) = \sum_n n \Theta_\lambda(n)$  is strictly increasing from 0 to  $\infty$ . If  $\lambda(r) = \psi^{-1}(r)$ , then  $\nu^r = \bigotimes_{u \in \mathbb{Z}^d} \Theta_{\lambda(r)} \cdot$   
Set

$$h(r) = \begin{cases} \lambda(r), & \text{zero range,} \\ r(1-r), & \text{simple exclusion,} \end{cases} \quad \gamma = \sum_{z \in \mathbb{Z}^d} zp(z), \quad \mathbf{F}(r) = \gamma h(r). \quad (24.8.16)$$

Thus  $h(r) = \mathbf{E}_{\nu^r} b(\eta(0), \eta(z))$  for  $z \neq 0$ , with  $b(n, m) = g(n)$  in the zero-range case and  $b(n, m) = n(1-m)$  for exclusion. On every bounded density interval  $h$  is Lipschitz. This also follows directly from the monotone equilibrium coupling: for  $r \geq c$ ,  $0 \leq h(r) - h(c) \leq \|g\|_\infty (r - c)$  in the zero-range case.

**Definition 8.3** (The printed local-equilibrium relation). Let  $\rho_0$  be a bounded measurable nonnegative function on  $\mathbb{R}^d$ ; in the exclusion case assume  $0 \leq \rho_0 \leq 1$ . A sequence of laws  $\mu^N$  satisfies  $\mu^N \sim \rho_0$  if it is a product law with one-site marginals

$$\mu^N \{\eta(u) = k\} = \nu^{\rho_{u,N}} \{\eta(u) = k\}, \quad (24.8.17)$$

and, for every  $K < \infty$ ,

$$\lim_{N \rightarrow \infty} \int_{|x| \leq K} |\rho_{[Nx],N} - \rho_0(x)| dx = 0. \quad (24.8.18)$$

Here  $[Nx]$  is the coordinatewise integer part. Condition (24.8.18), not pointwise sampling, is the source hypothesis. It permits sparse microscopic spikes in the zero-range parameters; these are removed in the proof by monotone truncation rather than by silently imposing a common  $L^\infty$  bound on all  $\rho_{u,N}$ .

For the process accelerated by  $N$ , define the empirical measure

$$\alpha_t^N = \frac{1}{N^d} \sum_{u \in \mathbb{Z}^d} \eta_{Nt}(u) \delta_{u/N}. \quad (24.8.19)$$

Its path law is taken in  $D([0, \infty), \mathcal{M}_+(\mathbb{R}^d))$  with the vague topology.

**Lemma 8.4** (Attractive coupling and invariant ordered components). *For either class, the basic coupling preserves coordinatewise order. The translation-invariant invariant laws of finite mean are mixtures of the  $\nu^r$ . Moreover, every translation-invariant invariant basic coupling is a mixture of components supported on one of the two orders  $\eta \leq \xi$  or  $\xi \leq \eta$ .*

*Proof.* For the common rate  $b(n, m)$ , the departure rate is nondecreasing in  $n$  and the arrival dependence is nonincreasing in  $m$ . Coupling common jumps at the minimum rate and using the excess rate in the appropriate coordinate therefore preserves order.

For completeness, the ordered-component argument is the multidimensional version of the discrepancy argument already used in the selected ASEP proof. If a translation-ergodic invariant coupling contained both signs of discrepancy with positive density, irreducibility would give a finite path along which an opposite-sign pair meets and annihilates with a fixed positive probability. In a cube of side  $R$  this would produce an annihilation rate of order  $R^d$ . Stationarity of the discrepancy count permits changes only through the boundary, of order  $R^{d-1}$ , a contradiction as  $R \rightarrow \infty$ . For exclusion the path may be traversed in either orientation, which is exactly why (24.8.14), rather than (24.8.12), is the required hypothesis.

An extremal translation-invariant stationary marginal can now be coupled invariantly with the product law of the same density. The joint law is ordered, but equal marginal densities force zero discrepancy density and hence equality almost surely. The marginal is therefore  $\nu^r$ . Ergodic decomposition proves the mixture statements.  $\square$

For  $\ell \geq 1$ , let  $T_\ell = [-\ell, \ell]^d \cap \mathbb{Z}^d$  and

$$M_{\ell, u}(\eta) = \frac{1}{|T_\ell|} \sum_{v \in T_\ell} \eta(u + v).$$

For  $c \in \mathbb{R}$  put

$$\mathbf{q}(r; c) = \text{sgn}(r - c)(\mathbf{F}(r) - \mathbf{F}(c)). \quad (24.8.20)$$

**Lemma 8.5** (Microscopic entropy inequality at block scale). *Suppose first that  $\mu^N \preceq \nu^{r*}$  for a finite  $r_*$ . For every  $c$  in the corresponding density domain,  $\varepsilon > 0$ , and nonnegative  $J \in C_c^\infty((0, \infty) \times \mathbb{R}^d)$ ,*

$$\begin{aligned} \lim_{\ell \rightarrow \infty} \liminf_{N \rightarrow \infty} P_{\mu^N}^N & \left( \int_0^\infty \frac{1}{N^d} \sum_u \partial_t J(t, u/N) |M_{\ell, u}(\eta_{Nt}) - c| dt \right. \\ & \left. + \int_0^\infty \frac{1}{N^d} \sum_u \nabla J(t, u/N) \cdot \mathbf{q}(M_{\ell, u}(\eta_{Nt}); c) dt \geq -\varepsilon \right) = 1. \end{aligned} \quad (24.8.21)$$

*Proof.* Couple  $\eta$  with a stationary process  $\xi$  of density  $c$ . Apply the coupled generator to the weighted discrepancy  $N^{-d} \sum_u J(t, u/N) |\eta(u) - \xi(u)|$ . Attractiveness gives a divergence of a coupled current plus a nonpositive annihilation term. Discrete summation by parts and finite range turn the divergence into the second term of (24.8.21), up to  $o(1)$ . The source quadratic-variation computation gives an  $O(N^{-d})$  bracket for the resulting martingale, hence its supremum vanishes in probability.

It remains to replace local discrepancy currents by block densities. Average the joint space-time laws over translates. Every limit is translation invariant and invariant for the coupled generator. By Lemma 8.4, its ergodic components are ordered product equilibrium couplings. On a component with first density  $r$ , the mean coupled current is

$$\text{sgn}(r - c) \gamma(h(r) - h(c)) = \mathbf{q}(r; c).$$

The block ergodic theorem identifies  $r$  with the limiting block density. This proves the two replacement limits for the absolute discrepancy and the current and yields (24.8.21).  $\square$

**Lemma 8.6** (Path tightness and the Young-measure limit). *Under a common stochastic upper bound  $\nu^{r*}$ , the laws of  $\alpha^N$  are tight. Every limit is concentrated on weakly continuous paths*

$\alpha_t(dx) = \rho(t, x)dx$ . Along the same subsequence the block densities generate a Young measure  $\pi_{t,x}(d\lambda)$  supported on  $[0, r_*]$ , with barycenter

$$\rho(t, x) = \int \lambda \pi_{t,x}(d\lambda), \quad (24.8.22)$$

and  $\pi$  satisfies the measure-valued Kruzhkov inequalities

$$\begin{aligned} \int_0^\infty \int_{\mathbb{R}^d} \left[ \partial_t J(t, x) \int |\lambda - c| \pi_{t,x}(d\lambda) \right. \\ \left. + \nabla J(t, x) \cdot \int \mathbf{q}(\lambda; c) \pi_{t,x}(d\lambda) \right] dx dt \geq 0 \end{aligned} \quad (24.8.23)$$

for every  $c$  and every nonnegative test  $J$  supported away from  $t = 0$ .

*Proof.* For  $G \in C_c^2(\mathbb{R}^d)$ , Dynkin's formula for  $\langle \alpha_t^N, G \rangle$  has uniformly bounded drift on compact time intervals. A jump changes it by  $O_G(N^{-d-1})$ , while the accelerated total rate meeting the support is  $O(N^{d+1})$  under the equilibrium upper bound. Thus the bracket is  $O_G(N^{-d-1})$ . Aldous' criterion for a countable dense set of tests gives tightness and continuity of limit paths. The same upper bound gives local uniform integrability and absolute continuity with density at most  $r_*$  in the stochastic-order sense.

Tightness of the block-density laws gives  $\pi$ . Testing the block mean linearly identifies its barycenter with the empirical density, proving (24.8.22). Finally pass successively  $N \rightarrow \infty$  and  $\ell \rightarrow \infty$  in Lemma 8.5; boundedness and the Young-measure convergence give (24.8.23).  $\square$

**Lemma 8.7** (Measure-valued entropy reduction and local initial trace). *Let  $\pi_{t,x}$  be supported on a bounded density interval and satisfy (24.8.23). Suppose its barycenter has the local initial trace  $\rho_0$  and*

$$\lim_{t \downarrow 0} \int_{|x| \leq K} \int |\lambda - \rho_0(x)| \pi_{t,x}(d\lambda) dx = 0 \quad (24.8.24)$$

for every  $K < \infty$ . Then

$$\pi_{t,x} = \delta_{\rho(t,x)} \text{quada.e.}, \quad (24.8.25)$$

where  $\rho$  is the unique bounded Kruzhkov solution of

$$\partial_t \rho + \nabla \cdot \mathbf{F}(\rho) = 0, \quad \rho(0, \cdot) = \rho_0. \quad (24.8.26)$$

*Proof.* First suppose that  $\rho_0 \in L^1 \cap L^\infty$ . Double the variables in (24.8.23) against the entropy inequalities for the Kruzhkov solution  $\rho$ . The standard diagonal mollifier limit gives

$$\frac{d}{dt} \int_{\mathbb{R}^d} \int |\lambda - \rho(t, x)| \pi_{t,x}(d\lambda) dx \leq 0$$

in the distributional sense. The initial value is zero by (24.8.24), so the nonnegative integral vanishes. Hence (24.8.25) holds. This is the measure-valued uniqueness step used in the primary proof, written here as the contraction argument rather than left as a name-only appeal.

For bounded nonintegrable data, insert a cutoff equal to one on  $|x| \leq K$  and let its exterior transition recede with speed larger than the Lipschitz constant of  $\mathbf{F}$  on the density range. The boundary term has the correct sign by the finite-speed local contraction. The same argument therefore proves the conclusion on each compact cylinder, and  $K$  is arbitrary.  $\square$

**Theorem 8.8** (Rezakhanlou hydrodynamic limit at exact source scope). *Choose either (24.8.11)–(24.8.12) or (24.8.13)–(24.8.14). Let  $\rho_0 \in L^\infty(\mathbb{R}^d)$  take values in the corresponding density domain and let  $\mu^N \sim \rho_0$  in the sense of (24.8.17)–(24.8.18). Then the path laws of (24.8.19) converge weakly to the point mass at the single weakly continuous path*

$$\rho(t, x) dx, \quad (24.8.27)$$

where  $\rho$  is the unique bounded Kruzhkov solution of (24.8.26), with vector flux (24.8.16). Equivalently, for every  $t > 0$ ,  $J \in C_c^\infty(\mathbb{R}^d)$ , and  $\delta > 0$ ,

$$\lim_{N \rightarrow \infty} P_{\mu^N}^N \left( \left| \frac{1}{N^d} \sum_u J(u/N) \eta_{Nt}(u) - \int J(x) \rho(t, x) dx \right| > \delta \right) = 0. \quad (24.8.28)$$

The solution obeys the local initial trace

$$\lim_{t \downarrow 0} \int_{|x| \leq K} |\rho(t, x) - \rho_0(x)| dx = 0 \quad (K < \infty) \quad (24.8.29)$$

and all entropy inequalities associated with (24.8.20).

*Proof.* Assume first that the microscopic parameters in (24.8.17) are bounded by a common  $r_*$ . The initial laws are then stochastically dominated by  $\nu^{r_*}$ . Lemmas 8.5 and 8.6 show that every subsequential limit produces a measure-valued entropy solution.

For a compactly supported Lipschitz profile, couple the initial process at each site with equilibrium processes whose densities are the nearby values  $\rho_0(u/N)$ . The basic coupling is contractive for the normalized total discrepancy. Uniform continuity of  $\rho_0$  then gives (24.8.24). For an integrable bounded profile, approximate in  $L^1$  by such profiles and use the same coupling contraction. For a merely bounded profile, agree with it on a growing cube and use the finite-propagation estimate for the graphical construction: a disagreement initially outside  $KN$  reaches an interior cube only with exponentially small probability when the cube recedes faster than a constant depending on the finite jump range and bounded rate. This establishes the local trace (24.8.24) in the full printed class.

Lemma 8.7 now makes every Young measure a Dirac mass at the same entropy solution. Hence every empirical-path limit is the deterministic path (24.8.27); tightness removes subsequences and proves path convergence and (24.8.28).

It remains to remove the temporary common microscopic bound. In the zero-range case replace  $\rho_{u,N}$  by  $\rho_{u,N} \wedge R$  and use the monotone one-site coupling. On each compact set the expected normalized discrepancy is

$$\frac{1}{N^d} \sum_{|u| \leq KN} (\rho_{u,N} - R)^+,$$

which tends to zero for every  $R > \|\rho_0\|_\infty$  by (24.8.18). Attractiveness does not increase the total discrepancy except for a vanishing boundary contribution on a slightly enlarged compact set. Letting first  $N \rightarrow \infty$  and then  $R \rightarrow \infty$  transfers the conclusion to the original laws. Exclusion needs no truncation. This proves the theorem without adding a hypothesis absent from the source.  $\square$

**Corollary 8.9** (Fixed-time local equilibrium at source strength). *Under the assumptions of the theorem, for every bounded cylinder function  $f$ , every  $J \in C_c(\mathbb{R}^d)$ , and every  $t > 0$ ,*

$$\lim_{N \rightarrow \infty} \mathbf{E}_{\mu^N} \left| \frac{1}{N^d} \sum_u J(u/N) (\tau_u f)(\eta_{Nt}) - \int_{\mathbb{R}^d} J(x) \bar{f}(\rho(t, x)) dx \right| = 0, \quad (24.8.30)$$

where  $\bar{f}(r) = \int f d\nu^r$ .

*Proof.* Take local space-time empirical laws around a fixed positive time. Their translation averages are tight under the same truncation and domination used above. Time averaging and the accelerated martingale problem make each local limit invariant for  $L$ ; the argument of Lemma 8.4 makes it a mixture of  $\nu^r$ . The block ergodic theorem identifies the mixing variable with the mesoscopic density. The hydrodynamic theorem and the measure-valued reduction make that density the deterministic value  $\rho(t, x)$ .

The source proof first obtains this statement after averaging over a short time interval. Equicontinuity at positive times follows by applying the coupled entropy estimate to the piecewise-linear functions  $r \mapsto |r - c|$  and then using uniform approximation of  $\bar{f}$  on the bounded density range. The time average can therefore be removed. Spatial simple-function approximation of  $J$  completes (24.8.30).  $\square$

## 8.2 Attractive systems and entropy solutions

**Definition 8.10** (Selected nearest-neighbour ASEP class). Fix  $p > q \geq 0$  with  $p + q = 1$  and  $\gamma = p - q$ . On  $\Omega = \{0, 1\}^{\mathbb{Z}}$  consider the nearest-neighbour ASEP

$$(LF)(\eta) = \sum_{x \in \mathbb{Z}} [p\eta(x)(1 - \eta(x+1)) + q\eta(x+1)(1 - \eta(x))] [F(\eta^{x, x+1}) - F(\eta)]. \quad (24.8.1)$$

Its Bernoulli products  $(\nu_r)_{0 \leq r \leq 1}$  are invariant and ergodic, and the basic graphical coupling is attractive. The signed equilibrium current and macroscopic flux are

$$j_x(\eta) = p\eta(x)(1 - \eta(x+1)) - q\eta(x+1)(1 - \eta(x)), \quad F(r) = \mathbf{E}_{\nu_r} j_0 = \gamma r(1 - r). \quad (24.8.2)$$

For a bounded Borel profile  $u_0 : \mathbb{R} \rightarrow [\delta, 1 - \delta]$ , put

$$u_{0,x}^N = N \int_{x/N}^{(x+1)/N} u_0(y) dy.$$

The selected initial law is the inhomogeneous Bernoulli product

$$\mu_{u_0}^N = \bigotimes_{x \in \mathbb{Z}} \text{Bernoulli}(u_{0,x}^N). \quad (24.8.3)$$

Cell averages are essential at this regularity: point samples of an arbitrary Borel representative need not converge to its Lebesgue class. This selected specialization is the internal theorem attached to the physical owner of SMX-24.04.02. It is not identified with the broader multidimensional attractive exclusion/zero-range source theorem.

**Definition 8.11** (Empirical density on Euler scale). For the ASEP above set

$$\pi_t^N = \frac{1}{N} \sum_{x \in \mathbb{Z}} \eta_{Nt}(x) \delta_{x/N}.$$

Convergence is in probability in  $D([0, T], \mathcal{M}_{\text{loc}}(\mathbb{R}))$ , tested against compactly supported  $C^1$  functions.

**Lemma 8.12** (Microscopic Kato inequality). *Let  $(\eta_t, \xi_t)$  evolve under the basic ASEP coupling. For every nonnegative  $\varphi \in C_c^1([0, T] \times \mathbb{R})$  the positive discrepancy satisfies the discrete entropy inequality*

$$\begin{aligned} & \frac{1}{N} \sum_x \varphi(t, x/N) (\eta_{Nt}(x) - \xi_{Nt}(x))^+ \\ & \leq \frac{1}{N} \sum_x \varphi(0, x/N) (\eta_0(x) - \xi_0(x))^+ + \int_0^t \frac{1}{N} \sum_x \partial_s \varphi(s, x/N) (\eta_{Ns}(x) - \xi_{Ns}(x))^+ ds \\ & \quad + \int_0^t \frac{1}{N} \sum_x (\nabla_N \varphi)(s, x/N) \mathbf{q}_x^+(\eta_{Ns}, \xi_{Ns}) ds + M_t^N, \end{aligned} \quad (24.8.4)$$

where  $\mathbf{q}_x^+$  is the bounded positive-discrepancy current and  $\sup_{t \leq T} |M_t^N| \rightarrow 0$  in probability. With the coordinates interchanged, the same assertion holds for  $(\xi - \eta)^+$  and a current  $\mathbf{q}_x^-$ . Adding the two inequalities gives the microscopic inequality for  $|\eta - \xi|$  with current  $\mathbf{q}_x^+ + \mathbf{q}_x^-$ .

*Proof.* Apply the coupled generator to  $(\eta(x) - \xi(x))^+$ . A coupled arrow either transports a positive discrepancy, leaves the positive-discrepancy field unchanged, or annihilates a neighbouring pair of opposite signs; it never creates such a pair. Transport contributes the difference of an incoming and an outgoing discrepancy current. Annihilation has a nonpositive compensator, which may be discarded in an upper bound. Summing against  $\varphi$  and performing discrete summation by parts gives (24.8.4).

Here and below  $\mathbf{q}_x^+$  denotes the signed rightward rate at which positive discrepancies cross the bond  $(x, x+1)$  in the graphical coupling;  $\mathbf{q}_x^-$  is defined after interchanging the two coordinates. This fixes the current convention pathwise, including away from ordered pairs, and also shows directly that both currents are bounded local functions.

We record the quadratic-variation count because Euler acceleration alone does not reduce the number of graphical rings. A transported discrepancy changes the weighted normalized sum by  $N^{-1}|\varphi(s, (x+1)/N) - \varphi(s, x/N)| = O(N^{-2})$ ; the total accelerated transport intensity meeting the support is  $O(N^2)$ , so these jumps contribute  $O(N^{-2})$  to the bracket. An annihilation can change the sum by  $O(N^{-1})$ , but the number of annihilations in the support is at most the initial number of discrepancies in a one-site enlargement plus the number entering through its two boundary bonds. Their expected total is  $O(N)$  on a fixed macroscopic time interval. Hence the annihilation contribution is  $O(N^{-1})$ . Thus  $\mathbf{E}\langle M^N \rangle_T = O(N^{-1})$ , and Doob's inequality gives the claimed uniform convergence in probability.  $\square$

For  $k, r \in [0, 1]$  put

$$\begin{aligned} Q_k^+(r) &= \mathbf{1}_{\{r > k\}} (F(r) - F(k)), \\ Q_k^-(r) &= \mathbf{1}_{\{r < k\}} (F(k) - F(r)), \\ Q_k(r) &= Q_k^+(r) + Q_k^-(r) = \text{sgn}(r - k) (F(r) - F(k)). \end{aligned} \quad (24.8.5)$$

**Lemma 8.13** (Invariant coupled limits for nearest-neighbour ASEP). *Let  $\bar{\mu}$  be invariant under spatial translations and under the basic coupled ASEP generator, and suppose that its second marginal is  $\nu_k$ . Then there is a probability measure  $\lambda$  on  $[0, 1]$  such that  $\bar{\mu}$  is a mixture of stationary couplings with marginal densities  $(r, k)$ ,  $r \sim \lambda$ , and each component is supported on  $\eta \geq \xi$  when  $r \geq k$  and on  $\eta \leq \xi$  when  $r \leq k$ . On the component of density  $r$ ,*

$$\mathbf{E} \mathbf{q}_0^\pm(\eta, \xi) = Q_k^\pm(r). \quad (24.8.5a)$$

*Proof.* We first record the one-dimensional no-coexistence argument. Under the basic coupling the total number of discrepancies in a finite interval can increase only through its two boundary bonds; in the bulk, discrepancies move or a pair of opposite signs annihilates. If an invariant translation-ergodic coupling gave positive density to both signs, some cylinder event containing an opposite-sign pair at finite separation would have positive probability. Nearest-neighbour irreducibility gives a fixed positive probability that the two discrepancies meet and annihilate in a fixed time while the finitely many relevant outside arrows are suppressed. Thus the expected bulk annihilation rate per unit length would be positive. On the other hand, stationarity of the discrepancy count in  $[-m, m]$  and the  $O(1)$  boundary flux make that rate vanish after division by  $2m + 1$  and passage to  $m \rightarrow \infty$ . This is a contradiction. Hence every invariant translation-ergodic basic coupling is supported on one of the two coordinatewise orders.

This also proves the marginal classification needed here. Let  $\mu$  be an extremal translation-invariant stationary ASEP law of density  $r$ . The set of translation-invariant invariant basic couplings of  $\mu$  and the Bernoulli law  $\nu_r$  is nonempty: start from any translation-invariant coupling, evolve by the basic coupling, and take a compact time average. Choose an extremal coupling in this set. Its two marginals remain  $\mu$  and  $\nu_r$ , and the no-coexistence argument makes it ordered. Since the two marginal densities are equal, the discrepancy density is zero; translation invariance then gives  $\eta = \xi$  almost surely. Thus  $\mu = \nu_r$ . Ergodic decomposition shows that every translation-invariant stationary ASEP law is a mixture of the  $\nu_r$ 's.

Now decompose  $\bar{\mu}$  into translation-ergodic invariant joint components. The second marginal is the extremal law  $\nu_k$ , so it remains  $\nu_k$  in almost every component; the first marginal is  $\nu_r$  for a component-dependent  $r$ . No coexistence gives  $\eta \geq \xi$  when  $r \geq k$  and  $\eta \leq \xi$  when  $r \leq k$ . On an ordered component the nonzero one-sided coupled discrepancy current is the corresponding difference of the two ordinary currents. Taking expectations and using  $\mathbf{E}_{\nu_r} j_0 = F(r)$  proves (24.8.5a).  $\square$

**Lemma 8.14** (ASEP coupled one-block entropy-flux replacement). *Couple the first coordinate from  $\mu_{u_0}^N$  with a stationary second coordinate of density  $k$  as follows. For i.i.d. uniform variables  $U_x$ , set  $\eta_0(x) = \mathbf{1}_{\{U_x \leq u_{0,x}^N\}}$  and  $\xi_0(x) = \mathbf{1}_{\{U_x \leq k\}}$ , and thereafter use the basic graphical coupling. Thus the two required marginal laws are unchanged, while the initial coupling is sitewise monotone. If  $\bar{\eta}_{x,\ell} = \ell^{-1} \sum_{y=x+1}^{x+\ell} \eta(y)$ , then for every bounded compactly supported deterministic weight  $a(s, x/N)$ ,*

$$\lim_{\ell \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathbf{E} \left| \int_0^T \frac{1}{N} \sum_x a(s, x/N) \{ \mathbf{q}_x^\sigma(\eta_{Ns}, \xi_{Ns}) - Q_k^\sigma(\bar{\eta}_{x,\ell}(Ns)) \} ds \right| = 0 \quad (24.8.6)$$

for each  $\sigma \in \{+, -\}$ .

*Proof.* This is the invariant-measure form of the one-block argument. We include the reduction because a bare finite-block spectral-gap sentence does not by itself control the nonstationary basic coupling. Average the translated joint laws of  $(\eta_{Ns}, \xi_{Ns})$  over a fixed compact macroscopic space-time box. The resulting random local empirical laws are tight. Dynkin's formula for a cylinder function  $g$  gives

$$\int L^c g d\Gamma_N = o_{\mathbb{P}}(1), \quad (24.8.6a)$$

because the microscopic generator is multiplied by  $N$ , whereas the time endpoints and the slowly varying spatial weight contribute only order one. Spatial summation by parts likewise makes every subsequential limit translation invariant. Here  $L^c$  is the basic coupled generator.

Consequently, almost every local space-time component of a subsequential limit is one of the invariant coupled laws classified in Lemma 8.13. If its first marginal has density  $r$ , the one-sided



current means are  $Q_k^+(r)$  and  $Q_k^-(r)$ . The block ergodic theorem identifies the same  $r$  as the limit of  $\bar{\eta}_{x,\ell}$  when first  $N \rightarrow \infty$  and then  $\ell \rightarrow \infty$ ; the stationary second block density converges to  $k$ . Thus every subsequential limit of the integral in (24.8.6) is zero in probability. The integral is uniformly bounded by a constant depending only on  $a$  and its support, so uniform integrability upgrades convergence in probability to the displayed  $L^1$  convergence. A partition of a general signed weight into positive and negative simple weights, followed by bounded approximation, completes the argument.  $\square$

**Theorem 8.15** (Selected one-dimensional ASEP specialization — independent proof). *Let  $\eta^N$  be the ASEP (24.8.1), started from (24.8.3), and let  $u$  be the unique Kruzhkov entropy solution of*

$$\partial_t u + \partial_x F(u) = 0, \quad F(u) = \gamma u(1 - u), \quad u(0, \cdot) = u_0. \quad (24.8.8)$$

Then for every  $T < \infty$ ,

$$\pi^N \xrightarrow[N \rightarrow \infty]{\mathbb{P}} u(t, x) dx \quad \text{in } D([0, T], \mathcal{M}_{\text{loc}}(\mathbb{R})). \quad (24.8.9)$$

*Proof.* The cell averages in (24.8.3) give the asserted mean empirical profile by Lebesgue differentiation, and independence gives convergence in probability by a variance estimate. We give the compactness step at the strength used below. For  $G \in C_c^2(\mathbb{R})$ , Dynkin's formula reads

$$\langle \pi_t^N, G \rangle = \langle \pi_0^N, G \rangle + \int_0^t \frac{1}{N} \sum_x (\nabla_N G)(x/N) j_x(\eta_{Ns}) ds + M_t^N(G). \quad (24.8.9a)$$

The drift has uniformly bounded variation on compact time intervals, and  $\mathbf{E}\langle M^N(G) \rangle_T = O_G(N^{-2})$ . Aldous' criterion for a countable dense family of tests therefore gives path tightness. Compact containment is deterministic because  $0 \leq \eta \leq 1$ . More precisely,  $\pi_t^N([a, b]) \leq b - a + 2/N$ ; hence every subsequential measure-valued limit is absolutely continuous, say  $u(t, x) dx$ , with  $0 \leq u \leq 1$ . Let a subsequence converge to such a limit.

Fix  $k \in [0, 1]$  and couple the process with a stationary ASEP of density  $k$ . Use (24.8.4) and its coordinate-swapped version with a smooth nonnegative test function. Apply (24.8.6) with  $\sigma = +$  and  $\sigma = -$ , respectively, and add the two inequalities. First  $N \rightarrow \infty$  and then  $\ell \rightarrow \infty$ . The stationary second empirical field converges to the constant  $k$ . Lebesgue differentiation and the domination  $0 \leq \pi^N \leq dx + o(1)$  identify the first block density with  $u$ . Since  $Q_k^+(u) + Q_k^-(u) = \text{sgn}(u - k)(F(u) - F(k))$ , for every rational  $k \in [0, 1]$  we obtain, in distributions,

$$\partial_t |u - k| + \partial_x [\text{sgn}(u - k)(F(u) - F(k))] \leq 0 \quad (24.8.10)$$

and continuity in  $k$  extends the inequality to all  $k$ . For the common-uniform initial coupling used in Lemma 8.14,

$$\mathbf{E}(\eta_0(x) - \xi_0(x))^+ = (u_{0,x}^N - k)^+, \quad \mathbf{E}(\xi_0(x) - \eta_0(x))^+ = (k - u_{0,x}^N)^+.$$

Independence across sites gives concentration of their weighted sums, while Lebesgue differentiation and dominated convergence identify their limits. Consequently the sum of the two initial terms in (24.8.4) gives the Kruzhkov initial trace  $|u_0 - k|$  in  $L_{\text{loc}}^1$ .

Thus every subsequential limit is a bounded Kruzhkov entropy solution of (24.8.8). SMX-24.04.03 gives uniqueness by the local  $L^1$  contraction, so all subsequences have the same deterministic limit. The entropy solution is continuous in time with values in  $L_{\text{loc}}^1$ ; hence the limiting measure-valued path is continuous. Tightness then yields (24.8.9) for the full sequence, and convergence in law to this deterministic path is convergence in probability.  $\square$

### 8.3 Second-order Boltzmann–Gibbs replacement

**Definition 8.16** (Static compressibility). For a translation-invariant equilibrium law  $\nu_\rho$  with occupation variables  $\eta(x)$  of mean  $\rho$ , the *static compressibility* (when the limit exists) is the asymptotic block-variance density

$$\chi(\rho) = \lim_{\ell \rightarrow \infty} \frac{1}{\ell} \text{Var}_{\nu_\rho} \left( \sum_{x=1}^{\ell} \eta(x) \right).$$

If the covariance series is absolutely summable, stationarity and the identity

$$\frac{1}{\ell} \text{Var} \left( \sum_{x=1}^{\ell} \eta(x) \right) = \sum_{|z| < \ell} \left( 1 - \frac{|z|}{\ell} \right) \text{Cov}_{\nu_\rho}(\eta(0), \eta(z))$$

show by dominated convergence that equivalently

$$\chi(\rho) = \sum_{z \in \mathbb{Z}} \text{Cov}_{\nu_\rho}(\eta(0), \eta(z)).$$

Thus the definition does not require absolute covariance summability; that stronger hypothesis is merely a convenient sufficient condition for the covariance-sum formula. The coefficient  $\chi(\rho)$  is the equilibrium density-fluctuation variance appearing in the Boltzmann–Gibbs replacement and in the limiting white-noise covariance.

**Definition 8.17** (Second-order Boltzmann–Gibbs class). Fix a density  $\rho$ . A one-dimensional stationary conservative process is in the *second-order Boltzmann–Gibbs class at  $\rho$*  if: (i)  $\nu_\rho$  is stationary and ergodic with finite static compressibility  $\chi(\rho)$ ; (ii) on every block  $\Lambda_\ell$ , the symmetric generator has spectral gap  $\text{Var}_{\nu_\rho}(g) \leq C\ell^2 \mathcal{D}_{\Lambda_\ell}(g)$ ; (iii) conditional expectations of local functions given the block density admit the second-order equivalence-of-ensembles expansion with an  $O(\ell^{-3/2})$  remainder in  $L^2(\nu_\rho)$ ; and (iv) the sector/ $H_{-1}$  bounds needed for the Kipnis–Varadhan estimate are uniform under block translation. These are the analytic inputs used in the dyadic replacement proof below. They abstract the hypotheses used by Gonçalves–Jara; the original speed-change exclusion theorem is recovered by taking the Bernoulli laws and their finite-volume spectral-gap/equivalence-of-ensembles estimates.

For a local observable  $f$ , write

$$\bar{f}(\sigma) = \mathbf{E}_{\nu_\sigma}[f]$$

whenever this equilibrium expectation is defined, and assume  $\bar{f}$  is  $C^2$  near the density under consideration. Let  $\nu_\rho$  be the invariant law of a process in the second-order Boltzmann–Gibbs class and let  $f$  satisfy

$$\bar{f}(\rho) = 0, \quad \bar{f}'(\rho) = 0. \quad (24.5)$$

Write  $\eta^\ell(x) = \ell^{-1} \sum_{y=x+1}^{x+\ell} \eta(y)$  and

$$\mathcal{Q}_\rho(\ell; x, \eta) = (\eta^\ell(x) - \rho)^2 - \frac{\chi(\rho)}{\ell}.$$

**Lemma 8.18** (Source-strength multiscale second-order replacement). *Let  $f$  satisfy (24.5) and have support contained in a fixed block. For every  $\ell$  larger than that support, every  $t \leq T$ , every  $N \geq 1$ ,*

and every deterministic measurable  $v : [0, T] \times \mathbb{Z} \rightarrow \mathbb{R}$  with  $\int_0^T \sum_x v_s(x)^2 ds < \infty$ , the process speeded by  $N^2$  satisfies

$$\begin{aligned} \mathbf{E}_{\nu_\rho} \left[ \left| \int_0^t \sum_x v_s(x) \left\{ f(\tau_x \eta_{sN^2}) - \frac{\bar{f}''(\rho)}{2} \mathcal{Q}_\rho(\ell; x, \eta_{sN^2}) \right\} ds \right|^2 \right] \\ \leq C \left( \frac{\ell}{N^2} + \frac{t}{\ell^2} \right) \int_0^t \sum_x v_s(x)^2 ds. \end{aligned} \quad (24.6)$$

The constant depends only on  $f, \rho$  and the constants in the class, not on  $N, \ell, t$  or block translates. In addition, for a time-independent finitely supported sequence  $v(x)$ ,

$$\begin{aligned} \mathbf{E}_{\nu_\rho} \left[ \sup_{0 \leq r \leq T} \left| \int_0^r \sum_x v(x) \left\{ f(\tau_x \eta_{sN^2}) - \frac{\bar{f}''(\rho)}{2} \mathcal{Q}_\rho(\ell; x, \eta_{sN^2}) \right\} ds \right|^2 \right] \\ \leq C \left( \frac{T\ell}{N^2} + \frac{T^2}{\ell^2} \right) \sum_x v(x)^2. \end{aligned} \quad (24.6m)$$

Thus (24.6) is the source-strength terminal-time estimate, while (24.6m) is the maximal fixed-weight extension used later in this volume.

*Proof.* We record both estimates because they serve different purposes. The terminal-time form permits time-dependent test weights, exactly as in the original second-order Boltzmann–Gibbs theorem; the maximal form is obtained from the same decomposition using the maximal Kipnis–Varadhan inequality.

*One block.* Put

$$\psi_f(\ell_0; \eta) = \mathbf{E}_{\nu_\rho}[f \mid \text{particle number in } \Lambda_{\ell_0}].$$

Split the sum over  $x$  into  $\ell_0$  residue classes so that translated blocks in each class are disjoint. Pointwise in time, the block spectral gap and the variational definition of the  $H_{-1}$  norm give

$$\left\| \sum_x v_s(x) \tau_x(f - \psi_f(\ell_0)) \right\|_{-1}^2 \leq C \ell_0^2 \sum_x v_s(x)^2. \quad (24.6a)$$

The time-dependent Kipnis–Varadhan inequality for the generator  $N^2 L$  therefore bounds the terminal-time contribution by  $C N^{-2} \ell_0^2 \int_0^t \sum_x v_s(x)^2 ds$ . For fixed  $v$ , its maximal version gives the corresponding  $CT N^{-2} \ell_0^2 \sum_x v(x)^2$  bound.

*Renormalization step.* Let  $\psi_f(r) = \mathbf{E}[f \mid \text{particle number in } \Lambda_r]$ . Conditioning successively on adjacent  $r$ -blocks and then on their union, the spectral gap and  $\bar{f}'(\rho) = 0$  give, again pointwise in time,

$$\left\| \sum_x v_s(x) \tau_x(\psi_f(r) - \psi_f(2r)) \right\|_{-1}^2 \leq C r \sum_x v_s(x)^2. \quad (24.6b)$$

Summing the dyadic scales  $r = 2^j \ell_0$  up to  $r \asymp \ell$  gives

$$\mathbf{E} \left| \int_0^t \sum_x v_s(x) \tau_x(\psi_f(\ell_0) - \psi_f(\ell))(\eta_{sN^2}) ds \right|^2 \leq C \frac{\ell}{N^2} \int_0^t \sum_x v_s(x)^2 ds. \quad (24.6c)$$

For fixed  $v$ , the maximal inequality gives  $CT \ell N^{-2} \sum_x v(x)^2$ . A final non-dyadic comparison satisfies the same estimates.

*Quadratic projection.* The second-order equivalence-of-ensembles assumption gives, uniformly in the block density,

$$\psi_f(\ell; \eta) = \frac{\bar{f}''(\rho)}{2} \left( (\eta^\ell - \rho)^2 - \frac{\chi(\rho)}{\ell} \right) + r_\ell(\eta), \quad \|r_\ell\|_{L^2(\nu_\rho)} \leq C\ell^{-3/2}. \quad (24.6d)$$

Split into  $\ell$  residue classes. Cauchy–Schwarz in time, stationarity and the block support of  $r_\ell$  imply

$$\mathbf{E} \left| \int_0^t \sum_x v_s(x) \tau_x r_\ell(\eta_{sN^2}) ds \right|^2 \leq C \frac{t}{\ell^2} \int_0^t \sum_x v_s(x)^2 ds. \quad (24.6e)$$

For fixed  $v$ , the same argument with the trivial maximal bound gives  $CT^2\ell^{-2} \sum_x v(x)^2$ . Combining (24.6a)–(24.6e) proves (24.6) and (24.6m).  $\square$

**Theorem 8.19** (Second-order Boltzmann–Gibbs principle). *Let  $\eta_t$  be a process in the second-order Boltzmann–Gibbs class at  $\rho$ , started from  $\nu_\rho$ , and let  $f$  satisfy (24.5). Then the source-strength quantitative replacement (24.6) holds for arbitrary deterministic  $L_t^2 \ell_x^2$  weights. In particular, for  $\varphi \in C_c^\infty(\mathbb{R})$  and  $\ell = \lfloor \varepsilon N \rfloor$ , set*

$$R_{N,\varepsilon}(t, \varphi) = \int_0^t \sum_x \nabla_N \varphi(x/N) \left[ f(\tau_x \eta_{sN^2}) - \frac{\bar{f}''(\rho)}{2} \mathcal{Q}_\rho(\ell; x, \eta_{sN^2}) \right] ds.$$

Then for every  $T < \infty$ ,

$$\lim_{\varepsilon \downarrow 0} \limsup_{N \rightarrow \infty} \mathbf{E}_{\nu_\rho} \left[ \sup_{0 \leq t \leq T} |R_{N,\varepsilon}(t, \varphi)|^2 \right] = 0. \quad (24.7)$$

Thus the centered quadratic block field, with the forced subtraction  $\chi(\rho)/\ell$ , is the unique second-order local-equilibrium contribution on the diffusive fluctuation scale.

*Proof.* The quantitative assertion is Lemma 8.18. For the fluctuation corollary apply (24.6m) with

$$v_N(x) = \nabla_N \varphi(x/N).$$

Because  $\varphi$  is smooth and compactly supported,

$$\frac{1}{N} \sum_x v_N(x)^2 = \frac{1}{N} \sum_x |\nabla_N \varphi(x/N)|^2 \longrightarrow \int_{\mathbb{R}} |\varphi'(u)|^2 du. \quad (24.7a)$$

Thus  $\sum_x v_N(x)^2 = O(N)$ . With  $\ell = \lfloor \varepsilon N \rfloor$ , (24.6m) is bounded by

$$C_\varphi \left( T\varepsilon + \frac{T^2}{\varepsilon^2 N} \right) + o_N(1).$$

Taking first  $N \rightarrow \infty$  and then  $\varepsilon \downarrow 0$  proves (24.7). Notice that there is deliberately no extra  $N^{-1/2}$  in front of the sum in  $R_{N,\varepsilon}$ : the asymmetric current term in the critical weak-asymmetry martingale decomposition is of order one precisely at this normalization.

**Source and scope.** Gonçalves–Jara’s Theorem 4.2 is the terminal-time estimate (24.6), with time-dependent weights and the error  $(\ell/N^2 + t/\ell^2) \int_0^t \|v_s\|_{\ell^2}^2 ds$ . The maximal estimate (24.6m) is the book’s explicit extension obtained from the same one-block/two-block decomposition using the maximal Kipnis–Varadhan inequality already owned in Chapter 5; it is not attributed to the source theorem itself.  $\square$

## 8.4 Weak asymmetry and the Burgers martingale problem

**Definition 8.20** (Stationary energy solution of stochastic Burgers). Fix coefficients  $D > 0$ ,  $\lambda \in \mathbb{R}$ , and  $\chi > 0$ . A pair  $(\mathcal{Y}, \mathcal{A})$  is a *stationary energy solution* of

$$d\mathcal{Y} = D\Delta\mathcal{Y} dt + \lambda\partial_x(\mathcal{Y}^2) dt + \sqrt{2D\chi}\partial_x dW \quad (24.7b)$$

if the following hold.

1.  $\mathcal{Y}$  has continuous  $\mathcal{S}'(\mathbb{R})$ -valued paths and, for every  $t$ ,  $\mathcal{Y}_t$  is spatial white noise of variance  $\chi$ .
2. For every  $\varphi \in \mathcal{S}(\mathbb{R})$ ,

$$\mathcal{Y}_t(\varphi) = \mathcal{Y}_0(\varphi) + D \int_0^t \mathcal{Y}_s(\Delta\varphi) ds + \mathcal{A}_t(\varphi) + M_t(\varphi), \quad (24.7c)$$

where  $M(\varphi)$  is a continuous martingale with bracket  $[M(\varphi)]_t = 2D\chi t \|\partial_x \varphi\|_2^2$ , and  $\mathcal{A}(\varphi)$  is continuous, starts from zero and has zero quadratic variation.

3. For an approximate identity  $\rho^L$ , the processes

$$\int_0^t ((\mathcal{Y}_s * \rho^L)^2 - \chi \|\rho^L\|_2^2) (-\partial_x \varphi) ds$$

converge uniformly on compact time intervals in probability, independently of the approximate identity, and  $\mathcal{A}$  equals  $\lambda$  times this limit.

4. For every  $T$ , the reversed pair  $\widehat{\mathcal{Y}}_t = \mathcal{Y}_{T-t}$  and  $\widehat{\mathcal{A}}_t = -(\mathcal{A}_T - \mathcal{A}_{T-t})$  satisfies the same Ornstein–Uhlenbeck martingale decomposition with a backward space–time white noise. Equivalently, the forward and reversed  $H_{-1}$  energy estimates used below both hold.

This is the controlled-process formulation needed by the uniqueness theorem; it is stronger than merely asking that a formal nonlinear martingale problem hold.

**Definition 8.21** (Equilibrium flux and characteristic frame). Consider a weakly asymmetric conservative family whose unaccelerated generator has the form

$$L_N = S + N^{-1/2}A,$$

where  $S$  is self-adjoint and  $A$  is anti-self-adjoint in  $L^2(\nu_\rho)$ . Thus the stationary time-reversed family has generator  $L_N^* = S - N^{-1/2}A$ . We require the spectral-gap, sector, equivalence-of-ensembles and martingale estimates used below uniformly for both signs of  $A$ . Let  $j_x^A$  be the local antisymmetric bond current for the forward family. After diffusive acceleration by  $N^2$ , the equilibrium asymmetric current across one bond has size  $N^{3/2}\mathbf{H}(\sigma)$ , where

$$\mathbf{H}(\sigma) = \mathbf{E}_{\nu_\sigma}[j_0^A]. \quad (24.7d)$$

We call  $\mathbf{H}$  the *equilibrium flux*. In the speed-change exclusion normalization of Gonçalves–Jara,  $\mathbf{H}(\sigma) = \alpha\chi(\sigma)D(\sigma)$ . At density  $\rho$ , the characteristic displacement in lattice coordinates is  $N^{3/2}\mathbf{H}'(\rho)t$ . Hence the characteristic-centered density field is

$$\mathcal{Y}_t^N(\varphi) = \frac{1}{\sqrt{N}} \sum_x \varphi\left(\frac{x - N^{3/2}\mathbf{H}'(\rho)t}{N}\right) (\eta_{tN^2}(x) - \rho). \quad (24.8)$$

When  $\mathbf{H}'(\rho) = 0$  this is the unshifted field. The coefficient predicted by the second-order Einstein relation is

$$\lambda_\rho = \frac{1}{2}\mathbf{H}''(\rho). \quad (24.9)$$

**Lemma 8.22** (Identification of the nonlinear current). *Assume the current decomposition above, the equilibrium fluctuation CLT, and the replacement hypotheses of SMX-24.06.01. In the characteristic frame, Dynkin's formula has the form*

$$\mathcal{Y}_t^N(\varphi) = \mathcal{Y}_0^N(\varphi) + M_t^N(\varphi) + D(\rho) \int_0^t \mathcal{Y}_s^N(\Delta\varphi) ds + \mathcal{A}_t^N(\varphi) + o_N(1). \quad (24.10)$$

Along every tight subsequence,

$$\mathcal{A}_t^N(\varphi) \implies \lambda_\rho \int_0^t \partial_x \mathcal{Y}_s^{\circ 2}(\varphi) ds, \quad (24.11)$$

and the martingales converge to Brownian martingales with bracket

$$2D(\rho)\chi(\rho)t\|\partial_x\varphi\|_2^2. \quad (24.12)$$

*Proof.* Expand the equilibrium asymmetric current around  $\rho$ :

$$H(\rho + z) = H(\rho) + H'(\rho)z + \frac{1}{2}H''(\rho)z^2 + o(z^2). \quad (24.13)$$

The constant term disappears after discrete summation by parts. Differentiating the moving test function in (24.8) produces exactly the negative of the linear-current term  $H'(\rho)z$ ; this is why the Galilean/characteristic recentering is part of the theorem rather than an optional change made after the limit. SMX-24.06.01 replaces the remaining centered local current by the quadratic block field, and (24.13) gives the coefficient  $\lambda_\rho = H''(\rho)/2$ . The uniform replacement estimate is precisely the energy estimate needed to define the limiting Wick square.

For the martingale, compute the predictable bracket from the microscopic jump rates. The first-order Boltzmann–Gibbs principle replaces the local bracket density by its equilibrium mean, and the equilibrium fluctuation-dissipation normalization gives  $2D(\rho)\chi(\rho)$ . The martingale functional CLT then gives (24.12).  $\square$

**Theorem 8.23** (Weakly asymmetric equilibrium systems converge subsequentially to energy Burgers). *Let the process start from its invariant law  $\nu_\rho$ . Assume the second-order Boltzmann–Gibbs hypotheses, the self-adjoint/anti-self-adjoint decomposition  $L_N = S + N^{-1/2}A$  above together with the same uniform estimates for the adjoint family  $L_N^* = S - N^{-1/2}A$ , the current decomposition (24.7d), the equilibrium fluctuation CLT, and the diffusive martingale/bracket bounds used in Lemma 8.22. Then  $(\mathcal{Y}_t^N)_{0 \leq t \leq T}$  defined in the characteristic frame (24.8) is tight in  $D([0, T], \mathcal{S}'(\mathbb{R}))$ . Every limit point is a stationary energy solution of*

$$d\mathcal{Y}_t = D(\rho)\Delta\mathcal{Y}_t dt + \frac{1}{2}H''(\rho)\partial_x(\mathcal{Y}_t^{\circ 2}) dt + \sqrt{2\chi(\rho)D(\rho)} \partial_x dW_t. \quad (24.14)$$

*In particular the invariant one-time law of the limit is spatial white noise of variance  $\chi(\rho)$ . This theorem is an equilibrium/stationary statement; no non-equilibrium initial-law extension is being claimed here.*

*Proof.* Mitoma's criterion reduces tightness to the real-valued processes  $\mathcal{Y}^N(\varphi)$ . The martingale bracket bound and the Kipnis–Varadhan estimate on the diffusive and nonlinear drifts give Aldous tightness in a negative Sobolev space. Pass to a limit in (24.10). Lemma 8.22 identifies the nonlinear term and the noise. Because the microscopic process starts from  $\nu_\rho$ , its one-time laws are stationary; the assumed equilibrium CLT therefore yields white-noise marginals of variance  $\chi(\rho)$  at every limiting time.

There is one additional point needed for the later uniqueness theorem. Gonçalves–Jara’s original energy-solution definition did not impose the time-reversal condition used by Gubinelli–Perkowski. For the selected class here, stationarity identifies the reversed microscopic path law with the adjoint dynamics generated by  $L_N^* = S - N^{-1/2}A$ . By hypothesis this adjoint family satisfies the same replacement and martingale estimates. Applying the preceding tightness/identification argument to the reversed sequence gives the backward Ornstein–Uhlenbeck martingale decomposition with the antisymmetric drift reversed. Passing to a joint subsequence therefore supplies exactly the forward–backward controlled-process condition in the definition above. Thus every limit belongs to the stronger Gubinelli–Perkowski energy class, not merely to the weaker 2014 class.

The source-level logical status of Gonçalves–Jara’s Theorem 2.8 remains tightness and identification of subsequential limits, not convergence of the full sequence. Full convergence below uses both this strengthened adjoint-stability bridge and the independent uniqueness theorem.  $\square$

**Corollary 8.24** (Edwards–Wilkinson side of the weak-asymmetry crossover). *Under the same uniform hypotheses, if  $H''(\rho) = 0$ , then the whole density-fluctuation sequence converges to the stationary Ornstein–Uhlenbeck field*

$$d\mathcal{Y} = D(\rho)\Delta\mathcal{Y}dt + \sqrt{2\chi(\rho)D(\rho)}\partial_x dW.$$

*The same conclusion holds if the antisymmetric perturbation is  $a_N N^{-1/2}A$  with  $a_N \rightarrow 0$  and the estimates above are uniform in  $N$ .*

*Proof.* In the first case the quadratic coefficient in (24.14) vanishes. In the second case the flux and its derivatives are multiplied by  $a_N$ , so the quadratic coefficient tends to zero. The stationary Ornstein–Uhlenbeck martingale problem is unique, hence subsequential identification promotes to full convergence. This is the crossover described after Gonçalves–Jara Theorem 2.8.  $\square$

## 8.5 Uniqueness of stationary energy solutions

**Lemma 8.25** (Anchored primitive of a stationary Burgers field). *Fix  $\sigma \in C_c^\infty(\mathbb{R})$  with  $\sigma \geq 0$  and  $\int \sigma = 1$ , and set*

$$\Theta_x(z) = \mathbf{1}_{(-\infty, x]}(z) - \int_z^\infty \sigma(y) dy.$$

*Then  $\Theta_x \in L^2(\mathbb{R})$ ,  $\partial_x \Theta_x = \delta_x$  in the parameter  $x$ , and for every stationary Burgers field  $u_t$  the random function*

$$h_t(x) := u_t(\Theta_x) + c_t$$

*is a spatial primitive of  $u_t$  for any scalar process  $c_t$ . Moreover, changing  $\sigma$  changes  $h_t$  only by an additive random constant independent of  $x$ .*

*Proof.* The two terms in  $\Theta_x$  have the same limits at  $\pm\infty$ , so their difference is square-integrable; differentiation with respect to the parameter  $x$  gives  $\delta_x$ . Hence for  $a < b$ ,

$$h_t(b) - h_t(a) = u_t(\Theta_b - \Theta_a) = u_t(\mathbf{1}_{(a,b]}),$$

first for smooth approximations and then in  $L^2$  by stationarity of white noise. This is precisely the distributional identity  $\partial_x h_t = u_t$ . If  $\tilde{\sigma}$  is another anchor, then  $\Theta_x^\sigma - \Theta_x^{\tilde{\sigma}}$  is independent of  $x$ ; applying  $u_t$  therefore changes the primitive by an  $x$ -independent scalar.  $\square$

**Lemma 8.26** (Forward-backward  $H_{-1}$  estimate). *Let  $(u, A)$  be a stationary energy solution and let  $L_0$  be the symmetric Ornstein-Uhlenbeck generator in the white-noise invariant measure. For every centered cylinder functional  $F$  in the form domain of  $(-L_0)^{-1/2}$ ,*

$$\mathbf{E} \left[ \sup_{0 \leq t \leq T} \left| \int_0^t F(u_s) ds \right|^2 \right] \leq CT \|(-L_0)^{-1/2} F\|_{L^2(\mu)}^2. \quad (24.10)$$

*The same bound holds for limits of cylinder functionals in the  $H_{-1}$  norm.*

*Proof.* For a cylinder  $G$  in the domain of  $L_0$ , the forward martingale problem gives

$$G(u_t) - G(u_0) = \int_0^t (L_0 G + AG)(u_s) ds + M_t^G,$$

where  $A$  is the antisymmetric Burgers part. The time-reversed energy martingale problem has the same  $L_0$  and the opposite  $A$ . Apply the reversed identity on  $[T-t, T]$ , add the two formulas, and cancel  $A$ . Thus  $2 \int_0^t (-L_0 G)(u_s) ds$  is the sum of two martingale increments. Doob's  $L^2$  inequality and the carré-du-champ identity give

$$\mathbf{E} \sup_{t \leq T} \left| \int_0^t (-L_0 G)(u_s) ds \right|^2 \leq CT \langle G, -L_0 G \rangle_{L^2(\mu)}.$$

Given centered  $F$ , choose  $G = (-L_0)^{-1} F$  first in a finite Wiener-chaos truncation and then approximate in the form norm. This yields (24.10). Completeness of the  $H_{-1}$  norm gives the extension to its closure.  $\square$

**Lemma 8.27** (Dirichlet-process Itô and product rules). *Let  $Y = M + A$  be continuous, where  $M$  is a continuous semimartingale and  $A$  has zero quadratic variation. If  $f \in C^2(\mathbb{R})$ , then*

$$f(Y_t) = f(Y_0) + \int_0^t f'(Y_s) dY_s + \frac{1}{2} \int_0^t f''(Y_s) d[M]_s, \quad (24.10a)$$

*where the integral against  $A$  is the forward integral obtained from the regularizing approximations defining the energy solution. If  $S$  is a continuous semimartingale, then*

$$d(SY) = S dY + Y dS + d[S, M], \quad (24.10b)$$

*and  $[S, A] = 0$ .*

*Proof.* For a partition  $\pi$ , Taylor's formula gives

$$f(Y_{t_{i+1}}) - f(Y_{t_i}) = f'(Y_{t_i}) \Delta_i Y + \frac{1}{2} f''(Y_{t_i}) (\Delta_i Y)^2 + o((\Delta_i Y)^2)$$

uniformly after localization. Since  $\sum_i (\Delta_i A)^2 \rightarrow 0$  in probability and

$$\left| \sum_i \Delta_i M \Delta_i A \right| \leq \left( \sum_i (\Delta_i M)^2 \right)^{1/2} \left( \sum_i (\Delta_i A)^2 \right)^{1/2} \rightarrow 0,$$

the quadratic term converges to  $[M]$ , proving (24.10a). The same partition identity  $\Delta(SY) = S \Delta Y + Y \Delta S + \Delta S \Delta Y$  shows that the only surviving cross variation is  $[S, M]$ ; the estimate above gives  $[S, A] = 0$  and hence (24.10b). The forward integrals are stable under the mollified approximations used below, so the identities pass from the regularized energy solutions to their limits.  $\square$



**Lemma 8.28** (Reduction to the unit Ornstein–Uhlenbeck normalization). *Let  $u$  solve the stationary Burgers martingale problem*

$$du = D\Delta u dt + \lambda \partial_x(u^{\circ 2}) dt + \sqrt{2D\chi} \partial_x dW, \quad D, \chi > 0.$$

Define

$$\tilde{u}_t = \chi^{-1/2} u_{t/D}, \quad \tilde{W}_t = \sqrt{D} W_{t/D}, \quad \tilde{\lambda} = \frac{\lambda\sqrt{\chi}}{D}.$$

Then  $\tilde{W}$  is space–time white noise and  $\tilde{u}$  is a stationary energy solution of

$$d\tilde{u} = \Delta \tilde{u} dt + \tilde{\lambda} \partial_x(\tilde{u}^{\circ 2}) dt + \sqrt{2} \partial_x d\tilde{W}, \quad (24.15)$$

with unit-variance white-noise marginals. The transformation is invertible. If the normalized KPZ primitive satisfies

$$\tilde{h}_t = \tilde{\lambda}^{-1} \log \tilde{Z}_t + \frac{\tilde{\lambda}^3}{12} t + C,$$

then in the original coefficients

$$h_t = \frac{D}{\lambda} \log Z_t + \frac{\lambda^3 \chi^2}{12D^2} t + C, \quad (24.16)$$

where

$$dZ = D\Delta Z dt + \lambda \sqrt{\frac{2\chi}{D}} Z dW. \quad (24.17)$$

*Proof.* The covariance identity  $\tilde{W}_t(\varphi) = \sqrt{D} W_{t/D}(\varphi)$  shows that  $\tilde{W}$  is again space–time white noise. Substitute  $u_s = \sqrt{\chi} \tilde{u}_{Ds}$  in the martingale problem and divide by  $\sqrt{\chi}$ : the diffusion coefficient becomes one, the noise coefficient becomes  $\sqrt{2}$ , and the nonlinear coefficient becomes  $\tilde{\lambda} = \lambda\sqrt{\chi}/D$ . White noise of variance  $\chi$  becomes unit white noise. The reverse substitution proves invertibility.

For the primitive,  $h_s = \sqrt{\chi} \tilde{h}_{Ds}$ . Hence the normalized drift becomes

$$\sqrt{\chi} \frac{\tilde{\lambda}^3}{12} Ds = \frac{\lambda^3 \chi^2}{12D^2} s.$$

The Cole–Hopf prefactor is  $\sqrt{\chi}/\tilde{\lambda} = D/\lambda$ . Rewriting the normalized SHE under the same time change gives (24.17).  $\square$

**Lemma 8.29** (Exact mollified exponential identity). *Work in the normalized stationary Burgers equation*

$$du = \Delta u dt + \lambda \partial_x(u^{\circ 2}) dt + \sqrt{2} \partial_x dW.$$

Let  $h_t(x) = u_t(\Theta_x) + c_t$  be the anchored primitive from Lemma 8.25, choose an even Schwartz mollifier  $\rho$  such that  $\hat{\rho} \in C_c^\infty$ ,  $\hat{\rho} = 1$  on a neighbourhood of the origin, and  $\int \rho = 1$ , set  $\rho^L(x) = L\rho(Lx)$ , and put  $\Theta_x^L = \Theta_x * \rho^L$  and  $h_t^L(x) = u_t(\Theta_x^L)$ , and define

$$\phi_t^L(x) = \exp\{\lambda h_t^L(x)\}.$$

Then, for every  $\varphi \in C_c^\infty(\mathbb{R})$ ,

$$\begin{aligned} d\phi_t^L(\varphi) &= \phi_t^L(\Delta\varphi) dt + \sqrt{2}\lambda \int_{\mathbb{R}} \phi_t^L(x) \varphi(x) dW_t(\rho_x^L) dx \\ &\quad + \lambda^2 dR_t^L(\varphi) + \lambda^2 \phi_t^L(K^L\varphi) dt + \lambda^2 \phi_t^L(\varphi) dQ_t^L \\ &\quad - 2\lambda^2 \int_{\mathbb{R}} \phi_t^L(x) \varphi(x) \langle \rho_x^L, \rho^L * \sigma \rangle dx dt - \sqrt{2}\lambda \phi_t^L(\varphi) dW_t(\rho^L * \sigma). \end{aligned} \quad (24.11)$$

Here  $K^L$  is deterministic, while

$$\begin{aligned} R_t^L(x) &= \int_0^t \phi_s^L(x) \left\{ \lambda^{-1} d\mathcal{A}_s(\Theta_x^L) - [(u_s^L(x))^2 - \langle (u_s^L)^2, \sigma \rangle] ds - K_x^L ds \right\}, \\ Q_t^L &= \int_0^t \left\{ -\langle (u_s^L)^2 - \|\rho^L\|_2^2, \sigma \rangle + \|\rho^L * \sigma\|_2^2 + \lambda^{-1} u_s(\rho^L * \partial_x \sigma) \right\} ds. \end{aligned}$$

The notation  $u^L(x) = u(\rho_x^L)$  is used throughout.

*Proof.* The identity  $h_t^L(x) = u_t(\Theta_x^L)$  follows from the evenness of  $\rho$ . Testing the controlled Burgers equation against  $\Theta_x^L$  gives

$$dh_t^L(x) = u_t(\Delta \Theta_x^L) dt + d\mathcal{A}_t(\Theta_x^L) + \sqrt{2} dW_t(-\partial_z \Theta_x^L).$$

Direct differentiation of the anchored kernel gives

$$\Delta_z \Theta_x^L(z) = \Delta_x \Theta_x^L(z) + \partial_z(\rho^L * \sigma)(z), \quad -\partial_z \Theta_x^L = \rho_x^L - \rho^L * \sigma.$$

Hence

$$u_t(\Delta \Theta_x^L) = \Delta_x h_t^L(x) + u_t(\rho^L * \partial_x \sigma).$$

Apply the Dirichlet-process Itô formula from Lemma 8.27 to  $e^{\lambda h_t^L(x)}$ . Its quadratic variation is

$$d[h^L(x)]_t = 2\|\rho_x^L - \rho^L * \sigma\|_2^2 dt,$$

and  $\lambda \phi^L \Delta h^L = \Delta \phi^L - \lambda^2 \phi^L (\partial_x h^L)^2$ , with  $\partial_x h^L = u^L$ . Expanding  $\|\rho_x^L - \rho^L * \sigma\|_2^2$  and grouping the centered spatial quadratic term with  $d\mathcal{A}(\Theta_x^L)$  gives precisely the three quantities  $R^L, K^L, Q^L$  displayed above. Integrating the pointwise identity against  $\varphi$  and integrating the Laplacian twice by parts yields (24.11). No limiting argument has been used at this stage.  $\square$

**Lemma 8.30** (Second-chaos kernel and partial Gaussian integration by parts). *For  $N \geq L$ , set  $\rho_z^N(y) = \rho^N(z - y)$  and*

$$\begin{aligned} g_x^{L,N}(y_1, y_2) &:= \int_{\mathbb{R}} (\rho_x^L(z) - \langle \rho_z^L, \sigma \rangle) \rho_z^N(y_1) \rho_z^N(y_2) dz \\ &\quad - \left( \rho_x^L(y_1) \rho_x^L(y_2) - \langle \rho_{y_1}^L \rho_{y_2}^L, \sigma \rangle \right). \end{aligned}$$

*Then the  $N$ -regularized discrepancy in the braces defining  $R^L$  is the second Wiener chaos  $W_2(g_x^{L,N})$ . If*

$$K_x^{L,N} = \lambda^2 \iint g_x^{L,N}(y_1, y_2) \Theta_x^L(y_1) \Theta_x^L(y_2) dy_1 dy_2,$$

*and  $r^{L,N}(u_0, \varphi)$  denotes the resulting centered integrand, then for every smooth cylinder functional  $F$ ,*

$$|\mathbf{E}[r^{L,N}(u_0, \varphi) F(u_0)]| \leq C \|F\|_1 (A_1^{L,N} + C_1^{L,N}), \quad (24.12)$$

*where  $\|F\|_1^2 = 2\mathbf{E}\|\partial_x DF\|_2^2$  and  $A_1^{L,N}, C_1^{L,N}$  are the two  $\dot{H}^{-1}$  norms obtained after one and two Gaussian integrations by parts, respectively.*

*Proof.* The identity for  $g_x^{L,N}$  follows by writing both centered squares as second homogeneous Wiener chaoses. Write  $W_1(g_x^{L,N}(y_1, \cdot)) = \int g_x^{L,N}(y_1, y_2) W(dy_2)$ . Gaussian integration by parts gives

$$\mathbf{E}[W_2(g_x^{L,N})\phi_0^L(x)F] = A^{L,N} + B^{L,N} + C^{L,N},$$

where  $A^{L,N}$  contains the partial Wick contraction

$$W_1(g_x^{L,N}(y_1, \cdot)) \diamond \phi_0^L(x) := W_1(g_x^{L,N}(y_1, \cdot))\phi_0^L(x) - \int g_x^{L,N}(y_1, y_2) D_{y_2}\phi_0^L(x) dy_2,$$

$B^{L,N}$  is the term in which both Malliavin derivatives hit  $\phi_0^L$ , and  $C^{L,N}$  is the mixed term in which one derivative hits  $\phi_0^L$  and one hits  $F$ . Since

$$D_y\phi_0^L(x) = \lambda\Theta_x^L(y)\phi_0^L(x), \quad D_{y_1,y_2}^2\phi_0^L(x) = \lambda^2\Theta_x^L(y_1)\Theta_x^L(y_2)\phi_0^L(x),$$

the definition of  $K_x^{L,N}$  cancels  $B^{L,N}$  exactly. Duality between  $\dot{H}^1$  and  $\dot{H}^{-1}$  in the remaining Malliavin variable, followed by Cauchy–Schwarz, gives  $|A^{L,N}| \leq A_1^{L,N}\|F\|_1$  and  $|C^{L,N}| \leq C_1^{L,N}\|F\|_1$ . This is (24.12).  $\square$

**Lemma 8.31** (The load-bearing  $\dot{H}^{-1}$  kernel estimate). *Define*

$$G_{x,x'}^{L,N}(y_1) = \int_{\mathbb{R}} \Theta_{x'}^L(y_2) g_x^{L,N}(y_1, y_2) dy_2.$$

There exists  $C < \infty$  such that, for every  $L$ , all sufficiently large  $N$  satisfy

$$\sup_{x,x'} \|G_{x,x'}^{L,N}\|_{\dot{H}^{-1}(\mathbb{R})} \leq CL^{-1/2}. \quad (24.13)$$

Consequently

$$\lim_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} A_1^{L,N} = 0, \quad \lim_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} C_1^{L,N} = 0. \quad (24.14)$$

*Proof.* Let  $f \in C_c^\infty(\mathbb{R})$ . Expanding the definition of  $g$  and integrating in  $y_1, y_2$  writes  $\langle G_{x,x'}^{L,N}, f \rangle$  as a difference of four expectations of products of translates of  $f$  and  $\Theta_{x'}^L$ . Introduce independent random variables  $R_i^L, R_i^N, Z$  with densities  $\rho^L, \rho^N, \sigma$ . Because  $\hat{\rho} = 1$  near the origin and has compact support, for every fixed  $L$  there is  $M(L)$  such that  $N \geq M(L)$  implies  $\rho^L * \rho^N = \rho^L$ , so the zero-order terms cancel. Every remaining term contains a difference  $f(a-r) - f(a)$  multiplied by a bounded difference of  $\Theta_{x'}^L$ . In one dimension

$$|f(a-r) - f(a)| \leq |r|^{1/2} \|f'\|_{L^2},$$

and  $\|\Theta_{x'}^L\|_\infty \leq 1$ . Since  $\mathbf{E}|R_i^L|^{1/2} = O(L^{-1/2})$  and  $\mathbf{E}|R_i^N|^{1/2} = O(N^{-1/2})$ , we obtain

$$|\langle G_{x,x'}^{L,N}, f \rangle| \leq C(L^{-1/2} + N^{-1/2}) \|f\|_{\dot{H}^1},$$

which proves (24.13) by duality.

For  $C_1^{L,N}$ , expand its square and recognize an integral of  $\langle G_{x,x}^{L,N}, G_{x',x'}^{L,N} \rangle_{\dot{H}^{-1}}$  against a bounded covariance kernel coming from  $\phi_0^L$  and  $\varphi$ ; (24.13) gives  $(C_1^{L,N})^2 \leq CL^{-1}$ . For  $A_1^{L,N}$ , the partial Wick contractions produce two terms. The first contains the same  $G$ -kernel twice and is  $O(L^{-1})$  by (24.13). In the second one uses duality once more directly on  $g_x^{L,N}(\cdot, y_2)$ ; the same translate-difference estimate yields an integrable  $O(L^{-1/2})$  bound, and after integration in  $y_2, x, x'$  its square is  $O(L^{-1})$ . Thus both limits in (24.14) hold.  $\square$

**Lemma 8.32** (Finite renormalization constant). *The constants  $K_x^{L,N}$  are uniformly bounded, the limit  $K_x^L = \lim_{N \rightarrow \infty} K_x^{L,N}$  exists, and*

$$K_x^L = -\lambda^2 \left( (\rho^L * \rho^L * \Theta_x)^2(x) - \langle (\rho^L * \rho^L * \Theta_x)^2, \sigma \rangle \right), \quad \lim_{L \rightarrow \infty} K_x^L = \frac{\lambda^2}{12}. \quad (24.15)$$

*The final limit is independent of  $x$  and of the even unit-mass mollifier.*

*Proof.* Substitute  $g_x^{L,N}$  in the definition of  $K_x^{L,N}$ . The part containing  $\rho_z^N(y_1)\rho_z^N(y_2)$  becomes

$$\lambda^2 \int (\rho_x^L(z) - \langle \rho_z^L, \sigma \rangle) (\rho^N * \rho^L * \Theta_x)^2(z) dz.$$

As  $N \rightarrow \infty$  this tends to  $\lambda^2 \int (\rho_x^L - \langle \rho_z^L, \sigma \rangle) (\rho^L * \Theta_x)^2$ . Since  $\rho_x^L(z) - \langle \rho_z^L, \sigma \rangle = -\partial_z(\rho^L * \Theta_x)(z)$ , this integral is  $-\lambda^2 \int \partial_z[(\rho^L * \Theta_x)^3]/3 = 0$ . The remaining term gives the displayed formula for  $K_x^L$ .

Put  $a = \int_x^\infty \sigma(y) dy$ . Symmetry of  $\rho * \rho$  gives  $(\rho^L * \rho^L * \Theta_x)(x) \rightarrow \frac{1}{2} - a$ , while convergence in every finite  $L^p$  gives

$$\langle (\rho^L * \rho^L * \Theta_x)^2, \sigma \rangle \longrightarrow \int \left( \mathbf{1}_{(-\infty, x]}(z) - \int_z^\infty \sigma(y) dy \right)^2 \sigma(z) dz.$$

Expanding the square, Fubini and symmetry of three independent variables with density  $\sigma$  give

$$\int \left( \mathbf{1}_{(-\infty, x]}(z) - \int_z^\infty \sigma(y) dy \right)^2 \sigma(z) dz = 1 - a - (1 - a^2) + \frac{1}{3} = a^2 - a + \frac{1}{3}.$$

Therefore

$$\left( \frac{1}{2} - a \right)^2 - (a^2 - a + \frac{1}{3}) = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12},$$

and the minus sign in the formula for  $K_x^L$  yields  $K_x^L \rightarrow \lambda^2/12$ .  $\square$

**Lemma 8.33** (Vanishing remainder and convergence of the anchor process). *For every  $T < \infty$  and  $\varphi \in C_c^\infty(\mathbb{R})$ ,*

$$\lim_{L \rightarrow \infty} \mathbf{E} \sup_{t \leq T} |R_t^L(\varphi)|^2 = 0. \quad (24.16)$$

*Moreover  $Q^L$  converges in probability, locally uniformly on  $[0, T]$ , to*

$$Q_t = - \int_0^t u_s^{\otimes 2}(\sigma) ds + \|\sigma\|_2^2 t + \lambda^{-1} \int_0^t u_s(\partial_x \sigma) ds. \quad (24.17)$$

*The limit  $Q$  has zero quadratic variation.*

*Proof.* Regularize first at scale  $N$ . By the definition of the energy nonlinearity, the regularized additive functional converges in  $L^2$  as  $N \rightarrow \infty$ . For fixed  $L, N$ , Lemma 8.30 and (24.14) give

$$\|r^{L,N}(\cdot, \varphi)\|_{-1} \leq C(A_1^{L,N} + C_1^{L,N}) \xrightarrow{N \rightarrow \infty, L \rightarrow \infty} 0.$$

Applying the forward-backward maximal estimate Lemma 8.26 to the time integral of  $r^{L,N}$  gives

$$\mathbf{E} \sup_{t \leq T} |R_t^{L,N}(\varphi)|^2 \leq CT \|r^{L,N}(\cdot, \varphi)\|_{-1}^2.$$

First let  $N \rightarrow \infty$ , using the  $L^2$  definition of  $d\mathcal{A}(\Theta_x^L)$ , and then let  $L \rightarrow \infty$ . This proves (24.16).

For  $Q^L$ , the first term is precisely the mollifier approximation used in the definition of  $u^{\diamond 2}(\sigma)$ , so the energy estimate gives uniform convergence in probability of its time integral. The deterministic term converges because  $\rho^L * \sigma \rightarrow \sigma$  in  $L^2$ , and stationarity plus  $\rho^L * \partial_x \sigma \rightarrow \partial_x \sigma$  in  $L^2$  gives

$$\mathbf{E} \sup_{t \leq T} \left| \int_0^t u_s(\rho^L * \partial_x \sigma - \partial_x \sigma) ds \right|^2 \leq T^2 \|\rho^L * \partial_x \sigma - \partial_x \sigma\|_2^2 \rightarrow 0.$$

This yields (24.17). The quadratic-current part of an energy solution and the time integral of  $u_s(\partial_x \sigma)$  both have zero quadratic variation; the deterministic term does as well. Hence  $Q$  has zero quadratic variation.  $\square$

**Lemma 8.34** (Weighted uniqueness for the multiplicative SHE). *Fix  $T, a > 0$ . In the class of predictable mild solutions of*

$$dZ = \Delta Z dt + \sqrt{2}\lambda Z dW$$

*for which*

$$\sup_{t \leq T, x \in \mathbb{R}} e^{-2a|x|} \mathbf{E}|Z_t(x)|^2 < \infty, \quad (24.18a)$$

*the initial condition determines the solution uniquely.*

*Proof.* Let  $Z, \tilde{Z}$  have the same initial condition and put  $D = Z - \tilde{Z}$ . By the mild equation and Itô isometry,

$$\mathbf{E}|D_t(x)|^2 = 2\lambda^2 \int_0^t \int_{\mathbb{R}} p_{t-s}(x-y)^2 \mathbf{E}|D_s(y)|^2 dy ds.$$

For the heat kernel,  $|y| \leq |x| + |x-y|$  and completing the square give, for  $0 < r \leq T$ ,

$$\int p_r(x-y)^2 e^{2a|y|} dy \leq C_{a,T} r^{-1/2} e^{2a|x|}.$$

Thus, with  $F(t) = \sup_x e^{-2a|x|} \mathbf{E}|D_t(x)|^2$ ,

$$F(t) \leq C \int_0^t (t-s)^{-1/2} F(s) ds. \quad (24.18b)$$

Iterating (24.18b) twice yields  $F(t) \leq C^2 \pi \int_0^t F(s) ds$  because  $\int_s^t (t-r)^{-1/2} (r-s)^{-1/2} dr = \pi$ . Gronwall's lemma and  $F(0) = 0$  imply  $F \equiv 0$ . Hence the two mild solutions are indistinguishable after taking continuous versions.  $\square$

**Lemma 8.35** (Finite-scale scalar cancellation and the stochastic heat equation). *Set  $a_L = \rho^L * \sigma$  and define, at every finite  $L$ ,*

$$X_t^L = \sqrt{2}\lambda W_t(a_L) + \left( \lambda^2 \|a_L\|_2^2 - \frac{\lambda^4}{12} \right) t - \lambda^2 Q_t^L, \quad Z_t^L(x) = e^{X_t^L} \phi_t^L(x). \quad (24.18)$$

*Then*

$$\begin{aligned} dZ_t^L(\varphi) &= Z_t^L(\Delta \varphi) dt + \sqrt{2}\lambda \int Z_t^L(x) \varphi(x) dW_t(\rho_x^L) dx \\ &\quad + \lambda^2 e^{X_t^L} dR_t^L(\varphi) + \lambda^2 Z_t^L((K^L - \lambda^2/12)\varphi) dt. \end{aligned} \quad (24.19)$$

*As  $L \rightarrow \infty$ ,  $Z^L$  converges locally in probability to the unique positive mild solution  $Z$  of*

$$dZ = \Delta Z dt + \sqrt{2}\lambda Z dW. \quad (24.20)$$

For the associated KPZ primitive,

$$h_t(x) = \lambda^{-1} \log Z_t(x) + \frac{\lambda^3}{12} t + C, \quad (24.21)$$

where  $C$  is fixed by the initial multiplicative normalization. Thus in the canonical  $\lambda = 1$  normalization the deterministic correction is  $t/12$ .

*Proof.* At finite  $L$ ,  $Q^L$  is a time integral of smooth cylinder functionals, hence has finite variation. The remaining energy part of  $R^L$  has zero quadratic variation. Therefore Lemma 8.27 applies to the product  $e^{X^L} \phi^L$ . From (24.18),

$$dX_t^L = \sqrt{2}\lambda dW_t(a_L) + \left( \lambda^2 \|a_L\|_2^2 - \frac{\lambda^4}{12} \right) dt - \lambda^2 dQ_t^L, \quad d[X^L]_t = 2\lambda^2 \|a_L\|_2^2 dt.$$

Multiply (24.11) by  $e^{X^L}$ . The  $+\lambda^2 \phi^L dQ^L$  term cancels  $-\lambda^2 Z^L dQ^L$ . The anchor noises  $-\sqrt{2}\lambda \phi^L dW(a_L)$  and  $+\sqrt{2}\lambda \phi^L dW(a_L)$  cancel as well. The cross variation between  $e^{X^L}$  and the spatial noise equals

$$2\lambda^2 e^{X^L} \left( \int \phi^L(x) \varphi(x) \langle \rho_x^L, a_L \rangle dx - \phi^L(\varphi) \|a_L\|_2^2 \right) dt.$$

Its first term cancels the explicit  $-2\lambda^2 \int \phi^L \varphi \langle \rho_x^L, a_L \rangle$  in (24.11). The second term, together with the Itô correction  $\frac{1}{2} d[X^L] = \lambda^2 \|a_L\|_2^2 dt$  and the drift chosen in  $X^L$ , leaves exactly  $-\lambda^4 Z^L(\varphi) dt/12$ . Combining this with  $+\lambda^2 Z^L(K^L \varphi) dt$  proves (24.19).

We now pass to the limit. Since the spatial law of the anchored primitive is Brownian modulo an additive constant, ordinary mollification gives  $h^L \rightarrow h$  locally uniformly in probability; after localization the same holds for  $\phi^L = e^{\lambda h^L}$ . Lemma 8.33 and  $a_L \rightarrow \sigma$  in  $L^2$  imply  $X^L \rightarrow X$  locally uniformly in probability, where

$$X_t = \sqrt{2}\lambda W_t(\sigma) + \left( \lambda^2 \|\sigma\|_2^2 - \frac{\lambda^4}{12} \right) t - \lambda^2 Q_t.$$

Consequently  $Z^L \rightarrow Z = e^X \phi$  locally after localization.

The remainder in (24.19) vanishes without any separate  $p$ -variation input. Indeed, integration by parts (the cross variation with  $R^L$  is zero) gives

$$\int_0^t e^{X_s^L} dR_s^L(\varphi) = e^{X_t^L} R_t^L(\varphi) - \int_0^t R_s^L(\varphi) e^{X_s^L} dX_s^L - \frac{1}{2} \int_0^t R_s^L(\varphi) e^{X_s^L} d[X^L]_s.$$

Stop when  $|X^L|$ , the finite variation of  $Q^L$ , or the quadratic variation of the Brownian part exceeds a fixed level. On the stopped interval the  $L^2$  maximal estimate (24.16), Itô isometry and Cauchy–Schwarz make all three terms tend to zero; the stopping level is then sent to infinity using tightness of  $X^L$  and  $Q^L$ .

For the potential term, Lemma 8.32 gives pointwise  $K_x^L \rightarrow \lambda^2/12$  and a uniform bound. Dominated convergence on the compact support of  $\varphi$ , followed by localization of  $Z^L$ , therefore shows

$$\int_0^t Z_s^L ((K^L - \lambda^2/12)\varphi) ds \rightarrow 0.$$

Finally, the stochastic convolution with  $\rho_x^L$  converges to that with space–time white noise by Itô isometry, because convolution by  $\rho^L$  converges strongly to the identity on  $L^2$ . Every limit of (24.19)

hence satisfies the mild SHE (24.20). The weighted second moments of the localized fields satisfy (24.18a); Lemma 8.34 therefore identifies the whole sequence.

To recover the height, substitute (24.17) into the limit  $X$ . The anchored KPZ martingale problem gives

$$dX_t = \lambda dh_t(\sigma) - \frac{\lambda^4}{12} dt.$$

Fix the initial normalization so that  $X_0 = \lambda h_0(\sigma)$ . Then  $\lambda^{-1} X_t = h_t(\sigma) - \lambda^3 t/12$ . Since  $u = \partial_x h = \lambda^{-1} \partial_x \log Z$  and  $X$  is independent of  $x$ , comparison at the anchor yields (24.21).  $\square$

**Theorem 8.36** (Uniqueness of stationary energy solutions). *For  $D, \chi > 0$ , stationary energy solutions of*

$$du = D\Delta u dt + \lambda \partial_x(u^{\circ 2}) dt + \sqrt{2D\chi} \partial_x dW, \quad u_0 \stackrel{\text{law}}{=} \sqrt{\chi} \eta, \quad (24.22)$$

*in the controlled forward-backward sense above are unique in law. If a space-time white noise  $W$  and initial white noise  $\eta$  are fixed, the corresponding strong stationary solution is unique up to indistinguishability. For  $\lambda \neq 0$  it is represented by*

$$u = \frac{D}{\lambda} \partial_x \log Z^{(\sigma)}, \quad (24.23)$$

*where, for any smooth compactly supported probability density  $\sigma$ ,  $Z^{(\sigma)}$  is the unique positive mild solution of*

$$dZ^{(\sigma)} = D\Delta Z^{(\sigma)} dt + \lambda \sqrt{\frac{2\chi}{D}} Z^{(\sigma)} dW, \quad Z_0^{(\sigma)}(x) = \exp\left\{\frac{\lambda\sqrt{\chi}}{D} \eta(\Theta_x^{(\sigma)})\right\}. \quad (24.24)$$

*For the associated KPZ primitive, modulo its spatial additive constant,*

$$h_t = \frac{D}{\lambda} \log Z_t + \frac{\lambda^3 \chi^2}{12D^2} t + C. \quad (24.25)$$

*In particular, in the normalized equation  $D = \chi = 1$ , the deterministic energy-solution correction to the Cole-Hopf height is  $\lambda^3 t/12$ , and is  $t/12$  when  $\lambda = 1$ .*

*Proof.* Lemma 8.28 reduces (24.22) bijectively to the normalized equation treated by Lemmas 8.25–8.35. Those lemmas construct from every normalized energy solution the multiplicative SHE solution, and Lemma 8.34 gives pathwise uniqueness in the weighted class containing it. Therefore the normalized strong solution is unique up to indistinguishability and normalized energy solutions are unique in law. Undoing the coefficient normalization gives (24.23)–(24.24).

For the height, Lemma 8.35 proves in unit normalization that  $\tilde{h} = \tilde{\lambda}^{-1} \log \tilde{Z} + \tilde{\lambda}^3 t/12 + C$ . The last part of Lemma 8.28 gives exactly (24.25). Thus the general Burgers representation is kept separate from the special unit normalization in which Gubinelli-Perkowski state the  $\lambda^3 t/12$  drift.  $\square$

**Theorem 8.37** (Weak KPZ universality for the selected equilibrium conservative class). *Under the hypotheses of SMX-24.06.02, the full characteristic-centered density-fluctuation sequence (24.8) converges in law, without subsequence extraction, to the unique stationary energy solution of (24.14). Thus the three coefficients are exactly*

$$D(\rho), \quad \chi(\rho), \quad \lambda_\rho = \frac{1}{2} H''(\rho). \quad (24.26)$$

If the microscopic model also carries the nearest-neighbour height/current field satisfying the continuity relation, define the centered height in the same characteristic coordinate by

$$\mathcal{H}_t^N\left(\frac{x - N^{3/2}\mathbf{H}'(\rho)t}{N}\right) = \frac{1}{\sqrt{N}}\left(h_t^N(x) - N^{3/2}\mathbf{H}(\rho)t + \rho x\right), \quad (24.27)$$

with linear interpolation between lattice points. Then the full height sequence converges in law to the unique almost-stationary KPZ energy solution whose spatial derivative is the Burgers limit. This is a stationary weak-KPZ universality theorem; it does not assert convergence from arbitrary non-equilibrium initial profiles.

*Proof.* SMX-24.06.02 gives relative compactness and identifies every subsequential density limit as an energy solution with the coefficients (24.26). SMX-24.06.03 gives uniqueness in law of that energy solution. Hence every subsequence has the same unique limit and the entire density sequence converges.

For the height, the microscopic continuity relation identifies its spatial derivative with the density fluctuation field. The mean current is  $N^{3/2}\mathbf{H}(\rho)$  and the moving coordinate in (24.27) uses precisely the characteristic displacement from (24.8); these are the two centerings that must be removed before taking a KPZ limit. The tightness/current argument of Gonçalves–Jara identifies every height limit as an almost-stationary KPZ energy solution. Gubinelli–Perkowski uniqueness, in the coefficient form of SMX-24.06.03, again makes all limit points equal in law, proving convergence of the full height sequence.  $\square$



# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 8.38** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 8.39** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 8.40** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 8.41** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 8.42** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

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Probability Foundations for Statistical Mechanics, SLE, and  
Liouville Quantum Gravity  
Supplementary Volume XXV: Singular SPDEs, KPZ, Stochastic  
Quantization, and Dynamic Universality

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# Scope and conventions

This supplementary volume develops the continuum dynamical objects toward which microscopic systems flow: KPZ, stochastic Burgers, stochastic quantization, regularity structures, and paracontrolled distributions. It also separates three notions often conflated under “KPZ universality”: weak convergence to the KPZ equation, convergence under 1:2:3 scaling to the KPZ fixed point, and the still-open general universality principle for nonintegrable growth models.



# Chapter 1

## Parabolic regularity, white noise, and renormalization

Singular SPDEs are not difficult merely because their solutions are rough. They are difficult because the nonlinearities ask us to multiply distributions whose classical products do not exist. The first task is therefore a scaling check.

**Definition 1.1** (Parabolic scaling and regularity exponent). In  $d$  spatial dimensions, parabolic scaling assigns weight 2 to time and weight 1 to each spatial coordinate:  $(t, x) \mapsto (\lambda^2 t, \lambda x)$ . Its homogeneous dimension is  $|\mathfrak{s}| = d + 2$ . A distribution has local parabolic Hölder–Besov regularity  $\alpha$  when its action on test functions rescaled at parabolic radius  $\lambda$  is  $O(\lambda^\alpha)$  after the standard moment/cancellation normalization. Every occurrence of  $\mathcal{C}^\alpha$  in this volume uses this parabolic convention unless explicitly stated otherwise.

*Worked example 1.2* (Why white noise has regularity below  $-(d + 2)/2$ ). Let  $\xi$  be space–time white noise and let  $\varphi_z^\lambda$  be an  $L^1$ -normalized test function supported in a parabolic ball of radius  $\lambda$ . Then

$$\mathbf{E}|\langle \xi, \varphi_z^\lambda \rangle|^2 = \|\varphi_z^\lambda\|_{L^2}^2 \asymp \lambda^{-(d+2)}.$$

A typical coefficient therefore has size  $\lambda^{-(d+2)/2}$ . This scaling calculation predicts regularity  $-(d + 2)/2$ ; Gaussian tail estimates and a wavelet/Besov characterization give almost-sure membership only at every strictly smaller exponent, as stated in the theorem below.

### 1.1 Parabolic regularity of white noise

**Theorem 1.3** (Parabolic regularity of space–time white noise).

*Under parabolic scaling in  $d$  spatial dimensions, space-time white noise has almost-sure local Holder-Besov regularity strictly below  $-(d + 2)/2$  and no regularity at the critical index.*

*Proof.* Test white noise against a parabolically rescaled smooth test function  $\varphi_z^\lambda$  normalized in  $L^1$ . Its variance is the squared  $L^2$  norm, which scales as  $\lambda^{-(d+2)}$ . Gaussian moments therefore scale as  $\lambda^{-p(d+2)/2}$ . A countable wavelet/Besov characterization plus Gaussian tail bounds and Borel–Cantelli gives membership in every exponent below  $-(d + 2)/2$ . Testing on disjoint wavelets at the critical scale shows that the critical norm diverges almost surely.  $\square$

## 1.2 Parabolic Schauder estimate

**Theorem 1.4** (Parabolic Schauder estimate).

*Convolution with the heat kernel improves parabolic Holder-Besov regularity by two derivatives away from integer thresholds: if  $f \in C^\alpha$ , then  $K * f \in C^{\alpha+2}$  locally, with the standard norm bound after low-frequency normalization.*

*Proof.* Decompose the heat kernel into dyadic pieces  $K_n$  supported at parabolic scale  $2^{-n}$ . Cancellation up to the required order gives  $\|K_n * f\|_\infty \lesssim 2^{-n(\alpha+2)} \|f\|_{C^\alpha}$ . The same estimate for derivatives gives the Holder increments. Summing over  $n$  proves the gain of two degrees; the smooth large-scale remainder is harmless.  $\square$

## 1.3 Wick renormalization in a mollifier scheme

**Theorem 1.5** (Wick renormalization in a mollifier scheme).

*Let  $X_\varepsilon$  be a centered Gaussian field mollification with variance  $C_\varepsilon = \mathbb{E}X_\varepsilon^2$ . Then the renormalized polynomials  $:X_\varepsilon^2 := X_\varepsilon^2 - C_\varepsilon$  and  $:X_\varepsilon^3 := X_\varepsilon^3 - 3C_\varepsilon X_\varepsilon$  are the second and third Hermite chaoses and have covariance kernels given by  $2K_\varepsilon^2$  and  $6K_\varepsilon^3$  respectively.*

*Proof.* For jointly Gaussian  $(X, Y)$ , Wick's formula gives  $\mathbb{E}[(X^2 - \mathbb{E}X^2)(Y^2 - \mathbb{E}Y^2)] = 2(\mathbb{E}XY)^2$  and the analogous cubic identity  $6(\mathbb{E}XY)^3$ . The subtraction coefficients are exactly the Hermite-polynomial coefficients that remove lower Wiener chaoses. Thus any distributional convergence reduces to integrability of powers of the covariance kernel, which is the renormalization mechanism used in the subsequent models.  $\square$

## Chapter 2

# The stochastic heat equation and Cole–Hopf KPZ

In one dimension the multiplicative stochastic heat equation is the canonical linear object behind KPZ. Taking a logarithm turns multiplicative noise into an additive rough forcing but creates an infinite Itô correction; that divergence is the first renormalization constant a student should see explicitly.

**Definition 2.1** (Cole–Hopf solution of KPZ). Whenever the multiplicative stochastic heat equation has a strictly positive solution  $Z$ , the *Cole–Hopf KPZ height* is  $h = \log Z$  after the deterministic centering convention fixed by the chosen approximation. Its spatial derivative  $u = \partial_x h$  is the associated stochastic Burgers solution. This is a definition through the linear SHE: the formal expression  $(\partial_x h)^2$  is not interpreted as a classical pointwise square.

*Worked example 2.2* (The logarithmic Itô correction in zero dimensions). Consider first the scalar stochastic differential equation

$$dZ_t = Z_t dB_t, \quad Z_0 = 1.$$

Itô's formula yields

$$d \log Z_t = \frac{dZ_t}{Z_t} - \frac{1}{2} \frac{d\langle Z \rangle_t}{Z_t^2} = dB_t - \frac{1}{2} dt,$$

so  $Z_t = \exp(B_t - t/2)$ . The deterministic correction is forced by quadratic variation. For the spatial stochastic heat equation the corresponding correction diverges as the noise is mollified, leading to the  $C_\varepsilon$  subtraction in the regularized KPZ equation below.

### 2.1 Mild well-posedness of multiplicative SHE

**Theorem 2.3** (Mild well-posedness of the one-dimensional multiplicative SHE).

*On the one-dimensional torus, let  $Z_0 \in L^2(\mathbb{T})$  be deterministic and let  $\xi$  be space-time white noise. Then*

$$Z_t = P_t Z_0 + \int_0^t P_{t-s}(Z_s \xi(ds))$$

*has a unique predictable mild solution satisfying  $\sup_{s \leq T} \sup_x \mathbb{E}|Z_s(x)|^2 < \infty$  for every  $T > 0$  when  $Z_0$  is bounded (and the corresponding integrated  $L^2$  bound for general  $L^2$  data). For every positive time the solution has a continuous version in  $(t, x)$ .*

*Proof.* Let  $p_t$  be the torus heat kernel and define Picard iterates by

$$Z_t^{(0)} = P_t Z_0, \quad Z_t^{(m+1)}(x) = P_t Z_0(x) + \int_0^t \int_{\mathbb{T}} p_{t-s}(x-y) Z_s^{(m)}(y) W(ds, dy).$$

For bounded initial data, the Walsh isometry and the estimate  $\int_{\mathbb{T}} p_r(z)^2 dz \leq C_T r^{-1/2}$  for  $0 < r \leq T$  give

$$D_{m+1}(t) \leq C_T \int_0^t (t-s)^{-1/2} D_m(s) ds, \quad D_m(t) = \sup_x \mathbb{E} |Z_t^{(m)}(x) - Z_t^{(m-1)}(x)|^2.$$

Iterating this Volterra inequality produces a summable sequence on every bounded time interval (equivalently one may iterate twice and compare with the exponential series). Hence the Picard sequence converges in the predictable  $L^2$  space and the limit satisfies the mild equation. Applying the same estimate to the difference of two solutions and iterating gives uniqueness.

For  $Z_0 \in L^2$  the same argument is carried out after spatial integration; the heat semigroup is an  $L^2$  contraction and the stochastic convolution is controlled by the Hilbert-space/Walsh isometry. Finally, for  $t \geq t_0 > 0$ , heat-kernel increment estimates give

$$\mathbb{E} |Z_t(x) - Z_s(y)|^2 \leq C_{t_0, T} (|t-s|^{1/2} + |x-y|),$$

and higher Gaussian/Walsh moment estimates improve the exponent sufficiently for Kolmogorov's continuity theorem. Thus a jointly continuous version exists away from the initial time.  $\square$

## 2.2 Positivity and comparison for SHE

**Theorem 2.4** (Comparison and nonnegativity for the multiplicative SHE).

*Let  $E = \mathbb{T}$  in the setting of the preceding theorem, or let  $E = \mathbb{R}$  and work in the corresponding exponentially weighted mild class. Couple two continuous mild solutions with the same white noise. If  $Z_0^{(1)} \leq Z_0^{(2)}$  almost everywhere, then*

$$Z_t^{(1)}(x) \leq Z_t^{(2)}(x)$$

*for every  $t > 0$  and  $x \in E$  almost surely. In particular nonnegative initial data produce a nonnegative solution.*

*Proof.* Mollify the noise in space and truncate it if necessary. For each approximation the equation is a classical random parabolic equation (or an Itô equation with smooth spatial covariance). If  $U = Z^{(2)} - Z^{(1)}$ , then  $U$  solves a linear parabolic equation with multiplicative coefficient driven by the same smooth noise and starts nonnegative. The parabolic comparison principle, equivalently the positive Feynman–Kac representation for the smooth equation, gives  $U_t \geq 0$  pointwise.

The Picard/Walsh estimates from the preceding theorem imply convergence of the spatially mollified solutions to the white-noise mild solutions in  $L^2(\Omega)$  locally in positive time and, after the standard Kolmogorov estimates, in probability in  $C([t_0, T] \times K)$  for every compact  $K \subset E$ . On  $\mathbb{R}$  the exponential weight controls the tails and the same argument is applied on an increasing compact exhaustion. Choose an almost-surely locally uniformly convergent subsequence. Pointwise order is closed under local uniform convergence, so the limiting continuous versions satisfy  $Z^{(1)} \leq Z^{(2)}$ . Taking  $Z^{(1)} \equiv 0$  proves nonnegativity.  $\square$

**Theorem 2.5** (Strict positivity for the one-dimensional multiplicative SHE). *Let  $E = \mathbb{T}$  or  $E = \mathbb{R}$ , let  $a > 0$  and  $b \in \mathbb{R}$ , and let  $Z$  be a continuous mild solution of*

$$dZ = a\Delta Z dt + bZ dW. \tag{25.6a}$$

On  $\mathbb{R}$  assume the exponentially weighted local moment bounds used for real-line mild solutions. Suppose that  $Z_0$  is nonnegative, continuous,  $\mathcal{F}_0$ -measurable, independent of the future white-noise increments, and is not identically zero almost surely. Then

$$\mathbf{P}(Z_t(x) > 0 \text{ for every } t > 0, x \in E) = 1. \quad (25.6b)$$

More quantitatively, for every compact  $K \Subset (0, \infty) \times E$ ,

$$\inf_{(t,x) \in K} Z_t(x) > 0 \quad \text{almost surely.} \quad (25.6c)$$

Consequently, if two solutions driven by the same white noise have  $Z_0^{(1)} \leq Z_0^{(2)}$  and the difference is not identically zero, then  $Z_t^{(1)}(x) < Z_t^{(2)}(x)$  for every positive time and every spatial point, almost surely.

*Proof.* The proof proves, in the linear Gaussian setting needed in this book, the short-time floor-propagation mechanism behind the classical strong comparison argument.

*Step 1: a uniform short-time floor.* Fix a bounded spatial window  $[-D, D]$  (on the torus this restriction is unnecessary). Let  $I = [c-d, c+d] \subset [-D, D]$  with  $d > 0$ , and let  $V$  solve (25.6a) with initial datum  $\mathbf{1}_I$ . There are numbers  $\beta \in (0, 1/8)$  and  $\tau_0 > 0$  such that, for every  $q > 0$ , a constant  $C_{q,D,d}$  exists with the following property. If  $0 < \tau \leq \tau_0$  and  $0 \leq h \leq \sqrt{a\tau}/8$ , then

$$\mathbf{P}\left(\inf_{\substack{\tau/2 \leq r \leq \tau \\ x \in I^h}} V_r(x) < \beta\right) \leq C_{q,D,d} \tau^q, \quad (25.6d)$$

where  $I^h$  is the  $h$ -neighbourhood of  $I$  (with periodic interpretation on  $\mathbb{T}$ ).

Here are the details of the estimate. Write

$$V_r = P_r \mathbf{1}_I + N_r, \quad N_r(x) = b \int_0^r \int_E p_{a(r-s)}(x-y) V_s(y) W(ds, dy).$$

For  $r \in [\tau/2, \tau]$  and  $h/\sqrt{a\tau} \leq 1/8$ , the Gaussian heat-kernel formula gives, after decreasing  $\tau_0$  if necessary,

$$\inf_{x \in I^h} P_r \mathbf{1}_I(x) \geq 4\beta. \quad (25.6e)$$

The Walsh–BDG inequality, the heat-kernel square bound  $\|p_r\|_2^2 \lesssim r^{-1/2}$ , and the usual Picard moment estimate give, for every  $p \geq 2$ ,

$$\sup_{r \leq \tau, |x| \leq D+1} \|N_r(x)\|_{L^p(\Omega)} \leq C_{p,D,d} \tau^{1/4}. \quad (25.6f)$$

The same calculation applied to heat-kernel differences yields, for every small  $\varepsilon > 0$ ,

$$\|N_r(x) - N_s(y)\|_{L^p} \leq C_{p,D,d,\varepsilon} (|r-s|^{1/4-\varepsilon} + |x-y|^{1/2-\varepsilon}). \quad (25.6g)$$

To pass from pointwise estimates to a supremum, cover  $[0, \tau] \times [-D-1, D+1]$  by a parabolic grid with temporal mesh  $2^{-2n}\tau$  and spatial mesh  $2^{-n}\sqrt{\tau}$ . At level  $n$  there are  $O_D(\tau^{-1/2}2^{3n})$  nearest-neighbour increments. Choose  $0 < \vartheta < 1/2 - 2\varepsilon$  and declare the level- $n$  increment bad when its absolute value exceeds  $\beta 2^{-n\vartheta}$ . From (25.6g) and Markov's inequality, the total level- $n$  bad-increment probability is at most

$$C_{p,D} \tau^{-1/2+p(1/4-\varepsilon)} 2^{-n\{p(1/2-2\varepsilon-\vartheta)-3\}}.$$

Choose first  $p$  so large that the exponent in braces exceeds 1 and that  $p(1/4 - \varepsilon) - 1/2 > q$ . Then the last display is bounded by  $C\tau^q 2^{-n}$ . On the complement of all bad-increment events, the dyadic chaining sum is at most  $\beta \sum_{n \geq 0} 2^{-n\vartheta}$ ; decreasing the numerical  $\beta$  in (25.6e) once makes this sum at most  $\beta$ . Adding the coarsest-grid bound from (25.6f) therefore gives

$$\mathbf{P} \left( \sup_{\substack{0 \leq r \leq \tau \\ |x| \leq D+1}} |N_r(x)| > 3\beta \right) \leq C_{q,D,d} \tau^q.$$

Together with (25.6e) this proves (25.6d). Notice that the equation is linear: if the initial floor is  $\theta \mathbf{1}_I$ , then the same conditional estimate holds with the lower floor  $\theta\beta$ .

*Step 2: propagate a random positive seed.* Because  $Z_0$  is continuous, nonnegative and nonzero, almost surely there are rational  $c, d, \theta > 0$  such that  $Z_0 \geq \theta \mathbf{1}_{[c-d, c+d]}$ . It is therefore enough to work on one of these countably many seed events. Comparison reduces the argument to the solution started from this deterministic floor.

Fix  $0 < t_0 < T < \infty$  and a compact spatial interval  $[-M, M]$  (for the torus, take one period). For a large integer  $m$  put  $\tau = t_0/m$  and choose a spatial enlargement  $h = L/m$ , where  $L$  is large enough that  $m$  successive enlargements of the seed interval contain  $[-M, M]$ . Since  $h/\sqrt{\tau} = O(m^{-1/2})$ , the hypothesis of (25.6d) holds for all large  $m$ . Start successive propagation steps at times separated by  $\tau/2$ . The output slab  $[s + \tau/2, s + \tau]$  of one step overlaps the next one, so the floor supplied by one step is a valid initial floor for the next step by the Markov property and independence of future white-noise increments. Up to time  $T$  there are at most  $C_{T,t_0} m$  steps. Conditional on all previous successful steps, (25.6d) bounds the next failure probability by  $C_q m^{-q}$ . Therefore, for  $q > 2$ ,

$$\mathbf{P}(\text{some propagation step fails}) \leq C_{T,t_0,q} m^{1-q} \longrightarrow 0. \quad (25.6h)$$

On the event that no step fails, after the first  $m$  spatial enlargements the solution has a (possibly very small but strictly positive) deterministic floor  $\theta\beta^{C_{T,t_0}m}$  throughout  $[t_0, T] \times [-M, M]$ . Hence every event in (25.6h)'s successful family is contained in

$$\left\{ \inf_{[t_0, T] \times [-M, M]} Z > 0 \right\}.$$

Since their probabilities tend to one, the latter event has probability one. Taking a countable exhaustion in  $t_0, T, M$  proves (25.6b)–(25.6c).

Finally, the difference of two solutions driven by the same noise solves the same linear equation with initial datum  $Z_0^{(2)} - Z_0^{(1)}$ . The comparison theorem makes that datum nonnegative and the result just proved makes the difference strictly positive whenever it is nontrivial. This is the asserted strong comparison statement.  $\square$

*Remark 2.6* (Source and role of the strict-positivity argument). The preceding proof is the one-dimensional linear-Gaussian specialization of the floor-propagation/strong-comparison mechanism introduced by Mueller and later sharpened quantitatively in strong-comparison work for stochastic heat equations. It is included here because comparison alone does not imply strict positivity, while strict positivity is load-bearing for both the periodic Cole–Hopf identification and the real-line Bertini–Giacomin logarithmic inversion.

## 2.3 Cole–Hopf transform of smooth approximations

**Theorem 2.7** (Cole–Hopf transform of smooth approximations).

For a smooth noise  $\xi_\varepsilon$ , if  $Z_\varepsilon > 0$  solves the multiplicative SHE and  $h_\varepsilon = \log Z_\varepsilon$ , then Itô/chain-rule computation gives a regularized KPZ equation

$$\partial_t h_\varepsilon = \frac{1}{2} \partial_x^2 h_\varepsilon + \frac{1}{2} (\partial_x h_\varepsilon)^2 + \xi_\varepsilon - C_\varepsilon,$$

where the diverging constant depends on the approximation convention.

*Proof.* Differentiate  $Z = e^h$ :  $Z_{xx}/Z = h_{xx} + h_x^2$ . For a temporally regularized approximation ordinary calculus gives the displayed identity with no stochastic correction; for an Itô semimartingale regularization the logarithm contributes  $-\frac{1}{2} d\langle Z \rangle / Z^2$ , which is spatially constant after stationarity and equals the counterterm  $C_\varepsilon$ . The divergence therefore comes from the square of the rough derivative, not from an arbitrary modification of the equation.  $\square$

## 2.4 Pathwise solution of KPZ

## Chapter 3

# Regularity structures and BPHZ renormalization

**Definition 3.1** (Regularity structure, model, and modelled distribution). A regularity structure is a triple  $(A, T, G)$ :  $A \subset \mathbb{R}$  is locally finite and bounded below,  $T = \bigoplus_{\alpha \in A} T_\alpha$  is the graded model space, and  $G$  is a structure group satisfying  $\Gamma\tau - \tau \in \bigoplus_{\beta < \alpha} T_\beta$  for  $\tau \in T_\alpha$ . A model is a pair  $(\Pi, \Gamma)$  with  $\Pi_x : T \rightarrow \mathcal{S}'$  and  $\Gamma_{xy} \in G$  satisfying  $\Pi_x \Gamma_{xy} = \Pi_y$  and the scale bounds used below. A modelled distribution  $f$  of regularity  $\gamma$  takes values in  $\bigoplus_{\alpha < \gamma} T_\alpha$  and satisfies

$$\|f(x) - \Gamma_{xy}f(y)\|_\alpha \lesssim |x - y|_s^{\gamma - \alpha}, \quad \alpha < \gamma.$$

The reconstruction theorem turns this compatible family of local abstract expansions into a single concrete distribution.

*Worked example 3.2* (Polynomial regularity structure and Taylor's theorem). Take  $A = \mathbb{N}_0$ , let  $T$  be spanned by monomials  $X^k$ , and let  $\Gamma_h X^k = (X - h)^k$ . For  $u \in C^2(\mathbb{R})$  define

$$f(x) = u(x)\mathbf{1} + u'(x)X.$$

Then

$$f(x) - \Gamma_{x-y}f(y) = (u(x) - u(y) - u'(y)(x - y))\mathbf{1} + (u'(x) - u'(y))X.$$

Taylor's theorem bounds the two coefficients by  $O(|x - y|^2)$  and  $O(|x - y|)$  respectively. Hence  $f$  is a modelled distribution of regularity 2, and reconstruction returns the ordinary function  $u$ . Singular regularity structures keep this overlap-compatibility principle while adding symbols for noise and renormalized products.

Regularity structures replace a Taylor polynomial by a problem-specific local expansion containing symbols for the noise, its convolutions, and their products. The reconstruction theorem then turns a coherent family of local symbolic descriptions back into a distribution.

### 3.1 Reconstruction theorem

**Theorem 3.3** (Reconstruction theorem).

*Let  $(A, T, G)$  be a regularity structure with an admissible model  $(\Pi, \Gamma)$ , and let  $f \in \mathcal{D}^\gamma$  with  $\gamma > 0$ . There exists a unique distribution  $\mathcal{R}f$  such that, for every compact  $K$ , every test function  $\varphi$  in a fixed bounded subset of  $C^r$  with  $r > |\min A|$ , every  $x \in K$ , and every  $0 < \lambda \leq 1$ ,*

$$|\langle \mathcal{R}f - \Pi_x f(x), \varphi_x^\lambda \rangle| \leq C_K \lambda^\gamma.$$



Moreover the reconstruction map is linear in  $f$ , locally continuous in  $(f, \Pi, \Gamma)$ , and satisfies the corresponding two-model stability estimate.

*Proof.* We give the multiresolution proof in the form needed later for singular SPDEs.

**1. Dyadic approximants.** Choose a compactly supported one-dimensional  $C^r$  scaling function and form the tensor-product anisotropic multiresolution associated with the parabolic scaling  $\mathfrak{s} = (s_1, \dots, s_d)$  (in the space-time applications below,  $\mathfrak{s} = (2, 1, \dots, 1)$ ). At level  $n$  use the anisotropic lattice

$$\Lambda_n^{\mathfrak{s}} = \{(2^{-ns_1}k_1, \dots, 2^{-ns_d}k_d) : k \in \mathbb{Z}^d\},$$

and the  $L^2$ -normalised scaling functions

$$\phi_x^n(y) = 2^{n|\mathfrak{s}|/2} \phi(2^{ns_1}(y_1 - x_1), \dots, 2^{ns_d}(y_d - x_d)), \quad x \in \Lambda_n^{\mathfrak{s}}.$$

The corresponding tensor-product wavelets span  $W_n$  and satisfy the same anisotropic scaling and moment bounds. Define

$$\mathcal{R}_n f = \sum_{x \in \Lambda_n^{\mathfrak{s}}} \langle \Pi_x f(x), \phi_x^n \rangle \phi_x^n.$$

For a fixed compact  $K$ , only grid points within a fixed enlargement of  $K$  enter when the resulting distribution is tested in  $K$ .

**2. Coherence estimate.** For neighbouring grid points  $x, y$  with  $\|x - y\|_{\mathfrak{s}} \lesssim 2^{-n}$ , the model identity  $\Pi_y = \Pi_x \Gamma_{xy}$  and the defining estimate for  $f \in \mathcal{D}^\gamma$  give, homogeneity by homogeneity,

$$|\langle \Pi_x f(x) - \Pi_y f(y), \phi_x^n \rangle| \lesssim \sum_{\alpha < \gamma} 2^{-n(\alpha + |\mathfrak{s}|/2)} \|x - y\|_{\mathfrak{s}}^{\gamma - \alpha}.$$

Since  $\|x - y\|_{\mathfrak{s}} \asymp 2^{-n}$  on the relevant stencil, every summand is  $O(2^{-n(\gamma + |\mathfrak{s}|/2)})$ . The same estimate holds with a wavelet  $\psi_x^n$  in place of  $\phi_x^n$ ; the vanishing moments of  $\psi$  remove the polynomial components above the order required by the model.

**3. Adjacent-scale summability.** The orthogonal decomposition  $V_{n+1} = V_n \oplus W_n$  shows that  $\mathcal{R}_{n+1}f - \mathcal{R}_n f$  has only  $W_n$  coefficients. By the preceding estimate, each such coefficient is  $O(2^{-n(\gamma + |\mathfrak{s}|/2)})$ . A test function at macroscopic scale meets  $O(2^{n|\mathfrak{s}|})$  wavelets, while its wavelet coefficients contribute the compensating factor  $2^{-n|\mathfrak{s}|/2}$  and the usual additional decay coming from smoothness. Thus the series

$$\mathcal{R}_0 f + \sum_{n \geq 0} (\mathcal{R}_{n+1} f - \mathcal{R}_n f)$$

converges in distributions. Call the limit  $\mathcal{R}f$ .

**4. Reconstruction bound.** Fix  $x$  and a test function  $\varphi_x^\lambda$  with  $2^{-(n+1)} < \lambda \leq 2^{-n}$ . Split

$$\mathcal{R}f - \Pi_x f(x) = (\mathcal{R}_n f - \Pi_x f(x)) + \sum_{m \geq n} (\mathcal{R}_{m+1} f - \mathcal{R}_m f).$$

For the first term, compare every grid coefficient entering  $\mathcal{R}_n f$  with the local model at  $x$  and use Step 2. Its action on  $\varphi_x^\lambda$  is  $O(2^{-n\gamma})$ . The tail is bounded by the geometric series from Step 3 and is again  $O(2^{-n\gamma})$ . Since  $2^{-n} \asymp \lambda$ , this gives the asserted  $C_K \lambda^\gamma$  estimate.

**5. Uniqueness.** Suppose  $\mathcal{R}f$  and  $\tilde{\mathcal{R}}f$  both satisfy the reconstruction estimate. Their difference  $u$  obeys  $|\langle u, \varphi_x^\lambda \rangle| \lesssim \lambda^\gamma$  uniformly on compacts. Approximate an arbitrary compactly supported smooth test function by a partition into  $O(\lambda^{-|\mathfrak{s}|})$  rescaled test functions with coefficients of size

$O(\lambda^{|s|})$ . Summing the local estimate gives  $|\langle u, \varphi \rangle| \lesssim \lambda^\gamma$ , which tends to zero because  $\gamma > 0$ . Hence  $u = 0$ .

**6. Stability.** For two models and modelled distributions, repeat Steps 2–4 for  $\Pi_x f(x) - \bar{\Pi}_x \bar{f}(x)$ . The coefficient bound is linear in the model distance and in the modelled-distribution distance, so the same summation yields the standard local Lipschitz estimate. This proves continuity in both the model and  $f$ .

The construction, the adjacent-scale estimate, summability, uniqueness, and two-model stability are all explicit above; no later BPHZ result is used.  $\square$

## 3.2 Abstract integration and fixed point

**Theorem 3.4** (Abstract Schauder estimate and selected local fixed-point theorem).

Let  $T = (A, T, G)$  be a regularity structure, let  $(\Pi, \Gamma)$  be a model, and let  $K$  be a  $\beta$ -regularising kernel with  $\beta > 0$ . Assume that  $K$  has the polynomial cancellation required in Hairer’s Assumption 5.4 up to the homogeneities used below, that an abstract integration map  $\mathcal{I}$  of order  $\beta$  acts on a sector  $V$ , and that the model realises  $K$  for  $\mathcal{I}$ .

For  $\gamma > 0$  with  $\gamma + \beta \notin \mathbb{N}$ , the lifted convolution operator  $\mathcal{K}_\gamma$  of Theorem 5.12 maps

$$\mathcal{D}^\gamma(V) \longrightarrow \mathcal{D}^{\gamma+\beta}, \quad \mathcal{R}\mathcal{K}_\gamma f = K * \mathcal{R}f,$$

and satisfies the corresponding two-model local Lipschitz estimate on compact sets.

For the time-zero singular fixed-point problem, let  $V, \bar{V}$  be sectors of regularities  $\zeta, \bar{\zeta}$ , let  $\gamma \geq \bar{\gamma} > 0$ , and let

$$F : \mathbb{R}^d \times V_\gamma \longrightarrow \bar{V}_{\bar{\gamma}}$$

be a smooth nonlinearity which maps the weighted space  $\mathcal{D}_P^{\gamma, \eta}$  into  $\mathcal{D}_P^{\bar{\gamma}, \bar{\eta}}$  and is locally Lipschitz there. Suppose

$$\zeta \leq \bar{\zeta} + \beta, \quad Q_{\bar{\gamma}}^- \mathcal{I} \bar{V}_{\bar{\gamma}} \subset V_\gamma,$$

and the strict exponent conditions

$$\eta < (\bar{\eta} \wedge \bar{\zeta}) + \beta, \quad \gamma < \bar{\gamma} + \beta, \quad \bar{\eta} \wedge \bar{\zeta} > -\beta \quad (25.4a)$$

hold. Assume also the usual non-anticipative parabolic decomposition of the Green kernel into the singular part  $K$  and a smooth remainder  $R$ . Then, for every admissible datum  $v \in \mathcal{D}_P^{\gamma, \eta}$ , the equation

$$U = (\mathcal{K}_{\bar{\gamma}} + \mathcal{R}_{\bar{\gamma}} \mathcal{R}) \mathbf{1}_{t>0} F(U) + v \quad (25.4b)$$

has a unique solution on a sufficiently short interval  $(0, T)$ . The solution is jointly continuous in the datum and the model; if  $F$  is strongly locally Lipschitz in Hairer’s sense, this dependence is locally Lipschitz.

*Proof.* We keep the Schauder and fixed-point statements separate, because their hypotheses are different and both are used later.

**1. Singular-kernel decomposition and the integration lift.** Write  $K = \sum_{n \geq 0} K_n$  with  $K_n$  supported at parabolic scale  $2^{-n}$ , with the derivative bounds of a  $\beta$ -regularising kernel and the polynomial cancellation in Assumption 5.4. Realisation of  $K$  by the model gives

$$\Pi_x \mathcal{I} \tau = K * \Pi_x \tau - \Pi_x \mathcal{J}(x) \tau.$$

For  $f \in \mathcal{D}^\gamma(V)$  set

$$(\mathcal{K}_\gamma f)(x) = \mathcal{I}f(x) + \mathcal{J}(x)f(x) + \mathcal{N}_\gamma f(x),$$

where  $\mathcal{N}_\gamma f$  is the polynomial remainder built from  $\mathcal{R}f - \Pi_x f(x)$ . The reconstruction estimate makes every coefficient of degree  $|k|_s < \gamma + \beta$  summable.

**2. Multi-level Schauder estimate.** The dyadic bounds on  $K_n$ , the model bounds and SMX-25.03.01 give, for compact  $\mathfrak{K}$  and its one-fattening  $\bar{\mathfrak{K}}$ ,

$$\|\mathcal{K}_\gamma f\|_{\gamma+\beta;\mathfrak{K}} \leq C\|f\|_{\gamma;\bar{\mathfrak{K}}}.$$

The same calculation for two models gives the standard two-model estimate. Cancellation of the polynomial jet and uniqueness of reconstruction yield  $\mathcal{R}\mathcal{K}_\gamma f = K * \mathcal{R}f$ . These are precisely the selected consequences of Hairer’s Theorem 5.12; the restriction  $\gamma + \beta \notin \mathbb{N}$  is retained rather than hidden in the phrase “away from integer homogeneities.”

**3. Weighted time-zero spaces.** For the fixed-point problem work in  $\mathcal{D}_P^{\gamma,\eta}$ , where  $P = \{t = 0\}$ . The weighted Schauder estimate sends the target regularity of  $F(U)$  back to the solution sector exactly when (25.4a) holds. The first inequality controls the time-zero weight after integration, the second leaves a strictly positive regularity margin, and the third is the integrability condition at the singular time slice. Thus no unspecified “local subcriticality” assertion is used at this stage.

**4. Short-time gain.** Let  $\Phi_Z$  denote the right-hand side of (25.4b) for a fixed model  $Z$ . The short-time estimates for the singular and smooth convolution pieces give some  $\kappa > 0$  such that, on every bounded ball,

$$\|\Phi_Z(U) - \Phi_Z(V)\|_{\gamma,\eta;T} \leq CT^\kappa \|F(U) - F(V)\|_{\bar{\gamma},\bar{\eta};T} \leq C_R T^\kappa \|U - V\|_{\gamma,\eta;T}. \quad (25.4c)$$

Likewise

$$\|\Phi_Z(U)\|_{\gamma,\eta;T} \leq C_R T^\kappa + \|v\|_{\gamma,\eta;T}.$$

For  $T$  sufficiently small the map preserves a closed ball and is a strict contraction. Banach’s theorem gives the unique local solution.

**5. Dependence on data and model.** For fixed  $Z$ , the same contraction estimate gives local Lipschitz dependence on  $v$ . If  $F$  is strongly locally Lipschitz, the two-model Schauder estimate gives directly

$$\|\Phi_Z(U); \Phi_Z(\bar{U})\|_{\gamma,\eta;T} \leq CT^\kappa (\|U; \bar{U}\|_{\gamma,\eta;T} + \|Z; \bar{Z}\|),$$

and hence joint local Lipschitz continuity. Under mere local Lipschitz continuity, Hairer’s iteration argument gives joint continuity of the solution map; the theorem does not silently promote this to Lipschitz dependence.

**6. From a concrete SPDE to this theorem.** Local subcriticality is used only to build a regularity structure with finitely many relevant symbols and to verify that its concrete nonlinearity  $F$  satisfies the sector mapping and strict inequalities above. Once those checks are made, the deterministic local solution theorem is exactly the contraction just proved. In particular the theorem does not claim that every semilinear SPDE is automatically covered. □

### 3.3 Algebraic BPHZ renormalization

### 3.4 Analytic BPHZ convergence

## Chapter 4

# Stochastic quantization and the dynamic Phi-four models

The stochastic quantization equation is the dynamical counterpart of the Euclidean  $\Phi^4$  field. In two dimensions Wick ordering plus a Da Prato–Debussche decomposition suffices. Three dimensions are genuinely more singular and require the modern theories.

*Worked example 4.1* (Why the quadratic Wick counterterm is forced). Let  $Z_\varepsilon$  be a centered Gaussian regularization with  $C_\varepsilon = \mathbf{E}[Z_\varepsilon(x)^2]$  and covariance  $K_\varepsilon(x, y) = \mathbf{E}[Z_\varepsilon(x)Z_\varepsilon(y)]$ . Wick’s formula gives

$$\mathbf{E}[Z_\varepsilon(x)^2 Z_\varepsilon(y)^2] = C_\varepsilon^2 + 2K_\varepsilon(x, y)^2.$$

Thus the raw square contains a diverging deterministic zeroth-chaos term. With

$$: Z_\varepsilon^2 : (x) = Z_\varepsilon(x)^2 - C_\varepsilon$$

we instead have

$$\mathbf{E}[: Z_\varepsilon^2 : (x) : Z_\varepsilon^2 : (y)] = 2K_\varepsilon(x, y)^2.$$

The counterterm is therefore forced by the Gaussian-chaos decomposition; it is not an arbitrary modification of the equation. The BPHZ counterterms later in the volume systematize this same removal of lower-order divergent substructures.

### 4.1 Da Prato–Debussche in two dimensions

**Theorem 4.2** (Da Prato–Debussche strong solution in two dimensions).

Let  $G = (\mathbb{R}/2\pi\mathbb{Z})^2$ , let  $A = \Delta - 1$ , and consider the cubic stochastic quantization equation

$$dX = (AX - : X^3 :) dt + dW_t, \quad X(0) = x. \quad (25.5)$$

Let  $\nu$  be the invariant  $\Phi_2^4$  Gibbs measure for this equation. Fix  $p > 3$ ,  $r \geq 1$ , and choose  $s < 0$  so that

$$0 > s > \max \left\{ -\frac{2}{7p}, -\frac{1}{2} \left( 1 - \frac{3}{p} \right) \right\}, \quad \alpha = \frac{2}{p} + 2s. \quad (25.5a)$$

Then for  $\nu$ -almost every initial condition  $x$ , and for every  $T > 0$ , there exists a unique strong solution  $X = Y + z$  of (25.5) such that

$$Y \in C([0, T], B_{p,r}^s(G)) \cap L^p(0, T; B_{p,r}^\alpha(G)), \quad (25.5b)$$

where

$$z(t) = \int_{-\infty}^t e^{(t-q)A} dW_q$$

is the stationary Ornstein–Uhlenbeck field. For arbitrary deterministic  $x \in B_{p,r}^s$  the same fixed-point argument gives a pathwise unique local solution up to a positive random time; the global statement above is only asserted for  $\nu$ -almost every  $x$ .

*Proof.* We give the cubic specialization of the source argument and keep the local and global steps distinct.

**1. Wick powers of the stationary field.** Let  $z_N$  be a Fourier cutoff of  $z$  and put

$$:z_N^2: = z_N^2 - C_N, \quad :z_N^3: = z_N^3 - 3C_N z_N.$$

Gaussian chaos estimates, hypercontractivity and the dyadic Besov bounds imply that for every  $q < \infty$  the Wick powers converge in  $L^q(\Omega; L^p(0, T; B_{p,r}^s))$  in the range (25.5a). Denote the limits by  $:z^2:$  and  $:z^3:$ .

**2. Da Prato–Debussche decomposition.** Writing  $X = Y + z$  turns (25.5) into the random deterministic equation

$$\partial_t Y = AY - Y^3 - 3Y^2 z - 3Y :z^2: - :z^3:, \quad Y(0) = x - z(0). \quad (25.5c)$$

All products are interpreted by the Besov multiplication estimates. The inequalities in (25.5a) are exactly the cubic specialization of the source conditions ensuring that multiplication by the negative-Besov Wick powers and heat smoothing close in the space below.

**3. Local fixed point.** Set

$$E_T = C([0, T], B_{p,r}^s) \cap L^p(0, T; B_{p,r}^\alpha).$$

The heat semigroup and Besov product estimates give, for the mild map associated with (25.5c),

$$\|\Phi(Y)\|_{E_T} \leq C\|x - z(0)\|_{B_{p,r}^s} + C(\omega, T)T^\varepsilon(1 + \|Y\|_{E_T}^3),$$

and, on every bounded ball,

$$\|\Phi(Y) - \Phi(\tilde{Y})\|_{E_T} \leq C_R(\omega, T)T^\varepsilon\|Y - \tilde{Y}\|_{E_T}.$$

For sufficiently small  $T = T_*(x, \omega)$  this is a contraction, proving local existence and pathwise uniqueness for every  $x \in B_{p,r}^s$ . This is the source’s Proposition 4.4 in the cubic case.

**4. The invariant-measure input.** The measure  $\nu$  is the classical finite-volume  $\Phi_2^4$  Gibbs measure, obtained from the Gaussian free field by Wick-ordering the polynomial potential; its integrability is the standard Nelson estimate used in the source. The present theorem uses this already-constructed invariant measure as an explicit input and does not disguise its construction as part of the contraction argument.

**5. A priori bound for stationary initial law.** Start the equation with initial law  $\nu$ . Invariance gives  $\mathcal{L}(X(t)) = \nu$  for each  $t$ . The source’s Besov/Wick estimates therefore imply, for the same admissible exponents and every finite  $T$ ,

$$\mathbf{E} \left[ \sup_{t \leq T} \|X(t)\|_{B_{p,r}^s} \right] < \infty \quad (25.5d)$$

after the factorisation estimate for  $z$  and the polynomial bounds for the remainder. In particular the local norm that controls continuation is finite on every bounded time interval for  $\nu$ -almost every initial point.

**6. Globalisation for  $\nu$ -almost every initial point.** Apply the local theorem at successive stopping times. If the maximal lifetime were finite, the  $B_{p,r}^s$  continuation norm would diverge. Estimate (25.5d), together with Fubini and invariance, excludes this on a full- $\nu$  set of initial data. Hence the local strong solution extends through every finite  $T$  for  $\nu$ -almost every  $x$ . Pathwise uniqueness on overlapping local intervals gives uniqueness of the global strong solution and proves (25.5b).  $\square$

## 4.2 Local construction of dynamic Phi-four-three

## 4.3 Global control for dynamic Phi-four-three

## 4.4 Discrete approximation of Phi-four-three

## 4.5 Weak universality of dynamic Phi-four-three

## Chapter 5

# Paracontrolled distributions

Paracontrolled calculus reaches many of the same equations through harmonic analysis rather than a bespoke symbolic algebra. It begins with Bony's observation that a product can be separated according to which factor oscillates at the larger frequency.

*Remark 5.1* (Spatial regularity convention in this chapter). Unlike Chapter 1, where  $\mathcal{C}^\alpha$  denotes a parabolic space–time regularity, the paraproduct calculus below is performed at each fixed time on the spatial torus. Throughout this chapter  $\mathcal{C}^\alpha = B_{\infty,\infty}^\alpha$  therefore means the *spatial* inhomogeneous Hölder–Besov space; time weights and time regularity are stated separately whenever they enter a parabolic fixed point.

### 5.1 Bony decomposition

**Proposition 5.2** (Bony paraproduct decomposition).

*Fix an inhomogeneous Littlewood–Paley partition  $(\Delta_j)_{j \geq -1}$  and write  $S_{j-1} = \sum_{i < j-1} \Delta_i$ . For smooth functions define*

$$f \prec g = \sum_j S_{j-1} f \Delta_j g, \quad f \succ g = g \prec f, \quad f \circ g = \sum_{|i-j| \leq 1} \Delta_i f \Delta_j g.$$

*Then*

$$fg = f \prec g + f \circ g + f \succ g.$$

*Whenever the three terms on the right extend continuously to a pair of Besov–Hölder distributions, this identity defines the corresponding product by density.*

*Proof.* For smooth  $f, g$  write  $f = \sum_i \Delta_i f$  and  $g = \sum_j \Delta_j g$ . The integer lattice of frequency pairs  $(i, j)$  is the disjoint union of  $i < j - 1$ ,  $j < i - 1$ , and  $|i - j| \leq 1$ . Summing the double series over these three regions gives respectively  $f \prec g$ ,  $f \succ g$ , and  $f \circ g$ . Smooth Littlewood–Paley series converge rapidly, so rearrangement is legitimate. The extension assertion follows by approximating distributions with their low frequency cutoffs and using the continuity estimates in the next proposition.  $\square$

**Proposition 5.3** (Basic paraproduct continuity estimates).

Write  $\mathcal{C}^\alpha = B_{\infty,\infty}^\alpha$ . For smooth  $f, g$  the following estimates hold, with constants depending only on the dyadic partition and the exponents:

$$\begin{aligned} \|f \prec g\|_{\mathcal{C}^\beta} &\lesssim \|f\|_{L^\infty} \|g\|_{\mathcal{C}^\beta}, \\ \|f \prec g\|_{\mathcal{C}^{\alpha+\beta}} &\lesssim \|f\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^\beta}, & \alpha < 0, \\ \|f \circ g\|_{\mathcal{C}^{\alpha+\beta}} &\lesssim \|f\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^\beta}, & \alpha + \beta > 0. \end{aligned}$$

Thus  $\prec$  and  $\circ$  extend uniquely to the indicated Hölder–Besov spaces; the estimate for  $\succ$  follows by symmetry.

*Proof.* In  $f \prec g = \sum_j S_{j-1} f \Delta_j g$ , the output of the  $j$ -th summand has frequency  $O(2^j)$ . If  $f \in L^\infty$ , then  $\|S_{j-1} f\|_\infty \leq C \|f\|_\infty$ , so multiplication by  $2^{j\beta}$  gives the first bound. If  $\alpha < 0$ , the geometric sum over  $i < j - 1$  gives

$$\|S_{j-1} f\|_\infty \lesssim 2^{-j\alpha} \|f\|_{\mathcal{C}^\alpha},$$

which gives the second bound. In the resonant product only  $|i - j| \leq 1$  appears, but the product of two comparable high frequencies can have lower output frequency. After applying  $\Delta_k$ , Fourier support therefore permits  $j \geq k - C$  rather than only  $j \simeq k$ . Hence

$$\|\Delta_k(f \circ g)\|_\infty \lesssim \sum_{j \geq k-C} 2^{-j(\alpha+\beta)} \|f\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^\beta}.$$

This geometric tail is summable exactly when  $\alpha + \beta > 0$  and is then  $O(2^{-k(\alpha+\beta)})$ , proving the third estimate. Density of smooth frequency cutoffs then yields the unique continuous extensions.  $\square$

**Theorem 5.4** (Paracontrolled commutator and resonant-product reduction).

Let  $\alpha \in (0, 1)$  and let  $\beta, \gamma \in \mathbb{R}$  satisfy

$$\beta + \gamma < 0, \quad \alpha + \beta + \gamma > 0.$$

For smooth inputs define

$$\mathcal{C}(f, g, h) = (f \prec g) \circ h - f(g \circ h).$$

Then the dyadic formula for  $\mathcal{C}$  admits a canonical bounded extension to a continuous trilinear map

$$\mathcal{C}^\alpha \times \mathcal{C}^\beta \times \mathcal{C}^\gamma \longrightarrow \mathcal{C}^{\alpha+\beta+\gamma},$$

with

$$\|\mathcal{C}(f, g, h)\|_{\alpha+\beta+\gamma} \lesssim \|f\|_\alpha \|g\|_\beta \|h\|_\gamma.$$

Consequently, if  $u = u' \prec X + u^\sharp$  and an enhancement contains the otherwise ill-defined resonant product  $X \circ \xi$ , then

$$u \circ \xi = u'(X \circ \xi) + \mathcal{C}(u', X, \xi) + u^\sharp \circ \xi$$

whenever the displayed classical remainder products have positive total regularity. This is the basic algebraic mechanism that turns an enhanced noise into a well-defined singular product.

*Proof.* We give the dyadic proof because both sign conditions in the theorem are load-bearing. Write  $\Delta_j$  for the Littlewood–Paley blocks and put  $R_i(a, g) = \Delta_i(a \prec g) - a\Delta_i g$ . The elementary commutator estimate

$$\|R_i(a, g)\|_\infty \lesssim 2^{-i(\alpha+\beta)} \|a\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^\beta} \tag{25.17a}$$



holds when  $a \in \mathcal{C}^\alpha$ ,  $0 < \alpha < 1$ . Indeed, decompose  $a \prec g$  into the finitely many  $g$ -blocks meeting the  $i$ -th annulus, commute  $\Delta_i$  through multiplication by  $a$ , and use  $\|[\Delta_i, a]v\|_\infty \lesssim 2^{-i\alpha} \|a\|_\alpha \|v\|_\infty$ . This is the dyadic remainder estimate that will supply the missing  $\alpha$  derivatives.

For smooth  $f, g, h$ , decompose  $f = \sum_k \Delta_k f$ . Fourier support implies that, for a fixed  $k$ ,  $\Delta_i(\Delta_k f \prec g)$  vanishes unless  $i \geq k - N$  for a fixed integer  $N$ . Therefore

$$\mathcal{C}(f, g, h) = \sum_{j,k} \sum_{|i-j| \leq 1} [\Delta_i(\Delta_k f \prec g) \Delta_j h - \Delta_k f \Delta_i g \Delta_j h] \quad (5.1)$$

$$= A - B, \quad (25.17b)$$

where

$$A = \sum_{j,k} \sum_{|i-j| \leq 1} \mathbf{1}_{i \geq k-N} R_i(\Delta_k f, g) \Delta_j h,$$

$$B = \sum_{j,k} \sum_{|i-j| \leq 1} \mathbf{1}_{i < k-N} \Delta_k f \Delta_i g \Delta_j h.$$

This is the cancellation that is hidden if one merely estimates the two singular products separately.

Consider first  $B$ . For fixed  $k$  its Fourier support is contained in a ball of radius  $C2^k$ . Since  $|i - j| \leq 1$ ,

$$\begin{aligned} \left\| \sum_j \sum_{|i-j| \leq 1} \mathbf{1}_{i < k-N} \Delta_k f \Delta_i g \Delta_j h \right\|_\infty &\lesssim 2^{-k\alpha} \|f\|_\alpha \sum_{i < k-N} 2^{-i(\beta+\gamma)} \|g\|_\beta \|h\|_\gamma \\ &\lesssim 2^{-k(\alpha+\beta+\gamma)} \|f\|_\alpha \|g\|_\beta \|h\|_\gamma. \end{aligned} \quad (25.17c)$$

Here the last geometric sum is dominated by its upper endpoint precisely because  $\beta + \gamma < 0$ . Since  $\delta := \alpha + \beta + \gamma > 0$ , the standard Besov summation lemma for a sequence spectrally supported in balls of radius  $C2^k$  yields  $\|B\|_{\mathcal{C}^\delta} \lesssim \|f\|_\alpha \|g\|_\beta \|h\|_\gamma$ .

For  $A$ , fix  $j$  and hence  $i \simeq j$ . By linearity of  $R_i$  in its first argument,

$$\sum_{k \leq i+N} R_i(\Delta_k f, g) = R_i \left( \sum_{k \leq i+N} \Delta_k f, g \right).$$

The truncated first factor has uniformly bounded  $\mathcal{C}^\alpha$  norm, so (25.17a) gives

$$\begin{aligned} \left\| \sum_k \sum_{|i-j| \leq 1} \mathbf{1}_{i \geq k-N} R_i(\Delta_k f, g) \Delta_j h \right\|_\infty &\lesssim 2^{-j(\alpha+\beta)} 2^{-j\gamma} \|f\|_\alpha \|g\|_\beta \|h\|_\gamma \\ &\lesssim 2^{-j\delta} \|f\|_\alpha \|g\|_\beta \|h\|_\gamma. \end{aligned} \quad (25.17d)$$

The  $j$ -th summand has Fourier support in a ball of radius  $C2^j$ , so the same positive-regularity summation lemma proves the identical  $\mathcal{C}^\delta$  bound for  $A$ . Combining (25.17c)–(25.17d) proves the claimed estimate for smooth inputs.

Smooth functions are not dense in  $B_{\infty,\infty}^s$ , so one must not claim uniqueness merely from density. Instead choose  $\alpha' < \alpha$ ,  $\beta' < \beta$ ,  $\gamma' < \gamma$  with  $\alpha' + \beta' + \gamma' > 0$  and  $\beta' + \gamma' < 0$ . Smooth truncations of  $(f, g, h)$  converge in the lowered exponents, the preceding dyadic formula therefore has a limit there, and the bound is uniform as the primed exponents increase to the original ones. Fatou's property

for Besov spaces yields a canonical element of  $\mathcal{C}^\delta$  satisfying the stated norm bound. This is the canonical bounded extension asserted in the theorem.

Finally, if  $u = u' \prec X + u^\sharp$ , bilinearity gives  $u \circ \xi = (u' \prec X) \circ \xi + u^\sharp \circ \xi$ . Applying the just-proved commutator identity to the first term yields

$$(u' \prec X) \circ \xi = u'(X \circ \xi) + \mathcal{C}(u', X, \xi),$$

so the only nonclassical datum is the enhanced resonant product  $X \circ \xi$ ; all remaining factors are classical whenever their total regularities are positive.  $\square$

**Lemma 5.5** (Time-smoothed paraproduct for the heat operator). *Let  $L = \partial_t - \Delta$  on  $[0, T] \times \mathbb{T}^2$ . Choose  $\varphi \in C_c^\infty(\mathbb{R})$ ,  $\varphi \geq 0$ ,  $\int \varphi = 1$ , and for  $j \geq -1$  set*

$$Q_j f(t) = \int_{\mathbb{R}} 2^{2j} \varphi(2^{2j}(t-s)) f((s \vee 0) \wedge T) ds, \quad f \prec g = \sum_j (S_{j-1} Q_j f) \Delta_j g. \quad (25.18)$$

If  $0 < \alpha < 1$ ,  $\beta \in \mathbb{R}$ ,  $f \in C([0, T]; \mathcal{C}^\alpha) \cap C^{\alpha/2}([0, T]; L^\infty)$  and  $g \in C([0, T]; \mathcal{C}^\beta)$ , then

$$\|L(f \prec g) - f \prec Lg\|_{C_T \mathcal{C}^{\alpha+\beta-2}} \lesssim (\|f\|_{C_T \mathcal{C}^\alpha} + \|f\|_{C_T^{\alpha/2} L^\infty}) \|g\|_{C_T \mathcal{C}^\beta}, \quad (25.19)$$

$$\|f \prec g - f \prec g\|_{C_T \mathcal{C}^{\alpha+\beta}} \lesssim \|f\|_{C_T^{\alpha/2} L^\infty} \|g\|_{C_T \mathcal{C}^\beta}. \quad (25.20)$$

The usual paraproduct estimate  $\|f \prec g\|_{C_T \mathcal{C}^\beta} \lesssim \|f\|_{C_T L^\infty} \|g\|_{C_T \mathcal{C}^\beta}$  also holds.

*Proof.* The time convolution occurs on the parabolic scale  $2^{-2j}$ . Since  $\int \varphi = 1$  and  $\int \varphi' = 0$ ,

$$\|Q_j f - f\|_{L^\infty} \lesssim 2^{-j\alpha} \|f\|_{C_T^{\alpha/2} L^\infty}, \quad \|\partial_t Q_j f\|_{L^\infty} \lesssim 2^{j(2-\alpha)} \|f\|_{C_T^{\alpha/2} L^\infty}. \quad (25.21)$$

The spatial low-frequency cutoff satisfies  $\|\Delta S_{j-1} f\|_\infty \lesssim 2^{j(2-\alpha)} \|f\|_{\mathcal{C}^\alpha}$ . Consequently the  $j$ -th block of  $L(f \prec g) - f \prec Lg$  is bounded by

$$2^{j(2-\alpha)} (\|f\|_{C_T \mathcal{C}^\alpha} + \|f\|_{C_T^{\alpha/2} L^\infty}) \cdot 2^{-j\beta} \|g\|_{C_T \mathcal{C}^\beta}.$$

Its Fourier support is contained in a fixed enlargement of the  $j$ -th annulus, so multiplication by  $2^{j(\alpha+\beta-2)}$  and taking the supremum gives (25.19). Similarly,

$$f \prec g - f \prec g = \sum_j (S_{j-1} f - S_{j-1} Q_j f) \Delta_j g,$$

and (25.21) bounds the  $j$ -th term by  $2^{-j(\alpha+\beta)} \|f\|_{C_T^{\alpha/2} L^\infty} \|g\|_{C_T \mathcal{C}^\beta}$ , which proves (25.20). The last estimate follows exactly as for the ordinary paraproduct because  $Q_j$  is an  $L^\infty$  contraction up to a fixed constant.  $\square$

**Lemma 5.6** (Paralinearization at the regularity used below). *Let  $0 < \alpha < 1$ ,  $F \in C_b^3(\mathbb{R})$  and  $u \in \mathcal{C}^\alpha$ . Then*

$$R_F(u) := F(u) - F'(u) \prec u \in \mathcal{C}^{2\alpha}, \quad (25.22)$$

locally Lipschitzly in  $u$ , with

$$\|R_F(u)\|_{2\alpha} \lesssim_{\|F\|_{C_b^3}, 1+\|u\|_\alpha} 1 + \|u\|_\alpha^2. \quad (25.23)$$

The same estimate holds uniformly in time for  $u \in C_T \mathcal{C}^\alpha$ .

*Proof.* Write  $u = S_{j-1}u + (u - S_{j-1}u)$  in the  $j$ -th block of  $F(u)$  and use Taylor's formula at  $S_{j-1}u$ . The linear term is precisely the contribution of  $F'(u) \prec u$  up to commutators whose two factors both carry an  $O(2^{-j\alpha})$  increment. Every remainder therefore has size  $O(2^{-2j\alpha})$  in  $L^\infty$ . The finite overlap of Littlewood–Paley blocks yields (25.23). Applying the same Taylor formula to two inputs and using the bounded third derivative of  $F$  gives the local Lipschitz estimate.  $\square$

**Theorem 5.7** (Selected paracontrolled generalized parabolic Anderson equation). *Let  $0 < \kappa < 1/3$  and put  $\alpha = 1 - \kappa \in (2/3, 1)$ . Let  $\xi \in \mathcal{C}^{\alpha-2}(\mathbb{T}^2) = \mathcal{C}^{-1-\kappa}$  and set*

$$\Pi\xi = \xi - \langle \xi, 1 \rangle, \quad X = (-\Delta)^{-1}\Pi\xi \in \mathcal{C}^\alpha. \quad (25.24)$$

*Assume that an enhanced resonant product  $\eta = X \diamond \xi \in \mathcal{C}^{2\alpha-2} = \mathcal{C}^{-2\kappa}$  is given. Let  $F \in C_b^3(\mathbb{R})$  and  $u_0 \in \mathcal{C}^\alpha$ .*

*Then there is a maximal time  $T_* > 0$  and a unique paracontrolled solution of*

$$Lu = F(u) \diamond \xi, \quad u(0) = u_0, \quad (25.25)$$

*in the following precise sense. On every  $T < T_*$ ,*

$$u \in C_T \mathcal{C}^\alpha \cap C_T^{\alpha/2} L^\infty, \quad u = F(u) \preceq X + u^\sharp, \quad (25.26)$$

*with  $u^\sharp$  in the corresponding time-weighted  $C_T \mathcal{C}^{2\alpha}$  class, and the right-hand side of (25.25) is the continuous paracontrolled product determined by  $(\xi, X, \eta)$ . The solution map  $(u_0, \xi, \eta) \mapsto u$  is locally Lipschitz on each interval  $[0, T]$  strictly below  $T_*$ .*

*More concretely, let  $\xi_\varepsilon$  be smooth, put  $X_\varepsilon = (-\Delta)^{-1}\Pi\xi_\varepsilon$ , and suppose that constants  $c_\varepsilon \in \mathbb{R}$  satisfy*

$$(\xi_\varepsilon, X_\varepsilon \circ \xi_\varepsilon - c_\varepsilon) \longrightarrow (\xi, \eta) \quad \text{in } \mathcal{C}^{\alpha-2} \times \mathcal{C}^{2\alpha-2}. \quad (25.27)$$

*If  $u_\varepsilon$  solves the renormalized smooth equation*

$$Lu_\varepsilon = F(u_\varepsilon)\xi_\varepsilon - c_\varepsilon F'(u_\varepsilon)F(u_\varepsilon), \quad u_\varepsilon(0) = u_0, \quad (25.28)$$

*then, for every  $T < T_*$ ,  $u_\varepsilon \rightarrow u$  in  $C_T \mathcal{C}^\alpha$ . In particular the limit depends only on the enhanced datum  $(\xi, \eta)$  and not on the chosen smooth approximation. For  $F \in C_b^3$  the maximal time is characterized by the  $L^\infty$  explosion criterion*

$$T_* = \inf\{t > 0 : \|u(t)\|_{L^\infty} = \infty\}. \quad (25.29)$$

*Proof.* We follow the proof architecture of Gubinelli–Imkeller–Perkowski, Theorem 5.4, specialized to  $\beta = \alpha$ . The choice  $\alpha > 2/3$  is exactly what makes the critical remainder exponent  $3\alpha - 2$  positive.

*Step 1: the paracontrolled ansatz and the heat commutator.* Because  $LX = \Pi\xi$ , insert (25.26) into (25.25). Lemma 5.5 gives

$$Lu^\sharp = -\{L(F(u) \preceq X) - F(u) \preceq LX\} + \{F(u) \prec \xi - F(u) \preceq LX\} \quad (5.2)$$

$$+ F(u) \succ \xi + F(u) \circ \xi. \quad (25.30)$$

The first two braces lie in  $\mathcal{C}^{2\alpha-2}$  by (25.19)–(25.20), and  $F(u) \succ \xi = \xi \prec F(u)$  has regularity  $\mathcal{C}^{2\alpha-2}$  because  $\alpha - 2 < 0$ . Hence only the resonant term needs a new meaning.

*Step 2: resolve the resonant product.* Using Lemma 5.6, the decomposition  $u = F(u) \preceq X + u^\sharp$ , the commutator theorem SMX-25.05.03, and (25.20), expand

$$F(u) \circ \xi = (F(u) - F'(u) \prec u) \circ \xi + (F'(u) \prec u^\sharp) \circ \xi \quad (5.3)$$

$$+ \mathcal{C}(F'(u), F(u) \prec X, \xi) + F'(u)((F(u) \preceq X - F(u) \prec X) \circ \xi) \quad (5.4)$$

$$+ F'(u)\mathcal{C}(F(u), X, \xi) + F'(u)F(u)\eta. \quad (25.31)$$

Every term except the last is a classical Besov product. Indeed,  $R_F(u)$  has regularity  $2\alpha$ , so  $R_F(u) \circ \xi$  has regularity  $3\alpha - 2 > 0$ ;  $u^\sharp$  has regularity  $2\alpha$  and gives the same gain; the two commutators have regularity  $3\alpha - 2$ ; and the difference between the two paraproducts has regularity  $2\alpha$ , again yielding a classical resonant product with  $\xi$ . The final term has regularity  $2\alpha - 2$  because  $F'(u)F(u) \in \mathcal{C}^\alpha$  and  $\eta \in \mathcal{C}^{2\alpha-2}$ . Thus (25.31) defines  $F(u) \diamond \xi$  continuously from  $(u, u^\sharp, \xi, X, \eta)$ .

*Step 3: contraction.* Let  $\mathcal{D}_T$  be the Banach space of pairs  $(u, u^\sharp)$  satisfying (25.26), with the norms

$$\|u\|_{C_T \mathcal{C}^\alpha} + \|u\|_{C_T^{\alpha/2} L^\infty} + \sup_{0 < t \leq T} t^{\alpha/2} \|u^\sharp(t)\|_{2\alpha}. \quad (25.32)$$

The parabolic Schauder estimate applied to (25.30)–(25.31) yields, on a ball of radius  $R$ ,

$$\|\mathcal{T}(u, u^\sharp)\|_{\mathcal{D}_T} \leq C_R(1 + \|u_0\|_\alpha + \|\xi\|_{\alpha-2} + \|X\|_\alpha + \|\eta\|_{2\alpha-2}) + C_R T^\theta \|(u, u^\sharp)\|_{\mathcal{D}_T}, \quad (25.33)$$

and the same calculation for two inputs gives

$$\|\mathcal{T}(U) - \mathcal{T}(V)\|_{\mathcal{D}_T} \leq C_R T^\theta \|U - V\|_{\mathcal{D}_T} \quad (25.34)$$

for some  $\theta > 0$ . The positive power is the strict gap  $3\alpha - 2 > 0$  together with the short-time Schauder gain. Choosing  $T$  so that  $C_R T^\theta < 1$  gives a unique local fixed point and local Lipschitz dependence on  $(u_0, \xi, \eta)$ .

*Step 4: renormalized smooth approximations.* For smooth data, replace  $\eta$  in (25.31) by  $X_\varepsilon \circ \xi_\varepsilon - c_\varepsilon$ . The last term becomes

$$F'(u_\varepsilon)F(u_\varepsilon)(X_\varepsilon \circ \xi_\varepsilon - c_\varepsilon).$$

Undoing the paracontrolled decomposition shows that this is exactly the ordinary smooth equation with the counterterm displayed in (25.28). The local Lipschitz estimate (25.34), now also in the enhanced-data variables, and (25.27) imply  $u_\varepsilon \rightarrow u$  on every common interval  $[0, T] \subset [0, T_*)$ .

*Step 5: maximal continuation.* For  $F \in C_b^3$ , the source estimate for the resonant term is linear in the high paracontrolled norm and only quadratic in  $\|u\|_{L^\infty}$ . Restarting the local problem at a positive time therefore gives a lifespan bounded below in terms of the enhanced-noise norm and an  $L^\infty$  bound for  $u$ , but not in terms of the full paracontrolled norm. Hence the solution can be continued as long as its  $L^\infty$  norm stays finite, proving (25.29).  $\square$

## Chapter 6

# KPZ universality: equation, energy solution, and fixed point

There are two different scaling limits called KPZ. Weak asymmetry on diffusive scales produces the KPZ equation. Strongly asymmetric growth under 1:2:3 scaling produces the KPZ fixed point. They are related by universality philosophy but are not the same theorem.

### 6.1 Bertini–Giacomin particle-to-KPZ theorem

### 6.2 Compatibility of KPZ solution frameworks

### 6.3 KPZ fixed point

### 6.4 Fixed-point structure and Airy processes

# Chapter 7

## Frontiers of dynamic universality

### 7.1 General one-dimensional strong KPZ universality

*Frontier statement 7.1.*

A broad nonintegrable class of one-dimensional local growth models should converge under 1:2:3 scaling, after model-dependent centering and scaling, to the KPZ fixed point.

*Remark 7.2. Frontier isolation.* The fixed point is rigorously obtained from major integrable models, while weak-KPZ equation limits cover different perturbative regimes; a theorem for arbitrary local nonintegrable growth rules is not known. This row is contextual only and is not available as a premise anywhere in the manuscript.

### 7.2 Two-plus-one-dimensional KPZ universality

*Frontier statement 7.3.*

Generic 2+1-dimensional local growth models with KPZ-type nonlinearity should share universal exponents and limiting fluctuation structures.

*Remark 7.4. Frontier isolation.* There is no theorem remotely comparable in generality to the one-dimensional theory, and even the correct universal limiting object is not characterized at the same level. This row is contextual only and is not available as a premise anywhere in the manuscript.

### 7.3 General dynamic critical-universality classification

*Frontier statement 7.5.*

Microscopic stochastic dynamics for equilibrium critical models should, after identifying conserved quantities and symmetries, converge to a small list of universal stochastic PDEs/dynamic fixed points.

*Remark 7.6. Frontier isolation.* Rigorous dynamic  $\Phi^4$  and KPZ weak-universality results are important examples, but no theorem supplies the full Hohenberg–Halperin-style classification from microscopic hypotheses. This row is contextual only and is not available as a premise anywhere in the manuscript.

## Chapter 8

# Pathwise KPZ, BPHZ, $\Phi_3^4$ , and the KPZ fixed point

This chapter develops the pathwise KPZ, BPHZ, dynamical  $\Phi_3^4$ , and KPZ-fixed-point results in a dependency order: algebraic renormalization first, analytic stochastic bounds second, deterministic fixed points third, and microscopic limits last. This order prevents the circular use of BPHZ convergence before the renormalized model and deterministic solution map have been constructed.

### 8.1 Hairer’s pathwise KPZ solution

The pathwise construction of the one-dimensional KPZ equation is the place where ordinary paracontrolled products are not quite sufficient. The remaining singular integral has to be interpreted as a rough spatial integral. We therefore separate the proof into five layers:

finite stochastic tree expansion  $\longrightarrow$  renormalized cross area  $\longrightarrow$  controlled rough integral,  
controlled rough integral  $\longrightarrow$  weighted fixed point  $\longrightarrow$  Cole–Hopf identification.

This is the proof architecture of Hairer’s pathwise construction; none of the five arrows below is replaced by a citation.

For  $\rho \in \mathbb{R}$ , write  $\bar{\mathcal{C}}^\rho$  for  $\mathcal{C}^\rho$  with one point  $\infty$  adjoined, with neighbourhoods  $\{f : \|f\|_{\mathcal{C}^\rho} > R\} \cup \{\infty\}$ . Thus a maximal solution which blows up in the  $\mathcal{C}^\rho$  norm becomes a continuous path after being set equal to  $\infty$  beyond its explosion time.

Fix the one-dimensional torus  $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ , let  $P_t$  denote the heat semigroup, and let  $\xi$  be space–time white noise. For a smooth spatial Fourier multiplier  $\varphi(\varepsilon D)$  write  $\xi_\varepsilon = \varphi(\varepsilon D)\xi$  and let  $X_\varepsilon$  be the stationary mean-zero solution of

$$(\partial_t - \partial_x^2)X_\varepsilon = \Pi_0^\perp \xi_\varepsilon. \quad (25.1)$$

Here  $\Pi_0^\perp$  removes the spatial zero Fourier mode. Constants are added back explicitly when the KPZ height, rather than its derivative, is reconstructed.

**Definition 8.1** (Finite KPZ tree family and counterterms). Let  $\mathcal{T}_{\text{KPZ}}$  be the finite binary-tree family needed in Hairer’s remainder equation. Starting from the one-leaf tree  $\bullet$ , put  $Y_\varepsilon^\bullet = X_\varepsilon$ . If  $\tau = [\tau_1, \tau_2]$  belongs to the chosen finite family, define

$$(\partial_t - \partial_x^2)Y_\varepsilon^\tau = \partial_x Y_\varepsilon^{\tau_1} \partial_x Y_\varepsilon^{\tau_2} - C_\varepsilon^\tau, \quad (25.2)$$

where  $C_\varepsilon^\tau$  is the spatially constant expectation of the right-hand side whenever that expectation diverges, and is 0 otherwise. The corresponding mean-zero field is denoted  $X_\varepsilon^\tau$ .

Assign the regularities

$$\alpha_\bullet = \frac{1}{2}, \quad \alpha_{[\tau_1, \tau_2]} = \min\{\alpha_{\tau_1}, \alpha_{\tau_2}\} + 1, \quad (25.3)$$

with an arbitrarily small common loss  $\kappa > 0$  understood at every borderline exponent. Only the finitely many trees appearing in the pathwise KPZ ansatz are retained.

**Lemma 8.2** (One-dimensional Fourier convolution). *Let  $a, b > 0$  and put*

$$\rho = \min\{a, b, a + b - 1\}.$$

*For every  $\delta > 0$  there is  $C = C(a, b, \delta)$  such that, for all  $k \in \mathbb{Z}$ ,*

$$\sum_{m \in \mathbb{Z}} \langle m \rangle^{-a} \langle k - m \rangle^{-b} \leq C \langle k \rangle^{-\rho + \delta}. \quad (25.4)$$

*If none of  $a, b, a + b - 1$  is zero, the loss  $\delta$  can be omitted away from the usual logarithmic endpoint.*

*Proof.* For  $|k| \leq 4$  the claim is the ordinary convergence estimate, with the logarithmic endpoint absorbed by  $\delta$ . Assume  $|k| > 4$  and split the sum into

$$A = \{|m| \leq |k|/2\}, \quad B = \{|k - m| \leq |k|/2\}, \quad C = (A \cup B)^c.$$

On  $A$ ,  $\langle k - m \rangle \asymp \langle k \rangle$ , hence

$$\sum_A \langle m \rangle^{-a} \langle k - m \rangle^{-b} \lesssim \langle k \rangle^{-b} \begin{cases} 1, & a > 1, \\ \log \langle k \rangle, & a = 1, \\ \langle k \rangle^{1-a}, & a < 1. \end{cases}$$

This is bounded by  $C \langle k \rangle^{-\min\{b, a+b-1\} + \delta}$ . The region  $B$  is symmetric. On  $C$  one has  $|m| \gtrsim |k|$  and  $|k - m| \gtrsim |k|$ ; summing over dyadic annuli gives  $C \langle k \rangle^{1-a-b}$  when  $a + b > 1$ , with the same harmless logarithmic loss at equality. Taking the weakest of the three estimates proves (25.4).  $\square$

**Lemma 8.3** (Finite stochastic-tree convergence). *Fix  $\kappa \in (0, 1/12)$ . For every  $\tau \in \mathcal{T}_{\text{KPZ}}$  the renormalized fields  $Y_\varepsilon^\tau$  of Definition 8.1 converge in probability, locally uniformly in time, in every spatial Hölder space  $\mathcal{C}^{\alpha_\tau - \kappa}$ . The same statement holds for the finitely many derivatives and heat convolutions used below. Every component belongs to a finite sum of Wiener chaoses of order at most four.*

*Proof.* We give the Fourier argument because it is the stochastic input used again for the cross area. For  $k \neq 0$ , the stationary linear field has representation

$$\widehat{X}_\varepsilon(t, k) = \int_{-\infty}^t e^{-k^2(t-s)} \varphi(\varepsilon k) d\widehat{W}_s(k),$$

so Itô isometry gives

$$\mathbf{E} |\widehat{X}_\varepsilon(t, k)|^2 \lesssim \langle k \rangle^{-2}. \quad (25.5)$$



Each binary vertex contributes two spatial derivatives and one heat resolvent. After Wick expansion, the covariance of a fixed tree is a finite sum of convolution diagrams whose edges carry powers of  $\langle k \rangle^{-1}$  or better. Applying Lemma 8.2 successively from the leaves to the root yields

$$\mathbf{E}|\widehat{Y_\varepsilon^\tau}(t, k)|^2 \lesssim \langle k \rangle^{-1-2\alpha_\tau+2\kappa}. \quad (25.6)$$

The only contractions for which the internal frequency sum has no external  $k$  are local loops. Their expectations are exactly the constants  $C_\varepsilon^\tau$  subtracted in (25.2). Once those constants are removed, all remaining diagrams satisfy (25.6).

For two mollification scales  $\varepsilon, \bar{\varepsilon}$ , at least one edge of every covariance difference carries  $\varphi(\varepsilon k) - \varphi(\bar{\varepsilon} k)$ . Smoothness of  $\varphi$  gives, for  $\theta \in (0, 1)$ ,

$$|\varphi(\varepsilon k) - \varphi(\bar{\varepsilon} k)| \lesssim |\varepsilon - \bar{\varepsilon}|^\theta |k|^\theta,$$

and therefore the same diagram estimate yields

$$\mathbf{E}|\widehat{Y_\varepsilon^\tau - Y_{\bar{\varepsilon}}^\tau}(t, k)|^2 \lesssim |\varepsilon - \bar{\varepsilon}|^{2\theta} \langle k \rangle^{-1-2\alpha_\tau+2\kappa+2\theta}. \quad (25.7)$$

Time increments are treated by  $|e^{-k^2 t} - e^{-k^2 s}| \lesssim |t - s|^\theta |k|^{2\theta}$ . Hypercontractivity converts the second-moment estimates into every finite moment because the chaos orders are bounded. The dyadic Littlewood–Paley criterion and Kolmogorov’s theorem then imply Cauchy convergence in  $C([0, T], \mathcal{C}^{\alpha_\tau - \kappa})$  after an arbitrarily small loss. The finite tree family lets the losses be chosen uniformly.  $\square$

The previous lemma is not yet the whole enhancement. One resonant spatial integral contains two distributions whose regularities sum to at most one, so we must retain an iterated integral.

**Definition 8.4** (KPZ reference path and renormalized cross area). Let  $Y_\varepsilon$  denote the tree field which appears as the rough integrator in Hairer’s remainder equation and define

$$\Phi_{\varepsilon, t} = \int_{-\infty}^t P_{t-s} \partial_x^2 Y_{\varepsilon, s} ds. \quad (25.8)$$

For smooth noise set

$$\mathbb{Y}_{\varepsilon, t}(x, y) = \int_x^y (\Phi_{\varepsilon, t}(z) - \Phi_{\varepsilon, t}(x)) dY_{\varepsilon, t}(z) - c_\varepsilon(t)(y - x), \quad (25.9)$$

where  $c_\varepsilon(t)$  removes the spatial zero mode of the resonant contraction. Then

$$\delta \mathbb{Y}_{\varepsilon, t}(x, y, z) := \mathbb{Y}_{\varepsilon, t}(x, z) - \mathbb{Y}_{\varepsilon, t}(x, y) - \mathbb{Y}_{\varepsilon, t}(y, z) = \delta \Phi_{\varepsilon, t}(x, y) \delta Y_{\varepsilon, t}(y, z). \quad (25.10)$$

**Lemma 8.5** (Renormalized cross-area convergence). *For every  $\gamma < 1$ , the family  $\mathbb{Y}_\varepsilon$  of Definition 8.4 converges in probability in  $C([0, T], \mathcal{C}_2^\gamma(\mathbb{T}))$ . Its limit  $\mathbb{Y}$  satisfies Chen’s relation (25.10) with the limiting pair  $(\Phi, Y)$ .*

*Proof.* The algebraic relation follows by splitting the integral over  $[x, z]$  at  $y$ ; the linear counterterm cancels under  $\delta$ . For the analytic estimate, expand (25.9) in Fourier modes. After Wick ordering, every second moment is a finite sum of diagrams of the form

$$\sum_{k_1 + \dots + k_r = 0} \frac{|e^{ik_1(y-x)} - 1|^2 Q(k_1, \dots, k_r)}{\prod_j \langle k_j \rangle^{q_j}}, \quad (25.11)$$

where  $Q$  has degree supplied by the two spatial derivatives in (25.8) and by the derivative  $dY$ , while each heat integration contributes two powers in the denominator. The local pairing with no external frequency is precisely  $c_\varepsilon(t)$  and has been subtracted. For the remaining diagrams, use  $|e^{ik\ell} - 1| \lesssim |k\ell|^{1-2\kappa}$  and successively apply Lemma 8.2; the finite list of diagrams then gives, uniformly in  $\varepsilon$ ,

$$\mathbf{E}|\mathbb{Y}_{\varepsilon,t}(x,y)|^2 \lesssim |x-y|^{2-4\kappa}. \quad (25.12)$$

If two mollifier scales are compared, one noise edge carries their multiplier difference, so the argument proving (25.7) yields

$$\mathbf{E}|\mathbb{Y}_{\varepsilon,t}(x,y) - \mathbb{Y}_{\bar{\varepsilon},t}(x,y)|^2 \lesssim |\varepsilon - \bar{\varepsilon}|^{2\theta} |x-y|^{2-4\kappa-2\theta} \quad (25.13)$$

for sufficiently small  $\theta > 0$ . The same Fourier computation with a heat increment gives positive temporal Hölder control. All terms lie in a finite chaos, hence hypercontractivity followed by Kolmogorov's criterion proves convergence in every  $\mathcal{C}_2^\gamma$  with  $\gamma < 1$ . Passing to the limit in (25.10) is then immediate.  $\square$

We next record the deterministic rough-integration result needed by the fixed-point argument. This is proved here from the sewing lemma so that the pathwise construction does not import rough integration as a hidden theorem.

**Lemma 8.6** (Sewing lemma on the circle). *Let  $A(x,y)$  be a two-point increment with  $A(x,x) = 0$ . Suppose that for some  $\theta > 1$ ,*

$$|\delta A(x,y,z)| \leq C|z-x|^\theta$$

*whenever  $x,y,z$  lie in a common coordinate interval of  $\mathbb{T}$ . Then there is a unique additive increment  $I(x,y)$  such that*

$$|I(x,y) - A(x,y)| \leq C_\theta C |y-x|^\theta.$$

*Moreover  $I$  depends Lipschitz-continuously on  $A$  in these seminorms.*

*Proof.* For the dyadic partition  $\pi_n$  of  $[x,y]$ , let  $I_n = \sum_{[u,v] \in \pi_n} A(u,v)$ . Replacing every interval by its two halves changes the sum by at most  $C2^n(2^{-n}|y-x|)^\theta$ . Hence

$$|I_{n+1} - I_n| \leq C2^{-n(\theta-1)} |y-x|^\theta,$$

and the geometric series converges since  $\theta > 1$ . Its limit is additive, and summing the displayed estimate gives the claimed error. The same calculation applied to a difference proves Lipschitz continuity. Uniqueness follows because an additive increment bounded by  $o(|y-x|)$  vanishes after subdivision.  $\square$

**Definition 8.7** (Controlled path). Let  $\Phi \in \mathcal{C}^\alpha$  and let  $\mathbb{Y} \in \mathcal{C}_2^{\alpha+\beta}$  satisfy (25.10) with  $Y \in \mathcal{C}^\beta$ . A function  $v$  is *controlled by  $\Phi$*  if there are  $v'$  and a remainder  $R^v$  such that

$$\delta v(x,y) = v'(x)\delta\Phi(x,y) + R^v(x,y), \quad R^v \in \mathcal{C}_2^{\bar{\alpha}}, \quad (25.14)$$

with  $\alpha + \beta + \min\{\alpha, \bar{\alpha}\} > 1$  in the applications below.

**Lemma 8.8** (Controlled rough integral). *Suppose  $(\Phi, Y, \mathbb{Y})$  satisfy (25.10),  $v$  satisfies (25.14), and the exponents obey*

$$\bar{\alpha} + \beta > 1, \quad \alpha + (\alpha + \beta) > 1. \quad (25.15)$$

Then the compensated sums

$$\sum_{[x_i, x_{i+1}] \in \pi} \left( v(x_i) \delta Y(x_i, x_{i+1}) + v'(x_i) \mathbb{Y}(x_i, x_{i+1}) \right) \quad (25.16)$$

converge as  $|\pi| \rightarrow 0$  to an integral denoted  $\int v dY$ . The integral is locally Lipschitz in the controlled-path and enhanced-path norms.

*Proof.* Set  $A(x, y) = v(x) \delta Y(x, y) + v'(x) \mathbb{Y}(x, y)$ . Using (25.10) and (25.14), the terms  $v'(x) \delta \Phi(x, y) \delta Y(y, z)$  cancel exactly, leaving

$$\delta A(x, y, z) = -R^v(x, y) \delta Y(y, z) - \delta v'(x, y) \mathbb{Y}(y, z).$$

The first term has order  $\bar{\alpha} + \beta > 1$  and the second has order  $\alpha + (\alpha + \beta) > 1$  for the derivative regularity used below. Apply Lemma 8.6. Its Lipschitz estimate gives the final continuity statement.  $\square$

**Lemma 8.9** (Controlled stochastic remainder). *Let  $\bar{Y}_\varepsilon$  be the stochastic field which appears in the last singular coefficient of Hairer's KPZ remainder equation and define*

$$R_{\varepsilon, t}^{\bar{Y}}(x, y) = \delta \bar{Y}_{\varepsilon, t}(x, y) - \bar{Y}_{\varepsilon, t}(x) \delta \Phi_{\varepsilon, t}(x, y). \quad (25.17)$$

*For every  $\delta > 0$  the pair  $(\bar{Y}_\varepsilon, R_\varepsilon^{\bar{Y}})$  converges in probability in  $C([0, T], \mathcal{C}^{1/2-\delta} \times \mathcal{C}_2^{1-\delta})$  after the same finite renormalizations as above.*

*Proof.* The smooth equation for  $\bar{Y}_\varepsilon$  can be written in Duhamel form as

$$\bar{Y}_{\varepsilon, t} = P_t \bar{Y}_{\varepsilon, 0} + \mathcal{M}_{\mathbb{Y}_\varepsilon}(\bar{Y}_\varepsilon)_t + F_{\varepsilon, t}, \quad (25.18)$$

where  $F_\varepsilon$  consists only of tree fields already covered by Lemma 8.3 and  $\mathcal{M}_{\mathbb{Y}}$  is a heat convolution of the rough integral in Lemma 8.8. The heat-kernel derivative bound

$$\|\partial_x P_t f\|_{\mathcal{C}^{\rho+1}} \lesssim t^{-1/2} \|f\|_{\mathcal{C}^\rho} \quad (25.19)$$

and the rough-integral estimate imply that  $\mathcal{M}$  maps the controlled space in (25.17) to itself and gains a factor  $T^\theta$  for some  $\theta > 0$ . For  $T$  small it is therefore a contraction uniformly in  $\varepsilon$  on a ball controlled by the enhanced-noise norm. Lemmas 8.3 and 8.5, together with the Lipschitz part of Lemma 8.8, make the right-hand side of (25.18) Cauchy. The fixed points are consequently Cauchy in the controlled norm. The local intervals concatenate because the estimates depend only on the controlled norm at the left endpoint, proving the claim on  $[0, T]$ .  $\square$

**Definition 8.10** (Pathwise KPZ enhancement space). For  $T > 0$ , let  $\mathcal{X}_T$  be the closure of smooth tuples

$$\Psi = (Y^\tau)_{\tau \in \mathcal{T}_{\text{KPZ}}} \oplus (\Phi, Y, \mathbb{Y}) \oplus (\bar{Y}, R^{\bar{Y}}) \quad (25.20)$$

in the product of the Hölder spaces specified in Lemmas 8.3, 8.5, and 8.9, subject to Chen's relation (25.10) and the controlled relation (25.17). Equip the projective limit over  $T \in \mathbb{N}$  with its product metric and call it  $\mathcal{X}$ . Then  $\mathcal{X}$  is Polish, and the limiting stochastic enhancement  $\Psi(\omega)$  is an  $\mathcal{X}$ -valued measurable random variable.

*Proof.* Each factor is separable after lowering the displayed Hölder exponent by a fixed arbitrarily small amount. Chen's relation and the controlled relation are closed under convergence in the product norm. Thus each  $\mathcal{X}_T$  is a closed subset of a separable Banach product and is Polish. The compatibility maps  $\mathcal{X}_{T+1} \rightarrow \mathcal{X}_T$  are continuous, so the projective-limit metric is complete and separable. Measurability follows because every approximating tuple is a finite Wiener-chaos functional of the noise and convergence holds in probability in the Polish target.  $\square$

**Definition 8.11** (Weighted controlled remainder norm). Fix  $\kappa \in (0, 1/12)$  and  $\beta > 2\kappa$ . For a remainder  $v$  controlled by  $\Phi$ , set schematically

$$\|v\|_{*,T} := \sup_{0 < t \leq T} t^{1-2\kappa} \left( \|v_t\|_{\mathcal{C}^{1/2-\kappa}} + \|v'_t\|_{\mathcal{C}^{3\kappa}} + \|R_t^v\|_{\mathcal{C}_2^{1/2+2\kappa}} \right. \quad (8.1)$$

$$\left. + t^{3\kappa-1/2} \|v_t\|_{\infty} \right) + [v]_{\text{time},T}, \quad (25.21)$$

where  $[v]_{\text{time},T}$  is the corresponding weighted time-Hölder seminorm from the heat-kernel scaling. Let  $\mathcal{B}_{*,T}$  be the completion of smooth controlled remainders in this norm.

**Lemma 8.12** (Rough heat operator and exponent table). *For  $\kappa < 1/12$  the rough heat-convolution terms in the KPZ remainder map are locally Lipschitz on  $\mathcal{B}_{*,T}$  and gain a factor  $T^\theta$  for some  $\theta > 0$ . One admissible source-compatible set of exponents is*

$$\alpha = 1 - 2\kappa, \quad \beta_1 = 1 - \frac{\kappa}{2}, \quad \gamma = 1 - 2\kappa, \quad \delta = 2\kappa. \quad (25.22)$$

*All sewing and time-convolution inequalities are then strict.*

*Proof.* The deterministic heat estimates give, for  $r \geq 0$  and  $\sigma < 2$ ,

$$\|P_t f\|_{\mathcal{C}^{r+\sigma}} \lesssim t^{-\sigma/2} \|f\|_{\mathcal{C}^r}, \quad \|\partial_x P_t f\|_{\mathcal{C}^{r+\sigma}} \lesssim t^{-(1+\sigma)/2} \|f\|_{\mathcal{C}^r}. \quad (25.23)$$

The singular rough term is estimated first in space by Lemma 8.8 and then convolved with  $\partial_x P_{t-s}$ . With (25.22), the worst spatial sewing exponent is

$$(1 - 2\kappa) + (1 - \kappa/2) - 1 = 1 - \frac{5}{2}\kappa > 0,$$

and the worst residual time singularity is strictly below 1 once the weight  $t^{1-2\kappa}$  in (25.21) is included. The terms containing the controlled remainder use  $(1/2 + 2\kappa) + (1/2 - \kappa) > 1$ , while the terms containing  $v'$  use Chen's second-level object of regularity  $1 - 2\kappa$ . The remaining polynomial terms are classical paraproducts and are easier. Integrating  $(t-s)^{-a} s^{-b}$  with  $a, b < 1$  yields  $T^{1-a-b}$  whenever  $a + b < 1$ ; the strict margin furnished by  $\kappa < 1/12$  supplies a common positive  $\theta$ . The same estimates on a difference give local Lipschitz continuity.  $\square$

**Lemma 8.13** (Weighted KPZ remainder fixed point). *For every  $\beta \in (0, 1/2)$ , after choosing  $\kappa < \min\{1/12, \beta/2\}$ , there is a lower-semicontinuous lifetime*

$$T_* : \mathcal{C}^\beta(\mathbb{T}) \times \mathcal{X} \rightarrow (0, \infty]$$

*and a solution map  $\mathcal{S}_R$  which is continuous jointly in the initial condition and enhancement on every set  $\{t < T_*\}$ . The solution is obtained by a unique fixed point in  $\mathcal{B}_{*,T}$  on short intervals and can be continued until its controlled norm diverges.*

*Proof.* Insert the finite tree ansatz

$$h = \sum_{\tau \in \mathcal{T}_{\text{KPZ}}} \lambda^{|\tau|} Y^\tau + u$$

into the smooth KPZ equation, differentiate the singular part when necessary, and express the unique nonclassical spatial integral by Lemma 8.8. The resulting map  $\widehat{\mathcal{M}}$  on  $\mathcal{B}_{*,T}$  is the sum of a

heat evolution of  $h_0$ , finitely many enhanced stochastic coefficients, ordinary para-products, and the rough heat operator of Lemma 8.12. Hence on a ball of radius  $R$ ,

$$\|\widehat{\mathcal{M}}(v)\|_{*,T} \leq C_0(h_0, \Psi) + CT^\theta P(R, \|\Psi\|),$$

while

$$\|\widehat{\mathcal{M}}(v) - \widehat{\mathcal{M}}(\tilde{v})\|_{*,T} \leq CT^\theta P(R, \|\Psi\|) \|v - \tilde{v}\|_{*,T}.$$

Choose  $R > 2C_0$  and then  $T$  so that the second constant is below  $1/2$ . Banach's theorem gives the unique local fixed point and the same inequalities give local Lipschitz dependence on  $(h_0, \Psi)$ . Restarting at every positive time gives a maximal lifetime. The continuation criterion makes  $\{T_* > t\}$  open, hence  $T_*$  is lower semicontinuous.  $\square$

**Lemma 8.14** (Smooth consistency and Cole–Hopf approximation). *If the enhancement is the canonical enhancement of a smooth forcing, the rough integral of Lemma 8.8 equals the ordinary Riemann integral and the fixed point of Lemma 8.13 is the classical solution of the correspondingly renormalized smooth KPZ equation. For the Fourier mollifications above, these classical solutions converge on every finite interval to the Cole–Hopf solution, and the canonical random pathwise solution cannot explode in finite time.*

*Proof.* For smooth paths the compensated Riemann sums in (25.16) converge to the ordinary integral by elementary calculus, so substitution into the fixed point recovers the classical mild equation. With the canonical constant  $C_\varepsilon$  furnished by the tree expansion, Itô's formula applied to  $Z_\varepsilon = e^{h_\varepsilon}$  gives the multiplicative stochastic heat equation with the matching Wick drift. The Walsh–Picard difference estimate proved in SMX-25.02.01, applied with the multiplier difference on the noise, gives Cauchy convergence of  $Z_\varepsilon$  in  $C([0, T], \mathcal{C}^{1/2-\beta})$  after lowering the exponent. The strict-positivity theorem 2.5, together with the mild well-posedness from Chapter 2, supplies the positive global SHE solution and hence a well-defined logarithm at every positive time. Thus the smooth KPZ solutions converge to the Cole–Hopf height. Because the same renormalization and constant-mode convention is used in the smooth equations and in the limiting enhancement, this identification is an exact equality in the convention fixed here.

On the other hand, Lemmas 8.3, 8.5, and 8.9 give  $\Psi_\varepsilon \rightarrow \Psi$  in probability in  $\mathcal{X}$ . Continuity from Lemma 8.13 therefore identifies the pathwise limit with the Cole–Hopf limit on every interval strictly below  $T_*(h_0, \Psi)$ . If  $T_* < \infty$  on an event of positive probability, choose a deterministic  $T$  for which this event has positive probability and use the blow-up criterion: along that event the pathwise controlled norm must leave every bounded set before  $T$ . But the same smooth approximations converge on  $[0, T]$  to the finite Cole–Hopf solution, while their enhancements converge to  $\Psi$ ; local continuity of the fixed point therefore keeps the controlled norm bounded on a slightly shorter interval and permits a restart across the putative explosion time, a contradiction. Hence  $T_* = \infty$  almost surely for the canonical white-noise enhancement.  $\square$

**Theorem 8.15** (Pathwise KPZ solution and approximation). *There exist a single Polish enhancement space  $\mathcal{X}$  and a measurable canonical white-noise enhancement  $\Psi : \Omega \rightarrow \mathcal{X}$  with the following property. For every  $\beta \in (0, 1/2)$  there are a lower-semicontinuous maximal lifetime*

$$T_* : \mathcal{C}^\beta(\mathbb{T}) \times \mathcal{X} \rightarrow (0, \infty],$$

*and a pathwise solution map*

$$\mathcal{S}_R : \mathcal{C}^\beta(\mathbb{T}) \times \mathcal{X} \longrightarrow C(\mathbb{R}_+, \bar{\mathcal{C}}^{1/2-\beta}(\mathbb{T})).$$

It is set equal to  $\infty$  after  $T_*$ . The map  $(t, h_0, \Psi) \mapsto \mathcal{S}_R(h_0, \Psi)(t)$  is continuous on  $\{t < T_*(h_0, \Psi)\}$ . For every deterministic  $h_0 \in \mathcal{C}^\beta$ , the canonical white-noise enhancement satisfies

$$T_*(h_0, \Psi) = \infty \quad \text{a.s.},$$

and, in the fixed counterterm and constant-mode convention of this volume,  $\mathcal{S}_R(h_0, \Psi)$  agrees exactly with the Cole–Hopf solution. Changing to another admissible mollifier-centering convention changes both representatives by the corresponding deterministic spatially constant height normalization; it does not weaken this equality within any fixed convention. Moreover the smooth renormalized KPZ solutions associated with  $\Psi_\varepsilon$  converge in probability to this solution on every compact time interval in the corresponding lowered Hölder topology. Finally,  $\mathcal{X}$  embeds in a separable Fréchet space  $\mathcal{W}$  so that every continuous linear functional  $\ell \in \mathcal{W}^*$  satisfies  $\ell(\Psi) \in \bigoplus_{j=0}^4 \mathcal{H}_j$ , the sum of the first four Wiener chaoses of the driving white noise.

*Proof.* Definition 8.10 and Lemmas 8.3, 8.5, and 8.9 construct the measurable enhancement and prove convergence of its smooth approximations. Every coordinate is a finite polynomial in Gaussian Wiener integrals of degree at most four; the same is true after closure for every continuous linear functional on the ambient product Fréchet space, proving the final chaos assertion. Lemmas 8.8 and 8.12 prove locally the only nonclassical deterministic operation. Lemma 8.13 supplies the continuous maximal solution map and lower-semicontinuous lifetime. Finally Lemma 8.14 proves smooth consistency, approximation, Cole–Hopf identification, and almost-sure nonexplosion. These are exactly the assertions of the theorem.  $\square$

**Lemma 8.16** (Periodic energy solution as a Cole–Hopf derivative). *Let  $W$  be periodic space–time white noise on  $\mathbb{T} = \mathbb{R}/\mathbb{Z}$  and let  $\eta$  be an  $\mathcal{F}_0$ -measurable mean-zero periodic spatial white noise. In the unit-diffusion normalization*

$$du = \Delta u \, dt + \lambda \partial_x(u^{\diamond 2}) \, dt + \sqrt{2} \partial_x dW, \quad u_0 = \eta, \quad (25.23p)$$

*strong stationary energy solutions are unique up to indistinguishability, and energy solutions are unique in law. If  $\lambda \neq 0$ , then*

$$u = \lambda^{-1} \partial_x \log Z \quad (25.23q)$$

*in  $\mathcal{D}'(\mathbb{T})$ , where  $Z$  is the unique mild solution of*

$$dZ = \Delta Z \, dt + \sqrt{2} \lambda Z \, dW, \quad Z_0 = e^{\lambda I \eta}, \quad (25.23r)$$

*and  $I\eta$  is the mean-zero periodic primitive.*

*Proof.* We give the periodic transfer of the already-owned forward–backward Cole–Hopf argument because one finite renormalization changes and is load-bearing. The Gaussian invariant law is the mean-zero periodic white noise. On a cylinder functional of the nonzero Fourier modes, the symmetric Ornstein–Uhlenbeck generator has Dirichlet form

$$\mathcal{E}(F) = 2 \int_{\mathbb{T}} |\partial_x D_x F|^2 \, dx.$$

Writing the controlled martingale problem forward and backward in time and adding the two identities cancels the antisymmetric Burgers drift. Doob’s inequality therefore gives the same maximal  $H_{-1}$  estimate as on the real line,

$$\mathbf{E} \sup_{t \leq T} \left| \int_0^t F(u_s) \, ds \right|^2 \leq CT \|(-L_0)^{-1/2} F\|_2^2, \quad (25.23s)$$

first for centered cylinders and then by form closure. The Dirichlet-process Itô and product rules use only this forward-backward structure and quadratic variation, so they also carry to  $\mathbb{T}$ .

Let  $\bar{\rho}_L$  be the periodization of an even unit-mass smooth mollifier at scale  $L^{-1}$  and let  $\Theta_L$  be its mean-zero periodic primitive,

$$\widehat{\Theta}_L(k) = \frac{\widehat{\bar{\rho}}_L(k)}{2\pi i k} \quad (k \neq 0), \quad \widehat{\Theta}_L(0) = 0.$$

For the translate  $\Theta_{L,x}(y) = \Theta_L(x - y)$  put  $h_t^L(x) = u_t(\Theta_{L,x})$  and  $\phi_t^L(x) = e^{\lambda h_t^L(x)}$ . Testing (25.23p) against  $\Theta_{L,x}$  and applying the Dirichlet-process Itô formula produces the periodic analogue of the finite-scale exponential identity: its nonconstant error is a centered second Wiener-chaos additive functional, while its scalar contraction is

$$K_L = \lambda^2 \|\bar{\rho}_L * \Theta_L\|_{L^2(\mathbb{T})}^2.$$

The Fourier calculation is explicit:

$$\lambda^{-2} K_L = \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{|\widehat{\bar{\rho}}_L(k)|^2}{(2\pi k)^2} \rightarrow \frac{1}{2\pi^2} \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{1}{12}. \quad (25.23t)$$

This is the sole periodic renormalization constant that is not obtained by literal replacement of integrals by Fourier series.

For completeness, the chaos remainder also vanishes in the required norm. After a second ultraviolet regularization at scale  $N \geq L$ , Gaussian integration by parts writes the centered discrepancy as a second-chaos kernel  $r^{L,N}$ . In Fourier variables every occurrence of the primitive contributes  $(2\pi k)^{-1}$  and every translate difference contributes  $e^{2\pi i k/L} - 1$ . Hence

$$\|r^{L,N}\|_{-1}^2 \leq C \sum_{k \neq 0} \frac{|e^{2\pi i k/L} - 1|^2}{k^2} |\widehat{\rho}(k/L)|^2 \omega_{L,N}(k) \leq \frac{C}{L}, \quad (25.23u)$$

where  $0 \leq \omega_{L,N} \leq 1$  is the harmless cutoff coming from the second regularization. The first inequality is the Fourier form of the same partial Gaussian integration-by-parts estimate used on the line; the second follows by splitting  $|k| \leq L$  and  $|k| > L$  and using the compact Fourier support (or rapid decay) of the chosen mollifier. Letting  $N \rightarrow \infty$  and then using (25.23s) shows that the time-integrated remainder tends to zero uniformly on compact time intervals in probability.

Multiplying  $\phi^L$  by the scalar exponential that cancels the zero Fourier mode and subtracts  $\lambda^4 t/12$ , the Itô product rule now gives

$$dZ^L = \Delta Z^L dt + \sqrt{2} \lambda Z^L dW^L + d\mathcal{R}^L + \lambda^2 (K_L - \lambda^2/12) Z^L dt, \quad (25.23v)$$

with  $\mathcal{R}^L \rightarrow 0$  by (25.23u). Ordinary mollification of the periodic Brownian primitive gives local uniform convergence of  $h^L$ ; hence the resulting  $Z = e^X e^{\lambda Iu}$  is strictly positive by construction. The noise mollifiers converge strongly in  $L^2(\mathbb{T})$  and (25.23t) removes the deterministic error, so every limit solves (25.23r). If two mild solutions have the same initial condition, Itô isometry for their difference  $D$  gives

$$\sup_x \mathbf{E} |D_t(x)|^2 \leq C \int_0^t (t-s)^{-1/2} \sup_x \mathbf{E} |D_s(x)|^2 ds;$$

iterating once and applying Gronwall yields uniqueness. Thus every energy solution maps to the same  $Z$ , and differentiating its logarithm proves (25.23q). The construction is measurable in the fixed driving noise, which gives pathwise uniqueness for strong stationary solutions and uniqueness in law for energy solutions.  $\square$

**Theorem 8.17** (Compatibility of Cole–Hopf, pathwise, and energy-solution frameworks).

*The compatibility statement has two source-exact parts, which must not be conflated.*

(a) Periodic three-way compatibility at Burgers level. *Work on the circle in the unit-diffusion normalization and rescale the driving white noise and  $\lambda$  once and for all so that the periodic stochastic Burgers equation is*

$$du = \Delta u dt + \lambda \partial_x (u^{\circ 2}) dt + \sqrt{2} \partial_x dW.$$

*Let  $\eta$  be an  $\mathcal{F}_0$ -measurable mean-zero periodic spatial white noise, let  $I\eta$  be its mean-zero periodic primitive,*

$$\widehat{I\eta}(k) = (2\pi i k)^{-1} \widehat{\eta}(k) \quad (k \neq 0), \quad \widehat{I\eta}(0) = 0,$$

*and take  $h_{\text{pw}}$  to be Hairer’s periodic pathwise KPZ solution started from  $I\eta$  in the matching renormalization convention. (The pathwise map is constructed for every initial condition in the relevant Hölder space; its almost-sure globality for this random initial datum follows by conditioning on  $I\eta$ , which is independent of the future white-noise increments.) Let  $Z^{\mathbb{T}}$  solve*

$$dZ^{\mathbb{T}} = \Delta Z^{\mathbb{T}} dt + \sqrt{2} \lambda Z^{\mathbb{T}} dW, \quad Z_0^{\mathbb{T}} = e^{\lambda I\eta},$$

*and let  $u_{\text{en}}$  be the periodic stationary energy solution driven by the same  $W$  with  $u_{\text{en}}(0) = \eta$ . On this common domain,*

$$\partial_x h_{\text{pw}} = u_{\text{en}} = \lambda^{-1} \partial_x \log Z^{\mathbb{T}} \quad \text{in } \mathcal{D}'(\mathbb{T}). \quad (25.24a)$$

*The identity is insensitive to every spatially constant height mode. No  $\lambda^3 t/12$  assertion is part of this periodic three-way statement.*

(b) Full-line energy-height normalization. *In the Gubinelli–Perkowski full-line almost-stationary normalization*

$$dh = \Delta h dt + \lambda((\partial_x h)^2 - \infty) dt + \sqrt{2} dW,$$

*if  $h_{\text{en}}$  is the strong almost-stationary energy height and  $Z^{\mathbb{R}}$  solves*

$$dZ^{\mathbb{R}} = \Delta Z^{\mathbb{R}} dt + \sqrt{2} \lambda Z^{\mathbb{R}} dW,$$

*with the matching initial condition, then*

$$h_{\text{en}}(t, \cdot) = \lambda^{-1} \log Z_t^{\mathbb{R}} + \frac{\lambda^3}{12} t \quad (\lambda \neq 0), \quad (25.24b)$$

*up to the chosen initial additive height constant. Consequently  $\partial_x h_{\text{en}} = \lambda^{-1} \partial_x \log Z^{\mathbb{R}}$ . Part (b) is a two-way energy/Cole–Hopf identification; Hairer’s 2013 theorem is periodic and is not used to turn it into a full-line three-way statement.*

*Proof.* For part (a), Theorem 8.15 and Lemma 8.14 identify Hairer’s periodic pathwise solution with the periodic Cole–Hopf construction under the matching smooth-noise renormalization convention. Conditioning on the independent initial primitive  $I\eta$  extends the deterministic-initial-data approximation statement to the stationary random initial condition used here. Differentiating gives  $\partial_x h_{\text{pw}} = \lambda^{-1} \partial_x \log Z^{\mathbb{T}}$ . Lemma 8.16 independently proves, inside this volume, that the periodic stationary energy solution is the same distributional derivative of the same SHE. This proves (25.24a). Since differentiation kills the zero Fourier mode, no height drift needs to be identified in the periodic three-way theorem.

For part (b), apply the full-line almost-stationary energy-height theorem in the normalization displayed above. Gubinelli–Perkowski’s Cole–Hopf computation gives exactly  $h_{\text{en}} = \lambda^{-1} \log Z^{\mathbb{R}} + \lambda^3 t/12$ . Differentiation removes the deterministic drift and yields the Burgers identity. The Volume XXIV coefficient-rescaling lemma records how this unit normalization pulls back to general  $D, \chi, \lambda$ ; no such rescaling is silently folded into (25.24b).  $\square$



**Normalization and parameter scope.** Hairer’s Theorem 1.1 uses one Polish  $\mathcal{X}$  and one measurable enhancement  $\Psi$  working simultaneously for every  $\beta \in (0, 1/2)$ , and it states the almost-sure identity  $\mathcal{S}_{CH}(h_0, \omega) = \mathcal{S}_R(h_0, \Psi(\omega))$  in its fixed normalization. The theorem above says exactly this: it does not let  $\mathcal{X}$  appear to depend on  $\beta$  and no longer weakens equality to “up to a deterministic normalization.” The normalization caveat is kept only as a comparison between different conventions, not inside the theorem’s fixed convention. Stochastic-tree convergence, renormalized area, rough integration, fixed point, nonexplosion, and the positivity needed by the Cole–Hopf logarithm are supplied by the preceding results, including Theorem 2.5.

*Remark 8.18. Primary-source note.* The theorem and proof architecture in this section are checked against M. Hairer, *Solving the KPZ equation*, Annals of Mathematics 178 (2013), 559–664, especially Theorem 1.1, Proposition 1.5, Theorems 2.3 and 2.9, the controlled rough-path construction in Section 3, the fixed-point estimates in Section 4, and the stochastic area/remainder estimates in Section 7.

## 8.2 The algebraic BPHZ renormalization group

The analytic convergence theorem in the next section uses an algebraic object which must already have been constructed correctly. We therefore separate the issues sharply. In this section all trees are finite, all random inputs are smooth, and no limit  $\varepsilon \downarrow 0$  is taken. The tasks are to build the two forest coactions, prove their cointeraction, prove that the negative character group acts on admissible models, and identify the canonical BPHZ character. The multiscale estimates needed for singular random models remain an independent obligation.

**Definition 8.19** (Typed, coloured and decorated forests). Let  $\mathfrak{L} = \mathfrak{L}_- \sqcup \mathfrak{L}_+$  be the finite set of noise and kernel types and let  $|\mathfrak{t}|_{\mathfrak{s}} \in \mathbb{R} \setminus \{0\}$  be their degrees for the fixed scaling  $\mathfrak{s}$ . An edge  $e$  has a type  $\mathfrak{t}(e) \in \mathfrak{L}$ . A *coloured forest* is a rooted forest  $F$ , together with a map

$$\widehat{F} : E_F \sqcup N_F \longrightarrow \{0, 1, 2\},$$

such that whenever an edge has colour  $i \in \{1, 2\}$ , both of its endpoints have colour  $i$ . Thus each positive colour identifies a union of connected subforests. Colour 1 records pieces already selected by the negative renormalization extraction and colour 2 records the rooted pieces used for positive reexpansion. Uncoloured edges and vertices have colour 0.

An *extended decorated forest* is a tuple

$$\tau = (F, \widehat{F}, \mathfrak{n}, \mathfrak{o}, \mathfrak{e})$$

with

$$\mathfrak{n} : N_F \rightarrow \mathbb{N}^d, \quad \mathfrak{e} : E_F \rightarrow \mathbb{N}^d, \quad \mathfrak{o} : N_F \rightarrow \mathbb{Z}^d \oplus \mathbb{Z}(\mathfrak{L}),$$

where  $\mathfrak{e}$  vanishes on coloured edges and  $\mathfrak{o}$  vanishes on uncoloured vertices. The decoration  $\mathfrak{n}$  carries polynomial powers,  $\mathfrak{e}$  carries kernel derivatives, and  $\mathfrak{o}$  remembers the multiindex and type-degree data of pieces contracted into a coloured vertex. For  $a = (k, v) \in \mathbb{Z}^d \oplus \mathbb{Z}(\mathfrak{L})$  put

$$|a|_{\mathfrak{s}} = |k|_{\mathfrak{s}} + \sum_{\mathfrak{t} \in \mathfrak{L}} v_{\mathfrak{t}} |\mathfrak{t}|_{\mathfrak{s}}.$$

Writing  $\widehat{E}$  for all coloured edges,  $\widehat{E}_2$  for the edges of colour 2, and  $\widehat{N}_2$  for the vertices of colour 2,

define

$$|\tau|_- = \sum_{e \in E_F \setminus \widehat{E}} (|\mathfrak{t}(e)|_{\mathfrak{s}} - |\mathfrak{c}(e)|_{\mathfrak{s}}) + \sum_{x \in N_F} |\mathfrak{n}(x)|_{\mathfrak{s}}, \quad (25.2a)$$

$$|\tau|_+ = \sum_{e \in E_F \setminus \widehat{E}_2} (|\mathfrak{t}(e)|_{\mathfrak{s}} - |\mathfrak{c}(e)|_{\mathfrak{s}}) + \sum_{x \in N_F} |\mathfrak{n}(x)|_{\mathfrak{s}} + \sum_{x \in N_F \setminus \widehat{N}_2} |\mathfrak{o}(x)|_{\mathfrak{s}}. \quad (25.2b)$$

The asymmetry is intentional: the negative degree ignores all coloured material, whereas the positive degree uses the extended decoration in order to remember the degree lost by negative contraction.

**Definition 8.20** (Rules and completion). A *node type* is a finite multiset of pairs  $(\mathfrak{t}, k)$  consisting of an edge type and derivative multiindex. A rule  $R$  assigns to each  $\mathfrak{t} \in \mathfrak{L}$  a collection  $R(\mathfrak{t})$  of allowed node types. It is *normal* if  $M \subset N \in R(\mathfrak{t})$  implies  $M \in R(\mathfrak{t})$ , and *subcritical* if there is a map  $\text{reg} : \mathfrak{L} \rightarrow \mathbb{R}$  with a strict margin  $\kappa > 0$  such that

$$\text{reg}(\mathfrak{t}) + \kappa \leq |\mathfrak{t}|_{\mathfrak{s}} + \inf_{N \in R(\mathfrak{t})} \text{reg}(N). \quad (25.2c)$$

Here  $\text{reg}(N)$  is the sum of the regularities of the entries of  $N$ , with derivative multiindices subtracted.

A normal subcritical rule is called *complete* if it is also closed under the following operation: whenever a negative rule-generated subtree is contracted inside an allowed node type, and boundary derivatives are redistributed with total scaled order smaller than the absolute value of that subtree's negative degree, the resulting node type is again allowed. We call the smallest rule obtained by iterating these substitutions the *renormalization completion*  $\overline{R}$ .

**Lemma 8.21** (Completion preserves local subcriticality). *Every normal subcritical rule  $R$  admits a normal subcritical complete extension  $\overline{R}$ . Moreover, below any fixed homogeneity cutoff only finitely many rule-generated tree shapes and derivative decorations occur. Thus replacing  $R$  by  $\overline{R}$  introduces no infinite algebraic sum in a bounded homogeneity window.*

*Proof.* Fix a strict subcriticality margin  $\kappa$  in (25.2c). For a planted tree  $\sigma$  with trunk of type  $(\mathfrak{t}, k)$ , induction on its number of edges gives

$$\text{reg}(\mathfrak{t}) - |k|_{\mathfrak{s}} \leq |\sigma|_{\mathfrak{s}} - \kappa |E_{\sigma}|. \quad (25.2d)$$

Indeed the inequality is (25.2c) for a one-edge tree, while at an internal vertex the induction hypotheses for the branches are inserted into (25.2c); every additional edge contributes another  $\kappa$  of slack. It follows that a tree of degree at most  $\gamma$  has a uniformly bounded number of edges. Because the type set is finite and the derivative decorations are also bounded by the same degree inequality, there are only finitely many such trees in every bounded homogeneity window.

Now enlarge  $R$  once by inserting every node type obtained by contracting a negative  $R$ -tree and redistributing a boundary multiindex  $m$  with  $|m|_{\mathfrak{s}} + |\tau|_{\mathfrak{s}} \leq 0$ , and then close downward under submultisets. Call this operation  $Q$ . The preceding finiteness shows that  $Q$  is well defined in every bounded window. Normality is preserved because deleting inputs commutes with the substitution. To check subcriticality, write the negative tree  $\tau$  as the product of planted branches  $\tau_1, \dots, \tau_r$  at its root, with trunk edge-types  $s_1, \dots, s_r$ . If  $M_i$  is an allowed replacement exposed by the  $i$ th branch, the same edge induction as in (25.2d) gives

$$\text{reg}(s_i) \leq |\tau_i|_{\mathfrak{s}} + \text{reg}(M_i).$$

If  $N$  is the original allowed node type and  $N_\tau = (s_1, \dots, s_r)$  is the node type occupied by the root of  $\tau$ , the substituted type  $N' = (N \setminus N_\tau) \sqcup M_1 \sqcup \dots \sqcup M_r$  therefore satisfies

$$\text{reg}(N') = \text{reg}(N) - \sum_i \text{reg}(s_i) + \sum_i \text{reg}(M_i) \geq \text{reg}(N) - |\tau|_s. \quad (25.2e)$$

Redistributing a total boundary derivative  $m$  replaces  $N'$  by a derivative node type of regularity  $\text{reg}(N') - |m|_s$ . Since completion only inserts such a type when  $|m|_s + |\tau|_s \leq 0$ , (25.2e) gives

$$\text{reg}(\partial^m N') \geq \text{reg}(N).$$

Thus adding these node types cannot decrease the infimum on the right-hand side of (25.2c), so the same  $\text{reg}$  and the same strict margin  $\kappa$  still work for  $QR$ .

Iterate  $Q$  and put  $\bar{R} = \bigcup_{j \geq 0} Q^j R$ . Increasing unions preserve normality and the same subcriticality inequality, and by construction  $Q\bar{R} = \bar{R}$ . Hence  $\bar{R}$  is complete. The edge bound (25.2d) remains uniform throughout the iteration, proving the final finiteness claim.  $\square$

From now on we silently replace a normal subcritical rule by its completion. This is not a new analytic assumption: it is the algebraic closure required so that contractions generated by renormalization remain inside the same rule-generated regularity structure.

**Definition 8.22** (Negative and positive extraction data). Let  $\tau = (F, \widehat{F}, \mathbf{n}, \mathbf{o}, \mathbf{e})$  be generated by the completed rule.

A *negative extraction datum* is a subforest  $A \subset F$  which contains the subforest already coloured 1, is disjoint from every component coloured 2, and whose newly selected connected components are rule-generated negative trees after their induced Taylor decorations are inserted. A *positive extraction datum* is a subforest  $P \subset F$  which contains the material already coloured 2 and which, in each connected component of  $F$ , is either empty or is connected and contains its root. Existing colour-1 components are treated as contracted vertices when testing this rooted condition.

For either datum, one chooses a vertex subdecoration  $\mathbf{n}_A \leq \mathbf{n}|_{N_A}$  and a boundary multiindex  $\varepsilon_A : \partial(A, F) \rightarrow \mathbb{N}^d$ . The *extracted factor*  $E_{A, \mathbf{n}_A, \varepsilon_A} \tau$  keeps the chosen subforest, places the selected polynomial decorations on it, and adds at each boundary vertex the Taylor multiindices entering from the complement. The *contracted factor*  $C_{A, \mathbf{n}_A, \varepsilon_A} \tau$  contracts each connected component of  $A$  to a coloured vertex, subtracts the chosen polynomial multiindices, transfers  $\varepsilon_A$  to the boundary-edge derivative decorations, and adds to the new extended decoration  $\mathbf{o}$  the total type and multiindex data of the contracted component. The coefficient  $c(A, \mathbf{n}_A, \varepsilon_A)$  is the product of the corresponding multinomial coefficients and factorial denominators from Taylor's formula.

**Lemma 8.23** (Extraction calculus and quotient stability). *For the completed rule-generated extraction families the following properties hold.*

1. *Extractions factor over disjoint unions.*
2. *Nested extraction is associative: extracting  $A$  and then the image of a larger  $B$  gives exactly the same decorated extracted/contracted pair and the same total coefficient as extracting  $B$  at once and recording the induced extraction inside  $B$ .*
3. *Negative extraction and positive rooted extraction satisfy a compatibility bijection: negative-first/positive-second data are in bijection with positive-first data together with compatible negative extractions in the two resulting factors.*

4. *The rule-completion relations, together with the negative and positive degree ideals, are coideals for the corresponding extraction maps. Hence the extraction maps descend to the quotient spaces used below.*

*Proof.* The first assertion is immediate because every connected extracted component belongs to exactly one component of a disjoint union. For nested extraction, the underlying graph after two contractions is the same quotient graph as for the single contraction. It remains to compare decorations. If a boundary multiindex  $k$  is split in two stages as  $k = k_1 + k_2 + k_3$ , then

$$\binom{k}{k_1} \binom{k - k_1}{k_2} = \frac{k!}{k_1! k_2! k_3!}, \quad (25.2f)$$

coordinatewise. Thus the Taylor coefficients agree. Vertex polynomial decorations add, edge derivatives add, and the extended decoration is additive because it records precisely the type/multiindex content removed by contraction. This proves (2).

For (3), begin with a negative extraction  $A$  and a rooted positive extraction  $P$  of the contracted forest. A connected component of  $A$  has exactly two possibilities: its contracted vertex lies in  $P$ , or it lies outside  $P$ . Taking inverse images partitions the negative data accordingly. Conversely, a rooted positive extraction of the original forest, together with negative extractions of the extracted and contracted factors that respect the existing colours, reconstructs the original pair. Formula (25.2f) again identifies all Taylor coefficients.

Finally, a contraction which would have left the original rule-generated space is exactly one of the substitutions inserted in the completion of Lemma 8.21; hence rule relations are stable under both coactions. The two degree projections are stable because (25.2a) ignores the coloured negative component while (25.2b), through  $\mathfrak{o}$ , restores the degree of material lost under negative contraction. Therefore the forbidden and wrong-degree spans are coideals. This proves (4).  $\square$

**Definition 8.24** (Negative and positive forest coactions). Let  $\widehat{\mathcal{T}}_-^{\text{ex}}$  be the completed rule-generated commutative algebra of colour-1 forests and let  $\widehat{\mathcal{T}}_+^{\text{ex}}$  be the corresponding rooted colour-2 algebra. Quotient by the respective degree ideals and denote the resulting spaces by  $\mathcal{T}_-^{\text{ex}}$  and  $\mathcal{T}_+^{\text{ex}}$ . The model space is denoted  $\mathcal{T}^{\text{ex}}$ .

Using Definition 8.22, define exactly

$$\Delta_- \tau = \sum_A \sum_{\mathbf{n}_A, \varepsilon_A} c(A, \mathbf{n}_A, \varepsilon_A) E_{A, \mathbf{n}_A, \varepsilon_A} \tau \otimes C_{A, \mathbf{n}_A, \varepsilon_A} \tau, \quad (25.3)$$

$$\Delta_+ \tau = \sum_P \sum_{\mathbf{n}_P, \varepsilon_P} c(P, \mathbf{n}_P, \varepsilon_P) C_{P, \mathbf{n}_P, \varepsilon_P} \tau \otimes E_{P, \mathbf{n}_P, \varepsilon_P} \tau. \quad (25.4)$$

The first sum is over negative extraction data and the second over rooted positive extraction data. Terms killed by the quotient projections are simply absent. Local finiteness from Lemma 8.21 makes these finite sums on every basis vector in a fixed cutoff.

**Lemma 8.25** (Hopf structures and twisted antipodes). *The positive coproduct makes  $\mathcal{T}_+^{\text{ex}}$  a connected graded Hopf algebra, while the negative coproduct makes  $\mathcal{T}_-^{\text{ex}}$  a connected graded Hopf algebra after the negative quotient. The same maps give the corresponding coactions on  $\mathcal{T}^{\text{ex}}$ . If  $\pi_-$  projects onto the strictly negative nonempty sector and  $\pi_+$  onto the positive non-polynomial sector, the recursive*

formulae

$$\tilde{\mathcal{A}}_- \tau = -\pi_- \left( \tau + \sum'_{(\tau)} \tilde{\mathcal{A}}_- (\tau^{(1)}) \tau^{(2)} \right), \quad (25.5)$$

$$\tilde{\mathcal{A}}_+ \tau = -\pi_+ \left( \tau + \sum'_{(\tau)} \tau^{(1)} \tilde{\mathcal{A}}_+ (\tau^{(2)}) \right) \quad (25.6)$$

define the negative and positive twisted antipodes. Here the primed sum is the reduced coproduct appropriate to the displayed coaction. Both recursions terminate.

*Proof.* Multiplicativity follows from factorization of extraction data over disjoint unions. Coassociativity is the nested-extraction assertion of Lemma 8.23, including its coefficient identity. The quotient-stability statement in that lemma shows that these maps are well-defined after passing to the positive and negative ideals.

Grade by the number of uncoloured edges, refined if necessary by total multiindex degree. Every nontrivial term in a reduced coproduct strictly lowers this grading in the factor on which the next recursive antipode is evaluated. The quotient bialgebras are therefore connected, so the usual antipode recursion terminates. Inserting  $\pi_-$  or  $\pi_+$  at each step yields the twisted antipodes (25.5)–(25.6). Induction on the grading proves the projected convolution identities. Algebraically, those identities say that a subtraction attached to a forest cancels not only the whole divergent component but also all nested divergent components with the correct inclusion–exclusion coefficients.  $\square$

**Lemma 8.26** (Negative–positive cointeraction). *Let  $M^{(13)(2)(4)}$  denote the map which multiplies the first and third negative factors and leaves the second and fourth factors in their original order. On  $\mathcal{T}^{\text{ex}}$  and on  $\mathcal{T}_+^{\text{ex}}$  one has*

$$M^{(13)(2)(4)}(\Delta_- \otimes \Delta_-)\Delta_+ = (\text{id} \otimes \Delta_+)\Delta_-. \quad (25.7)$$

Consequently, if a negative character acts on both the model and positive sectors, negative renormalization and positive reexpansion commute in the induced sense.

*Proof.* Expand the right-hand side by choosing a negative extraction first and then a rooted positive extraction of the contracted forest. By part (3) of Lemma 8.23, this is in bijection with choosing the positive extraction first and then compatible negative extractions in its two factors. Those two negative extracted factors are multiplied by  $M^{(13)(2)(4)}$ . The nested Taylor coefficient is unchanged by (25.2f). The  $\mathfrak{o}$ -decoration is essential here: it is precisely what makes the positive degree of the two descriptions agree after a negative component has been contracted. Summing the bijection proves (25.7). Applying characters to the indicated tensor factors gives the stated compatibility of actions.  $\square$

**Lemma 8.27** (Negative characters preserve admissible models). *Let  $G_-$  and  $G_+$  be the character groups of  $\mathcal{T}_-^{\text{ex}}$  and  $\mathcal{T}_+^{\text{ex}}$ . For  $g \in G_-$  define*

$$M_g = (g \otimes \text{id})\Delta_-.$$

*Let  $(\Pi, \Gamma)$  be a smooth admissible model and write its positive reexpansion through characters  $f_x \in G_+$ . Apply the negative coaction also to the positive sector and denote the induced positive character by  $f_x^g$ . Then*

$$\Pi_x^g = (\Pi M_g \otimes f_x^g)\Delta_+, \quad \Gamma_{xy}^g = (\text{id} \otimes (f_x^g)^{-1} \star f_y^g)\Delta_+$$

defines another smooth admissible model. The map  $g \mapsto M_g$  is a group action, is triangular in homogeneity, fixes polynomials, and respects the abstract integrations with their Taylor corrections.

*Proof.* Coassociativity of  $\Delta_-$  gives  $M_{g_1 \star g_2} = M_{g_1} M_{g_2}$ . A nontrivial negative contraction removes a strictly negative component; the homogeneity of the surviving symbol is therefore strictly larger, which proves triangularity.

Apply  $g$  to the negative tensor factors in (25.7). The resulting identity is exactly the statement that renormalization followed by positive reexpansion is the same as positive reexpansion with the induced positive character followed by renormalization. It yields the model identities  $\Pi_y^g = \Pi_x^g \Gamma_{xy}^g$  and  $\Gamma_{xz}^g = \Gamma_{xy}^g \Gamma_{yz}^g$ .

It remains to check the special admissibility identities. Pure polynomials have no nontrivial negative extraction, hence are fixed. For a planted integration symbol  $\mathcal{I}_k \tau$ , inspect the negative extraction in (25.3). An admissible negative component either lies entirely below the new trunk, in which case it is exactly an extraction of  $\tau$ , or it contains the new trunk/root. The second possibility is killed by the negative quotient for the planted kernel types used in the regularity structure. Hence, at the coaction level,

$$\Delta_- \mathcal{I}_k \tau = (\text{id} \otimes \mathcal{I}_k) \Delta_- \tau,$$

with the polynomial Taylor correction carried by the positive coproduct. After applying  $g$ , the kernel-convolution identity and the prescribed Taylor jet are therefore unchanged. These are precisely the admissibility relations for the smooth model.  $\square$

**Definition 8.28** (Stationary random lift and the negative expectation character). A smooth random admissible lift  $\Pi$  is *stationary* if spatial translations act simultaneously on its noise realization and on model base points without changing its law. In the source-strength algebraic BPHZ hypothesis, the model evaluations at the origin, and the finitely many base-point derivatives used to define the admissible lift, have moments of every order. (At any fixed homogeneity cutoff the proof below uses only finitely many of these moments.) Let  $B_\circ^\sharp$  be the finite set, in the chosen homogeneity cutoff, of connected negative generators that are not planted positive-type integration trees. The negative Hopf algebra is free commutative on these generators. Define  $g^-(\Pi)$  first on  $B_\circ^\sharp$  by

$$g^-(\Pi)(\iota_\circ \tau) = \mathbf{E}[(\Pi \tau)(0)]$$

and extend it multiplicatively to forests. No factorization of random expectations is asserted: multiplicativity is the definition of the character after its generator values have been fixed.

**Lemma 8.29** (BPHZ character, centering, and uniqueness). *For a stationary random lift as above set*

$$g_{\text{BPHZ}} = g^-(\Pi) \circ \tilde{\mathcal{A}}_-. \quad (25.8)$$

*Then for every negative basis symbol  $\tau$  and every base point  $z$ ,*

$$\mathbf{E}[(\Pi M_{g_{\text{BPHZ}}} \tau)(z)] = 0. \quad (25.9)$$

*Moreover  $g_{\text{BPHZ}}$  is the unique deterministic negative character whose renormalized stationary lift has property (25.9).*

*Proof.* First take a free generator  $\tau \in B_\circ^\sharp$  and  $z = 0$ . Evaluation of the negative coaction followed by expectation gives the convolution of  $g^-(\Pi)$  with the renormalization character. The defining twisted-antipode identity therefore yields

$$\mathbf{E}[(\Pi M_{g_{\text{BPHZ}}} \tau)(0)] = (g^-(\Pi) \circ \tilde{\mathcal{A}}_- \star g^-(\Pi))(\iota_\circ \tau) = 0. \quad (25.10)$$

A negative basis symbol not belonging to  $B_{\circ}^{\sharp}$  is a planted integration symbol  $\mathcal{I}_k\sigma$  of negative total degree. By the integration compatibility proved in Lemma 8.27,

$$\Delta_- \mathcal{I}_k \sigma = (\text{id} \otimes \mathcal{I}_k) \Delta_- \sigma.$$

Admissibility then represents its centered expectation at the origin as

$$\int D^k K(-y) \mathbf{E}[(\Pi M_{g_{\text{BPHZ}}} \sigma)(y)] dy. \quad (25.11)$$

Stationarity expresses the function under the integral as a finite polynomial combination of centered expectations of symbols obtained from  $\sigma$  by lowering vertex polynomial decorations. Terms whose polynomial degree is too small vanish against the kernel moments. Every remaining symbol is still of negative degree and has strictly fewer uncoloured kernel edges than  $\mathcal{I}_k\sigma$ . Induction on that number, starting from (25.10), makes (25.11) vanish. Thus (25.9) holds at the origin for all negative symbols. Since deterministic negative-character actions commute with translations by Lemma 8.26, the renormalized lift is stationary, and the same centering holds at every  $z$ .

For uniqueness, identify the negative Hopf algebra with the free commutative algebra on  $B_{\circ}^{\sharp}$ . If a nontrivial deterministic character  $h$  were to preserve the centering of the BPHZ lift, choose a generator  $\tau$  on which  $h$  is nonzero that is minimal first in negative degree and then in the number of uncoloured edges. Expand  $\Delta_- \tau$ . The full extraction gives the constant term  $h(\tau)$ . Every proper extraction either has a first factor on which  $h$  vanishes by minimality, or leaves a negative second factor whose expectation vanishes by (25.9). Taking expectation therefore leaves exactly  $h(\tau) \neq 0$ , contradicting centering. Hence  $h$  is the identity, proving uniqueness.  $\square$

**Theorem 8.30** (Algebraic BPHZ renormalization). *Let  $R$  be a normal locally subcritical rule and replace it by the normal subcritical completion  $\bar{R}$  of Lemma 8.21. The extended decorated forests generated by  $\bar{R}$  carry cointeracting negative and positive Hopf structures and their coactions on the model space. Negative characters act triangularly on the resulting regularity structure and send smooth admissible models to smooth admissible models. For every stationary random canonical lift satisfying the moment hypothesis of the preceding definition, the character  $g_{\text{BPHZ}}$  of Lemma 8.29 is the unique deterministic negative character such that*

$$\mathbf{E}[(\Pi M_{g_{\text{BPHZ}}} \tau)(z)] = 0 \quad (\tau \in B_{\circ}^-, z \in \mathbb{R}^d),$$

where  $B_{\circ}^-$  denotes the negative basis sector below the cutoff. Thus it produces the unique centered model in the negative-character orbit of the canonical lift.

All constructions are locally finite in homogeneity and compatible under enlargement of the cutoff. They therefore define the full algebraic BPHZ renormalization action for the locally subcritical rule. This theorem makes no claim about convergence of random models as a regularization is removed.

*Proof.* Lemma 8.21 supplies the complete rule required for contraction stability. Definitions 8.19 and 8.22, together with Lemma 8.23, construct the coloured/decorated quotient spaces and make the extraction operators well defined there. Lemma 8.25 gives the two Hopf structures and twisted antipodes. Lemma 8.26 gives their cointeraction, including the extended-decoration bookkeeping that is needed for the positive degree. Lemma 8.27 turns the negative character coaction into an action on admissible models and verifies the polynomial and integration identities. Lemma 8.29 then constructs the BPHZ character and proves centering and uniqueness.

Finally, local subcriticality bounds the number of edges and derivative multiindices below any fixed degree. Increasing a homogeneity cutoff therefore does not change any previously computed

extraction, coproduct, antipode, or character value; it merely adjoins finitely many new basis symbols. The finite-cutoff constructions are consequently compatible and exhaust the locally subcritical structure. The only missing ingredient for singular stochastic inputs is analytic control of the resulting random model, which is the separate root in the next section.  $\square$

**Algebraic proof scope.** The proof includes coloured decorated forests, the two exact degree functions, completion of a normal subcritical rule, extraction/contraction with Taylor decorations, quotient stability, both Hopf structures and twisted antipodes, negative–positive cointeraction, preservation of admissible models, and the stationary BPHZ centering/uniqueness argument. The forest/cut summability and cumulant estimates required for convergence are not consequences of this algebraic theorem; they are supplied separately in the analytic section which follows.

## 8.3 Analytic BPHZ convergence

The algebraic forest formula does not by itself imply convergence. The analytic problem is to show that the Taylor cancellations selected by the negative and positive coproducts can be harvested simultaneously at every scale, including in non-Gaussian moments. We isolate the analytic pieces of this argument. The first five are elementary multiscale/combinatorial statements; the sixth is the power-counting bottleneck; the seventh glues copies of a tree through cumulants; the eighth converts the resulting pointwise estimates into model convergence.

### 8.3.1 Analytic data and cumulant homogeneities

**Definition 8.31** (Kernel assignment and dyadic slices). For every integration type  $\mathfrak{t} \in \mathfrak{L}_+$  fix a compactly supported singular kernel  $K_{\mathfrak{t}}$  of order  $|\mathfrak{t}|_{\mathfrak{s}} > 0$ . A *dyadic kernel decomposition* is a representation

$$K_{\mathfrak{t}} = \sum_{n \geq 0} K_{\mathfrak{t}}^{(n)}$$

for which, for every multiindex  $k$  up to the finite order required by the chosen homogeneity cutoff,

$$\text{supp } K_{\mathfrak{t}}^{(n)} \subset \{z : \|z\|_{\mathfrak{s}} \leq C2^{-n}\}, \quad |D^k K_{\mathfrak{t}}^{(n)}(z)| \leq C_k 2^{n(|\mathfrak{s}| - |\mathfrak{t}|_{\mathfrak{s}} + |k|_{\mathfrak{s}})}. \quad (25.12)$$

The slices annihilate polynomials up to the order imposed by admissibility. Such a decomposition is obtained from a smooth annular partition of unity and Taylor subtraction at the origin.

**Definition 8.32** (Cumulant homogeneity and analytic noise norm). Let  $L$  be a finite set of noise leaves with type map  $\mathfrak{t} : L \rightarrow \mathfrak{L}_-$ . A clustering tree  $T$  on  $L$  is a rooted binary tree whose leaves are  $L$ ; an increasing scale labelling assigns an integer  $n_v$  to every internal vertex, with  $n_v \leq n_w$  whenever  $v$  is an ancestor of  $w$ . A *cumulant homogeneity* assigns numbers  $c_T(v)$  to the internal vertices, invariant under relabelling of equal noise types. It is *consistent* with the noise homogeneities when the following four conditions hold for every allowed cumulant block  $L$  and every clustering tree  $T$



on  $L$ :

$$\sum_{v \in T^\circ} c_T(v) = -|\mathbf{t}(L)|_s, \quad (25.13a)$$

$$\sum_{\substack{v \in T^\circ: \\ v \leq u \text{ for some } u \in A}} c_T(v) \geq -|\mathbf{t}(A)|_s \quad (A \subset L), \quad (25.13b)$$

$$\sum_{v \geq a} c_T(v) \leq -|\mathbf{t}(L_a)|_s \quad (a \in T^\circ), \quad (25.13c)$$

$$\sum_{v \geq a} c_T(v) < |s|(|L_a| - 1) \quad (|L| \geq 3, |L_a| \leq 3). \quad (25.13d)$$

Here  $L_a$  is the descendant leaf set of  $a$ . Thus (25.13b) controls the weight seen by an arbitrary subset, (25.13c) controls a whole cluster, and (25.13d) provides the strict small-cluster margin needed for non-Gaussian cumulants. We impose, in addition, the local-integrability conditions

$$|\mathbf{t}|_s > -|s| \quad (\mathbf{t} \in \mathfrak{L}_-), \quad |\mathbf{t}(B)|_s > (1 - |B|)|s| \quad (|B| \geq 3), \quad (25.13e)$$

for every allowed cumulant block  $B$ . These hypotheses are not cosmetic: they are exactly what makes singleton pieces created by a cumulant contraction integrable. Second cumulants are the only blocks which may require an additional local subtraction. We record its integer Taylor gain as

$$f(B) = \mathbf{1}_{\{|B|=2\}} \left( \lceil -|\mathbf{t}(B)|_s - |s| \rceil \vee 0 \right). \quad (25.13f)$$

Consequently every allowed block  $B$  satisfies the strict pair-or-cumulant integrability inequality

$$\left( |\mathbf{t}(B)|_{s,c,D} \wedge [f(B) + |\mathbf{t}(B)|_s] \right) + (|B| - 1)|s| > 0 \quad (25.13g)$$

for every finite external type set  $D$ . For  $|B| = 2$  this follows from the choice of  $f(B)$  (or from the stronger cumulant-homogeneity bound when available); for  $|B| \geq 3$  it follows from (25.13b), (25.13e). We will use (25.13g) repeatedly in the moment-gluing argument below.

For a stationary smooth noise  $\xi$  write  $\kappa_L(z_L) = \mathbf{E}_c[\xi_{\mathbf{t}(u)}(z_u) : u \in L]$ . We say  $\|\xi\|_{N,c} \leq M$  if all cumulants of order at most  $N$ , after localization to any clustering tree and differentiation up to the finite order needed below, satisfy the multiscale bounds dictated by  $c$ , with constant at most  $M$ . For two jointly defined noises  $\xi, \bar{\xi}$ ,  $\|\xi; \bar{\xi}\|_{N,c}$  is defined by the same bounds with at least one leaf replaced by  $\xi - \bar{\xi}$ .

**Definition 8.33** (External cumulant gains and super-regularity). Let  $A, D$  be finite leaf-typed sets. First define  $j_D(A)$  to be the minimum of

$$|\mathbf{t}(C)|_s + |C| |s| \quad (25.14a)$$

over nonempty leaf-typed sets  $C$  using only noise types occurring in  $D$  for which  $C \sqcup A$  admits a partition into allowed cumulant blocks and every block meets  $A$ ; if no such  $C$  exists set  $j_D(A) = +\infty$ . Next let  $|\mathbf{t}(A)|_{s,c,D}$  be the worst effective degree of  $A$  when its leaves are allowed to belong to an allowed cumulant together with additional leaves of types drawn from  $D$ : explicitly, take the infimum over all such extensions and all clustering trees of minus the  $c$ -weight above the cluster whose descendant set is exactly  $A$ . Put

$$h_{c,D}(A) = \begin{cases} 0, & A = \emptyset, \\ \min_{\emptyset \neq B \subset A} (|\mathbf{t}(B)|_{s,c,D} - |\mathbf{t}(B)|_s), & A \neq \emptyset. \end{cases} \quad (25.14b)$$

Thus  $h_{c,D}(A)$  is the smallest possible homogeneity gain caused by forcing at least one leaf of  $A$  into a cumulant with noise external to  $A$ .

A negative decorated tree  $\tau$  is *c-super-regular* if for every connected subtree  $S$  with more than one vertex,

$$|S|_{\mathfrak{s}} + \left( \frac{|S|}{2} \wedge h_{c,L(\tau)}(L(S)) \wedge j_{L(\tau)}(L(S)) \right) > 0. \quad (25.14c)$$

Here  $L(S)$  denotes the noise leaves contained in  $S$  and the minimum ignores an infinite entry. In the canonical choice of cumulant homogeneity for a Gaussian noise this reduces to the usual subtree super-regularity condition.

**Lemma 8.34** (Dyadic slicing and cumulant localization). *Fix a finite homogeneity cutoff. Kernel assignments satisfying the usual singular-kernel derivative and moment conditions admit a decomposition (25.12). Moreover every cumulant entering a moment of a basis tree can be partitioned into finitely overlapping regions indexed by clustering trees and increasing scale labels, on each of which the cumulant and all derivatives required by Taylor remainders obey the product bound prescribed by its cumulant homogeneity.*

*Proof.* Choose a smooth function  $\chi$  supported in the unit parabolic ball and equal to one on the ball of radius  $1/2$ . Put  $\chi_n(z) = \chi(2^{n5}z)$  and decompose a singular kernel into annuli with  $\chi_n - \chi_{n+1}$ . Subtract from each annular piece the finite Taylor jet whose moments must vanish. Rescaling gives (25.12), and the moment subtractions do not enlarge the support by more than a fixed factor.

For a cumulant in variables  $(z_u)_{u \in L}$ , construct the binary clustering tree by Kruskal's procedure: merge the two currently distinct clusters at the largest dyadic scale at which their mutual parabolic distance is  $O(2^{-n})$ . Tie breaking uses a fixed ordering of  $L$ . The resulting regions cover the configuration space with multiplicity bounded only by  $|L|$ . A smooth partition of unity subordinate to slightly enlarged regions turns this into an actual decomposition. On a region labelled by  $(T, n)$ , the definition of  $\|\xi\|_{N,c}$  is precisely the differentiated product estimate prescribed by  $c_T$ . Only finitely many orders and derivatives occur because the tree set below the cutoff is finite. This proves the assertion.  $\square$

### 8.3.2 Safe forests and compatible positive cuts

Let  $\tau$  be a fixed decorated tree, let  $\mathcal{F}_\tau$  denote its Zimmermann forests of divergent subtrees, and let  $n = (n_e)$  be a scale assignment to every kernel/cumulant edge in the multiscale representation. For  $S \in \mathcal{F}$  write  $A_{\mathcal{F}}(S)$  for the smallest member of  $\mathcal{F}$  properly containing  $S$ , or the ambient tree if there is none, and  $C_{\mathcal{F}}(S)$  for the children of  $S$  in  $\mathcal{F}$ . Define  $E_{\mathcal{F}}^{\text{int}}(S)$  to be the internal edges of  $S$  not already internal to one of its children, and  $E_{\mathcal{F}}^{\text{ext}}(S)$  the edges in the parent component touching  $S$ . Set

$$\text{int}_{\mathcal{F}}^n(S) = \min_{e \in E_{\mathcal{F}}^{\text{int}}(S)} n_e, \quad \text{ext}_{\mathcal{F}}^n(S) = \max_{e \in E_{\mathcal{F}}^{\text{ext}}(S)} n_e. \quad (25.14)$$

A subtree is *safe* when  $\text{int}_{\mathcal{F}}^n(S) \leq \text{ext}_{\mathcal{F}}^n(S)$ .

**Lemma 8.35** (Projection onto safe forests). *For*

$$P^n(\mathcal{F}) = \{S \in \mathcal{F} : \text{int}_{\mathcal{F}}^n(S) \leq \text{ext}_{\mathcal{F}}^n(S)\}, \quad (25.15)$$

*the internal and external scales of every member of  $\mathcal{F}$  are unchanged when computed relative to  $P^n(\mathcal{F})$ . Consequently  $P^n$  is idempotent and every fibre  $(P^n)^{-1}(S)$  is an interval  $[S, S \cup \mathcal{G}]$  in the forest poset.*

*Proof.* Removing an unsafe member  $T$  can only change the internal/external edge sets of an ancestor or a child. Follow an edge that would change status through the chain of removed ancestors. If it were internal before the removals and external afterwards, the successive unsafe inequalities would force a strict chain of scale inequalities in one direction. Following the same edge from the first surviving descendant to the first surviving ancestor forces the opposite inequality. Hence no extremizing internal or external scale can change, which proves the first claim.

Idempotence follows immediately. Fix now a projected forest  $\mathcal{S}$  and let  $\mathcal{G}$  consist of all divergent subtrees  $T$  compatible with  $\mathcal{S}$  for which  $\text{ext}_{\mathcal{S}}^n(T) < \text{int}_{\mathcal{S}}^n(T)$ . Two members of  $\mathcal{G}$  cannot overlap without being nested: otherwise choose an edge internal to the first and external to the second, and another with the reverse roles. The two strict unsafe inequalities would give  $n_{e_1} < n_{e_2} < n_{e_1}$ . Thus  $\mathcal{S} \cup \mathcal{G}$  is again a forest. If  $\mathcal{S} \subset \mathcal{F} \subset \mathcal{S} \cup \mathcal{G}$ , then every member of  $\mathcal{F} \setminus \mathcal{S}$  remains unsafe and every member of  $\mathcal{S}$  remains safe by the first part, so  $P^n(\mathcal{F}) = \mathcal{S}$ . Conversely, if  $P^n(\mathcal{F}) = \mathcal{S}$ , every removed member satisfies the strict unsafe inequality computed relative to  $\mathcal{S}$  and hence lies in  $\mathcal{G}$ . This proves the interval description.  $\square$

For a forest  $\mathcal{F}$ , delete from the ambient kernel graph all edges internal to members of  $\mathcal{F}$ . For vertices  $u, v$  define the bottleneck scale

$$n_{\mathcal{F}}(u, v) = \max_{\gamma: u \rightsquigarrow v} \min_{e \in \gamma} n_e, \quad (25.16)$$

the maximum being over paths in the remaining graph. For a positive Taylor-cut edge  $e$  with parent  $e_p$ , child  $e_c$ , and distinguished root  $\rho$ , set

$$G^n(\mathcal{F}) = \{e : n_{\mathcal{F}}(\rho, e_p) > n_{\mathcal{F}}(e_p, e_c)\}. \quad (25.17)$$

**Lemma 8.36** (Compatibility of the cut rule). *The rule  $G^n$  is compatible with the forest projection  $P^n$ : on each forest interval  $(P^n)^{-1}(\mathcal{S})$ , adding or deleting a cut selected by (25.17) does not change the interval of negative forests that contributes. Hence the simultaneous sum over negative forests and positive cut sets can be reorganized, for every fixed scale assignment, into finitely many products of one forest interval and one cut interval.*

*Proof.* Fix a forest interval  $[\mathcal{S}, \mathcal{S} \cup \mathcal{G}]$  and a candidate cut  $e$ . If inserting  $e$  does not change the maximal compatible forest, the claim is immediate. Suppose deleting  $e$  would enlarge that maximal forest by a subtree  $T$ . Then  $e$  is internal to  $T$ ,  $T$  lies in none of the surviving members of  $\mathcal{S}$ , and  $T$  is unsafe. Every path from  $\rho$  to  $e_p$  without entering a surviving forest must cross an external edge of  $T$ , while the edge  $e$  supplies a path from  $e_p$  towards  $e_c$  through the interior of  $T$ . Therefore

$$n_{\mathcal{S}}(\rho, e_p) \leq \text{ext}_{\mathcal{S}}^n(T) < \text{int}_{\mathcal{S}}^n(T) \leq n_{\mathcal{S}}(e_p, e_c),$$

so  $e \notin G^n(\mathcal{S})$ . Contrapositively, every selected cut can be inserted or removed without changing the negative-forest interval. Repeating one cut at a time proves the interval factorization.  $\square$

**Lemma 8.37** (Taylor gains harvested on the selected intervals). *On a scale assignment belonging to a fixed forest/cut interval, every dangerous negative subtree  $S$  with  $\text{int}^n(S) > \text{ext}^n(S)$  is renormalized, while a safe subtree is estimated without using its subtraction. Likewise every cut in  $G^n$  uses the positive Taylor remainder and every unselected cut is estimated before subtraction. If  $q_S$  is the subtraction order of a negative subtree and  $q_e$  the order of a positive Taylor jet, the harvested factors are respectively*

$$2^{-(q_S+1)(\text{int}^n(S)-\text{ext}^n(S))}, \quad 2^{-(q_e+1)(n_{\mathcal{F}}(\rho, e_p)-n_{\mathcal{F}}(e_p, e_c))}. \quad (25.18)$$

*Proof.* For the negative term write the integrand depending on the external variables as  $F(z)$  and the contracted subtree variables as  $z_0 + h$ . The twisted antipode replaces  $F(z_0 + h)$  by its Taylor remainder of order  $q_S$  at  $z_0$ . The integral form of the remainder contains  $q_S + 1$  powers of  $h$ . Internal localization gives  $|h|_{\mathfrak{s}} \lesssim 2^{-\text{int}^n(S)}$ , whereas differentiating the external factor at order  $q_S + 1$  costs at most  $2^{(q_S+1)\text{ext}^n(S)}$ . This is the first factor in (25.18). The positive remainder is identical with the base-point displacement in place of  $h$ ; the two relevant bottleneck scales are exactly those in (25.17). Compatibility from Lemma 8.36 ensures that harvesting one remainder does not remove the scale separation needed for another.  $\square$

### 8.3.3 The multiscale summation bottleneck

The preceding lemmas decide *which* cancellation is available at a given scale. What remains is a power-counting theorem showing that, after all such cancellations have been inserted, every scale label is summable.

**Lemma 8.38** (Fixed-scale coalescence integration). *Let  $X$  be a finite vertex set with one pinned vertex  $\star$ , and let  $T$  be a rooted binary coalescence tree with leaf set  $X$ . Assign an integer scale  $n_v$  to each internal vertex, increasing towards the leaves. Suppose a measurable integrand  $F_n$  is supported where, for every internal node  $v$ , all points indexed by its descendant set  $L_v$  have parabolic diameter at most  $C2^{-n_v}$ , and suppose*

$$|F_n(y)| \leq C_0 \prod_{v \in T^\circ} 2^{\varsigma_T(v)n_v}. \quad (25.19)$$

Then

$$\int |F_n(y)| dy_{X \setminus \{\star\}} \leq C_{T,C} C_0 \prod_{v \in T^\circ} 2^{(\varsigma_T(v) - |\mathfrak{s}|)n_v}. \quad (25.20)$$

The same estimate holds for differentiated integrands provided the derivative cost is included in  $\varsigma$ .

*Proof.* Choose recursively, for every nontrivial cluster  $L_v = L_{v_1} \sqcup L_{v_2}$ , one representative leaf  $r_i \in L_{v_i}$ . Keep the absolute coordinate of the pinned representative and use

$$z_v = y_{r_2} - y_{r_1}, \quad v \in T^\circ,$$

as the remaining coordinates, orienting the choice so that the pinned leaf is always retained in the branch containing it. Ordered from the leaves towards the root, this is a triangular linear change of variables with determinant  $\pm 1$ . The support condition gives  $\|z_v\|_{\mathfrak{s}} \lesssim 2^{-n_v}$ , whose volume is  $O(2^{-|\mathfrak{s}|n_v})$ . Integrating (25.19) in the  $z_v$  variables gives (25.20). Derivatives only alter the pointwise exponent and are handled in the same way.  $\square$

**Lemma 8.39** (Two-sided multiscale clustering summation). *In the setting of Lemma 8.38, let  $X_* \subset X$  be a nonempty set whose connecting hull in every admissible coalescence tree is connected. Sum over scale assignments for which the scales on this hull are at least  $r$ . Assume that the total homogeneity  $\varsigma$  has order*

$$\alpha := \sum_{v \in T^\circ} \varsigma_T(v) - (|X| - 1)|\mathfrak{s}| < 0, \quad (25.21)$$

and that for every proper cluster node  $a$

$$\sum_{v \geq a} \varsigma_T(v) < (|L_a| - 1)|\mathfrak{s}|, \quad (25.22)$$

while for every node  $a$  lying below the connecting hull of  $X_*$  for which the complementary part is nonempty,

$$\sum_{v \not\geq a} \varsigma_T(v) > |X \setminus L_a| |\mathfrak{s}|. \quad (25.23)$$

Then every adapted family satisfying the fixed-scale bound of Lemma 8.38 obeys

$$\sum_n \int |F_n(y)| dy_{X \setminus \{\star\}} \leq C C_0 2^{\alpha r}. \quad (25.24)$$

Polynomial factors in the scale labels are harmless if the strict inequalities (25.22)–(25.23) hold with a uniform positive margin.

*Proof.* After Lemma 8.38, the summand is a product of powers of two. Orient every internal edge of the coalescence tree towards the minimal connected hull of  $X_*$ . Relative to the reference level  $r$ , a free scale increment on an edge pointing away from the hull changes exactly the labels in one descendant cluster  $L_a$ . Its decay exponent is

$$(|L_a| - 1)|\mathfrak{s}| - \sum_{v \geq a} \varsigma_T(v) > 0$$

by (25.22). A free decrement on an edge pointing towards the hull changes the labels on the complementary side; its decay exponent is

$$\sum_{v \not\geq a} \varsigma_T(v) - |X \setminus L_a| |\mathfrak{s}| > 0$$

by (25.23). The common displacement of all hull labels from  $r$  has exponent  $-\alpha > 0$  by (25.21). Consequently the scale sum factors, after this edge increment parametrization, into finitely many geometric series. Their product is bounded by a constant times  $2^{\alpha r}$ . Strict margins absorb polynomial factors by decreasing each decay exponent slightly. There are only finitely many coalescence-tree shapes on  $X$ , so the constant can be chosen uniformly over the shapes.  $\square$

**Lemma 8.40** (One-copy small-scale power counting). *Fix one forest/cut interval in the expansion of a negative basis tree  $\tau$ . Let  $X$  be the resulting one-copy vertex set,  $\star$  its pinned test-function vertex, and  $L \subset X \setminus \{\star\}$  its noise leaves. For a cluster  $A = L_a$  in a coalescence tree, let  $L_A^{\text{ext}} \subset L \cap A$  be the noise leaves whose cumulant block is not wholly contained in  $A$ . If  $\varsigma^{(1)}$  denotes the deterministic one-copy total homogeneity after all safe-forest and compatible-cut Taylor gains have been inserted, then*

$$\begin{aligned} \sum_{v \geq a} \varsigma_T^{(1)}(v) &< (|A| - 1)|\mathfrak{s}| + |\mathfrak{t}(L_A^{\text{ext}})|_{\mathfrak{s}} \\ &+ \mathbf{1}_{\{\star \notin A\}} \left( h_{c,L}(L_A^{\text{ext}}) \wedge j_L(L_A^{\text{ext}}) \wedge \frac{|\mathfrak{s}|}{2} \right). \end{aligned} \quad (25.22a)$$

The strict margin can be chosen uniformly over all trees below a fixed homogeneity cutoff.

*Proof.* We give the cluster bookkeeping, since this is the first place where merely saying “subcriticality” hides a real argument. Contract every negative forest component already assigned to the interval. The remaining kernel graph is a rooted tree with some planted branches. For a connected vertex set  $M$  write  $\partial M$  for its unique edge towards the root (when it exists) and write  $M^{\text{ext}}$  for its external

noise leaves. Collecting all kernel orders, vertex integration dimensions, noise degrees and harvested Taylor orders defines the *small-scale defect*

$$\Delta_-(M) := [\text{homogeneity carried by } M] - (|M| - 1)|\mathfrak{s}| - |\mathfrak{t}(M^{\text{ext}})|_{\mathfrak{s}}. \quad (25.22b)$$

Because all decorations are additive,  $\Delta_-$  is additive over kernel-connected components, except for the single boundary contribution of  $\partial M$ . A negative Taylor subtraction contributes its full remainder order when the corresponding dangerous subtree is contained in  $M$ ; a positive cut contributes its remainder order precisely when the cut crosses  $\partial M$ . These are exactly the gains in (25.18). Internal cumulant blocks contribute their localized  $c$ -weight, while a block crossing the boundary contributes the effective subset degree from (25.14b); a second cumulant also contributes the correction (25.13f). Thus (25.13a)–(25.13g) account for every cumulant term exactly once, and no subtraction or cumulant reserve is counted twice.

We prove the required estimate by induction on the number of original vertices of a kernel-connected component.

1. *No pin and no external cumulant leaf.* If  $\star \notin M$  and  $M^{\text{ext}} = \emptyset$ , every component is an ordinary renormalized subtree. If it lies inside an unsafe negative component, the subtraction order was chosen strictly beyond its superficial degree and (25.18) gives  $\Delta_-(M) < 0$ . If it meets no such component, completeness of the subcritical rule says that the planted branch below its highest vertex has strictly positive renormalized homogeneity; after moving this branch term to the other side this is again  $\Delta_-(M) < 0$ . For disconnected  $M$ , sum over its connected components; one component contains the highest vertex and keeps the strict sign.

2. *No pin but external cumulant leaves.* Suppose  $\star \notin M$  and  $M^{\text{ext}} \neq \emptyset$ . If  $M$  is kernel-disconnected, apply Step 1 to components without external leaves and the present argument to the remaining components, and add. We may therefore assume that  $M$  is connected. Let  $S(M)$  be the smallest connected subtree of  $\tau$  containing  $M$ . The raw branch defect is the negative of  $|S(M)|_{\mathfrak{s}}$ . Forcing the external leaves to join cumulants outside  $M$  can improve this by at most  $h_{c,L}(M^{\text{ext}})$ , while completing those cumulants by extra leaves costs at least  $j_L(M^{\text{ext}})$ . A genuine merger of two branches also carries the universal reserve  $|\mathfrak{s}|/2$ . Hence

$$\Delta_-(M) < h_{c,L}(M^{\text{ext}}) \wedge j_L(M^{\text{ext}}) \wedge \frac{|\mathfrak{s}|}{2}, \quad (25.22c)$$

and the strict sign is exactly (25.14c) applied to  $S(M)$ . This is (25.22a) for clusters avoiding the pin.

3. *Components meeting the pin.* Let  $M$  contain  $\star$  and decompose  $M \setminus \{\star\}$  into its maximal kernel-connected branches  $M_1, \dots, M_q$ . For a terminal branch whose boundary edge is not a selected positive cut, the residual after adding that edge is bounded by  $-|\mathfrak{s}| - |\mathcal{B}|_{\mathfrak{s}}$ , where  $\mathcal{B}$  is the planted branch below the edge. This is strictly negative by super-regularity, or by (25.13e) when  $\mathcal{B}$  is a single noise leaf. If the boundary edge is selected, its Taylor jet is taken through the largest polynomial degree compatible with the branch, so the remainder in (25.18) leaves a strict negative residual. When the relevant degree is integral, the strictness comes from the super-regularity/local-integrability gap; we do not invent a fractional gain.

For the induction step remove the highest contracted negative subtree, or the highest vertex when there is none. What remains is a collection of smaller branches. Insert their inductive estimates. Every internal boundary-edge term cancels exactly once with the planted-branch term of its parent; this is just additivity of the decorated-tree homogeneity. Only the boundary term of the original branch remains, and the terminal calculation makes it strictly negative. Summing the branch estimates gives  $\Delta_-(M) < 0$ , which is stronger than (25.22a) for a cluster containing the pin.

Below a finite homogeneity cutoff only finitely many branch homogeneities, cumulant types and Taylor orders occur. The minimum of the positive gaps in the preceding three cases is therefore a uniform margin.  $\square$

**Lemma 8.41** (One-copy large-scale power counting). *In the setting of Lemma 8.40, let  $a$  lie below the connected hull of the pinned/test edges and assume  $X \setminus L_a \neq \emptyset$ . Then*

$$\sum_{v \not\geq a} \varsigma_T^{(1)}(v) > |X \setminus L_a| |\mathfrak{s}| + |\mathfrak{t}(L \setminus L_a^{\text{ext}})|_{\mathfrak{s}}. \quad (25.23a)$$

Again the margin is uniform below a fixed cutoff.

*Proof.* Set  $M = X \setminus L_a$  and split  $M$  into maximal kernel-connected components. The complementary defect is additive over these components, so treat one component  $C$ . It has a unique incoming kernel edge  $e$ . Collapse all lower contracted subtrees and expose the remaining vertices from the leaves toward the highest vertex. Every internal kernel order is paired once with the homogeneity of the planted branch below it, each integrated vertex contributes one factor  $-|\mathfrak{s}|$ , and the internal Taylor terms cancel between a child and its parent. Thus the only uncanceled deterministic contribution is the incoming edge. In particular the complementary defect has the form

$$\Delta_+(C) = B_e + q_e \mathbf{1}_{\{e \text{ selected}\}} + R_C, \quad (25.23b)$$

where  $B_e$  is the incoming-edge/branch balance,  $q_e$  is its positive Taylor remainder order, and  $R_C \geq 0$  is the cumulant-extension reserve.

If  $e$  is unselected, completeness and subcriticality say precisely that the planted branch below  $e$  has homogeneity strictly better than the singular order of  $e$ ; equivalently  $B_e > 0$ . For a one-noise branch this is (25.13e). If  $e$  is selected,  $q_e$  is the first Taylor order beyond the branch degree, so  $B_e + q_e > 0$ . Hence  $\Delta_+(C) > 0$  in either case. Adding the components of  $M$  proves (25.23a). Finiteness of the cutoff again makes the strict margin uniform.  $\square$

**Lemma 8.42** (Cumulant gluing preserves the clustering inequalities). *Take  $2p$  copies of the one-copy integrand and identify their pinned vertex. After internal BPHZ cumulants have been absorbed into the one-copy homogeneities, expand the remaining moment into cumulant partitions whose blocks meet at least two copies. Push the  $2p$  one-copy homogeneities to a common coalescence tree, add the cumulant homogeneities and the pair correction (25.13f), and call the result  $\tilde{\zeta}$ . Then*

$$\sum_{v \geq a} \tilde{\zeta}_T(v) < (|L_a| - 1) |\mathfrak{s}|, \quad (25.22d)$$

$$\sum_{v \not\geq a} \tilde{\zeta}_T(v) > |V \setminus L_a| |\mathfrak{s}| \quad (25.23c)$$

for the clusters required by Lemma 8.39. Moreover

$$\sum_{v \in T^\circ} \tilde{\zeta}_T(v) - (|V| - 1) |\mathfrak{s}| = 2p |\tau|_{\mathfrak{s}}. \quad (25.23d)$$

*Proof.* Let  $\alpha_\tau$  denote the deterministic one-copy order. By the definition of decorated-tree homogeneity,

$$\alpha_\tau - |\mathfrak{t}(L)|_{\mathfrak{s}} = |\tau|_{\mathfrak{s}}. \quad (25.23e)$$

The cumulant homogeneities in a partition sum to the homogeneity of all noise leaves by (25.13a). Therefore the noise term in (25.23e) cancels after  $2p$  copies are glued, and identifying their common pinned vertex gives (25.23d).

For the small-scale inequality fix a cluster  $A = L_a$  which avoids the common pin. For copy  $j$  let  $A_j$  be its vertices in  $A$  and put

$$J_1 = \{j : |A_j| = 1\}, \quad J_2 = \{j : |A_j| \geq 2\}.$$

The possible configurations are exhausted as follows.

(i)  $J_2 = \emptyset$ . The cluster contains only singleton pieces from individual copies, so it is connected entirely by cumulants. Blocks of size at least three have the strict local-integrability reserve (25.13e); a pair has the reserve supplied by  $f(B)$ . Summing (25.13g) over the blocks meeting  $A$  gives (25.22d).

(ii)  $J_1 = \emptyset$  and  $|J_2| = 1$ . Apply (25.22a) to the unique active copy. Cumulant consistency (25.13b)–(25.13c) cancels the external-noise term. If a cumulant crosses the cluster boundary it consumes no more than the  $h$ -reserve in (25.22a); if no block crosses, the external set is empty. The remaining inequality is strict and has volume  $(|A| - 1)|\mathfrak{s}|$ .

(iii)  $|J_2| = 1$  and  $J_1 \neq \emptyset$ . Every singleton copy is connected to the active copy through a crossing cumulant. The effective homogeneity change is controlled by  $h_{c,L}$  and the cost of completing the cumulants by  $j_L$ . Their minimum, already present in (25.22a), absorbs all singleton contributions, leaving a strict reserve.

(iv)  $|J_2| \geq 2$ . Apply (25.22a) to every active copy. The universal  $|\mathfrak{s}|/2$  reserve in each active copy, together with (25.13g) for singleton pieces, absorbs all cross-copy cumulant costs. Since  $\sum_{j \in J_2} |A_j| = |A| - |J_1|$ , the volume terms sum to at most  $(|A| - 1)|\mathfrak{s}|$ , and the presence of at least two active copies keeps the inequality strict.

If  $A$  contains the common pin, apply the pinned form of (25.22a) in every active copy and cancel external-noise terms by cumulant consistency. The identification of the  $2p$  pins removes exactly the duplicated volume factors, again leaving the strict bound  $(|A| - 1)|\mathfrak{s}|$ .

For the large-scale inequality, let  $Q_j$  be the vertices of copy  $j$  outside  $L_a$  and discard the empty ones. Apply (25.23a) to every nonempty  $Q_j$ . The only unmatched terms are the homogeneities of leaves whose cumulant block crosses the boundary. The subset consistency (25.13b) bounds the cumulant homogeneity on the complementary side by the negative of exactly these terms, so they cancel after summing over copies. At least one  $Q_j$  is nonempty and its one-copy estimate is strict; hence (25.23c) follows.

For fixed  $p$  and cutoff, the list of intersection patterns and cumulant types is finite, so all strict gaps admit a common positive lower bound.  $\square$

**Lemma 8.43** (Renormalized total-homogeneity inequalities). *Fix a complete subcritical rule, a consistent cumulant homogeneity satisfying (25.13e), and a  $c$ -super-regular negative tree  $\tau$ . For every forest/cut interval and every cumulant partition occurring in an even moment of the BPHZ evaluation, the glued total homogeneity satisfies the strict clustering conditions (25.22)–(25.23), with a uniform margin below a fixed cutoff, and has order  $2p|\tau|_{\mathfrak{s}}$  after  $2p$  copies are glued by cumulants.*

*Proof.* Lemma 8.40 proves the one-copy small-scale inequality with the external-cumulant correction needed before moment gluing. Lemma 8.41 proves the complementary large-scale inequality. Lemma 8.42 cancels the external corrections in the moment contraction and gives the global conditions (25.22)–(25.23), together with the exact  $2p$ -copy order. No scale-summation result is used in these three proofs, so the dependency is acyclic.  $\square$



**Proposition 8.44** (Single-tree multiscale moment bound). *For every negative basis tree  $\tau$ , test function  $\varphi_x^\lambda$ , and  $p \in \mathbb{N}$ ,*

$$\mathbf{E} \left| \langle \Pi_x^{\text{BPHZ}} \tau, \varphi_x^\lambda \rangle \right|^{2p} \leq C_{\tau,p} \|\xi\|_{N,c}^{2p} \lambda^{2p|\tau|_s}, \quad (25.25)$$

*with  $N = 2p$  times the number of noise leaves of  $\tau$ . If one noise in the joint construction is replaced by  $\xi - \bar{\xi}$ , the right-hand side gains one factor  $\|\xi; \bar{\xi}\|_{N,c}$ .*

*Proof.* Use the moment–cumulant formula to expand the  $2p$ -th moment as a finite sum over partitions of the  $2p$  copies of all noise leaves. Localize every kernel by Lemma 8.34; localize each cumulant block by its clustering-tree partition. Lemmas 8.35 and 8.36 reorganize, at each scale assignment, the BPHZ forest and cut sum into compatible intervals. Lemma 8.37 provides the factors (25.18). Lemma 8.42 supplies the cumulant-gluing power count, and Lemma 8.43 records its global consequence. Hence the remaining scale sum is of the form covered by Lemma 8.39; its root is pinned at  $r \simeq -\log_2 \lambda$ , so (25.20) gives the power  $\lambda^{2p|\tau|_s}$ . The number of tree shapes, partitions, forest intervals, and cut intervals is finite for fixed  $\tau$  and  $p$ .

For a difference, multilinearity gives a telescoping sum in which at least one noise leaf is replaced by  $\xi - \bar{\xi}$ . In every corresponding cumulant partition one block therefore carries the mixed norm  $\|\xi; \bar{\xi}\|_{N,c}$ ; all other blocks are controlled by the larger of the two ordinary noise norms. The same scale summation proves the difference estimate.  $\square$

### 8.3.4 From tree estimates to models

**Proposition 8.45** (Model-norm upgrade and approximation continuity). *Assume the single-tree estimate (25.25) and its difference version for all negative symbols below a fixed cutoff. Lower every model homogeneity by a small  $\kappa > 0$ . Then, for every compact parabolic set  $K$  and finite  $p$ ,*

$$\mathbf{E} \|Z^{\text{BPHZ}}\|_{K,\kappa}^p \leq C_{K,p} (1 + \|\xi\|_{N,c})^C, \quad (25.26)$$

*and two jointly defined noises satisfy the analogous local Lipschitz estimate in the mixed cumulant norm. Consequently any uniformly bounded family of smooth noises which is Cauchy in the weakened cumulant topology yields a Cauchy family of BPHZ models in  $L^p$ ; convergence in law of the noises gives convergence in law of the models by coupling/diagonalization.*

*Proof.* There are only finitely many negative symbols below the cutoff. Apply (25.25) to a dyadic grid of base points and test scales. The factor  $\lambda^\kappa$  gained by lowering the target homogeneity makes the sum over grid scales summable after Markov’s inequality. Positive reexpansion coefficients are evaluations of the same finitely many planted symbols and therefore satisfy identical estimates. The algebraic identities  $\Pi_y = \Pi_x \Gamma_{xy}$  and  $\Gamma_{xz} = \Gamma_{xy} \Gamma_{yz}$  transfer the grid estimates to arbitrary base points. This is the usual Kolmogorov/chaining argument for the local model metric and yields (25.26).

Apply the same argument to the difference estimate to obtain a local Lipschitz bound in  $\|\xi; \bar{\xi}\|_{N,c}$ . A common coupling therefore gives an  $L^p$ -Cauchy sequence of models whenever the mixed cumulant distance tends to zero. This already yields approximation independence inside the cumulant metric class: interlace two smooth approximation sequences and apply the same mixed estimate to consecutive terms.

For convergence in law no moment-determinacy assertion is needed. The space of distributions on a fixed compact set and the local model space are Polish in the metrizable topologies used here. Whenever a sequence can be coupled so that the noises converge in probability and the mixed cumulant distance tends to zero, the preceding estimate gives convergence in probability of the

models. If only convergence in law is given but the hypotheses permit a Skorokhod coupling with the same uniform cumulant bounds, apply the probability statement on that coupling and then forget the coupling. Thus the law statement is a consequence of the coupled estimate, not of an unproved assertion that moments determine the model law.  $\square$

**Lemma 8.46** (Uniform mollification gain and distributional continuity). *Fix the finite cutoff and assume that there is a number  $\vartheta > 0$  such that all super-regularity and power-counting inequalities used above remain strict after every noise homogeneity is lowered by  $\vartheta$  (kernel homogeneities are left unchanged). Denote this penalised assignment by  $c_{-\vartheta}$ . Let  $\rho_\delta$  be a smooth unit-mass parabolic mollifier and write  $\xi^\delta = \rho_\delta * \xi$ .*

*Suppose that  $\xi$  is a stationary random distribution and, for the finite cumulant order  $N$  required by the cutoff,*

$$\sup_{0 < \delta \leq 1} \|\xi^\delta\|_{N, c_{-\vartheta}} \leq M.$$

*Then for every  $0 < \vartheta' < \vartheta$ , every finite  $p$ , every compact parabolic set  $K$ , and every sufficiently small model-homogeneity loss  $\kappa > 0$ ,*

$$\mathbf{E} \|Z_{\text{BPHZ}}(\xi^\delta) - Z_{\text{BPHZ}}(\xi^{\bar{\delta}})\|_{K, \kappa}^p \leq C_{K, p, M} (\delta \vee \bar{\delta})^{p\vartheta'}. \quad (25.26a)$$

*Consequently the mollified BPHZ models have a limit, denoted  $Z_{\text{BPHZ}}(\xi)$ , independent of the mollifier.*

*More generally, let  $\xi_n \rightarrow \xi$  in probability in the product distribution topology of the finitely many noise types, and assume the preceding penalised cumulant bound uniformly for  $\xi_n$  and  $\xi$ . Then*

$$Z_{\text{BPHZ}}(\xi_n) \longrightarrow Z_{\text{BPHZ}}(\xi)$$

*in probability in every slightly weakened local model topology. If the noise convergence is only in law, the same conclusion holds in law.*

*Proof.* For two mollification scales, expand every difference of BPHZ evaluations by multilinearity exactly as in the difference part of Proposition 8.44. In each term at least one noise leaf carries the multiplier  $\widehat{\rho}(\delta\zeta) - \widehat{\rho}(\bar{\delta}\zeta)$ . On a dyadic block of parabolic scale  $2^{-j}$  this multiplier is bounded, for every  $0 < \vartheta' < \vartheta$ , by

$$C \min\{1, (2^j(\delta \vee \bar{\delta}))^{\vartheta'}\}.$$

On blocks coarser than the mollification scale this is the desired explicit gain. On finer blocks we spend  $\vartheta'$  of the reserve created by replacing  $c$  with  $c_{-\vartheta}$ . Since every clustering inequality had a strict margin before this expenditure, the forest/cut summation of Lemmas 8.43 and 8.39 is unchanged except for the global factor  $(\delta \vee \bar{\delta})^{\vartheta'}$ . The model-norm upgrade in Proposition 8.45 gives (25.26a). Completeness of the local model space gives the mollifier-independent limit.

Now suppose  $\xi_n \rightarrow \xi$  in probability as distributions. Fix  $\delta > 0$ . Convolution by  $\rho_\delta$  is continuous from distributions to smooth functions on compact sets, so every finite family of derivatives of  $\xi_n^\delta$  converges in probability to the corresponding family for  $\xi^\delta$ . The uniform penalised cumulant bounds imply uniform integrability of all moments used below the cutoff. Hence the mixed cumulants of the smooth noises converge, and the local Lipschitz estimate in Proposition 8.45 yields

$$Z_{\text{BPHZ}}(\xi_n^\delta) \longrightarrow Z_{\text{BPHZ}}(\xi^\delta) \quad \text{in probability}$$

for fixed  $\delta$ . Insert these two smooth models between  $Z_{\text{BPHZ}}(\xi_n)$  and  $Z_{\text{BPHZ}}(\xi)$ ; (25.26a) controls the outer two terms uniformly in  $n$ . First let  $n \rightarrow \infty$  and then  $\delta \downarrow 0$ . This proves convergence in probability.

For convergence in law, use a Skorokhod representation on each metrizable local distribution space, apply the probability statement there, and then forget the auxiliary coupling. A diagonal exhaustion of compact sets gives the asserted local-model convergence in law.  $\square$

**Theorem 8.47** (Analytic BPHZ theorem). *Fix a complete locally subcritical rule, a finite homogeneity cutoff, a consistent cumulant homogeneity, and assume that every negative basis tree below the cutoff is super-regular in the sense of Definition 8.33. Let  $(\xi_\varepsilon)$  be smooth stationary approximations with uniformly bounded analytic cumulant norms of the finite order required by the cutoff.*

*The associated BPHZ models have uniform moments in every slightly weakened local model topology. If the approximating noises converge in the corresponding cumulant topology, then the BPHZ models converge in probability under a common coupling and in law without one, and the limit is independent of the smooth approximation.*

*The same conclusion holds for stationary random distributions under the source-strength continuity hypothesis of Lemma 8.46: namely, a uniform penalised cumulant bound providing a positive homogeneity margin together with convergence of the noises in the product distribution topology. Thus, within the finite cutoff, the BPHZ lift extends continuously from smooth stationary noises to this distributional class.*

*Proof.* The algebraic BPHZ theorem in the preceding section gives the exact forest/cut expansion. Lemmas 8.34–8.37 provide the multiscale localization and all renormalization gains. Lemma 8.43, together with the geometric summation Lemma 8.39, then yields the single-tree and difference bounds of Proposition 8.44. Proposition 8.45 upgrades these finitely many estimates to the local model metric and proves continuity and approximation independence for smooth cumulant-Cauchy families. Lemma 8.46 uses the penalised homogeneity reserve to obtain a uniform mollification gain and then a diagonal argument, giving the source-strength distributional continuity clause. No deterministic SPDE fixed-point theorem is used in this analytic step.  $\square$

**Analytic BPHZ proof inventory.** The power-counting proof is split into three independent owners: the one-copy small-scale inequality (Lemma 8.40), the complementary large-scale inequality (Lemma 8.41), and the genuinely stochastic  $2p$ -copy cumulant-gluing step (Lemma 8.42). The local-integrability hypotheses for noise types and higher cumulants, together with the pair-cumulant Taylor correction, are explicit in (25.13e)–(25.13g). These estimates supply the hypotheses used by the analytic BPHZ convergence theorem below; they do not by themselves prove any later SPDE fixed-point statement.

## 8.4 Dynamical $\Phi_3^4$

Consider on the three-dimensional torus

$$(\partial_t - \Delta)\Phi = -\Phi^3 + m\Phi + \xi. \quad (25.7)$$

Parabolic scaling gives  $|\Xi| = -5/2 - \kappa$ ; local subcriticality leaves only finitely many negative trees.

**Theorem 8.48** (Local  $\Phi_3^4$  construction). *Fix  $m \in \mathbb{R}$ , choose  $\alpha := |\Xi|_s \in (-18/7, -5/2)$ , and let  $\Phi_0 \in \mathcal{C}^\eta(\mathbb{T}^3)$  with  $\eta > -2/3$ . Let  $Z = (\Pi, \Gamma)$  be an admissible renormalized model for the corresponding  $\Phi_3^4$  regularity structure. Assume, in addition, that the reconstructed noise  $\xi := \mathcal{R}\Xi$  belongs locally to  $\mathcal{C}_s^{\bar{\alpha}}$  for some  $\bar{\alpha} < \alpha$  arbitrarily close to  $\alpha$ , and that the heat convolution  $K * \xi$  has continuous positive-time trajectories in the spatial Hölder–Besov class used for the initial-data exponent below. (BPHZ models constructed from space–time white noise satisfy these hypotheses.)*

*Then there is a strictly positive maximal lifetime  $T_*$  and a unique modelled-distribution fixed point  $U$  on  $[0, T_*)$  satisfying*

$$U = \mathcal{P}\Phi_0 + \mathcal{K}(-U^3 + mU + \Xi).$$

Its reconstruction  $\Phi = \mathcal{R}U$  is the unique local renormalized mild solution associated with  $Z$ . For every  $t > 0$  before  $T_*$  and every  $\bar{\alpha} < -1/2$  sufficiently close to  $-1/2$ , one has  $\Phi_t \in \mathcal{C}^{\bar{\alpha}}(\mathbb{T}^3)$ , with the standard weighted behaviour at  $t = 0$ . On every interval strictly below the common lifetime, the solution map is locally Lipschitz in the initial condition and in the admissible model (including the two restart norms just stated).

*Proof.* We verify the exact deterministic interfaces rather than citing “local subcriticality” as a black box.

### 1. Choice of solution and initial-data exponents. Put

$$\zeta := \alpha + 2 \in (-4/7, -1/2).$$

If the given initial exponent happens to exceed  $\zeta$ , lower it without changing the datum: choose

$$\eta_* \in (-2/3, \min\{\eta, \zeta\}). \quad (25.7a)$$

On the compact torus  $\mathcal{C}^\eta \hookrightarrow \mathcal{C}^{\eta_*}$ . Choose furthermore  $\gamma > |2\alpha + 4|$  away from the discrete homogeneities. The solution sector generated by integrated noise has regularity  $\zeta$ . For the cubic nonlinearity the product theorem gives

$$U \mapsto U^3 : \quad \mathcal{D}^{\gamma, \eta_*}(V) \longrightarrow \mathcal{D}^{\gamma+2\alpha+4, 3\eta_*}(\bar{V}), \quad (25.7b)$$

locally Lipschitz on bounded sets. The lower-order term  $mU$  satisfies the same target bound after decreasing the target exponent if necessary.

**2. The subcritical inequalities.** After heat integration the right-hand side gains two parabolic degrees. The three inequalities needed by the weighted fixed-point theorem reduce to

$$\eta_* < 3\eta_* + 2, \quad \gamma < \gamma + 2\alpha + 6, \quad 3\eta_* > -2. \quad (25.7c)$$

They are respectively equivalent to  $\eta_* > -1$ ,  $\alpha > -3$ , and  $\eta_* > -2/3$ , all strict by (25.7a) and the chosen interval for  $\alpha$ . Moreover  $3(\alpha + 2) > -2$  follows from  $\alpha > -8/3$ , which is implied by  $\alpha > -18/7$ . Thus the cubic map, heat integration and the time-zero weight close on the same modelled-distribution ball. This also shows directly why only finitely many symbols lie below any fixed cutoff: each nonlinear Picard step replaces a solution-degree  $r$  by  $3r + 2$ , and near the fixed solution degree  $\zeta > -1$  this operation strictly increases degree. The narrower lower bound  $-18/7$  is the conventional window in which the finite negative-symbol list used for the  $\Phi_3^4$  renormalisation structure is independent of the precise choice of  $\alpha$ .

**3. Contraction and reconstruction.** Chapter 3, SMX-25.03.02, owns the weighted Schauder estimate and short-time contraction for an admissible model. Apply it to

$$\mathcal{T}(U) = \mathcal{P}\Phi_0 + \mathcal{K}(-U^3 + mU + \Xi).$$

By (25.7b)–(25.7c), on a ball of radius  $R$  there are  $C, \theta > 0$  such that

$$\|\mathcal{T}(U) - \mathcal{T}(V)\|_{\gamma, \eta_*; T} \leq CT^\theta \|U - V\|_{\gamma, \eta_*; T}.$$

For  $T$  small this is a contraction and maps the ball to itself. Reconstruction and the identity  $\mathcal{R}KF = K * \mathcal{R}F$  from SMX-25.03.02 give the mild renormalized equation. The two-model estimate gives local Lipschitz dependence on the datum and model.

**4. Maximal continuation and restart.** The additional assumptions on  $\xi$  and  $K * \xi$  are precisely the norms needed to restart the weighted problem away from the singular time slice. At any

$s > 0$  below the current lifetime, the stochastic convolution is an admissible new inhomogeneous term and the reconstructed solution is a valid initial datum for the same contraction argument. Concatenating these local intervals gives a maximal solution on  $[0, T_*)$ ; if  $T_* < \infty$ , the relevant weighted modelled-distribution / reconstructed spatial norm must diverge as  $t \uparrow T_*$ . The same restart argument for two solutions proves uniqueness and the claimed stability up to every common subcritical time.

For the probabilistic  $\Phi_3^4$  model, SMX-25.03.03–04 supplies the canonical BPHZ admissible model and its approximation continuity. This is an input for constructing the random model, not a hidden step in the deterministic fixed-point argument above.  $\square$

#### 8.4.1 The Mourrat–Weber dissipative package

The local regularity-structure theorem does not by itself rule out explosion. We therefore isolate the deterministic estimate that supplies the missing global control. The notation below is adapted to the BPHZ model constructed above, but the argument only uses the displayed Besov bounds on a finite collection of stochastic coefficients.

Fix an even integer  $p > 24$  and choose  $0 < \varepsilon \ll p^{-1}$ . On a time interval  $I = [s, s+1]$ , let  $\mathbb{X}_I$  denote the finite family of massive stationary stochastic diagrams entering the renormalized  $\Phi_3^4$  remainder system. Their spatial regularities are, up to an arbitrary loss  $\varepsilon$ ,

$$-\frac{1}{2} - \varepsilon, \quad -1 - \varepsilon, \quad \frac{1}{2} - \varepsilon, \quad -\varepsilon, \quad -\frac{1}{2} - \varepsilon, \quad -\varepsilon,$$

and one of the commutator coefficients is also controlled in time with exponent  $1/8$ . Write  $K_I \geq 1$  for one plus the maximum of the corresponding Besov norms and that time-increment norm. The Gaussian BPHZ construction gives  $\sup_s \mathbf{E} K_{[s, s+1]}^q < \infty$  for every  $q < \infty$ ; this stochastic fact is proved at the end of the subsection. Until then  $K := K_I$  is simply a deterministic parameter.

As in the renormalized paraproduct expansion, write on  $I$  (after translating  $s$  to 0)

$$\Phi = -X + v + w,$$

where  $X$  is the massive stationary stochastic convolution. Introduce an auxiliary damping constant  $c > 0$  and split the remainder equation as

$$\begin{aligned} (\partial_t - \Delta)v &= F(v + w) - cv, \\ (\partial_t - \Delta)w &= G(v, w) + cv. \end{aligned} \tag{8.2}$$

Here  $F$  contains precisely the rough low–high paraproducts, while  $G$  contains  $-(v + w)^3$ , the complementary paraproducts, the two renormalized commutators, and the degree-0, 1, 2 coefficient polynomial generated by the BPHZ diagrams. Changing  $c$  changes the splitting  $(v, w)$  but not  $v + w$  and hence not  $\Phi$ . At each restart we shall take  $v(0) = 0$  and put the entire incoming remainder in  $w(0)$ .

**Lemma 8.49** (Besov and heat-semigroup inequalities used below). *Let  $1 \leq r \leq q \leq \infty$ ,  $\beta \geq \alpha$ , and  $0 < t \leq 1$ . On  $\mathbb{T}^3$ ,*

$$\|e^{t\Delta} f\|_{B_r^\beta} \lesssim t^{-\frac{\beta-\alpha}{2} - \frac{3}{2}(\frac{1}{r} - \frac{1}{q})} \|f\|_{B_r^\alpha}, \tag{25.9a}$$

$$\|(e^{t\Delta} - 1)f\|_{B_r^{\alpha-2\theta}} \lesssim t^\theta \|f\|_{B_r^\alpha}, \quad 0 \leq \theta \leq 1, \tag{25.9b}$$

$$\|f\|_{B_r^{(1-\vartheta)\alpha + \vartheta\beta}} \lesssim \|f\|_{B_r^\alpha}^{1-\vartheta} \|f\|_{B_r^\beta}^\vartheta. \tag{25.9c}$$

If  $a < 0 < b$  and  $a + b > 0$ , the Bony decomposition gives

$$\|f \prec g\|_{B_r^{a+b}} \lesssim \|f\|_{B_{r_1}^a} \|g\|_{B_{r_2}^b}, \quad (8.3)$$

$$\|f \circ g\|_{B_r^{a+b}} \lesssim \|f\|_{B_{r_1}^a} \|g\|_{B_{r_2}^b}, \quad \frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2}, \quad (25.9d)$$

with the usual one-sided modification for  $\prec$  when only the low-frequency factor is bounded.

*Proof.* Let  $\Delta_j$  be the inhomogeneous dyadic blocks. The Fourier multiplier of  $e^{t\Delta}\Delta_j$  is bounded by  $Ce^{-ct2^{2j}}$ , while Bernstein gives  $\|\Delta_j f\|_q \lesssim 2^{3j(1/r-1/q)} \|\Delta_j f\|_r$ . Multiplication by  $2^{j\beta}$  and the elementary bound  $x^\rho e^{-cx} \lesssim_\rho 1$  with  $x = t2^{2j}$  give (25.9a). The multiplier  $e^{-t|\xi|^2} - 1$  is bounded on the  $j$ -th annulus by  $C \min\{1, t2^{2j}\} \leq C(t2^{2j})^\theta$ , which gives (25.9b). For (25.9c), apply Hölder to the sequence  $2^{j((1-\vartheta)\alpha+\vartheta\beta)} \|\Delta_j f\|_r$ . Finally, in  $f \prec g = \sum_{i < j-1} \Delta_i f \Delta_j g$  the output frequency is  $2^j$ ; summing the geometric series in  $i$  gives the first product bound. In  $f \circ g = \sum_{|i-j| \leq 1} \Delta_i f \Delta_j g$  the hypothesis  $a + b > 0$  makes the output-frequency sum convergent and gives the second. These are exactly the estimates used below; no unrecorded Besov multiplication rule is invoked.  $\square$

**Lemma 8.50** (Damped rough component). *There are  $C, \kappa < \infty$ , depending only on  $p, \varepsilon$ , such that if  $c \geq CK^{30p}$ , then every local solution of (8.2) satisfies, for  $0 < t \leq 1$ ,*

$$\|v(t)\|_{B_{2p}^{-3\varepsilon}} \lesssim e^{-c_* t} \|v(0)\|_{B_{2p}^{-3\varepsilon}} + K^\kappa \int_0^t e^{-c_*(t-r)} (t-r)^{-\sigma} (1 + \|w(r)\|_{L^p}) dr, \quad (8.4)$$

where  $\sigma < 7/8$  and  $c_* \geq c/2$ . For every  $\beta \in (\varepsilon, 1 - 4\varepsilon)$  the same argument gives

$$\|v(t) - v(s)\|_{L^p} \lesssim (c + K) |t - s|^{(\beta - \varepsilon)/2} \|v(s)\|_{B_p^\beta} + K^\kappa \int_s^t e^{-c_*(t-r)} (t-r)^{-1/2-\varepsilon} (1 + \|w(r)\|_{L^p}) dr. \quad (8.5)$$

*Proof.* Every term in  $F(v + w)$  is a low-high paraproduct with one stochastic coefficient of regularity at least  $-1 - \varepsilon$  and one factor linear in  $v + w$ . By Lemma 8.49, heat convolution maps such a term to  $B_{2p}^{-3\varepsilon}$  with kernel  $(t - r)^{-\sigma}$  for some  $\sigma < 7/8$  once  $p > 24$  and  $\varepsilon$  is sufficiently small. Separating the  $v$  and  $w$  factors gives

$$f(t) \leq g(t) + CK \int_0^t e^{-c(t-r)} (t-r)^{-\sigma} f(r) dr + CK \int_0^t e^{-c(t-r)} (t-r)^{-\sigma} (1 + \|w(r)\|_p) dr,$$

with  $f(t) = \|v(t)\|_{B_{2p}^{-3\varepsilon}}$  and  $g(t) \lesssim e^{-ct} \|v(0)\|_{B_{2p}^{-3\varepsilon}}$ . Iterating the Volterra kernel gives the resolvent series

$$e^{-ct} t^{-\sigma} \sum_{n \geq 0} \frac{(CK\Gamma(1-\sigma)t^{1-\sigma})^n}{\Gamma((n+1)(1-\sigma))}.$$

Stirling's formula bounds this by  $Ct^{-\sigma} \exp(-(c - CK^{1/(1-\sigma)})t)$ . Since  $1/(1-\sigma) \leq 8$  in the uses below,  $c \geq CK^{30p}$  leaves  $c_* \geq c/2$  and proves (8.4). For increments, write the mild formula from time  $s$ , use (25.9b) on the homogeneous term and (25.9a) on the integral term. The same resolvent estimate gives (8.5).  $\square$

**Lemma 8.51** (Positive regularity estimates for the damped component). *Under the hypotheses of Lemma 8.50, and after decreasing  $\varepsilon > 0$  in terms of  $p$ , one has for  $0 < t \leq 1$*

$$\|v(t)\|_{B_{\infty}^{1/2+2\varepsilon}} \lesssim t^{-a_1} \|v(0)\|_{B_p^{-3\varepsilon}} + K \int_0^t (t-r)^{-b_1} (K + \|w(r)\|_{L^{3p}}) dr, \quad (25.9e)$$

$$\|v(t)\|_{L^{\infty}} \lesssim t^{-a_2} \|v(0)\|_{B_p^{-3\varepsilon}} + K \int_0^t (t-r)^{-b_2} (K + \|w(r)\|_{L^{3p}}) dr, \quad (25.9f)$$

$$\|v(t)\|_{L^{3p}} \lesssim t^{-a_3} \|v(0)\|_{B_{2p}^{-3\varepsilon}} + K \int_0^t (t-r)^{-b_3} (K + \|w(r)\|_{L^{3p}}) dr, \quad (25.9g)$$

where

$$\begin{aligned} a_1 &= \frac{1}{4} + \frac{7\varepsilon}{2} + \frac{3}{2p}, & b_1 &= \frac{3}{4} + \frac{3\varepsilon}{2} + \frac{1}{2p}, \\ a_2 &= 2\varepsilon + \frac{3}{2p}, & b_2 &= \frac{1}{2} + \varepsilon + \frac{1}{2p}, & a_3 &= 2\varepsilon + \frac{1}{4p}, & b_3 &= \frac{1}{2} + \varepsilon. \end{aligned}$$

In particular  $a_i, b_i < 1$  for  $p > 24$  and sufficiently small  $\varepsilon$ .

*Proof.* The proof is the parameter-specialized form of the semigroup/Volterra argument in Lemma 8.50. We record it because these exponents are the load-bearing input in the quadratic estimate below.

Write  $F(v+w) = -3(v+w-X) \prec Z_{-1}$ , where  $\|Z_{-1}\|_{B_{\infty}^{-1-\varepsilon}} \leq K$ ; this is precisely the rough low-high term in the paracontrolled remainder system. Thus

$$\|(w-X) \prec Z_{-1}\|_{B_{3p}^{-1-\varepsilon}} \lesssim K(K + \|w\|_{L^{3p}}), \quad \|v \prec Z_{-1}\|_{B_q^{-1-4\varepsilon}} \lesssim K\|v\|_{B_q^{-3\varepsilon}}.$$

Applying (25.9a) to the mild equation and absorbing the second term by the same Volterra resolvent used in Lemma 8.50 gives, for every  $-3\varepsilon < \beta < 1 - 4\varepsilon$  and admissible  $q$ ,

$$\|v(t)\|_{B_q^{\beta}} \lesssim t^{-\frac{\beta+3\varepsilon}{2} - \frac{3}{2}(\frac{1}{p_0} - \frac{1}{q})} \|v(0)\|_{B_{p_0}^{-3\varepsilon}} + K \int_0^t (t-r)^{-\frac{\beta+1+\varepsilon}{2} - \frac{3}{2}(\frac{1}{3p} - \frac{1}{q})} (K + \|w(r)\|_{L^{3p}}) dr. \quad (25.9h)$$

Here replacing the damped heat kernel by the ordinary one only enlarges the right-hand side. Choose successively  $(\beta, q, p_0) = (1/2 + 2\varepsilon, \infty, p)$ ,  $(0, \infty, p)$ , and  $(0, 3p, 2p)$ . Formula (25.9h) gives slightly *better* homogeneous powers than the displayed  $a_i$  and slightly better forcing powers in the second and third choices. Since  $0 < t-r \leq 1$ , we may weaken those powers to the source-safe values  $a_i, b_i$  displayed above. For example the first homogeneous exponent produced directly by (25.9h) is  $1/4 + 5\varepsilon/2 + 3/(2p)$ , which is bounded by  $t^{-a_1}$  because  $a_1 = 1/4 + 7\varepsilon/2 + 3/(2p)$ . Likewise the direct bounds  $3\varepsilon/2 + 3/(2p)$  and  $3\varepsilon/2 - 1/(4p)$  may be enlarged to  $a_2$  and  $a_3$ , while the forcing exponents may be enlarged to  $b_2$  and  $b_3$ . This deliberate loss reproduces the powers used in the source proof of the quadratic estimate and avoids claiming an equality of exponents that the semigroup calculation does not give. The inequalities  $a_i, b_i < 1$  follow from  $p > 24$  after fixing  $\varepsilon$  sufficiently small.  $\square$

**Lemma 8.52** (Time increments of the smoother remainder). *Under the hypotheses of Lemma 8.50, for  $0 \leq s < t \leq 1$ ,*

$$\begin{aligned} \|w(t) - w(s)\|_{L^p} &\leq CcK^7 |t-s|^{1/8} \left[ 1 + \|v(0)\|_{B_p^{-3\varepsilon}}^3 + \|w(s)\|_{B_p^{1+4\varepsilon}} \right. \\ &\quad \left. + \left( \int_0^t \|w(r)\|_{B_p^{1+4\varepsilon}}^p dr \right)^{1/p} + \left( \int_0^t \|w(r)\|_{L^{3p}}^{3p} dr \right)^{1/p} \right]. \end{aligned} \quad (25.10a)$$

*Proof.* Write

$$\delta'_{st}w := w(t) - e^{(t-s)\Delta}w(s) = \int_s^t e^{(t-r)\Delta}[G(v, w) + cv](r) dr. \quad (25.10b)$$

We bound the terms in  $G$  according to the exact renormalized decomposition.

*Cubic term.* By  $L^p$  contractivity of the heat semigroup, Hölder in time, and Lemma 8.50,

$$\begin{aligned} \left\| \int_s^t e^{(t-r)\Delta}(v+w)^3(r) dr \right\|_p &\leq |t-s|^{1-1/p} \left( \int_0^t \|v+w\|_{3p}^{3p} dr \right)^{1/p} \\ &\lesssim |t-s|^{1-1/p} \left[ \|v(0)\|_{B_p^{-3\varepsilon}}^3 + K^6 + K^3 \left( \int_0^t \|w\|_{3p}^{3p} dr \right)^{1/p} \right]. \end{aligned} \quad (25.10c)$$

Since  $1 - 1/p > 1/8$ , this has more time gain than needed.

*First commutator.* Its defining heat-commutator identity and the  $1/8$  time regularity of the distinguished diagram yield

$$\begin{aligned} \|\mathfrak{C}_1(v, w)(r) - e^{r\Delta}v(0)\|_{B_p^{1+2\varepsilon}} &\lesssim K^3 + K(K+c)r^{-4\varepsilon}\|v(0)\|_{B_p^{-3\varepsilon}} \\ &\quad + K^2 \int_0^r (r-u)^{-3/4-\varepsilon} \|w(u)\|_{B_p^{1/2+2\varepsilon}} du \end{aligned} \quad (8.6)$$

$$+ K \int_0^r (r-u)^{-1-2\varepsilon} \|w(r) - w(u)\|_p du. \quad (25.10d)$$

To verify (25.10d), expand  $[e^{(r-u)\Delta}, f \prec]g$  dyadically. On a block of frequency  $2^j$  the commutator gains the factor  $\min\{1, (r-u)2^{2j}\}$ ; summing the low frequencies of  $f$  and using the  $1/8$  time increment of  $g$  gives the last two kernels. The terms containing  $v$  are then replaced by (8.4) and (8.5). The inequalities  $3/4 + \varepsilon < 1$  and  $1/8 - 2\varepsilon > 0$  make the resulting time convolutions finite. Combining (25.10d) with (25.9a) therefore bounds the heat convolution of the first commutator by the right-hand side of (25.10a) times  $K|t-s|^{1/8}$ .

*Direct rough and coefficient terms.* If  $Z_-$  has regularity  $-1 - \varepsilon$ , then  $\|uZ_-\|_p \lesssim K\|u\|_{B_p^{1+2\varepsilon}}$  by (25.9d). For  $a_j(v+w)^j$ ,  $j = 0, 1, 2$ , the coefficient  $a_j$  has regularity at least  $-1/2 - \varepsilon$  and heat smoothing to  $L^p$  costs strictly less than one time derivative. Applying Hölder in time gives a factor at least  $|t-s|^{1/8}$  and leaves only the two integrals displayed in (25.10a). The second commutator is bounded by  $K^2(K + \|v+w\|_{B_p^{3\varepsilon}})$  and is handled identically after Lemma 8.50.

These estimates prove the bound for  $\delta'_{st}w$ . Finally,

$$\|(e^{(t-s)\Delta} - 1)w(s)\|_p \lesssim |t-s|^{1/8} \|w(s)\|_{B_p^{1/4+\varepsilon}} \lesssim |t-s|^{1/8} \|w(s)\|_{B_p^{1+4\varepsilon}}$$

by (25.9b), and (25.10a) follows.  $\square$

**Lemma 8.53** (Integrated positive regularity of  $w$ ). *Under the same hypotheses there is  $\kappa < \infty$  such that*

$$\begin{aligned} \left( \int_0^t \|w(r)\|_{B_p^{1+7\varepsilon}}^p dr \right)^{1/p} &\lesssim \left( \int_0^t \|e^{r\Delta}w(0)\|_{B_p^{1+7\varepsilon}}^p dr \right)^{1/p} \\ &\quad + (cK)^\kappa \left[ 1 + \|v(0)\|_{B_{2p}^{-3\varepsilon}}^3 + \left( \int_0^t \|w(r)\|_{L_{3p}^{3p}}^{3p} dr \right)^{1/p} \right]. \end{aligned} \quad (25.10e)$$

*Proof.* Return to the mild equation for  $w$  from time zero and apply (25.9a) with target regularity  $1 + 7\varepsilon$ . We record the worst time powers, since their integrability is the point of the argument.



For the cubic forcing,  $L^p \rightarrow B_p^{1+7\varepsilon}$  costs  $(t-r)^{-(1+7\varepsilon)/2}$ , which is in  $L^{p/(p-1)}(dr)$  for  $p > 24$  and small  $\varepsilon$ . This gives  $K^\kappa(\int \|w\|_{3p}^{3p})^{1/p}$  after Lemma 8.50. For the first commutator, insert (25.10d); the heat kernel contributes at most  $(t-r)^{-1/2-O(\varepsilon)}$ , while the inner commutator convolution has kernel  $(r-u)^{-3/4-O(\varepsilon)}$  or the improved  $(r-u)^{-1-2\varepsilon}|r-u|^{1/8}$ . Their convolution exponents remain strictly below 1. Lemma 8.52 supplies the needed increment bound.

For the direct rough factor and the two commutator/coefficient remainders, (25.9d) places the forcing respectively in  $B_p^{-1+O(\varepsilon)}$  and  $B_p^{-1/2+O(\varepsilon)}$ . The corresponding heat kernels have powers strictly smaller than the cubic one. The quadratic coefficient term is estimated by

$$\|(v+w)^2 a_2\|_{B_p^{-1/2-\varepsilon}} \lesssim K(1 + \|v\|_{L^{2p}} + \|w\|_{L^{2p}})^2,$$

and Hölder in time followed by Lemma 8.50 again leaves only the  $L^{3p}$  integral. Summing the finitely many forcing classes yields (25.10e). This is the integrated form needed in the coercive estimate; unlike a pointwise Schauder bound it contains no uncontrolled positive norm of the rough initial datum.  $\square$

**Lemma 8.54** (Rigorous testing identity). *Let  $p > 24$  be even. With  $G = -w^3 + \tilde{G}$ , every local solution of (8.2) satisfies*

$$\begin{aligned} \frac{\|w(t)\|_{3p-2}^{3p-2} - \|w(s)\|_{3p-2}^{3p-2}}{3p-2} + (3p-3) \int_s^t \int |\nabla w|^2 |w|^{3p-4} \\ + \int_s^t \|w(r)\|_{3p}^{3p} dr = \int_s^t \langle \tilde{G}(v, w) + cv, w^{3p-3} \rangle dr. \end{aligned} \quad (25.11a)$$

*Proof.* Mollify  $w$  spatially and test the mollified equation with  $w_\delta^{3p-3}$ . The chain rule gives the first term and integration by parts gives  $(3p-3) \int |\nabla w_\delta|^2 |w_\delta|^{3p-4}$ . The separated drift  $-w^3$  gives  $-\|w\|_{3p}^{3p}$ . On compact subintervals of the local lifetime, Lemma 8.53 gives continuity of  $w$  in a positive Besov space and  $1/8$  time control in  $L^p$ ; hence  $w_\delta \rightarrow w$ ,  $\nabla w_\delta \rightarrow \nabla w$  in the integrability classes needed for the three left-hand terms. Each term in  $G$  is a continuous paraproduct / resonant expression in the same Besov spaces by (25.9d), so its pairing with  $w_\delta^{3p-3}$  converges as well. Letting  $\delta \downarrow 0$  proves (25.11a).  $\square$

**Lemma 8.55** (Absorption of the cubic cross terms). *For every  $\delta \in (0, 1]$  there is  $C_p$  such that, if  $c \geq C_p \delta^{-(5+15p)} K^{30p}$ , then*

$$\int_s^t |\langle v^3 + 3v^2 w + 3vw^2, w^{3p-3} \rangle| dr \leq \delta \left[ 1 + \|v(s)\|_{B_{2p}^{-3\varepsilon}}^{3p} + \int_s^t \|w(r)\|_{3p}^{3p} dr \right]. \quad (25.11b)$$

*Proof.* Hölder and Young give pointwise

$$\int (|v|^3 |w|^{3p-3} + 3|v|^2 |w|^{3p-2} + 3|v| |w|^{3p-1}) \leq \frac{\delta}{2} \|w\|_{3p}^{3p} + C_p \delta^{-3p} \|v\|_{3p}^{3p}.$$

Use Lemma 8.50 with the heat embedding  $B_{2p}^{-3\varepsilon} \rightarrow L^{3p}$  at positive time. The homogeneous piece obeys

$$\|e^{r(\Delta-c)} v(s)\|_{3p} \lesssim e^{-c(r-s)} (r-s)^{-2\varepsilon-1/(4p)} \|v(s)\|_{B_{2p}^{-3\varepsilon}}.$$

Its  $3p$ -th power is integrable for  $p > 24$  and  $\varepsilon$  small, with integral  $\lesssim c^{-1/4+6p\varepsilon} \|v(s)\|_{3p}^{3p}$ . For the forced part, Jensen's inequality applied to the convolution in (8.4) yields

$$\int_s^t \|v(r)\|_{3p}^{3p} dr \lesssim I(c) \left[ K^{3p} + K^{3p} \int_s^t \|w(r)\|_{3p}^{3p} dr \right] + I(c) \|v(s)\|_{B_{2p}^{-3\varepsilon}}^{3p},$$

where

$$I(c) \lesssim \int_0^\infty e^{-3pcu} u^{-3/4-6p\varepsilon} du \vee \int_0^\infty e^{-cu} u^{-1/2-\varepsilon} du \lesssim c^{-1/5}.$$

The last inequality follows after decreasing  $\varepsilon$  in terms of  $p$ . Therefore  $C_p \delta^{-3p} K^{3p} I(c) \leq \delta/2$  once  $c \geq C_p \delta^{-(5+15p)} K^{30p}$ , and (25.11b) follows.  $\square$

**Lemma 8.56** (First commutator contribution). *Let  $\mathfrak{C}_1(v, w)$  be the first commutator term in the renormalized remainder equation and let  $Z_-$  denote the rough diagram multiplying it. For every  $\delta \in (0, 1]$ ,*

$$\begin{aligned} \int_s^t |\langle \mathfrak{C}_1(v, w) Z_-, w^{3p-3} \rangle| dr &\leq C(\delta^{-1} c K)^\kappa \left[ 1 + \|v(s)\|_{B_p^{-3\varepsilon}}^{3p} + \int_s^t \|w(r)\|_{B_p^{1+4\varepsilon}}^p dr \right] \\ &\quad + \delta \left[ \sup_{s \leq r \leq t} \|w(r)\|_{3p-2}^{3p-2} + \int_s^t \|w(r)\|_{3p}^{3p} dr \right]. \end{aligned} \quad (25.11c)$$

*Proof.* We keep the two pieces of the source proof separate, because they use different strict exponent inequalities. Write

$$\mathfrak{C}_1(v, w)(r) = e^{(r-s)\Delta} v(s) + \mathfrak{C}_1^\circ(r).$$

By the product estimate (25.9d),

$$|\langle \mathfrak{C}_1 Z_-, w^{3p-3} \rangle| \lesssim K \|e^{(r-s)\Delta} v(s)\|_{B_{3p-2}^{1+2\varepsilon}} \|w\|_{3p-2}^{3p-3} + K \|\mathfrak{C}_1^\circ(r)\|_{B_p^{1+2\varepsilon}} \|w\|_{3p}^{3p-3}. \quad (25.11c1)$$

*The homogeneous heat term.* Heat smoothing gives

$$\|e^{(r-s)\Delta} v(s)\|_{B_{3p-2}^{1+2\varepsilon}} \lesssim (r-s)^{-\theta} \|v(s)\|_{B_p^{-3\varepsilon}}, \quad \theta := \frac{1+5\varepsilon}{2} + \frac{3}{2} \left( \frac{1}{p} - \frac{1}{3p-2} \right) < 1.$$

Hence, since  $t-s \leq 1$ ,

$$K \int_s^t (r-s)^{-\theta} \|v(s)\|_{B_p^{-3\varepsilon}} \|w(r)\|_{3p-2}^{3p-3} dr \lesssim K \|v(s)\|_{B_p^{-3\varepsilon}} \sup_{[s,t]} \|w\|_{3p-2}^{3p-3}.$$

First Young raises the initial norm to the power  $3p$  and leaves the power  $3p(3p-3)/(3p-1)$  of the supremum. The strict inequality

$$\frac{3p(3p-3)}{3p-1} < 3p-2 \quad (25.11c2)$$

then gives, for every  $\delta \in (0, 1]$ ,

$$K \int_s^t (r-s)^{-\theta} \|v(s)\|_{B_p^{-3\varepsilon}} \|w(r)\|_{3p-2}^{3p-3} dr \leq C(\delta^{-1} K)^\kappa (1 + \|v(s)\|_{B_p^{-3\varepsilon}}^{3p}) + \frac{\delta}{4} \sup_{[s,t]} \|w\|_{3p-2}^{3p-2}. \quad (25.11c3)$$

*The genuine commutator remainder.* Expanding its defining heat commutator exactly as in the proof of (25.10d) gives

$$\begin{aligned} E(r) := \|\mathfrak{C}_1^\circ(r)\|_{B_p^{1+2\varepsilon}} &\lesssim K^3 + K(K+c)(r-s)^{-4\varepsilon} \|v(s)\|_{B_p^{-3\varepsilon}} \\ &\quad + K^2 \int_s^r (r-u)^{-3/4-\varepsilon} \|w(u)\|_{B_p^{1/2+2\varepsilon}} du \\ &\quad + K \int_s^r (r-u)^{-1-2\varepsilon} \|w(r) - w(u)\|_p du. \end{aligned} \quad (25.11c4)$$

The first three terms are in  $L^p([s, t])$ :  $4p\varepsilon < 1$  after fixing  $\varepsilon$  sufficiently small, and the kernel  $u^{-3/4-\varepsilon}$  lies in  $L^1(0, 1)$ . Thus Young convolution, followed by interpolation of  $B_p^{1/2+2\varepsilon}$  between  $L^p$  and  $B_p^{1+4\varepsilon}$ , gives

$$\|E_{\text{first three}}\|_{L_r^p} \leq C(cK)^\kappa \left[ 1 + \|v(s)\|_{B_p^{-3\varepsilon}} + \left( \int_s^t \|w(u)\|_{B_p^{1+4\varepsilon}}^p du \right)^{1/p} \right]. \quad (25.11c5)$$

It remains to treat the last, nonintegrable-looking kernel in (25.11c4). Fix a splitting scale  $h \in (0, 1]$ . On  $r - u \geq h$  use the crude difference bound to obtain

$$I_{\text{far}}(r) \leq CKh^{-1-2\varepsilon} \int_s^r (\|w(u)\|_p + \|w(r)\|_p) du. \quad (25.11c6)$$

Consequently  $\|I_{\text{far}}\|_{L_r^p} \leq CKh^{-1-2\varepsilon} \|w\|_{L_r^p L_x^p}$ . On  $r - u < h$ , the time-translated version of Lemma 8.52 gives

$$\|w(r) - w(u)\|_p \leq CcK^7(r - u)^{1/8} \left[ N_{s,t} + \|w(u)\|_{B_p^{1+4\varepsilon}} \right],$$

where

$$N_{s,t} := 1 + \|v(s)\|_{B_p^{-3\varepsilon}}^3 + \left( \int_s^t \|w\|_{B_p^{1+4\varepsilon}}^p \right)^{1/p} + \left( \int_s^t \|w\|_{3p}^{3p} \right)^{1/p}.$$

Because  $1/8 - 2\varepsilon > 0$ ,

$$\|I_{\text{near}}\|_{L_r^p} \leq CcK^8 h^{1/8-2\varepsilon} \left[ N_{s,t} + \left( \int_s^t \|w\|_{B_p^{1+4\varepsilon}}^p \right)^{1/p} \right]. \quad (25.11c7)$$

Here Jensen followed by Fubini was used for the term depending on  $u$ . Choose  $h$  to be a sufficiently small fixed power of  $\delta/(1 + cK)$  so that, after (25.11c1) is integrated and Hölder is applied, the contribution of (25.11c7) is absorbed by  $\delta \int_s^t \|w\|_{3p}^{3p}$ . The price from (25.11c6) is then only a fixed power of  $\delta^{-1}cK$ ; since  $\|w\|_{L_r^p L_x^p}^p \lesssim 1 + \int_s^t \|w\|_{3p}^{3p}$ , a final Young inequality sends this contribution either into the same good integral or into  $C(\delta^{-1}cK)^\kappa$ .

Combining (25.11c3)–(25.11c7), then using Hölder in (25.11c1), yields exactly (25.11c). Notice that the two strict inequalities used in the proof are displayed explicitly:  $\theta < 1$  for the initial heat term and  $1/8 - 2\varepsilon > 0$  for the near-diagonal commutator kernel.  $\square$

**Lemma 8.57** (Direct rough factor and quadratic coefficient). *Let  $Z_-$  be a stochastic coefficient with  $\|Z_-\|_{B_\infty^{-1-\varepsilon}} \leq K$  and let  $a_2 \in B_\infty^{-1/2-\varepsilon}$  with norm at most  $K$ . Then*

$$\int_s^t |\langle wZ_-, w^{3p-3} \rangle| dr \lesssim K \left( \int_s^t \|w\|_{3p}^{3p} \right)^{(p-1)/p} \left( \int_s^t \|w\|_{B_p^{1+2\varepsilon}}^p \right)^{1/p}, \quad (25.11d)$$

$$\begin{aligned} \int_s^t |\langle a_2(v + w)^2, w^{3p-3} \rangle| dr &\leq C(\delta^{-1}K)^\kappa \left[ 1 + \int_s^t \|w\|_{B_p^{1+4\varepsilon}}^p dr \right] \\ &\quad + \delta \left[ \|v(s)\|_{B_{2p}^{-3\varepsilon}}^{3p} + \int_s^t \|w\|_{3p}^{3p} dr + \sup_{s \leq r \leq t} \|w(r)\|_{B_{3p-2}}^{3p-2} \right]. \end{aligned} \quad (25.11e)$$

*Proof.* The direct rough factor is immediate from the Besov product estimate:  $\|wZ_-\|_p \lesssim K\|w\|_{B_p^{1+2\varepsilon}}$ . Pairing with  $w^{3p-3}$  and applying Hölder first in space and then in time gives (25.11d).

We turn to the quadratic term and follow its three genuinely different branches. By time translation we may regard  $s$  as the new initial time in all estimates below; the diagram norm on every unit interval remains bounded by the same  $K$ . Thus the positive-regularity estimates of Lemma 8.51 apply with  $v(0)$  replaced by  $v(s)$  and  $t$  replaced by  $r - s$ . Throughout, constants denoted by  $\kappa$  may increase but depend only on  $p$  (and on the fixed small  $\varepsilon$ ).

*Step 1: the  $w^2$  contribution.* By duality and the dyadic product rule,

$$|\langle a_2 w^2, w^{3p-3} \rangle| \lesssim K \|w^{3p-1}\|_{B_1^{1/2+2\varepsilon}}.$$

Corollary-type Leibniz estimates obtained by placing the unique differentiated factor in  $B_{3p/2}^{1/2+2\varepsilon}$  give

$$\|w^{3p-1}\|_{B_1^{1/2+2\varepsilon}} \lesssim \|w\|_{3p}^{3p-2} \|w\|_{B_{3p/2}^{1/2+2\varepsilon}}.$$

Interpolating the last norm halfway between  $L^{3p}$  and  $B_p^{1+4\varepsilon}$  yields

$$\|w\|_{B_{3p/2}^{1/2+2\varepsilon}} \lesssim \|w\|_{3p}^{1/2} \|w\|_{B_p^{1+4\varepsilon}}^{1/2}.$$

Hence Young's inequality gives, for every  $\delta \in (0, 1]$ ,

$$\int_s^t |\langle a_2 w^2, w^{3p-3} \rangle| dr \leq \delta \int_s^t \|w\|_{3p}^{3p} dr + C(\delta^{-1} K)^\kappa \int_s^t \|w\|_{B_p^{1+4\varepsilon}}^p dr. \quad (25.11e1)$$

*Step 2: the  $v^2$  contribution.* The product estimate splits

$$\begin{aligned} \|v^2 w^{3p-3}\|_{B_1^{1/2+2\varepsilon}} &\lesssim \|v^2\|_{B_\infty^{1/2+2\varepsilon}} \|w^{3p-3}\|_{L^{(3p-2)/(3p-3)}} \\ &\quad + \|v^2\|_{L^{3p/2}} \|w^{3p-3}\|_{B_{3p/(3p-2)}^{1/2+2\varepsilon}}. \end{aligned} \quad (25.11e2)$$

For the first term use

$$\|w\|_{3p-2}^{3p-3} \leq \|w\|_{3p-2}^{(3p-6)/2} \|w\|_{3p}^{3p/2}, \quad (25.11e3)$$

which is just interpolation between the supremum-controlled  $L^{3p-2}$  norm and the time-integrated  $L^{3p}$  norm. Hölder in time and Young therefore reduce this branch to

$$C\delta^{-\kappa} + \delta \left[ J_v^{3p/4} + \sup_{s \leq r \leq t} \|w(r)\|_{3p-2}^{3p-2} + \int_s^t \|w(r)\|_{3p}^{3p} dr \right], \quad (25.11e4)$$

where  $J_v := \int_s^t \|v(r)\|_{B_\infty^{1/2+2\varepsilon}}^2 dr$ . By the elementary product estimate  $\|v^2\|_{B_\infty^{1/2+2\varepsilon}} \lesssim \|v\|_\infty \|v\|_{B_\infty^{1/2+2\varepsilon}}$  and (25.9e)–(25.9f), the pure initial-data part of  $J_v$  is integrable because

$$2(a_1 + a_2) = \frac{1}{2} + 11\varepsilon + \frac{6}{p} < 1$$

for  $p > 24$  and sufficiently small  $\varepsilon$ . Jensen and Fubini applied to the convolution parts (whose kernels have exponents  $b_1, b_2 < 1$ ) yield

$$J_v^{3p/4} \leq CK^\kappa \left[ 1 + \|v(s)\|_{B_{2p}^{-3\varepsilon}}^{3p} + \int_s^t \|w(r)\|_{3p}^{3p} dr \right]. \quad (25.11e5)$$

The mixed initial/convolution terms are controlled by Hölder; the worst integrability requirement is weaker than  $p > 18$ , hence follows from  $p > 24$ .

For the second term of (25.11e2), the power Leibniz estimate and interpolation give

$$\|w^{3p-3}\|_{B_{3p/(3p-2)}^{1/2+2\varepsilon}} \lesssim \|w\|_{3p}^{3p-7/2} \|w\|_{B_p^{1+4\varepsilon}}^{1/2}. \quad (25.11e6)$$

Using (25.9g), squaring its bracket, and applying Hölder in time, the initial-data contribution is bounded by

$$\|v(s)\|_{B_{2p}^{-3\varepsilon}}^2 \left( \int_s^t \|w\|_{3p}^{3p} \right)^{(6p-7)/(6p)} \left( \int_s^t \|w\|_{B_p^{1+4\varepsilon}}^p \right)^{1/(2p)}.$$

The Young exponents close because

$$\frac{1}{3p} \left( 2 + \frac{6p-7}{2} + \frac{3}{2} \right) = 1. \quad (25.11e7)$$

The convolution part of (25.9g) is estimated by the same Jensen–Fubini argument;  $b_3 < 1$  and  $p > 24$  make all time singularities integrable. Thus this second branch has the same right-hand side as (25.11e5), plus the  $B_p^{1+4\varepsilon}$  integral.

*Step 3: the vw contribution.* Similarly,

$$\begin{aligned} \|vw^{3p-2}\|_{B_1^{1/2+2\varepsilon}} &\lesssim \|v\|_{B_\infty^{1/2+2\varepsilon}} \|w^{3p-2}\|_{L^1} \\ &\quad + \|v\|_{L^{3p}} \|w^{3p-2}\|_{B_{3p/(3p-1)}^{1/2+2\varepsilon}}. \end{aligned} \quad (25.11e8)$$

For the first term interpolate  $\|w\|_{3p-2}^{3p-2} \leq \|w\|_{3p-2}^{(3p-4)/2} \|w\|_{3p}^{3p/2}$ . Set

$$J_+ := \int_s^t \|v(r)\|_{B_\infty^{1/2+2\varepsilon}}^2 dr.$$

By (25.9e),  $2a_1 < 1$ , and the  $L^1$ -integrability of the kernel  $u^{-b_1}$ ,

$$J_+ \leq CK^\kappa \left[ 1 + \|v(s)\|_{B_{2p}^{-3\varepsilon}}^2 + \left( \int_s^t \|w(r)\|_{3p}^{3p} dr \right)^{2/(3p)} \right]. \quad (25.11e9a)$$

Indeed the homogeneous part is integrated directly, while Young convolution in time sends  $K + \|w\|_{3p}$  from  $L^2$  to  $L^2$ ; on an interval of length at most one its  $L^2$  norm is controlled by the  $L^{3p}$  norm. Hölder in time therefore gives

$$\int_s^t \|v\|_{B_\infty^{1/2+2\varepsilon}} \|w\|_{3p-2}^{3p-2} dr \leq J_+^{1/2} \left( \sup_{[s,t]} \|w\|_{3p-2}^{(3p-4)/2} \right) \left( \int_s^t \|w\|_{3p}^{3p} dr \right)^{1/2}.$$

Young is strictly subcritical because

$$\frac{1}{3p} + \frac{3p-4}{6p-4} + \frac{1}{2} < 1, \quad (25.11e9)$$

so this is bounded by the right-hand side of (25.11e).

For the differentiated-power term use

$$\|w^{3p-2}\|_{B_{3p/(3p-1)}^{1/2+2\varepsilon}} \lesssim \|w\|_{3p}^{3p-5/2} \|w\|_{B_p^{1+4\varepsilon}}^{1/2}.$$

Writing the  $L^{3p}$  estimate (25.9g), with the time origin translated to  $s$ , as  $\|v(r)\|_{3p} \leq H(r) + R(r)$ , we have

$$H(r) = (r - s)^{-a_3} \|v(s)\|_{B_{2p}^{-3\varepsilon}}, \quad R(r) = K \int_s^r (r - u)^{-b_3} (K + \|w(u)\|_{3p}) du.$$

For  $H$  the pointwise Young exponents close exactly:

$$\frac{1}{3p} + \frac{3p - 5/2}{3p} + \frac{1/2}{p} = 1,$$

and  $3pa_3 = 6p\varepsilon + 3/4 < 1$ . Hence its time integral is at most

$$C(\delta^{-1}K)^\kappa + \delta \left[ \|v(s)\|_{B_{2p}^{-3\varepsilon}}^{3p} + \int_s^t \|w\|_{3p}^{3p} dr + \int_s^t \|w\|_{B_p^{1+4\varepsilon}}^p dr \right]. \quad (25.11e10)$$

For  $R$ , the same pointwise Young inequality reduces the problem to  $\int_s^t R(r)^{3p} dr$ . Since  $b_3 < 1$ , Jensen with respect to the finite measure  $(r - u)^{-b_3} du$  followed by Fubini gives

$$\int_s^t R(r)^{3p} dr \leq CK^\kappa \left( 1 + \int_s^t \|w(u)\|_{3p}^{3p} du \right). \quad (25.11e11)$$

Thus the differentiated branch also has the required form. Combining the three steps and multiplying by the coefficient norm  $\|a_2\|_{B_\infty^{-1/2-\varepsilon}} \leq K$  proves (25.11e).  $\square$

**Lemma 8.58** (Remaining lower-order renormalized terms). *Assume that the auxiliary damping has been fixed once and for all as  $c = C_* K^{30p}$ , with  $C_*$  sufficiently large depending only on  $p, \varepsilon$ . Let  $R(v, w)$  be the sum of the second commutator, the complementary rough paraproduct, the coefficient terms  $a_0 + a_1(v + w)$ , and  $cv$ . For every  $\delta \in (0, 1]$ ,*

$$\begin{aligned} \int_s^t |\langle R(v, w), w^{3p-3} \rangle| dr &\leq C(\delta^{-1}K)^\kappa \left[ 1 + \|v(s)\|_{B_{2p}^{-3\varepsilon}}^{3p} + \int_s^t \|w\|_{B_p^{1+6\varepsilon}}^p dr \right] \\ &\quad + \delta \left[ \int_s^t \|w\|_{3p}^{3p} dr + \sup_{s \leq r \leq t} \|w(r)\|_{3p-2}^{3p-2} \right]. \end{aligned} \quad (25.11f)$$

*Proof.* We estimate the four classes in the order in which they occur in the renormalized equation. The argument is written out because the fact that these terms are lower order is a statement about exponents, not a licence to omit them.

1. *The second commutator.* Its three-frequency definition and the commutator estimate already proved in Chapter 5 give

$$\|\mathfrak{C}_2(v + w)(r)\|_p \lesssim K^2 (K + \|v(r) + w(r)\|_{B_p^{3\varepsilon}}). \quad (25.11f1)$$

Consequently, pointwise Young with conjugate exponents  $p$  and  $p/(p - 1)$  gives

$$|\langle \mathfrak{C}_2(v + w), w^{3p-3} \rangle| \leq \frac{\delta}{8} \|w\|_{3p}^{3p} + C(\delta^{-1}K)^\kappa (1 + \|v + w\|_{B_p^{3\varepsilon}}^p). \quad (25.11f2)$$

For  $w$ , interpolation between  $L^p$  and  $B_p^{1+6\varepsilon}$  gives

$$\|w\|_{B_p^{3\varepsilon}}^p \leq C_\delta (1 + \|w\|_p^p) + \delta \|w\|_{B_p^{1+6\varepsilon}}^p.$$

For  $v$ , apply the semigroup estimate (25.9h) with target  $B_p^{3\varepsilon}$  (using the finite-volume embedding  $B_{2p}^{-3\varepsilon} \hookrightarrow B_p^{-3\varepsilon}$  before smoothing). Since the homogeneous singularity is  $3\varepsilon < 1/p$  after shrinking  $\varepsilon$  and the forcing kernel has exponent  $1/2 + O(\varepsilon) < 1$ , Young convolution in time yields

$$\int_s^t \|v(r)\|_{B_p^{3\varepsilon}}^p dr \leq CK^\kappa \left[ 1 + \|v(s)\|_{B_{2p}^{-3\varepsilon}}^p + \int_s^t \|w(r)\|_{3p}^p dr \right]. \quad (25.11f3)$$

On an interval of length at most one,  $\int \|w\|_{3p}^p \leq (\int \|w\|_{3p}^{3p})^{1/3}$ ; a final Young inequality therefore turns (25.11f2)–(25.11f3) into the required contribution to (25.11f).

2. *The complementary rough paraproduct.* Write its nonhomogeneous part as

$$Q = (v - e^{(r-s)(\Delta-c)}v(s) + w - X) \succ Z_\succ.$$

The stochastic diagram bound and the high–low paraproduct estimate give

$$\|Q(r)\|_{B_{3p/2}^{-1/2-2\varepsilon}} \lesssim K \left[ K + \|v - e^{(r-s)(\Delta-c)}v(s) + w\|_{B_{3p/2}^{1/2-\varepsilon}} \right]. \quad (25.11f4)$$

Duality pairs this with  $w^{3p-3}$  in  $B_{3p/(3p-2)}^{1/2+3\varepsilon}$ . The power Leibniz estimate gives

$$\|w^{3p-3}\|_{B_{3p/(3p-2)}^{1/2+3\varepsilon}} \lesssim \|w\|_{3p}^{3p-7/2} \|w\|_{B_p^{1+6\varepsilon}}^{1/2}. \quad (25.11f5)$$

For the  $w$  inside (25.11f4),

$$\|w\|_{B_{3p/2}^{1/2-\varepsilon}} \lesssim \|w\|_{B_{3p}^{-2\varepsilon}}^{1/2} \|w\|_{B_p^1}^{1/2} \lesssim \|w\|_{3p}^{1/2} \|w\|_{B_p^{1+6\varepsilon}}^{1/2}.$$

Multiplying by (25.11f5) produces a multiple of

$$K \|w\|_{3p}^{3p-3} \|w\|_{B_p^{1+6\varepsilon}},$$

and Young bounds this by

$$\delta \|w\|_{3p}^{3p} + C(\delta^{-1}K)^\kappa \|w\|_{B_p^{1+6\varepsilon}}^p.$$

The remaining  $v$ -piece has no singular homogeneous part. Indeed (25.9h), now with target  $B_{3p/2}^{1/2-\varepsilon}$ , gives

$$\|v(r) - e^{(r-s)(\Delta-c)}v(s)\|_{B_{3p/2}^{1/2-\varepsilon}} \lesssim K \int_s^r (r-u)^{-3/4+1/(2p)} (K + \|w(u)\|_{3p}) du, \quad (25.11f6)$$

where  $3/4 - 1/(2p) < 1$ . Young convolution in time and (25.11f5), followed by the same three-factor Young inequality as above, gives precisely the same right-hand side. The separated homogeneous contribution  $e^{(r-s)(\Delta-c)}v(s) \succ Z_\succ$  is the heat-initial term already estimated in (25.11c3), with  $e^{(r-s)\Delta}$  replaced by the smaller damped semigroup.

3. *The affine coefficient terms.* The coefficient norms satisfy the same or better  $B_\infty^{-1/2-\varepsilon}$  bound as  $a_2$ . Thus

$$\begin{aligned} |\langle a_0, w^{3p-3} \rangle| &\lesssim K \|w\|_{3p}^{3p-7/2} \|w\|_{B_p^{1+6\varepsilon}}^{1/2}, \\ |\langle a_1 w, w^{3p-3} \rangle| &\lesssim K \|w\|_{3p}^{3p-5/2} \|w\|_{B_p^{1+6\varepsilon}}^{1/2}. \end{aligned} \quad (25.11f7)$$

The normalized reciprocal exponent sums are respectively  $1 - 2/(3p)$  and  $1 - 1/(3p)$ , so generalized Young leaves a strictly positive constant exponent and yields the desired  $\delta$  good term plus  $C(\delta^{-1}K)^\kappa(1 + \|w\|_{B_p^{1+6\varepsilon}}^p)$ . For  $a_1v$ , split the differentiated product exactly as in Step 3 of Lemma 8.57. The branch with  $\|v\|_{B_\infty^{1/2+2\varepsilon}}$  is controlled by (25.11e9a), while the branch with  $\|v\|_{3p}$  has reciprocal exponent sum

$$\frac{1}{3p} + \frac{3p-7/2}{3p} + \frac{1/2}{p} = 1 - \frac{1}{3p} < 1.$$

Equations (25.9e) and (25.9g) therefore give the same bound as in (25.11f), with one fewer power of  $w$  than in the already verified quadratic case.

4. *The artificial damping transfer  $cv$ .* This term is regular enough to be treated without any stochastic duality. Hölder and Young give

$$c|\langle v, w^{3p-3} \rangle| \leq \frac{\delta}{8} \|w\|_{3p}^{3p} + C\delta^{-\kappa} c^p \|v\|_p^p. \quad (25.11f8)$$

Applying the damped semigroup estimate to the  $L^p$  target and integrating in time yields

$$\int_s^t \|v(r)\|_p^p dr \leq CK^\kappa \left[ 1 + \|v(s)\|_{B_{2p}^{-3\varepsilon}}^p + \int_s^t \|w(r)\|_{3p}^p dr \right]. \quad (25.11f9)$$

Here the exponential factor only improves the estimate. Because the damping has been fixed as  $c = C_* K^{30p}$  rather than allowed to vary independently,  $c^p$  is another fixed power of  $K$ . As in Step 1,  $\int \|w\|_{3p}^p \leq (\int \|w\|_{3p}^{3p})^{1/3}$ , and Young absorbs an arbitrarily small multiple of the  $L^{3p}$  good integral at the cost of a larger power of  $\delta^{-1}K$ . This proves the contribution of  $cv$ .

Adding the four classes and decreasing the fractions of  $\delta$  if necessary proves (25.11f).  $\square$

**Lemma 8.59** (Closed coercive inequality). *There are  $c_p, C, \kappa > 0$  such that, after fixing  $c = C_* K^{30p}$  with  $C_*$  sufficiently large depending only on  $p, \varepsilon$ , one has for  $0 \leq s < t \leq 1$*

$$\begin{aligned} & \|w(t)\|_{L^{3p-2}}^{3p-2} + c_p \int_s^t \|w(r)\|_{L^{3p}}^{3p} dr + c_p \int_s^t \|w(r)\|_{B_p^{1+7\varepsilon}}^p dr \\ & \leq CK^\kappa \left[ 1 + \|w(s)\|_{L^{3p-2}}^{3p-2} + \|v(s)\|_{B_{2p}^{-3\varepsilon}}^{3p} + \|w(s)\|_{B_p^{1+7\varepsilon}}^{(3p-2)/3} \right]. \end{aligned} \quad (25.11)$$

*Proof.* The proof has two estimates which must be coupled rather than used independently.

*Step 1: the energy inequality.* Insert Lemmas 8.55–8.58 into the testing identity (25.11a). Choose first a numerical  $\delta_0 > 0$  depending only on  $p$  so that every  $\delta_0$  multiple of the  $L^{3p}$  term and the gradient term is absorbed. Then choose the constant  $C_*$  in  $c = C_* K^{30p}$  large enough that Lemma 8.55 applies with this fixed  $\delta_0$ . The direct rough term (25.11d) is split by Young between  $\int \|w\|_{3p}^{3p}$  and  $\int \|w\|_{B_p^{1+2\varepsilon}}^p$ . The latter is bounded by the  $B_p^{1+6\varepsilon}$  norm plus a harmless  $L^p$  term. After also absorbing the supremum term arising in (25.11c) and (25.11f), we obtain

$$\|w(t)\|_{3p-2}^{3p-2} + c_0 \int_s^t \|w\|_{3p}^{3p} \leq CK^\kappa \left[ 1 + \|w(s)\|_{3p-2}^{3p-2} + \|v(s)\|_{B_{2p}^{-3\varepsilon}}^{3p} + \int_s^t \|w\|_{B_p^{1+6\varepsilon}}^p \right]. \quad (25.11g)$$

This is the manuscript analogue of the source energy theorem before the positive-regularity estimate is fed back into it.



*Step 2: remove the  $B_p^{1+6\varepsilon}$  term.* The time-translated form of Lemma 8.53, raised to the  $p$ -th power, gives

$$\begin{aligned} \int_s^t \|w(r)\|_{B_p^{1+7\varepsilon}}^p dr &\leq C \int_s^t \|e^{(r-s)\Delta} w(s)\|_{B_p^{1+7\varepsilon}}^p dr \\ &\quad + CK^\kappa \left[ 1 + \|v(s)\|_{B_{2p}^{-3\varepsilon}}^{3p} + \int_s^t \|w(r)\|_{3p}^{3p} dr \right]. \end{aligned} \quad (25.11h)$$

For the intermediate norm we use exactly the strict interpolation needed in the source. First

$$\|w\|_{B_p^{1+6\varepsilon}} \lesssim \|w\|_{B_p^{1+7\varepsilon}}^{\frac{1+6\varepsilon}{1+7\varepsilon}} \|w\|_{L^p}^{\frac{\varepsilon}{1+7\varepsilon}}.$$

Young, followed by  $\|w\|_p \lesssim \|w\|_{3p}$ , gives for every  $\rho \in (0, 1]$

$$\|w\|_{B_p^{1+6\varepsilon}}^p \leq C\rho^{-\kappa} \|w\|_{B_p^{1+7\varepsilon}}^{\sigma p} + \rho \|w\|_{3p}^{3p}, \quad \sigma := \frac{3+18\varepsilon}{3+20\varepsilon} < 1. \quad (25.11i)$$

Since  $t - s \leq 1$ ,

$$\int_s^t \|w\|_{B_p^{1+7\varepsilon}}^{\sigma p} \leq \left( \int_s^t \|w\|_{B_p^{1+7\varepsilon}}^p \right)^\sigma. \quad (25.11j)$$

Thus the first term on the right of (25.11i), after multiplication by the fixed polynomial  $K^\kappa$  from (25.11g), can be bounded by an arbitrarily small multiple of the  $B_p^{1+7\varepsilon}$  integral plus a larger power of  $K$ . The  $\rho \|w\|_{3p}^{3p}$  term is absorbed into the left of (25.11g) by choosing  $\rho$  to be a sufficiently small negative power of  $K$ .

It remains to evaluate the heat-initial term in (25.11h). Set

$$\gamma := (1 + 7\varepsilon) \left( 1 - \frac{1}{p} \right).$$

By (25.9a),

$$\int_s^t \|e^{(r-s)\Delta} w(s)\|_{B_p^{1+7\varepsilon}}^p dr \lesssim \|w(s)\|_{B_p^\gamma}^p, \quad (25.11k)$$

indeed the time singularity has exponent  $(1 + 7\varepsilon)/2 < 1$ . Interpolating  $B_p^\gamma$  between  $L^p$  and  $B_p^{1+7\varepsilon}$  yields

$$\|w(s)\|_{B_p^\gamma}^p \lesssim \|w(s)\|_{B_p^{1+7\varepsilon}}^{p-1} \|w(s)\|_{L^p} \leq C \left[ \|w(s)\|_{B_p^{1+7\varepsilon}}^{(3p-2)/3} + \|w(s)\|_{L^{3p-2}}^{3p-2} \right], \quad (25.11l)$$

where the last line is Young with conjugate exponents  $(3p-2)/(3p-3)$  after accounting for the factor  $p-1$ .

Finally multiply (25.11h) by a sufficiently large fixed power of  $K$ , add it to (25.11g), and use (25.11i)–(25.11l) with the choices of  $\rho$  just described. Both good integrals are then absorbed on the left, and the remaining initial terms are exactly those displayed in (25.11).  $\square$

**Lemma 8.60** (ODE descent from the coercive inequality). *Let*

$$F(t) := \|w(t)\|_{B_p^{1+7\varepsilon}}^{(3p-2)/3} + \|w(t)\|_{L^{3p-2}}^{3p-2}.$$

*If  $v(0) = 0$ , then Lemmas 8.50–8.59 imply, for  $0 < t \leq 1$ ,*

$$\|v(t)\|_{B_{2p}^{-3\varepsilon}} + \sqrt{t} \|w(t)\|_{L^{3p-2}} \leq CK^\kappa. \quad (8.7)$$

*Proof.* Set

$$\lambda := \frac{3p}{3p-2} > 1, \quad \tau := \inf\{t > 0 : F(t) \leq 1 \text{ or } \|v(t)\|_{B_{2p}^{-3\varepsilon}}^{3p} > F(t)\} \wedge 1. \quad (25.12a)$$

On  $[0, \tau)$  the stopping rule gives  $\|v(s)\|_{B_{2p}^{-3\varepsilon}}^{3p} \leq F(s)$  and  $F(s) > 1$ . Since

$$F(r)^\lambda \lesssim \|w(r)\|_{B_p^{1+7\varepsilon}}^p + \|w(r)\|_{L^{3p}}^{3p},$$

Lemma 8.59 yields, for every  $0 \leq s < t < \tau$ ,

$$\int_s^t F(r)^\lambda dr \leq C_0 K^{\kappa_0} F(s). \quad (25.12b)$$

We now prove the descent estimate rather than assuming it. Put  $A := C_0 K^{\kappa_0}$  and start with  $t_0 = 0$ . Given  $t_n < \tau$ , define the proposed endpoint

$$t_{n+1}^* := t_n + 2^\lambda A F(t_n)^{1-\lambda}.$$

If  $t_{n+1}^* \geq \tau$ , stop and put  $t_{n+1} = \tau$ . Otherwise, applying (25.12b) on  $[t_n, t_{n+1}^*]$  gives

$$(t_{n+1}^* - t_n) \min_{t_n \leq r \leq t_{n+1}^*} F(r)^\lambda \leq A F(t_n),$$

so the minimum is at most  $2^{-\lambda} F(t_n)^\lambda$ . Let  $t_{n+1}$  be the first point where that minimum is attained. Then

$$F(t_{n+1}) \leq \frac{1}{2} F(t_n), \quad t_{n+1} - t_n \leq 2^\lambda A F(t_n)^{1-\lambda}. \quad (25.12c)$$

Because  $1 - \lambda < 0$ , the proposed step lengths increase by at least the factor  $2^{\lambda-1} > 1$  after each successful halving; hence the construction reaches  $\tau$  after finitely many steps. Moreover (25.12c) implies, for  $n$  before the terminal step,

$$t_{n+1} \leq 2^\lambda A \sum_{j=0}^n F(t_j)^{1-\lambda} \leq C_\lambda A F(t_n)^{1-\lambda},$$

because the last term dominates the geometric sum. Therefore

$$F(t_n) \leq C_\lambda A^{1/(\lambda-1)} t_{n+1}^{-1/(\lambda-1)}. \quad (25.12d)$$

Since  $1/(\lambda-1) = (3p-2)/2$ , this is the source of the exponent  $1/2$ .

We pass from the selected times to arbitrary times in four steps.

1. *Times before  $\tau$ .* If  $t_n \leq t < t_{n+1} \leq \tau$ , apply Lemma 8.59 on  $[t_n, t]$ . The stopping rule and  $F(t_n) > 1$  give

$$\|w(t)\|_{3p-2}^{3p-2} \leq C K^\kappa F(t_n) \leq C K^\kappa t_{n+1}^{-(3p-2)/2} \leq C K^\kappa t^{-(3p-2)/2}. \quad (25.12e)$$

Thus  $\|w(t)\|_{3p-2} \leq C K^\kappa t^{-1/2}$  for  $t < \tau$ .

2. *Exit through  $F \leq 1$ .* Let

$$\tau' := \inf\{t > 0 : \|v(t)\|_{B_{2p}^{-3\varepsilon}}^{3p} > F(t)\} \wedge 1.$$

If  $\tau < \tau'$ , then  $F(\tau) \leq 1$  and the  $v$ -stopping inequality continues to hold on  $[\tau, \tau')$ . Applying Lemma 8.59 with initial time  $\tau$  gives  $\|w(t)\|_{3p-2} \leq CK^\kappa$  there, hence the same  $CK^\kappa t^{-1/2}$  bound for  $0 < t \leq 1$ .

3. *Exit through the  $v$ -barrier.* Assume  $\tau' < 1$ . By Steps 1–2,  $\|w(s)\|_{3p-2} \leq CK^\kappa s^{-1/2}$  for  $s \leq \tau'$ . Use the damped  $v$  estimate at target  $B_{2p}^{-3\varepsilon}$ . Since  $v(0) = 0$ , its time kernel can be taken with exponent  $1/2 - \varepsilon < 1$ , and the embedding  $L^{3p-2} \hookrightarrow L^{2p}$  gives

$$\|v(\tau')\|_{B_{2p}^{-3\varepsilon}} \leq CK^\kappa \int_0^{\tau'} (\tau' - s)^{-1/2+\varepsilon} s^{-1/2} ds \leq CK^\kappa. \quad (25.12f)$$

(The beta integral is finite at both endpoints.) By continuity and the definition of  $\tau'$ , equality holds at the barrier, so  $F(\tau') = \|v(\tau')\|^{3p} \leq CK^\kappa$ . Restarting Lemma 8.59 at  $\tau'$  therefore gives  $\|w(t)\|_{3p-2} \leq CK^\kappa$  for every later  $t \leq 1$ .

4. *Control of  $v$ .* Combining the previous steps gives  $\|w(t)\|_{3p-2} \leq CK^\kappa t^{-1/2}$  on  $(0, 1]$ . Insert this bound once more in the same  $v$  estimate. The convolution in (25.12f) is uniformly finite after scaling, so  $\sup_{0 < t \leq 1} \|v(t)\|_{B_{2p}^{-3\varepsilon}} \leq CK^\kappa$ . Together with the  $w$  bound this proves (8.7).  $\square$

**Lemma 8.61** (Stationary diagram moments and restart). *For the massive Gaussian BPHZ model of dynamical  $\Phi_3^4$ , the random controls  $K_{[s, s+1]}$  above satisfy*

$$\sup_{s \geq 0} \mathbf{E} K_{[s, s+1]}^q < \infty \quad (q < \infty).$$

Moreover, reinitializing the splitting at a deterministic time  $s$  by setting  $v(s) = 0$  and replacing  $w(s)$  by  $v(s) + w(s)$  leaves the reconstructed solution unchanged.

*Proof.* Use the massive kernel  $(\partial_t - \Delta + 1)^{-1}$  to define the stationary convolution. Every diagram entering  $K$  is a finite Wiener chaos after BPHZ centering. The covariance estimates already proved in the analytic BPHZ section give its  $L^2(\Omega)$  dyadic Besov bound uniformly under time translation; Gaussian hypercontractivity upgrades this to every finite moment. For the one diagram for which a time increment is required, apply the kernel difference estimate  $\|K(t - \cdot) - K(s - \cdot)\| \lesssim |t - s|^{1/8}$  after sacrificing the small spatial  $\varepsilon$  margin, and then Kolmogorov's criterion. Stationarity of the massive convolution makes the resulting constants independent of  $s$ .

For restart, add the two equations in (8.2). The terms  $\pm cv$  cancel, so the evolution equation for  $v + w$  is independent of the chosen splitting and of  $c$ . Therefore setting the new  $v$  datum to zero and the new  $w$  datum to their old sum preserves  $v + w$ , hence preserves  $\Phi = -X + v + w$ .  $\square$

**Theorem 8.62** (Global a priori control for  $\Phi_3^4$ ). *For the renormalized dynamical  $\Phi_3^4$  equation on  $\mathbb{T}^3$ , started from arbitrary  $\Phi_0 \in B_{\infty, \infty}^{-1/2-\varepsilon}$  with sufficiently small  $\varepsilon > 0$ , the maximal lifetime satisfies  $T_* = \infty$  almost surely. For every  $q < \infty$ ,*

$$\mathbf{E} \left[ \sup_{0 < t \leq 1} \sup_{\Phi_0 \in B_{\infty, \infty}^{-1/2-\varepsilon}} \left( \sqrt{t} \|\Phi(t)\|_{B_{\infty, \infty}^{-1/2-\varepsilon}} \right)^q \right] < \infty. \quad (8.8)$$

With the massive stationary realization, the stronger unit-window estimate

$$\sup_{s \geq 0} \mathbf{E} \left[ \sup_{s < t \leq s+1} \sup_{\Phi_0 \in B_{\infty, \infty}^{-1/2-\varepsilon}} \left( ((\sqrt{t}) \wedge 1) \|\Phi(t)\|_{B_{\infty, \infty}^{-1/2-\varepsilon}} \right)^q \right] < \infty \quad (25.13)$$

holds. In particular the equation comes down from infinity and has no finite-time blow-up.

*Proof.* Work first with a smooth approximation  $w_0^{(n)}$  of  $\Phi_0 + X(0)$  in  $B_{\infty,\infty}^{-1/2-\varepsilon}$ , set  $v(0) = 0$ , and take  $c = CK^{30p}$  in the auxiliary system. For each  $n$  all positive Besov norms appearing at the initial time are finite, so Lemma 8.60 applies. Its conclusion is independent of those positive norms and of  $\|w_0^{(n)}\|$  itself:

$$\|v^{(n)}(t)\|_{B_{2p}^{-3\varepsilon}} + \sqrt{t} \|w^{(n)}(t)\|_{L^{3p-2}} \leq CK^\kappa.$$

The local stability statement in Theorem 8.48 lets  $n \rightarrow \infty$  on every interval strictly below the common local lifetime, and lower semicontinuity of the displayed norms preserves the estimate. This recovers the rough initial datum without inserting an unowned positive regularity assumption at time zero. For  $p$  large, both  $B_{2p}^{-3\varepsilon}$  and  $L^{3p-2}$  embed continuously into  $B_{\infty,\infty}^{-1/2-\varepsilon}$ : by Bernstein the loss from  $L^r$  to  $L^\infty$  is  $3/r$ , and  $3/(3p-2) < 1/2 + \varepsilon$  while  $-3\varepsilon - 3/(2p) > -1/2 - \varepsilon$ . The massive convolution  $X$  has the same target regularity with moments of every order. Thus

$$\sqrt{t} \|\Phi(t)\|_{B_{\infty,\infty}^{-1/2-\varepsilon}} \leq C(1 + K^\kappa),$$

uniformly over the incoming initial condition. Lemma 8.61 and Hölder's inequality yield (8.8) for every finite  $q$  (first choose an even  $p > \max\{24, q\}$ ; smaller moments follow by Lyapunov).

At time  $1/2$  restart the splitting as in Lemma 8.61, then repeat on the translated interval  $[1/2, 3/2]$ , and subsequently at half-integers. The  $K$ -moments are uniform under these translations, so the same estimate holds on every unit window and gives (25.13). At every positive restart time the solution therefore lies in the local-theory space with finite norm. The continuation criterion of Theorem 8.48 can fail only if that norm diverges, which (25.13) excludes on every compact time interval. Hence  $T_* = \infty$  almost surely.  $\square$

#### 8.4.2 Hairer–Matetski lattice approximation: the discrete analytic package

The continuum reconstruction and Schauder theorems from Chapter 3 do not by themselves control a lattice approximation uniformly as the mesh tends to zero. The point of the Hairer–Matetski theory is that one stops the multiscale analysis at the mesh, keeps time continuous, and proves all constants uniformly in the cutoff. We first make the comparison topology explicit.

**Definition 8.63** (Discrete–continuum Hölder comparison norm). Let  $\Lambda_\varepsilon^3 = (\varepsilon\mathbb{Z}/\mathbb{Z})^3$  and

$$\langle f, g \rangle_\varepsilon := \varepsilon^3 \sum_{x \in \Lambda_\varepsilon^3} f(x)g(x).$$

For  $\alpha < 0$  and a distribution  $u$  together with a lattice function  $u^\varepsilon$ , set

$$\|u; u^\varepsilon\|_{C^\alpha}^{(\varepsilon)} := \sup_{\substack{x \in \Lambda_\varepsilon^3 \\ \rho \in [\varepsilon, 1] \\ \varphi \in \mathcal{B}_0^r}} \rho^{-\alpha} |\langle u, \varphi_x^\rho \rangle - \langle u^\varepsilon, \varphi_x^\rho \rangle_\varepsilon|,$$

where  $r > -\lfloor \alpha \rfloor$ ,  $\mathcal{B}_0^r$  is the unit ball of compactly supported  $C^r$  test functions, and  $\varphi_x^\rho(y) = \rho^{-3} \varphi((y-x)/\rho)$ . For  $\delta > 0$  and  $\bar{\eta} \leq 0$ , define the weighted parabolic comparison seminorm on  $(0, T]$  by

$$\begin{aligned} \|u; u^\varepsilon\|_{C_{\bar{\eta}, T}^{\delta, \alpha}}^{(\varepsilon)} &:= \sup_{0 < t \leq T} |t|_0^{-\bar{\eta}} \|u_t; u_t^\varepsilon\|_{C^\alpha}^{(\varepsilon)} \\ &+ \sup_{\substack{0 < s < t \leq T \\ t-s \leq |s, t|_0^2}} |s, t|_0^{-\bar{\eta}} \frac{\|(u_t - u_s); (u_t^\varepsilon - u_s^\varepsilon)\|_{C^{\alpha-\delta}}^{(\varepsilon)}}{|t-s|^{\delta/2}}, \end{aligned}$$

where  $|t|_0 = t^{1/2} \wedge 1$  and  $|s, t|_0 = |s|_0 \wedge |t|_0$ . The corresponding one-object norms are obtained by replacing the difference pair by either entry. This is the part of the Hairer–Matetski topology used below; changing the finite family of test functions or the equivalent weighted normalisation changes only constants.

**Lemma 8.64** (Uniform discrete reconstruction). *Let  $Z^\varepsilon = (\Pi^\varepsilon, \Gamma^\varepsilon, \Sigma^\varepsilon)$  be an inhomogeneous discrete model on a regularity structure with minimal homogeneity  $\alpha_0 < 0$ , with model norms bounded uniformly in  $\varepsilon = 2^{-N}$ . Let  $H^\varepsilon$  be a discrete modelled distribution of regularity  $\gamma > 0$  and time weight  $\eta$ . Define*

$$(\mathcal{R}_t^\varepsilon H_t^\varepsilon)(x) := (\Pi_x^{\varepsilon, t} H_t^\varepsilon(x))(x), \quad x \in \Lambda_\varepsilon^3.$$

Then, uniformly for  $\rho \in [\varepsilon, 1]$ ,

$$\left| \langle \mathcal{R}_t^\varepsilon H_t^\varepsilon - \Pi_x^{\varepsilon, t} H_t^\varepsilon(x), \varphi_x^\rho \rangle_\varepsilon \right| \leq C \rho^\gamma |t|_\varepsilon^{\eta-\gamma} \|H^\varepsilon\|_{\gamma, \eta; T}^{(\varepsilon)} \|\Pi^\varepsilon\|_{\gamma; T}^{(\varepsilon)}. \quad (25.14)$$

If two discrete models and modelled distributions are compared, the same estimate holds for their difference. If one entry is continuous, the mixed continuous/discrete reconstruction difference satisfies the analogous bound in the norm of Definition 8.63.

*Proof.* Choose a compactly supported  $C^r$  Daubechies scaling function with  $r > -\lfloor \alpha_0 \rfloor$  and its associated wavelets. At mesh  $\varepsilon = 2^{-N}$  the discrete multiresolution spaces are obtained by restricting the scaling functions to  $\Lambda_\varepsilon^3$  and using the inner product  $\langle \cdot, \cdot \rangle_\varepsilon$ . Orthogonality and the refinement identity are unchanged up to this normalisation.

Fix  $\rho \in [\varepsilon, 1]$  and take  $n_0$  minimal with  $2^{-n_0} \leq \rho$ . The reconstruction difference has the finite telescoping expansion

$$P_{n_0}^\varepsilon(\zeta_x^\varepsilon) - \mathcal{R}_t^\varepsilon H_t^\varepsilon = \sum_{n=n_0}^{N-1} (P_n^\varepsilon - P_{n+1}^\varepsilon)(\zeta_x^\varepsilon - \zeta_y^\varepsilon), \quad \zeta_y^\varepsilon := \Pi_y^{\varepsilon, t} H_t^\varepsilon(y). \quad (25.15)$$

For a wavelet  $\psi_z^n$  of scale  $2^{-n}$ , model coherence gives

$$|\langle \zeta_y^\varepsilon - \zeta_x^\varepsilon, \psi_z^n \rangle_\varepsilon| \leq C \sum_{\ell < \gamma} 2^{-n(3/2+\ell)} |x - y|^{\gamma-\ell} |t|_\varepsilon^{\eta-\gamma}.$$

Only  $O(1)$  wavelets at scale  $n$  meet the support of  $\varphi_x^\rho$ , while the vanishing moments of  $\psi_z^n$  imply  $|\langle \psi_z^n, \varphi_x^\rho \rangle_\varepsilon| \leq C 2^{-n(r+3/2)} \rho^{-r-3}$  when  $2^{-n} \leq \rho$ . Multiplying these bounds and summing first in the finite set of locations and then in  $n \geq n_0$  gives a geometric series with leading size  $\rho^\gamma$ . There is no scale below  $N$ : this is precisely why the constant is independent of  $\varepsilon$ . The coarse term at  $n_0$  is estimated directly from the model bound and has the same order. This proves (25.14).

For two models, subtract the two telescoping expansions. Every coefficient splits into a model difference plus a modelled-distribution difference, so the same summation yields the Lipschitz estimate. In the continuous/discrete case use the continuous expansion down to scale  $N$  and compare it with the discrete one there; the continuous tail below  $\varepsilon$  is  $O(\varepsilon^\theta)$  for some  $\theta > 0$  because the wavelets have more vanishing moments than the minimal homogeneity requires. This is the mixed estimate.  $\square$

**Lemma 8.65** (Uniform lattice heat-kernel decomposition and Schauder estimate). *Let  $\Delta_\varepsilon$  be the nearest-neighbour Laplacian on  $\Lambda_\varepsilon^3$  and  $G_t^\varepsilon = e^{t\Delta_\varepsilon}$ . There is a decomposition*

$$G^\varepsilon = K^\varepsilon + R^\varepsilon$$

where  $R^\varepsilon$  is smooth uniformly in  $\varepsilon$  and  $K^\varepsilon = \sum_{0 \leq n < N} K_n^\varepsilon + \tilde{K}^\varepsilon$  has parabolic scale  $2^{-n}$ , annihilates the required discrete polynomials, and satisfies for every multiindex  $k$

$$|D_\varepsilon^k K_n^\varepsilon(z)| \leq C_k 2^{n(3+2+|k|_s-2)}, \quad \text{supp } K_n^\varepsilon \subset \{\|z\|_s \leq C2^{-n}\}, \quad (25.16)$$

with the obvious mesh-scale version for  $\tilde{K}^\varepsilon$ . Consequently the lifted lattice convolution  $\mathcal{K}_\gamma^\varepsilon$  obeys the same weighted Schauder and two-model Lipschitz estimates as the continuum operator of SMX-25.03.02, with constants independent of  $\varepsilon$ .

*Proof.* On the Fourier side the nearest-neighbour symbol is

$$q_\varepsilon(k) = 4\varepsilon^{-2} \sum_{j=1}^3 \sin^2(\pi \varepsilon k_j).$$

For  $|k| \leq c\varepsilon^{-1}$ ,  $q_\varepsilon(k) \asymp |k|^2$  and every discrete derivative of the multiplier  $e^{-tq_\varepsilon(k)}$  satisfies the same parabolic estimates as its continuum counterpart; for frequencies comparable with  $\varepsilon^{-1}$  the exponential factor gives uniform rapid decay once  $t \gtrsim \varepsilon^2$ . Dyadically localising in  $\|(t, x)\|_s$  therefore gives (25.16). Subtracting finitely many discrete Taylor polynomials on each dyadic annulus forces the required moment cancellations; the corrections are smooth and are absorbed into  $R^\varepsilon$ . The remaining ball  $\|z\|_s \lesssim \varepsilon$  is  $\tilde{K}^\varepsilon$ .

Now repeat the proof of the abstract Schauder estimate in SMX-25.03.02. Every scale  $2^{-n} > \varepsilon$  is estimated with (25.16), while at the terminal mesh scale the Taylor remainder is bounded directly. The sums are again geometric and stop at  $n = N$ , so their constants are uniform. The identity

$$\mathcal{R}_t^\varepsilon(\mathcal{K}_\gamma^\varepsilon H)_t(x) = \int_0^t \langle \mathcal{R}_s^\varepsilon H_s, K_{t-s}^\varepsilon(x - \cdot) \rangle_\varepsilon ds \quad (25.17)$$

follows by the definition of admissibility. Subtracting two expansions gives the Lipschitz form. Combining this with Lemma 8.64 yields a contraction time depending only on bounds for the initial data and the model, not on  $\varepsilon$ .  $\square$

**Lemma 8.66** (Stationary-chaos criterion for uniform discrete model norms). *Let  $\tau$  be a negative-homogeneity basis symbol and suppose the stationary lattice realisation of  $\Pi^\varepsilon \tau$  has the finite chaos expansion*

$$\langle \Pi_0^{\varepsilon, t} \tau, \varphi \rangle_\varepsilon = \sum_{k \leq N_\tau} I_k^\varepsilon \left( \int_{\Lambda_\varepsilon^3} \varphi(x) W_\tau^{(\varepsilon, k)}(t, x) dx \right).$$

Write  $Q_\tau^{(\varepsilon, k)}(z_1, z_2)$  for the Hilbert-space inner product of the two kernels  $W_\tau^{(\varepsilon, k)}(z_i)$ . Assume that for some  $a_\tau > |\tau| \vee (-3/2)$  and  $\delta_\tau > 0$  one has uniformly in the mesh

$$\begin{aligned} |Q_\tau^{(\varepsilon, k)}((0, x_1), (0, x_2))| &\leq C \sum_{\zeta \in E_{au}} (\|(0, x_1)\|_{s, \varepsilon} + \|(0, x_2)\|_{s, \varepsilon})^\zeta \|(0, x_1 - x_2)\|_{s, \varepsilon}^{2a_\tau - \zeta}, \\ |\delta_{0, t} Q_\tau^{(\varepsilon, k)}(x_1, x_2)| &\leq C |t|^{2\delta_\tau/2} \sum_{\zeta \in E_{au}} (\|(t, x_1)\|_{s, \varepsilon} + \|(t, x_2)\|_{s, \varepsilon})^\zeta \|(0, x_1 - x_2)\|_{s, \varepsilon}^{2a_\tau - 2\delta_\tau - \zeta}, \end{aligned} \quad (25.18a)$$

for a finite set  $E_\tau$  such that every exponent of the last singular factor is  $> -3$ . Then, after an arbitrarily small loss of homogeneity,

$$\mathbf{E} |\langle \Pi_0^{\varepsilon, t} \tau, \varphi_x^\rho \rangle_\varepsilon|^2 \lesssim \rho^{2|\tau| + \kappa}, \quad \mathbf{E} |\langle (\Pi^{\varepsilon, t} - \Pi^{\varepsilon, s})_0 \tau, \varphi_x^\rho \rangle_\varepsilon|^2 \lesssim |t - s|^{\delta_\tau} \rho^{2(|\tau| - \delta_\tau) + \kappa}. \quad (25.18b)$$

If the same bounds for a difference of two kernel families carry an additional factor  $q^2$ , then the corresponding model distance has  $L^p$  norm  $O(q)$ .

*Proof.* Wiener isometry reduces the second moment of the  $k$ th chaos component to

$$\iint |\varphi_x^\rho(y_1) \varphi_x^\rho(y_2)| |Q_\tau^{(\varepsilon, k)}((t, y_1), (t, y_2))| dy_1 dy_2.$$

The test functions contribute  $\rho^{-6}$  and restrict both variables to a ball of radius  $O(\rho)$ . For one term of (25.18a), integrate first in  $h = y_1 - y_2$  and then in  $y_2$ . Since  $2a_\tau - \zeta > -3$ , the discrete radial sum obeys the same bound as the integral,

$$\varepsilon^3 \sum_{|h| \lesssim \rho} (|h| \vee \varepsilon)^{2a_\tau - \zeta} \lesssim \rho^{2a_\tau - \zeta + 3}.$$

The remaining factor is  $O(\rho^\zeta)$  and the  $y_2$  sum contributes  $O(\rho^3)$ ; after  $\rho^{-6}$  the result is  $O(\rho^{2a_\tau})$ . Choose  $a_\tau > |\tau|$  arbitrarily close to  $|\tau|$  and absorb the difference as the small exponent  $\kappa$ . The time-increment estimate is identical, using the second line of (25.18a).

Each random variable lies in a fixed finite Wiener chaos, so hypercontractivity turns the second-moment estimates into all finite moments. Kolmogorov's criterion in time and the discrete wavelet supremum used in Lemma 8.64 turn these pointwise estimates into uniform model norms. The algebraic identities for products and integrations then propagate the bounds from the generating negative symbols to the finite truncated structure. Repeating the calculation for a difference kernel gives the final claim.  $\square$

**Lemma 8.67** (Renormalised discrete Gaussian models). *Let white noise be coupled to the lattice by cell averaging,*

$$\xi^\varepsilon(t, x) = \varepsilon^{-3} \langle \xi(t, \cdot), \mathbf{1}_{\{| \cdot - x |_\infty \leq \varepsilon/2\}} \rangle,$$

*and let  $K^\varepsilon$  be the singular part of the lattice heat kernel from Lemma 8.65. Define*

$$C_1^{(\varepsilon)} := \int_{\mathbb{R} \times \Lambda_\varepsilon^3} (K^\varepsilon(z))^2 dz, \quad C_2^{(\varepsilon)} := 2 \int (K^\varepsilon \star_\varepsilon K^\varepsilon(z))^2 K^\varepsilon(z) dz. \quad (25.18)$$

*Then  $C_1^{(\varepsilon)} = O(\varepsilon^{-1})$  and  $C_2^{(\varepsilon)} = O(|\log \varepsilon|)$ , and the renormalisation constant for coupling  $\lambda > 0$  is*

$$C_\lambda^{(\varepsilon)} = 3\lambda C_1^{(\varepsilon)} - 9\lambda^2 C_2^{(\varepsilon)}. \quad (25.19)$$

*The corresponding stationary renormalised discrete models  $\widehat{Z}^\varepsilon$  satisfy, for every finite  $p$  and every fixed model regularity window,*

$$\sup_{\varepsilon \leq 1} \mathbf{E}[\|\widehat{Z}^\varepsilon\|^p] < \infty. \quad (25.20)$$

*Moreover, after coupling to the same white noise, they converge in probability to the continuum BPHZ model in the mixed discrete–continuum model metric.*

*Proof.* Every model component belongs to a finite Wiener chaos. Its second moment is a finite sum of integrals of products of the lattice kernels. By Lemma 8.65, each kernel satisfies the same dyadic singularity bounds as the continuum heat kernel uniformly above the mesh, and the terminal scale contributes exactly the value predicted by replacing the smallest distance by  $\varepsilon$ . The analytic BPHZ theorem SMX-25.03.04 applies to these dyadic kernels verbatim: its forest/cut power-counting inequalities are scale inequalities and are unaffected by terminating the scale tree at  $N$ . It gives the second-moment model bounds; Gaussian hypercontractivity upgrades them to every finite  $p$ .

For the two divergent local contractions the power counting is explicit. Since the heat kernel has parabolic degree  $-3$  in dimension  $3 + 1$ ,

$$\int_{\|z\|_s \geq \varepsilon} (K(z))^2 dz \asymp \int_{\varepsilon}^1 r^{-2} dr = O(\varepsilon^{-1}),$$

which is the first estimate. The sunset diagram has degree zero after the BPHZ subtractions, so its dyadic shells contribute  $O(1)$  each and hence  $O(|\log \varepsilon|)$ . The combinatorial coefficients of the cubic equation are  $3\lambda$  and  $-9\lambda^2$ , giving (25.19).

To prove convergence, introduce a second scale  $\bar{\varepsilon} \geq \varepsilon$  and mollify the common white noise at scale  $\bar{\varepsilon}$ . For every fixed  $\bar{\varepsilon}$ , the discrete heat kernels and all their finite differences converge uniformly on the support of the mollified diagrams to their continuum counterparts. Hence every fixed chaos kernel converges. The difference between mollified and unmollified models is bounded by the same BPHZ multiscale sum with one distinguished kernel difference; the cancellation of the mollifier gives an additional factor  $\bar{\varepsilon}^\theta$  for some  $\theta > 0$ . Thus, uniformly for  $\varepsilon \leq \bar{\varepsilon}$ ,

$$\mathbf{E}\|\widehat{Z}^\varepsilon; \widehat{Z}^{\varepsilon, \bar{\varepsilon}}\|^p \leq C\bar{\varepsilon}^{\theta p}, \quad \mathbf{E}\|\widehat{Z}; \widehat{Z}^{0, \bar{\varepsilon}}\|^p \leq C\bar{\varepsilon}^{\theta p}. \quad (25.21)$$

First send  $\varepsilon \rightarrow 0$  at fixed  $\bar{\varepsilon}$ , then send  $\bar{\varepsilon} \rightarrow 0$ . This proves (25.20) and convergence of the models.  $\square$

**Lemma 8.68** (Smooth two-scale lattice comparison). *Fix  $\bar{\varepsilon} > 0$  and replace the driving white noise by a smooth mollification at scale  $\bar{\varepsilon}$ . On every time interval on which the continuum and lattice solutions stay bounded in a fixed  $C^4$  norm, the nearest-neighbour lattice equation and its continuum counterpart satisfy*

$$\sup_{t \leq T} \|u_t - u_t^\varepsilon\|_{L^\infty} \leq C_{T, \bar{\varepsilon}, M} (\|u_0 - u_0^\varepsilon\|_{L^\infty} + \varepsilon). \quad (25.22)$$

*The same conclusion holds in each weaker Hölder comparison norm used in Definition 8.63.*

*Proof.* For a smooth periodic  $f$ , Taylor expansion in each coordinate gives

$$\|\Delta_\varepsilon f - \Delta f\|_{L^\infty} \leq C\varepsilon^2 \|f\|_{C^4}.$$

Cell averaging a  $C^1$  forcing changes it by at most  $C\varepsilon \|f\|_{C^1}$ . Subtract the two smooth equations. On a ball  $|u|, |u^\varepsilon| \leq M$ , the cubic drift is Lipschitz with constant  $C(1 + M^2)$ , while the renormalised coefficients at fixed  $\bar{\varepsilon}$  converge because the corresponding smooth kernel integrals are Riemann sums. Duhamel's formula and the  $L^\infty$  contraction of the heat semigroups therefore give

$$E(t) \leq E(0) + C\varepsilon + C \int_0^t E(s) ds.$$

Gronwall proves (25.22). Pairing with test functions and applying the same argument to time increments yields the weaker Hölder comparison norms.  $\square$

**Lemma 8.69** (Stopped solution transfer). *Suppose the initial data converge in the discrete-continuum  $C^\eta$  comparison norm and the renormalised discrete models converge as in Lemma 8.67. Let  $\Phi$  be the maximal continuum solution with lifetime  $T_*$  and let  $\Phi^\varepsilon$  be the lattice solution. For every  $K > 0$ , let  $\tau_K$  be the first time the continuum solution norm in Definition 8.63 reaches  $K$ . Then*

$$\lim_{\varepsilon \rightarrow 0} \mathbf{P} \left( \|\Phi; \Phi^\varepsilon\|_{C_{\bar{\eta}, \tau_K}^{\delta, \alpha}}^{(\varepsilon)} > c \right) = 0 \quad (c > 0). \quad (25.23)$$

*Consequently there is a diagonal sequence of stopping times  $T_\varepsilon \rightarrow T_*$  in probability for which the same convergence holds with  $\tau_K$  replaced by  $T_\varepsilon$ .*



*Proof.* Insert the intermediate solutions driven by noise mollified at a scale  $\bar{\varepsilon} \geq \varepsilon$ :

$$\Phi - \Phi^\varepsilon = (\Phi - \Phi^{0,\bar{\varepsilon}}) + (\Phi^{0,\bar{\varepsilon}} - \Phi^{\varepsilon,\bar{\varepsilon}}) + (\Phi^{\varepsilon,\bar{\varepsilon}} - \Phi^\varepsilon). \quad (25.24)$$

Stop additionally when the continuum and discrete model norms exceed a level  $\bar{K} > K$  or when one of the three differences in (25.24) exceeds 1. On that stopped event, Lemmas 8.64 and 8.65 make the continuum and discrete solution maps uniformly Lipschitz. The first and third terms of (25.24) therefore converge to zero in probability as  $\bar{\varepsilon} \downarrow 0$ , uniformly in the mesh, by (25.21). For fixed  $\bar{\varepsilon}$ , the middle term converges to zero as  $\varepsilon \downarrow 0$  by Lemma 8.68.

It remains to remove the auxiliary stopping. The model moment estimate (25.20) and Markov's inequality show that the probability of a model norm exceeding  $\bar{K}$  tends to zero as  $\bar{K} \rightarrow \infty$ , uniformly in  $\varepsilon$ . The three already-proved convergence statements imply that the probability of hitting a difference cutoff before the norm cutoff tends to zero in the order  $\varepsilon \rightarrow 0$ ,  $\bar{\varepsilon} \rightarrow 0$ ,  $\bar{K} \rightarrow \infty$ . This yields (25.23). Finally  $\tau_K \uparrow T_*$  in probability as  $K \rightarrow \infty$ ; choose  $K = K(\varepsilon) \uparrow \infty$  sufficiently slowly to obtain  $T_\varepsilon$ .  $\square$

**Theorem 8.70** (Hairer–Matetski convergence of lattice discretizations of  $\Phi_3^4$ ). *Let  $\xi$  be space-time white noise on  $\mathbb{R} \times \mathbb{T}^3$ ,  $a \in \mathbb{R}$ ,  $\lambda > 0$ , and let  $\Phi_0 \in C^\eta(\mathbb{T}^3)$  almost surely for some  $\eta > -2/3$ . Let  $\Phi$  be the unique maximal continuum dynamical  $\Phi_3^4$  solution on  $[0, T_*)$ . For  $\varepsilon = 2^{-N}$  let  $\Delta_\varepsilon$  be the nearest-neighbour lattice Laplacian, let  $\xi^\varepsilon$  be the cell-averaged noise from Lemma 8.67, and let  $\Phi_0^\varepsilon$  be lattice initial data, independent of the driving white noise  $\xi$ , such that*

$$\|\Phi_0; \Phi_0^\varepsilon\|_{C^\eta}^{(\varepsilon)} \longrightarrow 0 \quad \text{almost surely.} \quad (25.25)$$

Let  $\Phi^\varepsilon$  solve

$$\partial_t \Phi^\varepsilon = \Delta_\varepsilon \Phi^\varepsilon + (C_\lambda^{(\varepsilon)} - a)\Phi^\varepsilon - \lambda(\Phi^\varepsilon)^3 + \xi^\varepsilon, \quad (25.26)$$

with  $C_\lambda^{(\varepsilon)}$  given by (25.19). Then, for every  $\alpha < -1/2$ , every  $\bar{\eta} < \eta \wedge \alpha$ , and every sufficiently small  $\delta > 0$ , there are stopping times  $T_\varepsilon$  such that

$$T_\varepsilon \xrightarrow{\mathbf{P}} T_*, \quad \|\Phi; \Phi^\varepsilon\|_{C_{\bar{\eta}, T_\varepsilon}^{\delta, \alpha}}^{(\varepsilon)} \xrightarrow{\mathbf{P}} 0. \quad (25.27)$$

Moreover  $C_\lambda^{(\varepsilon)}$  is the sum of a term proportional to  $\lambda\varepsilon^{-1}$  and a term proportional to  $-\lambda^2 \log \varepsilon$ , up to finite parts depending on the chosen kernel normalisation.

In the present book, Theorem 8.62 gives  $T_* = \infty$  almost surely. Therefore for every deterministic  $T < \infty$  the convergence in (25.27) holds on  $[0, T]$  in probability after replacing  $T_\varepsilon$  by  $T_\varepsilon \wedge T$ ; equivalently, the stopping restriction disappears on every fixed compact time interval.

*Proof.* The renormalised discrete models are uniformly bounded and converge to the continuum BPHZ model by Lemma 8.67. The discrete reconstruction and Schauder estimates required by the fixed point are Lemmas 8.64 and 8.65; their constants are uniform in the mesh. The local solution map is therefore uniformly Lipschitz on every ball in initial-data/model norm. The stopped comparison and diagonal argument are exactly Lemma 8.69, which proves (25.27).

For fixed  $\varepsilon$  the lattice equation is a finite-dimensional SDE with a coercive cubic drift, hence has a unique global solution. The final assertion uses the independently proved continuum global estimate Theorem 8.62: on every deterministic  $[0, T]$  the continuum solution norm is finite almost surely, so  $\mathbf{P}(\tau_K < T) \rightarrow 0$  as  $K \rightarrow \infty$ . Apply (25.23), then let  $K \rightarrow \infty$ .  $\square$

### 8.4.3 Non-Gaussian weak universality: the Shen–Xu mechanism

The point of the next theorem is not merely that non-Gaussian noise satisfies a central limit theorem. The microscopic equation contains arbitrarily high odd powers, and these powers are multiplied by different powers of the scaling parameter. The proof has to show simultaneously that these apparently non-subcritical terms fit into a finite weighted regularity structure, that all non-Gaussian cumulant contractions are summable, and that the one logarithmic mass divergence can be removed by moving along the microscopic pitchfork parameter.

**Definition 8.71** (Shen–Xu microscopic data). Fix an integer  $m \geq 1$ . For each  $\varepsilon \in (0, 1)$  let  $\zeta$  be a continuous stationary random field on  $\mathbb{R} \times (\varepsilon^{-1}\mathbb{T})^3$  such that

1.  $\zeta \stackrel{\text{law}}{=} -\zeta$  and  $\mathbf{E}|\zeta(z)|^p < \infty$  for every  $p > 1$ ;
2. its covariance  $\varrho(z) = \mathbf{E}[\zeta(0)\zeta(z)]$  is normalized by  $\int \varrho(z) dz = 1$ ;
3. if compact sets  $K_1, K_2$  have parabolic distance at least 1, then the sigma-fields generated by  $\zeta|_{K_1}$  and  $\zeta|_{K_2}$  are independent.

Let  $\Psi$  be the stationary solution of  $(\partial_t - \Delta)\Psi = \zeta$ , let  $\mu$  be its one-point law, and put

$$\langle V_\theta \rangle(x) := \int_{\mathbb{R}} V_\theta(x+y) \mu(dy).$$

The potential  $V_\theta$  is an even polynomial in  $x$ , with coefficients smooth in  $\theta$ , and we write

$$\langle V_\theta \rangle'(x) = \sum_{j=0}^m \hat{a}_j(\theta) x^{2j+1}.$$

We impose the non-degenerate pitchfork conditions

$$\hat{a}_1(0) > 0, \quad \hat{a}_0(0) = 0, \quad \hat{a}'_0(0) < 0. \quad (25.28)$$

Finally

$$\zeta_\varepsilon(t, x) = \varepsilon^{-5/2} \zeta(t/\varepsilon^2, x/\varepsilon), \quad u_\varepsilon(t, x) = \varepsilon^{-1/2} u(t/\varepsilon^2, x/\varepsilon).$$

*Remark 8.72* (Cubic-coupling normalization). With the coefficient convention in Definition 8.71, the coefficient of  $x^3$  in  $\langle V_0 \rangle'$  is  $\hat{a}_1(0)$  and

$$\partial_x^4 \langle V \rangle(0, 0) = 6\hat{a}_1(0).$$

The source’s introductory Theorem 1.1 and the proof of Theorem 5.2 use  $\hat{a}_1$  as the cubic coupling, i.e. the coefficient of  $x^3$  in  $\langle V_0 \rangle'$ . The displayed statement of Theorem 5.2 writes  $\hat{a}_1 = \partial_x^4 \langle V \rangle(0, 0)$ . With the coefficient convention (1.8) used in the same paper, however,  $\partial_x^4 \langle V \rangle(0, 0) = 6\hat{a}_1(0)$ . Throughout this book we therefore keep the coefficient convention

$$\lambda_* := \hat{a}_1(0) > 0,$$

so that equivalently  $\lambda_* = \frac{1}{6} \partial_x^4 \langle V \rangle(0, 0)$ . The factor-six mismatch in the displayed endpoint is recorded explicitly rather than being absorbed into the definition of the limiting cubic coupling.

**Lemma 8.73** (Non-Gaussian Wick/Appell inversion). *Let  $\nu$  be a probability law with moments of every order. There is a unique monic polynomial  $W_{n,\nu}$  of degree  $n$  such that*

$$\int W_{n,\nu}(x+y) \nu(dy) = x^n. \quad (25.29)$$

*The family is Appell:  $W'_{n,\nu} = nW_{n-1,\nu}$ . Consequently, if  $\langle P \rangle(x) = \int P(x+y) \nu(dy) = \sum b_n x^n$ , then*

$$P(x) = \sum_n b_n W_{n,\nu}(x). \quad (25.30)$$

*If  $\nu_r$  is the law of  $r^{-1}Y$  for  $Y \sim \nu$ , then*

$$W_{n,\nu_r}(x) = r^{-n} W_{n,\nu}(rx). \quad (25.31)$$

*In particular, if  $\mu_\varepsilon$  is the one-point law of the stationary solution of  $(\partial_t - \Delta)\Psi_\varepsilon = \zeta_\varepsilon$ , the rescaled phase-coexistence equation is exactly*

$$(\partial_t - \Delta)u_\varepsilon = - \sum_{j=0}^m \hat{a}_j(\theta) \varepsilon^{j-1} W_{2j+1,\mu_\varepsilon}(u_\varepsilon) + \zeta_\varepsilon. \quad (25.32)$$

*Proof.* Write  $m_k = \int y^k \nu(dy)$ . The averaging operator  $AP(x) := \int P(x+y) \nu(dy)$  is triangular in the monomial basis and has unit diagonal:

$$A(x^n) = x^n + \sum_{k < n} \binom{n}{k} m_{n-k} x^k.$$

It is therefore an automorphism of the polynomials of degree at most  $n$ . Define  $W_{n,\nu} = A^{-1}(x^n)$ ; this proves existence, uniqueness, monicity and (25.29), and linearity gives (25.30). Since differentiation commutes with translation and hence with  $A$ ,  $A(W'_{n,\nu}) = nx^{n-1} = A(nW_{n-1,\nu})$ , proving the Appell identity. For (25.31), average the right-hand side against the law of  $r^{-1}Y$  and use (25.29); uniqueness then gives the claim.

Parabolic scaling of the stationary convolution gives  $\Psi_\varepsilon(t, x) = \varepsilon^{-1/2} \Psi(t/\varepsilon^2, x/\varepsilon)$ , hence  $\mu_\varepsilon$  is the dilation of  $\mu$  by  $\varepsilon^{-1/2}$ . Apply (25.30)–(25.31) to  $V'_\theta$  and to the rescaled equation. This gives (25.32) term by term; no Gaussian orthogonality has been used.  $\square$

**Lemma 8.74** (The  $\varepsilon$ -weighted regularity structure and classical renormalized equation). *Let  $|\Xi|_5 = -5/2 - \kappa$ ,  $|\mathcal{I}\tau| = |\tau| + 2$ , and introduce a bookkeeping operator  $\mathcal{E}$  of degree 1 whose model realization is multiplication by  $\varepsilon$ . For fixed  $m$  and sufficiently small  $\kappa > 0$ , the negative sector generated by*

$$\Xi, \quad \mathcal{E}^{k-1} \Psi^{2k+1-n} \quad (n = 0, 1, 2, 3), \quad (25.33)$$

*and by the second-order symbols generated when these expressions are inserted once into the heat integration map is finite. The only second-order family that needs a separate power-counting argument can be represented as*

$$\mathcal{E}^{\lfloor (k-1)/2 \rfloor} \left( \Psi^k \mathcal{I}(\mathcal{E}^{\lfloor \ell/2 \rfloor - 1} \Psi^\ell) \right), \quad (25.34)$$

*where the displayed expression is used only when  $\lfloor \ell/2 \rfloor - 1 \geq 0$  (in particular  $\ell \geq 2$ ), and with the source exclusion “ $k$  odd and  $\ell$  even”. These are exactly the second-order negative families occurring in the source enumeration.*

For the Wick/Appell renormalization relative to  $\mu_\varepsilon$  and local mass counterterms  $C_{k,\ell}^{(\varepsilon)}$ , reconstruction of the abstract fixed point solves

$$(\partial_t - \Delta)v_\varepsilon = - \sum_{j=1}^m \lambda_j^{(\varepsilon)} \varepsilon^{j-1} W_{2j+1,\mu_\varepsilon}(v_\varepsilon) - \lambda_0^{(\varepsilon)} v_\varepsilon - C_\varepsilon v_\varepsilon + \zeta_\varepsilon, \quad (25.35)$$

where

$$C_\varepsilon = \sum_{k,\ell=1}^m \lambda_k^{(\varepsilon)} \lambda_\ell^{(\varepsilon)} \left( (2k+1)(2k)C_{2k-1,2\ell+1}^{(\varepsilon)} + (2k+1)(2\ell+1)C_{2k,2\ell}^{(\varepsilon)} \right). \quad (25.36)$$

The local fixed-point map is uniformly Lipschitz on bounded sets of the  $\varepsilon$ -weighted model and initial-data norms for

$$1 < \gamma < 6/5, \quad -\frac{2m+2}{4m+3} < \eta < -1/2. \quad (25.37)$$

More precisely, let  $Z_\varepsilon$  be weighted models and  $Z$  a limiting model, let  $\phi_0^{(\varepsilon)} \rightarrow \phi_0$  in the  $\mathcal{C}_\varepsilon^{\gamma,\eta}$  comparison norm, and let  $\lambda_j^{(\varepsilon)} \rightarrow \lambda_j$ . If  $\|Z_\varepsilon; Z\|_{\varepsilon;0} \rightarrow 0$  and the limiting abstract fixed point exists on  $[0, T]$ , then for all sufficiently small  $\varepsilon$  the microscopic fixed point exists on the same interval and

$$\|\Phi_\varepsilon; \Phi\|_{\gamma,\eta;\varepsilon} + \sup_{t \leq T} \|\mathcal{R}^\varepsilon \Phi_\varepsilon(t) - \mathcal{R}\Phi(t)\|_{\mathcal{C}^\eta} \rightarrow 0. \quad (25.37a)$$

*Proof.* The homogeneity of a first-order symbol in (25.33) is

$$k-1 - (2k+1-n)(1/2 + \kappa) = -\frac{3-n}{2} - (2k+1-n)\kappa,$$

which is the source degree. Since  $m$  is fixed, only finitely many  $k$  occur. One insertion of the right-hand side into the integration map produces (25.34); further insertions either have positive homogeneity or belong to the ordinary  $\Phi_3^4$  sector. This gives the finite negative sector used by the fixed point.

The weighted spaces add to the ordinary modelled-distribution norm the two small-scale controls

$$\sup_z \sup_{\beta < \gamma} \frac{|U(z)|_\beta}{\varepsilon^{(\eta-\beta)\wedge 0}}, \quad \sup_{|z-z'| < \varepsilon \wedge \sqrt{|t|\wedge|t'|}} \sup_{\beta < \gamma} \frac{|U(z) - \Gamma_{zz'}U(z')|_\beta}{|z-z'|^{\gamma-\beta} \varepsilon^{\eta-\gamma}}.$$

Multiplication, integration and  $\mathcal{E}$  preserve these bounds because  $\mathcal{E}$  contributes exactly one factor  $\varepsilon$  and one unit of homogeneity. For the nonlinear map, a monomial  $\widehat{\mathcal{E}}^{j-1}\Phi^{2j+1}$  has abstract homogeneity obtained from that of  $\Phi^{2j+1}$  by adding  $j-1$ . At scales below  $\varepsilon$  this is exactly compensated by the model factor  $\varepsilon^{j-1}$ ; at larger scales the positive homogeneity of  $\mathcal{E}^{j-1}$  improves the ordinary product estimate. Near the initial-time hyperplane the worst weight is the  $j=m$  term. Substitution into the weighted heat-convolution estimate gives the strict condition  $\eta > -(2m+2)/(4m+3)$ , while  $\gamma < 6/5$  leaves a positive Schauder margin. Thus, on a ball of radius  $R$  in  $\mathcal{D}_\varepsilon^{\gamma,\eta}$ , the fixed-point map  $\mathcal{F}_\varepsilon$  satisfies, with constants uniform for bounded models and coefficients,

$$\|\mathcal{F}_\varepsilon(U)\|_{\gamma,\eta;\varepsilon} \leq C_R + C_R T^\theta, \quad \|\mathcal{F}_\varepsilon(U) - \mathcal{F}_\varepsilon(V)\|_{\gamma,\eta;\varepsilon} \leq C_R T^\theta \|U - V\|_{\gamma,\eta;\varepsilon}$$

for some  $\theta > 0$ . Choose  $T$  so that  $C_R T^\theta < 1/2$ ; this gives the uniform local fixed point. For two data sets, the same multilinear estimates give

$$\|\mathcal{F}_\varepsilon(U) - \mathcal{F}_\varepsilon(V)\| \leq \frac{1}{2} \|U - V\| + C_R (\|Z_\varepsilon; Z\|_{\varepsilon;0} + \|\phi_0^{(\varepsilon)}; \phi_0\|_{\gamma,\eta;\varepsilon} + \max_j |\lambda_j^{(\varepsilon)} - \lambda_j|).$$

Absorption yields convergence on the first local interval. If the limiting solution exists up to  $T$ , its modelled-distribution norm is bounded on each compact subinterval before  $T$ ; repeating the uniform contraction on finitely many overlapping intervals proves existence of the microscopic solutions on  $[0, T]$  and (25.37a). Reconstruction stability is included because the full comparison norm contains the stochastic-convolution term at time zero and the ordinary model distance thereafter.

Finally the Appell part of the renormalization turns ordinary powers into  $W_{n, \mu_\varepsilon}$ , while the only remaining local contractions are the two-vertex mass contractions indexed by  $(k, \ell)$ . Counting the place of the contracted leg gives respectively  $(2k+1)(2k)$  and  $(2k+1)(2\ell+1)$  possibilities, yielding (25.36). Reconstruction commutes with the heat integration map, so the abstract fixed point reconstructs to (25.35).  $\square$

**Lemma 8.75** (Specialized cumulant-graph summation criterion). *Work in parabolic dimension  $|\mathfrak{s}| = 5$ . Let  $(H, \mathcal{E})$  be a connected labelled graph with distinguished test vertex 0. Write  $H_0$  for the vertices other than 0,  $H_\star$  for the vertices allowed in the large-scale test, and decompose vertices into internal and external noise vertices. Ordinary kernel edges  $e$  have singularity degree  $a_e$  and Taylor label  $r_e$ , while a cumulant hyperedge joining  $q$  noise vertices carries scaling degree  $5q/2$ . For  $A \subset H$ , let  $E_0(A)$  be the ordinary edges wholly contained in  $A$ , and let  $E^\uparrow(A)$  and  $E^\downarrow(A)$  be the positively-renormalised outgoing and incoming edges in the source convention. Assume*

$$a_e + (r_e \wedge 0) < 5, \quad (25.38a)$$

$$\sum_{e \in E_0(A)} a_e < 5 \left( |A_{\text{in}}| + \frac{1}{2} (|A_{\text{ex}}| - 1 - \mathbf{1}_{A_{\text{ex}}=\emptyset}) \right) \quad (25.38)$$

for every  $A \subset H_0$  with  $|A| \geq 3$ ,

$$\sum_{e \in E_0(A)} a_e + \sum_{e \in E^\uparrow(A)} (a_e + r_e - 1) - \sum_{e \in E^\downarrow(A)} r_e < 5 \left( |A_{\text{in}}| + \frac{1}{2} |A_{\text{ex}}| \right) \quad (25.39)$$

for every  $A \subset H$  containing 0 with  $|A| \geq 2$ , and

$$\sum_{e \in E(A) \setminus E^\downarrow(A)} a_e + \sum_{e \in E^\uparrow(A)} r_e - \sum_{e \in E^\downarrow(A)} (r_e - 1) > 5 \left( |A_{\text{in}}| + \frac{1}{2} |A_{\text{ex}}| \right) \quad (25.40)$$

for every nonempty  $A \subset H_\star$ . Then every fixed moment  $p$  of the corresponding Wick-contracted random distribution satisfies

$$I_{p, \lambda} \lesssim \lambda^{\alpha p}, \quad \alpha = 5 |H_{\text{in}} \setminus H_\star| + \frac{5}{2} |H_{\text{ex}}| - \sum_e a_e, \quad (25.41)$$

uniformly in the microscopic cutoff, provided the cumulants have range  $O(\varepsilon)$  and the scale-invariant bounds implied by Definition 8.71.

*Proof.* Expand the  $p$ -th moment by the moment-cumulant formula and then Wick-contract between copies. Unit-range independence implies that after rescaling a  $q$ -cumulant vanishes unless its arguments form an  $O(\varepsilon)$  cluster. Its central-limit amplitude is  $\varepsilon^{-5q/2}$ , exactly the hyperedge degree assigned above.

Dyadically decompose every ordinary kernel and organise all endpoints by the coalescence tree of Lemma 8.34. For a cluster disjoint from the distinguished test vertex, the exponent of its scale is the right side minus the left side of (25.38). For a cluster containing the test vertex, the Taylor

remainder on every edge crossing the cluster boundary contributes the  $r_e$  correction in (25.39). For the complementary large-scale cluster, the same bookkeeping reverses sign and yields (25.40). These are precisely the small, pinned, and large cluster inequalities in Lemma 8.39; (25.38a) is the local edge-integrability condition used when a two-point cluster is collapsed. Thus every scale sum is geometric. The scales attached to the distinguished test cluster give the factor  $\lambda^{\alpha p}$ , and the finitely many cumulant partitions for fixed  $p$  and fixed graph are absorbed in the implicit constant.  $\square$

**Lemma 8.76** (Extremal-subgraph reduction for the Shen–Xu second-order graphs). *For a second-order graph write  $v^*, v_*$  for the two interaction vertices,  $\mathcal{P}, \mathcal{Q}$  for external noise leaves attached to them, and  $\mathcal{P}', \mathcal{Q}'$  for contracted leaves; put  $\mathcal{N} = \mathcal{P}' \cup \mathcal{Q}'$ . After the allocation in Lemma 8.77, verification of (25.38)–(25.40) reduces to the following all-or-none configurations:*

1. *for (25.38), each of  $\mathcal{P}, \mathcal{Q}$  and every connected component of  $\mathcal{N}$  is either entirely present or absent, except that when an interaction vertex is present its external block may also be reduced to the singleton interaction vertex;*
2. *for (25.39), the block attached to a present interaction vertex is wholly included and the block attached to an absent interaction vertex is omitted;  $\mathcal{N}$  is all or none;*
3. *for (25.40),  $\mathcal{Q}$  is omitted, while  $\mathcal{P} \cup \mathcal{N}$  is included exactly when  $v^*$  is included.*

*Proof.* Suppose first that a candidate set  $A$  contains only part of  $\mathcal{P}$ . If  $v^* \in A$ , adding another  $\mathcal{P}$  vertex increases the left side of (25.38) or (25.39) by 3 or  $5/2 + \delta$ , while the right side increases only by  $5/2$ ; the inequality is therefore worsened. If  $v^* \notin A$ , removing that vertex does not increase the left side and decreases the right side, again worsening the inequality. The same argument applies to  $\mathcal{Q}$ .

For (25.40) the direction is reversed. If  $v^* \in A$ , adding a  $\mathcal{P}$  vertex increases the right side by  $5/2$  but does not create a new internal kernel edge of larger cost, so the lower bound is worsened; if  $v^* \notin A$ , removing it is the worsening move. Since  $v_*$  is excluded in the large-scale test, including  $\mathcal{Q}$  strictly improves the left side relative to the right and hence the worst case omits it.

For a connected component of  $\mathcal{N}$ , adding all its  $n$  contracted vertices changes the right side of (25.38)–(25.39) by  $5n$ . If both interaction vertices are present, the left increment is

$$\sum_{e \in \mathcal{N}} a_e + \frac{5n}{2} > 5n.$$

If one interaction vertex is absent, at most  $n - 1$  of the incident ordinary edges remain internal; since there are at most five degree-3 edges in the whole second-order graph, the change differs from  $5n$  by only an arbitrarily small multiple of  $\delta$ . Thus one of the two all/none choices is always the worse one. The identical calculation with the reversed inequality gives the rule for (25.40): include  $\mathcal{N}$  exactly when  $v^*$  is included. Iterating component by component proves the reduction.  $\square$

**Lemma 8.77** (Allocation of positive  $\varepsilon$ -homogeneity). *Every negative first-order or second-order symbol in Lemma 8.74 admits an allocation of its explicit powers of  $\varepsilon$  to incident heat-kernel edges such that the specialized graph criterion holds with a strict margin, except for the finite zero/one-chaos local contractions treated in Lemma 8.79. More precisely:*

1. *for  $\mathcal{E}^{k-1}\Psi^{2k+1-n}$ , allocate  $\varepsilon^{1/2-\delta}$  to  $2k - 2$  of the noise edges; exactly  $3 - n \leq 3$  edges retain degree 3, while the others have effective degree  $5/2 + \delta$ ;*

2. for a second-order graph, put  $p = |\mathcal{P}|$ ,  $p' = |\mathcal{P}'|$ ,  $q = |\mathcal{Q}|$ ,  $q' = |\mathcal{Q}'|$ ,  $n = p' + q'$ , and  $\delta_\tau = \mathbf{1}_{k+\ell \text{ odd}}$ . Split  $p + q + n - 4 - \delta_\tau$  copies of  $\varepsilon^{1/2-\delta}$  as

$$p + p' - 2 - \mathbf{1}_{\ell \text{ odd}} \quad \text{and} \quad q + q' - 1 - \mathbf{1}_{k \text{ even}},$$

assigning contracted edges before external edges on each side. Then precisely  $2 + \mathbf{1}_{\ell \text{ odd}}$  left edges and  $1 + \mathbf{1}_{k \text{ even}}$  right edges retain degree 3; all remaining noise edges have degree  $5/2 + \delta$ . An external edge can receive the improved degree only after all contracted edges on its side have received it.

For every higher-chaos second-order term,  $p + q \geq 2$ , these allocations satisfy (25.38)–(25.40). The standard  $(k, \ell) = (1, 3), (2, 2), (2, 3)$   $\Phi_3^4$  graphs carry no positive  $\varepsilon$  allocation and are included in the same finite verification.

*Proof.* For a first-order graph the allocated graph homogeneity is

$$\frac{5}{2}(2k + 1 - n) - 3(3 - n) - (\frac{5}{2} + \delta)(2k - 2) = -\frac{3 - n}{2} - (2k - 2)\delta,$$

while the symbol has degree  $-(3 - n)/2 - (2k + 1 - n)\kappa$ . Choose  $\delta, \kappa > 0$  with the former strictly above the latter. If a candidate for (25.38) contains  $\ell_0$  noise vertices, of which  $r \leq 3$  have degree 3, and contains the interaction vertex, the required inequality is exactly

$$3r + (\frac{5}{2} + \delta)(\ell_0 - r) < 5(1 + \ell_0/2),$$

which holds for sufficiently small  $\delta$  because  $r \leq 3$ ; without the interaction vertex the internal-edge contribution vanishes. There are no Taylor labels, so (25.39) is weaker than (25.38). For (25.40), the left side contains  $\ell_0$  edges each of degree at least  $5/2 + \delta$ , whereas the right side is  $5\ell_0/2$ .

For a second-order graph the identity

$$p + q + n - 4 - \delta_\tau = (p + p' - 2 - \mathbf{1}_{\ell \text{ odd}}) + (q + q' - 1 - \mathbf{1}_{k \text{ even}})$$

follows from  $n = p' + q'$  and  $\mathbf{1}_{k \text{ even}} + \mathbf{1}_{\ell \text{ odd}} = 1 + \delta_\tau$ ; the excluded parity case is exactly what makes this identity valid. Assigning contracted edges first proves the asserted degree counts. The resulting graph homogeneity is

$$5|H_{\text{in}} \setminus H_\star| + \frac{5}{2}|H_{\text{ex}}| - \sum_e a_e = -\frac{1}{2}\delta_\tau - (p + q + n - 4 - \delta_\tau)\delta,$$

which is strictly above the symbol homogeneity for small  $\delta$ .

By Lemma 8.76, only the extremal configurations remain. For (25.38), first include both interaction vertices and all of  $\mathcal{N}$ . If both  $\mathcal{P}$  and  $\mathcal{Q}$  are included, the inequality reads

$$\sum_{e \in \mathcal{P} \cup \mathcal{Q} \cup \mathcal{N}} a_e + 3 + \frac{5n}{2} < 5\left(n + 2 + \frac{p + q - 1}{2}\right),$$

and it is strict because among the  $p + q + n$  displayed edges at most five have degree 3 and all others have degree  $5/2 + \delta$ . If neither external block is included, let  $r$  be the number of degree-3 contracted edges. The condition becomes

$$(\frac{5}{2} + \delta)n + (\frac{1}{2} - \delta)r + 3 + \frac{5n}{2} < 5(n + 1).$$

Higher chaos  $p + q \geq 2$  forces  $r \leq 3$ , hence this is strict for small  $\delta$ . Including exactly one of the external blocks increases the right side of (25.44) by an additional  $5/2$  relative to the worst edge increment, so is also safe. If neither interaction vertex is present the worst configuration has zero internal-edge contribution; if exactly one is present the same one-sided estimate applies.

For (25.39), the case with both interaction vertices follows from (25.38), since no Taylor edge crosses the boundary and the right side is larger. With neither interaction vertex, the left side is zero for the extremal candidate. If only  $v^*$  is present, the worst candidate includes  $\mathcal{P}$  and requires

$$\sum_{e \in \mathcal{P}} a_e + 3 < 5(1 + p/2),$$

which follows because at most three  $\mathcal{P}$  edges have degree 3; the  $v_*$ -only case is symmetric.

Finally consider (25.40). The extremal reduction leaves either no interaction vertex, which gives the limiting equality  $0 = 0$  before imposing nonemptiness, or the candidate  $\{v^*\} \cup \mathcal{P} \cup \mathcal{N}$ . The required strict inequality is

$$\sum_{e \in \mathcal{P} \cup \mathcal{N}} a_e + \frac{5n}{2} + 4 > 5\left(n + 1 + \frac{p}{2}\right).$$

If  $r$  of the  $p + n$  incident edges have degree 3, its left side equals

$$\left(\frac{5}{2} + \delta\right)(p + n) + \left(\frac{1}{2} - \delta\right)r + \frac{5n}{2} + 4.$$

The source degree allocation implies the strict inequality whenever the inner power  $\ell \geq 3$ , and also for  $\ell = 2$  with  $p + n \geq 3$ . The sole borderline higher-chaos configuration is the fourth chaos of  $\Psi^2\mathcal{I}(\Psi^2)$ , where  $p + n = r = 2$ ; assigning the barred heat edge the admissible degree  $3 + \delta$  supplies the missing strict margin. This exhausts the extremal cases and proves (25.38)–(25.40).  $\square$

**Lemma 8.78** (Finite-range cumulant integration). *Let  $q \geq 2$  and  $-5 < \alpha_i < 0$ . For the rescaled cumulant  $\mathfrak{C}_q^{(\varepsilon)}$  of Definition 8.71,*

$$\int \prod_{i=1}^q |y_i - x_i|_{\mathfrak{s}}^{\alpha_i} |\mathfrak{C}_q^{(\varepsilon)}(x_1, \dots, x_q)| \, d\mathbf{x} \lesssim \varepsilon^{5(q/2-1)} \int_{\mathbb{R}^4} \prod_{i=1}^q (|y_i - x|_{\mathfrak{s}} + \varepsilon)^{\alpha_i} \, dx.$$

*The same estimate holds with differentiated or smoothly truncated heat kernels, provided their singularity degrees stay below the parabolic dimension.*

*Proof.* By scaling,  $\mathfrak{C}_q^{(\varepsilon)}(x_1, \dots, x_q) = \varepsilon^{-5q/2} \mathfrak{C}_q(x_1/\varepsilon, \dots, x_q/\varepsilon)$ . Unit-range independence makes the unscaled cumulant vanish unless all its arguments lie in one bounded parabolic cluster. Fix  $x = x_1$  and write  $x_i = x + \varepsilon u_i$  for  $i \geq 2$ . The  $q - 1$  relative integrations contribute  $\varepsilon^{5(q-1)}$ , so the net scaling factor is  $\varepsilon^{5(q/2-1)}$ .

It remains to compare the singular weights. On the bounded support of the  $u_i$ , split according as  $|y_i - x|_{\mathfrak{s}} \geq 2C\varepsilon$  or not. In the first region  $|y_i - x_i|_{\mathfrak{s}} \asymp |y_i - x|_{\mathfrak{s}}$ ; in the second, direct integration of the locally integrable power  $|y_i - x_i|^{\alpha_i}$  over an  $O(\varepsilon)$  ball is bounded by a constant times  $(|y_i - x|_{\mathfrak{s}} + \varepsilon)^{\alpha_i}$  after the same relative-coordinate normalisation. Iterate this estimate for  $i = 2, \dots, q$ . Boundedness of the unscaled cumulant on its compact support follows from the assumed moments and the finite moment–cumulant formula. This yields the displayed estimate. Differentiated/truncated kernels are identical after replacing  $\alpha_i$  by their admissible singularity exponents.  $\square$



**Lemma 8.79** (Low-chaos contractions and the unique logarithmic divergence). *For  $k, \ell \geq 2$ , let a contraction pattern  $\pi = \{(k_i, \ell_i)\}_{i=1}^r$  partition the  $k$  noise leaves at one interaction vertex and the  $\ell$  leaves at the other into cross-cumulant blocks, and let  $C_{k, \ell, \pi}^{(\varepsilon)}$  be the corresponding local contraction, including the source prefactor  $\varepsilon^{(k+\ell-4)/2}$ . Define*

$$C_{k, \ell}^{(\varepsilon)} := \sum_{\pi} \pi! C_{k, \ell, \pi}^{(\varepsilon)}. \quad (25.42)$$

*Then all these constants are bounded as  $\varepsilon \downarrow 0$  except the pairwise  $(2, 2)$  contraction  $\pi = \{(1, 1), (1, 1)\}$ , for which*

$$C_{2, 2, \pi}^{(\varepsilon)} = C_{\log} |\log \varepsilon| + O(1). \quad (25.43)$$

*The contraction joining all four leaves has a finite limit. After subtracting (25.42), every remaining zero-chaos graph gains a positive power of  $\varepsilon$ ; every remaining one-chaos graph satisfies the strict graph criterion after one Taylor subtraction.*

*Proof.* Lemma 8.78 gives, for every contraction block of size  $r$ , the precise gain  $\varepsilon^{5(r/2-1)}$  after its relative coordinates are integrated. For a contraction pattern with blocks  $B_i$ , combine these gains with the explicit prefactor  $\varepsilon^{(k+\ell-4)/2}$ . If one block has size at least four, or the total number of contracted leaves is at least six, the result has a strictly positive power of  $\varepsilon$  after convolution with the two heat kernels. This proves decay of all such zero-chaos terms.

The exceptional four-leaf pair contraction consists of two covariance lines. Writing  $G_\varepsilon(z) = \mathbf{E}[\Psi_\varepsilon(0)\Psi_\varepsilon(z)]$ , its local contribution is

$$\int G_\varepsilon(z)^2 K(-z) dz.$$

In parabolic dimension five,  $G_\varepsilon(z) \asymp (|z|_s + \varepsilon)^{-1}$  and  $K(z) \asymp |z|_s^{-3}$ . Radial parabolic integration therefore contains  $r^4 r^{-2} r^{-3} dr = r^{-1} dr$ , giving the logarithm in (25.43). If all four leaves belong to one fourth cumulant, range localization provides an additional microscopic volume gain and the integral converges.

For one-chaos terms the source parity forces  $\ell$  odd,  $k$  even,  $\ell \geq 3$ ,  $k \geq 2$ , and there are exactly two contraction types. In the first, all  $\ell$  left leaves contract with  $k-1$  right leaves. Apply Lemma 8.78 block by block and convolve the resulting locally integrable powers with the adjacent heat kernels. The whole contracted subgraph collapses to a two-point kernel  $J_\varepsilon$  of degree  $2 + \delta$  satisfying

$$\|J_\varepsilon\|_{2+\delta; \beta} \lesssim \varepsilon^\theta$$

for some  $\theta > 0$ . If  $k + \ell \geq 7$ , the positive power comes from the unused explicit  $\varepsilon$ -weight; in the minimal case  $(k, \ell) = (2, 3)$ , the only contraction of this first type joins four points in one higher cumulant and the positive power comes from the finite-range cumulant gain. The remaining two-edge graph satisfies (25.38)–(25.40), hence is  $O(\varepsilon^\theta \lambda^{-1/2-\delta})$ .

In the second contraction type,  $\ell-1$  left leaves contract with all  $k$  right leaves. The collapsed kernel has to be interpreted as the renormalised distribution

$$Q_\varepsilon(\varphi) = \int Q_\varepsilon(z) (\varphi(z) - \varphi(0)) dz,$$

so the corresponding edge has negative-renormalisation label  $r_e = -1$ . If  $k + \ell \geq 7$ , or if  $(k, \ell) = (2, 3)$  and the four contracted points belong to one fourth cumulant, the same integration leaves a factor  $\varepsilon^\theta$  and the effective collapsed edge can be assigned degree  $5 + \delta$ ; the graph criterion then gives

$O(\varepsilon^\theta \lambda^{-1/2-\delta})$ . The only case with  $\theta = 0$  is  $(k, \ell) = (2, 3)$  with pairwise covariance contraction. It is the standard first-chaos  $\Phi_3^4$  symbol, not a new divergent local constant, and the  $r_e = -1$  subtraction gives the required strict large-scale inequality. Thus all nonstandard one-chaos terms vanish and the unique nonvanishing borderline term belongs to the limiting Gaussian  $\Phi_3^4$  model. Odd cumulants vanish by symmetry; without symmetry an additional  $\varepsilon^{-1/2}$  constant renormalisation would be present.  $\square$

**Proposition 8.80** (Uniform non-Gaussian model bounds). *Let  $\widehat{Z}_\varepsilon$  be the renormalized model obtained from  $\zeta_\varepsilon$  by Appell renormalization and the local constants (25.42), and let  $\widehat{Z}_{\varepsilon, \bar{\varepsilon}}$  be built after an additional smooth convolution at scale  $\bar{\varepsilon}$ . There exists  $\vartheta > 0$  such that, for every negative basis symbol  $\tau$  and every  $p > 1$ ,*

$$\mathbf{E}|\langle \widehat{\Pi}_z^\varepsilon \tau, \varphi_z^\lambda \rangle|^p \lesssim \lambda^{p(|\tau|+\vartheta)}, \quad (25.44)$$

$$\mathbf{E}|\langle (\widehat{\Pi}_z^\varepsilon - \widehat{\Pi}_z^{\varepsilon, \bar{\varepsilon}}) \tau, \varphi_z^\lambda \rangle|^p \lesssim \bar{\varepsilon}^\vartheta \lambda^{p(|\tau|+\vartheta)}, \quad (25.45)$$

uniformly in  $\varepsilon, \bar{\varepsilon} \in (0, 1)$ . Symbols containing a positive power of  $\mathcal{E}$  gain an additional positive power of  $\varepsilon$  and hence converge to zero.

*Proof.* First-order symbols are handled by the first part of Lemma 8.77. Higher-chaos second-order symbols are handled by its second part and Lemma 8.75. The remaining zero/one-chaos terms are exactly Lemma 8.79. Since the negative sector is finite, the minimum of the finitely many strict margins is a common  $\vartheta > 0$ , giving (25.44). Replacing one heat kernel by its mollified difference gains  $\bar{\varepsilon}^\vartheta$  by the dyadic heat-kernel estimate already proved in Lemma 8.34: on a dyadic block of scale  $2^{-j}$ , convolution with a unit-mass mollifier of radius  $\bar{\varepsilon}$  and the first-order Taylor formula give  $\|K_j - K_j * \rho_{\bar{\varepsilon}}\| \lesssim \min\{1, (2^j \bar{\varepsilon})^\vartheta\} 2^{j(3+\vartheta)}$ . Summing in  $j$  yields  $|K - K_{\bar{\varepsilon}}| \lesssim \bar{\varepsilon}^\vartheta |z|_{\mathfrak{s}}^{-3-\vartheta}$ , and the identical graph summation gives (25.45). Any unused  $\varepsilon^{1/2-\delta}$  factor from the allocation survives outside the graph sum and proves the final assertion.  $\square$

**Lemma 8.81** (The full Shen–Xu comparison norm). *Fix  $\eta < -1/2$ . For a weighted microscopic model  $Z_\varepsilon$  and a continuum model  $Z$ , define the source-strength comparison quantity*

$$\begin{aligned} \|Z_\varepsilon; Z\|_{\varepsilon; 0} &:= \|Z_\varepsilon; Z\| + \|Z_\varepsilon\|_{\mathcal{E}, \varepsilon} \\ &+ \sup_{t \in [0, 1]} \|\mathcal{R}^\varepsilon \mathcal{I}(\Xi)(t, \cdot) - \mathcal{R} \mathcal{I}(\Xi)(t, \cdot)\|_{\mathcal{C}^\eta}. \end{aligned} \quad (25.46)$$

Writing  $\mathcal{W}$  for the finite basis sector of the weighted structure,  $\mathcal{U} \subset \mathcal{W}$  for the sector tested below the microscopic scale, and  $\beta = 6/5$ , the extra weighted term is

$$\begin{aligned} \|Z_\varepsilon\|_{\mathcal{E}, \varepsilon} &:= \sup_{\tau \in \mathcal{W}, z, k, \ell} \varepsilon^{|\ell|-k-|\tau|} |f_z^\varepsilon(\mathcal{E}_\ell^k \tau)| \\ &+ \sup_{\tau \in \mathcal{U}, z, \psi, 0 < \lambda \leq \varepsilon} \lambda^{-\beta} \varepsilon^{\beta-|\tau|} |\langle \Pi_z^\varepsilon \tau, \psi_z^\lambda \rangle|, \end{aligned}$$

where the second supremum is over unit  $C^2$  test functions supported in the unit parabolic ball and annihilating affine polynomials. For the renormalized models  $\widehat{Z}_\varepsilon$  of Proposition 8.80,

$$\|\widehat{Z}_\varepsilon\|_{\mathcal{E}, \varepsilon} \longrightarrow 0 \quad (25.47)$$

in probability. Moreover, if  $Z_{\Phi_3^4}$  is the limiting Gaussian model, then

$$\sup_{t \in [0, 1]} \|\mathcal{R}^\varepsilon \mathcal{I}(\Xi)(t) - \mathcal{R} \mathcal{I}(\Xi)(t)\|_{\mathcal{C}^\eta} \longrightarrow 0 \quad (25.48)$$

in probability.

*Proof.* For (25.47), inspect the two pieces of the weighted model norm. If a symbol contains  $\mathcal{E}^k$  with  $k > 0$ , the allocation proof in Lemma 8.77 leaves a strictly positive power of  $\varepsilon$  after the graph sum. The same argument applied to the small-scale polynomial coefficient  $f_z^\varepsilon(\mathcal{E}_\ell^k \tau)$  gives

$$\mathbf{E}|f_z^\varepsilon(\mathcal{E}_\ell^k \tau)| \lesssim \varepsilon^{|\tau|+k-|\ell|+\vartheta}.$$

For the reconstruction test at scales  $\lambda < \varepsilon$ , use a test function annihilating affine polynomials and the dyadic kernel estimate in Lemma 8.34; with  $\beta = 6/5$  one obtains

$$\mathbf{E}|\langle \hat{\Pi}_z^\varepsilon \tau, \psi_z^\lambda \rangle| \lesssim \lambda^{\beta+\vartheta} \varepsilon^{|\tau|-\beta}.$$

The negative sector is finite, so Markov's inequality and a finite union bound prove (25.47).

For (25.48), fix a smooth space-time mollifier  $\rho$  and write  $\zeta_{\varepsilon, \bar{\varepsilon}} = \rho_{\bar{\varepsilon}} * \zeta_\varepsilon$  and  $\xi_{\bar{\varepsilon}} = \rho_{\bar{\varepsilon}} * \xi$ . Since  $\mathcal{R}^\varepsilon \mathcal{I}(\Xi) = K * \zeta_\varepsilon$  and  $\mathcal{R}\mathcal{I}(\Xi) = K * \xi$ , the triangle inequality gives

$$\begin{aligned} \|K * \zeta_\varepsilon - K * \xi\|_{C_t \mathcal{C}^\eta} &\leq \|K * \zeta_\varepsilon - K * \zeta_{\varepsilon, \bar{\varepsilon}}\|_{C_t \mathcal{C}^\eta} \\ &\quad + \|K * \zeta_{\varepsilon, \bar{\varepsilon}} - K * \xi_{\bar{\varepsilon}}\|_{C_t \mathcal{C}^\eta} \\ &\quad + \|K * \xi_{\bar{\varepsilon}} - K * \xi\|_{C_t \mathcal{C}^\eta}, \end{aligned}$$

where  $C_t$  denotes the supremum over  $t \in [0, 1]$ . The first term is  $O(\bar{\varepsilon}^\vartheta)$  uniformly in  $\varepsilon$ : on each heat-kernel block the mollifier difference gains  $\min\{1, (2^j \bar{\varepsilon})^\vartheta\}$ , while the cumulant estimate of Lemma 8.78 gives uniform moments in the slightly stronger space  $C_t^{\vartheta/2} \mathcal{C}_x^{\eta+\vartheta}$ . The third term obeys the same bound by the Gaussian heat-kernel estimate already owned in the analytic BPHZ construction.

For fixed  $\bar{\varepsilon}$ , all kernels are smooth. Joint cumulants of the finite family consisting of the fields and finitely many derivatives at finitely many space-time points have second cumulants converging to those of  $\xi_{\bar{\varepsilon}}$ , while every cumulant of order  $r \geq 3$  is  $O(\varepsilon^{5(r/2-1)})$  by Lemma 8.78. The moment-cumulant formula, uniform derivative moments, and the Kolmogorov criterion therefore give convergence in probability of the middle term in  $C_t \mathcal{C}^\eta$ . First let  $\varepsilon \downarrow 0$  at fixed  $\bar{\varepsilon}$ , then let  $\bar{\varepsilon} \downarrow 0$ ; this proves (25.48).  $\square$

**Proposition 8.82** (Identification of the limiting model). *Let  $Z_{\Phi_3^4}$  denote the Gaussian  $\Phi_3^4$  BPHZ model with cubic coupling convention fixed in Remark 8.72. Then, on the common diagonal coupling constructed in the proof, the renormalized models of Proposition 8.80 and a Gaussian  $\Phi_3^4$  BPHZ model satisfy*

$$\|\hat{Z}_\varepsilon; Z_{\Phi_3^4}\|_{\varepsilon; 0} \longrightarrow 0 \quad (25.49)$$

*in probability, and in every finite  $L^p$  comparison metric after the usual small homogeneity loss. In particular, without choosing a coupling, the laws of the renormalized models converge to the law of  $Z_{\Phi_3^4}$ .*

*Proof.* By Lemma 8.81, it remains only to prove convergence of the ordinary model distance. Fix a smooth mollification scale  $\bar{\varepsilon}$  and insert the intermediate models  $\hat{Z}_{\varepsilon, \bar{\varepsilon}}$  and  $Z_{0, \bar{\varepsilon}}$ :

$$\|\hat{Z}_\varepsilon; Z_{\Phi_3^4}\| \leq \|\hat{Z}_\varepsilon; \hat{Z}_{\varepsilon, \bar{\varepsilon}}\| + \|\hat{Z}_{\varepsilon, \bar{\varepsilon}}; Z_{0, \bar{\varepsilon}}\| + \|Z_{0, \bar{\varepsilon}}; Z_{\Phi_3^4}\|. \quad (25.50)$$

The first term is  $O(\bar{\varepsilon}^\vartheta)$  uniformly in  $\varepsilon$  by (25.45) and the model-norm upgrade Proposition 8.45. The last term tends to zero as  $\bar{\varepsilon} \downarrow 0$  by the already-owned Gaussian  $\Phi_3^4$  model construction.

For fixed  $\bar{\varepsilon}$ , every kernel in the middle term is smooth. For the finitely many smooth space-time integrals of  $\zeta_\varepsilon$  appearing in the finite model sector, compute joint cumulants directly. The second

joint cumulants converge, by the covariance normalization and dominated convergence, to the white-noise covariance. Lemma 8.78 shows that every joint cumulant of order  $r \geq 3$  is  $O(\varepsilon^{5(r/2-1)})$  and therefore vanishes. The moment–cumulant formula then shows that every joint moment tends to the corresponding Wick pairing moment. Uniform derivative moments and Kolmogorov compactness give joint convergence of the finitely many smooth model components, including base-point reexpansion coefficients. The finite local counterterms converge simultaneously; (25.43) is exactly the logarithmic term retained by the limiting  $\Phi_3^4$  renormalization. Hence the middle term in (25.50) tends to zero as  $\varepsilon \downarrow 0$  for fixed  $\bar{\varepsilon}$ . Send  $\varepsilon \downarrow 0$  first and then  $\bar{\varepsilon} \downarrow 0$  and combine with (25.47)–(25.48) to obtain (25.49).  $\square$

**Lemma 8.83** (Pitchfork tuning and cancellation of the mass divergence). *Choose in (25.35)*

$$\lambda_j^{(\varepsilon)} = \hat{a}_j(\theta) \quad (j \geq 1), \quad \lambda_0^{(\varepsilon)} = \varepsilon^{-1} \hat{a}_0(\theta) - C_\varepsilon. \quad (25.51)$$

*Then (25.35) is exactly the microscopic equation (25.32). Moreover*

$$C_\varepsilon = 9\lambda_*^2 C_{\log} |\log \varepsilon| + O(1), \quad (25.52)$$

*and the choice*

$$\theta(\varepsilon) = \frac{9\lambda_*^2 C_{\log}}{\hat{a}'_0(0)} \varepsilon |\log \varepsilon| + O(\varepsilon) \quad (25.53)$$

*makes  $\lambda_0^{(\varepsilon)}$  converge to a finite limit. Its leading coefficient is negative because  $\hat{a}'_0(0) < 0$ .*

*Proof.* The identity of the equations follows directly by comparing (25.32) and (25.35). In (25.36), every  $C_{k,\ell}^{(\varepsilon)}$  is bounded except the pairwise (2, 2) contraction. It can enter only with  $k = \ell = 1$  in the second term of (25.36), whose combinatorial factor is  $(2 \cdot 1 + 1)^2 = 9$ . Since  $\lambda_1^{(\varepsilon)} \rightarrow \lambda_*$ , Lemma 8.79 gives (25.52).

Taylor expansion at the pitchfork point gives  $\hat{a}_0(\theta) = \hat{a}'_0(0)\theta + O(\theta^2)$ . For  $\theta = O(\varepsilon |\log \varepsilon|)$ ,  $\varepsilon^{-1}O(\theta^2) = o(1)$ . Substituting (25.52) and (25.53) into (25.51) cancels the logarithmic term, leaving a finite limit. The sign follows from  $C_{\log} > 0$ ,  $\lambda_* > 0$ , and  $\hat{a}'_0(0) < 0$ .  $\square$

**Theorem 8.84** (Shen–Xu weak universality of dynamical  $\Phi_3^4$ ). *Assume Definition 8.71. Let  $1 < \gamma < 6/5$  and  $-(2m+2)/(4m+3) < \eta < -1/2$ , and let microscopic initial data  $\phi_0^{(\varepsilon)}$  converge to  $\phi_0 \in \mathcal{C}^\eta(\mathbb{T}^3)$  in the  $\varepsilon$ -weighted  $\mathcal{C}_\varepsilon^{\gamma,\eta}$  norm of Lemma 8.74. For every deterministic  $T < \infty$  there is a choice*

$$\theta(\varepsilon) = a\varepsilon |\log \varepsilon| + O(\varepsilon), \quad a = \frac{9\lambda_*^2 C_{\log}}{\hat{a}'_0(0)} < 0, \quad (25.54)$$

*such that the rescaled solutions converge in law*

$$u_\varepsilon \xrightarrow{\text{law}} \Phi \quad \text{in } C([0, T], \mathcal{C}^\eta(\mathbb{T}^3)), \quad (25.55)$$

*where  $\Phi$  belongs to the dynamical  $\Phi_3^4(\lambda_*)$  family; the finite  $O(\varepsilon)$  part in (25.54) selects the finite mass parameter in that family. Here  $\lambda_* = \hat{a}_1(0)$  is the coefficient convention of Remark 8.72.*

*Proof.* Lemma 8.73 converts the rescaled microscopic equation to the Appell form (25.32). Lemma 8.74 gives a finite locally subcritical weighted regularity structure, a uniform local solution map, and the reconstructed renormalized equation (25.35). Proposition 8.80 proves the required uniform non-Gaussian model estimates, and Proposition 8.82 identifies their limit with the Gaussian  $\Phi_3^4$  BPHZ model. Lemma 8.83 makes the reconstructed equation identical to the microscopic

equation while cancelling the sole logarithmic mass divergence. The coefficients  $\lambda_j^{(\varepsilon)}$  converge, in particular  $\lambda_1^{(\varepsilon)} \rightarrow \lambda_*$ , and every higher weighted nonlinearity with an explicit  $\mathcal{E}$  factor disappears in the limiting model.

On the auxiliary diagonal coupling of Proposition 8.82, continuity of the local fixed-point map yields convergence in probability up to the common model/solution stopping time. The global continuum estimate Theorem 8.62 gives almost-sure existence of the limiting  $\Phi_3^4$  solution on every deterministic  $[0, T]$ . Equivalently, the probability that the continuum modelled-distribution norm exits a ball before  $T$  tends to zero as the radius tends to infinity. Apply the stopped continuity statement on such a ball, then send the radius to infinity, and finally forget the auxiliary coupling. This gives the convergence in law asserted in (25.55). No Hairer–Matetski lattice theorem is needed here: the Shen–Xu microscopic equation is continuous in space and the convergence is through the weighted continuum regularity structure.  $\square$

## 8.5 Bertini–Giacomin weak asymmetry

This section gives a source-level proof of the classical weak-asymmetry limit. The point which is easy to miss is that the microscopic Gärtner transform has an *exact linear drift semimartingale equation*. Its martingale noise is still configuration dependent, so the transformed particle system is not itself a linear SPDE. The proof difficulty is not a drift remainder: it is identification of the martingale bracket. Bertini–Giacomin obtain that identification from a nearest-neighbour correlation estimate whose proof uses an exact cancellation for the discrete heat kernel. We retain that structure rather than replacing it by a generic replacement lemma.

Let  $\varepsilon \downarrow 0$ . In the single-step interface a local minimum is raised by two at rate

$$q_\varepsilon = \frac{1}{2} + \sqrt{\varepsilon},$$

and a local maximum is lowered by two at rate  $p_\varepsilon = 1/2$ . Write  $\zeta_s(j)$  for the microscopic height at  $j \in \mathbb{Z}$  and microscopic time  $s$ . Thus  $|\zeta_s(j+1) - \zeta_s(j)| = 1$ .

**Definition 8.85** (Bertini–Giacomin admissible initial laws). A family of microscopic initial laws  $\mu_\varepsilon$  is *BG-admissible* if the linearly interpolated profiles

$$H_0^\varepsilon(r) = \sqrt{\varepsilon} \zeta_0(\varepsilon^{-1}r)$$

converge in law in  $C_{\text{loc}}(\mathbb{R})$  to a continuous random profile  $H_0$ , and, for every integer  $m \geq 1$ , there are constants  $a_m, c_m, a'_m, c'_m > 0$ , independent of  $\varepsilon$ , such that

$$\sup_{j \in \mathbb{Z}} e^{-a_m \varepsilon |j|} \mathbf{E}_{\mu_\varepsilon} \exp\{-m\sqrt{\varepsilon} \zeta_0(j)\} \leq c_m, \quad (25.56)$$

$$\mathbf{E}_{\mu_\varepsilon} \left| \sqrt{\varepsilon} (\zeta_0(j) - \zeta_0(k)) \right|^{2m} \leq c'_m e^{a'_m \varepsilon (|j|+|k|)} (\varepsilon |j-k|)^m. \quad (25.57)$$

The source assumes these estimates for every  $m$ ; its proof in fact uses only finitely many moments. The exponential weight allows profiles of at most linear macroscopic growth.

Set

$$\gamma_\varepsilon = \frac{1}{2} \log(1 + 2\sqrt{\varepsilon}), \quad \lambda_\varepsilon = 1 + \sqrt{\varepsilon} - \sqrt{1 + 2\sqrt{\varepsilon}}, \quad (25.58)$$

and define first the *microscopic* Gärtner transform

$$\xi_s^\varepsilon(j) = \exp\{-\gamma_\varepsilon \zeta_s(j) + \lambda_\varepsilon s\}, \quad j \in \mathbb{Z}. \quad (25.59)$$

Its diffusively rescaled, linearly interpolated version is

$$Z_t^\varepsilon = \xi_{\varepsilon^{-2}t}^\varepsilon(\varepsilon^{-1}r), \quad t \in [0, T], \quad r \in \mathbb{R}. \quad (25.60)$$

At non-lattice arguments the right-hand side is understood by linear interpolation.

**Lemma 8.86** (Exact microscopic Gärtner equation). *Let*

$$a_\varepsilon = \frac{1}{2}e^{\gamma_\varepsilon}, \quad \nabla^+ f(j) = f(j+1) - f(j), \quad \nabla^- f(j) = f(j) - f(j-1), \quad \Delta f = \nabla^- \nabla^+ f.$$

*Then, with respect to the natural microscopic filtration,*

$$d\xi_s^\varepsilon(j) = a_\varepsilon \Delta \xi_s^\varepsilon(j) ds + dM_s^\varepsilon(j), \quad (25.61)$$

*where  $M^\varepsilon(j)$  and  $M^\varepsilon(k)$  are orthogonal for  $j \neq k$ , and*

$$\begin{aligned} d\langle M^\varepsilon(j) \rangle_s &= \frac{1}{2} \left[ (1 - e^{\gamma_\varepsilon})^2 (2(\xi_s^\varepsilon(j))^2 + \xi_s^\varepsilon(j) \Delta \xi_s^\varepsilon(j)) \right. \\ &\quad \left. - (1 + e^{2\gamma_\varepsilon}) \nabla^+ \xi_s^\varepsilon(j) \nabla^- \xi_s^\varepsilon(j) \right] ds. \end{aligned} \quad (25.62)$$

*In particular*

$$d\langle M^\varepsilon(j) \rangle_s \leq C\varepsilon (\xi_s^\varepsilon(j))^2 ds \quad (25.63)$$

*with  $C$  independent of  $\varepsilon \in (0, 1]$ .*

*Proof.* There are only two possible jumps at site  $j$ . At a local minimum the height changes by  $+2$ , hence  $\xi(j)$  is multiplied by  $e^{-2\gamma_\varepsilon}$ ; at a local maximum the height changes by  $-2$ , hence it is multiplied by  $e^{2\gamma_\varepsilon}$ . Applying the jump-generator formula to  $\xi(j)$ , and inserting  $q_\varepsilon = 1/2 + \sqrt{\varepsilon}$  and  $p_\varepsilon = 1/2$ , gives

$$\mathcal{L}_\varepsilon \xi(j) = \lambda_\varepsilon \xi(j) + q_\varepsilon \mathbf{1}_{\{j \text{ min}\}} (e^{-2\gamma_\varepsilon} - 1) \xi(j) + p_\varepsilon \mathbf{1}_{\{j \text{ max}\}} (e^{2\gamma_\varepsilon} - 1) \xi(j).$$

Because neighbouring height differences are  $\pm 1$ , the two indicators can be written algebraically through the neighbouring ratios  $\xi(j \pm 1)/\xi(j) = e^{\mp \gamma_\varepsilon(\zeta(j \pm 1) - \zeta(j))}$ . Substitution, followed by the two identities in (25.58), cancels the scalar drift and leaves  $a_\varepsilon \Delta \xi(j)$ . This proves (25.61).

For a pure-jump martingale the predictable bracket is rate times squared jump size. Using the same two local configurations gives

$$d\langle M(j), M(k) \rangle_s = \delta_{jk} \frac{\varepsilon}{2} (\xi(j))^2 \left[ (1 + e^{-2\gamma_\varepsilon})(1 - \sigma(j)\sigma(j+1)) + (1 - e^{-2\gamma_\varepsilon})(\sigma(j) - \sigma(j+1)) \right] ds,$$

where  $\sigma(j) = \zeta(j) - \zeta(j-1) \in \{\pm 1\}$ . Replacing the spins by logarithmic nearest-neighbour ratios of  $\xi$  gives exactly (25.62). Finally each allowed jump changes  $\xi(j)$  by at most  $C\sqrt{\varepsilon} \xi(j)$  and the total jump rate at a site is bounded by 2, which immediately yields (25.63).  $\square$

**Lemma 8.87** (Discrete heat-kernel package). *Let  $p_t^\varepsilon(j)$  be the transition kernel for the continuous-time symmetric walk with generator  $a_\varepsilon \Delta$ . Uniformly for  $0 < \varepsilon \leq 1$ ,  $0 < t \leq \varepsilon^{-2}T$ , and  $0 \leq \alpha \leq 1$ ,*

one has

$$\sum_j p_t^\varepsilon(j) = 1, \quad \sum_j p_t^\varepsilon(j) e^{a\varepsilon|j|} \leq C_{a,T}, \quad (25.64)$$

$$\sum_j |p_t^\varepsilon(j+\ell) - p_t^\varepsilon(j)|^2 \leq C_\alpha |\ell|^{2\alpha} t^{-1/2-\alpha}, \quad (25.65)$$

$$\sum_j |p_{t+h}^\varepsilon(j) - p_t^\varepsilon(j)|^2 \leq C_\alpha h^{2\alpha} t^{-1/2-2\alpha}, \quad (25.66)$$

$$\sup_j |\nabla^\pm p_t^\varepsilon(j)| \leq C(1 \wedge t^{-1}), \quad (25.67)$$

$$\sum_j |\nabla^\pm p_t^\varepsilon(j)|^2 e^{a\varepsilon|j|} \leq C_{a,T}(1 \wedge t^{-3/2}). \quad (25.68)$$

Moreover, for

$$K_t^\varepsilon(j) = \nabla^+ p_t^\varepsilon(j) \nabla^- p_t^\varepsilon(j),$$

$$\sum_j \int_0^\infty K_t^\varepsilon(j) dt = 0, \quad (25.69)$$

for every  $a, T > 0$  there exist  $\varepsilon_0 > 0$ ,  $\rho < 1$  such that

$$\sum_j \int_0^{\varepsilon^{-2}T} |K_t^\varepsilon(j)| e^{a\varepsilon|j|} dt \leq \rho, \quad 0 < \varepsilon < \varepsilon_0, \quad (25.70)$$

$$\sum_j \int_0^{\varepsilon^{-2}T} |K_t^\varepsilon(j)| e^{a\varepsilon|j|} (T - \varepsilon^2 t)^{-1/2} dt \leq C_{a,T}. \quad (25.71)$$

*Proof.* For the rate-one walk  $q_t$  with generator  $\frac{1}{2}\Delta$ ,

$$q_t(j) = \frac{1}{2\pi} \int_{-\pi}^\pi e^{ikj-t(1-\cos k)} dk, \quad p_t^\varepsilon(j) = q_{e^{\gamma\varepsilon}t}(j). \quad (25.72)$$

The moment-generating function of the walk is  $\exp\{e^{\gamma\varepsilon}t(\cosh\theta - 1)\}$ ; applying it with  $\theta = \pm a\varepsilon$  gives (25.64). Plancherel, together with  $|e^{ik\ell} - 1| \leq C_\alpha |k\ell|^\alpha$  and  $1 - \cos k \asymp k^2$  for  $|k| \leq 1$ , gives (25.65). The same calculation with  $|e^{-h(1-\cos k)} - 1| \leq C_\alpha h^\alpha |k|^{2\alpha}$  gives (25.66); differentiating the Fourier representation once gives (25.67). Squaring the differentiated Fourier multiplier and integrating gives the unweighted  $t^{-3/2}$  bound. The exponential weight is inserted by the random-walk moment estimate in (25.64), giving (25.68).

For (25.69), insert (25.70), integrate in  $t$ , and use the odd/even symmetry of the factors  $e^{\pm ik} - 1$ ; equivalently it is the Fourier identity proved by integrating the product of the two opposite nearest-neighbour gradients. For the strict contraction, first consider the unweighted kernel  $q_t$ . Cauchy–Schwarz gives

$$\sum_j \int_0^\infty |\nabla^+ q_t(j) \nabla^- q_t(j)| dt \leq \left( \sum_j \int_0^\infty |\nabla^+ q_t(j)|^2 dt \right)^{1/2} \left( \sum_j \int_0^\infty |\nabla^- q_t(j)|^2 dt \right)^{1/2} = 1.$$

The inequality is strict: the two gradient fields are not proportional on a set of positive  $(t, j)$ -measure. The equality of each square integral with 1 follows from

$$\frac{d}{dt} \sum_j q_t(j)^2 = - \sum_j |\nabla^+ q_t(j)|^2, \quad q_0 = \delta_0.$$

Choose a time cutoff so the unweighted integral is below some  $\rho_0 < 1$ . The Fourier gradient bound controls the tail uniformly, while  $e^{a\varepsilon|j|} \rightarrow 1$  on every fixed time window and is dominated by (25.64). Taking  $\varepsilon$  small therefore upgrades  $\rho_0$  to the weighted estimate (25.70) with some  $\rho < 1$ . Finally split the integral in (25.71) at  $\varepsilon^{-2}T/2$ . On the first half the factor  $(T - \varepsilon^2t)^{-1/2}$  is bounded and (25.70) applies; on the second half use  $|K_t| \leq \frac{1}{2}(|\nabla^+ p_t|^2 + |\nabla^- p_t|^2)$  and (25.68). The resulting integral is bounded uniformly in  $\varepsilon$ .  $\square$

The exact semimartingale has the mild form

$$\xi_t^\varepsilon(j) = \sum_k p_t^\varepsilon(j-k) \xi_0^\varepsilon(k) + \sum_k \int_0^t p_{t-s}^\varepsilon(j-k) dM_s^\varepsilon(k). \quad (25.73)$$

**Lemma 8.88** (Uniform weighted moments). *For every  $p \geq 2$  and  $T > 0$  there are  $a, C > 0$  such that*

$$\sup_{0 < \varepsilon \leq 1} \sup_{0 \leq t \leq \varepsilon^{-2}T} \sup_{j \in \mathbb{Z}} e^{-a\varepsilon|j|} \|\xi_t^\varepsilon(j)\|_{L^p} \leq C. \quad (25.74)$$

*Proof.* The initial term in (25.73) is bounded by (25.56) and (25.64). For the martingale term, Burkholder–Davis–Gundy and (25.63) yield, after increasing the exponential weight slightly,

$$F(t)^2 \leq C + C\varepsilon \int_0^t (t-s)^{-1/2} F(s)^2 ds, \quad (25.75)$$

where  $F(t) = \sup_j e^{-a\varepsilon|j|} \|\xi_t^\varepsilon(j)\|_{L^p}$ . Indeed  $\sum_k p_r^\varepsilon(k)^2 \leq Cr^{-1/2}$  by Plancherel. Putting  $t = \varepsilon^{-2}\tau$  turns (25.75) into

$$\tilde{F}(\tau)^2 \leq C + C \int_0^\tau (\tau-s)^{-1/2} \tilde{F}(s)^2 ds.$$

Iterate once. Since  $\int_0^t (t-s)^{-1/2} s^{-1/2} ds = \pi$ , the second iterate is bounded by an ordinary Grönwall inequality. Hence  $\sup_{\tau \leq T} \tilde{F}(\tau)$  is finite uniformly in  $\varepsilon$ .  $\square$

**Lemma 8.89** (Macroscopic space–time increments). *Let  $p \geq 2$ ,  $T, R < \infty$ . For every  $\alpha < 1/2$  and  $\beta < 1/4$  there is  $C = C(p, T, R, \alpha, \beta)$  such that, after spatial linear interpolation,*

$$\|Z_t^\varepsilon(x) - Z_t^\varepsilon(y)\|_{L^p} \leq C|x-y|^\alpha, \quad (25.76)$$

$$\|Z_t^\varepsilon(x) - Z_s^\varepsilon(x)\|_{L^p} \leq C|t-s|^\beta + C\varepsilon^\beta, \quad (25.77)$$

for  $|x|, |y| \leq R$  and  $s, t \in [0, T]$ .

*Proof.* Subtract the two mild formulas (25.73). The initial part is controlled by (25.57). For the martingale part use BDG, (25.63), Lemma 8.88, and (25.65). For a lattice displacement  $\ell$  this gives

$$\|\xi_u(j+\ell) - \xi_u(j)\|_{L^p}^2 \leq C(\varepsilon|\ell|)^{2\alpha} + C\varepsilon|\ell|^{2\alpha} \int_0^u (u-r)^{-1/2-\alpha} dr.$$

Since  $\alpha < 1/2$  and  $u \leq \varepsilon^{-2}T$ , the last expression is  $C(\varepsilon|\ell|)^{2\alpha}$ . Linear interpolation gives (25.76).

For time increments split the stochastic convolution at the earlier time. On  $[s, t]$  use  $\sum_j p_r(j)^2 \leq Cr^{-1/2}$ ; on  $[0, s]$  use (25.66). Taking any exponent below the endpoint  $1/4$  makes both singular integrals integrable after diffusive rescaling. The deterministic term is handled by the same kernel difference and (25.57). This yields (25.77).  $\square$



**Lemma 8.90** (Càdlàg field versus time interpolation). *Let  $\tilde{Z}^\varepsilon$  be obtained by linearly interpolating the microscopic transformed field between consecutive integer microscopic times. For every compact  $K \subset \mathbb{R}$ ,  $T > 0$ , and  $\delta > 0$ ,*

$$\mathbf{P} \left( \sup_{0 \leq t \leq T} \sup_{x \in K} |Z_t^\varepsilon(x) - \tilde{Z}_t^\varepsilon(x)| > \delta \right) \longrightarrow 0. \quad (25.78)$$

*Consequently the family  $Z^\varepsilon$  is tight in  $D([0, T]; C_{\text{loc}}(\mathbb{R}))$ , and every limit is supported on  $C([0, T]; C_{\text{loc}}(\mathbb{R}))$ .*

*Proof.* There are  $O(\varepsilon^{-3})$  unit microscopic space–time boxes meeting  $[0, \varepsilon^{-2}T] \times \varepsilon^{-1}K$ . During one unit microscopic time interval the number of jumps involving a fixed site is stochastically dominated by a Poisson variable of bounded mean. Each jump multiplies  $\xi(j)$  by  $e^{\pm 2\gamma_\varepsilon}$ , so its relative size is  $O(\sqrt{\varepsilon})$  for all jump counts below  $|\log \varepsilon|$ . Lemma 8.88 and Poisson tails therefore imply, for every  $m$ , a bound  $C_m \varepsilon^{m/2}$  for the  $m$ th moment of the maximum interpolation error on one box. Choose  $m > 6$  and sum over the  $O(\varepsilon^{-3})$  boxes; Markov’s inequality proves (25.78). Lemma 8.89 and Kolmogorov’s criterion give tightness of  $\tilde{Z}^\varepsilon$  in  $C([0, T] \times K)$  for every compact  $K$ . The displayed approximation then gives the asserted Skorokhod tightness and continuous support.  $\square$

**Lemma 8.91** (Bertini–Giacomin correlation cancellation). *For every  $T > 0$  and  $\delta > 0$  there are  $a, C > 0$  such that, for all  $\sqrt{\varepsilon} \leq s < t \leq T$ ,*

$$\sup_{j \in \mathbb{Z}} e^{-a\varepsilon|j|} \mathbf{E} \left| \nabla^+ \xi_{\varepsilon^{-2}t}^\varepsilon(j) \nabla^- \xi_{\varepsilon^{-2}s}^\varepsilon(j) \mid \mathcal{F}_{\varepsilon^{-2}s} \right| \leq C \varepsilon^{3/2-\delta} (t-s)^{-1/2}. \quad (25.79)$$

*Proof.* Put  $u = \varepsilon^{-2}s$  and  $v = \varepsilon^{-2}t$ . For  $r > 0$  write

$$N_r^v(j) = \sum_k \int_0^r p_{v-\tau}^\varepsilon(j-k) dM_\tau^\varepsilon(k).$$

Condition first at two times  $u \leq r_1 \leq r_2 < v$ . Orthogonality of the site martingales gives

$$\mathbf{E}(\nabla^+ N_{r_2}^v(j) \nabla^- N_{r_2}^v(j) \mid \mathcal{F}_{r_1}) = \nabla^+ N_{r_1}^v(j) \nabla^- N_{r_1}^v(j) + \mathbf{E} \left( \sum_k \int_{r_1}^{r_2} K_{v-\tau}^\varepsilon(j-k) d\langle M^\varepsilon(k) \rangle_\tau \mid \mathcal{F}_{r_1} \right). \quad (25.80)$$

Letting  $r_2 \uparrow v$  and expanding the mild formula at time  $u$  yields five terms:

$$\mathbf{E}(\nabla^+ \xi_v(j) \nabla^- \xi_v(j) \mid \mathcal{F}_u) = \nabla^+(p_v^\varepsilon * \xi_0)(j) \nabla^-(p_v^\varepsilon * \xi_0)(j) \quad (8.9)$$

$$+ \nabla^+(p_v^\varepsilon * \xi_0)(j) \nabla^- N_u^v(j) + \nabla^+ N_u^v(j) \nabla^-(p_v^\varepsilon * \xi_0)(j) \quad (8.10)$$

$$+ \nabla^+ N_u^v(j) \nabla^- N_u^v(j) \quad (8.11)$$

$$+ \mathbf{E} \left( \sum_k \int_u^v K_{v-r}^\varepsilon(j-k) d\langle M^\varepsilon(k) \rangle_r \mid \mathcal{F}_u \right). \quad (25.81)$$

All norms below are weighted by  $e^{-a\varepsilon|j|}$ . The initial-gradient term is controlled by Cauchy–Schwarz, (25.56), and (25.68):

$$\mathbf{E} |\nabla^\pm (p_r^\varepsilon * \xi_0)(j)|^2 \leq C e^{a\varepsilon|j|} \varepsilon^{-1} (1 \wedge r^{-3/2}). \quad (25.82)$$

For the martingale gradient, BDG, (25.63), Lemma 8.88, and (25.68) give

$$\mathbf{E} |\nabla^\pm N_u^v(j)|^2 \leq C e^{a\varepsilon|j|} \varepsilon (v-u)^{-1/2}. \quad (25.83)$$

Cauchy–Schwarz therefore controls the first four terms in (25.81) by the desired source size, after the harmless loss  $\varepsilon^{-\delta}$ .

It remains to analyze the bracket term. Insert (25.62); after factoring the coefficients it is a linear combination of

$$\begin{aligned} I_{v,u}^1(j) &= \sum_k \int_u^v K_{v-r}^\varepsilon(j-k) \mathbf{E}(\xi_r(k)^2 \mid \mathcal{F}_u) dr, \\ I_{v,u}^2(j) &= \sum_k \int_u^v K_{v-r}^\varepsilon(j-k) \mathbf{E}(\xi_r(k) \Delta \xi_r(k) \mid \mathcal{F}_u) dr, \\ I_{v,u}^3(j) &= - \sum_k \int_u^v K_{v-r}^\varepsilon(j-k) \mathbf{E}(\nabla^+ \xi_r(k) \nabla^- \xi_r(k) \mid \mathcal{F}_u) dr. \end{aligned} \quad (25.84)$$

The single-step constraint gives  $|\Delta \xi_r(k)| \leq C\sqrt{\varepsilon} \xi_r(k)$ ; hence (25.70) and Lemma 8.88 imply

$$\mathbf{E}|I_{v,u}^2(j)| \leq C e^{a\varepsilon|j|} \varepsilon^{1/2}. \quad (25.85)$$

The  $I^1$  term is where the exact cancellation enters. By (25.69),

$$\begin{aligned} I_{v,u}^1(j) &= \sum_k \int_u^v K_{v-r}^\varepsilon(j-k) \mathbf{E}(\xi_r(k)^2 - \xi_v(j)^2 \mid \mathcal{F}_u) dr \\ &\quad + \mathbf{E}(\xi_v(j)^2 \mid \mathcal{F}_u) \sum_k \int_{v-u}^\infty K_r^\varepsilon(j-k) dr. \end{aligned} \quad (25.86)$$

For the first line use  $|a^2 - b^2| \leq |a+b||a-b|$ , Cauchy–Schwarz, Lemma 8.88, and the space–time increment bounds of Lemma 8.89. Choosing any  $\alpha < 1/4$  gives

$$\mathbf{E}|\xi_r(k)^2 - \xi_v(j)^2| \leq C e^{a\varepsilon(|j|+|k|)} ((\varepsilon|j-k|)^{2\alpha} + (\varepsilon^2|v-r|)^\alpha + \varepsilon^{1/2}).$$

The weighted kernel integrals are finite because  $|K| \leq \frac{1}{2}(|\nabla^+ p|^2 + |\nabla^- p|^2)$  and (25.68); the second line of (25.86) is bounded by the same estimate using the tail integral in (25.71). We obtain, for every  $\delta > 0$ ,

$$\mathbf{E}|I_{v,u}^1(j)| \leq C e^{a\varepsilon|j|} (\varepsilon^{1/2-\delta} + (v-u)^{-1/2}). \quad (25.87)$$

Finally  $I^3$  has exactly the same conditional gradient product as the left side of the lemma. Define, for macroscopic  $\sqrt{\varepsilon} \leq s < t \leq T$ ,

$$f_{t,s}(j) = \varepsilon^{-1} e^{-a\varepsilon|j|} \mathbf{E} \left| \mathbf{E}(\nabla^+ \xi_{\varepsilon^{-2}t}(j) \nabla^- \xi_{\varepsilon^{-2}t}(j) \mid \mathcal{F}_{\varepsilon^{-2}s}) \right|, \quad F(t,s) = \sup_j f_{t,s}(j).$$

Equations (25.81)–(25.87) imply

$$\begin{aligned} F(t,s) &\leq C \varepsilon^{1/2-\delta} (1 + \varepsilon^{1/2}(t-s)^{-1/2}) \\ &\quad + C \varepsilon^{-2} \int_s^t \sum_j |K_{\varepsilon^{-2}(t-r)}^\varepsilon(j)| e^{a\varepsilon|j|} F(r,s) dr. \end{aligned} \quad (25.88)$$

After the change of variables  $q = \varepsilon^{-2}(t-r)$ , the Volterra operator in the second line has norm at most  $\rho < 1$  by (25.70); its action on the singular source  $(r-s)^{-1/2}$  is uniformly bounded by (25.71). Iterating (25.88) therefore produces a convergent Neumann series and gives

$$F(t,s) \leq C \varepsilon^{1/2-\delta} (t-s)^{-1/2}.$$

Multiplying back by  $\varepsilon$  proves (25.79). This is precisely why the mixed product  $\nabla^+ \xi \nabla^- \xi$  is usable whereas the square of one gradient would not have the required cancellation.  $\square$

**Lemma 8.92** (Weak martingale and bracket identification). *For  $\varphi \in C_c^\infty(\mathbb{R})$  write*

$$\langle f, \varphi \rangle_\varepsilon = \varepsilon \sum_{j \in \mathbb{Z}} f(\varepsilon j) \varphi(\varepsilon j)$$

and

$$\mathcal{M}_t^\varepsilon(\varphi) = \langle Z_t^\varepsilon, \varphi \rangle_\varepsilon - \langle Z_0^\varepsilon, \varphi \rangle_\varepsilon - a_\varepsilon \int_0^t \langle Z_s^\varepsilon, \Delta_\varepsilon \varphi \rangle_\varepsilon ds, \quad (25.89)$$

where  $\Delta_\varepsilon \varphi(x) = \varepsilon^{-2}[\varphi(x + \varepsilon) + \varphi(x - \varepsilon) - 2\varphi(x)]$ . Then  $\mathcal{M}^\varepsilon(\varphi)$  is a martingale and, for every fixed  $T$ ,

$$\sup_{t \leq T} \left| \langle \mathcal{M}^\varepsilon(\varphi), \mathcal{M}^\varepsilon(\psi) \rangle_t - \int_0^t \langle (Z_s^\varepsilon)^2, \varphi \psi \rangle_\varepsilon ds \right| \xrightarrow{L^1} 0. \quad (25.90)$$

*Proof.* Summation by parts in (25.61) gives (25.89). Since distinct-site microscopic martingales are orthogonal, rescaling time and inserting the exact bracket (25.62) gives the following finite algebraic decomposition:  $\mathcal{M}_t^\varepsilon(\varphi)^2$  minus

$$\int_0^t \langle (Z_s^\varepsilon)^2, \varphi^2 \rangle_\varepsilon ds - R_{1,t}^\varepsilon(\varphi) - R_{2,t}^\varepsilon(\varphi)$$

is a martingale, where

$$\begin{aligned} R_{1,t}^\varepsilon(\varphi) = \int_0^t ds \varepsilon \sum_j \varphi(\varepsilon j)^2 & \left[ (\varepsilon - (1 - e^{\gamma_\varepsilon})^2) (\xi_{\varepsilon-2s}^\varepsilon(j))^2 \right. \\ & \left. - \frac{1}{2} \varepsilon^{-1} (1 - e^{\gamma_\varepsilon})^2 \xi_{\varepsilon-2s}^\varepsilon(j) \Delta \xi_{\varepsilon-2s}^\varepsilon(j) \right], \end{aligned} \quad (25.91)$$

$$R_{2,t}^\varepsilon(\varphi) = \frac{1}{2} \int_0^t ds \varepsilon \sum_j \varphi(\varepsilon j)^2 \varepsilon^{-1} (1 + e^{2\gamma_\varepsilon}) \nabla^- \xi_{\varepsilon-2s}^\varepsilon(j) \nabla^+ \xi_{\varepsilon-2s}^\varepsilon(j). \quad (25.92)$$

This is simply (25.62) written in macroscopic variables; no replacement principle has been used.

Now  $\varepsilon - (1 - e^{\gamma_\varepsilon})^2 = O(\varepsilon^{3/2})$ , while the single-step constraint implies  $|\Delta \xi(j)| \leq C\sqrt{\varepsilon} \xi(j)$ . Lemma 8.88 therefore gives

$$\sup_{t \leq T} \|R_{1,t}^\varepsilon(\varphi)\|_{L^2} \leq C_{T,\varphi} \varepsilon^{1/2}. \quad (25.93)$$

For  $R_2$ , square the time integral, condition the later factor on the earlier filtration, and use  $|\nabla^\pm \xi(j)| \leq C\sqrt{\varepsilon} \xi(j)$ . The resulting double integral contains exactly the conditional expectation in Lemma 8.91. On the strip where the earlier macroscopic time is below  $\sqrt{\varepsilon}$ , the crude gradient bound and Lemma 8.88 give a quantity tending to zero. On the complement, (25.79) gives, for any fixed small  $\delta > 0$ ,

$$\mathbf{E} |R_{2,t}^\varepsilon(\varphi)|^2 \leq C \varepsilon^{1/2-\delta} \int_0^t \int_0^s (s-r)^{-1/2} dr ds + o(1) \longrightarrow 0. \quad (25.94)$$

Thus the predictable bracket differs from the Riemann-sum target by an  $L^1$ -vanishing quantity. This proves (25.90) for  $\varphi = \psi$ ; polarization gives the cross-variation statement.  $\square$

**Lemma 8.93** (Real-line SHE martingale problem). *Let  $Q$  be a probability law on  $C([0, T]; C_{\text{loc}}(\mathbb{R}))$  with canonical field  $f$ , initial law  $f_0 = e^{-H_0}$ , and an exponential weighted second-moment bound. Suppose that for every  $\varphi \in C_c^\infty(\mathbb{R})$ ,*

$$M_t(\varphi) = \langle f_t, \varphi \rangle - \langle f_0, \varphi \rangle - \frac{1}{2} \int_0^t \langle f_s, \varphi'' \rangle ds,$$

$$M_t(\varphi)^2 - \int_0^t \int_{\mathbb{R}} f_s(x)^2 \varphi(x)^2 dx ds$$

*are local martingales. Then  $Q$  is the law of the unique weighted mild solution of*

$$\partial_t Z = \frac{1}{2} \partial_x^2 Z + Z \xi, \quad Z(0) = e^{-H_0}. \quad (25.95)$$

*Proof.* Polarization shows  $\langle M(\varphi), M(\psi) \rangle_t = \int_0^t \int f_s^2 \varphi \psi$ . Hence the family  $M(\varphi)$  defines an orthogonal martingale measure  $\mathcal{M}(ds, dx)$  with covariance measure  $f_s(x)^2 ds dx$ . On an enlarged probability space set  $dW = f^{-1} d\mathcal{M}$  on  $\{f \neq 0\}$  and add an independent white noise on  $\{f = 0\}$ . Then  $W$  is space-time white noise and  $d\mathcal{M} = f dW$ . Test the weak equation against the backward heat solution  $P_{t-s}\varphi$  and use stochastic Fubini; one obtains

$$f_t = P_t f_0 + \int_0^t P_{t-s}(f_s dW_s),$$

so every martingale-problem solution is mild.

Conversely the Walsh isometry gives this martingale problem for every mild solution. To prove uniqueness, couple two mild solutions with the same  $W$  and let  $U$  be their difference. With  $D(t) = \sup_x e^{-2a|x|} \mathbf{E}|U_t(x)|^2$ , the heat-kernel square estimate and exponential-weight convolution give

$$D(t) \leq C \int_0^t (t-s)^{-1/2} D(s) ds.$$

Iterating once converts this to an ordinary Grönwall inequality with zero initial value, hence  $D \equiv 0$ . Picard iteration with the same estimate gives existence in the weighted class. This proves uniqueness of both the mild equation and the martingale problem.  $\square$

**Proposition 8.94** (Convergence of the microscopic Gärtner transform). *Under Definition 8.85, for every  $T < \infty$ ,*

$$Z^\varepsilon \implies Z \quad \text{in } D([0, T]; C_{\text{loc}}(\mathbb{R})), \quad (25.96)$$

*where  $Z$  is the weighted mild solution of (25.95). The limiting law is supported on  $C([0, T]; C_{\text{loc}}(\mathbb{R}))$ .*

*Proof.* Lemmas 8.89 and 8.90 give tightness and continuity of every limit. Lemma 8.88 and Fatou preserve the exponential weighted second-moment bound. Let a subsequence converge and use Skorokhod representation only to simplify the notation; all following statements are distributional. Since  $a_\varepsilon \rightarrow 1/2$  and  $\Delta_\varepsilon \varphi \rightarrow \varphi''$  uniformly, (25.89) passes to the weak drift equation. Uniform integrability follows from Lemma 8.88, while (25.90) passes the quadratic variation. The limit therefore solves the martingale problem of Lemma 8.93. The initial condition also passes correctly: on every compact set,  $H_0^\varepsilon \Rightarrow H_0$  and  $\gamma_\varepsilon/\sqrt{\varepsilon} \rightarrow 1$ , hence  $Z_0^\varepsilon = \exp\{-(\gamma_\varepsilon/\sqrt{\varepsilon})H_0^\varepsilon\} \Rightarrow e^{-H_0}$  by the continuous-mapping theorem. Its uniqueness identifies every subsequential limit and hence the whole sequence. Bertini–Giacomin write the limiting noise with a minus sign,  $dZ = \frac{1}{2}\Delta Z dt - Z dW$ ; replacing  $W$  by  $-W$  gives the plus-sign convention in (25.95) with the identical law.  $\square$

**Theorem 8.95** (Bertini–Giacomin particle-to-KPZ theorem). *Assume the weakly asymmetric single-step dynamics above and BG-admissible initial laws. Put*

$$v_\varepsilon = \frac{1}{2}\varepsilon^{-3/2} - \frac{1}{24}\varepsilon^{-1/2}, \quad \mathcal{H}_t^\varepsilon(r) = \sqrt{\varepsilon} \left( \zeta_{\varepsilon^{-2}t}(\varepsilon^{-1}r) - v_\varepsilon t \right). \quad (25.97)$$

Then, for every  $T < \infty$ ,

$$\mathcal{H}^\varepsilon \Longrightarrow H \quad \text{in } D([0, T]; C_{\text{loc}}(\mathbb{R})), \quad (25.98)$$

the limit is supported on  $C([0, T]; C_{\text{loc}}(\mathbb{R}))$ , and in the Bertini–Giacomin sign convention

$$H = -\log Z,$$

where  $Z$  is the strictly positive solution of (25.95). In particular the corresponding WASEP fluctuation field converges, after spatial distributional differentiation, to the stochastic Burgers field  $\partial_x H$  in  $D([0, T]; \mathcal{D}'(\mathbb{R}))$ .

*Proof.* Proposition 8.94 proves the transformed-process limit. The inverse microscopic Gärtner transform gives the exact identity

$$\mathcal{H}_t^\varepsilon(r) = -\frac{\sqrt{\varepsilon}}{\gamma_\varepsilon} \log Z_t^\varepsilon(r) + \sqrt{\varepsilon} \left( \frac{\lambda_\varepsilon}{\gamma_\varepsilon} \varepsilon^{-2} - v_\varepsilon \right) t. \quad (25.99)$$

Taylor expansion at  $\varepsilon = 0$  gives

$$\begin{aligned} \gamma_\varepsilon &= \sqrt{\varepsilon} - \varepsilon + \frac{4}{3}\varepsilon^{3/2} + O(\varepsilon^2), \\ \frac{\lambda_\varepsilon}{\gamma_\varepsilon} &= \frac{1}{2}\sqrt{\varepsilon} - \frac{1}{24}\varepsilon^{3/2} + O(\varepsilon^2). \end{aligned}$$

Consequently  $\sqrt{\varepsilon}/\gamma_\varepsilon \rightarrow 1$  and the second coefficient in (25.99) tends to zero.

The strict-positivity theorem 2.5, proved internally in Chapter 2, gives  $Z_t(x) > 0$  simultaneously for all  $t > 0$  and  $x \in \mathbb{R}$ . On every  $[\tau, T] \times [-R, R]$ ,  $\tau > 0$ , its compact lower-bound conclusion makes  $f \mapsto -\log f$  continuous in the uniform norm. At time zero the BG initial convergence and (25.56) give the matching initial value. Applying the continuous-mapping theorem to (25.96) and then letting  $\tau \downarrow 0$ ,  $R \uparrow \infty$ , proves (25.98).

Finally, for every test function  $\varphi$ ,

$$\langle \partial_x \mathcal{H}_t^\varepsilon, \varphi \rangle = - \int \mathcal{H}_t^\varepsilon(x) \varphi'(x) dx.$$

The map on the right is continuous from  $C_{\text{loc}}$  to  $\mathbb{R}$ . Applying it simultaneously to a countable dense family of tests yields the claimed distribution-valued Burgers convergence.  $\square$

**Microscopic KPZ proof inventory.** The argument supplies the exact microscopic Gärtner semimartingale and bracket, Fourier heat-kernel bounds, weighted moments, space–time Hölder estimates, interpolation control, the Bertini–Giacomin correlation cancellation, weak bracket identification, the real-line SHE martingale problem, and the inverse Gärtner centering. In particular there is no residual microscopic drift  $r^\varepsilon$  and no generic Boltzmann–Gibbs replacement in this proof. The strict-positivity input is Theorem 2.5, so the logarithmic inversion is justified within the selected one-dimensional scope.

*Remark 8.96. Source and scope.* The statement and proof architecture above are checked against L. Bertini and G. Giacomin, *Stochastic Burgers and KPZ Equations from Particle Systems*, Comm. Math. Phys. 183 (1997), 571–607: Definition 2.2, Theorems 2.3 and 3.3, the exact Gärtner equation (3.13)–(3.15), Lemmas 4.1–4.8, Definition 4.10, Proposition 4.11, Appendix A, and Theorem B.1. The source itself imports strict positivity from Mueller; this edition now proves the required one-dimensional linear-Gaussian positivity theorem in Chapter 2, while retaining Mueller as the historical primary source for that mechanism.

## 8.6 TASEP and the KPZ fixed point

For TASEP, exact determinantal formulas express the event that the height at several points lies below thresholds as a Fredholm determinant of an operator built from a one-dimensional random-walk hitting kernel and the initial height profile.

**Definition 8.97** (MQR upper-semicontinuous state space). Let UC denote the set of upper-semicontinuous functions  $h : \mathbb{R} \rightarrow [-\infty, \infty)$  which are not identically  $-\infty$  and satisfy an upper linear-growth bound  $h(x) \leq C(1 + |x|)$  for some  $C$ . We give UC the *local hypograph topology*:  $h_n \rightarrow h$  when the closed hypographs

$$\text{hypo}(h_n) = \{(x, y) : y \leq h_n(x)\}$$

converge locally in Hausdorff distance after compactifying the height coordinate by  $y \mapsto e^y$ , with a common upper linear-growth bound. Let LC = -UC, equipped with the corresponding local epigraph topology. This is the Matetski–Quastel–Remenik state space; on finite-valued continuous functions its topology agrees with uniform convergence on compact sets.

**Lemma 8.98** (Exact MQR biorthogonalization and hitting representation). *For right-finite TASEP data  $X_0(1) > X_0(2) > \dots$  set*

$$Q(x, y) = 2^{-(x-y)} \mathbf{1}_{x>y}, \quad R_t(x, y) = e^{-t} \frac{t^{x-y}}{2^{x-y}(x-y)!} \mathbf{1}_{x \geq y}.$$

Then

$$Q^m(x, y) = 2^{-(x-y)} \binom{x-y-1}{m-1} \mathbf{1}_{x \geq y+m}, \quad Q^{-1}(x, y) = 2 \mathbf{1}_{x=y-1} - \mathbf{1}_{x=y}, \quad (25.100)$$

and for  $0 \leq k < n$

$$\Psi_k^n(x) = R_t Q^{-k} \delta_{X_0(n-k)}(x) = \frac{1}{2\pi i} \oint_{\Gamma_0} \frac{(1-w)^k e^{t(w-1)}}{2^{x-X_0(n-k)} w^{x+k+1-X_0(n-k)}} dw. \quad (25.101)$$

For fixed  $0 \leq k < n$ , let  $h_k^n(\ell, z)$  be the unique solution of

$$(Q^*)^{-1} h_k^n(\ell, \cdot) = h_k^n(\ell + 1, \cdot), \quad \ell < k, \quad h_k^n(k, z) = 2^{z-X_0(n-k)}, \quad (25.102)$$

with boundary conditions

$$h_k^n(\ell, X_0(n-\ell)) = 0, \quad \ell < k. \quad (25.103)$$

Then

$$\Phi_k^n(z) = (R_t^*)^{-1} h_k^n(0, \cdot)(z) \quad (25.104)$$

is the required biorthogonal dual family. Moreover, if

$$\bar{Q}^{(m)}(x, y) = \frac{(x-y-1)_{m-1}}{2^{x-y}(m-1)!}$$

denotes the polynomial extension of  $Q^m$ , and  $B$  is the downward geometric walk with kernel  $Q$ , then with

$$\tau = \min\{r \geq 0 : B_r > X_0(r+1)\}$$

the boundary term satisfies

$$G_{0,n}(z_1, z_2) = \mathbf{E}_{B_0=z_1} \left[ \bar{Q}^{(n-\tau)}(B_\tau, z_2) \mathbf{1}_{\tau < n} \right]. \quad (25.105)$$

Consequently, with the MQR operators

$$\mathcal{S}_{-t,-n} = (e^{-t\nabla^-/2} Q^{-n})^*, \quad \bar{\mathcal{S}}_{-t,n} = \bar{Q}^{(n)} e^{t\nabla^-/2}, \quad (25.106)$$

and

$$\bar{\mathcal{S}}_{-t,n}^{\text{epi}(X_0)}(z_1, z_2) = \mathbf{E}_{B_0=z_1} \left[ \bar{\mathcal{S}}_{-t,n-\tau}(B_\tau, z_2) \mathbf{1}_{\tau < n} \right], \quad (25.107)$$

the right-finite extended kernel is

$$K_t^{\text{TASEP}}(n_i, \cdot; n_j, \cdot) = -Q^{n_j-n_i} \mathbf{1}_{n_i < n_j} + (\mathcal{S}_{-t,-n_i})^* \bar{\mathcal{S}}_{-t,n_j}^{\text{epi}(X_0)}. \quad (25.108)$$

*Proof.* The convolution identity for  $Q^m$  follows by counting compositions of  $x - y$  into  $m$  positive jumps. Multiplying the displayed kernel for  $Q^{-1}$  by  $Q$  gives the identity operator; translation invariance implies that  $Q$  commutes with  $R_t$ . Taking the residue of the contour formula for  $R_t Q^{-k} \delta_{X_0(n-k)}$  gives (25.101).

We next solve (25.102)–(25.103). On the vector space of all sequences,  $(Q^*)^{-1}$  has one-dimensional kernel spanned by  $z \mapsto 2^z$ . Thus, starting from level  $k$ , the equation determines the next lower level up to a multiple of  $2^z$ , and the single boundary value (25.103) fixes that multiple. This proves existence and uniqueness by downward induction.

Set  $\tilde{h}_k^n(\ell, z) = 2^{-z} h_k^n(\ell, z)$ . From (25.100),

$$\tilde{h}_k^n(\ell, y) = \tilde{h}_k^n(\ell - 1, y - 1) - \tilde{h}_k^n(\ell - 1, y). \quad (25.109)$$

At  $\ell = k$ ,  $\tilde{h}$  is constant. If  $\tilde{h}_k^n(\ell, \cdot)$  is a polynomial of degree at most  $k - \ell$ , then summing (25.109) from the boundary point  $X_0(n - \ell + 1)$  to  $x$  shows that  $\tilde{h}_k^n(\ell - 1, x)$  is a polynomial of degree at most  $k - \ell + 1$  on each side of the boundary. The two polynomial expressions agree because they satisfy the same finite-difference identity and the same boundary value. Hence  $2^{-z} h_k^n(0, z)$  is a polynomial of degree at most  $k$ .

Now substitute (25.101) and (25.104). Since  $R_t$  is invertible and commutes with  $Q$ ,

$$\sum_z \Psi_\ell^n(z) \Phi_k^n(z) = (Q^*)^{-\ell} h_k^n(0, X_0(n - \ell)). \quad (25.110)$$

If  $\ell \leq k$ , repeatedly use (25.102). For  $\ell < k$  the boundary value (25.103) gives zero, whereas for  $\ell = k$  the terminal value gives one. If  $\ell > k$ , the terminal sequence  $2^{z-X_0(n-k)}$  is killed by  $(Q^*)^{-1}$ , so the value is again zero. Thus (25.110) is  $\mathbf{1}_{\ell=k}$ .

It remains to check the polynomial condition on  $\Phi$ . Expanding the inverse Poisson kernel gives

$$2^{-x} \Phi_k^n(x) = \sum_{j \geq 0} e^t \frac{(-t)^j}{j!} 2^{-(x+j)} h_k^n(0, x + j), \quad (25.111)$$

which is a polynomial of degree at most  $k$ , since the exponential generating series acts termwise on a fixed polynomial. This completes the exact biorthogonalization.

For the hitting representation, first run the adjoint  $Q^*$ -walk upward from level  $\ell - 1$ . The geometric jump law is memoryless; therefore the solution of the boundary problem equals the probability that this walk first crosses the initial particle curve at the prescribed level. Substituting this first-hit expression into the biorthogonal sum, reversing the walk, and using the polynomial

continuation  $\bar{Q}^{(m)}$  yields (25.105). The continuation is essential:  $Q^m$  itself vanishes outside its natural support, while the boundary algebra requires evaluation at the hitting point. Finally conjugate by  $e^{-t\nabla^-/2}$  and insert (25.105). This gives (25.107) and hence the free-plus-hitting form (25.108), which is the exact right-finite TASEP kernel.  $\square$

**Lemma 8.99** (MQR cutoff propagation from the trace telescoping estimate). *Fix a positive scaled time and finitely many scaled observation points. Let  $h_0^{\varepsilon,L}$  be the right-finite cutoff of an admissible initial profile at scaled distance  $L$ . For every  $\delta > 0$  there is  $C < \infty$ , independent of  $\varepsilon$  and  $L$ , such that the difference between the cutoff and uncut multipoint distribution functions is bounded by*

$$C \sum_{\ell \geq L} e^{-(2/3-\delta)\ell^3}. \quad (25.112)$$

*In particular the cutoff error vanishes uniformly as  $L \rightarrow \infty$ , after which  $\varepsilon \downarrow 0$  may be taken. Thus right-finite asymptotics extend to general admissible data.*

*Proof.* MQR's cutoff is compared one unit at a time. For the one-point scattering operator write  $A_\ell^\varepsilon$  for the kernel obtained after truncation at level  $\ell$ . The exact hitting representation of Lemma 8.98 and the strong Markov property give the telescoping identity

$$A_L^\varepsilon - A_{L-1}^\varepsilon = \int H_L^\varepsilon(dz, ds, db) f_{L,z}^\varepsilon \otimes g_{L,s,b}^\varepsilon, \quad (25.113)$$

where  $H_L^\varepsilon$  is the first-hit measure on the newly exposed piece of the cutoff boundary. The rank-one trace norm is the product of the two  $L^2$  norms. The contour estimates recorded in Lemma 8.100 below imply, for every  $\delta > 0$ ,

$$\|A_L^\varepsilon - A_{L-1}^\varepsilon\|_1 \leq C e^{-(2/3-\delta)L^3}. \quad (25.114)$$

The constant is uniform in the microscopic mesh. Summing (25.114) from  $L$  to infinity gives a trace-norm difference bounded by the right-hand side of (25.112).

For one threshold, the standard determinant inequality

$$|\det(I - A) - \det(I - B)| \leq \|A - B\|_1 e^{1+\|A\|_1+\|B\|_1} \quad (25.115)$$

turns this trace bound into the same cubic cutoff bound for probabilities. For finitely many observation points, apply the one-point cutoff successively and use a union/telescoping bound; the number of points is fixed. This is the source-strength finite-propagation argument. In particular, the elementary Poisson-arrow light-cone argument is not used: at KPZ times it does not provide the uniform cubic cutoff estimate required by the Fredholm analysis.  $\square$

**Lemma 8.100** (Uniform MQR contour and trace-class estimates). *Let  $\mathbf{S}^\varepsilon$  and  $\bar{\mathbf{S}}^\varepsilon$  be the rescaled MQR kernels and put*

$$e^{2F_\varepsilon(x,u)} = \int_{-\infty}^0 |\mathbf{S}_{-1,x}^\varepsilon(u-\eta)|^2 d\eta, \quad e^{2\bar{F}_\varepsilon(x,u)} = \int_{-\infty}^0 |\bar{\mathbf{S}}_{-1,x}^\varepsilon(u-\eta)|^2 d\eta. \quad (25.116)$$

*Set*

$$x_\varepsilon = x - \frac{\varepsilon^{1/2}}{2}u - \frac{\varepsilon}{2}, \quad u_\varepsilon = u - \varepsilon^{1/2}, \quad \bar{x}_\varepsilon = x + \frac{\varepsilon^{1/2}}{2}u + \frac{3\varepsilon}{2}, \quad \bar{u}_\varepsilon = u + \varepsilon^{1/2}. \quad (25.117)$$

*The double-contour formulas for (25.116) have exponent*

$$\mathcal{F}(w; x, u) = \operatorname{arctanh} w - w - x \log(1 - w^2) - u \operatorname{arctanh} w. \quad (25.118)$$

*Uniformly in the microscopic mesh, the following estimates hold.*



1. If  $x_\varepsilon^2 + u_\varepsilon \geq 0$  and  $-x_\varepsilon + \sqrt{x_\varepsilon^2 + u_\varepsilon} \geq \varepsilon^{-1/2}$ , then  $F_\varepsilon(x, u) = -\infty$ .

2. There is  $d > 0$  such that if

$$-d \leq -x_\varepsilon + \sqrt{x_\varepsilon^2 + u_\varepsilon} < \varepsilon^{-1/2}, \quad \varepsilon^{1/2}x_\varepsilon \in (1 - \sqrt{5}, 1 + \sqrt{5}), \quad (25.119)$$

then  $F_\varepsilon(x, u) \leq \widehat{F}_\varepsilon(x_\varepsilon, u_\varepsilon)$ ; the same statement holds with barred variables.

3. If  $x, x_\varepsilon \geq 0$ , then  $F_\varepsilon(x, u) \leq C \log(2 + |x| + |u|)$ . If  $\bar{x}_\varepsilon \geq 0$  and  $u \geq -\varepsilon^{-1/2}x$ , the same bound holds for  $\bar{F}_\varepsilon$ .

4. If  $x_\varepsilon^2 + u_\varepsilon < 0$ ,  $-C\varepsilon^{-1/4} \leq x_\varepsilon < 0$  and  $|u_\varepsilon| \leq C\varepsilon^{-1/2}$ , then for every  $\delta > 0$ ,

$$F_\varepsilon(x, u) \leq -(2/3 - \delta)|x|^3 + C\delta^{-1}|u|^{3/2} + C', \quad (25.120)$$

where  $C'$  depends only on a bound for  $|u_\varepsilon|^{1/2}/|x_\varepsilon|$ .

5. If  $\bar{x}_\varepsilon^2 + \bar{u}_\varepsilon < 0$ ,  $-C < \bar{x}_\varepsilon < 0$  and  $|\bar{u}_\varepsilon| \leq C$ , then  $\bar{F}_\varepsilon(x, u) \leq C$ .

Consequently, for admissible initial profiles satisfying a common linear lower bound, the one-sided cutoff scattering operators are uniformly trace class. Moreover, if  $A_\ell^\varepsilon$  and  $A_{\ell-1}^\varepsilon$  are successive cutoff operators, then for every  $\delta > 0$

$$\|A_\ell^\varepsilon - A_{\ell-1}^\varepsilon\|_1 \leq Ce^{-(2/3-\delta)\ell^3}. \quad (25.121)$$

The rescaled scattering kernels, including the off-diagonal  $Q$  term, converge in trace norm to the Brownian-scattering kernels; the convergence is uniform on compact Hölder families of initial data.

*Proof.* We give the contour estimate rather than treating it as a black box. Squaring the single-contour kernels and summing the half-line variable gives double contours on  $C_\varepsilon^o$ , the circle of radius  $\varepsilon^{-1/2}$  centred at  $\varepsilon^{-1/2}$  with a small indentation keeping 0 outside. The unbarred exponent is  $\varepsilon^{-3/2} \sum_{j=1}^2 \mathcal{F}(\varepsilon^{1/2}w_j; \varepsilon^{1/2}x_\varepsilon, \varepsilon u_\varepsilon)$ ; the barred formula is identical with barred variables, up to bounded rational prefactors.

Write  $y = x^2 + u$  and  $w_+ = -x + \sqrt{y}$ . Evaluation at the critical point can be written

$$\widehat{F}_\varepsilon(x, u) = \Re \left[ (1 + \nu_1(\varepsilon^{1/2}w_+)) \left( xy - \frac{x^3}{3} - \frac{2y^{3/2}}{3} \right) + \nu_2(\varepsilon^{1/2}w_+) \left( xy - \frac{x^2y^{1/2}}{2} - \frac{y^{3/2}}{2} \right) \right], \quad (25.122)$$

where  $\nu_1, \nu_2$  are analytic off the cuts  $(-\infty, -1] \cup [1, \infty)$ , uniformly bounded there on the contours used below, satisfy  $\nu_i(w) = O(w^2)$  near zero, and are nonnegative on  $(-1, 1)$ . Their power series follow directly by expanding  $\operatorname{arctanh} w$  and  $\log(1 - w^2)$ .

*Case 1.* The hypothesis implies

$$\alpha_\varepsilon = \frac{1}{2}\varepsilon^{-3/2} + \varepsilon^{-1}x_\varepsilon - \frac{1}{2}\varepsilon^{-1/2}u_\varepsilon \leq 0.$$

This is precisely the condition under which the origin ceases to be a pole of the contour integrand. The contour may therefore be contracted and the kernel is zero.

*Case 2.* Put  $\tilde{w}_+ = -\tilde{x} + \sqrt{\tilde{x}^2 + \tilde{u}}$ , using either barred or unbarred variables. Deform  $C_\varepsilon^o$  through  $\tilde{w}_+$  and follow the rays  $\tilde{w}_+ + re^{\pm i\pi/4}$  until they meet the right arc of the original circle. No singularity is crossed. Since

$$\partial_w \mathcal{F}(w; x, u) = \frac{(w - w_+)(w - w_-)}{1 - w^2}, \quad (25.123)$$

a direct differentiation on the ray gives

$$\frac{d}{dr} \Re \left[ \varepsilon^{-3/2} \mathcal{F}(\varepsilon^{1/2}(\tilde{w}_+ + re^{i\pi/4}); \varepsilon^{1/2}\tilde{x}, \varepsilon\tilde{u}) \right] = -Kr^2. \quad (25.124)$$

With  $w = \varepsilon^{1/2}\tilde{w}_+$  and  $\tilde{r} = \varepsilon^{1/2}r$ , the explicit rational expression for  $K$  has numerator bounded below by  $\sqrt{2}(1 - w^2 + \tilde{r}^2)$  and denominator at most

$$10((1 - w^2)^2 + \tilde{r}^2 w^2 + \tilde{r}^2).$$

Using

$$1 - w^2 + \tilde{r}^2 \geq \frac{1}{2}(1 - w^2)(1 - w^2 - \tilde{r}^2) + \tilde{r}^2$$

gives  $K \geq \sqrt{2}/20$  for  $w \in [0, 1)$ , and by continuity for  $w \geq -d$  with fixed small  $d > 0$ . Hence the ray integral is bounded by a constant times its saddle value. On the right circular arc, differentiation with  $w = 1 + e^{i\theta}$  gives a factor

$$\frac{(4(1 + \cos \theta) + 2\varepsilon^{1/2}\tilde{x} + \varepsilon\tilde{u}) \sin \theta}{5 + 4 \cos \theta}. \quad (25.125)$$

It is nonnegative under (25.119), because  $\tilde{u} \geq -\tilde{x}^2$  and  $\varepsilon^{1/2}\tilde{x} \in (1 - \sqrt{5}, 1 + \sqrt{5})$ . The real part therefore continues to decrease. This proves case 2.

*Case 3.* When the critical point cannot be reached without crossing a pole, let  $q = (1 + |\tilde{u}| + |\tilde{x}|)^{-1}$ , deform through  $q$ , move vertically, then join the  $\pi/4$  ray above. On the vertical segment

$$\frac{d}{dr} \Re(\varepsilon^{-3/2} \mathcal{F}(\varepsilon^{1/2}(q + ir); \varepsilon^{1/2}\tilde{x}, \varepsilon\tilde{u})) = (-2(q + \tilde{x}) + O(\varepsilon))r, \quad (25.126)$$

so the real part decreases for nonnegative  $\tilde{x}$ . The ray estimate is case 2. On the circular arc, the unbarred exponent loses its  $u$  dependence and decreases for  $x \geq 0$ ; for the barred exponent the same derivative has  $x + \varepsilon^{1/2}u$ , so the extra condition  $u \geq -\varepsilon^{-1/2}x$  is exactly what is needed. The denominator at  $q$  contributes only  $\log(2 + |x| + |u|)$ , proving case 3.

*Cases 4 and 5.* The critical points are now complex conjugates. Join 0 to  $w_+$ , leave  $w_+$  along a  $\pi/4$  ray, then follow the circular arc and reflect. On the first segment, writing  $w_+ = Re^{i\theta}$  with  $\theta \in (0, \pi/2)$ , formula (25.123) shows that the real derivative has the correct sign; the perturbation from the denominator  $1 - \varepsilon r^2 w_+^2$  remains uniform because  $|w_+| \leq C\varepsilon^{-1/4}$ . Along the outgoing ray,

$$\partial_r(\varepsilon^{-3/2} \mathcal{F}) = -r(2|y|^{1/2} - re^{3\pi i/4})(1 - \varepsilon(w_+ + re^{i\pi/4})^2)^{-1}, \quad (25.127)$$

whose real part is negative for sufficiently small  $\varepsilon$ . Hence the integral is bounded by the saddle value (25.122). Since  $\nu_i(\varepsilon^{1/2}w_+) = O(\varepsilon^{1/2})$  and

$$|xy - x^3/3 - 2y^{3/2}/3| + |xy - x^2y^{1/2}/2 - y^{3/2}/2| \leq C(|x|^3 + |u|^{3/2}), \quad (25.128)$$

case 4 follows from

$$xu + \frac{2}{x}x^3 \leq -(2/3 - \delta)|x|^3 + C\delta^{-1}|u|^{3/2}, \quad x < 0. \quad (25.129)$$

Under the compact hypotheses of case 5, (25.128) is uniformly bounded and so is the saddle value. This proves all five estimates with constants uniform in  $\varepsilon$ .

We now prove the trace statements. The stopped scattering kernel is the Bochner integral of rank-one kernels over the first-hit measure. Thus the trace norm is bounded by

$$\int e^{F_\varepsilon(x_1, z) + \bar{F}_\varepsilon(-x_2 - s, b)} H^\varepsilon(dz, ds, db). \quad (25.130)$$

The admissible initial profile has a uniform linear lower barrier  $b \geq -\alpha - \gamma s$ . For  $z$  above the barrier, cases 1–2 give cubic Airy decay. For  $z$  below it, case 3 gives only logarithmic growth in the first factor, but the first-hit event is controlled exponentially by applying Doob's inequality to  $\exp\{\lambda(B_\varepsilon(s) + \alpha + \gamma s)\}$ ; its moment generating function is uniformly quadratic for bounded  $\lambda$ . The barred factor is bounded by cases 2 and 5 for bounded  $s$  and by a negative cubic in  $s$  for large  $s$ . Hence (25.130) is uniformly integrable, proving the  $L = 0$  trace bound.

For arbitrary cutoff  $L$ , translate the cutoff boundary and telescope in integer levels. At level  $\ell$  the unbarred factor has  $x = x_1 - \ell$ . The five-case split above yields

$$F_\varepsilon(x_1 - \ell, z) \leq -(2/3 - \delta)\ell^3 + C\delta^{-1}|z|^{3/2}\mathbf{1}_{z \leq -\alpha - \gamma\ell} - c|z|^{3/2}\mathbf{1}_{z \geq -\alpha - \gamma\ell} + C. \quad (25.131)$$

The barred factor is uniformly bounded or cubically decaying according to the same split, while the first-hit probability below the barrier carries a Gaussian exponential from the preceding martingale estimate. Substitution in (25.130) leaves an integrable  $z$ -integral independent of  $\ell$  and  $\varepsilon$ , proving (25.121). Summing the cubic tail proves the uniform cutoff trace bound.

Finally, on compact hitting variables Donsker convergence gives convergence of the first-hit measures and the contour kernels converge in the Hilbert–Schmidt factors. The complement is uniformly small by the trace estimates just proved. A bounded continuous Banach-valued function integrates continuously against weakly convergent probability measures (prove this by tightness plus Skorokhod coupling), so the stopped scattering operators converge in trace class. For the off-diagonal term split  $Q^n - e^{x\partial^2}$  into two products of half-time sum/difference factors; the weighted local central limit theorem gives Hilbert–Schmidt convergence of the difference factors and uniform Hilbert–Schmidt bounds of the sum factors. The inequality  $\|AB\|_1 \leq \|A\|_2\|B\|_2$  then gives trace-norm convergence. All constants used above are uniform on compact Hölder families, proving the last assertion.  $\square$

**Lemma 8.101** (Airy/Brownian limit and trace-class convergence). *Under the MQR scaling*

$$h^\varepsilon(t, x) = \varepsilon^{1/2} \left[ h_{2\varepsilon^{-3/2}t}(2\varepsilon^{-1}x) + \varepsilon^{-3/2}t \right], \quad (25.122)$$

*the rescaled free kernels converge to the Airy propagators, the discrete first-hit operators converge to Brownian scattering from the limiting initial profile, and the full threshold-restricted extended kernels converge in trace norm, uniformly on compact scaled parameter sets and compact families of initial profiles.*

*Proof.* At the characteristic critical point put  $w = w_c + \varepsilon^{1/2}z$ . The centering cancels the linear and quadratic terms, while the first surviving term is cubic. The contour in the proof of Lemma 8.100 localizes to  $|z| \leq \varepsilon^{-\eta}$ ; Taylor expansion there gives the Airy exponent, and the five-regime estimates dominate the complement uniformly.

For the hitting term, Donsker's theorem gives convergence of the geometric walk to Brownian motion. Local hypograph convergence of the initial profiles implies convergence of the first-hit time/location at every Brownian path which does not have a tangential first contact with the boundary; this exceptional set has Wiener measure zero. Therefore the stopped kernels converge pointwise on compact sets.

To upgrade the convergence, first truncate the hitting variables to a compact set. On that compact set, the hitting measures converge weakly and the kernel is an  $L^2$ -valued continuous function, so the corresponding factors converge in Hilbert–Schmidt norm. Outside the compact set, Lemma 8.100 makes the trace norm uniformly small. Thus, if  $A_\varepsilon B_\varepsilon$  denotes the standard factorization,

$$\|A_\varepsilon B_\varepsilon - AB\|_1 \leq \|A_\varepsilon - A\|_2 \|B_\varepsilon\|_2 + \|A\|_2 \|B_\varepsilon - B\|_2 \longrightarrow 0. \quad (25.123)$$

The off-diagonal  $Q$ -part is treated by the same half-convolution factorization. This proves trace-class convergence of the full restricted kernel.  $\square$

**Lemma 8.102** (Fredholm transition law and continuity in initial data). *For every  $t > 0$  and admissible  $h_0 \in \text{UC}$ , the Brownian-scattering kernel of Lemma 8.101 defines consistent multipoint threshold probabilities by a Fredholm determinant. These probabilities determine a probability law  $P_t(h_0, \cdot)$  on UC. If  $h_{0,n} \rightarrow h_0$  locally in hypograph topology with a common upper linear growth bound, then  $P_t(h_{0,n}, \cdot) \Rightarrow P_t(h_0, \cdot)$ .*

*Proof.* For the prelimit kernels the Fredholm determinant is an actual TASEP probability. Lemma 8.101 gives trace-norm convergence after the threshold projection, and

$$|\det(I - A) - \det(I - B)| \leq \|A - B\|_1 \exp\{1 + \|A\|_1 + \|B\|_1\} \quad (25.124)$$

passes positivity, normalization, and finite-dimensional consistency to the limit. The compact-containment estimate proved in the next lemma supplies a probability measure on the local hypograph state space.

If  $h_{0,n} \rightarrow h_0$ , the Brownian first-hit functionals converge away from the Wiener-null tangential-contact set. The uniform trace estimate of Lemma 8.100 permits compact truncation of the hitting variables uniformly in  $n$ . On the truncated set dominated convergence gives trace-norm convergence, hence (25.124) gives convergence of all multipoint threshold probabilities. Compact containment upgrades this to weak convergence in UC.  $\square$

**Lemma 8.103** (MQR positive-time Hölder estimate and compact containment). *Fix  $0 < \tau < T$  and  $R < \infty$ . For every  $\beta < 1/2$  the rescaled TASEP approximants obey a local Hölder compact-containment estimate, uniformly for initial profiles in compact admissible families:*

$$\lim_{M \rightarrow \infty} \sup_{\varepsilon} \mathbf{P}\left(\|h^\varepsilon(t, \cdot)\|_{C^\beta([-R, R])} > M\right) = 0, \quad t \in [\tau, T]. \quad (25.125)$$

*Together with one-point upper and lower tail control, this yields compact containment in the MQR state space at every positive intermediate time.*

*Proof.* It suffices by scaling to work at one fixed positive time. Let  $H_i = h^\varepsilon(x_i)$  with  $x_1 < x_2$ . For  $p \geq 2$ , Fubini gives the exact CDF identity

$$\mathbf{E}[|H_1 - H_2|^p \mathbf{1}_{H_1 > H_2}] = p(p-1) \iint_{a > b} |a - b|^{p-2} \mathbf{P}(H_1 > a, H_2 \leq b) da db. \quad (25.126)$$

Indeed, integrate  $p(p-1)(a-b)^{p-2}$  first over  $H_2 < b < a < H_1$ . The opposite ordering follows by spatial reflection.

First truncate the height variables to  $[-N, N]$ . One-point compact containment makes the discarded part arbitrarily small uniformly in  $\varepsilon$ : the upper tail follows from the line-profile comparison used in the Fredholm formula, and the lower tail from comparison with narrow-wedge TASEP and its Airy<sub>2</sub> limit.

Inside  $[-N, N]$ , the probability in (25.126) is a difference between a one-point and a two-point Fredholm determinant. By (25.124) it is bounded by the trace norm of the corresponding kernel difference. The path-integral algebra factors that difference into a scattering factor, uniformly bounded in trace norm by Lemma 8.100, and the off-diagonal rescaled geometric kernel

$$\bar{\chi}_{-a_2} \tilde{Q}_{x_2-x_1}^\varepsilon \chi_{-a_1}.$$

Schur's test gives

$$\|\bar{\chi}_{-a_2} \tilde{Q}_{x_2-x_1}^\varepsilon \chi_{-a_1}\|_{\text{op}} \leq \int_{|y| \geq |a_2-a_1|} \tilde{Q}_{x_2-x_1}^\varepsilon(y) dy. \quad (25.127)$$

Insert (25.127) into (25.126), exchange the order of integration, and use the  $(p-1)$ st moment of the centered rescaled geometric walk. Uniformly in the mesh,

$$\mathbf{E}|H_1 - H_2|^p \mathbf{1}_{|H_1|, |H_2| \leq N} \leq C_{N,p} |x_2 - x_1|^{(p-1)/2}. \quad (25.128)$$

Thus (with  $1 + \alpha = (p-1)/2$ )  $\alpha = (p-3)/2$ . Kolmogorov's criterion gives any

$$\beta < \frac{\alpha}{p} = \frac{1}{2} - \frac{3}{2p}.$$

Sending  $p \rightarrow \infty$  gives every  $\beta < 1/2$ . Finally send  $N \rightarrow \infty$  by the one-point compact-containment estimate. The constants are uniform on positive compact time windows by scaling and on compact admissible initial families by the uniform trace bounds. This proves (25.125).  $\square$

**Lemma 8.104** (MQR Chapman–Kolmogorov and Feller passage). *The limiting kernels satisfy*

$$P_{s+t}(h, A) = \int_{\text{UC}} P_s(h, dh') P_t(h', A), \quad (25.129)$$

*and  $P_t$  maps bounded continuous functions to bounded continuous functions. Hence  $(P_t)_{t \geq 0}$  is a Feller Markov semigroup on the MQR state space.*

*Proof.* Let  $F$  be a bounded continuous cylinder function and write the exact prelimit TASEP Chapman–Kolmogorov identity at times  $s$  and  $s+t$ . By Lemma 8.103, for every  $\eta > 0$  there is a compact  $C \subset \text{UC}$  which carries at least  $1 - \eta$  of the intermediate profile law, uniformly in the mesh. The contribution of  $C^c$  is bounded by  $2\|F\|_\infty \eta$ .

On  $C$ , Lemma 8.102 gives uniform convergence of the transition expectations: trace-norm convergence of the scattering kernels is uniform on compact families of initial profiles, and the Fredholm determinant is locally Lipschitz in trace norm. Therefore one may pass  $\varepsilon \downarrow 0$  inside the compact part of the Chapman–Kolmogorov integral. Letting  $\eta \downarrow 0$  yields (25.129).

For the Feller property take  $h_n \rightarrow h$ . Lemma 8.102 already gives the weak convergence  $P_t(h_n, \cdot) \Rightarrow P_t(h, \cdot)$  in the MQR state space. By the definition of weak convergence on a metrizable space, for every bounded continuous  $F$ ,

$$P_t F(h_n) = \int F(g) P_t(h_n, dg) \longrightarrow \int F(g) P_t(h, dg) = P_t F(h).$$

Thus  $P_t F$  is continuous. Constants are preserved because each  $P_t(h, \cdot)$  is a probability measure.  $\square$

**Theorem 8.105** (The KPZ fixed point). *Let the TASEP initial height profiles, after the 1:2:3 centering and scaling of Matetski–Quastel–Remenik, converge in UC to an admissible deterministic profile  $h_0$ , or converge in distribution in UC to an admissible random profile  $h_0$ . In the random*

case the initial profile is assumed independent of the Poisson randomness used to evolve each prelimit Markov process, exactly as in MQR Theorem 3.13. Then for every fixed  $t > 0$  the rescaled TASEP height profile converges in distribution in UC to  $P_t(h_0, \cdot)$ , whose multipoint threshold probabilities are the Brownian-scattering Fredholm determinants above. The kernels  $(P_t)$  form a Feller Markov semigroup on UC; its Markov process is the KPZ fixed point. For every finite collection  $0 < t_1 < \dots < t_m$ , the corresponding rescaled TASEP profiles converge jointly in  $\text{UC}^m$  to the fixed point at those times. No prelimit path-space convergence is asserted.

*Proof.* For right-finite data the one-time statement follows from Lemmas 8.98, 8.100, and 8.101; Lemma 8.99 removes the right-finite restriction. Lemma 8.102 upgrades the Fredholm limits to weak convergence in UC. Lemmas 8.103 and 8.104 prove compact containment, Chapman–Kolmogorov, and the Feller property.

For  $0 < t_1 < \dots < t_m$ , use the exact Markov property of TASEP and write the joint expectation of bounded continuous cylinder functions as an iterated action of prelimit transition operators. Positive-time compact containment permits replacement, one operator at a time, by the limiting operator. Induction on  $m$  gives the joint finite-positive-time law of  $(P_t)$ . This is exactly the transition-theory conclusion and does not claim convergence of the prelimit process as a path-valued random variable.  $\square$

**KPZ fixed-point proof inventory and scope.** The argument contains the exact  $Q$ ,  $R_t$ ,  $\Psi$ , backward boundary-value problem,  $\Phi$ , biorthogonality calculation, polynomial condition, polynomially continued  $\bar{Q}$ , first-hit formula, and right-finite scattering kernel. It also replaces the incorrect graphical finite-propagation shortcut by the source’s cutoff telescoping mechanism, and reconstructs the Appendix-C CDF-moment/Schur/Kolmogorov proof of every spatial Hölder exponent  $\beta < 1/2$  at positive time.

The trace-class owner has likewise been decompressed to the five MQR contour regimes, with the mesh-dependent variables displayed explicitly: the pole-removal case, the real saddle deformation with a uniform  $\pi/4$  descent constant, the logarithmic noncritical contour, the complex-conjugate saddle case, the saddle-value estimate, the rank-one first-hit trace integral, the Doob tail bound, the cubic cutoff increment, and the Hilbert–Schmidt half-kernel treatment of the off-diagonal  $Q$  term. The positive-time Hölder proof is then exactly the Appendix-C CDF-moment / Fredholm-difference / Schur / Kolmogorov argument, and the semigroup passage is the compact-set Markov-kernel lemma of Section 3.5.

The conclusion is deliberately limited to the statement proved by Matetski–Quastel–Remenik [18]: convergence in  $\text{UC}^m$  at finitely many positive times to the Feller Markov KPZ fixed point. No prelimit path-space convergence is asserted.

**Lemma 8.106** (Scaling, stationary Brownian increments, and preservation of maxima). *Let  $\mathfrak{h}(t, x; h_0)$  be the KPZ fixed point constructed in Theorem 8.105. Then, for every  $a > 0$ ,*

$$a \mathfrak{h}(a^{-3}t, a^{-2}x; a^{-1}h_0(a^2 \cdot)) \stackrel{d}{=} \mathfrak{h}(t, x; h_0). \quad (25.142)$$

*If  $B$  is a two-sided Brownian motion with diffusion coefficient 2, then for every  $t > 0$*

$$x \longmapsto \mathfrak{h}(t, x; B) - \mathfrak{h}(t, 0; B) \quad (25.143)$$

*is again a two-sided Brownian motion with diffusion coefficient 2. Finally, for  $f_1, f_2 \in \text{UC}$  there is a basic coupling for which*

$$\mathfrak{h}(t, \cdot; f_1 \vee f_2) = \mathfrak{h}(t, \cdot; f_1) \vee \mathfrak{h}(t, \cdot; f_2), \quad (25.144)$$

*and hence the same equality holds in distribution without specifying the coupling.*

*Proof.* We prove the three assertions from the same TASEP approximation used in SMX-25.06.03.

*Scaling.* Recall the exact rescaling (25.122):

$$h^\varepsilon(t, x) = \varepsilon^{1/2} \left[ h_{2\varepsilon^{-3/2}t}(2\varepsilon^{-1}x) + \varepsilon^{-3/2}t \right].$$

Put  $\varepsilon' = a^2\varepsilon$ . Then  $(\varepsilon')^{1/2} = a\varepsilon^{1/2}$ ,  $(\varepsilon')^{-3/2} = a^{-3}\varepsilon^{-3/2}$ , and  $(\varepsilon')^{-1} = a^{-2}\varepsilon^{-1}$ . Start with a microscopic family whose  $\varepsilon$ -rescaled initial profile converges to  $\tilde{h}_0 = a^{-1}h_0(a^2\cdot)$ . Reading the *same* microscopic profile with mesh  $\varepsilon'$  multiplies heights by  $a$  and contracts the macroscopic spatial coordinate by  $a^{-2}$ , so its new profile converges to

$$a\tilde{h}_0(a^{-2}x) = h_0(x).$$

Moreover the random field obtained by multiplying  $h^\varepsilon(a^{-3}t, a^{-2}x)$  by  $a$  is exactly this  $\varepsilon'$ -rescaled TASEP field at  $(t, x)$ . Both  $\varepsilon$  and  $\varepsilon'$  tend to zero. The uniqueness of the limiting transition law from Theorem 8.105 therefore identifies the two limits and gives (25.142). No scaling property of the limiting Fredholm determinant is being assumed here; it is inherited from the exact change of mesh in the approximating particle system.

*Brownian increments.* Start TASEP in its Bernoulli product equilibrium at density  $1/2$  and anchor the height at the origin. The height increments are then independent symmetric  $\{\pm 1\}$  variables. To verify stationarity rather than cite it, write the particle generator on a cylinder function  $F$  as

$$(LF)(\eta) = \sum_i \eta_i(1 - \eta_{i+1})[F(\eta^{i,i+1}) - F(\eta)]. \quad (25.143a)$$

Under Bernoulli( $1/2$ ) product measure, change variables  $\eta \leftrightarrow \eta^{i,i+1}$  in the gain term and then shift the index in the resulting finite sum (only finitely many summands meet the support of  $F$ ). The gain and loss expectations cancel, so  $\int LF d\nu_{1/2} = 0$ . Thus  $\nu_{1/2}$  is invariant. Under the spatial normalization in (25.122), Donsker's theorem gives a two-sided Brownian motion with variance  $2|x|$ . Applying the joint finite-positive-time convergence of SMX-25.06.03 to this stationary initial law shows that the increment field at time  $t$  has the same Brownian limit. The value at the origin is not stationary and must be subtracted; this is the nontrivial global height shift in (25.143). Thus we assert stationarity of increments, not invariance of the unanchored Brownian height.

*Preservation of maxima.* Use the standard graphical coupling of TASEP: one rate-one Poisson clock at each site, shared by all initial profiles. At a clock ring the height is lowered by 2 precisely when that site is a local maximum. If  $U_z$  denotes this local corner-flip update, an inspection of the four possible adjacent slope pairs shows

$$U_z(f \vee g) = U_z f \vee U_z g \quad (25.145)$$

for admissible height paths. Since every finite-time graphical evolution is a locally finite composition of the maps  $U_z$ , the discrete TASEP flow commutes with pointwise maximum. Couple the three rescaled processes started from  $f_1$ ,  $f_2$ , and  $f_1 \vee f_2$  by the same clocks. The positive-time Hölder/compact-containment estimate of Lemma 8.103 is uniform for this finite family, so a common subsequence converges jointly on every compact spatial window. Pointwise maximum is continuous under locally uniform convergence; the discrete identity therefore survives in the limit. Uniqueness of each fixed-point marginal then yields (25.144) for the canonical fixed-point coupling. The same graphical update plainly commutes with adding a constant, so passage to the limit also gives

$$\mathfrak{h}(t, x; f + c) = \mathfrak{h}(t, x; f) + c, \quad (25.145a)$$

which is the vertical-shift identity used below.  $\square$

**Lemma 8.107** (Narrow wedge, flat, and half-flat Airy specializations). *Let  $d_u$  be narrow-wedge data,  $d_u(u) = 0$  and  $d_u(x) = -\infty$  for  $x \neq u$ , and let*

$$h_{\text{hf}}(x) = \begin{cases} -\infty, & x < 0, \\ 0, & x \geq 0. \end{cases}$$

*In the normalization of this volume, at time  $t = 1$ ,*

$$\mathfrak{h}(1, x; d_u) + (x - u)^2 = \mathcal{A}_2(x), \quad (25.146)$$

$$\mathfrak{h}(1, x; 0) = 2^{1/3} \mathcal{A}_1(2^{-2/3}x), \quad (25.147)$$

*and*

$$\mathfrak{h}(1, x; h_{\text{hf}}) + x^2 \mathbf{1}_{\{x < 0\}} = \mathcal{A}_{2 \rightarrow 1}(x). \quad (25.148)$$

*Here the Airy processes are characterized by their standard extended Fredholm kernels, so the equalities are process-level equalities in the spatial variable  $x$ .*

*Proof.* We reduce the fixed-point Fredholm kernel to the classical extended Airy kernels. The calculation for half-flat data contains the only nontrivial reflection step and also gives the two limiting cases.

Write

$$S_{t,x} = \exp\left(x\partial^2 + \frac{t}{3}\partial^3\right), \quad S_{t,x}(z) = t^{-1/3} e^{2x^3/(3t^2) - zx/t} \text{Ai}(-t^{-1/3}z + t^{-4/3}x^2), \quad (25.149a)$$

for the Airy scattering propagator obtained as the trace-class limit in Lemma 8.101; it satisfies  $S_{s,x}S_{t,y} = S_{s+t,x+y}$  whenever the indicated kernels are defined. Let  $\bar{\chi}_0$  be multiplication by  $\mathbf{1}_{(0,\infty)}$ , and let  $\varrho f(v) = f(-v)$  be reflection. For half-flat data, the negative half-profile has hypograph hitting operator

$$S_{t,0}^{\text{hypo}(h_{\text{hf}}^-)} = \chi_0 S_{t,0}, \quad (25.149)$$

whereas reflection of Brownian motion at its first hit of zero gives, for  $v > 0$ ,

$$S_{t,0}^{\text{hypo}(h_{\text{hf}}^+)}(v, u) = S_{t,0}(-v, u). \quad (25.150)$$

Substitution into the two-sided Brownian-scattering formula therefore yields

$$K_t^{\text{hypo}(h_{\text{hf}})} = (S_{t,0})^*(I + \varrho)\bar{\chi}_0 S_{t,0}. \quad (25.151)$$

For observation points  $x_i$ , the associated extended kernel is

$$-e^{(x_j - x_i)\partial^2} \mathbf{1}_{\{x_i < x_j\}} + (S_{t,-x_i})^*(I + \varrho)\bar{\chi}_0 S_{t,x_j}. \quad (25.152)$$

At  $t = 1$ , after the threshold conjugation  $u_i \mapsto u_i - x_i^2 \mathbf{1}_{\{x_i < 0\}}$ , this is the standard extended  $\text{Airy}_{2 \rightarrow 1}$  kernel. Conjugation by a nonvanishing multiplication operator does not alter its Fredholm determinant, so (25.148) follows.

For completeness we also make the two endpoint initial data direct, rather than infer them from asymptotics of the half-flat process. By spatial translation it is enough to take the wedge at  $u = 0$ . In that case the one-sided first-hit term in the fixed-point kernel collapses at the wedge vertex, and the same Brownian-scattering formula gives

$$K_{1,\text{ext}}^{d_0}(x_i, \cdot; x_j, \cdot) = -e^{(x_j - x_i)\partial^2} \mathbf{1}_{\{x_i < x_j\}} + (S_{1,-x_i})^* \bar{\chi}_0 S_{1,x_j}. \quad (25.152a)$$



Since  $S_{1,0}(z) = \text{Ai}(-z)$ , conjugating by the parabolic threshold shift turns the second term into the usual Airy projection  $K_{\text{Ai}}(r, s) = \int_0^\infty \text{Ai}(r + \lambda) \text{Ai}(s + \lambda) d\lambda$ , with the heat-propagator term providing the standard extended-kernel continuation between distinct observation points. This is the extended Airy<sub>2</sub> kernel and proves (25.146).

For flat data  $h_0 \equiv 0$ , apply the reflection principle to the two half-lines before recombining the two-sided scattering formula. The identity pieces cancel against the free off-diagonal propagation, leaving

$$K_{1,\text{ext}}^0(x_i, \cdot; x_j, \cdot) = -e^{(x_j - x_i)\partial^2} \mathbf{1}_{\{x_i < x_j\}} + (S_{1,-x_i})^* \varrho S_{1,x_j}. \quad (25.152b)$$

The convolution identity for  $S_{1,0}$  gives the Airy<sub>1</sub> reflection kernel; a change of variables  $x = 2^{2/3}\xi$  and height scaling by  $2^{-1/3}$  turns (25.152b) into the standard extended Airy<sub>1</sub> kernel. Hence  $\mathfrak{h}(1, x; 0) = 2^{1/3}\mathcal{A}_1(2^{-2/3}x)$ , proving (25.147). These direct reductions are consistent with the two spatial ends of (25.152). All operator identities are taken after the same threshold projections as in Lemma 8.102; the cubic Airy bounds of Lemma 8.100 make the displayed products trace class and justify the Fredholm determinant manipulations.  $\square$

**Lemma 8.108** (Airy-sheet subsequential construction and max-plus variational formula). *Let  $\mathcal{A}^\varepsilon$  be the canonical 1:2:3 rescaled TASEP Airy-sheet approximation obtained by evolving, with one common graphical Poisson environment, the packed/narrow-wedge initial profiles based at every starting point. Then the laws of  $\mathcal{A}^\varepsilon$  are tight in  $C_{\text{loc}}(\mathbb{R}^2)$ . Every subsequential limit  $\mathcal{A}$  is a continuous Airy sheet in the following source-level sense:*

- (i) *for each fixed value of either coordinate, the other-coordinate process has the Airy<sub>2</sub> law in the normalization of (25.146);*
- (ii) *for every  $t > 0$  and admissible  $h_0 \in \text{UC}$ ,*

$$\mathfrak{h}(t, x; h_0) \stackrel{d}{=} \sup_{y \in \mathbb{R}} \left\{ t^{1/3} \mathcal{A}(t^{-2/3}x, t^{-2/3}y) - \frac{(x - y)^2}{t} + h_0(y) \right\} \quad (25.153)$$

*as a process in  $x$ ;*

- (iii) *the law in (25.153) is independent of which subsequential Airy-sheet limit is chosen, although MQR's Fredholm formula by itself does not determine the full joint law of  $\mathcal{A}$ ;*
- (iv) *for fixed  $t_1, t_2 > 0$  with  $t = t_1 + t_2$ , let  $\widehat{\mathcal{A}}^1, \widehat{\mathcal{A}}^2$  be independent Airy sheets with the parabolic term included as in MQR, and let  $\widehat{\mathcal{A}}$  denote any copy of the same Airy-sheet law. Then the max-plus composition obeys*

$$\sup_{z \in \mathbb{R}} \{ t_1^{1/3} \widehat{\mathcal{A}}^1(t_1^{-2/3}x, t_1^{-2/3}z) + t_2^{1/3} \widehat{\mathcal{A}}^2(t_2^{-2/3}z, t_2^{-2/3}y) \} \stackrel{d}{=} t^{1/3} \widehat{\mathcal{A}}(t^{-2/3}x, t^{-2/3}y). \quad (25.153s)$$

*This is a fixed-time semigroup identity for the Airy sheet; it is not a claim of a jointly coupled process-level identity as  $(t_1, t_2)$  vary.*

*Here our coordinate convention is the transpose of MQR's  $\mathcal{A}_{\text{MQR}}(x, y) = \mathfrak{h}(1, y; d_x) + (x - y)^2$ ; their permutation symmetry shows that this does not change either one-coordinate Airy<sub>2</sub> laws or the fixed-time variational statement.*

*Proof. Step 1: the microscopic two-parameter coupling.* Use the same family of rate-one Poisson clocks for TASEP started from every packed/narrow-wedge profile. For lattice-compatible starting and observation points define

$$\mathcal{A}^\varepsilon(x, y) = \mathfrak{h}^\varepsilon(1, x; d_y) + (x - y)^2 \quad (25.153a)$$

with the centering and interpolation of SMX-25.06.03. At the discrete level, reflection of the last-passage/TASEP picture interchanges the two spatial coordinates. Thus the positive-time spatial Hölder estimate already proved for the rescaled height profile applies uniformly to each coordinate of  $\mathcal{A}^\varepsilon$ .

Fix a compact square  $K = [-R, R]^2$  and choose any  $\beta < 1/2$ . The Appendix-C estimate used in Lemma 8.103, first in the observation coordinate and then, by the discrete permutation symmetry, in the starting coordinate, gives for every sufficiently large  $p$

$$\sup_\varepsilon \mathbf{E} \|\mathcal{A}^\varepsilon\|_{C^\beta(K)}^p < \infty. \quad (25.154)$$

Indeed the increment between  $(x_1, y_1)$  and  $(x_2, y_2)$  is bounded by the sum of an  $x$ -increment at fixed  $y_1$  and a  $y$ -increment at fixed  $x_2$ . Together with the one-point tightness from SMX-25.06.03, Kolmogorov–Chentsov and Arzelà–Ascoli imply tightness on  $K$ . A diagonal argument over  $R \in \mathbb{N}$  gives tightness in  $C_{\text{loc}}(\mathbb{R}^2)$  and a subsequence converging to a continuous field  $\mathcal{A}$ .

*Step 2: identify the slices.* For fixed  $y$ , SMX-25.06.03 with narrow-wedge initial data gives

$$x \mapsto \mathcal{A}(x, y)$$

the Airy<sub>2</sub> law. The discrete permutation symmetry passes through the joint  $C_{\text{loc}}$  convergence, so the same is true with the coordinates interchanged. This is the slice statement in MQR Example 4.17. Notice that we have proved existence of a continuous subsequential sheet, not uniqueness of its full two-parameter law.

*Step 3: finitely supported initial data.* For

$$h_0^{(n)} = \bigvee_{i=1}^n (c_i + d_{y_i}),$$

the graphical TASEP coupling gives the exact prelimit maximum identity. Passing to the joint subsequential limit and using the vertical-shift and 1:2:3 scaling symmetries therefore yields

$$\mathfrak{h}(t, x; h_0^{(n)}) = \max_{1 \leq i \leq n} \left\{ c_i + t^{1/3} \mathcal{A}(t^{-2/3}x, t^{-2/3}y_i) - \frac{(x - y_i)^2}{t} \right\} \quad (25.155)$$

in distribution as a process in  $x$ . This is precisely the repeated max-preservation step used by MQR.

*Step 4: arbitrary upper-semicontinuous initial data.* Choose finite-support hypograph approximations  $h_0^{(n)} \rightarrow h_0$  in the MQR state-space topology, first on growing compact intervals and then globally. The propagation estimate and trace-norm continuity in Lemma 8.102 imply

$$\mathfrak{h}(t, \cdot; h_0^{(n)}) \Longrightarrow \mathfrak{h}(t, \cdot; h_0)$$

in UC on every positive time. On the right-hand side of (25.155), continuity of  $\mathcal{A}$  on compacts and the same linear-growth barrier used in the MQR tightness argument allow the finite maximum to increase to the supremum in (25.153); points outside a sufficiently large compact interval cannot contribute with appreciable probability. Passing  $n \rightarrow \infty$  proves (25.153).

*Step 5: the fixed-time Airy-sheet semigroup law.* Take two independent graphical TASEP environments, evolve narrow-wedge data for the two macroscopic time increments  $t_1$  and  $t_2$ , and use the exact prelimit Markov/max-plus composition over the intermediate lattice location. The two Airy-sheet approximations are independent. Joint tightness from Step 1 and the same compact-tail barrier used in Step 4 allow the lattice maximum to pass to the continuum supremum. The one-step evolution over  $t_1 + t_2$  converges to the Airy-sheet approximation at total time  $t$ . Therefore the resulting continuous fields satisfy (25.153s) in distribution. This is precisely the “in particular” semigroup assertion (4.19) following MQR Theorem 4.18. Because the argument is made for a fixed pair  $(t_1, t_2)$ , it does not construct a common Airy-sheet process simultaneously for all time splittings.

The left-hand side of (25.153) is the uniquely defined KPZ fixed point. Hence every subsequential Airy sheet produced in Step 1 gives the same variational law, even though this argument does not identify the full two-parameter law of the sheet. This is exactly the distinction emphasized by MQR.  $\square$

**Remark 8.109. Source and scope.** The published source is Matetski–Quastel–Remenik, *The KPZ fixed point*, Acta Math. 227 (2021), 115–203. The Airy sheet is Example 4.17; the fixed-time variational formula and Airy-sheet semigroup law are Theorem 4.18, equations (4.18)–(4.19); and the canonical TASEP coupling plus  $C(\mathbb{R}^2)$  tightness construction is Remark 4.21. MQR Remark 4.19 explicitly warns that these identities hold for fixed times, not as a process in the time parameter, and their Fredholm formula does not determine the full joint distribution of the Airy sheet. Thus this owner asserts a continuous subsequential Airy sheet and the source’s fixed-time variational/semigroup laws, but not uniqueness from the Fredholm formula and not a four-parameter directed landscape.

**Theorem 8.110** (Scaling, variational structure, and Airy processes). *Let  $\mathfrak{h}(t, x; h_0)$  be the KPZ fixed point on the upper-semicontinuous state space of SMX-25.06.03. It obeys the exact 1:2:3 scaling law (25.142), the Brownian-increment stationarity (25.143), and preservation of maxima (25.144). It admits the fixed-time Airy-sheet variational representation (25.153) and, for each fixed time splitting  $t = t_1 + t_2$ , the Airy-sheet max-plus semigroup law (25.153s). At time  $t = 1$  the narrow-wedge, flat, and half-flat specializations are respectively (25.146), (25.147), and (25.148). The semigroup assertion is fixed-time in the precise sense of MQR Remark 4.19; no joint Airy-sheet process in the time-splitting parameter is claimed.*

*Proof.* The scaling, stationary-increment, and max-preservation statements are Lemma 8.106. The three classical Airy specializations are Lemma 8.107, and the Airy-sheet construction, variational formula, and fixed-time max-plus semigroup law are Lemma 8.108. These owners use only the already-established MQR transition theory SMX-25.06.03 and earlier probability/Fredholm infrastructure. In particular, no stationary Baik–Rains asymptotic is being hidden in the proof: two-sided Brownian data are used here only for the exact invariant-increments statement (25.143).  $\square$

**Remark 8.111** (Stationary one-point law versus Brownian stationarity). The invariant-increments assertion above is exactly what follows from the MQR symmetry theorem. Identifying the one-point law after the random global height shift with the classical stationary/Baik–Rains distribution requires the separate stationary-TASEP asymptotic analysis. That identification is therefore not claimed as a local proof consequence of SMX-25.06.04.

# Reader's proof-status guide and review problems

This series uses four kinds of mathematical support. A standard *Proof* is intended to establish the result from earlier material in the series. A *Reduction* checks that an earlier theorem applies with the same hypotheses, normalization, and topology. A *Proof sketch* explains the mechanism but does not replace the detailed estimates in the cited source. An *Imported theorem* is stated at the strength used here and is supported by a precise primary reference. Frontier statements are not proved inputs. When a proof-status label and the displayed argument disagree, the weaker status governs.

**Exercise 8.112** (Hypothesis check). Choose a major theorem from this volume. List every hypothesis, including dimension, boundary condition, parameter range, normalization, and topology of convergence. Identify where each hypothesis is used in the proof or reduction.

**Exercise 8.113** (Normalization check). Select one scaling-limit statement and derive the effect of multiplying the discrete observable, time coordinate, or metric by a deterministic constant. Explain why the normalization used in the theorem removes that ambiguity.

**Exercise 8.114** (Dependency trace). Trace one theorem backward to definitions or previously proved owners. For each step, record whether it is a direct proof, an internal reduction, a proof sketch, or an imported result.

**Exercise 8.115** (Toy model). Construct the simplest finite or Gaussian example in which the main mechanism of this volume can be computed explicitly. State which feature survives in the full theorem and which feature is an artifact of the toy model.

**Exercise 8.116** (Frontier discipline). Choose one frontier statement. Identify the strongest theorem actually proved in the series, then state the additional estimate or compactness, identification, or universality input that the frontier would require.

# Selected primary sources and further reading

The list below records primary entry points for the principal research-level results used across the series. It is deliberately selective; theorem-local citations govern whenever a statement is marked as imported.

# Bibliography

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